# From First Courses to the Modular Degree

A modular textbook on the arithmetic behind
"Local factors of the adjoint $L$-function and the $\ell$-adic valuations of the modular degree"

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# Preface

This book exists for one paper: the research note

> D. Soni, *Local factors of the adjoint L-function and the $\ell$-adic valuations of the modular degree*,

a short text that uses, without pausing, elliptic curves over local fields, Galois representations and their adjoints, modular forms and Hecke algebras, Galois cohomology and Bloch-Kato Selmer groups, the commutative algebra of congruence ideals, and large computations on Cremona's tables. The book takes a reader who has none of this, but who has the background listed below, to the point where every definition and every proof of the paper, and every table that its source prints, can be read with understanding. The paper's data tables are printed in it, and the data files and scripts behind them are in the repository that its appendix names. It also serves a second reader: the specialist who has seen all of this, has forgotten the exact form of one statement, and opens one chapter to find it at the top. Throughout, "the paper" means this note, and its results are named by the labels of its source.

## What the paper proves

Let $E$ be an elliptic curve over $\mathbb{Q}$ of conductor $N$. By the modularity theorem there is a newform $f = f_E$ of weight 2 on $\Gamma_0(N)$ with the same $L$-function as $E$, and a nonconstant map of curves $\varphi_E\colon X_0(N) \to E$ from the modular curve $X_0(N)$ to $E$. Among the curves isogenous to $E$ there is one, the $X_0(N)$-optimal curve, for which the degree of this map is smallest; the paper always takes $E$ to be that curve and calls $\deg \varphi_E$ the modular degree. The modular degree is an integer that can be computed, curve by curve, and the tables list it; what is not understood is its prime factorisation.

The paper proves that the $\ell$-adic valuation $v_\ell(\deg \varphi_E)$, for an odd prime $\ell$, is determined by a Selmer group and by invariants of $E$ at the primes of bad reduction, and is bounded below by the local invariants alone. Two local invariants enter.

- At a bad prime $q$, the adjoint $L$-function $L(\mathrm{ad}^0 f, s)$ has a local Euler factor $L_q(A_f, s)$, defined through the inertia invariants of the trace-zero adjoint $A_f = \mathrm{ad}^0 V_\ell E$. The naive adjoint $L$-function, the one built from the coefficients $a_n(f)$ alone, has the same factor when $E$ has good or multiplicative reduction at $q$ and the trivial factor 1 when the reduction is additive. The first invariant is

$$e_\ell(E, q) = v_\ell\Big(\frac{L_q(A_f, 1)^{-1}}{L_q^{\mathrm{nv}}(A_f, 1)^{-1}}\Big),$$

the $\ell$-adic valuation of the Euler factor at $s = 1$ that the naive $L$-function omits. It is zero at multiplicative primes. At an additive prime $q \geq 5$ it is read off the Kodaira type: $v_\ell(q - \chi_{-3}(q))$ for the types II, II*, IV, IV*; $v_\ell(q - \chi_{-4}(q))$ for III, III*; $v_\ell(q^2 - 1)$ for $\mathrm{I}_n^*$; and $v_\ell(q-1) + v_\ell((q+1)^2 - a_q(E')^2)$ for $\mathrm{I}_0^*$, where $E'$ is the quadratic twist of $E$ that has good reduction at $q$. Here $\chi_{-3}$ and $\chi_{-4}$ are the quadratic characters of $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(i)$. At $q = 2$ and $q = 3$ the factor is not determined by the Kodaira type alone; the paper shows that it has one of four shapes at every bad prime ([paper: Proposition prop:shapes]).

- At a prime $q$ where $E$ has potentially multiplicative reduction, that is, where $v_q(j_E) < 0$, the second invariant is $t_\ell(E, q) = v_\ell(-v_q(j_E))$, the $\ell$-part of the valuation of the Tate parameter; it is 0 at every other prime.

The main theorem ([paper: Theorem thm:main]) says: if $E$ is $X_0(N)$-optimal, $\ell$ is an odd prime not dividing $2N$, and the mod $\ell$ Galois representation $\bar{\rho}_{E,\ell}$ on $E[\ell]$ is absolutely irreducible when restricted to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, where $\ell^* = (-1)^{(\ell-1)/2}\ell$, then

$$v_\ell(\deg \varphi_E) \;=\; \mathrm{length}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell) + \sum_{q|N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) + \sum_{q|N} e_\ell(E, q).$$

Here $T = T_\ell E$ is the Tate module, $H^1_f$ is the Bloch-Kato Selmer group of the adjoint, and $\mathrm{Tam}_q$ is the Tamagawa factor of Fontaine and Perrin-Riou. The hypothesis says that $\ell$ is not in the exceptional set $S_f$ of Diamond, Flach and Guo; for $\ell \geq 5$ it only asks that $E[\ell]$ be irreducible ([paper: Proposition prop:Sf]). The Tamagawa factors are trivial at the potentially good primes ([paper: Lemma lem:tamadd]) and explicit at the potentially multiplicative ones ([paper: Lemma lem:tamfactor]): there their valuation is $v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$, where $n = -v_q(j_E)$ and $v_\ell(c(u))$ is the largest $k \leq v_\ell(q-1)$ such that the reduction of the unit part $u$ of the Tate parameter is an $\ell^k$-th power in $\mathbb{F}_q^\times$; in particular it is at least $t_\ell(E, q)$. Since the Selmer length is nonnegative, the identity gives

$$v_\ell(\deg \varphi_E) \;\geq\; \sum_{q|N} e_\ell(E, q) + \sum_{q|N} t_\ell(E, q).$$

*The assembly.* The proof of the main theorem is not a new argument but a reading, for elliptic curves, of theorems already in the literature, and the book is organised around those theorems. They are: the theorem of Diamond, Flach and Guo identifying the length of the $\Sigma$-imprimitive adjoint Selmer group with the valuation of a congruence ideal (their Theorem 3.7); their formula for how that ideal changes with $\Sigma$ (their Proposition 1.4); their exact sequence comparing the $\Sigma$-imprimitive and the Bloch-Kato Selmer groups (their Lemma 2.1), whose local terms are, by the formula of Fontaine and Perrin-Riou, the true Euler factors times the Tamagawa factors; and the observation that for the optimal curve the congruence ideal at level $N$ is generated by $\deg \varphi_E$, by the projection formula for $\varphi_E$ ([paper: Proposition prop:pullback]). The naive Euler factors at the multiplicative primes cancel, the true factors at the additive primes are the omitted ones, and the identity appears. One input of the assembly is not a published statement in the form needed: the identification of the integral structure of Diamond, Flach and Guo with the $\ell$-adic cohomology of a modular curve $X_0(M)$. The paper proves it, for every prime $\ell$ and with no localisation ([paper: Proposition prop:integral]), and Chapter 20 gives the proof in full.

Two further statements are proved by other means. The first, proved by level lowering (Ribet, Carayol, Diamond) and quadratic twisting, says that $\ell$ divides the modular degree as soon as $E$ has a prime of type IV or IV* ($q \geq 5$ or $q = 2$) or of type II or II* ($q \geq 5$), for $\ell = 3$, or an

odd prime of type $\mathrm{I}_n^*$ with $\ell \mid n$ ([paper: Theorem thm:qual]). The second, proved by a twisting argument and the level-raising formula of Diamond, Flach and Guo, is a quantitative version for the types $\mathrm{I}_0^*$ and $\mathrm{I}_n^*$ at the odd primes ([paper: Theorem thm:quant]); its inequality needs only the irreducibility of $E[\ell]$, so it holds also for the curves at the boundary described next, and it is an equality when $\ell \notin S_f$.

*The boundary.* The hypothesis on $\mathbb{Q}(\sqrt{\ell^*})$ is the hypothesis of Diamond, Flach and Guo, and the paper shows that it is sharp. For $\ell \geq 5$ it follows from the irreducibility of $E[\ell]$. For $\ell = 3$, with $E[3]$ irreducible, it fails exactly when the image of $\bar{\rho}_{E,3}$ is the normaliser of a split Cartan subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$ (the image labelled 3Ns in the tables); then $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$, and a correction term $h_3(E)$, the length of $H^0(\mathbb{Q}, \mathrm{ad}^0\, T(1) \otimes \mathbb{Q}_3/\mathbb{Z}_3)$, appears with the value 1 ([paper: Proposition prop:ns]). On the tables these curves violate the inequality, always by exactly one, in 31%, 23% and 18% of the cases on three ranges of conductors. The paper states a conjecture ([paper: Conjecture conj:euler]) with this correction and with a weight at the prime 3 itself when $3 \mid N$, valid also when $\ell$ divides $N$, where no theorem applies; the correction is the term that Lemma 2.1 of Diamond, Flach and Guo produces, provided a Selmer group of $V^D$ vanishes, which is assumed and not known for these curves. A second conjecture concerns the Eisenstein case, when $E$ has a rational $\ell$-isogeny ([paper: Conjecture conj:eis]); the paper proves that its bound at $\ell = 3$ follows from its first bound ([paper: Proposition prop:no9]).

*The computations.* Everything was tested on Cremona's tables of elliptic curves of conductor at most 500000, with a protocol fixed in advance: a working set ($N \leq 300000$) on which the statements were found, a hold-out ($300000 < N \leq 400000$) on which they were tested once, an extension ($400000 < N \leq 500000$), and 893 curves of conductor between 500000 and four million, generated and computed independently. The Euler factors at every bad prime, including the wild primes 2 and 3, are taken from PARI. At $\ell = 3, 5, 7, 11, 13$, for the curves without complex multiplication with $E[\ell]$ irreducible that the tables determine to be optimal, the conjecture holds with no exception; at $\ell = 3$ the bound is attained for 57% of the curves with a positive bound; and a calibration argument shows that the absence of violations is far from what independence of the two sides would predict. That shows that the statement is strong, not that it is true; Chapter 22 explains the difference and which statements were tested blind.

In several places the book proves a statement in greater generality than the paper needs; where it does, the chapter says so, and the paper's statement is recovered as a special case.

## Who the book is for, and the two ways to read it

The reader we have in mind has taken a first course in real analysis (sequences, series, continuity, differentiation, the Riemann integral, uniform convergence, metric spaces at an elementary level), a first course in linear algebra (vector spaces over $\mathbb{R}$ and $\mathbb{C}$, matrices, determinants, eigenvalues, diagonalisation, inner products, the Jordan form as a statement), an introduction to complex analysis (holomorphic functions, Cauchy's theorem, residues, Laurent series, Mobius transformations), an elementary course in statistics (means, variances, distributions, hypothesis tests), and has worked through Riley, Hobson and Bence, *Mathematical Methods for Physics and Engineering*: Fourier series and transforms, differential equations, special functions, tensors, finite groups and their representations, probability. Appendix A restates exactly what we assume from these courses, as a list of precise statements with references, and says where each is first used.

The reader has not seen abstract algebra beyond Riley, Hobson and Bence, nor Galois theory,

algebraic number theory, topology, algebraic geometry, homological algebra, modular forms, $p$-adic numbers or Galois cohomology. Every one of these is built from the beginning, with full proofs where the proofs are short enough to teach something, and with precise statements and references where they are not.

Two readings must both work, and the book is built so that they do.

*Linear mode.* The chapters are in dependency order. A reader with the background above can start at Chapter 1 and read to Chapter 22, in order, skipping nothing. This is a long road, and the paper is at the end of it.

*Refresher mode.* Every chapter opens with a card, *Core results at a glance*: a numbered list of the definitions, theorems, lemmas and constructions of the chapter, each stated precisely (not summarised), each tagged with its level of rigour, and each pointing to the place in the chapter where it is proved or discussed. A specialist who has forgotten the exact normalisation of the Fontaine-Perrin-Riou Tamagawa factor, or the exact hypotheses of the Atkin-Li twisting theorem, reads the card and stops. The cards together form a second, short book inside the first; the reading guide at the end of this preface says which card to open for which topic.

## How the book is organised

The book has six Parts and two appendices. One chapter is one topic, readable on its own given the cards of the chapters before it.

*Part I, Algebra.* Chapter 1 builds group theory from the axioms: cosets, quotients, finite abelian groups, inverse limits and profinite groups, and the matrix groups $\mathrm{GL}_2(\mathbb{F}_q)$, with $\mathrm{GL}_2(\mathbb{F}_3)$ in full detail: the normalisers of the split and nonsplit Cartan subgroups and their orbits on the four lines of $\mathbb{F}_3^2$ are the key to the boundary of the paper's hypothesis. Chapter 2 does rings, polynomials, finite fields, cyclotomic polynomials, quadratic reciprocity, the characters $\chi_{-3}$, $\chi_{-4}$, $\chi_{q^*}$ and $\chi_S$, and $\ell$-adic valuations. Chapter 3 is the commutative algebra the paper leans on: modules, exact sequences, tensor products, discrete valuation rings and $\mathbb{Z}_\ell$, lengths, Fitting ideals, Pontryagin duality, perfect pairings and Gram determinants, finite free algebras with their idempotents, congruence ideals, Gorenstein rings. Chapter 4 is the representation theory of finite groups over any field, including characteristic $\ell$, and over $\mathbb{Z}_\ell$: Schur, Maschke and its failure, Brauer-Nesbitt, induction and Clifford theory, the adjoint representation and its trace-zero part, $\mathrm{GL}_2(\mathbb{F}_3)$ on $\mathbb{F}_3^2$ and $\mathbb{F}_9^2$, lattices, and Dickson's list.

*Part II, Number fields and local fields.* Chapter 5 is Galois theory, finite and infinite, with Kummer theory, Chebotarev's theorem and the absolute Galois group $G_{\mathbb{Q}}$. Chapter 6 is algebraic number theory: rings of integers, Dedekind domains, the splitting of primes, decomposition and inertia groups, Frobenius elements, the quadratic and cyclotomic fields, the rings $\mathbb{Z}[\zeta_3]$ and $\mathbb{Z}_\ell[\zeta_3]$, class field theory over $\mathbb{Q}$, and the coefficient fields of eigenforms. Chapter 7 is the theory of $\mathbb{Q}_p$ and its extensions: Hensel's lemma, the structure of $\mathbb{Q}_p^\times$, unramified and tamely ramified extensions, the tame character, local Kummer theory, the Weil group, Weil-Deligne representations, local $L$-factors and conductors, and the local shapes that occur at a bad prime with their adjoints.

*Part III, Curves and elliptic curves.* Chapter 8 gives the language of algebraic curves and compact Riemann surfaces: divisors, Riemann-Roch, Riemann-Hurwitz, holomorphic differentials, the cohomology $H^1$ with its cup product, and correspondences with their adjoints. Chapter 9 is elliptic curves over fields: Weierstrass equations, the group law, isogenies, division polynomials, complex multiplication, curves over $\mathbb{C}$ and over finite fields, quadratic twists, the $L$-function, and

the Cremona labels. Chapter 10 is elliptic curves over $\mathbb{Q}_p$: minimal models, reduction types, Neron models and component groups, the Kodaira symbols and Tate's algorithm, the conductor and Kraus's semistability defect, the Tate curve, the effect of a quadratic twist on the reduction type (at $q = 3$ included), the wild primes 2 and 3, and how PARI and Cremona's tables record all this. Chapter 11 is the Galois representation $\rho_{E,\ell}$ and its reduction: the local behaviour at every kind of prime, the representation at $\ell$, the Serre conductor, the possible images, the Eisenstein case, and the paper's Propositions prop:ns and prop:Sf and Lemma lem:h0.

*Part IV, Modular forms and modular curves.* Chapter 12 is modular forms and the curves $X_0(N)$ as Riemann surfaces, with the Petersson product, Sturm's bound and the Atkin-Lehner involutions. Chapter 13 is Hecke operators, newforms, oldforms and their $U_p$-eigenvalues, depleted forms, the perfect pairing between integral cusp forms and the Hecke algebra, the Atkin-Li twisting theorem and the transport of congruences along a twist. Chapter 14 is the arithmetic of $X_0(N)$ and $J_0(N)$: the modularity theorem, the modular parametrisation and the modular degree, the optimal curve, the Manin constant and the formula of Zagier and Watkins, the paper's Proposition prop:pullback (the congruence ideal of Diamond, Flach and Guo at level $N$ and the modular degree), the congruence number and the theorem of Agashe, Ribet and Stein, and the Eisenstein ideal. Chapter 15 is the Galois representation of a newform, local-global compatibility, Serre's conjecture and the optimal-level theorem, level lowering and level raising, the paper's Lemmas lem:crit and lem:old, and the proof of the paper's Theorem thm:qual.

*Part V, Cohomology, Selmer groups and L-functions.* Chapter 16 is homological algebra and group cohomology, with the cohomology of a procyclic group and the model computation behind the paper's Lemma lem:tamfactor. Chapter 17 is Galois cohomology, local and global: Kummer theory, Tate local duality, unramified cohomology, the Kummer class of the Tate parameter, Selmer structures, Poitou-Tate and the Greenberg-Wiles formula. Chapter 18 is the Bloch-Kato Selmer group, the Fontaine-Perrin-Riou Tamagawa factor at every bad prime (the paper's Lemmas lem:fpr, lem:tamfactor and lem:tamadd and Corollary cor:explicit), and the Selmer half of the main theorem. Chapter 19 is $L$-functions: Euler products, the local factors of the adjoint and of the symmetric square, the naive versus the true adjoint $L$-function, the Rankin-Selberg method and the Petersson norm, the paper's Lemma lem:euler and its Proposition prop:shapes at every bad prime, and what PARI computes.

*Part VI, Congruences, the modular degree and the paper.* Chapter 20 is the theory of congruence ideals: Hecke algebras and their localisations, Wiles's numerical criterion and the paper's Lemma lem:ddt417, the theorems of Diamond, Flach and Guo, the comparison of their integral structure with the cohomology of the curves $X_0(M)$ (the paper's Proposition prop:integral), the Taylor-Wiles method in outline, the assembly of the paper's main theorem as an identity, the twist monotonicity theorem, and the proof of the paper's Theorem thm:quant with Step 1 as an equality. Chapter 21 is a guided reading of the paper, section by section in the paper's order: each result stated, each ingredient located in the book and each compressed step expanded; then a map of the paper (dependencies, hypotheses, imported statements) and worked verifications on curves. Chapter 22 is PARI/GP from zero, Cremona's tables and the `ecdata` files, the paper's function `rule(E, l, iii)` explained line by line, the paper's protocol and its statistics with the calibration argument, and the reproduction of the paper's tables on the curves of conductor below 10,000.

*Appendices.* Appendix A is the list of what we assume from the reader's first courses. Appendix B states the book-wide conventions and gives the index of every symbol of the paper and of the book, with the chapter where it is defined, and of the letters with several meanings.

## The dependency graph

The table lists, for each chapter, the earlier chapters that it uses directly; it also uses, through them, whatever they use. Everything else a chapter needs is proved inside it or is in Appendix A. No chapter uses a later one in a proof: where a chapter mentions a later result, in a remark or an example, it says so.

| Chapter | uses directly |
|---|---|
| 1. Groups and matrix groups over finite fields | nothing |
| 2. Rings, fields, polynomials and characters | 1 |
| 3. Modules and commutative algebra | 1, 2 |
| 4. Representations of finite groups | 1, 2, 3 |
| 5. Galois theory | 1 to 4 |
| 6. Algebraic number theory | 1 to 5 |
| 7. Local fields | 1 to 6 |
| 8. Curves and Riemann surfaces | 1 to 6, A |
| 9. Elliptic curves over fields | 1, 2, 4, 5, 6, 8 |
| 10. Reduction, Kodaira types, Tate curves | 7, 9 |
| 11. Galois representations of elliptic curves | 1, 4 to 7, 9, 10 |
| 12. Modular forms and modular curves | 1 to 4, 8, 9 |
| 13. Hecke operators, newforms and twists | 2, 3, 6, 12 |
| 14. Modular curves over $\mathbb{Q}$, the modular degree | 3, 8, 9, 12, 13 |
| 15. Galois representations of modular forms | 3, 4, 7, 11, 13, 14 |
| 16. Group cohomology | 1, 3, 7, 11 |
| 17. Galois cohomology, Selmer groups | 3, 5 to 7, 9 to 11, 16 |
| 18. Bloch-Kato Selmer groups, Tamagawa factors | 3, 7, 11, 16, 17 |
| 19. $L$-functions | 7, 10, 11, 13, 14, 18 |
| 20. Congruence ideals, Diamond-Flach-Guo | 3, 13 to 15, 18, 19 |
| 21. Reading the paper | everything before it |
| 22. Computation, data and protocol | 9 to 11, 14, 19, 21, A |

In words: there are three algebraic strands (groups to representations to Galois representations; rings to modules to cohomology to Galois cohomology to Bloch-Kato; rings to Galois theory to number fields to local fields to reduction), one geometric strand (curves to elliptic curves to reduction), and one modular strand (modular forms to Hecke operators to the modular degree to the Galois representations of newforms). They meet in Chapter 19 and Chapter 20, and everything meets in Chapter 21.

## How to read a chapter

Every chapter has the same shape.

1. *Core results at a glance.* The card: the numbered definitions, theorems, lemmas, propositions and constructions of the chapter, stated in full, each tagged with its level of rigour (see below), and each pointing to the numbered environment in the chapter where it lives. The card is complete: every numbered theorem, proposition, lemma, corollary and definition of the chapter is on it.

2. *Where the paper uses this chapter.* A list of the paper's lemmas, theorems, remarks, definitions and sections that depend on the chapter, and which results of the chapter they

use. The paper's results are named by the labels of the paper's own source, so that reader and paper agree on names: the paper's main theorem is [paper: Theorem thm:main], its computation of the Tamagawa factor is [paper: Lemma lem:tamfactor], its description of the boundary is [paper: Proposition prop:ns], and so on.

3. *The numbered sections.* Each section opens with one paragraph saying what it is for and where in the paper it is used. Inside the text, a box headed *In the paper* marks the exact place where a result of the chapter enters the paper, so that a reader going line by line through the paper can find the way back.

4. *Exercises.* Between eight and twenty, graded from routine to hard, with answers or hints. Some of them read "check the paper's claim that ..."; these are the ones to do.

5. *Notes and references.* Where to read more, which sources the chapter followed, which theorem numbers it cited, and which of them were checked against the source.

Chapters refer to one another only by chapter number, never by the number of a theorem in another chapter, so that each chapter stands on its own. The running examples are the curves of the paper, in Cremona's labels: `11a1`, `20a1`, `121a1`, `121c`, `121d`, `539c1`, `980i1`, `726a1` and `99A1`. They recur wherever a chapter's topic allows, and Chapter 22 shows how to reproduce every number quoted about them.

## Levels of rigour

Every result on a card, and every numbered theorem in the text, carries a tag.

**(proved here)** The proof is in the book, complete. This is the status of most results of Parts I to III and of the commutative algebra of Part VI.

**(proved here, given X)** The proof is complete, except that it uses the named input X, which is proved in another chapter, or is stated with a reference, or (in a few places, each said) is only sketched. This is the status of most of the paper's own lemmas and propositions: the paper's proofs are written for experts, and the book proves each, expanding every compressed step, given the statements that the paper imports.

**(proof sketched here)** The statement is precise, with all its hypotheses. The shape of the proof is explained, in a paragraph or a page, enough to see why the statement is true and what it costs; a reference with a theorem number carries the rest. A sketch is always labelled as a sketch and never presented as a proof.

**(stated; reference)** The statement is precise, with all its hypotheses, and no proof is given; the reference says where one is. This is the status of the deep theorems: modularity, Serre's conjecture, the level-lowering theorems, the theorems of Diamond, Flach and Guo, the formula of Fontaine and Perrin-Riou, Tate local duality, Poitou-Tate.

Definitions and conventions are tagged as such, and a statement established by a computation printed in the book is tagged *(proved here, by computation)*, with its range. The rule that governs the choice is that a card entry that is wrong is worse than one that is missing.

## How the book was checked

Every chapter was reviewed adversarially after it was written, by a review that looked for false statements, missing hypotheses, gaps in proofs and tags that claim more than the text proves, and the problems found were fixed. Every number the book prints was recomputed with PARI/GP 2.17.2 and Cremona's tables (the `ecdata` files at commit `25cec5e`, conductors below 10,000), and is printed with the range it covers; numbers that the paper reports for its larger ranges are quoted as the paper's and marked as such, since they cannot be rerun from those data. The cited literature was checked against the sources where the sources were available. A statement that was not checked against its source, or a theorem number that was not verified, is marked as such where it appears.

## Conventions

*Notation follows the paper.* The paper's symbols are the book's: $V = V_\ell E$, $T = T_\ell E$, $\rho = \rho_{E,\ell}$, $\bar\rho$, $A_f = \mathrm{ad}^0 V$ (the trace-zero adjoint, always), $\delta_p$, $L_p(A_f, s)$ and $L_p^{\mathrm{nv}}(A_f, s)$, $P_q(X)$, $e_\ell(E,q)$, $t_\ell(E,q)$, $n_q$, $q_E$ and its unit part $u$, $\mathrm{Tam}_p$, Fitt, $H^1_f$, $H^1_\Sigma$, $W$, $T^D$, $V^D$, $W^D$, $S_f$, $\eta^\Sigma_{f,\lambda}$, $M_{f,\lambda}$, $\hat\delta_!$, $\phi_f$, $I$, $J$, $r_E$, $\deg\varphi_E$, $\Phi_q^{\mathrm{geom}}$, $c_q$, $w_\ell(E,q)$, $w_3(E)$, $h_\ell(E)$, $\chi_{-3}$, $\chi_{-4}$, $\chi_{q^*}$, $\chi_S$, $q^*$, $\ell^*$, $E^{(q)}$, $E_S$, $N_s(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, 3Ns. Where the paper is ambiguous or uses one letter for two things, Appendix B fixes the book's reading; it lists every symbol with the chapter where it is defined.

*Valuations and lengths.* $v_\ell$ is the $\ell$-adic valuation normalised by $v_\ell(\ell) = 1$, extended to $\mathbb{Q}$ and to the fraction field of a discrete valuation ring; "length" means $\mathbb{Z}_\ell$-length (or $\mathcal{O}$-length when a ring $\mathcal{O}$ is named).

*Frobenius.* Arithmetic and geometric Frobenius are inverse to each other, and the literature uses both. In the book, as in the paper, $\mathrm{Frob}_p$ is an arithmetic Frobenius and the local $L$-factors use the geometric one: $L_p(M,s) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid M^{I_p})^{-1}$. A good prime has Frobenius eigenvalues $\alpha/\beta$, $1$, $\beta/\alpha$ on $\mathrm{ad}^0 V$, and $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$ relates the symmetric square to the adjoint.

*Names.* Double and triple names are hyphenated: Bloch-Kato, Atkin-Lehner, Atkin-Li, Brauer-Nesbitt, Diamond-Flach-Guo, Fontaine-Perrin-Riou, Taylor-Wiles, Riemann-Roch. A result is named by its authors on first appearance (the theorem of Agashe, Ribet and Stein) and by the book's label afterwards.

*Kodaira symbols and labels.* The Kodaira symbols are $\mathrm{I}_0$, $\mathrm{I}_n$, II, III, IV, $\mathrm{I}_0^*$, $\mathrm{I}_n^*$, IV$^*$, III$^*$, II$^*$; PARI's integer codes for them are explained in Chapter 10 and Chapter 22. Cremona labels are set in typewriter type, `121a1`; the paper's class labels `121c` and `121d` and the older capitalised label `99A1` are kept as the paper writes them.

*Typography.* Sentences are short. There are no dashes in the book: a parenthesis, a colon or a semicolon does the work. Number ranges are written "pages 663 to 727".

## A reading guide for the expert

The specialist who wants one statement should go straight to the card of the chapter that holds it. A dozen of the most likely requests:

- To recall the Tamagawa factor of Fontaine and Perrin-Riou,

$$\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}\big((H^1(I_p, M)_{\mathrm{tors}})^{G_{\mathbb{Q}_p}}\big),$$

its relation to the torsion of $H^1(\mathbb{Q}_p, T^D)$ and the Euler factor, its value at every bad prime, the exceptional set $S_f$, and the exact sequence of Lemma 2.1 of Diamond, Flach and Guo, open Chapter 18.

- For the Atkin-Li twisting theorem, the eigenvalues of $U_p$ on oldforms with the stabilisations and the depleted forms, the transport of congruences along a twist, and the perfect pairing between the integral cusp forms $S_2(\Gamma_0(N);\mathbb{Z})$ and the Hecke algebra $\mathbb{T}$, open Chapter 13.

- For Tate's algorithm, the Kodaira types with their geometric component groups, Kraus's semistability defect $12/\gcd(v_q(\Delta), 12)$, and the exchange of types under the twist by $\mathbb{Q}(\sqrt{q^*})$, at $q = 3$ included, open Chapter 10.

- For Theorem 3.7, Proposition 1.4, Lemma 2.1 and Theorem 2.7 of Diamond, Flach and Guo as read for weight 2, Wiles's numerical criterion in the form of Lemma 4.17 of Darmon, Diamond and Taylor, the lattice $M_{f,\lambda}$ and the pairing $\hat{\delta}_!$, the assembly of the main theorem, and the proof of Theorem thm:quant, open Chapter 20.

- For the theorem of Agashe, Ribet and Stein, the congruence number, the Manin constant, the formula of Zagier and Watkins for the modular degree, the projection formula behind Proposition prop:pullback, multiplicity one, and the comparison of Ribet and Takahashi, open Chapter 14.

- For the optimal-level theorem, Ribet's level lowering and level raising, local-global compatibility at a multiplicative prime, the description of $\bar{\rho}$ at $\ell$, and the congruence criterion and oldform lemma of the paper, open Chapter 15.

- For $\mathrm{GL}_2(\mathbb{F}_3)$, its subgroups $N_s(\mathbb{F}_3)$ and $N_{ns}(\mathbb{F}_3)$, their determinant-one subgroups and their orbits on the four lines of $\mathbb{F}_3^2$, open Chapter 1; for the images of $\bar{\rho}_{E,3}$, the Sutherland labels, and the paper's Propositions prop:ns and prop:Sf and Lemma lem:h0, open Chapter 11.

- For Brauer-Nesbitt, Clifford theory for an index-two subgroup, induced representations and Frobenius reciprocity, and the identity $\mathrm{ad}^0\,\rho = \mathrm{Sym}^2\,\rho \otimes \det\rho^{-1}$, open Chapter 4.

- For the tame character, the relation $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q$, the Kummer class of a uniformiser and its restriction to inertia, Weil-Deligne representations, and the conductor of a ramified twist of Steinberg, open Chapter 7.

- For inflation-restriction, $H^1(\langle t\rangle, M) = M/(t-1)M$ for a procyclic group, and the conjugation action of $G/I$ on $H^1(I, M)$, open Chapter 16; for Tate local duality, the Greenberg-Wiles formula and the comparison of Selmer groups for two sets of local conditions, open Chapter 17.

- For the table of Euler factors of the adjoint at the bad primes, the four shapes of the factor at every bad prime, the naive versus the true adjoint $L$-function, the Rankin-Selberg formula for the Petersson norm, and what PARI's `lfunsympow` returns, open Chapter 19.

- For Fitting ideals, lengths, Pontryagin duality, saturated sublattices and Gram determinants, open Chapter 3; for Chebotarev's theorem and $\mathbb{Q}(\sqrt{\ell^*})$ as the quadratic subfield of $\mathbb{Q}(\zeta_\ell)$, open Chapter 5; for $\mathbb{Z}_3[\zeta_3]$ as a discrete valuation ring and the Frobenius of $\mathbb{Q}(\sqrt{-3})$, open Chapter 6.

- For a guided reading of the paper section by section, with each result traced to its ingredients, and the table "hypothesis, where it is used, what fails without it", open Chapter 21; for the data protocol, the calibration argument and the PARI function `rule(E, l, iii)`, open Chapter 22.

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# Part I

# Algebra

# Chapter 1

# Groups, group actions and matrix groups over finite fields

The paper uses group theory in two ways. Most of the time it is the background language: Galois groups are profinite groups, inertia groups are stabilisers, Tamagawa numbers and component groups are finite abelian groups whose $\ell$-parts are taken, and the tame Galois group of $\mathbb{Q}_p$ is a semidirect product. At a few precise places, however, the paper's arguments are statements about specific finite groups: the proof of Proposition prop:ns is a classification of certain subgroups of the group $\mathrm{GL}_2(\mathbb{F}_3)$ of order 48, of their subgroups of determinant one and of their orbits on the four lines of $\mathbb{F}_3^2$; Lemma lem:IV and Remark rem:types are statements about unipotent elements of $\mathrm{GL}_2(\mathbb{F}_3)$; and Lemma lem:h0 uses that $\mathbb{F}_\ell^\times$ is cyclic of order $\ell-1$. This chapter develops the elementary theory from the definitions and then treats $\mathrm{GL}_2(\mathbb{F}_3)$ completely, so that every group-theoretic sentence of the paper can be checked line by line. It relies on no earlier chapter, only on linear algebra over a field and on the first notions of group theory as in Riley-Hobson-Bence.

## Core results at a glance

Throughout, $F$ is a field, $\mathbb{F}_q$ a finite field with $q$ elements ($q$ odd when Cartan subgroups appear; the reader may take $q=p$ prime), and the standard models of Convention 1.60 are used for the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$.

**1.1 Conventions.** Matrices act on column vectors on the left; $\mathbb{P}^1(F)$ is the set of lines $[v]=Fv$ of $F^2$, with $\infty=[1:0]=[e_1]$ and $x=[x:1]$, so that $\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)$ acts by $x\mapsto(ax+b)/(cx+d)$; $D_n$ has order $2n$; $q$ is a field size in this chapter and a bad prime in the paper; "the image is $X$" means "conjugate to the standard model $X$". Convention 1.1.

**1.2 Groups** (definition: group, abelian group, subgroup, the subgroup $\langle S\rangle$ generated by $S$, the order $\#G$ of a group and the order of an element, cyclic group). Definition 1.2.

**1.3 Lagrange.** For $H\le G$ finite, the left cosets of $H$ partition $G$ into $[G:H]$ sets of $\#H$ elements, so $\#G=[G:H]\,\#H$; the order of every element divides $\#G$. *(proved here)* Proposition 1.3.

**1.4 Cyclic groups.** Let $C=\langle t\rangle$ have order $n$. (i) For each $d\mid n$, $C$ has exactly one subgroup of order $d$, $C[d]=\{x:x^d=1\}=\langle t^{n/d}\rangle$, and every subgroup is of this form.

(ii) $x \mapsto x^m$ has kernel $C[\gcd(m,n)]$ and image $C^m = C^{\gcd(m,n)}$ of index $\gcd(m,n)$; so $[C : C^{\ell^k}] = \ell^{\min(k, v_\ell(n))}$ for a prime $\ell$. (iii) The generators are the $\varphi(n)$ elements of order $n$, and $\mathrm{Aut}(C) \simeq (\mathbb{Z}/n)^\times$ via $t \mapsto t^a$. (iv) A surjection $\chi\colon G \to C$ satisfies $\chi(g)^2 = 1$ for all $g$ iff $n \mid 2$. *(proved here)* Theorem 1.4.

**1.5 Normal subgroups and quotients** (definition: normal subgroup, quotient, kernel, centre $Z(G)$, centraliser $C_G(S)$, normaliser $N_G(S)$, commutator subgroup $[G,G]$). Definition 1.5.

**1.6 Isomorphism theorems.** First, $G/\ker f \simeq f(G)$. Second, $HN/N \simeq H/(H \cap N)$ for $H \le G$ and $N \trianglelefteq G$. Third, $(G/N)/(M/N) \simeq G/M$ for $N \le M$ normal. Correspondence: subgroups of $G/N$ correspond to subgroups of $G$ containing $N$, normal to normal. *(proved here)* Theorem 1.6.

**1.7 Index two; image of a kernel.** A subgroup of index 2 is normal. If $\rho\colon G \to \Gamma$ and $\delta\colon \Gamma \to A$ are homomorphisms, then $\rho(\ker(\delta \circ \rho)) = \rho(G) \cap \ker\delta$; in particular the image of $\ker(\det \circ \rho)$ is the determinant-one subgroup of $\rho(G)$. *(proved here)* Lemma 1.7.

**1.8 Conjugacy classes.** The class of $g$ has $[G : C_G(g)]$ elements; $H$ has $[G : N_G(H)]$ conjugates; a subgroup is normal iff it is a union of classes. *(proved here)* Proposition 1.9.

**1.9 $\ell$-primary parts.** For a finite abelian group $A$ and a prime $\ell$, $A[\ell^\infty] = \{a : \ell^k a = 0 \text{ for some } k\}$ has order $\ell^{v_\ell(\#A)}$, is the unique Sylow $\ell$-subgroup, and $A = \bigoplus_{\ell \mid \#A} A[\ell^\infty]$; Cauchy's theorem holds for $A$. *(proved here)* Proposition 1.10.

**1.10 Direct products.** If $H, K \trianglelefteq G$, $H \cap K = 1$, $HK = G$, then $H \times K \simeq G$ by $(h,k) \mapsto hk$. *(proved here)* Proposition 1.12.

**1.11 Chinese remainder theorem for groups.** $C_m \times C_n \simeq C_{mn}$ iff $\gcd(m,n) = 1$; $\mathbb{Z}/n \simeq \prod_{p \mid n} \mathbb{Z}/p^{v_p(n)}$. *(proved here)* Proposition 1.13.

**1.12 Semidirect products** (definition: $N \rtimes_\theta H$, with $(n,h)(n',h') = (n\,\theta(h)(n'), hh')$ and $hnh^{-1} = \theta(h)(n)$). Definition 1.14. **Recognition:** if $N \trianglelefteq G$, $H \le G$, $N \cap H = 1$, $NH = G$, then $G \simeq N \rtimes_\theta H$ with $\theta$ conjugation; a short exact sequence $1 \to N \to G \to Q \to 1$ splits iff $G \simeq N \rtimes Q$. *(proved here)* Proposition 1.15.

**1.13 Normal forms.** If $G = \{a^m b^e : 0 \le m < n,\ e \in \{0,1\}\}$ ($2n$ distinct elements) with $a^n = 1$, $bab^{-1} = a^k$, $b^2 = a^c$, the multiplication of $G$ is determined by $(n,k,c)$; two such groups with the same $(n,k,c)$ are isomorphic via $a^m b^e \mapsto a'^m b'^e$. *(proved here)* Lemma 1.17.

**1.14 Dihedral groups.** $D_n = \langle r, s \mid r^n = s^2 = 1,\ srs^{-1} = r^{-1}\rangle \simeq C_n \rtimes C_2$ has order $2n$; all elements outside $\langle r\rangle$ have order 2; a group of order $2n$ generated by $r$ of order $n$ and $s$ of order 2 with $srs^{-1} = r^{-1}$ is $\simeq D_n$; $D_3 \simeq S_3$, $D_6 \simeq C_2 \times S_3$; $D_4$ has five involutions. *(proved here)* Definition 1.18, Proposition 1.19.

**1.15 The quaternion group.** $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, $i^2 = j^2 = k^2 = ijk = -1$: $-1$ is the only involution, $Z(Q_8) = [Q_8, Q_8] = \{\pm 1\}$, every subgroup is normal, $Q_8 \not\simeq D_4$; a non-abelian group of order 8 with a unique involution is $\simeq Q_8$. *(proved here)* Definition 1.20, Proposition 1.21.

**1.16 The semidihedral group.** $SD_{16} = \langle a, x \mid a^8 = x^2 = 1,\ xax^{-1} = a^3\rangle$ has order 16; $(a^i x)^2 = a^{4i}$; it has $1, 5, 6, 4$ elements of orders $1, 2, 4, 8$; its index-2 subgroups are $\langle a\rangle \simeq C_8$,

$\langle a^2, x\rangle \simeq D_4$, $\langle a^2, ax\rangle \simeq Q_8$; a group of order 16 generated by $a$ of order 8 and $x$ of order 2 with $xax^{-1} = a^3$ is $\simeq SD_{16}$. *(proved here)* Definition 1.22, Proposition 1.23.

**1.17 Groups of order 8.** Every group of order 8 is isomorphic to exactly one of $C_8$, $C_4 \times C_2$, $C_2^3$, $D_4$, $Q_8$. *(proved here)* Theorem 1.25.

**1.18 Group actions** (definition: action, orbit $Gx$, stabiliser $G_x$, fixed set $X^G$, transitive, faithful). Definition 1.27.

**1.19 Orbit-stabiliser.** $gG_x \mapsto gx$ is a bijection $G/G_x \to Gx$; $G_{gx} = gG_xg^{-1}$; $\#X = \sum_i [G : G_{x_i}]$ over orbit representatives. *(proved here)* Theorem 1.28.

**1.20 Orbits of a subgroup.** If $G$ is transitive on $X$ and $H \leq G$, then $H$ is transitive iff $G = HG_x$; if $H \trianglelefteq G$, the $H$-orbits are permuted transitively by $G$ and have equal size; if $[G : H] = 2$, either $H$ is transitive or $X$ is the union of two $H$-orbits of size $\#X/2$, interchanged by $G \setminus H$. *(proved here)* Proposition 1.29.

**1.21 Orbit counting.** The number of orbits of a finite group on a finite set is $\frac{1}{\#G}\sum_g \#X^g$. *(proved here)* Lemma 1.30.

**1.22 Class equation and $p$-groups.** $\#G = \#Z(G) + \sum[G : C_G(g_i)]$; if a $p$-group acts on $X$, $\#X \equiv \#X^P \pmod p$; nontrivial $p$-groups have nontrivial centre; groups of order $p^2$ are abelian; a group of order $p^n$ has normal subgroups of every order $p^k$, and subgroups of index $p$ in a $p$-group are normal. *(proved here)* Theorem 1.31.

**1.23 Cauchy.** If a prime $p$ divides $\#G$, $G$ has an element of order $p$. *(proved here)* Theorem 1.32.

**1.24 Sylow.** Let $\#G = p^a m$, $p \nmid m$. Sylow $p$-subgroups (order $p^a$) exist; every $p$-subgroup lies in one; they are all conjugate; their number $n_p = [G : N_G(P)]$ divides $m$ and is $\equiv 1 \pmod p$; $P \trianglelefteq G$ iff $n_p = 1$. *(proved here)* Theorem 1.33.

**1.25 Finite abelian groups.** $A \simeq \bigoplus_i \mathbb{Z}/\ell_i^{e_i}$ with the multiset $\{\ell_i^{e_i}\}$ unique; equivalently $A \simeq \bigoplus \mathbb{Z}/d_i$ with $1 < d_1 \mid \cdots \mid d_r$ unique; the numbers $\#A[\ell^k]$ determine $A[\ell^\infty]$. *(proved here)* Theorem 1.35.

**1.26 Cyclicity criteria.** A finite abelian group is cyclic iff $\#A[\ell] \leq \ell$ for all primes $\ell$; every finite subgroup of $K^\times$, $K$ a field, is cyclic; so $\mathbb{F}_q^\times$ and $\mu_n(K)$ are cyclic. *(proved here)* Corollary 1.36.

**1.27 Counting torsion.** $\#A[n] = \#A/nA$; for $A \simeq \bigoplus_i \mathbb{Z}/\ell^{e_i}$, $\#A[\ell^k] = \ell^{\sum_i \min(k,e_i)}$. *(proved here)* Proposition 1.37.

**1.28 Linear groups** (definition: $\mathrm{GL}_2(F)$, $\mathrm{SL}_2(F)$, $\mathrm{PGL}_2(F) = \mathrm{GL}_2(F)/F^\times I$, $\mathrm{PSL}_2(F) = \mathrm{SL}_2(F)/\{\pm I\}$). Definition 1.40. $Z(\mathrm{GL}_2(F)) = F^\times I$, $Z(\mathrm{SL}_2(F)) = \{\pm I\}$; $\mathrm{GL}_2(F) \simeq \mathrm{SL}_2(F) \rtimes F^\times$; $\mathrm{SL}_2(F)$ is generated by elementary matrices; for $\mathrm{char}\, F \neq 2$, $-I$ is the only involution of $\mathrm{SL}_2(F)$; $[\mathrm{PGL}_2(F) : \mathrm{PSL}_2(F)] = [F^\times : (F^\times)^2]$. *(proved here)* Proposition 1.41.

**1.29 Companion form.** Two non-scalar matrices in $\mathrm{GL}_2(F)$ are conjugate iff they have the same characteristic polynomial; the centraliser of a non-scalar $g$ is $F[g]^\times$. *(proved here)* Lemma 1.42.

**1.30 Orders.** $\#\,\mathrm{GL}_2(\mathbb{F}_q) = q(q-1)^2(q+1)$, $\#\,\mathrm{SL}_2(\mathbb{F}_q) = \#\,\mathrm{PGL}_2(\mathbb{F}_q) = q(q^2-1)$, $\#\,\mathrm{PSL}_2(\mathbb{F}_q) = q(q^2-1)/\gcd(2,q-1)$ ($48, 24, 24, 12$ for $q=3$). *(proved here)* Proposition 1.43.

**1.31 The projective line.** $\mathrm{GL}_2(F)$ acts on $\mathbb{P}^1(F)$ with kernel $F^\times I$; $\mathrm{PGL}_2(F)$ is sharply 3-transitive; the stabiliser of $\infty$ is the upper triangular group $\mathrm{B}(F)$; $\mathrm{SL}_2(F)$ is 2-transitive; $\#\mathbb{P}^1(\mathbb{F}_q) = q+1$. *(proved here)* Proposition 1.44.

**1.32 Borel subgroups** (definition: the stabiliser $\mathrm{B}_L$ of a line $L$; U the upper unitriangular group, T the diagonal group). Definition 1.45. $\mathrm{B} = \mathrm{U} \rtimes \mathrm{T}$, $\#\mathrm{B}(\mathbb{F}_q) = q(q-1)^2$, Borel subgroups are conjugate and self-normalising, and a subgroup stabilises a line iff it lies in a Borel subgroup; $\mathrm{B}(\mathbb{F}_3) \simeq C_2 \times S_3 \simeq D_6$. *(proved here)* Proposition 1.46.

**1.33 Unipotent elements** (definition: $(u-1)^2 = 0$). Definition 1.47. A unipotent $u \neq 1$ is conjugate to $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and fixes exactly one line, pointwise; in characteristic $p$ the unipotents $\neq 1$ are exactly the elements of $p$-power order $\neq 1$, and have order $p$; $-u$ fixes no nonzero vector ($\mathrm{char}\, F \neq 2$), stabilises one line and has order $2p$; the stabiliser of a nonzero vector $v$ in $\mathrm{SL}_2(F)$ is the group $\mathrm{U}_{[v]} \simeq (F,+)$; in $\mathrm{GL}_2(\mathbb{F}_3)$ the 8 elements of order 3 are the nontrivial unipotents, form one class, lie in the four Sylow 3-subgroups $\mathrm{U}_L$, and generate $\mathrm{SL}_2(\mathbb{F}_3)$; the 8 elements of order 6 are the $-u$. *(proved here)* Proposition 1.48.

**1.34 Commutators in a Borel subgroup.** For $H \leq \mathrm{B}(K)$, $[H,H] \leq \mathrm{U}(K)$. If $\mathrm{char}\, K = p$: a subgroup of $\mathrm{B}(K)$ of order prime to $p$ is abelian, and a finite subgroup of $\mathrm{GL}_2(K)$ whose commutator subgroup is not a $p$-group stabilises no line of $K^2$. *(proved here)* Lemma 1.50.

**1.35 $\mathrm{PGL}_2$ in small characteristic.** $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$ and $\mathrm{PSL}_2(\mathbb{F}_3) \simeq A_4$ through the four lines; for $g \in \mathrm{GL}_2(\mathbb{F}_3)$, $\det g$ is the sign of the permutation of the lines. *(proved here)* $\mathrm{PGL}_2(\mathbb{F}_5) \simeq S_5$, $\mathrm{PSL}_2(\mathbb{F}_5) \simeq A_5$. *(proof sketched here)* $\mathrm{PSL}_2(\mathbb{F}_q)$ is simple for $q \geq 4$, and $\mathrm{PSL}_2(\mathbb{F}_7) \simeq \mathrm{GL}_3(\mathbb{F}_2)$. *(stated; reference)* Theorem 1.52.

**1.36 Cartan subgroups** (definition, $q$ odd, $\varepsilon$ a non-square: $C_s = \{\mathrm{diag}(a,d)\}$; $C_{ns} = \{\left(\begin{smallmatrix}a&\varepsilon b\\b&a\end{smallmatrix}\right) : (a,b) \neq (0,0)\} \simeq \mathbb{F}_q[\sqrt{\varepsilon}]^\times$). Definition 1.54. $N(C_s) = C_s \cup wC_s$ is the stabiliser of $\{[e_1],[e_2]\}$; $N(C_{ns}) = C_{ns} \cup C_{ns}\sigma$ with $\sigma = \mathrm{diag}(1,-1)$ and $\sigma c \sigma^{-1} = c^q$; $C_{ns}$ is cyclic of order $q^2-1$; all split, respectively all nonsplit, Cartan subgroups are conjugate. *(proved here)* Proposition 1.55.

**1.37 Cartan facts.** $\#C_s = (q-1)^2$, $\#N_s = 2(q-1)^2$, $\#C_{ns} = q^2-1$, $\#N_{ns} = 2(q^2-1)$; elements of $N_s \setminus C_s$ and of $N_{ns} \setminus C_{ns}$ have trace 0, with $(c\sigma)^2 = \det(c) I$; $\det(C_{ns}) = \mathbb{F}_q^\times$; the determinant-one subgroups have orders $q-1$, $2(q-1)$, $q+1$, $2(q+1)$; orbits on $\mathbb{P}^1(\mathbb{F}_q)$: $C_s$: $1, 1, q-1$; $N_s$: $2, q-1$; B: $1, q$; $C_{ns}$ and $N_{ns}$ transitive. *(proved here)* Proposition 1.56.

**1.38 Dickson.** For $\ell$ odd and $G \leq \mathrm{GL}_2(\mathbb{F}_\ell)$ with image $\bar{G}$ in $\mathrm{PGL}_2(\mathbb{F}_\ell)$: if $\ell \mid \#G$, then $G \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$ or $G$ lies in a Borel subgroup; if $\ell \nmid \#G$, then $\bar{G}$ is cyclic and $G$ lies in a Cartan subgroup, or $\bar{G}$ is dihedral and $G$ lies in the normaliser of a Cartan subgroup but not in the Cartan subgroup, or $\bar{G} \simeq A_4, S_4, A_5$. *(stated; reference; part (a) for $\ell = 3$ is proved in Proposition 1.61, and the case needed by the paper in Theorem 1.65)* Theorem 1.57.

**1.39 Conjugacy classes of $\mathrm{GL}_2(\mathbb{F}_3)$.** Order 48, centre $\{\pm I\}$, commutator subgroup $\mathrm{SL}_2(\mathbb{F}_3)$, eight classes, of sizes $1, 1, 12, 8, 6, 8, 6, 6$ with representatives $I, -I, \sigma, u, J, -u, \left(\begin{smallmatrix}1&-1\\1&1\end{smallmatrix}\right), \left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)$ of orders $1, 2, 2, 3, 4, 6, 8, 8$. *(proved here)* Proposition 1.59.

**1.40 Standard models.** $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (the paper's $J$), $\sigma = \mathrm{diag}(1,-1)$, $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$; $C_s(\mathbb{F}_3) = \{\mathrm{diag}(\pm 1, \pm 1)\}$, $N_s(\mathbb{F}_3) = \langle J, \sigma \rangle$, $C_{ns}(\mathbb{F}_3) = \{\left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)\}$, $N_{ns}(\mathbb{F}_3) = C_{ns}(\mathbb{F}_3) \cup C_{ns}(\mathbb{F}_3)\sigma$, $Q_8 = \{\pm I, \pm J, \pm\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right), \pm\left(\begin{smallmatrix} 1 & -1 \\ -1 & -1 \end{smallmatrix}\right)\}$, $C_4 = \langle J \rangle$. Convention 1.60.

**1.41 Subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$.** There are 55 subgroups in 16 conjugacy classes (table); 9 classes have surjective determinant, and all of these except $C_{ns}(\mathbb{F}_3)$ contain an element of determinant $-1$ and trace 0; a subgroup of order divisible by 3 contains $\mathrm{SL}_2(\mathbb{F}_3)$ or stabilises a line. *(proved here for the rows used; the table checked by computer)* Proposition 1.61.

**1.42 $\mathrm{SL}_2(\mathbb{F}_3)$.** Order 24; $-I$ its only involution; its elements of order dividing 4 form its unique Sylow 2-subgroup $Q_8$, normal in $\mathrm{GL}_2(\mathbb{F}_3)$; $\mathrm{SL}_2(\mathbb{F}_3) = Q_8 \rtimes \langle u \rangle$ with $u$ permuting $i, j, k$ cyclically; $[\mathrm{SL}_2(\mathbb{F}_3), \mathrm{SL}_2(\mathbb{F}_3)] = Q_8$; no subgroup of order 12; $\mathrm{GL}_2(\mathbb{F}_3)/Q_8 \simeq S_3$; $Q_8$ is transitive on the four lines. *(proved here)* Proposition 1.62.

**1.43 $N_s(\mathbb{F}_3)$.** $N_s(\mathbb{F}_3) = \langle J, \sigma \rangle \simeq D_4$; its elements of determinant $-1$ are $\pm\sigma, \pm w$, of trace 0; it has 3 conjugates; $N_s(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3) = \langle J \rangle \simeq C_4$; its orbits on the four lines are $\{\infty, 0\}$ and $\{1, -1\}$, and $\langle J \rangle$ has the same orbits; it stabilises no line of $\overline{\mathbb{F}}_3^2$, while $\langle J \rangle$ stabilises the two lines of $\mathbb{F}_9^2$ spanned by $(\pm i, 1)$; $N_s(\mathbb{F}_3)$ has index 2 in $N_{ns}(\mathbb{F}_3)$. *(proved here)* Proposition 1.63.

**1.44 $N_{ns}(\mathbb{F}_3)$.** $C_{ns}(\mathbb{F}_3) = \langle a \rangle \simeq C_8$ with $a = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$; $N_{ns}(\mathbb{F}_3) = \langle a, \sigma \rangle \simeq SD_{16}$, $\sigma a \sigma^{-1} = a^3$; it is a Sylow 2-subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$; $N_{ns}(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3) = Q_8$; its elements of determinant $-1$ and trace 0 are $\pm\sigma, \pm w$, and $C_{ns}(\mathbb{F}_3)$ has none; $C_{ns}(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ and $Q_8$ are transitive on the four lines. *(proved here)* Proposition 1.64.

**1.45 The three images.** If $H \leq \mathrm{GL}_2(\mathbb{F}_3)$ has $\det(H) = \mathbb{F}_3^\times$, stabilises no line of $\mathbb{F}_3^2$, and contains an element of determinant $-1$ and trace 0, then $H$ is conjugate to exactly one of $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, $N_s(\mathbb{F}_3)$; the determinant-one subgroups are $\mathrm{SL}_2(\mathbb{F}_3)$, $Q_8$, $\langle J \rangle$; the first two stabilise no line of $\overline{\mathbb{F}}_3^2$, the third is abelian. *(proved here)* Theorem 1.65.

**1.46 Orbit dichotomy.** For such $H$: $H$ is conjugate to $N_s(\mathbb{F}_3)$ iff it has two orbits of size 2 on the four lines, iff it is not transitive, iff $H \cap \mathrm{SL}_2(\mathbb{F}_3)$ is abelian, iff $H \cap \mathrm{SL}_2(\mathbb{F}_3)$ stabilises a line of $\overline{\mathbb{F}}_3^2$. *(proved here)* Corollary 1.66.

**1.47 Determinant $-1$, trace 0.** For $q$ odd, $g \in \mathrm{GL}_2(\mathbb{F}_q)$ has $\det g = -1$ and $\mathrm{tr}\, g = 0$ iff it is conjugate to $\sigma$; these $q(q+1)$ elements have order 2 and fix exactly two lines; in $\mathrm{GL}_2(\mathbb{F}_3)$, B, $N_s$, $N_{ns}$, $C_s$, $C_{ns}$, $\mathrm{SL}_2$ there are $12, 6, 4, 4, 2, 0, 0$ of them. *(proved here)* Proposition 1.68.

**1.48 Images in $S_4$.** Under $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$: $\mathrm{GL}_2, \mathrm{SL}_2, N_{ns}, Q_8, N_s, C_{ns}, C_s, \langle J \rangle, \mathrm{B}, \mathrm{U}$ map onto $S_4$, $A_4$, $D_4$, the normal $V_4$, a non-normal $V_4$, $C_4$, $\langle$transposition$\rangle$, $\langle$double transposition$\rangle$, $S_3$, $A_3$. *(proved here)* Proposition 1.71.

**1.49 Inverse limits** (definition: $\varprojlim G_n = \{(g_n) : \pi_n(g_{n+1}) = g_n\}$, profinite group). Definition 1.73. The inverse limit is a closed subgroup of $\prod G_n$, compact and Hausdorff, with the universal property. *(proved here, except Tychonoff's theorem: stated; reference)* Proposition 1.75.

**1.50 Profinite groups.** With $U_n = \ker(G \to G_n)$: the $U_n$ are a basis of neighbourhoods of 1; open subgroups are the closed subgroups of finite index; continuous maps to finite groups

factor through some $G/U_n$; closed subgroups and quotients by closed normal subgroups are profinite; continuous images in Hausdorff groups are closed; a continuous homomorphism from a pro-$p$ group to a finite group of order prime to $p$ is trivial. *(proved here)* Profinite iff compact, Hausdorff and totally disconnected. *(stated; reference)* Proposition 1.77.

**1.51** $\mathrm{GL}_2(\mathbb{Z}_\ell)$**.** $\mathrm{GL}_2(\mathbb{Z}_\ell) = \varprojlim \mathrm{GL}_2(\mathbb{Z}/\ell^n)$; the reduction maps are surjective, with kernels isomorphic to $(M_2(\mathbb{F}_\ell), +)$, so $\#\,\mathrm{GL}_2(\mathbb{Z}/\ell^n) = \ell^{4(n-1)}(\ell^2-1)(\ell^2-\ell)$; the kernel $\Gamma(\ell) = \ker(\mathrm{GL}_2(\mathbb{Z}_\ell) \to \mathrm{GL}_2(\mathbb{F}_\ell))$ is pro-$\ell$, and torsion-free if $\ell$ is odd, so a finite subgroup of $\mathrm{GL}_2(\mathbb{Z}_\ell)$ maps isomorphically into $\mathrm{GL}_2(\mathbb{F}_\ell)$ for $\ell$ odd. *(proved here)* Proposition 1.78.

## Where the paper uses this chapter

- **Proposition prop:ns** (the mod 3 image when the hypothesis on $\mathbb{Q}(\sqrt{-3})$ fails), and its proof: the determinant-one subgroup as the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, Lemma 1.7; the list of three images, Theorem 1.65, built on Proposition 1.62, Proposition 1.63, Proposition 1.64, Proposition 1.68; "The first two act absolutely irreducibly on $\mathbb{F}_3^2$", Lemma 1.50; the orbits on the four lines and the factorisation of the 3-division polynomial, Corollary 1.66, Proposition 1.29, Theorem 1.52(ii).
- **Lemma lem:IV** (types IV, IV$^*$): the stabiliser of a vector in $\mathrm{SL}_2(\mathbb{F}_3)$ is a group of order 3 of unipotents, Proposition 1.48; the kernel of $\mathrm{GL}_2(\mathbb{Z}_3) \to \mathrm{GL}_2(\mathbb{F}_3)$ is torsion-free, Proposition 1.78.
- **Remark rem:types** (the other additive types at $q \geq 5$): fixed vectors of $-I$, of $J$, of "minus a unipotent", Proposition 1.48, Proposition 1.59.
- **Lemma lem:h0** (the correction term $h_\ell(E)$): $\mathbb{F}_\ell^\times$ is cyclic of order $\ell-1$ and, for an odd prime $\ell$, a character onto it with square 1 forces $\ell = 3$, Theorem 1.4(iv), Corollary 1.36; $\rho$ mod $\ell^k$ factors through a finite quotient, Proposition 1.77; the coefficients $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell/\mathbb{Z}_\ell$, Example 1.76.
- **Section sec:anomaly and Remark rem:ns** (image 3Ns): Proposition 1.63, Corollary 1.66, Example 1.69.
- **Section sec:eis and Conjecture conj:eis** (the Eisenstein case): a rational isogeny means image in a Borel subgroup, Proposition 1.46; the labels 3Cs.1.1, 3B.1.1, 3B, Example 1.69.
- **Background** for Definition def:terms, Section 1.4 and Lemma lem:tamfactor (the $\ell$-part of a component group or a Tamagawa number, the count of Frobenius-fixed points in $(\mathbb{Z}/\ell^{v_\ell(n)})^2$): Proposition 1.10, Theorem 1.35, Proposition 1.37, Theorem 1.4(ii); for Lemma lem:crit ($\mathbb{T}/(I+J)$ cyclic): Theorem 1.4; for Lemma lem:tamagawa-exp ($\rho$ mod $\ell^t$): Proposition 1.78.
- **Proposition prop:shapes** (the shapes of the image of inertia at an additive prime): "For $q \geq 5$ the image is tame, hence cyclic", Example 1.80, Proposition 1.77(vi).
- **Theorem thm:main** at $\ell = 3$ (the hypothesis on the mod 3 image, through Proposition prop:ns): Corollary 1.66.
- **Section sec:data** (the intrinsic description of the image 3Ns: a 3-division polynomial that is a product of two quadratics): Corollary 1.66.

## 1.1 Groups, subgroups, cosets, quotients and cyclic groups

This section fixes the conventions of the chapter and proves the basic facts about cosets, quotients and cyclic groups. In the paper they appear in three places: the image of the absolute Galois group of $\mathbb{Q}(\sqrt{-3})$ is a determinant-one subgroup (Proposition prop:ns, through Lemma 1.7); for an odd prime $\ell$, a character of $G_{\mathbb{Q}}$ onto the cyclic group $\mathbb{F}_\ell^\times$ with square 1 forces $\ell = 3$ (Lemma lem:h0, through Theorem 1.4); and the "$\ell$-part" of a finite abelian group (Definition def:terms, Lemma lem:tamfactor, through Proposition 1.10).

**Convention 1.1.** The following conventions hold in this chapter and are inherited by the book.

(C1) Matrices act on column vectors on the left. For $v \in F^2 \setminus \{0\}$ the line through $v$ is $[v] = Fv$, and $\mathbb{P}^1(F)$ is the set of lines of $F^2$, with coordinates $[x : y]$. We write $\infty = [1 : 0] = [e_1]$ and identify $[x : 1]$ with $x \in F$. Then $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ acts by the Möbius formula $g \cdot x = (ax + b)/(cx + d)$, because $g\binom{x}{1} = \binom{ax+b}{cx+d}$.

(C2) Conjugation is $g \cdot h = ghg^{-1}$. In a semidirect product $N \rtimes H$ the group $H$ acts on $N$, and $hnh^{-1} = \theta(h)(n)$.

(C3) $C_n$, or $\mathbb{Z}/n$ when written additively, is the cyclic group of order $n$; $D_n$ is the dihedral group of order $2n$ (so $D_4$ has order 8; Dummit and Foote write $D_8$); $V_4 = C_2 \times C_2$; $Q_8$ is the quaternion group; $SD_{16}$ is the semidihedral group of order 16; $S_n$ and $A_n$ are the symmetric and alternating groups.

(C4) $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ is a field: an integer prime to $p$ is invertible modulo $p$ by Bezout's identity. $\mathbb{F}_q$ is a finite field with $q$ elements, of characteristic $p$ with $q$ a power of $p$ (Chapter 2 constructs it); the reader may take $q = p$ throughout. Whenever Cartan subgroups are discussed $q$ is odd. In this chapter $q$ is a field size; in the paper $q$ is a bad prime of $E$ (convention B-C1). The two meet only in $\mathrm{GL}_2(\mathbb{F}_3)$, where the field size is $\ell = 3$.

(C5) "The image is $N_s(\mathbb{F}_3)$" always means "the image is conjugate in $\mathrm{GL}_2(\mathbb{F}_3)$ to $N_s(\mathbb{F}_3)$"; the standard models of Convention 1.60 are fixed once and for all, and similarly for $N_{ns}(\mathbb{F}_3)$, $C_s$, $C_{ns}$, B, U.

**Definition 1.2.** A *group* is a set $G$ with an associative composition $G \times G \to G$, $(g, h) \mapsto gh$, an identity element 1 and inverses. It is *abelian* if $gh = hg$ for all $g, h$; abelian groups are often written additively, with 0 and $-a$. A *subgroup* $H \leq G$ is a nonempty subset closed under products and inverses. The subgroup $\langle S \rangle$ generated by a subset $S$ is the intersection of all subgroups containing $S$, that is, the set of finite products of elements of $S$ and their inverses. The *order* of $G$ is its cardinality $\#G$; the order of $g \in G$ is the least $n \geq 1$ with $g^n = 1$ (or $\infty$), which is $\#\langle g \rangle$. A group is *cyclic* if $G = \langle g \rangle$ for some $g$. Examples: $(\mathbb{Z}, +)$; $\mathbb{Z}/n$; the symmetric group $S_n$; the group $\mathrm{GL}_2(F)$ of invertible $2 \times 2$ matrices over a field $F$; the multiplicative group $F^\times$.

**Proposition 1.3** (Lagrange)**.** *Let $H \leq G$ with $G$ finite. The left cosets $gH$ partition $G$, each has $\#H$ elements, and $\#G = [G : H] \cdot \#H$, where $[G : H]$ is the number of cosets. Consequently the order of every element of $G$ divides $\#G$, and $g^{\#G} = 1$ for every $g \in G$.*

*Proof.* Every $g$ lies in $gH$. If $x \in gH \cap g'H$, say $x = gh = g'h'$, then $g' = ghh'^{-1} \in gH$, so $g'H \subseteq gH$, and symmetrically; so two cosets are equal or disjoint. The map $H \to gH$, $h \mapsto gh$, is bijective (its inverse is multiplication by $g^{-1}$). Counting gives $\#G = [G : H]\,\#H$. Applied to $H = \langle g \rangle$, the order of $g$ divides $\#G$, and then $g^{\#G} = 1$. □

**Theorem 1.4.** *Let $C = \langle t \rangle$ be cyclic of order $n$.*

*(i) Every subgroup of $C$ is cyclic. For each $d \mid n$ there is exactly one subgroup of order $d$, namely $C[d] = \{x \in C : x^d = 1\} = \langle t^{n/d} \rangle = C^{n/d}$, the set of $(n/d)$-th powers.*

*(ii) For $m \geq 1$ the map $x \mapsto x^m$ is an endomorphism of $C$ with kernel $C[\gcd(m,n)]$, of order $\gcd(m,n)$, and image $C^m = C^{\gcd(m,n)}$, of index $\gcd(m,n)$. In particular, for a prime $\ell$ and $k \geq 0$, $[C : C^{\ell^k}] = \ell^{\min(k, v_\ell(n))}$; and for $k \leq v_\ell(n)$, an element $x \in C$ is an $\ell^k$-th power iff its image in $C/C^{\ell^{v_\ell(n)}} \simeq \mathbb{Z}/\ell^{v_\ell(n)}$ is divisible by $\ell^k$.*

*(iii) The generators of $C$ are the $\varphi(n)$ elements $t^a$ with $\gcd(a,n) = 1$, which are its elements of order $n$. The map $(\mathbb{Z}/n)^\times \to \mathrm{Aut}(C)$, $a \mapsto (t \mapsto t^a)$, is an isomorphism. $\mathrm{Aut}(\mathbb{Z}) = \{\pm 1\}$.*

*(iv) Let $\chi \colon G \to C$ be a surjective homomorphism from any group. Then $\chi(g)^2 = 1$ for all $g \in G$ if and only if $n \mid 2$.*

*Proof.* The map $\mathbb{Z} \to C$, $a \mapsto t^a$, is a surjective homomorphism with kernel $n\mathbb{Z}$ (if $t^a = 1$, write $a = bn + r$ with $0 \leq r < n$; then $t^r = 1$ and $r = 0$ by minimality of $n$). So $t^a = t^b$ iff $a \equiv b \pmod{n}$, and the order of $t^a$ is $n/\gcd(a,n)$: indeed $(t^a)^k = 1$ iff $n \mid ak$ iff $\frac{n}{\gcd(a,n)} \mid k$.

(i) Let $H \leq C$, $H \neq 1$, and let $a$ be the least positive integer with $t^a \in H$. For $t^b \in H$ write $b = ca + r$, $0 \leq r < a$; then $t^r = t^b(t^a)^{-c} \in H$, so $r = 0$. Hence $H = \langle t^a \rangle$, and $a \mid n$ (take $b = n$). So $H = \langle t^a \rangle$ has order $n/a$: the subgroups are the $\langle t^{n/d} \rangle$ for $d \mid n$, one for each order $d$. The subgroup $\langle t^{n/d} \rangle$ consists of $(n/d)$-th powers and has order $d$, so its elements satisfy $x^d = 1$; conversely $x = t^b$ with $x^d = 1$ means $n \mid bd$, that is $\frac{n}{d} \mid b$. So $C[d] = \langle t^{n/d} \rangle = C^{n/d}$.

(ii) The map is a homomorphism because $C$ is abelian. Let $g = \gcd(m,n)$. Its kernel is $C[m]$; and $x^m = 1$ iff $x^g = 1$ (write $g = am + bn$, so $x^g = (x^m)^a (x^n)^b$), so the kernel is $C[g]$, of order $g$ by (i). By the first isomorphism theorem (Theorem 1.6, whose proof does not use this one) the image has order $n/g$; it is contained in $C^g$ (as $g \mid m$), which has order $n/g$ by (i) applied to $d = n/g$; so $C^m = C^g$, of index $g$. For $m = \ell^k$, $g = \gcd(\ell^k, n) = \ell^{\min(k, v_\ell(n))}$.

For the last statement put $e = v_\ell(n)$ and fix $k \leq e$. Since $\ell^k \mid \ell^e$, every $\ell^e$-th power is an $\ell^k$-th power: $C^{\ell^e} \subseteq C^{\ell^k}$. Let $\pi \colon C \to \bar{C} = C/C^{\ell^e}$ be the projection. By what was just proved, $\bar{C}$ has order $[C : C^{\ell^e}] = \ell^e$; it is cyclic, generated by $\pi(t)$, and $\pi(t) \mapsto 1$ identifies it with $\mathbb{Z}/\ell^e$. Under this identification $\pi(C^{\ell^k}) = \bar{C}^{\ell^k}$ (the image of an $\ell^k$-th power is an $\ell^k$-th power, and every $\ell^k$-th power $\pi(y)^{\ell^k} = \pi(y^{\ell^k})$ is such an image) becomes $\ell^k\mathbb{Z}/\ell^e$, the multiples of $\ell^k$. Because $C^{\ell^k}$ contains the kernel $C^{\ell^e}$ of $\pi$, it equals the full preimage $\pi^{-1}(\pi(C^{\ell^k}))$ (Theorem 1.6(iv)). Hence $x \in C^{\ell^k}$ iff $\pi(x) \in \ell^k\mathbb{Z}/\ell^e$, as claimed.

(iii) $t^a$ generates iff its order $n/\gcd(a,n)$ is $n$. An automorphism $\alpha$ is determined by $\alpha(t)$, which must be a generator $t^a$; conversely $x \mapsto x^a$ with $\gcd(a,n) = 1$ is a bijective endomorphism by (ii). Composition corresponds to multiplication of exponents. For $\mathbb{Z}$, an automorphism sends the generator 1 to a generator, $\pm 1$.

(iv) If $n \mid 2$, every element of $C$ satisfies $x^2 = 1$. Conversely, if $\chi(g)^2 = 1$ for all $g$, then, $\chi$ being surjective, every $x \in C$ satisfies $x^2 = 1$; in particular $t^2 = 1$ and $n \mid 2$. □

For example, let $C = \mathbb{Z}/12$ (written additively, so that "$m$-th powers" are multiples of $m$) and $\ell = 2$, so $v_2(12) = 2$. Then $2C = \{0, 2, \ldots, 10\}$ has index 2, $4C = \{0, 4, 8\}$ has index 4, and $8C = \gcd(8,12)C = 4C$ again has index $4 = 2^{\min(3,2)}$: once $k$ passes $v_\ell(n)$, raising to higher $\ell$-powers removes nothing more, because the $\ell$-part of $C$ has already been killed and the prime-to-$\ell$ part is unaffected. The quotient $C/4C \simeq \mathbb{Z}/4$ remembers exactly the 2-part of $C$, and

$x \in C$ is a multiple of 2 iff its image in $\mathbb{Z}/4$ is. Part (iv) is the form in which this is used: a character of $G_{\mathbb{Q}}$ onto $\mathbb{F}_\ell^\times$, cyclic of order $\ell - 1$ (Corollary 1.36), can be quadratic only if $\ell - 1 \mid 2$, that is, for an odd prime $\ell$, only if $\ell = 3$.

**Definition 1.5.** A subgroup $N \leq G$ is *normal*, $N \trianglelefteq G$, if $gNg^{-1} = N$ for all $g \in G$. Then the cosets $gN$ form a group $G/N$, the *quotient*, with $(gN)(g'N) = gg'N$, and $G \to G/N$ is the *canonical projection*. A *homomorphism* $f\colon G \to H$ satisfies $f(gg') = f(g)f(g')$; its *kernel* $\ker f = f^{-1}(1)$ is normal and its image $f(G)$ is a subgroup. A group is *simple* if it is nontrivial and has no normal subgroups other than 1 and itself. The *centre* is $Z(G) = \{z : zg = gz\ \forall g\}$; for $S \subseteq G$, the *centraliser* is $C_G(S) = \{g : gs = sg\ \forall s \in S\}$ and the *normaliser* is $N_G(S) = \{g : gSg^{-1} = S\}$. The *conjugacy class* of $g$ is $\{hgh^{-1} : h \in G\}$. The *commutator subgroup* $[G, G]$ is generated by the commutators $[g, h] = ghg^{-1}h^{-1}$; it is normal, $G/[G, G]$ is abelian (the *abelianisation*), and $[G, G] \leq N$ for a normal $N$ iff $G/N$ is abelian.

Normality is exactly what makes the coset product meaningful. Read $(gN)(g'N)$ as the set of all products $xy$ with $x \in gN$, $y \in g'N$. Using $Ng' = g'(g'^{-1}Ng') = g'N$ we get $gNg'N = gg'NN = gg'N$, so the product of two cosets is again a coset and does not depend on the representatives chosen. Without normality this fails: in $S_3$ take $N = \langle(12)\rangle$. Then $(13)N = \{(13), (123)\}$, and $(13)N \cdot (13)N$ contains both $(13)(13) = 1$ and $(123)(13) = (23)$ (permutations composed from right to left), which lie in the different cosets $N$ and $(23)N$. By contrast $A_3 = \langle(123)\rangle$ has index 2 in $S_3$, hence is normal (Lemma 1.7), and $S_3/A_3 \simeq C_2$ records the sign of a permutation. The kernel of a homomorphism is normal because $f(gkg^{-1}) = f(g)f(g)^{-1} = 1$ for $k \in \ker f$; conversely every normal $N$ is the kernel of $G \to G/N$, so normal subgroups and kernels are the same thing. The last statement of the definition holds because $G/N$ is abelian iff $ghN = hgN$ for all $g, h$, iff $g^{-1}h^{-1}gh \in N$ for all $g, h$, and as $g, h$ range over $G$ so do $g^{-1}, h^{-1}$; so this says that $N$ contains every commutator, hence the subgroup $[G, G]$ they generate.

**Theorem 1.6** (Isomorphism theorems)**.** *(i) A homomorphism $f\colon G \to H$ induces an isomorphism $G/\ker f \simeq f(G)$, $g \ker f \mapsto f(g)$.*

*(ii) If $H \leq G$ and $N \trianglelefteq G$, then $HN$ is a subgroup, $H \cap N \trianglelefteq H$, and $HN/N \simeq H/(H \cap N)$.*

*(iii) If $N \leq M$ are normal subgroups of $G$, then $M/N \trianglelefteq G/N$ and $(G/N)/(M/N) \simeq G/M$.*

*(iv) (Correspondence) For $N \trianglelefteq G$, $K \mapsto K/N$ is an inclusion-preserving bijection from the subgroups of $G$ containing $N$ to the subgroups of $G/N$, under which normal subgroups correspond to normal subgroups.*

*Proof.* (i) $f(g) = f(g')$ iff $g^{-1}g' \in \ker f$ iff $g \ker f = g' \ker f$; so the map is well defined and injective; it is a surjective homomorphism onto $f(G)$. (ii) $hnh'n' = hh'(h'^{-1}nh')n' \in HN$ and $(hn)^{-1} = h^{-1}(hn^{-1}h^{-1}) \in HN$, so $HN$ is a subgroup. The homomorphism $H \to HN/N$, $h \mapsto hN$, is surjective (as $hnN = hN$) with kernel $H \cap N$; apply (i). (iii) The map $G/N \to G/M$, $gN \mapsto gM$, is a well-defined surjective homomorphism with kernel $M/N$; apply (i). (iv) The inverse bijection sends $\bar{K} \leq G/N$ to its preimage under the projection $\pi$; $\pi(\pi^{-1}(\bar{K})) = \bar{K}$ as $\pi$ is surjective, and $\pi^{-1}(\pi(K)) = KN = K$ when $N \leq K$. Normality is preserved in both directions since $\pi(gKg^{-1}) = \pi(g)\pi(K)\pi(g)^{-1}$. □

**Lemma 1.7.** *(i) A subgroup $H$ of index 2 in $G$ is normal, and is the kernel of a surjection $G \to C_2$.*

*(ii) Let $\rho\colon G \to \Gamma$ and $\delta\colon \Gamma \to A$ be homomorphisms and $G_0 = \ker(\delta \circ \rho)$. Then $\rho(G_0) = \rho(G) \cap \ker \delta$. In particular, if $\rho\colon G \to \mathrm{GL}_2(F)$ and $G_0 = \ker(\det \circ \rho)$, then (with $\mathrm{SL}_2(F)$ the group of matrices of determinant 1, Definition 1.40) $\rho(G_0) = \rho(G) \cap \mathrm{SL}_2(F)$ is the determinant-one subgroup of the image.*

*Proof.* (i) There are two left cosets; one is $H$, so the other is the complement $G \setminus H$. The same holds for the right cosets. For $g \in H$ we have $gH = H = Hg$, and for $g \notin H$ both $gH$ and $Hg$ are cosets different from $H$, hence both equal $G \setminus H$. So $gH = Hg$, that is $gHg^{-1} = H$, for every $g$, and $H \trianglelefteq G$. The quotient $G/H$ has order 2, so it is $C_2$, and the projection $G \to G/H$ is the required surjection with kernel $H$. (ii) If $g \in G_0$, then $\rho(g) \in \rho(G)$ and $\delta(\rho(g)) = 1$. Conversely, if $\gamma \in \rho(G) \cap \ker \delta$, write $\gamma = \rho(g)$; then $\delta(\rho(g)) = 1$, so $g \in G_0$ and $\gamma \in \rho(G_0)$. □

*In the paper* 1.8. The proof of Proposition prop:ns begins: "Since $\det \bar\rho_{E,3} = \chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$, the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ is the subgroup of determinant 1 of the image $H$ of $\bar\rho_{E,3}$." This is Lemma 1.7(ii) with $G = G_{\mathbb{Q}}$, $\rho = \bar\rho_{E,3}$, $\delta = \det$: the kernel of $\det \circ \bar\rho_{E,3} = \chi_{-3}$ is $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ by Galois theory (Chapter 5), and the identity $\det \bar\rho_{E,3} = \bar\chi_3$, $\bar\chi_3 = \chi_{-3}$, is Chapter 9 and Chapter 11. Since $\mathbb{F}_3^\times = \{\pm 1\}$, the determinant-one subgroup has index 1 or 2 in $H$, and index 2 when $\det(H) = \mathbb{F}_3^\times$.

**Proposition 1.9.** *Conjugacy is an equivalence relation on $G$. For $G$ finite, the class of $g$ has $[G : C_G(g)]$ elements, and a subgroup $H$ has exactly $[G : N_G(H)]$ conjugates $gHg^{-1}$. A subgroup is normal iff it is a union of conjugacy classes.*

*Proof.* Reflexivity, symmetry and transitivity follow from $g = 1g1^{-1}$, $h = x^{-1}(xhx^{-1})x$ and $y(xgx^{-1})y^{-1} = (yx)g(yx)^{-1}$. The map $G \to \{xgx^{-1}\}$, $x \mapsto xgx^{-1}$, satisfies $xgx^{-1} = ygy^{-1}$ iff $y^{-1}x \in C_G(g)$ iff $xC_G(g) = yC_G(g)$; so the class is in bijection with $G/C_G(g)$. The same argument with $x \mapsto xHx^{-1}$ and $N_G(H)$ counts the conjugates of $H$. A subgroup $N$ satisfies $gNg^{-1} \subseteq N$ for all $g$ iff it contains the class of each of its elements; and $gNg^{-1} \subseteq N$ for all $g$ implies equality (apply it to $g^{-1}$). (This is the special case of Theorem 1.28 for the conjugation action.) □

**Proposition 1.10.** *Let $A$ be a finite abelian group, written additively, and $\ell$ a prime.*

*(i) (Cauchy for abelian groups) If $\ell \mid \#A$, then $A$ has an element of order $\ell$.*

*(ii) If $\#A = mn$ with $\gcd(m,n) = 1$, then $A = A[m] \oplus A[n]$, where $A[m] = \{a : ma = 0\}$, and $\#A[m] = m$, $\#A[n] = n$.*

*(iii) The $\ell$-primary part $A[\ell^\infty] = \{a : \ell^k a = 0 \text{ for some } k\}$ equals $A[\ell^{v_\ell(\#A)}]$, has order $\ell^{v_\ell(\#A)}$, and is the unique Sylow $\ell$-subgroup of $A$ (the unique subgroup of order $\ell^{v_\ell(\#A)}$; Sylow subgroups of arbitrary finite groups are treated in Theorem 1.33); every $\ell$-power-order subgroup lies in it.*

*(iv) $A = \bigoplus_{\ell \mid \#A} A[\ell^\infty]$, and the projection onto $A[\ell^\infty]$ is multiplication by an integer $e_\ell$ with $e_\ell \equiv 1 \pmod{\ell^{v_\ell(\#A)}}$ and $e_\ell \equiv 0$ modulo $\#A/\ell^{v_\ell(\#A)}$.*

*(v) The exponent of $A$ (the least $e \geq 1$ with $eA = 0$) divides $\#A$ and has the same prime divisors.*

*Proof.* (i) Induction on $\#A$. Pick $a \neq 0$ of order $k$. If $\ell \mid k$, then $(k/\ell)a$ has order $\ell$. Otherwise $\ell$ divides $\#(A/\langle a\rangle) = \#A/k < \#A$, and by induction there is $b \in A$ whose class $\bar{b}$ has order $\ell$. If $b$ has order $k'$, then $k'\bar{b} = 0$, so $\ell \mid k'$, and $(k'/\ell)b$ has order $\ell$.

(ii) Write $1 = xm + yn$ with $x, y \in \mathbb{Z}$. For $a \in A$, $a = xma + yna$, and $n(xma) = x(mn)a = 0$ by Lagrange, so $xma \in A[n]$; similarly $yna \in A[m]$. If $a \in A[m] \cap A[n]$, then $a = xma + yna = 0$. So $A = A[m] \oplus A[n]$ and $\#A[m] \cdot \#A[n] = \#A = mn$. By (i), no prime $p$ dividing $n$ divides $\#A[m]$: otherwise $A[m]$ would contain an element $a$ of order $p$, and then $na = 0$ as $p \mid n$, so $a \in A[m] \cap A[n] = 0$. Hence $\#A[m]$, which divides $mn$, is prime to $n$ and so divides $m$. Symmetrically $\#A[n]$ divides $n$. As the product of the two orders is $mn$, both divisibilities are equalities: $\#A[m] = m$ and $\#A[n] = n$.

(iii) Let $e = v_\ell(\#A)$ and $\#A = \ell^e m$. If $\ell^k a = 0$, then also $\#A \cdot a = 0$ (Lagrange), so $\gcd(\ell^k, \#A)a = 0$, because by Bezout $\gcd(\ell^k, \#A) = x\ell^k + y\#A$ for some integers $x, y$. As $\gcd(\ell^k, \#A) = \ell^{\min(k,e)}$ divides $\ell^e$, we get $\ell^e a = 0$; so $A[\ell^\infty] = A[\ell^e]$, of order $\ell^e$ by (ii). A subgroup of $\ell$-power order consists of elements killed by a power of $\ell$, so lies in $A[\ell^\infty]$; a subgroup of order $\ell^e$ therefore equals it.

(iv) Apply (ii) repeatedly to the factorisation of $\#A$ into prime powers. For the projector, take $e_\ell = yn$ in (ii) with $m = \ell^{v_\ell(\#A)}$: the proof shows $a \mapsto yna$ is the projection onto $A[m]$, and $yn \equiv 1 \pmod{m}$, $yn \equiv 0 \pmod{n}$.

(v) $\#A \cdot A = 0$ by Lagrange. Let $e$ be the exponent and write $\#A = ce + r$ with $0 \leq r < e$; then $rA = \#A \cdot A - c(eA) = 0$, so $r = 0$ by minimality of $e$, and $e \mid \#A$. Every prime $p$ dividing $\#A$ is the order of some element $a$ by (i); since $ea = 0$, the order $p$ divides $e$. Conversely every prime dividing $e$ divides $\#A$, as $e \mid \#A$. For example $A = \mathbb{Z}/2 \oplus \mathbb{Z}/4$ has order 8 and exponent 4. □

*In the paper* 1.11. Section 1.4 of the paper and the rule built on Definition def:terms use $v_\ell(\#\Phi_q^{\mathrm{geom}})$, the valuation of the order of a finite abelian group; by Proposition 1.10(iii) the $\ell$-part of a finite abelian group $\Phi$ is $\Phi[\ell^\infty]$, of order $\ell^{v_\ell(\#\Phi)}$. The proof of Lemma lem:tamfactor computes a finite abelian $\ell$-group, the torsion $(\mathbb{Z}/\ell^{v_\ell(n)})^2$ of $H^1(I_q, M)$, takes its Frobenius-fixed points and identifies them with "the $\ell$-part of the Tamagawa number $c_q(E)$"; the dictionary between the two is the same statement, with Proposition 1.37 for the count of fixed points.

## 1.2 Direct and semidirect products; dihedral, quaternion and semidihedral groups

This section builds the small groups that the paper names. Proposition prop:ns says that the image of $\bar{\rho}_{E,3}$ is $\mathrm{GL}_2(\mathbb{F}_3)$, the normaliser $N_{ns}(\mathbb{F}_3)$ of a nonsplit Cartan subgroup (order 16) or the normaliser $N_s(\mathbb{F}_3)$ of a split Cartan subgroup (order 8), and that their determinant-one subgroups are $\mathrm{SL}_2(\mathbb{F}_3)$, the quaternion group $Q_8$ and a cyclic group of order 4. To check such sentences one needs the groups $D_4$, $Q_8$ and $SD_{16}$ as abstract groups, together with criteria that recognise them from a few generators and relations; these are Proposition 1.19, Proposition 1.21 and Proposition 1.23, all resting on the elementary Lemma 1.17. Semidirect products also describe the Borel subgroup, the normalisers of Cartan subgroups and, in Chapter 7, the tame Galois group of $\mathbb{Q}_p$.

**Proposition 1.12** (Direct products)**.** *The* direct product $G \times H$ *of two groups is the set of pairs with* $(g, h)(g', h') = (gg', hh')$*. If* $H$ *and* $K$ *are normal subgroups of a group* $G$ *with* $H \cap K = 1$ *and*

*$HK = G$, then elements of $H$ commute with elements of $K$, and $(h,k) \mapsto hk$ is an isomorphism $H \times K \to G$.*

*Proof.* For $h \in H$ and $k \in K$ the commutator $hkh^{-1}k^{-1}$ equals $h \cdot (kh^{-1}k^{-1}) \in H$, as $H$ is normal, and equals $(hkh^{-1}) \cdot k^{-1} \in K$, as $K$ is normal; so it lies in $H \cap K = 1$ and $hk = kh$. Hence $(h,k)(h',k') = (hh', kk') \mapsto hh'kk' = hkh'k'$, and the map is a homomorphism. If $hk = 1$, then $h = k^{-1} \in H \cap K$, so $h = k = 1$: the map is injective. It is surjective because $HK = G$. □

**Proposition 1.13** (Chinese remainder theorem for groups)**.** *Let $m, n \geq 1$. Then $C_m \times C_n$ is cyclic (and so $\simeq C_{mn}$) if and only if* $\gcd(m,n) = 1$. *Consequently $\mathbb{Z}/n \simeq \prod_{p|n} \mathbb{Z}/p^{v_p(n)}$ as groups, and a finite cyclic group is the direct product of its $\ell$-primary parts.*

*Proof.* Let $s$, $t$ generate $C_m$, $C_n$. The order of $(s^a, t^b)$ is the least common multiple of the orders of $s^a$ and $t^b$, which divides $\operatorname{lcm}(m,n)$. If $\gcd(m,n) = 1$, the element $(s,t)$ has order $\operatorname{lcm}(m,n) = mn = \#(C_m \times C_n)$ and generates. If $d = \gcd(m,n) > 1$, every element has order dividing $\operatorname{lcm}(m,n) = mn/d < mn$, so no element generates. The decomposition of $\mathbb{Z}/n$ follows by induction on the number of distinct prime factors of $n$: if $p \mid n$, write $n = p^v n'$ with $v = v_p(n)$ and $p \nmid n'$; then $\gcd(p^v, n') = 1$, so $\mathbb{Z}/n \simeq \mathbb{Z}/p^v \times \mathbb{Z}/n'$, and $n'$ has one prime factor fewer. For instance $\mathbb{Z}/12 \simeq \mathbb{Z}/4 \times \mathbb{Z}/3$, whereas $\mathbb{Z}/2 \times \mathbb{Z}/6$, of order 12, is not cyclic, since $\gcd(2,6) = 2$ and every element is killed by 6. For a cyclic group $C$ of order $n$, the factor $\mathbb{Z}/p^{v_p(n)}$ is the unique subgroup of order $p^{v_p(n)}$ (Theorem 1.4(i)), which is $C[p^\infty]$ by Proposition 1.10(iii). □

**Definition 1.14** (Semidirect products)**.** Let $N$ and $H$ be groups and $\theta\colon H \to \operatorname{Aut}(N)$ a homomorphism. The *semidirect product* $N \rtimes_\theta H$ is the set $N \times H$ with the product

$$(n,h)(n',h') = \big(n\,\theta(h)(n'),\ hh'\big).$$

The product is associative: both $\big((n,h)(n',h')\big)(n'',h'')$ and $(n,h)\big((n',h')(n'',h'')\big)$ equal $\big(n\,\theta(h)(n')\,\theta(hh')(n''),\ hh'h''\big)$, because $\theta(h)(n'\theta(h')(n'')) = \theta(h)(n')\,\theta(h)\theta(h')(n'')$ and $\theta$ is a homomorphism. The identity is $(1,1)$ and the inverse of $(n,h)$ is $\big(\theta(h^{-1})(n^{-1}), h^{-1}\big)$. The maps $n \mapsto (n,1)$ and $h \mapsto (1,h)$ identify $N$ and $H$ with subgroups, and

$$(1,h)(n,1)(1,h)^{-1} = \big(\theta(h)(n), h\big)(1,h^{-1}) = \big(\theta(h)(n), 1\big),$$

so that $hnh^{-1} = \theta(h)(n)$, as in convention (C2). The projection $(n,h) \mapsto h$ is a homomorphism with kernel $N$, so $N$ is normal.

**Proposition 1.15** (Recognising semidirect products)**.** *(i) Let $N \trianglelefteq G$ and $H \leq G$ with $N \cap H = 1$ and $NH = G$, and let $\theta(h)(n) = hnh^{-1}$. Then $(n,h) \mapsto nh$ is an isomorphism $N \rtimes_\theta H \to G$.*

*(ii) A short exact sequence* $1 \to N \to G \xrightarrow{\pi} Q \to 1$ splits*, that is, has a homomorphism $s\colon Q \to G$ with $\pi \circ s = \mathrm{id}$, if and only if $G \simeq N \rtimes_\theta Q$ for some $\theta$, compatibly with the maps from $N$ and to $Q$.*

*(iii) In $N \rtimes_\theta H$ the subgroup $H$ is normal if and only if $\theta$ is trivial, in which case $N \rtimes_\theta H = N \times H$.*

*Proof.* (i) $nh \cdot n'h' = n(hn'h^{-1})hh'$, so the map is a homomorphism. If $nh = n'h'$, then $n'^{-1}n = h'h^{-1} \in N \cap H = 1$, so it is injective; it is surjective since $NH = G$. (ii) Given $s$, put $H = s(Q)$. If $s(x) \in N$, then $x = \pi(s(x)) = 1$, so $N \cap H = 1$; and every $g$ equals $\big(g\,s(\pi(g))^{-1}\big)\,s(\pi(g))$ with the first factor in $\ker\pi = N$, so $NH = G$; now apply (i), noting

$s\colon Q \to H$ is an isomorphism. Conversely $q \mapsto (1,q)$ is a section of $N \rtimes_\theta Q \to Q$. (iii) $(n,1)(1,h)(n,1)^{-1} = (n\,\theta(h)(n^{-1}),h)$ lies in $H$ for all $n$ iff $\theta(h)(n) = n$ for all $n$. If $\theta$ is trivial the product formula is the componentwise one. □

**Example 1.16.** The following semidirect products occur in this chapter and later in the book.

(a) $D_n = C_n \rtimes C_2$, with $C_2$ acting by inversion (Definition 1.18).

(b) $\mathrm{GL}_2(F) = \mathrm{SL}_2(F) \rtimes \{\mathrm{diag}(1,d) : d \in F^\times\}$ (Proposition 1.41(ii)), and $\mathrm{B} = \mathrm{U} \rtimes \mathrm{T}$ for the upper triangular group (Proposition 1.46).

(c) $N_s = C_s \rtimes \langle w\rangle$ and $N_{ns} = C_{ns} \rtimes \langle\sigma\rangle$ for the normalisers of Cartan subgroups (Proposition 1.55), and $\mathrm{SL}_2(\mathbb{F}_3) = Q_8 \rtimes C_3$ (Proposition 1.62).

(d) $S_4 = V_4 \rtimes S_3$, with $V_4 = \{1,(12)(34),(13)(24),(14)(23)\}$ normal and $S_3$ the stabiliser of the point 4: they meet trivially and $\#V_4 \cdot \#S_3 = 24$.

(e) (Foretaste of Chapter 7.) The Galois group of the maximal tamely ramified extension of $\mathbb{Q}_p$ is a semidirect product $\left(\prod_{\ell\neq p}\mathbb{Z}_\ell(1)\right) \rtimes \widehat{\mathbb{Z}}$, in which a Frobenius element acts on the tame inertia group by $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^p$. The paper uses this in the proofs of Lemma lem:tamfactor and Lemma lem:IV, in the form $\mathrm{Frob}_q\;\sigma\;\mathrm{Frob}_q^{-1} = \sigma^q$ for an element $\sigma$ of tame inertia. Here $\mathrm{Frob}_q$ is an arithmetic Frobenius (it induces $x \mapsto x^q$ on the residue field). The sign convention matters: for the geometric Frobenius $\mathrm{Frob}_q^{-1}$ the same relation reads $\mathrm{Frob}_q^{-1}\;\sigma\;\mathrm{Frob}_q = \sigma^{q^{-1}}$, the exponent $q^{-1}$ making sense because $q$ is invertible in each $\mathbb{Z}_\ell$, $\ell \neq q$. A typical consequence is insensitive to the choice: in a linear representation $\rho$, the matrices $\rho(\sigma)$ and $\rho(\sigma)^q$ are conjugate, so the set of eigenvalues of $\rho(\sigma)$ is stable under $\lambda \mapsto \lambda^q$ (and hence under $\lambda \mapsto \lambda^{q^{-1}}$).

The groups $D_n$, $Q_8$ and $SD_{16}$ each consist of the powers of one element $a$ and one coset of them. The following lemma, used three times below, says that such a group is determined by three numbers.

**Lemma 1.17** (Normal forms)**.** *Let $n \geq 1$, $k$ and $c$ be integers with $k^2 \equiv 1 \pmod n$ and $kc \equiv c \pmod n$. Let $G$ be a group with elements $a$, $b$ such that $a^n = 1$, $bab^{-1} = a^k$, $b^2 = a^c$, and such that the $2n$ elements $a^m b^e$ $(0 \leq m < n,\ e \in \{0,1\})$ are distinct and exhaust $G$. Then the product in $G$ is given by*

$$a^m b^e \cdot a^{m'} b^{e'} = \begin{cases} a^{m+k^e m'} b^{e+e'} & \text{if } e+e' \leq 1, \\ a^{m+km'+c} & \text{if } e = e' = 1. \end{cases}$$

*In particular, if $G'$, $a'$, $b'$ satisfy the same hypotheses with the same $(n,k,c)$, then $a^m b^e \mapsto a'^m b'^e$ is an isomorphism $G \to G'$.*

*Proof.* From $bab^{-1} = a^k$ we get $ba^{m'}b^{-1} = a^{km'}$, so $b^e a^{m'} = a^{k^e m'} b^e$, and $a^m b^e a^{m'} b^{e'} = a^{m+k^e m'} b^e b^{e'}$; when $e = e' = 1$, $b^e b^{e'} = b^2 = a^c$. Since exponents of $a$ only matter modulo $n$, the right-hand side is computed from $(n,k,c)$ alone, and the bijection $a^m b^e \mapsto a'^m b'^e$ respects products. (The congruences on $k$ and $c$ are forced: $a = b^2 a b^{-2} = a^{k^2}$ and $b$ commutes with $b^2 = a^c$; they are not used in the proof.) □

**Definition 1.18** (Dihedral groups)**.** For $n \geq 2$ the *dihedral group* of order $2n$ is $D_n = \mathbb{Z}/n \rtimes_\theta \mathbb{Z}/2$, with $\theta(1)$ the inversion $x \mapsto -x$. Writing $r = (1,0)$ and $s = (0,1)$, we have $r^n = s^2 = 1$ and

$srs^{-1} = r^{-1}$, and $D_n = \{r^m s^e\}$. For $n \geq 3$, $D_n$ is the group of symmetries of a regular $n$-gon in the plane, $r$ being the rotation by $2\pi/n$ and $s$ a reflection (the familiar picture from Riley-Hobson-Bence).

**Proposition 1.19.** *(i) (Recognition) Let $G$ be a group of order $2n$ generated by an element $r$ of order $n$ and an element $s$ of order 2 with $srs^{-1} = r^{-1}$. Then $G \simeq D_n$, with $r \mapsto (1,0)$ and $s \mapsto (0,1)$.*

*(ii) Every element of $D_n$ outside $\langle r \rangle$ has order 2.*

*(iii) For $n \geq 3$ the centre of $D_n$ is trivial if $n$ is odd and is $\{1, r^{n/2}\}$ if $n$ is even.*

*(iv) $D_2 \simeq V_4$, $D_3 \simeq S_3$ and $D_6 \simeq C_2 \times S_3$.*

*(v) $D_4$ has five elements of order 2 (namely $r^2$, $s$, $rs$, $r^2s$, $r^3s$) and two of order 4 (namely $r$, $r^3$).*

*Proof.* (i) Since $srs^{-1} = r^{-1}$, both generators normalise $\langle r \rangle$, which is therefore normal, and $\langle r \rangle \langle s \rangle$ is a subgroup (Theorem 1.6(ii)) containing $r$ and $s$, hence equal to $G$. Its order is computed from the second isomorphism theorem: with $N = \langle r \rangle$ and $H = \langle s \rangle$, $\#(HN) = \#N \cdot \#(HN/N) = \#N \cdot \#(H/(H \cap N)) = \#N \, \#H/\#(H \cap N)$, that is $n \cdot 2/\#(\langle r \rangle \cap \langle s \rangle)$. This must be $\#G = 2n$, so $\langle r \rangle \cap \langle s \rangle = 1$ and $s \notin \langle r \rangle$. Hence $G = \langle r \rangle \cup \langle r \rangle s$, the $2n$ elements $r^m s^e$ ($0 \leq m < n$, $e \in \{0,1\}$) exhaust $G$, and as there are $2n$ of them in a group of order $2n$ they are distinct. Apply Lemma 1.17 with $(n,k,c) = (n,-1,0)$ to $G$ and to $D_n$. (ii) $(r^m s)^2 = r^m (s r^m s^{-1}) = r^m r^{-m} = 1$, and $r^m s \neq 1$. (iii) $r^m$ commutes with $s$ iff $r^m = r^{-m}$ iff $n \mid 2m$, which gives $m = 0$, or $m = n/2$ when $n$ is even; $r^m$ commutes with $r$. An element $r^m s$ is not central: $r(r^m s) r^{-1} = r^{m+2} s \neq r^m s$ as $n \geq 3$. (iv) $D_2$ has order 4 and all its nontrivial elements have order 2 by (ii) and $r^2 = 1$; so it is $V_4$ (it is abelian, being generated by two commuting involutions, as $srs^{-1} = r^{-1} = r$). In $S_3$, $r = (123)$ and $s = (12)$ satisfy $srs^{-1} = (213) = r^{-1}$, so (i) gives $S_3 \simeq D_3$. In $D_6$, the centre $Z = \{1, r^3\}$ and the subgroup $K = \langle r^2, s \rangle = \{1, r^2, r^4, s, r^2 s, r^4 s\}$ are normal ($K$ has index 2), $Z \cap K = 1$ and $ZK = D_6$; so $D_6 \simeq C_2 \times K$ by Proposition 1.12, and $K \simeq D_3 \simeq S_3$ by (i). (v) follows from (ii) and the orders of the powers of $r$. □

**Definition 1.20** (The quaternion group)**.** Let $\mathbf{i} = \sqrt{-1} \in \mathbb{C}$. In $\mathrm{GL}_2(\mathbb{C})$ put

$$1 = I, \qquad i = \begin{pmatrix} \mathbf{i} & 0 \\ 0 & -\mathbf{i} \end{pmatrix}, \qquad j = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k = ij = \begin{pmatrix} 0 & \mathbf{i} \\ \mathbf{i} & 0 \end{pmatrix}.$$

The *quaternion group* is $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. Direct multiplication gives $i^2 = j^2 = k^2 = ijk = -1$, $ji = -k$, and $jij^{-1} = -i = i^{-1}$; so $Q_8$ is a subgroup of order 8, and $Q_8 = \{i^m j^e : 0 \leq m < 4,\ e \in \{0,1\}\}$ with $i^4 = 1$, $jij^{-1} = i^{-1}$, $j^2 = i^2$. (These are Hamilton's quaternion units.)

**Proposition 1.21.** *(i) $-1$ is the unique element of order 2 of $Q_8$, and the six elements $\pm i, \pm j, \pm k$ have order 4.*

*(ii) $Z(Q_8) = [Q_8, Q_8] = \{\pm 1\}$, and $Q_8/\{\pm 1\} \simeq V_4$.*

*(iii) Every subgroup of $Q_8$ is normal, and every nontrivial subgroup contains $-1$. In particular $Q_8$ is not a semidirect product of two nontrivial subgroups.*

*(iv) $Q_8 \not\simeq D_4$.*

*(v) (Recognition) A non-abelian group of order* 8 *with a unique element of order* 2 *is isomorphic to* $Q_8$.

*Proof.* (i) Each of $\pm i, \pm j, \pm k$ squares to $-1$. (ii) $-1 = -I$ is central; $i$ and $j$ do not commute ($ji = -ij$), and similarly no element other than $\pm 1$ is central (each of $\pm i$, $\pm j$, $\pm k$ anticommutes with one of the others). The commutator $iji^{-1}j^{-1} = ij(-i)(-j) = (ij)^2 = k^2 = -1$, so $-1 \in [Q_8, Q_8]$. The quotient $Q_8/\{\pm 1\}$ has order 4 and every element squares to 1 there, since $x^2 \in \{\pm 1\}$; so it is $V_4$, which is abelian, and $[Q_8, Q_8] \leq \{\pm 1\}$. (iii) Every $x \neq 1$ satisfies $x = -1$ or $x^2 = -1$, so a nontrivial subgroup contains $-1$. Subgroups of order 1, 8 are normal, $\{\pm 1\}$ is central, and subgroups of order 4 have index 2 (Lemma 1.7). Two nontrivial subgroups meet in $\{\pm 1\}$ at least, so $Q_8$ is not $N \rtimes H$ with $N, H \neq 1$. (iv) $D_4$ has five elements of order 2 (Proposition 1.19(v)), $Q_8$ one. (v) Let $G$ be non-abelian of order 8 with unique involution $z$. No element has order 8 ($G$ would be cyclic), so every element other than 1, $z$ has order 4. Choose $a$ of order 4; then $a^2$ has order 2, so $a^2 = z$. Choose $b \notin \langle a \rangle$; it has order 4 and $b^2 = z = a^2$. The subgroup $\langle a \rangle$ has index 2, hence is normal, so $bab^{-1}$ is an element of order 4 of $\langle a \rangle$, that is $a$ or $a^{-1}$. If $bab^{-1} = a$, then $G = \langle a \rangle \cup \langle a \rangle b$ is abelian, a contradiction. So $bab^{-1} = a^{-1}$, and Lemma 1.17 with $(n, k, c) = (4, -1, 2)$, applied to $(G, a, b)$ and $(Q_8, i, j)$, gives $G \simeq Q_8$. □

**Definition 1.22** (The semidihedral group)**.** The *semidihedral group* of order 16 is $SD_{16} = \mathbb{Z}/8 \rtimes_\theta \mathbb{Z}/2$ with $\theta(1)$ the automorphism $x \mapsto 3x$ (of order 2, as $9 \equiv 1 \pmod 8$). Writing $a = (1, 0)$, $x = (0, 1)$, we have $a^8 = x^2 = 1$, $xax^{-1} = a^3$ and $SD_{16} = \{a^m x^e\}$.

**Proposition 1.23.** *(i)* $(a^m x)^2 = a^{4m}$*; so* $a^m x$ *has order* 2 *for* $m$ *even and order* 4 *for* $m$ *odd.*

*(ii)* $SD_{16}$ *has* 1, 5, 6, 4 *elements of orders* 1, 2, 4, 8, *and* $Z(SD_{16}) = \{1, a^4\}$.

*(iii) The subgroups of index* 2 *of* $SD_{16}$ *are exactly three:* $\langle a \rangle \simeq C_8$, $\langle a^2, x \rangle \simeq D_4$ *and* $\langle a^2, ax \rangle \simeq Q_8$.

*(iv) (Recognition) A group of order* 16 *generated by an element* $a$ *of order* 8 *and an element* $x$ *of order* 2 *with* $xax^{-1} = a^3$ *is isomorphic to* $SD_{16}$.

*Proof.* (i) $(a^m x)^2 = a^m(xa^m x^{-1}) = a^m a^{3m} = a^{4m}$, which is 1 for $m$ even and $a^4 \neq 1$ for $m$ odd. (ii) In $\langle a \rangle$ there are $1, 1, 2, 4$ elements of orders $1, 2, 4, 8$; in the coset $\langle a \rangle x$ there are four of order 2 and four of order 4 by (i). An element $a^m$ is central iff $xa^m x^{-1} = a^{3m}$ equals $a^m$, iff $a^{2m} = 1$, iff $m \in \{0, 4\}$; no $a^m x$ is central, since $a(a^m x)a^{-1} = a^{m+1}a^{-3}x = a^{m-2}x$. (iii) A subgroup $K$ of index 2 is normal with quotient of order 2 (Lemma 1.7), so it contains every square: the image of $g^2$ in $SD_{16}/K \simeq C_2$ is the square of an element of $C_2$, hence trivial. The squares include $a^2$, so $K \supseteq \langle a^2 \rangle$. The subgroup $\langle a^2 \rangle$ of order 4 is normal ($xa^2x^{-1} = a^6$), and $SD_{16}/\langle a^2 \rangle$ has order 4 and is generated by the images of $a$ and $x$, which commute ($xax^{-1}a^{-1} = a^2$) and have order dividing 2; so it is $V_4$, which has exactly three subgroups of order 2. By the correspondence theorem (Theorem 1.6(iv)) there are exactly three subgroups of index 2, namely $\langle a \rangle$, $\langle a^2, x \rangle$ and $\langle a^2, ax \rangle$. The second is dihedral by Proposition 1.19(i): it has order 8, $a^2$ has order 4, $x$ has order 2, and $xa^2x^{-1} = a^6 = a^{-2}$. The third consists of $a^{2m}$ and $a^{2m+1}x$; by (i) the latter have order 4, so $a^4$ is its unique involution, and it is non-abelian ($(ax)a^2(ax)^{-1} = a \cdot a^6 \cdot a^{-1} = a^{-2} \neq a^2$); so it is $Q_8$ by Proposition 1.21(v). (iv) As in Proposition 1.19(i), $\langle a \rangle$ is normal, $G = \langle a \rangle \langle x \rangle$, and the order 16 forces $x \notin \langle a \rangle$; apply Lemma 1.17 with $(n, k, c) = (8, 3, 0)$. □

*Remark* 1.24. There are four non-abelian groups of order 16 with a cyclic subgroup $\langle a \rangle$ of order 8 and an element $x \notin \langle a \rangle$ with $xax^{-1} = a^k$, $x^2 = a^c$: the dihedral group $D_8$ ($k = -1$, $c = 0$), $SD_{16}$

($k = 3$, $c = 0$), the generalised quaternion group $Q_{16}$ ($k = -1$, $c = 4$) and the modular group $M_{16}$ ($k = 5$, $c = 0$). In each case $(a^m x)^2 = a^{m(1+k)+c}$, and counting as in Proposition 1.23 gives the numbers of elements of orders $1, 2, 4, 8$:

| group | order 1 | order 2 | order 4 | order 8 |
|---|---|---|---|---|
| $D_8$ | 1 | 9 | 2 | 4 |
| $SD_{16}$ | 1 | 5 | 6 | 4 |
| $Q_{16}$ | 1 | 1 | 10 | 4 |
| $M_{16}$ | 1 | 3 | 4 | 8 |

So these four groups are pairwise non-isomorphic, and a group of order 16 is recognised as $SD_{16}$ by its element orders among them. We use only $SD_{16}$, through Proposition 1.23(iv).

**Theorem 1.25** (Groups of order 8)**.** *Every group of order* 8 *is isomorphic to exactly one of* $C_8$, $C_4 \times C_2$, $C_2^3$, $D_4$, $Q_8$.

*Proof.* If $G$ is abelian, Theorem 1.35 (proved in Section 1.4, independently of this theorem) writes $G$ as a sum of cyclic groups $\mathbb{Z}/2^{e_i}$ with $\sum_i e_i = 3$; the multisets of exponents are $\{3\}$, $\{2, 1\}$, $\{1, 1, 1\}$, giving exactly the first three groups. By Proposition 1.37, $\#G[2] = 2^{\#\{i\}}$ is 2, 4, 8 for them, so they are pairwise non-isomorphic. Let $G$ be non-abelian. If every element had order at most 2, then $G$ would be abelian ($ab = (ab)^{-1} = b^{-1}a^{-1} = ba$); and no element has order 8. So there is $r$ of order 4, and $\langle r\rangle$ has index 2, hence is normal. Let $s \notin \langle r\rangle$. Then $srs^{-1} \in \{r, r^{-1}\}$ (the elements of order 4 of $\langle r\rangle$), and $srs^{-1} = r$ would make $G = \langle r, s\rangle$ abelian; so $srs^{-1} = r^{-1}$. Next $s^2 \in \langle r\rangle$ (the quotient has order 2) and $s^2$ commutes with $s$, so $s^2 = s(s^2)s^{-1}$, which is $(s^2)^{-1}$ if $s^2 \in \{r, r^{-1}\}$; that would force $r^2 = 1$. So $s^2 \in \{1, r^2\}$. If $s^2 = 1$, then $G \simeq D_4$ by Proposition 1.19(i); if $s^2 = r^2$, then $G \simeq Q_8$ by Lemma 1.17 with $(n, k, c) = (4, -1, 2)$. The five groups are pairwise non-isomorphic: the three abelian ones are distinguished by $\#G[2]$, and $D_4 \not\simeq Q_8$ by Proposition 1.21(iv). □

*In the paper* 1.26. The proof of Proposition prop:ns names an isomorphism type, "the quaternion group $Q_8$", and Section sec:open of the paper calls the case of image $N_s(\mathbb{F}_3)$ the dihedral case. In Section 1.6 these types are identified by the recognition criteria: $N_s(\mathbb{F}_3) \simeq D_4$ by Proposition 1.19(i), the determinant-one subgroup of $N_{ns}(\mathbb{F}_3)$ is $Q_8$ by Proposition 1.21(v), and $N_{ns}(\mathbb{F}_3) \simeq SD_{16}$ by Proposition 1.23(iv). Theorem 1.25 is context; nothing in the paper depends on it.

## 1.3 Group actions, orbit-stabiliser, $p$-groups and the Sylow theorems

Groups enter the paper through their actions: the Galois group acts on the torsion points of $E$, the decomposition and inertia groups are stabilisers (Chapter 6, Chapter 7), and the four roots of the 3-division polynomial correspond to the four lines of $E[3] \simeq \mathbb{F}_3^2$, permuted by the image of $\bar{\rho}_{E,3}$. The last sentence of Proposition prop:ns, on the factorisation of that polynomial, is a statement about the orbits of a subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$ on these four lines, and Proposition 1.29 is the tool that compares the orbits of a group and of a subgroup of index 2. The Sylow theorems are used to locate the 2-subgroups and the 3-subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ in Section 1.6.

**Definition 1.27** (Group actions)**.** A (left) *action* of a group $G$ on a set $X$ is a map $G \times X \to X$, $(g, x) \mapsto gx$, with $1x = x$ and $g(hx) = (gh)x$. Equivalently, it is a homomorphism $G \to \mathrm{Sym}(X)$ to the group of bijections of $X$ (the *permutation representation*). Its kernel is the kernel of this homomorphism; the action is *faithful* if the kernel is trivial. The *orbit* of $x$ is $Gx = \{gx : g \in G\}$, the *stabiliser* of $x$ is $G_x = \{g : gx = x\}$, a subgroup, and the *fixed set* is $X^G = \{x : gx = x \ \forall g\}$; for one element $g$ we write $X^g$ for its fixed points. The action is *transitive* if there is one orbit. An action restricts to every subgroup $H \leq G$. Examples: $G$ acts on $G/H$ by $g \cdot xH = gxH$; $G$ acts on itself and on the set of its subgroups by conjugation, with stabilisers $C_G(g)$ and $N_G(H)$; $\mathrm{GL}_2(F)$ acts on $F^2$, on $F^2 \setminus \{0\}$ and on the set $\mathbb{P}^1(F)$ of lines (Proposition 1.44).

**Theorem 1.28** (Orbit-stabiliser)**.** *Let $G$ act on $X$ and let $x \in X$. Then $gG_x \mapsto gx$ is a well-defined bijection $G/G_x \to Gx$, compatible with the actions of $G$; so $\#Gx = [G : G_x]$ when $G$ is finite. Moreover $G_{gx} = gG_xg^{-1}$. The orbits partition $X$, and if $X$ is finite and $x_1, \dots, x_r$ are representatives of the orbits, then $\#X = \sum_{i=1}^{r} [G : G_{x_i}]$.*

*Proof.* $gx = hx$ iff $h^{-1}g \in G_x$ iff $gG_x = hG_x$: so the map is well defined and injective; it is surjective by definition of $Gx$, and $h \cdot gG_x = hgG_x \mapsto hgx$. If $h \in G_x$, then $(ghg^{-1})(gx) = gx$, so $gG_xg^{-1} \subseteq G_{gx}$; applying this to $g^{-1}$ and $gx$ gives the reverse inclusion. The relation "$y \in Gx$" is an equivalence relation ($x = 1x$; $y = gx \Rightarrow x = g^{-1}y$; $z = hy$, $y = gx \Rightarrow z = hgx$), whose classes are the orbits. Summing the orbit sizes gives the last formula. □

A first example, which recurs throughout the chapter: $\mathrm{GL}_2(\mathbb{F}_3)$, of order 48 (Proposition 1.43), acts transitively on the four lines of $\mathbb{F}_3^2$ (Proposition 1.44), and the stabiliser of $\infty = [e_1]$ is the upper triangular group $\mathrm{B}(\mathbb{F}_3)$, of order $3 \cdot 2 \cdot 2 = 12$ (the entry $b$ is arbitrary and $a, d \in \{\pm 1\}$). Orbit-stabiliser gives $48/12 = 4$ lines, as it must. In the other direction, the formula is often used to compute a stabiliser from an orbit: the stabiliser of the pair $\{\infty, 0\}$ has order $48/6 = 8$, since $\mathrm{GL}_2(\mathbb{F}_3)$ is transitive on the six 2-element sets of lines (Proposition 1.44(v)); it is $N_s(\mathbb{F}_3)$ (Proposition 1.63(ii)).

**Proposition 1.29** (Orbits of a subgroup)**.** *Let $G$ act transitively on a finite set $X$, let $x \in X$ and let $H \leq G$.*

*(i) The map $HgG_x \mapsto H \cdot gx$ is a bijection from the double cosets $H\backslash G/G_x$ to the $H$-orbits on $X$. In particular $H$ is transitive on $X$ if and only if $G = HG_x$.*

*(ii) If $H \trianglelefteq G$, then $G$ permutes the $H$-orbits transitively; all $H$-orbits have the same size, which divides $\#X$.*

*(iii) If $[G : H] = 2$, then either $H$ is transitive on $X$, or $X$ is the union of exactly two $H$-orbits, each of size $\#X/2$, and every element of $G \setminus H$ interchanges them.*

*Proof.* (i) Every element of $X$ is $gx$ for some $g$. Now $Hgx = Hg'x$ iff $g'x = hgx$ for some $h \in H$ iff $g^{-1}h^{-1}g' \in G_x$ for some $h$ iff $g' \in HgG_x$. So the map is well defined and injective. It is surjective because every $H$-orbit is $Hy$ for some $y \in X$, and $y = gx$ for some $g$ by transitivity of $G$, so $Hy$ is the image of $HgG_x$. There is one $H$-orbit iff there is one double coset, iff $HG_x = G$ (the double coset of 1 is $HG_x$, and it is all of $G$ exactly when there are no others). (ii) For $g \in G$ and $y \in X$, $g(Hy) = gHg^{-1}gy = H(gy)$; so $g$ maps $H$-orbits to $H$-orbits, bijectively (with inverse $g^{-1}$). Given two orbits $Hy$, $Hy'$, choose $g$ with $gy = y'$ (transitivity of $G$); then $g(Hy) = Hy'$. Hence all orbits have the size of one of them, say $s$, and $\#X = s \cdot$ (number of orbits). (iii) $H$ is normal by Lemma 1.7. The group $G$ acts transitively on the set $\Omega$ of $H$-orbits by (ii), and $H$

acts trivially on $\Omega$; so the action factors through $G/H$, of order 2, and $\#\Omega \in \{1,2\}$. If $\#\Omega = 2$, each orbit has size $\#X/2$ by (ii); the stabiliser of one $H$-orbit is a subgroup containing $H$ and different from $G$ (the action on $\Omega$ is transitive), hence equal to $H$, so every element of $G \setminus H$ moves it to the other one. □

**Lemma 1.30** (Orbit counting)**.** *Let a finite group $G$ act on a finite set $X$. The number of orbits is* $\dfrac{1}{\#G}\displaystyle\sum_{g\in G} \#X^g$.

*Proof.* Count the pairs $(g,x)$ with $gx = x$ in two ways: $\sum_g \#X^g = \sum_x \#G_x$. By Theorem 1.28, $\#G_x = \#G/\#Gx$, so the elements of one orbit $O$ contribute $\#O \cdot \#G/\#O = \#G$, and the sum is $\#G$ times the number of orbits. □

For example, let $C_4 = \langle J\rangle = \{\pm I, \pm J\} \leq \mathrm{GL}_2(\mathbb{F}_3)$ act on the four lines. The scalars $\pm I$ fix all four lines; $\pm J$ fix none, since their characteristic polynomial $X^2+1$ has no root in $\mathbb{F}_3$, so they have no eigenvector over $\mathbb{F}_3$. The lemma gives $(4+4+0+0)/4 = 2$ orbits. Since the scalars act trivially, the action factors through $C_4/\{\pm I\}$, of order 2, so every orbit has size 1 or 2; there is no fixed line, so the two orbits both have size 2, recovering Proposition 1.63(iv) without listing the action.

**Theorem 1.31** (Class equation; $p$-groups)**.** *Let $G$ be finite. Then $\#G = \#Z(G) + \sum_i [G : C_G(g_i)]$, the sum over representatives $g_i$ of the conjugacy classes with more than one element. Let $p$ be a prime.*

*(i) If a finite $p$-group $P$ acts on a finite set $X$, then $\#X \equiv \#X^P \pmod p$.*

*(ii) A nontrivial finite $p$-group has nontrivial centre.*

*(iii) A group of order $p^2$ is abelian.*

*(iv) A group of order $p^n$ has a normal subgroup of order $p^k$ for every $0 \leq k \leq n$, and every subgroup of index $p$ in a finite $p$-group is normal.*

*Proof.* The class equation is Theorem 1.28 for the conjugation action: the fixed points are the central elements, and the class of $g_i$ has $[G : C_G(g_i)]$ elements (Proposition 1.9). (i) Each orbit has size $[P : P_x]$, a power of $p$, which is 1 exactly for the fixed points; so $\#X - \#X^P$ is a sum of positive powers of $p$. (ii) Apply (i) to the conjugation action of $P$ on itself: $\#Z(P) \equiv \#P \equiv 0 \pmod p$, and $\#Z(P) \geq 1$, so $\#Z(P) \geq p$. (iii) First, if $G/Z(G)$ is cyclic, generated by the class of $g$, then every element is $g^m z$ with $z$ central, and any two such elements commute; so $G$ is abelian. If $\#G = p^2$, then $\#Z(G) \in \{p, p^2\}$ by (ii); if it were $p$, then $G/Z(G)$ would have order $p$, hence be cyclic, and $G$ would be abelian, so $Z(G) = G$, a contradiction. (iv) Induction on $n$. For $k = 0$ the trivial subgroup does, which settles $n = 0$. For $n \geq 1$, $Z(G)$ is a nontrivial abelian $p$-group by (ii), so it has an element $z$ of order $p$ (Proposition 1.10(i)), and $\langle z\rangle$ is normal, being central. For $k \geq 1$, by induction $G/\langle z\rangle$ (order $p^{n-1}$) has a normal subgroup of order $p^{k-1}$, whose preimage is normal of order $p^k$ (Theorem 1.6(iv)). Finally let $H \leq P$ have index $p$. Then $H$ acts on $P/H$ by left multiplication; by (i), the number of fixed cosets is $\equiv p \equiv 0 \pmod p$, and the coset $H$ is fixed, so there are at least $p$ fixed cosets, that is, all of them. So $hgH = gH$ for all $h \in H$, $g \in P$, that is, $g^{-1}Hg \subseteq H$ for all $g$, and $H$ is normal. □

**Theorem 1.32** (Cauchy)**.** *If a prime $p$ divides the order of a finite group $G$, then $G$ has an element of order $p$.*

*Proof.* (McKay's argument.) Let $X = \{(g_1, \dots, g_p) \in G^p : g_1 g_2 \cdots g_p = 1\}$. The first $p-1$ entries may be chosen freely and determine the last, so $\#X = \#G^{p-1} \equiv 0 \pmod p$. The cyclic group $C_p$ acts on $X$ by rotation $(g_1, \dots, g_p) \mapsto (g_2, \dots, g_p, g_1)$; this preserves $X$, since $g_2 \cdots g_p g_1 = g_1^{-1}(g_1 \cdots g_p) g_1 = 1$. The fixed points are the constant tuples $(g, \dots, g)$ with $g^p = 1$. By Theorem 1.31(i) their number is $\equiv \#X \equiv 0 \pmod p$; it is at least 1 (take $g = 1$), hence at least $p$. So some $g \neq 1$ has $g^p = 1$, and it has order $p$. □

**Theorem 1.33** (Sylow)**.** *Let $G$ be a finite group of order $p^a m$ with $p$ prime and $p \nmid m$. A subgroup of order $p^a$ is a* Sylow $p$-subgroup.

*(i) $G$ has a Sylow $p$-subgroup.*

*(ii) Every $p$-subgroup of $G$ is contained in a Sylow $p$-subgroup, and any two Sylow $p$-subgroups are conjugate.*

*(iii) The number $n_p$ of Sylow $p$-subgroups equals $[G : N_G(P)]$ for any Sylow $p$-subgroup $P$; it divides $m$, and $n_p \equiv 1 \pmod p$.*

*(iv) A Sylow $p$-subgroup $P$ is normal if and only if $n_p = 1$; a normal Sylow $p$-subgroup contains every element of $p$-power order of $G$.*

*Proof.* (i) (Wielandt's argument.) First, $\binom{p^a m}{p^a} \equiv m \pmod p$: in $\mathbb{F}_p[X]$ we have $(1+X)^p = 1 + X^p$, since $p$ divides $\binom{p}{i}$ for $0 < i < p$; hence $(1+X)^{p^a m} = (1 + X^{p^a})^m$, and comparing the coefficients of $X^{p^a}$ gives the claim. Let $\Omega$ be the set of subsets of $G$ with $p^a$ elements, on which $G$ acts by left multiplication, $gS = \{gs : s \in S\}$. As $\#\Omega = \binom{p^a m}{p^a} \equiv m \not\equiv 0 \pmod p$ and $\#\Omega$ is the sum of the orbit sizes, some orbit $GS$ has size prime to $p$. Its stabiliser $G_S = \{g : gS = S\}$ has order $\#G/\#GS = p^a m/\#GS$ (Theorem 1.28), and since $\#GS$ is prime to $p$, all of $p^a$ survives: $p^a \mid \#G_S$. On the other hand, fix $s \in S$; for $g \in G_S$ we have $gs \in gS = S$, so the right coset $G_S s$ lies in $S$, and $\#G_S = \#G_S s \leq \#S = p^a$. Hence $\#G_S = p^a$, and $G_S$ is a Sylow $p$-subgroup. (ii) Let $Q$ be a $p$-subgroup and $P$ a Sylow $p$-subgroup. Then $Q$ acts on $G/P$ by left multiplication, and $\#(G/P) = m \not\equiv 0 \pmod p$; by Theorem 1.31(i) there is a fixed coset $gP$. Then $QgP = gP$, so $g^{-1}Qg \subseteq P$ and $Q \subseteq gPg^{-1}$, which is a Sylow $p$-subgroup. If $Q$ is itself Sylow, comparing orders gives $Q = gPg^{-1}$. (iii) By (ii), $G$ acts transitively by conjugation on the set $\mathrm{Syl}_p$ of Sylow $p$-subgroups, with stabiliser $N_G(P)$; so $n_p = [G : N_G(P)]$, which divides $[G : P] = m$ since $P \leq N_G(P)$. Let $P$ act on $\mathrm{Syl}_p$ by conjugation. If $Q$ is fixed, then $P \leq N_G(Q)$, and $P$, $Q$ are Sylow $p$-subgroups of $N_G(Q)$, in which $Q$ is normal; by (ii) applied to the group $N_G(Q)$ they are conjugate in $N_G(Q)$, and $Q$ is normal there, so $P = Q$. So $P$ has exactly one fixed point, and $n_p \equiv 1 \pmod p$ by Theorem 1.31(i). (iv) The first statement is $n_p = [G : N_G(P)]$. If $P$ is normal and $g$ has $p$-power order, then $\langle g \rangle$ lies in some Sylow $p$-subgroup by (ii), which is $P$. □

**Example 1.34.** (a) A group of order 15 is cyclic: $n_3$ divides 5 and is $\equiv 1 \pmod 3$, so $n_3 = 1$; $n_5$ divides 3 and is $\equiv 1 \pmod 5$, so $n_5 = 1$. The two Sylow subgroups $P_3 \simeq C_3$, $P_5 \simeq C_5$ are normal, meet trivially and $\#P_3P_5 = 15$; so $G \simeq C_3 \times C_5 \simeq C_{15}$ by Proposition 1.12 and Proposition 1.13.

(b) In $S_4$ (order $24 = 2^3 \cdot 3$): the eight 3-cycles lie in four subgroups of order 3, two in each, so $n_3 = 4$. The group $S_4$ permutes transitively the three partitions $\{\{1,2\},\{3,4\}\}$, $\{\{1,3\},\{2,4\}\}$, $\{\{1,4\},\{2,3\}\}$ of $\{1,2,3,4\}$ into two pairs, so each stabiliser has order $24/3 = 8$ by Theorem 1.28; it is a Sylow 2-subgroup. The stabiliser of $\{\{1,2\},\{3,4\}\}$ contains (1324), of order 4, and (12), with $(12)(1324)(12) = (1423) = (1324)^{-1}$, so it is dihedral by Proposition 1.19(i). This Sylow

2-subgroup is not normal (it contains (12) but not its conjugate (13), which sends $\{1,2\}$ to $\{3,2\}$), so $n_2 \neq 1$; as $n_2 \mid 3$, $n_2 = 3$, and the three stabilisers are the three Sylow 2-subgroups.

(c) In $\mathrm{GL}_2(\mathbb{F}_3)$ (order $48 = 2^4 \cdot 3$): $n_3 = 4$, the four subgroups $\mathrm{U}_L$ of Proposition 1.48(v), one for each line $L$ of $\mathbb{F}_3^2$; and $n_2 = 3$, the three conjugates of $N_{ns}(\mathbb{F}_3)$ (Proposition 1.64(iii)).

(d) In $\mathrm{SL}_2(\mathbb{F}_3)$ (order 24): $n_2 = 1$, the Sylow 2-subgroup being $Q_8$ (Proposition 1.62(ii)), and $n_3 = 4$.

## 1.4 Finite abelian groups

The paper's local invariants are orders of finite abelian groups and their $\ell$-parts: the geometric component groups $\Phi_q^{\mathrm{geom}}$, of the shapes $\mathbb{Z}/n$, $\mathbb{Z}/2$, $\mathbb{Z}/3$, $(\mathbb{Z}/2)^2$, $\mathbb{Z}/4$ (Chapter 10), the Tamagawa numbers, and the torsion subgroup $H^1(I_q, M)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2$ in the proof of Lemma lem:tamfactor, whose Frobenius-fixed points are counted there. This section proves the structure theorem and the counting rule $\#A[n] = \#A/nA$ used for that count, and shows that the multiplicative group of a finite field is cyclic, which Lemma lem:h0 uses for $\mathbb{F}_\ell^\times$. Groups are written additively; $A[n] = \{a \in A : na = 0\}$.

**Theorem 1.35** (Structure of finite abelian groups)**.** *Every finite abelian group $A$ is isomorphic to $\bigoplus_{i=1}^{s} \mathbb{Z}/\ell_i^{e_i}$ for primes $\ell_i$ and integers $e_i \geq 1$, and the multiset of prime powers $\ell_i^{e_i}$ (the* elementary divisors*) is determined by $A$. Equivalently, $A \simeq \mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_r$ with $1 < d_1 \mid d_2 \mid \cdots \mid d_r$, and the* invariant factors *$d_i$ are determined by $A$. For each prime $\ell$, the numbers $\#A[\ell^k]$, $k \geq 1$, determine $A[\ell^\infty]$ up to isomorphism: the number of $i$ with $\ell_i = \ell$ and $e_i \geq k$ is $\log_\ell(\#A[\ell^k]/\#A[\ell^{k-1}])$.*

*Proof.* By Proposition 1.10(iv), $A = \bigoplus_\ell A[\ell^\infty]$, so it suffices to treat a finite abelian $\ell$-group $A$.

*Existence.* We show by induction on $\#A$: if $a \in A$ has maximal order $\ell^e$, then $A = \langle a \rangle \oplus B$ for some subgroup $B$; applying this again to $B$ gives the decomposition. If $A = \langle a \rangle$, take $B = 0$. Otherwise choose $b \notin \langle a \rangle$ of least possible order. Then $\ell b$ has smaller order, so $\ell b \in \langle a \rangle$, say $\ell b = ta$. Since $\ell^{e-1}(ta) = \ell^e b = 0$ (the order of $b$ is at most $\ell^e$), the element $ta$ is not a generator of $\langle a \rangle$, so $\ell \mid t$, say $t = \ell t'$. Put $c = b - t'a$. Then $\ell c = 0$ and $c \notin \langle a \rangle$, so $C = \langle c \rangle$ has order $\ell$ and $C \cap \langle a \rangle = 0$. In $\bar{A} = A/C$ the image $\bar{a}$ has order $\ell^e$ (if $\ell^{e-1}a \in C$, then $\ell^{e-1}a \in C \cap \langle a \rangle = 0$), and this is the maximal order in $\bar{A}$, as orders can only drop in a quotient. By induction $\bar{A} = \langle \bar{a} \rangle \oplus \bar{B}$; let $B$ be the preimage of $\bar{B}$, which contains $C$. If $ma \in B$, then $m\bar{a} \in \langle \bar{a} \rangle \cap \bar{B} = 0$, so $ma \in C \cap \langle a \rangle = 0$; thus $\langle a \rangle \cap B = 0$. For $x \in A$ write $\bar{x} = m\bar{a} + \bar{y}$ with $y \in B$; then $x - ma - y \in C \subseteq B$, so $x \in \langle a \rangle + B$. Hence $A = \langle a \rangle \oplus B$.

*Uniqueness.* If $A \simeq \bigoplus_i \mathbb{Z}/\ell^{e_i}$, then $A[\ell^k] \simeq \bigoplus_i (\mathbb{Z}/\ell^{e_i})[\ell^k]$, and $(\mathbb{Z}/\ell^e)[\ell^k]$ has order $\ell^{\min(k,e)}$ (Theorem 1.4(i)). So $\#A[\ell^k] = \ell^{\sum_i \min(k,e_i)}$, and, since $\min(k,e_i) - \min(k-1,e_i)$ is 1 if $e_i \geq k$ and 0 otherwise, $\#A[\ell^k]/\#A[\ell^{k-1}] = \ell^{\#\{i : e_i \geq k\}}$. The left side is defined in terms of $A$ alone (an isomorphism $A \to A'$ maps $A[\ell^k]$ onto $A'[\ell^k]$), so the numbers $N_k = \#\{i : e_i \geq k\}$, $k \geq 1$, are the same for any two decompositions. They determine the multiset $\{e_i\}$: exactly $N_k - N_{k+1}$ of the exponents equal $k$.

*Invariant factors.* Given the elementary divisors, list for each $\ell$ the exponents in decreasing order $e_{\ell,1} \geq e_{\ell,2} \geq \cdots$ (padded with zeros), and put $d_r = \prod_\ell \ell^{e_{\ell,1}}$, $d_{r-1} = \prod_\ell \ell^{e_{\ell,2}}$, and so on. By Proposition 1.13, $\mathbb{Z}/d_j \simeq \bigoplus_\ell \mathbb{Z}/\ell^{e_{\ell,r+1-j}}$, so $A \simeq \bigoplus_j \mathbb{Z}/d_j$, with $d_1 \mid \cdots \mid d_r$. Conversely the elementary divisors of $\bigoplus_j \mathbb{Z}/d_j$ are the prime powers exactly dividing the $d_j$, so the $d_j$ are determined by the elementary divisors, hence by $A$. □

For example, $A = \mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/9$ has elementary divisors $2, 4, 3, 9$. Listing exponents in decreasing order gives $(2,1)$ for $\ell = 2$ and $(2,1)$ for $\ell = 3$, so $d_2 = 4 \cdot 9 = 36$, $d_1 = 2 \cdot 3 = 6$, and

$A \simeq \mathbb{Z}/6 \oplus \mathbb{Z}/36$. The counting rule recovers the 2-part: $\#A[2] = 2^2$ and $\#A[4] = 2^3$, so $N_1 = 2$, $N_2 = 1$, and there is one exponent equal to 1 and one equal to 2, that is, $A[2^\infty] \simeq \mathbb{Z}/2 \oplus \mathbb{Z}/4$. Note that the order alone does not determine $A$: $\mathbb{Z}/2 \oplus \mathbb{Z}/4$ and $\mathbb{Z}/8$ both have order 8, but $\#A[2]$ is 4 and 2 respectively.

**Corollary 1.36** (Cyclicity criteria)**.** *(i) For a finite abelian group $A$ the following are equivalent: $A$ is cyclic; $\#A[\ell] \le \ell$ for every prime $\ell$; $\#A[d] \le d$ for every $d \ge 1$.*

*(ii) Every finite subgroup of the multiplicative group $K^\times$ of a field $K$ is cyclic. In particular $\mathbb{F}_p^\times$ is cyclic of order $p-1$, more generally $\mathbb{F}_q^\times$ is cyclic of order $q-1$, the group $\mu_n(K)$ of $n$-th roots of unity in $K$ is cyclic, and the multiplicative group of the field $\mathbb{F}_q[\sqrt{\varepsilon}]$ of Definition 1.54 is cyclic of order $q^2-1$.*

*Proof.* (i) If $A \simeq \mathbb{Z}/n$, then $\#A[d] = \gcd(d,n) \le d$ by Theorem 1.4(ii). The third condition implies the second. If $\#A[\ell] \le \ell$ for all $\ell$, then in the decomposition of Theorem 1.35 each $A[\ell^\infty] \simeq \bigoplus_i \mathbb{Z}/\ell^{e_i}$ has $\#A[\ell] = \ell^{\#\{i\}}$, so it has at most one summand; hence $A \simeq \bigoplus_\ell \mathbb{Z}/\ell^{e_\ell}$, which is cyclic by Proposition 1.13. (ii) Let $A \le K^\times$ be finite. The elements of $A$ with $x^d = 1$ are roots of the polynomial $X^d - 1$, which has at most $d$ roots in the field $K$; so $\#A[d] \le d$ for all $d$, and $A$ is cyclic by (i). □

The proof of (ii) is not constructive: it does not exhibit a generator. In small cases one finds one by trial. In $\mathbb{F}_7^\times$, of order 6, the powers of 2 are $2, 4, 1$, so 2 has order 3, while the powers of 3 are $3, 2, 6, 4, 5, 1$, so 3 generates. By Theorem 1.4(iii) there are $\varphi(6) = 2$ generators, 3 and $3^5 = 5$. In $\mathbb{F}_3^\times = \{\pm 1\}$ the generator is $-1$, and this is the case $\ell - 1 = 2$ of Lemma lem:h0. The hypothesis that $K$ is a field is essential: in the ring $\mathbb{Z}/8$ the polynomial $X^2 - 1$ has the four roots $\pm 1, \pm 3$, and $(\mathbb{Z}/8)^\times \simeq V_4$ is not cyclic.

**Proposition 1.37** (Counting torsion)**.** *Let $A$ be a finite abelian group and $n \ge 1$. The sequence $0 \to A[n] \to A \xrightarrow{n} A \to A/nA \to 0$ is exact, and $\#A[n] = \#(A/nA)$. If $A \simeq \bigoplus_i \mathbb{Z}/\ell^{e_i}$ is an $\ell$-group, then $\#A[\ell^k] = \#(A/\ell^k A) = \ell^{\sum_i \min(k, e_i)}$.*

*Proof.* Exactness is the definition of $A[n]$ (the kernel of multiplication by $n$) and of $A/nA$ (its cokernel). By the first isomorphism theorem $\#nA = \#A/\#A[n]$, so $\#(A/nA) = \#A/\#nA = \#A[n]$. The formula for $\#A[\ell^k]$ was proved in the proof of Theorem 1.35. □

The two groups $A[n]$ and $A/nA$ have the same order but are in general different subquotients of $A$. For $A = \mathbb{Z}/4$ and $n = 2$, $A[2] = \{0, 2\} = 2A$ and $A/2A \simeq \mathbb{Z}/2$; for $A = \mathbb{Z}/2 \oplus \mathbb{Z}/4$, $A[2] = \{0,1\} \oplus \{0,2\}$ and $A/2A \simeq (\mathbb{Z}/2)^2$, both of order $4 = 2^{\min(1,1)+\min(1,2)}$. The equality of orders is special to finite groups: for $A = \mathbb{Z}$ one has $A[2] = 0$ but $A/2A \simeq \mathbb{Z}/2$.

*In the paper* 1.38. In the proof of Lemma lem:tamfactor the Frobenius-fixed part of $(\mathbb{Z}/\ell^{v_\ell(n)})^2$ is described by two congruences, $(q-1)y \equiv 0$ and $c'y \equiv 0 \pmod{\ell^{v_\ell(n)}}$, for $y \in \mathbb{Z}/\ell^{v_\ell(n)}$. The solutions form $(\mathbb{Z}/\ell^v)[\ell^w]$ with $v = v_\ell(n)$ and $w = \min(v_\ell(q-1), v_\ell(c'))$, a subgroup of order $\ell^{\min(v,w)} = \ell^{\min(v_\ell(n), v_\ell(q-1), v_\ell(c'))}$ by Theorem 1.4(i) and Proposition 1.37: indeed $\gamma y \equiv 0 \pmod{\ell^v}$ holds iff $\ell^{\min(v, v_\ell(\gamma))} y \equiv 0$, by Theorem 1.4(ii). The lemma itself is proved in Chapter 18; here only the count is at stake.

**Example 1.39.** The geometric component groups of an elliptic curve (Chapter 10) are $\mathbb{Z}/n$ (type $\mathrm{I}_n$), 0 (types II, II*), $\mathbb{Z}/2$ (types III, III*), $\mathbb{Z}/3$ (types IV, IV*), and $(\mathbb{Z}/2)^2$ or $\mathbb{Z}/4$ (types $\mathrm{I}_n^*$, according to the parity of $n$). Their $\ell$-parts are, by Proposition 1.10, of order $\ell^{v_\ell(n)}$ for $\mathbb{Z}/n$;

for odd $\ell$ only $\mathbb{Z}/n$ and $\mathbb{Z}/3$ (when $\ell = 3$) can contribute. The groups $(\mathbb{Z}/2)^2$ and $\mathbb{Z}/4$ have the same order but are not isomorphic, since $\#(\mathbb{Z}/2)^2[2] = 4$ and $\#(\mathbb{Z}/4)[2] = 2$. The paper's Section 1.4 and the rule built on Definition def:terms use the component groups only through the numbers $v_\ell(\#\Phi_q^{\mathrm{geom}})$.

## 1.5 The groups $\mathrm{GL}_2$, $\mathrm{SL}_2$, $\mathrm{PGL}_2$, $\mathrm{PSL}_2$; Borel, unipotent and Cartan subgroups

The Galois representation $\bar\rho_{E,\ell}$ takes values in $\mathrm{GL}_2(\mathbb{F}_\ell)$, and the paper's arguments at $\ell = 3$ are statements about its subgroups. This section introduces the subgroups that occur: the Borel subgroups (the image when $E$ has a rational $\ell$-isogeny, the Eisenstein case of Section sec:eis and Conjecture conj:eis), the unipotent elements (the image of inertia in Lemma lem:IV, and "minus a unipotent" in Remark rem:types), the split and nonsplit Cartan subgroups and their normalisers (Proposition prop:ns, Section sec:anomaly, Remark rem:ns), and the action on the projective line, which carries the four lines of $\mathbb{F}_3^2$ of Proposition prop:ns. Throughout, $F$ is a field and $\mathbb{F}_q$ a finite field with $q$ elements (convention (C4)).

**Definition 1.40** (Linear groups)**.** $\mathrm{GL}_2(F)$ is the group of invertible $2 \times 2$ matrices over $F$; $\mathrm{SL}_2(F) = \ker(\det\colon \mathrm{GL}_2(F) \to F^\times)$; $\mathrm{PGL}_2(F) = \mathrm{GL}_2(F)/F^\times I$; and $\mathrm{PSL}_2(F) = \mathrm{SL}_2(F)/\{\pm I\}$.

**Proposition 1.41.** *(i)* $Z(\mathrm{GL}_2(F)) = F^\times I$ *and* $Z(\mathrm{SL}_2(F)) = \{\pm I\}$.

*(ii)* $\det$ *is surjective,* $d \mapsto \mathrm{diag}(1, d)$ *is a homomorphic section, and* $\mathrm{GL}_2(F) \simeq \mathrm{SL}_2(F) \rtimes F^\times$.

*(iii)* $\mathrm{SL}_2(F)$ *is generated by the elementary matrices* $\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right)$ *and* $\left(\begin{smallmatrix}1 & 0\\ c & 1\end{smallmatrix}\right)$, $b, c \in F$.

*(iv) If* $\mathrm{char}\, F \neq 2$, *the only element of order* 2 *of* $\mathrm{SL}_2(F)$ *is* $-I$.

*(v) The image of* $\mathrm{SL}_2(F)$ *in* $\mathrm{PGL}_2(F)$ *is isomorphic to* $\mathrm{PSL}_2(F)$, *and its index is* $[F^\times : (F^\times)^2]$.

*Proof.* (i) Let $g = \left(\begin{smallmatrix}a & b\\ c & d\end{smallmatrix}\right)$ commute with $e = \left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$: $ge = \left(\begin{smallmatrix}a & a+b\\ c & c+d\end{smallmatrix}\right)$ and $eg = \left(\begin{smallmatrix}a+c & b+d\\ c & d\end{smallmatrix}\right)$, so $c = 0$ and $a = d$. Commuting with $\left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$ gives $b = 0$ in the same way. So $g$ is scalar; conversely scalars are central. Both elementary matrices lie in $\mathrm{SL}_2(F)$, so the same computation gives $Z(\mathrm{SL}_2(F)) = \{aI : a^2 = 1\} = \{\pm I\}$. (ii) $\det \mathrm{diag}(1, d) = d$, and $d \mapsto \mathrm{diag}(1, d)$ is a homomorphism; apply Proposition 1.15(ii). (iii) Let $g = \left(\begin{smallmatrix}a & b\\ c & d\end{smallmatrix}\right) \in \mathrm{SL}_2(F)$. If $c = 0$, multiplying on the left by $\left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$ replaces $c$ by $a \neq 0$. So assume $c \neq 0$. Left multiplication by $\left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right)$ with $x = (1 - a)/c$ makes the top left entry 1; then left multiplication by $\left(\begin{smallmatrix}1 & 0\\ -c & 1\end{smallmatrix}\right)$ clears the bottom left entry, leaving $\left(\begin{smallmatrix}1 & b'\\ 0 & d'\end{smallmatrix}\right)$ with $d' = 1$ (determinant 1); and left multiplication by $\left(\begin{smallmatrix}1 & -b'\\ 0 & 1\end{smallmatrix}\right)$ gives $I$. So $g$ is a product of elementary matrices. (iv) If $g^2 = I$, the minimal polynomial of $g$ divides $X^2 - 1 = (X - 1)(X + 1)$, which has distinct roots as $1 \neq -1$; so $g$ is diagonalisable with eigenvalues in $\{\pm 1\}$, and $\det g = 1$ forces them to be equal, so $g = \pm I$. (v) $\mathrm{SL}_2(F) \cap F^\times I = \{\pm I\}$, so the image is $\mathrm{SL}_2(F)F^\times/F^\times \simeq \mathrm{SL}_2(F)/\{\pm I\}$ by Theorem 1.6(ii). The index of this image in $\mathrm{PGL}_2(F) = \mathrm{GL}_2(F)/F^\times I$ is $[\mathrm{GL}_2(F) : \mathrm{SL}_2(F)F^\times I]$, by the correspondence theorem. Now compose $\det\colon \mathrm{GL}_2(F) \to F^\times$, which is surjective by (ii), with the projection $F^\times \to F^\times/(F^\times)^2$. The kernel of the composite consists of the $g$ with $\det g = \lambda^2$ for some $\lambda \in F^\times$; for such $g$ the matrix $\lambda^{-1}g$ has determinant $\lambda^{-2} \det g = 1$, so $g = (\lambda^{-1}g)(\lambda I) \in \mathrm{SL}_2(F)F^\times I$. Conversely $\det(s\lambda I) = \lambda^2$ for $s \in \mathrm{SL}_2(F)$. So the kernel is $\mathrm{SL}_2(F)F^\times I$, and the first isomorphism theorem gives $\mathrm{GL}_2(F)/\mathrm{SL}_2(F)F^\times I \simeq F^\times/(F^\times)^2$. For $F = \mathbb{F}_q$ with $q$ odd the squares form the subgroup

of index 2 of the cyclic group $\mathbb{F}_q^\times$ (Theorem 1.4(ii) with $m = 2$), so $\mathrm{PSL}_2(\mathbb{F}_q)$ has index 2 in $\mathrm{PGL}_2(\mathbb{F}_q)$; for $q = 3$ this is $A_4 \subset S_4$ (Theorem 1.52). For $F = \mathbb{C}$ every element is a square and $\mathrm{PSL}_2(\mathbb{C}) = \mathrm{PGL}_2(\mathbb{C})$. □

The next lemma is the linear algebra behind every conjugacy statement in this chapter.

**Lemma 1.42** (Companion form and centralisers)**.** *Let $g \in \mathrm{GL}_2(F)$ be non-scalar. Then there is a vector $v$ such that $(v, gv)$ is a basis, and in this basis $g$ has matrix $\left(\begin{smallmatrix} 0 & -\det g \\ 1 & \operatorname{tr} g \end{smallmatrix}\right)$. Hence two non-scalar matrices are conjugate in $\mathrm{GL}_2(F)$ if and only if they have the same characteristic polynomial. The centraliser of $g$ in $\mathrm{GL}_2(F)$ is $F[g]^\times$, the unit group of the two-dimensional commutative algebra $F[g] = F + Fg \simeq F[X]/(X^2 - \operatorname{tr}(g)X + \det g)$. Over $F = \mathbb{F}_q$: $\#C(g) = q^2 - 1$ if the characteristic polynomial is irreducible, $(q-1)^2$ if it has two distinct roots, and $q(q-1)$ if it has a double root.*

*Proof.* If every nonzero vector were an eigenvector, then $ge_1 = \alpha e_1$, $ge_2 = \beta e_2$ and $g(e_1 + e_2) = \gamma(e_1 + e_2)$ would give $\alpha = \gamma = \beta$, and $g$ would be scalar. So some $v$ is not an eigenvector, and $(v, gv)$ is a basis. By the Cayley-Hamilton theorem $g^2 v = \operatorname{tr}(g)\, gv - \det(g)\, v$, which gives the matrix. Two non-scalar matrices with the same characteristic polynomial are therefore both conjugate to the same companion matrix. If $h$ commutes with $g$, write $hv = \alpha v + \beta gv$; then $h(gv) = g(hv) = (\alpha + \beta g)gv$, so $h$ and $\alpha + \beta g$ agree on a basis: $h \in F[g]$. Conversely $F[g]$ commutes with $g$. As $g$ is not scalar, $I$ and $g$ are linearly independent and $F[g] \simeq F[X]/(\chi_g)$ with $\chi_g$ the characteristic polynomial (the minimal polynomial has degree 2). Over $\mathbb{F}_q$ the unit group of $\mathbb{F}_q[X]/(\chi)$ has order $q^2 - 1$ if $\chi$ is irreducible (a field with $q^2$ elements), $(q-1)^2$ if $\chi = (X - \lambda)(X - \mu)$ with $\lambda \neq \mu$ (the ring is $\mathbb{F}_q \times \mathbb{F}_q$ by the Chinese remainder theorem), and $q^2 - q$ if $\chi = (X - \lambda)^2$ (the non-units form the ideal generated by $X - \lambda$, with $q$ elements). □

Two remarks on the lemma. First, the hypothesis "non-scalar" matters: $I$ and $-I$ have characteristic polynomials $(X-1)^2$ and $(X+1)^2$, the same as $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $-u$, but a scalar matrix is conjugate only to itself. So for the polynomials with a double root there are two classes, the scalar one and the non-scalar one; this is why $\mathrm{GL}_2(\mathbb{F}_3)$ has 8 classes for 6 possible characteristic polynomials (Proposition 1.59). Second, the lemma is used as a counting tool: the class of a non-scalar $g$ in $\mathrm{GL}_2(\mathbb{F}_q)$ has $\#\,\mathrm{GL}_2(\mathbb{F}_q)/\#C(g)$ elements. For $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ in $\mathrm{GL}_2(\mathbb{F}_3)$ the centraliser is $\mathbb{F}_3[u]^\times = \{\pm u^m\}$, of order $q(q-1) = 6$, and the class has $48/6 = 8$ elements; for $\sigma = \mathrm{diag}(1, -1)$, with distinct eigenvalues, the centraliser is the diagonal group, of order $(q-1)^2 = 4$, and the class has 12 elements.

**Proposition 1.43** (Orders)**.** $\#\,\mathrm{GL}_2(\mathbb{F}_q) = (q^2-1)(q^2-q) = q(q-1)^2(q+1)$, $\#\,\mathrm{SL}_2(\mathbb{F}_q) = \#\,\mathrm{PGL}_2(\mathbb{F}_q) = q(q^2-1)$, *and* $\#\,\mathrm{PSL}_2(\mathbb{F}_q) = q(q^2-1)/\gcd(2, q-1)$. *More generally* $\#\,\mathrm{GL}_n(\mathbb{F}_q) = \prod_{i=0}^{n-1}(q^n - q^i)$. *For small $q$:*

| $q$ | $\#\,\mathrm{GL}_2(\mathbb{F}_q)$ | $\#\,\mathrm{SL}_2(\mathbb{F}_q)$ | $\#\,\mathrm{PGL}_2(\mathbb{F}_q)$ | $\#\,\mathrm{PSL}_2(\mathbb{F}_q)$ |
|---|---|---|---|---|
| 3 | 48 | 24 | 24 | 12 |
| 5 | 480 | 120 | 120 | 60 |
| 7 | 2016 | 336 | 336 | 168 |

*Proof.* An invertible matrix is an ordered basis of columns: the first column is any of the $q^2 - 1$ nonzero vectors, the second any of the $q^2 - q$ vectors outside the line of the first; for $\mathrm{GL}_n$ the $i$-th column avoids the $q^{i-1}$ vectors of the span of the previous ones. det is surjective

onto $\mathbb{F}_q^\times$ (Proposition 1.41(ii)), so $\#\,\mathrm{SL}_2 = \#\,\mathrm{GL}_2/(q-1)$; $\#F^\times I = q-1$ gives $\#\,\mathrm{PGL}_2$; and $\#\{\pm I\} = \gcd(2, q-1)$ gives $\#\,\mathrm{PSL}_2$. □

**Proposition 1.44** (The projective line)**.** *Let $\mathbb{P}^1(F)$ be the set of lines of $F^2$ (convention (C1)). Then $\#\mathbb{P}^1(\mathbb{F}_q) = q+1$, and:*

*(i)* $\mathrm{GL}_2(F)$ *acts on* $\mathbb{P}^1(F)$ *by* $g[v] = [gv]$*; in the coordinate* $x = [x:1]$, $\infty = [1:0]$, *the matrix* $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ *acts by* $x \mapsto (ax+b)/(cx+d)$, *with the usual conventions* $a/0 = \infty$ *for* $a \neq 0$ *and* $\infty \mapsto a/c$.

*(ii) The kernel of the action is* $F^\times I$, *so* $\mathrm{PGL}_2(F)$ *acts faithfully.*

*(iii)* $\mathrm{PGL}_2(F)$ *acts sharply* 3*-transitively: for two triples of distinct lines there is exactly one element of* $\mathrm{PGL}_2(F)$ *carrying the first to the second.*

*(iv) The stabiliser of* $\infty = [e_1]$ *is the upper triangular group* $\mathrm{B}(F)$*; so* $\mathbb{P}^1(F) \simeq \mathrm{GL}_2(F)/\mathrm{B}(F)$ *and* $[\mathrm{GL}_2(\mathbb{F}_q) : \mathrm{B}(\mathbb{F}_q)] = q+1$.

*(v)* $\mathrm{SL}_2(F)$ *acts* 2*-transitively, in particular transitively, on* $\mathbb{P}^1(F)$.

*Proof.* There are $q^2-1$ nonzero vectors, $q-1$ on each line, so $q+1$ lines. (i) $g(\lambda v) = \lambda gv$, so the action is well defined. $g\binom{x}{1} = \binom{ax+b}{cx+d}$, which is the line $(ax+b)/(cx+d)$ if $cx+d \neq 0$ and $\infty$ otherwise; and $g\binom{1}{0} = \binom{a}{c}$. (ii) An element acting trivially has every nonzero vector as an eigenvector, so it is scalar by the first step of the proof of Lemma 1.42. (iii) Let $L_1, L_2, L_3$ be distinct lines, $L_i = [v_i]$. Then $v_1, v_2$ is a basis and $v_3 = \alpha v_1 + \beta v_2$ with $\alpha, \beta \neq 0$ (as $L_3 \neq L_1, L_2$). The matrix $g$ with columns $\alpha v_1$, $\beta v_2$ is invertible and maps $[e_1], [e_2], [e_1+e_2]$, that is $\infty, 0, 1$, to $L_1, L_2, L_3$. If $h$ fixes $\infty$, 0 and 1, then $he_1 = \alpha e_1$, $he_2 = \beta e_2$ and $h(e_1+e_2) = \gamma(e_1+e_2)$, so $\alpha = \beta = \gamma$ and $h$ is scalar. So for each triple there is exactly one element of $\mathrm{PGL}_2(F)$ carrying $(\infty, 0, 1)$ to it, and composing gives (iii). (iv) $g[e_1] = [e_1]$ iff the first column $\binom{a}{c}$ is a multiple of $e_1$, iff $c = 0$. Then apply Theorem 1.28 and (iii) (which gives transitivity): $\#\mathbb{P}^1(\mathbb{F}_q) = [\mathrm{GL}_2(\mathbb{F}_q) : \mathrm{B}(\mathbb{F}_q)]$, and indeed $q(q-1)^2(q+1)/(q(q-1)^2) = q+1$. (v) Given two ordered pairs $P$, $P'$ of distinct lines, it suffices to find $g, g' \in \mathrm{SL}_2(F)$ moving $(\infty, 0)$ to $P$ and to $P'$, for then $g'g^{-1} \in \mathrm{SL}_2(F)$ moves $P$ to $P'$. To move $(\infty, 0)$ to $([v_1], [v_2])$: the vectors $v_1, v_2$ are a basis, so $\det(v_1, v_2) \neq 0$, and the matrix with columns $v_1$ and $\lambda v_2$, where $\lambda = \det(v_1, v_2)^{-1}$, has determinant 1, sends $e_1 \mapsto v_1$ and $e_2 \mapsto \lambda v_2$, hence $\infty \mapsto [v_1]$ and $0 \mapsto [v_2]$. (Unlike $\mathrm{PGL}_2(F)$, the group $\mathrm{SL}_2(F)$ need not be 3-transitive: for $F = \mathbb{F}_3$ there are $4 \cdot 3 \cdot 2 = 24$ ordered triples of distinct lines, but $\mathrm{SL}_2(\mathbb{F}_3)$ acts through $\mathrm{PSL}_2(\mathbb{F}_3)$, of order 12.) □

**Definition 1.45** (Borel subgroups)**.** A *Borel subgroup* of $\mathrm{GL}_2(F)$ is the stabiliser $\mathrm{B}_L$ of a line $L \in \mathbb{P}^1(F)$. We write $\mathrm{B} = \mathrm{B}_{[e_1]} = \{\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)\}$, the upper triangular group; $\mathrm{U} = \{\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right) : b \in F\}$, its *unipotent radical*; and $\mathrm{T} = \{\mathrm{diag}(a, d)\}$, the diagonal torus (which is the split Cartan subgroup $C_s$ of Definition 1.54).

**Proposition 1.46.** *(i)* $\mathrm{U} \simeq (F, +)$ *is normal in* B, *and* $\mathrm{B} = \mathrm{U} \rtimes \mathrm{T}$, *with* T *acting on* U *by*

$$\mathrm{diag}(a,d)\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)\mathrm{diag}(a,d)^{-1} = \left(\begin{smallmatrix} 1 & ab/d \\ 0 & 1 \end{smallmatrix}\right);$$

*and* $\#\mathrm{B}(\mathbb{F}_q) = q(q-1)^2$.

*(ii) The Borel subgroups are all conjugate:* $g\mathrm{B}_L g^{-1} = \mathrm{B}_{gL}$. *A Borel subgroup stabilises exactly one line, so* $L \mapsto \mathrm{B}_L$ *is a bijection from* $\mathbb{P}^1(F)$ *to the set of Borel subgroups, and* $N_{\mathrm{GL}_2(F)}(\mathrm{B}_L) = \mathrm{B}_L$.

*(iii) A subgroup of* $\mathrm{GL}_2(F)$ *stabilises a line of* $F^2$ *if and only if it is contained in a Borel subgroup.*

*(iv) For* $q = 3$*:* $\#\mathrm{B}(\mathbb{F}_3) = 12$*, and* $\mathrm{B}(\mathbb{F}_3) = \{\pm I\} \times \langle u, \sigma\rangle \simeq C_2 \times S_3 \simeq D_6$*, with* $u = \left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$ *and* $\sigma = \mathrm{diag}(1, -1)$*. Its image in* $\mathrm{PGL}_2(\mathbb{F}_3)$ *is the stabiliser of* $\infty$*, isomorphic to* $S_3$.

*Proof.* (i) The map $\left(\begin{smallmatrix}a & b\\ 0 & d\end{smallmatrix}\right) \mapsto \mathrm{diag}(a, d)$ is a homomorphism $\mathrm{B} \to \mathrm{T}$ (the diagonal of a product of upper triangular matrices is the product of the diagonals) with kernel U, and it restricts to the identity on T; so Proposition 1.15(ii) applies. The conjugation formula is a direct computation, and $\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right)\left(\begin{smallmatrix}1 & b'\\ 0 & 1\end{smallmatrix}\right) = \left(\begin{smallmatrix}1 & b+b'\\ 0 & 1\end{smallmatrix}\right)$. The order is $q \cdot (q-1)^2$. (ii) $g\mathrm{B}_L g^{-1}$ stabilises $gL$, and conversely; the action on lines is transitive. The element $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$ of B acts by $x \mapsto x + 1$, which fixes $\infty$ and no point of $F$; so B stabilises only $\infty$, and by conjugation $\mathrm{B}_L$ stabilises only $L$. If $g$ normalises $\mathrm{B}_L$, then $\mathrm{B}_L = g\mathrm{B}_L g^{-1} = \mathrm{B}_{gL}$, so $gL = L$ and $g \in \mathrm{B}_L$. (iii) A subgroup $H$ stabilises the line $L$ iff $hL = L$ for every $h \in H$, iff $H \leq \mathrm{B}_L$, by the definition of $\mathrm{B}_L$ as the stabiliser of $L$. (iv) $\#\mathrm{B}(\mathbb{F}_3) = 3 \cdot 4$. Since $\sigma u \sigma^{-1} = \left(\begin{smallmatrix}1 & -1\\ 0 & 1\end{smallmatrix}\right) = u^{-1}$, the subgroup $S = \langle u, \sigma\rangle = \{u^m \sigma^e\}$ has order 6 and is isomorphic to $D_3 \simeq S_3$ by Proposition 1.19(i),(iv). It has index 2 in $\mathrm{B}(\mathbb{F}_3)$, hence is normal; $\{\pm I\}$ is central; and $-I \notin S$ (the elements $u^m$ have trace 2, the elements $u^m\sigma$ determinant $-1$, while $-I$ has trace $-2 = 1$ and determinant 1). So $\mathrm{B}(\mathbb{F}_3) = \{\pm I\} \times S$ by Proposition 1.12, and $C_2 \times S_3 \simeq D_6$ by Proposition 1.19(iv). The image in $\mathrm{PGL}_2(\mathbb{F}_3)$ is $\mathrm{B}(\mathbb{F}_3)/\{\pm I\} \simeq S$, of order 6, contained in the stabiliser of $\infty$, which has order $24/4 = 6$ by Theorem 1.28. □

**Definition 1.47** (Unipotent elements)**.** An element $u \in \mathrm{GL}_2(F)$ is *unipotent* if $(u-1)^2 = 0$; equivalently, its characteristic polynomial is $(X-1)^2$ (if $(u-1)^2 = 0$, the only eigenvalue of $u$ over $\overline{F}$ is 1; conversely apply the Cayley-Hamilton theorem). For a line $L$, $\mathrm{U}_L$ denotes the group of unipotent elements fixing $L$ pointwise; $\mathrm{U}_{[e_1]} = \mathrm{U}$.

The name records that all eigenvalues equal 1: a unipotent element is $1 + N$ with $N$ nilpotent. Over $\mathbb{F}_p$ these are exactly the elements of $p$-power order (Proposition 1.48(ii)), and this is why they appear in the paper: in Lemma lem:IV the image of an inertia group in $\mathrm{GL}_2(\mathbb{F}_3)$ has order 3, so it consists of unipotent matrices. For example $u = \left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$ are unipotent, while $\sigma = \mathrm{diag}(1, -1)$, with eigenvalue $-1$, is not, and neither is $-u$, whose characteristic polynomial is $(X+1)^2$.

**Proposition 1.48.** *(i) A unipotent* $u \neq 1$ *is conjugate to* $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$*. Its fixed vectors form exactly one line,* $\ker(u-1) = \mathrm{im}(u-1)$*, and* $u$ *stabilises no other line of* $F^2$.

*(ii) If* $\mathrm{char}\, F = p > 0$*, a unipotent* $u \neq 1$ *has order* $p$*; conversely every element of* $p$*-power order of* $\mathrm{GL}_2(F)$ *is unipotent.*

*(iii) The unipotent elements of* B *are exactly the elements of* U*. For each line* $L$*,* $\mathrm{U}_L = g\mathrm{U}g^{-1}$ *for any* $g$ *with* $g[e_1] = L$*; it is the unipotent radical of* $\mathrm{B}_L$ *and is isomorphic to* $(F, +)$*. The stabiliser of a nonzero vector* $v$ *in* $\mathrm{SL}_2(F)$ *is* $\mathrm{U}_{[v]}$.

*(iv) (Minus a unipotent.) If* $u \neq 1$ *is unipotent,* $-u$ *has characteristic polynomial* $(X+1)^2$*; if* $\mathrm{char}\, F \neq 2$*, it has no nonzero fixed vector, it stabilises exactly one line (that of* $u$*), and in characteristic* $p$ *odd it has order* $2p$.

(v) *In* $\mathrm{GL}_2(\mathbb{F}_3)$ *the* 8 *elements of order* 3 *are the nontrivial unipotent elements and form one conjugacy class. They lie in the four subgroups* $\mathrm{U}_L$ *of order* 3*, one for each line* $L$*, which are the Sylow* 3*-subgroups;* $\mathrm{U}_L$ *fixes exactly the line* $L$*. The* 8 *elements of order* 6 *are the elements* $-u$ *with* $u \neq 1$ *unipotent. The* 8 *elements of order* 3 *generate* $\mathrm{SL}_2(\mathbb{F}_3)$*.*

*Proof.* (i) Let $N = u - 1 \neq 0$. As $N^2 = 0$, $\operatorname{im} N \subseteq \ker N$, and $N$ has rank 1 (rank 2 would make $N^2$ invertible); so $\ker N = \operatorname{im} N$ is a line $L$. Choose $v \notin L$ and put $w = Nv$, which spans $L$. In the basis $(w, v)$, $uw = w$ and $uv = v + w$, so $u$ has matrix $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. The fixed vectors are $\ker N = L$. A line stabilised by $u$ is spanned by an eigenvector, whose eigenvalue is 1, so it is $L$. (ii) Write $u = 1 + N$ with $N^2 = 0$; then $u^m = 1 + mN$, which is 1 iff $p \mid m$. Conversely if $x^{p^k} = 1$, then $(x-1)^{p^k} = x^{p^k} - 1 = 0$ (the binomial coefficients $\binom{p^k}{i}$, $0 < i < p^k$, vanish in characteristic $p$), so $x - 1$ is nilpotent; a nilpotent $2 \times 2$ matrix has characteristic polynomial $X^2$ and squares to 0 by Cayley-Hamilton. (iii) An upper triangular matrix with characteristic polynomial $(X-1)^2$ has diagonal entries 1. A unipotent element fixing $L = g[e_1]$ pointwise is $gu'g^{-1}$ with $u'$ unipotent fixing $e_1$, hence in B, hence in U; conversely U fixes $e_1$. So $\mathrm{U}_L = g\mathrm{U}g^{-1}$, the kernel of $\mathrm{B}_L \to \mathrm{T}$ transported by $g$. If $g' \in \mathrm{SL}_2(F)$ fixes $v \neq 0$, complete $v$ to a basis $(v, v')$; in it $g' = \left(\begin{smallmatrix} 1 & b \\ 0 & d \end{smallmatrix}\right)$ with $d = \det g' = 1$, so $g'$ is unipotent and fixes $v$; conversely elements of $\mathrm{U}_{[v]}$ fix $v$ and have determinant 1. (iv) $\det(X + u) = (X+1)^2$ follows from $\det(X - u) = (X-1)^2$. If $-uv = v$ then $uv = -v$, but $u$ has only the eigenvalue $1 \neq -1$. The lines stabilised by $-u$ are those stabilised by $u$. Finally $(-u)^m = (-1)^m u^m = 1$ iff $m$ is even and $p \mid m$, so the order is $2p$ for odd $p$. (v) By (ii) the elements of order 3 are the nontrivial unipotents; by (i) and Lemma 1.42 they form one class, of size $48/\#C(u) = 48/(3 \cdot 2) = 8$. Each lies in exactly one $\mathrm{U}_L$ (its fixed line), and $\#\mathrm{U}_L = 3$; so the four groups $\mathrm{U}_L$ contain $4 \cdot 2 = 8$ elements of order 3, and they are the subgroups of order 3, that is, the Sylow 3-subgroups ($48 = 16 \cdot 3$). If $x$ has order 6, then $x^3$ has order 2 and $x^{-2}$ has order 3, so $x^{-2} = u'$ is a nontrivial unipotent, and $x^3$ is an involution commuting with $u'$, hence in $C(u') = \mathbb{F}_3[u']^\times = \{\pm u'^m\}$; its only involution is $-I$ (the elements $-u'^m$ with $m \neq 0$ have order 6 by (iv)). So $x = x^3 x^{-2} = -u'$. Conversely the $-u$ have order 6 by (iv); there are 8. Finally let $S$ be the subgroup generated by the 8 elements of order 3. It lies in $\mathrm{SL}_2(\mathbb{F}_3)$ (unipotents have determinant 1), so $\#S$ divides 24, and $\#S \geq 9$; hence $\#S \in \{12, 24\}$. If $\#S = 12$, then $S$ has index 2 in $\mathrm{SL}_2(\mathbb{F}_3)$, hence contains the square of every element of $\mathrm{SL}_2(\mathbb{F}_3)$ (Lemma 1.7: the quotient has order 2); in particular $-I = J^2$ with $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and then also the 8 elements $-u$. So $\#S \geq 1 + 8 + 1 + 8 = 18$, a contradiction. Hence $S = \mathrm{SL}_2(\mathbb{F}_3)$. □

*In the paper* 1.49. The proof of Lemma lem:IV uses Proposition 1.48(iii) and (v) in the sentence "The elements of $\mathrm{SL}_2(\mathbb{F}_3)$ fixing a nonzero vector form a group of order 3 of unipotent elements; so $\bar{\rho}_{E,3}(I_q)$ is that group". The image of inertia lies in $\mathrm{SL}_2(\mathbb{F}_3)$ (the cyclotomic character is unramified at $q \neq 3$) and fixes a nonzero vector (since $E[3]^{I_q}$ is a line), so it lies in the stabiliser $\mathrm{U}_{[v]}$ of that vector, of order 3; it is nontrivial, so it equals $\mathrm{U}_{[v]}$, and $E[3]^{I_q}$ is exactly the line $[v]$ by Proposition 1.48(i). Remark rem:types speaks of "minus a unipotent, of order 6" (types II, II$^*$): by Proposition 1.48(iv) such an element fixes no nonzero vector of $\mathbb{F}_3^2$, which is the statement $E[3]^{I_q} = 0$ for these types. The other images listed there, $\{\pm I\}$ and a cyclic group of order 4 (conjugate to $\langle J \rangle$, $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$), also fix no nonzero vector: $-I$ and $J$ have no eigenvalue 1 (Proposition 1.59). For type $\mathrm{I}_n^*$ the image of inertia consists of a ramified quadratic character times unipotents; an element on which the character is $-1$ maps to $-I$ or to minus a nontrivial unipotent, which fixes no nonzero vector by Proposition 1.48(iv), so again $E[3]^{I_q} = 0$. The identification of the images of inertia with the Kodaira types is Chapter 11.

**Lemma 1.50** (Commutators in a Borel subgroup)**.** *Let $K$ be a field and $H \leq \mathrm{B}(K)$. Then $[H,H] \leq \mathrm{U}(K)$. If $\operatorname{char} K = p > 0$, then $\mathrm{U}(K)$ is an elementary abelian $p$-group (every nontrivial element has order $p$), and:*

*(i) a finite subgroup of $\mathrm{B}(K)$ of order prime to $p$ meets $\mathrm{U}(K)$ trivially and is abelian;*

*(ii) a finite subgroup of $\mathrm{GL}_2(K)$ whose commutator subgroup is not a $p$-group stabilises no line of $K^2$.*

*Proof.* The homomorphism $\mathrm{B}(K) \to \mathrm{T}(K) \simeq (K^\times)^2$ of Proposition 1.46(i) has abelian target and kernel $\mathrm{U}(K)$, so it kills all commutators: $[H,H] \leq [\mathrm{B},\mathrm{B}] \leq \mathrm{U}(K)$. In characteristic $p$, $\mathrm{U}(K) \simeq (K,+)$ and $p \cdot b = 0$. (i) $H \cap \mathrm{U}(K)$ is a $p$-group (each element has order 1 or $p$) whose order divides $\#H$, so it is trivial, and $[H,H] \leq H \cap \mathrm{U}(K) = 1$. (ii) If $H$ stabilises a line of $K^2$, then $gHg^{-1} \leq \mathrm{B}(K)$ for some $g \in \mathrm{GL}_2(K)$ (Proposition 1.46(ii),(iii)), so $g[H,H]g^{-1} \leq \mathrm{U}(K)$, and $[H,H]$ is a finite subgroup of a conjugate of $\mathrm{U}(K)$, hence a $p$-group. □

*In the paper* 1.51. In the proof of Proposition prop:ns the determinant-one subgroups $\mathrm{SL}_2(\mathbb{F}_3)$ and $Q_8$ are shown to act absolutely irreducibly on $\mathbb{F}_3^2$: "a subgroup of $\mathrm{GL}_2(\overline{\mathbb{F}}_3)$ fixing a line lies in a Borel subgroup, so its commutator subgroup is abelian of exponent 3, whereas the commutator subgroups $\{\pm 1\}$ of $Q_8$ and $Q_8$ of $\mathrm{SL}_2(\mathbb{F}_3)$ are not." This is Lemma 1.50(ii) with $K = \overline{\mathbb{F}}_3$ and $p = 3$; the commutator subgroups are computed in Proposition 1.21(ii) and Proposition 1.62(iv). Non-abelian and faithful would not be enough here: $\mathrm{B}(\mathbb{F}_3)$ is non-abelian and acts faithfully, yet stabilises a line. That "stabilises no line of $\overline{\mathbb{F}}_3^2$" is the same as "absolutely irreducible" is the definition of Chapter 4.

**Theorem 1.52** ($\mathrm{PGL}_2$ and $\mathrm{PSL}_2$ over small fields)**.** *(i) The action on the four lines of $\mathbb{F}_3^2$ induces isomorphisms $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$ and $\mathrm{PSL}_2(\mathbb{F}_3) \simeq A_4$.*

*(ii) For $g \in \mathrm{GL}_2(\mathbb{F}_3)$, $\det g \in \mathbb{F}_3^\times = \{\pm 1\}$ equals the sign of the permutation of the four lines induced by $g$.*

*(iii) $\mathrm{PGL}_2(\mathbb{F}_5) \simeq S_5$ and $\mathrm{PSL}_2(\mathbb{F}_5) \simeq A_5$.*

*(iv) $\mathrm{PSL}_2(\mathbb{F}_q)$ is simple for $q \geq 4$. In particular $\mathrm{PSL}_2(\mathbb{F}_7)$ is simple of order 168; it is isomorphic to $\mathrm{GL}_3(\mathbb{F}_2)$.*

*Proof.* (i) By Proposition 1.44(ii) the action gives an injective homomorphism $\mathrm{PGL}_2(\mathbb{F}_3) \to S_4$, and both groups have order 24. The group $\mathrm{SL}_2(\mathbb{F}_3)$ is generated by its elements of order 3 (Proposition 1.48(v)), whose images have order 3 in $S_4$, hence are 3-cycles, hence even; so the image of $\mathrm{SL}_2(\mathbb{F}_3)$, which is $\mathrm{PSL}_2(\mathbb{F}_3)$ by Proposition 1.41(v), lies in $A_4$, and it has order 12. (ii) Both det and $g \mapsto \operatorname{sgn}(\text{permutation})$ are homomorphisms $\mathrm{GL}_2(\mathbb{F}_3) \to \{\pm 1\}$. The kernel of det is $\mathrm{SL}_2(\mathbb{F}_3)$, which lies in the kernel of the sign by (i). The sign is not trivial: $\sigma = \operatorname{diag}(1,-1)$ acts by $x \mapsto -x$, fixing 0 and $\infty$ and swapping 1 and $-1$, a transposition. So both kernels have index 2 and contain $\mathrm{SL}_2(\mathbb{F}_3)$, hence are equal to it, and two homomorphisms onto $\{\pm 1\}$ with the same kernel are equal. □

We do not prove (iv); see [109, Chapter XIII, Section 8] for the simplicity of $\mathrm{PSL}_2(F)$ for every field with more than three elements (the proof uses the 2-transitivity of Proposition 1.44(v), the generation by the conjugates of U of Proposition 1.41(iii), and Iwasawa's criterion), and [58, Section 6.2] for the uniqueness of the simple group of order 168, which identifies $\mathrm{PSL}_2(\mathbb{F}_7)$ with $\mathrm{GL}_3(\mathbb{F}_2)$.

*Sketch of proof of (iii).* Granting the simplicity of $\mathrm{PSL}_2(\mathbb{F}_5)$ (order 60): the subgroup of $\mathrm{GL}_2(\mathbb{F}_5)$ generated by $N_s(\mathbb{F}_5)$ (Definition 1.54) and $\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$ has order 96 and contains the scalars (this is a finite check, done by machine; note $\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)^2 = 2I$), so its image $K$ in $\mathrm{PGL}_2(\mathbb{F}_5)$ has order 24 and index 5. The action of $\mathrm{PGL}_2(\mathbb{F}_5)$ on the five cosets of $K$ gives a homomorphism to $S_5$ whose kernel $M$ is normal and contained in $K$. Then $M \cap \mathrm{PSL}_2(\mathbb{F}_5)$ is normal in the simple group $\mathrm{PSL}_2(\mathbb{F}_5)$ and has order at most $24 < 60$, so it is trivial, and $\#M \le 2$; a normal subgroup of order 2 would be central, but the centre of $\mathrm{PGL}_2(\mathbb{F}_5)$ is trivial (a central element $z$ maps each point $x$ to a point fixed by the stabiliser of $x$, which is transitive on the other five points by Proposition 1.44(iii); so $zx = x$ for all $x$, and $z = 1$ by Proposition 1.44(ii)). So $\mathrm{PGL}_2(\mathbb{F}_5) \to S_5$ is injective, hence bijective, and it maps the index-2 subgroup $\mathrm{PSL}_2(\mathbb{F}_5)$ onto the unique index-2 subgroup $A_5$ of $S_5$. See [85, Chapter II, Section 8] for a full treatment of the small $\mathrm{PSL}_2(\mathbb{F}_q)$ and their exceptional isomorphisms. □

*In the paper* 1.53. Theorem 1.52(ii) is behind the parity of the permutation of the roots of the 3-division polynomial in Proposition prop:ns: the four roots correspond to the four lines of $E[3]$ (Chapter 9), and $\sigma \in G_{\mathbb{Q}}$ permutes them with sign $\det \bar{\rho}_{E,3}(\sigma) = \chi_{-3}(\sigma)$. So the discriminant of the 3-division polynomial is $-3$ times a square; this is taken up in Chapter 5 and Chapter 11.

**Definition 1.54** (Cartan subgroups and their normalisers)**.** Let $q$ be odd and fix a non-square $\varepsilon \in \mathbb{F}_q^\times$ (for $q = 3$ we take $\varepsilon = -1$; for $q = 5$, $\varepsilon = 2$; for $q = 7$, $\varepsilon = 3$). Put $w = \left(\begin{smallmatrix}0 & 1\\ 1 & 0\end{smallmatrix}\right)$ and $\sigma = \mathrm{diag}(1, -1)$.

(i) The *split Cartan subgroup* is $C_s(\mathbb{F}_q) = \{\mathrm{diag}(a, d) : a, d \in \mathbb{F}_q^\times\} \simeq (\mathbb{F}_q^\times)^2$, and $N_s(\mathbb{F}_q) = C_s(\mathbb{F}_q) \cup wC_s(\mathbb{F}_q)$.

(ii) Let $\mathbb{F}_q[\sqrt{\varepsilon}] = \{a + b\sqrt{\varepsilon} : a, b \in \mathbb{F}_q\}$, with $(\sqrt{\varepsilon})^2 = \varepsilon$, that is, $\mathbb{F}_q[X]/(X^2 - \varepsilon)$. The *nonsplit Cartan subgroup* is

$$C_{ns}(\mathbb{F}_q) = \left\{ \begin{pmatrix} a & \varepsilon b \\ b & a \end{pmatrix} : (a, b) \neq (0, 0) \right\},$$

the matrices of multiplication by $a + b\sqrt{\varepsilon}$ on $\mathbb{F}_q[\sqrt{\varepsilon}]$ in the basis $(1, \sqrt{\varepsilon})$, and $N_{ns}(\mathbb{F}_q) = C_{ns}(\mathbb{F}_q) \cup C_{ns}(\mathbb{F}_q)\sigma$.

A *Cartan subgroup* of $\mathrm{GL}_2(\mathbb{F}_q)$ is a conjugate of $C_s(\mathbb{F}_q)$ (split) or of $C_{ns}(\mathbb{F}_q)$ (nonsplit). We often drop $(\mathbb{F}_q)$ from the notation.

The definition is made because these are the centralisers of the "generic" semisimple elements. By Lemma 1.42, a matrix whose characteristic polynomial has two distinct roots in $\mathbb{F}_q$ is conjugate to a diagonal matrix with distinct entries, whose centraliser is $C_s$; a matrix whose characteristic polynomial is irreducible over $\mathbb{F}_q$ has centraliser $\mathbb{F}_q[g]^\times \simeq \mathbb{F}_{q^2}^\times$, a conjugate of $C_{ns}$ (Proposition 1.55(iv)). Consequently an abelian subgroup $H \le \mathrm{GL}_2(\mathbb{F}_q)$ of order prime to the characteristic $p$ lies in a Cartan subgroup. If $H$ consists of scalars, it lies in $C_s$. Otherwise choose a non-scalar $h \in H$. Its characteristic polynomial has no double root $\lambda$: otherwise $\lambda^{-1}h$ would be a nontrivial unipotent element, of order $p$ (Proposition 1.48(ii)), and $h^m = \lambda^m(\lambda^{-1}h)^m = 1$ would force $(\lambda^{-1}h)^m$ to be a scalar unipotent matrix, that is 1, so $p \mid m$ and $p$ would divide the order of $h$. So $C(h) = \mathbb{F}_q[h]^\times$ is a Cartan subgroup, and it contains $H$ because $H$ is abelian. Dickson's classification (Theorem 1.57) is phrased in terms of these groups and their normalisers. The choice of $\varepsilon$ only changes $C_{ns}$ by conjugation (Proposition 1.55(iv)). For $q = 3$ and $\varepsilon = -1$: $C_{ns}(\mathbb{F}_3)$ is the set of the 8 matrices $\left(\begin{smallmatrix}a & -b\\ b & a\end{smallmatrix}\right)$ with $(a, b) \neq (0, 0)$, the matrices of multiplication by $a + bi$ on $\mathbb{F}_9 = \mathbb{F}_3[i]$, and it contains $J = \left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)$ (take $a = 0$, $b = -1$).

**Proposition 1.55.** *Let $q$ be odd.*

*(i)* $N_s = N_{\mathrm{GL}_2(\mathbb{F}_q)}(C_s)$*, and it is the stabiliser of the set of two lines* $\{[e_1],[e_2]\}$*;* $C_s$ *is the subgroup fixing each of the two lines, and* $N_s = C_s \rtimes \langle w\rangle$*.*

*(ii)* $\mathbb{F}_q[\sqrt{\varepsilon}]$ *is a field with* $q^2$ *elements, and* $c \mapsto m_c$ *(multiplication by* $c$*) is an isomorphism* $\mathbb{F}_q[\sqrt{\varepsilon}]^\times \to C_{ns}$*, with* $\det m_c = a^2 - \varepsilon b^2$ *(the norm* $N(c) = c\bar{c}$*, where* $\bar{c} = a - b\sqrt{\varepsilon}$*) and* $\operatorname{tr} m_c = 2a$*. So* $C_{ns}$ *is cyclic of order* $q^2 - 1$*.*

*(iii) The conjugation* $c \mapsto \bar{c}$ *is the field automorphism* $c \mapsto c^q$*, its matrix is* $\sigma$*, and* $\sigma m_c \sigma^{-1} = m_{\bar{c}} = m_c^q$*. Moreover* $N_{ns} = N_{\mathrm{GL}_2(\mathbb{F}_q)}(C_{ns}) = C_{ns} \rtimes \langle\sigma\rangle$*.*

*(iv) All split Cartan subgroups are conjugate, and all nonsplit ones are conjugate. Intrinsically, a Cartan subgroup is the unit group* $A^\times$ *of a two-dimensional commutative subalgebra* $A \subset M_2(\mathbb{F}_q)$ *that is isomorphic to* $\mathbb{F}_q \times \mathbb{F}_q$ *(split) or is a field (nonsplit). Each Cartan subgroup has index* 2 *in its normaliser.*

*Proof.* (i) $w \operatorname{diag}(a,d)\, w^{-1} = \operatorname{diag}(d,a)$, so $w$ normalises $C_s$, and $N_s$ is a group containing $C_s$ with index 2 ($w \notin C_s$). An element fixing both $[e_1]$ and $[e_2]$ is diagonal; an element swapping them is $w$ times a diagonal matrix. So $N_s$ is the stabiliser of $\{[e_1],[e_2]\}$. Conversely let $g$ normalise $C_s$. As $q \geq 3$, there is $d \neq 1$ in $\mathbb{F}_q^\times$, and $\operatorname{diag}(1,d)$ has distinct eigenvalues, so the lines stabilised by every element of $C_s$ are exactly $[e_1]$ and $[e_2]$. Since $gC_sg^{-1} = C_s$, $g$ maps this set of two lines to itself, so $g \in N_s$. The semidirect product structure follows from Proposition 1.15(i).

(ii) $\mathbb{F}_q[\sqrt{\varepsilon}]$ is a commutative ring with $q^2$ elements. For $(a,b) \neq (0,0)$, $a^2 - \varepsilon b^2 \neq 0$: if $b \neq 0$ this would make $\varepsilon = (a/b)^2$ a square, and if $b = 0$ then $a \neq 0$. So $(a + b\sqrt{\varepsilon})^{-1} = (a - b\sqrt{\varepsilon})/(a^2 - \varepsilon b^2)$, and it is a field. Multiplication by $c = a + b\sqrt{\varepsilon}$ sends $1 \mapsto a + b\sqrt{\varepsilon}$ and $\sqrt{\varepsilon} \mapsto \varepsilon b + a\sqrt{\varepsilon}$, which gives the matrix $\left(\begin{smallmatrix} a & \varepsilon b \\ b & a \end{smallmatrix}\right)$, with the stated determinant and trace; $c \mapsto m_c$ is an injective homomorphism. Cyclicity is Corollary 1.36(ii).

(iii) $\bar{c}$ is a ring automorphism (it is the substitution $\sqrt{\varepsilon} \mapsto -\sqrt{\varepsilon}$, and $-\sqrt{\varepsilon}$ is also a root of $X^2 - \varepsilon$), with matrix $\sigma$ in the basis $(1, \sqrt{\varepsilon})$. In characteristic $p$, $(x+y)^p = x^p + y^p$, so $c \mapsto c^q$ is a ring endomorphism; it fixes $\mathbb{F}_q$ (as $a^{q-1} = 1$ for $a \in \mathbb{F}_q^\times$, by Lagrange) and sends $\sqrt{\varepsilon}$ to $\varepsilon^{(q-1)/2}\sqrt{\varepsilon}$. Now $\varepsilon^{(q-1)/2} = -1$ (Euler's criterion): in the cyclic group $\mathbb{F}_q^\times = \langle t\rangle$ (Corollary 1.36) of even order $q-1$, the squares are the even powers of $t$, since $q-1$ is even; so $\varepsilon = t^m$ with $m$ odd, and $\varepsilon^{(q-1)/2} = t^{m(q-1)/2} \neq 1$, as $q - 1 \nmid m(q-1)/2$. Its square is 1, and the only roots of $X^2 - 1$ in a field are $\pm 1$; so it is $-1$. Hence $c^q = \bar{c}$. Then $\sigma m_c \sigma^{-1}$ is the matrix of $x \mapsto \overline{c}\,\overline{x} = \bar{c}x$, that is $m_{\bar{c}}$. So $\sigma$ normalises $C_{ns}$; $\sigma \notin C_{ns}$ (it is not of the form $\left(\begin{smallmatrix} a & \varepsilon b \\ b & a \end{smallmatrix}\right)$, as $1 \neq -1$), and $N_{ns}$ is a group in which $C_{ns}$ has index 2. Conversely let $g$ normalise $C_{ns}$. The $\mathbb{F}_q$-span of $C_{ns}$ is the image $A$ of the field $\mathbb{F}_q[\sqrt{\varepsilon}]$, and $x \mapsto gxg^{-1}$ is a ring automorphism of $A$ fixing the scalars; it sends $m_{\sqrt{\varepsilon}}$ to a square root of $\varepsilon I$ in $A$, that is to $\pm m_{\sqrt{\varepsilon}}$ (a polynomial of degree 2 has at most two roots in a field). If it fixes $m_{\sqrt{\varepsilon}}$, then $g$ commutes with the non-scalar matrix $m_{\sqrt{\varepsilon}}$, so $g \in \mathbb{F}_q[m_{\sqrt{\varepsilon}}]^\times = C_{ns}$ by Lemma 1.42. Otherwise $g\sigma^{-1}$ commutes with $m_{\sqrt{\varepsilon}}$, so $g \in C_{ns}\sigma$.

(iv) Let $A \subset M_2(\mathbb{F}_q)$ be a two-dimensional commutative subalgebra (containing $I$). If $A \simeq \mathbb{F}_q \times \mathbb{F}_q$, its idempotent $e$ corresponding to $(1,0)$ satisfies $e^2 = e \neq 0, 1$, so $e$ is the projection onto a line $L_1$ along a line $L_2$ ($F^2 = \operatorname{im} e \oplus \ker e$, both of dimension 1), and $A = \{\alpha e + \beta(1-e)\}$ is the set of matrices acting by scalars on $L_1$ and on $L_2$; in a basis $v_1 \in L_1$, $v_2 \in L_2$, $A^\times$ becomes $C_s$. If $A$ is a field (with $q^2$ elements), then $\mathbb{F}_q^2$ is an $A$-module, and for $v \neq 0$ the map $A \to \mathbb{F}_q^2$, $x \mapsto xv$, is injective ($A$ is a field) and hence bijective by counting. The element $\varepsilon \in \mathbb{F}_q^\times \subset A^\times$ is a square in $A$: $A^\times$ is cyclic of order $q^2 - 1$, its subgroup $\mathbb{F}_q^\times$ of order $q-1$ consists of the $(q+1)$-th

powers (Theorem 1.4(i)), and $q+1$ is even. Choose $\delta \in A$ with $\delta^2 = \varepsilon I$; then in the basis $(v, \delta v)$ the elements $a + b\delta$ of $A$ have matrices $\left(\begin{smallmatrix} a & \varepsilon b \\ b & a \end{smallmatrix}\right)$, so $A^\times$ becomes $C_{ns}$. Conversely $C_s$ and $C_{ns}$ are of this form, with $A$ the diagonal matrices and the image of $\mathbb{F}_q[\sqrt{\varepsilon}]$. The index statement is (i) and (iii) transported by conjugation. □

**Proposition 1.56** (Facts about Cartan subgroups)**.** *Let $q$ be odd.*

*(i)* $\#C_s = (q-1)^2$, $\#N_s = 2(q-1)^2$, $\#C_{ns} = q^2-1$, $\#N_{ns} = 2(q^2-1)$. *For $q = 3, 5, 7$ these are $4, 16, 36$; $8, 32, 72$; $8, 24, 48$; $16, 48, 96$, and $\#\mathrm{B} = 12, 80, 252$.*

*(ii) Every element of $N_s \setminus C_s$ and of $N_{ns} \setminus C_{ns}$ has trace 0. In $N_s$, $\left(\begin{smallmatrix} 0 & b \\ c & 0 \end{smallmatrix}\right)^2 = bc\, I$ and $\det\left(\begin{smallmatrix} 0 & b \\ c & 0 \end{smallmatrix}\right) = -bc$. In $N_{ns}$, $(m_c\sigma)^2 = N(c)\, I$ and $\det(m_c\sigma) = -N(c)$.*

*(iii)* $\det(C_{ns}) = \mathbb{F}_q^\times$*: the norm $\mathbb{F}_q[\sqrt{\varepsilon}]^\times \to \mathbb{F}_q^\times$ is surjective, with kernel of order $q+1$.*

*(iv)* $C_s \cap \mathrm{SL}_2 \simeq C_{q-1}$, $\#(N_s \cap \mathrm{SL}_2) = 2(q-1)$, $C_{ns} \cap \mathrm{SL}_2 \simeq C_{q+1}$, $\#(N_{ns} \cap \mathrm{SL}_2) = 2(q+1)$.

*(v) The orbits on $\mathbb{P}^1(\mathbb{F}_q)$ (coordinate $x$ of (C1)) are: for $C_s$, $\{\infty\}$, $\{0\}$, $\mathbb{F}_q^\times$; for $N_s$, $\{0, \infty\}$, $\mathbb{F}_q^\times$; for* B*, $\{\infty\}$, $\mathbb{F}_q$. The group $C_{ns}$ acts simply transitively on $\mathbb{F}_q^2 \setminus \{0\}$, and $C_{ns}/\mathbb{F}_q^\times I \simeq C_{q+1}$ acts simply transitively on $\mathbb{P}^1(\mathbb{F}_q)$; hence $C_{ns}$ and $N_{ns}$ are transitive on $\mathbb{P}^1(\mathbb{F}_q)$ and stabilise no line of $\mathbb{F}_q^2$.*

*Proof.* (i) The cosets $wC_s$ and $C_{ns}\sigma$ are disjoint from $C_s$ and $C_{ns}$, so the normalisers have twice the order; the numbers follow (and were checked by enumeration). (ii) $wC_s$ consists of the matrices $\left(\begin{smallmatrix} 0 & b \\ c & 0 \end{smallmatrix}\right)$, and $m_c\sigma = \left(\begin{smallmatrix} a & -\varepsilon b \\ b & -a \end{smallmatrix}\right)$ has trace 0. The squares: $\left(\begin{smallmatrix} 0 & b \\ c & 0 \end{smallmatrix}\right)^2 = \left(\begin{smallmatrix} bc & 0 \\ 0 & bc \end{smallmatrix}\right)$, and $(m_c\sigma)^2 = m_c\,\sigma m_c \sigma^{-1}\,\sigma^2 = m_c m_{\bar{c}} = m_{c\bar{c}} = N(c)I$. The determinants are immediate. (iii) $N(c) = c\bar{c} = c^{q+1}$ by Proposition 1.55(iii). On the cyclic group $\mathbb{F}_q[\sqrt{\varepsilon}]^\times$ of order $q^2-1$ the $(q+1)$-th power map has kernel of order $\gcd(q+1, q^2-1) = q+1$ and image of order $q-1$ (Theorem 1.4(ii)), contained in $\mathbb{F}_q^\times$ (as $N(c) = a^2 - \varepsilon b^2 \in \mathbb{F}_q$), which has order $q-1$. (iv) $C_s \cap \mathrm{SL}_2 = \{\mathrm{diag}(a, a^{-1})\} \simeq \mathbb{F}_q^\times$; in $wC_s$ the determinant-one elements $\left(\begin{smallmatrix} 0 & b \\ c & 0 \end{smallmatrix}\right)$ have $bc = -1$, and there are $q-1$ of them. $C_{ns} \cap \mathrm{SL}_2$ is the kernel of the norm, cyclic (as a subgroup of a cyclic group) of order $q+1$; in $C_{ns}\sigma$ the determinant-one elements are the $m_c\sigma$ with $N(c) = -1$, a fibre of the surjective norm, with $q+1$ elements. (v) $\mathrm{diag}(a, d)$ acts by $x \mapsto ax/d$, fixing 0 and $\infty$ and acting transitively on $\mathbb{F}_q^\times$ (take $a = x'/x$, $d = 1$); $w$ acts by $x \mapsto 1/x$, swapping 0 and $\infty$; $\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$ acts by $x \mapsto (ax+b)/d$, fixing $\infty$ and transitive on $\mathbb{F}_q$. Identifying $\mathbb{F}_q^2$ with $\mathbb{F}_q[\sqrt{\varepsilon}]$ through the basis $(1, \sqrt{\varepsilon})$, $C_{ns}$ acts by multiplication by the nonzero elements of a field, which is simply transitive on the nonzero elements. If $m_c$ stabilises the line $\mathbb{F}_q v$, then $cv = \lambda v$ with $\lambda \in \mathbb{F}_q$, so $(c-\lambda)v = 0$ and $c = \lambda$: the stabiliser of every line is $\mathbb{F}_q^\times I$. By Theorem 1.28 the orbit of a line has $(q^2-1)/(q-1) = q+1$ elements, which is all of $\mathbb{P}^1(\mathbb{F}_q)$, and $C_{ns}/\mathbb{F}_q^\times I$ (cyclic of order $q+1$) acts simply transitively. Since $q+1 \geq 2$, no line is fixed. □

**Theorem 1.57** (Dickson)**.** *Let $\ell$ be an odd prime, $G \leq \mathrm{GL}_2(\mathbb{F}_\ell)$ a subgroup and $\bar{G}$ its image in $\mathrm{PGL}_2(\mathbb{F}_\ell)$.*

*(a) If $\ell \mid \#G$, then either $G \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$ or $G$ is contained in a Borel subgroup.*

*(b) If $\ell \nmid \#G$, then exactly one of the following holds: $\bar{G}$ is cyclic and $G$ is contained in a Cartan subgroup (split or nonsplit); $\bar{G}$ is dihedral of order $2n$ with $n \geq 2$, and $G$ is contained in the normaliser of a Cartan subgroup but not in the Cartan subgroup; $\bar{G}$ is isomorphic to $A_4$, $S_4$ or $A_5$ (the* exceptional *case).*

*Consequently a subgroup of* $\mathrm{GL}_2(\mathbb{F}_\ell)$ *not containing* $\mathrm{SL}_2(\mathbb{F}_\ell)$ *is contained in a Borel subgroup, in the normaliser of a split or a nonsplit Cartan subgroup, or in the preimage of an exceptional subgroup of* $\mathrm{PGL}_2(\mathbb{F}_\ell)$.

We do not prove this; see [171, Section 2, Propositions 15 and 16] and, for the original classification of the subgroups of $\mathrm{PSL}_2(\mathbb{F}_q)$, [54]. For $\ell = 3$ the part needed in the paper is proved below (Proposition 1.61 and Theorem 1.65). Its statement for $\mathrm{PGL}_2$ and its use for representations is in Chapter 4, and the list of the images that occur for elliptic curves over $\mathbb{Q}$ is in Chapter 11.

*Warning* 1.58. Theorem 1.57 is a list of containments, not a list of maximal subgroups. For $\ell = 3$ the standard models satisfy $N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$, with index 2 (Proposition 1.63(vii)). For $\ell = 5$, $N_s(\mathbb{F}_5)$ (order 32, a Sylow 2-subgroup of $\mathrm{GL}_2(\mathbb{F}_5)$) lies in a subgroup of order 96 whose image in $\mathrm{PGL}_2(\mathbb{F}_5)$ is isomorphic to $S_4$ (checked by enumeration; see the proof sketch of Theorem 1.52(iii)). So $N_s(\mathbb{F}_\ell)$ must not be called maximal for $\ell = 3$ or $\ell = 5$. The two cases of Proposition prop:ns are distinguished by orbits and by determinant-one subgroups, not by maximality.

## 1.6 The group $\mathrm{GL}_2(\mathbb{F}_3)$ in full, and $\mathrm{GL}_2(\mathbb{F}_5)$, $\mathrm{GL}_2(\mathbb{F}_7)$

This section is the group-theoretic core of Proposition prop:ns and of the boundary case studied in Section sec:anomaly. We determine the conjugacy classes and the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$, identify $N_s(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ and their determinant-one subgroups, and prove that the three properties used in the paper (surjective determinant, no stable line, an element of determinant $-1$ and trace 0) leave exactly the three images $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, $N_s(\mathbb{F}_3)$ (Theorem 1.65), distinguished by their orbits on the four lines (Corollary 1.66). The Galois-theoretic side (that the image of complex conjugation has determinant $-1$ and trace 0, that the roots of the 3-division polynomial correspond to lines, and the list of images that occur) is in Chapter 5, Chapter 9 and Chapter 11.

*Notation for this section.* $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (so $J^2 = -I$ and $J^{-1} = -J$), $\sigma = \mathrm{diag}(1, -1)$, $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) = \sigma J$, $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. This $J$ is the paper's (Proposition prop:ns and Section sec:anomaly), and the book uses it throughout. Note $J\sigma = -w$ and $J = m_{-i}$ in the notation of Proposition 1.55 (with $\varepsilon = -1$), so that $m_i = -J = J^{-1}$. The four lines of $\mathbb{F}_3^2$ are $\infty = [1 : 0]$, $0 = [0 : 1]$, $1 = [1 : 1]$ and $-1 = [-1 : 1]$ (convention (C1)).

**Proposition 1.59** (Conjugacy classes of $\mathrm{GL}_2(\mathbb{F}_3)$)**.** $\mathrm{GL}_2(\mathbb{F}_3)$ *has order* 48*, centre* $\{\pm I\}$*, commutator subgroup* $\mathrm{SL}_2(\mathbb{F}_3)$*, and exactly* 8 *conjugacy classes:*

| *representative* | *size* | *order* | det | tr | *char. polynomial* | *lines fixed* | *image in* $S_4$ |
|---|---|---|---|---|---|---|---|
| $I$ | 1 | 1 | 1 | $-1$ | $(X-1)^2$ | 4 | *identity* |
| $-I$ | 1 | 2 | 1 | 1 | $(X+1)^2$ | 4 | *identity* |
| $\sigma$ | 12 | 2 | $-1$ | 0 | $X^2-1$ | 2 | *transposition* |
| $u$ | 8 | 3 | 1 | $-1$ | $(X-1)^2$ | 1 | 3-*cycle* |
| $J$ | 6 | 4 | 1 | 0 | $X^2+1$ | 0 | *double transposition* |
| $-u$ | 8 | 6 | 1 | 1 | $(X+1)^2$ | 1 | 3-*cycle* |
| $\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ | 6 | 8 | $-1$ | $-1$ | $X^2+X-1$ | 0 | 4-*cycle* |
| $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ | 6 | 8 | $-1$ | 1 | $X^2-X-1$ | 0 | 4-*cycle* |

*The class of a non-scalar element is determined by its characteristic polynomial. The numbers of elements of orders* $1, 2, 3, 4, 6, 8$ *are* $1, 13, 8, 6, 8, 12$.

*Proof.* The order is Proposition 1.43 and the centre is Proposition 1.41(i). By Lemma 1.42 the non-scalar classes correspond to the characteristic polynomials $X^2-tX+d$ with $t \in \mathbb{F}_3$, $d \in \{\pm 1\}$, all six of which occur (companion matrices), and there are two scalar classes; that makes 8. The polynomial $X^2-1$ has distinct roots $\pm 1$; $(X \mp 1)^2$ are the two with a double root; and $X^2+1$, $X^2+X-1$, $X^2-X-1$ have no root in $\mathbb{F}_3$ (evaluate at $0, 1, -1$), so they are irreducible. By Lemma 1.42 the centralisers have orders 4, 6, 8 in these three cases, giving class sizes 12, 8, 6; and $1+1+12+8+6+8+6+6 = 48$. The representatives listed have the stated characteristic polynomials. Orders: $\sigma^2 = I$; $u$ and $-u$ have orders 3 and 6 by Proposition 1.48(ii),(iv); $J^2 = -I$. An element $g$ with irreducible characteristic polynomial generates a subgroup of $\mathbb{F}_3[g]^\times$, a cyclic group of order 8 (it is the unit group of a field with 9 elements, Corollary 1.36); if $g^4 = 1$, its eigenvalues in that field satisfy $\lambda^4 = 1$, so $\lambda \in \{\pm 1, \pm\sqrt{-1}\}$, and as $\lambda \notin \mathbb{F}_3$ the characteristic polynomial is $X^2+1$. For the last two classes the characteristic polynomial is $X^2 \pm X - 1 \neq X^2+1$, so the order divides 8 but not 4: these elements have order 8. (Directly: $a = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ has $a^2 = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) = J$, of order 4.) The lines fixed are the eigenlines over $\mathbb{F}_3$: two for $\sigma$, one for $\pm u$ (Proposition 1.48), none for an irreducible characteristic polynomial, all four for $\pm I$. The image in $S_4$ (Theorem 1.52(i)) has order equal to the order of the element modulo $\{\pm I\}$, and its number of fixed points is the number of fixed lines: order 2 with two fixed points is a transposition, order 3 is a 3-cycle, order 2 with no fixed point ($J$, since $J^2 = -I$) is a double transposition, and order 4 (for elements $g$ of order 8, $g^4 = -I$, $g^2$ not scalar) is a 4-cycle. Commutator subgroup: it lies in $\mathrm{SL}_2(\mathbb{F}_3)$, as det kills commutators; and $\sigma u \sigma^{-1} u^{-1} = u^{-2} = u$ is a commutator, so the normal subgroup $[\mathrm{GL}_2(\mathbb{F}_3), \mathrm{GL}_2(\mathbb{F}_3)]$ contains the whole class of $u$, which generates $\mathrm{SL}_2(\mathbb{F}_3)$ (Proposition 1.48(v)). The order statistics are read off the table: order 2 gives $1+12 = 13$ elements ($-I$ and the class of $\sigma$), order 8 gives $6+6 = 12$, and the others are single classes. (The check $\sigma u \sigma^{-1} = u^{-1}$: $\sigma u \sigma^{-1}$ replaces the off-diagonal entry 1 of $u$ by $-1$, and $u^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$; so $\sigma u \sigma^{-1} u^{-1} = u^{-2}$, which equals $u$ as $u^3 = I$.) □

**Convention 1.60** (Standard models)**.** The following models are fixed for the whole book (with $\varepsilon = -1$ in Definition 1.54, so that $\mathbb{F}_3[\sqrt{-1}] = \mathbb{F}_3[i]$ is the field with 9 elements and $J = m_{-i} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$):

$$C_s(\mathbb{F}_3) = \{\mathrm{diag}(\pm 1, \pm 1)\} \simeq V_4, \qquad N_s(\mathbb{F}_3) = C_s(\mathbb{F}_3) \cup wC_s(\mathbb{F}_3) = \langle J, \sigma \rangle,$$

$$C_{ns}(\mathbb{F}_3) = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : (a,b) \neq (0,0) \right\}, \qquad N_{ns}(\mathbb{F}_3) = C_{ns}(\mathbb{F}_3) \cup C_{ns}(\mathbb{F}_3)\sigma,$$

$$\mathrm{B}(\mathbb{F}_3) = \{\text{upper triangular}\}, \qquad \mathrm{U} = \langle u \rangle,$$

$$Q_8 = \left\{ \pm I, \ \pm J, \ \pm \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \ \pm \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix} \right\}, \qquad C_4 = \langle J \rangle = \{\pm I, \pm J\}.$$

"The image is $N_s(\mathbb{F}_3)$" always means "the image is conjugate to this model" (convention (C5)). With these models $N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$ and $C_4 = N_s(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3) = C_{ns}(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3)$ (Proposition 1.63, Proposition 1.64). The paper's macro 3Ns prints Sutherland's label of the image, not the group.

**Proposition 1.61** (Subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$)**.** $\mathrm{GL}_2(\mathbb{F}_3)$ *has* 55 *subgroups, in* 16 *conjugacy classes:*

| *order* | *type* | *conjugates* | *stable lines* | *orbits on* $\mathbb{P}^1(\mathbb{F}_3)$ | det *onto* | $\det -1$, $\mathrm{tr}\,0$ | *name* |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 4 | 1, 1, 1, 1 | *no* | *no* | |
| 2 | $C_2$ | 1 | 4 | 1, 1, 1, 1 | *no* | *no* | $\{\pm I\}$ |
| 2 | $C_2$ | 12 | 2 | 1, 1, 2 | *yes* | *yes* | $\langle\sigma\rangle$, *3Cs.1.1* |
| 3 | $C_3$ | 4 | 1 | 1, 3 | *no* | *no* | U |
| 4 | $V_4$ | 6 | 2 | 1, 1, 2 | *yes* | *yes* | $C_s$, *3Cs* |
| 4 | $C_4$ | 3 | 0 | 2, 2 | *no* | *no* | $\langle J\rangle$ |
| 6 | $S_3$ | 4 | 1 | 1, 3 | *yes* | *yes* | *3B.1.1* |
| 6 | $S_3$ | 4 | 1 | 1, 3 | *yes* | *yes* | *3B.1.2* |
| 6 | $C_6$ | 4 | 1 | 1, 3 | *no* | *no* | $\{\pm I\} \times \mathrm{U}$ |
| 8 | $C_8$ | 3 | 0 | 4 | *yes* | *no* | $C_{ns}$, *3Cn* |
| 8 | $D_4$ | 3 | 0 | 2, 2 | *yes* | *yes* | $N_s$, *3Ns* |
| 8 | $Q_8$ | 1 | 0 | 4 | *no* | *no* | $Q_8$ |
| 12 | $D_6$ | 4 | 1 | 1, 3 | *yes* | *yes* | B, *3B* |
| 16 | $SD_{16}$ | 3 | 0 | 4 | *yes* | *yes* | $N_{ns}$, *3Nn* |
| 24 | $\mathrm{SL}_2(\mathbb{F}_3)$ | 1 | 0 | 4 | *no* | *no* | $\mathrm{SL}_2(\mathbb{F}_3)$ |
| 48 | $\mathrm{GL}_2(\mathbb{F}_3)$ | 1 | 0 | 4 | *yes* | *yes* | $\mathrm{GL}_2(\mathbb{F}_3)$ |

*In particular the subgroups with surjective determinant form* 9 *conjugacy classes, and all of them except* $C_{ns}(\mathbb{F}_3)$ *contain an element of determinant* $-1$ *and trace* 0. *Every subgroup of order divisible by* 3 *either contains* $\mathrm{SL}_2(\mathbb{F}_3)$ *or stabilises a line of* $\mathbb{F}_3^2$ *(the case* $\ell = 3$ *of Theorem 1.57(a)).*

*Proof.* The table was obtained, and checked, by a complete computer enumeration of the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ (every subgroup is generated by two elements, so it suffices to close up all pairs). The rows that the paper needs are proved in Proposition 1.46(iv), Proposition 1.48, Proposition 1.62, Proposition 1.63, Proposition 1.64, Theorem 1.65 and Proposition 1.68. We prove the last statement. Let $3 \mid \#H$. A Sylow 3-subgroup of $H$ has order 3, so it is one of the groups $\mathrm{U}_L$ (Proposition 1.48(v)). If it is normal in $H$, then every $h \in H$ satisfies $h\mathrm{U}_L h^{-1} = \mathrm{U}_L$; since $h\mathrm{U}_L h^{-1} = \mathrm{U}_{hL}$ fixes exactly the line $hL$, we get $hL = L$, and $H$ stabilises $L$. Otherwise $n_3(H) > 1$, $n_3(H) \equiv 1 \pmod 3$, and $n_3(H) \le 4$ (there are only four subgroups of order 3 in $\mathrm{GL}_2(\mathbb{F}_3)$); so $n_3(H) = 4$, $H$ contains all four $\mathrm{U}_L$, hence all 8 elements of order 3, hence $\mathrm{SL}_2(\mathbb{F}_3)$ by Proposition 1.48(v). □

**Proposition 1.62** ($\mathrm{SL}_2(\mathbb{F}_3)$ and $Q_8$)**.** *(i)* $\mathrm{SL}_2(\mathbb{F}_3)$ *has order* 24, *and* 1, 1, 8, 6, 8 *elements of orders* 1, 2, 3, 4, 6; $-I$ *is its unique element of order* 2.

*(ii) The* 8 *elements of* $\mathrm{SL}_2(\mathbb{F}_3)$ *of order dividing* 4 *form the subgroup* $Q_8$ *of Convention 1.60. It is the unique Sylow* 2*-subgroup of* $\mathrm{SL}_2(\mathbb{F}_3)$, *it is normal in* $\mathrm{GL}_2(\mathbb{F}_3)$, *and it is isomorphic to the quaternion group, with* $i = J$, $j = \left(\begin{smallmatrix} 1 & -1 \\ -1 & -1 \end{smallmatrix}\right)$, $k = ij = \left(\begin{smallmatrix} -1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$.

*(iii)* $\mathrm{SL}_2(\mathbb{F}_3) = Q_8 \rtimes \mathrm{U} \simeq Q_8 \rtimes C_3$, *where* $u$ *permutes* $i, j, k$ *cyclically:* $uiu^{-1} = k$, $uku^{-1} = j$, $uju^{-1} = i$.

*(iv)* $[\mathrm{SL}_2(\mathbb{F}_3), \mathrm{SL}_2(\mathbb{F}_3)] = Q_8$, *and* $\mathrm{SL}_2(\mathbb{F}_3)$ *has no subgroup of order* 12.

*(v)* $\mathrm{SL}_2(\mathbb{F}_3)/\{\pm I\} \simeq A_4$, $\mathrm{SL}_2(\mathbb{F}_3) \not\simeq S_4$, *and* $\mathrm{GL}_2(\mathbb{F}_3)/Q_8 \simeq S_3$.

*(vi)* $Q_8$ *acts transitively on the four lines of* $\mathbb{F}_3^2$; *its image in* $S_4$ *is the normal subgroup* $V_4$.

*Proof.* (i) The classes of $\mathrm{GL}_2(\mathbb{F}_3)$ of determinant 1 are those of $I, -I, u, J, -u$ (Proposition 1.59), of sizes 1, 1, 8, 6, 8; and Proposition 1.41(iv). (ii) The elements of order dividing 4 are $\pm I$ and

the six elements of the class of $J$, which are the six matrices of trace 0 and determinant 1; these are $\pm J$, $\pm\left(\begin{smallmatrix}1&1\\1&-1\end{smallmatrix}\right)$, $\pm\left(\begin{smallmatrix}1&-1\\-1&-1\end{smallmatrix}\right)$, the set $Q_8$ of Convention 1.60. A Sylow 2-subgroup $P$ of $\mathrm{SL}_2(\mathbb{F}_3)$ has order 8, and its elements have order dividing 8, hence dividing 4 (there is no element of order 8 in $\mathrm{SL}_2(\mathbb{F}_3)$); so $P \subseteq Q_8$, and $P = Q_8$ by counting. Hence $Q_8$ is a subgroup, and the unique Sylow 2-subgroup. It is a union of conjugacy classes of $\mathrm{GL}_2(\mathbb{F}_3)$ (conjugation preserves order and determinant), hence normal in $\mathrm{GL}_2(\mathbb{F}_3)$. With $i = J$ and $j = \left(\begin{smallmatrix}1&-1\\-1&-1\end{smallmatrix}\right)$ one computes $k = ij = \left(\begin{smallmatrix}-1&-1\\-1&1\end{smallmatrix}\right)$, $ji = \left(\begin{smallmatrix}1&1\\1&-1\end{smallmatrix}\right) = -k$; so $Q_8$ is non-abelian with unique involution $-I$, and is the quaternion group by Proposition 1.21(v). (iii) $Q_8$ is normal, $\mathrm{U} \cap Q_8 = 1$ (orders 3 and 8), and $\#Q_8\mathrm{U} = 24$; apply Proposition 1.15(i). Direct computation with $u^{-1} = \left(\begin{smallmatrix}1&-1\\0&1\end{smallmatrix}\right)$ gives $uJu^{-1} = \left(\begin{smallmatrix}-1&2\\-1&1\end{smallmatrix}\right) = \left(\begin{smallmatrix}-1&-1\\-1&1\end{smallmatrix}\right) = k$ and $uku^{-1} = \left(\begin{smallmatrix}-2&2\\-1&2\end{smallmatrix}\right) = \left(\begin{smallmatrix}1&-1\\-1&-1\end{smallmatrix}\right) = j$, and then, as $j = ki$, $uju^{-1} = (uku^{-1})(uiu^{-1}) = jk = i$. (iv) $\mathrm{SL}_2(\mathbb{F}_3)/Q_8 \simeq C_3$ is abelian, so the commutator subgroup lies in $Q_8$. It contains $uiu^{-1}i^{-1} = ki^{-1} = -ki = -j$, hence $j = (-j)^{-1}$, and, being normal, the conjugates $uju^{-1} = i$ and $uiu^{-1} = k$; so it is $Q_8$. A subgroup of order 12 would have index 2, so it would contain every square of $\mathrm{SL}_2(\mathbb{F}_3)$: the 8 elements of order 3 (each is the square of its own square, $x = x^4 = (x^2)^2$), $-I = J^2$ and therefore the 8 elements $-u = (-I)u$; that is at least 18 elements, a contradiction. (v) The first isomorphism is Theorem 1.52(i). $S_4$ has nine elements of order 2 (six transpositions, three double transpositions) and $\mathrm{SL}_2(\mathbb{F}_3)$ has one. $\mathrm{GL}_2(\mathbb{F}_3)/Q_8$ has order 6; it is generated by the images of $u$ (order 3) and $\sigma$ (order 2, as $\sigma \notin Q_8$), with $\sigma u\sigma^{-1} = u^{-1}$; so it is $D_3 \simeq S_3$ by Proposition 1.19(i),(iv). (vi) $J$ acts by $x \mapsto -1/x$, that is as $(\infty\ 0)(1\ -1)$; $j$ acts by $x \mapsto (1-x)/(1+x)$, that is as $(\infty\ -1)(0\ 1)$; their product $k$ acts as $(\infty\ 1)(0\ -1)$. So the image of $Q_8$ is $\{1, (\infty\,0)(1\,{-1}), (\infty\,1)(0\,{-1}), (\infty\,{-1})(0\,1)\}$, the normal Klein four-group of $S_4$, which is transitive. □

**Proposition 1.63** (The normaliser of the split Cartan subgroup). *(i) The group $N_s(\mathbb{F}_3) = \{\pm I, \pm J, \pm\sigma, \pm w\}$, which equals $\langle J, \sigma\rangle$, has order 8 and is dihedral, $N_s(\mathbb{F}_3) \simeq D_4$, with $r = J$, $s = \sigma$. Its elements of determinant $-1$ are exactly $\pm\sigma$, $\pm w$; they have trace 0 and order 2.*

*(ii) It is the normaliser of $C_s(\mathbb{F}_3)$ and the stabiliser of the pair of lines $\{\infty, 0\}$; it has 3 conjugates, one for each partition of the four lines into two pairs.*

*(iii) Its determinant-one subgroup is $N_s(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3) = \langle J\rangle \simeq C_4$, cyclic, generated by $J$ and also by $J^{-1} = -J = \left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$.*

*(iv) Its orbits on the four lines are $\{\infty, 0\}$ (the two lines of the Cartan subgroup) and $\{1, -1\}$ (the other two), each of size 2; $\langle J\rangle$ has the same two orbits.*

*(v) $N_s(\mathbb{F}_3)$ stabilises no line of $\mathbb{F}_3^2$ and no line of $\overline{\mathbb{F}}_3^2$. The group $\langle J\rangle$ stabilises no line of $\mathbb{F}_3^2$, but stabilises the two lines of $\mathbb{F}_9^2$ spanned by $(i, 1)$ and $(-i, 1)$, where $i \in \mathbb{F}_9$, $i^2 = -1$; these are its eigenlines, with eigenvalues $-i$ and $i$ respectively, and $\sigma$ interchanges them.*

*(vi) Its image in $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$ is the non-normal Klein four-group generated by the transpositions $(\infty\,0)$ and $(1\,{-1})$.*

*(vii) $N_s(\mathbb{F}_3)$ is contained, with index 2, in the Sylow 2-subgroup $N_{ns}(\mathbb{F}_3)$.*

*Proof.* (i) $C_s(\mathbb{F}_3) = \{\pm I, \pm\sigma\}$, and $wC_s(\mathbb{F}_3) = \{\pm w, \pm w\sigma\} = \{\pm w, \pm J\}$ (as $w\sigma = \left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right) = -J$). So $N_s(\mathbb{F}_3) = \{\pm I, \pm J, \pm\sigma, \pm w\}$. Now $J$ has order 4, $\sigma$ order 2, $\sigma J\sigma^{-1} = \left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right) = J^{-1}$, and

$\langle J,\sigma\rangle \subseteq N_s(\mathbb{F}_3)$ has order at least 8 (it contains $\langle J\rangle$ and $\sigma \notin \langle J\rangle$); so $N_s(\mathbb{F}_3) = \langle J,\sigma\rangle \simeq D_4$ by Proposition 1.19(i). The determinants of $\pm I, \pm J$ are 1 and those of $\pm\sigma, \pm w$ are $-1$; the latter have trace 0 and square to $I$. (ii) is Proposition 1.55(i) (here $\infty = [e_1]$ and $0 = [e_2]$). The stabiliser of the 2-subset $\{\infty, 0\}$ also stabilises its complement $\{1, -1\}$, and by (iv) $N_s(\mathbb{F}_3)$ stabilises no other 2-subset. $\mathrm{GL}_2(\mathbb{F}_3)$ is transitive on the six 2-subsets (Proposition 1.44(v)), and $gN_s(\mathbb{F}_3)g^{-1}$ is the stabiliser of $g\{\infty, 0\}$; so the conjugates correspond to the three partitions into two pairs. (iii) The determinant-one elements are $\pm I, \pm J$. (iv) $\sigma$ acts by $x \mapsto -x$, as $(1\,{-1})$; $w$ by $x \mapsto 1/x$, as $(\infty\,0)$ (note $1/(-1) = -1$); $J$ by $x \mapsto -1/x$, as $(\infty\,0)(1\,{-1})$. (v) A stable line of $\mathbb{F}_3^2$ would be a fixed point on the four lines, and there is none by (iv). Over $\overline{\mathbb{F}}_3$: $[N_s(\mathbb{F}_3), N_s(\mathbb{F}_3)] \ni J\sigma J^{-1}\sigma^{-1} = J \cdot J = -I$, so the commutator subgroup contains an element of order 2 and is not a 3-group; apply Lemma 1.50(ii). For $\langle J\rangle$: $X^2 + 1$ has no root in $\mathbb{F}_3$, so $J$ has no eigenline over $\mathbb{F}_3$. Over $\mathbb{F}_9$, $J\binom{x}{y} = \binom{y}{-x}$, and $\binom{\lambda}{1}$ is an eigenvector, with eigenvalue $-\lambda$, iff $\lambda^2 = -1$; so the eigenlines are spanned by $(i, 1)$ and $(-i, 1)$, and they are stable under $\langle J\rangle$. Finally $\sigma\binom{i}{1} = \binom{i}{-1} = -\binom{-i}{1}$. (vi) follows from (iv). (vii) $J = m_{-i} \in C_{ns}(\mathbb{F}_3)$ and $\sigma \in N_{ns}(\mathbb{F}_3)$, so $\langle J,\sigma\rangle \subseteq N_{ns}(\mathbb{F}_3)$, which has order 16 and is a Sylow 2-subgroup by Proposition 1.64(iii). □

**Proposition 1.64** (The normaliser of the nonsplit Cartan subgroup)**.** *(i) The group $C_{ns}(\mathbb{F}_3)$ is cyclic of order 8, generated by $a = \left(\begin{smallmatrix}1 & -1\\ 1 & 1\end{smallmatrix}\right) = m_{1+i}$; $a^2 = J$ and $a^4 = -I$.*

*(ii) $N_{ns}(\mathbb{F}_3) = \langle a,\sigma\rangle$ has order 16, $\sigma a\sigma^{-1} = a^3$ (conjugation by $\sigma$ is the Frobenius $c \mapsto c^3$ of $\mathbb{F}_9$), and $N_{ns}(\mathbb{F}_3) \simeq SD_{16}$; it has $1, 5, 6, 4$ elements of orders $1, 2, 4, 8$.*

*(iii) It is a Sylow 2-subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$, with 3 conjugates.*

*(iv) Its determinant-one subgroup is $Q_8$: the four elements $\pm I, \pm J$ of $C_{ns}(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3)$ and the four elements $m_c\sigma$ with $N(c) = -1$. Its three subgroups of index 2 are $C_{ns}(\mathbb{F}_3) \simeq C_8$, $N_s(\mathbb{F}_3) = \langle a^2,\sigma\rangle \simeq D_4$ and $Q_8 = \langle a^2, a\sigma\rangle$.*

*(v) Its elements of determinant $-1$ and trace 0 are the four involutions $m_c\sigma$ with $N(c) = 1$, namely $\pm\sigma$, $\pm w$ (the same four as in $N_s(\mathbb{F}_3)$). $C_{ns}(\mathbb{F}_3)$ contains none: its elements of determinant $-1$ are its four elements of order 8, of trace $\pm 1$.*

*(vi) $C_{ns}(\mathbb{F}_3)$, hence $N_{ns}(\mathbb{F}_3)$, acts transitively on the four lines, $C_{ns}(\mathbb{F}_3)/\{\pm I\} \simeq C_4$ simply transitively; the image of $N_{ns}(\mathbb{F}_3)$ in $S_4$ is a Sylow 2-subgroup, isomorphic to $D_4$. $Q_8$ is also transitive.*

*Proof.* (i) $C_{ns}(\mathbb{F}_3) \simeq \mathbb{F}_9^\times$ is cyclic of order 8 (Proposition 1.55(ii)). In $\mathbb{F}_9$, $(1+i)^2 = 2i = -i$ and $(1+i)^4 = (-i)^2 = -1$, so $1+i$ has order 8; and $m_{1+i} = I + m_i = I - J = a$, $a^2 = m_{-i} = J$, $a^4 = -I$. (ii) By Proposition 1.55(iii), $\sigma a\sigma^{-1} = m_{1-i}$ and $(1+i)^3 = (1+i)(-i) = 1 - i$; so $\sigma a\sigma^{-1} = a^3$. The group $\langle a,\sigma\rangle$ contains $C_{ns}(\mathbb{F}_3)$ and $\sigma$, so it is $N_{ns}(\mathbb{F}_3)$, of order 16; by Proposition 1.23(iv) it is $SD_{16}$, and (ii) of that proposition gives the element orders. (iii) $48 = 16 \cdot 3$. The number of conjugates is $n_2 \in \{1, 3\}$, and $N_{ns}(\mathbb{F}_3)$ is not normal: its elements are the matrices $\left(\begin{smallmatrix}a & -b\\ b & a\end{smallmatrix}\right)$ and $m_c\sigma = \left(\begin{smallmatrix}a & b\\ b & -a\end{smallmatrix}\right)$, and $u\sigma u^{-1} = \left(\begin{smallmatrix}1 & 1\\ 0 & -1\end{smallmatrix}\right)$ is of neither form. So $n_2 = 3$. (iv) $C_{ns} \cap \mathrm{SL}_2$ is the kernel of the norm, of order 4, which is $\langle a^2\rangle = \langle J\rangle$; in the other coset $\det(m_c\sigma) = -N(c)$, and $N(c) = -1$ for four values of $c$ (Proposition 1.56(iv)). So the determinant-one subgroup has order 8; it is a 2-subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$ of order 8, hence the Sylow 2-subgroup $Q_8$ (Proposition 1.62(ii)). The index-2 subgroups are those of Proposition 1.23(iii): $\langle a\rangle = C_{ns}(\mathbb{F}_3)$; $\langle a^2,\sigma\rangle = \langle J,\sigma\rangle = N_s(\mathbb{F}_3)$; and $\langle a^2, a\sigma\rangle$, which lies in $\mathrm{SL}_2(\mathbb{F}_3)$ ($\det a = \det\sigma = -1$) and has order 8, so it is $Q_8$. (v) The

elements $m_c\sigma$ have trace 0, and determinant $-1$ iff $N(c) = 1$, that is $c \in \{\pm 1, \pm i\}$; these give $\pm\sigma$ and $\pm J\sigma = \pm w$. In $C_{ns}(\mathbb{F}_3)$, $\det a = -1$ and $\det a^2 = 1$, so the elements of determinant $-1$ are the odd powers of $a$, the elements of order 8; $a$ and $a^3 = m_{1-i}$ have trace $2 = -1$, and $a^5 = -a$, $a^7 = -a^3$ have trace 1. (vi) is Proposition 1.56(v) for $q = 3$, and the image has order $16/2 = 8$, a Sylow 2-subgroup of $S_4$, which is dihedral (Example 1.34(b)). For $Q_8$ see Proposition 1.62(vi). □

**Theorem 1.65** (The three images)**.** *Let $H \leq \mathrm{GL}_2(\mathbb{F}_3)$ satisfy:*

*(a)* $\det(H) = \mathbb{F}_3^\times$*;*

*(b) $H$ stabilises no line of $\mathbb{F}_3^2$;*

*(c) $H$ contains an element of determinant $-1$ and trace 0.*

*Then $H$ is conjugate to exactly one of $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ and $N_s(\mathbb{F}_3)$, of orders 48, 16, 8. Their determinant-one subgroups $H \cap \mathrm{SL}_2(\mathbb{F}_3)$, of index 2, are $\mathrm{SL}_2(\mathbb{F}_3)$, $Q_8$ and $C_4 = \langle J \rangle$ respectively. The first two stabilise no line of $\overline{\mathbb{F}}_3^2$; the third is abelian and stabilises two lines of $\mathbb{F}_9^2$. The subgroup $C_{ns}(\mathbb{F}_3)$ satisfies (a) and (b) but not (c).*

*Proof.* The three conditions are invariant under conjugation, and the three groups satisfy them: (a) and (c) because each contains $\sigma$; (b) because $\mathrm{GL}_2(\mathbb{F}_3)$ and $N_{ns}(\mathbb{F}_3)$ are transitive on the four lines and $N_s(\mathbb{F}_3)$ has two orbits of size 2 (Proposition 1.64(vi), Proposition 1.63(iv)). They have different orders, so $H$ is conjugate to at most one of them.

Suppose first that $3 \mid \#H$. By Proposition 1.61 (last statement) and (b), $H \supseteq \mathrm{SL}_2(\mathbb{F}_3)$; by (a), $H$ contains an element of determinant $-1$, so $H$ has index at most 2 in $\mathrm{GL}_2(\mathbb{F}_3)$ and properly contains $\mathrm{SL}_2(\mathbb{F}_3)$, whence $H = \mathrm{GL}_2(\mathbb{F}_3)$.

Suppose now that $3 \nmid \#H$. Then $\#H$ divides 16, $H$ is a 2-group, and by Theorem 1.33(ii) it is contained in a Sylow 2-subgroup, which is conjugate to $N_{ns}(\mathbb{F}_3)$. Replacing $H$ by a conjugate we may assume $H \leq N_{ns}(\mathbb{F}_3)$. By (a), $H \cap \mathrm{SL}_2(\mathbb{F}_3)$ has index 2 in $H$; in particular $\#H \geq 2$.

- If $\#H = 16$, then $H = N_{ns}(\mathbb{F}_3)$.
- If $\#H = 8$, $H$ is one of the three subgroups of index 2 of Proposition 1.64(iv). $Q_8$ fails (a), $C_{ns}(\mathbb{F}_3)$ fails (c) by Proposition 1.64(v), and $N_s(\mathbb{F}_3)$ satisfies all three. So $H = N_s(\mathbb{F}_3)$.
- If $\#H \in \{2, 4\}$, then by (c) $H$ contains an element $g$ of determinant $-1$ and trace 0, an involution with two eigenlines (characteristic polynomial $X^2 - 1$). Then $H \cap \mathrm{SL}_2(\mathbb{F}_3)$ has order $\#H/2 \leq 2$, so it is 1 or $\{\pm I\}$ ($-I$ being the only involution of $\mathrm{SL}_2(\mathbb{F}_3)$), and $H = (H \cap \mathrm{SL}_2(\mathbb{F}_3)) \cup g(H \cap \mathrm{SL}_2(\mathbb{F}_3))$ is $\{I, g\}$ or $\{\pm I, \pm g\}$. Both stabilise the eigenlines of $g$, contradicting (b).

So $H$ is conjugate to $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$. The determinant-one subgroups are computed in Proposition 1.64(iv) and Proposition 1.63(iii). The commutator subgroups of $\mathrm{SL}_2(\mathbb{F}_3)$ and of $Q_8$ are $Q_8$ and $\{\pm I\}$ (Proposition 1.62(iv), Proposition 1.21(ii)), which are not 3-groups, so by Lemma 1.50(ii) these groups stabilise no line of $\overline{\mathbb{F}}_3^2$. The group $\langle J \rangle$ is cyclic and stabilises the two eigenlines of $J$ in $\mathbb{F}_9^2$ (Proposition 1.63(v)). Finally $C_{ns}(\mathbb{F}_3)$ has surjective determinant and is transitive on the lines (Proposition 1.56(iii),(v)), but contains no element of determinant $-1$ and trace 0 (Proposition 1.64(v)). □

**Corollary 1.66** (Orbit dichotomy)**.** *For $H$ as in Theorem 1.65 the following are equivalent:*

*(i)* $H$ *is conjugate to* $N_s(\mathbb{F}_3)$*;*

*(ii)* $H$ *has two orbits, each of size* 2*, on the four lines of* $\mathbb{F}_3^2$*;*

*(iii)* $H$ *is not transitive on the four lines;*

*(iv)* $H \cap \mathrm{SL}_2(\mathbb{F}_3)$ *is abelian;*

*(v)* $H \cap \mathrm{SL}_2(\mathbb{F}_3)$ *stabilises a line of* $\overline{\mathbb{F}}_3^2$*.*

*Otherwise* $H$ *is transitive on the four lines.*

*Proof.* All five properties are invariant under conjugation, so by Theorem 1.65 we may take $H$ to be one of the three standard models. For $N_s(\mathbb{F}_3)$, (ii) holds by Proposition 1.63(iv), (iv) because $\langle J \rangle$ is cyclic, and (v) by Proposition 1.63(v); (ii) implies (iii). For $\mathrm{GL}_2(\mathbb{F}_3)$ and $N_{ns}(\mathbb{F}_3)$, (ii) and (iii) fail because they are transitive, (iv) fails because $\mathrm{SL}_2(\mathbb{F}_3)$ and $Q_8$ are non-abelian, and (v) fails by Theorem 1.65. □

*In the paper* 1.67. The proof of Proposition prop:ns states that when $E[3]$ is irreducible the image $H$ of $\bar{\rho}_{E,3}$ is conjugate to $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$, notes that the list is in [210], and derives it directly: "these are the subgroups with surjective determinant acting irreducibly on $\mathbb{F}_3^2$ that contain an element of determinant $-1$ and trace 0 (complex conjugation), the nonsplit Cartan subgroup itself being excluded by the last condition, and the list follows from these three properties by inspection of the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$". Theorem 1.65 is that inspection, carried out. The three hypotheses hold for $H = \bar{\rho}_{E,3}(G_{\mathbb{Q}})$: (a) because $\det \bar{\rho}_{E,3} = \bar{\chi}_3$ is surjective onto $\mathbb{F}_3^\times$; (b) because irreducibility of $E[3]$ means that no line is stable; (c) because the image of complex conjugation $c$ satisfies $\bar{\rho}(c)^2 = I$ and $\det \bar{\rho}(c) = \bar{\chi}_3(c) = -1$, so its eigenvalues are 1 and $-1$ and its trace is 0 (Chapter 11). As the paper says, the list holds with or without complex multiplication: none of (a), (b), (c) refers to it. The determinant-one subgroups are the paper's "$\mathrm{SL}_2(\mathbb{F}_3)$, the quaternion group $Q_8$ and the cyclic group $C_4$ of order 4". The paper's final sentence on the 3-division polynomial is Corollary 1.66(i)⇔(ii)⇔(iii): its four roots correspond to the four lines, the factors over $\mathbb{Q}$ correspond to the orbits of $H$ (Chapter 5), and so it factors as two irreducible quadratics exactly in the $N_s(\mathbb{F}_3)$ case and is irreducible otherwise. The equivalence (i)⇔(v) is the group-theoretic half of "$\bar{\rho}_{E,3}$ is absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ unless its image is conjugate to the normaliser $N_s(\mathbb{F}_3)$ of a split Cartan subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$"; the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ is $H \cap \mathrm{SL}_2(\mathbb{F}_3)$ by Lemma 1.7(ii). The consequences $\bar{\rho}_{E,3} \otimes \mathbb{F}_9 \simeq \mathrm{Ind}\,\psi$ and $\bar{\rho}_{E,3} \simeq \bar{\rho}_{E,3} \otimes \chi_{-3}$ are representation theory (Chapter 4), and $h_3(E) = 1$ is Lemma lem:h0, proved in Chapter 11.

**Proposition 1.68** (Determinant $-1$ and trace 0)**.** *Let* $q$ *be odd and* $g \in \mathrm{GL}_2(\mathbb{F}_q)$*. Then* $\det g = -1$ *and* $\operatorname{tr} g = 0$ *if and only if the characteristic polynomial of* $g$ *is* $X^2 - 1$*, if and only if* $g$ *is conjugate to* $\sigma = \mathrm{diag}(1, -1)$*. Such an element has order* 2 *and fixes exactly two lines (its eigenlines); for* $q = 3$ *it swaps the other two. These elements form one conjugacy class of* $q(q+1)$ *elements. For* $q = 3$ *their numbers in the standard subgroups are:* $\mathrm{GL}_2(\mathbb{F}_3)$*:* 12*;* B*:* 6*;* $N_s$*:* 4*;* $N_{ns}$*:* 4*;* $C_s$*:* 2*;* $C_{ns}$*:* 0*;* $\mathrm{SL}_2(\mathbb{F}_3)$*:* 0*.*

*Proof.* The characteristic polynomial is $X^2 - \operatorname{tr}(g)X + \det g$. A matrix with characteristic polynomial $X^2 - 1$ is not scalar (a scalar $\lambda I$ has $(X - \lambda)^2$, and $1 \neq -1$), so it is conjugate to $\sigma$ by Lemma 1.42. Its eigenvalues 1, $-1$ are distinct, which gives the order and the two eigenlines. For $q = 3$, $\sigma$ acts as the transposition $(1\,{-1})$ (Proposition 1.59). The centraliser of $\sigma$ is $C_s$, of

order $(q-1)^2$, so the class has $q(q-1)^2(q+1)/(q-1)^2 = q(q+1)$ elements. In $\mathrm{B}(\mathbb{F}_3)$: $\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$ with $a+d=0$ and $ad=-1$ means $(a,d) = \pm(1,-1)$ with $b$ arbitrary, 6 elements. The counts in $N_s$, $N_{ns}$ and $C_{ns}$ are Proposition 1.63(i) and Proposition 1.64(v); in $C_s$ they are $\pm\sigma$; and $\mathrm{SL}_2$ has no element of determinant $-1$. □

**Example 1.69** (Sutherland's labels at $\ell = 3$)**.** The paper names mod 3 images by the labels of [191]. Because a transposition (row vectors versus column vectors) interchanges some of the matrix descriptions, we define each label by an intrinsic property of the action on $E[3]$, which is convention-free; the groups are then the standard models up to conjugacy, and the orders are those of Proposition 1.61.

- **3Cs.1.1**: $\{I, \sigma\}$, of order 2: $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$, that is, a basis in which $G_{\mathbb{Q}}$ acts trivially on the first vector and by the determinant $\bar{\chi}_3$ on the second. (It is a subgroup of the split Cartan subgroup, not the split Cartan subgroup itself.)
- **3Cs**: $C_s(\mathbb{F}_3)$, of order 4: the full group of matrices stabilising each of two distinct lines.
- **3B.1.1**: $\{\left(\begin{smallmatrix} 1 & * \\ 0 & * \end{smallmatrix}\right)\}$, of order 6: a fixed nonzero vector, that is, a rational point of order 3.
- **3B.1.2**: $\{\left(\begin{smallmatrix} * & * \\ 0 & 1 \end{smallmatrix}\right)\}$, of order 6: a stable line, on which the group acts by the determinant, with trivial action on the quotient.
- **3B**: $\mathrm{B}(\mathbb{F}_3)$, of order 12: the full stabiliser of a line; a stable line means a rational 3-isogeny.
- **3Ns**: $N_s(\mathbb{F}_3)$, of order 8. **3Nn**: $N_{ns}(\mathbb{F}_3)$, of order 16.
- **3Cn**: $C_{ns}(\mathbb{F}_3)$, of order 8, in the same naming scheme. It contains no element of determinant $-1$ and trace 0 (Proposition 1.68), so it is never the mod 3 image of an elliptic curve over $\mathbb{Q}$ (Chapter 11).
- The surjective image $\mathrm{GL}_2(\mathbb{F}_3)$ carries no proper label.

These are the nine conjugacy classes of subgroups with surjective determinant in Proposition 1.61. The table there shows that 3B.1.1 and 3B.1.2 are not conjugate in $\mathrm{GL}_2(\mathbb{F}_3)$: they are distinguished by the existence of a nonzero fixed vector, which is the intrinsic property above. Which images occur for elliptic curves over $\mathbb{Q}$ is [210]; see Chapter 11.

*In the paper* 1.70. Section sec:anomaly and Remark rem:ns concern the curves with image 3Ns, that is, $H$ conjugate to $N_s(\mathbb{F}_3)$. By Corollary 1.66 these are exactly the curves (with $E[3]$ irreducible) whose 3-division polynomial factors as two quadratics, which is the intrinsic description of the label 3Ns that Section sec:data of the paper uses (Chapter 22). Section sec:eis and Conjecture conj:eis use the Borel-type labels: a rational 3-isogeny means image in a Borel subgroup (Proposition 1.46(iii)); a rational point of order 3 means image in 3B.1.1; and $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ means image 3Cs.1.1, the subgroup of order 2 of the split Cartan subgroup that fixes a nonzero vector.

**Proposition 1.71** (Images in $S_4$)**.** *Under* $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$ *(Theorem 1.52(i)) the standard subgroups map as follows:*

| *subgroup* | $\mathrm{GL}_2$ | $\mathrm{SL}_2$ | $N_{ns}$ | $Q_8$ | $N_s$ | $C_{ns}$ | $C_s$ | $\langle J \rangle$ | B | U |
|---|---|---|---|---|---|---|---|---|---|---|
| *image* | $S_4$ | $A_4$ | $D_4$ | $V_4 \trianglelefteq S_4$ | $V_4 \not\trianglelefteq S_4$ | $C_4$ | $\langle(1\,{-1})\rangle$ | $\langle(\infty\,0)(1\,{-1})\rangle$ | $S_3$ | $A_3$ |
| *order* | 24 | 12 | 8 | 4 | 4 | 4 | 2 | 2 | 6 | 3 |

*Here $D_4$ is a Sylow 2-subgroup of $S_4$, $S_3$ is the stabiliser of $\infty$, and $A_3$ is generated by the 3-cycle $(0\,1\,{-1})$.*

*Proof.* The image of $K$ is $K/(K\cap\{\pm I\})$, which gives the orders. $\mathrm{GL}_2$ and $\mathrm{SL}_2$: Theorem 1.52(i). $N_{ns}$: Proposition 1.64(vi). $Q_8$: Proposition 1.62(vi). $N_s$ and $\langle J\rangle$: Proposition 1.63(iv),(vi). $C_{ns}$ is cyclic of order 8 with $a^4=-I$, so its image is cyclic of order 4. $C_s=\{\pm I,\pm\sigma\}$ maps onto $\langle(1\,{-1})\rangle$. B: Proposition 1.46(iv). $u$ acts by $x\mapsto x+1$, that is as $(0\,1\,{-1})$, fixing $\infty$. □

**Example 1.72** ($\mathrm{GL}_2(\mathbb{F}_5)$ and $\mathrm{GL}_2(\mathbb{F}_7)$)**.** For $q=5$ and $q=7$ (with $\varepsilon=2$ and $\varepsilon=3$, non-squares modulo 5 and 7) the formulas of Proposition 1.43, Proposition 1.46 and Proposition 1.56 give:

| $q$ | $\mathrm{GL}_2$ | $\mathrm{SL}_2$ | $\mathrm{PGL}_2$ | $\mathrm{PSL}_2$ | B | $C_s$ | $N_s$ | $C_{ns}$ | $N_{ns}$ | $\#\mathbb{P}^1$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | 480 | 120 | 120 | 60 | 80 | 16 | 32 | 24 | 48 | 6 |
| 7 | 2016 | 336 | 336 | 168 | 252 | 36 | 72 | 48 | 96 | 8 |

(all checked by enumeration). By Theorem 1.52, $\mathrm{PGL}_2(\mathbb{F}_5)\simeq S_5$, $\mathrm{PSL}_2(\mathbb{F}_5)\simeq A_5$, and $\mathrm{PSL}_2(\mathbb{F}_7)\simeq\mathrm{GL}_3(\mathbb{F}_2)$ is simple. At $\ell=5$ the exceptional case $\bar G\simeq S_4$ of Theorem 1.57 occurs inside $\mathrm{PGL}_2(\mathbb{F}_5)\simeq S_5$ and contains the image of $N_s(\mathbb{F}_5)$ (Warning 1.58). The coincidence that makes $\ell=3$ special in the paper is not group-theoretic: for $\ell=3$ the reduction $\bar\chi_3$ of the cyclotomic character equals the quadratic character $\chi_{-3}$, while for $\ell\geq 5$ the character $\bar\chi_\ell$ has order $\ell-1>2$ and differs from $\chi_{\ell^*}$ (Theorem 1.4(iv)). This is why Lemma lem:h0 gives $h_\ell(E)=0$ for $\ell\geq 5$ (Chapter 4, Chapter 11).

## 1.7 Inverse limits and profinite groups

The Galois groups of the paper, $G_{\mathbb{Q}}=\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, the local groups $G_{\mathbb{Q}_p}$ and the inertia groups $I_p$, are infinite, and the representations $\rho_{E,\ell}$ take values in $\mathrm{GL}_2(\mathbb{Z}_\ell)$. Both are profinite groups: inverse limits of finite groups, with a compact topology in which "continuous" means "factors through a finite quotient at each finite level". This section gives the definitions and the facts used in the paper: a continuous representation modulo $\ell^k$ factors through a finite quotient (Lemma lem:h0, Lemma lem:tamagawa-exp), a pro-$p$ group has no nontrivial continuous image of order prime to $p$ (the wild inertia in Lemma lem:tamfactor), and the kernel of $\mathrm{GL}_2(\mathbb{Z}_3)\to\mathrm{GL}_2(\mathbb{F}_3)$ is torsion-free (Lemma lem:IV). The infinite Galois theory itself is Chapter 5.

**Definition 1.73** (Inverse limits; profinite groups)**.** An *inverse system* of groups (indexed by $\mathbb{N}$) is a sequence of groups $G_n$, $n\geq 1$, with homomorphisms $\pi_n\colon G_{n+1}\to G_n$. Its *inverse limit* is

$$\varprojlim G_n=\Big\{(g_n)_{n\geq 1}\in\prod_{n\geq 1}G_n:\ \pi_n(g_{n+1})=g_n\ \text{for all } n\Big\},$$

a subgroup of the product, with projections $p_n\colon\varprojlim G_n\to G_n$. When the $G_n$ are finite, each is given the discrete topology, $\prod G_n$ the product topology and $\varprojlim G_n$ the subspace topology. A *profinite group* is a topological group isomorphic, as a topological group, to such an inverse limit of finite groups. A *pro-$p$ group* is an inverse limit of finite $p$-groups.

An element of $\varprojlim G_n$ is a sequence of compatible approximations, one at each finite level. For $G_n=\mathbb{Z}/3^n$ with the reduction maps, the sequence $(2,8,26,80,\dots)$, with $n$-th term $3^n-1$, is compatible ($3^{n+1}-1\equiv 3^n-1\pmod{3^n}$) and is the element $-1$ of $\mathbb{Z}_3$; the sequence $(1,4,13,40,\dots)$ with $n$-th term $1+3+\cdots+3^{n-1}=(3^n-1)/2$ is compatible as well, and twice it is $-1$, so it is

$-1/2 \in \mathbb{Z}_3$ (Example 1.76). The topology says that two elements are close when they agree at many levels: the sets $gU_n$ of elements agreeing with $g$ in the first $n$ coordinates, $n \geq 1$, form a basis of open neighbourhoods of $g$ (Proposition 1.77(i)). A homomorphism to a finite group is continuous exactly when it only depends on finitely many coordinates (Proposition 1.77(iii)), which is the meaning of continuity for Galois representations modulo $\ell^k$.

*Remark* 1.74. For Galois groups one needs inverse systems indexed by a *directed* set $(I, \leq)$ (a partial order in which any two elements have an upper bound), with maps $\pi_{ij} \colon G_j \to G_i$ for $i \leq j$ satisfying $\pi_{ii} = \mathrm{id}$ and $\pi_{ij}\pi_{jk} = \pi_{ik}$; the limit is the set of $(g_i) \in \prod_i G_i$ with $\pi_{ij}(g_j) = g_i$. Everything below holds in this generality with the same proofs, replacing "$n$ large" by "$i$ large in $I$". For example $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) = \varprojlim \mathrm{Gal}(K/\mathbb{Q})$ over the finite Galois extensions $K/\mathbb{Q}$ inside $\overline{\mathbb{Q}}$, ordered by inclusion (Chapter 5).

**Proposition 1.75.** *Let $(G_n, \pi_n)$ be an inverse system of finite groups and $G = \varprojlim G_n$.*

*(i) (Universal property) For every group $H$ with homomorphisms $f_n \colon H \to G_n$ such that $\pi_n \circ f_{n+1} = f_n$, there is a unique homomorphism $f \colon H \to G$ with $p_n \circ f = f_n$ for all $n$.*

*(ii) $G$ is a closed subgroup of $\prod G_n$, and a Hausdorff topological group.*

*(iii) $G$ is compact.*

*Proof.* (i) Put $f(h) = (f_n(h))_n$, which lies in $G$ by the compatibility; it is a homomorphism, and it is the only map with $p_n \circ f = f_n$. (ii) For each $n$ the set $\{(g_m) : \pi_n(g_{n+1}) = g_n\}$ is the preimage of the closed subset $\{(x, y) : \pi_n(x) = y\}$ of the discrete space $G_{n+1} \times G_n$ under the continuous projection; so it is closed, and $G$ is the intersection of these closed sets. Multiplication and inversion are continuous coordinatewise, hence continuous on the product and on $G$. A product of Hausdorff spaces is Hausdorff. (iii) By Tychonoff's theorem a product of compact spaces is compact (we do not prove this; see [139, Theorem 37.3]); finite discrete spaces are compact, and a closed subset of a compact space is compact. □

**Example 1.76** (The $p$-adic integers)**.** Let $p$ be a prime. The rings $\mathbb{Z}/p^n$ with the reduction maps form an inverse system, and $\mathbb{Z}_p = \varprojlim \mathbb{Z}/p^n$ is a ring (the operations being coordinatewise) and a compact topological group; its analysis, as the completion of $\mathbb{Z}$ for the $p$-adic absolute value, is Chapter 7.

(a) $x = (x_n) \in \mathbb{Z}_p$ is a unit if and only if $x_1 \neq 0$ in $\mathbb{Z}/p$; and $\mathbb{Z}_p^\times = \varprojlim (\mathbb{Z}/p^n)^\times$. Indeed, if $x_1 \neq 0$, every $x_n$ is prime to $p$, hence a unit of $\mathbb{Z}/p^n$, and the inverses $x_n^{-1}$ are compatible (inverses are unique), giving $x^{-1}$; conversely $x_1$ is a unit if $x$ is.

(b) The kernel of $\mathbb{Z}_p \to \mathbb{Z}/p^n$ is $p^n\mathbb{Z}_p$. If $x_n = 0$, then for every $k$ the element $x_{n+k} \in \mathbb{Z}/p^{n+k}$ is divisible by $p^n$, $x_{n+k} = p^n y_k$ with $y_k$ well defined in $\mathbb{Z}/p^k$, and the $y_k$ are compatible, so $x = p^n y$. The converse is clear.

(c) $\mathbb{Z}_p$ is torsion-free. If $mx = 0$ with $m \geq 1$ and $v = v_p(m)$, then $mx_{n+v} = 0$ in $\mathbb{Z}/p^{n+v}$. Write $m = p^v m'$; as $m'$ is prime to $p$ it is invertible modulo $p^{n+v}$, so $p^v x_{n+v} = 0$ in $\mathbb{Z}/p^{n+v}$, that is $p^{n+v}$ divides $p^v$ times a representative of $x_{n+v}$, so $p^n \mid x_{n+v}$. Hence $x_n = x_{n+v} \bmod p^n = 0$ for every $n$. (By contrast every element of each finite level $\mathbb{Z}/p^n$ is torsion; the point is that $x_n$ is forced to vanish by looking at the deeper level $n + v$.)

(d) The open subgroups of $\mathbb{Z}_p$ are the $p^n\mathbb{Z}_p$, $n \geq 0$. By Proposition 1.77(i),(ii) (proved below) an open subgroup $V$ contains some $p^n\mathbb{Z}_p$ and corresponds to a subgroup of $\mathbb{Z}_p/p^n\mathbb{Z}_p \simeq \mathbb{Z}/p^n$, which is $p^k\mathbb{Z}/p^n$ for some $k \leq n$ (Theorem 1.4(i)); so $V = p^k\mathbb{Z}_p$.

(e) Let $\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n!$; every $\mathbb{Z}/m$ is a quotient of $\widehat{\mathbb{Z}}$ (as $m \mid n!$ for $n \geq m$). By Proposition 1.13 (in its ring form, Chapter 2) $\mathbb{Z}/n! \simeq \prod_p \mathbb{Z}/p^{v_p(n!)}$, compatibly in $n$, and taking limits gives $\widehat{\mathbb{Z}} \simeq \prod_p \mathbb{Z}_p$.

(f) (Foretaste.) For a prime $\ell$ and a field $K$ of characteristic $\neq \ell$ containing all $\ell$-power roots of unity, $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^n}(K)$, with the maps $\zeta \mapsto \zeta^\ell$. Choosing a compatible system of generators $\zeta_{\ell^n}$ ($\zeta_{\ell^{n+1}}^\ell = \zeta_{\ell^n}$) gives an isomorphism $\mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(1)$ of groups, but the Galois group acts on $\mathbb{Z}_\ell(1)$ through the cyclotomic character $\chi_\ell$ (Chapter 5, Chapter 7). The paper's coefficient modules $\mathbb{Z}_\ell$, $T_\ell E$, $\mathbb{Z}_\ell(1)$ and $\mathbb{Q}_\ell/\mathbb{Z}_\ell = \varinjlim \ell^{-n}\mathbb{Z}_\ell/\mathbb{Z}_\ell$ are all built from these.

**Proposition 1.77** (Profinite groups)**.** *Let $G = \varprojlim G_n$ be profinite, and $U_n = \ker(p_n \colon G \to G_n)$.*

*(i) The $U_n$ are open normal subgroups of finite index, $U_{n+1} \subseteq U_n$, $\bigcap_n U_n = 1$, and they form a basis of open neighbourhoods of 1.*

*(ii) A subgroup of $G$ is open if and only if it is closed of finite index.*

*(iii) A homomorphism $f$ from $G$ to a finite group $F$ that is continuous ($F$ discrete) has open kernel, and factors through $G/U_n$ for some $n$; if the maps $p_n$ are surjective, $f$ factors through $G_n$.*

*(iv) Closed subgroups of $G$, and quotients of $G$ by closed normal subgroups, are profinite. The image of a continuous homomorphism from $G$ to a Hausdorff topological group (such as $\mathrm{GL}_2(\mathbb{Z}_\ell)$) is closed.*

*(v) A topological group is profinite (for an inverse system indexed by a directed set, Remark 1.74) if and only if it is compact, Hausdorff and totally disconnected.*

*(vi) Closed subgroups and quotients by closed normal subgroups of a pro-$p$ group are pro-$p$; every continuous homomorphism from a pro-$p$ group to a finite group of order prime to $p$ is trivial.*

*Proof.* (i) $U_n = p_n^{-1}(1)$ is open ($p_n$ is continuous and $G_n$ discrete), normal, and of index at most $\#G_n$. If $g \in U_{n+1}$, then $g_n = \pi_n(g_{n+1}) = 1$; so $U_{n+1} \subseteq U_n$, and likewise $g \in U_n$ implies $g_m = 1$ for all $m \leq n$. An element of all $U_n$ is 1. A basic open neighbourhood of 1 in the product restricts finitely many coordinates, say $g_m \in S_m$ for $m \leq n$, with $1 \in S_m$; it contains $U_n$ by the previous remark. (ii) If $V$ is an open subgroup, its complement is the union of the open cosets $gV \neq V$, so $V$ is closed; the cosets of $V$ form a disjoint open cover of the compact space $G$, so there are finitely many. Conversely, if $V$ is closed of finite index, its complement is a finite union of closed cosets, so $V$ is open. (iii) $\ker f = f^{-1}(1)$ is open, so it contains some $U_n$ by (i), and $f$ factors through $G/U_n$, which is isomorphic to $p_n(G) \leq G_n$ by the first isomorphism theorem, and to $G_n$ if $p_n$ is surjective. (iv) Let $H \leq G$ be closed and $H_n = p_n(H)$; the $H_n$ form an inverse system (as $\pi_n(p_{n+1}(h)) = p_n(h)$, $\pi_n$ maps $H_{n+1}$ into $H_n$), and $h \mapsto (p_n(h))$ is a continuous injective homomorphism $H \to \varprojlim H_n$ (injective as $\bigcap U_n = 1$). It is surjective: let $(h_n)$ be compatible, and choose $x_n \in H$ with $h_n = p_n(x_n)$. The sets $C_n = H \cap x_n U_n$ are closed (a coset of the open

subgroup $U_n$ is closed, its complement being a union of cosets) and nonempty ($x_n \in C_n$). They are decreasing: $p_n(x_{n+1}) = \pi_n(p_{n+1}(x_{n+1})) = \pi_n(h_{n+1}) = h_n = p_n(x_n)$, so $x_{n+1} \in x_n U_n$, and $x_{n+1}U_{n+1} \subseteq x_{n+1}U_n = x_nU_n$. A decreasing sequence of nonempty closed subsets of the compact space $G$ has nonempty intersection, and any $h \in \bigcap C_n$ satisfies $p_n(h) = p_n(x_n) = h_n$ for all $n$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism, so $H \simeq \varprojlim H_n$. For a closed normal $N$, the same argument applied to the finite groups $G/NU_n$ shows $G/N \simeq \varprojlim G/NU_n$; injectivity now needs $\bigcap_n NU_n = N$, which holds because $N$ is closed: if $g \notin N$, the closed sets $N \cap gU_n$ are decreasing with empty intersection (as $\bigcap_n gU_n = \{g\}$), so one of them is empty by compactness, that is, $g \notin NU_n$ for some $n$. The image of a compact space under a continuous map is compact, and compact subsets of Hausdorff spaces are closed. (v) We do not prove this; see [143, Chapter I, Section 1] and [142, Chapter IV, Section 2]. (vi) In the proof of (iv) the groups $H_n \le G_n$ and $G/NU_n$ (a quotient of $G/U_n \simeq p_n(G)$) are $p$-groups when the $G_n$ are. If $f\colon G \to F$ is continuous with $p \nmid \#F$, then by (iii) $f(G)$ is a quotient of $G/U_n$, a $p$-group, and has order dividing $\#F$; so it is trivial. □

**Proposition 1.78** ($\mathrm{GL}_2(\mathbb{Z}_\ell)$ and its congruence subgroup)**.** *Let $\ell$ be a prime and $\mathrm{GL}_2(\mathbb{Z}_\ell)$ the group of $2 \times 2$ matrices over $\mathbb{Z}_\ell$ with determinant in $\mathbb{Z}_\ell^\times$.*

*(i)* $\mathrm{GL}_2(\mathbb{Z}_\ell) = \varprojlim \mathrm{GL}_2(\mathbb{Z}/\ell^n)$. *Each reduction map $\mathrm{GL}_2(\mathbb{Z}/\ell^{n+1}) \to \mathrm{GL}_2(\mathbb{Z}/\ell^n)$ ($n \ge 1$) is surjective with kernel $\{I + \ell^n X\} \simeq (M_2(\mathbb{F}_\ell), +)$ of order $\ell^4$; so $\#\,\mathrm{GL}_2(\mathbb{Z}/\ell^n) = \ell^{4(n-1)}(\ell^2 - 1)(\ell^2 - \ell)$. For $\ell = 3$, $n = 2$: $\#\,\mathrm{GL}_2(\mathbb{Z}/9) = 81 \cdot 48 = 3888$.*

*(ii) The kernel $\Gamma(\ell) = I + \ell M_2(\mathbb{Z}_\ell)$ of reduction modulo $\ell$ is a pro-$\ell$ group.*

*(iii) If $\ell$ is odd, $\Gamma(\ell)$ is torsion-free. Hence every finite subgroup of $\mathrm{GL}_2(\mathbb{Z}_\ell)$ maps isomorphically onto its image in $\mathrm{GL}_2(\mathbb{F}_\ell)$. For $\ell = 2$ this fails: $-I \in \Gamma(2)$.*

*Proof.* (i) $M_2(\mathbb{Z}_\ell) = \varprojlim M_2(\mathbb{Z}/\ell^n)$ entrywise, and a matrix is invertible over $\mathbb{Z}_\ell$ iff its determinant is a unit, iff the determinant is nonzero modulo $\ell$ (Example 1.76(a)), iff all its reductions are invertible. Given $\bar g \in \mathrm{GL}_2(\mathbb{Z}/\ell^n)$, any lift $g$ to $M_2(\mathbb{Z}/\ell^{n+1})$ has $\det g \equiv \det \bar g \not\equiv 0 \pmod{\ell}$, so it is invertible: the reduction is surjective. Its kernel consists of the $I + \ell^n X$, with $X$ well defined modulo $\ell$, and $(I + \ell^n X)(I + \ell^n Y) = I + \ell^n(X + Y) + \ell^{2n}XY \equiv I + \ell^n(X + Y) \pmod{\ell^{n+1}}$, as $2n \ge n + 1$; these matrices are invertible (determinant $\equiv 1 \bmod \ell$). So the kernel is isomorphic to $(M_2(\mathbb{F}_\ell), +)$, and the order follows by induction from Proposition 1.43. (ii) $\Gamma(\ell)$ is the kernel of the continuous map $p_1$, hence closed, and by the proof of Proposition 1.77(iv) it is the inverse limit of its images in $\mathrm{GL}_2(\mathbb{Z}/\ell^n)$, which lie in the kernels of $\mathrm{GL}_2(\mathbb{Z}/\ell^n) \to \mathrm{GL}_2(\mathbb{F}_\ell)$, groups of order $\ell^{4(n-1)}$. (iii) Let $g \in \Gamma(\ell)$, $g \ne I$, and write $g = I + \ell^k A$ with $k \ge 1$ and $A \in M_2(\mathbb{Z}_\ell)$, $A \not\equiv 0 \pmod{\ell}$. Then

$$g^\ell = I + \ell^{k+1}A + \sum_{i=2}^{\ell-1} \binom{\ell}{i} \ell^{ki} A^i + \ell^{k\ell} A^\ell.$$

(the binomial expansion is valid because $I$ commutes with $A$). For $2 \le i \le \ell - 1$ the prime $\ell$ divides $\binom{\ell}{i}$ and $ki \ge 2k$, so the term is divisible by $\ell^{1+2k}$, and $1 + 2k \ge k + 2$ as $k \ge 1$. The last term is divisible by $\ell^{k\ell}$, and $k\ell \ge k + 2$ because $k(\ell - 1) \ge 2$ for $\ell \ge 3$. So all terms after $\ell^{k+1}A$ are divisible by $\ell^{k+2}$, and $g^\ell = I + \ell^{k+1}A_1$ with $A_1 \equiv A \pmod{\ell}$. (For $\ell = 2$ and $k = 1$ the last term $\ell^{2k}A^2 = 4A^2$ is divisible by $\ell^{k+1}$ but not in general by $\ell^{k+2}$, and the argument breaks down, as it must since $-I = I + 2(-I)$ has order 2.) Since $A_1 \not\equiv 0 \pmod{\ell}$, the same step applies to $g^\ell$ with $k+1$ in place of $k$; by induction $g^{\ell^s} = I + \ell^{k+s}A_s$ with $A_s \equiv A \not\equiv 0 \pmod{\ell}$. Now let $m \ge 1$, $m = \ell^s m'$ with $\ell \nmid m'$, and $h = g^{\ell^s}$. Then $h^{m'} = I + m'\ell^{k+s}A_s + \sum_{i \ge 2} \binom{m'}{i} \ell^{(k+s)i} A_s^i$, and every

term of the sum is divisible by $\ell^{2(k+s)}$, hence by $\ell^{k+s+1}$; so $h^{m'} \equiv I + m'\ell^{k+s}A_s \pmod{\ell^{k+s+1}}$. As $m'A_s \not\equiv 0 \pmod{\ell}$, this is not $I$ modulo $\ell^{k+s+1}$, and $g^m = h^{m'} \neq I$. Thus $g$ has infinite order. If $H \leq \mathrm{GL}_2(\mathbb{Z}_\ell)$ is finite, $H \cap \Gamma(\ell)$ is a finite subgroup of a torsion-free group, hence trivial, and reduction is injective on $H$. Finally $-I = I + 2(-I)$ has order 2. □

*In the paper* 1.79. The proof of Lemma lem:IV uses Proposition 1.78(iii) with $\ell = 3$: "reduction modulo 3 is injective on it, the kernel of $\mathrm{GL}_2(\mathbb{Z}_3) \to \mathrm{GL}_2(\mathbb{F}_3)$ being torsion-free", applied to the finite group $\rho_{E,3}(I_q)$; together with Proposition 1.48(iii),(v) this shows that $\rho_{E,3}(I_q)$ is cyclic of order 3. Note the compatibility: an element of order 3 of $\mathrm{GL}_2(\mathbb{Z}_3)$ has eigenvalues the primitive cube roots of unity, characteristic polynomial $X^2 + X + 1$, and reduces to an element of characteristic polynomial $X^2 + X + 1 = (X-1)^2$ in $\mathbb{F}_3[X]$, a nontrivial unipotent; the structure of $T_3E$ as a module over $\mathbb{Z}_3[\zeta_3]$ is Chapter 4 and Chapter 11. Lemma lem:h0 and Lemma lem:tamagawa-exp consider $\rho$ mod $\ell^k$, the composite of $\rho$ with $\mathrm{GL}_2(\mathbb{Z}_\ell) \to \mathrm{GL}_2(\mathbb{Z}/\ell^k)$; it is a continuous homomorphism to a finite group, so it factors through a finite quotient of $G_{\mathbb{Q}}$ (Proposition 1.77(iii)).

**Example 1.80** (Profinite Galois groups; forward pointers)**.** The following statements are proved in Chapter 5 and Chapter 7; we record them to show where the definitions are used.

(a) $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) = \varprojlim \mathrm{Gal}(K/\mathbb{Q})$ over the finite Galois extensions $K/\mathbb{Q}$, a profinite group (Remark 1.74).

(b) $\mathrm{Gal}(\mathbb{Q}(\mu_{\ell^\infty})/\mathbb{Q}) \simeq \mathbb{Z}_\ell^\times$ through the cyclotomic character $\chi_\ell$.

(c) $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \simeq \widehat{\mathbb{Z}}$, topologically generated by the Frobenius $x \mapsto x^q$. A profinite group topologically generated by one element (procyclic) is a quotient of $\widehat{\mathbb{Z}}$.

(d) For a prime $p$, the inertia group $I_p \leq G_{\mathbb{Q}_p}$ has a normal pro-$p$ subgroup, the wild inertia group, with quotient the tame inertia group $\prod_{\ell \neq p} \mathbb{Z}_\ell(1)$, which is procyclic. So a finite image of inertia of order prime to $p$ is a quotient of the tame inertia group, hence cyclic (by Proposition 1.77(vi) the wild inertia group maps trivially). This is the step "For $q \geq 5$ the image is tame, hence cyclic" in the proof of Proposition prop:shapes, and the step "The order 3 is prime to $q$, so the image is a quotient of the tame inertia group" in the proof of Lemma lem:IV.

## 1.8 Exercises

The exercises are graded from routine to harder. Exercises 1.85 to 1.89 check the group-theoretic claims of Proposition prop:ns directly.

**Exercise 1.81.** Show that $\#\,\mathrm{GL}_2(\mathbb{Z}/9) = 3888$ and $\#\,\mathrm{SL}_2(\mathbb{Z}/9) = 648$. (Use Proposition 1.78(i) and the surjectivity of det onto $(\mathbb{Z}/9)^\times$, a group of order 6.)

**Exercise 1.82.** Compute the conjugacy classes of $D_4$ and of $Q_8$: both have five classes, of sizes $1, 1, 2, 2, 2$. Deduce from the formula $\#G = \sum_\chi \chi(1)^2$ for the degrees of the irreducible complex characters (Riley-Hobson-Bence, the chapter on representation theory; Chapter 4) that both groups have four characters of degree 1 and one of degree 2, although they are not isomorphic. (Hint: in both groups the centre has order 2 and every non-central element $g$ has centraliser of order 4, the subgroup generated by $g$ and the centre; the number of degree-one characters is $\#(G/[G,G]) = 4$, and $8 = 4 \cdot 1^2 + 1 \cdot 2^2$.)

**Exercise 1.83.** Prove that $C_8 \times C_2$, $D_8$, $SD_{16}$, $Q_{16}$ and $M_{16}$ (Remark 1.24) are pairwise non-isomorphic, by counting elements of each order and checking which are abelian. (Caution: counting alone does not separate $C_8 \times C_2$ from $M_{16}$. In $C_8 \times C_2$ the element $(a, b)$ has order the least common multiple of the orders of $a$ and $b$, which gives $1, 3, 4, 8$ elements of orders $1, 2, 4, 8$, the same as for $M_{16}$; these two are distinguished because $M_{16}$ is non-abelian. The other pairs are separated by the table of Remark 1.24.)

**Exercise 1.84.** Find all subgroups of $SD_{16}$ and identify each up to isomorphism. (There are 15: the trivial group, five of order 2, five of order 4 (three cyclic, two Klein), three of order 8, and $SD_{16}$.) Which are normal? (Answer: seven: the trivial group, the centre $\langle a^4 \rangle$, the cyclic group $\langle a^2 \rangle$ of order 4, the three subgroups of index 2, and $SD_{16}$. The other eight have order 2 or 4 and contain some $a^m x$. Such a subgroup $K$ meets $\langle a \rangle$ in a subgroup of $\{1, a^4\}$, so it meets the coset $\langle a \rangle x$ in at most $a^m x$ and $a^{m+4} x$; conjugation by $a$ sends $a^m x$ to $a^{m-2} x$, which is neither, so $K$ is not normal.)

**Exercise 1.85.** Verify the class table of Proposition 1.59 by listing the 48 elements of $\mathrm{GL}_2(\mathbb{F}_3)$. Then use the orbit-counting lemma (Lemma 1.30) and the column "lines fixed" to show that $N_s(\mathbb{F}_3)$ has two orbits on $\mathbb{P}^1(\mathbb{F}_3)$ and that $N_{ns}(\mathbb{F}_3)$ has one.

**Exercise 1.86.** Check the paper's claims that the determinant-one subgroup of $N_s(\mathbb{F}_3)$ is cyclic of order 4, generated by $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and that the determinant-one subgroup of $N_{ns}(\mathbb{F}_3)$ is the quaternion group.

**Exercise 1.87.** Check the paper's claim that $N_s(\mathbb{F}_3)$ has two orbits of size two on the four lines of $\mathbb{F}_3^2$ (the two lines of the Cartan subgroup and the two others), while $N_{ns}(\mathbb{F}_3)$ and $\mathrm{GL}_2(\mathbb{F}_3)$ are transitive. Find an element of $N_{ns}(\mathbb{F}_3)$ that maps $\infty$ to 1. (One answer: $a = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right) \in C_{ns}(\mathbb{F}_3)$ sends $e_1$ to $\binom{1}{1}$, that is $\infty \mapsto 1$; its powers move $\infty$ through all four lines.)

**Exercise 1.88.** Show that $C_{ns}(\mathbb{F}_3)$ contains no element of determinant $-1$ and trace 0, and conclude that the image of $\bar{\rho}_{E,3}$ for an elliptic curve over $\mathbb{Q}$ is never conjugate to $C_{ns}(\mathbb{F}_3)$ (the label 3Cn), using that complex conjugation maps to an element of determinant $-1$ and order 2.

**Exercise 1.89.** Show that every subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$ of order 6 stabilises a line. Classify these subgroups up to conjugacy: two classes isomorphic to $S_3$ (3B.1.1 and 3B.1.2) and one cyclic class, $\{\pm I\} \times \mathrm{U}$. Show that 3B.1.1 and 3B.1.2 are interchanged by $g \mapsto (g^{-1})^{\mathsf{T}}$, which is an automorphism of $\mathrm{GL}_2(\mathbb{F}_3)$ but not an inner one. (Hint: a subgroup of order 6 contains an element of order 3, so its Sylow 3-subgroup $\mathrm{U}_L$ is normal in it, and the argument in the proof of Proposition 1.61 shows that it stabilises $L$. The map $g \mapsto (g^{-1})^{\mathsf{T}}$ is a homomorphism because both inversion and transposition reverse products. It is not inner: an inner automorphism maps a subgroup with a nonzero fixed vector to a conjugate subgroup, which again has one, whereas 3B.1.2 has none.)

**Exercise 1.90.** Prove Theorem 1.52(ii) by direct computation: for each of the eight class representatives of Proposition 1.59, compare the determinant with the sign of the induced permutation of the four lines.

**Exercise 1.91.** Let $q$ be odd. Show that $N_{ns}(\mathbb{F}_q) \cap \mathrm{SL}_2(\mathbb{F}_q)$ has a unique element of order 2, namely $-I$, and that its elements outside $C_{ns}(\mathbb{F}_q)$ have order 4. (Use $(m_c \sigma)^2 = N(c) I$.) For $q = 3$ this recovers $Q_8$; what is the group for $q = 5$? (Answer: a group of order $2(q+1) = 12$ containing the cyclic group $C_{ns}(\mathbb{F}_5) \cap \mathrm{SL}_2(\mathbb{F}_5) \simeq C_6$ with index 2, whose six other elements have order 4

and square to $-I$. It has $1, 1, 2, 6, 2$ elements of orders $1, 2, 3, 4, 6$ (checked by enumeration); it is the dicyclic group $C_3 \rtimes C_4$, in which a generator of $C_4$ inverts $C_3$: among the five groups of order 12 ($C_{12}$, $C_6 \times C_2$, $A_4$, $D_6$ and this one; we do not prove the classification here) it is the only one with a unique involution and no element of order 12.)

**Exercise 1.92.** (Harder.) Show that $N_s(\mathbb{F}_5)$ is a Sylow 2-subgroup of $\mathrm{GL}_2(\mathbb{F}_5)$, and that its image in $\mathrm{PGL}_2(\mathbb{F}_5) \simeq S_5$, a dihedral group of order 8, lies in a subgroup isomorphic to $S_4$. (Show that $N_s(\mathbb{F}_5)$ and $\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ generate a subgroup of order 96 containing the scalars.)

**Exercise 1.93.** Show that $I + 3M_2(\mathbb{Z}_3)$ is torsion-free, and that $I + 2M_2(\mathbb{Z}_2)$ contains elements of order 2 (for instance $-I$ and $\mathrm{diag}(1, -1)$) but no element of order 4. (Hint: an element $g$ of order 4 of $\mathrm{GL}_2(\mathbb{Z}_2)$ has eigenvalues $\pm\sqrt{-1}$, so its characteristic polynomial is $X^2 + 1$ and $g^2 = -I$; if $g = I + 2A$ and $g^2 = -I$, then $4(A + A^2) = -2I$.)

**Exercise 1.94.** Let $p \neq \ell$ be primes. Show that the image of a continuous homomorphism $\mathbb{Z}_p \to \mathrm{GL}_2(\mathbb{F}_\ell)$ is cyclic of $p$-power order, and deduce that it is trivial when $p \nmid \#\,\mathrm{GL}_2(\mathbb{F}_\ell)$. (This is a first instance of the tame inertia arguments of Chapter 7.) (Hint: the kernel is open, so contains some $p^n\mathbb{Z}_p$ by Example 1.76(d), and the image is a quotient of $\mathbb{Z}_p/p^n\mathbb{Z}_p \simeq \mathbb{Z}/p^n$; apply Proposition 1.77(vi) for the last part, $\mathbb{Z}_p$ being pro-$p$.)

## 1.9 Notes and references

The elementary group theory of Section 1.1 to Section 1.4 follows [58] (Chapters 1 to 6; in particular Section 3.2 for Lagrange, Section 3.3 for the isomorphism theorems, Sections 4.1 to 4.3 for actions and the class equation, Section 4.5 for the Sylow theorems, Section 5.2 for finite abelian groups and Section 5.5 for semidirect products). Dummit and Foote write $D_{2n}$ for our $D_n$. The proof of Cauchy's theorem is McKay's, and the existence proof in Theorem 1.33 is Wielandt's; both are standard. The proof of the structure theorem in Theorem 1.35 is the elementary one; the general theorem for finitely generated modules over a principal ideal domain is proved in Chapter 3.

The matrix groups of Section 1.5 are treated in [109, Chapter XIII] (the simplicity of $\mathrm{PSL}_2$ is in Section 8 there) and, for the subgroup structure relevant to Galois representations, in [171, Section 2], where Propositions 15 and 16 give the classification we quote as Theorem 1.57; Dickson's original treatment is [54], and [85, Chapter II, Section 8] is a thorough modern account of the groups $\mathrm{PSL}_2(\mathbb{F}_q)$. The uniqueness of the simple group of order 168 is in [58, Section 6.2].

The tables of Section 1.6 were computed by hand and checked by a complete enumeration of the elements and subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$; the orders for $q = 5, 7$ and the subgroup of order 96 of $\mathrm{GL}_2(\mathbb{F}_5)$ were checked in the same way. Sutherland's labels are defined in [191], and the list of mod $\ell$ images of elliptic curves over $\mathbb{Q}$ is [210]; both are discussed in Chapter 11. The paper cites [210] for the list of three images in Proposition prop:ns and derives it from three properties by inspection; Theorem 1.65 carries out that inspection.

For profinite groups we follow [142, Chapter IV, Sections 1 and 2] and [143, Chapter I, Section 1]; Tychonoff's theorem is [139, Theorem 37.3]. The torsion-freeness of the congruence subgroup $\Gamma(\ell)$ for odd $\ell$ is elementary; for its role in the study of $\ell$-adic images see [171, Section 4].

# Chapter 2

# Rings, fields, polynomials and characters

The paper is written for readers who handle rings, finite fields and quadratic characters without thinking about them. This chapter makes that background explicit, with complete proofs, and organises it around the seven places where the paper actually uses it: the $\ell$-adic valuations of rational numbers in Definition def:terms and the table of Section 1; the quadratic characters $\chi_{-3}$, $\chi_{-4}$, $\chi_{q^*}$ and $\chi_S$ as Dirichlet characters; quadratic reciprocity, which makes $\chi_{q^*}$ the quadratic character of conductor $q$; the $\ell$-th powers in $\mathbb{F}_q^\times$ and the class $c$ of Lemma lem:tamfactor; the ring theory behind Lemma lem:crit ($\mathbb{T}/(I+J)$ is cyclic, and its order is divisible by the residue characteristic of every maximal ideal containing $I+J$); the congruence $a^2 \equiv 1 \pmod{\ell^k}$ behind Lemma lem:h0; and the foundations (ideals, unique factorisation, Gauss's lemma, Eisenstein's criterion, finite fields, cyclotomic polynomials, Gauss sums) that Chapters 3, 5, 6 and 7 build on. The group theory it needs (cyclic groups, finite abelian groups, finite subgroups of $K^\times$) is in Chapter 1 and is used, not reproved.

## Core results at a glance

Rings are commutative with 1. The letters $p$, $q$, $\ell$ denote primes, except in Sections 2.6 and 2.7, where $q = p^r$ is the size of a finite field; in the paper $q$ is a prime, so $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$ there. Quadratic characters are named by their discriminant: $\chi_D = \left(\frac{D}{\cdot}\right)$ for a fundamental discriminant $D$ (for other discriminants the Kronecker symbol $\left(\frac{D}{\cdot}\right)$ is written out), so $\chi_{q^*}$ (never $\chi_q$) is the character of conductor $q$, and $\chi_\ell$ is the $\ell$-adic cyclotomic character.

**2.1 Conventions.** Rings, homomorphisms and subrings are unital; $p, q, \ell$ are primes ($q$ a prime power in Sections 6 and 7); $v_p(0) = +\infty$, $|x|_p = p^{-v_p(x)}$, and for $c \in \mathbb{Z}/\ell^m$, $v_\ell(c) = \min(v_\ell(\tilde{c}), m)$; the Kronecker symbol at 2 and at $-1$; Dirichlet characters and the names $\chi_D$ ($D$ fundamental), $\chi_{q^*}$, $\omega_\ell$, $\chi_0$, $\chi^{(p)}$; $\zeta_n = e^{2\pi i/n}$ and $\sqrt{-3} = 1 + 2\zeta_3$. Conventions 2.1 to 2.6.

**2.2 Rings** (definition: ring, unit group $R^\times$, zero divisor, nilpotent, field, integral domain, subring, homomorphism, kernel, image). Definition 2.7.

**2.3 Characteristic.** For every ring $R$ there is a unique homomorphism $\mathbb{Z} \to R$; its kernel is $n\mathbb{Z}$ for a unique $n \geq 0$, the characteristic. A domain has characteristic 0 or a prime, and a

finite domain is a field. *(proved here)* Proposition 2.9.

**2.4 Ideals** (definition: ideal, the ideal generated by a set, $I+J$, $IJ$, $I\cap J$, the annihilator $\mathrm{Ann}_R(S)$, comaximal ideals, the quotient ring $R/I$). Definition 2.10.

**2.5 Ideals and kernels.** Kernels of homomorphisms are exactly the ideals; $IJ\subseteq I\cap J$; every ideal of $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique $n\geq 0$; $m\mathbb{Z}+n\mathbb{Z}=\gcd(m,n)\mathbb{Z}$ and $m\mathbb{Z}\cap n\mathbb{Z}=\mathrm{lcm}(m,n)\mathbb{Z}$. *(proved here)* Proposition 2.11.

**2.6 Isomorphism theorems.** (i) $R/\ker\phi\simeq\phi(R)$. (ii) Ideals of $R/I$ correspond to ideals $J\supseteq I$ of $R$, with $(R/I)/(J/I)\simeq R/J$. (iii) If $\phi\colon R\to S$ is surjective with kernel $I$ and $J$ is an ideal of $R$, then $\phi(J)$ is an ideal and $R/(I+J)\simeq S/\phi(J)$. (iv) Every quotient ring of $\mathbb{Z}$ is $\mathbb{Z}/r\mathbb{Z}$, cyclic as a group. *(proved here)* Theorem 2.12.

**2.7 Two ideals.** The sequence $0\to R/(I\cap J)\to R/I\times R/J\to R/(I+J)\to 0$, $x\mapsto(x,x)$, $(y,z)\mapsto y-z$, is exact. If $I+J=R$ then $I\cap J=IJ$ and $R/IJ\simeq R/I\times R/J$; if $I_1,\dots,I_k$ are pairwise comaximal, $R/I_1\cdots I_k\simeq\prod R/I_i$. *(proved here)* Proposition 2.13.

**2.8 Idempotents.** If $e^2=e$ then $R\simeq Re\times R(1-e)$; $R$ is a product of two nonzero rings iff it has an idempotent $e\neq 0,1$. The ideals of $R_1\times R_2$ are the $I_1\times I_2$, its prime ideals the $P_1\times R_2$ and $R_1\times P_2$. *(proved here)* Proposition 2.14.

**2.9 Frobenius in characteristic $p$.** $(x+y)^{p^k}=x^{p^k}+y^{p^k}$, so $x\mapsto x^p$ is a ring endomorphism; it is injective on a domain and bijective on a finite field; $a^p\equiv a\pmod p$. *(proved here)* Proposition 2.16.

**2.10 Prime and maximal ideals.** $R/P$ is a domain iff $P$ is prime, $R/M$ a field iff $M$ is maximal; maximal ideals are prime; preimages of primes are prime; the kernel of a map onto a field (or to a field with finite image) is maximal; if $R/P$ is finite, $P$ is prime iff maximal; the primes of $\mathbb{Z}$ are 0 and $p\mathbb{Z}$; every non-unit lies in a maximal ideal (Zorn). *(proved here)* Theorem 2.17.

**2.11 Residue characteristic.** If $\mathfrak{m}$ is maximal with $R/\mathfrak{m}$ of characteristic $\ell>0$ and $\mathfrak{a}\subseteq\mathfrak{m}$ is an ideal with $R/\mathfrak{a}$ finite, then $\ell\mid\#R/\mathfrak{a}$; a homomorphism $\mathbb{Z}/r\to k$ to a field of characteristic $\ell$ exists iff $\ell\mid r$. *(proved here)* Proposition 2.18.

**2.12 Field of fractions.** A domain $R$ embeds in a field $\mathrm{Frac}(R)$, and every injective homomorphism from $R$ to a field extends uniquely to $\mathrm{Frac}(R)$. *(proved here)* Theorem 2.20.

**2.13 The ring $\mathbb{Z}_{(p)}$.** $\mathbb{Z}_{(p)}=\{a/b : p\nmid b\}$ has units $\{a/b : p\nmid ab\}$, nonzero ideals $p^k\mathbb{Z}_{(p)}$, a unique maximal ideal $p\mathbb{Z}_{(p)}$, and a surjective reduction map $\mathbb{Z}_{(p)}\to\mathbb{F}_p$ with kernel $p\mathbb{Z}_{(p)}$; $\mathbb{Z}_{(p)}=\{x : v_p(x)\geq 0\}$ and $\mathbb{Z}=\bigcap_p\mathbb{Z}_{(p)}$. *(proved here)* Proposition 2.21.

**2.14 Divisibility** (definition: associates, irreducible and prime elements, Euclidean domain, PID, UFD, gcd). Definition 2.23.

**2.15 Primes and irreducibles.** In a domain a prime element is irreducible, and $(a)$ is a nonzero prime ideal iff $a$ is a prime element; a factorisation into prime elements is unique up to order and units. *(proved here)* Lemma 2.24.

**2.16 Euclid, PID, UFD.** A Euclidean domain is a PID; a PID is Noetherian and a UFD, with Bezout's identity and nonzero primes maximal; in a UFD irreducibles are prime. $\mathbb{Z}$ and $F[x]$ are Euclidean; $\mathbb{Z}[x]$ and $F[x,y]$ are UFDs, not PIDs. *(proved here)* Theorem 2.25.

**2.17 Arithmetic of $\mathbb{Z}$.** Bezout, Euclid's lemma, $n = \pm\prod p^{v_p(n)}$, gcd and lcm by min and max of exponents. *(proved here)* Proposition 2.26.

**2.18 $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$.** Both are Euclidean for $N(z) = |z|^2$ (remainders of norm at most $\frac{1}{2}N(\beta)$, resp. $\frac{1}{3}N(\beta)$), hence PIDs and UFDs; $\mathbb{Z}[i]^\times = \mu_4$, $\mathbb{Z}[\zeta_3]^\times = \mu_6$; elements of prime norm are prime. *(proved here)* Theorem 2.27.

**2.19 Polynomials.** Over a domain $\deg fg = \deg f + \deg g$ and $R[x]^\times = R^\times$; division by a monic polynomial over any ring; $F[x]$ is Euclidean, its maximal ideals are the $(f)$, $f$ irreducible; factor theorem; at most $d$ roots in a domain. *(proved here)* Proposition 2.30.

**2.20 Repeated roots.** $a$ is a multiple root of $f$ iff $f(a) = f'(a) = 0$; $f$ has no repeated root in any extension iff $\gcd(f, f') = 1$; $x^{p^r} - x$ and, when $\operatorname{char} F \nmid n$, $x^n - 1$ are separable. *(proved here)* Proposition 2.31.

**2.21 Gauss's lemma.** Over a UFD $R$ with fraction field $K$: contents multiply; a factorisation in $K[x]$ descends to $R[x]$; a primitive polynomial is irreducible in $R[x]$ iff in $K[x]$; $R[x]$ is a UFD; rational root test. *(proved here)* Theorem 2.32.

**2.22 Eisenstein's criterion.** If $a_n \notin P$, $a_i \in P$ for $i < n$, $a_0 \notin P^2$ ($P$ prime), then $f$ is not a product of two polynomials of positive degree; over a UFD a primitive such $f$ is irreducible in $R[x]$ and $K[x]$. *(proved here)* Theorem 2.33.

**2.23 Reduction modulo $p$.** If $p$ does not divide the leading coefficient of a primitive $f \in \mathbb{Z}[x]$ and $\bar{f}$ is irreducible in $\mathbb{F}_p[x]$, then $f$ is irreducible in $\mathbb{Q}[x]$; the degrees of a rational factorisation are sums of degrees of the factors of $\bar{f}$. *(proved here)* Proposition 2.34.

**2.24 Kronecker's construction.** For $f \in F[x]$ irreducible of degree $d$, $F[x]/(f)$ is a field of degree $d$ in which $\bar{x}$ is a root of $f$; $F[x]/(f) \simeq F[\alpha]$ for any root $\alpha$; every polynomial splits in a finite extension. *(proved here)* Theorem 2.38.

**2.25 The ring $\mathbb{Z}/n\mathbb{Z}$.** $a$ is a unit iff $\gcd(a,n) = 1$; $\mathbb{Z}/n$ is a field iff $n$ is prime; $\#(\mathbb{Z}/n)^\times = \varphi(n)$, $a^{\varphi(n)} \equiv 1$, $\varphi(p^k) = p^{k-1}(p-1)$, $\sum_{d\mid n}\varphi(d) = n$; $(\mathbb{Z}/n)^\times = \operatorname{Aut}(\mathbb{Z}/n)$. *(proved here)* Proposition 2.40.

**2.26 Chinese remainder theorem.** $\mathbb{Z}/mn \simeq \mathbb{Z}/m \times \mathbb{Z}/n$ iff $\gcd(m,n) = 1$; $\mathbb{Z}/n \simeq \prod \mathbb{Z}/p_i^{k_i}$, $(\mathbb{Z}/n)^\times \simeq \prod(\mathbb{Z}/p_i^{k_i})^\times$, $\varphi$ is multiplicative. *(proved here)* Theorem 2.41.

**2.27 Lifting the exponent.** For odd $\ell$, $\ell \mid a-b$, $\ell \nmid ab$: $v_\ell(a^n - b^n) = v_\ell(a-b) + v_\ell(n)$; for odd $a$ and even $n$, $v_2(a^n-1) = v_2(a-1) + v_2(a+1) + v_2(n) - 1$; $1+\ell$ has order $\ell^{k-1}$ modulo $\ell^k$; $v_\ell(q^f - 1)$ for $q \neq \ell$; $v_\ell(q^2-1) = v_\ell(q-1) + v_\ell(q+1)$. *(proved here)* Lemma 2.42.

**2.28 Primitive roots.** $(\mathbb{Z}/p^k)^\times$ is cyclic for odd $p$; $(\mathbb{Z}/2^k)^\times = \{\pm 1\} \times \langle 5\rangle$ for $k \geq 3$; $(\mathbb{Z}/n)^\times$ is cyclic iff $n \in \{1, 2, 4, p^k, 2p^k\}$, $p$ odd. *(proved here)* Theorem 2.43.

**2.29 Squares congruent to one.** For odd $\ell$ and $k \geq 1$, $a^2 \equiv 1 \pmod{\ell^k}$ iff $a \equiv \pm 1$; every unit squares to 1 modulo $\ell^k$ iff $\ell^k = 3$; a homomorphism onto $(\mathbb{Z}/\ell^k)^\times$ (or one whose image contains 2) with $\chi^2 \equiv 1$ forces $\ell^k = 3$. *(proved here)* Corollary 2.44.

**2.30 Finite fields.** A finite field has prime characteristic $p$ and $p^r$ elements; $F^\times$ is cyclic; $x^{p^r} - x = \prod_{a\in F}(x-a)$; $F = \mathbb{F}_p[\alpha]$ for a generator $\alpha$ of $F^\times$. *(proved here)* Proposition 2.46.

**2.31 Existence and uniqueness.** For every prime power $q = p^r$ there is a field with $q$ elements, unique up to isomorphism, isomorphic to $\mathbb{F}_p[x]/(f)$ for every irreducible $f$ of degree $r$. *(proved here)* Theorem 2.47.

**2.32 Frobenius and subfields.** $\mathrm{Frob}_p\colon x \mapsto x^p$ is an automorphism of $\mathbb{F}_{p^r}$ of order $r$ with fixed field $\mathbb{F}_p$; $\mathbb{F}_{p^r}$ has a subfield with $p^d$ elements iff $d \mid r$, and then exactly one; the fixed field of $\mathrm{Frob}_p^s$ is $\mathbb{F}_{p^{\gcd(s,r)}}$. *(proved here)* Proposition 2.48.

**2.33 Irreducible polynomials over $\mathbb{F}_q$.** $x^{q^r} - x$ is the product of the monic irreducibles of degree dividing $r$; $q^r = \sum_{d|r} dN_d$ and $N_r = \frac{1}{r}\sum_{d|r}\mu(d)q^{r/d} > 0$. *(proved here)* Proposition 2.49.

**2.34 Roots of unity in $\mathbb{F}_q$.** $\mu_n(\mathbb{F}_q)$ is cyclic of order $\gcd(n, q-1)$; the least $f$ with $\mu_n \subset \mathbb{F}_{q^f}$ is the order of $q$ modulo $n$; no nontrivial $p$-power roots of unity; $-1$ is a square iff $q \equiv 1 \pmod 4$, $-3$ (for $p \neq 2, 3$) iff $q \equiv 1 \pmod 3$; for $\sigma$ of order 3 (resp. 4) and $m$ prime to 3 (resp. odd), $\sigma^m = \sigma^{\pm 1}$ according to $m$ modulo 3 (resp. 4). *(proved here)* Proposition 2.50.

**2.35 Powers in $\mathbb{F}_q^\times$.** For $d = \gcd(m, q-1)$, $(\mathbb{F}_q^\times)^m = (\mathbb{F}_q^\times)^d$ has index $d$; $u$ is an $m$-th power iff $u^{(q-1)/d} = 1$; $[\mathbb{F}_q^\times : (\mathbb{F}_q^\times)^{\ell^k}] = \ell^{\min(k, v_\ell(q-1))}$. *(proved here)* Theorem 2.52.

**2.36 The class $c(u)$.** For $\ell \nmid q$ and $v = v_\ell(q-1)$, $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$ is cyclic of order $\ell^v$; a generator identifies it with $\mathbb{Z}/\ell^v$, and $v_\ell(c(u))$ does not depend on the generator; for $k \le v$, $u$ is an $\ell^k$-th power iff $v_\ell(c(u)) \ge k$; $\min(v_\ell(n), v, v_\ell(c(u))) \ge 1$ iff $\ell \mid n$, $q \equiv 1 \pmod \ell$ and $u \in (\mathbb{F}_q^\times)^\ell$. *(proved here)* Proposition 2.53.

**2.37 Roots of unity** (definition: $\mu_n(K)$, $\mu(K)$, primitive $n$-th root of unity). Definition 2.56.

**2.38 Roots of unity in fields.** $\mu_n(K)$ is cyclic of order dividing $n$, and of order $n$ only if $\mathrm{char}\, K \nmid n$; $\mu_n(\mathbb{C}) = \langle\zeta_n\rangle$; $\mu(\mathbb{Q}) = \{\pm 1\}$, $\mu(\mathbb{Q}(i)) = \mu_4$, $\mu(\mathbb{Q}(\zeta_3)) = \mu_6$. *(proved here)* Proposition 2.57.

**2.39 Cyclotomic polynomials.** $x^n - 1 = \prod_{d|n}\Phi_d$; $\Phi_n \in \mathbb{Z}[x]$ is monic of degree $\varphi(n)$; $\Phi_n = \prod_{d|n}(x^d-1)^{\mu(n/d)}$; formulas for $\Phi_{p^k}$, $\Phi_{2m}$, $\Phi_n(1)$, $\Phi_n(0)$; $\Phi_{p^k}$ is irreducible over $\mathbb{Q}$. *(proved here)* Theorem 2.58.

**2.40 Cyclotomic polynomials modulo $p$.** For $p \nmid n$ the roots of $\Phi_n$ in characteristic $p$ are the elements of order $n$; $p \mid \Phi_n(a)$ iff $a$ has order $n$ modulo $p$, so $p \equiv 1 \pmod n$; there are infinitely many primes $\equiv 1 \pmod n$; $\Phi_n$ splits modulo $p$ iff $p \equiv 1 \pmod n$. *(proved here)* Proposition 2.59.

**2.41 Legendre symbol** (definition). Definition 2.60.

**2.42 Euler's criterion.** $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod p$; the symbol is the unique character of order 2 of $(\mathbb{Z}/p)^\times$; $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$; $\left(\frac{a}{3}\right) \equiv a \pmod 3$. *(proved here)* Proposition 2.61.

**2.43 The quadratic Gauss sum.** In a field of characteristic $\neq q$ with a primitive $q$-th root of unity $\zeta$, $g = \sum\left(\frac{a}{q}\right)\zeta^a$ satisfies $g^2 = q^* = (-1)^{(q-1)/2}q$ and $\sum\left(\frac{a}{q}\right)\zeta^{ab} = \left(\frac{b}{q}\right)g$; so $\sqrt{q^*} \in \mathbb{Q}(\zeta_q)$. *(proved here; the sign of $g$ over $\mathbb{C}$ is stated in Remark 2.63)* Proposition 2.62.

**2.44 Quadratic reciprocity.** For odd primes $p \neq q$: $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$, i.e. $\left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right)$; $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$; $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$. *(proved here)* Theorem 2.64.

**2.45 Jacobi symbol** (definition). Definition 2.65.

**2.46 Jacobi reciprocity.** The Jacobi symbol is multiplicative in both entries, periodic in the top one, and satisfies reciprocity and both supplements; $\left(\frac{a}{n}\right) = 1$ does not make $a$ a square modulo $n$. *(proved here)* Proposition 2.66.

**2.47 Kronecker symbol, discriminants** (definition: the Kronecker symbol $\left(\frac{a}{n}\right)$ for all $a, n \in \mathbb{Z}$; discriminant; fundamental and prime discriminants; $\chi_D = \left(\frac{D}{\cdot}\right)$ for fundamental $D$). Definition 2.67.

**2.48 Kronecker characters.** For a discriminant $D$, $n \mapsto \left(\frac{D}{n}\right)$ is completely multiplicative, vanishes iff $\gcd(D, n) > 1$, and is a real Dirichlet character modulo $|D|$ with value $\operatorname{sgn} D$ at $-1$; values at primes; $\left(\frac{DD'}{\cdot}\right) = \left(\frac{D}{\cdot}\right)\left(\frac{D'}{\cdot}\right)$ for discriminants $D, D'$, and $\left(\frac{Df^2}{n}\right) = \left(\frac{D}{n}\right)$ for $\gcd(n, f) = 1$. *(proved here)* Proposition 2.68.

**2.49 The character $\chi_{q^*}$.** $q^* \equiv 1 \pmod 4$, and $\left(\frac{q^*}{n}\right) = \left(\frac{n}{q}\right)$ for every $n \in \mathbb{Z}$; $\chi_{q^*}$ is the unique character of order 2 and conductor $q$; the same for $\ell^*$; $3^* = -3$, $5^* = 5$, $7^* = -7$, $11^* = -11$, $13^* = 13$. *(proved here)* Proposition 2.69.

**2.50 Dirichlet characters** (definition: character modulo $n$, principal character, parity, the group $X(n)$, characters with values in another field, $\omega_\ell$). Definition 2.71.

**2.51 Basic properties.** Complete multiplicativity, periodicity, values in $\mu_{\varphi(n)}$, $\bar\chi = \chi^{-1}$, $\chi(-1) = \pm 1$; $X(n) \simeq (\mathbb{Z}/n)^\times$ non-canonically. *(proved here)* Proposition 2.72.

**2.52 Orthogonality.** $\sum_a \chi(a) = \varphi(n)[\chi = \chi_0]$, $\sum_\chi \chi(a) = \varphi(n)[a \equiv 1]$, where the Iverson bracket $[P]$ is 1 if the statement $P$ holds and 0 otherwise; the characters are an orthonormal basis of the functions on $(\mathbb{Z}/n)^\times$. *(proved here)* Theorem 2.73.

**2.53 Conductor** (definition: $\chi$ defined modulo $d$, conductor $f_\chi$, primitive character). Definition 2.74.

**2.54 Conductors.** If $\chi$ is defined modulo $d_1$ and $d_2$, it is defined modulo $\gcd(d_1, d_2)$; the conductor divides every such modulus and $\chi$ is induced by a unique primitive $\chi^{\text{prim}}$; $f_\chi = \prod_p f_{\chi^{(p)}}$ for the $p$-components $\chi^{(p)}$; products of primitive characters of coprime conductors are primitive; $\chi_{-4}$, $\chi_{\pm 8}$ have conductor 4, 8, and $\chi_S$ has conductor $\prod_{q \in S} q$. *(proved here)* Proposition 2.75.

**2.55 Fundamental discriminants.** A fundamental discriminant $D \neq 1$ is uniquely a product of coprime prime discriminants $-4, 8, -8, q^*$; $\chi_D$ is primitive of conductor $|D|$. *(proved here)* Corollary 2.76.

**2.56 Gauss sums.** $\tau(\chi, b) = \bar\chi(b)\tau(\chi)$ for $b$ prime to $n$, and for all $b$ if $\chi$ is primitive; then $|\tau(\chi)|^2 = n$ and $\tau(\chi)\tau(\bar\chi) = \chi(-1)n$; $\tau(\chi_{-4}) = 2i$, $\tau(\chi_{-3}) = \sqrt{-3}$; $\tau(\chi) = \mu(n/f)\chi^{\text{prim}}(n/f)\tau(\chi^{\text{prim}})$ when $\gcd(n/f, f) = 1$. *(proved here)* Theorem 2.77.

**2.57 The paper's characters.** The table of $\chi_{-4}$, $\chi_{-3}$, $\chi_{q^*}$, $\chi_S$, $\chi_{\ell^*}$, $\omega_\ell$ (Kronecker symbol, conductor, parity, values at primes); $p \equiv \chi_{-3}(p) \pmod 3$ for $p \neq 3$ and $p \equiv \chi_{-4}(p) \pmod 4$ for odd $p$; $\left(\frac{\cdot}{\ell}\right) = \omega_\ell^{(\ell-1)/2}$, equal to $\omega_\ell$ iff $\ell = 3$; $3 \mid q - \chi_{-4}(q)$ iff $q \equiv \pm 1 \pmod{12}$. *(proved here)* Proposition 2.78.

**2.58 Valuations** (definition: $v_p$ on $\mathbb{Q}$, $v_p(0) = +\infty$). Definition 2.81.

**2.59 Properties of $v_p$.** $v_p(xy) = v_p(x) + v_p(y)$; $v_p(x+y) \geq \min$, with equality if the valuations differ; $v_p \geq 0$ on $\mathbb{Z}_{(p)}$ exactly; $p^k \mid n$ iff $v_p(n) \geq k$; Legendre's formula for $v_p(n!)$. *(proved here)* Proposition 2.82.

**2.60 The $p$-adic absolute value** (definition: $|x|_p = p^{-v_p(x)}$). Definition 2.83.

**2.61 Absolute values on $\mathbb{Q}$.** $|\cdot|_p$ is an ultrametric absolute value; the product formula $\prod_{p \leq \infty} |x|_p = 1$; $\mathbb{Q}$ is not complete for $|\cdot|_p$. *(proved here)* Proposition 2.84.

**2.62 Ostrowski's theorem.** Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_\infty$ or to exactly one $|\cdot|_p$. *(stated; reference)* Theorem 2.85.

**2.63 $\ell$-integrality.** For primes $q \neq \ell$, $q$ is an $\ell$-adic unit, $P(q^{-k}) \in \mathbb{Z}_{(\ell)}$ for $P \in \mathbb{Z}[X]$, and $v_\ell(\frac{q-1}{q}) = v_\ell(q-1)$, $v_\ell(1 \mp q^{-1}) = v_\ell(q \mp 1)$ for either sign, $v_\ell(1-q^{-2}) = v_\ell(q-1) + v_\ell(q+1)$. *(proved here)* Lemma 2.86.

**2.64 The $\mathrm{I}_0^*$ identity.** If $\alpha\beta = q$, $\alpha + \beta = a$, then $(1 - \frac{\alpha}{\beta q})(1 - \frac{1}{q})(1 - \frac{\beta}{\alpha q}) = \frac{q-1}{q} \cdot \frac{(q+1)^2 - a^2}{q^2}$, and $(q+1)^2 - a^2 \geq (q-1)^2$ if $a^2 \leq 4q$. *(proved here)* Proposition 2.87.

**2.65 The paper's local terms.** For primes $q \geq 5$, odd $\ell \neq q$ and $a \in \mathbb{Z}$ with $a^2 \leq 4q$: (i) $v_\ell(q - \chi_{-3}(q)) \geq 1$ always at $\ell = 3$ and iff $q \equiv \chi_{-3}(q) \pmod \ell$ otherwise; (ii) $v_\ell(q - \chi_{-4}(q)) \geq 1$ iff $q \equiv \pm 1 \pmod{12}$ at $\ell = 3$, iff $q \equiv \chi_{-4}(q) \pmod \ell$ otherwise; (iii) $v_\ell(q^2 - 1) \geq 1$ always at $\ell = 3$, iff $q \equiv \pm 1 \pmod \ell$ otherwise; (iv) $v_\ell(q-1) + v_\ell((q+1)^2 - a^2) \geq 1$ iff $q \equiv 1 \pmod \ell$ or $a \equiv \pm(q+1) \pmod \ell$; (v) for $n \geq 1$ and $\bar{u} \in \mathbb{F}_q^\times$, $v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c(\bar{u}))) > v_\ell(n)$ iff $\ell \mid n$, $q \equiv 1 \pmod \ell$ and $\bar{u} \in (\mathbb{F}_q^\times)^\ell$. *(proved here)* Proposition 2.88.

## Where the paper uses this chapter

- **Definition def:terms and the table of Section 1.** The terms $e_\ell(E,q)$ and $t_\ell(E,q)$ are $\ell$-adic valuations of integers and of rationals with denominator a power of $q$: Propositions 2.82, 2.87 and 2.88 and Lemma 2.86 and Example 2.89; the same results give Remark rem:afterquant(1) ($e_3 > 0$ at every prime $q \geq 5$ of type $\mathrm{I}_n^*$). The characters $\chi_{-3}$, $\chi_{-4}$ of the table: Proposition 2.78.

- **Lemma lem:euler (the omitted factors at tame primes) and Lemma lem:IV.** "Frobenius centralises $\sigma$ if $q \equiv 1 \pmod 3$ and inverts it if $q \equiv 2$": Proposition 2.50(v) and Proposition 2.78(ii); the identity for type $\mathrm{I}_0^*$: Proposition 2.87; $e_\ell = v_\ell(1-q^{-2}) = v_\ell(q^2-1)$: Lemma 2.86.

- **Proposition prop:forms.** "$e_3(E,q) = v_3(q - \chi_{-3}(q)) \geq 1 = w_3(E,q)$ since $q \equiv \chi_{-3}(q) \pmod 3$": Proposition 2.78(i), Proposition 2.88(i).

- **Lemma lem:twist, Theorems thm:twist, thm:quant, thm:qual and Example ex:qual.** The character $\chi_{q^*}$ of conductor $q$, the field $\mathbb{Q}(\sqrt{q^*})$, the character $\chi_S$ of conductor $D = \prod_{q\in S} q$ and the vanishing of $\chi_S(n)$ for $\gcd(n, D) > 1$: Theorem 2.64 and Propositions 2.69, 2.75 and 2.78.

- **Lemma lem:tamfactor, Remark rem:ko(1) and Section sec:tam.** $\ell$-th powers in $\mathbb{F}_q^\times$, the class $c$ of the reduction of the unit part $u$, and the reading of $v_\ell(c)$: Theorem 2.52 and Proposition 2.53 and Example 2.54; reduction of units: Proposition 2.21(iii).

- **Lemma lem:crit (congruence criterion), Lemma lem:hecke and Step 1 of the proof of Theorem thm:quant.** $\mathbb{T}/(I + J) \simeq \mathbb{Z}/\phi_f(J)$ is cyclic, the cokernel of $\mathbb{T} \subset \mathbb{T}/I \times \mathbb{T}/J$ is $\mathbb{T}/(I+J)$, the decomposition $\mathbb{T}\otimes\mathbb{Q} = \mathbb{Q}\times\mathbb{T}'$, and "$\ell$ divides $\#\mathbb{T}/(I+J) = r$": Theorems 2.12 and 2.17 and Propositions 2.13, 2.14 and 2.18.

- **Lemma lem:fpr.** "$L_p(A_f, 1)^{-1} = L_p(A_f(1), 0)^{-1}$ is a nonzero rational number which is an $\ell$-adic integer": Proposition 2.21 and Lemma 2.86.

- **Remark rem:wild.** $P_2(X) = 1 + 2X$ gives $e_3 = v_3(\frac{3}{2}) = 1$; the cubic factors at 2: Example 2.89(e).

- **Section sec:anomaly.** The type III primes with $e_3 > 0$ are those with $q \equiv \pm 1 \pmod{12}$: Proposition 2.78(iv).

- **Lemma lem:h0 and Proposition prop:ns.** $\chi_\ell^2 \equiv 1 \pmod{\ell^k}$ on a group mapping onto $\mathbb{Z}_\ell^\times$ forces $k = 0$ or $\ell^k = 3$: Corollary 2.44; $\bar{\chi}_3^{\mathrm{cyc}} = \chi_{-3}$ on the Dirichlet side: Proposition 2.78(iii); $\mathbb{F}_9 = \mathbb{F}_3[i]$: Theorem 2.38; the 3-division polynomial test: Proposition 2.34 and Example 2.36.

## 2.1 Rings, ideals, quotients and products

This section fixes the language of rings and ideals and proves the isomorphism theorems, the exact sequence of two ideals and the decomposition of a ring by idempotents. In the paper these are exactly the steps of Lemma lem:crit: the Hecke algebra $\mathbb{T}$ is a ring, $I = \ker\phi_f$ and $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$ are ideals with $I \cap J = 0$, the quotient $\mathbb{T}/(I + J)$ is identified with $\mathbb{Z}/\phi_f(J)$, and $\mathbb{T}\otimes\mathbb{Q}$ splits as $\mathbb{Q}\times\mathbb{T}'$. We begin with the conventions of the whole chapter.

**Convention 2.1** (rings)**.** Rings are commutative with an identity 1; ring homomorphisms send 1 to 1; subrings contain 1. The zero ring (in which $1 = 0$) is allowed, but fields and integral domains are nonzero by definition. $R^\times$ is the group of units of $R$. The paper's Hecke algebra $\mathbb{T}$ is such a ring. Matrix rings appear only as examples and are then flagged as non-commutative.

**Convention 2.2** (primes and the letter $q$)**.** The letters $p$, $q$, $\ell$ denote primes unless stated otherwise. In Sections 2.6 and 2.7, and wherever $\mathbb{F}_q$ is written with $q$ not declared prime, $q = p^r$ is a prime power and $\mathbb{F}_q$ is a field with $q$ elements. In the paper $q$ is always a prime (a bad prime of $E$), so $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$ there and primitive roots modulo $q$ exist. The coefficient prime $\ell$ is odd in every use of the paper. In one place, Paper link 2.80, $q$ is also the variable $q = e^{2\pi iz}$ of $q$-expansions (Appendix B, B-C1), and this is said there. No two of these three meanings of $q$ occur in one display.

**Convention 2.3** (valuations)**.** For a prime $p$, $v_p\colon \mathbb{Q} \to \mathbb{Z} \cup \{+\infty\}$ is the $p$-adic valuation (Definition 2.81), with $v_p(0) = +\infty$ and $v_p(p) = 1$, and $|x|_p = p^{-v_p(x)}$, $|0|_p = 0$. For a nonzero ideal $r\mathbb{Z}$, $v_p(r\mathbb{Z}) := v_p(r)$. For a class $c \in \mathbb{Z}/\ell^m\mathbb{Z}$ put

$$v_\ell(c) := \min\bigl(v_\ell(\tilde{c}), m\bigr) \qquad \text{for any lift } \tilde{c} \in \mathbb{Z} \text{ of } c.$$

This does not depend on the lift: two lifts differ by a multiple of $\ell^m$, and $\min(v_\ell(x), m) = \min(v_\ell(x + \ell^m y), m)$ by Proposition 2.82(ii). So $v_\ell(0) = m$, and for $0 \le k \le m$, $v_\ell(c) \ge k$ if and only if $c \in \ell^k\mathbb{Z}/\ell^m\mathbb{Z}$. With this reading the paper's $\min(v_\ell(n), v_\ell(q-1), v_\ell(c))$ in Lemma lem:tamfactor makes sense also when $c = 0$, and also when $v_\ell(q-1) = 0$ (then $c$ lies in the zero group $\mathbb{Z}/1$ and the minimum is 0).

**Convention 2.4** (symbols)**.** $\left(\frac{a}{p}\right)$ is the Legendre symbol for an odd prime $p$ (Definition 2.60), $\left(\frac{a}{n}\right)$ the Jacobi symbol for an odd $n > 0$ (Definition 2.65), and $\left(\frac{a}{n}\right)$ the Kronecker symbol for arbitrary $a, n \in \mathbb{Z}$ (Definition 2.67). The three agree where more than one is defined. The Kronecker symbol has $\left(\frac{a}{2}\right) = 0$ for even $a$, 1 for $a \equiv \pm 1 \pmod 8$ and $-1$ for $a \equiv \pm 3 \pmod 8$; $\left(\frac{a}{-1}\right) = 1$ for $a \ge 0$ and $-1$ for $a < 0$; $\left(\frac{a}{1}\right) = 1$; and $\left(\frac{a}{0}\right) = 1$ if $a = \pm 1$, 0 otherwise.

**Convention 2.5** (characters and their names)**.** A Dirichlet character modulo $n$ is a function $\chi\colon \mathbb{Z} \to \mathbb{C}$ coming from a homomorphism $(\mathbb{Z}/n)^\times \to \mathbb{C}^\times$, with $\chi(a) = 0$ when $\gcd(a, n) > 1$ (Definition 2.71). Quadratic characters are named by a fundamental discriminant (Definition 2.67): for such $D$, $\chi_D(a) = \left(\frac{D}{a}\right)$ is (for $D \ne 1$) the character of $\mathbb{Q}(\sqrt{D})$ (Appendix B, B-C6(b)), primitive of conductor $|D|$ (Corollary 2.76); so $\chi_{-3} = \left(\frac{-3}{\cdot}\right)$ and $\chi_{-4} = \left(\frac{-4}{\cdot}\right)$. For a discriminant $D$ that is not fundamental the Kronecker symbol $\left(\frac{D}{\cdot}\right)$ is always written out and never called $\chi_D$: for instance $\left(\frac{-12}{\cdot}\right)$ is an imprimitive character modulo 12 that agrees with $\chi_{-3}$ at odd integers, whereas under B-C6(b) the symbol $\chi_{-12}$ would mean $\chi_{-3}$ itself, of conductor 3; the chapter does not use the symbol $\chi_{-12}$. The letter $D$ denotes a discriminant; as in the paper, $D = \prod_{q\in S} q$ is also the conductor of $\chi_S$ (the absolute value of the fundamental discriminant $\prod_{q\in S} q^*$). For an odd prime $q$ put $q^* = (-1)^{(q-1)/2}q$; the quadratic character of conductor $q$ is $\chi_{q^*}$, never $\chi_q$. Thus $\chi_{-7}$, $\chi_{-11}$ are the characters of conductor 7, 11, while $\chi_5$, $\chi_{13}$ have conductor 5, 13. A prime-letter subscript ($\chi_\ell$, $\chi_p$) is reserved for the cyclotomic character ($\chi_D$, $\chi_S$, $\chi_{q^*}$ name quadratic characters): $\chi_\ell$ is the $\ell$-adic cyclotomic character of Chapter 5, $\bar{\chi}_\ell$ its reduction, written $\chi_3^{\mathrm{cyc}}$, $\bar{\chi}_3^{\mathrm{cyc}}$ at the prime 3; and $\omega_\ell\colon a \mapsto a \bmod \ell \in \mathbb{F}_\ell^\times$ is the Dirichlet character that corresponds to $\bar{\chi}_\ell$ (Chapters 5 and 6 make the correspondence). Two further names are not of this kind: $\chi_0$ is the principal character modulo $n$, the standard name (the numeral 0 is not a discriminant, so it does not collide with B-C6(b)); and the $p$-component of a character $\chi$, a character modulo a power of $p$, is written $\chi^{(p)}$ with a superscript (Proposition 2.75(v)), never $\chi_p$, and members of a family of characters are written $\chi^{(i)}$ in the same way. The paper follows the same rules.

**Convention 2.6** (roots of unity)**.** $\mu_n(K) = \{x \in K : x^n = 1\}$; $\zeta_n = e^{2\pi i/n} \in \mathbb{C}$; $\zeta_3 = (-1 + \sqrt{-3})/2$ with $\sqrt{-3} := 1 + 2\zeta_3 = i\sqrt{3}$. A primitive cube root of unity is always $\zeta_3$, never $\omega$ (the letter $\omega$ is reserved for differentials and for the characters $\omega_\ell$). In this chapter $\pi$ is the real number $3.14159\ldots$; the element $1 - \zeta_3$ is not given a name.

**Definition 2.7.** Let $R$ be a ring. An element $a \in R$ is a *unit* if $ab = 1$ for some $b \in R$; the units form the group $R^\times$. A *zero divisor* is an $a \ne 0$ with $ab = 0$ for some $b \ne 0$; $a$ is *nilpotent* if $a^n = 0$ for some $n \ge 1$. $R$ is a *field* if $R \ne 0$ and $R^\times = R \setminus \{0\}$, and an *integral domain* (or *domain*) if $R \ne 0$ and $R$ has no zero divisors. A *subring* is a subset containing 0 and 1 and closed under $+$, $-$, $\times$. A *homomorphism* $\phi\colon R \to S$ is a map with $\phi(a + b) = \phi(a) + \phi(b)$, $\phi(ab) = \phi(a)\phi(b)$, $\phi(1) = 1$; its *kernel* is $\phi^{-1}(0)$ and its *image* $\phi(R)$, a subring of $S$.

**Example 2.8.** $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$; $\mathbb{Z}/n\mathbb{Z}$; the polynomial ring $R[x]$ (Section 2.4); the product $R_1 \times R_2$ with componentwise operations, which has the zero divisors $(1,0)$ and $(0,1)$; the subring $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ of $\mathbb{C}$. If $A$ is an abelian group, $\mathrm{End}_{\mathbb{Z}}(A)$ is a ring under addition and composition, in general non-commutative; a set of commuting endomorphisms generates a commutative subring. This is the shape of the Hecke algebra: the paper's $\mathbb{T}$ is "the subring of $\mathrm{End}\, S_2(\Gamma_0(N))$ generated over $\mathbb{Z}$ by all $T_n$" (Chapter 13). The element $x$ of $\mathbb{Z}[x]/(x^2)$ is nilpotent; a domain has no nonzero nilpotents.

**Proposition 2.9.** *Let $R$ be a ring.*

*(i) There is exactly one homomorphism $\mathbb{Z} \to R$, namely $n \mapsto n \cdot 1$. Its kernel is $n\mathbb{Z}$ for a unique $n \geq 0$, called the* characteristic $\mathrm{char}\, R$*; its image is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.*

*(ii) If $R$ is a domain,* $\mathrm{char}\, R$ *is $0$ or a prime.*

*(iii) A finite domain is a field.*

*Proof.* (i) A homomorphism $\psi\colon \mathbb{Z} \to R$ has $\psi(1) = 1$, hence $\psi(n) = n \cdot 1$ for $n \geq 0$ by additivity and $\psi(-n) = -\psi(n)$; conversely $n \mapsto n \cdot 1$ is a homomorphism by the distributive law. The kernel is an ideal of $\mathbb{Z}$, hence of the form $n\mathbb{Z}$ with a unique $n \geq 0$ (Proposition 2.11(iii) below, whose proof uses only division with remainder), and the image is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ by Theorem 2.12(i). (ii) Let $n = \mathrm{char}\, R > 0$. Since $R \neq 0$, $n \neq 1$. If $n = ab$ with $1 < a, b < n$, then $(a \cdot 1)(b \cdot 1) = 0$ with both factors nonzero (by minimality of $n$), which is impossible in a domain. So $n$ is prime. (iii) Let $R$ be a finite domain and $a \neq 0$. The map $x \mapsto ax$ is injective, because $ax = ay$ gives $a(x - y) = 0$, so $x = y$. An injective map of a finite set to itself is surjective, so $ab = 1$ for some $b$. □

For example $\mathrm{char}\, \mathbb{Z}/6\mathbb{Z} = 6$, and indeed $\mathbb{Z}/6\mathbb{Z}$ is not a domain: $2 \cdot 3 = 0$, exactly the factorisation used in (ii). The rings $\mathbb{F}_p[x]$ and $\mathbb{F}_p(x)$ have characteristic $p$ and are infinite; $\mathbb{F}_p[x]$ is a domain that is not a field ($x$ has no inverse, by degrees), so the finiteness in (iii) cannot be dropped. In the paper the Hecke algebra $\mathbb{T}$ has characteristic 0 (it contains $\mathbb{Z} \cdot T_1 = \mathbb{Z}$), while its quotients $\mathbb{T}/\mathfrak{m}$ by maximal ideals containing $\ell$ have characteristic $\ell$.

**Definition 2.10.** An *ideal* of $R$ is an additive subgroup $I$ with $ra \in I$ for all $r \in R$, $a \in I$. The ideal generated by $a_1, \ldots, a_n$ is $(a_1, \ldots, a_n) = \{\sum r_i a_i\}$; an ideal $(a)$ is *principal*. For ideals $I, J$: the sum $I + J = \{a + b\}$, the product $IJ$ (the set of finite sums $\sum a_i b_i$, $a_i \in I$, $b_i \in J$) and the intersection $I \cap J$ are ideals; $I$ and $J$ are *comaximal* if $I + J = R$. For an $R$-module $M$ (modules are Chapter 3; here $M$ may be any abelian group on which $R$ acts) and $S \subseteq M$, the *annihilator* is $\mathrm{Ann}_R(S) = \{r \in R : rs = 0 \text{ for all } s \in S\}$, an ideal. The *quotient ring* $R/I$ is the quotient group with $(a + I)(b + I) = ab + I$; the projection $R \to R/I$ is a surjective homomorphism with kernel $I$.

The multiplication on $R/I$ is well defined: if $a' = a + x$ and $b' = b + y$ with $x, y \in I$, then $a'b' - ab = ay + xb + xy \in I$. This computation is the reason for the definition: an additive subgroup $I$ can serve as the set of elements declared to be 0 in a quotient ring exactly when it absorbs multiplication by all of $R$, since $ay$ and $xb$ must lie in $I$ for arbitrary $a, b$. Small examples: in $\mathbb{Z}$, $(4, 6) = 2\mathbb{Z}$ and $4\mathbb{Z} \cdot 6\mathbb{Z} = 24\mathbb{Z} \subsetneq 4\mathbb{Z} \cap 6\mathbb{Z} = 12\mathbb{Z}$; the ideal $(2, x)$ of $\mathbb{Z}[x]$ is the set of polynomials with even constant term; $\mathrm{Ann}_{\mathbb{Z}}(\mathbb{Z}/6\mathbb{Z}) = 6\mathbb{Z}$, and for the $\mathbb{Z}$-module $M = \mathbb{Z}/4 \times \mathbb{Z}/6$ one has $\mathrm{Ann}_{\mathbb{Z}}(M) = 12\mathbb{Z}$; $2\mathbb{Z}$ and $3\mathbb{Z}$ are comaximal since $1 = 3 - 2$. The ideal $J = \mathrm{Ann}_{\mathbb{T}}(f^{\perp})$ of Lemma lem:crit is an annihilator of exactly this kind, with $R = \mathbb{T}$ acting on the module $f^{\perp}$.

**Proposition 2.11.** *(i) The kernel of a homomorphism is an ideal, and every ideal is the kernel of a homomorphism (the projection $R \to R/I$). An ideal $I$ equals $R$ iff it contains a unit. A ring is a field iff it is nonzero and its only ideals are $0$ and $R$.*

*(ii) $IJ \subseteq I \cap J$ for ideals $I, J$.*

*(iii) Every ideal of $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique $n \geq 0$. For $m, n \in \mathbb{Z}$, $m\mathbb{Z} + n\mathbb{Z} = \gcd(m,n)\mathbb{Z}$, and $m\mathbb{Z} \cap n\mathbb{Z} = \operatorname{lcm}(m,n)\mathbb{Z}$.*

*Proof.* (i) If $\phi(a) = \phi(b) = 0$ then $\phi(a-b) = 0$ and $\phi(ra) = \phi(r)\phi(a) = 0$. The projection has kernel $I$. If $u \in I$ is a unit, then $r = (ru^{-1})u \in I$ for every $r$. If $R$ is a field, a nonzero ideal contains a unit, so it is $R$; conversely, if $R \neq 0$ has only the ideals $0$ and $R$ and $a \neq 0$, then $(a) = R \ni 1$, so $a$ is a unit. (ii) Each $a_i b_i$ lies in $I$ (as $a_i \in I$) and in $J$ (as $b_i \in J$). (iii) Let $I \neq 0$ be an ideal of $\mathbb{Z}$ and $n$ its least positive element. For $a \in I$ write $a = bn + r$ with $0 \leq r < n$; then $r = a - bn \in I$, so $r = 0$ by minimality. Hence $I = n\mathbb{Z}$; $n$ is unique as the least positive element. Now $m\mathbb{Z} + n\mathbb{Z} = d\mathbb{Z}$ for some $d \geq 0$. Since $m, n \in d\mathbb{Z}$, $d$ is a common divisor; since $d = um + vn$ for some $u, v$, every common divisor divides $d$. So $d = \gcd(m,n)$. Next, $m\mathbb{Z} \cap n\mathbb{Z}$ is an ideal (an intersection of ideals), so it is $e\mathbb{Z}$ for a unique $e \geq 0$. Since $e \in m\mathbb{Z} \cap n\mathbb{Z}$, $e$ is a common multiple of $m$ and $n$; and a common multiple $c$ lies in $m\mathbb{Z} \cap n\mathbb{Z} = e\mathbb{Z}$, so $e \mid c$. A nonnegative common multiple that divides every common multiple is $\operatorname{lcm}(m,n)$ (with $\operatorname{lcm}(m,n) = 0$ when $m$ or $n$ is $0$), so $e = \operatorname{lcm}(m,n)$. □

**Theorem 2.12** (isomorphism theorems)**.** *Let $\phi\colon R \to S$ be a homomorphism and $I$ an ideal of $R$.*

*(i) $\phi$ induces an isomorphism $R/\ker\phi \simeq \phi(R)$.*

*(ii) (Correspondence.) $J \mapsto J/I$ is an inclusion-preserving bijection from the ideals of $R$ containing $I$ to the ideals of $R/I$, with inverse $\bar{J} \mapsto$ its preimage, and $(R/I)/(J/I) \simeq R/J$. Under it, prime ideals correspond to prime ideals and maximal ideals to maximal ideals (Theorem 2.17).*

*(iii) If $\phi$ is surjective with kernel $I$ and $J$ is any ideal of $R$, then $\phi(J)$ is an ideal of $S$ and $\phi$ induces an isomorphism $R/(I+J) \simeq S/\phi(J)$.*

*(iv) Every quotient ring of $\mathbb{Z}$ is $\mathbb{Z}/r\mathbb{Z}$ for a unique $r \geq 0$; as an additive group it is cyclic, generated by the class of $1$.*

*Proof.* (i) The map $a + \ker\phi \mapsto \phi(a)$ is well defined and injective because $\phi(a) = \phi(b)$ iff $a - b \in \ker\phi$; it is additive, multiplicative and sends 1 to 1, and it is onto $\phi(R)$. (ii) For an ideal $\bar{J}$ of $R/I$ its preimage $J$ under the projection $\pi$ is an ideal containing $I$ with $\pi(J) = \bar{J}$, since $\pi$ is onto. Conversely for $J \supseteq I$, $\pi(J) = J/I$ is an ideal of $R/I$ (an additive subgroup, stable under multiplication by $\pi(r)$, as $\pi$ is onto), and its preimage is $J + I = J$. Both maps preserve inclusions. The composite $R \to R/I \to (R/I)/(J/I)$ is onto with kernel $\pi^{-1}(J/I) = J$, so (i) gives $(R/I)/(J/I) \simeq R/J$. The statement on prime and maximal ideals follows from this isomorphism and Theorem 2.17(i): $J$ is prime (maximal) iff $R/J$ is a domain (a field) iff $(R/I)/(J/I)$ is. (iii) $\phi(J)$ is an additive subgroup, and for $s = \phi(r) \in S$ (every $s$ is of this form) and $\phi(a) \in \phi(J)$, $s\phi(a) = \phi(ra) \in \phi(J)$. The composite $R \to S \to S/\phi(J)$ is onto, and $a$ lies in its kernel iff $\phi(a) = \phi(b)$ for some $b \in J$, iff $a - b \in I$ for some $b \in J$, iff $a \in I + J$. Now apply (i). (iv) A quotient ring of $\mathbb{Z}$ is $\mathbb{Z}/I$ with $I = r\mathbb{Z}$, $r \geq 0$ unique (Proposition 2.11(iii)); every class is a multiple of the class of 1. □

**Proposition 2.13.** *Let $I, J$ be ideals of a ring $R$. The sequence of additive groups (indeed of $R$-modules)*

$$0 \longrightarrow R/(I \cap J) \xrightarrow{x \mapsto (x,x)} R/I \times R/J \xrightarrow{(y,z) \mapsto y-z} R/(I+J) \longrightarrow 0 \tag{2.1}$$

*is exact. Consequently:*

*(i) if $I \cap J = 0$, then $R \to R/I \times R/J$ is injective, and its cokernel is isomorphic to $R/(I+J)$;*

*(ii) (Chinese remainder theorem for two ideals) if $I + J = R$, then $I \cap J = IJ$ and $R/IJ \to R/I \times R/J$ is a ring isomorphism;*

*(iii) (general Chinese remainder theorem) if $I_1, \dots, I_k$ are pairwise comaximal, then the natural map $R/(I_1 \cdots I_k) \to \prod_i R/I_i$ is a ring isomorphism.*

*Proof.* In (2.1) the first map is well defined (if $x \in I \cap J$ both components vanish) and the second also (the class of $y - z$ modulo $I + J$ does not change if $y$ is moved by $I$ or $z$ by $J$). The first map is injective: $(x, x) = 0$ means $x \in I$ and $x \in J$. The composite is zero. If $y - z \in I + J$, say $y - z = a + b$ with $a \in I$, $b \in J$, then $x := y - a = z + b$ satisfies $x \equiv y \pmod I$ and $x \equiv z \pmod J$; so the kernel of the second map is the image of the first. The second map is onto, since $(y, 0) \mapsto y$. (i) is the case $I \cap J = 0$ read off from exactness. (ii) If $I + J = R$ write $1 = a + b$ with $a \in I$, $b \in J$. For $x \in I \cap J$, $x = xa + xb \in IJ$ (as $xa = ax \in JI$ and $xb \in IJ$); with Proposition 2.11(ii), $I \cap J = IJ$. Exactness with $R/(I + J) = 0$ shows that $R/IJ \to R/I \times R/J$ is bijective; it is a ring homomorphism because both coordinates are. (iii) Induction on $k$, the case $k = 2$ being (ii). For $k \geq 3$, $I_1$ and $I_2 \cdots I_k$ are comaximal: writing $1 = a_j + b_j$ with $a_j \in I_1$, $b_j \in I_j$ for $j \geq 2$, the product $\prod_{j \geq 2}(a_j + b_j) = 1$ is $b_2 \cdots b_k \in I_2 \cdots I_k$ plus terms in $I_1$. By (ii) and the induction hypothesis, $R/I_1 \cdots I_k \simeq R/I_1 \times R/I_2 \cdots I_k \simeq R/I_1 \times \prod_{j \geq 2} R/I_j$. □

The maps in (2.1) commute with multiplication by elements of $R$, which is why the sequence is one of $R$-modules and not only of groups. A worked case in $R = \mathbb{Z}$: for $I = 4\mathbb{Z}$, $J = 6\mathbb{Z}$ one has $I \cap J = 12\mathbb{Z}$ and $I + J = 2\mathbb{Z}$ (Proposition 2.11(iii)), and the sequence is

$$0 \longrightarrow \mathbb{Z}/12\mathbb{Z} \longrightarrow \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0,$$

with maps $x \mapsto (x \bmod 4, x \bmod 6)$ and $(y, z) \mapsto (y - z) \bmod 2$. The orders are consistent, $12 \cdot 2 = 24 = 4 \cdot 6$, and the cokernel $\mathbb{Z}/2\mathbb{Z}$ measures the failure of the Chinese remainder theorem: a pair $(y, z)$ comes from an integer iff $y \equiv z \pmod 2$. In (ii), the decomposition is computed by idempotents: for $I = 3\mathbb{Z}$, $J = 5\mathbb{Z}$, Bezout gives $1 = 2 \cdot 5 - 3 \cdot 3$, and $a = -9 \in I$, $b = 10 \in J$. The class of $b = 10$ is 1 modulo 3 and 0 modulo 5, and $-9 \equiv 6 \pmod{15}$ is 0 modulo 3 and 1 modulo 5; both are idempotents of $\mathbb{Z}/15\mathbb{Z}$ ($10^2 = 100 \equiv 10$, $6^2 = 36 \equiv 6$). So the pair $(y \bmod 3, z \bmod 5)$ comes from $10y + 6z \bmod 15$; for instance $(2, 3)$ comes from $38 \equiv 8 \pmod{15}$. This is the situation of Lemma lem:crit with the roles reversed: there $I \cap J = 0$ and the interesting group is the cokernel $R/(I + J)$, here it is $I + J = R$ and the cokernel vanishes.

**Proposition 2.14.** *An* idempotent *of $R$ is an element $e$ with $e^2 = e$.*

*(i) If $e$ is idempotent, so is $1 - e$, and $e(1 - e) = 0$. The set $Re$ is a ring with identity $e$ (not a subring of $R$ unless $e = 1$), and $x \mapsto (xe, x(1-e))$ is a ring isomorphism $R \to Re \times R(1-e)$.*

*(ii) $R$ is isomorphic to a product of two nonzero rings iff it has an idempotent $e \neq 0, 1$.*

*(iii) The ideals of $R_1 \times R_2$ are the products $I_1 \times I_2$ of ideals; its prime ideals are the $P_1 \times R_2$ and $R_1 \times P_2$ with $P_i$ prime, and the same holds for maximal ideals.*

*(iv) If $f = f_1 \cdots f_k \in K[x]$ ($K$ a field) with pairwise coprime $f_i$, then $K[x]/(f) \simeq \prod_i K[x]/(f_i)$. For instance $\mathbb{Q}[x]/(x^2 - x) \simeq \mathbb{Q} \times \mathbb{Q}$, $x \mapsto (0, 1)$.*

*Proof.* (i) $(1-e)^2 = 1 - 2e + e^2 = 1 - e$ and $e(1-e) = e - e^2 = 0$. $Re$ is closed under addition and multiplication, and $e \cdot xe = xe^2 = xe$, so $e$ is its identity. The map $x \mapsto (xe, x(1-e))$ is additive and multiplicative ($xye = xe \cdot ye$ as $e^2 = e$) and sends 1 to $(e, 1-e)$, the identity of the product. It is injective, since $xe = x(1-e) = 0$ gives $x = xe + x(1-e) = 0$, and surjective, since $(ae, b(1-e))$ is the image of $ae + b(1-e)$ (use $e(1-e) = 0$). (ii) If $R = R_1 \times R_2$ with $R_i \neq 0$, then $e = (1, 0)$ is an idempotent different from 0 and 1. Conversely, given such $e$, (i) gives $R \simeq Re \times R(1-e)$ with $Re \ni e \neq 0$ and $R(1-e) \ni 1 - e \neq 0$. (iii) If $I$ is an ideal of $R_1 \times R_2$, let $I_1$, $I_2$ be its projections; they are ideals. If $(a, b) \in I$ then $(a, 0) = (1, 0)(a, b) \in I$ and $(0, b) \in I$; so $I = I_1 \times I_2$. The quotient $(R_1 \times R_2)/(I_1 \times I_2) \simeq R_1/I_1 \times R_2/I_2$ is a domain (resp. a field) iff one factor is zero and the other a domain (resp. a field), because $(1, 0)(0, 1) = 0$; by Theorem 2.17(i) this gives the prime and maximal ideals. (iv) This part uses two results proved later: $K[x]$ is a principal ideal domain (PID, Definition 2.23), by Proposition 2.30(ii), and in a PID $(a) + (b) = (\gcd(a, b))$ (Theorem 2.25(ii)); neither proof uses this proposition. So coprime $f_i$, $f_j$ satisfy $(f_i) + (f_j) = (\gcd(f_i, f_j)) = K[x]$, and $(f_1) \cdots (f_k) = (f)$; apply Proposition 2.13(iii). For $x^2 - x = x(x-1)$, the map is $g \mapsto (g(0), g(1))$, and the class of $x$ is the idempotent $(0, 1)$. □

For example, in $R = \mathbb{Z}/6\mathbb{Z}$ the element $e = 3$ is idempotent ($9 \equiv 3$), with $1 - e = -2 \equiv 4$. Then $Re = \{0, 3\}$, a ring with identity 3 isomorphic to $\mathbb{Z}/2\mathbb{Z}$, and $R(1-e) = \{0, 4, 2\}$, a ring with identity 4 isomorphic to $\mathbb{Z}/3\mathbb{Z}$; the map of (i) is $x \mapsto (3x, 4x)$, and it recovers $\mathbb{Z}/6\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. By (iii) the prime ideals of $\mathbb{Z}/6\mathbb{Z}$ are the two ideals $0 \times \mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times 0$, that is $2\mathbb{Z}/6\mathbb{Z} = \{0, 2, 4\}$ and $3\mathbb{Z}/6\mathbb{Z} = \{0, 3\}$, in accordance with the correspondence Theorem 2.12(ii). In the paper the idempotents are those of $\mathbb{T} \otimes \mathbb{Q}$ and $\mathbb{T} \otimes \mathbb{Z}_\ell$: $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ corresponds to the idempotent $e$ projecting onto the factor $\mathbb{Q}$ (the action on the line spanned by $f$). The idempotent $e$ itself need not lie in $\mathbb{T}$: in the ring $R = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod r\}$ of Exercise 2.91, the idempotent $e = (1, 0)$ of $R \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{Q}$ is not in $R$ for $r > 1$, and the least positive integer $m$ with $me = (m, 0) \in R$ is $m = r$, the order of $R/(I + J)$.

*In the paper* 2.15. Lemma lem:crit (congruence criterion) is ring theory once one fact about Hecke algebras is granted. Let $\phi_f \colon \mathbb{T} \to \mathbb{Z}$, $T_n \mapsto a_n(f)$, let $I = \ker \phi_f$ and $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$. The paper writes $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$, where the first factor is the action on the line $\mathbb{Q}f$ and $\mathbb{T}'$ the (faithful) action on $f^\perp$; that this decomposition exists is a statement about Hecke operators and the Petersson product (Chapter 13). Granting it, $\mathbb{T}$ embeds in $\mathbb{T} \otimes \mathbb{Q}$ ($\mathbb{T}$ is torsion-free) and $t \mapsto (\phi_f(t), t')$. Then $t \in I$ iff its first coordinate vanishes, so $I = \mathbb{T} \cap (0 \times \mathbb{T}')$; and $t \in J$ iff $t$ kills $f^\perp$ iff $t' = 0$, so $J = \mathbb{T} \cap (\mathbb{Q} \times 0)$. Hence $I \cap J = 0$. The map $\phi_f$ is onto $\mathbb{Z}$ because $T_1 = 1$ and $\phi_f(T_1) = a_1(f) = 1$; by Theorem 2.12(iii),

$$\mathbb{T}/(I + J) \simeq \mathbb{Z}/\phi_f(J),$$

and this is cyclic (Theorem 2.12(iv)): "$\mathbb{T}/(I + J)$ is cyclic of order $r$; equivalently, $r$ generates the congruence ideal $\phi_f(J) = \phi_f(\mathrm{Ann}_{\mathbb{T}} I)$". By Proposition 2.13(i), the inclusion $\mathbb{T} \subset \mathbb{T}/I \times \mathbb{T}/J$ has cokernel $\mathbb{T}/(I + J)$, which is the sentence "Dualising the inclusion $\mathbb{T} \subset \mathbb{T}/I \times \mathbb{T}/J$, whose cokernel is $\mathbb{T}/(I + J)$" of the proof. The module version of the same sequence, $L[\wp] \oplus L[I] \subseteq L$ in Lemma lem:ddt417, is Chapter 3. In Step 1 of the proof of Theorem thm:quant, and in Lemma

lem:hecke, $\mathbb{T}\otimes\mathbb{Z}_\ell$ is a product of local rings cut out by idempotents (Proposition 2.14; the general statement for finite $\mathbb{Z}_\ell$-algebras is Chapter 3).

**Proposition 2.16.** *Let $R$ be a ring of prime characteristic $p$.*

*(i)* $p \mid \binom{p}{i}$ *for* $0 < i < p$*; hence* $(x+y)^p = x^p + y^p$ *and* $(x+y)^{p^k} = x^{p^k} + y^{p^k}$ *for* $x, y \in R$*,* $k \geq 0$*.*

*(ii)* $F\colon x \mapsto x^p$ *is a ring endomorphism of $R$ (the* absolute Frobenius*). It is injective if $R$ is a domain and bijective if $R$ is a finite domain, that is (Proposition 2.9(iii)) a finite field.*

*(iii) (Fermat)* $a^p \equiv a \pmod{p}$ *for every* $a \in \mathbb{Z}$*.*

*Proof.* (i) $\binom{p}{i} = \frac{p!}{i!(p-i)!}$ is an integer; $p$ divides the numerator and not the denominator when $0 < i < p$ (a prime dividing a product of integers smaller than $p$ would divide one of them: Euclid's lemma, Proposition 2.26(ii) below, whose proof does not use this proposition). So the middle terms of the binomial expansion vanish in $R$. The case $p^k$ follows by induction, applying the case $p$ to $x^{p^{k-1}}$ and $y^{p^{k-1}}$. (ii) $F$ is additive by (i) and multiplicative by commutativity, and $F(1) = 1$. In a domain $x^p = 0$ forces $x = 0$; an injective map of a finite set to itself is bijective. (iii) In $R = \mathbb{Z}/p\mathbb{Z}$, $F$ is additive and fixes 1, so it fixes every $n \cdot 1$: $n^p \equiv n \pmod{p}$. □

## 2.2 Prime and maximal ideals, fractions, and the ring $\mathbb{Z}_{(p)}$

Prime and maximal ideals are how the paper speaks about reduction: a maximal ideal $\mathfrak{m}$ of $\mathbb{T}\otimes\mathbb{Z}_\ell$ "containing $\ell$ and $\ker\phi_f$" (Lemma lem:hecke, Theorem thm:quant) is a congruence modulo $\ell$ between $f$ and other eigenforms, and the last step of Lemma lem:crit is that a maximal ideal of residue characteristic $\ell$ containing $I + J$ forces $\ell$ to divide $\#\mathbb{T}/(I+J)$. The ring $\mathbb{Z}_{(p)}$ of rationals with denominator prime to $p$ is the first local ring of the book; it is where the paper's "nonzero rational number which is an $\ell$-adic integer" (Lemma lem:fpr) lives.

**Theorem 2.17.** *An ideal $P \neq R$ is* prime *if $ab \in P$ implies $a \in P$ or $b \in P$; an ideal $M \neq R$ is* maximal *if the only ideals containing $M$ are $M$ and $R$.*

*(i) $R/P$ is a domain iff $P$ is prime; $R/M$ is a field iff $M$ is maximal. Every maximal ideal is prime.*

*(ii) The preimage of a prime ideal under a homomorphism is prime. The kernel of a homomorphism $\phi\colon R \to K$ to a field is prime; it is maximal if $\phi$ is surjective, or more generally if $\phi(R)$ is finite.*

*(iii) If $R/P$ is finite, then $P$ is prime iff it is maximal.*

*(iv) The prime ideals of $\mathbb{Z}$ are $0$ and the $p\mathbb{Z}$; the maximal ideals are the $p\mathbb{Z}$.*

*(v) Every nonzero ring has a maximal ideal; every non-unit of $R$ lies in a maximal ideal.*

*Proof.* (i) $R/P \neq 0$ iff $P \neq R$, and $\bar{a}\bar{b} = 0$ in $R/P$ means $ab \in P$; so "no zero divisors" is "prime". By the correspondence Theorem 2.12(ii) the ideals of $R/M$ are the $J/M$ with $M \subseteq J$, so $R/M$ has exactly two ideals iff $M$ is maximal, and Proposition 2.11(i) says this means $R/M$ is a field. A field is a domain. (ii) If $Q \subset S$ is prime and $\phi\colon R \to S$, then $\phi^{-1}(Q) \neq R$ (as $\phi(1) = 1 \notin Q$), and $ab \in \phi^{-1}(Q)$ gives $\phi(a)\phi(b) \in Q$. The zero ideal of a field is prime, so $\ker\phi$ is prime. By

(i) of Theorem 2.12, $R/\ker\phi \simeq \phi(R)$, a subring of a field, hence a domain; if it is all of $K$ or finite, it is a field (Proposition 2.9(iii)), and $\ker\phi$ is maximal by (i). (iii) $R/P$ is a finite ring; it is a domain iff it is a field (Proposition 2.9(iii)). (iv) A nonzero ideal of $\mathbb{Z}$ is $n\mathbb{Z}$ with $n \geq 1$ (Proposition 2.11); $\mathbb{Z}/n$ is a domain iff $n$ is prime (Proposition 2.40(i) below, proved from Bezout's identity and independent of this theorem), and is then finite, hence a field. $\mathbb{Z}/0 = \mathbb{Z}$ is a domain and not a field. (v) Let $a$ be a non-unit (for instance $a = 0$ in a nonzero ring). The set $\Sigma$ of ideals $I \neq R$ containing $a$ is nonempty, as $(a) \neq R$, and is partially ordered by inclusion. The union of a chain in $\Sigma$ is an ideal containing $a$, and it does not contain 1 because no member does; so it lies in $\Sigma$. By Zorn's lemma $\Sigma$ has a maximal element, which is a maximal ideal containing $a$. (For Noetherian rings, Definition 2.23 and Theorem 2.25(ii), Zorn's lemma can be avoided: a nonempty set of ideals then has a maximal element.) □

Examples: in $\mathbb{Z}[x]$, $(x)$ is prime but not maximal, since $\mathbb{Z}[x]/(x) \simeq \mathbb{Z}$ (evaluation at 0) is a domain but not a field; it lies in the maximal ideal $(2, x)$, with $\mathbb{Z}[x]/(2, x) \simeq \mathbb{F}_2$. In $\mathbb{Z}$, $(4)$ is not prime ($2 \cdot 2 \in (4)$). Part (ii) in its form "$\phi(R)$ finite" is the one used in the paper: the map $\mathbb{T} \to \mathcal{O}/\lambda$ of Paper link 2.19 need not be surjective, but its image is a subring of a finite field, hence a finite domain, hence a field.

**Proposition 2.18.** *Let $\mathfrak{m}$ be a maximal ideal of $R$ such that the field $R/\mathfrak{m}$ has characteristic $\ell > 0$ (we say $\mathfrak{m}$ has* residue characteristic $\ell$*).*

*(i)* $\ell \cdot 1 \in \mathfrak{m}$.

*(ii) If $\mathfrak{a} \subseteq \mathfrak{m}$ is an ideal with $R/\mathfrak{a}$ finite, then $\ell \mid \#(R/\mathfrak{a})$.*

*(iii) For $r \geq 1$, there is a ring homomorphism $\mathbb{Z}/r\mathbb{Z} \to k$ into a field $k$ of characteristic $\ell$ iff $\ell \mid r$.*

*Proof.* (i) The image of $\ell \cdot 1$ in $R/\mathfrak{m}$ is $\ell \cdot 1 = 0$. (ii) The projection $R/\mathfrak{a} \to R/\mathfrak{m}$ is onto (Theorem 2.12(ii)), so $\#(R/\mathfrak{a}) = \#(\mathfrak{m}/\mathfrak{a}) \cdot \#(R/\mathfrak{m})$ by Lagrange's theorem for the additive group. The field $R/\mathfrak{m}$ contains $\mathbb{F}_\ell$ (the image of $\mathbb{Z}$) and is a vector space over it of dimension $\geq 1$, finite as $R/\mathfrak{a}$ is; so $\#(R/\mathfrak{m})$ is a positive power of $\ell$. (iii) A homomorphism $\mathbb{Z}/r \to k$ composed with $\mathbb{Z} \to \mathbb{Z}/r$ is the unique map $\mathbb{Z} \to k$ (Proposition 2.9), whose kernel is $\ell\mathbb{Z}$; it factors through $\mathbb{Z}/r$ iff $r\mathbb{Z} \subseteq \ell\mathbb{Z}$, i.e. iff $\ell \mid r$. □

*In the paper* 2.19. The last step of the proof of Lemma lem:crit reads: "Thus $I + J$ lies in the kernel of $\mathbb{T} \to \mathcal{O}/\lambda$, a maximal ideal of residue characteristic $\ell$, and $\ell$ divides $\#\mathbb{T}/(I + J) = r$." Here the map $\mathbb{T} \to \mathcal{O}/\lambda$ is $t \mapsto \phi_f(t) \bmod \lambda$; its image is a subring of the finite field $\mathcal{O}/\lambda$ (whose characteristic is $\ell$, as $\lambda \mid \ell$), so its kernel $\mathfrak{m}$ is maximal by Theorem 2.17(ii), of residue characteristic $\ell$. It contains $I = \ker\phi_f$, and $J$ because $\phi_g$ kills $J$ and $\phi_g \equiv \phi_f \pmod{\lambda}$. Since $\mathbb{T}/(I + J) \simeq \mathbb{Z}/r$ is finite (if $r \neq 0$), Proposition 2.18(ii) gives $\ell \mid r$; equivalently, by (iii), $\mathbb{Z}/r$ maps to a field of characteristic $\ell$. The same reasoning is behind "after enlarging $\mathcal{O}$ and $\lambda$" in Lemma lem:old.

**Theorem 2.20.** *Let $R$ be a domain. On pairs $(a, b) \in R \times (R \setminus \{0\})$ put $(a, b) \sim (c, d)$ iff $ad = bc$, and write $a/b$ for the class. Then* $\mathrm{Frac}(R) = \{a/b\}$ *with $a/b + c/d = (ad + bc)/bd$ and $(a/b)(c/d) = ac/bd$ is a field, $a \mapsto a/1$ is an injective homomorphism $R \to \mathrm{Frac}(R)$, and every injective homomorphism $\iota\colon R \to K$ into a field extends uniquely to a homomorphism* $\mathrm{Frac}(R) \to K$*, namely $a/b \mapsto \iota(a)\iota(b)^{-1}$. Examples:* $\mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}$, $\mathrm{Frac}(F[x]) = F(x)$, $\mathrm{Frac}(\mathbb{Z}[i]) = \mathbb{Q}(i)$, $\mathrm{Frac}(\mathbb{Z}[\zeta_3]) = \mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$.

*Proof.* $\sim$ is reflexive and symmetric; if $ad = bc$ and $cf = de$ then $adf = bcf = bde$, so $d(af - be) = 0$ and $af = be$ as $d \neq 0$: it is transitive. The operations are well defined: if $ab' = a'b$, then $(ad + bc)b'd = a'bd^2 + bb'cd = (a'd + b'c)bd$ and $ac \cdot b'd = a'c \cdot bd$, so replacing $(a, b)$ by $(a', b')$ does not change the classes of $(ad + bc, bd)$ and $(ac, bd)$ (by symmetry the same holds in the second argument). The ring axioms are checked by computing both sides, using that $(x, y) \sim (tx, ty)$ for $t \neq 0$; for instance for distributivity, $\frac{a}{b}(\frac{c}{d} + \frac{e}{f}) = \frac{a(cf+de)}{bdf}$ and $\frac{ac}{bd} + \frac{ae}{bf} = \frac{acbf+bdae}{b^2df} = \frac{b\cdot a(cf+de)}{b\cdot bdf}$, which are equal; associativity of the addition holds because both bracketings give $(adf + bcf + bde)/bdf$. Here $bd \neq 0$ because $R$ is a domain, which is where the hypothesis is used: in $\mathbb{Z}/6\mathbb{Z}$ the would-be fraction $1/2 \cdot 1/3$ has denominator 0. A nonzero $a/b$ has $a \neq 0$ and inverse $b/a$. The map $a \mapsto a/1$ is a homomorphism, injective since $a/1 = 0/1$ means $a = 0$. If $\iota$ is injective, $\iota(b) \neq 0$ for $b \neq 0$, so $a/b \mapsto \iota(a)\iota(b)^{-1}$ makes sense; it is well defined ($ad = bc$ gives $\iota(a)\iota(d) = \iota(b)\iota(c)$) and a homomorphism. Uniqueness: any extension $\tilde{\iota}$ satisfies $\tilde{\iota}(a/b)\iota(b) = \tilde{\iota}(a/b \cdot b/1) = \iota(a)$. For the examples: $\mathbb{Q}(i)$ and $\mathbb{Q}(\zeta_3)$ are fields containing $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ in which every element is a quotient (for $z = x + yi \in \mathbb{Q}(i)$ clear the denominators of $x, y$); $\zeta_3 = (-1 + \sqrt{-3})/2$ gives $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$. □

**Proposition 2.21.** *Let $p$ be a prime and $\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : a, b \in \mathbb{Z},\ p \nmid b\}$.*

*(i) $\mathbb{Z}_{(p)}$ is a subring of $\mathbb{Q}$, and $\mathbb{Z}_{(p)}^\times = \{a/b : p \nmid ab\}$; an integer prime to $p$ is a unit of $\mathbb{Z}_{(p)}$.*

*(ii) Every nonzero element is $p^k u$ with $k \geq 0$ and $u \in \mathbb{Z}_{(p)}^\times$, uniquely; the nonzero ideals are the $p^k\mathbb{Z}_{(p)}$, $k \geq 0$, and $p\mathbb{Z}_{(p)}$ is the only maximal ideal (so $\mathbb{Z}_{(p)}$ is a* local ring*; it is the first example of a discrete valuation ring, Chapter 3).*

*(iii) The reduction map $\mathbb{Z}_{(p)} \to \mathbb{F}_p$, $a/b \mapsto \bar{a}\bar{b}^{-1}$, is a well-defined surjective ring homomorphism with kernel $p\mathbb{Z}_{(p)}$; it maps $\mathbb{Z}_{(p)}^\times$ onto $\mathbb{F}_p^\times$.*

*(iv) $\mathbb{Z}_{(p)} = \{x \in \mathbb{Q} : v_p(x) \geq 0\}$ and $\mathbb{Z}_{(p)}^\times = \{x \in \mathbb{Q} : v_p(x) = 0\}$.*

*(v) $\mathbb{Z} = \bigcap_p \mathbb{Z}_{(p)}$.*

*Proof.* Every rational is $a/b$ in lowest terms with $b > 0$, and $x \in \mathbb{Z}_{(p)}$ iff $p$ does not divide this $b$ (if $x = a'/b'$ with $p \nmid b'$, the reduced denominator divides $b'$). (i) Sums and products of fractions with denominators prime to $p$ have denominators $bd$ prime to $p$ (Euclid's lemma, Proposition 2.26). In lowest terms, $a/b$ has an inverse $b/a$ in $\mathbb{Z}_{(p)}$ iff $p \nmid a$. (ii) Write $a = p^k a'$ with $p \nmid a'$ (unique factorisation in $\mathbb{Z}$); then $a/b = p^k \cdot a'/b$ with $a'/b$ a unit, and $k = v_p(a/b)$ is determined (Definition 2.81). If $I \neq 0$ is an ideal, let $k$ be the least exponent of its elements; then $p^k = (p^k u)u^{-1} \in I$ and every element $p^{k'}u'$ of $I$, $k' \geq k$, lies in $p^k\mathbb{Z}_{(p)}$; so $I = p^k\mathbb{Z}_{(p)}$. These ideals form a chain, and the largest proper one is $p\mathbb{Z}_{(p)}$, which is the set of non-units. (iii) If $p \nmid b$ then $\bar{b} \in \mathbb{F}_p^\times$; if $a/b = a'/b'$ then $ab' = a'b$, so $\bar{a}\bar{b}^{-1} = \bar{a}'\bar{b}'^{-1}$. Additivity and multiplicativity follow from the formulas for fractions. It is onto because it restricts to $\mathbb{Z} \to \mathbb{F}_p$. Its kernel is $\{a/b : p \mid a\} = p\mathbb{Z}_{(p)}$, and units map to units. (iv) is (i) and (ii) rewritten with $v_p$. (v) A rational in lowest terms $a/b$ with $b > 1$ is not in $\mathbb{Z}_{(p)}$ for any prime $p \mid b$. □

For example, in $\mathbb{Z}_{(5)}$ the element $\frac{1}{2}$ is a unit and reduces to $\bar{2}^{-1} = 3 \in \mathbb{F}_5$ (as $2 \cdot 3 = 6 \equiv 1$); the element $\frac{10}{3} = 5 \cdot \frac{2}{3}$ has $v_5 = 1$ and reduces to 0; and $\frac{1}{5} \notin \mathbb{Z}_{(5)}$, so it has no reduction. The word *local* refers to the fact that $\mathbb{Z}_{(p)}$ only sees the prime $p$: every other prime has become a unit, and the non-units form the single ideal $p\mathbb{Z}_{(p)}$. This is the setting of every statement of the form "$x$ is an $\ell$-adic integer": it means $x \in \mathbb{Z}_{(\ell)}$, i.e. $\ell$ does not divide the reduced denominator of $x$.

*In the paper* 2.22. In Lemma lem:fpr, "$L_p(A_f,1)^{-1} = L_p(A_f(1),0)^{-1}$ is a nonzero rational number which is an $\ell$-adic integer" means: it lies in $\mathbb{Z}_{(\ell)}$ (Chapters 3 and 7 replace $\mathbb{Z}_{(\ell)}$ by $\mathbb{Z}_\ell$, and $\mathbb{Z}_{(\ell)} = \mathbb{Q} \cap \mathbb{Z}_\ell$). Indeed $L_p(A_f,1)^{-1} = P_p(p^{-2})$ with $P_p \in \mathbb{Z}[X]$, a rational whose denominator is a power of $p \neq \ell$, and $p$ is a unit of $\mathbb{Z}_{(\ell)}$ by Proposition 2.21(i); see Lemma 2.86. The same remark gives sense to Definition def:terms ($q$ is an $\ell$-adic unit, so $v_\ell(1 - a_q^2 q^{-2})$ and $v_\ell(P_q(q^{-2}))$ are valuations of elements of $\mathbb{Z}_{(\ell)}$). In Lemma lem:tamfactor, the class $c(u)$ of the reduction of $u$ is computed from the image of the unit $u = q_E q^{-n} \in \mathbb{Z}_q^\times$ under the reduction map, of which Proposition 2.21(iii) is the rational version; Chapter 7 gives the reduction $\mathbb{Z}_q \to \mathbb{F}_q$.

## 2.3 Euclidean domains, PIDs and UFDs; the rings $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$

This section proves unique factorisation where the book needs it: in $\mathbb{Z}$ (the $\ell$-part of an integer, used in every statement of the paper about $v_\ell(\deg \varphi_E)$), in $F[x]$ (minimal polynomials, finite fields), and in the rings $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ of Gaussian and Eisenstein integers, whose characters $\chi_{-4}$ and $\chi_{-3}$ appear in the table of Section 1 and whose $\ell$-adic versions $\mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ (a product of two copies of $\mathbb{Z}_\ell$, so not a domain, when $\ell \equiv 1 \pmod 3$) and $\mathbb{Z}_3[\zeta_3]$ appear in the proofs of Lemmas lem:euler and lem:IV. The primes of $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$, the element $1 - \zeta_3$ as a uniformiser and the ring $\mathbb{Z}_3[\zeta_3]$ are treated in Chapter 6.

**Definition 2.23.** Let $R$ be a domain. For $a, b \in R$, $a \mid b$ if $b = ac$ for some $c$; $a$ and $b$ are *associates* if $b = ua$ with $u \in R^\times$. A nonzero non-unit $a$ is *irreducible* if $a = bc$ implies that $b$ or $c$ is a unit, and *prime* if $a \mid bc$ implies $a \mid b$ or $a \mid c$. A *greatest common divisor* of $a$ and $b$ is a common divisor divisible by every common divisor (unique up to units if it exists). $R$ is

- a *Euclidean domain* if there is $N\colon R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ such that for all $a$ and all $b \neq 0$ there are $\kappa, \rho \in R$ with $a = \kappa b + \rho$ and $\rho = 0$ or $N(\rho) < N(b)$;
- a *principal ideal domain* (PID) if every ideal is principal;
- a *unique factorisation domain* (UFD) if every nonzero non-unit is a product of irreducibles, uniquely up to order and associates.

A ring is *Noetherian* if every ascending chain of ideals $I_1 \subseteq I_2 \subseteq \cdots$ is eventually constant.

**Lemma 2.24.** *Let $R$ be a domain.*

*(i) A prime element is irreducible.*

*(ii) For $a \neq 0$, the ideal $(a)$ is prime iff $a$ is a prime element.*

*(iii) If $p_1 \cdots p_r = u q_1 \cdots q_s$ with $p_i$, $q_j$ prime elements and $u$ a unit, then $r = s$ and, after renumbering, $q_i$ is an associate of $p_i$ for every $i$.*

*Proof.* (i) If $p = bc$ is prime then $p \mid bc$, say $p \mid b$, $b = pt$; then $p = ptc$, so $tc = 1$ and $c$ is a unit. (ii) $ab \in (a')$ means $a' \mid ab$; $(a') \neq R$ means $a'$ is not a unit. So the definitions coincide. (iii) Induction on $r$. If $r = 0$ the right side is a unit, so $s = 0$ (a prime is not a unit). Otherwise $p_1$ divides $u q_1 \cdots q_s$, hence some $q_j$ (it does not divide $u$); $q_j = p_1 t$ with $t$ a unit by (i). Renumber so $j = 1$, cancel $p_1$ (in a domain $p_1 x = p_1 y$ gives $x = y$), and apply induction to $p_2 \cdots p_r = (ut) q_2 \cdots q_s$. □

**Theorem 2.25.** *(i) A Euclidean domain is a PID.*

*(ii) A PID is Noetherian and a UFD. In a PID:* $\gcd(a,b)$ *exists, with* $(a,b)=(\gcd(a,b))$ *(Bezout's identity); irreducible elements are prime; nonzero prime ideals are maximal.*

*(iii) In a UFD irreducible elements are prime, and greatest common divisors exist.*

*(iv)* $\mathbb{Z}$ *(with* $N(a)=|a|$*) and* $F[x]$ *for a field* $F$ *(with* $N=\deg$*, Proposition 2.30) are Euclidean.* $\mathbb{Z}[x]$ *and* $F[x,y]$ *are UFDs (Theorem 2.32) but not PIDs:* $(2,x)\subset\mathbb{Z}[x]$ *and* $(x,y)\subset F[x,y]$ *are not principal.*

*Proof.* (i) Let $I\neq 0$ be an ideal and $b\in I\setminus\{0\}$ with $N(b)$ minimal. For $a\in I$ write $a=\kappa b+\rho$; then $\rho=a-\kappa b\in I$, so $\rho=0$ by minimality. Hence $I=(b)$. (ii) Let $I_1\subseteq I_2\subseteq\cdots$ be ideals of a PID. The union is an ideal, $(a)$ say; $a$ lies in some $I_n$, and then $I_m=(a)$ for $m\geq n$. For $a,b$ write $(a,b)=(d)$; $d$ divides $a$ and $b$, and $d=ua+vb$ is divisible by every common divisor, so $d=\gcd(a,b)$. Let $p$ be irreducible with $p\mid ab$ and $p\nmid a$. Then $\gcd(p,a)$ divides $p$, so it is a unit or an associate of $p$; the latter would give $p\mid a$. So $1=up+va$ and $b=upb+vab$ is divisible by $p$: $p$ is prime. If $(p)\neq 0$ is prime and $(p)\subseteq(m)\neq R$, then $p=mt$, and as $p$ is irreducible (Lemma 2.24) and $m$ is not a unit, $t$ is a unit and $(m)=(p)$; so $(p)$ is maximal. Existence of factorisations: suppose a nonzero non-unit $a_0$ is not a product of irreducibles. Then $a_0$ is not irreducible, so $a_0=bc$ with $b,c$ non-units, and one of them, $a_1$ say, is not a product of irreducibles; $(a_0)\subsetneq(a_1)$ because the cofactor is not a unit. Repeating gives a strictly increasing chain $(a_0)\subsetneq(a_1)\subsetneq\cdots$, which is impossible. Uniqueness follows from Lemma 2.24(iii), since irreducibles are prime. (iii) Let $p$ be irreducible with $p\mid ab$, $ab=pc$. Factoring $a$, $b$, $c$ into irreducibles, the two factorisations (factors of $a$)(factors of $b$) and $p$(factors of $c$) of $ab$ agree up to order and units, so $p$ is an associate of an irreducible factor of $a$ or of $b$. For gcds, fix one irreducible $p$ in each class of associates, write $a=u\prod p^{e_p(a)}$, $b=u'\prod p^{e_p(b)}$, and take $\prod p^{\min(e_p(a),e_p(b))}$: by uniqueness, $d\mid a$ iff $e_p(d)\leq e_p(a)$ for all $p$. (iv) Division with remainder in $\mathbb{Z}$ is $a=\kappa b+\rho$ with $0\leq\rho<|b|$; in $F[x]$ it is Proposition 2.30(i), as every nonzero polynomial over a field is a unit times a monic one. If $(2,x)=(d)$ in $\mathbb{Z}[x]$, then $d\mid 2$ forces $\deg d=0$, and $d\mid x$ forces $d=\pm1$; but then $1=2g+xh$, and evaluating at $x=0$ gives $1=2g(0)$ in $\mathbb{Z}$, impossible. If $(x,y)=(d)$ in $F[x,y]$, then $d$ divides $x$ and $y$, so it is a nonzero constant, and $1=xg+yh$ evaluated at $(0,0)$ gives $1=0$. □

**Proposition 2.26.** *(i) (Bezout) For* $m,n\in\mathbb{Z}$ *there are* $u,v\in\mathbb{Z}$ *with* $um+vn=\gcd(m,n)$.

*(ii) (Euclid's lemma) If a prime* $p$ *divides* $ab$*, it divides* $a$ *or* $b$.

*(iii) (Fundamental theorem of arithmetic) Every* $n\neq 0$ *is* $n=\pm\prod_p p^{v_p(n)}$ *with unique exponents* $v_p(n)\geq 0$*, almost all zero.*

*(iv)* $d\mid n$ *iff* $v_p(d)\leq v_p(n)$ *for all* $p$*; hence* $\gcd(m,n)=\prod_p p^{\min(v_p(m),v_p(n))}$ *and* $\operatorname{lcm}(m,n)=\prod_p p^{\max(v_p(m),v_p(n))}$ *for* $m,n\neq 0$*. The* $p$-part *of* $n\neq 0$ *is* $p^{v_p(n)}$.

*Proof.* $\mathbb{Z}$ is Euclidean, hence a PID (Theorem 2.25). (i) is Bezout's identity there (or Proposition 2.11(iii)); (ii) says that irreducible elements of $\mathbb{Z}$, the $\pm p$, are prime; (iii) is unique factorisation, the units being $\pm1$ and the positive primes a system of representatives of the irreducibles. (iv) follows from (iii) as in the proof of Theorem 2.25(iii). □

**Theorem 2.27.** *Let* $R$ *be* $\mathbb{Z}[i]=\{a+bi:a,b\in\mathbb{Z}\}$ *or* $\mathbb{Z}[\zeta_3]=\{a+b\zeta_3:a,b\in\mathbb{Z}\}$*, subrings of* $\mathbb{C}$ *(note* $\zeta_3^2=-1-\zeta_3$*), and* $N(z)=z\bar{z}=|z|^2$*. Then* $N(a+bi)=a^2+b^2$ *and* $N(a+b\zeta_3)=a^2-ab+b^2$ *are nonnegative integers, zero only at* $0$*, and* $N(zw)=N(z)N(w)$.

*(i)* *For $\alpha, \beta \in R$, $\beta \neq 0$, there is $\kappa \in R$ with $N(\alpha - \kappa\beta) \leq \frac{1}{2}N(\beta)$ if $R = \mathbb{Z}[i]$, and $\leq \frac{1}{3}N(\beta)$ if $R = \mathbb{Z}[\zeta_3]$. Hence both rings are Euclidean for $N$, so PIDs and UFDs.*

*(ii)* *$z \in R$ is a unit iff $N(z) = 1$; $\mathbb{Z}[i]^\times = \{\pm 1, \pm i\} = \mu_4$ and $\mathbb{Z}[\zeta_3]^\times = \{\pm 1, \pm\zeta_3, \pm\zeta_3^2\} = \mu_6$.*

*(iii)* *An element of prime norm is irreducible, hence prime.*

*(iv)* *$2 = -i(1+i)^2$, $5 = (2+i)(2-i)$, $3 = -\zeta_3^2(1-\zeta_3)^2$, $7 = (3+\zeta_3)(3+\zeta_3^2)$, $13 = (4+\zeta_3)(4+\zeta_3^2)$, the factors being primes.*

*Proof.* Both sets are closed under multiplication ($i^2 = -1$; $\zeta_3^2 = -1 - \zeta_3$ because $\zeta_3 \neq 1$ is a root of $x^3 - 1 = (x-1)(x^2+x+1)$) and under complex conjugation ($\bar{\zeta_3} = \zeta_3^2$). $N(a + b\zeta_3) = a^2 + ab(\zeta_3 + \bar{\zeta_3}) + b^2 = a^2 - ab + b^2$. Multiplicativity is that of $|\cdot|^2$. (i) Put $\gamma = \alpha/\beta \in \mathbb{C}$. It suffices to find $\kappa \in R$ with $|\gamma - \kappa|^2 \leq \frac{1}{2}$, resp. $\frac{1}{3}$, and then $N(\alpha - \kappa\beta) = |\gamma - \kappa|^2 N(\beta)$. In the first case $R$ is the square lattice, and $\mathbb{C}$ is tiled by the closed unit squares with vertices in $R$. In the second, the parallelograms with vertices $\kappa, \kappa+1, \kappa+\zeta_3, \kappa+1+\zeta_3$ ($\kappa \in R$) tile $\mathbb{C}$, and the diagonal from $\kappa$ to $\kappa + 1 + \zeta_3 = \kappa - \zeta_3^2$ cuts each into two equilateral triangles of side 1 (all sides have length $|1| = |\zeta_3| = |1 + \zeta_3| = 1$). So $\gamma$ lies in a square of side 1, resp. an equilateral triangle of side 1, with vertices in $R$. Join the centre $c$ (of the square, resp. the centroid of the triangle) to the midpoints of the sides: this cuts the polygon into convex quadrilaterals, one at each vertex $w$, with corners $w$, two half-sides and $c$. The distance to $w$ is a convex function, so on such a quadrilateral it is maximal at a corner; the corners are at distance 0, $\frac{1}{2}$, $\frac{1}{2}$ and $|c - w|$, which is $\frac{\sqrt{2}}{2}$ for the square and the circumradius $\frac{1}{\sqrt{3}}$ for the triangle. Hence $|\gamma - w|^2 \leq \frac{1}{2}$, resp. $\frac{1}{3}$, for the vertex $w$ of the piece containing $\gamma$; take $\kappa = w$. The bound is attained at the centre, resp. the centroid. By Theorem 2.25, both rings are PIDs and UFDs. (ii) If $zw = 1$ then $N(z)N(w) = 1$ with both positive integers, so $N(z) = 1$; conversely $N(z) = 1$ gives $z \cdot \bar{z} = 1$ with $\bar{z} \in R$. Now $a^2 + b^2 = 1$ has the solutions $(\pm 1, 0)$, $(0, \pm 1)$; and $a^2 - ab + b^2 = 1$, i.e. $(2a - b)^2 + 3b^2 = 4$, forces $b \in \{0, \pm 1\}$ and gives $(a, b) = (\pm 1, 0)$, $(0, 1)$, $(1, 1)$, $(0, -1)$, $(-1, -1)$, that is $\pm 1$, $\zeta_3$, $1 + \zeta_3 = -\zeta_3^2$, $-\zeta_3$, $-1 - \zeta_3 = \zeta_3^2$. These are the six sixth roots of unity, $-\zeta_3$ having order 6. (iii) If $z = xy$ with $N(z)$ prime, one of $N(x)$, $N(y)$ is 1. In a PID irreducibles are prime. (iv) Direct multiplication: $(1 + i)^2 = 2i$; $(1 - \zeta_3)^2 = 1 - 2\zeta_3 + \zeta_3^2 = -3\zeta_3$, so $-\zeta_3^2(1 - \zeta_3)^2 = 3\zeta_3^3 = 3$; $(3 + \zeta_3)(3 + \zeta_3^2) = 9 + 3(\zeta_3 + \zeta_3^2) + 1 = 7$; $(4 + \zeta_3)(4 + \zeta_3^2) = 16 - 4 + 1 = 13$. The norms of the factors are $2, 5, 5, 3, 7, 7, 13, 13$, prime, so the factors are prime by (iii). □

**Example 2.28** (the Euclidean algorithm in $\mathbb{Z}[i]$)**.** To compute $\gcd(11 + 3i, 1 + 8i)$ (norms 130 and 65): $\frac{11+3i}{1+8i} = \frac{(11+3i)(1-8i)}{65} = \frac{35-85i}{65}$, nearest Gaussian integer $1 - i$, and $11 + 3i = (1 - i)(1 + 8i) + (2 - 4i)$ with $N(2 - 4i) = 20 < 65$. Next $\frac{1+8i}{2-4i} = \frac{(1+8i)(2+4i)}{20} = \frac{-30+20i}{20}$, take $-2 + i$: $1 + 8i = (-2 + i)(2 - 4i) + (1 - 2i)$, $N(1 - 2i) = 5 < 20$. Finally $2 - 4i = 2(1 - 2i)$. So $\gcd(11 + 3i, 1 + 8i) = 1 - 2i$, of norm 5, up to units. Although 13 divides both norms, the factorisations $11 + 3i = (1 - 2i)(1 + i)(3 + 2i)$ and $1 + 8i = -(1 - 2i)(3 - 2i)$ show that different primes of norm 13 occur: $13 = (3 + 2i)(3 - 2i)$, and $3 - 2i$ is not an associate of $3 + 2i$.

**Example 2.29** (unique factorisation fails in general)**.** In $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5}\}$, with $N(a + b\sqrt{-5}) = a^2 + 5b^2$, one has $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. The four factors have norms 4, 9, 6, 6; no element has norm 2 or 3, so each factor is irreducible; the units are $\pm 1$ (norm 1), so no two of the factors are associates. Thus $\mathbb{Z}[\sqrt{-5}]$ is not a UFD, hence not a PID; explicitly, $(2, 1 + \sqrt{-5})$ is not principal: it lies in the kernel of the homomorphism $\mathbb{Z}[\sqrt{-5}] \to \mathbb{F}_2$, $a + b\sqrt{-5} \mapsto a + b \bmod 2$ (well defined since $\mathbb{Z}[\sqrt{-5}] \simeq \mathbb{Z}[x]/(x^2 + 5)$ and $1^2 + 5 = 0$ in $\mathbb{F}_2$), so it is proper, while a generator

$d$ would have $N(d) \mid \gcd(4,6) = 2$, so $N(d) = 1$ and the ideal would be everything. Similarly $4 = 2 \cdot 2 = (1+\sqrt{-3})(1-\sqrt{-3})$ in $\mathbb{Z}[\sqrt{-3}] \subsetneq \mathbb{Z}[\zeta_3]$, with three non-associate irreducibles of norm 4; in $\mathbb{Z}[\zeta_3]$ the anomaly disappears because $1+\sqrt{-3} = 2+2\zeta_3 = -2\zeta_3^2$ is an associate of 2. The ideal-theoretic repair is Chapter 6.

## 2.4 Polynomial rings and irreducibility

Polynomials enter the paper as characteristic polynomials of Frobenius ($X^2 - a_p X + p$), as the Euler factors $P_q(X) \in \mathbb{Z}[X]$ of Definition def:terms, as the 3-division polynomial of Proposition prop:ns, and, behind the scenes, through finite fields and cyclotomic polynomials. This section proves division with remainder, the criterion for repeated roots, Gauss's lemma, the criteria of Eisenstein and of reduction modulo $p$, and Kronecker's construction of a field in which a given polynomial has a root. The input that a polynomial of degree $d$ over a field has at most $d$ roots (Proposition 2.30(iv)) is what Chapter 1 uses to prove that finite subgroups of $K^\times$ are cyclic.

**Proposition 2.30.** *Let $R$ be a ring and $R[x]$ the polynomial ring;* $\deg 0 = -\infty$.

*(i) (Division by a monic polynomial.) If $g \in R[x]$ is monic of degree $m$, every $f \in R[x]$ is uniquely $f = hg + r$ with* $\deg r < m$.

*(ii) If $R$ is a domain,* $\deg fg = \deg f + \deg g$, *$R[x]$ is a domain and $R[x]^\times = R^\times$. If $F$ is a field, $F[x]$ is Euclidean for the degree, hence a PID and a UFD, and its maximal ideals are the $(f)$ with $f$ irreducible.*

*(iii) (Factor theorem.) For $a \in R$, $f(a) = 0$ iff $(x-a) \mid f$.*

*(iv) A nonzero polynomial of degree $d$ over a domain $R$ has at most $d$ roots in $R$.*

*(v) $x^p - x \in \mathbb{F}_p[x]$ is nonzero but vanishes at every point of $\mathbb{F}_p$; over an infinite domain a polynomial is determined by its values.*

*Proof.* (i) Existence by induction on $\deg f$: if $\deg f = n \geq m$ with leading coefficient $a$, then $f - ax^{n-m}g$ has degree $< n$. Uniqueness: if $hg + r = h'g + r'$ with $h \neq h'$, then $(h-h')g = r' - r$; as $g$ is monic, the left side has degree $\deg(h-h') + m \geq m > \deg(r'-r)$. (ii) The leading coefficient of $fg$ is the product of those of $f$ and $g$, nonzero in a domain. If $fg = 1$ then $\deg f = \deg g = 0$. Over a field every nonzero $g$ is a unit times a monic polynomial, so (i) gives the Euclidean property; then Theorem 2.25 applies. A nonzero prime ideal of a PID is maximal, $(f)$ is prime iff $f$ is irreducible (Lemma 2.24, Theorem 2.25(ii)), and 0 is not maximal since $F[x]$ is not a field. (iii) Divide by the monic $x - a$: $f = h \cdot (x-a) + r$ with $r$ constant; evaluating at $a$ gives $r = f(a)$. (iv) Induction on $d$. If $f(a) = 0$ then $f = (x-a)h$ with $\deg h = d - 1$, and a root $b \neq a$ of $f$ satisfies $(b-a)h(b) = 0$, so $h(b) = 0$. (v) $a^p = a$ for $a \in \mathbb{F}_p$ (Proposition 2.16(iii)). Over an infinite domain, two polynomials with the same values differ by a polynomial with infinitely many roots, which is 0 by (iv). □

**Proposition 2.31.** *The formal derivative of $f = \sum a_i x^i$ is $f' = \sum i a_i x^{i-1}$; it is additive and $(fg)' = f'g + fg'$. Let $F$ be a field, $f \in F[x]$ nonzero and $K \supseteq F$ a field.*

*(i) $a \in K$ is a multiple root of $f$ (that is, $(x-a)^2 \mid f$ in $K[x]$) iff $f(a) = f'(a) = 0$.*

*(ii) The gcd of $f$ and $f'$ is the same in $F[x]$ and in $K[x]$; $f$ has no repeated root in any extension field iff* $\gcd(f, f') = 1$ *(we then call $f$* separable*).*

*(iii) In characteristic $p$, $x^{p^r}-x$ is separable. For $n\geq 1$, $x^n-1$ is separable over $F$ iff* $\operatorname{char} F\nmid n$.

*Proof.* The Leibniz rule holds for monomials, $(x^{m+n})'=(m+n)x^{m+n-1}=mx^{m-1}x^n+x^m nx^{n-1}$, and extends by bilinearity. (i) If $f(a)=0$ write $f=(x-a)g$; then $f'=g+(x-a)g'$ and $f'(a)=g(a)$, which vanishes iff $(x-a)\mid g$. (ii) Let $d=\gcd(f,f')$ in $F[x]$, $d=uf+vf'$. Then $d$ divides $f$, $f'$ in $K[x]$, and every common divisor in $K[x]$ divides $d$; so $d$ is also the gcd in $K[x]$. If $f$ has a multiple root $a$ in some $K$, then $x-a$ divides $f$ and $f'$ in $K[x]$, so $d\neq 1$. If $d$ is nonconstant, it has a root $a$ in some extension $K$ (Theorem 2.38(iii)), and $f(a)=f'(a)=0$. (iii) $(x^{p^r}-x)'=p^r x^{p^r-1}-1=-1$. If $p=\operatorname{char} F\nmid n$, a common root of $x^n-1$ and $nx^{n-1}$ would be 0, which is not a root of $x^n-1$. If $p\mid n$, the derivative is 0 and $x^n-1=(x^{n/p}-1)^p$ by Proposition 2.16. □

**Theorem 2.32** (Gauss's lemma)**.** *Let $R$ be a UFD with fraction field $K$. The* content *$c(f)$ of a nonzero $f\in R[x]$ is a gcd of its coefficients (defined up to units), and $f$ is* primitive *if $c(f)$ is a unit. Every nonzero $g\in K[x]$ is $g=cg_1$ with $c\in K^\times$ and $g_1\in R[x]$ primitive, and $c$ is unique up to $R^\times$.*

*(i) $c(fg)=c(f)c(g)$ up to units; a product of primitive polynomials is primitive.*

*(ii) If $f\in R[x]$ factors as $f=gh$ in $K[x]$, then $f=g_1h_1$ in $R[x]$ with $g_1\in K^\times g$ and $h_1\in K^\times h$. Hence a primitive $f$ of positive degree is irreducible in $R[x]$ iff it is irreducible in $K[x]$.*

*(iii) $R[x]$ is a UFD. Its prime elements are the primes of $R$ and the primitive polynomials of positive degree irreducible in $K[x]$. Consequently $\mathbb{Z}[x_1,\dots,x_n]$ and $F[x_1,\dots,x_n]$ ($F$ a field) are UFDs.*

*(iv) A monic $f\in\mathbb{Z}[x]$ which is a product of monic polynomials in $\mathbb{Q}[x]$ has those factors in $\mathbb{Z}[x]$. If $a/b$ in lowest terms is a root of $a_nx^n+\cdots+a_0\in\mathbb{Z}[x]$, then $b\mid a_n$ and $a\mid a_0$; a rational root of a monic integer polynomial is an integer.*

*Proof.* For $g\in K[x]$, multiply by a common denominator $b\in R$ and write $bg=c(bg)g_1$; then $g=(c(bg)/b)g_1$. If $cg_1=c'g_1'$ with $g_1,g_1'$ primitive, write $c/c'=a/b$ with $a,b\in R$; then $ag_1=bg_1'$, and taking contents $a=b\cdot$unit, so $c/c'\in R^\times$. (i) Suppose $f$, $g$ are primitive and $fg$ is not. Some irreducible, hence prime (Theorem 2.25(iii)), $p\in R$ divides every coefficient of $fg$. Reduce modulo $p$: $R/p$ is a domain (Lemma 2.24(ii)), so $(R/p)[x]$ is a domain, and $\bar f\bar g=0$ with $\bar f\neq 0\neq\bar g$, a contradiction. In general write $f=c(f)f_1$, $g=c(g)g_1$ with $f_1,g_1$ primitive; then $fg=c(f)c(g)f_1g_1$ with $f_1g_1$ primitive, so the gcd of the coefficients of $fg$ is $c(f)c(g)$ times the gcd of those of $f_1g_1$, a unit: $c(fg)=c(f)c(g)$ up to units. (For instance $(2x+2)(3x+3)=6x^2+12x+6$ has content $6=2\cdot 3$.) (ii) Write $g=cg_1$, $h=c'h_1$ with $g_1,h_1$ primitive. Then $f=(cc')g_1h_1$ with $g_1h_1$ primitive by (i), and also $f=c(f)f_1$; by uniqueness, $cc'=c(f)u$ with $u\in R^\times$, so $cc'\in R$. Put $g_1':=cc'g_1\in R[x]$; then $f=g_1'h_1$. For the last claim: if $f$ is primitive and $f=gh$ in $R[x]$ with $g$ a non-unit, then $g$ is not a constant (a constant factor divides $c(f)$, a unit), so a factorisation in $R[x]$ into non-units has both factors of positive degree; and a factorisation in $K[x]$ into factors of positive degree gives one in $R[x]$ with the same degrees. (iii) Existence: write $f=c(f)f_1$ and factor $c(f)$ in $R$. If the primitive $f_1$ is not irreducible, $f_1=gh$ with $g,h$ non-units, necessarily of positive degree as just seen and primitive (as $c(g)c(h)=c(f_1)$ is a unit); induction on the degree. Primes: if $p$ is prime in $R$, then $R[x]/pR[x]\simeq(R/p)[x]$ is a domain, so $p$ is prime in $R[x]$. If $f$ is primitive of positive degree and irreducible in $R[x]$, it is irreducible in $K[x]$ by (ii), hence prime there ($K[x]$ is a PID). If $f\mid gh$

in $R[x]$, then $f$ divides $g$, say, in $K[x]$: $g = fk$ with $k = ck_1$, $k_1$ primitive; by (i) $fk_1$ is primitive, so $c$ equals $c(g)$ up to a unit and lies in $R$; hence $k \in R[x]$ and $f \mid g$ in $R[x]$. These two kinds exhaust the irreducibles of $R[x]$: an irreducible of degree 0 is an irreducible element of $R$ (a factorisation in $R$ is one in $R[x]$, and $R[x]^\times = R^\times$), hence prime in $R$; an irreducible $f$ of positive degree is primitive, since otherwise $f = c(f)f_1$ would be a factorisation into two non-units. So every irreducible is prime and uniqueness follows from Lemma 2.24(iii). The polynomial rings in several variables follow by induction, $R[x_1, \dots, x_n] = R[x_1, \dots, x_{n-1}][x_n]$. (iv) By (ii), $f = g_1 h_1$ in $\mathbb{Z}[x]$ with $g_1 = \lambda g$, $h_1 = \lambda' h$; comparing leading coefficients, $\lambda\lambda' = 1$ with $\lambda, \lambda' \in \mathbb{Z}$ the leading coefficients of $g_1, h_1$, so $\lambda = \pm 1$ and $g = \pm g_1 \in \mathbb{Z}[x]$. If $f(a/b) = 0$ then $\sum a_i a^i b^{n-i} = 0$, so $b \mid a_n a^n$ and $a \mid a_0 b^n$; since $\gcd(a, b) = 1$, Euclid's lemma gives $b \mid a_n$ and $a \mid a_0$. □

**Theorem 2.33** (Eisenstein's criterion)**.** *Let $R$ be a domain, $P$ a prime ideal and $f = a_n x^n + \cdots + a_0 \in R[x]$, $n \geq 1$, with $a_n \notin P$, $a_i \in P$ for $i < n$, and $a_0 \notin P^2$. Then $f$ is not a product of two polynomials of positive degree in $R[x]$. If $R$ is a UFD with fraction field $K$ and $f$ is primitive, then $f$ is irreducible in $R[x]$ and in $K[x]$. Examples: $x^n - p$; $x^2 + 3x + 3 = \Phi_3(x + 1)$; $\Phi_p(x + 1) = ((x + 1)^p - 1)/x$ and $\Phi_{p^k}(x + 1)$ (Theorem 2.58(vi)).*

*Proof.* Suppose $f = gh$ with $\deg g = r \geq 1$, $\deg h = s \geq 1$, $r + s = n$. Let $k$ be the fraction field of the domain $R/P$ and reduce modulo $P$: $\bar{g}\bar{h} = \bar{a}_n x^n$ in $k[x]$ with $\bar{a}_n \neq 0$. Then $\deg \bar{g} = r$ and $\deg \bar{h} = s$ (their degrees are at most $r$, $s$ and add up to $n$). In the UFD $k[x]$ the prime $x$ and units are the only factors of $\bar{a}_n x^n$, so $\bar{g} = bx^r$ and $\bar{h} = cx^s$. As $r, s \geq 1$, the constant terms $g(0)$, $h(0)$ lie in $P$, and $a_0 = g(0)h(0) \in P^2$, a contradiction. If $R$ is a UFD and $f$ is primitive, a factorisation $f = gh$ into non-units has both factors of positive degree (a constant non-unit factor would divide $c(f)$), so $f$ is irreducible in $R[x]$, and in $K[x]$ by Theorem 2.32(ii). The examples: $x^n - p$ at $P = p\mathbb{Z}$; $\Phi_p(x + 1) = \sum_{i=1}^{p} \binom{p}{i} x^{i-1}$ has leading coefficient 1, the other coefficients divisible by $p$ (Proposition 2.16(i)) and constant term $\binom{p}{1} = p$. □

For $p = 5$: $\Phi_5(x + 1) = \frac{(x+1)^5 - 1}{x} = x^4 + 5x^3 + 10x^2 + 10x + 5$, Eisenstein at 5. The condition $a_0 \notin P^2$ cannot be dropped: $x^2 - 4 = (x - 2)(x + 2)$ has all lower coefficients in $2\mathbb{Z}$ but $a_0 = -4 \in 4\mathbb{Z}$. The substitution $x \mapsto x + 1$ is often needed: $\Phi_5$ itself has all coefficients equal to 1 and no prime divides them, and the shift replaces the roots $\zeta_5^a$ by the numbers $\zeta_5^a - 1$, which are all multiples of $\zeta_5 - 1$ in $\mathbb{Z}[\zeta_5]$, since $\zeta_5^a - 1 = (\zeta_5 - 1)(1 + \zeta_5 + \cdots + \zeta_5^{a-1})$; their product is $\Phi_5(1) = 5$ up to sign.

**Proposition 2.34** (reduction modulo $p$)**.** *Let $f \in \mathbb{Z}[x]$ have degree $n$ and let $p$ be a prime not dividing the leading coefficient. If $f = gh$ in $\mathbb{Z}[x]$, then $\bar{f} = \bar{g}\bar{h}$ in $\mathbb{F}_p[x]$ with $\deg \bar{g} = \deg g$ and $\deg \bar{h} = \deg h$. Hence:*

*(i) if $f$ is primitive and $\bar{f}$ is irreducible in $\mathbb{F}_p[x]$, then $f$ is irreducible in $\mathbb{Q}[x]$ (and in $\mathbb{Z}[x]$);*

*(ii) the degrees of the factors in any factorisation of $f$ in $\mathbb{Q}[x]$ are sums of degrees of irreducible factors of $\bar{f}$ (disjoint sub-multisets); so if two primes give incompatible patterns, $f$ is irreducible over $\mathbb{Q}$.*

*Proof.* Reduction $\mathbb{Z}[x] \to \mathbb{F}_p[x]$ is a ring homomorphism. The leading coefficients of $g$ and $h$ multiply to that of $f$, which is nonzero modulo $p$, so neither leading coefficient vanishes modulo $p$. (i) A factorisation in $\mathbb{Q}[x]$ into factors of positive degree gives one in $\mathbb{Z}[x]$ with the same degrees (Theorem 2.32(ii)), and then one of $\bar{f}$ with the same degrees. (ii) Write $f = g_1 \cdots g_k$ in $\mathbb{Q}[x]$, hence (Gauss) in $\mathbb{Z}[x]$ with the same degrees; then $\bar{f} = \bar{g}_1 \cdots \bar{g}_k$, $\deg \bar{g}_i = \deg g_i$, and by unique factorisation in $\mathbb{F}_p[x]$ the irreducible factors of $\bar{f}$ are distributed among the $\bar{g}_i$. □

*Warning* 2.35 ($x^4+1$). The test works in one direction only. $x^4+1$ is irreducible over $\mathbb{Q}$: $(x+1)^4+1 = x^4+4x^3+6x^2+4x+2$ is Eisenstein at 2. But it is reducible modulo every prime $p$. For $p=2$, $x^4+1=(x+1)^4$. For odd $p$, $8 \mid p^2-1$, so the cyclic group $\mathbb{F}_{p^2}^\times$ (Theorem 2.47 and Proposition 2.46) contains an element $\zeta$ of order 8, and $\zeta^4=-1$. If $x^4+1$ were irreducible over $\mathbb{F}_p$, then $\mathbb{F}_p[x]/(x^4+1) \simeq \mathbb{F}_p[\zeta] \subseteq \mathbb{F}_{p^2}$ (Theorem 2.38(ii)) would have $p^4 \le p^2$ elements. (Checked with PARI for all $p<500$: every factor has degree at most 2.)

**Example 2.36** (the 3-division polynomial of 11a1)**.** For $E =$ `11a1`: $y^2+y=x^3-x^2-10x-20$, the 3-division polynomial, whose roots are the $x$-coordinates of the points of order 3 (Chapter 9), is $\psi_3 = 3x^4-4x^3-60x^2-237x-21$. It is primitive, 5 does not divide its leading coefficient, and modulo 5 it is $3(x^4+2x^3+x+3)$, which is irreducible in $\mathbb{F}_5[x]$: the monic factor has no root (its values at $0,1,2,3,4$ are $3,2,2,1,1$), and it is not $(x^2+ax+b)(x^2+cx+d)$, since $bd=3$ leaves $(b,d) \in \{(1,3),(3,1),(2,4),(4,2)\}$, and in each case $a+c=2$, $ad+bc=1$ determine $a,c$ and give $b+d+ac \in \{4,4,2,2\}$, never 0. By Proposition 2.34(i), $\psi_3$ is irreducible over $\mathbb{Q}$.

```
E = ellinit([0,-1,1,-10,-20]);      \\ 11a1
f = elldivpol(E, 3)                  \\ 3*x^4 - 4*x^3 - 60*x^2 - 237*x - 21
factormod(f, 5)                      \\ a single factor x^4 + 2*x^3 + x + 3
polisirreducible(f)                  \\ 1
```

*In the paper* 2.37. Proposition prop:ns ends: "The 3-division polynomial of $E$ then factors as a product of two irreducible quadratics, and otherwise it is irreducible", and Section sec:anomaly checks that the 3-division polynomial of every violating curve (all of image 3Ns) factors as two quadratics. Proposition 2.34 is how such a statement is tested for a given curve: one prime $p$ with $\bar{\psi}_3$ irreducible proves irreducibility, as for 11a1, whose mod 3 image is $\mathrm{GL}_2(\mathbb{F}_3)$. The converse direction (that a factorisation over $\mathbb{Q}$ is detected) needs the factorisation over $\mathbb{Q}$ itself, which PARI computes; the Galois-theoretic meaning of the pattern is Chapter 5.

**Theorem 2.38** (Kronecker's construction)**.** *Let $F$ be a field and $f \in F[x]$ irreducible of degree $d$.*

*(i) $F[x]/(f)$ is a field containing (a copy of) $F$, of dimension $d$ over $F$ with basis $1, \bar{x}, \dots, \bar{x}^{d-1}$, and $f(\bar{x})=0$.*

*(ii) If $\alpha$ is a root of $f$ in a field $K \supseteq F$, then $x \mapsto \alpha$ induces an isomorphism $F[x]/(f) \simeq F[\alpha]$; in particular $F[\alpha]$ is a field of dimension $d$ over $F$.*

*(iii) For every nonconstant $g \in F[x]$ there is a field $K \supseteq F$, of finite dimension over $F$, in which $g$ is a product of linear factors.*

*Proof.* (i) $(f)$ is maximal (Proposition 2.30(ii)), so the quotient is a field. $F \to F[x]/(f)$ is injective, as a nonzero constant is not divisible by $f$. Division by $f$ (made monic) shows that $1, \bar{x}, \dots, \bar{x}^{d-1}$ span, and they are independent because a nonzero polynomial of degree $< d$ is not divisible by $f$. Clearly $f(\bar{x}) = \overline{f(x)} = 0$. (ii) Evaluation $F[x] \to K$, $x \mapsto \alpha$, has image $F[\alpha]$ and a kernel containing $f$, not equal to $F[x]$; as $(f)$ is maximal, the kernel is $(f)$. Apply Theorem 2.12(i). (iii) Induction on $\deg g$. Let $f$ be an irreducible factor of $g$ and $K_1 = F[x]/(f)$, in which $f$, hence $g$, has a root $\alpha_1$; then $g=(x-\alpha_1)g_1$ in $K_1[x]$ (Proposition 2.30(iii)). By induction $g_1$ splits in some $K \supseteq K_1$ with $\dim_{K_1} K$ finite, and $\dim_F K = \dim_{K_1} K \cdot \dim_F K_1$ (a basis of $K$ over $F$ is given by the products of bases). □

In (i) the inverse of a nonzero class is computed by Bezout's identity: if $g \notin (f)$ then $\gcd(f,g)=1$ ($f$ is irreducible), so $uf+vg=1$ for some $u,v \in F[x]$, and $\bar v = \bar g^{-1}$. For $F=\mathbb{Q}$ and $f=x^2-2$ this gives the familiar formula: in $\mathbb{Q}[x]/(x^2-2) \simeq \mathbb{Q}(\sqrt{2})$, with $t=\bar x$,

$$(a+bt)(a-bt)=a^2-2b^2, \qquad \text{so} \qquad (a+bt)^{-1}=\frac{a-bt}{a^2-2b^2},$$

and $a^2-2b^2 \neq 0$ for rational $(a,b)\neq(0,0)$ precisely because $x^2-2$ has no rational root. The point of the construction is that it needs no ambient field such as $\mathbb{C}$: over $\mathbb{F}_p$, where there is no such field to start from, it is the only way to produce a root.

**Example 2.39** ($\mathbb{F}_9$ and the paper's matrix $J$)**.** $x^2+1$ has no root in $\mathbb{F}_3$ (its values at $0,1,2$ are $1,2,2$), so it is irreducible and $\mathbb{F}_9 := \mathbb{F}_3[x]/(x^2+1) = \mathbb{F}_3[i]$, $i=\bar x$, is a field with 9 elements (by Theorem 2.47 every field with 9 elements is isomorphic to it). In the basis $1, i$ of $\mathbb{F}_9$ over $\mathbb{F}_3$, multiplication by $-i$ sends $1 \mapsto -i$ and $i \mapsto 1$, so its matrix is the paper's $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ of Proposition prop:ns: $J = m_{-i}$, as in the standard models of Chapter 1. Its characteristic polynomial $X^2+1$ is irreducible over $\mathbb{F}_3$, so $J$ has no eigenvalue in $\mathbb{F}_3$ and the eigenvalues $\pm i$ in $\mathbb{F}_9$. This is why the character $\psi$ of order 4 in Proposition prop:ns takes values in $\mathbb{F}_9^\times$ and why the paper writes $\bar\rho_{E,3} \otimes \mathbb{F}_9 \simeq \operatorname{Ind}\psi$ (over $\mathbb{F}_3$ itself $\bar\rho_{E,3}$ is irreducible). More generally $\mathbb{F}_q[x]/(x^2-c)$, $c$ a non-square, is the field $\mathbb{F}_q[\sqrt{c}]$ with $q^2$ elements used in Chapter 1.

## 2.5 The ring $\mathbb{Z}/n\mathbb{Z}$, the Chinese remainder theorem, units and primitive roots

The groups $(\mathbb{Z}/n\mathbb{Z})^\times$ are where Dirichlet characters live (Section 2.10), and the groups $(\mathbb{Z}/\ell^k\mathbb{Z})^\times = \mathbb{Z}_\ell^\times/(1+\ell^k\mathbb{Z}_\ell)$ ($k \geq 1$) are the finite levels of $\mathbb{Z}_\ell^\times$, the image of the cyclotomic character. The paper uses one fact about them in an essential way: in Lemma lem:h0, an isomorphism $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$ forces $\chi_\ell^2 \equiv 1 \pmod{\ell^k}$, and this is possible for $k \geq 1$ only when $\ell^k = 3$ (Corollary 2.44). The lifting-the-exponent lemma computes valuations such as $v_\ell(q^2-1)$ in the table of Section 1.

**Proposition 2.40.** *Let $n \geq 1$.*

*(i) $a \bmod n$ is a unit of $\mathbb{Z}/n\mathbb{Z}$ iff $\gcd(a,n)=1$. For $n \geq 2$, $\mathbb{Z}/n\mathbb{Z}$ is a field iff it is a domain iff $n$ is prime; we write $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.*

*(ii) $\#(\mathbb{Z}/n\mathbb{Z})^\times = \varphi(n)$, the number of $1 \leq a \leq n$ prime to $n$, and $a^{\varphi(n)} \equiv 1 \pmod n$ for $\gcd(a,n)=1$ (Euler). Moreover $\varphi(p^k) = p^{k-1}(p-1)$ and $\sum_{d\mid n}\varphi(d) = n$.*

*(iii) The map $a \mapsto (x \mapsto ax)$ is an isomorphism $(\mathbb{Z}/n\mathbb{Z})^\times \simeq \operatorname{Aut}(\mathbb{Z}/n\mathbb{Z})$ (automorphisms of the additive group).*

*For $k \geq 2$, $\mathbb{Z}/p^k\mathbb{Z}$ is not a field ($p$ is nilpotent in it), and in particular it is not $\mathbb{F}_{p^k}$.*

*Proof.* (i) If $\gcd(a,n)=1$, Bezout gives $ua+vn=1$, so $ua \equiv 1$. If $d=\gcd(a,n)>1$ then $a\cdot(n/d) \equiv 0$ with $n/d \not\equiv 0$, so $a$ is a zero divisor and not a unit. If $n=p$ is prime, every nonzero class is a unit; if $n=ab$ with $1<a,b<n$, then $\bar a\bar b = 0$ and $\mathbb{Z}/n$ is not a domain. A field is a domain. (ii) The first claim is (i); Euler's congruence is Lagrange's theorem in the group $(\mathbb{Z}/n)^\times$ (Chapter 1). Among $1,\dots,p^k$ exactly the $p^{k-1}$ multiples of $p$ are not prime to $p^k$. For

the sum, $\mathbb{Z}/n$ is cyclic of order $n$ and has, for each $d \mid n$, exactly $\varphi(d)$ elements of order $d$ (the generators of its unique subgroup of order $d$, Chapter 1); count the elements by their order. (iii) An endomorphism of the cyclic group $\mathbb{Z}/n$ is determined by the image $a$ of 1, and it is $x \mapsto ax$; it is bijective iff $a$ generates, iff $\gcd(a, n) = 1$. Composition corresponds to multiplication. □

For $n = 12$: $(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$, so $\varphi(12) = 4$, and each of these units squares to 1 ($25 = 2 \cdot 12 + 1$, $49 = 4 \cdot 12 + 1$, $121 = 10 \cdot 12 + 1$), so $(\mathbb{Z}/12\mathbb{Z})^\times$ is not cyclic. The divisors $1, 2, 3, 4, 6, 12$ of 12 have $\varphi = 1, 1, 2, 2, 2, 4$, with sum 12. The class 2 is a zero divisor, as in the proof of (i): $2 \cdot 6 \equiv 0$.

**Theorem 2.41** (Chinese remainder theorem)**.** *For $m, n \geq 1$ the reduction map $\mathbb{Z}/mn\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ is a ring isomorphism iff $\gcd(m, n) = 1$. Hence for $n = \prod_i p_i^{k_i}$ the reduction maps give ring isomorphisms*

$$\mathbb{Z}/n\mathbb{Z} \simeq \prod_i \mathbb{Z}/p_i^{k_i}\mathbb{Z}, \qquad (\mathbb{Z}/n\mathbb{Z})^\times \simeq \prod_i (\mathbb{Z}/p_i^{k_i}\mathbb{Z})^\times,$$

*and $\varphi(mn) = \varphi(m)\varphi(n)$ for coprime $m, n$, so $\varphi(n) = \prod_i p_i^{k_i - 1}(p_i - 1)$.*

*Proof.* If $\gcd(m, n) = 1$ then $m\mathbb{Z} + n\mathbb{Z} = \mathbb{Z}$ and $m\mathbb{Z} \cdot n\mathbb{Z} = mn\mathbb{Z}$ (Proposition 2.11(iii)), and Proposition 2.13(ii) applies; (iii) of that proposition gives the decomposition over the prime powers $p_i^{k_i}$, which are pairwise coprime. If $d = \gcd(m, n) > 1$, every element of $\mathbb{Z}/m \times \mathbb{Z}/n$ has additive order dividing $\operatorname{lcm}(m, n) = mn/d < mn$, while $1 \in \mathbb{Z}/mn$ has order $mn$. The unit group of a product of rings is the product of the unit groups. (The group-theoretic version is in Chapter 1.) □

The inverse of the reduction map is explicit once Bezout's identity $um + vn = 1$ is known: the pair ($y \bmod m, z \bmod n$) comes from $x = y \cdot vn + z \cdot um$, since $vn \equiv 1 \pmod m$, $vn \equiv 0 \pmod n$, and symmetrically for $um$. For $m = 3$, $n = 5$ this is the computation after Proposition 2.13. For $\gcd(m, n) > 1$ the failure is visible already for $m = n = 2$: $\mathbb{Z}/4\mathbb{Z}$ is cyclic of order 4, while every element of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has order at most 2. The unit statement gives, for example, $(\mathbb{Z}/15\mathbb{Z})^\times \simeq (\mathbb{Z}/3\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times$, of order $2 \cdot 4 = 8 = \varphi(15)$.

**Lemma 2.42** (lifting the exponent)**.** *(i) Let $\ell$ be an odd prime and $a, b \in \mathbb{Z}$ with $\ell \mid a - b$ and $\ell \nmid ab$. Then $v_\ell(a^n - b^n) = v_\ell(a - b) + v_\ell(n)$ for every $n \geq 1$.*

*(ii) If $a$ is odd and $n \geq 2$ is even, then $v_2(a^n - 1) = v_2(a - 1) + v_2(a + 1) + v_2(n) - 1$.*

*(iii) For $\ell$ odd and $j \geq 0$, $v_\ell((1 + \ell)^{\ell^j} - 1) = j + 1$; hence $1 + \ell$ has order $\ell^{k-1}$ in $(\mathbb{Z}/\ell^k\mathbb{Z})^\times$.*

*(iv) Let $q \neq \ell$ be primes, $\ell$ odd, and $d$ the order of $q$ modulo $\ell$. Then $v_\ell(q^f - 1) = 0$ if $d \nmid f$, and $v_\ell(q^f - 1) = v_\ell(q^d - 1) + v_\ell(f/d)$ if $d \mid f$.*

*(v) $v_\ell(q^2 - 1) = v_\ell(q - 1) + v_\ell(q + 1)$, and for $\ell$ odd at most one of the two terms is positive.*

*Proof.* (i) If $\ell \nmid n$: $a^n - b^n = (a - b)S$ with $S = \sum_{i=0}^{n-1} a^{n-1-i}b^i \equiv na^{n-1} \not\equiv 0 \pmod \ell$, as $a \equiv b$. If $n = \ell$: write $a = b + t$ with $\ell \mid t$. By the binomial theorem, $a^j = (b + t)^j = b^j + jtb^{j-1} + (\text{terms divisible by } t^2)$, so modulo $t^2$, $a^{\ell-1-i}b^i \equiv b^{\ell-1} + (\ell - 1 - i)tb^{\ell-2}$, and summing over $i$ (with $\sum_{i=0}^{\ell-1}(\ell - 1 - i) = 0 + 1 + \cdots + (\ell - 1)$),

$$S = \sum_{i=0}^{\ell-1} a^{\ell-1-i}b^i \equiv \ell b^{\ell-1} + tb^{\ell-2}\sum_{i=0}^{\ell-1}(\ell - 1 - i) = \ell b^{\ell-1} + tb^{\ell-2}\frac{\ell(\ell-1)}{2} \pmod{t^2}.$$

Since $\ell$ is odd, $\ell \mid \frac{\ell(\ell-1)}{2}$, so the second term and $t^2$ are divisible by $\ell^2$, and $S \equiv \ell b^{\ell-1} \pmod{\ell^2}$: $v_\ell(S) = 1$. For general $n = \ell^s m$, $\ell \nmid m$, apply the first case to $a, b$ and $m$: $v_\ell(a^m - b^m) = v_\ell(a-b)$. Then apply the second case $s$ times, to the pairs $a^{m\ell^j}$, $b^{m\ell^j}$ for $j = 0, \dots, s-1$. These still satisfy the hypotheses: $a^{m\ell^j} \equiv b^{m\ell^j} \pmod{\ell}$ because $a \equiv b$, and neither is divisible by $\ell$ because $\ell \nmid ab$. Each application adds exactly 1 to the valuation, since $a^{m\ell^{j+1}} - b^{m\ell^{j+1}} = (a^{m\ell^j})^\ell - (b^{m\ell^j})^\ell$; so $v_\ell(a^n - b^n) = v_\ell(a-b) + s$. (ii) Write $n = 2^s m$ with $m$ odd and $s \geq 1$. As in (i), $a^m - 1 = (a-1)(a^{m-1} + \cdots + 1)$ and $a^m + 1 = (a+1)(a^{m-1} - a^{m-2} + \cdots + 1)$, the second factors being sums of $m$ odd terms, hence odd; so $v_2(a^m - 1) = v_2(a-1)$, $v_2(a^m+1) = v_2(a+1)$, and $v_2(a^{2m} - 1) = v_2((a^m-1)(a^m+1)) = v_2(a-1) + v_2(a+1)$. For odd $x$, $x^2 \equiv 1 \pmod 8$ (as $x^2 - 1 = (x-1)(x+1)$ is a product of two consecutive even numbers, one divisible by 4), so $x^2 + 1 \equiv 2 \pmod 8$ and $v_2(x^2+1) = 1$. For $j \geq 2$ put $x = a^{2^{j-2}m}$, which is odd; then $a^{2^j m} - 1 = (x^2-1)(x^2+1)$, so each further squaring adds exactly 1: $v_2(a^{2^j m} - 1) = v_2(a^{2^{j-1}m} - 1) + 1$. After the $s-1$ squarings from $2m$ to $2^s m = n$ the valuation is $v_2(a-1) + v_2(a+1) + (s-1)$. (iii) Apply (i) with $a = 1+\ell$, $b = 1$: $v_\ell((1+\ell)^{\ell^j} - 1) = 1 + j$. So $(1+\ell)^{\ell^{k-1}} \equiv 1 \pmod{\ell^k}$ but $(1+\ell)^{\ell^{k-2}} \not\equiv 1 \pmod{\ell^k}$ for $k \geq 2$; as the order divides $\ell^{k-1}$, it equals $\ell^{k-1}$. (iv) $\ell \mid q^f - 1$ iff $q^f \equiv 1 \pmod{\ell}$ iff $d \mid f$. If $d \mid f$, apply (i) to $a = q^d$, $b = 1$, $n = f/d$. (v) $q^2 - 1 = (q-1)(q+1)$, and an odd prime cannot divide both $q-1$ and $q+1$, whose difference is 2. □

Two checks. With $\ell = 3$, $a = 4$, $b = 1$, $n = 9$: (i) predicts $v_3(4^9 - 1) = v_3(3) + v_3(9) = 3$, and indeed $4^9 - 1 = 262143 = 3^3 \cdot 7 \cdot 19 \cdot 73$. At $\ell = 2$ the naive formula fails, which is why (ii) has the extra term: $v_2(3^2 - 1) = v_2(8) = 3$, not $v_2(3-1) + v_2(2) = 2$; (ii) gives $1 + 2 + 1 - 1 = 3$, and for $n = 4$ it gives $v_2(3^4 - 1) = v_2(80) = 1 + 2 + 2 - 1 = 4$. The failure comes from the step $n = \ell$ of the proof: for $\ell = 2$ the number $\frac{\ell(\ell-1)}{2} = 1$ is not divisible by $\ell$.

**Theorem 2.43** (primitive roots). *(i)* $(\mathbb{Z}/p\mathbb{Z})^\times = \mathbb{F}_p^\times$ *is cyclic of order* $p-1$*; a generator is a* primitive root *modulo* $p$.

*(ii) For* $p$ *odd and* $k \geq 1$, $(\mathbb{Z}/p^k\mathbb{Z})^\times$ *is cyclic of order* $p^{k-1}(p-1)$. *More precisely, if* $g$ *is a primitive root modulo* $p$ *with* $g^{p-1} \not\equiv 1 \pmod{p^2}$ *(true for* $g$ *or for* $g+p$*), then* $g$ *generates* $(\mathbb{Z}/p^k\mathbb{Z})^\times$ *for every* $k$.

*(iii)* $(\mathbb{Z}/2\mathbb{Z})^\times$ *and* $(\mathbb{Z}/4\mathbb{Z})^\times$ *are cyclic. For* $k \geq 3$, $(\mathbb{Z}/2^k\mathbb{Z})^\times = \{\pm 1\} \times \langle 5 \rangle \simeq \mathbb{Z}/2 \times \mathbb{Z}/2^{k-2}$, *which is not cyclic.*

*(iv)* $(\mathbb{Z}/n\mathbb{Z})^\times$ *is cyclic iff* $n \in \{1, 2, 4, p^k, 2p^k\}$ *with* $p$ *an odd prime and* $k \geq 1$.

*Proof.* (i) A finite subgroup of the multiplicative group of a field is cyclic (Chapter 1; the proof uses Proposition 2.30(iv)). (ii) Let $g$ be a primitive root modulo $p$. If $g^{p-1} \equiv 1 \pmod{p^2}$, then $(g+p)^{p-1} \equiv g^{p-1} + (p-1)pg^{p-2} \equiv 1 - pg^{p-2} \not\equiv 1 \pmod{p^2}$, and $g+p$ is also a primitive root modulo $p$. So assume $g^{p-1} = 1 + pt$ with $p \nmid t$. The order $o$ of $g$ modulo $p^k$ is a multiple of $p-1$ (its order modulo $p$) and divides $\varphi(p^k) = p^{k-1}(p-1)$, so $o = (p-1)p^j$ with $j \leq k-1$. By Lemma 2.42(i), $v_p(g^{(p-1)p^j} - 1) = v_p(g^{p-1} - 1) + j = 1 + j$, which is $\geq k$ only if $j \geq k-1$. So $o = \varphi(p^k)$. (iii) $(\mathbb{Z}/2)^\times = \{1\}$ and $(\mathbb{Z}/4)^\times = \{\pm 1\}$. Let $k \geq 3$. By Lemma 2.42(ii), $v_2(5^{2^j} - 1) = 2 + 1 + j - 1 = j + 2$ for $j \geq 1$, and $v_2(5 - 1) = 2$; so $5^{2^j} \equiv 1 \pmod{2^k}$ iff $j \geq k-2$, and 5 has order $2^{k-2}$. All powers of 5 are $\equiv 1 \pmod 4$, so $-1 \notin \langle 5 \rangle$; the subgroup $\{\pm 1\} \times \langle 5 \rangle$ has $2^{k-1} = \varphi(2^k)$ elements and is everything. It has the two distinct elements $-1$ and $5^{2^{k-3}}$ of order 2, while a cyclic group has at most one. (iv) By Theorem 2.41, $(\mathbb{Z}/n)^\times \simeq \prod_i (\mathbb{Z}/p_i^{k_i})^\times$. For odd $p$ the factor has even order $p^{k-1}(p-1)$, and $(\mathbb{Z}/2^k)^\times$ has even order for $k \geq 2$. If two factors have even order, each

contains an element of order 2, and the product contains a subgroup $\mathbb{Z}/2 \times \mathbb{Z}/2$, impossible in a cyclic group; if $8 \mid n$, the factor $(\mathbb{Z}/2^k)^\times$ is not cyclic and neither is a group containing it. This excludes every $n$ outside the list. Conversely, $(\mathbb{Z}/2p^k)^\times \simeq (\mathbb{Z}/2)^\times \times (\mathbb{Z}/p^k)^\times \simeq (\mathbb{Z}/p^k)^\times$, and the other cases are (ii), (iii). (Checked numerically for all $n < 200$.) □

The proviso in (ii) is real. The class of 7 is a primitive root modulo 5 (it is $\equiv 2$, of order 4), but $7^4 = 2401 = 96 \cdot 25 + 1 \equiv 1 \pmod{25}$, so 7 has order 4 modulo 25 and does not generate the group of order 20; the replacement $7 + 5 = 12$ has $12^4 \equiv 11 \pmod{25}$ and is a generator. On the other hand 2 is a primitive root modulo 3 with $2^2 = 4 \not\equiv 1 \pmod 9$, so 2 generates $(\mathbb{Z}/3^k\mathbb{Z})^\times$ for every $k$: its orders modulo 9 and 27 are 6 and 18. (Orders computed with PARI's `znorder`.)

**Corollary 2.44.** *Let $\ell$ be an odd prime and $k \geq 1$.*

*(i) The solutions of $a^2 \equiv 1 \pmod{\ell^k}$ are $a \equiv \pm 1$.*

*(ii) $a^2 \equiv 1 \pmod{\ell^k}$ holds for every $a$ prime to $\ell$ iff $\varphi(\ell^k) = 2$ iff $\ell^k = 3$.*

*(iii) If $\chi\colon G \to (\mathbb{Z}/\ell^k)^\times$ is a homomorphism from a group with $\chi(g)^2 = 1$ for all $g$, and either $\chi$ is surjective or its image contains the class of 2, then $\ell^k = 3$. Example: $2^2 = 4 \not\equiv 1 \pmod 9$, and $4 \not\equiv 1 \pmod \ell$ for $\ell \geq 5$.*

*Proof.* (i) $(\mathbb{Z}/\ell^k)^\times$ is cyclic of even order (Theorem 2.43(ii)), so it has exactly $\gcd(2, \varphi(\ell^k)) = 2$ elements with $x^2 = 1$ (Chapter 1): 1 and $-1$, distinct as $\ell^k \geq 3$. (Directly: $\ell^k \mid (a-1)(a+1)$, and the odd prime $\ell$ cannot divide both factors, whose difference is 2, so $\ell^k$ divides one of them.) (ii) By (i), every unit squares to 1 iff there are only two units. And $\varphi(\ell^k) = \ell^{k-1}(\ell - 1) = 2$ iff $\ell = 3$, $k = 1$. (iii) Every element of the image squares to 1, so it is $\pm 1$ by (i). If $\chi$ is surjective, $(\mathbb{Z}/\ell^k)^\times = \{\pm 1\}$ and $\ell^k = 3$ by (ii). If 2 is in the image, $2 \equiv \pm 1 \pmod{\ell^k}$, so $\ell^k \mid 3$. □

*In the paper* 2.45. Lemma lem:h0: an element of exact order $\ell^k$ of $H^0(\mathbb{Q}, \mathrm{ad}^0\, T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ gives an isomorphism $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$, and comparing determinants, $\chi_\ell(g)^2 \equiv 1 \pmod{\ell^k}$ for all $g \in G_\mathbb{Q}$. The cyclotomic character maps $G_\mathbb{Q}$ onto $\mathbb{Z}_\ell^\times$ (Chapter 5), hence onto $(\mathbb{Z}/\ell^k)^\times$; so Corollary 2.44(iii) gives $\ell^k \in \{1, 3\}$: "Since $\chi_\ell(G_\mathbb{Q}) = \mathbb{Z}_\ell^\times$ contains 2, and $4 \not\equiv 1$ modulo $\ell$ for $\ell \geq 5$ and modulo 9, this forces $k = 0$ for $\ell \geq 5$ and $k \leq 1$ for $\ell = 3$." Hence $h_\ell(E) = 0$ for $\ell \geq 5$ and $h_3(E) \leq 1$. The case $\ell^k = 3$ is the identity $\bar\chi_3^{\mathrm{cyc}} = \chi_{-3}$, whose Dirichlet side is Proposition 2.78(iii).

## 2.6 Finite fields

In this section and the next $q = p^r$ is a prime power (Convention 2.2). Finite fields are the residue fields of everything in the book: $\mathbb{F}_q$ at a bad prime $q$ of $E$, $\mathbb{F}_\ell$ and $\mathbb{F}_9$ as coefficient fields of $\bar\rho$, and $\mathbb{F}_{q^f}$ in the unramified extensions of Chapter 7. We prove existence and uniqueness, the Frobenius automorphism and the subfields, count irreducible polynomials, and determine the roots of unity in $\mathbb{F}_q$; the last item is the arithmetic input of Lemma lem:euler. That $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ is cyclic generated by $x \mapsto x^q$ is proved in Chapter 5.

**Proposition 2.46.** *Let $F$ be a finite field.*

*(i) $\mathrm{char}\, F = p$ is a prime, $F$ contains the image $\mathbb{F}_p$ of $\mathbb{Z}$, and $\#F = p^r$ with $r = \dim_{\mathbb{F}_p} F$.*

*(ii) $F^\times$ is cyclic of order $p^r - 1$.*

*(iii)* *Every $a \in F$ satisfies $a^{p^r} = a$, and $x^{p^r} - x = \prod_{a \in F}(x - a)$ in $F[x]$.*

*(iv)* *$F = \mathbb{F}_p[\alpha]$ for every generator $\alpha$ of $F^\times$.*

*Proof.* (i) $\mathbb{Z} \to F$ is not injective, as $F$ is finite, so $\operatorname{char} F$ is a prime $p$ (Proposition 2.9); the image is $\mathbb{Z}/p = \mathbb{F}_p$, and $F$ is a finite-dimensional $\mathbb{F}_p$-vector space. (ii) is the cyclicity of finite subgroups of $K^\times$ (Chapter 1). (iii) For $a \neq 0$, Lagrange gives $a^{p^r-1} = 1$; and $0^{p^r} = 0$. So the $p^r$ distinct linear polynomials $x - a$ divide $x^{p^r} - x$; they are pairwise coprime, so their product, monic of degree $p^r$, divides it and equals it. (iv) $\mathbb{F}_p[\alpha]$ contains 0 and all powers of $\alpha$. □

**Theorem 2.47** (existence and uniqueness of finite fields)**.** *Let $q = p^r$.*

*(i)* *There is a field with $q$ elements.*

*(ii)* *Any two fields with $q$ elements are isomorphic. We write $\mathbb{F}_q$ for any of them.*

*(iii)* *$\mathbb{F}_q \simeq \mathbb{F}_p[x]/(f)$ for every irreducible $f \in \mathbb{F}_p[x]$ of degree $r$ (such $f$ exist by Proposition 2.49).*

*(iv)* *The isomorphism in (ii) is not unique when $r > 1$ (compose with powers of the Frobenius, Proposition 2.48); inside a fixed algebraic closure of $\mathbb{F}_p$ the subfield with $q$ elements is unique (Chapter 5).*

*Proof.* (i) Let $K \supseteq \mathbb{F}_p$ be a field in which $x^q - x$ splits (Theorem 2.38(iii)). Its roots are $q$ distinct elements, since $x^q - x$ is separable (Proposition 2.31(iii)). They form a subfield: they contain 0, 1; if $a^q = a$ and $b^q = b$ then $(a \pm b)^q = a^q \pm b^q = a \pm b$ (Proposition 2.16; for $p = 2$, $-b = b$; for odd $p$, $(-b)^q = -b^q$), $(ab)^q = ab$, and $(a^{-1})^q = a^{-1}$. (ii) Let $K$, $K'$ have $q$ elements. Write $K = \mathbb{F}_p[\alpha]$ (Proposition 2.46(iv)). The kernel of $\mathbb{F}_p[x] \to K$, $x \mapsto \alpha$, is a nonzero ($\mathbb{F}_p[x]$ is infinite) prime ideal, so it is $(f)$ with $f$ monic irreducible, and $K \simeq \mathbb{F}_p[x]/(f)$. Since $\alpha^q = \alpha$, $f \mid x^q - x$. In $K'[x]$, $x^q - x = \prod_{b \in K'}(x - b)$, so by unique factorisation in $K'[x]$ the divisor $f$ is a product of some of these linear factors and has a root $\beta \in K'$. By Theorem 2.38(ii), $\mathbb{F}_p[\beta] \simeq \mathbb{F}_p[x]/(f) \simeq K$, so $\mathbb{F}_p[\beta]$ has $q$ elements and equals $K'$. (iii) $\mathbb{F}_p[x]/(f)$ is a field with $p^r$ elements (Theorem 2.38(i)); apply (ii). (iv) If $\sigma\colon K \to K'$ is an isomorphism, so is $\sigma \circ \operatorname{Frob}_p^s$ for every $s$, and these are distinct for $0 \leq s < r$ by Proposition 2.48(i). □

**Proposition 2.48** (Frobenius and subfields)**.** *Let $q = p^r$.*

*(i)* *$\operatorname{Frob}_p\colon x \mapsto x^p$ is a field automorphism of $\mathbb{F}_q$, its fixed points are exactly $\mathbb{F}_p$, and it has order exactly $r$.*

*(ii)* *For $d \mid r$, $\{x \in \mathbb{F}_q : x^{p^d} = x\}$ is a subfield with $p^d$ elements, and it is the only subfield with $p^d$ elements. A subfield with $p^e$ elements exists only if $e \mid r$.*

*(iii)* *The fixed field of $\operatorname{Frob}_p^s$ is the subfield with $p^{\gcd(s,r)}$ elements.*

*The fact that every automorphism of $\mathbb{F}_q$ is a power of $\operatorname{Frob}_p$, so that $\operatorname{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ is cyclic of order $n$ generated by $x \mapsto x^q$, is proved in Chapter 5.*

*Proof.* (i) $\operatorname{Frob}_p$ is an injective endomorphism of the finite field $\mathbb{F}_q$, hence bijective (Proposition 2.16(ii)). Its fixed points are the roots of $x^p - x$, at most $p$ of them (Proposition 2.30(iv)), and every element of $\mathbb{F}_p$ is one (Fermat). By Proposition 2.46(iii), $\operatorname{Frob}_p^r = \mathrm{id}$. If $\operatorname{Frob}_p^s = \mathrm{id}$ with $1 \leq s < r$, all $p^r$ elements would be roots of $x^{p^s} - x$, of degree $p^s < p^r$. (ii) The set is the fixed set of the automorphism $\operatorname{Frob}_p^d$, hence a subfield. As $d \mid r$, $p^d - 1$ divides $p^r - 1$ (write

$r = dt$ and $y = p^d$; then $y^t - 1 = (y-1)(y^{t-1} + \cdots + 1)$), and for the same reason $x^{p^d-1} - 1$ divides $x^{p^r-1} - 1$; multiplying by $x$, $x^{p^d} - x$ divides $x^{p^r} - x$; since $x^{p^r} - x$ splits into distinct linear factors over $\mathbb{F}_q$, so does its divisor $x^{p^d} - x$, which therefore has exactly $p^d$ roots in $\mathbb{F}_q$. A subfield $L$ with $p^d$ elements satisfies $x^{p^d} = x$ on $L$ (Proposition 2.46(iii)), so it lies in this set, and equals it by counting. If $L$ has $p^e$ elements, $\mathbb{F}_q$ is an $L$-vector space of some dimension $t$, and $p^r = p^{et}$. (iii) Let $g = \gcd(s, r) = as + br$ with $a, b \in \mathbb{Z}$. In the cyclic group generated by $\mathrm{Frob}_p$ (of order $r$), $\mathrm{Frob}_p^g = (\mathrm{Frob}_p^s)^a$, since $\mathrm{Frob}_p^{br} = \mathrm{id}$. So an element fixed by $\mathrm{Frob}_p^s$ is fixed by $\mathrm{Frob}_p^g$; the converse holds as $g \mid s$. Apply (ii) to $g \mid r$. □

**Proposition 2.49.** *Let $q$ be a prime power and $r \geq 1$. In $\mathbb{F}_q[x]$,*

$$x^{q^r} - x = \prod_{d \mid r} \prod_{f \text{ monic irreducible, } \deg f = d} f,$$

*each factor occurring once. Hence $q^r = \sum_{d \mid r} d N_d$, where $N_d$ is the number of monic irreducible polynomials of degree $d$ over $\mathbb{F}_q$, and*

$$N_r = \frac{1}{r} \sum_{d \mid r} \mu(d)\, q^{r/d} > 0.$$

*Over $\mathbb{F}_2$, $N_1, \ldots, N_8 = 2, 1, 2, 3, 6, 9, 18, 30$; over $\mathbb{F}_3$, $N_1, \ldots, N_5 = 3, 3, 8, 18, 48$, and the three monic irreducible quadratics are $x^2 + 1$, $x^2 + x + 2$, $x^2 + 2x + 2$.*

*Proof.* The polynomial $x^{q^r} - x$ is separable, so no irreducible factor occurs twice ($f^2 \mid g$ implies $f \mid g'$). Let $f$ be monic irreducible of degree $d$, $K = \mathbb{F}_q[x]/(f)$, a field with $q^d$ elements, and $\alpha = \bar{x}$; $f$ is the minimal polynomial of $\alpha$, so $f \mid x^{q^r} - x$ iff $\alpha^{q^r} = \alpha$. The map $\phi\colon y \mapsto y^q$ is an automorphism of $K$ fixing $\mathbb{F}_q$ (Proposition 2.46(iii) for $\mathbb{F}_q$), and it has order exactly $d$ by the argument of Proposition 2.48(i). If $\alpha^{q^r} = \alpha$, then $\phi^r$ fixes $\mathbb{F}_q$ and $\alpha$, hence all of $K = \mathbb{F}_q[\alpha]$, so $d \mid r$. If $d \mid r$, then $\alpha^{q^d} = \alpha$ and so $\alpha^{q^r} = \alpha$ (apply $y \mapsto y^{q^d}$ repeatedly, $r/d$ times). Comparing degrees gives $q^r = \sum_{d \mid r} d N_d$. For Moebius inversion recall $\sum_{e \mid m} \mu(e) = 0$ for $m > 1$: if $p_1, \ldots, p_s$ ($s \geq 1$) are the distinct primes dividing $m$, only squarefree $e$ contribute, and the sum is $\sum_{T \subseteq \{1,\ldots,s\}} (-1)^{\#T} = (1-1)^s = 0$. Hence

$$\sum_{d \mid r} \mu(d) q^{r/d} = \sum_{d \mid r} \mu(d) \sum_{e \mid r/d} e N_e = \sum_{e \mid r} e N_e \sum_{d \mid r/e} \mu(d) = r N_r,$$

since the inner sum is 1 for $e = r$ and 0 otherwise. For $r \geq 2$ the term $d = 1$ is $q^r$, and the others have absolute value $q^{r/d}$ with $r/d \leq r/2$ distinct integers; so

$$r N_r \geq q^r - \sum_{j=0}^{\lfloor r/2 \rfloor} q^j > q^r - q^{\lfloor r/2 \rfloor + 1} \geq 0,$$

using $\sum_{j=0}^{M} q^j = (q^{M+1} - 1)/(q-1) < q^{M+1}$ for $q \geq 2$, and $\lfloor r/2 \rfloor + 1 \leq r$; and $N_1 = q$. The numerical values follow from the formula; for the quadratics over $\mathbb{F}_3$, each of the three listed has no root in $\mathbb{F}_3$ and $N_2 = (9-3)/2 = 3$. □

For $q = 2$, $r = 2$ the factorisation reads $x^4 - x = x(x+1)(x^2 + x + 1)$ in $\mathbb{F}_2[x]$: the two linear polynomials and the unique irreducible quadratic, and $4 = 1 \cdot 2 + 2 \cdot 1$. The formula for $N_r$ is the finite-field analogue of the prime number theorem: $N_r = \frac{q^r}{r} + O(\frac{q^{r/2}}{r})$, so a random monic polynomial of degree $r$ is irreducible with probability about $1/r$.

**Proposition 2.50** (roots of unity in $\mathbb{F}_q$)**.** *Let $q = p^r$ and $n \geq 1$.*

*(i) $\mu_n(\mathbb{F}_q)$ is cyclic of order $\gcd(n, q-1)$; it has $n$ elements iff $n \mid q-1$.*

*(ii) If $p \nmid n$, the least $f \geq 1$ with $\#\mu_n(\mathbb{F}_{q^f}) = n$ is the order of $q$ in $(\mathbb{Z}/n\mathbb{Z})^\times$.*

*(iii) In characteristic $p$, $x^{p^k m} - 1 = (x^m - 1)^{p^k}$; there are no roots of unity of order divisible by $p$.*

*(iv) Let $q$ be odd. Then $-1$ is a square in $\mathbb{F}_q$ iff $q \equiv 1 \pmod 4$. If moreover $p \neq 3$, then $-3$ is a square in $\mathbb{F}_q$ iff $\mathbb{F}_q$ contains a primitive cube root of unity iff $q \equiv 1 \pmod 3$; indeed $(1+2\zeta)^2 = -3$ for a primitive cube root of unity $\zeta$, and $\zeta = (-1+s)/2$ is one if $s^2 = -3$.*

*(v) Let $\sigma$ be an element of order 3 of any group and $m$ an integer prime to 3. Then $\sigma^m = \sigma$ if $m \equiv 1 \pmod 3$ and $\sigma^m = \sigma^{-1}$ if $m \equiv 2 \pmod 3$. Likewise if $\sigma$ has order 4 (or 2) and $m$ is odd, $\sigma^m = \sigma$ or $\sigma^{-1}$ according as $m \equiv 1$ or $3 \pmod 4$. In particular this applies to $\mu_3(K)$ and $\mu_4(K)$ for any field $K$ and to $\zeta \mapsto \zeta^q$; in the notation of Proposition 2.78, $\sigma^m = \sigma^{\chi_{-3}(m)}$, resp. $\sigma^m = \sigma^{\chi_{-4}(m)}$.*

*Proof.* (i) $\mu_n(\mathbb{F}_q)$ is the kernel of $x \mapsto x^n$ on the cyclic group $\mathbb{F}_q^\times$ of order $q-1$, of order $\gcd(n, q-1)$ (Chapter 1). (ii) By (i), $\#\mu_n(\mathbb{F}_{q^f}) = n$ iff $n \mid q^f - 1$ iff $q^f \equiv 1 \pmod n$. (iii) $(x^m-1)^{p^k} = x^{p^k m} - 1$ by Proposition 2.16. If $\zeta$ has order $p^k m$ with $k \geq 1$, then $\eta = \zeta^m$ satisfies $\eta^{p^k} = 1$, so $(\eta - 1)^{p^k} = 0$ and $\eta = 1$, contradicting the order. (iv) An element $y$ with $y^2 = -1$ has order 4 (as $-1 \neq 1$), and conversely if $y$ has order 4 then $y^2$ has order 2, so $y^2 = -1$, the unique element of order 2 of the cyclic group $\mathbb{F}_q^\times$. Such $y$ exists iff $4 \mid q-1$. Next, a primitive cube root of unity $\zeta$ satisfies $\zeta^2 + \zeta + 1 = 0$ (as $\zeta \neq 1$ and $x^3 - 1 = (x-1)(x^2+x+1)$), so $(1+2\zeta)^2 = 1 + 4(\zeta + \zeta^2) = -3$. Conversely if $s^2 = -3$ and $\zeta = (-1+s)/2$ (2 is invertible), then $4(\zeta^2 + \zeta + 1) = (1 - 2s + s^2) + (-2 + 2s) + 4 = s^2 + 3 = 0$, and $\zeta \neq 1$ because $1+1+1 = 3 \neq 0$ ($p \neq 3$); so $\zeta$ has order 3. By (i), such $\zeta$ exists iff $3 \mid q-1$. (v) $\sigma^m$ depends only on $m$ modulo the order of $\sigma$, and $\sigma^2 = \sigma^{-1}$, resp. $\sigma^3 = \sigma^{-1}$. The value $\chi_{-3}(m) \equiv m \pmod 3$, resp. $\chi_{-4}(m) \equiv m \pmod 4$, is Proposition 2.78(i). □

*In the paper* 2.51. In Lemma lem:IV the paper writes "$\rho(\mathrm{Frob}_q)\sigma\rho(\mathrm{Frob}_q)^{-1} = \sigma^q = \sigma^{\chi_{-3}(q)}$", for an element $\sigma$ of order 3 generating the image of inertia, and in the proof of Lemma lem:euler: "Frobenius centralises $\sigma$ if $q \equiv 1 \pmod 3$ and inverts it if $q \equiv 2$, for either choice of Frobenius, so it acts on the line by $\chi_{-3}(q)$"; for types III, III$^*$ the same with an element $\tau$ of order 4 and $\chi_{-4}(q)$. The Galois input (Frobenius acts on the tame quotient of inertia by $x \mapsto x^q$) is Chapters 6 and 7; the arithmetic input is Proposition 2.50(v): $\sigma^q = \sigma^{\pm 1}$ according to $q$ modulo 3, and $\pm 1 = \chi_{-3}(q)$. Conjugation by Frobenius on the invariant line spanned by $\sigma - \sigma^{-1}$ (resp. by $\tau$) is then multiplication by $\chi_{-3}(q)$ (resp. $\chi_{-4}(q)$), and it does not matter whether the arithmetic or the geometric Frobenius is used, since $\pm 1$ is its own inverse.

## 2.7 Powers in $\mathbb{F}_q^\times$

This section is the arithmetic of Lemma lem:tamfactor (the Tamagawa factor at a multiplicative prime). There $q$ is a prime, $u \in \mathbb{Z}_q^\times$ is the unit part of the Tate parameter, and the Tamagawa factor of $\mathrm{ad}^0 T$ exceeds $t_\ell(E,q) = v_\ell(n)$ by $\min(v_\ell(n), v_\ell(q-1), v_\ell(c))$, where $c$ is the class of the reduction $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$. We prove that this group is cyclic of order $\ell^{v_\ell(q-1)}$, that $v_\ell(c)$

does not depend on the identification with $\mathbb{Z}/\ell^{v_\ell(q-1)}$, and that $v_\ell(c) \geq k$ iff $\bar{u}$ is an $\ell^k$-th power. The statements hold for any prime power $q$; the group theory is that of cyclic groups (Chapter 1).

**Theorem 2.52.** *Let $q$ be a prime power, $m \geq 1$ and $d = \gcd(m, q-1)$.*

*(i) The $m$-th power map of $\mathbb{F}_q^\times$ has kernel $\mu_d(\mathbb{F}_q)$, of order $d$, and image $(\mathbb{F}_q^\times)^m = (\mathbb{F}_q^\times)^d$, the unique subgroup of index $d$.*

*(ii) (Generalised Euler criterion.) $u \in \mathbb{F}_q^\times$ is an $m$-th power iff $u^{(q-1)/d} = 1$.*

*(iii) For a prime $\ell$ and $k \geq 0$,*

$$\left[\mathbb{F}_q^\times : (\mathbb{F}_q^\times)^{\ell^k}\right] = \ell^{\min(k, v_\ell(q-1))}.$$

*In particular, if $q \not\equiv 1 \pmod{\ell}$, the $\ell$-th power map is bijective and every element is an $\ell^k$-th power for every $k$; if $q \equiv 1 \pmod{\ell}$, exactly a fraction $1/\ell$ of the elements are $\ell$-th powers.*

*(iv) ($m = 2$, $q$ odd.) The squares form the subgroup of index 2, and $u^{(q-1)/2} = \pm 1$, with $+1$ iff $u$ is a square.*

*Proof.* $\mathbb{F}_q^\times$ is cyclic of order $q-1$ (Proposition 2.46). (i) and (iii) are the statements on cyclic groups of Chapter 1: for $C$ cyclic of order $N$, $x \mapsto x^m$ has kernel $C[\gcd(m,N)]$ and image $C^{\gcd(m,N)}$ of index $\gcd(m,N)$; and $\gcd(\ell^k, q-1) = \ell^{\min(k, v_\ell(q-1))}$. (ii) The image $C^d$ is the unique subgroup of order $(q-1)/d$, which is $C[(q-1)/d] = \{u : u^{(q-1)/d} = 1\}$. (iv) is (i), (ii) with $m = 2$, $d = 2$: $u^{(q-1)/2}$ squares to 1, so it is $\pm 1$ (the roots of $x^2 - 1$). □

The hypothesis is $d = \gcd(m, q-1)$, not $m$: every element of $\mathbb{F}_{11}^\times$ is a cube, since $\gcd(3, 10) = 1$, although "$u^{10/3}$" is meaningless.

**Proposition 2.53** (the class of a unit modulo $\ell$-power powers)**.** *Let $q$ be a prime power, $\ell \nmid q$ a prime, $v = v_\ell(q-1)$, and $A = \mathbb{F}_q^\times / (\mathbb{F}_q^\times)^{\ell^v}$.*

*(i) $A$ is cyclic of order $\ell^v$. The projection restricted to $\mu_{\ell^v}(\mathbb{F}_q)$, the $\ell$-primary part of $\mathbb{F}_q^\times$, is an isomorphism onto $A$, and $u \mapsto u^{(q-1)/\ell^v}$ induces an isomorphism $A \simeq \mu_{\ell^v}(\mathbb{F}_q)$.*

*(ii) A generator $g$ of $\mathbb{F}_q^\times$ identifies $A$ with $\mathbb{Z}/\ell^v$: $u = g^a \mapsto c(u) = a \bmod \ell^v$. Replacing $g$ by another generator multiplies $c(u)$ by a unit of $\mathbb{Z}/\ell^v$; so $v_\ell(c(u))$, read as in Convention 2.3, does not depend on $g$.*

*(iii) For $0 \leq k \leq v$: $u \in (\mathbb{F}_q^\times)^{\ell^k}$ iff $v_\ell(c(u)) \geq k$. For $k \geq v$: $(\mathbb{F}_q^\times)^{\ell^k} = (\mathbb{F}_q^\times)^{\ell^v}$.*

*(iv) For $n \in \mathbb{Z} \setminus \{0\}$,*

$$\min(v_\ell(n), v, v_\ell(c(u))) = \max\{k \leq \min(v_\ell(n), v) : u \in (\mathbb{F}_q^\times)^{\ell^k}\},$$

*and this is $\geq 1$ iff $\ell \mid n$, $q \equiv 1 \pmod{\ell}$ and $u$ is an $\ell$-th power in $\mathbb{F}_q^\times$.*

*(v) If $q$ is prime and $u \in \mathbb{Z}_{(q)}^\times$ (or $u \in \mathbb{Z}_q^\times$, Chapter 7), the class of $u$ means the class of its reduction $\bar{u} \in \mathbb{F}_q^\times$ (Proposition 2.21(iii)); we write $c(u) = c(\bar{u})$.*

*Proof.* Write $C = \mathbb{F}_q^\times$, cyclic of order $q-1 = \ell^v m$ with $\ell \nmid m$. (i) $A = C/C^{\ell^v}$ has order $\gcd(\ell^v, q-1) = \ell^v$ and is cyclic, as a quotient of a cyclic group. $C^{\ell^v}$ is the unique subgroup of order $m$, i.e. $C[m]$, and $C = C[\ell^v] \times C[m]$ (Chapter 1), so $C[\ell^v] = \mu_{\ell^v}(\mathbb{F}_q)$ maps isomorphically onto $C/C[m] = A$. The map $u \mapsto u^m$ has kernel $C[m] = C^{\ell^v}$ and image $C^m = C[\ell^v]$. (ii) Every $u$ is $g^a$ with $a$ well defined modulo $q-1$, hence modulo $\ell^v$; the map $u \mapsto a \bmod \ell^v$ is a surjective homomorphism $C \to \mathbb{Z}/\ell^v$ whose kernel is $\{g^a : \ell^v \mid a\} = \langle g^{\ell^v}\rangle = C^{\ell^v}$. Another generator is $g' = g^b$ with $\gcd(b, q-1) = 1$; then $u = g'^{a'} = g^{ba'}$, so $c_g(u) = b\, c_{g'}(u)$ with $b$ a unit modulo $\ell^v$. Finally $\min(v_\ell(\tilde{b}\tilde{c}), v) = \min(v_\ell(\tilde{c}), v)$ when $\ell \nmid \tilde{b}$. (iii) For $k \le v$, $\gcd(\ell^k, q-1) = \ell^k$, so $C^{\ell^k} = \langle g^{\ell^k}\rangle = \{g^a : \ell^k \mid a\}$ (this condition on $a$ is well defined modulo $q-1$), and $\ell^k \mid a$ iff $c(u) \in \ell^k\mathbb{Z}/\ell^v\mathbb{Z}$ iff $v_\ell(c(u)) \ge k$. For $k \ge v$, $C^{\ell^k} = C^{\gcd(\ell^k, q-1)} = C^{\ell^v}$. (iv) By (iii) the set of $k \in [0, v]$ with $u \in C^{\ell^k}$ is $\{0, 1, \dots, v_\ell(c(u))\}$, as $v_\ell(c(u)) \le v$. Intersecting with $k \le v_\ell(n)$, the maximum is $\min(v_\ell(n), v_\ell(c(u))) = \min(v_\ell(n), v, v_\ell(c(u)))$. It is $\ge 1$ iff $v_\ell(n) \ge 1$, $v \ge 1$ and $v_\ell(c(u)) \ge 1$, the last meaning $u \in C^\ell$ by (iii). (v) is a definition, justified by Proposition 2.21(iii). □

When $v = 0$ the group $A$ is trivial, $c(u) = 0 \in \mathbb{Z}/1$ and $v_\ell(c(u)) = 0$; the minimum in (iv) is then 0, as it should be. The identification $A \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(1)/(q-1)$ used in the proof of Lemma lem:tamfactor is Chapters 3 and 17; the Kummer interpretation $\bar{u}^{(q-1)/\ell^k}$ = the value of a Kummer character at Frobenius is Chapter 5.

**Example 2.54.** (a) $q = 7$, $\ell = 3$: $v = 1$; the cubes in $\mathbb{F}_7^\times$ are $\{1, 6\}$. So $\bar{u} = -1$ is a cube and $\bar{u} = 2$ is not. A multiplicative prime $q = 7$ with $3 \mid n_7$ and $u \equiv -1 \pmod 7$ therefore contributes $v_3(n_7) + 1$ to $v_3(\mathrm{Tam}_7(\mathrm{ad}^0 T))$ by Lemma lem:tamfactor. (b) $q = 19$, $\ell = 3$: $v = 2$, and 2 is a primitive root modulo 19. The cubes are $\{1, 7, 8, 11, 12, 18\}$ and the ninth powers $\{1, 18\}$. With $g = 2$: $c(2) = 1$, $c(8) = 3$, $c(-1) = c(2^9) = 9 \bmod 9 = 0$, so $v_3(c) = 0, 1, 2$ respectively: 2 is not a cube, 8 is a cube but not a ninth power, and $-1$ is a ninth power. (c) $q = 13$, $\ell = 3$: $v = 1$, $(q-1)/3 = 4$, and $u$ is a cube iff $u^4 = 1$; the cubes are $\{1, 5, 8, 12\}$, the fourth roots of unity. (d) $q = 5$, $\ell = 3$: $v = 0$, every element is a cube. $q = 11$, $\ell = 5$: $v = 1$, the fifth powers are $\{\pm 1\}$. (All lists recomputed with PARI's `znprimroot` and `znlog`.)

*In the paper* 2.55. Lemma lem:tamfactor states: "$c(u)$ the class of the reduction of $u$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$, $v = v_\ell(q-1)$, identified with an element of $\mathbb{Z}/\ell^v$ by the choice of a generator; $v_\ell(c(u))$ does not depend on that choice, $v_\ell(0)$ is read as $v$, and $v_\ell(c(u)) \ge k$ if and only if the reduction of $u$ is an $\ell^k$-th power in $\mathbb{F}_q^\times$, for $k \le v$." (The book writes $c$ for $c(u)$.) This is Proposition 2.53(i) to (iii) and (v), with Convention 2.3. The "In particular" clause ("with equality unless $q \equiv 1 \pmod \ell$, $\ell \mid n$ and the reduction of $u$ is an $\ell$-th power") is (iv). Remark rem:ko(1) and Section sec:tam test exactly this condition on the tables; Example 2.54 shows the three cases at $q = 19$. Here $q$ is prime, so $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$ and "the class of $u$" is computed as a discrete logarithm of $u$ mod $q$ to a primitive root.

## 2.8 Roots of unity and cyclotomic polynomials

Roots of unity appear in the paper as $\zeta_3$ and $i$ (the images of inertia of types II to IV$^*$ have eigenvalues in $\mu_3$, $\mu_4$, $\mu_6$), as $\mu_{\ell^k}$ (the cyclotomic character, the Tate module $\mathbb{Z}_\ell(1)$), and through the condition $q \equiv 1 \pmod \ell$ of Lemma lem:tamfactor, which says that $\mathbb{F}_q$ contains the $\ell$-th roots of unity. This section proves what is needed about cyclotomic polynomials over $\mathbb{Z}$ and modulo $p$; the irreducibility of all $\Phi_n$ and $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \simeq (\mathbb{Z}/n)^\times$ are Chapter 5.

**Definition 2.56.** For a field $K$, $\mu_n(K) = \{x \in K : x^n = 1\}$ and $\mu(K) = \bigcup_{n\geq 1} \mu_n(K)$, the group of roots of unity of $K$. A *primitive $n$-th root of unity* is an element of order exactly $n$ in $K^\times$.

**Proposition 2.57.** *(i) $\mu_n(K)$ is cyclic of order dividing $n$; it has order $n$ iff $K$ contains a primitive $n$-th root of unity, and this requires* $\operatorname{char} K \nmid n$.

*(ii) $\mu_n(\mathbb{C}) = \langle \zeta_n \rangle$; its primitive elements are the $\zeta_n^a$ with $\gcd(a,n) = 1$, $\varphi(n)$ in number.*

*(iii) $\mu(\mathbb{Q}) = \mu(\mathbb{R}) = \{\pm 1\}$, $\mu(\mathbb{Q}(i)) = \mu_4$ and $\mu(\mathbb{Q}(\zeta_3)) = \mu_6$.*

*Proof.* (i) $\mu_n(K)$ is a finite subgroup of $K^\times$ ($x^n - 1$ has at most $n$ roots), hence cyclic (Chapter 1), and its elements have order dividing $n$; a cyclic group of order dividing $n$ has order $n$ iff it has an element of order $n$. If $\operatorname{char} K = p \mid n$, then $x^n - 1 = (x^{n/p} - 1)^p$ and $\mu_n(K) = \mu_{n/p}(K)$ has order at most $n/p$. (ii) The roots of $x^n - 1$ in $\mathbb{C}$ are the $n$ numbers $e^{2\pi i a/n} = \zeta_n^a$, $0 \leq a < n$; the order of $\zeta_n^a$ is $n/\gcd(a,n)$. (iii) A real root of unity has absolute value 1, so it is $\pm 1$. Let $K$ be $\mathbb{Q}(i)$ or $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$, a two-dimensional $\mathbb{Q}$-vector space, and $z \in \mu(K)$ of order $m$. The elements $1, z, z^2$ are linearly dependent over $\mathbb{Q}$, so the minimal polynomial $h$ of $z$ over $\mathbb{Q}$ (the monic generator of the kernel of $\mathbb{Q}[x] \to \mathbb{C}$, $x \mapsto z$) has degree at most 2, and it divides $x^m - 1$. By Theorem 2.32(iv), $h \in \mathbb{Z}[x]$. If $\deg h = 2$, $h = x^2 + bx + c$ with roots $z$ and $\bar{z}$ (both roots of $x^m - 1$, of absolute value 1): $|c| = |z\bar{z}| = 1$ and $|b| = |z + \bar{z}| \leq 2$. If $c = -1$ the roots are real, hence $\pm 1$, and $h = x^2 - 1$ is not irreducible; if $c = 1$, $b = \pm 2$ gives $(x \pm 1)^2$, not irreducible. So $h$ is $x \mp 1$, $x^2 + 1$ or $x^2 \pm x + 1$, and $m \in \{1, 2, 4, 3, 6\}$. Now $\mathbb{Q}(i)$ does not contain $\zeta_3$: if $(a + bi)^2 = -3$ with $a, b \in \mathbb{Q}$ then $ab = 0$, $a^2 - b^2 = -3$, so $a = 0$ and $b^2 = 3$, impossible. And $\mathbb{Q}(\sqrt{-3})$ does not contain $i$: $(a + b\sqrt{-3})^2 = -1$ forces $a = 0$, $3b^2 = 1$. Hence $\mu(\mathbb{Q}(i)) = \mu_4$ and $\mu(\mathbb{Q}(\zeta_3)) = \{\pm\zeta_3^j\} = \mu_6$. □

**Theorem 2.58** (cyclotomic polynomials)**.** *For $n \geq 1$ let $\Phi_n(x) = \prod_{1\leq a\leq n,\ \gcd(a,n)=1} (x - \zeta_n^a) \in \mathbb{C}[x]$.*

*(i) $x^n - 1 = \prod_{d\mid n} \Phi_d(x)$.*

*(ii) $\Phi_n \in \mathbb{Z}[x]$, monic of degree $\varphi(n)$.*

*(iii) $\Phi_n(x) = \prod_{d\mid n} (x^d - 1)^{\mu(n/d)}$, with $\mu$ the Moebius function.*

*(iv) $\Phi_p = x^{p-1} + \cdots + x + 1$; $\Phi_{p^k}(x) = \Phi_p(x^{p^{k-1}})$; $\Phi_{2m}(x) = \Phi_m(-x)$ for odd $m > 1$.*

*(v) $\Phi_1(1) = 0$; $\Phi_n(1) = p$ if $n = p^k$, $k \geq 1$; $\Phi_n(1) = 1$ if $n$ has at least two distinct prime factors; $\Phi_n(0) = 1$ for $n \geq 2$.*

*(vi) $\Phi_{p^k}$ is irreducible over $\mathbb{Q}$.*

*Proof.* (i) Group the $n$-th roots of unity by their order $d \mid n$; those of order $d$ are the primitive $d$-th roots, the roots of $\Phi_d$ (Proposition 2.57(ii)). (ii) Induction on $n$: $\Phi_1 = x - 1$. By (i), $\Phi_n = (x^n - 1)/g$ with $g = \prod_{d\mid n, d<n} \Phi_d$ monic in $\mathbb{Z}[x]$ by induction. Division of $x^n - 1$ by the monic $g$ in $\mathbb{Z}[x]$ (Proposition 2.30(i)) gives a quotient in $\mathbb{Z}[x]$ and a remainder, which must vanish by uniqueness of division in $\mathbb{C}[x]$. The degree is the number of primitive $n$-th roots of unity, $\varphi(n)$. (iii) In the field $\mathbb{C}(x)$, using (i), $\prod_{d\mid n} (x^d - 1)^{\mu(n/d)} = \prod_{d\mid n} \prod_{e\mid d} \Phi_e^{\mu(n/d)} = \prod_{e\mid n} \Phi_e^{\sum_{e\mid d\mid n} \mu(n/d)}$, and $\sum_{e\mid d\mid n} \mu(n/d) = \sum_{k\mid n/e} \mu(k)$ (put $k = n/d$, which runs over the divisors of $n/e$) is 1 if $e = n$ and 0 otherwise (the identity $\sum_{k\mid m} \mu(k) = 0$ for $m > 1$ is proved

in Proposition 2.49). For example $\Phi_6 = \frac{(x^6-1)(x-1)}{(x^3-1)(x^2-1)} = \frac{x^3+1}{x+1} = x^2 - x + 1$. (iv) By (iii), $\Phi_{p^k} = (x^{p^k}-1)/(x^{p^{k-1}}-1) = \Phi_p(x^{p^{k-1}})$, and $\Phi_p = (x^p-1)/(x-1)$. For odd $m > 1$ the divisors of $2m$ are the $d$ and $2d$ with $d \mid m$, and $\mu(2m/d) = -\mu(m/d)$, $\mu(2m/2d) = \mu(m/d)$; so $\Phi_{2m} = \prod_{d|m}((x^{2d}-1)/(x^d-1))^{\mu(m/d)} = \prod_{d|m}(x^d+1)^{\mu(m/d)}$. On the other hand $\Phi_m(-x) = \prod_{d|m}(-(x^d+1))^{\mu(m/d)}$ ($d$ is odd) $= (-1)^{\sum_{d|m}\mu(m/d)}\prod_{d|m}(x^d+1)^{\mu(m/d)}$, and the sign is $+1$ as $m > 1$. (v) $\Phi_1(1) = 0$ and $\Phi_{p^k}(1) = \Phi_p(1) = p$ by (iv). For $n > 1$, (i) gives $1 + x + \cdots + x^{n-1} = \prod_{d|n,\, d>1}\Phi_d(x)$, and at $x = 1$: $n = \prod_{d|n,d>1}\Phi_d(1)$. The prime power divisors $d = p^j > 1$ contribute $\prod_p p^{v_p(n)} = n$, so the product of $\Phi_d(1)$ over the divisors $d > 1$ that are not prime powers is 1; by induction on $n$ all of them except $d = n$ are 1, hence so is $\Phi_n(1)$ when $n$ is not a prime power. Similarly $x^n - 1$ at $x = 0$ gives $-1 = \Phi_1(0)\prod_{d|n,d>1}\Phi_d(0) = -\prod_{d|n,d>1}\Phi_d(0)$, and induction gives $\Phi_n(0) = 1$. (vi) Put $g(x) = \Phi_{p^k}(x+1)$, monic of degree $\varphi(p^k)$. Modulo $p$, $\Phi_{p^k} = (x^{p^k}-1)/(x^{p^{k-1}}-1) \equiv (x-1)^{p^k}/(x-1)^{p^{k-1}} = (x-1)^{\varphi(p^k)}$ (Proposition 2.16; division by a monic polynomial commutes with reduction), so $g \equiv x^{\varphi(p^k)} \pmod p$, and $g(0) = \Phi_{p^k}(1) = p$. By Eisenstein (Theorem 2.33) $g$ is irreducible, and so is $\Phi_{p^k}$, since $x \mapsto x+1$ is an automorphism of $\mathbb{Q}[x]$ preserving degrees. □

| $n$ | $\Phi_n(x)$ | $\Phi_n(1)$ | $n$ | $\Phi_n(x)$ | $\Phi_n(1)$ |
|---|---|---|---|---|---|
| 1 | $x-1$ | 0 | 7 | $x^6+x^5+x^4+x^3+x^2+x+1$ | 7 |
| 2 | $x+1$ | 2 | 8 | $x^4+1$ | 2 |
| 3 | $x^2+x+1$ | 3 | 9 | $x^6+x^3+1$ | 3 |
| 4 | $x^2+1$ | 2 | 10 | $x^4-x^3+x^2-x+1$ | 1 |
| 5 | $x^4+x^3+x^2+x+1$ | 5 | 11 | $x^{10}+x^9+\cdots+x+1$ | 11 |
| 6 | $x^2-x+1$ | 1 | 12 | $x^4-x^2+1$ | 1 |

**Proposition 2.59.** *Let $p$ be a prime with $p \nmid n$, and write $\bar\Phi_n$ for the reduction of $\Phi_n$ modulo $p$.*

*(i) $\bar\Phi_n$ divides $x^n - 1$ in $\mathbb{F}_p[x]$, so it is separable; in any field of characteristic $p$ its roots are exactly the elements of order $n$.*

*(ii) For $a \in \mathbb{Z}$: $p \mid \Phi_n(a)$ iff $a \bmod p$ has order exactly $n$ in $\mathbb{F}_p^\times$. Hence every prime divisor $p \nmid n$ of $\Phi_n(a)$ satisfies $p \equiv 1 \pmod n$.*

*(iii) There are infinitely many primes $p \equiv 1 \pmod n$.*

*(iv) $\bar\Phi_n$ is a product of distinct linear factors in $\mathbb{F}_p[x]$ iff $p \equiv 1 \pmod n$.*

*The complete factorisation pattern ($\varphi(n)/f$ irreducible factors of degree $f$, $f$ the order of $p$ modulo $n$) is Chapter 5.*

*Proof.* (i) Reduce Theorem 2.58(i) modulo $p$: $x^n - 1 = \prod_{d|n}\bar\Phi_d$, and $x^n - 1$ is separable over $\mathbb{F}_p$ (Proposition 2.31(iii)). Let $K$ have characteristic $p$; enlarging $K$ (Theorem 2.38(iii)) we may assume $x^n - 1$ splits in $K$, which does not change orders of elements of $K$. The roots of $x^n - 1$ are the $n$ elements of $\mu_n(K)$, those of order $d$ being, by induction on $n$, the roots of $\bar\Phi_d$ for $d < n$; since the roots of the $\bar\Phi_d$ are pairwise distinct, the roots of $\bar\Phi_n$ are the remaining elements, of order exactly $n$. (ii) $p \mid \Phi_n(a)$ iff $\bar a$ is a root of $\bar\Phi_n$ in $\mathbb{F}_p$, iff $\bar a$ has order $n$ in $\mathbb{F}_p^\times$ by (i) ($\bar a \neq 0$ because $\bar\Phi_n(0) = 1$ for $n \geq 2$; for $n = 1$ the claim is clear). Then $n \mid p - 1$. (iii) For $n = 1$ this is Euclid's theorem; let $n \geq 2$ and $S$ a finite set of primes, $N = n\prod_{p\in S} p$. For an integer

$t$, $\Phi_n(tN) \equiv \Phi_n(0) = 1 \pmod N$, so $\Phi_n(tN)$ is prime to $n$ and to every $p \in S$. For $t$ large, $|\Phi_n(tN)| > 1$ ($\Phi_n$ is monic of positive degree), so it has a prime divisor $p \notin S$, $p \nmid n$, and $p \equiv 1 \pmod n$ by (ii). (iv) If $p \equiv 1 \pmod n$, the cyclic group $\mathbb{F}_p^\times$ contains $\varphi(n)$ elements of order $n$, all roots of $\bar\Phi_n$, which has degree $\varphi(n)$. Conversely a root in $\mathbb{F}_p$ has order $n$, so $n \mid p-1$. □

For example $\Phi_{12}(2) = 16 - 4 + 1 = 13 \equiv 1 \pmod{12}$ and $\Phi_9(2) = \Phi_3(8) = 73 \equiv 1 \pmod 9$; and 2 has order 12 modulo 13 and order 9 modulo 73. In the paper the primes $q \equiv 1 \pmod \ell$ of Lemma lem:tamfactor are exactly those with $\mu_\ell \subset \mathbb{F}_q$ (Proposition 2.50(i)).

## 2.9 Quadratic residues and quadratic reciprocity

The paper's quadratic characters are Kronecker symbols: $\chi_{-3} = \left(\frac{-3}{\cdot}\right)$, $\chi_{-4} = \left(\frac{-4}{\cdot}\right)$, and $\chi_{q^*} = \left(\frac{q^*}{\cdot}\right)$, the character of $\mathbb{Q}(\sqrt{q^*})$ by which Lemma lem:twist twists. That $\chi_{q^*}$ is "the quadratic character of conductor $q$", i.e. that $\left(\frac{q^*}{n}\right) = \left(\frac{n}{q}\right)$ for all $n$, is quadratic reciprocity (Proposition 2.69). We prove reciprocity by Gauss sums in characteristic $p$, extend it to Jacobi and Kronecker symbols, and show that $\left(\frac{D}{\cdot}\right)$ is a Dirichlet character modulo $|D|$ for every discriminant $D$. Chapter 6 gives a second proof and the meaning of $\chi_D(p)$ ($D$ fundamental) as a splitting law.

**Definition 2.60.** For an odd prime $p$ and $a \in \mathbb{Z}$ the *Legendre symbol* is $\left(\frac{a}{p}\right) = 0$ if $p \mid a$, 1 if $a$ is a nonzero square modulo $p$ (a *quadratic residue*), and $-1$ otherwise (a *non-residue*).

**Proposition 2.61.** *Let $p$ be an odd prime.*

*(i) (Euler's criterion)* $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod p$.

*(ii)* $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$; *$a \mapsto \left(\frac{a}{p}\right)$ is the unique homomorphism of order 2 from $(\mathbb{Z}/p)^\times$ to $\{\pm 1\}$; there are $(p-1)/2$ residues and $(p-1)/2$ non-residues.*

*(iii)* $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.

*(iv) For $p = 3$:* $\left(\frac{a}{3}\right) \equiv a \pmod 3$ *for all $a$.*

*Proof.* (i) For $p \mid a$ both sides are 0; otherwise it is Theorem 2.52(iv) with $q = p$. (ii) Multiplicativity follows from (i), as $1 \not\equiv -1 \pmod p$. A homomorphism onto $\{\pm 1\}$ has a kernel of index 2, and the cyclic group $(\mathbb{Z}/p)^\times$ has only one subgroup of index 2, the squares (Theorem 2.52(i)). (iii) is (i) for $a = -1$. (iv) The nonzero squares modulo 3 are $\{1\}$, so $\left(\frac{2}{3}\right) = -1 \equiv 2$. □

For $p = 7$ the squares of $1, \dots, 6$ are $1, 4, 2, 2, 4, 1$, so the residues are $\{1, 2, 4\}$ and the non-residues $\{3, 5, 6\}$. Euler's criterion agrees: $3^3 = 27 \equiv -1$ and $2^3 = 8 \equiv 1 \pmod 7$. In particular $\left(\frac{-1}{7}\right) = -1$, as (iii) says for $7 \equiv 3 \pmod 4$.

**Proposition 2.62** (the quadratic Gauss sum)**.** *Let $q$ be an odd prime, $K$ a field of characteristic $\neq q$ containing a primitive $q$-th root of unity $\zeta$, and $g = \sum_{a=1}^{q-1} \left(\frac{a}{q}\right)\zeta^a \in K$.*

*(i) $g^2 = q^* = (-1)^{(q-1)/2} q$ in $K$; in particular $g \neq 0$.*

*(ii) For $b$ prime to $q$, $\sum_{a=1}^{q-1} \left(\frac{a}{q}\right)\zeta^{ab} = \left(\frac{b}{q}\right) g$.*

*(iii) $\sqrt{q^*} \in \mathbb{Q}(\zeta_q)$: over $\mathbb{C}$ with $\zeta = \zeta_q$, $g \in \mathbb{Z}[\zeta_q]$ and $g^2 = q^*$.*

*Proof.* For $b \in \mathbb{Z}$, $\sum_{a=0}^{q-1} \zeta^{ab}$ is $q$ if $q \mid b$ and 0 otherwise: if $q \nmid b$, $\zeta^b \neq 1$ is a root of $x^q - 1 = (x-1)(1 + x + \cdots + x^{q-1})$, hence of the second factor. So $\sum_{a=1}^{q-1} \zeta^{ab}$ is $q-1$ if $q \mid b$ and $-1$ otherwise. Also $\sum_{c=1}^{q-1} \left(\frac{c}{q}\right) = 0$ by Proposition 2.61(ii). (ii) As $a$ runs over $(\mathbb{Z}/q)^\times$ so does $c = ab$, and $\left(\frac{a}{q}\right) = \left(\frac{c}{q}\right)\left(\frac{b}{q}\right)$ since $\left(\frac{b}{q}\right)^{-1} = \left(\frac{b}{q}\right)$. (i) $g^2 = \sum_{a,b} \left(\frac{ab}{q}\right) \zeta^{a+b}$. Substitute $b = ac$ ($c$ runs over $(\mathbb{Z}/q)^\times$ for fixed $a$): $\left(\frac{a \cdot ac}{q}\right) = \left(\frac{a^2}{q}\right)\left(\frac{c}{q}\right) = \left(\frac{c}{q}\right)$. The inner sum over $a$ is $\sum_{a=1}^{q-1} \zeta^{a(1+c)}$, which by the first paragraph is $q-1$ for the one value $c \equiv -1$ and $-1$ for the others; then $\sum_{c \neq -1} \left(\frac{c}{q}\right) = 0 - \left(\frac{-1}{q}\right)$. So

$$g^2 = \sum_{c=1}^{q-1} \left(\frac{c}{q}\right) \sum_{a=1}^{q-1} \zeta^{a(1+c)} = \left(\frac{-1}{q}\right)(q-1) - \sum_{c \neq -1} \left(\frac{c}{q}\right) = \left(\frac{-1}{q}\right)(q-1) + \left(\frac{-1}{q}\right) = \left(\frac{-1}{q}\right) q,$$

which is $q^*$ by Proposition 2.61(iii). It is nonzero because $\operatorname{char} K \neq q$. (iii) is (i) over $\mathbb{C}$. □

For $q = 3$ over $\mathbb{C}$: $g = \zeta_3 - \zeta_3^2 = 1 + 2\zeta_3 = \sqrt{-3}$, and $g^2 = -3 = 3^*$. For $q = 5$: $g = \zeta_5 - \zeta_5^2 - \zeta_5^3 + \zeta_5^4 = 2\cos\frac{2\pi}{5} - 2\cos\frac{4\pi}{5} = \sqrt{5}$ (Exercise 2.105). The point of allowing an arbitrary field $K$ is the proof of reciprocity below, where $K$ is a finite field of characteristic $p \neq q$ and the identity $g^2 = q^*$ is used inside $K$.

*Remark* 2.63 (the sign of the Gauss sum). Over $\mathbb{C}$ with $\zeta = \zeta_q$, Gauss proved $g = \sqrt{q}$ if $q \equiv 1 \pmod 4$ and $g = i\sqrt{q}$ if $q \equiv 3 \pmod 4$. We do not prove this and do not use it; see [87, Chapter 6, Section 6.4]. (Checked numerically for all $q < 60$.)

**Theorem 2.64** (quadratic reciprocity)**.** *Let $p \neq q$ be odd primes.*

*(i)* $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$*; equivalently* $\left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right)$.

*(ii) (First supplement)* $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.

*(iii) (Second supplement)* $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$*, that is, 2 is a square modulo $p$ iff* $p \equiv \pm 1 \pmod 8$.

*Proof.* (i) Let $f$ be the order of $p$ modulo $q$. By Proposition 2.50(ii) (and Theorem 2.47) the field $K = \mathbb{F}_{p^f}$ contains a primitive $q$-th root of unity $\zeta$; its characteristic $p$ is different from $q$. Let $g = \sum_a \left(\frac{a}{q}\right)\zeta^a \in K$. By Proposition 2.62(i), $g^2 = q^*$, an element of $\mathbb{F}_p \subseteq K$. By Proposition 2.16, and because $\left(\frac{a}{q}\right)^p = \left(\frac{a}{q}\right)$ ($p$ odd),

$$g^p = \sum_a \left(\frac{a}{q}\right)^p \zeta^{ap} = \sum_a \left(\frac{a}{q}\right)\zeta^{ap} = \left(\frac{p}{q}\right) g$$

by Proposition 2.62(ii). As $g \neq 0$, $g^{p-1} = \left(\frac{p}{q}\right)$. But $g^{p-1} = (g^2)^{(p-1)/2} = (q^*)^{(p-1)/2} = \left(\frac{q^*}{p}\right)$ in $\mathbb{F}_p$ by Euler's criterion. So $\left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right)$ in $\mathbb{F}_p$, hence as integers ($1 \neq -1$ in $\mathbb{F}_p$). Finally $\left(\frac{q^*}{p}\right) = \left(\frac{-1}{p}\right)^{(q-1)/2}\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}\left(\frac{q}{p}\right)$. (ii) is Proposition 2.61(iii). (iii) As $p$ is odd, $8 \mid p^2 - 1$, so $\mathbb{F}_{p^2}$ contains an element $\zeta$ of order 8 (Proposition 2.50(i)). Then $\zeta^4 = -1$ and $\zeta^{-2} = \zeta^6 = -\zeta^2$. Put $\tau = \zeta + \zeta^{-1}$; then $\tau^2 = \zeta^2 + 2 + \zeta^{-2} = 2$, so $\tau \neq 0$. Now $\tau^p = \zeta^p + \zeta^{-p}$, which is $\tau$ if $p \equiv \pm 1 \pmod 8$, and if $p \equiv \pm 3 \pmod 8$ it is $\zeta^3 + \zeta^{-3} = \zeta^3 + \zeta^5 = \zeta^4(\zeta^{-1} + \zeta) = -\tau$. Hence $2^{(p-1)/2} = \tau^{p-1} = \pm 1$ accordingly, and Euler's criterion concludes. Note that $(-1)^{(p^2-1)/8}$ is 1 for $p \equiv \pm 1$ and $-1$ for $p \equiv \pm 3 \pmod 8$. □

**Definition 2.65.** For an odd integer $n = \prod_i p_i^{e_i} > 0$ and $a \in \mathbb{Z}$, the *Jacobi symbol* is $\left(\frac{a}{n}\right) = \prod_i \left(\frac{a}{p_i}\right)^{e_i}$ (Legendre symbols; $\left(\frac{a}{1}\right) = 1$).

**Proposition 2.66.** *Let $m, n > 0$ be odd.*

*(i)* $\left(\frac{a}{n}\right)$ *is multiplicative in $a$ and in $n$, depends only on $a \bmod n$, and vanishes iff* $\gcd(a,n) > 1$.

*(ii) (Reciprocity and supplements.) If* $\gcd(m,n) = 1$, *then* $\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}}$. *Moreover* $\left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}$ *and* $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$.

*(iii)* $\left(\frac{a}{n}\right) = 1$ *does not imply that $a$ is a square modulo $n$:* $\left(\frac{2}{15}\right) = \left(\frac{2}{3}\right)\left(\frac{2}{5}\right) = (-1)(-1) = 1$, *but 2 is not a square modulo 3, hence not modulo 15. (If* $\left(\frac{a}{n}\right) = -1$, *then $a$ is not a square modulo $n$.)*

*(iv) Reciprocity computes* $\left(\frac{a}{n}\right)$ *without factoring $n$: reduce $a$ modulo $n$, remove the factors $-1$ and 2 with (ii), and invert the remaining odd symbol. For instance, 1009 being prime,* $\left(\frac{713}{1009}\right) = \left(\frac{1009}{713}\right) = \left(\frac{296}{713}\right) = \left(\frac{2}{713}\right)^3\left(\frac{37}{713}\right) = \left(\frac{713}{37}\right) = \left(\frac{10}{37}\right) = \left(\frac{2}{37}\right)\left(\frac{5}{37}\right) = (-1)\left(\frac{37}{5}\right) = -\left(\frac{2}{5}\right) = 1$, *so 713 is a square modulo 1009 (PARI's* `kronecker(713,1009)` *returns 1).*

*Proof.* (i) is inherited from the Legendre symbol factor by factor. (ii) Let $e(n) = \frac{n-1}{2}$ and $w(n) = \frac{n^2-1}{8}$ modulo 2, for odd $n$. Both are additive: $mn-1 = (m-1)(n-1)+(m-1)+(n-1)$ with $4 \mid (m-1)(n-1)$, and $m^2n^2-1 = (m^2-1)(n^2-1)+(m^2-1)+(n^2-1)$ with $64 \mid (m^2-1)(n^2-1)$. Write $m = \prod p_i$, $n = \prod q_j$ (with repetitions). By Theorem 2.64, $\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = \prod_{i,j}\left(\frac{p_i}{q_j}\right)\left(\frac{q_j}{p_i}\right) = (-1)^{\sum_{i,j} e(p_i)e(q_j)} = (-1)^{e(m)e(n)}$, and likewise $\left(\frac{-1}{n}\right) = (-1)^{\sum_j e(q_j)} = (-1)^{e(n)}$, $\left(\frac{2}{n}\right) = (-1)^{\sum_j w(q_j)} = (-1)^{w(n)}$. (iii) A square modulo 15 is a square modulo 3. If $\left(\frac{a}{n}\right) = -1$, some $\left(\frac{a}{p_i}\right) = -1$. (iv) In the example $713 \equiv 1 \pmod 8$, $37 \equiv 5 \pmod 8$, and $713 \equiv 1$, $37 \equiv 1 \pmod 4$, so no signs arise from (ii) in the inversions. □

**Definition 2.67.** The *Kronecker symbol* $\left(\frac{a}{n}\right)$, for $a, n \in \mathbb{Z}$, extends the Jacobi symbol using Convention 2.4: for $n = s2^e\prod_i p_i^{e_i} \neq 0$ with a sign $s = \pm 1$ and odd primes $p_i$,

$$\left(\frac{a}{n}\right) = \left(\frac{a}{s}\right)\left(\frac{a}{2}\right)^e \prod_i \left(\frac{a}{p_i}\right)^{e_i},$$

and $\left(\frac{a}{0}\right) = 1$ if $a = \pm 1$, 0 otherwise. A *discriminant* is an integer $D \neq 0$ with $D \equiv 0$ or $1 \pmod 4$. A *fundamental discriminant* is a discriminant $D$ with either $D \equiv 1 \pmod 4$ squarefree, or $D = 4m$ with $m \equiv 2, 3 \pmod 4$ squarefree. The *prime discriminants* are $-4$, $8$, $-8$ and $q^* = (-1)^{(q-1)/2}q$ for odd primes $q$. For a fundamental discriminant $D$ put $\chi_D(a) := \left(\frac{D}{a}\right)$ for all $a \in \mathbb{Z}$; for a discriminant that is not fundamental no name $\chi_D$ is introduced, and the function $\left(\frac{D}{\cdot}\right)$ is written as a Kronecker symbol (Convention 2.5).

The definition is made so that $n \mapsto \left(\frac{D}{n}\right)$, a function of the lower entry, is a character; for the Jacobi symbol it is the upper entry that varies. For $D = -3$ and $n = 1, \dots, 6$ the values are $1, -1, 0, 1, -1, 0$: at $n = 2$ because $-3 \equiv 5 \pmod 8$, at $n = 4$ because $\left(\frac{-3}{2}\right)^2 = 1$, and at $n = 5$ because $-3 \equiv 2$ is not a square modulo 5. These are the values of $\left(\frac{n}{3}\right)$, a first instance of Proposition 2.69. The congruence conditions on fundamental discriminants say that $D$ is the discriminant of the quadratic field $\mathbb{Q}(\sqrt{D})$ (Chapter 6): $-3$, $-4$, $5$, $8$, $-8$, $12$ are fundamental, while $-12 = -3 \cdot 2^2$ and $20 = 5 \cdot 2^2$ are not.

**Proposition 2.68.** *Let $D$ be a discriminant.*

*(i)* $n \mapsto \left(\frac{D}{n}\right)$ *is completely multiplicative on* $\mathbb{Z} \setminus \{0\}$ *and vanishes exactly when* $\gcd(D,n) > 1$. *For nonzero $a, b$,* $\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right)\left(\frac{b}{n}\right)$.

*(ii)* *If* $D \equiv 1 \pmod 4$, *then* $\left(\frac{D}{n}\right) = \left(\frac{n}{|D|}\right)$ *(Jacobi symbol) for every* $n > 0$.

*(iii)* *For* $n, n' > 0$ *with* $n \equiv n' \pmod{|D|}$, $\left(\frac{D}{n}\right) = \left(\frac{D}{n'}\right)$; *and* $\left(\frac{D}{-n}\right) = \operatorname{sgn}(D)\left(\frac{D}{n}\right)$. *Hence* $\left(\frac{D}{\cdot}\right)$ *is a Dirichlet character modulo* $|D|$ *(Definition 2.71), real (with values in* $\{0, \pm 1\}$*), with value* $\operatorname{sgn} D$ *at* $-1$. *For fundamental* $D$ *this character is* $\chi_D$.

*(iv)* *Values at primes:* $\left(\frac{D}{p}\right)$ *is the Legendre symbol for odd* $p$; $\left(\frac{D}{2}\right) = 0$ *if* $4 \mid D$, $1$ *if* $D \equiv 1 \pmod 8$, $-1$ *if* $D \equiv 5 \pmod 8$.

*(v)* *For discriminants* $D, D'$, $\left(\frac{DD'}{n}\right) = \left(\frac{D}{n}\right)\left(\frac{D'}{n}\right)$ *for all* $n \in \mathbb{Z}$; *and for* $f \neq 0$, $\left(\frac{Df^2}{n}\right) = \left(\frac{D}{n}\right)$ *for* $\gcd(n, f) = 1$. *So for fundamental* $D$ *the character* $\left(\frac{Df^2}{\cdot}\right)$ *modulo* $|D|f^2$ *is* $\chi_D$ *made imprimitive at the primes dividing* $f$ *(Proposition 2.75), and in particular* $\chi_{DD'} = \chi_D\chi_{D'}$ *when* $D$, $D'$ *and* $DD'$ *are fundamental.*

*Proof.* (i) Multiplicativity in $n$ holds by construction (for the sign, $\left(\frac{a}{-1}\right)^2 = 1$ when $a \neq 0$). Multiplicativity in nonzero $a$ holds for each factor: Legendre symbols, $\left(\frac{\cdot}{2}\right)$ (on odd $a$ it is the character of $(\mathbb{Z}/8)^\times$ with kernel $\{\pm 1\}$, and it is 0 on even $a$), and $\left(\frac{\cdot}{-1}\right) = \operatorname{sgn}$. A factor vanishes iff $p_i \mid D$, or $e \geq 1$ and $D$ is even; that is, iff a prime dividing $n$ divides $D$. (ii) Let $n > 0$, $n = 2^e m$ with $m$ odd. If $\gcd(D, n) > 1$ both sides vanish; assume not. For the odd part, write $D = s|D|$, $s = \operatorname{sgn} D$. By (i) and Proposition 2.66(ii), $\left(\frac{D}{m}\right) = \left(\frac{s}{m}\right)\left(\frac{|D|}{m}\right) = \left(\frac{s}{m}\right)(-1)^{\frac{m-1}{2}\frac{|D|-1}{2}}\left(\frac{m}{|D|}\right)$. If $s = 1$ then $|D| \equiv 1 \pmod 4$ and the sign is 1. If $s = -1$ then $|D| \equiv 3 \pmod 4$ and $(-1)^{\frac{m-1}{2}} = \left(\frac{-1}{m}\right)$ cancels $\left(\frac{s}{m}\right)$. So $\left(\frac{D}{m}\right) = \left(\frac{m}{|D|}\right)$. For the factor 2: $\left(\frac{D}{2}\right) = 1$ iff $D \equiv 1 \pmod 8$ iff $|D| \equiv \pm 1 \pmod 8$ (as $D \equiv 1 \pmod 4$), iff $\left(\frac{2}{|D|}\right) = 1$ (Proposition 2.66(ii)). Multiply. (iii) For $D = 1$ the symbol $\left(\frac{1}{n}\right)$ is 1 for every $n$ (each factor is), and everything is trivial; let $D \neq 1$, so that $|D| \geq 3$ and $n = |D| - 1$ is positive. If $D \equiv 1 \pmod 4$, (ii) shows that $\left(\frac{D}{n}\right)$ depends only on $n \bmod |D|$ for $n > 0$, and $\left(\frac{D}{|D|-1}\right) = \left(\frac{-1}{|D|}\right) = (-1)^{(|D|-1)/2} = \operatorname{sgn} D$. If $D = 2^t D'$ with $D'$ odd and $t \geq 2$: for even $n$ both $\left(\frac{D}{n}\right)$ and $\left(\frac{D}{n'}\right)$ vanish ($n' \equiv n \pmod 4$ is even). For odd $n > 0$, as in (ii), $\left(\frac{D}{n}\right) = \left(\frac{2}{n}\right)^t\left(\frac{\operatorname{sgn} D'}{n}\right)(-1)^{\frac{n-1}{2}\frac{|D'|-1}{2}}\left(\frac{n}{|D'|}\right)$. Here $\left(\frac{2}{n}\right)$ depends on $n \bmod 8$, the next two factors on $n \bmod 4$, and the last on $n \bmod |D'|$; if $t \geq 3$ the period $8|D'|$ divides $|D|$, and if $t = 2$ the factor $\left(\frac{2}{n}\right)^2 = 1$ disappears and the period $4|D'| = |D|$. Evaluating at $n = |D| - 1$ (so $n \equiv -1$ modulo 4, modulo $|D'|$, and modulo 8 if $t \geq 3$): $\left(\frac{2}{n}\right)^t = 1$; $\left(\frac{n}{|D'|}\right) = (-1)^{(|D'|-1)/2}$ cancels the sign $(-1)^{\frac{n-1}{2}\frac{|D'|-1}{2}} = (-1)^{(|D'|-1)/2}$; and $\left(\frac{\operatorname{sgn} D'}{n}\right) = \operatorname{sgn} D'$. So in all cases $\left(\frac{D}{|D|-1}\right) = \operatorname{sgn} D$. Now $\left(\frac{D}{-n}\right) = \left(\frac{D}{-1}\right)\left(\frac{D}{n}\right) = \operatorname{sgn}(D)\left(\frac{D}{n}\right)$ by (i). Hence for $n > 0$, $\left(\frac{D}{-n}\right) = \left(\frac{D}{|D|-1}\right)\left(\frac{D}{n}\right) = \left(\frac{D}{(|D|-1)n}\right)$, and $(|D| - 1)n \equiv -n \pmod{|D|}$: the symbol is periodic modulo $|D|$ on $\mathbb{Z} \setminus \{0\}$; at multiples of $|D|$ (including 0) it vanishes, as $|D| > 1$. It is completely multiplicative and vanishes exactly off $(\mathbb{Z}/|D|)^\times$, so it comes from the homomorphism $(\mathbb{Z}/|D|)^\times \to \{\pm 1\}$ it induces. (iv) is the definition. (v) By (i) for $n \neq 0$; at $n = 0$ both sides vanish unless $D = D' = 1$. And $\left(\frac{Df^2}{n}\right) = \left(\frac{D}{n}\right)\left(\frac{f}{n}\right)^2 = \left(\frac{D}{n}\right)$ when $\gcd(f, n) = 1$ and $n \neq 0$; for $n = 0$ the condition forces $f = \pm 1$. The conductor statement is Proposition 2.75(i) and (iii), the two characters agreeing at the integers prime to $Df$. □

**Proposition 2.69.** *Let* $q$ *be an odd prime and* $q^* = (-1)^{(q-1)/2}q$.

*(i)* $q^* \equiv 1 \pmod 4$*; it is the prime discriminant attached to* $q$.

*(ii)* *For every* $n \in \mathbb{Z}$, $\left(\frac{q^*}{n}\right) = \left(\frac{n}{q}\right)$ *(the Legendre symbol of* $n \bmod q$*). So* $\chi_{q^*} = \left(\frac{\cdot}{q}\right)$ *as functions on* $\mathbb{Z}$.

*(iii)* $\chi_{q^*}$ *is the unique Dirichlet character of order* 2 *and conductor* $q$*; this is why the paper calls it "the quadratic character of conductor* $q$*". By contrast, for* $q \equiv 3 \pmod 4$ *the function* $n \mapsto \left(\frac{q}{n}\right)$ *agrees on odd* $n$ *with the character* $\chi_{4q} = \chi_{-4}\chi_{q^*}$*, of conductor* $4q$*, but on all of* $\mathbb{Z}$ *it is not periodic modulo* $4q$*:* $\left(\frac{q}{4q+2}\right) = -\left(\frac{q}{2}\right) \neq 0$*. For* $q = 3$*:* $\left(\frac{3}{2}\right) = -1$ *but* $\left(\frac{3}{14}\right) = 1$*.*

*(iv) The same holds for the coefficient prime* $\ell$*:* $\ell^* \equiv 1 \pmod 4$*, and for primes* $p \neq \ell$*,* $\ell^*$ *is a square modulo* $p$ *(for odd* $p$*), resp.* $\ell^* \equiv 1 \pmod 8$ *(for* $p = 2$*), iff* $p$ *is a square modulo* $\ell$*.*

*(v)* $3^* = -3$*,* $5^* = 5$*,* $7^* = -7$*,* $11^* = -11$*,* $13^* = 13$*; so the characters of conductor* 3*,* 5*,* 7*,* 11*,* 13 *are* $\chi_{-3}$*,* $\chi_5$*,* $\chi_{-7}$*,* $\chi_{-11}$*,* $\chi_{13}$*.*

*Proof.* (i) If $q \equiv 1 \pmod 4$, $q^* = q$; if $q \equiv 3$, $q^* = -q \equiv 1 \pmod 4$. (ii) Both sides are completely multiplicative in $n \neq 0$ (Proposition 2.68(i), Proposition 2.61(ii)), so it suffices to compare them at $n = -1$, 2, odd primes $p$, and to check $n = 0$. At $-1$: $\left(\frac{q^*}{-1}\right) = \operatorname{sgn} q^* = (-1)^{(q-1)/2} = \left(\frac{-1}{q}\right)$. At 2: $\left(\frac{q^*}{2}\right) = 1$ iff $q^* \equiv 1 \pmod 8$ iff $q \equiv 1 \pmod 8$ (when $q \equiv 1 \pmod 4$) or $q \equiv 7 \pmod 8$ (when $q \equiv 3 \pmod 4$), i.e. iff $q \equiv \pm 1 \pmod 8$ iff $\left(\frac{2}{q}\right) = 1$ (Theorem 2.64(iii)). At odd $p \neq q$ it is Theorem 2.64(i); at $p = q$ and at $n = 0$ both sides vanish. (Checked numerically for all $q < 200$ and $0 < |n| \leq 300$.) (iii) By (ii), $\chi_{q^*}$ is a nontrivial character modulo the prime $q$, so its conductor is $q$ (the only other divisor, 1, belongs to the trivial character). The characters modulo $q$ form a group isomorphic to the cyclic group $(\mathbb{Z}/q)^\times$ (Proposition 2.72), which has a unique element of order 2. For $q \equiv 3 \pmod 4$, $q = (-1) \cdot q^*$, so for odd $n > 0$, $\left(\frac{q}{n}\right) = \left(\frac{-1}{n}\right)\left(\frac{q^*}{n}\right) = \chi_{-4}(n)\chi_{q^*}(n)$ (for odd $n > 0$, $\left(\frac{-1}{n}\right) = (-1)^{(n-1)/2} = \left(\frac{-4}{n}\right)$), and $\chi_{-4}\chi_{q^*} = \chi_{4q}$ is primitive of conductor $4q$ by Corollary 2.76. For odd $n < 0$ both sides equal their values at $-n$: $\left(\frac{q}{-1}\right) = \operatorname{sgn} q = 1$, and $\chi_{4q}(-1) = \chi_{-4}(-1)\chi_{q^*}(-1) = (-1)(-1) = 1$ since $q^* < 0$. Non-periodicity: $2q + 1$ is odd, positive, prime to $q$ and $\equiv 3 \pmod 4$, so by Proposition 2.66(ii) $\left(\frac{q}{2q+1}\right) = -\left(\frac{2q+1}{q}\right) = -\left(\frac{1}{q}\right) = -1$, and $\left(\frac{q}{4q+2}\right) = \left(\frac{q}{2}\right)\left(\frac{q}{2q+1}\right) = -\left(\frac{q}{2}\right)$, which is nonzero and differs from $\left(\frac{q}{2}\right)$. For $q = 3$: $3 \equiv 3 \pmod 8$ gives $\left(\frac{3}{2}\right) = -1$, and $\left(\frac{3}{7}\right) = -\left(\frac{7}{3}\right) = -1$ gives $\left(\frac{3}{14}\right) = 1$. (iv) is (ii) with $q = \ell$, read at $n = p$. (v) is arithmetic. □

*In the paper* 2.70. Lemma lem:twist begins: "Let $q$ be an odd prime, $\chi = \chi_{q^*}$ the quadratic character of conductor $q$ and $E^{(q)}$ the twist of $E$ by $\chi$, that is, by $\mathbb{Q}(\sqrt{q^*})$." The sign in $q^*$ is forced: by Proposition 2.69, $\chi_{q^*} = \left(\frac{\cdot}{q}\right)$ is ramified only at $q$, while for $q \equiv 3 \pmod 4$ the field $\mathbb{Q}(\sqrt{q})$ has discriminant $4q$ and its character $\chi_{4q}$ is ramified at 2 as well; the twist by it would in general change the reduction at 2 and violate part (i) of the lemma. The same character appears in Theorem thm:main through $\ell^*$ (the field $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$, by Proposition 2.62(iii)), in Theorems thm:twist and thm:quant through $\chi_S = \prod_{q \in S} \chi_{q^*}$, and in Example ex:qual, where the twists of 121a1 and 539c1 are by $\chi_{-11}$ and $\chi_{-7}$ ($11^* = -11$, $7^* = -7$).

## 2.10 Dirichlet characters, conductors and Gauss sums; the paper's characters

The paper twists newforms by Dirichlet characters, $a_n(g \otimes \chi) = \chi(n)a_n(g)$ (Lemma lem:twist(iii), Theorems thm:twist and thm:quant, Example ex:qual), and the level of the twist depends on the conductor of $\chi$ (Atkin-Li). This section defines Dirichlet characters and their conductors, proves orthogonality and the basic properties of Gauss sums, and assembles the paper's characters in one table (Proposition 2.78). Their interpretation as characters of $G_{\mathbb{Q}}$, with $\chi(\mathrm{Frob}_p) = \chi(p)$, is Chapter 6; here they are functions on $\mathbb{Z}$.

**Definition 2.71.** A *Dirichlet character modulo* $n \geq 1$ is a function $\chi\colon \mathbb{Z} \to \mathbb{C}$ of the form $\chi(a) = \tilde{\chi}(a \bmod n)$ for $\gcd(a,n) = 1$ and $\chi(a) = 0$ for $\gcd(a,n) > 1$, where $\tilde{\chi}\colon (\mathbb{Z}/n)^\times \to \mathbb{C}^\times$ is a homomorphism. The *principal character* $\chi_0$ modulo $n$ has $\tilde{\chi} = 1$. $\chi$ is *even* or *odd* as $\chi(-1) = 1$ or $-1$. The characters modulo $n$ form a group $X(n)$ under pointwise multiplication (of the $\tilde{\chi}$). Characters with values in another field $k$ (for instance $\mathbb{F}_\ell$, $\mathbb{F}_9$) are defined in the same way with $\tilde{\chi}\colon (\mathbb{Z}/n)^\times \to k^\times$. The example used later is $\omega_\ell\colon \mathbb{Z} \to \mathbb{F}_\ell$, $\omega_\ell(a) = a \bmod \ell$, an $\mathbb{F}_\ell^\times$-valued character modulo $\ell$ of order $\ell - 1$: the Dirichlet character of the reduced cyclotomic character $\bar{\chi}_\ell$ (Chapters 5 and 6).

**Proposition 2.72.** *Let $\chi$ be a Dirichlet character modulo $n$.*

*(i) $\chi$ is completely multiplicative and $n$-periodic; for $\gcd(a,n) = 1$, $\chi(a)$ is a root of unity whose order divides the exponent of $(\mathbb{Z}/n)^\times$ (in particular divides $\varphi(n)$), and $\overline{\chi(a)} = \chi(a)^{-1}$; $\chi(-1) = \pm 1$.*

*(ii) $X(n)$ is a finite abelian group isomorphic (non-canonically) to $(\mathbb{Z}/n)^\times$; in particular $\#X(n) = \varphi(n)$. For $a \in (\mathbb{Z}/n)^\times$, $a \neq 1$, there is $\chi \in X(n)$ with $\chi(a) \neq 1$.*

*Proof.* (i) $\chi(ab) = \chi(a)\chi(b)$ when both are units modulo $n$, and both sides vanish otherwise ($\gcd(ab,n) > 1$ iff $\gcd(a,n) > 1$ or $\gcd(b,n) > 1$). If $a^e \equiv 1$ then $\chi(a)^e = 1$, so $|\chi(a)| = 1$ and $\bar{z} = z^{-1}$. $\chi(-1)^2 = \chi(1) = 1$. (ii) For a finite abelian group $A$ let $\hat{A} = \mathrm{Hom}(A, \mathbb{C}^\times)$. If $A = \mathbb{Z}/m$, a homomorphism is determined by the image of 1, an element of $\mu_m(\mathbb{C})$, so $\hat{A} \simeq \mu_m(\mathbb{C}) \simeq \mathbb{Z}/m$. For $A = A_1 \times A_2$, $\hat{A} \simeq \hat{A}_1 \times \hat{A}_2$ by restriction. By the structure theorem (Chapter 1), $A$ is a product of cyclic groups, so $\hat{A} \simeq A$. If $a \neq 1$, some coordinate $a_i \in \mathbb{Z}/m_i$ of $a$ is nonzero, and the character $x \mapsto \zeta_{m_i}^{x_i}$ is $\neq 1$ at $a$. Apply this to $A = (\mathbb{Z}/n)^\times$. □

For $n = 5$: $(\mathbb{Z}/5\mathbb{Z})^\times$ is cyclic generated by 2 ($2, 4, 3, 1$ are its powers), so a character is determined by $\chi(2) \in \mu_4(\mathbb{C})$ and there are four. The choice $\chi(2) = i$ gives $\chi(4) = -1$, $\chi(3) = \chi(2^3) = -i$, $\chi(1) = 1$: a character of order 4, odd since $\chi(-1) = \chi(4) = -1$. The choice $\chi(2) = -1$ gives the Legendre symbol $\left(\frac{\cdot}{5}\right) = \chi_5$, which is even. This is the non-canonical isomorphism of (ii): it depends on the choice of the generator 2 and of $i$ among the primitive fourth roots of unity.

**Theorem 2.73** (orthogonality)**.** *For Dirichlet characters modulo $n$:*

*(i) $\sum_{a \bmod n} \chi(a) = \varphi(n)$ if $\chi = \chi_0$, and 0 otherwise;*

*(ii) $\sum_{\chi \in X(n)} \chi(a) = \varphi(n)$ if $a \equiv 1 \pmod n$, and 0 otherwise;*

*(iii) for $\gcd(a,n) = 1$, $\frac{1}{\varphi(n)} \sum_\chi \overline{\chi(a)}\chi(b)$ is 1 if $b \equiv a \pmod n$ and 0 otherwise;*

*(iv) the characters modulo $n$ (restricted to $(\mathbb{Z}/n)^\times$) form an orthonormal basis of the space of functions on $(\mathbb{Z}/n)^\times$ for $\langle f, h\rangle = \varphi(n)^{-1} \sum_{a \in (\mathbb{Z}/n)^\times} f(a)\overline{h(a)}$.*

*Proof.* (i) If $\chi \neq \chi_0$, pick $b$ with $\chi(b) \neq 1$; $a \mapsto ab$ permutes $(\mathbb{Z}/n)^\times$, so $S = \sum \chi(a) = \sum \chi(ab) = \chi(b)S$, and $S = 0$. (ii) If $a$ is not a unit every term vanishes. If $a \not\equiv 1$ is a unit, pick $\psi$ with $\psi(a) \neq 1$ (Proposition 2.72(ii)); $\chi \mapsto \psi\chi$ permutes $X(n)$, so $T = \sum \chi(a) = \psi(a)T$ and $T = 0$. (iii) Apply (ii) to $a^{-1}b$, using $\overline{\chi(a)} = \chi(a^{-1})$. (iv) $\langle \chi, \psi\rangle = \varphi(n)^{-1} \sum (\chi\bar{\psi})(a)$, which is 1 or 0 by (i) applied to $\chi\psi^{-1}$. Orthonormal vectors are independent, and there are $\varphi(n)$ of them in a space of dimension $\varphi(n)$. □

This is the character table of the finite abelian group $(\mathbb{Z}/n)^\times$, with its row and column orthogonality as in Riley-Hobson-Bence; the expansion of a function in characters is finite Fourier analysis on $(\mathbb{Z}/n)^\times$.

**Definition 2.74.** Let $\chi$ be a character modulo $n$ and $d \mid n$. $\chi$ is *defined modulo* $d$ (induced from a character $\chi'$ modulo $d$) if $\chi(a) = \chi'(a)$ for all $a$ with $\gcd(a,n) = 1$. The *conductor* $f_\chi$ is the least $d \mid n$ modulo which $\chi$ is defined; $\chi$ is *primitive* if $f_\chi = n$.

A modulus is a choice, the conductor is not: the same function on the integers prime to 12 can be presented as a character modulo 4, 12 or 24, and the conductor records the least modulus that is really needed. For example the character modulo 12 given by $a \mapsto (-1)^{(a-1)/2}$ on units ($1, 5 \mapsto 1$; $7, 11 \mapsto -1$) is induced from $\chi_{-4}$ modulo 4, so its conductor is 4. The conductor is what matters in the paper: the level of a twist $g \otimes \chi$ and the ramification of the corresponding Galois character are governed by $f_\chi$, not by the modulus used to write $\chi$ down.

**Proposition 2.75.** *Let $\chi$ be a character modulo $n$.*

*(i) For $d \mid n$ the reduction $(\mathbb{Z}/n)^\times \to (\mathbb{Z}/d)^\times$ is surjective, and $\chi$ is defined modulo $d$ iff $\chi(a) = 1$ for all $a$ with $\gcd(a,n) = 1$ and $a \equiv 1 \pmod d$; the inducing $\chi'$ is then unique.*

*(ii) If $\chi$ is defined modulo $d_1$ and modulo $d_2$, it is defined modulo $\gcd(d_1, d_2)$.*

*(iii) $f_\chi$ divides every $d$ modulo which $\chi$ is defined. $\chi$ is induced by a unique character $\chi^{\mathrm{prim}}$ modulo $f_\chi$, and $\chi^{\mathrm{prim}}$ is primitive.*

*(iv) $\chi$ and $\chi^{\mathrm{prim}}$ agree at integers prime to $n$ but may differ at $a$ with $\gcd(a,n) > 1 = \gcd(a, f_\chi)$: $\chi_{-3}$ regarded modulo 6 vanishes at 2, while $\chi_{-3}(2) = -1$.*

*(v) Write $n = \prod_p p^{k_p}$. Then $\chi = \prod_p \chi^{(p)}$ with unique characters $\chi^{(p)}$ modulo $p^{k_p}$ (the $p$-components of $\chi$), and $f_\chi = \prod_p f_{\chi^{(p)}}$. A product of primitive characters with pairwise coprime conductors is primitive, of conductor the product.*

*(vi) The principal character has conductor 1; the nontrivial character modulo 4, $\chi_{-4}$, has conductor 4 (not 2); $\left(\frac{\cdot}{q}\right)$ has conductor $q$; $\chi_8$ ($= 1$ iff $n \equiv \pm 1 \pmod 8$) and $\chi_{-8}$ ($= 1$ iff $n \equiv 1, 3 \pmod 8$) have conductor 8; for a finite set $S$ of odd primes, $\chi_S = \prod_{q \in S} \chi_{q^*}$ is primitive of conductor $D = \prod_{q \in S} q$ and vanishes exactly at the $n$ with $\gcd(n, D) > 1$.*

*Proof.* (i) Given $b$ prime to $d$, let $n'$ be the product of the primes dividing $n$ but not $d$; by the Chinese remainder theorem there is $a$ with $a \equiv b \pmod d$ and $a \equiv 1 \pmod{n'}$, and $a$ is prime to every prime dividing $n$. So the reduction is onto, with kernel $\{a \equiv 1 \pmod d\}$. $\chi$ is defined modulo $d$ iff $\tilde\chi$ factors through this surjection, iff it is trivial on the kernel; the factorisation is unique because the map is onto. (ii) Let $g = \gcd(d_1, d_2)$, $L = \mathrm{lcm}(d_1, d_2)$, and $a$ prime to $n$ with $a \equiv 1 \pmod g$. Since $1 \equiv a \pmod g$ and $d_1\mathbb{Z} + d_2\mathbb{Z} = g\mathbb{Z}$, write $a - 1 = d_1 u + d_2 w$; then $b_0 = 1 + d_1 u$ satisfies $b_0 \equiv 1 \pmod{d_1}$ and $b_0 \equiv a \pmod{d_2}$. Adjust $b_0$ by a multiple of $L$ so that $b \equiv 1$ modulo every prime dividing $n$ but not $L$; then $b$ is prime to $n$ (primes dividing $d_1$ do not divide $b \equiv 1$, primes dividing $d_2$ do not divide $b \equiv a$). Now $\chi(b) = 1$ as $b \equiv 1 \pmod{d_1}$, and $\chi(ab^{-1}) = 1$ as $ab^{-1} \equiv 1 \pmod{d_2}$ ($b^{-1}$ an inverse modulo $n$). So $\chi(a) = 1$, and (i) applies. (iii) If $\chi$ is defined modulo $d$, it is defined modulo $\gcd(d, f_\chi) \le f_\chi$, which must equal $f_\chi$, so $f_\chi \mid d$. $\chi^{\mathrm{prim}}$ is the $\chi'$ of (i) for $d = f_\chi$. If $\chi^{\mathrm{prim}}$ were defined modulo $d' \mid f_\chi$, $d' < f_\chi$, so would be $\chi$ (integers prime to $n$ are prime to $f_\chi$). (iv) By definition, and $2 \mid 6$ while $\chi_{-3}(2) = \left(\frac{-3}{2}\right) = -1$ as $-3 \equiv 5$

(mod 8). (v) By the Chinese remainder theorem $(\mathbb{Z}/n)^\times = \prod_p (\mathbb{Z}/p^{k_p})^\times$, and $\tilde\chi$ is the product of its restrictions $\tilde\chi^{(p)}$ to the factors. Under this identification $\{a \equiv 1 \pmod d\}$ corresponds to $\prod_p \{a_p \equiv 1 \pmod{p^{v_p(d)}}\}$, and $\chi$ is trivial on it iff each $\chi^{(p)}$ is trivial on its factor. So $\chi$ is defined modulo $d$ iff each $\chi^{(p)}$ is defined modulo $p^{v_p(d)}$, and the least such $d$ is $\prod_p f_{\chi^{(p)}}$. If $\chi = \prod_i \chi^{(i)}$ with $\chi^{(i)}$ primitive of pairwise coprime conductors $f_i$, regard it modulo $\prod f_i$; its $p$-component for $p \mid f_i$ is the $p$-component $(\chi^{(i)})^{(p)}$ of $\chi^{(i)}$, so $f_\chi = \prod_i \prod_{p\mid f_i} f_{(\chi^{(i)})^{(p)}} = \prod_i f_{\chi^{(i)}} = \prod_i f_i$ (the second equality is the first statement of (v) applied to $\chi^{(i)}$). (vi) The trivial character is defined modulo 1. $(\mathbb{Z}/2)^\times$ is trivial, so a nontrivial character modulo 4 is not defined modulo 2. $(\frac{\cdot}{q})$ is nontrivial modulo the prime $q$. $\chi_8(5) = (\frac{8}{5}) = (\frac{3}{5}) = -1$ and $\chi_{-8}(5) = (\frac{-8}{5}) = (\frac{2}{5}) = -1$ although $5 \equiv 1 \pmod 4$, so neither is defined modulo 4; their values at odd $n > 0$ are $(\frac{2}{n})^3 = (\frac{2}{n})$ and $(\frac{-1}{n})(\frac{2}{n})$, read off from Proposition 2.66(ii). For $\chi_S$ apply (v) to the primitive characters $\chi_{q^*}$ of the distinct primes $q \in S$ (Proposition 2.69); a product of characters vanishes where one factor does. □

**Corollary 2.76.** *A fundamental discriminant $D \neq 1$ is, uniquely, a product of pairwise coprime prime discriminants $(-4,\ 8,\ -8,\ q^*)$, and $\chi_D$ is the product of the corresponding characters. It is primitive of conductor $|D|$.*

*Proof.* If $D \equiv 1 \pmod 4$ is squarefree, $D = \pm\prod q_i$ with distinct odd primes; $\prod q_i^* = \pm\prod q_i$ is $\equiv 1 \pmod 4$, and of the two numbers $\pm\prod q_i$ only one is $\equiv 1 \pmod 4$, so $D = \prod q_i^*$. If $D = 4m$ with $m \equiv 3 \pmod 4$ squarefree, $-m \equiv 1 \pmod 4$ is squarefree and $D = (-4)\cdot(-m)$. If $D = 4m$ with $m = 2m'$, $m'$ odd squarefree, then $D = 8m'$ if $m' \equiv 1 \pmod 4$ and $D = (-8)(-m')$ if $m' \equiv 3 \pmod 4$. The odd prime discriminants occurring are the $q^*$ for the odd $q \mid D$, and the even one is then determined. By Proposition 2.68(i) (multiplicativity in the upper entry, all entries nonzero), $\chi_D$ is the product of the characters of the factors; these are primitive of conductors 4, 8, 8, $q$ (Proposition 2.75(vi)), pairwise coprime, so $\chi_D$ is primitive of conductor $|D|$ by Proposition 2.75(v). (For the 39 fundamental discriminants with $|D| \le 60$ the conductor was also found by brute force to be $|D|$.) □

**Theorem 2.77** (Gauss sums)**.** *For a character $\chi$ modulo $n$ and $b \in \mathbb{Z}$ put*

$$\tau(\chi, b) = \sum_{a \bmod n} \chi(a)\zeta_n^{ab}, \qquad \tau(\chi) = \tau(\chi, 1).$$

*(i) If $\gcd(b, n) = 1$, $\tau(\chi, b) = \bar\chi(b)\tau(\chi)$. Always $\overline{\tau(\chi)} = \chi(-1)\tau(\bar\chi)$.*

*(ii) If $\chi$ is primitive, $\tau(\chi, b) = \bar\chi(b)\tau(\chi)$ for all $b$ (both sides vanish when $\gcd(b, n) > 1$), $|\tau(\chi)|^2 = n$, and $\tau(\chi)\tau(\bar\chi) = \chi(-1)n$.*

*(iii) $\tau((\frac{\cdot}{q}))^2 = q^*$ for odd primes $q$; $\tau(\chi_{-4}) = i - (-i) = 2i$; $\tau(\chi_{-3}) = \zeta_3 - \zeta_3^2 = 1 + 2\zeta_3 = \sqrt{-3}$.*

*(iv) If $\chi$ modulo $n$ is induced from the primitive $\chi^{\mathrm{prim}}$ modulo $f$ and $\gcd(n/f, f) = 1$, then $\tau(\chi) = \mu(n/f)\chi^{\mathrm{prim}}(n/f)\tau(\chi^{\mathrm{prim}})$. For instance $\chi_{-3}$ modulo 6 has $\tau = (-1)(-1)\sqrt{-3} = \sqrt{-3}$, and the principal character modulo $n$ has $\tau(\chi_0) = \mu(n)$, the sum of the primitive $n$-th roots of unity.*

*Proof.* (i) Substitute $c = ab$, which runs over $(\mathbb{Z}/n)^\times$ with $a$: $\chi(a) = \chi(c)\bar\chi(b)$. Then $\overline{\tau(\chi)} = \sum \bar\chi(a)\zeta_n^{-a} = \tau(\bar\chi, -1) = \chi(-1)\tau(\bar\chi)$. (ii) Let $\gcd(b, n) = g > 1$, $n = gn'$, $b = gb'$; then $\zeta_n^{ab} = \zeta_{n'}^{ab'}$ depends only on $a$ mod $n'$. As $\chi$ is not defined modulo $n'$, there is $c$ prime to $n$ with $c \equiv 1 \pmod{n'}$

and $\chi(c)\neq 1$ (Proposition 2.75(i)). Substituting $a\mapsto ac$: $\tau(\chi,b)=\sum\chi(ac)\zeta_{n'}^{acb'}=\chi(c)\tau(\chi,b)$, so $\tau(\chi,b)=0=\bar\chi(b)\tau(\chi)$. Then

$$\varphi(n)|\tau(\chi)|^2=\sum_{b \bmod n}|\tau(\chi,b)|^2=\sum_{a,a'}\chi(a)\overline{\chi(a')}\sum_b\zeta_n^{(a-a')b}=n\sum_a|\chi(a)|^2=n\varphi(n),$$

using $\sum_b\zeta_n^{cb}=n[n\mid c]$. Finally $n=\tau(\chi)\overline{\tau(\chi)}=\chi(-1)\tau(\chi)\tau(\bar\chi)$ by (i), and $\chi(-1)^2=1$. (iii) The first is Proposition 2.62(i) with $\zeta=\zeta_q$. $\chi_{-4}$ is 1 at 1 and $-1$ at 3, and $\zeta_4=i$. $\chi_{-3}=\left(\frac{\cdot}{3}\right)$ is 1 at 1 and $-1$ at 2, and $\zeta_3-\zeta_3^2=\zeta_3+1+\zeta_3=1+2\zeta_3$. (iv) Put $m=n/f$, choose $u,w$ with $um+wf=1$, so that $\zeta_n=\zeta_n^{um+wf}=\zeta_f^u\zeta_m^w$. For $\gcd(a,n)=1$ we have $\chi(a)=\chi^{\rm prim}(a)$, and $\gcd(a,n)=1$ iff $\gcd(a,f)=\gcd(a,m)=1$; so $\chi(a)=\chi^{\rm prim}(a)\cdot[\gcd(a,m)=1]$ for all $a$. By the Chinese remainder theorem $a \bmod n\leftrightarrow(a \bmod f, a \bmod m)$,

$$\tau(\chi)=\sum_{x \bmod f}\chi^{\rm prim}(x)\zeta_f^{ux}\cdot\sum_{y\in(\mathbb{Z}/m)^\times}\zeta_m^{wy}=\tau(\chi^{\rm prim},u)\cdot c_m,$$

where $c_m$ is the sum of the primitive $m$-th roots of unity ($w$ is prime to $m$, as $wf\equiv 1$, so $y\mapsto wy$ permutes $(\mathbb{Z}/m)^\times$). Since $\sum_{d\mid m}c_d$ is the sum of all $m$-th roots of unity, which is 1 for $m=1$ and 0 otherwise (the coefficient of $x^{m-1}$ in $x^m-1$), Moebius inversion gives $c_m=\mu(m)$. And $um\equiv 1 \pmod f$, so $u$ is prime to $f$ and $\tau(\chi^{\rm prim},u)=\overline{\chi^{\rm prim}(u)}\tau(\chi^{\rm prim})=\chi^{\rm prim}(m)\tau(\chi^{\rm prim})$ by (i). For $\chi_{-3}$ modulo 6: $f=3$, $m=2$, $\mu(2)=-1$, $\chi_{-3}(2)=-1$. For $\chi_0$: $f=1$, $\chi^{\rm prim}=1$ with $\tau=1$, $m=n$. □

All of (i) and (ii) were also checked numerically for the 278 primitive characters of modulus $3\le n<40$. Gauss sums are the finite Fourier transforms of characters; they reappear in the root numbers of twisted $L$-functions (Chapter 19).

**Proposition 2.78** (the paper's characters)**.** *The characters used in the paper are the following Dirichlet characters ($q$, $\ell$ odd primes, $S$ a finite set of odd primes):*

| *name* | *Kronecker symbol* | *conductor* | $\chi(-1)$ | *value at a prime $p$* |
|---|---|---|---|---|
| $\chi_{-4}$ | $\left(\frac{-4}{\cdot}\right)$ | 4 | $-1$ | 0 *at $p=2$;* $(-1)^{(p-1)/2}$, *i.e.* 1 *iff* $p\equiv 1 \pmod 4$ |
| $\chi_{-3}$ | $\left(\frac{-3}{\cdot}\right)=\left(\frac{\cdot}{3}\right)$ | 3 | $-1$ | 0 *at $p=3$;* 1 *iff* $p\equiv 1 \pmod 3$ *(so* $\chi_{-3}(2)=-1$*)* |
| $\chi_{q^*}$ | $\left(\frac{q^*}{\cdot}\right)=\left(\frac{\cdot}{q}\right)$ | $q$ | $(-1)^{(q-1)/2}$ | $\left(\frac{p}{q}\right)$ *(0 at $p=q$; at $p=2$ also* $\left(\frac{2}{q}\right)$*)* |
| $\chi_S=\prod_{q\in S}\chi_{q^*}$ | $\left(\frac{\prod_{q\in S}q^*}{\cdot}\right)$ | $\prod_{q\in S}q$ | $\operatorname{sgn}\prod_{q\in S}q^*$ | $\prod_{q\in S}\left(\frac{p}{q}\right)$ |
| $\chi_{\ell^*}$ | $\left(\frac{\ell^*}{\cdot}\right)=\left(\frac{\cdot}{\ell}\right)$ | $\ell$ | $(-1)^{(\ell-1)/2}$ | $\left(\frac{p}{\ell}\right)$ |
| $\omega_\ell$ *($\mathbb{F}_\ell^\times$-valued)* | *none* | $\ell$ | $-1$ | $p \bmod \ell$ |

*Moreover:*

*(i)* $\chi_{-3}(a)\equiv a \pmod 3$ *for every* $a\in\mathbb{Z}$, *and* $\chi_{-4}(a)\equiv a \pmod 4$ *for every odd* $a$. *In particular* $p\equiv\chi_{-3}(p) \pmod 3$ *for every prime* $p\neq 3$, *and* $p\equiv\chi_{-4}(p) \pmod 4$ *for every odd prime* $p$.

*(ii) For an element $\sigma$ of order 3 of any group and $m$ prime to 3, $\sigma^m=\sigma^{\chi_{-3}(m)}$; for $\sigma$ of order dividing 4 and $m$ odd, $\sigma^m=\sigma^{\chi_{-4}(m)}$.*

*(iii) As $\{\pm1\}\subset\mathbb{F}_\ell^\times$-valued functions, $\left(\frac{\cdot}{\ell}\right)=\omega_\ell^{(\ell-1)/2}$, and $\left(\frac{\cdot}{\ell}\right)=\omega_\ell$ iff $\ell=3$; so $\chi_{-3}=\omega_3$.*

*(iv) For primes $q \geq 5$: $3 \mid q - \chi_{-4}(q)$ iff $q \equiv \pm 1 \pmod{12}$.*

*Proof.* The rows: $\chi_{-4}$ and $\chi_{\pm 8}$ are discussed in Proposition 2.75(vi); for odd $a > 0$, $\chi_{-4}(a) = \left(\frac{-1}{a}\right)\left(\frac{2}{a}\right)^2 = (-1)^{(a-1)/2}$ by Proposition 2.66(ii), and $\chi_{-4}(2) = 0$; parity by Proposition 2.68(iii). The rows $\chi_{-3}$, $\chi_{q^*}$, $\chi_{\ell^*}$ are Proposition 2.69 ($3^* = -3$) with Proposition 2.61(iv); $\chi_S$ is Proposition 2.75(vi), with $\chi_S = \chi_{\prod q^*}$ by Proposition 2.68(v); $\omega_\ell$ is defined on $(\mathbb{Z}/\ell)^\times = \mathbb{F}_\ell^\times$, which it maps isomorphically to itself, so it has order $\ell - 1$, conductor $\ell$ and $\omega_\ell(-1) = -1$. (i) $\chi_{-3} = \left(\frac{\cdot}{3}\right)$ and Proposition 2.61(iv); $\chi_{-4}(a) = (-1)^{(a-1)/2} \equiv a \pmod 4$ for odd $a > 0$, and both sides are periodic modulo 4. (ii) Proposition 2.50(v) and (i). (iii) Euler's criterion (Proposition 2.61(i)) is $\left(\frac{a}{\ell}\right) = \omega_\ell(a)^{(\ell-1)/2}$ in $\mathbb{F}_\ell$. The character $\omega_\ell$ has order $\ell - 1$ and $\left(\frac{\cdot}{\ell}\right)$ has order 2, so they are equal only if $\ell - 1 = 2$; for $\ell = 3$, $\omega_3(a) = a \bmod 3 = \left(\frac{a}{3}\right)$ by (i). (iv) If $q \equiv 1 \pmod 4$, $\chi_{-4}(q) = 1$ and the condition is $q \equiv 1 \pmod 3$, i.e. $q \equiv 1 \pmod{12}$; if $q \equiv 3 \pmod 4$, $\chi_{-4}(q) = -1$ and the condition is $q \equiv 2 \pmod 3$, i.e. $q \equiv 11 \pmod{12}$. □

*Warning* 2.79 (names of characters). (a) $\chi_\ell$, with a letter subscript, is the $\ell$-adic cyclotomic character, not a quadratic character. At $\ell = 3$ the two readings of "$\chi_3$" would collide: the quadratic $\chi_{-3}$ and the cyclotomic $\chi_3^{\mathrm{cyc}}$, which agree only modulo 3 ($\bar{\chi}_3^{\mathrm{cyc}} = \chi_{-3}$, the Galois form of (iii) above). The book therefore never writes $\chi_3$, $\chi_7$ or $\chi_{11}$. (b) $\omega_\ell$ and $\left(\frac{\cdot}{\ell}\right) = \chi_{\ell^*}$ are different characters for $\ell \geq 5$: the first has order $\ell - 1$, the second order 2. This is why, in Lemma lem:h0 and in Section 1.3 of the paper, the relevant condition is $\bar{\rho} \simeq \bar{\rho} \otimes \bar{\chi}_\ell$, not $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{\ell^*}$. (c) The character of a quadratic field $\mathbb{Q}(\sqrt{d})$ is $\chi_D$ with $D$ the discriminant of the field: the character of $\mathbb{Q}(i)$ is $\chi_{-4}$ and that of $\mathbb{Q}(\sqrt{2})$ is $\chi_8$ (Appendix B, B-C6(b), also writes them with the radicand as index; this chapter does not) (Chapter 6). (d) Do not confuse the Kronecker symbol $\left(\frac{D}{2}\right)$ (lower entry 2) with the Jacobi symbol $\left(\frac{2}{D}\right)$: $\chi_{-3}(2) = \left(\frac{-3}{2}\right) = -1$ because $-3 \equiv 5 \pmod 8$, consistent with $2 \equiv -1 \pmod 3$.

*In the paper* 2.80. The table of Section 1 uses $\chi_{-3}$ and $\chi_{-4}$ "the quadratic characters of $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(i)$", whose values at the bad prime $q$ are read from the table above. The proof of Proposition prop:forms says "$e_3(E,q) = v_3(q - \chi_{-3}(q)) \geq 1 = w_3(E,q)$ since $q \equiv \chi_{-3}(q) \pmod 3$": this is (i). Theorem thm:twist uses $\chi_S$ of conductor $D = \prod_{q \in S} q$ and, in its proof, that the twist map $\iota\colon h \mapsto \sum \chi_S(n) a_n(h) q^n$ kills the coefficients at the $n$ with $\gcd(n, D) > 1$: Proposition 2.75(vi) (here $q^n$ is the $q$-expansion variable, a different meaning of $q$). Section sec:anomaly observes: "The residue classes of the type III primes of the violating curves are $q \equiv 1$ and $q \equiv 11 \pmod{12}$ in comparable numbers"; by (iv) these are exactly the classes with $e_3(E,q) > 0$. Lemma lem:h0 and Proposition prop:ns use $\bar{\chi}_3^{\mathrm{cyc}} = \chi_{-3}$, whose Dirichlet side is (iii). In Example ex:qual, $a_p(121\mathrm{d}) = \chi_{-11}(p) a_p(11\mathrm{a})$ for $p \neq 11$.

## 2.11 Valuations and the paper's local terms

Every number in the paper's main statements is an $\ell$-adic valuation: $v_\ell(\deg \varphi_E)$, $t_\ell(E,q) = v_\ell(-v_q(j_E))$, and the Euler terms $e_\ell(E,q) = v_\ell(P_q(q^{-2})/(1 - a_q(E)^2 q^{-2}))$ of Definition def:terms, which the table of Section 1 evaluates as $v_\ell(q - \chi_{-3}(q))$, $v_\ell(q - \chi_{-4}(q))$, $v_\ell(q^2 - 1)$ and $v_\ell(q - 1) + v_\ell((q+1)^2 - a_q(E')^2)$. This section defines $v_p$ and $|\cdot|_p$ on $\mathbb{Q}$ (the completions $\mathbb{Q}_p$, $\mathbb{Z}_p$ are Chapter 7), shows that the rationals in question are $\ell$-integral, proves the algebraic identity behind the $\mathrm{I}_0^*$ row, and works out when each entry of the table is positive. Chapter 1 used $v_\ell(n)$ for integers informally; the formal definition is the following.

**Definition 2.81.** For a prime $p$ and $x \in \mathbb{Q}^\times$ write $x = p^k a/b$ with $a, b \in \mathbb{Z}$ prime to $p$ and $k \in \mathbb{Z}$; then $v_p(x) := k$. Put $v_p(0) = +\infty$. (The exponent $k$ is well defined: if $p^k a/b = p^{k'} a'/b'$ then $p^k ab' = p^{k'} a' b$ in $\mathbb{Z}$, and unique factorisation gives $k = k'$.)

For example $v_3(\frac{18}{5}) = 2$, $v_5(\frac{18}{5}) = -1$ and $v_2(\frac{18}{5}) = 1$. The valuation measures divisibility, and it is used throughout because the paper's quantities are rationals whose numerators and denominators matter prime by prime: $v_\ell(x) > 0$ says that $\ell$ divides $x$ in $\mathbb{Z}_{(\ell)}$.

**Proposition 2.82.** *Let $p$ be a prime and $x, y \in \mathbb{Q}$.*

*(i)* $v_p(xy) = v_p(x) + v_p(y)$.

*(ii)* $v_p(x+y) \geq \min(v_p(x), v_p(y))$, *with equality if* $v_p(x) \neq v_p(y)$.

*(iii)* $v_p(x) \geq 0$ *iff* $x \in \mathbb{Z}_{(p)}$, *and* $v_p(x) = 0$ *iff* $x \in \mathbb{Z}_{(p)}^\times$ *(Proposition 2.21).*

*(iv) On* $\mathbb{Z}$: $p^k \mid n$ *iff* $v_p(n) \geq k$; $v_p(\gcd(m,n)) = \min(v_p(m), v_p(n))$ *and* $v_p(\operatorname{lcm}(m,n)) = \max(v_p(m), v_p(n))$ *(Proposition 2.26).*

*(v) For ideals,* $v_p(r\mathbb{Z}) = v_p(r)$ *is well defined; the $p$-part of $n \neq 0$ is $p^{v_p(n)}$; and $v_p(\ell) = 1$ if $p = \ell$, 0 otherwise.*

*(vi) (Legendre's formula)* $v_p(n!) = \sum_{k \geq 1} \lfloor n/p^k \rfloor$.

*Proof.* (i) Multiply the representations. (ii) If $x = p^k a/b$, $y = p^{k'} a'/b'$ with $k \leq k'$, then $x + y = p^k(ab' + p^{k'-k} a' b)/(bb')$, whose numerator is an integer and denominator prime to $p$; if $k < k'$ the numerator is $\equiv ab' \not\equiv 0 \pmod p$. (The cases with $x$ or $y$ zero are trivial.) (iii) is Proposition 2.21(iv). (iv) is Proposition 2.26(iv). (v) $r\mathbb{Z} = r'\mathbb{Z}$ iff $r' = \pm r$. (vi) $v_p(n!) = \sum_{m=1}^n v_p(m) = \sum_{m=1}^n \sum_{k \geq 1} [p^k \mid m]$, and exchanging the sums, $\#\{m \leq n : p^k \mid m\} = \lfloor n/p^k \rfloor$. □

The inequality in (ii) can be strict when the valuations agree: $v_3(\frac{1}{3}) = v_3(\frac{2}{3}) = -1$, but $\frac{1}{3} + \frac{2}{3} = 1$ has $v_3 = 0$. Legendre's formula gives, for instance, $v_3(10!) = 3 + 1 = 4$, and $10! = 3628800 = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

**Definition 2.83.** The *$p$-adic absolute value* on $\mathbb{Q}$ is $|x|_p = p^{-v_p(x)}$, $|0|_p = 0$; $|x|_\infty$ is the usual absolute value. An *absolute value* on a field is a function $|\cdot| \colon K \to \mathbb{R}_{\geq 0}$ with $|x| = 0$ iff $x = 0$, $|xy| = |x||y|$, and $|x + y| \leq |x| + |y|$; two are *equivalent* if one is a positive power of the other.

**Proposition 2.84.** *(i) $|\cdot|_p$ is an absolute value satisfying the ultrametric inequality $|x+y|_p \leq \max(|x|_p, |y|_p)$; $d_p(x,y) = |x-y|_p$ is a metric on $\mathbb{Q}$, in which $p^k \to 0$.*

*(ii) (Product formula)* $\prod_{p \leq \infty} |x|_p = 1$ *for $x \in \mathbb{Q}^\times$ (the product over all primes and $\infty$, almost all factors being 1).*

*(iii)* $\mathbb{Q}$ *is not complete for $d_p$.*

*Proof.* (i) is Proposition 2.82(i), (ii); the ultrametric inequality implies the triangle inequality, and $|p^k|_p = p^{-k}$. (ii) Write $x = \pm \prod_p p^{v_p(x)}$ (Proposition 2.26(iii) applied to numerator and denominator); then $|x|_\infty = \prod_p p^{v_p(x)}$ and $\prod_p |x|_p = \prod_p p^{-v_p(x)}$, the inverse. For instance, for $x = -\frac{12}{5}$: $|x|_\infty = \frac{12}{5}$, $|x|_2 = \frac{1}{4}$, $|x|_3 = \frac{1}{3}$, $|x|_5 = 5$, and the product is $\frac{12}{5} \cdot \frac{1}{12} \cdot 5 = 1$. (iii) Choose $m$ and $c$ as follows: for odd $p$, $m = 2$ and $c = 1 + 2p$, which is not a square (if $1 + 2p = s^2$ then $2p = (s-1)(s+1)$ with both factors even, so $4 \mid 2p$); for $p = 2$, $m = 3$ and $c = 3$, not a cube (a

rational cube root of an integer is an integer, Theorem 2.32(iv), and $1 < 3 < 8$). In both cases $c \equiv 1 \pmod p$ and $p \nmid m$. Starting from $x_1 = 1$, define integers $x_k \equiv 1 \pmod p$ with $x_k^m \equiv c \pmod{p^k}$: given $x_k$, choose $t \in \mathbb{Z}$ with $m x_k^{m-1} t \equiv (c - x_k^m)/p^k \pmod p$ (possible as $m x_k^{m-1} \not\equiv 0 \pmod p$) and put $x_{k+1} = x_k + t p^k$; the binomial expansion gives $x_{k+1}^m \equiv x_k^m + m x_k^{m-1} t p^k \equiv c \pmod{p^{k+1}}$. Then $|x_{k+1} - x_k|_p \le p^{-k}$, so $(x_k)$ is Cauchy by the ultrametric inequality. If $x_k \to x \in \mathbb{Q}$, then $|x_k^m - x^m|_p \le |x_k - x|_p \max(1, |x|_p)^{m-1} \to 0$ and $|x_k^m - c|_p \le p^{-k} \to 0$, so $x^m = c$, a contradiction. (The completion $\mathbb{Q}_p$ contains the limit; Chapter 7.) □

**Theorem 2.85** (Ostrowski)**.** *Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_\infty$ or to $|\cdot|_p$ for exactly one prime $p$.*

We do not prove this; see [142, Chapter II, Section 3]. It is not used in the paper; it explains why the places of $\mathbb{Q}$ are the primes and $\infty$.

**Lemma 2.86.** *Let $q \neq \ell$ be primes.*

*(i)* $v_\ell(q^k) = 0$ *for all $k \in \mathbb{Z}$: $q$ is a unit of $\mathbb{Z}_{(\ell)}$ (an "$\ell$-adic unit").*

*(ii) For $P \in \mathbb{Z}[X]$ of degree at most $d$ and $k \ge 0$, $P(q^{-k}) \in \mathbb{Z}_{(\ell)}$, and $v_\ell(P(q^{-k})) = v_\ell(q^{kd} P(q^{-k}))$ is the valuation of the integer $q^{kd}P(q^{-k})$; it is finite iff $P(q^{-k}) \neq 0$.*

*(iii) For $a \in \mathbb{Z}$ and either choice of sign:*

$$v_\ell\Big(\frac{q-1}{q}\Big) = v_\ell(q-1), \qquad v_\ell(1 \mp q^{-1}) = v_\ell(q \mp 1),$$
$$v_\ell(1 - q^{-2}) = v_\ell(q^2 - 1) = v_\ell(q-1) + v_\ell(q+1),$$
$$v_\ell\Big(\frac{q-1}{q} \cdot \frac{(q+1)^2 - a^2}{q^2}\Big) = v_\ell(q-1) + v_\ell((q+1)^2 - a^2) \quad \text{when } (q+1)^2 \neq a^2.$$

*All these valuations are $\ge 0$.*

*(iv) For $q = 2$, $\ell = 3$: $v_3(1 + 2 \cdot 2^{-2}) = v_3(3/2) = 1$.*

*Proof.* (i) $v_\ell(q) = 0$ and Proposition 2.82(i). (ii) $q^{kd}P(q^{-k}) = \sum_i a_i q^{k(d-i)} \in \mathbb{Z}$ for $P = \sum_{i \le d} a_i X^i$, and $v_\ell(q^{-kd}) = 0$. (iii) Each rational is an integer divided by a power of $q$; apply (i), Proposition 2.82(i) and Lemma 2.42(v). (iv) $1 + \frac12 = \frac32$. □

**Proposition 2.87** (the $\mathrm{I}_0^*$ identity)**.** *Let $q \neq 0$ and $a$ be complex numbers and $\alpha, \beta$ the roots of $X^2 - aX + q$, so $\alpha + \beta = a$ and $\alpha\beta = q$. Then*

$$\Big(1 - \frac{\alpha}{\beta q}\Big)\Big(1 - \frac{1}{q}\Big)\Big(1 - \frac{\beta}{\alpha q}\Big) = \frac{q-1}{q} \cdot \frac{(q+1)^2 - a^2}{q^2},$$

*and $(q+1)^2 - a^2 = (q+1-a)(q+1+a)$. Equivalently, $(1 - qX)(1 - (a^2 - 2q)X + q^2X^2)$ at $X = q^{-2}$ equals the right side. If $q > 0$ and $a$ is real with $a^2 \le 4q$, then $(q+1)^2 - a^2 \ge (q-1)^2$, which is $> 0$ for $q \neq 1$.*

*Proof.* $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{a^2 - 2q}{q}$ and $\frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 1$, so

$$\Big(1 - \frac{\alpha}{\beta q}\Big)\Big(1 - \frac{\beta}{\alpha q}\Big) = 1 - \frac{a^2 - 2q}{q^2} + \frac{1}{q^2} = \frac{q^2 + 2q + 1 - a^2}{q^2} = \frac{(q+1)^2 - a^2}{q^2}.$$

Multiply by $1 - \frac1q = \frac{q-1}{q}$. The polynomial form: $1 - (a^2 - 2q)q^{-2} + q^2 q^{-4}$ is the same middle factor, and $1 - q \cdot q^{-2} = 1 - q^{-1}$. Finally $(q+1)^2 - a^2 \ge (q+1)^2 - 4q = (q-1)^2$. □

For an elliptic curve $E'$ over $\mathbb{F}_q$ with $a = q+1-\#E'(\mathbb{F}_q)$ and $|a| \le 2\sqrt{q}$ (Hasse's bound, Chapter 9), the two factors $q+1-a$ and $q+1+a$ are $\#E'(\mathbb{F}_q)$ and the number of points of its quadratic twist over $\mathbb{F}_q$; so $(q+1)^2 - a^2$ is a nonzero integer, and the last valuation in Lemma 2.86(iii) is finite.

**Proposition 2.88** (the paper's local terms)**.** *Let $q \ge 5$ be a prime, $\ell \ne q$ an odd prime, and $a \in \mathbb{Z}$ with $a^2 \le 4q$. Put*

$$e_{\mathrm{II}} = v_\ell(q - \chi_{-3}(q)), \qquad e_{\mathrm{III}} = v_\ell(q - \chi_{-4}(q)),$$
$$e_{\mathrm{I}^*} = v_\ell(q^2-1), \qquad e_{\mathrm{I}_0^*}(a) = v_\ell(q-1) + v_\ell((q+1)^2 - a^2).$$

*These are finite nonnegative integers, equal to the valuations of $1-\chi_{-3}(q)q^{-1}$, $1-\chi_{-4}(q)q^{-1}$, $1-q^{-2}$ and of the product in Proposition 2.87, and:*

*(i)* $e_{\mathrm{II}} = v_\ell(q-1)$ *if* $q \equiv 1 \pmod 3$ *and* $v_\ell(q+1)$ *if* $q \equiv 2 \pmod 3$. *For* $\ell = 3$, $e_{\mathrm{II}} \ge 1$ *always, and it is unbounded ($q = 53$: $v_3(54) = 3$; $q = 163$: $v_3(162) = 4$; these are the least primes with $e_{\mathrm{II}} \ge 3$, resp. $\ge 4$). For $\ell \ge 5$, $e_{\mathrm{II}} \ge 1$ iff $q \equiv \chi_{-3}(q) \pmod \ell$.*

*(ii) For $\ell = 3$, $e_{\mathrm{III}} \ge 1$ iff $q \equiv \pm 1 \pmod{12}$; for $\ell \ge 5$, iff $q \equiv \chi_{-4}(q) \pmod \ell$.*

*(iii) For $\ell = 3$, $e_{\mathrm{I}^*} \ge 1$ always; for $\ell \ge 5$, $e_{\mathrm{I}^*} \ge 1$ iff $q \equiv \pm 1 \pmod \ell$.*

*(iv) For $\ell = 3$, $e_{\mathrm{I}_0^*}(a) \ge 1$ iff $q \equiv 1 \pmod 3$ or $3 \mid a$; for $\ell \ge 5$, iff $q \equiv 1 \pmod \ell$ or $a \equiv \pm(q+1) \pmod \ell$.*

*(v) (Type $\mathrm{I}_n$.) For $n \ge 1$ and $\bar{u} \in \mathbb{F}_q^\times$, the Tamagawa refinement $v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c(\bar{u})))$ of Lemma lem:tamfactor exceeds $t = v_\ell(n)$ iff $\ell \mid n$, $q \equiv 1 \pmod \ell$ and $\bar{u} \in (\mathbb{F}_q^\times)^\ell$.*

*Proof.* Finiteness: $q - \chi(q) \ge q - 1 > 0$, $q^2 - 1 > 0$, and $(q+1)^2 - a^2 \ge (q-1)^2 > 0$ by Proposition 2.87. The identification with valuations of rationals is Lemma 2.86(iii). (i) By Proposition 2.78, $\chi_{-3}(q) = 1$ iff $q \equiv 1 \pmod 3$, and $q \equiv \chi_{-3}(q) \pmod 3$. The least primes: $e_{\mathrm{II}} \ge 3$ means $q \equiv \pm 1 \pmod{27}$, and the candidates $26, 28, 53$ give the prime $53$ (and $53 \equiv 2 \pmod 3$, so $e_{\mathrm{II}} = v_3(54) = 3$); $e_{\mathrm{II}} \ge 4$ means $q \equiv \pm 1 \pmod{81}$, and $80, 82, 161 = 7 \cdot 23$ are not prime, while $163$ is, with $163 \equiv 1 \pmod 3$. (ii) Proposition 2.78(iv), and the definition for $\ell \ge 5$. (iii) $q^2 \equiv 1 \pmod 3$ for $q \ne 3$; for $\ell \ge 5$, $\ell \mid (q-1)(q+1)$ iff $\ell$ divides a factor. (iv) $e_{\mathrm{I}_0^*}(a) \ge 1$ iff $\ell \mid q-1$ or $\ell \mid (q+1-a)(q+1+a)$. For $\ell = 3$ and $q \equiv 2 \pmod 3$, $q+1 \equiv 0$, so the second condition is $3 \mid a^2$. (v) is Proposition 2.53(iv). (Parts (i) to (iii) were also checked for all primes $5 \le q < 10^5$.) □

**Example 2.89** (the paper's numbers)**.** The curve data below (Kodaira types, modular degrees, Euler factors at 2) were recomputed with PARI and Cremona's tables; the arithmetic is that of this section.

(a) `121a1`, type II at 11: $\chi_{-3}(11) = -1$, so $e_3 = v_3(11+1) = v_3(12) = 1$; $\deg \varphi_E = 6$ and $v_3(6) = 1$: the bound is attained (Example ex:qual(b)).

(b) `539c1`, type $\mathrm{I}_3^*$ at 7 (and $\mathrm{I}_2$ at 11): $e_3 = v_3(7^2-1) = v_3(48) = 1$, $t_3 = v_3(3) = 1$; $\deg \varphi_E = 288 = 2^5 \cdot 3^2$, and $v_3 = 2 = e_3 + t_3$.

(c) `980i1`, type $\mathrm{I}_5^*$ at 7, $\ell = 5$: $e_5 = v_5(48) = 0$, $t_5 = v_5(5) = 1$; $\deg \varphi_E = 2880 = 2^6 \cdot 3^2 \cdot 5$. (Here $5 \mid N$, so the example illustrates the congruence, not the theorem.)

(d) `726a1`, type III at 11: $\chi_{-4}(11) = -1$ and $e_3 = v_3(12) = 1$; $11 \equiv 11 \pmod{12}$, as Proposition 2.78(iv) requires.

(e) At $q = 2$ the Kodaira type does not determine $P_2$, and $e_3 = v_3(P_2(\frac{1}{4}))$ (the denominator $1 - a_2^2/4$ is 1, as $a_2 = 0$ at an additive prime). `52a1` (type IV* at 2) and `336f1` (type II at 2) have $P_2 = 1 + 2X$, so $e_3 = v_3(\frac{3}{2}) = 1$; $\deg \varphi_E = 3$, resp. 24. `123211` (type II* at 2) has $P_2 = 1 + 2X - 4X^2 - 8X^3 = (1 - 2X)(1 + 2X)^2$, so $P_2(\frac{1}{4}) = \frac{1}{2} \cdot \frac{9}{4} = \frac{9}{8}$ and $e_3 = 2$, with $\deg \varphi_E = 288$; and `592c1` (type II* at 2) has $P_2 = (1 - 2X)(1 + 4X^2)$, $P_2(\frac{1}{4}) = \frac{5}{8}$, $e_5 = 1$, $\deg \varphi_E = 80$. The two cubic factors are the polynomial of Proposition 2.87 with $q = 2$ and $a = 0$, resp. $a = -2$: $\frac{1}{2} \cdot \frac{9 - a^2}{4}$ is $\frac{9}{8}$, resp. $\frac{5}{8}$.

(f) `539d1`, type $\mathrm{I}_0^*$ at 7; its twist by $\chi_{-7}$ is $E' =$ `11a3`, with good reduction at 7 and $a_7(E') = -2$ (so $\#E'(\mathbb{F}_7) = 10$). Then $e_3 = v_3(6) + v_3(64 - 4) = 1 + 1 = 2$, and $\deg \varphi_E = 72 = 2^3 \cdot 3^2$.

(g) `11a1` has type $\mathrm{I}_5$ at 11 and $e_3 = 0$ there; its twist by $\chi_{-11}$ is `121d2`, of type $\mathrm{I}_5^*$ at 11, with $e_3 = v_3(11^2 - 1) = v_3(120) = 1$. So $e_\ell(E, q)$ is not invariant under a quadratic twist ramified at $q$, as Definition def:terms says.

*In the paper* 2.90. Definition def:terms sets

$$e_\ell(E, q) = v_\ell(L_q(A_f, 1)^{-1} / L_q^{\mathrm{nv}}(A_f, 1)^{-1}) = v_\ell(P_q(q^{-2})/(1 - a_q(E)^2 q^{-2})).$$

By Lemma 2.86(ii) both numerator and denominator lie in $\mathbb{Z}_{(\ell)}$, and the valuation of the quotient is the difference of their valuations. At a multiplicative prime $a_q^2 = 1$ and $P_q(X) = 1 - X$, so the quotient is 1 and $e_\ell = 0$; at an additive prime $a_q = 0$ and the denominator is 1. The proof of Lemma lem:euler then reads "$e_\ell = v_\ell(1 - q^{-2}) = v_\ell(q^2 - 1)$" (type $\mathrm{I}_n^*$) and "$e_\ell = v_\ell(1 - \chi_{-3}(q)q^{-1}) = v_\ell(q - \chi_{-3}(q))$" (types II to IV*): Lemma 2.86(iii); its displayed identity for type $\mathrm{I}_0^*$ is Proposition 2.87, and the same polynomial is shape (ii) of Proposition prop:shapes. Remark rem:wild ("$P_2(X) = 1 + 2X$ and $e_3 = 1$") is Lemma 2.86(iv). Remark rem:afterquant(1) and the abstract's description of the terms rest on Proposition 2.88: under the hypotheses of Theorem thm:quant ($3 \nmid 2N$, $E[3]$ irreducible) a single prime of type $\mathrm{I}_n^*$ forces $3 \mid \deg \varphi_E$, because $e_3 = v_3(q^2 - 1) \geq 1$ for every $q \geq 5$, and so does a prime of type $\mathrm{I}_0^*$ with $q \equiv 1 \pmod 3$, because then $v_3(q - 1) \geq 1$. That the table is right, i.e. that these expressions are the $e_\ell(E, q)$, is Lemma lem:euler, proved in Chapters 19 and 21.

## 2.12 Exercises

**Exercise 2.91** (a toy Hecke algebra)**.** Let $r \geq 1$ and $R = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod r\}$, a subring of $\mathbb{Z} \times \mathbb{Z}$; let $\phi(a, b) = a$, $I = \ker \phi$ and $J = \{(a, 0) \in R\}$. Show that $I \cap J = 0$, that $R/(I + J) \simeq \mathbb{Z}/\phi(J) = \mathbb{Z}/r$, and that the cokernel of $R \to R/I \times R/J$ is $\mathbb{Z}/r$ (Theorem 2.12 and Proposition 2.13). Interpret $R$ as the Hecke algebra of two "eigenforms" whose eigenvalues are congruent modulo $r$, and $r$ as their congruence number, as in Lemma lem:crit.

**Exercise 2.92.** Show that a ring homomorphism $\mathbb{Z}/r\mathbb{Z} \to k$, $k$ a field of characteristic $\ell$, exists iff $\ell \mid r$. Deduce the last step of Lemma lem:crit: if $I + J$ is contained in a maximal ideal of $\mathbb{T}$ of residue characteristic $\ell$, then $\ell$ divides $\#\mathbb{T}/(I + J)$ (Proposition 2.18).

**Exercise 2.93.** In $\mathbb{Z}[\zeta_3]$, check $5+\zeta_3 = (2-\zeta_3)(2+\zeta_3)$ and $7 = (3+\zeta_3)(2-\zeta_3)$ (note $3+\zeta_3^2 = 2-\zeta_3$), and compute $\gcd(7, 5+\zeta_3)$ by the Euclidean algorithm (answer: $2-\zeta_3$, of norm 7, up to units). Verify by listing the solutions of $a^2-ab+b^2=1$ that $\mathbb{Z}[\zeta_3]^\times = \mu_6$.

**Exercise 2.94.** Check the values in Example ex:qual of the paper: $e_3(\texttt{121a1}, 11) = v_3(12) = 1$; $e_3(\texttt{539c1}, 7) = v_3(48) = 1$ and $t_3(\texttt{539c1}, 7) = 1$; $e_5(\texttt{980i1}, 7) = 0$ and $t_5(\texttt{980i1}, 7) = 1$. Compare with $v_3(6)$, $v_3(2^5\cdot 3^2)$ and $v_5(2^6\cdot 3^2\cdot 5)$ (Example 2.89).

**Exercise 2.95.** Check the claim in the proof of Proposition prop:forms that $q \equiv \chi_{-3}(q) \pmod 3$ for every prime $q \neq 3$, so that $e_3(E,q) \geq 1$ at every prime $q \geq 5$ of type II, II*, IV, IV*. Find the least prime $q$ with $v_3(q-\chi_{-3}(q)) \geq 3$ and with $v_3(q-\chi_{-3}(q)) \geq 4$ (answers 53 and 163).

**Exercise 2.96.** Check the statement of Section sec:anomaly that the type III primes $q \geq 5$ with $e_3 > 0$ are those with $q \equiv 1$ or $11 \pmod{12}$. For $\ell = 5$, show that the type III primes with $e_5 > 0$ are those with $q \equiv \pm 1 \pmod{20}$.

**Exercise 2.97.** Prove Proposition 2.87 by expanding the product. Then check part (1) of Remark rem:afterquant: every prime $q \geq 5$ of type $\mathrm{I}_n^*$ has $e_3 \geq 1$, and a prime of type $\mathrm{I}_0^*$ has $e_3 \geq 1$ iff $q \equiv 1 \pmod 3$ or $3 \mid a_q(E')$. Compute $e_3$ for type $\mathrm{I}_0^*$ at $q=7$ with $a_7(E') = -2$, and at $q=5$ and at $q=11$ with $a_q(E') = 0$ (answers: 2, 2, 2).

**Exercise 2.98.** Check Remark rem:wild: with $P_2(X) = 1+2X$, $e_3 = v_3(P_2(1/4)) = 1$; with $P_2 = 1$, $e_3 = 0$. Explain why the denominator $1-a_q^2 q^{-2}$ of Definition def:terms is 1 at an additive prime and $1-q^{-2} = P_q(q^{-2})$ at a multiplicative one, so that $e_\ell = 0$ there.

**Exercise 2.99.** For $q = 7, 13, 19, 37$ and $\ell = 3$, list the cubes and the ninth powers in $\mathbb{F}_q^\times$, and compute $v_3(c(u))$ for $u = 2, 3, -1$ (answers: $0,0,1$ for $q=7$ and 13; $0,0,2$ for $q=19$ and 37). Check the statement of Lemma lem:tamfactor that $\min(v_\ell(n), v_\ell(q-1), v_\ell(c)) > 0$ iff $q \equiv 1 \pmod \ell$, $\ell \mid n$ and $u$ is an $\ell$-th power.

**Exercise 2.100.** Prove Theorem 2.52(iii) directly from a generator of $\mathbb{F}_q^\times$, without quoting Chapter 1. Show that the $\ell$-th power map of $\mathbb{F}_q^\times$ is bijective iff $q \not\equiv 1 \pmod \ell$. Find the least $f$ with $\mu_9 \subset \mathbb{F}_{2^f}$ (answer 6) and show $v_3(2^{6m}-1) = 2 + v_3(m)$ (Lemma 2.42).

**Exercise 2.101.** Show that the only solutions of $a^2 \equiv 1 \pmod{\ell^k}$ ($\ell$ odd) are $a \equiv \pm 1$, and deduce that a homomorphism $\chi$ onto $(\mathbb{Z}/\ell^k)^\times$ with $\chi^2 = 1$ forces $\ell^k = 3$. Explain how this gives $h_\ell(E) = 0$ for $\ell \geq 5$ and $h_3(E) \leq 1$ in Lemma lem:h0 (the Galois side is Chapter 5).

**Exercise 2.102.** Construct $\mathbb{F}_{16} = \mathbb{F}_2[x]/(x^4+x+1)$ (check irreducibility: no root, and $x^4+x+1 \neq (x^2+x+1)^2$, the only irreducible quadratic being $x^2+x+1$). Show that $\bar{x}$ generates $\mathbb{F}_{16}^\times$ ($\bar{x}^3 \neq 1$ and $\bar{x}^5 = \bar{x}^2+\bar{x} \neq 1$). List the subfields, and show that the Frobenius has orbits of sizes $1,1,2,4,4,4$ on $\mathbb{F}_{16}$; deduce $N_4 = 3$ over $\mathbb{F}_2$.

**Exercise 2.103** (Eisenstein's proof of reciprocity)**.** Let $p \neq q$ be odd primes. Prove Eisenstein's lemma $\left(\frac{q}{p}\right) = (-1)^{\sum_{a=1}^{(p-1)/2} \lfloor 2qa/p \rfloor}$ and show that the exponent has the parity of $\sum_{u=1}^{(p-1)/2} \lfloor qu/p \rfloor$. By counting the lattice points in the rectangle $0 < x < p/2$, $0 < y < q/2$ on either side of the diagonal, show $\sum_{u=1}^{(p-1)/2} \lfloor qu/p \rfloor + \sum_{v=1}^{(q-1)/2} \lfloor pv/q \rfloor = \frac{p-1}{2}\cdot\frac{q-1}{2}$, and deduce Theorem 2.64(i). (Both identities were checked for all odd primes below 200.)

**Exercise 2.104.** Verify that $\left(\frac{D}{\cdot}\right)$ is periodic modulo $|D|$ for $D = -3, -4, -7, 5, 8, 12$. Tabulate $\chi_{-3}, \chi_{-4}, \chi_{-7}, \chi_{-11}, \chi_5$ at the primes below 30. Show that on integers prime to 6 the function $\left(\frac{3}{\cdot}\right)$ agrees with $\chi_{12}$, of conductor 12, while $\left(\frac{-3}{\cdot}\right) = \chi_{-3}$ has conductor 3: the sign in $q^*$ matters.

**Exercise 2.105.** Compute $\tau(\chi_{-4}) = 2i$, $\tau(\chi_{-3}) = \sqrt{-3}$ and $\tau((\frac{\cdot}{5})) = \sqrt{5}$ (use $\cos(2\pi/5) = (\sqrt{5}-1)/4$). Show that $\chi_{-3}$ regarded modulo 6 has Gauss sum $\sqrt{-3}$ and that the principal character modulo $n$ has $\tau(\chi_0) = \mu(n)$.

**Exercise 2.106.** Show that $x^4+1$ is irreducible over $\mathbb{Q}$ but reducible modulo every prime (Warning 2.35). Check the claim of Proposition prop:ns for `11a1`, whose mod 3 image is $\mathrm{GL}_2(\mathbb{F}_3)$: its 3-division polynomial $3x^4 - 4x^3 - 60x^2 - 237x - 21$ is irreducible (Example 2.36).

**Exercise 2.107.** Show that if $p \nmid n$ and $p \mid \Phi_n(a)$, then $p \equiv 1 \pmod{n}$, and deduce that there are infinitely many primes $\equiv 1 \pmod{n}$ (Proposition 2.59). Check with $\Phi_{12}(2) = 13$ and $\Phi_9(2) = 73$.

## 2.13 Notes and references

Sections 1 to 4, 6 and 8 follow Dummit and Foote [58]: Sections 7.1 to 7.6 (rings, ideals, quotients, fractions, the Chinese remainder theorem), 8.1 to 8.3 (Euclidean domains, PIDs, UFDs), 9.1 to 9.5 (polynomial rings, Gauss's lemma, Eisenstein's criterion), 13.1, 13.4 to 13.6 (field extensions, splitting fields, separability, cyclotomic polynomials) and 14.3 (finite fields); Lang [109, Chapter IV, Section 2, and Chapter V, Section 5] gives Gauss's lemma and finite fields in greater generality, and Atiyah and Macdonald [7, Chapter 1] the existence of maximal ideals and the ideal theory of products. The Gaussian and Eisenstein integers are treated in Ireland and Rosen [87, Chapter 1], together with unique factorisation in $\mathbb{Z}$.

Sections 5 to 10 are classical elementary number theory: Ireland and Rosen [87], Chapter 4 for $(\mathbb{Z}/n)^\times$ and primitive roots, Chapter 5 for quadratic reciprocity and the Jacobi symbol, Chapter 6 for the quadratic Gauss sum (Section 6.4 for Gauss's determination of its sign, Remark 2.63), Chapter 7 for finite fields, Chapter 8 for Gauss sums of general characters; Serre [172, Chapter I] proves reciprocity by the same computation in characteristic $p$ as Theorem 2.64 and the second supplement with an eighth root of unity. Among the many other proofs of reciprocity, Eisenstein's lattice-point proof is Exercise 2.103, and the proof through the splitting of $p$ in $\mathbb{Q}(\zeta_q)$ is given in Chapter 6. The lifting-the-exponent lemma is folklore; the form used here, with its proof by the binomial theorem, is the standard one in the proof that $(\mathbb{Z}/p^k)^\times$ is cyclic.

The Kronecker symbol follows Cohen [30, Section 1.4], and fundamental discriminants Cohen [30, Section 5.1]; Dirichlet characters, conductors and primitivity follow Washington [201, Chapter 3], and orthogonality is Serre [172, Chapter VI, Section 1]. The formula for the Gauss sum of an imprimitive character (Theorem 2.77(iv)) is proved here by the Chinese remainder theorem and the Ramanujan sum $\sum_{(y,m)=1} \zeta_m^y = \mu(m)$. Dirichlet characters as characters of $G_{\mathbb{Q}}$ are Chapter 6, twisting of modular forms by them Chapter 13.

Valuations and the $p$-adic absolute value are in Serre [172, Chapter II] and Neukirch [142, Chapter II], where Ostrowski's theorem (Theorem 2.85) is proved in Section 3. The arithmetic of the paper's local terms (Lemma 2.86 and Propositions 2.87 and 2.88) is elementary; the identification of these expressions with the Euler factors of the adjoint $L$-function is Lemma lem:euler of the paper, treated in Chapters 19 and 21. Curve data were computed with PARI/GP [148] and Cremona's tables [35].

The following are proved in full here although a statement with a reference would have sufficed: the existence of maximal ideals (from Zorn's lemma), the unique factorisation of $R[x]$ over a UFD, the primitivity of $\chi_D$ for fundamental $D$ (Corollary 2.76) and the Gauss sum of an imprimitive character. Only Ostrowski's theorem and the sign of the quadratic Gauss sum are stated without proof, and neither is used. No theorem numbers of the sources are cited. The

chapter and section numbers above are given as pointers to the relevant part of each book; they were not verified against the printed editions.

# Chapter 3

# Modules and commutative algebra

The paper's arithmetic is carried by modules: Selmer groups and Tamagawa factors are finite $\mathbb{Z}_\ell$-modules measured by length and by Fitting ideals; the Hecke algebra $\mathbb{T}$ is a finite free $\mathbb{Z}$-algebra whose completion $\mathbb{T}\otimes\mathbb{Z}_\ell$ splits into local factors $\mathbb{T}_\mathfrak{m}$; the cohomology $H^1(X_0(N),\mathbb{Z}_\ell)$ is a lattice with a perfect alternating pairing for which $\mathbb{T}$ is self-adjoint; and the inequality of Lemma lem:ddt417 compares a Gram determinant with a congruence ideal. This chapter develops, with complete proofs except where a reference is given, the commutative algebra behind all of this: exact sequences and the snake lemma, Hom and $\otimes$, Nakayama's lemma and localisation, modules over principal ideal domains and saturation, discrete valuation rings and $\mathbb{Z}_\ell$, length (with the Jordan-Holder theorem) and Fitting ideals, Pontryagin duality, pairings and Gram determinants, finite free algebras over a discrete valuation ring with their idempotents and generalised eigenspaces, congruence ideals and Gorenstein algebras. It uses the ring theory of Chapter 2 and the finite abelian groups of Chapter 1, and does not reprove them.

## Core results at a glance

Rings are commutative with 1 (except in item **3.36**); $\mathcal{O}$ is a discrete valuation ring with uniformiser $\varpi$, residue field $k$, fraction field $K$ and valuation $v$; $M^\vee=\mathrm{Hom}(M,R)$ is the linear dual and $M^*$ the Pontryagin dual; "length" is $\mathrm{length}_{\mathbb{Z}_\ell}$; $\mathbb{T}_\mathfrak{m}$ is a factor of $\mathbb{T}\otimes\mathbb{Z}_\ell$.

**3.1 Conventions.** Appendix B (B-C4, B-C10, B-C13, B-C15); rings commutative; $\mathcal{O}$, $\varpi$, $k$, $K$, $v$ with $v(\varpi)=1$; $M[\mathfrak{a}]$; lattices; pairings, left and right orthogonals, self-adjointness; lengths in place of the order $\#\mathcal{O}/\Delta$; relations are rows; the letters $I$, $J$ of Lemma lem:crit; $\mathbb{T}_\mathfrak{m}=e_\mathfrak{m}(\mathbb{T}\otimes\mathbb{Z}_\ell)$, never the algebraic localisation $\mathbb{T}_{(\mathfrak{m})}$. Conventions 3.1 and 3.2.

**3.2 Modules** (definition: module, submodule, quotient, finitely generated, cyclic, homomorphism, kernel, image, cokernel, $\mathrm{Hom}_R$, $\mathrm{End}_R$; action of an algebra, faithful action). Definitions 3.3 and 3.6.

**3.3 Basic facts.** The set $\mathrm{Hom}_R(M,N)$ is an $R$-module, the ring $\mathrm{End}_R(M)$ is an $R$-algebra, and $Rm\simeq R/\operatorname{Ann}_R(m)$ for $m\in M$. *(proved here)* Proposition 3.4.

**3.4 Isomorphism theorems.** $M/\ker f\simeq f(M)$; $(A+B)/B\simeq A/(A\cap B)$; $(M/B)/(A/B)\simeq M/A$ with the correspondence of submodules; for $f$ surjective with kernel $K_0$, $M/(K_0+A)\simeq N/f(A)$. *(proved here)* Theorem 3.8.

**3.5 Exact sequences; splitting lemma** (definition: exactness, short exact sequences). For $0 \to M' \to M \to M'' \to 0$: a section of $M \to M''$ exists iff a retraction of $M' \to M$ exists iff $M \simeq M' \oplus M''$ compatibly. *(proved here)* Definition 3.10, Lemma 3.11.

**3.6 Two submodules.** $0 \to M/(A \cap B) \to M/A \oplus M/B \to M/(A+B) \to 0$ is exact; $A + B$ is direct iff $A \cap B = 0$. *(proved here)* Proposition 3.13.

**3.7 Snake lemma.** A commutative diagram with exact rows $M' \to M \to M'' \to 0$ and $0 \to N' \to N \to N''$ gives an exact sequence $\ker f' \to \ker f \to \ker f'' \xrightarrow{\delta} \operatorname{coker} f' \to \operatorname{coker} f \to \operatorname{coker} f''$, with $0 \to$ at the left if $M' \to M$ is injective and $\to 0$ at the right if $N \to N''$ is surjective. *(proved here)* Lemma 3.15.

**3.8 Multiplication by $n$.** For $0 \to A \to B \to C \to 0$: $0 \to A[n] \to B[n] \to C[n] \to A/n \to B/n \to C/n \to 0$ is exact; $\#\ker\phi = \#\operatorname{coker}\phi$ for an endomorphism of a finite abelian group. *(proved here)* Corollary 3.16.

**3.9 Five lemma.** In a commutative ladder of exact rows of five terms, if the four outer maps are isomorphisms so is the middle one (with the two one-sided halves). *(proved here)* Lemma 3.17.

**3.10 Sums and products.** Hom turns direct sums in the first variable, and products in the second, into products; $M = A \oplus B$ iff $A + B = M$ and $A \cap B = 0$ iff $A = eM$, $B = (1-e)M$ for an idempotent $e \in \operatorname{End} M$. *(proved here)* Definition 3.18, Proposition 3.19.

**3.11 Free modules.** Universal property; every module has a presentation $F_1 \to F_0 \to M \to 0$; $R^m \simeq R^n$ implies $m = n$ $(R \neq 0)$; $A \in \mathrm{M}_n(R)$ is invertible iff $\det A \in R^\times$. *(proved here)* Definition 3.20, Proposition 3.21.

**3.12 Projective modules.** Equivalent: every surjection onto $P$ splits; $P$ is a summand of a free module; $\mathrm{Hom}(P, -)$ is exact; maps from $P$ lift along surjections. $eR$ is projective for an idempotent $e$; $\mathbb{Z}/2$ over $\mathbb{Z}/6$ is projective, not free. *(proved here)* Definition 3.22, Proposition 3.23.

**3.13 Hom.** $\mathrm{Hom}(R/\mathfrak{a}, M) = M[\mathfrak{a}]$; $\mathrm{Hom}(N, -)$ and $\mathrm{Hom}(-, N)$ are left exact, not right exact; $M \simeq M^{\vee\vee}$ for $M$ free of finite rank; over a domain $M^\vee$ is torsion-free. *(proved here)* Proposition 3.25.

**3.14 Tensor product.** Existence and universal property; $(R/\mathfrak{a}) \otimes M = M/\mathfrak{a}M$; distributivity; adjunction $\mathrm{Hom}(M \otimes N, P) \simeq \mathrm{Hom}(M, \mathrm{Hom}(N, P))$; base change $S \otimes_R M$. Examples: $\mathbb{Z}/m \otimes \mathbb{Z}/n = \mathbb{Z}/\gcd(m,n)$, $\mathbb{Z}_\ell \otimes \mathbb{Z}/n = \mathbb{Z}/\ell^{v_\ell(n)}$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell \otimes A = 0$ for $A$ torsion. *(proved here)* Theorem 3.26, Example 3.27.

**3.15 Right exactness.** $- \otimes N$ is right exact and not left exact; $N$ is flat if $- \otimes N$ is exact; projective modules are flat. *(proved here)* Proposition 3.28.

**3.16 Flatness.** $S^{-1}R$ is flat; over a principal ideal domain flat equals torsion-free (so $\mathbb{Z}_\ell$, $\mathbb{Q}$ are flat over $\mathbb{Z}$); for $R \to S$ flat, $S \otimes -$ commutes with kernels, images, cokernels, finite intersections, $M[\mathfrak{a}]$ ($\mathfrak{a}$ finitely generated) and Ann of finitely generated modules; saturation commutes with flat extensions of principal ideal domains. *(proved here)* Proposition 3.29.

**3.17 Noetherian modules.** Ascending chain condition, maximal condition, every submodule finitely generated: equivalent; closed under submodules, quotients, extensions; finitely generated modules over Noetherian rings are finitely presented; in a nonzero Noetherian ring every proper ideal lies in a maximal ideal; rings finite over Noetherian rings are Noetherian. Hilbert's basis theorem. *(proved here, except Hilbert's theorem: stated; reference)* Proposition 3.31.

**3.18 Determinant trick.** If $M$ has $n$ generators and $\phi(M) \subseteq \mathfrak{a}M$, then $\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0$ with $a_i \in \mathfrak{a}^i$; if $\mathfrak{a}M = M$ some $x \equiv 1 \bmod \mathfrak{a}$ kills $M$; surjective endomorphisms of finitely generated modules are bijective. *(proved here)* Lemma 3.32.

**3.19 Local rings.** $R$ is local iff its non-units form an ideal $\mathfrak{m}$; $1 + x$ is a unit for $x$ in the Jacobson radical; examples $\mathbb{Z}/p^n$, $k[x]/(x^2)$, $\mathbb{T}_{\mathfrak{m}}$; non-examples $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$, $\mathbb{Z}_{(11)}[x]/(x^2 + x - 1)$. *(proved here)* Definition 3.33, Proposition 3.34.

**3.20 Nakayama's lemma.** Over a local ring $(R, \mathfrak{m}, k)$ and $M$ finitely generated: $\mathfrak{m}M = M \Rightarrow M = 0$ (also for ideals in the Jacobson radical); $N + \mathfrak{m}M = M \Rightarrow N = M$; $x_i$ generate $M$ iff they span $M/\mathfrak{m}M$, and the minimal number of generators is $\dim_k M/\mathfrak{m}M$; surjectivity and invertibility can be tested modulo $\mathfrak{m}$; finitely generated projective modules are free. *(proved here)* Theorem 3.35.

**3.21 Localisation.** $S^{-1}R$ has the universal property for inverting $S$; $S^{-1}M = S^{-1}R \otimes M$ and localisation is exact; primes of $S^{-1}R$ are the primes of $R$ missing $S$; $R_{\mathfrak{p}}$ is local; $M = 0$ iff all $M_{\mathfrak{m}} = 0$; localisation commutes with sums, quotients and Ann of finitely generated modules (and with Hom out of finitely presented modules: stated); $(\mathbb{Z}/n)_{(p)} = \mathbb{Z}/p^{v_p(n)}$; localising $R_1 \times R_2$ at $\mathfrak{p}_1 \times R_2$ gives $(R_1)_{\mathfrak{p}_1}$. *(proved here, except the statement on* Hom*: stated; reference)* Theorem 3.38.

**3.22 Submodules of free modules.** Over a principal ideal domain a submodule of $R^n$ is free of rank $\leq n$; finitely generated projective modules are free. *(proved here)* Theorem 3.40.

**3.23 Smith normal form.** Over a principal ideal domain, $UAV = \operatorname{diag}(d_1, \ldots, d_s, 0, \ldots)$ with $d_1 \mid \cdots \mid d_s$, $U$, $V$ invertible; $(d_1 \cdots d_k)$ is the ideal of $k \times k$ minors, so the $(d_i)$ are unique. *(proved here)* Theorem 3.41.

**3.24 Structure theorem.** Over a principal ideal domain, $M \simeq R^r \oplus \bigoplus R/(d_i)$ ($d_1 \mid \cdots \mid d_s$ non-units) $\simeq R^r \oplus \bigoplus R/(p_j^{a_j})$, with $r$, the $(d_i)$ and the multiset of $(p_j^{a_j})$ unique. *(proved here)* Theorem 3.43.

**3.25 Torsion.** $M_{\mathrm{tor}} = \ker(M \to M \otimes K)$; over a principal ideal domain finitely generated torsion-free modules are free and $M \simeq M_{\mathrm{tor}} \oplus M/M_{\mathrm{tor}}$; if $R \subseteq R'$ with $R'$ a principal ideal domain free of rank $e$ over $R$, a finitely generated $R'$-module that is torsion-free of $R$-rank $n$ is $R'$-free of rank $n/e$ (so $T_3E$ is free of rank one over $\mathbb{Z}_3[\zeta_3]$ in Lemma lem:IV). *(proved here)* Definition 3.45, Proposition 3.46.

**3.26 Saturation.** For $L$ in a lattice $M$ over a principal ideal domain: $M/L$ torsion-free iff $L = M \cap KL$ iff $L$ is a direct summand iff (over $\mathcal{O}$) $L \cap \varpi M = \varpi L$; $\operatorname{sat}(L) = M \cap KL$ is the least saturated submodule containing $L$; kernels to torsion-free modules, $M \cap W$, $M[\mathfrak{a}]$ and orthogonals are saturated. *(proved here)* Definition 3.48, Proposition 3.49.

**3.27 Stacked bases.** For $L \subseteq M \simeq R^n$ of rank $d$ there is a basis $e_i$ of $M$ with $d_1 e_1, \dots, d_d e_d$ a basis of $L$, $d_1 \mid \cdots \mid d_d$ unique; $L$ is saturated iff all $d_i$ are units; for $d = n$, $[M : L] = |\det B|$ over $\mathbb{Z}$ and $\operatorname{length}(M/L) = v(\det B)$ over $\mathcal{O}$. *(proved here)* Theorem 3.50.

**3.28 Dual of an inclusion.** For lattices $L \subseteq M$ of the same rank, $M^\vee \hookrightarrow L^\vee$ and $L^\vee/M^\vee \simeq \operatorname{Hom}(M/L, K/R)$, which is $(M/L)^*$ over $\mathbb{Z}$ or $\mathbb{Z}_\ell$; so $[L^\vee : M^\vee] = [M : L]$. *(proved here)* Proposition 3.51.

**3.29 Congruence module of a decomposition.** If $M \otimes K = V_1 \oplus V_2$ and $L_i = M \cap V_i$, then $M/(L_1 \oplus L_2) \simeq p_1(M)/L_1 \simeq p_2(M)/L_2$; if $V_1 = Kf$ with $f$ primitive, the quotient is cyclic $\simeq R/(r)$, $rh = f + g_0$ with $p_1(h) = f/r$, $g_0 \in L_2$, and $f \equiv g \pmod{r'M}$ for some $g \in L_2$ iff $r' \mid r$. *(proved here)* Proposition 3.53.

**3.30 Discrete valuation rings.** For a Noetherian local domain $\mathcal{O}$, not a field: $\mathfrak{m}$ principal iff $\mathcal{O}$ is a principal ideal domain iff the nonzero ideals are the $\mathfrak{m}^k$ iff $\mathcal{O}$ is the ring of a discrete valuation iff $\mathcal{O}$ is integrally closed of dimension one; then $x = u\varpi^{v(x)}$ uniquely. $\mathcal{O}$ is *complete* if $\mathcal{O} \to \varprojlim \mathcal{O}/\varpi^n$ is bijective. *(proved here, except the equivalence with integral closure: stated; reference)* Theorem 3.55.

**3.31 Modules over a discrete valuation ring.** $M \simeq \mathcal{O}^r \oplus \bigoplus_{i \le s} \mathcal{O}/\varpi^{a_i}$ uniquely; torsion-free implies free; $\dim_k M/\varpi M = r + s$, $\dim_k M[\varpi] = s$; formulas for $M[\varpi^k]$, $M/\varpi^k M$; cyclic iff $r + s \le 1$; $L$ saturated iff $L \cap \varpi M = \varpi L$. *(proved here)* Proposition 3.57.

**3.32 The ring $\mathbb{Z}_\ell$.** $\mathbb{Z}_\ell = \varprojlim \mathbb{Z}/\ell^n$ is a discrete valuation ring with uniformiser $\ell$ and residue field $\mathbb{F}_\ell$; $\mathbb{Z}_\ell/\ell^n = \mathbb{Z}/\ell^n$; integers prime to $\ell$ are units, so a $\mathbb{Z}_\ell$-module killed by such an integer is 0; $\mathbb{Q}_\ell = \mathbb{Z}_\ell[1/\ell]$ and $\mathbb{Q} \cap \mathbb{Z}_\ell = \mathbb{Z}_{(\ell)}$; $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$ and $A \otimes \mathbb{Z}_\ell = \mathbb{Z}_\ell^r \oplus A_{\mathrm{tor}}[\ell^\infty]$; finitely generated $\mathbb{Z}_\ell$-modules are complete; completions of Noetherian rings are flat. *(proved here, except the statement on completions: stated; reference)* Theorem 3.58.

**3.33 Baer's criterion.** Over a principal ideal domain divisible modules are injective, and $\operatorname{Hom}(-, D)$ is exact. *(proved here)* Lemma 3.59.

**3.34 The module $\mathbb{Q}_\ell/\mathbb{Z}_\ell$.** $\mathbb{Q}_\ell/\mathbb{Z}_\ell = \mathbb{Z}[1/\ell]/\mathbb{Z} = (\mathbb{Q}/\mathbb{Z})[\ell^\infty] = \varinjlim \mathbb{Z}/\ell^k$; divisible, with $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)[\ell^k] \simeq \mathbb{Z}/\ell^k$ its only proper submodules; $\operatorname{End}_{\mathbb{Z}_\ell}(\mathbb{Q}_\ell/\mathbb{Z}_\ell) = \mathbb{Z}_\ell$; $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ and $\mathbb{Q}/\mathbb{Z}$ are injective. *(proved here)* Proposition 3.60.

**3.35 Topology.** Finitely generated $\mathbb{Z}_\ell$-modules are compact, Hausdorff and complete; linear maps are continuous, submodules and images closed; $\mathrm{GL}_n(\mathbb{Z}_\ell)$ is profinite; a compact $\mathbb{Z}_\ell$-module $A$ with $A/\ell A$ finite is finitely generated. *(proved here, except topological Nakayama: proof sketched here)* Proposition 3.61.

**3.36 Jordan-Holder and length.** Over any ring, any two composition series have the same length and factors; submodules and quotients of finite-length modules have finite length; length is additive in exact sequences; finite length iff Noetherian and Artinian (stated). Over a local ring simple modules are $\simeq k$, $\operatorname{length} M = \sum \dim_k \mathfrak{m}^i M/\mathfrak{m}^{i+1} M$ when $\mathfrak{m}^N M = 0$, and finitely generated modules over a Noetherian local ring have finite length iff killed by a power of $\mathfrak{m}$; $\operatorname{length}_R = f \operatorname{length}_{R'}$ for a finite local extension with residue degree $f$. *(proved here, except "Noetherian and Artinian": stated; reference)* Definition 3.64, Theorem 3.65.

**3.37 Length over a discrete valuation ring.** $\operatorname{length} \mathcal{O}/\varpi^a = a$; finite length iff torsion; $\#M = (\#k)^{\operatorname{length} M}$ for finite $k$, so $\operatorname{length}_{\mathbb{Z}_\ell} M = v_\ell(\#M)$; $\operatorname{length} M[\varpi^k] = \operatorname{length} M/\varpi^k M$ for finite length. *(proved here)* Proposition 3.66.

**3.38 Annihilators.** $\operatorname{Ann}(R/\mathfrak{a}) = \mathfrak{a}$; $\operatorname{Ann}(M \oplus N) = \operatorname{Ann} M \cap \operatorname{Ann} N$; $M[\mathfrak{a}] = \operatorname{Hom}(R/\mathfrak{a}, M)$; $\mathbb{T}[\mathfrak{a}] = \operatorname{Ann}_{\mathbb{T}}(\mathfrak{a})$ and $L[\mathfrak{a}] = \operatorname{Ann}_{\mathbb{T}}(\mathfrak{a})^r$ for $L \simeq \mathbb{T}^r$; flat base change; $\operatorname{Ann}(\bigoplus \mathcal{O}/\varpi^{a_i}) = \varpi^{\max a_i}$. *(proved here)* Definition 3.67, Proposition 3.68.

**3.39 Fitting ideals** (definition: for generators $x_1, \dots, x_n$, the ideal generated by the determinants of $n$-tuples of relations, equivalently by the $n \times n$ minors of a presentation matrix; it does not depend on the choices). *(proved here)* Definition 3.69, Lemma 3.70.

**3.40 Properties of Fitting ideals.** $\operatorname{Fitt}(R/\mathfrak{a}) = \mathfrak{a}$ and $\operatorname{Fitt}(R^n) = 0$; the Fitting ideal of a direct sum is the product, $\operatorname{Fitt}(M \oplus N) = \operatorname{Fitt}(M)\operatorname{Fitt}(N)$; Fitt grows along surjections; for $M$ with $n$ generators, $\operatorname{Ann}(M)^n \subseteq \operatorname{Fitt}(M) \subseteq \operatorname{Ann}(M)$; Fitt commutes with every base change; for a short exact sequence with $M'$ finitely generated, $\operatorname{Fitt}(M')\operatorname{Fitt}(M'') \subseteq \operatorname{Fitt}(M)$, with strict inclusion for the module $\mathbb{Z}[x]/(4, x)$; over a domain, $\operatorname{Fitt}(M) \neq 0$ if and only if $M$ is torsion. *(proved here)* Proposition 3.71.

**3.41 Fitting equals $\varpi^{\text{length}}$.** Over a principal ideal domain $\operatorname{Fitt}(\bigoplus R/(d_i)) = (\prod d_i)$ (and 0 with a free summand); over $\mathcal{O}$, $\operatorname{Fitt}(M) = \varpi^{\operatorname{length} M}\mathcal{O}$ for torsion $M$, so $\operatorname{Fitt}_{\mathbb{Z}_\ell}(M) = \#M \cdot \mathbb{Z}_\ell$, multiplicative in exact sequences; $\operatorname{Fitt}_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m) = \ell^{v+m}\mathbb{Z}_\ell$, while $\operatorname{Ann} = \ell^{\max(v,m)}\mathbb{Z}_\ell$. *(proved here)* Theorem 3.72, Example 3.74.

**3.42 Pontryagin duality for finite groups.** $A^* = \operatorname{Hom}(A, \mathbb{Q}/\mathbb{Z}) \simeq A$; $A \simeq A^{**}$; duality is exact; $B^\perp \simeq (A/B)^*$, $B^* \simeq A^*/B^\perp$; for $\ell$-primary $A$, $A^* = \operatorname{Hom}(A, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ and $\operatorname{length} A^* = \operatorname{length} A$. *(proved here)* Theorem 3.76.

**3.43 Pontryagin duality for $\mathbb{Z}_\ell$-modules.** $X^* = \operatorname{Hom}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is an exact duality between finitely generated $\mathbb{Z}_\ell$-modules and cofinitely generated $\ell$-primary modules, with $\mathbb{Z}_\ell^* = \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^* = \mathbb{Z}_\ell$, $X \simeq X^{**}$, and $W^* = \varprojlim W[\ell^k]^*$ profinite; every cofinitely generated $W$ is $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r \oplus F$ with $F$ finite and $r = \operatorname{rank} W^*$; lengths are preserved. *(proved here; the extension to all compact and discrete modules: stated; reference)* Theorem 3.77.

**3.44 $T$ and $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$.** For $T$ free of rank $r$: $0 \to T \to V \to W \to 0$ is exact; $W \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$; $W[\ell^k] \simeq T/\ell^k T$; $T = \varprojlim W[\ell^k]$; $W^* \simeq T^\vee$; all functorially; the same over any complete $\mathcal{O}$. *(proved here)* Proposition 3.78.

**3.45 Cyclic $\ell$-primary modules.** An $\ell$-primary torsion module with $\#A[\ell] \leq \ell$ is 0, $\mathbb{Z}/\ell^h$ or $\mathbb{Q}_\ell/\mathbb{Z}_\ell$; if finite its length is the largest $k$ with an element of exact order $\ell^k$. *(proved here)* Corollary 3.79.

**3.46 Perfect pairings and Gram determinants.** A pairing between free modules of rank $n$ over any ring is perfect iff its Gram determinant is a unit (over a domain, nondegenerate iff it is nonzero); $\det(P^{\mathsf{T}} G Q) = \det P \det Q \det G$, so $\Delta R$, and over $\mathcal{O}$ the valuation $v(\Delta)$, are basis independent; perfectness survives base change. *(proved here)* Definition 3.81, Proposition 3.82.

**3.47 Orthogonals.** For a perfect pairing on a lattice of rank $n$, orthogonals are saturated, $N^\perp$ has rank $n - \operatorname{rank} N$, and ${}^\perp(N^\perp) = \operatorname{sat}(N)$. *(proved here)* Proposition 3.83.

**3.48 Gram determinant of a saturated sublattice.** For a perfect pairing on $L$ over $\mathcal{O}$ and a *saturated* $L' \subseteq L$ with Gram determinant $\Delta \neq 0$: $L/L'^{\perp} \simeq L'^{\vee}$, $L' \cap L'^{\perp} = 0$, and $L/(L' \oplus L'^{\perp}) \simeq \operatorname{coker}(L' \to L'^{\vee})$ has length $v(\Delta)$ (order $\#\mathcal{O}/\Delta$ for finite $k$); $L'^{\perp}$ has the same Gram valuation; saturation is necessary. *(proved here)* Theorem 3.84.

**3.49 Alternating pairings.** Alternating implies antisymmetric (and conversely if 2 is a nonzerodivisor); odd-rank alternating Gram determinants vanish; in rank two $\Delta = a^2$ with $a\mathcal{O}$ the ideal of values; $\det = \mathrm{Pf}^2$ (stated, checked in rank 4); symplectic normal form. *(proved here, except the Pfaffian identity: stated; reference; and the normal form: proof sketched here)* Proposition 3.87.

**3.50 Adjoints.** For a perfect pairing $A^{\dagger} = G^{-1}A^{\mathsf{T}}G$; for a commutative algebra $\mathbb{T}$ of self-adjoint operators $(\mathfrak{a}L)^{\perp} = L[\mathfrak{a}]$, idempotents of $\mathbb{T}$ (or $\mathbb{T} \otimes S$) give orthogonal decompositions on whose summands the pairing stays perfect; if $A^{\dagger} = wAw^{-1}$ on $\mathbb{T}$ then $\langle x, wy\rangle$ is perfect with $\mathbb{T}$ self-adjoint, and alternating if $\langle\ ,\ \rangle$ is antisymmetric and $w^{\dagger} = w$. *(proved here)* Proposition 3.89.

**3.51 An algebra paired with a module.** If $S \times \mathbb{T} \to R$ is perfect with $\langle ts, u\rangle = \langle s, tu\rangle$, then $S \simeq \mathbb{T}^{\vee}$ as $\mathbb{T}$-modules, $S[\mathfrak{a}] \simeq \operatorname{Hom}(\mathbb{T}/\mathfrak{a}, R)$ is saturated, $\mathbb{T}$ acts faithfully, and for $\mathfrak{a} \cap \mathfrak{b} = 0$ with $\mathbb{T}/\mathfrak{a}$, $\mathbb{T}/\mathfrak{b}$ torsion-free and complementary ranks, $S/(S[\mathfrak{a}] \oplus S[\mathfrak{b}]) \simeq (\mathbb{T}/(\mathfrak{a}+\mathfrak{b}))^*$ over $\mathbb{Z}$ or $\mathbb{Z}_{\ell}$. *(proved here)* Proposition 3.91.

**3.52 Finite-dimensional commutative algebras.** Finitely many maximal ideals, all primes maximal, nilpotent Jacobson radical; $A \simeq \prod A_{\mathfrak{m}_i}$ via idempotents; reduced iff a product of fields; $V = \bigoplus e_i V$ with $e_i V = V_{\mathfrak{m}_i}$. *(proved here)* Theorem 3.93.

**3.53 Finite free algebras over $\mathcal{O}$.** Noetherian; maximal ideals contain $\varpi$ and correspond to those of $\mathbb{T}/\varpi\mathbb{T}$; $\mathbb{T} \subseteq \mathbb{T}_K$; $\mathbb{T}$ reduced iff $\mathbb{T}_K$ is; for $\pi\colon \mathbb{T} \twoheadrightarrow \mathcal{O}$, $\mathbb{T} = \mathcal{O} \oplus \wp$, $\mathfrak{m}_{\pi} = \wp + \varpi\mathbb{T}$, and $\pi_K$ factors through exactly one local factor of $\mathbb{T}_K$. *(proved here)* Theorem 3.94.

**3.54 Idempotent lifting.** Over a complete $\mathcal{O}$, idempotents of $\mathbb{T}/\varpi\mathbb{T}$ lift uniquely (Newton iteration $e \mapsto 3e^2 - 2e^3$); $\mathbb{T} = \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$ with $\mathbb{T}_{\mathfrak{m}} = e_{\mathfrak{m}}\mathbb{T}$ local and finite free; $L = \bigoplus e_{\mathfrak{m}}L$; a map to a local ring factors through one $\mathbb{T}_{\mathfrak{m}}$; completeness is necessary ($\mathbb{Z}_{(11)}[x]/(x^2+x-1)$). *(proved here)* Theorem 3.95.

**3.55 Hecke-type algebras.** If $\mathbb{T}$ is free of finite rank over $\mathbb{Z}$, then $\mathbb{T} \otimes \mathbb{Z}_{\ell} = \prod_{\mathfrak{m} \ni \ell} \mathbb{T}_{\mathfrak{m}}$ over the maximal ideals of $\mathbb{T}$ containing $\ell$. Example: the level-23 algebra $\mathbb{Z}[x]/(x^2+x-1)$ splits over $\mathbb{Z}_{11}$, and is local over $\mathbb{Z}_2$, $\mathbb{Z}_3$, $\mathbb{Z}_5$. *(proved here)* Corollary 3.96, Example 3.98.

**3.56 Generalised eigenspaces.** For a commutative $\mathbb{T} \subseteq \mathrm{End}_F(V)$ with split characteristic polynomials, $V = \bigoplus_{\phi} V(\phi)$ over the systems of eigenvalues, with projections the idempotents of $\mathbb{T}$, a common eigenvector in each $V(\phi)$, and $W = \bigoplus(W \cap V(\phi))$ for stable $W$; integrally, over a complete $\mathcal{O}$, systems of eigenvalues are $\mathcal{O}$-valued, $\mathfrak{m}_{\phi} = \mathfrak{m}_{\psi}$ iff $\phi \equiv \psi \bmod \varpi$, and $e_{\mathfrak{m}}L \otimes K = \bigoplus_{\mathfrak{m}_{\phi} = \mathfrak{m}} V(\phi)$. *(proved here)* Theorem 3.100.

**3.57 Projecting a congruence.** $e_{\mathfrak{m}}$ preserves $L$ and $\mathbb{T}$-stable subspaces and fixes $V(\phi)$ for $\mathfrak{m}_{\phi} = \mathfrak{m}$; so $x \equiv y \pmod{\varpi^t L}$ with $x \in V(\phi)$ gives $x \equiv e_{\mathfrak{m}}y$. *(proved here)* Proposition 3.102.

**3.58 Semisimple operators.** $A$ is semisimple iff $\mu_A$ is squarefree iff $F[A]$ is reduced (iff diagonalisable over $\overline{F}$ for perfect $F$); a cyclic vector makes $\mu_A$ the characteristic polynomial;

a commutative $\mathbb{T}$ is reduced iff all its elements are semisimple; Jordan-Chevalley (stated). The shift operator ($U_p$ on oldforms) has characteristic polynomial $X^{e-1}(X^2 - aX + p)$, is not semisimple for $e \geq 3$, yet every eigenvalue has an eigenvector. *(proved here, except Jordan-Chevalley: stated; reference)* Proposition 3.104, Example 3.105.

**3.59 Common eigenvectors.** Over an algebraically closed field, a nonzero joint generalised eigenspace of commuting operators contains a common eigenvector of every commutative algebra containing them; for operators acting on the factors of a tensor product, $V(\lambda) = \bigotimes V_i(\lambda_i)$. *(proved here)* Lemma 3.107.

**3.60 Congruence ideal** (definition: for $\pi\colon \mathbb{T} \twoheadrightarrow \mathcal{O}$, $\wp = \ker \pi$, $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$, $\eta = \pi(I)$; congruence module $\mathbb{T}/(\wp + I) \simeq \mathcal{O}/\eta$). Definition 3.109.

**3.61 Congruence ideals.** $I \neq 0$; $\eta \neq 0$ iff $\wp \cap I = 0$ iff the factor of $\pi$ in $\mathbb{T}_K$ is $K$ (proof: $\pi(t)x = tx + (\pi(t) - t)x = 0$); then $I \simeq \eta$, $\mathrm{Ann}_{\mathbb{T}}(I) = \wp$, $\eta$ is the denominator of $e_\pi$; over complete $\mathcal{O}$, $\mathrm{Ann}_{\mathbb{T}}(\wp) = \mathrm{Ann}_{\mathbb{T}_\mathfrak{m}}(\wp_\mathfrak{m}) \times 0$, so $\eta$ is computed in $\mathbb{T}_\mathfrak{m}$ (the other factors lie in $\wp$); $\eta$ commutes with flat base change. *(proved here)* Proposition 3.110, Example 3.112.

**3.62 Ingredients of Lemma lem:ddt417.** For $L$ free over $\mathcal{O}$ with a perfect pairing making $\mathbb{T}$ self-adjoint and $\eta \neq 0$: $L[\wp]$, $L[I]$ are saturated, $L[\wp] \cap L[I] = 0$, $L[\wp]^\perp = L[I]$, $Q = L/(L[\wp] \oplus L[I])$ is killed by $\eta$, and $Q \simeq (\mathcal{O}/\eta)^r$ if $L \simeq \mathbb{T}^r$. *(proved here)* Proposition 3.113, Remark 3.118.

**3.63 Gorenstein algebras** (definition: $\mathrm{Hom}_{\mathcal{O}}(\mathbb{T}, \mathcal{O})$ free of rank one over $\mathbb{T}$). Equivalent to a perfect pairing $\lambda(st)$; over complete $\mathcal{O}$ local by local; for local $\mathbb{T}$, iff the socle of $\mathbb{T}/\varpi\mathbb{T}$ is a line; $\mathcal{O}[x]/(f)$ and $\mathbb{T}_r$ are Gorenstein, $\mathbb{T}_3^\flat$ is not; complete intersections are Gorenstein (stated). *(proved here, except the last statement: stated; reference)* Definition 3.114, Proposition 3.115.

**3.64 Mazur-style freeness.** For $\mathbb{T}$ local finite free over $\mathcal{O}$ and $L$ free over $\mathcal{O}$ with a perfect pairing making $\mathbb{T}$ self-adjoint: if $\dim L/\mathfrak{m}L = d$ and $\mathrm{rank}\, L = d\,\mathrm{rank}\,\mathbb{T}$, then $L \simeq \mathbb{T}^d$ and $\mathbb{T}$ is Gorenstein. *(proved here)* Theorem 3.116.

## Where the paper uses this chapter

- **Section 1.2** (the Tamagawa factor $\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}})$): Fitting ideals and $\mathrm{Fitt} = \ell^{\mathrm{length}}$ (Definition 3.69 and Theorem 3.72).
- **Lemma lem:tamfactor** (the Tamagawa factor at a multiplicative prime): the structure $H^1(I, M) = (\mathbb{Z}_\ell/n')^2 \oplus \mathbb{Z}_\ell$ and its torsion (Theorem 3.43); the Fitting ideal of $\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m$ (Theorem 3.72, Paper link 3.75).
- **Lemma lem:fpr** ($\mathrm{Fitt}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}_p, T^D) = L_p(A_f, 1)^{-1}\,\mathrm{Tam}_p(T^D)$): Fitting ideals as lengths (Theorem 3.72); $\mathbb{Q} \cap \mathbb{Z}_\ell = \mathbb{Z}_{(\ell)}$ (Theorem 3.58(iv)).
- **Theorem thm:main, Theorem thm:dfg37, Remark rem:ns, Conjecture conj:euler** (lengths of Selmer groups and of $H^0(\mathbb{Q}, W^D)$): additivity of length (Theorem 3.65), length over $\mathbb{Z}_\ell$ (Proposition 3.66), Pontryagin duality preserves length (Theorem 3.77, Paper link 3.80).

- **Lemma lem:dfg21** (the four-term sequence with Pontryagin duals) and the paragraph after it ($H^1(\mathbb{R}, T^D)$ killed by 2): Theorems 3.76 and 3.77, Theorem 3.58(iii).
- **Lemma lem:h0** ($W^D$, its $\ell^k$-torsion, cyclicity, Nakayama): Proposition 3.78 and Corollary 3.79, Theorem 3.35.
- **Lemma lem:crit** (congruence criterion): $\mathbb{T}/(I+J) \simeq \mathbb{Z}/\phi_f(J)$ (Theorem 3.8); $\mathbb{T} \hookrightarrow \mathbb{T}/I \times \mathbb{T}/J$ (Proposition 3.13); dualising it (Propositions 3.51 and 3.91); the congruence module of $S_2 = \mathbb{Z}f + L$ (Proposition 3.53); $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ and $I \cap J = 0$ (Theorem 3.93, Proposition 3.110); the passage to $\mathbb{Z}_\ell$ (Proposition 3.29). The lemma itself is proved in full in Chapter 15.
- **Lemma lem:old** (oldforms): the non-semisimple $U_p$ and its eigenvectors (Example 3.105); common eigenforms in a tensor product (Lemma 3.107).
- **Lemma lem:IV** and Lemma lem:tamadd ($T_3E$ free of rank one over $\mathbb{Z}_3[\zeta_3]$): Proposition 3.46(iii).
- **Section 4.1** (the pairing $\hat{\delta}_!$, perfect, alternating, with self-adjoint Hecke operators; Gram determinant $(\eta_{f,\lambda}^{\Sigma})^2$) and **Proposition prop:pullback**: Propositions 3.87 and 3.89.
- **Lemma lem:ddt417** (inequality half of DDT Lemma 4.17): every ingredient, Propositions 3.110 and 3.113 and Theorem 3.84, Proposition 3.49, summarised in Remark 3.118; the lemma itself is proved in Chapter 20.
- **Lemma lem:hecke** and the paragraph after it ($\mathbb{T}_\mathfrak{m}$, the congruence ideal computed in $\mathbb{T}_\mathfrak{m}$, $e_\mathfrak{m}$ self-adjoint): Convention 3.2, Theorem 3.95 and Corollary 3.96, Proposition 3.110(v), Proposition 3.89(iii).
- **Remark rem:free** (Gorenstein, freeness): Proposition 3.115 and Theorem 3.116, Theorem 3.35.
- **Theorem thm:twist** ($g^\perp$ as a sum of generalised eigenspaces) and **Lemma lem:step1** and **Theorem thm:quant**, Steps 1 and 2 (idempotents of $\mathbb{T}_\ell$, generalised eigenspaces, projecting a congruence, the $\mathbb{Z}_\ell$-version of Lemma lem:crit, Lemma lem:ddt417 without freeness): Theorem 3.100, Proposition 3.102, Theorem 3.95, Proposition 3.29.

## 3.1 Modules, submodules, quotients and homomorphisms

This section fixes the language. Every object of the paper's commutative algebra is a module over a commutative ring: the space $S_2(\Gamma_0(N);\mathbb{Z})$ of integral cusp forms and the cohomology $H^1(X_0(N),\mathbb{Z}_\ell)$ are modules over the Hecke algebra $\mathbb{T}$; the Tate module $T = T_\ell E$ is a module over $\mathbb{Z}_\ell$, and in Lemma lem:IV over $\mathbb{Z}_3[\zeta_3]$; the lattice $L$ of Lemma lem:ddt417 is a module over an $\mathcal{O}$-algebra $\mathbb{T}$. We first record the conventions of the chapter.

**Convention 3.1** (notation of the chapter)**.** The book-wide conventions of Appendix B apply, in particular B-C4 (one symbol for each dual: the linear dual $M^\vee = \mathrm{Hom}_R(M,R)$, the Pontryagin dual $M^*$, the Kummer dual $T^D$, which is not used here, and the adjoint $A^\dagger$ of an operator, never $A^*$; a star after the name of a map is a pullback), B-C10 ("length" without subscript is $\mathrm{length}_{\mathbb{Z}_\ell}$, and $\#X = \ell^{\mathrm{length}\,X}$ for finite $\mathbb{Z}_\ell$-modules), B-C13(a), (c) (the paper's $\mathbb{T}_\mathfrak{m}$ and its two uses of the

letter $I$) and B-C15(a), (h) (uniformisers are $\varpi$; the paper's $\pi\colon \mathbb{T} \to \mathcal{O}$ and Gram determinant $\Delta$ keep their names). In addition:

(a) Rings are commutative with 1, ring homomorphisms preserve 1 and modules are unital, as in Chapter 2, except in the Jordan-Holder theorem (Theorem 3.65). An $R$-algebra is a ring $A$ with a ring homomorphism $R \to A$. The algebras $\mathbb{T}$ acting on modules, like the Hecke algebra, are commutative subrings of the (noncommutative) ring $\mathrm{End}_R(M)$.

(b) $\mathcal{O}$ is a discrete valuation ring (Theorem 3.55) with uniformiser $\varpi$, residue field $k = \mathcal{O}/\varpi\mathcal{O}$, fraction field $K$ and valuation $v\colon K \to \mathbb{Z} \cup \{+\infty\}$, $v(\varpi) = 1$; $v(a\mathcal{O}) = v(a)$ for a nonzero ideal. For $\mathcal{O} = \mathbb{Z}_\ell$: $\varpi = \ell$, $k = \mathbb{F}_\ell$, $v = v_\ell$. The paper writes $\lambda$ and $v_\lambda$.

(c) $M[\mathfrak{a}] = \{m \in M : am = 0 \text{ for all } a \in \mathfrak{a}\}$, $M[x] = M[xR]$ and $M[\ell^\infty] = \bigcup_k M[\ell^k]$; these are the paper's $L[\wp]$ and $L[I]$. The Pontryagin dual is $M^* = \mathrm{Hom}(M, \mathbb{Q}/\mathbb{Z})$ for a finite abelian group and $M^* = \mathrm{Hom}_{\mathbb{Z}_\ell}(M, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ for a $\mathbb{Z}_\ell$-module that is finitely generated or $\ell$-primary torsion (the paper's $^*$ in Lemma lem:dfg21).

(d) Over a principal ideal domain $R$ with fraction field $K$, a *lattice* is a finitely generated free $R$-module, and a lattice in a finite-dimensional $K$-vector space $V$ is a finitely generated $R$-submodule spanning $V$ (free of rank $\dim V$ by Theorem 3.40).

(e) A *pairing* is an $R$-bilinear map $\langle\ ,\ \rangle\colon L \times M \to R$; its Gram matrix in bases $(e_i)$, $(f_j)$ is $G = (\langle e_i, f_j\rangle)_{i,j}$. For $S' \subseteq L$ and $S \subseteq M$, $S'^{\perp} = \{y \in M : \langle S', y\rangle = 0\}$ is the *right* and ${}^{\perp}S = \{x \in L : \langle x, S\rangle = 0\}$ the *left* orthogonal; for a symmetric or antisymmetric pairing on $L = M$ they agree and we write $S^{\perp}$. An endomorphism $t$ of $L$ is *self-adjoint* if $\langle tx, y\rangle = \langle x, ty\rangle$ for all $x, y$.

(f) The paper's order $\#\mathcal{O}/\Delta$ is $(\#k)^{v(\Delta)}$ and makes sense only for finite $k$; the book writes the length $\mathrm{length}_{\mathcal{O}}(\mathcal{O}/\Delta\mathcal{O}) = v(\Delta)$.

(g) Relations are rows: a presentation $R^m \xrightarrow{A} R^n \to M \to 0$ means $M = R^n/(\text{row space of } A)$. The letter $\varepsilon$ is not used (dual numbers are $k[x]/(x^2)$), and no Frobenius element occurs.

(h) Where the text quotes Lemma lem:crit (in the list above and in paperlinks), $I = \ker \phi_f$ and $J = \mathrm{Ann}_{\mathbb{T}}(f^{\perp})$ are that lemma's letters ($J$ equals $\mathrm{Ann}_{\mathbb{T}}(I)$, Proposition 3.91 and its paperlink); everywhere else $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$, as in Lemma lem:ddt417.

**Convention 3.2** (the paper's $\mathbb{T}_{\mathfrak{m}}$)**.** Let $\mathbb{T}$ be a commutative ring that is free of finite rank as a $\mathbb{Z}$-module, and $\mathfrak{m}$ a maximal ideal of $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$ (equivalently, by Corollary 3.96, a maximal ideal of $\mathbb{T}$ containing $\ell$). Then $\mathbb{T}_{\mathfrak{m}}$ *always* means $(\mathbb{T} \otimes \mathbb{Z}_\ell)_{\mathfrak{m}}$, the direct factor $e_{\mathfrak{m}}\mathbb{T}_\ell$ of $\mathbb{T}_\ell$ cut out by an idempotent $e_{\mathfrak{m}}$ (Theorem 3.95); likewise $L_{\mathfrak{m}} = e_{\mathfrak{m}}(L \otimes \mathbb{Z}_\ell)$ for a $\mathbb{T}$-module $L$. The algebraic localisation of $\mathbb{T}$ itself at a maximal ideal is written $\mathbb{T}_{(\mathfrak{m})}$; it is not complete and is not a direct factor of anything. This is the paper's usage in Lemma lem:hecke, Remark rem:free and Theorem thm:quant.

**Definition 3.3** (modules)**.** An *$R$-module* is an abelian group $M$ with a map $R \times M \to M$, $(r, m) \mapsto rm$, such that $r(m + m') = rm + rm'$, $(r + r')m = rm + r'm$, $(rr')m = r(r'm)$ and $1m = m$. A *submodule* is a subgroup $N \subseteq M$ with $RN \subseteq N$, and the *quotient* $M/N$ is the module of cosets with $r(m + N) = rm + N$. The submodule generated by a family $(m_i)$ is $\sum_i Rm_i$, the set of finite sums $\sum r_i m_i$; $M$ is *finitely generated* (f.g.) if $M = \sum_{i=1}^n Rm_i$ for some $m_1, \ldots, m_n$,

and *cyclic* if $M = Rm$. A *homomorphism* ($R$-linear map) $f\colon M \to N$ is a group homomorphism with $f(rm) = rf(m)$; its kernel $\ker f$, image $f(M) = \operatorname{im} f$ and cokernel $\operatorname{coker} f = N/f(M)$ are modules. $\operatorname{Hom}_R(M,N)$ is the set of homomorphisms and $\operatorname{End}_R(M) = \operatorname{Hom}_R(M,M)$.

**Proposition 3.4.** *(i)* $\operatorname{Hom}_R(M,N)$ *is an* $R$*-module under* $(f+g)(m) = f(m)+g(m)$ *and* $(rf)(m) = rf(m)$*, and* $\operatorname{End}_R(M)$ *is a ring under composition, with an* $R$*-algebra structure* $R \to \operatorname{End}_R(M)$*,* $r \mapsto (m \mapsto rm)$*.*

*(ii) For* $m \in M$ *let* $\operatorname{Ann}_R(m) = \{r \in R : rm = 0\}$*. Then* $\operatorname{Ann}_R(m)$ *is an ideal and* $R/\operatorname{Ann}_R(m) \simeq Rm$*,* $r \mapsto rm$*.*

*Proof.* (i) $rf$ is $R$-linear because $R$ is commutative: $(rf)(sm) = rsf(m) = s(rf)(m)$. This is the only place where commutativity matters; over a noncommutative ring $rf$ would in general not be linear. The module axioms for $\operatorname{Hom}_R(M,N)$ are inherited pointwise from those of $N$: for instance $((r+r')f)(m) = (r+r')f(m) = rf(m)+r'f(m) = (rf+r'f)(m)$. Composition of linear maps is linear and distributes over the sum, so $\operatorname{End}_R(M)$ is a ring with unit $\mathrm{id}_M$. The map $r \mapsto (m \mapsto rm)$ preserves sums and products by the module axioms ($(rr')m = r(r'm)$), sends 1 to $\mathrm{id}_M$, and its image is central: for $g \in \operatorname{End}_R(M)$, $g(rm) = rg(m)$ is exactly the statement that $g$ commutes with multiplication by $r$. (ii) The map $R \to Rm$, $r \mapsto rm$, is $R$-linear and surjective with kernel $\operatorname{Ann}_R(m)$; apply Theorem 3.8(i) below (whose proof does not use this proposition). □

**Example 3.5.** A $\mathbb{Z}$-module is an abelian group, with $nm = m + \cdots + m$; a module over a field $K$ is a $K$-vector space. A module over the polynomial ring $K[x]$ is a $K$-vector space $V$ together with one endomorphism (the action of $x$); submodules are the subspaces stable under it. This dictionary underlies Section 3.10. A module over $R/\mathfrak{a}$ is the same as an $R$-module killed by $\mathfrak{a}$. The submodules of $R$ are its ideals.

**Definition 3.6** (algebras acting on modules)**.** Let $\phi\colon R \to S$ be a ring homomorphism. Every $S$-module $M$ is an $R$-module by $rm = \phi(r)m$ (*restriction of scalars*). If $A$ is an $R$-algebra and $M$ an $R$-module, an *action* of $A$ on $M$ is a homomorphism of $R$-algebras $A \to \operatorname{End}_R(M)$; equivalently, an $A$-module structure on $M$ whose restriction to $R$ is the given one. The action is *faithful* if $A \to \operatorname{End}_R(M)$ is injective.

**Example 3.7** (running examples)**.** (a) A module over $R_1 \times R_2$ is the same as a pair of modules: with the idempotents $e_1 = (1,0)$ and $e_2 = (0,1)$ (Chapter 2), $M = e_1M \oplus e_2M$, where $e_iM$ is an $R_i$-module, and every pair $(M_1, M_2)$ arises from $M_1 \times M_2$. This is how a module over $\mathbb{T} \otimes \mathbb{Z}_\ell = \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$ splits (Theorem 3.95).

(b) The Hecke algebra $\mathbb{T} \subset \operatorname{End}_{\mathbb{Z}} S_2(\Gamma_0(N);\mathbb{Z})$ acts faithfully on the integral cusp forms, by definition (Chapters 12 and 13); it also acts on $H^1(X_0(N),\mathbb{Z}_\ell)$ (Chapter 14).

(c) Let $T$ be a free $\mathbb{Z}_3$-module of rank 2 and $\sigma \in \operatorname{Aut}(T)$ with $\sigma^2 + \sigma + 1 = 0$ (the generator of the image of inertia in Lemma lem:IV). Then $T$ is a module over $\mathbb{Z}_3[X]/(X^2+X+1)$, with $X$ acting as $\sigma$; this ring is $\mathbb{Z}_3[\zeta_3]$, a discrete valuation ring (Chapter 6). Proposition 3.46(iii) shows that $T$ is free of rank one over it.

(d) The toy Hecke algebras of Example 3.112, which are the smallest algebras exhibiting a congruence.

**Theorem 3.8** (isomorphism theorems). *(i) For $f\colon M\to N$, $m+\ker f\mapsto f(m)$ is an isomorphism $M/\ker f\simeq f(M)$.*

*(ii) For submodules $A,B\subseteq M$, $(A+B)/B\simeq A/(A\cap B)$.*

*(iii) For submodules $B\subseteq A\subseteq M$, $(M/B)/(A/B)\simeq M/A$, and $A\mapsto A/B$ is an inclusion-preserving bijection between submodules of $M$ containing $B$ and submodules of $M/B$.*

*(iv) If $f\colon M\to N$ is surjective with kernel $K_0$ and $A\subseteq M$ is a submodule, then $f$ induces an isomorphism $M/(K_0+A)\simeq N/f(A)$.*

*Proof.* (i) The map is well defined and injective because $f(m)=f(m')$ iff $m-m'\in\ker f$; it is $R$-linear and onto $f(M)$. (ii) Apply (i) to $A\to(A+B)/B$, $a\mapsto a+B$, which is surjective (every $a+b+B$ equals $a+B$) with kernel $A\cap B$. (iii) Apply (i) to $M/B\to M/A$, $m+B\mapsto m+A$, surjective with kernel $A/B$. A submodule $X\subseteq M/B$ is the image of its preimage, which contains $B$, and $A=\{m : m+B\in A/B\}$ for $A\supseteq B$. (iv) The composite $M\to N\to N/f(A)$ is surjective; its kernel is $f^{-1}(f(A))$. This equals $A+K_0$: if $f(m)=f(a)$ with $a\in A$, then $m-a\in K_0$, so $m\in A+K_0$; conversely $f(a+k)=f(a)$ for $k\in K_0$. Apply (i). □

Part (iii) is used constantly in the form "a submodule of a quotient is the image of a submodule containing what was divided out"; for instance the subgroups of $\mathbb{Z}/n$ are the $d\mathbb{Z}/n\mathbb{Z}$ for $d\mid n$, the images of the subgroups $d\mathbb{Z}\supseteq n\mathbb{Z}$. Part (iv) says that dividing first by the kernel of $f$ and then by $f(A)$ is the same as dividing by $K_0+A$ at once.

*In the paper* 3.9. Lemma lem:crit writes "$\mathbb{T}/(I+J)\simeq\mathbb{Z}/\phi_f(J)$". This is Theorem 3.8(iv) for the surjection $\phi_f\colon\mathbb{T}\to\mathbb{Z}$ with kernel $I$ and the submodule $A=J$. In the same way the congruence module of Definition 3.109 is $\mathbb{T}/(\wp+I)\simeq\mathcal{O}/\pi(I)$ for $\pi\colon\mathbb{T}\twoheadrightarrow\mathcal{O}$ with kernel $\wp$. A toy case: for $\mathbb{T}=\mathbb{Z}[x]/(x^2-4)$, $\phi(x)=2$, the kernel is $I=(x-2)$; take $J=(x+2)$. In $\mathbb{T}/(I+J)$ one has $x\equiv 2$ and $x\equiv -2$, hence $4\equiv 0$, and $\mathbb{T}/(I+J)\simeq\mathbb{Z}/4$; on the other side $\phi(J)=\phi(x+2)\phi(\mathbb{T})=4\mathbb{Z}$, so $\mathbb{Z}/\phi(J)=\mathbb{Z}/4$ as (iv) predicts. This algebra returns in Example 3.112(b) as the smallest example of a congruence modulo 4.

## 3.2 Exact sequences, the snake lemma, sums, free and projective modules

Exact sequences are the bookkeeping device of the whole book. The paper's identity (eq:identity) is obtained by adding lengths along the four-term exact sequence of Lemma lem:dfg21, and every connecting map of Chapter 16 is built from the snake lemma proved here. The sequence of two submodules (Proposition 3.13) is the exact form of two sentences of the paper: "the inclusion $\mathbb{T}\subset\mathbb{T}/I\times\mathbb{T}/J$, whose cokernel is $\mathbb{T}/(I+J)$" in Lemma lem:crit, and the directness of the sum $L[\wp]+L[I]$ in Lemma lem:ddt417. Projective modules enter because the local factors $e\mathbb{T}$ of a finite free $\mathbb{Z}_\ell$-algebra are again free over $\mathbb{Z}_\ell$.

**Definition 3.10** (exact sequences). A sequence of homomorphisms $\cdots\to M_{i-1}\xrightarrow{f_{i-1}}M_i\xrightarrow{f_i}M_{i+1}\to\cdots$ is *exact at $M_i$* if $\ker f_i=\operatorname{im} f_{i-1}$, and *exact* if it is exact at every interior term. A *short exact sequence* is an exact sequence $0\to M'\xrightarrow{i}M\xrightarrow{p}M''\to 0$: $i$ is injective, $p$ is surjective and $\operatorname{im} i=\ker p$, so $M''\simeq M/i(M')$. It *splits* if the equivalent conditions of Lemma 3.11 hold.

**Lemma 3.11** (splitting lemma)**.** *For a short exact sequence* $0 \to M' \xrightarrow{i} M \xrightarrow{p} M'' \to 0$ *the following are equivalent: (a) there is* $s\colon M'' \to M$ *with* $ps = \mathrm{id}$*; (b) there is* $r\colon M \to M'$ *with* $ri = \mathrm{id}$*; (c) there is an isomorphism* $\theta\colon M \to M' \oplus M''$ *with* $\theta i(x) = (x, 0)$ *and* $p\theta^{-1}(x, y) = y$*.*

*Proof.* (a)⇒(c): for $m \in M$, $m - sp(m) \in \ker p = \operatorname{im} i$, so $m - sp(m) = i(r(m))$ for a unique $r(m) \in M'$; put $\theta(m) = (r(m), p(m))$, with inverse $(x, y) \mapsto i(x) + s(y)$. Indeed $r$ is linear by the uniqueness of $r(m)$, and $i(r(m)) + s(p(m)) = m$ by construction; conversely, for $m = i(x) + s(y)$ one has $p(m) = ps(y) = y$ and $m - sp(m) = i(x)$, so $r(m) = x$. The two compatibilities hold: $\theta i(x) = (x, 0)$ because $pi = 0$ and $r(i(x)) = x$, and $p\theta^{-1}(x, y) = p(i(x) + s(y)) = y$. (b)⇒(c): put $\theta(m) = (r(m), p(m))$. It is injective: if $p(m) = 0$ then $m = i(x)$ and $r(m) = x$, so $r(m) = 0$ forces $m = 0$. It is surjective: given $(x, y)$, choose $m$ with $p(m) = y$; then $m' = m + i(x - r(m))$ has $\theta(m') = (x, y)$. (c)⇒(a),(b): take $s(y) = \theta^{-1}(0, y)$ and $r$ the first component of $\theta$. □

**Example 3.12.** The sequence $0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}/n \to 0$ does not split for $n \geq 2$, since $\mathbb{Z}$ has no element of order $n$. Likewise $0 \to \mathbb{Z}_\ell \xrightarrow{\ell^k} \mathbb{Z}_\ell \to \mathbb{Z}/\ell^k \to 0$ does not split for $k \geq 1$. If $T \neq 0$ is a free $\mathbb{Z}_\ell$-module of finite rank, $V = T \otimes \mathbb{Q}_\ell$ and $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, the sequence $0 \to T \to V \to W \to 0$ (Proposition 3.78) does not split: a splitting would embed the torsion module $W \neq 0$ into $V$, which has no nonzero torsion.

**Proposition 3.13** (two submodules)**.** *For submodules* $A, B$ *of* $M$ *the sequence*

$$0 \to M/(A \cap B) \xrightarrow{x \mapsto (x,x)} M/A \oplus M/B \xrightarrow{(y,z) \mapsto y - z} M/(A + B) \to 0$$

*is exact. In particular, if* $A \cap B = 0$ *then* $M \hookrightarrow M/A \oplus M/B$ *with cokernel* $M/(A + B)$*. The sum* $A + B$ *is direct (that is,* $A \oplus B \to A + B$*,* $(a, b) \mapsto a + b$*, is an isomorphism) if and only if* $A \cap B = 0$*.*

*Proof.* The first map is well defined and injective: $(x + A, x + B) = 0$ iff $x \in A \cap B$. The second is surjective, and the composite is zero. If $y - z \in A + B$, say $y - z = a + b$, then $x = y - a = z + b$ satisfies $(x + A, x + B) = (y + A, z + B)$, which is exactness in the middle. The last statement: the kernel of $(a, b) \mapsto a + b$ is $\{(c, -c) : c \in A \cap B\}$. This is the module form of the two-ideal sequence of Chapter 2. □

*In the paper* 3.14. In Lemma lem:crit, $I \cap J = 0$ in $\mathbb{T}$, so Proposition 3.13 gives $\mathbb{T} \hookrightarrow \mathbb{T}/I \times \mathbb{T}/J$ with cokernel $\mathbb{T}/(I + J)$; Proposition 3.91(iii) dualises this inclusion. In Lemma lem:ddt417 the directness of $L[\wp] + L[I]$ is the statement $L[\wp] \cap L[I] = 0$, proved in Proposition 3.113(ii).

**Lemma 3.15** (the snake lemma)**.** *Consider a commutative diagram with exact rows*

$$\begin{array}{cccccccc} & M' & \xrightarrow{i} & M & \xrightarrow{p} & M'' & \to 0 \\ & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f''} & \\ 0 \to & N' & \xrightarrow{j} & N & \xrightarrow{q} & N'' & \end{array}$$

*(so* $fi = jf'$ *and* $f''p = qf$*). There is an* $R$*-linear map* $\delta\colon \ker f'' \to \operatorname{coker} f'$ *defined as follows: for* $x'' \in \ker f''$ *choose* $x \in M$ *with* $p(x) = x''$*; then* $q(f(x)) = f''(x'') = 0$*, so* $f(x) = j(y')$ *for a unique* $y' \in N'$*; put* $\delta(x'') = y' + f'(M')$*. It is well defined, and the sequence*

$$\ker f' \xrightarrow{i} \ker f \xrightarrow{p} \ker f'' \xrightarrow{\delta} \operatorname{coker} f' \xrightarrow{\bar{j}} \operatorname{coker} f \xrightarrow{\bar{q}} \operatorname{coker} f''$$

*is exact. If* $i$ *is injective, so is* $\ker f' \to \ker f$*; if* $q$ *is surjective, so is* $\operatorname{coker} f \to \operatorname{coker} f''$*.*

*Proof.* The restrictions of $i$, $p$ map kernels to kernels, and $j$, $q$ induce maps of cokernels, by commutativity. *$\delta$ is well defined.* The element $y'$ is unique because $j$ is injective. If $x_1$ is another lift of $x''$, then $x_1 - x = i(m')$ with $m' \in M'$, and $f(x_1) - f(x) = fi(m') = j(f'(m'))$, so $y'$ changes by $f'(m')$, which does not change its class. Linearity follows from the same uniqueness: if $x$ lifts $x''$ and $\tilde{x}$ lifts $\tilde{x}''$, with $f(x) = j(y')$ and $f(\tilde{x}) = j(\tilde{y}')$, then $x + r\tilde{x}$ lifts $x'' + r\tilde{x}''$ and $f(x + r\tilde{x}) = j(y' + r\tilde{y}')$, so $\delta(x'' + r\tilde{x}'') = \delta(x'') + r\delta(\tilde{x}'')$. *Exactness at* $\ker f$. $pi = 0$. If $x \in \ker f$ and $p(x) = 0$, then $x = i(m')$ and $jf'(m') = f(x) = 0$, so $f'(m') = 0$ as $j$ is injective: $m' \in \ker f'$. *Exactness at* $\ker f''$. If $x'' = p(x)$ with $x \in \ker f$, take this $x$ as the lift: $f(x) = 0$, so $\delta(x'') = 0$. Conversely, if $\delta(x'') = 0$ then $f(x) = j(f'(m')) = f(i(m'))$ for some $m'$, so $x - i(m') \in \ker f$ maps to $x''$. *Exactness at* $\operatorname{coker} f'$. $\bar{j}\delta(x'')$ is the class of $j(y') = f(x)$, which is 0 in $\operatorname{coker} f$. Conversely, if $j(y') = f(x)$ for some $x \in M$, then $x'' = p(x)$ satisfies $f''(x'') = qf(x) = qj(y') = 0$, and $\delta(x'') = [y']$ by construction. *Exactness at* $\operatorname{coker} f$. $qj = 0$. If $q(y) = f''(x'')$, write $x'' = p(x)$; then $q(y - f(x)) = 0$, so $y - f(x) = j(y')$ and $[y] = \bar{j}[y']$. The last two assertions: $\ker f' \to \ker f$ is a restriction of $i$, so it is injective when $i$ is; and if $q$ is surjective, every class $[z] \in \operatorname{coker} f''$ is $[q(y)] = \bar{q}[y]$. □

The way to remember $\delta$ is the path it follows through the diagram: up from $\ker f''$ to $M''$, back along $p$ to $M$, down by $f$ to $N$, and back along $j$ to $N'$. Each backward step is possible only up to an ambiguity (the choice of a lift along $p$) or only because of an exactness property ($f(x)$ lands in $\ker q = \operatorname{im} j$), and the proof checks that the ambiguity dies in $\operatorname{coker} f'$. Every connecting homomorphism of group cohomology in Chapter 16 is built this way.

**Corollary 3.16.** *(a) For a short exact sequence $0 \to A \to B \to C \to 0$ and $n \in R$ there is an exact sequence*

$$0 \to A[n] \to B[n] \to C[n] \to A/nA \to B/nB \to C/nC \to 0.$$

*(b) If $\phi$ is an endomorphism of a finite abelian group $A$, then $\#\ker\phi = \#\operatorname{coker}\phi$.*

*Proof.* (a) Apply Lemma 3.15 to the diagram whose two rows are the given sequence and whose vertical maps are multiplication by $n$. The squares commute because the maps of the sequence are $R$-linear. The kernel of multiplication by $n$ on $A$ is $A[n]$ and its cokernel is $A/nA$, and likewise for $B$ and $C$; both end maps behave as stated because $A \to B$ is injective and $B \to C$ surjective. (b) By Theorem 3.8(i), $A/\ker\phi \simeq \operatorname{im}\phi$, so $\#A = \#\ker\phi \cdot \#\operatorname{im}\phi$; and $\operatorname{coker}\phi = A/\operatorname{im}\phi$, so $\#A = \#\operatorname{im}\phi \cdot \#\operatorname{coker}\phi$. Compare the two. □

For example, multiplication by 2 on $\mathbb{Z}/4$ has kernel $\{0, 2\}$ and cokernel $(\mathbb{Z}/4)/(2\mathbb{Z}/4) \simeq \mathbb{Z}/2$, both of order 2. Part (b) fails for infinite groups: multiplication by 2 on $\mathbb{Z}$ has kernel 0 and cokernel $\mathbb{Z}/2$. Part (a) with $n = \ell^k$ is how the $\ell^k$-torsion of the coefficient module $W$ is related to $T/\ell^k T$ in Proposition 3.78(ii).

**Lemma 3.17** (the five lemma)**.** *Consider a commutative diagram with exact rows $M_1 \to M_2 \to M_3 \to M_4 \to M_5$ and $N_1 \to N_2 \to N_3 \to N_4 \to N_5$ and vertical maps $f_i\colon M_i \to N_i$. If $f_2$, $f_4$ are injective and $f_1$ is surjective, then $f_3$ is injective; if $f_2, f_4$ are surjective and $f_5$ is injective, then $f_3$ is surjective. In particular, if $f_1, f_2, f_4, f_5$ are isomorphisms, so is $f_3$.*

*Proof.* Write $a_i\colon M_i \to M_{i+1}$ and $b_i\colon N_i \to N_{i+1}$. *Injectivity.* Let $f_3(x) = 0$. Then $f_4a_3(x) = b_3f_3(x) = 0$, so $a_3(x) = 0$ and $x = a_2(y)$. Now $b_2f_2(y) = f_3(x) = 0$, so $f_2(y) = b_1(z)$, and $z = f_1(w)$. Then $f_2(y - a_1(w)) = b_1(z) - b_1f_1(w) = 0$, so $y = a_1(w)$ and $x = a_2a_1(w) = 0$. *Surjectivity.* Let $u \in N_3$. Write $b_3(u) = f_4(x)$. Then $f_5a_4(x) = b_4b_3(u) = 0$, so $a_4(x) = 0$ and $x = a_3(y)$. Now $b_3(u - f_3(y)) = 0$, so $u - f_3(y) = b_2(v)$ with $v = f_2(w)$, and $u = f_3(y + a_2(w))$. □

**Definition 3.18** (direct sums and products)**.** For a family $(M_i)_{i\in I}$ the *direct product* $\prod_i M_i$ is the set of all families $(m_i)$ with componentwise operations, and the *direct sum* $\bigoplus_i M_i$ is the submodule of families with finitely many nonzero terms. If the $M_i$ are submodules of $M$, the sum $\sum M_i$ is *direct*, written $\bigoplus M_i$ (internal direct sum), if the map $\bigoplus_i M_i \to M$ is injective.

**Proposition 3.19.** *(i) By restriction to the summands and composition with the projections,*

$$\operatorname{Hom}_R\Big(\bigoplus_i M_i, N\Big) = \prod_i \operatorname{Hom}_R(M_i, N), \qquad \operatorname{Hom}_R\Big(N, \prod_i M_i\Big) = \prod_i \operatorname{Hom}_R(N, M_i).$$

*For finite $I$, $\bigoplus_i M_i = \prod_i M_i$.*

*(ii) For submodules $A, B \subseteq M$ the following are equivalent: $M = A \oplus B$ internally; $A + B = M$ and $A \cap B = 0$; there is an idempotent $e \in \operatorname{End}_R(M)$ with $eM = A$ and $(1-e)M = B$.*

*Proof.* (i) A map out of $\bigoplus M_i$ is determined by its restrictions, which may be arbitrary because each element is a finite sum; a map into $\prod M_i$ is the family of its components. (ii) The first two conditions agree by Proposition 3.13. If $M = A \oplus B$, every $m$ is uniquely $a + b$, so $e(a + b) = a$ is well defined, and it is linear because the decomposition of $rm + m'$ is the corresponding combination of decompositions; then $e^2 = e$, $eM = A$, and $(1 - e)(a + b) = b$ gives $(1 - e)M = B$. Conversely, given $e$, every $m = em + (1 - e)m$, and $x \in A \cap B$ gives $x = ex'$, $x = (1 - e)x''$, so $x = ex = e(1 - e)x'' = 0$. □

**Definition 3.20** (free modules)**.** A family $(e_i)_{i\in I}$ in $M$ is a *basis* if every $m \in M$ is uniquely a finite sum $\sum r_i e_i$. A module with a basis is *free*; $R^{(I)} = \bigoplus_{i\in I} R$ is free with the standard basis, and $R^n = R^{(\{1,\dots,n\})}$. A *presentation* of $M$ is an exact sequence $F_1 \to F_0 \to M \to 0$ with $F_0$, $F_1$ free; $M$ is *finitely presented* if $F_0$ and $F_1$ can be taken of finite rank.

**Proposition 3.21.** *(i) (Universal property.) For any family $(m_i)_{i\in I}$ in $N$ there is a unique homomorphism $R^{(I)} \to N$ with $e_i \mapsto m_i$. Hence every module is a quotient of a free module and has a presentation.*

*(ii) (Rank.) If $R \neq 0$ and $R^m \simeq R^n$, then $m = n$.*

*(iii) A matrix $A \in \mathrm{M}_n(R)$ is invertible if and only if $\det A \in R^\times$; an endomorphism of $R^n$ is an automorphism if and only if the determinant of its matrix is a unit.*

*Proof.* (i) Send $\sum r_i e_i$ to $\sum r_i m_i$. Taking all elements of $M$ as the family gives $R^{(M)} \twoheadrightarrow M$; its kernel is again a quotient of a free module, which gives $F_1$. (ii) Let $\mathfrak{m}$ be a maximal ideal of $R$; it exists by Zorn's lemma (Chapter 2), or by Proposition 3.31(iii) when $R$ is Noetherian. By Theorem 3.26(ii), $R^m \simeq R^n$ gives $(R/\mathfrak{m})^m \simeq (R/\mathfrak{m})^n$ as vector spaces over the field $R/\mathfrak{m}$, so $m = n$ by linear algebra. (We use the tensor product before proving its properties only to keep the statement here; the proof of Theorem 3.26 does not use (ii).) (iii) If $AB = 1$ then $\det A \det B = 1$. Conversely, the adjugate satisfies $A \operatorname{adj}(A) = \operatorname{adj}(A) A = \det(A) \cdot 1$ over any commutative ring (the identity is a polynomial identity in the entries, true over $\mathbb{Z}[x_{ij}]$), so $A^{-1} = \det(A)^{-1} \operatorname{adj}(A)$. □

**Definition 3.22** (projective modules)**.** A module $P$ is *projective* if the equivalent conditions of Proposition 3.23 hold.

**Proposition 3.23.** *For an $R$-module $P$ the following are equivalent: (a) every surjection $M \twoheadrightarrow P$ splits; (b) $P$ is a direct summand of a free module; (c) $\mathrm{Hom}_R(P,-)$ is exact, that is, carries short exact sequences to short exact sequences; (d) for every surjection $g\colon M \twoheadrightarrow N$ and every $h\colon P \to N$ there is $\tilde h\colon P \to M$ with $g\tilde h = h$. Free modules are projective, and so is $eR$ for every idempotent $e \in R$; so each factor $R_i$ of $R = \prod R_i$ is a projective $R$-module. The $\mathbb{Z}/6$-module $\mathbb{Z}/2 \simeq 3\mathbb{Z}/6$ is projective and not free.*

*Proof.* (a)⇒(b): split a surjection $R^{(P)} \twoheadrightarrow P$ and use Lemma 3.11. (b)⇒(d): let $F = P \oplus Q$ be free with basis $(e_i)$; extend $h$ by 0 on $Q$, lift each image $h(e_i)$ to $M$ and use the universal property of $F$; restrict the lift to $P$. (d)⇒(a): lift $\mathrm{id}_P$ along $M \twoheadrightarrow P$. (c)⇔(d): $\mathrm{Hom}(P,-)$ is always left exact (Proposition 3.25(ii)), and (d) is exactly surjectivity of $\mathrm{Hom}(P,M) \to \mathrm{Hom}(P,N)$. For an idempotent $e$, $R = eR \oplus (1-e)R$. Under $\mathbb{Z}/6 \simeq \mathbb{Z}/2 \times \mathbb{Z}/3$ the module $\mathbb{Z}/2$ is the first factor, so it is projective; it is not free because a nonzero free $\mathbb{Z}/6$-module has at least 6 elements. □

*Remark* 3.24. Over a local ring every finitely generated projective module is free (Theorem 3.35(v)), and over a principal ideal domain every submodule of a free module is free (Theorem 3.40). Consequently, if $\mathbb{T}$ is a commutative $\mathbb{Z}_\ell$-algebra that is finite and free as a $\mathbb{Z}_\ell$-module and $e \in \mathbb{T}$ is an idempotent, then $e\mathbb{T}$ is a direct summand of $\mathbb{T}$, hence a finitely generated free $\mathbb{Z}_\ell$-module. This is why the local factors $\mathbb{T}_{\mathfrak{m}}$ in Lemma lem:hecke and Theorem thm:quant are again finite free $\mathbb{Z}_\ell$-algebras, to which Lemma lem:ddt417 applies.

## 3.3 Hom and tensor product; flatness

The paper changes coefficients constantly: $\mathbb{T} \otimes \mathbb{Z}_\ell$ and $\mathbb{T} \otimes \mathbb{Q}$ in Lemmas lem:crit and lem:hecke and in Theorem thm:quant, $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ in Theorem thm:main and Lemma lem:h0, and the sentence "The same statements hold with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$ and $r$ by its $\ell$-part" at the end of Lemma lem:crit. This section proves what makes such changes harmless: Hom is left exact, the tensor product is right exact, and tensoring with a flat module such as $\mathbb{Z}_\ell$ or $\mathbb{Q}$ (over $\mathbb{Z}$) commutes with kernels, annihilators and saturation.

**Proposition 3.25** (Hom)**.** *(i)* $\mathrm{Hom}_R(R,M) \simeq M$ *by* $f \mapsto f(1)$*;* $\mathrm{Hom}_R(R^n,M) \simeq M^n$*; and* $\mathrm{Hom}_R(R/\mathfrak{a},M) \simeq M[\mathfrak{a}]$ *by* $f \mapsto f(1)$*.*

*(ii)* $\mathrm{Hom}_R(N,-)$ *and* $\mathrm{Hom}_R(-,N)$ *are left exact: if* $0 \to M' \xrightarrow{i} M \xrightarrow{p} M''$ *is exact then so is* $0 \to \mathrm{Hom}(N,M') \to \mathrm{Hom}(N,M) \to \mathrm{Hom}(N,M'')$*, and if* $M' \xrightarrow{i} M \xrightarrow{p} M'' \to 0$ *is exact then so is* $0 \to \mathrm{Hom}(M'',N) \to \mathrm{Hom}(M,N) \to \mathrm{Hom}(M',N)$*.*

*(iii) Neither is right exact in general: applying* $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2,-)$ *to* $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$ *gives* $0 \to 0 \to 0 \to \mathbb{Z}/2$*, and applying* $\mathrm{Hom}_{\mathbb{Z}}(-,\mathbb{Z})$ *gives* $0 \to 0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z}$*; in both the last map is not surjective.*

*(iv) For $M$ free with basis $e_1,\dots,e_n$, $M^\vee = \mathrm{Hom}_R(M,R)$ is free with the* dual basis *$e_i^\vee$, $e_i^\vee(e_j) = \delta_{ij}$, and the evaluation map $M \to M^{\vee\vee}$, $m \mapsto (\varphi \mapsto \varphi(m))$, is an isomorphism. Over a domain $M^\vee$ is torsion-free for every $M$.*

*Proof.* (i) A homomorphism out of $R$ is determined by the image of 1, which may be arbitrary; out of $R/\mathfrak{a}$ the image of 1 must be killed by $\mathfrak{a}$, and every such element occurs. (ii) For the first sequence: if $if = 0$ then $f = 0$ since $i$ is injective; if $pf = 0$ then $f(N) \subseteq \ker p = i(M')$ and

$f = ig$ with $g = i^{-1}f$. For the second: if $fp = 0$ then $f = 0$ as $p$ is surjective; if $fi = 0$ then $f$ vanishes on $\ker p$ and factors through $M/\ker p \simeq M''$. (iii) $\mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}) = 0$, and $\mathrm{Hom}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}$ with the induced map multiplication by 2. (iv) $\varphi = \sum \varphi(e_i) e_i^\vee$, and the evaluation map sends $e_i$ to the dual basis of $(e_i^\vee)$. Over a domain, $r\varphi = 0$ with $r \neq 0$ gives $r\varphi(m) = 0$, so $\varphi(m) = 0$ for all $m$. □

**Theorem 3.26** (tensor product)**.** *(i) For $R$-modules $M, N$ there is an $R$-module $M \otimes_R N$ with an $R$-bilinear map $M \times N \to M \otimes_R N$, $(m, n) \mapsto m \otimes n$, such that every bilinear map $b\colon M \times N \to P$ factors uniquely as $b(m, n) = \beta(m \otimes n)$ with $\beta$ linear. It is unique up to unique isomorphism, and generated by the elements $m \otimes n$.*

*(ii) $R \otimes M \simeq M$; $(\bigoplus_i M_i) \otimes N \simeq \bigoplus_i (M_i \otimes N)$; $R^m \otimes R^n \simeq R^{mn}$; $(R/\mathfrak{a}) \otimes M \simeq M/\mathfrak{a}M$; $M \otimes N \simeq N \otimes M$ and $(M \otimes N) \otimes P \simeq M \otimes (N \otimes P)$.*

*(iii) (Adjunction.) $\mathrm{Hom}_R(M \otimes N, P) \simeq \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$, $\beta \mapsto (m \mapsto (n \mapsto \beta(m \otimes n)))$.*

*(iv) (Base change.) For an $R$-algebra $S$, $S \otimes_R M$ is an $S$-module by $s(s' \otimes m) = ss' \otimes m$; $S \otimes_R R^n \simeq S^n$ and $S \otimes_R (R/\mathfrak{a}) \simeq S/\mathfrak{a}S$. If $A$, $B$ are $R$-algebras, $A \otimes_R B$ is an $R$-algebra with $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$.*

*Proof.* (i) Let $F$ be the free module on the set $M \times N$ and $F_0$ its submodule generated by the elements $(m + m', n) - (m, n) - (m', n)$, $(m, n + n') - (m, n) - (m, n')$, $(rm, n) - r(m, n)$ and $(m, rn) - r(m, n)$. Put $M \otimes N = F/F_0$ and $m \otimes n$ the class of $(m, n)$; the map is bilinear by construction. A bilinear $b$ defines a linear map $F \to P$ (Proposition 3.21(i)) that kills $F_0$ because $b$ is bilinear, hence $\beta$; it is unique because the $m \otimes n$ generate. Uniqueness up to unique isomorphism is the usual consequence of a universal property. (ii) Each isomorphism is produced by the universal property from an evident bilinear map, with inverse written down directly: $r \otimes m \mapsto rm$ with inverse $m \mapsto 1 \otimes m$; $(m_i) \otimes n \mapsto (m_i \otimes n)$ with inverse given on each summand; $R^m \otimes R^n$ follows, with basis $e_i \otimes e_j$; for $R/\mathfrak{a}$, the bilinear map $(\bar{r}, m) \mapsto rm + \mathfrak{a}M$ gives $(R/\mathfrak{a}) \otimes M \to M/\mathfrak{a}M$, with inverse $m + \mathfrak{a}M \mapsto \bar{1} \otimes m$, well defined because $\bar{1} \otimes am = \bar{a} \otimes m = 0$ for $a \in \mathfrak{a}$. Commutativity and associativity are proved in the same way. The bilinear map $(m, n) \mapsto n \otimes m$ gives $M \otimes N \to N \otimes M$, and $(n, m) \mapsto m \otimes n$ its inverse. For associativity fix $p \in P$; then $(m, n) \mapsto m \otimes (n \otimes p)$ is bilinear, giving $\gamma_p\colon M \otimes N \to M \otimes (N \otimes P)$, and $(z, p) \mapsto \gamma_p(z)$ is bilinear on $(M \otimes N) \times P$, giving $(M \otimes N) \otimes P \to M \otimes (N \otimes P)$ with $(m \otimes n) \otimes p \mapsto m \otimes (n \otimes p)$; the inverse is built symmetrically, and the two composites are the identity on the generators, hence everywhere. (iii) Bilinear maps $M \times N \to P$ are exactly the linear maps $M \to \mathrm{Hom}(N, P)$. (iv) For fixed $s$, multiplication by $s$ on $S$ is $R$-linear and induces the action on $S \otimes M$; the rest follows from (ii). The product on $A \otimes B$ is induced by the $R$-multilinear map $(a, b, a', b') \mapsto aa' \otimes bb'$. □

**Example 3.27.** Over $\mathbb{Z}$ (all isomorphisms by Theorem 3.26(ii)):

| tensor product | value |
|---|---|
| $\mathbb{Z}/m \otimes \mathbb{Z}/n$ | $\mathbb{Z}/(m\mathbb{Z} + n\mathbb{Z}) = \mathbb{Z}/\gcd(m, n)$ |
| $\mathbb{Q} \otimes \mathbb{Z}/n$ | $\mathbb{Q}/n\mathbb{Q} = 0$ |
| $\mathbb{Z}_\ell \otimes \mathbb{Z}/n$ | $\mathbb{Z}_\ell/n\mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^{v_\ell(n)}$ (Theorem 3.58(iii): $n = \ell^{v_\ell(n)}u$ with $u \in \mathbb{Z}_\ell^\times$) |
| $\mathbb{Q}_\ell/\mathbb{Z}_\ell \otimes A$, $A$ torsion | 0 |
| $\mathbb{Q}_\ell/\mathbb{Z}_\ell \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ | 0 |

The first row is $(R/\mathfrak{a}) \otimes M \simeq M/\mathfrak{a}M$ with $\mathfrak{a} = m\mathbb{Z}$ and $M = \mathbb{Z}/n$: the quotient of $\mathbb{Z}/n$ by $m(\mathbb{Z}/n) = (m\mathbb{Z} + n\mathbb{Z})/n\mathbb{Z}$ is $\mathbb{Z}/(m\mathbb{Z} + n\mathbb{Z})$, and $m\mathbb{Z} + n\mathbb{Z} = \gcd(m, n)\mathbb{Z}$. The second is the same rule with the roles exchanged, $\mathbb{Q} \otimes \mathbb{Z}/n \simeq \mathbb{Q}/n\mathbb{Q}$, and $n\mathbb{Q} = \mathbb{Q}$. For the fourth row, if $na = 0$ write $x = ny$ in the divisible group $\mathbb{Q}_\ell/\mathbb{Z}_\ell$; then $x \otimes a = y \otimes na = 0$. The fifth row is a case of the fourth. The same argument shows $D \otimes A = 0$ whenever $D$ is divisible and $A$ is torsion. These rows also show that a tensor product can be much smaller than either factor, and that elements $m \otimes n$ can vanish without $m$ or $n$ vanishing: $1 \otimes 1 = 0$ in $\mathbb{Z}/2 \otimes \mathbb{Z}/3$.

**Proposition 3.28** (right exactness)**.** *If $M' \xrightarrow{i} M \xrightarrow{p} M'' \to 0$ is exact, so is $M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$. The functor $- \otimes N$ is not left exact: tensoring the injection $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ with $\mathbb{Z}/2$ gives the zero map $\mathbb{Z}/2 \to \mathbb{Z}/2$. A module $N$ is* flat *if $- \otimes_R N$ is exact (carries injections to injections). Free modules, and more generally projective modules, are flat.*

*Proof.* First a criterion: a sequence $A \xrightarrow{a} B \xrightarrow{b} C \to 0$ is exact if $0 \to \mathrm{Hom}(C, P) \to \mathrm{Hom}(B, P) \to \mathrm{Hom}(A, P)$ is exact for every module $P$. Indeed, with $P = \operatorname{coker} b$ the projection $C \to P$ restricts to 0 on $B$, so it is 0, and $b$ is surjective; with $P = C$, the image of $\mathrm{id}_C$ in $\mathrm{Hom}(A, C)$ is $ba$, which is 0 because the composite of the two maps of Hom sequences is 0; with $P = B/a(A)$, the projection $B \to P$ kills $a(A)$, so it factors through $b$, and $\ker b \subseteq a(A)$. By the adjunction (Theorem 3.26(iii)) the sequence $0 \to \mathrm{Hom}(M'' \otimes N, P) \to \mathrm{Hom}(M \otimes N, P) \to \mathrm{Hom}(M' \otimes N, P)$ is the sequence $0 \to \mathrm{Hom}(M'', Q) \to \mathrm{Hom}(M, Q) \to \mathrm{Hom}(M', Q)$ with $Q = \mathrm{Hom}(N, P)$, which is exact by Proposition 3.25(ii). The criterion gives right exactness. For flatness of free modules: $M \otimes R \simeq M$ naturally, so $- \otimes R$ is exact, and for $F = R^{(I)}$ the isomorphism $M \otimes F \simeq \bigoplus_{i \in I} M$ of Theorem 3.26(ii) turns $f \otimes \mathrm{id}_F$ into the map $\bigoplus f$, applied componentwise, which is injective when $f$ is. If $F = P \oplus Q$, then $M \otimes F = (M \otimes P) \oplus (M \otimes Q)$ and $f \otimes \mathrm{id}_F = (f \otimes \mathrm{id}_P) \oplus (f \otimes \mathrm{id}_Q)$; a direct sum of two maps is injective only if each is, so $f \otimes \mathrm{id}_P$ is injective. Hence projective modules, the direct summands of free ones, are flat. □

The failure of left exactness comes from torsion: under $\mathbb{Z} \otimes \mathbb{Z}/2 \simeq \mathbb{Z}/2$ the map $2 \otimes \mathrm{id}$ sends $1 \otimes 1$ to $2 \otimes 1 = 1 \otimes 2 = 0$, so multiplication by 2, injective on $\mathbb{Z}$, becomes zero once the second factor is killed by 2. Over a principal ideal domain, flatness means exactly torsion-freeness (Proposition 3.29(ii)), which is why tensoring with $\mathbb{Z}_\ell$ or $\mathbb{Q}$ never loses information about kernels.

**Proposition 3.29** (flatness)**.** *(i) For a multiplicative set $S \subseteq R$, the ring $S^{-1}R$ is a flat $R$-module.*

*(ii) Over a principal ideal domain a module is flat if and only if it is torsion-free. Hence $\mathbb{Q}$, $\mathbb{Z}_{(\ell)}$, $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell$ are flat over $\mathbb{Z}$, and $\mathbb{Q}_\ell$ is flat over $\mathbb{Z}_\ell$.*

*(iii) Let $R \to S$ be flat. Then $S \otimes_R -$ commutes with kernels, images, cokernels and finite intersections of submodules. If $\mathfrak{a} = (a_1, \dots, a_k)$ is finitely generated, $M[\mathfrak{a}] \otimes_R S = (M \otimes_R S)[\mathfrak{a}S]$ inside $M \otimes_R S$. If $M$ is finitely generated, $\mathrm{Ann}_S(M \otimes_R S) = \mathrm{Ann}_R(M)S$.*

*(iv) If $R \subseteq S$ are principal ideal domains with $S$ flat over $R$, and $L' \subseteq L$ with $L$ a lattice over $R$, then the saturation of $L' \otimes S$ in $L \otimes S$ is $\mathrm{sat}(L') \otimes S$ (Definition 3.48).*

*Proof.* (i) By Theorem 3.38(ii) below, $S^{-1}R \otimes_R M \simeq S^{-1}M$ and $M \mapsto S^{-1}M$ is exact; that proof does not use this proposition. (ii) If $N$ is flat and $r \neq 0$, tensoring the injection $R \xrightarrow{r} R$ with $N$ shows that $r$ is injective on $N$. Conversely let $N$ be torsion-free. If $N$ is finitely generated it is free (Proposition 3.46(ii)), hence flat. In general let $f\colon M' \to M$ be injective and

$z = \sum_i m'_i \otimes n_i \in M' \otimes N$ with $(f \otimes 1)(z) = 0$. In the construction of Theorem 3.26(i), this says that $\sum_i (f(m'_i), n_i)$ is a finite combination of the defining relations of $M \otimes N$; these involve finitely many elements of $N$. Let $N_0 \subseteq N$ be the submodule generated by them and the $n_i$. The same relations show that $z_0 = \sum m'_i \otimes n_i \in M' \otimes N_0$ maps to 0 in $M \otimes N_0$. As $N_0$ is finitely generated and torsion-free, it is flat, so $z_0 = 0$, and $z$, the image of $z_0$, is 0. (iii) Kernels, images and cokernels: tensor the exact sequences $0 \to \ker g \to M \to M''$ and $M \to M'' \to \operatorname{coker} g \to 0$ with $S$. Intersections: $A \cap B = \ker(M \to M/A \oplus M/B)$; after tensoring, flatness identifies $A \otimes S$ with a submodule of $M \otimes S$, right exactness identifies $(M/A) \otimes S$ with $(M \otimes S)/(A \otimes S)$, and the kernel of $M \otimes S \to (M \otimes S)/(A \otimes S) \oplus (M \otimes S)/(B \otimes S)$ is $(A \otimes S) \cap (B \otimes S)$, while by exactness it is also $(A \cap B) \otimes S$. Torsion: $M[\mathfrak{a}]$ is the kernel of $M \to M^k$, $m \mapsto (a_i m)$, and after tensoring the kernel of the same map is $(M \otimes S)[\mathfrak{a}S]$. Annihilators: for generators $m_1, \dots, m_n$, $\operatorname{Ann}_R(M)$ is the kernel of $R \to M^n$, $r \mapsto (rm_i)$, and after tensoring the kernel of $S \to (M \otimes S)^n$ is $\operatorname{Ann}_S(M \otimes S)$, because the $m_i \otimes 1$ generate $M \otimes S$ over $S$. (iv) Write $L/L' \simeq F \oplus C$ with $F$ free and $C$ torsion (Theorem 3.43); then $\operatorname{sat}(L')$ is the kernel of $L \to L/L' \to F$. After tensoring, $\operatorname{sat}(L') \otimes S$ is the kernel of $L \otimes S \to F \otimes S$, a free target, so it is saturated in $L \otimes S$; it contains $L' \otimes S$ with quotient $C \otimes S$, which is killed by the nonzero element of $R$ killing $C$, hence torsion. A saturated submodule containing $L' \otimes S$ with torsion quotient is its saturation (Proposition 3.49). □

*In the paper* 3.30. "The same statements hold with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$ and $r$ by its $\ell$-part" (Lemma lem:crit) and "Every step holds verbatim over $\mathbb{Z}_\ell$" (the end of its proof) are Proposition 3.29: $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$, so $\ker \phi_f \otimes \mathbb{Z}_\ell$ is the kernel of $\phi_f \otimes \mathbb{Z}_\ell$, the annihilator $\operatorname{Ann}_{\mathbb{T}}(\ker \phi_f) \otimes \mathbb{Z}_\ell$ is the annihilator in $\mathbb{T} \otimes \mathbb{Z}_\ell$, and the congruence ideal $\phi_f(\operatorname{Ann}_{\mathbb{T}} \ker \phi_f)$ becomes $\phi_f(\operatorname{Ann}_{\mathbb{T}} \ker \phi_f)\mathbb{Z}_\ell$, whose generator is the $\ell$-part of $r$. The same proposition is used in Lemma lem:step1 and in the proof of Theorem thm:quant, which work over $\mathbb{Z}_\ell$ from the start.

## 3.4 Noetherian rings, Nakayama's lemma, local rings and localisation

Every ring in the paper is Noetherian: $\mathbb{Z}$, $\mathbb{Z}_\ell$, the Hecke algebra $\mathbb{T}$ (a finitely generated $\mathbb{Z}$-module) and its completions $\mathbb{T}_{\mathfrak{m}}$. Nakayama's lemma enters the paper twice: in Remark rem:free, where the two-dimensionality of $J_0(N)[\mathfrak{m}]$ gives two generators of the Tate module over $\mathbb{T}_{\mathfrak{m}}$ (Theorem 3.116), and in Lemma lem:h0, by name, where a homomorphism $T/\ell^k \to (T/\ell^k)(1)$ that is nonzero modulo $\ell$ is an isomorphism. Local rings and localisation give the meaning of $\mathbb{T}_{\mathfrak{m}}$.

**Proposition 3.31** (Noetherian modules)**.** *(i) For an $R$-module $M$ the following are equivalent: every ascending chain of submodules is eventually constant; every nonempty set of submodules has a maximal element; every submodule is finitely generated. Such $M$ is* Noetherian*; $R$ is a Noetherian ring if it is Noetherian as an $R$-module, that is, if every ideal is finitely generated.*

*(ii) Submodules and quotients of Noetherian modules are Noetherian, and if $0 \to M' \to M \to M'' \to 0$ is exact with $M'$, $M''$ Noetherian then $M$ is Noetherian. Over a Noetherian ring every finitely generated module is Noetherian; hence submodules of finitely generated modules are finitely generated, and finitely generated modules are finitely presented.*

*(iii) In a nonzero Noetherian ring every proper ideal is contained in a maximal ideal.*

*(iv) (Hilbert's basis theorem.) If $R$ is Noetherian, so is $R[x]$.*

(v) *Fields, principal ideal domains, $\mathbb{Z}_\ell$ (Theorem 3.58), and every ring that is finitely generated as a module over a Noetherian ring (such as the Hecke algebra $\mathbb{T}$ over $\mathbb{Z}$, or $\mathbb{T}\otimes\mathbb{Z}_\ell$ over $\mathbb{Z}_\ell$) are Noetherian.*

*Proof.* (i) If chains stabilise and a nonempty set $\Sigma$ of submodules had no maximal element, we could choose a strictly increasing infinite chain in $\Sigma$. If maximal elements exist and $N \subseteq M$, a maximal element of the set of finitely generated submodules of $N$ is $N$ itself (otherwise add an element of $N$ outside it). If every submodule is finitely generated and $N_1 \subseteq N_2 \subseteq \cdots$, then $\bigcup N_i$ is generated by finitely many elements, all in some $N_k$, so the chain is constant from $k$ on. (ii) Submodules: by the third condition. Quotients: by the first, via Theorem 3.8(iii). Extensions: if $N_1 \subseteq N_2 \subseteq \cdots$ in $M$, both $N_i \cap M'$ and the images of $N_i$ in $M''$ stabilise, and then so does $N_i$ (if $N \subseteq N'$ have the same intersection with $M'$ and the same image in $M''$, then $N = N'$: for $x \in N'$ choose $y \in N$ with the same image in $M''$; then $x - y \in N' \cap M' = N \cap M'$, so $x \in N$). By induction $R^n$ is Noetherian, and a finitely generated module is a quotient of some $R^n$. The kernel of $R^n \twoheadrightarrow M$ is then finitely generated. (iii) The set of proper ideals containing a given proper ideal is nonempty and has a maximal element by (i), which is a maximal ideal. (iv) We do not prove this; see [7, Theorem 7.5] (number not verified against the printed edition). (v) Fields and principal ideal domains: every ideal is generated by one element. A ring $A$ finite over a Noetherian $R$ is a Noetherian $R$-module by (ii), and its ideals are $R$-submodules. □

**Lemma 3.32** (the determinant trick)**.** *Let $M$ be generated by $n$ elements, $\mathfrak{a} \subseteq R$ an ideal and $\phi \in \mathrm{End}_R(M)$ with $\phi(M) \subseteq \mathfrak{a}M$. Then $\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0$ in $\mathrm{End}_R(M)$ for some $a_i \in \mathfrak{a}^i$. Consequently:*

*(a) if $M$ is finitely generated and $\mathfrak{a}M = M$, there is $x \in R$ with $x \equiv 1 \pmod{\mathfrak{a}}$ and $xM = 0$;*

*(b) a surjective endomorphism of a finitely generated module is an isomorphism.*

*Proof.* Let $m_1, \ldots, m_n$ generate $M$ and write $\phi(m_i) = \sum_j a_{ij} m_j$ with $a_{ij} \in \mathfrak{a}$. Work in the commutative ring $R[\phi] \subseteq \mathrm{End}_R(M)$ and let $B = (\delta_{ij}\phi - a_{ij})$, a matrix over it. Then $B$ applied to the column $(m_j)$ is 0. Multiplying by the adjugate of $B$ gives $\det(B)m_j = 0$ for all $j$, so $\det B = 0$ in $\mathrm{End}_R(M)$. Expanding $\det(\delta_{ij}\phi - a_{ij})$ as a sum over permutations, each term is a product of $n$ entries; multiplying out the diagonal entries $\phi - a_{jj}$, every resulting monomial is $\pm\phi^{n-i}$ times a product of $i$ entries $a_{jk}$; so the coefficient $a_i$ of $\phi^{n-i}$ is a sum of products of $i$ elements of $\mathfrak{a}$, which lies in $\mathfrak{a}^i$, and the coefficient of $\phi^n$ is 1 (from the identity permutation only). This is the stated polynomial; it is the characteristic polynomial of the matrix $(a_{ij})$, evaluated at $\phi$, and for $M = R^n$ free it is Cayley-Hamilton. (a) Take $\phi = \mathrm{id}$, which maps $M$ into $\mathfrak{a}M = M$: then $x = 1 + a_1 + \cdots + a_n$ kills $M$, and $x \equiv 1 \pmod{\mathfrak{a}}$. (b) Let $\phi$ be surjective and regard $M$ as an $R[X]$-module, $X$ acting as $\phi$; it is finitely generated over $R[X]$ (by the same generators), and $(X)M = \phi(M) = M$. By (a) over the ring $R[X]$ with the ideal $(X)$ there is $x \equiv 1 \pmod{X}$, that is, $x = 1 - Xg(X)$ with $g \in R[X]$, with $xM = 0$; that is, $\phi \circ g(\phi) = \mathrm{id}$. If $\phi(m) = 0$ then $m = g(\phi)\phi(m) = 0$ (the two commute); so $\phi$ is injective. □

**Definition 3.33** (local rings)**.** A ring is *local* if it has exactly one maximal ideal $\mathfrak{m}$; we write $(R, \mathfrak{m}, k)$ with $k = R/\mathfrak{m}$ the *residue field*. The *Jacobson radical* $\mathfrak{J}(R)$ is the intersection of all maximal ideals.

**Proposition 3.34.** *(i) $R$ is local if and only if its non-units form an ideal; then this ideal is $\mathfrak{m}$, and $R^\times = R \setminus \mathfrak{m}$.*

*(ii)* *If $x \in \mathfrak{J}(R)$ then $1+x$ is a unit.*

*(iii)* *Examples of local rings: fields; $\mathbb{Z}_{(p)}$ (Chapter 2); discrete valuation rings (Theorem 3.55); $\mathbb{Z}/p^n$; $k[x]/(x^2)$; $\mathbb{Z}_\ell[x]/(x^2)$; each factor $\mathbb{T}_\mathfrak{m}$ of a finite free $\mathbb{Z}_\ell$-algebra (Theorem 3.95). Non-examples: $\mathbb{Z}$; $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$ (maximal ideals $\ell\mathbb{Z}_\ell \times \mathbb{Z}_\ell$ and $\mathbb{Z}_\ell \times \ell\mathbb{Z}_\ell$); $\mathbb{Z}_{(11)}[x]/(x^2+x-1)$, which has two maximal ideals (Theorem 3.95(v)).*

*Proof.* (i) A non-unit $x$ generates a proper ideal, which lies in a maximal ideal (Chapter 2, by Zorn's lemma; or Proposition 3.31(iii)). If $R$ is local this ideal is $\mathfrak{m}$, so the non-units are exactly $\mathfrak{m}$. Conversely if the non-units form an ideal $\mathfrak{n}$, every proper ideal consists of non-units and lies in $\mathfrak{n}$, which is therefore the unique maximal ideal. (ii) If $1+x$ were not a unit it would lie in a maximal ideal $\mathfrak{m}'$; so would $x$, and then $1 \in \mathfrak{m}'$. (iii) In $\mathbb{Z}/p^n$ an integer prime to $p$ is invertible modulo $p^n$, and a multiple of $p$ is nilpotent, so the non-units are the multiples of $p$. In $k[x]/(x^2)$ and $\mathbb{Z}_\ell[x]/(x^2)$ write an element as $a+bx$; then $(a+bx)(a-bx) = a^2$, so $a+bx$ is a unit as soon as $a$ is a unit of $k$, respectively of $\mathbb{Z}_\ell$, while if $a = 0$ in $k$ (respectively $a \in \ell\mathbb{Z}_\ell$) the element lies in the proper ideal $(x)$ (respectively $(\ell, x)$, with quotient $\mathbb{F}_\ell$). So the non-units are the multiples of $x$, respectively the ideal $(\ell, x)$; these form ideals. The others are proved where cited. □

The elements of the Jacobson radical are the "small" elements of $R$: in a local ring $\mathfrak{J}(R) = \mathfrak{m}$, and (ii) is the algebraic shadow of the geometric series $1/(1+x) = 1 - x + x^2 - \cdots$. In $\mathbb{Z}_\ell$, for example, $1+\ell y$ is a unit for every $y \in \mathbb{Z}_\ell$. Nakayama's lemma, next, exploits exactly this.

**Theorem 3.35** (Nakayama's lemma)**.** *Let $(R, \mathfrak{m}, k)$ be a local ring and $M$ a finitely generated $R$-module.*

*(i)* *If $\mathfrak{m}M = M$ then $M = 0$. More generally, for any ring $R$ and any ideal $\mathfrak{a} \subseteq \mathfrak{J}(R)$, $\mathfrak{a}M = M$ implies $M = 0$.*

*(ii)* *If $N \subseteq M$ is a submodule with $N + \mathfrak{m}M = M$, then $N = M$.*

*(iii)* *Elements $x_1, \dots, x_r$ generate $M$ if and only if their images span the $k$-vector space $M/\mathfrak{m}M$. The minimal number of generators of $M$ is $\dim_k M/\mathfrak{m}M$.*

*(iv)* *A homomorphism $f\colon N' \to M$ is surjective if and only if $f \otimes k\colon N'/\mathfrak{m}N' \to M/\mathfrak{m}M$ is. A matrix in $\mathrm{M}_n(R)$ is invertible if and only if its reduction modulo $\mathfrak{m}$ is; an endomorphism of $R^n$ is an automorphism if and only if it is so modulo $\mathfrak{m}$.*

*(v)* *Every finitely generated projective $R$-module is free.*

*Proof.* (i) By Lemma 3.32(a) some $x = 1+a$ with $a \in \mathfrak{a}$ kills $M$; $x$ is a unit by Proposition 3.34(ii), so $M = 0$. (In a local ring $\mathfrak{J}(R) = \mathfrak{m}$, so the two forms of (i) agree.) (ii) The quotient $M/N$ is finitely generated, and $\mathfrak{m}(M/N) = (\mathfrak{m}M + N)/N = M/N$; by (i), $M/N = 0$. (iii) Apply (ii) to $N = \sum Rx_i$; conversely generators of $M$ span $M/\mathfrak{m}M$. A minimal generating set therefore has $\dim_k M/\mathfrak{m}M$ elements. (iv) The first statement is (ii) for $N = f(N')$. A matrix is invertible iff its determinant is a unit (Proposition 3.21(iii)), iff the determinant is not in $\mathfrak{m}$, iff the reduced determinant is nonzero. (v) Let $P$ be finitely generated projective and $n = \dim_k P/\mathfrak{m}P$. By (iii) there is a surjection $R^n \twoheadrightarrow P$; it splits, so $R^n \simeq P \oplus Q$ with $Q$ finitely generated (a quotient of $R^n$). Reducing modulo $\mathfrak{m}$ gives $k^n \simeq P/\mathfrak{m}P \oplus Q/\mathfrak{m}Q$, so $Q/\mathfrak{m}Q = 0$ and $Q = 0$ by (i). □

**Example 3.36.** A finitely generated $\mathbb{Z}_\ell$-module $M$ with $M/\ell M \simeq \mathbb{F}_\ell^2$ is generated by two elements. An endomorphism of $(\mathbb{Z}/\ell^k)^n$ whose reduction modulo $\ell$ is invertible is invertible (apply (iv) over the local ring $\mathbb{Z}/\ell^k$).

The hypothesis "finitely generated" cannot be dropped: $\mathbb{Q}$ is a module over the local ring $\mathbb{Z}_{(p)}$ with $p\mathbb{Q}=\mathbb{Q}$, and $\mathbb{Q}\neq 0$. In (iii) the point is that one reduction suffices: over $\mathbb{Z}_\ell$, to know that $M$ is cyclic it is enough to know that $M/\ell M$ is a line, while over $\mathbb{Z}$ one would have to look at $M/pM$ for every prime $p$ (and over a general ring even that is not enough). This is how the two-dimensionality of $J_0(N)[\mathfrak{m}]$ in Remark rem:free yields two generators (Paper link 3.37).

*In the paper* 3.37. In the proof of Lemma lem:h0 the paper writes: "An element of exact order $\ell^k$ is a map $\phi\colon T/\ell^k \to (T/\ell^k)(1)$ whose reduction modulo $\ell$ is nonzero, hence an isomorphism, so $\phi$ is an isomorphism by Nakayama's lemma". The underlying module map is an endomorphism of $(\mathbb{Z}/\ell^k)^2$ (the twist does not change the module), invertible modulo $\ell$ because a nonzero $G_{\mathbb{Q}}$-map between irreducible representations is an isomorphism, its kernel and image being subrepresentations (Chapter 4); it is therefore invertible by Example 3.36. In Remark rem:free, the two-dimensionality of $J_0(N)[\mathfrak{m}]$ is turned into two generators by Theorem 3.35(iii); see Theorem 3.116.

**Theorem 3.38** (localisation)**.** *Let $S\subseteq R$ be multiplicative ($1\in S$, $SS\subseteq S$). Let $S^{-1}M$ be the set of fractions $m/s$ ($m\in M$, $s\in S$), where $m/s=m'/s'$ if and only if $u(s'm-sm')=0$ for some $u\in S$; with the evident operations $S^{-1}R$ is a ring and $S^{-1}M$ an $S^{-1}R$-module.*

*(i) The map $R\to S^{-1}R$, $r\mapsto r/1$, sends $S$ to units, and every ring homomorphism $R\to A$ sending $S$ to units factors uniquely through it.*

*(ii) $S^{-1}R\otimes_R M\simeq S^{-1}M$, $(a/s)\otimes m\mapsto am/s$, and $M\mapsto S^{-1}M$ is exact.*

*(iii) $\mathfrak{b}\mapsto \mathfrak{b}\cap R$ (preimage) and $\mathfrak{p}\mapsto S^{-1}\mathfrak{p}$ are inverse bijections between the prime ideals of $S^{-1}R$ and the prime ideals of $R$ not meeting $S$.*

*(iv) For a prime $\mathfrak{p}$, $R_{\mathfrak{p}}=(R\setminus\mathfrak{p})^{-1}R$ is local with maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$; we write $M_{\mathfrak{p}}$ for the localised module.*

*(v) $M=0$ if and only if $M_{\mathfrak{m}}=0$ for every maximal ideal $\mathfrak{m}$; a homomorphism is injective (surjective) if and only if it is so after localising at every maximal ideal.*

*(vi) Localisation commutes with finite direct sums, quotients, and annihilators of finitely generated modules: $S^{-1}\operatorname{Ann}_R(M)=\operatorname{Ann}_{S^{-1}R}(S^{-1}M)$. It commutes with* Hom *out of finitely presented modules (stated; see [7, Chapter 3]).*

*(vii) Examples: $\mathbb{Z}_{(p)}$; $\mathbb{Q}=\mathbb{Z}_{(0)}$; $\mathbb{Z}[1/\ell]$; $(\mathbb{Z}/n)_{(p)}\simeq\mathbb{Z}/p^{v_p(n)}$; for $R=R_1\times R_2$ and the prime $\mathfrak{p}=\mathfrak{p}_1\times R_2$, $R_{\mathfrak{p}}\simeq(R_1)_{\mathfrak{p}_1}$.*

*Proof.* The relation is an equivalence relation. Reflexivity and symmetry are clear. For transitivity, let $u(s'm-sm')=0$ and $u'(s''m'-s'm'')=0$. Multiply the first by $u's''$ and the second by $us$ and add: the terms in $m'$ cancel and $uu's'(s''m-sm'')=0$, with $uu's'\in S$. (The factor $u$ cannot be omitted: without it the relation need not be transitive when $M$ has elements killed by elements of $S$.) The operations $m/s+m'/s'=(s'm+sm')/ss'$ and $(r/t)(m/s)=rm/ts$ are well defined by computations of the same kind. (i) $s/1$ has inverse $1/s$. If $\phi\colon R\to A$ inverts $S$, put $\psi(r/s)=\phi(r)\phi(s)^{-1}$; if $u(s'r-sr')=0$ then $\phi(u)$ is a unit and $\phi(r)\phi(s)^{-1}=\phi(r')\phi(s')^{-1}$. Uniqueness is forced by $\psi(r/s)\psi(s/1)=\psi(r/1)$. (ii) The bilinear map $(a/s,m)\mapsto am/s$ gives $\alpha\colon S^{-1}R\otimes M\to S^{-1}M$. Every element of $S^{-1}R\otimes M$ has the form $(1/s)\otimes m$ (bring the finitely many fractions to a common denominator). Define $\beta(m/s)=(1/s)\otimes m$; it is well defined: if $u(s'm-sm')=0$ then $(1/s)\otimes m=(us'/uss')\otimes m=(1/uss')\otimes us'm=$

$(1/uss')\otimes usm' = (1/s')\otimes m'$. The two composites are identities: $\alpha\beta(m/s) = \alpha((1/s)\otimes m) = m/s$, and $\beta\alpha((a/s)\otimes m) = \beta(am/s) = (1/s)\otimes am = (a/s)\otimes m$. Exactness: let $M' \xrightarrow{f} M \xrightarrow{g} M''$ be exact at $M$ and $g(m)/s = 0$. Then $ug(m) = 0$ for some $u \in S$, so $um = f(m')$ and $m/s = f(m')/us$. (iii) If $\mathfrak{b}$ is prime in $S^{-1}R$, its preimage is prime and misses $S$ (it contains no unit), and $\mathfrak{b} = S^{-1}(\mathfrak{b}\cap R)$ because $r/s \in \mathfrak{b}$ iff $r/1 \in \mathfrak{b}$. If $\mathfrak{p}$ is prime with $\mathfrak{p}\cap S = \emptyset$, then $S^{-1}\mathfrak{p}$ is a proper ideal, it is prime (if $(a/s)(b/t) \in S^{-1}\mathfrak{p}$ then $uab \in \mathfrak{p}$ for some $u \in S$, and $u \notin \mathfrak{p}$), and its preimage is $\mathfrak{p}$ (if $r/1 = p/s$ then $u(sr-p) = 0$, so $usr \in \mathfrak{p}$ and $r \in \mathfrak{p}$). (iv) By (iii) the primes of $R_{\mathfrak{p}}$ correspond to the primes contained in $\mathfrak{p}$, so $\mathfrak{p}R_{\mathfrak{p}}$ is the unique maximal ideal. (v) If $0 \neq m \in M$, then $\operatorname{Ann}(m)$ is proper and lies in a maximal ideal $\mathfrak{m}$; then $m/1 \neq 0$ in $M_{\mathfrak{m}}$, since $um = 0$ would put $u \in \operatorname{Ann}(m) \subseteq \mathfrak{m}$. Apply this to kernels and cokernels, which commute with localisation by (ii). (vi) Direct sums and quotients: by (ii) and Theorem 3.26(ii). Annihilators: $S^{-1}R$ is flat by (ii), so Proposition 3.29(iii) applies. (vii) The first three are the definitions (with $S = \mathbb{Z}\setminus p\mathbb{Z}$, $\mathbb{Z}\setminus\{0\}$, $\{\ell^k\}$). For $\mathbb{Z}/n = \mathbb{Z}/p^a \times \mathbb{Z}/m$ with $p \nmid m$: $m$ is a unit of $\mathbb{Z}_{(p)}$ and kills $\mathbb{Z}/m$, so $(\mathbb{Z}/m)_{(p)} = 0$, while every element of $\mathbb{Z}\setminus p\mathbb{Z}$ acts invertibly on $\mathbb{Z}/p^a$. For the product, $(1,0) \notin \mathfrak{p}$ becomes a unit in $R_{\mathfrak{p}}$ and $(1,0)(0,1) = 0$, so $(0,1) = 0$ in $R_{\mathfrak{p}}$, and $R_{\mathfrak{p}} = (R_1 \times 0)_{\mathfrak{p}} \simeq (R_1)_{\mathfrak{p}_1}$. □

*Remark* 3.39. The last example of (vii) is the reason why, for a finite algebra over a *complete* discrete valuation ring, the localisation at a maximal ideal is a direct factor (Theorem 3.95): completeness produces the decomposition into a product, and localisation then picks out one factor. Without completeness there is in general no such decomposition (Theorem 3.95(v)), which is why the paper always works with $\mathbb{T}\otimes\mathbb{Z}_\ell$ (Convention 3.2).

With the general machinery of exactness, tensor products, Nakayama and localisation in place, the next section specialises to principal ideal domains, where finitely generated modules can be classified completely. That classification is what turns statements about lattices of modular forms into statements about integers: indices, invariant factors and congruence moduli.

## 3.5 Finitely generated modules over a principal ideal domain; lattices and saturation

All the lattices of the paper live over the principal ideal domains $\mathbb{Z}$ and $\mathbb{Z}_\ell$: $S_2(\Gamma_0(N);\mathbb{Z})$ and its sublattices $\mathbb{Z}f$ and $L = f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z})$ in Lemma lem:crit; $T_3E$ in Lemma lem:IV; the lattice $L$ and its saturated sublattices $L[\wp]$, $L[I]$ in Lemma lem:ddt417; the module $H^1(I,M) \simeq (\mathbb{Z}_\ell/n')^2 \oplus \mathbb{Z}_\ell$ in Lemma lem:tamfactor. This section proves the structure theorem through the Smith normal form, introduces saturation, and ends with the two results that carry the second paragraph of the proof of Lemma lem:crit: the dual of an inclusion of lattices (Proposition 3.51) and the congruence module of a decomposition (Proposition 3.53). Throughout, $R$ is a principal ideal domain with fraction field $K$.

**Theorem 3.40** (submodules of free modules)**.** *A submodule of a free $R$-module of rank $n$ is free of rank at most $n$. Consequently finitely generated $R$-modules are finitely presented, and finitely generated projective $R$-modules are free.*

*Proof.* Induction on $n$, the case $n = 0$ being trivial. Let $N \subseteq R^n$ and let $\pi_n\colon R^n \to R$ be the last coordinate. Then $\pi_n(N)$ is an ideal, $aR$ say. If $a = 0$, $N \subseteq R^{n-1}$ and we are done. Otherwise choose $x \in N$ with $\pi_n(x) = a$. Then $N = (N \cap R^{n-1}) \oplus Rx$: every $y \in N$ has $\pi_n(y) = ra$, and $y - rx \in N \cap R^{n-1}$; the sum is direct because $\pi_n(rx) = ra = 0$ forces $r = 0$ in the domain $R$. By induction $N \cap R^{n-1}$ is free of rank at most $n-1$, and $Rx \simeq R$. The consequences: the kernel of

$R^n \twoheadrightarrow M$ is free of finite rank, and a projective module is a direct summand, hence a submodule, of a free module. □

**Theorem 3.41** (Smith normal form)**.** *For $A \in \mathrm{M}_{m\times n}(R)$ there are $U \in \mathrm{GL}_m(R)$ and $V \in \mathrm{GL}_n(R)$ such that $UAV$ is the matrix with diagonal entries $d_1, \dots, d_s, 0, \dots, 0$ and zeros elsewhere, where $d_i \neq 0$ and $d_1 \mid d_2 \mid \cdots \mid d_s$. For each $k$ the ideal $D_k(A)$ generated by the $k \times k$ minors of $A$ (the $k$-th* determinantal ideal*) satisfies $D_k(UAV) = D_k(A)$; hence $D_k(A) = (d_1 \cdots d_k)$ for $k \leq s$ and $0$ for $k > s$, and the ideals $(d_i)$, the* invariant factors *of $A$, are uniquely determined.*

*Proof.* For $a \neq 0$ let $\lambda(a)$ be the number of prime factors of $a$ counted with multiplicity ($R$ is a unique factorisation domain, Chapter 2); units have $\lambda = 0$. We use row and column operations by invertible matrices: permutations, adding a multiple of one row (column) to another, and the following *Bezout operation*: if $g = \gcd(a, b) = xa + yb$ with $a, b$ not both 0, then $\begin{pmatrix} x & y \\ -b/g & a/g \end{pmatrix}$ has determinant 1 and sends the column $(a, b)^{\mathsf{T}}$ to $(g, 0)^{\mathsf{T}}$; acting on two rows (or, transposed, on two columns) it replaces $a$ by $g$. If $A = 0$ there is nothing to do. Otherwise move a nonzero entry with $\lambda$ minimal to position $(1, 1)$; call it $a$. If $a$ does not divide some entry $b$ of the first column, the Bezout operation on the two rows replaces $a$ by $g$ with $\lambda(g) < \lambda(a)$; similarly for the first row. Since $\lambda$ cannot decrease forever, after finitely many steps $a$ divides every entry of the first row and column, which we clear by elementary operations. Now $A = \mathrm{diag}(a, A')$. If $a$ does not divide an entry $b$ of $A'$, add the row of $b$ to the first row: the first row now contains $b$, and the previous step lowers $\lambda(a)$ again. Eventually $a$ divides every entry of $A'$, and this persists under operations on $A'$, whose entries are combinations of the old ones. Induction on the size finishes the existence, with $d_1 = a$. For invariance: each $k \times k$ minor of $UA$ is an $R$-combination of $k \times k$ minors of $A$ (Cauchy-Binet, or expand the rows of $UA$ as combinations of rows of $A$ by multilinearity), so $D_k(UA) \subseteq D_k(A)$, and equality follows by applying the same to $U^{-1}$; columns likewise. For a diagonal matrix with $d_1 \mid \cdots \mid d_s$ the nonzero $k \times k$ minors are products of $k$ of the $d_i$, all divisible by $d_1 \cdots d_k$. Hence $(d_1 \cdots d_k)$ is determined by $A$, and so is $(d_k) = (d_1 \cdots d_k)/(d_1 \cdots d_{k-1})$. □

**Example 3.42.** Computed by hand and checked with PARI's `matsnf`, over $\mathbb{Z}$: $\begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix} \sim \mathrm{diag}(2, 4)$; the relation matrix with rows $(2, 4, 4)$, $(-6, 6, 12)$, $(10, -4, -16)$ (from [58, Section 12.1]) $\sim \mathrm{diag}(2, 6, 12)$; $\begin{pmatrix} 3 & 6 \\ 9 & 3 \end{pmatrix} \sim \mathrm{diag}(3, 15)$ (its determinant is $-45$); the $3 \times 2$ matrix with rows $(6, 0)$, $(0, 4)$, $(2, 2) \sim \mathrm{diag}(2, 2)$ (the gcd of its entries is 2 and its $2 \times 2$ minors are 24, 12, $-8$, with gcd 4). Over $\mathbb{Z}_3$ the matrix $\begin{pmatrix} 3 & 6 \\ 9 & 3 \end{pmatrix}$ is equivalent to $\mathrm{diag}(3, 3)$, since $15 = 3 \cdot 5$ and $5 \in \mathbb{Z}_3^{\times}$.

```
matsnf([2,4;6,8])                    \\ [4, 2]
matsnf([2,4,4;-6,6,12;10,-4,-16])    \\ [12, 6, 2]
matsnf([3,6;9,3])                    \\ [15, 3]
matsnf([6,0,2;0,4,2])                \\ [2, 2] (transpose of the 3x2 matrix)
```

PARI lists the invariant factors in decreasing order. Transposing a matrix does not change its invariant factors, since $A$ and $A^{\mathsf{T}}$ have the same $k \times k$ minors; so it does not matter that the book writes relations as rows (Convention 3.1(g)). By hand, for the first matrix: subtracting 3 times the first row from the second gives rows $(2, 4)$, $(0, -4)$; subtracting 2 times the first column from the second gives $\mathrm{diag}(2, -4)$; changing the sign of the second row gives $\mathrm{diag}(2, 4)$. The

invariance statement checks this: the gcd of the entries $2,4,6,8$ is $2=d_1$, and the determinant $-8$ gives $d_1d_2=8$ up to sign, so $d_2=4$.

**Theorem 3.43** (structure of finitely generated modules over a principal ideal domain)**.** *Let $M$ be a finitely generated $R$-module.*

*(i) (Invariant factors.) $M\simeq R^r\oplus R/(d_1)\oplus\cdots\oplus R/(d_s)$ with $d_i$ nonzero non-units and $d_1\mid\cdots\mid d_s$; the integer $r$ and the ideals $(d_i)$ are uniquely determined.*

*(ii) (Elementary divisors.) $M\simeq R^r\oplus\bigoplus_j R/(p_j^{a_j})$ with $p_j$ primes and $a_j\geq 1$; the multiset of the ideals $(p_j^{a_j})$ is uniquely determined.*

*(iii) Special cases: for $R=\mathbb{Z}$ this is the structure of finitely generated abelian groups (for finite groups, Chapter 1); for a discrete valuation ring, $M\simeq\mathcal{O}^r\oplus\bigoplus_i\mathcal{O}/\varpi^{a_i}$ (Proposition 3.57); for $R=F[x]$ and $M$ a finite-dimensional vector space with an endomorphism, the rational canonical form, and over an algebraically closed $F$ the Jordan form.*

*Proof. Existence.* By Theorem 3.40 there is a presentation $R^m\xrightarrow{A}R^n\to M\to 0$. Replacing $A$ by $UAV$ changes the basis of the relations by $U$ and the basis of $R^n$ by $V^{-1}$, so $M\simeq R^n/(\text{row space of }UAV)$. For $UAV=\operatorname{diag}(d_1,\dots,d_s,0,\dots)$ this is $\bigoplus_{i\leq s}R/(d_i)\oplus R^{n-s}$; drop the summands with $d_i$ a unit. (ii) follows from (i) by the Chinese remainder theorem $R/(d)\simeq\bigoplus_p R/(p^{v_p(d)})$ (Chapter 2). *Uniqueness.* The torsion submodule $M_{\mathrm{tor}}$ (Definition 3.45) is intrinsic, and $M/M_{\mathrm{tor}}\simeq R^r$, so $r=\dim_K(M\otimes K)$ is determined. For a prime $p$ the $p$-primary part $M(p)=M_{\mathrm{tor}}[p^\infty]$ is intrinsic and is the sum of the summands $R/(p^{a_j})$ with $p_j=p$ (up to units). Since $p^kR/(p^a)/p^{k+1}R/(p^a)$ is $R/(p)$ if $k<a$ and 0 otherwise, $\dim_{R/(p)}(p^kM(p)/p^{k+1}M(p))=\#\{j:p_j=p,\ a_j>k\}$, and these numbers determine the exponents $a_j$ belonging to $p$. So the elementary divisors are unique, and the invariant factors, which are recovered from them ($d_s$ is the product of the largest prime powers for each prime, $d_{s-1}$ of the next ones, and so on), are unique too. □

**Example 3.44.** The abelian group with relation rows $(2,4,4)$, $(-6,6,12)$, $(10,-4,-16)$ is $\mathbb{Z}/2\oplus\mathbb{Z}/6\oplus\mathbb{Z}/12$ by Example 3.42; its elementary divisors are $2,2,4$ and $3,3$, that is, it is $\mathbb{Z}/2\oplus\mathbb{Z}/2\oplus\mathbb{Z}/4\oplus\mathbb{Z}/3\oplus\mathbb{Z}/3$.

The uniqueness proof can be watched at work on $M=\mathbb{Z}/2\oplus\mathbb{Z}/4$ and $p=2$: $\dim_{\mathbb{F}_2}M/2M=2$ (both summands have exponent $\geq 1$), $\dim_{\mathbb{F}_2}2M/4M=1$ (only $\mathbb{Z}/4$ has exponent $\geq 2$), and $\dim_{\mathbb{F}_2}4M/8M=0$. So exactly two exponents are $\geq 1$ and one is $\geq 2$, which forces the exponents 1 and 2; no rearrangement such as $\mathbb{Z}/8$ or $(\mathbb{Z}/2)^3$ gives the same numbers.

**Definition 3.45** (torsion)**.** Let $R$ be a domain with fraction field $K$. An element $m\in M$ is *torsion* if $rm=0$ for some $r\neq 0$. The torsion elements form a submodule $M_{\mathrm{tor}}$; $M$ is *torsion-free* if $M_{\mathrm{tor}}=0$ and *torsion* if $M_{\mathrm{tor}}=M$. The *rank* of $M$ is $\operatorname{rank}M=\dim_K(M\otimes_R K)$.

**Proposition 3.46.** *(i) $M_{\mathrm{tor}}$ is the kernel of $M\to M\otimes_R K$, $m\mapsto m\otimes 1$, and $M/M_{\mathrm{tor}}$ is torsion-free.*

*(ii) Over a principal ideal domain a finitely generated torsion-free module is free, and $M\simeq M_{\mathrm{tor}}\oplus M/M_{\mathrm{tor}}$ for every finitely generated $M$ (the splitting is not canonical).*

*(iii) Let $R\subseteq R'$ be domains with $R'$ a principal ideal domain that is free of rank $e$ as an $R$-module. A finitely generated $R'$-module $X$ that is torsion-free over $R$, of $R$-rank $n$, is free over $R'$ of rank $n/e$. For instance, $\mathbb{Z}_3[\zeta_3]$ is free of rank 2 over $\mathbb{Z}_3$, with basis $1,\zeta_3$.*

*Proof.* (i) $M \otimes K = S^{-1}M$ with $S = R \setminus \{0\}$ (Theorem 3.38(ii)), and $m/1 = 0$ iff $um = 0$ for some $u \neq 0$. If $rm \in M_{\text{tor}}$ with $r \neq 0$, then $r'rm = 0$ with $r' \neq 0$, so $m \in M_{\text{tor}}$. (ii) By Theorem 3.43, a finitely generated torsion-free module has no summands $R/(d_i)$; and $M/M_{\text{tor}}$ is free, so $M \twoheadrightarrow M/M_{\text{tor}}$ splits (Proposition 3.23). (iii) First, $X$ is torsion-free over $R'$. Let $0 \neq r' \in R'$ and let $\mu$ be multiplication by $r'$ on $R' \simeq R^e$; it is injective ($R'$ is a domain), so $\delta = \det \mu \neq 0$ (it is injective on $R' \otimes K \simeq K^e$, hence invertible there). By Cayley-Hamilton, $\mu^e + c_1\mu^{e-1} + \cdots + c_e = 0$ with $c_e = \pm\delta$, so $\delta = r'y$ for some $y \in R'$. If $r'x = 0$ then $\delta x = 0$ with $0 \neq \delta \in R$, so $x = 0$. By (ii) over the principal ideal domain $R'$, $X \simeq R'^m$, which is free of rank $me$ over $R$; so $n = me$. □

*In the paper* 3.47. The paragraph after the proof of Lemma lem:IV says that "$T_3E$ is a torsion-free module of $\mathbb{Z}_3$-rank 2 over the discrete valuation ring $\mathbb{Z}_3[X]/(X^2+X+1) \simeq \mathbb{Z}_3[\zeta_3]$, with $X$ acting as $\sigma$, hence free of rank one over it". Apply Proposition 3.46(iii) with $R = \mathbb{Z}_3$, $R' = \mathbb{Z}_3[\zeta_3]$ (a discrete valuation ring, hence a principal ideal domain, by Chapter 6), $e = 2$, and $X = T_3E$, which is a free $\mathbb{Z}_3$-module of rank 2 and an $R'$-module through $\sigma$ (Example 3.7(c)): it is free of rank $2/2 = 1$ over $R'$. Lemma lem:tamadd uses the same statement. Exercise 3.120 works out the consequences for $T/3T$.

**Definition 3.48** (saturation)**.** Let $M$ be a lattice and $L \subseteq M$ a submodule. $L$ is *saturated* in $M$ if $M/L$ is torsion-free. The *saturation* of $L$ is $\text{sat}(L) = M \cap KL$, computed inside $M \otimes K$ (where $KL$ is the $K$-span of $L$).

The definition separates two ways in which a sublattice can be "small". A sublattice of smaller rank, such as $L = \mathbb{Z}(1,1) \subset \mathbb{Z}^2$, misses whole directions; a sublattice such as $\mathbb{Z}(2,2)$ misses in addition the points $(1,1)$ that lie on its own line. Saturation adds back exactly these points: $\text{sat}(\mathbb{Z}(2,2)) = \mathbb{Z}^2 \cap \mathbb{Q}(1,1) = \mathbb{Z}(1,1)$. The submodules that occur naturally in the paper, such as the forms killed by an ideal of the Hecke algebra or the lattice of forms orthogonal to $f$, are cut out by linear conditions and are automatically saturated (Proposition 3.49(iii)); a non-saturated sublattice would encode a congruence that is not visible in its own span, and the index $[\text{sat}(L) : L]$ measures it.

**Proposition 3.49.** *Let $M$ be a lattice and $L \subseteq M$ a submodule.*

*(i) The following are equivalent: (a) $L$ is saturated; (b) $L = M \cap KL$; (c) $L$ is a direct summand of $M$; and, when $R = \mathcal{O}$ is a discrete valuation ring, (d) $L \cap \varpi M = \varpi L$, that is, $L/\varpi L \to M/\varpi M$ is injective.*

*(ii) $\text{sat}(L)$ is the smallest saturated submodule containing $L$; it has the same rank as $L$, and $\text{sat}(L)/L = (M/L)_{\text{tor}}$. If $L \subseteq N \subseteq M$ with $N$ saturated and $N/L$ torsion, then $N = \text{sat}(L)$.*

*(iii) The following are saturated: kernels of homomorphisms $M \to N$ with $N$ torsion-free; $M \cap W$ for a $K$-subspace $W \subseteq M \otimes K$; $M[\mathfrak{a}]$ for any action of a ring $\mathbb{T}$ on $M$ and any ideal $\mathfrak{a} \subseteq \mathbb{T}$; and orthogonals for a pairing (Proposition 3.83). Non-examples: $3\mathbb{Z} \subset \mathbb{Z}$; $\mathbb{Z}(3,3) \subset \mathbb{Z}^2$; while $\mathbb{Z}(1,3) \subset \mathbb{Z}^2$ is saturated.*

*Proof.* (i) (a)⇒(c): $M/L$ is finitely generated and torsion-free, hence free (Proposition 3.46(ii)), so $M \twoheadrightarrow M/L$ splits. (c)⇒(a): $M/L$ is isomorphic to a complement, a submodule of the torsion-free $M$. (a)⇔(b): $x \in M \cap KL$ iff $rx \in L$ for some $r \neq 0$, iff the class of $x$ in $M/L$ is torsion. (a)⇒(d): if $\varpi m \in L$ then $m \in L$ by torsion-freeness of $M/L$. (d)⇒(a): suppose $rx \in L$ with $r = u\varpi^k$, $u$ a unit; we show $x \in L$ by induction on $k$. If $k \geq 1$, then $\varpi(\varpi^{k-1}x) \in L \cap \varpi M = \varpi L$, so

$\varpi(\varpi^{k-1}x - l) = 0$ for some $l \in L$, and $\varpi^{k-1}x = l \in L$ because $M$ is torsion-free. (ii) $M/\operatorname{sat}(L)$ embeds in $(M \otimes K)/KL$, a $K$-vector space, so it is torsion-free. A saturated $N \supseteq L$ satisfies $N = M \cap KN \supseteq M \cap KL$. $KL$ and $K\operatorname{sat}(L)$ coincide, so the ranks agree, and $\operatorname{sat}(L)/L$ consists of the classes killed by some $r \neq 0$. If $N/L$ is torsion, $KN = KL$ and $N = M \cap KN = \operatorname{sat}(L)$. (iii) $M/\ker f \simeq f(M) \subseteq N$; $M/(M \cap W)$ embeds in $(M \otimes K)/W$; $M/M[\mathfrak{a}]$ embeds in $\prod_{a\in\mathfrak{a}} M$ by $x \mapsto (ax)_a$ (the $\mathbb{T}$-action is $R$-linear). In $\mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}^2/\mathbb{Z}(3,3) \simeq \mathbb{Z} \oplus \mathbb{Z}/3$ there is torsion; $\mathbb{Z}(1,3)$ is a direct summand with complement $\mathbb{Z}(0,1)$. □

**Theorem 3.50** (stacked bases)**.** *Let $M$ be free of rank $n$ and $L \subseteq M$ a submodule of rank $d$. There are a basis $e_1, \dots, e_n$ of $M$ and nonzero $d_1 \mid \cdots \mid d_d$ such that $d_1e_1, \dots, d_de_d$ is a basis of $L$. The ideals $(d_i)$, the* elementary divisors of $L$ in $M$*, are unique. Hence $M/L \simeq R^{n-d} \oplus \bigoplus_i R/(d_i)$; $L$ is saturated if and only if every $d_i$ is a unit; and if $d = n$ and $B$ is the matrix whose rows are the coordinates of a basis of $L$ in a basis of $M$, then $M/L$ is torsion, of order $|\det B|$ (the* index $[M : L]$*) when $R = \mathbb{Z}$, and of length $v(\det B)$ when $R$ is a discrete valuation ring.*

*Proof.* $L$ is free of rank $d$ (Theorem 3.40); let $B \in \mathrm{M}_{d\times n}(R)$ be the matrix of a basis of $L$ in a basis $f$ of $M$. Its rows are independent, so in Theorem 3.41, $UBV = \operatorname{diag}(d_1, \dots, d_d)$ with all $d_i \neq 0$. The rows of $UB$ form another basis of $L$, and in the basis $e = V^{-1}f$ of $M$ (that is, $e_i = \sum_j (V^{-1})_{ij} f_j$) their coordinates are the rows of $UBV$. So $d_ie_i$ is a basis of $L$. Then $M/L \simeq R^{n-d} \oplus \bigoplus R/(d_i)$, and uniqueness follows from Theorem 3.43 (the number of unit $d_i$ is $d$ minus the number of nonzero torsion summands). The other statements follow. $M/L \simeq R^{n-d} \oplus \bigoplus R/(d_i)$ is torsion-free iff no summand $R/(d_i)$ is nonzero, that is, iff every $d_i$ is a unit. If $d = n$, the free summand disappears and $M/L$ is torsion; from $UBV = \operatorname{diag}(d_1, \dots, d_n)$ we get $\det B = \det(U^{-1}) \det(V^{-1}) \prod d_i$, with $\det U$, $\det V$ units. Over $\mathbb{Z}$ the units are $\pm 1$, so $|\det B| = \prod |d_i| = \#\bigoplus \mathbb{Z}/d_i$; over a discrete valuation ring $v(\det B) = \sum v(d_i) = \operatorname{length} \bigoplus \mathcal{O}/(d_i)$ by Proposition 3.66. □

For example, in $M = \mathbb{Z}^2$ the sublattice $L$ spanned by $(2,0)$ and $(1,3)$ has $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$, $\det B = 6$, and gcd of the entries 1; so its invariant factors are $1, 6$ (PARI: `matsnf([2,0;1,3])` returns `[6,1]`), and $\mathbb{Z}^2/L \simeq \mathbb{Z}/6$ is cyclic. The theorem says that in a suitable basis $e_1, e_2$ of $\mathbb{Z}^2$ the lattice is $\mathbb{Z}e_1 \oplus \mathbb{Z}\,6e_2$; one such basis is $e_1 = (1,3)$, $e_2 = (0,1)$, since $(2,0) = 2e_1 - 6e_2$.

**Proposition 3.51** (the dual of an inclusion of lattices)**.** *Let $L \subseteq M$ be lattices of the same rank $n$, so $M/L$ is torsion. Then restriction $M^\vee \to L^\vee$ is injective and $L^\vee/M^\vee \simeq \bigoplus_i R/(d_i)$, with the same invariant factors as $M/L$. More precisely, there is a canonical isomorphism*

$$L^\vee/M^\vee \xrightarrow{\sim} \operatorname{Hom}_R(M/L, K/R), \qquad \varphi \mapsto (m + L \mapsto \varphi_K(m) + R),$$

*where $\varphi_K$ is the $K$-linear extension of $\varphi$ to $M \otimes K = L \otimes K$. For $R = \mathbb{Z}$ or $\mathbb{Z}_\ell$ the right-hand side is the Pontryagin dual $(M/L)^*$ (Theorem 3.76), so $[L^\vee : M^\vee] = [M : L]$. Equivalently: if $A \hookrightarrow B$ is an injection of lattices of the same rank with finite cokernel $C$, then $B^\vee \hookrightarrow A^\vee$ with cokernel $\simeq C^*$.*

*Proof.* The displayed map is well defined: $\varphi_K(L) = \varphi(L) \subseteq R$, and if $\varphi \in M^\vee$ the image is 0. Its kernel is $\{\varphi : \varphi_K(M) \subseteq R\} = M^\vee$. For surjectivity take stacked bases (Theorem 3.50): $M = \bigoplus Re_i$, $L = \bigoplus Rd_ie_i$. A homomorphism $\psi\colon M/L \to K/R$ is determined by the values $\psi(e_i)$, which are killed by $d_i$, so $\psi(e_i) = c_i/d_i + R$ with $c_i \in R$; the functional $\varphi = \sum_i c_i e_i^\vee/d_i$ is integral on $L$ and maps to $\psi$. In these bases $L^\vee$ has basis $e_i^\vee/d_i$ and $M^\vee$ has basis $e_i^\vee$, which gives the

invariant factors. For a finite group, $\mathrm{Hom}(X, \mathbb{Q}/\mathbb{Z})$ and, for an $\ell$-primary one, $\mathrm{Hom}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ are the Pontryagin duals, and $\#X^* = \#X$ (Theorem 3.76). □

The direction of the inclusion is reversed by duality: the smaller lattice has the larger dual. In rank one, $L = 2\mathbb{Z} \subset M = \mathbb{Z}$; the functional $e^\vee$ $(1 \mapsto 1)$ generates $M^\vee$, while $L^\vee$ is generated by the functional $2 \mapsto 1$, which is $\frac{1}{2}e^\vee$ extended to $\mathbb{Q}$. So $L^\vee = \frac{1}{2}\mathbb{Z}e^\vee \supset M^\vee = \mathbb{Z}e^\vee$, of index $2 = [M : L]$, and the isomorphism of the proposition sends $\frac{1}{2}e^\vee$ to the character $1 + 2\mathbb{Z} \mapsto \frac{1}{2} + \mathbb{Z}$ of $\mathbb{Z}/2$.

*Remark* 3.52. In the language of Chapter 16, $\mathrm{Hom}_R(M/L, K/R) \simeq \mathrm{Ext}^1_R(M/L, R)$, and Proposition 3.51 is the long exact Ext sequence of $0 \to L \to M \to M/L \to 0$. We do not need this.

**Proposition 3.53** (the congruence module of a decomposition)**.** *Let $M$ be a lattice, $M \otimes K = V_1 \oplus V_2$ a decomposition into $K$-subspaces with projections $p_1, p_2$, and $L_i = M \cap V_i$ (saturated).*

*(i) $L_1 \oplus L_2 \subseteq M$ has torsion quotient, and $x \mapsto p_i(x)$ induces isomorphisms $M/(L_1 \oplus L_2) \simeq p_1(M)/L_1 \simeq p_2(M)/L_2$.*

*(ii) Suppose $V_1 = Kf$ with $f \in M$ and $L_1 = Rf$ (that is, $f$ is primitive). Then $M/(L_1 \oplus L_2)$ embeds in $Kf/Rf \simeq K/R$, so it is cyclic, $\simeq R/(r)$, generated by the class of any $h \in M$ with $p_1(h) = f/r$; then $rh = f + g_0$ with $g_0 \in L_2$, so $f \equiv -g_0 \pmod{rM}$. For $0 \neq r' \in R$ there is $g \in L_2$ with $f \equiv g \pmod{r'M}$ if and only if $r' \mid r$. So $(r)$ is the* congruence modulus *of $f$ against $L_2$: the largest modulus of a congruence between $f$ and an element of $L_2$.*

*(iii) Example: $M = \mathbb{Z}^2$, $f = (1, a)$, $g = (1, b)$ with $a \neq b$, $V_1 = \mathbb{Q}f$, $V_2 = \mathbb{Q}g$. Then $L_1 = \mathbb{Z}f$, $L_2 = \mathbb{Z}g$, $M/(\mathbb{Z}f \oplus \mathbb{Z}g) \simeq \mathbb{Z}/(a - b)$, and $f \equiv g \pmod{(a - b)}$ coordinatewise.*

*Proof.* (i) Every $v \in V_1$ has a nonzero multiple in $M$ (which spans $M \otimes K$), hence in $L_1$; so $L_1$ spans $V_1$, likewise $L_2$, and the quotient has rank 0. The map $M \to p_1(M)/L_1$ is onto; $x$ lies in its kernel iff $p_1(x) \in L_1 \subseteq M$, and then $p_2(x) = x - p_1(x) \in M \cap V_2 = L_2$; so the kernel is $L_1 \oplus L_2$. Symmetrically for $p_2$. (ii) $p_1(M)$ is a finitely generated submodule of $Kf$ containing $Rf$, hence free of rank one, $p_1(M) = R(f/r)$ for some $r \neq 0$ (a generator $u$ satisfies $f = su$, so $u = f/s$). By (i) the quotient is $R(f/r)/Rf \simeq R/(r)$, generated by any $h$ with $p_1(h) = f/r$. Then $rh - f = rp_2(h) \in M \cap V_2 = L_2$. If $f - g = r'm$ with $g \in L_2$ and $m \in M$, then $p_1(m) = f/r' \in R(f/r)$, so $r = sr'$; conversely $f \equiv -g_0 \pmod{rM}$ implies the congruence modulo every divisor $r'$ of $r$. (iii) $f$ and $g$ are primitive, so $L_1 = \mathbb{Z}f$ and $L_2 = \mathbb{Z}g$, and the index of $\mathbb{Z}f \oplus \mathbb{Z}g$ is $|\det \begin{pmatrix} 1 & a \\ 1 & b \end{pmatrix}| = |b - a|$ (Theorem 3.50); the quotient is cyclic by (ii), and $f - g = (0, a - b)$. □

Here is (ii) in the case $a = 0$, $b = 4$ of (iii), so $f = (1, 0)$, $g = (1, 4)$ and $r = 4$. To project $h = (0, -1)$ onto $\mathbb{Q}f$ along $\mathbb{Q}g$, solve $(0, -1) = \alpha(1, 0) + \beta(1, 4)$: $\beta = -\frac{1}{4}$, $\alpha = \frac{1}{4}$, so $p_1(h) = f/4$, and $h$ generates the quotient $\mathbb{Z}^2/(\mathbb{Z}f \oplus \mathbb{Z}g) \simeq \mathbb{Z}/4$. Then $4h = (0, -4) = f + g_0$ with $g_0 = (-1, -4) = -g \in L_2$, and $f \equiv -g_0 = g \pmod{4\mathbb{Z}^2}$, the congruence of the two vectors modulo 4. No congruence modulo 8 exists: $f - mg = (1 - m, -4m)$ lies in $8\mathbb{Z}^2$ only if $m \equiv 1 \pmod 8$ and $4m \equiv 0 \pmod 8$, which is impossible. This is the pattern of the second paragraph of the proof of Lemma lem:crit, with $f$ a newform and $g$ ranging over the other forms.

*In the paper* 3.54. The second paragraph of the proof of Lemma lem:crit is Propositions 3.51 and 3.53 with $M = S_2(\Gamma_0(N);\mathbb{Z})$, $V_1 = \mathbb{Q}f$, $V_2 = f^\perp$, $L_1 = \mathbb{Z}f$ (as $a_1(f) = 1$, $f$ is primitive) and $L_2 = L = f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z})$. "Dualising the inclusion $\mathbb{T} \subset \mathbb{T}/I \times \mathbb{T}/J$, whose cokernel is $\mathbb{T}/(I+J)$, shows that $\mathbb{Z}f \oplus L$ has index $r'$" is Proposition 3.91(iii), which rests on Proposition 3.51. That the quotient $S_2(\Gamma_0(N);\mathbb{Z})/(\mathbb{Z}f \oplus L)$ "embeds in $\mathbb{Q}f/\mathbb{Z}f$ by projection onto $\mathbb{Q}f$ along $f^\perp$, so it is cyclic, generated by the class of a form $h$ with projection $f/r'$; then $r'h = f + g_0$ with $g_0 \in L$" is Proposition 3.53(ii), whose last clause is the paper's "$r' \le r$" and "$r \mid r'$" in one statement: the congruence number is the congruence modulus. Lemma lem:ddt417 uses the same structure, with $L \otimes K = L[\wp]_K \oplus L[I]_K$ and $Q = L/(L[\wp] \oplus L[I])$.

## 3.6 Discrete valuation rings, $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$ and $\mathbb{Q}_\ell/\mathbb{Z}_\ell$

The coefficient rings of the paper are discrete valuation rings: $\mathbb{Z}_\ell$ everywhere, $\mathcal{O}$ in Lemma lem:ddt417 and Section 4.1, $\mathbb{Z}_3[\zeta_3]$ in Lemma lem:IV. Over such a ring finitely generated modules are classified by a rank and a list of exponents, which is what makes lengths and Fitting ideals computable (Section 3.7). The module $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ carries the discrete coefficients $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ of the paper's Selmer groups, and its injectivity makes Pontryagin duality exact (Section 3.8). Here $\mathbb{Z}_\ell$ is built algebraically; the analytic theory is Chapter 7 (Warning 3.62).

**Theorem 3.55** (discrete valuation rings)**.** *Let $\mathcal{O}$ be a Noetherian local domain that is not a field, with maximal ideal $\mathfrak{m}$ and fraction field $K$. The following are equivalent:*

*(a) $\mathfrak{m}$ is principal;*

*(b) $\mathcal{O}$ is a principal ideal domain;*

*(c) every nonzero ideal is a power of $\mathfrak{m}$;*

*(d) there is a surjective homomorphism $v\colon K^\times \to \mathbb{Z}$ with $v(x+y) \ge \min(v(x), v(y))$ and $\mathcal{O} = \{x \in K^\times : v(x) \ge 0\} \cup \{0\}$;*

*(e) $\mathcal{O}$ is integrally closed in $K$ (and of dimension one).*

*Such a ring is a* discrete valuation ring*. A generator $\varpi$ of $\mathfrak{m}$ is a* uniformiser*; every $x \in K^\times$ is uniquely $x = u\varpi^{v(x)}$ with $u \in \mathcal{O}^\times$; $\mathcal{O}^\times = \{v = 0\}$; and the nonzero ideals are the $\varpi^k\mathcal{O}$, $k \ge 0$.*

*Proof.* (a)⇒(c),(d),(b). Let $\mathfrak{m} = \varpi\mathcal{O}$. First, $\bigcap_k \mathfrak{m}^k = 0$: if $x = \varpi^k y_k$ for every $k$, then $\varpi^k y_k = \varpi^{k+1} y_{k+1}$ gives $y_k = \varpi y_{k+1}$ ($\mathcal{O}$ is a domain), so $(y_0) \subseteq (y_1) \subseteq \cdots$; this chain stabilises, $y_{k+1} = u y_k$ for some $k$ and $u \in \mathcal{O}$, so $(1 - \varpi u)y_k = 0$, and $1 - \varpi u$ is a unit (Proposition 3.34); hence $y_k = 0$ and $x = 0$. So every $x \ne 0$ lies in $\mathfrak{m}^k$ but not $\mathfrak{m}^{k+1}$ for a unique $k$; then $x = \varpi^k u$ with $u \notin \mathfrak{m}$, a unit. The exponent is unique: $\varpi^k u = \varpi^j u'$ with $k < j$ would give $u = \varpi^{j-k}u' \in \mathfrak{m}$. Put $v(x) = k$. On $\mathcal{O} \setminus \{0\}$ it is additive: $x = \varpi^a u$, $y = \varpi^b u'$ give $xy = \varpi^{a+b}uu'$ with $uu'$ a unit, so $v(xy) = v(x) + v(y)$. Extend by $v(x/y) = v(x) - v(y)$; this is well defined, since $x/y = x'/y'$ means $xy' = x'y$, whence $v(x) + v(y') = v(x') + v(y)$, and it is a homomorphism $K^\times \to \mathbb{Z}$, onto because $v(\varpi) = 1$. Finally $v(x+y) \ge \min(v(x), v(y))$ because $\varpi^{\min}$ divides both (clear denominators first for elements of $K$). A nonzero ideal $\mathfrak{a}$ contains an element of least valuation $k$, and then $\mathfrak{a} = \varpi^k\mathcal{O}$. This proves (c), (d) and (b). (b)⇒(a): trivial. (c)⇒(a): $\mathfrak{m} \ne \mathfrak{m}^2$, since otherwise $\mathfrak{m} = 0$ by Nakayama ($\mathfrak{m}$ is finitely generated); pick $\varpi \in \mathfrak{m} \setminus \mathfrak{m}^2$; then $\varpi\mathcal{O} = \mathfrak{m}^k$ for some $k$, and $k = 1$: $k = 0$ would make $\varpi$ a unit, and $k \ge 2$ would put $\varpi$ in $\mathfrak{m}^2$. (d)⇒(a): pick

$\varpi$ with $v(\varpi) = 1$. If $x \in \mathfrak{m}$ then $v(x) \geq 1$ (if $v(x) = 0$ then $v(x^{-1}) = 0$ and $x^{-1} \in \mathcal{O}$), so $x/\varpi \in \mathcal{O}$. (e): we do not prove the equivalence with (e); see [7, Proposition 9.2] (number not verified against the printed edition) or [174, Chapter I, Section 2]. It is used only in Chapter 6. The last statements were proved along the way. □

A discrete valuation ring $\mathcal{O}$ is *complete* if the map $\mathcal{O} \to \varprojlim_n \mathcal{O}/\varpi^n\mathcal{O}$ is an isomorphism. Then every finitely generated $\mathcal{O}$-module $M$, in particular every free one of finite rank, is complete: $M \to \varprojlim_n M/\varpi^n M$ is an isomorphism, by the proof of Theorem 3.58(vi), which uses only completeness of the ring. $\mathbb{Z}_\ell$ is complete (Theorem 3.58(vi)); $\mathbb{Z}_{(\ell)}$ is not, since $\varprojlim \mathbb{Z}_{(\ell)}/\ell^n = \varprojlim \mathbb{Z}/\ell^n = \mathbb{Z}_\ell$ is uncountable.

**Example 3.56.** Discrete valuation rings: $\mathbb{Z}_{(p)}$ (Chapter 2); $F[[x]]$ for a field $F$ (every nonzero series is $x^k u$ with $u$ a unit, since a series with nonzero constant term is invertible, so the ideals are the $(x^k)$); $\mathbb{Z}_\ell$ (Theorem 3.58); and, with proofs in Chapters 6 and 7, $\mathbb{Z}_3[\zeta_3]$, $\mathbb{Z}_\ell[\zeta_3]$ for $\ell \equiv 2 \pmod 3$, and the completions $\mathcal{O}_\lambda$ of rings of integers. Not discrete valuation rings: $\mathbb{Z}$ (not local); $\mathbb{Z}/p^2$ and $\mathcal{O}[x]/(x^2)$ (not domains); $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$ (not local); $\mathbb{Z}_\ell[x]$ (not local: $(\ell, x)$ and $(x-1, \ell)$ are distinct maximal ideals); $\mathbb{Z}_{(11)}[x]/(x^2+x-1)$ (a domain with two maximal ideals, Theorem 3.95(v)).

**Proposition 3.57** (modules over a discrete valuation ring)**.** *Let $M$ be a finitely generated $\mathcal{O}$-module.*

*(i)* $M \simeq \mathcal{O}^r \oplus \bigoplus_{i=1}^s \mathcal{O}/\varpi^{a_i}$ *with $a_i \geq 1$; $r$ and the multiset $\{a_i\}$ are unique.*

*(ii) If $M$ is torsion-free, it is free.*

*(iii)* $\dim_k M/\varpi M = r + s$ *is the minimal number of generators, and* $\dim_k M[\varpi] = s$.

*(iv) If $r = 0$ then $M[\varpi^k] \simeq \bigoplus_i \mathcal{O}/\varpi^{\min(k,a_i)}$; in general $M/\varpi^k M \simeq (\mathcal{O}/\varpi^k)^r \oplus \bigoplus_i \mathcal{O}/\varpi^{\min(k,a_i)}$.*

*(v) $M$ is cyclic if and only if $r + s \leq 1$; a torsion $M$ with $\dim_k M[\varpi] \leq 1$ is cyclic.*

*(vi) A submodule $L$ of a lattice $M$ is saturated if and only if $L \cap \varpi M = \varpi L$.*

*Proof.* (i) is Theorem 3.43, as $\varpi$ is the only prime up to units; (ii) is Proposition 3.46(ii). For (iii) and (iv) compute summand by summand. For the summand $\mathcal{O}$: $\mathcal{O}/\varpi\mathcal{O} = k$ and $\mathcal{O}[\varpi] = 0$. For $\mathcal{O}/\varpi^a$: the quotient by $\varpi$ is $k$, the $\varpi$-torsion is $\varpi^{a-1}\mathcal{O}/\varpi^a \simeq k$, the $\varpi^k$-torsion is $\varpi^{\max(a-k,0)}\mathcal{O}/\varpi^a \simeq \mathcal{O}/\varpi^{\min(k,a)}$, and the quotient by $\varpi^k$ is $\mathcal{O}/\varpi^{\min(k,a)}$. The minimal number of generators is Theorem 3.35(iii). (v) By (iii) the minimal number of generators is $r + s$, so $M$ is cyclic iff $r + s \leq 1$; if $M$ is torsion then $r = 0$, and $s = \dim_k M[\varpi] \leq 1$. (vi) is Proposition 3.49(i)(d). □

For instance $M = \mathcal{O}/\varpi^2 \oplus \mathcal{O}/\varpi$ has $r = 0$, $s = 2$: it needs two generators, $M[\varpi] = \varpi\mathcal{O}/\varpi^2 \oplus \mathcal{O}/\varpi \simeq k^2$, and $M[\varpi^2] = M$, $M/\varpi M \simeq k^2$, in accordance with (iv). In general the two numbers $\dim_k M/\varpi M$ and $\dim_k M[\varpi]$ differ exactly by the rank $r$, so together they determine $r$ and the number $s$ of torsion summands, though not the exponents $a_i$.

**Theorem 3.58** (the ring $\mathbb{Z}_\ell$)**.** *Let $\mathbb{Z}_\ell = \varprojlim_n \mathbb{Z}/\ell^n$, the ring of compatible sequences $x = (x_n)_{n\geq 1}$, $x_n \in \mathbb{Z}/\ell^n$, $x_{n+1} \equiv x_n \pmod{\ell^n}$, with componentwise operations (as a profinite group it is constructed in Chapter 1).*

*(i) $x$ is a unit if and only if $x_1 \neq 0$ in $\mathbb{F}_\ell$.*

*(ii) Every nonzero $x$ is uniquely $\ell^k u$ with $k \geq 0$ and $u \in \mathbb{Z}_\ell^\times$. Hence $\mathbb{Z}_\ell$ is a discrete valuation ring with uniformiser $\ell$, residue field $\mathbb{F}_\ell$ (via $x \mapsto x_1$) and ideals $\ell^k\mathbb{Z}_\ell$, and its valuation extends $v_\ell$ on $\mathbb{Z}$.*

*(iii) $\mathbb{Z} \to \mathbb{Z}_\ell$, $n \mapsto (n \bmod \ell^k)_k$, is injective; $\mathbb{Z}_{(\ell)} \subseteq \mathbb{Z}_\ell$; and $x \mapsto x_n$ induces $\mathbb{Z}_\ell/\ell^n\mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^n$. An integer prime to $\ell$ is a unit of $\mathbb{Z}_\ell$, so a $\mathbb{Z}_\ell$-module killed by an integer prime to $\ell$ is $0$.*

*(iv) $\mathbb{Q}_\ell := \operatorname{Frac}(\mathbb{Z}_\ell) = \mathbb{Z}_\ell[1/\ell]$, and $\mathbb{Q} \cap \mathbb{Z}_\ell = \mathbb{Z}_{(\ell)}$ inside $\mathbb{Q}_\ell$.*

*(v) $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$. For a finitely generated abelian group $A \simeq \mathbb{Z}^r \oplus A_{\mathrm{tor}}$, $A \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \simeq \mathbb{Z}_\ell^r \oplus A_{\mathrm{tor}}[\ell^\infty]$; for finite $A$, $\#(A \otimes \mathbb{Z}_\ell) = \ell^{v_\ell(\#A)}$, the $\ell$-part of $\#A$.*

*(vi) Every finitely generated $\mathbb{Z}_\ell$-module $M$ is complete: $M \to \varprojlim_n M/\ell^n M$ is an isomorphism.*

*(vii) More generally, for a Noetherian ring $R$ and an ideal $I$, the $I$-adic completion $\hat{R} = \varprojlim R/I^n$ is Noetherian and flat over $R$, and $\hat{M} = \varprojlim M/I^n M \simeq M \otimes_R \hat{R}$ for finitely generated $M$.*

*Proof.* (i) If $x_1 \neq 0$, then $\ell \nmid x_n$ for all $n$, so each $x_n$ is a unit of $\mathbb{Z}/\ell^n$, and the inverses are compatible (inverses are unique); conversely $x_1$ is the image of $x$ in the field $\mathbb{F}_\ell$. (ii) Let $x \neq 0$ and let $k$ be the largest index with $x_k = 0$ (with $x_0 = 0$ by convention); it exists because $x_n = 0$ forces $x_j = 0$ for $j \leq n$. For $n \geq 1$, $x_{n+k} \equiv x_k = 0 \pmod{\ell^k}$, so $x_{n+k} = \ell^k y_n$ with $y_n \in \mathbb{Z}/\ell^n$ well defined; the $y_n$ are compatible, $y_1 \neq 0$ because $x_{k+1} \neq 0$, so $u = (y_n)$ is a unit with $x = \ell^k u$. Then $k = v(x)$ is determined by $x$. Since $\ell^k \neq 0$ (its component at $k+1$ is nonzero), the product $\ell^{a+b}uu'$ of two nonzero elements is nonzero: $\mathbb{Z}_\ell$ is a domain. A nonzero ideal is $\ell^k\mathbb{Z}_\ell$ for its least valuation $k$, so $\mathbb{Z}_\ell$ is a principal ideal domain, local with maximal ideal $\ell\mathbb{Z}_\ell$ (the non-units, by (i)): a discrete valuation ring by Theorem 3.55. The kernel of $x \mapsto x_1$ is $\{v \geq 1\} = \ell\mathbb{Z}_\ell$. (iii) $n \neq 0$ has nonzero image in $\mathbb{Z}/\ell^k$ for $\ell^k > |n|$. By (ii), $x_n = 0$ iff $v(x) \geq n$, so the kernel of $x \mapsto x_n$ is $\ell^n\mathbb{Z}_\ell$; it is onto because $\mathbb{Z}$ maps onto $\mathbb{Z}/\ell^n$. Integers prime to $\ell$ are units by (i), which gives $\mathbb{Z}_{(\ell)} \subseteq \mathbb{Z}_\ell$. (iv) $x/y = x\ell^{-v(y)}u^{-1}$ for $y = \ell^{v(y)}u$. By (iii), $\mathbb{Q} \subseteq \mathbb{Q}_\ell$ and $\mathbb{Z}_{(\ell)} \subseteq \mathbb{Z}_\ell$. Conversely, if $a/b$ is in lowest terms with $\ell \mid b$, then $\ell \nmid a$ and $v(a/b) = -v_\ell(b) < 0$ by (ii), so $a/b \notin \mathbb{Z}_\ell$. (v) $\mathbb{Z}_\ell$ is torsion-free over $\mathbb{Z}$, hence flat (Proposition 3.29(ii)). $\mathbb{Z}/n \otimes \mathbb{Z}_\ell = \mathbb{Z}_\ell/n\mathbb{Z}_\ell = \mathbb{Z}_\ell/\ell^{v_\ell(n)}\mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^{v_\ell(n)}$ by (ii), (iii); sum over the cyclic summands of $A$. (vi) For $M = \mathbb{Z}_\ell$ the map $\mathbb{Z}_\ell \to \varprojlim \mathbb{Z}_\ell/\ell^n\mathbb{Z}_\ell = \varprojlim \mathbb{Z}/\ell^n = \mathbb{Z}_\ell$ is the identity; for $M = \mathbb{Z}/\ell^a$ the system is eventually constant; and both sides commute with finite direct sums. Conclude by Proposition 3.57(i). (vii) We do not prove this; see [7, Chapter 10] (the Artin-Rees lemma) or [62, Chapter 7]. □

**Lemma 3.59** (Baer's criterion over a principal ideal domain)**.** *Let $R$ be a principal ideal domain and $D$ a* divisible *$R$-module ($dD = D$ for every $d \neq 0$). Then $D$ is* injective*: for every submodule $N \subseteq M$, every homomorphism $N \to D$ extends to $M$. Consequently $\operatorname{Hom}_R(-, D)$ is exact.*

*Proof.* Let $f\colon N \to D$. By Zorn's lemma there is a maximal extension $f_0\colon N_0 \to D$ ($N \subseteq N_0 \subseteq M$), the union of a chain of extensions being an extension. Suppose $m \in M \setminus N_0$. The ideal $\{r : rm \in N_0\}$ is $dR$. If $d = 0$, extend by $f(n + rm) = f_0(n)$. If $d \neq 0$, choose $y \in D$ with $dy = f_0(dm)$ and put $f(n + rm) = f_0(n) + ry$. This is well defined: if $n + rm = 0$ then $rm \in N_0$, so $r = sd$, and $f_0(n) + ry = f_0(n) + sf_0(dm) = f_0(n + rm) = 0$. This contradicts maximality, so $N_0 = M$. Exactness of $\operatorname{Hom}(-, D)$: by Proposition 3.25(ii) only surjectivity of $\operatorname{Hom}(M, D) \to \operatorname{Hom}(N, D)$ for $N \subseteq M$ remains, and that is injectivity of $D$. □

The choice of $y$ is the only place where divisibility is used, and it cannot be avoided: $\mathbb{Z}$ is not divisible and not injective, since the map $2\mathbb{Z} \to \mathbb{Z}$, $2 \mapsto 1$, has no extension to $\mathbb{Z}$ (it would send 1

to an integer $y$ with $2y = 1$). Over $\mathbb{Z}$ the divisible groups $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$ and $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ are injective, and these are the coefficient groups of duality.

**Proposition 3.60** (the module $\mathbb{Q}_\ell/\mathbb{Z}_\ell$)**.** *(i) $\mathbb{Q}_\ell/\mathbb{Z}_\ell \simeq \mathbb{Z}[1/\ell]/\mathbb{Z} \simeq (\mathbb{Q}/\mathbb{Z})[\ell^\infty] \simeq \varinjlim_k \mathbb{Z}/\ell^k$, the transition maps being multiplication by $\ell$. As an abstract group it is isomorphic to $\mu_{\ell^\infty}$ (no Galois action is involved here).*

*(ii) It is divisible and $\ell$-primary torsion; $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)[\ell^k] = \ell^{-k}\mathbb{Z}_\ell/\mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^k$; its proper $\mathbb{Z}_\ell$-submodules are exactly these.*

*(iii) The ring $\mathrm{End}_{\mathbb{Z}_\ell}(\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is $\mathbb{Z}_\ell$, acting by multiplication; moreover $\mathrm{Hom}_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^k, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \simeq \mathbb{Z}/\ell^k$, and $\mathrm{Hom}_{\mathbb{Z}_\ell}(\mathbb{Z}_\ell, \mathbb{Q}_\ell/\mathbb{Z}_\ell) = \mathbb{Q}_\ell/\mathbb{Z}_\ell$.*

*(iv) $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is an injective $\mathbb{Z}_\ell$-module and $\mathbb{Q}/\mathbb{Z}$ an injective $\mathbb{Z}$-module; so $\mathrm{Hom}(-, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ and $\mathrm{Hom}(-, \mathbb{Q}/\mathbb{Z})$ are exact.*

*Proof.* (i) Every element of $\mathbb{Q}_\ell$ is $x = \ell^{-k}u$ with $u \in \mathbb{Z}_\ell$; choosing $a \in \mathbb{Z}$ with $u \equiv a \pmod{\ell^k}$ gives $x \equiv a/\ell^k$ modulo $\mathbb{Z}_\ell$. So $\mathbb{Z}[1/\ell] \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$ is onto, with kernel $\mathbb{Z}[1/\ell] \cap \mathbb{Z}_\ell = \mathbb{Z}$. A class $a/b + \mathbb{Z}$ in reduced form is killed by a power of $\ell$ iff $b$ is a power of $\ell$, which identifies $\mathbb{Z}[1/\ell]/\mathbb{Z}$ with $(\mathbb{Q}/\mathbb{Z})[\ell^\infty]$; and $\mathbb{Z}/\ell^k \simeq \ell^{-k}\mathbb{Z}/\mathbb{Z}$. The map $\mathbb{Z}[1/\ell]/\mathbb{Z} \to \mu_{\ell^\infty}$, $x \mapsto e^{2\pi i x}$, is an isomorphism. (ii) Divisibility: $x = \ell^{-k}u$ is $\ell^j u'$ times $\ell^{-k-j}u'^{-1}u$ for any $j$ and unit $u'$. The $\ell^k$-torsion is $\ell^{-k}\mathbb{Z}_\ell/\mathbb{Z}_\ell$, cyclic of order $\ell^k$. A submodule containing elements of every order contains every $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)[\ell^k]$, since an element of exact order $\ell^k$ generates this cyclic group; otherwise it lies in some $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)[\ell^k]$ and is one of its subgroups $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)[\ell^j]$. (iii) An endomorphism $\phi$ preserves each $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)[\ell^k] \simeq \mathbb{Z}/\ell^k$, where it is multiplication by some $a_k \in \mathbb{Z}/\ell^k$; the $a_k$ are compatible, so $\phi$ is multiplication by $a = (a_k) \in \mathbb{Z}_\ell$, and $a \neq 0$ acts nontrivially. The other two are Proposition 3.25(i). (iv) Both are divisible (for $\mathbb{Q}/\mathbb{Z}$ over $\mathbb{Z}$, $x/n$ divides $x$), so Lemma 3.59 applies. □

**Proposition 3.61** (the topology of finitely generated $\mathbb{Z}_\ell$-modules)**.** *Give every finitely generated $\mathbb{Z}_\ell$-module $M$ the $\ell$-adic topology, in which the $\ell^n M$ form a basis of neighbourhoods of 0.*

*(i) $M$ is compact, Hausdorff and complete.*

*(ii) Every $\mathbb{Z}_\ell$-linear map between finitely generated modules is continuous; every submodule of $M$ is finitely generated and closed, and every linear map has closed image.*

*(iii) $\mathrm{GL}_n(\mathbb{Z}_\ell) = \mathrm{Aut}_{\mathbb{Z}_\ell}(\mathbb{Z}_\ell^n) = \varprojlim_k \mathrm{GL}_n(\mathbb{Z}/\ell^k)$ is profinite (Chapter 1).*

*(iv) (Topological Nakayama.) A compact Hausdorff $\mathbb{Z}_\ell$-module $A$ on which $\mathbb{Z}_\ell$ acts continuously and with $A/\ell A$ finite is finitely generated.*

*Proof.* (i) $M \simeq \mathbb{Z}_\ell^r \oplus F$ with $F$ finite, and $\ell^n M$ corresponds to $(\ell^n\mathbb{Z}_\ell)^r \oplus \ell^n F$, so the $\ell$-adic topology is the product topology. $\mathbb{Z}_\ell = \varprojlim \mathbb{Z}/\ell^n$ is a closed subset of the compact product $\prod \mathbb{Z}/\ell^n$ of finite discrete sets, and its subspace topology has the basis $\ell^n\mathbb{Z}_\ell$ (the kernels of the projections). Completeness is Theorem 3.58(vi). (ii) $f(\ell^n M) \subseteq \ell^n N$, so $f$ is continuous. A submodule $N$ is finitely generated (Proposition 3.31), hence the image of a continuous map $\mathbb{Z}_\ell^s \to M$, hence compact, hence closed. The same applies to $f(M)$. (iii) A matrix over $\mathbb{Z}_\ell$ is invertible iff it is invertible modulo $\ell$ (Theorem 3.35(iv)), and $\mathbb{Z}_\ell^{n\times n} = \varprojlim (\mathbb{Z}/\ell^k)^{n\times n}$. □

*Sketch of proof of (iv).* By Pontryagin duality for compact and discrete modules, which is Theorem 3.77(v), the dual $A^*$ is a discrete $\ell$-primary module with $A^*[\ell] = (A/\ell A)^*$ finite, hence cofinitely generated, so $A = A^{**}$ is finitely generated by Theorem 3.77(ii). The duality theory for compact modules is in [143, Chapter I, Section 1, and Chapter V, Section 2] and [167, Appendix B] (section numbers not verified against the printed editions). □

*Warning* 3.62. Theorem 3.58 constructs $\mathbb{Z}_\ell$ as a ring and proves only what is needed for modules. The absolute value, the description of $\mathbb{Q}_p$ as the completion of $\mathbb{Q}$, Hensel's lemma and the structure of $\mathbb{Z}_p^\times$ and $\mathbb{Q}_p^\times$ are proved in Chapter 7. Idempotent lifting (Theorem 3.95) is proved below by a Newton iteration that uses only completeness, not Hensel's lemma.

*In the paper* 3.63. After Lemma lem:dfg21 the paper says: "At $\infty$, $H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for $\ell$ odd". The group $H^1(\mathbb{R}, T^D)$ is a $\mathbb{Z}_\ell$-module (Chapter 16) killed by the integer 2, which is a unit of $\mathbb{Z}_\ell$ for $\ell$ odd (Theorem 3.58(iii)); so it is 0, and so is its submodule $H^1_f(\mathbb{R}, T^D)$. In Lemma lem:fpr, "a nonzero rational number which is an $\ell$-adic integer" means an element of $\mathbb{Q} \cap \mathbb{Z}_\ell = \mathbb{Z}_{(\ell)}$ (Theorem 3.58(iv)).

## 3.7 Length, annihilators and Fitting ideals

The paper measures its finite modules in two ways. Selmer groups and $H^0$ are measured by their length over $\mathbb{Z}_\ell$ (Theorem thm:main, Theorem thm:dfg37, Remark rem:ns, Lemma lem:h0). Tamagawa factors are *Fitting ideals*: Section 1.2 defines $\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}\big((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}}\big)$, and Lemma lem:fpr reads $\mathrm{Fitt}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}_p, T^D) = L_p(A_f, 1)^{-1}\,\mathrm{Tam}_p(T^D)$. The main result of this section, Theorem 3.72, says that over a discrete valuation ring the two measurements agree: $\mathrm{Fitt} = \varpi^{\mathrm{length}}$. It is the reason why the paper can add valuations of Tamagawa factors to lengths of Selmer groups. The Jordan-Holder theorem is proved here for modules over any ring; Chapter 4 uses it for composition factors of representations.

**Definition 3.64** (composition series and length)**.** In this definition and in Theorem 3.65, $R$ is any ring with 1 (not necessarily commutative) and modules are left modules. A module is *simple* if it is nonzero and has no submodules other than 0 and itself. A *composition series* of $M$ is a chain $0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$ with every $M_i/M_{i-1}$ simple; $n$ is its *length* and the $M_i/M_{i-1}$ are its *factors*. The *length* $\mathrm{length}_R(M)$ is the length of any composition series (well defined by Theorem 3.65), and $\mathrm{length}_R(M) = \infty$ if $M$ has none.

**Theorem 3.65** (Jordan-Holder; length)**.** *Let $R$ be any ring.*

*(i) If $M$ has a composition series, then every chain of submodules of $M$ is finite and can be refined to a composition series, and any two composition series of $M$ have the same length and the same factors up to order and isomorphism (Jordan-Holder). Submodules and quotients of $M$ have composition series.*

*(ii) A module has finite length if and only if it is Noetherian and Artinian (descending chains stabilise).*

*(iii) (Additivity.) For an exact sequence $0 \to M' \to M \to M'' \to 0$, $\mathrm{length}(M) = \mathrm{length}(M') + \mathrm{length}(M'')$ (in $\mathbb{Z}_{\geq 0} \cup \{\infty\}$). For an exact sequence $0 \to M_1 \to \cdots \to M_k \to 0$ of modules of finite length, $\sum_i (-1)^i \mathrm{length}(M_i) = 0$.*

*Now let $R$ be commutative.*

*(iv) Let $R$ be local, with maximal ideal $\mathfrak{m}$ and residue field $k$. Every simple module is isomorphic to $k$. If $\mathfrak{m}^N M = 0$, then $\operatorname{length}_R(M) = \sum_{i<N} \dim_k \mathfrak{m}^i M/\mathfrak{m}^{i+1}M$. If $R$ is Noetherian local, a finitely generated module has finite length if and only if it is killed by a power of $\mathfrak{m}$.*

*(v) If $R \to R'$ is a local homomorphism of local rings ($\mathfrak{m}_R$ maps into $\mathfrak{m}_{R'}$), $R'$ is finitely generated as an $R$-module and $f = [k' : k]$ is the degree of the residue field extension, then $\operatorname{length}_R(M) = f \cdot \operatorname{length}_{R'}(M)$ for every $R'$-module $M$ of finite length.*

*Proof.* (i) For a module $X$ with a composition series let $\lambda(X)$ be the least length of a composition series of $X$. *Step 1: if $N \subsetneq M$ then $N$ has a composition series and $\lambda(N) < \lambda(M)$.* Let $(M_i)_{i\le n}$ be a composition series with $n = \lambda(M)$ and put $N_i = N \cap M_i$. Then $N_i/N_{i-1} = N_i/(N_i \cap M_{i-1}) \simeq (N_i + M_{i-1})/M_{i-1} \subseteq M_i/M_{i-1}$ (Theorem 3.8(ii), which holds verbatim over any ring), so it is 0 or simple. Deleting repetitions gives a composition series of $N$ of length at most $n$. If the length were $n$, every $N_i/N_{i-1}$ would be nonzero, hence equal to $M_i/M_{i-1}$ inside it; by induction on $i$, $N_{i-1} = M_{i-1}$ and $N_i + M_{i-1} = M_i$ give $N_i = M_i$, so $N = M$, a contradiction. *Step 2: lengths.* A strictly decreasing chain $M = K_0 \supsetneq K_1 \supsetneq \cdots \supsetneq K_j$ gives $\lambda(M) > \lambda(K_1) > \cdots > \lambda(K_j) \ge 0$ by Step 1, so $j \le \lambda(M)$. Hence every chain is finite of length at most $\lambda(M)$, every composition series has length exactly $\lambda(M)$ (it is at most $\lambda(M)$ and at least $\lambda(M)$ by minimality), and a chain can be refined by inserting terms until all quotients are simple, which must stop. *Step 3: factors.* Induction on $n = \lambda(M)$, the case $n \le 1$ being clear. Let $(M_i)$ and $(M'_i)$ be composition series, both of length $n$ by Step 2. If $M_{n-1} = M'_{n-1}$, apply the induction hypothesis to $M_{n-1}$, of length $n-1$. Otherwise $M_{n-1} + M'_{n-1} = M$ (it strictly contains $M_{n-1}$, and $M/M_{n-1}$ is simple). Put $K = M_{n-1} \cap M'_{n-1}$. By Theorem 3.8(ii), $M/M_{n-1} \simeq M'_{n-1}/K$ and $M/M'_{n-1} \simeq M_{n-1}/K$, both simple. $K$ has a composition series $(K_j)$ (Step 1), of length $n-2$, because $(K_j)$ followed by $M_{n-1}$ is a composition series of $M_{n-1}$, of length $n-1$. By induction, the factors of $(M_i)_{i\le n-1}$ are those of $(K_j)$ together with $M_{n-1}/K \simeq M/M'_{n-1}$; so the factors of $(M_i)$ are those of $(K_j)$ together with $M/M'_{n-1}$ and $M/M_{n-1}$. The same computation for $(M'_i)$ gives the factors of $(K_j)$ together with $M'_{n-1}/K \simeq M/M_{n-1}$ and $M/M'_{n-1}$: the same list. Quotients: the image in $M/N$ of a composition series of $M$ has successive quotients that are quotients of simple modules, so 0 or simple. (ii) We do not prove this; see [7, Proposition 6.8] (number not verified). We use only the direction "finite length implies Noetherian", which is Step 2. (iii) If $M$ has finite length so do $M'$ and $M''$ by (i); conversely, the preimage in $M$ of a composition series of $M''$, placed after a composition series of $M'$, is a composition series of $M$. Its length is $\operatorname{length} M' + \operatorname{length} M''$. For the long sequence, cut it into short exact sequences $0 \to Z_i \to M_i \to Z_{i+1} \to 0$ with $Z_i$ the images, and add. (iv) A simple module is $Rs$ for any $s \ne 0$, so $\simeq R/\operatorname{Ann}(s)$ with $\operatorname{Ann}(s)$ maximal (Proposition 3.4), hence $\simeq k$. A module killed by $\mathfrak{m}$ is a $k$-vector space whose $R$-submodules are its subspaces, so its length is its dimension; apply (iii) to the filtration $M \supseteq \mathfrak{m}M \supseteq \cdots \supseteq \mathfrak{m}^N M = 0$. If $R$ is Noetherian and $M$ finitely generated, the $\mathfrak{m}^i M/\mathfrak{m}^{i+1}M$ are finitely generated over $k$, which gives one direction. Conversely if $\operatorname{length} M = n$, the chain $(\mathfrak{m}^i M)_i$ has at most $n$ strict steps, so $\mathfrak{m}^j M = \mathfrak{m}^{j+1}M$ for some $j$, and $\mathfrak{m}^j M = 0$ by Nakayama. (v) By (iii) along an $R'$-composition series it suffices to treat the simple $R'$-module $k'$. It is killed by $\mathfrak{m}_R$, so by (iv) its $R$-length is $\dim_k k' = f$, which is finite as $k'$ is a quotient of the finitely generated $R$-module $R'$. □

For finite abelian groups the theorem is the statement that the prime factorisation of the order can be read off any composition series. In $\mathbb{Z}/12$ the chains $0 \subset 6\mathbb{Z}/12 \subset 2\mathbb{Z}/12 \subset \mathbb{Z}/12$ and $0 \subset 4\mathbb{Z}/12 \subset 2\mathbb{Z}/12 \subset \mathbb{Z}/12$ have factors $\mathbb{Z}/2, \mathbb{Z}/3, \mathbb{Z}/2$ and $\mathbb{Z}/3, \mathbb{Z}/2, \mathbb{Z}/2$: the same multiset, in a different order, and $\operatorname{length}_{\mathbb{Z}}(\mathbb{Z}/12) = 3$, the number of prime factors of 12 with multiplicity. Over $\mathbb{Z}_2$ the same group becomes $\mathbb{Z}/12 \otimes \mathbb{Z}_2 \simeq \mathbb{Z}/4$ (Theorem 3.58(v)), of length $2 = v_2(12)$: the

length over $\mathbb{Z}_\ell$ counts only the factors $\mathbb{Z}/\ell$. Part (v) is the reason for the factor $f$ when lengths over a ring $\mathcal{O}_\lambda$ with residue field $\mathbb{F}_{\ell^f}$ are compared with lengths over $\mathbb{Z}_\ell$: the residue field $\mathbb{F}_{\ell^f}$, simple over $\mathcal{O}_\lambda$, has a composition series of length $f$ over $\mathbb{Z}_\ell$.

**Proposition 3.66** (length over a discrete valuation ring)**.** *Let $\mathcal{O}$ be a discrete valuation ring.*

*(i)* $\operatorname{length}(\mathcal{O}/\varpi^a) = a$. *A finitely generated $\mathcal{O}$-module has finite length if and only if it is torsion, and then* $\operatorname{length}(\bigoplus_i \mathcal{O}/\varpi^{a_i}) = \sum_i a_i$.

*(ii) If $k$ is finite, $\#M = (\#k)^{\operatorname{length}_{\mathcal{O}} M}$. Over $\mathbb{Z}_\ell$, $\operatorname{length}_{\mathbb{Z}_\ell}(M) = v_\ell(\#M)$ for finite $M$; over a ring $\mathcal{O}_\lambda$ with residue field $\mathbb{F}_{\ell^f}$ (finite over $\mathbb{Z}_\ell$), $v_\ell(\#M) = f \operatorname{length}_{\mathcal{O}_\lambda}(M)$.*

*(iii) For a finite-length $\mathcal{O}$-module $M$ and $k \geq 0$, $\operatorname{length} M[\varpi^k] = \operatorname{length} M/\varpi^k M$.*

*For example,* $\operatorname{length}_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m) = v + m$ *and* $\operatorname{length}_{\mathbb{Z}_\ell}(\mathbb{Z}_\ell/n'\mathbb{Z}_\ell) = v_\ell(n')$ *for $n' \in \mathbb{Z}_\ell \setminus \{0\}$.*

*Proof.* (i) $0 \subsetneq \varpi^{a-1}\mathcal{O}/\varpi^a \subsetneq \cdots \subsetneq \mathcal{O}/\varpi^a$ has factors $\simeq k$. $\mathcal{O}$ itself has the infinite chain $\varpi^i\mathcal{O}$, so it has infinite length (Theorem 3.65(i)); now use Proposition 3.57(i) and additivity. (ii) Orders are multiplicative in exact sequences and $\#k$ is the order of each factor; for $\mathcal{O}_\lambda$ use Theorem 3.65(v). (iii) The sequence $0 \to M[\varpi^k] \to M \xrightarrow{\varpi^k} M \to M/\varpi^k M \to 0$ is exact; apply Theorem 3.65(iii). □

**Definition 3.67** (annihilators)**.** $\operatorname{Ann}_R(M) = \{r \in R : rM = 0\}$, an ideal; $\operatorname{Ann}_R(m)$ was defined in Proposition 3.4; for an ideal $\mathfrak{b}$ of a ring $\mathbb{T}$, $\operatorname{Ann}_{\mathbb{T}}(\mathfrak{b})$ is the annihilator of $\mathfrak{b}$ viewed as a $\mathbb{T}$-module. "$M$ is killed by $\eta$" means $\eta \subseteq \operatorname{Ann}(M)$.

**Proposition 3.68.** *(i)* $\operatorname{Ann}(R/\mathfrak{a}) = \mathfrak{a}$; $\operatorname{Ann}(M \oplus N) = \operatorname{Ann}(M) \cap \operatorname{Ann}(N)$; *for generators $m_1, \ldots, m_n$,* $\operatorname{Ann}(M)$ *is the kernel of $R \to M^n$, $r \mapsto (rm_i)$.*

*(ii)* $M[\mathfrak{a}] \simeq \operatorname{Hom}_R(R/\mathfrak{a}, M)$; $M[\mathfrak{a}] \cap M[\mathfrak{b}] = M[\mathfrak{a}+\mathfrak{b}]$; $M[\mathfrak{a}] + M[\mathfrak{b}] \subseteq M[\mathfrak{a}\cap\mathfrak{b}]$.

*(iii) For a ring $\mathbb{T}$ and an ideal $\mathfrak{a}$, $\mathbb{T}[\mathfrak{a}] = \operatorname{Ann}_{\mathbb{T}}(\mathfrak{a})$; if $L \simeq \mathbb{T}^r$ as $\mathbb{T}$-modules, then $L[\mathfrak{a}] \simeq \operatorname{Ann}_{\mathbb{T}}(\mathfrak{a})^r$.*

*(iv) For a flat $R \to S$, finitely generated $M$ and finitely generated $\mathfrak{a}$:* $\operatorname{Ann}_S(M \otimes S) = \operatorname{Ann}_R(M)S$ *and* $M[\mathfrak{a}] \otimes S = (M \otimes S)[\mathfrak{a}S]$.

*(v) Over a discrete valuation ring,* $\operatorname{Ann}(\bigoplus_{i=1}^s \mathcal{O}/\varpi^{a_i}) = \varpi^{\max_i a_i}\mathcal{O}$ *(for $s \geq 1$), and* $\operatorname{Ann}(M) = 0$ *if $M$ has a free summand.*

*Proof.* (i) $r$ kills $R/\mathfrak{a}$ iff it kills the generator $1 + \mathfrak{a}$, iff $r \in \mathfrak{a}$. An element kills $M \oplus N$ iff it kills both summands. It kills $M$ iff it kills each generator $m_i$, iff it lies in the kernel of $r \mapsto (rm_i)_i$. (ii) The first statement is Proposition 3.25(i). An element is killed by $\mathfrak{a}$ and by $\mathfrak{b}$ iff it is killed by every $a+b$, that is, by $\mathfrak{a}+\mathfrak{b}$. If $m$ is killed by $\mathfrak{a}$ and $m'$ by $\mathfrak{b}$, then $m+m'$ is killed by $\mathfrak{a}\cap\mathfrak{b}$, which lies in both. (Over a principal ideal domain the last inclusion is an equality for finitely generated $M$, as one checks on cyclic summands; in general it can be strict: for $R = k[x,y]$, $M = R/(x,y)^2$ with basis $1, x, y$, $\mathfrak{a} = (x)$ and $\mathfrak{b} = (y)$, one has $M[\mathfrak{a}] = M[\mathfrak{b}] = kx + ky$, while $\mathfrak{a}\cap\mathfrak{b} = (xy)$ kills all of $M$.) (iii) $\mathbb{T}[\mathfrak{a}]$ and $\operatorname{Ann}_{\mathbb{T}}(\mathfrak{a})$ are both the set of $t \in \mathbb{T}$ with $at = 0$ for all $a \in \mathfrak{a}$. For $L = \mathbb{T}^r$, an element $(t_1, \ldots, t_r)$ is killed by $\mathfrak{a}$ iff each coordinate is, so $L[\mathfrak{a}] = \operatorname{Ann}_{\mathbb{T}}(\mathfrak{a})^r$; an isomorphism $L \simeq \mathbb{T}^r$ of $\mathbb{T}$-modules carries $L[\mathfrak{a}]$ to $\mathbb{T}^r[\mathfrak{a}]$. (iv) is Proposition 3.29(iii). (v) $\varpi^a$ kills $\mathcal{O}/\varpi^b$ iff $a \geq b$, so it kills the sum iff $a \geq \max_i b_i$; and $\mathcal{O}$ is torsion-free. □

**Definition 3.69** (Fitting ideals)**.** Let $M$ be a finitely generated $R$-module with generators $x = (x_1, \dots, x_n)$, and let $\mathrm{Rel}(x) = \{r \in R^n : \sum r_i x_i = 0\}$ be the module of relations (row vectors). The (zeroth) *Fitting ideal* $\mathrm{Fitt}_R(M)$ is the ideal generated by the determinants $\det(r^{(1)}, \dots, r^{(n)})$ of all $n$-tuples of relations (the $n \times n$ matrices with these rows). If $R^m \xrightarrow{A} R^n \to M \to 0$ is a presentation (Convention 3.1(g)), $\mathrm{Fitt}_R(M)$ is the ideal generated by the $n \times n$ minors of $A$ (and is 0 if $m < n$); see Lemma 3.70. More generally $\mathrm{Fitt}_i(M)$ is generated by the $(n-i) \times (n-i)$ minors; we use only $\mathrm{Fitt} = \mathrm{Fitt}_0$.

The Fitting ideal is an ideal-valued replacement for the order of a finite abelian group. For a finite abelian group with a square presentation matrix $A$, the order is $|\det A|$ (Theorem 3.50), and $\mathrm{Fitt}_{\mathbb{Z}}(M) = (\det A) = \#M \cdot \mathbb{Z}$. Over a general ring there is no order to take, and a presentation need not be square, so one takes all maximal minors at once. The definition through determinants of $n$-tuples of relations, rather than through a fixed matrix, is what makes independence of the choices provable. A small check: $\mathbb{Z}/6$ with the single generator 1 has relations $6\mathbb{Z}$, so $\mathrm{Fitt} = (6)$. With the two generators $x_1 = 2$, $x_2 = 3$, a pair $(a, b)$ is a relation iff $2a + 3b \equiv 0 \pmod 6$, iff $3 \mid a$ and $2 \mid b$; so the relations are generated by $(3, 0)$ and $(0, 2)$, and the ideal of $2 \times 2$ minors is again $(6)$, as Lemma 3.70 guarantees.

**Lemma 3.70** (the Fitting ideal is well defined)**.** *(i) If the rows of $A \in \mathrm{M}_{m\times n}(R)$ generate $\mathrm{Rel}(x)$, the ideal of $n \times n$ minors of $A$ equals the ideal $F(x)$ generated by the determinants of $n$-tuples of relations.*

*(ii) $F(x)$ does not depend on the generating family $x$.*

*Proof.* (i) Every $n \times n$ minor of $A$ is the determinant of $n$ relations. Conversely, if $r^{(i)} = \sum_j c_{ij} a_j$ with $a_j$ the rows of $A$, then by multilinearity

$$\det(r^{(1)}, \dots, r^{(n)}) = \sum_{j_1, \dots, j_n} c_{1j_1} \cdots c_{nj_n} \det(a_{j_1}, \dots, a_{j_n}),$$

and each term is 0 (two equal rows) or $\pm$ a minor of $A$. (ii) Permuting the generators permutes the columns of the relation matrices and changes determinants by a sign. Next let $x' = (x_1, \dots, x_n, y)$ with $y = \sum_i a_i x_i$. If the rows of $A$ generate $\mathrm{Rel}(x)$, then $\mathrm{Rel}(x')$ is generated by the rows $(r, 0)$, $r$ a row of $A$, and the row $(-a_1, \dots, -a_n, 1)$: given $(s, t) \in \mathrm{Rel}(x')$, subtracting $t$ times the last row leaves $(s + ta, 0)$ with $s + ta \in \mathrm{Rel}(x)$. An $(n+1) \times (n+1)$ minor of this matrix that omits the last row has a zero last column; one that contains it equals, expanding along the last column, $\pm$ an $n \times n$ minor of $A$, and every such minor occurs. By (i), $F(x') = F(x)$. Finally, for two generating families $x$ and $z = (z_1, \dots, z_p)$, add the $z_j$ to $x$ one at a time (each is a combination of the $x_i$, hence of the current family), and the $x_i$ to $z$ likewise; both processes end at the same family up to order. So $F(x) = F(x \cup z) = F(z)$. □

**Proposition 3.71** (properties of Fitting ideals)**.** *Let $M$, $N$ be finitely generated.*

*(i)* $\mathrm{Fitt}(R/\mathfrak{a}) = \mathfrak{a}$; $\mathrm{Fitt}(0) = R$; $\mathrm{Fitt}(R^n) = 0$ *for* $n \geq 1$.

*(ii)* $\mathrm{Fitt}(M \oplus N) = \mathrm{Fitt}(M)\,\mathrm{Fitt}(N)$.

*(iii) If* $M \twoheadrightarrow N$ *then* $\mathrm{Fitt}(M) \subseteq \mathrm{Fitt}(N)$.

*(iv)* $\mathrm{Fitt}(M) \subseteq \mathrm{Ann}(M)$, *and* $\mathrm{Ann}(M)^n \subseteq \mathrm{Fitt}(M)$ *if $M$ is generated by $n$ elements.*

*(v)* $\mathrm{Fitt}_S(S\otimes_R M)=\mathrm{Fitt}_R(M)S$ *for every ring homomorphism* $R\to S$ *(no flatness is needed).*

*(vi) For an exact sequence* $0\to M'\to M\to M''\to 0$ *with* $M'$ *finitely generated (automatic if* $R$ *is Noetherian, Proposition 3.31(ii)),* $\mathrm{Fitt}(M')\,\mathrm{Fitt}(M'')\subseteq\mathrm{Fitt}(M)$*. Equality can fail: over* $R=\mathbb{Z}[x]$*, with* $M=R/(4,x)\simeq\mathbb{Z}/4$*,* $M'=2M\simeq R/(2,x)$ *and* $M''=M/2M\simeq R/(2,x)$*, one has* $\mathrm{Fitt}(M)=(4,x)$ *but* $\mathrm{Fitt}(M')\,\mathrm{Fitt}(M'')=(2,x)^2=(4,2x,x^2)$*.*

*(vii) Over a domain,* $\mathrm{Fitt}(M)\neq 0$ *if and only if* $M$ *is torsion.*

*Proof.* (i) $R/\mathfrak{a}$ has the generator 1 with relations $\mathfrak{a}$, and its $1\times 1$ "determinants" generate $\mathfrak{a}$. For $M=0$ use the generator 0 with the relation (1). $R^n$ with its basis has no nonzero relation. (ii) Use generators $x$ of $M$ and $y$ of $N$; $\mathrm{Rel}(x,y)=\mathrm{Rel}(x)\oplus\mathrm{Rel}(y)$ is generated by the rows of $\mathrm{diag}(A,B)$, with $A$ of size $m\times n$ and $B$ of size $m'\times n'$. Consider $n+n'$ of these rows, $j$ from the first block and $j'$ from the second, so $j+j'=n+n'$. If $j>n$, the $j$ rows from the first block are zero outside the first $n$ columns, and the Laplace expansion along these $j$ rows involves only $j\times j$ minors of them, each of which uses at least $j-n\geq 1$ of the last $n'$ columns, where these rows vanish; so every such minor, and the determinant, is 0. Likewise if $j'>n'$; and since $j+j'=n+n'$, the only remaining case is $j=n$, $j'=n'$. If $j=n$ and $j'=n'$ the determinant is the product of an $n\times n$ minor of $A$ and an $n'\times n'$ minor of $B$. So the ideal is $\mathrm{Fitt}(M)\,\mathrm{Fitt}(N)$. (iii) Generators of $M$ map to generators of $N$, and relations among them in $M$ remain relations in $N$. (iv) Let $B$ be an $n\times n$ matrix of relations, so $B$ kills the column $x$ of generators; multiplying by the adjugate gives $\det(B)x_i=0$ for all $i$. For $a_1,\dots,a_n\in\mathrm{Ann}(M)$ the rows $a_ie_i$ are relations, with determinant $a_1\cdots a_n$. (v) Tensoring the presentation gives a presentation of $S\otimes M$ with the image of the same matrix (Proposition 3.28). (vi) Let $x'$ generate $M'$ (viewed in $M$) and let $y\in M^{n''}$ lift generators of $M''$. For $r'\in\mathrm{Rel}(x')$, $(r',0)$ is a relation of $(x',y)$; for $r''\in\mathrm{Rel}(\bar{y})$, $\sum r''_jy_j\in M'$, say $=\sum c_ix'_i$, and $(-c,r'')$ is a relation. The determinant of $n'$ relations of the first kind followed by $n''$ of the second is block triangular, equal to $\det(r'\text{s})\det(r''\text{s})$. In the example: $M'=2\mathbb{Z}/4$ is killed by 2 and $x$, so it is $R/(2,x)$, as is $M/2M$; $\mathrm{Fitt}(R/(4,x))=(4,x)$ by (i); $(2,x)^2=(4,2x,x^2)$; and $x\notin(4,2x,x^2)$ (every element of that ideal has the coefficient of $x$ even). (vii) If $M$ is torsion, $\mathrm{Ann}(M)\neq 0$ (the product of nonzero annihilators of the generators), so $\mathrm{Fitt}(M)\supseteq\mathrm{Ann}(M)^n\neq 0$. If not, $M\otimes K\simeq K^r$ with $r\geq 1$ (Proposition 3.46(i)), and $\mathrm{Fitt}(M)K=\mathrm{Fitt}_K(K^r)=0$ by (i) and (v). □

**Theorem 3.72** (Fitting ideals and length)**.** *(i) Over a principal ideal domain, the module* $M=R^r\oplus\bigoplus_i R/(d_i)$ *has* $\mathrm{Fitt}(M)=0$ *if* $r>0$*, and* $\mathrm{Fitt}(M)=(\prod_i d_i)$ *if* $r=0$*.*

*(ii) Over a discrete valuation ring* $\mathcal{O}$*, every finitely generated torsion module* $M$ *has* $\mathrm{Fitt}_{\mathcal{O}}(M)=\varpi^{\mathrm{length}(M)}\mathcal{O}$*, so* $v(\mathrm{Fitt}\,M)=\mathrm{length}\,M$*. Over* $\mathbb{Z}_\ell$*:* $\mathrm{Fitt}_{\mathbb{Z}_\ell}(M)=\#M\cdot\mathbb{Z}_\ell=\ell^{v_\ell(\#M)}\mathbb{Z}_\ell$ *for finite* $M$*.*

*(iii) Consequently, over a discrete valuation ring Fitting ideals of finite-length modules are multiplicative in short exact sequences, and* $\mathrm{Fitt}(N)\supseteq\mathrm{Fitt}(M)$ *for every submodule or quotient* $N$ *of* $M$*.*

*(iv) The Fitting ideal is not the annihilator:* $(\mathbb{Z}/\ell)^2$ *and* $\mathbb{Z}/\ell^2$ *both have Fitting ideal* $\ell^2\mathbb{Z}_\ell$*, with annihilators* $\ell\mathbb{Z}_\ell$ *and* $\ell^2\mathbb{Z}_\ell$*.*

*Proof.* (i) By Proposition 3.71(i), (ii). (ii) By (i), $\mathrm{Fitt}(\bigoplus\mathcal{O}/\varpi^{a_i})=\varpi^{\sum a_i}\mathcal{O}$, and $\sum a_i=\mathrm{length}\,M$ (Proposition 3.66). Over $\mathbb{Z}_\ell$, $\#M=\ell^{\mathrm{length}\,M}$. (iii) For $0\to M'\to M\to M''\to 0$ of finite length, additivity of length (Theorem 3.65(iii)) gives $\varpi^{\mathrm{length}\,M}=\varpi^{\mathrm{length}\,M'}\varpi^{\mathrm{length}\,M''}$, that is,

$\mathrm{Fitt}(M) = \mathrm{Fitt}(M')\,\mathrm{Fitt}(M'')$ by (ii). A submodule or quotient $N$ of $M$ has length $N \le$ length $M$, again by additivity, so $\mathrm{Fitt}(N) = \varpi^{\text{length } N}\mathcal{O} \supseteq \varpi^{\text{length } M}\mathcal{O} = \mathrm{Fitt}(M)$. (iv) By (ii), $\mathrm{Fitt} = \ell^2\mathbb{Z}_\ell$ for both modules, as both have length 2; the annihilators are given by Proposition 3.68(v). □

*Warning* 3.73. Over $\mathbb{Z}_\ell$ the Fitting ideal of a finite module sees its whole order, while the annihilator sees only its exponent. The Tamagawa factor is a Fitting ideal; reading it as an annihilator would replace $v_\ell(n) + \min(\dots)$ in Lemma lem:tamfactor by $\max(v_\ell(n), \min(\dots)) = v_\ell(n)$. Over a general ring the Fitting ideal is not determined by lengths either (Proposition 3.71(vi)).

**Example 3.74.** Over $\mathbb{Z}$ (relations as rows), and then over $\mathbb{Z}_3$ and $\mathbb{Z}_\ell$; all entries were computed by hand and the Smith forms checked with PARI (Example 3.42).

| relations | module | Fitt | Ann |
|---|---|---|---|
| rows $(2,4)$, $(6,8)$ | $\mathbb{Z}/2 \oplus \mathbb{Z}/4$ | $(8)$ | $(4)$ |
| rows $(2,4,4)$, $(-6,6,12)$, $(10,-4,-16)$ | $\mathbb{Z}/2 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}/12$ | $(144)$ | $(12)$ |
| rows $(3,6)$, $(9,3)$ | $\mathbb{Z}/3 \oplus \mathbb{Z}/15$ | $(45)$ | $(15)$ |
| the same over $\mathbb{Z}_3$ | $\mathbb{Z}/3 \oplus \mathbb{Z}/3$ | $9\mathbb{Z}_3$ | $3\mathbb{Z}_3$ |
| rows $(6,0)$, $(0,4)$, $(2,2)$ | $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ | $(4)$ | $(2)$ |
| one generator, rows $(2)$, $(4)$ | $\mathbb{Z}/2$ | $(2)$ | $(2)$ |
| over $\mathbb{Z}[x]$: one generator, rows $(2)$, $(x)$ | $\mathbb{Z}[x]/(2,x)$ | $(2,x)$ | $(2,x)$ |
| over $\mathbb{Z}[x]$: the square of the previous | $(\mathbb{Z}[x]/(2,x))^2$ | $(4,2x,x^2)$ | $(2,x)$ |
| over $\mathbb{Z}_\ell$: the paper's module | $\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m$ | $\ell^{v+m}\mathbb{Z}_\ell$ | $\ell^{\max(v,m)}\mathbb{Z}_\ell$ |
| $\ell = 3$, $v = 2$, $m = 1$ | $\mathbb{Z}/9 \oplus \mathbb{Z}/3$ | $27\mathbb{Z}_3$ | $9\mathbb{Z}_3$ |

In the fifth row the three $2 \times 2$ minors are 24, 12, $-8$, with gcd 4, although there are more relations than generators. In the eighth row the four relation rows $(2,0)$, $(x,0)$, $(0,2)$, $(0,x)$ have $2 \times 2$ minors $0, 4, 2x, 2x, x^2, 0$, and the Fitting ideal $(2,x)^2$ is strictly smaller than the annihilator, as Proposition 3.71(iv) allows.

*In the paper* 3.75. Section 1.2 defines $\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}})$, the Fitting ideal *of the invariants* (Appendix B, convention B-C10). This is a finite $\mathbb{Z}_\ell$-module, so by Theorem 3.72(ii) $v_\ell(\mathrm{Tam}_p(M))$ is its length. In Lemma lem:tamfactor the invariants are the $x\bar{e} + y\bar{h}$ in $(\mathbb{Z}/\ell^{v_\ell(n)})\bar{e} \oplus (\mathbb{Z}/\ell^{v_\ell(n)})\bar{h}$ with $(q-1)y \equiv 0$ and $c'y \equiv 0$, a module isomorphic to $\mathbb{Z}/\ell^{v_\ell(n)} \oplus \mathbb{Z}/\ell^{\min(v_\ell(n), v_\ell(q-1), v_\ell(c'))}$, of length $v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c'))$: this is the count in the paper's proof, where $v_\ell(c')$ may be replaced by $v_\ell(c(u))$, and the Fitting ideal of $\mathbb{Z}/\ell^v \oplus \mathbb{Z}/\ell^m$ is $\ell^{v+m}\mathbb{Z}_\ell$ by the last rows of Example 3.74. In Lemma lem:fpr the identity of Fitting ideals is an identity of lengths, $\text{length}\, H^1_f(\mathbb{Q}_p, T^D) = v_\ell(L_p(A_f,1)^{-1}) + v_\ell(\mathrm{Tam}_p(T^D))$, which is how the proof of Theorem thm:main uses it. The definitions of $\mathrm{Tam}_p$ and $H^1_f$ are Chapter 18.

## 3.8 Pontryagin duality

Lemma lem:dfg21 is an exact sequence

$$0 \to H^1_f(\mathbb{Q}, W) \to H^1_\Sigma(\mathbb{Q}, W) \to \bigoplus_{p \in \Sigma \cup \{\infty\}} H^1_f(\mathbb{Q}_p, T^D)^* \to H^0(\mathbb{Q}, W^D)^* \to 0$$

in which $^*$ is the Pontryagin dual, and the proof of Theorem thm:main adds lengths along it. This needs two facts proved here: Pontryagin duality is exact, and it preserves the length of finite

modules. The coefficient module $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ is the basic example of a discrete module dual to a finitely generated one (Proposition 3.78), and Lemma lem:h0 identifies $H^0(\mathbb{Q}, W^D)$ by its $\ell^k$-torsion, using $W[\ell^k] = T/\ell^k T$ and the cyclicity criterion Corollary 3.79.

**Theorem 3.76** (Pontryagin duality for finite abelian groups)**.** *For a finite abelian group $A$ let $A^* = \mathrm{Hom}(A, \mathbb{Q}/\mathbb{Z})$.*

*(i)* $(\mathbb{Z}/n)^* \simeq \mathbb{Z}/n$, *and* $A^* \simeq A$ *(non-canonically).*

*(ii) The evaluation map $A \to A^{**}$, $a \mapsto (\xi \mapsto \xi(a))$, is an isomorphism.*

*(iii) $A \mapsto A^*$ is an exact contravariant functor, and $\#A^* = \#A$.*

*(iv) For a subgroup $B \subseteq A$ let $B^\perp = \{\xi \in A^* : \xi(B) = 0\}$. Then $B^\perp \simeq (A/B)^*$ and $B^* \simeq A^*/B^\perp$, and $B \mapsto B^\perp$ is an inclusion-reversing bijection between the subgroups of $A$ and those of $A^*$.*

*(v) If $A$ is $\ell$-primary, then $A^* = \mathrm{Hom}_{\mathbb{Z}_\ell}(A, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ and $\mathrm{length}_{\mathbb{Z}_\ell} A^* = \mathrm{length}_{\mathbb{Z}_\ell} A$; dualising an exact sequence of finite $\mathbb{Z}_\ell$-modules gives an exact sequence with the arrows reversed.*

*(vi) Via $x \mapsto e^{2\pi i x}$, $A^*$ is the group of characters $A \to \mathbb{C}^\times$ (Chapter 2); in the language of Riley, Hobson and Bence, Pontryagin duality is the discrete Fourier transform on $A$.*

*Proof.* (i) $\mathrm{Hom}(\mathbb{Z}/n, \mathbb{Q}/\mathbb{Z}) \simeq (\mathbb{Q}/\mathbb{Z})[n] = \frac{1}{n}\mathbb{Z}/\mathbb{Z} \simeq \mathbb{Z}/n$ by Proposition 3.25(i); and $A$ is a finite direct sum of cyclic groups (Chapter 1), and Hom turns finite sums into sums. (ii) Injectivity: for $a \neq 0$ of order $d$, the map $\langle a \rangle \to \mathbb{Q}/\mathbb{Z}$, $a \mapsto 1/d$, extends to $A$ because $\mathbb{Q}/\mathbb{Z}$ is injective (Proposition 3.60(iv)); so some $\xi$ has $\xi(a) \neq 0$. Both sides have order $\#A$ by (i). (iii) Exactness is the injectivity of $\mathbb{Q}/\mathbb{Z}$: $\mathrm{Hom}(-, \mathbb{Q}/\mathbb{Z})$ is left exact by Proposition 3.25(ii), and surjectivity of $A^* \to B^*$ for $B \subseteq A$ is the statement that characters of $B$ extend to $A$ (Lemma 3.59). $\#A^* = \#A$ is (i). (iv) The exact sequence $0 \to B \to A \to A/B \to 0$ dualises to $0 \to (A/B)^* \to A^* \to B^* \to 0$, with the image of the first map equal to $B^\perp$. Hence $\#B^\perp = \#A/\#B$. Under $A = A^{**}$, $B \subseteq (B^\perp)^\perp$ and $\#(B^\perp)^\perp = \#A^*/\#B^\perp = \#B$, so $B = (B^\perp)^\perp$; and symmetrically for subgroups of $A^*$. (v) $A$ is killed by some $\ell^k$, so every homomorphism lands in $(\mathbb{Q}/\mathbb{Z})[\ell^k] \subseteq (\mathbb{Q}/\mathbb{Z})[\ell^\infty] = \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (Proposition 3.60(i)); it is $\mathbb{Z}_\ell$-linear because $\mathbb{Z}_\ell$ acts on a module killed by $\ell^k$ through $\mathbb{Z}/\ell^k$. Lengths agree because orders do (Proposition 3.66(ii)). (vi) $x \mapsto e^{2\pi i x}$ is an isomorphism from $\mathbb{Q}/\mathbb{Z}$ onto the roots of unity in $\mathbb{C}^\times$. □

For $A = \mathbb{Z}/n$ the duality is explicit: the character $\xi_a$ with $\xi_a(1) = a/n + \mathbb{Z}$ corresponds to $a \in \mathbb{Z}/n$, and in the language of (vi) it is $x \mapsto e^{2\pi i a x/n}$, the $a$-th Fourier mode. For the subgroup $B = d\mathbb{Z}/n\mathbb{Z}$ (with $d \mid n$), $B^\perp$ consists of the $\xi_a$ with $ad/n \in \mathbb{Z}$, that is, $a \in (n/d)\mathbb{Z}/n\mathbb{Z}$, a group of order $d = \#(A/B)$, as (iv) predicts. The isomorphism $A \simeq A^*$ depends on the choice of the generator, whereas the double duality $A \simeq A^{**}$ of (ii) is canonical and therefore compatible with any group action on $A$; only the canonical statements survive when $A$ carries a Galois action.

**Theorem 3.77** (Pontryagin duality for $\mathbb{Z}_\ell$-modules)**.** *Let $\mathcal{C}$ be the category of finitely generated $\mathbb{Z}_\ell$-modules and $\mathcal{D}$ that of $\ell$-primary torsion $\mathbb{Z}_\ell$-modules $W$ with $W[\ell]$ finite (*cofinitely generated *modules), with the discrete topology. For $X$ in either, put $X^* = \mathrm{Hom}_{\mathbb{Z}_\ell}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$. (For $M \in \mathcal{C}$ every such homomorphism is continuous for the $\ell$-adic topology, since its image is a finitely generated, hence finite, submodule of $\mathbb{Q}_\ell/\mathbb{Z}_\ell$, killed by some $\ell^k$; so the homomorphism kills the open submodule $\ell^k M$.) For $W \in \mathcal{D}$, $W^* = \varprojlim_k (W[\ell^k])^*$ along the restriction maps, and we give*

*it this inverse-limit topology, which is profinite; by (ii) below $W^*$ is finitely generated, and the topology is its $\ell$-adic one, because $\ell^k W^*$ kills $W[\ell^k]$ and a continuous bijection from a compact space onto a Hausdorff space is a homeomorphism.*

*(i)* $\mathbb{Z}_\ell^* = \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^* = \mathbb{Z}_\ell$, *and* $(\mathbb{Z}/\ell^a)^* \simeq \mathbb{Z}/\ell^a$.

*(ii)* $X \mapsto X^*$ *is exact, maps* $\mathcal{C}$ *to* $\mathcal{D}$ *and* $\mathcal{D}$ *to* $\mathcal{C}$, *and the evaluation maps* $M \to M^{**}$, $W \to W^{**}$ *are isomorphisms.*

*(iii) Every* $W \in \mathcal{D}$ *is isomorphic to* $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r \oplus F$ *with* $F$ *finite;* $r$, *the* corank *of* $W$, *equals* $\operatorname{rank}_{\mathbb{Z}_\ell} W^*$. $W$ *is finite iff* $r = 0$, *and divisible iff* $F = 0$.

*(iv) Lengths of finite modules are preserved:* $\operatorname{length} X^* = \operatorname{length} X$.

*(v) More generally,* $X \mapsto \operatorname{Hom}_{\mathrm{cont}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ *is an exact anti-equivalence between compact and discrete* $\ell$*-primary* $\mathbb{Z}_\ell$*-modules (stated; see [143, Chapter I, Section 1]).*

*Proof.* (iii) first. By induction $W[\ell^{k+1}]$ is finite: multiplication by $\ell$ maps it to $W[\ell^k]$ with kernel $W[\ell]$. Let $D = \bigcap_k \ell^k W$. *$D$ is divisible.* Let $x \in D$; for each $k$ write $x = \ell^{k+1} w_k$ and put $y_k = \ell^k w_k$, so $\ell y_k = x$ and $y_k \in \ell^k W$. The $y_k$ all lie in the finite set $y_0 + W[\ell]$, so one value $y$ occurs for infinitely many $k$; then $y \in \ell^k W$ for all $k$, so $y \in D$ and $\ell y = x$. Divisibility by units is automatic. $D \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$, $r = \dim_{\mathbb{F}_\ell} D[\ell]$. Choose an isomorphism $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r[\ell] \simeq D[\ell]$ and extend it to $\phi\colon (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r \to D$ using the injectivity of $D$ (Lemma 3.59). The kernel of $\phi$ has no $\ell$-torsion, and an $\ell$-primary module without $\ell$-torsion is 0; so $\phi$ is injective. Its image is divisible, hence injective, hence a direct summand: $D = \phi((\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r) \oplus C$, where $C[\ell] = 0$ because $D[\ell]$ lies in the image; so $C = 0$. *$W = D \oplus F$ with $F$ finite.* $D$ is injective, so $W = D \oplus F$. Then $\ell^k W = D \oplus \ell^k F$, so $\bigcap_k \ell^k F = 0$. The subgroups $\ell^k F \cap F[\ell]$ of the finite group $F[\ell]$ decrease and have intersection 0, so $\ell^k F \cap F[\ell] = 0$ for some $k$; then $\ell^k F$ has no $\ell$-torsion, so $\ell^k F = 0$, and $F = F[\ell^k]$ is finite. $W$ is finite iff $r = 0$; it is divisible iff $F = 0$ (a finite divisible $\ell$-primary module is 0). (i) $\operatorname{Hom}(\mathbb{Z}_\ell, X) = X$; $\operatorname{End}(\mathbb{Q}_\ell/\mathbb{Z}_\ell) = \mathbb{Z}_\ell$ and $\operatorname{Hom}(\mathbb{Z}/\ell^a, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \simeq \mathbb{Z}/\ell^a$ (Proposition 3.60(iii)). (ii) Exactness: $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is injective (Proposition 3.60(iv)). By Proposition 3.57(i) and (iii) above, every object of $\mathcal{C}$ or $\mathcal{D}$ is a finite direct sum of copies of $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ and $\mathbb{Z}/\ell^a$, and by (i) duality exchanges $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ and preserves $\mathbb{Z}/\ell^a$. The evaluation map is natural and compatible with finite sums, so it suffices to check it on these three modules: for $\mathbb{Z}_\ell$, identify $\mathbb{Z}_\ell^* = \mathbb{Q}_\ell/\mathbb{Z}_\ell$ by $\xi \mapsto \xi(1)$; then the evaluation at $a \in \mathbb{Z}_\ell$ is multiplication by $a$ on $\mathbb{Q}_\ell/\mathbb{Z}_\ell$, which is the element $a$ of $\operatorname{End}(\mathbb{Q}_\ell/\mathbb{Z}_\ell) = \mathbb{Z}_\ell$. For $\mathbb{Q}_\ell/\mathbb{Z}_\ell$, identify $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^* = \mathbb{Z}_\ell$; the evaluation at $x$ is $a \mapsto ax$, which is the element $x$ of $\operatorname{Hom}(\mathbb{Z}_\ell, \mathbb{Q}_\ell/\mathbb{Z}_\ell) = \mathbb{Q}_\ell/\mathbb{Z}_\ell$. For $\mathbb{Z}/\ell^a$ it is Theorem 3.76(ii). Finally $\operatorname{rank}_{\mathbb{Z}_\ell} W^* = r$ by the decomposition. (iv) is Theorem 3.76(v). (v) We do not prove this. □

**Proposition 3.78** (the duality between $T$ and $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$)**.** *Let $T$ be a free $\mathbb{Z}_\ell$-module of rank $r$, $V = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ and $W = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell/\mathbb{Z}_\ell$.*

*(i)* $0 \to T \to V \to W \to 0$ *is exact, and* $W \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$ *is divisible of corank $r$.*

*(ii)* $W[\ell^k] \simeq T/\ell^k T$, *by* $t \otimes \ell^{-k} \mapsto t \bmod \ell^k$. *Under these isomorphisms the inclusion* $W[\ell^k] \subseteq W[\ell^{k+1}]$ *becomes multiplication by* $\ell\colon T/\ell^k \to T/\ell^{k+1}$, *and multiplication by* $\ell\colon W[\ell^{k+1}] \to W[\ell^k]$ *becomes reduction* $T/\ell^{k+1} \to T/\ell^k$.

*(iii)* $T \simeq \varprojlim_k W[\ell^k]$, *the transition maps being multiplication by $\ell$: $T$ is the* Tate module *of $W$.*

*(iv)* $W^* = \mathrm{Hom}(T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathrm{End}(\mathbb{Q}_\ell/\mathbb{Z}_\ell)) = \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathbb{Z}_\ell) = T^\vee$, *and* $T^* \simeq T^\vee \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$.

*(v) Everything is functorial in* $T$*: a group acting* $\mathbb{Z}_\ell$*-linearly on* $T$ *acts on* $V$, $W$, $W[\ell^k] = T/\ell^k T$ *compatibly with the maps above, and on duals by* $(g\xi)(x) = \xi(g^{-1}x)$.

*(vi) The same holds over any complete discrete valuation ring* $\mathcal{O}$ *with fraction field* $K$*, with* $K/\mathcal{O}$ *in place of* $\mathbb{Q}_\ell/\mathbb{Z}_\ell$.

*Proof.* (i) Tensor the exact sequence $0 \to \mathbb{Z}_\ell \to \mathbb{Q}_\ell \to \mathbb{Q}_\ell/\mathbb{Z}_\ell \to 0$ with the free, hence flat, module $T$. (ii) Apply Corollary 3.16(a) with $n = \ell^k$: $V[\ell^k] = 0$ and $V/\ell^k V = 0$, so the connecting map $W[\ell^k] \to T/\ell^k T$ is an isomorphism. It sends the class of $v \in V$ (with $\ell^k v \in T$) to $\ell^k v \bmod \ell^k T$, which on $t \otimes \ell^{-k}$ is $t \bmod \ell^k$. If $w$ is the class of $v$ with $\ell^k v = t$, then in $W[\ell^{k+1}]$ it corresponds to $\ell^{k+1}v = \ell t$; and $\ell w$, the class of $\ell v$, corresponds in $W[\ell^k]$ to $\ell^k(\ell v) = \ell^{k+1}v$ reduced modulo $\ell^k$. (iii) By (ii) the system $(W[\ell^k], \cdot\ell)$ is $(T/\ell^k T, \text{reduction})$, whose limit is $T$ (Theorem 3.58(vi)). (iv) The adjunction (Theorem 3.26(iii)) and Proposition 3.60(iii). For the second, the canonical map $T^\vee \otimes X \to \mathrm{Hom}(T, X)$, $\varphi \otimes x \mapsto (t \mapsto \varphi(t)x)$, is an isomorphism for $T$ free of finite rank (check on a basis). (v) All maps above are natural in $T$. (vi) The proofs use only that $\mathcal{O}$ is a discrete valuation ring, that $K/\mathcal{O}$ is divisible (hence injective, Lemma 3.59), and, for (iii) and (iv), completeness of $\mathcal{O}$: $\mathrm{End}_{\mathcal{O}}(K/\mathcal{O}) = \varprojlim \mathcal{O}/\varpi^k = \mathcal{O}$ as in Proposition 3.60(iii). □

**Corollary 3.79.** *An* $\ell$*-primary torsion* $\mathbb{Z}_\ell$*-module* $A$ *with* $\#A[\ell] \le \ell$ *is* $0$, $\mathbb{Z}/\ell^h$ *for some* $h \ge 1$, *or* $\mathbb{Q}_\ell/\mathbb{Z}_\ell$*. If it is finite, its length is the largest* $k$ *such that* $A$ *has an element of exact order* $\ell^k$.

*Proof.* $A \in \mathcal{D}$, so $A \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r \oplus F$ by Theorem 3.77(iii), and $\dim_{\mathbb{F}_\ell} A[\ell] = r + \dim F[\ell] \le 1$. If $r = 1$ then $F[\ell] = 0$, so $F = 0$. If $r = 0$, $F$ is finite with $\dim F[\ell] \le 1$, hence cyclic (Proposition 3.57(v)), $F \simeq \mathbb{Z}/\ell^h$, of length $h$, and an element of exact order $\ell^k$ exists iff $k \le h$. □

*In the paper* 3.80. In the proof of Theorem thm:main the paper takes lengths in the sequence of Lemma lem:dfg21. All four terms are finite there: $H^1_\Sigma(\mathbb{Q}, W)$ by Theorem thm:dfg37, the local terms $H^1_f(\mathbb{Q}_p, T^D)$ as torsion submodules of finitely generated $\mathbb{Z}_\ell$-modules (the paragraph after Lemma lem:dfg21), and $H^0(\mathbb{Q}, W^D)$ by Lemma lem:h0. By Theorem 3.77(iv) the duals have the same lengths, and by Theorem 3.65(iii)

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma \cup \{\infty\}} \operatorname{length} H^1_f(\mathbb{Q}_p, T^D) - \operatorname{length} H^0(\mathbb{Q}, W^D),$$

which is the paper's display once $H^0(\mathbb{Q}, W^D) = 0$ and $H^1_f(\mathbb{R}, T^D) = 0$ (Paper link 3.63); Remark rem:ns keeps the term $\operatorname{length} H^0(\mathbb{Q}, W^D) = h_\ell(E)$. In Lemma lem:h0, $W^D = \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, so $W^D[\ell^k] \simeq \mathrm{ad}^0 T(1)/\ell^k$ by Proposition 3.78(ii) (functorially, hence as Galois modules), which is the paper's "$\ell^k$-torsion of $H^0(\mathbb{Q}, W^D)$ is $H^0(\mathbb{Q}, \mathrm{ad}^0(T/\ell^k)(1))$". The $\ell$-torsion of $H^0(\mathbb{Q}, W^D)$ has $\mathbb{F}_\ell$-dimension at most one, so Corollary 3.79 shows that $H^0(\mathbb{Q}, W^D)$ is 0, cyclic of order $\ell^h$, or $\mathbb{Q}_\ell/\mathbb{Z}_\ell$; the last is excluded by the determinant argument of the lemma, and $h_\ell(E)$ is the largest $k$ for which an element of exact order $\ell^k$ exists.

Pontryagin duality pairs a module with its characters, with values in the torsion module $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. The next section studies the other duality of the paper, $L \mapsto L^\vee = \mathrm{Hom}(L, \mathcal{O})$ for lattices, realised by a bilinear pairing with values in $\mathcal{O}$; the two are linked by Proposition 3.78(iv), $W^* \simeq T^\vee$, and by Proposition 3.51, where the cokernel of $M^\vee \hookrightarrow L^\vee$ is a Pontryagin dual.

## 3.9 Pairings: Gram determinants, saturated sublattices, alternating pairings, adjoints

The heart of Lemma lem:ddt417 is a statement about a perfect pairing on a lattice $L$ and a saturated sublattice $L[\wp]$: "The pairing identifies $L/L[\wp]^{\perp}$ with $\mathrm{Hom}(L[\wp],\mathcal{O})$, because $L[\wp]$ is saturated", and "$Q$ is the cokernel of $L[\wp]\to\mathrm{Hom}(L[\wp],\mathcal{O})$, of order $\#\mathcal{O}/\Delta$". Section 4.1 of the paper adds that the pairing $\hat{\delta}_!$ is perfect and alternating, that the Hecke operators are self-adjoint for it, and that "the Gram determinant of the pairing on $M_{f^\Sigma,\lambda}$ is $(\eta^{\Sigma}_{f,\lambda})^2$ up to a unit". Lemma lem:crit uses the pairing $S_2(\Gamma_0(N);\mathbb{Z})\times\mathbb{T}\to\mathbb{Z}$. This section proves each of these statements in the abstract. Throughout, except in Proposition 3.82, which holds over any commutative ring, $R$ is a principal ideal domain (in practice $\mathbb{Z}$, $\mathbb{Z}_\ell$ or $\mathcal{O}$) with fraction field $K$, and pairings follow Convention 3.1(e). To a reader of Riley, Hobson and Bence a Gram matrix is a metric tensor $g_{ij}=\langle e_i,e_j\rangle$; the novelty is that over $\mathbb{Z}$ or $\mathbb{Z}_\ell$ the determinant is meaningful only up to units, and its valuation is what is measured.

**Definition 3.81** (perfect pairings)**.** For a pairing $\langle\ ,\ \rangle\colon L\times M\to R$ let $\alpha\colon L\to M^\vee$, $x\mapsto\langle x,-\rangle$, and $\beta\colon M\to L^\vee$, $y\mapsto\langle -,y\rangle$. The pairing is *perfect* if $\alpha$ and $\beta$ are isomorphisms and *nondegenerate* if they are injective. For free modules $L$, $M$ of the same finite rank with bases $(e_i)$, $(f_j)$, the *Gram determinant* is $\Delta=\det(\langle e_i,f_j\rangle)$.

**Proposition 3.82.** *Let $R$ be any commutative ring, $L$, $M$ free $R$-modules of rank $n$ and $G$ the Gram matrix of a pairing $L\times M\to R$.*

*(i) The pairing is perfect iff $\det G\in R^\times$, iff one of $\alpha$, $\beta$ is an isomorphism. If $R$ is a domain, it is nondegenerate iff $\det G\neq 0$. For example, $\langle x,y\rangle=2xy$ on $\mathbb{Z}$ is nondegenerate and not perfect; over $\mathbb{Z}_\ell$ with $\ell$ odd it is perfect.*

*(ii) If the bases are changed by $e'=eP$, $f'=fQ$ (that is, $e'_j=\sum_i P_{ij}e_i$), then $G'=P^{\mathsf{T}}GQ$ and $\det G'=\det P\det Q\det G$; for $L=M$ with the same change of basis on both sides, $\det G'=(\det P)^2\det G$. So the ideal $\Delta R$ does not depend on the bases; over a discrete valuation ring $v(\Delta)$ is a basis-independent invariant.*

*(iii) Perfectness is preserved by base change: if $R\to S$ is a ring homomorphism, the $S$-bilinear extension to $(L\otimes S)\times(M\otimes S)$ has Gram matrix the image of $G$, so it is perfect if the original is.*

*Proof.* (i) In the dual basis $(f_j^\vee)$, $\alpha(e_i)=\sum_j\langle e_i,f_j\rangle f_j^\vee$, so $\alpha$ has matrix $G$ and $\beta$ has matrix $G^{\mathsf{T}}$, both of determinant $\det G$. Proposition 3.21(iii), valid over any commutative ring, gives the first claim; over a domain $R$ with fraction field $K$, a square matrix is injective iff its determinant is nonzero (compute over $K$). (ii) $\langle e'_i,f'_j\rangle=\sum_{a,b}P_{ai}\langle e_a,f_b\rangle Q_{bj}$, and $\det P$, $\det Q$ are units. (iii) The elements $e_i\otimes 1$ and $f_j\otimes 1$ are bases of $L\otimes S$ and $M\otimes S$ (Theorem 3.26(iv)), and $\langle e_i\otimes 1,f_j\otimes 1\rangle$ is the image of $\langle e_i,f_j\rangle$ in $S$. The determinant of the image matrix is the image of $\det G$, and a ring homomorphism maps units to units; apply (i) over $S$. □

The example in (i) is the typical way in which perfectness depends on the coefficients: the Gram determinant 2 is a nonzero non-unit in $\mathbb{Z}$ and in $\mathbb{Z}_2$, and a unit in $\mathbb{Z}_\ell$ for $\ell$ odd. Over $\mathbb{Z}_\ell$ only $v_\ell(\Delta)$ matters, by (ii); this is the number that Lemma lem:ddt417 compares with the congruence ideal.

**Proposition 3.83** (orthogonals)**.** *Let $\langle\ ,\ \rangle\colon L\times L\to R$ be a perfect pairing on a lattice of rank $n$, not necessarily symmetric.*

*(i) For every subset $S$, $S^{\perp}$ and ${}^{\perp}S$ are saturated.*

*(ii) For a submodule $N$ of rank $d$, $N^{\perp}$ and ${}^{\perp}N$ have rank $n-d$, and ${}^{\perp}(N^{\perp})=\operatorname{sat}(N)=({}^{\perp}N)^{\perp}$. In particular ${}^{\perp}(N^{\perp})=N=({}^{\perp}N)^{\perp}$ if $N$ is saturated.*

*(iii) If the pairing is symmetric or antisymmetric, left and right orthogonals agree.*

*Proof.* (i) If $ry\in S^{\perp}$ with $r\neq 0$, then $r\langle s,y\rangle=0$, so $\langle s,y\rangle=0$ for $s\in S$. (ii) Since $\langle rx,y\rangle=0$ iff $\langle x,y\rangle=0$ for $r\neq 0$, $N^{\perp}=\operatorname{sat}(N)^{\perp}$, and we may assume $N$ saturated, hence a direct summand (Proposition 3.49). Then the map $L\to N^{\vee}$, $y\mapsto\langle -,y\rangle|_N$, is surjective: a functional on $N$ extends by 0 on a complement to an element of $L^{\vee}$, which is $\langle -,y\rangle$ for some $y$ because $\beta$ is onto. Its kernel is $N^{\perp}$, so $\operatorname{rank}N^{\perp}=n-d$; likewise for ${}^{\perp}N$. Now ${}^{\perp}(N^{\perp})\supseteq N$, both are saturated (by (i)), and both have rank $d$; a saturated module containing another of the same rank equals its saturation (Proposition 3.49(ii)). The same argument with the roles of the two arguments exchanged gives $({}^{\perp}N)^{\perp}=\operatorname{sat}(N)$. (iii) If $\langle x,y\rangle=\pm\langle y,x\rangle$, then $\langle s,y\rangle=0$ iff $\langle y,s\rangle=0$, so the two orthogonals of any subset coincide. □

**Theorem 3.84** (the Gram determinant of a saturated sublattice)**.** *Let $\mathcal{O}$ be a discrete valuation ring, $L$ a lattice of rank $n$ with a perfect pairing, $L'\subseteq L$ a* saturated *sublattice of rank $d$, and $\Delta$ the Gram determinant of the restriction of the pairing to $L'\times L'$ (in any basis).*

*(i) The map $\rho\colon L\to\operatorname{Hom}_{\mathcal{O}}(L',\mathcal{O})=L'^{\vee}$, $y\mapsto\langle -,y\rangle|_{L'}$, is surjective with kernel $L'^{\perp}$; so $L/L'^{\perp}\simeq L'^{\vee}$.*

*(ii) If $\Delta\neq 0$, then $L'\cap L'^{\perp}=0$, the restriction $\rho|_{L'}\colon L'\to L'^{\vee}$ is injective, and*

$$L/(L'\oplus L'^{\perp})\ \simeq\ \operatorname{coker}(L'\xrightarrow{\rho}L'^{\vee})\ \simeq\ \bigoplus_{i=1}^{d}\mathcal{O}/(\delta_i),$$

*where the $(\delta_i)$ are the invariant factors of the Gram matrix of $L'$. Its length is $v(\Delta)$; when $k$ is finite its order is $\#\mathcal{O}/\Delta\mathcal{O}=(\#k)^{v(\Delta)}$.*

*(iii) In (ii), $L'^{\perp}$ is saturated of rank $n-d$, and the Gram determinant $\Delta^{\perp}$ of the restriction of the pairing to $L'^{\perp}$ satisfies $v(\Delta^{\perp})=v(\Delta)$.*

*(iv) Saturation is necessary: in $L=\mathbb{Z}_\ell^2$ with the standard pairing $\langle x,y\rangle=x_1y_1+x_2y_2$, the sublattice $L'=\mathbb{Z}_\ell\cdot\ell e_1$ has $\Delta=\ell^2$, while $L/(L'\oplus L'^{\perp})=\mathbb{Z}_\ell^2/(\ell\mathbb{Z}_\ell\oplus\mathbb{Z}_\ell)\simeq\mathbb{Z}/\ell$ has length 1.*

*Proof.* (i) is the surjectivity shown in the proof of Proposition 3.83(ii) (saturation makes $L'$ a direct summand); the kernel is $L'^{\perp}$ by definition. (ii) In a basis of $L'$ and the dual basis of $L'^{\vee}$, $\rho|_{L'}$ has matrix the transpose of the Gram matrix of $L'$, of determinant $\Delta\neq 0$; so it is injective, and its kernel $L'\cap L'^{\perp}$ is 0. By (i), $\rho$ induces $L/L'^{\perp}\simeq L'^{\vee}$, under which $(L'+L'^{\perp})/L'^{\perp}$ corresponds to $\rho(L')$; hence $L/(L'\oplus L'^{\perp})\simeq L'^{\vee}/\rho(L')$. By Theorem 3.41 this cokernel is $\bigoplus\mathcal{O}/(\delta_i)$ with $\prod\delta_i=\Delta$ up to a unit, of length $\sum v(\delta_i)=v(\Delta)$ (Proposition 3.66). (iii) $L'^{\perp}$ is saturated of rank $n-d$ by Proposition 3.83. If $\Delta^{\perp}=0$, some $0\neq y\in L'^{\perp}\otimes K$ is right-orthogonal to $L'^{\perp}\otimes K$; it is also right-orthogonal to $L'\otimes K$ (as $y\in L'^{\perp}\otimes K$); and $L'\otimes K+L'^{\perp}\otimes K=L\otimes K$ by (ii) (the quotient is torsion), so $y$ is right-orthogonal to everything, contradicting perfectness.

So $\Delta^\perp \neq 0$. Apply (ii) to the transposed pairing $\langle x,y\rangle^{\mathsf T} = \langle y,x\rangle$, which is perfect, and to the saturated $N = L'^\perp$: its Gram determinant is $\Delta^\perp$ (transposed), its right orthogonal for the transposed pairing is the left orthogonal ${}^\perp(L'^\perp) = L'$ (Proposition 3.83(ii)), so $L/(L'^\perp \oplus L')$ has length $v(\Delta^\perp)$. It also has length $v(\Delta)$. (iv) $L'^\perp = \{y : \langle \ell e_1, y\rangle = \ell y_1 = 0\} = \mathbb{Z}_\ell e_2$, so $L' \oplus L'^\perp = \ell\mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$ and the quotient is $\mathbb{Z}_\ell/\ell\mathbb{Z}_\ell \simeq \mathbb{Z}/\ell$, of length 1; while $\Delta = \langle \ell e_1, \ell e_1\rangle = \ell^2$ has valuation 2. The proof of (ii) breaks at (i): the restriction map $\rho\colon L \to L'^\vee$ has image the functionals with $\ell e_1 \mapsto \ell y_1$, which is $\ell L'^\vee$, not $L'^\vee$. □

In words: for a saturated $L'$, the pairing gives as much room as possible in the quotient $L/L'^\perp$, namely all of $L'^\vee$, and the failure of $L' \oplus L'^\perp$ to fill $L$ is then measured exactly by how far $L'$ is from being self-dual, which is what $\Delta$ measures. For a non-saturated $L'$ the Gram determinant also records the index of $L'$ in its saturation, twice (Exercise 3.124(a)).

**Example 3.85.** On $\mathbb{Z}^2$ take the Gram matrix $\begin{pmatrix} 9 & 1 \\ 1 & 0 \end{pmatrix}$ (perfect: determinant $-1$) and $L' = \mathbb{Z}e_1$, saturated, with $\Delta = 9$. Then $L'^\perp = \{y : 9y_1 + y_2 = 0\} = \mathbb{Z}(1,-9)$, and $L' \oplus L'^\perp$ has basis $(1,0)$, $(1,-9)$, so index $|\det \begin{pmatrix} 1 & 0 \\ 1 & -9 \end{pmatrix}| = 9 = \Delta$, as Theorem 3.84(ii) predicts; the Gram determinant on $L'^\perp$ is $\langle (1,-9),(1,-9)\rangle = 9 - 9 - 9 = -9$, of the same valuation. With the standard pairing and the non-saturated $L' = \mathbb{Z}\cdot 3e_1$ the Gram determinant is 9 but the index of $L' \oplus L'^\perp = 3\mathbb{Z} \oplus \mathbb{Z}$ is 3.

*In the paper* 3.86. In the proof of Lemma lem:ddt417, "The pairing identifies $L/L[\wp]^\perp$ with $\mathrm{Hom}(L[\wp], \mathcal{O})$, because $L[\wp]$ is saturated" is Theorem 3.84(i) with $L' = L[\wp]$, saturated by Proposition 3.49(iii); "$Q$ is the cokernel of $L[\wp] \to \mathrm{Hom}(L[\wp], \mathcal{O})$, of order $\#\mathcal{O}/\Delta$" is part (ii), once $L[\wp]^\perp = L[I]$ is known (Proposition 3.113). Part (iv) shows that the word "saturated" cannot be dropped. For a residue field that is not finite the order must be read as the length $v(\Delta)$ (Convention 3.1(f)); the paper assumes $k$ finite.

**Proposition 3.87** (alternating pairings)**.** *A pairing $L \times L \to R$ is* alternating *if $\langle x,x\rangle = 0$ for all $x$. An alternating pairing is antisymmetric; conversely an antisymmetric pairing is alternating if 2 is a nonzerodivisor in $R$ (for instance $R$ a domain of characteristic 0).*

*(i) The Gram matrix of an alternating pairing is antisymmetric with zero diagonal; its determinant vanishes in odd rank, so a perfect alternating pairing on a nonzero lattice has even rank. In even rank $\det G = \mathrm{Pf}(G)^2$ for a polynomial $\mathrm{Pf}$ in the entries (the Pfaffian); in rank 4, $\mathrm{Pf}(G) = g_{12}g_{34} - g_{13}g_{24} + g_{14}g_{23}$.*

*(ii) Rank two: $G = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}$ with $a = \langle e_1, e_2\rangle$, so $\Delta = a^2$; and $\langle x,y\rangle = a(x_1y_2 - x_2y_1)$, so the ideal of values $\langle L,L\rangle = \{\langle x,y\rangle\}$ is $aR$. Hence, without reference to a basis, $\Delta R = (\text{ideal of values})^2$.*

*(iii) (Symplectic normal form.) Over a principal ideal domain an alternating pairing has a basis in which $G$ is block diagonal with blocks $\begin{pmatrix} 0 & d_i \\ -d_i & 0 \end{pmatrix}$, $d_1 \mid d_2 \mid \cdots$, followed by zeros; it is perfect iff all $d_i$ are units and there is no zero block.*

*Proof.* $0 = \langle x+y, x+y\rangle = \langle x,y\rangle + \langle y,x\rangle$; conversely antisymmetry gives $2\langle x,x\rangle = 0$. (i) The diagonal entries are $\langle e_i, e_i\rangle = 0$. For odd $n$, let $X$ be the generic alternating matrix over the

polynomial ring $\mathbb{Z}[x_{ij} : i < j]$; then $\det X = \det X^{\mathsf{T}} = \det(-X) = -\det X$, and as $\mathbb{Z}[x_{ij}]$ is a domain of characteristic 0, $\det X = 0$; specialising the variables gives $\det G = 0$ over any $R$. A perfect pairing has $\det G$ a unit, and units are nonzero when $R \neq 0$. The identity $\det G = \mathrm{Pf}(G)^2$ in general is classical and we do not prove it (see [109, Chapter XV, Section 9], section number not verified); in rank 4 it is a direct expansion, which we checked with PARI. (ii) Expand $\langle x_1 e_1 + x_2 e_2, y_1 e_1 + y_2 e_2\rangle$; the value $x_1 y_2 - x_2 y_1$ is 1 for $x = e_1$, $y = e_2$. □

*Sketch of proof of (iii).* Choose $x, y$ with $\langle x, y\rangle = d_1$ a generator of the ideal of values (if it is 0 the pairing is zero). The map $z \mapsto z - \frac{\langle z,y\rangle}{d_1} x + \frac{\langle z,x\rangle}{d_1} y$ is defined over $R$ because $d_1$ divides every value, and it projects $L$ onto the orthogonal of $\{x, y\}$, giving $L = (Rx \oplus Ry) \oplus \{x, y\}^{\perp}$; induct, noting that the ideal of values only shrinks. See [109, Chapter XV, Section 8] (section number not verified). □

*In the paper* 3.88. Section 4.1: "The restriction of $\hat{\delta}_!$ to the rank-two lattice $M_{f^\Sigma,\lambda}$ takes its values in an ideal of $\mathbb{Z}_\ell$, the image of $\wedge^2 M_{f^\Sigma,\lambda}$, which is $\eta^\Sigma_{f,\lambda}$, ...; equivalently the Gram determinant of the pairing on $M_{f^\Sigma,\lambda}$ is $(\eta^\Sigma_{f,\lambda})^2$ up to a unit". This is Proposition 3.87(ii), as $\hat{\delta}_!$ is alternating. The same proposition is behind "the Gram determinant of a rank-two alternating form is the square of its ideal of values" in the proof of Proposition prop:pullback, and behind $d = 2$, $\Delta = (\eta^\emptyset_{f,\lambda})^2$ when Lemma lem:ddt417 is applied in Step 2 of the proof of Theorem thm:quant.

**Proposition 3.89** (adjoints)**.** *Let $\langle\ ,\ \rangle$ be a perfect pairing on a lattice $L$, with Gram matrix $G$ in a basis, coordinates written as columns so that $\langle x, y\rangle = x^{\mathsf{T}} G y$.*

*(i) Every $A \in \mathrm{End}_R(L)$ has a unique* adjoint *$A^\dagger$ with $\langle Ax, y\rangle = \langle x, A^\dagger y\rangle$; in matrices $A^\dagger = G^{-1} A^{\mathsf{T}} G$. Moreover $(AB)^\dagger = B^\dagger A^\dagger$, and $(A^\dagger)^\dagger = A$ if the pairing is symmetric or antisymmetric.*

*(ii) Let $\mathbb{T} \subseteq \mathrm{End}_R(L)$ be a commutative subring of self-adjoint endomorphisms. For every ideal $\mathfrak{a} \subseteq \mathbb{T}$, $(\mathfrak{a}L)^\perp = L[\mathfrak{a}]$.*

*(iii) In (ii), every idempotent $e$ of $\mathbb{T}$, or of $\mathbb{T} \otimes_R S$ acting on $L \otimes_R S$ for an $R$-algebra $S$ that is a principal ideal domain, is self-adjoint; $L = eL \oplus (1-e)L$ is an orthogonal decomposition (on both sides), and the restriction of the pairing to $eL$ is perfect.*

*(iv) (Twisted pairings.) Let $w \in \mathrm{Aut}_R(L)$ satisfy $A^\dagger = wAw^{-1}$ for all $A \in \mathbb{T}$, and put $[x, y] = \langle x, wy\rangle$. Then $[\ ,\ ]$ is perfect and every $A \in \mathbb{T}$ is self-adjoint for it. If $\langle\ ,\ \rangle$ is antisymmetric and $w^\dagger = w$ (for instance if $w$ is an involution preserving the pairing, so that $w^\dagger = w^{-1} = w$), then $[\ ,\ ]$ is antisymmetric, hence alternating when 2 is a nonzerodivisor.*

*Proof.* (i) For fixed $y$, $x \mapsto \langle Ax, y\rangle$ is in $L^\vee$, so equals $\langle -, z\rangle$ for a unique $z$ ($\beta$ is bijective); put $A^\dagger y = z$. In matrices $x^{\mathsf{T}} A^{\mathsf{T}} G y = x^{\mathsf{T}} G A^\dagger y$ for all $x, y$, so $GA^\dagger = A^{\mathsf{T}} G$. Then $\langle ABx, y\rangle = \langle Bx, A^\dagger y\rangle = \langle x, B^\dagger A^\dagger y\rangle$. If $\langle x, y\rangle = \pm\langle y, x\rangle$, then $\langle A^\dagger x, y\rangle = \pm\langle y, A^\dagger x\rangle = \pm\langle Ay, x\rangle = \langle x, Ay\rangle$. (ii) $y \in (\mathfrak{a}L)^\perp$ iff $\langle ax, y\rangle = \langle x, ay\rangle = 0$ for all $a \in \mathfrak{a}$ and $x \in L$, iff $ay = 0$ for all $a \in \mathfrak{a}$ (perfectness). (iii) Elements of $\mathbb{T}$ are self-adjoint by hypothesis, and self-adjointness is preserved by $S$-linear combinations, so every element of $\mathbb{T} \otimes S$ is self-adjoint for the extended pairing (which is perfect, Proposition 3.82(iii)). For $x \in eL$ and $y \in (1-e)L$, $\langle x, y\rangle = \langle ex, y\rangle = \langle x, ey\rangle = 0$ since $ey = e(1-e)y = 0$; likewise $\langle y, x\rangle = 0$. The summands are free (direct summands of a lattice over a principal ideal domain), and in a basis adapted to the decomposition the Gram matrix is block diagonal, with unit determinant equal to the product of the two block determinants; so

each block determinant is a unit. (iv) $x \mapsto [x,-]$ is the composite of $\alpha$ with the isomorphism induced by $w$, so $[\ ,\ ]$ is perfect. For $A \in \mathbb{T}$, $[Ax,y] = \langle x, A^\dagger wy\rangle = \langle x, wAw^{-1}wy\rangle = [x,Ay]$. If $\langle\ ,\ \rangle$ is antisymmetric and $w^\dagger = w$, then $[y,x] = \langle y,wx\rangle = -\langle wx,y\rangle = -\langle x,w^\dagger y\rangle = -[x,y]$. An involution preserving the pairing satisfies $\langle wx,y\rangle = \langle w^2x,wy\rangle = \langle x,wy\rangle$. □

Why part (iv) is needed is already visible in rank two. Take the alternating pairing $\langle x,y\rangle = x_1y_2 - x_2y_1$, whose Gram matrix is the matrix $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ of the paper. For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the formula of (i) gives

$$A^\dagger = J^{-1}A^{\mathsf{T}}J = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \operatorname{adj}(A),$$

the adjugate (checked with PARI). So $A$ is self-adjoint iff $a = d$ and $2b = 2c = 0$, that is, over a ring in which 2 is a nonzerodivisor, iff $A$ is a scalar. An alternating pairing therefore makes almost no operator self-adjoint; instead it satisfies $A^\dagger = \det(A)A^{-1}$ for invertible $A$. This is the situation of the cup product on $H^1(X_0(N))$, for which the adjoint of $T_n$ is not $T_n$ but $wT_nw^{-1}$; twisting by the Atkin-Lehner involution $w$, as in (iv), restores self-adjointness, and the twisted pairing is still alternating because $w^\dagger = w$.

*In the paper* 3.90. Section 4.1: the cup product on $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$ is perfect and alternating, the Atkin-Lehner involution $w = w_{M^\Sigma}$ preserves it, and $wT_nw^{-1}$ is the adjoint of $T_n$ for the cup product (Chapters 8 and 14). By Proposition 3.89(iv) the pairing $\hat\delta_!$, which is $\frac12\langle x,wy\rangle$ on classes from $X_0(M^\Sigma)$, is perfect (the factor $\frac12$ being a unit as $\ell$ is odd), alternating (antisymmetric over the domain $\mathbb{Z}_\ell$ of characteristic 0, Proposition 3.87) and makes every Hecke operator, $U_p$ included, self-adjoint: "this is what the composition with $w$ is for". In the paragraph after Lemma lem:hecke (and in Step 2 of the proof of Theorem thm:quant) the paper uses Proposition 3.89(iii): "the idempotent $e_{\mathfrak{m}}$ is self-adjoint for $\hat\delta_!$, so $H^1(X_0(N),\mathbb{Z}_\ell)$ splits orthogonally into its local components and the pairing is perfect on each". The idempotent $e_{\mathfrak{m}}$ lies in $\mathbb{T}\otimes\mathbb{Z}_\ell$, not in $\mathbb{T}$, which is why (iii) allows $\mathbb{T}\otimes S$.

**Proposition 3.91** (an algebra paired with a module)**.** *Let $\mathbb{T}$ be a commutative $R$-algebra that is free of finite rank as an $R$-module, $S$ a $\mathbb{T}$-module free of finite rank over $R$, and $\langle\ ,\ \rangle\colon S\times\mathbb{T}\to R$ a perfect $R$-bilinear pairing with $\langle ts,u\rangle = \langle s,tu\rangle$ for $s\in S$ and $t,u\in\mathbb{T}$.*

*(i) $s\mapsto\langle s,-\rangle$ is an isomorphism of $\mathbb{T}$-modules $S\simeq\mathbb{T}^\vee = \operatorname{Hom}_R(\mathbb{T},R)$, where $(t\varphi)(u) = \varphi(tu)$.*

*(ii) For every ideal $\mathfrak{a}\subseteq\mathbb{T}$, $S[\mathfrak{a}]$ corresponds to $\operatorname{Hom}_R(\mathbb{T}/\mathfrak{a},R)$, the functionals vanishing on $\mathfrak{a}$; it is saturated in $S$, of rank $\operatorname{rank}_R(\mathbb{T}/\mathfrak{a})$.*

*(iii) Let $\mathfrak{a},\mathfrak{b}\subseteq\mathbb{T}$ be ideals with $\mathfrak{a}\cap\mathfrak{b} = 0$, $\mathbb{T}/\mathfrak{a}$ and $\mathbb{T}/\mathfrak{b}$ torsion-free, and $\operatorname{rank}\mathbb{T}/\mathfrak{a} + \operatorname{rank}\mathbb{T}/\mathfrak{b} = \operatorname{rank}\mathbb{T}$. Then $\mathbb{T}\hookrightarrow\mathbb{T}/\mathfrak{a}\times\mathbb{T}/\mathfrak{b}$ has torsion cokernel $\mathbb{T}/(\mathfrak{a}+\mathfrak{b})$ (Proposition 3.13), $S[\mathfrak{a}]\oplus S[\mathfrak{b}]\subseteq S$, and $S/(S[\mathfrak{a}]\oplus S[\mathfrak{b}])\simeq\operatorname{Hom}_R(\mathbb{T}/(\mathfrak{a}+\mathfrak{b}),K/R)$; for $R=\mathbb{Z}$ or $\mathbb{Z}_\ell$ this is $(\mathbb{T}/(\mathfrak{a}+\mathfrak{b}))^*$, of the same order as $\mathbb{T}/(\mathfrak{a}+\mathfrak{b})$.*

*(iv) $\mathbb{T}$ acts faithfully on $S$.*

*Proof.* (i) It is an $R$-isomorphism because the pairing is perfect, and $\mathbb{T}$-linear: $\langle ts,u\rangle = \langle s,tu\rangle = (t\langle s,-\rangle)(u)$. (ii) $s\in S[\mathfrak{a}]$ iff $\langle as,u\rangle = \langle s,au\rangle = 0$ for all $a\in\mathfrak{a}$, $u\in\mathbb{T}$, iff $\langle s,-\rangle$ vanishes on the ideal $\mathfrak{a}$. Such functionals are $\operatorname{Hom}_R(\mathbb{T}/\mathfrak{a},R) = \operatorname{Hom}_R((\mathbb{T}/\mathfrak{a})/(\mathbb{T}/\mathfrak{a})_{\mathrm{tor}},R)$, of rank $\operatorname{rank}(\mathbb{T}/\mathfrak{a})$; saturation is Proposition 3.49(iii). (iii) The cokernel of $\mathbb{T}\to\mathbb{T}/\mathfrak{a}\times\mathbb{T}/\mathfrak{b}$ is $\mathbb{T}/(\mathfrak{a}+\mathfrak{b})$, and the

two lattices have the same rank by hypothesis, so it is torsion. Apply Proposition 3.51 to this injection of lattices: $(\mathbb{T}/\mathfrak{a}\times\mathbb{T}/\mathfrak{b})^\vee\hookrightarrow\mathbb{T}^\vee$, with cokernel $\mathrm{Hom}(\mathbb{T}/(\mathfrak{a}+\mathfrak{b}),K/R)$. Under (i) and (ii) the image of $(\mathbb{T}/\mathfrak{a})^\vee\oplus(\mathbb{T}/\mathfrak{b})^\vee$ in $\mathbb{T}^\vee$ is $S[\mathfrak{a}]+S[\mathfrak{b}]$; this sum is direct because the source is a direct sum and the map is injective. (iv) If $tS=0$ then $\langle s,tu\rangle=\langle ts,u\rangle=0$ for all $s,u$, so $tu=0$ for all $u$ (perfectness); with $u=1$, $t=0$. □

*In the paper* 3.92. In Lemma lem:crit, $R=\mathbb{Z}$, $\mathbb{T}$ is the Hecke algebra, $S=S_2(\Gamma_0(N);\mathbb{Z})$ and $\langle g,t\rangle=a_1(tg)$ is perfect (Chapter 13); it satisfies $\langle tg,u\rangle=a_1(utg)=\langle g,tu\rangle$. Take $\mathfrak{a}=\ker\phi_f$ and $\mathfrak{b}=\mathrm{Ann}_{\mathbb{T}}(\mathfrak{a})$, the paper's $I$ and $J$ (the paper defines $J=\mathrm{Ann}_{\mathbb{T}}(f^\perp)$; it equals $\mathrm{Ann}_{\mathbb{T}}(I)$ because $\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$ with $\mathbb{T}'$ acting faithfully on $f^\perp$). Then $S[\mathfrak{a}]=\mathbb{Z}f$ ($f$ is the primitive form killed by $\mathfrak{a}$), and $S[\mathfrak{b}]=L$ is the orthogonal complement lattice. By Proposition 3.110(iii), $\mathfrak{a}=\mathbb{T}\cap(0\times\mathbb{T}')$ and $\mathfrak{b}=\mathbb{T}\cap(\mathbb{Q}\times0)$, so $\mathbb{T}/\mathfrak{a}$ and $\mathbb{T}/\mathfrak{b}$ are torsion-free (Proposition 3.49(iii)), and $\mathfrak{b}$ is a lattice in $\mathbb{Q}\times0$, so $\operatorname{rank}\mathbb{T}/\mathfrak{b}=\operatorname{rank}\mathbb{T}-1$ and the ranks are complementary. So the paper's dualising step (Paper link 3.54) is Proposition 3.91(iii). The paper leaves these hypotheses implicit; its assertion that $\mathbb{T}/(I+J)\simeq\mathbb{Z}/\phi_f(J)$ is cyclic *of order* $r'$ presupposes them: as $I\cap J=0$, the quotient is finite iff $\phi_f(J)\neq0$, iff $J\neq0$, which is the rank condition.

## 3.10 Finite free algebras over a discrete valuation ring; idempotents; generalised eigenspaces

The Hecke algebra $\mathbb{T}$ is a commutative ring that is free of finite rank over $\mathbb{Z}$. The paper uses three of its structural features. After tensoring with $\mathbb{Q}$ it is a product of local Artinian algebras, one of which is $\mathbb{Q}$, the factor of $f$ ("$\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$", Lemma lem:crit). After tensoring with $\mathbb{Z}_\ell$ it is a product of local rings $\mathbb{T}_\mathfrak{m}$ cut out by idempotents (Lemma lem:hecke, Remark rem:free, Theorem thm:quant). And its action on forms decomposes into generalised eigenspaces, on which the operators need not be semisimple (Lemma lem:old, Theorem thm:twist, Lemma lem:step1). In the language of linear algebra this section is the Jordan decomposition of a commuting family of matrices, done over $\mathbb{Z}$ and $\mathbb{Z}_\ell$ rather than over $\mathbb{C}$.

**Theorem 3.93** (finite-dimensional commutative algebras)**.** *Let $A$ be a commutative algebra of finite dimension over a field $F$.*

*(i) Every prime ideal of $A$ is maximal; $A$ has finitely many maximal ideals $\mathfrak{m}_1,\dots,\mathfrak{m}_s$, with $s\le\dim_F A$; and the Jacobson radical $\mathfrak{J}=\bigcap_i\mathfrak{m}_i$ is nilpotent.*

*(ii) $A\simeq\prod_{i=1}^s A_{\mathfrak{m}_i}$, each factor a local ring with nilpotent maximal ideal; the factors are cut out by idempotents $e_1,\dots,e_s$ with $\sum e_i=1$ and $e_ie_j=0$ for $i\neq j$, and $A_{\mathfrak{m}_i}\simeq e_iA\simeq A/\mathfrak{m}_i^N$ for $N$ large.*

*(iii) $A$ is reduced (has no nonzero nilpotents) iff $\mathfrak{J}=0$ iff $A$ is a product of fields.*

*(iv) Every $A$-module $V$ is $\bigoplus_i e_iV$, and $e_iV=V_{\mathfrak{m}_i}$.*

*Proof.* (i) If $P$ is prime, $A/P$ is a finite-dimensional domain, hence a field (multiplication by $a\neq0$ is an injective, hence surjective, linear map). If $\mathfrak{m}_1,\dots,\mathfrak{m}_s$ are distinct maximal ideals, they are pairwise comaximal and $A\to\prod A/\mathfrak{m}_i$ is surjective by the Chinese remainder theorem (Chapter 2), so $s\le\dim A$. The powers $\mathfrak{J}^k$ form a decreasing chain of subspaces, so $\mathfrak{J}^N=\mathfrak{J}^{N+1}=\mathfrak{J}\cdot\mathfrak{J}^N$ for some $N$; by Theorem 3.35(i) (for an ideal in the Jacobson radical), $\mathfrak{J}^N=0$: its hypotheses hold because $\mathfrak{J}^N$ is a finite-dimensional $F$-space, hence a finitely generated $A$-module, and $\mathfrak{J}\cdot\mathfrak{J}^N=\mathfrak{J}^N$

with $\mathfrak{J} = \mathfrak{J}(A)$. (ii) The ideals $\mathfrak{m}_i^N$ are pairwise comaximal (if $\mathfrak{m}_i^N + \mathfrak{m}_j^N$ were proper it would lie in a maximal ideal containing both $\mathfrak{m}_i$ and $\mathfrak{m}_j$), and $\prod_i \mathfrak{m}_i^N \subseteq \mathfrak{J}^N = 0$. By the Chinese remainder theorem $A \simeq \prod_i A/\mathfrak{m}_i^N$. The maximal ideals of $A/\mathfrak{m}_i^N$ correspond to maximal ideals containing $\mathfrak{m}_i^N$, hence $\mathfrak{m}_i$; so $A/\mathfrak{m}_i^N$ is local with maximal ideal $\mathfrak{m}_i/\mathfrak{m}_i^N$, nilpotent. The idempotents are the unit vectors of the product, and localisation at $\mathfrak{m}_i$ picks out the $i$-th factor (Theorem 3.38(vii)). (iii) If $A$ is reduced, the nilpotent $\mathfrak{J}$ is 0, and $A \simeq \prod A/\mathfrak{m}_i$ by (ii) with $N = 1$. A product of fields is reduced. (iv) $v = \sum e_i v$, and $e_i V \cap \sum_{j\neq i} e_j V = 0$ by applying $e_i$. In $A_{\mathfrak{m}_i}$, $e_i$ becomes a unit ($e_i \notin \mathfrak{m}_i$) and $e_j = 0$ for $j \neq i$ (as $e_i e_j = 0$); so $(e_j V)_{\mathfrak{m}_i} = 0$, while elements outside $\mathfrak{m}_i$ act invertibly on $e_i V$, a module over the local ring $e_i A$; hence $V_{\mathfrak{m}_i} = e_i V$. □

A small example: $A = \mathbb{Q}[x]/(x^2(x-1))$ has the two maximal ideals $(x)$ and $(x-1)$. In $A$ one has $x^3 = x^2$, so $x^2$ is idempotent ($x^4 = x{\cdot}x^3 = x^3 = x^2$), and $A = x^2A \times (1-x^2)A \simeq \mathbb{Q}\times\mathbb{Q}[x]/(x^2)$: on the first factor $x$ acts as 1, on the second it is nilpotent. So $A$ is not reduced, $\mathfrak{J} = (x(x-1))$ with $\mathfrak{J}^2 = 0$ (as $x^2(x-1)^2 = x^2(x-1)\cdot(x-1) = 0$), and the idempotent $x^2$ is a polynomial in $x$ found by the Chinese remainder theorem: it is $\equiv 0 \pmod{x^2}$ and $\equiv 1 \pmod{x-1}$. The integral version of this algebra is Example 3.112(e).

**Theorem 3.94** (finite free algebras over a discrete valuation ring)**.** *Let $\mathbb{T}$ be a commutative $\mathcal{O}$-algebra that is finite and free as an $\mathcal{O}$-module (a* finite free $\mathcal{O}$*-algebra).*

*(i) $\mathbb{T}$ is Noetherian. Every maximal ideal $\mathfrak{M}$ of $\mathbb{T}$ contains $\varpi$, so the maximal ideals of $\mathbb{T}$ correspond to those of the finite $k$-algebra $\mathbb{T}/\varpi\mathbb{T}$; they are finite in number, and their residue fields are finite extensions of $k$.*

*(ii) $\mathbb{T}$ is torsion-free and $\mathbb{T} \subseteq \mathbb{T}_K := \mathbb{T} \otimes_{\mathcal{O}} K$, a finite-dimensional $K$-algebra. $\mathbb{T}$ is reduced iff $\mathbb{T}_K$ is, iff $\mathbb{T}$ embeds in a product of fields.*

*(iii) Let $\pi\colon \mathbb{T} \to \mathcal{O}$ be a surjective homomorphism of $\mathcal{O}$-algebras, with kernel $\wp$. Then $\wp$ is a prime ideal, $\mathbb{T} = \mathcal{O}\cdot 1 \oplus \wp$ as $\mathcal{O}$-modules, and $\mathfrak{m}_\pi = \pi^{-1}(\varpi\mathcal{O}) = \wp + \varpi\mathbb{T}$ is a maximal ideal with residue field $k$. The extension $\pi_K\colon \mathbb{T}_K \to K$ is a surjective homomorphism; it factors through exactly one of the local factors of $\mathbb{T}_K$ (Theorem 3.93).*

*Proof.* (i) $\mathbb{T}$ is a finitely generated module over the Noetherian $\mathcal{O}$ (Proposition 3.31). Let $\mathfrak{M}$ be maximal; $\mathbb{T}/\mathfrak{M}$ is a field which is a finitely generated module over the image $D = \mathcal{O}/(\mathfrak{M}\cap\mathcal{O})$ of $\mathcal{O}$, a domain. Then $D$ is a field: for $0 \neq x \in D$, the element $y = x^{-1} \in \mathbb{T}/\mathfrak{M}$ satisfies a monic equation $y^n + d_1 y^{n-1} + \cdots + d_n = 0$ over $D$ (Lemma 3.32 with $\mathfrak{a} = D$, applied to multiplication by $y$), and multiplying by $x^{n-1}$ gives $y = -(d_1 + d_2 x + \cdots + d_n x^{n-1}) \in D$. So $\mathfrak{M}\cap\mathcal{O}$ is maximal, that is, $\varpi\mathcal{O}$, and $\varpi \in \mathfrak{M}$. The rest follows from Theorem 3.8(iii) and Theorem 3.93(i), as $\mathbb{T}/\mathfrak{M}$ is a quotient of the finite-dimensional $k$-algebra $\mathbb{T}/\varpi\mathbb{T}$. (ii) Free modules are torsion-free and embed in their tensor product with $K$ (Proposition 3.46(i)). If $x \in \mathbb{T}_K$ is nilpotent, so is $\varpi^k x \in \mathbb{T}$ for $k$ large; so $\mathbb{T}$ is reduced iff $\mathbb{T}_K$ is, and then $\mathbb{T} \subseteq \mathbb{T}_K$, a product of fields (Theorem 3.93(iii)). Conversely a subring of a product of fields is reduced. (iii) $\mathbb{T}/\wp \simeq \mathcal{O}$ is a domain. $t = \pi(t)\cdot 1 + (t - \pi(t)\cdot 1)$ with $t - \pi(t)\cdot 1 \in \wp$, and $\mathcal{O}\cdot 1 \cap \wp = 0$ because $\pi(a\cdot 1) = a$. If $\pi(t) = \varpi a$ then $t = \varpi a \cdot 1 + (t - \pi(t)\cdot 1) \in \varpi\mathbb{T} + \wp$; the reverse inclusion is clear, and $\mathbb{T}/\mathfrak{m}_\pi \simeq \mathcal{O}/\varpi = k$. The kernel of $\pi_K$ is a maximal ideal of $\mathbb{T}_K = \prod_i A_i$, hence of the form $A_1 \times \cdots \times \mathfrak{m}_{A_i} \times \cdots \times A_s$ for one $i$ (Chapter 2: ideals of a product); so $\pi_K(e_j) = 0$ for $j \neq i$ and $\pi_K(e_i) = 1$. □

**Theorem 3.95** (idempotent lifting and the decomposition $\mathbb{T} = \prod \mathbb{T}_{\mathfrak{m}}$)**.** *Let $\mathcal{O}$ be a* complete *discrete valuation ring (for example $\mathbb{Z}_\ell$, or $\mathcal{O}_\lambda$ of Chapter 7), and $\mathbb{T}$ a finite free $\mathcal{O}$-algebra.*

*(i) Every idempotent $\bar{e}$ of $\mathbb{T}/\varpi\mathbb{T}$ lifts to a unique idempotent of $\mathbb{T}$.*

*(ii) $\mathbb{T} \simeq \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$, the product over the maximal ideals of $\mathbb{T}$, where $\mathbb{T}_{\mathfrak{m}} = e_{\mathfrak{m}}\mathbb{T}$ for an idempotent $e_{\mathfrak{m}}$, and $\mathbb{T}_{\mathfrak{m}}$ is the localisation of $\mathbb{T}$ at $\mathfrak{m}$; each $\mathbb{T}_{\mathfrak{m}}$ is a local ring and a finite free $\mathcal{O}$-algebra. Moreover $e_{\mathfrak{m}} \notin \mathfrak{m}$ and $e_{\mathfrak{m}'} \in \mathfrak{m}$ for $\mathfrak{m}' \neq \mathfrak{m}$.*

*(iii) For every $\mathbb{T}$-module $L$, $L = \bigoplus_{\mathfrak{m}} e_{\mathfrak{m}}L$ and $L_{\mathfrak{m}} = e_{\mathfrak{m}}L$; if $L$ is finitely generated and free over $\mathcal{O}$, so is each $L_{\mathfrak{m}}$.*

*(iv) A ring homomorphism $\phi$ from $\mathbb{T}$ to a local ring factors through exactly one $\mathbb{T}_{\mathfrak{m}}$. For $\pi\colon \mathbb{T} \to \mathcal{O}$ as in Theorem 3.94(iii): $\pi(e_{\mathfrak{m}_\pi}) = 1$ and $\pi(e_{\mathfrak{m}'}) = 0$ for $\mathfrak{m}' \neq \mathfrak{m}_\pi$.*

*(v) Completeness is necessary: $\mathbb{Z}_{(11)}[x]/(x^2 + x - 1)$ is a finite free $\mathbb{Z}_{(11)}$-algebra with two maximal ideals, $(11, x - 3)$ and $(11, x - 7)$, but no idempotent other than 0 and 1.*

*Proof.* (i) *Existence.* Let $e_0 \in \mathbb{T}$ lift $\bar{e}$, and define $e_{j+1} = 3e_j^2 - 2e_j^3$. Put $d_j = e_j^2 - e_j$. A direct computation gives

$$e_{j+1} - e_j = (1 - 2e_j)d_j, \qquad e_{j+1}^2 - e_{j+1} = d_j^2\,(4d_j - 3).$$

(Expand both sides as polynomials in $e = e_j$, with $d = e^2 - e$: the first is $3e^2 - 2e^3 - e$ on both sides, and the second is $4e^6 - 12e^5 + 9e^4 + 2e^3 - 3e^2$ on both sides; both identities were also checked with PARI.) The iteration is Newton's method for the equation $e^2 - e = 0$: the Newton step is $e - (2e - 1)^{-1}d$, and since $(2e - 1)^2 = 4d + 1$, the inverse $(2e - 1)^{-1}$ is congruent to $2e - 1$ modulo $d$; replacing it by $2e - 1$ gives $e - (2e - 1)d = 3e^2 - 2e^3$, and keeps the quadratic convergence. Since $d_0 \in \varpi\mathbb{T}$, induction gives $d_j \in \varpi^{2^j}\mathbb{T}$, and $e_{j+1} - e_j \in \varpi^{2^j}\mathbb{T}$. As $\mathbb{T} \simeq \mathcal{O}^n$ is complete (Theorem 3.58(vi) holds verbatim over any complete $\mathcal{O}$), the $e_j$ converge to some $e$, with $e \equiv e_0 \pmod{\varpi}$ and $e^2 - e = \lim d_j = 0$. (Here $d_j \in \varpi^{2^j}\mathbb{T}$ follows from $d_{j+1} = d_j^2(4d_j - 3)$ by induction, starting from $d_0 \in \varpi\mathbb{T}$ because $\bar{e}$ is idempotent.) *Uniqueness.* If $e$, $e'$ are idempotents with $e \equiv e' \pmod{\varpi}$, then $x = e - e'$ satisfies $x^3 = x$ (expand, using $e^2 = e$, $e'^2 = e'$ and commutativity: $x^2 = e - 2ee' + e'$ and $x^3 = e - e'$). So $x = x^3 = x^9 = \cdots \in \bigcap_k \varpi^k\mathbb{T} = 0$: since $x \in \varpi\mathbb{T}$, the power $x^{3^k}$ lies in $\varpi^{3^k}\mathbb{T}$, and $\bigcap_k \varpi^k\mathbb{T} = 0$ because $\mathbb{T}$ is free over $\mathcal{O}$ and $\bigcap_k \varpi^k\mathcal{O} = 0$. (ii) By Theorem 3.93, $\mathbb{T}/\varpi\mathbb{T} = \prod_{\bar{\mathfrak{m}}} A_{\bar{\mathfrak{m}}}$ with orthogonal idempotents $\bar{e}_{\mathfrak{m}}$ summing to 1; the $\bar{\mathfrak{m}}$ correspond to the maximal ideals $\mathfrak{m}$ of $\mathbb{T}$ (Theorem 3.94(i)). Lift them to idempotents $e_{\mathfrak{m}}$. For $\mathfrak{m} \neq \mathfrak{m}'$, $e_{\mathfrak{m}}e_{\mathfrak{m}'}$ is an idempotent lifting 0, hence 0; and $\sum e_{\mathfrak{m}}$ is an idempotent lifting 1, hence 1. So $\mathbb{T} = \prod e_{\mathfrak{m}}\mathbb{T}$. Each $e_{\mathfrak{m}}\mathbb{T}$ is a direct summand of $\mathbb{T}$, so finite free over $\mathcal{O}$ (Theorem 3.40), and a finite free $\mathcal{O}$-algebra with identity $e_{\mathfrak{m}}$; its maximal ideals contain $\varpi$ and correspond to those of $e_{\mathfrak{m}}\mathbb{T}/\varpi e_{\mathfrak{m}}\mathbb{T} = A_{\bar{\mathfrak{m}}}$, which is local. So $e_{\mathfrak{m}}\mathbb{T}$ is local, and the maximal ideal of $\mathbb{T}$ corresponding to it is the product of its maximal ideal with the other factors, which is $\mathfrak{m}$; it contains $e_{\mathfrak{m}'}$ for $\mathfrak{m}' \neq \mathfrak{m}$ and not $e_{\mathfrak{m}}$. Localisation at $\mathfrak{m}$ picks out the factor $e_{\mathfrak{m}}\mathbb{T}$ (Theorem 3.38(vii)). (iii) As in Theorem 3.93(iv). The summands of a free $\mathcal{O}$-module of finite rank are free. (iv) The image of an idempotent in a local ring is 0 or 1 ($e(1-e) = 0$ and one of $e$, $1-e$ is a unit). The images of the $e_{\mathfrak{m}}$ are orthogonal and sum to 1, so exactly one is 1 and $\phi$ factors through that factor. For $\pi$, reduce modulo $\varpi$: $\bar{\pi}\colon \mathbb{T} \to k$ has kernel $\mathfrak{m}_\pi$, which contains $e_{\mathfrak{m}'}$ for $\mathfrak{m}' \neq \mathfrak{m}_\pi$ and not $e_{\mathfrak{m}_\pi}$, by (ii). (v) The ring is a subring of $\mathbb{Q}(\sqrt{5})$, a domain (the polynomial is irreducible over $\mathbb{Q}$, having discriminant 5), so $e(1 - e) = 0$ forces $e \in \{0, 1\}$. It is free with basis $1, x$ over $\mathbb{Z}_{(11)}$, so its maximal ideals contain 11 (Theorem 3.94(i)) and correspond to those of $\mathbb{F}_{11}[x]/((x - 3)(x - 7)) \simeq \mathbb{F}_{11} \times \mathbb{F}_{11}$, since $x^2 + x - 1 \equiv (x - 3)(x - 7) \pmod{11}$. □

**Corollary 3.96** (Hecke-type algebras)**.** *Let $\mathbb{T}$ be a commutative ring that is free of finite rank as a $\mathbb{Z}$-module. Then $\mathbb{T}\otimes\mathbb{Z}_\ell$ is a finite free $\mathbb{Z}_\ell$-algebra; its maximal ideals correspond to the maximal ideals $\mathfrak{m}$ of $\mathbb{T}$ containing $\ell$ (as $\mathfrak{m}\otimes\mathbb{Z}_\ell$); and $\mathbb{T}\otimes\mathbb{Z}_\ell=\prod_{\mathfrak{m}\ni\ell}\mathbb{T}_\mathfrak{m}$ with $\mathbb{T}_\mathfrak{m}=(\mathbb{T}\otimes\mathbb{Z}_\ell)_\mathfrak{m}$, the convention of Convention 3.2.*

*Proof.* $\mathbb{T}\otimes\mathbb{Z}_\ell\simeq\mathbb{Z}_\ell^n$. Its maximal ideals correspond to those of $(\mathbb{T}\otimes\mathbb{Z}_\ell)/\ell=\mathbb{T}/\ell\mathbb{T}$ (Theorem 3.94(i) and Theorem 3.58(iii)), which are the maximal ideals of $\mathbb{T}$ containing $\ell$. Apply Theorem 3.95. □

*Remark* 3.97. One can show that $\mathbb{T}_\mathfrak{m}$ is also the $\mathfrak{m}$-adic completion $\varprojlim\mathbb{T}/\mathfrak{m}^k$ of $\mathbb{T}$; we do not use this, and do not prove it (see [62, Chapter 7]).

**Example 3.98** (the Hecke algebra of level 23)**.** Let $\mathbb{T}=\mathbb{Z}[x]/(x^2+x-1)=\mathbb{Z}[\frac{1+\sqrt{5}}{2}]$. This is the Hecke algebra of $S_2(\Gamma_0(23))$, with $x=T_2$ (Chapter 6): the newforms of level 23 have $a_2=(-1\pm\sqrt{5})/2$, the roots of $x^2+x-1$ (PARI: `charpoly(mfheckemat(mfinit([23,2],1),2))`); $\mathbb{T}_{23}\otimes\mathbb{Q}$ has dimension $\dim S_2(\Gamma_0(23))=2$ (Proposition 3.91(i)), so it is $\mathbb{Q}(\sqrt{5})$; and $\mathbb{Z}[T_2]\subseteq\mathbb{T}_{23}$ is an order of discriminant 5, squarefree, hence the maximal order, so $\mathbb{T}_{23}=\mathbb{Z}[T_2]$.

- $\ell=11$: $x^2+x-1\equiv(x-3)(x-7)\pmod{11}$, so $\mathbb{T}\otimes\mathbb{Z}_{11}\simeq\mathbb{Z}_{11}\times\mathbb{Z}_{11}$, by $x\mapsto(r_1,r_2)$ with $r_1\equiv7$, $r_2\equiv3$ the roots in $\mathbb{Z}_{11}$. Modulo $11^6=1771561$ they are $r_1=728020$ and $r_2=1043540$ (PARI's `polrootspadic`; $r_1+r_2=11^6-1\equiv-1$). The idempotent of the first factor is $e=(x-r_2)/(r_1-r_2)$, which is $\equiv784x+1058\pmod{11^3}$ and $\equiv1708457x+854229\pmod{11^6}$; it is also the limit of the iteration $e\mapsto3e^2-2e^3$ started at $(x-3)/4$ (five steps give it modulo $11^6$).
- $\ell=5$: $x^2+x-1\equiv(x-2)^2\pmod5$, so $\mathbb{T}\otimes\mathbb{Z}_5$ is local; it is the ring of integers of $\mathbb{Q}_5(\sqrt{5})$, a ramified discrete valuation ring.
- $\ell=2,3$: $x^2+x-1$ is irreducible modulo 2 and modulo 3, so $\mathbb{T}\otimes\mathbb{Z}_2$ and $\mathbb{T}\otimes\mathbb{Z}_3$ are local, unramified discrete valuation rings with residue fields $\mathbb{F}_4$ and $\mathbb{F}_9$.

A second example: $\mathbb{T}=\mathbb{Z}[x]/(x^2-4)\simeq\{(a,b)\in\mathbb{Z}^2:a\equiv b\pmod4\}$ (by $x\mapsto(2,-2)$). Then $\mathbb{T}\otimes\mathbb{Z}_3\simeq\mathbb{Z}_3\times\mathbb{Z}_3$ with idempotent $(x+2)/4$, while $\mathbb{T}\otimes\mathbb{Z}_2$ is local ($x^2-4\equiv x^2\pmod2$) and not a domain ($(x-2)(x+2)=0$).

```
polrootspadic(x^2+x-1, 11, 6)      \\ roots = 3 + ... and 7 + ... in Z_11
e = Mod(Mod(1,11^6)*(x-3)/4, x^2+x-1);
for(i=1,5, e = 3*e^2 - 2*e^3); e   \\ Mod(1708457*x + 854229, x^2+x-1)
factormod(x^2+x-1, 5)              \\ (x+3)^2, that is (x-2)^2
```

*In the paper* 3.99. In the paragraph after Lemma lem:hecke, $\mathfrak{m}$ is "the maximal ideal of $\mathbb{T}\otimes\mathbb{Z}_\ell$ containing $\ell$ and $\ker\phi_f$"; it is $\mathfrak{m}_{\phi_f}$ of Theorem 3.94(iii), for $\phi_f\otimes\mathbb{Z}_\ell\colon\mathbb{T}\otimes\mathbb{Z}_\ell\to\mathbb{Z}_\ell$. By Theorem 3.95(iv), $\phi_f$ factors through $\mathbb{T}_\mathfrak{m}$ and kills the other local factors; in the proof of Lemma lem:step1, "The idempotent $e_\mathfrak{m}$ of $\mathbb{T}_\ell$ fixes $g^S$ (as $\pi$ factors through $\mathbb{T}_\mathfrak{m}$)" uses the idempotent $e_\mathfrak{m}$ of (ii). Remark rem:free is about $H^1(X_0(N),\mathbb{Z}_\ell)_\mathfrak{m}=e_\mathfrak{m}H^1(X_0(N),\mathbb{Z}_\ell)$, a free $\mathbb{Z}_\ell$-module by (iii).

**Theorem 3.100** (generalised eigenspaces)**.** *(i) Let $V$ be a nonzero finite-dimensional vector space over a field $F$ and $\mathbb{T}\subseteq\mathrm{End}_F(V)$ a commutative subalgebra such that the characteristic polynomial of every $t\in\mathbb{T}$ splits into linear factors over $F$. For an $F$-algebra homomorphism $\phi\colon\mathbb{T}\to F$ (a* system of eigenvalues*) put*

$$V(\phi)=\{v\in V:(t-\phi(t))^{\dim V}v=0\text{ for all }t\in\mathbb{T}\}.$$

*Then $V = \bigoplus_\phi V(\phi)$, the sum over all systems of eigenvalues, and each $V(\phi) \neq 0$. The projection onto $V(\phi)$ is an idempotent $e_\phi \in \mathbb{T}$, the idempotent of the local factor of $\mathbb{T}$ at $\ker\phi$ (Theorem 3.93). Each $V(\phi)$ contains a common eigenvector $v \neq 0$, with $tv = \phi(t)v$ for all $t \in \mathbb{T}$. Every $\mathbb{T}$-stable subspace $W$ satisfies $W = \bigoplus_\phi (W \cap V(\phi))$.*

*(ii) (Integral version.) Let $\mathcal{O}$ be a complete discrete valuation ring, $L$ a lattice, and $\mathbb{T} \subseteq \mathrm{End}_{\mathcal{O}}(L)$ a commutative subalgebra, such that the characteristic polynomials of the elements of $\mathbb{T}$ on $V = L \otimes K$ split over $K$. Then every system of eigenvalues $\phi\colon \mathbb{T} \to K$ takes values in $\mathcal{O}$; $\mathfrak{m}_\phi = \phi^{-1}(\varpi\mathcal{O})$ is a maximal ideal of $\mathbb{T}$; $\mathfrak{m}_\phi = \mathfrak{m}_\psi$ iff $\phi \equiv \psi \pmod{\varpi}$ on $\mathbb{T}$; and for each maximal ideal $\mathfrak{m}$,*

$$e_{\mathfrak{m}} L \otimes K = \bigoplus_{\phi:\ \mathfrak{m}_\phi = \mathfrak{m}} V(\phi).$$

*(If the eigenvalues are not in $K$, replace $\mathcal{O}$ by the ring of integers of a finite extension, Chapter 7.)*

*Proof.* (i) $\mathbb{T}$ is a finite-dimensional commutative $F$-algebra, so $\mathbb{T} = \prod_i \mathbb{T}_i$ with local factors $\mathbb{T}_i = e_i\mathbb{T}$ (Theorem 3.93). The residue field $\mathbb{T}/\mathfrak{m}_i$ is generated over $F$ by the images of the $t \in \mathbb{T}$. Each such image is a root of the minimal polynomial of $t$ as an element of $\mathbb{T}$, which is the minimal polynomial of $t$ as an operator ($\mathbb{T} \subseteq \mathrm{End}\, V$), a divisor of the characteristic polynomial, hence a product of linear factors over $F$; so the image lies in $F$, and $\mathbb{T}/\mathfrak{m}_i = F$. This gives a system of eigenvalues $\phi_i\colon \mathbb{T} \to \mathbb{T}/\mathfrak{m}_i = F$, and conversely every $\phi$ has a maximal kernel; so the $\phi$ correspond to the $\mathfrak{m}_i$. Put $V_i = e_iV$; then $V = \bigoplus V_i$, and $V_i \neq 0$ because $e_i \neq 0$ in $\mathbb{T} \subseteq \mathrm{End}\, V$. On $V_i$, $t - \phi_i(t)$ acts through the nilpotent maximal ideal of $\mathbb{T}_i$, so it is a nilpotent operator and $(t-\phi_i(t))^{\dim V} = 0$ there: $V_i \subseteq V(\phi_i)$. Conversely, if $v \in V(\phi_i)$ and $j \neq i$, take $t = e_j$: $\phi_i(e_j) = 0$, so $e_j^{\dim V} v = e_j v = 0$; hence $v = e_iv \in V_i$. So $V(\phi_i) = V_i$ and $e_{\phi_i} = e_i$. A common eigenvector: let $\mathfrak{n}$ be the maximal ideal of $\mathbb{T}_i$ and $c \geq 0$ the largest integer with $\mathfrak{n}^c V_i \neq 0$ (with $\mathfrak{n}^0 = \mathbb{T}_i$; it exists as $V_i \neq 0$ and $\mathfrak{n}$ is nilpotent); then $\mathfrak{n}^{c+1}V_i = 0$, so a nonzero $v \in \mathfrak{n}^c V_i$ is killed by $\mathfrak{n}$, and $tv = \phi_i(t)v$. Finally $e_iW \subseteq W \cap V_i$ for $W$ stable, and $w = \sum e_i w$. (ii) $\mathbb{T}$ is a finitely generated $\mathcal{O}$-submodule of $\mathrm{End}_{\mathcal{O}}(L) \simeq \mathcal{O}^{n^2}$, hence finite free, and complete $\mathcal{O}$ makes Theorem 3.95 available. For $\phi$ a system of eigenvalues on $V$, $\phi(t)$ is an eigenvalue of $t$ (by (i) over $K$), hence a root of the characteristic polynomial of $t$ on $L$, which is monic with coefficients in $\mathcal{O}$; a root $x \in K$ of such a polynomial has $v(x) \geq 0$ (if $v(x) < 0$ the leading term has strictly smaller valuation than all others). So $\phi(\mathbb{T}) \subseteq \mathcal{O}$, and $\phi$ is an $\mathcal{O}$-algebra map, so $\mathbb{T} \to \mathcal{O} \to k$ is onto with kernel $\mathfrak{m}_\phi$, maximal. If $\mathfrak{m}_\phi = \mathfrak{m}_\psi = \mathfrak{m}$, the reductions $\bar\phi, \bar\psi\colon \mathbb{T} \to k$ both induce $\mathcal{O}$-algebra isomorphisms $\mathbb{T}/\mathfrak{m} \to k$, which coincide since $\mathbb{T}/\mathfrak{m}$ is generated by the image of $\mathcal{O}$ ($\mathbb{T}/\mathfrak{m} \simeq k$ via either, and $k$ is generated by $\mathcal{O}$); conversely congruent systems have the same kernel modulo $\varpi$. For the last statement: $e_{\mathfrak{m}}$ acts on $V(\phi)$ as an idempotent whose only eigenvalue is $\phi(e_{\mathfrak{m}}) \in \{0,1\}$, so as the scalar $\phi(e_{\mathfrak{m}})$; and $\phi(e_{\mathfrak{m}}) \equiv 1$ iff $e_{\mathfrak{m}} \notin \mathfrak{m}_\phi$ iff $\mathfrak{m}_\phi = \mathfrak{m}$ (Theorem 3.95(ii)). Sum over $\phi$ using (i). □

A small example of (i): $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ on $F^3$ generates $\mathbb{T} = F[A]$ with two systems of eigenvalues, $A \mapsto 2$ and $A \mapsto 3$. Here $V(2) = Fe_1 + Fe_2$ and $V(3) = Fe_3$, and $A$ is not diagonalisable on $V(2)$. The projection onto $V(2)$ is a polynomial in $A$, as the theorem says: $p(X) = 1 - (X-2)^2$ satisfies $p(3) = 0$, $p(2) = 1$ and $p'(2) = 0$, so $p(A)$ is 1 on $V(2)$ (where $(A-2)^2 = 0$) and 0 on $V(3)$; indeed $p(A) = \mathrm{diag}(1,1,0)$ (checked with PARI). The common eigenvector in $V(2)$ is $e_1$, spanning $(A-2)V(2)$, the last nonzero power of the maximal ideal

applied to $V(2)$, as in the proof. For (ii), replace the entry 3 by 5 and work on $L=\mathbb{Z}_3^3$. The systems $A\mapsto 2$ and $A\mapsto 5$ are now congruent modulo 3, so they have the same maximal ideal of $\mathbb{Z}_3[A]\simeq\mathbb{Z}_3[X]/((X-2)^2(X-5))$, which is local because $(X-2)^2(X-5)\equiv(X-2)^3 \pmod 3$; its only idempotents are 0 and 1, and $e_{\mathfrak{m}}L\otimes\mathbb{Q}_3=V(2)\oplus V(5)$ is everything. Over $\mathbb{Z}_\ell$ with $\ell\neq 3$ the reductions 2 and 5 differ, and the two generalised eigenspaces are separated by an idempotent of $\mathbb{Z}_\ell[A]$. Congruences between eigenvalues are exactly what glues generalised eigenspaces together integrally.

*In the paper* 3.101. In Lemma lem:crit, "$\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$ with $\mathbb{T}'$ acting faithfully on $f^{\perp}$" is Theorem 3.93 for $\mathbb{T}\otimes\mathbb{Q}$ together with the fact that the local factor through which $\phi_f$ factors is $\mathbb{Q}$ itself, which is multiplicity one for the newform $f$ (Chapter 13; compare Proposition 3.110(ii)). In Theorem thm:twist, "$g^{\perp}$, the $\mathbb{T}$-stable complement of the $g$-line (the sum of the generalised eigenspaces of the other systems of eigenvalues)" is $\bigoplus_{\phi\neq\phi_g}V(\phi)$ of Theorem 3.100(i), applied to the algebra generated by the $T_p$, $p\nmid N_S$, over $\overline{\mathbb{Q}}$. In the proof of Lemma lem:step1, "The form $h'$ lies in the sum of the generalised eigenspaces of the eigencharacters $\phi\neq\pi$ of $\mathbb{T}$ congruent to $\pi$ modulo the maximal ideal above $\ell$" is part (ii).

**Proposition 3.102** (projecting a congruence to a local component)**.** *In Theorem 3.100(ii) let $\mathfrak{m}$ be a maximal ideal of $\mathbb{T}$ and $e=e_{\mathfrak{m}}$.*

*(i) $e$ maps $L$ into $L$ and every $\mathbb{T}$-stable subspace $W\subseteq V$ into itself, and fixes $V(\phi)$ pointwise when $\mathfrak{m}_\phi=\mathfrak{m}$.*

*(ii) If $x\in V(\phi)\cap L$ with $\mathfrak{m}_\phi=\mathfrak{m}$, $y\in L\cap W$ and $x\equiv y \pmod{\varpi^tL}$, then $x\equiv ey \pmod{\varpi^tL}$, and $ey\in L\cap W\cap eV$.*

*Proof.* (i) $e\in\mathbb{T}$ and $W$ is $\mathbb{T}$-stable; $e$ acts on $V(\phi)$ by $\phi(e)=1$ (proof of Theorem 3.100(ii)). (ii) $ex=x$, and $e(x-y)\in e\varpi^tL\subseteq\varpi^tL$. □

*In the paper* 3.103. The proof of Lemma lem:step1 (the inequality $\geq$): "The idempotent $e_{\mathfrak{m}}$ of $\mathbb{T}_\ell$ fixes $g^S$ (as $\pi$ factors through $\mathbb{T}_{\mathfrak{m}}$) and preserves $C$ and the integral forms, so $h'=e_{\mathfrak{m}}h$ lies in $C\cap S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ and $g^S\equiv h' \pmod{\ell^t}$". This is Proposition 3.102(ii) with $x=g^S$, $W=C$ the complement of the line of $g^S$, and $y=h$.

**Proposition 3.104** (semisimple and non-semisimple operators)**.** *Let $A\in\mathrm{End}_F(V)$ with $V$ finite-dimensional over a field $F$, and let $\mu_A$ be its minimal polynomial.*

*(i) $A$ is* semisimple *if every $A$-stable subspace has an $A$-stable complement. This holds iff $\mu_A$ is squarefree, iff $F[A]$ is reduced; for $F$ perfect (of characteristic 0, or finite) it holds iff $A$ is diagonalisable over an algebraic closure $\overline{F}$.*

*(ii) If $A$ has a cyclic vector ($v$ with $v,Av,\dots,A^{n-1}v$ a basis), then $\mu_A$ equals the characteristic polynomial; so $A$ is not semisimple as soon as the characteristic polynomial has a repeated root.*

*(iii) A commutative subalgebra $\mathbb{T}\subseteq\mathrm{End}_F(V)$ is reduced iff each of its elements is semisimple. If some $t\in\mathbb{T}$ is not semisimple, $\mathbb{T}$ contains a nonzero nilpotent element.*

*(iv) (Jordan-Chevalley.) For $F$ perfect, $A=A_s+A_n$ with $A_s$ semisimple, $A_n$ nilpotent, $A_sA_n=A_nA_s$, and $A_s$, $A_n$ polynomials in $A$; the decomposition is unique.*

*Proof.* (i) $F[A] \simeq F[X]/(\mu_A)$. If $\mu_A = g^2 h$ with $g$ nonconstant then $gh(A) \neq 0$ and $(gh)^2(A) = 0$; and if $\mu_A = \prod g_i$ with distinct irreducible $g_i$ then $F[X]/(\mu_A) \simeq \prod F[X]/(g_i)$ is a product of fields (Chinese remainder theorem). So $\mu_A$ squarefree iff $F[A]$ reduced. If $F[A] = \prod F_i$ is a product of fields with idempotents $e_i$, an $A$-stable $W$ is an $F[A]$-submodule, $W = \bigoplus e_i W$ with $e_i W$ an $F_i$-subspace of the $F_i$-vector space $e_i V$; choose $F_i$-complements $C_i$; then $\bigoplus C_i$ is an $A$-stable complement. If $\mu_A$ is not squarefree, let $N = gh(A)$ as above, $N \neq 0 = N^2$, and $W = \ker N \supseteq \operatorname{im} N \neq 0$. If $V = W \oplus C$ with $C$ stable, then $N(C) \subseteq C \cap \operatorname{im} N \subseteq C \cap W = 0$, so $C \subseteq \ker N = W$, $C = 0$ and $N = 0$: a contradiction. For perfect $F$, an irreducible polynomial has distinct roots in $\overline{F}$; so $\mu_A$ is squarefree over $F$ iff it has distinct roots in $\overline{F}$, iff $A$ is diagonalisable over $\overline{F}$. (The minimal polynomial does not change under field extension, being determined by the linear dependences among $1, A, A^2, \ldots$.) (ii) No nonzero polynomial of degree $< n$ kills $v$, so $\deg \mu_A \geq n$, and $\mu_A$ divides the characteristic polynomial (Cayley-Hamilton). A repeated root makes $\mu_A$ non-squarefree. (iii) If $\mathbb{T}$ is reduced so is each $F[t] \subseteq \mathbb{T}$; conversely a nonzero nilpotent $t$ has $\mu_t = X^k$ with $k \geq 2$. If $t$ is not semisimple, $F[t]$ contains the nonzero nilpotent $gh(t)$. (iv) We do not prove this; see [58, Section 12.3] (section number not verified) or any text on linear algebra. □

**Example 3.105** (the shift operator: $U_p$ on oldforms)**.** Let $V$ have basis $v_0, \ldots, v_e$ ($e \geq 1$) over a field $F$ and let $Uv_j = v_{j-1}$ for $j \geq 1$.

(a) If $Uv_0 = av_0 - pv_1$ with $p \neq 0$, the characteristic polynomial of $U$ is $X^{e-1}(X^2 - aX + p)$, and $v_e$ is a cyclic vector ($U^j v_e = v_{e-j}$). So for $e \geq 3$ the eigenvalue 0 has a single Jordan block, of size $e - 1 \geq 2$, and $U$ is not semisimple. Nevertheless every eigenvalue has an eigenvector: for $e \geq 2$, $U(v_0 - av_1 + pv_2) = (av_0 - pv_1) - av_0 + pv_1 = 0$; and if $\alpha, \beta$ are the roots of $X^2 - aX + p$, $U(v_0 - \beta v_1) = (a - \beta)v_0 - pv_1 = \alpha(v_0 - \beta v_1)$.

(b) If $Uv_0 = av_0$ with $a \neq 0$, the characteristic polynomial is $X^e(X - a)$, $v_e$ is again cyclic, $U$ is not semisimple for $e \geq 2$, and $U(v_0 - av_1) = 0$.

The characteristic polynomials (for $e \leq 4$, with $a$, $p$ as indeterminates) and the cyclicity of $v_e$ were checked with PARI. With $v_j = h(p^j z)$ for a newform $h$ of level $M$, (a) is $U_p$ on the span of the $h(p^j z)$ when $p \nmid M$, and (b) when $p \parallel M$ (Chapter 13); for instance at level 88 with $h = f_{11}$, $p = 2$, $a = a_2(f_{11}) = -2$ and $e = 3$, the characteristic polynomial is $X^2(X^2 + 2X + 2)$ and the algebra generated by $U_2$ on this span is not reduced (Proposition 3.104(iii)).

*In the paper* 3.106. Lemma lem:old: "The operator $U_p$ is not semisimple on this span when 0 is a repeated eigenvalue, but each listed eigenvalue has an eigenvector" is Example 3.105, with the eigenvectors the $p$-stabilisations $h - \beta_p h(pz)$, $h - \alpha_p h(pz)$ and the depleted forms. The same example shows that the full Hecke algebra need not be reduced, which matters in Lemma lem:ddt417 (Proposition 3.110(iv)).

**Lemma 3.107** (common eigenvectors)**.** *Let $F$ be algebraically closed and $V \neq 0$ finite-dimensional over $F$.*

*(i) Let $A_1, \ldots, A_k$ be commuting operators on $V$ and $\lambda = (\lambda_1, \ldots, \lambda_k) \in F^k$, with joint generalised eigenspace $V(\lambda) = \bigcap_i \ker(A_i - \lambda_i)^{\dim V}$. If $V(\lambda) \neq 0$, it contains a common eigenvector of the $A_i$, which can be chosen to be an eigenvector of every element of any commutative algebra $\mathbb{T} \subseteq \operatorname{End}_F(V)$ containing the $A_i$.*

*(ii) If $V = V_1 \otimes \cdots \otimes V_k$ and $A_i$ acts on the $i$-th factor (as $1 \otimes \cdots \otimes A_i \otimes \cdots \otimes 1$), then $V(\lambda) = V_1(\lambda_1) \otimes \cdots \otimes V_k(\lambda_k)$, which is nonzero iff each $\lambda_i$ is an eigenvalue of $A_i$ on $V_i$.*

*Proof.* (i) $V(\lambda)$ is stable under $\mathbb{T}$, since $\mathbb{T}$ commutes with the $A_i$. Apply Theorem 3.100(i) to the image of $\mathbb{T}$ in $\operatorname{End} V(\lambda)$ (characteristic polynomials split as $F$ is algebraically closed): $V(\lambda)$ contains a common eigenvector $v$ of $\mathbb{T}$, and $A_i v = \lambda_i v$ because $A_i - \lambda_i$ is nilpotent on $V(\lambda)$. (ii) Decompose each $V_i = \bigoplus_\mu V_i(\mu)$ into generalised eigenspaces of $A_i$; then $V$ is the direct sum of the $V_1(\mu_1) \otimes \cdots \otimes V_k(\mu_k)$. On such a summand $A_i - \mu_i$ is nilpotent, and if $\mu_i \neq \lambda_i$ then $A_i - \lambda_i$ is invertible on it. So the summands in $V(\lambda)$ are exactly those with $\mu = \lambda$. □

*In the paper* 3.108. Lemma lem:old: "Since $V$ is the tensor product of these local spaces and $\mathbb{T}$ is commutative, for every choice of one such eigenvalue $\mu_p$ at each $p \mid N/M$ there is a common eigenform $g \in V \otimes \overline{\mathbb{Q}}$ of all of $\mathbb{T}$". Here $V = \bigotimes_{p|N/M} \operatorname{span}\{h(p^j z) : 0 \le j \le e_p\}$, with $U_p$ acting on the $p$-th factor, and Lemma 3.107(ii) then (i) give $g$.

## 3.11 Congruence ideals, Gorenstein algebras, and the ingredients of Lemma lem:ddt417

This section assembles the previous ones into the commutative algebra of congruences. A homomorphism $\pi\colon \mathbb{T} \to \mathcal{O}$ (the eigencharacter of a form) has a *congruence ideal* $\eta = \pi(\operatorname{Ann}_{\mathbb{T}} \ker \pi)$, which measures how far the factor of $\pi$ is congruent to the rest of $\mathbb{T}$; it is the paper's $r$ in Lemma lem:crit and the $\eta$ of Lemma lem:ddt417. We prove every module-theoretic step of the proof of Lemma lem:ddt417 and stop short of the lemma itself, which Chapter 20 proves; we then define Gorenstein algebras and prove the freeness criterion behind Remark rem:free. Throughout, $\mathcal{O}$ is a discrete valuation ring with fraction field $K$, and $\mathbb{T}$ is a finite free $\mathcal{O}$-algebra.

**Definition 3.109** (congruence ideal and congruence module)**.** Let $\pi\colon \mathbb{T} \to \mathcal{O}$ be a surjective $\mathcal{O}$-algebra homomorphism, $\wp = \ker \pi$ and $I = \operatorname{Ann}_{\mathbb{T}}(\wp)$. The *congruence ideal* of $\pi$ is $\eta = \pi(I) \subseteq \mathcal{O}$, and the *congruence module* is $\mathbb{T}/(\wp + I)$, which is isomorphic to $\mathcal{O}/\eta$ by Theorem 3.8(iv). Let $\pi_K\colon \mathbb{T}_K \to K$ be the extension of $\pi$ and $\mathbb{T}_K = A_\pi \times \mathbb{T}'_K$ the decomposition in which $A_\pi$ is the local factor through which $\pi_K$ factors (Theorem 3.94(iii)), with idempotent $e_\pi$.

The idea behind the definition: over $K$ the factor of $\pi$ is cut out by the idempotent $e_\pi$, but $e_\pi$ need not lie in $\mathbb{T}$. The ideal $I = \operatorname{Ann}_{\mathbb{T}}(\wp)$ consists of the elements of $\mathbb{T}$ that "live only on the factor of $\pi$" (they are killed by everything that vanishes there), and when $\eta \neq 0$ they are exactly the integral multiples $ae_\pi \in \mathbb{T}$ (Proposition 3.110(iii)). So $\eta$ is the ideal of denominators of $e_\pi$, and the congruence module $\mathcal{O}/\eta$ measures how far the factor of $\pi$ is from splitting off integrally. In $\mathbb{T}_r = \{(a,b) : a \equiv b \bmod \varpi^r\}$ with $\pi(a,b) = a$, the idempotent $(1,0)$ is not in $\mathbb{T}_r$ but $\varpi^r(1,0)$ is, and $\eta = \varpi^r\mathcal{O}$: the two eigencharacters $(a,b) \mapsto a$ and $(a,b) \mapsto b$ of $\mathbb{T}_r$ are congruent modulo $\varpi^r$ and not modulo $\varpi^{r+1}$ (Example 3.112(a)).

**Proposition 3.110** (congruence ideals)**.** *In the setting of Definition 3.109:*

*(i) $I \neq 0$; indeed $I = \mathbb{T} \cap \operatorname{Ann}_{\mathbb{T}_K}(\wp_K)$, where $\wp_K = \wp \otimes K = \ker \pi_K$.*

*(ii) The following are equivalent: (a) $\eta \neq 0$; (b) $\wp \cap I = 0$; (c) $A_\pi = K$, that is, $\mathbb{T}_K = K \times \mathbb{T}'_K$ with $\pi_K$ the first projection.*

*(iii)* *If $\eta \neq 0$: $\pi|_I : I \to \eta$ is an isomorphism, so $I$ is free of rank one over $\mathcal{O}$, generated by any $t \in I$ with $\pi(t)\mathcal{O} = \eta$; $\mathrm{Ann}_{\mathbb{T}}(I) = \wp$; $I = \mathbb{T} \cap (K \times 0)$ and $\wp = \mathbb{T} \cap (0 \times \mathbb{T}'_K)$; and $\eta = \{a \in \mathcal{O} : ae_\pi \in \mathbb{T}\}$, the "denominator" of the idempotent $e_\pi$.*

*(iv)* *If $\mathbb{T}_K$ is reduced, (c) holds automatically. $\mathbb{T}_K$ need not be reduced (Example 3.105); then (c) must be checked directly. For the eigencharacter of a newform it is multiplicity one (Chapter 13).*

*(v)* *(Compatibility.) If $\mathcal{O}$ is complete and $\mathfrak{m} = \mathfrak{m}_\pi$, write $\mathbb{T} = \mathbb{T}_\mathfrak{m} \times \mathbb{T}''$ with $\mathbb{T}'' = \prod_{\mathfrak{m}' \neq \mathfrak{m}} \mathbb{T}_{\mathfrak{m}'}$ (Theorem 3.95). Then $\wp = \wp_\mathfrak{m} \times \mathbb{T}''$ with $\wp_\mathfrak{m} = \ker(\pi|_{\mathbb{T}_\mathfrak{m}})$, and $\mathrm{Ann}_{\mathbb{T}}(\wp) = \mathrm{Ann}_{\mathbb{T}_\mathfrak{m}}(\wp_\mathfrak{m}) \times 0$; so $\eta(\mathbb{T}, \pi) = \eta(\mathbb{T}_\mathfrak{m}, \pi|_{\mathbb{T}_\mathfrak{m}})$: the congruence ideal is computed in $\mathbb{T}_\mathfrak{m}$. The other local factors lie in $\wp$, and their component of the annihilator is 0. For a flat extension $\mathcal{O} \to \mathcal{O}'$ of discrete valuation rings (for example $\mathbb{Z}_{(\ell)} \to \mathbb{Z}_\ell$), $\eta(\mathbb{T} \otimes \mathcal{O}', \pi \otimes \mathcal{O}') = \eta\mathcal{O}'$.*

*Proof.* (i) For $t \in \mathbb{T}$, $t\wp = 0$ iff $t\wp_K = 0$, as $\wp_K = K\wp$. Write $\mathbb{T}_K = A_\pi \times \mathbb{T}'_K$; then $\wp_K = \mathfrak{m}_{A_\pi} \times \mathbb{T}'_K$ and $\mathrm{Ann}_{\mathbb{T}_K}(\wp_K) = \mathrm{Ann}_{A_\pi}(\mathfrak{m}_{A_\pi}) \times 0$. In the local Artinian ring $A_\pi$ the maximal ideal is nilpotent (Theorem 3.93); if $\mathfrak{m}_{A_\pi}^{c+1} = 0 \neq \mathfrak{m}_{A_\pi}^{c}$ (with $\mathfrak{m}^0 = A_\pi$), then $0 \neq \mathfrak{m}_{A_\pi}^{c} \subseteq \mathrm{Ann}(\mathfrak{m}_{A_\pi})$. Every nonzero $y \in \mathbb{T}_K$ has a nonzero multiple $\varpi^s y \in \mathbb{T}$; so $I \neq 0$. (ii) (a)⇒(b): choose $t \in I$ with $\pi(t)$ a generator of $\eta$. For $x \in \wp \cap I$ write $\pi(t)x = tx + (\pi(t) - t)x$. Here $tx \in I\wp = 0$, and $(\pi(t) - t)x \in \wp I = 0$ because $\pi(t) - t \in \wp$. So $\pi(t)x = 0$, and $x = 0$ because $\mathbb{T}$ is torsion-free and $\pi(t) \neq 0$. (b)⇒(a): if $\eta = 0$ then $I \subseteq \wp$, so $\wp \cap I = I \neq 0$ by (i). (a)⇔(c): if $A_\pi$ is not a field, then $\mathfrak{m}_{A_\pi} \neq 0$ and $\mathrm{Ann}_{A_\pi}(\mathfrak{m}_{A_\pi}) \subseteq \mathfrak{m}_{A_\pi}$ (a unit cannot kill a nonzero ideal), so $I \subseteq \mathbb{T} \cap (\mathfrak{m}_{A_\pi} \times 0) \subseteq \wp_K$ and $\eta = \pi(I) = 0$. If $A_\pi$ is a field, it is $K$ ($\pi_K$ maps it onto $K$ as a $K$-algebra). Then $e_\pi = (1, 0)$ has a nonzero multiple $ae_\pi \in \mathbb{T}$ with $a \in \mathcal{O}$; it kills $\wp_K = 0 \times \mathbb{T}'_K$, so lies in $I$, and $\pi(ae_\pi) = a \neq 0$. (iii) The kernel of $\pi|_I$ is $\wp \cap I = 0$, and its image is $\eta$ by definition; $\eta \simeq \mathcal{O}$. If $xI = 0$ then $\pi(x)\pi(t) = \pi(xt) = 0$, so $\pi(x) = 0$; conversely $\wp I = 0$. By (c), $\wp_K = 0 \times \mathbb{T}'_K$ and $\mathrm{Ann}_{\mathbb{T}_K}(\wp_K) = K \times 0$, so (i) gives $I = \mathbb{T} \cap (K \times 0)$, and $\wp = \mathbb{T} \cap \ker \pi_K = \mathbb{T} \cap (0 \times \mathbb{T}'_K)$. An element of $\mathbb{T} \cap (K \times 0)$ is $ce_\pi$ with $c = \pi_K(ce_\pi)$; so $\eta = \{c : ce_\pi \in \mathbb{T}\}$ (such $c$ lie in $\mathcal{O}$ automatically). (iv) If $\mathbb{T}_K$ is reduced its local factors are fields (Theorem 3.93(iii)), and $A_\pi = K$ as in (ii). (v) $\pi$ kills $\mathbb{T}''$ (Theorem 3.95(iv)), so $\wp = \wp_\mathfrak{m} \times \mathbb{T}''$. An element $(a, b)$ kills $\wp$ iff $a\wp_\mathfrak{m} = 0$ and $b\mathbb{T}'' = 0$, that is, $b = 0$; so $\mathrm{Ann}_{\mathbb{T}}(\wp) = \mathrm{Ann}_{\mathbb{T}_\mathfrak{m}}(\wp_\mathfrak{m}) \times 0$, and $\pi(\mathrm{Ann}_{\mathbb{T}} \wp) = \pi(\mathrm{Ann}_{\mathbb{T}_\mathfrak{m}} \wp_\mathfrak{m})$. For a flat $\mathcal{O} \to \mathcal{O}'$, $\ker(\pi \otimes \mathcal{O}') = \wp \otimes \mathcal{O}'$ and $\mathrm{Ann}(\wp \otimes \mathcal{O}') = I \otimes \mathcal{O}'$ (Proposition 3.29(iii); $\wp$ is finitely generated), so $\eta(\mathbb{T} \otimes \mathcal{O}') = \pi(I)\mathcal{O}'$. □

*In the paper* 3.111. Lemma lem:crit: "$I = \mathbb{T} \cap (0 \times \mathbb{T}')$ and $J = \mathbb{T} \cap (\mathbb{Q} \times 0)$, so $I \cap J = 0$ and $\mathbb{T}/(I + J) \simeq \mathbb{Z}/\phi_f(J)$" is Proposition 3.110(iii) over $\mathcal{O} = \mathbb{Z}_{(\ell)}$ (or over $\mathbb{Z}$ verbatim, the proofs using only that $\mathbb{Z}$ is a principal ideal domain), with $\pi = \phi_f$, $\wp = \ker \phi_f$ (the paper's $I$ there) and $\mathrm{Ann}(\wp)$ (the paper's $J$); Appendix B, convention B-C13(c), explains the clash of letters between Lemmas lem:crit and lem:ddt417. In Lemma lem:ddt417 the paper proves $\wp \cap I = 0$ by the argument of (ii), and remarks that reducedness would not do. The paragraph after Lemma lem:hecke says (and Step 1 of the proof of Theorem thm:quant says the same with $\ker \pi$) that "the congruence ideal is computed in $\mathbb{T}_\mathfrak{m}$: the other local factors of $\mathbb{T} \otimes \mathbb{Z}_\ell$ lie in the kernel $\wp$, $\phi_f$ factors through $\mathbb{T}_\mathfrak{m}$, and $\mathrm{Ann}(\wp)$ decomposes along the idempotents": this is (v).

**Example 3.112** (toy congruences)**.** All entries were computed by hand from the definitions (and, for $\varpi = 3$, $r = 1, 2$, checked by computer).

| $\mathbb{T}$, $\pi$ | $\wp$ | $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$ | $\eta$ | Gorenstein? |
|---|---|---|---|---|
| (a) $\mathbb{T}_r$, $\pi(a,b) = a$ | $0 \times \varpi^r\mathcal{O}$ | $\varpi^r\mathcal{O} \times 0$ | $\varpi^r\mathcal{O}$ | yes |
| (b) $\mathbb{Z}[x]/(x^2-4)$, $\pi(x) = 2$ | $\mathbb{Z}(x-2)$ | $\mathbb{Z}(x+2)$ | $4\mathbb{Z}$ | yes |
| (c) $\mathbb{T}_3^\flat$, $\pi(a,b,c) = a$ | $0 \times \varpi\mathcal{O} \times \varpi\mathcal{O}$ | $\varpi\mathcal{O} \times 0 \times 0$ | $\varpi\mathcal{O}$ | no |
| (d) $\mathcal{O}[x]/(x^2)$, $\pi(x) = 0$ | $\mathcal{O}x$ | $\mathcal{O}x$ | $0$ | yes |
| (e) $\mathbb{Z}[x]/(x^2(x-1))$, $\pi(x) = 0$ | $(x)$ | $\mathbb{Z}(x-x^2)$ | $0$ | yes |
| (e′) $\mathbb{Z}[x]/(x^2(x-1))$, $\pi(x) = 1$ | $(x-1)$ | $\mathbb{Z}x^2$ | $\mathbb{Z}$ | yes |

Here $\mathbb{T}_r = \{(a,b) \in \mathcal{O}^2 : a \equiv b \bmod \varpi^r\}$ and $\mathbb{T}_3^\flat = \{(a,b,c) \in \mathcal{O}^3 : a \equiv b \equiv c \bmod \varpi\}$. (a) $\mathbb{T}_r \simeq \mathcal{O}[x]/(x(x-\varpi^r))$ with $x = (\varpi^r, 0)$; $e_\pi = (1,0) = x/\varpi^r$, with denominator $\varpi^r$; the congruence module is $\mathcal{O}/\varpi^r$: "two eigenforms congruent modulo $\varpi^r$". (b) $(a+bx)(x-2) = (a-2b)x+(4b-2a)$ vanishes iff $a = 2b$, so $I = \mathbb{Z}(x+2)$, $\pi(x+2) = 4$, $e_\pi = (x+2)/4$, congruence module $\mathbb{Z}/4$; under $\mathbb{T} \simeq \{(a,b) : a \equiv b \bmod 4\}$ this is (a) with $\varpi^r = 4$. (c) An element $(a,b,c)$ kills $\wp$ iff $b = c = 0$; three pairwise congruent eigenforms, congruence ideal $\varpi\mathcal{O}$. (d) Here $\wp \cap I = \mathcal{O}x \neq 0$ and $\eta = 0$: the factor of $\pi$ in $\mathbb{T}_K = K[x]/(x^2)$ is not a field. (e) For $\pi(x) = 0$ the factor of $\mathbb{T}_{\mathbb{Q}}$ is $\mathbb{Q}[x]/(x^2)$; the annihilator of $x$ consists of the $c(x^2-x)$ (use $x^3 = x^2$), and $\pi(x-x^2) = 0$. For $\pi(x) = 1$ the factor is $\mathbb{Q}$; $I = \mathbb{Z}x^2$ with $x^2$ an idempotent of $\mathbb{T}$, so $\eta = \mathbb{Z}$ and the congruence module is 0: no congruence. The Gorenstein column is Proposition 3.115(iii).

**Proposition 3.113** (the module ingredients of Lemma lem:ddt417)**.** *In the setting of Definition 3.109 with $\eta \neq 0$, choose $t \in I$ with $\pi(t)\mathcal{O} = \eta$. Let $L$ be a $\mathbb{T}$-module that is free of finite rank over $\mathcal{O}$, with a perfect $\mathcal{O}$-bilinear pairing for which every element of $\mathbb{T}$ is self-adjoint. Then:*

*(i)* $L[\wp]$ *and* $L[I]$ *are saturated in* $L$.

*(ii)* $L[\wp] \cap L[I] = 0$*, so the sum* $L[\wp] + L[I]$ *is direct.*

*(iii)* $L[\wp]^\perp = L[I] = {}^\perp L[\wp]$ *and* $L[I]^\perp = L[\wp] = {}^\perp L[I]$.

*(iv) More generally* $(\mathfrak{a}L)^\perp = L[\mathfrak{a}]$ *for every ideal* $\mathfrak{a} \subseteq \mathbb{T}$.

*(v) For every* $x \in L$, $\pi(t)x = tx + (\pi(t)-t)x$ *with* $tx \in L[\wp]$ *and* $(\pi(t)-t)x \in L[I]$*; so* $Q = L/(L[\wp] \oplus L[I])$ *is killed by* $\eta$.

*(vi) If* $L \simeq \mathbb{T}^r$ *as* $\mathbb{T}$*-modules, then* $L[\wp] = I^r$ *(free of rank* $r$ *over* $\mathcal{O}$*),* $L[I] = \wp^r$*, and* $Q \simeq (\mathbb{T}/(\wp+I))^r \simeq (\mathcal{O}/\eta)^r$.

*Proof.* (i) Proposition 3.49(iii), applied to the action of $\mathbb{T}$ on the lattice $L$ and the ideals $\wp$ and $I$. (ii) $\pi(t) = t + (\pi(t)-t) \in I + \wp$ (here $\pi(t)$ stands for $\pi(t) \cdot 1 \in \mathbb{T}$, and $\pi(\pi(t)-t) = 0$), so an element killed by $\wp$ and by $I$ is killed by $\pi(t) \neq 0$, hence is 0 ($L$ is torsion-free over $\mathcal{O}$). Directness is Proposition 3.13. (iii) Let $x \in L[\wp]$ and $y \in L[I]$. Since $(\pi(t)-t)x \in \wp x = 0$, $\pi(t)\langle x,y\rangle = \langle tx,y\rangle = \langle x,ty\rangle = 0$, and $\langle x,y\rangle = 0$; likewise $\langle y,x\rangle = 0$. So $L[I] \subseteq L[\wp]^\perp \cap {}^\perp L[\wp]$. Conversely let $y \in L[\wp]^\perp$. For $s \in I$ and $z \in L$, $sz \in L[\wp]$ (as $\wp s = 0$), so $\langle z, sy\rangle = \langle sz, y\rangle = 0$; by perfectness $sy = 0$, and $y \in L[I]$. The same argument with the arguments exchanged gives ${}^\perp L[\wp] = L[I]$. Then $L[I]^\perp = ({}^\perp L[\wp])^\perp = L[\wp]$ and ${}^\perp L[I] = {}^\perp(L[\wp]^\perp) = L[\wp]$ by Proposition 3.83(ii), $L[\wp]$ being saturated. (iv) is Proposition 3.89(ii). (v) $\wp tx \subseteq \wp Ix = 0$; and for $s \in I$, $s(\pi(t)-t) \in I\wp = 0$. (vi) Componentwise, $\mathbb{T}^r[\mathfrak{a}] = \mathrm{Ann}_{\mathbb{T}}(\mathfrak{a})^r$ (Proposition 3.68(iii)); $\mathrm{Ann}_{\mathbb{T}}(\wp) = I$ and $\mathrm{Ann}_{\mathbb{T}}(I) = \wp$ (Proposition 3.110(iii)). So $Q = \mathbb{T}^r/(I^r \oplus \wp^r) = (\mathbb{T}/(I+\wp))^r$, and $\mathbb{T}/(\wp+I) \simeq \mathcal{O}/\eta$. □

**Definition 3.114** (Gorenstein algebras)**.** Let $\mathbb{T}^\vee = \mathrm{Hom}_{\mathcal{O}}(\mathbb{T}, \mathcal{O})$, a $\mathbb{T}$-module by $(t\varphi)(u) = \varphi(tu)$. The finite free $\mathcal{O}$-algebra $\mathbb{T}$ is *Gorenstein* if $\mathbb{T}^\vee$ is free of rank one as a $\mathbb{T}$-module.

The module $\mathbb{T}^\vee$ is the abstract version of a space of forms: for the Hecke algebra over $\mathbb{Z}$ the pairing $\langle g,t\rangle = a_1(tg)$ identifies $S_2(\Gamma_0(N);\mathbb{Z})$ with $\mathbb{T}^\vee$ (Proposition 3.91(i) and Paper link 3.92). Freeness of rank one says that this space of forms is a copy of $\mathbb{T}$ itself, generated by one element, a property closely tied to multiplicity one (Theorem 3.116). It fails for the algebra $\mathbb{T}_3^\flat$ of three pairwise congruent eigenforms (Proposition 3.115(iii)), whose reduction $k[u,w]/(u^2,uw,w^2)$ has a two-dimensional socle.

**Proposition 3.115** (Gorenstein criteria)**.** *(i) $\mathbb{T}$ is Gorenstein iff there is a perfect $\mathcal{O}$-bilinear pairing $\mathbb{T}\times\mathbb{T}\to\mathcal{O}$ with $\langle st,u\rangle = \langle s,tu\rangle$; one may take $\langle s,t\rangle = \lambda(st)$ for a generator $\lambda$ of $\mathbb{T}^\vee$.*

*(ii) If $\mathcal{O}$ is complete, $\mathbb{T}$ is Gorenstein iff each local factor $\mathbb{T}_\mathfrak{m}$ is. If $\mathbb{T}$ is local, with maximal ideal $\mathfrak{m}$ and residue field $k_\mathbb{T}$, the following are equivalent: $\mathbb{T}$ is Gorenstein; $\dim_{k_\mathbb{T}}(\mathbb{T}/\varpi\mathbb{T})[\mathfrak{m}] = 1$ (the* socle *of $\mathbb{T}/\varpi\mathbb{T}$ is a line); $\dim_{k_\mathbb{T}} \mathbb{T}^\vee/\mathfrak{m}\mathbb{T}^\vee = 1$.*

*(iii) Examples: $\mathcal{O}[x]/(f)$ for monic $f$ is Gorenstein; $\mathbb{T}_r$ of Example 3.112(a) is Gorenstein; $\mathbb{T}_3^\flat$ of (c) is not, although $\mathbb{T}_3^\flat\otimes K = K^3$ is reduced; $\mathcal{O}[x]/(x^2)$ is Gorenstein and not reduced.*

*(iv) Complete intersections (quotients $\mathcal{O}[[x_1,\dots,x_n]]/(f_1,\dots,f_n)$ that are finite free over $\mathcal{O}$) are Gorenstein; and for $\mathbb{T}$ local Gorenstein and $\pi$ as in Definition 3.109, Wiles's numerical criterion compares $\eta$ with $\wp/\wp^2$ (stated; see Chapter 20).*

*Proof.* (i) If $\lambda$ generates $\mathbb{T}^\vee$ freely, $s\mapsto s\lambda = \lambda(s\cdot)$ is an isomorphism $\mathbb{T}\to\mathbb{T}^\vee$, which says that $\lambda(st)$ is perfect. Conversely, given the pairing, $s\mapsto\langle s,-\rangle$ is a $\mathbb{T}$-linear isomorphism $\mathbb{T}\to\mathbb{T}^\vee$ (Proposition 3.91(i) with $S=\mathbb{T}$). (ii) $\mathbb{T}^\vee = \prod_\mathfrak{m}\mathbb{T}^\vee_\mathfrak{m}$ compatibly with $\mathbb{T} = \prod\mathbb{T}_\mathfrak{m}$, and a module over a product is free of rank one iff each component is. Let $\mathbb{T}$ be local and $A = \mathbb{T}/\varpi\mathbb{T}$, a local $k$-algebra with maximal ideal $\bar{\mathfrak{m}} = \mathfrak{m}/\varpi\mathbb{T}$ and $A/\bar{\mathfrak{m}} = k_\mathbb{T}$. As $\mathbb{T}$ is free, $\mathbb{T}^\vee/\varpi\mathbb{T}^\vee = \mathrm{Hom}_k(A,k) =: A'$. Under the perfect pairing $A\times A'\to k$, $(x,\varphi)\mapsto\varphi(x)$, the orthogonal of the subspace $\bar{\mathfrak{m}}A'$ is $\{x : \varphi(ax) = 0\ \forall a\in\bar{\mathfrak{m}},\varphi\in A'\} = A[\bar{\mathfrak{m}}]$. So $\dim_k \mathbb{T}^\vee/\mathfrak{m}\mathbb{T}^\vee = \dim_k A'/\bar{\mathfrak{m}}A' = \dim_k A[\bar{\mathfrak{m}}]$, and dividing by $[k_\mathbb{T}:k]$ gives the equality of $k_\mathbb{T}$-dimensions. By Nakayama (Theorem 3.35(iii)), $\dim_{k_\mathbb{T}}\mathbb{T}^\vee/\mathfrak{m}\mathbb{T}^\vee = 1$ iff $\mathbb{T}^\vee$ is cyclic, $\mathbb{T}\twoheadrightarrow\mathbb{T}^\vee$. Both are free $\mathcal{O}$-modules of the same rank, so this surjection is an isomorphism (compose with an $\mathcal{O}$-isomorphism $\mathbb{T}^\vee\simeq\mathbb{T}$ and use Lemma 3.32(b)), and $\mathbb{T}^\vee$ is free of rank one. (iii) For $f$ monic of degree $n$, with basis $1,x,\dots,x^{n-1}$, let $\lambda$ be the coefficient of $x^{n-1}$. Then $\lambda(x^ix^j) = 0$ for $i+j<n-1$ and $=1$ for $i+j=n-1$, so the Gram matrix of $\lambda(st)$ is triangular with respect to the antidiagonal, with 1's on it, and has determinant $\pm1$; apply (i). This covers $\mathcal{O}[x]/(x^2)$ and $\mathbb{T}_r\simeq\mathcal{O}[x]/(x(x-\varpi^r))$. Directly for $\mathbb{T}_r$: $\lambda(a,b) = (a-b)/\varpi^r$ is $\mathcal{O}$-valued on $\mathbb{T}_r$, and in the basis $(1,1)$, $(\varpi^r,0)$ the Gram matrix of $\lambda(st)$ is $\begin{pmatrix}0&1\\1&\varpi^r\end{pmatrix}$, of determinant $-1$. For $\mathbb{T}_3^\flat$, local with basis $(1,1,1)$, $u=(\varpi,0,0)$, $w=(0,\varpi,0)$: modulo $\varpi$, $u^2=\varpi u\equiv0$, $uw=0$, $w^2\equiv0$, so $\mathbb{T}_3^\flat/\varpi = k[u,w]/(u^2,uw,w^2)$, whose socle $ku+kw$ has dimension 2. (iv) We do not prove this; see [62, Chapter 21] and [38, Section 4]. □

**Theorem 3.116** (a freeness criterion in the style of Mazur)**.** *Let $\mathbb{T}$ be a local finite free $\mathcal{O}$-algebra, with maximal ideal $\mathfrak{m}$ and residue field $k_\mathbb{T}$, and let $L\neq0$ be a $\mathbb{T}$-module, free of finite rank over $\mathcal{O}$, with a perfect $\mathcal{O}$-bilinear pairing for which every element of $\mathbb{T}$ is self-adjoint. Suppose*

*(a) $\dim_{k_\mathbb{T}} L/\mathfrak{m}L = d$, and*

*(b) $\mathrm{rank}_\mathcal{O} L = d\cdot\mathrm{rank}_\mathcal{O}\mathbb{T}$.*

*Then $L \simeq \mathbb{T}^d$ as $\mathbb{T}$-modules, and $\mathbb{T}$ is Gorenstein.*

*Proof.* $L$ is a finitely generated $\mathbb{T}$-module (it is even finitely generated over $\mathcal{O}$), so by Nakayama (Theorem 3.35(iii)) any $d$ elements whose images form a $k_{\mathbb{T}}$-basis of $L/\mathfrak{m}L$ generate it, and there is a surjection $\mathbb{T}^d \twoheadrightarrow L$. Both are free over $\mathcal{O}$ of rank $d \operatorname{rank} \mathbb{T}$, by (b), so it is an isomorphism: composing with an $\mathcal{O}$-isomorphism $L \simeq \mathbb{T}^d$ gives a surjective endomorphism of a finitely generated module, which is bijective by Lemma 3.32(b) (as in the proof of Proposition 3.115(ii)). The pairing gives a $\mathbb{T}$-linear isomorphism $L \simeq L^\vee$, $y \mapsto \langle -, y\rangle$ (linear because $\langle -, ty\rangle = \langle t\cdot, y\rangle$), and $L^\vee \simeq (\mathbb{T}^d)^\vee = (\mathbb{T}^\vee)^d$ (Hom out of a finite direct sum, Proposition 3.19(i)). So $(\mathbb{T}^\vee)^d \simeq \mathbb{T}^d$ as $\mathbb{T}$-modules, and reducing modulo $\mathfrak{m}$, $(\mathbb{T}^\vee/\mathfrak{m}\mathbb{T}^\vee)^d \simeq k_{\mathbb{T}}^d$; comparing $k_{\mathbb{T}}$-dimensions, $d \dim_{k_{\mathbb{T}}} \mathbb{T}^\vee/\mathfrak{m}\mathbb{T}^\vee = d$, and since $d \geq 1$ ($L \neq 0$, so $L/\mathfrak{m}L \neq 0$ by Nakayama), $\dim_{k_{\mathbb{T}}} \mathbb{T}^\vee/\mathfrak{m}\mathbb{T}^\vee = 1$ and $\mathbb{T}$ is Gorenstein by Proposition 3.115(ii). □

*In the paper* 3.117. Remark rem:free: "Freeness of $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ over $\mathbb{T}_{\mathfrak{m}}$ holds when $J_0(N)[\mathfrak{m}]$ is two-dimensional, since the localised Hecke algebra is then Gorenstein and the Tate module is free of rank two". In Theorem 3.116 take $\mathcal{O} = \mathbb{Z}_\ell$, $\mathbb{T} = \mathbb{T}_{\mathfrak{m}}$, $L = H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ with the pairing $\hat{\delta}_!$ (perfect on the $\mathfrak{m}$-component by Proposition 3.89(iii)), and $d = 2$. Hypothesis (a) is the two-dimensionality of $J_0(N)[\mathfrak{m}]$, transported to $L/\mathfrak{m}L$ by duality, and (b) is a rank count from the Eichler-Shimura isomorphism; both are Chapter 14. The conclusion $L \simeq \mathbb{T}_{\mathfrak{m}}^2$ is the hypothesis of the equality clause of Lemma lem:ddt417. After Proposition prop:pullback the paper does not need this for Theorems thm:main and thm:quant.

*Remark* 3.118 (the statement of Lemma lem:ddt417, and where each step is proved). In the book's notation the paper's Lemma lem:ddt417 reads: *let $\mathbb{T}$ be a finite free $\mathcal{O}$-algebra acting on a module $L$ that is free of finite rank over $\mathcal{O}$, with a perfect pairing for which $\mathbb{T}$ is self-adjoint; let $\pi\colon \mathbb{T} \twoheadrightarrow \mathcal{O}$, $\wp = \ker \pi$, $I = \operatorname{Ann}_{\mathbb{T}}(\wp)$ and $\eta = \pi(I)$; suppose $L[\wp]$ has rank $d$ and Gram determinant $\Delta \neq 0$. Then $v(\Delta) \leq d\, v(\eta)$, that is, $\eta^d \subseteq \Delta\mathcal{O}$, with equality if $L$ is free over $\mathbb{T}$.* It is proved in Chapter 20. Every step except the final count is proved in this chapter:

| step of the paper's proof | proved in |
|---|---|
| $\wp \cap I = 0$ when $\eta \neq 0$ | Proposition 3.110(ii) |
| $L[\wp] + L[I]$ is direct | Proposition 3.113(ii) |
| $Q = L/(L[\wp] \oplus L[I])$ is killed by $\eta$ | Proposition 3.113(v) |
| $L[\wp]$ is saturated | Proposition 3.49(iii), Proposition 3.113(i) |
| $L/L[\wp]^\perp \simeq \operatorname{Hom}(L[\wp], \mathcal{O})$ | Theorem 3.84(i) |
| $L[\wp]^\perp = L[I]$ | Proposition 3.113(iii) |
| $Q \simeq \operatorname{coker}(L[\wp] \to L[\wp]^\vee)$, of length $v(\Delta)$ | Theorem 3.84(ii) |
| $Q$ is generated by $d$ elements | Theorem 3.84(ii) ($L[\wp]^\vee \simeq \mathcal{O}^d$ surjects onto $Q$) |
| if $L \simeq \mathbb{T}^d$, $Q \simeq (\mathcal{O}/\eta)^d$ | Proposition 3.113(vi) |
| the Gram determinant in rank two is $\eta^2$ | Proposition 3.87(ii) |

The remaining step is that a module generated by $d$ elements and killed by $\eta$ is a quotient of $(\mathcal{O}/\eta)^d$, of length at most $d\, v(\eta)$. *Checks on the toy algebra* $\mathbb{T}_r$ (all computed): with $L = \mathbb{T}_r$ and $\langle s, u\rangle = \lambda(su)$, $\lambda(a, b) = (a - b)/\varpi^r$, one has $L[\wp] = I = \mathcal{O}(\varpi^r, 0)$, $d = 1$ and $\Delta = \lambda((\varpi^r, 0)^2) = \varpi^r$, so $v(\Delta) = r = d\, v(\eta)$ (free case, equality). With $L = \mathcal{O}^2$, $\mathbb{T}_r$ acting diagonally and the standard pairing, $L[\wp] = \mathcal{O}e_1$ and $\Delta = 1$, so $0 = v(\Delta) < r$ (strict inequality; $L$ is not free over $\mathbb{T}_r$). With $L = \mathbb{T}_r^2$ and the alternating pairing $\langle x, y\rangle = \lambda(x_1y_2 - x_2y_1)$ (perfect, with $\mathbb{T}_r$ self-adjoint), $L[\wp] = I^2$ has basis $(u, 0)$, $(0, u)$ with $u = (\varpi^r, 0)$, $d = 2$, and $\Delta = \lambda(u^2)^2 = \varpi^{2r} = \eta^2$: the shape of Lemma lem:hecke and of Step 2 of the proof of Theorem thm:quant. Finally, when $\eta = 0$ the

inequality is empty, and if moreover $L \simeq \mathbb{T}^r$, then $I \subseteq \wp$ gives $I \cdot I = 0$, so the pairing vanishes on $L[\wp] = I^r$ ($\langle \sum a_i e_i, \sum b_j e_j \rangle = \sum \langle b_j a_i e_i, e_j \rangle = 0$) and $\Delta = 0$: the hypothesis $\Delta \neq 0$ excludes this case.

*The paper's proof.* The paper works over a discrete valuation ring with finite residue field and counts orders; the book's lengths make sense over any discrete valuation ring (Convention 3.1(f)), and the lemma is proved in that form in Chapter 20. In the converse inclusion "if $y \perp L[\wp]$ then $\dots$ $ty = 0$, and $y \in L[I]$ because $I = \mathcal{O}t$", the single element $t$ suffices because $I = \mathcal{O}t$ (Proposition 3.110(iii)); Proposition 3.113(iii) uses every $s \in I$ instead. The directness of $L[\wp] + L[I]$ rests, in the paper as in Proposition 3.113(ii), on the nonzero scalar $\pi(t) \in \wp + I$.

## 3.12 Exercises

The exercises are graded from routine to harder; those marked "check the paper" verify a sentence of the paper with the tools of this chapter.

**Exercise 3.119.** Compute $\mathbb{Z}/m \otimes \mathbb{Z}/n$, $\mathbb{Q} \otimes \mathbb{Z}/n$, $\mathbb{Z}_\ell \otimes \mathbb{Z}/n$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell \otimes \mathbb{Z}/n$, $\mathrm{Hom}(\mathbb{Z}/m, \mathbb{Z}/n)$ and $\mathrm{Hom}_{\mathbb{Z}_\ell}(\mathbb{Q}_\ell/\mathbb{Z}_\ell, \mathbb{Z}/\ell^k)$. Show that $\mathrm{Hom}(\mathbb{Z}/2, -)$ is not right exact and $- \otimes \mathbb{Z}/2$ is not left exact on $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$. (Answers: $\mathbb{Z}/\gcd(m,n)$, $0$, $\mathbb{Z}/\ell^{v_\ell(n)}$, $0$, $\mathbb{Z}/\gcd(m,n)$, $0$.)

**Exercise 3.120** (check the paper: Lemma lem:IV)**.** Let $T$ be free of rank 2 over $\mathbb{Z}_3$ with an automorphism $\sigma$ such that $\sigma^2 + \sigma + 1 = 0$. Using that $\mathbb{Z}_3[\zeta_3]$ is a discrete valuation ring with uniformiser $1 - \zeta_3$ (Chapter 6), show that $T$ is free of rank one over $\mathbb{Z}_3[X]/(X^2 + X + 1) \simeq \mathbb{Z}_3[\zeta_3]$ (Proposition 3.46(iii)), that $T/3T \simeq \mathbb{Z}_3[\zeta_3]/(1-\zeta_3)^2$ (note $(1-\zeta_3)^2 = -3\zeta_3$), and that the vectors of $T/3T$ fixed by $\sigma$ form a line. Deduce that in a suitable basis $\sigma = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$.

**Exercise 3.121.** Show that Fitting ideals are multiplicative in short exact sequences of finite $\mathbb{Z}_\ell$-modules. Show that they are not over $\mathbb{Z}[x]$, using $0 \to R/(2,x) \to R/(4,x) \to R/(2,x) \to 0$ (Proposition 3.71(vi)). Over $R = k[x]/(x^2)$, show that $\mathrm{Fitt}(k) = (x)$ and that $0 \to k \to R \to k \to 0$ and $0 \to k \to k \oplus k \to k \to 0$ both satisfy $\mathrm{Fitt}(M) = \mathrm{Fitt}(M')\,\mathrm{Fitt}(M'')$ (both sides are $0$). Compute Fitt and Ann of $\mathbb{Z}[x]/(2,x)$ and of its square.

**Exercise 3.122** (higher Fitting ideals)**.** For $M$ with $n$ generators and a relation matrix $A$, let $\mathrm{Fitt}_i(M)$ be the ideal generated by the $(n-i) \times (n-i)$ minors of $A$, with $\mathrm{Fitt}_i(M) = R$ for $i \geq n$ (so $\mathrm{Fitt}_0$ is the Fitting ideal of Definition 3.69). (a) Show that $\mathrm{Fitt}_i(M)$ does not depend on the presentation, and that $\mathrm{Fitt}_0(M) \subseteq \mathrm{Fitt}_1(M) \subseteq \cdots$. (Hint: the proof of Lemma 3.70 works for minors of size $n - i$; for an added generator, column operations turn the new relation matrix into $\mathrm{diag}(A, 1)$, and Laplace expansion puts its minors of size $n + 1 - i$ in the ideal of minors of size $n - i$ of $A$.) (b) Over a discrete valuation ring let $M = \bigoplus_{j=1}^{s} \mathcal{O}/\varpi^{a_j}$ with $1 \leq a_1 \leq \cdots \leq a_s$. Show that $\mathrm{Fitt}_i(M) = \varpi^{a_1 + \cdots + a_{s-i}}\mathcal{O}$ for $i < s$ and $\mathrm{Fitt}_i(M) = \mathcal{O}$ for $i \geq s$. Deduce that the $\mathrm{Fitt}_i$ determine a torsion $M$ up to isomorphism, that $\dim_k M[\varpi]$ is the least $i$ with $\mathrm{Fitt}_i(M) = \mathcal{O}$, and that $M$ is cyclic iff $\mathrm{Fitt}_1(M) = \mathcal{O}$. (c) Over $\mathbb{Z}_\ell$, compute $\mathrm{Fitt}_0$ and $\mathrm{Fitt}_1$ of $\mathbb{Z}/\ell^2$ and of $(\mathbb{Z}/\ell)^2$. (Answers: $\mathrm{Fitt}_0 = \ell^2\mathbb{Z}_\ell$ for both; $\mathrm{Fitt}_1 = \mathbb{Z}_\ell$ and $\ell\mathbb{Z}_\ell$. So $\mathrm{Fitt}_1$ separates the two modules that Theorem 3.72(iv) shows $\mathrm{Fitt}_0$ cannot.)

**Exercise 3.123** (cofinitely generated modules)**.** (a) Show that a quotient of a divisible module is divisible. Let $W$ be cofinitely generated of corank $r$ and $F \subseteq W$ a finite submodule; show that $W/F$ is cofinitely generated of corank $r$, and that $W/F \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$ if $W$ is divisible.

(Hint: dualising, $(W/F)^*$ is the kernel of $W^* \to F^*$, a submodule of finite index of $W^*$; use Theorem 3.77.) (b) Show that multiplication by $\ell$ identifies $\mathbb{Q}_\ell/\mathbb{Z}_\ell/(\mathbb{Q}_\ell/\mathbb{Z}_\ell)[\ell]$ with $\mathbb{Q}_\ell/\mathbb{Z}_\ell$, and that under this identification the dual of the quotient map $\mathbb{Q}_\ell/\mathbb{Z}_\ell \to \mathbb{Q}_\ell/\mathbb{Z}_\ell/(\mathbb{Q}_\ell/\mathbb{Z}_\ell)[\ell]$ is the inclusion $\ell\mathbb{Z}_\ell \subseteq \mathbb{Z}_\ell$. (c) For $W = \mathbb{Q}_\ell/\mathbb{Z}_\ell \oplus \mathbb{Z}/\ell$ compute $W[\ell]$, the divisible part $\bigcap_k \ell^k W$, the corank and $W^*$. (Answers: $(\mathbb{Z}/\ell)^2$; $\mathbb{Q}_\ell/\mathbb{Z}_\ell \oplus 0$; 1; $\mathbb{Z}_\ell \oplus \mathbb{Z}/\ell$.)

**Exercise 3.124** (the Gram determinant of a sublattice of finite index; the prime 2)**.** (a) Let $L' \subseteq L$ be lattices of the same rank over $\mathcal{O}$, $B$ the matrix whose rows are the coordinates of a basis of $L'$ in a basis of $L$, and $G$ the Gram matrix of a pairing on $L$. Show that the Gram matrix of $L'$ is $BGB^{\mathsf{T}}$, so that $v(\Delta_{L'}) = v(\Delta_L) + 2\,\mathrm{length}(L/L')$ (Theorem 3.50). (b) On $L = \mathbb{Z}_2^2$ with the perfect alternating pairing $\langle x, y\rangle = x_1y_2 - x_2y_1$, let $L' = \mathbb{Z}_2 e_1 \oplus \mathbb{Z}_2 \cdot 2e_2$, of index 2. Compute the Gram determinant and the ideal of values of $L'$, and check (a) and Proposition 3.87(ii). (Answer: $\Delta = 4$, values $2\mathbb{Z}_2$.) (c) Show that $\frac{1}{2}\langle\ ,\ \rangle$ is perfect on $L'$ but not $\mathbb{Z}_2$-valued on $L$, while for $\ell$ odd $\frac{1}{2}\langle\ ,\ \rangle$ is perfect on $\mathbb{Z}_\ell^2$. Deduce that passing between $\hat{\delta}_!$ and the cup product, which differ by the factor $\frac{1}{2}$ (Paper link 3.90), preserves perfectness and Gram determinants up to units only for $\ell$ odd: over $\mathbb{Z}_2$ the factor $\frac{1}{2}$ changes the ideal of values by 2 and the Gram determinant of a rank-two lattice by 4.

**Exercise 3.125** (the saturation of $\wp L$; reduction modulo $\varpi$)**.** In the setting of Proposition 3.113: (a) Show that $\mathrm{sat}(\wp L) = L[I]$ and $\mathrm{sat}(IL) = L[\wp]$. (Hint: Proposition 3.89(ii), Proposition 3.83(ii) and Proposition 3.113(iii).) (b) Let $\mathbb{T}_r$ ($r \geq 1$, Example 3.112) act on $L = \mathcal{O}^2$ coordinatewise, with the standard pairing and $\pi(a, b) = a$. Show that $\wp L = 0 \oplus \varpi^r\mathcal{O}$ is not saturated, that $L[I] = 0 \oplus \mathcal{O}$, and that the map $L[\wp]/\varpi L[\wp] \to (L/\varpi L)[\wp]$, injective for every $L$ because $L[\wp]$ is saturated (Proposition 3.49(i)(d)), is not surjective here: the reduction of $L[\wp]$ is a line, while $\wp$ kills all of $L/\varpi L = k^2$.

**Exercise 3.126** (check the paper: Lemma lem:crit)**.** Let $\mathbb{T}$ act on $S$ with a perfect pairing $S \times \mathbb{T} \to \mathbb{Z}$ as in Proposition 3.91, and let $\mathfrak{a}$, $\mathfrak{b}$ be ideals as in its part (iii). Show that $S/(S[\mathfrak{a}] \oplus S[\mathfrak{b}])$ has the same order as $\mathbb{T}/(\mathfrak{a} + \mathfrak{b})$. Work it out for $\mathbb{T} = \mathbb{Z}[x]/(x^2 - 4)$ acting on $S = \mathbb{T}^\vee$, $\mathfrak{a} = (x - 2)$, $\mathfrak{b} = (x + 2)$ (answer: 4), and interpret the result as a congruence modulo 4 between two "eigenforms", the elements of $S[\mathfrak{a}]$ and $S[\mathfrak{b}]$. Show that the rank hypothesis in (iii) cannot be dropped. (Hint: $\mathfrak{a} = \mathfrak{b} = 0$.)

**Exercise 3.127** (three eigenvalues)**.** Let $\mathbb{T} = \mathbb{Z}[x]/(x(x - 2)(x - 6))$, which embeds in $\mathbb{Z}^3$ by $x \mapsto (0, 2, 6)$, and $\pi(x) = 0$. (a) Show that $\wp = (x)$, $I = \mathrm{Ann}_{\mathbb{T}}(\wp) = \mathbb{Z}\,(x-2)(x-6)$ and $\eta = 12\mathbb{Z}$, and that 12 is the denominator of the idempotent $e_\pi = (x - 2)(x - 6)/12$ (Proposition 3.110(iii); its proof works verbatim over $\mathbb{Z}$, Paper link 3.111). (b) Show that $\mathbb{T} \otimes \mathbb{Z}_2$ is local and that $\mathbb{T} \otimes \mathbb{Z}_3 = \mathbb{T}_{\mathfrak{m}} \times \mathbb{T}_{\mathfrak{m}'}$ with $\mathfrak{m} = (3, x)$ and $\mathfrak{m}' = (3, x - 2)$, with idempotent $e_{\mathfrak{m}'} = x(x - 6)/(-8)$. Compute the congruence ideal in $\mathbb{T}_{\mathfrak{m}}$ and in $\mathbb{T} \otimes \mathbb{Z}_2$, and compare with Proposition 3.110(v). (Answers: $3\mathbb{Z}_3$ and $4\mathbb{Z}_2$.) (c) 2-adically, the eigenvalue 0 is congruent to 2 and to 6 only modulo 2, yet $v_2(\eta) = 2$. Explain this: $\eta$ is generated by $\pi((x - 2)(x - 6)) = (0 - 2)(0 - 6)$, and it measures the congruence of $\pi$ with the whole complementary factor $\mathbb{Q} \times \mathbb{Q}$ of $\mathbb{T} \otimes \mathbb{Q}$, not with each other eigenvalue separately (Proposition 3.53).

**Exercise 3.128** (check the paper: the congruence ideal is computed in $\mathbb{T}_{\mathfrak{m}}$)**.** For $\mathbb{T} \otimes \mathbb{Z}_\ell = \prod_{\mathfrak{m}'} \mathbb{T}_{\mathfrak{m}'}$ and $\pi$ factoring through $\mathbb{T}_{\mathfrak{m}}$, show that $\mathrm{Ann}(\ker \pi) = \mathrm{Ann}_{\mathbb{T}_{\mathfrak{m}}}(\wp_{\mathfrak{m}}) \times 0$. Do the other local factors lie in the annihilator or in the kernel? Compute both for $\mathbb{T} = \mathbb{Z}[x]/(x^2 - 4)$, $\ell = 3$ and $\pi(x) = 2$.

**Exercise 3.129** (check the paper: non-semisimple $U_p$ in Lemma lem:old)**.** For the operator $U$ of Example 3.105(a) compute the characteristic polynomial, show that $v_e$ is a cyclic vector, and find an eigenvector for each eigenvalue; deduce that for $e \geq 3$ the algebra generated by $U$ is not reduced. With Chapter 13: carry this out for the full Hecke algebra of level 88 acting on the span of $f_{11}(z)$, $f_{11}(2z)$, $f_{11}(4z)$, $f_{11}(8z)$, where $a_2(f_{11}) = -2$, and check with PARI that $U_2$ has characteristic polynomial $X^7(X^2 + 2X + 2)$ on all of $S_2(\Gamma_0(88))$.

**Exercise 3.130** (Gorenstein algebras and faithful modules)**.** (a) Show that $\mathbb{T}_r$ ($r \geq 1$) and $\mathbb{Z}_\ell[x]/(x^2)$ are local and Gorenstein, and that $\mathbb{T}_3^\flat = \{(a,b,c) \in \mathbb{Z}_\ell^3 : a \equiv b \equiv c \bmod \ell\}$ is local and not, by computing the socles of their reductions modulo $\ell$. For which monic $f$ is $\mathbb{Z}_\ell[x]/(f)$ local? (Answer: when $f \bmod \ell$ is a power of one irreducible polynomial.) (b) Let $\mathbb{T}$ be a local finite free $\mathbb{Z}_\ell$-algebra. Show that there is a *faithful* $\mathbb{T}$-module $L$, free of finite rank over $\mathbb{Z}_\ell$, with a perfect pairing for which $\mathbb{T}$ is self-adjoint and with $\dim_{k_\mathbb{T}} L/\mathfrak{m}L = 1$, if and only if $\mathbb{T}$ is Gorenstein. (Hint: by Nakayama $L$ is cyclic, so $L \simeq \mathbb{T}/\operatorname{Ann}_\mathbb{T}(L) = \mathbb{T}$, and Theorem 3.116 applies with $d = 1$; conversely take $L = \mathbb{T}$ with the pairing of Proposition 3.115(i).) (c) Show that faithfulness cannot be dropped in (b): $\mathbb{T}_3^\flat$ acting on $L = \mathbb{Z}_\ell$ through $\pi(a,b,c) = a$, with the pairing $xy$, has $\dim L/\mathfrak{m}L = 1$. Show that $\mathbb{T}_3^\flat$ acts faithfully on $L = \mathbb{Z}_\ell^3$ coordinatewise, self-adjointly for the standard pairing, with $\dim_{\mathbb{F}_\ell} L/\mathfrak{m}L = 3$, so that hypothesis (b) of Theorem 3.116 fails ($\operatorname{rank} L = 3 \neq 3 \operatorname{rank} \mathbb{T}_3^\flat$).

## 3.13 Notes and references

The material of Sections 1 to 4 is standard and follows Atiyah and Macdonald [7, Chapters 1 to 3, 6, 7] and Dummit and Foote [58, Chapters 10 and 12]; Lang [109, Chapters III and X] and Eisenbud [62, Chapters 2 to 4] give the same results in greater generality. Our proof of right exactness of the tensor product through the adjunction is the one in Atiyah and Macdonald; the snake and five lemmas are in Weibel [205, Section 1.3], whose numbering Chapter 16 uses. Hilbert's basis theorem, the equivalence of finite length with "Noetherian and Artinian", and the characterisation of discrete valuation rings as integrally closed one-dimensional Noetherian local domains are cited, not proved. Every chapter, section or theorem number quoted from a book in this chapter (Atiyah-Macdonald, Dummit-Foote, Lang, Eisenbud, Weibel, Jacobson, Matsumura, Serre, Neukirch-Schmidt-Wingberg, Rubin, Mazur, Darmon-Diamond-Taylor) is a pointer, not a verified citation.

The Smith normal form over a principal ideal domain (Theorem 3.41) is proved here by induction on the number of prime factors, which avoids the Euclidean hypothesis; Jacobson [88, Section 3.7] gives the same argument. The structure theorem and stacked bases are in Dummit and Foote [58, Section 12.1]. Saturated submodules are the pure submodules of Bourbaki. The congruence module of a decomposition (Proposition 3.53) and the duality of Proposition 3.91 are the commutative algebra behind Hida's congruence modules [82] and Ribet's treatment of the congruence number [159]; Agashe, Ribet and Stein [3] use them for the modular degree.

Discrete valuation rings and the ring $\mathbb{Z}_\ell$ follow Serre [174, Chapters I and II]; the analytic theory is Chapter 7. Baer's criterion and the injectivity of divisible modules are in Weibel [205, Section 2.3]. Pontryagin duality for finite groups is in Serre [172, Chapter VI, Section 1]; for compact and discrete $\mathbb{Z}_\ell$-modules see Neukirch, Schmidt and Wingberg [143, Chapter I, Section 1] and Rubin [167, Appendix B]. Our proof of the structure of cofinitely generated modules (Theorem 3.77(iii)) is algebraic and avoids compactness.

Fitting ideals were introduced by Fitting in 1936. We follow Northcott [144] and Eisenbud [62, Section 20.2]; the proof of Lemma 3.70 is the standard one through the ideal of determinants of all tuples of relations. Mazur and Wiles [127, Appendix] use Fitting ideals of $\mathbb{Z}_\ell$-modules as here, and the Tamagawa factors of the paper are Fitting ideals because of Fontaine and Perrin-Riou [69, I.4.2.2], as recalled by Diamond, Flach and Guo [49].

Sections 10 and 11 follow Darmon, Diamond and Taylor [38, Section 4], especially the discussion of congruence modules leading to their Lemma 4.17, whose inequality half is the paper's Lemma lem:ddt417; the toy algebras $\mathbb{T}_r$ and $\mathbb{T}_3^\flat$ are the standard examples of a congruence and of a non-Gorenstein algebra. The structure of Artinian rings is [7, Chapter 8], and idempotent lifting over complete rings is in Eisenbud [62, Section 7.5]; our Newton iteration $e \mapsto 3e^2 - 2e^3$ needs no Hensel lemma. Gorenstein rings are treated in Matsumura [124, Chapter 18] and Eisenbud [62, Chapter 21]; the socle criterion (Proposition 3.115(ii)) is the form used for Hecke algebras by Mazur [125, Chapter II, Section 15] and Tilouine [199]. Theorem 3.116 abstracts Mazur's argument that multiplicity one gives freeness and the Gorenstein property; the geometric input for the Hecke algebra is Chapter 14, with Ribet and Stein [163].

*Rigour.* Everything in this chapter is proved except: Hilbert's basis theorem; "finite length iff Noetherian and Artinian"; criterion (e) for discrete valuation rings; completions in general (Artin-Rees); localisation of Hom; topological Nakayama (sketched); Pontryagin duality for general compact modules; the Pfaffian identity in general rank and the symplectic normal form (sketched); Jordan-Chevalley; complete intersections are Gorenstein, and Wiles's numerical criterion; and the identification of $\mathbb{T}_{\mathfrak{m}}$ with an $\mathfrak{m}$-adic completion. The computations quoted (Smith forms, 11-adic roots and idempotents, characteristic polynomials of the shift operator and of $U_2$ at level 88, the Pfaffian in rank 4) were checked with PARI/GP [148].

# Chapter 4

# Representations of finite groups and the mod $\ell$ world

The objects of the paper are representations: $V = V_\ell E$, $T = T_\ell E$, $\rho = \rho_{E,\ell}$ and its reduction $\bar\rho$ are representations of the absolute Galois group $G_{\mathbb{Q}}$, and the adjoint $A_f = \operatorname{ad}^0 V$ is built from them by linear algebra. Most of what the paper needs about them is not Galois theory at all but the representation theory of a group, often a finite one, over a field of characteristic 0 or $\ell$, or over $\mathbb{Z}_\ell$. This chapter supplies that theory: irreducibility and absolute irreducibility, Schur's lemma, Maschke's theorem and its failure in characteristic $\ell$, the theorem of Brauer and Nesbitt, induced representations and Clifford theory for a subgroup of index 2, twisting, the adjoint representation and its trace form, the representations of $\mathrm{GL}_2(\mathbb{F}_3)$ and of its subgroups on $\mathbb{F}_3^2$ and $\mathbb{F}_9^2$, the action of finite cyclic groups and of unipotent elements on rank-two spaces and lattices, and lattices over a discrete valuation ring. Every statement is proved for an abstract group; the Galois input (continuity, Chebotarev, complex conjugation, the cyclotomic character) is added in Chapter 5 and Chapter 11, which cite the results proved here. The chapter uses Chapter 1 for the group theory of $\mathrm{GL}_2(\mathbb{F}_3)$, Chapter 2 for finite fields, and Chapter 3 for modules over a discrete valuation ring and the Jordan-Hölder theorem.

## Core results at a glance

Throughout, $G$ is a group, $k$ a field, $\bar k$ an algebraic closure, $R$ a commutative ring, and representations are finite-dimensional (Convention 4.1). $\mathcal{O}$ is a discrete valuation ring with uniformiser $\varpi$, fraction field $K$ and residue field $\mathcal{O}/\varpi$. The matrices $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $\sigma = \operatorname{diag}(1,-1)$, $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $a = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ and the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ are the standard models of Chapter 1; $\mathbb{F}_9 = \mathbb{F}_3[i]$.

**4.1 Conventions.** Matrices act on column vectors on the left; $V^\vee = \operatorname{Hom}_k(V,k)$ with $(g\lambda)(v) = \lambda(g^{-1}v)$ (the symbol $^*$ is reserved for Pontryagin duals); $\operatorname{End}(V) = V \otimes V^\vee$ with $g \cdot x = \rho(g) x \rho(g)^{-1}$, $\operatorname{ad}^0 V$ its trace-zero part, with basis $e = e_1 \otimes e_2^\vee$, $h = e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee$, $f = e_2 \otimes e_1^\vee$ when $\dim V = 2$; a character is a homomorphism $\chi\colon G \to R^\times$, $M(\chi) = M \otimes_R R(\chi)$; $\operatorname{tr}_V$ is the trace function. Convention 4.1.

**4.2 Representations** (definition): a homomorphism $\rho\colon G \to \mathrm{GL}(V)$, $V$ a finite-dimensional $k$-vector space, equivalently a $k[G]$-module of finite dimension; matrix representations are

isomorphic iff conjugate by one $P \in \mathrm{GL}_n(k)$; examples: characters, permutation and regular representations, the standard representation of $H \leq \mathrm{GL}_2(k)$, inflation. Definition 4.2.

**4.3 Constructions** (definition): subrepresentations, quotients, $V \oplus W$, $V \otimes W$, $V^\vee$, $\mathrm{Hom}_k(V, W)$ with $(g\phi)(v) = g\phi(g^{-1}v)$, invariants $V^G$, $\det \rho$. Definition 4.3.

**4.4 Basic facts.** $\mathrm{tr}(g \mid V^\vee) = \mathrm{tr}(g^{-1} \mid V)$, $\mathrm{tr}(g \mid V \otimes W) = \mathrm{tr}(g \mid V)\,\mathrm{tr}(g \mid W)$, $\det(V \otimes \chi) = \chi^n \det V$; $V^{\vee\vee} \simeq V$, $(V \otimes W)^\vee \simeq V^\vee \otimes W^\vee$; if $\dim V = 2$ (or $V$ free of rank 2 over $R$), $v \mapsto \det[v \mid -]$ is a $G$-isomorphism $V \simeq V^\vee \otimes \det$, so $V^\vee \simeq V \otimes \det^{-1}$. *(proved here)* Proposition 4.4.

**4.5 $\mathrm{Hom}_G$ as invariants.** $V^\vee \otimes W \to \mathrm{Hom}(V, W)$, $\lambda \otimes w \mapsto \lambda(\cdot)w$, is a $G$-isomorphism (also for free modules of finite rank over $R$), and $\mathrm{Hom}_G(V, W) = \mathrm{Hom}(V, W)^G = (V^\vee \otimes W)^G$; for a character $\chi$, $(\mathrm{End}(V) \otimes \chi)^G = \mathrm{Hom}_G(V, V \otimes \chi)$. *(proved here)* Proposition 4.5.

**4.6 Extension of scalars.** For a field extension $K/k$: (i) $\mathrm{Hom}_{K[G]}(V_K, W_K) = K \otimes_k \mathrm{Hom}_{k[G]}(V, W)$ and $(V_K)^G = K \otimes_k V^G$; (ii) if $V, W$ are irreducible and $V_K \simeq W_K$ then $V \simeq W$; (iii) if $V_K$ is irreducible and $W_K \simeq V_K$ then $W \simeq V$; (iv) (Noether-Deuring) $V_K \simeq W_K$ implies $V \simeq W$ for all $V, W$. *(proved here for (i) to (iii); (iv) stated; reference)* Proposition 4.7.

**4.7 Irreducible, indecomposable** (definition): $V \neq 0$ with no subrepresentations but 0, $V$; not a direct sum of two nonzero subrepresentations; for $\dim V = 2$, irreducible iff no line is stable. Definition 4.9.

**4.8 Schur's lemma.** For irreducible $V, W$ over any field: (i) a $G$-map $V \to W$ is 0 or an isomorphism, and $\mathrm{End}_G(V)$ is a division algebra; (ii) if $k = \bar{k}$, $\mathrm{End}_G(V) = k$ and $\dim \mathrm{Hom}_G(V, W) \in \{0, 1\}$, with 1 iff $V \simeq W$; (iii) the same over any $k$ for absolutely irreducible $V, W$; (iv) the centre acts on an absolutely irreducible $V$ by scalars, absolutely irreducible representations of abelian groups are one-dimensional, and $\mathrm{End}_G(V)$ is a field for $V$ irreducible and $G$ abelian. *(proved here)* Theorem 4.10.

**4.9 Absolutely irreducible** (definition): $V_{\bar{k}}$ is irreducible. Definition 4.11.

**4.10 Criteria for absolute irreducibility.** (i) $V$ is absolutely irreducible iff $k[G] \to \mathrm{End}_k(V)$ is surjective, and then $V_K$ is irreducible for every extension $K/k$; (ii) absolutely irreducible implies irreducible with $\mathrm{End}_G(V) = k$; (iii) for $\dim V = 2$: iff no line of $\bar{k}^2$ is stable; (iv) over any field, an irreducible $V$ is absolutely irreducible iff $\mathrm{End}_G(V) = k$. *(proved here)* Proposition 4.12.

**4.11 Distinct eigenvalues.** If some $g \in G$ acts on $V$ with $\dim V$ distinct eigenvalues in $k$, every $G$-stable subspace of $V_{\bar{k}}$ is defined over $k$; so $V$ is irreducible iff it is absolutely irreducible. In particular this holds for $\dim V = 2$, $\mathrm{char}\, k \neq 2$, if some $c \in G$ has $\rho(c)^2 = 1$, $\det \rho(c) = -1$. *(proved here)* Proposition 4.13.

**4.12 $C_4$ on $\mathbb{F}_3^2$.** $C_4 = \langle J \rangle$ acts irreducibly but not absolutely irreducibly on $\mathbb{F}_3^2$, with $\mathrm{End}_{C_4}(\mathbb{F}_3^2) = \mathbb{F}_3[J] \simeq \mathbb{F}_9$; over $\mathbb{F}_9$, $\mathbb{F}_9^2 = \mathbb{F}_9 v \oplus \mathbb{F}_9 \bar{v}$ with $v = (1, i)^t$, $\bar{v} = (1, -i)^t$, $Jv = iv$, $J\bar{v} = -i\bar{v}$: the sum $\psi \oplus \psi^{-1}$, $\psi(J) = i$ of order 4. *(proved here)* Example 4.14.

**4.13 Semisimple** (definition): a sum of irreducible subrepresentations. Definition 4.16.

**4.14 Semisimple modules.** For finite-dimensional $V$ and any $G$: $V$ is a sum of irreducibles iff a direct sum of irreducibles iff every subrepresentation has a stable complement; subrepresentations and quotients of semisimple representations are semisimple; $V = \bigoplus_S V_S$ with $V_S \simeq S^{m_S}$, and $m_S = \dim \mathrm{Hom}_G(S, V)$ if $S$ is absolutely irreducible. *(proved here)* Proposition 4.17.

**4.15 Maschke.** (i) If $G$ is finite and $\mathrm{char}\, k \nmid \#G$, every representation of $G$ over $k$ is semisimple. (ii) If $\#G \in R^\times$ and $M$ is an $R[G]$-module, $e_G = \frac{1}{\#G}\sum_g g$ is an idempotent with $e_G M = M^G$, $M = M^G \oplus (1 - e_G)M$, and $(M/IM)^G = M^G/IM^G$ for every ideal $I$: invariants commute with reduction. *(proved here)* Theorem 4.18.

**4.16 Failures of semisimplicity.** $\mathrm{U} = \langle u \rangle$ on $\mathbb{F}_3^2$ is indecomposable and not semisimple; $\mathbb{F}_p[C_p]$ is uniserial; $\mathbb{Z}$ acting on $\mathbb{Q}_\ell^2$ through $\left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$, $n' \neq 0$, is not semisimple; $\mathrm{B}(\mathbb{F}_3)$ on $\mathbb{F}_3^2$ has $\mathrm{End}_{\mathrm{B}} = \mathbb{F}_3$ but is reducible. *(proved here)* Example 4.20.

**4.17 Jordan-Hölder.** A module of finite length over any ring (here $k[G]$ or $\mathcal{O}[G]$) has a composition series, and the composition factors with multiplicity, and the length, do not depend on the series. *(stated; reference: proved in Chapter 3)* Theorem 4.21.

**4.18 Semisimplification** (definition): $V^{ss} = \bigoplus_i M_i/M_{i-1}$ for a composition series $(M_i)$. Definition 4.22.

**4.19 Properties of $V^{ss}$.** (i) $V^{ss}$ is well defined, $V^{ss} \simeq V$ iff $V$ is semisimple, and $(\cdot)^{ss}$ is additive in short exact sequences; (ii) $\det(X - g \mid V) = \det(X - g \mid V^{ss})$; (iii) over a perfect $k$, $(V^{ss})_K \simeq (V_K)^{ss}$ for algebraic $K/k$. *(proved here for (i), (ii); (iii) stated; reference)* Proposition 4.23.

**4.20 Extensions of two characters.** Let $\dim V = 2$ with a stable line $L$ on which $G$ acts by $\eta_1$, and $\eta_2$ on $V/L$, $\rho = \left(\begin{smallmatrix} \eta_1 & b \\ 0 & \eta_2 \end{smallmatrix}\right)$. (i) $V$ is semisimple iff $V \simeq \eta_1 \oplus \eta_2$ iff $V$ has a second stable line iff $b(g) = c(\eta_2(g) - \eta_1(g))$ for some $c \in k$; (ii) if $V$ is not semisimple, $L$ is its only stable line, and an isomorphism of two such extensions matches the sub-characters and the quotient characters; (iii) if $\eta_1 \neq \eta_2$ and $V$ is semisimple, $V$ has exactly two stable lines. *(proved here)* Proposition 4.24.

**4.21 Absolute irreducibility in dimension 2.** (i) If $\rho(G)$ is abelian, $\rho$ is absolutely reducible; (ii) if $\rho(G)$ is finite of order prime to $\mathrm{char}\, k$ and $\rho$ is absolutely reducible, then $\rho(G)$ is abelian; so a finite non-abelian subgroup of $\mathrm{GL}_2(k)$ of order prime to $\mathrm{char}\, k$ acts absolutely irreducibly; (iii) if $\mathrm{Res}_H \rho$ is absolutely irreducible, so is $\rho$; (iv) $Q_8$, $\mathrm{SL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, $\mathrm{GL}_2(\mathbb{F}_3)$ and $N_s(\mathbb{F}_3)$ act absolutely irreducibly on $\mathbb{F}_3^2$; $C_4$, $C_{ns}(\mathbb{F}_3)$, $\mathrm{B}(\mathbb{F}_3)$ do not. *(proved here)* Proposition 4.26.

**4.22 Trace function** (definition): $\mathrm{tr}_V(g) = \mathrm{tr}\, \rho(g)$, a class function (the "character" of Riley-Hobson-Bence). Definition 4.28.

**4.23 Orthogonality.** $G$ finite, $k = \mathbb{C}$, $\langle \phi, \psi \rangle = \frac{1}{\#G}\sum_g \phi(g)\overline{\psi(g)}$: (i) for irreducible $V, W$, $\langle \mathrm{tr}_V, \mathrm{tr}_W \rangle$ is 1 if $V \simeq W$ and 0 otherwise; (ii) $\dim \mathrm{Hom}_G(V, W) = \langle \mathrm{tr}_W, \mathrm{tr}_V \rangle$, $V$ is irreducible iff $\langle \mathrm{tr}_V, \mathrm{tr}_V \rangle = 1$, and the trace function determines $V$; (iii) the irreducible trace functions are an orthonormal basis of the class functions, their number is the number of conjugacy classes, and $\sum_i (\dim V_i)^2 = \#G$; (iv) $\dim V_i$ divides $\#G$. *(proved here for (i) to (iii); (iv) stated; reference)* Theorem 4.29.

**4.24 Traces in characteristic 0.** If $\operatorname{char} k = 0$, two semisimple representations of any group with the same trace function are isomorphic. *(proved here)* Corollary 4.30.

**4.25 $D_4$ and $Q_8$.** They have the same character table and are not isomorphic; the two-dimensional representation of $D_4$ is defined over $\mathbb{Q}$ and reduces modulo 3 to the standard representation of $N_s(\mathbb{F}_3)$. *(proved here)* Example 4.31.

**4.26 Traces fail in characteristic $p$.** Over $\mathbb{F}_3$, $\mathbf{1}^{\oplus 3}$ and $\operatorname{sgn}^{\oplus 3}$ of $C_2$ have the same traces and are not isomorphic; traces suffice in dimension $< p$. *(proved here)* Example 4.32.

**4.27 Density theorem (Jacobson, Burnside).** For pairwise non-isomorphic irreducible $S_1, \dots, S_r$ with $D_i = \operatorname{End}_G(S_i)$: (a) the image of $k[G] \to \prod_i \operatorname{End}_k(S_i)$ is $\prod_i \operatorname{End}_{D_i}(S_i)$; (b) if $k = \bar{k}$, it is $\prod_i \operatorname{End}_k(S_i)$. *(proved here)* Theorem 4.34.

**4.28 Brauer-Nesbitt.** If $\det(X - \rho_1(g)) = \det(X - \rho_2(g))$ in $k[X]$ for all $g \in G$ (any group, any field), then $V_1^{ss} \simeq V_2^{ss}$. Equal traces suffice if $\operatorname{char} k = 0$ or $\dim V_1, \dim V_2 < \operatorname{char} k$. *(proved here)* Theorem 4.35.

**4.29 Consequences.** (i) In dimension 2, equal traces and equal determinants give $V_1^{ss} \simeq V_2^{ss}$; (ii) if $V_1$ is over $k$, $V_2$ over $k' \supseteq k$, they have the same characteristic polynomials, and $V_1 \otimes_k k'$ is irreducible, then $V_2 \simeq V_1 \otimes_k k'$; this fails when $V_1$ is only irreducible over $k$. *(proved here)* Corollary 4.36.

**4.30 Induction** (definition): $\operatorname{Ind}_H^G W = k[G] \otimes_{k[H]} W$, $\operatorname{Res}_H^G V$. Definition 4.38.

**4.31 Induced representations.** For $G = \bigsqcup_i s_i H$ of finite index $m$: $\operatorname{Ind} W = \bigoplus_i s_i \otimes W$, $\dim \operatorname{Ind} W = m \dim W$, $\operatorname{tr}(g \mid \operatorname{Ind} W) = \sum_{s_i^{-1} g s_i \in H} \operatorname{tr}(s_i^{-1} g s_i \mid W)$; for $[G : H] = 2$, $s \notin H$ and a character $\psi$ of $H$: $h \mapsto \operatorname{diag}(\psi(h), \psi^s(h))$, $s \mapsto \left(\begin{smallmatrix} 0 & \psi(s^2) \\ 1 & 0 \end{smallmatrix}\right)$, with $\psi^s(h) = \psi(s^{-1}hs)$. *(proved here)* Proposition 4.39.

**4.32 Frobenius reciprocity.** $\operatorname{Hom}_G(\operatorname{Ind}_H^G W, V) \simeq \operatorname{Hom}_H(W, \operatorname{Res}_H V)$, and for finite index $\operatorname{Hom}_G(V, \operatorname{Ind}_H^G W) \simeq \operatorname{Hom}_H(\operatorname{Res}_H V, W)$, over a field or a commutative ring. *(proved here)* Theorem 4.40.

**4.33 Projection formula.** $\operatorname{Ind}(W \otimes \operatorname{Res} U) \simeq \operatorname{Ind} W \otimes U$; so $\operatorname{Ind} W \otimes \chi \simeq \operatorname{Ind} W$ if $\chi|_H = 1$; $\operatorname{Ind}_H^G \mathbf{1} = k[G/H]$, $\simeq \mathbf{1} \oplus \chi_H$ for index 2 and $\operatorname{char} k \neq 2$; transitivity, additivity, extension of scalars. *(proved here)* Proposition 4.41.

**4.34 Mackey.** (a) $\operatorname{Res}_K \operatorname{Ind}_H^G W \simeq \bigoplus_{s \in K \backslash G / H} \operatorname{Ind}_{K \cap sHs^{-1}}^K W^s$; (b) in characteristic 0, $G$ finite, $\operatorname{Ind} W$ is irreducible iff $W$ is irreducible and $W^s$, $W$ have no common constituent on $sHs^{-1} \cap H$ for $s \notin H$; (c) for $[G : H] = 2$, any characteristic, $W$ absolutely irreducible: $\operatorname{Res}_H \operatorname{Ind} W \simeq W \oplus W^s$, and $\operatorname{Ind} W$ is absolutely irreducible iff $W_{\bar{k}} \not\simeq W^s_{\bar{k}}$. *(stated; reference for (a); proof sketched here for (b); proved here for (c))* Theorem 4.42.

**4.35 Clifford.** For $N \trianglelefteq G$ of finite index and $V$ irreducible over any field, $\operatorname{Res}_N V$ is semisimple, its constituents form one $G$-orbit and have equal multiplicities. *(proved here)* Theorem 4.43.

**4.36 Clifford theory for index 2.** $[G : H] = 2$, $\chi_H$ the character with kernel $H$, $\operatorname{char} k \neq 2$, $V$ absolutely irreducible of dimension $n$. Either (A) $\operatorname{Res}_H V$ is absolutely irreducible and $V \not\simeq V \otimes \chi_H$; or (B) $n = 2d$, $\operatorname{Res}_H V_{\bar{k}} \simeq W \oplus W^s$ with $W \not\simeq W^s$ absolutely irreducible,

$V_{\bar{k}} \simeq \operatorname{Ind}_H^G W$, and $V \simeq V \otimes \chi_H$ over $k$, with $\operatorname{Hom}_G(V, V \otimes \chi_H)$ a line spanned by an isomorphism of trace 0 with eigenspaces $W$, $W^s$. For $n = 2$: (B) iff $V_{\bar{k}} \simeq \operatorname{Ind} \psi$ with $\psi^s \neq \psi$. *(proved here)* Theorem 4.44.

**4.37 Dihedral representations.** For $\rho\colon G \to \mathrm{GL}_2(k)$ absolutely irreducible, $\operatorname{char} k \neq 2$, with finite projective image $P$: $P$ is not cyclic; $P$ is dihedral (including $V_4$) iff $\rho \simeq \rho \otimes \chi$ for a character of order 2 iff $\rho_{\bar{k}}$ is induced from an index-two subgroup; the characters $\chi \neq 1$ with $\rho \simeq \rho \otimes \chi$ correspond to the cyclic subgroups of index 2 of $P$ (one if $\#P \geq 6$, three if $P \simeq V_4$). *(proved here)* Proposition 4.46.

**4.38 Twisting.** $\operatorname{tr}(\rho \otimes \chi) = \chi \operatorname{tr} \rho$, $\det(\rho \otimes \chi) = \chi^n \det \rho$; (absolute) irreducibility and $\operatorname{Hom}_G$ are unchanged; $\operatorname{End}(V \otimes \chi) = \operatorname{End}(V)$ and $\operatorname{ad}^0(V \otimes \chi) = \operatorname{ad}^0 V$; $V \simeq V \otimes \chi$ forces $\chi^n = 1$. *(proved here)* Proposition 4.48.

**4.39 Trace-zero lemma.** If $\phi\colon V \to V \otimes \chi$ is a $G$-map then $(1 - \chi(g)) \operatorname{tr} \phi = 0$ for all $g$; so $\operatorname{tr} \phi = 0$ if $\chi \neq 1$ (over a field) or if $1 - \chi(g) \in R^\times$ for some $g$. *(proved here)* Lemma 4.49.

**4.40 Adjoint** (definition): $\operatorname{End}(V)$ with $g \cdot x = \rho(g) x \rho(g)^{-1}$, and $\operatorname{ad}^0 V = \ker \operatorname{tr}$. Definition 4.51.

**4.41 The adjoint.** (i) $\operatorname{End}(V) = k \oplus \operatorname{ad}^0 V$ if $n \in k^\times$; (ii) $\operatorname{Ad} \rho(g)$ has eigenvalues $\lambda_i/\lambda_j$, and on $\operatorname{ad}^0 V$ for $n = 2$: $\lambda_1/\lambda_2, 1, \lambda_2/\lambda_1$; (iii) in the basis $e, h, f$: $\operatorname{Ad} \operatorname{diag}(x, y) = \operatorname{diag}(x/y, 1, y/x)$; for $t = \left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$, $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$; for $F = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$, $FeF^{-1} = qe$, $FhF^{-1} = h - 2c'\varepsilon^{-1}e$, $FfF^{-1} = q^{-1}f + c'\varepsilon^{-1}q^{-1}h - c'^2\varepsilon^{-2}q^{-1}e$. *(proved here)* Proposition 4.52.

**4.42 Invariants of twisted adjoints.** (i) $(\operatorname{End} V \otimes \chi)^G = \operatorname{Hom}_G(V, V \otimes \chi)$; (ii) if $n \in R^\times$, $(\operatorname{ad}^0 V \otimes \chi)^G$ is its trace-zero part; (iii) if $\chi$ (over a local $R$: its reduction) is nontrivial, every such map has trace 0; (iv) for $V$ absolutely irreducible, $(\operatorname{ad}^0 V)^G = 0$, and for $\chi \neq 1$, $(\operatorname{ad}^0 V \otimes \chi)^G \neq 0$ iff $V \simeq V \otimes \chi$, and then it is a line. *(proved here)* Proposition 4.54.

**4.43 Symmetric square.** For $\dim V = 2$, $\Xi(xy)(v) = \omega(v, x)y + \omega(v, y)x$ is a $G$-map $\operatorname{Sym}^2 V \otimes \det^{-1} \to \operatorname{ad}^0 V$ with $\Xi(e_1^2) = -2e$, $\Xi(e_1 e_2) = h$, $\Xi(e_2^2) = 2f$, an isomorphism iff 2 is invertible (index 4 over $\mathbb{Z}$); so $\operatorname{ad}^0 V \simeq \operatorname{Sym}^2 V \otimes \det^{-1}$; for $V = V_\ell E$, $\operatorname{Sym}^2(V^\vee) \simeq \operatorname{ad}^0 V(-1)$ and $L_p(\operatorname{Sym}^2 f, s) = L_p(A_f, s - 1)$. *(proved here)* Proposition 4.56.

**4.44 Trace form.** $B(x, y) = \operatorname{tr}(xy)$ on $\operatorname{End}_R(T)$ is $G$-invariant and perfect; on $\operatorname{ad}^0 T$ ($n = 2$) its Gram matrix in $e, h, f$ has determinant $-2$, so for $\ell$ odd $\operatorname{ad}^0 T \simeq (\operatorname{ad}^0 T)^\vee$ and $\operatorname{Hom}_{\mathbb{Z}_\ell}(\operatorname{ad}^0 T, \mathbb{Z}_\ell(\chi)) \simeq \operatorname{ad}^0 T(\chi)$; for $\ell = 2$ the index is 2. *(proved here)* Proposition 4.58.

**4.45 The adjoint ignores twists, duals and homotheties.** $\operatorname{ad}^0(V \otimes \chi) = \operatorname{ad}^0 V$; $x \mapsto -x^t$ is a $G$-isomorphism $\operatorname{ad}^0 V \simeq \operatorname{ad}^0(V^\vee)$; for lattices, $\operatorname{ad}^0(cL) = \operatorname{ad}^0 L$ and $\operatorname{ad}^0(L^\vee)$ corresponds to $\operatorname{ad}^0 L$. *(proved here)* Proposition 4.60.

**4.46 The subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ on $\mathbb{F}_3^2$.** The table of irreducibility, absolute irreducibility, $\dim \operatorname{End}_H$, $\dim(\operatorname{ad}^0 \rho)^H$, $\dim \operatorname{Hom}_H(\rho, \rho \otimes \det)$, the twist $\rho \simeq \rho \otimes \det$, and projective images, for $\mathrm{GL}_2, \mathrm{SL}_2, N_{ns}, Q_8, N_s, C_{ns}, C_4, C_s, \mathrm{B}, \mathrm{U}$. *(proved here; checked by computer)* Proposition 4.62.

**4.47 $N_s(\mathbb{F}_3)$ is induced over $\mathbb{F}_9$.** The standard representation $\rho$ of $N_s(\mathbb{F}_3)$ is absolutely irreducible; in the basis $v = (1, i)^t$, $\sigma v = (1, -i)^t$ of $\mathbb{F}_9^2$, $\rho(J) = \operatorname{diag}(i, -i)$ and $\rho(\sigma) = w$,

so $\rho \otimes \mathbb{F}_9 \simeq \mathrm{Ind}_{C_4}^{N_s} \psi$, $\psi(J) = i$; over $\mathbb{F}_3$, $\rho$ is not induced from $C_4$; $\rho \simeq \rho \otimes \chi$ for the three characters with kernels $C_4$, $C_s$, $\{\pm 1, \pm w\}$, with intertwiners $J$, $\sigma$, $w$; $(\mathrm{ad}^0 \rho \otimes \det)^{N_s} = \mathbb{F}_3 J$. *(proved here)* Proposition 4.63.

**4.48** $N_{ns}(\mathbb{F}_3)$**.** Its standard representation and its restriction to $Q_8$ are absolutely irreducible; $\rho \not\simeq \rho \otimes \det$ and $(\mathrm{ad}^0 \rho \otimes \det)^{N_{ns}} = 0$; $\rho \otimes \mathbb{F}_9 \simeq \mathrm{Ind}_{C_{ns}}^{N_{ns}} \theta$, $\theta(a) = 1 + i$, and $\rho \simeq \rho \otimes \chi$ only for the character with kernel $C_{ns}(\mathbb{F}_3)$. *(proved here)* Proposition 4.64.

**4.49** $\mathrm{SL}_2(\mathbb{F}_3)$ **and** $\mathrm{GL}_2(\mathbb{F}_3)$**.** Their standard representations are absolutely irreducible; for $\mathrm{GL}_2(\mathbb{F}_3)$, $\rho \not\simeq \rho \otimes \det$ and $(\mathrm{ad}^0 \rho \otimes \det)^{\mathrm{GL}_2(\mathbb{F}_3)} = 0$; projective images $A_4$, $S_4$. *(proved here)* Proposition 4.65.

**4.50 The three images.** For $H$ conjugate to $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$ and $H_1 = H \cap \mathrm{SL}_2(\mathbb{F}_3)$, equivalent: (a) $H$ is conjugate to $N_s(\mathbb{F}_3)$; (b) $\mathrm{Res}_{H_1} \rho$ is not absolutely irreducible; (c) $\rho \simeq \rho \otimes \det$; (d) $(\mathrm{ad}^0 \rho \otimes \det)^H \neq 0$; (e) $\rho \otimes \mathbb{F}_9 \simeq \mathrm{Ind}_{H_1}^H \psi$ with $\psi$ of order 4; (f) $H_1$ is abelian. Then $(\mathrm{ad}^0 \rho \otimes \det)^H$ is a line spanned by a trace-zero isomorphism; otherwise it is 0. *(proved here)* Theorem 4.66.

**4.51 The letter** $\sigma$**.** In Section 4.9 $\sigma$ is the paper's element of order 3 and $\mathrm{diag}(1, -1)$ is $h$. Convention 4.68.

**4.52 Unipotent elements.** For $u \neq 1$ unipotent in $\mathrm{GL}_2(k)$ and $\xi \in k^\times \setminus \{1\}$: $V^u$ is a line, $V^{\xi u} = 0$; if $\mathrm{char}\, k \neq 2$, $(\mathrm{ad}^0 V)^u = (\mathrm{ad}^0 V)^{\xi u} = k(u-1)$, a line; if $FuF^{-1} = u^q$ then $F$ acts on it by $q$ (and $F^{-1}$ by $q^{-1}$). *(proved here)* Proposition 4.69.

**4.53 Commutant lemma.** If $x \in \mathrm{End}_k(V)$ has $\dim V$ distinct eigenvalues in $\bar{k}$, or $\dim V = 2$ and $x$ is not scalar, its commutant is $k[x] \simeq k[X]/(\det(X - x))$; for $\dim V = 2$ and $\mathrm{char}\, k \neq 2$ its trace-zero part is the line $k(2x - \mathrm{tr}\, x)$; it is a field iff $\det(X - x)$ is irreducible. *(proved here)* Lemma 4.70.

**4.54 Cyclic groups of order** $2, 3, 4, 6$**.** An element $g \in \mathrm{SL}_2(K)$ of order $m \in \{3, 4, 6\}$, $\mathrm{char}\, K \nmid m$, has characteristic polynomial $\Phi_m$; order 2 means $g = -1$. Fixed spaces on $V$ and $\mathrm{ad}^0 V$ (table), the invariant line of $\mathrm{ad}^0 V$ spanned by $g - g^{-1}$, and an $F$ with $FgF^{-1} = g^r$ acts on it by $\pm 1$ according as $r \equiv \pm 1 \pmod m$. *(proved here)* Proposition 4.71.

**4.55 Reduction of finite cyclic subgroups of** $\mathrm{SL}_2(\mathbb{Z}_\ell)$**.** For $g$ of order $m \in \{2, 3, 4, 6\}$: if $\ell \nmid m$, invariants on $T$ and $\mathrm{ad}^0 T$ commute with reduction; if $\ell = 3$, $m = 3$, then $\bar{g}$ is a nontrivial unipotent, $(T/3T)^g$ is a line while $T^g = 0$, and $(\mathrm{ad}^0 T)^g = \mathbb{Z}_3(2g + 1)$ reduces onto $(\mathrm{ad}^0 T/3)^g$; for $m = 6, 4, 2$ and $\ell = 3$, $(T/3T)^g = 0$; the integral commutant is $\mathbb{Z}_\ell[g] \simeq \mathbb{Z}_\ell[X]/(\Phi_m)$. *(proved here; the freeness of $T$ over $\mathbb{Z}_\ell[g]$ stated; reference)* Proposition 4.72.

**4.56 Lattices** (definition): finitely generated $\mathcal{O}$-submodules spanning $V$; homothety $L' = cL$. Definition 4.74.

**4.57 Stable lattices.** Any two lattices satisfy $\varpi^a L \subseteq L' \subseteq \varpi^{-b} L$; for finite $G$, $\sum_g \rho(g) L_0$ is stable; $\varpi^m L$, $L^\vee$, $L \otimes \mathcal{O}(\chi)$, $\mathrm{End}_{\mathcal{O}}(L)$, $\mathrm{ad}^0 L$ are stable, and $\mathrm{ad}^0 L/\varpi^m = \mathrm{ad}^0(L/\varpi^m L)$. *(proved here)* Proposition 4.75.

**4.58 Reduction.** $\det(X - \bar{\rho}_L(g))$ is the reduction of $\det(X - \rho(g))$; $\bar{\rho}_L^{ss}$ does not depend on $L$; if $\bar{\rho}_L$ is irreducible for one $L$, $\bar{\rho}_{L'} \simeq \bar{\rho}_L$ for all $L'$. *(proved here)* Proposition 4.76.

**4.59 Homothety lemma.** If $\bar\rho_L$ is irreducible, every $G$-stable lattice is $\varpi^m L$ for a unique $m \in \mathbb{Z}$. *(proved here)* Theorem 4.77.

**4.60 Non-homothetic lattices.** For the upper triangular group, $\mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$ and $\mathbb{Z}_\ell e_1 \oplus \ell\mathbb{Z}_\ell e_2$ are stable, not homothetic, with non-isomorphic reductions of equal semisimplification. *(proved here)* Example 4.79.

**4.61 Ribet's lemma.** Let $K$ be finite over $\mathbb{Q}_\ell$ with ring of integers $\mathcal{O}$, $G$ compact, and $\rho\colon G \to \mathrm{GL}_2(K)$ continuous and irreducible. If $\bar\rho^{ss} \simeq \eta_1 \oplus \eta_2$, some stable lattice reduces to a non-split extension with sub-character $\eta_1$ (and one with sub-character $\eta_2$). *(stated; reference)* Theorem 4.80.

**4.62 Twisted self-isomorphisms modulo $\varpi^m$.** If $T/\varpi T$ is absolutely irreducible, the module of $G$-maps $T/\varpi^m \to (T/\varpi^m)(\chi)$ is cyclic, its elements of exact order $\varpi^m$ are isomorphisms, and the length of the invariants of $X = \mathrm{End}_{\mathcal{O}}(T)(\chi) \otimes K/\mathcal{O}$ is the largest $h$ with $T/\varpi^h \simeq (T/\varpi^h)(\chi)$; the same for $\mathrm{ad}^0 T(\chi)$ if $n \in \mathcal{O}^\times$, $\bar\chi \neq 1$; $\chi^n \equiv 1 \pmod{\varpi^h}$; for $n = 2$, $\ell$ odd, $\chi(G) = \mathbb{Z}_\ell^\times$: $h = 0$ if $\ell \geq 5$, $h \leq 1$ if $\ell = 3$. *(proved here)* Proposition 4.81.

**4.63 The mod $\ell$ dictionary.** For $\bar\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F}_\ell)$ continuous, $\ell$ odd, $\det\bar\rho = \bar\chi_\ell$: irreducible equals absolutely irreducible; for $\bar\rho$ irreducible, the hypothesis of Theorem thm:main fails iff $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$, iff $\bar\rho \otimes \overline{\mathbb{F}}_\ell$ is induced from $G_{\mathbb{Q}(\sqrt{\ell^*})}$ (for $\bar\rho$ reducible it always fails); Lemma lem:h0 concerns $\bar\rho \simeq \bar\rho \otimes \bar\chi_\ell$, possible only for $\ell = 3$. *(proved here from the cited Galois input)* Remark 4.83.

**4.64 Dickson.** For $\ell$ odd and $G \leq \mathrm{GL}_2(\mathbb{F}_\ell)$: if $\ell \mid \#G$, $G \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$ or $G$ lies in a Borel subgroup; if $\ell \nmid \#G$, the image in $\mathrm{PGL}_2(\mathbb{F}_\ell)$ is cyclic, dihedral, or $A_4$, $S_4$, $A_5$. *(stated; reference)* Theorem 4.84.

## Where the paper uses this chapter

- **Proposition prop:ns** (the image $N_s(\mathbb{F}_3)$): the three images and their determinant-one subgroups, Theorem 4.66; absolute irreducibility of $Q_8$ and $\mathrm{SL}_2(\mathbb{F}_3)$, Proposition 4.26; $\psi$ of order 4 and $\bar\rho \otimes \mathbb{F}_9 \simeq \mathrm{Ind}\,\psi$, Example 4.14, Proposition 4.63, Theorem 4.40; $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$, Proposition 4.41, Theorem 4.44, descended to $\mathbb{F}_3$ by Proposition 4.7; $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho(1)) \neq 0$, Proposition 4.54.

- **Lemma lem:h0** (the term $h_\ell(E)$): $H^0$ of a twisted adjoint as trace-zero $G$-maps, Proposition 4.5, Proposition 4.54; the trace-zero lemma, Lemma 4.49; Schur, Theorem 4.10; cyclicity, isomorphisms modulo $\ell^k$ and the determinant bound, Proposition 4.81; Clifford theory for $\mathbb{Q}(\sqrt{-3})$, Theorem 4.44.

- **Lemma lem:IV, Remark rem:types, Lemma lem:euler, Proposition prop:ns-structure**: unipotent and minus-unipotent elements, Proposition 4.69; the commutant and its trace-zero line, Lemma 4.70; elements of order $2, 3, 4, 6$ and the action of Frobenius on the invariant line, Proposition 4.71; reduction modulo 3, Proposition 4.72; Maschke over $\mathbb{Z}_\ell$, Theorem 4.18.

- **Lemma lem:old** (oldforms): Brauer-Nesbitt over a common finite field, Theorem 4.35, Corollary 4.36; absolute irreducibility from complex conjugation, Proposition 4.13; the unique stable line of a non-split extension, Proposition 4.24.

- **Section 4.2** (the theorem of Diamond, Flach and Guo) **and Lemma lem:fpr**: $A = \mathrm{ad}^0 T$, $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ by the trace form, Proposition 4.58; the adjoint of $M_{f,\lambda} \simeq T_\ell E(-1)$ is $\mathrm{ad}^0 T$, Proposition 4.60; the homothety lemma as an alternative route, Theorem 4.77.

- **Lemma lem:tamfactor** (the Tamagawa factor at a multiplicative prime): the basis $e, h, f$ and the action of $t$ and of the Frobenius matrix, Proposition 4.52; the invariants of a unipotent, Proposition 4.69; the shape $\left(\begin{smallmatrix} \chi_\ell \varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$, Proposition 4.24.

- **Section 1.1 and the Introduction**: $A_f = \mathrm{ad}^0 V$, the eigenvalues $\alpha/\beta, 1, \beta/\alpha$, Proposition 4.52; $\mathrm{ad}^0 \simeq \mathrm{Sym}^2 \otimes \det^{-1}$ and $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$, Proposition 4.56.

- **Proofs of Theorems thm:main and thm:quant, Lemma lem:euler (types $\mathrm{I}_n^*$, $\mathrm{I}_0^*$)**: $\mathrm{ad}^0 T$ is unchanged by twists, Proposition 4.48, Proposition 4.60.

- **The hypothesis of Theorem thm:main, Proposition prop:Sf, DDT Theorem 3.15**: absolute irreducibility on $G_{\mathbb{Q}(\sqrt{\ell^*})}$, Definition 4.11, Theorem 4.44, Proposition 4.46, Remark 4.83.

- **Conjecture conj:euler, Remark rem:ns, Section sec:anomaly** (the image 3Ns and $h_3(E) = 1$): Proposition 4.63, Theorem 4.66, Proposition 4.81.

## 4.1 Representations over a field

This section fixes the language: representations, the linear constructions on them, and the behaviour of spaces of $G$-maps under extension of the field of scalars. In the paper these are used silently everywhere. Three uses are explicit: the identification of $H^0$ of a twisted adjoint with a space of $G$-maps in Lemma lem:h0 (Proposition 4.5), the descent from $\mathbb{F}_9$ to $\mathbb{F}_3$ of the isomorphism $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ in Proposition prop:ns, and the phrase "over a common finite extension of $\mathbb{F}_\ell$" in Lemma lem:old (Proposition 4.7). The words "irreducible" and "absolutely irreducible" and Schur's lemma, defined and proved in Section 4.2, are used here by forward reference; there is no circularity, since Proposition 4.7(ii) uses only Theorem 4.10(i), whose proof uses nothing from this section.

**Convention 4.1.** The following conventions hold in this chapter.

(C1) *Representations.* $G$ is a group (finite only where this is said), $k$ a field and $R$ a commutative ring. A representation of $G$ over $k$ is a homomorphism $\rho\colon G \to \mathrm{GL}(V)$ with $V$ a *finite-dimensional* $k$-vector space; over $R$, $V$ is a free $R$-module of finite rank. Matrices act on column vectors on the left (as in Chapter 1). A *$G$-map* (intertwiner) is a $k[G]$-linear map; $\mathrm{Hom}_G(V, W)$ is the space of $G$-maps and $V \simeq W$ means isomorphic as representations. We often write $gv$ for $\rho(g)v$.

(C2) *Duals.* The linear dual is $V^\vee = \mathrm{Hom}_k(V, k)$ with $(g\lambda)(v) = \lambda(g^{-1}v)$; in the dual basis $(e_i^\vee)$, $g$ acts by the matrix $\rho(g^{-1})^t$. The symbol $^*$ is reserved for Pontryagin duals (convention B-C4 of Appendix B).

(C3) *Adjoints.* $\mathrm{End}(V) = \mathrm{End}_k(V)$ is a representation by $g \cdot x = \rho(g) x \rho(g)^{-1}$; written $\mathrm{Ad}\, V$ when the action is to be stressed. $\mathrm{ad}^0 V$ is its subspace of trace zero (the paper's notation;

the superscript 0 is part of the symbol). $\operatorname{Sym}^2 V$ is the symmetric square. The matrix units are $E_{ij} = e_i \otimes e_j^\vee$, and for $\dim V = 2$ the basis of $\mathrm{ad}^0 V$ is the paper's

$$e = E_{12} = e_1 \otimes e_2^\vee, \qquad h = E_{11} - E_{22}, \qquad f = E_{21} = e_2 \otimes e_1^\vee,$$
$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

(C4) *Characters and twists.* A character of $G$ with values in $R$ is a homomorphism $\chi\colon G \to R^\times$. $R(\chi)$ is $R$ with $G$ acting through $\chi$, and $M(\chi) = M \otimes_R R(\chi)$, also written $M \otimes \chi$ or $\rho \otimes \chi$. The trace function of $V$ is written $\mathrm{tr}_V$, never $\chi_V$, so that the letter $\chi$ is free for characters such as $\chi_{-3}$ and the cyclotomic $\chi_\ell$.

(C5) *Fields.* $\bar{k}$ is an algebraic closure of $k$, and $V_K = K \otimes_k V$ for an extension $K/k$. $\mathbb{F}_9 = \mathbb{F}_3[i]$ with $i^2 = -1$ (Chapter 2).

(C6) *Standard models.* $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $\sigma = \mathrm{diag}(1, -1)$, $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $a = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$, and the subgroups $C_s$, $N_s$, $C_{ns}$, $N_{ns}$, $Q_8$, $C_4 = \langle J \rangle$, B, U $= \langle u \rangle$ of $\mathrm{GL}_2(\mathbb{F}_3)$ are the standard models of Chapter 1. Thus $J = m_{-i}$ is multiplication by $-i$ on $\mathbb{F}_9 = \mathbb{F}_3 \oplus \mathbb{F}_3 i$, $a = m_{1+i}$, $a^2 = J$, $w = \sigma J$, $N_s(\mathbb{F}_3) = \langle J, \sigma \rangle$, $N_{ns}(\mathbb{F}_3) = \langle a, \sigma \rangle$, and $\psi\colon C_4 \to \mathbb{F}_9^\times$ is the character $\psi(J) = i$.

(C7) *Lattices.* $\mathcal{O}$ is a discrete valuation ring with uniformiser $\varpi$, fraction field $K$ and residue field $\mathcal{O}/\varpi$; in the paper $\mathcal{O} = \mathbb{Z}_\ell$ and $\varpi = \ell$. A lattice in a $K$-vector space of dimension $n$ is a finitely generated $\mathcal{O}$-submodule that spans it; it is free of rank $n$ (Chapter 3).

(C8) *No Galois groups.* This chapter never uses Frobenius elements. Where the paper's Frobenius appears it is an abstract element $F$ with $FgF^{-1} = g^q$ for $g$ in an inertia image, which is the relation satisfied by the *arithmetic* Frobenius (convention B-C2); the geometric Frobenius $F^{-1}$ has the inverse eigenvalues. The letter $q$ is then the paper's bad prime.

(C9) *Typography.* Double names take plain hyphens: Brauer-Nesbitt, Jordan-Hölder.

**Definition 4.2** (Representations)**.** A *representation* of $G$ over $k$ is a homomorphism $\rho\colon G \to \mathrm{GL}(V)$, $V$ a finite-dimensional $k$-vector space; equivalently, a left $k[G]$-module structure on $V$, where $k[G]$ is the group algebra (finite sums $\sum_g a_g g$). Its *degree* is $n = \dim V$; it is *faithful* if $\ker \rho = 1$. A basis of $V$ turns $\rho$ into a homomorphism $G \to \mathrm{GL}_n(k)$, and two such matrix representations $\rho_1$, $\rho_2$ are isomorphic iff there is one $P \in \mathrm{GL}_n(k)$ with $\rho_2(g) = P\rho_1(g)P^{-1}$ for all $g$ (the matrix of an isomorphism). Examples:

(a) the trivial representation $\mathbf{1}$ ($V = k$, $g \mapsto 1$);

(b) one-dimensional representations, which are the characters $\chi\colon G \to k^\times$; they factor through the abelianisation $G^{\mathrm{ab}}$;

(c) the permutation representation $k[X]$ of a $G$-set $X$ (basis $X$, $g$ permuting it), and the regular representation $k[G]$;

(d) the *standard representation* of a subgroup $H \leq \mathrm{GL}_2(k)$ on $k^2$ (the inclusion);

(e) inflation: if $\pi\colon G \to G'$ is a homomorphism and $\rho'$ a representation of $G'$, then $\rho' \circ \pi$ is one of $G$. A representation of a finite quotient of a profinite group is thus one of the group (Chapter 5).

The representations of Riley-Hobson-Bence [165] are the case $k = \mathbb{C}$, $G$ finite.

The two descriptions in Definition 4.2 are used side by side. The homomorphism $\rho$ is what one computes with (matrices, traces, eigenvalues). The module language is what makes general theorems available: subrepresentations, quotients, composition series and $G$-maps are submodules, quotient modules, composition series and module maps over the ring $k[G]$, so the results of Chapter 3 apply without change. A small example already shows the phenomenon that organises Section 4.3. Let $S_3$ permute the coordinates of $k^3$. The line $D = k(1,1,1)^t$ and the plane $P = \{x : x_1 + x_2 + x_3 = 0\}$ are stable. If $\operatorname{char} k \neq 3$, then $D \cap P = 0$ (the vector $(c,c,c)^t$ lies in $P$ iff $3c = 0$), and $k^3 = D \oplus P$ by counting dimensions. If $\operatorname{char} k = 3$, then $(1,1,1)^t \in P$, so $D \subset P$ and this decomposition is lost; we return to the example after Maschke's theorem.

**Definition 4.3** (Constructions)**.** Let $V, W$ be representations of $G$ over $k$. A *subrepresentation* is a $G$-stable subspace $U \subseteq V$, and $V/U$ is the quotient representation. $G$ acts on $V \oplus W$ componentwise and on $V \otimes W$ by $g(v \otimes w) = gv \otimes gw$; on $V^\vee$ as in (C2); on $\operatorname{Hom}_k(V,W)$ by $(g\phi)(v) = g\,\phi(g^{-1}v)$. The *invariants* are $V^G = \{v : gv = v \text{ for all } g\}$, and $\det\rho\colon G \to k^\times$ is the character $g \mapsto \det\rho(g)$, the representation on $\wedge^n V$. The same definitions apply to free modules of finite rank over $R$.

**Proposition 4.4.** *Let $V, W$ be representations of $G$ over $k$, $n = \dim V$, $\chi$ a character.*

*(i)* $\operatorname{tr}(g \mid V \oplus W) = \operatorname{tr}(g \mid V) + \operatorname{tr}(g \mid W)$, $\operatorname{tr}(g \mid V \otimes W) = \operatorname{tr}(g \mid V)\operatorname{tr}(g \mid W)$, $\operatorname{tr}(g \mid V^\vee) = \operatorname{tr}(g^{-1} \mid V)$ *and* $\det(V \otimes \chi) = \chi^n \det V$.

*(ii) The canonical maps $V \to V^{\vee\vee}$ and $V^\vee \otimes W^\vee \to (V \otimes W)^\vee$, $\lambda \otimes \mu \mapsto (v \otimes w \mapsto \lambda(v)\mu(w))$, are $G$-isomorphisms.*

*(iii) If $n = 2$ (or $V$ is free of rank 2 over $R$), let $\omega(x,y) = \det[x\,|\,y]$. Then $\omega(gx,gy) = \det(g)\omega(x,y)$, and $v \mapsto \omega(v,-)$ is a $G$-isomorphism $V \to V^\vee \otimes \det$. Hence $V^\vee \simeq V \otimes \det^{-1}$.*

*Proof.* (i) The first is clear in a basis adapted to $V \oplus W$; the second holds because the matrix of $g$ on $V \otimes W$ in the basis $e_i \otimes f_j$ is the Kronecker product of the two matrices, whose trace is the product of the traces; the third because $g$ acts on $V^\vee$ by $\rho(g^{-1})^t$; the fourth because $\det(\chi(g)\rho(g)) = \chi(g)^n \det\rho(g)$. (ii) Both maps are isomorphisms of vector spaces (compare bases). For the first, $v \mapsto (\lambda \mapsto \lambda(v))$, and $(g \cdot \mathrm{ev}_v)(\lambda) = \mathrm{ev}_v(g^{-1}\lambda) = (g^{-1}\lambda)(v) = \lambda(gv)$, so $g \cdot \mathrm{ev}_v = \mathrm{ev}_{gv}$. For the second, write $\Lambda$ for the image of $\lambda \otimes \mu$, the form $v \otimes w \mapsto \lambda(v)\mu(w)$. The image of $g(\lambda \otimes \mu) = g\lambda \otimes g\mu$ is the form $v \otimes w \mapsto (g\lambda)(v)\,(g\mu)(w) = \lambda(g^{-1}v)\mu(g^{-1}w)$, while by (C2) for the representation $V \otimes W$,

$$(g\Lambda)(v \otimes w) = \Lambda(g^{-1}(v \otimes w)) = \Lambda(g^{-1}v \otimes g^{-1}w) = \lambda(g^{-1}v)\mu(g^{-1}w).$$

The two forms agree on pure tensors, which span $V \otimes W$, so the map is equivariant. It is bijective because it sends the basis $e_i^\vee \otimes f_j^\vee$ to the basis of $(V \otimes W)^\vee$ dual to $(e_i \otimes f_j)$. (iii) $\omega(gx,gy) = \det(\rho(g)[x\,|\,y]) = \det(g)\omega(x,y)$. Put $\Psi(v) = \omega(v,-)$. As $\omega$ is a perfect pairing ($\omega(e_1,e_2) = 1$), $\Psi$ is an isomorphism of vector spaces (or of free $R$-modules). For $y \in V$,

$$(g \cdot \Psi(v))(y) = \Psi(v)(g^{-1}y) = \omega(v, g^{-1}y) = \det(g)^{-1}\omega(gv, y),$$

so $\Psi(gv) = \det(g)\,(g \cdot \Psi(v))$, which says that $\Psi$ is $G$-equivariant from $V$ to $V^\vee \otimes \det$.

In coordinates, $\Psi(e_1) = \omega(e_1, -) = e_2^\vee$ and $\Psi(e_2) = \omega(e_2, -) = -e_1^\vee$, so the matrix of $\Psi$ from the basis $(e_1, e_2)$ to $(e_1^\vee, e_2^\vee)$ is $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = J^{-1}$, and the equivariance of $\Psi$ becomes the matrix identity

$$\rho(g^{-1})^t = \det(g)^{-1}\, J\rho(g)J^{-1}, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

which for $\rho(g) = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ says that both sides equal $\frac{1}{ad-bc}\left(\begin{smallmatrix} d & -c \\ -b & a \end{smallmatrix}\right)$. So a two-dimensional representation and its dual differ only by the twist by the determinant; for $V_\ell E$, whose determinant is the cyclotomic character, this is used in Proposition 4.56(iii). □

**Proposition 4.5** ($G$-maps as invariants)**.** *For representations $V, W$ over $k$ (or free $R$-modules of finite rank with a $G$-action), the map*

$$\Theta\colon V^\vee \otimes W \to \operatorname{Hom}(V, W), \qquad \lambda \otimes w \mapsto (v \mapsto \lambda(v)w)$$

*is a $G$-isomorphism, and $\operatorname{Hom}_G(V, W) = \operatorname{Hom}(V, W)^G = (V^\vee \otimes W)^G$. In particular $\operatorname{End}(V)^G = \operatorname{End}_G(V)$, and for a character $\chi$, $(\operatorname{End}(V) \otimes \chi)^G = \operatorname{Hom}_G(V, V \otimes \chi)$, the space of $\phi \in \operatorname{End}(V)$ with $\phi\rho(g) = \chi(g)\rho(g)\phi$ for all $g$.*

*Proof.* $\Theta$ sends $e_i^\vee \otimes f_j$ to the matrix unit mapping $e_i$ to $f_j$, so it is bijective. It is equivariant: $\Theta(g\lambda \otimes gw)(v) = \lambda(g^{-1}v)\, gw = g(\Theta(\lambda \otimes w)(g^{-1}v)) = (g \cdot \Theta(\lambda \otimes w))(v)$. A linear map $\phi$ is fixed by $G$ iff $g\phi g^{-1} = \phi$ for all $g$, that is, iff $\phi$ is a $G$-map. For the last statement, $\operatorname{Hom}(V, V \otimes \chi)$ is $\operatorname{End}(V)$ with $g$ acting by $\phi \mapsto \chi(g)\rho(g)\phi\rho(g)^{-1}$, that is, $\operatorname{End}(V) \otimes \chi$; it is fixed iff $\phi\rho(g) = \chi(g)\rho(g)\phi$. □

*In the paper* 4.6. In the proof of Lemma lem:h0, the sentence "The $\ell^k$-torsion of $H^0(\mathbb{Q}, W^D)$ is $H^0(\mathbb{Q}, \mathrm{ad}^0(T/\ell^k)(1))$, the group of trace-zero $G_\mathbb{Q}$-maps $T/\ell^k \to (T/\ell^k)(1)$" is Proposition 4.5 for the free $\mathbb{Z}/\ell^k$-module $T/\ell^k$ and $\chi = \chi_\ell \bmod \ell^k$, together with Proposition 4.75(iii) for $W^D[\ell^k] = \mathrm{ad}^0(T/\ell^k)(1)$. In Lemma lem:tamfactor the basis $e = e_1 \otimes e_2^\vee$ of $\mathrm{ad}^0 T$ is the image under $\Theta$ of a vector of $V^\vee \otimes V$, as in (C3).

**Proposition 4.7** (Extension of scalars)**.** *Let $K/k$ be a field extension and $V, W$ representations of $G$ over $k$.*

*(i)* $\operatorname{Hom}_{K[G]}(V_K, W_K) = K \otimes_k \operatorname{Hom}_{k[G]}(V, W)$ *inside* $\operatorname{Hom}_K(V_K, W_K) = K \otimes_k \operatorname{Hom}_k(V, W)$. *In particular the two spaces of $G$-maps have the same dimension, and $(V_K)^G = K \otimes_k V^G$.*

*(ii) If $V$ and $W$ are irreducible (Definition 4.9) and $V_K \simeq W_K$, then $V \simeq W$.*

*(iii) If $V_K$ is irreducible (for instance if $V$ is absolutely irreducible, Proposition 4.12) and $W_K \simeq V_K$, then $W \simeq V$.*

*(iv) (Noether-Deuring) If $V_K \simeq W_K$, then $V \simeq W$, for arbitrary $V, W$.*

*Proof.* (i) Being a $G$-map is a system of linear equations, with coefficients in $k$, on the entries of a matrix, and the point is that such a system has, over $K$, exactly the solutions obtained from those over $k$ by extending scalars. Precisely, for $g \in G$ let $\delta_g\colon \operatorname{Hom}_k(V, W) \to \operatorname{Hom}_k(V, W)$ be $\phi \mapsto \rho_W(g)\phi - \phi\rho_V(g)$; then $\operatorname{Hom}_{k[G]}(V, W) = \bigcap_g \ker \delta_g$. As $\operatorname{Hom}_k(V, W)$ is finite-dimensional, the descending chain of finite intersections stabilises, so there are $g_1, \dots, g_r$ with $\bigcap_g \ker \delta_g = \ker \delta$ for the single linear map $\delta = (\delta_{g_1}, \dots, \delta_{g_r})$. A linear map defined over $k$ has the same rank over $K$ (rank is the size of a nonvanishing minor of its matrix), so $\ker(\delta \otimes K) = K \otimes \ker \delta$. Hence

$$\textstyle\bigcap_g \ker(\delta_g \otimes K) \subseteq \ker(\delta \otimes K) = K \otimes \bigcap_g \ker \delta_g \subseteq \bigcap_g \ker(\delta_g \otimes K),$$

and the left side is $\mathrm{Hom}_{K[G]}(V_K, W_K)$. The statement on invariants is the case $V = \mathbf{1}$. (ii) By (i), $\mathrm{Hom}_{k[G]}(V, W) \neq 0$; a nonzero $G$-map between irreducible representations is an isomorphism (Theorem 4.10(i)). (iii) If $U \subseteq W$ is a subrepresentation, $U_K$ is a subrepresentation of $W_K \simeq V_K$, so $U_K \in \{0, W_K\}$ and $U \in \{0, W\}$; thus $W$ is irreducible (it is nonzero). The same argument applied to $V$ shows that $V$ is irreducible, since $V_K$ is. So (ii) applies. (iv) We do not prove this; see [37, Section 29]. It is not needed in the paper: wherever an isomorphism is descended, (ii) or (iii) applies. □

*In the paper* 4.8. Proposition prop:ns: $\bar\rho_{E,3} \otimes \mathbb{F}_9 \simeq \mathrm{Ind}\,\psi \simeq (\bar\rho_{E,3} \otimes \chi_{-3}) \otimes \mathbb{F}_9$, and both $\bar\rho_{E,3}$ and $\bar\rho_{E,3} \otimes \chi_{-3}$ are irreducible over $\mathbb{F}_3$; so (ii) gives $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \chi_{-3}$ over $\mathbb{F}_3$ ("the isomorphism descending to $\mathbb{F}_3$"; the same sentence ends the proof of Lemma lem:h0, where it is (i)). Lemma lem:old: the residual representations of $f$ and of $h$ are compared over a common finite extension $k'$ of $\mathbb{F}_\ell$; by (iii), an isomorphism there between $\bar\rho \otimes k'$ and $\bar\rho_h$ is all that is needed.

## 4.2 Irreducibility, Schur's lemma, absolute irreducibility

This section introduces irreducible and absolutely irreducible representations and proves Schur's lemma, the tool behind every step of the paper at which a nonzero map is concluded to be an isomorphism. Absolute irreducibility is the hypothesis of Theorem thm:main (on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$) and of DDT Theorem 3.15; before using Schur in the form $\mathrm{End} = k$ (Lemmas lem:h0 and lem:old) the paper passes from "irreducible" to "absolutely irreducible", and for $\bar\rho_{E,\ell}$ with $\ell$ odd this is automatic by the distinct-eigenvalue criterion (Proposition 4.13) applied to complex conjugation.

**Definition 4.9.** A representation $V \neq 0$ is *irreducible* (or simple) if its only subrepresentations are 0 and $V$, and *reducible* otherwise; it is *indecomposable* if it is not the direct sum of two nonzero subrepresentations. Irreducible implies indecomposable, not conversely (Example 4.20). If $\dim V = 2$, $V$ is irreducible iff no line of $V$ is $G$-stable, iff $\rho(G)$ lies in no Borel subgroup of $\mathrm{GL}(V)$ (the stabiliser of a line, Chapter 1). Examples: every one-dimensional representation is irreducible. The standard representations of $\mathrm{GL}_2(\mathbb{F}_3)$ and of $N_s(\mathbb{F}_3)$ on $\mathbb{F}_3^2$ are irreducible (they stabilise no line, Chapter 1), and so is that of $C_4 = \langle J \rangle$: a stable line would be spanned by an eigenvector of $J$, but the characteristic polynomial $X^2 + 1$ of $J$ has no root in $\mathbb{F}_3$. The standard representations of $\mathrm{U} = \langle u \rangle$ and of $\mathrm{B}(\mathbb{F}_3)$ are reducible: they stabilise $\mathbb{F}_3 e_1$.

**Theorem 4.10** (Schur's lemma)**.** *Let $V, W$ be irreducible representations of a group $G$ over a field $k$.*

*(i) Every $G$-map $V \to W$ is 0 or an isomorphism. Hence $D = \mathrm{End}_G(V)$ is a division algebra, of finite dimension over $k$.*

*(ii) If $k$ is algebraically closed, then $\mathrm{End}_G(V) = k$ (the scalars), and $\dim_k \mathrm{Hom}_G(V, W)$ is 1 if $V \simeq W$ and 0 otherwise.*

*(iii) The conclusions of (ii) hold over any $k$ if $V$ and $W$ are absolutely irreducible (Definition 4.11).*

*(iv) If $V$ is absolutely irreducible, every central element of $G$ acts on $V$ by a scalar; every absolutely irreducible representation of an abelian group is one-dimensional; and if $G$ is abelian and $V$ irreducible, $\mathrm{End}_G(V)$ is a field, equal to the image of $k[G]$ in $\mathrm{End}_k(V)$.*

*Proof.* (i) If $\phi\colon V \to W$ is a $G$-map, $\ker\phi$ and $\operatorname{im}\phi$ are subrepresentations. If $\phi \neq 0$, then $\ker\phi \neq V$, so $\ker\phi = 0$, and $\operatorname{im}\phi \neq 0$, so $\operatorname{im}\phi = W$. A nonzero element of $\operatorname{End}_G(V)$ is therefore invertible, and its inverse is again a $G$-map. (ii) Let $\phi \in \operatorname{End}_G(V)$. As $V \neq 0$ is finite-dimensional and $k$ algebraically closed, $\phi$ has an eigenvalue $c$; then $\phi - c$ is a $G$-map that is not injective, so $\phi - c = 0$ by (i). If $V \simeq W$ via $\alpha$, then $\operatorname{Hom}_G(V,W) = \alpha \operatorname{End}_G(V) = k\alpha$; if not, $\operatorname{Hom}_G(V,W) = 0$ by (i). (iii) Now $V_{\bar k}$ and $W_{\bar k}$ are irreducible, so (ii) applies to them. By Proposition 4.7(i), $\dim_k \operatorname{End}_G(V) = \dim_{\bar k} \operatorname{End}_G(V_{\bar k}) = 1$, so $\operatorname{End}_G(V) = k$ (it contains the scalars); and $\dim_k \operatorname{Hom}_G(V,W) = \dim_{\bar k} \operatorname{Hom}_G(V_{\bar k}, W_{\bar k})$, which is 1 if $V_{\bar k} \simeq W_{\bar k}$ and 0 otherwise. It remains to see that $V_{\bar k} \simeq W_{\bar k}$ iff $V \simeq W$. One direction is clear. For the other, $V$ and $W$ are irreducible (a nonzero proper subrepresentation $U$ of $V$ would give the nonzero proper subrepresentation $U_{\bar k}$ of $V_{\bar k}$), so Proposition 4.7(ii) applies. (iv) A central $z$ satisfies $\rho(z)\rho(g) = \rho(g)\rho(z)$ for all $g$, so $\rho(z) \in \operatorname{End}_G(V) = k$ by (iii). If $G$ is abelian, every element is central, so every $\rho(g)$ is a scalar, every line is stable, and $\dim V = 1$. For the last statement let $A \subseteq \operatorname{End}_k(V)$ be the image of $k[G]$; it is a commutative subalgebra, and $\operatorname{End}_G(V)$ is the commutant of $A$, which contains $A$. For $0 \neq x \in A$, $\ker x$ is $A$-stable ($A$ is commutative), hence $G$-stable and different from $V$, so $\ker x = 0$: $A$ has no zero divisors, and a finite-dimensional commutative $k$-algebra without zero divisors is a field. Then $V$ is an $A$-vector space whose $A$-subspaces are $G$-stable, so $\dim_A V = 1$, $V \simeq A$, and the commutant of $A$ acting on itself by multiplication is $A$. □

Over a field that is not algebraically closed, $\operatorname{End}_G(V)$ can be a proper extension field of $k$ (Example 4.14 below: $\mathbb{F}_9$ over $\mathbb{F}_3$), or even non-commutative. Let $\mathbb{H}$ be Hamilton's quaternions, and let $Q_8 = \{\pm 1, \pm\mathbf{i}, \pm\mathbf{j}, \pm\mathbf{k}\} \subset \mathbb{H}^\times$ act on $V = \mathbb{H} \simeq \mathbb{R}^4$ by left multiplication. As $Q_8$ spans $\mathbb{H}$ over $\mathbb{R}$, a $Q_8$-stable subspace is a left ideal of the division ring $\mathbb{H}$, hence 0 or $\mathbb{H}$: $V$ is irreducible. A $G$-map is an $\mathbb{R}$-linear map $\phi$ commuting with left multiplication by all of $\mathbb{H}$, so $\phi(x) = \phi(x \cdot 1) = x\phi(1)$ is right multiplication by $\phi(1)$. Hence $\operatorname{End}_G(V) \simeq \mathbb{H}^{\mathrm{op}}$, a non-commutative division algebra of dimension 4 over $\mathbb{R}$, as (i) allows.

**Definition 4.11.** A representation $V$ of $G$ over $k$ is *absolutely irreducible* if $V_{\bar k}$ is irreducible. It is *absolutely reducible* otherwise.

**Proposition 4.12.** *Let $V$ be a representation of $G$ over $k$, $n = \dim V$.*

*(i) $V$ is absolutely irreducible iff the map $k[G] \to \operatorname{End}_k(V)$ is surjective. If so, $V_K$ is irreducible for every extension $K/k$.*

*(ii) An absolutely irreducible $V$ is irreducible and $\operatorname{End}_G(V) = k$.*

*(iii) If $n = 2$, $V$ is absolutely irreducible iff $\rho(G)$ stabilises no line of $\bar k^2$, iff $\rho(G)$ is not conjugate in $\mathrm{GL}_2(\bar k)$ into the upper triangular group.*

*(iv) If $V$ is irreducible, $V$ is absolutely irreducible iff $\operatorname{End}_G(V) = k$. This holds over every field $k$.*

*Proof.* (i) Let $A \subseteq \operatorname{End}_k(V)$ be the image of $k[G]$. The image of $\bar k[G]$ in $\operatorname{End}_{\bar k}(V_{\bar k}) = \bar k \otimes_k \operatorname{End}_k(V)$ is $\bar k \otimes_k A$, of $\bar k$-dimension $\dim_k A$. If $V_{\bar k}$ is irreducible, the density theorem (Theorem 4.34(b), whose proof does not use this proposition) says that $\bar k \otimes_k A = \operatorname{End}(V_{\bar k})$, so $\dim_k A = n^2$ and $A = \operatorname{End}_k(V)$. Conversely, if $A = \operatorname{End}_k(V)$, then for every extension $K/k$ the image of $K[G]$ is $\operatorname{End}_K(V_K)$, and a subspace stable under all of $\operatorname{End}_K(V_K)$ is 0 or $V_K$. (ii) If $U \subseteq V$ is a subrepresentation, $U_{\bar k}$ is one of $V_{\bar k}$; so $V$ is irreducible. $\operatorname{End}_G(V) = k$ is Theorem 4.10(iii). (iii) A

nonzero proper subrepresentation of the two-dimensional $V_{\bar{k}}$ is a stable line; and $\rho(G)$ stabilises the line $\bar{k}e_1'$ iff it is upper triangular in a basis $e_1', e_2'$. (iv) One direction is (ii). Conversely let $V$ be irreducible with $\mathrm{End}_G(V) = k$. The density theorem (Theorem 4.34(a), with $r = 1$) says that the image of $k[G]$ is $\mathrm{End}_{\mathrm{End}_G(V)}(V) = \mathrm{End}_k(V)$, and (i) applies. □

**Proposition 4.13** (Distinct eigenvalues)**.** *Let $V$ be a representation of any group $G$ over $k$, $n = \dim V$, and suppose that some $g \in G$ (or some element of the image of $k[G]$) acts on $V$ with $n$ distinct eigenvalues $\lambda_1, \dots, \lambda_n \in k$. Then every $G$-stable subspace $U$ of $V_{\bar{k}}$ is a sum of eigenlines of $g$, and $U = (U \cap V)_{\bar{k}}$ with $U \cap V$ a subrepresentation of $V$. In particular $V$ is irreducible iff it is absolutely irreducible. This applies when $n = 2$, $\mathrm{char}\, k \neq 2$ and some $c \in G$ has $\rho(c)^2 = 1$, $\det \rho(c) = -1$: the eigenvalues of $\rho(c)$ are then $1$ and $-1$, which are distinct.*

*Proof.* Let $v_1, \dots, v_n \in V$ be eigenvectors of $g$ for $\lambda_1, \dots, \lambda_n$ (they exist over $k$ because the $\lambda_i$ lie in $k$); they form a basis of $V$ and of $V_{\bar{k}}$. Let $u = \sum_i c_i v_i \in U$. Applying $\prod_{j \neq i}(g - \lambda_j)$, which preserves $U$, gives $c_i \prod_{j \neq i}(\lambda_i - \lambda_j) v_i \in U$, and the product is nonzero; so $c_i v_i \in U$ for every $i$. Hence $U$ is spanned by the $v_i$ it contains, $U = (U')_{\bar{k}}$ for the $k$-span $U'$ of these $v_i$, and $U' = U \cap V$ is $G$-stable. If $V$ is irreducible, $U \cap V \in \{0, V\}$, so $U \in \{0, V_{\bar{k}}\}$. The case $n = 2$: $\rho(c)^2 = 1$ gives eigenvalues in $\{\pm 1\}$, with product $-1$. □

**Example 4.14** ($C_4$ on $\mathbb{F}_3^2$)**.** Let $C_4 = \langle J \rangle$, $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, act on $\mathbb{F}_3^2$. It is irreducible (Definition 4.9) but not absolutely irreducible: over $\mathbb{F}_9 = \mathbb{F}_3[i]$,

$$J\begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} i \\ -1 \end{pmatrix} = i\begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad J\begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} -i \\ -1 \end{pmatrix} = -i\begin{pmatrix} 1 \\ -i \end{pmatrix},$$

so with $v = (1, i)^t$ and $\bar{v} = (1, -i)^t$ we get $\mathbb{F}_9^2 = \mathbb{F}_9 v \oplus \mathbb{F}_9 \bar{v} \simeq \psi \oplus \psi^{-1}$, where $\psi(J) = i$ is a character of order 4 with values in $\mathbb{F}_9^\times$ (these are the two lines spanned by $(\mp i, 1)^t$ of Chapter 1). The vector $\bar{v}$ is obtained from $v$ by the Frobenius $x \mapsto x^3$ of $\mathbb{F}_9$ on coordinates, and $\psi^{-1} = \psi^3$ is the Frobenius conjugate of $\psi$. The commutant is $\mathrm{End}_{C_4}(\mathbb{F}_3^2) = \mathbb{F}_3[J] \simeq \mathbb{F}_3[X]/(X^2 + 1) \simeq \mathbb{F}_9$ (Lemma 4.70), a field of dimension 2 over $\mathbb{F}_3$, as Proposition 4.12(iv) and Theorem 4.10(iv) predict. In contrast, $C_{ns}(\mathbb{F}_3) = \langle a \rangle \simeq \mathbb{F}_9^\times$ also acts irreducibly and not absolutely irreducibly, with commutant $\mathbb{F}_3[a] = \mathbb{F}_9$; and U fixes exactly one line of $\mathbb{F}_3^2$ and is not a direct sum of two lines (Example 4.20).

*In the paper* 4.15. The hypothesis of Theorem thm:main is that $\bar{\rho}_{E,\ell}$ is absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$; in DFG04 this is $\ell \notin S_f$. Example 4.14 is the reason why "absolutely" cannot be dropped: in the $N_s(\mathbb{F}_3)$ case the restriction to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ has image $C_4$, which is irreducible over $\mathbb{F}_3$ and splits as $\psi \oplus \psi^{-1}$ over $\mathbb{F}_9$ (Proposition prop:ns). In Lemmas lem:h0 and lem:old and in Proposition prop:Sf the paper deduces absolute irreducibility from irreducibility because complex conjugation has the distinct eigenvalues $\pm 1$: this is the case $n = 2$ of Proposition 4.13, with $c$ complex conjugation and $\det \bar{\rho}(c) = \bar{\chi}_\ell(c) = -1$ (Chapter 5).

## 4.3 Semisimplicity, Maschke's theorem, Jordan-Hölder

This section explains when a representation is a direct sum of irreducibles. Over a finite group of order prime to the characteristic it always is (Maschke); in characteristic dividing the order it need not be, and this failure is exactly what the paper's Lemma lem:IV records: an element of order 3 in $\mathrm{SL}_2(\mathbb{Z}_3)$ fixes no vector of $T$ but a line of $T/3T$. The section also proves the facts about

extensions of two characters used in Lemma lem:old (a non-split extension has a unique stable line) and gives a second proof that $Q_8$ acts absolutely irreducibly on $\mathbb{F}_3^2$, a step of Proposition prop:ns.

**Definition 4.16.** A representation is *semisimple* if it is a sum of irreducible subrepresentations. The zero representation is semisimple.

**Proposition 4.17.** *Let $V$ be a finite-dimensional representation of any group $G$ over $k$.*

*(i) The following are equivalent: (a) $V$ is a sum of irreducible subrepresentations; (b) $V$ is a direct sum of irreducible subrepresentations; (c) every subrepresentation of $V$ has a $G$-stable complement.*

*(ii) Subrepresentations and quotients of a semisimple representation are semisimple.*

*(iii) (Isotypic decomposition.) For an irreducible $S$ let $V_S$ be the sum of the subrepresentations of $V$ isomorphic to $S$. If $V$ is semisimple, $V = \bigoplus_S V_S$ over the isomorphism classes of irreducible $S$, and $V_S \simeq S^{m_S}$, where $m_S \dim_k \operatorname{End}_G(S) = \dim_k \operatorname{Hom}_G(S, V)$. In particular $m_S = \dim \operatorname{Hom}_G(S, V)$ if $S$ is absolutely irreducible.*

*Proof.* (i) Suppose (a), $V = \sum_{i\in I} S_i$, and let $U$ be a subrepresentation. Choose $J \subseteq I$ maximal such that the sum $U + \sum_{j\in J} S_j$ is direct. For every $i$, $S_i \cap (U \oplus \bigoplus_J S_j)$ is a subrepresentation of the irreducible $S_i$; if it were 0, $J \cup \{i\}$ would contradict maximality; so $S_i \subseteq U \oplus \bigoplus_J S_j$. Hence $V = U \oplus \bigoplus_J S_j$, which gives (c), and (b) for $U = 0$. (b) implies (a) trivially. Suppose (c), and let $V'$ be the sum of all irreducible subrepresentations. If $V' \neq V$, a complement $C \neq 0$ exists; $C$ contains a nonzero subrepresentation of minimal dimension, which is irreducible and lies in $V' \cap C = 0$, a contradiction. (ii) By the proof of (i), $V = U \oplus \bigoplus_J S_j$, so $V/U \simeq \bigoplus_J S_j$ is semisimple. Applying this to a complement $C$ of $U$, with $V = C \oplus U$, shows that $U \simeq V/C$ is semisimple. (iii) Write $V = \bigoplus_i S_i$ by (b), and let $W_S$ be the sum of the $S_i$ isomorphic to $S$. If $T \subseteq V$ is irreducible and $T \simeq S$, its projection to any $S_i \not\simeq S$ is 0 by Theorem 4.10(i); so $T \subseteq W_S$. Hence $V_S = W_S$, $V = \bigoplus_S V_S$, and $V_S \simeq S^{m_S}$ with $m_S$ the number of $i$ with $S_i \simeq S$. Then $\operatorname{Hom}_G(S, V) = \bigoplus_i \operatorname{Hom}_G(S, S_i)$ has dimension $m_S \dim \operatorname{End}_G(S)$; the number $m_S$ does not depend on the decomposition. □

**Theorem 4.18** (Maschke)**.** *(i) Let $G$ be finite and $\operatorname{char} k \nmid \#G$. Every finite-dimensional representation of $G$ over $k$ is semisimple.*

*(ii) Let $G$ be finite, $R$ a commutative ring with $\#G \in R^\times$, and $M$ an $R[G]$-module. Then $e_G = \frac{1}{\#G}\sum_{g\in G} g$ is a central idempotent of $R[G]$ with $e_G M = M^G$, $M = M^G \oplus (1 - e_G)M$ as $R[G]$-modules, and $(M/IM)^G = M^G/IM^G$ for every ideal $I$ of $R$. In particular, for $R = \mathbb{Z}_\ell$ with $\ell \nmid \#G$ and a $\mathbb{Z}_\ell[G]$-lattice $T$, $(T/\ell^m T)^G = T^G/\ell^m T^G$: invariants commute with reduction.*

*Proof.* (i) Let $U \subseteq V$ be a subrepresentation and $\pi\colon V \to U$ any $k$-linear projection ($\pi|_U = \mathrm{id}$; extend a basis of $U$ to one of $V$). Such a $\pi$ need not commute with $G$, and the idea is to average it over the group. Put $\pi' = \frac{1}{\#G}\sum_g g\pi g^{-1}$, which makes sense because $\#G \in k^\times$. Each term maps $V$ into $gU = U$, so $\pi'(V) \subseteq U$. For $u \in U$, $g^{-1}u \in U$, so $g\pi g^{-1}u = gg^{-1}u = u$, and $\pi' u = \frac{1}{\#G} \cdot \#G \cdot u = u$. For $h \in G$,

$$h\pi' h^{-1} = \frac{1}{\#G}\sum_g (hg)\pi(hg)^{-1} = \pi',$$

since $g \mapsto hg$ permutes $G$. So $\pi'$ is a $G$-equivariant projection onto $U$; then $V = U \oplus \ker \pi'$ (write $v = \pi' v + (v - \pi' v)$), and $\ker \pi'$ is $G$-stable because $\pi'$ commutes with $G$. It is a stable complement; apply Proposition 4.17(i). (ii) $he_G = e_G = e_G h$ for $h \in G$, so $e_G$ is central and $e_G^2 = \frac{1}{\#G} \sum_g g e_G = e_G$. If $m \in M^G$, $e_G m = m$; and $he_G m = e_G m$, so $e_G M = M^G$. Since $e_G$ is a central idempotent, $M = e_G M \oplus (1 - e_G)M$ as $R[G]$-modules. Finally $M/IM$ is an $R[G]$-module, so what was just proved, applied to it, gives $(M/IM)^G = e_G(M/IM)$; as $e_G$ applied to the class of $m$ is the class of $e_G m$, this is the image of $e_G M = M^G$, so $M^G \to (M/IM)^G$ is onto. The kernel of $M^G \to M/IM$ is $M^G \cap IM$, and if $x = \sum r_i m_i \in M^G$ with $r_i \in I$, then $x = e_G x = \sum r_i e_G m_i \in IM^G$. □

Two comments on the proof. First, the hypothesis $\operatorname{char} k \nmid \#G$ enters only through the factor $\frac{1}{\#G}$. Without it the sum $\sum_g g\pi g^{-1}$ is still $G$-equivariant with image in $U$, but on $U$ it is multiplication by $\#G$, which is then 0. Second, the conclusion really fails then. In the permutation representation $k^3$ of $S_3$ (after Definition 4.2) with $\operatorname{char} k = 3$, the line $D = k(1,1,1)^t$ has no stable complement. Indeed, a stable complement $C$ would give the $G$-equivariant projection $k^3 \to D$ along $C$; composed with $D \simeq k$, $c(1,1,1)^t \mapsto c$, it is a linear form $\lambda(x) = a_1x_1 + a_2x_2 + a_3x_3$ with $\lambda(gx) = \lambda(x)$ for all $g$ (as $G$ acts trivially on $D$) and $\lambda(1,1,1) = 1$. Invariance under the transposition exchanging $x_i$ and $x_j$ gives $a_i = a_j$, so $\lambda = a(x_1 + x_2 + x_3)$ and $\lambda(1,1,1) = 3a = 0$, a contradiction. Here 3 divides $\#S_3 = 6$.

*In the paper* 4.19. In Lemma lem:euler and Remark rem:types, for $\ell \geq 5$ the finite images of inertia (of order $2, 3, 4, 6$) have order prime to $\ell$, so by (ii) their invariants on $T$ and on $\operatorname{ad}^0 T$ commute with reduction modulo $\ell^m$. At $\ell = 3$ this fails only for the elements of order 3 acting on $T$: $T^g = 0$ while $(T/3T)^g$ is a line (Proposition 4.72(ii)); that failure is the content of Lemma lem:IV. For an element of order 6, $T^g = 0 = (T/3T)^g$ (Proposition 4.72(iii)), and on $\operatorname{ad}^0 T$ it acts as $-g$, of order 3, whose invariant line $\mathbb{Z}_3(1 - 2g)$ reduces onto $(\operatorname{ad}^0 T/3)^g$ (Proposition 4.72(ii)): there the invariants still commute with reduction. In Proposition prop:ns the groups $Q_8$ and $C_4$ have order prime to 3, which is used in Proposition 4.26.

**Example 4.20** (Failures of semisimplicity)**.** (a) $\mathrm{U} = \langle u \rangle \simeq C_3$ acting on $\mathbb{F}_3^2$. The line $\mathbb{F}_3 e_1$ is stable. A stable line is spanned by an eigenvector of $u$, and $\ker(u - 1) = \mathbb{F}_3 e_1$ is the only eigenspace; so $\mathbb{F}_3 e_1$ has no stable complement, $\mathbb{F}_3^2$ is indecomposable and not semisimple, and its composition factors (Theorem 4.21) are $\mathbf{1}, \mathbf{1}$. Maschke does not apply: $3 \mid \#\mathrm{U}$.

(b) The regular representation $\mathbb{F}_p[C_p] \simeq \mathbb{F}_p[x]/(x^p - 1) = \mathbb{F}_p[x]/((x-1)^p)$ ($x$ a generator of $C_p$). Its subrepresentations are the ideals, which are the $((x-1)^j)$, $0 \leq j \leq p$, a chain: the representation is uniserial, with the single composition factor $\mathbf{1}$ of multiplicity $p$.

(c) For an infinite group Maschke fails even in characteristic 0: $\mathbb{Z}$ acting on $\mathbb{Q}_\ell^2$ by $m \mapsto \left(\begin{smallmatrix} 1 & mn' \\ 0 & 1 \end{smallmatrix}\right)$ with $n' \neq 0$ stabilises only the line $\mathbb{Q}_\ell e_1$ (as in (a)), so it is not semisimple. This is the shape of tame inertia at a multiplicative prime (Lemma lem:tamfactor).

(d) $\mathrm{B}(\mathbb{F}_3)$ on $\mathbb{F}_3^2$: it contains $u$, so $\mathbb{F}_3 e_1$ is its only stable line, and the representation is indecomposable and reducible. Its commutant is $\mathbb{F}_3$: a matrix commuting with $\sigma = \operatorname{diag}(1, -1)$ is diagonal, and $\operatorname{diag}(x, y)$ commutes with $u$ iff $x = y$. So "$\operatorname{End}_G(V) = k$" does not imply irreducibility without semisimplicity.

**Theorem 4.21** (Jordan-Hölder)**.** *Let $A$ be a ring (here $k[G]$ or $\mathcal{O}[G]$) and $M$ an $A$-module of finite length, for example a finite-dimensional representation or a finite $\mathcal{O}[G]$-module. Then $M$ has a composition series $0 = M_0 \subset M_1 \subset \cdots \subset M_r = M$ with simple quotients $M_i/M_{i-1}$, and*

*the length $r$ and the multiset of isomorphism classes of the quotients (the* composition factors*, with multiplicity) do not depend on the series.*

*Proof.* This is proved in Chapter 3 for modules of finite length over any ring; we do not prove it again. A finite-dimensional representation has finite length because a strictly increasing chain of subspaces has at most $\dim V + 1$ terms. □

**Definition 4.22.** The *semisimplification* of a finite-dimensional representation $V$ is $V^{ss} = \bigoplus_{i=1}^{r} M_i/M_{i-1}$ for a composition series $(M_i)$ of $V$.

**Proposition 4.23.** *(i) $V^{ss}$ is well defined up to isomorphism; $V^{ss} \simeq V$ iff $V$ is semisimple; and if $0 \to U \to V \to Q \to 0$ is exact, $V^{ss} \simeq U^{ss} \oplus Q^{ss}$.*

*(ii) $\det(X - g \mid V) = \det(X - g \mid V^{ss})$ and $\operatorname{tr}(g \mid V) = \operatorname{tr}(g \mid V^{ss})$ for all $g \in G$.*

*(iii) If $k$ is perfect and $K/k$ algebraic, $(V^{ss})_K \simeq (V_K)^{ss}$.*

*Proof.* (i) Well defined by Theorem 4.21. If $V = \bigoplus_i S_i$ with $S_i$ irreducible, $0 \subset S_1 \subset S_1 \oplus S_2 \subset \cdots$ is a composition series with quotients $S_i$, so $V^{ss} \simeq V$; conversely $V^{ss}$ is semisimple by definition. For the exact sequence, a composition series of $U$ followed by the preimages of one of $Q$ is a composition series of $V$. (ii) In a basis adapted to a composition series the matrix of $g$ is block upper triangular with the matrices of $g$ on the $M_i/M_{i-1}$ on the diagonal. (iii) We do not prove this; see [37, Section 29] (over a perfect field an irreducible representation stays semisimple after an algebraic extension of scalars). It is not used below: Theorem 4.35 is proved over an arbitrary field without it. □

For example, for $\mathrm{U} = \langle u \rangle$ acting on $\mathbb{F}_3^2$ the series $0 \subset \mathbb{F}_3 e_1 \subset \mathbb{F}_3^2$ has quotients $\mathbf{1}$ and $\mathbf{1}$ ($u$ acts on $\mathbb{F}_3^2/\mathbb{F}_3 e_1$ through its lower right entry 1), so $(\mathbb{F}_3^2)^{ss} \simeq \mathbf{1} \oplus \mathbf{1}$, while $\mathbb{F}_3^2$ itself is not semisimple. The semisimplification forgets the upper right entry of $u$, and (ii) says that it keeps everything the characteristic polynomials see. This is why $\bar{\rho}^{ss}$ is the part of a residual representation that congruences between Fourier coefficients can detect (Section 4.5).

**Proposition 4.24** (Extensions of two characters)**.** *Let $V$ be a two-dimensional representation of any group $G$ over $k$ with a stable line $L$, on which $G$ acts by $\eta_1$, and let $G$ act on $V/L$ by $\eta_2$. In a basis $e_1 \in L$, $e_2$,*

$$\rho(g) = \begin{pmatrix} \eta_1(g) & b(g) \\ 0 & \eta_2(g) \end{pmatrix}, \qquad b(gh) = \eta_1(g)b(h) + b(g)\eta_2(h).$$

*(i) The following are equivalent: $V$ is semisimple; $V \simeq \eta_1 \oplus \eta_2$; $V$ has a stable line other than $L$; there is $c \in k$ with $b(g) = c\,(\eta_2(g) - \eta_1(g))$ for all $g$ (then $k(ce_1 + e_2)$ is stable).*

*(ii) If $V$ is not semisimple (a* non-split extension*), $L$ is its only stable line. Hence if $V'$ is another non-split extension, with sub-character $\eta_1'$ and quotient character $\eta_2'$, every isomorphism $V \simeq V'$ maps $L$ onto $L'$, and $\eta_1 = \eta_1'$, $\eta_2 = \eta_2'$.*

*(iii) If $\eta_1 \neq \eta_2$ and $V$ is semisimple, $V$ has exactly two stable lines.*

*Proof.* The cocycle identity is the upper right entry of $\rho(gh) = \rho(g)\rho(h)$. (i) If $V$ is semisimple, $L$ has a stable complement $L'$, a second stable line, and $L' \simeq V/L$, so $V = L \oplus L' \simeq \eta_1 \oplus \eta_2$; if $V \simeq \eta_1 \oplus \eta_2$ it is semisimple. A line other than $L$ is $k(ce_1 + e_2)$ for a unique $c$, and

$g(ce_1+e_2)=(c\eta_1(g)+b(g))e_1+\eta_2(g)e_2$ lies in $k(ce_1+e_2)$ iff $c\eta_1(g)+b(g)=c\eta_2(g)$. A second stable line is a complement of $L$. (ii) A second stable line would make $V$ semisimple by (i). An isomorphism maps stable lines to stable lines, so $L$ to $L'$, and induces isomorphisms $L\simeq L'$ and $V/L\simeq V'/L'$ of one-dimensional representations, which are equalities of characters. (iii) In a basis $v_1,v_2$ of stable lines, $V=kv_1\oplus kv_2$ with characters $\eta_1\neq\eta_2$, a vector $xv_1+yv_2$ with $xy\neq 0$ spans a stable line only if $\eta_1(g)=\eta_2(g)$ for all $g$. □

For example, for U on $\mathbb{F}_3^2$ one has $\eta_1=\eta_2=\mathbf{1}$ and $b(u^m)=m$; the condition of (i) would require $b=c(\eta_2-\eta_1)=0$, so the extension is non-split, as found in Example 4.20(a). In general the condition of (i) says that $b$ has the special form $c(\eta_2-\eta_1)$, determined by one constant. Dividing by $\eta_2$, the identity for $b$ becomes

$$\frac{b(gh)}{\eta_2(gh)}=\frac{\eta_1(g)}{\eta_2(g)}\cdot\frac{b(h)}{\eta_2(h)}+\frac{b(g)}{\eta_2(g)},$$

which says that $b/\eta_2$ is a 1-cocycle of $G$ with values in $k(\eta_1\eta_2^{-1})$, and the split case $b/\eta_2=-c\,(\eta_1\eta_2^{-1}-1)$ is a coboundary. This is the beginning of the cohomological reading of extensions, developed in Chapter 16.

*In the paper* 4.25. Lemma lem:old, case $p\,\|\,N$ with $\bar\rho$ ramified at $p$: "A non-split extension of two characters has a unique stable line, so the isomorphism $\bar\rho|_{D_p}\simeq\bar\rho_h|_{D_p}$ matches the sublines and hence the quotients" is (ii), applied over a common finite extension of $\mathbb{F}_\ell$. Lemma lem:tamfactor uses the shape $\left(\begin{smallmatrix}\chi_\ell\varepsilon & \varepsilon\kappa\\ 0 & \varepsilon\end{smallmatrix}\right)$ of the Tate representation: the corner $\varepsilon\kappa$ satisfies the cocycle identity above with $\eta_1=\chi_\ell\varepsilon$, $\eta_2=\varepsilon$ (convention B-C14; the cohomological reading is in Chapter 16).

**Proposition 4.26** (Absolute irreducibility in dimension 2)**.** *Let $\rho\colon G\to\mathrm{GL}_2(k)$ be a representation.*

*(i) If $\rho(G)$ is abelian, $\rho$ is absolutely reducible.*

*(ii) If $\rho(G)$ is finite of order prime to $\operatorname{char}k$ and $\rho$ is absolutely reducible, then $\rho_{\bar k}\simeq\eta_1\oplus\eta_2$ and $\rho(G)$ is abelian. Hence a finite non-abelian subgroup of $\mathrm{GL}_2(k)$ of order prime to $\operatorname{char}k$ acts absolutely irreducibly on $k^2$.*

*(iii) If $H\le G$ and $\operatorname{Res}_H\rho$ is absolutely irreducible, so is $\rho$.*

*(iv) On $\mathbb{F}_3^2$: $Q_8$ acts absolutely irreducibly, and hence so do $\mathrm{SL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ and $\mathrm{GL}_2(\mathbb{F}_3)$, which contain it; $N_s(\mathbb{F}_3)\simeq D_4$ acts absolutely irreducibly; $C_4$ and $C_{ns}(\mathbb{F}_3)$ do not. The Borel subgroup $\mathrm{B}(\mathbb{F}_3)$ is non-abelian and acts faithfully but reducibly: its order 12 is divisible by 3, and (ii) does not apply.*

*Proof.* (i) Commuting operators on $\bar k^2$ have a common eigenvector: if all $\rho(g)$ are scalars any line is stable; otherwise take $g_0$ with $\rho(g_0)$ not scalar and an eigenvalue $c$ of it; its eigenspace $\ker(\rho(g_0)-c)$ is a line (it is not everything), stable under every $\rho(g)$, which commutes with $\rho(g_0)$. (ii) $\rho_{\bar k}$ has a stable line $L$. The finite group $\rho(G)$ acts on $\bar k^2$ and has order prime to $\operatorname{char}\bar k=\operatorname{char}k$, so by Maschke (Theorem 4.18(i), applied to $\rho(G)$) $L$ has a stable complement $L'$, again a line. In a basis $v\in L$, $v'\in L'$ every $\rho(g)$ is diagonal, so $\rho_{\bar k}\simeq\eta_1\oplus\eta_2$ with $\eta_1,\eta_2$ the characters by which $G$ acts on $L$ and $L'$, and $\rho(G)$ is abelian. The second sentence of (ii) is the contrapositive, for $\rho$ the inclusion of the subgroup. (iii) A stable line of $\bar k^2$ for $G$ is one for $H$. (iv) $Q_8$ is non-abelian of order 8, prime to 3, and acts faithfully; apply (ii) and then (iii) to the groups containing $Q_8$ (Chapter 1). $N_s(\mathbb{F}_3)$ is non-abelian of order 8. $C_4$ and $C_{ns}(\mathbb{F}_3)$ are cyclic; apply (i). $\mathrm{B}(\mathbb{F}_3)$ stabilises $\mathbb{F}_3e_1$. □

*In the paper* 4.27. Proposition prop:ns: "The first two act absolutely irreducibly on $\mathbb{F}_3^2$". The paper's argument is the commutator one of Chapter 1: a subgroup of $\mathrm{GL}_2(\overline{\mathbb{F}}_3)$ fixing a line lies in a Borel subgroup, whose commutator subgroup is unipotent, so abelian of exponent 3, while the commutator subgroups $\{\pm 1\}$ of $Q_8$ and $Q_8$ of $\mathrm{SL}_2(\mathbb{F}_3)$ are not. Proposition 4.26(ii), (iii) give a second proof: $Q_8$ is non-abelian of order prime to 3, and $\mathrm{SL}_2(\mathbb{F}_3)$ contains $Q_8$. A group that is merely non-abelian and acts faithfully need not act irreducibly, by the example of $\mathrm{B}(\mathbb{F}_3)$; this is why an argument of one of these two kinds is needed. "The third is abelian, hence absolutely reducible" is (i).

## 4.4 Characters in characteristic zero

This short section connects the chapter with the character theory of Riley-Hobson-Bence: over $\mathbb{C}$ a representation of a finite group is determined by its trace function, and the trace functions of the irreducible representations are orthonormal. The paper uses none of this directly; it is here for two reasons. It shows the reader where the familiar theory stops: in characteristic $\ell$ traces no longer determine semisimple representations (Example 4.32), which is why Brauer-Nesbitt in Lemma lem:old is applied with characteristic polynomials. And the two groups of order 8 of Proposition prop:ns, $N_s(\mathbb{F}_3) \simeq D_4$ and $Q_8$, are a classical pair with the same character table.

**Definition 4.28.** The *trace function* of a representation $V$ is $\mathrm{tr}_V \colon G \to k$, $\mathrm{tr}_V(g) = \mathrm{tr}\,\rho(g)$. It is a class function ($\mathrm{tr}_V(xgx^{-1}) = \mathrm{tr}_V(g)$), $\mathrm{tr}_V(1) = \dim V$ (in $k$), and by Proposition 4.4 $\mathrm{tr}_{V\oplus W} = \mathrm{tr}_V + \mathrm{tr}_W$, $\mathrm{tr}_{V\otimes W} = \mathrm{tr}_V\,\mathrm{tr}_W$, $\mathrm{tr}_{V^\vee}(g) = \mathrm{tr}_V(g^{-1})$. For $k = \mathbb{C}$ and $G$ finite, $\rho(g)$ has finite order, its eigenvalues are roots of unity, and $\mathrm{tr}_V(g^{-1}) = \overline{\mathrm{tr}_V(g)}$. In Riley-Hobson-Bence, and in most books on finite groups, $\mathrm{tr}_V$ is called the *character* of $V$; we avoid the word for trace functions because the paper's characters $\chi_{-3}$, $\chi_\ell$, $\psi$ are homomorphisms to $k^\times$ (one-dimensional representations, whose trace function is the character itself).

**Theorem 4.29** (Orthogonality)**.** *Let $G$ be finite, $k = \mathbb{C}$, and $\langle \phi, \psi\rangle = \frac{1}{\#G}\sum_{g\in G}\phi(g)\overline{\psi(g)}$ on class functions.*

*(i) For irreducible $V, W$: $\langle \mathrm{tr}_V, \mathrm{tr}_W\rangle = 1$ if $V \simeq W$ and $0$ otherwise.*

*(ii) For all $V, W$: $\dim \mathrm{Hom}_G(V,W) = \langle \mathrm{tr}_W, \mathrm{tr}_V\rangle$. A representation $V$ is irreducible iff $\langle \mathrm{tr}_V, \mathrm{tr}_V\rangle = 1$, and two representations with the same trace function are isomorphic.*

*(iii) The trace functions $\mathrm{tr}_{V_1}, \dots, \mathrm{tr}_{V_r}$ of the irreducible representations (up to isomorphism) form an orthonormal basis of the space of class functions; so $r$ is the number of conjugacy classes; and $\sum_i (\dim V_i)^2 = \#G$.*

*(iv) Each $\dim V_i$ divides $\#G$.*

*Proof.* (ii) By Proposition 4.5, $\mathrm{Hom}_G(V,W) \simeq (V^\vee \otimes W)^G$, which is the image of the idempotent $e_G$ of Theorem 4.18(ii) acting on $V^\vee \otimes W$. An idempotent $p$ satisfies $p(p-1) = 0$, so the space is $\mathrm{im}\,p \oplus \ker p$ with $p$ acting by 1 and by 0, and $\mathrm{tr}\,p = \dim \mathrm{im}\,p$ (an integer, read in $\mathbb{C}$). By linearity of the trace, Proposition 4.4(i) and Definition 4.28 ($\mathrm{tr}_{V^\vee}(g) = \mathrm{tr}_V(g^{-1}) = \overline{\mathrm{tr}_V(g)}$),

$$\dim \mathrm{Hom}_G(V,W) = \mathrm{tr}(e_G \mid V^\vee \otimes W) = \frac{1}{\#G}\sum_g \mathrm{tr}_{V^\vee}(g)\,\mathrm{tr}_W(g)$$
$$= \frac{1}{\#G}\sum_g \mathrm{tr}_W(g)\overline{\mathrm{tr}_V(g)} = \langle \mathrm{tr}_W, \mathrm{tr}_V\rangle.$$

(i) follows from (ii) and Theorem 4.10(ii); the value is real, so the order of the arguments does not matter. By Maschke, $V \simeq \bigoplus_i V_i^{m_i}$, and by (i) $\langle \mathrm{tr}_V, \mathrm{tr}_V\rangle = \sum_i m_i^2$, which is 1 iff $V$ is irreducible; and $m_i = \langle \mathrm{tr}_V, \mathrm{tr}_{V_i}\rangle$ is determined by $\mathrm{tr}_V$. (iii) Orthonormality is (i). The regular representation $\mathbb{C}[G]$ has $\mathrm{tr}(1) = \#G$ and $\mathrm{tr}(g) = 0$ for $g \neq 1$ (a nontrivial $g$ moves every basis vector), so $\langle \mathrm{tr}_{\mathbb{C}[G]}, \mathrm{tr}_{V_i}\rangle = \dim V_i$, $\mathbb{C}[G] \simeq \bigoplus_i V_i^{\dim V_i}$, and $\#G = \sum_i (\dim V_i)^2$. Let $\phi$ be a class function orthogonal to every $\mathrm{tr}_{V_i}$. For each irreducible $V$, the operator $\Phi_V = \sum_g \overline{\phi(g)}\rho_V(g)$ commutes with $G$ ($\phi$ is a class function), so it is a scalar $c$ by Schur, and $c \dim V = \mathrm{tr}\,\Phi_V = \#G\,\langle \mathrm{tr}_V, \phi\rangle = 0$. So $\Phi$ vanishes on every irreducible, hence on $\mathbb{C}[G]$, and applied to $1 \in \mathbb{C}[G]$ it gives $\sum_g \overline{\phi(g)} g = 0$, so $\phi = 0$. Hence the $\mathrm{tr}_{V_i}$ span the class functions, whose dimension is the number of conjugacy classes. (iv) We do not prove this; see [173, Section 6.5]. □

**Corollary 4.30.** *If* $\mathrm{char}\, k = 0$*, two semisimple finite-dimensional representations of any group* $G$ *with the same trace function are isomorphic.*

*Proof.* This is the last statement of Theorem 4.35 (traces suffice in characteristic 0), together with Proposition 4.23(i); the proof of that theorem does not use this corollary. □

**Example 4.31** (The character tables of $D_4$ and $Q_8$)**.** Take $D_4 = N_s(\mathbb{F}_3) = \langle J, \sigma\rangle$ (rotation $J$, reflection $\sigma$; Chapter 1) and $Q_8 = \{\pm 1, \pm J, \pm j, \pm k\}$ with $j = \left(\begin{smallmatrix} 1 & -1 \\ -1 & -1 \end{smallmatrix}\right)$, $k = Jj$. Both have five conjugacy classes, and their irreducible trace functions over $\mathbb{C}$ are:

| $D_4$ | 1 | $-1$ | $\pm J$ | $\pm\sigma$ | $\pm w$ |
|---|---|---|---|---|---|
| $Q_8$ | 1 | $-1$ | $\pm J$ | $\pm j$ | $\pm k$ |
| $\lambda_0$ | 1 | 1 | 1 | 1 | 1 |
| $\lambda_1$ | 1 | 1 | 1 | $-1$ | $-1$ |
| $\lambda_2$ | 1 | 1 | $-1$ | 1 | $-1$ |
| $\lambda_3$ | 1 | 1 | $-1$ | $-1$ | 1 |
| $\mathrm{tr}_2$ | 2 | $-2$ | 0 | 0 | 0 |

The four characters factor through the quotient by $\{\pm 1\}$, which is $V_4$ for both groups; for $D_4$, $\lambda_1 = \det$ (kernel $C_4$), $\lambda_2$ has kernel $C_s(\mathbb{F}_3) = \{\pm 1, \pm\sigma\}$ and $\lambda_3$ kernel $\{\pm 1, \pm w\}$. The orthogonality relations and $4 \cdot 1^2 + 2^2 = 8$ show the table is complete. In detail: the classes have $1, 1, 2, 2, 2$ elements, so for instance $\langle \mathrm{tr}_2, \mathrm{tr}_2\rangle = \frac{1}{8}(2^2 + (-2)^2) = 1$ and $\langle \lambda_1, \lambda_2\rangle = \frac{1}{8}(1 + 1 - 2 - 2 + 2) = 0$. For $D_4$ the row $\mathrm{tr}_2$ is read off the matrices ($\mathrm{tr}\, J = \mathrm{tr}\,\sigma = \mathrm{tr}\, w = 0$, $\mathrm{tr}(-1) = -2$). For $Q_8$ it is forced: by the proof of Theorem 4.29(iii), $\sum_i \dim V_i\ \mathrm{tr}_{V_i}(g) = \mathrm{tr}(g \mid \mathbb{C}[G]) = 0$ for $g \neq 1$, so the remaining irreducible representation, of degree $\sqrt{8-4} = 2$, has $\mathrm{tr}_2(g) = -\frac{1}{2}\sum_{j=0}^{3} \lambda_j(g)$, which is $-2$ at $-1$ and 0 on the other three classes $\neq \{1\}$. Yet $D_4 \not\simeq Q_8$ ($D_4$ has five involutions, $Q_8$ one). The two-dimensional representation of $D_4$ is given by the integral matrices $J$, $\sigma$, so it is defined over $\mathbb{Q}$, and its reduction modulo 3 is the standard representation of $N_s(\mathbb{F}_3)$. That of $Q_8$ is not defined over $\mathbb{R}$: its Frobenius-Schur indicator $\frac{1}{8}\sum_g \mathrm{tr}_2(g^2)$ is $\frac{1}{8}(2 \cdot 2 + 6 \cdot (-2)) = -1$, which for an irreducible representation means that it is not realisable over $\mathbb{R}$ (we do not prove this; see [173, Section 13.2]). Over $\mathbb{F}_3$, however, it is realised by the standard representation of $Q_8 \subset \mathrm{SL}_2(\mathbb{F}_3)$, whose trace function is the reduction of $\mathrm{tr}_2$.

**Example 4.32** (Traces do not suffice in characteristic $p$)**.** Let $G = C_2 = \{1, s\}$ and $k = \mathbb{F}_3$. The semisimple representations $\mathbf{1}^{\oplus 3}$ and $\mathrm{sgn}^{\oplus 3}$ ($s \mapsto -1$) have trace functions $(3, 3)$ and $(3, -3)$, both $(0, 0)$ in $\mathbb{F}_3$, but they are not isomorphic: $\det(X - s)$ is $(X-1)^3$, respectively $(X+1)^3$. In general, if $n < p$ (or $p = 0$), the traces of $g, g^2, \ldots, g^n$ determine the characteristic polynomial of

$g$ on an $n$-dimensional space by Newton's identities, $je_j = \sum_{i=1}^{j}(-1)^{i-1}e_{j-i}p_i$ for the elementary symmetric functions $e_j$ and power sums $p_i = \operatorname{tr}(g^i)$ of the eigenvalues, which can be solved for the $e_j$ because $1,\dots,n$ are invertible. For $n=2$ Newton's identities read $e_1 = p_1$ and $2e_2 = e_1p_1 - p_2$, so $\det g = \frac{1}{2}(\operatorname{tr}(g)^2 - \operatorname{tr}(g^2))$ if $2 \in k^\times$; in characteristic 2, $\operatorname{tr}(g^2) = \lambda_1^2 + \lambda_2^2 = \operatorname{tr}(g)^2$ carries no information about $\det g$. In any case the characteristic polynomial $X^2 - \operatorname{tr}(g)X + \det(g)$ is determined by the trace and the determinant in every characteristic.

*Warning* 4.33. In characteristic $\ell$ the right invariant of a representation is the family of characteristic polynomials, not the trace function. The paper's representations have dimension 2 and determinant $\bar{\chi}_\ell$, so for them traces do suffice (Corollary 4.36(i)): "$a_p(h) \equiv a_p(f) \pmod{\lambda}$ for all primes $p \nmid \ell N$" is a statement about traces of Frobenius elements, and it determines the semisimplification.

## 4.5 The theorem of Brauer and Nesbitt

This section proves that a semisimple representation over any field is determined by the characteristic polynomials of the elements of the group (Brauer and Nesbitt). With the Chebotarev density theorem (Chapter 5) it is the reason why a congruence $a_p(h) \equiv a_p(f) \pmod{\lambda}$ for almost all $p$ gives $\bar{\rho}_h \simeq \bar{\rho}$, which is the first step of the paper's Lemma lem:old. The proof needs the density theorem of Jacobson and Burnside, proved first.

**Theorem 4.34** (Density theorem of Jacobson and Burnside)**.** *Let $G$ be any group, $S_1,\dots,S_r$ pairwise non-isomorphic irreducible finite-dimensional representations over $k$, and $D_i = \operatorname{End}_G(S_i)$.*

*(a) The image of $k[G] \to \prod_i \operatorname{End}_k(S_i)$ is $\prod_i \operatorname{End}_{D_i}(S_i)$.*

*(b) If $k$ is algebraically closed, the image is $\prod_i \operatorname{End}_k(S_i)$. In particular, for each $i$ there is $x \in k[G]$ acting on $S_i$ as a projector of rank one and on every $S_j$, $j \neq i$, as 0.*

*Proof.* (a) Let $A$ be the image. Each element of $k[G]$ commutes with $D_i$ on $S_i$, so $A \subseteq \prod_i \operatorname{End}_{D_i}(S_i)$. By Theorem 4.10(i), $D_i$ is a division algebra and $S_i$ is a finite-dimensional (left) $D_i$-vector space; let $x_{i1},\dots,x_{in_i}$ be a $D_i$-basis. Put $M = \bigoplus_i S_i^{n_i}$, a semisimple representation, and $x = (x_{ij})_{i,j} \in M$. The subrepresentation $k[G]x = Ax$ of the semisimple $M$ has a $G$-stable complement $C$ (Proposition 4.17(i)), and the projection $\pi$ of $M = Ax \oplus C$ onto $Ax$ along $C$ is a $G$-map: $\pi \in \operatorname{End}_G(M)$. We describe $\operatorname{End}_G(M)$. A $G$-map $M \to M$ has components $S_i \to S_j$ between the summands, and those with $i \neq j$ vanish by Schur, since $S_i \not\simeq S_j$. So a $G$-endomorphism of $M$ is a family, over $i$, of $G$-endomorphisms of $S_i^{n_i}$, and each of these is an $n_i \times n_i$ matrix $(d_{ab})$ with entries $d_{ab} \in D_i$, acting by $(s_b)_b \mapsto (\sum_b d_{ab}s_b)_a$. Thus $\operatorname{End}_G(M) = \prod_i \mathrm{M}_{n_i}(D_i)$. Now let $\phi = (\phi_i) \in \prod_i \operatorname{End}_{D_i}(S_i)$ and let $\Phi$ act on $M$ by $\phi_i$ on each copy of $S_i$. As $\phi_i$ commutes with every $d \in D_i$, the map $\Phi$, which acts by $\phi_i$ on each coordinate of $S_i^{n_i}$, commutes with every such matrix: $\Phi((\sum_b d_{ab}s_b)_a) = (\sum_b d_{ab}\phi_i(s_b))_a$. In particular $\Phi$ commutes with $\pi$. Hence $\Phi(x) = \Phi(\pi x) = \pi\Phi(x) \in Ax$: there is $\alpha \in A$ with $\alpha x_{ij} = \phi_i(x_{ij})$ for all $i,j$. On $S_i$, both $\alpha$ and $\phi_i$ are $D_i$-linear and agree on a $D_i$-basis, so they are equal. Thus $\phi = \alpha \in A$. (b) If $k = \bar{k}$ then $D_i = k$ by Theorem 4.10(ii), so $\operatorname{End}_{D_i}(S_i) = \operatorname{End}_k(S_i)$; take the element of $A$ that is a rank-one projector on $S_i$ and 0 on the other factors. □

For example, let $C_4$ act on $S = \mathbb{F}_3^2$ (Example 4.14). Then $D = \operatorname{End}_{C_4}(S) = \mathbb{F}_3[J] \simeq \mathbb{F}_9$, and $S$ is a one-dimensional $D$-vector space, so $\operatorname{End}_D(S) = D$: (a) says that the image of $\mathbb{F}_3[C_4]$ in the 4-dimensional $\operatorname{End}_{\mathbb{F}_3}(S)$ is the 2-dimensional $\mathbb{F}_3[J]$, which is visible directly (the image is

spanned by the powers of $J$). Over $\mathbb{F}_9$, $S_{\mathbb{F}_9} \simeq \psi \oplus \psi^{-1}$ is a sum of two non-isomorphic characters, and (b) provides an element of $\mathbb{F}_9[C_4]$ acting by 1 on $\psi$ and by 0 on $\psi^{-1}$; one is $x = \frac{1}{2}(1 - iJ)$, since $J$ acts by $i$ on $\psi$ and by $-i$ on $\psi^{-1}$, and $1 - i \cdot i = 2$, $1 - i \cdot (-i) = 0$.

**Theorem 4.35** (Brauer-Nesbitt)**.** *Let $G$ be any group, $k$ any field, and $V_1, V_2$ finite-dimensional representations of $G$ over $k$ with*

$$\det(X - \rho_1(g)) = \det(X - \rho_2(g)) \quad \text{in } k[X] \text{ for every } g \in G.$$

*Then $V_1$ and $V_2$ have the same composition factors with multiplicities, that is, $V_1^{ss} \simeq V_2^{ss}$. If $\operatorname{char} k = 0$, or if $\dim V_1 < \operatorname{char} k$ and $\dim V_2 < \operatorname{char} k$, the equality $\operatorname{tr} \rho_1(g) = \operatorname{tr} \rho_2(g)$ for all $g$ suffices.*

*Proof.* We prove both statements together; "the hypothesis" means equal characteristic polynomials, or, in the two special cases, equal traces.

*Step 0.* By Proposition 4.23(ii) we may replace $V_1, V_2$ by $V_1^{ss}, V_2^{ss}$.

*Step 1: reduction to $k = \bar{k}$.* Let $S, S'$ be non-isomorphic irreducible representations over $k$. By Theorem 4.34(a) with $r = 2$ there is $x \in k[G]$ acting as 1 on $S$ and as 0 on $S'$. Then $x$ acts as the identity on $S_{\bar{k}}$, hence on all its subquotients, and as 0 on $S'_{\bar{k}}$; so $x$ acts as 1 on every composition factor of $S_{\bar{k}}$ and as 0 on every composition factor of $S'_{\bar{k}}$, so $S_{\bar{k}}$ and $S'_{\bar{k}}$ have no common composition factor. If $V$ has composition factors $S$ with multiplicities $m_S(V)$, a composition series of $V$ gives a filtration of $V_{\bar{k}}$ with quotients $S_{\bar{k}}$, so the multiset of composition factors of $V_{\bar{k}}$ is $\sum_S m_S(V)\, c(S_{\bar{k}})$, where the multisets $c(S_{\bar{k}})$ of composition factors of the $S_{\bar{k}}$ are nonempty with pairwise disjoint supports. Hence the composition factors of $V_{1,\bar{k}}$ and $V_{2,\bar{k}}$ agree iff $m_S(V_1) = m_S(V_2)$ for all $S$. As the hypothesis is unchanged by extending scalars, we may assume $k = \bar{k}$, and (Step 0) that $V_1, V_2$ are semisimple.

*Step 2: cancellation.* Write $V_1 \simeq C \oplus B_1$ and $V_2 \simeq C \oplus B_2$ where $C = \bigoplus_S S^{\min(m_S(V_1), m_S(V_2))}$; then $B_1$ and $B_2$ are semisimple without a common irreducible constituent. We show $B_1 = B_2 = 0$. The hypothesis passes to $B_1, B_2$: traces by subtraction, and characteristic polynomials because $\det(X - g \mid V_i) = \det(X - g \mid C) \det(X - g \mid B_i)$ and $k[X]$ is a domain. We argue by induction on $\dim B_1 + \dim B_2$; if it is 0 there is nothing to prove.

*Step 3: multiplicities modulo $p$.* The traces of $B_1$ and $B_2$ agree on $G$ (they are coefficients of the characteristic polynomials), hence on all of $k[G]$ by linearity. Let $S$ be an irreducible constituent of $B_1$, of multiplicity $m \geq 1$. By Theorem 4.34(b) applied to the distinct irreducible constituents of $B_1 \oplus B_2$, there is $x \in k[G]$ acting on $S$ as a projector of rank one and as 0 on all the other constituents. Write $B_1 \simeq S^m \oplus B_1''$ with $S$ not a constituent of $B_1''$; then $x$ acts as a rank-one projector on each of the $m$ copies of $S$ and as 0 on $B_1''$, so $\operatorname{tr}(x \mid B_1) = m \cdot 1$. All constituents of $B_2$ differ from $S$, so $x$ acts on $B_2$ as 0. Hence $m \cdot 1 = \operatorname{tr}(x \mid B_1) = \operatorname{tr}(x \mid B_2) = 0$ in $k$. If $\operatorname{char} k = 0$ this is impossible, so $B_1 = 0$, and symmetrically $B_2 = 0$. If $p = \operatorname{char} k > 0$, then $p \mid m$; if moreover $\dim V_1 < p$, then $m \leq \dim B_1 \leq \dim V_1 < p$ is impossible, so $B_1 = 0$; exchanging the roles of $B_1$ and $B_2$, $\dim V_2 < p$ gives $B_2 = 0$. This proves the statements about traces.

*Step 4: dividing by $p$.* In the remaining case ($p > 0$, characteristic polynomials), Step 3, applied to each constituent of $B_1$ and, with the roles exchanged, to each constituent of $B_2$, shows that every multiplicity in $B_1$ and in $B_2$ is divisible by $p$, so $B_i \simeq B_i'^{\oplus p}$ with $B_i'$ semisimple, and $\det(X - g \mid B_i) = \det(X - g \mid B_i')^p$. The map $f \mapsto f^p$ is an injective ring endomorphism of $k[X]$ (it is additive in characteristic $p$, and $k[X]$ is a domain), so $\det(X - g \mid B_1') = \det(X - g \mid B_2')$ for all $g$. The representations $B_1', B_2'$ have no common constituent and $\dim B_1' + \dim B_2' < \dim B_1 + \dim B_2$ unless both are 0; by induction $B_1' = B_2' = 0$, so $B_1 = B_2 = 0$. $\square$

**Corollary 4.36.** *(i) If* $\dim V_1 = \dim V_2 = 2$, $\det\rho_1 = \det\rho_2$ *and* $\operatorname{tr}\rho_1 = \operatorname{tr}\rho_2$ *on* $G$, *then* $V_1^{ss} \simeq V_2^{ss}$ *(in every characteristic).*

*(ii) Let* $k \subseteq k'$ *be fields,* $V_1$ *a representation over* $k$ *and* $V_2$ *one over* $k'$, *with* $\det(X-\rho_1(g)) = \det(X-\rho_2(g))$ *for all* $g$. *If* $V_1 \otimes_k k'$ *is irreducible (for instance if* $V_1$ *is absolutely irreducible), then* $V_2 \simeq V_1 \otimes_k k'$; *no semisimplicity of* $V_2$ *is assumed.*

*(iii) The irreducibility of* $V_1 \otimes_k k'$ *cannot be weakened to that of* $V_1$. *First, the irreducibility of* $V_1$ *does not make* $V_2$ *irreducible: for* $C_4$ *acting on* $V_1 = \mathbb{F}_3^2$ *and* $V_2 = \psi \oplus \psi^{-1}$ *over* $\mathbb{F}_9$ *(Example 4.14),* $V_1$ *is irreducible and* $V_2$ *is not. This example does not contradict the conclusion of (ii), since here* $V_2 \simeq V_1 \otimes \mathbb{F}_9$. *A genuine counterexample to the conclusion is the following. For the group* $G = \left\{ \left(\begin{smallmatrix} x & y \\ 0 & x^3 \end{smallmatrix}\right) : x \in \mathbb{F}_9^\times, y \in \mathbb{F}_9 \right\}$, *its standard representation* $V_2$ *on* $\mathbb{F}_9^2$ *and the representation* $V_1$ *on* $\mathbb{F}_9 = \mathbb{F}_3^2$ *by multiplication by* $x$ *have the same characteristic polynomials* $(X-x)(X-x^3)$ *and* $V_1$ *is irreducible over* $\mathbb{F}_3$, *but* $V_2$ *is not semisimple while* $V_1 \otimes \mathbb{F}_9$ *is, so* $V_2 \not\simeq V_1 \otimes \mathbb{F}_9$.

*Proof.* (i) The characteristic polynomial of a $2\times 2$ matrix is $X^2 - \operatorname{tr}(g)X + \det(g)$; apply Theorem 4.35. (ii) Apply Theorem 4.35 over $k'$ to $V_1 \otimes_k k'$ and $V_2$: $V_2^{ss} \simeq V_1 \otimes_k k'$ is irreducible, so $V_2$ has a single composition factor, is irreducible, and equals $V_2^{ss}$. (iii) In the second example, an element $x \in \mathbb{F}_9^\times$ of order 8 acts on $\mathbb{F}_9$ by multiplication with eigenvalues $x, x^3$ over $\mathbb{F}_9$; the $\mathbb{F}_3$-subspaces of $\mathbb{F}_9$ stable under multiplication by $\mathbb{F}_9^\times$ are 0 and $\mathbb{F}_9$, so $V_1$ is irreducible, and $V_1 \otimes \mathbb{F}_9$ is the sum of the two characters $g \mapsto x$, $g \mapsto x^3$. In $V_2$ the line $\mathbb{F}_9 e_1$ is stable and is the only stable line, because $G$ contains $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$; so $V_2$ is not semisimple (Proposition 4.24). □

*In the paper* 4.37. Lemma lem:old: "$\bar\rho_h \simeq \bar\rho$ over a common finite extension of $\mathbb{F}_\ell$, by Chebotarev and Brauer-Nesbitt, $\bar\rho$ being irreducible, hence absolutely irreducible". The residual representation $\bar\rho_h$ of the newform $h$ lives over the residue field $k'$ of the coefficient ring of $h$, and $\bar\rho$ over $\mathbb{F}_\ell$. For $p \nmid \ell N$ both have characteristic polynomial $X^2 - a_p X + p$ at $\mathrm{Frob}_p$, with $a_p(h) \equiv a_p(f)$; by Chebotarev these elements are dense, so the characteristic polynomials agree on all of $G_{\mathbb{Q}}$ (Chapter 5). Then Corollary 4.36(ii), with $\bar\rho$ absolutely irreducible (Proposition 4.13), gives $\bar\rho_h \simeq \bar\rho \otimes k'$; with only $\bar\rho$ irreducible over $\mathbb{F}_\ell$ one would get $\bar\rho_h^{ss} \simeq (\bar\rho \otimes k')^{ss}$ and nothing more, by (iii).

## 4.6 Restriction, induction, Frobenius reciprocity, Mackey and Clifford

This section develops induced representations and their behaviour under restriction to a subgroup of index 2. In the paper, the image $N_s(\mathbb{F}_3)$ of Proposition prop:ns gives a representation induced, over $\mathbb{F}_9$, from a character of order 4 of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, and for $\bar\rho$ irreducible the condition of Theorem thm:main, absolute irreducibility on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, is, by Clifford theory (Theorem 4.44), the condition that $\bar\rho$ is not induced from $\mathbb{Q}(\sqrt{\ell^*})$, equivalently $\bar\rho \not\simeq \bar\rho \otimes \chi_{\ell^*}$ (Lemma lem:h0 for $\ell = 3$, Proposition prop:Sf).

**Definition 4.38.** Let $H \le G$ and let $W$ be a representation of $H$ over $k$ (or an $R[H]$-module). The *induced representation* is $\mathrm{Ind}_H^G W = k[G] \otimes_{k[H]} W$, with $G$ acting on the left factor. For a representation $V$ of $G$, $\mathrm{Res}_H^G V$ (or $\mathrm{Res}_H V$) is $V$ viewed as a representation of $H$. For $s \in G$ and a representation $W$ of a subgroup $H$, $W^s$ is the representation of $sHs^{-1}$ on the space $W$

given by $x \mapsto \rho_W(s^{-1}xs)$. If $H$ is normal, $W^s$ is again a representation of $H$; for a character $\psi$ of $H$, $\psi^s(x) = \psi(s^{-1}xs)$.

Concretely (Proposition 4.39 below), for $[G:H] = m$ the space $\operatorname{Ind}_H^G W$ is a direct sum of $m$ copies $s_i \otimes W$ of $W$, one for each coset $s_iH$; $G$ permutes the copies as it permutes the cosets, and acts between them through elements of $H$. The tensor product over $k[H]$ is what makes this independent of the choice of representatives: $sx \otimes w = s \otimes xw$ for $x \in H$. For example, $\operatorname{Ind}_H^G \mathbf{1}$ is the permutation representation on $G/H$. The convention for $W^s$ matters for inverses below: with $W^s(x) = \rho_W(s^{-1}xs)$, the subspace $s \otimes W$ of $\operatorname{Ind}_H^G W$ is a copy of $W^s$ (Proposition 4.39(i)); some books use $\rho_W(sxs^{-1})$, which is our $W^{s^{-1}}$. For a normal subgroup $H$ of index 2 the two agree up to isomorphism: $W^s$ is the conjugate of $W^{s^{-1}}$ by $s^2 \in H$, and for any representation $U$ of $H$ and $y \in H$ the operator $\rho_U(y)$ is an isomorphism $U^y \simeq U$.

**Proposition 4.39.** *Let $H \le G$ have finite index $m$ and choose $G = \bigsqcup_{i=1}^m s_iH$ with $s_1 = 1$.*

*(i)* $\operatorname{Ind}_H^G W = \bigoplus_i s_i \otimes W$, $\dim \operatorname{Ind}_H^G W = m \dim W$, *and if $gs_i = s_jx$ with $x \in H$, then $g(s_i \otimes w) = s_j \otimes xw$. The subspace $s_i \otimes W$ is $s_iHs_i^{-1}$-stable and isomorphic to $W^{s_i}$.*

*(ii)* $\operatorname{tr}(g \mid \operatorname{Ind}_H^G W) = \sum_{i:\, s_i^{-1}gs_i \in H} \operatorname{tr}(s_i^{-1}gs_i \mid W)$.

*(iii) Let $[G:H] = 2$, $s \notin H$. Then $\operatorname{tr}(x \mid \operatorname{Ind} W) = \operatorname{tr}(x \mid W) + \operatorname{tr}(s^{-1}xs \mid W)$ for $x \in H$ and $\operatorname{tr}(g \mid \operatorname{Ind} W) = 0$ for $g \notin H$. For a character $\psi$ of $H$, in the basis $1 \otimes 1$, $s \otimes 1$ of $\operatorname{Ind}_H^G \psi$,*

$$x \mapsto \begin{pmatrix} \psi(x) & 0 \\ 0 & \psi^s(x) \end{pmatrix} \ (x \in H), \qquad s \mapsto \begin{pmatrix} 0 & \psi(s^2) \\ 1 & 0 \end{pmatrix}.$$

*Proof.* (i) $k[G]$ is a free right $k[H]$-module with basis $s_1, \dots, s_m$, so $k[G] \otimes_{k[H]} W = \bigoplus_i s_i \otimes W$, and $g(s_i \otimes w) = s_jx \otimes w = s_j \otimes xw$. For $y = s_ixs_i^{-1}$, $y(s_i \otimes w) = s_ix \otimes w = s_i \otimes xw$, so $w \mapsto s_i \otimes w$ is an isomorphism $W^{s_i} \to s_i \otimes W$. (ii) By (i), $g$ maps the block $s_i \otimes W$ into $s_j \otimes W$; the diagonal blocks are those with $j = i$, that is, $s_i^{-1}gs_i = x \in H$, where $g$ acts as $x$ on $W$. (iii) With $s_1 = 1$, $s_2 = s$: for $x \in H$ both blocks are diagonal ($s^{-1}xs \in H$ as $H$ is normal), and for $g \notin H$ neither is. For $\psi$: $x(1 \otimes 1) = 1 \otimes \psi(x)$, $x(s \otimes 1) = s(s^{-1}xs) \otimes 1 = \psi^s(x)\, s \otimes 1$, $s(1 \otimes 1) = s \otimes 1$ and $s(s \otimes 1) = s^2 \otimes 1 = \psi(s^2)\, 1 \otimes 1$. □

**Theorem 4.40** (Frobenius reciprocity)**.** *Let $H \le G$, $W$ a representation of $H$ and $V$ one of $G$, over a field or a commutative ring.*

*(i) The map $\phi \mapsto \phi(1 \otimes -)$ is an isomorphism $\operatorname{Hom}_G(\operatorname{Ind}_H^G W, V) \simeq \operatorname{Hom}_H(W, \operatorname{Res}_H V)$.*

*(ii) If $[G:H]$ is finite, with $\pi\colon \operatorname{Ind}_H^G W \to W$ the projection onto $1 \otimes W$ along $\bigoplus_{i \ge 2} s_i \otimes W$, the map $\phi \mapsto \pi \circ \phi$ is an isomorphism $\operatorname{Hom}_G(V, \operatorname{Ind}_H^G W) \simeq \operatorname{Hom}_H(\operatorname{Res}_H V, W)$.*

*In characteristic 0 with $G$ finite, (i) and Theorem 4.29(ii) give $\langle \operatorname{tr}_{\operatorname{Ind} W}, \operatorname{tr}_V \rangle_G = \langle \operatorname{tr}_W, \operatorname{tr}_{\operatorname{Res} V} \rangle_H$.*

*Proof.* (i) $w \mapsto \phi(1 \otimes w)$ is $H$-linear because $\phi(1 \otimes xw) = \phi(x \otimes w) = x\phi(1 \otimes w)$. The inverse sends $\psi \in \operatorname{Hom}_H(W, V)$ to $y \otimes w \mapsto y\psi(w)$, which is well defined on $k[G] \otimes_{k[H]} W$ ($yx \otimes w$ and $y \otimes xw$ have the same image, $\psi$ being $H$-linear) and $G$-linear. The two constructions are inverse to each other: starting from $\psi$ one recovers $w \mapsto 1 \cdot \psi(w) = \psi(w)$, and starting from $\phi$ one obtains $y \otimes w \mapsto y\phi(1 \otimes w) = \phi(y \otimes w)$. (ii) $\pi$ is $H$-linear: $H$ preserves $1 \otimes W$, and permutes the blocks $s_i \otimes W$, $i \ge 2$ (if $xs_i \in s_1H = H$ then $s_i \in H$). Given $\psi \in \operatorname{Hom}_H(V, W)$ put $\Psi(v) = \sum_i s_i \otimes \psi(s_i^{-1}v)$.

This does not depend on the representatives, since $s_i x \otimes \psi(x^{-1}s_i^{-1}v) = s_i \otimes \psi(s_i^{-1}v)$. It is $G$-linear: $g\Psi(v) = \sum_i gs_i \otimes \psi(s_i^{-1}v)$, and $t_i = gs_i$ is another system of representatives with $t_i^{-1}gv = s_i^{-1}v$, so $g\Psi(v) = \sum_i t_i \otimes \psi(t_i^{-1}gv) = \Psi(gv)$. Then $\pi\Psi = \psi$ (the term $i = 1$). Conversely, for a $G$-map $\phi$ write $\phi(v) = \sum_i s_i \otimes w_i$; then $\pi(\phi(s_i^{-1}v)) = \pi(s_i^{-1}\phi(v)) = w_i$, because $s_i^{-1}(s_j \otimes w_j) = s_i^{-1}s_j \otimes w_j$ lies in the block $1 \otimes W$ for $j = i$ and in another block for $j \neq i$ ($s_i^{-1}s_j \in H$ only for $j = i$); so the map $\Psi$ built from $\psi = \pi\phi$ is $\phi$. □

**Proposition 4.41** (Projection formula)**.** *Let $H \leq G$ have finite index.*

*(i) For a representation $W$ of $H$ and $U$ of $G$, $y \otimes (w \otimes u) \mapsto (y \otimes w) \otimes yu$ is a $G$-isomorphism $\operatorname{Ind}_H^G(W \otimes \operatorname{Res}_H U) \simeq \operatorname{Ind}_H^G W \otimes U$. In particular, if $\chi$ is a character of $G$ trivial on $H$, then $\operatorname{Ind}_H^G W \otimes \chi \simeq \operatorname{Ind}_H^G W$.*

*(ii) $\operatorname{Ind}_H^G \mathbf{1} = k[G/H]$, the permutation representation on the cosets. If $[G : H] = 2$ and $\operatorname{char} k \neq 2$, then $\operatorname{Ind}_H^G \mathbf{1} \simeq \mathbf{1} \oplus \chi_H$, where $\chi_H \colon G \to \{\pm 1\}$ is the character with kernel $H$.*

*(iii) $\operatorname{Ind}_K^G \operatorname{Ind}_H^K \simeq \operatorname{Ind}_H^G$ for $H \leq K \leq G$; $\operatorname{Ind}(W \oplus W') \simeq \operatorname{Ind} W \oplus \operatorname{Ind} W'$; and $(\operatorname{Ind}_H^G W)_K \simeq \operatorname{Ind}_H^G(W_K)$ for a field extension $K/k$.*

*Proof.* (i) The formula is compatible with $yx \otimes (w \otimes u) = y \otimes (xw \otimes xu)$ ($x \in H$), since both go to $(yx \otimes w) \otimes yxu$; it is $G$-linear. On the block $s_i \otimes (W \otimes U)$ it is $w \otimes u \mapsto (s_i \otimes w) \otimes s_i u$, a bijection onto $(s_i \otimes W) \otimes U$; so it is bijective. If $\chi|_H = 1$, then $\operatorname{Res}_H \chi = \mathbf{1}$ and $W \otimes \mathbf{1} = W$. (ii) $k[G] \otimes_{k[H]} k$ has basis $s_i \otimes 1$, permuted by $G$ as the cosets $s_i H$. For index 2 the vectors $1 \otimes 1 \pm s \otimes 1$ are fixed, respectively multiplied by $\chi_H$, and they form a basis when $2 \neq 0$. (iii) Associativity of the tensor product, $k[G] \otimes_{k[K]} (k[K] \otimes_{k[H]} W) = k[G] \otimes_{k[H]} W$; additivity of $\otimes$; and $K \otimes_k (k[G] \otimes_{k[H]} W) = K[G] \otimes_{K[H]} W_K$. □

**Theorem 4.42** (Mackey)**.** *Let $H \leq G$ have finite index and $W$ a representation of $H$.*

*(a) For $K \leq G$, $\operatorname{Res}_K \operatorname{Ind}_H^G W \simeq \bigoplus_{s \in K\backslash G/H} \operatorname{Ind}_{K \cap sHs^{-1}}^K \operatorname{Res}_{K \cap sHs^{-1}} W^s$, the sum over a set of representatives of the double cosets.*

*(b) (Mackey's irreducibility criterion.) Let $G$ be finite and $k = \mathbb{C}$. Then $\operatorname{Ind}_H^G W$ is irreducible iff $W$ is irreducible and, for every $s \in G\backslash H$, the representations $W^s$ and $W$ of $H_s = sHs^{-1} \cap H$ have no common irreducible constituent.*

*(c) (Index two, any characteristic.) Let $[G : H] = 2$, $s \in G \setminus H$, and $W$ absolutely irreducible. Then $\operatorname{Res}_H \operatorname{Ind}_H^G W \simeq W \oplus W^s$, and $\operatorname{Ind}_H^G W$ is absolutely irreducible iff $W_{\bar{k}} \not\simeq W^s_{\bar{k}}$.*

*Proof.* (a) We do not prove this; see [173, Section 7.3]. (The summand for the double coset $KsH$ is the sum of the blocks $t \otimes W$ with $t$ running over representatives of the cosets $tH \subseteq KsH$.) (c) The restriction is Proposition 4.39(i) with $s_1 = 1$, $s_2 = s$ ($H$ is normal). Work over $\bar{k}$, which is allowed by Proposition 4.41(iii); $W$ and $W^s$ are irreducible. Suppose $W \not\simeq W^s$. The $H$-representation $W \oplus W^s$ is semisimple with non-isomorphic summands, so by Proposition 4.17(iii) its subrepresentations are 0, $W = 1 \otimes W$, $W^s = s \otimes W$ and the whole space (a subrepresentation $U$ is semisimple, $U = \bigoplus_S U_S$, and each isotypic component $U_S$ lies in the corresponding isotypic component of the whole space, which is the irreducible $1 \otimes W$ or $s \otimes W$). A nonzero $G$-stable subspace is $H$-stable and $s$-stable, and $s$ exchanges $1 \otimes W$ and $s \otimes W$; so it is everything, and $\operatorname{Ind} W$ is irreducible over $\bar{k}$. Suppose $W \simeq W^s$. By Theorem 4.40(i) and Schur over $\bar{k}$, $\operatorname{End}_G(\operatorname{Ind} W) = \operatorname{Hom}_H(W, W \oplus W^s)$ has dimension 2; an irreducible representation over $\bar{k}$ has endomorphism ring $\bar{k}$ (Theorem 4.10(ii)), so $\operatorname{Ind} W$ is reducible. □

*Sketch of proof of (b).* By (i) of Theorem 4.40 and (a) with $K = H$,

$$\operatorname{End}_G(\operatorname{Ind} W) \simeq \operatorname{Hom}_H(W, \operatorname{Res}_H \operatorname{Ind} W) = \bigoplus_{s \in H\backslash G/H} \operatorname{Hom}_H(W, \operatorname{Ind}_{H_s}^H W^s),$$

and by Frobenius reciprocity again the summand for $s$ is $\operatorname{Hom}_{H_s}(\operatorname{Res}_{H_s} W, W^s)$. The summand for $s = 1$ is $\operatorname{End}_H(W)$. In characteristic 0 all representations are semisimple, so $\operatorname{Ind} W$ is irreducible iff $\dim \operatorname{End}_G(\operatorname{Ind} W) = 1$, iff $\operatorname{End}_H(W) = \mathbb{C}$ (that is, $W$ irreducible) and the other summands vanish. See [173, Section 7.4]. In characteristic $p$ the criterion fails as stated, because $\operatorname{End}_G(V) = k$ does not force irreducibility without semisimplicity (Example 4.20(d)); only (c) is needed here. □

**Theorem 4.43** (Clifford)**.** *Let $N \trianglelefteq G$ have finite index and $V$ an irreducible representation of $G$ over any field. Then $\operatorname{Res}_N V$ is semisimple, its irreducible constituents are the conjugates $W^g$ ($g \in G$) of any one of them, and they all occur with the same multiplicity.*

*Proof.* Let $W \subseteq V$ be an irreducible $N$-subrepresentation (one of minimal positive dimension). For $g \in G$, $gW$ is $N$-stable, since $n\,gW = g(g^{-1}ng)W = gW$, and $w \mapsto gw$ is an isomorphism $W^g \to gW$ of representations of $N$; so $gW$ is irreducible. The sum $\sum_g gW$ is $G$-stable and nonzero, hence equal to $V$. So $\operatorname{Res}_N V$ is a sum of irreducibles of the form $W^g$, semisimple by Proposition 4.17, and every irreducible constituent is isomorphic to some $W^g$ (Theorem 4.21). For an irreducible $S$ of $N$, the element $g$ maps the isotypic component $V_S$ bijectively onto $V_{S^g}$, so $\dim V_S = \dim V_{S^g}$, and the multiplicities agree. □

**Theorem 4.44** (Clifford theory for a subgroup of index 2)**.** *Let $G$ be any group, $H \leq G$ of index 2, $s \in G \setminus H$, $\chi_H : G \to \{\pm 1\}$ the character with kernel $H$, $k$ a field with $\operatorname{char} k \neq 2$, and $V$ an absolutely irreducible representation of $G$ over $k$ of dimension $n$. Exactly one of the following holds.*

*(A)* $\operatorname{Res}_H V$ *is absolutely irreducible. Then $V \not\simeq V \otimes \chi_H$, and $\operatorname{Ind}_H^G \operatorname{Res}_H V \simeq V \oplus (V \otimes \chi_H)$.*

*(B)* $\operatorname{Res}_H V$ *is not absolutely irreducible. Then $n = 2d$, there is an absolutely irreducible representation $W$ of $H$ over $\bar{k}$ of dimension $d$ with $W \not\simeq W^s$, $\operatorname{Res}_H V_{\bar{k}} \simeq W \oplus W^s$ and $V_{\bar{k}} \simeq \operatorname{Ind}_H^G W$, and $V \simeq V \otimes \chi_H$ over $k$.*

*Hence $V \simeq V \otimes \chi_H$ iff (B) holds. In that case $\operatorname{Hom}_G(V, V \otimes \chi_H)$ is a line over $k$, spanned by an isomorphism $\phi$ with $\phi^2 \in k^\times$ and $\operatorname{tr} \phi = 0$, whose two eigenspaces over $\bar{k}$ are $W$ and $W^s$. For $n = 2$, (B) holds iff $V_{\bar{k}} \simeq \operatorname{Ind}_H^G \psi$ for a character $\psi$ of $H$ with $\psi^s \neq \psi$.*

*Proof.* (A) An isomorphism $\phi\colon V \to V \otimes \chi_H$ commutes with $\rho(H)$, so it is a scalar $c$ (Theorem 4.10(iii) for $\operatorname{Res}_H V$). Then $c\rho(s) = \chi_H(s)\rho(s)c = -c\rho(s)$, so $2c = 0$ and $c = 0$, a contradiction. The induced statement is $\operatorname{Ind}_H^G \operatorname{Res}_H V \simeq V \otimes \operatorname{Ind}_H^G \mathbf{1} \simeq V \otimes (\mathbf{1} \oplus \chi_H)$ (Proposition 4.41(i), (ii)).

(B) $V_{\bar{k}}$ is irreducible and $\operatorname{Res}_H V_{\bar{k}}$ is reducible. By Theorem 4.43, $\operatorname{Res}_H V_{\bar{k}}$ is semisimple and its constituents are $W$ and $W^s$ for an irreducible $W$ (absolutely irreducible, as $\bar{k}$ is algebraically closed), each of dimension $d$; being reducible, it has at least two constituents, so $n \geq 2d$. By Theorem 4.40(i) the inclusion $W \subseteq \operatorname{Res}_H V_{\bar{k}}$ gives a nonzero $G$-map $\operatorname{Ind}_H^G W \to V_{\bar{k}}$, surjective as $V_{\bar{k}}$ is irreducible; so $n \leq 2d$. Hence $n = 2d$ and the map is an isomorphism $\operatorname{Ind}_H^G W \simeq V_{\bar{k}}$. As $\operatorname{Ind} W$ is irreducible, $W \not\simeq W^s$ by Theorem 4.42(c), which also gives $\operatorname{Res}_H V_{\bar{k}} \simeq W \oplus W^s$. By

Proposition 4.41(i), $V_{\bar{k}} \otimes \chi_H \simeq \mathrm{Ind}_H^G(W \otimes \mathrm{Res}_H \chi_H) = \mathrm{Ind}_H^G W \simeq V_{\bar{k}}$, and since $V_{\bar{k}}$ is irreducible, Proposition 4.7(iii) gives $V \simeq V \otimes \chi_H$ over $k$.

The two cases are exclusive and exhaustive, so $V \simeq V \otimes \chi_H$ iff (B). In case (B), $V \otimes \chi_H$ is absolutely irreducible (Proposition 4.48(ii)), so $\mathrm{Hom}_G(V, V \otimes \chi_H)$ is a line (Theorem 4.10(iii)) spanned by an isomorphism $\phi$. The composite $\phi \circ \phi$ is a $G$-isomorphism $V \to V \otimes \chi_H^2 = V$ (the linear map $\phi$ is also a $G$-map $V \otimes \chi_H \to V \otimes \chi_H^2$, by Proposition 4.48(ii)), hence a nonzero scalar (Theorem 4.10(iii)). By Lemma 4.49, $\mathrm{tr}\,\phi = 0$ ($\chi_H(s) = -1 \neq 1$). Over $\bar{k}$, $\phi$ commutes with $H$, so it preserves the isotypic components $W$ and $W^s$ of $\mathrm{Res}_H V_{\bar{k}}$ (Proposition 4.17(iii)) and acts on each by a scalar (Schur); the two scalars differ, since otherwise $\phi$ would be a scalar $c \neq 0$, and $c\rho(s) = \phi\rho(s) = \chi_H(s)\rho(s)\phi = -c\rho(s)$ would force $c = 0$, as in (A). For $n = 2$, $d = 1$ and $W = \psi$ is a character; conversely if $V_{\bar{k}} \simeq \mathrm{Ind}\,\psi$ then $\mathrm{Res}_H V_{\bar{k}} \simeq \psi \oplus \psi^s$ is reducible. □

A small example over $\mathbb{C}$: $G = S_3$, $H = A_3 = \langle r \rangle$ with $r$ a 3-cycle, $s$ a transposition, $\chi_H = \mathrm{sgn}$. The two-dimensional irreducible representation $V$ of $S_3$ (the plane $P$ of the example after Definition 4.2) restricts to $A_3$ as $\omega \oplus \omega^{-1}$, where $\omega(r) = e^{2\pi i/3}$: the permutation matrix of $r$ has eigenvalues $1, e^{\pm 2\pi i/3}$ on $\mathbb{C}^3$, and the eigenvalue 1 belongs to $D$. Since $srs^{-1} = r^{-1}$, $\omega^s = \omega^{-1} \neq \omega$, so we are in case (B): $V \simeq \mathrm{Ind}_{A_3}^{S_3} \omega$ and $V \simeq V \otimes \mathrm{sgn}$. If $v$ is an eigenvector of $r$ for $\omega(r)$, then $sv$ is one for $\omega(r)^{-1}$, and in the basis $v, sv$ the intertwiner is $\phi = \mathrm{diag}(1, -1)$: it commutes with $r$, which is diagonal, and anticommutes with $s = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$; its trace is 0. The one-dimensional representations $\mathbf{1}$ and sgn are in case (A), and the twist by sgn exchanges them.

*In the paper* 4.45. Proposition prop:ns: "by Frobenius reciprocity $\bar{\rho} \otimes \mathbb{F}_9 \simeq \mathrm{Ind}\,\psi$, so $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$". In the language above: $G = G_{\mathbb{Q}}$, $H = G_{\mathbb{Q}(\sqrt{-3})}$, $\chi_H = \chi_{-3}$; the eigenline of $\bar{\rho}(H) = C_4$ with character $\psi$ (Example 4.14) is a nonzero $H$-map $\psi \to \mathrm{Res}_H(\bar{\rho} \otimes \mathbb{F}_9)$, which by Theorem 4.40(i) gives a nonzero $G$-map $\mathrm{Ind}_H^G \psi \to \bar{\rho} \otimes \mathbb{F}_9$, surjective because $\bar{\rho} \otimes \mathbb{F}_9$ is irreducible, hence an isomorphism (both have dimension 2). Then $\mathrm{Ind}\,\psi \otimes \chi_{-3} \simeq \mathrm{Ind}\,\psi$ (Proposition 4.41(i)), and the isomorphism descends to $\mathbb{F}_3$ (Proposition 4.7(ii)). The last equivalence of Lemma lem:h0 ($\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$ iff $\bar{\rho}$ is not absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$) and the first sentence of the proof of Proposition prop:Sf (the same with $\chi_{\ell^*}$) are Theorem 4.44; it needs $\bar{\rho}$ absolutely irreducible, which complex conjugation provides (Proposition 4.13). The representation $\bar{\rho}$ is induced from $H$ only after extending scalars: over $\mathbb{F}_3$, $\mathrm{Res}_H \bar{\rho}$ is irreducible (Proposition 4.63).

**Proposition 4.46** (Dihedral representations)**.** *Let* $\mathrm{char}\, k \neq 2$ *and* $\rho\colon G \to \mathrm{GL}_2(k)$ *absolutely irreducible, with finite projective image* $P$*, the image of* $G$ *in* $\mathrm{PGL}_2(\bar{k})$*.*

*(i)* $P$ *is not cyclic.*

*(ii) The following are equivalent: (a)* $P$ *is dihedral of order* $2m$ *for some* $m \geq 2$ *(for* $m = 2$*,* $P \simeq V_4$*); (b)* $\rho \simeq \rho \otimes \chi$ *for a character* $\chi$ *of order* 2*; (c)* $\rho_{\bar{k}} \simeq \mathrm{Ind}_H^G \psi$ *for a subgroup* $H$ *of index* 2 *and a character* $\psi$ *of* $H$*.*

*(iii) The characters* $\chi \neq 1$ *with* $\rho \simeq \rho \otimes \chi$ *correspond bijectively, by* $\chi \mapsto$ *the image of* $\ker \chi$ *in* $P$*, to the cyclic subgroups of index* 2 *of* $P$*: there is exactly one if* $P$ *is dihedral of order at least* 6*, and there are three if* $P \simeq V_4$*.*

*Proof.* Let $Z = \rho^{-1}(\bar{k}^\times)$ be the preimage of the scalars, so $P = G/Z$. (i) If $P$ is generated by the image of $g_0$, then $\rho(G) \subseteq \bar{k}^\times \langle \rho(g_0) \rangle$ is abelian, and $\rho$ is absolutely reducible by Proposition 4.26(i). First a claim: *for* $H \leq G$ *of index* 2*,* $\mathrm{Res}_H \rho$ *is absolutely reducible iff the image of* $H$ *in* $P$ *is cyclic.* If the image is cyclic, $\rho(H)$ is abelian as in (i) and Proposition 4.26(i) applies. If $\mathrm{Res}_H \rho_{\bar{k}}$ has a stable line, then by Theorem 4.44(B) it is $\psi \oplus \psi^s$, so $\rho(H)$ is diagonal in a suitable basis,

and its image in $\mathrm{PGL}_2(\bar{k})$ is a finite subgroup of the group of classes of $\mathrm{diag}(x,1)$, which is isomorphic to $\bar{k}^\times$; finite subgroups of $\bar{k}^\times$ are cyclic. (ii) (b)$\Leftrightarrow$(c): a character of order 2 is $\chi_H$ for $H=\ker\chi$ of index 2, and Theorem 4.44 says that $\rho\simeq\rho\otimes\chi_H$ iff (B) holds for $H$, iff $\rho_{\bar{k}}$ is induced from a character of $H$. (c)$\Rightarrow$(a): in the basis of Proposition 4.39(iii), $\rho(H)$ is diagonal and $\rho(G\setminus H)$ antidiagonal. The image $C$ of $H$ in $P$ is cyclic by the claim, since $\mathrm{Res}_H\,\rho_{\bar{k}}\simeq\psi\oplus\psi^s$ is reducible; it has index 1 or 2 in $P$; by (i), $P\neq C$. An antidiagonal matrix $\left(\begin{smallmatrix}0&a\\b&0\end{smallmatrix}\right)$ squares to the scalar $ab$, and conjugating $\mathrm{diag}(x,y)$ by it gives $\mathrm{diag}(y,x)$, which is the inverse of $\mathrm{diag}(x,y)$ modulo scalars. So $P=C\rtimes C_2$ with the involution inverting $C$, and $\#C=m\geq 2$ by (i): $P$ is dihedral of order $2m$ (for $m=2$, $V_4$). (a)$\Rightarrow$(c): a dihedral group of order $2m$ has a cyclic subgroup of index 2 (for $V_4$, any subgroup of order 2); its preimage $H$ in $G$ has index 2 and cyclic image, so $\mathrm{Res}_H\,\rho$ is absolutely reducible by the claim, and Theorem 4.44(B) gives (c). (iii) If $\rho\simeq\rho\otimes\chi$ with $\chi\neq 1$, then $\chi^2=1$ (Proposition 4.48(iv)), so $\chi=\chi_H$ with $H=\ker\chi$ of index 2. For $z\in Z$, $\mathrm{tr}\,\rho(z)=\chi(z)\,\mathrm{tr}\,\rho(z)$ and $\mathrm{tr}\,\rho(z)=2c\neq 0$ for the scalar $c=\rho(z)$; so $\chi(z)=1$ and $Z\subseteq H$. Hence such $H$ correspond to the subgroups of index 2 of $P$, and by the claim and Theorem 4.44 exactly those with cyclic image occur. In a dihedral group of order $2m$, $m\geq 3$, the only cyclic subgroup of index 2 is the rotation subgroup (the others, which exist for $m$ even, are dihedral of order $m\geq 4$, hence not cyclic); $V_4$ has three subgroups of index 2, all cyclic. □

*In the paper* 4.47. Among irreducible $\bar{\rho}$ (a reducible one fails it trivially), the hypothesis of Theorem thm:main excludes exactly the representations induced from $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, not all dihedral ones (Remark 4.83(ii)). At $\ell=3$, the image $N_s(\mathbb{F}_3)$ has projective image $V_4$ and is induced in three ways, one of them from the determinant-one subgroup (Proposition 4.63); the image $N_{ns}(\mathbb{F}_3)$ has projective image $D_4$ and is induced only from $C_{ns}(\mathbb{F}_3)$, whose fixed field is not $\mathbb{Q}(\sqrt{-3})$ (Proposition 4.64), so it satisfies the hypothesis.

## 4.7 Twisting, the adjoint, the symmetric square and the trace form

This section treats the operations the paper performs on $\rho$: twisting by a character, passing to the adjoint $\mathrm{End}(V)$ and to its trace-zero part $\mathrm{ad}^0 V$, and the identification of $\mathrm{ad}^0 V$ with a twist of the symmetric square and with its own dual. In the paper, $A_f=\mathrm{ad}^0 V$ (Section 1.1), $A=\mathrm{ad}^0 T$ and $T^D=\mathrm{Hom}(A,\mathbb{Z}_\ell(1))\simeq A(1)$ (Section 4.2), $H^0(\mathbb{Q},W^D)$ is a space of trace-zero $G$-maps $\bar{\rho}\to\bar{\rho}\otimes\bar{\chi}_\ell$ (Lemma lem:h0), and the proof of Lemma lem:tamfactor computes the adjoint action of two triangular matrices in the basis $e,h,f$.

**Proposition 4.48** (Twisting)**.** *Let $V$ be a representation of $G$ over $k$ (or a free $R$-module) of rank $n$ and $\chi\colon G\to k^\times$ (or $R^\times$) a character.*

*(i)* $\mathrm{tr}(\rho\otimes\chi)(g)=\chi(g)\,\mathrm{tr}\,\rho(g)$, $\det(X-(\rho\otimes\chi)(g))=\det(X-\chi(g)\rho(g))$ *and* $\det(\rho\otimes\chi)=\chi^n\det\rho$.

*(ii) $V$ is irreducible (absolutely irreducible) iff $V\otimes\chi$ is;* $\mathrm{Hom}_G(V\otimes\chi,W\otimes\chi)=\mathrm{Hom}_G(V,W)$; $(V\otimes\chi)^\vee=V^\vee\otimes\chi^{-1}$; *and* $\mathrm{Res}_{\ker\chi}(V\otimes\chi)=\mathrm{Res}_{\ker\chi}V$.

*(iii)* $\mathrm{End}(V\otimes\chi)=\mathrm{End}(V)$ *as representations (the same space with the same action), hence* $\mathrm{ad}^0(V\otimes\chi)=\mathrm{ad}^0 V$.

*(iv) If $V\simeq V\otimes\chi$, then $\chi^n=1$ on $G$; for $n=2$, $\chi^2=1$.*

*Proof.* On the space $V$ the twist acts by $\chi(g)\rho(g)$; this gives (i) and the last statement of (ii). The $G$-stable subspaces of $V$ and of $V\otimes\chi$ coincide (a scalar preserves every subspace), also after extending scalars. A linear map $\phi$ satisfies $\phi\,\chi(g)\rho_V(g)=\chi(g)\rho_W(g)\phi$ iff $\phi\rho_V(g)=\rho_W(g)\phi$. The dual of $\chi(g)\rho(g)$ is $\chi(g)^{-1}\rho(g^{-1})^t$. (iii) $(\chi(g)\rho(g))x(\chi(g)\rho(g))^{-1}=\rho(g)x\rho(g)^{-1}$. (iv) An isomorphism $V\to V\otimes\chi$ is an invertible $P$ with $P\rho(g)P^{-1}=\chi(g)\rho(g)$ for all $g$. Taking determinants, $\det\rho(g)=\chi(g)^n\det\rho(g)$, and $\det\rho(g)$ is a unit, so $\chi(g)^n=1$. (Equivalently: $\det(V\otimes\chi)=\chi^n\det V$ by (i), and isomorphic representations have the same determinant.) □

**Lemma 4.49** (The trace-zero lemma)**.** *Let $\phi\colon V\to V\otimes\chi$ be a $G$-map, that is, $\phi\rho(g)=\chi(g)\rho(g)\phi$ for all $g\in G$. Then $(1-\chi(g))\operatorname{tr}\phi=0$ for every $g\in G$. Hence $\operatorname{tr}\phi=0$ if $\chi\neq 1$ (over a field), or, over a ring $R$, if $1-\chi(g)\in R^\times$ for some $g$. If moreover $V$ is absolutely irreducible, $\operatorname{Hom}_G(V,V\otimes\chi)$ has dimension at most 1, and its nonzero elements are isomorphisms.*

*Proof.* $\operatorname{tr}\phi=\operatorname{tr}(\rho(g)^{-1}\phi\rho(g))=\operatorname{tr}(\rho(g)^{-1}\chi(g)\rho(g)\phi)=\chi(g)\operatorname{tr}\phi$. The last statement is Theorem 4.10(iii) for the absolutely irreducible representations $V$ and $V\otimes\chi$ (Proposition 4.48(ii)). □

*In the paper* 4.50. Lemma lem:h0: "every such map has trace 0 automatically, since $\operatorname{tr}\phi=\chi_\ell(g)\operatorname{tr}\phi$ for every $g$ and $1-\chi_\ell(c)=2$ is a unit for complex conjugation $c$" is Lemma 4.49 over $R=\mathbb{Z}/\ell^k$, $\ell$ odd. "Taking determinants" (the step $\chi_\ell^2\equiv 1$) is Proposition 4.48(iv) modulo $\ell^k$ (see Proposition 4.81(v)). "$\mathrm{ad}^0T$ is unchanged by the twist" (proofs of Theorems thm:main and thm:quant, Lemma lem:tamfactor at an additive potentially multiplicative prime, Lemma lem:euler for $\mathrm{I}_n^*$, $\mathrm{I}_0^*$) is (iii).

**Definition 4.51.** The *adjoint representation* of $V$ is $\operatorname{End}(V)=V\otimes V^\vee$ with $g\cdot x=\rho(g)x\rho(g)^{-1}$, also written $\operatorname{Ad}V$ or $\operatorname{Ad}\rho$; the *trace-zero adjoint* is $\mathrm{ad}^0V=\{x\in\operatorname{End}(V):\operatorname{tr}x=0\}$, a subrepresentation since the trace is conjugation invariant. The paper writes $A_f=\mathrm{ad}^0V$ for $V=V_\ell E$ and $A=\mathrm{ad}^0T$ for $T=T_\ell E$.

The adjoint enters number theory because it governs the deformations of $\rho$. If $k[\epsilon]=k[X]/(X^2)$ and $\rho_\epsilon\colon G\to\mathrm{GL}_n(k[\epsilon])$ reduces to $\rho$ modulo $\epsilon$, write $\rho_\epsilon(g)=(1+\epsilon c(g))\rho(g)$ with $c(g)\in\operatorname{End}(V)$. Since $\epsilon^2=0$,

$$\rho_\epsilon(g)\rho_\epsilon(h)=\big(1+\epsilon(c(g)+\rho(g)c(h)\rho(g)^{-1})\big)\rho(gh),$$

so $\rho_\epsilon$ is a homomorphism iff $c(gh)=c(g)+g\cdot c(h)$ for the adjoint action: $c$ is a 1-cocycle with values in $\operatorname{End}(V)$. As $\det(1+\epsilon c)=1+\epsilon\operatorname{tr}c$, the deformations with unchanged determinant correspond to cocycles with values in $\mathrm{ad}^0V$. This is the origin of the Selmer groups of $\mathrm{ad}^0V$ studied in Chapter 18. A first example of the representation itself: if $\rho=\eta_1\oplus\eta_2$ is diagonal, conjugation by $\operatorname{diag}(\eta_1(g),\eta_2(g))$ multiplies $e$ by $\eta_1(g)/\eta_2(g)$, fixes $h$ and multiplies $f$ by $\eta_2(g)/\eta_1(g)$ (Proposition 4.52(iii)), so $\mathrm{ad}^0V\simeq\eta_1\eta_2^{-1}\oplus\mathbf{1}\oplus\eta_2\eta_1^{-1}$.

**Proposition 4.52.** *Let $V$ be a representation of $G$ over $k$ (or a free $R$-module) of rank $n$.*

*(i) If $n$ is invertible in $k$ (or $R$), $\operatorname{End}(V)=k\cdot 1\oplus\mathrm{ad}^0V$ as representations, $x=\frac{\operatorname{tr}x}{n}\cdot 1+(x-\frac{\operatorname{tr}x}{n}\cdot 1)$. For $n=2$ this needs $\operatorname{char}k\neq 2$, that is, $\ell$ odd.*

*(ii) If $\rho(g)$ has eigenvalues $\lambda_1,\dots,\lambda_n$ in $\bar{k}$ (with multiplicity), then $\operatorname{Ad}\rho(g)$ has the eigenvalues $\lambda_i/\lambda_j$ on $\operatorname{End}(V)$. For $n=2$ it has the eigenvalues $\lambda_1/\lambda_2$, 1, $\lambda_2/\lambda_1$ on $\mathrm{ad}^0V$, in every characteristic, and $\det(X-\operatorname{Ad}\rho(g)\mid\mathrm{ad}^0V)=(X-1)(X^2-(\operatorname{tr}(g)^2/\det(g)-2)X+1)$.*

*(iii) In the basis $e, h, f$ of $\mathrm{ad}^0 V$ (convention (C3)):*

$$\begin{aligned}
&\mathrm{Ad}\,\mathrm{diag}(x,y): \quad e \mapsto \tfrac{x}{y}\,e, \quad h \mapsto h, \quad f \mapsto \tfrac{y}{x}\,f;\\
&t = \begin{pmatrix} 1 & n' \\ 0 & 1 \end{pmatrix}: \quad (t-1)e = 0, \quad (t-1)h = -2n'e, \quad (t-1)f = n'h - n'^2 e;\\
&F = \begin{pmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{pmatrix}: \quad FeF^{-1} = qe, \quad FhF^{-1} = h - 2c'\varepsilon^{-1}e,\\
&\qquad\qquad FfF^{-1} = q^{-1}f + c'\varepsilon^{-1}q^{-1}h - c'^2\varepsilon^{-2}q^{-1}e,
\end{aligned}$$

*for $x, y, \varepsilon \in R^\times$, $q \in R^\times$, $n', c' \in R$. In the paper $\varepsilon = \pm 1$ is the value at Frobenius of the unramified character of the Tate data (convention B-C14), so $\varepsilon^{-1} = \varepsilon$.*

*Proof.* (i) The map $x \mapsto \frac{\mathrm{tr}\,x}{n} \cdot 1$ is a projection onto the scalars with kernel $\mathrm{ad}^0 V$, and it is $G$-equivariant because $\mathrm{tr}(gxg^{-1}) = \mathrm{tr}\,x$ and $g1g^{-1} = 1$; so both summands are subrepresentations. (ii) Over $\bar{k}$, conjugate $\rho(g)$ to an upper triangular matrix with diagonal $\lambda_1, \dots, \lambda_n$. On $V^\vee$, $g$ acts by $\rho(g^{-1})^t$, with eigenvalues $\lambda_j^{-1}$, and the eigenvalues on $V \otimes V^\vee$ are the products (triangularise both factors). For $n = 2$ and $g = \left(\begin{smallmatrix} \lambda_1 & b \\ 0 & \lambda_2 \end{smallmatrix}\right)$, the matrix of $\mathrm{Ad}\,g$ in the ordered basis $e, h, f$ is upper triangular with diagonal $\lambda_1/\lambda_2, 1, \lambda_2/\lambda_1$: $geg^{-1} = \frac{\lambda_1}{\lambda_2}e$, $ghg^{-1} \in h + ke$ and $gfg^{-1} \in \frac{\lambda_2}{\lambda_1}f + kh + ke$, as the formulas of (iii) for $F$ show (with $\varepsilon q = \lambda_1$, $\varepsilon = \lambda_2$). The factor $X^2 - (\mathrm{tr}(g)^2/\det(g) - 2)X + 1$ has roots $\lambda_1/\lambda_2$, $\lambda_2/\lambda_1$, since their sum is $(\lambda_1^2 + \lambda_2^2)/\lambda_1\lambda_2$. (iii) Here $(t-1)x$ means $t \cdot x - x = txt^{-1} - x$, the adjoint action minus the identity, not the matrix product of $t - 1$ and $x$: the matrix product $(t-1)h$ is $\left(\begin{smallmatrix} 0 & -n' \\ 0 & 0 \end{smallmatrix}\right) = -n'e$, while $tht^{-1} - h = -2n'e$. The formulas follow by direct multiplication. For instance $th = \left(\begin{smallmatrix} 1 & -n' \\ 0 & -1 \end{smallmatrix}\right)$, so $tht^{-1} = \left(\begin{smallmatrix} 1 & -n' \\ 0 & -1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & -n' \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & -2n' \\ 0 & -1 \end{smallmatrix}\right) = h - 2n'e$; and $tf = \left(\begin{smallmatrix} n' & 0 \\ 1 & 0 \end{smallmatrix}\right)$, $tft^{-1} = \left(\begin{smallmatrix} n' & -n'^2 \\ 1 & -n' \end{smallmatrix}\right) = f + n'h - n'^2 e$. With $F^{-1} = \frac{1}{\varepsilon^2 q}\left(\begin{smallmatrix} \varepsilon & -c' \\ 0 & \varepsilon q \end{smallmatrix}\right)$: $Fe = \left(\begin{smallmatrix} 0 & \varepsilon q \\ 0 & 0 \end{smallmatrix}\right)$ and $FeF^{-1} = qe$; $Fh = \left(\begin{smallmatrix} \varepsilon q & -c' \\ 0 & -\varepsilon \end{smallmatrix}\right)$ and $FhF^{-1} = \frac{1}{\varepsilon^2 q}\left(\begin{smallmatrix} \varepsilon^2 q & -2\varepsilon q c' \\ 0 & -\varepsilon^2 q \end{smallmatrix}\right) = h - 2c'\varepsilon^{-1}e$; $Ff = \left(\begin{smallmatrix} c' & 0 \\ \varepsilon & 0 \end{smallmatrix}\right)$ and $FfF^{-1} = \frac{1}{\varepsilon^2 q}\left(\begin{smallmatrix} \varepsilon c' & -c'^2 \\ \varepsilon^2 & -\varepsilon c' \end{smallmatrix}\right)$, which is the stated combination. These identities were also checked symbolically. □

*In the paper* 4.53. Section 1.1: $A_f = \mathrm{ad}^0 V$, and at a good prime the Frobenius has the eigenvalues $\alpha/\beta, 1, \beta/\alpha$ on $\mathrm{ad}^0 V$ by (ii), for either Frobenius (the set is stable under inversion; convention B-C3), which gives the good factor of $L(A_f, s)$ (Chapter 19). Lemma lem:tamfactor: "$(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$, and as 2 is a unit, $(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$", and "the adjoint action sends $e$ to $qe$ and $h$ to $h - 2c'\varepsilon e$", are (iii) with $\varepsilon^{-1} = \varepsilon$; $F = \rho(\mathrm{Frob})$ there is the arithmetic Frobenius (convention B-C2(c)). The cohomology built on these formulas is in Chapter 16.

**Proposition 4.54** (Invariants of twisted adjoints)**.** *Let $V$ be a representation over $k$, or a free module of rank $n$ over a local ring $R$, and $\chi$ a character.*

*(i)* $(\mathrm{End}(V) \otimes \chi)^G = \mathrm{Hom}_G(V, V \otimes \chi)$.

*(ii) If $n$ is invertible,* $(\mathrm{End}(V) \otimes \chi)^G = (\mathrm{ad}^0 V \otimes \chi)^G \oplus R(\chi)^G$, *and* $(\mathrm{ad}^0 V \otimes \chi)^G = \{\phi \in \mathrm{Hom}_G(V, V \otimes \chi) : \mathrm{tr}\,\phi = 0\}$.

*(iii)* *If $\chi \neq 1$ (over a local ring $R$: if the reduction $\bar\chi$ of $\chi$ modulo the maximal ideal is nontrivial), then $R(\chi)^G = 0$, every $G$-map $V \to V \otimes \chi$ has trace $0$, and $(\mathrm{ad}^0 V \otimes \chi)^G = \mathrm{Hom}_G(V, V \otimes \chi)$.*

*(iv)* *If $V$ is absolutely irreducible over $k$ and $n \in k^\times$, then $(\mathrm{ad}^0 V)^G = 0$; and for $\chi \neq 1$, $(\mathrm{ad}^0 V \otimes \chi)^G \neq 0$ iff $V \simeq V \otimes \chi$, in which case it is a line spanned by an isomorphism.*

*Proof.* (i) is Proposition 4.5. (ii) Twist the decomposition of Proposition 4.52(i) by $\chi$ and take invariants. (iii) If $\bar\chi(g) \neq 1$ then $\chi(g) - 1$ is a unit, and $\chi(g)x = x$ forces $x = 0$; the rest is Lemma 4.49 and (ii). (iv) $\mathrm{End}_G(V) = k$, and a scalar $c$ has trace $nc$, which vanishes only for $c = 0$. For $\chi \neq 1$, by (iii) $(\mathrm{ad}^0 V \otimes \chi)^G = \mathrm{Hom}_G(V, V \otimes \chi)$, which is described by Lemma 4.49. □

*In the paper* 4.55. Lemma lem:h0, case $k = 1$ of the sentence quoted after Proposition 4.5: the $\ell$-torsion of $H^0(\mathbb{Q}, W^D)$ is $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho \otimes \bar\chi_\ell)$, the space of trace-zero $G_\mathbb{Q}$-maps $\bar\rho \to \bar\rho \otimes \bar\chi_\ell$; by (iii) the trace condition is automatic because $\bar\chi_\ell \neq 1$ for $\ell$ odd. Proposition prop:ns: "$H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho_{E,3}(1)) \neq 0$" is (iv) with $\chi = \bar\chi_3 = \chi_{-3}$ and the isomorphism $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$.

**Proposition 4.56** (Adjoint and symmetric square)**.** *Let $V$ be free of rank $2$ over $R$ with basis $e_1, e_2$, $\omega(x, y) = \det[x \,|\, y]$, and let $\mathrm{Sym}^2 V$ be the quotient of $V \otimes V$ by the span of the $x \otimes y - y \otimes x$, with basis $e_1^2, e_1e_2, e_2^2$. If $\rho(g)$ has eigenvalues $\lambda, \mu$, then $\mathrm{Sym}^2 \rho(g)$ has eigenvalues $\lambda^2, \lambda\mu, \mu^2$.*

*(i)* *The map $\Xi\colon \mathrm{Sym}^2 V \to \mathrm{End}(V)$, $\Xi(xy)(v) = \omega(v, x)y + \omega(v, y)x$, is well defined, takes values in $\mathrm{ad}^0 V$, and satisfies $\Xi(g\cdot m) = \det(g)\,\rho(g)\Xi(m)\rho(g)^{-1}$. So $\Xi$ is a $G$-map $\mathrm{Sym}^2 V \otimes \det^{-1} \to \mathrm{ad}^0 V$. In coordinates*

$$\Xi(e_1^2) = -2e, \qquad \Xi(e_1e_2) = h, \qquad \Xi(e_2^2) = 2f,$$

*so $\Xi$ is an isomorphism iff $2 \in R^\times$; over $\mathbb{Z}$ its image has index $4$ in $\mathrm{ad}^0 \mathbb{Z}^2$, and over $\mathbb{Z}_\ell$ it is an isomorphism of lattices iff $\ell$ is odd.*

*(ii)* *Hence, if $2 \in R^\times$, $\mathrm{ad}^0 \rho \simeq \mathrm{Sym}^2 \rho \otimes \det^{-1}$ and $\mathrm{Sym}^2 \rho \simeq \mathrm{ad}^0 \rho \otimes \det$; moreover $V \otimes V \simeq \mathrm{Sym}^2 V \oplus \det$ and $V \otimes V^\vee \simeq \mathrm{ad}^0 V \oplus \mathbf{1}$.*

*(iii)* *(Normalisations.) Let $V = V_\ell E$, with $\det = \chi_\ell$ (Chapter 9). Then $\mathrm{Sym}^2 V \simeq \mathrm{ad}^0 V(1)$ and $\mathrm{Sym}^2(V^\vee) \simeq \mathrm{ad}^0 V(-1)$. The classical symmetric square $L$-function is that of $\mathrm{Sym}^2(V^\vee) = \mathrm{Sym}^2 H^1$, so with the convention $L_p(M(n), s) = L_p(M, s + n)$ (B-C2(d)), $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s - 1)$.*

*Proof.* The eigenvalue statement follows by triangularising $\rho(g)$: in the basis $e_1^2, e_1e_2, e_2^2$ the matrix of $\mathrm{Sym}^2 \rho(g)$ is then triangular with diagonal $\lambda^2, \lambda\mu, \mu^2$. (i) The expression is bilinear and symmetric in $(x, y)$, so it factors through $\mathrm{Sym}^2 V$. The trace of $v \mapsto \lambda(v)y$ is $\lambda(y)$, so $\mathrm{tr}\,\Xi(xy) = \omega(y, x) + \omega(x, y) = 0$. Using $\omega(v, gx) = \det(g)\omega(g^{-1}v, x)$ (Proposition 4.4(iii)),

$$\Xi(gx\,gy)(v) = \det(g)(\omega(g^{-1}v, x)gy + \omega(g^{-1}v, y)gx) = \det(g)\,\rho(g)\Xi(xy)\rho(g)^{-1}(v).$$

In coordinates, $\omega(v, e_1) = -v_2$ and $\omega(v, e_2) = v_1$ for $v = v_1e_1 + v_2e_2$, so $\Xi(e_1^2)(v) = -2v_2e_1$, $\Xi(e_1e_2)(v) = v_1e_1 - v_2e_2$ and $\Xi(e_2^2)(v) = 2v_1e_2$, which are $-2e$, $h$, $2f$. The matrix of $\Xi$ in these bases is $\mathrm{diag}(-2, 1, 2)$, of determinant $-4$. (ii) The first two statements are (i). For the others, $x \otimes y \mapsto (xy,\ \omega(x, y))$ is a $G$-map $V \otimes V \to \mathrm{Sym}^2 V \oplus \det$, bijective when $2 \in R^\times$ (inverse $(xy, c) \mapsto \frac{1}{2}(x \otimes y + y \otimes x) + \frac{c}{2}(e_1 \otimes e_2 - e_2 \otimes e_1)$); and $V \otimes V^\vee = \mathrm{End}(V) = \mathrm{ad}^0 V \oplus \mathbf{1}$ by Proposition 4.52(i). (iii) $\mathrm{Sym}^2 V \simeq \mathrm{ad}^0 V \otimes \det = \mathrm{ad}^0 V(1)$ by (ii). By Proposition 4.4(iii), $V^\vee \simeq V \otimes \chi_\ell^{-1} = V(-1)$, so $\mathrm{Sym}^2(V^\vee) \simeq \mathrm{Sym}^2 V(-2) \simeq \mathrm{ad}^0 V(-1)$. Then $L_p(\mathrm{Sym}^2(V^\vee), s) = L_p(\mathrm{ad}^0 V(-1), s) = L_p(\mathrm{ad}^0 V, s - 1)$; the definitions of the local factors are in Chapter 19 (convention B-C7). □

*In the paper* 4.57. Section 1.1: "In the normalisation of the symmetric square, $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$" is (iii); the Introduction's comparison of Flach's theorem on $\mathrm{Sym}^2 f$ with the adjoint uses (ii). Over $\mathbb{Z}_2$ the map $\Xi$ is not an isomorphism, one of the places where $\ell$ odd is needed for lattice-level statements.

**Proposition 4.58** (The trace form)**.** *Let $T$ be free of rank $n$ over a commutative ring $R$, with an action of $G$.*

*(i)* $B(x,y) = \mathrm{tr}(xy)$ *on* $\mathrm{End}_R(T)$ *is $G$-invariant,* $B(gxg^{-1}, gyg^{-1}) = B(x,y)$*, and perfect:* $B(E_{ij}, E_{kl}) = \delta_{jk}\delta_{il}$*, a permutation Gram matrix, of determinant* $\pm 1$*.*

*(ii) For $n = 2$ the Gram matrix of $B$ on $\mathrm{ad}^0 T$ in the basis $e, h, f$ is* $\left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right)$*, of determinant $-2$, so $B$ is perfect on $\mathrm{ad}^0 T$ iff $2 \in R^\times$.*

*(iii) Let $R = \mathbb{Z}_\ell$ with $\ell$ odd and $A = \mathrm{ad}^0 T$. Then $x \mapsto B(x, -)$ is a $G$-isomorphism $A \simeq A^\vee = \mathrm{Hom}_{\mathbb{Z}_\ell}(A, \mathbb{Z}_\ell)$, and for every character $\chi\colon G \to \mathbb{Z}_\ell^\times$, $\mathrm{Hom}_{\mathbb{Z}_\ell}(A, \mathbb{Z}_\ell(\chi)) \simeq A(\chi)$; the same holds with $\mathbb{Q}_\ell$ or $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ in place of $\mathbb{Z}_\ell$. For $\ell = 2$ the image of $A$ in $A^\vee$ has index 2.*

*Proof.* (i) $\mathrm{tr}(gxg^{-1}gyg^{-1}) = \mathrm{tr}(gxyg^{-1}) = \mathrm{tr}(xy)$; and $E_{ij}E_{kl} = \delta_{jk}E_{il}$ has trace $\delta_{jk}\delta_{il}$. (ii) $e^2 = f^2 = 0$, $eh = -e$, $he = e$, $hf = -f$, so $\mathrm{tr}(e^2) = \mathrm{tr}(f^2) = \mathrm{tr}(eh) = \mathrm{tr}(hf) = 0$, while $\mathrm{tr}(ef) = \mathrm{tr}(E_{11}) = 1$ and $\mathrm{tr}(h^2) = 2$. A bilinear form on a free module of finite rank is perfect iff its Gram determinant is a unit (Chapter 3). (iii) By (ii) the map is an isomorphism; it is equivariant because $B(g \cdot x, y) = \mathrm{tr}(gxg^{-1}y) = \mathrm{tr}(x\, g^{-1}yg) = B(x, g^{-1} \cdot y)$ for the conjugation action, which says $B(gx, -) = g \cdot B(x, -)$ in $A^\vee$. Twisting by $\chi$ gives $\mathrm{Hom}(A, \mathbb{Z}_\ell(\chi)) = A^\vee(\chi) \simeq A(\chi)$; tensor with $\mathbb{Q}_\ell$ or $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ for the other coefficients. For $\ell = 2$ the cokernel of the Gram matrix has order $|\det|_2^{-1} = 2$. □

*In the paper* 4.59. Section 4.2: "$T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ (via the trace form)" is (iii) with $\chi = \chi_\ell$; it is one of the places where $\ell$ odd is used. Then $V^D = A(1) \otimes \mathbb{Q}_\ell$ and $W^D = \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, the module of Lemma lem:h0 and Lemma lem:dfg21 (Chapter 18).

**Proposition 4.60** (Invariance of the adjoint)**.** *Let $V$ be a representation of $G$ over a field and $\chi$ a character of $G$.*

*(i) Twisting does not change the adjoint:* $\mathrm{End}(V \otimes \chi)$ *is* $\mathrm{End}(V)$*, and* $\mathrm{ad}^0(V \otimes \chi)$ *is* $\mathrm{ad}^0 V$*, by Proposition 4.48(iii).*

*(ii) The transpose $x \mapsto x^t$, where $x^t(\lambda) = \lambda \circ x$, is a $G$-isomorphism* $\mathrm{End}(V) \to \mathrm{End}(V^\vee)$ *preserving traces, so it restricts to a $G$-isomorphism* $\mathrm{ad}^0 V \simeq \mathrm{ad}^0(V^\vee)$*; the map $x \mapsto -x^t$ has the same properties and also respects commutators.*

*(iii) Let $L \subseteq V$ be an $\mathcal{O}$-lattice in a $K$-vector space, and put* $\mathrm{ad}^0 L = \mathrm{End}_{\mathcal{O}}(L) \cap \mathrm{ad}^0 V$*. For $c \in K^\times$,* $\mathrm{End}_{\mathcal{O}}(cL) = \mathrm{End}_{\mathcal{O}}(L)$ *and* $\mathrm{ad}^0(cL) = \mathrm{ad}^0 L$ *inside* $\mathrm{End}_K(V)$*. For $L^\vee = \mathrm{Hom}_{\mathcal{O}}(L, \mathcal{O}) \subseteq V^\vee$, the transpose maps* $\mathrm{End}_{\mathcal{O}}(L)$ *onto* $\mathrm{End}_{\mathcal{O}}(L^\vee)$ *and* $\mathrm{ad}^0 L$ *onto* $\mathrm{ad}^0(L^\vee)$*.*

*(iv) Consequently, if $L'$ is a stable lattice homothetic to $L$, to $L^\vee$, or to a twist $L \otimes \mathcal{O}(\chi)$, then* $\mathrm{ad}^0 L'$ *is canonically isomorphic to* $\mathrm{ad}^0 L$ *as a representation.*

*Proof.* (ii) On $V^\vee$, $g$ acts by $\rho^\vee(g) = \rho(g^{-1})^t$. Then

$$(\rho(g)x\rho(g)^{-1})^t = \rho(g)^{-t}x^t\rho(g)^t = \rho^\vee(g)x^t\rho^\vee(g)^{-1},$$

and $\mathrm{tr}(x^t) = \mathrm{tr}\, x$. Also $[x,y]^t = -[x^t, y^t]$, so $-x^t$ respects commutators. (iii) $x(cL) \subseteq cL$ iff $xL \subseteq L$. A $K$-linear $x$ preserves $L$ iff $x^t$ preserves $L^\vee$, because $L = (L^\vee)^\vee$ (both are free and dual bases correspond). (iv) Combine (i) to (iii); a twist does not change the underlying module. □

*In the paper* 4.61. Section 4.2: the lattice $M_{f,\lambda}$ of DFG04 is identified by $\varphi_E^*$ with $H^1(E, \mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$ (Proposition prop:pullback; convention B-C5(b)), "so its adjoint is $\mathrm{ad}^0 T$": this is (ii) and (i), with no irreducibility needed. When $\bar\rho$ is irreducible there is a second route: $M_{f,\lambda}$ is homothetic to the dual $T_\ell E^\vee$ (Theorem 4.77), and then (iv) identifies the adjoints. Theorem thm:quant, Step 3 ("$A_g = A_f$" for a twist) is (i).

## 4.8 $\mathrm{GL}_2(\mathbb{F}_3)$ and its subgroups on $\mathbb{F}_3^2$ and $\mathbb{F}_9^2$

This section applies the theory to the standard representations of the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ that occur as mod 3 images. It is the representation-theoretic core of the paper's Proposition prop:ns: for the three possible images $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, $N_s(\mathbb{F}_3)$ of an irreducible $\bar\rho_{E,3}$, the restriction to the determinant-one subgroup (the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$) is absolutely irreducible except for $N_s(\mathbb{F}_3)$, where it is $\psi \oplus \psi^{-1}$ over $\mathbb{F}_9$ and the representation is isomorphic to its twist by det (Theorem 4.66). The group theory (subgroups, orders, orbits on lines, the three images) is Chapter 1's; only the representation-theoretic reading is added here. Throughout, $\rho$ is the standard representation (the inclusion $H \subseteq \mathrm{GL}_2(\mathbb{F}_3)$), and since det takes values in $\{\pm 1\}$, $\rho \otimes \det$ is the twist by the character $\det|_H$.

**Proposition 4.62** (The standard representations of the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$)**.** *For the standard models, the standard representation $\rho$ of $H$ on $\mathbb{F}_3^2$ has the following properties. In the first table, "irr." means irreducible over $\mathbb{F}_3$, "abs." absolutely irreducible, and the last column is the image of $H$ in $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$.*

| $H$ | *order* | det *onto* $\mathbb{F}_3^\times$ | *irr.* | *abs.* | # *of* $H/(H \cap \{\pm 1\})$ | *projective image* |
|---|---|---|---|---|---|---|
| $\mathrm{GL}_2(\mathbb{F}_3)$ | 48 | *yes* | *yes* | *yes* | 24 | $S_4$ |
| $\mathrm{SL}_2(\mathbb{F}_3)$ | 24 | *no* | *yes* | *yes* | 12 | $A_4$ |
| $N_{ns}(\mathbb{F}_3)$ | 16 | *yes* | *yes* | *yes* | 8 | $D_4$ |
| $Q_8$ | 8 | *no* | *yes* | *yes* | 4 | $V_4$ *(normal in* $S_4$*)* |
| $N_s(\mathbb{F}_3)$ | 8 | *yes* | *yes* | *yes* | 4 | $V_4$ *(not normal)* |
| $C_{ns}(\mathbb{F}_3)$ | 8 | *yes* | *yes* | *no* | 4 | $C_4$ |
| $C_4 = \langle J \rangle$ | 4 | *no* | *yes* | *no* | 2 | $C_2$ |
| $C_s(\mathbb{F}_3)$ | 4 | *yes* | *no* | *no* | 2 | $C_2$ |
| $\mathrm{B}(\mathbb{F}_3)$ | 12 | *yes* | *no* | *no* | 6 | $S_3$ |
| U | 3 | *no* | *no* | *no* | 3 | $C_3$ |

*In the second table,* Hom *is* $\mathrm{Hom}_H(\rho, \rho \otimes \det)$*, with the dimension of its trace-zero part in parentheses, and the last column says whether* $\rho \simeq \rho \otimes \det$ *("triv." when* $\det|_H = 1$*, so that the question is empty).*

| $H$ | $\dim \mathrm{End}_H(\rho)$ | $\dim(\mathrm{ad}^0\,\rho)^H$ | $\dim \mathrm{Hom}$ | $\rho \simeq \rho \otimes \det$ | *trace-zero part* |
|---|---|---|---|---|---|
| $\mathrm{GL}_2(\mathbb{F}_3)$ | 1 | 0 | 0 (0) | *no* | |
| $\mathrm{SL}_2(\mathbb{F}_3)$ | 1 | 0 | 1 (0) | *triv.* | |
| $N_{ns}(\mathbb{F}_3)$ | 1 | 0 | 0 (0) | *no* | |
| $Q_8$ | 1 | 0 | 1 (0) | *triv.* | |
| $N_s(\mathbb{F}_3)$ | 1 | 0 | 1 (1) | ***yes*** | $J$ |
| $C_{ns}(\mathbb{F}_3)$ | 2 | 1 | 0 (0) | *no* | |
| $C_4$ | 2 | 1 | 2 (1) | *triv.* | $J$ |
| $C_s(\mathbb{F}_3)$ | 2 | 1 | 2 (2) | *yes* | $w$, $J = \sigma w$ |
| $\mathrm{B}(\mathbb{F}_3)$ | 1 | 0 | 1 (1) | *no* | $e$ *(not invertible)* |
| U | 2 | 1 | 2 (1) | *triv.* | $e = u - 1$ |

*Proof.* Orders, surjectivity of det and the stable lines over $\mathbb{F}_3$ (hence the column "irr.") are in Chapter 1. Absolute irreducibility is Proposition 4.26(iv). The order of $H/(H \cap \{\pm 1\})$ is $\#H/2$ when $-1 \in H$, which is the case for all rows but U; the identifications $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$, $\mathrm{PSL}_2(\mathbb{F}_3) \simeq A_4$ and of the smaller images are in Chapter 1 (a non-abelian group of order 6 is $S_3$; $N_{ns}(\mathbb{F}_3)/\{\pm 1\}$ is non-abelian of order 8 with a cyclic subgroup $C_{ns}/\{\pm 1\}$ of order 4 and the involution $\sigma$ outside it, so it is $D_4$). For the second table: in the absolutely irreducible rows $\mathrm{End}_H = \mathbb{F}_3$ (Theorem 4.10(iii)); for the cyclic groups $C_{ns} = \langle a \rangle$, $C_4 = \langle J \rangle$, $\mathrm{U} = \langle u \rangle$ the commutant of the non-scalar generator is 2-dimensional (Lemma 4.70); for $C_s$ it is the diagonal matrices (the commutant of $\sigma$); for B it is $\mathbb{F}_3$ (Example 4.20(d)). As $2 \in \mathbb{F}_3^\times$, $\dim(\mathrm{ad}^0\,\rho)^H = \dim \mathrm{End}_H - 1$ (Proposition 4.52(i)). When $\det|_H = 1$, $\mathrm{Hom} = \mathrm{End}_H$ and its trace-zero part is $(\mathrm{ad}^0\,\rho)^H$; for the cyclic groups $C_4$ and U the commutant is $\{\alpha + \beta g\}$ and its trace-zero part is spanned by $2g - \mathrm{tr}\, g$ (Lemma 4.70(ii)), which is $2J = -J$ for $g = J$ and $2u - 2 = -(u-1)$ for $g = u$, giving the last column. When $\det|_H \neq 1$, every element of Hom has trace 0 (Lemma 4.49). For $\mathrm{GL}_2(\mathbb{F}_3)$ and $N_{ns}(\mathbb{F}_3)$, $\mathrm{Hom} = 0$ by Proposition 4.65 and Proposition 4.64; for $N_s(\mathbb{F}_3)$ it is $\mathbb{F}_3 J$ by Proposition 4.63. For $C_{ns}$: if $\phi a = -a\phi$ with $\phi \neq 0$, then $\ker \phi$ is $a$-stable, hence 0 ($C_{ns}$ is irreducible), so $\phi$ is invertible and $\phi a \phi^{-1} = -a$: $a$ and $-a$ would be conjugate, hence have the same eigenvalues. But the characteristic polynomial of $a$ is $X^2 - 2X + 2$, with roots $1 \pm i$ in $\mathbb{F}_9$, while that of $-a$ is $X^2 + 2X + 2$, with roots $-1 \pm i$. For $C_s$ and B: $\phi$ must anticommute with $\sigma = \mathrm{diag}(1, -1)$ (of determinant $-1$), so $\phi = \left(\begin{smallmatrix} 0 & x \\ y & 0 \end{smallmatrix}\right)$, and the other generator $\mathrm{diag}(-1, 1)$ of $C_s$ imposes nothing new; in B, commuting with $u$ forces $y = 0$. Finally $\rho \simeq \rho \otimes \det$ iff Hom contains an invertible element. Every entry was also checked by enumerating $\mathrm{GL}_2(\mathbb{F}_3)$, for instance with the following PARI/GP script, which prints, for each group, its order, $\dim \mathrm{End}_H$, $\dim(\mathrm{ad}^0\,\rho)^H$, $\dim \mathrm{Hom}$ and the dimension of its trace-zero part. □

```
red(M) = lift(Mod(M, 3));
gen(L) = my(S = Set([matid(2)]), T);\
  while(1, T = S; foreach(T, x, foreach(L, g, S = setunion(S, [red(x*g)])));\
    if(#S == #T, return(S)));
mats = concat(vector(81, k, my(d = digits(k-1+81, 3)); [[d[2],d[3];d[4],d[5]]]));
sols(H, tw, t0) = #select(X -> (!t0 || trace(X)%3 == 0) &&\
  #select(g -> red(X*g - matdet(g)^tw*g*X) != 0, H) == 0, mats);
row(H) = [#H, logint(sols(H,0,0),3), logint(sols(H,0,1),3),\
  logint(sols(H,1,0),3), logint(sols(H,1,1),3)];
J = [0,1;-1,0]; s = [1,0;0,-1]; a = [1,-1;1,1]; u = [1,1;0,1];
{
```

```
  G = [["Ns",[J,s]], ["Nns",[a,s]], ["Q8",[J,[1,-1;-1,-1]]], ["C4",[J]],
       ["Cns",[a]], ["U",[u]], ["B",[u,s,-matid(2)]]];
  foreach(G, x, print(x[1], ": ", row(gen(x[2]))));
}
\\ Ns: [8, 1, 0, 1, 1]    Nns: [16, 1, 0, 0, 0]   Q8: [8, 1, 0, 1, 0]
\\ C4: [4, 2, 1, 2, 1]    Cns: [8, 2, 1, 0, 0]    U: [3, 2, 1, 2, 1]
\\ B: [12, 1, 0, 1, 1]
```

**Proposition 4.63** ($N_s(\mathbb{F}_3)$: induced over $\mathbb{F}_9$, not over $\mathbb{F}_3$)**.** *Let $\rho$ be the standard representation of $N_s(\mathbb{F}_3) = \langle J, \sigma\rangle$ and $C_4 = \langle J\rangle = N_s(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3)$, the kernel of* det.

*(i) $\rho$ is absolutely irreducible, but $\mathrm{Res}_{C_4}\rho \otimes \mathbb{F}_9 \simeq \psi \oplus \psi^{-1}$ with $\psi(J) = i$, the eigenvectors being $v = (1, i)^t$ and $\sigma v = (1, -i)^t$. Since $\sigma J \sigma^{-1} = J^{-1}$, $\psi^\sigma = \psi^{-1} \neq \psi$.*

*(ii) In the basis $(v, \sigma v)$ of $\mathbb{F}_9^2$, $\rho(J) = \mathrm{diag}(i, -i)$ and $\rho(\sigma) = w$. This is the matrix form of $\mathrm{Ind}_{C_4}^{N_s}\psi$ of Proposition 4.39(iii) (with $s = \sigma$, $\sigma^2 = 1$), so $\rho \otimes \mathbb{F}_9 \simeq \mathrm{Ind}_{C_4}^{N_s}\psi$. Over $\mathbb{F}_3$, $\rho$ is not induced from a character of $C_4$: $\mathrm{Res}_{C_4}\rho$ is irreducible over $\mathbb{F}_3$.*

*(iii) $\rho \simeq \rho \otimes \det$ over $\mathbb{F}_3$, with intertwiner $J$: $J\rho(g) = \det(g)\rho(g)J$ for all $g$. Moreover $\mathrm{tr}\, J = 0$, $J^2 = -1$, and the eigenlines of $J$ over $\mathbb{F}_9$ are the two summands of (i), as Theorem 4.44(B) predicts.*

*(iv) The projective image is $V_4$, and $\rho \simeq \rho \otimes \chi$ for each of the three nontrivial characters $\chi$ of $N_s(\mathbb{F}_3)/\{\pm 1\}$: their kernels are $C_4$ ($\chi = \det$, intertwiner $J$), $C_s(\mathbb{F}_3) = \{\pm 1, \pm\sigma\}$ (intertwiner $\sigma$) and $\{\pm 1, \pm w\}$ (intertwiner $w$). So $\rho \otimes \mathbb{F}_9$ is induced from each of these three subgroups.*

*(v) $(\mathrm{ad}^0\rho \otimes \det)^{N_s} = \mathbb{F}_3 J$, a line, and $(\mathrm{ad}^0\rho)^{N_s} = 0$.*

*Proof.* (i) Absolute irreducibility is Proposition 4.26(iv), and the decomposition over $\mathbb{F}_9$ is Example 4.14; $\sigma v = (1, -i)^t$ is the $(-i)$-eigenvector. $\sigma J \sigma^{-1} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = J^{-1}$, so $\psi^\sigma(J) = \psi(J^{-1}) = -i$. (ii) $Jv = iv$ and $J(\sigma v) = -i\,\sigma v$; $\sigma(v) = \sigma v$ and $\sigma(\sigma v) = v$. The matrix of Proposition 4.39(iii) for $\psi$ and $s = \sigma$ is $J \mapsto \mathrm{diag}(\psi(J), \psi^\sigma(J)) = \mathrm{diag}(i, -i)$ and $\sigma \mapsto \left(\begin{smallmatrix} 0 & \psi(1) \\ 1 & 0 \end{smallmatrix}\right) = w$. Over $\mathbb{F}_3$, $\mathrm{Res}_{C_4}\rho$ is irreducible (Example 4.14), whereas the restriction of a representation induced from a character of $C_4$ would contain that character as a line. (iii) It suffices to check the generators: $\det J = 1$ and $J$ commutes with $J$; $\det\sigma = -1$ and $J\sigma = \left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right) = -\sigma J$. And $\mathrm{tr}\, J = 0$, $J^2 = -1$, with eigenlines $\mathbb{F}_9 v$, $\mathbb{F}_9 \sigma v$. (iv) $N_s(\mathbb{F}_3)/\{\pm 1\}$ has order 4 and its three involutions $\bar{J}, \bar{\sigma}, \bar{w}$ commute, so it is $V_4$. The relations $wJ = -Jw$, $\sigma J = -J\sigma$, $w\sigma = -\sigma w$ show that $x \in \{J, \sigma, w\}$ commutes with $\pm 1$ and $\pm x$ and anticommutes with the other four elements, so $x\rho(g) = \chi(g)\rho(g)x$ for the character $\chi$ with kernel $\{\pm 1, \pm x\}$. Induction from each kernel is Theorem 4.44 (or Proposition 4.46(iii)). (v) is Proposition 4.54(iii), (iv) with $\chi = \det \neq 1$ and (iii). □

**Proposition 4.64** ($N_{ns}(\mathbb{F}_3)$)**.** *Let $\rho$ be the standard representation of $N_{ns}(\mathbb{F}_3) = \langle a, \sigma\rangle$, with $Q_8 = N_{ns}(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3)$ the kernel of* det.

*(i) $\rho$ and $\mathrm{Res}_{Q_8}\rho$ are absolutely irreducible.*

*(ii) $\rho \not\simeq \rho \otimes \det$, and $(\mathrm{ad}^0\rho \otimes \det)^{N_{ns}} = 0$.*

*(iii) Over $\mathbb{F}_9$, $C_{ns}(\mathbb{F}_3) = \langle a \rangle$ acts on the lines spanned by $(1,-i)^t$ and $(1,i)^t$ by the characters $\theta$ and $\theta^3$, where $\theta(a) = 1+i$ has order 8; $\theta^\sigma = \theta^3 \neq \theta$, and $\rho \otimes \mathbb{F}_9 \simeq \operatorname{Ind}_{C_{ns}}^{N_{ns}} \theta$. The projective image is $D_4$, and $\rho \simeq \rho \otimes \chi$ for exactly one nontrivial character $\chi$, the character $\chi_{ns}$ with kernel $C_{ns}(\mathbb{F}_3)$, with intertwiner $J$.*

*Proof.* (i) is Proposition 4.26(iv). (ii) If $\phi$ were an isomorphism, $\phi\rho(g)\phi^{-1} = \det(g)\rho(g)$ would give $\operatorname{tr}\rho(g) = \det(g)\operatorname{tr}\rho(g)$, so $\operatorname{tr}\rho(g) = 0$ whenever $\det g = -1$ (as $2 \neq 0$). But $\det a = 2 = -1$ and $\operatorname{tr} a = 2 \neq 0$. The invariants vanish by Proposition 4.54(iv). (iii) $a\binom{1}{-i} = \binom{1+i}{1-i} = (1+i)\binom{1}{-i}$, since $(1+i)(-i) = 1-i$; and $a\binom{1}{i} = (1-i)\binom{1}{i}$, with $1-i = (1+i)^3$. The element $1+i$ has order 8 in $\mathbb{F}_9^\times$ ($(1+i)^2 = 2i = -i$). From $\sigma a \sigma^{-1} = a^3$ (Chapter 1), $\theta^\sigma(a) = \theta(a^3) = \theta(a)^3$, and $\theta^3 \neq \theta$. So $\operatorname{Res}_{C_{ns}} \rho \otimes \mathbb{F}_9 \simeq \theta \oplus \theta^3$ is reducible, case (B) of Theorem 4.44 for $H = C_{ns}$, and $\rho \otimes \mathbb{F}_9 \simeq \operatorname{Ind}\theta$, $\rho \simeq \rho \otimes \chi_{ns}$. Directly, $J = a^2$ commutes with $a$ and anticommutes with $\sigma$, so it is an intertwiner $\rho \to \rho \otimes \chi_{ns}$. The projective image $N_{ns}(\mathbb{F}_3)/\{\pm 1\}$ is $D_4$ (Proposition 4.62), of order 8, so by Proposition 4.46(iii) there is exactly one such character. □

**Proposition 4.65** ($\mathrm{SL}_2(\mathbb{F}_3)$ and $\mathrm{GL}_2(\mathbb{F}_3)$)**.** *The standard representations of $\mathrm{SL}_2(\mathbb{F}_3)$ and $\mathrm{GL}_2(\mathbb{F}_3)$ are absolutely irreducible. For $\mathrm{GL}_2(\mathbb{F}_3)$, $\rho \not\simeq \rho \otimes \det$ and $(\mathrm{ad}^0\rho \otimes \det)^{\mathrm{GL}_2(\mathbb{F}_3)} = 0$. The projective images are $A_4$ and $S_4$, exceptional in Dickson's list (Theorem 4.84), so neither representation is induced from a subgroup of index 2.*

*Proof.* Absolute irreducibility: both contain $Q_8$ (Proposition 4.26(iii), (iv)). For $\mathrm{GL}_2(\mathbb{F}_3)$ argue as in Proposition 4.64(ii) with $a \in \mathrm{GL}_2(\mathbb{F}_3)$ (or with $\left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)$, of determinant $-1$ and trace 1). Neither $A_4$ nor $S_4$ is dihedral, so Proposition 4.46(ii) applies. □

**Theorem 4.66** (The three images)**.** *Let $H \leq \mathrm{GL}_2(\mathbb{F}_3)$ be conjugate to $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$ (by Chapter 1 these are exactly the subgroups with surjective determinant, no stable line in $\mathbb{F}_3^2$, and an element of determinant $-1$ and trace 0). Let $H_1 = H \cap \mathrm{SL}_2(\mathbb{F}_3)$, the kernel of $\det|_H$, of index 2, and $\rho$ the standard representation. The following are equivalent:*

*(a) $H$ is conjugate to $N_s(\mathbb{F}_3)$;*

*(b) $\operatorname{Res}_{H_1}\rho$ is not absolutely irreducible;*

*(c) $\rho \simeq \rho \otimes \det$;*

*(d) $(\mathrm{ad}^0\rho \otimes \det)^H \neq 0$;*

*(e) $\rho \otimes \mathbb{F}_9 \simeq \operatorname{Ind}_{H_1}^H \psi$ for a character $\psi$ of $H_1$ of order 4;*

*(f) $H_1$ is abelian.*

*When they hold, $(\mathrm{ad}^0\rho \otimes \det)^H$ is the line spanned by a trace-zero isomorphism $\rho \to \rho \otimes \det$; when they fail, $(\mathrm{ad}^0\rho \otimes \det)^H = 0$.*

*Proof.* All statements are invariant under conjugation, so let $H$ be one of the three models; then $H_1$ is $\mathrm{SL}_2(\mathbb{F}_3)$, $Q_8$ or $C_4$ (Chapter 1). (a)⇔(f): of these three only $C_4$ is abelian. (a)⇔(b): $\mathrm{SL}_2(\mathbb{F}_3)$ and $Q_8$ act absolutely irreducibly and $C_4$ does not (Proposition 4.26(iv)). (b)⇔(c): Theorem 4.44 with $\chi_H = \det|_H$ (its kernel is $H_1$, of index 2 because det is onto $\{\pm 1\}$), $\operatorname{char}\mathbb{F}_3 \neq 2$ and $\rho$ absolutely irreducible. (c)⇔(d) and the description of the invariants: Proposition 4.54(iii), (iv). (a)⇒(e): Proposition 4.63(ii), $\psi(J) = i$ of order 4. (e)⇒(b): $\operatorname{Res}_{H_1}\operatorname{Ind}_{H_1}^H \psi \simeq \psi \oplus \psi^s$ is reducible over $\mathbb{F}_9$. □

*In the paper* 4.67. Proposition prop:ns and its proof. As $\det \bar\rho_{E,3} = \bar\chi_3 = \chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$ (Chapter 5), the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ is $H_1$, and Theorem 4.66 gives at once: $\bar\rho_{E,3}$ is absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ unless the image is $N_s(\mathbb{F}_3)$; in that case $\bar\rho \otimes \mathbb{F}_9$ is induced from a character of order 4, $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ and $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho(1)) \neq 0$. The value $h_3(E) = 1$ then comes from Lemma lem:h0 (Proposition 4.81). This is also the reading of the hypothesis of Theorem thm:main at $\ell = 3$ (Proposition prop:Sf), and of Remark rem:ns and Section sec:anomaly, which concern exactly the image 3Ns. That $N_s(\mathbb{F}_3)$ is the only one of the three images whose determinant-one subgroup is abelian is the orbit dichotomy of Chapter 1 in another form.

## 4.9 Unipotent elements and cyclic groups of order 2, 3, 4, 6

This section computes the fixed vectors of the groups that occur as images of inertia at a bad prime $q \neq \ell$: unipotent groups and "minus unipotent" ones (potentially multiplicative reduction), and cyclic groups of order 2, 3, 4, 6 in $\mathrm{SL}_2$ (potentially good reduction), on $V$, on $\mathrm{ad}^0 V$, and modulo 3. In the paper these computations are Lemma lem:IV, Remark rem:types, Lemma lem:euler, the table of Section 1 and Proposition prop:ns-structure. Which shape occurs for which Kodaira type is a theorem about elliptic curves (Serre-Tate, Kraus) proved in Chapter 10 and Chapter 11; here it is only recorded in the table of Proposition 4.71.

**Convention 4.68.** In this section $\sigma$ denotes an element of order 3 of $\mathrm{SL}_2$, as in the paper's Lemmas lem:IV and lem:euler, and $\tau$ an element of order 4; the matrix $\mathrm{diag}(1,-1)$, called $\sigma$ in Chapter 1 and in Section 4.8, is written $h$ here (it is the basis vector $h$ of $\mathrm{ad}^0 V$ viewed as a matrix). A Frobenius element is an abstract $F$ satisfying $FgF^{-1} = g^q$ on the inertia image (convention (C8)).

**Proposition 4.69** (Unipotent elements)**.** *Let $k$ be a field, $V = k^2$, $u \in \mathrm{GL}_2(k)$ unipotent (characteristic polynomial $(X-1)^2$) with $u \neq 1$, $\nu = u - 1$, and $\xi \in k^\times$, $\xi \neq 1$.*

*(i) $\nu \neq 0$ and $\nu^2 = 0$; in a basis $e_1 \in \ker\nu$, $e_2 \notin \ker\nu$ one has $u = \left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$ with $n' \neq 0$. $V^u = \ker\nu = \mathrm{im}\,\nu$ is a line, and $V^{\xi u} = 0$. In particular "minus a unipotent" ($\xi = -1$, $\mathrm{char}\,k \neq 2$) fixes no nonzero vector.*

*(ii) If $\mathrm{char}\,k \neq 2$, then $(\mathrm{ad}^0 V)^u = (\mathrm{ad}^0 V)^{\xi u} = k\nu = k\,e$, a line. If $\mathrm{char}\,k = 2$, $(\mathrm{ad}^0 V)^u = ke + kh$. The same holds for any group $H$ acting through matrices $\xi' u^m$ ($\xi'$ scalar, $m \in \mathbb{Z}$) one of which has $u^m \neq 1$.*

*(iii) If $F \in \mathrm{GL}_2(k)$ satisfies $FuF^{-1} = u^q$ with $q \in \mathbb{Z}$, then $F$ preserves $(\mathrm{ad}^0 V)^u$ and acts on it by $q$: $F\nu F^{-1} = u^q - 1 = q\nu$. Hence $F^{-1}$ acts by $q^{-1}$ when $q \in k^\times$. The same holds for $q \in \mathbb{Z}_\ell$ when $k \supseteq \mathbb{Q}_\ell$ and $u^q$ is defined by continuity, $u^q = 1 + q\nu$.*

*Proof.* (i) By Cayley-Hamilton $\nu^2 = 0$, and $\nu \neq 0$ as $u \neq 1$. Then $\mathrm{im}\,\nu \subseteq \ker\nu$, both are lines, so they are equal. For $e_2 \notin \ker\nu$, $\nu e_2 \in \ker\nu \setminus \{0\}$, say $\nu e_2 = n' e_1$. A vector fixed by $\xi u$ would be an eigenvector of $u$ with eigenvalue $\xi^{-1} \neq 1$, but 1 is the only eigenvalue of $u$. (ii) Scalars act trivially on $\mathrm{ad}^0 V$, so $(\mathrm{ad}^0 V)^{\xi u} = (\mathrm{ad}^0 V)^u$. By Proposition 4.52(iii) with $t = u$, for $x = \alpha e + \beta h + \gamma f$, $(u-1)x = (-2n'\beta - n'^2\gamma)e + n'\gamma h$, which vanishes iff $\gamma = 0$ and $2\beta = 0$. For $H$: $u^m = 1 + m\nu$, and if $u^m \neq 1$ it is again a nontrivial unipotent with $(\mathrm{ad}^0 V)^{u^m} = ke$; every element of $H$ fixes $e$. (iii) $F\nu F^{-1} = FuF^{-1} - 1 = u^q - 1$, and $u^q = (1+\nu)^q = 1 + q\nu$ for every $q \in \mathbb{Z}$ (binomial theorem with $\nu^2 = 0$, and $(1+\nu)^{-1} = 1 - \nu$). By continuity the same holds for $q \in \mathbb{Z}_\ell$. □

**Lemma 4.70** (The commutant of one matrix)**.** *Let $x \in \mathrm{End}_k(V)$, $n = \dim V$.*

*(i) If the characteristic polynomial of $x$ has $n$ distinct roots in $\bar{k}$, the commutant $\{y : xy = yx\}$ is $k[x] \simeq k[X]/(\det(X - x))$, of dimension $n$.*

*(ii) If $n = 2$ and $x$ is not a scalar, the commutant is $k \cdot 1 \oplus k \cdot x = k[x]$; if moreover $\mathrm{char}\, k \neq 2$, its trace-zero part is the line spanned by $2x - \mathrm{tr}\, x$.*

*(iii) In (i) and (ii) the commutant is a field iff $\det(X - x)$ is irreducible over $k$.*

*Examples: if $\sigma^2+\sigma+1 = 0$ ($\sigma$ of order 3 in $\mathrm{SL}_2$), the trace-zero line is spanned by $2\sigma+1 = \sigma-\sigma^{-1}$, and $(2\sigma+1)^2 = -3$; if $\tau^2 = -1$, it is $k\tau$. For $k = \mathbb{Q}_\ell$ and $\sigma$ of order 3 in $\mathrm{SL}_2(\mathbb{Q}_\ell)$ the commutant $\mathbb{Q}_\ell[\sigma] \simeq \mathbb{Q}_\ell[X]/(X^2 + X + 1)$ is a field iff $\ell \not\equiv 1 \pmod 3$, and for $\tau$ of order 4, $\mathbb{Q}_\ell[\tau]$ is a field iff $\ell \not\equiv 1 \pmod 4$ (the roots of unity of $\mathbb{Q}_\ell$, Chapter 7).*

*Proof.* (i) The minimal polynomial of $x$ vanishes at all $n$ eigenvalues, so it has degree $n$ and equals $\det(X - x)$; hence $k[x] \simeq k[X]/(\det(X - x))$ has dimension $n$, and $k[x]$ lies in the commutant. Over $\bar{k}$, $x$ is diagonal with distinct entries in a suitable basis, and the matrices commuting with it are the diagonal ones, of dimension $n$. The commutant is the solution space of a linear system with coefficients in $k$, so its dimension over $k$ is also $n$ (as in Proposition 4.7(i)); it therefore equals $k[x]$. (ii) Choose $v$ that is not an eigenvector of $x$ (the eigenvectors lie in at most two lines, or in one line if $x$ has a single eigenvalue and is not scalar), while $k^2$ contains at least the three lines spanned by $(1,0)^t$, $(0,1)^t$, $(1,1)^t$, so such a $v$ exists over every field. Then $v, xv$ is a basis (if $xv$ were a multiple of $v$, $v$ would be an eigenvector). If $y$ commutes with $x$, write $yv = \alpha v + \beta xv$; then $y$ and $\alpha + \beta x$ agree on $v$ and on $xv$ ($y\,xv = x\,yv = (\alpha + \beta x)xv$), so $y = \alpha + \beta x$. The trace of $\alpha + \beta x$ is $2\alpha + \beta\,\mathrm{tr}\, x$, which vanishes iff $\alpha + \beta x = \frac{\beta}{2}(2x - \mathrm{tr}\, x)$. (iii) $k[X]/(P)$ is a field iff $P$ is irreducible. For the examples: $\mathrm{tr}\,\sigma = -1$, so $2\sigma - \mathrm{tr}\,\sigma = 2\sigma + 1$, $\sigma^{-1} = \sigma^2 = -1 - \sigma$, and $(2\sigma + 1)^2 = 4(\sigma^2 + \sigma) + 1 = -3$; $\mathrm{tr}\,\tau = 0$. □

**Proposition 4.71** (Cyclic groups of order $2, 3, 4, 6$)**.** *Let $K$ be a field, $V = K^2$, and $g \in \mathrm{SL}_2(K)$ of finite order $m \in \{2, 3, 4, 6\}$ with $\mathrm{char}\, K \nmid m$.*

*(i) If $m = 2$ then $g = -1$. If $m \geq 3$, $g$ has the distinct eigenvalues $\zeta, \zeta^{-1}$ with $\zeta$ a primitive $m$-th root of unity, and characteristic polynomial $\Phi_m$: $X^2 + X + 1$, $X^2 + 1$, $X^2 - X + 1$ for $m = 3, 4, 6$. For $m = 6$, $\sigma = -g$ has order 3.*

*(ii) $V^g = 0$. If $m = 2$, $(\mathrm{ad}^0 V)^g = \mathrm{ad}^0 V$. If $m \geq 3$, $(\mathrm{ad}^0 V)^g$ is the line spanned by $g - g^{-1}$, which is $2g + 1$ for $m = 3$, $2g$ for $m = 4$, and $2g - 1 = -(2\sigma + 1)$ for $m = 6$.*

*(iii) Let $F \in \mathrm{GL}_2(K)$ with $FgF^{-1} = g^r$, $r \in \mathbb{Z}$ prime to $m$. If $m \geq 3$, then $F$ preserves $(\mathrm{ad}^0 V)^g$ and acts on it by $+1$ if $r \equiv 1 \pmod m$ and by $-1$ if $r \equiv -1 \pmod m$, and every $r$ prime to $m$ is $\equiv \pm 1 \pmod m$. For $r = q$ a prime not dividing $m$ the sign is $\chi_{-3}(q)$ if $m \in \{3, 6\}$ and $\chi_{-4}(q)$ if $m = 4$; it is the same for $F^{-1}$.*

*Proof.* (i) $X^m - 1$ is separable since $\mathrm{char}\, K \nmid m$, so $g$ is diagonalisable over $\bar{K}$ with eigenvalues $m$-th roots of unity of product $\det g = 1$. If the two eigenvalues are equal, to $\zeta$ say, then $\zeta^2 = 1$ and $g = \zeta = \pm 1$, of order at most 2; this is the case $m = 2$, $g = -1$. Otherwise they are $\zeta \neq \zeta^{-1}$, the order of $g$ is that of $\zeta$, so $\zeta$ is a primitive $m$-th root, and $(X - \zeta)(X - \zeta^{-1}) = \Phi_m$ because $\Phi_m$ has degree 2 for these $m$. For $m = 6$, $g^3$ has the eigenvalues $\zeta^{\pm 3} = -1$, so $g^3 = -1$ and $(-g)^3 = 1$, with $-g \neq 1$. (ii) 1 is not an eigenvalue. Conjugation by $-1$ is trivial. For $m \geq 3$,

Lemma 4.70(ii) gives the line $K(2g - \operatorname{tr} g)$; and $g - g^{-1}$ has trace 0 and lies in $K[g]$, and is nonzero since $g^2 \neq 1$. The three expressions follow from $g^{-1} = -1 - g$, $-g$, $1 - g$ respectively. (iii) $F$ normalises $\langle g \rangle$, so it preserves the invariants, and $F(g - g^{-1})F^{-1} = g^r - g^{-r}$. The units modulo 3, 4, 6 are $\pm 1$, and $g^{\pm 1} - g^{\mp 1} = \pm(g - g^{-1})$. Finally $q \equiv \chi_{-3}(q) \pmod 3$ for a prime $q \neq 3$ (including $q = 2$, where both sides are $-1$ modulo 3) and $q \equiv \chi_{-4}(q) \pmod 4$ for an odd prime $q$ (Chapter 2); and for $m = 6$ a prime $q \geq 5$ is $\equiv \pm 1 \pmod 6$ with the sign of $q$ mod 3. $F^{-1}$ acts by the inverse sign, which is the same. □

The following table collects the fixed spaces for the shapes of the image of inertia $I = I_q$ at a prime $q \neq \ell$, $q \geq 5$, on $V = V_\ell E$ (dimension 2 over $\mathbb{Q}_\ell$). The first column is Chapter 10 and Chapter 11's (Serre-Tate, Kraus), the other columns are Proposition 4.69, Proposition 4.71 and Proposition 4.72. In the unipotent rows the image of $I$ is infinite, topologically generated, up to the finite part, by $t = \left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$; $\xi$ is a ramified quadratic character. The column "$F$" gives the action of an arithmetic Frobenius $F$ ($FgF^{-1} = g^q$) on $(\mathrm{ad}^0 V)^I$; the geometric Frobenius $F^{-1}$ gives the inverse, which matters only in the unipotent rows ($q$ against $q^{-1}$; convention B-C3(d)). The last column is $\dim(T/3T)^I$ for $T = T_3 E$.

| image of $I$ | types | $\dim V^I$ | $\dim(\mathrm{ad}^0 V)^I$ | generator | $F$ on $(\mathrm{ad}^0 V)^I$ | $\dim(T/3T)^I$ |
|---|---|---|---|---|---|---|
| trivial | $\mathrm{I}_0$ | 2 | 3 | $\mathrm{ad}^0 V$ | $\alpha/\beta, 1, \beta/\alpha$ | 2 |
| $\{\pm 1\}$ | $\mathrm{I}_0^*$ | 0 | 3 | $\mathrm{ad}^0 V$ | as for the twist | 0 |
| unipotent, $t \neq 1$ | $\mathrm{I}_n$ | 1 | 1 | $e$ | $q$ | 1 or 2 |
| $\xi$·unipotent | $\mathrm{I}_n^*$ | 0 | 1 | $e$ | $q$ | 0 |
| order 3 | IV, IV$^*$ | 0 | 1 | $2\sigma + 1$ | $\chi_{-3}(q)$ | 1 |
| order 4 | III, III$^*$ | 0 | 1 | $\tau$ | $\chi_{-4}(q)$ | 0 |
| order 6, $-\sigma$ | II, II$^*$ | 0 | 1 | $2\sigma + 1$ | $\chi_{-3}(q)$ | 0 |

In the row $\mathrm{I}_n$, $\dim(T/3T)^I$ is 1 if $3 \nmid n'$ and 2 if $3 \mid n'$; for the Tate curve $v_3(n') = v_3(n)$ (Chapter 10). In the row $\mathrm{I}_0^*$ the eigenvalues of $F$ on $\mathrm{ad}^0 V$ are those of the unramified twist $V'$ (Proposition 4.48(iii)). The degree of $L_q(A_f, s)^{-1}$ is $\dim(\mathrm{ad}^0 V)^I$, that is, $3, 3, 1, 1, 1, 1, 1$, and $\delta_q = \dim V^I$ is $2, 0, 1, 0, 0, 0, 0$ (Definition def:terms). The row $\mathrm{I}_n$ holds at every $q \neq \ell$, the rows $\mathrm{I}_0^*$, $\mathrm{I}_n^*$ at every odd $q \neq \ell$; for the finite images the tame relation $FgF^{-1} = g^q$ needs the image to be tame, which holds for $q \geq 5$ (Chapter 7).

**Proposition 4.72** (Reduction of finite cyclic subgroups of $\mathrm{SL}_2(\mathbb{Z}_\ell)$)**.** *Let $\ell$ be a prime, $T = \mathbb{Z}_\ell^2$, $g \in \mathrm{SL}_2(\mathbb{Z}_\ell)$ of finite order $m \in \{2, 3, 4, 6\}$, and $\bar{g}$ its reduction modulo $\ell$.*

*(i) If $\ell \nmid m$, then $\bar{g}$ has order $m$, and the invariants of $\langle g \rangle$ on $T$ and on $\mathrm{ad}^0 T$ commute with reduction: $(T/\ell^j T)^g = T^g/\ell^j T^g$ and $(\mathrm{ad}^0 T/\ell^j)^g = (\mathrm{ad}^0 T)^g/\ell^j$. In particular $(T/\ell T)^g = 0$.*

*(ii) Let $\ell = 3$, $m = 3$. Then $\bar{g}$ is a nontrivial unipotent element of $\mathrm{SL}_2(\mathbb{F}_3)$, so $(T/3T)^g$ is a line while $T^g = 0$: invariants do not commute with reduction. On the adjoint they do: $(\mathrm{ad}^0 T)^g = \mathbb{Z}_3(2g + 1)$ and $2g + 1 \not\equiv 0 \pmod 3$ spans $(\mathrm{ad}^0 T/3)^g$.*

*(iii) Let $\ell = 3$, $m = 6$. Then $\bar{g} = -\bar{u}$ with $\bar{u}$ a nontrivial unipotent, and $(T/3T)^g = 0$.*

*(iv) Let $\ell = 3$, $m \in \{2, 4\}$. Then $(T/3T)^g = 0$; for $m = 4$, $\bar{g}$ has the irreducible characteristic polynomial $X^2 + 1$ over $\mathbb{F}_3$ and eigenvalues $\pm i \in \mathbb{F}_9$.*

*(v) Let $\ell$ be odd and $m \geq 3$. Then $\bar{g}$ is not a scalar, and the commutant of $g$ in $\mathrm{End}_{\mathbb{Z}_\ell}(T)$ is $\mathbb{Z}_\ell[g] = \mathbb{Z}_\ell \oplus \mathbb{Z}_\ell g \simeq \mathbb{Z}_\ell[X]/(\Phi_m)$. Moreover $T$ is free of rank one over $\mathbb{Z}_\ell[g]$; for $\ell = 3$, $m = 3$, this gives $T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$, $\pi = 1 - \zeta_3$, with $(T/3T)^g = \pi T/3T$.*

*Proof.* First, the kernel $1+\ell\mathrm{M}_2(\mathbb{Z}_\ell)$ of reduction contains no element $\gamma \neq 1$ of finite order prime to $\ell$, and for $\ell$ odd no element of order $\ell$. Write $\gamma = 1+\ell^a y$ with $a \geq 1$ and $y \not\equiv 0 \pmod{\ell}$. For $r$ prime to $\ell$, $\gamma^r \equiv 1 + r\ell^a y \pmod{\ell^{a+1}}$, so $\gamma^r \neq 1$. For $\ell$ odd, $\gamma^\ell = 1 + \ell^{a+1}y + \binom{\ell}{2}\ell^{2a}y^2 + \cdots \equiv 1 + \ell^{a+1}y \pmod{\ell^{a+2}}$, because $\ell \mid \binom{\ell}{j}$ for $0 < j < \ell$, $2a+1 \geq a+2$ and $\ell a \geq a+2$ (using $\ell \geq 3$); so $\gamma^\ell \neq 1$ (this is also in Chapter 1). A nontrivial finite group contains an element of prime order, so the finite subgroups of the kernel are trivial when $\ell$ is odd, and the finite subgroups of order prime to $\ell$ are trivial for every $\ell$. Hence reduction modulo $\ell$ is injective on $\langle g\rangle$ if $\ell$ is odd or $\ell \nmid m$. (i) $\bar{g}$ has order $m$ by the above, and $\#\langle g\rangle = m \in \mathbb{Z}_\ell^\times$; apply Theorem 4.18(ii) to the lattices $T$ and $\mathrm{ad}^0 T$. $T^g = 0$ by Proposition 4.71(ii). (ii) $\bar{g} \neq 1$ by injectivity, and its characteristic polynomial is $X^2 + X + 1 \equiv (X-1)^2 \pmod 3$; so $\bar{g}$ is a nontrivial unipotent and $(T/3T)^g$ is a line (Proposition 4.69(i)), while $T^g = 0$. On the adjoint, $(\mathrm{ad}^0 T)^g = \mathrm{ad}^0 T \cap \mathbb{Q}_3(2g+1)$ by Proposition 4.71(ii). If $2g+1 \equiv 0 \pmod 3$ then $g \equiv -2^{-1}\cdot 1 = 1 \pmod 3$, which is false; so $2g+1$ is not divisible by 3 in $\mathrm{End}(T)$, and $\mathrm{ad}^0 T \cap \mathbb{Q}_3(2g+1) = \mathbb{Z}_3(2g+1)$. Its reduction $2\bar{g}+1 = -(\bar{g}-1)$ spans $(\mathrm{ad}^0 T/3)^g = \mathbb{F}_3(\bar{g}-1)$ (Proposition 4.69(ii)). (iii) $g = -\sigma$ with $\sigma = -g$ of order 3; by (ii) $\bar{\sigma}$ is a nontrivial unipotent, and $-\bar{\sigma}$ fixes no nonzero vector (Proposition 4.69(i)). (iv) is (i), as $3 \nmid m$; for $m = 4$ the characteristic polynomial is $\Phi_4 = X^2 + 1$, without roots in $\mathbb{F}_3$. (v) $\bar{g}$ is not a scalar: if $\ell \nmid m$, $\bar{g}$ has order $m \geq 3$ while a scalar of determinant 1 is $\pm 1$; if $\ell = 3$, $\bar{g}$ is $\pm$ a nontrivial unipotent. So 1 and $g$ are linearly independent modulo $\ell$. Extend $\bar{1}, \bar{g}$ to a basis of $\mathrm{End}(T/\ell T)$ and lift it; by Nakayama's lemma the lift, which contains 1 and $g$, is a $\mathbb{Z}_\ell$-basis of the free module $\mathrm{End}_{\mathbb{Z}_\ell}(T) \simeq \mathbb{Z}_\ell^4$. Hence $\mathbb{Z}_\ell \oplus \mathbb{Z}_\ell g$ is a direct summand of $\mathrm{End}_{\mathbb{Z}_\ell}(T)$: an element $\alpha + \beta g$ with $\alpha, \beta \in \mathbb{Q}_\ell$ lies in $\mathrm{End}_{\mathbb{Z}_\ell}(T)$ iff $\alpha, \beta \in \mathbb{Z}_\ell$. So the commutant $\mathrm{End}_{\mathbb{Z}_\ell}(T) \cap (\mathbb{Q}_\ell \oplus \mathbb{Q}_\ell g)$ (Lemma 4.70(ii) over $\mathbb{Q}_\ell$) is $\mathbb{Z}_\ell \oplus \mathbb{Z}_\ell g$. The freeness of $T$ over $\mathbb{Z}_\ell[g]$ and the description of $T/3T$ are proved in Chapter 6; we do not prove them here. (The fixed space $\pi T/3T$ is the kernel of $g-1$, which acts as multiplication by $\zeta_3 - 1 = -\pi$.) □

*In the paper* 4.73. Lemma lem:IV (types IV, IV* at $q \neq 3$): "its image in $\mathrm{Aut}(E[3])$ is a nontrivial unipotent subgroup. Consequently $E[3]^{I_q}$ is a line" is Proposition 4.72(ii), and "reduction modulo 3 is injective on it, the kernel of $\mathrm{GL}_2(\mathbb{Z}_3) \to \mathrm{GL}_2(\mathbb{F}_3)$ being torsion-free" is the first paragraph of its proof; the route through $T_3E$ free of rank one over $\mathbb{Z}_3[X]/(X^2+X+1)$, in the unnumbered paragraph after the lemma in the paper, is (v). "the trace-zero endomorphisms commuting with $\sigma$ form the line spanned by $\sigma - \sigma^{-1} = 2\sigma + 1$" is Lemma 4.70, and "$\mathrm{Frob}_q$, and therefore also $\mathrm{Frob}_q^{-1}$, acts on the line by $\chi_{-3}(q)$" is Proposition 4.71(iii). In the exclusion of potentially multiplicative reduction, "$\bar{\rho}_{E,3}(I_q)$ would contain $-1$, which fixes no line" is trivial: the scalar $-1$ fixes no nonzero vector of $E[3]$, so $E[3]^{I_q}$ would be 0. Remark rem:types ("$E[3]^{I_q} = 0$" for $\mathrm{I}_0^*$, III, III*, II, II*, $\mathrm{I}_n^*$) is (iii), (iv) and Proposition 4.69(i). Lemma lem:euler is the table above: types II, IV and their duals use $2\sigma + 1$ inside $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[\sigma]$, which is not a field for $\ell \equiv 1 \pmod 3$ ((v) and Lemma 4.70); types III, III* use the line $\mathbb{Q}_\ell\tau$, with no need to diagonalise $\tau$ over $\mathbb{Q}_\ell$; type $\mathrm{I}_n$ uses Proposition 4.69(iii) with the geometric Frobenius, eigenvalue $q^{-1}$. Proposition prop:ns-structure (ii): "The trace-zero endomorphisms commuting with $\tau$ form the line spanned by $\tau$" and "the sign can be read modulo 3" are Lemma 4.70 and the injectivity of reduction on $\langle\tau\rangle$ (order 4, prime to 3).

## 4.10 Representations over $\mathbb{Z}_\ell$: lattices, reduction, homothety

This section passes from representations over $\mathbb{Q}_\ell$ to representations over $\mathbb{Z}_\ell$ and modulo $\ell^m$. In the paper $T = T_\ell E$ is a $G_{\mathbb{Q}}$-stable lattice in $V = V_\ell E$, $\bar{\rho}$ is its reduction, $\rho \bmod \ell^k$ its reduction

modulo $\ell^k$, $A = \mathrm{ad}^0 T$ is a lattice in $A_f$, and the term $h_\ell(E)$ of Lemma lem:h0 is "the largest $k$ such that $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$". Everything is proved for an abstract group and a discrete valuation ring $\mathcal{O}$ (convention (C7)).

**Definition 4.74.** Let $V$ be a representation of $G$ over $K = \mathrm{Frac}\,\mathcal{O}$. A *lattice* in $V$ is a finitely generated $\mathcal{O}$-submodule $L$ spanning $V$; it is *$G$-stable* if $\rho(g)L = L$ for all $g$. Then $\rho_L \colon G \to \mathrm{GL}(L) \simeq \mathrm{GL}_n(\mathcal{O})$, conjugate in $\mathrm{GL}_n(K)$ to $\rho$, and $\bar\rho_L$ is the representation of $G$ on $L/\varpi L$ over the residue field; $\rho_L \bmod \varpi^m$ is the representation on $L/\varpi^m L$. Two lattices are *homothetic* if $L' = cL$ for some $c \in K^\times$, equivalently $L' = \varpi^m L$ for some $m \in \mathbb{Z}$.

**Proposition 4.75** (Stable lattices)**.** *(i) For any two lattices $L, L'$ there are $a, b \geq 0$ with $\varpi^a L \subseteq L' \subseteq \varpi^{-b} L$.*

*(ii) If $G$ is finite, or more generally if the stabiliser of a lattice $L_0$ has finite index in $G$, then $L = \sum_g \rho(g) L_0$ (a finite sum over coset representatives) is a $G$-stable lattice.*

*(iii) If $L$ is stable, so are $\varpi^m L$, $L^\vee = \mathrm{Hom}_{\mathcal{O}}(L, \mathcal{O}) \subseteq V^\vee$, $L \otimes \mathcal{O}(\chi)$ for a character $\chi \colon G \to \mathcal{O}^\times$, $\mathrm{End}_{\mathcal{O}}(L) \subseteq \mathrm{End}_K(V)$, and $\mathrm{ad}^0 L = \mathrm{End}_{\mathcal{O}}(L) \cap \mathrm{ad}^0 V$. Moreover $\mathrm{End}_{\mathcal{O}}(L) = \mathrm{ad}^0 L \oplus \mathcal{O} E_{11}$ as $\mathcal{O}$-modules, so $\mathrm{ad}^0 L/\varpi^m \mathrm{ad}^0 L = \mathrm{ad}^0(L/\varpi^m L)$, the trace-zero endomorphisms of $L/\varpi^m L$.*

*Proof.* (i) Each of the finitely many generators of $L'$ is a $K$-combination of a basis of $L$, so lies in $\varpi^{-b} L$ for $b$ large; exchange the roles for $a$. (ii) Let $G_0$ be the stabiliser of $L_0$ and $s_1, \dots, s_r$ representatives of the cosets $gG_0$. Then $\rho(g)L_0$ depends only on the coset $gG_0$, so $\sum_g \rho(g)L_0 = \sum_i \rho(s_i)L_0$ is a finite sum of lattices; it is finitely generated and spans $V$. For $h \in G$, $\rho(h)\sum_g \rho(g)L_0 = \sum_g \rho(hg)L_0$ is the same sum, so it is stable. This covers a continuous representation of a compact group $G$ on $V$, $K$ a finite extension of $\mathbb{Q}_\ell$: the stabiliser of $L_0$ is the preimage of the open subgroup $\mathrm{GL}(L_0)$ of $\mathrm{GL}(V)$, hence open, hence of finite index. (iii) $(g\lambda)(L) = \lambda(g^{-1}L) = \lambda(L) \subseteq \mathcal{O}$; $\chi(g)$ is a unit; $\rho(g)x\rho(g)^{-1}$ preserves $L$ if $x$ does; the trace is invariant. The trace $\mathrm{End}_{\mathcal{O}}(L) \to \mathcal{O}$ is onto ($\mathrm{tr}\, E_{11} = 1$) and split by $c \mapsto cE_{11}$, so its kernel $\mathrm{ad}^0 L$ is a direct summand, and the split exact sequence $0 \to \mathrm{ad}^0 L \to \mathrm{End}_{\mathcal{O}}(L) \to \mathcal{O} \to 0$ stays exact modulo $\varpi^m$, with $\mathrm{End}_{\mathcal{O}}(L)/\varpi^m = \mathrm{End}(L/\varpi^m L)$ since $L$ is free. □

**Proposition 4.76** (Reduction and semisimplification)**.** *Let $L$ be a $G$-stable lattice in $V$.*

*(i) $\det(X - \bar\rho_L(g))$ is the reduction of $\det(X - \rho(g)) \in \mathcal{O}[X]$; it does not depend on $L$.*

*(ii) Hence (Brauer-Nesbitt, Theorem 4.35) $\bar\rho_L^{ss}$ does not depend on $L$; it is written $\bar\rho^{ss}$.*

*(iii) If $\bar\rho_L$ is irreducible for one $L$, then $\bar\rho_{L'} \simeq \bar\rho_L$ for every stable lattice $L'$.*

*Proof.* (i) The characteristic polynomial of $\rho_L(g) \in \mathrm{GL}_n(\mathcal{O})$ is that of the conjugate $\rho(g)$, with coefficients in $\mathcal{O}$, and reduction commutes with taking characteristic polynomials. (ii) is immediate. (iii) $\bar\rho_{L'}^{ss} \simeq \bar\rho_L$ is irreducible of full dimension, so $\bar\rho_{L'}$ has one composition factor and $\bar\rho_{L'} \simeq \bar\rho_L$ (Corollary 4.36(ii) with $k' = k$). □

**Theorem 4.77** (The homothety lemma)**.** *If $\bar\rho_L$ is irreducible, every $G$-stable lattice $L' \subseteq V$ is homothetic to $L$: $L' = \varpi^m L$ for a unique $m \in \mathbb{Z}$.*

*Proof.* By Proposition 4.75(i) the set of $m \in \mathbb{Z}$ with $L' \subseteq \varpi^m L$ is nonempty and bounded above (as $L' \neq 0$ and $\bigcap_m \varpi^m L = 0$); let $m$ be its maximum and replace $L'$ by $\varpi^{-m} L'$. Then $L' \subseteq L$ and $L' \not\subseteq \varpi L$, so the image of $L'$ in $L/\varpi L$ is a nonzero $G$-stable subspace, hence all of $L/\varpi L$ since $\bar{\rho}_L$ is irreducible. Thus $L' + \varpi L = L$, and $L' = L$ by Nakayama's lemma (Chapter 3). Uniqueness: $\varpi^m L = L$ with $m \neq 0$ would give $L = \varpi L$, hence $L = 0$ by Nakayama. □

*In the paper* 4.78. Section 4.2. Proposition prop:pullback identifies the lattice $M_{f,\lambda}$ of DFG04 with $H^1(E, \mathbb{Z}_\ell)$ itself, so the paper does not need the homothety lemma there. It gives a second route when $\bar{\rho}$ is irreducible: as $M_{f,\lambda} \subseteq H^1$ is a lattice in the dual $V_\ell E^\vee$, Theorem 4.77 applies to $M_{f,\lambda}$ and $T_\ell E^\vee$ inside $V_\ell E^\vee$ (both stable, and $T_\ell E^\vee/\ell \simeq \bar{\rho}^\vee$ is irreducible), after which Proposition 4.60(iv) identifies $\mathrm{ad}^0 M_{f,\lambda}$ with $\mathrm{ad}^0 T$. The homothety lemma is also the reason why, when $E[\ell]$ is irreducible, every curve in the isogeny class has the same $T_\ell$ up to homothety, hence the same $\mathrm{ad}^0 T$ (Section 1, "at such an $\ell$ the choice of the optimal curve does not matter"; Chapter 9).

**Example 4.79** (Non-homothetic lattices)**.** Let $G \subseteq \mathrm{GL}_2(\mathbb{Z}_\ell)$ consist of upper triangular matrices and contain $t = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Then $L = \mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$ and $L' = \mathbb{Z}_\ell e_1 \oplus \ell\mathbb{Z}_\ell e_2$ are $G$-stable and not homothetic ($L'$ has index $\ell$ in $L$, while $\varpi^m L$ has index $\ell^{2m}$). In the basis $e_1, \ell e_2$ of $L'$ the matrix of $g = \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$ is $\left(\begin{smallmatrix} a & \ell b \\ 0 & d \end{smallmatrix}\right)$, so $\bar{\rho}_{L'}$ is the direct sum of the two characters $g \mapsto \bar{a}$, $g \mapsto \bar{d}$, while $\bar{\rho}_L$ is the non-split extension of the same two characters (Proposition 4.24: $\bar{t}$ is a nontrivial unipotent, so $\mathbb{F}_\ell e_1$ is the only stable line). Same semisimplification (Proposition 4.76), non-isomorphic reductions. For elliptic curves, the Tate modules of two $\ell$-isogenous curves give such a pair (Chapter 9, Chapter 11); this is why the Eisenstein case of the paper (Section sec:eis, Conjecture conj:eis) behaves differently.

**Theorem 4.80** (Ribet's lemma)**.** *Let $K$ be a finite extension of $\mathbb{Q}_\ell$ with ring of integers $\mathcal{O}$, $G$ a compact group and $\rho\colon G \to \mathrm{GL}_2(K)$ a continuous irreducible representation such that $\bar{\rho}^{ss} \simeq \eta_1 \oplus \eta_2$ for characters $\eta_1, \eta_2$. Then there is a $G$-stable lattice $L$ whose reduction $\bar{\rho}_L$ is a non-split extension with sub-character $\eta_1$ (and one with sub-character $\eta_2$).*

*Proof.* We do not prove this; see [157, Section 2]. (The idea: among the stable lattices, which up to homothety form a bounded convex subset (a finite subtree) of the tree of $\mathrm{PGL}_2(K)$ because $\rho$ is irreducible, an extremal one has a reduction with the required non-split shape.) In the literature "Ribet's lemma" usually means this statement; the name is sometimes also attached to Theorem 4.77, which concerns the opposite situation of an irreducible reduction. □

**Proposition 4.81** (Twisted self-isomorphisms modulo $\varpi^m$)**.** *Let $G$ be any group, $T$ free of rank $n$ over $\mathcal{O}$ with a $G$-action such that $\bar{T} = T/\varpi T$ is absolutely irreducible, and $\chi\colon G \to \mathcal{O}^\times$ a character, with reduction $\bar{\chi}$. For $m \geq 1$ put $M_m = \mathrm{Hom}_G(T/\varpi^m, (T/\varpi^m)(\chi))$.*

*(i) For $\phi \in \mathrm{End}(T/\varpi^m)$, $\varpi^{m-1}\phi \neq 0$ iff $\phi(T/\varpi^m) \not\subseteq \varpi(T/\varpi^m)$. The $\varpi$-torsion of $M_m$ is isomorphic to $\mathrm{Hom}_G(\bar{T}, \bar{T}(\bar{\chi}))$, of dimension at most 1, so $M_m \simeq \mathcal{O}/\varpi^j$ for some $j \leq m$.*

*(ii) An element $\phi \in M_m$ with $\varpi^{m-1}\phi \neq 0$ is an isomorphism $T/\varpi^m \simeq (T/\varpi^m)(\chi)$. Conversely an isomorphism has $\varpi^{m-1}\phi \neq 0$, so it generates $M_m \simeq \mathcal{O}/\varpi^m$.*

*(iii) Let $h = \sup\{m \geq 0 : T/\varpi^m \simeq (T/\varpi^m)(\chi)\} \in \{0, 1, \ldots, \infty\}$ and $X = \mathrm{End}_{\mathcal{O}}(T)(\chi) \otimes_{\mathcal{O}} K/\mathcal{O}$. Then $X[\varpi^m] = \mathrm{End}(T/\varpi^m)(\chi)$, $X^G[\varpi^m] = M_m \simeq \mathcal{O}/\varpi^{\min(m,h)}$, and $X^G$ has length $h$; if $h < \infty$, $X^G \simeq \mathcal{O}/\varpi^h$. If $h = \infty$, then $T \simeq T(\chi)$ (no completeness of $\mathcal{O}$ is needed).*

*(iv) If $n \in \mathcal{O}^\times$ and $\bar\chi \neq 1$, the same holds for* $\mathrm{ad}^0 T(\chi) \otimes K/\mathcal{O}$: $(\mathrm{ad}^0 T(\chi) \otimes K/\mathcal{O})^G = X^G$.

*(v) If $T/\varpi^m \simeq (T/\varpi^m)(\chi)$ then $\chi^n \equiv 1 \pmod{\varpi^m}$ on $G$. In particular, if $n = 2$, $\mathcal{O} = \mathbb{Z}_\ell$ with $\ell$ odd and $\chi(G) = \mathbb{Z}_\ell^\times$, then $h = 0$ for $\ell \geq 5$ and $h \leq 1$ for $\ell = 3$.*

*Proof.* (i) $T/\varpi^m$ is free over $\mathcal{O}/\varpi^m$, so its $\varpi^{m-1}$-torsion is $\varpi(T/\varpi^m)$; this gives the first statement. If $\varpi\phi = 0$, then $\phi$ kills $\varpi(T/\varpi^m)$ and takes values in the $\varpi$-torsion $\varpi^{m-1}T/\varpi^m T$, which multiplication by $\varpi^{m-1}$ identifies with $\bar T$ (with the twist by $\bar\chi$ for the target); so $M_m[\varpi] \simeq \mathrm{Hom}_G(\bar T, \bar T(\bar\chi))$, of dimension at most one by Lemma 4.49 ($\bar T(\bar\chi)$ is absolutely irreducible). $M_m$ is a finitely generated $\mathcal{O}/\varpi^m$-module whose $\varpi$-torsion has length at most one, hence cyclic: by the structure theorem (Chapter 3) $M_m \simeq \bigoplus_i \mathcal{O}/\varpi^{a_i}$ with $1 \leq a_i \leq m$, its $\varpi$-torsion is $\bigoplus_i \varpi^{a_i-1}\mathcal{O}/\varpi^{a_i}$, of length equal to the number of summands, so there is at most one summand. (ii) By (i), $\phi \bmod \varpi$ is a nonzero $G$-map $\bar T \to \bar T(\bar\chi)$, an isomorphism by Schur. So $\phi$ is surjective modulo $\varpi$, hence surjective by Nakayama, hence bijective, the two modules having the same finite length. Conversely, an isomorphism is nonzero modulo $\varpi$. (iii) $X[\varpi^m] = \mathrm{End}(T)(\chi) \otimes \varpi^{-m}\mathcal{O}/\mathcal{O} \simeq \mathrm{End}(T/\varpi^m)(\chi)$, and taking invariants gives $M_m$. Let $M_m \simeq \mathcal{O}/\varpi^j$ with generator $\phi$. If $j = m$, $\phi$ is an isomorphism by (ii), so $m \leq h$. In general $\varpi^j\phi = 0$ means that $\phi$ takes values in $\varpi^{m-j}(T/\varpi^m)$, so $\phi = \varpi^{m-j}\phi'$ for a $G$-map $\phi' : T/\varpi^j \to (T/\varpi^j)(\chi)$ (defined by dividing the values by $\varpi^{m-j}$, well defined modulo $\varpi^j$), with $\varpi^{j-1}\phi' \neq 0$ because $\varpi^{j-1}\phi \neq 0$. By (ii) $\phi'$ is an isomorphism, so $j \leq h$. Conversely, for $m \leq h$ the reduction of an isomorphism at a level $\geq m$ is an isomorphism at level $m$, which generates $M_m \simeq \mathcal{O}/\varpi^m$ by (ii); and for $m > h$ with $h$ finite, if $\psi$ is an isomorphism at level $h$ and $\tilde\psi \in \mathrm{End}_\mathcal{O}(T)$ lifts it, then $x \mapsto \varpi^{m-h}\tilde\psi(x) \bmod \varpi^m$ is an element of $M_m$ (it is $G$-equivariant because $\tilde\psi g - \chi(g) g\tilde\psi$ has values in $\varpi^h T$) of exact order $\varpi^h$. Hence $j = \min(m, h)$, and $X^G = \bigcup_m X^G[\varpi^m]$ has length $h$, and equals $X^G[\varpi^h] \simeq \mathcal{O}/\varpi^h$ when $h$ is finite.

Now let $h = \infty$. *Compatible isomorphisms.* The reduction map $M_{m+1} \to M_m$ is surjective: $M_{m+1}$ contains an isomorphism, whose reduction is an isomorphism at level $m$ and so generates $M_m$ by (ii). Choose an isomorphism $\phi_1 \in M_1$ and, inductively, $\phi_{m+1} \in M_{m+1}$ reducing to $\phi_m$. Each $\phi_m$ reduces modulo $\varpi$ to $\phi_1 \neq 0$, so $\varpi^{m-1}\phi_m \neq 0$ by (i), and $\phi_m$ is an isomorphism by (ii). *Passage to the completion.* Let $\widehat{\mathcal{O}} = \varprojlim \mathcal{O}/\varpi^m$ be the $\varpi$-adic completion, a discrete valuation ring with uniformiser $\varpi$ and $\widehat{\mathcal{O}}/\varpi^m = \mathcal{O}/\varpi^m$, containing $\mathcal{O}$, with fraction field $\widehat{K} \supseteq K$ (for $\mathcal{O} = \mathbb{Z}_\ell$ nothing changes). In a basis of $T$, the matrices of the $\phi_m$ are compatible, so they define a matrix $\hat\phi \in \mathrm{M}_n(\widehat{\mathcal{O}})$ with $\hat\phi \equiv \phi_m \pmod{\varpi^m}$ for all $m$. For $g \in G$, every entry of $\hat\phi\rho(g) - \chi(g)\rho(g)\hat\phi$ lies in $\varpi^m\widehat{\mathcal{O}}$ for all $m$, hence is 0; and $\hat\phi \neq 0$ since $\hat\phi \equiv \phi_1 \pmod{\varpi}$. So $\mathrm{Hom}_{\widehat{K}[G]}(V_{\widehat{K}}, V_{\widehat{K}}(\chi)) \neq 0$, where $V = T \otimes_\mathcal{O} K$. *Descent.* By Proposition 4.7(i) for the field extension $\widehat{K}/K$, $\mathrm{Hom}_{K[G]}(V, V(\chi)) \neq 0$. Let $x$ be a nonzero element. Multiplying $x$ by a power of $\varpi$, we may assume that $x(T) \subseteq T$ and $x(T) \not\subseteq \varpi T$ (the entries of its matrix lie in $\mathcal{O}$, not all in $\varpi\mathcal{O}$). Then $\bar x = x \bmod \varpi$ is a nonzero $G$-map $\bar T \to \bar T(\bar\chi)$ between irreducible representations, an isomorphism by Theorem 4.10(i). So $\det x \in \mathcal{O}$ has nonzero reduction, $\det x \in \mathcal{O}^\times$, and $x$ is an isomorphism $T \simeq T(\chi)$ of $\mathcal{O}[G]$-modules. (iv) $\mathrm{End}(T) = \mathrm{ad}^0 T \oplus \mathcal{O}$ (Proposition 4.52(i)), and $(K/\mathcal{O}(\chi))^G = 0$ because $\chi(g) - 1$ is a unit for some $g$; so the $\mathcal{O}$-summand contributes nothing. (v) Determinants (Proposition 4.48(iv) modulo $\varpi^m$). For the last statement, $\chi^2 \equiv 1 \pmod{\ell^m}$ on $G$ means $x^2 \equiv 1 \pmod{\ell^m}$ for every $x \in \mathbb{Z}_\ell^\times$; with $x = 2$, $\ell^m \mid 3$, so $m = 0$ for $\ell \geq 5$ and $m \leq 1$ for $\ell = 3$ (Chapter 2). □

*In the paper* 4.82. Lemma lem:h0 with $G = G_\mathbb{Q}$, $\mathcal{O} = \mathbb{Z}_\ell$, $T = T_\ell E$, $\chi = \chi_\ell$: $W^D = \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and by (iv) and (iii), $H^0(\mathbb{Q}, W^D)$ has length $h_\ell(E)$, "the largest $k$ such that $\rho \bmod \ell^k \simeq$

$(\rho \otimes \chi_\ell) \bmod \ell^k$", and is cyclic when this is finite. Absolute irreducibility of $\bar\rho$ comes from complex conjugation (Proposition 4.13), $\bar\chi_\ell \neq 1$ for $\ell$ odd, and $\chi_\ell(G_\mathbb{Q}) = \mathbb{Z}_\ell^\times$ (Chapter 5); so (v) gives $h_\ell(E) = 0$ for $\ell \geq 5$ and $h_3(E) \leq 1$ (in particular $H^0(\mathbb{Q}, W^D)$ is finite and cyclic), with equality iff $\bar\rho \simeq \bar\rho \otimes \bar\chi_3 = \bar\rho \otimes \chi_{-3}$, for every $E$ with $\bar\rho$ irreducible, with or without complex multiplication. In Lemma lem:h0, "a submodule of $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^3$ with $\ell$-torsion of rank at most one, is finite cyclic or isomorphic to $\mathbb{Q}_\ell/\mathbb{Z}_\ell$" is (iii), and the determinant bound (v) excludes the second case. Remark rem:ns and Section sec:anomaly use $h_3(E) = 1$ for the image 3Ns (Theorem 4.66).

## 4.11 The mod $\ell$ world: a foretaste, and Dickson's list

This last section reads the chapter's results in the setting of the paper: a continuous representation $\bar\rho\colon G_\mathbb{Q} \to \mathrm{GL}_2(\mathbb{F}_\ell)$ has finite image, so everything above applies to it, and the Galois input needed is collected in Chapter 5 and Chapter 11. It also records Dickson's classification, which organises the possible images.

*Remark* 4.83 (The mod $\ell$ dictionary). Let $\bar\rho\colon G_\mathbb{Q} \to \mathrm{GL}_2(\mathbb{F}_\ell)$ be continuous, $\ell$ odd, with $\det\bar\rho = \bar\chi_\ell$ (as for $\bar\rho_{E,\ell}$, Chapter 9).

(i) Complex conjugation $c$ has $\bar\rho(c)^2 = 1$ and $\det\bar\rho(c) = \bar\chi_\ell(c) = -1$, so $\bar\rho$ is irreducible iff it is absolutely irreducible (Proposition 4.13).

(ii) The hypothesis of Theorem thm:main concerns $\mathrm{Res}_H\,\bar\rho$ for the subgroup $H = G_{\mathbb{Q}(\sqrt{\ell^*})}$ of index 2, with $\chi_H = \chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$. If $\bar\rho$ is reducible, so is $\mathrm{Res}_H\,\bar\rho$, and the hypothesis fails. If $\bar\rho$ is irreducible, it is absolutely irreducible by (i), and Theorem 4.44 applies: the hypothesis fails iff $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$, iff $\bar\rho \otimes \overline{\mathbb{F}}_\ell$ is induced from a character of $H$. Irreducibility cannot be dropped here: for $\ell \geq 5$, $\bar\rho = \mathbf{1} \oplus \bar\chi_\ell$ has reducible restriction to $H$, but $\bar\rho \otimes \chi_{\ell^*} = \chi_{\ell^*} \oplus \bar\chi_\ell^{(\ell+1)/2} \not\simeq \bar\rho$ (its constituents are not $\mathbf{1}$, as $\chi_{\ell^*} \neq 1$ and $\bar\chi_\ell^{(\ell+1)/2} \neq 1$ for $\ell \geq 5$), and $\bar\rho \otimes \overline{\mathbb{F}}_\ell$ is not induced from $H$ (Proposition 4.41: an induced representation is isomorphic to its twist by $\chi_H$).

(iii) Lemma lem:h0 concerns $\bar\rho \simeq \bar\rho \otimes \bar\chi_\ell$, which forces $\bar\chi_\ell^2 = 1$ (Proposition 4.48(iv)), that is $\ell = 3$; there $\bar\chi_3 = \chi_{-3} = \det\bar\rho$ and Theorem 4.66 applies. For $\ell \geq 5$ the two conditions differ: for $\bar\rho = \bar\rho_{E,\ell}$, $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$ can occur (only when $\ell \mid N$, by Proposition prop:Sf), while $\bar\rho \simeq \bar\rho \otimes \bar\chi_\ell$ never does.

(iv) The objects of the paper and where they are built:

| object | meaning | here |
|---|---|---|
| $V = V_\ell E$, $\rho$ | two-dimensional $\mathbb{Q}_\ell$-representation of $G_\mathbb{Q}$ | Definition 4.2 |
| $T = T_\ell E$ | a $G_\mathbb{Q}$-stable lattice in $V$ | Definition 4.74 |
| $\bar\rho$, $\rho \bmod \ell^k$ | $T/\ell T$, $T/\ell^k T$ | Proposition 4.76, Proposition 4.81 |
| $A_f = \mathrm{ad}^0 V$ | trace-zero adjoint | Definition 4.51, Proposition 4.52 |
| $A = \mathrm{ad}^0 T$ | adjoint lattice, a direct summand of $\mathrm{End}(T)$ | Proposition 4.75(iii) |
| $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ | Kummer dual, via the trace form | Proposition 4.58 |
| $W = A \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $W^D = A(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ | discrete coefficients | Proposition 4.81(iii), (iv) |
| $H^0(\mathbb{Q}, W^D)$, $h_\ell(E)$ | twisted self-isomorphisms of $\rho \bmod \ell^k$ | Proposition 4.81 |
| $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T$ | adjoint of a dual, twisted lattice | Proposition 4.60 |

**Theorem 4.84** (Dickson)**.** *Let $\ell$ be an odd prime and $G \le \mathrm{GL}_2(\mathbb{F}_\ell)$, with image $P$ in $\mathrm{PGL}_2(\mathbb{F}_\ell)$. If $\ell \mid \#G$, then $G \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$ or $G$ lies in a Borel subgroup. If $\ell \nmid \#G$, then $P$ is cyclic (and $G$ lies in a Cartan subgroup), or dihedral (and $G$ lies in the normaliser of a Cartan subgroup, not in the Cartan subgroup), or $P$ is isomorphic to $A_4$, $S_4$ or $A_5$.*

*Proof.* We do not prove this; see [171, Section 2] and Chapter 1, where the statement is the same and the case $\ell = 3$ is checked by enumeration. □

For an absolutely irreducible $\bar\rho$ with projective image $P$, Proposition 4.46 reads Dickson's list as follows: $P$ is never cyclic; $P$ is dihedral iff $\bar\rho \otimes \overline{\mathbb{F}}_\ell$ is induced from a subgroup of index 2; the exceptional images $A_4$, $S_4$, $A_5$ and the large images ($P \supseteq \mathrm{PSL}_2(\mathbb{F}_\ell)$) are not induced (for $\ell \ge 5$, $\mathrm{PSL}_2(\mathbb{F}_\ell)$ is simple and non-abelian, so it is not contained in a dihedral group, whose subgroups are cyclic or dihedral). At $\ell = 3$: $\mathrm{GL}_2(\mathbb{F}_3) \mapsto S_4$, $N_{ns}(\mathbb{F}_3) \mapsto D_4$, $N_s(\mathbb{F}_3) \mapsto V_4$ (Proposition 4.62). The subgroup $N_s(\mathbb{F}_3)$ is not a maximal subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$: it has index 2 in $N_{ns}(\mathbb{F}_3)$ (Chapter 1).

## 4.12 Exercises

**Exercise 4.85.** Show that $C_4 = \langle J \rangle$ acting on $\mathbb{F}_3^2$ is irreducible but not absolutely irreducible, that its commutant $\mathbb{F}_3[J]$ is a field with 9 elements, and decompose $\mathbb{F}_9^2$ into eigenlines of $J$. Which eigenline is spanned by a vector with coordinates in $\mathbb{F}_3$? *Hint.* $X^2 + 1$ has no root in $\mathbb{F}_3$, since $0, 1, 2$ have squares $0, 1, 1$; the commutant is $\mathbb{F}_3[J] \simeq \mathbb{F}_3[X]/(X^2+1)$ by Lemma 4.70. Over $\mathbb{F}_9$ the eigenlines are $\mathbb{F}_9(1, i)^t$ and $\mathbb{F}_9(1, -i)^t$. Neither contains a nonzero vector of $\mathbb{F}_3^2$: such a vector would be an eigenvector of $J$ with eigenvalue $\pm i \notin \mathbb{F}_3$.

**Exercise 4.86.** Prove directly (without Theorem 4.10) that every absolutely irreducible representation of a finite abelian group is one-dimensional. Show that $C_{ns}(\mathbb{F}_3) = \langle a \rangle \simeq C_8$ gives an irreducible two-dimensional representation of $C_8$ over $\mathbb{F}_3$, and determine all irreducible representations of $C_8$ over $\mathbb{F}_3$ (their degrees and their number). *Hint.* Commuting operators on a nonzero finite-dimensional space over $\bar k$ have a common eigenvector: an eigenspace of one is stable under the others; induct on the dimension. For $C_8$: as $3 \nmid 8$, $\mathbb{F}_3[C_8] \simeq \mathbb{F}_3[X]/(X^8 - 1)$ is the product of the fields $\mathbb{F}_3[X]/(f)$ over the irreducible factors $f$ of $X^8 - 1$, and these fields

are the irreducible representations. The roots of $X^8-1$ are the elements of $\mathbb{F}_9^\times$, and over $\mathbb{F}_3$, $X^8-1=(X-1)(X+1)(X^2+1)(X^2+X+2)(X^2-X+2)$; so there are 5 irreducible representations, two of degree 1 and three of degree 2. That of $\langle a\rangle$ is the one for $X^2+X+2$, the characteristic polynomial $X^2-2X+2$ of $a$.

**Exercise 4.87.** Show that the standard representation of $\mathrm{U}=\langle u\rangle$ on $\mathbb{F}_3^2$ is indecomposable but not irreducible, and that $\mathbb{F}_p[C_p]$ is uniserial. Why does this not contradict Maschke's theorem? Show that every representation of $C_p$ over $\mathbb{F}_p$ is a direct sum of the quotients $\mathbb{F}_p[x]/(x-1)^j$, $1\le j\le p$. *Hint.* There is no contradiction: $\#\mathrm{U}=3$ is the characteristic. With $y=x-1$, a representation of $C_p$ over $\mathbb{F}_p$ is a finitely generated $\mathbb{F}_p[y]$-module killed by $y^p=x^p-1$; apply the structure theorem over the principal ideal domain $\mathbb{F}_p[y]$ (Chapter 3). Each summand $\mathbb{F}_p[y]/(y^j)$ is uniserial, its subrepresentations being the $y^i\mathbb{F}_p[y]/(y^j)$, $0\le i\le j$.

**Exercise 4.88.** Give two non-isomorphic semisimple representations of $C_2$ over $\mathbb{F}_3$ with the same trace function. Show that two two-dimensional representations with the same determinant and the same trace function have isomorphic semisimplifications, in every characteristic. *Hint.* Example 4.32; then Theorem 4.35, since the characteristic polynomial of a $2\times 2$ matrix is $X^2-\mathrm{tr}(g)X+\det(g)$.

**Exercise 4.89.** Check the claim of Proposition prop:ns that the determinant-one subgroup $C_4$ of $N_s(\mathbb{F}_3)$ acts on $\mathbb{F}_9^2$ as $\psi\oplus\psi^{-1}$ with $\psi$ of order 4, and that the standard representation of $N_s(\mathbb{F}_3)$ over $\mathbb{F}_9$ is isomorphic to $\mathrm{Ind}_{C_4}^{N_s}\psi$: compute the matrices of $J$ and $\sigma$ in the basis $v=(1,i)^t$, $\sigma v$. Show that over $\mathbb{F}_3$ it is not induced from a character of $C_4$, and that it is nevertheless isomorphic over $\mathbb{F}_3$ to its twist by det. *Hint.* Example 4.14 and Proposition 4.63. A character of $C_4$ with values in $\mathbb{F}_3^\times=\{\pm1\}$ sends $J$ to $\pm1$; if $\rho$ were induced from it over $\mathbb{F}_3$, $\mathrm{Res}_{C_4}\rho$ would contain a line on which $J$ acts by $\pm1$, but $J$ has no eigenvalue in $\mathbb{F}_3$. For the twist, check $J\rho(g)=\det(g)\rho(g)J$ on the generators $J$ and $\sigma$.

**Exercise 4.90.** Check the claim of Proposition prop:ns that $Q_8$ and $\mathrm{SL}_2(\mathbb{F}_3)$ act absolutely irreducibly on $\mathbb{F}_3^2$ in two ways: by Maschke's theorem (Proposition 4.26) and by the commutator argument (a subgroup of a Borel subgroup of $\mathrm{GL}_2(\overline{\mathbb{F}}_3)$ has abelian commutator subgroup of exponent dividing 3). Explain why "non-abelian and faithful" does not suffice, using $\mathrm{B}(\mathbb{F}_3)$. *Hint.* The commutator $xyx^{-1}y^{-1}$ of two upper triangular matrices is upper triangular with diagonal entries $1,1$, so the commutator subgroup of a Borel subgroup of $\mathrm{GL}_2(\overline{\mathbb{F}}_3)$ lies in $\{\left(\begin{smallmatrix}1&b\\0&1\end{smallmatrix}\right)\}\simeq(\overline{\mathbb{F}}_3,+)$, abelian of exponent 3. The commutator subgroup of $Q_8$ is $\{\pm1\}$, and $-1$ is not unipotent; that of $\mathrm{SL}_2(\mathbb{F}_3)$ is $Q_8$, which is not abelian (Chapter 1). $\mathrm{B}(\mathbb{F}_3)$ is non-abelian, acts faithfully, and fixes the line $\mathbb{F}_3e_1$.

**Exercise 4.91.** Show that the standard representation of $N_s(\mathbb{F}_3)$ is isomorphic to its twist by each of the three nontrivial characters of $N_s(\mathbb{F}_3)/\{\pm1\}$, with intertwiners $J$, $\sigma$, $w$, and that these intertwiners have trace 0. Show that the standard representation of $N_{ns}(\mathbb{F}_3)$ is induced, over $\mathbb{F}_9$, from $C_{ns}(\mathbb{F}_3)$, but is not isomorphic to its twist by det. Which of the two satisfies the hypothesis "absolutely irreducible on the determinant-one subgroup"? *Hint.* Proposition 4.63(iii), (iv) and Proposition 4.64. The hypothesis holds for $N_{ns}(\mathbb{F}_3)$, whose determinant-one subgroup $Q_8$ acts absolutely irreducibly, and fails for $N_s(\mathbb{F}_3)$, whose determinant-one subgroup $C_4$ does not.

**Exercise 4.92.** Check the claim of Lemma lem:h0 that an isomorphism $\phi\colon\bar\rho\to\bar\rho\otimes\chi_{-3}$ has trace 0. More generally, show that $(\mathrm{ad}^0V\otimes\chi)^G=\mathrm{Hom}_G(V,V\otimes\chi)$ for every nontrivial character $\chi$ when $\dim V$ is invertible in $k$, and give an example with $\chi=1$ where the two spaces differ. *Hint.*

Lemma 4.49 with $g$ such that $\chi_{-3}(g) = -1$, as $1-(-1) = 2 \neq 0$ in $\mathbb{F}_3$; then Proposition 4.54(ii), (iii). For $\chi = 1$, $\mathrm{Hom}_G(V,V)$ contains $\mathrm{id}_V$, of trace $\dim V \neq 0$, which is not in $(\mathrm{ad}^0 V)^G$; for instance, for $V = k$ with trivial action, $\mathrm{Hom}_G(V,V) = k$ and $\mathrm{ad}^0 V = 0$.

**Exercise 4.93.** Let $H \leq G$ have index 2, $\mathrm{char}\, k \neq 2$, and $V$ absolutely irreducible of dimension 2. Prove that $V \simeq V \otimes \chi_H$ iff $\mathrm{tr}\,\rho(g) = 0$ for all $g \notin H$. (For "if", compare traces and determinants of $V$ and $V \otimes \chi_H$ and use Corollary 4.36.) Deduce again that the standard representation of $\mathrm{GL}_2(\mathbb{F}_3)$ is not isomorphic to its twist by det, and that that of $N_s(\mathbb{F}_3)$ is.

**Exercise 4.94.** Prove $\mathrm{ad}^0 V \simeq \mathrm{Sym}^2 V \otimes \det^{-1}$ for $\dim V = 2$ by the explicit map $\Xi$ of Proposition 4.56, and show that over $\mathbb{Z}$ the image of $\mathrm{Sym}^2 \mathbb{Z}^2$ has index 4 in $\mathrm{ad}^0 \mathbb{Z}^2$. Deduce the eigenvalues $\alpha/\beta$, 1, $\beta/\alpha$ of $\mathrm{Ad}\, g$ on $\mathrm{ad}^0 V$ from the eigenvalues $\alpha^2, \alpha\beta, \beta^2$ of $\mathrm{Sym}^2 g$. *Hint.* In the bases $e_1^2, e_1e_2, e_2^2$ and $e, h, f$ the matrix of $\Xi$ is $\mathrm{diag}(-2,1,2)$, and the image of a map $\mathbb{Z}^3 \to \mathbb{Z}^3$ with nonzero determinant has index equal to the absolute value of the determinant. On $\mathrm{Sym}^2 V \otimes \det^{-1}$ the eigenvalues are $\alpha^2, \alpha\beta, \beta^2$ divided by $\alpha\beta$.

**Exercise 4.95.** Compute the Gram matrix of the trace form on $\mathrm{ad}^0 \mathbb{Z}_\ell^2$ in the basis $e, h, f$, and check the identification $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ of the paper's Section 4.2 for $\ell$ odd. What goes wrong for $\ell = 2$? Is the trace form on all of $\mathrm{End}(\mathbb{Z}_2^2)$ perfect? *Hint.* Proposition 4.58(ii), (iii): the Gram matrix has determinant $-2$, a unit in $\mathbb{Z}_\ell$ for $\ell$ odd. For $\ell = 2$, $B(x,h) = 2\beta$ is even for every $x = \alpha e + \beta h + \gamma f \in \mathrm{ad}^0 \mathbb{Z}_2^2$, so the linear form taking the values $0, 1, 0$ on $e, h, f$ is not of the form $B(x,-)$, and $A \to A^\vee$ is not onto. On $\mathrm{End}(\mathbb{Z}_2^2)$ the Gram matrix in the basis $(E_{ij})$ is a permutation matrix, so the trace form there is perfect: the defect comes from restricting to trace zero.

**Exercise 4.96.** Check the formulas in the proof of Lemma lem:tamfactor: for $t = 1 + n'e$, $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$; and for $F = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$ with $\varepsilon = \pm 1$, $FeF^{-1} = qe$ and $FhF^{-1} = h - 2c'\varepsilon e$. Deduce that for $\ell$ odd $(t-1)\,\mathrm{ad}^0 T = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$. *Hint.* Proposition 4.52(iii), with $\varepsilon^{-1} = \varepsilon$. The submodule $(t-1)\,\mathrm{ad}^0 T$ is generated by $(t-1)e = 0$, $(t-1)h = -2n'e$ and $(t-1)f = n'h - n'^2 e$; it lies in $n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$, and since $2 \in \mathbb{Z}_\ell^\times$ it contains $n'e$, hence also $n'h$.

**Exercise 4.97.** For $g = -1$, $g = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ (order 3), $g = J$ (order 4), $g = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (order 6), $g = u$ and $g = -u$ in $\mathrm{SL}_2(\mathbb{Z})$, compute $\dim V^g$ and $\dim(\mathrm{ad}^0 V)^g$ for $V = \mathbb{Q}^2$ and for $V = \mathbb{F}_3^2$, and a generator of $(\mathrm{ad}^0 V)^g$. Check the claims of Lemma lem:IV and Remark rem:types about $E[3]^{I_q}$. *Answer.* Over $\mathbb{Q}$: for $-1$, $\dim V^g = 0$ and $(\mathrm{ad}^0 V)^g = \mathrm{ad}^0 V$; for the elements of order 3, 4, 6, $\dim V^g = 0$ and $(\mathrm{ad}^0 V)^g$ is the line spanned by $2g+1$, $J$, $2g-1$ respectively (the first and the last are both $\left(\begin{smallmatrix} 1 & -2 \\ 2 & -1 \end{smallmatrix}\right)$); for $u$, $\dim V^u = 1$ and $(\mathrm{ad}^0 V)^u = \mathbb{Q}e$; for $-u$, $\dim V^{-u} = 0$ and $(\mathrm{ad}^0 V)^{-u} = \mathbb{Q}e$. Over $\mathbb{F}_3$ the only change is for the element of order 3, whose reduction is a nontrivial unipotent: $\dim V^g = 1$ (Proposition 4.72(ii)). The generators of $(\mathrm{ad}^0 V)^g$ over $\mathbb{F}_3$ are the reductions of those over $\mathbb{Q}$. These values were checked in PARI/GP.

**Exercise 4.98.** Let $u \neq 1$ be unipotent and $FuF^{-1} = u^q$. Show that $F$ acts on $(\mathrm{ad}^0 V)^u$ by $q$. Explain why the factor $1 - q^{-1-s}$ of $L_q(A_f, s)^{-1}$ at a multiplicative prime requires the geometric Frobenius $F^{-1}$, and why at the primes of types II, III, IV and their duals the choice of Frobenius does not matter. *Hint.* Proposition 4.69(iii). The factor is $\det(1 - F^{-1}q^{-s} \mid (\mathrm{ad}^0 V)^I)$, and $F^{-1}$ acts on the line by $q^{-1}$, which gives $1 - q^{-1-s}$; with $F$ one would get $1 - q^{1-s}$. For the types II, III, IV and their duals, $F$ acts by $\pm 1$ (Proposition 4.71(iii)), which is its own inverse.

**Exercise 4.99.** Show that for a finite group of order prime to $\ell$ acting on a $\mathbb{Z}_\ell$-lattice $T$, invariants commute with reduction modulo $\ell^m$. Show that this fails for $g \in \mathrm{SL}_2(\mathbb{Z}_3)$ of order 3 acting on $T = \mathbb{Z}_3^2$ (Lemma lem:IV), although it holds for the invariants of $g$ on $\mathrm{ad}^0 T$. Does it hold for $g$ of order 6 on $T$? On $\mathrm{ad}^0 T$? *Hint.* Theorem 4.18(ii) and Proposition 4.72. For $g$ of order 6, $T^g = 0 = (T/3T)^g$, so it holds on $T$; on $\mathrm{ad}^0 T$, $g$ acts as $-g$, of order 3, and the case of order 3 applies: it holds.

**Exercise 4.100.** Let $G \subseteq \mathrm{GL}_2(\mathbb{Z}_\ell)$ be the closed subgroup generated by $t = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Show that the lattices $L_k = \mathbb{Z}_\ell e_1 \oplus \ell^k \mathbb{Z}_\ell e_2$, $k \geq 0$, are $G$-stable and pairwise non-homothetic, that their reductions have the same semisimplification, and that the reduction of $L_k$ is split iff $k \geq 1$. Explain why this does not contradict Theorem 4.77, and prove that theorem again in your own words. *Hint.* In the basis $e_1, \ell^k e_2$ of $L_k$ the matrix of $t$ is $\left(\begin{smallmatrix} 1 & \ell^k \\ 0 & 1 \end{smallmatrix}\right)$. If $L_k = \ell^m L_j$, intersect with $\mathbb{Q}_\ell e_1$: $\mathbb{Z}_\ell e_1 = \ell^m \mathbb{Z}_\ell e_1$, so $m = 0$, and then $k = j$. The theorem does not apply because the reduction of $L_0$ is reducible.

**Exercise 4.101.** Let $T$ be free of rank 2 over $\mathbb{Z}_\ell$, $\ell$ odd, with an action of a group $G$ such that $T/\ell T$ is absolutely irreducible, and let $\chi\colon G \to \mathbb{Z}_\ell^\times$ be surjective. Show that $\mathrm{Hom}_G(T/\ell^m, (T/\ell^m)(\chi))$ is cyclic for every $m$, that $H^0(G, \mathrm{ad}^0 T(\chi) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is finite of length at most 1, and that it is 0 unless $\ell = 3$ (the corrected Lemma lem:h0). For $\ell = 3$, show that the length is 1 iff $T/3T \simeq (T/3T)(\bar\chi)$. *Hint.* Proposition 4.81 with $\mathcal{O} = \mathbb{Z}_\ell$: (i) gives the cyclicity; (iv) applies because $\bar\chi$ maps onto $\mathbb{F}_\ell^\times \neq 1$; (v) gives $h = 0$ for $\ell \geq 5$ and $h \leq 1$ for $\ell = 3$; and for $\ell = 3$, $h \geq 1$ iff $T/3T \simeq (T/3T)(\bar\chi)$, by the definition of $h$.

## 4.13 Notes and references

The representation theory of finite groups over $\mathbb{C}$ in Section 4.1 to Section 4.4 follows [173]: Maschke's theorem is in its Section 1.3, Schur's lemma in Section 2.2, the orthogonality relations in Sections 2.3 to 2.5, the divisibility of degrees in Section 6.5, induction, Frobenius reciprocity and Mackey's criterion in Sections 7.1 to 7.4, Clifford's theorem in Section 8.1, and the Frobenius-Schur indicator in Section 13.2. These section numbers are recalled, not checked against a printed copy, and should be read as pointers; no theorem number of [173] is relied upon. The proofs of Schur's lemma, Maschke's theorem and Frobenius reciprocity given here are characteristic-free, as is the averaging proof of $\dim \mathrm{Hom}_G(V, W) = \langle \mathrm{tr}_W, \mathrm{tr}_V \rangle$. The treatment of representations over arbitrary fields (absolute irreducibility, extension of scalars, the density theorem, the theorem of Brauer and Nesbitt, the Noether-Deuring theorem) is classical and can be found in [37], Sections 27 to 30 and 29 for Noether-Deuring, and in [109, Chapter XVII], where semisimple modules and the density theorem are treated; again these are unverified pointers. The proof of Brauer-Nesbitt in Section 4.5 is the standard one by reduction to an algebraically closed field, cancellation, and the Frobenius $f \mapsto f^p$; the reduction step uses only the density theorem, so no perfectness of $k$ is needed. The criterion Proposition 4.12(iv), that an irreducible representation with commutant $k$ is absolutely irreducible over every field, is a consequence of the same density theorem. Riley-Hobson-Bence [165] treats character tables over $\mathbb{C}$ in its chapter on representation theory (Chapter 29 of the third edition; an unverified pointer); [58, Chapter 18] is an algebraic introduction at the same level.

Clifford theory for a subgroup of index 2 (Theorem 4.44) and the characterisation of dihedral representations (Proposition 4.46) are standard; for the Galois representations of elliptic curves they go back to [171, Section 2], which also contains the classification of subgroups of $\mathrm{GL}_2(\mathbb{F}_\ell)$

stated as Theorem 4.84 (Propositions 15 and 16 there, as in Chapter 1). Lattices in $\ell$-adic representations, their reductions and the independence of $\bar{\rho}^{ss}$ from the lattice are in [178, Chapter I]; the homothety lemma Theorem 4.77 is the elementary half of that circle of ideas, and the converse construction is Ribet's lemma [157, Section 2], where it appears as a proposition about reducible residual representations (we recall it as Proposition 2.1 but have not checked the number). The use of Brauer-Nesbitt together with Chebotarev for Galois representations, as in Lemma lem:old, is standard; see [38] and Chapter 5.

The table of Proposition 4.62 and the facts of Proposition 4.63 and Proposition 4.64 were computed by hand and checked by enumerating $\mathrm{GL}_2(\mathbb{F}_3)$ with the script of Section 4.8 and with an independent Python enumeration (stable lines over $\mathbb{F}_3$ and $\mathbb{F}_9$, commutants, intertwiners, projective images). The formulas of Proposition 4.52(iii), the map $\Xi$ of Proposition 4.56 and the Gram matrix of Proposition 4.58 were checked symbolically in PARI/GP [148], and the fixed spaces of Section 4.9 on the six integral matrices of Exercise 4.97 over $\mathbb{Q}$ and over $\mathbb{F}_3$.

# Part II

# Number fields and local fields

# Chapter 5

# Galois theory, finite and infinite

This chapter builds Galois theory from the definition of a field extension to the absolute Galois group $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, its continuous representations and the Chebotarev density theorem. The paper uses it in five places: the hypothesis of Theorem thm:main (absolute irreducibility on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$), Proposition prop:ns (the determinant cuts out $\mathbb{Q}(\sqrt{-3})$, and the 3-division polynomial factors according to the orbits on the four lines), Lemma lem:old (Chebotarev and Brauer-Nesbitt), Lemmas lem:tamagawa-exp and lem:tamfactor (Kummer theory and the Kummer cocycle), and Lemma lem:h0 (the cyclotomic character $\chi_\ell$ against the quadratic character $\chi_{\ell^*}$). Group theory, profinite groups, rings, finite fields and representation theory are taken from Chapter 1, Chapter 2, Chapter 3 and Chapter 4; prime ideals and Frobenius elements in number fields are Chapter 6, local fields Chapter 7.

## Core results at a glance

Throughout, $F$ is a field, $\overline{\mathbb{F}}$ an algebraic closure, $F^{\mathrm{sep}} \subseteq \overline{\mathbb{F}}$ the separable closure and $G_F = \mathrm{Gal}(F^{\mathrm{sep}}/F)$; $\overline{\mathbb{Q}} \subset \mathbb{C}$ is the field of algebraic numbers, $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and $G_K = \mathrm{Gal}(\overline{\mathbb{Q}}/K)$ for $K \subseteq \overline{\mathbb{Q}}$; $\ell$ is a prime, $\ell^* = (-1)^{(\ell-1)/2}\ell$ for odd $\ell$; $\mathcal{O}$ is the ring of integers of a finite extension of $\mathbb{Q}_\ell$, with uniformiser $\varpi$ and maximal ideal $\lambda = \varpi\mathcal{O}$.

**5.1 Conventions.** Automorphisms compose as maps, $(\sigma\tau)(x) = \sigma(\tau(x))$; $\mathrm{Frob}_q$ is $x \mapsto x^q$ on $\overline{\mathbb{F}}_q$ and $\mathrm{Frob}_p$ an arithmetic Frobenius, so $\chi_\ell(\mathrm{Frob}_p) = p$; $\chi_d$ ($d \in \mathbb{Q}^\times$ a non-square) is the character of $\mathbb{Q}(\sqrt{d})$, $\chi_{q^*}$ that of conductor $q$ (never $\chi_q$); $\chi_\ell$ is the $\ell$-adic cyclotomic character, $\bar{\chi}_\ell$ its reduction, $\chi_3^{\mathrm{cyc}}$ at the prime 3; Kummer characters take values in $\mu_n$; $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$. Conventions 5.1 to 5.4.

**5.2 Extensions and the tower law** (definition and proposition). $[L : F] = \dim_F L$; for $F \subseteq L \subseteq M$, $[M : F] = [M : L][L : F]$, products of bases being bases. *(proved here)* Definition 5.5, Proposition 5.6.

**5.3 Minimal polynomial.** If $\alpha$ is algebraic over $F$, the kernel of $F[x] \to L$, $x \mapsto \alpha$, is generated by a unique monic irreducible $m_{\alpha,F}$; $F(\alpha) = F[\alpha] \simeq F[x]/(m_{\alpha,F})$, $[F(\alpha) : F] = \deg m_{\alpha,F}$, and $g(\alpha) = 0$ iff $m_{\alpha,F} \mid g$; $\alpha$ is algebraic iff $[F(\alpha) : F] < \infty$. *(proved here)* Definition 5.7, Proposition 5.8.

**5.4 Algebraic extensions.** Finite extensions are algebraic; $F(\alpha_1, \ldots, \alpha_r)$ with $\alpha_i$ algebraic

is finite of degree at most $\prod \deg m_{\alpha_i,F}$; algebraic elements form a subfield; algebraic over algebraic is algebraic. *(proved here)* Proposition 5.9.

**5.5 Algebraic closure.** (a) Every field has an algebraic closure, unique up to (non-unique) isomorphism. (b) An embedding $F \to \Omega$, $\Omega$ algebraically closed, extends to any algebraic $L/F$. (c) $\overline{\mathbb{Q}} = \{z \in \mathbb{C} : z \text{ algebraic over } \mathbb{Q}\}$ is an algebraic closure of $\mathbb{Q}$, countable. (d) Every $F$-embedding $\overline{\mathbb{F}} \to \overline{\mathbb{F}}$ is an automorphism, so every $F$-embedding $K \to \overline{\mathbb{F}}$ of a subextension $K \subseteq \overline{\mathbb{F}}$ extends to an element of $\mathrm{Aut}(\overline{\mathbb{F}}/F)$. *((a) and (b) for infinite $L$ proof sketched here; (b) for finite $L$, (c) and (d) given (b) proved here)* Theorem 5.10.

**5.6 Splitting fields.** A splitting field of $f \in F[x]$ exists, of degree at most $n!$; an isomorphism $F \to F'$ extends to an isomorphism of splitting fields of $f$ and $\phi(f)$, in at most $[L : F]$ ways, exactly $[L : F]$ if $f$ is separable. *(proved here; families of polynomials sketched)* Theorem 5.12.

**5.7 Normal extensions.** For $L/F$ algebraic inside $\overline{\mathbb{F}}$, equivalent: every irreducible polynomial over $F$ with a root in $L$ splits in $L$; $L$ is a splitting field of a family; every $F$-embedding $L \to \overline{\mathbb{F}}$ has image $L$. Normal closures exist; normality is not transitive. *(proved here)* Definition 5.13, Proposition 5.14.

**5.8 Separability.** An irreducible $f$ is inseparable iff $f' = 0$ iff $\mathrm{char}\, F = p > 0$ and $f \in F[x^p]$; $F$ is perfect iff $\mathrm{char}\, F = 0$ or $F = F^p$; $[L : F]_s = \#\mathrm{Emb}_F(L, \overline{\mathbb{F}})$ divides $[L : F]$, with equality iff $L/F$ is separable, and is multiplicative in towers. *(proved here)* Definition 5.15, Proposition 5.16.

**5.9 Primitive element theorem.** A finite separable extension is simple; a finite extension is simple iff it has finitely many intermediate fields. *(proved here)* Theorem 5.17.

**5.10 Separable closure.** $F^{\mathrm{sep}}$ is a field, normal and separable over $F$, equal to $\overline{\mathbb{F}}$ when $F$ is perfect. *(proved here)* Proposition 5.18.

**5.11 Dedekind's lemma.** Distinct homomorphisms $G \to L^\times$ from a monoid are linearly independent over $L$. *(proved here)* Theorem 5.19.

**5.12 Artin's theorem.** For a finite group $G$ of automorphisms of $L$, $L/L^G$ is finite, normal and separable of degree $\#G$, and $\mathrm{Aut}(L/L^G) = G$. For any finite $L/F$, $\#\,\mathrm{Aut}(L/F) \le [L : F]$, with equality iff $L/F$ is normal and separable. *(proved here)* Theorem 5.20.

**5.13 Galois extensions** (definition and theorem). $L/F$ algebraic is Galois if normal and separable; for $L/F$ finite: Galois iff splitting field of a separable polynomial iff $L^{\mathrm{Aut}(L/F)} = F$ iff $\#\,\mathrm{Aut}(L/F) = [L : F]$. *(proved here)* Definition 5.21, Theorem 5.22.

**5.14 Fundamental theorem.** For $L/F$ finite Galois with group $G$, $H \mapsto L^H$ and $M \mapsto \mathrm{Gal}(L/M)$ are inverse inclusion-reversing bijections; $[M : F]$ is the index of $\mathrm{Gal}(L/M)$; $\mathrm{Gal}(L/\sigma M) = \sigma\,\mathrm{Gal}(L/M)\sigma^{-1}$; $M/F$ is Galois iff $\mathrm{Gal}(L/M)$ is normal in $G$, and then $G/\,\mathrm{Gal}(L/M) \simeq \mathrm{Gal}(M/F)$. *(proved here)* Theorem 5.23.

**5.15 Translation.** For $L/F$ Galois and any extension $M/F$, the extension $LM/M$ is Galois and restriction is an isomorphism $\mathrm{Gal}(LM/M) \simeq \mathrm{Gal}(L/L\cap M)$; $\mathrm{Gal}(L_1L_2/F) \to \mathrm{Gal}(L_1/F) \times \mathrm{Gal}(L_2/F)$ is injective, bijective iff $L_1 \cap L_2 = F$. *(proved here; the infinite case in Corollary 5.57)* Proposition 5.24.

**5.16 Finite fields.** Inside $\overline{\mathbb{F}}_p$, $\mathbb{F}_{q^n}$ is the unique extension of $\mathbb{F}_q$ of degree $n$; it is Galois with cyclic group generated by $\mathrm{Frob}_q$; its subfields are the $\mathbb{F}_{q^d}$, $d \mid n$; an irreducible polynomial of degree $d$ over $\mathbb{F}_q$ has the roots $\alpha^{q^i}$, $0 \le i < d$. *(proved here)* Theorem 5.25.

**5.17 Cyclotomic polynomials modulo $p$.** For $p \nmid n$ and $f$ the order of $p$ in $(\mathbb{Z}/n)^\times$, $\Phi_n \bmod p$ is the product of $\varphi(n)/f$ distinct irreducible polynomials of degree $f$, and $\mathbb{F}_p(\mu_n) = \mathbb{F}_{p^f}$. *(proved here)* Proposition 5.26.

**5.18 Quadratic extensions.** If $\mathrm{char}\, F \neq 2$, the quadratic extensions are the $F(\sqrt{d})$, in bijection with the nontrivial elements of $F^\times/F^{\times 2}$; two independent classes give a $V_4$-extension with three quadratic subfields. *(proved here)* Proposition 5.27.

**5.19 Cyclotomic fields.** $\Phi_n$ is irreducible over $\mathbb{Q}$; $a \mapsto \sigma_a$, $\sigma_a(\zeta) = \zeta^a$ on $\mu_n$, is an isomorphism $(\mathbb{Z}/n)^\times \simeq \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$; $\mathbb{Q}(\zeta_m) \cap \mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{\gcd(m,n)})$, $\mathbb{Q}(\zeta_m)\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{\mathrm{lcm}(m,n)})$; $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$. *(proved here)* Theorem 5.28.

**5.20 $\mathbb{Q}(\sqrt{\ell^*})$.** For odd $\ell$, $\mathbb{Q}(\sqrt{\ell^*})$ is the unique quadratic subfield of $\mathbb{Q}(\zeta_\ell)$, $\sigma_a(\sqrt{\ell^*}) = (\frac{a}{\ell})\sqrt{\ell^*}$, so $\chi_{\ell^*} = \bar{\chi}_\ell^{(\ell-1)/2}$; $\chi_{\ell^*} = \bar{\chi}_\ell$ iff $\ell = 3$; every quadratic field lies in $\mathbb{Q}(\zeta_{|D|})$, $D$ its discriminant. *(proved here)* Proposition 5.29.

**5.21 Norm and trace** (definition and proposition). $\mathrm{N}_{L/F}$, $\mathrm{Tr}_{L/F}$ are the determinant and trace of multiplication, the product and sum of the conjugates; multiplicative, additive, transitive. *(proved here)* Definition 5.32, Proposition 5.33.

**5.22 Hilbert's Theorem 90.** For $L/F$ cyclic with generator $\sigma$: $\mathrm{N}(b) = 1$ iff $b = c/\sigma(c)$; $\mathrm{Tr}(b) = 0$ iff $b = c - \sigma(c)$; for $L/F$ finite Galois every $(a_\sigma)$ with $a_{\sigma\tau} = a_\sigma \sigma(a_\tau)$ is $a_\sigma = \sigma(c)/c$. *(proved here)* Theorem 5.34.

**5.23 Kummer characters.** If $\mathrm{char}\, F \nmid n$ and $\mu_n \subset F$, $\chi_a(\sigma) = \sigma(\alpha)/\alpha$ $(\alpha^n = a)$ is a homomorphism to $\mu_n$ independent of $\alpha$, with kernel $\mathrm{Gal}(L/F(\alpha))$; $[F(\alpha) : F]$ is the order of $a$ in $F^\times/F^{\times n}$; $\chi_a = 1$ iff $a \in F^{\times n}$; $x^n - a$ is irreducible iff $a \notin F^{\times p}$ for every prime $p \mid n$. *(proved here; Capelli's theorem stated)* Proposition 5.35.

**5.24 Cyclic extensions.** If $\mathrm{char}\, F \nmid n$ and $\mu_n \subset F$, every cyclic extension of degree $n$ is $F(\alpha)$ with $\alpha^n \in F$. *(proved here)* Theorem 5.36.

**5.25 Kummer theory.** If $\mathrm{char}\, F \nmid n$ and $\mu_n \subset F$, $\Delta \mapsto F(\Delta^{1/n})$ is a bijection from the groups $F^{\times n} \subseteq \Delta \subseteq F^\times$ with $\Delta/F^{\times n}$ finite to the finite abelian extensions of exponent dividing $n$, and $\mathrm{Gal}(F(\Delta^{1/n})/F) \times \Delta/F^{\times n} \to \mu_n$ is a perfect pairing. *(proved here)* Theorem 5.37.

**5.26 The affine shape.** Without $\mu_n \subset F$, $\sigma \mapsto \left(\begin{smallmatrix} u(\sigma) & b(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ embeds $\mathrm{Gal}(F(\mu_n, a^{1/n})/F)$ in $\mathbb{Z}/n \rtimes (\mathbb{Z}/n)^\times$. *(proved here)* Proposition 5.38.

**5.27 Kummer theory over $\mathbb{F}_q$.** For $\ell \nmid q$, $v = v_\ell(q-1)$, $u \in \mathbb{F}_q^\times$, $k \le v$: $\chi_u(\mathrm{Frob}_q) = u^{(q-1)/\ell^k}$; $u \mapsto u^{(q-1)/\ell^v}$ identifies $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$ with $\mu_{\ell^v}(\mathbb{F}_q)$; the $\ell$-adic value $\kappa_u(\mathrm{Frob}_q) \in \mathbb{Z}_\ell(1)/(q-1)$ is the class of $u$, of valuation $\ge k$ iff $u \in (\mathbb{F}_q^\times)^{\ell^k}$. *(proved here)* Proposition 5.39.

**5.28 Artin-Schreier theory.** In characteristic $p$, $x^p - x - a$ is split or irreducible with cyclic group of order $p$, and every cyclic extension of degree $p$ is of this form; $\mathrm{Hom}_{\mathrm{cont}}(G_F, \mathbb{Z}/p) \simeq F/\wp(F)$. *((a), (b) proved here; (c) proof sketched here)* Theorem 5.41.

**5.29 Galois group of a polynomial.** $G_f$ acts faithfully on the roots; its orbits are the root sets of the irreducible factors, of sizes $[F(r):F]$; over an intermediate field $M$ of a Galois $\Omega/F$, the irreducible factors over $M$ are the $\mathrm{Gal}(\Omega/M)$-orbits. *(proved here)* Proposition 5.42.

**5.30 Discriminant** (definition and proposition). For $f = a\prod_i (x - r_i)$,

$$\operatorname{disc}(f) = a^{2n-2}\prod_{i<j}(r_i - r_j)^2 = (-1)^{n(n-1)/2}a^{n-2}\prod_i f'(r_i);$$

if $\operatorname{char} F \neq 2$, $G_f \subseteq A_n$ iff $\operatorname{disc} f \in F^{\times 2}$, and $F(\sqrt{\operatorname{disc} f})$ is cut out by the sign character; $\operatorname{disc}(x^3 + px + q) = -4p^3 - 27q^2$, $\operatorname{disc}(x^n - a) = (-1)^{(n-1)(n-2)/2}n^n a^{n-1}$, $\operatorname{disc}(\Phi_\ell) = (-1)^{(\ell-1)/2}\ell^{\ell-2}$. *(proved here)* Definition 5.43, Proposition 5.44.

**5.31 Cubics.** An irreducible separable cubic has group $A_3$ or $S_3$ according as its discriminant is a square or not. *(proved here)* Proposition 5.45.

**5.32 Quartics.** The resolvent cubic $R_f = y^3 - by^2 + (ac - 4d)y - (a^2d - 4bd + c^2)$ has $\operatorname{disc} R_f = \operatorname{disc} f$; an irreducible quartic has group $S_4$, $A_4$, $V_4$, $D_4$ or $C_4$, read off from $R_f$, $\operatorname{disc} f$, and the factorisation of $f$ over $F(\sqrt{\operatorname{disc} f})$. *(proved here)* Proposition 5.46.

**5.33 Quartics and the four lines.** If the roots of a separable quartic $f$ correspond $G_F$-equivariantly to the four lines of $V = \mathbb{F}_3^2$ under $\rho\colon G_F \to \mathrm{GL}(V)$ with image $H$, then $G_f$ is the image of $H$ in $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$, the factors of $f$ are the orbits of $H$ on the lines, $F(\sqrt{\operatorname{disc} f})$ is cut out by $\det\rho$; the image is $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$ (when $H$ fixes no line and contains an element of determinant $-1$ and trace 0) according as $f$ is irreducible with group $S_4$, irreducible with group $D_4$, or a product of two irreducible quadratics. *(proved here)* Proposition 5.47.

**5.34 Nonempty inverse limits.** An inverse limit, over a directed set, of nonempty finite sets is nonempty. *(proved here)* Lemma 5.50.

**5.35 Infinite Galois extensions.** $\mathrm{Gal}(L/F) = \varprojlim \mathrm{Gal}(K/F)$ over the finite Galois $K \subseteq L$, a profinite group in the Krull topology. *(proved here)* Proposition 5.51.

**5.36 Krull's fundamental theorem.** Closed subgroups correspond to intermediate fields; $\mathrm{Gal}(L/L^H) = \overline{H}$; open subgroups correspond to finite subextensions, normal closed subgroups to Galois subextensions, with $G/\mathrm{Gal}(L/M) \simeq \mathrm{Gal}(M/F)$ topologically. *(proved here)* Theorem 5.52.

**5.37 The absolute Galois group** (definition and proposition). $G_K \subseteq G_F$ is open of index $[K:F]$ for $K/F$ finite separable, every open subgroup is one, $G_K \trianglelefteq G_F$ iff $K/F$ is Galois; $G_{\mathbb{F}_q} \simeq \widehat{\mathbb{Z}}$ with $1 \mapsto \mathrm{Frob}_q$; $G_{\mathbb{Q}}^{\mathrm{ab}} \simeq \widehat{\mathbb{Z}}^\times$. *(proved here, except the last statement, stated; reference)* Definition 5.53, Proposition 5.54.

**5.38 Absolute Kummer theory.** If $\operatorname{char} F \nmid n$ and $\mu_n \subset F$, $a \mapsto \chi_a$ is an isomorphism $F^\times/F^{\times n} \simeq \mathrm{Hom}_{\mathrm{cont}}(G_F, \mu_n)$; for $n = 2$, $\operatorname{char} F \neq 2$: $F^\times/F^{\times 2} \simeq \mathrm{Hom}_{\mathrm{cont}}(G_F, \{\pm 1\})$. *(proved here)* Corollary 5.56.

**5.39 Translation, infinite case.** The isomorphism $\mathrm{Gal}(LM/M) \simeq \mathrm{Gal}(L/L \cap M)$ holds for infinite $L/F$, as topological groups. *(proved here)* Corollary 5.57.

**5.40 The cyclotomic character** (definition and proposition). $\chi_\ell\colon G_{\mathbb{Q}} \to \mathbb{Z}_\ell^\times$, $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ on $\mu_{\ell^\infty}$, is continuous and surjective with kernel $G_{\mathbb{Q}(\mu_{\ell^\infty})}$; $\bar\chi_\ell$ is surjective onto $\mathbb{F}_\ell^\times$ with kernel $G_{\mathbb{Q}(\zeta_\ell)}$; over any $F$ of characteristic $\neq \ell$, $G_F$ acts on $\mathbb{Z}_\ell(1)$ through $\chi_\ell$, and $\chi_\ell(\mathrm{Frob}_q) = q$ on $G_{\mathbb{F}_q}$. *(proved here)* Definition 5.58, Proposition 5.59.

**5.41 Complex conjugation.** $c \in G_{\mathbb{Q}}$ has order 2, $\chi_\ell(c) = -1$, $\chi_d(c) = -1$ iff $d < 0$; if $\det\rho(c) = -1$ for $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(k)$, $\mathrm{char}\, k \neq 2$, then $\rho(c) \sim \mathrm{diag}(1,-1)$ has trace 0 and $\rho$ is irreducible iff absolutely irreducible. Every nontrivial element of finite order of $G_{\mathbb{Q}}$ is a complex conjugation. *(proved here, except the last statement, stated; reference)* Proposition 5.60.

**5.42 Continuous representations** (definition and proposition). Over a discrete field, continuous means open kernel, and the field cut out is finite Galois with group the image; over $\mathcal{O}$, continuous means continuous modulo every $\varpi^m$; every continuous representation over a finite extension $E_\lambda$ of $\mathbb{Q}_\ell$ preserves a lattice; continuous complex representations of profinite groups have finite image. *(proved here)* Definition 5.61, Proposition 5.62.

**5.43 Quadratic characters of $G_{\mathbb{Q}}$.** $d \mapsto \chi_d$ is an isomorphism $\mathbb{Q}^\times/\mathbb{Q}^{\times 2} \simeq \mathrm{Hom}_{\mathrm{cont}}(G_{\mathbb{Q}}, \{\pm 1\})$, $\ker\chi_d = G_{\mathbb{Q}(\sqrt{d})}$, $\chi_{dd'} = \chi_d\chi_{d'}$; for $p \nmid D$, $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$ and $\chi_{\ell^*}(\mathrm{Frob}_p) = \left(\frac{p}{\ell}\right)$. *((a), (b) proved here; (c), (d) stated; reference)* Proposition 5.63.

**5.44 Twists.** $\det(\rho\otimes\chi) = \chi^n \det\rho$; if $\rho \simeq \rho\otimes\chi$ in dimension 2 then $\chi^2 = 1$; for $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$, $\ell$ odd, $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$ with $k \geq 1$ forces $\ell = 3$, $k = 1$. *(proved here)* Proposition 5.64.

**5.45 Restriction to open subgroups.** $\rho(G_K)$ has index at most $[K:F]$ in $\rho(G_F)$; if $K$ is cut out by $\delta \circ \rho$ then $\rho(G_K) = \rho(G_F) \cap \ker\delta$; for $E/\mathbb{Q}$, $\bar\rho_{E,\ell}(G_{\mathbb{Q}(\sqrt{\ell^*})})$ is the set of elements of the image with square determinant, the determinant-one subgroup when $\ell = 3$. *(proved here)* Proposition 5.66.

**5.46 Index two.** For $H \subseteq G$ open of index 2 with character $\chi$ and $\rho$ two-dimensional, absolutely irreducible, $\mathrm{char}\, k \neq 2$: $\rho|_H$ is not absolutely irreducible iff $\rho \simeq \rho \otimes \chi$ iff $\rho \otimes \bar k \simeq \mathrm{Ind}_H^G \psi$ for a continuous character $\psi$ of $H$. *(proved here)* Proposition 5.67.

**5.47 Reduction.** Reductions of two stable lattices have isomorphic semisimplifications; if one is irreducible the lattices are homothetic. *(proved here)* Proposition 5.69.

**5.48 The $\ell$-adic Kummer cocycle.** For $a \in F^\times$ and $\ell \neq \mathrm{char}\, F$, $\kappa_a(\sigma) = (\sigma(\alpha_k)/\alpha_k)_k \in \mathbb{Z}_\ell(1)$ satisfies $\kappa_a(\sigma\tau) = \kappa_a(\sigma) + \chi_\ell(\sigma)\kappa_a(\tau)$, so, writing $\kappa_a = b \cdot e$ for a generator $e$ of $\mathbb{Z}_\ell(1)$, $\sigma \mapsto \left(\begin{smallmatrix} \chi_\ell(\sigma) & b(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ is a continuous representation; it changes by a coboundary with the root system; $\kappa_a(\sigma\tau\sigma^{-1}) = \chi_\ell(\sigma)\kappa_a(\tau)$ for $\tau \in \ker\chi_\ell$; if $\mu_{\ell^k} \subset K$, $\kappa_a \bmod \ell^k$ vanishes on $G_K$ iff $a \in K^{\times \ell^k}$. *(proved here)* Proposition 5.70.

**5.49 Frobenius elements.** For $K/\mathbb{Q}$ finite Galois unramified at $p$, the Frobenius elements at the primes above $p$ form a conjugacy class $\mathrm{Frob}_p$, trivial iff $p$ splits completely, compatible with quotients; $\mathrm{Frob}_p = \sigma_p$ in $\mathbb{Q}(\zeta_n)$; $\mathrm{Frob}_p$ acts on the roots of $f$ with the cycle type of the factorisation of $f \bmod p$ (Dedekind). *(stated; reference)* Theorem 5.72.

**5.50 Density** (definition). Dirichlet and natural density; density one; almost all. Definition 5.74.

**5.51 Chebotarev's density theorem.** For $K/\mathbb{Q}$ finite Galois with group $G$ and a conjugacy class $C$, the primes with $\mathrm{Frob}_p = C$ have density $\#C/\#G$. *(stated; reference)* Theorem 5.75.

**5.52 Density of Frobenius elements.** For $L/\mathbb{Q}$ Galois unramified outside a finite $S$ and $X$ of density one, the Frobenius elements at primes over $X \setminus S$ are dense in $\mathrm{Gal}(L/\mathbb{Q})$. *(proved here, from Chebotarev)* Corollary 5.76.

**5.53 Chebotarev and Brauer-Nesbitt.** Two continuous semisimple $n$-dimensional representations of $G_{\mathbb{Q}}$, unramified outside a finite set, with the same Frobenius characteristic polynomials on a set of density one, are isomorphic. *(proved here)* Corollary 5.77.

**5.54 Congruent traces.** If $\rho_1, \rho_2 \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathcal{O})$ have congruent traces and determinants of Frobenius modulo $\lambda$ on a set of density one, then $\bar\rho_1^{ss} \simeq \bar\rho_2^{ss}$, and $\bar\rho_2 \simeq \bar\rho_1$ if $\bar\rho_1$ is absolutely irreducible; (d) for a newform $h$ of weight 2 on $\Gamma_0(M)$ and $\ell$ odd, $a_p(h) \equiv a_p(E) \pmod\lambda$ on a set of density one gives $\bar\rho_h \simeq \bar\rho_{E,\ell} \otimes k$ when $\bar\rho_{E,\ell}$ is irreducible. *((a) to (c) proved here; the newform statement (d) proved here, given the representations of newforms of Chapter 15)* Corollary 5.78.

## Where the paper uses this chapter

- **The hypothesis of Theorem thm:main** ("$\bar\rho_{E,\ell}$ absolutely irreducible when restricted to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$"), **the set $S_f$ of Section 4.2, Proposition prop:Sf and the quoted DDT Theorem 3.15**: $G_{\mathbb{Q}}$ and its open subgroups, Proposition 5.54; the field $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$, Proposition 5.29; restriction, Proposition 5.66; the index-two equivalence, Proposition 5.67; irreducible equals absolutely irreducible, Proposition 5.60.

- **Proposition prop:ns** (the image $N_s(\mathbb{F}_3)$): $\det\bar\rho_{E,3} = \chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$ and the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ is the determinant-one subgroup, Proposition 5.66; the induced case, Proposition 5.67; the factorisation of the 3-division polynomial, Proposition 5.42, Proposition 5.44, Proposition 5.47, Example 5.49; complex conjugation, Proposition 5.60.

- **Lemma lem:h0** (the term $h_\ell(E)$) **and Section 1.3 (Conjecture conj:euler)**: $\chi_\ell(G_{\mathbb{Q}}) = \mathbb{Z}_\ell^\times$, Proposition 5.59; the determinant argument forcing $\ell = 3$, $k = 1$, Proposition 5.64; $\bar\chi_3 = \chi_{-3}$ but $\bar\chi_\ell \neq \chi_{\ell^*}$ for $\ell \geq 5$, Proposition 5.29; induction from $\mathbb{Q}(\sqrt{-3})$, Proposition 5.67.

- **Lemma lem:old** ("by Chebotarev and Brauer-Nesbitt"): Theorem 5.72, Theorem 5.75, Corollary 5.76, Corollary 5.77, Corollary 5.78, Proposition 5.69.

- **Lemma lem:tamagawa-exp** (Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$): Proposition 5.35, Theorem 5.37, Proposition 5.70(d).

- **Lemma lem:tamfactor** (the Kummer cocycle of the Tate parameter): Proposition 5.70, Proposition 5.38, Proposition 5.39, Proposition 5.59.

- **Section 1 (the table), Lemma lem:euler, Lemma lem:twist, Theorem thm:twist**: the characters $\chi_{-3}$, $\chi_{-4}$, $\chi_{q^*}$ and $\chi_S$, Proposition 5.63, Proposition 5.27; $a_p(f \otimes \chi) = \chi(p)a_p(f)$ as a twist, Proposition 5.64.

- **Section 1.1 and Section 4** ($\rho \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$, $\bar\rho$, $\mathbb{Z}_\ell(1)$, $T(1)$): Proposition 5.62, Proposition 5.59.

- **Section sec:data and the appendix** (the mod 3 image labels checked against the 3-division polynomial, the test `ns3`): Proposition 5.47, Example 5.73.

## 5.1 Field extensions and algebraic elements

This section fixes the conventions of the chapter and introduces the basic objects: field extensions, their degree, algebraic elements and their minimal polynomials, and the algebraic closures $\overline{\mathbb{F}}$ and $\overline{\mathbb{Q}}$. Everything in the paper takes place inside $\overline{\mathbb{Q}}$ or inside an algebraic closure of $\mathbb{Q}_q$ or $\mathbb{F}_q$: the Galois group $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ of Section 1.1 of the paper, the fields $\mathbb{Q}(\sqrt{\ell^*})$ and $\mathbb{Q}(\sqrt{-3})$ of Theorem thm:main and Proposition prop:ns, and the unramified extension $\mathbb{Q}_q^{\mathrm{ur}}$ of Lemma lem:tamagawa-exp. Fields, rings, ideals, polynomial rings, Gauss's lemma, Kronecker's construction $F[x]/(f)$ and finite fields are Chapter 2.

**Convention 5.1** (Fields, closures, composition)**.** $\overline{\mathbb{Q}} \subset \mathbb{C}$ is the field of complex numbers algebraic over $\mathbb{Q}$ (Theorem 5.10); every algebraic extension of $\mathbb{Q}$ is taken inside $\overline{\mathbb{Q}}$, $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and $G_K = \mathrm{Gal}(\overline{\mathbb{Q}}/K) \subseteq G_{\mathbb{Q}}$ for $K \subseteq \overline{\mathbb{Q}}$. For a general field $F$ we fix an algebraic closure $\overline{\mathbb{F}}$ and the separable closure $F^{\mathrm{sep}} \subseteq \overline{\mathbb{F}}$, and put $G_F = \mathrm{Gal}(F^{\mathrm{sep}}/F)$ (Definition 5.53). Automorphisms are composed as maps, $(\sigma\tau)(x) = \sigma(\tau(x))$, and groups act on the left. As in the rest of the book (Appendix B, convention B-C1), $p$, $q$, $\ell$ are primes, except that in statements about finite fields $q$ is a power of a prime $p$; in the paper's applications $q$ is a bad prime of $E$ and $\mathbb{F}_q$ its residue field, and the two meanings never occur in one display.

**Convention 5.2** (Frobenius is arithmetic)**.** For a finite field $\mathbb{F}_q$, $\mathrm{Frob}_q \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ is $x \mapsto x^q$. A Frobenius element $\mathrm{Frob}_p$ at a prime $p$ (Theorem 5.72, made precise in Chapter 6) is *arithmetic*: it induces $x \mapsto x^p$ on residue fields, so it acts on $\mu_n$ $(p \nmid n)$ by $\zeta \mapsto \zeta^p$ and $\chi_\ell(\mathrm{Frob}_p) = p$ for $\ell \neq p$. This is the convention of the proof of Lemma lem:tamfactor; the geometric Frobenius $\mathrm{Frob}_p^{-1}$ enters only through $L$-factors, in later chapters.

**Convention 5.3** (Characters)**.** (a) For $d \in \mathbb{Q}^\times$ not a square, $\chi_d\colon G_{\mathbb{Q}} \to \{\pm 1\}$ is $\sigma \mapsto \sigma(\sqrt{d})/\sqrt{d}$. It depends only on the class of $d$ modulo squares, and its kernel is $G_{\mathbb{Q}(\sqrt{d})}$. For an odd prime $q$, $q^* = (-1)^{(q-1)/2}q$ and $\chi_{q^*}$ is the quadratic character of conductor $q$, the character of $\mathbb{Q}(\sqrt{q^*})$; the book never writes $\chi_q$ for it, and neither does the paper. $\chi_{-3}$ and $\chi_{-4}$ are the characters of $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(i)$. A numerical subscript always means such a quadratic character, written with the sign of $d$.

(b) $\chi_\ell\colon G_{\mathbb{Q}} \to \mathbb{Z}_\ell^\times$ is the $\ell$-adic cyclotomic character, $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ on $\mu_{\ell^\infty}$ (Definition 5.58); $\bar{\chi}_\ell\colon G_{\mathbb{Q}} \to \mathbb{F}_\ell^\times$ is its reduction. A prime-letter subscript ($\chi_\ell$, $\chi_p$) always means the cyclotomic character; at the number 3 it is written $\chi_3^{\mathrm{cyc}}$, and its reduction $\bar{\chi}_3$. We shall prove $\chi_{\ell^*} = \bar{\chi}_\ell^{(\ell-1)/2}$, which equals $\bar{\chi}_\ell$ only for $\ell = 3$ (Proposition 5.29).

(c) A Kummer character $\chi_a$ (Proposition 5.35) takes values in the group $\mu_n$ of roots of unity, not in $\mathbb{Z}/n$; the two are identified only after a primitive $n$-th root of unity is chosen. $\mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}$, with transition maps $\zeta \mapsto \zeta^\ell$, is written multiplicatively as a set of compatible systems $(\zeta_k)_k$ and additively when it receives cocycles: $(\zeta_k) + (\zeta_k') = (\zeta_k\zeta_k')$ and $a \cdot (\zeta_k) = (\zeta_k^a)$ for $a \in \mathbb{Z}_\ell$ (with $\zeta_k^a$ meaning $\zeta_k^{a \bmod \ell^k}$).

**Convention 5.4** (Representations)**.** A representation of a profinite group $G$ is a continuous homomorphism $\rho\colon G \to \mathrm{GL}_n(k)$ (Definition 5.61), where $k$ is a finite field or $\overline{\mathbb{F}}_\ell$ with the discrete

topology, or a finite extension $E_\lambda$ of $\mathbb{Q}_\ell$ with its $\ell$-adic topology, or the ring of integers $\mathcal{O}$ of such an $E_\lambda$. For $\rho\colon G \to \mathrm{GL}_n(\mathcal{O})$, $\rho \bmod \varpi^m$ is the action on $\mathcal{O}^n/\varpi^m\mathcal{O}^n$ and $\bar\rho = \rho \bmod \varpi$. Irreducibility, absolute irreducibility, $\rho^{ss}$, $\rho\otimes\chi$, Ind and Res are as in Chapter 4. "$\rho$ is absolutely irreducible on $G_K$" means that $(\rho\otimes\bar k)|_{G_K}$ is irreducible.

**Definition 5.5** (Field extensions)**.** A *field extension* $L/F$ is a field $L$ together with a subfield $F$. Then $L$ is an $F$-vector space, and its dimension $[L:F] = \dim_F L \in \{1,2,3,\dots\}\cup\{\infty\}$ is the *degree* of $L/F$; the extension is *finite* if $[L:F]<\infty$. A field $M$ with $F\subseteq M\subseteq L$ is an *intermediate field*. For $S\subseteq L$, $F(S)$ is the smallest subfield of $L$ containing $F$ and $S$, and $F[S]$ the smallest subring; $F(\alpha_1,\dots,\alpha_r)$ and $F[\alpha_1,\dots,\alpha_r]$ are the usual abbreviations. For two subfields $L_1, L_2$ of a field, the *compositum* $L_1L_2$ is the smallest subfield containing both. An *$F$-embedding* of $L$ into a field $\Omega\supseteq F$ is a ring homomorphism $L\to\Omega$ that is the identity on $F$ (it is automatically injective, $L$ being a field); $\mathrm{Emb}_F(L,\Omega)$ is the set of them, and $\mathrm{Aut}(L/F)$ the group of $F$-automorphisms of $L$.

The degree is the first invariant of an extension, and it needs nothing beyond a first course in linear algebra: $L$ is simply a vector space over $F$. For example $[\mathbb{C}:\mathbb{R}]=2$, with basis $1, i$. Next, $[\mathbb{Q}(\sqrt2):\mathbb{Q}]=2$, with basis $1,\sqrt2$: the numbers $x+y\sqrt2$ $(x,y\in\mathbb{Q})$ form a ring, and a field because $(x+y\sqrt2)^{-1} = (x-y\sqrt2)/(x^2-2y^2)$ for $(x,y)\neq(0,0)$ (the denominator is nonzero as $\sqrt2\notin\mathbb{Q}$); and $1,\sqrt2$ are independent over $\mathbb{Q}$, again because $\sqrt2\notin\mathbb{Q}$. Finally $[\mathbb{R}:\mathbb{Q}]=\infty$, since a $\mathbb{Q}$-vector space of finite dimension $d$ is in bijection with $\mathbb{Q}^d$, hence countable, while $\mathbb{R}$ is not. The tower law below turns statements about degrees into divisibility statements, and most impossibility arguments of the chapter are of that kind.

**Proposition 5.6** (Tower law)**.** *Let $F\subseteq L\subseteq M$ be fields. If $(x_i)_{i\in I}$ is an $F$-basis of $L$ and $(y_j)_{j\in J}$ an $L$-basis of $M$, then $(x_iy_j)_{(i,j)\in I\times J}$ is an $F$-basis of $M$. Hence $[M:F]=[M:L]\,[L:F]$, with the convention $\infty\cdot n = n\cdot\infty=\infty$; in particular $M/F$ is finite iff $M/L$ and $L/F$ are, and then $[L:F]$ and $[M:L]$ divide $[M:F]$.*

*Proof.* Spanning: an element $z\in M$ is $z=\sum_j b_jy_j$ with $b_j\in L$ (finitely many nonzero), and each $b_j=\sum_i a_{ij}x_i$ with $a_{ij}\in F$; so $z=\sum_{i,j}a_{ij}x_iy_j$. Independence: if $\sum_{i,j}a_{ij}x_iy_j=0$ with $a_{ij}\in F$, almost all zero, then $\sum_j(\sum_i a_{ij}x_i)y_j=0$ with coefficients in $L$, so $\sum_i a_{ij}x_i=0$ for every $j$, and then $a_{ij}=0$ for all $i,j$. The degree formula is the count of $I\times J$. □

A typical use: $\sqrt2\notin\mathbb{Q}(2^{1/3})$. Indeed $[\mathbb{Q}(\sqrt2):\mathbb{Q}]=2$ and $[\mathbb{Q}(2^{1/3}):\mathbb{Q}]=3$ (the minimal polynomials are $x^2-2$ and $x^3-2$, Proposition 5.8 below), and if $\mathbb{Q}(\sqrt2)\subseteq\mathbb{Q}(2^{1/3})$ the tower law would make 2 divide 3. In the same way $[\mathbb{Q}(\sqrt2,2^{1/3}):\mathbb{Q}]=6$: it is divisible by 2 and by 3, and it is at most $2\cdot3$, because $2^{1/3}$ is a root of $x^3-2$ over $\mathbb{Q}(\sqrt2)$, so $[\mathbb{Q}(\sqrt2,2^{1/3}):\mathbb{Q}(\sqrt2)]\le3$ (Proposition 5.8(ii), (iv)).

**Definition 5.7** (Algebraic elements)**.** Let $L/F$ be an extension. An element $\alpha\in L$ is *algebraic* over $F$ if $g(\alpha)=0$ for some nonzero $g\in F[x]$, and *transcendental* otherwise. The extension $L/F$ is *algebraic* if every element of $L$ is algebraic over $F$.

**Proposition 5.8** (Minimal polynomial)**.** *Let $L/F$ be an extension and $\alpha\in L$ algebraic over $F$.*

*(i) The kernel of the evaluation map $F[x]\to L$, $g\mapsto g(\alpha)$, is generated by a unique monic irreducible polynomial $m_{\alpha,F}$, the* minimal polynomial *of $\alpha$ over $F$. For $g\in F[x]$, $g(\alpha)=0$ iff $m_{\alpha,F}\mid g$; and $m_{\alpha,F}$ is the unique monic irreducible polynomial in $F[x]$ with root $\alpha$.*

*(ii)* $F[\alpha] = F(\alpha) \simeq F[x]/(m_{\alpha,F})$, *and* $1, \alpha, \dots, \alpha^{d-1}$ *is an* $F$*-basis of* $F(\alpha)$, *where* $d = \deg m_{\alpha,F}$; *so* $[F(\alpha) : F] = \deg m_{\alpha,F}$.

*(iii) Conversely, if* $\beta \in L$ *and* $[F(\beta) : F] < \infty$, *then* $\beta$ *is algebraic over* $F$.

*(iv) If* $F \subseteq M \subseteq L$, *then* $m_{\alpha,M}$ *divides* $m_{\alpha,F}$ *in* $M[x]$.

*Proof.* (i) The kernel $\mathfrak{p}$ is an ideal of the principal ideal domain $F[x]$ (Chapter 2); it is nonzero as $\alpha$ is algebraic, and prime because $F[x]/\mathfrak{p}$ embeds in the field $L$. A nonzero prime ideal of $F[x]$ is generated by an irreducible polynomial, unique up to a unit, hence unique when monic. If $h$ is monic irreducible with $h(\alpha) = 0$, then $m_{\alpha,F} \mid h$, and irreducibility forces $h = m_{\alpha,F}$. (ii) The image of evaluation is $F[\alpha]$, so $F[\alpha] \simeq F[x]/(m_{\alpha,F})$, which is a field because $(m_{\alpha,F})$ is maximal (Kronecker's construction, Chapter 2). For the basis: by division with remainder every $g \in F[x]$ is $g = s\, m_{\alpha,F} + r$ with $\deg r < d$, so $g(\alpha) = r(\alpha)$ is an $F$-linear combination of $1, \alpha, \dots, \alpha^{d-1}$; and a relation $\sum_{i<d} c_i \alpha^i = 0$ is a polynomial of degree $< d$ vanishing at $\alpha$, hence divisible by $m_{\alpha,F}$ by (i), hence zero. A subring of $L$ containing $F$ and $\alpha$ that is a field contains $F(\alpha)$; so $F[\alpha] = F(\alpha)$. (iii) If $[F(\beta) : F] = n < \infty$, the $n+1$ elements $1, \beta, \dots, \beta^n$ are linearly dependent over $F$, which is a nonzero polynomial relation. (iv) $m_{\alpha,F} \in M[x]$ has the root $\alpha$; apply (i) over $M$. □

For example $m_{\sqrt{2},\mathbb{Q}} = x^2 - 2$, $m_{2^{1/3},\mathbb{Q}} = x^3 - 2$ (Eisenstein, Chapter 2), and $m_{\zeta_n,\mathbb{Q}} = \Phi_n$ (Theorem 5.28). The number $\pi$ is transcendental over $\mathbb{Q}$ (Lindemann's theorem; we do not prove this, see [135, Chapter 1] for references); the book never uses transcendental elements except through the countability argument of Theorem 5.10(c).

**Proposition 5.9** (Algebraic extensions)**.** *Let* $L/F$ *be an extension.*

*(a) If* $L/F$ *is finite, it is algebraic.*

*(b) If* $\alpha_1, \dots, \alpha_r \in L$ *are algebraic over* $F$, *then* $F(\alpha_1, \dots, \alpha_r) = F[\alpha_1, \dots, \alpha_r]$ *is finite over* $F$, *of degree at most* $\prod_i \deg m_{\alpha_i,F}$.

*(c) The elements of* $L$ *algebraic over* $F$ *form a subfield of* $L$.

*(d) If* $M/L$ *and* $L/F$ *are algebraic, then* $M/F$ *is algebraic.*

*(e) A finite extension is generated by finitely many algebraic elements.*

*Proof.* (a) For $\alpha \in L$, $F(\alpha) \subseteq L$ has finite degree; apply Proposition 5.8(iii). (b) Induction on $r$. Put $F_i = F(\alpha_1, \dots, \alpha_i)$. The element $\alpha_{i+1}$ is a root of $m_{\alpha_{i+1},F} \in F_i[x]$, so it is algebraic over $F_i$ and $[F_{i+1} : F_i] = \deg m_{\alpha_{i+1},F_i} \le \deg m_{\alpha_{i+1},F}$ by Proposition 5.8(ii), (iv); also $F_{i+1} = F_i[\alpha_{i+1}]$. The tower law gives the bound, and by induction $F_r = F[\alpha_1, \dots, \alpha_r]$. (c) If $\alpha, \beta$ are algebraic, $F(\alpha, \beta)$ is finite by (b), so $\alpha \pm \beta$, $\alpha\beta$ and $\alpha/\beta$ ($\beta \neq 0$), which lie in it, are algebraic by (a). (d) Let $\gamma \in M$, with $m_{\gamma,L} = x^m + c_{m-1}x^{m-1} + \cdots + c_0$. The $c_i$ are algebraic over $F$, so $F' = F(c_0, \dots, c_{m-1})$ is finite over $F$ by (b), and $\gamma$ is algebraic over $F'$, so $F'(\gamma)$ is finite over $F'$. By the tower law $F(\gamma) \subseteq F'(\gamma)$ is finite over $F$, and $\gamma$ is algebraic. (e) An $F$-basis of $L$ generates $L$, and its elements are algebraic by (a). □

**Theorem 5.10** (Algebraic closure)**.** *A field* $\Omega$ *is* algebraically closed *if every nonconstant polynomial in* $\Omega[x]$ *has a root in* $\Omega$ *(equivalently, splits into linear factors). An* algebraic closure *of* $F$ *is an algebraic extension of* $F$ *that is algebraically closed.*

*(a) Every field $F$ has an algebraic closure $\overline{\mathbb{F}}$, and any two are $F$-isomorphic (by an isomorphism that is in general not unique).*

*(b) Let $L/F$ be algebraic, $\Omega$ algebraically closed and $\sigma\colon F \to \Omega$ a field embedding. Then $\sigma$ extends to an embedding $L \to \Omega$. If $L = F(\alpha)$, the extensions correspond bijectively to the distinct roots of $\sigma(m_{\alpha,F})$ in $\Omega$, through $\tau \mapsto \tau(\alpha)$.*

*(c) $\overline{\mathbb{Q}} = \{z \in \mathbb{C} : z$ is algebraic over $\mathbb{Q}\}$ is an algebraic closure of $\mathbb{Q}$. It is countable; hence $\mathbb{C}$ is not algebraic over $\mathbb{Q}$.*

*(d) Every $F$-embedding $\overline{\mathbb{F}} \to \overline{\mathbb{F}}$ is an automorphism. Hence, for every subextension $K/F$ of $\overline{\mathbb{F}}/F$, every $F$-embedding $K \to \overline{\mathbb{F}}$ extends to an element of* $\mathrm{Aut}(\overline{\mathbb{F}}/F)$.

*Proof of (b) for finite $L$, and of (c).* (b), $L = F(\alpha)$. If $\tau$ extends $\sigma$, then $0 = \tau(m_{\alpha,F}(\alpha)) = \sigma(m_{\alpha,F})(\tau(\alpha))$, so $\tau(\alpha)$ is a root of $\sigma(m_{\alpha,F})$, and $\tau$ is determined by $\tau(\alpha)$ because $L = F[\alpha]$. Conversely, let $\beta \in \Omega$ be a root of $\sigma(m_{\alpha,F})$ (one exists as $\Omega$ is algebraically closed). The homomorphism $F[x] \to \Omega$, $\sum a_i x^i \mapsto \sum \sigma(a_i)\beta^i$, kills $m_{\alpha,F}$, so it factors through $F[x]/(m_{\alpha,F}) \simeq F(\alpha)$ and gives $\tau$ with $\tau(\alpha) = \beta$. For a finite $L$, write $L = F(\alpha_1, \dots, \alpha_r)$ (Proposition 5.9(e)) and extend step by step through $F(\alpha_1, \dots, \alpha_i)$.

(c) $\overline{\mathbb{Q}}$ is a subfield of $\mathbb{C}$ by Proposition 5.9(c), algebraic over $\mathbb{Q}$ by definition. Let $g \in \overline{\mathbb{Q}}[x]$ be nonconstant. By the fundamental theorem of algebra $g$ has a root $z \in \mathbb{C}$. The coefficients of $g$ generate a finite extension $F'$ of $\mathbb{Q}$, and $z$ is algebraic over $F'$, so $F'(z)/\mathbb{Q}$ is finite (tower law) and $z$ is algebraic over $\mathbb{Q}$: $z \in \overline{\mathbb{Q}}$. Countability: $\mathbb{Q}[x]$ is countable and each nonzero polynomial has finitely many roots in $\mathbb{C}$; as $\mathbb{C}$ is uncountable, some complex number is transcendental. □

*Sketch of proof of (a), and of (b) for infinite $L$.* For (b) one applies Zorn's lemma to the set of pairs $(M, \tau)$ with $F \subseteq M \subseteq L$ and $\tau\colon M \to \Omega$ extending $\sigma$, ordered by extension: a maximal pair has $M = L$, since otherwise the case $M(\alpha)$ above extends it. For (a), one constructs an algebraically closed field containing $F$ (Artin's construction: adjoin a root of every polynomial at once, in the quotient of a large polynomial ring by a maximal ideal, and repeat countably many times, or once with an extra argument), and takes the algebraic elements in it; uniqueness follows from (b), an $F$-embedding $\overline{\mathbb{F}} \to \overline{\mathbb{F}}'$ having algebraically closed image over which $\overline{\mathbb{F}}'$ is algebraic, hence being surjective. We do not give the details; see [109, Chapter V, Section 2] or [58, Section 13.4]. □

*Proof of (d), given (b).* Let $\sigma\colon \overline{\mathbb{F}} \to \overline{\mathbb{F}}$ be an $F$-embedding and $x \in \overline{\mathbb{F}}$. The set $R$ of roots of $m_{x,F}$ in $\overline{\mathbb{F}}$ is finite, and $\sigma$ maps $R$ injectively into $R$ (as $\sigma(m_{x,F}) = m_{x,F}$); so $\sigma(R) = R$ and $x \in \sigma(\overline{\mathbb{F}})$. Thus $\sigma$ is surjective. If $K \subseteq \overline{\mathbb{F}}$ and $\tau\colon K \to \overline{\mathbb{F}}$ is an $F$-embedding, then $\overline{\mathbb{F}}/K$ is algebraic, so by (b) $\tau$ extends to an embedding $\overline{\mathbb{F}} \to \overline{\mathbb{F}}$, which is an automorphism by what we have shown. □

*Remark* 5.11. The non-uniqueness in (a) is the whole subject of this chapter: for perfect $F$ the $F$-automorphisms of $\overline{\mathbb{F}}$ are the elements of the absolute Galois group $G_F$ (Definition 5.53). For $F = \mathbb{Q}$ we avoid all choices by working inside $\mathbb{C}$; the price is that one embedding $\overline{\mathbb{Q}} \subset \mathbb{C}$ is singled out, and with it one element of $G_{\mathbb{Q}}$, complex conjugation (Proposition 5.60).

## 5.2 Splitting fields, normality, separability

A Galois extension is one that is normal (closed under conjugation) and separable (no repeated roots). This section develops the two notions and the tools used everywhere afterwards: the

extension of isomorphisms to splitting fields, the separable degree, and the primitive element theorem. In the paper, every field "cut out" by a character or a representation (the proof of Proposition prop:ns: "$\det \bar\rho_{E,3} = \chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$") is a normal extension of $\mathbb{Q}$; the warning examples $\mathbb{Q}(2^{1/3})$ and $\mathbb{Q}(2^{1/4})$ below show that an extension generated by one root need not be.

**Theorem 5.12** (Splitting fields)**.** *Let $f \in F[x]$ have degree $n \geq 1$. A* splitting field *of $f$ over $F$ is an extension $L = F(r_1, \dots, r_n)$ such that $f = c\prod_{i=1}^{n}(x - r_i)$ in $L[x]$, $c \in F^\times$.*

*(a) A splitting field exists, and $[L : F] \leq n!$.*

*(b) Let $\phi\colon F \to F'$ be an isomorphism of fields, $L$ a splitting field of $f$ over $F$ and $L'$ a splitting field of $\phi(f)$ over $F'$. Then $\phi$ extends to an isomorphism $L \to L'$. The number of extensions is at most $[L : F]$, and equal to $[L : F]$ if $f$ has $n$ distinct roots in $L$.*

*(c) Any two splitting fields of $f$ over $F$ are $F$-isomorphic; inside a fixed $\overline{\mathbb{F}}$ the splitting field $F(\{r \in \overline{\mathbb{F}} : f(r) = 0\})$ is unique.*

*The same holds for a family $\mathcal{S} \subseteq F[x]$ of nonconstant polynomials: a splitting field is an extension generated by roots of the members of $\mathcal{S}$ over which each of them splits; inside $\overline{\mathbb{F}}$ it is unique.*

*Proof.* (a) Induction on $n$, the case $n = 1$ being $L = F$. By Kronecker's construction (Chapter 2) there is an extension $F(r_1)$ in which $f$ has a root $r_1$, with $[F(r_1) : F] \leq n$ (the degree of the minimal polynomial of $r_1$, a factor of $f$). Then $f = (x - r_1)g$ with $g \in F(r_1)[x]$ of degree $n - 1$, and a splitting field $L$ of $g$ over $F(r_1)$, of degree at most $(n - 1)!$, is one of $f$ over $F$. The tower law gives $[L : F] \leq n \cdot (n - 1)!$.

(b) Induction on $[L : F]$. If $[L : F] = 1$, $f$ splits over $F$, so $\phi(f)$ splits over $F'$, $L' = F'$, and $\phi$ is the only extension. Otherwise some root $r \in L$ of $f$ is not in $F$; let $g = m_{r,F}$, of degree $d > 1$, a factor of $f$. Then $\phi(g)$ is an irreducible factor of $\phi(f)$, so it splits in $L'$; let $r'_1, \dots, r'_e$ be its distinct roots in $L'$, with $e \leq d$, and $e = d$ if $f$ (hence $g$) has distinct roots. Exactly as in the proof of Theorem 5.10(b), $\phi$ has exactly $e$ extensions $\phi_j\colon F(r) \to L'$, given by $r \mapsto r'_j$, and each is an isomorphism onto $F'(r'_j)$. Now $L$ is a splitting field of $f$ over $F(r)$, $L'$ one of $\phi(f) = \phi_j(f)$ over $F'(r'_j)$, and $[L : F(r)] = [L : F]/d < [L : F]$. By induction each $\phi_j$ extends to an isomorphism $L \to L'$, in at most $[L : F(r)]$ ways, exactly $[L : F(r)]$ if the roots are distinct. Every extension of $\phi$ to $L$ sends $r$ to some $r'_j$, so it is one of these. The total number is at most $e[L : F(r)] \leq d[L : F(r)] = [L : F]$, with equality in the separable case. (That an extension $\tau\colon L \to L'$ is surjective: $\tau(f) = \phi(f) = \phi(c)\prod(x - \tau(r_i))$, so the $\tau(r_i)$ are all the roots of $\phi(f)$ in $L'$, and they generate $L'$.)

(c) The first statement is (b) with $\phi = \mathrm{id}$. Inside $\overline{\mathbb{F}}$, $f$ splits and any splitting field contained in $\overline{\mathbb{F}}$ is generated by all the roots of $f$ in $\overline{\mathbb{F}}$.

For a family, the splitting field inside $\overline{\mathbb{F}}$ is the subfield generated by all roots in $\overline{\mathbb{F}}$ of all members of $\mathcal{S}$; existence and uniqueness up to $F$-isomorphism of abstract ones follow from Theorem 5.10 (embed into $\overline{\mathbb{F}}$). For finite families, take the product polynomial. □

The bound in (a) improves to $[L : F] \mid n!$ (Exercise 5.85).

**Definition 5.13** (Normal extensions)**.** An algebraic extension $L/F$ with $L \subseteq \overline{\mathbb{F}}$ is *normal* if it satisfies the equivalent conditions of Proposition 5.14. For $L/F$ finite, a *normal closure* of $L$ is a smallest normal extension of $F$ inside $\overline{\mathbb{F}}$ containing $L$.

**Proposition 5.14** (Normality)**.** *For an algebraic extension $F \subseteq L \subseteq \overline{\mathbb{F}}$ the following are equivalent:*

*(i) every irreducible polynomial in $F[x]$ with a root in $L$ splits into linear factors in $L[x]$;*

*(ii) $L$ is the splitting field over $F$ of a family of polynomials;*

*(iii) every $F$-embedding $\tau\colon L \to \overline{\mathbb{F}}$ satisfies $\tau(L) = L$; that is,* $\mathrm{Emb}_F(L, \overline{\mathbb{F}}) = \mathrm{Aut}(L/F)$.

*Moreover: if $L/F$ is normal and $F \subseteq M \subseteq L$, then $L/M$ is normal. A finite $L = F(\alpha_1, \dots, \alpha_r)$ has a unique normal closure inside $\overline{\mathbb{F}}$, the splitting field of $\prod_i m_{\alpha_i,F}$, finite over $F$. Every extension of degree 2 is normal, but normality is not transitive: $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(2^{1/4})$, and $\mathbb{Q}(2^{1/3})/\mathbb{Q}$ is not normal.*

*Proof.* (i)⇒(ii): $L$ is the splitting field of the family $\{m_{\alpha,F} : \alpha \in L\}$, each member splitting in $L$ by (i). (ii)⇒(iii): let $L$ be the splitting field of $\mathcal{S}$, so $L$ is generated by the set $R$ of roots in $\overline{\mathbb{F}}$ of the members of $\mathcal{S}$, and $R \subseteq L$. An $F$-embedding $\tau$ maps a root of $g \in \mathcal{S}$ to a root of $g$, so $\tau(R_g) \subseteq R_g$ for the finite set $R_g$ of roots of $g$, with equality because $\tau$ is injective. Hence $\tau(L)$ is generated by $\tau(R) = R$, and $\tau(L) = L$. (iii)⇒(i): let $g \in F[x]$ be irreducible with a root $\alpha \in L$, and $\beta \in \overline{\mathbb{F}}$ another root. By Theorem 5.10(b) there is an $F$-embedding $F(\alpha) \to \overline{\mathbb{F}}$ with $\alpha \mapsto \beta$, and it extends to an $F$-embedding $\tau\colon L \to \overline{\mathbb{F}}$. By (iii) $\tau(L) = L$, so $\beta \in L$; all roots of $g$ lie in $L$.

If $L/F$ is normal and $F \subseteq M \subseteq L$, every $M$-embedding $L \to \overline{\mathbb{F}}$ is an $F$-embedding, so (iii) holds for $L/M$. For the normal closure: the splitting field $N$ of $\prod_i m_{\alpha_i,F}$ in $\overline{\mathbb{F}}$ is normal by (ii), finite by Theorem 5.12(a), and contains $L$; a normal extension of $F$ containing $L$ contains all roots of each $m_{\alpha_i,F}$ by (i), hence contains $N$.

Let $[L : F] = 2$ and let $g \in F[x]$ be irreducible with a root $\alpha \in L$. If $\alpha \in F$, $g$ is linear. Otherwise $L = F(\alpha)$ and $g$ is (up to a constant) $m_{\alpha,F} = x^2 + bx + c$, whose second root $-b - \alpha$ lies in $L$; so (i) holds. Thus $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ and $\mathbb{Q}(2^{1/4})/\mathbb{Q}(\sqrt{2})$ are normal ($2^{1/4}$ has minimal polynomial $x^2 - \sqrt{2}$ over $\mathbb{Q}(\sqrt{2})$, as $[\mathbb{Q}(2^{1/4}) : \mathbb{Q}] = 4$ by Eisenstein). But $x^4 - 2$ is irreducible over $\mathbb{Q}$ with the root $2^{1/4} \in \mathbb{Q}(2^{1/4}) \subset \mathbb{R}$ and the non-real root $i2^{1/4}$, so $\mathbb{Q}(2^{1/4})/\mathbb{Q}$ is not normal; likewise $x^3 - 2$ has one real root and two non-real ones. □

**Definition 5.15** (Separability)**.** A polynomial $f \in F[x]$ is *separable* if it has no repeated root in $\overline{\mathbb{F}}$. An algebraic element is *separable* over $F$ if its minimal polynomial is; an algebraic extension is separable if all its elements are. $F$ is *perfect* if every irreducible polynomial over $F$ is separable. For $L/F$ finite, the *separable degree* is $[L : F]_s = \#\mathrm{Emb}_F(L, \overline{\mathbb{F}})$. The *separable closure* of $F$ in $\overline{\mathbb{F}}$ is $F^{\mathrm{sep}} = \{\alpha \in \overline{\mathbb{F}} : \alpha \text{ is separable over } F\}$.

The formal derivative $f'$ satisfies the usual rules (Chapter 2). Since the Euclidean algorithm computes $\gcd(f, f')$ inside $F[x]$, the gcd is the same whether computed over $F$ or over $\overline{\mathbb{F}}$; and $r \in \overline{\mathbb{F}}$ is a repeated root of $f$ iff $f(r) = f'(r) = 0$ (write $f = (x - r)h$, so $f' = h + (x - r)h'$). Hence *$f$ is separable iff* $\gcd(f, f') = 1$.

**Proposition 5.16** (Separability)**.** *(a) An irreducible $f \in F[x]$ is inseparable iff $f' = 0$, iff* $\mathrm{char}\, F = p > 0$ *and $f \in F[x^p]$. If* $\mathrm{char}\, F = p$*, every irreducible $f$ can be written $f(x) = g(x^{p^e})$ with $e \geq 0$ and $g$ irreducible and separable; then $f$ has exactly* $\deg g$ *distinct roots, each of multiplicity $p^e$.*

*(b) $F$ is perfect iff* $\mathrm{char}\, F = 0$*, or* $\mathrm{char}\, F = p$ *and $F = F^p$. Fields of characteristic 0, finite fields and algebraically closed fields are perfect; $\mathbb{F}_p(t)$ is not ($x^p - t$ is irreducible and inseparable).*

*(c) Let $F \subseteq K \subseteq L$ be finite extensions inside $\overline{\mathbb{F}}$. Every $F$-embedding $K \to \overline{\mathbb{F}}$ has exactly $[L : K]_s$ extensions to $L$. Hence $[L : F]_s = [L : K]_s [K : F]_s$.*

*(d) For $L/F$ finite, $[L:F]_s$ divides $[L:F]$, and $[L:F]_s = [L:F]$ iff $L/F$ is separable. In particular $L = F(\alpha_1, \dots, \alpha_r)$ with all $\alpha_i$ separable is separable over $F$, and for $F \subseteq K \subseteq L$ finite, $L/F$ is separable iff $L/K$ and $K/F$ are.*

*(e) Over a perfect field every algebraic extension is separable.*

*Proof.* (a) As $f$ is irreducible, $\gcd(f, f')$ is 1 or $f$ (up to units); it is $f$ iff $f \mid f'$, iff $f' = 0$ since $\deg f' < \deg f$. Writing $f = \sum a_i x^i$, $f' = \sum i a_i x^{i-1}$ vanishes iff $ia_i = 0$ for all $i$; in characteristic 0 this forces $f$ constant, and in characteristic $p$ it means $a_i = 0$ unless $p \mid i$. For the second statement let $e$ be the largest integer with $f \in F[x^{p^e}]$ and $f(x) = g(x^{p^e})$. A factorisation of $g$ would give one of $f$, so $g$ is irreducible, and $g \notin F[x^p]$ by maximality of $e$, so $g$ is separable. If $g = \prod_j (x - \gamma_j)$ with distinct $\gamma_j$ (leading coefficient omitted) and $\beta_j \in \overline{\mathbb{F}}$ with $\beta_j^{p^e} = \gamma_j$, then $x^{p^e} - \gamma_j = (x - \beta_j)^{p^e}$ (the $p$-th power map is additive), so $f = \prod_j (x - \beta_j)^{p^e}$ with distinct $\beta_j$.

(b) In characteristic 0 apply (a). In characteristic $p$ with $F = F^p$: an inseparable irreducible $f = \sum a_i x^{pi}$ would be $\left(\sum b_i x^i\right)^p$ with $b_i^p = a_i$, not irreducible. If $a \in F \setminus F^p$, let $\alpha^p = a$ in $\overline{\mathbb{F}}$; then $x^p - a = (x - \alpha)^p$, and a monic factor in $F[x]$ is $(x - \alpha)^k$ with $0 < k < p$, whose coefficient of $x^{k-1}$ is $-k\alpha \notin F$ (as $\alpha \notin F$ and $k \in \mathbb{F}_p^\times$); so $x^p - a$ is irreducible and inseparable. For a finite field the Frobenius $x \mapsto x^p$ is injective, hence surjective (Chapter 2); over an algebraically closed field irreducible polynomials have degree 1. For $\mathbb{F}_p(t)$, $t \notin \mathbb{F}_p(t)^p$ (compare degrees in $t$).

(c) Fix an $F$-embedding $\sigma\colon K \to \overline{\mathbb{F}}$. If $L = K(\beta)$, the extensions of $\sigma$ correspond to the distinct roots of $\sigma(m_{\beta,K})$ (Theorem 5.10(b)). Write $m_{\beta,K} = g(x^{p^e})$ as in (a) ($e = 0$ in characteristic 0). Then $\sigma(m_{\beta,K}) = \sigma(g)(x^{p^e})$, and $\sigma(g)$ is separable because $\gcd(\sigma(g), \sigma(g)') = \sigma(\gcd(g, g')) = 1$; by the argument of (a), $\sigma(m_{\beta,K})$ has exactly $\deg g$ distinct roots, a number independent of $\sigma$. Taking $\sigma$ the inclusion, it is $[L:K]_s$. For general $L = K(\beta_1, \dots, \beta_s)$ extend in steps. For the formula: every $F$-embedding of $L$ restricts to an $F$-embedding of $K$, and each of the $[K:F]_s$ embeddings of $K$ has exactly $[L:K]_s$ extensions, so $\#\mathrm{Emb}_F(L, \overline{\mathbb{F}}) = [K:F]_s[L:K]_s$.

(d) For $L = F(\alpha)$: $[L:F]_s$ is the number of distinct roots of $m_{\alpha,F}$, which is $\deg m_{\alpha,F} = [L:F]$ if $\alpha$ is separable and, by (a), $\deg g = [L:F]/p^e$ with $e \geq 1$ otherwise. So $[F(\alpha):F]_s$ divides $[F(\alpha):F]$, with equality iff $\alpha$ is separable. In general write $L = F(\alpha_1, \dots, \alpha_r)$ and use (c) and the tower law along $F_i = F(\alpha_1, \dots, \alpha_i)$: $[L:F]_s = \prod [F_{i+1}:F_i]_s$ divides $\prod [F_{i+1}:F_i] = [L:F]$. If $L/F$ is separable, each $\alpha_{i+1}$ is separable over $F$, hence over $F_i$ (its minimal polynomial over $F_i$ divides that over $F$), so equality holds factor by factor. Conversely, if $[L:F]_s = [L:F]$ and $\alpha \in L$, then $[L:F(\alpha)]_s[F(\alpha):F]_s = [L:F(\alpha)][F(\alpha):F]$ with each factor on the left at most the corresponding one on the right, so $[F(\alpha):F]_s = [F(\alpha):F]$ and $\alpha$ is separable. The last two statements follow from the equality criterion and multiplicativity.

(e) Every irreducible polynomial over a perfect field is separable. □

For the paper, separability is never an obstacle. Every field that occurs there ($\mathbb{Q}$, number fields, $\mathbb{Q}_q$ and its finite extensions such as the unramified ones, the finite residue fields $\mathbb{F}_q$, and the algebraic closures of these) has characteristic 0 or is finite, hence is perfect by (b), so every algebraic extension in sight is separable and $F^{\mathrm{sep}} = \overline{\mathbb{F}}$. The general notion is kept so that the theorems below can be stated in their natural generality. To see what goes wrong without it, let $F = \mathbb{F}_p(t)$ and $\alpha \in \overline{\mathbb{F}}$ with $\alpha^p = t$. The minimal polynomial of $\alpha$ is $x^p - t = (x - \alpha)^p$ (irreducible by (b)), so $[F(\alpha):F] = p$, while $\alpha$ is the only root and $[F(\alpha):F]_s = 1$: there is a single $F$-embedding $F(\alpha) \to \overline{\mathbb{F}}$, and $\mathrm{Aut}(F(\alpha)/F)$ is trivial although the degree is $p$.

**Theorem 5.17** (Primitive element)**.** *(a) A finite separable extension $L/F$ is simple: $L = F(\theta)$ for some $\theta \in L$. If $F$ is infinite and $L = F(\alpha, \beta)$, one can take $\theta = \alpha + c\beta$ for all but finitely many $c \in F$.*

*(b) A finite extension $L/F$ is simple iff it has only finitely many intermediate fields.*

*Proof.* (a) If $F$ is finite, so is $L$, and $L^\times$ is cyclic (Chapter 2); a generator $\theta$ of $L^\times$ satisfies $L = F(\theta)$. Let $F$ be infinite. A finite separable $L$ is generated by finitely many separable elements (for instance an $F$-basis), and by induction on their number it suffices to treat $L = F(\alpha, \beta)$: if $F(\alpha_1, \alpha_2) = F(\theta_1)$, then $F(\alpha_1, \alpha_2, \alpha_3) = F(\theta_1, \alpha_3)$ has one generator fewer. In the two-generator case we only need $\beta$ separable. Let $\alpha = \alpha_1, \dots, \alpha_n$ be the roots of $m_{\alpha,F}$ and $\beta = \beta_1, \dots, \beta_m$ the distinct roots of $g = m_{\beta,F}$ in $\overline{\mathbb{F}}$. For each $i$ and each $j \neq 1$ the equation $\alpha_i + c\beta_j = \alpha + c\beta$ has at most one solution $c$; choose $c \in F$ avoiding these finitely many values and put $\theta = \alpha + c\beta$. The polynomials $g(x)$ and $h(x) = m_{\alpha,F}(\theta - cx)$ lie in $F(\theta)[x]$ and both vanish at $\beta$. If $h(\beta_j) = 0$ with $j \neq 1$, then $\theta - c\beta_j = \alpha_i$ for some $i$, which $c$ was chosen to avoid. So $\beta$ is the only common root of $g$ and $h$ in $\overline{\mathbb{F}}$, and it is a simple root of $g$; hence $\gcd(g, h) = x - \beta$ in $\overline{\mathbb{F}}[x]$. The gcd of two polynomials of $F(\theta)[x]$ is computed by the Euclidean algorithm inside $F(\theta)[x]$ and does not change when the field is enlarged (the remark before Proposition 5.16); so $x - \beta \in F(\theta)[x]$, that is $\beta \in F(\theta)$, and then $\alpha = \theta - c\beta \in F(\theta)$.

(b) Suppose $L = F(\theta)$ and let $M$ be intermediate, $g = m_{\theta,M}$, and $M_0 \subseteq M$ the field generated over $F$ by the coefficients of $g$. Then $g$ is irreducible over $M_0$ (a factorisation over $M_0$ would be one over the larger field $M$), $g(\theta) = 0$ and $L = M_0(\theta)$, so $[L : M_0] = \deg g = [L : M]$, and $M_0 \subseteq M$ gives $M = M_0$. Thus $M$ is determined by $g$, a monic divisor of $m_{\theta,F}$ in $L[x]$, of which there are finitely many: in $\overline{\mathbb{F}}[x]$, $m_{\theta,F} = \prod_{i=1}^{n}(x - \theta_i)$ with $n = [L : F]$, and by unique factorisation every monic divisor is the product of the factors $x - \theta_i$ over a sub-multiset, so there are at most $2^n$ of them. Conversely, suppose there are finitely many intermediate fields. If $F$ is finite, (a) applies. If $F$ is infinite and $L = F(\alpha, \beta)$, the fields $F(\alpha + c\beta)$, $c \in F$, cannot all differ, so $F(\alpha + c\beta) = F(\alpha + c'\beta) =: M$ for some $c \neq c'$; then $M$ contains $(c - c')\beta$, hence $\beta$ and $\alpha$, and $L = M$ is simple. Induction on the number of generators finishes the proof. □

For example, take $L = \mathbb{Q}(2^{1/3}, \zeta_3)$, the splitting field of $x^3 - 2$ (Example 5.31), with $\alpha = 2^{1/3}$ and $\beta = \zeta_3$, whose conjugates are $\alpha_i = \zeta_3^i\alpha$ and $\beta_2 = \zeta_3^2$. The values of $c$ to avoid are the solutions of $\alpha_i + c\beta_2 = \alpha + c\beta$, namely $c = (\zeta_3^i - 1)\alpha/(\zeta_3 - \zeta_3^2)$, $i = 0, 1, 2$. For $i = 0$ this is $c = 0$; for $i = 1, 2$ it is not real, since $\zeta_3 - \zeta_3^2 = i\sqrt{3}$ and $(\zeta_3^{\pm 1} - 1)/(i\sqrt{3}) = (1 \pm i\sqrt{3})/2$. So every $c \in \mathbb{Q}^\times$ works, and $\theta = 2^{1/3} + \zeta_3$ generates $L$. Its minimal polynomial is the resultant $\operatorname{res}_y\big((x-y)^3 - 2,\ y^2 + y + 1\big) = x^6 + 3x^5 + 6x^4 + 3x^3 + 9x + 9$, which PARI confirms is irreducible.

**Proposition 5.18** (Separable closure)**.** *$F^{\mathrm{sep}}$ is a subfield of $\overline{\mathbb{F}}$, normal and separable over $F$, and every separable algebraic extension of $F$ inside $\overline{\mathbb{F}}$ lies in it. If $F$ is perfect then $F^{\mathrm{sep}} = \overline{\mathbb{F}}$; in particular $\mathbb{Q}^{\mathrm{sep}} = \overline{\mathbb{Q}}$, $\mathbb{F}_q^{\mathrm{sep}} = \overline{\mathbb{F}}_q$ and $\mathbb{Q}_p^{\mathrm{sep}} = \overline{\mathbb{Q}}_p$.*

*Proof.* If $\alpha, \beta \in \overline{\mathbb{F}}$ are separable over $F$, then $F(\alpha, \beta)/F$ is separable by Proposition 5.16(d), so $\alpha \pm \beta$, $\alpha\beta$, $\alpha/\beta$ are separable: $F^{\mathrm{sep}}$ is a field, separable over $F$ by definition. If $\alpha \in F^{\mathrm{sep}}$, all roots of $m_{\alpha,F}$ have the same separable minimal polynomial, so they lie in $F^{\mathrm{sep}}$: condition (i) of Proposition 5.14 holds. The rest is the definition and Proposition 5.16(e). □

## 5.3 Galois extensions and the fundamental theorem

This section proves the fundamental theorem of Galois theory for finite extensions: subgroups of the Galois group correspond to intermediate fields. Every "fixed field", "cut out by" and "subgroup of index two" in the paper is an instance: the first sentence of the proof of Proposition prop:ns, that the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ is the determinant-one subgroup combines the fundamental

theorem with Proposition 5.66. The translation theorem at the end is what makes restriction of representations to $G_K$ well behaved, and in Chapter 7 it says that an unramified base change does not change the inertia group (the first sentence of the proof of Lemma lem:tamagawa-exp).

**Theorem 5.19** (Dedekind's lemma)**.** *Let $G$ be a monoid, $L$ a field and $\chi_1, \dots, \chi_n \colon G \to L^\times$ distinct monoid homomorphisms. If $a_1, \dots, a_n \in L$ satisfy $\sum_i a_i \chi_i(g) = 0$ for all $g \in G$, then $a_1 = \cdots = a_n = 0$. In particular distinct field embeddings $\sigma_1, \dots, \sigma_n \colon K \to L$ are linearly independent over $L$ (apply the lemma to $G = K^\times$).*

*Proof.* Suppose not, and take a nontrivial relation $\sum_i a_i \chi_i = 0$ with the least number of nonzero coefficients; renumber so that $a_1, \dots, a_m \neq 0$ and the others vanish. Then $m \geq 2$: for $m = 1$ the relation would read $a_1 \chi_1(g) = 0$ for all $g$, and $\chi_1(g) \in L^\times$ would force $a_1 = 0$. Choose $h$ with $\chi_1(h) \neq \chi_2(h)$. For all $g$, $\sum_{i \leq m} a_i \chi_i(hg) = \sum_{i \leq m} a_i \chi_i(h) \chi_i(g) = 0$, and $\chi_1(h) \sum_{i \leq m} a_i \chi_i(g) = 0$. Subtracting, $\sum_{2 \leq i \leq m} a_i (\chi_i(h) - \chi_1(h)) \chi_i(g) = 0$, a relation with fewer nonzero coefficients whose coefficient at $i = 2$ is nonzero: a contradiction. □

For two embeddings the lemma is a two-line computation: if $a \cdot 1 + b \cdot \sigma = 0$ on $\mathbb{Q}(\sqrt{2})$, $\sigma(\sqrt{2}) = -\sqrt{2}$, then evaluating at 1 and at $\sqrt{2}$ gives $a + b = 0$ and $(a - b)\sqrt{2} = 0$, so $a = b = 0$. The lemma enters below through Hilbert's Theorem 90 (Theorem 5.34) and the non-vanishing of the trace (Proposition 5.33(d)); in both cases the $\chi_i$ are the elements of a Galois group, restricted to $L^\times$.

**Theorem 5.20** (Artin)**.** *(a) Let $L$ be a field, $G \subseteq \mathrm{Aut}(L)$ a finite group and $F = L^G = \{x \in L : \sigma(x) = x \text{ for all } \sigma \in G\}$. Then $L/F$ is finite, normal and separable, $[L : F] = \#G$ and $\mathrm{Aut}(L/F) = G$. For $\alpha \in L$, $m_{\alpha,F} = \prod_{\beta \in G\alpha}(x - \beta)$, the product over the orbit.*

*(b) For every finite extension $L/F$, $\#\mathrm{Aut}(L/F) \leq [L : F]_s \leq [L : F]$, and $\#\mathrm{Aut}(L/F) = [L : F]$ iff $L/F$ is normal and separable.*

*Proof.* (a) Let $\alpha \in L$ with orbit $G\alpha = \{\beta_1, \dots, \beta_k\}$, $k \leq \#G$. The polynomial $P = \prod_j (x - \beta_j)$ has distinct roots, and every $\sigma \in G$ permutes them, so its coefficients are fixed by $G$: $P \in F[x]$. If $g \in F[x]$ vanishes at $\alpha$, it vanishes at each $\sigma(\alpha)$, so $P \mid g$; hence $P = m_{\alpha,F}$. Thus every $\alpha \in L$ is algebraic and separable over $F$ of degree at most $\#G$, and its minimal polynomial splits in $L$: $L/F$ is algebraic, separable and normal. Every finite subextension $K/F$ of $L$ is separable, hence $K = F(\theta)$ (Theorem 5.17), and $[K : F] \leq \#G$. Choose such a $K$ of maximal degree; for $\beta \in L$, $K(\beta)$ is again finite over $F$, so $K(\beta) = K$. Hence $L = K$ and $[L : F] \leq \#G$. On the other hand $G \subseteq \mathrm{Aut}(L/F)$, and by (b) $\#\mathrm{Aut}(L/F) \leq [L : F]$. So $\#G \leq \#\mathrm{Aut}(L/F) \leq [L : F] \leq \#G$, all are equal and $G = \mathrm{Aut}(L/F)$.

(b) Embed $L$ in $\overline{\mathbb{F}}$. Composing with the inclusion $L \subseteq \overline{\mathbb{F}}$ makes $\mathrm{Aut}(L/F)$ a subset of $\mathrm{Emb}_F(L, \overline{\mathbb{F}})$, of size $[L : F]_s \leq [L : F]$ (Proposition 5.16(d)). If $L/F$ is normal and separable, every $F$-embedding has image $L$ (Proposition 5.14), so $\mathrm{Aut}(L/F) = \mathrm{Emb}_F(L, \overline{\mathbb{F}})$ has $[L : F]_s = [L : F]$ elements. Conversely, if $\#\mathrm{Aut}(L/F) = [L : F]$, then $[L : F]_s = [L : F]$, so $L/F$ is separable, and $\mathrm{Aut}(L/F) = \mathrm{Emb}_F(L, \overline{\mathbb{F}})$, so every embedding has image $L$ and $L/F$ is normal. □

Part (a) manufactures Galois extensions out of groups. For example, let $k$ be a field, $L = k(t)$ the field of rational functions and $G = \{1, \iota\}$ with $\iota(r(t)) = r(1/t)$, an automorphism of order 2. The element $u = t + 1/t$ lies in $L^G$, and $t$ is a root of $x^2 - ux + 1 \in k(u)[x]$, so $[L : k(u)] \leq 2$. As $k(u) \subseteq L^G$ and $[L : L^G] = \#G = 2$, the tower law gives $L^G = k(u)$, and (a) gives $m_{t,L^G} = (x - t)(x - 1/t) = x^2 - ux + 1$, the product over the orbit of $t$.

**Definition 5.21** (Galois extensions)**.** An algebraic extension $L/F$ is *Galois* if it is normal and separable. Its *Galois group* is $\mathrm{Gal}(L/F) = \mathrm{Aut}(L/F)$.

The two conditions fail independently. The extension $\mathbb{Q}(2^{1/3})/\mathbb{Q}$ is separable but not normal, and the identity is its only automorphism: an automorphism sends $2^{1/3}$ to a root of $x^3-2$ in the real field $\mathbb{Q}(2^{1/3})$, and $2^{1/3}$ is the only real root. $\mathbb{F}_p(t^{1/p})/\mathbb{F}_p(t)$ is normal (the only root of $x^p - t$ is $t^{1/p}$) but not separable, and again its automorphism group is trivial. In both cases the automorphism group is too small to see the extension; the next theorem shows that being Galois is exactly the condition $\#\,\mathrm{Aut}(L/F) = [L:F]$.

**Theorem 5.22** (Characterisations of finite Galois extensions)**.** *For a finite extension $L/F$ the following are equivalent:*

*(i)* $L/F$ *is Galois;*

*(ii)* $L$ *is the splitting field over* $F$ *of a separable polynomial;*

*(iii)* $L^{\mathrm{Aut}(L/F)} = F$*;*

*(iv)* $\#\,\mathrm{Aut}(L/F) = [L:F]$.

*If $L/F$ is Galois and $F \subseteq M \subseteq L$, then $L/M$ is Galois.*

*Proof.* (i)⇔(iv) is Theorem 5.20(b). (i)⇒(ii): $L = F(\theta)$ (Theorem 5.17); $m_{\theta,F}$ is separable, splits in $L$ by normality, and its roots generate $L$. (ii)⇒(iv): by Theorem 5.12(b) with $\phi = \mathrm{id}$, $L = L'$, the identity of $F$ has exactly $[L:F]$ extensions to automorphisms of $L$. (i)⇒(iii): put $G = \mathrm{Aut}(L/F)$ and $F' = L^G \supseteq F$. By Theorem 5.20(a), $[L:F'] = \#G = [L:F]$, so $F' = F$. (iii)⇒(i): $G = \mathrm{Aut}(L/F)$ is finite (Theorem 5.20(b)), and Theorem 5.20(a) shows that $L/L^G = L/F$ is normal and separable. Finally, if $L$ is the splitting field over $F$ of a separable $f$, it is the splitting field of $f$ over $M$. □

**Theorem 5.23** (Fundamental theorem of Galois theory)**.** *Let $L/F$ be a finite Galois extension with group $G = \mathrm{Gal}(L/F)$.*

*(a) The maps $H \mapsto L^H$ and $M \mapsto \mathrm{Gal}(L/M)$ are mutually inverse, inclusion-reversing bijections between the subgroups $H \leq G$ and the intermediate fields $F \subseteq M \subseteq L$.*

*(b)* $[L:M] = \#\,\mathrm{Gal}(L/M)$ *and* $[M:F] = [G:\mathrm{Gal}(L/M)]$.

*(c) For $\sigma \in G$,* $\mathrm{Gal}(L/\sigma M) = \sigma\,\mathrm{Gal}(L/M)\sigma^{-1}$.

*(d) $M/F$ is Galois iff* $\mathrm{Gal}(L/M)$ *is normal in $G$. Then restriction $\sigma \mapsto \sigma|_M$ is a surjection $G \to \mathrm{Gal}(M/F)$ with kernel* $\mathrm{Gal}(L/M)$*, so* $G/\,\mathrm{Gal}(L/M) \simeq \mathrm{Gal}(M/F)$.

*(e) The compositum $M_1M_2$ corresponds to* $\mathrm{Gal}(L/M_1) \cap \mathrm{Gal}(L/M_2)$*, and $M_1 \cap M_2$ to the subgroup generated by* $\mathrm{Gal}(L/M_1)$ *and* $\mathrm{Gal}(L/M_2)$.

*Proof.* (a), (b). For $H \leq G$, Theorem 5.20(a) gives $[L:L^H] = \#H$ and $\mathrm{Gal}(L/L^H) = H$. For $M$ intermediate, $L/M$ is Galois (Theorem 5.22), so $L^{\mathrm{Gal}(L/M)} = M$ by (iii) there. Both maps reverse inclusions. The tower law gives $[M:F] = [L:F]/[L:M] = \#G/\#\,\mathrm{Gal}(L/M)$. (c) For $m \in M$, $\tau(\sigma(m)) = \sigma(m)$ iff $\sigma^{-1}\tau\sigma(m) = m$; so $\tau$ fixes $\sigma M$ pointwise iff $\sigma^{-1}\tau\sigma \in \mathrm{Gal}(L/M)$, that is iff $\tau \in \sigma\,\mathrm{Gal}(L/M)\sigma^{-1}$. (d) The extension $M/F$ is separable. Every $F$-embedding

$M \to \overline{\mathbb{F}}$ (with $L \subseteq \overline{\mathbb{F}}$) extends to an $F$-embedding $L \to \overline{\mathbb{F}}$ (Theorem 5.10(b)), which has image $L$ because $L/F$ is normal, so it is an element of $G$. Hence the $F$-embeddings of $M$ are the $\sigma|_M$, $\sigma \in G$, and by Proposition 5.14(iii) $M/F$ is normal iff $\sigma M = M$ for all $\sigma \in G$, iff (by (a) and (c)) $\sigma \operatorname{Gal}(L/M)\sigma^{-1} = \operatorname{Gal}(L/M)$ for all $\sigma$. In that case $\sigma \mapsto \sigma|_M$ is a homomorphism $G \to \operatorname{Gal}(M/F)$, surjective because every automorphism of $M/F$ is an embedding and so extends, as just shown, with kernel $\operatorname{Gal}(L/M)$. (e) $\sigma$ fixes $M_1M_2$ iff it fixes $M_1$ and $M_2$ (the fixed field of $\sigma$ is a field). An element fixed by $H_1 = \operatorname{Gal}(L/M_1)$ and by $H_2 = \operatorname{Gal}(L/M_2)$ is fixed by every product of their elements, so the fixed field of the subgroup generated by $H_1$ and $H_2$ is $L^{H_1} \cap L^{H_2} = M_1 \cap M_2$; apply (a). □

Example 5.31 below works out the whole correspondence for a group of order 6, including the conjugate subgroups of (c) and the normal subgroup of (d); the reader may want to look at it now.

**Proposition 5.24** (Translation)**.** *Let $L/F$ be finite Galois and $M/F$ any extension, with $L$ and $M$ contained in a common field. Then $LM/M$ is finite Galois, and restriction* $\operatorname{Gal}(LM/M) \to \operatorname{Gal}(L/F)$, $\sigma \mapsto \sigma|_L$, *is injective with image* $\operatorname{Gal}(L/L \cap M)$. *Consequently:*

*(a) $[LM : M] = [L : L \cap M]$; in particular $[LM : M]$ divides $[L : F]$.*

*(b) If $L_1, L_2$ are finite Galois over $F$, so are $L_1L_2$ and $L_1 \cap L_2$, and* $\operatorname{Gal}(L_1L_2/F) \to \operatorname{Gal}(L_1/F) \times \operatorname{Gal}(L_2/F)$, $\sigma \mapsto (\sigma|_{L_1}, \sigma|_{L_2})$, *is injective, with image the pairs that agree on $L_1 \cap L_2$; it is bijective iff $L_1 \cap L_2 = F$.*

*Proof.* Let $L$ be the splitting field over $F$ of a separable $f$ with roots $r_1, \dots, r_n$. Then $LM = M(r_1, \dots, r_n)$ is the splitting field of $f$ over $M$, so $LM/M$ is Galois. An element $\sigma \in \operatorname{Gal}(LM/M)$ permutes the $r_i$, so $\sigma(L) = L$ and $\sigma|_L \in \operatorname{Gal}(L/F)$; if $\sigma|_L = \mathrm{id}$, then $\sigma$ fixes $M$ and $L$, hence $LM$. Let $H$ be the image. An element $x \in L$ fixed by $H$ is fixed by $\operatorname{Gal}(LM/M)$, so $x \in M$ (Theorem 5.22(iii)); hence $L^H = L \cap M$ (the inclusion $\supseteq$ being clear) and $H = \operatorname{Gal}(L/L \cap M)$ by Theorem 5.23(a). (a) Count. (b) $L_1L_2$ is the splitting field of the product of the distinct irreducible factors of $f_1f_2$, which is separable; $L_1 \cap L_2$ is normal by Proposition 5.14(i) (an irreducible polynomial over $F$ with a root in $L_1 \cap L_2$ splits in $L_1$ and in $L_2$, so all its roots lie in $L_1 \cap L_2$), and separable as a subfield of $L_1$. The map is injective as $L_1L_2$ is generated by $L_1$ and $L_2$. Its image lies in the set of pairs agreeing on $L_1 \cap L_2$, which has $\#\operatorname{Gal}(L_1/F) \cdot \#\operatorname{Gal}(L_2/L_1 \cap L_2)$ elements: given $\sigma_1 \in \operatorname{Gal}(L_1/F)$, the $\sigma_2 \in \operatorname{Gal}(L_2/F)$ with $\sigma_2|_{L_1 \cap L_2} = \sigma_1|_{L_1 \cap L_2}$ form one coset of $\operatorname{Gal}(L_2/L_1 \cap L_2)$, nonempty because restriction $\operatorname{Gal}(L_2/F) \to \operatorname{Gal}(L_1 \cap L_2/F)$ is surjective (Theorem 5.23(d)). By (a), $\#\operatorname{Gal}(L_1L_2/F) = [L_1L_2 : L_1][L_1 : F] = [L_2 : L_1 \cap L_2][L_1 : F]$ is the same number, so the injective map is onto this set. □

The hypothesis that $L/F$ be Galois cannot be dropped. Let $\alpha = 2^{1/3}$, $L = \mathbb{Q}(\alpha)$ and $M = \mathbb{Q}(\zeta_3\alpha)$. Then $L \neq M$ ($L \subset \mathbb{R}$, $M \not\subset \mathbb{R}$), so $L \cap M$, whose degree over $\mathbb{Q}$ divides 3 and is not 3, is $\mathbb{Q}$. But $LM$ contains $\zeta_3 = \zeta_3\alpha/\alpha$, so it is the splitting field of $x^3 - 2$, of degree 6, and $[LM : M] = 2 \neq 3 = [L : L \cap M]$. With the Galois extension $L = \mathbb{Q}(\zeta_3)$ and $M = \mathbb{Q}(\alpha)$ instead, the proposition applies: $L \cap M = \mathbb{Q}$ and $[LM : M] = 6/3 = 2 = [L : \mathbb{Q}]$.

## 5.4 Finite fields, quadratic and cyclotomic extensions

The three families of Galois extensions that the paper uses constantly are worked out here. Finite fields give the residue fields $\mathbb{F}_q$ of Lemmas lem:tamfactor and lem:euler and the honest

model of a Frobenius element; quadratic fields give the characters $\chi_d$ of the table of Section 1, of Lemma lem:twist and of Theorem thm:twist; cyclotomic fields give $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ (Proposition prop:ns, Lemma lem:h0) and the field $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$ of the hypothesis of Theorem thm:main. The section ends with the splitting field of $x^3 - 2$, the model of a non-abelian Galois group and of a quadratic subfield cut out by a sign.

**Theorem 5.25** (Galois theory of finite fields)**.** *Let $q = p^r$ and $n \geq 1$, and work inside a fixed algebraic closure $\overline{\mathbb{F}}_p$; $\mathbb{F}_q$ is the unique subfield with $q$ elements.*

*(a) $\mathbb{F}_{q^n} = \{x \in \overline{\mathbb{F}}_p : x^{q^n} = x\}$ is the splitting field of $x^{q^n} - x$ over $\mathbb{F}_q$, it is the unique extension of $\mathbb{F}_q$ of degree $n$ inside $\overline{\mathbb{F}}_p$, and every extension of $\mathbb{F}_q$ of degree $n$ is $\mathbb{F}_q$-isomorphic to it.*

*(b) $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Galois, and $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ is cyclic of order $n$, generated by $\mathrm{Frob}_q \colon x \mapsto x^q$.*

*(c) The intermediate fields of $\mathbb{F}_{q^n}/\mathbb{F}_q$ are the $\mathbb{F}_{q^d}$ with $d \mid n$, and $\mathbb{F}_{q^d}$ is the fixed field of $\langle \mathrm{Frob}_q^d \rangle$.*

*(d) If $g \in \mathbb{F}_q[x]$ is irreducible of degree $d$ with a root $\alpha$, then $\mathbb{F}_q(\alpha) = \mathbb{F}_{q^d}$ and $g$ has the $d$ distinct roots $\alpha, \alpha^q, \ldots, \alpha^{q^{d-1}}$ in $\mathbb{F}_{q^d}$. Every finite extension of a finite field is Galois with cyclic group.*

The smallest example to keep in mind: $x^2 + x + 1$ is the only irreducible quadratic over $\mathbb{F}_2$ (each of $x^2$, $x^2 + 1 = (x+1)^2$, $x^2 + x$ has a root), so $\mathbb{F}_4 = \mathbb{F}_2(\omega) = \{0, 1, \omega, \omega + 1\}$ with $\omega^2 = \omega + 1$. Here $\mathrm{Frob}_2(\omega) = \omega^2 = \omega + 1$ is the other root of $x^2 + x + 1$, as (d) says; $\mathrm{Frob}_2$ swaps $\omega$ and $\omega + 1$, fixes exactly $\mathbb{F}_2$, and generates $\mathrm{Gal}(\mathbb{F}_4/\mathbb{F}_2) \simeq \mathbb{Z}/2$.

*Proof.* (a) The polynomial $x^{q^n} - x$ has derivative $-1$, so it is separable, and its $q^n$ roots in $\overline{\mathbb{F}}_p$ form a field (Chapter 2: the roots of $x^{p^m} - x$ are closed under the field operations). It contains $\mathbb{F}_q$, whose elements satisfy $x^q = x$. A subextension $K \subseteq \overline{\mathbb{F}}_p$ of degree $n$ over $\mathbb{F}_q$ has $q^n$ elements, all satisfying $x^{q^n} = x$ (Lagrange in $K^\times$), so $K = \mathbb{F}_{q^n}$ by counting. An abstract extension of degree $n$ embeds over $\mathbb{F}_q$ into $\overline{\mathbb{F}}_p$ (Theorem 5.10(b)), with image $\mathbb{F}_{q^n}$. (b) $\mathrm{Frob}_q$ is an injective ring endomorphism of the finite field $\mathbb{F}_{q^n}$, hence an automorphism; it fixes $\mathbb{F}_q$. Its powers $\mathrm{Frob}_q^k$, $1 \leq k < n$, are not the identity, since $x^{q^k} = x$ has at most $q^k < q^n$ solutions. So $\langle \mathrm{Frob}_q \rangle \subseteq \mathrm{Aut}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ has order $n = [\mathbb{F}_{q^n} : \mathbb{F}_q]$, and Theorem 5.20(b) gives equality and that the extension is Galois. (c) The subgroups of the cyclic group of order $n$ are the $\langle \mathrm{Frob}_q^d \rangle$, $d \mid n$, of order $n/d$; the fixed field of $\mathrm{Frob}_q^d$ is $\{x : x^{q^d} = x\} = \mathbb{F}_{q^d}$. Apply Theorem 5.23. (d) $\mathbb{F}_q(\alpha)$ has degree $d$, so it is $\mathbb{F}_{q^d}$ by (a), Galois over $\mathbb{F}_q$. The roots of $g$ form the orbit of $\alpha$ under $\mathrm{Gal}(\mathbb{F}_{q^d}/\mathbb{F}_q) = \langle \mathrm{Frob}_q \rangle$ (Theorem 5.20(a)), namely the $\mathrm{Frob}_q^i(\alpha) = \alpha^{q^i}$, $0 \leq i < d$, which are distinct since the orbit has $\deg g = d$ elements. □

**Proposition 5.26** (Cyclotomic polynomials modulo $p$)**.** *Let $n \geq 1$, $p$ a prime with $p \nmid n$, $f$ the order of $p$ in $(\mathbb{Z}/n)^\times$, and $\bar{\Phi}_n \in \mathbb{F}_p[x]$ the reduction of $\Phi_n$. Then $\bar{\Phi}_n = \prod_\zeta (x - \zeta)$, the product over the $\varphi(n)$ elements of order exactly $n$ in $\overline{\mathbb{F}}_p^\times$; it is the product of $\varphi(n)/f$ distinct monic irreducible polynomials of degree $f$; and $\mathbb{F}_p(\mu_n) = \mathbb{F}_{p^f}$.*

*Proof.* As $p \nmid n$, $x^m - 1$ is separable over $\mathbb{F}_p$ for every $m \mid n$ (its derivative $mx^{m-1}$ is prime to it), so $\mu_n(\overline{\mathbb{F}}_p)$ is a group of order $n$, cyclic (Chapter 2), with exactly $\varphi(n)$ elements of order $n$. Reducing $x^m - 1 = \prod_{e \mid m} \Phi_e$ (Chapter 2) modulo $p$, the roots of $\prod_{e \mid m} \bar{\Phi}_e$ are the elements of order dividing $m$, each once. The roots of $\bar{\Phi}_n$ are therefore the roots of $x^n - 1$ that are not roots of $\bar{\Phi}_e$ for any proper divisor $e$ of $n$, that is, which have order dividing no proper divisor of $n$:

the elements of order exactly $n$. Let $\zeta$ be one. By Theorem 5.25(d) its minimal polynomial over $\mathbb{F}_p$ has the roots $\zeta^{p^i}$, $0 \le i < d$, where $d$ is the least integer with $\zeta^{p^d} = \zeta$, that is, with $p^d \equiv 1 \pmod{n}$; so $d = f$ for every $\zeta$. Grouping the $\varphi(n)$ roots into these orbits gives $\varphi(n)/f$ distinct irreducible factors of degree $f$, and $\mathbb{F}_p(\mu_n) = \mathbb{F}_p(\zeta) = \mathbb{F}_{p^f}$. □

This is the finite-field shadow of the decomposition of $p$ in $\mathbb{Q}(\zeta_n)$: in Chapter 6 it gives that $p \nmid n$ splits into $\varphi(n)/f$ primes of residue degree $f$, and that $\mathrm{Frob}_p$ is $\sigma_p$ (Theorem 5.72).

For instance, take $n = 5$, $\Phi_5 = x^4 + x^3 + x^2 + x + 1$. The order of 2 and of 3 modulo 5 is 4, so $\Phi_5$ is irreducible modulo 2 and modulo 3. As $11 \equiv 1 \pmod{5}$, $\Phi_5$ splits into four linear factors modulo 11; its roots are $3, 4, 5, 9$, the elements of order 5 of $\mathbb{F}_{11}^\times$. As $19 \equiv 4 \pmod{5}$ has order 2, modulo 19 it is a product of two irreducible quadratics, $(x^2 + 5x + 1)(x^2 + 15x + 1)$ (PARI). Since every element of $(\mathbb{Z}/8)^\times$ has order at most 2, $\Phi_8 = x^4 + 1$, irreducible over $\mathbb{Q}$, is reducible modulo every odd prime.

**Proposition 5.27** (Quadratic extensions)**.** *Let* $\operatorname{char} F \neq 2$.

*(a) Every extension $L/F$ of degree 2 is Galois and equals $F(\sqrt{d})$ for some $d \in F^\times \setminus F^{\times 2}$; $\mathrm{Gal}(L/F) = \{1, \sigma\}$ with $\sigma(\sqrt{d}) = -\sqrt{d}$.*

*(b) Inside $\overline{\mathbb{F}}$, $F(\sqrt{d}) = F(\sqrt{d'})$ iff $d/d' \in F^{\times 2}$. So $d \mapsto F(\sqrt{d})$ is a bijection from the nontrivial elements of $F^\times/F^{\times 2}$ to the quadratic extensions of $F$ in $\overline{\mathbb{F}}$.*

*(c) The quadratic fields are the $\mathbb{Q}(\sqrt{d})$, $d \neq 0, 1$ a squarefree integer, pairwise distinct.*

*(d) If the classes of $a, b \in F^\times$ in $F^\times/F^{\times 2}$ are independent (that is, none of $a$, $b$, $ab$ is a square), then $F(\sqrt{a}, \sqrt{b})/F$ is Galois with group $\{\pm 1\}^2 \simeq V_4$, the element $(\epsilon_1, \epsilon_2)$ acting by $\sqrt{a} \mapsto \epsilon_1\sqrt{a}$, $\sqrt{b} \mapsto \epsilon_2\sqrt{b}$; its quadratic subfields are exactly $F(\sqrt{a})$, $F(\sqrt{b})$ and $F(\sqrt{ab})$.*

*Proof.* (a) Let $\alpha \in L \setminus F$, so $L = F(\alpha)$ and $m_{\alpha,F} = x^2 + bx + c$. Then $(\alpha + b/2)^2 = b^2/4 - c =: d$, and $d$ is not a square in $F$ (otherwise $\alpha \in F$), so $L = F(\sqrt{d})$ with $\sqrt{d} = \alpha + b/2$. The roots $\pm\sqrt{d}$ are distinct as $d \neq 0$ and $2 \neq 0$, so $L$ is the splitting field of the separable $x^2 - d$, Galois of degree 2, and the nontrivial automorphism sends $\sqrt{d}$ to the other root. (b) If $d' = dt^2$, then $\sqrt{d'} = \pm t\sqrt{d}$. Conversely if $\sqrt{d'} = x + y\sqrt{d}$ with $x, y \in F$, squaring gives $d' = x^2 + y^2 d + 2xy\sqrt{d}$, so $xy = 0$ (as $1, \sqrt{d}$ is a basis and $2 \neq 0$); $y = 0$ would make $d'$ a square, so $x = 0$ and $d' = y^2 d$. With (a) this is the bijection. (c) Every class in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ has a unique squarefree integer representative. (d) $\sqrt{b} \notin F(\sqrt{a})$: otherwise $F(\sqrt{b}) = F(\sqrt{a})$ (both of degree 2), and $ab = a^2(b/a)$ would be a square by (b). So $[F(\sqrt{a}, \sqrt{b}) : F] = 4$; the field is the splitting field of the separable $(x^2 - a)(x^2 - b)$, hence Galois, and an automorphism is determined by the signs $\epsilon_1, \epsilon_2$, all four occurring by counting: the group has $[F(\sqrt{a}, \sqrt{b}) : F] = 4$ elements (Theorem 5.22) and there are four sign pairs. The three subgroups of order 2 are generated by $(1, -1)$, $(-1, 1)$ and $(-1, -1)$, which fix $\sqrt{a}$, $\sqrt{b}$ and $\sqrt{ab} = \pm\sqrt{a}\sqrt{b}$ respectively; by Theorem 5.23 the three quadratic subfields are $F(\sqrt{a})$, $F(\sqrt{b})$, $F(\sqrt{ab})$. □

**Theorem 5.28** (Cyclotomic fields)**.** *Let $n \ge 1$, $\zeta_n = e^{2\pi i/n}$, $\mu_n = \langle \zeta_n \rangle \subset \overline{\mathbb{Q}}$, and $\Phi_n = \prod_{1 \le a \le n,\ (a,n)=1} (x - \zeta_n^a) \in \mathbb{Z}[x]$ (Chapter 2).*

*(a) $\Phi_n$ is irreducible over $\mathbb{Q}$; so $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$.*

*(b) $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is Galois, and $a \mapsto \sigma_a$, where $\sigma_a(\zeta) = \zeta^a$ for all $\zeta \in \mu_n$, is an isomorphism $(\mathbb{Z}/n)^\times \to \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$, which does not depend on the choice of a primitive $n$-th root of unity.*

(c) *For $m \mid n$, restriction* $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \to \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ *is reduction* $(\mathbb{Z}/n)^\times \to (\mathbb{Z}/m)^\times$. *For all $m, n$:* $\mathbb{Q}(\zeta_m)\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{\mathrm{lcm}(m,n)})$ *and* $\mathbb{Q}(\zeta_m) \cap \mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{\gcd(m,n)})$*;* $\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{2n})$ *for odd $n$. Every subfield of* $\mathbb{Q}(\zeta_n)$ *is Galois over* $\mathbb{Q}$ *with abelian group.*

(d) $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$, *since* $(2\zeta_3+1)^2 = -3$*;* $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$*;* $\mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$, *since* $\zeta_8 + \zeta_8^{-1} = \sqrt{2}$, *with the three quadratic subfields* $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-2})$.

*Proof.* (a) Let $\zeta$ be a primitive $n$-th root of unity and $g$ its minimal polynomial over $\mathbb{Q}$, a monic factor of the monic $\Phi_n \in \mathbb{Z}[x]$; by Gauss's lemma (Chapter 2) $g \in \mathbb{Z}[x]$ and $\Phi_n = gh$ with $h \in \mathbb{Z}[x]$ monic. We show: *if $p \nmid n$ is prime, then $g(\zeta^p) = 0$.* If not, $\zeta^p$ (a primitive $n$-th root of unity, as $p \nmid n$) is a root of $h$, so $\zeta$ is a root of $h(x^p)$, and $g \mid h(x^p)$ in $\mathbb{Q}[x]$, hence in $\mathbb{Z}[x]$ by Gauss's lemma: $h(x^p) = g(x)k(x)$ with $k \in \mathbb{Z}[x]$. Reduce modulo $p$: since $\bar{h}$ has coefficients in $\mathbb{F}_p$, $\bar{h}(x^p) = \bar{h}(x)^p$. So $\bar{g}\bar{k} = \bar{h}^p$, and any irreducible factor $\pi$ of $\bar{g}$ in $\mathbb{F}_p[x]$ divides $\bar{h}$. Then $\pi^2$ divides $\bar{g}\bar{h} = \bar{\Phi}_n$, which divides $x^n - 1$ in $\mathbb{F}_p[x]$; but $x^n - 1$ is separable over $\mathbb{F}_p$ because $p \nmid n$. This contradiction proves the claim. Now every primitive root is $\zeta^a$ with $(a, n) = 1$, $a = p_1 \cdots p_k$ a product of primes not dividing $n$; applying the claim successively to $\zeta, \zeta^{p_1}, \zeta^{p_1 p_2}, \ldots$ (all primitive, all roots of the same $g$) gives $g(\zeta^a) = 0$. So all $\varphi(n)$ roots of $\Phi_n$ are roots of $g$, and $g = \Phi_n$. (b) $\mathbb{Q}(\zeta_n)$ contains all primitive $n$-th roots of unity (powers of $\zeta_n$), so it is the splitting field of the separable $\Phi_n$, Galois of degree $\varphi(n)$. An automorphism $\sigma$ sends $\zeta_n$ to a root of $\Phi_n$, so $\sigma(\zeta_n) = \zeta_n^a$ with $a \in (\mathbb{Z}/n)^\times$ determined by $\sigma$, and then $\sigma(\zeta_n^k) = \zeta_n^{ak}$: $\sigma = \sigma_a$ acts on all of $\mu_n$ by $\zeta \mapsto \zeta^a$, which does not mention $\zeta_n$. As $\sigma_a\sigma_b(\zeta) = \sigma_a(\zeta^b) = \zeta^{ab}$, $a \mapsto \sigma_a$ is an injective homomorphism, bijective by counting. (c) $\zeta_m = \zeta_n^{n/m}$, so $\sigma_a(\zeta_m) = \zeta_m^a = \zeta_m^{a \bmod m}$. Let $g = \gcd(m, n)$, $s = \mathrm{lcm}(m, n)$ and $um + vn = g$ with $u, v \in \mathbb{Z}$. Then $\zeta_m^v \zeta_n^u = e^{2\pi i(vn+um)/mn} = e^{2\pi i g/mn} = \zeta_s$, so $\mathbb{Q}(\zeta_m)\mathbb{Q}(\zeta_n) \supseteq \mathbb{Q}(\zeta_s)$, and the converse inclusion is clear. By Proposition 5.24(a), $[\mathbb{Q}(\zeta_m) \cap \mathbb{Q}(\zeta_n) : \mathbb{Q}] = [\mathbb{Q}(\zeta_m) : \mathbb{Q}]\,[\mathbb{Q}(\zeta_n) : \mathbb{Q}]/[\mathbb{Q}(\zeta_s) : \mathbb{Q}] = \varphi(m)\varphi(n)/\varphi(s)$, and $\varphi(m)\varphi(n) = \varphi(s)\varphi(g)$ (from $\varphi(k) = k\prod_{p|k}(1 - 1/p)$ and $mn = sg$, the primes dividing $s$ being those dividing $m$ or $n$, those dividing $g$ those dividing both). So the intersection has degree $\varphi(g)$ and contains $\mathbb{Q}(\zeta_g)$: they are equal. For odd $n$, $\mathbb{Q}(\zeta_n) \subseteq \mathbb{Q}(\zeta_{2n})$ and $\varphi(2n) = \varphi(n)$. Subgroups of the abelian group $(\mathbb{Z}/n)^\times$ are normal, so Theorem 5.23(d) applies. (d) $(2\zeta_3+1)^2 = 4(\zeta_3^2 + \zeta_3 + 1) - 3 = -3$, and both fields have degree 2. $\zeta_8 + \zeta_8^{-1} = 2\cos(\pi/4) = \sqrt{2}$; $\mathbb{Q}(i, \sqrt{2}) \subseteq \mathbb{Q}(\zeta_8)$ and both have degree 4. The quadratic subfields follow from Proposition 5.27(d) with $a = -1$, $b = 2$. □

**Proposition 5.29** (The field $\mathbb{Q}(\sqrt{\ell^*})$)**.** *Let $\ell$ be an odd prime and $\ell^* = (-1)^{(\ell-1)/2}\ell$.*

(a) *The Gauss sum $g = \sum_{a=1}^{\ell-1}\left(\frac{a}{\ell}\right)\zeta_\ell^a$ satisfies $g^2 = \ell^*$ and $\sigma_a(g) = \left(\frac{a}{\ell}\right)g$ for $a \in (\mathbb{Z}/\ell)^\times$. In particular $\sqrt{\ell^*} = \pm g \in \mathbb{Q}(\zeta_\ell)$.*

(b) $\mathbb{Q}(\sqrt{\ell^*})$ *is the unique quadratic subfield of $\mathbb{Q}(\zeta_\ell)$, the fixed field of the subgroup of squares of $(\mathbb{Z}/\ell)^\times$, and $\sigma_a(\sqrt{\ell^*}) = \left(\frac{a}{\ell}\right)\sqrt{\ell^*}$. In terms of characters of $G_\mathbb{Q}$ (Definition 5.58): for $\sigma \in G_\mathbb{Q}$,*

$$\chi_{\ell^*}(\sigma) = \left(\frac{\bar{\chi}_\ell(\sigma)}{\ell}\right) = \bar{\chi}_\ell(\sigma)^{(\ell-1)/2},$$

*the last equality in $\mathbb{F}_\ell^\times \supseteq \{\pm 1\}$ (Euler's criterion). Briefly, $\chi_{\ell^*} = \bar{\chi}_\ell^{(\ell-1)/2}$.*

(c) *For $\ell = 3$:* $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$ *and $\chi_{-3} = \bar{\chi}_3$. For $\ell \geq 5$: $\chi_{\ell^*} \neq \bar{\chi}_\ell$, since $\bar{\chi}_\ell$ has order $\ell - 1 > 2$.*

*(d) Every quadratic field $\mathbb{Q}(\sqrt{d})$, $d$ squarefree, lies in $\mathbb{Q}(\zeta_{|D|})$, where $D = d$ if $d \equiv 1 \pmod 4$ and $D = 4d$ otherwise is its discriminant. In particular, for a finite set $S$ of odd primes, $\mathbb{Q}(\sqrt{\prod_{q\in S} q^*}) \subseteq \mathbb{Q}(\zeta_{\prod_{q\in S} q})$.*

*Proof.* (a) $g^2 = \ell^*$ is Chapter 2 (the quadratic Gauss sum). For $a$ prime to $\ell$, $\sigma_a(g) = \sum_b (\frac{b}{\ell})\zeta_\ell^{ab} = (\frac{a}{\ell})\sum_b (\frac{ab}{\ell})\zeta_\ell^{ab} = (\frac{a}{\ell})g$, as $b \mapsto ab$ permutes $(\mathbb{Z}/\ell)^\times$ and $(\frac{a}{\ell})^2 = 1$. (b) $(\mathbb{Z}/\ell)^\times$ is cyclic of even order $\ell - 1$ (Chapter 2), so it has exactly one subgroup of index 2, the squares; by Theorem 5.23 $\mathbb{Q}(\zeta_\ell)$ has exactly one quadratic subfield, the fixed field of the squares. By (a), $g$ is fixed exactly by the $\sigma_a$ with $(\frac{a}{\ell}) = 1$, and $g \notin \mathbb{Q}$ (it is moved by $\sigma_a$ for a non-residue $a$), so that subfield is $\mathbb{Q}(g) = \mathbb{Q}(\sqrt{\ell^*})$. For $\sigma \in G_\mathbb{Q}$, $\sigma|_{\mathbb{Q}(\zeta_\ell)} = \sigma_a$ with $a \equiv \bar\chi_\ell(\sigma)$ (Definition 5.58), so $\chi_{\ell^*}(\sigma) = \sigma(g)/g = (\frac{a}{\ell})$, and $(\frac{a}{\ell}) \equiv a^{(\ell-1)/2} \pmod \ell$. (c) $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$ is Theorem 5.28(d); with $\ell = 3$, $(\ell-1)/2 = 1$. For $\ell \geq 5$, $\bar\chi_\ell$ is surjective onto the cyclic group $\mathbb{F}_\ell^\times$ of order $\ell - 1 \geq 4$ (Proposition 5.59), while $\chi_{\ell^*}^2 = 1$. (d) Write $d = \pm 2^e \prod_i q_i$ with $e \in \{0, 1\}$ and distinct odd primes $q_i$; then $d = \epsilon 2^e \prod_i q_i^*$ with $\epsilon = \pm 1$, and $\sqrt{q_i^*} \in \mathbb{Q}(\zeta_{q_i})$ by (b). If $d \equiv 1 \pmod 4$, then $e = 0$ and $\epsilon = 1$, because $\prod q_i^* \equiv 1 \pmod 4$ while $-\prod q_i^* \equiv 3 \pmod 4$; so $\sqrt{d} \in \mathbb{Q}(\zeta_{|d|})$. If $d \equiv 3 \pmod 4$, $e = 0$ and $\sqrt{d} = i\prod \sqrt{q_i^*}$ up to sign, in $\mathbb{Q}(\zeta_{4|d|})$ ($i \in \mathbb{Q}(\zeta_4)$, Theorem 5.28(c)). If $d$ is even, $d = 2d_1$ with $d_1$ odd and squarefree; $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$ lies in $\mathbb{Q}(\zeta_8)$, and $\sqrt{d_1} \in \mathbb{Q}(\zeta_{4|d_1|})$ by the two odd cases (or trivially if $d_1 = \pm 1$), so $\sqrt{d} = \pm\sqrt{2}\sqrt{d_1} \in \mathbb{Q}(\zeta_8)\mathbb{Q}(\zeta_{4|d_1|}) = \mathbb{Q}(\zeta_{8|d_1|}) = \mathbb{Q}(\zeta_{4|d|})$ by Theorem 5.28(c). The last statement is the case $d = \prod_{q\in S} q^* \equiv 1 \pmod 4$, up to the squarefree representative, which here is $d$ itself. □

The two smallest cases make (a) concrete. For $\ell = 3$ (with $\zeta_3 = e^{2\pi i/3}$), $g = \zeta_3 - \zeta_3^2 = i\sqrt{3}$, a square root of $-3$. For $\ell = 5$, the squares modulo 5 are $1, 4$, and $g = (\zeta_5 + \zeta_5^4) - (\zeta_5^2 + \zeta_5^3) = 2\cos(2\pi/5) - 2\cos(4\pi/5) = \frac{\sqrt{5}-1}{2} + \frac{\sqrt{5}+1}{2} = \sqrt{5}$. In both cases $g$ is the square root of $\ell^*$ that is positive or has positive imaginary part; that this holds for every $\ell$ is Gauss's determination of the sign, which the book does not need.

*In the paper* 5.30. The hypothesis of Theorem thm:main concerns $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*})) = \ker \chi_{\ell^*}$, the subgroup of index 2 of $G_\mathbb{Q}$ on which $\bar\chi_\ell$ takes square values. For $\ell = 3$ it is $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\zeta_3)) = \ker \bar\chi_3$; this coincidence $\chi_{-3} = \bar\chi_3$ is why Lemma lem:h0 can phrase the exceptional case both as $\bar\rho \simeq \bar\rho \otimes \bar\chi_3$ and as $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$. For $\ell \geq 5$ the two characters differ, and Section 1.3 of the paper states the criterion with $\bar\chi_\ell$ (see Proposition 5.64(c) and Proposition 5.67).

**Example 5.31** (The splitting field of $x^3 - 2$)**.** Let $\alpha = 2^{1/3} \in \mathbb{R}$, $\zeta = \zeta_3$ and $L = \mathbb{Q}(\alpha, \zeta)$, the splitting field of $x^3 - 2$ (roots $\alpha, \zeta\alpha, \zeta^2\alpha$). As $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$ and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 2$ divide $[L : \mathbb{Q}] \leq 6$, $[L : \mathbb{Q}] = 6$. Let $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$ fix $\zeta$ and send $\alpha \mapsto \zeta\alpha$, and let $\tau$ be complex conjugation, which fixes $\alpha$ and sends $\zeta \mapsto \zeta^2$ (both exist by Theorem 5.12(b): the extensions of $\mathbb{Q}(\zeta) \to \mathbb{Q}(\zeta)$, identity or conjugation, to $L$ are determined by the image of $\alpha$, any of the three roots). Then $\sigma^3 = \tau^2 = 1$ and $\tau\sigma\tau(\alpha) = \tau(\zeta\alpha) = \zeta^2\alpha = \sigma^{-1}(\alpha)$, $\tau\sigma\tau(\zeta) = \zeta = \sigma^{-1}(\zeta)$, so $\tau\sigma\tau = \sigma^{-1}$ and $\mathrm{Gal}(L/\mathbb{Q}) \simeq S_3$, acting faithfully on the three roots. One computes $\sigma\tau(\zeta^2\alpha) = \zeta^4 \cdot \zeta\alpha = \zeta^2\alpha$ and $\sigma^2\tau(\zeta\alpha) = \zeta^2 \cdot \zeta^2\alpha = \zeta\alpha$; so each subgroup of order 2 is the stabiliser of the root it fixes, and the correspondence of Theorem 5.23 is:

| subgroup | fixed field | degree over $\mathbb{Q}$ |
|---|---|---|
| $\{1\}$ | $L$ | 6 |
| $\langle\tau\rangle$ | $\mathbb{Q}(\alpha)$ | 3 |
| $\langle\sigma\tau\rangle$ | $\mathbb{Q}(\zeta^2\alpha)$ | 3 |
| $\langle\sigma^2\tau\rangle$ | $\mathbb{Q}(\zeta\alpha)$ | 3 |
| $\langle\sigma\rangle = A_3$ | $\mathbb{Q}(\zeta) = \mathbb{Q}(\sqrt{-3})$ | 2 |
| $S_3$ | $\mathbb{Q}$ | 1 |

The three cubic fields are conjugate (Theorem 5.23(c)) and not normal; $\mathbb{Q}(\sqrt{-3})$ is the only normal subfield other than $\mathbb{Q}$ and $L$, cut out by the sign of the permutation of the roots, and $\operatorname{disc}(x^3-2) = -108 = -3\cdot 6^2$ (Proposition 5.44). This is the pattern of Proposition prop:ns in miniature: a group of permutations of roots, and the quadratic subfield cut out by the sign, which there is $\det\bar\rho_{E,3} = \chi_{-3}$. The group is also the affine group $\mathbb{Z}/3 \rtimes (\mathbb{Z}/3)^\times$ of Proposition 5.38.

## 5.5 Cyclic extensions: Hilbert 90, Kummer, Artin-Schreier

Kummer theory describes the abelian extensions of exponent dividing $n$ of a field containing the $n$-th roots of unity: they are obtained by adjoining $n$-th roots, and the Galois action on an $n$-th root $\alpha$ of $a$ is recorded by the character $\sigma \mapsto \sigma(\alpha)/\alpha$. The paper uses it twice. In Lemma lem:tamagawa-exp the $\ell^t$-torsion of a Tate curve over $\mathbb{Q}_q^{\mathrm{ur}}$ is generated by $\mu_{\ell^t}$ and an $\ell^t$-th root of the Tate parameter $q_E$, and "by Kummer theory" inertia acts trivially iff $q_E$ is an $\ell^t$-th power; in Lemma lem:tamfactor the Galois action on $T_\ell E$ is given by the Kummer cocycle $\kappa$ of $q_E$, whose unramified part is the class of the unit part of $q_E$ in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$. The field-theoretic statements are proved here; the $\ell$-adic cocycle is Proposition 5.70, and the cohomological form $H^1(K,\mu_n) \simeq K^\times/K^{\times n}$ is Chapter 17.

**Definition 5.32** (Norm and trace)**.** Let $L/F$ be finite separable of degree $n$ and $\sigma_1,\dots,\sigma_n$ its $F$-embeddings into $\overline{\mathbb{F}}$. For $a \in L$, $\mathrm{N}_{L/F}(a) = \prod_i \sigma_i(a)$ and $\mathrm{Tr}_{L/F}(a) = \sum_i \sigma_i(a)$.

**Proposition 5.33** (Norm and trace)**.** *Let $L/F$ be finite separable of degree $n$ and $a \in L$.*

*(a) The characteristic polynomial of the $F$-linear map $m_a\colon x \mapsto ax$ of $L$ is $m_{a,F}^{[L:F(a)]}$; hence $\mathrm{N}_{L/F}(a) = \det m_a$ and $\mathrm{Tr}_{L/F}(a) = \operatorname{tr} m_a$, and both lie in $F$.*

*(b) $\mathrm{N}_{L/F}$ is multiplicative and $\mathrm{Tr}_{L/F}$ is $F$-linear; for $L/F$ Galois, $\mathrm{N}_{L/F}(a) = \prod_{\sigma\in\mathrm{Gal}(L/F)}\sigma(a)$.*

*(c) For $F \subseteq K \subseteq L$: $\mathrm{N}_{L/F} = \mathrm{N}_{K/F}\circ\mathrm{N}_{L/K}$ and $\mathrm{Tr}_{L/F} = \mathrm{Tr}_{K/F}\circ\mathrm{Tr}_{L/K}$.*

*(d) $\mathrm{Tr}_{L/F}$ is not identically zero.*

*Proof.* (a) Let $K = F(a)$, $d = [K:F]$, $m = [L:K]$, and $y_1,\dots,y_m$ a $K$-basis of $L$. In the $F$-basis $(a^i y_j)$ of $L$ (Proposition 5.6) the matrix of $m_a$ is block diagonal with $m$ copies of the companion matrix of $m_{a,F}$, whose characteristic polynomial is $m_{a,F}$. On the other side, each of the $d$ embeddings of $K$ (sending $a$ to the $d$ distinct roots $a_1,\dots,a_d$ of $m_{a,F}$) has exactly $m$ extensions to $L$ (Proposition 5.16(c), (d)), so $\prod_i\sigma_i(a) = (\prod_k a_k)^m$ and $\sum_i\sigma_i(a) = m\sum_k a_k$, which are the determinant and the trace of the block matrix. (b) Multiplicativity and linearity are clear from the definition (or from (a)). If $L/F$ is Galois, the embeddings are the elements of $\mathrm{Gal}(L/F)$. (c) Every $F$-embedding of $L$ is obtained, exactly once, by choosing an $F$-embedding

$\tau$ of $K$, extending it to an automorphism $\tilde\tau$ of $\overline{\mathbb{F}}$ (Theorem 5.10(d)), and composing with one of the $K$-embeddings $\rho_1,\dots,\rho_m$ of $L$; so $\mathrm{N}_{L/F}(a)=\prod_\tau\prod_j\tilde\tau\rho_j(a)=\prod_\tau\tau(\mathrm{N}_{L/K}(a))$, and similarly for the trace. (d) $\mathrm{Tr}_{L/F}=\sum_i\sigma_i$ is a nontrivial combination of distinct embeddings, nonzero by Theorem 5.19. □

Two examples show both descriptions at work. For $L=\mathbb{Q}(\sqrt{d})$ and $a=x+y\sqrt{d}$, the matrix of $m_a$ in the basis $1,\sqrt{d}$ is $\left(\begin{smallmatrix}x & dy\\ y & x\end{smallmatrix}\right)$, so $\mathrm{N}(a)=x^2-dy^2=(x+y\sqrt{d})(x-y\sqrt{d})$ and $\mathrm{Tr}(a)=2x$. For $a=2^{1/3}\in\mathbb{Q}(2^{1/3})$, $m_a$ sends the basis $1,a,a^2$ to $a,a^2,2$; its matrix has determinant 2 and trace 0, and indeed the conjugates $a,\zeta_3a,\zeta_3^2a$ have product 2 and sum 0.

**Theorem 5.34** (Hilbert's Theorem 90)**.** *(a) Let $L/F$ be finite Galois with group $G$, and let $(a_\sigma)_{\sigma\in G}$ be elements of $L^\times$ with $a_{\sigma\tau}=a_\sigma\,\sigma(a_\tau)$ for all $\sigma,\tau$. Then there is $c\in L^\times$ with $a_\sigma=\sigma(c)/c$ for all $\sigma$.*

*(b) Let $L/F$ be cyclic of degree $n$ with $\mathrm{Gal}(L/F)=\langle\sigma\rangle$. For $b\in L^\times$: $\mathrm{N}_{L/F}(b)=1$ iff $b=c/\sigma(c)$ for some $c\in L^\times$.*

*(c) In the same situation, for $b\in L$: $\mathrm{Tr}_{L/F}(b)=0$ iff $b=c-\sigma(c)$ for some $c\in L$.*

*Proof.* (a) By Theorem 5.19 the map $\sum_{\tau\in G}a_\tau\tau$ is not identically zero; choose $\theta$ with $c_0=\sum_\tau a_\tau\tau(\theta)\neq0$. For $\sigma\in G$, $\sigma(c_0)=\sum_\tau\sigma(a_\tau)\,\sigma\tau(\theta)=\sum_\tau a_\sigma^{-1}a_{\sigma\tau}\,\sigma\tau(\theta)=a_\sigma^{-1}c_0$. So $a_\sigma=c_0/\sigma(c_0)=\sigma(c)/c$ with $c=c_0^{-1}$. (b) If $b=c/\sigma(c)$, then $\mathrm{N}(b)=\mathrm{N}(c)/\mathrm{N}(\sigma c)=1$, the norm being the product over the group. Conversely, put $a_{\sigma^i}=b\,\sigma(b)\cdots\sigma^{i-1}(b)$ for $0\le i<n$ ($a_1=1$); this defines $a$ on all of $G$, each element being $\sigma^i$ for exactly one such $i$. For $0\le i,j<n$,

$$a_{\sigma^i}\,\sigma^i(a_{\sigma^j})=\prod_{k=0}^{i-1}\sigma^k(b)\cdot\prod_{k=i}^{i+j-1}\sigma^k(b)=\prod_{k=0}^{i+j-1}\sigma^k(b).$$

If $i+j<n$ this is $a_{\sigma^{i+j}}$ by definition. If $i+j\ge n$, then $\sigma^{i+j}=\sigma^{i+j-n}$ with $0\le i+j-n<n$, and as $\sigma^n=1$,

$$\prod_{k=0}^{i+j-1}\sigma^k(b)=\prod_{k=0}^{n-1}\sigma^k(b)\cdot\prod_{k=0}^{i+j-n-1}\sigma^k(b)=\mathrm{N}_{L/F}(b)\,a_{\sigma^{i+j-n}}=a_{\sigma^{i+j}}.$$

So the cocycle relation holds. By (a), $b=a_\sigma=\sigma(d)/d$ for some $d$; put $c=d^{-1}$. (c) If $b=c-\sigma(c)$, $\mathrm{Tr}(b)=0$. Conversely, choose $\theta$ with $\mathrm{Tr}\,\theta\neq0$ (Proposition 5.33(d)), put $s_i=b+\sigma(b)+\cdots+\sigma^i(b)$ (so $s_{n-1}=\mathrm{Tr}\,b=0$ and $\sigma(s_i)=s_{i+1}-b$) and $c=(\mathrm{Tr}\,\theta)^{-1}\sum_{i=0}^{n-2}s_i\sigma^{i+1}(\theta)$. Then

$$(\mathrm{Tr}\,\theta)\,\sigma(c)=\sum_{i=0}^{n-2}(s_{i+1}-b)\sigma^{i+2}(\theta)=\sum_{j=1}^{n-2}s_j\sigma^{j+1}(\theta)-b\sum_{j=2}^{n}\sigma^j(\theta),$$

using $s_{n-1}=0$. Subtracting from $(\mathrm{Tr}\,\theta)c=s_0\sigma(\theta)+\sum_{j=1}^{n-2}s_j\sigma^{j+1}(\theta)$ and using $s_0=b$ gives $(\mathrm{Tr}\,\theta)(c-\sigma(c))=b\sum_{j=1}^{n}\sigma^j(\theta)=b\,\mathrm{Tr}\,\theta$. □

For $L=\mathbb{Q}(i)$ and $\sigma$ complex conjugation, (b) says that the points $x+iy$ with $x,y\in\mathbb{Q}$ on the circle $x^2+y^2=1$ are the quotients $c/\bar c$, $c=m+ni$ with $m,n\in\mathbb{Q}$ not both zero; and $c/\bar c=c^2/(c\bar c)=\frac{m^2-n^2}{m^2+n^2}+\frac{2mn}{m^2+n^2}\,i$ is the classical parametrisation of Pythagorean triples ($c=2+i$ gives $(3+4i)/5$). The shape of the condition in (a), $a_{\sigma\tau}=a_\sigma\,\sigma(a_\tau)$, depends on the convention that automorphisms compose as maps and act on the left (Convention 5.1); the same shape recurs in the cocycles $b$ of Proposition 5.38 and $\kappa_a$ of Proposition 5.70.

**Proposition 5.35** (Kummer characters)**.** *Let $n \geq 1$, $\operatorname{char} F \nmid n$, and assume $\mu_n \subset F$ (so $\mu_n$ is cyclic of order $n$). Let $a \in F^\times$, $\alpha \in \overline{\mathbb{F}}$ with $\alpha^n = a$, and $L/F$ a Galois extension containing $\alpha$.*

*(a) $\chi_a\colon \operatorname{Gal}(L/F) \to \mu_n$, $\sigma \mapsto \sigma(\alpha)/\alpha$, is a homomorphism that does not depend on the choice of $\alpha$; its kernel is $\operatorname{Gal}(L/F(\alpha))$. Hence $F(\alpha)/F$ is Galois with cyclic group $\simeq \chi_a(\operatorname{Gal}(L/F))$, of order $d$ equal to the order of the class of $a$ in $F^\times/F^{\times n}$; in particular $d \mid n$.*

*(b) $\chi_{ab} = \chi_a\chi_b$, and $\chi_a = 1$ iff $a \in F^{\times n}$. So if $L$ contains an $n$-th root of every element of $F^\times$ (for instance $L = F^{\mathrm{sep}}$), $a \mapsto \chi_a$ is an injective homomorphism $F^\times/F^{\times n} \to \operatorname{Hom}(\operatorname{Gal}(L/F), \mu_n)$.*

*(c) The minimal polynomial of $\alpha$ is $x^d - \alpha^d$, and $x^n - a$ is irreducible over $F$ iff $a \notin F^{\times p}$ for every prime $p \mid n$.*

*Without the assumption $\mu_n \subset F$ the irreducibility criterion is Capelli's theorem: $x^n - a$ is irreducible iff $a \notin F^{\times p}$ for every prime $p \mid n$ and $a \notin -4F^{\times 4}$ when $4 \mid n$ (for example $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$). We do not prove this; see [109, Chapter VI, Theorem 9.1].*

*Proof.* (a) The polynomial $x^n - a$ is separable ($a \neq 0$ and $n \neq 0$ in $F$), with the $n$ roots $\zeta\alpha$, $\zeta \in \mu_n$, all in $F(\alpha)$; so $F(\alpha)$ is its splitting field, Galois over $F$. For $\sigma \in \operatorname{Gal}(L/F)$, $\sigma(\alpha)^n = \sigma(a) = a$, so $\chi_a(\sigma) = \sigma(\alpha)/\alpha \in \mu_n$. Another root is $\alpha' = \zeta\alpha$ with $\zeta \in \mu_n \subset F$ fixed by $\sigma$, so $\sigma(\alpha')/\alpha' = \sigma(\alpha)/\alpha$. Since $\chi_a(\tau) \in F$,

$$\chi_a(\sigma\tau) = \frac{\sigma(\tau(\alpha))}{\alpha} = \frac{\sigma(\chi_a(\tau)\alpha)}{\alpha} = \chi_a(\tau)\frac{\sigma(\alpha)}{\alpha} = \chi_a(\sigma)\chi_a(\tau).$$

The kernel consists of the $\sigma$ fixing $\alpha$, that is $\operatorname{Gal}(L/F(\alpha))$, and

$$\operatorname{Gal}(F(\alpha)/F) \simeq \operatorname{Gal}(L/F)/\operatorname{Gal}(L/F(\alpha)) \simeq \chi_a(\operatorname{Gal}(L/F)),$$

a subgroup of the cyclic group $\mu_n$, of some order $d \mid n$. Then $\chi_a(\sigma)^d = 1$ for all $\sigma$, so $\alpha^d$ is fixed by $\operatorname{Gal}(L/F)$, $\alpha^d \in F$, and $a^d = (\alpha^d)^n \in F^{\times n}$. Conversely, if $a^k = b^n$ with $b \in F^\times$, then $(\alpha^k/b)^n = 1$, so $\alpha^k \in \mu_n b \subseteq F$ and $\chi_a(\sigma)^k = \sigma(\alpha^k)/\alpha^k = 1$ for all $\sigma$, whence $d \mid k$. So $d$ is the order of $a$ modulo $F^{\times n}$. (b) With $\alpha^n = a$, $\beta^n = b$, $(\alpha\beta)^n = ab$ and $\sigma(\alpha\beta)/\alpha\beta = \chi_a(\sigma)\chi_b(\sigma)$. $\chi_a = 1$ iff $\alpha \in F$ iff $a \in F^{\times n}$ (if $a = b^n$, then $\alpha \in \mu_n b \subseteq F$). (c) $\alpha$ is a root of $x^d - \alpha^d \in F[x]$, of degree $d = [F(\alpha) : F]$. So $x^n - a$ is irreducible iff $d = n$, iff the class of $a$ in the group $F^\times/F^{\times n}$ of exponent $n$ has order exactly $n$, iff $a^{n/p} \notin F^{\times n}$ for every prime $p \mid n$ (an element $x$ with $x^n = 1$ has order exactly $n$ iff $x^{n/p} \neq 1$ for every prime $p \mid n$, since every proper divisor of $n$ divides some $n/p$). Finally $a^{n/p} \in F^{\times n}$ iff $a \in F^{\times p}$: if $a = b^p$, $a^{n/p} = b^n$; if $a^{n/p} = b^n$, then $(a/b^p)^{n/p} = 1$, so $a/b^p \in \mu_{n/p} = \{\zeta^p : \zeta \in \mu_n\}$ ($\mu_n$ being cyclic of order $n$), and $a$ is a $p$-th power. □

For $n = 2$ the hypothesis $\mu_2 = \{\pm 1\} \subset F$ always holds (characteristic $\neq 2$), $\chi_a$ is the quadratic character $\sigma \mapsto \sigma(\sqrt{a})/\sqrt{a}$ of $F(\sqrt{a})$, and (c) says that $x^2 - a$ is irreducible iff $a$ is not a square. For $n = 3$, $F = \mathbb{Q}(\zeta_3)$ and $a = 2$: 2 is not a cube in $F$, because a cube root of 2 has degree 3 over $\mathbb{Q}$ ($x^3 - 2$ is irreducible) and cannot lie in a field of degree 2. So $x^3 - 2$ is irreducible over $\mathbb{Q}(\zeta_3)$, $F(2^{1/3})$ is the field $L$ of Example 5.31, and $\chi_2$ maps $\operatorname{Gal}(L/\mathbb{Q}(\zeta_3)) = \langle\sigma\rangle$ isomorphically onto $\mu_3$, with $\chi_2(\sigma) = \zeta_3$.

**Theorem 5.36** (Cyclic extensions)**.** *Let $\operatorname{char} F \nmid n$ and $\mu_n \subset F$. Every cyclic extension $L/F$ of degree $n$ is of the form $L = F(c)$ with $c^n \in F^\times$.*

*Proof.* Let $\mathrm{Gal}(L/F) = \langle\sigma\rangle$ and $\zeta$ a primitive $n$-th root of unity. As $\zeta \in F$, $\mathrm{N}_{L/F}(\zeta^{-1}) = \zeta^{-n} = 1$, so by Theorem 5.34(b) $\zeta^{-1} = c/\sigma(c)$ for some $c \in L^\times$: $\sigma(c) = \zeta c$. Then $\sigma^i(c) = \zeta^i c$, $0 \le i < n$, are $n$ conjugates of $c$, distinct because $\zeta$ has order $n$ and $c \neq 0$; all are roots of $m_{c,F}$, so $[F(c) : F] \ge n = [L : F]$ and $F(c) = L$; and $\sigma(c^n) = c^n$, so $c^n \in F$. □

**Theorem 5.37** (Kummer theory)**.** *Let* $\mathrm{char}\, F \nmid n$ *and* $\mu_n \subset F$*. For a subgroup* $F^{\times n} \subseteq \Delta \subseteq F^\times$ *let* $F_\Delta = F(\Delta^{1/n})$ *be the subfield of* $F^{\mathrm{sep}}$ *generated by all* $\alpha$ *with* $\alpha^n \in \Delta$*.*

*(a) If* $\Delta/F^{\times n}$ *is finite,* $F_\Delta/F$ *is a finite abelian extension of exponent dividing* $n$*, and*

$$\mathrm{Gal}(F_\Delta/F) \times \Delta/F^{\times n} \to \mu_n, \qquad (\sigma, a) \mapsto \sigma(\alpha)/\alpha \quad (\alpha^n = a),$$

*is a well defined bilinear pairing which is perfect: it induces isomorphisms* $\mathrm{Gal}(F_\Delta/F) \simeq \mathrm{Hom}(\Delta/F^{\times n}, \mu_n)$ *and* $\Delta/F^{\times n} \simeq \mathrm{Hom}(\mathrm{Gal}(F_\Delta/F), \mu_n)$*. In particular* $[F_\Delta : F] = [\Delta : F^{\times n}]$*.*

*(b)* $\Delta \mapsto F_\Delta$ *is an inclusion-preserving bijection from the groups* $\Delta$ *with* $\Delta/F^{\times n}$ *finite to the finite abelian extensions of* $F$ *inside* $F^{\mathrm{sep}}$ *of exponent dividing* $n$*; the inverse is* $L \mapsto L^{\times n} \cap F^\times$*. The cyclic extensions of degree* $d$ *correspond to the* $\Delta$ *with* $\Delta/F^{\times n}$ *cyclic of order* $d$*.*

*The form without finiteness,* $F^\times/F^{\times n} \simeq \mathrm{Hom}_{\mathrm{cont}}(G_F, \mu_n)$*, is Corollary 5.56.*

*Proof.* (a) Let $a_1, \dots, a_r$ represent generators of $\Delta/F^{\times n}$ and $\alpha_i^n = a_i$. If $\alpha^n = a \in \Delta$, write $a = b^n \prod a_i^{k_i}$ with $b \in F^\times$; then $\alpha/(b\prod \alpha_i^{k_i}) \in \mu_n \subset F$, so $F_\Delta = F(\alpha_1, \dots, \alpha_r)$, the splitting field of the separable $\prod_i (x^n - a_i)$, finite Galois. The pairing is well defined by Proposition 5.35(a) (independence of $\alpha$) and (b) ($\chi_a$ depends only on $a$ modulo $F^{\times n}$), and bilinear by the homomorphism property in each variable. If $\sigma$ pairs trivially with all $a$, it fixes every $\alpha_i$, so $\sigma = 1$; if $a$ pairs trivially with all $\sigma$, then $\chi_a = 1$ and $a \in F^{\times n}$. So $G = \mathrm{Gal}(F_\Delta/F) \to \mathrm{Hom}(\Delta/F^{\times n}, \mu_n)$ and $\Delta/F^{\times n} \to \mathrm{Hom}(G, \mu_n)$ are injective. The first shows that $G$ is abelian of exponent dividing $n$. For a finite abelian group $A$ of exponent dividing $n$, $\#\,\mathrm{Hom}(A, \mu_n) = \#A$ (write $A$ as a product of cyclic groups $\mathbb{Z}/m$ with $m \mid n$, Chapter 1, and note $\mathrm{Hom}(\mathbb{Z}/m, \mu_n) \simeq \mu_m$). Hence $\#G \le \#(\Delta/F^{\times n}) \le \#G$, both injections are bijections, and $[F_\Delta : F] = \#G = [\Delta : F^{\times n}]$.

(b) Let $L/F$ be finite abelian of exponent dividing $n$ and $\Delta_L = L^{\times n} \cap F^\times$. If $\beta \in L$ with $\beta^n = a \in F$, all $n$-th roots of $a$ are $\zeta\beta \in L$; so $F_{\Delta_L} \subseteq L$, and in particular $\Delta_L/F^{\times n}$ is finite by (a) (it injects into $\mathrm{Hom}(\mathrm{Gal}(F_{\Delta_L}/F), \mu_n)$, the injectivity argument of (a) not using finiteness). Conversely, $A = \mathrm{Gal}(L/F)$ is a finite abelian group, and the intersection of the subgroups $B$ with $A/B$ cyclic is trivial (by the structure theorem, Chapter 1, for $1 \neq x \in A$ some projection to a cyclic factor is nonzero at $x$). By Theorem 5.23(e), $L$ is the compositum of the fixed fields $L^B$, each cyclic of some degree $e \mid n$, hence $L^B = F(\beta)$ with $\beta^e \in F$ (Theorem 5.36, as $\mu_e \subseteq \mu_n \subset F$). Then $\beta^n = (\beta^e)^{n/e} \in F$, so $\beta \in F_{\Delta_L}$; hence $L = F_{\Delta_L}$. Finally $\Delta_{F_\Delta} = \Delta$: the inclusion $\supseteq$ is clear; if $a \in F^\times \cap F_\Delta^{\times n}$, the group $\Delta'$ generated by $\Delta$ and $a$ has $F_{\Delta'} = F_\Delta$, so $[\Delta' : F^{\times n}] = [\Delta : F^{\times n}]$ by (a) and $a \in \Delta$. The two maps preserve inclusions. By (a), $\mathrm{Gal}(F_\Delta/F)$ is cyclic of order $d$ iff its dual $\Delta/F^{\times n}$ is. □

For example, let $F = \mathbb{Q}$, $n = 2$ and $\Delta$ the group generated by $\mathbb{Q}^{\times 2}$, 2 and 3. Then $\Delta/\mathbb{Q}^{\times 2}$ consists of the classes of $1, 2, 3, 6$, $F_\Delta = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, and the pairing identifies $\mathrm{Gal}(F_\Delta/\mathbb{Q}) \simeq V_4$ with the dual of $\Delta/\mathbb{Q}^{\times 2} \simeq (\mathbb{Z}/2)^2$. The three subgroups of order 2 of $\Delta/\mathbb{Q}^{\times 2}$, generated by 2, 3 and 6, correspond under (b) to the three quadratic subfields $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{6})$ found in Proposition 5.27(d).

**Proposition 5.38** (The affine shape without roots of unity)**.** *Let* $\operatorname{char} F \nmid n$, $a \in F^\times$, $\alpha^n = a$, $L = F(\mu_n, \alpha)$, *and fix a primitive* $n$*-th root of unity* $\zeta_n$*. For* $\sigma \in \mathrm{Gal}(L/F)$ *define* $u(\sigma) \in (\mathbb{Z}/n)^\times$ *and* $b(\sigma) \in \mathbb{Z}/n$ *by* $\sigma(\zeta_n) = \zeta_n^{u(\sigma)}$ *and* $\sigma(\alpha) = \zeta_n^{b(\sigma)}\alpha$*. Then*

$$\sigma \longmapsto \begin{pmatrix} u(\sigma) & b(\sigma) \\ 0 & 1 \end{pmatrix}$$

*is an injective homomorphism from* $\mathrm{Gal}(L/F)$ *to the affine group* $\left\{ \left(\begin{smallmatrix} u & b \\ 0 & 1 \end{smallmatrix}\right) : u \in (\mathbb{Z}/n)^\times,\ b \in \mathbb{Z}/n \right\} \simeq \mathbb{Z}/n \rtimes (\mathbb{Z}/n)^\times$*. Equivalently,* $b(\sigma\tau) = b(\sigma) + u(\sigma)b(\tau)$*:* $b$ *is a* 1*-cocycle for the action of* $\mathrm{Gal}(L/F)$ *on* $\mathbb{Z}/n$ *through* $u$*. Its restriction to* $\mathrm{Gal}(L/F(\mu_n))$ *(where* $u = 1$*) is the Kummer character* $\chi_a$*, read in* $\mathbb{Z}/n$ *through* $\zeta_n$*.*

*Proof.* $L$ is the splitting field of the separable $x^n - a$ (its roots $\zeta\alpha$ generate $\mu_n$ and $\alpha$), so it is Galois. We have $\sigma\tau(\zeta_n) = \sigma(\zeta_n^{u(\tau)}) = \zeta_n^{u(\sigma)u(\tau)}$ and $\sigma\tau(\alpha) = \sigma(\zeta_n^{b(\tau)}\alpha) = \zeta_n^{u(\sigma)b(\tau)}\zeta_n^{b(\sigma)}\alpha$, which is the matrix product $\left(\begin{smallmatrix} u(\sigma) & b(\sigma) \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} u(\tau) & b(\tau) \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} u(\sigma)u(\tau) & b(\sigma)+u(\sigma)b(\tau) \\ 0 & 1 \end{smallmatrix}\right)$. The map is injective since $\sigma$ is determined by $\sigma(\zeta_n)$ and $\sigma(\alpha)$. □

The order of the factors reflects the convention $(\sigma\tau)(x) = \sigma(\tau(x))$: with automorphisms composed in the opposite order the same formulas would define an anti-homomorphism, and the cocycle relation would read $b(\sigma\tau) = b(\tau) + u(\tau)b(\sigma)$.

For $x^3 - 2$ over $\mathbb{Q}$ (Example 5.31) the Galois group has order 6 and embeds in the affine group of order 6, so $\mathrm{Gal} \simeq \mathbb{Z}/3 \rtimes (\mathbb{Z}/3)^\times \simeq S_3$. The $\ell$-adic limit of this matrix, $\left(\begin{smallmatrix} \chi_\ell & \kappa_a \\ 0 & 1 \end{smallmatrix}\right)$, is Proposition 5.70; twisted by an unramified character $\varepsilon$ of order at most 2 it becomes the matrix $\left(\begin{smallmatrix} \chi_\ell\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$ of the proof of Lemma lem:tamfactor (Chapter 10).

**Proposition 5.39** (Kummer theory over a finite field)**.** *Let* $q$ *be a power of a prime* $p$*,* $\ell \neq p$ *a prime,* $v = v_\ell(q-1)$*, and* $u \in \mathbb{F}_q^\times$*.*

*(a) For* $1 \le k \le v$*,* $\mu_{\ell^k} \subset \mathbb{F}_q$*, and the Kummer character* $\chi_u$ *(with* $n = \ell^k$*) satisfies* $\chi_u(\mathrm{Frob}_q) = u^{(q-1)/\ell^k}$*. Hence* $u \in (\mathbb{F}_q^\times)^{\ell^k}$ *iff* $u^{(q-1)/\ell^k} = 1$*, and* $u \mapsto u^{(q-1)/\ell^v}$ *induces an isomorphism of cyclic groups of order* $\ell^v$

$$\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v} \xrightarrow{\ \sim\ } \mu_{\ell^v}(\mathbb{F}_q);$$

*the left side is also* $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$*.*

*(b) Let* $(\alpha_k)_{k\ge 0}$ *be a compatible system of* $\ell$*-power roots of* $u$ *in* $\overline{\mathbb{F}}_q$ *(*$\alpha_0 = u$*,* $\alpha_{k+1}^\ell = \alpha_k$*), and put* $\kappa_u(\mathrm{Frob}_q) = (\mathrm{Frob}_q(\alpha_k)/\alpha_k)_k = (\alpha_k^{q-1})_k \in \mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}(\overline{\mathbb{F}}_q)$*. Another compatible system changes* $\kappa_u(\mathrm{Frob}_q)$ *by an element of* $(q-1)\mathbb{Z}_\ell(1) = \ell^v\mathbb{Z}_\ell(1)$*. The class of* $\kappa_u(\mathrm{Frob}_q)$ *in* $\mathbb{Z}_\ell(1)/\ell^v\mathbb{Z}_\ell(1) \simeq \mu_{\ell^v}(\mathbb{F}_q)$ *is* $u^{(q-1)/\ell^v}$*, and for* $k \le v$ *it lies in* $\ell^k\mathbb{Z}_\ell(1)/\ell^v\mathbb{Z}_\ell(1)$ *iff* $u \in (\mathbb{F}_q^\times)^{\ell^k}$*.*

*Proof.* (a) $\mathbb{F}_q^\times$ is cyclic of order $q-1$ (Chapter 2), divisible by $\ell^k$, so it contains $\mu_{\ell^k}$. If $\alpha^{\ell^k} = u$, then $\chi_u(\mathrm{Frob}_q) = \alpha^q/\alpha = \alpha^{q-1} = (\alpha^{\ell^k})^{(q-1)/\ell^k} = u^{(q-1)/\ell^k}$. As $\mathrm{Frob}_q$ generates $\mathrm{Gal}(\mathbb{F}_q(\alpha)/\mathbb{F}_q)$ (Theorem 5.25), $\chi_u = 1$ iff $\chi_u(\mathrm{Frob}_q) = 1$, and $\chi_u = 1$ iff $u \in (\mathbb{F}_q^\times)^{\ell^k}$ (Proposition 5.35(b)). The map $u \mapsto u^{(q-1)/\ell^v}$ is a homomorphism to $\mu_{\ell^v}(\mathbb{F}_q)$ with kernel $(\mathbb{F}_q^\times)^{\ell^v}$ (case $k = v$), surjective because a generator $g$ of $\mathbb{F}_q^\times$ maps to an element of order $\ell^v$. Writing $q - 1 = \ell^v m$ with $\ell \nmid m$, $\mathbb{F}_q^\times \simeq \mathbb{Z}/\ell^v \times \mathbb{Z}/m$ and $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^v$. (b) $(\alpha_k^{q-1})^{\ell^k} = u^{q-1} = 1$ and $(\alpha_{k+1}^{q-1})^\ell = \alpha_k^{q-1}$, so $(\alpha_k^{q-1})_k \in \mathbb{Z}_\ell(1)$. Another system is $(\epsilon_k\alpha_k)$ with $\epsilon = (\epsilon_k) \in \mathbb{Z}_\ell(1)$, and $(\epsilon_k\alpha_k)^{q-1} = \epsilon_k^{q-1}\alpha_k^{q-1}$: in additive notation the value changes by $(q-1)\epsilon$; and $(q-1)\mathbb{Z}_\ell(1) = \ell^v\mathbb{Z}_\ell(1)$ as $(q-1)/\ell^v \in \mathbb{Z}_\ell^\times$.

The projection to the $v$-th component identifies $\mathbb{Z}_\ell(1)/\ell^v\mathbb{Z}_\ell(1)$ with $\mu_{\ell^v}$, and the $v$-th component of $\kappa_u(\mathrm{Frob}_q)$ is $\alpha_v^{q-1} = u^{(q-1)/\ell^v}$ by (a). Finally, $\ell^k\mathbb{Z}_\ell(1)/\ell^v\mathbb{Z}_\ell(1)$ corresponds to $\mu_{\ell^{v-k}} = \mu_{\ell^v}^{\ell^k}$, and $u^{(q-1)/\ell^v} \in \mu_{\ell^{v-k}}$ iff $u^{(q-1)/\ell^k} = 1$, iff $u \in (\mathbb{F}_q^\times)^{\ell^k}$ by (a). □

For $\ell = 2$, $q$ odd and $k = 1$, (a) is Euler's criterion: $u$ is a square in $\mathbb{F}_q$ iff $u^{(q-1)/2} = 1$. A cubic example: $q = 7$, $\ell = 3$, so $v = 1$ and $(q-1)/3 = 2$. Then $u \in \mathbb{F}_7^\times$ is a cube iff $u^2 = 1$, that is $u = \pm 1$; indeed the cubes of $1, \dots, 6$ are $1, 1, 6, 1, 6, 6$. The class of $u = 3$ is $3^2 = 2$, a primitive cube root of unity in $\mathbb{F}_7$; so 3 is not a cube, $x^3 - 3$ is irreducible over $\mathbb{F}_7$ (Proposition 5.35(c)), and $\mathrm{Frob}_7$ multiplies each cube root $\alpha$ of 3 by $\alpha^6 = 3^2 = 2$.

*In the paper* 5.40. Lemma lem:tamfactor defines $c(u)$ as the class of the reduction $\bar{u}$ of the unit part $u = q_E q^{-n}$ of the Tate parameter in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$, identified with $\mathbb{Z}/\ell^v$ by a choice of generator, and states that $v_\ell(c(u)) \geq k$ iff $\bar{u}$ is an $\ell^k$-th power, for $k \leq v$, with $v_\ell(0)$ read as $v$. That is Proposition 5.39(a), (b): the valuation of the class does not depend on the generator, and the class is the value at $\mathrm{Frob}_q$ of the Kummer cocycle of $\bar{u}$, an element of $\mathbb{Z}_\ell(1)/(q-1) = \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$. In the proof of the lemma this class is the unramified part $\kappa_u$ of the Kummer class $\kappa = n\kappa_q + \kappa_u$ of $q_E$; passing from $\mathbb{F}_q$ to $\mathbb{Q}_q$ needs Hensel's lemma and the inflation from $G_{\mathbb{Q}_q}/I_q$, which are Chapter 7 and Chapter 17. In Lemma lem:tamagawa-exp the field is $\mathbb{Q}_q^{\mathrm{ur}}$, which contains $\mu_{\ell^t}$ because $q \neq \ell$ (Chapter 7), so Proposition 5.35(b) applies over it: inertia, which is $G_{\mathbb{Q}_q^{\mathrm{ur}}}$, fixes an $\ell^t$-th root of $q_E$ iff $q_E$ is an $\ell^t$-th power in $\mathbb{Q}_q^{\mathrm{ur}}$.

**Theorem 5.41** (Artin-Schreier theory)**.** *Let* $\mathrm{char}\, F = p > 0$ *and* $\wp(b) = b^p - b$*, an additive map* $F \to F$ *with kernel* $\mathbb{F}_p$*.*

*(a) For* $a \in F$ *the polynomial* $x^p - x - a$ *is separable; if* $\alpha$ *is a root, its roots are* $\alpha + i$*,* $i \in \mathbb{F}_p$*. It splits over* $F$ *if* $a \in \wp(F)$*, and is irreducible otherwise; then* $F(\alpha)/F$ *is cyclic of degree* $p$*, with group generated by* $\alpha \mapsto \alpha + 1$*.*

*(b) Every cyclic extension of* $F$ *of degree* $p$ *is* $F(\alpha)$ *with* $\alpha^p - \alpha \in F$*.*

*(c)* $\mathrm{Hom}_{\mathrm{cont}}(G_F, \mathbb{Z}/p) \simeq F/\wp(F)$*, the class of* $a$ *corresponding to* $\sigma \mapsto \sigma(\alpha) - \alpha$ *(*$\wp(\alpha) = a$*).*

*For example* $x^p - x - 1$ *is irreducible over* $\mathbb{F}_p$ *(no element of* $\mathbb{F}_p$ *is a root, as* $\wp(\mathbb{F}_p) = 0$*), with splitting field* $\mathbb{F}_{p^p}$*.*

*Proof of (a) and (b).* (a) The derivative is $-1$, so the polynomial is separable; $(\alpha+i)^p-(\alpha+i)-a = \alpha^p-\alpha-a+i^p-i = 0$ for $i \in \mathbb{F}_p$. If one root lies in $F$, all do. Otherwise $F(\alpha)$ contains all roots, so it is Galois over $F$, and $\sigma \mapsto \sigma(\alpha)-\alpha \in \mathbb{F}_p$ is an injective homomorphism (as in Proposition 5.35(a), $\mathbb{F}_p$ being fixed) into $\mathbb{Z}/p$ with nontrivial image; so the group is $\mathbb{Z}/p$, $[F(\alpha) : F] = p$, and $x^p - x - a$ is irreducible. (b) Let $\mathrm{Gal}(L/F) = \langle\sigma\rangle$ of order $p$. Then $\mathrm{Tr}_{L/F}(-1) = -p = 0$, so Theorem 5.34(c) gives $c$ with $-1 = c - \sigma(c)$, that is $\sigma(c) = c + 1$. Then $c \notin F$, so $L = F(c)$, and $\sigma(\wp(c)) = (c+1)^p - (c+1) = \wp(c)$, so $\wp(c) \in F$. □

*Sketch of proof of (c).* As in the proof of Corollary 5.56, with (a), (b) and the additive pairing $(\sigma, a) \mapsto \sigma(\alpha) - \alpha$ in place of the multiplicative one; see [109, Chapter VI, Section 8]. □

The name is shared with the Artin-Schreier theorem on real closed fields (Proposition 5.60(c)); the two are unrelated. Artin-Schreier extensions are the cyclic $p$-extensions that Kummer theory cannot reach in characteristic $p$; over the $p$-adic fields of Chapter 7, their analogues are the wildly ramified extensions behind Remark rem:wild of the paper.

## 5.6 The Galois group of a polynomial; discriminants, cubics, quartics and the four lines

The Galois group of a polynomial is the group of permutations of its roots induced by field automorphisms, and its orbits are the irreducible factors. This is the dictionary behind the last sentence of Proposition prop:ns: the four roots of the 3-division polynomial $\psi_3$ of $E$ are the $x$-coordinates of the four lines of $E[3]$, so the factorisation of $\psi_3$ over $\mathbb{Q}$ records the orbits of the mod 3 image on the lines, and its discriminant records the determinant. It is also what the test `ns3` of the paper's appendix and the consistency check of Section sec:data compute. The discriminant, cubics and quartics are developed only as far as this application needs.

**Proposition 5.42** (The Galois group of a polynomial)**.** *Let $f \in F[x]$ be separable of degree $n \geq 1$, $R \subseteq F^{\mathrm{sep}}$ its set of $n$ roots, $L = F(R)$ its splitting field and $G_f = \mathrm{Gal}(L/F)$, the* Galois group of $f$*.*

*(a) $G_f$ acts faithfully on $R$; numbering the roots gives an injective homomorphism $G_f \to S_n$, whose image is well defined up to conjugation in $S_n$; $\#G_f = [L : F]$.*

*(b) The $G_f$-orbits on $R$ are the root sets of the monic irreducible factors of $f$ in $F[x]$: the orbit of $r$ is the set of roots of $m_{r,F}$; its size is $[F(r) : F]$, and the stabiliser of $r$ is $\mathrm{Gal}(L/F(r))$. In particular $f$ is irreducible iff $G_f$ is transitive on $R$, and then $n \mid \#G_f$.*

*(c) Let $\Omega/F$ be a Galois extension containing $L$ (for instance $\Omega = F^{\mathrm{sep}}$). Then $\mathrm{Gal}(\Omega/F)$ acts on $R$ through its quotient $G_f$, with the same orbits; and for every intermediate field $F \subseteq M \subseteq \Omega$, the irreducible factors of $f$ in $M[x]$ correspond to the orbits of $\mathrm{Gal}(\Omega/M)$ on $R$.*

*Proof.* (a) An automorphism of $L/F$ permutes the roots of $f$ and is determined by its values on the generators $R$. Renumbering conjugates the image; $\#G_f = [L : F]$ as $L/F$ is Galois (Theorem 5.22). (b) Since $f$ is separable, $f = c\prod_j g_j$ with distinct monic irreducible $g_j$, whose root sets partition $R$. For $r \in R$ and $\sigma \in G_f$, $\sigma(r)$ is a root of $m_{r,F}$; conversely, by Theorem 5.20(a) applied to $G_f$ (with $L^{G_f} = F$), $m_{r,F} = \prod_{\beta \in G_f r}(x - \beta)$. So the orbit of $r$ is the root set of $m_{r,F}$, of size $\deg m_{r,F} = [F(r) : F]$; $\sigma$ fixes $r$ iff it fixes $F(r)$. A transitive group has order divisible by the orbit size $n$. (c) Restriction $\mathrm{Gal}(\Omega/F) \to G_f$ is surjective (Theorem 5.23(d) for finite $\Omega$; Theorem 5.52(d) in general), so the orbits are the same. Over $M$: $f$ is separable in $M[x]$, $\Omega/M$ is Galois, and the first part applied over $M$ (with splitting field $M(R) \subseteq \Omega$) gives the factors over $M$. □

For example, let $f = (x^2-2)(x^2-3)$ over $\mathbb{Q}$. Its splitting field is $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $G_f \simeq V_4$ (Proposition 5.27(d)), with the two orbits $\{\pm\sqrt{2}\}$ and $\{\pm\sqrt{3}\}$ on $R$, the root sets of the two factors. Numbering the roots $\sqrt{2}, -\sqrt{2}, \sqrt{3}, -\sqrt{3}$, the image of $G_f$ in $S_4$ is $\langle (12), (34) \rangle$, the Klein group that is not normal in $S_4$; it reappears for $N_s(\mathbb{F}_3)$ in Proposition 5.47. Over $M = \mathbb{Q}(\sqrt{2})$ the group shrinks to $\mathrm{Gal}(L/M) = \langle (34) \rangle$, and $f = (x - \sqrt{2})(x + \sqrt{2})(x^2 - 3)$ in $M[x]$, as (c) predicts.

**Definition 5.43** (Discriminant)**.** Let $f = a\prod_{i=1}^{n}(x - r_i)$ with $a \neq 0$ and roots $r_i$ in $\overline{\mathbb{F}}$. Put $\delta_f = \prod_{i<j}(r_i - r_j)$ (which depends on the numbering up to sign) and $\mathrm{disc}(f) = a^{2n-2}\delta_f^2$. For monic $g = \prod_i (x - r_i)$ and $h = \prod_j (x - s_j)$, the resultant is $\mathrm{res}(g, h) = \prod_{i,j}(r_i - s_j)$ (written res to avoid confusion with the restriction Res).

**Proposition 5.44** (Discriminants)**.** *Let $f = a\prod(x - r_i) \in F[x]$ have degree $n$.*

*(a)* $\operatorname{disc}(f) = (-1)^{n(n-1)/2}a^{n-2}\prod_{i=1}^{n} f'(r_i)$*; it lies in $F$, and it is nonzero iff $f$ is separable.*

*(b) If $f$ is separable and $\sigma \in G_f \subseteq S_n$, then $\sigma(\delta_f) = \operatorname{sgn}(\sigma)\delta_f$.*

*(c) Let* $\operatorname{char} F \neq 2$ *and $f$ separable. Then $G_f \subseteq A_n$ iff* $\operatorname{disc}(f) \in F^{\times 2}$*. The fixed field of $G_f \cap A_n$ is $F(\delta_f) = F(\sqrt{\operatorname{disc} f})$, and the quadratic character of* $\operatorname{Gal}(F^{\mathrm{sep}}/F)$ *attached to it by Proposition 5.27 is the sign of the permutation of the roots: $\sigma(\sqrt{\operatorname{disc} f}) = \operatorname{sgn}(\sigma|_R)\sqrt{\operatorname{disc} f}$.*

*(d) For monic $g, h$:* $\operatorname{disc}(gh) = \operatorname{disc}(g)\operatorname{disc}(h)\operatorname{res}(g,h)^2$*.*

*(e)* $\operatorname{disc}(x^2 + bx + c) = b^2 - 4c$*;* $\operatorname{disc}(x^3 + px + q) = -4p^3 - 27q^2$*;* $\operatorname{disc}(x^3 + bx^2 + cx + d) = b^2c^2 - 4c^3 - 4b^3d - 27d^2 + 18bcd$*;* $\operatorname{disc}(x^n - a) = (-1)^{(n-1)(n-2)/2}n^n a^{n-1}$*; and for an odd prime $\ell$,* $\operatorname{disc}(\Phi_\ell) = (-1)^{(\ell-1)/2}\ell^{\ell-2} = \ell^* \cdot \ell^{\ell-3}$*.*

*Proof.* (a) Differentiating, $f'(r_i) = a\prod_{j\neq i}(r_i - r_j)$, so

$$\prod_i f'(r_i) = a^n \prod_{i\neq j}(r_i - r_j) = a^n(-1)^{n(n-1)/2}\delta_f^2$$

(each pair $i < j$ contributes $(r_i - r_j)(r_j - r_i) = -(r_i - r_j)^2$, and there are $n(n-1)/2$ pairs). Hence $a^{2n-2}\delta_f^2 = (-1)^{n(n-1)/2}a^{n-2}\prod f'(r_i)$. It vanishes iff two roots coincide. It lies in $F$: if $f$ is separable, $\delta_f^2$ is fixed by $G_f$ (by (b)), hence in $L^{G_f} = F$; in general $\operatorname{disc}(f)$ is the value at the coefficients of $f$ of a universal polynomial with integer coefficients (a symmetric polynomial in the roots, expressed through the elementary symmetric functions, which are $\pm$ the coefficients divided by $a$; the powers of $a$ clear the denominators), and we do not need the inseparable case. (b) $\sigma$ permutes the factors $r_i - r_j$ $(i < j)$ up to sign, and the factor $r_i - r_j$ goes to $r_{\sigma(i)} - r_{\sigma(j)}$, which is $-(r_{\sigma(j)} - r_{\sigma(i)})$ exactly when $\sigma$ inverts the pair; the number of inversions has the parity of $\operatorname{sgn}\sigma$. (c) As $\operatorname{char} F \neq 2$ and $\delta_f \neq 0$, $\delta_f \neq -\delta_f$. By (b), $\delta_f$ is fixed by $\sigma$ iff $\sigma$ is even; so $G_f \subseteq A_n$ iff $\delta_f \in F$ (fundamental theorem), iff $\operatorname{disc} f = (a^{n-1}\delta_f)^2$ is a square in $F$ (if $\operatorname{disc} f = b^2$ with $b \in F$, then $a^{n-1}\delta_f = \pm b$). The fixed field of $G_f \cap A_n$, of index 1 or 2, contains $\delta_f$, and equals $F(\delta_f)$ by degrees. The last formula is (b) with $\sqrt{\operatorname{disc} f} = a^{n-1}\delta_f$. (d) The roots of $gh$ are those of $g$ and of $h$; the squared differences between roots of $g$ and roots of $h$ contribute $\prod_{i,j}(r_i - s_j)^2$. (e) The quadratic case is $(r_1 - r_2)^2 = (r_1 + r_2)^2 - 4r_1r_2$. For $f = x^3 + px + q$, by (a) $\operatorname{disc} f = -\prod_i(3r_i^2 + p)$. With $e_1 = \sum r_i = 0$, $e_2 = p$, $e_3 = r_1r_2r_3 = -q$: $\sum r_i^2 = e_1^2 - 2e_2 = -2p$, $\sum_{i<j} r_i^2r_j^2 = e_2^2 - 2e_1e_3 = p^2$, $\prod r_i^2 = q^2$. Expanding the product of the three factors $3r_i^2 + p$ by powers of $p$,

$$\prod_i(3r_i^2 + p) = 27\prod_i r_i^2 + 9p\sum_{i<j} r_i^2r_j^2 + 3p^2\sum_i r_i^2 + p^3,$$

which is $27q^2 + 9p \cdot p^2 + 3p^2(-2p) + p^3 = 27q^2 + 4p^3$. For the general cubic $f = x^3 + bx^2 + cx + d$ over a field of characteristic 0, the substitution $x \mapsto x - b/3$ does not change the differences of the roots and gives $f(x - b/3) = x^3 + px + q$ with $p = c - b^2/3$ and $q = d - bc/3 + 2b^3/27$; expanding $-4p^3 - 27q^2$ gives $b^2c^2 - 4c^3 - 4b^3d - 27d^2 + 18bcd$ (a routine expansion, which we also checked by computer algebra). This proves the formula over every field of characteristic 0. For the generic monic cubic, $\prod_{i<j}(r_i - r_j)^2$ is a symmetric polynomial in the roots with integer coefficients, hence a polynomial with integer coefficients in $b, c, d$ (the theorem on symmetric polynomials, which holds over $\mathbb{Z}$; we do not prove it, see [109, Chapter IV, Section 6]); the computation just made shows that this polynomial is $b^2c^2 - 4c^3 - 4b^3d - 27d^2 + 18bcd$ in $\mathbb{Q}[b, c, d]$, hence in $\mathbb{Z}[b, c, d]$,

and specialising gives the formula over every field, of characteristic 3 included. For $f = x^n - a$ with roots $r_i$: $f'(r_i) = nr_i^{n-1}$ and $\prod r_i = (-1)^{n+1}a$, so $\prod f'(r_i) = n^n(-1)^{(n+1)(n-1)}a^{n-1}$ and $\operatorname{disc} f = (-1)^{n(n-1)/2+n^2-1}n^n a^{n-1}$; the exponent differs from $(n-1)(n-2)/2$ by $n^2+n-2 = (n+2)(n-1)$, which is even. For $\Phi_\ell$ (degree $\ell-1$, roots $\zeta^a$, $1 \le a \le \ell-1$, $\zeta = \zeta_\ell$): differentiating $(x-1)\Phi_\ell(x) = x^\ell - 1$ at $\zeta^a$ gives $(\zeta^a - 1)\Phi'_\ell(\zeta^a) = \ell\zeta^{-a}$. Now $\prod_a \zeta^{-a} = \zeta^{-\ell(\ell-1)/2} = 1$ ($\ell$ odd) and $\prod_a(\zeta^a - 1) = \prod_a(1-\zeta^a) = \Phi_\ell(1) = \ell$ (an even number of sign changes), so $\prod_a \Phi'_\ell(\zeta^a) = \ell^{\ell-1}/\ell = \ell^{\ell-2}$, and $\operatorname{disc}\Phi_\ell = (-1)^{(\ell-1)(\ell-2)/2}\ell^{\ell-2} = (-1)^{(\ell-1)/2}\ell^{\ell-2}$, since $\ell-2$ is odd. □

Since $\ell^{\ell-3}$ is a square, $\operatorname{disc}(\Phi_\ell) \in \ell^*\mathbb{Q}^{\times 2}$, and (c) gives a second proof that $\sqrt{\ell^*} \in \mathbb{Q}(\zeta_\ell)$: the Galois group $(\mathbb{Z}/\ell)^\times$ of $\Phi_\ell$ acts on the roots $\zeta^a$ by multiplication, and the sign of $x \mapsto ax$ on $(\mathbb{Z}/\ell)^\times$ is the Legendre symbol (Zolotarev's lemma). The values were also checked with PARI for $\ell \le 13$.

**Proposition 5.45** (Cubics)**.** *Let* $\operatorname{char} F \neq 2$ *and* $f \in F[x]$ *a separable cubic. If* $f$ *is irreducible,* $G_f = A_3 \simeq C_3$ *when* $\operatorname{disc} f \in F^{\times 2}$ *and* $G_f = S_3$ *otherwise. If* $f$ *is a linear factor times an irreducible quadratic* $g$*,* $G_f \simeq C_2$*, with splitting field* $F(\sqrt{\operatorname{disc} g}) = F(\sqrt{\operatorname{disc} f})$*. Examples over* $\mathbb{Q}$*:* $x^3 - 2$ *has* $\operatorname{disc} = -108$ *and* $G_f = S_3$ *(Example 5.31);* $x^3 - 3x + 1$ *has* $\operatorname{disc} = 81$ *and* $G_f = C_3$*, its roots are* $2\cos(2\pi k/9)$ *for* $k = 1, 2, 4$*, and its splitting field is the real cubic subfield* $\mathbb{Q}(\zeta_9 + \zeta_9^{-1})$ *of* $\mathbb{Q}(\zeta_9)$*.*

*Proof.* An irreducible cubic has transitive $G_f$ (Proposition 5.42(b)), so $G_f \in \{A_3, S_3\}$, distinguished by Proposition 5.44(c). In the reducible case $L$ is the splitting field of $g$, of degree 2, and $\operatorname{disc} f = \operatorname{disc} g \cdot \operatorname{res}^2$ with $\operatorname{res} \in F^\times$ by Proposition 5.44(d) ($f$ made monic). For $x^3 - 3x + 1$: it has no rational root ($\pm 1$ give $-1$ and 3), so it is irreducible; its discriminant is $-4(-3)^3 - 27 = 81 = 9^2$. With $z = \zeta_9 + \zeta_9^{-1} = 2\cos(2\pi/9)$, $z^3 = \zeta_9^3 + \zeta_9^{-3} + 3z = -1 + 3z$ (as $\zeta_3 + \zeta_3^{-1} = -1$), so $z$ is a root; the others are its images $\zeta_9^2 + \zeta_9^{-2}$ and $\zeta_9^4 + \zeta_9^{-4}$ under $\sigma_2$ and $\sigma_4$. Hence the splitting field is $\mathbb{Q}(z)$, the fixed field of $\{\sigma_{\pm 1}\}$ in $\mathbb{Q}(\zeta_9)$, of degree $\varphi(9)/2 = 3$. □

**Proposition 5.46** (Quartics)**.** *Let* $\operatorname{char} F \neq 2$ *and* $f = x^4 + ax^3 + bx^2 + cx + d \in F[x]$ *separable with roots* $r_1, \dots, r_4$*. Put* $\theta_1 = r_1r_2 + r_3r_4$*,* $\theta_2 = r_1r_3 + r_2r_4$*,* $\theta_3 = r_1r_4 + r_2r_3$*, one for each of the three partitions of the roots into two pairs, and* $R_f(y) = \prod_{i=1}^3 (y - \theta_i)$*, the* resolvent cubic*.*

*(a)* $R_f(y) = y^3 - by^2 + (ac - 4d)y - (a^2d - 4bd + c^2)$ *and* $\operatorname{disc} R_f = \operatorname{disc} f$*; in particular the* $\theta_i$ *are distinct.*

*(b)* $S_4$ *acts on the three partitions through a surjection* $S_4 \to S_3$ *with kernel*

$$V_4 = \{1, (12)(34), (13)(24), (14)(23)\};$$

*for any separable* $f$*,* $\theta_i \in F$ *iff* $G_f$ *stabilises the* $i$*-th partition, and the degrees of the irreducible factors of* $f$ *are the orbit sizes of* $G_f$*.*

*(c) If* $f$ *is irreducible,* $G_f$ *is conjugate to one of* $S_4$*,* $A_4$*,* $D_4$*,* $C_4$*,* $V_4$*, and:* $R_f$ *irreducible and* $\operatorname{disc} f \notin F^{\times 2}$ *gives* $S_4$*;* $R_f$ *irreducible and* $\operatorname{disc} f \in F^{\times 2}$ *gives* $A_4$*;* $R_f$ *split into linear factors gives* $V_4$*;* $R_f$ *with exactly one root in* $F$ *gives* $D_4$ *or* $C_4$*, and* $C_4$ *iff* $f$ *is reducible over* $F(\sqrt{\operatorname{disc} f})$ *(the criterion of Kappe and Warren [90]).*

*Proof.* (a) With $e_1 = -a$, $e_2 = b$, $e_3 = -c$, $e_4 = d$ the elementary symmetric functions of the roots, direct expansion gives $\theta_1 + \theta_2 + \theta_3 = e_2$, $\sum_{i<j}\theta_i\theta_j = e_1e_3 - 4e_4$ and $\theta_1\theta_2\theta_3 = e_1^2e_4 + e_3^2 - 4e_2e_4$

(each is a short computation; we also checked them by computer algebra), which is the formula. The first identity holds because each of the six products $r_ir_j$ occurs in exactly one $\theta_i$. For the second, $\theta_1\theta_2 = (r_1r_2 + r_3r_4)(r_1r_3 + r_2r_4) = r_1^2r_2r_3 + r_1r_2^2r_4 + r_1r_3^2r_4 + r_2r_3r_4^2$; the three products $\theta_i\theta_j$ together contain each of the twelve monomials $r_i^2r_jr_k$ ($j < k$, both different from $i$) exactly once, while $e_1e_3 = \sum_i r_i \cdot \sum_{j<k<m} r_jr_kr_m$ contains each of them once and $r_1r_2r_3r_4$ four times. Next $\theta_1 - \theta_2 = (r_1 - r_4)(r_2 - r_3)$, $\theta_1 - \theta_3 = (r_1 - r_3)(r_2 - r_4)$, $\theta_2 - \theta_3 = (r_1 - r_2)(r_3 - r_4)$, whose product is $\pm\delta_f$; so $\operatorname{disc} R_f = \delta_f^2 = \operatorname{disc} f$. (b) A permutation of the roots permutes the partitions; the kernel consists of the permutations preserving each partition, which is $V_4$, and the image has order $24/4 = 6$. By (a) the $\theta_i$ are distinct, so $\sigma \in G_f$ fixes $\theta_i$ iff it stabilises the $i$-th partition, and $\theta_i \in F$ iff all of $G_f$ does. The last statement is Proposition 5.42(b). (c) $G_f$ is transitive, so $4 \mid \#G_f$. The subgroups of $S_4$ of order 24, 12, 8 are $S_4$, $A_4$ (the only subgroup of index 2) and the Sylow 2-subgroups $D_4$ (Chapter 1); a subgroup of order 4 is cyclic, generated by a 4-cycle (transitive), or a Klein group, either the normal $V_4$ (transitive) or $\langle (ij), (kl) \rangle$ for disjoint transpositions (not transitive), since two commuting transpositions are disjoint. The image of $G_f$ in $S_3$ is $S_3$, $A_3$, trivial for $S_4$, $A_4$, $V_4$, and of order 2 for $D_4$ and $C_4$ (which contain $V_4$, respectively $\{1, (13)(24)\}$ for $C_4 = \langle (1234) \rangle$, and stabilise one partition). By (b), $R_f$ is irreducible iff the image is transitive on the partitions ($S_4$, $A_4$, told apart by the discriminant, $A_4 = S_4 \cap A_4$), splits iff the image is trivial ($V_4$), and has exactly one root iff $G_f \in \{D_4, C_4\}$. In the last case $G_f \cap A_4$ is $V_4$ for $D_4$ (transitive) and $\langle (13)(24) \rangle$ for $C_4$ (not transitive); by Proposition 5.44(c) and Proposition 5.42(c), $G_f \cap A_4$ is the group of $f$ over $F(\sqrt{\operatorname{disc} f})$, so $f$ stays irreducible there iff $G_f = D_4$. □

The next proposition is the Galois-theoretic form of the last sentence of Proposition prop:ns. Its group theory is Chapter 1: $\mathrm{GL}_2(\mathbb{F}_3)$ acts on the set $\mathbb{P}(V)$ of the four lines of $V = \mathbb{F}_3^2$, the kernel of the action is the group of scalars $\{\pm 1\}$, the induced map $\mathrm{PGL}_2(\mathbb{F}_3) \to S_4$ is an isomorphism, the sign of the permutation induced by $g$ is $\det g \in \{\pm 1\} = \mathbb{F}_3^\times$, and the three properties of the paper (surjective determinant, no stable line, an element of determinant $-1$ and trace 0) leave exactly the three images $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$, $N_s(\mathbb{F}_3)$. The table below was also checked by enumerating the 55 subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ in 16 conjugacy classes.

**Proposition 5.47** (Quartics with group acting on the four lines)**.** *Let $V$ be a two-dimensional $\mathbb{F}_3$-vector space, $\rho\colon G_F \to \mathrm{GL}(V) \simeq \mathrm{GL}_2(\mathbb{F}_3)$ a continuous homomorphism with image $H$, and $f \in F[x]$ a separable quartic whose root set $R$ is in a $G_F$-equivariant bijection with $\mathbb{P}(V)$.*

*(a) $G_F$ acts on $R$ through $H \to \mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$, so $G_f$ is isomorphic to the image $\bar{H}$ of $H$ in $\mathrm{PGL}_2(\mathbb{F}_3)$, and the splitting field of $f$ is the fixed field of $\rho^{-1}(\{\pm 1\})$.*

*(b) The irreducible factors of $f$ over $F$ correspond to the orbits of $H$ on the four lines, with degrees the orbit sizes.*

*(c) The sign of $\sigma \in G_F$ on $R$ is $\det \rho(\sigma)$. Hence, if $\operatorname{char} F \neq 2$, $F(\sqrt{\operatorname{disc} f})$ is the field cut out by $\det \circ \rho$, and $\operatorname{disc} f \in F^{\times 2}$ iff $\det(H) = 1$.*

*(d) For the subgroups $H$ with $\det(H) = \mathbb{F}_3^\times$, up to conjugacy, the orbit sizes (which by (b) are the degrees of the irreducible factors of $f$) and the group $G_f$ are:*

| $H$ | *label* | $\#H$ | *orbits on lines* | $G_f$ |
|---|---|---|---|---|
| $\mathrm{GL}_2(\mathbb{F}_3)$ | | 48 | 4 | $S_4$ |
| $N_{ns}(\mathbb{F}_3)$ | *3Nn* | 16 | 4 | $D_4$ |
| $C_{ns}(\mathbb{F}_3)$ | *3Cn* | 8 | 4 | $C_4$ |
| $N_s(\mathbb{F}_3)$ | *3Ns* | 8 | $2+2$ | $V_4$, *not normal in* $S_4$ |
| $B(\mathbb{F}_3)$ | *3B* | 12 | $1+3$ | $S_3$ |
| *two of order* 6 | *3B.1.1, 3B.1.2* | 6 | $1+3$ | $S_3$ |
| $C_s(\mathbb{F}_3)$ | *3Cs* | 4 | $1+1+2$ | $C_2$ |
| $\langle \mathrm{diag}(1,-1)\rangle$ | *3Cs.1.1* | 2 | $1+1+2$ | $C_2$ |

*So $f$ is irreducible in the first three cases, a product of two irreducible quadratics for $N_s(\mathbb{F}_3)$, a linear factor times an irreducible cubic in the Borel cases, and two linear factors times an irreducible quadratic in the last two.*

*(e) If $H = N_s(\mathbb{F}_3)$, the two quadratic factors have distinct splitting fields $K_1 \neq K_2$, and the third quadratic subfield of $K_1K_2$ is $F(\sqrt{\mathrm{disc}\, f})$, the field cut out by $\det\circ\rho$. The resolvent cubic of $f$ (made monic) has exactly one root in $F$ when $H$ is $N_s(\mathbb{F}_3)$ or $N_{ns}(\mathbb{F}_3)$, and none when $H = \mathrm{GL}_2(\mathbb{F}_3)$.*

*(f) If $H$ fixes no line and contains an element of determinant $-1$ and trace 0, then $H$ is conjugate to $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$, and $f$ is irreducible in the first two cases and a product of two irreducible quadratics in the third.*

*Proof.* (a) The action of $\sigma$ on $R$ corresponds to the action of $\rho(\sigma)$ on $\mathbb{P}(V)$, which depends only on the image of $\rho(\sigma)$ in $\mathrm{PGL}_2(\mathbb{F}_3)$, and $\mathrm{PGL}_2(\mathbb{F}_3)$ acts faithfully. So the kernel of $G_F \to \mathrm{Sym}(R)$ is $\rho^{-1}(\{\pm1\})$, whose fixed field is the splitting field $F(R)$ (an element of $G_F$ fixes $F(R)$ iff it fixes every root), and $G_f \simeq G_F/\rho^{-1}(\{\pm1\}) \simeq \bar{H}$. (b) Proposition 5.42(c) with $\Omega = F^{\mathrm{sep}}$. (c) The sign of the permutation of $\mathbb{P}(V)$ induced by $g$ is $\det g$ (Chapter 1). Apply Proposition 5.44(c). (d) The orders, orbits and images are the group theory of Chapter 1 (and our enumeration). $G_f$ is the image $\bar{H} = H/(H \cap \{\pm1\})$; all these $H$ contain $-1$ except the two groups of order 6 (which contain an element of order 3 and one of determinant $-1$, and meet $\{\pm1\}$ trivially) and $\langle \mathrm{diag}(1,-1)\rangle$, so $\#\bar{H} = \#H/2$ in the other cases. The transitive images of orders 24, 8, 4 are $S_4$, $D_4$ and $C_4$ (for $C_{ns}(\mathbb{F}_3)$, cyclic of order 8, the image is cyclic of order 4); for $N_s(\mathbb{F}_3)$ the image has order 4 with two orbits of size two, hence is $\langle (12),(34)\rangle$ for a suitable numbering, the Klein group that is not normal in $S_4$. The factorisations follow from (b). (e) Number the lines so that $\bar{H} = \langle (12),(34)\rangle$ and $G_f = \bar{H}$ acts on $R$ in the same way. $K_2 = F(r_3)$, the splitting field of the factor with roots $r_3, r_4$, is the fixed field of the stabiliser $\langle (12)\rangle$ of $r_3$; likewise $K_1 = F(r_1)$ is the fixed field of $\langle (34)\rangle$. These are two of the three subgroups of order 2 of $\bar{H} \simeq V_4$, so $K_1 \neq K_2$; the third, $\langle (12)(34)\rangle = \bar{H} \cap A_4$, has fixed field $F(\sqrt{\mathrm{disc}\, f})$ by Proposition 5.44(c). By Proposition 5.46(b) the roots of $R_f$ in $F$ correspond to the partitions stabilised by $G_f$: $\langle (12),(34)\rangle$ stabilises $\{12|34\}$ and swaps the other two; $D_4$ stabilises exactly one partition; $S_4$ none. (f) The group-theoretic statement is Chapter 1; the determinant is then surjective (it takes the values 1 and $-1$), and (d) applies. □

*In the paper* 5.48. In Proposition prop:ns, $F = \mathbb{Q}$, $V = E[3]$, $\rho = \bar{\rho}_{E,3}$, and $f = \psi_3$ is the 3-division polynomial (leading coefficient 3; irreducibility is meant up to this constant). The equivariant bijection $R \leftrightarrow \mathbb{P}(E[3])$ is $\{\pm P\} \mapsto x(P)$, which is injective because $x(P) = x(Q)$ iff $Q = \pm P$

(Chapter 9). The inputs $\det\bar\rho_{E,3}=\bar\chi_3=\chi_{-3}$ (Chapter 9, Weil pairing, and Proposition 5.29) and the fact that complex conjugation has determinant $-1$ and trace 0 (Proposition 5.60) give the hypotheses of (f) when $E[3]$ is irreducible. Then (f) is the paper's last sentence: $\psi_3$ is a product of two irreducible quadratics exactly for the image $N_s(\mathbb{F}_3)$, and irreducible otherwise. By (c), $\operatorname{disc}\psi_3\in -3\cdot\mathbb{Q}^{\times 2}$ for every $E/\mathbb{Q}$; for 11a1, $\operatorname{disc}\psi_3=-3^3\cdot 11^{10}$, and for 121a1, $-3^3\cdot 11^4$. Reducible $E[3]$ always gives a rational line, hence a linear factor of $\psi_3$ (the patterns $1+3$ and $1+1+2$); the pattern $2+2$ of $N_s(\mathbb{F}_3)$ occurs although $E[3]$ is irreducible, because the factorisation detects the orbits of the image, not reducibility. The argument uses neither Zywina's classification nor the absence of complex multiplication. The function `ns3` of the paper's appendix returns 1 iff PARI's factorisation of `elldivpol(E,3)` has exactly two factors, the first of degree 2; as PARI lists the factors by increasing degree, the possible patterns are those of (d), and $\det\bar\rho_{E,3}$ is surjective for every $E/\mathbb{Q}$, the test returns 1 exactly when the image of $\bar\rho_{E,3}$ is $N_s(\mathbb{F}_3)$, with no hypothesis on $E[3]$.

**Example 5.49** (An image $N_s(\mathbb{F}_3)$, and a surjective image)**.** Let $E$ be 6912d1, $y^2=x^3-6x-8$ ($j=-1728$, which is not a CM $j$-invariant; $\Delta=-2^9\cdot 3^3$). PARI gives

$$\psi_3=3x^4-36x^2-96x-36=3(x^2-4x-2)(x^2+4x+6),$$

two irreducible quadratics with discriminants 24 and $-8$, so $K_1=\mathbb{Q}(\sqrt{6})$, $K_2=\mathbb{Q}(\sqrt{-2})$, and the third quadratic subfield of $K_1K_2$ is $\mathbb{Q}(\sqrt{-12})=\mathbb{Q}(\sqrt{-3})$, as Proposition 5.47(e) predicts; the resolvent cubic of $\psi_3/3$ is $(y-4)(y^2+16y+112)$, with exactly one rational root. Since $\det\bar\rho_{E,3}$ is surjective and only $N_s(\mathbb{F}_3)$ has the pattern $2+2$ in the table of (d), the image is $N_s(\mathbb{F}_3)$ (Sutherland's 3Ns, as in Cremona's tables). The group $N_s(\mathbb{F}_3)$ acts absolutely irreducibly on $\overline{\mathbb{F}}_3^2$: a line stable under it is an eigenline of $\operatorname{diag}(1,-1)$, hence a coordinate axis, and the antidiagonal elements swap the two axes. The image of $G_{\mathbb{Q}(\sqrt{-3})}$ is the determinant-one subgroup $\{\pm 1,\pm J\}$ (Proposition 5.66(c)), which is abelian, so $\bar\rho|_{G_{\mathbb{Q}(\sqrt{-3})}}$ is not absolutely irreducible, and $\bar\rho\simeq\bar\rho\otimes\chi_{-3}$ by Proposition 5.67. For the primes $p\neq 3$ of good reduction (here $p\geq 5$) one has $\operatorname{tr}\bar\rho_{E,3}(\operatorname{Frob}_p)\equiv a_p \pmod 3$ (Chapter 9, Chapter 11). So $a_p\equiv\chi_{-3}(\operatorname{Frob}_p)a_p \pmod 3$, and $a_p\equiv 0 \pmod 3$ whenever $\chi_{-3}(\operatorname{Frob}_p)=-1$, that is for $p\equiv 2 \pmod 3$ (Proposition 5.63(c)). This part of the example looks ahead: it uses Proposition 5.66, Proposition 5.67 and Proposition 5.63 of Section 8 and the Frobenius elements of Section 9 (Theorem 5.72); nothing in Sections 6 to 8 depends on it. Indeed $a_5=3$, $a_{11}=3$, $a_{17}=-6$, $a_{23}=6$, $a_{29}=6$, $a_{41}=6$, $a_{47}=12$, $a_{53}=-3$, $a_{59}=12$, $a_{71}=-6$; note $9\nmid a_5$. By contrast 11a1, $y^2+y=x^3-x^2-10x-20$, has

$$\psi_3=3x^4-4x^3-60x^2-237x-21,$$

irreducible, with irreducible resolvent cubic and $\operatorname{disc}\psi_3=-3^3\cdot 11^{10}\notin\mathbb{Q}^{\times 2}$: $G_{\psi_3}=S_4$ and the image of $\bar\rho_{E,3}$ is $\mathrm{GL}_2(\mathbb{F}_3)$. The paper's own example 726a1 is Exercise 5.91. The computations are reproduced in PARI/GP by:

```
E = ellinit([0,0,0,-6,-8]); f = elldivpol(E,3);
factor(f)                          \\ [x^2-4*x-2,1; x^2+4*x+6,1]
[poldisc(x^2-4*x-2), poldisc(x^2+4*x+6)]          \\ [24, -8]
[[p, ellap(E,p)] | p <- primes([5,71]), p%3 == 2]
F = elldivpol(ellinit([0,-1,1,-10,-20]),3);  \\ 11a1
[factor(F), factor(poldisc(F))]
```

## 5.7 Infinite Galois theory, $G_{\mathbb{Q}}$, the cyclotomic character, complex conjugation

The Galois groups of the paper are infinite: $G_{\mathbb{Q}}$ in Section 1.1 and Section 4, the decomposition and inertia groups $G_{\mathbb{Q}_q} \supseteq I_q$ in Lemmas lem:tamagawa-exp and lem:tamfactor. This section extends the fundamental theorem to infinite Galois extensions, where subgroups must be *closed* in the Krull topology, defines the absolute Galois group and the cyclotomic character $\chi_\ell$, proves $\chi_\ell(G_{\mathbb{Q}}) = \mathbb{Z}_\ell^\times$ (the fact that forces $\ell = 3$ in Lemma lem:h0), and studies complex conjugation, the element of $G_{\mathbb{Q}}$ that makes "irreducible" and "absolutely irreducible" coincide for $\bar\rho_{E,\ell}$. Inverse limits, profinite groups and their basic properties (open subgroups are closed of finite index, open normal subgroups form a basis at 1, continuous images of compact groups are closed) are Chapter 1. One compactness statement is used repeatedly, here and in Chapter 6, so we isolate it first.

**Lemma 5.50** (Inverse limits of nonempty finite sets)**.** *Let $(I, \le)$ be a directed set (a partially ordered set in which any two elements have a common upper bound), and let $(X_i)_{i\in I}$ be nonempty finite sets with maps $f_{ij}\colon X_j \to X_i$ for $i \le j$, such that $f_{ii}$ is the identity and $f_{ij} \circ f_{jk} = f_{ik}$ for $i \le j \le k$. Then the inverse limit*

$$\varprojlim_{i\in I} X_i = \Big\{ (x_i)_{i\in I} \in \prod_{i\in I} X_i : \ f_{ij}(x_j) = x_i \text{ for all } i \le j \Big\}$$

*is nonempty.*

*Proof.* Give each $X_i$ the discrete topology; it is compact, being finite, so $P = \prod_i X_i$ is compact by Tychonoff's theorem (Chapter 1). For $i \le j$ the set $C_{ij} = \{x \in P : f_{ij}(x_j) = x_i\}$ is closed: it is the preimage of a subset of the finite discrete space $X_i \times X_j$ under the continuous projection $P \to X_i \times X_j$. The inverse limit is $\bigcap_{i\le j} C_{ij}$. By compactness it suffices to show that finitely many of the $C_{ij}$, say $C_{i_1j_1}, \dots, C_{i_rj_r}$, have a common point. As $I$ is directed, there is $m \in I$ with $j_s \le m$ (hence $i_s \le m$) for $s = 1, \dots, r$. Choose $y \in X_m$ (it is nonempty), put $x_i = f_{im}(y)$ for every $i$ among the $i_s$, $j_s$, and choose $x_i \in X_i$ arbitrarily for the other $i$. Then $f_{i_sj_s}(x_{j_s}) = f_{i_sj_s} f_{j_sm}(y) = f_{i_sm}(y) = x_{i_s}$ for each $s$, so $x$ lies in all $r$ sets. Hence $\bigcap_{i\le j} C_{ij} \neq \emptyset$. □

Finiteness matters: an inverse limit of nonempty *infinite* sets can be empty (take $X_n = \{n, n+1, n+2, \dots\}$, $n \in \mathbb{N}$, with the inclusions as maps).

**Proposition 5.51** (Infinite Galois extensions)**.** *Let $L/F$ be a Galois extension of arbitrary degree and $\mathcal{K}$ the set of finite Galois subextensions $K/F$ of $L$.*

*(a) $L = \bigcup_{K\in\mathcal{K}} K$; $\mathcal{K}$ is directed by inclusion (any two members lie in a third); and restriction is a group isomorphism*

$$\mathrm{Gal}(L/F) \xrightarrow{\sim} \varprojlim_{K\in\mathcal{K}} \mathrm{Gal}(K/F), \qquad \sigma \mapsto (\sigma|_K)_K.$$

*(b) The* Krull topology *on $\mathrm{Gal}(L/F)$ is the one in which the subgroups $\mathrm{Gal}(L/K)$, $K \in \mathcal{K}$, form a basis of open neighbourhoods of 1 (and their cosets a basis of the topology). It is the inverse-limit topology transported by (a), so $\mathrm{Gal}(L/F)$ is a profinite group: compact, Hausdorff and totally disconnected. The subgroups $\mathrm{Gal}(L/M)$ with $M \subseteq L$ finite over $F$ are open.*

(c) *The restriction maps* $\mathrm{Gal}(L/F) \to \mathrm{Gal}(K/F)$, $K \in \mathcal{K}$, *are surjective. If* $L/F$ *is finite, the Krull topology is discrete.*

*Proof.* (a) For $\alpha \in L$, $m_{\alpha,F}$ is separable and splits in $L$ (normality), so its splitting field is a member of $\mathcal{K}$ containing $\alpha$. If $K_1, K_2 \in \mathcal{K}$, then $K_1K_2 \in \mathcal{K}$ (Proposition 5.24(b)). Each $\sigma \in \mathrm{Gal}(L/F)$ maps $K \in \mathcal{K}$ onto itself (Proposition 5.14(iii)), so $\sigma|_K \in \mathrm{Gal}(K/F)$, and the restrictions are compatible. The map is injective because $L$ is the union of the $K$. Surjective: a compatible family $(\sigma_K)$ defines a map $\sigma\colon L \to L$, $\sigma(x) = \sigma_K(x)$ for any $K \ni x$. It is well defined: if $x \in K \cap K'$, choose $K'' \in \mathcal{K}$ containing $K$ and $K'$; then $\sigma_K(x) = \sigma_{K''}(x) = \sigma_{K'}(x)$ by compatibility. It is a field homomorphism over $F$: for $x, y \in L$ choose one $K$ containing both, and $\sigma(x+y) = \sigma_K(x+y) = \sigma(x) + \sigma(y)$, and similarly for products. Its inverse is given by the family $(\sigma_K^{-1})$. (b) In $\varprojlim \mathrm{Gal}(K/F)$ with the topology induced by the product of discrete groups, a basis of neighbourhoods of 1 is given by the kernels of the projections to finitely many factors $\mathrm{Gal}(K_1/F), \dots, \mathrm{Gal}(K_r/F)$; under (a) such a kernel is $\mathrm{Gal}(L/K_1 \cdots K_r)$, and $K_1 \cdots K_r \in \mathcal{K}$. So the two topologies agree, and the inverse limit of finite groups is profinite (Chapter 1). If $M/F$ is finite, its normal closure $M'$ in $L$ is in $\mathcal{K}$ (Proposition 5.14), and $\mathrm{Gal}(L/M) \supseteq \mathrm{Gal}(L/M')$ is a union of cosets of an open subgroup. (c) Let $K_0 \in \mathcal{K}$ and $\tau \in \mathrm{Gal}(K_0/F)$. The set $\mathcal{K}_0$ of $K \in \mathcal{K}$ containing $K_0$ is directed by inclusion (by (a)). For $K \in \mathcal{K}_0$ the set $X_K$ of $\sigma \in \mathrm{Gal}(K/F)$ with $\sigma|_{K_0} = \tau$ is finite and nonempty (Theorem 5.23(d)), and for $K' \subseteq K''$ in $\mathcal{K}_0$ restriction maps $X_{K''}$ to $X_{K'}$, compatibly with composition. By Lemma 5.50 there is a compatible family $(\sigma_K)_{K \in \mathcal{K}_0}$. Every $K \in \mathcal{K}$ lies in $KK_0 \in \mathcal{K}_0$, so the members of $\mathcal{K}_0$ exhaust $L$, and the family defines as in (a) an element $\sigma \in \mathrm{Gal}(L/F)$, with $\sigma|_{K_0} = \tau$. (Alternatively, extend $\tau$ to an $F$-embedding $L \to \overline{\mathbb{F}}$ by Theorem 5.10(b); its image is $L$ by normality.) For $L/F$ finite, $\{1\} = \mathrm{Gal}(L/L)$ is open. □

**Theorem 5.52** (Krull's fundamental theorem)**.** *Let* $L/F$ *be Galois with group* $G = \mathrm{Gal}(L/F)$ *in the Krull topology.*

(a) *For every intermediate field* $M$, $L/M$ *is Galois,* $\mathrm{Gal}(L/M)$ *is a closed subgroup of* $G$, *and* $L^{\mathrm{Gal}(L/M)} = M$.

(b) *For every subgroup* $H \leq G$, $\mathrm{Gal}(L/L^H) = \overline{H}$, *the closure of* $H$.

(c) *Hence* $H \mapsto L^H$ *and* $M \mapsto \mathrm{Gal}(L/M)$ *are mutually inverse inclusion-reversing bijections between the* closed *subgroups of* $G$ *and the intermediate fields.*

(d) *Open subgroups correspond to the intermediate fields* $M$ *finite over* $F$, *and then* $[G : \mathrm{Gal}(L/M)] = [M : F]$.

(e) $M/F$ *is Galois iff* $\mathrm{Gal}(L/M)$ *is normal in* $G$; *then restriction induces an isomorphism of topological groups* $G/\mathrm{Gal}(L/M) \simeq \mathrm{Gal}(M/F)$ *(quotient topology on the left, Krull topology on the right).*

(f) $\mathrm{Gal}(L/\sigma M) = \sigma\,\mathrm{Gal}(L/M)\sigma^{-1}$ *for* $\sigma \in G$.

*Proof.* (a) $L/M$ is separable, and normal by Proposition 5.14. For $x \in L$, the stabiliser of $x$ in $G$ contains the open subgroup $\mathrm{Gal}(L/K)$ for any $K \in \mathcal{K}$ containing $x$, so it is open, hence closed; $\mathrm{Gal}(L/M)$ is the intersection of the stabilisers of the $x \in M$, hence closed. If $x \in L \setminus M$, then $m_{x,M}$ has degree $> 1$, is separable and splits in $L$, so it has a root $y \neq x$ in $L$. Let $K \subseteq L$ be the splitting field of $m_{x,M}$ over $M$, finite Galois over $M$; by Proposition 5.42(b) some $\tau \in \mathrm{Gal}(K/M)$ has $\tau(x) = y$, and by Proposition 5.51(c), applied to $L/M$, $\tau$ extends to an

element of $\mathrm{Gal}(L/M)$, which moves $x$. So $L^{\mathrm{Gal}(L/M)} = M$. (b) $H \subseteq \mathrm{Gal}(L/L^H)$, which is closed by (a), so $\overline{H} \subseteq \mathrm{Gal}(L/L^H)$. Conversely, let $\sigma \in \mathrm{Gal}(L/L^H)$ and let $\sigma\,\mathrm{Gal}(L/K)$, $K \in \mathcal{K}$, be a basic neighbourhood of $\sigma$. Let $H_K \leq \mathrm{Gal}(K/F)$ be the image of $H$ under restriction. Its fixed field in $K$ is $K \cap L^H$ (an element $x \in K$ satisfies $h|_K(x) = h(x)$, so it is fixed by $H_K$ iff it is fixed by $H$), which $\sigma$ fixes; so $\sigma|_K \in \mathrm{Gal}(K/K^{H_K}) = H_K$ by the finite fundamental theorem. Thus some $h \in H$ has $h|_K = \sigma|_K$, that is $h \in \sigma\,\mathrm{Gal}(L/K)$. Every neighbourhood of $\sigma$ meets $H$, so $\sigma \in \overline{H}$. (c) follows from (a) and (b). (d) If $[M : F] < \infty$, $\mathrm{Gal}(L/M)$ is open (Proposition 5.51(b)). If $\mathrm{Gal}(L/M)$ is open, it contains some $\mathrm{Gal}(L/K)$, $K \in \mathcal{K}$, so $M \subseteq K$ is finite over $F$. Then $\mathrm{Gal}(L/M)$ is the preimage of $\mathrm{Gal}(K/M)$ under the surjection $G \to \mathrm{Gal}(K/F)$, and its index is $[\mathrm{Gal}(K/F) : \mathrm{Gal}(K/M)] = [M : F]$. (e) and (f). (f) is immediate. As in the finite case, every $F$-embedding $M \to \overline{\mathbb{F}}$ extends to one of $L$ (Theorem 5.10(b)), which lies in $G$, so $M/F$ is normal iff $\sigma M = M$ for all $\sigma \in G$, iff $\mathrm{Gal}(L/M)$ is normal by (f) and (c). Then restriction $G \to \mathrm{Gal}(M/F)$ is a homomorphism, surjective (extension of automorphisms), with kernel $\mathrm{Gal}(L/M)$. It is continuous: the preimage of a basic open subgroup $\mathrm{Gal}(M/K)$ ($K \subseteq M$ finite Galois over $F$) is $\mathrm{Gal}(L/K)$, open. The induced bijection $G/\,\mathrm{Gal}(L/M) \to \mathrm{Gal}(M/F)$ is continuous from a compact space to a Hausdorff space, hence a homeomorphism. □

**Definition 5.53** (The absolute Galois group)**.** The *absolute Galois group* of $F$ is $G_F = \mathrm{Gal}(F^{\mathrm{sep}}/F)$ with the Krull topology (Proposition 5.18). For $K \subseteq F^{\mathrm{sep}}$, $G_K = \mathrm{Gal}(F^{\mathrm{sep}}/K) \subseteq G_F$. In particular $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. A homomorphism from $G_F$ *cuts out* the fixed field of its kernel when the kernel is closed.

**Proposition 5.54** (The absolute Galois group)**.** *(a) For $K/F$ finite inside $F^{\mathrm{sep}}$, $G_K$ is an open subgroup of $G_F$ of index $[K : F]$; every open subgroup is of this form; $\sigma G_K \sigma^{-1} = G_{\sigma K}$; and $G_K$ is normal iff $K/F$ is Galois, in which case $G_F/G_K \simeq \mathrm{Gal}(K/F)$.*

*(b) $G_{\mathbb{F}_q} = \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \simeq \widehat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n$, by an isomorphism sending $\mathrm{Frob}_q$ to 1; $\mathrm{Frob}_q$ generates a dense subgroup, and the open subgroups are the $n\widehat{\mathbb{Z}} = G_{\mathbb{F}_{q^n}}$, $n \geq 1$.*

*(c) For a number field $K \subseteq \overline{\mathbb{Q}}$, $G_K$ is open of index $[K : \mathbb{Q}]$ in $G_{\mathbb{Q}}$. So $G_{\mathbb{Q}(\sqrt{d})}$ has index 2 for $d$ not a square, and $G_{\mathbb{Q}(\zeta_n)}$ is open and normal of index $\varphi(n)$, with $G_{\mathbb{Q}}/G_{\mathbb{Q}(\zeta_n)} \simeq (\mathbb{Z}/n)^\times$.*

*(d) $G_{\mathbb{Q}}$ is uncountable and non-abelian. Its maximal abelian quotient is $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \simeq \widehat{\mathbb{Z}}^\times$, where $\mathbb{Q}^{\mathrm{ab}} = \mathbb{Q}(\mu_\infty)$ is the union of the cyclotomic fields (the Kronecker-Weber theorem).*

*Proof of (a) to (c), and of the first sentence of (d).* (a) is Theorem 5.52(d), (e), (f) with $L = F^{\mathrm{sep}}$. (b) Every finite extension of $\mathbb{F}_q$ in $\overline{\mathbb{F}}_q$ is an $\mathbb{F}_{q^n}$, Galois with group $\mathbb{Z}/n$ generated by $\mathrm{Frob}_q$ (Theorem 5.25); $\mathbb{F}_{q^m} \subseteq \mathbb{F}_{q^n}$ iff $m \mid n$, with restriction $\mathbb{Z}/n \to \mathbb{Z}/m$ the reduction. By Proposition 5.51(a), $G_{\mathbb{F}_q} \simeq \varprojlim_n \mathbb{Z}/n$ (divisibility order), which is $\widehat{\mathbb{Z}}$ of Chapter 1 (the factorials are cofinal), with $\mathrm{Frob}_q \mapsto (1 \bmod n)_n = 1$. The image of $\langle \mathrm{Frob}_q \rangle$ in every finite quotient is everything, so it is dense. The open subgroups are the $G_{\mathbb{F}_{q^n}}$ by (a), and $G_{\mathbb{F}_{q^n}}$ is the kernel of $\widehat{\mathbb{Z}} \to \mathbb{Z}/n$, which is $n\widehat{\mathbb{Z}}$. (c) is (a) with $F = \mathbb{Q}$, and Theorem 5.28. (d) By (c) and Proposition 5.27(c), $G_{\mathbb{Q}}$ has infinitely many distinct subgroups of index 2; a profinite group with infinitely many open subgroups is infinite, and an infinite profinite group is uncountable (it is a compact Hausdorff space without isolated points, the open subgroups having finite index). It is non-abelian since $\mathrm{Gal}(\mathbb{Q}(2^{1/3}, \zeta_3)/\mathbb{Q}) \simeq S_3$ is a quotient (Example 5.31). The rest of (d) is the Kronecker-Weber theorem, which we do not prove; see Chapter 6 and [201, Chapter 14]. □

**Example 5.55** (Two warnings)**.** (1) The subgroup $\mathrm{Frob}_p^{\mathbb{Z}}$ of $G_{\mathbb{F}_p} \simeq \widehat{\mathbb{Z}}$ is dense and not closed: it is countable, while $\widehat{\mathbb{Z}}$ is uncountable. Its fixed field is $\mathbb{F}_p$, the same as that of $G_{\mathbb{F}_p}$; this is why Theorem 5.52(c) requires closed subgroups. (2) A subgroup of finite index need not be open. Let $L = \mathbb{Q}(\sqrt{-1}, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \dots)$, the compositum of all quadratic fields. The fields $\mathbb{Q}(\Delta^{1/2})$ with $\Delta/\mathbb{Q}^{\times 2}$ finite are finite Galois and cofinal among the finite subextensions of $L$ (each of these lies in a compositum of finitely many quadratic fields), and $\mathrm{Gal}(\mathbb{Q}(\Delta^{1/2})/\mathbb{Q}) \simeq \mathrm{Hom}(\Delta/\mathbb{Q}^{\times 2}, \{\pm 1\})$ compatibly with restriction (Theorem 5.37(a) with $n = 2$). Passing to the limit (Proposition 5.51(a), where the limit may be taken over this cofinal family) gives $\mathrm{Gal}(L/\mathbb{Q}) \simeq \varprojlim_{\Delta} \mathrm{Hom}(\Delta/\mathbb{Q}^{\times 2}, \{\pm 1\}) = \mathrm{Hom}(\mathbb{Q}^{\times}/\mathbb{Q}^{\times 2}, \{\pm 1\}) \simeq \prod_{\mathbb{N}} \mathbb{Z}/2$, as $\mathbb{Q}^{\times}/\mathbb{Q}^{\times 2}$ is the union of the $\Delta/\mathbb{Q}^{\times 2}$ and is an $\mathbb{F}_2$-vector space with basis $-1$ and the primes. The open subgroups of index 2 are the kernels of the countably many continuous characters, which are the evaluations at the elements of $\mathbb{Q}^{\times}/\mathbb{Q}^{\times 2}$; but $\prod_{\mathbb{N}} \mathbb{Z}/2$ has uncountably many subgroups of index 2 (kernels of linear forms on an $\mathbb{F}_2$-vector space of uncountable dimension, whose existence uses the axiom of choice), so most are not open. We do not use this.

**Corollary 5.56** (Absolute Kummer theory)**.** *Let* $\mathrm{char}\, F \nmid n$ *and* $\mu_n \subset F$*. Then* $a \mapsto \chi_a$*,* $\chi_a(\sigma) = \sigma(\alpha)/\alpha$ *(*$\alpha^n = a$*), is an isomorphism*

$$F^{\times}/F^{\times n} \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{cont}}(G_F, \mu_n),$$

*$\mu_n$ being discrete. After a primitive $\zeta_n$ is chosen (so $\mu_n \simeq \mathbb{Z}/n$), the surjective continuous homomorphisms $G_F \to \mathbb{Z}/n$ correspond to the elements of exact order $n$ of $F^{\times}/F^{\times n}$. For $n = 2$ and* $\mathrm{char}\, F \neq 2$*: $F^{\times}/F^{\times 2} \simeq \mathrm{Hom}_{\mathrm{cont}}(G_F, \{\pm 1\})$, the class of $d$ corresponding to the character of $F(\sqrt{d})$.*

*Proof.* $\chi_a$ is continuous: its kernel $G_{F(\alpha)}$ is open. It is a homomorphism in $a$ and injective modulo $F^{\times n}$ by Proposition 5.35(b) with $L = F^{\mathrm{sep}}$. Surjective: a continuous $\chi\colon G_F \to \mu_n$ has open kernel $G_K$, $K/F$ finite Galois with group a subgroup of $\mu_n$, abelian of exponent dividing $n$. By Theorem 5.37(b), $K = F_{\Delta}$ with $\Delta/F^{\times n}$ finite, and by (a) of that theorem the character of $\mathrm{Gal}(K/F)$ induced by $\chi$ is $\sigma \mapsto \sigma(\alpha)/\alpha$ for some $a = \alpha^n \in \Delta$; so $\chi = \chi_a$. The surjective homomorphisms are those of order $n$ in the group $\mathrm{Hom}_{\mathrm{cont}}(G_F, \mu_n)$, which correspond to the elements of order $n$. For $n = 2$, $\mu_2 = \{\pm 1\} \subset F$ always, and $\chi_d(\sigma) = \sigma(\sqrt{d})/\sqrt{d}$. □

**Corollary 5.57** (Translation, infinite case)**.** *Let $L/F$ be Galois (of any degree) and $M/F$ any extension, inside a common field. Then $LM/M$ is Galois and restriction* $\mathrm{Gal}(LM/M) \to \mathrm{Gal}(L/L \cap M)$ *is an isomorphism of topological groups. In particular, for $K/F$ finite separable and $L \subseteq F^{\mathrm{sep}}$ Galois over $F$,* $\mathrm{Gal}(LK/K) \simeq \mathrm{Gal}(L/L \cap K)$*.*

*Proof.* $LM$ is generated over $M$ by roots of separable polynomials over $F$ that split in $L$, so $LM/M$ is normal and separable. Restriction is a continuous homomorphism $\mathrm{Gal}(LM/M) \to \mathrm{Gal}(L/F)$ (the preimage of $\mathrm{Gal}(L/K)$ is $\mathrm{Gal}(LM/MK)$), injective as before. Its image $H$ is compact, hence closed, and as in Proposition 5.24 $L^H = L \cap M$; by Theorem 5.52(c), $H = \mathrm{Gal}(L/L \cap M)$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. □

**Definition 5.58** (The cyclotomic character)**.** Let $\ell$ be a prime and $F$ a field of characteristic $\neq \ell$. For $k \geq 1$ the group $\mu_{\ell^k} = \mu_{\ell^k}(F^{\mathrm{sep}})$ is cyclic of order $\ell^k$ ($x^{\ell^k} - 1$ is separable), and $G_F$ acts on it by automorphisms, $\sigma(\zeta) = \zeta^{a_k(\sigma)}$ with $a_k(\sigma) \in (\mathbb{Z}/\ell^k)^{\times}$. The *$\ell$-adic cyclotomic character* is

$$\chi_\ell\colon G_F \to \mathbb{Z}_\ell^{\times} = \varprojlim_k (\mathbb{Z}/\ell^k)^{\times}, \qquad \chi_\ell(\sigma) = (a_k(\sigma))_k,$$

so that $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ for every $\zeta \in \mu_{\ell^\infty} = \bigcup_k \mu_{\ell^k}$; $\bar\chi_\ell = \chi_\ell \bmod \ell \colon G_F \to \mathbb{F}_\ell^\times$. The *Tate module of* $\mu$ is $\mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}$ (transition maps $\zeta \mapsto \zeta^\ell$), and for a $\mathbb{Z}_\ell$-module $M$ with $G_F$-action, $M(1) = M \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(1)$.

**Proposition 5.59** (The cyclotomic character)**.** *(a) $\chi_\ell$ is a well defined continuous homomorphism, and it is the unique map $G_F \to \mathbb{Z}_\ell^\times$ with $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ on $\mu_{\ell^\infty}$. $\chi_\ell \bmod \ell^k$ is trivial iff $\mu_{\ell^k} \subset F$, and $\ker \chi_\ell = G_{F(\mu_{\ell^\infty})}$.*

*(b) For $F = \mathbb{Q}$: $\chi_\ell$ is surjective, and $\mathrm{Gal}(\mathbb{Q}(\mu_{\ell^\infty})/\mathbb{Q}) \simeq \mathbb{Z}_\ell^\times$ as topological groups; $\chi_\ell(\sigma) \bmod \ell^k$ is the $a$ with $\sigma|_{\mathbb{Q}(\zeta_{\ell^k})} = \sigma_a$.*

*(c) For $F = \mathbb{Q}$: $\bar\chi_\ell$ is surjective onto $\mathbb{F}_\ell^\times$ with kernel $G_{\mathbb{Q}(\zeta_\ell)}$, and for odd $\ell$, $\chi_{\ell^*} = \left(\frac{\bar\chi_\ell}{\ell}\right) = \bar\chi_\ell^{(\ell-1)/2}$ (Proposition 5.29).*

*(d) For $F = \mathbb{F}_q$ with $\ell \nmid q$: $\chi_\ell(\mathrm{Frob}_q) = q$.*

*(e) $\mathbb{Z}_\ell(1)$ is a free $\mathbb{Z}_\ell$-module of rank one, on which $\sigma \in G_F$ acts by multiplication by $\chi_\ell(\sigma)$; $\mathbb{Z}_\ell(1)/\ell^k\mathbb{Z}_\ell(1) \simeq \mu_{\ell^k}$, by projection to the $k$-th component.*

*Proof.* (a) If $\zeta \in \mu_{\ell^{k+1}}$ then $\zeta^\ell \in \mu_{\ell^k}$ and $\sigma(\zeta^\ell) = \zeta^{\ell a_{k+1}(\sigma)}$, so $a_{k+1}(\sigma) \equiv a_k(\sigma) \pmod{\ell^k}$: the $a_k(\sigma)$ form an element of $\mathbb{Z}_\ell^\times$. Each $a_k$ is multiplicative in $\sigma$ and constant on cosets of the open subgroup $G_{F(\mu_{\ell^k})}$; as the subgroups $1 + \ell^k\mathbb{Z}_\ell$ form a basis of neighbourhoods of 1 in $\mathbb{Z}_\ell^\times$, $\chi_\ell$ is continuous. Uniqueness is clear, and the statements on kernels are the definitions. (b) By Theorem 5.28, $G_\mathbb{Q} \to \mathrm{Gal}(\mathbb{Q}(\zeta_{\ell^k})/\mathbb{Q}) \simeq (\mathbb{Z}/\ell^k)^\times$ is surjective for each $k$, and this composite is $\chi_\ell \bmod \ell^k$. So $\chi_\ell(G_\mathbb{Q})$ is dense in $\mathbb{Z}_\ell^\times$; being compact, it is closed, hence equal to $\mathbb{Z}_\ell^\times$. By Theorem 5.52(e), $\mathrm{Gal}(\mathbb{Q}(\mu_{\ell^\infty})/\mathbb{Q}) \simeq G_\mathbb{Q}/\ker\chi_\ell \simeq \mathbb{Z}_\ell^\times$, topologically (continuous bijection of compact Hausdorff groups). (c) The case $k = 1$ of (b), and Proposition 5.29(b). (d) $\mathrm{Frob}_q(\zeta) = \zeta^q$. (e) Choose $\zeta_{\ell^k}$ inductively: $\zeta_\ell$ a generator of $\mu_\ell$, and $\zeta_{\ell^{k+1}}$ an $\ell$-th root of $\zeta_{\ell^k}$, which has order $\ell^{k+1}$. Then $a \mapsto (\zeta_{\ell^k}^a)_k$ is an isomorphism $\mathbb{Z}_\ell \to \mathbb{Z}_\ell(1)$ (its components $\mathbb{Z}/\ell^k \to \mu_{\ell^k}$ are isomorphisms), and $\sigma(\zeta_{\ell^k}^a) = \zeta_{\ell^k}^{a\chi_\ell(\sigma)}$. The kernel of the projection to $\mu_{\ell^k}$ corresponds to $\ell^k\mathbb{Z}_\ell$. □

In the additive notation of Convention 5.3(c), $\sigma \cdot z = \chi_\ell(\sigma)z$ for $z \in \mathbb{Z}_\ell(1)$; so $\mathrm{Frob}_q$ acts on $\mathbb{Z}_\ell(1)$ by $q$, which is the statement "Frob acts on $\mathbb{Z}_\ell(1)$ by $q$" in the proof of Lemma lem:tamfactor (with $F = \mathbb{Q}_q$ and Frob acting through the residue field, Chapter 7).

The module $\mathbb{Z}_\ell(1)$ is isomorphic to $\mathbb{Z}_\ell$ as a $\mathbb{Z}_\ell$-module, but only after a compatible system of roots of unity has been chosen, and never as a $G_F$-module (unless $\chi_\ell$ is trivial): $G_F$ acts trivially on $\mathbb{Z}_\ell$ and through $\chi_\ell$ on $\mathbb{Z}_\ell(1)$. Over $\mathbb{Q}$, for instance, complex conjugation acts on $\mathbb{Z}_\ell(1)$ by $-1$ (Proposition 5.60(b)) and an arithmetic Frobenius at $p \neq \ell$ by $p$ (Theorem 5.72(c)). This is the whole content of the "(1)" in the paper's $T(1)$ and $\mathrm{ad}^0 T(1)$: the twist changes the Galois action, not the underlying module.

**Proposition 5.60** (Complex conjugation)**.** *Let $c \in G_\mathbb{Q}$ be the restriction to $\overline{\mathbb{Q}}$ of complex conjugation (Convention 5.1).*

*(a) $c$ is an element of order 2. Every embedding $\overline{\mathbb{Q}} \to \mathbb{C}$ is $\iota \circ \sigma$ with $\iota$ the inclusion and $\sigma \in G_\mathbb{Q}$, and complex conjugation transported by it is $\sigma^{-1}c\sigma$. A* complex conjugation *is an element of the conjugacy class of $c$.*

*(b) $\chi_\ell(c) = -1$ for every $\ell$; $\chi_d(c) = -1$ iff $d < 0$. So $\chi_{\ell^*}(c) = (-1)^{(\ell-1)/2}$ and $\chi_{-3}(c) = \chi_{-4}(c) = -1$.*

(c) *(Artin and Schreier.) Every element of $G_{\mathbb{Q}}$ of finite order other than 1 is a complex conjugation; in particular it has order 2.*

(d) *Let $k$ be a field with $\operatorname{char} k \neq 2$ and $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(k)$ a homomorphism with $\det\rho(c) = -1$ (for instance $\det\rho = \bar{\chi}_\ell$ with $\ell$ odd, or $\det\rho = \chi_\ell$). Then $\rho(c)$ is conjugate to $\mathrm{diag}(1,-1)$, so it has determinant $-1$ and trace 0, and $\rho$ is irreducible over $k$ iff it is absolutely irreducible.*

*Proof.* (a) The conjugate of an algebraic number is algebraic (it is a root of the same polynomial with rational coefficients), so complex conjugation maps $\overline{\mathbb{Q}}$ to itself; it has order 2 because $c(i) = -i$. An embedding $\tau\colon \overline{\mathbb{Q}} \to \mathbb{C}$ has image algebraic over $\mathbb{Q}$, so contained in $\overline{\mathbb{Q}}$, and equal to it ($\overline{\mathbb{Q}}/\tau(\overline{\mathbb{Q}})$ is algebraic and $\tau(\overline{\mathbb{Q}})$ is algebraically closed); so $\tau = \iota\circ\sigma$ with $\sigma \in G_{\mathbb{Q}}$, and $x \mapsto \tau^{-1}(\tau(x))$ is $\sigma^{-1}c\sigma$. (b) For $\zeta \in \mu_{\ell^\infty} \subset \mathbb{C}$, $c(\zeta) = \bar{\zeta} = \zeta^{-1}$. If $d > 0$, $\sqrt{d}$ is real and fixed; if $d < 0$, $\sqrt{d}$ is purely imaginary and $c(\sqrt{d}) = -\sqrt{d}$. Apply to $\ell^*$, $-3$, $-4$. (d) $\rho(c)^2 = 1$, so the minimal polynomial of $\rho(c)$ divides $X^2 - 1 = (X-1)(X+1)$, which has distinct roots as $\operatorname{char} k \neq 2$; hence $\rho(c)$ is diagonalisable with eigenvalues in $\{\pm 1\}$, and $\det\rho(c) = -1$ forces the eigenvalues 1 and $-1$. The equivalence is the case $n = 2$ of the distinct-eigenvalue criterion of Chapter 4: a line of $\bar{k}^2$ stable under $\rho(G_{\mathbb{Q}})$ is an eigenline of $\rho(c)$, and both eigenlines are defined over $k$. For $\det\rho = \bar{\chi}_\ell$, $\bar{\chi}_\ell(c) = -1 \neq 1$ in $\mathbb{F}_\ell$ as $\ell$ is odd. □

We do not prove (c). It follows from the theorem of Artin and Schreier that a field $K$ with $[\bar{K} : K]$ finite and greater than 1 has characteristic 0 and $\bar{K} = K(i)$ ([109, Chapter VI, Section 9]), together with the theory of real closed fields ([109, Chapter XI]), which shows that such a $K$ is real closed; both locations are pointers we have not re-verified. It is not used in the paper; what is used, in Proposition prop:ns and Lemmas lem:h0, lem:old and lem:dfg21, is (d) and the fact that $\mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1, c\}$ has order 2 (so $H^1(\mathbb{R}, M)$ is killed by 2, Chapter 17).

## 5.8 Continuous representations of profinite groups

The paper's objects are continuous representations of $G_{\mathbb{Q}}$ and of its local subgroups: $\rho = \rho_{E,\ell}\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ on $T = T_\ell E$, its reductions $\rho \bmod \ell^k$ and $\bar{\rho}$, and the characters $\chi_\ell$, $\chi_d$, $\varepsilon$. This section translates the representation theory of Chapter 4 (which is about abstract groups) to profinite groups: what continuity means, the field cut out by a representation, twists, restriction to an open subgroup such as $G_{\mathbb{Q}(\sqrt{\ell^*})}$ (the hypothesis of Theorem thm:main), the index-two criterion used in Lemma lem:h0 and Proposition prop:ns, independence of $\bar{\rho}^{ss}$ from the lattice (Lemma lem:old), and the $\ell$-adic Kummer cocycle (Lemma lem:tamfactor).

**Definition 5.61** (Continuous representations)**.** Let $G$ be a profinite group. A finite field, $\overline{\mathbb{F}}_\ell$, or any field regarded as discrete, gives $\mathrm{GL}_n(k)$ the discrete topology. A finite extension $E_\lambda$ of $\mathbb{Q}_\ell$, with ring of integers $\mathcal{O}$ and uniformiser $\varpi$ (Chapter 7), gives $\mathrm{GL}_n(E_\lambda) \subseteq E_\lambda^{n^2}$ the subspace topology of the $\ell$-adic topology; the subgroups $1 + \varpi^m \mathrm{M}_n(\mathcal{O})$, $m \geq 1$, form a basis of neighbourhoods of 1, and $\mathrm{GL}_n(\mathcal{O})$ is an open subgroup. A *representation* of $G$ is a continuous homomorphism $\rho\colon G \to \mathrm{GL}_n(k)$ for such a $k$, or $\rho\colon G \to \mathrm{GL}_n(\mathcal{O})$. For $G = G_F$, the field *cut out* by $\rho$ is $F_\rho = (F^{\mathrm{sep}})^{\ker\rho}$; $\rho$ *factors through* $\mathrm{Gal}(F_\rho/F)$.

**Proposition 5.62** (Continuous representations)**.** *Let $G$ be profinite.*

(a) *If $k$ is discrete, a homomorphism $\rho\colon G \to \mathrm{GL}_n(k)$ is continuous iff $\ker\rho$ is open, iff $\rho$ factors through a finite quotient $G/U$, $U$ open normal; then $\rho(G)$ is finite. For $G = G_F$, $F_\rho/F$ is then finite Galois and $\mathrm{Gal}(F_\rho/F) \simeq \rho(G_F)$. In particular a representation of a finite quotient of $G$ is, by inflation, a representation of $G$.*

(b) *A homomorphism* $\rho\colon G \to \mathrm{GL}_n(\mathcal{O})$ *is continuous iff every* $\rho \bmod \varpi^m$ *is. Then* $\rho(G)$ *is compact, hence closed,* $\ker\rho$ *is a closed normal subgroup, and for* $G = G_F$, $F_\rho/F$ *is Galois (in general infinite) with* $\mathrm{Gal}(F_\rho/F) \simeq \rho(G_F)$ *as topological groups.*

(c) *Every continuous* $\rho\colon G \to \mathrm{GL}_n(E_\lambda)$ *stabilises an* $\mathcal{O}$*-lattice* $\Lambda \subset E_\lambda^n$ *(a free* $\mathcal{O}$*-submodule of rank* $n$ *spanning* $E_\lambda^n$*); in a basis of* $\Lambda$, $\rho$ *is a continuous* $\rho\colon G \to \mathrm{GL}_n(\mathcal{O})$*, and its reduction* $\bar\rho_\Lambda$ *on* $\Lambda/\varpi\Lambda$ *is continuous.*

(d) *Every continuous* $\rho\colon G \to \mathrm{GL}_n(\mathbb{C})$ *has finite image. In particular continuous characters* $G_\mathbb{Q} \to \mathbb{C}^\times$ *have finite order.*

(e) *Subrepresentations, quotients, direct sums, tensor products, duals,* Hom *and* $\mathrm{ad}^0$ *of continuous representations, the semisimplification* $\rho^{ss}$*, and twists* $\rho \otimes \chi$ *by continuous characters are continuous.*

*Proof.* (a) $\{1\}$ is open in $\mathrm{GL}_n(k)$, so continuity implies $\ker\rho$ open; conversely if $\ker\rho$ is open, the preimage of any set is a union of cosets of $\ker\rho$, hence open. An open subgroup has finite index and contains an open normal subgroup (Chapter 1), and $\ker\rho$ is itself normal. The statement about $F_\rho$ is Theorem 5.52(d), (e) with the open normal subgroup $\ker\rho$. For inflation: $\rho' \circ \pi$ with $\pi\colon G \to G/U$ has open kernel. (b) A homomorphism of topological groups is continuous iff it is continuous at 1, and $\rho^{-1}(1 + \varpi^m \mathrm{M}_n(\mathcal{O})) = \ker(\rho \bmod \varpi^m)$. The image of a compact group is compact; the kernel is the intersection of the closed subgroups $\ker(\rho \bmod \varpi^m)$. Then Theorem 5.52(e) gives $\mathrm{Gal}(F_\rho/F) \simeq G_F/\ker\rho$, and the continuous bijection $G_F/\ker\rho \to \rho(G_F)$ is a homeomorphism (compact to Hausdorff). (c) $\mathrm{GL}_n(\mathcal{O})$ is open in $\mathrm{GL}_n(E_\lambda)$, so $U = \rho^{-1}(\mathrm{GL}_n(\mathcal{O}))$, the stabiliser of $\mathcal{O}^n$, is an open subgroup, of finite index. Put $\Lambda = \sum_{gU \in G/U} \rho(g)\mathcal{O}^n$, a finite sum (well defined since $\rho(u)\mathcal{O}^n = \mathcal{O}^n$ for $u \in U$). It is a finitely generated $\mathcal{O}$-submodule of $E_\lambda^n$ containing $\mathcal{O}^n$ and stable under $G$; finitely generated and torsion-free over the principal ideal domain $\mathcal{O}$, it is free (Chapter 3), of rank $n$ as it contains $\mathcal{O}^n$ and lies in $\varpi^{-m}\mathcal{O}^n$ for some $m$ (the finitely many matrices $\rho(g)$, $gU \in G/U$, have entries with bounded denominators: choose $m$ with $\varpi^m\rho(g) \in \mathrm{M}_n(\mathcal{O})$ for all of them). In a basis of $\Lambda$ the matrices $\rho(g)$ lie in $\mathrm{GL}_n(\mathcal{O})$, and continuity is unchanged by the change of basis. The reduction is continuous by (b) and (a). (d) Let $\|\cdot\|$ be the operator norm and $W = \{A \in \mathrm{GL}_n(\mathbb{C}) : \|A - 1\| < 1\}$. We claim that $W$ contains no subgroup other than $\{1\}$. Let $A$ generate a subgroup inside $W$. For an eigenvalue $\nu$ of $A$ and every $k \in \mathbb{Z}$, $\nu^k - 1$ is an eigenvalue of $A^k - 1$, so $|\nu^k - 1| \le \|A^k - 1\| < 1$. Then $|\nu| = 1$ (otherwise $|\nu^k|$ is unbounded for $k \to \infty$ or $k \to -\infty$), and all powers of $\nu$ lie on the arc $|z - 1| < 1$ of the unit circle, that is in $\{e^{i\theta} : |\theta| < \pi/3\}$; if $\nu = e^{i\theta} \ne 1$ with $0 < |\theta| < \pi/3$, some multiple $k\theta$ lies in $[\pi/3, \pi]$ modulo $2\pi$, a contradiction. So all eigenvalues are 1 and $A = 1 + N$ with $N$ nilpotent. If $N \ne 0$, let $m \ge 1$ be maximal with $N^m \ne 0$; then $A^k - 1 = \sum_{j=1}^m \binom{k}{j} N^j$, and as $k \to \infty$ the coefficient $\binom{k}{m}$ dominates, so $\|A^k - 1\| \to \infty$ (compare $\|\cdot\|$ with the norm of the coefficients in the basis $N, \dots, N^m$ of their span), a contradiction. Hence $A = 1$, proving the claim. Now $\rho^{-1}(W)$ is an open neighbourhood of 1, so it contains an open normal subgroup $U$ (Chapter 1); $\rho(U)$ is a subgroup inside $W$, so $\rho(U) = 1$ and $\rho$ factors through the finite group $G/U$. (e) Each construction is given in suitable bases by matrices whose entries are polynomials in the entries of $\rho(g)$, $\rho'(g)$ and $\det\rho(g)^{-1}$; these are continuous. A subrepresentation (or quotient) is given, in a basis adapted to it, by a block of the matrix. The semisimplification is the direct sum of the composition factors (Jordan-Hölder, Chapter 3 and Chapter 4), which are subquotients. □

By (a), the characters $\chi_d$ and $\bar\chi_\ell$ are continuous, their kernels $G_{\mathbb{Q}(\sqrt{d})}$ and $G_{\mathbb{Q}(\zeta_\ell)}$ being open; $\chi_\ell$ is continuous by Proposition 5.59(a). Continuity is a genuine condition: if $U$ is one of the

subgroups of index 2 of $\mathrm{Gal}(L/\mathbb{Q})$ that are not open (Example 5.55(2), $L$ the compositum of all quadratic fields), the composite $G_{\mathbb{Q}} \to \mathrm{Gal}(L/\mathbb{Q}) \to \mathrm{Gal}(L/\mathbb{Q})/U \simeq \{\pm 1\}$ is a homomorphism whose kernel is not open, hence not closed (a closed subgroup of finite index is open), so it is not of the form $G_K$ and the character cuts out no field.

**Proposition 5.63** (Quadratic characters of $G_{\mathbb{Q}}$)**.** *(a) $d \mapsto \chi_d$ is an isomorphism $\mathbb{Q}^\times/\mathbb{Q}^{\times 2} \simeq \mathrm{Hom}_{\mathrm{cont}}(G_{\mathbb{Q}}, \{\pm 1\})$, with $\ker \chi_d = G_{\mathbb{Q}(\sqrt{d})}$; the nontrivial continuous quadratic characters of $G_{\mathbb{Q}}$ correspond to the quadratic fields.*

*(b) $\chi_{dd'} = \chi_d \chi_{d'}$. For example $\chi_{-15} = \chi_{-3}\chi_5$, and $\chi_{-4}\chi_{-3} = \chi_{12}$ is the character of $\mathbb{Q}(\sqrt{3})$ (written by its discriminant 12; the symbol $\chi_3$ is never used, Convention 5.3); and for a finite set $S$ of odd primes $\chi_S = \prod_{q\in S} \chi_{q^*}$ is the character of $\mathbb{Q}(\sqrt{\prod_{q\in S} q^*})$.*

*(c) For a prime $p \nmid D$, $D$ the discriminant of $\mathbb{Q}(\sqrt{d})$: $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$ (Kronecker symbol), which is 1 iff $p$ splits in $\mathbb{Q}(\sqrt{d})$. In particular $\chi_{-3}(\mathrm{Frob}_p) = 1$ iff $p \equiv 1 \pmod 3$, $\chi_{-4}(\mathrm{Frob}_p) = 1$ iff $p \equiv 1 \pmod 4$, and for odd $\ell \neq p$, $\chi_{\ell^*}(\mathrm{Frob}_p) = \left(\frac{p}{\ell}\right)$.*

*(d) As a Dirichlet character (Chapter 2), $\chi_d$ has conductor $|D|$: $\chi_{q^*}$ has conductor $q$, $\chi_{-3}$ conductor 3, $\chi_{-4}$ conductor 4.*

*Proof of (a), (b) and of the last formula of (c); (c), (d) otherwise stated.* (a) is Corollary 5.56 with $F = \mathbb{Q}$, $n = 2$, and Proposition 5.27(c). (b) $\sqrt{dd'} = \pm\sqrt{d}\sqrt{d'}$, and $\chi_{dd'}$ does not depend on the choice of square root. The last formula of (c) is a direct consequence of $\chi_{\ell^*} = \left(\frac{\bar\chi_\ell}{\ell}\right)$ (Proposition 5.29) and $\bar\chi_\ell(\mathrm{Frob}_p) = p \bmod \ell$ (Theorem 5.72(c)). The general formula in (c) and (d) are proved in Chapter 6 and Chapter 2; we do not prove them here. Comparing $\chi_{\ell^*}(\mathrm{Frob}_p) = \left(\frac{p}{\ell}\right)$ with $\chi_{\ell^*}(\mathrm{Frob}_p) = \left(\frac{\ell^*}{p}\right)$ from (c) is quadratic reciprocity (Chapter 2). □

The formulas of (c) were also checked numerically: for $5 \le p < 2000$, $x^2+3$ has a root modulo $p$ iff $p \equiv 1 \pmod 3$, $x^2+1$ iff $p \equiv 1 \pmod 4$, and for $\ell \in \{3,5,7,11,13\}$, $\ell^*$ is a square modulo $p$ iff $p$ is a square modulo $\ell$.

**Proposition 5.64** (Twists)**.** *Let $\rho\colon G \to \mathrm{GL}_n(k)$ and $\chi\colon G \to k^\times$ be continuous ($k$ a field or $\mathcal{O}$ as above).*

*(a) $\rho \otimes \chi\colon g \mapsto \chi(g)\rho(g)$ is continuous, $\mathrm{tr}(\rho\otimes\chi)(g) = \chi(g)\,\mathrm{tr}\,\rho(g)$ and $\det(\rho\otimes\chi) = \chi^n \det\rho$. For $G = G_F$ the field cut out by $\rho\otimes\chi$ lies in $F_\rho F_\chi$.*

*(b) If $n = 2$ and $\rho \simeq \rho\otimes\chi$, then $\chi^2 = 1$. If moreover $k$ is a field, $\chi \neq 1$ and $\rho$ is absolutely irreducible, every isomorphism $\rho \to \rho\otimes\chi$ has trace 0, and these isomorphisms form a line together with 0.*

*(c) Let $\ell$ be odd, $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ continuous and $k \ge 1$. If $\rho \bmod \ell^k \simeq (\rho\otimes\chi_\ell) \bmod \ell^k$ (conjugate by an element of $\mathrm{GL}_2(\mathbb{Z}/\ell^k)$), then $\chi_\ell(g)^2 \equiv 1 \pmod{\ell^k}$ for all $g \in G_{\mathbb{Q}}$, and this forces $\ell = 3$ and $k = 1$. For $\ell = 3$, $k = 1$ the condition reads $\bar\rho \simeq \bar\rho \otimes \bar\chi_3 = \bar\rho \otimes \chi_{-3}$.*

*Proof.* (a) The formulas are immediate; continuity is Proposition 5.62(e). The kernel of $\rho\otimes\chi$ contains $\ker\rho \cap \ker\chi = G_{F_\rho F_\chi}$ (Theorem 5.52(c) and Theorem 5.23(e) in the limit). (b) Taking determinants of $P\rho(g)P^{-1} = \chi(g)\rho(g)$ gives $\det\rho(g) = \chi(g)^2 \det\rho(g)$, and $\det\rho(g)$ is a unit. The rest is the trace-zero lemma of Chapter 4: a $G$-map $\phi\colon \rho \to \rho\otimes\chi$ satisfies $(1-\chi(g))\,\mathrm{tr}\,\phi = 0$, and $\mathrm{Hom}_G(\rho, \rho\otimes\chi)$ has dimension at most one with nonzero elements isomorphisms when $\rho$

is absolutely irreducible. (c) Let $P \in \mathrm{GL}_2(\mathbb{Z}/\ell^k)$ with $P\rho(g)P^{-1} \equiv \chi_\ell(g)\rho(g) \pmod{\ell^k}$ for all $g$. Taking determinants in $\mathbb{Z}/\ell^k$ gives $\det\rho(g) \equiv \chi_\ell(g)^2 \det\rho(g)$, and $\det\rho(g) \in \mathbb{Z}_\ell^\times$ is a unit modulo $\ell^k$; so $\chi_\ell(g)^2 \equiv 1 \pmod{\ell^k}$ for all $g$. (This is the argument of (b), which works over any commutative ring.) By Proposition 5.59(b) there is $g$ with $\chi_\ell(g) = 2$ ($2 \in \mathbb{Z}_\ell^\times$ as $\ell$ is odd), so $\ell^k \mid 4-1=3$: $\ell = 3$ and $k = 1$. For $\ell = 3$, $k = 1$ there is no contradiction: $\bar\chi_3$ takes the values $\pm1$, whose squares are 1. Finally $\bar\chi_3 = \chi_{-3}$ by Proposition 5.29(c). □

*In the paper* 5.65. Lemma lem:h0 identifies $h_\ell(E)$, the length of $H^0(\mathbb{Q}, \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$, with the largest $k$ such that $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$ (the module-theoretic part, Nakayama and cyclicity, is Chapter 4). Proposition 5.64(c) is then the step "Taking determinants, $\chi_\ell(g)^2 \equiv 1 \pmod{\ell^k}$ for all $g \in G_\mathbb{Q}$. Since $\chi_\ell(G_\mathbb{Q}) = \mathbb{Z}_\ell^\times$ contains 2, ... this forces $k = 0$ for $\ell \geq 5$ and $k \leq 1$ for $\ell = 3$": $h_\ell(E) = 0$ for $\ell \geq 5$ and $h_3(E) \leq 1$, for every $E$ with $\bar\rho$ irreducible, with or without complex multiplication. It holds for every continuous $\rho\colon G_\mathbb{Q} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$, irreducible or not. The argument needs the cyclotomic character $\chi_\ell$, not the quadratic character $\chi_{\ell^*}$, which differs from $\bar\chi_\ell$ for $\ell \geq 5$ (Proposition 5.29).

**Proposition 5.66** (Restriction to open subgroups)**.** *Let $K/F$ be a finite separable extension inside $F^{\mathrm{sep}}$ and $\rho\colon G_F \to \mathrm{GL}_n(k)$ continuous.*

*(a)* $\mathrm{Res}\,\rho = \rho|_{G_K}$ *is continuous, and* $[\rho(G_F) : \rho(G_K)] \leq [K : F]$. *If $K/F$ is Galois, $\rho(G_K)$ is normal in $\rho(G_F)$ and the index divides $[K : F]$.*

*(b) (Image of a kernel.) If $\delta\colon \mathrm{GL}_n(k) \to A$ is a homomorphism to an abelian group and $G_K = \ker(\delta \circ \rho)$ (that is, $K$ is cut out by $\delta \circ \rho$), then $\rho(G_K) = \rho(G_F) \cap \ker\delta$.*

*(c) Let $E/\mathbb{Q}$ be an elliptic curve, $\ell$ odd and $\bar\rho = \bar\rho_{E,\ell}$, with $\det\bar\rho = \bar\chi_\ell$ (Chapter 9). Then $\bar\rho(G_{\mathbb{Q}(\zeta_\ell)}) = \bar\rho(G_\mathbb{Q}) \cap \mathrm{SL}_2(\mathbb{F}_\ell)$ and $\bar\rho(G_{\mathbb{Q}(\sqrt{\ell^*})}) = \{g \in \bar\rho(G_\mathbb{Q}) : \det g \in (\mathbb{F}_\ell^\times)^2\}$. For $\ell = 3$ both are the determinant-one subgroup of the image.*

*(d) The field cut out by* $\mathrm{Res}\,\rho$ *is $KF_\rho$, and* $\mathrm{Gal}(KF_\rho/K) \simeq \mathrm{Gal}(F_\rho/F_\rho \cap K)$.

*(e) $\rho$ is absolutely irreducible on $G_K$ iff the subgroup $\rho(G_K)$ fixes no proper nonzero subspace of $\bar k^n$; this depends only on $\rho(G_K)$.*

*Proof.* (a) Restriction of a continuous map. The cosets of $G_K$ (there are $[K : F]$ of them, Proposition 5.54(a)) map onto the cosets of $\rho(G_K)$. If $K/F$ is Galois, $G_K$ is normal, so $\rho(G_K)$ is normal in $\rho(G_F)$, and $\rho(G_F)/\rho(G_K)$ is a quotient of $G_F/G_K \simeq \mathrm{Gal}(K/F)$. (b) If $g \in G_K$ then $\delta(\rho(g)) = 1$. Conversely if $\rho(g) \in \ker\delta$ then $g \in \ker(\delta\circ\rho) = G_K$; so every element of $\rho(G_F) \cap \ker\delta$ is in $\rho(G_K)$. (c) $\mathbb{Q}(\zeta_\ell)$ is cut out by $\bar\chi_\ell = \det\circ\bar\rho$ (Proposition 5.59(c)): take $\delta = \det$. $\mathbb{Q}(\sqrt{\ell^*})$ is cut out by $\chi_{\ell^*} = (\frac{\bar\chi_\ell}{\ell}) = (\frac{\det\bar\rho}{\ell})$ (Proposition 5.29): take $\delta = (\frac{\det}{\ell})$. For $\ell = 3$, $(\mathbb{F}_3^\times)^2 = \{1\}$. (d) The kernel of $\mathrm{Res}\,\rho$ is $G_K \cap \ker\rho = G_K \cap G_{F_\rho} = G_{KF_\rho}$ (Theorem 5.52). The isomorphism is Corollary 5.57. (e) Definition (Convention 5.4); the stable subspaces of $\bar k^n$ under $G_K$ are those under $\rho(G_K)$. □

**Proposition 5.67** (Index two)**.** *Let $G$ be a profinite group, $H \subseteq G$ an open subgroup of index 2, $\chi\colon G \to G/H \simeq \{\pm1\}$, $s \in G \setminus H$, $k$ a field with* $\mathrm{char}\,k \neq 2$ *(discrete, or a finite extension of $\mathbb{Q}_\ell$), and $\rho\colon G \to \mathrm{GL}_2(k)$ continuous and absolutely irreducible. The following are equivalent:*

*(i) $\rho|_H$ is not absolutely irreducible;*

*(ii) $\rho \simeq \rho \otimes \chi$ (over $k$; equivalently over $\bar k$);*

*(iii)* $\rho \otimes \bar{k} \simeq \operatorname{Ind}_H^G \psi$ *for a continuous character* $\psi\colon H \to \bar{k}^\times$.

*When they hold,* $\rho|_H \otimes \bar{k} \simeq \psi \oplus \psi^s$ *with* $\psi^s(h) = \psi(s^{-1}hs)$ *and* $\psi^s \neq \psi$*, and every isomorphism* $\rho \to \rho \otimes \chi$ *has trace* 0*. For* $G = G_{\mathbb{Q}}$ *and* $H = G_K$*,* $K$ *quadratic,* $\chi = \chi_d$ *is the character of* $K = \mathbb{Q}(\sqrt{d})$.

*Proof.* This is the Clifford theory for a subgroup of index two of Chapter 4, applied to the abstract group $G$; the equivalence (ii) over $k$ and over $\bar{k}$ holds because spaces of $G$-maps commute with extension of scalars (Chapter 4). Only continuity needs comment. In (iii), $\psi$ is the character by which $H$ acts on a line $\bar{k}w \subset \bar{k}^2$ stable under $\rho(H)$. The coordinates of $w$ generate a finite extension $k'$ of $k$, so $\psi(h) = \rho(h)w/w \in k'$, and in a basis $w, w'$ of $k'^2$ the value $\psi(h)$ is the upper left entry of the matrix of $\rho(h)$, a continuous function of $h$ ($k'$ discrete if $k$ is, and with its $\ell$-adic topology if $k$ is a finite extension of $\mathbb{Q}_\ell$). Conversely, $\operatorname{Ind}_H^G \psi$ for continuous $\psi$ is continuous: in the basis $\{v, sv\}$ its matrix entries are values of $\psi$ at $h$, $s^{-1}hs$ or products with $\psi(s^2)$, all continuous functions of $h \in H$ and $g \in G$ ($H$ being open and closed). □

The mechanism behind (i)⇒(iii)⇒(ii) is short enough to recall. Let $\bar{k}w$ be a line stable under $\rho(H)$, on which $H$ acts by $\psi$. For $h \in H$, $s^{-1}hs \in H$ ($H$ is normal, having index 2), so $\rho(h)\rho(s)w = \rho(s)\rho(s^{-1}hs)w = \psi^s(h)\rho(s)w$: the line $\bar{k}\rho(s)w$ is stable under $H$, with character $\psi^s$. It differs from $\bar{k}w$, since otherwise $\bar{k}w$ would be stable under $H \cup sH = G$, against absolute irreducibility. In the basis $w, \rho(s)w$, every $\rho(h)$ is $\operatorname{diag}(\psi(h), \psi^s(h))$ and $\rho(s) = \left(\begin{smallmatrix} 0 & \psi(s^2) \\ 1 & 0 \end{smallmatrix}\right)$ swaps the two lines: this is the matrix form of $\operatorname{Ind}_H^G \psi$. The matrix $P = \operatorname{diag}(1, -1)$ commutes with the diagonal $\rho(h)$, and $P\rho(s)P^{-1} = -\rho(s)$; as $\chi(h) = 1$ and $\chi(s) = -1$, $P\rho(g)P^{-1} = \chi(g)\rho(g)$ for all $g \in H \cup sH$, so $P$ is an isomorphism $\rho \otimes \bar{k} \to (\rho \otimes \chi) \otimes \bar{k}$, of trace 0.

*In the paper* 5.68. For $\bar{\rho} = \bar{\rho}_{E,\ell}$, irreducible, hence absolutely irreducible by Proposition 5.60(d), and $K = \mathbb{Q}(\sqrt{\ell^*})$: condition (i) is the failure of the hypothesis of Theorem thm:main (DFG04's $\ell \in S_f$). For $\ell = 3$, $\chi_{-3} = \bar{\chi}_3$, and (ii) says that the space of $G_{\mathbb{Q}}$-maps $\bar{\rho} \to \bar{\rho} \otimes \bar{\chi}_3$, which are automatically of trace zero, is nonzero: $H^0(\mathbb{Q}, \operatorname{ad}^0 \bar{\rho}(1)) \neq 0$, the condition $h_3(E) = 1$ of Lemma lem:h0; and by (iii) it happens exactly when $\bar{\rho} \otimes \mathbb{F}_9$ is induced from $G_{\mathbb{Q}(\sqrt{-3})}$, the case $N_s(\mathbb{F}_3)$ of Proposition prop:ns (Proposition 5.47(f) and Chapter 4: the determinant-one subgroup $C_4$ of $N_s(\mathbb{F}_3)$ is abelian, so $\bar{\rho}|_{G_{\mathbb{Q}(\sqrt{-3})}} \otimes \mathbb{F}_9 = \psi \oplus \psi^{-1}$ with $\psi$ of order 4). For $\ell \geq 5$, (ii) for $\chi_{\ell^*}$ can hold (the dihedral case excluded by $S_f$; by Proposition prop:Sf only when $\ell \mid N$), but it is not the condition $\bar{\rho} \simeq \bar{\rho} \otimes \bar{\chi}_\ell$ of Lemma lem:h0, which never holds for $\ell \geq 5$ (Proposition 5.64(b): it would force $\bar{\chi}_\ell^2 = 1$).

**Proposition 5.69** (Reduction and semisimplification)**.** *Let* $E_\lambda$ *be a finite extension of* $\mathbb{Q}_\ell$ *with ring of integers* $\mathcal{O}$*, uniformiser* $\varpi$ *and residue field* $k$*,* $G$ *a profinite group, and* $\rho\colon G \to \mathrm{GL}_n(E_\lambda)$ *continuous. For two* $G$*-stable lattices* $\Lambda, \Lambda'$*, the reductions* $\bar{\rho}_\Lambda$*,* $\bar{\rho}_{\Lambda'}$ *have the same characteristic polynomials, namely the reductions of the characteristic polynomials of* $\rho$*, which lie in* $\mathcal{O}[X]$*; hence* $\bar{\rho}_\Lambda^{ss} \simeq \bar{\rho}_{\Lambda'}^{ss}$*, and* $\bar{\rho}^{ss}$ *is well defined. If* $\bar{\rho}_\Lambda$ *is irreducible, then* $\Lambda' = \varpi^m \Lambda$ *for some* $m \in \mathbb{Z}$.

*Proof.* In a basis of $\Lambda$ the matrix of $\rho(g)$ lies in $\mathrm{GL}_n(\mathcal{O})$, so $\det(X - \rho(g)) \in \mathcal{O}[X]$, and it does not depend on the basis of $E_\lambda^n$; its reduction is the characteristic polynomial of $\bar{\rho}_\Lambda(g)$, and likewise for $\Lambda'$. Brauer-Nesbitt (Chapter 4, valid for any group and any field) gives $\bar{\rho}_\Lambda^{ss} \simeq \bar{\rho}_{\Lambda'}^{ss}$. The last statement is the homothety lemma of Chapter 4. □

**Proposition 5.70** (The $\ell$-adic Kummer cocycle)**.** *Let* $\ell \neq \operatorname{char} F$ *be a prime and* $a \in F^\times$*. Choose a compatible system* $(\alpha_k)_{k \geq 0}$ *in* $F^{\mathrm{sep}}$ *with* $\alpha_0 = a$ *and* $\alpha_{k+1}^\ell = \alpha_k$*, and define, in the additive*

*notation of Convention 5.3(c),*

$$\kappa_a\colon G_F \to \mathbb{Z}_\ell(1), \qquad \kappa_a(\sigma) = \big(\sigma(\alpha_k)/\alpha_k\big)_{k\ge 0}.$$

(a) *$\kappa_a$ is continuous and $\kappa_a(\sigma\tau) = \kappa_a(\sigma) + \chi_\ell(\sigma)\kappa_a(\tau)$: a continuous 1-cocycle (Chapter 16). Equivalently, writing $\kappa_a(\sigma) = b(\sigma)e$ for a generator $e$ of $\mathbb{Z}_\ell(1)$,*

$$\sigma \longmapsto \begin{pmatrix} \chi_\ell(\sigma) & b(\sigma) \\ 0 & 1 \end{pmatrix}$$

*is a continuous homomorphism $G_F \to \mathrm{GL}_2(\mathbb{Z}_\ell)$: the action on $\mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$ with $e_1 = e$, $\sigma e_1 = \chi_\ell(\sigma)e_1$, $\sigma e_2 = e_2 + b(\sigma)e_1$.*

(b) *Another compatible system changes $\kappa_a$ by the coboundary $\sigma \mapsto (\chi_\ell(\sigma) - 1)z$ for some $z \in \mathbb{Z}_\ell(1)$. For the product systems, $\kappa_{aa'} = \kappa_a + \kappa_{a'}$ for $a, a' \in F^\times$. The $k$-th component of $\kappa_a$, restricted to $G_{F(\mu_{\ell^k})}$, is the Kummer character $\chi_a$ with $n = \ell^k$.*

(c) *On $\ker\chi_\ell = G_{F(\mu_{\ell^\infty})}$, $\kappa_a$ is a homomorphism, and for $\sigma \in G_F$, $\tau \in \ker\chi_\ell$: $\kappa_a(\sigma\tau\sigma^{-1}) = \chi_\ell(\sigma)\kappa_a(\tau)$.*

(d) *If $K \supseteq F$ contains $\mu_{\ell^k}$, then $\kappa_a \bmod \ell^k$ vanishes on $G_K$ iff $a \in (K^\times)^{\ell^k}$.*

*Proof.* (a) $(\sigma(\alpha_k)/\alpha_k)^{\ell^k} = \sigma(a)/a = 1$, and $(\sigma(\alpha_{k+1})/\alpha_{k+1})^{\ell} = \sigma(\alpha_k)/\alpha_k$, so $\kappa_a(\sigma) \in \mathbb{Z}_\ell(1)$. Multiplicatively,

$$\frac{\sigma\tau(\alpha_k)}{\alpha_k} = \sigma\Big(\frac{\tau(\alpha_k)}{\alpha_k}\Big)\cdot\frac{\sigma(\alpha_k)}{\alpha_k},$$

and $\sigma$ acts on $\mu_{\ell^k}$ by raising to the power $\chi_\ell(\sigma)$; additively this is the cocycle relation. The $k$-th component is constant on the cosets of the open subgroup $G_{F(\mu_{\ell^k},\alpha_k)}$ (for $\tau$ in it, $\kappa_a(\sigma\tau)_k = \kappa_a(\sigma)_k + \chi_\ell(\sigma)\kappa_a(\tau)_k = \kappa_a(\sigma)_k$), so $\kappa_a$ is continuous. The matrix identity $\left(\begin{smallmatrix}\chi(\sigma) & b(\sigma)\\ 0 & 1\end{smallmatrix}\right)\left(\begin{smallmatrix}\chi(\tau) & b(\tau)\\ 0 & 1\end{smallmatrix}\right) = \left(\begin{smallmatrix}\chi(\sigma\tau) & b(\sigma)+\chi(\sigma)b(\tau)\\ 0 & 1\end{smallmatrix}\right)$ is the cocycle relation. (b) Another system is $(\epsilon_k\alpha_k)$ with $\epsilon = (\epsilon_k) \in \mathbb{Z}_\ell(1)$, and $\sigma(\epsilon_k\alpha_k)/\epsilon_k\alpha_k = (\sigma(\epsilon_k)/\epsilon_k)\cdot\sigma(\alpha_k)/\alpha_k$; additively the change is $\sigma(\epsilon) - \epsilon = (\chi_\ell(\sigma) - 1)\epsilon$. Additivity in $a$ is clear. On $G_{F(\mu_{\ell^k})}$ the $k$-th component is $\sigma(\alpha_k)/\alpha_k = \chi_a(\sigma)$ by definition (Proposition 5.35, over $F(\mu_{\ell^k})$, with $n = \ell^k$, $\alpha = \alpha_k$). (c) For $\tau, \tau' \in \ker\chi_\ell$ the cocycle relation reads $\kappa_a(\tau\tau') = \kappa_a(\tau) + \kappa_a(\tau')$. From $0 = \kappa_a(1) = \kappa_a(\sigma\sigma^{-1}) = \kappa_a(\sigma) + \chi_\ell(\sigma)\kappa_a(\sigma^{-1})$ and $\chi_\ell(\tau) = 1$:

$$\kappa_a(\sigma\tau\sigma^{-1}) = \kappa_a(\sigma) + \chi_\ell(\sigma)\big(\kappa_a(\tau) + \kappa_a(\sigma^{-1})\big) = \chi_\ell(\sigma)\kappa_a(\tau).$$

(d) $G_K \subseteq G_{F(\mu_{\ell^k})}$, so by (b) $\kappa_a \bmod \ell^k$ on $G_K$ is the Kummer character of $a$ over $K$ with $n = \ell^k$; it is trivial iff $a \in (K^\times)^{\ell^k}$ by Proposition 5.35(b). □

The Kummer representation also illustrates Proposition 5.69. Let $\ell$ be odd, $F = \mathbb{Q}$, $a = 2$, and $\rho$ the representation of (a) on $\Lambda = \mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$. The lattice $\Lambda' = \mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell\, \ell e_2$ is also stable, since $\sigma(\ell e_2) = \ell e_2 + \ell b(\sigma)e_1$; in the basis $e_1, \ell e_2$ the matrix is $\left(\begin{smallmatrix}\chi_\ell & \ell b\\ 0 & 1\end{smallmatrix}\right)$, with reduction $\mathrm{diag}(\bar\chi_\ell, 1)$, which is semisimple. On $\Lambda$ the reduction is $\left(\begin{smallmatrix}\bar\chi_\ell & \bar b\\ 0 & 1\end{smallmatrix}\right)$, and $\bar b = b \bmod \ell$ is not identically zero on $G_{\mathbb{Q}(\zeta_\ell)}$ by (d), because 2 is not an $\ell$-th power in $\mathbb{Q}(\zeta_\ell)$ ($x^\ell - 2$ is Eisenstein, so an $\ell$-th root of 2 has degree $\ell$ over $\mathbb{Q}$, which does not divide $\ell - 1 = [\mathbb{Q}(\zeta_\ell) : \mathbb{Q}]$). An element of $G_{\mathbb{Q}(\zeta_\ell)}$ with $\bar b \neq 0$ acts by a nontrivial unipotent matrix, whose only stable line is $\mathbb{F}_\ell e_1$; so the reduction on $\Lambda$ is not

semisimple. Both reductions have semisimplification $\bar{\chi}_\ell \oplus 1$, as the proposition says, but they are not isomorphic, and $\Lambda'$ is not homothetic to $\Lambda$; this does not contradict the last statement of the proposition because neither reduction is irreducible.

*In the paper* 5.71. In the proof of Lemma lem:tamfactor, $F = \mathbb{Q}_q$ with $q \neq \ell$, $a = q_E$ is the Tate parameter, and over the unramified quadratic extension the action on $T_\ell E$ in the basis $e_1 = (\zeta_{\ell^k})_k$, $e_2 = (q_E^{1/\ell^k})_k$ is the matrix of (a) (Chapter 10); for the nonsplit curve it is twisted by the unramified character $\varepsilon$, giving $\left(\begin{smallmatrix} \chi_\ell\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$. Part (c) with $\sigma = \mathrm{Frob}$ and $\tau = t$ a generator of the pro-$\ell$ tame inertia quotient (on which $\chi_\ell$ is trivial, Chapter 7) gives $\kappa(\mathrm{Frob}\; t\; \mathrm{Frob}^{-1}) = q\kappa(t)$, the abstract form of the tame relation $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q$, equivalently $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$, which the paper uses. By (b), $\kappa_{q_E} = n\kappa_q + \kappa_u$ for $q_E = q^n u$; the restriction of $\kappa_u$ to inertia vanishes (the unit $u$ has $\ell$-power roots in $\mathbb{Q}_q^{\mathrm{ur}}$, by Hensel, Chapter 7), and the restriction of $\kappa_q$ to inertia is the tame character, whence "the restriction of $\kappa$ to $I$ is $n$ times the tame character". In Lemma lem:tamagawa-exp, (d) with $K = \mathbb{Q}_q^{\mathrm{ur}}$ and $G_K = I_q$ says that inertia acts trivially on $E[\ell^t]$ iff $q_E \in (\mathbb{Q}_q^{\mathrm{ur},\times})^{\ell^t}$; Chapter 7 turns this into $\ell^t \mid n_q$. The value $c' = \varepsilon\kappa(\mathrm{Frob})$ depends on the choices (by (b) and on the lift of Frob), and only its class modulo $(n, q-1)\mathbb{Z}_\ell$ is meaningful, which does not affect the minimum in the lemma (Chapter 17).

## 5.9 Frobenius elements and the Chebotarev density theorem

The step "$\bar{\rho}_h \simeq \bar{\rho}$ by Chebotarev and Brauer-Nesbitt" in Lemma lem:old compares two residual representations through the characteristic polynomials of Frobenius elements, which are known only at the primes not dividing $\ell N$. This section states what Frobenius elements are (they are constructed in Chapter 6; no theorem of Sections 1 to 7 depends on them, and in Sections 6 and 8 they enter only through the stated Theorem 5.72, in Example 5.49 and Proposition 5.63(c)), states the Chebotarev density theorem, and proves the consequences used in the paper: Frobenius elements are dense, so a semisimple representation is determined by their characteristic polynomials, and congruent traces give isomorphic residual representations. A worked example checks the factorisation patterns of $\psi_3$ against Frobenius, as the test of Section sec:data does.

**Theorem 5.72** (Frobenius elements)**.** *Let $K/\mathbb{Q}$ be a finite Galois extension with ring of integers $\mathcal{O}_K$, and $p$ a prime unramified in $K$ (Chapter 6).*

*(a) For each prime $\mathfrak{P}$ of $\mathcal{O}_K$ above $p$ there is a unique $\mathrm{Frob}_\mathfrak{P} \in \mathrm{Gal}(K/\mathbb{Q})$ with $\mathrm{Frob}_\mathfrak{P}(x) \equiv x^p \pmod{\mathfrak{P}}$ for all $x \in \mathcal{O}_K$. The $\mathrm{Frob}_\mathfrak{P}$, $\mathfrak{P} \mid p$, form one conjugacy class $\mathrm{Frob}_p$ ($\mathrm{Frob}_{\sigma\mathfrak{P}} = \sigma\,\mathrm{Frob}_\mathfrak{P}\,\sigma^{-1}$). The order of $\mathrm{Frob}_\mathfrak{P}$ is the residue degree $[\mathcal{O}_K/\mathfrak{P} : \mathbb{F}_p]$, and $p$ splits completely in $K$ iff $\mathrm{Frob}_p = 1$.*

*(b) For $K' \subseteq K$ Galois over $\mathbb{Q}$, $\mathrm{Frob}_\mathfrak{P}$ restricts to $\mathrm{Frob}_{\mathfrak{P}\cap K'}$; so $\mathrm{Frob}_p(K/\mathbb{Q})$ maps to $\mathrm{Frob}_p(K'/\mathbb{Q})$.*

*(c) In $\mathbb{Q}(\zeta_n)$ with $p \nmid n$, $\mathrm{Frob}_p = \sigma_p$; hence $\bar{\chi}_\ell(\mathrm{Frob}_p) = p \bmod \ell$ and $\chi_\ell(\mathrm{Frob}_p) = p$ for $p \neq \ell$. In $\mathbb{Q}(\sqrt{d})$ with $p \nmid D$, $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$.*

*(d) For $L/\mathbb{Q}$ Galois of any degree, unramified at $p$ (that is, every finite subextension is), the Frobenius elements at the primes of $L$ above $p$ (compatible systems of primes of the finite subextensions) are the elements $\mathrm{Frob}_\mathfrak{P} \in \mathrm{Gal}(L/\mathbb{Q})$ restricting to $\mathrm{Frob}_{\mathfrak{P}\cap K}$ for each finite Galois $K \subseteq L$; they form one conjugacy class. In $G_\mathbb{Q}$ a Frobenius element at $p$ is well defined up to conjugation and up to the inertia group at a prime above $p$; so for a continuous*

*representation $\rho$ of $G_{\mathbb{Q}}$ unramified at $p$ (trivial on the inertia groups at $p$), the conjugacy class of $\rho(\mathrm{Frob}_p)$, and in particular its characteristic polynomial, is well defined.*

*(e) (Dedekind.) Let $f \in \mathbb{Z}[x]$ be separable with leading coefficient $a$, and $p \nmid a\,\mathrm{disc}(f)$. If $f \bmod p$ has irreducible factors of degrees $d_1, \dots, d_r$, then $p$ is unramified in the splitting field of $f$ and $\mathrm{Frob}_p$ acts on the roots of $f$ with cycle type $(d_1, \dots, d_r)$.*

We do not prove this here; see Chapter 6, and [142, Chapter I, Section 9] (decomposition groups and Frobenius) and [142, Chapter I, Section 8] (the factorisation of $p$ through $f \bmod p$); the section numbers in this book are pointers we have not re-verified. The finite-field part is proved here: the residue extension $(\mathcal{O}_K/\mathfrak{P})/\mathbb{F}_p$ is cyclic generated by $\mathrm{Frob}_p$ (Theorem 5.25), and $\mathrm{Frob}_{\mathfrak{P}}$ is the lift of that generator, which exists and is unique when $p$ is unramified. Note the convention (Convention 5.2): $\mathrm{Frob}_p$ is arithmetic.

For example, in $\mathbb{Q}(i) = \mathbb{Q}(\zeta_4)$ and for odd $p$, $\mathrm{Frob}_p = \sigma_p$ sends $i$ to $i^p$, which is $i$ for $p \equiv 1 \pmod 4$ and $-i$ for $p \equiv 3 \pmod 4$. By (a), $p$ splits completely in $\mathbb{Q}(i)$ exactly when $p \equiv 1 \pmod 4$, and $\chi_{-4}(\mathrm{Frob}_p) = (-1)^{(p-1)/2}$, in accordance with (c).

**Example 5.73** (Frobenius on the four lines)**.** Let $E/\mathbb{Q}$ have good reduction at a prime $p \neq 3$ with $p \nmid \mathrm{disc}(\psi_3)$. Then $\bar\rho_{E,3}$ is unramified at $p$ and $\bar\rho_{E,3}(\mathrm{Frob}_p)$ has characteristic polynomial $X^2 - a_p X + p \bmod 3$ (Chapter 9, Chapter 11). By Theorem 5.72(e) and Proposition 5.47, the degrees of the irreducible factors of $\psi_3 \bmod p$ are the orbit sizes of $g = \bar\rho_{E,3}(\mathrm{Frob}_p)$ on the four lines. These depend only on the characteristic polynomial, except in the last case:

- $X^2 \pm X - 1$: the eigenvalues $\lambda, \lambda^3 \in \mathbb{F}_9 \setminus \mathbb{F}_3$ satisfy $\lambda^4 = \det g = -1$, so $\lambda$ has order 8 and $g$ has order 4 in $\mathrm{PGL}_2(\mathbb{F}_3)$ with no fixed line: a 4-cycle, pattern (4);
- $X^2 + 1$: $g^2 = -1$, so $g$ is an involution on the lines, without fixed line (no eigenvalue in $\mathbb{F}_3$): pattern $(2,2)$;
- $X^2 - 1$: $g \sim \mathrm{diag}(1,-1)$ fixes its two eigenlines and swaps the other two: pattern $(1,1,2)$;
- $(X \mp 1)^2$: $g = \pm(1+N)$ with $N$ nilpotent; if $N \neq 0$, $g$ fixes one line and has order 3 in $\mathrm{PGL}_2(\mathbb{F}_3)$: pattern $(1,3)$; if $g = \pm 1$: $(1,1,1,1)$.

For 11a1 ($a_2 = -2$, $a_5 = 1$, $a_7 = -2$, $a_{13} = 4$, $a_{17} = -2$, $a_{19} = 0$, $a_{23} = -1$, $a_{29} = 0$, $a_{31} = 7$, $a_{37} = 3$; $\mathrm{disc}\,\psi_3 = -3^3 \cdot 11^{10}$) we checked with PARI that $\psi_3 \bmod p$ has the pattern (4) for $p = 2, 5, 17, 23$, $(1,3)$ for $p = 7, 13, 31$, $(2,2)$ for $p = 19, 37$ and $(1,1,2)$ for $p = 29$, as predicted. The pattern $(1,1,2)$ at $p = 29$ means that $\mathrm{Frob}_{29}$ fixes two lines, although the image is $\mathrm{GL}_2(\mathbb{F}_3)$: Frobenius elements see conjugacy classes, not the image.

**Definition 5.74** (Density)**.** Let $X$ be a set of primes. Its *Dirichlet density* is

$$\lim_{s \to 1^+} \frac{\sum_{p \in X} p^{-s}}{\log \frac{1}{s-1}},$$

and its *natural density* is $\lim_{x\to\infty} \#\{p \le x : p \in X\}/\#\{p \le x\}$, when these limits exist. A set with a natural density has the same Dirichlet density (we do not use the converse, which is false). Finite sets have density 0, a subset of a set of density 0 has density 0 (the sums are nonnegative), and densities are additive on disjoint unions of sets that have densities. $X$ has *density one* if its complement has density 0. "*Almost all $p$*" means all but finitely many. Unless stated otherwise, density means Dirichlet density.

The sets of density one used in the paper are the simplest ones: for a fixed integer $N$, the primes $p \nmid N$ form a set of density one, its complement being finite. So a hypothesis "for all $p$ in a set of density one" in the corollaries below is satisfied as soon as the statement holds for all $p \nmid N$; the extra generality costs nothing and allows one to discard any set of density 0.

For the comparison of the two densities see [172, Chapter VI, Section 4]; the normalisation uses $\sum_p p^{-s} \sim \log \frac{1}{s-1}$ as $s \to 1^+$, which is proved there.

**Theorem 5.75** (Chebotarev's density theorem)**.** *Let $K/\mathbb{Q}$ be a finite Galois extension with group $G$ and $C \subseteq G$ a conjugacy class. The set of primes $p$, unramified in $K$, with $\mathrm{Frob}_p = C$ has Dirichlet density $\#C/\#G$; it also has natural density $\#C/\#G$. The same holds for a finite Galois extension of number fields, with the primes of the base field. In particular every conjugacy class is $\mathrm{Frob}_p$ for infinitely many $p$.*

We do not prove this; see [142, Chapter VII, Section 13] for the Dirichlet density and [178, Chapter I, Section 2.2] for the statement with natural density; the book uses only the positivity of the Dirichlet density. Special cases: for $K = \mathbb{Q}(\zeta_n)$ (Theorem 5.72(c)) it is Dirichlet's theorem, each class $a \in (\mathbb{Z}/n)^\times$ containing primes of density $1/\varphi(n)$; for $K = \mathbb{Q}(\sqrt{d})$, half of the primes split. For the splitting field of $x^3 - 2$ (Example 5.31, group $S_3$) and $p \nmid 6$, Theorem 5.72(e) says that $x^3 - 2$ has exactly one root modulo $p$ iff $\mathrm{Frob}_p$ is a transposition (density $1/2$), three roots iff $\mathrm{Frob}_p = 1$ (density $1/6$), and none iff $\mathrm{Frob}_p$ is a 3-cycle (density $1/3$). A transposition is odd, and the sign character cuts out $\mathbb{Q}(\sqrt{-108}) = \mathbb{Q}(\sqrt{-3})$, so "exactly one root" happens iff $\chi_{-3}(\mathrm{Frob}_p) = -1$, iff $p \equiv 2 \pmod 3$. Among the 428 primes $5 \le p < 3000$ the observed proportions are 0.516, 0.159 and 0.325.

**Corollary 5.76** (Frobenius elements are dense)**.** *Let $L/\mathbb{Q}$ be a Galois extension (possibly infinite) unramified outside a finite set $S$ of primes, and $X$ a set of primes of density one. Then the Frobenius elements $\mathrm{Frob}_{\mathfrak{P}} \in \mathrm{Gal}(L/\mathbb{Q})$ at the primes $\mathfrak{P}$ of $L$ above the $p \in X \setminus S$ form a dense subset of $\mathrm{Gal}(L/\mathbb{Q})$.*

*Proof.* A basic open set of $\mathrm{Gal}(L/\mathbb{Q})$ is $\sigma\,\mathrm{Gal}(L/K)$ with $K \subseteq L$ finite Galois over $\mathbb{Q}$ (Proposition 5.51(b)). Let $C$ be the conjugacy class of $\sigma|_K$ in $\mathrm{Gal}(K/\mathbb{Q})$. By Theorem 5.75 the set $Y$ of primes unramified in $K$ with $\mathrm{Frob}_p(K/\mathbb{Q}) = C$ has density $\#C/[K:\mathbb{Q}] > 0$. The complement of $X \setminus S$ has density 0 (the complement of $X$ has density 0 and $S$ is finite), so $Y \cap (X \setminus S)$ has the positive density of $Y$, and in particular contains a prime $p$. As $\mathrm{Frob}_p(K/\mathbb{Q}) = C$ is the set of $\mathrm{Frob}_{\mathfrak{p}}$ for the primes $\mathfrak{p}$ of $K$ above $p$, there is $\mathfrak{p}$ with $\mathrm{Frob}_{\mathfrak{p}} = \sigma|_K$; choose a prime $\mathfrak{P}$ of $L$ above $\mathfrak{p}$ (Chapter 6). By Theorem 5.72(d), $\mathrm{Frob}_{\mathfrak{P}}$ restricts to $\mathrm{Frob}_{\mathfrak{p}} = \sigma|_K$ on $K$, that is $\mathrm{Frob}_{\mathfrak{P}} \in \sigma\,\mathrm{Gal}(L/K)$. □

**Corollary 5.77** (Chebotarev and Brauer-Nesbitt)**.** *Let $k$ be a finite field, $\overline{\mathbb{F}}_\ell$, or a finite extension of $\mathbb{Q}_\ell$, and $\rho_1, \rho_2 \colon G_{\mathbb{Q}} \to \mathrm{GL}_n(k)$ continuous, semisimple, and unramified outside a finite set $S$ of primes. If $\det(X - \rho_1(\mathrm{Frob}_p)) = \det(X - \rho_2(\mathrm{Frob}_p))$ for all $p$ in a set $X$ of density one, $p \notin S$, then $\rho_1 \simeq \rho_2$.*

*Proof.* Let $L$ be the field cut out by $\rho_1 \oplus \rho_2$, the fixed field of $\ker\rho_1 \cap \ker\rho_2$ (Proposition 5.62); it is Galois over $\mathbb{Q}$ and unramified outside $S$, since the inertia groups at $p \notin S$ act trivially through both $\rho_i$. Both $\rho_i$ factor through $\mathrm{Gal}(L/\mathbb{Q})$. The map

$$\mathrm{Gal}(L/\mathbb{Q}) \to k^n, \qquad g \mapsto \text{coefficients of } \det(X - \rho_1(g)) - \det(X - \rho_2(g)),$$

is continuous (polynomial in the matrix entries), so its zero set is closed; it contains the Frobenius elements at the primes above $X \setminus S$, which are dense by Corollary 5.76. Hence $\rho_1(g)$ and $\rho_2(g)$ have the same characteristic polynomial for every $g \in G_{\mathbb{Q}}$. By Brauer-Nesbitt (Chapter 4, for an arbitrary group over an arbitrary field) $\rho_1^{ss} \simeq \rho_2^{ss}$, that is $\rho_1 \simeq \rho_2$ as both are semisimple. (For finite $k$ the image is finite and the finite-group form suffices.) □

**Corollary 5.78** (Congruent traces)**.** *Let $\mathcal{O}$ be the ring of integers of a finite extension of $\mathbb{Q}_\ell$, $\lambda = \varpi\mathcal{O}$ its maximal ideal, $k = \mathcal{O}/\lambda$, and $\rho_1, \rho_2 \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathcal{O})$ continuous and unramified outside a finite set $S$. Suppose $\operatorname{tr}\rho_1(\mathrm{Frob}_p) \equiv \operatorname{tr}\rho_2(\mathrm{Frob}_p)$ and $\det\rho_1(\mathrm{Frob}_p) \equiv \det\rho_2(\mathrm{Frob}_p) \pmod{\lambda}$ for all $p$ in a set of density one, $p \notin S$.*

*(a) $\bar\rho_1^{ss} \simeq \bar\rho_2^{ss}$ over $k$.*

*(b) If $\bar\rho_1$ is absolutely irreducible, then $\bar\rho_2 \simeq \bar\rho_1$.*

*(c) Conversely, if $\bar\rho_1^{ss} \simeq \bar\rho_2^{ss}$, the congruences hold for every $p \notin S$.*

*(d) (Newforms; given the representations attached to newforms, Chapter 15.) Let $\ell$ be odd, $E/\mathbb{Q}$ an elliptic curve of conductor $N$ with $\bar\rho_{E,\ell}$ irreducible, and $h$ a newform of weight 2 on $\Gamma_0(M)$ (trivial character) whose coefficients lie in $\mathcal{O}$ (for a chosen embedding of its coefficient field). If $a_p(h) \equiv a_p(E) \pmod{\lambda}$ for all $p \nmid \ell MN$ in a set of density one, then $\bar\rho_h \simeq \bar\rho_{E,\ell} \otimes_{\mathbb{F}_\ell} k$, for every choice of the stable lattice defining $\bar\rho_h$.*

*Proof.* (a) The characteristic polynomial of $\bar\rho_i(\mathrm{Frob}_p)$ is $X^2 - \operatorname{tr} X + \det$ reduced modulo $\lambda$, and it is also that of $\bar\rho_i^{ss}(\mathrm{Frob}_p)$ (characteristic polynomials are multiplicative along a composition series). The $\bar\rho_i^{ss}$ are continuous, semisimple and unramified outside $S$ (inertia acts trivially on $\bar\rho_i$, hence on its subquotients); apply Corollary 5.77. (b) $\bar\rho_2^{ss} \simeq \bar\rho_1^{ss} = \bar\rho_1$ is irreducible of dimension 2, so the two-dimensional $\bar\rho_2$ has an irreducible composition factor of dimension 2, hence is irreducible and equal to its semisimplification. (c) Characteristic polynomials of $\bar\rho_i(\mathrm{Frob}_p)$ and of $\bar\rho_i^{ss}(\mathrm{Frob}_p)$ agree. (d) The black box of Chapter 15 gives a continuous $\rho_{h,\lambda} \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathcal{O})$ (after choosing a stable lattice, Proposition 5.62(c)), unramified outside $M\ell$, with $\det\rho_{h,\lambda} = \chi_\ell$ and $\operatorname{tr}\rho_{h,\lambda}(\mathrm{Frob}_p) = a_p(h)$ for $p \nmid M\ell$. Take $\rho_1 = \rho_{E,\ell} \otimes_{\mathbb{Z}_\ell} \mathcal{O}$, which is unramified outside $\ell N$ with $\det\rho_1 = \chi_\ell$ and $\operatorname{tr}\rho_1(\mathrm{Frob}_p) = a_p(E)$ for $p \nmid \ell N$ (Chapter 9, Chapter 11), and $\rho_2 = \rho_{h,\lambda}$, with $S$ the primes dividing $\ell MN$. The determinants of Frobenius agree and the traces are congruent on a set of density one. Since $\bar\rho_{E,\ell}$ is irreducible and $\det\bar\rho_{E,\ell} = \bar\chi_\ell$ with $\ell$ odd, it is absolutely irreducible (Proposition 5.60(d)), and so is $\bar\rho_1 = \bar\rho_{E,\ell} \otimes_{\mathbb{F}_\ell} k$. By (b), $\bar\rho_h = \bar\rho_2 \simeq \bar\rho_1$; the argument applies to every stable lattice (and Proposition 5.69 gives the independence of $\bar\rho_h^{ss}$ directly). □

*In the paper* 5.79. Lemma lem:old, case $p \nmid \ell N$: "$\bar\rho_h \simeq \bar\rho$ over a common finite extension of $\mathbb{F}_\ell$, by Chebotarev and Brauer-Nesbitt, $\bar\rho$ being irreducible, hence absolutely irreducible". This is Corollary 5.78(d), after enlarging $\mathcal{O}$ so that it contains the coefficients of the newform $h$ of the lemma; it rests on the black box of Chapter 15 (the representation $\rho_{h,\lambda}$ with $\operatorname{tr}\rho_{h,\lambda}(\mathrm{Frob}_p) = a_p(h)$). The two residual representations a priori live over different finite fields ($k$ and $\mathbb{F}_\ell$), so they are compared over $k$, as $\bar\rho_{E,\ell} \otimes_{\mathbb{F}_\ell} k$; and "irreducible, hence absolutely irreducible" is Proposition 5.60(d). The equivalence quoted in the proof of Theorem thm:qual (Section sec:proof: "$\bar\rho_g \simeq \bar\rho$, that is, $a_p(g) \equiv a_p(f) \pmod{\lambda}$ for all $p \nmid 3N$"), and in the Example of that section, is (c) in one direction and (d) in the other.

## 5.10 Exercises

The exercises are graded from routine to harder. Exercises 5.91 and 5.92 check the paper's statements on the 3-division polynomial on curves not treated in the text; Exercise 5.93 generalises the determinant argument of Lemma lem:h0.

**Exercise 5.80.** Show that $\mathbb{Q}(\sqrt{2},\sqrt{3}) = \mathbb{Q}(\sqrt{2}+\sqrt{3})$, that the minimal polynomial of $\sqrt{2}+\sqrt{3}$ is $x^4 - 10x^2 + 1$, and list the intermediate fields with the corresponding subgroups of $\mathrm{Gal}(\mathbb{Q}(\sqrt{2},\sqrt{3})/\mathbb{Q}) \simeq V_4$. Deduce that $x^4 - 10x^2 + 1$ is irreducible over $\mathbb{Q}$ but reducible modulo every prime $p$. (Hint: modulo $p$ one of 2, 3, 6 is a square, and then the polynomial factors over $\mathbb{F}_p$; compare with Proposition 5.46(c), case $V_4$.)

**Exercise 5.81.** Let $\alpha = 2^{1/4} \in \mathbb{R}$. Show that the splitting field of $x^4 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\alpha, i)$, of degree 8, with Galois group $D_4$, and identify the three quadratic subfields. Show that over $\mathbb{Q}(i)$ the polynomial $x^4 - 2$ stays irreducible with cyclic Galois group of order 4 (use Proposition 5.35(c) with $n = 4$, $\mu_4 \subset \mathbb{Q}(i)$, after checking that 2 is not a square in $\mathbb{Q}(i)$). Confirm the result with Proposition 5.46(c): the resolvent cubic of $x^4 - 2$ is $y^3 + 8y = y(y^2 + 8)$.

**Exercise 5.82.** Let $z = \zeta_7 + \zeta_7^{-1} = 2\cos(2\pi/7)$. Show that $z$ is a root of $x^3 + x^2 - 2x - 1$, that this cubic has discriminant 49 and roots $2\cos(2\pi k/7)$, $k = 1, 2, 3$, and that its splitting field is the real cubic subfield of $\mathbb{Q}(\zeta_7)$. Which primes $p$ split completely in it? (Answer: $p \equiv \pm 1 \pmod 7$.)

**Exercise 5.83.** Let $p$ be an odd prime and $k \geq 1$. Using $(x^{p^{k-1}} - 1)\Phi_{p^k}(x) = x^{p^k} - 1$ and the method of Proposition 5.44(e), show that

$$\mathrm{disc}(\Phi_{p^k}) = (-1)^{\varphi(p^k)/2}\, p^{\,p^{k-1}(pk-k-1)}.$$

Check the cases $p^k = 9$ $(-3^9)$ and 25 $(5^{35})$. Deduce that the unique quadratic subfield of $\mathbb{Q}(\zeta_{p^k})$ is $\mathbb{Q}(\sqrt{p^*})$.

**Exercise 5.84.** Show that $\mathbb{Q}(\sqrt{-5}) \subseteq \mathbb{Q}(\zeta_{20})$ and $\mathbb{Q}(\sqrt{6}) \subseteq \mathbb{Q}(\zeta_{24})$. Show that if $\mathbb{Q}(\sqrt{-5}) \subseteq \mathbb{Q}(\zeta_m)$ then $20 \mid m$. (Hint: by Theorem 5.28(c), $\mathbb{Q}(\sqrt{-5}) \subseteq \mathbb{Q}(\zeta_m) \cap \mathbb{Q}(\zeta_{20}) = \mathbb{Q}(\zeta_{\gcd(m,20)})$; list the quadratic subfields of $\mathbb{Q}(\zeta_g)$ for the proper divisors $g$ of 20.) Prove the analogous statement for $\mathbb{Q}(\sqrt{6})$ and 24.

**Exercise 5.85.** Prove that the degree of a splitting field of a polynomial of degree $n$ divides $n!$. (Hint: induction on $n$; if $f$ is reducible, $f = gh$, use that $\deg g!\,\deg h!$ divides $n!$; if $f$ is irreducible, adjoin one root.)

**Exercise 5.86.** Let $q = p^r$ and $a \in \mathbb{F}_q$. Show that $x^p - x - a$ is irreducible over $\mathbb{F}_q$ iff $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a) \neq 0$. (Hint: $\wp\colon \mathbb{F}_q \to \mathbb{F}_q$ is $\mathbb{F}_p$-linear with kernel $\mathbb{F}_p$, so its image has index $p$; show that $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(\wp(b)) = 0$, and compare sizes, using Proposition 5.33(d).)

**Exercise 5.87.** Let $u \in \mathbb{F}_{13}^\times$ and $\ell = 3$, so $v = v_3(12) = 1$. Show that $u$ is a cube in $\mathbb{F}_{13}$ iff $u^4 = 1$, that the cubes are $\{1, 5, 8, 12\}$, and that the class of $u = 2$ in $\mathbb{F}_{13}^\times/(\mathbb{F}_{13}^\times)^3 \simeq \mu_3(\mathbb{F}_{13})$ is $2^4 = 3$. Compute $\mathrm{Frob}_{13}(\alpha)/\alpha$ for a cube root $\alpha$ of 2 in $\mathbb{F}_{13^3}$, and explain why $x^3 - 2$ is irreducible over $\mathbb{F}_{13}$.

**Exercise 5.88.** Let $\widehat{\mathbb{Z}} \simeq G_{\mathbb{F}_p}$. Show that the closed subgroups of $\widehat{\mathbb{Z}} = \prod_q \mathbb{Z}_q$ are the products $\prod_q q^{a_q}\mathbb{Z}_q$ with $a_q \in \{0, 1, 2, \ldots, \infty\}$ $(q^\infty\mathbb{Z}_q = 0)$, and that the closed subgroup topologically generated by $\mathrm{Frob}_p^n$ is $n\widehat{\mathbb{Z}}$. Describe the fixed fields: for instance, the union of the fields $\mathbb{F}_{p^{\ell^k}}$, $k \geq 1$, has Galois group $\mathbb{Z}_\ell$ over $\mathbb{F}_p$ and is the fixed field of $\prod_{q \neq \ell} \mathbb{Z}_q$.

**Exercise 5.89.** Let $\ell$ be odd. Show that $\chi_\ell(G_{\mathbb{Q}}) = \mathbb{Z}_\ell^\times$ is infinite, and that every neighbourhood of 1 in $\mathrm{GL}_1(\mathbb{Q}_\ell) = \mathbb{Q}_\ell^\times$ contains a nontrivial closed subgroup (namely $1 + \ell^k\mathbb{Z}_\ell$ for $k$ large). Explain why the argument of Proposition 5.62(d) fails over $\mathbb{Q}_\ell$. Show also that a continuous homomorphism $\widehat{\mathbb{Z}} \to \mathbb{C}^\times$ is determined by the image of 1, which is a root of unity, and that every root of unity occurs.

**Exercise 5.90.** Let $\ell \geq 5$ and $\bar\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F}_\ell)$ continuous and absolutely irreducible, with image contained in the normaliser $N_s(\mathbb{F}_\ell)$ of the diagonal split Cartan subgroup $C_s(\mathbb{F}_\ell)$ but not in $C_s(\mathbb{F}_\ell)$. Show: (a) the composite $G_{\mathbb{Q}} \to N_s(\mathbb{F}_\ell)/C_s(\mathbb{F}_\ell) \simeq \{\pm 1\}$ is $\chi_d$ for a unique quadratic field $\mathbb{Q}(\sqrt{d})$; (b) $\bar\rho \simeq \bar\rho \otimes \chi_d$ and $\bar\rho \simeq \mathrm{Ind}_{G_{\mathbb{Q}(\sqrt{d})}}^{G_{\mathbb{Q}}} \psi$ for a character $\psi$ of $G_{\mathbb{Q}(\sqrt{d})}$ with values in $\mathbb{F}_\ell^\times$; (c) if $d = \ell^*$, the hypothesis of Theorem thm:main fails for $\bar\rho$, although $\bar\rho \not\simeq \bar\rho \otimes \bar\chi_\ell$.

**Exercise 5.91.** The paper's example 726a1 is $y^2 + xy = x^3 + x^2 - 35x - 51$, with mod 3 image 3Ns. Check with PARI (`elldivpol(E,3)`, `factor`) that

$$\psi_3 = 3x^4 + 5x^3 - 210x^2 - 612x - 1480 = (x^2 - x - 74)(3x^2 + 8x + 20),$$

that the two factors have discriminants $297 = 3^3 \cdot 11$ and $-176 = -2^4 \cdot 11$, so that $K_1 = \mathbb{Q}(\sqrt{33})$, $K_2 = \mathbb{Q}(\sqrt{-11})$ and the third quadratic subfield of $K_1K_2$ is $\mathbb{Q}(\sqrt{-3})$, as Proposition 5.47(e) requires. Verify that $a_p \equiv 0 \pmod 3$ for the primes $p \equiv 2 \pmod 3$, $p \nmid 726$, $p < 100$, and find the primes $p < 60$, $p \nmid 726$, at which $\psi_3 \bmod p$ splits into four distinct linear factors (answer: 31, 37). (At $p = 2$ and $p = 11$ it also splits into linear factors, with repetitions; at $p = 3$ its degree drops.)

**Exercise 5.92.** For 121a1, $y^2 + xy + y = x^3 + x^2 - 30x - 76$, check that $\psi_3 = 3x^4 + 5x^3 - 177x^2 - 909x - 1249$ is irreducible with $\operatorname{disc}\psi_3 = -3^3 \cdot 11^4$ and irreducible resolvent cubic, so that its group is $S_4$ and the image of $\bar\rho_{E,3}$ is $\mathrm{GL}_2(\mathbb{F}_3)$. With $a_{19} = -6$ and $a_{29} = -9$, predict the factorisation patterns of $\psi_3$ modulo 19 and 29 from Example 5.73, and verify them.

**Exercise 5.93.** Let $\ell$ be odd, $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ continuous and $m \geq 1$. Show that if $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell^m) \bmod \ell^k$ with $k \geq 1$, then $\ell^{k-1}(\ell - 1)$ divides $2m$. (Hint: $\chi_\ell^{2m} \equiv 1 \pmod{\ell^k}$ on $G_{\mathbb{Q}}$, and $(\mathbb{Z}/\ell^k)^\times$ is cyclic of order $\ell^{k-1}(\ell-1)$, Chapter 2.) Recover Proposition 5.64(c) as the case $m = 1$, and show that for $\ell = 5$, $m = 2$ the condition allows only $k = 1$.

**Exercise 5.94.** Using Theorem 5.72 and Theorem 5.75, show that $x^3 - 3x + 1$ has either three roots or none modulo every prime $p \neq 3$, that it has three roots iff $p \equiv \pm 1 \pmod 9$, and that the set of these primes has density $1/3$. Count the primes $5 \leq p < 3000$ with three roots (there are 142 of 428).

**Exercise 5.95.** Let $E/\mathbb{Q}$ be an elliptic curve with $\bar\rho_{E,5}$ surjective onto $\mathrm{GL}_2(\mathbb{F}_5)$. Compute the images of $G_{\mathbb{Q}(\sqrt{5})}$ and of $G_{\mathbb{Q}(\zeta_5)}$ under $\bar\rho_{E,5}$ and their orders (240 and 120). Show that $\bar\rho_{E,5}$ is absolutely irreducible on $G_{\mathbb{Q}(\sqrt{5})}$. (Use Proposition 5.66(c) and the fact that $\mathrm{SL}_2(\mathbb{F}_5)$ acts irreducibly on $\overline{\mathbb{F}}_5^2$, Chapter 1.)

## 5.11 Notes and references

The finite theory (Sections 1 to 6) follows [109, Chapters V and VI], [58, Chapters 13 and 14] and [135]; the proofs of the primitive element theorem, of Artin's theorem and of Hilbert's Theorem 90 are the classical ones. The irreducibility proof of $\Phi_n$ is Dedekind's, as in [109, Chapter VI,

Section 3] and [201, Chapter 2]; Gauss sums and $g^2 = \ell^*$ are Chapter 2 and [87, Chapter 6]. Kummer theory and Artin-Schreier theory are [109, Chapter VI, Sections 6 and 8]; Capelli's criterion is [109, Chapter VI, Section 9]. The resolvent cubic and the Galois groups of quartics are [58, Section 14.6]; the $C_4$ versus $D_4$ test is due to Kappe and Warren [90]. Infinite Galois theory and the Krull topology follow [142, Chapter IV, Section 1], [143, Chapter I, Section 1] and [135, Chapter 7]. Continuous $\ell$-adic representations, lattices, the independence of $\bar{\rho}^{ss}$ from the lattice and the use of Chebotarev's theorem with Brauer-Nesbitt are in [178, Chapter I] and [38, Section 1]; the Chebotarev density theorem itself is [142, Chapter VII, Section 13], and the comparison of densities [172, Chapter VI, Section 4]. The theorem of Artin and Schreier is in [109, Chapter VI, Section 9], and the real closed fields behind the statement on elements of finite order in [109, Chapter XI]. The Kummer cocycle of a Tate parameter and its matrix form are [178, Chapter IV] and Chapter 10; its cohomological meaning is [143, Chapter VI] and Chapter 17.

The chapter and section numbers of the books cited in this chapter (Lang, Dummit-Foote, Neukirch, Neukirch-Schmidt-Wingberg, Serre, Washington, Ireland-Rosen, Milne) are pointers recalled from the editions listed in the bibliography and have not been checked against the printed books; the only theorem number given, Lang's Theorem VI.9.1 for Capelli's criterion, is unverified in the same sense. The numerical statements (discriminants, factorisations of $\psi_3$, Frobenius patterns, traces $a_p$, the subgroup table of $\mathrm{GL}_2(\mathbb{F}_3)$, the prime counts) were recomputed for this chapter with PARI/GP 2.17 through cypari2 and a short enumeration of $\mathrm{GL}_2(\mathbb{F}_3)$; the curve data agree with Cremona's tables (6912d1 and 726a1 have mod 3 image 3Ns there).

Notation: the quadratic characters $\chi_{q^*}$ are written as in the paper, and the paper's $\bar{\chi}_3^{\mathrm{cyc}}$ is the book's $\bar{\chi}_3$. Lemma lem:h0 and Section 1.3 of the paper state the exceptional case with $\bar{\chi}_\ell$; Proposition 5.64(c) and Proposition 5.29(c) are the facts behind this choice.

# Chapter 6

# Algebraic number theory

The paper lives over $\mathbb{Q}$, but its arguments constantly leave $\mathbb{Q}$: the coefficients of a congruent eigenform lie in a ring of integers $\mathcal{O}$ with a prime $\lambda \mid \ell$; the Galois group acts through decomposition and inertia groups and Frobenius elements; the quadratic fields $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{q^*})$ and $\mathbb{Q}(\sqrt{\ell^*})$ supply the characters $\chi_{-4}$, $\chi_{-3}$, $\chi_{q^*}$ and $\chi_{\ell^*}$; and the rings $\mathbb{Z}_3[\zeta_3]$ and $\mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ carry the action of inertia at the additive primes of types IV and II. This chapter supplies that algebraic number theory, from rings of integers and unique factorisation of ideals to the splitting of primes, the Frobenius element, cyclotomic fields, the dictionary between Dirichlet characters and characters of $G_{\mathbb{Q}}$, the Chebotarev density theorem and complex multiplication. It uses Chapter 1 for group actions and profinite groups, Chapter 2 for rings, finite fields, Kronecker symbols, quadratic reciprocity and Dirichlet characters, Chapter 3 for modules over principal ideal domains and discrete valuation rings, Chapter 4 for representations, and Chapter 5 for Galois theory; local fields are in Chapter 7.

## Core results at a glance

Throughout, $K$, $L$ are number fields, $\mathcal{O}_K$ the ring of integers, $\mathfrak{p}$, $\mathfrak{P}$, $\lambda$ nonzero primes, $N\mathfrak{p} = \#\mathcal{O}_K/\mathfrak{p}$, $D$ the discriminant of a quadratic field and $\chi_D$ its character; $\pi = 1 - \zeta_3$; Frob is the arithmetic Frobenius and $\mathrm{Frob}^{-1}$ the geometric one.

**6.1 Conventions.** $\overline{\mathbb{Q}} \subset \mathbb{C}$, $\bar{\mathbb{Z}}$ the algebraic integers, $\zeta_n = e^{2\pi i/n}$; $D$ is a quadratic discriminant and $D_{\mathfrak{P}}$ a decomposition group; $\chi_D$ is indexed by the fundamental discriminant ($\chi_{-4}$, $\chi_{-3}$, $\chi_8$, $\chi_{q^*}$); $\chi_\ell$ is cyclotomic; the geometric Frobenius is written $\mathrm{Frob}_p^{-1}$. Convention 6.1.

**6.2 Number fields** (definition): $[K : \mathbb{Q}] = n < \infty$, $K = \mathbb{Q}(\theta)$, $n$ embeddings into $\mathbb{C}$, $n = r_1 + 2r_2$. Definition 6.2.

**6.3 Algebraic integers.** $\alpha$ is integral iff its minimal polynomial is in $\mathbb{Z}[X]$ iff $\mathbb{Z}[\alpha]$ is a finitely generated $\mathbb{Z}$-module iff $\alpha$ lies in a subring finitely generated as a $\mathbb{Z}$-module; $\mathcal{O}_K$ is a ring, $\mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$, $\operatorname{Frac} \mathcal{O}_K = K$, and $\mathcal{O}_K$ is integrally closed. *(proved here)* Proposition 6.3.

**6.4 Trace and norm** (definition): trace and determinant of $x \mapsto \alpha x$. Definition 6.4.

**6.5 Properties of trace and norm.** $\mathrm{Tr}(\alpha) = \sum \sigma_i(\alpha)$, $N(\alpha) = \prod \sigma_i(\alpha)$; integers on $\mathcal{O}_K$; $\alpha \in \mathcal{O}_K^\times$ iff $N(\alpha) = \pm 1$; transitivity in towers. *(proved here)* Proposition 6.5.

**6.6 Integral bases and the discriminant.** The trace form is nondegenerate; $\mathcal{O}_K \simeq \mathbb{Z}^n$; $d_K = \det(\mathrm{Tr}(\beta_i\beta_j)) = \det(\sigma_k\beta_j)^2$ is independent of the basis; $\mathrm{disc}(\alpha_1,\dots,\alpha_n) = [\mathcal{O}_K : \sum \mathbb{Z}\alpha_i]^2 d_K$; $\mathrm{disc}(1,\theta,\dots,\theta^{n-1}) = \mathrm{disc}(g) = (-1)^{n(n-1)/2}N(g'(\theta))$. *(proved here)* Theorem 6.6.

**6.7 Quadratic fields.** For $d$ squarefree, $\mathcal{O} = \mathbb{Z}[\sqrt{d}]$, $d_K = 4d$ if $d \equiv 2,3 \pmod 4$, and $\mathcal{O} = \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$, $d_K = d$ if $d \equiv 1 \pmod 4$; quadratic fields correspond to fundamental discriminants; $\mathbb{Z}[i]$ ($D=-4$), $\mathbb{Z}[\zeta_3]$ ($D=-3$). *(proved here)* Proposition 6.7.

**6.8 Units.** (i) $\mathcal{O}_K^\times \simeq \mu(K)\times\mathbb{Z}^{r_1+r_2-1}$ *(stated; reference)*; (ii) for $K$ imaginary quadratic $\mathcal{O}_K^\times = \mu(K)$, which is $\{\pm1\}$ except $\mu_4$ for $\mathbb{Q}(i)$ and $\mu_6$ for $\mathbb{Q}(\sqrt{-3})$; (iii) $\mathbb{Z}[\sqrt{2}]^\times = \pm(1+\sqrt{2})^{\mathbb{Z}}$ *((ii), (iii) proved here).* Theorem 6.9.

**6.9 Dedekind domains** (definition): Noetherian, integrally closed, not a field, nonzero primes maximal. Definition 6.10.

**6.10 $\mathcal{O}_K$ is a Dedekind domain**; $\mathfrak{p}\cap\mathbb{Z} = p\mathbb{Z}$ and $\mathcal{O}_K/\mathfrak{p}$ is a finite field; PIDs (not fields) are Dedekind; a Noetherian domain (not a field) is Dedekind iff all its localisations at nonzero primes are discrete valuation rings. *(proved here)* Theorem 6.11.

**6.11 Inverse of a prime.** Every nonzero ideal contains a product of primes; $\mathfrak{p}^{-1} = \{x : x\mathfrak{p} \subseteq R\}$ satisfies $\mathfrak{a}\mathfrak{p}^{-1} \neq \mathfrak{a}$ and $\mathfrak{p}\mathfrak{p}^{-1} = R$. *(proved here)* Lemma 6.12.

**6.12 Unique factorisation of ideals.** In a Dedekind domain every nonzero proper ideal is uniquely a product of primes; fractional ideals form a group, free on the primes; to divide is to contain; $v_\mathfrak{p}$ is a discrete valuation. *(proved here)* Theorem 6.13.

**6.13 Norm of an ideal.** $N\mathfrak{a} = \#\mathcal{O}_K/\mathfrak{a}$ is multiplicative, $N(\alpha) = |N_{K/\mathbb{Q}}\alpha|$, $N\mathfrak{p} = p^f$ with $\mathcal{O}_K/\mathfrak{p} \simeq \mathbb{F}_{p^f}$, $\mathfrak{p}^k/\mathfrak{p}^{k+1} \simeq \mathcal{O}_K/\mathfrak{p}$; ideals of prime norm are prime; disc of a basis of $\mathfrak{a}$ is $(N\mathfrak{a})^2 d_K$. *(proved here)* Proposition 6.14.

**6.14 Chinese remainder theorem, two generators, localisation.** $R/\mathfrak{a} \simeq \prod R/\mathfrak{p}_i^{a_i}$; every ideal is $(\alpha,\beta)$ with $\alpha$ arbitrary; finitely many primes implies PID; UFD iff PID; $R_\mathfrak{p}$ is a discrete valuation ring with residue field $R/\mathfrak{p}$ and valuation $v_\mathfrak{p}$. *(proved here)* Proposition 6.15.

**6.15 Class group.** $\mathrm{Cl}(K) = J_K/P_K$; $h_K = 1$ iff PID iff UFD *(proved here)*; Minkowski: $\mathrm{Cl}(K)$ is finite, each class contains an ideal of norm $\le (4/\pi)^{r_2}(n!/n^n)|d_K|^{1/2}$, and $|d_K| > 1$ for $K \neq \mathbb{Q}$ *(stated; reference)*. Examples: $h = 1$ for $D = -3,-4,-7,5,8,-11$, $h = 2$ for $D = -20$. Theorem 6.16.

**6.16 $\sum e_i f_i = n$.** $\mathfrak{p}\mathcal{O}_L = \prod \mathfrak{P}_i^{e_i}$ with residue degrees $f_i$ satisfies $\sum e_i f_i = [L:K]$; ramified, inert, split; $e$ and $f$ are multiplicative in towers. *(proved here)* Theorem 6.19.

**6.17 Dedekind-Kummer.** If $p \nmid [\mathcal{O}_K : \mathbb{Z}[\theta]]$ and $g \equiv \prod \bar{g}_i^{e_i} \pmod p$, then $p\mathcal{O}_K = \prod (p, g_i(\theta))^{e_i}$ with $f_i = \deg \bar{g}_i$ *(proved here)*; relative version *(stated; reference)*. Theorem 6.20.

**6.18 Dedekind's discriminant theorem.** $p$ ramifies in $K$ iff $p \mid d_K$ *(proved here)*; relative version *(stated; reference)*. Theorem 6.22.

**6.19 Decomposition and inertia groups.** For $L/K$ Galois, $G$ is transitive on the primes above $\mathfrak{p}$; $efg = n$; $D_\mathfrak{P}$ is the stabiliser, $\#D_\mathfrak{P} = ef$, $I_\mathfrak{P} \trianglelefteq D_\mathfrak{P}$, both conjugate along the orbit. *(proved here)* Proposition 6.23.

**6.20 Residue field extension.** $D_{\mathfrak{P}} \to \mathrm{Gal}(\kappa(\mathfrak{P})/\kappa(\mathfrak{p}))$ is surjective with kernel $I_{\mathfrak{P}}$, so $\#I_{\mathfrak{P}} = e$ and $D/I$ is cyclic of order $f$; behaviour in towers; split completely iff $D = 1$, unramified iff $I = 1$. *(proved here)* Theorem 6.24.

**6.21 Arithmetic and geometric Frobenius** (convention): $\mathrm{Frob}_p$ is arithmetic, $\chi_\ell(\mathrm{Frob}_p) = p$; $L_p(V, s) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid V^{I_p})^{-1}$; the two conventions agree when the eigenvalues are inversion-stable, and differ at a multiplicative prime, where only the geometric one gives $1 - q^{-1-s}$. Convention 6.25, Example 6.26.

**6.22 The Frobenius element.** For $\mathfrak{p}$ unramified there is a unique $\mathrm{Frob}_{\mathfrak{P}}$ with $\mathrm{Frob}_{\mathfrak{P}}(x) \equiv x^{N\mathfrak{p}} \pmod{\mathfrak{P}}$; it generates $D_{\mathfrak{P}}$; its conjugacy class is $\mathrm{Frob}_{\mathfrak{p}}$; it is 1 iff $\mathfrak{p}$ splits completely; functoriality; at ramified primes only a coset of inertia. *(proved here)* Proposition 6.27.

**6.23 Frobenius and subfields.** For $M = L^H$, the primes of $M$ above $p$ correspond to the $\langle \mathrm{Frob}_{\mathfrak{P}} \rangle$-orbits on $G/H$, with residue degrees the orbit lengths. *(proved here)* Proposition 6.28.

**6.24 Dedekind's theorem on cycle types.** If $p \nmid a\,\mathrm{disc}(f)$, $p$ is unramified in the splitting field of $f$ and $\mathrm{Frob}_{\mathfrak{P}}$ permutes the roots with cycle type the degrees of the irreducible factors of $f$ mod $p$. *(proved here)* Corollary 6.29.

**6.25 Frobenius elements in $G_{\mathbb{Q}}$.** Primes of $\bar{\mathbb{Z}}$ above $p$; $D_p$, $I_p$ closed, conjugate, with $1 \to I_p \to D_p \to \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \to 1$ exact and surjective projections to finite levels; $\mathrm{Frob}_p$ defined up to $I_p$ and conjugation; unramified representations and $\rho(\mathrm{Frob}_p)$ *(proved here)*; $D_p \simeq G_{\mathbb{Q}_p}$ *(stated; Chapter 7)*. Proposition 6.30.

**6.26 Tame inertia.** If $\#I_{\mathfrak{P}}$ is prime to $p$, $I_{\mathfrak{P}}$ is cyclic and $F\tau F^{-1} = \tau^{N\mathfrak{p}}$ for every Frobenius $F$ and $\tau \in I_{\mathfrak{P}}$. *(proof sketched here; proved in Chapter 7)* Proposition 6.31.

**6.27 Frobenius on a tame image of inertia.** For $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$, $\ell$ odd, $\ell \neq q$, with $\rho(I_q)$ finite of order prime to $q$: $\rho(I_q)$ is cyclic and $\rho(F)\rho(\tau)\rho(F)^{-1} = \rho(\tau)^q$; so Frobenius centralises or inverts an element of order 3 according as $q \equiv 1$ or $2 \pmod 3$, and one of order 4 according as $q \equiv 1$ or $3 \pmod 4$. *(proved here, given Proposition 6.31)* Corollary 6.32.

**6.28 Splitting in quadratic fields.** $p$ ramifies, splits or is inert in $\mathbb{Q}(\sqrt{d})$ according as $(\frac{D}{p}) = 0, 1, -1$, with explicit primes, including $p = 2$. *(proved here)* Theorem 6.34.

**6.29 The character of a quadratic field.** $\chi_D(\mathrm{Frob}_p) = (\frac{D}{p})$ for $p \nmid D$; $\chi_D$ is ramified exactly at $p \mid D$; $\chi_{-4}(\mathrm{Frob}_p) = (-1)^{(p-1)/2}$, $\chi_{-3}(\mathrm{Frob}_p) = 1$ iff $p \equiv 1 \pmod 3$, $\chi_{q^*}(\mathrm{Frob}_p) = (\frac{p}{q})$ for $p \neq q$ including $p = 2$. *(proved here)* Proposition 6.35.

**6.30 Ramified only at $q$.** $\mathbb{Q}(\sqrt{q^*})$ is the unique quadratic field ramified only at the odd prime $q$; fundamental discriminants are products of prime discriminants; $\chi_S = \prod_{q \in S} \chi_{q^*}$ is the character of $\mathbb{Q}(\sqrt{\prod q^*})$, of conductor $\prod q$, ramified exactly at $S$. *(proved here)* Proposition 6.36.

**6.31 Gaussian and Eisenstein primes.** $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$ are PIDs; their primes are $1 + i$, $a \pm bi$ ($p = a^2 + b^2 \equiv 1 \bmod 4$), $p \equiv 3 \pmod 4$; resp. $1 - \zeta_3$, $a + b\zeta_3^{\pm 1}$ ($p = a^2 - ab + b^2 \equiv 1 \bmod 3$), $p \equiv 2 \pmod 3$. *(proved here)* Proposition 6.38.

**6.32 The prime above 3.** $\sqrt{-3} = 1 + 2\zeta_3 = \zeta_3\pi$, $\pi^2 = -3\zeta_3$, $3 = -\zeta_3^2\pi^2$; $\mathbb{Z}[\zeta_3]/(\pi) = \mathbb{F}_3$; on $\mathbb{Z}[\zeta_3]/(\pi^2)$ multiplication by $\zeta_3 = 1 - \pi$ is a nontrivial unipotent with fixed line $(\pi)/(\pi^2)$; at 2 in $\mathbb{Z}[i]$, $(2) = (1+i)^2$ and $i$ acts unipotently on $\mathbb{Z}[i]/(2)$. *(proved here)* Proposition 6.39.

**6.33 Quadratic algebras over $\mathbb{Z}_\ell$.** $\mathbb{Z}_\ell[X]/(g)$ is an unramified discrete valuation ring if $\bar{g}$ is irreducible, a ramified one with uniformiser $x - c$ if $g(X + c)$ is Eisenstein, and $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$ if $\bar{g}$ has distinct roots; lattices $T$ with $T \otimes \mathbb{Q}_\ell \simeq R \otimes \mathbb{Q}_\ell$ are free of rank one. *(proved here)* Lemma 6.40.

**6.34 $\mathbb{Z}_3[\zeta_3]$.** $\mathbb{Z}_3[X]/(X^2 + X + 1)$ is a discrete valuation ring with uniformiser $\pi = 1 - \zeta_3$ and residue field $\mathbb{F}_3$, $3 = \pi^2 \cdot$unit, and it is $\mathbb{Z}_3 \otimes \mathbb{Z}[\zeta_3]$ *(proved here)*; it is the ring of integers of $\mathbb{Q}_3(\sqrt{-3})$ *(stated; Chapter 7)*. Theorem 6.41.

**6.35 The lattice of Lemma lem:IV.** If $\sigma \in \mathrm{Aut}_{\mathbb{Z}_3}(\mathbb{Z}_3^2)$ satisfies $\sigma^2 + \sigma + 1 = 0$, then $T = \mathbb{Z}_3^2$ is free of rank one over $\mathbb{Z}_3[\sigma] \simeq \mathbb{Z}_3[\zeta_3]$, $T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$ with $\sigma$ a nontrivial unipotent fixing the line $\pi T/3T$, and $T^\sigma = 0$. *(proved here)* Corollary 6.42.

**6.36 $\mathbb{Z}_\ell[\zeta_3]$ and $\mathbb{Z}_\ell[i]$.** $\mathbb{Z}_\ell[\zeta_3] := \mathbb{Z}_\ell[X]/(X^2 + X + 1)$ is ramified for $\ell = 3$, $\simeq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$ (not a domain) for $\ell \equiv 1 \pmod 3$, an unramified discrete valuation ring for $\ell \equiv 2 \pmod 3$; likewise $\mathbb{Z}_\ell[i]$ according to $\ell \bmod 4$; $\zeta_3 \mapsto \zeta_3^2$ negates $\sqrt{-3}$, a unit iff $\ell \neq 3$. *(proved here)* Proposition 6.43.

**6.37 Lattices with an automorphism of order 3, 4, 6.** For $\ell$ odd and $g \in \mathrm{Aut}(\mathbb{Z}_\ell^2)$ with characteristic polynomial $\Phi_m$, $m \in \{3, 4, 6\}$ (e.g. $g \in \mathrm{SL}_2(\mathbb{Z}_\ell)$ of order $m$): $\mathbb{Z}_\ell[g] \simeq \mathbb{Z}_\ell[X]/(\Phi_m)$, the lattice is free of rank one over it, and the commutant of $g$ is $\mathbb{Z}_\ell[g]$. *(proved here)* Proposition 6.44.

**6.38 The commutant and the invariant line.** For $\sigma$ with characteristic polynomial $X^2+X+1$ on a plane over a field $F$ of characteristic 0: the commutant is $F[\sigma] \simeq F \otimes \mathbb{Q}(\zeta_3)$, its trace-zero part is $F(2\sigma + 1)$ with $(2\sigma + 1)^2 = -3$, and $A\sigma A^{-1} = \sigma^q$, $q \neq 3$ prime, makes $A^{\pm 1}$ act on it by $\chi_{-3}(q)$; for $\tau$ with characteristic polynomial $X^2 + 1$ (for example of order 4 in $\mathrm{SL}_2$) and $q$ an odd prime, the line $F\tau$ and $\chi_{-4}(q)$; integrally, for $\ell$ odd, $\mathrm{End}_{\langle\sigma\rangle}(T) = \mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$. *(proved here)* Proposition 6.45.

**6.39 Cyclotomic integers.** $\mathcal{O} = \mathbb{Z}[\zeta_n]$; for $n = \ell^a$, $\ell = u(1-\zeta)^{\varphi(\ell^a)}$ is totally ramified; the discriminant formula; $p$ ramifies in $\mathbb{Q}(\zeta_n)$ iff $p \mid n$ ($n \not\equiv 2 \bmod 4$). *((i), (ii) proved here for prime powers, (iii) for odd primes, (iv) in general; the rest stated with reference)* Theorem 6.48.

**6.40 Splitting in cyclotomic fields.** For $n = p^a m$: $e = \varphi(p^a)$, $f =$ order of $p$ modulo $m$, $g = \varphi(m)/f$; inertia $(\mathbb{Z}/p^a)^\times \times 1$, decomposition $(\mathbb{Z}/p^a)^\times \times \langle p\rangle$; $\mathrm{Frob}_p = \sigma_p$ for $p \nmid n$. *(proved here)* Theorem 6.49.

**6.41 $\mathbb{Q}(\sqrt{\ell^*})$.** The quadratic subfield of $\mathbb{Q}(\zeta_\ell)$, ramified only at $\ell$; $\chi_{\ell^*}(\mathrm{Frob}_p) = \left(\frac{p}{\ell}\right)$ for all $p \neq \ell$; imaginary iff $\ell \equiv 3 \pmod 4$. *(proved here)* Proposition 6.51.

**6.42 Quadratic reciprocity**, with both supplements, from Proposition 6.35 and Proposition 6.51. *(proved here)* Corollary 6.52.

**6.43 Kronecker-Weber.** Every abelian $K/\mathbb{Q}$ lies in some $\mathbb{Q}(\zeta_n)$; conductor; $G_{\mathbb{Q}}^{\mathrm{ab}} \simeq \hat{\mathbb{Z}}^\times$. *(stated; reference)* Theorem 6.53.

**6.44 Artin map over $\mathbb{Q}$.** For $K \subseteq \mathbb{Q}(\zeta_n)$, $\mathrm{Frob}_p(K/\mathbb{Q}) = \sigma_p|_K$ depends only on $p \bmod n$ *(proved here)*; idelic form and its sign conventions *(stated; reference)*. Theorem 6.54.

**6.45 Dirichlet characters and characters of $G_\mathbb{Q}$.** $\chi \mapsto \psi_\chi$, $\psi_\chi(\mathrm{Frob}_p) = \chi(p)$, is a bijection from primitive Dirichlet characters to continuous characters of $G_\mathbb{Q}$; ramification and conductor match; $\chi_\ell$ and $\bar{\chi}_\ell = \psi_{\omega_\ell}$ *(proved here, given Kronecker-Weber and Chebotarev)*; twisting newforms *(stated; Chapter 13)*. Theorem 6.55.

**6.46 Dedekind zeta and densities.** $\zeta_K$ converges for $\mathrm{Re}\, s > 1$ *(proved here)*; simple pole at 1 *(stated; reference)*; Dirichlet density; primes of degree one have density one *(proved here, given the pole)*. Proposition 6.59.

**6.47 Chebotarev.** For $L/K$ Galois and a class $C$, the primes with $\mathrm{Frob}_\mathfrak{p} = C$ have density $\#C/\#G$. *(stated; reference)* Theorem 6.60.

**6.48 Consequences of Chebotarev.** Every element is a Frobenius; split primes have density $1/[L : K]$; Bauer's theorem; Dirichlet's theorem; inert primes of $\mathbb{Q}(\sqrt{\ell^*})$ have density $\frac{1}{2}$. *(proved here, given Theorem 6.60)* Corollary 6.61.

**6.49 Detecting a quadratic twist.** For $\rho$ semisimple, unramified outside $S$, in characteristic $\neq 2$: $\rho \simeq \rho \otimes \chi_K$ iff $\mathrm{tr}\,\rho(\mathrm{Frob}_p) = 0$ for the primes $p \notin S$ inert in $K$ (outside density zero). *(proved here, given Chapter 5)* Proposition 6.62.

**6.50 Complex multiplication.** For $E/\mathbb{Q}$ with CM by $K$: class number one, 13 values of $j$ *(stated; reference)*; $\rho \simeq \rho \otimes \chi_K$, $a_p = 0$ at good inert primes, and primes ramified in $K$ are bad *(proof sketched here)*; conversely $a_p = 0$ at almost all inert primes forces CM by $K$, and without CM $\{p : a_p = 0\}$ has density 0 *(stated; reference)*; so CM by $\mathbb{Q}(\sqrt{\ell^*})$ forces $\ell \mid N$. Theorem 6.63.

**6.51 Coefficient fields.** The image of a ring finite over $\mathbb{Z}$ in $\mathbb{C}$ is an order in a number field; $K_f$ is a number field and $a_n(f) \in \mathcal{O}_{K_f}$ *(proved here)*; conjugates are newforms, $K_f$ totally real *(stated; Chapter 13)*. Proposition 6.66.

**6.52 Primes above $\ell$.** Reduction modulo $\lambda$; congruences of integers modulo $\lambda$ and $\ell$; enlarging $\mathcal{O}$ and $\lambda$; lying over for $\mathcal{O}[\alpha]$; the root of $X^2 - aX + p$ congruent to a given root modulo a prime above $\lambda$. *(proved here)* Proposition 6.67.

**6.53 Maximal ideals from congruences.** $\ker(R \to \mathcal{O} \to \mathcal{O}/\lambda)$ is maximal of residue characteristic $\ell$, the same for congruent homomorphisms, and an ideal $J$ inside it with $R/J \simeq \mathbb{Z}/r$ has $\ell \mid r$ *(proved here)*; $R \otimes \mathbb{Z}_\ell = \prod R_\mathfrak{m}$ *(stated; Chapter 3)*. Proposition 6.68.

**6.54** $\mathcal{O}_K \otimes \mathbb{Z}_\ell = \prod_{\lambda|\ell} \mathcal{O}_{K,\lambda}$, each a complete discrete valuation ring of rank $e_\lambda f_\lambda$ over $\mathbb{Z}_\ell$. *(stated; Chapter 7)* Proposition 6.69.

## Where the paper uses this chapter

- **Lemma lem:IV** (types IV, IV$^*$ at $q \neq 3$): $\mathbb{Z}_3[\zeta_3]$ a discrete valuation ring with uniformiser $\pi$, Theorem 6.41; $T_3E$ free of rank one over it and $E[3] \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$ with a unipotent action, Corollary 6.42, Proposition 6.39; Frobenius conjugating $\sigma$ to $\sigma^q = \sigma^{\chi_{-3}(q)}$, Corollary 6.32, Proposition 6.35; the line of $\mathrm{ad}^0 V$ and its sign, Proposition 6.45.

- **Lemma lem:euler** (the omitted factors at tame primes) **and the table of Section 1**: the tame relation, Proposition 6.31, Corollary 6.32; $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$, not a field for $\ell \equiv 1 \pmod 3$, Proposition 6.43, Proposition 6.44, Proposition 6.45; $\chi_{-3}(q)$, $\chi_{-4}(q)$, Proposition 6.35, Example 6.56; the geometric Frobenius in $L_q(A_f, s)$ and the row $\mathrm{I}_n$, Convention 6.25, Example 6.26.

- **Section 1.1** (the local terms): $I_p$, $\mathrm{Frob}_p$, $V^{I_p}$, Proposition 6.30; $L_p(A_f, s) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid A_f^{I_p})^{-1}$, Convention 6.25. **Lemma lem:fpr**: $L_p(A_f, 1) = L_p(A_f(1), 0)$, Convention 6.25. **Lemma lem:tamfactor**: the arithmetic Frobenius, with $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ on tame inertia, Convention 6.25, Proposition 6.31.

- **Section 1.4, Lemma lem:twist, Theorems thm:twist and thm:quant**: $\chi_{q^*}$, the character of conductor $q$ of $\mathbb{Q}(\sqrt{q^*})$, ramified only at $q$, Proposition 6.36, Proposition 6.35, Corollary 6.52; $\chi_S$ and its conductor $\prod_{q \in S} q$ (the paper's $D$), Proposition 6.36(iii); characters as Dirichlet and as Galois characters, Theorem 6.55, Example 6.56.

- **Theorem thm:main** (the hypothesis on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$): $\mathbb{Q}(\sqrt{\ell^*}) \subseteq \mathbb{Q}(\zeta_\ell)$ and its splitting law, Proposition 6.51; the condition $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$ (reached in the paper by Clifford theory, Chapter 4; its trace form Proposition 6.62 is not used by the paper), Warning 6.57.

- **Lemma lem:h0 and Proposition prop:ns**: $\bar\chi_3^{\mathrm{cyc}} = \chi_{-3}$ and $\chi_{\ell^*} \neq \bar\chi_\ell$ for $\ell \geq 5$, Example 6.56, Warning 6.57; the isomorphism $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ comes from induction from $\mathbb{Q}(\sqrt{-3})$, $\bar\rho \otimes \overline{\mathbb{F}}_3 \simeq \mathrm{Ind}\, \psi$ with $\mathrm{Ind}\, \psi \otimes \chi_{-3} \simeq \mathrm{Ind}\, \psi$ (Chapter 4), not from Frobenius traces; Proposition 6.62 gives the equivalent trace criterion ($3 \mid a_p$ for almost all $p \equiv 2 \pmod 3$), which the book uses in an example of Chapter 5 and the paper does not use. Lemma lem:h0 holds "for every $E$, with or without complex multiplication", and its proof uses neither complex multiplication nor $a_p = 0$ at inert primes (Theorem 6.63, Corollary 6.61, Paper link 6.65).

- **Proposition prop:Sf**: Clifford theory for $\bar\rho|_{G_K}$, $K = \mathbb{Q}(\sqrt{\ell^*})$ (Chapter 4), and the restriction to $I_\ell$ of $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$, a power of $\bar\chi_\ell|_{I_\ell}$ (the paper's fundamental character $\omega$ of level one), Example 6.56, Warning 6.57.

- **Proposition prop:ns-structure**(ii) and (iii): the trace-zero commutant of an element $\tau$ of order 4 of $\rho(I_p)$ is the line $F\tau$, on which Frobenius acts by $\pm 1$, Proposition 6.45(iv); in (iii) the tame relation $\tau \mapsto \tau^p$ (Corollary 6.32) makes the sign $\chi_{-4}(p)$, and $\chi_{-4}(p) = \chi_{-3}(p)$ iff $p \equiv \pm 1 \pmod{12}$, Proposition 6.35(iii).

- **Lemma lem:tamadd** (potentially good primes, $\ell = 3$): $T$ free of rank one over $\mathbb{Z}_3[X]/(X^2 + X + 1)$, with $X$ acting as an element $\sigma$ of order 3 of $\rho(I_q)$, at every $q \neq 3$, $q = 2$ included, Corollary 6.42(i), and the basis in which $\sigma$ is the companion matrix, Paper link 6.46.

- **Proposition prop:forms**: $q \equiv \chi_{-3}(q) \pmod 3$ for $q \neq 3$, Proposition 6.35(iii).

- **Lemma lem:crit** (congruence criterion): coefficients in the ring of integers $\mathcal{O}$ of a number field, Proposition 6.66; $\lambda \mid \ell$ and the maximal ideal of residue characteristic $\ell$, $\ell \mid r$, Proposition 6.67, Proposition 6.68; over $\mathbb{Z}_\ell$, Proposition 6.69.

- **Lemma lem:old** (oldforms): Frobenius at unramified primes and Chebotarev, Proposition 6.27, Proposition 6.30, Theorem 6.60; "after enlarging $\mathcal{O}$ and $\lambda$" and "a prime above $\lambda$ in $\mathcal{O}[\alpha_p]$", Proposition 6.67(iii) to (v).

- **Section 3 and Example ex:qual**: the twisting fields $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{-11})$, Example 6.37; the congruences $a_p(g) \equiv a_p(f) \pmod{\lambda}$ at all $p \nmid 3N$ are traces of Frobenius of $\bar{\rho}_g \simeq \bar{\rho}$, Proposition 6.30(iv) (Chebotarev is needed only for the converse, as in Lemma lem:old).

## 6.1 Number fields and their rings of integers

This section sets up the objects behind every "$\mathcal{O}$" and every "prime $\lambda \mid \ell$" of the paper: number fields, their rings of integers, traces and norms, integral bases and the discriminant, and the units. The paper's Lemmas lem:crit and lem:old take the coefficients of a newform in the ring of integers $\mathcal{O}$ of a number field; Lemma lem:IV and Lemma lem:euler work inside $\mathbb{Z}[\zeta_3]$ and $\mathbb{Z}[i]$, the rings of integers of $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(i)$, whose discriminants $-3$ and $-4$ name the characters $\chi_{-3}$ and $\chi_{-4}$ of the table of Section 1. Nothing here is specific to elliptic curves.

**Convention 6.1.** (a) We fix an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ inside $\mathbb{C}$ (as in Chapter 5); every number field is a subfield of $\overline{\mathbb{Q}}$, and $\bar{\mathbb{Z}}$ denotes the ring of all algebraic integers of $\overline{\mathbb{Q}}$. We write $\zeta_n = e^{2\pi i/n}$, and $\zeta_3 = (-1+\sqrt{-3})/2$. (b) The letters $p$, $q$, $\ell$ denote rational primes, as in the paper ($\ell$ the coefficient prime, $q$ a bad prime of $E$); the size of a residue field is written $N\mathfrak{p}$, never $q$. Uniformisers of discrete valuation rings are written $\varpi$; the element $\pi = 1-\zeta_3$ of $\mathbb{Z}[\zeta_3]$ (written $1-\zeta_3$ in the paper) is the one exception, and in Section 6.4 the letter $\pi$ means nothing else. (c) $D$ denotes the discriminant of a quadratic field, and $D_{\mathfrak{P}}$, $D_p$ decomposition groups; no other meaning of $D$ occurs. The paper's $D$ of Theorem thm:twist, the conductor of $\chi_S$, is written out here as $\prod_{q\in S} q$. (d) The quadratic character of a quadratic field $K$ of discriminant $D$ is written $\chi_D$ (convention B-C6 of Appendix B): $\chi_{-4}$ for $\mathbb{Q}(i)$, $\chi_{-3}$ for $\mathbb{Q}(\sqrt{-3})$, $\chi_8$ for $\mathbb{Q}(\sqrt{2})$, $\chi_{q^*}$ for $\mathbb{Q}(\sqrt{q^*})$; the index is always a fundamental discriminant, and $\chi_\ell$ with a letter index is the $\ell$-adic cyclotomic character. (e) Frob is the arithmetic Frobenius and $\mathrm{Frob}^{-1}$ the geometric one (Convention 6.25). Since the cyclotomic polynomials are written $\Phi_n$ in this chapter, the geometric Frobenius is always written $\mathrm{Frob}_p^{-1}$ here (Appendix B also writes it $\Phi_p$).

**Definition 6.2.** A *number field* is a field $K \supseteq \mathbb{Q}$ with $n = [K:\mathbb{Q}] < \infty$. By the primitive element theorem (Chapter 5) $K = \mathbb{Q}(\theta)$ for some $\theta$, and there are exactly $n$ field embeddings $\sigma_1, \dots, \sigma_n \colon K \to \mathbb{C}$, sending $\theta$ to the $n$ roots of its minimal polynomial. If $r_1$ of them have image in $\mathbb{R}$, the other $2r_2$ come in complex conjugate pairs and $n = r_1 + 2r_2$. The field is *totally real* if $r_2 = 0$ and *imaginary quadratic* if $n = 2$ and $r_1 = 0$.

**Proposition 6.3** (algebraic integers)**.** *Let $K$ be a number field and $\alpha \in K$. The following are equivalent, and $\alpha$ is then called an* algebraic integer*: (o) $\alpha$ is a root of a monic polynomial in $\mathbb{Z}[X]$; (i) the minimal polynomial of $\alpha$ over $\mathbb{Q}$ lies in $\mathbb{Z}[X]$; (ii) $\mathbb{Z}[\alpha]$ is a finitely generated $\mathbb{Z}$-module; (iii) $\alpha$ lies in a subring of $K$ that is finitely generated as a $\mathbb{Z}$-module. The algebraic integers of $K$ form a subring $\mathcal{O}_K$, the* ring of integers*; $\mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$; for every $\alpha \in K$ there is an integer $m \geq 1$ with $m\alpha \in \mathcal{O}_K$, so $\mathrm{Frac}(\mathcal{O}_K) = K$; and $\mathcal{O}_K$ is* integrally closed *in $K$: an element of $K$ that is a root of a monic polynomial with coefficients in $\mathcal{O}_K$ lies in $\mathcal{O}_K$.*

*Proof.* (o)⇒(i): if $\alpha$ is a root of a monic $g \in \mathbb{Z}[X]$, its minimal polynomial $m_\alpha$ divides $g$ in $\mathbb{Q}[X]$, and a monic factor in $\mathbb{Q}[X]$ of a monic polynomial of $\mathbb{Z}[X]$ lies in $\mathbb{Z}[X]$ by Gauss's lemma (Chapter 2). (i)⇒(ii): if $m_\alpha \in \mathbb{Z}[X]$ is monic of degree $d$, division by $m_\alpha$ shows $\mathbb{Z}[\alpha] = \mathbb{Z} + \mathbb{Z}\alpha + \cdots + \mathbb{Z}\alpha^{d-1}$. (ii)⇒(iii) is trivial. (iii)⇒(o): let $M \ni \alpha$ be a subring generated as a $\mathbb{Z}$-module by $m_1, \dots, m_r$. Write $\alpha m_i = \sum_j a_{ij} m_j$ with $a_{ij} \in \mathbb{Z}$, so $(\alpha I - A)\underline{m} = 0$ for $A = (a_{ij})$

and the column $\underline{m}$. Multiplying by the adjugate of $\alpha I - A$ gives $\det(\alpha I - A)m_i = 0$ for all $i$; as $1 \in M$ is a $\mathbb{Z}$-combination of the $m_i$, $\det(\alpha I - A) = 0$, and $\det(XI - A)$ is monic in $\mathbb{Z}[X]$.

If $\alpha, \beta$ are algebraic integers, roots of monic integer polynomials of degrees $a$ and $b$, the ring $\mathbb{Z}[\alpha, \beta]$ is generated as a $\mathbb{Z}$-module by the $\alpha^i\beta^j$, $i < a$, $j < b$ (every element is a $\mathbb{Z}$-combination of monomials $\alpha^r\beta^s$, and the monic equation of degree $a$ rewrites $\alpha^a$, hence by induction every $\alpha^r$ with $r \geq a$, as a $\mathbb{Z}$-combination of $1, \alpha, \dots, \alpha^{a-1}$; likewise for $\beta$); it contains $\alpha \pm \beta$ and $\alpha\beta$, which are therefore integral by (iii). A rational root $x = u/v$ (in lowest terms) of a monic $X^d + c_{d-1}X^{d-1} + \cdots + c_0 \in \mathbb{Z}[X]$ satisfies $u^d = -v(c_{d-1}u^{d-1} + \cdots + c_0v^{d-1})$, so $v \mid u^d$ and $v = \pm 1$: hence $\mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$. If $\alpha$ is a root of $a_dX^d + \cdots + a_0 \in \mathbb{Z}[X]$ with $a_d \neq 0$, then $a_d\alpha$ is a root of the monic polynomial $X^d + a_{d-1}X^{d-1} + a_{d-2}a_dX^{d-2} + \cdots + a_0a_d^{d-1}$. Finally, if $x \in K$ satisfies $x^d + c_{d-1}x^{d-1} + \cdots + c_0 = 0$ with $c_i \in \mathcal{O}_K$, then $\mathbb{Z}[c_0, \dots, c_{d-1}, x]$ is a finitely generated $\mathbb{Z}$-module (each $c_i$ is integral, and powers of $x$ beyond $x^{d-1}$ are reduced by the equation), so $x \in \mathcal{O}_K$ by (iii). □

**Definition 6.4.** For $\alpha \in K$ let $m_\alpha \colon K \to K$, $x \mapsto \alpha x$, a $\mathbb{Q}$-linear map. The *trace* and *norm* of $\alpha$ are $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = \mathrm{tr}(m_\alpha)$ and $N_{K/\mathbb{Q}}(\alpha) = \det(m_\alpha)$. For a tower $L \supseteq K$ of number fields, $\mathrm{Tr}_{L/K}$ and $N_{L/K}$ are defined in the same way with $L$ viewed as a $K$-vector space.

The trace and the norm turn an element of $K$ into rational numbers by linear algebra, without choosing an embedding into $\mathbb{C}$; this is what makes them useful for integrality questions. For example, in $K = \mathbb{Q}(i)$ with basis $1, i$, multiplication by $\alpha = a+bi$ sends $1 \mapsto a+bi$ and $i \mapsto -b+ai$, so its matrix is $\left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$, with trace $2a$ and determinant $a^2 + b^2$: $\mathrm{Tr}(a + bi) = (a + bi) + (a - bi)$ and $N(a + bi) = (a + bi)(a - bi)$, the sum and the product of the two conjugates, which is the general pattern of Proposition 6.5(i).

**Proposition 6.5** (trace and norm)**.** *Let $K$ be a number field of degree $n$ with embeddings $\sigma_1, \dots, \sigma_n$, and $\alpha \in K$.*

*(i) The characteristic polynomial of $m_\alpha$ is $\prod_i(X - \sigma_i(\alpha)) = m_\alpha(X)^{n/d}$, $d = [\mathbb{Q}(\alpha) : \mathbb{Q}]$, $m_\alpha$ the minimal polynomial. Hence $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = \sum_i \sigma_i(\alpha)$ and $N_{K/\mathbb{Q}}(\alpha) = \prod_i \sigma_i(\alpha)$; Tr is $\mathbb{Q}$-linear, $N$ is multiplicative, and $N(\alpha) = 0$ iff $\alpha = 0$. If $K = \mathbb{Q}(\alpha)$ with minimal polynomial $X^n + c_{n-1}X^{n-1} + \cdots + c_0$, then $\mathrm{Tr}(\alpha) = -c_{n-1}$ and $N(\alpha) = (-1)^n c_0$.*

*(ii) If $\alpha \in \mathcal{O}_K$ then $\mathrm{Tr}(\alpha), N(\alpha) \in \mathbb{Z}$, and $\alpha \in \mathcal{O}_K^\times$ if and only if $N(\alpha) = \pm 1$.*

*(iii) For number fields $L \supseteq K$: $\mathrm{Tr}_{L/\mathbb{Q}} = \mathrm{Tr}_{K/\mathbb{Q}} \circ \mathrm{Tr}_{L/K}$ and $N_{L/\mathbb{Q}} = N_{K/\mathbb{Q}} \circ N_{L/K}$.*

*Proof.* (i) If $\beta_1, \dots, \beta_{n/d}$ is a basis of $K$ over $\mathbb{Q}(\alpha)$, the elements $\beta_j\alpha^i$ $(0 \leq i < d)$ form a basis of $K$ over $\mathbb{Q}$ in which $m_\alpha$ is block diagonal with $n/d$ copies of the companion matrix of $m_\alpha(X)$; so its characteristic polynomial is $m_\alpha(X)^{n/d}$. Each of the $d$ embeddings of $\mathbb{Q}(\alpha)$, which send $\alpha$ to the $d$ distinct roots of $m_\alpha$, extends to exactly $n/d$ embeddings of $K$ (Chapter 5), so $\prod_i(X - \sigma_i(\alpha)) = m_\alpha(X)^{n/d}$ as well. The trace of a matrix is the sum of the roots of its characteristic polynomial and the determinant their product, which gives the two formulas. The remaining statements are properties of the trace and the determinant: $m_{\alpha+\beta} = m_\alpha + m_\beta$ and $m_{c\alpha} = cm_\alpha$ for $c \in \mathbb{Q}$ give linearity of Tr; $m_{\alpha\beta} = m_\alpha m_\beta$ gives multiplicativity of $N$; $m_\alpha$ is invertible for $\alpha \neq 0$ (with inverse $m_{\alpha^{-1}}$) and $m_0 = 0$. If $K = \mathbb{Q}(\alpha)$, the characteristic polynomial is the minimal polynomial $X^n + c_{n-1}X^{n-1} + \cdots + c_0$, so the sum of its roots is $-c_{n-1}$ and their product is $(-1)^n c_0$. (ii) By Proposition 6.3(i), $m_\alpha(X) \in \mathbb{Z}[X]$, so the characteristic polynomial has integer coefficients. If $\alpha\beta = 1$ with $\beta \in \mathcal{O}_K$, then $N(\alpha)N(\beta) = 1$ in $\mathbb{Z}$. Conversely, let

$\chi(X) = X^n + \cdots + c_1 X + c_0$ be the characteristic polynomial of $m_\alpha$, with $c_0 = (-1)^n N(\alpha) = \pm 1$; by Cayley-Hamilton $\chi(\alpha) = 0$, so $\alpha^{-1} = -c_0^{-1}(\alpha^{n-1} + \cdots + c_1) \in \mathbb{Z}[\alpha] \subseteq \mathcal{O}_K$. (iii) By (i) applied over $K$ (with the same proof), $\mathrm{Tr}_{L/K}(\alpha) = \sum_\sigma \sigma(\alpha)$ over the $[L:K]$ embeddings $\sigma\colon L \to \overline{\mathbb{Q}}$ that are the identity on $K$. Fix an embedding $\tau\colon K \to \overline{\mathbb{Q}}$ and extend it to an automorphism $\tilde{\tau}$ of $\overline{\mathbb{Q}}$ (Chapter 5); the embeddings of $L$ extending $\tau$ are the $\tilde{\tau} \circ \sigma$, so their sum at $\alpha$ is $\tau(\mathrm{Tr}_{L/K}\,\alpha)$. Summing over the $[K:\mathbb{Q}]$ choices of $\tau$ gives $\mathrm{Tr}_{L/\mathbb{Q}}(\alpha) = \mathrm{Tr}_{K/\mathbb{Q}}(\mathrm{Tr}_{L/K}\,\alpha)$; the norm is the same with products. □

**Theorem 6.6** (integral bases and the discriminant)**.** *Let $K$ be a number field of degree $n$.*

*(i) The trace form $(x,y) \mapsto \mathrm{Tr}_{K/\mathbb{Q}}(xy)$ is a nondegenerate symmetric bilinear form on $K$.*

*(ii) $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of rank $n$. For a $\mathbb{Z}$-basis $\beta_1, \ldots, \beta_n$ (an* integral basis*) the integer $\mathrm{disc}(\beta_1, \ldots, \beta_n) = \det(\mathrm{Tr}(\beta_i\beta_j)) = \det(\sigma_k(\beta_j))^2$ is nonzero and independent of the basis; it is the* discriminant $d_K$ *of $K$.*

*(iii) If $\alpha_1, \ldots, \alpha_n \in \mathcal{O}_K$ are linearly independent over $\mathbb{Q}$ and $M = \mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_n$, then $\mathrm{disc}(\alpha_1, \ldots, \alpha_n) = [\mathcal{O}_K : M]^2\, d_K$. In particular, if $\mathrm{disc}(\alpha_1, \ldots, \alpha_n)$ is squarefree then $M = \mathcal{O}_K$.*

*(iv) If $K = \mathbb{Q}(\theta)$ with $\theta \in \mathcal{O}_K$ of minimal polynomial $g$, with roots $\theta_1, \ldots, \theta_n$, then*

$$\mathrm{disc}(1, \theta, \ldots, \theta^{n-1}) = \prod_{i<j} (\theta_i - \theta_j)^2 = \mathrm{disc}(g) = (-1)^{n(n-1)/2} N_{K/\mathbb{Q}}(g'(\theta)).$$

*Proof.* (i) Symmetry and bilinearity come from the linearity of Tr (Proposition 6.5(i)). Nondegeneracy means that no $x \neq 0$ is orthogonal to all of $K$; and if $x \neq 0$, the element $y = x^{-1}$ gives $\mathrm{Tr}(xy) = \mathrm{Tr}(1) = n \neq 0$ ($m_1$ is the identity of an $n$-dimensional space). (ii) By Proposition 6.3 a $\mathbb{Q}$-basis of $K$ can be multiplied by integers to lie in $\mathcal{O}_K$; let $\alpha_1, \ldots, \alpha_n$ be such a basis and $\alpha_1^\vee, \ldots, \alpha_n^\vee$ the dual basis for the trace form, which exists by (i). For $x \in \mathcal{O}_K$, $x = \sum_i \mathrm{Tr}(x\alpha_i)\alpha_i^\vee$ with $\mathrm{Tr}(x\alpha_i) \in \mathbb{Z}$ (Proposition 6.5(ii)). So $\bigoplus_i \mathbb{Z}\alpha_i \subseteq \mathcal{O}_K \subseteq \bigoplus_i \mathbb{Z}\alpha_i^\vee$, and $\mathcal{O}_K$, a subgroup of a free abelian group of rank $n$ containing one of rank $n$, is free of rank $n$ (Chapter 3). Since $\mathrm{Tr}(\beta_i\beta_j) = \sum_k \sigma_k(\beta_i)\sigma_k(\beta_j)$, the Gram matrix is $S^{\mathsf{T}}S$ with $S = (\sigma_k(\beta_j))_{k,j}$, which gives the second expression; it is nonzero by (i). If $\beta' = C\beta$ is another basis, $C \in \mathrm{GL}_n(\mathbb{Z})$, the Gram matrix becomes $C(\cdot)C^{\mathsf{T}}$ and the determinant is multiplied by $\det(C)^2 = 1$. (iii) Write $\alpha = C\beta$ with $C \in \mathrm{M}_n(\mathbb{Z})$; then $\mathrm{disc}(\alpha) = \det(C)^2 d_K$, and $|\det C| = [\mathcal{O}_K : M]$ by the elementary divisor theorem (Chapter 3). If $\mathrm{disc}(\alpha)$ is squarefree, the square $[\mathcal{O}_K : M]^2$ dividing it is 1. (iv) With $\sigma_k(\theta) = \theta_k$, the matrix $(\sigma_k(\theta^{j-1}))$ is the Vandermonde matrix of the $\theta_k$, of determinant $\prod_{i<j}(\theta_j - \theta_i)$; its square is $\prod_{i<j}(\theta_i - \theta_j)^2$, which is $\mathrm{disc}(g)$ by definition. Writing $g(X) = \prod_k (X - \theta_k)$, the product rule gives $g'(\theta_i) = \prod_{k \neq i}(\theta_i - \theta_k)$. In the product $\prod_i \prod_{k \neq i}(\theta_i - \theta_k)$ each unordered pair $\{i,j\}$, $i < j$, contributes the two factors $(\theta_i - \theta_j)(\theta_j - \theta_i) = -(\theta_i - \theta_j)^2$; there are $n(n-1)/2$ pairs, so the product is $(-1)^{n(n-1)/2} \prod_{i<j}(\theta_i - \theta_j)^2$. Finally $\prod_i g'(\theta_i) = \prod_i \sigma_i(g'(\theta)) = N(g'(\theta))$ by Proposition 6.5(i). For example, for $g = X^2 - d$ the formula gives $(-1)^1 N(2\sqrt{d}) = -(2\sqrt{d})(-2\sqrt{d}) = 4d$, the discriminant of $1, \sqrt{d}$ found again in Proposition 6.7. □

**Proposition 6.7** (quadratic fields)**.** *Let $d \in \mathbb{Z}$ be squarefree, $d \neq 0, 1$, and $K = \mathbb{Q}(\sqrt{d})$. If $d \equiv 2, 3 \pmod 4$ then $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ and $d_K = 4d$; if $d \equiv 1 \pmod 4$ then $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ and $d_K = d$. A* fundamental discriminant *is an integer $D \neq 1$ that is either squarefree with $D \equiv 1 \pmod 4$, or $D = 4m$ with $m$ squarefree and $m \equiv 2, 3 \pmod 4$; the map $K \mapsto d_K$ is a bijection*

*from quadratic fields to fundamental discriminants, with inverse* $D \mapsto \mathbb{Q}(\sqrt{D})$. *The norm forms are* $N(a+b\sqrt{d}) = a^2 - db^2$ *and, for* $d \equiv 1 \pmod 4$ *and* $\theta_d = \frac{1+\sqrt{d}}{2}$, $N(a+b\theta_d) = a^2 + ab + \frac{1-d}{4}b^2$.

*Proof.* Let $\alpha = a + b\sqrt{d}$ with $a, b \in \mathbb{Q}$, $b \neq 0$. Its minimal polynomial is $X^2 - 2aX + (a^2 - db^2)$, so $\alpha \in \mathcal{O}_K$ iff $2a \in \mathbb{Z}$ and $a^2 - db^2 \in \mathbb{Z}$. If $a \in \mathbb{Z}$ then $db^2 \in \mathbb{Z}$, and $b \in \mathbb{Z}$ because $d$ is squarefree (if $b = u/v$ in lowest terms, $v^2 \mid du^2$, so $v^2 \mid d$). If $a = A/2$ with $A$ odd, then $4(a^2 - db^2) = A^2 - d(2b)^2 \in 4\mathbb{Z}$, so $d(2b)^2 \in \mathbb{Z}$, $B := 2b \in \mathbb{Z}$ as before, and $A^2 \equiv dB^2 \pmod 4$ with $A$ odd forces $B$ odd and $d \equiv 1 \pmod 4$. This gives $\mathcal{O}_K$. For the basis $1, \sqrt{d}$ the Gram matrix of the trace form is $\operatorname{diag}(2, 2d)$, of determinant $4d$; for $1, \theta_d$ it is $\left(\begin{smallmatrix} 2 & 1 \\ 1 & (1+d)/2 \end{smallmatrix}\right)$, of determinant $d$. The fundamental discriminants are exactly these values: $d$ for $d \equiv 1 \pmod 4$ and $4d$ for $d \equiv 2, 3 \pmod 4$, and $\mathbb{Q}(\sqrt{d_K}) = K$. The norm forms follow from $N(x) = x\bar{x}$. □

**Example 6.8.** $\mathbb{Q}(i)$: $\mathcal{O} = \mathbb{Z}[i]$, $D = -4$, $N(a+bi) = a^2 + b^2$. $\mathbb{Q}(\sqrt{-3})$: $\mathcal{O} = \mathbb{Z}[\zeta_3]$ since $\zeta_3 = \frac{-1+\sqrt{-3}}{2} = \theta_{-3} - 1$, $D = -3$, and $N(a + b\zeta_3) = a^2 - ab + b^2$; the subring $\mathbb{Z}[\sqrt{-3}]$ has index 2 (its discriminant is $-12 = 2^2 \cdot (-3)$, Theorem 6.6(iii)) and is not integrally closed, which is why unique factorisation fails in it (Chapter 2, where $4 = 2 \cdot 2 = (1 + \sqrt{-3})(1 - \sqrt{-3})$). $\mathbb{Q}(\sqrt{5})$: $\mathcal{O} = \mathbb{Z}[\frac{1+\sqrt{5}}{2}]$, $D = 5$. $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{-11})$: $D = -7, -11$. $\mathbb{Q}(\sqrt{-5})$, $\mathbb{Q}(\sqrt{2})$: $D = -20, 8$.

**Theorem 6.9** (units)**.** *(i) (Dirichlet's unit theorem.)* $\mathcal{O}_K^\times \simeq \mu(K) \times \mathbb{Z}^{r_1+r_2-1}$, *where* $\mu(K)$, *the group of roots of unity in* $K$, *is finite and cyclic. (ii) If* $K$ *is imaginary quadratic,* $\mathcal{O}_K^\times = \mu(K)$, *which is* $\{\pm 1\}$ *except for* $\mu(\mathbb{Q}(i)) = \{\pm 1, \pm i\} = \mu_4$ *and* $\mu(\mathbb{Q}(\sqrt{-3})) = \{\pm 1, \pm\zeta_3, \pm\zeta_3^2\} = \mu_6$. *(iii)* $\mathbb{Z}[\sqrt{2}]^\times = \{\pm(1+\sqrt{2})^k : k \in \mathbb{Z}\}$.

*Proof.* (i) We do not prove this; see [142, Chapter I, Section 7] or [123, Chapter 5]. (ii) A unit has norm $\pm 1$ (Proposition 6.5(ii)), and norms are $\geq 0$ in an imaginary quadratic field, so units are the elements of norm 1. For $d \equiv 2, 3 \pmod 4$, $d < 0$: $a^2 + |d|b^2 = 1$ has the solutions $(\pm 1, 0)$, and also $(0, \pm 1)$ when $d = -1$. For $d \equiv 1 \pmod 4$, $d < 0$: with $\alpha = (A + B\sqrt{d})/2$, $A \equiv B \pmod 2$, the equation is $A^2 + |d|B^2 = 4$; for $|d| \geq 7$ only $B = 0$, $A = \pm 2$; for $d = -3$ also $A = \pm 1$, $B = \pm 1$, which are the six elements $\pm 1, \pm\zeta_3, \pm\zeta_3^2$. (iii) Let $u_0 = 1 + \sqrt{2}$, a unit of norm $1 - 2 = -1$. If $u = a + b\sqrt{2} > 1$ is a unit ($a, b \in \mathbb{Z}$), its conjugate $\bar{u} = a - b\sqrt{2}$ satisfies $u\bar{u} = N(u) = \pm 1$, so $\bar{u} = \pm u^{-1}$ has $|\bar{u}| < 1$; hence $a = (u + \bar{u})/2 > (1 - 1)/2 = 0$ and $b\sqrt{2} = (u - \bar{u})/2 > 0$. So $a$ and $b$ are positive integers, $a, b \geq 1$, and $u \geq 1 + \sqrt{2} = u_0$: $u_0$ is the smallest unit greater than 1. Given any unit $u > 0$, choose $k$ with $u_0^k \leq u < u_0^{k+1}$; then $u\,u_0^{-k}$ is a unit in $[1, u_0)$, so it is 1. Units $u < 0$ are $-1$ times positive ones. □

The units of $\mathbb{Z}[\zeta_3]$ enter the paper through the relation $(1 - \zeta_3)^2 = -3$ up to a unit: $3 = -\zeta_3^2(1 - \zeta_3)^2$ and $\zeta_3^2 \in \mu_6$ (Proposition 6.39). It makes $1 - \zeta_3$ a uniformiser of the discrete valuation ring $\mathbb{Z}_3[\zeta_3]$ of the paragraph after Lemma lem:IV, and gives $T/(1 - \zeta_3)T \simeq \mathbb{F}_3$ in the proof of Lemma lem:tamadd.

So far $\mathcal{O}_K$ is known as an abelian group ($\mathbb{Z}^n$) and through its units. What it lacks in general is unique factorisation of elements, and the next section replaces it by unique factorisation of ideals; that is the structure behind the primes $\lambda$, the valuations and the residue fields used everywhere later.

## 6.2 Dedekind domains, ideals, norms and class groups

Unique factorisation of elements fails in rings of integers, but unique factorisation of ideals holds, and it is what gives meaning to "a prime $\lambda \mid \ell$", to the valuation $v_\lambda$ and to the residue field

$\mathcal{O}/\lambda$ in Lemmas lem:crit and lem:old of the paper. This section proves it (Theorem 6.13), with the norm of an ideal, the Chinese remainder theorem and the localisations, which are discrete valuation rings in the sense of Chapter 3. The class group is described but not used by the paper: the rings $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ that the paper needs are principal ideal domains.

**Definition 6.10.** A *Dedekind domain* is a Noetherian integral domain $R$, not a field, which is integrally closed in its fraction field and in which every nonzero prime ideal is maximal.

The three conditions are exactly what the proof of unique factorisation of ideals uses, and each can fail on its own. Noetherian guarantees that the inductions over ideals terminate. Integral closedness fails for $\mathbb{Z}[\sqrt{-3}]$, which misses the integral element $\zeta_3$, and Exercise 6.73 shows that ideals cannot be cancelled there. "Nonzero primes are maximal" (dimension one) fails for $\mathbb{Z}[X]$, which is Noetherian and integrally closed (a UFD) but has the chain of primes $0 \subsetneq (X) \subsetneq (2, X)$; and in $\mathbb{Z}[X]$ the ideal $(4, X)$ is not a product of primes at all: a prime factor would contain $(4, X)$, hence 2 and $X$, hence be the maximal ideal $(2, X)$, but $(2, X) \neq (4, X)$ (as $2 \notin (4, X)$) and $(2, X)^k \subseteq (4, 2X, X^2) \not\ni X$ for $k \geq 2$. The rings of the paper, $\mathcal{O}_K$, $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$, $\mathbb{Z}_3[\zeta_3]$, satisfy all three.

**Theorem 6.11.** *(i) For every number field $K$, $\mathcal{O}_K$ is a Dedekind domain; for a nonzero prime $\mathfrak{p}$ of $\mathcal{O}_K$, $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ for a prime $p$, and $\mathcal{O}_K/\mathfrak{p}$ is a finite field of characteristic $p$. (ii) A principal ideal domain that is not a field is a Dedekind domain. (iii) A Noetherian domain $R$ that is not a field is a Dedekind domain if and only if the localisation $R_\mathfrak{p}$ is a discrete valuation ring for every nonzero prime $\mathfrak{p}$.*

*Proof.* (i) $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of finite rank (Theorem 6.6), so every ideal, a subgroup, is finitely generated: $\mathcal{O}_K$ is Noetherian. It is integrally closed (Proposition 6.3) and not a field ($\frac{1}{2} \notin \mathcal{O}_K$). Let $\mathfrak{p} \neq 0$ be prime and $0 \neq \alpha \in \mathfrak{p}$, with minimal polynomial $X^d + \cdots + c_1X + c_0$; then $c_0 \neq 0$ by irreducibility, and $c_0 = -\alpha(\alpha^{d-1} + \cdots + c_1) \in \mathfrak{p} \cap \mathbb{Z}$. So $\mathfrak{p} \cap \mathbb{Z}$ is a nonzero prime ideal of $\mathbb{Z}$, equal to $p\mathbb{Z}$. Then $\mathcal{O}_K/\mathfrak{p}$ is a quotient of $\mathcal{O}_K/p\mathcal{O}_K \simeq (\mathbb{Z}/p)^n$, a finite integral domain, hence a field: $\mathfrak{p}$ is maximal. (ii) A PID is Noetherian, is a UFD and hence integrally closed (Chapter 2), and a nonzero prime $(\varpi)$ is maximal: if $(\varpi) \subseteq (a)$ then $\varpi = ab$, and $\varpi \mid a$ or $\varpi \mid b$, so $(a) = (\varpi)$ or $(a) = R$. (iii) "Only if" is Proposition 6.15(iv) below, whose proof uses only Theorem 6.13. "If": a discrete valuation ring is integrally closed (Chapter 3). If $x \in K$ lies in $R_\mathfrak{m}$ for every maximal ideal $\mathfrak{m}$, the ideal $\{r \in R : rx \in R\}$ is contained in no maximal ideal (for each $\mathfrak{m}$ write $x = a/s$ with $a \in R$, $s \notin \mathfrak{m}$; then $s$ lies in the ideal but not in $\mathfrak{m}$), so it is $R$, it contains 1, and $x \in R$. Hence $R = \bigcap_\mathfrak{m} R_\mathfrak{m}$. Now if $x \in K$ is a root of a monic polynomial with coefficients in $R$, it is integral over each $R_\mathfrak{m} \supseteq R$, so it lies in each $R_\mathfrak{m}$, hence in $R$: $R$ is integrally closed. If $0 \neq \mathfrak{q} \subsetneq \mathfrak{m}$ were primes, $\mathfrak{q}R_\mathfrak{m} \subsetneq \mathfrak{m}R_\mathfrak{m}$ would be two distinct nonzero primes of the discrete valuation ring $R_\mathfrak{m}$, which has only one. □

**Lemma 6.12.** *Let $R$ be a Dedekind domain with fraction field $K$, and for a nonzero prime $\mathfrak{p}$ put $\mathfrak{p}^{-1} = \{x \in K : x\mathfrak{p} \subseteq R\}$. (i) Every nonzero ideal of $R$ contains a product of nonzero prime ideals. (ii) For every nonzero ideal $\mathfrak{a}$, $\mathfrak{a}\mathfrak{p}^{-1} \neq \mathfrak{a}$; in particular $\mathfrak{p}^{-1} \neq R$. (iii) $\mathfrak{p}\mathfrak{p}^{-1} = R$.*

*Proof.* (i) If not, the set of nonzero ideals not containing such a product is nonempty and has a maximal element $\mathfrak{a}$ ($R$ is Noetherian). It is not prime (a nonzero prime contains the product of one prime, itself), and not $R$ (which contains every product), so there are $b, c \notin \mathfrak{a}$ with $bc \in \mathfrak{a}$; the ideals $\mathfrak{a} + (b)$ and $\mathfrak{a} + (c)$ are strictly larger, so by maximality they contain products of primes, and their product $(\mathfrak{a} + (b))(\mathfrak{a} + (c)) \subseteq \mathfrak{a} + (bc) = \mathfrak{a}$ contains the product of these

two products: a contradiction. (ii) First $\mathfrak{p}^{-1} \neq R$: take $0 \neq a \in \mathfrak{p}$ and, by (i), primes with $\mathfrak{p}_1 \cdots \mathfrak{p}_r \subseteq (a) \subseteq \mathfrak{p}$, $r$ minimal. Some $\mathfrak{p}_i \subseteq \mathfrak{p}$: otherwise choose $x_i \in \mathfrak{p}_i \setminus \mathfrak{p}$, and $x_1 \cdots x_r \in \mathfrak{p}$ contradicts the primality of $\mathfrak{p}$. Say $\mathfrak{p}_1 \subseteq \mathfrak{p}$; as $\mathfrak{p}_1$ is a nonzero prime, it is maximal, so $\mathfrak{p}_1 = \mathfrak{p}$. By minimality of $r$, $\mathfrak{p}_2 \cdots \mathfrak{p}_r \not\subseteq (a)$ (for $r = 1$ read $\mathfrak{p}_2 \cdots \mathfrak{p}_r = R$), so there is $b \in \mathfrak{p}_2 \cdots \mathfrak{p}_r$ with $b \notin (a)$. Then $b\mathfrak{p} \subseteq \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_r \subseteq (a)$, so $(b/a)\mathfrak{p} \subseteq R$, that is $b/a \in \mathfrak{p}^{-1}$, while $b/a \notin R$ because $b \notin aR$. Now let $\mathfrak{a} = (\alpha_1, \ldots, \alpha_m) \neq 0$ with $\mathfrak{a}\mathfrak{p}^{-1} = \mathfrak{a}$. For $x \in \mathfrak{p}^{-1}$, $x\alpha_i \in \mathfrak{a}$, so $x\alpha_i = \sum_j a_{ij}\alpha_j$ with $a_{ij} \in R$, and the adjugate argument of Proposition 6.3 gives $\det(xI - (a_{ij}))\alpha_j = 0$ for all $j$, hence $\det(xI - (a_{ij})) = 0$ (as $R$ is a domain and some $\alpha_j \neq 0$): $x$ is a root of a monic polynomial over $R$, and $x \in R$ because $R$ is integrally closed. So $\mathfrak{p}^{-1} \subseteq R$, contradicting the first step. (iii) $\mathfrak{p}\mathfrak{p}^{-1}$ is an ideal of $R$ (by the definition of $\mathfrak{p}^{-1}$), it contains $\mathfrak{p}$ (as $1 \in \mathfrak{p}^{-1}$), and $\mathfrak{p}\mathfrak{p}^{-1} \neq \mathfrak{p}$ by (ii) with $\mathfrak{a} = \mathfrak{p}$. Since $\mathfrak{p}$ is maximal, $\mathfrak{p}\mathfrak{p}^{-1} = R$. □

**Theorem 6.13** (unique factorisation of ideals)**.** *Let $R$ be a Dedekind domain with fraction field $K$.*

*(i) Every nonzero ideal $\mathfrak{a} \neq R$ is a product $\mathfrak{a} = \mathfrak{p}_1^{a_1} \cdots \mathfrak{p}_r^{a_r}$ of distinct nonzero primes, $a_i \geq 1$, uniquely up to order. We write $v_\mathfrak{p}(\mathfrak{a})$ for the exponent of $\mathfrak{p}$ (0 if $\mathfrak{p}$ does not occur).*

*(ii) A* fractional ideal *is a nonzero finitely generated $R$-submodule of $K$, equivalently a set $x\mathfrak{b}$ with $x \in K^\times$ and $\mathfrak{b}$ a nonzero ideal. The fractional ideals form an abelian group $J_R$ under multiplication, free on the nonzero primes; the inverse of $\mathfrak{a}$ is $\mathfrak{a}^{-1} = \{x \in K : x\mathfrak{a} \subseteq R\}$.*

*(iii) For nonzero ideals, $\mathfrak{a} \mid \mathfrak{b}$ (that is, $\mathfrak{b} = \mathfrak{a}\mathfrak{c}$ for an ideal $\mathfrak{c}$) if and only if $\mathfrak{a} \supseteq \mathfrak{b}$; $\mathfrak{a} + \mathfrak{b}$ and $\mathfrak{a} \cap \mathfrak{b}$ have exponents* min *and* max *of those of $\mathfrak{a}$, $\mathfrak{b}$.*

*(iv) For $\alpha \in K^\times$ put $v_\mathfrak{p}(\alpha) = v_\mathfrak{p}(\alpha R)$. Then $v_\mathfrak{p}(\alpha\beta) = v_\mathfrak{p}(\alpha) + v_\mathfrak{p}(\beta)$ and $v_\mathfrak{p}(\alpha + \beta) \geq \min(v_\mathfrak{p}(\alpha), v_\mathfrak{p}(\beta))$: $v_\mathfrak{p}$ is a discrete valuation of $K$.*

*Proof.* (i) Existence: otherwise let $\mathfrak{a}$ be maximal among the nonzero proper ideals that are not products of primes, and $\mathfrak{p} \supseteq \mathfrak{a}$ a maximal ideal. Then $\mathfrak{a} \subseteq \mathfrak{a}\mathfrak{p}^{-1} \subseteq \mathfrak{p}\mathfrak{p}^{-1} = R$, with $\mathfrak{a} \neq \mathfrak{a}\mathfrak{p}^{-1}$ (Lemma 6.12). If $\mathfrak{a}\mathfrak{p}^{-1} = R$ then $\mathfrak{a} = \mathfrak{a}\mathfrak{p}^{-1}\mathfrak{p} = \mathfrak{p}$, a prime; otherwise $\mathfrak{a}\mathfrak{p}^{-1} = \mathfrak{p}_1 \cdots \mathfrak{p}_s$ by maximality, and $\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_s\mathfrak{p}$. Either way a contradiction. Uniqueness: if $\mathfrak{p}_1 \cdots \mathfrak{p}_r = \mathfrak{q}_1 \cdots \mathfrak{q}_s$, then $\mathfrak{p}_1$ contains the right side, hence some $\mathfrak{q}_j$, and $\mathfrak{p}_1 = \mathfrak{q}_j$ by maximality; multiply by $\mathfrak{p}_1^{-1}$ (Lemma 6.12(iii)) and induct. (ii) A fractional ideal $\mathfrak{a}$ with generators $x_i = a_i/s$ satisfies $s\mathfrak{a} \subseteq R$, so $\mathfrak{a} = s^{-1}\mathfrak{b}$; writing $sR$ and $\mathfrak{b}$ as products of primes, $\mathfrak{a} = \prod \mathfrak{p}^{a_\mathfrak{p}}$ with $a_\mathfrak{p} \in \mathbb{Z}$, almost all zero, and the exponents are unique by (i) after multiplying by a common denominator. Put $\mathfrak{a}' = \prod \mathfrak{p}^{-a_\mathfrak{p}}$; then $\mathfrak{a}\mathfrak{a}' = R$, so $\mathfrak{a}' \subseteq \mathfrak{c} := \{x : x\mathfrak{a} \subseteq R\}$, and $\mathfrak{c} = \mathfrak{c}\mathfrak{a}\mathfrak{a}' \subseteq \mathfrak{a}'$. (iii) If $\mathfrak{a} \supseteq \mathfrak{b}$, then $\mathfrak{c} = \mathfrak{a}^{-1}\mathfrak{b} \subseteq \mathfrak{a}^{-1}\mathfrak{a} = R$ is an ideal with $\mathfrak{b} = \mathfrak{a}\mathfrak{c}$; the converse is clear. So $\mathfrak{a} \supseteq \mathfrak{b}$ iff $v_\mathfrak{p}(\mathfrak{a}) \leq v_\mathfrak{p}(\mathfrak{b})$ for all $\mathfrak{p}$; $\mathfrak{a} + \mathfrak{b}$ is the smallest ideal containing both, $\mathfrak{a} \cap \mathfrak{b}$ the largest contained in both, whence the exponents. (iv) The first identity is (i); and $(\alpha + \beta) \subseteq (\alpha) + (\beta)$, whose exponent at $\mathfrak{p}$ is the minimum, by (iii) extended to fractional ideals by scaling. □

Fractional ideals are introduced so that ideals can be inverted, as rational numbers invert integers. For $R = \mathbb{Z}$ the fractional ideals are the groups $r\mathbb{Z}$, $r \in \mathbb{Q}_{>0}$, and $J_\mathbb{Z} \simeq \mathbb{Q}_{>0}^\times$, free on the primes: $\frac{12}{5}\mathbb{Z} = (2)^2(3)(5)^{-1}$. In $\mathbb{Z}[\sqrt{-5}]$, with the primes of Example 6.18, $\mathfrak{p}_2^{-1} = \frac{1}{2}\mathfrak{p}_2$ (because $\mathfrak{p}_2 \cdot \frac{1}{2}\mathfrak{p}_2 = \frac{1}{2}(2) = R$), a fractional ideal that is not of the form $xR$. The valuation $v_\mathfrak{p}$ of (iv) is the exponent of $\mathfrak{p}$ in $\alpha R$; for $\mathcal{O}_K = \mathbb{Z}[i]$ and $\mathfrak{p} = (1 + i)$, for instance, $v_\mathfrak{p}(2) = 2$ since $2 = -i(1 + i)^2$, and $v_\mathfrak{p}(3) = 0$.

**Proposition 6.14** (the norm of an ideal)**.** *Let $K$ be a number field. For a nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ the* norm *$N\mathfrak{a} = \#(\mathcal{O}_K/\mathfrak{a})$ is finite, and:*

*(i) $N(\mathfrak{ab}) = N\mathfrak{a}\,N\mathfrak{b}$; for a prime $\mathfrak{p}$ and $k \geq 0$, $\mathfrak{p}^k/\mathfrak{p}^{k+1} \simeq \mathcal{O}_K/\mathfrak{p}$ as $\mathcal{O}_K$-modules;*

*(ii) $N(\alpha\mathcal{O}_K) = |N_{K/\mathbb{Q}}(\alpha)|$ for $\alpha \neq 0$;*

*(iii) if $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ then $\mathcal{O}_K/\mathfrak{p} \simeq \mathbb{F}_{p^f}$, where $f = f(\mathfrak{p} \mid p) = [\mathcal{O}_K/\mathfrak{p} : \mathbb{F}_p]$, and $N\mathfrak{p} = p^f$;*

*(iv) an ideal of prime norm is prime; $N\mathfrak{a} \in \mathfrak{a}$; and if $\gcd(N\mathfrak{a}, m) = 1$ then $\mathfrak{a} + m\mathcal{O}_K = \mathcal{O}_K$;*

*(v) the discriminant of a $\mathbb{Z}$-basis of $\mathfrak{a}$ is $(N\mathfrak{a})^2 d_K$.*

*Proof.* $\mathfrak{a}$ contains a nonzero integer $m$ (the constant term $c_0$ of the proof of Theorem 6.11), and $\mathcal{O}_K/m\mathcal{O}_K \simeq (\mathbb{Z}/m)^n$ is finite. (i) The module $M = \mathfrak{a}/\mathfrak{ap}$ is a vector space over $\mathcal{O}_K/\mathfrak{p}$; it is nonzero (Theorem 6.13) and has no subspace other than 0 and $M$, because an ideal $\mathfrak{b}$ with $\mathfrak{ap} \subseteq \mathfrak{b} \subseteq \mathfrak{a}$ satisfies $\mathfrak{a} \mid \mathfrak{b} \mid \mathfrak{ap}$, so $\mathfrak{b} \in \{\mathfrak{a}, \mathfrak{ap}\}$. Hence $M$ is one-dimensional, $M \simeq \mathcal{O}_K/\mathfrak{p}$, and $N(\mathfrak{ap}) = N\mathfrak{a} \cdot \#(\mathfrak{a}/\mathfrak{ap}) = N\mathfrak{a}\,N\mathfrak{p}$; induction on the number of prime factors of $\mathfrak{b}$ gives the claim. (ii) $[\mathcal{O}_K : \alpha\mathcal{O}_K]$ is $|\det|$ of the matrix of $m_\alpha$ on a $\mathbb{Z}$-basis (Chapter 3), which is $|N(\alpha)|$. (iii) $\mathcal{O}_K/\mathfrak{p}$ is a finite field (Theorem 6.11) containing $\mathbb{Z}/p$. (iv) If $N\mathfrak{a} = p$ is prime, $\mathfrak{a} \neq \mathcal{O}_K$, and a factorisation $\mathfrak{a} = \mathfrak{bc}$ into proper ideals would give $N\mathfrak{a} = N\mathfrak{b}\,N\mathfrak{c}$ composite. The additive group $\mathcal{O}_K/\mathfrak{a}$ has order $N\mathfrak{a}$, so $N\mathfrak{a} \cdot 1 \in \mathfrak{a}$; if $uN\mathfrak{a} + vm = 1$, then $1 \in \mathfrak{a} + m\mathcal{O}_K$. (v) is Theorem 6.6(iii), with $[\mathcal{O}_K : \mathfrak{a}] = N\mathfrak{a}$. □

**Proposition 6.15** (Chinese remainder theorem, generators, localisation)**.** *In (ii) to (iv), $R$ is a Dedekind domain.*

*(i) In a commutative ring $R$, let $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$ be pairwise coprime ideals, that is, $\mathfrak{a}_i + \mathfrak{a}_j = R$ for $i \neq j$. Then*

$$\mathfrak{a}_1 \cdots \mathfrak{a}_r = \bigcap_i \mathfrak{a}_i \quad \text{and} \quad R/\mathfrak{a}_1 \cdots \mathfrak{a}_r \simeq \prod_i R/\mathfrak{a}_i.$$

*In a Dedekind domain, powers of distinct primes are coprime, so $R/\mathfrak{a} \simeq \prod_i R/\mathfrak{p}_i^{a_i}$ for $\mathfrak{a} = \prod \mathfrak{p}_i^{a_i}$, and $R/\mathfrak{p}^a$ has length $a$, with composition factors $\mathfrak{p}^k/\mathfrak{p}^{k+1} \simeq R/\mathfrak{p}$.*

*(ii) In a Dedekind domain every nonzero ideal $\mathfrak{a}$ is generated by two elements, the first an arbitrary nonzero $\alpha \in \mathfrak{a}$.*

*(iii) A Dedekind domain with finitely many primes is a principal ideal domain; a Dedekind domain is a UFD if and only if it is a PID.*

*(iv) For a nonzero prime $\mathfrak{p}$ of a Dedekind domain $R$, $R_\mathfrak{p}$ is a discrete valuation ring with maximal ideal $\mathfrak{p}R_\mathfrak{p}$, residue field $R/\mathfrak{p}$, and valuation $v_\mathfrak{p}$.*

*Proof.* (i) For two ideals, $1 = a + b$ with $a \in \mathfrak{a}_1$, $b \in \mathfrak{a}_2$; then $\mathfrak{a}_1 \cap \mathfrak{a}_2 = (\mathfrak{a}_1 \cap \mathfrak{a}_2)(a+b) \subseteq \mathfrak{a}_1\mathfrak{a}_2$, and $x \mapsto (x \bmod \mathfrak{a}_1, x \bmod \mathfrak{a}_2)$ is surjective since $(x, y)$ is the image of $xb + ya$. Induct, using that $\mathfrak{a}_1$ is coprime to $\mathfrak{a}_2 \cdots \mathfrak{a}_r$ (expand $1 = \prod_{j \geq 2}(a_j + b_j)$). In a Dedekind domain, a maximal ideal containing $\mathfrak{p}^a + \mathfrak{q}^b$ contains $\mathfrak{p}$ and $\mathfrak{q}$, so equals both; hence distinct primes give coprime powers. The filtration $R \supseteq \mathfrak{p} \supseteq \cdots \supseteq \mathfrak{p}^a$ has the quotients of Proposition 6.14(i), whose proof holds in any Dedekind domain. (ii) Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be the primes dividing $\alpha R$, $e_i = v_{\mathfrak{p}_i}(\mathfrak{a})$, and choose $x_i \in \mathfrak{p}_i^{e_i} \setminus \mathfrak{p}_i^{e_i+1}$. By (i) there is $\beta \in R$ with $\beta \equiv x_i \pmod{\mathfrak{p}_i^{e_i+1}}$ for all $i$; then $v_{\mathfrak{p}_i}(\beta) = e_i$. Every prime dividing $\mathfrak{a}$ divides $\alpha R$, so $\beta \in \mathfrak{a}$ (Theorem 6.13(iii)), and $(\alpha, \beta) = \alpha R + \beta R$ has exponent

$\min(v_{\mathfrak{p}_i}(\alpha), e_i) = e_i$ at $\mathfrak{p}_i$ and 0 elsewhere: it is $\mathfrak{a}$. (iii) If the primes are $\mathfrak{p}_1, \dots, \mathfrak{p}_r$, choose by (i) $\pi_i \equiv x_i \pmod{\mathfrak{p}_i^2}$ with $x_i \in \mathfrak{p}_i \setminus \mathfrak{p}_i^2$ and $\pi_i \equiv 1 \pmod{\mathfrak{p}_j}$ for $j \neq i$ (the ideals $\mathfrak{p}_i^2$ and $\mathfrak{p}_j$, $j \neq i$, are pairwise coprime). Then $\pi_i \in \mathfrak{p}_i$, $\pi_i \notin \mathfrak{p}_i^2$ (since $\pi_i - x_i \in \mathfrak{p}_i^2$ and $x_i \notin \mathfrak{p}_i^2$) and $\pi_i \notin \mathfrak{p}_j$; so $v_{\mathfrak{p}_i}(\pi_i) = 1$ and $v_{\mathfrak{p}_j}(\pi_i) = 0$, and as there are no other primes, $\pi_i R = \mathfrak{p}_i$ by Theorem 6.13(i). Every ideal, a product of primes, is then principal. A PID is a UFD (Chapter 2). Conversely, in a UFD a nonzero prime $\mathfrak{p}$ contains a nonzero element, hence one of its irreducible factors $\varpi$, which is a prime element; $(\varpi) \subseteq \mathfrak{p}$ is a nonzero prime ideal, hence maximal, so $\mathfrak{p} = (\varpi)$, and all ideals are principal. (iv) A nonzero ideal $A$ of $R_{\mathfrak{p}}$ is $(A \cap R)R_{\mathfrak{p}}$, and for a prime $\mathfrak{q} \neq \mathfrak{p}$, $\mathfrak{q}R_{\mathfrak{p}} = R_{\mathfrak{p}}$ ($\mathfrak{q} \not\subseteq \mathfrak{p}$). So $A = \mathfrak{p}^a R_{\mathfrak{p}}$ with $a = v_{\mathfrak{p}}(A \cap R)$. If $\varpi \in \mathfrak{p} \setminus \mathfrak{p}^2$ then $\varpi R = \mathfrak{p}\mathfrak{b}$ with $\mathfrak{b}$ prime to $\mathfrak{p}$, so $\varpi R_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}$, and every nonzero ideal is $\varpi^a R_{\mathfrak{p}}$: $R_{\mathfrak{p}}$ is a local principal ideal domain, not a field, that is, a discrete valuation ring (Chapter 3). Its valuation of $\alpha$ is the exponent $a$ with $\alpha R_{\mathfrak{p}} = \varpi^a R_{\mathfrak{p}}$, which is $v_{\mathfrak{p}}(\alpha)$. The map $R/\mathfrak{p} \to R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$ is injective ($\mathfrak{p}R_{\mathfrak{p}} \cap R = \mathfrak{p}$) and surjective (an $s \notin \mathfrak{p}$ is invertible modulo the maximal ideal $\mathfrak{p}$). □

**Theorem 6.16** (the class group)**.** *Let $K$ be a number field, $J_K$ its group of fractional ideals and $P_K$ the subgroup of principal fractional ideals $\alpha\mathcal{O}_K$. The* class group *is $\mathrm{Cl}(K) = J_K/P_K$ and the* class number *is $h_K = \#\,\mathrm{Cl}(K)$; $h_K = 1$ iff $\mathcal{O}_K$ is a PID iff it is a UFD (Proposition 6.15(iii)). (Minkowski.) $\mathrm{Cl}(K)$ is finite: every class contains an integral ideal of norm at most $M_K = (4/\pi)^{r_2}\,(n!/n^n)\,|d_K|^{1/2}$. Moreover $|d_K| > 1$ for $K \neq \mathbb{Q}$.*

*Reference.* The definitions and the equivalences are proved above. We do not prove Minkowski's theorem; see [142, Chapter I, Sections 5 and 6] or [123, Chapter 5]. The idea: the embeddings $(\sigma_1, \dots, \sigma_{r_1+r_2})$ map an ideal $\mathfrak{a}$ to a lattice in $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \simeq \mathbb{R}^n$ of covolume $2^{-r_2}|d_K|^{1/2} N\mathfrak{a}$ (the Gram determinant of Theorem 6.6, up to the factor $2^{-r_2}$ from the coordinates of $\mathbb{C}$), and Minkowski's convex body theorem gives $\alpha \in \mathfrak{a}$ with $|N(\alpha)| \leq M_K N\mathfrak{a}$. The inequality $|d_K| > 1$ follows: an integral ideal has norm at least 1, so $M_K \geq 1$, that is $|d_K|^{1/2} \geq (\pi/4)^{r_2} n^n/n! \geq (\pi/4)^{n/2} n^n/n!$, and the last expression is $\pi/2 > 1$ for $n = 2$ and increases with $n$ (the ratio of consecutive terms is $(\pi/4)^{1/2}(1 + \frac{1}{n})^n \geq 2(\pi/4)^{1/2} > 1$). □

**Example 6.17** (class numbers of small quadratic fields)**.** For a quadratic field $n!/n^n = \frac{1}{2}$, so $M_K = \frac{2}{\pi}|D|^{1/2}$ if $D < 0$ and $M_K = \frac{1}{2}D^{1/2}$ if $D > 0$. Hence (all values computed with PARI's `quadclassunit`, which also confirms the class numbers):

| $K$ | $\mathbb{Q}(\sqrt{-3})$ | $\mathbb{Q}(i)$ | $\mathbb{Q}(\sqrt{-7})$ | $\mathbb{Q}(\sqrt{5})$ | $\mathbb{Q}(\sqrt{2})$ | $\mathbb{Q}(\sqrt{-11})$ | $\mathbb{Q}(\sqrt{-5})$ |
|---|---|---|---|---|---|---|---|
| $D$ | $-3$ | $-4$ | $-7$ | $5$ | $8$ | $-11$ | $-20$ |
| $M_K$ | 1.10 | 1.27 | 1.68 | 1.12 | 1.41 | 2.11 | 2.85 |
| $h_K$ | 1 | 1 | 1 | 1 | 1 | 1 | 2 |

In the first five cases $M_K < 2$, so every class contains the ideal of norm 1, which is $\mathcal{O}_K$. For $\mathbb{Q}(\sqrt{-11})$, $M_K > 2$, but there is no ideal of norm 2, because 2 is inert (Theorem 6.34: $-11 \equiv 5 \pmod 8$); so again $h = 1$. For $\mathbb{Q}(\sqrt{-5})$ the ideals of norm $\leq 2$ are $\mathcal{O}_K$ and $\mathfrak{p}_2$ of Example 6.18 (the norm 2 ideals are prime, Proposition 6.14(iv), and $\mathfrak{p}_2$ is the only prime above the ramified prime 2), which is not principal, and $\mathfrak{p}_2^2 = (2)$. Every class contains one of these two ideals, and they lie in different classes, so there are exactly two classes, and $h = 2$. Also $h(\mathbb{Q}(\sqrt{-23})) = 3$ (stated; PARI). For $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ the class number one is also a consequence of the Euclidean algorithm (Chapter 2).

**Example 6.18** ($\mathbb{Z}[\sqrt{-5}]$)**.** $R = \mathbb{Z}[\sqrt{-5}]$ is the ring of integers of $\mathbb{Q}(\sqrt{-5})$ ($-5 \equiv 3 \pmod 4$), with $D = -20$. In $R$, $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ are two factorisations into irreducibles,

no factor of one being associate to a factor of the other (Chapter 2). Put $\mathfrak{p}_2 = (2, 1+\sqrt{-5})$, $\mathfrak{p}_3 = (3, 1+\sqrt{-5})$, $\bar{\mathfrak{p}}_3 = (3, 1-\sqrt{-5})$. The ring homomorphisms $R \to \mathbb{F}_2$, $a+b\sqrt{-5} \mapsto a+b$, and $R \to \mathbb{F}_3$, $a+b\sqrt{-5} \mapsto a \mp b$ (well defined because $(\pm 1)^2 + 5 \equiv 0$ modulo 2, resp. 3), are surjective with kernels $\mathfrak{p}_2$, $\mathfrak{p}_3$, $\bar{\mathfrak{p}}_3$ (an element $a+b\sqrt{-5}$ with $a \equiv b \pmod 3$ is $b(1+\sqrt{-5})+(a-b) \in \mathfrak{p}_3$, and similarly for the others). So these are primes of norms $2, 3, 3$. Next, $\mathfrak{p}_2^2 = (4, 2+2\sqrt{-5}, -4+2\sqrt{-5})$ contains $6 = (2+2\sqrt{-5}) - (-4+2\sqrt{-5})$, hence 2, and is contained in (2): $\mathfrak{p}_2^2 = (2)$. Similarly $\mathfrak{p}_3\bar{\mathfrak{p}}_3 = (9, 3+3\sqrt{-5}, 3-3\sqrt{-5}, 6) = (3)$. Finally $1+\sqrt{-5} \in \mathfrak{p}_2 \cap \mathfrak{p}_3 = \mathfrak{p}_2\mathfrak{p}_3$ and both ideals have norm 6, so $(1+\sqrt{-5}) = \mathfrak{p}_2\mathfrak{p}_3$, and likewise $(1-\sqrt{-5}) = \mathfrak{p}_2\bar{\mathfrak{p}}_3$. Thus

$$(6) = \mathfrak{p}_2^2\,\mathfrak{p}_3\,\bar{\mathfrak{p}}_3,$$

and the two factorisations of 6 are two groupings of the same four prime ideals. None of $\mathfrak{p}_2, \mathfrak{p}_3, \bar{\mathfrak{p}}_3$ is principal: a generator would have norm $a^2+5b^2 \in \{2, 3\}$, which is impossible. This is the ideal-theoretic repair of Chapter 2, and the pattern is the one predicted by Theorem 6.20: $X^2+5 \equiv (X+1)^2 \pmod 2$ and $\equiv (X-1)(X+1) \pmod 3$.

The example already shows the three ways a rational prime can behave in a quadratic ring: 2 becomes a square of a prime, 3 a product of two distinct primes, and 11 stays prime (as $X^2+5$ is irreducible modulo 11: $-5 \equiv 6$ is not among the squares $1, 3, 4, 5, 9$ modulo 11). The next section turns this into a general theory: how $p\mathcal{O}_K$ factors, how to read the factorisation off a polynomial, which primes ramify, and how a Galois group acts on the primes above $p$.

## 6.3 Splitting of primes, ramification and Frobenius

This section explains how a prime of $K$ factors in an extension $L$, defines the decomposition and inertia groups and the Frobenius element, and proves, given the tame relation (sketched here, proved in Chapter 7), the form of it that the paper uses. The paper's Section 1.1 is written in terms of $I_p$, $\mathrm{Frob}_p$ and $V^{I_p}$ (Proposition 6.30); its local $L$-factors use the geometric Frobenius (Convention 6.25); Lemma lem:old uses the Frobenius at unramified primes and Chebotarev (Proposition 6.27, Section 6.6); Lemmas lem:IV and lem:euler use that a Frobenius at a tamely ramified prime $q$ conjugates inertia to its $q$-th power (Corollary 6.32). The results on Frobenius elements are the black boxes that Chapter 5 states.

**Theorem 6.19** ($\sum e_i f_i = n$)**.** *Let $L/K$ be an extension of number fields of degree $n$, $\mathfrak{p}$ a nonzero prime of $\mathcal{O}_K$, and $\mathfrak{p}\mathcal{O}_L = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_g^{e_g}$ with distinct primes $\mathfrak{P}_i$ (Theorem 6.13). The $\mathfrak{P}_i$ are exactly the primes $\mathfrak{P}$ of $\mathcal{O}_L$ with $\mathfrak{P} \cap \mathcal{O}_K = \mathfrak{p}$ ("$\mathfrak{P}$ lies above $\mathfrak{p}$", $\mathfrak{P} \mid \mathfrak{p}$). With the* ramification index *$e_i = e(\mathfrak{P}_i \mid \mathfrak{p})$ and the* residue degree *$f_i = f(\mathfrak{P}_i \mid \mathfrak{p}) = [\mathcal{O}_L/\mathfrak{P}_i : \mathcal{O}_K/\mathfrak{p}]$,*

$$\sum_{i=1}^{g} e_i f_i = n.$$

*$\mathfrak{p}$ is* ramified *in $L$ if some $e_i > 1$ and* unramified *otherwise;* totally ramified *if $g = 1$, $e_1 = n$;* inert *if $g = 1$, $f_1 = n$;* split completely *if $g = n$. For a quadratic $L/K$ each $\mathfrak{p}$ is exactly one of ramified, inert, split. In a tower $K \subseteq M \subseteq L$ with $\mathfrak{P} \mid \mathfrak{q} \mid \mathfrak{p}$: $e(\mathfrak{P} \mid \mathfrak{p}) = e(\mathfrak{P} \mid \mathfrak{q})e(\mathfrak{q} \mid \mathfrak{p})$ and $f(\mathfrak{P} \mid \mathfrak{p}) = f(\mathfrak{P} \mid \mathfrak{q})f(\mathfrak{q} \mid \mathfrak{p})$.*

*Proof.* A prime $\mathfrak{P}$ of $\mathcal{O}_L$ contains $\mathfrak{p}\mathcal{O}_L$ iff $\mathfrak{P} \cap \mathcal{O}_K \supseteq \mathfrak{p}$ iff $\mathfrak{P} \cap \mathcal{O}_K = \mathfrak{p}$ (the left side is a prime of $\mathcal{O}_K$, nonzero as it contains $\mathfrak{P} \cap \mathbb{Z}$, hence maximal), and $\mathfrak{P} \supseteq \mathfrak{p}\mathcal{O}_L$ iff $\mathfrak{P}$ divides $\mathfrak{p}\mathcal{O}_L$ (Theorem 6.13(iii)).

Let $\kappa = \mathcal{O}_K/\mathfrak{p}$. By Proposition 6.15(i), $\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \simeq \prod_i \mathcal{O}_L/\mathfrak{P}_i^{e_i}$, and $\mathcal{O}_L/\mathfrak{P}_i^{e_i}$ has a filtration with $e_i$ quotients $\mathfrak{P}_i^k/\mathfrak{P}_i^{k+1} \simeq \mathcal{O}_L/\mathfrak{P}_i$, each of dimension $f_i$ over $\kappa$; so $\dim_\kappa \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L = \sum e_i f_i$. To see that this dimension is $n$, let $S = \mathcal{O}_K \setminus \mathfrak{p}$ and $A = S^{-1}\mathcal{O}_K$, a discrete valuation ring (Proposition 6.15(iv)), hence a PID. $B = S^{-1}\mathcal{O}_L$ is a finitely generated torsion-free $A$-module ($\mathcal{O}_L$ is finitely generated over $\mathbb{Z}$), hence free (Chapter 3), of rank $n$ because $B \otimes_A K = L$ (Proposition 6.3). So $B/\mathfrak{p}B \simeq \kappa^n$. Finally $\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \to B/\mathfrak{p}B$ is an isomorphism: every $s \in S$ has an inverse $s'$ modulo $\mathfrak{p}$, so $x/s \equiv s'x$ (surjectivity), and if $x \in \mathcal{O}_L$ and $sx \in \mathfrak{p}\mathcal{O}_L$ with $s \in S$, then $x = s'sx - (ss'-1)x \in \mathfrak{p}\mathcal{O}_L$ (injectivity). For $K = \mathbb{Q}$ this is simply $\mathcal{O}_L \simeq \mathbb{Z}^n$. A quadratic $L/K$ has $\sum e_i f_i = 2$, so $(e, f, g)$ is $(2,1,1)$, $(1,2,1)$ or $(1,1,2)$. In a tower, substitute $\mathfrak{q}\mathcal{O}_L = \prod \mathfrak{P}^{e(\mathfrak{P}|\mathfrak{q})}$ into $\mathfrak{p}\mathcal{O}_M = \prod \mathfrak{q}^{e(\mathfrak{q}|\mathfrak{p})}$ and use uniqueness; the residue degrees multiply as degrees of the tower of residue fields. □

**Theorem 6.20** (Dedekind-Kummer)**.** *Let $K = \mathbb{Q}(\theta)$ with $\theta \in \mathcal{O}_K$ of minimal polynomial $g \in \mathbb{Z}[X]$, and let $p$ be a prime with $p \nmid [\mathcal{O}_K : \mathbb{Z}[\theta]]$ (for instance $p^2 \nmid \operatorname{disc}(g)$, by Theorem 6.6(iii)). Let $\bar{g} = \bar{g}_1^{e_1} \cdots \bar{g}_r^{e_r}$ be the factorisation of $g \bmod p$ into distinct monic irreducibles of $\mathbb{F}_p[X]$, and $g_i \in \mathbb{Z}[X]$ monic lifts of $\bar{g}_i$. Then $\mathfrak{P}_i = (p, g_i(\theta))$ are distinct primes, $f(\mathfrak{P}_i \mid p) = \deg \bar{g}_i$, and*

$$p\mathcal{O}_K = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}.$$

*The same holds for a relative extension $L = K(\theta)$, $\theta \in \mathcal{O}_L$, and a prime $\mathfrak{p}$ of $\mathcal{O}_K$ prime to the conductor $\{x \in \mathcal{O}_L : x\mathcal{O}_L \subseteq \mathcal{O}_K[\theta]\}$ (stated; [142, Chapter I, Proposition 8.3]).*

*Proof.* Let $m = [\mathcal{O}_K : \mathbb{Z}[\theta]]$, prime to $p$. The inclusion induces $\mathbb{Z}[\theta]/p\mathbb{Z}[\theta] \to \mathcal{O}_K/p\mathcal{O}_K$. It is injective: if $x \in \mathbb{Z}[\theta]$ and $x = py$ with $y \in \mathcal{O}_K$, then $my \in \mathbb{Z}[\theta]$, and with $um + vp = 1$, $y = u(my) + vx \in \mathbb{Z}[\theta]$. Both sides have $p^n$ elements, so it is an isomorphism, and $\mathbb{Z}[\theta]/p \simeq \mathbb{Z}[X]/(p, g) \simeq \mathbb{F}_p[X]/(\bar{g})$. By the Chinese remainder theorem in $\mathbb{F}_p[X]$,

$$A := \mathcal{O}_K/p\mathcal{O}_K \simeq \mathbb{F}_p[X]/(\bar{g}) \simeq \prod_i \mathbb{F}_p[X]/(\bar{g}_i^{e_i}),$$

whose maximal ideals are the $(\bar{g}_i)$, with residue fields $\mathbb{F}_p[X]/(\bar{g}_i)$ of degree $\deg \bar{g}_i$. The primes of $\mathcal{O}_K$ containing $p$ correspond to the maximal ideals of $A$; the one corresponding to $(\bar{g}_i)$ is the preimage of $(\bar{g}_i(X))$, which is $\mathfrak{P}_i = (p, g_i(\theta))$. So these are the primes above $p$, distinct, with $f(\mathfrak{P}_i \mid p) = \deg \bar{g}_i$. In $A$, $\prod_i (\bar{g}_i)^{e_i} = (\bar{g}) = 0$, so $\prod_i \mathfrak{P}_i^{e_i} \subseteq p\mathcal{O}_K$; writing $p\mathcal{O}_K = \prod_i \mathfrak{P}_i^{a_i}$, this says $a_i \le e_i$ (Theorem 6.13(iii)), while $\sum a_i f_i = n = \deg g = \sum e_i f_i$ (Theorem 6.19). Hence $a_i = e_i$. □

**Example 6.21** (worked factorisations)**.** All of the following factorisations were checked with PARI. (a) $\mathbb{Q}(i)$, $g = X^2 + 1$, index 1. Modulo 2, $g \equiv (X+1)^2$: $2\mathbb{Z}[i] = (2, 1+i)^2 = (1+i)^2$, and indeed $2 = -i(1+i)^2$. Modulo 5, $g \equiv (X-2)(X+2)$: $5\mathbb{Z}[i] = (5, i-2)(5, i+2) = (2-i)(2+i)$. Modulo 3, $g$ is irreducible: 3 is inert, with residue field $\mathbb{F}_9$. (b) $\mathbb{Q}(\zeta_3)$, $g = X^2 + X + 1$, index 1. Modulo 3, $g \equiv (X-1)^2$: $3\mathbb{Z}[\zeta_3] = (3, \zeta_3 - 1)^2 = (1-\zeta_3)^2$. Modulo 7, $g \equiv (X-2)(X-4)$: $7 = (7, \zeta_3 - 2)(7, \zeta_3 - 4)$, and these ideals are $(2-\zeta_3)$ and $(2-\zeta_3^2)$, of norm $4+2+1 = 7$, with $(2-\zeta_3)(2-\zeta_3^2) = 7$ (note $\zeta_3^2 = -1-\zeta_3 \equiv 4$ when $\zeta_3 \equiv 2$). Modulo 2 and 5, $g$ is irreducible: 2 and 5 are inert, with residue fields $\mathbb{F}_4$ and $\mathbb{F}_{25}$. (c) $\mathbb{Q}(\sqrt{-5})$: Example 6.18. (d) The index matters. In $K = \mathbb{Q}(\sqrt{-7})$, $\mathcal{O}_K = \mathbb{Z}[\theta]$ with $\theta = \frac{1+\sqrt{-7}}{2}$, $g = X^2 - X + 2$, and $[\mathcal{O}_K : \mathbb{Z}[\sqrt{-7}]] = 2$. Modulo 2, $X^2 + 7 \equiv (X+1)^2$ would wrongly suggest that 2 ramifies; but $X^2 - X + 2 \equiv X(X-1) \pmod 2$, so 2 splits: $2 = \frac{1+\sqrt{-7}}{2} \cdot \frac{1-\sqrt{-7}}{2}$, and indeed $2 \nmid d_K = -7$ (Theorem 6.22). (e) A non-Galois cubic. $K = \mathbb{Q}(2^{1/3})$ has $\mathcal{O}_K = \mathbb{Z}[2^{1/3}]$ and $d_K = \operatorname{disc}(X^3 - 2) = -108 = -2^2 \cdot 3^3$ (stated; PARI's

`nfbasis`; the index is prime to 5, 7 and 31 since disc is divisible only by 2 and 3). Modulo 5, $X^3-2\equiv(X+2)(X^2+3X+4)$ with the quadratic irreducible, so $5\mathcal{O}_K=\mathfrak{P}_1\mathfrak{P}_2$ with $f=1$ and $f=2$; modulo 7, $X^3-2$ is irreducible (2 is not a cube modulo 7), so 7 is inert; modulo 31 it has three roots $(4,7,20)$, so 31 splits completely. At 2 and 3 PARI finds $2\mathcal{O}_K=\mathfrak{P}^3$ and $3\mathcal{O}_K=\mathfrak{P}^3$, totally ramified ($X^3-2\equiv X^3$, resp. $(X+1)^3$, consistently with this, although the theorem applies at 2 and 3 only once the index is known to be 1). Unequal residue degrees above 5 cannot occur in a Galois extension (Proposition 6.23).

**Theorem 6.22** (Dedekind's discriminant theorem)**.** *A prime $p$ ramifies in the number field $K$ if and only if $p\mid d_K$. Hence only finitely many primes ramify in $K$, and at least one does if $K\neq\mathbb{Q}$ (Theorem 6.16). For an extension $L/K$, a prime $\mathfrak{p}$ of $K$ ramifies in $L$ iff $\mathfrak{p}$ divides the relative discriminant ideal $\mathfrak{d}_{L/K}$ (stated; [142, Chapter III, Section 2]).*

*Proof.* Let $\beta_1,\dots,\beta_n$ be an integral basis and $A=\mathcal{O}_K/p\mathcal{O}_K$, with $\mathbb{F}_p$-basis $\bar\beta_i$. The matrix of multiplication by $\bar x$ on $A$ is the reduction of the matrix of $m_x$, so $\mathrm{Tr}_{A/\mathbb{F}_p}(\bar x)=\mathrm{Tr}_{K/\mathbb{Q}}(x) \bmod p$ and $d_K \bmod p = \det(\mathrm{Tr}_{A/\mathbb{F}_p}(\bar\beta_i\bar\beta_j))$. Thus $p\mid d_K$ iff the trace form of the $\mathbb{F}_p$-algebra $A$ is degenerate. By Proposition 6.15, $A\simeq\prod_i A_i$ with $A_i=\mathcal{O}_K/\mathfrak{P}_i^{e_i}$, and the trace form of $A$ is the orthogonal sum of those of the $A_i$. If some $e_i>1$, $A_i$ contains a nonzero nilpotent $x$ (the image of an element $\xi\in\mathfrak{P}_i\setminus\mathfrak{P}_i^{e_i}$, nonzero in $A_i$, with $\xi^{e_i}\in\mathfrak{P}_i^{e_i}$); then $xy$ is nilpotent for all $y$ ($A$ is commutative), multiplication by a nilpotent element is a nilpotent endomorphism, a nilpotent endomorphism has trace 0 (its only eigenvalue is 0), and $x$ lies in the kernel of the trace form. If all $e_i=1$, each $A_i=\mathbb{F}_{p^{f_i}}$ is a field, whose trace form over $\mathbb{F}_p$ is nondegenerate: $\mathrm{Tr}(x)=x+x^p+\cdots+x^{p^{f-1}}$ (the sum of the conjugates, as in Proposition 6.5(i), the Galois group being generated by $x\mapsto x^p$, Chapter 2) is a polynomial function of degree $p^{f-1}<p^f$, hence not identically zero (a nonzero polynomial of degree $<p^f$ cannot vanish at all $p^f$ elements of the field), and if $\mathrm{Tr}(xy)=0$ for all $y$ with $x\neq0$ then, as $xy$ runs over the whole field when $y$ does, Tr would vanish identically. □

For example, for $K=\mathbb{Q}(i)$ and $p=2$ the algebra is $A=\mathbb{F}_2[X]/(X+1)^2$, the nilpotent is $x=1+i$, and the Gram matrix of the trace form in the basis $1,i$ is $\left(\begin{smallmatrix}2&0\\0&-2\end{smallmatrix}\right)$, zero modulo 2; for $p=3$, $A=\mathbb{F}_9$ and the same matrix is invertible modulo 3. The theorem says that this mechanism, nilpotents killing the trace form, is the whole story.

From now on in this section $L/K$ is a *Galois* extension of number fields with group $G$, $\mathfrak{p}$ is a nonzero prime of $K$ and $\mathfrak{P}\mid\mathfrak{p}$. Write $\kappa(\mathfrak{p})=\mathcal{O}_K/\mathfrak{p}$ and $\kappa(\mathfrak{P})=\mathcal{O}_L/\mathfrak{P}$, finite fields with $\#\kappa(\mathfrak{p})=N\mathfrak{p}$. Every $\sigma\in G$ maps $\mathcal{O}_L$ to itself and primes above $\mathfrak{p}$ to primes above $\mathfrak{p}$.

**Proposition 6.23** (decomposition and inertia groups)**.** *(i) $G$ acts transitively on the primes of $L$ above $\mathfrak{p}$. Hence all $e(\mathfrak{P}\mid\mathfrak{p})=e$ are equal, all $f(\mathfrak{P}\mid\mathfrak{p})=f$ are equal, and $efg=n=\#G$. (ii) The* decomposition group $D_\mathfrak{P}=\{\sigma\in G:\sigma\mathfrak{P}=\mathfrak{P}\}$ *is the stabiliser of $\mathfrak{P}$ (Chapter 1); each $\sigma\in D_\mathfrak{P}$ induces an automorphism $\bar\sigma$ of $\kappa(\mathfrak{P})$ over $\kappa(\mathfrak{p})$, and the* inertia group $I_\mathfrak{P}=\{\sigma\in D_\mathfrak{P}:\sigma(x)\equiv x \pmod{\mathfrak{P}}$ *for all $x\in\mathcal{O}_L$}$\}$ *is the kernel of $\sigma\mapsto\bar\sigma$, a normal subgroup of $D_\mathfrak{P}$. Moreover $[G:D_\mathfrak{P}]=g$, $\#D_\mathfrak{P}=ef$, $D_{\sigma\mathfrak{P}}=\sigma D_\mathfrak{P}\sigma^{-1}$ and $I_{\sigma\mathfrak{P}}=\sigma I_\mathfrak{P}\sigma^{-1}$; if $G$ is abelian, $D_\mathfrak{P}$ and $I_\mathfrak{P}$ depend only on $\mathfrak{p}$.*

*Proof.* (i) Suppose $\mathfrak{P}'\mid\mathfrak{p}$ and $\mathfrak{P}'\neq\sigma\mathfrak{P}$ for all $\sigma\in G$. By the Chinese remainder theorem there is $x\in\mathcal{O}_L$ with $x\in\mathfrak{P}'$ and $x\equiv1 \pmod{\sigma\mathfrak{P}}$ for all $\sigma$. The norm $N_{L/K}(x)=\prod_{\sigma\in G}\sigma(x)$ is fixed by $G$ and integral, so lies in $\mathcal{O}_K$; it lies in $\mathfrak{P}'$ (the factor $x$ does), so in $\mathfrak{P}'\cap\mathcal{O}_K=\mathfrak{p}\subseteq\mathfrak{P}$. As $\mathfrak{P}$ is prime, some $\sigma(x)\in\mathfrak{P}$, that is $x\in\sigma^{-1}\mathfrak{P}$, contradicting $x\equiv1 \pmod{\sigma^{-1}\mathfrak{P}}$. Each $\sigma$ induces an

isomorphism $\mathcal{O}_L/\mathfrak{P} \simeq \mathcal{O}_L/\sigma\mathfrak{P}$ over $\kappa(\mathfrak{p})$, and applying $\sigma$ to $\mathfrak{p}\mathcal{O}_L = \prod \mathfrak{P}_i^{e_i}$ permutes the factors keeping exponents; so $e$ and $f$ are constant on the orbit, which is everything, and Theorem 6.19 gives $efg = n$. (ii) Orbit-stabiliser (Chapter 1) gives $[G : D_{\mathfrak{P}}] = g$, so $\#D_{\mathfrak{P}} = n/g = ef$. A kernel is normal. $\tau \in I_{\mathfrak{P}}$ iff $\tau x - x \in \mathfrak{P}$ for all $x$ iff $(\sigma\tau\sigma^{-1})(\sigma x) - \sigma x \in \sigma\mathfrak{P}$ for all $x$ iff $\sigma\tau\sigma^{-1} \in I_{\sigma\mathfrak{P}}$; the same for $D$. If $G$ is abelian, conjugation is trivial, and transitivity gives the last claim. □

The smallest example shows all three behaviours. Let $L = \mathbb{Q}(i)$, $K = \mathbb{Q}$, $G = \{1, c\}$ with $c$ complex conjugation (Example 6.21(a)). At $p = 5$, $5 = (2+i)(2-i)$ and $c$ exchanges the two primes, so $g = 2$ and $D_{\mathfrak{P}} = I_{\mathfrak{P}} = 1$. At $p = 3$, $(3)$ is prime with residue field $\mathbb{F}_9$, so $g = 1$, $f = 2$ and $D_{\mathfrak{P}} = G$; and $c$ induces the nontrivial automorphism of $\mathbb{F}_9 = \mathbb{F}_3(i)$, as $c(i) = -i \neq i$ modulo 3, so $I_{\mathfrak{P}} = 1$. At $p = 2$, $(2) = (1+i)^2$, so $e = 2$, $D_{\mathfrak{P}} = G$, and $c$ acts trivially on $\mathbb{Z}[i]/(1+i) = \mathbb{F}_2$, so $I_{\mathfrak{P}} = G$. In each case $\#D_{\mathfrak{P}} = ef$.

**Theorem 6.24** (the residue field extension)**.** *In this situation:*

*(i) The homomorphism $D_{\mathfrak{P}} \to \mathrm{Gal}(\kappa(\mathfrak{P})/\kappa(\mathfrak{p}))$, $\sigma \mapsto \bar{\sigma}$, is surjective with kernel $I_{\mathfrak{P}}$. Hence $\#I_{\mathfrak{P}} = e$, and $D_{\mathfrak{P}}/I_{\mathfrak{P}}$ is cyclic of order $f$, generated by the class of any $\sigma$ with $\bar{\sigma} = (x \mapsto x^{N\mathfrak{p}})$.*

*(ii) For $K \subseteq M \subseteq L$ with $H = \mathrm{Gal}(L/M)$ and $\mathfrak{q} = \mathfrak{P} \cap M$: $D_{\mathfrak{P}}(L/M) = D_{\mathfrak{P}} \cap H$ and $I_{\mathfrak{P}}(L/M) = I_{\mathfrak{P}} \cap H$; if $M/K$ is Galois, the images of $D_{\mathfrak{P}}$ and $I_{\mathfrak{P}}$ in $G/H = \mathrm{Gal}(M/K)$ are $D_{\mathfrak{q}}$ and $I_{\mathfrak{q}}$.*

*(iii) $\mathfrak{p}$ splits completely in $L$ iff $D_{\mathfrak{P}} = 1$, and $\mathfrak{p}$ is unramified in $L$ iff $I_{\mathfrak{P}} = 1$.*

*Proof.* (i) $\kappa(\mathfrak{P})/\kappa(\mathfrak{p})$ is an extension of finite fields of degree $f$, Galois with cyclic group generated by $x \mapsto x^{N\mathfrak{p}}$ (Chapter 2). Choose $\bar{\theta}$ generating the cyclic group $\kappa(\mathfrak{P})^\times$, so $\kappa(\mathfrak{P}) = \kappa(\mathfrak{p})(\bar{\theta})$, and by the Chinese remainder theorem a lift $\theta \in \mathcal{O}_L$ of $\bar{\theta}$ lying in every $\tau\mathfrak{P} \neq \mathfrak{P}$. The polynomial $h(X) = \prod_{\sigma \in G}(X - \sigma\theta)$ has coefficients in $\mathcal{O}_K$, and its reduction $\bar{h} = \prod_\sigma (X - \overline{\sigma\theta})$ modulo $\mathfrak{P}$ has coefficients in $\kappa(\mathfrak{p})$ and the root $\bar{\theta}$; so $\bar{\theta}^{N\mathfrak{p}}$ is also a root: $\bar{\theta}^{N\mathfrak{p}} = \overline{\sigma\theta}$ for some $\sigma$. Since $\bar{\theta}^{N\mathfrak{p}} \neq 0$, $\sigma\theta \notin \mathfrak{P}$, so $\theta \notin \sigma^{-1}\mathfrak{P}$, which forces $\sigma^{-1}\mathfrak{P} = \mathfrak{P}$: $\sigma \in D_{\mathfrak{P}}$, and $\bar{\sigma}(\bar{\theta}) = \bar{\theta}^{N\mathfrak{p}}$, so $\bar{\sigma}$ is the generator $x \mapsto x^{N\mathfrak{p}}$. The image is therefore everything; the kernel is $I_{\mathfrak{P}}$ by definition, and $\#I_{\mathfrak{P}} = ef/f = e$. (ii) The first two equalities hold by definition. Restriction maps $D_{\mathfrak{P}}$ into $D_{\mathfrak{q}}$ and $I_{\mathfrak{P}}$ into $I_{\mathfrak{q}}$, with kernels $D_{\mathfrak{P}} \cap H$ and $I_{\mathfrak{P}} \cap H$, of orders $e(\mathfrak{P} \mid \mathfrak{q})f(\mathfrak{P} \mid \mathfrak{q})$ and $e(\mathfrak{P} \mid \mathfrak{q})$ by (i) for $L/M$. So the images have orders $e(\mathfrak{q} \mid \mathfrak{p})f(\mathfrak{q} \mid \mathfrak{p}) = \#D_{\mathfrak{q}}$ and $e(\mathfrak{q} \mid \mathfrak{p}) = \#I_{\mathfrak{q}}$ (Theorem 6.19, and (i) for $M/K$). (iii) $D_{\mathfrak{P}} = 1$ iff $ef = 1$ iff $g = n$; $I_{\mathfrak{P}} = 1$ iff $e = 1$. □

**Convention 6.25** (arithmetic and geometric Frobenius)**.** For $\mathfrak{P}$ unramified over $\mathfrak{p}$, the *arithmetic Frobenius* $\mathrm{Frob}_{\mathfrak{P}}$ is the element of Proposition 6.27, characterised by $\mathrm{Frob}_{\mathfrak{P}}(x) \equiv x^{N\mathfrak{p}} \pmod{\mathfrak{P}}$; its inverse is the *geometric Frobenius*. The book writes $\mathrm{Frob}_p$ for an arithmetic Frobenius at $p$ (Proposition 6.30), as Chapter 5 does and as the paper does (Section 1.1: "$\mathrm{Frob}_p$ is an arithmetic Frobenius at $p$ and $\mathrm{Frob}_p^{-1}$ a geometric one"). So $\mathrm{Frob}_p$ acts on $\mu_n$ ($p \nmid n$) by $\zeta \mapsto \zeta^p$, $\chi_\ell(\mathrm{Frob}_p) = p$, $\bar{\chi}_\ell(\mathrm{Frob}_p) = p \bmod \ell$, and for $E/\mathbb{Q}$ with good reduction at $p \neq \ell$ the characteristic polynomial of $\rho_{E,\ell}(\mathrm{Frob}_p)$ on $V_\ell E$ is $X^2 - a_p X + p$ (Chapters 9 and 11). Local $L$-factors of a representation $V$ of $G_{\mathbb{Q}}$ on a finite-dimensional $\mathbb{Q}_\ell$-vector space are written with the geometric Frobenius:

$$L_p(V, s) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid V^{I_p})^{-1} \qquad (p \neq \ell),$$

so that $L_p(\mathbb{Q}_\ell(1), s) = (1 - p^{-1-s})^{-1}$ and $L_p(V(1), s) = L_p(V, s+1)$, as $\chi_\ell(\mathrm{Frob}_p^{-1}) = p^{-1}$. On $V_\ell E$ the geometric Frobenius has the eigenvalues $\alpha^{-1}, \beta^{-1}$ (with $\alpha + \beta = a_p$, $\alpha\beta = p$), and on

the dual $V_\ell E^\vee$ it has $\alpha, \beta$, so $L_p(V_\ell E^\vee, s) = (1 - a_p p^{-s} + p^{1-2s})^{-1}$ is the Euler factor of $L(E, s)$. *Dictionary.* When the multiset of Frobenius eigenvalues on $V^{I_p}$ is stable under inversion, the two conventions give the same factor: this is the case for $\mathrm{ad}^0 V$ at a good prime (eigenvalues $\alpha/\beta, 1, \beta/\alpha$) and on the lines on which Frobenius acts by $\pm 1$ (types II, III, IV and their duals). When it is not stable they differ: if $E$ has multiplicative reduction at $q$, $(\mathrm{ad}^0 V)^{I_q}$ is a line on which the arithmetic Frobenius acts by $q$ and the geometric one by $q^{-1}$ (Example 6.26), and the factor $1 - q^{-1-s}$ of the paper's table requires the geometric one.

The two identities in the convention are short computations. On $V(1) = V \otimes \chi_\ell$ the geometric Frobenius acts as $p^{-1}\,\mathrm{Frob}_p^{-1}$, and twisting by the character $\chi_\ell$, unramified at $p \neq \ell$, does not change the inertia invariants; so $\det(1 - p^{-1}\,\mathrm{Frob}_p^{-1}\,p^{-s} \mid V^{I_p}) = \det(1 - \mathrm{Frob}_p^{-1}\,p^{-(s+1)} \mid V^{I_p})$, which is $L_p(V(1), s) = L_p(V, s+1)$. On the dual $V_\ell E^\vee$ an element $g$ acts by the inverse transpose of $\rho(g)$; so $\mathrm{Frob}_p^{-1}$ acts by the transpose of $\rho(\mathrm{Frob}_p)$, whose eigenvalues are $\alpha, \beta$, and $\det(1 - \mathrm{Frob}_p^{-1}\,p^{-s} \mid V_\ell E^\vee) = (1 - \alpha p^{-s})(1 - \beta p^{-s}) = 1 - a_p p^{-s} + p^{1-2s}$. With the arithmetic Frobenius on $V_\ell E^\vee$ one would get $(1 - \alpha^{-1} p^{-s})(1 - \beta^{-1} p^{-s}) = 1 - \frac{a_p}{p} p^{-s} + p^{-1-2s}$ instead, which is not the Euler factor of $L(E, s)$: this is why the choice matters.

**Example 6.26** (the multiplicative case)**.** Let $E$ have split multiplicative reduction at $q \neq \ell$. In a suitable basis $e_1, e_2$ of $T_\ell E$ (Chapter 10), inertia acts through $\left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right)$ with a nonzero corner, and an arithmetic Frobenius through $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} q & c' \\ 0 & 1 \end{smallmatrix}\right)$. On $\mathrm{ad}^0 V$, with basis $e = e_1 \otimes e_2^\vee$, $h$, $f$ (Chapter 4), the invariants of a nontrivial unipotent $u = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$, $t \neq 0$, form the line $\mathbb{Q}_\ell e$. Indeed, with $e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, a direct multiplication gives

$$ueu^{-1} = e, \qquad uhu^{-1} = h - 2te, \qquad ufu^{-1} = f + th - t^2 e,$$

so $xe + yh + zf$ is sent to $(x - 2ty - t^2 z)e + (y + tz)h + zf$, and it is fixed iff $tz = 0$ and $2ty + t^2 z = 0$, that is $z = y = 0$ (as $t \neq 0$ and 2 is invertible). The image of inertia consists of such $u$, and contains one with $t \neq 0$, so $(\mathrm{ad}^0 V)^{I_q} = \mathbb{Q}_\ell e$. Next $\rho(\mathrm{Frob})\, e\, \rho(\mathrm{Frob})^{-1} = qe$: with $F = \left(\begin{smallmatrix} q & c' \\ 0 & 1 \end{smallmatrix}\right)$ and $F^{-1} = \left(\begin{smallmatrix} q^{-1} & -c'q^{-1} \\ 0 & 1 \end{smallmatrix}\right)$, $Fe = \left(\begin{smallmatrix} 0 & q \\ 0 & 0 \end{smallmatrix}\right)$ and $FeF^{-1} = \left(\begin{smallmatrix} 0 & q \\ 0 & 0 \end{smallmatrix}\right) = qe$ (in general conjugation by $\left(\begin{smallmatrix} a & * \\ 0 & d \end{smallmatrix}\right)$ scales $e$ by $a/d$). So Frob acts on $(\mathrm{ad}^0 V)^{I_q}$ by $q$, and $\mathrm{Frob}^{-1}$ by $q^{-1}$: $\det(1 - \mathrm{Frob}_q^{-1}\, q^{-s} \mid (\mathrm{ad}^0 V)^{I_q}) = 1 - q^{-1-s}$, the entry of the paper's table, whereas the arithmetic Frobenius would give $1 - q^{1-s}$. The nonsplit case is the same, the unramified quadratic character $\pm 1$ cancelling in the adjoint.

**Proposition 6.27** (the Frobenius element)**.** *Let $L/K$ be Galois with group $G$, $\mathfrak{p}$ unramified in $L$ and $\mathfrak{P} \mid \mathfrak{p}$.*

*(i) There is a unique* $\mathrm{Frob}_\mathfrak{P} \in G$ *with* $\mathrm{Frob}_\mathfrak{P}(x) \equiv x^{N\mathfrak{p}} \pmod{\mathfrak{P}}$ *for all $x \in \mathcal{O}_L$; it lies in $D_\mathfrak{P}$, generates it, and has order $f$.*

*(ii)* $\mathrm{Frob}_{\sigma\mathfrak{P}} = \sigma\,\mathrm{Frob}_\mathfrak{P}\,\sigma^{-1}$*; so* $\{\mathrm{Frob}_\mathfrak{P} : \mathfrak{P} \mid \mathfrak{p}\}$ *is a conjugacy class of $G$, the* Frobenius class $\mathrm{Frob}_\mathfrak{p}$.

*(iii) $\mathfrak{p}$ splits completely in $L$ iff* $\mathrm{Frob}_\mathfrak{p} = \{1\}$.

*(iv) If $G$ is abelian,* $\mathrm{Frob}_\mathfrak{P}$ *depends only on $\mathfrak{p}$ and is written* $\mathrm{Frob}_\mathfrak{p} \in G$.

*(v) If $K \subseteq M \subseteq L$ with $M/K$ Galois, then* $\mathrm{Frob}_\mathfrak{P}|_M = \mathrm{Frob}_{\mathfrak{P} \cap M}$. *If $K \subseteq K' \subseteq L$ and $\mathfrak{p}' = \mathfrak{P} \cap K'$ has residue degree $f'$ over $\mathfrak{p}$, then* $\mathrm{Frob}_\mathfrak{P}(L/K') = \mathrm{Frob}_\mathfrak{P}(L/K)^{f'}$.

*(vi) If $\mathfrak{p}$ is ramified, the elements of $D_{\mathfrak{P}}$ inducing $x \mapsto x^{N\mathfrak{p}}$ on $\kappa(\mathfrak{P})$ form a coset of $I_{\mathfrak{P}}$; each of them is called* a Frobenius at $\mathfrak{P}$*, and there is no distinguished one.*

*Proof.* (i) By Theorem 6.24, $I_{\mathfrak{P}} = 1$ and $D_{\mathfrak{P}} \simeq \mathrm{Gal}(\kappa(\mathfrak{P})/\kappa(\mathfrak{p}))$, cyclic of order $f$; let $\mathrm{Frob}_{\mathfrak{P}}$ be the preimage of $x \mapsto x^{N\mathfrak{p}}$. If $\sigma \in G$ satisfies the congruence, then $\sigma(\mathfrak{P}) \subseteq \mathfrak{P}$ (take $x \in \mathfrak{P}$), so $\sigma\mathfrak{P} = \mathfrak{P}$ and $\sigma \in D_{\mathfrak{P}}$ induces $x \mapsto x^{N\mathfrak{p}}$: $\sigma = \mathrm{Frob}_{\mathfrak{P}}$. (ii) $\sigma\,\mathrm{Frob}_{\mathfrak{P}}\,\sigma^{-1}(x) = \sigma(\mathrm{Frob}_{\mathfrak{P}}(\sigma^{-1}x)) \equiv \sigma((\sigma^{-1}x)^{N\mathfrak{p}}) = x^{N\mathfrak{p}} \pmod{\sigma\mathfrak{P}}$, and use uniqueness; all primes above $\mathfrak{p}$ are conjugate. (iii) $\mathrm{Frob}_{\mathfrak{P}}$ generates $D_{\mathfrak{P}}$; use Theorem 6.24(iii). (iv) follows from (ii). (v) $\mathrm{Frob}_{\mathfrak{P}}|_M$ satisfies the defining congruence modulo $\mathfrak{P} \cap \mathcal{O}_M$ on $\mathcal{O}_M$, and $\mathfrak{P} \cap M$ is unramified over $\mathfrak{p}$ (Theorem 6.19); use uniqueness in $\mathrm{Gal}(M/K)$. For $K'$: both elements lie in $D_{\mathfrak{P}}(L/K)$ (as $D_{\mathfrak{P}}(L/K') \subseteq D_{\mathfrak{P}}(L/K)$) and induce $x \mapsto x^{(N\mathfrak{p})^{f'}} = x^{N\mathfrak{p}'}$ on $\kappa(\mathfrak{P})$; $D_{\mathfrak{P}}(L/K) \to \mathrm{Gal}(\kappa(\mathfrak{P})/\kappa(\mathfrak{p}))$ is injective. (vi) is Theorem 6.24(i). □

Examples, proved in Theorem 6.49 and Proposition 6.35: in $\mathbb{Q}(\zeta_n)$, for $p \nmid n$, $\mathrm{Frob}_p = \sigma_p\colon \zeta_n \mapsto \zeta_n^p$; in $\mathbb{Q}(\sqrt{d})$ of discriminant $D$, for $p \nmid D$, $\mathrm{Frob}_p(\sqrt{d}) = (\frac{D}{p})\sqrt{d}$. The defining congruence can be checked by hand in $\mathbb{Q}(i)$: for an odd prime $p$ and $x = a + bi \in \mathbb{Z}[i]$, $x^p \equiv a^p + b^p i^p \equiv a + b\,i^p \pmod{p}$ (binomial theorem modulo $p$ and Fermat), and $i^p = (-1)^{(p-1)/2}i$. So $\mathrm{Frob}_p$ is the identity for $p \equiv 1 \pmod 4$ and complex conjugation for $p \equiv 3 \pmod 4$, matching the decomposition groups found after Proposition 6.23.

**Proposition 6.28** (Frobenius and subfields)**.** *Let $L/\mathbb{Q}$ be finite Galois with group $G$, $H \leq G$, $M = L^H$, $p$ a prime unramified in $L$, $\mathfrak{P} \mid p$ and $F = \mathrm{Frob}_{\mathfrak{P}}$. The map $\sigma H \mapsto \sigma^{-1}\mathfrak{P} \cap M$ induces a bijection from the orbits of $\langle F\rangle$ acting on $G/H$ by left multiplication to the primes of $M$ above $p$; the residue degree of the prime is the length of the orbit, and all ramification indices are* 1*.*

*Proof.* Every prime of $M$ above $p$ lies under a prime of $L$ above $p$ (Theorem 6.19 for $L/M$), and these are the $\tau\mathfrak{P}$, $\tau \in G$. Two of them, $\tau\mathfrak{P}$ and $\tau'\mathfrak{P}$, lie over the same prime of $M$ iff they are conjugate under $H = \mathrm{Gal}(L/M)$ (Proposition 6.23(i) for $L/M$), iff $\tau' \in H\tau D$ with $D = D_{\mathfrak{P}} = \langle F\rangle$. So the primes of $M$ above $p$ correspond to the double cosets $H\backslash G/D$, and inversion $H\tau D \mapsto D\tau^{-1}H$ identifies these with the $D$-orbits on $G/H$; with $\sigma = \tau^{-1}$ this is the stated map. For $\mathfrak{q} = \tau\mathfrak{P} \cap M$: $D_{\tau\mathfrak{P}}(L/M) = \tau D\tau^{-1} \cap H$ has order $f(\tau\mathfrak{P} \mid \mathfrak{q})$ (all ramification is trivial), so $f(\mathfrak{q} \mid p) = f/\#(\tau D\tau^{-1} \cap H) = \#D/\#(D \cap \tau^{-1}H\tau)$, which is the length of the $D$-orbit of $\tau^{-1}H$ (its stabiliser in $D$ is $D \cap \tau^{-1}H\tau$). □

**Corollary 6.29** (Dedekind's theorem on cycle types)**.** *Let $f \in \mathbb{Z}[X]$ be separable of degree $n$ with leading coefficient $a$, $p$ a prime with $p \nmid a\,\mathrm{disc}(f)$, $L$ the splitting field of $f$, and $d_1, \dots, d_r$ the degrees of the irreducible factors of $f$ mod $p$ in $\mathbb{F}_p[X]$. Then $p$ is unramified in $L$, and for $\mathfrak{P} \mid p$ the permutation of the roots of $f$ induced by $\mathrm{Frob}_{\mathfrak{P}}$ has cycle type $(d_1, \dots, d_r)$.*

*Proof.* Replacing $f$ by $a^{n-1}f(X/a)$, whose roots are $a$ times those of $f$, whose discriminant is a power of $a$ times $\mathrm{disc}(f)$, and whose factorisation modulo $p$ has the same degrees ($a$ is invertible modulo $p$), we may assume $f$ monic. Its roots $\beta_1, \dots, \beta_n$ lie in $\mathcal{O}_L$, and $\mathrm{disc}(f) = \prod_{i<j}(\beta_i - \beta_j)^2 \notin \mathfrak{P}$, so their reductions are distinct roots of $\bar{f}$ in $\kappa(\mathfrak{P})$: reduction is a bijection from the roots of $f$ to the roots of $\bar{f}$, compatible with the action of $D_{\mathfrak{P}}$. An element of $I_{\mathfrak{P}}$ acts trivially on $\kappa(\mathfrak{P})$, so fixes each reduction, hence each $\beta_i$, hence $L$: $I_{\mathfrak{P}} = 1$ and $p$ is unramified. $\mathrm{Frob}_{\mathfrak{P}}$ acts on the reductions as $x \mapsto x^p$, and the orbit of a root of $\bar{f}$ under $x \mapsto x^p$ consists of the roots of its irreducible factor over $\mathbb{F}_p$ (Chapter 2); so the cycles have lengths $d_1, \dots, d_r$. □

For example, in $\mathbb{Q}(2^{1/3})$ (Example 6.21(e)), $X^3-2 \equiv (X+2)(X^2+3X+4) \pmod 5$, so $\mathrm{Frob}_5$ in $\mathrm{Gal}(\mathbb{Q}(2^{1/3},\zeta_3)/\mathbb{Q}) \simeq S_3$ is a transposition: it fixes one root and swaps the other two, and the primes above 5 in $\mathbb{Q}(2^{1/3})$ have residue degrees 1 and 2, the orbit lengths of Proposition 6.28. To see the dictionary concretely, take $H = \mathrm{Gal}(L/\mathbb{Q}(2^{1/3}))$, the stabiliser of the real root; then $G/H$ is identified with the set of the three cube roots of 2, and $\langle \mathrm{Frob}_{\bar{\mathfrak{P}}} \rangle$-orbits on $G/H$ are orbits on the roots. At $p = 31$, $X^3 - 2$ has three roots modulo 31, Frob is trivial, there are three orbits of length 1, and 31 splits completely in $\mathbb{Q}(2^{1/3})$; at $p = 7$ it is irreducible, Frob is a 3-cycle, there is one orbit of length 3, and 7 is inert, in agreement with Example 6.21(e).

**Proposition 6.30** (Frobenius elements in $G_{\mathbb{Q}}$)**.** *Let $p$ be a prime. A* prime of $\bar{\mathbb{Z}}$ above $p$ *is a maximal ideal $\bar{\mathfrak{P}}$ of $\bar{\mathbb{Z}}$ containing $p$.*

*(i) $\bar{\mathfrak{P}} \mapsto (\bar{\mathfrak{P}} \cap \mathcal{O}_L)_L$ is a bijection from the primes of $\bar{\mathbb{Z}}$ above $p$ to the compatible families of primes above $p$ in the finite Galois extensions $L/\mathbb{Q}$; $G_{\mathbb{Q}}$ acts transitively on them, and $\bar{\mathbb{Z}}/\bar{\mathfrak{P}} \simeq \overline{\mathbb{F}}_p$.*

*(ii) The* decomposition group $D_{\bar{\mathfrak{P}}} = \{\sigma \in G_{\mathbb{Q}} : \sigma\bar{\mathfrak{P}} = \bar{\mathfrak{P}}\}$ *and the* inertia group $I_{\bar{\mathfrak{P}}}$ *(the kernel of the action of $D_{\bar{\mathfrak{P}}}$ on $\bar{\mathbb{Z}}/\bar{\mathfrak{P}}$) are closed subgroups, equal to $\varprojlim D_{\mathfrak{P}_L}$ and $\varprojlim I_{\mathfrak{P}_L}$, with surjective projections to $D_{\mathfrak{P}_L}$ and $I_{\mathfrak{P}_L}$ for every finite Galois $L$; for another choice $\sigma\bar{\mathfrak{P}}$ they are conjugated by $\sigma$. Once $\bar{\mathfrak{P}}$ is fixed we write $D_p$, $I_p$. The sequence $1 \to I_p \to D_p \to \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \to 1$ is exact, and $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \simeq \hat{\mathbb{Z}}$ is topologically generated by $x \mapsto x^p$.*

*(iii)* A Frobenius at $p$ *is any $\mathrm{Frob}_p \in D_p$ mapping to $x \mapsto x^p$; it is well defined up to $I_p$ and, when $\bar{\mathfrak{P}}$ varies, up to conjugation in $G_{\mathbb{Q}}$. Its image in $\mathrm{Gal}(L/\mathbb{Q})$, for $L$ finite Galois and unramified at $p$, lies in the class $\mathrm{Frob}_p(L/\mathbb{Q})$.*

*(iv) A continuous representation $\rho$ of $G_{\mathbb{Q}}$ is* unramified at $p$ *if $\rho(I_p) = 1$ for one, equivalently every, $\bar{\mathfrak{P}} \mid p$; then the conjugacy class of $\rho(\mathrm{Frob}_p)$, and in particular $\det(X - \rho(\mathrm{Frob}_p))$, is well defined. If $\rho$ has finite image, with $L$ the field cut out by $\rho$ (the fixed field of $\ker\rho$), then $\rho$ is unramified at $p$ iff $p$ is unramified in $L$, and $\rho(\mathrm{Frob}_p)$ is the image of the class $\mathrm{Frob}_p(L/\mathbb{Q})$.*

*(v) (Stated; Chapter 7.) An embedding $\overline{\mathbb{Q}} \to \overline{\mathbb{Q}}_p$ identifies $D_{\bar{\mathfrak{P}}}$, for the prime it defines, with $G_{\mathbb{Q}_p}$ and $I_{\bar{\mathfrak{P}}}$ with the inertia subgroup of $G_{\mathbb{Q}_p}$.*

*Proof.* (i) $\bar{\mathfrak{P}} \cap \mathcal{O}_L$ is a prime of $\mathcal{O}_L$ containing $p$, and the family is compatible. Conversely the union of a compatible family is an ideal of $\bar{\mathbb{Z}} = \bigcup_L \mathcal{O}_L$ (a directed union) which is prime and contains $p$, and every nonzero prime of $\bar{\mathbb{Z}}$ is maximal (its trace on each $\mathcal{O}_L$ is). Given $\bar{\mathfrak{P}}, \bar{\mathfrak{P}}'$, the sets $S_L = \{\sigma \in \mathrm{Gal}(L/\mathbb{Q}) : \sigma\mathfrak{P}_L = \mathfrak{P}'_L\}$ are nonempty (Proposition 6.23) and finite, and restriction maps $S_{L'}$ into $S_L$ for $L \subseteq L'$; an inverse limit of nonempty finite sets is nonempty (Chapter 5), and an element of it is a $\sigma \in G_{\mathbb{Q}}$ with $\sigma\bar{\mathfrak{P}} = \bar{\mathfrak{P}}'$. The residue ring $\bigcup_L \kappa(\mathfrak{P}_L)$ is an algebraic extension of $\mathbb{F}_p$; it contains every $\mathbb{F}_{p^f}$, since a root in $\bar{\mathbb{Z}}$ of a monic lift of an irreducible polynomial of degree $f$ over $\mathbb{F}_p$ reduces to a root of that polynomial. (ii) By definition $D_{\bar{\mathfrak{P}}}$ is the set of $\sigma$ whose restrictions lie in all $D_{\mathfrak{P}_L}$, which is the inverse limit, a closed subgroup; likewise for $I$. Surjectivity of the projections: for $L \subseteq L'$ the restriction maps $D_{\mathfrak{P}_{L'}} \to D_{\mathfrak{P}_L}$ and $I_{\mathfrak{P}_{L'}} \to I_{\mathfrak{P}_L}$ are surjective (Theorem 6.24(ii)), and the fibres over a given element form an inverse system of nonempty finite sets. The exact sequence is the inverse limit of those of Theorem 6.24(i), surjectivity again by compactness; and $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) = \varprojlim \mathbb{Z}/f \simeq \hat{\mathbb{Z}}$ (Chapter 5). (iii) follows from (ii) and Proposition 6.27. (iv) Inertia groups for different $\bar{\mathfrak{P}}$ are conjugate.

If $\rho(I_p) = 1$, two Frobenius elements differ by $I_p$ and conjugation, so $\rho(\mathrm{Frob}_p)$ is defined up to conjugacy. If $\rho$ factors through $\mathrm{Gal}(L/\mathbb{Q})$ injectively, then $\rho(I_p)$ is the image of $I_{\mathfrak{P}_L}$ by (ii), trivial iff $p$ is unramified in $L$, and the image of $\mathrm{Frob}_p$ is $\mathrm{Frob}_{\mathfrak{P}_L}$. (v) We do not prove this here; see Chapter 7. □

**Proposition 6.31** (tame inertia)**.** *Let $L/K$ be finite Galois, $\mathfrak{P} \mid \mathfrak{p}$, $p$ the residue characteristic, and suppose that $\#I_{\mathfrak{P}}$ is prime to $p$ ($\mathfrak{P}$ is* tamely ramified*). Then $I_{\mathfrak{P}}$ is cyclic, and for every $F \in D_{\mathfrak{P}}$ inducing $x \mapsto x^{N\mathfrak{p}}$ on $\kappa(\mathfrak{P})$ and every $\tau \in I_{\mathfrak{P}}$,*

$$F\tau F^{-1} = \tau^{N\mathfrak{p}}.$$

*Sketch of proof.* Let $\varpi$ be a uniformiser of the discrete valuation ring $\mathcal{O}_{L,\mathfrak{P}}$ (Proposition 6.15(iv)), on which $D_{\mathfrak{P}}$ acts. For $\tau \in I_{\mathfrak{P}}$, $\tau(\varpi)$ is again a uniformiser, so $\tau(\varpi)/\varpi$ is a unit; put $\theta(\tau) = \tau(\varpi)/\varpi \bmod \mathfrak{P} \in \kappa(\mathfrak{P})^\times$. Since $\tau$ acts trivially modulo $\mathfrak{P}$ on $\mathcal{O}_{L,\mathfrak{P}}$ (the congruence $\tau x \equiv x$ extends from $\mathcal{O}_L$ to fractions with denominators prime to $\mathfrak{P}$), $\theta$ does not depend on $\varpi$ (replacing $\varpi$ by $u\varpi$ multiplies by $\tau(u)/u \equiv 1$) and is a homomorphism: for $\sigma, \tau \in I_{\mathfrak{P}}$, $\sigma\tau(\varpi)/\varpi = \sigma(\tau(\varpi)/\varpi) \cdot \sigma(\varpi)/\varpi$, and $\sigma$ fixes the unit $\tau(\varpi)/\varpi$ modulo $\mathfrak{P}$, so $\theta(\sigma\tau) \equiv \theta(\tau)\theta(\sigma)$. The kernel of $\theta$ is the first ramification group, which is a $p$-group; this is the step we do not prove (it uses the filtration of $I_{\mathfrak{P}}$ by the higher ramification groups, whose successive quotients embed in the additive groups $\mathfrak{P}^i/\mathfrak{P}^{i+1}$; [174, Chapter IV, Section 2]). So when $\#I_{\mathfrak{P}}$ is prime to $p$, $\theta$ is injective and $I_{\mathfrak{P}}$, a subgroup of the cyclic group $\kappa(\mathfrak{P})^\times$, is cyclic. For $F$ as stated, $\varpi' = F^{-1}(\varpi)$ is a uniformiser ($F \in D_{\mathfrak{P}}$ preserves $\mathfrak{P}$ and its powers), and $F\tau F^{-1}(\varpi)/\varpi = F(\tau(\varpi'))/F(\varpi') = F(\tau(\varpi')/\varpi')$. Reducing modulo $\mathfrak{P}$: the unit $\tau(\varpi')/\varpi'$ reduces to $\theta(\tau)$ (computed with the uniformiser $\varpi'$, which is allowed), and $F$ acts on $\kappa(\mathfrak{P})$ as $x \mapsto x^{N\mathfrak{p}}$, so $\theta(F\tau F^{-1}) = \theta(\tau)^{N\mathfrak{p}} = \theta(\tau^{N\mathfrak{p}})$. Injectivity of $\theta$ gives $F\tau F^{-1} = \tau^{N\mathfrak{p}}$. The profinite form, $\mathrm{Frob}\ t\ \mathrm{Frob}^{-1} = t^q$ on the tame inertia quotient of $G_{\mathbb{Q}_q}$, is proved in Chapter 7. □

**Corollary 6.32** (Frobenius on a tame image of inertia)**.** *Let $q$ be a prime, $\ell \neq q$ an odd prime, and $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ continuous with $\rho(I_q)$ finite of order prime to $q$ (for $\rho = \rho_{E,\ell}$ at a prime $q \geq 5$ of additive potentially good reduction the order is 2, 3, 4 or 6, by Kraus's theorem, Chapters 10 and 11). Then $\rho(I_q)$ is cyclic, and for every Frobenius $F \in D_q$ (Proposition 6.30) and every $\tau \in I_q$,*

$$\rho(F)\rho(\tau)\rho(F)^{-1} = \rho(\tau)^q.$$

*In particular, if $\rho(I_q)$ is generated by $\sigma$ of order 3 (or by $-\sigma$, of order 6), then $\rho(F)$ commutes with $\sigma$ if $q \equiv 1 \pmod 3$ and conjugates it to $\sigma^{-1}$ if $q \equiv 2 \pmod 3$; if $\rho(I_q)$ is generated by $\tau$ of order 4, $\rho(F)$ commutes with $\tau$ if $q \equiv 1 \pmod 4$ and conjugates it to $\tau^{-1}$ if $q \equiv 3 \pmod 4$. None of this depends on the choice of $F$.*

*Proof.* The kernel $1 + \ell\,\mathrm{M}_2(\mathbb{Z}_\ell)$ of reduction modulo $\ell$ contains no nontrivial element of finite order when $\ell$ is odd (Chapters 1 and 4). For the reader's convenience, the argument: write $g \neq 1$ as $g = 1 + \ell^k x$ with $k \geq 1$ and $x \not\equiv 0 \pmod \ell$. For $m$ prime to $\ell$, $g^m \equiv 1 + m\ell^k x \not\equiv 1 \pmod{\ell^{k+1}}$. For $m = \ell$, $g^\ell = 1 + \ell^{k+1}x + \binom{\ell}{2}\ell^{2k}x^2 + \sum_{j\geq 3}\binom{\ell}{j}\ell^{jk}x^j$, where $\ell \mid \binom{\ell}{2}$ ($\ell$ odd) and $jk \geq k+2$ for $j \geq 3$; so $g^\ell = 1 + \ell^{k+1}x'$ with $x' \equiv x \not\equiv 0 \pmod \ell$, again of the same shape. By induction $g^{\ell^a} = 1 + \ell^{k+a}x_a$ with $x_a \not\equiv 0$, and then $g^{\ell^a m'} \neq 1$ for $m'$ prime to $\ell$; so $g$ has infinite order. (For $\ell = 2$ the step $m = \ell$ fails at $k = 1$: $-1 \in 1 + 2\,\mathrm{M}_2(\mathbb{Z}_2)$.) Hence reduction $\rho \mapsto \bar\rho$ is injective on the finite group $\rho(I_q)$. Let $L$ be the field cut out by $\bar\rho$, a finite Galois extension of $\mathbb{Q}$ with $\bar\rho$: $\mathrm{Gal}(L/\mathbb{Q}) \simeq \bar\rho(G_{\mathbb{Q}})$, and $\mathfrak{P} = \bar{\mathfrak{P}} \cap \mathcal{O}_L$. By Proposition 6.30(ii), $I_{\mathfrak{P}}$ is the image of $I_q$, so $\bar\rho(I_{\mathfrak{P}}) = \bar\rho(I_q) \simeq \rho(I_q)$ has order prime to $q$, and $F|_L \in D_{\mathfrak{P}}$ induces $x \mapsto x^q$ on $\kappa(\mathfrak{P})$.

Proposition 6.31 gives that $I_{\mathfrak{P}}$ is cyclic and $F|_L\,\tau|_L\,F|_L^{-1} = (\tau|_L)^q$, that is, $\bar\rho(F\tau F^{-1}) = \bar\rho(\tau^q)$. Both $F\tau F^{-1}$ and $\tau^q$ lie in $I_q$, which is normal in $D_q$; so $\rho(F\tau F^{-1})$ and $\rho(\tau)^q$ lie in $\rho(I_q)$ and have the same reduction, hence are equal. The special cases: $\sigma^q = \sigma$ or $\sigma^2 = \sigma^{-1}$ according to $q \bmod 3$ ($q \neq 3$ since $3 \mid \#\rho(I_q)$ is prime to $q$); $(-\sigma)^q = -\sigma^q$ for $q$ odd, and for $q = 2$ the order 6 is not prime to $q$; $\tau^q = \tau$ or $\tau^3 = \tau^{-1}$ according to $q \bmod 4$. Two Frobenius elements differ by an element of $I_q$, whose image commutes with the cyclic group $\rho(I_q)$. □

*In the paper* 6.33. Lemma lem:IV: "The arithmetic Frobenius acts on the tame quotient of $I_q$ by $x \mapsto x^q$, so $\rho(\mathrm{Frob}_q)\sigma\rho(\mathrm{Frob}_q)^{-1} = \sigma^q = \sigma^{\chi_{-3}(q)}$" (proof of Lemma lem:IV) is Corollary 6.32, at every prime $q \neq 3$ at which the image of inertia has order 3, including $q = 2$ (the order 3 is prime to 2); $\sigma^{\chi_{-3}(q)} = \sigma^q$ because $\chi_{-3}(q) = 1$ iff $q \equiv 1 \pmod 3$ (Proposition 6.35). Lemma lem:euler: "Frobenius centralises $\sigma$ if $q \equiv 1 \pmod 3$ and inverts it if $q \equiv 2$, for either choice of Frobenius" is the same corollary, and the types III, III$^*$ use its order-4 case. The action on the invariant line of the adjoint is Proposition 6.45.

The general theory is now in place. The next section applies it to the quadratic fields, where everything can be computed by hand, and then passes from $\mathbb{Z}[\zeta_3]$ and $\mathbb{Z}[i]$ to their $\ell$-adic versions, the rings that the image of inertia generates in the paper's lemmas on additive reduction.

## 6.4 Quadratic fields, $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$ and the rings $\mathbb{Z}_\ell[\zeta_3]$

Quadratic fields carry all the characters of the paper: $\chi_{-3}$ and $\chi_{-4}$ in the table of Section 1 and in Lemma lem:euler, $\chi_{q^*}$ in the twists of Section 1.4 and Lemma lem:twist, $\chi_S$ in Theorems thm:twist and thm:quant, and $\mathbb{Q}(\sqrt{\ell^*})$ in the hypothesis of Theorem thm:main. This section proves the splitting law (Theorem 6.34), identifies the character of a quadratic field with a Kronecker symbol (Proposition 6.35), and then studies the two rings $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$ and their $\ell$-adic versions, which carry the algebra of Lemma lem:IV ($\mathbb{Z}_3[\zeta_3]$ a discrete valuation ring with uniformiser $\pi = 1-\zeta_3$, $T_3E$ free of rank one over it, $E[3] \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$) and of Lemma lem:euler (the line spanned by $\sqrt{-3} = 2\sigma + 1$ and the sign $\chi_{-3}(q)$). The Kronecker symbol $(\frac{D}{p})$ is that of Chapter 2: the Legendre symbol for odd $p$, and for $p = 2$ it is 0 if $2 \mid D$, 1 if $D \equiv \pm 1 \pmod 8$ and $-1$ if $D \equiv \pm 3 \pmod 8$.

**Theorem 6.34** (splitting in quadratic fields)**.** *Let $K = \mathbb{Q}(\sqrt{d})$, $d \neq 0, 1$ squarefree, of discriminant $D$, and $p$ a prime. Then $p$ ramifies in $K$ iff $(\frac{D}{p}) = 0$ (iff $p \mid D$), splits iff $(\frac{D}{p}) = 1$, and is inert iff $(\frac{D}{p}) = -1$. Explicitly: for odd $p \mid D$, $p\mathcal{O}_K = (p, \sqrt{d})^2$; for odd $p \nmid D$ with $a^2 \equiv d \pmod p$, $p\mathcal{O}_K = (p, \sqrt{d} - a)(p, \sqrt{d} + a)$ with distinct factors; for $p = 2$: if $d \equiv 3 \pmod 4$, $(2) = (2, 1 + \sqrt{d})^2$; if $d \equiv 2 \pmod 4$, $(2) = (2, \sqrt{d})^2$; if $d \equiv 1 \pmod 8$, $(2) = (2, \frac{1+\sqrt{d}}{2})(2, \frac{1-\sqrt{d}}{2})$; if $d \equiv 5 \pmod 8$, 2 is inert.*

*Proof.* Apply Theorem 6.20. If $p$ is odd, take $\theta = \sqrt{d}$, $g = X^2 - d$: the index $[\mathcal{O}_K : \mathbb{Z}[\sqrt{d}]]$ is 1 or 2, prime to $p$. If $p \mid d$, $\bar g = X^2$ and $p = (p, \sqrt{d})^2$; if $d$ is a nonzero square $a^2$ modulo $p$, $\bar g = (X - a)(X + a)$ with $a \not\equiv -a$; otherwise $\bar g$ is irreducible. For odd $p$, $p \mid D$ iff $p \mid d$, and $(\frac{D}{p}) = (\frac{d}{p})$ since $D/d \in \{1, 4\}$. If $p = 2$ and $d \equiv 2, 3 \pmod 4$, take $\theta = \sqrt{d}$ (index 1): $X^2 - d \equiv X^2$ or $(X + 1)^2 \pmod 2$, and $2 \mid D = 4d$. If $d \equiv 1 \pmod 4$ take $\theta_d = \frac{1+\sqrt{d}}{2}$, $g = X^2 - X + \frac{1-d}{4}$ (index 1): modulo 2, $g \equiv X(X + 1)$ if $\frac{1-d}{4}$ is even, that is $d \equiv 1 \pmod 8$, and $g \equiv X^2 + X + 1$, irreducible, if $d \equiv 5 \pmod 8$; here $D = d$ is odd and $(\frac{D}{2}) = 1$, resp. $-1$. The primes above 2 for $d \equiv 1 \pmod 8$ are $(2, \theta_d)$ and $(2, \theta_d - 1) = (2, \frac{1-\sqrt{d}}{2})$, as $\theta_d - 1 = -\frac{1-\sqrt{d}}{2}$. □

**Proposition 6.35** (the character of a quadratic field)**.** *Let $K=\mathbb{Q}(\sqrt{d})$ have discriminant $D$ and let $\chi_D\colon G_{\mathbb{Q}}\to\{\pm1\}$ be its character, $\sigma(\sqrt{d})=\chi_D(\sigma)\sqrt{d}$ (Chapter 5).*

*(i) For every prime $p\nmid D$, $\chi_D(\mathrm{Frob}_p)=\left(\frac{D}{p}\right)$, the value at $p$ of the primitive real Dirichlet character $n\mapsto\left(\frac{D}{n}\right)$ of conductor $|D|$ (Chapter 2).*

*(ii) $\chi_D$ is unramified exactly at the primes $p\nmid D$; for $p\mid D$, $\chi_D(I_p)=\{\pm1\}$.*

*(iii) In particular: $\chi_{-4}(\mathrm{Frob}_p)=(-1)^{(p-1)/2}$ for odd $p$ (conductor 4); $\chi_{-3}(\mathrm{Frob}_p)=\left(\frac{p}{3}\right)$, which is 1 iff $p\equiv1 \pmod 3$, for $p\neq3$, including $\chi_{-3}(\mathrm{Frob}_2)=-1$ (conductor 3); for an odd prime $q$, $\chi_{q^*}(\mathrm{Frob}_p)=\left(\frac{q^*}{p}\right)=\left(\frac{p}{q}\right)$ for every prime $p\neq q$, including $p=2$ (conductor $q$); $\chi_8$ and $\chi_{-8}$, the characters of $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$, have conductor 8.*

*Proof.* (i) $p$ is unramified (Theorem 6.22, $d_K=D$), and $\mathrm{Frob}_p\in\mathrm{Gal}(K/\mathbb{Q})$ is trivial iff $D_{\mathfrak{P}}=1$ iff $p$ splits (Proposition 6.27(iii)) iff $\left(\frac{D}{p}\right)=1$ (Theorem 6.34); since both sides are $\pm1$ they agree. That $n\mapsto\left(\frac{D}{n}\right)$ is a primitive Dirichlet character of conductor $|D|$ for a fundamental discriminant $D$ is proved in Chapter 2. (ii) For $p\mid D$, $p$ ramifies, the inertia group in $\mathrm{Gal}(K/\mathbb{Q})$ has order $e=2$, and $\chi_D(I_p)$ is its image (Proposition 6.30(ii)). (iii) is (i) for $D=-4,-3,q^*,\pm8$, together with $\left(\frac{-4}{p}\right)=\left(\frac{-1}{p}\right)=(-1)^{(p-1)/2}$, $\left(\frac{-3}{p}\right)=\left(\frac{p}{3}\right)$ and $\left(\frac{q^*}{p}\right)=\left(\frac{p}{q}\right)$ (quadratic reciprocity with its supplements, Chapter 2, or Corollary 6.52); at $p=2$, $\left(\frac{q^*}{2}\right)=1$ iff $q^*\equiv1 \pmod 8$ iff $q\equiv\pm1 \pmod 8$ iff $\left(\frac{2}{q}\right)=1$. All these congruence conditions were also checked numerically for $p<3000$. □

**Proposition 6.36** (quadratic fields unramified outside a set of primes)**.** *(i) For an odd prime $q$, $\mathbb{Q}(\sqrt{q^*})$, $q^*=(-1)^{(q-1)/2}q$, is the unique quadratic field ramified only at $q$; its discriminant is $q^*$. (ii) Every fundamental discriminant is uniquely a product $D=D_0\prod_{q\in T}q^*$ with $T$ a finite set of odd primes and $D_0\in\{1,-4,8,-8\}$ (the* prime discriminants*). A quadratic field is unramified outside a finite set $S$ of primes iff its discriminant has this form with $T\subseteq S$ and $D_0=1$ unless $2\in S$. (iii) For a finite set $S$ of odd primes, $\chi_S:=\prod_{q\in S}\chi_{q^*}$ is the character of $\mathbb{Q}(\sqrt{\prod_{q\in S}q^*})$, of discriminant $\prod_{q\in S}q^*$ and conductor $\prod_{q\in S}q$; it is ramified exactly at the primes of $S$. (iv) Hence twisting a representation of $G_{\mathbb{Q}}$ by $\chi_{q^*}$ or by $\chi_S$ does not change it on $I_p$ for $p\notin S$.*

*Proof.* (i) $q^*\equiv1 \pmod 4$ and is squarefree, so it is a fundamental discriminant, and $\mathbb{Q}(\sqrt{q^*})$ is ramified exactly at $q$ (Theorem 6.22). Conversely, a quadratic field ramified only at $q$ has odd discriminant $D$ divisible only by $q$, so $D=\pm q$; an odd fundamental discriminant is $\equiv1 \pmod 4$, which fixes the sign: $D=q^*$. (ii) If $D$ is odd, it is squarefree and $\equiv1 \pmod 4$; writing $D=\pm\prod_{q\mid D}q$, the product $\prod_{q\mid D}q^*$ has the same absolute value and is $\equiv1 \pmod 4$, so equals $D$. If $D=4m$ with $m\equiv3 \pmod 4$, then $-m\equiv1 \pmod 4$ is an odd fundamental discriminant (or 1) and $D=-4\cdot(-m)$; if $m\equiv2 \pmod 4$, $m=2m'$ with $m'$ odd, and $D=8m'=\pm8\cdot(\pm m')$ with the sign chosen so that $\pm m'\equiv1 \pmod 4$. Uniqueness: the odd part determines $T$, the sign and the 2-part then determine $D_0$. The second statement follows from Theorem 6.22. (iii) $\prod_{q\in S}\chi_{q^*}(\sigma)$ is the sign by which $\sigma$ multiplies $\prod_{q\in S}\sqrt{q^*}$, a square root of $\prod q^*$; the product is a fundamental discriminant by (ii), whose conductor is its absolute value. (iv) $\chi_S$ is trivial on $I_p$ for $p\notin S$ by (iii) and Proposition 6.35(ii). □

**Example 6.37** (splitting tables)**.** By Theorem 6.34 and quadratic reciprocity:

| field | $D$ | ramified | $p$ splits iff |
|---|---|---|---|
| $\mathbb{Q}(i)$ | $-4$ | 2 | $p \equiv 1 \pmod 4$ |
| $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$ | $-3$ | 3 | $p \equiv 1 \pmod 3$ (2 is inert) |
| $\mathbb{Q}(\sqrt{-7})$ | $-7$ | 7 | $p \equiv 1, 2, 4 \pmod 7$ (2 splits, $-7 \equiv 1 \pmod 8$) |
| $\mathbb{Q}(\sqrt{-11})$ | $-11$ | 11 | $p$ a square modulo 11 (2, 7 inert; 3, 5 split) |
| $\mathbb{Q}(\sqrt{5})$ | 5 | 5 | $p \equiv \pm 1 \pmod 5$ |
| $\mathbb{Q}(\sqrt{-5})$ | $-20$ | 2, 5 | $p \equiv 1, 3, 7, 9 \pmod{20}$ (3 and 7 split) |

The paper's twists by these fields: 121a1 and its twist by $\chi_{-11}$ ($\mathbb{Q}(\sqrt{-11}) = \mathbb{Q}(\sqrt{11^*})$), 539c1 and 980i1 twisted by $\chi_{-7}$ ($\mathbb{Q}(\sqrt{-7}) = \mathbb{Q}(\sqrt{7^*})$), in the Example after the proof of Theorem thm:qual (Example ex:qual).

**Proposition 6.38** (Gaussian and Eisenstein primes)**.** *(i) $\mathbb{Z}[i]$ and $\mathbb{Z}[\zeta_3]$ are Euclidean for the norms $a^2 + b^2$ and $a^2 - ab + b^2$ (Chapter 2), hence principal ideal domains; their unit groups are $\mu_4$ and $\mu_6$ (Theorem 6.9). (ii) Up to units, the primes of $\mathbb{Z}[i]$ are: $1 + i$, with $2 = -i(1 + i)^2$; for each $p \equiv 1 \pmod 4$, the two non-associate factors $a \pm bi$ of $p = a^2 + b^2$; for each $p \equiv 3 \pmod 4$, $p$ itself. For example $5 = (2 + i)(2 - i)$, $13 = (3 + 2i)(3 - 2i)$. (iii) Up to units, the primes of $\mathbb{Z}[\zeta_3]$ are: $\pi = 1 - \zeta_3$ (Proposition 6.39); for each $p \equiv 1 \pmod 3$, the two non-associate factors $a + b\zeta_3$, $a + b\zeta_3^2$ of $p = a^2 - ab + b^2$; for each $p \equiv 2 \pmod 3$, $p$ itself. For example $7 = (3 + \zeta_3)(3 + \zeta_3^2)$, $13 = (4 + \zeta_3)(4 + \zeta_3^2)$. (iv) A prime $p$ is a sum of two squares iff $p = 2$ or $p \equiv 1 \pmod 4$ (Fermat); $p = a^2 - ab + b^2$ for integers $a, b$ iff $p = 3$ or $p \equiv 1 \pmod 3$, and then $p = x^2 + 3y^2$ for some integers $x, y$.*

*Proof.* (ii) Every prime $\varpi$ of $\mathbb{Z}[i]$ lies above a unique rational $p$ (Theorem 6.11), and the primes above $p$ are given by Example 6.37: $p = 2$ is ramified, $p \equiv 1 \pmod 4$ split, $p \equiv 3 \pmod 4$ inert. In the split case $p\mathbb{Z}[i] = \mathfrak{P}\bar{\mathfrak{P}}$ with $\mathfrak{P} = (a + bi)$ principal of norm $p$, so $a^2 + b^2 = p$, and $\bar{\mathfrak{P}} = (a - bi) \neq \mathfrak{P}$. In the inert case $(p)$ is prime. (iii) The same with Example 6.21(b) and $N(a + b\zeta_3) = a^2 - ab + b^2$, $\overline{a + b\zeta_3} = a + b\zeta_3^2$. (iv) $p$ is a norm from $\mathbb{Z}[i]$ (resp. $\mathbb{Z}[\zeta_3]$) iff some prime above $p$ has norm $p$, iff $p$ is ramified or split. If $p = a^2 - ab + b^2$, then $p$ is odd (it is 3 or $\equiv 1 \pmod 3$), so not both $a, b$ are even, and one of $a$, $b$, $a - b$ is even; by the symmetries $(a, b) \mapsto (b, a)$ and $(a, b) \mapsto (a - b, -b)$, which preserve the form, we may assume $b = 2y$ even, and then $p = (a - y)^2 + 3y^2$. The small cases were also checked by computer for $p < 300$. □

**Proposition 6.39** (the prime above 3)**.** *In $\mathbb{Z}[\zeta_3]$ put $\pi = 1 - \zeta_3$ and $\sqrt{-3} := 1 + 2\zeta_3$.*

*(i) $(\sqrt{-3})^2 = -3$, $\sqrt{-3} = \zeta_3 - \zeta_3^2 = \zeta_3\pi$, $\pi^2 = -3\zeta_3$ and $3 = -\zeta_3^2\pi^2$. So $\pi$ and $\sqrt{-3}$ are associates, $N(\pi) = 3$, $3\mathbb{Z}[\zeta_3] = (\pi)^2$ (3 is totally ramified), and $\pi^2 = -3$ up to the unit $\zeta_3$.*

*(ii) $\mathbb{Z}[\zeta_3]/(\pi) = \mathbb{F}_3$, and $\zeta_3 \equiv 1 \pmod \pi$.*

*(iii) $\mathbb{Z}[\zeta_3]/(3) = \mathbb{Z}[\zeta_3]/(\pi^2)$ is a local ring with nine elements, with $\mathbb{F}_3$-basis $1, \pi$; multiplication by $\zeta_3 = 1 - \pi$ has the matrix $\left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right)$ in this basis. It is a nontrivial unipotent map, and its fixed points form the line $\pi\mathbb{Z}[\zeta_3]/(\pi^2)$.*

*(iv) The analogue at 2 in $\mathbb{Z}[i]$: $(2) = (1 + i)^2$, $\mathbb{Z}[i]/(2) \simeq \mathbb{F}_2[x]/(x^2)$ with $x = 1 + i$, and multiplication by $i = 1 + x$ is a nontrivial unipotent map.*

*Proof.* All identities follow from $\zeta_3^2 = -1 - \zeta_3$: $(1 + 2\zeta_3)^2 = 1 + 4\zeta_3 + 4\zeta_3^2 = -3$; $\zeta_3 - \zeta_3^2 = 1 + 2\zeta_3$; $\zeta_3\pi = \zeta_3 - \zeta_3^2$; $\pi^2 = 1 - 2\zeta_3 + \zeta_3^2 = -3\zeta_3$; $-\zeta_3^2\pi^2 = 3\zeta_3^3 = 3$. $N(\pi) = 1 + 1 + 1 = 3$. (ii) $\mathbb{Z}[\zeta_3]/(\pi)$ has $N(\pi) = 3$ elements, and $\zeta_3 = 1 - \pi$. (iii) $(3) = (\pi^2)$ by (i); the quotient has 9 elements, its

ideals are 0, $(\pi)/(\pi^2)$ and the whole ring (Proposition 6.15(i)), and $1, \pi$ is a basis since $\pi \notin (\pi^2)$. $\zeta_3 \cdot 1 = 1 - \pi$ and $\zeta_3 \pi = \pi - \pi^2 \equiv \pi$, giving the matrix (columns are images). The fixed points of $x \mapsto \zeta_3 x$ are the $x$ with $\pi x \equiv 0$, that is $x \in (\pi)$. (iv) $(1+i)^2 = 2i$, so $(2) = (1+i)^2$; $\mathbb{Z}[i]/(2)$ has basis $1, x$ with $x^2 = 2i \equiv 0$, and $i \cdot 1 = 1 + x$, $i \cdot x = x + x^2 = x$. □

**Lemma 6.40** (quadratic algebras over $\mathbb{Z}_\ell$)**.** *Let $\ell$ be a prime, $g \in \mathbb{Z}_\ell[X]$ monic of degree 2, $R = \mathbb{Z}_\ell[X]/(g)$ (free over $\mathbb{Z}_\ell$ with basis $1, x$, $x$ the class of $X$) and $\bar{g} = g \bmod \ell$.*

*(i) If $\bar{g}$ is irreducible over $\mathbb{F}_\ell$, $R$ is a discrete valuation ring with uniformiser $\ell$ and residue field $\mathbb{F}_{\ell^2}$.*

*(ii) If $g(X + c)$ is an Eisenstein polynomial at $\ell$ for some $c \in \mathbb{Z}_\ell$, $R$ is a discrete valuation ring with uniformiser $x - c$ and residue field $\mathbb{F}_\ell$, and $\ell R = (x - c)^2 R$.*

*(iii) If $\bar{g} = (X - \bar{a})(X - \bar{b})$ with $\bar{a} \neq \bar{b}$, then $g$ has roots $a, b \in \mathbb{Z}_\ell$ lifting $\bar{a}, \bar{b}$, and $R \simeq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$, $x \mapsto (a, b)$.*

*In each case $R$ is a principal ideal ring. If $T$ is an $R$-module which is free of rank 2 over $\mathbb{Z}_\ell$ and satisfies $T \otimes \mathbb{Q}_\ell \simeq R \otimes \mathbb{Q}_\ell$ as $R \otimes \mathbb{Q}_\ell$-modules (automatic in cases (i) and (ii)), then $T$ is free of rank one over $R$.*

*Proof.* $R$ is Noetherian (finite over $\mathbb{Z}_\ell$) and $\ell$-adically complete, so an element $1 - \ell z$ is a unit ($\sum_k (\ell z)^k$ converges). (i) $R/\ell R = \mathbb{F}_\ell[X]/(\bar{g}) = \mathbb{F}_{\ell^2}$ is a field, so $\ell R$ is maximal. If $y \notin \ell R$, $\bar{y}$ is invertible in $R/\ell R$: $yy' = 1 - \ell z$, and $y$ is a unit. So $R$ is local with maximal ideal $\ell R$. Every nonzero $y$ is $\ell^k y_0$ with $y_0 \notin \ell R$ (as $\bigcap_k \ell^k R = 0$, $R$ being free over $\mathbb{Z}_\ell$), so a product of nonzero elements is $\ell^{k+k'}$ times a unit, nonzero: $R$ is a domain. A Noetherian local domain, not a field, with principal maximal ideal is a discrete valuation ring (Chapter 3). (ii) Write $y = x - c$, so $R = \mathbb{Z}_\ell[Y]/(Y^2 + uY + v)$ with $u = \ell u'$, $u' \in \mathbb{Z}_\ell$, and $v = \ell w_0$, $w_0 \in \mathbb{Z}_\ell^\times$. Then $R/(y) = \mathbb{Z}_\ell/(v) = \mathbb{F}_\ell$, so $(y)$ is maximal. If $z = s + ty$ ($s, t \in \mathbb{Z}_\ell$) is not in $(y)$, then $\ell \nmid s$, and the determinant of multiplication by $z$ in the basis $1, y$ is $s^2 - stu + t^2 v \equiv s^2 \not\equiv 0 \pmod{\ell}$; so $z$ is a unit (Cayley-Hamilton, as in Proposition 6.5(ii)), and $R$ is local with maximal ideal $(y)$. Next $y^2 = -\ell(u'y + w_0)$ with $u'y + w_0 \notin (y)$ a unit, so $\ell R = y^2 R$. Hence $\bigcap_k y^k R = \bigcap_k \ell^k R = 0$, every nonzero $z$ is $y^k w$ with $w$ a unit ($k$ the largest exponent with $z \in y^k R$), and $y^k \neq 0$ for all $k$; so a product of nonzero elements is nonzero, $R$ is a domain, and it is a discrete valuation ring with uniformiser $y$ (Chapter 3). (iii) $R/\ell R \simeq \mathbb{F}_\ell[X]/(X - \bar{a}) \times \mathbb{F}_\ell[X]/(X - \bar{b}) = \mathbb{F}_\ell \times \mathbb{F}_\ell$ by the Chinese remainder theorem. Its idempotent $(1, 0)$ lifts to an idempotent $e \in R$ (Chapter 3, idempotent lifting over the complete ring $\mathbb{Z}_\ell$). Then $R = Re \times R(1 - e)$, each factor is a direct summand of $R \simeq \mathbb{Z}_\ell^2$, hence free over $\mathbb{Z}_\ell$, and nonzero modulo $\ell$, hence of rank 1; as $Re/\ell Re \simeq \mathbb{F}_\ell$ is spanned by $\bar{e}$, Nakayama gives $Re = \mathbb{Z}_\ell e$, and similarly $R(1-e) = \mathbb{Z}_\ell(1-e)$. So $R \simeq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$ as rings, and the image of $x$ is a pair $(a, b)$ of roots of $g$ in $\mathbb{Z}_\ell$ reducing to $\bar{a}, \bar{b}$. Modules: a discrete valuation ring and $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$ are principal ideal rings. Let $T$ be as stated. If $R$ is a discrete valuation ring, $T$ is finitely generated and torsion-free over $R$ (if $zt = 0$ with $z \neq 0$, put $z' = \mathrm{Tr}(z) - z$, with Tr and $N$ the trace and determinant of multiplication by $z$ on $R$; by Cayley-Hamilton $zz' = N(z)$, which is nonzero as $R$ is a domain, so $N(z)t = z'zt = 0$ and $t = 0$), hence free (Chapter 3), of rank $\mathrm{rank}_{\mathbb{Z}_\ell} T / \mathrm{rank}_{\mathbb{Z}_\ell} R = 1$; in this case $R \otimes \mathbb{Q}_\ell$ is a field and the hypothesis on $T \otimes \mathbb{Q}_\ell$ holds automatically. If $R = \mathbb{Z}_\ell e \times \mathbb{Z}_\ell(1 - e)$, then $T = eT \oplus (1 - e)T$, each summand free over $\mathbb{Z}_\ell$, of ranks $\dim(e(T \otimes \mathbb{Q}_\ell)) = 1$ and 1 by the hypothesis; a generator $t_1$ of $eT$ and $t_2$ of $(1 - e)T$ give $T = R(t_1 + t_2)$. □

**Theorem 6.41** ($\mathbb{Z}_3[\zeta_3]$)**.** *Let $\mathbb{Z}_3[\zeta_3] := \mathbb{Z}_3[X]/(X^2 + X + 1)$, $\zeta_3$ the class of $X$, $\pi = 1 - \zeta_3$. Then:*

*(i)* $\mathbb{Z}_3[\zeta_3]$ *is an integral domain, free over* $\mathbb{Z}_3$ *with basis* $1, \zeta_3$ *(or* $1, \pi$*), and* $\mathbb{Z}_3[\zeta_3] \simeq \mathbb{Z}_3 \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_3]$*;*

*(ii) it is a discrete valuation ring with maximal ideal* $(\pi)$*, uniformiser* $\pi$ *and residue field* $\mathbb{F}_3$*; every nonzero element is* $u\pi^k$ *with* $u$ *a unit and* $k \geq 0$*;*

*(iii)* $\pi^2 = -3\zeta_3$*, so* $3\mathbb{Z}_3[\zeta_3] = \pi^2\mathbb{Z}_3[\zeta_3]$ *and* $v_\pi(3) = 2$*;*

*(iv) (stated; Chapter 7) it is the* $\pi$*-adic completion of* $\mathbb{Z}[\zeta_3]$ *and the ring of integers of* $\mathbb{Q}_3(\zeta_3) = \mathbb{Q}_3(\sqrt{-3})$*, a totally ramified quadratic extension of* $\mathbb{Q}_3$*.*

*Proof.* With $g = X^2+X+1$ and $c = 1$, $g(X+1) = X^2+3X+3$ is Eisenstein at 3; Lemma 6.40(ii) gives (i) except the last isomorphism, (ii) with uniformiser $\zeta_3 - 1 = -\pi$, and $3\mathbb{Z}_3[\zeta_3] = \pi^2\mathbb{Z}_3[\zeta_3]$; the identity $\pi^2 = -3\zeta_3$ is Proposition 6.39(i). The isomorphism $\mathbb{Z}_3 \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_3] = \mathbb{Z}_3 \otimes_{\mathbb{Z}} \mathbb{Z}[X]/(X^2 + X + 1) \simeq \mathbb{Z}_3[X]/(X^2 + X + 1)$ is right exactness of the tensor product. A direct check of locality: if $a + b\zeta_3 \notin (\pi)$, i.e. $3 \nmid a + b$, then $N(a + b\zeta_3) = a^2 - ab + b^2 \equiv (a + b)^2 \not\equiv 0 \pmod 3$ and $(a + b\zeta_3)^{-1} = (a + b\zeta_3^2)/(a^2 - ab + b^2)$. (iv) We do not prove this here; see Chapter 7 and [174, Chapter I, Section 6]. □

**Corollary 6.42** (the lattice of Lemma lem:IV)**.** *Let* $T$ *be a free* $\mathbb{Z}_3$*-module of rank* 2 *and* $\sigma \in \mathrm{Aut}_{\mathbb{Z}_3}(T)$ *with* $\sigma^2 + \sigma + 1 = 0$ *(for* $\sigma \in \mathrm{SL}_2(\mathbb{Z}_3)$ *of order* 3 *this holds, its eigenvalues being the primitive cube roots of unity, Chapter 4). Make* $T$ *a* $\mathbb{Z}_3[\zeta_3]$*-module with* $\zeta_3$ *acting as* $\sigma$*. Then:*

*(i)* $T$ *is free of rank one over* $\mathbb{Z}_3[\zeta_3]$*, and* $\mathbb{Z}_3[\sigma] \simeq \mathbb{Z}_3[X]/(X^2 + X + 1)$*;*

*(ii)* $T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$*, and* $\sigma$ *acts on* $T/3T$ *as the nontrivial unipotent map of Proposition 6.39(iii), with fixed line* $\pi T/3T$*;*

*(iii) for every* $k \geq 1$ *the* $\sigma$*-fixed vectors of* $T/\pi^k T$ *form the line* $\pi^{k-1}T/\pi^k T$ *over* $\mathbb{F}_3$*, and* $T^\sigma = 0$*.*

*Proof.* (i) $\sigma$ is not a scalar: a scalar $s \in \mathbb{Z}_3$ with $s^2 + s + 1 = 0$ does not exist (Theorem 6.41(i); directly, $s^2 + s + 1 \equiv (s - 1)^2 \pmod 3$ forces $s = 1 + 3t$, and then $s^2 + s + 1 = 3 + 9t + 9t^2 \equiv 3 \pmod 9$ is not 0). So the polynomials over $\mathbb{Z}_3$ killing $\sigma$ are exactly the multiples of $X^2 + X + 1$ (division by the monic $X^2 + X + 1$ leaves a remainder $a + bX$ with $a + b\sigma = 0$, and $b \neq 0$ would make $\sigma = -a/b$ a scalar in $\mathbb{Q}_3$ satisfying $s^2 + s + 1 = 0$, hence in $\mathbb{Z}_3$ since $\mathbb{Z}_3$ is integrally closed, which was just excluded), and $\mathbb{Z}_3[\sigma] \simeq \mathbb{Z}_3[X]/(X^2 + X + 1)$. $T$ is torsion-free over the domain $\mathbb{Z}_3[\zeta_3]$ by the norm argument of Lemma 6.40, hence free, of rank $2/2 = 1$. (ii) $T \simeq \mathbb{Z}_3[\zeta_3]$ gives $T/3T \simeq \mathbb{Z}_3[\zeta_3]/(3) = \mathbb{Z}_3[\zeta_3]/(\pi^2) \simeq \mathbb{Z}[\zeta_3]/(\pi^2)$, and Proposition 6.39(iii) applies. (iii) In $\mathbb{Z}_3[\zeta_3]/(\pi^k)$, $\zeta_3 x = x$ iff $\pi x \equiv 0 \pmod{\pi^k}$ iff $x \in (\pi^{k-1})$ (write $x = u\pi^j$ with $u$ a unit, Theorem 6.41(ii)); and in $T$ itself $\pi x = 0$ forces $x = 0$, as $T$ is torsion-free over the domain $\mathbb{Z}_3[\zeta_3]$. □

A worked case: let $T = \mathbb{Z}_3^2$ and let $\sigma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ be the companion matrix of $X^2 + X + 1$ ($\sigma e_1 = e_2$, $\sigma e_2 = -e_1 - e_2$). Then $e_1$ generates $T$ over $\mathbb{Z}_3[\sigma]$, as $e_1, \sigma e_1 = e_2$ is a basis, and the isomorphism $\mathbb{Z}_3[\zeta_3] \to T$, $z \mapsto z \cdot e_1$, sends $1, \zeta_3$ to $e_1, e_2$. Modulo 3, $\sigma - 1 \equiv \left(\begin{smallmatrix} 2 & 2 \\ 1 & 1 \end{smallmatrix}\right)$ has rank one and $(\sigma - 1)^2 \equiv \left(\begin{smallmatrix} 6 & 6 \\ 3 & 3 \end{smallmatrix}\right) \equiv 0$, so $\sigma$ is a nontrivial unipotent on $T/3T$; its fixed line is the kernel of $\left(\begin{smallmatrix} 2 & 2 \\ 1 & 1 \end{smallmatrix}\right)$, spanned by $e_1 - e_2 = (1 - \sigma)e_1 = \pi e_1$, which is the line $\pi T/3T$ of (ii).

**Proposition 6.43** ($\mathbb{Z}_\ell[\zeta_3]$ and $\mathbb{Z}_\ell[i]$)**.** *For a prime* $\ell$ *put* $\mathbb{Z}_\ell[\zeta_3] := \mathbb{Z}_\ell[X]/(X^2+X+1) \simeq \mathbb{Z}_\ell \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_3]$ *and* $\mathbb{Z}_\ell[i] := \mathbb{Z}_\ell[X]/(X^2 + 1) \simeq \mathbb{Z}_\ell \otimes_{\mathbb{Z}} \mathbb{Z}[i]$*. These are defined as these quotient rings, not as subrings of an algebraic closure of* $\mathbb{Q}_\ell$*.*

*(i)* $\ell = 3$*:* $\mathbb{Z}_3[\zeta_3]$ *is the ramified discrete valuation ring of Theorem 6.41.*

*(ii)* $\ell \equiv 1 \pmod 3$*:* $X^2 + X + 1$ *has two distinct roots in* $\mathbb{Z}_\ell$*, and* $\mathbb{Z}_\ell[\zeta_3] \simeq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$*, which is not a domain.*

*(iii)* $\ell \equiv 2 \pmod 3$*:* $\mathbb{Z}_\ell[\zeta_3]$ *is an unramified discrete valuation ring with uniformiser* $\ell$ *and residue field* $\mathbb{F}_{\ell^2}$*.*

*(iv) Likewise* $\mathbb{Z}_\ell[i] \simeq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$ *for* $\ell \equiv 1 \pmod 4$*, an unramified discrete valuation ring with residue field* $\mathbb{F}_{\ell^2}$ *for* $\ell \equiv 3 \pmod 4$*, and a ramified discrete valuation ring with uniformiser* $1 + i$ *for* $\ell = 2$*.*

*(v) In all cases* $\mathbb{Q}_\ell[\zeta_3] := \mathbb{Z}_\ell[\zeta_3][1/\ell] = \mathbb{Q}_\ell \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_3)$ *carries the* $\mathbb{Q}_\ell$*-linear involution* $\zeta_3 \mapsto \zeta_3^2$ *(the nontrivial element of* $\mathrm{Gal}(\mathbb{Q}(\zeta_3)/\mathbb{Q})$*, extended by scalars), which negates* $\sqrt{-3} = 1+2\zeta_3$*; and* $\sqrt{-3}$ *is a unit of* $\mathbb{Z}_\ell[\zeta_3]$ *iff* $\ell \neq 3$*. The same holds for* $\mathbb{Q}_\ell[i]$ *with* $i \mapsto -i$*, and* $2i$ *is a unit of* $\mathbb{Z}_\ell[i]$ *iff* $\ell \neq 2$*.*

*The trichotomy is the splitting of* $\ell$ *in* $\mathbb{Q}(\zeta_3)$ *and in* $\mathbb{Q}(i)$ *(Example 6.37), and is the case* $K = \mathbb{Q}(\zeta_3)$*,* $\mathbb{Q}(i)$ *of Proposition 6.69.*

*Proof.* (i) is Theorem 6.41. For $\ell \neq 3$, $X^2 + X + 1$ is separable modulo $\ell$ (its discriminant is $-3$), and it has a root in $\mathbb{F}_\ell$ iff $\mathbb{F}_\ell^\times$ contains an element of order 3 iff $3 \mid \ell - 1$ (Chapter 2). So (ii) and (iii) are Lemma 6.40(iii) and (i) (for $\ell = 2$, $X^2 + X + 1$ is irreducible over $\mathbb{F}_2$, and $2 \equiv 2 \pmod 3$). (iv) The same with $X^2 + 1$, of discriminant $-4$: for odd $\ell$ it has a root modulo $\ell$ iff $4 \mid \ell - 1$; for $\ell = 2$, $(X + 1)^2 + 1 = X^2 + 2X + 2$ is Eisenstein. (v) The tensor product with $\mathbb{Q}_\ell$ of the automorphism $\zeta_3 \mapsto \zeta_3^2$ of $\mathbb{Z}[\zeta_3]$ preserves $\mathbb{Z}_\ell[\zeta_3]$ and sends $1 + 2\zeta_3$ to $1 + 2\zeta_3^2 = -(1 + 2\zeta_3)$. $(\sqrt{-3})^2 = -3$ is a unit iff $\ell \neq 3$, and then $\sqrt{-3}$ is a unit; for $\ell = 3$ it lies in the maximal ideal $(\pi)$. Likewise $(2i)^2 = -4$. □

**Proposition 6.44** (lattices with an automorphism of order 3, 4 or 6)**.** *Let* $\ell$ *be an odd prime,* $T$ *a free* $\mathbb{Z}_\ell$*-module of rank* 2*,* $m \in \{3, 4, 6\}$*, and* $g \in \mathrm{Aut}_{\mathbb{Z}_\ell}(T)$ *with characteristic polynomial the cyclotomic polynomial* $\Phi_m$ *(*$X^2 + X + 1$*,* $X^2 + 1$*,* $X^2 - X + 1$*). This holds for every* $g \in \mathrm{SL}_2(\mathbb{Z}_\ell)$ *of order* $m$ *(its eigenvalues are* $\zeta$*,* $\zeta^{-1}$ *with* $\zeta$ *a primitive* $m$*-th root of unity). Then* $\mathbb{Z}_\ell[g] \simeq \mathbb{Z}_\ell[X]/(\Phi_m)$*, which is* $\mathbb{Z}_\ell[\zeta_3]$ *for* $m = 3, 6$ *(via* $X \mapsto \zeta_3$*, resp.* $X \mapsto -\zeta_3$*) and* $\mathbb{Z}_\ell[i]$ *for* $m = 4$*;* $T$ *is free of rank one over* $\mathbb{Z}_\ell[g]$*; and the commutant of* $g$ *in* $\mathrm{End}_{\mathbb{Z}_\ell}(T)$ *is* $\mathbb{Z}_\ell[g]$*. For* $\ell = 3$ *and* $m = 3$ *this is Corollary 6.42.*

*Proof.* $g$ is not a scalar (its characteristic polynomial has distinct roots), so $\Phi_m$ is its minimal polynomial and $\mathbb{Z}_\ell[g] \simeq \mathbb{Z}_\ell[X]/(\Phi_m) =: R$; and $\Phi_6(X) = \Phi_3(-X)$. By Proposition 6.43, $R$ is a discrete valuation ring or $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$. Over $\mathbb{Q}_\ell$, $T \otimes \mathbb{Q}_\ell$ is a cyclic $\mathbb{Q}_\ell[g]$-module (a non-scalar endomorphism of a plane has a cyclic vector: if every nonzero vector were an eigenvector, two independent eigenvectors and their sum would have to share one eigenvalue and $g$ would be a scalar; so some $v$ is not an eigenvector, and $v, gv$ is a basis), so $T \otimes \mathbb{Q}_\ell \simeq R \otimes \mathbb{Q}_\ell$ via $z \mapsto zv$, and Lemma 6.40 gives $T \simeq R$. Then the commutant of $g$ is $\mathrm{End}_R(T) = \mathrm{End}_R(R) = R$, since an $R$-linear endomorphism of $R$ is multiplication by the image of 1. (In the split case the idempotents are explicit: if $\zeta \in \mathbb{Z}_\ell$ is a root of $\Phi_m$, the other root is $\zeta^{-1}$, and $e = (g - \zeta^{-1})/(\zeta - \zeta^{-1})$ is 1 on the $\zeta$-eigenline and 0 on the $\zeta^{-1}$-eigenline. It lies in $\mathbb{Z}_\ell[g]$ because its denominator is a unit: $(\zeta - \zeta^{-1})^2 = (\zeta + \zeta^{-1})^2 - 4$ is $1 - 4 = -3$ for $m = 3, 6$ and $0 - 4 = -4$ for $m = 4$, units for $\ell$ odd, $\ell \neq 3$; and the split case requires $\ell \neq 3$.) □

**Proposition 6.45** (the commutant and the invariant line of the adjoint)**.** *Let $F$ be a field of characteristic 0, $V$ a two-dimensional $F$-vector space, and $\sigma \in \mathrm{GL}(V)$ with characteristic polynomial $X^2+X+1$ (so $\sigma^3=1$, $\det\sigma=1$).*

*(i) The commutant $\{x \in \mathrm{End}_F(V) : x\sigma = \sigma x\}$ is $F[\sigma] \simeq F[X]/(X^2+X+1) = F \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_3)$, of dimension 2; it is a field iff $F$ contains no primitive cube root of unity (for $F=\mathbb{Q}_\ell$: iff $\ell \not\equiv 1 \pmod 3$). The same holds for $-\sigma$.*

*(ii) The trace-zero part of $F[\sigma]$ is the line $F\cdot(2\sigma+1)$, and $(2\sigma+1)^2 = -3$: it is the line spanned by a square root of $-3$.*

*(iii) If $A \in \mathrm{GL}(V)$ satisfies $A\sigma A^{-1} = \sigma^q$ for a prime $q \neq 3$, then conjugation by $A$ preserves $F[\sigma]$ and acts on it as $\mathrm{id}_F \otimes \mathrm{Frob}_q$, where $\mathrm{Frob}_q \in \mathrm{Gal}(\mathbb{Q}(\zeta_3)/\mathbb{Q})$ is $\zeta_3 \mapsto \zeta_3^q$; hence it acts on the line $F(2\sigma+1)$ by $\chi_{-3}(\mathrm{Frob}_q) = \chi_{-3}(q)$, and so does conjugation by $A^{-1}$.*

*(iv) If $\tau \in \mathrm{GL}(V)$ has characteristic polynomial $X^2+1$, the commutant is $F[\tau] \simeq F \otimes_{\mathbb{Q}} \mathbb{Q}(i)$, its trace-zero part is $F\tau$, and $A\tau A^{-1} = \tau^q$ with $q$ an odd prime gives the action of conjugation by $A^{\pm 1}$ on $F\tau$ by $\chi_{-4}(q)$.*

*(v) Integral form: if $F=\mathbb{Q}_\ell$, $\ell$ odd, and $T \subset V$ is a $\sigma$-stable $\mathbb{Z}_\ell$-lattice, the commutant of $\sigma$ in $\mathrm{End}_{\mathbb{Z}_\ell}(T)$ is $\mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_3]$, and $T$ is free of rank one over it (Proposition 6.44); $\mathrm{End}_{\langle\sigma\rangle}(V) = \mathbb{Q}_\ell[\sigma] \simeq \mathbb{Q}_\ell \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_3)$. For $\ell \equiv 1 \pmod 3$ neither is a domain.*

*Proof.* (i) $\sigma$ is not a scalar, so $V$ has a cyclic vector $v$ ($v, \sigma v$ a basis); an $x$ commuting with $\sigma$ is determined by $x(v) = av + b\sigma v$, so $x = a + b\sigma$. $F[X]/(X^2+X+1)$ is a field iff $X^2+X+1$ has no root in $F$; for $\mathbb{Q}_\ell$ use Proposition 6.43. (ii) $\mathrm{tr}(a+b\sigma) = 2a-b$, which vanishes iff $a+b\sigma = \frac{b}{2}(2\sigma+1)$; and $(2\sigma+1)^2 = 4(\sigma^2+\sigma+1) - 3 = -3$. (iii) $AF[\sigma]A^{-1} = F[\sigma^q] = F[\sigma]$, and conjugation is an $F$-algebra automorphism sending $\sigma \mapsto \sigma^q$, which under $\sigma \leftrightarrow 1\otimes\zeta_3$ is $\mathrm{id}\otimes(\zeta_3 \mapsto \zeta_3^q)$. Explicitly $A(2\sigma+1)A^{-1} = 2\sigma^q + 1$, which is $2\sigma+1$ if $q \equiv 1 \pmod 3$ and $2\sigma^2+1 = -(2\sigma+1)$ if $q \equiv 2 \pmod 3$; and $\chi_{-3}(q) = \pm 1$ accordingly (Proposition 6.35(iii)). Since $\pm 1$ is its own inverse, conjugation by $A^{-1}$ acts by the same sign. (iv) The same with $\mathrm{tr}(a+b\tau) = 2a$ and $\tau^q = \tau$ or $\tau^3 = -\tau$ according to $q$ mod 4. (v) is Proposition 6.44 with $m=3$, tensored with $\mathbb{Q}_\ell$ for $V$; and Proposition 6.43(ii) for $\ell \equiv 1 \pmod 3$. □

*In the paper* 6.46. Lemma lem:IV and the paragraph after it: "$T_3E$ is a torsion-free module of $\mathbb{Z}_3$-rank 2 over the discrete valuation ring $\mathbb{Z}_3[X]/(X^2+X+1) \simeq \mathbb{Z}_3[\zeta_3]$, with $X$ acting as $\sigma$, hence free of rank one over it" is Corollary 6.42(i), with $\sigma$ the generator of $\rho_{E,3}(I_q)$, of order 3 and determinant 1 (Chapter 11); here $\mathbb{Z}_3[\sigma]$, the subring of $\mathrm{End}(T)$ generated by $\sigma$, is isomorphic to $\mathbb{Z}_3[X]/(X^2+X+1)$ because $\sigma$ is not a scalar. "its image in $\mathrm{Aut}(E[3])$ is a nontrivial unipotent subgroup. Consequently $E[3]^{I_q}$ is a line" (statement of Lemma lem:IV) is Corollary 6.42(ii): $E[3] = T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$, $\sigma$ acts as multiplication by $\zeta_3 = 1-\pi$, and its fixed space is $\pi T/3T$. That $\mathbb{Z}_3[\zeta_3]$ is a discrete valuation ring with uniformiser $\pi = 1-\zeta_3$ and residue field $\mathbb{F}_3$ is Theorem 6.41; $\pi^2 = -3$ up to a unit is Proposition 6.39(i). The same freeness is used in Lemma lem:tamadd, with the basis in which $\sigma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$, the companion matrix of $X^2+X+1$ acting on the basis $t, \sigma t$ for a generator $t$. The paper states the freeness at every prime $q \neq 3$ at which $\rho_{E,3}(I_q)$ is cyclic of order 3, $q=2$ included; it holds wherever an element of order 3 generates the 3-Sylow subgroup of $\rho(I_q)$, as in Lemma lem:tamadd, since Corollary 6.42 needs only $\sigma^2+\sigma+1=0$. Only the identification of the order 3 with Kraus's semistability defect needs $q \geq 5$.

*In the paper* 6.47. Lemma lem:euler, types II, II*, IV, IV*: "the invariants form the line spanned by $2\sigma+1$, whose square is $-3$, inside $\mathrm{End}_{I_q}(T)=\mathbb{Z}_\ell[\sigma]\simeq\mathbb{Z}_\ell\otimes\mathbb{Z}[\zeta_3]$ (not a field when $\ell\equiv 1 \pmod 3$)". The invariants of $I_q$ on $\mathrm{ad}^0 V$ are the trace-zero elements of $\mathrm{End}_{I_q}(V)$, the commutant of $\rho(I_q)=\langle\pm\sigma\rangle$, which is the line $\mathbb{Q}_\ell(2\sigma+1)$ of Proposition 6.45(ii); the integral statement is Proposition 6.45(v). "Frobenius ... acts on the line by $\chi_{-3}(q)$" is Proposition 6.45(iii) with $A=\rho(F)$ supplied by Corollary 6.32. Types III, III*: the line spanned by $\tau$ and the sign $\chi_{-4}(q)$, Proposition 6.45(iv); no diagonalisation of $\tau$ over $\mathbb{Q}_\ell$ is needed (it is impossible for $\ell\equiv 3 \pmod 4$, $\mathbb{Q}_\ell[i]$ being a field). The table entries $1-\chi_{-3}(q)q^{-s}$ and $1-\chi_{-4}(q)q^{-s}$ follow with either Frobenius (Convention 6.25). The two rings are kept apart: $\mathrm{End}_{I_q}(V)$ is the $\mathbb{Q}_\ell$-algebra $\mathbb{Q}_\ell\otimes\mathbb{Q}(\zeta_3)$, and $\mathbb{Z}_\ell\otimes\mathbb{Z}[\zeta_3]$ is the endomorphism ring $\mathrm{End}_{I_q}(T)$ of the lattice.

The quadratic characters met so far were defined through quadratic fields. The next section places them, together with the cyclotomic characters $\chi_\ell$ and $\bar\chi_\ell$, inside the cyclotomic fields, and proves that the two descriptions of a character of finite order, as a Dirichlet character and as a character of $G_\mathbb{Q}$, are the same.

## 6.5 Cyclotomic fields and class field theory over $\mathbb{Q}$

The cyclotomic fields are where the characters of the paper live: $\chi_\ell$ and $\bar\chi_\ell$ describe the action on $\mu_{\ell^\infty}$, the quadratic characters $\chi_{-3}$, $\chi_{-4}$, $\chi_{q^*}$, $\chi_S$ factor through $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})\simeq(\mathbb{Z}/n)^\times$, and the theorem of Kronecker and Weber says that every character of $G_\mathbb{Q}$ of finite order does. This section computes the splitting of primes in $\mathbb{Q}(\zeta_n)$, shows that $\mathbb{Q}(\sqrt{\ell^*})$ is the quadratic subfield of $\mathbb{Q}(\zeta_\ell)$ with its splitting law, and sets up the dictionary between Dirichlet characters and characters of $G_\mathbb{Q}$ that the paper uses in both directions: as Galois characters ($\rho\otimes\chi$ in Proposition prop:ns and Lemma lem:h0) and as Dirichlet characters ($a_n(g\otimes\chi)=\chi(n)a_n(g)$ in Lemma lem:twist(iii) and Theorems thm:twist and thm:qual). The Galois theory of $\mathbb{Q}(\zeta_n)$, $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})\simeq(\mathbb{Z}/n)^\times$, $a\mapsto\sigma_a$ with $\sigma_a(\zeta_n)=\zeta_n^a$, is in Chapter 5; the cyclotomic polynomials $\Phi_n$ and their factorisation modulo $p$ are in Chapter 2.

**Theorem 6.48** (rings of integers of cyclotomic fields)**.** *Let $n\geq 1$ and $K=\mathbb{Q}(\zeta_n)$, of degree $\varphi(n)$.*

*(i)* $\mathcal{O}_K=\mathbb{Z}[\zeta_n]$.

*(ii) If $n=\ell^a$ with $\ell$ prime and $\zeta=\zeta_{\ell^a}$, then $\prod_{b\in(\mathbb{Z}/\ell^a)^\times}(1-\zeta^b)=\Phi_{\ell^a}(1)=\ell$, each $1-\zeta^b$ is a unit times $1-\zeta$, and $\ell\mathcal{O}_K=(1-\zeta)^{\varphi(\ell^a)}\mathcal{O}_K$: $\ell$ is totally ramified, and $(1-\zeta)$ is a prime with residue field $\mathbb{F}_\ell$.*

*(iii)* $d_K=(-1)^{\varphi(n)/2}n^{\varphi(n)}/\prod_{p\mid n}p^{\varphi(n)/(p-1)}$*; in particular* $d(\mathbb{Q}(\zeta_\ell))=(-1)^{(\ell-1)/2}\ell^{\ell-2}$ *for an odd prime $\ell$,* $d(\mathbb{Q}(\zeta_3))=-3$, $d(\mathbb{Q}(\zeta_4))=-4$, $d(\mathbb{Q}(\zeta_8))=256$.

*(iv) For $n\not\equiv 2 \pmod 4$, a prime $p$ ramifies in $\mathbb{Q}(\zeta_n)$ iff $p\mid n$. (Recall $\mathbb{Q}(\zeta_n)=\mathbb{Q}(\zeta_{2n})$ for $n$ odd.)*

*Proof.* We prove (i) and (ii) for $n=\ell^a$, (iii) for $n$ an odd prime, and (iv) after Theorem 6.49; the general case of (i) and (iii) is stated, see [201, Chapter 2] or [142, Chapter I, Section 10] (the formula of (iii) was checked against PARI's `nfdisc` for $n\in\{3,4,5,7,8,9,12,15,16,20\}$). (ii) $\Phi_{\ell^a}(X)=\sum_{j=0}^{\ell-1}X^{j\ell^{a-1}}=\prod_b(X-\zeta^b)$, so $\prod_b(1-\zeta^b)=\Phi_{\ell^a}(1)=\ell$. For $\ell\nmid b$, $(1-\zeta^b)/(1-\zeta)=1+\zeta+\cdots+\zeta^{b-1}\in\mathbb{Z}[\zeta]$, and its inverse $(1-\zeta)/(1-\zeta^b)$ is of the same form since $\zeta=(\zeta^b)^{b'}$

with $bb' \equiv 1$. So $\ell = u(1-\zeta)^{\varphi(\ell^a)}$ with $u$ a unit. Writing $(1-\zeta) = \prod \mathfrak{P}^{a_\mathfrak{P}}$, $\ell\mathcal{O}_K = \prod \mathfrak{P}^{\varphi(\ell^a)a_\mathfrak{P}}$ and $\sum_\mathfrak{P} \varphi(\ell^a)a_\mathfrak{P} f_\mathfrak{P} = \varphi(\ell^a)$ (Theorem 6.19) forces a single $\mathfrak{P}$ with $a_\mathfrak{P} = f_\mathfrak{P} = 1$. (i) for $n = \ell^a$: differentiating $(X^{\ell^{a-1}} - 1)\Phi_{\ell^a}(X) = X^{\ell^a} - 1$ and evaluating at $\zeta$, where $\Phi_{\ell^a}(\zeta) = 0$ kills the term containing $\Phi_{\ell^a}$, gives $(\xi - 1)\Phi'_{\ell^a}(\zeta) = \ell^a\zeta^{\ell^a-1} = \ell^a\zeta^{-1}$ with $\xi = \zeta^{\ell^{a-1}}$ a primitive $\ell$-th root of unity, whose norm $N_{K/\mathbb{Q}}(\xi - 1)$ is $\pm$ a power of $\ell$ by (ii). So $\operatorname{disc}(\Phi_{\ell^a}) = \pm N(\Phi'_{\ell^a}(\zeta))$ is $\pm$ a power of $\ell$ (Theorem 6.6(iv)), and the index $[\mathcal{O}_K : \mathbb{Z}[\zeta]]$ is a power of $\ell$. Suppose $\ell$ divides it. Then some $x \in \mathcal{O}_K \setminus \mathbb{Z}[\zeta]$ has $\ell x \in \mathbb{Z}[\zeta]$. With $\pi_\ell = 1 - \zeta$, the elements $1, \pi_\ell, \dots, \pi_\ell^{\varphi-1}$ ($\varphi = \varphi(\ell^a)$) form a $\mathbb{Z}$-basis of $\mathbb{Z}[\zeta]$; write $\ell x = \sum_i c_i\pi_\ell^i$ with not all $c_i$ divisible by $\ell$, and let $j$ be the least index with $\ell \nmid c_j$. Then $\sum_{i\ge j} c_i\pi_\ell^i = \ell x - \sum_{i<j} c_i\pi_\ell^i$ lies in $\ell\mathcal{O}_K = \pi_\ell^\varphi\mathcal{O}_K$ (the terms with $i < j$ have $c_i \in \ell\mathbb{Z}$, and $\ell x \in \ell\mathcal{O}_K$), so its valuation $v_{\pi_\ell}$, the valuation at the prime $(\pi_\ell)$, is at least $\varphi$. Here $j < \varphi$: otherwise all $c_i$ ($i < \varphi$) would be divisible by $\ell$. But $v_{\pi_\ell}(c_j\pi_\ell^j) = j$, because an integer prime to $\ell$ lies in no prime above $\ell$, and $(\pi_\ell)$ is the only such prime, so $v_{\pi_\ell}(c_j) = 0$; while the terms with $i > j$ have valuation $\ge i > j$. If $v(a) < v(b)$ then $v(a+b) = v(a)$, since $v(a+b) \ge v(a)$ and $v(a) = v((a+b)-b) \ge \min(v(a+b), v(b))$ (Theorem 6.13(iv)); so the sum has valuation exactly $j < \varphi$: a contradiction. Hence $\mathcal{O}_K = \mathbb{Z}[\zeta]$. (iii) for an odd prime $\ell$: with $a = 1$, $(\zeta-1)\Phi'_\ell(\zeta) = \ell\zeta^{\ell-1}$, $N(\zeta - 1) = \prod_b(\zeta^b - 1) = \ell$ (as $\ell - 1$ is even) and $N(\ell\zeta^{\ell-1}) = \ell^{\ell-1}N(\zeta)^{\ell-1} = \ell^{\ell-1}$ ($\zeta$ is a unit of norm $\prod_b \zeta^b = \zeta^{\ell(\ell-1)/2} = 1$), so by multiplicativity $N(\Phi'_\ell(\zeta)) = \ell^{\ell-1}/\ell = \ell^{\ell-2}$, and Theorem 6.6(iv) with $n = \ell - 1$ gives $d = (-1)^{(\ell-1)(\ell-2)/2}\ell^{\ell-2}$. Since $(\ell-1)(\ell-2)/2 = \frac{\ell-1}{2}(\ell-2)$ with $\ell - 2$ odd, the sign is $(-1)^{(\ell-1)/2}$. For $\ell = 3, 5, 7$ this gives $-3$, $125$ and $-7^5 = -16807$. □

**Theorem 6.49** (splitting in cyclotomic fields)**.** *Let $n = p^a m$ with $p \nmid m$, and let $f$ be the order of $p$ in $(\mathbb{Z}/m)^\times$. In $\mathbb{Q}(\zeta_n)$ the prime $p$ has $e = \varphi(p^a)$, residue degree $f$, and $g = \varphi(m)/f$ primes above it. Under $(\mathbb{Z}/n)^\times = (\mathbb{Z}/p^a)^\times \times (\mathbb{Z}/m)^\times$ the inertia group of $p$ is $(\mathbb{Z}/p^a)^\times \times \{1\} = \operatorname{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}(\zeta_m))$ and the decomposition group is $(\mathbb{Z}/p^a)^\times \times \langle p\rangle$. For $p \nmid n$, $\operatorname{Frob}_p = \sigma_p$; so $p$ splits completely iff $p \equiv 1 \pmod n$, and $p$ is inert iff $p$ generates $(\mathbb{Z}/n)^\times$ (possible only if this group is cyclic).*

*Proof.* First let $p \nmid n$. $\operatorname{disc}(\Phi_n)$ divides $\operatorname{disc}(X^n - 1) = \pm n^n$ (the discriminant of a product is the product of the discriminants times a square of a resultant), so $p \nmid \operatorname{disc}(\Phi_n)$; then $p$ is unramified (Theorem 6.22, as $d_K \mid \operatorname{disc}(\Phi_n)$) and $p \nmid [\mathcal{O}_K : \mathbb{Z}[\zeta_n]]$ (Theorem 6.6(iii)). Modulo $p$, $\Phi_n$ is a product of $\varphi(n)/f$ distinct irreducible factors of degree $f$, the order of $p$ modulo $n$ (Chapter 2); Theorem 6.20 gives the splitting. For $x = \sum_i a_i\zeta_n^i \in \mathbb{Z}[\zeta_n]$, $\sigma_p(x) = \sum_i a_i\zeta_n^{pi} \equiv (\sum_i a_i\zeta_n^i)^p \pmod p$ (the binomial theorem modulo $p$ and $a_i^p \equiv a_i$). A general $x \in \mathcal{O}_K$ has $kx \in \mathbb{Z}[\zeta_n]$ for the index $k$, which is prime to $p$, and then $k\sigma_p(x) \equiv (kx)^p \equiv kx^p \pmod{\mathfrak{P}}$; so $\sigma_p(x) \equiv x^p \pmod{\mathfrak{P}}$ for every $\mathfrak{P} \mid p$, and $\sigma_p = \operatorname{Frob}_p$ (Proposition 6.27). It has order $f$; it is trivial iff $p \equiv 1 \pmod n$, and generates the group iff $p$ is inert. Now let $p \mid n$. In $\mathbb{Q}(\zeta_m)$, $p$ has $(e, f, g) = (1, f, \varphi(m)/f)$; in $\mathbb{Q}(\zeta_{p^a})$ it has $(\varphi(p^a), 1, 1)$ (Theorem 6.48(ii)). Write $(e', f', g')$ for the invariants of $p$ in $\mathbb{Q}(\zeta_n)$, which contains both fields. By multiplicativity in towers (Theorem 6.19), $e'$ is a multiple of $\varphi(p^a)$ (pass through $\mathbb{Q}(\zeta_{p^a})$) and $f'$ a multiple of $f$ (pass through $\mathbb{Q}(\zeta_m)$), and $g' \ge \varphi(m)/f$ because each of the $\varphi(m)/f$ primes of $\mathbb{Q}(\zeta_m)$ above $p$ has at least one prime of $\mathbb{Q}(\zeta_n)$ above it, and these are distinct. As $e'f'g' = \varphi(n) = \varphi(p^a)\varphi(m) = \varphi(p^a) \cdot f \cdot \varphi(m)/f$, all three inequalities are equalities: $e' = \varphi(p^a)$, $f' = f$, $g' = \varphi(m)/f$. The inertia group has order $\varphi(p^a)$, and its image in $\operatorname{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ is the inertia group there, which is trivial (Theorem 6.24(ii)); so it is $\operatorname{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}(\zeta_m))$, of order $\varphi(p^a)$. The decomposition group contains it and maps onto the decomposition group $\langle p\rangle$ of $\mathbb{Q}(\zeta_m)$, so it is the full preimage $(\mathbb{Z}/p^a)^\times \times \langle p\rangle$. □

*Proof of Theorem 6.48(iv).* By Theorem 6.49, $p$ ramifies iff $p \mid n$ and $\varphi(p^a) > 1$, and $\varphi(p^a) = 1$ only for $p^a = 2$, excluded when $n \not\equiv 2 \pmod 4$. □

**Example 6.50.** (Checked with PARI.) $n=3$: $p\equiv 1 \pmod 3$ splits, $p\equiv 2 \pmod 3$ is inert, 3 is ramified. $n=5$: 11 splits completely; 19 and 29 (order 2 modulo 5) give $f=g=2$; 2 and 3 (order 4) are inert. $n=7$: 2 has order 3 modulo 7, so $f=3$, $g=2$; 13 has order 2, so $f=2$, $g=3$; $29\equiv 1$ splits completely. $n=8$: 2 is totally ramified; 3, 5, 7 have $f=g=2$; 17 splits completely; $(\mathbb{Z}/8)^\times$ is not cyclic, so no prime is inert.

**Proposition 6.51** ($\mathbb{Q}(\sqrt{\ell^*})$)**.** *Let $\ell$ be an odd prime and $\ell^* = (-1)^{(\ell-1)/2}\ell$.*

*(i) $\mathbb{Q}(\sqrt{\ell^*})$ is the unique quadratic subfield of $\mathbb{Q}(\zeta_\ell)$ (Chapter 5); it is ramified only at $\ell$, with discriminant $\ell^*$.*

*(ii) For every prime $p\neq\ell$, including $p=2$, $\chi_{\ell^*}(\mathrm{Frob}_p) = \left(\frac{p}{\ell}\right)$: $p$ splits in $\mathbb{Q}(\sqrt{\ell^*})$ iff $p$ is a square modulo $\ell$, and $p$ is* inert *iff $p$ is a non-square modulo $\ell$. The inert primes have density $\frac{1}{2}$ (Corollary 6.61).*

*(iii) For $\ell=3$, $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$, and the inert primes are the $p\equiv 2 \pmod 3$.*

*(iv) $\mathbb{Q}(\sqrt{\ell^*})$ is imaginary iff $\ell\equiv 3 \pmod 4$.*

*Proof.* (i) $\mathrm{Gal}(\mathbb{Q}(\zeta_\ell)/\mathbb{Q}) \simeq \mathbb{F}_\ell^\times$ is cyclic of even order $\ell-1$, so it has a unique subgroup of index 2 and $\mathbb{Q}(\zeta_\ell)$ has a unique quadratic subfield $M$ (Chapter 5). A prime $p\neq\ell$ is unramified in $\mathbb{Q}(\zeta_\ell)$ (Theorem 6.48(iv)), hence in $M$ (ramification indices multiply in towers, Theorem 6.19); and some prime ramifies in $M\neq\mathbb{Q}$ (Theorem 6.22). So $M$ is ramified exactly at $\ell$, and Proposition 6.36(i) gives $M=\mathbb{Q}(\sqrt{\ell^*})$, of discriminant $\ell^*$. For example $\mathbb{Q}(\zeta_5)\supset\mathbb{Q}(\sqrt{5})$, as $\zeta_5+\zeta_5^{-1} = \frac{-1+\sqrt{5}}{2}$, and $\mathbb{Q}(\zeta_7)\supset\mathbb{Q}(\sqrt{-7})$. (ii) In $\mathrm{Gal}(\mathbb{Q}(\zeta_\ell)/\mathbb{Q})\simeq\mathbb{F}_\ell^\times$, $\mathrm{Frob}_p=\sigma_p$ corresponds to $p$ mod $\ell$ (Theorem 6.49). Its restriction to $\mathbb{Q}(\sqrt{\ell^*})$ is $\mathrm{Frob}_p$ there (Proposition 6.27(v)), which is trivial iff $\sigma_p$ lies in the unique subgroup of index 2 of the cyclic group $\mathbb{F}_\ell^\times$, the squares. (iii) $\zeta_3 = (-1+\sqrt{-3})/2$. (iv) $\ell^*<0$ iff $(\ell-1)/2$ is odd. □

**Corollary 6.52** (quadratic reciprocity, second proof)**.** *For distinct odd primes $p,q$: $\left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right)$, equivalently $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$. Moreover $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$ and $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$.*

*Proof.* $\chi_{q^*}(\mathrm{Frob}_p)$ equals $\left(\frac{q^*}{p}\right)$ by Proposition 6.35(i) together with Theorem 6.34 (whose proof uses only the meaning of the Legendre symbol, not reciprocity), and $\left(\frac{p}{q}\right)$ by Proposition 6.51(ii). Since $\left(\frac{q^*}{p}\right) = \left(\frac{-1}{p}\right)^{(q-1)/2}\left(\frac{q}{p}\right)$, the second form follows from the first supplement. For the supplements: $\mathbb{Q}(i)=\mathbb{Q}(\zeta_4)$, and $\sigma_p(i) = i^p = (-1)^{(p-1)/2}i$, so $\left(\frac{-4}{p}\right) = \chi_{-4}(\mathrm{Frob}_p) = (-1)^{(p-1)/2}$; and $\sqrt{2}=\zeta_8+\zeta_8^{-1}\in\mathbb{Q}(\zeta_8)$, with $\sigma_p(\sqrt{2}) = \zeta_8^p+\zeta_8^{-p}$, which is $\sqrt{2}$ if $p\equiv\pm1 \pmod 8$ and $-\sqrt{2}$ if $p\equiv\pm3 \pmod 8$; so $\left(\frac{8}{p}\right) = \left(\frac{2}{p}\right) = \chi_8(\mathrm{Frob}_p) = (-1)^{(p^2-1)/8}$. □

Chapter 2 proves reciprocity by Gauss sums; the proof above explains why the character of $\mathbb{Q}(\sqrt{q^*})$ is the Legendre symbol modulo $q$, which is what Lemma lem:twist means by $\chi_{q^*}$, "the quadratic character of conductor $q$".

**Theorem 6.53** (Kronecker-Weber)**.** *Every finite abelian extension $K/\mathbb{Q}$ is contained in $\mathbb{Q}(\zeta_n)$ for some $n$. The least such $n$ with $n\not\equiv 2 \pmod 4$ is the* conductor *$f_K$ of $K$, and the primes dividing $f_K$ are exactly the primes ramified in $K$. Equivalently, the maximal abelian extension of $\mathbb{Q}$ is $\mathbb{Q}^{\mathrm{ab}} = \mathbb{Q}(\mu_\infty) = \bigcup_n \mathbb{Q}(\zeta_n)$, and the cyclotomic characters give $G_{\mathbb{Q}}^{\mathrm{ab}} = \mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \simeq \varprojlim_n (\mathbb{Z}/n)^\times = \hat{\mathbb{Z}}^\times = \prod_\ell \mathbb{Z}_\ell^\times$. For a quadratic field of discriminant $D$, $f_K = |D|$.*

*Reference.* We do not prove this; see [201, Chapter 14] or [142, Chapter V, Section 1, and Chapter VI] (chapter-level pointers). For a quadratic field the inclusion $K \subseteq \mathbb{Q}(\zeta_{|D|})$ is elementary (Chapter 5), and $f_K = |D|$ is Theorem 6.55(i) applied to the primitive character $n \mapsto (\frac{D}{n})$ of conductor $|D|$ (Chapter 2), which corresponds to $\chi_D$ by Proposition 6.35(i). □

**Theorem 6.54** (Artin reciprocity over $\mathbb{Q}$, explicit form)**.** *Let $K/\mathbb{Q}$ be abelian with $K \subseteq \mathbb{Q}(\zeta_n)$.*

*(i) The* Artin map $\mathrm{Art}_K\colon (\mathbb{Z}/n)^\times \to \mathrm{Gal}(K/\mathbb{Q})$, $a \mapsto \sigma_a|_K$, *is surjective with kernel $H_K$, and $K \mapsto H_K$ is an inclusion-reversing bijection between the subfields of $\mathbb{Q}(\zeta_n)$ and the subgroups of $(\mathbb{Z}/n)^\times$.*

*(ii) For every prime $p \nmid n$, $\mathrm{Frob}_p(K/\mathbb{Q}) = \mathrm{Art}_K(p \bmod n)$: the Frobenius of an unramified prime in an abelian extension depends only on $p \bmod n$. So $p \nmid n$ splits completely in $K$ iff $p \bmod n \in H_K$, and the residue degree of $p$ in $K$ is the order of $p$ in $(\mathbb{Z}/n)^\times/H_K$.*

*(iii) (Stated.) In idelic form, the global Artin map $\mathbb{Q}^\times\backslash\mathbb{A}_\mathbb{Q}^\times \to G_\mathbb{Q}^{\mathrm{ab}}$ is surjective with kernel the connected component of the identity, and $\mathbb{A}_\mathbb{Q}^\times = \mathbb{Q}^\times \cdot (\mathbb{R}_{>0} \times \hat{\mathbb{Z}}^\times)$. In the normalisation in which a uniformiser at $p$ maps to an arithmetic Frobenius, an idele $(1, \dots, 1, u_p, 1, \dots)$ with $u_p \in \mathbb{Z}_p^\times$ maps to an element of the inertia group at $p$ acting on $\mu_{p^\infty}$ by $u_p^{-1}$. Deligne's normalisation sends uniformisers to geometric Frobenius elements, and then the sign is reversed.*

*Proof.* (i) is Galois theory (Chapter 5) for $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \simeq (\mathbb{Z}/n)^\times$. (ii) By Proposition 6.27(v) and Theorem 6.49, $\mathrm{Frob}_p(K/\mathbb{Q}) = \mathrm{Frob}_p(\mathbb{Q}(\zeta_n)/\mathbb{Q})|_K = \sigma_p|_K$; the residue degree is the order of $\mathrm{Frob}_p$ (Proposition 6.27(i)). (iii) We do not prove this; see [142, Chapter VI, Sections 5 and 6] or [133, Chapter V]; nothing in the paper needs it. □

**Theorem 6.55** (Dirichlet characters and characters of $G_\mathbb{Q}$)**.** *(i) Let $\chi$ be a primitive Dirichlet character, of conductor $f_\chi$ (Chapter 2), and let $\psi_\chi$ be the composite*

$$G_\mathbb{Q} \to \mathrm{Gal}(\mathbb{Q}(\zeta_{f_\chi})/\mathbb{Q}) \simeq (\mathbb{Z}/f_\chi)^\times \xrightarrow{\chi} \mathbb{C}^\times.$$

*Then $\psi_\chi(\mathrm{Frob}_p) = \chi(p)$ for every prime $p \nmid f_\chi$, and $\chi \mapsto \psi_\chi$ is a bijection from primitive Dirichlet characters to continuous homomorphisms $G_\mathbb{Q} \to \mathbb{C}^\times$ (these have finite image, Chapter 5). Each $\psi$ is determined by its values at the $\mathrm{Frob}_p$, $p \nmid f_\chi$. The field cut out by $\psi_\chi$ has conductor $f_\chi$, and $\psi_{\chi\chi'} = \psi_\chi\psi_{\chi'}$ when $\chi\chi'$ denotes the primitive character inducing the product.*

*(ii) $\psi_\chi$ is unramified at $p$ iff $p \nmid f_\chi$. More precisely, if $f_\chi = p^a m$ with $p \nmid m$, then $\psi_\chi(I_p) = \chi((\mathbb{Z}/p^a)^\times \times \{1\})$.*

*(iii) The same holds for characters with values in $\overline{\mathbb{Q}}_\ell^\times$ or $\overline{\mathbb{F}}_\ell^\times$ of finite order (through a fixed embedding of roots of unity), and every continuous $\psi\colon G_\mathbb{Q} \to \mathbb{Z}_\ell^\times$ factors through $G_\mathbb{Q}^{\mathrm{ab}} \simeq \hat{\mathbb{Z}}^\times$; for instance $\chi_\ell$ is the projection $\hat{\mathbb{Z}}^\times \to \mathbb{Z}_\ell^\times$, and $\bar{\chi}_\ell = \psi_{\omega_\ell}$ for the character $\omega_\ell\colon a \mapsto a \bmod \ell$ of $(\mathbb{Z}/\ell)^\times$.*

*(iv) (Stated; Chapters 13 and 15.) Under this dictionary, twisting a newform's coefficients, $a_n \mapsto \chi(n)a_n$, corresponds to tensoring its Galois representation with $\psi_\chi$.*

*Proof.* (i) $\psi_\chi(\mathrm{Frob}_p) = \chi(p)$ by Theorem 6.49. Injectivity: $G_\mathbb{Q} \to (\mathbb{Z}/f_\chi)^\times$ is surjective, so $\psi_\chi$ determines $\chi$ on $(\mathbb{Z}/f_\chi)^\times$, and $\psi_\chi$ determines $f_\chi$ as below. Surjectivity: a continuous $\psi\colon G_\mathbb{Q} \to \mathbb{C}^\times$ has finite abelian image, and its kernel cuts out a finite abelian extension, contained in some $\mathbb{Q}(\zeta_n)$ (Theorem 6.53); so $\psi$ factors as $G_\mathbb{Q} \to (\mathbb{Z}/n)^\times \xrightarrow{\chi'} \mathbb{C}^\times$, and $\psi = \psi_\chi$ for the primitive $\chi$ inducing $\chi'$ (restriction from $\mathbb{Q}(\zeta_n)$ to $\mathbb{Q}(\zeta_{f_\chi})$ is reduction $(\mathbb{Z}/n)^\times \to (\mathbb{Z}/f_\chi)^\times$). That $\psi$ is determined by its values at Frobenius elements outside a finite set follows from the density of Frobenius elements (Corollary 6.61), or directly from Dirichlet's theorem on primes in progressions. The field $K$ cut out by $\psi_\chi$ lies in $\mathbb{Q}(\zeta_{f_\chi})$; if it lay in $\mathbb{Q}(\zeta_{f'})$ with $f' \mid f_\chi$, $f' < f_\chi$, then $\chi$ would be trivial on the kernel of $(\mathbb{Z}/f_\chi)^\times \to (\mathbb{Z}/f')^\times$, hence induced from a character modulo $f'$, contradicting primitivity; and if $K \subseteq \mathbb{Q}(\zeta_n)$ then $K \subseteq \mathbb{Q}(\zeta_n) \cap \mathbb{Q}(\zeta_{f_\chi}) = \mathbb{Q}(\zeta_{\gcd(n,f_\chi)})$ (the intersection contains $\mathbb{Q}(\zeta_{\gcd})$ and has degree $\varphi(n)\varphi(f_\chi)/\varphi(\mathrm{lcm}) = \varphi(\gcd)$, by Galois theory of the compositum $\mathbb{Q}(\zeta_{\mathrm{lcm}})$). So the least $n$ is $f_\chi$, which is $\not\equiv 2 \pmod 4$ for a primitive character. Multiplicativity is clear on a common modulus. (ii) By Theorem 6.49 the image of $I_p$ in $\mathrm{Gal}(\mathbb{Q}(\zeta_{f_\chi})/\mathbb{Q})$ is $(\mathbb{Z}/p^a)^\times \times \{1\}$; $\chi$ is nontrivial on it when $a \geq 1$, by primitivity (otherwise $\chi$ would factor through $(\mathbb{Z}/m)^\times$). (iii) The same proofs; the factorisation through $\hat{\mathbb{Z}}^\times$ is Theorem 6.53, since a continuous homomorphism to the abelian group $\mathbb{Z}_\ell^\times$ factors through $G_\mathbb{Q}^{\mathrm{ab}}$. $\bar\chi_\ell(\mathrm{Frob}_p) = p \bmod \ell = \omega_\ell(p)$. (iv) We do not prove this here; see Chapters 13 and 15. □

**Example 6.56** (the paper's characters in both languages)**.** For $p$ unramified (arithmetic Frobenius):

| Galois character | field cut out | Dirichlet character | conductor | value at $\mathrm{Frob}_p$ |
|---|---|---|---|---|
| $\chi_{-4}$ | $\mathbb{Q}(i) = \mathbb{Q}(\zeta_4)$ | the nontrivial character mod 4 | 4 | $(-1)^{(p-1)/2}$ |
| $\chi_{-3} = \bar\chi_3^{\mathrm{cyc}}$ | $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$ | $\left(\frac{\cdot}{3}\right)$ | 3 | 1 iff $p \equiv 1 \pmod 3$ |
| $\chi_{q^*}$, $q$ odd | $\mathbb{Q}(\sqrt{q^*}) \subseteq \mathbb{Q}(\zeta_q)$ | $\left(\frac{\cdot}{q}\right)$ | $q$ | $\left(\frac{p}{q}\right)$ |
| $\chi_S = \prod_{q\in S} \chi_{q^*}$ | $\mathbb{Q}\left(\sqrt{\prod_{q\in S} q^*}\right)$ | $\prod_{q\in S}\left(\frac{\cdot}{q}\right)$ | $\prod_{q\in S} q$ | $\prod_{q\in S}\left(\frac{p}{q}\right)$ |
| $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$ | $\mathbb{Q}(\sqrt{\ell^*})$ | $\left(\frac{\cdot}{\ell}\right)$ | $\ell$ | $\left(\frac{p}{\ell}\right)$ |
| $\bar\chi_\ell$ (values in $\mathbb{F}_\ell^\times$) | $\mathbb{Q}(\zeta_\ell)$ | $\omega_\ell\colon a \mapsto a \bmod \ell$ | $\ell$ | $p \bmod \ell$ |

The identity $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$ means: the quadratic character $\left(\frac{\cdot}{\ell}\right)$ equals $a \mapsto a^{(\ell-1)/2} \bmod \ell$ (Euler's criterion), read in $\{\pm 1\} \subset \mathbb{F}_\ell^\times$. The Dirichlet character $\left(\frac{\cdot}{q}\right)$ equals $\left(\frac{q^*}{\cdot}\right)$ by quadratic reciprocity (Chapter 2), so the two descriptions of the column agree.

*Warning* 6.57. $\chi_{\ell^*}$ and $\bar\chi_\ell$ coincide only for $\ell = 3$: for $\ell \geq 5$, $\bar\chi_\ell$ has order $\ell - 1 > 2$. The cyclotomic character $\chi_\ell$ itself has infinite image $\mathbb{Z}_\ell^\times$. So "$\rho \simeq \rho \otimes \chi_\ell$" (impossible for every two-dimensional $\rho$: taking determinants, $\chi_\ell^2 = 1$, while $\chi_\ell(G_\mathbb{Q}) = \mathbb{Z}_\ell^\times$ is infinite), "$\bar\rho \simeq \bar\rho \otimes \bar\chi_\ell$" (possible only for $\ell = 3$, by determinants: $\bar\chi_\ell^2 = 1$ forces $\ell - 1 \mid 2$) and "$\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$" (a statement about the quadratic field $\mathbb{Q}(\sqrt{\ell^*})$, Proposition 6.62) are three different conditions. The paper's term $h_\ell(E)$ (Lemma lem:h0) is governed by the second, and the hypothesis of Theorem thm:main by the third.

*In the paper* 6.58. Section 1.4 and Lemma lem:twist: $E^{(q)}$ is the twist by $\chi_{q^*}$, "the quadratic character of conductor $q$", that is by $\mathbb{Q}(\sqrt{q^*})$; it is the character $\left(\frac{\cdot}{q}\right)$ of Example 6.56, ramified only at $q$ (Proposition 6.36), which is why Lemma lem:twist(i) holds at every $p \neq q$. Theorems thm:twist and thm:quant: $\chi_S = \prod_{q\in S} \chi_{q^*}$ and $\prod_{q\in S} q$ (the paper's $D$) are the character and the conductor of $\mathbb{Q}(\sqrt{\prod_{q\in S} q^*})$ (Proposition 6.36(iii)), and "the twist is unramified outside $S$" is Proposition 6.36(iv). The map $\iota\colon h \mapsto \sum \chi_S(n) a_n(h) q^n$ uses $\chi_S$ as a Dirichlet character, and

$\rho_{E_S} \simeq \rho_E \otimes \chi_S$ uses it as a Galois character; the two are the same character by Theorem 6.55. The table of Section 1 uses $\chi_{-3}(q)$ and $\chi_{-4}(q)$ as values of Dirichlet characters, equal to the Galois values at $\mathrm{Frob}_q$ (Proposition 6.35(iii)), which is how they enter the proof of Lemma lem:euler. Proposition prop:forms uses $q \equiv \chi_{-3}(q) \pmod 3$, which holds for every prime $q \neq 3$ by Proposition 6.35(iii).

Up to now every statement about Frobenius elements has concerned one prime at a time. The next section asks how the Frobenius classes are distributed as $p$ varies; the answer, Chebotarev's theorem, is what allows a representation to be recognised from its traces of Frobenius.

## 6.6 Densities, the Chebotarev density theorem, complex multiplication

The paper uses Chebotarev's theorem to compare Galois representations through their Frobenius traces (Lemma lem:old, "by Chebotarev and Brauer-Nesbitt"). The opposite direction needs no density theorem: in Section 3 of the paper the optimal-level theorem gives $\bar\rho_g \simeq \bar\rho$, and $a_p(g) \equiv a_p(f)$ for all $p \nmid 3N$ follows by taking traces of Frobenius (Proposition 6.30(iv)). This section states the theorem with the Frobenius of Proposition 6.27 and over any number field, proves the corollaries the book needs, gives the criterion for a representation to be isomorphic to its twist by a quadratic character, and explains why $a_p(E) = 0$ at the inert primes of a quadratic field happens only for curves with complex multiplication by that field. The form of Chebotarev over $\mathbb{Q}$ and its Brauer-Nesbitt corollaries are in Chapter 5.

**Proposition 6.59** (the Dedekind zeta function and densities)**.** *Let $K$ be a number field of degree $n$.*

*(i)* $\zeta_K(s) = \sum_{\mathfrak{a}} N\mathfrak{a}^{-s} = \prod_{\mathfrak{p}} (1 - N\mathfrak{p}^{-s})^{-1}$ *(sum over the nonzero ideals, product over the nonzero primes) converges absolutely for* $\operatorname{Re} s > 1$.

*(ii) (Stated.)* $\zeta_K$ *extends meromorphically to* $\operatorname{Re} s > 1 - 1/n$ *with a simple pole at* $s = 1$*, whose residue is given by the analytic class number formula.*

*(iii) Given (ii):* $\sum_{\mathfrak{p}} N\mathfrak{p}^{-s} \sim \log \frac{1}{s-1}$ *as* $s \to 1^+$*. So the* Dirichlet density *of a set $X$ of primes of $K$,* $\delta(X) = \lim_{s\to 1^+} \sum_{\mathfrak{p}\in X} N\mathfrak{p}^{-s} / \log \frac{1}{s-1}$ *when the limit exists, is defined as in Chapter 5 with $N\mathfrak{p}$ in place of $p$. The primes $\mathfrak{p}$ with $f(\mathfrak{p} \mid p) \geq 2$ have density* 0*, so the primes of degree one have density* 1*; finite sets have density* 0.

*Proof.* (i) By unique factorisation (Theorem 6.13) and multiplicativity of the norm (Proposition 6.14), the expansion of the finite products $\prod_{N\mathfrak{p}\leq x}(1 - N\mathfrak{p}^{-s})^{-1}$ contains each $N\mathfrak{a}^{-s}$ with $\mathfrak{a}$ composed of these primes exactly once. For real $s = \sigma > 1$, at most $n$ primes lie above each rational $p$, each with $N\mathfrak{p} \geq p$ (Theorem 6.19), so $\prod_{\mathfrak{p}}(1-N\mathfrak{p}^{-\sigma})^{-1} \leq \prod_p (1-p^{-\sigma})^{-n} = \zeta(\sigma)^n < \infty$; this bounds all partial sums of the series of positive terms. (ii) We do not prove this; see [142, Chapter VII, Section 5] or [108, Chapter VIII]. (iii) $\log \zeta_K(s) = \sum_{\mathfrak{p}} \sum_{m\geq 1} N\mathfrak{p}^{-ms}/m$ (take logarithms in the Euler product, $-\log(1-x) = \sum_m x^m/m$), and for real $s > 1$ the terms with $m \geq 2$ are bounded by $n \sum_p \sum_{m\geq 2} p^{-m} = n \sum_p \frac{p^{-2}}{1-p^{-1}} \leq n \sum_p \frac{2}{p^2} < \infty$; so $\log \zeta_K(s) = \sum_{\mathfrak{p}} N\mathfrak{p}^{-s} + O(1)$. By (ii) $\zeta_K(s) = \frac{c}{s-1}(1 + o(1))$ as $s \to 1^+$ with $c > 0$ the residue (positive, as $\zeta_K(s) > 0$ for real $s > 1$), so $\log \zeta_K(s) = \log \frac{1}{s-1} + \log c + o(1)$, which gives the asymptotic. The primes with $f \geq 2$ have $N\mathfrak{p} \geq p^2$, and $\sum_{f(\mathfrak{p})\geq 2} N\mathfrak{p}^{-s} \leq n \sum_p p^{-2}$ stays bounded. □

**Theorem 6.60** (Chebotarev density theorem)**.** *Let $L/K$ be a finite Galois extension of number fields with group $G$ and $C \subseteq G$ a conjugacy class. The set of primes $\mathfrak{p}$ of $K$, unramified in $L$, with $\mathrm{Frob}_\mathfrak{p} = C$ (Proposition 6.27(ii)) has Dirichlet density $\#C/\#G$. It also has natural density $\#C/\#G$ (counting the primes with $N\mathfrak{p} \le x$). For $G$ abelian the classes are elements, and the statement contains Dirichlet's theorem on primes in arithmetic progressions.*

*Reference.* We do not prove this; see [142, Chapter VII, Section 13] (the theorem is numbered 13.4 there; this number and the others in this chapter are pointers not checked against a printed copy) or [178, Chapter I, Section 2.2]. The idea: for $G$ cyclic it follows from the nonvanishing at $s = 1$ of the Hecke $L$-functions of $K$ (abelian class field theory), and the general case reduces to the cyclic one by restricting to the cyclic subgroup generated by an element of $C$. Only the Dirichlet density is used in this book; the natural density is not used in any proof. □

**Corollary 6.61.** *In the setting of Theorem 6.60:*

*(i) every $\sigma \in G$ is $\mathrm{Frob}_\mathfrak{P}$ for infinitely many $\mathfrak{P}$ (lying over a set of primes $\mathfrak{p}$ of positive density);*

*(ii) the primes of $K$ that split completely in $L$ have density $1/[L : K]$; if $M/K$ is a finite extension, not necessarily Galois, the primes of $K$ having a prime of degree one above them in $M$ have positive density;*

*(iii) (Bauer) for $L$, $L'$ Galois over $K$: $L \subseteq L'$ iff every prime of $K$ that splits completely in $L'$, outside a set of density zero, splits completely in $L$; so a Galois extension of $K$ is determined by its set of split primes;*

*(iv) (Dirichlet) for $K = \mathbb{Q}$, $L = \mathbb{Q}(\zeta_n)$: each class $a \in (\mathbb{Z}/n)^\times$ contains a set of primes of density $1/\varphi(n)$;*

*(v) for $L/\mathbb{Q}$ quadratic, the split and the inert primes each have density $\frac{1}{2}$; in particular the primes inert in $\mathbb{Q}(\sqrt{\ell^*})$ (the $p$ with $\left(\frac{p}{\ell}\right) = -1$) have density $\frac{1}{2}$, and for $\ell = 3$ these are the $p \equiv 2 \pmod 3$;*

*(vi) if $L/\mathbb{Q}$ is Galois, possibly infinite, and unramified outside a finite set, the Frobenius elements at the primes outside any set of density zero are dense in $\mathrm{Gal}(L/\mathbb{Q})$ (Chapter 5).*

*Proof.* (i) Take the class of $\sigma$; a set of positive density is infinite (finite sets have density 0), and for each such $\mathfrak{p}$ some $\mathfrak{P} \mid \mathfrak{p}$ has $\mathrm{Frob}_\mathfrak{P} = \sigma$ (Proposition 6.27(ii)). (ii) $C = \{1\}$ and Proposition 6.27(iii). For $M$, let $L$ be a Galois closure of $M/K$, $G = \mathrm{Gal}(L/K)$, $H = \mathrm{Gal}(L/M)$; by Proposition 6.28 (whose proof is the same over $K$), $\mathfrak{p}$ unramified in $L$ has a degree-one prime in $M$ iff $\langle \mathrm{Frob}_\mathfrak{P} \rangle$ fixes a point of $G/H$, which holds when $\mathrm{Frob}_\mathfrak{P} = 1$; so the set contains the split primes, of density $1/[L : K] > 0$. (iii) If $L \subseteq L'$, a prime split completely in $L'$ is split completely in $L$ (Theorem 6.19). Conversely, consider the compositum $LL'$, Galois over $K$, and the injection $\mathrm{Gal}(LL'/K) \to \mathrm{Gal}(L/K) \times \mathrm{Gal}(L'/K)$ by restriction, which sends $\mathrm{Frob}_\mathfrak{P}$ to the pair of Frobenius elements (Proposition 6.27(v)). So a prime unramified in $LL'$ splits completely in $LL'$ iff it splits completely in $L$ and in $L'$; by hypothesis this is the set of primes splitting completely in $L'$, up to density zero. By (ii), $1/[LL' : K] = 1/[L' : K]$, so $LL' = L'$ and $L \subseteq L'$. (iv) $\mathrm{Frob}_p = \sigma_p$ corresponds to $p \bmod n$ (Theorem 6.49), and $\#G = \varphi(n)$. (v) $G$ has two classes of size 1; for $\mathbb{Q}(\sqrt{\ell^*})$ use Proposition 6.51(ii). (vi) is proved in Chapter 5 from Theorem 6.60. □

**Proposition 6.62** (detecting a quadratic twist)**.** *Let $k$ be a field of characteristic $\neq 2$ (a finite field, $\overline{\mathbb{F}}_\ell$, or a finite extension of $\mathbb{Q}_\ell$), $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(k)$ continuous, semisimple, unramified outside a finite set $S$, and $K$ a quadratic field with character $\chi_K$. The following are equivalent: (a) $\rho \simeq \rho \otimes \chi_K$; (b) $\operatorname{tr}\rho(\mathrm{Frob}_p) = 0$ for every prime $p \notin S$ inert in $K$; (c) $\operatorname{tr}\rho(\mathrm{Frob}_p) = 0$ for all primes inert in $K$ outside a set of density zero.*

*Proof.* (a)⇒(b): for $p \notin S$ unramified in $K$ and inert, $\operatorname{tr}\rho(\mathrm{Frob}_p) = \operatorname{tr}(\rho\otimes\chi_K)(\mathrm{Frob}_p) = \chi_K(\mathrm{Frob}_p)\operatorname{tr}\rho(\mathrm{Frob}_p) = -\operatorname{tr}\rho(\mathrm{Frob}_p)$, and 2 is invertible in $k$; a prime inert in $K$ is unramified in $K$. (b)⇒(c) is trivial. (c)⇒(a): $\rho\otimes\chi_K$ is semisimple (it has the same invariant subspaces as $\rho$, $\chi_K$ acting by scalars) and unramified outside $S$ and the primes ramified in $K$. For $p$ outside this finite set, $\det(X - (\rho\otimes\chi_K)(\mathrm{Frob}_p)) = X^2 - \chi_K(\mathrm{Frob}_p)\operatorname{tr}\rho(\mathrm{Frob}_p)X + \det\rho(\mathrm{Frob}_p)$, which equals $\det(X-\rho(\mathrm{Frob}_p))$ when $p$ splits, and when $p$ is inert and $\operatorname{tr}\rho(\mathrm{Frob}_p)=0$. By (c) the characteristic polynomials agree at all primes outside a set of density zero, so $\rho \simeq \rho\otimes\chi_K$ by the Chebotarev and Brauer-Nesbitt theorem of Chapter 5 (both being semisimple). □

Two applications. (a) If $\bar\rho_{E,3}$ is irreducible, then $\bar\rho_{E,3} \simeq \bar\rho_{E,3}\otimes\chi_{-3}$ iff $3 \mid a_p(E)$ for all primes $p \equiv 2 \pmod 3$ of good reduction (outside a density zero set), since $\operatorname{tr}\bar\rho_{E,3}(\mathrm{Frob}_p) = a_p \bmod 3$ and the inert primes of $\mathbb{Q}(\sqrt{-3})$ are the $p \equiv 2 \pmod 3$ (Chapter 5 gives the example $y^2 = x^3 - 6x - 8$). (b) If $\rho_{E,\ell}$ is semisimple (it always is, by Faltings's theorem, stated in Chapter 11; and it is irreducible when $\bar\rho_{E,\ell}$ is), then $\rho_{E,\ell} \simeq \rho_{E,\ell}\otimes\chi_K$ iff $a_p(E) = 0$ for almost all $p$ inert in $K$ (outside a density zero set). The criterion concerns twisting by a quadratic character; it says nothing about $\rho\otimes\chi_\ell$, which is never isomorphic to $\rho$ (Warning 6.57).

**Theorem 6.63** (complex multiplication)**.** *Let $E/\mathbb{Q}$ be an elliptic curve with complex multiplication, that is $\mathrm{End}_{\overline{\mathbb{Q}}}(E) \neq \mathbb{Z}$. Then $\mathrm{End}_{\overline{\mathbb{Q}}}(E)$ is an order $\mathcal{O}$ in an imaginary quadratic field $K$, the endomorphisms are defined over $K$, and $\sigma \in G_{\mathbb{Q}}$ acts on them through $\mathrm{Gal}(K/\mathbb{Q})$, by complex conjugation of $\mathcal{O} \subset K$ when $\chi_K(\sigma) = -1$ (Chapter 9).*

*(i) (Stated.) $\mathcal{O}$ has class number one; so $K = \mathbb{Q}(\sqrt{-d})$ with $d \in \{1,2,3,7,11,19,43,67,163\}$, and $j(E)$ is one of 13 rational values.*

*(ii) For every prime $\ell$, $\rho_{E,\ell} \simeq \rho_{E,\ell}\otimes\chi_K$, and $a_p(E) = 0$ for every prime $p$ of good reduction inert in $K$.*

*(iii) Every prime ramified in $K$ is a prime of bad reduction of $E$.*

*(iv) (Stated.) Conversely, if $a_p(E) = 0$ for all primes $p$ inert in a quadratic field $K$ outside a set of density zero, then $E$ has complex multiplication by $K$; in particular $K$ is imaginary.*

*(v) (Stated.) If $E$ has no complex multiplication, the set of primes $p$ with $a_p(E) = 0$ has density 0.*

*(vi) Consequently, for an odd prime $\ell$, the condition "$a_p(E) = 0$ for almost all $p$ inert in $\mathbb{Q}(\sqrt{\ell^*})$" holds if and only if $E$ has complex multiplication by $\mathbb{Q}(\sqrt{\ell^*})$; this needs $\ell \equiv 3 \pmod 4$ and $\ell \in \{3,7,11,19,43,67,163\}$, and then $\ell$ divides the conductor of $E$ by (iii). It never happens when $\ell \nmid N$.*

*Sketch of proof.* The structure of $\mathrm{End}_{\overline{\mathbb{Q}}}(E)$ and the Galois action, $\rho_{E,\ell}(\sigma)\alpha\rho_{E,\ell}(\sigma)^{-1} = \sigma(\alpha)$ on $V_\ell E$ for $\alpha \in \mathrm{End}_{\overline{\mathbb{Q}}}(E)$, are quoted from Chapter 9 and [184, Chapter II, Section 2]. (ii) Choose $\alpha \in \mathcal{O}$ with $\bar\alpha = -\alpha$, $\alpha \neq 0$ (for instance $\alpha = f\sqrt{-d}$ for a suitable integer $f$); it acts

invertibly on $V_\ell E$, and $\rho(\sigma)\alpha\rho(\sigma)^{-1} = \sigma(\alpha) = \chi_K(\sigma)\alpha$. So $\alpha\rho(\sigma)\alpha^{-1} = \chi_K(\sigma)\rho(\sigma)$: $\alpha$ is an isomorphism $\rho \to \rho \otimes \chi_K$. For $p \neq \ell$ of good reduction and inert in $K$, $\chi_K(\mathrm{Frob}_p) = -1$ gives $a_p = \operatorname{tr}\rho(\mathrm{Frob}_p) = -a_p$, so $a_p = 0$ (choose $\ell \neq p$, and $a_p \in \mathbb{Z}$). (iii) If $p$ is ramified in $K$ and $E$ had good reduction at $p$, take $\ell \neq p$: $\rho$ is unramified at $p$, and there is $\tau \in I_p$ with $\chi_K(\tau) = -1$ (Proposition 6.35(ii)); then $\alpha = \rho(\tau)\alpha\rho(\tau)^{-1} = \chi_K(\tau)\alpha = -\alpha$, a contradiction. (i) is the theorem of Heegner, Baker and Stark on imaginary quadratic fields of class number one, together with the fact that $j(E) \in \mathbb{Q}$ generates the ring class field of $\mathcal{O}$; see [184, Chapter II and Appendix A] and [34, Sections 7, 11 and 12]. (iv) We only indicate the argument. Let $L$ be the compositum of $K$ and the field cut out by $\rho_{E,\ell}$, unramified outside $\ell N d_K$. By Corollary 6.61(vi) the Frobenius elements of $\mathrm{Gal}(L/\mathbb{Q})$ at the primes outside the exceptional set of density zero are dense, so those at the inert such primes are dense in the open coset $\mathrm{Gal}(L/\mathbb{Q}) \setminus \mathrm{Gal}(L/K)$; the trace of $\rho_{E,\ell}$ is continuous and vanishes at them, so it vanishes on $G_\mathbb{Q} \setminus G_K$. If $E$ had no complex multiplication, the image of $\rho_{E,\ell}$ would be open in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ (Serre's open image theorem [171]), and the trace cannot vanish on a coset of an open subgroup of index 2 of an open subgroup of $\mathrm{GL}_2(\mathbb{Z}_\ell)$ (such a coset contains elements $g$ and $(1+\ell^k x)g$ for all $x \in \mathrm{M}_2(\mathbb{Z}_\ell)$ and $k$ large, and $\operatorname{tr}(xg)$ is not identically zero). So $E$ has complex multiplication by some $K'$. If $K' \neq K$, a set of primes of positive density splits in $K'$ and is inert in $K$ (Corollary 6.61(i) in $KK'$); at such a prime of good reduction the reduction is ordinary (Deuring's criterion, [34, Section 14]), so $a_p \not\equiv 0 \pmod p$ and $a_p \neq 0$, a contradiction. (v) is due to Serre [175, Section 7]. (vi) The forward implication is (iv); the converse is (ii). The class number one list and $\ell^* < 0$ (needed for $K$ imaginary, Proposition 6.51(iv)) give the list of $\ell$, and $\ell$ is ramified in $\mathbb{Q}(\sqrt{\ell^*})$. □

**Example 6.64** (curves with complex multiplication)**.** (Checked with PARI: $a_p = 0$ at every good prime $p$ inert in the field among the first 70 primes, and $a_p \neq 0$ at the split primes listed.) $y^2 = x^3 - x$, conductor 32, CM by $\mathbb{Z}[i]$: $a_p = 0$ for $p \equiv 3 \pmod 4$, and $a_5 = -2$, $a_{13} = 6$, $a_{17} = 2$, $a_{29} = -10$, $a_{37} = -2$. $y^2 = x^3 + 1$ (36a1) and $y^2 + y = x^3 - 7$ (27a1), $j = 0$, CM by $\mathbb{Z}[\zeta_3]$: $a_p = 0$ for $p \equiv 2 \pmod 3$; for 36a1, $a_{13} = 2$, $a_{37} = -10$. 121b1 $= [0, -1, 1, -7, 10]$, $j = -2^{15}$, CM by the maximal order of $\mathbb{Q}(\sqrt{-11}) = \mathbb{Q}(\sqrt{11^*})$: $a_p = 0$ when $\left(\frac{p}{11}\right) = -1$, for example $a_2 = a_7 = 0$, while $a_5 = -3$, $a_{37} = 7$. In each case the primes ramified in $K$ (2, 3, 3, 11) divide the conductor, as Theorem 6.63(iii) requires.

*In the paper* 6.65. Lemma lem:h0: $H^0(\mathbb{Q}, W^D)$ is finite and cyclic, of length $h_\ell(E)$, "Moreover $h_\ell(E) = 0$ for every $\ell \geq 5$, and $h_3(E) \leq 1$", and "This holds for every $E$, with or without complex multiplication". The proof is the determinant argument (Chapters 4 and 5): an isomorphism $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$ forces $\chi_\ell^2 \equiv 1 \pmod{\ell^k}$; it uses neither complex multiplication nor the distribution of inert primes. The condition $\rho \simeq \rho \otimes \chi_\ell$ is not the condition $\rho \simeq \rho \otimes \chi_{\ell^*}$ (Warning 6.57); the statements about $a_p = 0$ at inert primes belong to the second and are Proposition 6.62 and Theorem 6.63, and by Theorem 6.63(vi) the complex multiplication case of the second cannot occur when $\ell \nmid N$. The equality $h_3(E) = 1$ holds "if and only if $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \chi_{-3}$", which uses $\bar\chi_3^{\mathrm{cyc}} = \chi_{-3}$ (Example 6.56); its link with the reducibility of $\bar\rho|_{G_{\mathbb{Q}(\sqrt{-3})}}$, like Proposition prop:ns, obtains $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ from $\bar\rho \otimes \overline{\mathbb{F}}_3 \simeq \operatorname{Ind}\psi$ for a character $\psi$ of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, by Frobenius reciprocity and $\operatorname{Ind}\psi \otimes \chi_{-3} \simeq \operatorname{Ind}\psi$ (Chapter 4), not from Frobenius traces. Application (a) of Proposition 6.62 gives the equivalent trace criterion ($3 \mid a_p$ for almost all $p \equiv 2 \pmod 3$), used in the example of Chapter 5, not in the paper. Section sec:anomaly takes the values of $h_3(E)$ from Lemma lem:h0 and Proposition prop:ns.

The last section returns from Galois groups to rings: the coefficients of an eigenform, the primes $\lambda$ above $\ell$ in its coefficient ring, and the maximal ideals of a Hecke algebra that congruences

between eigenforms produce.

## 6.7 Coefficient fields and primes above $\ell$

Lemmas lem:crit and lem:old of the paper work with eigenforms whose coefficients lie in "the ring of integers $\mathcal{O}$ of a number field", with "a prime $\lambda \mid \ell$ of $\mathcal{O}$", with congruences modulo $\lambda$, with "a maximal ideal of residue characteristic $\ell$" of the Hecke algebra, and with the operation "after enlarging $\mathcal{O}$ and $\lambda$". This section makes each of these precise. The Hecke algebra $\mathbb{T}$ itself, and the facts that its eigencharacters correspond to eigenforms, are in Chapters 12 and 13; here only the algebra of a commutative ring that is finitely generated as a $\mathbb{Z}$-module is used.

**Proposition 6.66** (coefficient fields)**.** *Let $R$ be a commutative ring, finitely generated as a $\mathbb{Z}$-module, and $\phi\colon R \to \mathbb{C}$ a ring homomorphism. Then every $\phi(t)$ is an algebraic integer, $K_\phi = \mathbb{Q}(\phi(R))$ is a number field, and $\phi(R)$ is a subring of $\mathcal{O}_{K_\phi}$ of finite index (an* order*). For $\sigma \in G_\mathbb{Q}$, $\sigma \circ \phi$ is again such a homomorphism (a* Galois conjugate*). For the eigencharacter $\phi_f\colon \mathbb{T} \to \mathbb{C}$, $T_n \mapsto a_n(f)$, of a newform $f$ of weight 2 on $\Gamma_0(N)$: the* coefficient field $K_f = \mathbb{Q}(a_n(f) : n \geq 1)$ *is a number field, every $a_n(f)$ lies in $\mathcal{O}_{K_f}$, and (stated; Chapter 13) the conjugates $f^\sigma = \sum \sigma(a_n)q^n$ are newforms and $K_f$ is totally real. For $f = f_E$, $K_f = \mathbb{Q}$, $\mathcal{O} = \mathbb{Z}$, and the only prime above $\ell$ is $\lambda = \ell$.*

*Proof.* $\phi(R)$ is a subring of $\mathbb{C}$ that is finitely generated as a $\mathbb{Z}$-module, so each of its elements is integral over $\mathbb{Z}$ (Proposition 6.3(iii), whose proof works in any field). The field $K_\phi$ is generated by the images of finitely many $\mathbb{Z}$-module generators, each algebraic, so it is a number field, and $\phi(R) \subseteq \mathcal{O}_{K_\phi}$. The $\mathbb{Q}$-span of $\phi(R)$ is a subring of $\mathbb{C}$ of finite dimension over $\mathbb{Q}$, a domain, hence a field, so it is $K_\phi$; thus $\phi(R)$ and $\mathcal{O}_{K_\phi}$ are free abelian groups of the same rank $[K_\phi : \mathbb{Q}]$ (Theorem 6.6), and the index is finite. For $\sigma \in G_\mathbb{Q}$, $\sigma \circ \phi$ is a ring homomorphism. For $\phi_f$, $\mathbb{T}$ is finitely generated as a $\mathbb{Z}$-module (Chapter 13), $\phi_f(\mathbb{T}) = \mathbb{Z}[a_n(f) : n \geq 1]$, which spans $K_f$, and $a_1(f) = 1$. □

**Proposition 6.67** (primes above $\ell$ and reduction modulo $\lambda$)**.** *Let $K$ be a number field, $\mathcal{O} = \mathcal{O}_K$ and $\ell$ a prime.*

*(i) The primes $\lambda \mid \ell$ of $\mathcal{O}$ are the maximal ideals containing $\ell$; there are at most $[K : \mathbb{Q}]$ of them; $k_\lambda = \mathcal{O}/\lambda \simeq \mathbb{F}_{\ell^f}$ with $f = f(\lambda \mid \ell)$; reduction modulo $\lambda$ is the ring homomorphism $\mathcal{O} \to k_\lambda$, $a \mapsto \bar{a}$.*

*(ii) For $a, b \in \mathbb{Z}$: $a \equiv b \pmod{\lambda}$ iff $a \equiv b \pmod{\ell}$. For $a \in \mathcal{O}$: $\ell \mid N_{K/\mathbb{Q}}(a)$ iff some $\lambda \mid \ell$ contains $a$; and $a$ lies in every $\lambda \mid \ell$ iff $a$ lies in the radical of $\ell\mathcal{O}$, iff $a^m \in \ell\mathcal{O}$ for some $m \geq 1$.*

*(iii) (Enlarging.) If $K \subseteq K'$ then $\lambda\mathcal{O}_{K'} \neq \mathcal{O}_{K'}$, every prime $\lambda'$ of $\mathcal{O}_{K'}$ dividing $\lambda\mathcal{O}_{K'}$ has $\lambda' \cap \mathcal{O} = \lambda$, and reduction modulo $\lambda'$ restricts on $\mathcal{O}$ to reduction modulo $\lambda$ followed by $k_\lambda \subseteq k_{\lambda'}$.*

*(iv) (Lying over, for orders.) Let $\alpha$ be integral over $\mathcal{O}$ and $\mathcal{O}' = \mathcal{O}[\alpha]$. For every prime $\lambda$ of $\mathcal{O}$ there is a maximal ideal $\mathfrak{q}$ of $\mathcal{O}'$ with $\mathfrak{q} \cap \mathcal{O} = \lambda$, namely $\mathfrak{q} = \lambda' \cap \mathcal{O}'$ for a prime $\lambda'$ above $\lambda$ in the ring of integers of $K(\alpha)$; its residue field is a finite field containing $k_\lambda$.*

(v) *(The step of Lemma lem:old.) Let $p \neq \ell$ be a prime, $a \in \mathcal{O}$, and $\alpha_p, \beta_p$ the roots of $X^2 - aX + p$; let $\lambda'$ be a prime of the ring of integers of $K(\alpha_p)$ above $\lambda$. If $X^2 - aX + p \equiv (X-u)(X-w) \pmod{\lambda}$ with $u, w \in \mathcal{O}$, then $\alpha_p \equiv u$ or $\alpha_p \equiv w \pmod{\lambda'}$, and $\beta_p$ is congruent to the other one. In particular, if $u = sp$ and $w = s$ with $s = \pm 1$, one of $\alpha_p, \beta_p$ is congruent to $s$ modulo $\lambda'$.*

*Proof.* (i) A prime containing $\ell$ is nonzero, hence maximal; it lies above $\ell$ (Theorem 6.19), and there are at most $\sum e_i f_i = [K:\mathbb{Q}]$ of them. $k_\lambda$ is a finite field of characteristic $\ell$ (Proposition 6.14(iii)). (ii) $\lambda \cap \mathbb{Z} = \ell\mathbb{Z}$. $N_{K/\mathbb{Q}}(a\mathcal{O}) = \prod_{\mathfrak{p}} N\mathfrak{p}^{v_{\mathfrak{p}}(a)}$ (Proposition 6.14), divisible by $\ell$ iff some $\mathfrak{p} \mid \ell$ has $v_{\mathfrak{p}}(a) > 0$. The intersection of the $\lambda \mid \ell$ is $\prod_{\lambda|\ell} \lambda$ (Proposition 6.15(i)), and $a$ lies in it iff $v_\lambda(a) \geq 1$ for all $\lambda \mid \ell$ iff some power of $a$ lies in $\ell\mathcal{O} = \prod \lambda^{e_\lambda}$. (iii) Theorem 6.19 for $K'/K$ and $\mathfrak{p} = \lambda$. (iv) Let $K'' = K(\alpha)$ and $\lambda'$ a prime of $\mathcal{O}_{K''}$ above $\lambda$, which exists by (iii). Then $\mathfrak{q} = \lambda' \cap \mathcal{O}'$ is a prime ideal of $\mathcal{O}' \subseteq \mathcal{O}_{K''}$ with $\mathfrak{q} \cap \mathcal{O} = \lambda$, and $\mathcal{O}'/\mathfrak{q}$ embeds in the finite field $\mathcal{O}_{K''}/\lambda'$, so it is a finite domain, hence a field. (v) $\alpha_p, \beta_p$ are integral over $\mathcal{O}$, so they lie in the ring of integers of $K(\alpha_p)$, whose quotient by $\lambda'$ is a field $k$. In $k[X]$, $(X - \bar{\alpha}_p)(X - \bar{\beta}_p) = X^2 - \bar{a}X + \bar{p} = (X - \bar{u})(X - \bar{w})$, and unique factorisation in $k[X]$ gives $\{\bar{\alpha}_p, \bar{\beta}_p\} = \{\bar{u}, \bar{w}\}$ as multisets. □

**Proposition 6.68** (maximal ideals from congruences)**.** *Let $R$ be a commutative ring, finitely generated as a $\mathbb{Z}$-module, $\mathcal{O} = \mathcal{O}_K$, $\lambda \mid \ell$ a prime of $\mathcal{O}$, and $\phi\colon R \to \mathcal{O}$ a ring homomorphism.*

*(i) $\mathfrak{m} = \ker(R \xrightarrow{\phi} \mathcal{O} \to k_\lambda)$ is a maximal ideal of $R$, and $R/\mathfrak{m}$ is a subfield of $k_\lambda$, of characteristic $\ell$.*

*(ii) If $\phi'\colon R \to \mathcal{O}$ is another ring homomorphism with $\phi'(t) \equiv \phi(t) \pmod{\lambda}$ for all $t$ in a set of ring generators of $R$ (for $R = \mathbb{T}$: $a_n(g) \equiv a_n(f)$ for all $n$), then $\phi$ and $\phi'$ define the same $\mathfrak{m}$.*

*(iii) If $J \subseteq R$ is an ideal contained in $\mathfrak{m}$ such that $R/J$ is a cyclic group of order $r$, then $\ell \mid r$ (with $\ell \mid 0$).*

*(iv) (Stated; Chapter 3.) $R \otimes \mathbb{Z}_\ell$ is the product of its localisations at its maximal ideals, which are the maximal ideals of $R$ containing $\ell$; and $\phi$ extends to $R \otimes \mathbb{Z}_\ell \to \mathcal{O}_\lambda$ (Proposition 6.69), which factors through the factor $(R \otimes \mathbb{Z}_\ell)_{\mathfrak{m}}$.*

*Proof.* (i) $R/\mathfrak{m}$ embeds in the finite field $k_\lambda$; a subring of a finite field is a finite domain, hence a field. (ii) The two compositions $R \to k_\lambda$ are ring homomorphisms agreeing on generators, hence equal, so they have the same kernel. (iii) $R/\mathfrak{m}$ is a nonzero quotient of the cyclic group $R/J \simeq \mathbb{Z}/r$, so it is $\mathbb{Z}/s$ with $s \mid r$; it has characteristic $\ell$, so $s = \ell$. (iv) We do not prove this here; the decomposition is the idempotent-lifting theorem of Chapter 3, and the extension of $\phi$ is $\phi \otimes \mathbb{Z}_\ell$ followed by the projection of $\mathcal{O} \otimes \mathbb{Z}_\ell$ onto its factor $\mathcal{O}_\lambda$ (Proposition 6.69), which kills the idempotents of the other factors of $R \otimes \mathbb{Z}_\ell$. □

**Proposition 6.69** ($\mathcal{O}_K \otimes \mathbb{Z}_\ell$)**.** *For a number field $K$ and a prime $\ell$,*

$$\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \simeq \prod_{\lambda|\ell} \mathcal{O}_{K,\lambda}, \qquad K \otimes_{\mathbb{Q}} \mathbb{Q}_\ell \simeq \prod_{\lambda|\ell} K_\lambda,$$

*where $\mathcal{O}_{K,\lambda} = \varprojlim_k \mathcal{O}_K/\lambda^k$ is the $\lambda$-adic completion, a complete discrete valuation ring, free over $\mathbb{Z}_\ell$ of rank $e_\lambda f_\lambda$ ($\sum_\lambda e_\lambda f_\lambda = [K:\mathbb{Q}]$), with uniformiser any $\varpi \in \lambda \setminus \lambda^2$ and residue field $k_\lambda$; and $K_\lambda$, its fraction field, is an extension of $\mathbb{Q}_\ell$ of degree $e_\lambda f_\lambda$. The cases $K = \mathbb{Q}(\zeta_3)$ and $\mathbb{Q}(i)$ are Proposition 6.43.*

*Reference.* We do not prove this here; it is proved in Chapter 7, and see [142, Chapter II, Section 8] or [174, Chapter II]. The algebraic core is the Chinese remainder theorem: $\mathcal{O}_K/\ell^k\mathcal{O}_K \simeq \prod_{\lambda\mid\ell}\mathcal{O}_K/\lambda^{ke_\lambda}$ for every $k$ (Proposition 6.15), and $\mathcal{O}_K\otimes\mathbb{Z}_\ell = \varprojlim_k \mathcal{O}_K/\ell^k\mathcal{O}_K$ because $\mathcal{O}_K$ is free of finite rank over $\mathbb{Z}$. □

**Example 6.70** (level 23)**.** The space $S_2(\Gamma_0(23))$ is two-dimensional and spanned by a newform $f$ and its Galois conjugate (stated; PARI's `mfinit([23,2],0)`, and LMFDB 23.2.a.a). Its coefficient field is $K_f=\mathbb{Q}(\sqrt{5})$, and $a_2(f)$ is a root $\theta$ of $X^2+X-1$, the characteristic polynomial of $T_2$ (computed with PARI's `mfheckemat`); the first coefficients are

$$f = q+\theta q^2+(-2\theta-1)q^3+(-\theta-1)q^4+2\theta q^5+(\theta-2)q^6+(2\theta+2)q^7+\cdots, \qquad \theta^2+\theta-1=0,$$

and $a_{11}=-2\theta-4$, $a_{13}=3$ (PARI's `mfeigenbasis`). The ring of integers is $\mathcal{O}_{K_f}=\mathbb{Z}[\frac{1+\sqrt{5}}{2}]=\mathbb{Z}[\theta]$, as $\operatorname{disc}(X^2+X-1)=5$ is squarefree. Primes: $5=(\sqrt{5})^2$ is ramified, with $\theta\equiv 2 \pmod{\sqrt{5}}$ since $X^2+X-1\equiv(X-2)^2 \pmod 5$; 11 splits ($(\frac{5}{11})=1$) into $\lambda=(11,\theta-3)$ and $\lambda'=(11,\theta-7)$, as $X^2+X-1\equiv(X-3)(X-7) \pmod{11}$, both with residue field $\mathbb{F}_{11}$; 2 is inert ($5\equiv 5 \pmod 8$), with residue field $\mathbb{F}_4$, in which $\theta$ reduces to a root of $X^2+X+1$, a primitive cube root of unity. So the reductions of $f$ modulo $\lambda$ and $\lambda'$ are different: modulo $\lambda$, $(a_2,a_3,a_5,a_7)\equiv(3,4,6,8)$, and modulo $\lambda'$, $(7,7,3,5)$; they are exchanged by $\mathrm{Gal}(K_f/\mathbb{Q})$. A congruence "$a_n(g)\equiv a_n(f) \pmod\lambda$" as in Lemma lem:crit depends on $\lambda$, not only on $\ell$. The Hecke algebra of level 23 is $\mathbb{T}=\mathbb{Z}[T_2]\simeq\mathbb{Z}[\theta]$. Indeed, let $\mathbb{T}$ be the subring of $\mathrm{End}(S_2(\Gamma_0(23)))$ generated by all the $T_n$, and $\phi_f\colon\mathbb{T}\to\mathcal{O}_{K_f}$ the eigencharacter of $f$ (Proposition 6.66). It is injective: if $\phi_f(t)=0$, then also $\phi_{f^\sigma}(t)=\sigma(\phi_f(t))=0$ for the conjugate $f^\sigma$, so $t$ kills $f$ and $f^\sigma$, which span $S_2(\Gamma_0(23))$, and $t=0$. Its image is a subring of $\mathcal{O}_{K_f}=\mathbb{Z}[\theta]$ containing $\phi_f(T_2)=a_2(f)=\theta$, so it is $\mathbb{Z}[\theta]$, and $\mathbb{T}=\mathbb{Z}[T_2]$ since $\phi_f(T_2)$ generates the image. (Checked with PARI: `mfdim([23,2],1)` is 2 and `nfbasis(x^2+x-1)` is $[1,x]$, so $\mathbb{Z}[\theta]$ is the full ring of integers of $\mathbb{Q}(\sqrt{5})$.)

*In the paper* 6.71. Lemma lem:crit: "an eigenform $g$ ... with coefficients in the ring of integers $\mathcal{O}$ of a number field, and a prime $\lambda\mid\ell$ of $\mathcal{O}$ such that $a_n(g)\equiv a_n(f) \pmod\lambda$ for every $n\geq 1$"; then "$I+J$ lies in the kernel of $\mathbb{T}\to\mathcal{O}/\lambda$, a maximal ideal of residue characteristic $\ell$, and $\ell$ divides $\#\mathbb{T}/(I+J)=r$". The ring $\mathcal{O}$ exists by Proposition 6.66 (the eigencharacter $\phi_g$ of $g$ takes values in a number field and in its ring of integers); $\phi_g\equiv\phi_f \pmod\lambda$ on the generators $T_n$ gives the same kernel $\mathfrak{m}$ (Proposition 6.68(ii)), so $I=\ker\phi_f\subseteq\mathfrak{m}$ and $J\subseteq\ker\phi_g\subseteq\mathfrak{m}$; and $\ell\mid r$ is Proposition 6.68(iii) applied to $I+J$. "The same statements hold with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$" uses Proposition 6.68(iv). Lemma lem:old: "after enlarging $\mathcal{O}$ and $\lambda$" is Proposition 6.67(iii): the eigenform $g$ built from $h$ has coefficients $\mu_p\in\{\alpha_p,\beta_p\}$ in a larger field, and a prime above $\lambda$ there restricts to $\lambda$; "one of $\alpha_p,\beta_p$ is congruent to $a_p(f)$ modulo a prime above $\lambda$ in $\mathcal{O}[\alpha_p]$" is Proposition 6.67(iv) and (v), with $X^2-a_p(h)X+p\equiv(X-a_p(f)p)(X-a_p(f)) \pmod\lambda$ and $a_p(f)=\pm1$. In Section 3 of the paper all forms have rational coefficients and $\lambda=\ell$; Lemma lem:old is needed for general $h$, whose coefficient field can be larger, as in Example 6.70.

## 6.8 Exercises

**Exercise 6.72.** Let $K=\mathbb{Q}(\sqrt{13})$. Show that $\mathcal{O}_K=\mathbb{Z}[\theta_{13}]$ with $\theta_{13}=\frac{1+\sqrt{13}}{2}$, that $d_K=13$, and that $\frac{3+\sqrt{13}}{2}$ is a unit of norm $-1$. Factor 2, 3 and 13 in $\mathcal{O}_K$ (answer: 2 inert, 3 split, 13 ramified). Show that Dedekind-Kummer applied to the generator $\sqrt{13}$ (that is, to $\mathbb{Z}[\sqrt{13}]$) at $p=2$ would predict that 2 ramifies, and explain why the theorem does not apply there.

**Exercise 6.73.** In the non-integrally closed ring $A=\mathbb{Z}[\sqrt{-3}]$ let $\mathfrak{a}=(2,1+\sqrt{-3})$. Show that $\mathfrak{a}^2=2\mathfrak{a}$ but $\mathfrak{a}\neq(2)$, so ideals cannot be cancelled in $A$; deduce that $A$ is not a Dedekind domain and identify the hypothesis of Definition 6.10 that fails. What is $\mathfrak{a}\mathbb{Z}[\zeta_3]$?

**Exercise 6.74.** Factor 65, 91 and $1729=7\cdot13\cdot19$ into primes in $\mathbb{Z}[i]$ and in $\mathbb{Z}[\zeta_3]$. Deduce the number of ways of writing 65 as $a^2+b^2$ with $0<a<b$, and explain the answer by counting the elements of norm 65 in $\mathbb{Z}[i]$.

**Exercise 6.75.** Show that a prime $p\neq2,5$ splits in $\mathbb{Q}(\sqrt{-5})$ iff $p\equiv1,3,7,9 \pmod{20}$. Show that the primes above 3 and 7 are not principal, that the primes above 3 lie in the class of $\mathfrak{p}_2$ of Example 6.18, and, using Theorem 6.16, that $h(\mathbb{Q}(\sqrt{-5}))=2$. Which primes $p$ are of the form $a^2+5b^2$?

**Exercise 6.76.** Let $L=\mathbb{Q}(2^{1/3},\zeta_3)$, with $\mathrm{Gal}(L/\mathbb{Q})\simeq S_3$ acting on the three cube roots of 2. For $p\geq5$, show using Corollary 6.29: if $p\equiv2 \pmod 3$, $\mathrm{Frob}_p$ is a transposition and $X^3-2$ has exactly one root modulo $p$; if $p\equiv1 \pmod 3$, $\mathrm{Frob}_p$ is trivial or a 3-cycle according as 2 is or is not a cube modulo $p$. Determine $\mathrm{Frob}_p$ for $p=5,7,31$, and deduce from Theorem 6.60 that 2 is a cube modulo $p$ for a set of primes of density $\frac{2}{3}$.

**Exercise 6.77.** Without using Theorem 6.22, show that an odd prime $p$ ramifies in $\mathbb{Q}(\sqrt{d})$ ($d$ squarefree) iff $p\mid d$: compute $(p,\sqrt{d})^2$ when $p\mid d$, and when $p\nmid d$ show that $\mathcal{O}_K/p\mathcal{O}_K$ is reduced.

**Exercise 6.78.** Factor 2, 13 and 29 in $\mathbb{Z}[\zeta_7]$ and check $efg=6$ against Theorem 6.49. The quadratic subfield of $\mathbb{Q}(\zeta_7)$ is $\mathbb{Q}(\sqrt{-7})$; check that the splitting of each of these primes in it agrees with $(\frac{p}{7})$ and with the image of $\mathrm{Frob}_p=\sigma_p$ in $(\mathbb{Z}/7)^\times/(\mathbb{Z}/7)^{\times2}$.

**Exercise 6.79.** Determine the decomposition and inertia groups in $\mathrm{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q})\simeq(\mathbb{Z}/8)^\times$ of 2, 3, 5, 7 and 17. Identify the three quadratic subfields $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-2})$ with the subgroups $\{1,5\}$, $\{1,7\}$, $\{1,3\}$, and deduce that $(\frac{-2}{p})=1$ iff $p\equiv1,3 \pmod 8$.

**Exercise 6.80.** (a) List the quadratic fields unramified outside $\{3,7\}$ (answer: $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{21})$). Which of them is cut out by $\chi_S$ for $S=\{3,7\}$, and what is its conductor? (b) List the seven quadratic fields unramified outside $\{2,3\}$, with their discriminants.

**Exercise 6.81.** In $\mathbb{Z}_3[\zeta_3]$ compute $v_\pi$ of $1+\zeta_3$, $2+\zeta_3$, $3$, $\sqrt{-3}$, $1-\zeta_3^2$ and $4+\zeta_3$ (answers $0,1,2,1,1,0$). Show that the unit group of $\mathbb{Z}_3[\zeta_3]/(\pi^2)$ is cyclic of order 6, generated by the class of $-\zeta_3$, and that every unit of $\mathbb{Z}_3[\zeta_3]$ is $\pm\zeta_3^k$ times a unit congruent to 1 modulo $\pi^2$.

**Exercise 6.82.** Let $\sigma=\left(\begin{smallmatrix}1&-3\\1&-2\end{smallmatrix}\right)\in\mathrm{GL}_2(\mathbb{Z}_3)$, acting on $T=\mathbb{Z}_3^2$. Check that $\sigma$ has order 3 and determinant 1, that its reduction modulo 3 is a nontrivial unipotent, and that $e_1$ generates $T$ over $\mathbb{Z}_3[\sigma]$. Show more generally that $v\in T$ generates $T$ over $\mathbb{Z}_3[\sigma]$ iff $v\notin(1-\sigma)T$, and find $(1-\sigma)T$ and the fixed line of $\sigma$ on $T/3T$ explicitly.

**Exercise 6.83.** Describe $\mathbb{Z}_\ell[\zeta_3]$ for $\ell=2,5,7$. For $\ell=7$, find the roots of $X^2+X+1$ modulo 7 and modulo 49 (answers $2,4$ and $30,18$), and show that the idempotents of $\mathbb{Z}_7[\zeta_3]\simeq\mathbb{Z}_7\times\mathbb{Z}_7$ are congruent modulo 49 to $4\zeta_3+27$ and $45\zeta_3+23$.

**Exercise 6.84.** (a) Let $\sigma$ have characteristic polynomial $X^2+X+1$ and let $g=-\sigma$ (order 6, types II, II$^*$). Show that the commutant of $g$ and its trace-zero line are those of $\sigma$, and that $AgA^{-1}=g^q$ for an odd prime $q\neq3$ gives the same sign $\chi_{-3}(q)$ as in Proposition 6.45(iii). (b) Using the table of Section 1 of the paper, compute $e_\ell(E,q)$ for $q\in\{5,7,11,13\}$, every odd $\ell\neq q$, and the types IV and III. Show that $e_3(E,q)\geq1$ for every $q\geq5$ of type II, II$^*$, IV, IV$^*$, and decide for which $q$ the same holds for type III.

**Exercise 6.85.** Let $E$ have nonsplit multiplicative reduction at $q \neq \ell$, with $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} -q & c' \\ 0 & -1 \end{smallmatrix}\right)$ for an arithmetic Frobenius. Show that the arithmetic Frobenius acts on $(\mathrm{ad}^0 V)^{I_q}$ by $q$ and on $(\mathrm{ad}^0 V(1))^{I_q}$ by $q^2$, and the geometric one by $q^{-1}$ and $q^{-2}$. Deduce that $\det(1 - \mathrm{Frob}_q^{-1} \mid (\mathrm{ad}^0 V(1))^{I_q}) = 1 - q^{-2} = L_q(\mathrm{ad}^0 V, 1)^{-1}$, whereas the arithmetic Frobenius gives $1 - q^2$: a different number with the same $\ell$-adic valuation.

**Exercise 6.86.** Let $E\colon y^2 = x^3 - x$. Show that $a_p = 0$ for every prime $p \equiv 3 \pmod 4$, using the substitution $x \mapsto -x$ and $(\frac{-1}{p}) = -1$. Compute $a_5 = -2$, $a_{13} = 6$ and $a_{17} = 2$ by counting points, and check that $p = (a_p/2)^2 + b^2$ with $b \in \mathbb{Z}$ for $p = 5, 13, 17$ and $29$ ($a_{29} = -10$). Relate the first statement to Theorem 6.63(ii), and the second to the factorisation $p = \varpi\bar{\varpi}$ in $\mathbb{Z}[i]$ with $a_p = \varpi + \bar{\varpi}$ for a suitable prime $\varpi$ above $p$ (stated; Chapter 9).

**Exercise 6.87.** (a) For 11a1, $a_2 = -2$. Using Proposition 6.62 ((a)⇒(b) needs no semisimplicity), show that $\bar{\rho}_{E,3} \not\cong \bar{\rho}_{E,3} \otimes \chi_{-3}$. (b) For $E\colon y^2 = x^3 - 6x - 8$, compute $a_5$, $a_{11}$, $a_{17}$ (answers 3, 3, $-6$) and explain, using Proposition 6.62 and Chapter 5, why $3 \mid a_p$ for every prime $p \equiv 2 \pmod 3$ of good reduction.

**Exercise 6.88.** Let $f$ be the newform of level 23 of Example 6.70, with coefficients $a_n = x_n + y_n\theta$, $x_n, y_n \in \mathbb{Z}$, and $f^\sigma$ its conjugate. Show that $a_n(f) \equiv a_n(f^\sigma) \pmod{\sqrt{5}}$ for all $n$, and that $(\sqrt{5})$ is the only prime of $\mathcal{O}_{K_f}$ modulo which all coefficients of $f$ and $f^\sigma$ agree. Compute the reduction of $f$ modulo the prime above 2 up to $q^7$, in $\mathbb{F}_4$.

**Exercise 6.89.** Let $K = \mathbb{Q}(\sqrt{5})$ and $K' = \mathbb{Q}(\sqrt{5}, i)$. Determine $e$, $f$, $g$ for 2, 3 and 5 in $K$ and in $K'$ (answers in $K'$: $(2,2,1)$, $(1,2,2)$, $(2,1,2)$), and describe, for each prime $\lambda$ of $K$ above these, the primes $\lambda'$ of $K'$ above it and the inclusions $k_\lambda \subseteq k_{\lambda'}$ of Proposition 6.67(iii).

**Exercise 6.90.** Show that $\mathbb{Z}[\frac{1+\sqrt{5}}{2}]^\times = \{\pm u^k : k \in \mathbb{Z}\}$ with $u = \frac{1+\sqrt{5}}{2}$, by the method of Theorem 6.9(iii). Show that the reduction of $u$ modulo the prime above 2 has order 3 in $\mathbb{F}_4^\times$ and modulo each prime above 11 has order dividing 10; which orders occur?

## 6.9 Notes and references

The algebraic number theory of Sections 6.1 to 6.3 follows Neukirch [142], Chapter I (Sections 2 to 3 for integrality, integral bases and Dedekind domains, Sections 8 and 9 for the splitting of primes, decomposition and inertia groups, Section 10 for cyclotomic fields), and Marcus [123], Chapters 2 to 4, which is the gentlest complete treatment and has many exercises of the kind given here. The proof of unique factorisation of ideals (Lemma 6.12, Theorem 6.13) is Neukirch's; the proof of transitivity in Proposition 6.23 and of the surjectivity onto the residue Galois group in Theorem 6.24 are the standard ones (Marcus, Chapter 4). Atiyah and Macdonald [7], Chapters 5 and 9, treat integral extensions, lying over and Dedekind domains in the language of commutative algebra. Lang [108] is a further reference for the Dedekind zeta function and densities (Chapter VIII). Minkowski's theorem, Dirichlet's unit theorem, the analytic continuation of $\zeta_K$ and Chebotarev's theorem are stated, with references to Neukirch, Chapter I, Sections 5 to 7, and Chapter VII, Sections 5 and 13.

The splitting of primes in quadratic fields and the Gaussian and Eisenstein integers are in Ireland and Rosen [87], Chapters 1 and 9 and Chapter 13, and in Cox [34], Sections 5 and 7, where prime discriminants (Proposition 6.36) are also discussed. Cyclotomic fields, the discriminant formula, Kronecker-Weber and the dictionary between Dirichlet characters and characters of $G_{\mathbb{Q}}$

follow Washington [201], Chapters 2, 3 and 14. The tame relation (Proposition 6.31) is in Serre [174], Chapter IV, Sections 1 and 2, where the higher ramification groups are defined; Chapter 7 gives the local version. Frobenius elements in $G_{\mathbb{Q}}$ and their use for $\ell$-adic representations are in Serre [178], Chapter I. Class field theory in idelic form, with its two sign conventions, is in Neukirch, Chapter VI, and in Milne's notes [133].

For complex multiplication, Theorem 6.63 quotes Silverman [184], Chapter II and Appendix A (the thirteen rational $j$-invariants), and Cox [34], Sections 7 to 14 (class number one and Deuring's reduction criterion); the open image theorem is Serre [171], and the density zero of $\{p : a_p(E) = 0\}$ for $E$ without complex multiplication is Serre [175]. The argument that a prime ramified in the CM field divides the conductor (Theorem 6.63(iii)) is the elementary one given here; it can also be read off the conductor of an induced representation.

All theorem, section and chapter numbers of books cited in this chapter (Neukirch, Marcus, Washington, Ireland-Rosen, Cox, Serre, Lang, Silverman) are recalled and were not checked against printed copies; they should be read as pointers, and no proof in the book depends on them. Every numerical example (class numbers, discriminants, factorisations of primes, Frobenius classes, the coefficients of the level 23 newform, the traces of Frobenius of the curves of Example 6.64 and Exercise 6.87) was checked with PARI/GP [148], for instance with the following commands.

```
quadclassunit(-20)[1]                        \\ class number 2
K = nfinit(x^3 - 2); idealprimedec(K, 5)  \\ residue degrees 1 and 2
nfdisc(polcyclo(7))                          \\ -16807 = -7^5
mf = mfinit([23,2],0); mfcoefs(mfeigenbasis(mf)[1], 13)
E = ellinit([0,-1,1,-7,10]); [ellap(E,p) | p <- [2,7,5,37]]   \\ 0,0,-3,7
```

# Chapter 7

# Local fields

The paper works at a bad prime $q \neq \ell$ with the restriction of $\rho = \rho_{E,\ell}$ to a decomposition group $G_{\mathbb{Q}_q}$, and every local statement it makes rests on a small amount of local field theory: the fields $\mathbb{Q}_q$, $\mathbb{Q}_q^{\mathrm{ur}}$ and $\mathbb{Z}_\ell$; Hensel's lemma and $\ell$-th powers; the inertia group, its wild part and its tame quotient; the tame relation Frob $t$ Frob$^{-1} = t^q$; the Kummer cocycle of the Tate parameter on inertia and on Frobenius; quadratic characters locally; Grothendieck's monodromy theorem and Weil-Deligne representations; and the inertia invariants, local $L$-factors and conductors of the representations that occur at a bad prime, together with those of their adjoints. This chapter supplies all of it, with complete proofs except for local class field theory, the identification of Kummer classes with Galois cohomology (proved in Chapter 17), the integrality of the Swan conductor, upper numbering, epsilon factors and a few classification statements, which are stated with references. It uses Chapter 1 for profinite groups, Chapter 2 for $v_p$, $|\cdot|_p$, finite fields and powers in $\mathbb{F}_q^\times$, Chapter 3 for discrete valuation rings and the algebraic $\mathbb{Z}_\ell$, Chapter 4 for the linear algebra of inertia actions, Chapter 5 for infinite Galois theory and Kummer theory, and Chapter 6 for number fields, decomposition groups and quadratic characters. Elliptic curves appear only in examples and in quoted statements from Chapters 10 and 11.

## Core results at a glance

Throughout, $K$ is $\mathbb{Q}_p$ or a finite extension, with $\mathcal{O}_K$, $\mathfrak{m}_K$, uniformiser $\varpi$, residue field $\mathbb{F}_q$; $\ell \neq p$; Frob is an arithmetic and $\Phi = \mathrm{Frob}^{-1}$ a geometric Frobenius; $I_K \supseteq P_K$ are the inertia and wild inertia groups; $\|w\| = q^{d(w)}$ on the Weil group.

**7.1 Conventions.** $K$, $\mathcal{O}_K$, $\varpi_K$, $k_K = \mathbb{F}_q$, $v_K$, $|\cdot|_K = q^{-v_K}$; $q$ is both $\#k_K$ and the paper's bad prime; $K^{\mathrm{ur}} \subseteq K^{\mathrm{tame}}$, $I_K$, $P_K$; $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$ and $t_\zeta$; $\varepsilon$ unramified of order $\leq 2$, $\theta_m$, $\eta_{\mathrm{ur}}$, $[a]$, $N$, $\chi_d$ by discriminant. Convention 7.1.

**7.2 Absolute values** (definition): nonarchimedean, equivalent, discrete; valuation ring, maximal ideal, residue field. Definition 7.2.

**7.3 Basic properties.** Nonarchimedean iff $|n| \leq 1$ on $\mathbb{Z}$; isosceles triangles; $\mathcal{O}$ local with $\mathcal{O}^\times = \{|x| = 1\}$; balls are open and closed; $\sum a_n$ converges iff $a_n \to 0$ (complete case); discrete implies $|x| = c^{v(x)}$ with $\mathcal{O}$ a discrete valuation ring; equivalence iff $|\cdot|_1 = |\cdot|_2^s$. *(proved here)* Proposition 7.3.

**7.4 Unconditional summation.** In a complete nonarchimedean field a countable family with only finitely many terms above each $\delta > 0$ can be summed in any order and grouping. *(proved here)* Lemma 7.4.

**7.5 Completion.** Exists, unique up to unique isometry; the value group and the rings $\mathcal{O}/\varpi^n$ do not change; $\hat{\mathcal{O}} \simeq \varprojlim \mathcal{O}/\varpi^n$ topologically. *(proved here)* Theorem 7.6.

**7.6 $\mathbb{Q}_p$ and $\mathbb{Z}_p$** (definition): completion of $\mathbb{Q}$ for $|\cdot|_p$, and its valuation ring. Definition 7.7.

**7.7 Structure of $\mathbb{Z}_p$.** $\mathbb{Z}_p$ is the closure of $\mathbb{Z}$, a complete discrete valuation ring with residue field $\mathbb{F}_p$; $\mathbb{Z}_p \simeq \varprojlim \mathbb{Z}/p^n$ as topological rings (so it is the algebraic $\mathbb{Z}_\ell$ of Chapter 3); $\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mathbb{Z}_p^\times$; $\mathbb{Z}_p$ compact, $\mathbb{Q}_p$ locally compact and totally disconnected; $q^{-1} \in \mathbb{Z}_p$ for primes $q \neq p$. *(proved here)* Theorem 7.8.

**7.8 $p$-adic expansions.** $x = \sum_{i \geq v} a_i p^i$ uniquely, $a_i \in \{0, \dots, p-1\}$; partial sums are the residues; $x \in \mathbb{Q}$ iff the digits are eventually periodic. *(proved here)* Proposition 7.10.

**7.9 Hensel's lemma.** If $|f(a_0)| < |f'(a_0)|^2$, Newton's iteration converges to the unique root $a$ with $|a - a_0| < |f'(a_0)|$, and $|a - a_0| = |f(a_0)/f'(a_0)|$; a simple root modulo $\mathfrak{m}$ lifts uniquely. *(proved here)* Theorem 7.12.

**7.10 Hensel, factorisation form.** Over a complete discrete valuation ring, $\bar{f} = \bar{g}\bar{h}$ with $\bar{g}$ monic and $\bar{g}, \bar{h}$ coprime lifts to $f = gh$, $g$ monic of degree $\deg \bar{g}$. *(proved here)* Theorem 7.14.

**7.11 Coefficients of irreducible polynomials.** If $f = \sum a_i X^i \in K[X]$ is irreducible with $a_0 a_n \neq 0$, $\max|a_i| = \max(|a_0|, |a_n|)$; a monic irreducible polynomial with $f(0) \in \mathcal{O}$ lies in $\mathcal{O}[X]$. *(proved here)* Corollary 7.15.

**7.12 Teichmüller lifts.** $[a] = \lim \tilde{a}^{q^n}$ is the unique lift of $a \in \mathbb{F}_q$ with $[a]^q = [a]$; $k^\times \simeq \mu_{q-1}(K)$; $\mathcal{O}^\times = \mu_{q-1}(K) \times (1 + \mathfrak{m})$. *(proved here)* Theorem 7.17.

**7.13 Units.** For $p$ odd, exp and log give $p^n\mathbb{Z}_p \simeq 1 + p^n\mathbb{Z}_p$ $(n \geq 1)$, $1 + p\mathbb{Z}_p \simeq \mathbb{Z}_p$ generated by $1 + p$, and $\mathbb{Q}_p^\times \simeq \mathbb{Z} \times \mathbb{Z}/(p-1) \times \mathbb{Z}_p$; for $p = 2$, $\mathbb{Z}_2^\times = \{\pm 1\} \times (1 + 4\mathbb{Z}_2)$, $1 + 4\mathbb{Z}_2 \simeq \mathbb{Z}_2$ generated by 5; in general $K^\times = \varpi^{\mathbb{Z}} \times \mu_{q-1}(K) \times U^{(1)}$ with $U^{(1)}$ pro-$p$ and uniquely $m$-divisible for $p \nmid m$. *(proved here)* Theorem 7.18.

**7.14 Roots of unity.** $\mu(\mathbb{Q}_p) = \mu_{p-1}$ ($p$ odd), $\mu(\mathbb{Q}_2) = \{\pm 1\}$; $\mu_m(K) = \mu_{\gcd(m, q-1)}$ for $p \nmid m$; $\mu_{\ell^k} \subset \mathbb{Q}_p$ iff $\ell^k \mid p - 1$; $i \in \mathbb{Q}_\ell$ iff $\ell \equiv 1 \pmod 4$. *(proved here)* Corollary 7.19.

**7.15 $\ell^k$-th powers ($\ell \neq p$).** A unit is an $\ell^k$-th power iff its reduction is one in $\mathbb{F}_q^\times$; $\varpi^a u$ is one iff moreover $\ell^k \mid a$; $K^\times/(K^\times)^{\ell^k} \simeq \mathbb{Z}/\ell^k \times \mathbb{Z}/\ell^{\min(k, v_\ell(q-1))}$, of order $\ell^k \# \mu_{\ell^k}(K)$; $\varprojlim K^\times/(K^\times)^{\ell^k} \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^{v_\ell(q-1)}$ with torsion $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$; square classes of $\mathbb{Q}_p$ and $\mathbb{Q}_2$. *(proved here)* Theorem 7.20.

**7.16 Equivalence of norms.** All norms on a finite-dimensional space over a complete field are equivalent, and the space is complete. *(proved here)* Lemma 7.23.

**7.17 Extension of the absolute value.** $|x|_L = |N_{L/K} x|^{1/[L:K]}$ is the unique extension; $L$ is complete; $\mathcal{O}_L$ is the integral closure; conjugates have equal absolute values. *(proved here)* Theorem 7.24.

**7.18** $e$ **and** $f$**.** $\mathcal{O}_L$ is free over $\mathcal{O}_K$ with basis $\alpha_i \varpi_L^j$, and $ef = [L : K]$; multiplicative in towers. *(proved here)* Theorem 7.25.

**7.19 Unramified, totally, tamely, wildly ramified** (definition): $e = 1$, $f = 1$, $p \nmid e$, $p \mid e$. Definition 7.26.

**7.20 Eisenstein.** $L/K$ is totally ramified of degree $e$ iff it is generated by a root $\pi$ of an Eisenstein polynomial of degree $e$; then $\pi$ is a uniformiser and $\mathcal{O}_L = \mathcal{O}_K[\pi]$. *(proved here)* Theorem 7.27.

**7.21 Examples;** $\mathbb{Z}_3[\zeta_3]$**.** $\mathbb{Q}_p(\zeta_{p^n})$ is totally ramified with group $(\mathbb{Z}/p^n)^\times$; $\mathbb{Q}_3(\zeta_3) = \mathbb{Q}_3(\sqrt{-3})$ is tamely totally ramified with ring of integers $\mathbb{Z}_3[\zeta_3] = \mathbb{Z}_3 \otimes \mathbb{Z}[\zeta_3]$, uniformiser $1-\zeta_3$, $(1-\zeta_3)^2 = -3\zeta_3$, the completion of $\mathbb{Z}[\zeta_3]$; $\mathbb{Q}_p(p^{1/m})$, $\mathbb{Q}_p(\sqrt{p^*})$. *(proved here)* Proposition 7.28.

**7.22 Completions of number fields.** The extensions of $|\cdot|_p$ to $F$ correspond to the $\mathfrak{p} \mid p$; $F_\mathfrak{p}$ has $e_\mathfrak{p}$, $f_\mathfrak{p}$ and $\mathcal{O}_{F_\mathfrak{p}} = \varprojlim \mathcal{O}_F/\mathfrak{p}^n$; $\mathcal{O}_F \otimes \mathbb{Z}_p \simeq \prod_{\mathfrak{p}|p} \mathcal{O}_{F_\mathfrak{p}}$. *(proved here)* Theorem 7.30.

**7.23 Krasner's lemma.** If $|\beta - \alpha| < |\alpha - \alpha_i|$ for the other conjugates, $K(\alpha) \subseteq K(\beta)$; nearby polynomials define the same fields. *(proved here)* Lemma 7.31.

**7.24 Decomposition groups.** $\mathrm{Gal}(L_\mathfrak{P}/F_\mathfrak{p}) \simeq D_\mathfrak{P}$ with inertia to inertia; an embedding $\overline{\mathbb{Q}} \to \overline{\mathbb{Q}}_p$ identifies $G_{\mathbb{Q}_p}$ with $D_{\bar{\mathfrak{P}}} \subseteq G_\mathbb{Q}$, $I_p$ with $I_{\bar{\mathfrak{P}}}$, Frobenius with Frobenius. *(proved here)* Proposition 7.32.

**7.25 Tame totally ramified extensions.** $L = K(\varpi^{1/e})$; Galois iff $e \mid q-1$, and then cyclic, $\sigma \mapsto \sigma(\alpha)/\alpha$. *(proved here)* Proposition 7.33.

**7.26 Unramified extensions.** Exactly one of each degree, $K_f = K(\mu_{q^f-1})$; $\mathrm{Gal}(K_f/K) \simeq \mathrm{Gal}(\mathbb{F}_{q^f}/\mathbb{F}_q)$, cyclic, generated by Frob; largest unramified subextension $L_0$ with $L/L_0$ totally ramified. *(proved here)* Theorem 7.34.

**7.27** $K^{\mathrm{ur}}$ (definition): $\bigcup K_f = K(\mu_m : p \nmid m)$. Definition 7.35.

**7.28 Properties of** $K^{\mathrm{ur}}$**.** $\mathrm{Gal}(K^{\mathrm{ur}}/K) \simeq \hat{\mathbb{Z}}$, generated by Frob; discrete, not complete; Hensel holds; units are $m$-th powers for $p \nmid m$, and $x$ is an $m$-th power iff $m \mid v_K(x)$; $\mu_{\ell^\infty} \subset \mathbb{Q}_q^{\mathrm{ur}}$. *(proved here)* Proposition 7.36.

**7.29 Frobenius convention.** Frob arithmetic, $\Phi = \mathrm{Frob}^{-1}$ geometric; $\chi_\ell(\mathrm{Frob}) = q$; group theory uses Frob, $L$-factors use $\Phi$. Convention 7.37.

**7.30 Unramified base change.** For $K/\mathbb{Q}_p$ unramified: $K^{\mathrm{ur}} = \mathbb{Q}_p^{\mathrm{ur}}$, $I_K = I_p$, $P_K = P_p$; images of inertia, invariants, monodromy and conductors are unchanged. In general $I_K = I_p \cap G_K$ has index $e(K/\mathbb{Q}_p)$. *(proved here)* Proposition 7.38.

**7.31 The cyclotomic character locally.** $\chi_\ell$ is unramified at $p \neq \ell$ with $\chi_\ell(\mathrm{Frob}_p) = p$; $\chi_p(I_p) = \mathbb{Z}_p^\times$. *(proved here)* Proposition 7.39.

**7.32 Ramification groups** (definition): $G_i = \{\sigma : v_L(\sigma x - x) \geq i+1 \ \forall x \in \mathcal{O}_L\}$; $G_0$ inertia, $G_1$ wild inertia. Definition 7.41.

**7.33 Structure of the $G_i$.** $\#G_0 = e$; $G_0/G_1 \hookrightarrow k_L^\times$ by $\sigma \mapsto \sigma\pi/\pi$; $G_i/G_{i+1} \hookrightarrow (k_L, +)$ for $i \geq 1$; so $G_1$ is the Sylow $p$-subgroup of $G_0$, and $L/K$ is tame iff $G_1 = 1$. Upper numbering, Herbrand and Hasse-Arf are stated. *(proved here, except (iv): stated; reference)* Proposition 7.42.

**7.34 The filtration $G_K \supset I_K \supset P_K$.** $G_K/I_K \simeq \hat{\mathbb{Z}}$; $K^{\mathrm{tame}} = \bigcup_{p\nmid m} K^{\mathrm{ur}}(\varpi^{1/m})$; $P_K = \mathrm{Gal}(\overline{\mathbb{Q}}_p/K^{\mathrm{tame}})$ is normal, pro-$p$, maps onto every $G_1(L/K)$, and is the unique pro-$p$ Sylow subgroup of $I_K$. *(proved here, except the upper filtration (iv): stated; reference)* Theorem 7.45.

**7.35 Semidirect product.** $G_K = I_K \rtimes \overline{\langle \mathrm{Frob} \rangle}$ with $\overline{\langle \mathrm{Frob} \rangle} \simeq \hat{\mathbb{Z}}$; a Frobenius-invariant character of $I_K$ extends to $G_K$. *(proved here)* Corollary 7.46.

**7.36 The tame character.** $\theta_m(\sigma) = \sigma(\varpi^{1/m})/\varpi^{1/m}$ is a surjection $I_K \to \mu_m$ independent of choices; $I_K/P_K \simeq \prod_{\ell\neq p} \mathbb{Z}_\ell(1)$, procyclic of order prime to $p$; $t_\ell\colon I_K \to \mathbb{Z}_\ell(1)$ identifies the maximal pro-$\ell$ quotient of $I_K$ with $\mathbb{Z}_\ell(1)$; maps from $I_K$ to pro-$\ell$ groups factor through $t_\ell$; $t_{\ell,K}|_{I_L} = e(L/K)t_{\ell,L}$. *(proved here)* Theorem 7.47.

**7.37 Open subgroups of inertia.** Every open $U \subseteq I_K$ is $I_L$ for a finite $L/K$; its maximal pro-$\ell$ quotient is $\mathbb{Z}_\ell(1)$. *(proved here)* Corollary 7.48.

**7.38 A Frobenius fixing $\varpi^{1/m}$.** Some arithmetic Frobenius lift fixes all $\varpi^{1/m}$, $p \nmid m$; two differ by $P_K$. *(proved here)* Corollary 7.49.

**7.39 The tame relation.** $t_\ell(\tau\sigma\tau^{-1}) = \chi_\ell(\tau)t_\ell(\sigma)$. On $I_K/P_K$ one has $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^q$, equivalently $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$, and for the geometric Frobenius $\Phi\, t\, \Phi^{-1} = t^{q^{-1}}$. The group $\mathrm{Gal}(K^{\mathrm{tame}}/K)$ is $\prod_{\ell\neq p} \mathbb{Z}_\ell(1) \rtimes \hat{\mathbb{Z}}$, with 1 acting by $q$. *(proved here; the presentation by two generators is stated; reference)* Theorem 7.50.

**7.40 Finite tame images of inertia.** If $\rho(I_K)$ is finite of order prime to $p$, it is cyclic and $\rho(\mathrm{Frob})h\rho(\mathrm{Frob})^{-1} = h^q$; Frobenius centralises (inverts) a generator of order $m$ iff $q \equiv 1$ $(q \equiv -1) \pmod m$. *(proved here)* Corollary 7.52.

**7.41 Wild inertia in $\ell$-adic representations.** $\rho(P_K)$ is a finite $p$-group embedding in $\mathrm{GL}_n(\mathcal{O}/\varpi_\lambda)$; on an open normal $I_1 \subseteq I_K$, $\rho$ factors through $t_\ell$; $\rho$ is tame iff $\rho(I_K)$ has no element of order $p$. *(proved here)* Proposition 7.53.

**7.42 Kummer theory over $K$.** The Kummer cocycle $\kappa_x$ and its properties; $K^\times/(K^\times)^{\ell^k} \simeq H^1(K, \mu_{\ell^k})$ and $\widehat{K^\times} \simeq H^1(K, \mathbb{Z}_\ell(1)) \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^{v_\ell(q-1)}$. *(proved here, except the isomorphisms with $H^1$: stated; Chapter 17)* Proposition 7.57.

**7.43 The Kummer cocycle on inertia and Frobenius.** $\kappa_x = v_K(x)\kappa_\varpi + \kappa_u$; $\kappa_x|_{I_K} = v_K(x)\, t_\ell$; $\kappa_u$ is unramified with class the class of $\bar{u}$ in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$; $c' = \kappa_x(\mathrm{Frob})$ is defined modulo $(v_K(x), q-1)$; $I_K$ fixes $x^{1/\ell^k}$ iff $\ell^k \mid v_K(x)$. *(proved here)* Theorem 7.58.

**7.44 Quadratic extensions of $\mathbb{Q}_p$.** Three for $p$ odd (one unramified, two tame), restricting to the unique order-two character $\theta_2$ of $I_p$ when ramified; seven for $p = 2$, with $\mathbb{Q}_2(\sqrt{5}) = \mathbb{Q}_2(\sqrt{-3})$ unramified and jumps 1 for $\mathbb{Q}_2(\sqrt{-1})$, $\mathbb{Q}_2(\sqrt{3})$, 2 for $\mathbb{Q}_2(\sqrt{\pm 2})$, $\mathbb{Q}_2(\sqrt{\pm 10})$. *(proved here)* Proposition 7.60.

**7.45 Global quadratic characters locally.** $\chi_d$ is unramified at $p \nmid d$ with $\chi_d(\mathrm{Frob}_p) = (d/p)$; $\chi_{q^*}$ is unramified outside $q$ and equals $\theta_2$ on $I_q$; twisting by $\chi_{q^*}$ changes nothing on inertia at $p \neq q$; every quadratic character of $G_{\mathbb{Q}_p}$ is a restriction of a product of $\chi_d$'s. *(proved here)* Proposition 7.61.

**7.46 Local class field theory for $\mathbb{Q}_p$.** $\mathbb{Q}_p^{\mathrm{ab}} = \mathbb{Q}_p^{\mathrm{ur}}\mathbb{Q}_p(\mu_{p^\infty})$; the Artin map ($p \mapsto \mathrm{Frob}$, $u \mapsto (\zeta \mapsto \zeta^{u^{-1}})$); $a(\chi)$ is the least $n$ with $\chi(\mathrm{Art}(U^{(n)})) = 1$. *(stated; reference)* Theorem 7.62.

**7.47 Weil-Deligne relation.** $\|w\| = q^{d(w)}$, $r(w)Nr(w)^{-1} = \|w\|N$: $r(\mathrm{Frob})Nr(\mathrm{Frob})^{-1} = qN$, $r(\Phi)Nr(\Phi)^{-1} = q^{-1}N$. Convention 7.64.

**7.48 The Weil group** (definition): $W_K = \{w \in G_K : w|_{K^{\mathrm{ur}}} \in \mathrm{Frob}^{\mathbb{Z}}\}$, with $I_K$ open. Definition 7.65.

**7.49 Properties of $W_K$.** Dense in $G_K$, $W_K/I_K \simeq \mathbb{Z}$, $\|w\| = \chi_\ell(w)$; unramified characters are free on $\Phi$; $W_{\mathbb{Q}_p}^{\mathrm{ab}} \simeq \mathbb{Q}_p^\times$ (stated). *(proved here, except (iv): stated; reference)* Proposition 7.66.

**7.50 Matrix logarithm.** exp and log are inverse bijections $\ell^2\,\mathrm{M}_n(\mathcal{O}) \leftrightarrows 1 + \ell^2\,\mathrm{M}_n(\mathcal{O})$, with $g^a = \exp(a \log g)$. *(proved here)* Lemma 7.67.

**7.51 Grothendieck's monodromy theorem.** For $\rho\colon G_K \to \mathrm{GL}(V)$ $\ell$-adic, there is a unique nilpotent $N$ with $\rho(\sigma) = \exp(t_\zeta(\sigma)N)$ on an open normal $I_1 \subseteq I_K$, and $\rho(w)N\rho(w)^{-1} = \|w\|N$; $N = 0$ iff $\rho(I_K)$ is finite. *(proved here)* Theorem 7.68.

**7.52 Weil-Deligne representations** (definition): pairs $(r, N)$, with $r$ a representation of $W_K$ such that $r(I_K)$ is finite and $N$ nilpotent, satisfying $r(w)Nr(w)^{-1} = \|w\|N$; Frobenius-semisimplicity. Definition 7.69.

**7.53 Operations.** Sums, tensor products, duals ($-N^{\mathsf{T}}$), Hom, $\mathrm{ad}^0$ ($[N, \cdot\,]$), twists. *(proved here; Frobenius-semisimplification stated)* Proposition 7.70.

**7.54 WD($\rho$).** $r(\Phi^n\sigma) = \rho(\Phi^n\sigma)\exp(-t_\zeta(\sigma)N)$ is a Weil-Deligne representation, independent of $\Phi$ and $\zeta$ up to isomorphism; equivalence of categories and independence of $\ell$ for elliptic curves are stated. *(proved here, except (iii), (iv): stated; reference)* Theorem 7.71.

**7.55** $\mathrm{sp}(n)$ (definition): $r(w)e_i = \|w\|^i e_i$, $Ne_i = e_{i+1}$; twists $\mathrm{sp}(n) \otimes \mu$; unramified and ramified twists of Steinberg. Definition 7.72.

**7.56 Two-dimensional Weil-Deligne representations.** $\mathrm{sp}(n) \otimes \mu$ is indecomposable; an $(r, N)$ of dimension 2 with $N \neq 0$ is Frobenius-semisimple and isomorphic to $\mathrm{sp}(2) \otimes \mu$ for a unique $\mu$; the case $N = 0$ is stated. *(proved here, except (iii): stated; reference)* Proposition 7.73.

**7.57 Inertia invariants.** $V^{\rho(I_K)} = (\ker N)^{r(I_K)}$, with $\rho(\Phi) = r(\Phi)$ there. *(proved here)* Proposition 7.76.

**7.58 Local $L$-factor** (definition): $L(V, s) = \det(1 - \Phi q^{-s} \mid V^{I_K})^{-1}$, $\Phi$ geometric. Definition 7.77.

**7.59 Properties of $L$-factors.** Well defined, additive; $L(\mathbb{Q}_\ell(1), s) = (1-q^{-1-s})^{-1}$, $L(V(1), s) = L(V, s+1)$ (false with Frob); $L(\mathrm{sp}(2) \otimes \mu, s)$; $L_p(V_\ell E, s) = L_p(E, s+1)$ and $L_p(V_\ell E^\vee, s) = L_p(E, s)$. *(proved here)* Proposition 7.78.

**7.60 Swan and Artin conductors** (definition): $\mathrm{sw}(V) = \sum_{i\geq 1} \frac{g_i}{g_0} \dim V/V^{G_i}$, $a(V) = \dim V - \dim V^{I_K} + \mathrm{sw}(V)$. Definition 7.81.

**7.61 Properties of the conductors.** Independence of $L$ and integrality (stated); $\mathrm{sw} = 0$ iff tame; additivity, invariance under unramified twists and base change; $a(\chi) = 1$ for tame $\chi$, $a(\chi_{-4}) = 2$, $a(\chi_{\pm 8}) = 3$ at 2; the same for mod $\ell$ representations. *(proved here, except (i), (ii): stated; reference)* Theorem 7.82.

**7.62 Conductor of a Weil-Deligne representation** (definition): the exponent is $a(r, N) = \dim V - \dim(\ker N)^{r(I_K)} + \mathrm{sw}(r)$. Definition 7.83.

**7.63 Properties.** $a(\rho) = \dim V - \dim V^{I_K} + \mathrm{sw}(\rho)$; $a(V \otimes \chi) = \dim V \cdot a(\chi)$ for $V$ unramified; $a(\mathrm{sp}(2) \otimes \mu) = 1$, $a(\mathrm{sp}(2) \otimes \mu\chi) = 2a(\chi)$; the conductor exponent of $E$ at $p$ is $a(V_\ell E)$ (definition, Chapter 10), equal to $v_p(\Delta_{\min}) + 1 - m$ with $m$ the number of components of the special fibre of the minimal regular model (Ogg, Saito). *(proved here; independence of $\ell$ and Ogg's formula stated; reference)* Proposition 7.84.

**7.64 Epsilon factors.** Existence and uniqueness (Langlands, Deligne) and the Weil-Deligne correction factor. *(stated; reference)* Theorem 7.85.

**7.65 Unramified shape.** $A$ has eigenvalues $\alpha/\beta, 1, \beta/\alpha$ for both Frobenii; $L(A,1)^{-1} = \frac{q-1}{q} \cdot \frac{(q+1)^2 - a^2}{q^2}$. *(proved here)* Proposition 7.86.

**7.66 Finite cyclic tame inertia.** Order 2: $V \otimes \chi$ unramified; orders $3, 4, 6$: $V^{I_q} = 0$, $a(V) = 2$, $A^{I_q} = \Omega(2g - \mathrm{tr}\, g)$ with Frob and $\Phi$ acting by $\chi(q)$, $\chi = \chi_{-3}$ or $\chi_{-4}$, $L(A,s)^{-1} = 1 - \chi(q)q^{-s}$, $a(A) = 2$. *(proved here)* Proposition 7.87.

**7.67 Steinberg and its twists.** $a(\mathrm{sp}(2) \otimes \mu) = 1$, $a(\mathrm{sp}(2) \otimes \mu\chi) = 2a(\chi)$; $\mathrm{ad}^0 \mathrm{sp}(2) \simeq \mathrm{sp}(3) \otimes \|\cdot\|^{-1}$, $A^{I_q} = \Omega N$ with $\Phi$-eigenvalue $q^{-1}$, $L(\mathrm{ad}^0 V, s) = (1 - q^{-1-s})^{-1}$ for both. *(proved here)* Proposition 7.88.

**7.68 The table of shapes at a tame prime.** $\rho(I_q)$, $\dim V^{I_q}$, $a(V)$, $L(V,s)$, $\dim A^{I_q}$, $L(A,s)$ for the six shapes, for $q \geq 5$ (and partly for $q = 2, 3$). *(proved here; the Kodaira column stated, Chapters 10 and 11)* Theorem 7.89.

**7.69 The primes 2 and 3.** An inertia image of order divisible by 3 is wild at 3 ($a \geq 3$); order 3 is tame at 2 with $L(A,s)^{-1} = 1 + 2^{-s}$; non-abelian finite images have $A^{I_q} = 0$; $a(\mathrm{sp}(2) \otimes \mu\chi) \in \{4, 6\}$ at 2. *(proved here)* Proposition 7.91.

## Where the paper uses this chapter

- **Section 1.1** (the local terms): $I_p$ and $D_p \simeq G_{\mathbb{Q}_p}$, Proposition 7.32; $\delta_p = \dim V^{I_p}$, Proposition 7.76; $L_p(A_f, s) = \det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid A_f^{I_p})^{-1}$ and the naive factor, Definition 7.77, Convention 7.37, Paper link 7.80.
- **Definition def:terms and the table of Section 1**: the shapes and their factors, Theorem 7.89; $e_\ell = 0$ at multiplicative primes, Proposition 7.88; potentially good $\mathrm{I}_n^*$ at 2, Remark 7.92.

- **Lemma lem:tamagawa-exp** (unramified at $q$ modulo $\ell^t$ iff $\ell^t \mid n_q$): unramified base change, Proposition 7.38; $\mu_{\ell^\infty} \subset \mathbb{Q}_q^{\mathrm{ur}}$ and $\ell^t$-th powers in $\mathbb{Q}_q^{\mathrm{ur}}$, Proposition 7.36(iv); Kummer theory on inertia, Theorem 7.58(v); Paper link 7.40, Paper link 7.59.

- **Lemma lem:IV and Remark rem:types**: $\mathbb{Z}_3[\zeta_3]$ as the ring of integers of $\mathbb{Q}_3(\sqrt{-3})$, Proposition 7.28; tame inertia images are cyclic and Frobenius acts by the $q$-th power, Corollary 7.52; wild inertia, Proposition 7.53; conductor exponents and the Serre conductor, Theorem 7.82(v), Proposition 7.84, Proposition 7.87; the value at $q = 2$, Proposition 7.91(ii).

- **Lemma lem:twist**: (i) unramified base change and the local triviality of $\chi_{q^*}$ at $p \neq q$, Proposition 7.38, Proposition 7.61(v); (ii) tame conductor exponents for the exchanges listed, Proposition 7.60, Proposition 7.84; (iii) the ramified twist of Steinberg has conductor exponent 2, Proposition 7.88(ii).

- **Lemma lem:euler**: Frobenius centralises or inverts the generator of inertia, Corollary 7.52; the invariant lines of $\mathrm{ad}^0 V$ and their eigenvalues, Proposition 7.87, Proposition 7.88; the rings $\mathbb{Z}_\ell[\zeta_3]$, $\mathbb{Z}_\ell[i]$, Example 7.16(g); Paper link 7.90.

- **Proposition prop:shapes**: the four shapes of $P_q$, Theorem 7.89, Proposition 7.91(iii); $G_{\mathbb{Q}_q} = I_q \rtimes \langle \phi \rangle$ and the extension of $\psi$, Corollary 7.46; quadratic characters of $G_{\mathbb{Q}_q}$ as restrictions of Dirichlet characters, Proposition 7.61(vi).

- **Remark rem:wild and Section sec:wild**: wild inertia at 2 and 3, Remark 7.54, Proposition 7.91, Remark 7.92.

- **Lemma lem:tamfactor**: the tame character and the maximal pro-$\ell$ quotient of inertia, Theorem 7.47; $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$, Theorem 7.50; the Kummer cocycle of $q_E$, its restriction $n\, t_\ell$ to inertia, the class of $u$ and the indeterminacy of $c'$, Theorem 7.58, Corollary 7.49; $\varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$ and $\mathbb{Z}_\ell(1)/(q-1)$, Theorem 7.20, Proposition 7.57; the matrices with corner $\varepsilon\kappa$, Example 7.74; the class $c(u)$ of the reduction of $u$, Paper link 7.21; Paper link 7.56.

- **Lemma lem:tamadd**: the maximal pro-$\ell$ quotient of an open subgroup of inertia, Corollary 7.48.

- **Lemma lem:fpr and the paragraph after Lemma lem:dfg21**: $L_p(M(1), s) = L_p(M, s+1)$ with the geometric Frobenius, Proposition 7.78(iii), Warning 7.79.

- **Lemmas lem:crit, lem:old, lem:ddt417**: the completions $\mathcal{O}_\lambda$ and $\mathcal{O} \otimes \mathbb{Z}_\ell = \prod \mathcal{O}_\lambda$, Theorem 7.30.

- **Theorems thm:twist, thm:quant, thm:qual**: $\chi_S$ and $\chi_{q^*}$ locally, Proposition 7.61.

## 7.1 Absolute values, completions, $\mathbb{Q}_p$ and $\mathbb{Z}_p$

This section builds the fields in which every local statement of the paper lives: $\mathbb{Q}_q$ at a bad prime $q$, and $\mathbb{Q}_\ell$, $\mathbb{Z}_\ell$ for the coefficients. Chapter 2 defined $v_p$ and $|\cdot|_p$ on $\mathbb{Q}$, and Chapter 3 built $\mathbb{Z}_\ell$ algebraically as an inverse limit; here we complete $\mathbb{Q}$ analytically and show that the two constructions agree (Theorem 7.8(ii)). The paper uses these fields without comment in Sections 2 and 4; nothing in this section is a statement of the paper, but everything after it depends on it.

**Convention 7.1** (local fields)**.** (a) $p$ is a prime and $K$ denotes $\mathbb{Q}_p$ or a finite extension of $\mathbb{Q}_p$ (Section 7.3), with valuation ring $\mathcal{O}_K$, maximal ideal $\mathfrak{m}_K$, uniformiser $\varpi_K$, residue field $k_K = \mathbb{F}_q$, $q = p^f$, normalised valuation $v_K$ ($v_K(\varpi_K) = 1$) and normalised absolute value $|x|_K = q^{-v_K(x)}$. On $\mathbb{Q}_p$ we write $v_p$, $|\cdot|_p$ as in Chapter 2. The letter $q$ is both $\#k_K$ and, when the paper's prime is meant, the bad prime $q$; the two agree for $K = \mathbb{Q}_q$, the field over which the paper works (when its unramified quadratic extension appears, its residue field is written $\mathbb{F}_{q^2}$). In the general theory the residue characteristic is $p$ and $\ell \neq p$ is another prime; at the paper's prime, $\ell \neq q$.

(b) $\overline{\mathbb{Q}}_p$ is a fixed algebraic closure of $\mathbb{Q}_p$ and $G_K = \mathrm{Gal}(\overline{\mathbb{Q}}_p/K)$. Inside $\overline{\mathbb{Q}}_p$ we have $K \subseteq K^{\mathrm{ur}} \subseteq K^{\mathrm{tame}}$, the maximal unramified and the maximal tamely ramified extensions (Definition 7.35, Theorem 7.45); $I_K = \mathrm{Gal}(\overline{\mathbb{Q}}_p/K^{\mathrm{ur}})$ is the inertia group and $P_K = \mathrm{Gal}(\overline{\mathbb{Q}}_p/K^{\mathrm{tame}})$ the wild inertia group; for $K = \mathbb{Q}_p$ we write $I_p$, $P_p$.

(c) $\mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}$ (transition maps $\zeta \mapsto \zeta^\ell$), on which $G_K$ acts through the cyclotomic character $\chi_\ell$; it is written additively when it receives the values of cocycles. Where a $\mathbb{Z}_\ell$-valued version is needed we fix a generator $\zeta = (\zeta_{\ell^k})_k$ of $\mathbb{Z}_\ell(1)$ and identify $\mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$ by $\zeta \mapsto 1$.

(d) Frob is an *arithmetic* Frobenius and $\Phi = \mathrm{Frob}^{-1}$ the *geometric* one (Convention 7.37); $\Phi$ without subscript is always the geometric Frobenius of the local field in question, and the geometric Frobenius at a named prime $p$ is written $\mathrm{Frob}_p^{-1}$. With a numerical subscript, as in $\Phi_{p^n}$ and $\Phi_3$, the letter denotes a cyclotomic polynomial (Chapter 2); $\Phi_q^{\mathrm{geom}}$ is the geometric component group of the Néron model at $q$ (Chapter 10).

(e) Other letters: $\varepsilon$ is an unramified character of order at most 2 (the paper's Tate data), $\epsilon$ an epsilon factor; $\theta_m$ are the tame characters of Theorem 7.47, $\theta_2$ the character of order two of inertia, and $\eta_{\mathrm{ur}}$ the unramified quadratic character; $[a]$ is the Teichmüller lift of $a$; $N$ is a monodromy operator (Sections 7.7 to 7.9), never a conductor; $e$ is a ramification index, except in Section 7.9, where it is also the paper's basis vector $e = e_1 \otimes e_2^\vee$ of $\mathrm{ad}^0 T$, and we say which. Quadratic characters are $\chi_d$, indexed by the fundamental discriminant $d$ ($\chi_{-3}$, $\chi_{-4}$, $\chi_8$, $\chi_{q^*}$); a letter subscript $\chi_\ell$ is the cyclotomic character.

### 7.1.1 Absolute values

**Definition 7.2** (absolute values)**.** An *absolute value* on a field $F$ is a map $|\cdot| : F \to \mathbb{R}_{\geq 0}$ with $|x| = 0$ iff $x = 0$, $|xy| = |x||y|$ and $|x+y| \leq |x|+|y|$. It is *nonarchimedean* if $|x+y| \leq \max(|x|, |y|)$ for all $x, y$ (the ultrametric inequality), and *trivial* if $|x| = 1$ for all $x \neq 0$. Two absolute values are *equivalent* if they define the same topology. For a nonarchimedean $|\cdot|$ the *valuation ring* is $\mathcal{O} = \{|x| \leq 1\}$, its *maximal ideal* $\mathfrak{m} = \{|x| < 1\}$, and $k = \mathcal{O}/\mathfrak{m}$ the *residue field*. The absolute value is *discrete* if $|F^\times|$ is a discrete subgroup of $\mathbb{R}_{>0}$.

The definition abstracts the one example that matters here. On $\mathbb{Q}$, $|x|_p = p^{-v_p(x)}$ (Chapter 2) measures how divisible $x$ is by $p$: $|12|_2 = \frac{1}{4}$, $|1/12|_2 = 4$, $|12|_3 = \frac{1}{3}$ and $|12|_5 = 1$. It is nonarchimedean, because a sum of two multiples of $p^k$ is again a multiple of $p^k$, that is $v_p(x+y) \geq \min(v_p(x), v_p(y))$; and the inequality can be strict, as $|2+6|_2 = \frac{1}{8} < \frac{1}{2}$ shows. Its value group is $p^{\mathbb{Z}}$, so it is discrete; its valuation ring is $\mathbb{Z}_{(p)}$, the fractions $a/b$ with $p \nmid b$, with maximal ideal $p\mathbb{Z}_{(p)}$ and residue field $\mathbb{F}_p$. Two numbers are $p$-adically close when their difference is divisible by a high power of $p$; completing $\mathbb{Q}$ for this notion of closeness is what produces $\mathbb{Q}_p$ below.

**Proposition 7.3.** *Let $|\cdot|$ be an absolute value on a field $F$.*

*(i) $|\cdot|$ is nonarchimedean iff $|n\cdot 1|\le 1$ for all $n\in\mathbb{Z}$.*

*(ii) If $|\cdot|$ is nonarchimedean: $|x+y|=\max(|x|,|y|)$ whenever $|x|\neq|y|$; $\mathcal{O}$ is a subring of $F$, $\mathfrak{m}$ its unique maximal ideal and $\mathcal{O}^\times=\{|x|=1\}$; every point of an open ball $B(a,r)=\{|x-a|<r\}$ is a centre of it, and open balls are closed. If $F$ is complete, a series $\sum a_n$ converges iff $a_n\to 0$.*

*(iii) If $|\cdot|$ is nonarchimedean, discrete and nontrivial, then $|F^\times|=c^{\mathbb{Z}}$ for a unique $c\in(0,1)$, there is a valuation $v\colon F^\times\to\mathbb{Z}$ with $|x|=c^{v(x)}$, and $\mathcal{O}$ is a discrete valuation ring with maximal ideal $\varpi\mathcal{O}$ for any $\varpi$ with $v(\varpi)=1$.*

*(iv) Two nontrivial absolute values $|\cdot|_1$, $|\cdot|_2$ are equivalent iff $|\cdot|_1=|\cdot|_2^s$ for some $s>0$.*

*Proof.* (i) If the ultrametric inequality holds, $|n\cdot 1|=|1+\cdots+1|\le 1$ by induction. Conversely, if $|n\cdot 1|\le 1$ for all $n$, then for every $k\ge 1$

$$|x+y|^k=\Big|\sum_{i=0}^{k}\tbinom{k}{i}x^iy^{k-i}\Big|\le (k+1)\max(|x|,|y|)^k,$$

since each binomial coefficient is an integer, hence of absolute value at most 1, and each $|x^iy^{k-i}|$ is at most $\max(|x|,|y|)^k$. Taking $k$-th roots gives $|x+y|\le (k+1)^{1/k}\max(|x|,|y|)$, and $(k+1)^{1/k}\to 1$ as $k\to\infty$, so $|x+y|\le\max(|x|,|y|)$.

(ii) If $|x|<|y|$ then $|y|=|(x+y)-x|\le\max(|x+y|,|x|)$ forces $|y|\le|x+y|$, and $|x+y|\le|y|$ always. The ultrametric inequality and multiplicativity show that $\mathcal{O}$ is a ring and $\mathfrak{m}$ an ideal; an element of $\mathcal{O}\setminus\mathfrak{m}$ has absolute value 1, so its inverse lies in $\mathcal{O}$; hence $\mathfrak{m}$ is the set of nonunits, the unique maximal ideal. If $|b-a|<r$ and $|x-a|<r$ then $|x-b|<r$, so $B(a,r)=B(b,r)$. If $c\notin B(a,r)$, then $B(c,r)\cap B(a,r)=\emptyset$ (a common point would make both balls equal), so the complement of $B(a,r)$ is open. Finally $|\sum_{i=n+1}^{m}a_i|\le\max_{n<i\le m}|a_i|$, so the partial sums are Cauchy iff $a_i\to 0$; in a complete field Cauchy sequences converge, and conversely if the partial sums $s_n$ converge then $a_n=s_n-s_{n-1}\to 0$. (Over $\mathbb{R}$ the harmonic series shows that $a_n\to 0$ is not enough; the ultrametric inequality is what makes it so.)

(iii) A nontrivial discrete subgroup of $\mathbb{R}_{>0}$ (isomorphic by log to a discrete subgroup of $\mathbb{R}$) is infinite cyclic; let $c<1$ be its generator below 1 and $v(x)$ the integer with $|x|=c^{v(x)}$. Then $v$ is a valuation: $v(xy)=v(x)+v(y)$ is multiplicativity, and $v(x+y)\ge\min(v(x),v(y))$ is the ultrametric inequality, $c$ being less than 1. Moreover $\mathcal{O}=\{v\ge 0\}$, and every nonzero $x\in\mathcal{O}$ is $\varpi^{v(x)}u$ with $u=x\varpi^{-v(x)}$ of absolute value 1, a unit. If $I\neq 0$ is an ideal and $n$ the least value of $v$ on $I\setminus\{0\}$, then $I$ contains some $\varpi^n u$, hence $\varpi^n$; and every $x\in I$ is $\varpi^n\cdot x\varpi^{-n}$ with $x\varpi^{-n}\in\mathcal{O}$. So $I=\varpi^n\mathcal{O}$, and $\mathcal{O}$ is a principal ideal domain with the single nonzero prime ideal $\varpi\mathcal{O}$, a discrete valuation ring (Chapter 3).

(iv) If $|\cdot|_1=|\cdot|_2^s$ the open balls coincide. Conversely, in the topology of $|\cdot|_i$ one has $|x|_i<1$ iff $x^n\to 0$; so equivalent absolute values have the same set $\{|x|<1\}$. Fix $y$ with $|y|_1>1$ (it exists, the absolute value being nontrivial), so $|y|_2>1$ too, and put $s=\log|y|_1/\log|y|_2$. For $x\neq 0$ write $|x|_1=|y|_1^{\alpha}$. For integers $m$, $n>0$ with $m/n>\alpha$ we have $|x^n/y^m|_1<1$, hence $|x^n/y^m|_2<1$, that is $|x|_2<|y|_2^{m/n}$; likewise $m/n<\alpha$ gives $|x|_2>|y|_2^{m/n}$. So $|x|_2=|y|_2^{\alpha}$ and $|x|_1=|x|_2^s$. □

**Lemma 7.4** (summation in a complete nonarchimedean field)**.** *Let $F$ be complete for a nonarchimedean absolute value and $(a_i)_{i\in I}$ a countable family in $F$ such that for every $\delta>0$ only finitely*

*many $|a_i|$ exceed $\delta$. Then $\sum_i a_i$ converges for every enumeration of $I$, to a sum independent of the enumeration; and for every partition $I = \bigsqcup_j I_j$ the partial sums $s_j = \sum_{i\in I_j} a_i$ satisfy the same hypothesis and $\sum_j s_j = \sum_i a_i$.*

*Proof.* For an enumeration, the terms tend to 0, so the series converges by Proposition 7.3(ii). Given $\delta > 0$, let $I_\delta$ be the finite set of indices with $|a_i| > \delta$. Any partial sum of any enumeration that contains $I_\delta$ differs from $\sum_{i\in I_\delta} a_i$ by a finite sum of terms of absolute value at most $\delta$, hence by at most $\delta$; so two limits differ by at most $\delta$, for every $\delta$. For a partition, $|s_j| \le \max_{i\in I_j}|a_i|$, so only finitely many $|s_j|$ exceed $\delta$, and $\sum_j s_j$ differs from $\sum_{i\in I_\delta} a_i$ by at most $\delta$ once the finitely many $j$ meeting $I_\delta$ are included. □

*Remark* 7.5 (Ostrowski, product formula, weak approximation). By Ostrowski's theorem (Chapter 2) every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_\infty$ or to $|\cdot|_p$ for exactly one prime $p$, and $\prod_{v\le\infty}|x|_v = 1$ for $x \in \mathbb{Q}^\times$ (the product formula, also Chapter 2). For finitely many pairwise inequivalent absolute values $|\cdot|_1, \dots, |\cdot|_r$ on a field $F$ and elements $a_1, \dots, a_r$, there is for every $\delta > 0$ an $x \in F$ with $|x - a_i|_i < \delta$ for all $i$ (weak approximation). We do not prove this; see Neukirch [142, Chapter II, Section 3]. It is not used in the book.

### 7.1.2 Completions

**Theorem 7.6** (completion). *Let $F$ be a field with an absolute value $|\cdot|$. There is a field $\hat F$ containing $F$, with an absolute value extending $|\cdot|$, which is complete and in which $F$ is dense; it is unique up to a unique isometric $F$-isomorphism. If $|\cdot|$ is nonarchimedean, then:*

*(i) $|\hat F^\times| = |F^\times|$, and the extended absolute value is nonarchimedean;*

*(ii) the valuation ring $\hat{\mathcal{O}}$ of $\hat F$ is the closure of $\mathcal{O}$, and if $|\cdot|$ is discrete with uniformiser $\varpi$, then $\mathcal{O}/\varpi^n\mathcal{O} \to \hat{\mathcal{O}}/\varpi^n\hat{\mathcal{O}}$ is an isomorphism for every $n \ge 1$ (so the residue field does not change);*

*(iii) if $|\cdot|$ is discrete, $\hat{\mathcal{O}} \to \varprojlim_n \mathcal{O}/\varpi^n\mathcal{O}$, $x \mapsto (x \bmod \varpi^n)_n$, is an isomorphism of topological rings, the right side carrying the inverse limit of the discrete topologies.*

*Proof.* Let $C$ be the ring of Cauchy sequences in $F$ (termwise operations) and $\mathfrak{n} \subset C$ the ideal of sequences tending to 0. If $(x_n) \in C \setminus \mathfrak{n}$, then $|x_n|$ is bounded below by some $\delta > 0$ for $n \ge n_0$ (otherwise a subsequence tends to 0 and so does the Cauchy sequence); then $(x_n^{-1})_{n\ge n_0}$ is Cauchy, since $|x_n^{-1} - x_m^{-1}| \le \delta^{-2}|x_m - x_n|$. So $\mathfrak{n}$ is a maximal ideal and $\hat F = C/\mathfrak{n}$ is a field; $F$ embeds by constant sequences. Put $|(x_n)| = \lim|x_n|$, which exists since $||x_n| - |x_m|| \le |x_n - x_m|$, and depends only on the class modulo $\mathfrak{n}$. It is an absolute value extending $|\cdot|$: multiplicativity and the triangle inequality hold term by term, $|x_ny_n| = |x_n||y_n|$ and $|x_n + y_n| \le |x_n| + |y_n|$, and survive the limit; $\lim|x_n| = 0$ means $(x_n) \in \mathfrak{n}$, the zero class; and a constant sequence has the absolute value of its term. The class of $(x_n)$ is the limit of the constants $x_n$, so $F$ is dense; and a Cauchy sequence of classes converges to the class of a diagonal sequence of approximating elements of $F$, so $\hat F$ is complete. If $F'$ is another such field, the identity of $F$ extends uniquely to an isometry $\hat F \to F'$ by continuity (send $\lim x_n$ to $\lim x_n$), which is a field isomorphism.

(i) Let $\hat x = \lim x_n \ne 0$ with $x_n \in F$. For $n$ large, $|x_n - \hat x| < |\hat x|$, so $|x_n| = |\hat x|$ by Proposition 7.3(ii) applied in $\hat F$, where the ultrametric inequality holds by passing to the limit. Hence the value groups agree.

(ii) If $|\hat{x}| \le 1$ and $x_n \to \hat{x}$ with $x_n \in F$, then $|x_n| \le 1$ for $n$ large; so $\hat{\mathcal{O}}$ is the closure of $\mathcal{O}$, and $\hat{\mathcal{O}} \cap F = \mathcal{O}$. By (i), $\varpi$ is a uniformiser of $\hat{F}$. The map $\mathcal{O} \to \hat{\mathcal{O}}/\varpi^n\hat{\mathcal{O}}$ is surjective, because every $\hat{x} \in \hat{\mathcal{O}}$ has an $x \in F$ with $|x - \hat{x}| \le |\varpi|^n$, and then $x \in \mathcal{O}$; its kernel is $\mathcal{O} \cap \varpi^n\hat{\mathcal{O}} = \varpi^n\mathcal{O}$.

(iii) By (ii), $\hat{\mathcal{O}}/\varpi^n\hat{\mathcal{O}} = \mathcal{O}/\varpi^n\mathcal{O}$. The map is injective because $\bigcap_n \varpi^n\hat{\mathcal{O}} = 0$. It is surjective: given a compatible family $(x_n \bmod \varpi^n)$ with $x_n \in \mathcal{O}$, the sequence $(x_n)$ is Cauchy, since $x_m - x_n \in \varpi^n\mathcal{O}$ for $m \ge n$, and its limit maps to the family. The ideals $\varpi^n\hat{\mathcal{O}}$ are a basis of neighbourhoods of 0 on the left and correspond to the kernels of the projections on the right, so the bijection is a homeomorphism. □

### 7.1.3 The field $\mathbb{Q}_p$

**Definition 7.7.** $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ for $|\cdot|_p$, and $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \le 1\}$ is the ring of *$p$-adic integers*. The valuation $v_p$ extends to $\mathbb{Q}_p^\times$ by $|x|_p = p^{-v_p(x)}$ (Theorem 7.6(i)).

**Theorem 7.8.** *(i) $\mathbb{Z}_p$ is the closure of $\mathbb{Z}$, and of $\mathbb{Z}_{(p)}$, in $\mathbb{Q}_p$. It is a complete discrete valuation ring with uniformiser $p$, residue field $\mathbb{F}_p$ and nonzero ideals $p^n\mathbb{Z}_p$.*

*(ii) The map $\mathbb{Z}_p \to \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$, $x \mapsto (x \bmod p^n)_n$, is an isomorphism of topological rings. So the $\mathbb{Z}_p$ of this chapter is the ring $\mathbb{Z}_\ell$ built algebraically in Chapter 3 (at $\ell = p$), and the profinite group of Chapter 1.*

*(iii) $\mathbb{Z}_p^\times = \{|x|_p = 1\}$, and $x \in \mathbb{Z}_p$ is a unit iff $x \not\equiv 0 \pmod p$. Every $x \in \mathbb{Q}_p^\times$ is uniquely $x = p^{v_p(x)}u$ with $u \in \mathbb{Z}_p^\times$; so $\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mathbb{Z}_p^\times$ and $\mathbb{Q}_p = \mathbb{Z}_p[1/p] = \operatorname{Frac}\mathbb{Z}_p$.*

*(iv) $\mathbb{Z}_p$ is compact; $\mathbb{Q}_p$ is locally compact and totally disconnected.*

*(v) For a prime $q \ne p$, $q \in \mathbb{Z}_p^\times$, so $q^{-1} \in \mathbb{Z}_p$, and $q^a$ is defined for every $a \in \mathbb{Z}_p$ when $q \in 1+p\mathbb{Z}_p$ (Theorem 7.18).*

*Proof.* (i) The absolute value $|\cdot|_p$ on $\mathbb{Q}$ is discrete with value group $p^{\mathbb{Z}}$ and valuation ring $\mathbb{Z}_{(p)}$, whose residue field is $\mathbb{F}_p$ (Chapter 2). By Theorem 7.6, $\mathbb{Z}_p$ is the closure of $\mathbb{Z}_{(p)}$ and a complete discrete valuation ring with uniformiser $p$ and residue field $\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} = \mathbb{F}_p$, and its nonzero ideals are $p^n\mathbb{Z}_p$ (Proposition 7.3(iii)). $\mathbb{Z}$ is dense in $\mathbb{Z}_{(p)}$: for $a/b \in \mathbb{Z}_{(p)}$ and $n \ge 1$, $b$ is invertible modulo $p^n$, and an integer $c$ with $bc \equiv a \pmod{p^n}$ has $|a/b - c|_p \le p^{-n}$. (ii) is Theorem 7.6(iii) with $\mathcal{O} = \mathbb{Z}_{(p)}$, $\varpi = p$, since $\mathbb{Z}_{(p)}/p^n\mathbb{Z}_{(p)} = \mathbb{Z}/p^n\mathbb{Z}$. (iii) Units of the valuation ring are the elements of absolute value 1 (Proposition 7.3(ii)), that is, those not in $p\mathbb{Z}_p$. For $x \ne 0$, $u = xp^{-v_p(x)}$ has absolute value 1. (iv) By (ii), $\mathbb{Z}_p$ is homeomorphic to a closed subset of the product $\prod_n \mathbb{Z}/p^n$ of finite discrete sets, which is compact (Tychonoff; Chapter 1). $\mathbb{Z}_p$ is an open neighbourhood of 0, and $x + \mathbb{Z}_p$ of $x$, so $\mathbb{Q}_p$ is locally compact. Balls are open and closed (Proposition 7.3(ii)) and arbitrarily small, so the connected components are points: if a subset $S$ contains $x \ne y$, the ball $B = B(x, |x-y|_p)$ contains $x$ but not $y$, and $S = (S \cap B) \sqcup (S \setminus B)$ writes $S$ as a disjoint union of two nonempty relatively open sets. (v) $q$ is an integer not divisible by $p$. □

*Warning* 7.9. Chapter 3 defines $\mathbb{Z}_\ell = \varprojlim \mathbb{Z}/\ell^n$ and proves that it is a complete discrete valuation ring; Theorem 7.8(ii) shows that this is the closure of $\mathbb{Z}$ in the completion $\mathbb{Q}_\ell$ of $\mathbb{Q}$. Every statement about $\mathbb{Z}_\ell$ in either chapter therefore applies to the other construction. The two topologies agree as well, so "continuous" means the same thing in both.

**Proposition 7.10** ($p$-adic expansions)**.** *Every $x \in \mathbb{Q}_p^\times$ has a unique expansion*

$$x = \sum_{i \geq v} a_i p^i, \qquad v = v_p(x), \quad a_i \in \{0, 1, \dots, p-1\}, \quad a_v \neq 0,$$

*converging in $\mathbb{Q}_p$. $x \in \mathbb{Z}_p$ iff $v \geq 0$, and for $x \in \mathbb{Z}_p$ the partial sum $\sum_{i<n} a_i p^i$ is the integer in $[0, p^n)$ congruent to $x$ modulo $p^n$. The number $x$ lies in $\mathbb{Q}$ iff its digit sequence $(a_i)$ is eventually periodic.*

*Proof.* Multiplying by a power of $p$ we may assume $x \in \mathbb{Z}_p^\times$. Let $s_n \in [0, p^n)$ be the integer congruent to $x$ modulo $p^n$ (Theorem 7.8(ii)). Then $s_{n+1} \equiv s_n \pmod{p^n}$, so $s_{n+1} = s_n + a_n p^n$ with $a_n \in \{0, \dots, p-1\}$, and $s_n \to x$; this gives the expansion, and uniqueness holds because the partial sums of any expansion are the $s_n$. A sum $\sum a_i p^i$ with integer digits converges by Proposition 7.3(ii). If the digits are periodic of period $k$ from the index $m$ on, then $x = A + p^m B/(1 - p^k)$ with integers $A$, $B$, a rational number. Conversely let $x = a/b \in \mathbb{Z}_p \cap \mathbb{Q}$ with $b \geq 1$, $p \nmid b$, and put $x_n = (x - s_n)/p^n$, the number whose digits are $a_n, a_{n+1}, \dots$. Since $b s_n \equiv a \pmod{p^n}$, $b x_n = (a - b s_n)/p^n$ is an integer, and as $0 \leq s_n < p^n$ it lies in the interval $(a/p^n - b,\ a/p^n]$, which for all $n$ is contained in $(-|a| - b, |a|]$. So the $x_n$ take finitely many values, $x_n = x_m$ for some $n < m$, and the digit sequence is periodic from $n$ on, with period $m - n$. □

**Example 7.11.** $-1 = \sum_{i \geq 0} (p-1) p^i$, since $s_n = p^n - 1$; and $1/(1-p) = \sum_{i \geq 0} p^i$. For readers who know Laurent series from Riley, Hobson and Bence: a $p$-adic number is a "Laurent series in $p$" with digits in $\{0, \dots, p-1\}$, but arithmetic carries from one digit to the next, so the digits of a sum are not the sums of the digits. In $\mathbb{Z}_5$, the digits of $\sqrt{-1}$ (a root of $X^2 + 1$ congruent to 2 modulo 5, Example 7.16) begin $2, 1, 2, 1$: that root is $\equiv 182 \pmod{5^4}$.

## 7.2 Hensel's lemma; units, roots of unity and powers

This section answers the questions the paper asks about elements of $\mathbb{Q}_q$ and $\mathbb{Z}_\ell$: when a polynomial has a root (Hensel's lemma), which roots of unity lie in $\mathbb{Q}_p$, what $\mathbb{Z}_p^\times$ and $\mathbb{Q}_p^\times$ look like, and when an element is an $\ell^k$-th power. The paper uses the last answer in Lemma lem:tamfactor, where "the reduction of $u$ is an $\ell^k$-th power in $\mathbb{F}_q^\times$" is equivalent to $u$ being an $\ell^k$-th power in $\mathbb{Z}_q^\times$ (Theorem 7.20(ii)), and in Lemma lem:tamagawa-exp, through the units of $\mathbb{Q}_q^{\mathrm{ur}}$ (Proposition 7.36(iv)). The rings $\mathbb{Z}_\ell[\zeta_3]$ and $\mathbb{Z}_\ell[i]$ of Lemma lem:euler are decided here too (Example 7.16(g)).

### 7.2.1 Hensel's lemma

**Theorem 7.12** (Hensel's lemma)**.** *Let $K$ be a field complete for a nonarchimedean absolute value, with valuation ring $\mathcal{O}$ and maximal ideal $\mathfrak{m}$, and let $f \in \mathcal{O}[X]$.*

*(a) (Newton form.) If $a_0 \in \mathcal{O}$ satisfies $|f(a_0)| < |f'(a_0)|^2$, then the Newton sequence $a_{n+1} = a_n - f(a_n)/f'(a_n)$ is defined, lies in $\mathcal{O}$ and converges to a root $a$ of $f$. It is the unique root of $f$ with $|a - a_0| < |f'(a_0)|$; moreover $|a - a_0| = |f(a_0)/f'(a_0)|$ and $|f'(a)| = |f'(a_0)|$.*

*(b) (Simple-root form.) If $a_0 \in \mathcal{O}$ satisfies $f(a_0) \equiv 0 \pmod{\mathfrak{m}}$ and $f'(a_0) \not\equiv 0 \pmod{\mathfrak{m}}$, there is a unique $a \in \mathcal{O}$ with $f(a) = 0$ and $a \equiv a_0 \pmod{\mathfrak{m}}$.*

*Proof.* (a) Taylor's formula in $\mathcal{O}[x,h]$ reads $f(x+h)=f(x)+f'(x)h+h^2g(x,h)$ and $f'(x+h)=f'(x)+hg_1(x,h)$ with $g,g_1\in\mathcal{O}[x,h]$: for $f=\sum c_kX^k$, expand each $(x+h)^k=x^k+kx^{k-1}h+\sum_{j\geq 2}\binom{k}{j}x^{k-j}h^j$ by the binomial theorem; the integer binomial coefficients keep all coefficients in $\mathcal{O}$, and no division by $j!$ occurs. The idea of the proof is the one of Newton's method over $\mathbb{R}$: the error is squared at each step. Put $c=|f'(a_0)|$ and $\delta=|f(a_0)|/c^2<1$. We show by induction that $a_n\in\mathcal{O}$, $|f'(a_n)|=c$ and $|f(a_n)|\leq c^2\delta^{2^n}$. This holds for $n=0$. Given it for $n$, the step $h_n=-f(a_n)/f'(a_n)$ has $|h_n|=|f(a_n)|/c\leq c\,\delta^{2^n}<c\leq 1$, so $a_{n+1}=a_n+h_n\in\mathcal{O}$; $|f'(a_{n+1})-f'(a_n)|=|h_ng_1(a_n,h_n)|\leq|h_n|<c$ gives $|f'(a_{n+1})|=c$ (Proposition 7.3(ii)); and since $h_n$ was chosen so that $f(a_n)+f'(a_n)h_n=0$, Taylor's formula leaves $f(a_{n+1})=h_n^2g(a_n,h_n)$, which gives $|f(a_{n+1})|\leq|h_n|^2\leq c^2\delta^{2^{n+1}}$. Hence $|a_{n+1}-a_n|=|h_n|\to 0$, the sequence converges to some $a\in\mathcal{O}$, and $f(a)=\lim f(a_n)=0$, $|f'(a)|=c$. If $f(a_0)=0$, every $h_n$ is 0, $a=a_0$ and $|a-a_0|=0=|f(a_0)/f'(a_0)|$; so assume $\delta>0$. For $n\geq 1$, $|h_n|\leq c\delta^{2^n}<c\delta=|h_0|$, so $|a-a_0|=|h_0|=|f(a_0)/f'(a_0)|<c$. If $b$ is a root with $|b-a_0|<c$, then $h=b-a$ has $|h|<c$, and $0=f(b)=f'(a)h+h^2g(a,h)$; if $h\neq 0$ this gives $c=|f'(a)|=|hg(a,h)|\leq|h|<c$, which is absurd. (b) is (a) with $|f'(a_0)|=1$, since $|f(a_0)|<1$ and $|a-a_0|<1$ means $a\equiv a_0 \pmod{\mathfrak{m}}$. □

*Warning* 7.13. The simple-root form needs $f'(a_0)$ to be a unit. $X^2-p$ has the root 0 modulo $p$ but no root in $\mathbb{Q}_p$, since $v_p(x^2)$ is even. On the other hand $X^2+7$ has no simple root modulo 2 (its derivative $2X$ vanishes there), yet $-7$ is a square in $\mathbb{Z}_2$: with $a_0=1$, $|f(1)|_2=|8|_2=\frac{1}{8}<\frac{1}{4}=|f'(1)|_2^2$, so the Newton form applies, and the root is $\equiv 1 \pmod 4$. The iteration can be followed by hand: $a_1=1-8/2=-3$ with $f(-3)=16$, then $a_2=-3-16/(-6)=-\frac{1}{3}$ with $f(-\frac{1}{3})=\frac{64}{9}$. So $|f(a_n)|_2=\frac{1}{8},\frac{1}{16},\frac{1}{64}$, exactly the bound $c^2\delta^{2^n}$ of the proof with $c=\frac{1}{2}$, $\delta=\frac{1}{2}$. The root is $\equiv 181 \pmod{2^8}$, and indeed $-\frac{1}{3}\equiv 85 \pmod{2^8}$ agrees with it modulo $2^5$, as $|a-a_2|_2=|f(a_2)/f'(a_2)|_2=2^{-5}$ predicts.

**Theorem 7.14** (Hensel's lemma, factorisation form)**.** *Let $\mathcal{O}$ be a complete discrete valuation ring with uniformiser $\varpi$ and residue field $k$, and let $f\in\mathcal{O}[X]$. Suppose that its reduction is $\bar f=\bar g\bar h$ in $k[X]$ with $\bar g$ monic and $\bar g$, $\bar h$ coprime. Then $f=gh$ with $g,h\in\mathcal{O}[X]$, $g$ monic of degree* $\deg\bar g$*, and $g$, $h$ reducing to $\bar g$, $\bar h$.*

*Proof.* Choose $a,b\in\mathcal{O}[X]$ with $\bar a\bar g+\bar b\bar h=1$, a monic lift $g_1$ of $\bar g$ of degree $m=\deg\bar g$, and a lift $h_1$ of $\bar h$. We construct $g_k$ monic of degree $m$ and $h_k$ with $f\equiv g_kh_k \pmod{\varpi^k}$, $g_{k+1}\equiv g_k$ and $h_{k+1}\equiv h_k \pmod{\varpi^k}$. Given $g_k$, $h_k$, write $f-g_kh_k=\varpi^kr$. In $k[X]$, divide $\bar r\bar b$ by the monic $\bar g$: $\bar r\bar b=\bar s\bar g+\bar u$ with $\deg\bar u<m$; then $\bar r=\bar r(\bar a\bar g+\bar b\bar h)=\bar g(\bar r\bar a+\bar s\bar h)+\bar h\bar u$. Lift $\bar u$ to $u$ of degree $<m$ and $\bar r\bar a+\bar s\bar h$ to $w$, and put $g_{k+1}=g_k+\varpi^ku$, $h_{k+1}=h_k+\varpi^kw$. Then $g_{k+1}$ is monic of degree $m$ and $f-g_{k+1}h_{k+1}\equiv\varpi^k(r-g_kw-h_ku)\equiv 0 \pmod{\varpi^{k+1}}$, since $\bar g_k=\bar g$, $\bar h_k=\bar h$ and $\bar r=\bar g\bar w+\bar h\bar u$ (the term $\varpi^{2k}uw$ is divisible by $\varpi^{k+1}$). As $g_{k+1}\equiv g_k \pmod{\varpi^k}$, the coefficients of the $g_k$ are Cauchy sequences, so they converge, and $g=\lim g_k$ is monic of degree $m$. Divide $f$ by $g$ in $\mathcal{O}[X]$: $f=gh+\rho$ with $\deg\rho<m$. Since $g\equiv g_k \pmod{\varpi^k}$, we have $gh_k\equiv f \pmod{\varpi^k}$, so $f-gh_k=\varpi^kt$ for some $t\in\mathcal{O}[X]$; dividing $t$ by the monic $g$ shows that the remainder of $f$ on division by $g$, which is $\rho$, is $\equiv 0 \pmod{\varpi^k}$ (uniqueness of division by a monic polynomial). This holds for every $k$, so $\rho=0$ and $f=gh$. Finally $\bar g\bar h=\bar f=\bar g\,\bar h'$ for the reduction $\bar h'$ of $h$, and $k[X]$ is a domain, so $h$ reduces to $\bar h$. □

**Corollary 7.15.** *Let $K$ be complete for a discrete valuation, with valuation ring $\mathcal{O}$, and let $f=a_nX^n+\cdots+a_0\in K[X]$ be irreducible with $a_0a_n\neq 0$. Then* $\max_i|a_i|=\max(|a_0|,|a_n|)$*. In particular a monic irreducible polynomial over $K$ with constant term in $\mathcal{O}$ lies in $\mathcal{O}[X]$.*

*Proof.* Scale $f$ so that $\max_i|a_i| = 1$, and suppose $|a_0| < 1$ and $|a_n| < 1$. Let $r$ be the least index with $|a_r| = 1$; then $0 < r < n$ and $\bar{f} = X^r\bar{h}$ with $\bar{h}(0) = \bar{a}_r \neq 0$, so $X^r$ and $\bar{h}$ are coprime. Theorem 7.14 gives $f = gh$ with $g$ monic of degree $r$; then $\deg h = n - r \geq 1$, contradicting irreducibility. For the last statement, $|a_n| = 1 \geq |a_0|$, so $\max_i|a_i| = \max(|a_0|, 1) = 1$, that is, every coefficient lies in $\mathcal{O}$. □

Read in reverse, the corollary is a reducibility test. For $f = X^2 + p^{-1}X + 1$ over $\mathbb{Q}_p$ the middle coefficient has $|p^{-1}|_p = p$, larger than $|a_0| = |a_2| = 1$; so $f$ is reducible and has two roots in $\mathbb{Q}_p$. Directly: $pf = pX^2 + X + p$ reduces to $X$ modulo $p$, and Theorem 7.14 splits off a linear factor. The roots have valuations 1 and $-1$ (their product is 1 and their sum $-p^{-1}$); for $p = 3$, PARI's `polrootspadic` gives $2 \cdot 3 + 2 \cdot 3^2 + \dots$ and $2 \cdot 3^{-1} + 2 + \dots$.

**Example 7.16** (Hensel's lemma at work)**.** All residues below were computed by Newton's iteration and checked with PARI/GP (`polrootspadic`).

(a) $X^2 + 1$ over $\mathbb{Z}_5$ has the simple roots $2, 3$ modulo 5; the two roots in $\mathbb{Z}_5$ are $\equiv 182$ and $\equiv 443 \pmod{5^4}$, and $182^2 + 1 = 53 \cdot 5^4$.

(b) $\sqrt{-3} \in \mathbb{Z}_7$: the roots of $X^2 + 3$ are $\equiv 37, 306 \pmod{7^3}$. $\sqrt{2} \in \mathbb{Z}_7$: the roots of $X^2 - 2$ are $\equiv 108, 235 \pmod{7^3}$.

(c) $X^2 + X + 1$ has the simple roots $2, 4$ modulo 7, and its roots in $\mathbb{Z}_7$ are $\equiv 30, 18 \pmod{49}$. For instance, one Newton step from $a_0 = 2$, where $f(2) = 7$ and $f'(2) = 5$, gives $a_1 = 2 - 7 \cdot 5^{-1} \equiv 2 - 7 \cdot 10 = -68 \equiv 30 \pmod{49}$, since $5 \cdot 10 \equiv 1 \pmod{49}$; and $30^2 + 30 + 1 = 931 = 19 \cdot 49$. Hence $\mathbb{Z}_7[\zeta_3] := \mathbb{Z}_7[X]/(X^2 + X + 1) \simeq \mathbb{Z}_7 \times \mathbb{Z}_7$ by the Chinese remainder theorem, the two roots differing by a unit.

(d) (Squares in $\mathbb{Z}_2$.) $u \in \mathbb{Z}_2^\times$ is a square iff $u \equiv 1 \pmod 8$. If $u \equiv 1 \pmod 8$, apply Theorem 7.12(a) to $X^2 - u$ with $a_0 = 1$: $|1 - u|_2 \leq \frac{1}{8} < \frac{1}{4} = |2|_2^2$. Conversely an odd square is $\equiv 1 \pmod 8$. So $-7$, 17 are squares in $\mathbb{Z}_2$, and $-1$, 3, 5 are not.

(e) (Squares in $\mathbb{Z}_p$, $p$ odd.) $u \in \mathbb{Z}_p^\times$ is a square iff $\bar{u}$ is a square in $\mathbb{F}_p$: the roots of $X^2 - \bar{u}$ are simple as $2\bar{u} \neq 0$.

(f) (Roots of $X^m - u$.) More generally, for $p \nmid m$ and $u \in \mathbb{Z}_p^\times$, every root of $X^m - \bar{u}$ in $\mathbb{F}_p$ is simple and lifts uniquely.

(g) (The rings $\mathbb{Z}_\ell[\zeta_3]$ and $\mathbb{Z}_\ell[i]$.) Let $g = X^2 + X + 1$ (resp. $X^2 + 1$) and $R = \mathbb{Z}_\ell[X]/(g)$. If $\ell \equiv 1 \pmod 3$ (resp. $\pmod 4$), $\bar{g}$ has two simple roots in $\mathbb{F}_\ell$, they lift by (b) of the theorem, and $R \simeq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$, which is not a domain. If $\ell \equiv 2 \pmod 3$ (resp. $\ell \equiv 3 \pmod 4$), $\bar{g}$ is irreducible over $\mathbb{F}_\ell$, so $g$ is irreducible over $\mathbb{Q}_\ell$ (a factorisation over $\mathbb{Q}_\ell$ would be into monic factors in $\mathbb{Z}_\ell[X]$ by Corollary 7.15, and would reduce to one of $\bar{g}$); then $R$ is a domain, $R/\ell R = \mathbb{F}_\ell[X]/(\bar{g}) \simeq \mathbb{F}_{\ell^2}$ is a field, and an element of $R$ outside $\ell R$ is a unit (it is invertible modulo $\ell$, and $R$ is $\ell$-adically complete), so $R$ is a discrete valuation ring with uniformiser $\ell$ and residue field $\mathbb{F}_{\ell^2}$, the ring of integers of the unramified quadratic extension of $\mathbb{Q}_\ell$ (Theorem 7.34). For $\ell = 3$ (resp. $\ell = 2$), $R$ is the ring of integers of a ramified quadratic extension: of $\mathbb{Q}_3(\sqrt{-3})$ by Proposition 7.28(ii), and of $\mathbb{Q}_2(i)$ because $1 + i$ is a root of the Eisenstein polynomial $X^2 - 2X + 2$, so that $\mathbb{Z}_2[i] = \mathbb{Z}_2[1 + i]$ is that ring of integers by Theorem 7.27 (Example 7.43(a)). This is the trichotomy that Chapter 6 stated for $\mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ and $\mathbb{Z}_\ell \otimes \mathbb{Z}[i]$, now with the local input proved; it is why the paper's $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ is not a field for $\ell \equiv 1 \pmod 3$.

### 7.2.2 Teichmüller lifts and the structure of the units

**Theorem 7.17** (Teichmüller lifts)**.** *Let $K$ be complete for a discrete valuation with finite residue field $k = \mathbb{F}_q$, $q = p^f$.*

*(i) For $a \in k$ and any lift $\tilde{a} \in \mathcal{O}$, the limit $[a] = \lim_{n\to\infty} \tilde{a}^{q^n}$ exists and does not depend on $\tilde{a}$; $[a]$ is the unique lift of $a$ with $[a]^q = [a]$.*

*(ii) $a \mapsto [a]$ is multiplicative, $[0] = 0$, and it restricts to an isomorphism $k^\times \to \mu_{q-1}(K)$ inverse to reduction. It is not additive.*

*(iii) $\mathcal{O}^\times = \mu_{q-1}(K) \times U^{(1)}$ with $U^{(1)} = 1 + \mathfrak{m}$, the projection to the first factor being $u \mapsto [\bar{u}]$.*

*Proof.* First, if $x \equiv y \pmod{\mathfrak{m}^j}$ with $j \geq 1$, then $x^p \equiv y^p \pmod{\mathfrak{m}^{j+1}}$: indeed $x^p - y^p = (x-y)\sum_{i=0}^{p-1} x^i y^{p-1-i}$ and the sum is $\equiv p y^{p-1} \equiv 0 \pmod{\mathfrak{m}}$. Iterating, $x^q \equiv y^q \pmod{\mathfrak{m}^{j+1}}$. (i) Since $a^q = a$ in $k$, $\tilde{a}^q \equiv \tilde{a} \pmod{\mathfrak{m}}$, hence $\tilde{a}^{q^{n+1}} \equiv \tilde{a}^{q^n} \pmod{\mathfrak{m}^{n+1}}$; the sequence is Cauchy, its limit $[a]$ satisfies $[a]^q = [a]$ and reduces to $a$. Two lifts are congruent modulo $\mathfrak{m}$, so their $q^n$-th powers are congruent modulo $\mathfrak{m}^{n+1}$, and the limits agree. The polynomial $X^q - X$ has derivative $qX^{q-1} - 1 \equiv -1$, a unit, so by Theorem 7.12(b) it has a unique root lifting $a$. (ii) $[a][b]$ is a root of $X^q - X$ lifting $ab$, hence equals $[ab]$. For $a \neq 0$, $[a]^{q-1} = 1$; conversely every $\zeta \in \mu_{q-1}(K)$ is a root of $X^q - X$, so $\zeta = [\bar{\zeta}]$. In $\mathbb{Z}_5$, $[1] + [1] = 2 \neq [2] \equiv 182 \pmod{5^4}$. (iii) For $u \in \mathcal{O}^\times$, $u = [\bar{u}] \cdot u[\bar{u}]^{-1}$ with the second factor in $U^{(1)}$, and $\mu_{q-1}(K) \cap U^{(1)} = \{1\}$ by (ii). □

For example, in $\mathbb{Z}_5$ one has $[1], [2], [3], [4] \equiv 1, 182, 443, 624 \pmod{5^4}$; so $[2]$ and $[3]$ are the two square roots of $-1$ of Example 7.16(a), and $[4] = -1$.

**Theorem 7.18** (structure of the units)**.** *(i) Let $p$ be odd. The series*

$$\exp(x) = \sum_{n\geq 0} \frac{x^n}{n!}, \qquad \log(1+x) = \sum_{n\geq 1} (-1)^{n+1} \frac{x^n}{n}$$

*converge for $x \in p\mathbb{Z}_p$ and give mutually inverse isomorphisms of topological groups $p\mathbb{Z}_p \simeq 1 + p\mathbb{Z}_p$, carrying $p^n\mathbb{Z}_p$ onto $1 + p^n\mathbb{Z}_p$ for every $n \geq 1$. Hence $1 + p\mathbb{Z}_p \simeq \mathbb{Z}_p$, topologically generated by $1 + p$, $\mathbb{Z}_p^\times = \mu_{p-1} \times (1 + p\mathbb{Z}_p)$, and*

$$\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mu_{p-1} \times (1 + p\mathbb{Z}_p) \simeq \mathbb{Z} \times \mathbb{Z}/(p-1) \times \mathbb{Z}_p.$$

*(ii) For $p = 2$ the same series give $4\mathbb{Z}_2 \simeq 1 + 4\mathbb{Z}_2$, carrying $2^n\mathbb{Z}_2$ onto $1 + 2^n\mathbb{Z}_2$ for $n \geq 2$; $1 + 4\mathbb{Z}_2 \simeq \mathbb{Z}_2$ is topologically generated by $5$, $\mathbb{Z}_2^\times = \{\pm 1\} \times (1 + 4\mathbb{Z}_2)$ and $\mathbb{Q}_2^\times \simeq \mathbb{Z} \times \mathbb{Z}/2 \times \mathbb{Z}_2$.*

*(iii) These are the inverse limits of the finite statements of Chapter 2: $(\mathbb{Z}/p^k)^\times$ is cyclic for $p$ odd, and $(\mathbb{Z}/2^k)^\times = \{\pm 1\} \times \langle 5 \rangle$ for $k \geq 2$.*

*(iv) Let $K$ be complete for a discrete valuation with residue field $\mathbb{F}_q$. Then $K^\times = \varpi^{\mathbb{Z}} \times \mu_{q-1}(K) \times U^{(1)}$; the quotients $U^{(i)}/U^{(i+1)}$, $U^{(i)} = 1 + \mathfrak{m}^i$, are isomorphic to the additive group of $\mathbb{F}_q$; $U^{(1)}$ is an abelian pro-$p$ group, hence a $\mathbb{Z}_p$-module; and $U^{(1)}$ is uniquely $m$-divisible for every $m$ prime to $p$.*

*Proof.* We use $v = v_p$ and two estimates for $n \geq 1$: $v(n!) \leq (n-1)/(p-1)$ (Legendre's formula $v(n!) = (n - s_p(n))/(p-1)$, $s_p(n) \geq 1$ the digit sum, Chapter 2) and $v(n) \leq (n-1)/(p-1)$ (if

$v(n) = k$ then $n \geq p^k$ and $p^k - 1 = (p-1)(1 + p + \cdots + p^{k-1}) \geq (p-1)k$). Let $c = 1$ if $p$ is odd and $c = 2$ if $p = 2$, so that $c - 1/(p-1) > 0$, and let $v(x) \geq c$. Then for $n \geq 2$

$$v\Big(\frac{x^n}{n!}\Big) - v(x) \geq (n-1)\Big(v(x) - \frac{1}{p-1}\Big) > 0, \qquad v\Big(\frac{x^n}{n}\Big) - v(x) \geq (n-1)\Big(v(x) - \frac{1}{p-1}\Big) > 0.$$

So both series converge, $v(\exp(x) - 1) = v(x)$ and $v(\log(1+x)) = v(x)$; exp maps $p^n\mathbb{Z}_p$ into $1 + p^n\mathbb{Z}_p$ and log maps $1 + p^n\mathbb{Z}_p$ into $p^n\mathbb{Z}_p$ for $n \geq c$, and both preserve $v$ of the deviation from 0, resp. 1.

The identities $\exp(X+Y) = \exp(X)\exp(Y)$, $\log((1+X)(1+Y)) = \log(1+X) + \log(1+Y)$, $\exp(\log(1+X)) = 1 + X$ and $\log(\exp X) = X$ hold in $\mathbb{Q}[[X, Y]]$. They hold numerically for $v(x), v(y) \geq c$ by Lemma 7.4. Take $\exp(\log(1+x))$: expanding $\sum_m \frac{1}{m!}\big(\sum_n (-1)^{n+1}x^n/n\big)^m$ gives a family of terms $\pm x^N/(n_1 \cdots n_m\, m!)$ with $N = n_1 + \cdots + n_m \geq m$, of valuation

$$\geq Nc - \frac{N-m}{p-1} - \frac{m-1}{p-1} = N\Big(c - \frac{1}{p-1}\Big) + \frac{1}{p-1},$$

and for each $N$ there are finitely many terms; so the family is summable. Grouping by $m$ gives $\exp(\log(1+x))$, and grouping by $N$ gives $\sum_N \alpha_N x^N$ where $\alpha_N$ is the coefficient of $X^N$ in $\exp(\log(1+X)) = 1 + X$. The other three identities are proved in the same way (the terms of $\log(\exp x)$ are $\pm\prod x^{n_i}/(n_i!\, m)$, with the same bound; those of $\log((1+x)(1+y))$ come from $(x + y + xy)^n/n$; those of $\exp(x+y)$ are $x^a y^b/(a!\, b!)$). Hence exp and log are inverse continuous homomorphisms $p^c\mathbb{Z}_p \leftrightarrows 1 + p^c\mathbb{Z}_p$, matching $p^n\mathbb{Z}_p$ and $1 + p^n\mathbb{Z}_p$ for $n \geq c$.

(i) $p$ odd: $1 + p\mathbb{Z}_p \simeq p\mathbb{Z}_p \simeq \mathbb{Z}_p$ $(x \mapsto x/p)$; since $v(\log(1+p)) = 1$, $\log(1+p)$ generates $p\mathbb{Z}_p$ as a $\mathbb{Z}_p$-module, so $1 + p$ generates $1 + p\mathbb{Z}_p$ topologically. With Theorem 7.17(iii) and Theorem 7.8(iii) this gives the structure of $\mathbb{Q}_p^\times$. (ii) $p = 2$: every $u \in \mathbb{Z}_2^\times$ is $\equiv \pm 1 \pmod 4$, so exactly one of $\pm u$ lies in $1 + 4\mathbb{Z}_2$; and $v(\log 5) = v(4) = 2$, so 5 generates. (iii) Reduce modulo $p^k$: $(\mathbb{Z}/p^k)^\times = \mathbb{Z}_p^\times/(1+p^k\mathbb{Z}_p) \simeq \mu_{p-1} \times \mathbb{Z}_p/p^{k-1}\mathbb{Z}_p$, cyclic for $p$ odd, and $\{\pm 1\} \times \mathbb{Z}_2/2^{k-2}$ for $p = 2$. (iv) The first statement is Theorem 7.17(iii) together with $K^\times = \varpi^{\mathbb{Z}} \times \mathcal{O}^\times$. The map $1 + \varpi^i a \mapsto \bar{a}$ is a homomorphism $U^{(i)} \to (\mathbb{F}_q, +)$ with kernel $U^{(i+1)}$, since $(1+\varpi^i a)(1+\varpi^i b) = 1 + \varpi^i(a+b) + \varpi^{2i}ab$; so $U^{(1)}/U^{(n)}$ is a finite abelian $p$-group, and $U^{(1)} = \varprojlim_n U^{(1)}/U^{(n)}$ ($U^{(1)}$ is compact and the $U^{(n)}$ form a basis of open neighbourhoods of 1). An abelian pro-$p$ group is a $\mathbb{Z}_p$-module ($u^a = \lim u^{a_k}$ for integers $a_k \to a$; the limit exists because $u^{p^N} \in U^{(n)}$ for $N$ large). For $p \nmid m$, multiplication by $m$ is an automorphism of each finite $p$-group $U^{(1)}/U^{(n)}$, compatibly in $n$, hence of $U^{(1)}$. □

**Corollary 7.19** (roots of unity). *$\mu(\mathbb{Q}_p) = \mu_{p-1}$ for $p$ odd and $\mu(\mathbb{Q}_2) = \{\pm 1\}$. For $K$ as in Theorem 7.18(iv) and $m$ prime to $p$, $\mu_m(K) = \mu_{\gcd(m,q-1)}$, and reduction is injective on it. In particular, for primes $\ell \neq p$: $\mu_{\ell^k} \subset \mathbb{Q}_p$ iff $\ell^k \mid p-1$, $\#\mu_{\ell^\infty}(\mathbb{Q}_p) = \ell^{v_\ell(p-1)}$, and $i \in \mathbb{Q}_\ell$ iff $\ell \equiv 1 \pmod 4$ (so $\mathbb{Q}_\ell(i)$ is a field iff $\ell \not\equiv 1 \pmod 4$). Also $\mu_{p^n} \subset \mathbb{Q}_p$ only for $p^n \in \{1, 2\}$.*

*Proof.* The torsion of $\mathbb{Z} \times \mathbb{Z}/(p-1) \times \mathbb{Z}_p$ is $\mathbb{Z}/(p-1)$, and that of $\mathbb{Z} \times \mathbb{Z}/2 \times \mathbb{Z}_2$ is $\mathbb{Z}/2$. For $K$: an element of $U^{(1)}$ of order prime to $p$ is trivial by unique divisibility, so $\mu_m(K)$ injects into $\mu_{q-1}(K) \simeq \mathbb{F}_q^\times$, a cyclic group of order $q-1$, whose elements of order dividing $m$ form $\mu_{\gcd(m,q-1)}$. The remaining statements are special cases ($\ell = 2$: $\mu_2 \subset \mathbb{Q}_p$ always; $i \in \mathbb{Q}_\ell$ iff $4 \mid \ell - 1$ for $\ell$ odd, and $i \notin \mathbb{Q}_2$). □

### 7.2.3 Powers

**Theorem 7.20** ($\ell^k$-th powers, $\ell \neq p$). *Let $\ell \neq p$ be primes, $k \geq 1$, and $K$ complete for a discrete valuation with residue field $\mathbb{F}_q$ of characteristic $p$ and uniformiser $\varpi$ (for $K = \mathbb{Q}_p$: $q = p$, $\varpi = p$). Put $v = v_\ell(q-1)$.*

*(i)* $U^{(1)}$ *is uniquely* $\ell^k$*-divisible.*

*(ii)* $u \in \mathcal{O}_K^\times$ *is an* $\ell^k$*-th power in* $K$ *iff its reduction* $\bar{u}$ *is an* $\ell^k$*-th power in* $\mathbb{F}_q^\times$*; this holds iff* $\bar{u}^{(q-1)/d} = 1$*, where* $d = \gcd(\ell^k, q-1)$*.*

*(iii)* $x = \varpi^a u \in K^\times$ *(*$u$ *a unit) is an* $\ell^k$*-th power iff* $\ell^k \mid a$ *and* $\bar{u} \in (\mathbb{F}_q^\times)^{\ell^k}$*.*

*(iv) There is an exact sequence* $1 \to \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^k} \to K^\times/(K^\times)^{\ell^k} \xrightarrow{v_K} \mathbb{Z}/\ell^k \to 0$*, split by* $\varpi$*, so*

$$K^\times/(K^\times)^{\ell^k} \simeq \mathbb{Z}/\ell^k \times \mathbb{Z}/\ell^{\min(k,v)}, \qquad \#K^\times/(K^\times)^{\ell^k} = \ell^k \cdot \#\mu_{\ell^k}(K).$$

*(v) In the limit,* $\widehat{K^\times} := \varprojlim_k K^\times/(K^\times)^{\ell^k}$ *sits in an exact sequence* $0 \to \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \to \widehat{K^\times} \xrightarrow{v_K} \mathbb{Z}_\ell \to 0$*, split by* $\varpi$*; so* $\widehat{K^\times} \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^v$*, and* $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell = \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v} \simeq \mu_{\ell^v}(\mathbb{F}_q)$*, via* $y \mapsto y^{(q-1)/\ell^v}$*.*

*(vi) (*$\ell = 2$*,* $K = \mathbb{Q}_p$*.) For* $p$ *odd,* $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$ *has order* 4*, with representatives* $1, u_0, p, u_0 p$ *for a non-square unit* $u_0$*;* $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$ *has order* 8*, with representatives* $\pm 1, \pm 5, \pm 2, \pm 10$*.*

*Proof.* (i) is Theorem 7.18(iv); alternatively, Theorem 7.12(b) applies to $X^{\ell^k} - u$ at $a_0 = 1$, since $\ell^k$ is a unit. (ii) Write $u = [\bar{u}]u_1$ with $u_1 \in U^{(1)}$ (Theorem 7.17). If $\bar{u} = \bar{b}^{\ell^k}$, then $[\bar{u}] = [\bar{b}]^{\ell^k}$ and $u_1$ is an $\ell^k$-th power by (i). Conversely, reduce an equation $u = y^{\ell^k}$ (then $y$ is a unit). The last equivalence is the criterion for powers in the cyclic group $\mathbb{F}_q^\times$ (Chapter 2). (iii) If $x = y^{\ell^k}$ then $a = \ell^k v_K(y)$ and $u = (y\varpi^{-v_K(y)})^{\ell^k}$; conversely apply (ii). (iv) By (iii) the kernel of $v_K$ mod $\ell^k$ is the image of $\mathcal{O}_K^\times$, and $\mathcal{O}_K^\times \cap (K^\times)^{\ell^k} = (\mathcal{O}_K^\times)^{\ell^k}$; by (ii) $\mathcal{O}_K^\times/(\mathcal{O}_K^\times)^{\ell^k} \simeq \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^k}$, cyclic of order $\gcd(\ell^k, q-1) = \ell^{\min(k,v)} = \#\mu_{\ell^k}(K)$ (Corollary 7.19). (v) The sequences of (iv) form an inverse system of finite groups, and inverse limits of finite groups are exact (Chapter 1). For $k \geq v$ the left term is $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$, which is $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$; and $y \mapsto y^{(q-1)/\ell^v}$ maps the cyclic group $\mathbb{F}_q^\times$ onto its subgroup of order $\ell^v$ with kernel $(\mathbb{F}_q^\times)^{\ell^v}$. (vi) For $p$ odd, (iv) with $\ell = 2$ gives order 4. For $p = 2$: $\mathbb{Q}_2^\times = 2^{\mathbb{Z}} \times \{\pm 1\} \times (1 + 4\mathbb{Z}_2)$, and the squares of $1 + 4\mathbb{Z}_2$ form $1 + 8\mathbb{Z}_2$ (Example 7.16(d)), of index 2 with representatives $1, 5$. □

*In the paper* 7.21. Lemma lem:tamfactor defines $c(u)$ as "the class of the reduction of $u$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$" and uses that $v_\ell(c(u)) \geq k$ iff the reduction of $u$ is an $\ell^k$-th power in $\mathbb{F}_q^\times$ ($k \leq v$). By Theorem 7.20(ii) this is the same as $u \in (\mathbb{Z}_q^\times)^{\ell^k}$, since $\ell \neq q$. The equivalence fails for $\ell = q$ (for $q$ odd, $1 + q$ is a $q$-th power in $\mathbb{F}_q$ but not in $\mathbb{Z}_q$, since $1 + q\mathbb{Z}_q \simeq \mathbb{Z}_q$ carries the $q$-th powers onto $1 + q^2\mathbb{Z}_q$). The group $\varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k} \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^v$ of (v) is the paper's $H^1(G, \mathbb{Z}_\ell(1))$ (Proposition 7.57), with torsion $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(1)/(q-1)$.

**Example 7.22.** (a) $\mathbb{Q}_7^\times/(\mathbb{Q}_7^\times)^3 \simeq \mathbb{Z}/3 \times \mathbb{Z}/3$. The cubes of $\mathbb{F}_7^\times$ are $\{1, 6\}$; so 6 is a cube in $\mathbb{Q}_7$, while 2 is not, 14 is not ($v_7(14) = 1$), and $7^3 \cdot 6$ is.

(b) $\mathbb{Q}_{19}^\times/(\mathbb{Q}_{19}^\times)^9 \simeq \mathbb{Z}/9 \times \mathbb{Z}/9$, as $v_3(18) = 2$. Since 2 generates $\mathbb{F}_{19}^\times$, the classes of 2, $8 = 2^3$ and $-1 = 2^9$ in $\mathbb{F}_{19}^\times/(\mathbb{F}_{19}^\times)^9 \simeq \mathbb{Z}/9$ are 1, 3 and 0, of 3-adic valuations 0, 1 and (by the convention of Lemma lem:tamfactor, $v_\ell(0) = v$) 2.

(c) For the curve `11a1`, $j = -2^{12}31^3/11^5$, and the Tate parameter $q_E = j^{-1} + 744j^{-2} + 750420j^{-3} + \dots$ (Chapter 10) satisfies $v_{11}(q_E) = 5$ and $q_E 11^{-5} \equiv -1 \pmod{11}$ (by hand: $q_E 11^{-5} \equiv (j11^5)^{-1} = -(2^{12}31^3)^{-1}$, with $2^{12} \equiv 4$, $31^3 \equiv 3$, $4 \cdot 3 \equiv 1$; PARI gives $q_E = 10 \cdot 11^5 + 2 \cdot 11^6 + \dots$). The fifth powers of $\mathbb{F}_{11}^\times$ are $\{\pm 1\}$, so $q_E \in (\mathbb{Q}_{11}^\times)^5$.

(d) $\mathbb{Q}_5^\times/(\mathbb{Q}_5^\times)^3 \simeq \mathbb{Z}/3$: every unit of $\mathbb{Z}_5$ is a cube, as $3 \nmid 4$.

## 7.3 Finite extensions of $\mathbb{Q}_p$

The coefficient rings of the paper ($\mathbb{Z}_\ell[\zeta_3]$, the completions $\mathcal{O}_\lambda$ of rings of integers at primes $\lambda \mid \ell$) and the fields over which the paper's curves acquire good or multiplicative reduction are finite extensions of $\mathbb{Q}_\ell$ or $\mathbb{Q}_q$. This section extends the absolute value to them, defines the ramification index and residue degree, identifies totally ramified extensions with Eisenstein polynomials, compares completions of number fields with the global theory of Chapter 6, and proves that the decomposition group of a prime of $\overline{\mathbb{Q}}$ above $p$ is $G_{\mathbb{Q}_p}$. In the paper these facts are used in Lemma lem:IV ($\mathbb{Z}_3[\zeta_3]$ as a discrete valuation ring, now the ring of integers of $\mathbb{Q}_3(\sqrt{-3})$), in Lemma lem:crit ("The same statements hold with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$ and $r$ by its $\ell$-part") and Lemma lem:old, and in every local argument at a bad prime $q$, which is an argument about $G_{\mathbb{Q}_q} \subset G_{\mathbb{Q}}$.

Throughout, $K$ is complete for a discrete valuation with finite residue field, as in Convention 7.1.

**Lemma 7.23** (equivalence of norms)**.** *Let $K$ be complete for an absolute value and $V$ a finite-dimensional $K$-vector space. A* norm *on $V$ is a map $\|\cdot\|\colon V \to \mathbb{R}_{\geq 0}$ with $\|x\| = 0$ iff $x = 0$, $\|ax\| = |a|\|x\|$ and $\|x+y\| \leq \|x\| + \|y\|$. Any two norms $\|\cdot\|_1, \|\cdot\|_2$ on $V$ are equivalent: $c\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1$ for constants $c, C > 0$. In particular $V$ is complete for every norm.*

*Proof.* Fix a basis $e_1, \dots, e_n$ and the sup norm $\|\sum x_i e_i\|_\infty = \max|x_i|$; it suffices to compare a norm $\|\cdot\|$ with it. Clearly $\|x\| \leq C\|x\|_\infty$ with $C = \sum\|e_i\|$, since $\|\sum x_i e_i\| \leq \sum|x_i|\|e_i\| \leq \max_i|x_i| \cdot \sum_i\|e_i\|$. For the other inequality we argue by induction on $n$; for $n = 1$, $\|x_1 e_1\| = |x_1|\|e_1\|$, so $c = \|e_1\|$ works. If no $c > 0$ works, there are $v_k = \sum_i x_i^{(k)} e_i$ with $\|v_k\|_\infty = 1$ and $\|v_k\| \to 0$. For each $k$ some coordinate has absolute value 1, and one index occurs for infinitely many $k$; passing to a subsequence and renumbering, $|x_n^{(k)}| = 1$ for all $k$. Put $w_k = v_k/x_n^{(k)} = u_k + e_n$ with $u_k \in W = \langle e_1, \dots, e_{n-1}\rangle$. Then $\|w_k\| = \|v_k\| \to 0$, so $(u_k)$ is Cauchy for $\|\cdot\|$ on $W$ (as $\|u_k - u_j\| = \|w_k - w_j\| \leq \|w_k\| + \|w_j\|$), hence for $\|\cdot\|_\infty$ by induction, and converges in the sup norm (as $K$ is complete) to some $u \in W$, hence also for $\|\cdot\|$. Then $\|u + e_n\| = \lim\|w_k\| = 0$, so $e_n = -u \in W$, which is absurd. Completeness follows: a $\|\cdot\|$-Cauchy sequence is $\|\cdot\|_\infty$-Cauchy, and converges coordinatewise. □

**Theorem 7.24** (extension of the absolute value)**.** *Let $L/K$ be a finite extension of degree $n$. Then*

$$|x|_L = |N_{L/K}(x)|^{1/n}$$

*is the unique absolute value on $L$ extending $|\cdot|$; it is discrete, $L$ is complete for it, and $\mathcal{O}_L = \{|x|_L \leq 1\}$ is the integral closure of $\mathcal{O}_K$ in $L$. Consequently every algebraic extension of $K$ (for instance $\overline{\mathbb{Q}}_p$) carries a unique absolute value extending $|\cdot|$; automorphisms over $K$ are isometries, and conjugate elements over $K$ have the same absolute value.*

*Proof.* Let $x \in L$ have minimal polynomial $m$ over $K$, of degree $d$; then the norm of $x$ is $N_{L/K}(x) = ((-1)^d m(0))^{n/d}$. We claim: $x$ is integral over $\mathcal{O}_K$ iff $m \in \mathcal{O}_K[X]$ iff $|N_{L/K}(x)| \leq 1$. If $x$ is integral, so are its conjugates in a splitting field, and $m(0)$, a product of them up to sign, is an element of $K$ integral over the integrally closed $\mathcal{O}_K$, so $m(0) \in \mathcal{O}_K$, and $m \in \mathcal{O}_K[X]$ by Corollary 7.15. If $m \in \mathcal{O}_K[X]$, then $x$ is integral (it is a root of the monic $m$), and $|N_{L/K}(x)| = |m(0)|^{n/d} \leq 1$. Finally, if $|N_{L/K}(x)| \leq 1$, then $|m(0)| \leq 1$, and the monic irreducible

$m$ lies in $\mathcal{O}_K[X]$ by the last statement of Corollary 7.15. So $\{|x|_L \le 1\}$ is the integral closure, a ring. Now $|\cdot|_L$ is multiplicative (the norm is), vanishes only at 0, and extends $|\cdot|$ since $N_{L/K}(x) = x^n$ for $x \in K$. For the ultrametric inequality, let $x, y \in L$ with, say, $|y|_L \le |x|_L$ and $x \neq 0$; then $|x+y|_L = |x|_L|1 + y/x|_L$, so it suffices to show $|1+z|_L \le 1$ when $|z|_L \le 1$; and this holds because $z$ and 1 lie in the integral closure, which is a ring. The values lie in $|K^\times|^{1/n}$, a discrete group. Uniqueness: two extensions are norms on the $K$-vector space $L$, hence equivalent by Lemma 7.23: $c|x|_1 \le |x|_2 \le C|x|_1$; applied to $x^k$ this gives $c^{1/k}|x|_1 \le |x|_2 \le C^{1/k}|x|_1$ for all $k$, so $|x|_1 = |x|_2$. Completeness is Lemma 7.23. An algebraic extension is the union of its finite subextensions, on which the extensions are compatible by uniqueness. If $\sigma$ is a $K$-automorphism, $x \mapsto |\sigma x|$ is another extension, hence equals $|x|$; conjugates are exchanged by such automorphisms of a normal closure. □

The field $\overline{\mathbb{Q}}_p$ is not complete; its completion $\mathbb{C}_p$ is algebraically closed, but we shall not need it.

**Theorem 7.25** (ramification index and residue degree)**.** *Let $L/K$ be finite of degree $n$, with normalised valuation $v_L$ and residue field $k_L$. Put $e = e(L/K) = v_L(\varpi_K) = [|L^\times| : |K^\times|]$ and $f = f(L/K) = [k_L : k_K]$. Let $\alpha_1, \dots, \alpha_f \in \mathcal{O}_L$ lift a basis of $k_L$ over $k_K$ and let $\varpi_L$ be a uniformiser of $L$. Then $\{\alpha_i \varpi_L^j : 1 \le i \le f,\ 0 \le j < e\}$ is a basis of $\mathcal{O}_L$ over $\mathcal{O}_K$; so $\mathcal{O}_L$ is free of rank $n$ and $ef = n$. Both $e$ and $f$ are multiplicative in towers.*

*Proof.* First, for $c_1, \dots, c_f \in K$ not all 0, $v_L(\sum_i c_i\alpha_i) = e \min_i v_K(c_i)$: dividing by $c_{i_0}$ with $v_K(c_{i_0})$ minimal, the sum becomes an element of $\mathcal{O}_L$ whose reduction is a nontrivial $k_K$-combination of the $\bar\alpha_i$, hence a unit. Consequently, for $c_{ij} \in K$ not all 0,

$$v_L\Big(\sum_{i,j} c_{ij}\alpha_i\varpi_L^j\Big) = \min_j\Big(e \min_i v_K(c_{ij}) + j\Big),$$

because the inner sums $s_j = \sum_i c_{ij}\alpha_i\varpi_L^j$ have valuations in distinct classes modulo $e$ ($0 \le j < e$), so the minimum is attained once. Hence the $\alpha_i\varpi_L^j$ are linearly independent over $K$ (in particular $f \le n$ and $e \le n$ are finite), and an element of their $\mathcal{O}_K$-span $M$ that lies in $\varpi_K^k\mathcal{O}_L$ has all coefficients in $\varpi_K^k\mathcal{O}_K$ (if $e\,v_K(c_{ij}) + j \ge ek$ with $j < e$, then $v_K(c_{ij}) \ge k$). Next, $\mathcal{O}_L = M + \varpi_K\mathcal{O}_L$: the quotient $\mathcal{O}_L/\varpi_K\mathcal{O}_L = \mathcal{O}_L/\mathfrak{m}_L^e$ is filtered by the $\mathfrak{m}_L^j/\mathfrak{m}_L^{j+1}$, $0 \le j < e$, each a one-dimensional $k_L$-space spanned by $\varpi_L^j$, hence a $k_K$-space with basis the $\bar\alpha_i\varpi_L^j$. Iterating, $\mathcal{O}_L = M + \varpi_K^k\mathcal{O}_L$ for all $k$. Given $x \in \mathcal{O}_L$, write $x = m_k + \varpi_K^k y_k$ with $m_k \in M$; then $m_{k+1} - m_k \in M \cap \varpi_K^k\mathcal{O}_L$ has coefficients in $\varpi_K^k\mathcal{O}_K$, so the coefficients converge in the complete $\mathcal{O}_K$, and $m_k \to m \in M$ with $x - m \in \bigcap_k \varpi_K^k\mathcal{O}_L = 0$. So $\mathcal{O}_L = M$ is free with the stated basis. Every $x \in L$ has $\varpi_K^k x \in \mathcal{O}_L$ for $k$ large, so the basis is also a $K$-basis of $L$, and $ef = n$. Multiplicativity in towers $K \subseteq L \subseteq M$: writing $\varpi_K = \varpi_L^{e(L/K)}u$ with $u \in \mathcal{O}_L^\times$ and applying $v_M$, which is $e(M/L)$ on $\varpi_L$ and 0 on units, gives $e(M/K) = e(M/L)e(L/K)$; and $[k_M : k_K] = [k_M : k_L][k_L : k_K]$ is the tower law for the residue fields. □

The two numbers measure different things: $e$ counts how much the value group grows, $f$ how much the residue field grows. In the two extreme cases one of them is 1: $\mathbb{Q}_3(\sqrt{-3})$ has $e = 2$, $f = 1$ (Proposition 7.28(ii)), and $\mathbb{Q}_2(\sqrt{5})$ has $e = 1$, $f = 2$ (Proposition 7.60(iii)); the next definition names these cases.

**Definition 7.26.** A finite extension $L/K$ is *unramified* if $e(L/K) = 1$, *totally ramified* if $f(L/K) = 1$, *tamely ramified* if $p \nmid e(L/K)$ and *wildly ramified* if $p \mid e(L/K)$, $p$ the residue

characteristic. (Residue fields are finite, hence perfect, so no separability condition on $k_L/k_K$ is needed.) For $K$ a completion of a number field, this is the global notion of Chapter 6 for the one prime above $\mathfrak{p}$ in the completion (Theorem 7.30).

**Theorem 7.27** (Eisenstein polynomials)**.** *An extension $L/K$ is totally ramified of degree $e$ iff $L = K(\pi)$ for a root $\pi$ of an Eisenstein polynomial $X^e + c_{e-1}X^{e-1} + \cdots + c_0$ ($c_i \in \mathfrak{m}_K$, $c_0 \notin \mathfrak{m}_K^2$). Then the polynomial is irreducible, $\pi$ is a uniformiser of $L$ and $\mathcal{O}_L = \mathcal{O}_K[\pi]$. Conversely the minimal polynomial over $K$ of any uniformiser of a totally ramified $L$ is Eisenstein.*

*Proof.* Let $\pi$ be a root, $L = K(\pi)$, $d = [L : K] \le e$, and $w = v_L$. As $\pi$ is integral, $w(\pi) \ge 0$; if $w(\pi) = 0$, then $\pi^e = -\sum_{i<e} c_i\pi^i$ has positive valuation, which is absurd; so $w(\pi) \ge 1$. On $K$, $w = e(L/K)\, v_K$; as $v_K(c_0) = 1$ and $v_K(c_i) \ge 1$, we get $w(c_0) = e(L/K)$ and $w(c_i\pi^i) \ge e(L/K) + i > e(L/K)$ for $i \ge 1$. In $\pi^e = -c_0 - \sum_{1\le i<e} c_i\pi^i$ the term $c_0$ therefore has strictly the smallest valuation, so $ew(\pi) = w(\pi^e) = e(L/K) \le d \le e$ (Proposition 7.3(ii) and Theorem 7.25). Hence $w(\pi) = 1$, $e(L/K) = d = e$, the polynomial is irreducible and $L/K$ is totally ramified; and by Theorem 7.25 (with $f = 1$, $\alpha_1 = 1$) $1, \pi, \dots, \pi^{e-1}$ is a basis of $\mathcal{O}_L$. Conversely, if $L/K$ is totally ramified of degree $e$ and $\pi$ is a uniformiser, $1, \pi, \dots, \pi^{e-1}$ is a basis (Theorem 7.25), so $L = K(\pi)$ has degree $e$. The conjugates of $\pi$ in a normal closure have absolute value $|\pi| < 1$ (Theorem 7.24), so the non-leading coefficients of the minimal polynomial, which are sums of products of conjugates, lie in $\mathfrak{m}_K$; and $v_K(c_0) = v_K(N_{L/K}\pi) = v_L(\pi) = 1$, because $|N_{L/K}x| = |x|_L^e$ and $v_L = e\, v_K$ on $K$. □

**Proposition 7.28** (examples of totally ramified extensions)**.** *(i) For $n \ge 1$, $\Phi_{p^n}(X+1)$ is Eisenstein, so $\mathbb{Q}_p(\zeta_{p^n})/\mathbb{Q}_p$ is totally ramified of degree $\varphi(p^n)$, with uniformiser $1-\zeta_{p^n}$, ring of integers $\mathbb{Z}_p[\zeta_{p^n}]$, and $\mathrm{Gal}(\mathbb{Q}_p(\zeta_{p^n})/\mathbb{Q}_p) \simeq (\mathbb{Z}/p^n)^\times$; the residue field is $\mathbb{F}_p$, so the whole group acts trivially on it (it is all inertia, in the language of Section 7.5).*

*(ii) $\mathbb{Q}_3(\zeta_3) = \mathbb{Q}_3(\sqrt{-3})$ is a totally ramified quadratic extension, tamely ramified ($3 \nmid 2$). Its ring of integers is $\mathbb{Z}_3[\zeta_3] \simeq \mathbb{Z}_3[X]/(X^2+X+1) \simeq \mathbb{Z}_3 \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_3]$, a discrete valuation ring with uniformiser $\pi = 1 - \zeta_3$, residue field $\mathbb{F}_3$ and $\pi^2 = -3\zeta_3$; it is the $\pi$-adic completion of $\mathbb{Z}[\zeta_3]$.*

*(iii) For $m \ge 1$, $X^m - p$ is Eisenstein, so $\mathbb{Q}_p(p^{1/m})/\mathbb{Q}_p$ is totally ramified of degree $m$.*

*(iv) For $p$ odd and $p^* = (-1)^{(p-1)/2}p$, $\mathbb{Q}_p(\sqrt{p^*})/\mathbb{Q}_p$ is totally and tamely ramified of degree 2.*

*Proof.* (i) $\Phi_{p^n}(X) = \sum_{i=0}^{p-1} X^{ip^{n-1}}$, so modulo $p$, $\Phi_{p^n}(X+1) = ((X+1)^{p^n}-1)/((X+1)^{p^{n-1}}-1) \equiv X^{p^n}/X^{p^{n-1}} = X^{\varphi(p^n)}$, and its constant term is $\Phi_{p^n}(1) = p$. The root $\zeta_{p^n} - 1$ is then a uniformiser and $\mathcal{O} = \mathbb{Z}_p[\zeta_{p^n} - 1]$ by Theorem 7.27. The Galois group embeds in $(\mathbb{Z}/p^n)^\times$ (Chapter 5) and has order $\varphi(p^n)$; as $f = 1$ the residue field is $\mathbb{F}_p$, on which every automorphism is trivial (so, once Section 7.5 defines inertia, the whole group is the inertia group, Proposition 7.42(i)). (ii) $\Phi_3(X+1) = (X+1)^2 + (X+1) + 1 = X^2 + 3X + 3$ is Eisenstein at 3, and $\sqrt{-3} = 2\zeta_3 + 1$, because $(2\zeta_3+1)^2 = 4(\zeta_3^2 + \zeta_3) + 1 = -4 + 1 = -3$. So $\mathbb{Q}_3(\zeta_3) = \mathbb{Q}_3(\sqrt{-3})$, of degree 2, $X^2+X+1$ is irreducible over $\mathbb{Q}_3$, the ring of integers is $\mathbb{Z}_3[\zeta_3 - 1] = \mathbb{Z}_3[\zeta_3] \simeq \mathbb{Z}_3[X]/(X^2+X+1)$, with uniformiser $\zeta_3 - 1 = -\pi$; and $\pi^2 = 1 - 2\zeta_3 + \zeta_3^2 = -3\zeta_3$. Right exactness of $\otimes$ gives $\mathbb{Z}_3 \otimes \mathbb{Z}[\zeta_3] \simeq \mathbb{Z}_3[X]/(X^2+X+1)$, and Theorem 7.30(ii) identifies this with the $\pi$-adic completion of $\mathbb{Z}[\zeta_3]$, $(\pi)$ being the only prime above 3. (iii), (iv) are Theorem 7.27: all non-leading coefficients of $X^m - p$ and of $X^2 - p^*$ are divisible by $p$, and the constant terms $-p$ and $-p^* = \mp p$ are not divisible by $p^2$. In (iv) the ramification index 2 is prime to the odd $p$, so the extension is tame. □

*In the paper* 7.29. The paragraph after Lemma lem:IV says that $T_3E$ "is a torsion-free module of $\mathbb{Z}_3$-rank 2 over the discrete valuation ring $\mathbb{Z}_3[X]/(X^2+X+1) \simeq \mathbb{Z}_3[\zeta_3]$, with $X$ acting as $\sigma$, hence free of rank one over it", and the proof of Lemma lem:tamadd uses the same ring. Chapter 6 proved that this ring is a discrete valuation ring and equals $\mathbb{Z}_3 \otimes \mathbb{Z}[\zeta_3]$; Proposition 7.28(ii) identifies it as the ring of integers of $\mathbb{Q}_3(\sqrt{-3})$, a discrete valuation ring with uniformiser $\pi = 1-\zeta_3$ and residue field $\mathbb{F}_3$, and as the completion of $\mathbb{Z}[\zeta_3]$ at $(1-\zeta_3)$, which is what Chapter 6 stated with a pointer here.

### 7.3.1 Completions of number fields

**Theorem 7.30** (completions of a number field)**.** *Let $F$ be a number field and $p$ a prime with $p\mathcal{O}_F = \prod_{\mathfrak{p}\mid p} \mathfrak{p}^{e_\mathfrak{p}}$ and residue degrees $f_\mathfrak{p}$ (Chapter 6).*

*(i) The absolute values on $F$ extending $|\cdot|_p$ are exactly the maps $|x|_\mathfrak{p} = p^{-v_\mathfrak{p}(x)/e_\mathfrak{p}}$, one for each $\mathfrak{p} \mid p$, and distinct primes give inequivalent absolute values.*

*(ii) The completion $F_\mathfrak{p}$ is a finite extension of $\mathbb{Q}_p$ with $e(F_\mathfrak{p}/\mathbb{Q}_p) = e_\mathfrak{p}$ and $f(F_\mathfrak{p}/\mathbb{Q}_p) = f_\mathfrak{p}$, and its ring of integers is $\mathcal{O}_{F_\mathfrak{p}} = \varprojlim_n \mathcal{O}_F/\mathfrak{p}^n$.*

*(iii) $\mathcal{O}_F \otimes_\mathbb{Z} \mathbb{Z}_p \simeq \prod_{\mathfrak{p}\mid p} \mathcal{O}_{F_\mathfrak{p}}$ and $F \otimes_\mathbb{Q} \mathbb{Q}_p \simeq \prod_{\mathfrak{p}\mid p} F_\mathfrak{p}$; each $\mathcal{O}_{F_\mathfrak{p}}$ is free over $\mathbb{Z}_p$ of rank $e_\mathfrak{p}f_\mathfrak{p}$, and $\sum_\mathfrak{p} e_\mathfrak{p}f_\mathfrak{p} = [F:\mathbb{Q}]$.*

*Proof.* (i) Let $|\cdot|$ extend $|\cdot|_p$. It is nonarchimedean (Proposition 7.3(i)). Every $x \in \mathcal{O}_F$ satisfies a monic equation over $\mathbb{Z}$, and $|x| \le 1$ follows as in the proof of Theorem 7.24 (if $|x| > 1$ the leading term of the equation dominates). So $\mathfrak{p} = \{x \in \mathcal{O}_F : |x| < 1\}$ is a prime ideal containing $p$, and elements of $\mathcal{O}_F \setminus \mathfrak{p}$ have absolute value 1; hence the valuation ring of $|\cdot|$ contains the localisation $\mathcal{O}_{F,\mathfrak{p}}$, a discrete valuation ring (Chapter 6). A ring strictly between a discrete valuation ring and its fraction field contains $\varpi^{-1}$ and is the whole field; so the valuation ring is $\mathcal{O}_{F,\mathfrak{p}}$ and $|x| = c^{v_\mathfrak{p}(x)}$; the normalisation $|p| = p^{-1}$ and $v_\mathfrak{p}(p) = e_\mathfrak{p}$ give $c = p^{-1/e_\mathfrak{p}}$. Different primes give different valuation rings, hence inequivalent absolute values. (ii) By Theorem 7.6, $\mathcal{O}_{F_\mathfrak{p}} = \varprojlim \mathcal{O}_{F,\mathfrak{p}}/\mathfrak{p}^n = \varprojlim \mathcal{O}_F/\mathfrak{p}^n$, with residue field $\mathcal{O}_F/\mathfrak{p}$ and $v_\mathfrak{p}(p) = e_\mathfrak{p}$; finiteness over $\mathbb{Q}_p$ follows from (iii). (iii) As $\mathcal{O}_F \simeq \mathbb{Z}^n$, $\mathcal{O}_F \otimes \mathbb{Z}_p = \varprojlim_k \mathcal{O}_F/p^k\mathcal{O}_F$, and by the Chinese remainder theorem $\mathcal{O}_F/p^k\mathcal{O}_F \simeq \prod_\mathfrak{p} \mathcal{O}_F/\mathfrak{p}^{ke_\mathfrak{p}}$ compatibly in $k$; the limit is $\prod_\mathfrak{p} \mathcal{O}_{F_\mathfrak{p}}$. Each factor is a direct summand of the free module $\mathbb{Z}_p^n$, so it is finitely generated and torsion free over the discrete valuation ring $\mathbb{Z}_p$, hence free (Chapter 3); its rank is $e_\mathfrak{p}f_\mathfrak{p}$ by Theorem 7.25, and the ranks add up to $n$. Inverting $p$ gives the statement for $F \otimes \mathbb{Q}_p$. □

For example $\mathbb{Z}[\zeta_3] \otimes \mathbb{Z}_3 = \mathbb{Z}_3[\zeta_3]$ (ramified), $\mathbb{Z}[\zeta_3] \otimes \mathbb{Z}_7 \simeq \mathbb{Z}_7 \times \mathbb{Z}_7$ (split, Example 7.16(c)), and $\mathbb{Z}[\zeta_3] \otimes \mathbb{Z}_2$ is the ring of integers of the unramified quadratic extension of $\mathbb{Q}_2$ (inert). In each case $\mathbb{Z}[\zeta_3] \otimes \mathbb{Z}_p = \mathbb{Z}_p[X]/(X^2+X+1)$, and the three behaviours are read off from the reduction of $X^2+X+1$ modulo $p$: it is $(X-1)^2$ for $p=3$ (one prime, $e=2$, $f=1$), it has the two distinct roots 2, 4 for $p=7$ (two primes, $e=f=1$), and it is irreducible for $p=2$, having no root in $\mathbb{F}_2$ (one prime, $e=1$, $f=2$, by Example 7.16(g)). In all three, $\sum e_\mathfrak{p}f_\mathfrak{p} = 2$, as (iii) requires. For the coefficient field of a newform and a prime $\lambda \mid \ell$ the theorem gives the complete discrete valuation ring $\mathcal{O}_\lambda$ used in Chapters 13 and 15. This is the result that Chapter 6 stated and deferred here.

**Lemma 7.31** (Krasner)**.** *Let $\alpha \in \overline{K}$ have distinct conjugates $\alpha = \alpha_1, \alpha_2, \dots, \alpha_d$ over $K$.*

*(i) If $\beta \in \overline{K}$ satisfies $|\beta - \alpha| < |\alpha - \alpha_i|$ for $2 \le i \le d$, then $K(\alpha) \subseteq K(\beta)$.*

*(ii) Let $f$ be the minimal polynomial of $\alpha$. There is $\delta > 0$ such that every monic $g \in K[X]$ of degree $d$ whose coefficients differ from those of $f$ by less than $\delta$ has a root $\beta$ with $K(\beta) = K(\alpha)$.*

*Proof.* (i) Let $\sigma \in \mathrm{Gal}(\overline{K}/K(\beta))$. Since $\sigma$ is an isometry (Theorem 7.24), $|\sigma\alpha - \beta| = |\sigma(\alpha - \beta)| = |\alpha - \beta|$, so $|\sigma\alpha - \alpha| \le \max(|\sigma\alpha - \beta|, |\beta - \alpha|) = |\alpha - \beta| < |\alpha - \alpha_i|$ for $i \ge 2$; as $\sigma\alpha$ is a conjugate, $\sigma\alpha = \alpha$. So $\alpha$ is fixed by $\mathrm{Gal}(\overline{K}/K(\beta))$ and lies in $K(\beta)$ (characteristic 0). (ii) If the coefficients of $g - f$ are smaller than $\delta$, then $|g(\alpha)| = |g(\alpha) - f(\alpha)| < \delta \max(1, |\alpha|)^d$, because $g - f = \sum_{i<d} b_i X^i$ with $|b_i| < \delta$ and each $|b_i \alpha^i|$ is at most $|b_i| \max(1, |\alpha|)^d$. Writing $g = \prod_{j=1}^{d}(X - \beta_j)$ over $\overline{K}$ we have $|g(\alpha)| = \prod_j |\alpha - \beta_j|$, so the smallest factor satisfies $|\alpha - \beta_j| \le |g(\alpha)|^{1/d} < \left(\delta \max(1, |\alpha|)^d\right)^{1/d}$, which is less than $\min_{i \ge 2} |\alpha - \alpha_i|$ once $\delta$ is small enough. By (i), $K(\alpha) \subseteq K(\beta_j)$, and $[K(\beta_j) : K] \le d$ gives equality. □

(Here $\overline{K}$ denotes $\overline{\mathbb{Q}}_p$, an algebraic closure of $K$.)

**Proposition 7.32** (decomposition groups are local Galois groups)**.** *(i) Let $L/F$ be a finite Galois extension of number fields and $\mathfrak{P} \mid \mathfrak{p}$ primes. Then $L_{\mathfrak{P}} = L \cdot F_{\mathfrak{p}}$, the extension $L_{\mathfrak{P}}/F_{\mathfrak{p}}$ is Galois, and restriction to $L$ is an isomorphism $\mathrm{Gal}(L_{\mathfrak{P}}/F_{\mathfrak{p}}) \to D_{\mathfrak{P}}$, carrying the inertia subgroup onto $I_{\mathfrak{P}}$ and Frobenius elements to Frobenius elements.*

*(ii) Let $\iota \colon \overline{\mathbb{Q}} \to \overline{\mathbb{Q}}_p$ be an embedding and $\bar{\mathfrak{P}} = \{x \in \bar{\mathbb{Z}} : |\iota x|_p < 1\}$ the prime of $\bar{\mathbb{Z}}$ it defines. Then $\sigma \mapsto \iota^{-1}\sigma\iota$ is an isomorphism of profinite groups $G_{\mathbb{Q}_p} \to D_{\bar{\mathfrak{P}}} \subseteq G_{\mathbb{Q}}$, carrying $I_p$ onto $I_{\bar{\mathfrak{P}}}$ and arithmetic Frobenius lifts to Frobenius elements. Another embedding $\iota \circ \tau$ ($\tau \in G_{\mathbb{Q}}$) composes this map with conjugation by $\tau^{-1}$.*

*(iii) Hence for a continuous representation $\rho$ of $G_{\mathbb{Q}}$ the restriction $\rho|_{G_{\mathbb{Q}_p}}$, the image $\rho(I_p)$, the invariants $V^{I_p}$ and, if $\rho$ is unramified at $p$, $\rho(\mathrm{Frob}_p)$, are well defined up to conjugation.*

*Proof.* (i) The closure of $F$ in $L_{\mathfrak{P}}$ is a completion of $F$ for $|\cdot|_{\mathfrak{p}}$ (up to equivalence, by Theorem 7.30(i)), so $F_{\mathfrak{p}} \subseteq L_{\mathfrak{P}}$. The compositum $LF_{\mathfrak{p}}$ is finite over $F_{\mathfrak{p}}$, hence complete (Theorem 7.24) and closed, and contains the dense subfield $L$; so $L_{\mathfrak{P}} = LF_{\mathfrak{p}}$. Write $L = F(\theta)$; the minimal polynomial $g$ of $\theta$ splits in $L$, and $L_{\mathfrak{P}} = F_{\mathfrak{p}}(\theta)$ is generated by roots of $g$, so it is a splitting field of $g$ over $F_{\mathfrak{p}}$, hence Galois. An element $\tau$ of $\mathrm{Gal}(L_{\mathfrak{P}}/F_{\mathfrak{p}})$ permutes the roots of $g$, so maps $L$ to $L$; it is an isometry, so $\tau|_L$ preserves $\mathfrak{P} = \{x \in \mathcal{O}_L : |x| < 1\}$, i.e. $\tau|_L \in D_{\mathfrak{P}}$; and $\tau$ is determined by $\tau|_L$ by continuity. Conversely, for $\sigma \in D_{\mathfrak{P}}$, $x \mapsto |\sigma x|_{\mathfrak{P}}$ is the absolute value of $\sigma^{-1}\mathfrak{P} = \mathfrak{P}$, so $\sigma$ is an isometry of $L$ and extends by continuity to an automorphism of $L_{\mathfrak{P}}$ over $F_{\mathfrak{p}}$. The residue fields of $L$ at $\mathfrak{P}$ and of $L_{\mathfrak{P}}$ coincide (Theorem 7.6(ii)), compatibly with the actions; so inertia corresponds to inertia and Frobenius to Frobenius. (ii) $\iota(\overline{\mathbb{Q}})$ is the set of elements of $\overline{\mathbb{Q}}_p$ algebraic over $\mathbb{Q}$, so it is stable under $G_{\mathbb{Q}_p}$, and $\sigma \mapsto \iota^{-1}\sigma\iota$ is a continuous homomorphism into $G_{\mathbb{Q}}$, with image in $D_{\bar{\mathfrak{P}}}$ since $\sigma$ is an isometry. It is injective: given $\alpha \in \overline{\mathbb{Q}}_p$ with minimal polynomial $f$ over $\mathbb{Q}_p$, choose $g \in \mathbb{Q}[X]$ monic, with coefficients close to those of $f$ ($\mathbb{Q}$ is dense in $\mathbb{Q}_p$); by Lemma 7.31(ii) some root $\beta$ of $g$ has $\mathbb{Q}_p(\beta) = \mathbb{Q}_p(\alpha)$, and $\beta \in \iota(\overline{\mathbb{Q}})$. So $\overline{\mathbb{Q}}_p$ is generated over $\mathbb{Q}_p$ by $\iota(\overline{\mathbb{Q}})$, and an element of $G_{\mathbb{Q}_p}$ trivial on $\iota(\overline{\mathbb{Q}})$ is trivial. It is surjective: for each finite Galois $L/\mathbb{Q}$, with $\mathfrak{P}_L$ the prime below $\bar{\mathfrak{P}}$, $G_{\mathbb{Q}_p}$ maps onto $\mathrm{Gal}(\iota(L)\mathbb{Q}_p/\mathbb{Q}_p)$, which (i) identifies with $D_{\mathfrak{P}_L}$; so the image is dense in $D_{\bar{\mathfrak{P}}} = \varprojlim D_{\mathfrak{P}_L}$ (Chapter 6), and closed, being compact. A continuous bijection of compact Hausdorff groups is a homeomorphism. Inertia and Frobenius correspond at every finite level by (i), hence in the limit. Finally $(\iota\tau)^{-1}\sigma(\iota\tau) = \tau^{-1}(\iota^{-1}\sigma\iota)\tau$. (iii) follows, since the primes above $p$ are conjugate (Chapter 6). □

From now on a decomposition group at $p$ is identified with $G_{\mathbb{Q}_p}$ in this way, and we speak of $\rho|_{G_{\mathbb{Q}_p}}$ and $\rho(I_p)$; this is the statement $D_p \simeq G_{\mathbb{Q}_p}$ that Chapter 6 deferred here.

**Proposition 7.33** (tame totally ramified extensions)**.** *Let $L/K$ be totally ramified of degree $e$ with $p \nmid e$. Then $L = K(\alpha)$ with $\alpha^e = \varpi$ for some uniformiser $\varpi$ of $K$. The extension is Galois iff $\mu_e \subset K$, that is iff $e \mid q-1$; then $\mathrm{Gal}(L/K) \to \mu_e$, $\sigma \mapsto \sigma(\alpha)/\alpha$, is an isomorphism, and $\mathrm{Gal}(L/K)$ is cyclic.*

*Proof.* Let $\pi_L$ be a uniformiser of $L$; then $\pi_L^e = \varpi_K w$ with $w \in \mathcal{O}_L^\times$. As $k_L = k_K$, choose $w_0 \in \mathcal{O}_K^\times$ with the same reduction as $w$; then $w_1 = w/w_0 \in U_L^{(1)}$, which is uniquely $e$-divisible (Theorem 7.18(iv)). So $\alpha = \pi_L w_1^{-1/e} \in L$ is a uniformiser of $L$ with $\alpha^e = \varpi_K w_0 =: \varpi$, a uniformiser of $K$. By Theorem 7.27, $[K(\alpha):K] = e = [L:K]$, so $L = K(\alpha)$. The conjugates of $\alpha$ are the $\zeta\alpha$, $\zeta \in \mu_e$. If $L/K$ is Galois they all lie in $L$, so $\mu_e \subset L$, and then $e \mid \#k_L^\times = q-1$ (Corollary 7.19) and $\mu_e \subset K$. Conversely, if $\mu_e \subset K$, $L$ is the splitting field of $X^e - \varpi$, and $\sigma \mapsto \sigma(\alpha)/\alpha$ is an injective homomorphism into $\mu_e$ (as $\sigma$ fixes $\mu_e$) between groups of order $e$. □

## 7.4 Unramified extensions, $\mathbb{Q}_p^{\mathrm{ur}}$ and Frobenius

The unramified extensions of a local field are generated by roots of unity of order prime to $p$, are cyclic, and are classified by their residue fields; their union $K^{\mathrm{ur}}$ has Galois group $\hat{\mathbb{Z}}$, generated by the Frobenius. This is where the paper's Frobenius lives, and where its two conventions (arithmetic and geometric) must be fixed once and for all (Convention 7.37). The paper uses the section in Lemmas lem:tamagawa-exp and lem:tamfactor ("Over the unramified quadratic extension of $\mathbb{Q}_q$ (which does not change the inertia group) $E$ is a Tate curve"; Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$, whose units are $\ell^t$-th powers), in Lemma lem:twist(i) (invariance under unramified base change), and in Section 1.1, where $\mathrm{Frob}_p$ and its inverse appear.

**Theorem 7.34** (unramified extensions)**.** *Let $K$ have residue field $\mathbb{F}_q$.*

*(i) For each $f \geq 1$ there is exactly one unramified extension $K_f$ of $K$ of degree $f$ inside $\overline{\mathbb{Q}}_p$, namely $K_f = K(\mu_{q^f-1})$.*

*(ii) $K_f/K$ is Galois, and reduction is an isomorphism $\mathrm{Gal}(K_f/K) \to \mathrm{Gal}(\mathbb{F}_{q^f}/\mathbb{F}_q)$. So $\mathrm{Gal}(K_f/K)$ is cyclic of order $f$, generated by the unique $\mathrm{Frob}_{K_f/K}$ with $\mathrm{Frob}(x) \equiv x^q \pmod{\mathfrak{m}_{K_f}}$ for all $x \in \mathcal{O}_{K_f}$; on $\mu_{q^f-1}$ it is $\zeta \mapsto \zeta^q$.*

*(iii) A finite $L/K$ is unramified iff $L = K_f$ for some $f$. Composita and subextensions of unramified extensions are unramified. Every finite $L/K$ has a largest unramified subextension $L_0$; $L_0 = K_{f(L/K)}$ and $L/L_0$ is totally ramified.*

*(iv) If $L/K$ is unramified and $\alpha \in \mathcal{O}_L$ reduces to a generator of $k_L$ over $k_K$, then $L = K(\alpha)$.*

*Proof.* (i) Let $\zeta \in \overline{\mathbb{Q}}_p$ be a primitive $(q^f-1)$-th root of unity and $L = K(\zeta)$. Reduction is injective on roots of unity of order prime to $p$ (Corollary 7.19, applied in $L$), so $\bar{\zeta}$ has order $q^f-1$ and generates $\mathbb{F}_{q^f}$ over $\mathbb{F}_q$; hence $f(L/K) \geq f$. The minimal polynomial $g$ of $\zeta$ over $K$ lies in $\mathcal{O}_K[X]$ (Corollary 7.15) and divides $X^{q^f-1}-1$, so $\bar{g}$ divides the separable polynomial $X^{q^f-1}-1$ and is separable. Let $\bar{h}$ be the minimal polynomial of $\bar{\zeta}$ over $\mathbb{F}_q$, of degree $f$; it divides $\bar{g}$. If $\bar{g} \neq \bar{h}$, then $\bar{g} = \bar{h}\bar{r}$ with $\bar{h}$, $\bar{r}$ coprime, and Theorem 7.14 would factor $g$. So $\bar{g} = \bar{h}$, $[L:K] = f \leq f(L/K)$, and therefore $f(L/K) = f$, $e(L/K) = 1$. Uniqueness: if $L'/K$

is unramified of degree $f$, then $k_{L'} = \mathbb{F}_{q^f}$, and the Teichmüller lifts (Theorem 7.17, applied in $L'$) give $\mu_{q^f-1} \subset L'$; so $K_f \subseteq L'$, and the degrees agree. (ii) $K_f$ is the splitting field of $X^{q^f-1} - 1$, so it is Galois. Every $\sigma \in \mathrm{Gal}(K_f/K)$ is an isometry, preserves $\mathcal{O}_{K_f}$ and $\mathfrak{m}_{K_f}$, and induces an automorphism $\bar\sigma$ of $\mathbb{F}_{q^f}$ over $\mathbb{F}_q$. If $\bar\sigma = 1$, then $\sigma$ fixes $\mu_{q^f-1}$ (by injectivity of reduction on it), so $\sigma = 1$. Both groups have order $f$, so $\sigma \mapsto \bar\sigma$ is an isomorphism, and Frob is the preimage of $x \mapsto x^q$; it maps $\zeta \in \mu_{q^f-1}$ to the root of unity reducing to $\bar\zeta^q$, which is $\zeta^q$. (iii) If $L$ is unramified with $f(L/K) = f$, the argument for uniqueness in (i) shows $L = K_f$. Subextensions are unramified since $e$ is multiplicative; and $K_f, K_{f'} \subseteq K_{ff'}$, so composita are unramified. For any finite $L$, the Teichmüller lifts give $\mu_{q^{f(L/K)}-1} \subset L$, so $K_{f(L/K)} \subseteq L$; if $K_{f'} \subseteq L$ then $\mathbb{F}_{q^{f'}} \subseteq k_L$, so $f' \mid f(L/K)$ and $\mu_{q^{f'}-1} \subseteq \mu_{q^{f(L/K)}-1}$; hence $L_0 = K_{f(L/K)}$, and $f(L/L_0) = 1$. (iv) Let $f = [L : K] = f(L/K)$. The residue field of $K(\alpha)$ contains $\bar\alpha$, hence $k_K(\bar\alpha) = k_L$, so $[K(\alpha) : K] \geq f(K(\alpha)/K) \geq [k_K(\bar\alpha) : k_K] = f = [L : K]$ (the first inequality is $ef = n$); as $K(\alpha) \subseteq L$, equality holds and $K(\alpha) = L$. □

**Definition 7.35** (maximal unramified extension)**.** $K^{\mathrm{ur}} = \bigcup_{f\geq 1} K_f \subseteq \overline{\mathbb{Q}}_p$. Equivalently $K^{\mathrm{ur}} = K(\mu_m : p \nmid m)$, since $\mu_m \subset K_f$ when $m \mid q^f - 1$, which holds for $f$ the order of $q$ modulo $m$. For $K = \mathbb{Q}_p$ we write $\mathbb{Q}_p^{\mathrm{ur}}$.

The point of passing to $K^{\mathrm{ur}}$ is that it removes the residue field from the picture: in $K^{\mathrm{ur}}$ every unit is an $m$-th power for every $m$ prime to $p$ (Proposition 7.36(iv) below), so the only obstruction left to extracting roots is the valuation, and what remains of the Galois group is inertia. For example $\mathbb{Q}_7^{\mathrm{ur}}$ contains $\sqrt{3}$, since 3 is not a square modulo 7 but is one in $\mathbb{F}_{49}$, so $\mathbb{Q}_7(\sqrt{3})$ is the quadratic unramified extension $K_2$ of $\mathbb{Q}_7$; and $\mathbb{Q}_2^{\mathrm{ur}}$ contains $\zeta_3$, with $f = 2$ the order of 2 modulo 3, so $\mathbb{Q}_2(\zeta_3) = \mathbb{Q}_2(\sqrt{-3})$ is the unramified quadratic extension of $\mathbb{Q}_2$ (Proposition 7.60(iii)).

**Proposition 7.36.** *(i) The residue field of $K^{\mathrm{ur}}$ is $\overline{\mathbb{F}}_q$, and restriction gives isomorphisms $\mathrm{Gal}(K^{\mathrm{ur}}/K) \simeq \varprojlim_f \mathrm{Gal}(K_f/K) \simeq \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \simeq \hat{\mathbb{Z}}$, topologically generated by the arithmetic Frobenius.*

*(ii) The absolute value of $K^{\mathrm{ur}}$ is discrete with the same value group as $K$, and $\varpi_K$ is a uniformiser; but $K^{\mathrm{ur}}$ is not complete.*

*(iii) Hensel's lemma (Theorem 7.12) holds for polynomials over $\mathcal{O}_{K^{\mathrm{ur}}}$.*

*(iv) For every $m$ prime to $p$: $\mu_m \subset K^{\mathrm{ur}}$; every unit of $\mathcal{O}_{K^{\mathrm{ur}}}$ is an $m$-th power in $K^{\mathrm{ur}}$; and $x \in (K^{\mathrm{ur}})^\times$ is an $m$-th power in $K^{\mathrm{ur}}$ iff $m \mid v_K(x)$. In particular $\mu_{\ell^\infty} \subset \mathbb{Q}_q^{\mathrm{ur}}$ for every prime $\ell \neq q$.*

*Proof.* (i) The residue field is $\bigcup_f \mathbb{F}_{q^f} = \overline{\mathbb{F}}_q$. Restriction $\mathrm{Gal}(K^{\mathrm{ur}}/K) \to \varprojlim \mathrm{Gal}(K_f/K)$ is an isomorphism by infinite Galois theory (Chapter 5), and the isomorphisms of Theorem 7.34(ii) are compatible; $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \simeq \hat{\mathbb{Z}}$ with generator $x \mapsto x^q$ (Chapter 5). (ii) Each $K_f$ is unramified, so has the value group of $K$. Not complete: choose $\omega_n \in \mu_{q^{n!}-1}$ whose reduction generates $\mathbb{F}_{q^{n!}}$, and let $x = \sum_n \omega_n \varpi_K^n$ in the completion. If $x$ lay in some $K_f$, its expansion with Teichmüller digits in $K_f$ (obtained as in Proposition 7.10, the digit of $y \in \mathcal{O}_{K_f}$ being $[\bar y]$) would be the expansion above, by uniqueness of digits; but $\omega_n \notin K_f$ once $n! \nmid f$. (iii) The coefficients of $f$ and the approximate root $a_0$ lie in some $K_f$, which is complete; the Newton sequence stays in $K_f$, and so does the root. (iv) $\mu_m \subset K^{\mathrm{ur}}$ by definition. For a unit $u$, $\bar u \in \mathbb{F}_{q^f}^\times$ for some $f$; choose $f'$ with $m(q^f - 1) \mid q^{f'} - 1$ (possible since $p \nmid m(q^f - 1)$); in the cyclic group $\mathbb{F}_{q^{f'}}^\times$ of order $q^{f'} - 1$,

the element $\bar{u}$, of order dividing $q^f-1$, is an $m$-th power. The root of $X^m-\bar{u}$ so found is simple ($m$ is a unit), and lifts by (iii). Finally $x=\varpi_K^{am}u$ is then an $m$-th power, and conversely $v_K(y^m)=mv_K(y)$. □

**Convention 7.37** (Frobenius)**.** Frob $=$ Frob$_K$ denotes an *arithmetic Frobenius* of $K$: any element of $G_K$ whose restriction to $K^{\mathrm{ur}}$ is the generator of Proposition 7.36(i), that is, which induces $x\mapsto x^q$ on $\overline{\mathbb{F}}_q$. It is well defined modulo $I_K$ (and, in a global setting, up to conjugation). The *geometric Frobenius* is $\Phi=\mathrm{Frob}^{-1}$. The book's Frob$_p$ is always arithmetic (Appendix B, convention B-C2).

(a) Frob acts on $\mu_m$ ($p\nmid m$) by $\zeta\mapsto\zeta^q$, so $\chi_\ell(\mathrm{Frob})=q$ and $\chi_\ell(\Phi)=q^{-1}$ for $\ell\neq p$ (Proposition 7.39).

(b) Since Frob is defined only modulo $I_K$, a statement about $\rho(\mathrm{Frob})$ for a ramified $\rho$ must be checked to be independent of the lift, as in Corollary 7.52 and Theorem 7.58(iv).

(c) An eigenvalue $\lambda$ of Frob is an eigenvalue $\lambda^{-1}$ of $\Phi$. When the multiset of eigenvalues on the relevant space is stable under inversion (as on $\mathrm{ad}^0V$ at a good prime, or on a line where Frobenius acts by $\pm1$), the two conventions give the same $L$-factor; otherwise they do not (the Steinberg case, Proposition 7.88).

(d) Group-theoretic statements (the tame relation, conjugation actions, the matrices in the proof of Lemma lem:tamfactor, Frobenius on Kummer classes) use Frob; local $L$-factors use $\Phi$ (Definition 7.77). The paper fixes the same convention in Section 1.1: Frob$_p$ is an arithmetic Frobenius and Frob$_p^{-1}$ a geometric one; its table, Lemma lem:euler and Lemma lem:fpr use the geometric Frobenius, and the proof of Lemma lem:tamfactor the arithmetic one.

**Proposition 7.38** (unramified base change)**.** *Let $K/\mathbb{Q}_p$ be finite unramified of degree $f$. Then $K^{\mathrm{ur}}=\mathbb{Q}_p^{\mathrm{ur}}$ and $K^{\mathrm{tame}}=\mathbb{Q}_p^{\mathrm{tame}}$, so $I_K=I_p$ and $P_K=P_p$ as subgroups of $G_K\subseteq G_{\mathbb{Q}_p}$, and* $\mathrm{Frob}_K\equiv\mathrm{Frob}_{\mathbb{Q}_p}^f \pmod{I_p}$. *Consequently, for a representation $\rho$ of $G_{\mathbb{Q}_p}$, the image $\rho(I_p)$, the invariants $V^{I_p}$ and the property of being tamely or wildly ramified are unchanged by restriction to $G_K$; so are the monodromy operator and the Swan and Artin conductors, which are defined later (forward references: Theorem 7.68, Definition 7.81, Definition 7.83). For an arbitrary finite $K/\mathbb{Q}_p$, $K^{\mathrm{ur}}=K\mathbb{Q}_p^{\mathrm{ur}}$ and $I_K=I_p\cap G_K$ has index $e(K/\mathbb{Q}_p)$ in $I_p$.*

*Proof.* For an arbitrary finite $K$: $\mathbb{Q}_p^{\mathrm{ur}}=\mathbb{Q}_p(\mu_m : p\nmid m)\subseteq K^{\mathrm{ur}}$, and $K^{\mathrm{ur}}=K(\mu_m : p\nmid m)\subseteq K\mathbb{Q}_p^{\mathrm{ur}}$; so $K^{\mathrm{ur}}=K\mathbb{Q}_p^{\mathrm{ur}}$ and $I_K=\mathrm{Gal}(\overline{\mathbb{Q}}_p/K\mathbb{Q}_p^{\mathrm{ur}})=G_K\cap I_p$. Its index in $I_p$ is $[K\mathbb{Q}_p^{\mathrm{ur}}:\mathbb{Q}_p^{\mathrm{ur}}]$ (Chapter 5, translation theorem). Let $K_0$ be the largest unramified subextension, of degree $f(K/\mathbb{Q}_p)$, so that $K=K_0(\pi)$ with $\pi$ a root of an Eisenstein polynomial of degree $e=e(K/\mathbb{Q}_p)$ over $K_0$ (Theorem 7.34(iii), Theorem 7.27). That polynomial stays Eisenstein over every finite unramified extension of $K_0$, hence irreducible over $K_0^{\mathrm{ur}}=\mathbb{Q}_p^{\mathrm{ur}}$; so the index is $e$. If $K$ is unramified, $K=K_0\subseteq\mathbb{Q}_p^{\mathrm{ur}}$, so $K^{\mathrm{ur}}=\mathbb{Q}_p^{\mathrm{ur}}$. A finite extension of $K$ is tamely ramified over $K$ iff it is over $\mathbb{Q}_p$, $e$ being multiplicative and $e(K/\mathbb{Q}_p)=1$; so $K^{\mathrm{tame}}=\mathbb{Q}_p^{\mathrm{tame}}$ (Theorem 7.45(ii)). Frob$_K$ induces $x\mapsto x^{p^f}$ on the residue field. The image $\rho(I_p)$, the invariants $V^{I_p}$ and tameness ($\rho(P_p)=1$) depend only on the subgroups $I_p$, $P_p$, which do not change. The monodromy operator is determined by $\rho$ on an open subgroup of $I_p$ and by the tame character, which is the same for $K$ and $\mathbb{Q}_p$ because $p$ is a uniformiser of both (Theorem 7.47(i)). For the conductors see

Theorem 7.82(iii). (These two forward references do not use the present proposition, so there is no circularity.) □

**Proposition 7.39** (the cyclotomic character locally)**.** *(i) For a prime $\ell \neq p$, $\mathbb{Q}_p(\mu_{\ell^k})/\mathbb{Q}_p$ is unramified of degree the order of $p$ in $(\mathbb{Z}/\ell^k)^\times$, and* Frob *acts on $\mu_{\ell^\infty}$ by $\zeta \mapsto \zeta^p$. So $\chi_\ell|_{G_{\mathbb{Q}_p}}$ is unramified with $\chi_\ell(\mathrm{Frob}) = p$; more generally $\chi_\ell(\mathrm{Frob}_K) = q$ for $K$ with residue field $\mathbb{F}_q$.*

*(ii) For $\ell = p$, $\mathbb{Q}_p(\mu_{p^n})/\mathbb{Q}_p$ is totally ramified with group $(\mathbb{Z}/p^n)^\times$, so $\chi_p(I_p) = \mathbb{Z}_p^\times$, and $\mathbb{Q}_p(\mu_{p^\infty}) \cap \mathbb{Q}_p^{\mathrm{ur}} = \mathbb{Q}_p$.*

*(iii) Globally, through Proposition 7.32: $\chi_\ell(\mathrm{Frob}_p) = p$ for every prime $p \neq \ell$.*

*Proof.* (i) Write $\mathbb{Q}_{p,f}$ for the unramified extension of degree $f$. Its residue field is $\mathbb{F}_{p^f}$, so $\mu_{\ell^k} \subset \mathbb{Q}_{p,f}$ iff $\ell^k \mid p^f - 1$ (Corollary 7.19), that is iff $f$ is a multiple of the order $f_0$ of $p$ in $(\mathbb{Z}/\ell^k)^\times$. The field $\mathbb{Q}_p(\mu_{\ell^k})$ is contained in $\mathbb{Q}_{p,f_0} = \mathbb{Q}_p(\mu_{p^{f_0}-1})$, hence unramified (Theorem 7.34(iii)), so it is some $\mathbb{Q}_{p,f}$ with $f \mid f_0$; it contains $\mu_{\ell^k}$, so $f_0 \mid f$, and $f = f_0$. By Theorem 7.34(ii) Frob acts on roots of unity of order prime to $p$ by $\zeta \mapsto \zeta^p$; as $\chi_\ell$ is defined by $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ on $\mu_{\ell^\infty}$, $\chi_\ell(\mathrm{Frob}) = p$, and $\chi_\ell$ is trivial on $I_p$, which fixes $\mathbb{Q}_p^{\mathrm{ur}} \supset \mu_{\ell^\infty}$. For $K$ with residue field $\mathbb{F}_q$ the same argument gives $\zeta \mapsto \zeta^q$. (ii) Proposition 7.28(i): the image of $I_p$ in $\mathrm{Gal}(\mathbb{Q}_p(\mu_{p^n})/\mathbb{Q}_p)$ is the inertia group, which is everything; so $\chi_p(I_p)$ is dense in $\mathbb{Z}_p^\times$, and closed. An extension that is both unramified and totally ramified is trivial. (iii) follows from (i). □

*In the paper* 7.40. The proof of Lemma lem:tamagawa-exp passes to the unramified quadratic extension $K_2$ of $\mathbb{Q}_q$, "which does not change the inertia group": this is Proposition 7.38. It then uses that $\mathbb{Q}_q^{\mathrm{ur}}$ contains $\mu_{\ell^t}$ $(q \neq \ell)$ and that "the units of its valuation ring are $\ell^t$-th powers by Hensel's lemma": Proposition 7.36(iv), where Hensel's lemma is applied in a finite, complete, subextension because $\mathbb{Q}_q^{\mathrm{ur}}$ itself is not complete. The conclusion, that $q_E$ is an $\ell^t$-th power in $\mathbb{Q}_q^{\mathrm{ur}}$ if and only if $\ell^t \mid n_q$, is the last statement of Proposition 7.36(iv); Theorem 7.58(v) puts it in terms of inertia.

## 7.5 Ramification groups, inertia and the tame quotient

This section is the heart of the chapter for the paper. It shows that the inertia group $I_K$ has a normal pro-$p$ subgroup $P_K$ with procyclic quotient $\prod_{\ell \neq p} \mathbb{Z}_\ell(1)$, defines the tame character $t_\ell \colon I_K \to \mathbb{Z}_\ell(1)$, and proves the tame relation $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^q$. The paper uses these facts in Lemma lem:tamfactor (inertia acts through its maximal pro-$\ell$ quotient, generated by $t$, and $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$), in Lemmas lem:IV and lem:euler and Proposition prop:shapes (a finite image of inertia of order prime to $q$ is cyclic, and Frobenius acts on it by the $q$-th power), in Lemma lem:tamadd (the maximal pro-$\ell$ quotient of an open subgroup of inertia), and in Remark rem:wild (wild inertia at $q = 2, 3$). Chapter 6 proved the finite-level relation for tamely ramified extensions of number fields modulo one step, that the kernel of $\tau \mapsto \tau(\varpi)/\varpi$ is a $p$-group; that step is Proposition 7.42(iii).

### 7.5.1 Ramification groups of a finite Galois extension

**Definition 7.41** (lower ramification groups)**.** Let $L/K$ be finite Galois with group $G$. For an integer $i \geq -1$ put

$$G_i = \{\sigma \in G : \ v_L(\sigma(x) - x) \geq i + 1 \text{ for all } x \in \mathcal{O}_L\}.$$

$G_0 = I(L/K)$ is the *inertia group* and $G_1 = P(L/K)$ the *wild inertia group* of $L/K$.

The index $i$ measures how close $\sigma$ is to the identity: $\sigma \in G_i$ means that $\sigma$ moves every integer of $L$ by something divisible by $\pi^{i+1}$. Every $\sigma$ lies in $G_{-1}$; $\sigma \in G_0$ means that $\sigma$ acts trivially on the residue field $\mathcal{O}_L/\mathfrak{m}_L$; and the higher $G_i$ refine this. For $L = \mathbb{Q}_3(\sqrt{-3})$ and the nontrivial $\sigma$, the uniformiser $\pi = \sqrt{-3}$ is moved by $\sigma\pi - \pi = -2\sqrt{-3}$, of valuation 1; by Proposition 7.42(i) below this places $\sigma$ in $G_0$ but not in $G_1$. Wild ramification is the case in which some $\sigma \neq 1$ moves everything by at least $\pi^2$, and it can happen only when $p$ divides $e$.

**Proposition 7.42.** *Let $L/K$ be finite Galois with group $G$, $\pi$ a uniformiser of $L$, $L_0$ the largest unramified subextension and $e = e(L/K)$.*

*(i) $G_{-1} = G$; the $G_i$ are normal in $G$, decreasing, and $G_i = 1$ for $i$ large. $G_0$ is the kernel of the surjection $G \to \mathrm{Gal}(k_L/k_K)$, $G_0 = \mathrm{Gal}(L/L_0)$ and $\#G_0 = e$. For $\sigma \in G_0$ and $i \geq 0$: $\sigma \in G_i$ iff $v_L(\sigma(\pi) - \pi) \geq i + 1$. For a subgroup $H$, $H_i = H \cap G_i$.*

*(ii) $\theta_0(\sigma) = \sigma(\pi)/\pi \bmod \mathfrak{m}_L$ defines a homomorphism $G_0 \to k_L^\times$, independent of $\pi$, with kernel $G_1$. So $G_0/G_1$ is cyclic of order prime to $p$.*

*(iii) For $i \geq 1$, $\sigma \mapsto \sigma(\pi)/\pi - 1 \bmod \mathfrak{m}_L^{i+1}$ defines a homomorphism $G_i \to \mathfrak{m}_L^i/\mathfrak{m}_L^{i+1} \simeq (k_L, +)$ with kernel $G_{i+1}$. So $G_1$ is a $p$-group, the unique Sylow $p$-subgroup of $G_0$, and $L/K$ is tamely ramified iff $G_1 = 1$.*

*(iv) (Stated.) Put $G_t = G_{\lceil t \rceil}$ for real $t \geq -1$ and $\varphi_{L/K}(u) = \int_0^u dt/[G_0 : G_t]$, with inverse $\psi_{L/K}$; the* upper numbering *is $G^v = G_{\psi_{L/K}(v)}$. If $H$ is normal in $G$ then $(G/H)^v = G^v H/H$ (Herbrand), and if $G$ is abelian the jumps of the upper filtration are integers (Hasse-Arf). We do not prove this; see Serre [174, Chapter IV, Section 3, and Chapter V, Section 7].*

*Proof.* (i) $G_{-1} = G$ trivially. For $\tau \in G$, $v_L(\tau\sigma\tau^{-1}x - x) = v_L(\sigma(\tau^{-1}x) - \tau^{-1}x)$ since $\tau$ is an isometry, and $\tau^{-1}$ permutes $\mathcal{O}_L$; so $G_i$ is normal. Every element of $L$ is $x/\varpi_K^k$ with $x \in \mathcal{O}_L$, so each $\sigma \neq 1$ moves some $x \in \mathcal{O}_L$, and $G_i = 1$ once $i + 1$ exceeds the finitely many values $v_L(\sigma x - x)$ so obtained. By definition $G_0$ is the kernel of the action on $k_L$. An element acts trivially on $k_L$ iff it fixes the Teichmüller lifts $\mu_{q^f-1} \subset L$, $f = f(L/K)$ (reduction is injective on them), iff it fixes $L_0 = K(\mu_{q^f-1})$ (Theorem 7.34); so $G_0 = \mathrm{Gal}(L/L_0)$, of order $[L : L_0] = e$, and $G/G_0 = \mathrm{Gal}(L_0/K) \simeq \mathrm{Gal}(k_L/k_K)$. As $L/L_0$ is totally ramified, $\mathcal{O}_L = \mathcal{O}_{L_0}[\pi]$ (Theorem 7.27); for $\sigma \in G_0$, which fixes $\mathcal{O}_{L_0}$, $\sigma(\sum c_j\pi^j) - \sum c_j\pi^j = \sum c_j(\sigma(\pi)^j - \pi^j)$ is divisible by $\sigma(\pi) - \pi$, so $\min_x v_L(\sigma x - x) = v_L(\sigma\pi - \pi)$. The last assertion is immediate from the definition. (ii) For $\sigma \in G_0$, $\sigma(\pi)$ is a uniformiser, so $\sigma(\pi)/\pi$ is a unit. If $\pi' = u\pi$, then $\sigma(\pi')/\pi' = (\sigma(\pi)/\pi)(\sigma(u)/u)$ and $\sigma(u) \equiv u \pmod{\mathfrak{m}_L}$, so $\theta_0$ does not depend on $\pi$. Then $\sigma\tau(\pi)/\pi = \big(\sigma(\tau\pi)/\tau\pi\big)\big(\tau\pi/\pi\big)$, and the first factor is $\theta_0(\sigma)$ computed with the uniformiser $\tau\pi$; so $\theta_0$ is a homomorphism. Its kernel is $\{v_L(\sigma\pi - \pi) \geq 2\} = G_1$ by (i). As $k_L^\times$ is cyclic of order $\#k_L - 1$, prime to $p$, so is $G_0/G_1$. (iii) For $\sigma \in G_i$ $(i \geq 1)$ write $\sigma(\pi) = \pi(1 + a_\sigma)$, $v_L(a_\sigma) \geq i$. First, $\sigma \in G_i$ and $x \in \mathfrak{m}_L^j$ give $v_L(\sigma x - x) \geq i + j$: write $x = \pi^j y$, so $\sigma x - x = \sigma(\pi^j)(\sigma y - y) + y(\sigma(\pi^j) - \pi^j)$, with $v_L(\sigma y - y) \geq i + 1$ and $\sigma(\pi^j) - \pi^j = \pi^j((1 + a_\sigma)^j - 1)$ of valuation $\geq i + j$. Now for $\sigma, \tau \in G_i$, $1 + a_{\sigma\tau} = \frac{\sigma(\tau\pi)}{\tau\pi} \cdot \frac{\tau\pi}{\pi}$ and $\tau\pi = \pi u$ with $u = 1 + a_\tau$; then $\sigma(\tau\pi)/\tau\pi = (1 + a_\sigma)\sigma(u)/u$ and $v_L(\sigma(u) - u) = v_L(\sigma a_\tau - a_\tau) \geq 2i \geq i + 1$. Hence $1 + a_{\sigma\tau} \equiv (1 + a_\sigma)(1 + a_\tau)$ and $a_{\sigma\tau} \equiv a_\sigma + a_\tau \pmod{\mathfrak{m}_L^{i+1}}$. The kernel is $\{v_L(a_\sigma) \geq i + 1\} = G_{i+1}$ by (i). The quotient $\mathfrak{m}_L^i/\mathfrak{m}_L^{i+1}$ is a one-dimensional $k_L$-vector space, of exponent $p$; so each $G_i/G_{i+1}$ $(i \geq 1)$ is an elementary abelian $p$-group, and $G_1$ is a $p$-group. It is normal in $G_0$ with quotient of order prime to $p$ by (ii), so it

is the unique Sylow $p$-subgroup. If $p \nmid e = \#G_0$ then $G_1 = 1$; conversely, if $G_1 = 1$ then $\#G_0$ divides $\#k_L^\times$, prime to $p$. □

**Example 7.43** (ramification groups of quadratic extensions)**.** Let $L/K$ be a ramified quadratic extension with $\mathcal{O}_L = \mathcal{O}_K[\pi]$, $G = \{1, \sigma\}$. By Proposition 7.42(i), $\sigma \in G_i$ iff $v_L(\sigma\pi - \pi) \geq i + 1$. In (a) to (c), $K = \mathbb{Q}_2$ and $e = 2$, so $v_L = 2v_2$ on $\mathbb{Q}_2$; in particular $v_L(2) = 2$, and $v_L(\sqrt{d}) = \frac{1}{2}v_L(d) = v_2(d)$: $\sqrt{d}$ is a unit if $d$ is odd and a uniformiser if $v_2(d) = 1$.

(a) $\mathbb{Q}_2(i)$: $\pi = 1 + i$ (a root of the Eisenstein polynomial $(X-1)^2 + 1 = X^2 - 2X + 2$), $\sigma\pi - \pi = (1 - i) - (1 + i) = -2i$ has $v_L = v_L(2) = 2$, $i$ being a unit; so $G_0 = G_1 = G$, $G_2 = 1$.

(b) $\mathbb{Q}_2(\sqrt{3}) = \mathbb{Q}_2(\sqrt{-5})$ (since $-5/3 = -15/9$ and $-15 \equiv 1 \pmod 8$ is a square in $\mathbb{Q}_2$, Example 7.16(d)): $\pi = 1 + \sqrt{3}$ (root of $(X-1)^2 - 3 = X^2 - 2X - 2$), $\sigma\pi - \pi = -2\sqrt{3}$ has $v_L = 2$; again $G_1 = G$, $G_2 = 1$.

(c) $\mathbb{Q}_2(\sqrt{d})$ for $d \in \{\pm 2, \pm 10\}$: $\pi = \sqrt{d}$ (root of the Eisenstein polynomial $X^2 - d$, as $v_2(d) = 1$), $\sigma\pi - \pi = -2\sqrt{d}$ has $v_L = 2 + 1 = 3$; so $G_0 = G_1 = G_2 = G$, $G_3 = 1$.

(d) $\mathbb{Q}_p(\sqrt{p^*})$, $p$ odd, and $\mathbb{Q}_3(\sqrt{-3})$: $e = 2$ is prime to $p$, so $G_1 = 1$. Likewise $\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$ ($e = p - 1$) is tame.

These are the wildly ramified quadratic extensions of $\mathbb{Q}_2$ (Proposition 7.60(iii)); their conductor exponents are computed in Theorem 7.82(iv).

*Remark* 7.44 (the tame relation of Chapter 6). Let $L/F$ be a finite Galois extension of number fields and $\mathfrak{P} \mid \mathfrak{p}$. By Proposition 7.32(i), $I_\mathfrak{P}$ is the inertia group $G_0$ of $L_\mathfrak{P}/F_\mathfrak{p}$, and the map $\tau \mapsto \tau(\varpi)/\varpi \bmod \mathfrak{P}$ used in Chapter 6 is $\theta_0$ (a uniformiser of the localisation $\mathcal{O}_{L,\mathfrak{P}}$ is one of the completion). By Proposition 7.42(ii), (iii), its kernel is $G_1$, a $p$-group. This is the step that Chapter 6 left unproved; so the tame relation $F\tau F^{-1} = \tau^{N\mathfrak{p}}$ for tamely ramified primes is now proved in full.

### 7.5.2 The inertia group of $G_K$ and the tame quotient

**Theorem 7.45** (the filtration $G_K \supset I_K \supset P_K$)**.** *(i) $I_K$ is a closed normal subgroup of $G_K$ with $G_K/I_K \simeq \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \simeq \hat{\mathbb{Z}}$; for every finite Galois $L/K$, the image of $I_K$ in $\mathrm{Gal}(L/K)$ is $G_0 = I(L/K)$.*

*(ii) Let $\varpi$ be a uniformiser of $K$ and fix a compatible system of roots $\varpi^{1/m} \in \overline{\mathbb{Q}}_p$ ($p \nmid m$). A finite extension $L/K$ is tamely ramified iff $L \subseteq K^{\mathrm{ur}}(\varpi^{1/m})$ for some $m$ prime to $p$. Hence the union $K^{\mathrm{tame}}$ of all finite tamely ramified extensions of $K$ in $\overline{\mathbb{Q}}_p$ is the field $\bigcup_{p\nmid m} K^{\mathrm{ur}}(\varpi^{1/m})$, Galois over $K$.*

*(iii) $P_K = \mathrm{Gal}(\overline{\mathbb{Q}}_p/K^{\mathrm{tame}})$ is a closed normal subgroup of $G_K$ contained in $I_K$; its image in $\mathrm{Gal}(L/K)$ is $G_1(L/K)$ for every finite Galois $L/K$; it is a pro-$p$ group, and every pro-$p$ subgroup of $I_K$ is contained in it (it is the unique pro-$p$ Sylow subgroup of $I_K$).*

*(iv) (Stated.) The upper numbering of Proposition 7.42(iv) is compatible with quotients, so it defines a decreasing filtration $(G_K^u)_{u \geq -1}$ by closed normal subgroups, with $G_K^{-1} = G_K$, $G_K^0 = I_K$ and $P_K$ the closure of $\bigcup_{u>0} G_K^u$. We do not prove this; see Serre [174, Chapter IV, Section 3].*

*Proof.* (i) $I_K$ is the kernel of the continuous restriction map $G_K \to \mathrm{Gal}(K^{\mathrm{ur}}/K) \simeq \hat{\mathbb{Z}}$ (Proposition 7.36(i)). Its image in $\mathrm{Gal}(L/K)$ is $\mathrm{Gal}(L/L \cap K^{\mathrm{ur}})$ (Chapter 5); $L \cap K^{\mathrm{ur}}$ is unramified and contains every unramified subextension, so it is $L_0$, and $\mathrm{Gal}(L/L_0) = G_0$ by Proposition 7.42(i). (ii) Suppose $p \nmid e(L/K)$. $L/L_0$ is totally and tamely ramified of degree $e = e(L/K)$, so $L = L_0(\alpha)$ with $\alpha^e = \varpi_0$ a uniformiser of $L_0$ (Proposition 7.33). Write $\varpi_0 = \varpi u$ with $u \in \mathcal{O}_{L_0}^\times \subseteq \mathcal{O}_{K^{\mathrm{ur}}}^\times$; then $u = \beta^e$ with $\beta \in K^{\mathrm{ur}}$ (Proposition 7.36(iv)), so $(\alpha/\beta)^e = \varpi$ and $\alpha/\beta = \zeta\varpi^{1/e}$ with $\zeta \in \mu_e \subset K^{\mathrm{ur}}$. Hence $L \subseteq K^{\mathrm{ur}}(\varpi^{1/e})$. Conversely, if $L \subseteq K^{\mathrm{ur}}(\varpi^{1/m})$ then $L \subseteq M = K_f(\varpi^{1/m})$ for some $f$; $X^m - \varpi$ is Eisenstein over $K_f$, so $e(M/K) = m$ is prime to $p$, and $e(L/K) \mid e(M/K)$. The fields $K^{\mathrm{ur}}(\varpi^{1/m})$ form a directed family ($K^{\mathrm{ur}}(\varpi^{1/m})$ and $K^{\mathrm{ur}}(\varpi^{1/m'})$ lie in $K^{\mathrm{ur}}(\varpi^{1/mm'})$), so their union is a field, equal to $K^{\mathrm{tame}}$; it is normal over $K$ because the conjugates of $\varpi^{1/m}$ are the $\zeta\varpi^{1/m}$, $\zeta \in \mu_m \subset K^{\mathrm{ur}}$. (iii) $P_K$ is closed and normal since $K^{\mathrm{tame}}/K$ is Galois, and $P_K \subseteq I_K$ since $K^{\mathrm{ur}} \subseteq K^{\mathrm{tame}}$. Its image in $\mathrm{Gal}(L/K)$ is $\mathrm{Gal}(L/L_t)$ with $L_t = L \cap K^{\mathrm{tame}}$, the largest tamely ramified subextension. Let $L_1 = L^{G_1}$. The inertia group of $L_1/K$ is the image $G_0/G_1$ of $G_0$ (by (i), applied to $L$ and to $L_1$), of order prime to $p$; so $e(L_1/K)$ is prime to $p$ and $L_1 \subseteq L_t$. Conversely the image of the $p$-group $G_1$ in $\mathrm{Gal}(L_t/K)$ lies in the inertia group of $L_t/K$, of order $e(L_t/K)$ prime to $p$; so it is trivial, and $L_t \subseteq L_1$. Hence the image of $P_K$ is $G_1$, a $p$-group, and $P_K = \varprojlim_L G_1(L/K)$ is pro-$p$. If $Q \subseteq I_K$ is pro-$p$, its image in each $\mathrm{Gal}(L/K)$ is a $p$-subgroup of $G_0$, hence lies in the unique Sylow $p$-subgroup $G_1$, the image of $P_K$; as $P_K$ is closed, $Q \subseteq P_K$. □

**Corollary 7.46** ($G_K$ is a semidirect product)**.** *Let* $\mathrm{Frob} \in G_K$ *be an arithmetic Frobenius lift and* $Z$ *the closed subgroup it generates. Then* $Z \simeq \hat{\mathbb{Z}}$, $Z \cap I_K = 1$ *and* $G_K = I_K \rtimes Z$. *Consequently a continuous character* $\psi$ *of* $I_K$ *with* $\psi(\mathrm{Frob}\,\sigma\,\mathrm{Frob}^{-1}) = \psi(\sigma)$ *for all* $\sigma \in I_K$ *extends to a continuous character* $\chi$ *of* $G_K$, *by* $\chi(\sigma z) = \psi(\sigma)$ *(*$\sigma \in I_K$, $z \in Z$*).*

*Proof.* $Z$ is procyclic, so $n \mapsto \mathrm{Frob}^n$ extends to a continuous surjection $\hat{\mathbb{Z}} \to Z$ (each finite quotient of $Z$ is cyclic, generated by the image of Frob). The composite $\hat{\mathbb{Z}} \to Z \to G_K/I_K \simeq \hat{\mathbb{Z}}$ sends 1 to 1, hence is the identity; so $\hat{\mathbb{Z}} \to Z$ is injective, $Z \simeq \hat{\mathbb{Z}}$, $Z \cap I_K = 1$, and $Z$ maps onto $G_K/I_K$, so $G_K = I_K Z$. The map $I_K \times Z \to G_K$ is a continuous bijection of compact spaces, hence a homeomorphism. For the extension, $(\sigma z)(\sigma' z') = \sigma(z\sigma' z^{-1})zz'$ and $\psi(z\sigma' z^{-1}) = \psi(\sigma')$ ($z$ is a limit of powers of Frob and $\psi$ is continuous), so $\chi$ is a homomorphism, continuous by the homeomorphism. □

**Theorem 7.47** (the tame character)**.** *Let* $m$ *be prime to* $p$ *and* $\varpi$ *a uniformiser of* $K$.

*(i)* $\theta_m \colon I_K \to \mu_m$, $\sigma \mapsto \sigma(\varpi^{1/m})/\varpi^{1/m}$, *is a continuous surjective homomorphism. It does not depend on the root* $\varpi^{1/m}$ *or on the uniformiser* $\varpi$, *and it does not change when* $K$ *is replaced by a finite unramified extension.*

*(ii) For compatible roots,* $\theta_{mm'}^{m'} = \theta_m$. *The limit* $\theta \colon I_K \to \varprojlim_{p\nmid m} \mu_m = \prod_{\ell \neq p} \mathbb{Z}_\ell(1)$ *is surjective with kernel* $P_K$*:*

$$I_K/P_K \;\simeq\; \prod_{\ell \neq p} \mathbb{Z}_\ell(1).$$

*This group is abelian and procyclic, and each of its finite quotients has order prime to* $p$.

*(iii) For a prime* $\ell \neq p$, *the* tame character $t_\ell \colon I_K \to \mathbb{Z}_\ell(1)$, $\sigma \mapsto \left(\sigma(\varpi^{1/\ell^k})/\varpi^{1/\ell^k}\right)_k$, *is the* $\ell$*-component of* $\theta$. *It is surjective, every finite quotient of its kernel has order prime to* $\ell$, *and* $t_\ell$ *identifies the maximal pro-*$\ell$ *quotient* $I_K^{(\ell)}$ *of* $I_K$ *with* $\mathbb{Z}_\ell(1)$. *A topological generator* $t$ *of* $I_K^{(\ell)}$ *(or of* $I_K/P_K$*) is called a* generator of tame inertia.

(iv) *Every continuous homomorphism from $I_K$ to a pro-$\ell$ group (for instance to $1+\ell\,\mathrm{M}_n(\mathbb{Z}_\ell)$, or to a group of unipotent matrices over $\mathbb{Z}_\ell$) factors through $t_\ell$.*

(v) *If $L/K$ is finite with ramification index $e$, then $I_L \subseteq I_K$ and $\theta_{m,K}|_{I_L} = \theta_{m,L}^{e}$; so $t_{\ell,K}|_{I_L} = e\,t_{\ell,L}$, and the two have the same kernel.*

*Proof.* (i) $\mu_m \subset K^{\mathrm{ur}}$ is fixed by $I_K$. So for another root $\zeta\varpi^{1/m}$ the quotient is unchanged, and $\theta_m(\sigma\tau) = \sigma(\theta_m(\tau)\varpi^{1/m})/\varpi^{1/m} = \theta_m(\tau)\theta_m(\sigma)$. If $\varpi' = u\varpi$ with $u \in \mathcal{O}_K^\times$, then $u^{1/m}$ can be chosen in $K^{\mathrm{ur}}$ (Proposition 7.36(iv)), where it is fixed by $I_K$, and $(\varpi')^{1/m} = \zeta u^{1/m}\varpi^{1/m}$ gives the same character. A finite unramified extension $K'$ has $I_{K'} = I_K$ and $\varpi$ as a uniformiser (Proposition 7.38). $\theta_m$ factors through $\mathrm{Gal}(K^{\mathrm{ur}}(\varpi^{1/m})/K^{\mathrm{ur}})$, so it is continuous. Since $X^m - \varpi$ is Eisenstein over every $K_f$, it is irreducible over $K^{\mathrm{ur}}$, so that Galois group has order $m$, and it maps injectively to $\mu_m$ by $\theta_m$; as $I_K$ maps onto it, $\theta_m$ is surjective. (ii) $(\varpi^{1/mm'})^{m'} = \varpi^{1/m}$ gives the compatibility. The kernel of $\theta$ is the subgroup fixing every $\varpi^{1/m}$, that is $\mathrm{Gal}(\overline{\mathbb{Q}}_p/K^{\mathrm{tame}}) = P_K$ by Theorem 7.45(ii). The image is closed (compact) and surjects onto every $\mu_m$, so it is everything. By the Chinese remainder theorem $\mu_m = \prod_\ell \mu_{\ell^{v_\ell(m)}}$, so the limit is $\prod_{\ell\neq p}\mathbb{Z}_\ell(1) \simeq \prod_{\ell\neq p}\mathbb{Z}_\ell$, which is procyclic. An open subgroup contains the kernel of some projection to $\mu_m$, so finite quotients are quotients of some $\mu_m$, of order prime to $p$. (iii) Surjectivity is (ii). The kernel of $t_\ell$ is an extension of $\prod_{\ell'\neq\ell,p}\mathbb{Z}_{\ell'}(1)$ by $P_K$; a finite quotient of it is an extension of a quotient of the former (order prime to $\ell$) by a quotient of $P_K$ (a $p$-group), so it has order prime to $\ell$. If $f\colon I_K \to H$ is a continuous homomorphism to a pro-$\ell$ group, the image of $\ker t_\ell$ in each finite quotient of $H$ is an $\ell$-group of order prime to $\ell$, hence trivial; so $f$ kills $\ker t_\ell$ and factors through $I_K/\ker t_\ell \simeq \mathbb{Z}_\ell(1)$, which is itself pro-$\ell$. This proves (iii) and (iv); $1+\ell\,\mathrm{M}_n(\mathbb{Z}_\ell)$ is pro-$\ell$ because its quotients by $1+\ell^k\,\mathrm{M}_n(\mathbb{Z}_\ell)$ have order a power of $\ell$ (Chapter 1). (v) $L^{\mathrm{ur}} = LK^{\mathrm{ur}} \supseteq K^{\mathrm{ur}}$ (as in Proposition 7.38), so $I_L \subseteq I_K$. Write $\varpi_K = \varpi_L^e u$ with $u \in \mathcal{O}_L^\times$ and choose $u^{1/m} \in L^{\mathrm{ur}}$; then $\varpi_K^{1/m} = \zeta(\varpi_L^{1/m})^e u^{1/m}$ for some $\zeta \in \mu_m$, and $\sigma \in I_L$ fixes $\zeta$ and $u^{1/m}$. Since $\mathbb{Z}_\ell(1)$ is torsion free, $e\,t_{\ell,L}$ and $t_{\ell,L}$ have the same kernel. □

**Corollary 7.48** (open subgroups of inertia)**.** *Every open subgroup $U$ of $I_K$ is $I_L$ for some finite extension $L/K$. Its maximal pro-$\ell$ quotient ($\ell \neq p$) is $\mathbb{Z}_\ell(1)$, through $t_{\ell,L}$; equivalently, every continuous homomorphism from $U$ to a pro-$\ell$ group factors through $t_{\ell,K}|_U$.*

*Proof.* The fixed field $F$ of $U$ is a finite extension of $K^{\mathrm{ur}}$ (Chapter 5), generated by finitely many elements, each algebraic over $K$; let $L$ be the extension of $K$ they generate. Then $L^{\mathrm{ur}} = LK^{\mathrm{ur}} = F$ (Theorem 7.47(v)), so $I_L = \mathrm{Gal}(\overline{\mathbb{Q}}_p/F) = U$. Now apply Theorem 7.47(iii), (iv) to $L$ and (v). □

**Corollary 7.49** (a Frobenius fixing the roots of a uniformiser)**.** *Let $\varpi$ be a uniformiser of $K$, $(\varpi^{1/m})_{p\nmid m}$ a compatible system of roots and $K_\varpi = K(\varpi^{1/m} : p \nmid m)$. There is an arithmetic Frobenius lift* $\mathrm{Frob} \in G_K$ *fixing $K_\varpi$; two such lifts differ by an element of $P_K$.*

*Proof.* Each $K(\varpi^{1/m})$ is totally ramified over $K$ (Eisenstein), so $K_\varpi \cap K^{\mathrm{ur}} = K$. By the translation theorem (Chapter 5), restriction $G_{K_\varpi} \to \mathrm{Gal}(K^{\mathrm{ur}}/K)$ is surjective, and any preimage of the Frobenius generator is the required lift. Two lifts differ by an element of $G_{K_\varpi} \cap I_K$, which fixes every $\varpi^{1/m}$, so has $\theta = 1$ and lies in $P_K$. □

**Theorem 7.50** (the tame relation)**.** *(i) For $\tau \in G_K$ and $\sigma \in I_K$: $\theta_m(\tau\sigma\tau^{-1}) = \tau(\theta_m(\sigma))$, and $t_\ell(\tau\sigma\tau^{-1}) = \chi_\ell(\tau)\,t_\ell(\sigma)$ (in the additive notation for $\mathbb{Z}_\ell(1)$).*

*(ii) For every arithmetic Frobenius lift* $\mathrm{Frob} \in G_K$ *and every* $t \in I_K/P_K$,

$$\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^{q}, \qquad \textit{equivalently} \qquad \mathrm{Frob}^{-1}\; t\; \mathrm{Frob} = t^{q^{-1}},$$

*where* $q^{-1}$ *is the inverse of* $q$ *in the ring* $\prod_{\ell \neq p} \mathbb{Z}_\ell$ *of exponents of the procyclic group* $I_K/P_K$. *For the geometric Frobenius,* $\Phi\, t\, \Phi^{-1} = t^{q^{-1}}$. *The same relations hold in* $I_K^{(\ell)} \simeq \mathbb{Z}_\ell(1)$, *with* $q^{-1} \in \mathbb{Z}_\ell$.

*(iii) Let* $\mathrm{Frob} \in G_K$ *be any arithmetic Frobenius lift. Then* $\mathrm{Gal}(K^{\mathrm{tame}}/K) = (I_K/P_K) \rtimes \overline{\langle \mathrm{Frob} \rangle} \simeq \left(\prod_{\ell \neq p} \mathbb{Z}_\ell(1)\right) \rtimes \hat{\mathbb{Z}}$, *the generator* $1 \in \hat{\mathbb{Z}}$ *acting by multiplication by* $q$. *(Stated.) As a profinite group it is generated by* $t$ *and* Frob *subject to the single relation* $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^{q}$; *see [143, Chapter VII, Section 5].*

*Proof.* (i) Put $\alpha = \varpi^{1/m}$. Since $\tau(\alpha)$ is a root of $X^m - \varpi$, $\tau(\alpha) = \zeta_\tau^{-1}\alpha$ for some $\zeta_\tau \in \mu_m$, i.e. $\tau^{-1}(\alpha) = \tau^{-1}(\zeta_\tau)\alpha$. As $\sigma$ fixes $\mu_m \subset K^{\mathrm{ur}}$,

$$\tau\sigma\tau^{-1}(\alpha) = \tau\sigma\big(\tau^{-1}(\zeta_\tau)\alpha\big) = \tau\big(\tau^{-1}(\zeta_\tau)\theta_m(\sigma)\alpha\big) = \zeta_\tau\, \tau(\theta_m(\sigma))\, \zeta_\tau^{-1}\alpha = \tau(\theta_m(\sigma))\, \alpha.$$

Passing to the limit over $m = \ell^k$ gives the formula for $t_\ell$, since $\tau$ acts on $\mathbb{Z}_\ell(1)$ through $\chi_\ell$. (ii) Frob acts on $\mu_m$ by $\zeta \mapsto \zeta^q$ (Convention 7.37(a)), so $\theta(\mathrm{Frob}\; \sigma\; \mathrm{Frob}^{-1}) = \theta(\sigma)^q$; and $\mathrm{Frob}^{-1}$ acts by the inverse map $\zeta \mapsto \zeta^{q'}$, $qq' \equiv 1 \pmod m$, so $\theta(\mathrm{Frob}^{-1}\, \sigma\, \mathrm{Frob}) = \theta(\sigma)^{q^{-1}}$. As $\theta$ identifies $I_K/P_K$ with $\prod \mathbb{Z}_\ell(1)$, these are the stated relations; $\Phi = \mathrm{Frob}^{-1}$ gives the last one, and projecting to the $\ell$-component gives the relations in $I_K^{(\ell)}$. (iii) The proof of Corollary 7.46 applies verbatim to the profinite group $\mathrm{Gal}(K^{\mathrm{tame}}/K) = G_K/P_K$, its closed normal subgroup $\mathrm{Gal}(K^{\mathrm{tame}}/K^{\mathrm{ur}}) = I_K/P_K$ (with quotient $G_K/I_K \simeq \hat{\mathbb{Z}}$) and the image of Frob: the closure of $\langle \mathrm{Frob} \rangle$ in $G_K/P_K$ is isomorphic to $\hat{\mathbb{Z}}$, meets $I_K/P_K$ trivially, and $\mathrm{Gal}(K^{\mathrm{tame}}/K)$ is the semidirect product of $I_K/P_K$ by it; the action is (ii). □

*Warning* 7.51 (the sign in the tame relation). With the arithmetic Frobenius, conjugation by Frob raises tame inertia to the power $q$; with the geometric one, $\Phi\, t\, \Phi^{-1} = t^{q^{-1}}$. The paper writes "$\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$" in the proof of Lemma lem:tamfactor, with Frob arithmetic: this is the second form in Theorem 7.50(ii), and it is correct. Every $q$ prime to $m \in \{3, 4, 6\}$ satisfies $q \equiv \pm 1 \pmod m$, hence $q^{-1} \equiv q \pmod m$; so the distinction is invisible on inertia images of order 3, 4 or 6 (Corollary 7.52), but not on $\mathbb{Z}_\ell(1)$, where it decides whether Frob acts on $H^1(I_q, \cdot)$ through $q$ or through $q^{-1}$.

**Corollary 7.52** (finite tame images of inertia)**.** *Let* $\rho\colon G_K \to H$ *be a continuous homomorphism to a Hausdorff topological group with* $\rho(I_K)$ *finite of order prime to* $p$ *(for example* $\rho\colon G_K \to \mathrm{GL}_n(\mathbb{Q}_\ell)$ *continuous with such an image of inertia). Then* $\rho(P_K) = 1$; $\rho|_{I_K}$ *factors through* $I_K/P_K$, *so* $\rho(I_K)$ *is cyclic, generated by* $\rho(t)$ *for any generator* $t$ *of* $I_K/P_K$; *and for every arithmetic Frobenius lift and every* $h \in \rho(I_K)$,

$$\rho(\mathrm{Frob})\, h\, \rho(\mathrm{Frob})^{-1} = h^{q}.$$

*If* $\rho(I_K) = \langle g \rangle$ *has order* $m$, *then* $\rho(\mathrm{Frob})$ *commutes with* $g$ *iff* $q \equiv 1 \pmod m$, *and it conjugates* $g$ *to* $g^{-1}$ *iff* $q \equiv -1 \pmod m$.

*Proof.* The kernel of $\rho|_{I_K}$ is closed of finite index, hence open, so $N = \ker \rho|_{I_K}$ is an open normal subgroup of $I_K$. The image of $P_K$ in the finite group $I_K/N$ is a quotient of $P_K$ by an

open subgroup, a $p$-group (Theorem 7.45(iii)); its order divides $\#\rho(I_K)$, which is prime to $p$; so $\rho(P_K) = 1$. Then $\rho(I_K)$ is a quotient of the procyclic $I_K/P_K$, and the image of the dense subgroup $\langle t\rangle$ is a dense subgroup of a finite discrete group, hence everything. For $h = \rho(\sigma)$, $\rho(\mathrm{Frob})h\rho(\mathrm{Frob})^{-1} = \rho(\mathrm{Frob}\ \sigma\ \mathrm{Frob}^{-1}) = \rho(\sigma^q) = h^q$ by Theorem 7.50(ii) (the relation holds modulo $P_K$, which $\rho$ kills). Another lift is $\mathrm{Frob}\,\sigma'$ with $\sigma' \in I_K$, and $\rho(\sigma')$ commutes with $h$, $\rho(I_K)$ being abelian. Finally $g^q = g$ iff $m \mid q-1$, and $g^q = g^{-1}$ iff $m \mid q+1$. □

**Proposition 7.53** (the image of wild inertia)**.** *Let $\ell \neq p$, $E_\lambda$ a finite extension of $\mathbb{Q}_\ell$ with ring of integers $\mathcal{O}$ and uniformiser $\varpi_\lambda$, and $\rho\colon G_K \to \mathrm{GL}_n(\mathcal{O})$ continuous. Then $\rho(P_K)$ is a finite $p$-group, mapping injectively to $\mathrm{GL}_n(\mathcal{O}/\varpi_\lambda)$. Moreover:*

*(i) $I_1 = I_K \cap \rho^{-1}(1 + \ell^2\,\mathrm{M}_n(\mathcal{O}))$ is an open subgroup of $I_K$, normal in $G_K$, and $\rho|_{I_1}$ factors through $t_\ell|_{I_1}$;*

*(ii) $\rho$ is tamely ramified (that is, $\rho(P_K) = 1$) iff $\rho(I_K)$ contains no element of order $p$; in particular if $\rho(I_K)$ is finite of order prime to $p$.*

*Proof.* $\Gamma_1 = 1 + \varpi_\lambda\,\mathrm{M}_n(\mathcal{O})$ is an open normal pro-$\ell$ subgroup of $\mathrm{GL}_n(\mathcal{O})$: its quotients by $1 + \varpi_\lambda^k\,\mathrm{M}_n(\mathcal{O})$ are successive extensions of copies of the additive group $\mathrm{M}_n(\mathcal{O}/\varpi_\lambda)$. The group $P' = P_K \cap \rho^{-1}(\Gamma_1)$ is open in $P_K$, hence pro-$p$, and $\rho(P')$ is a compact subgroup of $\Gamma_1$, so pro-$\ell$, and a continuous image of a pro-$p$ group, so pro-$p$ (its finite quotients are quotients of finite quotients of $P'$). Hence $\rho(P') = 1$, $\rho(P_K) \cap \Gamma_1 = 1$, and $\rho(P_K)$ embeds in $\mathrm{GL}_n(\mathcal{O}/\varpi_\lambda)$; being a finite continuous image of a pro-$p$ group, it is a $p$-group. (i) $1 + \ell^2\,\mathrm{M}_n(\mathcal{O})$ is open, normal and pro-$\ell$, so $I_1$ is open in $I_K$ and normal in $G_K$, and $I_1 = I_L$ for a finite $L/K$ (Corollary 7.48); $\rho|_{I_1}$ is a homomorphism to a pro-$\ell$ group, so it factors through $t_{\ell,K}|_{I_1}$ by Corollary 7.48. (ii) If $\rho(P_K) \neq 1$ it is a nontrivial $p$-group, which contains an element of order $p$. Conversely, if $\rho(P_K) = 1$ then $\rho(I_K)$ is a continuous image of $I_K/P_K$, a profinite abelian group all of whose finite quotients have order prime to $p$; so is $\rho(I_K)$, and an element of order $p$ would generate a finite subgroup of order $p$, whose image in a suitable finite quotient of $\rho(I_K)$ is nontrivial. □

*Remark* 7.54. Nothing in Proposition 7.53 forbids wild ramification for a general $\rho$ at $p \geq 5$: $\mathrm{GL}_2(\mathbb{Z}_\ell)$ contains elements of order 5 when $\ell \equiv 1 \pmod 5$. For $\rho = \rho_{E,\ell}$ with potentially good reduction, the characteristic polynomials of the elements of $\rho(I_K)$ have integer coefficients independent of $\ell$, so their orders lie in $\{1,2,3,4,6\}$, and the image is tame for $p \geq 5$ (Serre and Tate [179, Section 2]; Chapter 10). At $p = 2, 3$ the image can be wild: $Q_8$ or $\mathrm{SL}_2(\mathbb{F}_3)$ at $p = 2$, cyclic of order 3 or 6 at $p = 3$; an image of order 3 is wild at $p = 3$, because $I_K/P_K$ has no quotient of order 3, and tame at $p = 2$. These are the facts behind Remark rem:wild.

**Example 7.55** (the tame relation at a finite level)**.** Let $\ell \neq q$ be primes, $m = \ell^k$, and $L = \mathbb{Q}_q(\mu_m, q^{1/m})$. Fix a primitive $\zeta \in \mu_m$. An element $\sigma \in \mathrm{Gal}(L/\mathbb{Q}_q)$ is determined by $(a,b)$ with $\sigma(\zeta) = \zeta^a$, $\sigma(q^{1/m}) = \zeta^b q^{1/m}$, and $\sigma_{a,b}\sigma_{a',b'} = \sigma_{aa',ab'+b}$, because $\sigma_{a,b}(\zeta^{b'}q^{1/m}) = \zeta^{ab'}\zeta^b q^{1/m}$: the group is a subgroup of the affine group of $\mathbb{Z}/m$ (Chapter 5). Since $\mathbb{Q}_q(\mu_m)$ is unramified with group generated by $\zeta \mapsto \zeta^q$, the values of $a$ are exactly the powers of $q$; and $X^m - q$ is Eisenstein over $\mathbb{Q}_q(\mu_m)$, hence irreducible, so each automorphism of $\mathbb{Q}_q(\mu_m)$ has $m$ extensions to $L$, one for each value of $b$. So the group is $\{(a,b) : a \in \langle q\rangle \subseteq (\mathbb{Z}/m)^\times,\ b \in \mathbb{Z}/m\}$, of order $m \cdot \#\langle q\rangle$. The inertia group fixes the unramified field $\mathbb{Q}_q(\mu_m)$, so it is contained in $\{a = 1\}$, and it has order $e = m$ (Proposition 7.42(i)); so it is $\{a = 1\}$, generated by $t = \sigma_{1,1}$, and $t_\ell \bmod m$ is $\sigma_{1,b} \mapsto \zeta^b$ (as $\sigma_{1,b}^n = \sigma_{1,nb}$). The element $\mathrm{Frob} = \sigma_{q,0}$ is a Frobenius fixing $q^{1/m}$. Then

$$\mathrm{Frob}\ t\ \mathrm{Frob}^{-1} = \sigma_{q,q}\sigma_{q',0} = \sigma_{1,q} = t^q, \qquad \mathrm{Frob}^{-1}\ t\ \mathrm{Frob} = \sigma_{q',q'}\sigma_{q,0} = \sigma_{1,q'} = t^{q'},$$

where $qq' \equiv 1 \pmod m$; here $\mathrm{Frob}^{-1} = \sigma_{q',0}$, since $\sigma_{q,0}\sigma_{q',0} = \sigma_{1,0}$, and the first product is $\mathrm{Frob}\, t = \sigma_{q,0}\sigma_{1,1} = \sigma_{q,q}$ followed by $\sigma_{q,q}\sigma_{q',0} = \sigma_{qq',q\cdot 0+q} = \sigma_{1,q}$. For $(q,\ell,k) = (7,3,2)$ these are $t^7$ and $t^4$; for $(13,3,2)$, $t^4$ and $t^7$; for $(2,3,2)$, $t^2$ and $t^5$ (all checked by composing the pairs).

*In the paper* 7.56. In the proof of Lemma lem:tamfactor: "The inertia group $I = I_q$ acts through its maximal pro-$\ell$ tame quotient, generated by an element $t$" is Theorem 7.47(iii), (iv), since $I_q$ acts on $T_\ell E$ through unipotent matrices over $\mathbb{Z}_\ell$ (a pro-$\ell$ group); "the kernel of $I \to \langle t\rangle$ has trivial pro-$\ell$ quotient" is the statement that every finite quotient of $\ker t_\ell$ has order prime to $\ell$; and "$\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$" is Theorem 7.50(ii) for the arithmetic Frobenius. In the proof of Lemma lem:IV, "The order 3 is prime to $q$, so the image is a quotient of the tame inertia group", and in the proof of Lemma lem:euler, "Frobenius centralises $\sigma$ if $q \equiv 1 \pmod 3$ and inverts it if $q \equiv 2$", are Corollary 7.52, which needs only $q \nmid \#\rho(I_q)$, not $q \geq 5$. In Proposition prop:shapes(ii), the extension of the $G_{\mathbb{Q}_q}$-invariant quadratic character $\psi$ of $I_q$ to $G_{\mathbb{Q}_q} = I_q \rtimes \langle\phi\rangle$ is Corollary 7.46. In Lemma lem:tamadd, the fact that the maximal pro-$\ell$ abelian quotient of the open subgroup $K \subseteq I_q$ is $\mathbb{Z}_\ell(1)$ is Corollary 7.48.

## 7.6 Kummer theory over local fields; quadratic characters

The Tate curve makes the Kummer extension of a single element $q_E \in \mathbb{Q}_q^\times$ appear inside $T_\ell E$, and the paper reads off the action of inertia and Frobenius on $T_\ell E$ from the Kummer cocycle of $q_E$. This section computes that cocycle on inertia ($v_K(x)$ times the tame character) and on Frobenius (the class of the reduction of the unit part), with all the choices made explicit. It then describes the quadratic characters of $G_{\mathbb{Q}_p}$ and the local behaviour of the global characters $\chi_d$, and states local class field theory in the form needed for conductors. The paper uses this in Lemmas lem:tamagawa-exp and lem:tamfactor (Kummer classes), in Lemma lem:twist and Theorem thm:twist (the twist by $\chi_{q^*}$ is unramified outside $q$ and tame at odd $q$), in Lemma lem:euler and Proposition prop:shapes ($V = V' \otimes \chi$ locally) and in Theorem thm:qual. The cohomological statements (Hilbert 90, $H^1(K,\mathbb{Z}_\ell(1)) = \widehat{K^\times}$) are proved in Chapter 17; here we prove the field-theoretic half.

**Proposition 7.57** (Kummer theory over $K$)**.** *Let $\ell \neq p$ and $x \in K^\times$.*

*(i) (Chapter 5, recalled.) For a compatible system of roots $(x^{1/\ell^k})_k$ in $\overline{\mathbb{Q}}_p$, the* Kummer cocycle *$\kappa_x(\sigma) = \left(\sigma(x^{1/\ell^k})/x^{1/\ell^k}\right)_k \in \mathbb{Z}_\ell(1)$ is continuous and satisfies $\kappa_x(\sigma\tau) = \kappa_x(\sigma) + \chi_\ell(\sigma)\kappa_x(\tau)$. Changing the system changes $\kappa_x$ by the coboundary $\sigma \mapsto (\chi_\ell(\sigma) - 1)z$, $z \in \mathbb{Z}_\ell(1)$; $\kappa_{xy} = \kappa_x + \kappa_y$ for the product systems; and for any extension $K' \supseteq K(\mu_{\ell^k})$, $\kappa_x \bmod \ell^k$ vanishes on $G_{K'}$ iff $x \in (K'^\times)^{\ell^k}$.*

*(ii) (Stated.) The Kummer maps $K^\times/(K^\times)^{\ell^k} \to H^1(K,\mu_{\ell^k})$ and $\widehat{K^\times} = \varprojlim_k K^\times/(K^\times)^{\ell^k} \to H^1(K,\mathbb{Z}_\ell(1))$ are isomorphisms. We do not prove this here; it is proved in Chapter 17 from Hilbert's Theorem 90 (see also [174, Chapter XIV]).*

*(iii) With Theorem 7.20: $H^1(K,\mathbb{Z}_\ell(1)) \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^v$, $v = v_\ell(q-1)$; the projection to $\mathbb{Z}_\ell$ is induced by $v_K$, the torsion is the image of the units, $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1)$; and $\#H^1(K,\mu_{\ell^k}) = \ell^k \cdot \#\mu_{\ell^k}(K)$.*

*Proof.* (i) Write $\alpha_k = x^{1/\ell^k}$. Then $\sigma(\alpha_k)/\alpha_k \in \mu_{\ell^k}$, compatible in $k$, and $\sigma\tau(\alpha_k)/\alpha_k = \sigma\big(\tau(\alpha_k)/\alpha_k\big) \cdot \sigma(\alpha_k)/\alpha_k$, which is the cocycle identity written multiplicatively. Each component

factors through the finite extension $K(\mu_{\ell^k}, \alpha_k)$, so $\kappa_x$ is continuous. Another system is $(\zeta_k \alpha_k)$ with $z = (\zeta_k) \in \mathbb{Z}_\ell(1)$, which adds $\sigma(\zeta_k)/\zeta_k$, that is $(\chi_\ell(\sigma) - 1)z$. The product rule is clear. For $\sigma \in G_{K'}$, $\sigma$ fixes $\mu_{\ell^k}$, and $\kappa_x(\sigma) \equiv 0 \pmod{\ell^k}$ means $\sigma(\alpha_k) = \alpha_k$; this holds for all such $\sigma$ iff $\alpha_k \in K'$; and if $x = y^{\ell^k}$ with $y \in K'$, then $\alpha_k = \zeta y \in K'$ for some $\zeta \in \mu_{\ell^k}$. (iii) combines (ii) with Theorem 7.20(iv), (v). □

**Theorem 7.58** (the Kummer cocycle on inertia and on Frobenius)**.** *Let $\ell \neq p$, $\varpi$ a uniformiser of $K$, $x \in K^\times$, $a = v_K(x)$ and $x = \varpi^a u$ with $u \in \mathcal{O}_K^\times$. Choose compatible systems of $\ell$-power roots of $\varpi$ and $u$, and give $x$ the product system. Put $v = v_\ell(q-1)$.*

*(i)* $\kappa_x = a\,\kappa_\varpi + \kappa_u$.

*(ii)* $\kappa_\varpi|_{I_K} = t_\ell$ *and* $\kappa_u|_{I_K} = 0$*; hence*

$$\kappa_x|_{I_K} = v_K(x) \cdot t_\ell,$$

*independently of all choices.*

*(iii)* $\kappa_u$ *is unramified: it factors through $G_K/I_K$, so it is determined by* $\kappa_u(\mathrm{Frob})$*, which is well defined modulo $(q-1)\mathbb{Z}_\ell(1)$. Under $\mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1) = \mu_{\ell^v}$ its class is the Teichmüller lift of $\bar{u}^{(q-1)/\ell^v}$; that is, it is the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v} = \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$. In particular $\kappa_u$ is a coboundary iff $\bar{u} \in (\mathbb{F}_q^\times)^{\ell^v}$.*

*(iv) If* Frob *fixes the chosen roots of $\varpi$ (Corollary 7.49), then* $\kappa_\varpi(\mathrm{Frob}) = 0$ *and* $\kappa_x(\mathrm{Frob}) = \kappa_u(\mathrm{Frob})$*. For any other lift* $\mathrm{Frob}\,\sigma$ *($\sigma \in I_K$),* $\kappa_x(\mathrm{Frob}\,\sigma) = \kappa_x(\mathrm{Frob}) + q\,a\,t_\ell(\sigma)$*. Hence, after identifying $\mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$, the value $c' = \kappa_x(\mathrm{Frob})$ is well defined modulo the ideal $(a, q-1)\mathbb{Z}_\ell = \ell^{\min(v_\ell(a),v)}\mathbb{Z}_\ell$ when the lift and the roots vary, and* $\min(v_\ell(a), v, v_\ell(c'))$ *does not depend on any choice.*

*(v) For $k \geq 1$: $I_K$ fixes the $\ell^k$-th roots of $x$ iff $\ell^k \mid a$; the image of $I_K$ in* $\mathrm{Gal}(K^{\mathrm{ur}}(x^{1/\ell^k})/K^{\mathrm{ur}}) \subseteq \mu_{\ell^k}$ *has order $\ell^{k-\min(k,v_\ell(a))}$. Equivalently, $x$ is an $\ell^k$-th power in $K^{\mathrm{ur}}$ iff $\ell^k \mid v_K(x)$.*

*Proof.* (i) is the product rule. (ii) On $I_K$, $\kappa_\varpi$ is by definition $t_\ell$ (Theorem 7.47(iii)). The $\ell^k$-th roots of $u$ lie in $K^{\mathrm{ur}}$ (Proposition 7.36(iv), with $\mu_{\ell^k} \subset K^{\mathrm{ur}}$), so $I_K$ fixes them and $\kappa_u|_{I_K} = 0$. A cocycle restricted to $I_K$, on which $\chi_\ell = 1$, is a homomorphism, and coboundaries $(\chi_\ell - 1)z$ vanish on $I_K$; so the restriction does not depend on the choices. (iii) For $g \in G_K$ and $\sigma \in I_K$, $\kappa_u(g\sigma) = \kappa_u(g) + \chi_\ell(g)\kappa_u(\sigma) = \kappa_u(g)$, so $\kappa_u$ factors through $G_K/I_K$; there Frob is a topological generator, and the coboundaries are $(\chi_\ell(\mathrm{Frob}) - 1)z = (q-1)z$. To compute the class, write $u = [\bar{u}]u_1$ with $u_1 \in U^{(1)}$ (Theorem 7.17). Changing roots changes $\kappa_u$ by a coboundary, so we may take the unique roots of $u_1$ in $U^{(1)}$ (Theorem 7.20(i)), for which $\kappa_{u_1} = 0$, and roots $\omega_k = [\bar{u}]^{1/\ell^k}$, which are roots of unity of order prime to $p$. Since Frob acts on such roots of unity by $y \mapsto y^q$, $\kappa_u(\mathrm{Frob}) = (\omega_k^{q-1})_k$. The projection $\mathbb{Z}_\ell(1) \to \mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1) = \mathbb{Z}_\ell(1)/\ell^v\mathbb{Z}_\ell(1) = \mu_{\ell^v}$ is the component at $k = v$, which is $\omega_v^{q-1} = \omega_v^{\ell^v N} = [\bar{u}]^N$ with $N = (q-1)/\ell^v$. Its reduction is $\bar{u}^N$. Finally $y \mapsto y^N$ maps the cyclic group $\mathbb{F}_q^\times$ onto $\mu_{\ell^v}(\mathbb{F}_q)$ with kernel $(\mathbb{F}_q^\times)^{\ell^v}$ (Theorem 7.20(v)). (iv) If Frob fixes the roots of $\varpi$, $\kappa_\varpi(\mathrm{Frob}) = 0$ and (i) gives $\kappa_x(\mathrm{Frob}) = \kappa_u(\mathrm{Frob})$. The cocycle identity and (ii) give $\kappa_x(\mathrm{Frob}\,\sigma) = \kappa_x(\mathrm{Frob}) + \chi_\ell(\mathrm{Frob})\,a\,t_\ell(\sigma) = \kappa_x(\mathrm{Frob}) + q\,a\,t_\ell(\sigma)$, and $t_\ell$ is surjective; changing roots adds $(q-1)z$. As $q$ is a unit in $\mathbb{Z}_\ell$, the ambiguity is the ideal $(a, q-1)$, generated by $\ell^{\min(v_\ell(a),v)}$, and changing $c'$ by an element of it does not change $\min(v_\ell(a), v, v_\ell(c'))$. Changing the identification $\mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$ multiplies $c'$ by a unit. (v) By (ii), $\sigma \in I_K$ moves $x^{1/\ell^k}$

by $\zeta^{a\,t(\sigma)}$, with $\zeta$ a generator of $\mu_{\ell^k}$ and $t(\sigma) \in \mathbb{Z}/\ell^k$ the image of $t_\ell(\sigma)$; as $t_\ell$ is surjective, the image of $I_K$ is the subgroup of $\mu_{\ell^k}$ generated by $\zeta^a$, of order $\ell^{k-\min(k,v_\ell(a))}$, which is trivial iff $\ell^k \mid a$. The last statement is also Proposition 7.36(iv). □

A small case shows (iii) at work. Take $K = \mathbb{Q}_7$, $\ell = 3$, so $v = v_3(6) = 1$ and $N = (q-1)/\ell^v = 2$, and $x = u = 2$. Then $\bar{u}^N = 4$, a primitive cube root of unity in $\mathbb{F}_7$ ($4^3 = 64 \equiv 1$), so the class of $\kappa_2$ in $\mathbb{Z}_3(1)/(q-1)\mathbb{Z}_3(1) \simeq \mu_3$ is nontrivial: $\kappa_2$ is not a coboundary, and 2 is not a cube in $\mathbb{Q}_7$, in agreement with Example 7.22(a). For $x = u = 6$ instead, $\bar{u}^N = 36 \equiv 1$, and $\kappa_6$ is a coboundary, matching the fact that $6 \equiv -1 = (-1)^3$ is a cube.

*In the paper* 7.59. *Lemma lem:tamagawa-exp.* Over $\mathbb{Q}_q^{\mathrm{ur}}$ (or already over the unramified quadratic extension, where the curve is a Tate curve) the $\ell^t$-torsion is generated by $\mu_{\ell^t}$ and an $\ell^t$-th root of $q_E$, and inertia acts trivially on $E[\ell^t]$ iff it fixes that root, iff $\ell^t \mid v_q(q_E) = n_q$ by Theorem 7.58(v). This is the precise form of the paper's statement that $q_E$ is an $\ell^t$-th power in $\mathbb{Q}_q^{\mathrm{ur}}$ if and only if $\ell^t \mid n_q$. *Lemma lem:tamfactor.* "the restriction of $\kappa$ to $I$ is $n$ times the tame character" is Theorem 7.58(ii) with $x = q_E$, $a = n$. "$\kappa = n\kappa_q + \kappa_u$, where $\kappa_u$ is unramified, inflated from the class of $u$" is (i) and (iii) with $\varpi = q$. "$c' = \varepsilon\kappa(\mathrm{Frob})$ is a cocycle value: it is $\varepsilon$ times the class of $u$ for the Frobenius lift fixing a compatible system of $\ell$-power roots of $q$, and in general it is well defined only modulo $\ell^{\min(v_\ell(n),v_\ell(q-1))}$" is (iii) and (iv), together with Corollary 7.49; the factor $\varepsilon$ is the twist of Example 7.74. The identification of the class in $H^1(G/I, \mathbb{Z}_\ell(1))$ with the class of $\bar{u}$ is (iii); its cohomological meaning is Chapter 17.

### 7.6.1 Quadratic extensions and quadratic characters

**Proposition 7.60** (quadratic extensions of $\mathbb{Q}_p$)**.** *(i) Let $p$ be odd and $u_0 \in \mathbb{Z}_p^\times$ a non-square. $\mathbb{Q}_p$ has exactly three quadratic extensions: the unramified $\mathbb{Q}_p(\sqrt{u_0}) = \mathbb{Q}_p(\mu_{p^2-1})$ and the ramified $\mathbb{Q}_p(\sqrt{p})$, $\mathbb{Q}_p(\sqrt{u_0 p})$, both tamely ramified. The field $\mathbb{Q}_p(\sqrt{p^*})$ is $\mathbb{Q}_p(\sqrt{p})$ if $p \equiv 1 \pmod 4$ and $\mathbb{Q}_p(\sqrt{-p}) = \mathbb{Q}_p(\sqrt{u_0 p})$ if $p \equiv 3 \pmod 4$.*

*(ii) Let $p$ be odd. The quadratic characters of $G_{\mathbb{Q}_p}$ are $1$, the unramified $\eta_{\mathrm{ur}}$ ($\eta_{\mathrm{ur}}(\mathrm{Frob}) = -1$) and two ramified ones $\chi$, $\chi\eta_{\mathrm{ur}}$. Every ramified one restricts on $I_p$ to the unique character of order 2 of $I_p$, which is $\theta_2 \colon \sigma \mapsto \sigma(\sqrt{p})/\sqrt{p}$.*

*(iii) $\mathbb{Q}_2$ has seven quadratic extensions, $\mathbb{Q}_2(\sqrt{d})$ for $d \in \{-1, \pm 5, \pm 2, \pm 10\}$. The unramified one is $\mathbb{Q}_2(\sqrt{5}) = \mathbb{Q}_2(\sqrt{-3})$; the others are wildly ramified, $\mathbb{Q}_2(\sqrt{-1})$ and $\mathbb{Q}_2(\sqrt{-5}) = \mathbb{Q}_2(\sqrt{3})$ with lower ramification jump $1$, and $\mathbb{Q}_2(\sqrt{\pm 2})$, $\mathbb{Q}_2(\sqrt{\pm 10})$ with jump $2$ (Example 7.43).*

*Proof.* By Kummer theory for $n = 2$ (Chapter 5), quadratic extensions of $\mathbb{Q}_p$ correspond to the nontrivial elements of $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$, through $d \mapsto \mathbb{Q}_p(\sqrt{d})$, and quadratic characters to the same group (the character of $\mathbb{Q}_p(\sqrt{d})$ being $\sigma \mapsto \sigma(\sqrt{d})/\sqrt{d}$). (i) By Theorem 7.20(vi) the nontrivial classes are $u_0$, $p$, $u_0 p$. The residue field $\mathbb{F}_{p^2}$ of $\mathbb{Q}_p(\mu_{p^2-1})$ (Theorem 7.34) contains a square root of $\bar{u}_0$, so $u_0$ is a square there (Theorem 7.20(ii)), and $\mathbb{Q}_p(\sqrt{u_0}) = \mathbb{Q}_p(\mu_{p^2-1})$ is unramified. $X^2 - p$ and $X^2 - u_0 p$ are Eisenstein. If $p \equiv 3 \pmod 4$, $-1$ is not a square modulo $p$, so $-p \in u_0 p(\mathbb{Q}_p^\times)^2$. Ramification index 2 is prime to $p$. (ii) A quadratic character is unramified iff its field is, and $\eta_{\mathrm{ur}}$ is the character of the unramified extension, on whose group Frob acts nontrivially. A ramified quadratic character is nontrivial on $I_p$. A character of order 2 of $I_p$ is continuous with finite image, so it kills the pro-$p$ group $P_p$ ($p$ odd), and factors through the procyclic $I_p/P_p \simeq \prod_{\ell \neq p} \mathbb{Z}_\ell(1)$, which has exactly one quotient of order 2; and $\theta_2$ (Theorem 7.47, $m = 2$, $\varpi = p$) is such a character. (iii) $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$ has order 8 with representatives $\pm 1, \pm 5, \pm 2, \pm 10$ (Theorem 7.20(vi)).

$\alpha = (1+\sqrt{5})/2$ is a root of $X^2 - X - 1 \in \mathbb{Z}_2[X]$, so it lies in the ring of integers of $L = \mathbb{Q}_2(\sqrt{5})$, and its reduction is a root of $X^2 + X + 1$, which is irreducible over $\mathbb{F}_2$. So the residue field of $L$ contains $\mathbb{F}_4$, $f(L/\mathbb{Q}_2) = 2 = [L : \mathbb{Q}_2]$, $e(L/\mathbb{Q}_2) = 1$ and $L$ is unramified (Theorem 7.25); and $-3 \cdot 5 = -15 \equiv 1 \pmod 8$ and $3 \cdot (-5) \equiv 1 \pmod 8$ are squares, so $\mathbb{Q}_2(\sqrt{-3}) = \mathbb{Q}_2(\sqrt{5})$ and $\mathbb{Q}_2(\sqrt{3}) = \mathbb{Q}_2(\sqrt{-5})$. There is only one unramified quadratic extension, so the other six are ramified, of degree 2, hence wild; the jumps are Example 7.43. □

**Proposition 7.61** (global quadratic characters locally)**.** *Let $d$ be a fundamental discriminant and $\chi_d$ the character of $\mathbb{Q}(\sqrt{d})$ (Chapter 5). Through Proposition 7.32, $\chi_d|_{G_{\mathbb{Q}_p}}$ is the character of $\mathbb{Q}_p(\sqrt{d})$.*

*(i) $\chi_d$ is unramified at $p$ iff $p \nmid d$, and then $\chi_d(\mathrm{Frob}_p) = \left(\frac{d}{p}\right)$ (Chapter 6).*

*(ii) For an odd prime $q$: at every $p \neq q$, including $p = 2$, $\chi_{q^*}$ is unramified, with $\chi_{q^*}(\mathrm{Frob}_p) = \left(\frac{p}{q}\right)$; at $q$ it is ramified and tame, and $\chi_{q^*}|_{I_q} = \theta_2$.*

*(iii) $\chi_{-3}$ at 3 is tamely ramified; $\chi_{-4}$ at 2 and $\chi_8$, $\chi_{-8}$ at 2 are wildly ramified, with ramification jumps 1 and 2.*

*(iv) For a finite set $S$ of odd primes, $\chi_S = \prod_{q\in S} \chi_{q^*}$ is unramified outside $S$ and restricts to $\theta_2$ on $I_q$ for each $q \in S$.*

*(v) For any representation $\rho$ of $G_{\mathbb{Q}}$ and odd $q$: $(\rho \otimes \chi_{q^*})|_{I_p} = \rho|_{I_p}$ for $p \neq q$, and $(\rho \otimes \chi_{q^*})|_{I_q} = \rho|_{I_q} \otimes \theta_2$. So twisting by $\chi_{q^*}$ changes nothing on inertia at $p \neq q$ (the image of inertia, $V^{I_p}$, the conductor exponent), multiplies Frobenius there by the sign $\chi_{q^*}(\mathrm{Frob}_p)$, and at $q$ multiplies the action of inertia by the character $\theta_2$.*

*(vi) Every quadratic character of $G_{\mathbb{Q}_p}$ is the restriction of a product of characters $\chi_d$: for $p$ odd, of 1, $\chi_{d_0}$, $\chi_{p^*}$ or $\chi_{d_0}\chi_{p^*}$, where $d_0$ is a fundamental discriminant prime to $p$ with $\left(\frac{d_0}{p}\right) = -1$; for $p = 2$, of a product of $\chi_{-4}$, $\chi_8$ and $\chi_5$.*

*Proof.* The local statement follows from the definitions: $\sigma(\sqrt{d}) = \chi_d(\sigma)\sqrt{d}$ for $\sigma \in G_{\mathbb{Q}}$, and the embedding $\iota$ of Proposition 7.32 transports this to $\mathbb{Q}_p(\sqrt{d})$. (i) For $p$ odd, $d$ is $p$ times a unit or a unit, according as $p \mid d$ or not; in the first case $X^2 - d$ is Eisenstein, in the second $\mathbb{Q}_p(\sqrt{d})$ is trivial or unramified (Proposition 7.60(i)). For $p = 2$, a fundamental discriminant is $\equiv 1 \pmod 4$ (odd) or divisible by 4; if $d \equiv 1 \pmod 8$ the field is trivial, if $d \equiv 5 \pmod 8$ it is $\mathbb{Q}_2(\sqrt{5})$, unramified, and if $4 \mid d$ then $d/4 \equiv 2, 3 \pmod 4$ and $\mathbb{Q}_2(\sqrt{d}) = \mathbb{Q}_2(\sqrt{d/4})$ is one of the six ramified fields (Proposition 7.60(iii)). The value at $\mathrm{Frob}_p$ is Chapter 6. (ii) $q^* \equiv 1 \pmod 4$, so (i) applies; $(q^*/p) = (p/q)$ is quadratic reciprocity (Chapter 6). At $q$, $\chi_{q^*}$ is ramified and $q$ is odd, so Proposition 7.60(ii) gives $\theta_2$. (iii) $\mathbb{Q}_3(\sqrt{-3})$ is tame (Proposition 7.28(ii)); $\mathbb{Q}_2(\sqrt{-1})$, $\mathbb{Q}_2(\sqrt{\pm 2})$ are Example 7.43. (iv) and (v) follow from (ii), since a character is trivial on $I_p$ where it is unramified. (vi) By Proposition 7.60 the classes in $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$ are represented by $1, d_0, p^*, d_0p^*$ for $p$ odd ($d_0$ is a non-square unit; such a $d_0$ exists: $-4$ if $p \equiv 3 \pmod 4$, 8 if $p \equiv 5 \pmod 8$, and in general a prime $d_0 \equiv 1 \pmod 4$ that is a quadratic non-residue modulo $p$, which exists by Dirichlet's theorem, a case of Chebotarev's, Chapter 6), and are generated by $-1$, 2, 5 for $p = 2$; the characters of $\mathbb{Q}_2(\sqrt{-1})$, $\mathbb{Q}_2(\sqrt{2})$, $\mathbb{Q}_2(\sqrt{5})$ are the restrictions of $\chi_{-4}$, $\chi_8$, $\chi_5$. □

**Theorem 7.62** (local class field theory for $\mathbb{Q}_p$)**.** *(i) (Local Kronecker-Weber.) The maximal abelian extension of $\mathbb{Q}_p$ is $\mathbb{Q}_p^{\mathrm{ab}} = \mathbb{Q}_p^{\mathrm{ur}} \cdot \mathbb{Q}_p(\mu_{p^\infty})$.*

*(ii) There is a unique continuous homomorphism* $\mathrm{Art}\colon \mathbb{Q}_p^\times \to \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{ab}}/\mathbb{Q}_p)$ *such that* $\mathrm{Art}(p)$ *is the arithmetic Frobenius on* $\mathbb{Q}_p^{\mathrm{ur}}$ *and the identity on* $\mathbb{Q}_p(\mu_{p^\infty})$*, and, for* $u \in \mathbb{Z}_p^\times$*,* $\mathrm{Art}(u)$ *is the identity on* $\mathbb{Q}_p^{\mathrm{ur}}$ *and acts on* $\mu_{p^\infty}$ *by* $\zeta \mapsto \zeta^{u^{-1}}$*. It is injective with dense image,* $\mathrm{Art}(\mathbb{Z}_p^\times)$ *is the image of* $I_p$*, and* $\mathrm{Art}(1+p^n\mathbb{Z}_p)$ *is the image of the upper ramification group* $G^n_{\mathbb{Q}_p}$ *(*$n \geq 1$*).*

*(iii) For a character* $\chi$ *of* $G_{\mathbb{Q}_p}$ *of finite order, its Artin conductor exponent* $a(\chi)$ *(Definition 7.81) is the least* $n \geq 0$ *with* $\chi(\mathrm{Art}(U^{(n)})) = 1$*, where* $U^{(0)} = \mathbb{Z}_p^\times$ *and* $U^{(n)} = 1+p^n\mathbb{Z}_p$*. For a Dirichlet character* $\chi$ *(Chapter 6) the product of the* $p^{a(\chi|_{G_{\mathbb{Q}_p}})}$ *is the conductor of* $\chi$*.*

*We do not prove this; see Serre's chapter "Local class field theory" in Cassels and Fröhlich [26], Milne [133, Chapter I], and Neukirch [142, Chapter V].*

*Warning* 7.63 (normalisations of the Artin map). The book's Artin map sends a uniformiser to an *arithmetic* Frobenius, as in Chapter 6; units then act on $\mu_{p^\infty}$ through $u^{-1}$. Deligne [41] and many sources on Weil-Deligne representations send uniformisers to geometric Frobenius elements, and units act through $u$. The book never needs the other normalisation, and uses local class field theory only for the conductors of characters, Theorem 7.62(iii). For example, $\chi_{-4}$ is trivial on $\mathrm{Art}(1+4\mathbb{Z}_2)$ but not on $\mathrm{Art}(\mathbb{Z}_2^\times)$, so $a(\chi_{-4}|_{G_{\mathbb{Q}_2}}) = 2$; $\chi_8$ is trivial on $\mathrm{Art}(1+8\mathbb{Z}_2)$ but not on $\mathrm{Art}(1+4\mathbb{Z}_2)$, so $a = 3$; these agree with Theorem 7.82(iv) and with the conductors 4 and 8.

## 7.7 The Weil group and Weil-Deligne representations

An $\ell$-adic representation of $G_K$ ($\ell \neq p$) can have infinite image of inertia, as the Tate module of a curve with multiplicative reduction does; Grothendieck's theorem says that this infinite part is always unipotent, given by a nilpotent "monodromy" operator $N$. Recording the finite part $r$ and $N$ separately gives the Weil-Deligne representation, in terms of which the local invariants ($V^I$, $L$-factors, conductors) are computed in Sections 7.8 and 7.9, independently of $\ell$. The paper speaks of "local components" and the "unramified twist of the Steinberg representation" in Lemmas lem:twist(iii) and lem:euler; on the Galois side these are the Weil-Deligne representations $\mathrm{sp}(2) \otimes \mu$ of this section (the automorphic side is Chapter 15).

**Convention 7.64** (the norm on the Weil group)**.** For $w$ in the Weil group $W_K$ (Definition 7.65), $d(w) \in \mathbb{Z}$ is defined by $w|_{K^{\mathrm{ur}}} = \mathrm{Frob}^{d(w)}$, and $\|w\| = q^{d(w)}$. So $\|\mathrm{Frob}\| = q$, $\|\Phi\| = q^{-1}$, $\|\cdot\|$ is trivial on $I_K$, and $\|w\| = \chi_\ell(w)$ for every $\ell \neq p$. A Weil-Deligne representation satisfies

$$r(w)\,N\,r(w)^{-1} = \|w\|\,N, \qquad \text{that is} \qquad r(\mathrm{Frob})\,N\,r(\mathrm{Frob})^{-1} = q\,N, \quad r(\Phi)\,N\,r(\Phi)^{-1} = q^{-1}N.$$

Some sources write $\mathrm{Frob}\,N\,\mathrm{Frob}^{-1} = q^{-1}N$ with Frob geometric; here Frob is arithmetic (Appendix B, convention B-C2(f)). This $\|\cdot\|$ agrees with Tate's normalisation in [194].

**Definition 7.65** (the Weil group)**.** The *Weil group* of $K$ is $W_K = \{w \in G_K : \ w|_{K^{\mathrm{ur}}} \in \mathrm{Frob}^{\mathbb{Z}}\}$, the preimage of the subgroup $\mathbb{Z} \subset \hat{\mathbb{Z}} \simeq \mathrm{Gal}(K^{\mathrm{ur}}/K)$. It is given the topology in which $I_K$, with its profinite topology, is an open subgroup.

**Proposition 7.66.** *(i)* $W_K$ *is a dense subgroup of* $G_K$ *containing* $I_K$*, with* $W_K/I_K \simeq \mathbb{Z}$ *generated by the image of* Frob*; its topology is finer than the topology induced from* $G_K$*, and every* $w \in W_K$ *is uniquely* $w = \Phi^n\sigma$ *with* $n \in \mathbb{Z}$*,* $\sigma \in I_K$*, for a fixed geometric Frobenius lift* $\Phi \in W_K$*.*

*(ii)* $d\colon W_K \to \mathbb{Z}$ *is a continuous homomorphism,* $\|\mathrm{Frob}\| = q$, $\|\Phi\| = q^{-1}$, *and* $\|w\| = \chi_\ell(w)$ *for* $\ell \neq p$.

*(iii) A homomorphism* $\mu\colon W_K \to \Omega^\times$ *trivial on* $I_K$ *(an* unramified character*) is determined by* $\mu(\Phi)$*, which may be any element of* $\Omega^\times$*;* $\|\cdot\|$ *is the unramified character with* $\Phi \mapsto q^{-1}$*. A character of* $G_K$ *with values in* $\mathbb{Q}_\ell^\times$ *takes values in* $\mathbb{Z}_\ell^\times$*, but a character of* $W_K$ *need not.*

*(iv) (Stated.) The Artin map of Theorem 7.62 identifies* $\mathbb{Q}_p^\times$ *with* $W_{\mathbb{Q}_p}^{\mathrm{ab}}$*, and* $\|\mathrm{Art}(x)\| = |x|_p^{-1}$.

*Proof.* (i) The restriction $G_K \to \hat{\mathbb{Z}}$ is open and surjective, and $\mathbb{Z}$ is dense in $\hat{\mathbb{Z}}$, so the preimage $W_K$ is dense; $I_K$ is the kernel. Open subsets of $I_K$ are open in $W_K$ by definition, and a subset of $W_K$ open in the induced topology is a union of cosets of open subsets of $I_K$; so the topology is finer. A lift $\Phi$ maps to $-1 \in \mathbb{Z}$, a generator. (ii) $d$ is the composite $W_K \to W_K/I_K \simeq \mathbb{Z}$, continuous because $I_K$ is open; $\chi_\ell$ is unramified with $\chi_\ell(\mathrm{Frob}) = q$ (Proposition 7.39). (iii) $W_K/I_K \simeq \mathbb{Z}$ is free on the image of $\Phi$. A continuous character of the compact group $G_K$ has compact image in $\mathbb{Q}_\ell^\times$; its image under $v_\ell$ is a compact subgroup of $\mathbb{Z}$, hence 0, so the image lies in $\mathbb{Z}_\ell^\times$. The unramified character of $W_K$ with $\Phi \mapsto \ell$ does not. □

**Lemma 7.67** (matrix logarithm)**.** *Let* $\mathcal{O}$ *be the ring of integers of a finite extension* $E_\lambda$ *of* $\mathbb{Q}_\ell$ *and* $\mathfrak{g} = \ell^2\,\mathrm{M}_n(\mathcal{O})$*. Then* $\exp(X) = \sum X^k/k!$ *converges for* $X \in \mathfrak{g}$ *and* $\log(g) = \sum_{k\geq 1}(-1)^{k+1}(g-1)^k/k$ *for* $g \in 1+\mathfrak{g}$*; they are mutually inverse bijections* $\mathfrak{g} \leftrightarrows 1+\mathfrak{g}$*;* $\exp(X+Y) = \exp X \exp Y$ *when* $XY = YX$*; and* $g^a = \exp(a \log g)$ *for* $a \in \mathbb{Z}_\ell$*, where* $g^a = \lim g^{a_k}$ *for integers* $a_k \to a$*. For nilpotent* $N$ *the series* $\exp(tN)$ *is a polynomial in* $t$ *and* $\log\exp(tN) = tN$ *for all* $t \in E_\lambda$.

*Proof.* Use $\|X\| = \max|x_{ij}|_\ell$, which satisfies $\|XY\| \leq \|X\|\|Y\|$ (ultrametric inequality), and the estimates in the proof of Theorem 7.18 with $c = 2$: the terms are of valuation $\geq 2k - (k-1) = k+1 \geq 2$, so both series converge into $\mathfrak{g}$, resp. $1+\mathfrak{g}$. The identities hold in $\mathbb{Q}[[X,Y]]$ (for commuting variables) and transfer, entry by entry, by Lemma 7.4 with the same bounds, all the matrices involved commuting. For $a \in \mathbb{Z}$, $\exp(a\log g) = \exp(\log g)^a = g^a$; both sides are continuous in $a$ (the left side by the bounds, the right side because $g^{\ell^k} \to 1$ in the pro-$\ell$ group $1+\mathfrak{g}$), so they agree on $\mathbb{Z}_\ell$. For nilpotent $N$ the series are finite and the identity is a polynomial identity. □

**Theorem 7.68** (Grothendieck's $\ell$-adic monodromy theorem)**.** *Let* $\ell \neq p$*,* $E_\lambda$ *a finite extension of* $\mathbb{Q}_\ell$*,* $V$ *a finite-dimensional* $E_\lambda$*-vector space and* $\rho\colon G_K \to \mathrm{GL}(V)$ *continuous. Fix a generator* $\zeta$ *of* $\mathbb{Z}_\ell(1)$ *and let* $t_\zeta\colon I_K \to \mathbb{Z}_\ell$ *be the tame character followed by* $\zeta \mapsto 1$*. There are an open subgroup* $I_1 \subseteq I_K$*, normal in* $G_K$*, and a unique nilpotent* $N \in \mathrm{End}(V)$ *with*

$$\rho(\sigma) = \exp\bigl(t_\zeta(\sigma)N\bigr) \qquad \text{for all } \sigma \in I_1,$$

*and* $\rho(w)\,N\,\rho(w)^{-1} = \|w\|\,N$ *for all* $w \in W_K$*; in particular* $\rho(\mathrm{Frob})N\rho(\mathrm{Frob})^{-1} = qN$*. Moreover* $N = 0$ *iff* $\rho(I_K)$ *is finite, and replacing* $\zeta$ *by* $\zeta^a$ *(*$a \in \mathbb{Z}_\ell^\times$*) replaces* $N$ *by* $aN$.

*Proof.* $\rho(G_K)$ is compact, so it stabilises a lattice $T$ (the stabiliser of any lattice $T_0$ is open, and the sum of the finitely many translates $\rho(g)T_0$ is stable); in a basis of $T$, $\rho\colon G_K \to \mathrm{GL}_n(\mathcal{O})$. Let $I_1 = I_K \cap \rho^{-1}(1+\ell^2\,\mathrm{M}_n(\mathcal{O}))$, open in $I_K$ and normal in $G_K$. By Proposition 7.53(i), $\rho|_{I_1} = \phi \circ t_\zeta|_{I_1}$ for a homomorphism $\phi\colon t_\zeta(I_1) \to 1+\ell^2\,\mathrm{M}_n(\mathcal{O})$, continuous because $t_\zeta|_{I_1}$ is a quotient map of compact groups. The image $t_\zeta(I_1)$ is a closed subgroup of finite index of $\mathbb{Z}_\ell$, so $t_\zeta(I_1) = \ell^m\mathbb{Z}_\ell$. Choose $\sigma_1 \in I_1$ with $t_\zeta(\sigma_1) = \ell^m$, put $g = \rho(\sigma_1)$ and $N = \ell^{-m}\log g$ (Lemma 7.67). For $\sigma \in I_1$

with $t_\zeta(\sigma) = \ell^m b$, $b \in \mathbb{Z}_\ell$, continuity of $\phi$ gives $\rho(\sigma) = \phi(\ell^m)^b = g^b = \exp(b \log g) = \exp(t_\zeta(\sigma)N)$. For $w \in W_K$, $w\sigma_1 w^{-1} \in I_1$ and $t_\zeta(w\sigma_1 w^{-1}) = \chi_\ell(w)\ell^m = \|w\|\ell^m$ (Theorem 7.50(i)); hence

$$\exp(\rho(w) \log g\, \rho(w)^{-1}) = \rho(w) g \rho(w)^{-1} = \rho(w\sigma_1 w^{-1}) = \exp(\|w\| \log g).$$

Both arguments of exp lie in $\ell^2 \operatorname{M}_n(\mathcal{O})$ ($\rho(w) \in \operatorname{GL}_n(\mathcal{O})$, and $\|w\| \in \mathbb{Z}_\ell^\times$), where exp is injective; so $\rho(w) N \rho(w)^{-1} = \|w\| N$. In particular $N$ is conjugate to $qN$, so the finite set of eigenvalues of $N$ (in an algebraic closure) is stable under multiplication by $q$; as $q^i \neq q^j$ for $i \neq j$, all eigenvalues are 0 and $N$ is nilpotent. Uniqueness: if $N'$ is nilpotent with $\rho = \exp(t_\zeta N')$ on an open $I_1'$, pick $\sigma \in I_1 \cap I_1'$ with $t_\zeta(\sigma) = \ell^{m'} \neq 0$; then $\ell^{m'} N = \log \rho(\sigma) = \ell^{m'} N'$ by Lemma 7.67. If $N = 0$ then $\rho(I_1) = 1$ and $\rho(I_K)$ is finite. If $\rho(I_K)$ is finite, $g$ has finite order and $\log(g^k) = k \log g$, so $\log g = 0$ and $N = 0$. Finally $t_{\zeta^a} = a^{-1} t_\zeta$, so $\exp(t_\zeta(\sigma)N) = \exp(t_{\zeta^a}(\sigma)\, aN)$. □

The theorem says that $N$ is the logarithm of the monodromy: just as a unipotent matrix $u$ is $\exp(\log u)$ with $\log u$ nilpotent, the action of an open subgroup of inertia is exp of a multiple of $N$. The same statement holds for continuous representations of $W_K$ (Deligne [41, Section 8]); we need only $G_K$.

**Definition 7.69** (Weil-Deligne representations)**.** Let $\Omega$ be a field of characteristic 0. A *Weil-Deligne representation* of $W_K$ over $\Omega$ is a pair $(r, N)$ on a finite-dimensional $\Omega$-vector space $V$: a homomorphism $r\colon W_K \to \operatorname{GL}(V)$ whose restriction to $I_K$ has open kernel (so $r(I_K)$ is finite), and a nilpotent $N \in \operatorname{End}(V)$ with $r(w) N r(w)^{-1} = \|w\| N$ for all $w \in W_K$. A morphism is a linear map commuting with $r$ and $N$. $(r, N)$ is *Frobenius-semisimple* if $r(\Phi)$ is semisimple.

**Proposition 7.70** (operations on Weil-Deligne representations)**.** *If $(r, N, V)$ and $(r', N', V')$ are Weil-Deligne representations, so are: $V \oplus V'$ with $N \oplus N'$; $V \otimes V'$ with $N \otimes 1 + 1 \otimes N'$; the dual $V^\vee$ with $r^\vee(w) = r(w^{-1})^{\mathsf{T}}$ and $-N^{\mathsf{T}}$; $\operatorname{Hom}(V, V')$ with $f \mapsto N'f - fN$; in particular $\operatorname{End} V$ and its trace-zero part $\operatorname{ad}^0 V$ with $N_{\operatorname{ad}^0} = [N, \cdot\,]$; and the twist $V \otimes \mu$ by a character $\mu$ of $W_K$ (with $N$ unchanged). The kernel $\ker N$ and the invariants $V^{r(I_K)}$ are stable under $r(W_K)$. (Stated.) Every $(r, N)$ has a Frobenius-semisimplification, obtained by replacing $r(\Phi^n\sigma)$ by $r(\Phi)^n_{\mathrm{ss}} r(\sigma)$; see [41, Section 8], [194, Section 4.1].*

*Proof.* Each relation is a one-line check. For example $(r \otimes r')(w)(N \otimes 1 + 1 \otimes N')(r \otimes r')(w)^{-1} = \|w\| N \otimes 1 + 1 \otimes \|w\| N'$; for the dual, $r(w^{-1})^{\mathsf{T}}(-N^{\mathsf{T}}) r(w)^{\mathsf{T}} = -(r(w) N r(w)^{-1})^{\mathsf{T}}$; and for $\operatorname{ad}^0$, $r(w)[N, x] r(w)^{-1} = \|w\| [N, r(w) x r(w)^{-1}]$. Nilpotency is clear in each case ($N \otimes 1$ and $1 \otimes N'$ commute). If $Nv = 0$ then $N r(w) v = \|w\|^{-1} r(w) N v = 0$; and $I_K$ is normal in $W_K$. □

**Theorem 7.71** (the Weil-Deligne representation of an $\ell$-adic representation)**.** *Let $\rho$, $\zeta$, $N$ be as in Theorem 7.68, and fix a geometric Frobenius lift $\Phi \in W_K$. Put*

$$r(\Phi^n \sigma) = \rho(\Phi^n \sigma) \exp(-t_\zeta(\sigma) N) \qquad (n \in \mathbb{Z},\ \sigma \in I_K).$$

*(i)* $\operatorname{WD}(\rho) = (r, N)$ *is a Weil-Deligne representation over $E_\lambda$; $r = \rho$ on $\Phi$, and $r$ is trivial on $I_1$.*

*(ii) Its isomorphism class does not depend on $\Phi$ or on $\zeta$.*

*(iii) (Stated.) $\rho \mapsto \operatorname{WD}(\rho)$ is an equivalence between continuous representations of $W_K$ on finite-dimensional $E_\lambda$-spaces and Weil-Deligne representations over $E_\lambda$; those coming from $G_K$ are the ones for which the eigenvalues of $r(\Phi)$ are $\ell$-adic units. It is compatible with $\oplus$, $\otimes$, duals, $\operatorname{ad}^0$ and twists by characters; $\mathbb{Q}_\ell(1)$ goes to $(\|\cdot\|, 0)$.*

*(iv) (Stated.) For the Tate module $V_\ell E$ of an elliptic curve over $K$, the Frobenius semisimplification of $\mathrm{WD}(V_\ell E)$, viewed over an algebraic closure of $\mathbb{Q}_\ell$, is defined over $\mathbb{Q}$ and does not depend on $\ell \neq p$ (Chapter 11, Chapter 15).*

*Proof.* (i) $\exp(-tN)$ is defined for all $t \in \mathbb{Z}_\ell$ ($N$ nilpotent). Let $w = \Phi^n\sigma$, $w' = \Phi^{n'}\sigma'$; then $ww' = \Phi^{n+n'}\sigma''\sigma'$ with $\sigma'' = \Phi^{-n'}\sigma\Phi^{n'}$, and $t_\zeta(\sigma'') = \chi_\ell(\Phi^{-n'})t_\zeta(\sigma) = q^{n'}t_\zeta(\sigma)$ by Theorem 7.50(i). The relation $\rho(w')^{-1}N\rho(w') = \|w'\|^{-1}N = q^{n'}N$ gives $\exp(-tN)\rho(w') = \rho(w')\exp(-tq^{n'}N)$, so

$$r(w)r(w') = \rho(w)\rho(w')\exp(-q^{n'}t_\zeta(\sigma)N)\exp(-t_\zeta(\sigma')N) = \rho(ww')\exp(-t_\zeta(\sigma''\sigma')N) = r(ww').$$

On $I_1$, $\rho(\sigma) = \exp(t_\zeta(\sigma)N)$, so $r(\sigma) = 1$; the kernel of $r|_{I_K}$ is open. $r(w)Nr(w)^{-1} = \rho(w)N\rho(w)^{-1} = \|w\|N$, as $\exp(tN)$ commutes with $N$. (ii) Changing $\zeta$ to $\zeta^a$ leaves $r$ unchanged and replaces $N$ by $aN$. Let $k \geq 1$ be such that conjugation by $r(\Phi^k)$ is trivial on the finite group $r(I_K)$ (it is an automorphism of finite order). Then $r(\Phi^k)$ commutes with $r(W_K)$, so its generalised eigenspaces $V_\lambda$ are $r(W_K)$-stable, and $N(V_\lambda) \subseteq V_{q^{-k}\lambda}$ because $r(\Phi^k)N = q^{-k}Nr(\Phi^k)$. Choose a representative $\lambda_0$ in each orbit of $q^{k\mathbb{Z}}$ on the eigenvalues and let $A$ act on $V_{\lambda_0 q^{-kj}}$ by $a^j$. Then $A$ commutes with $r(W_K)$ and $ANA^{-1} = aN$, so $(r, N) \simeq (r, aN)$. Here the eigenvalues lie in a finite Galois extension $F$ of $E_\lambda$, and $A$ is first defined on $V \otimes F$; it is defined over $E_\lambda$ if the representatives are chosen compatibly with $\mathrm{Gal}(F/E_\lambda)$. This is possible: if $\gamma \in \mathrm{Gal}(F/E_\lambda)$ maps an orbit to itself, then $\gamma(\lambda_0) = \lambda_0 q^{-kj}$ for some $j$, and since $\gamma$ has finite order $m$, $\lambda_0 = \gamma^m(\lambda_0) = \lambda_0 q^{-kjm}$, so $j = 0$. Choosing one $\lambda_0$ in each $\mathrm{Gal}(F/E_\lambda)$-orbit of orbits and $\gamma(\lambda_0)$ for its transforms is therefore well defined, and then $\gamma$ maps $V_{\lambda_0 q^{-kj}} \otimes F$ semilinearly onto $V_{\gamma(\lambda_0)q^{-kj}} \otimes F$, so $A$ commutes with $\mathrm{Gal}(F/E_\lambda)$ and descends to $E_\lambda$ (as $a \in \mathbb{Z}_\ell^\times \subseteq E_\lambda$). If $\Phi$ is replaced by $\Phi' = \Phi\sigma_0$ ($\sigma_0 \in I_K$), write $\Phi'^n = \Phi^n\tau_n$ with $\tau_n \in I_K$ and put $c_n = t_\zeta(\tau_n)$. From $\Phi'^{n+1} = \Phi^n\tau_n\Phi\sigma_0 = \Phi^{n+1}(\Phi^{-1}\tau_n\Phi)\sigma_0$ and $t_\zeta(\Phi^{-1}\tau_n\Phi) = \chi_\ell(\Phi^{-1})c_n = qc_n$ (Theorem 7.50(i)) we get $c_{n+1} = qc_n + t_\zeta(\sigma_0)$, with $c_0 = 0$; this recursion determines $c_n$ for all $n \in \mathbb{Z}$ (it can be run backwards), and $c_n = t_\zeta(\sigma_0)\frac{q^n-1}{q-1}$ satisfies it. For $w = \Phi'^n\sigma = \Phi^n\tau_n\sigma$ the two definitions give $r'(w) = \rho(w)\exp(-t_\zeta(\sigma)N)$ and $r(w) = \rho(w)\exp(-(c_n + t_\zeta(\sigma))N)$, so $r'(w) = r(w)\exp(t_\zeta(\sigma_0)\frac{q^n-1}{q-1}N)$. Put $c = t_\zeta(\sigma_0)/(q-1) \in \mathbb{Q}_\ell$ and $B = \exp(-cN)$, which commutes with $N$. Since $\|w\| = q^{-n}$, $r(w)^{-1}Nr(w) = q^nN$, hence $r(w)^{-1}\exp(cN)r(w) = \exp(cq^nN)$ and

$$B^{-1}r(w)B = r(w)\cdot r(w)^{-1}\exp(cN)r(w)\cdot\exp(-cN) = r(w)\exp\big(c(q^n-1)N\big) = r'(w);$$

so $B$ is an isomorphism $(r', N) \simeq (r, N)$. (iii), (iv) We do not prove these; see Deligne [41, Section 8], Tate [194, Section 4.2] and Rohrlich [166]; (iv) for elliptic curves follows from the description of $V_\ell E$ in Chapters 10 and 11. □

**Definition 7.72** (the special representations)**.** For $n \geq 1$, $\mathrm{sp}(n)$ is the Weil-Deligne representation on $V = \bigoplus_{i=0}^{n-1}\Omega e_i$ with $r(w)e_i = \|w\|^i e_i$ and $Ne_i = e_{i+1}$ ($e_n := 0$). For a character $\mu$ of $W_K$, $\mathrm{sp}(n)\otimes\mu$ is the *special* (or *Steinberg*) representation twisted by $\mu$; for $n = 2$ and $\mu$ unramified it is an *unramified twist of Steinberg*, and for $\mu$ ramified a *ramified twist*.

**Proposition 7.73.** *(i) $\mathrm{sp}(n)\otimes\mu$ is an indecomposable Weil-Deligne representation. On $\mathrm{sp}(2)$, $\Phi$ acts on $e_0, e_1$ by $1, q^{-1}$ and Frob by $1, q$, and $\ker N = \Omega e_1$.*

*(ii) If $\dim V = 2$ and $N \neq 0$, then $(r, N)$ is automatically Frobenius-semisimple, and $(r, N) \simeq \mathrm{sp}(2)\otimes\mu$ for a unique character $\mu$ of $W_K$, namely the character by which $W_K$ acts on $V/\ker N$.*

*(iii) (Stated.) A Frobenius-semisimple $(r,0)$ of dimension 2 is a sum of two characters or irreducible; an irreducible one is induced from a character of $W_{K'}$ for a quadratic $K'/K$, except possibly when $p=2$. See [194, Section 4.1] and [166].*

*Proof.* (i) $r(w)Nr(w)^{-1}e_i = \|w\|^{-i}r(w)e_{i+1} = \|w\|e_{i+1}$, so the relation holds. $\ker N = \Omega e_{n-1}$ is a line. If $V = V' \oplus V''$ as Weil-Deligne representations, $\ker N = \ker N|_{V'} \oplus \ker N|_{V''}$, and a nilpotent operator on a nonzero space has a nonzero kernel; so $V'$ or $V''$ is 0. (ii) $N^2 = 0$ and $\ker N = \operatorname{im} N$ is a line, $r(W_K)$-stable (Proposition 7.70). Let $\mu'$ be the character of $W_K$ on $\ker N$ and $\mu$ that on $V/\ker N$. From $Nr(w) = \|w\|^{-1}r(w)N$ applied to $v \notin \ker N$ we get $\mu(w)Nv = \|w\|^{-1}\mu'(w)Nv$, so $\mu' = \|\cdot\|\mu$. Since $\mu'(\Phi) = q^{-1}\mu(\Phi) \neq \mu(\Phi)$, the operator $r(\Phi)$ has two distinct eigenvalues, so it is semisimple, and there is an eigenvector $e_0 \notin \ker N$ with $r(\Phi)e_0 = \mu(\Phi)e_0$. For $\sigma \in I_K$, $r(\sigma)$ has finite order, hence is semisimple, with the eigenvalue $\mu(\sigma) = \mu'(\sigma)$ on $\ker N$ and on $V/\ker N$; so $r(\sigma) = \mu(\sigma)$ is a scalar. Hence $r(w)e_0 = \mu(w)e_0$ for all $w = \Phi^n\sigma$. Put $e_1 = Ne_0$; then $r(w)e_1 = \|w\|Nr(w)e_0 = \|w\|\mu(w)e_1$. In the basis $e_0, e_1$, $(r,N) = \mathrm{sp}(2) \otimes \mu$; and $\mu$ is determined as the character on $V/\ker N$. □

**Example 7.74** (the Tate shape)**.** Let $\ell \neq p$, $x \in K^\times$ with $a = v_K(x)$, $\varepsilon\colon G_K \to \{\pm 1\}$ unramified, and let $\rho$ act on $\mathbb{Q}_\ell e_1 \oplus \mathbb{Q}_\ell e_2$ by

$$\rho(g) = \begin{pmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa_x(g) \\ 0 & \varepsilon(g) \end{pmatrix},$$

with $\kappa_x$ read in $\mathbb{Z}_\ell$ through $\zeta$. This is a homomorphism: the upper right entry of $\rho(g)\rho(h)$ is $\varepsilon(gh)(\chi_\ell(g)\kappa_x(h) + \kappa_x(g)) = \varepsilon(gh)\kappa_x(gh)$. (It is the twist by $\varepsilon$ of the Kummer representation $\left(\begin{smallmatrix}\chi_\ell & \kappa_x \\ 0 & 1\end{smallmatrix}\right)$ of Chapter 5; the corner of the twist is $\varepsilon\kappa_x$, and the matrix with corner $\kappa_x$ and diagonal $\chi_\ell\varepsilon, \varepsilon$ is not a homomorphism unless $\varepsilon = 1$.)

(i) On $I_K$, $\rho(\sigma) = \left(\begin{smallmatrix}1 & a\,t_\zeta(\sigma) \\ 0 & 1\end{smallmatrix}\right) = \exp(t_\zeta(\sigma)N)$ with $N = aE_{12}$ ($Ne_2 = ae_1$, $Ne_1 = 0$), by Theorem 7.58(ii); here $I_1 = I_K$, and $N \neq 0$ iff $a \neq 0$. With $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix}\varepsilon q & c' \\ 0 & \varepsilon\end{smallmatrix}\right)$ one checks $\rho(\mathrm{Frob})N\rho(\mathrm{Frob})^{-1} = qN$: for an upper triangular $g = \left(\begin{smallmatrix}x & y \\ 0 & z\end{smallmatrix}\right)$ one has $gE_{12} = xE_{12}$ and $E_{12}g = zE_{12}$, so $gE_{12}g^{-1} = (x/z)E_{12}$, and here $x/z = \varepsilon q/\varepsilon = q$.

(ii) If $a \neq 0$: $r$ is unramified, $r(\Phi) = \rho(\mathrm{Frob})^{-1}$ has the distinct eigenvalues $\varepsilon q^{-1}$ (on $e_1$) and $\varepsilon$ (on the quotient), with $\varepsilon = \varepsilon(\mathrm{Frob})$, so $\mathrm{WD}(\rho) \simeq \mathrm{sp}(2) \otimes \mu_\varepsilon$ with $\mu_\varepsilon$ unramified, $\mu_\varepsilon(\Phi) = \varepsilon(\mathrm{Frob})$ (Proposition 7.73(ii)). Moreover $V^{I_K} = \mathbb{Q}_\ell e_1$, on which $\Phi$ acts by $\varepsilon q^{-1}$ and Frob by $\varepsilon q$.

(iii) If $a = 0$: $\rho$ is unramified, $N = 0$, and Frob has the distinct eigenvalues $\varepsilon q$, $\varepsilon$.

(iv) On the lattice $\mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$, $I_K$ acts trivially modulo $\ell^k$ iff $\ell^k \mid a$ (Theorem 7.58(v)).

Chapters 10 and 11 show that $T_\ell E$, at a prime of multiplicative reduction, has exactly this shape, with $x = q_E$ the Tate parameter and $\varepsilon$ trivial (split) or the unramified quadratic character (nonsplit).

*In the paper* 7.75. In Lemma lem:euler (type $\mathrm{I}_n$) and in the proof of Lemma lem:twist(iii), the paper uses that the local component is an unramified twist of the Steinberg representation. On the Galois side this means $\mathrm{WD}(V_\ell E|_{G_{\mathbb{Q}_q}}) \simeq \mathrm{sp}(2) \otimes \mu$ with $\mu$ unramified, which is Example 7.74(ii) for the Tate shape; the passage between the automorphic and the Galois side is Chapter 15. The proof of Lemma lem:tamfactor uses exactly the matrices of Example 7.74, with corner $\varepsilon\kappa$ (the paper notes that $\varepsilon\kappa$, not $\kappa$, satisfies the cocycle relation for the twisted matrices), $te_2 = e_2 + n'e_1$ with $n' = n\,t_\zeta(t)$, and $v_\ell(n') = v_\ell(n)$ for a generator $t$ of tame inertia.

## 7.8 Inertia invariants, local $L$-factors and conductors

The paper's local terms are built from three invariants of $V = V_\ell E$ and of $A_f = \mathrm{ad}^0 V$ at a bad prime: the inertia invariants (the paper's $\delta_p = \dim V^{I_p}$), the local $L$-factor, read with the geometric Frobenius, and the conductor exponent. This section defines them, proves that they depend only on the Weil-Deligne representation, and fixes the Frobenius convention for $L$-factors. The paper uses them in Section 1.1 (the definitions of $\delta_p$, $L_p(A_f, s)$ and $L_p^{\mathrm{nv}}$), in Definition def:terms, Lemma lem:euler and the table of Section 1, in Lemma lem:fpr ($L_p(A_f, 1) = L_p(A_f(1), 0)$), in the paragraph after Lemma lem:dfg21, and in Lemmas lem:twist and lem:IV and Remark rem:types (conductor exponents).

**Proposition 7.76** (inertia invariants)**.** *Let $\rho\colon G_K \to \mathrm{GL}(V)$ be continuous, $\ell \neq p$, and $\mathrm{WD}(\rho) = (r, N)$ as in Theorem 7.71. Then*

$$V^{\rho(I_K)} = (\ker N)^{r(I_K)},$$

*and $\rho(\Phi) = r(\Phi)$ on this space. So $\dim V^{I_K}$ and the characteristic polynomial of $\Phi$ on $V^{I_K}$ depend only on the isomorphism class of $\mathrm{WD}(\rho)$.*

*Proof.* For $\sigma \in I_K$, $\rho(\sigma) = r(\sigma)\exp(t_\zeta(\sigma)N)$. If $Nv = 0$ and $r(\sigma)v = v$ for all $\sigma$, then $\rho(\sigma)v = v$. Conversely let $v$ be fixed by $\rho(I_K)$. Take $\sigma_1 \in I_1$ with $t_\zeta(\sigma_1) = \ell^m \neq 0$; then $v = \exp(\ell^m N)v$, and $\exp(\ell^m N) - 1 = \ell^m N U$ with $U = \sum_{k\geq 0}(\ell^m N)^k/(k+1)!$ unipotent and commuting with $N$, so $NUv = UNv = 0$ and $Nv = 0$. Then $r(\sigma)v = \rho(\sigma)\exp(-t_\zeta(\sigma)N)v = \rho(\sigma)v = v$. Finally $r(\Phi) = \rho(\Phi)$ by definition. An isomorphism of Weil-Deligne representations carries $(\ker N)^{r(I_K)}$ to the corresponding space and intertwines $r(\Phi)$. □

For example, $(\ker N)^{r(I_K)}$ is $\Omega e_1$ for $\mathrm{sp}(2) \otimes \mu$ with $\mu$ unramified, and 0 for $\mathrm{sp}(2) \otimes \mu$ with $\mu$ ramified ($r(I_K)$ acts on $\ker N = \Omega e_1$ by the nontrivial $\mu|_{I_K}$); for $(r, 0)$ it is $V^{r(I_K)}$. The paper's $\delta_p$ is $\dim V^{I_p}$ for $V = V_\ell E$ (Section 1.1).

**Definition 7.77** (local $L$-factors)**.** For $\rho$ as in Proposition 7.76, or a Weil-Deligne representation $(r, N)$, the *local $L$-factor* is

$$L(V, s) = \det\bigl(1 - \Phi\, q^{-s} \mid V^{I_K}\bigr)^{-1} = \det\bigl(1 - r(\Phi)\, q^{-s} \mid (\ker N)^{r(I_K)}\bigr)^{-1},$$

with $\Phi = \mathrm{Frob}^{-1}$ the geometric Frobenius. For $K = \mathbb{Q}_p$ and a representation $M$ of $G_{\mathbb{Q}}$ we write $L_p(M, s)$; this is the paper's $L_p(\cdot, s)$ of Section 1.1 (Appendix B, convention B-C2(d)).

**Proposition 7.78.** *(i) $L(V, s)$ is well defined (any two geometric Frobenius lifts agree on $V^{I_K}$) and depends only on the isomorphism class of $\mathrm{WD}(V)$.*

*(ii) $L(V \oplus V', s) = L(V, s)L(V', s)$.*

*(iii) For $\mu$ unramified with $\mu(\Phi) = \lambda$, $L(\mu, s) = (1 - \lambda q^{-s})^{-1}$. In particular $L(\mathbb{Q}_\ell(1), s) = (1 - q^{-1-s})^{-1}$, $L(V(1), s) = L(V, s+1)$ and $L(V \otimes \|\cdot\|^k, s) = L(V, s+k)$.*

*(iv) With the arithmetic* Frob *in place of $\Phi$ one would get $(1-q^{1-s})^{-1}$ for $\mathbb{Q}_\ell(1)$ and $L(V(1), s) = L(V, s-1)$.*

(v) $L(\mathrm{sp}(2)\otimes\mu,s)=(1-\mu(\Phi)q^{-1-s})^{-1}$ *for* $\mu$ *unramified, and* $=1$ *for* $\mu$ *ramified. If* $E/\mathbb{Q}$ *has good reduction at* $p\neq\ell$ *and* $\alpha,\beta$ *are the roots of* $X^2-a_pX+p$*, the arithmetic Frobenius on* $V_\ell E$ *has eigenvalues* $\alpha,\beta$ *(Chapter 9), the geometric one* $\alpha^{-1},\beta^{-1}$*, and*

$$L_p(V_\ell E,s)=(1-a_pp^{-1-s}+p^{-1-2s})^{-1}=L_p(E,s+1),$$
$$L_p(V_\ell E^\vee,s)=(1-a_pp^{-s}+p^{1-2s})^{-1}=L_p(E,s).$$

*Proof.* (i) Two lifts differ by an element of $I_K$, which acts trivially on $V^{I_K}$; $V^{I_K}$ is $\Phi$-stable because $I_K$ is normal. The second expression and Proposition 7.76 give the independence. (ii) Invariants and determinants are additive. (iii) $I_K$ acts trivially on $\mathbb{Q}_\ell(1)$ and $\Phi$ by $\chi_\ell(\Phi)=q^{-1}$; $V(1)^{I_K}=V^{I_K}(1)$, on which $\Phi$ acts by $q^{-1}$ times its action on $V^{I_K}$, so $\det(1-\Phi q^{-s}\mid V(1)^{I_K})=\det(1-\Phi q^{-s-1}\mid V^{I_K})$. The same holds for $\|\cdot\|^k$. (iv) Frob acts on $\mathbb{Q}_\ell(1)$ by $q$. (v) The first statement is Proposition 7.76 and the example after it, $\Phi$ acting on $e_1$ by $\mu(\Phi)\|\Phi\|=\mu(\Phi)q^{-1}$. For $E$: $\det(1-\Phi X\mid V_\ell E)=(1-X/\alpha)(1-X/\beta)=1-a_pX/p+X^2/p$; with $X=p^{-s}$ this is $1-a_pp^{-1-s}+p^{-1-2s}$, which is $1-a_pp^{-(s+1)}+p^{1-2(s+1)}$. On $V^\vee$ the geometric Frobenius has the inverse eigenvalues $\alpha,\beta$, giving $1-a_pX+pX^2$. □

*Warning* 7.79 (the paper's $L$-factors are geometric). The paper's $L_p(A_f,s)$, the table of Section 1, Lemma lem:euler, Lemma lem:fpr ($L_p(A_f,1)^{-1}=L_p(A_f(1),0)^{-1}$), the values of PARI's `lfunsympow` and the conventions of DFG04 [49, Section 1.7.2] all use the geometric Frobenius: by Proposition 7.78(iii), (iv) the identity $L_p(M(1),s)=L_p(M,s+1)$ of Lemma lem:fpr holds for $\Phi$ and fails for Frob, and at a multiplicative prime the factor $1-q^{-1-s}$ of the table requires $\Phi$ (Proposition 7.88(iii)). On $\mathrm{ad}^0V$ at a good prime the eigenvalues $\alpha/\beta,1,\beta/\alpha$ are stable under inversion, so there the choice is invisible.

*In the paper* 7.80. Section 1.1 defines $L_p(A_f,s)=\det(1-\mathrm{Frob}_p^{-1}p^{-s}\mid A_f^{I_p})^{-1}$: this is Definition 7.77 for $M=A_f$, the geometric Frobenius at $p$ being $\mathrm{Frob}_p^{-1}$. The naive factor $L_p^{\mathrm{nv}}(A_f,s)$ is $L_p(A_f,s)$ when $\delta_p=\dim V^{I_p}>0$ and 1 when $\delta_p=0$. The paragraph after Lemma lem:dfg21 writes "$\det(1-\mathrm{Frob}_p^{-1}\mid(V^D)^{I_p})=L_p(A_f,1)^{-1}\neq0$" with $V^D=A_f(1)$: the left side is $L_p(A_f(1),0)^{-1}$, which equals $L_p(A_f,1)^{-1}$ by Proposition 7.78(iii).

### 7.8.1 Swan and Artin conductors

**Definition 7.81** (Swan and Artin conductors)**.** Let $V$ be a finite-dimensional representation over a field $\Omega$ of $G_K$, $W_K$ or $I_K$, on which $I_K$ acts through a finite quotient. Choose a finite Galois $L/K$ such that $I_K\cap G_L$ acts trivially (possible because the kernel is open in $I_K$), so that $I_K$ acts through its image $G_0=I(L/K)$ in $\mathrm{Gal}(L/K)$ (Theorem 7.45(i)); let $G_i=G_i(L/K)$ and $g_i=\#G_i$. The *Swan conductor* and the *Artin conductor* (exponent) are

$$\mathrm{sw}(V)=\sum_{i\geq1}\frac{g_i}{g_0}\dim(V/V^{G_i}),\qquad a(V)=\sum_{i\geq0}\frac{g_i}{g_0}\dim(V/V^{G_i})=\dim V-\dim V^{I_K}+\mathrm{sw}(V).$$

For a continuous $\ell$-adic $\rho$ ($\ell\neq p$), $P_K$ acts through a finite quotient (Proposition 7.53), and $\mathrm{sw}(\rho)$ is defined by the same sum, which involves only the $G_i$ with $i\geq1$, the images of $P_K$.

**Theorem 7.82** (properties of the conductors)**.** *(i) (Stated.)* $\mathrm{sw}(V)$ *and* $a(V)$ *do not depend on the choice of* $L$ *(by Herbrand's theorem), and in upper numbering*

$$\mathrm{sw}(V)=\int_0^\infty\dim(V/V^{G_K^u})\,du.$$

*(ii) (Stated; Artin.)* $\mathrm{sw}(V)$ *and* $a(V)$ *are nonnegative integers.*

*(iii)* $\mathrm{sw}(V) = 0$ *iff* $P_K$ *acts trivially on* $V$*;* $\mathrm{sw}(V)$ *depends only on* $V|_{P_K}$*; both are additive in direct sums, unchanged by twisting by an unramified character and by restriction to* $G_{K'}$ *for* $K'/K$ *finite unramified; and if* $V$ *is tamely ramified,* $a(V) = \dim V - \dim V^{I_K}$*.*

*(iv) For a character* $\chi$*:* $a(\chi) = 0$ *if* $\chi$ *is unramified,* $a(\chi) = 1$ *if it is tamely ramified, and* $a(\chi) = 1 + (\text{the largest } i \text{ with } \chi(G_i) \neq 1)$ *for the quadratic characters of Example 7.43. Hence* $a(\chi_{-3}|_{G_{\mathbb{Q}_3}}) = 1$*,* $a(\chi_{q^*}|_{G_{\mathbb{Q}_q}}) = 1$ *for* $q$ *odd,* $a(\chi_{-4}|_{G_{\mathbb{Q}_2}}) = 2$ *and* $a(\chi_{\pm 8}|_{G_{\mathbb{Q}_2}}) = 3$*, in agreement with Theorem 7.62(iii) and with the conductors* $3$*,* $q$*,* $4$*,* $8$*. (Stated: for a general wildly ramified character,* $a(\chi) = 1 + c$ *with* $c$ *the largest upper jump.)*

*(v) The same definitions apply verbatim to representations over* $\mathbb{F}_\ell$*, where the image of inertia is finite; this gives the local factor at* $q$ *of the Serre conductor of* $\bar{\rho}_{E,\ell}$ *(Chapter 11).*

*Proof of (iii), (iv); references for (i), (ii).* (iii) If $P_K$ acts trivially, so does each $G_i$, $i \geq 1$, and all terms vanish. If $\mathrm{sw}(V) = 0$, then either $G_1 = 1$, or the term $i = 1$ (with $g_1/g_0 > 0$) vanishes and $G_1$ acts trivially; in both cases $P_K$, which maps onto $G_1$, acts trivially. The sum for $i \geq 1$ involves only $V|_{G_1}$, the image of $V|_{P_K}$. Additivity holds because $(V \oplus V')^{G_i} = V^{G_i} \oplus V'^{G_i}$. An unramified character is trivial on $I_K$, so it does not change any $V^{G_i}$. If $K'/K$ is unramified and $L' = LK'$, the inertia group of $L'/K$ fixes $K'$, so the ramification groups of $L'/K'$, which are $G_i(L'/K) \cap \mathrm{Gal}(L'/K')$ (Proposition 7.42(i)), equal $G_i(L'/K)$ for $i \geq 0$; by (i) one may compute both conductors with $L'$. If $V$ is tame, $P_K$ acts trivially and $\mathrm{sw} = 0$. (iv) For a character, $\dim V/V^{G_i}$ is 1 if $\chi(G_i) \neq 1$ and 0 otherwise. If $\chi$ is tamely ramified, $\chi(G_0) \neq 1 = \chi(G_1)$ and $a = 1$. For the quadratic characters of Example 7.43 the field $L$ is the quadratic extension itself, $g_i/g_0 = 1$ while $G_i = G$; so $a = 1 + 1 = 2$ for the jump 1 ($\mathbb{Q}_2(i)$, $\mathbb{Q}_2(\sqrt{3})$) and $a = 1 + 2 = 3$ for the jump 2 ($\mathbb{Q}_2(\sqrt{\pm 2})$, $\mathbb{Q}_2(\sqrt{\pm 10})$). The values for the global characters follow from Proposition 7.61(ii), (iii). (i) and (ii) are Serre [174, Chapter IV, Section 3, and Chapter VI, Section 2]; the integrality is Artin's theorem, proved there through the Artin character (see also [173, Chapter 19]). (v) is a definition. □

**Definition 7.83** (conductor of a Weil-Deligne representation)**.** The conductor exponent of a Weil-Deligne representation $(r, N, V)$ is

$$a(r, N) = a(r) + \dim V^{r(I_K)} - \dim(\ker N)^{r(I_K)} = \dim V - \dim(\ker N)^{r(I_K)} + \mathrm{sw}(r),$$

and for a continuous $\ell$-adic $\rho$, $a(\rho) = a(\mathrm{WD}(\rho))$.

**Proposition 7.84.** *(i)* $a(\rho) = \dim V - \dim V^{\rho(I_K)} + \mathrm{sw}(\rho)$*, where* $\mathrm{sw}(\rho) = \mathrm{sw}(\rho|_{P_K})$*.*

*(ii)* $a$ *is additive in direct sums, unchanged by unramified twists and by unramified base change,* $a(\rho) = 0$ *iff* $\rho$ *is unramified, and* $a(\rho) = \dim V - \dim V^{I_K}$ *if* $\rho$ *is tamely ramified.*

*(iii) If* $V$ *is unramified and* $\chi$ *is a character,* $a(V \otimes \chi) = \dim V \cdot a(\chi)$*. If* $\chi$ *is ramified and* $\mu$ *unramified,* $a(\mathrm{sp}(2) \otimes \mu) = 1$ *and* $a(\mathrm{sp}(2) \otimes \mu\chi) = 2a(\chi)$*.*

*(iv) For an elliptic curve* $E/\mathbb{Q}$ *and* $\ell \neq p$*, Chapter 10 defines the exponent of* $p$ *in the conductor of* $E$ *to be* $a(V_\ell E|_{G_{\mathbb{Q}_p}})$*; it does not depend on* $\ell$*. (Stated.) Ogg's formula (Saito in general) gives it as* $v_p(\Delta_{\min}) + 1 - m$*, where* $m$ *is the number of irreducible components over* $\overline{\mathbb{F}}_p$ *of the special fibre of the minimal regular model of* $E$ *over* $\mathbb{Z}_p$ *(Chapter 10).*

*Proof.* (i) By Proposition 7.76, $(\ker N)^{r(I_K)} = V^{\rho(I_K)}$; and $r = \rho$ on $P_K$, because $t_\zeta$ vanishes on $P_K$; so $\mathrm{sw}(r) = \mathrm{sw}(\rho)$. (ii) follows from (i), Theorem 7.82(iii) and Proposition 7.38; $a(\rho) = 0$ forces $V^{I_K} = V$. (iii) $I_K$ acts on $V \otimes \chi$ by the scalars $\chi(\sigma)$, so $(V \otimes \chi)^{G_i}$ is $V \otimes \chi$ or $0$ according as $\chi(G_i) = 1$ or not, and each term of the sum is $\dim V$ times that for $\chi$. For $\mathrm{sp}(2) \otimes \mu$, $r$ is unramified and $\dim(\ker N)^{r(I_K)} = 1$, so $a = 2 - 1 + 0 = 1$. For $\mathrm{sp}(2) \otimes \mu\chi$, $r = (\mu\chi \oplus \|\cdot\|\mu\chi)$ has $a(r) = 2a(\chi)$ by the previous case, and $(\ker N)^{r(I_K)} = V^{r(I_K)} = 0$, so $a = 2a(\chi)$. (iv) The first sentence is a definition. We do not prove the independence of $\ell$ or Ogg's formula; see [179, Section 2], Ogg [145] and Chapter 10, where Saito's general case is cited. □

**Theorem 7.85** (epsilon factors)**.** *(Langlands, Deligne.) Fix a nontrivial additive character $\psi$ of $K$ and a Haar measure $dx$ on $K$. There is a unique way to attach to each representation $r$ of $W_K$ over $\mathbb{C}$ with $r(I_K)$ finite a constant $\epsilon(r, \psi, dx) \in \mathbb{C}^\times$ that is multiplicative in short exact sequences, inductive in degree $0$, and equal to Tate's local constant for characters. For a Weil-Deligne representation one puts*

$$\epsilon(r, N, \psi, dx) = \epsilon(r, \psi, dx)\, \det\!\big(-r(\Phi) \mid V^{r(I_K)}/(\ker N)^{r(I_K)}\big).$$

*We do not prove this; see Deligne [41, Sections 4, 5 and 8] and Tate [194, Sections 3 and 4] (Deligne's normalisation of the Artin map is the geometric one, Warning 7.63). The paper does not use epsilon factors; Chapters 15 and 19 recall this statement for root numbers and functional equations.*

## 7.9 The local shapes and their adjoints

This section computes, for each shape of local representation that the paper meets, the inertia invariants, the conductor exponent and the $L$-factor of $V$ and of its trace-zero adjoint $A = \mathrm{ad}^0 V$. It is the Galois-theoretic content of Lemma lem:euler, of the table of Section 1, of Proposition prop:shapes and of the conductor statements in Lemmas lem:IV and lem:twist. The linear algebra of the inertia actions (unipotent elements, commutants, the table of cyclic inertia shapes) is Chapter 4; the dictionary between shapes and Kodaira types is proved in Chapters 10 and 11 and only quoted here.

Standing hypotheses: $K = \mathbb{Q}_q$, $\ell \neq q$, $V$ a two-dimensional representation of $G_{\mathbb{Q}_q}$ over a finite extension $\Omega$ of $\mathbb{Q}_\ell$ with $\det V|_{I_q} = 1$ (true for $V_\ell E$, whose determinant $\chi_\ell$ is unramified at $q$), $A = \mathrm{ad}^0 V$ with $g \cdot x = \rho(g) x \rho(g)^{-1}$, and $\Phi$ the geometric Frobenius. In the basis $e_1, e_2$ of Example 7.74, $e = e_1 \otimes e_2^\vee$, $h = e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee$, $f = e_2 \otimes e_1^\vee$ is the paper's basis of $\mathrm{ad}^0 T$ (here $e$ is a vector, not a ramification index).

**Proposition 7.86** (unramified $V$)**.** *If $V$ is unramified with arithmetic Frobenius eigenvalues $\alpha, \beta$, then $\mathrm{WD}(V) = (r, 0)$, $a(V) = 0$, $L(V, s)^{-1} = (1 - \alpha^{-1}q^{-s})(1 - \beta^{-1}q^{-s})$, and $A$ is unramified with eigenvalues $\alpha/\beta, 1, \beta/\alpha$ for both* Frob *and $\Phi$. Hence*

$$L(A, s)^{-1} = \Big(1 - \frac{\alpha}{\beta}q^{-s}\Big)(1 - q^{-s})\Big(1 - \frac{\beta}{\alpha}q^{-s}\Big), \qquad L(A, 1)^{-1} = \frac{q-1}{q} \cdot \frac{(q+1)^2 - a^2}{q^2},$$

*the last formula when $\alpha + \beta = a$ and $\alpha\beta = q$.*

*Proof.* $N = 0$ since $\rho(I_q) = 1$ is finite, and $a = 0$ by Proposition 7.84(ii). The eigenvalues of $\Phi$ on $V$ are $\alpha^{-1}, \beta^{-1}$. If Frob has eigenvectors $v_1, v_2$ (over an extension of $\Omega$, or after triangularising), it acts on $v_i \otimes v_j^\vee$ by $\alpha_i/\alpha_j$; the trace-zero part has eigenvalues $\alpha/\beta$, $1$, $\beta/\alpha$, a set stable under

inversion. For $s=1$, $(1-\frac{\alpha}{\beta q})(1-\frac{\beta}{\alpha q}) = 1-\frac{\alpha^2+\beta^2}{\alpha\beta q}+\frac{1}{q^2} = 1-\frac{a^2-2q}{q^2}+\frac{1}{q^2} = \frac{(q+1)^2-a^2}{q^2}$ (Chapter 2), using $\alpha^2+\beta^2 = (\alpha+\beta)^2-2\alpha\beta = a^2-2q$; the middle factor contributes $1-q^{-1} = \frac{q-1}{q}$. □

For example, `11a1` has good reduction at 2 with $a_2=-2$ (PARI, `ellap`), so $L_2(A,1)^{-1} = \frac{1}{2}\cdot\frac{9-4}{4} = \frac{5}{8}$. Note that $(q+1)^2-a^2 = (q+1-a)(q+1+a)$, and $q+1-a$ is the number of points of the reduced curve over $\mathbb{F}_q$ (Chapter 9); here $\#\tilde{E}(\mathbb{F}_2)=5$, and the factor 5 in $\frac{5}{8}$ is that count.

**Proposition 7.87** (finite cyclic tame image of inertia)**.** *Suppose $\rho(I_q)$ is cyclic of order $m \in \{2,3,4,6\}$ with $q \nmid m$, generated by $g$. Then $N=0$, the ramification is tame, and:*

*(i) If $m=2$, then $g=-1$, $V^{I_q}=0$, $a(V)=2$ and $L(V,s)=1$. For $q$ odd and any ramified quadratic character $\chi$ of $G_{\mathbb{Q}_q}$, $V\otimes\chi$ is unramified, and $A=\mathrm{ad}^0(V\otimes\chi)$ is unramified, with $L(A,s)$ as in Proposition 7.86 for $V\otimes\chi$.*

*(ii) If $m\in\{3,4,6\}$, $g$ has the eigenvalues $\zeta,\zeta^{-1}$ with $\zeta$ a primitive $m$-th root of unity; $V^{I_q}=0$, $a(V)=2$ and $L(V,s)=1$. Put $m'=3$, $\chi=\chi_{-3}$ if $m\in\{3,6\}$, and $m'=4$, $\chi=\chi_{-4}$ if $m=4$. Then $\rho(\mathrm{Frob})g\rho(\mathrm{Frob})^{-1}=g^{\chi(q)}$. If $\chi(q)=1$, $V\otimes\Omega(\zeta)$ is a sum of two characters of $G_{\mathbb{Q}_q}$ restricting to $\zeta^{\pm1}$ on $g$; if $\chi(q)=-1$, $\rho(\mathrm{Frob})$ exchanges the two eigenlines of $g$, and $V\otimes\Omega(\zeta)$ is induced from a character of $G_{K_2}$, $K_2$ the unramified quadratic extension of $\mathbb{Q}_q$.*

*(iii) In case (ii), $A^{I_q}$ is the line spanned by $2g-\operatorname{tr} g$ (which is $\pm(2\sigma+1)$, of square $-3$, for $m\in\{3,6\}$ with $\sigma$ the element of order 3 in $\langle g\rangle$, and $g$ itself for $m=4$), and both* Frob *and* $\Phi$ *act on it by $\chi(q)\in\{\pm1\}$. Hence*

$$L(A,s)^{-1} = 1-\chi(q)\,q^{-s}, \qquad a(A)=2.$$

*Proof.* $\rho(I_q)$ is finite, so $N=0$ (Theorem 7.68), and of order prime to $q$, so $\rho(P_q)=1$ (Corollary 7.52). Since $\det g=1$, the eigenvalues of $g$ (semisimple, of finite order in characteristic 0) are $\lambda,\lambda^{-1}$ with $\lambda$ of order $m$. (i) For $m=2$, $\lambda=\lambda^{-1}$ forces $\lambda=-1$ and $g=-1$. In all cases 1 is not an eigenvalue, so $V^{I_q}=0$, $L(V,s)=1$ and $a(V)=2-0=2$ (tame). If $q$ is odd and $\chi$ ramified quadratic, then $\chi|_{I_q}=\theta_2$ (Proposition 7.60(ii)), and $\rho|_{I_q}$ is a character of $I_q$ of order 2 times the identity, so it is $\theta_2\cdot 1$; hence $(V\otimes\chi)|_{I_q}=\theta_2^2=1$. Twisting by a character does not change $\mathrm{ad}^0$. (ii) By Corollary 7.52, $\rho(\mathrm{Frob})g\rho(\mathrm{Frob})^{-1}=g^q$. For $m\in\{3,4\}$, $q\equiv\chi(q) \pmod{m'}$ and $g^q=g^{\chi(q)}$, as $g$ has order $m=m'$. For $m=6$, $q$ is prime to 6, so $q\equiv\pm1 \pmod 6$ and again $g^q=g^{\pm1}$ with the sign $\chi_{-3}(q)$. If $\rho(\mathrm{Frob})$ commutes with $g$, it preserves the two eigenlines of $g$ in $V\otimes\Omega(\zeta)$, and so does $\rho(I_q)=\langle g\rangle$; by Corollary 7.46 these lines are stable under $G_{\mathbb{Q}_q}$. If $\rho(\mathrm{Frob})g\rho(\mathrm{Frob})^{-1}=g^{-1}$, $\rho(\mathrm{Frob})$ maps the $\zeta$-eigenline $D$ to the $\zeta^{-1}$-eigenline; $D$ is stable under the index-two subgroup $G_{K_2}=I_q\rtimes\overline{\langle\mathrm{Frob}^2\rangle}$, and $V\otimes\Omega(\zeta)=D\oplus\rho(\mathrm{Frob})D$ is induced from the character of $G_{K_2}$ on $D$. (iii) The commutant of $g$ in $\mathrm{End}\,V$ is $\Omega+\Omega g$, since $g$ has distinct eigenvalues (Chapter 4), so its trace-zero part is spanned by $2g-\mathrm{tr}(g)$; and $A^{I_q}$ is that trace-zero commutant. By Cayley-Hamilton, $g^{-1}=\mathrm{tr}(g)-g$, so $2g^{-1}-\operatorname{tr} g=-(2g-\operatorname{tr} g)$; therefore Frob, which conjugates $g$ to $g^{\chi(q)}$, acts on the line by $\chi(q)$, and so does $\Phi=\mathrm{Frob}^{-1}$. For $m=3$, $\operatorname{tr}\sigma=-1$ and $(2\sigma+1)^2=4(\sigma^2+\sigma)+1=-3$; for $m=6$, $g=-\sigma$; for $m=4$, $\operatorname{tr} g=0$. Finally $A$ is tame (its inertia image is a quotient of $\langle g\rangle$), so $a(A)=3-1=2$. □

**Proposition 7.88** (Steinberg and its twists)**.** *Let $\mu$ be an unramified and $\chi$ a ramified quadratic character of $G_{\mathbb{Q}_q}$.*

*(i)* $V = \mathrm{sp}(2) \otimes \mu$*:* $V^{I_q} = \ker N$ *is a line on which* $\Phi$ *acts by* $\mu(\Phi)q^{-1}$ *(and* Frob *by* $\mu(\mathrm{Frob})q$*);* $a(V) = 1$*;* $L(V,s) = (1 - \mu(\Phi)q^{-1-s})^{-1}$*.*

*(ii)* $V = \mathrm{sp}(2) \otimes \mu\chi$*:* $V^{I_q} = 0$*,* $L(V,s) = 1$ *and* $a(V) = 2a(\chi)$*, which is* 2 *for* $q$ *odd and* 4 *or* 6 *for* $q = 2$*.*

*(iii) For* $V = \mathrm{sp}(2) \otimes \mu$ *and* $V = \mathrm{sp}(2) \otimes \mu\chi$*,* $\mathrm{ad}^0 V = \mathrm{ad}^0 \mathrm{sp}(2) \simeq \mathrm{sp}(3) \otimes \|\cdot\|^{-1}$*. In the basis* $E_{01}$*,* $h = E_{00} - E_{11}$*,* $E_{10} = N$ *(with* $E_{ij} = e_i \otimes e_j^\vee$ *in the basis of Definition 7.72),* $\Phi$ *acts by* $q, 1, q^{-1}$*, and* $[N, E_{01}] = -h$*,* $[N,h] = 2N$*,* $[N,N] = 0$*. So* $A^{I_q} = \ker[N,\cdot\,] = \Omega N$*,* $\Phi$ *acts on it by* $q^{-1}$ *(*Frob *by* $q$*),* $a(A) = 2$*, and*

$$L(\mathrm{ad}^0 V, s) = (1 - q^{-1-s})^{-1}.$$

*With the arithmetic Frobenius one would get* $(1 - q^{1-s})^{-1}$*.*

*(iv) The line* $\Omega N$ *is the line of* $e = e_1 \otimes e_2^\vee$ *in the basis of Example 7.74, the line* $\mathbb{Z}_\ell e$ *of the proof of Lemma lem:tamfactor.*

*Proof.* (i) and (ii) are Proposition 7.76, Proposition 7.78(v) and Proposition 7.84(iii), with $a(\chi) = 1$ for $q$ odd and $a(\chi) \in \{2,3\}$ for $q = 2$ (Theorem 7.82(iv)). (iii) Twisting by a character does not change $\mathrm{ad}^0$. $r(\Phi)$ acts on $e_0, e_1$ by $1, q^{-1}$, hence on $E_{ij}$ by the ratio of the eigenvalues of $e_i$ and $e_j$: $q$ on $E_{01}$, 1 on $h$, $q^{-1}$ on $E_{10}$. The brackets: $E_{10}E_{01} - E_{01}E_{10} = E_{11} - E_{00} = -h$; $E_{10}h - hE_{10} = E_{10} + E_{10} = 2E_{10}$. So $f_0 = E_{01}$, $f_1 = -h$, $f_2 = -2E_{10}$ satisfy $N_{\mathrm{ad}^0} f_i = f_{i+1}$, with $\Phi$-eigenvalues $q^{1-i} = \|\Phi\|^{i-1}$: this is $\mathrm{sp}(3) \otimes \|\cdot\|^{-1}$. In characteristic 0, $\ker N_{\mathrm{ad}^0} = \Omega E_{10} = \Omega N$, and $r(I_q)$ acts trivially on $\mathrm{ad}^0$ (it acts on $V$ by scalars), so $A^{I_q} = \Omega N$ (Proposition 7.76), with $\Phi$-eigenvalue $q^{-1}$. $a(A) = 3 - 1 + 0 = 2$. (iv) In Example 7.74, $N = aE_{12}$ maps $e_2$ to $e_1$, so $N$ is a multiple of $e_1 \otimes e_2^\vee$. □

**Theorem 7.89** (the shapes at a tame prime)**.** *Under the standing hypotheses, suppose that* $q \geq 5$*. (Columns* 1 *to* 7*, which do not involve Kodaira types, hold more generally, with the same proofs: rows* 1 *and* 3 *for every* $q$*, rows* 2 *and* 4 *for every odd* $q$*, and rows* 5 *and* 6 *whenever* $q \nmid \#\rho(I_q)$*. The Kodaira column is asserted only for* $q \geq 5$*.) The following table lists, for each shape of* $\mathrm{WD}(V)$*, the image of inertia,* $\dim V^{I_q}$*,* $a(V)$*,* $L(V,s)^{-1}$*,* $\dim A^{I_q}$ *and* $L(A,s)^{-1}$ *(*$\Phi$ *geometric); the last column, the Kodaira types of an elliptic curve with* $V = V_\ell E$*, is quoted from Chapters 10 and 11.*

| $\mathrm{WD}(V)$ | $\rho(I_q)$ | $\delta_q$ | $a(V)$ | $L(V,s)^{-1}$ | $\dim A^{I_q}$ | $L(A,s)^{-1}$ | *types* |
|---|---|---|---|---|---|---|---|
| *unramified* | 1 | 2 | 0 | $(1 - \frac{X}{\alpha})(1 - \frac{X}{\beta})$ | 3 | $(1 - \frac{\alpha}{\beta}X)(1 - X)(1 - \frac{\beta}{\alpha}X)$ | $\mathrm{I}_0$ |
| *unram.* $\otimes\, \chi$ | $\{\pm 1\}$ | 0 | 2 | 1 | 3 | *as in row* 1 | $\mathrm{I}_0^*$ |
| $\mathrm{sp}(2) \otimes \mu$ | *unipotent* | 1 | 1 | $1 - \mu(\Phi)q^{-1}X$ | 1 | $1 - q^{-1}X$ | $\mathrm{I}_n$ |
| $\mathrm{sp}(2) \otimes \mu\chi$ | $\pm$*unipotent* | 0 | 2 | 1 | 1 | $1 - q^{-1}X$ | $\mathrm{I}_n^*$ |
| *order* 3 *or* 6 | $\langle\sigma\rangle$, $\langle -\sigma\rangle$ | 0 | 2 | 1 | 1 | $1 - \chi_{-3}(q)X$ | $\mathrm{IV}, \mathrm{IV}^*$; $\mathrm{II}, \mathrm{II}^*$ |
| *order* 4 | $\langle\tau\rangle$ | 0 | 2 | 1 | 1 | $1 - \chi_{-4}(q)X$ | $\mathrm{III}, \mathrm{III}^*$ |

*Here* $X = q^{-s}$*,* $\delta_q = \dim V^{I_q}$*,* $\alpha, \beta$ *are the arithmetic Frobenius eigenvalues of the unramified representation (in the second row, of* $V \otimes \chi$*),* $\mu$ *is unramified,* $\chi$ *is a ramified quadratic character,* $N = 0$ *in the last two rows, and* $n \geq 1$*. So* $\delta_q$ *is* $2, 0, 1, 0, 0, 0$ *down the rows; the naive factor is* $L(A,s)$ *when* $\delta_q > 0$ *and* 1 *when* $\delta_q = 0$*; and the factor that the naive L-function omits is* $L(A,s)^{-1}$ *in rows* 2*,* 4*,* 5*,* 6 *and nothing in rows* 1*,* 3*.*

*Proof.* Every column but the last is Propositions 7.86 to 7.88. That these are all the shapes with $q \nmid \#\rho(I_q)$ when $\rho(I_q)$ is finite of order in $\{1,2,3,4,6\}$ follows from Proposition 7.73(ii) when $N \neq 0$ (and $\det V|_{I_q} = 1$ forces $\mu^2$ to be trivial on $I_q$, so $\mu$ is unramified or an unramified twist of a ramified quadratic character, Corollary 7.46), and from Chapter 4 when $N = 0$. The orders $1,2,3,4,6$ and the Kodaira column are Chapters 10 and 11 (Serre and Tate [179], Kraus [107]); we do not prove them here. □

*In the paper* 7.90. Lemma lem:euler and the table of Section 1 are the last column of Theorem 7.89 read through the Kodaira types: $L_q(A_f,s)^{-1}$ is $1-q^{-1-s}$ for $\mathrm{I}_n$ and $\mathrm{I}_n^*$, $1-\chi_{-3}(q)q^{-s}$ for $\mathrm{II},\mathrm{II}^*,\mathrm{IV},\mathrm{IV}^*$, $1-\chi_{-4}(q)q^{-s}$ for $\mathrm{III},\mathrm{III}^*$ and the good factor of the twist for $\mathrm{I}_0^*$; $e_\ell(E,q) = v_\ell(L_q(A_f,1)^{-1})$ in the rows with $\delta_q = 0$ gives $v_\ell(q^2-1)$, $v_\ell(q-\chi_{-3}(q))$, $v_\ell(q-\chi_{-4}(q))$ and $v_\ell(q-1)+v_\ell((q+1)^2-a_q(E')^2)$, and $e_\ell = 0$ in the rows with $\delta_q > 0$. "Frobenius centralises $\sigma$ if $q \equiv 1 \pmod 3$ and inverts it if $q \equiv 2$, for either choice of Frobenius" is Proposition 7.87(ii), (iii). "$(\mathrm{ad}^0 V)^{I_q}$ is a line on which the geometric Frobenius has eigenvalue $q^{-1}$" (type $\mathrm{I}_n$) is Proposition 7.88(iii). Proposition prop:shapes(i) to (iii) is the same computation organised by $P_q(X) = L_q(A_f, s-1)^{-1}$, $X = q^{-s}$: $1-X$ for the Steinberg rows, $(1-qX)(1-(a^2-2q)X+q^2X^2)$ for row 2, $1-cqX$ with $c = \chi(q)$ for rows 5 and 6. In Lemma lem:twist(iii), the fact that the twist of an unramified twist of the Steinberg representation by a ramified quadratic character has conductor exponent 2 ($q$ odd) is Proposition 7.88(ii). In Lemma lem:twist(ii), "the conductor exponents are the tame ones, $\dim V - \dim V^{I_q}$" holds for the exchanges listed there because twisting a tame representation by the tame $\chi_{q^*}$ keeps it tame (Proposition 7.61(v)); at $q = 3$ a general additive curve can be wild (Remark 7.54). In Lemma lem:IV, "the conductor exponent of $E$ is $\dim V - \dim V^{I_q} = 2$, there being no wild part" is row 5 with Proposition 7.84(ii), and "The Serre conductor exponent of $\bar\rho_{E,3}$ at $q$ is $\dim E[3] - \dim E[3]^{I_q} = 1$" is Theorem 7.82(v) for the tame mod 3 representation.

**Proposition 7.91** (the primes 2 and 3)**.** *Keep the standing hypotheses, with $q \in \{2,3\}$.*

*(i) If $q = 3$ and $\rho(I_3)$ is finite of order divisible by 3, then $\rho$ is wildly ramified and $a(V) \geq 3$. If moreover $\rho(I_3)$ is cyclic of order 3 or 6, $A^{I_3}$ is still the line spanned by $2\sigma+1$, and* Frob *still acts on it by $+1$ or $-1$; but the sign is no longer determined by $q$, since the tame relation is not available on wild inertia.*

*(ii) If $q = 2$ and $\rho(I_2)$ is cyclic of order 3, then Proposition 7.87 applies: $a(V) = 2$, $L(A,s)^{-1} = 1-\chi_{-3}(2)2^{-s} = 1+2^{-s}$, and $L(A,1)^{-1} = \frac{3}{2}$.*

*(iii) If $\rho(I_q)$ is finite and not abelian, then $V$ is an absolutely irreducible representation of $\rho(I_q)$, $A^{I_q} = 0$ and $L(A,s) = 1$.*

*(iv) For $q = 2$, $a(\mathrm{sp}(2) \otimes \mu\chi)$ is 4 or 6 according as $\chi$ has ramification jump 1 or 2.*

*Proof.* (i) $I_3/P_3$ has no quotient of order 3 (Theorem 7.47(ii)), so an element of order 3 in $\rho(I_3)$ comes from $P_3$ (Proposition 7.53(ii)), and $\mathrm{sw}(V) \geq \frac{g_1}{g_0}\dim(V/V^{G_1}) > 0$. An element of order 3 and determinant 1 has the eigenvalues $\zeta_3^{\pm 1}$ and fixes no vector, so $V^{I_3} = 0$; as sw is an integer (Theorem 7.82(ii)), $a(V) = 2+\mathrm{sw}(V) \geq 3$. The computation of the commutant in Proposition 7.87(iii) uses only the cyclicity. Frob normalises $\rho(I_3)$, hence its unique subgroup $\langle\sigma\rangle$ of order 3, so $\rho(\mathrm{Frob})\sigma\rho(\mathrm{Frob})^{-1} = \sigma^{\pm 1}$; as $2\sigma^{-1}+1 = 2\sigma^2+1 = -(2\sigma+1)$ (from $\sigma^2+\sigma+1=0$), Frob acts on the line by $\pm 1$. In the tame case the sign was read off from $g \mapsto g^q$, which used the tame relation; that is not available on wild inertia. (ii) $2 \nmid 3$, so the image is tame; $\chi_{-3}(2) = -1$.

(iii) A two-dimensional faithful representation of a finite non-abelian group in characteristic 0 that fixed a line over $\overline{\Omega}$ would be triangular, hence (being semisimple) diagonal and abelian; so it is absolutely irreducible, its commutant is the scalars (Schur, Chapter 4), and the trace-zero commutant is 0. (iv) is Proposition 7.84(iii) with Theorem 7.82(iv). □

*Remark* 7.92 (what the paper takes from this at 2 and 3). (a) For a curve of type IV or IV* at 2, the image of inertia on $T_\ell E$ is cyclic of order exactly 3; this is a theorem proved in Chapter 11 from $E[3]^{I_2} \simeq \Phi_2^{\mathrm{geom}}[3]$ and in Chapter 10 from the conductor exponent. With Proposition 7.91(ii) and $L(\mathrm{Sym}^2 f, s) = L(A_f, s-1)$ (Chapter 19), $P_2(X) = 1 + 2X$ and $e_3(E,2) = v_3(3/2) = 1$, as Lemma lem:IV and Remark rem:wild state.

(b) For type II or II* at 2 the Kodaira type does not determine the image of inertia. When the image is non-abelian ($Q_8$ or $\mathrm{SL}_2(\mathbb{F}_3)$, acting absolutely irreducibly), $P_2 = 1$ by Proposition 7.91(iii); but `336f1` has type II at 2 and $P_2 = 1 + 2X$, and `123211` has type II* at 2 and a cubic $P_2$. Remark rem:wild says this.

(c) At $q = 2$ a curve of type $\mathrm{I}_n^*$ may have potentially good reduction, so the Kodaira type does not determine the Weil-Deligne shape (Definition def:terms: $t_\ell = 0$ then); and in the potentially multiplicative case the conductor exponent is 4 or 6 by Proposition 7.91(iv).

(d) At $q = 3$ the image of inertia for types IV, IV* has order divisible by 3 (Chapters 10 and 11), so by (i) these types are wild; the paper says "At $q = 3$ the types IV, IV* are wildly ramified, and the lemma does not apply." This is why Lemma lem:IV excludes $q = 3$ and Lemma lem:euler assumes $q \geq 5$ for the potentially good rows.

## 7.10 Exercises

**Exercise 7.93.** Find the first six 5-adic digits of $1/3$ and of $-1/3$, and check, carry by carry, that their sum is 0. Which rational number has the 7-adic expansion $1 + 2\cdot 7 + 1\cdot 7^2 + 2\cdot 7^3 + \cdots$ (digits $1, 2$ repeated)? (Answers: $2, 3, 1, 3, 1, 3$ and $3, 1, 3, 1, 3, 1$; $-5/16$.)

**Exercise 7.94.** (a) Show that $\mathbb{Q}_p$ is not algebraically closed. (b) Show that $u \in \mathbb{Q}_p^\times$ lies in $\mathbb{Z}_p^\times$ iff $u$ is an $m$-th power for infinitely many integers $m \geq 1$ (use Theorem 7.20 and Theorem 7.18). (c) Deduce that every field isomorphism $\mathbb{Q}_p \to \mathbb{Q}_{p'}$ maps $\mathbb{Z}_p^\times$ onto $\mathbb{Z}_{p'}^\times$, and that $\mathbb{Q}_p \not\simeq \mathbb{Q}_{p'}$ as fields for $p \neq p'$. (Hints: (a) $X^2 - p$. (b) A unit is $[\bar{u}]u_1$, and $u_1$ is an $m$-th power for every $m$ prime to $p$; an $m$-th power $x$ has $m \mid v_p(x)$. (c) The characterisation in (b) uses only the field structure. A field isomorphism is the identity on $\mathbb{Q}$, and $p \notin \mathbb{Z}_p^\times$ while $p \in \mathbb{Z}_{p'}^\times$.)

**Exercise 7.95.** (a) Find the two square roots of $-1$ in $\mathbb{Z}_{13}$ modulo $13^3$. (b) Show that 2 has exactly one cube root in $\mathbb{Z}_5$ and find it modulo $5^3$. (c) Show that $X^3 - 3$ has no root in $\mathbb{Z}_7$. (d) Compute the Teichmüller lift $[2] \in \mathbb{Z}_7$ modulo $7^3$ and check that it is a root of $X^2 + X + 1$. (Answers: 239, 1958; 53; 324.)

**Exercise 7.96.** For $d \in \{-7, 6, 14, -15, 17, -3, 13\}$ decide whether $\mathbb{Q}_2(\sqrt{d})$ is $\mathbb{Q}_2$, the unramified quadratic extension, or ramified, and in the last case give the conductor exponent of its character. Compare with the 2-adic valuation of the discriminant of $\mathbb{Q}(\sqrt{d})$. (Answer: $\mathbb{Q}_2$ for $-7$, $-15$, $17$ ($d \equiv 1 \pmod 8$); unramified for $-3$, $13$ ($d \equiv 5 \pmod 8$); ramified with conductor exponent 3 for $6 \in -10(\mathbb{Q}_2^\times)^2$ and $14 \in -2(\mathbb{Q}_2^\times)^2$. The discriminants $-7, 24, 56, -15, 17, -3, 13$ have 2-adic valuations $0, 3, 3, 0, 0, 0, 0$: in every case the conductor exponent.)

**Exercise 7.97.** Let $p$ be odd. (a) Show that $(1+p\mathbb{Z}_p)^p = 1+p^2\mathbb{Z}_p$ and that $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^p \simeq (\mathbb{Z}/p)^2$. (b) Compare with the formula $\#K^\times/(K^\times)^{\ell^k} = \ell^k \cdot \#\mu_{\ell^k}(K)$ of Theorem 7.20(iv), valid for $\ell \neq p$; which factor is missing when $\ell = p$? (c) Show that log is not injective on $1+2\mathbb{Z}_2$, and explain why this does not contradict Theorem 7.18(ii). (Hints: (a) log carries $1+p\mathbb{Z}_p$ onto $p\mathbb{Z}_p$ and $p$-th powers to $p\mathbb{Z}_p \cdot p = p^2\mathbb{Z}_p$. (b) The factor coming from the $\mathbb{Z}_p$ in $\mathbb{Z}_p^\times \simeq \mathbb{Z}/(p-1)\times\mathbb{Z}_p$; for $\ell \neq p$ the corresponding $U^{(1)}$ is $\ell$-divisible. (c) $\log(-1) = \frac{1}{2}\log((-1)^2) = 0$; the theorem concerns $1+4\mathbb{Z}_2$ only.)

**Exercise 7.98.** Determine $\mathbb{Q}_{13}^\times/(\mathbb{Q}_{13}^\times)^3$ and $\mathbb{Q}_{13}^\times/(\mathbb{Q}_{13}^\times)^9$, and decide which of 2, 3, 5, 8, 65 are cubes in $\mathbb{Q}_{13}$. (Answer: $\mathbb{Z}/3\times\mathbb{Z}/3$ and $\mathbb{Z}/9\times\mathbb{Z}/3$, as $v_3(12) = 1$. The cubes of $\mathbb{F}_{13}^\times$ are $\{1,5,8,12\}$, so 5 and 8 are cubes; 2 and 3 are not; $65 = 13\cdot 5$ is not, as $3 \nmid v_{13}(65)$.)

**Exercise 7.99.** Let $E =$ `11a1`, which has split multiplicative reduction at 11 with $n_{11} = v_{11}(\Delta_E) = 5$. Using Theorem 7.58(v) and Example 7.74, show that $I_{11}$ acts trivially on $E[5]$ but not on $E[25]$ (the paper's Lemma lem:tamagawa-exp). Using Example 7.22(c), show more: $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ as a $G_{\mathbb{Q}_{11}}$-module. (Hint: $E[\ell^k]$ is an extension of $\mathbb{Z}/\ell^k$ by $\mu_{\ell^k}$ whose class is $\kappa_{q_E} \bmod \ell^k$; $q_E$ is a fifth power in $\mathbb{Q}_{11}$, so by Proposition 7.57(i) the cocycle modulo 5 is a coboundary, and the extension splits.)

**Exercise 7.100.** Redo Example 7.55 for $(q,\ell,k) = (5,3,2)$ and $(11,3,2)$: give the order of $\mathrm{Gal}(\mathbb{Q}_q(\mu_9, q^{1/9})/\mathbb{Q}_q)$, and compute $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1}$ and $\mathrm{Frob}^{-1}\, t\; \mathrm{Frob}$ as powers of $t$. Then show directly, in the affine model, that $\Phi\, t\, \Phi^{-1} = t^{q'}$ for the geometric Frobenius $\Phi = \mathrm{Frob}^{-1}$. (Answers: 5 and $11 \equiv 2$ both have order 6 modulo 9, so both groups have order 54; for $q=5$, $t^5$ and $t^2$; for $q=11$, $t^2$ and $t^5$.)

**Exercise 7.101.** The curve $E =$ `14a1`, $y^2+xy+y = x^3+4x-6$, has split multiplicative reduction of type $\mathrm{I}_3$ at 7, and PARI gives the Tate parameter $q_E = 6\cdot 7^3 + 4\cdot 7^4 + 6\cdot 7^5 + \cdots$. Compute $n$, the reduction $\bar{u}$ of the unit part, the class $c$ of Lemma lem:tamfactor at $\ell = 3$ and $v_3(c)$, and deduce $v_3(\mathrm{Tam}_7(\mathrm{ad}^0\, T_3 E))$ from the lemma. Do the same for `11a1` at $q = 11$, $\ell = 5$. (Answers: 2 in both cases.)

**Exercise 7.102.** Let $q \neq \ell$ be primes, $x \in \mathbb{Q}_q^\times$ with $v_q(x) = n \geq 1$, and $\kappa$ the Kummer cocycle of $x$ for a chosen compatible system of $\ell$-power roots, with $\mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$. Working directly from the definition of $\kappa$ (without quoting Theorem 7.58(iv)): (a) show that replacing the system of roots by another one changes $\kappa$ by the coboundary $g \mapsto (\chi_\ell(g)-1)z$ for some $z \in \mathbb{Z}_\ell$, so that $c' = \kappa(\mathrm{Frob})$ changes by $(q-1)z$; (b) show that replacing Frob by $\mathrm{Frob}\,\sigma$ ($\sigma \in I_q$) changes $c'$ by $q\, n\, t_\ell(\sigma)$; (c) deduce that $\min(v_\ell(n), v_\ell(q-1), v_\ell(c'))$ does not depend on any choice. This is the quantity in the formula of the paper's Lemma lem:tamfactor. (d) For $q=7$, $\ell=3$ and $x = 7^3 u$ with $u \equiv 3 \pmod 7$, compute this minimum. (Answer: 0, since 3 is not a cube in $\mathbb{F}_7^\times$; contrast Exercise 7.101, where $\bar{u} = 6 = (-1)^3$ is a cube.)

**Exercise 7.103.** Let $q$ be odd, $K = \mathbb{Q}_q(\sqrt{q^*})$ and $\ell \neq q$. (a) Show that $I_K$ has index 2 in $I_q$ and that $t_{\ell,\mathbb{Q}_q}|_{I_K} = 2t_{\ell,K}$. (b) For $\rho$ the Tate shape of Example 7.74 with $x \in \mathbb{Q}_q^\times$, $a = v_q(x)$, show that $\rho \bmod \ell^k$ is unramified on $G_K$ iff $\ell^k \mid 2a$. (c) Conclude that the criterion of Lemma lem:tamagawa-exp is insensitive to this ramified base change for $\ell$ odd but not for $\ell = 2$, in contrast with the unramified base change of its proof. (Hints: (a) Proposition 7.38 and Theorem 7.47(v) with $e=2$. (b) On $I_K$, $\kappa_x = a\, t_{\ell,\mathbb{Q}_q} = 2a\, t_{\ell,K}$, and $t_{\ell,K}$ is surjective. (c) For $\ell$ odd, $\ell^k \mid 2a$ iff $\ell^k \mid a$.)

**Exercise 7.104.** Let $\rho$ be the Tate shape of Example 7.74 with $a \neq 0$ and $\varepsilon = 1$. Show that $\mathrm{WD}(\mathrm{Sym}^2 \rho) \simeq \mathrm{sp}(3)$, and compute $L(\mathrm{Sym}^2 V, s)$, $\dim(\mathrm{Sym}^2 V)^{I_K}$ and $a(\mathrm{Sym}^2 V)$. Check your answer against $\mathrm{Sym}^2 V \simeq \mathrm{ad}^0 V \otimes \det V$ and Proposition 7.88(iii). (Answer: $L(\mathrm{Sym}^2 V, s) = (1 - q^{-2-s})^{-1}$, $\dim(\mathrm{Sym}^2 V)^{I_K} = 1$, $a(\mathrm{Sym}^2 V) = 2$; indeed $\det V = \chi_\ell$ corresponds to $\|\cdot\|$, and $\mathrm{sp}(3) \otimes \|\cdot\|^{-1} \otimes \|\cdot\| = \mathrm{sp}(3)$.)

**Exercise 7.105.** The curve `14a1` has multiplicative reduction at 2 and 7. Using Proposition 7.84 and Proposition 7.61, predict the conductors of its quadratic twists by $\chi_{-4}$, $\chi_8$, $\chi_{-3}$ and $\chi_{-7}$. (PARI gives 112, 448, 126 and 98.)

**Exercise 7.106.** `11a1` has type $\mathrm{I}_5$ at 11, and its twist by $\chi_{-11}$ is `121d2`, of type $\mathrm{I}_5^*$ at 11. Using Theorem 7.89, compute $L_{11}(A_f, 1)^{-1}$ and the naive factor for both curves, and $e_3(E, 11)$, $e_5(E, 11)$. Explain why the true factor is invariant under the twist and the naive one is not (Definition def:terms). (Answer: $L_{11}(A_f, 1)^{-1} = 1 - 11^{-2} = \frac{120}{121}$ for both curves (rows 3 and 4); the naive factor is $\frac{120}{121}$ for `11a1` ($\delta_{11} = 1$) and 1 for `121d2` ($\delta_{11} = 0$); $e_3 = e_5 = 0$ for `11a1` and $e_3 = v_3(120) = 1$, $e_5 = v_5(120) = 1$ for `121d2`. PARI's `elllocalred` confirms the Kodaira type $\mathrm{I}_5^*$ of `121d2` at 11.)

**Exercise 7.107.** For a prime $q \neq 3$ at which $\rho(I_q)$ is cyclic of order 3, $e_3(E, q) = v_3(q - \chi_{-3}(q))$ (Proposition 7.87). Find all such $q < 60$ with $e_3(E, q) \geq 2$, and the largest value of $e_3$ among them. (Answer: $e_3 \geq 2$ iff $q \equiv \pm 1 \pmod 9$; for $q < 60$ these are 17, 19, 37, 53, with $e_3 = 2, 2, 2, 3$; the largest is $e_3 = 3$ at $q = 53$, as $53 + 1 = 2 \cdot 3^3$.)

**Exercise 7.108.** Let $L = \mathbb{Q}_3(\zeta_9)$, $G = \mathrm{Gal}(L/\mathbb{Q}_3) \simeq (\mathbb{Z}/9)^\times$. (a) Using $v_L(\zeta^{a-1} - 1) = 3^{v_3(a-1)}$ for $a \not\equiv 1 \pmod 9$, show that $G_0 = G$, $G_1 = G_2$ is the subgroup of order 3, and $G_3 = 1$. (b) Compute the Swan and Artin conductors of the characters of $G$ of orders 2, 3 and 6, and compare with the conductors of the Dirichlet characters modulo 9. (c) Which of them is $\chi_{-3}$? (Answer: (b) the character of order 2 is trivial on $G_1$, so it is tame with $\mathrm{sw} = 0$, $a = 1$; those of orders 3 and 6 are nontrivial on $G_1 = G_2$ and trivial on $G_3$, so $\mathrm{sw} = \frac{3}{6} + \frac{3}{6} = 1$ and $a = 2$; the conductors 3 and 9 of the Dirichlet characters agree. (c) The one of order 2.)

**Exercise 7.109.** Suppose $\rho(I_q) = \langle \tau \rangle$ is cyclic of order 4 with $\det \rho|_{I_q} = 1$ and $q \equiv 3 \pmod 4$. Show that $\mathrm{tr}\, \rho(\mathrm{Frob}) = 0$ for every Frobenius lift, that $\rho(\mathrm{Frob})^2$ commutes with $\tau$, and that $V \otimes \Omega(i)$ is irreducible as a representation of $G_{\mathbb{Q}_q}$. (Hint: $\rho(\mathrm{Frob})$ conjugates $\tau$ to $\tau^q = \tau^{-1}$, so it exchanges the $i$- and $(-i)$-eigenlines of $\tau$; a $G_{\mathbb{Q}_q}$-stable line would be an eigenline of $\tau$ fixed by $\rho(\mathrm{Frob})$.)

**Exercise 7.110.** (a) Show that for $\ell \neq p$ every continuous character $G_K \to \mathbb{Q}_\ell^\times$ has finite image on $I_K$. (b) Show that the cyclotomic character $\chi_p$ of $G_{\mathbb{Q}_p}$ has infinite image on $I_p$, and explain which step of the proof of Theorem 7.68 fails for $\ell = p$. (Hints: (a) the target is abelian, so $\psi(\mathrm{Frob}\, \sigma\, \mathrm{Frob}^{-1}) = \psi(\sigma)$. Let $U \subseteq I_K$ be the open subgroup mapping into the torsion-free pro-$\ell$ part of $\mathbb{Z}_\ell^\times$ ($1 + \ell\mathbb{Z}_\ell$, or $1 + 4\mathbb{Z}_2$); $U$ is normal in $G_K$, and $\psi|_U$ factors through $t_\ell$ (Corollary 7.48), so the tame relation gives $\psi(\sigma)^q = \psi(\sigma)$ on $U$. Hence $\psi(U)$ is torsion free and killed by $q - 1$, so trivial. (b) $\chi_p(I_p) = \mathbb{Z}_p^\times$ (Proposition 7.39(ii)); the step that fails is the factorisation through $t_\ell$, as $I_p$ has the huge pro-$p$ quotient $P_p$.)

**Exercise 7.111.** (a) For $p$ odd, show that $\mathbb{Q}_p(\sqrt{p + p^3}) = \mathbb{Q}_p(\sqrt{p})$, once by Hensel's lemma and once by Lemma 7.31(i). (b) Show that $\mathbb{Q}_2(\sqrt{10}) \neq \mathbb{Q}_2(\sqrt{2})$ although $|\sqrt{10} - \sqrt{2}|_2 = |\sqrt{2} - (-\sqrt{2})|_2$: the strict inequality in Krasner's lemma cannot be relaxed. (Hints: (a) $(p + p^3)/p = 1 + p^2$

is a square in $\mathbb{Z}_p$; and choosing the sign with $\sqrt{p+p^3}/\sqrt{p} \equiv 1 \pmod p$, $|\sqrt{p+p^3} - \sqrt{p}|_p = |p^3|_p/|\sqrt{p+p^3} + \sqrt{p}|_p = p^{-5/2}$, less than $|\sqrt{p} - (-\sqrt{p})|_p = p^{-1/2}$. (b) $10/2 = 5$ is not a square in $\mathbb{Q}_2$.)

## 7.11 Notes and references

Absolute values, completions, $\mathbb{Q}_p$, Hensel's lemma and the structure of $\mathbb{Q}_p^\times$ follow Serre's *A Course in Arithmetic* [172, Chapter II] and Neukirch [142, Chapter II, Sections 1 to 5]; the factorisation form of Hensel's lemma and its corollary on coefficients of irreducible polynomials are in Neukirch, Chapter II, Section 4, and many variants are collected in Keith Conrad's notes [33]. Cassels' chapter "Global fields" in Cassels and Fröhlich [26] is a compact alternative. Extensions of the absolute value, $e$ and $f$, Eisenstein polynomials, unramified and tamely ramified extensions, and Krasner's lemma follow Serre's *Local Fields* [174, Chapters I to IV] and Neukirch, Chapter II, Sections 6 to 10; the comparison of completions of number fields with the global theory (Theorem 7.30, Proposition 7.32) is Neukirch, Chapter II, Sections 8 and 9.

The ramification groups, their quotients and the tame character are Serre [174, Chapter IV, Sections 1 and 2]; Herbrand's theorem, upper numbering and Hasse-Arf are Chapter IV, Section 3, and Chapter V, Section 7 there; the Artin and Swan conductors and Artin's integrality theorem are Chapter VI, Section 2, and the representation-theoretic side is in [173, Chapter 19]. The structure of $\mathrm{Gal}(K^{\mathrm{tame}}/K)$ as a profinite group with two generators is in Neukirch, Schmidt and Wingberg [143, Chapter VII, Section 5]. The use of the tame quotient and of Kummer classes for $\ell$-adic representations, and the Tate curve, are in Serre [178, Chapters I and IV] and Silverman [184, Chapter V]; Serre and Tate [179, Sections 1 and 2] prove Grothendieck's theorem in the form used here (their appendix), the criterion of Néron, Ogg and Shafarevich, and the independence of $\ell$ of the image of inertia for potentially good reduction. Local class field theory is stated after Serre's chapter in [26] and Milne [133, Chapter I].

Weil-Deligne representations, local $L$-factors, conductors and epsilon factors follow Deligne [41, Section 8] and Tate [194, Section 4]; Rohrlich [166] is a gentle account written for elliptic curves, with the classification of two-dimensional representations. The convention that local $L$-factors use the geometric Frobenius is the one of [49, Section 1.7.2], which the paper follows.

All chapter and section numbers of books cited in this chapter (Serre, Neukirch, Neukirch-Schmidt-Wingberg, Milne, Cassels-Fröhlich, Silverman, Deligne, Tate, Rohrlich) are recalled and were not checked against printed copies; they are pointers, and no proof in the chapter depends on them. No theorem numbers from these sources are cited. Every numerical example and the answers to the exercises were checked with PARI/GP [148] through `cypari2`, with commands such as the following.

```
polrootspadic(x^2+1, 5, 4)            \\ 182 and 443 modulo 5^4
lift(teichmuller(2 + O(5^4)))         \\ 182
polrootspadic(x^2+x+1, 7, 2)          \\ 30 and 18 modulo 49
E = ellinit([0,-1,1,-10,-20], O(11^8)); E.tate[3]   \\ q_E of 11a1: 10*11^5 + ...
E = ellinit([1,0,1,4,-6], O(7^6)); E.tate[3]        \\ q_E of 14a1 at 7: 6*7^3 + ...
ellglobalred(ellinit(elltwist(ellinit([1,0,1,4,-6]), -4)))[1]   \\ 112
valuation(coredisc(-2), 2)            \\ 3: conductor exponent of chi_{-8} at 2
```

# Part III

# Curves and elliptic curves

# Chapter 8

# Algebraic curves, function fields and Riemann surfaces

The paper's objects live on curves: the modular curve $X_0(N)$, the elliptic curve $E$, the modular parametrisation $\varphi_E\colon X_0(N) \to E$ and its degree, the cohomology $H^1(X_0(N), \mathbb{Z}_\ell)$ with its cup product, and the Jacobian $J_0(N)$ with its torsion and Tate modules. This chapter supplies the theory of curves that these words presuppose, in two languages: algebraic (varieties, function fields, divisors, differentials, Riemann-Roch, morphisms and their degree, Riemann-Hurwitz), in the classical style of Silverman's book, and analytic (compact Riemann surfaces, holomorphic forms, periods, $H_1$, $H^1$ and the cup product, the Jacobian as a complex torus), joined by Riemann's existence theorem. It ends with correspondences and the theorem that the adjoint of a correspondence for the cup product is its transpose, which is what makes the Hecke operators self-adjoint for the pairing of Section 4.1 of the paper. It uses Chapter 1 (inverse limits, $\mathbb{Z}_\ell$, Moebius maps), Chapter 2 (polynomial rings, finite fields), Chapter 3 (Nakayama, discrete valuation rings, lattices, saturation, perfect and alternating pairings, adjoints, idempotents), Chapter 4 (contragredients), Chapter 5 (separability, Hilbert 90) and Chapter 6 (the identity $\sum e_i f_i = n$), and the complex analysis of Appendix A.

## Core results at a glance

Throughout, $k$ is a perfect field, $C$ a smooth projective curve over $k$ of genus $g$, $X$, $Y$ compact Riemann surfaces, $\varphi\colon X \to Y$ a nonconstant morphism of degree $n$, $\langle\ ,\ \rangle$ the cup product, $A^\dagger$ the adjoint of an operator, and $\mathcal{C} = (C, \alpha, \beta)$ a correspondence with transpose $\mathcal{C}^t = (C, \beta, \alpha)$.

**8.1 Conventions.** $G_k$ acts on points coordinatewise; $\varphi^*$ is pullback, $\varphi_*$ pushforward, $A^\dagger$ an adjoint; cup product $\int_X \theta \wedge \eta$ with the complex orientation; $\pi_q$ the Frobenius morphism; $l(D)$ the dimension of $\mathcal{L}(D)$. Convention 8.1.

**8.2 Affine varieties** (definition): zero sets $V(S) \subseteq \mathbb{A}^n(\bar{k})$ with $I(V)$ prime; coordinate ring, function field, dimension = transcendence degree; defined over $k$. Definition 8.2.

**8.3 Nullstellensatz.** $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$; for $k$ perfect, $V$ is defined over $k$ iff $\sigma(V) = V$ for all $\sigma \in G_k$. *(stated; reference)* Theorem 8.3.

**8.4 Projective varieties** (definition): $\mathbb{P}^n = (\bar{k}^{n+1} \setminus 0)/\bar{k}^\times$, homogeneous ideals, charts $U_i \simeq \mathbb{A}^n$, projective closure, function field of quotients of forms of equal degree. Definition 8.4.

**8.5 Points over subfields.** $\mathbb{P}^n(\bar{k})^{G_L} = \mathbb{P}^n(L)$; $\mathbb{P}^n = \mathbb{A}^n \sqcup \mathbb{P}^{n-1}$; the function field of a projective variety is that of any affine chart. *(proved here)* Proposition 8.5.

**8.6 Curves, rational maps, morphisms** (definition): a curve is a projective variety of dimension one defined over $k$; morphisms are rational maps regular everywhere; defined over $k$ iff Galois-equivariant. Definition 8.6.

**8.7 Plane curves** (definition): $V(F)$, $F$ homogeneous and irreducible over $\bar{k}$; $P$ singular iff $F_X = F_Y = F_Z = 0$ at $P$. Definition 8.8.

**8.8 Affine criterion.** $XF_X + YF_Y + ZF_Z = dF$; an affine point is singular iff $f = f_x = f_y = 0$; tangent line $F_X(P)X + F_Y(P)Y + F_Z(P)Z = 0$. *(proved here)* Proposition 8.9.

**8.9 Smooth points are discrete valuations.** $P$ smooth iff $\dim \mathfrak{m}_P/\mathfrak{m}_P^2 = 1$ iff $\bar{k}[C]_P$ is a discrete valuation ring; a line other than the tangent gives a uniformiser; $\mathrm{ord}_P$ is the order of vanishing. *(proved here for plane models; general case stated)* Theorem 8.11.

**8.10 Bezout.** Plane curves of degrees $m$, $n$ without common component meet in $mn$ points with multiplicity. *(stated; reference)* Theorem 8.12.

**8.11 Smooth models.** Every curve is birational over $k$ to a unique smooth projective curve; its points are the valuations of the function field. *(stated; reference)* Theorem 8.13.

**8.12 Function fields** (definition): finitely generated of transcendence degree one with $k$ algebraically closed in it. Definition 8.14.

**8.13 Valuations.** Each point gives $\mathrm{ord}_P$; a function has finitely many zeros and poles; a function without poles is constant. *(proved here, given the finiteness of proper closed subsets)* Proposition 8.15.

**8.14 Extension of morphisms.** A rational map from a smooth curve to a projective variety is a morphism; every $f \in \bar{k}(C)$ is a morphism $C \to \mathbb{P}^1$. *(proved here)* Theorem 8.16.

**8.15 Curves and function fields.** $C \mapsto k(C)$ is a contravariant equivalence between smooth projective curves with nonconstant morphisms over $k$ and function fields of one variable over $k$; a morphism of degree one is an isomorphism. *(stated; reference)* Theorem 8.17.

**8.16 Galois action.** $C(\bar{k})^{G_L} = C(L)$; $(\sigma f)(\sigma P) = \sigma(f(P))$ and $\mathrm{ord}_{\sigma P}\, \sigma f = \mathrm{ord}_P f$; morphisms over $k$ commute with $G_k$. *(proved here; the fixed field $k(C)$ stated)* Proposition 8.18.

**8.17 Divisors** (definition): $\mathrm{Div}(C)$, degree, $\mathrm{Div}_k(C) = \mathrm{Div}(C)^{G_k}$, $\mathrm{div}(f) = \sum \mathrm{ord}_P(f)(P)$. Definition 8.21.

**8.18 Principal divisors.** $\deg \mathrm{div}\, f = 0$ iff $f$ is constant; $\deg \mathrm{div}\, f = 0$, with $\deg \mathrm{div}_0 f = \deg \mathrm{div}_\infty f = [\bar{k}(C) : \bar{k}(f)]$; on $\mathbb{P}^1$, $\mathrm{div}(x - a) = (a) - (\infty)$. *(proved here, given Theorem 8.38)* Proposition 8.22.

**8.19 Picard group** (definition): $\mathrm{Pic}(C) = \mathrm{Div}(C)/\mathrm{Prin}(C)$, $\mathrm{Pic}^0(C)$ the degree-zero part. Definition 8.23.

**8.20 The Picard group.** $\mathrm{Pic}^0(\mathbb{P}^1) = 0$; $G_k$ acts; $\mathrm{Div}_k^0(C)/\mathrm{div}(k(C)^\times) \to \mathrm{Pic}^0(C)^{G_k}$ is injective *(proved here)*, and bijective if $C(k) \neq \emptyset$ *(stated; reference)*. Proposition 8.24.

**8.21 Differentials** (definition): $\Omega_C$, $\operatorname{ord}_P \omega$, $\operatorname{div}\omega$, regular differentials. Definition 8.25.

**8.22 Differentials and the canonical class.** $\Omega_C$ is one-dimensional, $\operatorname{ord}_P$ is well defined *(stated; reference)*; all $\operatorname{div}\omega$ form one class $K_C$; on $\mathbb{P}^1$, $\operatorname{div}(dx) = -2(\infty)$; on $y^2 = h(x)$, $\operatorname{div}(dx/y) = 0$ *(proved here)*. Proposition 8.26.

**8.23 Riemann-Roch spaces** (definition): $\mathcal{L}(D) = \{f : \operatorname{div} f + D \geq 0\} \cup \{0\}$, $l(D) = \dim \mathcal{L}(D)$. Definition 8.27.

**8.24 Riemann-Roch spaces.** $\mathcal{L}(0) = \bar{k}$; $\mathcal{L}(D) = 0$ if $\deg D < 0$; $l(D + (P)) \leq l(D) + 1$, so $l(D) \leq \deg D + 1$ for $D \geq 0$; $l$ depends on the class; for $D$ over $k$, $\mathcal{L}(D)$ has a $k$-rational basis *(proved here; the rational basis stated)*. Proposition 8.28.

**8.25 Riemann-Roch.** $l(D) - l(K_C - D) = \deg D - g + 1$. *(stated; reference, with the shape of the proof)* Theorem 8.29.

**8.26 Consequences.** $l(K_C) = g$; $\deg K_C = 2g - 2$; $l(D) = \deg D - g + 1$ if $\deg D > 2g - 2$; genus 0 with a rational point means $\mathbb{P}^1$; for $g = 1$, $l(D) = \deg D$ ($\deg D \geq 1$), $K_C \sim 0$, and regular differentials have no zeros. *(proved here)* Corollary 8.30.

**8.27 Plane curves.** A smooth plane curve of degree $d$ has genus $(d-1)(d-2)/2$, in any characteristic. *(proof sketched here)* Theorem 8.31.

**8.28 Weierstrass equations.** A genus-one curve $C$ with $O \in C(k)$ is isomorphic over $k$, by $[x : y : 1]$ with $x \in \mathcal{L}(2(O))$, $y \in \mathcal{L}(3(O))$, to a smooth Weierstrass cubic $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, $a_i \in k$, with $O \mapsto [0 : 1 : 0]$; conversely. *(proved here, given that images of projective varieties are closed; the converse in characteristic 2 rests on the sketched Theorem 8.31)* Theorem 8.33.

**8.29 Genus one is $\operatorname{Pic}^0$.** For $g = 1$ and $O \in C(k)$, $P \mapsto [(P) - (O)]$ is a $G_k$-equivariant bijection $C(\bar{k}) \to \operatorname{Pic}^0(C)$. *(proved here)* Proposition 8.34.

**8.30 Nonconstant morphisms.** A morphism of curves is constant or surjective *(stated; reference)*; $\varphi^*$ embeds $\bar{k}(C_2)$ in $\bar{k}(C_1)$ with finite index *(proved here)*. Theorem 8.36.

**8.31 Degree, ramification, pullback** (definition): $\deg\varphi = [\bar{k}(C_1) : \varphi^*\bar{k}(C_2)]$; $e_\varphi(P) = \operatorname{ord}_P(\varphi^* t_{\varphi(P)})$; $\varphi^*(Q) = \sum_{\varphi(P)=Q} e_\varphi(P)(P)$; $\varphi_*(P) = (\varphi(P))$. Definition 8.37.

**8.32 Degree formula.** $\sum_{P\mapsto Q} e_\varphi(P) = \deg\varphi$ for every $Q$. *(proof sketched here)* Theorem 8.38.

**8.33 The degree in three ways.** Field degree; $\#\varphi^{-1}(Q) = \deg\varphi$ off the branch points for separable $\varphi$; $\deg\varphi^* D = \deg\varphi \deg D$, $\varphi_*\varphi^* = \deg\varphi$, $\varphi^* \operatorname{div} f = \operatorname{div}\varphi^* f$, so $\varphi_*\varphi^* = [\deg\varphi]$ on $\operatorname{Pic}^0$; degrees multiply; degree one means isomorphism; a function without poles is constant. *(proved here, given Theorem 8.38; the norm formula and the inseparable case stated)* Corollary 8.39.

**8.34 Separability and Frobenius.** Characteristic 0 morphisms are separable *(proved here)*; separable iff $\varphi^*\omega \neq 0$, and the factorisation through Frobenius *(stated; reference)*; $\pi_q$ is a purely inseparable bijection of degree $q$ with $e = q$ everywhere, fixed points $C(\mathbb{F}_q)$, acting on points as $\operatorname{Frob}_q$ *(proved here; the degree sketched)*. Proposition 8.40.

**8.35 Ramification** (definition): unramified, branch points, tame and wild. Definition 8.42.

**8.36 Local computation.** For separable $\varphi$, $\operatorname{ord}_P \varphi^*\omega = e \operatorname{ord}_Q \omega + \operatorname{ord}_P(d(\varphi^* t_Q)/dt_P)$ with the last term $\geq e-1$, equality iff tame; finitely many ramified points. *(proved here, given Proposition 8.40(b) and Theorem 8.38)* Lemma 8.43.

**8.37 Riemann-Hurwitz.** $2g_1 - 2 \geq \deg\varphi(2g_2-2) + \sum_P (e_\varphi(P)-1)$ for separable $\varphi$, with equality iff all ramification is tame. *(proved here, given Theorem 8.38 and Riemann-Roch)* Theorem 8.44.

**8.38 Consequences.** $g_1 \geq g_2$ if $g_2 \geq 1$; morphisms of genus-one curves are unramified; hyperelliptic and Fermat genera; for $\varphi\colon X \to E$, $\sum(e_P - 1) = 2g(X) - 2$ (for $\varphi_{\texttt{121a1}}$: 10; $\varphi_{\texttt{539c1}}$: 96). *(proved here, given Theorem 8.44)* Corollary 8.45.

**8.39 Riemann surfaces** (definition): charts with holomorphic transitions; meromorphic functions; holomorphic maps. Definition 8.46.

**8.40 Holomorphic maps.** Local normal form $w = z^e$; open, surjective, finite fibres; constant degree $\sum_{P\mapsto Q} \operatorname{mult}_P$; finitely many ramification points; orientation-preserving. *(proved here)* Theorem 8.48.

**8.41 Riemann existence and GAGA.** Compact Riemann surfaces and smooth projective complex curves are equivalent, compatibly with functions, orders, multiplicities, degrees, differentials, divisors and genus. *(stated; reference)* Theorem 8.49.

**8.42 Topology.** A compact Riemann surface is a sphere with $g$ handles, $\chi = 2-2g$ *(stated; reference)*; $\chi(X) = d\chi(Y) - \sum(\operatorname{mult}_P - 1)$ *(proof sketched here)*. Theorem 8.50.

**8.43 1-forms** (definition): meromorphic and holomorphic 1-forms, residues, $H^0(X,\Omega^1)$, smooth forms and Stokes. Definition 8.51.

**8.44 Residue theorem.** $\sum_P \operatorname{res}_P \omega = 0$. *(proved here)* Theorem 8.52.

**8.45 Argument principle.** $\deg\operatorname{div} f = \sum \operatorname{res}(df/f) = 0$. *(proved here)* Corollary 8.53.

**8.46 Genus and forms.** $\dim H^0(X,\Omega^1) = g$; $\deg\operatorname{div}\omega = 2g-2$. *(proved here from Riemann-Roch and GAGA, both stated)* Theorem 8.54.

**8.47 Bilinear relations.** $\int \omega\wedge\eta = 0$ and $\frac{i}{2}\int \omega\wedge\bar\omega > 0$ for holomorphic forms; $\int \theta\wedge\eta = \sum_j (A_j(\theta)B_j(\eta) - B_j(\theta)A_j(\eta))$ *(proof sketched here)*; Riemann's relations; the periods form a lattice $\Lambda_X$ *(proved here, given the sketched identity)*. Theorem 8.55.

**8.48 First homology** (definition): $H_1(X,\mathbb{Z})$, $\varphi_*$, $H_1(X,A) = H_1 \otimes A$. Definition 8.56.

**8.49 Symplectic bases.** $H_1(X,\mathbb{Z}) \simeq \mathbb{Z}^{2g}$ with a symplectic basis; the intersection pairing is perfect and alternating, $a_i \cdot b_j = \delta_{ij}$. *(stated; reference)* Theorem 8.57.

**8.50 First cohomology** (definition): $H^1(X,A) = \operatorname{Hom}(H_1(X,\mathbb{Z}),A)$; $\varphi^*$ the transpose of $\varphi_*$; de Rham classes; $H^2$. Definition 8.58.

**8.51 Cohomology.** $H^1(X,\mathbb{Z}_\ell) = H^1(X,\mathbb{Z}) \otimes \mathbb{Z}_\ell$ free of rank $2g$ *(proved here)*; de Rham and $H^2(X,\mathbb{Z}) = \mathbb{Z}$ *(stated; reference)*; Hodge $H^1(X,\mathbb{C}) = H^0(\Omega^1) \oplus \overline{H^0(\Omega^1)}$ *(proof sketched here)*; $H^1(X,M) = H^1(\pi_1(X),M)$ *(stated; reference)*. Proposition 8.59.

**8.52 Cup product.** $\langle\theta,\eta\rangle = \int_X \theta\wedge\eta$ is integral, alternating and perfect on $H^1(X,\mathbb{Z})$ with Gram matrix $\left(\begin{smallmatrix} 0 & I_g \\ -I_g & 0\end{smallmatrix}\right)$, and after any base change, in particular on $H^1(X,\mathbb{Z}_\ell)$; Poincare duality $\mathrm{PD}(a_j^\vee) = b_j$, $\mathrm{PD}(b_j^\vee) = -a_j$ is an isometry. *(proved here, given Theorem 8.57 and Theorem 8.55(c))* Theorem 8.60.

**8.53 Transfer and projection formula.** $\varphi^!\colon H_1(Y)\to H_1(X)$ with $\varphi_*\varphi^! = n$ *(proof sketched here)*; $\varphi_*$ on $H^1$ (transpose of $\varphi^!$, the trace of forms); $\langle\varphi_*\eta,\xi\rangle = \langle\eta,\varphi^*\xi\rangle$, $\varphi_*\varphi^* = n$, $\langle\varphi^*x,\varphi^*y\rangle = n\langle x,y\rangle$, $\varphi^* = n$ on $H^2$ *(proved here)*. Proposition 8.62.

**8.54 Which is dual to which.** $H_1$, $\Lambda_X$, $J$, $J[n]$, $T_\ell J$ covariant; $H^1 = \mathrm{Hom}(T_\ell J,\mathbb{Z}_\ell)$, $H^0(\Omega^1)$ contravariant; PD turns $\mathcal{C}^*$ into $(\mathcal{C}^t)_*$; $H^1_{\mathrm{et}} \simeq T_\ell J(-1)$. Convention 8.63.

**8.55 Jacobian** (definition): $J(C)(\bar k) = \mathrm{Pic}^0(C)$; $J_{\mathrm{an}}(X) = H^0(X,\Omega^1)^\vee/\Lambda_X$. Definition 8.65.

**8.56 Abel-Jacobi.** AJ: $\mathrm{Div}^0(X)\to J_{\mathrm{an}}(X)$ is a well-defined homomorphism and $\iota\colon X\to J_{\mathrm{an}}$ is a holomorphic injection inducing the period isomorphism on $H_1$ *(proved here, injectivity given Abel's theorem)*; Abel's theorem and Jacobi inversion, $\mathrm{Pic}^0(X)\simeq J_{\mathrm{an}}(X)$ *(stated; reference)*. Theorem 8.66.

**8.57 Jacobian variety.** $J$ is an abelian variety of dimension $g$ over $k$ with $J(\bar k) = \mathrm{Pic}^0(C)$, $J(\mathbb{C}) = J_{\mathrm{an}}$, the Albanese universal property, and Mordell-Weil. *(stated; reference)* Theorem 8.67.

**8.58 Functoriality.** $\varphi_*\colon J(X)\to J(Y)$ and $\varphi^*\colon J(Y)\to J(X)$ with $\varphi_*\varphi^* = [n]$; over $\mathbb{C}$, $\varphi_*$ is induced by $(\varphi^*)^\vee$ and $\varphi_*$ on $H_1$ *(proved here)*, $\varphi^*$ by the trace and $\varphi^!$ *(proof sketched here)*; for $Y = E$, $\varphi = \varphi_*\circ\iota_X$. Proposition 8.68.

**8.59 Tori.** For a surjective map of complex tori, $\pi_0(\ker)\simeq M/A(\Lambda)$; for $B\colon W\to V$ $\mathbb{C}$-linear and injective with $B(M)\subseteq\Lambda$, $\ker\simeq \mathrm{sat}(B(M))/B(M)$; maps of one-dimensional tori are $z\mapsto cz$ of degree $[\Lambda' : c\Lambda]$. *(proved here)* Lemma 8.69.

**8.60 Degree and cup product.** $\langle\varphi^*x,\varphi^*y\rangle = n\langle x,y\rangle$; $\varphi_*\varphi^* = n$; $\varphi^*H^1(Y,\mathbb{Z})$ saturated iff $\varphi_*$ onto on $H_1$ iff $\ker(J(X)\to J(Y))$ connected iff $\varphi^*$ injective on $J$, the indices all equal; for $Y = E$, the cup product on $\varphi^*H^1(E)$ has ideal of values $n\mathbb{Z}$, Gram determinant $n^2$, $[H^1 : M\oplus M^\perp] = n^2$ if saturated, and ideal $(n/m_0)\mathbb{Z}$ on the saturation in general, $m_0 = \#\pi_0(\ker(J(X)\to E))$; an isogeny of degree $m$ destroys saturation. *(proved here; the identification of $\varphi^*$ on $J$ sketched)* Proposition 8.70.

**8.61 Torsion.** $J(X)[n] = \frac{1}{n}\Lambda_X/\Lambda_X \simeq H_1(X,\mathbb{Z}/n)\simeq(\mathbb{Z}/n)^{2g}$; $J[\mathfrak{m}]$ is a $\mathbb{T}/\mathfrak{m}$-vector space *(proved here)*; $J[n](\bar k)\simeq(\mathbb{Z}/n)^{2g}$ over $k$, $\mathrm{char}\,k\nmid n$ *(stated; reference)*. Proposition 8.71.

**8.62 Tate modules** (definition): $T_\ell J = \varprojlim J[\ell^m]$, $V_\ell J = T_\ell J\otimes\mathbb{Q}_\ell$. Definition 8.72.

**8.63 Tate module and $H^1$.** $T_\ell J\simeq H_1(X,\mathbb{Z}_\ell)$, free of rank $2g$; $H^1(X,\mathbb{Z}_\ell) = \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell J,\mathbb{Z}_\ell)$; PD gives $H^1\simeq T_\ell J$, not Galois-equivariant, turning $\mathcal{C}^*$ into $(\mathcal{C}^t)_*$ *(proved here)*; over $k$, $T_\ell J$ free of rank $2g$ with $G_k$-action *(stated; reference)*. Proposition 8.73.

**8.64 Weil pairing and etale cohomology.** $e_\ell\colon T_\ell J\times T_\ell J\to\mathbb{Z}_\ell(1)$ perfect alternating equivariant; $H^1_{\mathrm{et}} = (T_\ell J)^\vee\simeq T_\ell J(-1)$, comparison with $H^1(C(\mathbb{C}),\mathbb{Z})\otimes\mathbb{Z}_\ell$; for $E$, $H^1(E,\mathbb{Z}_\ell)\simeq T_\ell E(-1)$ with determinant $\chi_\ell^{-1}$; Weil's theorem over $\mathbb{F}_q$ and $a_q = \mathrm{tr}(\mathrm{Frob}_q \mid T_\ell E) = \mathrm{tr}(\mathrm{Frob}_q^{-1}\mid H^1)$. *(stated; reference)* Theorem 8.74.

**8.65 Correspondences** (definition): $\mathcal{C} = (C, \alpha, \beta)$, transpose $(C, \beta, \alpha)$, graphs, $u\mathcal{C}v$. Definition 8.78.

**8.66 Actions** (definition): $\mathcal{C}_* = \beta_*\alpha^*$ on divisors and $J$, $\mathcal{C}_* = \beta_*\alpha^!$ on $H_1$, $\mathcal{C}^* = \alpha_*\beta^*$ on $H^1$ and on differentials. Definition 8.79.

**8.67 Compatibilities.** $\mathcal{C}^*$ on $H^1$ is the transpose of $\mathcal{C}_*$ on $H_1$; graphs act by $\varphi^*$, $\varphi_*$, $\varphi^!$; $u\mathcal{C}v$ acts by $u_*\mathcal{C}_*v_*$ and $v^*\mathcal{C}^*u^*$ *(proved here)*; the action on $J$ and $T_\ell J$ is $\mathcal{C}_*$ *(proof sketched here)*. Proposition 8.80.

**8.68 The adjoint is the transpose.** $\langle \mathcal{C}^*\xi, \eta\rangle_X = \langle \xi, (\mathcal{C}^t)^*\eta\rangle_Y$, that is $(\mathcal{C}^*)^\dagger = (\mathcal{C}^t)^*$; automorphisms are isometries; $\mathrm{PD} \circ \mathcal{C}^* = (\mathcal{C}^t)_* \circ \mathrm{PD}$. *(proved here, given Proposition 8.62)* Theorem 8.81.

**8.69 Twisted pairing.** If $t^t = w^*tw^*$ for every $t$ in a commutative $\mathbb{Z}_\ell$-subalgebra $\mathbb{T}$ of $\mathrm{End}_{\mathbb{Z}_\ell}(H^1(X, \mathbb{Z}_\ell))$ spanned by actions of correspondences, $w$ an involution of $X$, then $\langle x, w^*y\rangle$ is perfect and alternating on $H^1(X, \mathbb{Z}_\ell)$, every $t \in \mathbb{T}$ is self-adjoint for it, and $H^1(X, \mathbb{Z}_\ell) = \bigoplus_{\mathfrak{m}} H^1(X, \mathbb{Z}_\ell)_{\mathfrak{m}}$ is orthogonal with the pairing perfect on each summand. *(proved here)* Corollary 8.82.

## Where the paper uses this chapter

- **Abstract and Section 1** (the modular parametrisation $\varphi_E\colon X_0(N) \to E$, the $X_0(N)$-optimal curve "for which the modular parametrisation $\varphi_E\colon X_0(N) \to E$ has minimal degree", the modular degree $\deg \varphi_E$): morphisms and their degree in every sense, Definition 8.37, Theorem 8.38, Corollary 8.39, Table 8.1, Theorem 8.49; Riemann-Hurwitz checks of the quoted degrees, Corollary 8.45(e); why the optimal curve has minimal degree, Proposition 8.70(iv), (v) and Paper link 8.76.

- **Remark rem:degree (Section 4, sec:quant)** (the formula of Zagier and Watkins: Petersson norm over the area of the period lattice): positivity and periods, Theorem 8.55, Example 8.61.

- **Section 1.1 and Proposition prop:ns** ($\rho_{E,\ell}$ and the action of $G_{\mathbb{Q}}$ on $E[3]$ and on the roots of the 3-division polynomial): the Galois action on points, Proposition 8.18, Paper link 8.20.

- **Appendix app:repro** (curves given by Weierstrass coefficients, `ellinit`): Theorem 8.33, Proposition 8.34, Paper link 8.35.

- **Section 4.1 (sec:dfg)** ("perfect and alternating on $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$, and the Hecke operators, $U_p$ included, are self-adjoint for it, because $w_{M^\Sigma}T_n w_{M^\Sigma}^{-1}$ is the adjoint of $T_n$ for the cup product"): Theorem 8.60, Definition 8.79, Theorem 8.81, Corollary 8.82, Paper link 8.64, Paper link 8.84.

- **Proposition prop:pullback** ($\varphi^*$ injective with image $H^1(X_0(N), \mathbb{Z}_\ell)[I_f]$; $\langle \varphi^*a, w_N^*\varphi^*b\rangle = \varepsilon_N(f)\deg\varphi_E\langle a, b\rangle_E$; the congruence ideal is $\deg\varphi_E\,\mathbb{Z}_\ell$): optimality as connectedness, saturation and surjectivity on $H_1$, Lemma 8.69, Proposition 8.70(iii); the projection formula, Proposition 8.62(iv); Hodge, Proposition 8.59(iii); unimodularity, Theorem 8.60; step-by-step reading in Paper link 8.76.

- **The paragraph after Proposition prop:pullback and Section 4.2** ($M_{f,\lambda} = H^1(E, \mathbb{Z}_\ell)$, which is $\mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$; "its adjoint is $\mathrm{ad}^0 T_\ell E$"): Proposition 8.73(ii); Theorem 8.74(b), (c); Warning 8.75; Paper link 8.77.
- **Lemma lem:hecke, Lemma lem:ddt417 as applied, and Step 2 of the proof of Theorem thm:quant** ($L = H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ "with a perfect $\mathcal{O}$-bilinear pairing for which $\mathbb{T}$ is self-adjoint"; "the idempotent $e_{\mathfrak{m}}$ is self-adjoint for $\hat{\delta}_!$, so $H^1(X_0(N), \mathbb{Z}_\ell)$ splits orthogonally into its local components and the pairing is perfect on each"): Corollary 8.82(iii).
- **Remark rem:free** ($J_0(N)[\mathfrak{m}]$ two-dimensional, the Tate module free of rank two, $T_\ell J_0(N)$ versus $H^1(X_0(N), \mathbb{Z}_\ell)$ through Poincare duality composed with $w_N$, not Galois equivariant): Proposition 8.71, Proposition 8.73, Theorem 8.81(c), Warning 8.75, Paper link 8.77.

## 8.1 Affine and projective varieties

This section fixes the language. A reader who knows complex analysis and linear algebra meets here algebraic geometry and algebraic topology at once, so the chapter keeps two languages and one dictionary. The *algebraic* side is the classical language of Silverman [185, Chapters I and II]: a variety is the set of its points over an algebraic closure $\bar{k}$, cut out by polynomials, with the Galois group $G_k$ acting on coordinates; no schemes, no sheaves and no sheaf cohomology are used. The *analytic* side (Sections 8.8 to 8.10) is the theory of compact Riemann surfaces with charts, as in a first course in complex analysis. The dictionary between the two (Riemann's existence theorem and GAGA) is stated once, in Theorem 8.49, and every other comparison is proved from it. What is proved: every statement that is a computation with polynomials, valuations, local normal forms of holomorphic maps, change of variables in integrals, or linear algebra of lattices. What is sketched: the formula $\sum e_P = \deg \varphi$, the genus of plane curves, the degree of the Frobenius morphism, the topological form of Riemann-Hurwitz, Riemann's bilinear relations, the Hodge decomposition, the transfer, the description of $\varphi^*$ on the Jacobian by the trace and the transfer, and the action of correspondences on the Jacobian. What is stated with a reference: the Nullstellensatz, Bezout's theorem, the equivalence between curves and function fields, Riemann-Roch, Riemann existence and GAGA, the classification of surfaces, Abel's theorem and Jacobi inversion, the algebraicity of the Jacobian, the Weil pairing and etale cohomology. In the paper the objects of this section are the background of every sentence that mentions $X_0(N)$, $E$ or $E(\bar{\mathbb{Q}})$.

**Convention 8.1** (conventions of this chapter)**.**

(C1) *Fields.* $k$ is a perfect field ($\mathbb{Q}$, $\mathbb{Q}_p$, a number field, a finite field $\mathbb{F}_q$ with $q$ a power of a prime $p$: here $q$ is always the size of a finite field, Appendix B, B-C1); $\bar{k}$ is an algebraic closure and $G_k = \mathrm{Gal}(\bar{k}/k)$ (Chapter 5). Over $\mathbb{C}$ we write $X$ for a compact Riemann surface and identify it with the smooth projective curve it defines (Theorem 8.49).

(C2) *Curves.* A *curve over* $k$ is a projective variety of dimension one defined over $k$; "variety" includes "irreducible over $\bar{k}$". Unless said otherwise curves are smooth and projective; a singular or affine curve birational to $C$ is called a *model* of $C$. Morphisms are defined over $k$ when $k$ is named, over $\bar{k}$ otherwise. The genus is $g$.

(C3) *Points and Galois.* $C(L)$ is the set of points with coordinates in $L$, for $k \subseteq L \subseteq \bar{k}$; $\sigma \in G_k$ acts on $C(\bar{k})$ coordinatewise, on the left, and $C(L) = C(\bar{k})^{G_L}$. This is the action behind $\rho_{E,\ell}$ (Chapters 9 and 11).

(C4) *Pullback, pushforward, transpose, adjoint.* For a nonconstant morphism $\varphi\colon X \to Y$, $\varphi^*$ is pullback (of functions, divisors, divisor classes, differentials, classes in $H^1$) and $\varphi_*$ is pushforward (of divisors, divisor classes, classes in $H_1$, and on $H^1$ the transpose of the transfer, Proposition 8.62). A correspondence is $\mathcal{C} = (C, \alpha, \beta)$ and its transpose $\mathcal{C}^t = (C, \beta, \alpha)$. The adjoint of an endomorphism $A$ for a pairing is $A^\dagger$, never $A^*$, because a star after the name of a map is a pullback (Appendix B, B-C4).

(C5) *Duals.* $M^\vee = \mathrm{Hom}(M, \mathbb{Z})$ or $\mathrm{Hom}_{\mathbb{Z}_\ell}(M, \mathbb{Z}_\ell)$ for lattices; $V^\vee$ for linear duals and contragredients (Chapter 4). Pontryagin duals do not occur in this chapter.

(C6) *Orientation and the cup product.* A Riemann surface is oriented by its complex structure: in a chart $z = x + iy$ the form $dx \wedge dy = \frac{i}{2} dz \wedge d\bar{z}$ is positive. The cup product of classes in $H^1$ is $\langle \theta, \eta \rangle = \int_X \theta \wedge \eta$ (de Rham representatives). The intersection number $a \cdot b$ of transverse oriented loops counts crossings with sign $+1$ when (tangent of $a$, tangent of $b$) is a positive basis. Poincare duality $\mathrm{PD}\colon H^1 \to H_1$ is fixed by $\int_{\mathrm{PD}(\theta)} \eta = \langle \theta, \eta \rangle$. Other books use other signs; the paper needs only properties that do not change under a global sign (perfectness, alternation, adjointness, ideals of values, Gram determinants in even rank), Appendix B, B-C5(d).

(C7) *Frobenius.* The arithmetic Frobenius $\mathrm{Frob}_q$ of $G_{\mathbb{F}_q}$ raises coordinates of $\bar{\mathbb{F}}_q$-points to the $q$-th power; the $q$-power Frobenius *morphism* of a curve over $\mathbb{F}_q$ is $\pi_q$; the geometric Frobenius is $\mathrm{Frob}_q^{-1}$ (Appendix B, B-C2).

(C8) *Notation.* We write $l(D)$, not $\ell(D)$, for the dimension of a Riemann-Roch space, because $\ell$ is a prime; $K_C$ is a canonical divisor and $K$ a field; $D$ is always a divisor. "Riemann-Roch", "Riemann-Hurwitz", "Abel-Jacobi" take plain hyphens.

**Definition 8.2** (affine varieties)**.** Let $\mathbb{A}^n = \mathbb{A}^n(\bar{k}) = \bar{k}^n$, with $G_k$ acting coordinatewise, so that $\mathbb{A}^n(L) = \mathbb{A}^n(\bar{k})^{G_L}$. For a set $S \subseteq \bar{k}[x_1, \dots, x_n]$ its zero set $V(S) \subseteq \mathbb{A}^n$ is an *algebraic set*; for $Y \subseteq \mathbb{A}^n$, $I(Y)$ is the ideal of polynomials vanishing on $Y$. The algebraic sets are the closed sets of a topology, the *Zariski topology* (finite unions and arbitrary intersections of algebraic sets are algebraic: $V(S) \cup V(S') = V(SS')$, $\bigcap V(S_i) = V(\bigcup S_i)$). An *affine variety* is an algebraic set $V$ with $I(V)$ prime. It is *defined over* $k$ if $I(V)$ is generated by polynomials in $k[x]$. Its *coordinate ring* is the domain $\bar{k}[V] = \bar{k}[x]/I(V)$, its *function field* $\bar{k}(V) = \mathrm{Frac}\,\bar{k}[V]$; if $V$ is defined over $k$, $k[V] = k[x]/(I(V) \cap k[x])$ and $k(V) = \mathrm{Frac}\,k[V]$. A function $f \in \bar{k}(V)$ is *regular at* $P \in V$ if $f = g/h$ with $g, h \in \bar{k}[V]$ and $h(P) \neq 0$; then $f(P) = g(P)/h(P)$ does not depend on the choice. The *dimension* of $V$ is the transcendence degree of $\bar{k}(V)$ over $\bar{k}$ ([109, Chapter VIII, Section 1]; Chapter 5 treats algebraic and transcendental elements).

**Theorem 8.3** (Hilbert's Nullstellensatz; stated)**.** *For every ideal $\mathfrak{a}$ of $\bar{k}[x_1, \dots, x_n]$, $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. Hence $V \mapsto I(V)$ is a bijection from algebraic sets to radical ideals, under which affine varieties correspond to prime ideals and points to maximal ideals. For $k$ perfect, an affine variety $V$ is defined over $k$ iff $\sigma(V) = V$ for all $\sigma \in G_k$.*

We do not prove this; see [72, Chapter 1] and [79, Chapter I, Section 1] for the Nullstellensatz, and [185, Chapter I, Section 1] with its exercises for the Galois criterion. Examples: $\mathbb{A}^1$, with $\bar{k}[\mathbb{A}^1] = \bar{k}[x]$; the parabola $y = x^2$, with $\bar{k}[V] = \bar{k}[x, y]/(y - x^2) \simeq \bar{k}[x]$ and $\dim V = 1$; the affine Weierstrass cubics of Example 8.10.

Why points over $\bar{k}$ and not over $k$? Over a field that is not algebraically closed a polynomial can have too few zeros to be recovered from them ($x^2 + y^2 + 1$ has no zero in $\mathbb{R}^2$, and neither

has 1), so the dictionary between ideals and zero sets of the Nullstellensatz fails; over $\bar{k}$ it holds, and the arithmetic over $k$ is then recorded by the action of $G_k$. Two small examples over $k=\mathbb{Q}$ show how the definitions interact. The set $V(x^2-2)\subseteq\mathbb{A}^1$ is $\{\sqrt{2},-\sqrt{2}\}$; its ideal $(x^2-2)\bar{k}[x]=(x-\sqrt{2})(x+\sqrt{2})\bar{k}[x]$ is not prime, so it is an algebraic set but not a variety, although $x^2-2$ is irreducible over $\mathbb{Q}$. Each of its two points is a variety, $V(x\mp\sqrt{2})$, but neither is defined over $\mathbb{Q}$, and indeed $\sigma(\sqrt{2})=-\sqrt{2}$ for some $\sigma\in G_{\mathbb{Q}}$, so the Galois criterion of the theorem fails for each point while the union is $G_{\mathbb{Q}}$-stable. In the same way $V(x^2+1)\subseteq\mathbb{A}^2$ is the union of the two lines $x=\pm i$, exchanged by complex conjugation: the polynomial $x^2+1$ is irreducible over $\mathbb{Q}$ but not over $\bar{\mathbb{Q}}$. This is why "variety" means "irreducible over $\bar{k}$" in this chapter (Convention 8.1(C2)).

**Definition 8.4** (projective space and projective varieties)**.** $\mathbb{P}^n=\mathbb{P}^n(\bar{k})=(\bar{k}^{n+1}\setminus\{0\})/\bar{k}^\times$, with homogeneous coordinates $[x_0:\cdots:x_n]$; for $n=1$ this is the set of lines of Chapter 1, on which $\mathrm{PGL}_2$ acts by Moebius transformations. $G_k$ acts coordinatewise, and $\mathbb{P}^n(L)$ is the set of points with a representative in $L^{n+1}$. An ideal of $\bar{k}[x_0,\dots,x_n]$ is *homogeneous* if it is generated by homogeneous polynomials; for a set $S$ of homogeneous polynomials $V(S)\subseteq\mathbb{P}^n$ is well defined (whether $F(x)=0$ does not depend on the representative $x$) and is a *projective algebraic set*; a *projective variety* is a projective algebraic set whose homogeneous ideal $I(V)$ is prime, *defined over $k$* if $I(V)$ is generated by polynomials with coefficients in $k$. The sets $U_i=\{x_i\neq 0\}$ are the standard charts, $U_i\simeq\mathbb{A}^n$ by dividing by $x_i$. The *projective closure* $\bar{V}$ of an affine variety $V\subseteq\mathbb{A}^n=U_0$ is the zero set of the homogenisations of the elements of $I(V)$; conversely $V\cap U_i$ is obtained by dehomogenising. The *function field* $\bar{k}(V)$ of a projective variety is the field of quotients $F/G$ of homogeneous polynomials of the same degree with $G\notin I(V)$, two quotients being equal when $FG'-F'G\in I(V)$; such an $f$ is *regular at $P\in V$* if $f=F/G$ with $G(P)\neq 0$, and then $f(P)=F(P)/G(P)$.

**Proposition 8.5** (points over subfields)**.** *Let $k\subseteq L\subseteq\bar{k}$.*

*(i)* $\mathbb{P}^n(\bar{k})^{G_L}=\mathbb{P}^n(L)$.

*(ii)* $\mathbb{P}^n=\mathbb{A}^n\sqcup\mathbb{P}^{n-1}$ *(the chart $U_0$ and the hyperplane $x_0=0$); in particular* $\mathbb{P}^1=\mathbb{A}^1\sqcup\{\infty\}$.

*(iii) For a projective variety $V$ meeting $U_i$, the map $F/G\mapsto(F/G)(x_0,\dots,1,\dots,x_n)$ (the 1 in position $i$) is an isomorphism of $\bar{k}(V)$ onto the function field of the affine variety $V\cap U_i$; for $P\in V\cap U_i$, a function is regular at $P$ on $V$ iff its image is regular at $P$ on $V\cap U_i$, with the same value.*

*Proof.* (i) A point with a representative in $L^{n+1}$ is fixed by $G_L$. Conversely let $P=[x_0:\cdots:x_n]$ be fixed by $G_L$; choose $i$ with $x_i\neq 0$ and scale so that $x_i=1$. For $\sigma\in G_L$, $\sigma P=P$ means $\sigma(x_j)=cx_j$ for all $j$ with one $c\in\bar{k}^\times$; the coordinate $j=i$ gives $c=1$. So every $x_j$ is fixed by $G_L$, hence lies in $L$ ($\bar{k}^{G_L}=L$ by Galois theory, Chapter 5, $k$ being perfect). (ii) A point $[x_0:\cdots:x_n]$ either has $x_0\neq 0$, and then it has exactly one representative with $x_0=1$, namely $(1,x_1/x_0,\dots,x_n/x_0)$, which identifies $U_0$ with $\mathbb{A}^n$; or it has $x_0=0$, and then $[x_1:\cdots:x_n]$ is a well-defined point of $\mathbb{P}^{n-1}$ (not all $x_j$ vanish, and scaling acts on both sides in the same way). For $n=1$ the second set is the single point $[0:1]$, which is called $\infty$. (iii) A quotient $F/G$ of forms of degree $d$ equals $(F/x_i^d)/(G/x_i^d)$, a quotient of polynomials in the affine coordinates $x_j/x_i$; conversely a quotient of polynomials in the affine coordinates is homogenised by multiplying numerator and denominator by a power of $x_i$. The two constructions are inverse, and both respect the ideals. If $G(P)\neq 0$ then $(G/x_i^d)(P)\neq 0$, as $x_i(P)\neq 0$; conversely, if $f=g/h$ with

$h(P) \neq 0$ on $V \cap U_i$, homogenising $g$ and $h$ to forms $F$, $G$ of the same degree $d$ (multiplying the one of lower degree by a power of $x_i$) gives $f = F/G$ with $G(P) = x_i(P)^d h(P) \neq 0$. The values agree, since both are computed from the same representative of $P$ with $x_i = 1$. □

**Definition 8.6** (curves, rational maps, morphisms)**.** A *curve over $k$* is a projective variety of dimension one defined over $k$ (Convention 8.1). Let $V_1$, $V_2 \subseteq \mathbb{P}^n$ be projective varieties. A *rational map* $\varphi = [f_0 : \cdots : f_n] \colon V_1 \dashrightarrow V_2$ is given by $f_i \in \bar{k}(V_1)$, not all zero, such that $\varphi(P) = [f_0(P) : \cdots : f_n(P)] \in V_2$ wherever all $f_i$ are defined and not all zero; $[f_0 : \cdots : f_n]$ and $[gf_0 : \cdots : gf_n]$ define the same map for $g \in \bar{k}(V_1)^\times$. It is *regular* at $P$ if there is $g$ with all $gf_i$ regular at $P$ and some $(gf_i)(P) \neq 0$; a *morphism* is a rational map regular at every point; an *isomorphism* is a morphism with an inverse morphism. $\varphi$ is *defined over $k$* if it can be written with all $f_i \in k(V_1)$; for $k$ perfect this holds iff $\varphi(\sigma P) = \sigma(\varphi(P))$ for all $\sigma \in G_k$ and all $P$ ([185, Chapter I, Section 3]; stated).

**Example 8.7** (a conic is a line)**.** The map $\psi = [s^2 : st : t^2] \colon \mathbb{P}^1 \to \mathbb{P}^2$ is a morphism onto the conic $C \colon XZ = Y^2$, defined over every $k$. Its inverse is $[X : Y] = [Y : Z]$, a rational map on $C$ that is regular everywhere: at points with $X \neq 0$ use $[X : Y]$, at points with $Z \neq 0$ use $[Y : Z]$; the two agree on $C$ because $XZ = Y^2$, and $X = Z = 0$ forces $Y = 0$, which is not a point. So $C \simeq \mathbb{P}^1$ over $k$.

## 8.2 Plane curves and smoothness

The curves of the paper are given by equations: the elliptic curves by Weierstrass equations (the Appendix of the paper calls `ellinit([1,1,1,-30,-76])`), and $X_0(N)$ by the analytic construction of Chapter 12, whose smooth projective model is the curve of Theorem 8.49. This section says when an equation defines a smooth curve and proves the local fact behind every order of a zero or a pole in the rest of the book: at a smooth point the local ring is a discrete valuation ring (Theorem 8.11). Bezout's theorem is stated for the degree of the divisor of a linear form.

**Definition 8.8** (plane curves; singular points)**.** A *plane curve of degree $d$* is $C = V(F) \subseteq \mathbb{P}^2$ for a homogeneous $F \in k[X, Y, Z]$ of degree $d$ that is irreducible in $\bar{k}[X, Y, Z]$; its affine part is $f(x, y) = F(x, y, 1) = 0$. A point $P \in C(\bar{k})$ is *singular* if $F_X(P) = F_Y(P) = F_Z(P) = 0$ (subscripts are partial derivatives) and *smooth* otherwise; $C$ is *smooth* if it has no singular point in $C(\bar{k})$. For a general variety $V \subseteq \mathbb{P}^n$ with $I(V) = (F_1, \ldots, F_m)$ of dimension one, $P$ is smooth if the Jacobian matrix $(\partial F_i / \partial x_j(P))$ has rank $n - 1$ [185, Chapter I, Section 1].

**Proposition 8.9** (the affine criterion; tangent lines)**.** *Let $C = V(F)$ be a plane curve of degree $d$.*

*(i) (Euler) $XF_X + YF_Y + ZF_Z = dF$. Hence a point $P = (a, b)$ of the affine part is singular iff $f(P) = f_x(P) = f_y(P) = 0$; and if $\operatorname{char} k \nmid d$, every common zero in $\mathbb{P}^2$ of $F_X$, $F_Y$, $F_Z$ lies on $C$.*

*(ii) At a smooth point $P$ the* tangent line *is $F_X(P)X + F_Y(P)Y + F_Z(P)Z = 0$; on the affine part it is $f_x(P)(x - a) + f_y(P)(y - b) = 0$.*

*Proof.* (i) Both sides of Euler's identity are additive in $F$, and for a monomial $X^i Y^j Z^m$ with $i + j + m = d$ the left side is $(i + j + m)X^i Y^j Z^m$. On the affine part $f_x(a, b) = F_X(a, b, 1)$ and

$f_y(a,b) = F_Y(a,b,1)$, and at a point of $C$, Euler gives $F_Z(a,b,1) = -aF_X(a,b,1) - bF_Y(a,b,1)$; so the three partials vanish iff $f_x$ and $f_y$ do. If $\operatorname{char} k \nmid d$ and the three partials vanish at $P$, Euler gives $dF(P) = 0$, so $F(P) = 0$. (ii) The projective equation is a definition. On the affine part put $Z = 1$, $X = x$, $Y = y$ and $P = (a,b,1)$; using $F_Z(P) = -af_x(P) - bf_y(P)$ from (i), the projective equation becomes

$$f_x(P)x + f_y(P)y - af_x(P) - bf_y(P) = f_x(P)(x-a) + f_y(P)(y-b) = 0,$$

the affine tangent line of calculus. □

So for an affine equation one never needs the homogenisation away from infinity: smoothness at $(a,b)$ is the familiar condition that the gradient $(f_x, f_y)$ does not vanish there, and (in the real picture) the tangent line is orthogonal to that gradient. Only the points at infinity require the projective partials, and Example 8.10 shows how they are handled.

**Example 8.10** (the paper's curves are smooth; the node and the cusp are not)**.**

(a) The *nodal cubic* $y^2 = x^3 + x^2$ and the *cuspidal cubic* $y^2 = x^3$ have the single singular point $(0,0)$: for $f = y^2 - x^3 - x^2$, $f_y = 2y$ and $f_x = -3x^2 - 2x$ vanish together on the curve only at $(0,0)$ when $\operatorname{char} k \neq 2$ (at $x = -1$, $f_x = -1$), and at infinity ($[0:1:0]$, where $F_Z = Y^2 - X^2 \neq 0$) the curve is smooth; likewise for $y^2 = x^3$. Over $\mathbb{R}$ the node is a loop crossing itself at the origin with two tangent lines $y = \pm x$, and the cusp is a curve with a sharp point at the origin with the single tangent $y = 0$ counted twice.

(b) The *Fermat curve* $X^d + Y^d + Z^d = 0$ is smooth iff $\operatorname{char} k \nmid d$. If $\operatorname{char} k \nmid d$, the partials $dX^{d-1}$, $dY^{d-1}$, $dZ^{d-1}$ vanish together only at $X = Y = Z = 0$. If $d = p = \operatorname{char} k$ the polynomial is $(X+Y+Z)^p$, which is not even irreducible; for $p \mid d$ in general all partials vanish identically.

(c) The Weierstrass cubics

$$\texttt{11a1}\colon y^2 + y = x^3 - x^2 - 10x - 20, \qquad \texttt{20a1}\colon y^2 = x^3 + x^2 + 4x + 4,$$
$$\texttt{121a1}\colon y^2 + xy + y = x^3 + x^2 - 30x - 76$$

are smooth over $\bar{\mathbb{Q}}$, including the point $[0:1:0]$ at infinity. At infinity: for $F = Y^2Z + a_1XYZ + a_3YZ^2 - X^3 - a_2X^2Z - a_4XZ^2 - a_6Z^3$ one has $F_Z(0,1,0) = 1$, so $[0:1:0]$ is always smooth. On the affine part of `11a1`, $f_y = 2y+1$ and $f_x = -3x^2+2x+10$; a singular point would have $y = -\frac{1}{2}$ and then $x$ would be a double root of $4x^3 - 4x^2 - 40x - 79$, whose discriminant is $-2^4 \cdot 11^5 \neq 0$ (computed with PARI). The general criterion (a Weierstrass cubic is smooth iff its discriminant $\Delta$ is nonzero; $\Delta = -11^5$ for `11a1`, $-2^8 \cdot 5^2$ for `20a1`, $-11^2$ for `121a1`) is proved in Chapter 9.

**Theorem 8.11** (smooth points and discrete valuation rings)**.** *Let $C$ be a curve, $P \in C(\bar{k})$, $\bar{k}[C]_P = \{f \in \bar{k}(C) : f$ regular at $P\}$ (Definition 8.4) its local ring and $\mathfrak{m}_P$ the maximal ideal of functions vanishing at $P$. The following are equivalent: (a) $P$ is smooth; (b) $\dim_{\bar{k}} \mathfrak{m}_P/\mathfrak{m}_P^2 = 1$; (c) $\bar{k}[C]_P$ is a discrete valuation ring. At a smooth point $P = (a,b)$ of an affine plane model $f = 0$, the class of a linear form $\lambda = u(x-a) + v(y-b)$ is a uniformiser iff the line $\lambda = 0$ is not the tangent line; for instance $x - a$ is a uniformiser if $f_y(P) \neq 0$. The valuation* $\operatorname{ord}_P\colon \bar{k}(C)^\times \to \mathbb{Z}$ *of $\bar{k}[C]_P$ is the* order of vanishing *at $P$, and $g \in \bar{k}(C)$ is regular at $P$ iff $g = 0$ or $\operatorname{ord}_P g \geq 0$.*

*Proof.* We prove the theorem for a point of an affine plane model $f(x,y)=0$, $f$ irreducible; for a point of a curve in $\mathbb{P}^n$ the same argument with the Jacobian matrix gives the general case [185, Chapter II, Proposition 1.1]. Translate so that $P=(0,0)$. Let $A=\bar{k}[x,y]$, $\mathfrak{M}=(x,y)$, $R=\bar{k}[C]=A/(f)$ and $\mathfrak{m}=\mathfrak{M}/(f)$. The local ring $\bar{k}[C]_P$ is the localisation $R_{\mathfrak{m}}$ (Chapter 3; regularity may be tested on an affine chart by Proposition 8.5(iii)), with maximal ideal $\mathfrak{m}_P=\mathfrak{m}R_{\mathfrak{m}}$. Since $\mathfrak{m}/\mathfrak{m}^2$ is killed by $\mathfrak{m}$, it is a vector space over $R/\mathfrak{m}=\bar{k}$, and an element $s\in R\setminus\mathfrak{m}$ acts on it as the nonzero scalar $s(P)$ (because $s-s(P)\in\mathfrak{m}$), hence invertibly. So inverting the elements of $R\setminus\mathfrak{m}$ changes nothing: $\mathfrak{m}_P/\mathfrak{m}_P^2=\mathfrak{m}/\mathfrak{m}^2=\mathfrak{M}/(\mathfrak{M}^2+(f))$. Now $\mathfrak{M}/\mathfrak{M}^2$ is the two-dimensional space with basis the classes of $x$, $y$, and by Taylor's formula $f\equiv f_x(P)x+f_y(P)y \pmod{\mathfrak{M}^2}$ (as $f(P)=0$). So $\mathfrak{m}_P/\mathfrak{m}_P^2$ is the quotient of $\bar{k}x\oplus\bar{k}y$ by the span of $f_x(P)x+f_y(P)y$: of dimension 1 if $P$ is smooth and 2 if not (Proposition 8.9(i)). This proves (a)$\Leftrightarrow$(b) and shows that a linear form is nonzero in $\mathfrak{m}_P/\mathfrak{m}_P^2$ iff it is not proportional to the tangent form.

The ring $\bar{k}[C]_P$ is a Noetherian local domain (a localisation of a quotient of the Noetherian ring $A$), and not a field: $C$ has dimension one, so $x$ and $y$ are not both constant on $C$, and a nonconstant one of them is a nonzero element of $\mathfrak{m}_P$. (b)$\Rightarrow$(c): if $t\in\mathfrak{m}_P$ spans $\mathfrak{m}_P/\mathfrak{m}_P^2$, then $\mathfrak{m}_P=t\bar{k}[C]_P+\mathfrak{m}_P^2$, and Nakayama's lemma (Chapter 3) gives $\mathfrak{m}_P=(t)$; a Noetherian local domain that is not a field and has principal maximal ideal is a discrete valuation ring with uniformiser $t$ (the criterion of Chapter 3). (c)$\Rightarrow$(b): if $\mathfrak{m}_P=(t)$ then $\mathfrak{m}_P/\mathfrak{m}_P^2$ is spanned by the class of $t$, and it is nonzero because $\mathfrak{m}_P\neq\mathfrak{m}_P^2$ by Nakayama. The last sentence is the description of a discrete valuation ring by its valuation (Chapter 3). □

Two computations show the theorem at work. On the smooth cubic $y^2=h(x)$, $h$ squarefree of degree 3, $\operatorname{char}k\neq 2$, take a root $e$ of $h$ and $P=(e,0)$. There $f_y=2y$ vanishes and $f_x=-h'(e)\neq 0$, so the tangent line is the vertical line $x=e$, and $x-e$ is *not* a uniformiser; $y$ is one. Indeed $x-e=y^2/q(x)$ with $q(x)=h(x)/(x-e)$, and $q(e)=h'(e)\neq 0$, so $q(x)$ is a unit at $P$ and $\operatorname{ord}_P(x-e)=2$: the vertical tangent meets the curve doubly at $P$. At the node of $y^2=x^3+x^2$ the computation of the proof gives $\mathfrak{m}_P/\mathfrak{m}_P^2=\bar{k}x\oplus\bar{k}y$ (the gradient vanishes), of dimension two, so the local ring at the node is not a discrete valuation ring and no single order of vanishing can be attached to a function there; the smooth model of Theorem 8.13 replaces the node by two points, each with its own valuation.

**Theorem 8.12** (Bezout; stated)**.** *Two plane curves of degrees $m$ and $n$ without a common component meet in exactly $mn$ points of $\mathbb{P}^2(\bar{k})$ counted with intersection multiplicity. In particular a line not contained in a plane curve of degree $d$ meets it in $d$ points counted with multiplicity, and a line meets a cubic in three points (the chord-tangent construction of Chapter 9).*

We do not prove this; see [72, Chapter 5, Section 3] and [79, Chapter I, Section 7]. On a smooth plane curve the intersection multiplicity of the curve with a line $L=0$ at $P$ is $\operatorname{ord}_P(L/M)$ for any linear form $M$ with $M(P)\neq 0$ [72, Chapter 3], so the divisor of zeros of $L/M$ (Definition 8.21) has degree $d$. For example, on a Weierstrass cubic the line at infinity $Z=0$ meets the curve only at $O=[0:1:0]$ (put $Z=0$ in the equation: $X^3=0$), and in the chart $Y=1$ the function $Z/Y=v$ has $\operatorname{ord}_O v=3$ (computed for $y^2=h(x)$ in the proof of Proposition 8.26(iv)): the line at infinity is a flex tangent, meeting the cubic at $O$ with multiplicity $3=d$, as Bezout requires.

**Theorem 8.13** (smooth projective models; stated)**.** *Every curve, possibly singular or affine, is birational over $k$ to a smooth projective curve, which is unique up to isomorphism over $k$. Its points over $\bar{k}$ correspond bijectively to the discrete valuations of $\bar{k}(C)$ trivial on $\bar{k}$.*

We do not prove this; it follows from Theorem 8.17, see [79, Chapter I, Section 6] and [72, Chapter 7]. Example (proved): the nodal cubic $y^2 = x^3 + x^2$ is birational to $\mathbb{P}^1$ through $t = y/x$, with inverse $x = t^2 - 1$, $y = t(t^2 - 1)$: substituting $y = tx$ in the equation gives $x^2(t^2 - x - 1) = 0$. The two branches through the node correspond to $t = \pm 1$: the smooth model separates them. In the paper a curve given by an equation always means the smooth projective model.

## 8.3 Function fields, points over extensions, the Galois action

This section turns a curve into its field of rational functions, and back. Its uses in the paper are two. The modular parametrisation $\varphi_E \colon X_0(N) \to E$ is the same thing as the field embedding $\varphi_E^* \colon \mathbb{Q}(E) \to \mathbb{Q}(X_0(N))$ (Theorem 8.17), and its degree is the degree of that field extension (Definition 8.37). And the Galois representation $\rho_{E,\ell}$ of Section 1.1 of the paper, and the action of $G_\mathbb{Q}$ on $E[3]$ in Proposition prop:ns, are the action of $G_k$ on points of Proposition 8.18.

**Definition 8.14** (function fields of one variable)**.** A *function field of one variable over $k$* is a finitely generated field extension $F/k$ of transcendence degree one in which $k$ is algebraically closed. For a curve $C$ over $k$, $k(C)$ is one, and $\bar{k}(C) = \bar{k} \cdot k(C)$ is one over $\bar{k}$ [185, Chapter II, Section 1].

The examples to keep in mind are $k(\mathbb{P}^1) = k(x)$, and, for a Weierstrass cubic $E$, $k(E) = k(x)[y]/(y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6)$, a quadratic extension of $k(x)$. "Transcendence degree one" says that the curve has dimension one. The condition that $k$ be algebraically closed in $F$ excludes, for $k = \mathbb{Q}$, a field such as $\mathbb{Q}(i)(y)$: it is finitely generated of transcendence degree one over $\mathbb{Q}$, and it is the function field over $\mathbb{Q}$ of the plane set $x^2 + 1 = 0$, which over $\bar{\mathbb{Q}}$ falls apart into the two conjugate lines $x = \pm i$ (the second example after Theorem 8.3). A function field of one variable over $k$ is thus the function field of a curve that stays irreducible over $\bar{k}$.

**Proposition 8.15** (valuations of a function field)**.** *Let $C$ be a smooth curve.*

*(i) Each $P \in C(\bar{k})$ gives a discrete valuation $\operatorname{ord}_P$ of $\bar{k}(C)$ with $\operatorname{ord}_P(c) = 0$ for $c \in \bar{k}^\times$.*

*(ii) For $f \in \bar{k}(C)^\times$, $\operatorname{ord}_P f = 0$ for all but finitely many $P$.*

*(iii) A rational function without poles is constant (proved in Corollary 8.39(vi)).*

*Proof.* (i) is Theorem 8.11; a nonzero constant is a unit of every local ring. (ii) Cover $C$ by finitely many affine pieces (the charts $U_i$). On one piece write $f = g/h$ with $g$, $h$ regular functions, not zero. Off the zero sets $V(g)$ and $V(h)$ of the piece, $f$ is a unit of the local ring, so $\operatorname{ord}_P f = 0$. The zero set of a nonzero regular function on an irreducible curve is a proper closed subset, hence finite (a proper closed subset of an irreducible variety of dimension one has dimension zero; [79, Chapter I, Section 1], stated). (iii) is proved after Theorem 8.38. □

**Theorem 8.16** (rational maps from smooth curves are morphisms)**.** *Let $C$ be a smooth curve, $V \subseteq \mathbb{P}^n$ a projective variety and $\varphi = [f_0 : \cdots : f_n] \colon C \dashrightarrow V$ a rational map. Then $\varphi$ is a morphism. In particular a rational map between smooth projective curves is a morphism, and every $f \in \bar{k}(C)$ defines a morphism $[f : 1] \colon C \to \mathbb{P}^1$ (constant if $f$ is), with $\infty = [1 : 0]$ the image of the poles.*

*Proof.* Let $P \in C$, $t$ a uniformiser at $P$ (Theorem 8.11) and $m = \min\{\operatorname{ord}_P f_i : f_i \neq 0\}$. Then each $t^{-m} f_i$ has $\operatorname{ord}_P \geq 0$, so is regular at $P$, and for an index with $\operatorname{ord}_P f_i = m$ the function $t^{-m} f_i$ is a unit at $P$, nonzero at $P$. So $g = t^{-m}$ shows that $\varphi$ is regular at $P$. For $f \in \bar{k}(C)$ nonconstant this applies to $[f : 1]$; at a pole $m = \operatorname{ord}_P f < 0$ and $[t^{-m} f : t^{-m}](P) = [1 : 0]$. □

Smoothness of the source is essential. On the nodal cubic $y^2 = x^3 + x^2$ the rational map $t = y/x$ to $\mathbb{P}^1$ is not regular at the node. The parametrisation $\psi(t) = (t^2 - 1, t(t^2 - 1))$ of the example after Theorem 8.13 is a morphism $\mathbb{P}^1 \to C$ (extended to $\infty$ by this theorem, since $\mathbb{P}^1$ is smooth), and $(y/x) \circ \psi$ is the identity of $\mathbb{P}^1$. If $y/x$ were regular at the node $S$ with some value $c$, then evaluating $(y/x) \circ \psi$ at $t = 1$ and at $t = -1$, both of which map to $S$, would give $c = 1$ and $c = -1$. The two branches through the node ask for two different values, which a single point cannot provide; the smooth model has two points there.

**Theorem 8.17** (curves and function fields; stated)**.** *The functor $C \mapsto k(C)$, $(\varphi\colon C_1 \to C_2) \mapsto (\varphi^*\colon k(C_2) \to k(C_1),\ f \mapsto f \circ \varphi)$ is a contravariant equivalence between the category of smooth projective curves over $k$ with nonconstant morphisms defined over $k$, and the category of function fields of one variable over $k$ with $k$-embeddings. In particular two smooth projective curves with $k$-isomorphic function fields are isomorphic over $k$, and a nonconstant morphism of degree one (Definition 8.37) is an isomorphism.*

We do not prove this; see [185, Chapter II, Theorem 2.4] and [79, Chapter I, Corollary 6.12]. The idea: the points of the smooth projective model are the discrete valuations of the function field trivial on the constants (Theorem 8.13), and a field embedding pulls back valuations.

**Proposition 8.18** (the Galois action on points and functions)**.** *Let $C$ be a curve over $k$.*

*(i) $G_k$ acts on $C(\bar{k})$, and $C(\bar{k})^{G_L} = C(L)$ for $k \subseteq L \subseteq \bar{k}$.*

*(ii) $G_k$ acts on $\bar{k}(C)$ through the coefficients: for $f = F/G$, $\sigma f = F^\sigma / G^\sigma$. For $f$ regular at $P$, $\sigma f$ is regular at $\sigma P$ and $(\sigma f)(\sigma P) = \sigma(f(P))$; for $C$ smooth, $\operatorname{ord}_{\sigma P}(\sigma f) = \operatorname{ord}_P f$. The fixed field of $G_k$ on $\bar{k}(C)$ is $k(C)$ (stated).*

*(iii) A morphism $\varphi$ defined over $k$ satisfies $\varphi(\sigma P) = \sigma(\varphi(P))$; hence $\varphi(C_1(L)) \subseteq C_2(L)$, and the fibre $\varphi^{-1}(Q)$ over $Q \in C_2(k)$ is $G_k$-stable.*

*Proof.* (i) $C(\bar{k})$ is $G_k$-stable: if $F(P) = 0$ for every $F$ in a set of generators of $I(C)$ with coefficients in $k$, then $F(\sigma P) = \sigma(F(P)) = 0$. The fixed points are given by Proposition 8.5(i). (ii) $F^\sigma$ (apply $\sigma$ to the coefficients) is well defined on $\bar{k}(C)$ because $I(C)$ is generated by polynomials with coefficients in $k$, so $\sigma$ maps $I(C) \otimes \bar{k}$ to itself. For $f = F/G$ with $G(P) \neq 0$, $F^\sigma(\sigma P) = \sigma(F(P))$ and $G^\sigma(\sigma P) = \sigma(G(P)) \neq 0$. So $f \mapsto \sigma f$ is a field automorphism of $\bar{k}(C)$ (semilinear over $\sigma$) that maps the local ring at $P$ onto the local ring at $\sigma P$ and $\mathfrak{m}_P$ onto $\mathfrak{m}_{\sigma P}$; it therefore maps a uniformiser at $P$ to one at $\sigma P$ and preserves valuations. That the fixed field is $k(C)$ is Hilbert's Theorem 90 for the vector space $\bar{k}(C)$ over $\bar{k}$ [185, Chapter II, Section 5]. (iii) If $\varphi = [f_0 : \cdots : f_n]$ with $f_i \in k(C_1)$, then $\sigma f_i = f_i$, and by (ii) $\sigma(\varphi(P)) = [\sigma f_0(\sigma P) : \ldots] = \varphi(\sigma P)$. If some $g \in \bar{k}(C_1)^\times$ is needed to evaluate at $P$, that is $\varphi(P) = [(gf_0)(P) : \cdots : (gf_n)(P)]$, then $\sigma(\varphi(P)) = [(\sigma g\, f_0)(\sigma P) : \cdots : (\sigma g\, f_n)(\sigma P)]$, and $\sigma g$ does the same job at $\sigma P$, so this is $\varphi(\sigma P)$. The rest follows from (i). □

*Warning* 8.19 (a curve need not have a rational point). The conic $X^2 + Y^2 + Z^2 = 0$ over $\mathbb{Q}$ has no point over $\mathbb{R}$, so $C(\mathbb{Q}) = \emptyset$; but $[1 : i : 0] \in C(\mathbb{Q}(i))$. It is a smooth curve of genus 0 (Example 8.32)

that is isomorphic to $\mathbb{P}^1$ over $\mathbb{Q}(i)$ (project from $[1:i:0]$, as in Corollary 8.30(d)) but not over $\mathbb{Q}$, since an isomorphism over $\mathbb{Q}$ would map $\mathbb{P}^1(\mathbb{Q}) \neq \emptyset$ into $C(\mathbb{Q})$ (Proposition 8.18(iii)). Selmer's cubic $3X^3 + 4Y^3 + 5Z^3 = 0$ is smooth of genus one with no point over $\mathbb{Q}$, although it has points over $\mathbb{R}$ and over every $\mathbb{Q}_p$ (stated; see [185, Chapter X], section not verified). So Theorem 8.33 and Proposition 8.34 assume a rational point, and the paper's curves have one: $O$ on $E$, and the cusp $\infty$ on $X_0(N)$.

*In the paper* 8.20. Section 1.1 of the paper: $\rho_{E,\ell}$ is the action of $G_{\mathbb{Q}}$ on $E[\ell^n] \subseteq E(\overline{\mathbb{Q}})$ of Proposition 8.18(i), extended to $T_\ell E = \varprojlim E[\ell^n]$ (Chapters 9 and 11). This action respects the group law because the group law is given by rational functions with coefficients in $\mathbb{Q}$ (Proposition 8.34). Proposition prop:ns: $G_{\mathbb{Q}}$ acts on the eight points of $E[3]\setminus\{O\}$ and hence on the four roots of the 3-division polynomial, the $x$-coordinates of those points, by Proposition 8.18(ii), (iii) applied to the function $x$, which is defined over $\mathbb{Q}$: $x(\sigma P) = \sigma(x(P))$.

## 8.4 Divisors, differentials, the Picard group

Divisors record zeros and poles; the Picard group is the group of divisors modulo divisors of functions, and its degree-zero part is the Jacobian of Section 8.11. Differentials give the canonical class and, through Riemann-Roch, the genus. In the paper these objects stand behind the group law of $E$ (Proposition 8.34), behind the Jacobian $J_0(N)$ of Remark rem:free, and behind the differential $\omega_E$ whose pullback $\varphi_E^*\omega_E = c_E \cdot 2\pi i\, f(z)\, dz$ appears in the proof of Proposition prop:pullback (Chapter 14). Throughout, $C$ is a smooth projective curve over $k$.

**Definition 8.21** (divisors)**.** $\mathrm{Div}(C)$ is the free abelian group on $C(\bar{k})$; its elements are finite sums $D = \sum n_P(P)$. The *degree* is $\deg D = \sum n_P$; $D \geq 0$ (*effective*) means all $n_P \geq 0$; $\mathrm{supp}\, D = \{P : n_P \neq 0\}$. $G_k$ acts by $\sigma D = \sum n_P(\sigma P)$, and $\mathrm{Div}_k(C) = \mathrm{Div}(C)^{G_k}$ is the group of divisors *defined over* $k$; for instance the sum of the points of a $G_k$-orbit. The *divisor* of $f \in \bar{k}(C)^\times$ is $\mathrm{div}(f) = \sum_P \mathrm{ord}_P(f)(P)$, a finite sum by Proposition 8.15(ii); its positive and negative parts are the divisors of zeros $\mathrm{div}_0(f)$ and of poles $\mathrm{div}_\infty(f)$. Divisors of functions are *principal*; they form a subgroup $\mathrm{Prin}(C)$, since $\mathrm{div}(fg) = \mathrm{div}(f) + \mathrm{div}(g)$.

**Proposition 8.22** (principal divisors)**.** *Let $f \in \bar{k}(C)^\times$.*

*(i)* $\mathrm{div}(f) = 0$ *iff* $f \in \bar{k}^\times$.

*(ii)* $\deg \mathrm{div}(f) = 0$*; more precisely, for $f$ nonconstant,* $\deg \mathrm{div}_0(f) = \deg \mathrm{div}_\infty(f) = [\bar{k}(C) : \bar{k}(f)]$.

*(iii)* $\mathrm{div}(\sigma f) = \sigma\, \mathrm{div}(f)$ *for* $\sigma \in G_k$.

*(iv) On $\mathbb{P}^1$ with coordinate $x$:* $\mathrm{div}(x - a) = (a) - (\infty)$*, and* $\mathrm{div}\big(c\prod (x - a_i)^{n_i}\big) = \sum n_i(a_i) - (\sum n_i)(\infty)$.

*Proof.* (i) If $\mathrm{div}(f) = 0$, $f$ has no pole, so it is constant (Proposition 8.15(iii)); constants have divisor 0. (ii) is Corollary 8.39(vi), which uses the degree formula of Section 8.6; over $\mathbb{C}$ a second proof is Corollary 8.53. (iii) is Proposition 8.18(ii). (iv) At $a \in \bar{k}$ the function $x - a$ is a uniformiser; at $\infty$ the function $t = 1/x$ is one (in the chart $[1:t]$), and $x - a = t^{-1}(1 - at)$ with $1 - at$ a unit at $\infty$, so $\mathrm{ord}_\infty(x - a) = -1$. The second formula follows by additivity. □

*Example: divisors on a cubic.* Let $C$ be the smooth cubic $y^2 = h(x) = (x-e_1)(x-e_2)(x-e_3)$, $e_i$ distinct, $\operatorname{char} k \neq 2$, $O = [0:1:0]$ and $T_i = (e_i, 0)$. The orders needed are computed in the remarks after Theorem 8.11 and in the proof of Proposition 8.26(iv) below: $y$ is a uniformiser at each $T_i$, $\operatorname{ord}_O y = -3$, $\operatorname{ord}_O x = -2$, and at an affine point with $y \neq 0$ the function $x - a$ is a uniformiser. Hence

$$\operatorname{div}(y) = (T_1) + (T_2) + (T_3) - 3(O), \qquad \operatorname{div}(x - e_1) = 2(T_1) - 2(O),$$
$$\operatorname{div}(x - a) = (P) + (P') - 2(O),$$

the last for $a$ not a root of $h$, with $P, P' = (a, \pm b)$ and $b^2 = h(a) \neq 0$. All three have degree zero, as Proposition 8.22(ii) requires, and the degrees of the zero divisors, 3 and 2, are the field degrees $[\bar{k}(C) : \bar{k}(y)]$ and $[\bar{k}(C) : \bar{k}(x)]$ (the second because $y$ satisfies a quadratic equation over $\bar{k}(x)$ and is not in $\bar{k}(x)$, $h$ not being a square). Note also $\operatorname{div}(y^2) = 2\operatorname{div}(y) = \sum_i \operatorname{div}(x - e_i)$, as it must be, since $y^2 = \prod(x - e_i)$.

**Definition 8.23** (linear equivalence; the Picard group)**.** $D \sim D'$ (*linearly equivalent*) iff $D - D'$ is principal. $\operatorname{Pic}(C) = \operatorname{Div}(C)/\operatorname{Prin}(C)$. By Proposition 8.22(ii) the degree induces $\deg\colon \operatorname{Pic}(C) \to \mathbb{Z}$, with kernel $\operatorname{Pic}^0(C) = \operatorname{Div}^0(C)/\operatorname{Prin}(C)$, $\operatorname{Div}^0$ the divisors of degree zero.

**Proposition 8.24** (the Picard group)**.**

*(i)* $\operatorname{Pic}^0(\mathbb{P}^1) = 0$, *and* $\deg\colon \operatorname{Pic}(\mathbb{P}^1) \to \mathbb{Z}$ *is an isomorphism.*

*(ii)* $G_k$ *acts on* $\operatorname{Pic}(C)$ *and* $\operatorname{Pic}^0(C)$.

*(iii) The natural map* $\operatorname{Div}^0_k(C)/\operatorname{div}(k(C)^\times) \to \operatorname{Pic}^0(C)^{G_k}$ *is injective; it is bijective when* $C(k) \neq \emptyset$ *(stated).*

*Proof.* (i) Every $f \in \bar{k}(\mathbb{P}^1)^\times = \bar{k}(x)^\times$ is $c\prod(x - a_i)^{n_i}$ (factor numerator and denominator over $\bar{k}$), so by Proposition 8.22(iv) a divisor $\sum n_i(a_i) + n_\infty(\infty)$ of degree zero, that is with $n_\infty = -\sum n_i$, is $\operatorname{div}\prod(x - a_i)^{n_i}$. (ii) $\sigma$ preserves degrees and, by Proposition 8.22(iii), principal divisors. (iii) Let $D \in \operatorname{Div}^0_k(C)$ with $D = \operatorname{div}(f)$ for some $f \in \bar{k}(C)^\times$. Choose a finite Galois extension $L/k$ containing the coefficients of $f$. For $\sigma \in G_k$, $\operatorname{div}(\sigma f) = \sigma D = D$, so $\sigma f = c_\sigma f$ with $c_\sigma \in \bar{k}^\times$ by Proposition 8.22(i); in fact $c_\sigma \in L^\times$ and $c_\sigma$ depends only on $\sigma|_L$. Indeed, $\sigma f$ is computed by applying $\sigma|_L$ to the coefficients of $f$, which lie in $L$, so it depends only on $\sigma|_L$ and again has coefficients in $L$ ($L/k$ being Galois). So $c_\sigma = \sigma f/f$ is a constant lying in $L(C)$, and the constants in $L(C)$ are the elements of $L$, since $L$ is algebraically closed in $L(C)$ (Definition 8.14, applied over $L$). From $\sigma\tau f = \sigma(c_\tau)c_\sigma f$ we get $c_{\sigma\tau} = \sigma(c_\tau)c_\sigma$: a 1-cocycle of $\operatorname{Gal}(L/k)$ with values in $L^\times$. By Hilbert's Theorem 90 (Chapter 5) there is $b \in L^\times$ with $c_\sigma = \sigma(b)/b$; then $f/b$ is fixed by $G_k$, so lies in $k(C)$ (Proposition 8.18(ii)), and $D = \operatorname{div}(f/b)$. This is injectivity. Surjectivity when $C(k) \neq \emptyset$ uses the vanishing of an obstruction in the Brauer group of $k$; we do not prove it, see [130, Section 1] (section number not verified) and Chapter 17 for Hilbert 90 in cohomological form. □

**Definition 8.25** (differentials)**.** The space $\Omega_C$ of (meromorphic) *differentials* is the $\bar{k}(C)$-vector space generated by symbols $df$, $f \in \bar{k}(C)$, subject to $d(f+g) = df + dg$, $d(fg) = f\,dg + g\,df$ and $da = 0$ for $a \in \bar{k}$. For a uniformiser $t$ at $P$, every $\omega \in \Omega_C$ is $\omega = g\,dt$ with a unique $g = \omega/dt \in \bar{k}(C)$, and $\operatorname{ord}_P \omega := \operatorname{ord}_P g$. Then $\operatorname{div}(\omega) = \sum_P \operatorname{ord}_P(\omega)(P)$; $\omega$ is *regular* (or holomorphic) if $\operatorname{div}(\omega) \geq 0$, and the regular differentials form a $\bar{k}$-vector space.

The definition is the algebraic shadow of the expressions $g(z)\,dz$ of complex analysis: the rules are those of the differential calculus, and $\omega/dt$ is "the coefficient of $dt$" in a local coordinate. Differentials are introduced because, unlike functions, they can be regular everywhere on a projective curve without being constant multiples of each other, and the number of independent regular ones turns out to be the genus (Corollary 8.30(a)). For instance on $\mathbb{P}^1$, $d(x^2) = 2x\,dx$, and at $a \neq 0$ with uniformiser $x - a$, $d(x^2)/d(x-a) = 2x$ has order 0 (if $\operatorname{char} k \neq 2$), while at 0 it has order 1.

**Proposition 8.26** (differentials; the canonical class)**.**

*(i)* $\Omega_C$ *is one-dimensional over* $\bar{k}(C)$*, and* $df$ *is a basis iff* $\bar{k}(C)/\bar{k}(f)$ *is a finite separable extension (in characteristic* $p$*,* $d(f^p) = pf^{p-1}df = 0$*). In particular* $dt$ *is a basis for a uniformiser* $t$ *(stated).*

*(ii)* $\operatorname{ord}_P \omega$ *does not depend on the uniformiser* $t$*; if* $f$ *is regular at* $P$*, so is* $df/dt$*; and* $\operatorname{ord}_P \omega = 0$ *for all but finitely many* $P$ *(stated).*

*(iii)* $\operatorname{div}(f\omega) = \operatorname{div}(f) + \operatorname{div}(\omega)$*. Hence the divisors of nonzero differentials form one linear equivalence class, the* canonical class*; any* $\operatorname{div}(\omega)$*,* $\omega \neq 0$*, is a* canonical divisor $K_C$*.*

*(iv) Examples. On* $\mathbb{P}^1$*,* $\operatorname{div}(dx) = -2(\infty)$*, so* $\mathbb{P}^1$ *has no nonzero regular differential. On the smooth cubic* $y^2 = h(x)$ *with* $h$ *squarefree of degree* 3 *and* $\operatorname{char} k \neq 2$*,* $\operatorname{div}(dx/y) = 0$*.*

*Proof.* (i) and (ii) are [185, Chapter II, Propositions 4.2 and 4.3]; we do not prove them. (iii) $f\omega = (fg)\,dt$, and $\operatorname{ord}_P(fg) = \operatorname{ord}_P f + \operatorname{ord}_P g$. If $\omega, \omega' \neq 0$ then $\omega' = f\omega$ for some $f \in \bar{k}(C)^\times$ by (i), so $\operatorname{div}\omega' - \operatorname{div}\omega = \operatorname{div} f$; conversely every divisor linearly equivalent to $\operatorname{div}\omega$ is $\operatorname{div}(f\omega)$.

(iv) On $\mathbb{P}^1$, at $a \in \bar{k}$, $dx = d(x-a)$ has order 0; at $\infty$, with $t = 1/x$, $dx = d(t^{-1}) = -t^{-2}dt$ has order $-2$. A regular differential $f\,dx$ would need $\operatorname{div} f \geq 2(\infty)$, of positive degree, impossible by Proposition 8.22(ii) unless $f = 0$.

On $y^2 = h(x)$ let $P = (a, b)$. If $b \neq 0$, then $f_y = 2b \neq 0$, so $x - a$ is a uniformiser (Theorem 8.11), $\operatorname{ord}_P dx = 0$ and $y$ is a unit: $\operatorname{ord}_P(dx/y) = 0$. If $b = 0$, then $h(a) = 0$ and $f_x(P) = -h'(a) \neq 0$ as $h$ is squarefree, so $y$ is a uniformiser; differentiating $y^2 = h(x)$ gives $2y\,dy = h'(x)\,dx$, so $dx = \frac{2y}{h'(x)}dy$ with $h'(x)$ a unit at $P$: $\operatorname{ord}_P dx = 1$ and $\operatorname{ord}_P(dx/y) = 0$. At the point $O = [0:1:0]$ at infinity (write $h = x^3 + c_2x^2 + c_1x + c_0$ after scaling) use the chart $Y = 1$, with coordinates $u = x/y$, $v = 1/y$; the equation $Y^2Z = X^3 + c_2X^2Z + c_1XZ^2 + c_0Z^3$ becomes $v(1 - c_2u^2 - c_1uv - c_0v^2) = u^3$. Its partial derivative in $v$ at $(0,0)$ is 1, so $O$ is smooth and $u$ is a uniformiser (the tangent line is $v = 0$). The bracket is a unit at $O$, so $\operatorname{ord}_O v = 3$; hence $\operatorname{ord}_O y = -3$ and $\operatorname{ord}_O x = \operatorname{ord}_O(u/v) = -2$. Write $x = u^{-2}w$ with $w = u^3/v$ a unit, $w(O) = 1$; then $dx = (-2u^{-3}w + u^{-2}w')du$ with $w' = dw/du$ regular (by (ii)), and $-2w + uw'$ takes the value $-2 \neq 0$ at $O$. So $\operatorname{ord}_O dx = -3$ and $\operatorname{ord}_O(dx/y) = 0$. □

The translation invariance of $dx/y$, the *invariant differential*, is proved in Chapter 9; for a general Weierstrass equation the differential is $dx/(2y + a_1x + a_3)$, with the same divisor (Example 8.32(b)).

## 8.5 Riemann-Roch and its consequences

Riemann-Roch is the one deep theorem of the algebraic side; it is stated with a reference and the shape of its proof, and everything after it in this section is proved. Its consequences give the

genus as the number of independent regular differentials, show that a genus-one curve with a rational point is a Weierstrass cubic (Theorem 8.33), and identify such a curve with its own $\mathrm{Pic}^0$ (Proposition 8.34). In the paper these are the facts that an elliptic curve over $\mathbb{Q}$ is given by a Weierstrass equation (as in the Appendix app:repro) and that $E = \mathrm{Pic}^0(E)$ is its own Jacobian, which underlies the factorisation $X_0(N) \to J_0(N) \to E$ used in Proposition prop:pullback.

**Definition 8.27** (Riemann-Roch spaces)**.** For $D \in \mathrm{Div}(C)$, $\mathcal{L}(D) = \{f \in \bar{k}(C)^\times : \mathrm{div}(f) + D \geq 0\} \cup \{0\}$, the functions whose poles are bounded by $D$ and which vanish where $D$ requires; $l(D) = \dim_{\bar{k}} \mathcal{L}(D)$.

**Proposition 8.28** (Riemann-Roch spaces)**.**

*(i)* $\mathcal{L}(D)$ *is a* $\bar{k}$*-vector space;* $\mathcal{L}(0) = \bar{k}$*;* $\mathcal{L}(D) = 0$ *if* $\deg D < 0$*; if* $D' = D + \mathrm{div}\, h$ *then* $f \mapsto fh^{-1}$ *is an isomorphism* $\mathcal{L}(D) \to \mathcal{L}(D')$*.*

*(ii)* $l(D + (P)) \leq l(D) + 1$*. Hence* $l(D) \leq \deg D + 1$ *for* $D \geq 0$*, and* $l(D)$ *is finite for every* $D$*.*

*(iii)* $l(D)$ *depends only on the class of* $D$ *in* $\mathrm{Pic}(C)$*; so does* $l(K_C - D)$*.*

*(iv) If* $D$ *is defined over* $k$*,* $\mathcal{L}(D)$ *has a basis in* $k(C)$*, and* $\mathcal{L}(D) \cap k(C)$ *is a* $k$*-space of dimension* $l(D)$ *(stated, except the second clause).*

*Proof.* (i) If $\mathrm{div}\, f \geq -D$ and $\mathrm{div}\, g \geq -D$, then $\mathrm{ord}_P(f+g) \geq \min(\mathrm{ord}_P f, \mathrm{ord}_P g) \geq -n_P$ for every $P$; scalars do not change divisors. $\mathcal{L}(0)$ is the space of functions without poles, the constants (Proposition 8.15(iii)). If $f \in \mathcal{L}(D)$, $f \neq 0$, then $\mathrm{div}\, f + D \geq 0$ has degree $\deg D$ (Proposition 8.22(ii)), so $\deg D \geq 0$. Finally $\mathrm{div}(fh^{-1}) + D' = \mathrm{div}\, f + D$.

(ii) Let $n$ be the coefficient of $P$ in $D$ and $t$ a uniformiser at $P$. For $f \in \mathcal{L}(D + (P))$, $\mathrm{ord}_P f \geq -n-1$, so $t^{n+1}f$ is regular at $P$, and $\lambda(f) = (t^{n+1}f)(P)$ is a linear map $\mathcal{L}(D+(P)) \to \bar{k}$. Its kernel is $\{f : \mathrm{ord}_P f \geq -n\} = \mathcal{L}(D)$. So $l(D+(P)) \leq l(D)+1$. For $D \geq 0$ add the points one at a time starting from $l(0) = 1$. For general $D$, $\mathcal{L}(D) \subseteq \mathcal{L}(D^+)$, where $D^+$ keeps the positive coefficients.

(iii) is (i), applied to $D$ and to $K_C - D$. (iv) The first clause is [185, Chapter II, Proposition 5.8] (Hilbert 90 for vector spaces). Given a basis $f_1, \dots, f_m$ in $k(C)$, if $f = \sum c_i f_i \in k(C)$ then $\sigma f = f$ gives $\sum(\sigma c_i - c_i) f_i = 0$, so $c_i \in k$: the $k$-span of the $f_i$ is $\mathcal{L}(D) \cap k(C)$. □

**Theorem 8.29** (Riemann-Roch; stated)**.** *Let* $C$ *be a smooth projective curve over* $k$ *and* $K_C$ *a canonical divisor. There is an integer* $g \geq 0$*, the* genus *of* $C$*, such that for every* $D \in \mathrm{Div}(C)$

$$l(D) - l(K_C - D) = \deg D - g + 1.$$

We do not prove this; see [185, Chapter II, Theorem 5.4], which refers the proof to Hartshorne and Lang, [79, Chapter IV, Theorem 1.3], [72, Chapter 8] and [136, Chapter VI] (chapter numbers of Miranda not verified). *Shape of the proof.* By Proposition 8.28(ii), $\deg D + 1 - l(D)$ is bounded above as $D$ grows (one compares $\mathcal{L}(D)$ with $\mathcal{L}(D + n(P))$ and uses a function with a single pole); the maximum of this defect is the genus, which gives Riemann's inequality $l(D) \geq \deg D + 1 - g$. The correction term $l(K_C - D)$ is the dimension of the space of obstructions to finding a function with prescribed principal parts bounded by $D$ (the algebraic Mittag-Leffler problem, in the language of repartitions), and residues of differentials make that space dual to $\mathcal{L}(K_C - D)$; this duality is Serre duality. Over $\mathbb{C}$ the obstruction space is the sheaf cohomology group $H^1(X, \mathcal{O}(D))$, the only appearance of sheaf cohomology in the book; it is never used in a proof.

**Corollary 8.30** (consequences of Riemann-Roch)**.** *Let $C$ have genus $g$.*

*(a)* $l(K_C) = g$*: the genus is the dimension of the space of regular differentials.*

*(b)* $\deg K_C = 2g - 2$*.*

*(c) If* $\deg D > 2g - 2$ *then* $l(D) = \deg D - g + 1$*.*

*(d)* $g(\mathbb{P}^1) = 0$*. Conversely, a curve $C$ of genus 0 with a point $P \in C(k)$ is isomorphic over $k$ to $\mathbb{P}^1$. A curve with two distinct points $P$, $Q$ such that $(P) \sim (Q)$ has genus 0.*

*(e) If* $g = 1$*:* $l(D) = \deg D$ *whenever* $\deg D \geq 1$*;* $K_C \sim 0$*; and a nonzero regular differential has no zeros.*

*Proof.* (a) Take $D = 0$: $1 - l(K_C) = 1 - g$. The map $f \mapsto f\omega$ identifies $\mathcal{L}(\operatorname{div}\omega)$ with the regular differentials, because $\operatorname{div}(f\omega) = \operatorname{div} f + \operatorname{div}\omega$ (Proposition 8.26(iii)) and every differential is $f\omega$. (b) Take $D = K_C$: $g - 1 = \deg K_C - g + 1$. (c) $\deg(K_C - D) < 0$, so $l(K_C - D) = 0$ by Proposition 8.28(i).

(d) $\deg \operatorname{div}(dx) = -2$ on $\mathbb{P}^1$ (Proposition 8.26(iv)), so $2g - 2 = -2$ by (b). Let $g(C) = 0$ and $P \in C(k)$. By (c), $l((P)) = 1 - 0 + 1 = 2$, and by Proposition 8.28(iv) there is a nonconstant $f \in \mathcal{L}((P)) \cap k(C)$. Its divisor of poles is at most $(P)$ and is nonzero (Proposition 8.15(iii)), so it is exactly $(P)$. By Proposition 8.22(ii) the morphism $f\colon C \to \mathbb{P}^1$, defined over $k$, has degree $[\bar{k}(C) : \bar{k}(f)] = \deg \operatorname{div}_\infty f = 1$, so it is an isomorphism (Theorem 8.17). If $(P) - (Q) = \operatorname{div} f$ with $P \neq Q$, the same argument applies to $f$, whose divisor of poles is $(Q)$: $C \simeq \mathbb{P}^1$ over $\bar{k}$, and the genus, defined through $l$ and degrees, is invariant under isomorphism, so $g = 0$.

(e) By (b), $\deg K_C = 0$, so (c) gives $l(D) = \deg D$ for $\deg D \geq 1$. By (a), $l(K_C) = 1$: there is $f \neq 0$ with $\operatorname{div} f + K_C \geq 0$; this effective divisor has degree 0, so it is 0, and $K_C \sim 0$. If $\omega$ is regular and nonzero, $\operatorname{div}\omega$ is effective of degree 0, hence 0. □

**Theorem 8.31** (genus of a smooth plane curve)**.** *A smooth plane curve of degree $d$ over $k$, in any characteristic, has genus*

$$g = \frac{(d-1)(d-2)}{2}.$$

*Sketch of proof.* Choose coordinates in general position: so that $f_y$ is not identically zero, the line at infinity $Z = 0$ is tangent to $C$ nowhere (so it meets $C$ in $d$ distinct points, by Theorem 8.12), and $[0 : 1 : 0] \notin C$. On the affine part $f(x, y) = 0$ put $\omega_0 = dx/f_y = -dy/f_x$ (the two agree because $f_x\,dx + f_y\,dy = d(f) = 0$ on $C$). At an affine point with $f_y \neq 0$, $x - a$ is a uniformiser and $\operatorname{ord}\omega_0 = 0$; at an affine point with $f_y = 0$, $f_x \neq 0$ by smoothness, $y - b$ is a uniformiser and $\operatorname{ord}\omega_0 = \operatorname{ord}(-dy/f_x) = 0$. So $\omega_0$ is regular and nowhere zero on the affine part. At infinity we compute in a chart, as in Proposition 8.26(iv). Since $[0 : 1 : 0] \notin C$, every point $P$ of $C$ on the line $Z = 0$ has $X \neq 0$, so it lies in the chart $X = 1$, with affine coordinates $u = Y/X$, $v = Z/X$ and equation $g(u, v) = F(1, u, v) = 0$. In these coordinates $x = X/Z = 1/v$ and $y = Y/Z = u/v$. The line $Z = 0$ is $v = 0$; it is not the tangent at $P$, so $g_u(P) \neq 0$ (the tangent is $g_u(P)(u - u(P)) + g_v(P)v = 0$, Proposition 8.9(ii)), and $v$ is a uniformiser at $P$ (Theorem 8.11). Now $F_Y$ is homogeneous of degree $d - 1$, so

$$f_y = F_Y(x, y, 1) = F_Y(v^{-1}, uv^{-1}, 1) = v^{1-d}F_Y(1, u, v) = v^{1-d}g_u(u, v),$$

and $g_u$ is a unit at $P$: $\operatorname{ord}_P f_y = 1 - d$. Also $dx = d(v^{-1}) = -v^{-2}dv$ has order $-2$. Hence $\operatorname{ord}_P \omega_0 = -2 - (1 - d) = d - 3$ at each of the $d$ points at infinity. As $Z/M$ (for a linear form $M$

not vanishing at these points) has a simple zero at each of them and no other zero, this reads $\operatorname{div}(\omega_0) = (d-3)\cdot\operatorname{div}_0(Z/M)$, a divisor of degree $d(d-3)$ (Theorem 8.12). By Corollary 8.30(b), $2g-2 = d(d-3)$, so $g = \frac{d^2-3d+2}{2} = \frac{(d-1)(d-2)}{2}$. For $d=3$ the form $\omega_0$ has no zeros or poles at all, as for $dx/(2y+a_1x+a_3)$ on a Weierstrass cubic. One also finds that the regular differentials are exactly the $h(x,y)\,\omega_0$ with $\deg h \le d-3$, a space of dimension $\binom{d-1}{2}$, in agreement with Corollary 8.30(a). See [72, Chapter 8] and [79, Chapter IV, Section 1, and Chapter I, Section 7] (sections not verified). The argument needs no hypothesis on the characteristic, because such coordinates always exist: the tangent lines of $C$ form at most a one-dimensional family in the two-dimensional family of lines, so a general line is tangent nowhere; and $F_X$, $F_Y$, $F_Z$ are not all identically zero ($C$ being smooth), so after a general linear change of coordinates $F_Y$, hence $f_y$, is not identically zero (in characteristic $p$ this can fail for special coordinates, when $f \in k[x,y^p]$). □

**Example 8.32** (genera)**.**

(a) Lines and smooth conics have genus 0 (Theorem 8.31 with $d = 1, 2$; for the conic, also Example 8.7).

(b) Smooth plane cubics have genus 1; in particular every smooth Weierstrass cubic, such as `11a1`, `20a1`, `121a1`. If $\operatorname{char} k \neq 2$ the substitution $y \mapsto y - \frac{1}{2}(a_1x+a_3)$ turns $y^2 + a_1xy + a_3y = x^3 + \dots$ into $y^2 = h(x)$ and $dx/(2y+a_1x+a_3)$ into $\frac{1}{2}dx/y$; so the differential $\omega = dx/(2y+a_1x+a_3)$ has divisor 0 by Proposition 8.26(iv) (smoothness makes $h$ squarefree), and is regular and nowhere zero. This gives $g = 1$ again, from Corollary 8.30(b).

(c) The Fermat quartic $X^4 + Y^4 + Z^4 = 0$ has genus 3 ($\operatorname{char} k \neq 2$): it is smooth by Example 8.10(b), and $\frac{(4-1)(4-2)}{2} = 3$.

(d) Hyperelliptic curves: the smooth projective model of $y^2 = h(x)$, $h$ squarefree of degree $2g+1$ or $2g+2$, $\operatorname{char} k \neq 2$, has genus $g$ (Corollary 8.45(c)); for instance $y^2 = x^5 - x$ has genus 2, since $x^5 - x = x(x-1)(x+1)(x^2+1)$ has degree $5 = 2\cdot 2 + 1$ and discriminant $-2^8$, nonzero for $\operatorname{char} k \neq 2$. For $g \ge 2$ the projective closure in $\mathbb{P}^2$ is singular at infinity, so Theorem 8.31 does not apply directly. Indeed, for $\deg h = n \ge 4$ the homogenised equation $Y^2Z^{n-2} = Z^n h(X/Z)$ meets $Z = 0$ only at $[0:1:0]$, and in the chart $Y = 1$ it reads $Z^{n-2} = (\text{leading coefficient})X^n + (\text{terms divisible by } Z)$, whose lowest-degree part at the origin has degree $\min(n-2, n) = n-2 \ge 2$: all partial derivatives vanish there. Were the plane formula applicable it would give $\frac{(n-1)(n-2)}{2}$, which for $n = 5$ is 6, not 2.

(e) Singular plane models have smaller genus: the nodal and cuspidal cubics are rational (Theorem 8.13 and Example 8.10). In general $g = \frac{(d-1)(d-2)}{2} - \sum_P \delta_P$, with a positive local invariant $\delta_P$ at each singular point (stated; [72, Chapter 7], [79, Chapter IV, Section 1], sections not verified).

(f) The modular curves are not given as smooth plane curves; their genera come from Riemann-Hurwitz (Corollary 8.45(f) and Chapter 12): $g(X_0(11)) = g(X_0(20)) = 1$, $g(X_0(121)) = 6$, $g(X_0(539)) = 49$. We computed these from the genus formula of Chapter 12 and checked them with PARI's `mfdim([N,2],1)`, which is $\dim S_2(\Gamma_0(N)) = g(X_0(N))$ (Theorem 8.54).

**Theorem 8.33** (a genus-one curve with a rational point is a Weierstrass cubic)**.** *Let $C$ be a smooth projective curve of genus one over $k$ and $O \in C(k)$.*

*(i)* *There are* $x \in \mathcal{L}(2(O))$ *and* $y \in \mathcal{L}(3(O))$, *in* $k(C)$, *with* $\mathcal{L}(2(O)) = \langle 1, x\rangle$, $\mathcal{L}(3(O)) = \langle 1, x, y\rangle$, $\operatorname{ord}_O x = -2$, $\operatorname{ord}_O y = -3$.

*(ii)* *After rescaling* $x$ *and* $y$ *by elements of* $k^\times$ *they satisfy a Weierstrass equation*

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in k.$$

*(iii)* *The morphism* $\varphi = [x : y : 1]\colon C \to \mathbb{P}^2$ *is an isomorphism onto the projective cubic* $E$ *defined by this equation,* $E$ *is smooth, and* $\varphi(O) = [0 : 1 : 0]$.

*(iv)* *Conversely, a smooth Weierstrass cubic has genus one, and* $[0 : 1 : 0]$ *is a point of it defined over* $k$.

*Proof.* (i) By Corollary 8.30(e), $l(n(O)) = n$ for $n \geq 1$; so $\mathcal{L}((O)) = \bar{k}$ and each step $\mathcal{L}((n-1)(O)) \subset \mathcal{L}(n(O))$, $n \geq 2$, adds exactly one dimension, that is, a function with a pole of order exactly $n$ at $O$ and no other pole. By Proposition 8.28(iv) the $k$-rational parts $\mathcal{L}(n(O)) \cap k(C)$ have $k$-dimension $n$, so we can choose $x \in \mathcal{L}(2(O)) \cap k(C)$ not in $\mathcal{L}((O))$ and $y \in \mathcal{L}(3(O)) \cap k(C)$ not in $\mathcal{L}(2(O))$.

(ii) The seven functions $1, x, y, x^2, xy, y^2, x^3$ have poles at $O$ only, of orders $0, 2, 3, 4, 5, 6, 6$, so they lie in $\mathcal{L}(6(O)) \cap k(C)$, a $k$-space of dimension 6. They are linearly dependent over $k$. In a nontrivial relation the coefficients $A$ of $y^2$ and $B$ of $x^3$ are both nonzero: if one of them were zero, the other term would be the only one with a pole of order 6, so its coefficient would vanish too, and then $1, x, y, x^2, xy$, which have distinct pole orders, would be linearly independent, forcing all coefficients to be zero. Write the relation as $Ay^2 + b_1xy + b_3y = Bx^3 + b_2x^2 + b_4x + b_6$ and substitute $x = ABx'$, $y = AB^2y'$. Then $Ay^2 = A^3B^4y'^2$ and $Bx^3 = A^3B^4x'^3$, so after dividing by $A^3B^4$ both leading terms have coefficient 1, and the relation becomes

$$y'^2 + \frac{b_1}{AB}x'y' + \frac{b_3}{A^2B^2}y' = x'^3 + \frac{b_2}{AB^2}x'^2 + \frac{b_4}{A^2B^3}x' + \frac{b_6}{A^3B^4},$$

a Weierstrass equation with coefficients in $k$. The new functions $x'$, $y'$ still have poles of orders exactly 2 and 3 at $O$ and no other pole, so (i) holds for them; we rename them $x$, $y$. (A scaling $x = \alpha x'$, $y = \beta y'$ makes both leading coefficients equal exactly when $A\beta^2 = B\alpha^3$; the choice $\alpha = AB$, $\beta = AB^2$ is one solution with $\alpha, \beta \in k^\times$.)

(iii) $\varphi$ is a morphism (Theorem 8.16) defined over $k$, with image in $E$. Writing $\varphi = [x/y : 1 : 1/y]$ and using $\operatorname{ord}_O(x/y) = 1$, $\operatorname{ord}_O(1/y) = 3$ gives $\varphi(O) = [0 : 1 : 0]$. The Weierstrass polynomial is irreducible over $\bar{k}$: it is monic of degree two in $y$, so no nonconstant polynomial in $x$ alone divides it, and a factorisation would be $(y - p(x))(y - r(x))$ with $p + r = -a_1x - a_3$ and $pr = -(x^3 + a_2x^2 + a_4x + a_6)$. Then $\deg p + \deg r = 3$ is odd, so $\deg p \neq \deg r$, and $\deg(p+r) = \max(\deg p, \deg r) \geq 2$, contradicting $\deg(p+r) \leq 1$. The image of $\varphi$ is an irreducible closed subset of $E$ (stated: images of projective varieties are closed) and not a point (as $x$ is not constant), so it is all of $E$. By Proposition 8.22(ii), $[\bar{k}(C) : \bar{k}(x)] = \deg \operatorname{div}_\infty x = 2$ and $[\bar{k}(C) : \bar{k}(y)] = 3$; hence $[\bar{k}(C) : \bar{k}(x, y)]$ divides 2 and 3, so $\bar{k}(C) = \bar{k}(x, y) = \varphi^*\bar{k}(E)$ and $\varphi$ is birational. If $E$ were singular, it would be birational to $\mathbb{P}^1$: projecting from the singular point $S$, a line through $S$ meets $E$ in $S$ with multiplicity at least two and in one further point (Theorem 8.12), which gives a birational map $E \dashrightarrow \mathbb{P}^1$, as for the node in Theorem 8.13. Then $C$ would be birational to $\mathbb{P}^1$, hence isomorphic to it (Theorem 8.17), and of genus 0, a contradiction. So $E$ is smooth, $\varphi$ is a morphism of degree one between smooth projective curves, and it is an isomorphism by Theorem 8.17.

(iv) Genus one is Example 8.32(b); $[0 : 1 : 0]$ has coordinates in $k$. □

**Proposition 8.34** (a genus-one curve is its own $\mathrm{Pic}^0$)**.** *Let $C$ have genus one and $O \in C(k)$. The map*

$$\kappa\colon C(\bar{k}) \to \mathrm{Pic}^0(C), \qquad P \mapsto \text{the class of } (P)-(O),$$

*is a bijection, and $\kappa(\sigma P) = \sigma\kappa(P)$ for $\sigma \in G_k$. Hence $C(\bar{k})$ carries a unique group structure, with identity $O$, for which $\kappa$ is an isomorphism, and $G_k$ acts by group automorphisms.*

*Proof.* Surjective: for $D \in \mathrm{Div}^0(C)$, $l(D+(O)) = 1$ by Corollary 8.30(e), so there is $f \neq 0$ with $\operatorname{div} f + D + (O) \geq 0$; this effective divisor has degree one, so it is $(P)$ for a point $P$, and $D \sim (P)-(O)$. Injective: if $(P)-(O) \sim (Q)-(O)$ then $(P) \sim (Q)$, and $P \neq Q$ would force genus 0 (Corollary 8.30(d)). Equivariance: $\sigma((P)-(O)) = (\sigma P)-(O)$ because $\sigma O = O$, and $\sigma$ preserves principal divisors (Proposition 8.22(iii)). □

Chapter 9 identifies this group law with the chord-tangent law of a Weierstrass cubic: if $P$, $Q$, $R$ are the three intersections of a line $L = 0$ with $E$, then $(P)+(Q)+(R)-3(O) = \operatorname{div}(L/Z)$, so $\kappa(P)+\kappa(Q)+\kappa(R) = 0$. Associativity of the chord-tangent law, which is awkward to check directly, is then automatic.

*In the paper* 8.35. Appendix app:repro of the paper: every elliptic curve is given by a Weierstrass equation, as `ellinit([1,1,1,-30,-76])` for `121a1`; Theorem 8.33 is why this loses nothing. Proposition 8.34 says $E = \mathrm{Pic}^0(E) = J(E)$: an elliptic curve is its own Jacobian (Definition 8.65), which is what makes $\varphi_E$ factor through $J_0(N)$ (Proposition 8.68(iv)) and lets the proof of Proposition prop:pullback speak of "the map $J_0(N) \to E$ induced by $\varphi$".

## 8.6 Morphisms and their degree; separability; the Frobenius morphism

The modular degree of the paper is the degree of the morphism $\varphi_E\colon X_0(N) \to E$. This section defines the degree of a nonconstant morphism of curves and proves that its descriptions agree: the degree of the extension of function fields, the number of preimages of a general point, the factor by which pullback multiplies degrees of divisors, and (over $\mathbb{C}$, in Sections 8.8 and 8.10) the topological degree on $H^2$. It also proves $\varphi_*\varphi^* = [\deg\varphi]$ on divisor classes, the identity behind the fact that $E \to J_0(N) \to E$ is multiplication by $\deg\varphi_E$ (Chapter 14), and introduces the Frobenius morphism used in Chapters 9 and 14.

**Theorem 8.36** (nonconstant morphisms)**.** *A morphism $\varphi\colon C_1 \to C_2$ of curves is constant or surjective. If it is nonconstant, $\varphi^*\colon \bar{k}(C_2) \to \bar{k}(C_1)$, $f \mapsto f\circ\varphi$, is a well-defined injective homomorphism of fields fixing $\bar{k}$, and $\bar{k}(C_1)$ is a finite extension of $\varphi^*\bar{k}(C_2)$; if $\varphi$ is defined over $k$, the same holds for $k(C_2) \to k(C_1)$.*

*Proof.* Surjectivity is [185, Chapter II, Theorem 2.3] (it uses that the image of a projective curve is closed); we do not prove it. Let $\varphi$ be nonconstant, hence surjective. For $f \in \bar{k}(C_2)$, $f$ is regular off a finite set $S$ (Proposition 8.15(ii)), and $\varphi^{-1}(S)$ is a proper closed subset of $C_1$ (it is not all of $C_1$, as $\varphi$ is surjective and $C_2 \neq S$), so $f\circ\varphi$ is a rational function on $C_1$. The map $f \mapsto f\circ\varphi$ respects sums and products, so it is a field homomorphism, injective as every field homomorphism is. Both fields are finitely generated of transcendence degree one over $\bar{k}$, and $\varphi^*\bar{k}(C_2)$ contains an element transcendental over $\bar{k}$ (the pullback of a nonconstant function); so $\bar{k}(C_1)$ is algebraic and finitely generated over $\varphi^*\bar{k}(C_2)$, hence finite. Over $k$ the argument is the same. □

**Definition 8.37** (degree, ramification index, pullback and pushforward)**.** Let $\varphi\colon C_1 \to C_2$ be a nonconstant morphism of smooth curves.

(i) $\deg\varphi = [\bar{k}(C_1) : \varphi^*\bar{k}(C_2)]$; $\deg_s\varphi$ and $\deg_i\varphi$ are the separable and inseparable degrees of this extension (Chapter 5), so $\deg\varphi = \deg_s\varphi \cdot \deg_i\varphi$; $\varphi$ is *separable* (*purely inseparable*) if the extension is. A constant map has degree 0.

(ii) The *ramification index* of $\varphi$ at $P$ is $e_\varphi(P) = \operatorname{ord}_P(\varphi^* t_{\varphi(P)}) \geq 1$, for a uniformiser $t_{\varphi(P)}$ at $\varphi(P)$; it does not depend on the choice, since two uniformisers differ by a unit $u$ and $\varphi^*u$ is a unit at $P$.

(iii) $\varphi^*\colon \operatorname{Div}(C_2) \to \operatorname{Div}(C_1)$, $(Q) \mapsto \sum_{\varphi(P)=Q} e_\varphi(P)(P)$; $\varphi_*\colon \operatorname{Div}(C_1) \to \operatorname{Div}(C_2)$, $(P) \mapsto (\varphi(P))$; both extended linearly.

**Theorem 8.38** (the degree formula)**.** *For a nonconstant morphism $\varphi\colon C_1 \to C_2$ of smooth projective curves and every $Q \in C_2(\bar{k})$,*

$$\sum_{P \in \varphi^{-1}(Q)} e_\varphi(P) = \deg\varphi.$$

*Sketch of proof.* Let $A = \bar{k}[C_2]_Q$, a discrete valuation ring with uniformiser $t = t_Q$, and $B$ the integral closure of $A$ in $\bar{k}(C_1)$ (identifying $\bar{k}(C_2)$ with its image under $\varphi^*$). Two facts are taken from the literature: $B$ is a finitely generated $A$-module (finiteness of integral closure for finitely generated algebras over a field; [7, Chapter 5]), and the maximal ideals of $B$ are the $\mathfrak{m}_P \cap B$ for the points $P \in \varphi^{-1}(Q)$, with $B_{\mathfrak{m}_P \cap B} = \bar{k}[C_1]_P$ (Theorem 8.13). Then $B$ is a semilocal Dedekind domain, free over the principal ideal domain $A$ (torsion-free and finitely generated, Chapter 3) of rank $n = \deg\varphi$, and $tB = \prod_P (\mathfrak{m}_P \cap B)^{e_\varphi(P)}$ by the definition of $e_\varphi(P)$. The argument of Chapter 6 for $\sum e_i f_i = n$ now applies: $\dim_{\bar{k}} B/tB = n$ by freeness, and by the Chinese remainder theorem $B/tB \simeq \prod_P B/(\mathfrak{m}_P \cap B)^{e_\varphi(P)}$, a product of spaces of dimensions $e_\varphi(P)$, because every residue field is $\bar{k}$ (all residue degrees are 1). See [185, Chapter II, Proposition 2.6] and [79, Chapter II, Proposition 6.9]. In the language of physics: the number of sheets of a branched cover, counted with multiplicity, does not jump. □

The simplest example shows every feature. Let $\varphi\colon \mathbb{P}^1 \to \mathbb{P}^1$, $z \mapsto z^n$, with $n \geq 1$. Then $\varphi^*\bar{k}(w) = \bar{k}(z^n)$, and $z$ is a root of $T^n - z^n$, which is irreducible over $\bar{k}(z^n)$ (Eisenstein at the prime $z^n$ of $\bar{k}[z^n]$), so $\deg\varphi = n$. Suppose $\operatorname{char} k \nmid n$. Over $a \neq 0, \infty$ the fibre is the $n$ distinct roots of $z^n = a$, and at each of them $\varphi^*(w - a) = z^n - a$ has a simple zero: $n$ points with $e = 1$. Over 0 the fibre is the single point 0, with $e_\varphi(0) = \operatorname{ord}_0(z^n) = n$; over $\infty$, with uniformiser $1/w$, $e_\varphi(\infty) = \operatorname{ord}_\infty(z^{-n}) = n$. In each case $\sum e_\varphi(P) = n$. If instead $n = p = \operatorname{char} k$, then $z^p - a = (z - a^{1/p})^p$, so every fibre is a single point with $e = p$: the map is a bijection on points, of degree $p$, and $\deg_s\varphi = 1$; this is the Frobenius morphism of Proposition 8.40.

**Corollary 8.39** (the degree in three ways)**.** *Let $\varphi\colon C_1 \to C_2$ be a nonconstant morphism of smooth projective curves.*

*(i)* $\deg\varphi = [\bar{k}(C_1) : \varphi^*\bar{k}(C_2)]$ *(the definition).*

*(ii)* $\#\varphi^{-1}(Q) = \deg_s\varphi$ *for all but finitely many $Q \in C_2(\bar{k})$; if $\varphi$ is separable (always in characteristic 0), $\#\varphi^{-1}(Q) = \deg\varphi$ for every $Q$ outside the finite set of branch points (Definition 8.42).*

*(iii)* $\deg \varphi^* D = \deg\varphi \cdot \deg D$; $\varphi_*\varphi^* D = (\deg\varphi)D$; $\varphi^* \operatorname{div}(f) = \operatorname{div}(\varphi^* f)$; *and* $\varphi_* \operatorname{div}(h) = \operatorname{div}(\mathrm{N}\, h)$, *with* N *the norm from* $\bar{k}(C_1)$ *to* $\varphi^*\bar{k}(C_2)$ *(stated). Hence* $\varphi^*$ *and* $\varphi_*$ *induce homomorphisms* $\varphi^*\colon \operatorname{Pic}(C_2) \to \operatorname{Pic}(C_1)$ *and* $\varphi_*\colon \operatorname{Pic}(C_1) \to \operatorname{Pic}(C_2)$, *preserving* $\operatorname{Pic}^0$, *and* $\varphi_*\varphi^* = [\deg\varphi]$ *on* $\operatorname{Pic}^0(C_2)$.

*(iv) Over* $\mathbb{C}$: $\deg\varphi$ *is the analytic degree of the holomorphic map* $\varphi$ *(Theorem 8.48(c), Theorem 8.49), and* $\varphi^*$ *on* $H^2(\cdot,\mathbb{Z}) = \mathbb{Z}$ *is multiplication by* $\deg\varphi$ *(Proposition 8.62(iv)).*

*(v)* $\deg(\psi\circ\varphi) = \deg\psi \cdot \deg\varphi$; *and* $\deg\varphi = 1$ *iff* $\varphi$ *is an isomorphism.*

*(vi) For nonconstant* $f \in \bar{k}(C)$, *the morphism* $f\colon C \to \mathbb{P}^1$ *has degree* $[\bar{k}(C) : \bar{k}(f)] = \deg \operatorname{div}_0(f) = \deg \operatorname{div}_\infty(f)$. *Hence* $\deg \operatorname{div}(f) = 0$, *and a function without poles is constant.*

*Proof.* (iii) The first two identities follow from Theorem 8.38: $\deg \varphi^*(Q) = \sum_{P\mapsto Q} e_\varphi(P) = \deg\varphi$, and $\varphi_*\varphi^*(Q) = \sum_{P\mapsto Q} e_\varphi(P)(Q) = (\deg\varphi)(Q)$. For the third, let $Q = \varphi(P)$, $e = e_\varphi(P)$, $m = \operatorname{ord}_Q f$, and write $f = u t_Q^m$ with $u$ a unit at $Q$. Then $\varphi^* f = (\varphi^* u)(\varphi^* t_Q)^m$, with $\varphi^* u$ a unit at $P$ and $\operatorname{ord}_P \varphi^* t_Q = e$, so $\operatorname{ord}_P \varphi^* f = em$: the coefficient of $P$ in $\varphi^* \operatorname{div}(f)$. The norm formula is [185, Chapter II, Section 3] (stated). So $\varphi^*$ and $\varphi_*$ map principal divisors to principal divisors and multiply, respectively preserve, degrees; they descend to Pic and $\operatorname{Pic}^0$, and the divisor identity $\varphi_*\varphi^* = \deg\varphi$ descends.

(vi) For $f$ nonconstant, $f^*\bar{k}(\mathbb{P}^1) = f^*\bar{k}(t) = \bar{k}(f)$, so the degree is $[\bar{k}(C) : \bar{k}(f)]$. The coordinate $t$ is a uniformiser at $0 \in \mathbb{P}^1$, so $e_f(P) = \operatorname{ord}_P(f^* t) = \operatorname{ord}_P f$ at a zero $P$ of $f$; and $1/t$ is a uniformiser at $\infty$, so $e_f(P) = -\operatorname{ord}_P f$ at a pole. Hence $f^*((0)) = \operatorname{div}_0(f)$ and $f^*((\infty)) = \operatorname{div}_\infty(f)$, both of degree $\deg f$ by Theorem 8.38. So $\deg \operatorname{div} f = 0$, and a nonconstant function has a pole divisor of degree $\deg f \geq 1$. This proves Proposition 8.15(iii) and Proposition 8.22(ii); neither was used in the proof of Theorem 8.38, which rests on the sketched facts named there.

(ii) For separable $\varphi$ only finitely many points are ramified (Lemma 8.43), and over every other $Q$ all $e_\varphi(P) = 1$, so $\#\varphi^{-1}(Q) = \deg\varphi$ by Theorem 8.38. In general $\varphi$ factors as a purely inseparable morphism, which is bijective on points, followed by a separable one of degree $\deg_s\varphi$ (Proposition 8.40(b), stated).

(v) The first statement is the tower law for degrees of field extensions. If $\deg\varphi = 1$, $\varphi^*$ is an isomorphism of function fields and $\varphi$ is an isomorphism (Theorem 8.17); an isomorphism induces an isomorphism of function fields. (iv) is proved where cited. □

| description of $\deg\varphi$ | statement | where |
|---|---|---|
| field degree | $[\bar{k}(C_1) : \varphi^*\bar{k}(C_2)]$ | Definition 8.37 |
| counting preimages | $\sum_{P\mapsto Q} e_\varphi(P)$ for every $Q$; $\#\varphi^{-1}(Q)$ for general $Q$ ($\varphi$ separable) | Theorem 8.38, Corollary 8.39(ii) |
| divisors | $\deg\varphi^* D = \deg\varphi \deg D$; $\varphi_*\varphi^* = \deg\varphi$ | Corollary 8.39(iii) |
| analytic (over $\mathbb{C}$) | local multiplicities summed over a fibre | Theorem 8.48(c) |
| cohomology (over $\mathbb{C}$) | $\varphi^*$ on $H^2 = \mathbb{Z}$; $\langle\varphi^* x, \varphi^* y\rangle = \deg\varphi\,\langle x, y\rangle$ | Proposition 8.62, Proposition 8.70 |

Table 8.1: The degree of a nonconstant morphism of smooth projective curves.

**Proposition 8.40** (separability and the Frobenius morphism)**.**

(a) *In characteristic* 0 *every nonconstant morphism of curves is separable.*

(b) *In characteristic* $p$, $\varphi$ *is separable iff* $\varphi^*\omega \neq 0$ *for one (equivalently every) nonzero differential* $\omega$ *on* $C_2$, *where* $\varphi^*(f\,dg) = (\varphi^* f)\,d(\varphi^* g)$. *For a power* $r$ *of* $p$ *and a curve* $C \subseteq \mathbb{P}^n$, *let* $C^{[r]}$ *(not to be confused with the twist notation* $E^{(q)}$ *of Appendix B) be the curve whose ideal is generated by the polynomials obtained from generators of* $I(C)$ *by raising their coefficients to the* $r$*-th power, and* $\pi_r\colon C \to C^{[r]}$, $[x_0 : \cdots : x_n] \mapsto [x_0^r : \cdots : x_n^r]$, *the* $r$-power Frobenius morphism *(it maps* $C$ *into* $C^{[r]}$ *because* $F^{[r]}(x_0^r, \dots, x_n^r) = F(x_0, \dots, x_n)^r$, *where* $F^{[r]}$ *has the coefficients of* $F$ *raised to the* $r$*-th power). Every nonconstant* $\varphi$ *factors as* $C_1 \xrightarrow{\pi_r} C_1^{[r]} \xrightarrow{\lambda} C_2$ *with* $r = \deg_i \varphi$ *and* $\lambda$ *separable (stated). For* $C$ *defined over* $\mathbb{F}_q$, $C^{[q]} = C$ *and* $\pi_q$ *is the morphism of the next item.*

(c) *Let* $k = \mathbb{F}_q$ *and* $C \subseteq \mathbb{P}^n$ *a curve defined over* $\mathbb{F}_q$. *The* $q$-power Frobenius morphism $\pi_q\colon [x_0 : \cdots : x_n] \mapsto [x_0^q : \cdots : x_n^q]$ *maps* $C$ *to itself and is a bijection on* $C(\bar{\mathbb{F}}_q)$; $\pi_q^* \bar{\mathbb{F}}_q(C) = \bar{\mathbb{F}}_q(C)^q$; $d(\pi_q^* f) = 0$ *for every* $f$; $\pi_q$ *is purely inseparable of degree* $q$ *and* $e_{\pi_q}(P) = q$ *at every point; its fixed points are* $C(\mathbb{F}_q)$; *and on* $C(\bar{\mathbb{F}}_q)$ *it coincides with the action of the arithmetic Frobenius* $\mathrm{Frob}_q \in G_{\mathbb{F}_q}$.

*Proof.* (a) Fields of characteristic 0 are perfect, and every algebraic extension of a perfect field is separable (Chapter 5); here $\varphi^* \bar{k}(C_2)$ has characteristic 0. (b) is [185, Chapter II, Proposition 4.2(c) and Corollary 2.12]; we do not prove it.

(c) If $F$ is a homogeneous polynomial with coefficients in $\mathbb{F}_q$, then $F(x_0^q, \dots, x_n^q)$ equals $F(x_0, \dots, x_n)^q$ (the freshman's dream, Chapter 2, and $a^q = a$ for $a \in \mathbb{F}_q$); so $\pi_q$ maps $C$ into $C$. On $\bar{\mathbb{F}}_q$ the map $a \mapsto a^q$ is a bijection, so $\pi_q$ is a bijection of $\mathbb{P}^n(\bar{\mathbb{F}}_q)$ preserving $C(\bar{\mathbb{F}}_q)$; in particular it is nonconstant. For $f = F/G \in \bar{\mathbb{F}}_q(C)$ the pullback $\pi_q^* f = F(x^q)/G(x^q)$ equals $(F'/G')^q$, where the coefficients of $F'$, $G'$ are the $q$-th roots of those of $F$, $G$ ($\bar{\mathbb{F}}_q$ is perfect); conversely every $(F'/G')^q$ is of this form. So $\pi_q^* \bar{\mathbb{F}}_q(C) = \bar{\mathbb{F}}_q(C)^q$, and $d(h^q) = q h^{q-1} dh = 0$. The degree $[\bar{\mathbb{F}}_q(C) : \bar{\mathbb{F}}_q(C)^q]$ is $q$: for a separating transcendental element $t$ (one with $\bar{\mathbb{F}}_q(C)/\bar{\mathbb{F}}_q(t)$ separable, which exists, [185, Chapter II, Section 1]), $\bar{\mathbb{F}}_q(C) = \bar{\mathbb{F}}_q(C)^q(t)$ and $t$ has minimal polynomial $X^q - t^q$ over $\bar{\mathbb{F}}_q(C)^q$ (sketched; [185, Chapter II, Proposition 2.11]). Every point has exactly one preimage, so $e_{\pi_q}(P) = q$ by Theorem 8.38. A point $[x_0 : \cdots : x_n]$, scaled so that some $x_i = 1$, is fixed iff $x_j^q = x_j$ for all $j$, that is, iff all $x_j \in \mathbb{F}_q$. Finally $\mathrm{Frob}_q$ acts on coordinates by $a \mapsto a^q$ (Convention 8.1(C7)). □

On $\mathbb{P}^1$ over $\mathbb{F}_q$ the sketched degree computation is immediate: $\pi_q^* \bar{\mathbb{F}}_q(t) = \bar{\mathbb{F}}_q(t^q)$, and $t$ is a root of $X^q - t^q = (X - t)^q$, irreducible over $\bar{\mathbb{F}}_q(t^q)$ by Eisenstein, so $\deg \pi_q = q$, while $d(t^q) = 0$ shows that the extension is purely inseparable. Two features of (c) matter later. First, the morphism $\pi_q$ and the Galois element $\mathrm{Frob}_q$ have the same effect on points but act differently on functions: $\pi_q^* f = f \circ \pi_q$ is $\bar{\mathbb{F}}_q$-linear in $f$, whereas $\mathrm{Frob}_q$ acts on $\bar{\mathbb{F}}_q(C)$ through the coefficients (Proposition 8.18(ii)), semilinearly. Second, because they agree on points, the arithmetic Frobenius is what $\pi_q$ induces on torsion points and on $T_\ell J$, and its inverse, the geometric Frobenius, is what appears on $H^1$ (Theorem 8.74(d), Convention 8.63(e)). Since $H^1$ is the dual of $T_\ell J$, at a prime $p$ of good reduction $\det(1 - \mathrm{Frob}_p^{-1} X \mid H^1) = \det(1 - \mathrm{Frob}_p X \mid T_\ell J)$; using $\mathrm{Frob}_p$ itself on $H^1$ would replace each eigenvalue $\alpha$ by $\alpha^{-1}$. This is the convention behind the book's form $\det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid H^1)$ of the $L$-factors.

*In the paper* 8.41. Section 1 of the paper: "the $X_0(N)$-optimal curve, for which the modular parametrisation $\varphi_E\colon X_0(N) \to E$ has minimal degree" and "$\deg \varphi_E$ its *modular degree*". The

degree is that of Definition 8.37, and by Corollary 8.39 and Table 8.1 it is at once $[\mathbb{Q}(X_0(N)) : \varphi_E^*\mathbb{Q}(E)]$ (degrees of field extensions do not change under $\otimes_{\mathbb{Q}}\overline{\mathbb{Q}}$ for function fields of curves defined over $\mathbb{Q}$), the number of points of $X_0(N)(\mathbb{C})$ over a general point of $E(\mathbb{C})$, and the factor $\deg\varphi_E$ in $\varphi_{E*}\varphi_E^* = [\deg\varphi_E]$ on $E = \mathrm{Pic}^0(E)$. The proof of Proposition prop:pullback uses the cohomological form, "$\varphi^*$ multiplies the fundamental class of $H^2$ by $\deg\varphi_E$", which is Proposition 8.62(iv). PARI's `ellmoddegree` computes this number (Chapter 22).

## 8.7 Ramification and the Riemann-Hurwitz formula

The Riemann-Hurwitz formula relates the genera of two curves joined by a morphism. In the book it gives the genus of $X_0(N)$ (Chapter 12), shows that isogenies are unramified (Chapter 9), and checks the modular degrees quoted in the paper against the genera of the modular curves (Corollary 8.45(e)).

**Definition 8.42** (ramification)**.** Let $\varphi\colon C_1 \to C_2$ be nonconstant. $\varphi$ is *unramified* at $P$ if $e_\varphi(P) = 1$ and *ramified* otherwise, and $\varphi(P)$ is then a *branch point*. The ramification at $P$ is *tame* if $\mathrm{char}\,k \nmid e_\varphi(P)$ and *wild* otherwise. $\varphi$ is *unramified* if it is unramified at every point.

**Lemma 8.43** (the local computation)**.** *Let $\varphi\colon C_1 \to C_2$ be nonconstant and separable, $\omega \neq 0$ a differential on $C_2$, $P \in C_1$, $Q = \varphi(P)$, $e = e_\varphi(P)$, and $t_P$, $t_Q$ uniformisers. Then $\varphi^*\omega \neq 0$ and*

$$\mathrm{ord}_P(\varphi^*\omega) = e\,\mathrm{ord}_Q(\omega) + \mathrm{ord}_P\Big(\frac{d(\varphi^*t_Q)}{dt_P}\Big), \qquad \mathrm{ord}_P\Big(\frac{d(\varphi^*t_Q)}{dt_P}\Big) \geq e-1,$$

*with equality iff $\mathrm{char}\,k \nmid e$. In particular only finitely many points of $C_1$ are ramified.*

*Proof.* $\varphi^*\omega \neq 0$ by Proposition 8.40(b) (in characteristic 0 by Proposition 8.26(i): $d$ of a separating element is nonzero). Write $\omega = g\,dt_Q$ and $\varphi^*t_Q = ut_P^e$ with $u$ a unit at $P$. Then $\varphi^*\omega = (\varphi^*g)\,d(ut_P^e)$ and

$$\frac{d(ut_P^e)}{dt_P} = eut_P^{e-1} + t_P^e\frac{du}{dt_P},$$

where $du/dt_P$ is regular at $P$ (Proposition 8.26(ii)). If $\mathrm{char}\,k \nmid e$, the first term has order exactly $e-1$ and the second order at least $e$, so the sum has order $e-1$. If $\mathrm{char}\,k \mid e$, the first term vanishes and the order is at least $e$. Since $\mathrm{ord}_P\,\varphi^*g = e\,\mathrm{ord}_Q\,g$ (Corollary 8.39(iii)), the formula follows. At a ramified $P$ the second term is $\geq e-1 \geq 1$; but it equals $\mathrm{ord}_P\,\varphi^*\omega - e\,\mathrm{ord}_Q\,\omega$, which is nonzero only at the finitely many zeros and poles of $\varphi^*\omega$ and points over zeros and poles of $\omega$ (Proposition 8.26(ii), Theorem 8.36: fibres are finite by Theorem 8.38). □

**Theorem 8.44** (Riemann-Hurwitz)**.** *Let $\varphi\colon C_1 \to C_2$ be a nonconstant separable morphism of smooth projective curves of genera $g_1$, $g_2$. Then*

$$2g_1 - 2 \;\geq\; \deg\varphi\,(2g_2-2) + \sum_{P\in C_1}(e_\varphi(P)-1),$$

*with equality iff every ramification is tame; in particular with equality in characteristic 0.*

*Proof.* Take $\omega \neq 0$ on $C_2$ and sum the formula of Lemma 8.43 over $P \in C_1$. The first terms give $\sum_Q \mathrm{ord}_Q(\omega)\sum_{P\mapsto Q} e_\varphi(P) = \deg\varphi\cdot\deg\mathrm{div}(\omega)$ by Theorem 8.38, that is $\deg\varphi\,(2g_2-2)$ by Corollary 8.30(b); the left side is $\deg\mathrm{div}(\varphi^*\omega) = 2g_1-2$. The second terms are $\geq\sum(e_\varphi(P)-1)$, with equality iff equality holds at every $P$, that is, iff all ramification is tame. □

A topological proof over $\mathbb{C}$ is sketched in Theorem 8.50(c): triangulate $C_2$ with the branch points among the vertices, lift the triangulation and count $V - E + F$.

For $z \mapsto z^n$ on $\mathbb{P}^1$ with $\operatorname{char} k \nmid n$ (the example after Theorem 8.38) the formula reads $-2 = n(-2) + (n-1) + (n-1)$, with the two ramification points 0 and $\infty$. Wild ramification is genuinely different. In characteristic $p$ let $\varphi\colon \mathbb{P}^1 \to \mathbb{P}^1$ be $y \mapsto x = y^p - y$ (the Artin-Schreier map). It has degree $p$ ($y$ is a root of $T^p - T - x$, irreducible over $\bar{k}(x)$ because the map is totally ramified at $\infty$, as we now see) and is separable, because $d(y^p - y) = -dy \neq 0$ (Proposition 8.40(b)). At finite points $dx/dy = -1$ is a unit, so $\varphi$ is unramified there. At $\infty$, with uniformisers $t_P = 1/y$ and $t_Q = 1/x$,

$$\varphi^* t_Q = \frac{1}{y^p - y} = \frac{t_P^p}{1 - t_P^{p-1}}, \qquad \frac{d(\varphi^* t_Q)}{dt_P} = \frac{p t_P^{p-1}}{1 - t_P^{p-1}} + \frac{(p-1)t_P^{2p-2}}{(1 - t_P^{p-1})^2} = \frac{-t_P^{2p-2}}{(1 - t_P^{p-1})^2},$$

so $e_\varphi(\infty) = p$, wild, and the order in Lemma 8.43 is $2p-2$, strictly larger than $e - 1 = p - 1$. With $2p-2$ the exact formula of the proof gives $-2 = p(-2) + (2p-2)$, which is correct, while the tame count $\sum(e_\varphi(P) - 1) = p - 1$ gives the strict inequality $-2 > -2p + (p-1)$ of the theorem. (The degree claim: the fibre over $\infty$ is one point with $e = p$, so $\deg \varphi = p$ by Theorem 8.38, and $T^p - T - x$, having $y$ as a root and degree $p$, is the minimal polynomial.)

**Corollary 8.45** (consequences and examples)**.** *Let $\varphi\colon C_1 \to C_2$ be nonconstant and separable.*

*(a) If $g_2 \geq 1$ then $g_1 \geq g_2$. If $g_1 = 0$ then $g_2 = 0$ (Lueroth's theorem, for separable maps).*

*(b) If $g_1 = g_2 = 1$ then $\varphi$ is unramified, and $\#\varphi^{-1}(Q) = \deg \varphi$ for every $Q$. In characteristic 0 this applies to every nonconstant morphism between genus-one curves, in particular to every isogeny (Chapter 9).*

*(c) The double cover $x\colon C \to \mathbb{P}^1$ of the hyperelliptic curve $C$ given by $y^2 = h(x)$, $h$ squarefree of degree $2g+1$ or $2g+2$, $\operatorname{char} k \neq 2$, is branched exactly at the $2g+2$ points consisting of the roots of $h$ and, if $\deg h$ is odd, $\infty$; so $g(C) = g$.*

*(d) The projection $[X : Y : Z] \mapsto [X : Y]$ of the Fermat curve $X^d + Y^d + Z^d = 0$, $\operatorname{char} k \nmid d$, has degree $d$ and is totally ramified ($e = d$) over the $d$ roots of $X^d + Y^d = 0$ and unramified elsewhere; so $2g - 2 = -2d + d(d-1)$ and $g = \frac{(d-1)(d-2)}{2}$, as in Theorem 8.31.*

*(e) Over $\mathbb{C}$, for a nonconstant $\varphi\colon X \to E$ with $g(E) = 1$: $\sum_P (e_\varphi(P) - 1) = 2g(X) - 2$, whatever $\deg \varphi$. For the paper's curves (genera from Example 8.32(f), degrees from Cremona's tables, checked with PARI's `ellmoddegree`):*

| $\varphi_E$ | $g(X_0(N))$ | $\deg \varphi_E$ | $\sum_P (e_\varphi(P) - 1)$ |
|---|---|---|---|
| $X_0(11) \to$ `11a1` | 1 | 1 | 0 |
| $X_0(20) \to$ `20a1` | 1 | 1 | 0 |
| $X_0(121) \to$ `121a1` | 6 | 6 | 10 |
| $X_0(539) \to$ `539c1` | 49 | $288 = 2^5 \cdot 3^2$ | 96 |

*A morphism of degree one is an isomorphism, so $X_0(11) \simeq$ `11a1` and $X_0(20) \simeq$ `20a1` over $\mathbb{Q}$: these modular curves are elliptic curves.*

*(f) (Forward pointer.) The genus formula for $X_0(N)$ in Chapter 12 is Theorem 8.44 for the morphism $X_0(N) \to X_0(1) \simeq \mathbb{P}^1$, of degree $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$, ramified only over $j = 0$, 1728, $\infty$.*

*Proof.* (a) If $g_2 \geq 1$ the theorem gives $2g_1 - 2 \geq \deg\varphi(2g_2-2) \geq 2g_2 - 2$. If $g_1 = 0$ and $g_2 \geq 1$, the right side is $\geq 0 > -2$, a contradiction. (b) $0 \geq 0 + \sum(e_\varphi(P) - 1)$, so every $e_\varphi(P) = 1$, and Theorem 8.38 gives $\#\varphi^{-1}(Q) = \deg\varphi$ for every $Q$.

(c) $y \notin \bar{k}(x)$ because $h$ is not a square, so $[\bar{k}(C) : \bar{k}(x)] = 2$, and the map is separable as $\operatorname{char} k \neq 2$. Over $a \in \bar{k}$ the points of $C$ are $(a, \pm\sqrt{h(a)})$: two points, hence $e = 1$ at both, if $h(a) \neq 0$, and one point, hence $e = 2$, if $h(a) = 0$. Over $\infty$: the smooth projective model is covered by the affine curve $y^2 = h(x)$ and the affine curve $w^2 = H(s)$, $H(s) = s^{2g+2}h(1/s)$, glued by $s = 1/x$, $w = ys^{g+1}$ ([185, Chapter II, Exercise 2.14], exercise number not verified); $H$ is squarefree, $H(0)$ is the leading coefficient of $h$ if $\deg h = 2g+2$, and $H$ has a simple zero at 0 if $\deg h = 2g+1$. So $\infty$ is a branch point iff $\deg h$ is odd. All ramification is tame, and the theorem gives $2g(C) - 2 = 2(-2) + (2g+2)$.

(d) The map is defined everywhere ($X = Y = 0$ forces $Z = 0$). The fibre over $[a : b]$ is $\{[a : b : z] : z^d = -(a^d + b^d)\}$: $d$ distinct points if $a^d + b^d \neq 0$, and the single point $[a : b : 0]$ otherwise. The extension $\bar{k}(C)/\bar{k}(X/Y)$ is generated by $Z/Y$, a root of $T^d + (X/Y)^d + 1$, which is irreducible (Eisenstein at a simple root of $s^d + 1$) and separable ($\operatorname{char} k \nmid d$); so the degree is $d$, and by Theorem 8.38 the single point over a root of $X^d + Y^d$ has $e = d$. All ramification is tame. (e) is the theorem with $g_2 = 1$; the table uses $2g - 2 = 0, 0, 10, 96$. □

## 8.8 Compact Riemann surfaces and smooth projective complex curves

$X_0(N)$ is built analytically in Chapter 12, as a compact Riemann surface, and used algebraically in Chapter 14 and in the paper, as a curve over $\mathbb{Q}$; the modular parametrisation $\varphi_E$ is at once a holomorphic map and a morphism. This section introduces compact Riemann surfaces, proves the local normal form of holomorphic maps (which gives the analytic degree), and states the dictionary between the two languages (Theorem 8.49), summarised in Table 8.2. The prerequisites are those of Appendix A: holomorphic functions, zeros of finite order, the inverse function theorem, compactness and connectedness.

**Definition 8.46** (Riemann surfaces)**.** A *Riemann surface* is a connected Hausdorff topological space $X$ with a countable basis, together with a maximal atlas of *charts* $z\colon U \to z(U) \subseteq \mathbb{C}$ (homeomorphisms from open sets of $X$ onto open subsets of $\mathbb{C}$, the $U$ covering $X$) whose transition maps $z_2 \circ z_1^{-1}$ are holomorphic where defined. A function $f\colon U \to \mathbb{C}$ on an open subset is *holomorphic* (*meromorphic*) if $f \circ z^{-1}$ is, for every chart $z$; $\mathcal{M}(X)$ is the field of meromorphic functions on $X$ (for $X$ connected, Appendix A). A continuous map $F\colon X \to Y$ of Riemann surfaces is *holomorphic* if $w \circ F \circ z^{-1}$ is holomorphic for all charts $z$ of $X$ and $w$ of $Y$.

**Example 8.47** (Riemann surfaces)**.**

(a) $\mathbb{C}$, and every connected open subset of $\mathbb{C}$, with the identity chart.

(b) The Riemann sphere $\mathbb{P}^1(\mathbb{C}) = \mathbb{C} \cup \{\infty\}$, with the charts $z$ on $\mathbb{C}$ and $w = 1/z$ on $\mathbb{P}^1 \setminus \{0\}$; the transition map $w = 1/z$ is holomorphic on $\mathbb{C}^\times$. It is compact.

(c) $\mathbb{C}/\Lambda$ for a lattice $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ ($\omega_1$, $\omega_2$ linearly independent over $\mathbb{R}$): the charts are local inverses of the projection $\mathbb{C} \to \mathbb{C}/\Lambda$ on discs of radius less than half the shortest nonzero vector of $\Lambda$; the transition maps are translations $z \mapsto z + \lambda$. It is compact (the image of a closed parallelogram) and, topologically, a torus. Chapter 9 proves that every elliptic curve over $\mathbb{C}$ is of this form.

(d) The smooth points of an affine plane curve $f(x,y)=0$ over $\mathbb{C}$: near a point with $f_y \neq 0$, the holomorphic implicit function theorem writes the curve as a graph $y=y(x)$, and $x$ is a chart; near a point with $f_x \neq 0$, $y$ is a chart; the transition maps are holomorphic.

(e) Hence $C(\mathbb{C})$ is a compact Riemann surface for every smooth projective curve $C$ over $\mathbb{C}$: it is covered by the affine charts, it is closed in the compact space $\mathbb{P}^n(\mathbb{C})$, and it is connected because $C$ is irreducible (stated; [79, Appendix B], [136, Chapter VII], chapter number not verified).

(f) (Forward pointer, Chapter 12.) $\Gamma_0(N)\backslash\mathcal{H}$, compactified by adding the finitely many cusps, with special charts at the cusps and at the elliptic points, is the compact Riemann surface $X_0(N)(\mathbb{C})$.

**Theorem 8.48** (holomorphic maps of compact Riemann surfaces)**.** *Let $F\colon X \to Y$ be a nonconstant holomorphic map of compact Riemann surfaces.*

*(a) (Local normal form.) For each $P \in X$ there are charts $z$ at $P$ and $w$ at $F(P)$, centred at 0, in which $F$ is $w=z^e$. The integer $e=\operatorname{mult}_P(F)\geq 1$ does not depend on the charts.*

*(b) $F$ is open and surjective, and its fibres are finite.*

*(c) The number $d=\sum_{P\in F^{-1}(Q)}\operatorname{mult}_P(F)$ does not depend on $Q\in Y$; it is the* degree *of $F$.*

*(d) $\operatorname{mult}_P(F)=1$ except at finitely many points of $X$; so $\#F^{-1}(Q)=d$ except for finitely many $Q$.*

*(e) $F$ preserves orientation: in local coordinates its real Jacobian determinant is $|h'(z)|^2\geq 0$.*

*Proof.* (a) Take any charts $z$ at $P$ and $w$ at $F(P)$, centred at 0, and put $h=w\circ F\circ z^{-1}$, holomorphic near 0 with $h(0)=0$. $h$ is not identically zero near 0: otherwise, by the identity theorem and connectedness of $X$, $F$ would be constant. So $h(z)=z^e g(z)$ with $e\geq 1$ and $g(0)\neq 0$. On a small disc $g$ has a holomorphic $e$-th root $g_1$ (Appendix A: a nonvanishing function on a disc has a holomorphic logarithm), and $\zeta=zg_1(z)$ has $\zeta'(0)=g_1(0)\neq 0$, so it is a chart near $P$ by the inverse function theorem. In the charts $\zeta$, $w$ the map is $w=\zeta^e$. The exponent $e$ is the order of vanishing at 0 of $w\circ F$ in any chart at $P$; changing the chart at $P$ or at $F(P)$ composes with a biholomorphism fixing 0, whose derivative at 0 is nonzero, and this does not change the order of a zero.

(b) The map $\zeta\mapsto\zeta^e$ is open, so by (a) $F$ is open. Then $F(X)$ is open, and compact, hence closed; $Y$ is connected, so $F(X)=Y$. By (a), $P$ is the only preimage of $F(P)$ near $P$; so fibres are discrete, and they are closed in the compact $X$, hence finite.

(c) Let $F^{-1}(Q)=\{P_1,\dots,P_r\}$, with multiplicities $e_i$ and normal-form charts on pairwise disjoint neighbourhoods $U_i$ of $P_i$. There is a neighbourhood $V$ of $Q$ with $F^{-1}(V)\subseteq\bigcup U_i$: otherwise there are $x_n\notin\bigcup U_i$ with $F(x_n)\to Q$; a subsequence converges (compactness) to some $x\notin\bigcup U_i$ (the complement is closed) with $F(x)=Q$, which is absurd. Shrinking $V$, every $Q'\in V\setminus\{Q\}$ has exactly $e_i$ preimages in $U_i$ (the $e_i$-th roots of $w(Q')\neq 0$), each of multiplicity one since $\frac{d}{d\zeta}\zeta^{e_i}\neq 0$ for $\zeta\neq 0$. So $Q'\mapsto\sum_{F(P)=Q'}\operatorname{mult}_P(F)$ is constant on $V$; it is locally constant, hence constant on the connected $Y$.

(d) In the normal form at $P$, every point near $P$ other than $P$ has multiplicity one. So the set of points of multiplicity $\geq 2$ is discrete; it is also closed (its complement is open by the same remark), hence finite since $X$ is compact. Over $Q$ outside its finite image, $\#F^{-1}(Q)=d$ by (c).

(e) For $h = u + iv$ holomorphic, $\det\left(\begin{smallmatrix} u_x & u_y \\ v_x & v_y \end{smallmatrix}\right) = u_x^2 + v_x^2 = |h'|^2$ by the Cauchy-Riemann equations. □

The proofs of (a) and the openness in (b) use only that the target is a Riemann surface, not that it is compact. Hence *a holomorphic function $f\colon X \to \mathbb{C}$ on a compact Riemann surface is constant*: otherwise $f(X)$ would be open in $\mathbb{C}$, and also compact, hence closed and bounded, which is impossible for a nonempty open subset of $\mathbb{C}$. (The same follows from the maximum modulus principle, Appendix A.)

**Theorem 8.49** (Riemann existence and GAGA for curves; stated)**.**

*(a) (Riemann.) On a compact Riemann surface $X$ the meromorphic functions separate points and give local coordinates at every point, and $\mathcal{M}(X)$ is a function field of one variable over $\mathbb{C}$.*

*(b) (Equivalence.) $X \mapsto$ the smooth projective curve with function field $\mathcal{M}(X)$, and $C \mapsto C(\mathbb{C})$, are inverse equivalences between compact Riemann surfaces with nonconstant holomorphic maps and smooth projective curves over $\mathbb{C}$ with nonconstant morphisms. Under them: meromorphic functions are rational functions; the order of a zero or pole is* $\operatorname{ord}_P$*; $\operatorname{mult}_P(F) = e_F(P)$; the analytic degree of Theorem 8.48(c) is* $\deg F$ *of Definition 8.37; meromorphic and holomorphic* 1*-forms (Definition 8.51) are the differentials and the regular differentials of Definition 8.25; divisors, linear equivalence and* Pic *agree.*

*(c) The algebraic genus $g = l(K_C)$ equals the topological genus of Theorem 8.50.*

We do not prove this; see [136, Chapters VI and VII] (Riemann existence and the algebraic curve of a Riemann surface), [79, Appendix B] (GAGA), and [70, Sections 14 and 16] (all section and chapter numbers not verified). Once (b) is granted, $\operatorname{mult}_P(F) = e_F(P)$ is a tautology: the multiplicity is the order of vanishing of $w \circ F$ at $P$ for a local coordinate $w$ at $F(P)$, which is $\operatorname{ord}_P(F^* t_{F(P)})$. In particular $\deg \varphi_E$ means the same in the analytic and the algebraic sense.

| smooth projective curve $C$ over $\mathbb{C}$ | compact Riemann surface $X = C(\mathbb{C})$ |
|---|---|
| function field $\mathbb{C}(C)$ | meromorphic functions $\mathcal{M}(X)$ |
| $\operatorname{ord}_P f$ (valuation of $\bar{k}[C]_P$) | order of zero or pole in a chart |
| ramification index $e_\varphi(P)$ | multiplicity $\operatorname{mult}_P(\varphi)$, normal form $w = z^e$ |
| $\deg \varphi = [\mathbb{C}(C_1) : \varphi^* \mathbb{C}(C_2)]$ | number of sheets; degree on $H^2$ |
| regular differentials | holomorphic 1-forms $H^0(X, \Omega^1)$ |
| genus $g = l(K_C)$ | $\dim H^0(X, \Omega^1)$; number of handles; $\frac{1}{2}\operatorname{rank} H_1(X, \mathbb{Z})$ |
| $\operatorname{Pic}^0(C)$ | $H^0(X, \Omega^1)^\vee / H_1(X, \mathbb{Z})$ (Abel-Jacobi) |

Table 8.2: The dictionary of Theorems 8.49, 8.50, 8.54 and 8.66.

**Theorem 8.50** (topology of compact Riemann surfaces)**.**

*(a) A compact Riemann surface is an orientable compact surface, homeomorphic to a sphere with $g$ handles for a unique $g \geq 0$, its* topological genus*; its Euler characteristic is $\chi(X) = 2 - 2g$ (stated).*

*(b) This $g$ is the genus of Theorem 8.29 (stated; Theorem 8.49(c)).*

(c) *(Topological Riemann-Hurwitz.) For* $F\colon X \to Y$ *holomorphic of degree* $d$, $\chi(X) = d\,\chi(Y) - \sum_P (\mathrm{mult}_P(F) - 1)$.

*Sketch of proof of (c).* Triangulate $Y$ so that the finitely many branch points are vertices. Away from the branch points $F$ is a $d$-sheeted covering (Theorem 8.48(c), (d)), so each open face and each open edge has exactly $d$ lifts, and these lifts triangulate $X$. A vertex $Q$ has $\#F^{-1}(Q) = d - \sum_{F(P)=Q}(\mathrm{mult}_P(F)-1)$ lifts. Counting, $V_X - E_X + F_X = d(V_Y - E_Y + F_Y) - \sum_P(\mathrm{mult}_P(F)-1)$. For (a) and (b) see [136, Chapters I and II] and [70, Section 17] (not verified). This is Euler's $V - E + F$ for a branched cover; with $\chi = 2 - 2g$ it is Theorem 8.44 over $\mathbb{C}$. □

For example, triangulate the sphere $Y = \mathbb{P}^1(\mathbb{C})$ as a double pyramid over a triangle, with apexes 0 and $\infty$ and the equator $|w| = 1$ divided into three edges: $V = 5$, $E = 9$, $F = 6$, $\chi = 2$. The map $z \mapsto z^2$ is branched only over 0 and $\infty$, each with one preimage of multiplicity 2. The three equatorial vertices have two lifts each, every edge and face has two lifts, and the apexes have one lift each: $V = 8$, $E = 18$, $F = 12$, so $\chi(X) = 2 = 2 \cdot 2 - (1 + 1)$, in agreement with (c) and with $X \simeq \mathbb{P}^1$.

## 8.9 Holomorphic differentials, residues, periods

Holomorphic 1-forms are the analytic face of regular differentials. In the book they are the weight-two cusp forms ($f \mapsto f(z)\,dz$ identifies $S_2(\Gamma_0(N))$ with $H^0(X_0(N), \Omega^1)$, Chapter 12), and their periods give the period lattice $\Lambda_E$ and the Jacobian. In the paper, positivity (Theorem 8.55(b)) is the positivity of the Petersson norm $\langle f, f\rangle$ and of the area $\mathrm{vol}(\Lambda_E)$ in the formula of Zagier and Watkins (Remark rem:degree; Chapter 14), and $\varphi^* H^1(E, \mathbb{C})$ in the proof of Proposition prop:pullback is spanned by $\varphi^*\omega_E$ and its conjugate (Proposition 8.59(iii)).

**Definition 8.51** (1-forms on a Riemann surface)**.** A *meromorphic* (*holomorphic*) 1-form $\omega$ on $X$ is given in each chart $z$ by $f(z)\,dz$ with $f$ meromorphic (holomorphic), the expressions in two charts being related by $f(z)\,dz = f(z(w))z'(w)\,dw$. $\mathrm{ord}_P\,\omega = \mathrm{ord}_P\,f$ in a chart at $P$; $\mathrm{div}\,\omega = \sum \mathrm{ord}_P(\omega)(P)$; the *residue* $\mathrm{res}_P\,\omega$ is the coefficient of $z^{-1}$ in the Laurent expansion of $f$ in a chart centred at $P$. $H^0(X, \Omega^1)$ is the complex vector space of holomorphic 1-forms. Smooth real or complex 1-forms $p\,dx + q\,dy$ and 2-forms $r\,dx \wedge dy$, the exterior derivative $d$, the wedge product and $\int_X$ of 2-forms (with the orientation of Convention 8.1(C6)) are defined chartwise as on open subsets of $\mathbb{R}^2$, using a partition of unity for $\int_X$; Stokes' theorem $\int_M d\theta = \int_{\partial M}\theta$ holds for a compact surface $M$ with smooth boundary (Appendix A; [136, Chapter IV]). A 1-form $\theta$ is *closed* if $d\theta = 0$, *exact* if $\theta = dg$ for a smooth function $g$; line integrals $\int_\gamma \theta$ along piecewise smooth paths are defined chartwise.

The residue does not depend on the chart: $\mathrm{res}_P\,\omega = \frac{1}{2\pi i}\oint_\gamma \omega$ for a small positively oriented loop $\gamma$ around $P$ in a chart, by the residue theorem of complex analysis, and the line integral is chart-independent. A holomorphic 1-form is closed: $d(f\,dz) = \frac{\partial f}{\partial \bar z}\,d\bar z \wedge dz = 0$. Examples: $dz$ on $\mathbb{C}/\Lambda$ is a holomorphic 1-form without zeros (the transition maps are translations, with derivative 1); on $\mathbb{P}^1$, $dz = -w^{-2}dw$ has a double pole at $\infty$, and there is no nonzero holomorphic 1-form, as in Proposition 8.26(iv).

**Theorem 8.52** (the residue theorem on a compact Riemann surface)**.** *For a meromorphic* 1*-form* $\omega$ *on a compact Riemann surface* $X$, $\sum_{P \in X} \mathrm{res}_P\,\omega = 0$.

*Proof.* $\omega$ has finitely many poles $P_1,\dots,P_r$ (they are isolated in the compact $X$). Choose pairwise disjoint closed coordinate discs $\Delta_j$ around $P_j$ and put $M = X \setminus \bigcup_j \operatorname{int}\Delta_j$, a compact surface with boundary $\partial M = \bigcup \partial\Delta_j$, each circle oriented clockwise as seen from its disc (the boundary orientation of $M$). On $M$, $\omega$ is holomorphic, hence closed, so Stokes gives $0 = \int_M d\omega = \int_{\partial M}\omega = -\sum_j \oint_{\partial\Delta_j}\omega = -2\pi i\sum_j \operatorname{res}_{P_j}\omega$, the circles $\partial\Delta_j$ now oriented counterclockwise. □

On $\mathbb{P}^1$ the form $dz/z$ has residue 1 at 0, and in the chart $w = 1/z$ at $\infty$ it is $-w^{-2}dw/w^{-1} = -dw/w$, with residue $-1$: the sum is 0. Compactness is what makes the theorem work: on $\mathbb{C}$ alone, $dz/z$ has the single residue 1. The point at infinity carries the missing residue, just as a closed surface admits no net flux out of it.

**Corollary 8.53** (the argument principle)**.** *For $f \in \mathcal{M}(X)^\times$, $\deg\operatorname{div}(f) = \sum_P \operatorname{res}_P(df/f) = 0$.*

*Proof.* In a chart at $P$ write $f = z^m u$ with $m = \operatorname{ord}_P f$ and $u$ holomorphic and nonzero near 0; then $df/f = m\,dz/z + du/u$, and $du/u$ is holomorphic, so $\operatorname{res}_P(df/f) = m$. Apply Theorem 8.52. □

This is a second proof of Proposition 8.22(ii) over $\mathbb{C}$, independent of Theorem 8.38. In the language of physics: the total charge on a closed surface is zero.

**Theorem 8.54** (the genus counts holomorphic forms)**.** *For a compact Riemann surface $X$ of genus $g$, $\dim_{\mathbb{C}} H^0(X,\Omega^1) = g$, and $\deg\operatorname{div}\omega = 2g-2$ for every meromorphic 1-form $\omega \neq 0$.*

*Proof.* By Theorem 8.49(b), holomorphic 1-forms are regular differentials and meromorphic ones are differentials, so this is Corollary 8.30(a), (b) (proved there from the stated Riemann-Roch theorem). □

Examples: $H^0(\mathbb{C}/\Lambda,\Omega^1) = \mathbb{C}\,dz$ (a holomorphic form is $f\,dz$ with $f$ holomorphic on the compact $\mathbb{C}/\Lambda$, hence constant); $H^0(\mathbb{P}^1,\Omega^1) = 0$. Forward pointer: $\dim S_2(\Gamma_0(N)) = g(X_0(N))$ (Chapter 12), for instance $\dim S_2(\Gamma_0(121)) = 6$.

**Theorem 8.55** (Riemann's bilinear relations)**.** *Let $X$ be a compact Riemann surface of genus $g \geq 1$ and $a_1,\dots,a_g,b_1,\dots,b_g$ a symplectic basis of $H_1(X,\mathbb{Z})$ (Theorem 8.57). For a closed 1-form $\theta$ put $A_j(\theta) = \int_{a_j}\theta$ and $B_j(\theta) = \int_{b_j}\theta$.*

*(a) For $\omega,\eta \in H^0(X,\Omega^1)$, $\int_X \omega\wedge\eta = 0$.*

*(b) For $\omega \in H^0(X,\Omega^1)$, $\omega \neq 0$: $\frac{i}{2}\int_X \omega\wedge\bar\omega > 0$.*

*(c) For closed 1-forms $\theta$, $\eta$:* $$\int_X \theta\wedge\eta = \sum_{j=1}^{g}\bigl(A_j(\theta)B_j(\eta) - B_j(\theta)A_j(\eta)\bigr).$$

*(d) For $\omega,\eta \in H^0(X,\Omega^1)$, and for $\omega \neq 0$ respectively,*

$$\sum_j \bigl(A_j(\omega)B_j(\eta) - B_j(\omega)A_j(\eta)\bigr) = 0, \qquad \sum_j \operatorname{Im}\bigl(\overline{A_j(\omega)}B_j(\omega)\bigr) > 0.$$

*(e) A holomorphic 1-form with all $A$-periods zero is zero. The period map $H_1(X,\mathbb{Z}) \to H^0(X,\Omega^1)^\vee$, $\gamma \mapsto (\omega \mapsto \int_\gamma\omega)$, is injective, and its image $\Lambda_X$ is a lattice in the $g$-dimensional complex vector space $H^0(X,\Omega^1)^\vee$.*

*Proof.* (a) Locally $\omega \wedge \eta = fg\, dz \wedge dz = 0$. (b) Locally $\omega = f\, dz$ and $\frac{i}{2}\omega \wedge \bar{\omega} = \frac{i}{2}|f|^2 dz \wedge d\bar{z} = |f|^2 dx \wedge dy$, since $dz \wedge d\bar{z} = -2i\, dx \wedge dy$; the integral of this nonnegative density is positive unless $f \equiv 0$. (In physical terms $\frac{i}{2}\int \omega \wedge \bar{\omega} = \int |f|^2\, dx\, dy$ is a squared norm; the Petersson product of Chapter 12 is $\langle f, g\rangle = \frac{i}{2}\int_{X_0(N)} f\, dz \wedge \overline{g\, dz}$, Appendix B, B-C8.) (d) The first identity is (c) with (a). For the second apply (c) to $\theta = \omega$, $\eta = \bar{\omega}$, using $A_j(\bar{\omega}) = \overline{A_j(\omega)}$: $\frac{i}{2}\int \omega \wedge \bar{\omega} = \frac{i}{2}\sum_j (A_j \bar{B}_j - \overline{A_j \bar{B}_j}) = \frac{i}{2} \cdot 2i \operatorname{Im} \sum_j A_j \bar{B}_j = \sum_j \operatorname{Im}(\bar{A}_j B_j)$, positive by (b). (e) If all $A_j(\omega) = 0$, the sum in (d) is zero, so $\omega = 0$. If $\int_\gamma \omega = 0$ for all $\omega$, then also $\int_\gamma \bar{\omega} = 0$, so $\gamma$ pairs to zero with $H^0(X, \Omega^1) \oplus \overline{H^0(X, \Omega^1)}$, which is all of $H^1(X, \mathbb{C})$ (Proposition 8.59(iii)); as $H_1(X, \mathbb{Z})$ is free and $H^1(X, \mathbb{C}) = \operatorname{Hom}(H_1(X, \mathbb{Z}), \mathbb{C})$, $\gamma = 0$. The same argument with real coefficients shows that the $\mathbb{R}$-linear extension $H_1(X, \mathbb{Z}) \otimes \mathbb{R} \to H^0(X, \Omega^1)^\vee$ is injective; both sides have real dimension $2g$ (Theorem 8.57, Theorem 8.54), so it is an isomorphism, and the image of $H_1(X, \mathbb{Z})$ is a lattice. □

*Sketch of proof of (c).* Cut $X$ along loops representing the symplectic basis, all through one point; the result is a $4g$-gon $\Pi$ whose sides are identified in the pattern $a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$. On the simply connected $\Pi$, $\theta = dF$ for a smooth function $F$, and Stokes gives $\int_X \theta \wedge \eta = \int_\Pi d(F\eta) = \int_{\partial\Pi} F\eta$, because $d(F\eta) = dF \wedge \eta + F\, d\eta = \theta \wedge \eta$ ($\eta$ is closed); here $\partial\Pi$ is traversed counterclockwise, which for the torus of Example 8.61 is the word $aba^{-1}b^{-1}$ with $a \cdot b = +1$. Fix $j$ and label the vertices of the block $a_j b_j a_j^{-1} b_j^{-1}$ as $v_0 \xrightarrow{a_j} v_1 \xrightarrow{b_j} v_2 \xrightarrow{a_j^{-1}} v_3 \xrightarrow{b_j^{-1}} v_4$. A point $p(s)$ on the first side and the point $p'(s)$ on the third side that is glued to it lie over the same point of $a_j$, the third side running from $v_3$ ($s = 0$) to $v_2$ ($s = 1$). The difference $F(p'(s)) - F(p(s))$ is constant in $s$: its derivative is $\theta$ evaluated on the same tangent vector of $a_j$, minus itself. At $s = 1$ it is $F(v_2) - F(v_1) = \int_{v_1}^{v_2} dF = B_j(\theta)$. As the third side is traversed against $a_j$, the two sides contribute

$$\int_{a_j} (F(p) - F(p'))\eta = -B_j(\theta) A_j(\eta).$$

In the same way a point $q(s)$ of the second side and the glued point $q'(s)$ of the fourth (running from $v_4$ at $s = 0$ to $v_3$ at $s = 1$) have $F(q') - F(q)$ constant, equal at $s = 1$ to $F(v_3) - F(v_2) = -A_j(\theta)$, and the two sides contribute $\int_{b_j} (F(q) - F(q'))\eta = A_j(\theta) B_j(\eta)$. Summing over $j$ gives (c). On the torus, with $\theta = ds$, $\eta = dt$ and $F = s$, the two contributions are 0 and 1. What is not proved here is that cutting along the loops of a symplectic basis gives such a polygon. See [136, Chapter VIII] and [75, Chapter 2, Section 2] (sections not verified). On the torus (Example 8.61) the formula is checked directly. □

In the language of physics: for $X = \mathbb{C}/\Lambda$ and $\omega = dz$, (b) is the area of a period parallelogram, $\frac{i}{2}\int dz \wedge d\bar{z} = \operatorname{Im}(\bar{\omega}_1 \omega_2)$, and the lattice $\Lambda_X$ of (e) is the crystal lattice of periods (Example 8.61).

## 8.10 $H_1$, $H^1$ and the cup product

This section is the main input of the chapter to Section 4.1 of the paper. The paper works with $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$ and states that the cup product is "perfect and alternating" on it and that the Hecke operators are self-adjoint for the cup product composed with $w_{M^\Sigma}$; Proposition prop:pullback computes the cup product on $\varphi_E^* H^1(E, \mathbb{Z}_\ell)$. We define $H_1$ and $H^1$ of a compact Riemann surface without sheaves ($H^1 := \operatorname{Hom}(H_1, \mathbb{Z})$, with de Rham forms to compute), prove that the cup product is perfect and alternating with Gram matrix $\left(\begin{smallmatrix} 0 & I_g \\ -I_g & 0 \end{smallmatrix}\right)$ in a symplectic basis, and set up pullback, pushforward and the transfer with the projection formula, which is the

cohomological meaning of the degree. The convention box Convention 8.63 says which object is dual to which. Throughout, $X$ and $Y$ are compact Riemann surfaces of genera $g_X$, $g_Y$ (or $g$).

**Definition 8.56** (first homology)**.** $H_1(X,\mathbb{Z})$ is the abelianisation of the fundamental group $\pi_1(X)$; equivalently, singular homology in degree one. For the reader: an element is represented by a finite formal sum of closed oriented loops, two sums being identified when their difference bounds a (sum of oriented) surface pieces; a loop's class depends only on its free homotopy class. A holomorphic map $\varphi\colon X \to Y$ induces $\varphi_*\colon H_1(X,\mathbb{Z}) \to H_1(Y,\mathbb{Z})$, $[\gamma] \mapsto [\varphi\circ\gamma]$. For an abelian group $A$, $H_1(X,A) = H_1(X,\mathbb{Z}) \otimes A$; in particular $H_1(X,\mathbb{Z}_\ell) = H_1(X,\mathbb{Z}) \otimes \mathbb{Z}_\ell$ and $H_1(X,\mathbb{Z}/n) = H_1(X,\mathbb{Z})/n$.

**Theorem 8.57** (symplectic bases; stated)**.** *Let $X$ have genus $g$.*

*(i) $H_1(X,\mathbb{Z})$ is free of rank $2g$, with a* symplectic basis *$a_1,\dots,a_g,b_1,\dots,b_g$: on the model "sphere with $g$ handles", $a_j$ goes around the $j$-th handle and $b_j$ through it.*

*(ii) The intersection pairing $H_1(X,\mathbb{Z}) \times H_1(X,\mathbb{Z}) \to \mathbb{Z}$, which on transverse oriented loops is the signed count of crossings of Convention 8.1(C6), is well defined, bilinear, alternating and perfect, with $a_i \cdot b_j = \delta_{ij}$ and $a_i \cdot a_j = b_i \cdot b_j = 0$.*

*(iii) Every class in $H_1(X,\mathbb{Z})$ is represented by a sum of loops avoiding any given finite set, and the kernel of $H_1(X\setminus S,\mathbb{Z}) \to H_1(X,\mathbb{Z})$, for a finite set $S$, is generated by small loops around the points of $S$.*

We do not prove this; see [136, Chapter VIII, Section 1], [70, Section 21], [64, Chapter III, Section 1] and, for (iii), the Mayer-Vietoris sequence in [80, Chapter 2] (all section numbers not verified). In the language of physics, the intersection number is a signed crossing count, like a winding or linking number.

**Definition 8.58** (first cohomology)**.** $H^1(X,\mathbb{Z}) = \mathrm{Hom}(H_1(X,\mathbb{Z}),\mathbb{Z})$, and for an abelian group (or ring) $A$, $H^1(X,A) = \mathrm{Hom}(H_1(X,\mathbb{Z}),A)$. A holomorphic map $\varphi\colon X \to Y$ induces $\varphi^*\colon H^1(Y,A) \to H^1(X,A)$, the transpose of $\varphi_*$: $(\varphi^*\xi)(\gamma) = \xi(\varphi_*\gamma)$. A closed 1-form $\theta$ defines the class $[\theta] \in H^1(X,\mathbb{C})$, $\gamma \mapsto \int_\gamma \theta$ (well defined by Stokes: the integral over a boundary is zero). $H^2(X,\mathbb{R})$ is the space of smooth 2-forms modulo exact ones, and $H^2(X,\mathbb{Z})$ the subgroup of classes with integral integral.

**Proposition 8.59** (cohomology of a compact Riemann surface)**.**

*(i) $H^1(X,\mathbb{Z})$ is free of rank $2g$, and $H^1(X,A) = H^1(X,\mathbb{Z}) \otimes A$ for every abelian group $A$; in particular $H^1(X,\mathbb{Z}_\ell) = H^1(X,\mathbb{Z}) \otimes \mathbb{Z}_\ell$ is free of rank $2g$ over $\mathbb{Z}_\ell$. With the dual basis of a symplectic basis, written $a_1^\vee,\dots,a_g^\vee,b_1^\vee,\dots,b_g^\vee$, $H^1(X,\mathbb{Z}) = \bigoplus \mathbb{Z}a_j^\vee \oplus \bigoplus \mathbb{Z}b_j^\vee$.*

*(ii) (de Rham; stated.) Every class in $H^1(X,\mathbb{R})$ is $[\theta]$ for a closed real 1-form $\theta$, unique up to an exact form; $H^1(X,\mathbb{Z})$ is the lattice of classes with integral periods. $\int_X\colon H^2(X,\mathbb{R}) \to \mathbb{R}$ is an isomorphism, so $H^2(X,\mathbb{Z}) \simeq \mathbb{Z}$, generated by the class of integral 1; this fixes the isomorphism $H^2(X,\mathbb{Z}) = \mathbb{Z}$.*

*(iii) (Hodge.) $H^1(X,\mathbb{C}) = H^0(X,\Omega^1) \oplus \overline{H^0(X,\Omega^1)}$: every class is $[\omega+\bar\eta]$ for unique holomorphic $\omega$, $\eta$.*

*(iv) (Stated.) For $g \geq 1$ the universal cover of $X$ is $\mathbb{C}$ or the unit disc, so $X$ is a $K(\pi_1(X),1)$, and $H^1(X,M) = H^1(\pi_1(X),M)$ in the sense of Chapter 16 for every abelian group $M$ with trivial action, compatibly with cup products. This is the bridge to the group cohomology of $\Gamma_0(N)$ used by Diamond, Flach and Guo (Chapter 14).*

*Proof.* (i) $H_1(X,\mathbb{Z}) \simeq \mathbb{Z}^{2g}$ (Theorem 8.57), and $\mathrm{Hom}(\mathbb{Z}^{2g},A) = A^{2g} = \mathrm{Hom}(\mathbb{Z}^{2g},\mathbb{Z}) \otimes A$. The universal coefficient theorem, with $H_0$ free, identifies this with singular cohomology (stated, [80, Section 3.1]). (ii) is [136, Chapter VIII] and [70, Sections 10 and 19] (not verified); we do not prove it.

(iii) (Sketch, given (ii) and Theorem 8.54.) The map $(\omega,\eta) \mapsto [\omega + \bar{\eta}]$ from $H^0(X,\Omega^1) \oplus H^0(X,\Omega^1)$ to $H^1(X,\mathbb{C})$ is injective: if $\omega + \bar{\eta} = dF$, then $\int_X (\omega+\bar{\eta}) \wedge \bar{\omega} = \int_X d(F\bar{\omega}) = 0$ by Stokes ($\bar{\omega}$ is closed), while $\int_X \bar{\eta} \wedge \bar{\omega} = \overline{\int_X \eta \wedge \omega} = 0$ by Theorem 8.55(a); so $\int_X \omega \wedge \bar{\omega} = 0$ and $\omega = 0$ by Theorem 8.55(b); then $\bar{\eta} = dF$ and the same argument gives $\eta = 0$. Both sides have complex dimension $2g$, by Theorem 8.54 and (i) with (ii). (The proof is complete given the stated inputs; it is labelled a sketch only because de Rham's theorem is.) (iv) We do not prove this; see [80, Section 1.B] (not verified). □

**Theorem 8.60** (the cup product)**.** *The* cup product

$$\langle\ ,\ \rangle \colon H^1(X,\mathbb{Z}) \times H^1(X,\mathbb{Z}) \to H^2(X,\mathbb{Z}) = \mathbb{Z}, \qquad \langle \theta,\eta\rangle = \int_X \theta \wedge \eta$$

*(de Rham representatives) is well defined, integral, bilinear, alternating and perfect. In the basis dual to a symplectic basis,*

$$\langle a_i^\vee, b_j^\vee\rangle = \delta_{ij}, \qquad \langle a_i^\vee, a_j^\vee\rangle = \langle b_i^\vee, b_j^\vee\rangle = 0,$$

*so the Gram matrix is $\left(\begin{smallmatrix} 0 & I_g \\ -I_g & 0 \end{smallmatrix}\right)$, of determinant* 1*. The map* PD$\colon H^1(X,\mathbb{Z}) \to H_1(X,\mathbb{Z})$ *defined by $\xi(\mathrm{PD}(\theta)) = \langle\theta,\xi\rangle$ for all $\xi$ is an isomorphism (*Poincare duality*) with $\mathrm{PD}(a_j^\vee) = b_j$ and $\mathrm{PD}(b_j^\vee) = -a_j$, and an isometry: $\langle\theta,\eta\rangle = \mathrm{PD}(\theta)\cdot\mathrm{PD}(\eta)$. For every commutative ring $R$ (in particular $\mathbb{Z}_\ell$, $\mathbb{F}_\ell$, $\mathbb{Q}_\ell$, $\mathbb{C}$) the $R$-bilinear extension to $H^1(X,R)$ is perfect and alternating.*

*Proof.* Changing $\theta$ by $dF$ changes $\int \theta\wedge\eta$ by $\int d(F\eta) = 0$ (Stokes, $\eta$ closed), so the pairing is well defined; it is bilinear, and alternating because $\theta\wedge\theta = 0$. By Theorem 8.55(c), $\langle\theta,\eta\rangle = \sum_j (\theta(a_j)\eta(b_j) - \theta(b_j)\eta(a_j))$, an integer for integral classes; on the dual basis this gives the displayed values and the Gram matrix, whose determinant is 1. So the pairing is perfect (Chapter 3: perfect iff the Gram determinant is a unit). For PD: $\xi \mapsto \langle\theta,\xi\rangle$ is an element of $\mathrm{Hom}(H^1(X,\mathbb{Z}),\mathbb{Z})$, which is $H_1(X,\mathbb{Z})$ (a free module of finite rank is its double dual); this defines PD, an isomorphism by perfectness. The class $b_j$ has $a_i^\vee(b_j) = 0$ and $b_i^\vee(b_j) = \delta_{ij}$, which are the values $\langle a_j^\vee, a_i^\vee\rangle = 0$ and $\langle a_j^\vee, b_i^\vee\rangle = \delta_{ij}$; so $\mathrm{PD}(a_j^\vee) = b_j$, and likewise $\mathrm{PD}(b_j^\vee) = -a_j$. Then $\mathrm{PD}(a_i^\vee)\cdot\mathrm{PD}(b_j^\vee) = b_i\cdot(-a_j) = a_j\cdot b_i = \delta_{ij} = \langle a_i^\vee, b_j^\vee\rangle$, and the other basis pairs give 0 on both sides; by bilinearity PD is an isometry. Base change: the Gram matrix over $R$ is the image of the integral one, of determinant 1 (Chapter 3), and $\langle x,x\rangle = 0$ holds for $x \in H^1(X,R)$ because it holds on a basis and $\langle x,y\rangle = -\langle y,x\rangle$. □

The input that is only stated is the existence of a symplectic basis and the bilinear identity of Theorem 8.55(c); perfectness is Poincare duality for surfaces, which in general is [80, Section 3.3]. Over $\mathbb{R}$ nondegeneracy also follows from positivity: by Proposition 8.59(iii), $\langle[\omega],[\bar{\omega}]\rangle \neq 0$ for $\omega \neq 0$.

**Example 8.61** (the torus)**.** Let $X = \mathbb{C}/\Lambda$, $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\operatorname{Im}(\omega_2/\omega_1) > 0$, and real coordinates $z = s\omega_1 + t\omega_2$, $s, t \in \mathbb{R}/\mathbb{Z}$. From $dz = \omega_1 ds + \omega_2 dt$ one computes

$$dz \wedge d\bar{z} = (\omega_1 ds + \omega_2 dt) \wedge (\bar{\omega}_1 ds + \bar{\omega}_2 dt) = (\omega_1\bar{\omega}_2 - \omega_2\bar{\omega}_1)\, ds \wedge dt = -2i \operatorname{Im}(\bar{\omega}_1\omega_2)\, ds \wedge dt,$$

so $dx \wedge dy = \frac{i}{2} dz \wedge d\bar{z} = \operatorname{Im}(\bar{\omega}_1\omega_2)\, ds \wedge dt$, and $\operatorname{Im}(\bar{\omega}_1\omega_2) = |\omega_1|^2 \operatorname{Im}(\omega_2/\omega_1) > 0$ (as $\bar{\omega}_1\omega_2 = |\omega_1|^2\, \omega_2/\omega_1$); so $ds \wedge dt$ is positive and $\int_X ds \wedge dt = 1$ (the unit square in $(s,t)$ is a fundamental domain).

(a) $H_1(X, \mathbb{Z}) = \Lambda$ (a path from 0 to $\lambda$ in $\mathbb{C}$ projects to a loop), with $a$ the image of $[0, \omega_1]$ and $b$ that of $[0, \omega_2]$. They meet only at 0, with tangents $\partial_s$, $\partial_t$, a positive basis: $a \cdot b = +1$.

(b) $H^1(X, \mathbb{Z})$ has basis $[ds]$, $[dt]$, dual to $a$, $b$; $\langle ds, dt\rangle = \int ds \wedge dt = 1$; $\mathrm{PD}(ds) = b$, $\mathrm{PD}(dt) = -a$; and Theorem 8.55(c) reads $1 = A(ds)B(dt) - B(ds)A(dt) = 1 \cdot 1 - 0 \cdot 0$.

(c) The holomorphic form $dz = \omega_1 ds + \omega_2 dt$ has periods $A = \omega_1$, $B = \omega_2$, and $\frac{i}{2}\int dz \wedge d\bar{z} = \operatorname{Im}(\bar{\omega}_1\omega_2)$ is the area of the period parallelogram: Theorem 8.55(b), (d). For $\Lambda = \mathbb{Z} + \tau\mathbb{Z}$ the inequality of (d) is $\operatorname{Im}\tau > 0$.

(d) $[-1]\colon z \mapsto -z$ acts by $-1$ on $H_1$ and $H^1$, and is an isometry: $\langle -x, -y\rangle = \langle x, y\rangle$. Multiplication by an integer $m \neq 0$ acts by $m$ on $H_1$ and on $H^1$ (it pulls $ds$ back to $m\, ds$), and by $m^2$ on $H^2$ (it pulls $ds \wedge dt$ back to $m^2 ds \wedge dt$), so it has degree $m^2$ (Proposition 8.62(iv)); directly, the fibre over 0 is $\frac{1}{m}\Lambda/\Lambda$, with $m^2$ points, all unramified.

In the language of physics, $\Lambda$ is a crystal lattice, $H^1(X, \mathbb{Z}) = \operatorname{Hom}(\Lambda, \mathbb{Z})$ its reciprocal lattice, and $ds \wedge dt$ the normalised volume form of the periodic box.

**Proposition 8.62** (pullback, pushforward, transfer; the projection formula)**.** *Let $\varphi\colon X \to Y$ be a nonconstant holomorphic map of compact Riemann surfaces, $n = \deg\varphi$, $B \subseteq Y$ its finite set of branch points, $Y' = Y \setminus B$, $X' = \varphi^{-1}(Y')$.*

*(i) On de Rham classes $\varphi^*[\theta] = [\varphi^*\theta]$: $\int_\gamma \varphi^*\theta = \int_{\varphi_*\gamma} \theta$ (change of variables). $\varphi\colon X' \to Y'$ is an $n$-sheeted covering map, locally the disjoint union of $n$ biholomorphisms, each orientation-preserving.*

*(ii) (Transfer.) There is a unique homomorphism $\varphi^!\colon H_1(Y, \mathbb{Z}) \to H_1(X, \mathbb{Z})$ sending the class of a loop $\gamma$ in $Y'$ to the class of $\varphi^{-1}(\gamma)$, the sum of the closed loops formed by the $n$ lifts of $\gamma$. It satisfies $\varphi_*\varphi^! = n$.*

*(iii) (Pushforward on $H^1$.) Define $\varphi_*\colon H^1(X, A) \to H^1(Y, A)$ as the transpose of $\varphi^!$: $(\varphi_*\eta)(\gamma) = \eta(\varphi^!\gamma)$. Every class in $H^1(X, \mathbb{R})$ is represented by a closed form $\theta$ vanishing near $\varphi^{-1}(B)$, and then $\varphi_*[\theta]$ is represented by the trace $\operatorname{Tr}\theta$: on $Y'$, $\operatorname{Tr}\theta = \sum_i s_i^*\theta$ over the local sections $s_1, \dots, s_n$ of $\varphi$, extended by 0 near $B$.*

*(iv) (Projection formula.) For $\eta \in H^1(X, \mathbb{Z})$ and $\xi, \xi' \in H^1(Y, \mathbb{Z})$:*

$$\langle \varphi_*\eta, \xi\rangle_Y = \langle \eta, \varphi^*\xi\rangle_X, \qquad \varphi_*\varphi^* = n \text{ on } H^1(Y, \mathbb{Z}), \qquad \langle \varphi^*\xi, \varphi^*\xi'\rangle_X = n\, \langle \xi, \xi'\rangle_Y,$$

*and $\int_X \varphi^*\rho = n \int_Y \rho$ for every smooth 2-form $\rho$ on $Y$: $\varphi^*$ is multiplication by $n$ on $H^2(\cdot, \mathbb{Z}) = \mathbb{Z}$. In particular $\varphi^*$ is injective on $H^1(Y, \mathbb{Z})$ and $\varphi_*\colon H_1(X, \mathbb{Z}) \to H_1(Y, \mathbb{Z})$ has finite cokernel, killed by $n$.*

*(v)* $(\psi\circ\varphi)_* = \psi_*\varphi_*$, $(\psi\circ\varphi)^* = \varphi^*\psi^*$, $(\psi\circ\varphi)^! = \varphi^!\psi^!$*; for an automorphism* $w$, $w^! = (w^{-1})_*$ *and* $w_* = (w^{-1})^*$ *on* $H^1$.

*Proof.* (i) The integral identity is the change of variables formula for line integrals (on each chart piece of $\gamma$). By Theorem 8.48(c), (d), every point of $Y'$ has exactly $n$ preimages, each of multiplicity one, at which $\varphi$ is a local biholomorphism; the proof of Theorem 8.48(c) gives a neighbourhood $V$ with $\varphi^{-1}(V)$ a disjoint union of $n$ open sets mapped biholomorphically onto $V$. Orientation is preserved by Theorem 8.48(e).

(ii) (Sketch.) By Theorem 8.57(iii), classes in $H_1(Y,\mathbb{Z})$ are represented by loops in $Y'$. For a covering, the preimage of a loop is a union of closed loops covering it $n$ times in total, and the covering transfer $H_1(Y',\mathbb{Z})\to H_1(X',\mathbb{Z})$ is well defined ([80, Section 3.G], not verified). By Theorem 8.57(iii), the kernel of $H_1(Y',\mathbb{Z})\to H_1(Y,\mathbb{Z})$ is generated by small loops around the points of $B$; the preimage of such a loop is a sum of small loops around the points of $\varphi^{-1}(B)$ (by the normal form $w=z^e$, a loop around $Q$ lifts to loops around the points over $Q$), which are null-homologous in $X$. So the transfer descends to $\varphi^!$. Pushing forward the $n$ lifts of $\gamma$ gives $n\gamma$, so $\varphi_*\varphi^! = n$.

(iii) Let $\theta$ be a closed form on $X$ and $P\in\varphi^{-1}(B)$. On a coordinate disc $U$ around $P$, $\theta = dF$ for a smooth function $F$ (Poincare lemma on a disc). With $\rho$ a smooth function equal to 1 near $P$ and supported in $U$, $\theta - d(\rho F)$ is in the same class and vanishes near $P$. Doing this at the finitely many points of $\varphi^{-1}(B)$ gives a representative vanishing near $\varphi^{-1}(B)$. For such $\theta$, $\operatorname{Tr}\theta$ is smooth and closed on $Y'$ (locally a finite sum of pullbacks of closed forms by biholomorphisms) and vanishes near $B$, so it extends by 0 to a closed form on $Y$. For a loop $\gamma$ in $Y'$, $\int_\gamma \operatorname{Tr}\theta = \sum_i \int_{s_i\circ\gamma}\theta = \int_{\varphi^{-1}(\gamma)}\theta$, which is $[\theta](\varphi^!\gamma)$; as loops in $Y'$ represent all of $H_1(Y,\mathbb{Z})$, $[\operatorname{Tr}\theta] = \varphi_*[\theta]$.

(iv) For a 2-form $\psi$ on $X$, let $\operatorname{Tr}\psi = \sum_i s_i^*\psi$ on $Y'$. On $V$ as in (i), $\int_V \operatorname{Tr}\psi = \sum_i \int_{s_i(V)}\psi = \int_{\varphi^{-1}(V)}\psi$ (each $s_i$ is an orientation-preserving diffeomorphism onto its image); covering $Y'$ by countably many such $V$ and using a partition of unity, $\int_{Y'}\operatorname{Tr}\psi = \int_{X'}\psi$. Finite sets have measure zero, so $\int_{X'} = \int_X$ and $\int_{Y'} = \int_Y$ for smooth forms. Now take $\theta$ as in (iii) representing $\eta$, and $\xi$ represented by a closed form, also written $\xi$. On $Y'$, $\operatorname{Tr}(\theta)\wedge\xi = \sum_i s_i^*\theta\wedge s_i^*\varphi^*\xi = \operatorname{Tr}(\theta\wedge\varphi^*\xi)$, since $\varphi\circ s_i = \mathrm{id}$. Hence $\langle\varphi_*\eta,\xi\rangle = \int_Y \operatorname{Tr}(\theta)\wedge\xi = \int_X \theta\wedge\varphi^*\xi = \langle\eta,\varphi^*\xi\rangle$. For a 2-form $\rho$ on $Y$, $\operatorname{Tr}(\varphi^*\rho) = n\rho$ on $Y'$, so $\int_X\varphi^*\rho = n\int_Y\rho$. Choosing the representative $\xi$ of a class in $H^1(Y)$ to vanish near $B$ (as in (iii), with $Y$ in place of $X$), $\varphi^*\xi$ vanishes near $\varphi^{-1}(B)$ and $\operatorname{Tr}(\varphi^*\xi) = n\xi$; so $\varphi_*\varphi^* = n$ by (iii). Then $\langle\varphi^*\xi,\varphi^*\xi'\rangle = \langle\varphi_*\varphi^*\xi,\xi'\rangle = n\langle\xi,\xi'\rangle$ (alternatively, $\int_X\varphi^*(\xi\wedge\xi') = n\int_Y\xi\wedge\xi'$). Since $H^1(Y,\mathbb{Z})$ is torsion-free and $\varphi_*\varphi^* = n$, $\varphi^*$ is injective; dually $\varphi_*\varphi^! = n$ on $H_1$ shows that $nH_1(Y,\mathbb{Z})\subseteq\varphi_*H_1(X,\mathbb{Z})$.

(v) Functoriality of $\varphi_*$ on loops is clear, $\varphi^*$ is its transpose, and $\varphi^!$ is compatible with composition on loops avoiding the branch points of both maps. An automorphism $w$ is unramified of degree one, so $w^{-1}(\gamma) = (w^{-1})\circ\gamma$; transposing gives the last statement. □

The transfer is easiest to see on the torus $X = Y = \mathbb{C}/\Lambda$ with $\varphi = [m]$, $z\mapsto mz$, of degree $n = m^2$ (Example 8.61(d)), which is unramified, so no loop has to be moved. The loop $a$, the image of $s\mapsto s\omega_1$, $s\in[0,1]$, has the $m^2$ lifts $s\mapsto(s\omega_1+\lambda)/m$, one for each class $\lambda\in\Lambda/m\Lambda$. The lift indexed by $\lambda$ ends where the lift indexed by $\lambda+\omega_1$ begins, so the lifts link up in chains of $m$ (indexed by $\lambda,\lambda+\omega_1,\dots,\lambda+(m-1)\omega_1$), each chain forming a closed loop from $\lambda/m$ to $\lambda/m+\omega_1$, whose class in $H_1=\Lambda$ is $\omega_1$, that is $a$. There are $m^2/m = m$ such loops, so $\varphi^!a = m\,a$, and likewise $\varphi^!b = m\,b$. As $\varphi_* = m$ on $H_1$, $\varphi_*\varphi^! = m^2 = n$, as (ii) says; on $H^1$, $\varphi^* = m$ and $\varphi_*$ (the transpose of $\varphi^!$) is also $m$, and $\varphi_*\varphi^* = m^2$ as in (iv). For a general $\varphi$ the transfer is not a

multiple of the identity, and it is the transfer, not $\varphi_*$, that is injective on $H_1$.

**Convention 8.63** (which is dual to which)**.**

(a) *Covariant objects*: $H_1(X,\mathbb{Z})$, the period lattice $\Lambda_X$, the Jacobian $J(X) = H^0(X,\Omega^1)^\vee/\Lambda_X$, its torsion $J[n] = H_1(X,\mathbb{Z}/n)$ and its Tate module $T_\ell J = H_1(X,\mathbb{Z}_\ell)$ (Section 8.11). A morphism $\varphi$ acts by $\varphi_*$, a correspondence $\mathcal{C} = (C,\alpha,\beta)$ by $\mathcal{C}_* = \beta_*\alpha^!$ (Proposition 8.80).

(b) *Contravariant objects*: $H^1(X,\mathbb{Z}) = \mathrm{Hom}(H_1,\mathbb{Z})$, $H^1(X,\mathbb{Z}_\ell) = \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell J,\mathbb{Z}_\ell)$, $H^0(X,\Omega^1)$. A morphism acts by $\varphi^*$, a correspondence by $\mathcal{C}^* = \alpha_*\beta^*$, and $\mathcal{C}^*$ is the transpose of $\mathcal{C}_*$ for the evaluation pairing.

(c) The two sides are identified by Poincare duality (Theorem 8.60), an isometry for the cup product and the intersection pairing, which turns $\mathcal{C}^*$ into $(\mathcal{C}^t)_*$ (Theorem 8.81(c)); it is canonical up to the global orientation sign.

(d) With Galois action (Theorem 8.74): $H^1_{\text{et}} = (T_\ell J)^\vee \simeq T_\ell J(-1)$. For an elliptic curve, $V = V_\ell E$ (the paper's $V$) has determinant $\chi_\ell$, and $H^1(E,\mathbb{Q}_\ell) = V^\vee \simeq V(-1)$ has determinant $\chi_\ell^{-1}$; $\mathrm{ad}^0 V$ is the same for $V$, $V^\vee$ and $V(n)$ (Chapter 4).

(e) Frobenius (Convention 8.1(C7)): the arithmetic Frobenius on $T_\ell J$ and the geometric Frobenius on $H^1$ have the same characteristic polynomial $P(T)$, with $\#C(\mathbb{F}_q) = q + 1 - \mathrm{tr}$.

*In the paper* 8.64. Section 4.1 of the paper (sec:dfg): the pairing $\hat{\delta}_!$ "is perfect and alternating on $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$", being $\frac{1}{2}$ times the cup product composed with $w_{M^\Sigma}$ on classes from $X_0(M^\Sigma)$. The cup product is perfect and alternating on $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$ by Theorem 8.60 with $R = \mathbb{Z}_\ell$; the factor $\frac{1}{2}$ is a unit for odd $\ell$; composing with the automorphism $w^*_{M^\Sigma}$ preserves perfectness, and alternation needs $w^*$ to be an isometric involution (Corollary 8.82). Proposition prop:pullback: "$\varphi^*(a\cup b) = \varphi^*a\cup\varphi^*b$ and $\varphi^*$ multiplies the fundamental class of $H^2$ by $\deg\varphi_E$; so $\langle\varphi^*a,\varphi^*b\rangle_{X_0(N)} = \deg\varphi_E\,\langle a,b\rangle_E$, the projection formula" is Proposition 8.62(iv); "The cup product on $H^1(E,\mathbb{Z})$ is unimodular" is Theorem 8.60 for $g = 1$ (Example 8.61).

## 8.11 The Jacobian: $\mathrm{Pic}^0$, complex torus, abelian variety; torsion and Tate modules

The Jacobian $J(X)$ of a curve is $\mathrm{Pic}^0$, made into a complex torus by integrating holomorphic forms and into an abelian variety by a theorem. The paper uses it three times: the modular parametrisation factors as $X_0(N)\to J_0(N)\to E$, and optimality of $E$ means that the kernel of $J_0(N)\to E$ is connected (Section 1, Proposition prop:pullback); Remark rem:free speaks of $J_0(N)[\mathfrak{m}]$ and of the Tate module of $J_0(N)$; and the paragraph after Proposition prop:pullback identifies the lattice $M_{f,\lambda}$ with $H^1(E,\mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell E,\mathbb{Z}_\ell)\simeq T_\ell E(-1)$. This section proves the facts behind these sentences for a general curve, in particular Proposition 8.70, which turns "connected kernel" into "saturated image of $\varphi^*$ on $H^1$" and computes the cup product on that image.

**Definition 8.65** (the Jacobian)**.** For a smooth projective curve $C$ over $k$, $J(C)(\bar{k}) = \mathrm{Pic}^0(C)$, with the action of $G_k$ of Proposition 8.24(ii). For a compact Riemann surface $X$ of genus $g\geq 1$, the *analytic Jacobian* is $J_{\text{an}}(X) = H^0(X,\Omega^1)^\vee/\Lambda_X$, with $\Lambda_X$ the period lattice of Theorem 8.55(e), a complex torus of dimension $g$. Examples: $J(\mathbb{P}^1) = 0$ (Proposition 8.24(i)); for $g = 1$ and $O\in C(k)$, $J(C)\simeq C$ by Proposition 8.34.

**Theorem 8.66** (Abel-Jacobi). *Let $X$ be a compact Riemann surface of genus $g \geq 1$.*

*(i) The* Abel-Jacobi map $\mathrm{AJ}\colon \mathrm{Div}^0(X) \to J_{\mathrm{an}}(X)$, $\sum_i((P_i)-(Q_i)) \mapsto (\omega \mapsto \sum_i \int_{Q_i}^{P_i} \omega) \bmod \Lambda_X$, *is a well-defined homomorphism.*

*(ii) (Abel; stated.)* $\ker \mathrm{AJ} = \mathrm{Prin}(X)$. *(Jacobi inversion; stated.)* AJ *is surjective. Hence* $\mathrm{AJ}\colon \mathrm{Pic}^0(X) \to J_{\mathrm{an}}(X)$ *is an isomorphism.*

*(iii) For a base point $P_0$, $\iota\colon X \to J_{\mathrm{an}}(X)$, $P \mapsto \mathrm{AJ}((P)-(P_0))$, is holomorphic and injective, and $\iota_*\colon H_1(X,\mathbb{Z}) \to H_1(J_{\mathrm{an}}(X),\mathbb{Z}) = \Lambda_X$ is the period isomorphism of Theorem 8.55(e). For $g = 1$, $\iota\colon X \to \mathbb{C}/\Lambda_X$ is an isomorphism.*

*Proof.* (i) Two paths from $Q_i$ to $P_i$ differ by a closed loop, so the integrals differ by a period, an element of $\Lambda_X$; the map is additive. (ii) We do not prove this; see [136, Chapter VIII], [70, Sections 20 and 21] and [75, Chapter 2, Section 2] (not verified). The "only if" half of Abel's theorem ($\mathrm{AJ}(\operatorname{div} f) = 0$) is a trace computation: for nonconstant $f\colon X \to \mathbb{P}^1$, $\operatorname{div} f = f^*((0)-(\infty))$, and as in the proof of Proposition 8.68(iii), $\mathrm{AJ}(\operatorname{div} f)(\omega) \equiv \int_\infty^0 \mathrm{Tr}_f\,\omega$ modulo periods; $\mathrm{Tr}_f\,\omega$ is a holomorphic form on $\mathbb{P}^1$, hence 0 (Proposition 8.26(iv)). (iii) In a chart near $P_1$, $\iota(P) = \iota(P_1) + (\omega \mapsto \int_{P_1}^P \omega)$, and $P \mapsto \int_{P_1}^P \omega$ is a primitive of a holomorphic function, so $\iota$ is holomorphic. If $\iota(P) = \iota(Q)$ then $\mathrm{AJ}((P)-(Q)) = 0$, so $(P) \sim (Q)$ by Abel's theorem, and $P = Q$ since $g \geq 1$ (Corollary 8.30(d)). For a loop $\gamma$ based at $P_0$, the path $t \mapsto (\omega \mapsto \int_{\gamma|[0,t]} \omega)$ in $H^0(X,\Omega^1)^\vee$ lifts $\iota \circ \gamma$ and ends at the period vector of $\gamma$; and $H_1(V/\Lambda,\mathbb{Z}) = \Lambda$ for a torus, by the endpoint of the lift of a loop through 0 (as in Example 8.61). So $\iota_*$ is the period map. For $g = 1$, $\iota$ is an injective holomorphic map between compact Riemann surfaces, hence surjective and of degree one (Theorem 8.48), hence an isomorphism. This is the analytic form of Proposition 8.34; Chapter 9 makes it explicit with the Weierstrass function $\wp$. □

**Theorem 8.67** (the Jacobian variety; stated). *Let $C$ be a smooth projective curve of genus $g \geq 1$ over $k$.*

*(i) There is an abelian variety $J = \mathrm{Jac}(C)$ over $k$ (a smooth projective connected group variety) of dimension $g$ with $J(\bar{k}) = \mathrm{Pic}^0(C)$ compatibly with $G_k$; and $J(L) \supseteq \mathrm{Div}^0_L(C)/\operatorname{div}(L(C)^\times)$, with equality when $C(L) \neq \emptyset$ (Proposition 8.24(iii)).*

*(ii) Over $\mathbb{C}$, $J(\mathbb{C}) \simeq J_{\mathrm{an}}(C(\mathbb{C}))$ as complex Lie groups, via* AJ*; the torus is algebraic because the cup product, with the positivity of Theorem 8.55, is a Riemann form on $\Lambda_X$ (Riemann, Lefschetz).*

*(iii) (Universal property.) For $P_0 \in C(k)$, $\iota\colon C \to J$, $P \mapsto [(P)-(P_0)]$, is a closed immersion defined over $k$, and every morphism $C \to A$ to an abelian variety over $k$ sending $P_0$ to 0 factors uniquely as $h \circ \iota$ with $h\colon J \to A$ a homomorphism over $k$.*

*(iv) (Mordell-Weil.) $J(K)$ is finitely generated for a number field $K$; in particular $J(K)_{\mathrm{tors}}$ is finite.*

We do not prove this; see [130, Sections 1 to 6], [129], [138, Section 24] and [15, Chapter 9] (section numbers not verified).

**Proposition 8.68** (functoriality of the Jacobian). *Let $\varphi\colon X \to Y$ be a nonconstant morphism of smooth projective curves, $n = \deg \varphi$.*

*(i)* $\varphi_*\colon J(X)\to J(Y)$ *(pushforward of divisor classes, "Albanese") and* $\varphi^*\colon J(Y)\to J(X)$ *(pullback, "Picard") are homomorphisms,* $G_k$*-equivariant if* $\varphi$ *is defined over* $k$*, with* $\varphi_*\varphi^* = [n]$ *on* $J(Y)$*.*

*(ii) Over* $\mathbb{C}$*, under* AJ*,* $\varphi_*$ *is induced by the transpose* $(\varphi^*)^\vee\colon H^0(X,\Omega^1)^\vee\to H^0(Y,\Omega^1)^\vee$ *of the pullback of forms, and on lattices by* $\varphi_*\colon H_1(X,\mathbb{Z})\to H_1(Y,\mathbb{Z})$*.*

*(iii) Over* $\mathbb{C}$*,* $\varphi^*$ *on* $J$ *is induced by the transpose of the trace* $\operatorname{Tr}\colon H^0(X,\Omega^1)\to H^0(Y,\Omega^1)$ *and on lattices by the transfer* $\varphi^!$*.*

*(iv) If* $Y = E$ *has genus one, with origin* $O=\varphi(P_0)$ *and* $J(E)=E$ *(Proposition 8.34), then* $\varphi = \varphi_*\circ\iota_X$ *with* $\iota_X(P) = [(P)-(P_0)]$*, and* $\varphi_*\circ\varphi^*\colon E\to J(X)\to E$ *is multiplication by* $n$*.*

*Proof.* (i) is Corollary 8.39(iii) and Proposition 8.18(iii). (ii) For $D=\sum((P_i)-(Q_i))$ and paths $c_i$ from $Q_i$ to $P_i$, $\mathrm{AJ}_Y(\varphi_*D)(\omega) = \sum\int_{\varphi\circ c_i}\omega = \sum\int_{c_i}\varphi^*\omega = \mathrm{AJ}_X(D)(\varphi^*\omega)$, by change of variables (Proposition 8.62(i)); on closed loops this is the statement about lattices. (iii) (Sketch.) For a holomorphic form $\omega$ on $X$, $\operatorname{Tr}\omega = \sum s_i^*\omega$ is holomorphic on $Y'$, and it extends holomorphically over each branch point: in the normal form $w=z^e$ at a point over it, the $e$ local sections are $z=\zeta_e^k w^{1/e}$, and for $\omega=\sum_m c_m z^m\,dz$ one finds $\sum_k s_k^*\omega = \sum_{e\mid m+1} c_m w^{(m+1)/e-1}\,dw$, with nonnegative exponents. For $D=(P)-(Q)$ and a path $c$ from $Q$ to $P$ through $Y'$ except at its ends, the $n$ lifts of $c$ run from the points of $\varphi^*(Q)$ to those of $\varphi^*(P)$, with multiplicities $e_\varphi$, so $\mathrm{AJ}_X(\varphi^*D)(\omega)=\int_c\operatorname{Tr}\omega = \mathrm{AJ}_Y(D)(\operatorname{Tr}\omega)$. On loops this is $\varphi^!$. See [130, Section 6] (not verified). (iv) $\varphi_*\iota_X(P) = [(\varphi(P))-(O)]$, which is $\varphi(P)$ under $\kappa^{-1}$ of Proposition 8.34; and $\varphi_*\varphi^*=[n]$ by (i). □

This is the universal property of Theorem 8.67(iii) made explicit: $\varphi$ factors through $J(X)$, and for $X=X_0(N)$, $P_0=\infty$, the factor $\varphi_*\colon J_0(N)\to E$ is "the map $J_0(N)\to E$ induced by $\varphi$" of the paper. $E$ is its own dual, so the statement that $E^\vee\to J_0(N)\to E$ is multiplication by $\deg\varphi_E$ (Chapter 14) is (iv).

**Lemma 8.69** (homomorphisms of complex tori)**.** *Let* $V$*,* $W$ *be finite-dimensional complex vector spaces with lattices* $\Lambda\subset V$*,* $M\subset W$*.*

*(a) Let* $A\colon V\to W$ *be* $\mathbb{C}$*-linear and surjective with* $A(\Lambda)\subseteq M$*, and* $\pi\colon V/\Lambda\to W/M$ *the induced homomorphism. Then* $\pi$ *is surjective,* $A(\Lambda)$ *has finite index in* $M$*, the identity component of* $\ker\pi$ *is* $(\ker A+\Lambda)/\Lambda$*, and*

$$\pi_0(\ker\pi) = \ker\pi/(\ker\pi)^0 \;\simeq\; M/A(\Lambda) = \operatorname{coker}\bigl(H_1(\pi)\bigr).$$

*(b) Let* $B\colon W\to V$ *be* $\mathbb{C}$*-linear and injective with* $B(M)\subseteq\Lambda$*, and* $\beta\colon W/M\to V/\Lambda$ *the induced homomorphism. Then* $\ker\beta\simeq(\Lambda\cap B(W))/B(M) = \operatorname{sat}(B(M))/B(M)$*, the saturation being taken in* $\Lambda$*; so* $\beta$ *is injective iff* $B(M)$ *is saturated in* $\Lambda$*.*

*(c) A holomorphic map* $\psi\colon\mathbb{C}/\Lambda\to\mathbb{C}/\Lambda'$ *with* $\psi(0)=0$ *is* $z\mapsto cz$ *for a* $c\in\mathbb{C}$ *with* $c\Lambda\subseteq\Lambda'$*; if* $c\neq 0$*, it is unramified of degree* $[\Lambda':c\Lambda]$*, and* $\psi_*\colon H_1=\Lambda\to H_1=\Lambda'$ *is the inclusion* $c\Lambda\subseteq\Lambda'$ *(after multiplication by* $c$*).*

*Proof.* (a) $\pi$ is surjective because $A$ is. $\Lambda$ spans $V$ over $\mathbb{R}$, so $A(\Lambda)$ spans $W$ over $\mathbb{R}$; it is a subgroup of the lattice $M$ of full rank, so of finite index. Now $\ker\pi = A^{-1}(M)/\Lambda$. The map $A$

induces $A^{-1}(M)/(\ker A+\Lambda)\to M/A(\Lambda)$; it is surjective (every $m\in M$ is $A(v)$ for some $v$, and $v\in A^{-1}(M)$) and injective (if $A(v)=A(\lambda)$ with $\lambda\in\Lambda$ then $v\in\lambda+\ker A$). So $H=(\ker A+\Lambda)/\Lambda$ has finite index in $\ker\pi$. $H$ is connected, as the image of the vector space $\ker A$. $H$ is closed: $\ker A+\Lambda=A^{-1}(A(\Lambda))$, and $A(\Lambda)$ is discrete, hence closed, in $W$. So $\ker\pi$ is a finite disjoint union of closed cosets of $H$, each therefore open in $\ker\pi$, and $H$ is the identity component. Finally $H_1(V/\Lambda,\mathbb{Z})=\Lambda$ and $H_1(\pi)=A|_\Lambda$, as in the proof of Theorem 8.66(iii). (b) $\ker\beta=B^{-1}(\Lambda)/M$, and $B$ maps it isomorphically onto $(\Lambda\cap B(W))/B(M)$. Since $B(M)$ spans the real space $B(W)$, $\Lambda\cap B(W)$ is the intersection of $\Lambda$ with the real span of $B(M)$, which is the saturation (Chapter 3). (c) $\psi$ lifts to a holomorphic $\tilde\psi\colon\mathbb{C}\to\mathbb{C}$ with $\tilde\psi(0)=0$ (the plane is simply connected). For $\lambda\in\Lambda$, $\tilde\psi(z+\lambda)-\tilde\psi(z)$ is a continuous function with values in the discrete $\Lambda'$, hence constant; so $\tilde\psi'$ is $\Lambda$-periodic and entire, hence bounded and constant (Liouville), and $\tilde\psi(z)=cz$ with $c\Lambda\subseteq\Lambda'$. If $c\neq0$, $\psi$ is a local biholomorphism, so unramified, and its fibre over 0 is $c^{-1}\Lambda'/\Lambda\simeq\Lambda'/c\Lambda$; its degree is the order of the fibre (Theorem 8.48(c)). The last claim is the lift of loops. □

**Proposition 8.70** (the degree and the cup product; saturation)**.** *Let $\varphi\colon X\to Y$ be a nonconstant holomorphic map of compact Riemann surfaces of genus $\geq1$, $n=\deg\varphi$.*

*(i)* $\langle\varphi^*x,\varphi^*y\rangle_X=n\,\langle x,y\rangle_Y$ *for $x,y\in H^1(Y,\mathbb{Z})$.*

*(ii)* $\varphi_*\varphi^*=n$ *on $H^1(Y,\mathbb{Z})$, $\varphi_*\varphi^!=n$ on $H_1(Y,\mathbb{Z})$ and $\varphi_*\varphi^*=[n]$ on $J(Y)$. Moreover* $\mathrm{PD}_X\circ\varphi^*=\varphi^!\circ\mathrm{PD}_Y$.

*(iii) The following are equivalent: (a) $\varphi^*H^1(Y,\mathbb{Z})$ is saturated in $H^1(X,\mathbb{Z})$; (b) $\varphi_*\colon H_1(X,\mathbb{Z})\to H_1(Y,\mathbb{Z})$ is surjective; (c) $\ker(\varphi_*\colon J(X)\to J(Y))$ is connected; (d) $\varphi^*\colon J(Y)\to J(X)$ is injective. More precisely*

$$\begin{aligned}[\mathrm{sat}(\varphi^*H^1(Y,\mathbb{Z})):\varphi^*H^1(Y,\mathbb{Z})]&=\#\operatorname{coker}(\varphi_*\colon H_1(X,\mathbb{Z})\to H_1(Y,\mathbb{Z}))\\&=\#\pi_0(\ker(\varphi_*\colon J(X)\to J(Y)))\\&=\#\ker(\varphi^*\colon J(Y)\to J(X)),\end{aligned}$$

*a divisor of $n^{2g_Y}$.*

*(iv) Let $Y=E$ have genus one and $M=\varphi^*H^1(E,\mathbb{Z})$. The cup product on $M$ has ideal of values $n\mathbb{Z}$ and Gram determinant $n^2$. If (iii) holds, $[H^1(X,\mathbb{Z}):M\oplus M^\perp]=n^2$, and for every prime $\ell$, $[H^1(X,\mathbb{Z}_\ell):(M\otimes\mathbb{Z}_\ell)\oplus(M\otimes\mathbb{Z}_\ell)^\perp]=\ell^{2v_\ell(n)}$. In general, with $m_0=\#\pi_0(\ker(J(X)\to E))$, $m_0$ divides $n$ and the cup product on $\mathrm{sat}(M)$ has ideal of values $(n/m_0)\mathbb{Z}$.*

*(v) (Isogenies.) Let $Y=E$, $\psi\colon E\to E'$ an isogeny of degree $m$ (a nonconstant holomorphic map of genus-one surfaces with $\psi(O)=O'$) and $\varphi'=\psi\circ\varphi$. Then $(\varphi')^*H^1(E',\mathbb{Z})$ has index $m$ in $\varphi^*H^1(E,\mathbb{Z})$. Hence if $\varphi^*H^1(E,\mathbb{Z})$ is saturated and $m>1$, then $(\varphi')^*H^1(E',\mathbb{Z})$ is not.*

*Proof.* (i) and the first two identities of (ii) are Proposition 8.62(ii), (iv); the third is Proposition 8.68(i). For the last, let $\xi\in H^1(Y,\mathbb{Z})$ and $\eta\in H^1(X,\mathbb{Z})$. Then $\eta(\mathrm{PD}_X\,\varphi^*\xi)=\langle\varphi^*\xi,\eta\rangle_X$, while $\eta(\varphi^!\,\mathrm{PD}_Y\,\xi)=(\varphi_*\eta)(\mathrm{PD}_Y\,\xi)=\langle\xi,\varphi_*\eta\rangle_Y=-\langle\varphi_*\eta,\xi\rangle_Y=-\langle\eta,\varphi^*\xi\rangle_X=\langle\varphi^*\xi,\eta\rangle_X$, by the projection formula and alternation.

(iii) Put $L=H_1(X,\mathbb{Z})$, $L'=H_1(Y,\mathbb{Z})$, $u=\varphi_*\colon L\to L'$. Its cokernel $Q$ is finite and killed by $n$ (Proposition 8.62(iv)). By the Smith normal form (Chapter 3) there are bases of $L$ and

$L'$ in which $u$ is $(D \mid 0)$ with $D = \operatorname{diag}(d_1, \dots, d_r)$, $r = 2g_Y$, all $d_i \neq 0$, and $Q \simeq \bigoplus \mathbb{Z}/d_i$. In the dual bases $\varphi^* = u^{\mathsf{T}} = \binom{D}{0}$, whose image $\bigoplus d_i\mathbb{Z} \oplus 0$ has saturation $\mathbb{Z}^r \oplus 0$, of index $\prod d_i = \#Q$. This gives the first equality and (a)⇔(b). By Proposition 8.68(ii), $\varphi_*$ on $J$ is the map of Lemma 8.69(a) with $A = (\varphi^*)^\vee$, surjective because $\varphi^*$ is injective on holomorphic forms ($\varphi$ is a local biholomorphism off a finite set), and with $H_1(\varphi_*) = u$; so $\pi_0(\ker) \simeq \operatorname{coker} u$, which is the second equality and (b)⇔(c). By Proposition 8.68(iii) (sketched), $\varphi^*$ on $J$ is the map of Lemma 8.69(b) with lattice map $\varphi^!$ (injective, as $\varphi_*\varphi^! = n$); so $\#\ker(\varphi^*)$ is the index of $\varphi^! H_1(Y,\mathbb{Z})$ in its saturation in $H_1(X,\mathbb{Z})$. Poincare duality is an isomorphism carrying $\varphi^* H^1(Y,\mathbb{Z})$ to $\varphi^! H_1(Y,\mathbb{Z})$, by (ii), so this index is the first term. Hence the last equality and (a)⇔(d). As $nL' \subseteq u(L)$, $\#Q$ divides $n^{2g_Y}$.

(iv) The cup product on $H^1(E,\mathbb{Z})$ is perfect and alternating of rank two, so its ideal of values is $\mathbb{Z}$ (Chapter 3: a rank-two alternating form with Gram matrix $\left(\begin{smallmatrix} 0 & a \\ -a & 0 \end{smallmatrix}\right)$ has ideal of values $a\mathbb{Z}$ and determinant $a^2$). By (i) and the injectivity of $\varphi^*$, the cup product on $M$ is $n$ times a unimodular alternating form: ideal of values $n\mathbb{Z}$, Gram determinant $n^2$. If $M$ is saturated, then $M \otimes \mathbb{Z}_\ell$ is saturated in $H^1(X,\mathbb{Z}_\ell)$, the cup product is perfect there (Theorem 8.60), and the theorem on Gram determinants of saturated sublattices (Chapter 3) gives $[H^1(X,\mathbb{Z}_\ell) : (M \otimes \mathbb{Z}_\ell) \oplus (M \otimes \mathbb{Z}_\ell)^\perp] = \ell^{v_\ell(n^2)}$. The orthogonal commutes with $\otimes\mathbb{Z}_\ell$ (it is a kernel, and $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$), and an index of lattices is the product of its $\ell$-parts, which gives $n^2$. In general $[\operatorname{sat}(M) : M] = m_0$ by (iii), so the Gram determinant of $\operatorname{sat}(M)$ is $n^2/m_0^2$, an integer square, and its ideal of values is $(n/m_0)\mathbb{Z}$; in particular $m_0 \mid n$.

(v) By Lemma 8.69(c), in suitable coordinates $E = \mathbb{C}/\Lambda$, $E' = \mathbb{C}/\Lambda'$, $\psi(z) = z$ with $\Lambda \subseteq \Lambda'$ of index $m$, and $\psi_*$ on $H_1$ is this inclusion. Its transpose $\psi^*\colon H^1(E',\mathbb{Z}) \to H^1(E,\mathbb{Z})$ has image of index $m$ (the determinant of a matrix and of its transpose agree). As $\varphi^*$ is injective on $H^1(E,\mathbb{Z})$ with image $\varphi^* H^1(E,\mathbb{Z})$, the image $\varphi^*\psi^* H^1(E',\mathbb{Z})$ has index $m$ in it. If $\varphi^* H^1(E,\mathbb{Z})$ is saturated and $m > 1$, then $(\varphi')^* H^1(E',\mathbb{Z})$ is a proper sublattice of finite index of a saturated lattice of the same rank, so it is not saturated. □

All the numbers of the proposition can be seen on the least saturated example, $\varphi = [m]\colon E \to E$, $m \geq 2$, of degree $n = m^2$. Here $\varphi^* = m$ on $H^1(E,\mathbb{Z})$, so $M = \varphi^* H^1(E,\mathbb{Z}) = mH^1(E,\mathbb{Z})$, whose saturation is $H^1(E,\mathbb{Z})$, of index $m^2$. On the other side, $\varphi_* = m$ on $H_1(E,\mathbb{Z}) \simeq \mathbb{Z}^2$ has cokernel $(\mathbb{Z}/m)^2$, of order $m^2$; the kernel of $[m]$ on $J(E) = E$ is $E[m]$, finite of order $m^2$ (Proposition 8.71(i)), so it has $m^2$ components; and $\varphi^* = [m]$ on $J$ has the same kernel. All four quantities of (iii) are $m^2$, a divisor of $n^{2g_Y} = m^4$. In (iv), the Gram matrix on $M = mH^1$ is $m^2$ times the unimodular one, so the ideal of values is $m^2\mathbb{Z} = n\mathbb{Z}$ and the determinant $m^4 = n^2$; here $m_0 = m^2$ and the ideal of values on $\operatorname{sat}(M) = H^1(E,\mathbb{Z})$ is $(n/m_0)\mathbb{Z} = \mathbb{Z}$. At the other extreme, a parametrisation of degree one ($X_0(11) \to$ `11a1`) is an isomorphism and every index is 1.

**Proposition 8.71** (torsion of Jacobians)**.**

*(i) For a complex torus $V/\Lambda$ of dimension $g$ and $n \geq 1$, $(V/\Lambda)[n] = \frac{1}{n}\Lambda/\Lambda \simeq \Lambda/n\Lambda \simeq (\mathbb{Z}/n)^{2g}$, the middle map being $x \mapsto nx$. Hence $J(X)[n] \simeq H_1(X,\mathbb{Z}/n)$ for a compact Riemann surface $X$, and $[n]\colon J(X) \to J(X)$ has $n^{2g}$ points in each fibre.*

*(ii) For a curve $C$ over $k$ with $\operatorname{char} k \nmid n$, $J[n](\bar{k}) \simeq (\mathbb{Z}/n)^{2g}$ as groups, with a continuous action of $G_k$ (stated).*

*(iii) For a commutative ring $\mathbb{T}$ of endomorphisms of $J$ and an ideal $\mathfrak{a} \subseteq \mathbb{T}$, put $J[\mathfrak{a}] = \{P : tP = 0 \text{ for all } t \in \mathfrak{a}\}$. Then $J[\ell]$ is a $\mathbb{T}/\ell\mathbb{T}$-module, and for a maximal ideal $\mathfrak{m} \ni \ell$, $J[\mathfrak{m}]$ is a vector space over the field $\mathbb{T}/\mathfrak{m}$.*

*Proof.* (i) $x \in V$ maps to an $n$-torsion point iff $nx \in \Lambda$. Multiplication by $n$ maps $\frac{1}{n}\Lambda$ isomorphically onto $\Lambda$ and $\Lambda$ onto $n\Lambda$, so $\frac{1}{n}\Lambda/\Lambda \simeq \Lambda/n\Lambda$; and $\Lambda \simeq \mathbb{Z}^{2g}$. With Theorem 8.66 and $\Lambda_X \simeq H_1(X,\mathbb{Z})$ this gives the second statement; $[n]$ is a surjective homomorphism, so each fibre is a coset of $J[n]$. (ii) is [129, Section 8] (not verified); we do not prove it. (iii) $\mathbb{T}$ acts on $J[\mathfrak{a}]$ because $\mathbb{T}$ is commutative; $\ell\mathbb{T}$ kills $J[\ell]$ and $\mathfrak{m}$ kills $J[\mathfrak{m}]$. □

**Definition 8.72** (Tate modules)**.** For a prime $\ell$, $T_\ell J = \varprojlim_m J[\ell^m]$, with transition maps multiplication by $\ell$, and $V_\ell J = T_\ell J \otimes \mathbb{Q}_\ell$ (Chapter 1 for inverse limits and $\mathbb{Z}_\ell$).

**Proposition 8.73** (the Tate module and $H^1$)**.** *Let $X$ be a compact Riemann surface of genus $g$ and $J = J(X)$.*

*(i) The isomorphisms $J[\ell^m] = \ell^{-m}\Lambda_X/\Lambda_X \to \Lambda_X/\ell^m\Lambda_X$, $x \mapsto \ell^m x$, are compatible with the transition maps (multiplication by $\ell$ on the left, reduction on the right) and give a canonical isomorphism $T_\ell J \simeq \Lambda_X \otimes \mathbb{Z}_\ell = H_1(X,\mathbb{Z}_\ell)$; $T_\ell J$ is free of rank $2g$ and $T_\ell J/\ell^m T_\ell J = J[\ell^m]$.*

*(ii) (Duality.) $H^1(X,\mathbb{Z}_\ell) = \mathrm{Hom}(H_1(X,\mathbb{Z}),\mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell J,\mathbb{Z}_\ell)$ canonically: $H^1$ is the $\mathbb{Z}_\ell$-dual of the Tate module.*

*(iii) Composing with Poincare duality, $H^1(X,\mathbb{Z}_\ell) \simeq H_1(X,\mathbb{Z}_\ell) \simeq T_\ell J$ is an isomorphism of $\mathbb{Z}_\ell$-modules. It depends on the orientation, it is not Galois-equivariant when $X$ comes from a curve over a number field (Warning 8.75), and it carries the action $\mathcal{C}^*$ of a correspondence on $H^1$ to the action $(\mathcal{C}^t)_*$ of the transpose on $T_\ell J$ (Theorem 8.81(c)).*

*(iv) For a curve $C$ over $k$ and $\ell \neq \mathrm{char}\, k$, $T_\ell J$ is free of rank $2g$ over $\mathbb{Z}_\ell$ with a continuous action of $G_k$ (stated). For $g = 1$, $T_\ell J(E) = T_\ell E$ (Chapter 9).*

*Proof.* (i) If $x \in \ell^{-m-1}\Lambda_X$, then $\ell x \in \ell^{-m}\Lambda_X$ is sent to $\ell^m(\ell x) = \ell^{m+1}x$, the image of the class of $\ell^{m+1}x$ under reduction modulo $\ell^m$: the squares commute. So $T_\ell J \simeq \varprojlim \Lambda_X/\ell^m\Lambda_X = \Lambda_X \otimes \mathbb{Z}_\ell$ (for a free module of finite rank, $\varprojlim_m \mathbb{Z}^{2g}/\ell^m = \mathbb{Z}_\ell^{2g}$), and $\Lambda_X \simeq H_1(X,\mathbb{Z})$ by Theorem 8.66(iii). (ii) $\mathrm{Hom}(H_1(X,\mathbb{Z}),\mathbb{Z}_\ell) = \mathrm{Hom}_{\mathbb{Z}_\ell}(\varprojlim_m H_1(X,\mathbb{Z}) \otimes \mathbb{Z}_\ell, \mathbb{Z}_\ell)$. (iii) is Theorem 8.60 tensored with $\mathbb{Z}_\ell$, and the last clause is proved in Theorem 8.81(c). (iv) is [129, Section 15] and [138, Section 18] (not verified); we do not prove it. □

**Theorem 8.74** (the Weil pairing and etale cohomology; stated)**.** *Let $C$ be a smooth projective curve of genus $g$ over $k$, $J$ its Jacobian and $\ell \neq \mathrm{char}\, k$.*

*(a) (Weil pairing.) There is a perfect, alternating, $G_k$-equivariant $\mathbb{Z}_\ell$-bilinear pairing $e_\ell\colon T_\ell J \times T_\ell J \to \mathbb{Z}_\ell(1)$ (from the canonical principal polarisation). Over $\mathbb{C}$, after identifying $\mathbb{Z}_\ell(1)$ with $\mathbb{Z}_\ell$ by $\zeta_{\ell^m} = e^{2\pi i/\ell^m}$, it is the intersection pairing on $H_1(X,\mathbb{Z}_\ell)$ up to a universal sign. For $g = 1$ it is the Weil pairing of Chapter 9, and $\det \rho_{E,\ell} = \chi_\ell$.*

*(b) (Etale cohomology.) $H^1_{\mathrm{et}}(C_{\bar{k}},\mathbb{Z}_\ell)$ is free of rank $2g$ with a continuous action of $G_k$, and $H^1_{\mathrm{et}}(C_{\bar{k}},\mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell J,\mathbb{Z}_\ell)$ as $G_k$-modules (contragredient action, Chapter 4). For $k \subseteq \mathbb{C}$ there is a comparison isomorphism $H^1_{\mathrm{et}}(C_{\bar{k}},\mathbb{Z}_\ell) \simeq H^1(C(\mathbb{C}),\mathbb{Z}) \otimes \mathbb{Z}_\ell$, compatible with Proposition 8.73(ii), with cup products, and with pullbacks and pushforwards along morphisms.*

*(c) Hence $H^1_{\mathrm{et}}(C_{\bar{k}},\mathbb{Z}_\ell) \simeq T_\ell J(-1) = T_\ell J \otimes \mathbb{Z}_\ell(-1)$ as $G_k$-modules, and the cup product $H^1 \times H^1 \to H^2_{\mathrm{et}} = \mathbb{Z}_\ell(-1)$ is perfect, alternating and $G_k$-equivariant. For $g = 1$: $H^1(E,\mathbb{Z}_\ell) \simeq (T_\ell E)^\vee \simeq T_\ell E(-1)$, with determinant $\chi_\ell^{-1}$.*

*(d) (Weil.) For $k = \mathbb{F}_q$: the arithmetic Frobenius acts on $T_\ell J$ as the Frobenius endomorphism $\pi_q$ of $J$; $P(T) = \det(T - \pi_q \mid T_\ell J)$ has integer coefficients independent of $\ell$, degree $2g$ and roots $\alpha_i$ with $|\alpha_i| = \sqrt{q}$; $\#C(\mathbb{F}_{q^m}) = q^m + 1 - \sum_i \alpha_i^m$; and on $H^1 = (T_\ell J)^\vee$ the geometric Frobenius has the same eigenvalues $\alpha_i$. For $g = 1$, $a_q = q + 1 - \#E(\mathbb{F}_q) = \operatorname{tr}(\mathrm{Frob}_q \mid T_\ell E) = \operatorname{tr}(\mathrm{Frob}_q^{-1} \mid H^1(E))$.*

We do not prove this; see [129, Sections 16 and 19] and [130] for (a), (d); [138, Section 20] for the Weil pairing; [132] for (b) and the comparison theorem (all section numbers not verified); and [185, Chapter III, Section 8, and Chapter V, Section 2] for $g = 1$. Part (c) follows from (a) and (b): the Weil pairing gives $T_\ell J \simeq \operatorname{Hom}(T_\ell J, \mathbb{Z}_\ell(1)) = (T_\ell J)^\vee(1)$, so $(T_\ell J)^\vee \simeq T_\ell J(-1)$. In (d), under the contragredient action $\mathrm{Frob}_q$ acts on $(T_\ell J)^\vee$ with eigenvalues $\alpha_i^{-1}$, so the geometric Frobenius $\mathrm{Frob}_q^{-1}$ acts with eigenvalues $\alpha_i$ (Appendix B, B-C3).

*Warning* 8.75 ($H^1 \simeq T_\ell J$ is not Galois-equivariant). The isomorphism $H^1(X, \mathbb{Z}_\ell) \simeq T_\ell J$ of Proposition 8.73(iii) uses Poincare duality, which over a number field is the Weil pairing and carries a Tate twist: the Galois-equivariant statement is $H^1_{\text{ét}} \simeq T_\ell J(-1)$ (Theorem 8.74(c)). For an elliptic curve, $T_\ell E$ has determinant $\chi_\ell$ and $H^1(E, \mathbb{Z}_\ell)$ has determinant $\chi_\ell^{-1}$; they are not isomorphic as Galois modules. They have the same adjoint: $\mathrm{ad}^0(T^\vee) \simeq \mathrm{ad}^0 T = \mathrm{ad}^0(T(-1))$ (Chapter 4, Appendix B, B-C12(b)). Moreover $H^1$ carries the action $\mathcal{C}^*$ of correspondences and $T_\ell J$ the action $\mathcal{C}_*$, and Poincare duality turns $\mathcal{C}^*$ into $(\mathcal{C}^t)_*$, not $\mathcal{C}_*$.

*In the paper* 8.76. Proposition prop:pullback of the paper is Proposition 8.70 applied to $\varphi = \varphi_E \colon X_0(N) \to E$, with two inputs from Chapter 14. Step by step:

(a) "By optimality the kernel $B$ of the map $J_0(N) \to E$ induced by $\varphi$ is connected; the exact sequence $H_1(J_0(N), \mathbb{Z}) \to H_1(E, \mathbb{Z}) \to \pi_0(B) = 0$ shows that $\varphi_* \colon H_1(X_0(N), \mathbb{Z}) = H_1(J_0(N), \mathbb{Z}) \to H_1(E, \mathbb{Z})$ is surjective, so its dual $\varphi^* \colon H^1(E, \mathbb{Z}) \to H^1(X_0(N), \mathbb{Z})$ is injective with torsion-free cokernel": Proposition 8.70(iii), (c)⇒(b)⇒(a), with Lemma 8.69(a) for the exact sequence and Proposition 8.68(iv) for the factorisation; "the same holds after tensoring with $\mathbb{Z}_\ell$" because a saturated sublattice is a direct summand.

(b) "$\varphi^* H^1(E, \mathbb{C})$, which is spanned by $\varphi^*\omega_E$ and its conjugate, is $\mathbb{C} f\,dz \oplus \mathbb{C}\overline{f\,dz}$": Proposition 8.59(iii) for $E$, and $\varphi_E^*\omega_E = c_E \cdot 2\pi i f(z)\,dz$ (Chapter 14). "It is defined over $\mathbb{Q}$: $H^1(X_0(N), \mathbb{Q})[I_f]$ is a two-dimensional $\mathbb{Q}$-form of it, and by flatness $H^1(X_0(N), \mathbb{Z}_\ell)[I_f] = H^1(X_0(N), \mathbb{Z})[I_f] \otimes \mathbb{Z}_\ell$ is a saturated lattice of rank two in $H^1(X_0(N), \mathbb{Q}_\ell)[I_f]$. The image of $\varphi^*$ is a saturated lattice of rank two in the same $\mathbb{Q}_\ell$-plane; so the two coincide": the Hecke operators act on $H^1(X_0(N), \mathbb{Z})$, so kernels commute with the field extension $\mathbb{Q} \to \mathbb{C}$ and with the flat extension $\mathbb{Z} \to \mathbb{Z}_\ell$; and two saturated sublattices with the same rational span are equal.

(c) "$\varphi^*$ multiplies the fundamental class of $H^2$ by $\deg \varphi_E$; so $\langle \varphi^* a, \varphi^* b \rangle_{X_0(N)} = \deg \varphi_E \, \langle a, b \rangle_E$": Proposition 8.62(iv).

(d) "$w_N$ preserves the plane, on which it acts by $\varepsilon_N(f)$": the plane is also the eigenspace of the anemic Hecke algebra for the eigenvalues of $f$ (strong multiplicity one for the newform $f$, Chapter 13), which $w_N$ preserves; then $w_N^*(f\,dz) = (f|_2 W_N)\,dz = \varepsilon_N(f) f\,dz$ (Chapter 12), and likewise on the conjugate since $\varepsilon_N(f) = \pm 1$ is real. Hence $\langle \varphi^* a, w_N^* \varphi^* b \rangle = \varepsilon_N(f) \deg \varphi_E \langle a, b \rangle_E$.

(e) "the Gram determinant of a rank-two alternating form is the square of its ideal of values": Chapter 3; this is Proposition 8.70(iv).

Proposition 8.70(iv), (v) also explain "minimal degree" in the definition of the optimal curve: for a non-optimal parametrisation the image of $\varphi^*$ is not saturated, and the ideal of values on its saturation is $(n/m_0)\mathbb{Z}$, with $m_0$ the order of the component group of Proposition 8.70(iv). Writing $\varphi = \psi \circ \varphi_{\mathrm{opt}}$ with $\psi$ an isogeny, the kernel of $\varphi_{\mathrm{opt},*}$ is connected, so $m_0 = \#\ker\psi = \deg\psi$ and $n/m_0 = \deg\varphi_{\mathrm{opt}}$ is the degree of the optimal parametrisation through which $\varphi$ factors (Chapter 14). Over $\mathbb{Z}_\ell$ only the $\ell$-part of $m_0$ matters, which is why, when $E[\ell]$ is irreducible, the choice of curve in the class does not change $v_\ell$ (Section 1 of the paper).

*In the paper* 8.77. The paragraph after Proposition prop:pullback: "The lattice $M_{f,\lambda}$ of [49] is therefore $H^1(E,\mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$, by Galois-equivariance of $\varphi^*$ in étale cohomology, and its adjoint is $\mathrm{ad}^0\, T_\ell E$". This is Proposition 8.73(ii) and Theorem 8.74(b), (c): the comparison isomorphism is compatible with $\varphi_E^*$, and $\varphi_E$ is defined over $\mathbb{Q}$. Remark rem:free: "$J_0(N)[\mathfrak{m}]$ is two-dimensional" is a statement about the $\mathbb{T}/\mathfrak{m}$-vector space of Proposition 8.71(iii); "the Tate module is free of rank two" concerns $T_\ell J_0(N)_{\mathfrak{m}}$ with the action $t_*$; and "The identification of $T_\ell J_0(N)$ with $H^1(X_0(N),\mathbb{Z}_\ell)$ as Hecke modules uses Poincaré duality composed with $w_N$, which is not Galois equivariant" is Proposition 8.73(iii) and Warning 8.75: PD turns $t^*$ into $(t^t)_* = (w_N t w_N)_*$ (Theorem 8.81(c) and Chapter 14), so $x \mapsto w_{N*}\,\mathrm{PD}(x)$ is Hecke-equivariant.

## 8.12 Correspondences, their actions, and adjoints

Hecke operators are correspondences: for $p \nmid N$, $T_p$ on $X_0(N)$ is the pair of maps $X_0(Np) \rightrightarrows X_0(N)$ given by $z \mapsto z$ and $z \mapsto pz$ (Chapter 14). This section defines correspondences between curves, their actions on divisors, the Jacobian, $H_1$, $H^1$ and holomorphic differentials, and proves the general theorem behind the sentence of Section 4.1 of the paper "the Hecke operators, $U_p$ included, are self-adjoint for it, because $w_{M^\Sigma} T_n w_{M^\Sigma}^{-1}$ is the adjoint of $T_n$ for the cup product": the adjoint of a correspondence for the cup product is its transpose (Theorem 8.81). Corollary 8.82 then turns "the transpose of every $t$ is $wtw$" into "every $t$ is self-adjoint for the perfect alternating pairing $\langle x, w^*y\rangle$", and gives the orthogonal splitting into local components used in Lemma lem:hecke and in Step 2 of the proof of Theorem thm:quant. The specific identity $T_n^t = w_N T_n w_N$ on $X_0(N)$ is proved in Chapter 14.

**Definition 8.78** (correspondences)**.** Let $X$, $Y$ be smooth projective curves over $k$ (or compact Riemann surfaces). A *correspondence* $\mathcal{C}$ from $X$ to $Y$ is a triple $(C,\alpha,\beta)$ with $C$ a finite disjoint union of smooth projective curves and $\alpha\colon C \to X$, $\beta\colon C \to Y$ morphisms nonconstant on every component of $C$. Its *transpose* is $\mathcal{C}^t = (C,\beta,\alpha)$, from $Y$ to $X$. The *sum* of two correspondences is the disjoint union. For automorphisms $u$ of $Y$ and $v$ of $X$ put $u\mathcal{C}v = (C, v^{-1}\circ\alpha, u\circ\beta)$.

Examples. (a) The *graph* $\Gamma_\varphi = (X,\mathrm{id},\varphi)$ of a nonconstant morphism $\varphi\colon X \to Y$, and its transpose $\Gamma_\varphi^t = (X,\varphi,\mathrm{id})$. (b) An automorphism $w$ of $X$ gives $\Gamma_w$, whose transpose acts as $\Gamma_{w^{-1}}$ (Proposition 8.80(c) and Proposition 8.62(v)). (c) (Forward pointer, Chapter 14.) For a prime $p \nmid N$, the Hecke correspondence $T_p = (X_0(Np),\alpha_1,\alpha_p)$ on $X_0(N)$, with $\alpha_1(z) = z$ and $\alpha_p(z) = pz$ on $\mathcal{H}$. For $p \mid N$ the same triple is $U_p^t = w_N U_p w_N$, not $U_p$, and $U_p$ is $(X_0(Np),\alpha_p,\alpha_1)$ (Chapter 14). The image of $(\alpha,\beta)\colon C \to X\times Y$ is a divisor on the surface $X\times Y$, and correspondences up to the obvious equivalence form a ring under composition (fibre products); we need neither, because the Hecke algebra is defined as the ring of operators that the correspondences generate (Chapter 13).

**Definition 8.79** (actions of a correspondence)**.** Let $\mathcal{C} = (C,\alpha,\beta)$ be a correspondence from $X$ to $Y$ (over $\mathbb{C}$ for (ii) to (iv)).

(i) On divisors and the Jacobian: $\mathcal{C}_* = \beta_*\alpha^*\colon \operatorname{Div}(X) \to \operatorname{Div}(Y)$; it multiplies degrees by $\deg\alpha$ and maps principal divisors to principal divisors (Corollary 8.39(iii)), so it induces $\mathcal{C}_*\colon J(X) \to J(Y)$.

(ii) On homology: $\mathcal{C}_* = \beta_*\alpha^!\colon H_1(X,\mathbb{Z}) \to H_1(Y,\mathbb{Z})$, with the transfer $\alpha^!$ of Proposition 8.62(ii).

(iii) On cohomology: $\mathcal{C}^* = \alpha_*\beta^*\colon H^1(Y,\mathbb{Z}) \to H^1(X,\mathbb{Z})$, with $\alpha_*$ the pushforward of Proposition 8.62(iii).

(iv) On holomorphic differentials: $\mathcal{C}^*\omega = \operatorname{Tr}_\alpha(\beta^*\omega)$, the trace along $\alpha$ of Proposition 8.68(iii), which is holomorphic.

The same definitions apply with coefficients in $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$ or $\mathbb{C}$, and componentwise when $C$ is disconnected.

**Proposition 8.80** (compatibilities)**.** *Let $\mathcal{C} = (C,\alpha,\beta)$ be a correspondence from $X$ to $Y$.*

*(a) $(\mathcal{C}^*\xi)(\gamma) = \xi(\mathcal{C}_*\gamma)$ for $\xi \in H^1(Y,\mathbb{Z})$ and $\gamma \in H_1(X,\mathbb{Z})$: $\mathcal{C}^*$ on $H^1$ is the transpose of $\mathcal{C}_*$ on $H_1$.*

*(b) Under Abel-Jacobi, $\mathcal{C}_*$ on $J(X)$ is induced by the transpose of $\mathcal{C}^*$ on holomorphic differentials and, on lattices, by $\mathcal{C}_*$ on $H_1$. Hence on $T_\ell J(X) \simeq H_1(X,\mathbb{Z}_\ell)$ a correspondence acts by $\mathcal{C}_*$, and on $H^1(X,\mathbb{Z}_\ell) = \operatorname{Hom}(T_\ell J(X),\mathbb{Z}_\ell)$ by the transpose.*

*(c) For a graph: $\Gamma_\varphi^* = \varphi^*$ on $H^1$ and on differentials; $(\Gamma_\varphi)_* = \varphi_*$ on $H_1$ and on $J$ (Albanese); $(\Gamma_\varphi^t)_* = \varphi^!$ on $H_1$ and $\varphi^*$ on $J$ (Picard); $(\Gamma_\varphi^t)^* = \varphi_*$ on $H^1$.*

*(d) The actions are additive in $\mathcal{C}$, and $u\mathcal{C}v$ acts by $u_*\mathcal{C}_*v_*$ on $H_1$ and by $v^*\mathcal{C}^*u^*$ on $H^1$; moreover $(u\mathcal{C}v)^t = v^{-1}\mathcal{C}^t u^{-1}$.*

*Proof.* (a) $(\alpha_*\beta^*\xi)(\gamma) = (\beta^*\xi)(\alpha^!\gamma) = \xi(\beta_*\alpha^!\gamma)$, by the definitions of $\alpha_*$ on $H^1$ as the transpose of $\alpha^!$ and of $\beta^*$ as the transpose of $\beta_*$. (b) (Sketch.) Compose Proposition 8.68(iii) for $\alpha$ (pullback along $\alpha$: transpose of the trace, transfer on lattices) with (ii) for $\beta$ (pushforward: transpose of $\beta^*$, $\beta_*$ on lattices). The statement on $T_\ell J$ follows from Proposition 8.73(i), which is natural for homomorphisms of tori, and the one on $H^1$ from (a). (c) Substitute $\alpha = \mathrm{id}$, $\beta = \varphi$, respectively $\alpha = \varphi$, $\beta = \mathrm{id}$; on $J$, $(\Gamma_\varphi^t)_*D = \mathrm{id}_*\varphi^*D = \varphi^*D$. (d) Additivity is clear. By Proposition 8.62(v), $(u\beta)_*(v^{-1}\alpha)^! = u_*\beta_*\alpha^!(v^{-1})^! = u_*\beta_*\alpha^!v_*$, and $(v^{-1}\alpha)_*(u\beta)^* = (v^{-1})_*\alpha_*\beta^*u^* = v^*\alpha_*\beta^*u^*$ on $H^1$. The transpose of $(C, v^{-1}\alpha, u\beta)$ is $(C, u\beta, v^{-1}\alpha) = v^{-1}(C,\beta,\alpha)u^{-1}$ in the notation of Definition 8.78. □

**Theorem 8.81** (the adjoint of a correspondence is its transpose)**.** *Let $\mathcal{C} = (C,\alpha,\beta)$ be a correspondence from $X$ to $Y$ (compact Riemann surfaces). For all $\xi \in H^1(Y,\mathbb{Z})$ and $\eta \in H^1(X,\mathbb{Z})$,*

$$\langle \mathcal{C}^*\xi, \eta\rangle_X = \langle \xi, (\mathcal{C}^t)^*\eta\rangle_Y,$$

*that is, for the cup products, $(\mathcal{C}^*)^\dagger = (\mathcal{C}^t)^*$. Consequently:*

*(a) for an automorphism $w$ of $X$, $\langle w^*x, w^*y\rangle = \langle x, y\rangle$: $w^*$ is an isometry, with adjoint $(w^{-1})^*$;*

*(b) for a nonconstant $\varphi\colon X \to Y$, $\langle \varphi^*\xi, \eta\rangle_X = \langle \xi, \varphi_*\eta\rangle_Y$, and Proposition 8.70(i), (ii) follow again;*

(c) $\mathrm{PD}_X \circ \mathcal{C}^* = (\mathcal{C}^t)_* \circ \mathrm{PD}_Y$: *under Poincare duality the action* $\mathcal{C}^*$ *on* $H^1$ *corresponds to the action of the transpose on* $H_1$*, hence on* $T_\ell J$*;*

(d) *the same holds with coefficients in* $\mathbb{Z}_\ell$*,* $\mathbb{Q}_\ell$ *or* $\mathbb{C}$*, and for* $\mathbb{Z}_\ell$*-linear combinations of correspondences.*

*Proof.* The projection formula (Proposition 8.62(iv)) and alternation give, for a nonconstant $\varphi\colon X_1 \to Y_1$, $\theta \in H^1(X_1)$, $\zeta \in H^1(Y_1)$, both

$$\langle \varphi_*\theta, \zeta\rangle_{Y_1} = \langle \theta, \varphi^*\zeta\rangle_{X_1} \quad \text{and} \quad \langle \zeta, \varphi_*\theta\rangle_{Y_1} = -\langle \varphi_*\theta, \zeta\rangle_{Y_1} = -\langle \theta, \varphi^*\zeta\rangle_{X_1} = \langle \varphi^*\zeta, \theta\rangle_{X_1}.$$

Apply the first to $\varphi = \alpha$ (componentwise on $C$) and the second to $\varphi = \beta$:

$$\langle \alpha_*\beta^*\xi, \eta\rangle_X = \langle \beta^*\xi, \alpha^*\eta\rangle_C = \langle \xi, \beta_*\alpha^*\eta\rangle_Y.$$

The identity holds on integral classes, hence with any coefficients by linearity. (a) Take $\mathcal{C} = \Gamma_w$: $\Gamma_w^* = w^*$ and $(\Gamma_w^t)^* = w_* = (w^{-1})^*$ (Proposition 8.62(v)); so $\langle w^*x, w^*y\rangle = \langle x, (w^{-1})^*w^*y\rangle = \langle x, y\rangle$. (b) Take $\mathcal{C} = \Gamma_\varphi$. (c) For $\eta \in H^1(X, \mathbb{Z})$: $\eta(\mathrm{PD}_X\, \mathcal{C}^*\xi) = \langle \mathcal{C}^*\xi, \eta\rangle_X = \langle \xi, (\mathcal{C}^t)^*\eta\rangle_Y = ((\mathcal{C}^t)^*\eta)(\mathrm{PD}_Y\, \xi) = \eta((\mathcal{C}^t)_*\, \mathrm{PD}_Y\, \xi)$, by Proposition 8.80(a) for $\mathcal{C}^t$. With Proposition 8.73(i) and Proposition 8.80(b) this is the statement on $T_\ell J$. (d) is linearity. □

**Corollary 8.82** (twisted pairings and self-adjoint Hecke algebras)**.** *Let* $X$ *be a compact Riemann surface,* $w$ *an automorphism of* $X$ *with* $w^2 = \mathrm{id}$*, and* $\mathbb{T}$ *a commutative* $\mathbb{Z}_\ell$*-subalgebra of* $\mathrm{End}_{\mathbb{Z}_\ell}(H^1(X, \mathbb{Z}_\ell))$ *(so a* $\mathbb{Z}_\ell$*-submodule, not merely a subring), each of whose elements is of the form* $t = \sum_i c_i\mathcal{C}_i^*$ *with* $c_i \in \mathbb{Z}_\ell$ *and* $\mathcal{C}_i$ *correspondences from* $X$ *to* $X$*. Put* $t^t = \sum_i c_i(\mathcal{C}_i^t)^*$ *(well defined, being the adjoint* $t^\dagger$ *by Theorem 8.81), and assume* $t^t = w^*tw^*$ *for all* $t \in \mathbb{T}$*. Put* $[x, y] = \langle x, w^*y\rangle$*. Then:*

(i) $[\ ,\ ]$ *is a perfect alternating pairing on* $H^1(X, \mathbb{Z}_\ell)$*;*

(ii) *every* $t \in \mathbb{T}$ *is self-adjoint for* $[\ ,\ ]$*:* $[tx, y] = [x, ty]$*;*

(iii) *every idempotent* $e$ *of* $\mathbb{T}$ *is self-adjoint,* $H^1(X, \mathbb{Z}_\ell) = eH^1 \oplus (1 - e)H^1$ *is an orthogonal decomposition and* $[\ ,\ ]$ *is perfect on each summand; in particular,* $\mathbb{T}$ *being a finite* $\mathbb{Z}_\ell$*-algebra,* $H^1(X, \mathbb{Z}_\ell) = \bigoplus_{\mathfrak{m}} H^1(X, \mathbb{Z}_\ell)_{\mathfrak{m}}$ *over the maximal ideals of* $\mathbb{T}$ *is orthogonal for* $[\ ,\ ]$*, with* $[\ ,\ ]$ *perfect on each* $H^1(X, \mathbb{Z}_\ell)_{\mathfrak{m}}$*.*

*Proof.* By Theorem 8.81, $t^\dagger = t^t = w^*t(w^*)^{-1}$ for the cup product, which is perfect and antisymmetric on $H^1(X, \mathbb{Z}_\ell)$ (Theorem 8.60). By Theorem 8.81(a), $w^*$ is an isometry, and $w^*$ is an involution, so $(w^*)^\dagger = (w^*)^{-1} = w^*$. The proposition on adjoints and twisted pairings of Chapter 3 (its parts on adjoints, idempotents and twisted pairings) now gives: $[\ ,\ ]$ is perfect, as the composite of a perfect pairing with the automorphism $w^*$; for $t \in \mathbb{T}$, $[tx, y] = \langle x, t^\dagger w^*y\rangle = \langle x, w^*tw^*w^*y\rangle = [x, ty]$; and $[y, x] = \langle y, w^*x\rangle = -\langle w^*x, y\rangle = -\langle x, w^*y\rangle = -[x, y]$, so $[\ ,\ ]$ is antisymmetric, hence alternating since 2 is a nonzerodivisor in $\mathbb{Z}_\ell$. For (iii), the same proposition of Chapter 3 shows that idempotents of a commutative ring of self-adjoint endomorphisms are self-adjoint, that the decomposition is orthogonal and that the pairing is perfect on each summand. $\mathbb{T}$ is a $\mathbb{Z}_\ell$-submodule of the finite free module $\mathrm{End}_{\mathbb{Z}_\ell}(H^1(X, \mathbb{Z}_\ell))$ by hypothesis, so it is finite and free over $\mathbb{Z}_\ell$ (a submodule of a finite free module over the principal ideal domain $\mathbb{Z}_\ell$), and it is the product of its localisations $\mathbb{T}_{\mathfrak{m}}$, cut out by idempotents $e_{\mathfrak{m}}$ (Chapter 3); $H^1(X, \mathbb{Z}_\ell)_{\mathfrak{m}} = e_{\mathfrak{m}}H^1(X, \mathbb{Z}_\ell)$. □

**Example 8.83** (adjoints of correspondences)**.**

(a) On $X=\mathbb{C}/\Lambda$ the automorphism $[-1]$ acts by $-1$ on $H^1$; it is an isometry and its own adjoint (Example 8.61).

(b) For $\varphi\colon X\to E$ of degree $n$, the correspondence $\Gamma_\varphi^t$ followed by $\Gamma_\varphi$ acts on $H^1(E)$ by $\varphi_*\varphi^*=n$ and on $E=J(E)$ by $[n]$ (Proposition 8.70(ii)). On $H^1(X,\mathbb{Q})$ the operator $p=\frac{1}{n}\varphi^*\varphi_*$ satisfies $p^2=\frac{1}{n^2}\varphi^*(\varphi_*\varphi^*)\varphi_*=p$, has image $\varphi^*H^1(E,\mathbb{Q})$ and is the identity there; and $p^\dagger=\frac{1}{n}\varphi_*^\dagger\varphi^{*\dagger}=\frac{1}{n}\varphi^*\varphi_*=p$ by Theorem 8.81(b) (for an antisymmetric pairing $(A^\dagger)^\dagger=A$). So $p$ is the orthogonal projector onto $\varphi^*H^1(E,\mathbb{Q})$ for the cup product; it is integral on $H^1(X,\mathbb{Z}_\ell)$ for $\ell\nmid n$.

(c) On a genus-one curve $E$, for an endomorphism $\psi$ (a nonconstant morphism fixing $O$) the adjoint of $\psi^*$ on $H^1(E,\mathbb{Z})$ is $\psi_*=(\deg\psi)(\psi^*)^{-1}$ (on $H^1(E,\mathbb{Q})$; by Theorem 8.81(b) and Proposition 8.70(ii), $\psi_*\psi^*=\deg\psi$). For $\psi=[m]$, $\psi^*=\psi_*=m$ is self-adjoint. For $E=\mathbb{C}/\mathbb{Z}[i]$ and $\psi(z)=iz$: with $z=s+it$, $\psi^*s=-t$ and $\psi^*t=s$, so in the basis $ds$, $dt$ the matrix of $\psi^*$ is $\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$ (which is the matrix $J$ of Chapter 1), a rotation of order 4; its adjoint for the Gram matrix $G=\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$ is $G^{-1}(\psi^*)^{\mathsf T}G=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)=(\psi^*)^{-1}$, the inverse rotation. So a correspondence need not be self-adjoint for the cup product.

(d) (Forward pointer, Chapter 14.) On $X_0(N)$, $T_n^t=w_NT_nw_N$ for all $n$, $U_p$ included. For $p\nmid N$, $w_N$ commutes with $T_p$, so $T_p^t$ acts as $T_p$ and $T_p^*$ is self-adjoint for the cup product; for $p\mid N$, $U_p^t=w_NU_pw_N$ is in general not $U_p$, and $U_p^*$ is not self-adjoint for the cup product. This is why the paper composes the cup product with $w$: by Corollary 8.82 every $t$ is self-adjoint for $\langle x,w_N^*y\rangle$.

*In the paper* 8.84. In Section 4.1 of the paper, the pairing $\hat\delta_!$ "is perfect and alternating on $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$, and the Hecke operators, $U_p$ included, are self-adjoint for it, because $w_{M^\Sigma}T_nw_{M^\Sigma}^{-1}$ is the adjoint of $T_n$ for the cup product", which is what the composition with $w$ is for. With the input $T_n^t=w_{M^\Sigma}T_nw_{M^\Sigma}$ from Chapter 14, Theorem 8.81 says that $w^*T_n^*w^*$ is the adjoint of $T_n^*$, and Corollary 8.82(i), (ii) is the sentence, with $X=X_0(M^\Sigma)$, $w=w_{M^\Sigma}$ and $\mathbb{T}$ the image of the Hecke algebra tensored with $\mathbb{Z}_\ell$ in $\mathrm{End}_{\mathbb{Z}_\ell}(H^1(X_0(M^\Sigma),\mathbb{Z}_\ell))$, a commutative $\mathbb{Z}_\ell$-subalgebra. Every element of $\mathbb{T}$ is a $\mathbb{Z}_\ell$-combination of the $T_n^*$ ($\mathbb{T}$ is the $\mathbb{Z}$-span of the $T_n$, by the multiplicativity relations of Chapter 13), so the identity $t^t=w^*tw^*$ follows from the case $t=T_n$ by linearity; the factor $\frac12$ in $\hat\delta_!$ is a unit. The paragraph after Lemma lem:hecke and Step 2 of the proof of Theorem thm:quant: "the idempotent $e_\mathfrak{m}$ is self-adjoint for $\hat\delta_!$, so $H^1(X_0(N),\mathbb{Z}_\ell)$ splits orthogonally into its local components and the pairing is perfect on each" is Corollary 8.82(iii). This supplies the hypothesis of Lemma lem:ddt417 for the module $L=H^1(X_0(N),\mathbb{Z}_\ell)_\mathfrak{m}$: a perfect $\mathbb{Z}_\ell$-bilinear pairing for which $\mathbb{T}_\mathfrak{m}$ is self-adjoint.

## 8.13 Exercises

**Exercise 8.85.** (a) Show that $y^2=x^3+ax+b$ ($\mathrm{char}\,k\neq2,3$) is singular iff $4a^3+27b^2=0$, and that the singular point is then $(x_0,0)$ with $x_0$ the double root. (b) For the cusp $y^2=x^3$ show that $t\mapsto(t^2,t^3)$ is a bijection $\mathbb{A}^1\to C$ which is not an isomorphism, by computing $\dim\mathfrak{m}_P/\mathfrak{m}_P^2$ at $P=(0,0)$. (c) Show that the Fermat curve $X^d+Y^d+Z^d=0$ is singular at every point when $p=\mathrm{char}\,k$ divides $d$, and that for $d=2p$ it is not even irreducible. *Hints.* (a) A singular point has $f_y=2y=0$, and then $x$ is a common root of $x^3+ax+b$ and $3x^2+a$; the

discriminant of $x^3+ax+b$ is $-(4a^3+27b^2)$. (b) $f=y^2-x^3$ has no linear term at $(0,0)$. (c) $X^{2p}+Y^{2p}+Z^{2p}=(X^2+Y^2+Z^2)^p$.

**Exercise 8.86.** For `121a1` ($y^2+xy+y=x^3+x^2-30x-76$) and `20a1` ($y^2=x^3+x^2+4x+4$) show that the three partial derivatives of the homogenised equation have no common zero on the curve over $\overline{\mathbb{Q}}$. For which primes $p$ is the reduction of the model of `20a1` modulo $p$ singular? (Answer: 2 and 5; at 2 find the singular point.) *Hint.* The discriminant of the model of `20a1` is $-2^8\cdot 5^2=-6400$. Modulo 2 the equation is $y^2=x^3+x^2$, with $f_y\equiv 0$ and $f_x\equiv x^2$, singular at $(0,0)$; modulo 5, $x^3+x^2+4x+4\equiv(x+1)^2(x-1)$, singular at $(-1,0)$.

**Exercise 8.87.** On $\mathbb{P}^1$ show that $l(D)=\deg D+1$ for every divisor of degree $\geq -1$, by reducing to $D=n(\infty)$ with Proposition 8.24(i). Deduce Riemann-Roch for $\mathbb{P}^1$ with $g=0$ and $K=-2(\infty)$, including the cases $\deg D\leq -2$. *Hint.* $\mathcal{L}(n(\infty))$ is the space of polynomials in $x$ of degree at most $n$.

**Exercise 8.88.** Let $h$ be squarefree of degree 4, $\operatorname{char}k\neq 2$, and $C$ the smooth projective model of $y^2=h(x)$ (Corollary 8.45(c)). Show that $C$ has two points over $\infty$, that $\operatorname{div}(dx/y)=0$, and that $g(C)=1$. If $h$ has no root in $k$ and its leading coefficient is not a square, $C$ may have no $k$-rational point at infinity: what does Theorem 8.33 need before it applies?

**Exercise 8.89.** Let $C$ have genus one, $O\in C(k)$, and $x$, $y$ as in Theorem 8.33. Show that $\{x^iy^j : j\in\{0,1\},\ 2i+3j\leq n\}$ is a basis of $\mathcal{L}(n(O))$ for every $n\geq 1$. Explain why the construction cannot be started on the conic $X^2+Y^2+Z^2=0$ over $\mathbb{Q}$ or on Selmer's cubic.

**Exercise 8.90.** On a Weierstrass cubic with $O=[0:1:0]$, use Proposition 8.34 to show: (a) if a line meets $E$ in $P$, $Q$, $R$ (with multiplicity), then $\kappa(P)+\kappa(Q)+\kappa(R)=0$; (b) the inverse of $P=(x_0,y_0)$ is $(x_0,-y_0-a_1x_0-a_3)$; (c) $P$ has order 2 iff $2y_0+a_1x_0+a_3=0$.

**Exercise 8.91.** For a Weierstrass cubic with $\operatorname{char}k\neq 2,3$, show that $x\colon C\to\mathbb{P}^1$ has degree 2 and is ramified exactly at $O$ and at the three points with $2y+a_1x+a_3=0$, and verify Riemann-Hurwitz. Show that $y\colon C\to\mathbb{P}^1$ has degree 3, that $e_y(O)=3$, and that the total ramification $\sum(e_y(P)-1)$ is 6.

**Exercise 8.92.** (a) Show that the plane quartic $y^3=x^4-1$ (projectively $X^4-Y^3Z-Z^4=0$) is smooth in characteristic 0, and compute its genus twice: by Theorem 8.31, and by Riemann-Hurwitz for $x\colon C\to\mathbb{P}^1$ (degree 3, totally ramified over the four roots of $x^4=1$ and over $\infty$). (b) For the Fermat quartic, show that $\omega_0$, $x\omega_0$, $y\omega_0$ ($\omega_0=dx/f_y$) are regular, by checking the order at the four points at infinity. *Answer to (a):* $g=3$ both times; Riemann-Hurwitz reads $2g-2=3(-2)+5\cdot(3-1)=4$.

**Exercise 8.93.** Let $\varphi\colon X\to E$ be a nonconstant morphism over $\mathbb{C}$ from a curve of genus 2 to a curve of genus 1. Show that $\varphi$ has either two ramification points with $e=2$, or one with $e=3$, and give the possible ramification patterns over the branch points when $\deg\varphi=2$. Show that a nonconstant morphism from a genus-one curve to $\mathbb{P}^1$ has degree at least 2. *Hint.* Riemann-Hurwitz gives $\sum(e_\varphi(P)-1)=2$. If $\deg\varphi=2$, a point with $e=2$ is its whole fibre, so there are exactly two branch points, each with a single preimage. A morphism of degree one is an isomorphism (Corollary 8.39(v)).

**Exercise 8.94.** The optimal curves `37a1` and `99a1` have modular degrees 2 and 4 (Cremona's tables), and $g(X_0(37))=2$, $g(X_0(99))=9$. Compute the total ramification of their modular

parametrisations. Show that for every optimal $E$ of conductor $N$, $\deg\varphi_E = 1$ forces $g(X_0(N)) = 1$, and that $g(X_0(N)) \geq 2$ forces $\deg\varphi_E \geq 2$. (Answers: 2 and 16; we checked the genera with `mfdim` and the degrees with `ellmoddegree`.)

**Exercise 8.95.** For $X = \mathbb{C}/\Lambda$ and a new basis $(\omega_1', \omega_2') = (a\omega_1 + b\omega_2, c\omega_1 + d\omega_2)$ of $\Lambda$, $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z})$, compute the cup product of the new dual basis $ds'$, $dt'$ and show $\langle ds', dt' \rangle = \det\gamma \cdot \langle ds, dt \rangle$. Deduce that $\mathrm{Im}(\bar{\omega}_1'\omega_2') > 0$ iff $\det\gamma = 1$, and that the convention "$a \cdot b = +1$" of Example 8.61 singles out the bases oriented by the complex structure.

**Exercise 8.96.** Let $T$ be a free $\mathbb{Z}_\ell$-module of rank two with a continuous action of a group $G$ and a perfect alternating $G$-equivariant pairing $e\colon T \times T \to \mathbb{Z}_\ell(1)$. Show that $\det T = \chi_\ell$ (as a character of $G$), that $x \mapsto e(x, -)$ is a $G$-isomorphism $T \to T^\vee(1)$, and hence $T^\vee \simeq T(-1)$ with $\det T^\vee = \chi_\ell^{-1}$. Apply this to $T = T_\ell E$ and explain why the paper's lattice $M_{f,\lambda} = \varphi_E^* H^1(E, \mathbb{Z}_\ell)$ is isomorphic to the dual of $T_\ell E$, and not to $T_\ell E$, as a Galois module, and why $\mathrm{ad}^0 M_{f,\lambda} \simeq \mathrm{ad}^0 T_\ell E$ anyway.

**Exercise 8.97.** Let $E = \mathbb{C}/\mathbb{Z}[\zeta_3]$ and $\psi(z) = \zeta_3 z$. Write the matrix of $\psi^*$ on $H^1(E, \mathbb{Z})$ in the basis dual to $1$, $\zeta_3$, and of its adjoint for the cup product. Show that $\psi^*$ is an isometry with $(\psi^*)^\dagger = (\psi^*)^2$ and that $\psi^* + (\psi^*)^\dagger = -1$. Which elements of $\mathbb{Z}[\psi^*]$ are self-adjoint? *Hint.* With $z = s + t\zeta_3$ and $\zeta_3^2 = -1 - \zeta_3$, $\psi(z) = -t + (s - t)\zeta_3$, so $\psi^* ds = -dt$ and $\psi^* dt = ds - dt$. Use the formula $G^{-1}P^{\mathsf{T}}G$ of Example 8.83(c). (Answer to the last question: only the integers.)

**Exercise 8.98.** Let $L = \mathbb{Z}_\ell^2$ with $\langle x, y \rangle = x_1 y_2 - x_2 y_1$, $\ell$ odd. (a) For $w = \mathrm{diag}(2, \frac{1}{2})$, an isometry of infinite order, show that $[x, y] = \langle x, wy \rangle$ is perfect but not alternating. (b) For $w = \mathrm{diag}(1, -1)$, an involution, show that $w$ is not an isometry and that $[\ ,\ ]$ is symmetric. (c) Conclude that both hypotheses on $w$ in Corollary 8.82 (an isometric involution, which an automorphism of $X$ of order two provides) are needed for alternation.

**Exercise 8.99.** Let $H = \mathbb{Z}^4$ with basis $a_1^\vee, a_2^\vee, b_1^\vee, b_2^\vee$ and the symplectic cup product of Theorem 8.60, and $M$ the span of $u = a_1^\vee$ and $v = nb_1^\vee + b_2^\vee$. Show that $M$ is saturated, that the pairing on $M$ has ideal of values $n\mathbb{Z}$, compute $M^\perp$ and show $[H : M \oplus M^\perp] = n^2$, as Proposition 8.70(iv) predicts for $M = \varphi^* H^1(E, \mathbb{Z})$ when $g(X) = 2$. Do the same for $M' = \mathbb{Z}u \oplus \mathbb{Z} \cdot 2v$ and compare with the non-saturated case. (Answers: $M^\perp = \mathbb{Z}(a_1^\vee - na_2^\vee) \oplus \mathbb{Z}b_2^\vee$; for $M'$ the ideal of values is $2n\mathbb{Z}$, $M'^\perp = M^\perp$, and the index is $2n^2$.)

**Exercise 8.100.** Let $V = \mathbb{C}^2$, $\Lambda = \mathbb{Z}[i]^2$, $W = \mathbb{C}$, $M = \mathbb{Z}[i]$ and $A(z_1, z_2) = 2z_1 + (1+i)z_2$. Show that $A$ induces a surjective homomorphism $\pi$ of complex tori, compute $A(\Lambda)$ and show that $\ker\pi$ has two connected components. Replace $A$ by $z_1 + (1+i)z_2$ and show that the kernel becomes connected. *Hint.* $2 = -i(1+i)^2$, so $A(\Lambda) = (1+i)\mathbb{Z}[i]$, of index 2 in $\mathbb{Z}[i]$; apply Lemma 8.69(a).

**Exercise 8.101.** Let $E\colon y^2 = x^3 - x$ over $\mathbb{F}_3$. Count $E(\mathbb{F}_3)$, the fixed points of $\pi_3$, and deduce $a_3$ and the roots $\alpha$, $\beta$ of $T^2 - a_3 T + 3$. Predict $\#E(\mathbb{F}_9)$ from Theorem 8.74(d) and verify it by counting the fixed points of $\pi_3^2$ (answers: 4, $a_3 = 0$, 16). Show that $\pi_3$ is a bijection on $E(\bar{\mathbb{F}}_3)$ of degree 3, and explain why Theorem 8.44 does not apply to it.

**Exercise 8.102.** (a) On $\mathbb{P}^1$ compute all residues of $\omega = dz/(z^3 - z)$, including at $\infty$, check Theorem 8.52, and deduce the partial fraction decomposition of $1/(z^3 - z)$. (b) Show that on a compact Riemann surface no meromorphic 1-form has a single simple pole and no other pole. (c) On a Weierstrass cubic, show that $x\,dx/(2y + a_1 x + a_3)$ has a double pole at $O$ and no other pole, and deduce that its residue at $O$ is 0. (Answer to (a): the residues are $-1$ at 0, $\frac{1}{2}$ at $\pm 1$ and 0 at $\infty$.)

## 8.14 Notes and references

The algebraic side (Sections 8.1 to 8.7) follows Silverman [185], Chapter I (Sections 1 to 3: varieties, Galois action, maps) and Chapter II (Sections 1 to 5: curves, divisors, differentials, Riemann-Roch), and Chapter III, Section 3 for Weierstrass equations. The numbered results of Silverman cited above (Propositions II.1.1, II.2.1, II.2.6, II.2.11, II.4.2, II.4.3, II.5.8, Theorems II.2.3, II.2.4, II.5.4, Corollary II.2.12) are cited from the second edition as we recall them; the reader should treat the numbers as pointers and check them against the printed edition. Fulton [72] is the gentlest complete account of plane curves, smoothness, intersection multiplicities and Bezout (Chapters 3, 5), the smooth model (Chapter 7) and Riemann-Roch (Chapter 8); Hartshorne [79] (Chapter I, Sections 1 to 7; Chapter II, Section 6; Chapter IV, Sections 1 and 2; Appendix B) gives the scheme-theoretic versions; Corollary I.6.12, Proposition II.6.9, Theorem IV.1.3 and Corollary IV.2.4 are cited as recalled. The finiteness of integral closure is in Atiyah and Macdonald [7], Chapter 5 (chapter not verified).

The analytic side (Sections 8.8 to 8.11) follows Miranda [136] (Riemann surfaces and holomorphic maps in Chapters I and II, differentials and residues in Chapter IV, Riemann-Roch and Riemann existence in Chapters V to VII, Abel's theorem, Jacobi inversion and the bilinear relations in Chapter VIII) and Forster [70] (Sections 1, 2 and 10 for the local theory, 14 to 17 for Riemann-Roch and its consequences, 19 to 21 for de Rham, periods and Abel-Jacobi). Griffiths and Harris [75], Chapter 2, Section 2, and Farkas and Kra [64], Chapter III, give the period relations and the Jacobian with the symplectic picture; Hatcher [80] is the reference for singular (co)homology, the universal coefficient theorem, transfers of coverings (Section 3.G) and Poincare duality (Section 3.3). All chapter and section numbers in this paragraph are recalled pointers, not checked against the books.

The Jacobian as an abelian variety, its universal property, its torsion and Tate modules and the Weil pairing are in Milne's two articles in Cornell-Silverman [130, 129] and in Mumford [138] (Sections 18 to 24); the Picard functor is in [15, Chapter 9]. Etale cohomology and the comparison theorem are in Milne's notes [132]. Correspondences and the Hecke correspondences on modular curves are in Shimura [180], Chapter 7, and Diamond and Shurman [52], Section 6.3; the self-adjointness argument of Corollary 8.82 is the one used by Darmon, Diamond and Taylor [38, Section 4.4]; Diamond, Flach and Guo state the corresponding facts without proof [49, Sections 1.6.1 and 1.8.2] (checked: Proposition 1.3 there makes the Hecke action compatible with their duality morphisms, and Section 1.8.2 says that "the Hecke operators are self-adjoint with respect to the pairing of (25)"). The saturation criterion and the cup product on $\varphi^* H^1(E)$ (Proposition 8.70) are elementary once Theorem 8.60 and Lemma 8.69 are available; compare Agashe, Ribet and Stein [3, Section 3] (section not verified) for the modular degree and optimal quotients. Section numbers for Milne, Mumford, Shimura and Diamond-Shurman are recalled pointers. The genera and modular degrees quoted were computed with PARI/GP [148] and Cremona's tables [35].

Three choices differ from some sources. The cup product is $\int \theta \wedge \eta$ with the complex orientation; the sign of Poincare duality and of the Weil pairing then differ from some books, but the paper uses only sign-invariant properties. The adjoint of an operator is $A^\dagger$, the star being reserved for pullbacks. And the transfer $\varphi^!$ and the pushforward $\varphi_*$ on $H^1$ are defined topologically, with the trace of forms as a computational device (Proposition 8.62), rather than through sheaf cohomology.

# Chapter 9

# Elliptic curves over fields

Every object in the paper is built from an elliptic curve $E/\mathbb{Q}$: its Weierstrass invariants and $v_q(j_E)$, its torsion points and isogenies, its Tate module $T_\ell E$ with the Galois representation $\rho_{E,\ell}$, the numbers $a_p$ and the roots $\alpha, \beta$ of $X^2 - a_p X + p$, its quadratic twists, its period lattice, its 3-division polynomial. This chapter develops these objects over an arbitrary field, then over $\mathbb{C}$, over finite fields and over $\mathbb{Q}$, proving what the paper uses and stating the deep theorems (Mordell-Weil, Mazur, Kenku, Faltings, Deuring, modularity) precisely with references. It uses Chapter 1 (finite abelian groups, $\mathrm{GL}_2$, inverse limits), Chapter 2 (finite fields, quadratic characters), Chapter 4 (twists, contragredients, adjoints), Chapter 5 (Galois theory of polynomials, profinite groups), Chapter 6 (Frobenius elements, imaginary quadratic orders), and above all Chapter 8 (curves, divisors, Riemann-Roch, Weierstrass models, morphisms and their degree, the Abel-Jacobi theorem). Local questions (minimal models, reduction types, the Tate curve) are Chapter 10; the images of the Galois representations are Chapter 11.

## Core results at a glance

Throughout, $K$ is a field, $\bar{K}$ an algebraic closure, $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$, $E$ an elliptic curve over $K$ given by a Weierstrass equation, and $\ell \neq \mathrm{char}\, K$ a prime.

**9.1 Conventions.** Elliptic curves are Weierstrass cubics with $O = [0:1:0]$; over $\mathbb{Q}$, global minimal models as in Cremona's tables; $\chi_d$ is the character of $K(\sqrt{d})$, $\chi_{q^*}$ the quadratic character of conductor $q$ (never $\chi_q$), $\chi_\ell$ the cyclotomic character; $e_m$ is Silverman's Weil pairing. Convention 9.1.

**9.2 Weierstrass invariants** (definition): $b_2, b_4, b_6, b_8$, $c_4 = b_2^2 - 24b_4$, $c_6 = -b_2^3 + 36b_2b_4 - 216b_6$, $\Delta$, $j = c_4^3/\Delta$, $\omega = dx/(2y + a_1x + a_3)$. Definition 9.2.

**9.3 Identities.** $4b_8 = b_2b_6 - b_4^2$, $1728\Delta = c_4^3 - c_6^2$; $(2y + a_1x + a_3)^2 = 4x^3 + b_2x^2 + 2b_4x + b_6$, a cubic of discriminant $16\Delta$; for $\mathrm{char}\, K \neq 2, 3$ the short form $Y^2 = X^3 - 27c_4X - 54c_6$. *(proved here)* Lemma 9.3.

**9.4 Smoothness.** A Weierstrass cubic is smooth iff $\Delta \neq 0$; if $\Delta = 0$ it has one singular point, a node if $c_4 \neq 0$, a cusp if $c_4 = 0$; every elliptic curve is a smooth Weierstrass cubic. *(proved here for* $\mathrm{char}\, K \neq 2$*; reference in characteristic* 2*)* Proposition 9.4.

**9.5 Changes of coordinates.** $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ give $u^4c_4' = c_4$, $u^6c_6' = c_6$, $u^{12}\Delta' = \Delta$, $j' = j$, $\omega' = u\omega$; every isomorphism fixing $O$ is of this form; over $\mathbb{Q}$, $v_q(j) = 3v_q(c_4) - v_q(\Delta)$ is independent of the model and $v_q(\Delta)$ changes by multiples of 12. *(proved here)* Proposition 9.5.

**9.6 The $j$-invariant.** Curves are isomorphic over $\bar{K}$ iff they have the same $j$; every $j_0 \in K$ occurs. *(proved here for* $\operatorname{char} K \neq 2, 3$*; reference in general)* Theorem 9.6.

**9.7 Automorphisms.** For $\operatorname{char} K \neq 2, 3$, $\#\operatorname{Aut}_{\bar{K}}(E) = 2, 4, 6$ for $j \neq 0, 1728$, $j = 1728$, $j = 0$; $[-1](x, y) = (x, -y - a_1x - a_3)$. *(proved here)* Proposition 9.7.

**9.8 The chord-tangent law** (construction): $P+Q+R = O$ for the three intersections of a line; explicit formulas for $P_1 + P_2$ and $x(2P) = (x^4 - b_4x^2 - 2b_6x - b_8)/(4x^3 + b_2x^2 + 2b_4x + b_6)$; for $\operatorname{char} K \neq 2$, $\#E[2] = 4$. *(proved here)* Construction 9.10, Proposition 9.11.

**9.9 Chord-tangent $= \operatorname{Pic}^0$.** $P \mapsto [(P) - (O)]$ is additive for the chord-tangent law. *(proved here, given Chapter 8)* Proposition 9.12.

**9.10 The group law.** $E(\bar{K})$ is an abelian group, $E(L)$ a subgroup for $K \subseteq L \subseteq \bar{K}$, $G_K$ acts by automorphisms, addition and translations are morphisms. *(proved here; the morphism property at the excluded points by reference)* Theorem 9.13.

**9.11 Principal divisors.** $\sum n_P(P)$ is principal iff $\sum n_P = 0$ and $\sum [n_P]P = O$. *(proved here)* Corollary 9.14.

**9.12 The invariant differential.** $\operatorname{div}\omega = 0$, $\tau_Q^*\omega = \omega$, $(\varphi + \psi)^*\omega = \varphi^*\omega + \psi^*\omega$, $[m]^*\omega = m\omega$. *(proved here, in every characteristic; one polynomial identity checked by machine; a second proof in characteristic* 0 *in Remark 9.54)* Proposition 9.15.

**9.13 Isogenies** (definition): morphisms with $\varphi(O) = O$; degree $[\bar{K}(E_1) : \varphi^*\bar{K}(E_2)]$; isogenous over $K$. Definition 9.17.

**9.14 First properties.** Isogenies are 0 or surjective; Hom is a group, deg is multiplicative; $[m] \neq 0$ for $m \neq 0$, so Hom is torsion-free. *(proved here; characteristic* 2 *by reference)* Proposition 9.18.

**9.15 Kernels.** Isogenies are homomorphisms; $\#\ker\varphi = \deg_s\varphi$; a separable isogeny is unramified, with $\#\ker\varphi = \deg\varphi$ and Galois function field extension with group $\ker\varphi$. *(proof sketched here)* Theorem 9.19.

**9.16 Quotients.** For a finite subgroup $H$ there is a separable $E \to E/H$ with kernel $H$, over $K$ if $H$ is $G_K$-stable (Vélu); an isogeny factors through any separable isogeny whose kernel is contained in its kernel. *(stated; reference)* Theorem 9.20.

**9.17 The dual isogeny.** $\hat{\varphi} \circ \varphi = [\deg\varphi]$, $\varphi \circ \hat{\varphi} = [\deg\varphi]$, $\widehat{\varphi + \psi} = \hat{\varphi} + \hat{\psi}$, $\widehat{[m]} = [m]$, $\deg\hat{\varphi} = \deg\varphi$; deg is a positive definite quadratic form on Hom. *(proof sketched here)* Theorem 9.21.

**9.18 Torsion.** $\deg[m] = m^2$; if $\operatorname{char} K \nmid m$, $[m]$ is separable and $E[m] \simeq (\mathbb{Z}/m)^2$; in characteristic $p$, $E[p^e]$ is 0 or $\mathbb{Z}/p^e$. *(proved here, given Theorem 9.21; the last statement stated, reference)* Corollary 9.22.

**9.19 Cyclic factorisation** (char $K = 0$). $\ker\varphi \simeq \mathbb{Z}/a \times \mathbb{Z}/ab$ and $\varphi = \psi \circ [a]$ with $\psi$ cyclic over $K$; a cyclic $n$-isogeny over $K$ has $G_K$-stable kernels of every order $d \mid n$, and factors into cyclic isogenies of prime degree over $K$. *(proved here)* Proposition 9.23.

**9.20 Division polynomials** (definition): $\psi_2 = 2y + a_1x + a_3$, $\psi_3 = 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8$, $\psi_4$, and the recursions. Definition 9.25.

**9.21 Degrees.** For odd $m$, $\psi_m \in \mathbb{Z}[a_i][x]$ has degree $(m^2-1)/2$ and leading coefficient $m$; for even $m$, $\psi_m/\psi_2$ has degree $(m^2-4)/2$ and leading coefficient $m/2$. *(proved here)* Proposition 9.26.

**9.22 Multiplication by $m$.** For $\operatorname{char} K \nmid m$: $mP = O$ iff $\psi_m(P) = 0$, and $x(mP) = x - \psi_{m-1}\psi_{m+1}/\psi_m^2$. *(proof sketched here)* Theorem 9.27.

**9.23 Roots of $\psi_m$.** For odd $m$, $\operatorname{char} K \nmid m$: the roots of $\psi_m$ are the $(m^2-1)/2$ distinct values $x(P)$, $P \in E[m] \smallsetminus \{O\}$, $G_K$-equivariantly. *(proved here, given Theorem 9.27)* Corollary 9.28.

**9.24 The 3-division polynomial.** $x(2P) - x(P) = -\psi_3/\psi_2^2$; for $\operatorname{char} K \neq 2, 3$ the four roots of $\psi_3$ are the $x$-coordinates of the four lines of $E[3]$, $G_K$-equivariantly, so the factorisation pattern of $\psi_3$ is the orbit pattern of $\bar\rho_{E,3}(G_K)$ on $\mathbb{P}^1(\mathbb{F}_3)$, and a root in $K$ means a $K$-rational 3-isogeny; over $\mathbb{Q}$, $\operatorname{disc}\psi_3 \in -3(\mathbb{Q}^\times)^2$. *(proved here)* Proposition 9.29.

**9.25 The Weil pairing.** $e_m\colon E[m] \times E[m] \to \mu_m$ is bilinear, alternating, nondegenerate, Galois-equivariant, compatible in $m$, and $e_m(S, \hat\varphi T) = e_m(\varphi S, T)$. *(proof sketched here)* Theorem 9.32.

**9.26 The determinant.** $G_K$ acts on $E[m]$ with determinant $\chi_m^{\mathrm{cyc}}$; $E[m] \subseteq E(K)$ forces $\mu_m \subseteq K$; for a stable line $L \subset E[\ell]$, $E[\ell]/L \simeq \operatorname{Hom}(L, \mu_\ell)$, so a rational point of order $\ell$ gives $0 \to \mathbb{Z}/\ell \to E[\ell] \to \mu_\ell \to 0$. *(proved here)* Corollary 9.33.

**9.27 Tate modules** (definition and structure): $T_\ell E = \varprojlim E[\ell^n]$ is free of rank 2 over $\mathbb{Z}_\ell$, $T_\ell E/\ell^n = E[\ell^n]$, $V_\ell E/T_\ell E = E[\ell^\infty]$; the Weil pairings give a perfect alternating $G_K$-equivariant $e\colon T_\ell E \times T_\ell E \to \mathbb{Z}_\ell(1)$; in characteristic $p$, $T_pE$ is 0 or $\mathbb{Z}_p$. *(proved here; the last statement stated, reference)* Definition 9.34, Proposition 9.35.

**9.28 The Galois representations** (definition and properties): $\rho_{E,\ell}\colon G_K \to \mathrm{GL}_2(\mathbb{Z}_\ell)$, $\bar\rho_{E,\ell}$ on $E[\ell]$, $\rho \bmod \ell^k$ on $E[\ell^k]$; continuous, $\det\rho_{E,\ell} = \chi_\ell$; an isogeny over $K$ gives $T_\ell E \hookrightarrow T_\ell E'$ with cokernel $(\ker\varphi)[\ell^\infty]$, so $V_\ell E \simeq V_\ell E'$, and $T_\ell E \simeq T_\ell E'$ if $\ell \nmid \deg\varphi$. *(proved here)* Definition 9.36, Proposition 9.37.

**9.29 Self-duality.** $T_\ell E \simeq (T_\ell E)^\vee(1)$, so $H^1(E, \mathbb{Z}_\ell) = (T_\ell E)^\vee \simeq T_\ell E(-1)$ and $\mathrm{ad}^0$ of $T_\ell E$ and of its dual agree. *(proved here)* Corollary 9.38.

**9.30 Complex conjugation.** For $E/\mathbb{Q}$ and $\ell$ odd, $\rho_{E,\ell}(c) \sim \operatorname{diag}(1, -1)$: eigenvalues $\pm1$, trace 0, determinant $-1$. *(proved here)* Corollary 9.39.

**9.31 Frobenius conventions.** $\mathrm{Frob}_p$ arithmetic; $\rho_{E,\ell}(\mathrm{Frob}_p)$ has eigenvalues $\alpha_p, \beta_p$; at good $p$, $L_p(E,s) = \det(1 - \mathrm{Frob}_p\, p^{-s} \mid V_\ell E)^{-1} = L_p(V_\ell E^\vee, s)$ and $L_p(V_\ell E, s) = L_p(E, s+1)$, with $L$-factors defined by the geometric Frobenius. Convention 9.40.

**9.32 Endomorphism rings.** $\mathrm{End}_{\bar{K}}(E)$ is $\mathbb{Z}$, an order in an imaginary quadratic field, or (characteristic $p$ only) a maximal order in the quaternion algebra ramified at $p$, $\infty$. *(stated; reference)* Theorem 9.42.

**9.33 Normalised endomorphisms.** In characteristic 0, $\alpha \mapsto c_\alpha$ ($\alpha^*\omega = c_\alpha\omega$) embeds $\mathrm{End}(E)$ in $\bar{K}$; $\mathrm{End}_{\mathbb{Q}}(E) = \mathbb{Z}$; for $E/\mathbb{Q}$ with CM by $\mathcal{O} \subset F$, $\sigma[c]\sigma^{-1} = [\sigma(c)]$, endomorphisms are defined over $F$, and $\rho(\sigma)[c]_\ell\rho(\sigma)^{-1} = [\sigma c]_\ell$. *(proved here, given Proposition 9.15 and Theorem 9.42)* Proposition 9.43.

**9.34 Complex multiplication** (definition): $\mathrm{End}_{\bar{K}}(E) \neq \mathbb{Z}$; it is an isogeny invariant. *(the invariance proved here)* Definition 9.44, Lemma 9.45.

**9.35 The thirteen CM $j$-invariants.** $E/\mathbb{Q}$ with CM by $\mathcal{O}$ has $h(\mathcal{O}) = 1$, $D_{\mathcal{O}}$ is one of $-3$, $-4$, $-7$, $-8$, $-11$, $-12$, $-16$, $-19$, $-27$, $-28$, $-43$, $-67$, $-163$, and $j(E)$ is the corresponding value of the table. *(stated; reference)* Theorem 9.46.

**9.36 CM and $a_p$.** For $E/\mathbb{Q}$ with CM by $F$: $\rho \simeq \rho\otimes\chi_F$; $a_p = 0$ at good $p$ inert in $F$; $a_p = \pi_p + \bar{\pi}_p$ with $\pi_p\bar{\pi}_p = p$ at good split $p$; every prime ramified in $F$ divides $N$ (so CM by $\mathbb{Q}(\sqrt{\ell^*})$ forces $\ell \mid N$); conversely $a_p = 0$ at almost all inert $p$ forces CM by $F$. *(proof sketched here; (i), (ii), (iv) proved in Chapter 6)* Theorem 9.47.

**9.37 Lattice sums.** $\sum'|\omega|^{-s} < \infty$ for $s > 2$; $G_{2k}$, $g_2 = 60G_4$, $g_3 = 140G_6$, $\Delta(\Lambda)$, $j(\Lambda)$ are homogeneous of weights $2k, 4, 6, 12, 0$. *(proved here)* Definition 9.50, Lemma 9.51.

**9.38 The Weierstrass function.** $\wp$ is even, $\Lambda$-periodic, with double poles at $\Lambda$; $\wp = z^{-2} + \sum(2k+1)G_{2k+2}z^{2k}$; $\wp'^2 = 4\wp^3 - g_2\wp - g_3$; $\Delta(\Lambda) \neq 0$; periodic functions with poles of order $\leq 2$ (resp. $\leq 3$) at $\Lambda$ only are $a\wp + b$ (resp. $a\wp' + b\wp + c$). *(proved here)* Theorem 9.52.

**9.39 Uniformisation.** $z \mapsto (\wp, \wp')$ is a biholomorphic group isomorphism $\mathbb{C}/\Lambda \to E_\Lambda(\mathbb{C})$ with $dx/y \leftrightarrow dz$; every $E/\mathbb{C}$ is $E_{\Lambda_E}$ for its period lattice; for $4A^3 + 27B^2 \neq 0$ there is a unique $\Lambda$ with $g_2 = -4A$, $g_3 = -4B$. *(proved here, given the Abel-Jacobi theorem of Chapter 8)* Theorem 9.53.

**9.40 Maps of tori.** Holomorphic maps $\mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$ fixing 0 are $z \mapsto \alpha z$, $\alpha\Lambda_1 \subseteq \Lambda_2$, of degree $[\Lambda_2 : \alpha\Lambda_1]$; $E[m] = \frac{1}{m}\Lambda/\Lambda$; $\mathrm{End}(\mathbb{C}/\Lambda)$ is $\mathbb{Z}$ or an imaginary quadratic order, the latter iff $\tau$ is imaginary quadratic; $e_m = \zeta^{\det}$ on $\frac{1}{m}\Lambda/\Lambda$. *(proved here; the sign of $\zeta$ by reference)* Proposition 9.55.

**9.41 Moduli over $\mathbb{C}$.** Lattices up to homothety = curves over $\mathbb{C}$ up to isomorphism = $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$; $j(\tau)$ is holomorphic and invariant, $j(i) = 1728$, $j(\zeta_3) = 0$; $j$ is a bijection $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H} \to \mathbb{C}$. *(proved here; the bijectivity and the $q$-expansion in Chapter 12)* Theorem 9.56.

**9.42 The period lattice** (definition and properties): $\Lambda_E$ of the Néron differential of the global minimal model; conjugation-stable, rectangular iff $\Delta > 0$; $\mathrm{vol}(\Lambda_E) = \frac{i}{2}\int \omega \wedge \bar{\omega}$; computed by the AGM; commensurable for isogenous curves. *(proof sketched here)* Definition 9.57, Proposition 9.58.

**9.43 Frobenius endomorphism** (definition and properties): $\pi(x, y) = (x^{\mathfrak{q}}, y^{\mathfrak{q}})$, purely inseparable of degree $\mathfrak{q}$, acting as $\mathrm{Frob}_{\mathfrak{q}}$; $E(\mathbb{F}_{\mathfrak{q}^n}) = \ker(1-\pi^n)$; $1-\pi$ separable, $\#E(\mathbb{F}_{\mathfrak{q}}) = \deg(1-\pi)$; $a_{\mathfrak{q}} = \mathfrak{q} + 1 - \#E(\mathbb{F}_{\mathfrak{q}})$. *(proved here, given Chapter 8)* Definition 9.60, Proposition 9.61.

**9.44 Hasse.** $|a_{\mathfrak{q}}| \le 2\sqrt{\mathfrak{q}}$. *(proved here)* Theorem 9.62.

**9.45 Characteristic polynomial.** $\det \varphi_\ell = \deg \varphi$ on $T_\ell E$; $\pi_\ell$ has characteristic polynomial $X^2 - a_{\mathfrak{q}} X + \mathfrak{q}$, roots $\alpha, \beta$ with $\alpha\beta = \mathfrak{q}$, $|\alpha| = |\beta| = \sqrt{\mathfrak{q}}$; $\#E(\mathbb{F}_{\mathfrak{q}^n}) = \mathfrak{q}^n + 1 - \alpha^n - \beta^n$; the zeta function. *(proved here)* Theorem 9.63.

**9.46 Supersingular curves** (definition and criteria): $E[p] = 0$; iff $p \mid a_{\mathfrak{q}}$; for $\mathfrak{q} = p \ge 5$ iff $a_p = 0$. *(the first criterion stated, reference; the second proved here)* Definition 9.64, Proposition 9.65.

**9.47 Twists over $\mathbb{F}_{\mathfrak{q}}$.** $a_{\mathfrak{q}}(E^{\mathrm{tw}}) = -a_{\mathfrak{q}}(E)$ and $(\mathfrak{q}+1)^2 - a_{\mathfrak{q}}^2 = \#E(\mathbb{F}_{\mathfrak{q}})\#E^{\mathrm{tw}}(\mathbb{F}_{\mathfrak{q}})$. *(proved here)* Proposition 9.66.

**9.48 Good reduction.** For $E/\mathbb{Q}$ good at $p \neq \ell$: $\rho_{E,\ell}$ is unramified at $p$ and $\rho_{E,\ell}(\mathrm{Frob}_p)$ has characteristic polynomial $X^2 - a_p X + p$; $\operatorname{tr} \bar{\rho}_{E,\ell}(\mathrm{Frob}_p) \equiv a_p$. *(proof sketched here, using Chapter 10)* Theorem 9.67.

**9.49 Mordell-Weil.** $E(\mathbb{Q}) \simeq \mathbb{Z}^r \oplus E(\mathbb{Q})_{\mathrm{tors}}$. *(stated; reference)* Theorem 9.70.

**9.50 Torsion and reduction.** Reduction is injective on prime-to-$p$ torsion at good $p$ (on all torsion if $p \ge 3$); Nagell-Lutz; $E(\mathbb{Q})_{\mathrm{tors}}$ of `11a1`, `20a1`. *(stated, reference; the examples proved here)* Proposition 9.71.

**9.51 Mazur's torsion theorem.** $E(\mathbb{Q})_{\mathrm{tors}}$ is $\mathbb{Z}/n$, $n \le 10$ or 12, or $\mathbb{Z}/2 \times \mathbb{Z}/2n$, $n \le 4$. *(stated; reference)* Theorem 9.72.

**9.52 Isogeny classes** (definition and properties): an equivalence relation; isogenous curves have the same $V_\ell$, $a_p$, $L$-function, rank and conductor; Faltings's converse; at most 8 curves (Kenku). *(the first two proved here, except the conductor; the rest stated, reference)* Definition 9.73, Theorem 9.74.

**9.53 Rational $\ell$-isogenies.** $K$-rational $\ell$-isogenies $\leftrightarrow$ $G_K$-stable lines of $E[\ell]$; $E[\ell]$ reducible iff a rational $\ell$-isogeny exists; then the two characters multiply to $\bar{\chi}_\ell$. *(proved here, given Theorem 9.20)* Proposition 9.75.

**9.54 Mazur and Kenku.** Rational $\ell$-isogenies only for $\ell = 2$, 3, 5, 7, 11, 13, 17, 19, 37, 43, 67, 163; cyclic $n$-isogenies only for $n \le 19$ or $n = 21$, 25, 27, 37, 43, 67, 163, with 19, 27, 43, 67, 163 only for CM curves. *(stated; reference)* Theorem 9.76.

**9.55 $\mathbb{Z}/\ell \oplus \mu_\ell$.** With a rational point of order $\ell$ (odd), $\bar{\rho} = \left(\begin{smallmatrix} 1 & * \\ 0 & \bar{\chi}_\ell \end{smallmatrix}\right)$; splitting $\Leftrightarrow$ image $\{\mathrm{diag}(1,d)\}$ $\Leftrightarrow$ two rational $\ell$-isogenies; $a_p \equiv 1 + p \pmod{\ell}$. *(proved here)* Proposition 9.77.

**9.56 No point of order** 9 (the paper's Proposition prop:no9). A non-CM $E/\mathbb{Q}$ with $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ has no rational point of order 9; so $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3] \le 2$. *(proved here, given Theorem 9.76)* Proposition 9.78.

**9.57 Isogenies prime to $\ell$.** If $E[\ell]$ is irreducible, every rational isogeny is $\psi \circ [a]$ with $\psi$ cyclic of degree prime to $\ell$; all curves of the class have isomorphic $T_\ell$, the same $\mathrm{ad}^0 T_\ell$ and $h_\ell$, and the same $t_\ell(E,q)$. *(proved here; the last statement sketched, using Chapter 10)* Lemma 9.80.

**9.58 Trivial isogeny classes.** Then $E[\ell]$ is irreducible for every $\ell$ and $E$ is optimal. *(proved here)* Corollary 9.81.

**9.59 Quadratic twists** (definition and properties): $E^{(d)}\colon y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6$, for a general model $Y^2 = X^3 + db_2X^2 + 8d^2b_4X + 16d^3b_6$; depends on $d$ modulo squares; $\psi\colon E^{(d)} \to E$ over $K(\sqrt{d})$ with $\psi^\sigma = [\chi_d(\sigma)]\psi$; the twisted Galois action. *(proved here; the classification of all twists stated)* Definition 9.83, Proposition 9.84.

**9.60 Invariants of twists.** $d^2c_4$, $d^3c_6$, $d^6\Delta$, same $j$; conversely equal $j \neq 0, 1728$ means a quadratic twist by $d = c_6'c_4/(c_6c_4')$; the twists by $q$ and $q^*$ differ by the twist by $-1$. *(proved here)* Proposition 9.85.

**9.61 Galois representation of a twist.** $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi_d$; same $\mathrm{ad}^0$, same det, same $E[2]$, irreducibility preserved. *(proved here)* Theorem 9.86.

**9.62 $a_p$ of a twist.** $a_p(E^{(d)}) = \chi_d(p)a_p(E)$ for good $p \nmid D$. *(proved here)* Corollary 9.87.

**9.63 The twist of good reduction.** At a type $\mathrm{I}_0^*$ prime $q$, $E' = E^{(q^*)}$ is good at $q$, $V_\ell E \simeq V_\ell E' \otimes \chi_{q^*}$, and $a_q(E')^2$ does not depend on the choice of twist. *(proved here, given Chapter 10)* Proposition 9.88.

**9.64 Composite twists** (definition): $\chi_S = \prod_{q\in S}\chi_{q^*}$, $E_S = E^{(d_S)}$, $d_S = \prod q^*$; $a_p(E_S) = \chi_S(p)a_p(E)$. Definition 9.89.

**9.65 What a twist changes.** $j$, $t_\ell$, $\mathrm{ad}^0\,\rho$ are twist-invariant; $e_\ell(E,q)$ is invariant only under twists unramified at $q$. Remark 9.90.

**9.66 The $L$-function** (definition and properties): $L(E,s) = \prod_{p\nmid N}(1-a_pp^{-s}+p^{1-2s})^{-1}\prod_{p|N}(1-a_pp^{-s})^{-1}$, absolutely convergent for $\mathrm{Re}\,s > 3/2$, with multiplicative coefficients and the recursion $a_{p^{k+1}} = a_pa_{p^k} - pa_{p^{k-1}}$; continuation and functional equation by modularity. *(proved here, except (iv): stated, reference)* Definition 9.93, Proposition 9.94.

**9.67 Labels.** Cremona's $N\,c\,k$, curve 1 optimal except in `990h`; LMFDB labels differ. Convention 9.95, Example 9.96.

## Where the paper uses this chapter

- **Section 1.1** ($V = V_\ell E$, $T = T_\ell E$, $\rho$, $\bar\rho$, the Frobenius conventions): Definition 9.34, Proposition 9.35, Definition 9.36, Proposition 9.37, Convention 9.40, Paper link 9.41.
- **Definition def:terms** ($j_E$, $v_q(j_E)$, the twist-invariance of $t_\ell$ and the failure for $e_\ell$): Proposition 9.5(iii), Paper link 9.9, Remark 9.90.
- **The table of Section 1 and Lemma lem:euler** (the twist $E'$ of good reduction at a type $\mathrm{I}_0^*$ prime, $\alpha,\beta$ with $\alpha\beta = q$, $\alpha+\beta = a_q(E')$, $V = V'\otimes\chi_{q^*}$, $\mathrm{ad}^0\,V = \mathrm{ad}^0\,V'$): Theorem 9.63, Proposition 9.66, Theorem 9.86, Proposition 9.88, Paper link 9.69, Paper link 9.92.
- **Lemma lem:twist (ii)** (the model and the invariants $(q^*)^2c_4$, $(q^*)^3c_6$, $(q^*)^6\Delta$ of the twist; $v_q(\Delta)$ modulo 12): Definition 9.83, Proposition 9.85, Proposition 9.5(iii), Warning 9.91.
- **Section 1.4** (the twist-symmetrised weight $w_\ell(E,q)$, which compares $E$ with its twist by $\chi_{q^*}$; the paper writes this twist $E^{(q)}$, the book $E^{(q^*)}$): Definition 9.83(c), Proposition 9.85(iii), Warning 9.91(a).

- **Theorems thm:twist and thm:quant** ($E_S$, $\chi_S$, $a_p(f) = \chi_S(p)a_p(g)$, irreducibility of $\bar\rho \otimes \chi_S$): Definition 9.89, Theorem 9.86, Corollary 9.87.
- **Theorem thm:qual and the Example of Section 3** (ex:qual: twists of `121a1`, `539c1`, `980i1`; labels; $a_p$): Corollary 9.87, Warning 9.91, Example 9.68, Example 9.96.
- **Lemma lem:h0** (complex conjugation with eigenvalues $\pm 1$, $\det\rho = \chi_\ell$, $\bar\chi_3 = \chi_{-3}$): Corollary 9.39, Proposition 9.37, Theorem 9.47, Warning 9.48.
- **Proposition prop:ns** (the 3-division polynomial, $\det\bar\rho_{E,3} = \chi_{-3}$, complex conjugation): Proposition 9.29, Example 9.30, Corollary 9.33, Corollary 9.39, Paper link 9.31.
- **Conjecture conj:eis, Section sec:eis and Proposition prop:no9** (torsion, rational $\ell$-isogenies, $\mathbb{Z}/3 \oplus \mu_3$, no point of order 9, "not CM"): Theorem 9.72, Proposition 9.75, Theorem 9.76, Proposition 9.77, Proposition 9.78, Lemma 9.45, Paper link 9.79, Paper link 9.49.
- **Section 1 (optimality and $\ell$; the area of the period lattice) and Section sec:oot (trivial isogeny classes)**: Lemma 9.80, Corollary 9.81, Theorem 9.74, Definition 9.57, Example 9.59, Paper link 9.82.
- **Section 4.2 and Proposition prop:pullback** ($M_{f,\lambda} = H^1(E,\mathbb{Z}_\ell) = (T_\ell E)^\vee \simeq T_\ell E(-1)$, with the same adjoint): Corollary 9.38, Paper link 9.41.
- **Lemma lem:old** ($X^2 - a_p(h)X + p$, multiplicativity and recursion of the coefficients): Theorem 9.67, Proposition 9.94.
- **Section sec:two** (Calegari-Emerton; CM curves with $N \in \{27, 32, 49, 243\}$): Theorem 9.46, Paper link 9.49.
- **The appendix and Section sec:anomaly** (`ellinit`, `elldivpol`, `ns3`, `valuation(j, q)` with `j = E.j`, `121a1`, `726a1`): Definition 9.2, Proposition 9.29, Example 9.30, Convention 9.95, Example 9.96.

## 9.1 Weierstrass equations and their invariants

Every elliptic curve in the paper is given by a Weierstrass equation, in the form that PARI's `ellinit` expects, and every local quantity the paper reads off at a bad prime $q$ starts from the invariants $c_4$, $c_6$, $\Delta$ and $j$ of such an equation. This section defines them, proves that a Weierstrass cubic is smooth exactly when $\Delta \neq 0$, records how the invariants change under a change of coordinates, and proves that $j$ classifies curves over an algebraically closed field. The paper uses these facts in Definition def:terms (the valuation $v_q(j_E)$, which must not depend on the model), in the proof of Lemma lem:twist (ii) (the invariants of a twisted model, and the rule that $v_q(\Delta)$ changes by multiples of 12), and in its appendix (`valuation(E.j, q)`).

**Convention 9.1** (conventions of this chapter)**.**

(a) *Fields and curves.* $K$ is a field, $\bar K$ a fixed algebraic closure, $K^{\mathrm{sep}}$ the separable closure in it and $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$. An *elliptic curve over* $K$ is a pair $(E, O)$ of a smooth projective curve $E$ of genus one over $K$ and a point $O \in E(K)$. By Chapter 8 (the theorem

that a genus-one curve with a rational point is a Weierstrass cubic) every elliptic curve is isomorphic to a smooth Weierstrass cubic with $O \mapsto [0:1:0]$, and we always work with such a model. Whenever a prime $\ell$ appears, $\ell \neq \operatorname{char} K$.

(b) *Over* $\mathbb{Q}$. "$E/\mathbb{Q}$" means a curve given by a Weierstrass equation with rational coefficients; from Section 9.9 on, by its global minimal model (Chapter 10), which is the model printed in Cremona's tables [35, 36]. Such models have $a_1, a_3 \neq 0$ in general (`121a1`, `726a1`, `99a1`), so "a model" always means a general Weierstrass equation.

(c) *Letters.* $p$, $q$, $\ell$ are primes; as in the paper, $q$ is usually a bad prime of $E/\mathbb{Q}$, $p$ any prime and $\ell$ the coefficient prime. The number of elements of a finite field that need not be prime is written $\mathfrak{q} = p^r$ (Section 9.8); for $\mathbb{F}_p$ we write $a_p$. In two places only, Theorem 9.56(iii) and the last sentence of Example 9.68, $q = e^{2\pi i\tau}$ is the variable of $q$-expansions, as in Chapter 12; no prime $q$ occurs there.

(d) *Characters.* For $d \in K^\times$ and $\operatorname{char} K \neq 2$, $\chi_d\colon G_K \to \{\pm 1\}$ is the character of $K(\sqrt{d})/K$: $\sigma(\sqrt{d}) = \chi_d(\sigma)\sqrt{d}$. For $K = \mathbb{Q}$ and $d$ not a square, $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$ (Kronecker symbol) for $p \nmid D$, where $D$ is the discriminant of $\mathbb{Q}(\sqrt{d})$ (Chapters 2 and 6). For an odd prime $q$, $q^* = (-1)^{(q-1)/2}q$ and $\chi_{q^*}$ is the quadratic character of conductor $q$. The paper's $\chi_q$ is written $\chi_{q^*}$; the symbol $\chi_\ell$ with a letter subscript is the $\ell$-adic cyclotomic character, $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ on $\mu_{\ell^\infty}$, and $\bar{\chi}_\ell$ is its reduction modulo $\ell$; at a numerical prime we write $\chi_3^{\mathrm{cyc}}$, so that $\bar{\chi}_3 = \chi_{-3}$. For an integer $m$ prime to $\operatorname{char} K$, $\chi_m^{\mathrm{cyc}}\colon G_K \to (\mathbb{Z}/m)^\times$ is defined by $\sigma(\zeta) = \zeta^{\chi_m^{\mathrm{cyc}}(\sigma)}$ for $\zeta \in \mu_m$.

(e) *Labels.* Cremona's labels are written as the paper writes them: `11a1` is a curve, `121c` an isogeny class. The capital letters of Cremona's book and of [3] (`99A1`) denote the same classes as the lower-case letters of the data files. LMFDB labels are written `11.a2` (Section 9.12).

(f) *The Weil pairing* is Silverman's $e_m$ [185, Section III.8]. Some books use its inverse; the only statement of this chapter that depends on the choice is the comparison with $e^{2\pi i/m}$ over $\mathbb{C}$ in Proposition 9.55(iv).

(g) *Duals and maps.* $M^\vee = \operatorname{Hom}(M, R)$ is the linear dual; $\varphi^*$ and $\varphi_*$ are pullback and pushforward along a map $\varphi$ of curves; $\hat{\varphi}$ is the dual isogeny. Maps of curves are written $\varphi$, $\psi$; the letter $\psi_m$ is also used for division polynomials (Section 9.4), which is never ambiguous in context.

The Frobenius conventions are fixed in Convention 9.40.

**Definition 9.2** (Weierstrass equations and their invariants)**.** A *Weierstrass equation* over $K$ is

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_1, a_2, a_3, a_4, a_6 \in K. \tag{9.1}$$

The curve it defines is its projective closure $Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3$ in $\mathbb{P}^2$, which has exactly one point on the line $Z = 0$, namely $O = [0:1:0]$. Put

$$b_2 = a_1^2 + 4a_2, \qquad b_4 = 2a_4 + a_1a_3, \qquad b_6 = a_3^2 + 4a_6,$$
$$b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2, \qquad c_4 = b_2^2 - 24b_4, \qquad c_6 = -b_2^3 + 36b_2b_4 - 216b_6,$$

$$\Delta = -b_2^2 b_8 - 8b_4^3 - 27b_6^2 + 9b_2 b_4 b_6, \qquad j = \frac{c_4^3}{\Delta} \text{ when } \Delta \neq 0.$$

$\Delta$ is the *discriminant* and $j$ the *$j$-invariant*. The *invariant differential* is

$$\omega = \frac{dx}{2y + a_1 x + a_3} = \frac{dy}{3x^2 + 2a_2 x + a_4 - a_1 y}.$$

The two expressions agree because differentiating (9.1) gives $(2y + a_1x + a_3)\,dy = (3x^2 + 2a_2x + a_4 - a_1y)\,dx$. These are the conventions of [185, Section III.1], and PARI's `ellinit` returns $a_1, \dots, a_6, b_2, b_4, b_6, b_8, c_4, c_6, \Delta, j$ in this order (Chapter 22).

The invariants are less arbitrary than they look. The $b_i$ are what remains after completing the square in $y$: the equation becomes $(2y + a_1x + a_3)^2 = 4x^3 + b_2x^2 + 2b_4x + b_6$ (Lemma 9.3(ii) below), and $b_8 = (b_2b_6 - b_4^2)/4$, which is a polynomial in the $a_i$ with integer coefficients, is the combination needed to write $\frac{1}{16}$ of the discriminant of this cubic without denominators. The $c_4$ and $c_6$ are what remains after also removing the $x^2$ term, which gives the short form $Y^2 = X^3 - 27c_4X - 54c_6$ (Lemma 9.3(iii)). The discriminant $\Delta$ vanishes exactly when the cubic has a repeated root, that is, when the curve is singular (Proposition 9.4), and $j$ is the combination of $c_4$ and $\Delta$ that does not change under any change of coordinates (Proposition 9.5). The point of working with the general equation, and not only with short models, is that over $\mathbb{Q}$ the global minimal models of Cremona's tables, which the paper uses, often have $a_1, a_3 \neq 0$, and the short form is not available in characteristics 2 and 3, which matter for reduction modulo 2 and 3 (Chapter 10).

For a first example take $y^2 = x^3 - x$, so $a_1 = a_2 = a_3 = a_6 = 0$ and $a_4 = -1$. Then $b_2 = 0$, $b_4 = -2$, $b_6 = 0$, $b_8 = -1$, so $c_4 = -24b_4 = 48$, $c_6 = 0$, $\Delta = -8b_4^3 = 64$ and $j = 48^3/64 = 1728$. The cubic $4x^3 - 4x$ has the three roots $0, \pm 1$, and its discriminant is $16 \cdot 64 = 1024 = 4^4 \cdot (0-1)^2(0+1)^2(1+1)^2$, in accordance with Lemma 9.3(ii).

**Lemma 9.3** (identities)**.** *For every Weierstrass equation:*

*(i)* $4b_8 = b_2b_6 - b_4^2$ *and* $1728\Delta = c_4^3 - c_6^2$.

*(ii) (Completing the square.) With* $\eta = 2y + a_1x + a_3$ *the equation* (9.1) *is equivalent, when* $\operatorname{char} K \neq 2$*, to*

$$\eta^2 = 4x^3 + b_2x^2 + 2b_4x + b_6.$$

*The cubic on the right is written $F(x)$ below; its discriminant (as a polynomial in $x$) is* $16\Delta$.

*(iii) (Short form.) If* $\operatorname{char} K \neq 2, 3$*, the substitution* $X = 36x + 3b_2$, $Y = 108(2y + a_1x + a_3)$ *transforms* (9.1) *into*

$$Y^2 = X^3 - 27c_4X - 54c_6.$$

*Proof.* All three are identities between polynomials in $a_1, \dots, a_6$ (and $x$, $y$). (i) Expanding $b_2b_6 - b_4^2 = (a_1^2 + 4a_2)(a_3^2 + 4a_6) - (2a_4 + a_1a_3)^2$ gives $a_1^2a_3^2 + 4a_1^2a_6 + 4a_2a_3^2 + 16a_2a_6 - 4a_4^2 - 4a_1a_3a_4 - a_1^2a_3^2$, which is $4b_8$. For the second identity expand both cubes as polynomials in $b_2, b_4, b_6$: the terms $b_2^6$ and $b_2^4b_4$ cancel, and

$$\begin{aligned} c_4^3 - c_6^2 &= 432b_2^2b_4^2 - 432b_2^3b_6 - 13824b_4^3 + 15552b_2b_4b_6 - 46656b_6^2 \\ &= 1728\Big(-\tfrac{1}{4}b_2^2(b_2b_6 - b_4^2) - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6\Big), \end{aligned}$$

which is $1728\Delta$ by the first identity. (ii) $\eta^2 = 4y^2 + 4a_1xy + 4a_3y + (a_1x + a_3)^2$, and replacing $4(y^2 + a_1xy + a_3y)$ by $4(x^3 + a_2x^2 + a_4x + a_6)$ gives $4x^3 + (a_1^2 + 4a_2)x^2 + (2a_1a_3 + 4a_4)x + (a_3^2 + 4a_6)$. Conversely, when 2 is invertible, $y = (\eta - a_1x - a_3)/2$ recovers $y$. The discriminant of $4x^3 + b_2x^2 + 2b_4x + b_6$ is, by the formula for the discriminant of a cubic (Chapter 5), $b_2^2(2b_4)^2 - 4\cdot 4(2b_4)^3 - 4b_2^3b_6 - 27\cdot 16\,b_6^2 + 18\cdot 4\,b_2(2b_4)b_6$, which equals $16\Delta$ after replacing $b_2^2b_4^2 - b_2^3b_6$ by $-4b_2^2b_8$. (iii) Here 2 and 3 are invertible, so $x = (X - 3b_2)/36$ and $\eta = Y/108$ recover the old coordinates, and by (ii) the equation is equivalent to $Y^2 = 108^2F(x)$. Since $108^2 = 11664 = 4\cdot 36^3/16$, one has $108^2\cdot 4x^3 = (X - 3b_2)^3$, $108^2b_2x^2 = 9b_2(X - 3b_2)^2$, $108^2\cdot 2b_4x = 648b_4(X - 3b_2)$ and $108^2b_6 = 11664b_6$. Adding,

$$\begin{aligned} 108^2F(x) &= (X^3 - 9b_2X^2 + 27b_2^2X - 27b_2^3) + (9b_2X^2 - 54b_2^2X + 81b_2^3) \\ &\quad + 648b_4X - 1944b_2b_4 + 11664b_6 \\ &= X^3 - 27(b_2^2 - 24b_4)X + 54(b_2^3 - 36b_2b_4 + 216b_6) = X^3 - 27c_4X - 54c_6. \end{aligned}$$

In the language of Proposition 9.5 this is the change of coordinates with $u = 1/6$, $r = -b_2/12$ (and suitable $s$, $t$), since $x = \frac{1}{36}X - \frac{b_2}{12}$ and $Y = 216y + \dots$. Each expansion was also checked by machine (PARI/GP, with $a_1, \dots, a_6$ as indeterminates). □

**Proposition 9.4** (smoothness)**.** *Let $E$ be the curve defined by a Weierstrass equation over $K$.*

*(i) The point $O$ is smooth, and $E$ is smooth if and only if $\Delta \neq 0$.*

*(ii) If $\Delta = 0$, $E$ has exactly one singular point. It is a* node *(two distinct tangent lines) if $c_4 \neq 0$ and a* cusp *(one tangent line) if $c_4 = 0$.*

*(iii) A smooth Weierstrass cubic, with the point $O$, is an elliptic curve; every elliptic curve over $K$ is isomorphic over $K$ to one, with its origin going to $O$.*

*Proof.* (i) That $O$ is smooth is checked in Chapter 8 (the partial derivative of the homogeneous equation with respect to $Z$ is 1 at $O$). Assume $\operatorname{char} K \neq 2$. By Lemma 9.3(ii) we may use the coordinates $(x, \eta)$, in which the equation is $G = \eta^2 - F(x) = 0$. A point $(x_0, \eta_0)$ is singular iff $G = \partial G/\partial\eta = \partial G/\partial x = 0$ there, that is, iff $\eta_0 = 0$ and $F(x_0) = F'(x_0) = 0$, iff $x_0$ is a double root of $F$. The cubic $F$ has leading coefficient $4 \neq 0$, roots $r_1, r_2, r_3 \in \bar{K}$, and discriminant $4^4\prod_{i<j}(r_i - r_j)^2$ (Chapter 5); so it has a double root in $\bar{K}$ iff its discriminant $16\Delta$ vanishes, iff $\Delta = 0$ (as $2 \neq 0$ in $K$). Every point other than $O$ is affine, since $O$ is the only point on $Z = 0$, so this covers all points. In characteristic 2 the statement is [185, Section III.1]; we do not prove it.

(ii) Let $S$ be a singular point; it is the only one, for a line through two singular points would meet the cubic with multiplicity at least 4 and so lie in it (Chapter 8, Bezout), while a Weierstrass cubic is irreducible (Chapter 8). Move $S$ to $(0,0)$ by $x \mapsto x + x_0$, $y \mapsto y + y_0$ (a change of coordinates with $u = 1$, $s = 0$, which by Proposition 9.5 below does not change $c_4$ or $\Delta$). Since $(0,0)$ lies on the curve and both partial derivatives vanish there, $a_6 = a_4 = a_3 = 0$, and the equation is $y^2 + a_1xy - a_2x^2 = x^3$. Now $b_4 = b_6 = b_8 = 0$, so $\Delta = 0$ and $c_4 = b_2^2 = (a_1^2 + 4a_2)^2$. The tangent lines at $(0,0)$ are the factors of the quadratic part $y^2 + a_1xy - a_2x^2$, which are distinct iff its discriminant $a_1^2 + 4a_2 = b_2$ is nonzero (in characteristic 2: iff $a_1 \neq 0$, iff $b_2 = a_1^2 \neq 0$). So the point is a node iff $c_4 \neq 0$.

(iii) A smooth Weierstrass cubic has genus one and $O$ is a $K$-point (Chapter 8); the converse is the theorem of Chapter 8 quoted in Convention 9.1(a). □

**Proposition 9.5** (changes of coordinates)**.**

*(i) For $u \in K^\times$ and $r, s, t \in K$, the substitution*

$$x = u^2x' + r, \qquad y = u^3y' + su^2x' + t$$

*transforms a Weierstrass equation into a Weierstrass equation, with*

$$\begin{aligned} ua_1' &= a_1 + 2s, & u^2a_2' &= a_2 - sa_1 + 3r - s^2, \\ u^3a_3' &= a_3 + ra_1 + 2t, & u^4a_4' &= a_4 - sa_3 + 2ra_2 - (t + rs)a_1 + 3r^2 - 2st, \end{aligned}$$

$$\begin{aligned} u^6a_6' &= a_6 + ra_4 + r^2a_2 + r^3 - ta_3 - t^2 - rta_1, \\ u^2b_2' &= b_2 + 12r, \qquad u^4b_4' = b_4 + rb_2 + 6r^2, \qquad u^6b_6' = b_6 + 2rb_4 + r^2b_2 + 4r^3, \\ u^8b_8' &= b_8 + 3rb_6 + 3r^2b_4 + r^3b_2 + 3r^4, \qquad u^4c_4' = c_4, \qquad u^6c_6' = c_6, \end{aligned}$$

$$u^{12}\Delta' = \Delta, \qquad j' = j, \qquad \omega' = u\,\omega.$$

*These substitutions form a group under composition.*

*(ii) Every isomorphism between two smooth Weierstrass cubics over $K$ that sends $O$ to $O$ is given by such a substitution.*

*(iii) Let $K = \mathbb{Q}$ and $q$ a prime. For any two models of $E$, $v_q(\Delta') - v_q(\Delta) \in 12\mathbb{Z}$, and $v_q(j) = 3v_q(c_4) - v_q(\Delta)$ is the same for both. In particular $v_q(j_E)$ depends only on the isomorphism class of $E$ over $\mathbb{Q}$.*

*Proof.* (i) Substitute and compare coefficients: with $W(x, y)$ the left side minus the right side of (9.1) and $W'$ the same for the primed coefficients, one has $W(u^2x' + r, u^3y' + su^2x' + t) = u^6W'(x', y')$, which defines the $a_i'$ and is equivalent to the five formulas displayed. The formulas for the $b_i'$, $c_4'$, $c_6'$, $\Delta'$ follow by substitution into Definition 9.2; all were checked by machine with $a_i, u, r, s, t$ indeterminate. Then $j' = c_4'^3/\Delta' = u^{-12}c_4^3/(u^{-12}\Delta) = j$. Finally $dx = u^2dx'$ and $2y + a_1x + a_3 = u^3(2y' + a_1'x' + a_3')$ (from the formulas for $a_1'$ and $a_3'$), so $\omega = u^{-1}\omega'$. The composite of two substitutions is the substitution with $u = u_1u_2$, $r = r_1 + u_1^2r_2$, $s = s_1 + u_1s_2$, $t = t_1 + u_1^2s_1r_2 + u_1^3t_2$. Indeed, substituting $x' = u_2^2x'' + r_2$, $y' = u_2^3y'' + s_2u_2^2x'' + t_2$ into $x = u_1^2x' + r_1$, $y = u_1^3y' + s_1u_1^2x' + t_1$ gives $x = (u_1u_2)^2x'' + (r_1 + u_1^2r_2)$ and

$$y = (u_1u_2)^3y'' + (s_1 + u_1s_2)(u_1u_2)^2x'' + \big(t_1 + u_1^2s_1r_2 + u_1^3t_2\big),$$

which is of the same shape. With these formulas, composing $(u, r, s, t)$ with the substitution

$$(u^{-1},\ -ru^{-2},\ -su^{-1},\ (rs - t)u^{-3})$$

gives $u = 1$, $r + u^2(-ru^{-2}) = 0$, $s + u(-su^{-1}) = 0$ and $t + u^2s(-ru^{-2}) + u^3(rs - t)u^{-3} = 0$, the identity substitution; so every substitution has an inverse of the same kind.

(ii) Let $\varphi\colon E \to E'$ be such an isomorphism, with coordinates $x, y$ on $E$ and $x', y'$ on $E'$. By Chapter 8, $\mathcal{L}(2(O))$ has basis $1, x$ and $\mathcal{L}(3(O))$ has basis $1, x, y$, and likewise on $E'$. Since $\varphi$ is an isomorphism with $\varphi(O) = O$, $\varphi^*x'$ has a double pole at $O$ and no other pole, and $\varphi^*y'$ a triple pole; so $\varphi^*x' = \alpha x + \rho$ and $\varphi^*y' = \beta y + \gamma x + \tau$ with $\alpha, \beta \in K^\times$, $\rho, \gamma, \tau \in K$ (all in $K$ when $\varphi$ is defined over $K$). Substituting these into the equation of $E'$ gives a relation among $1, x, y, x^2, xy, y^2, x^3$ in $K(E)$. These seven functions lie in $\mathcal{L}(6(O))$, of dimension 6, and have pole orders $0, 2, 3, 4, 5, 6, 6$ at $O$; the first six have distinct pole orders and are therefore linearly

independent. So the relations among the seven form a one-dimensional space, spanned by the equation of $E$ (Chapter 8, proof of the Weierstrass form). Comparing the coefficients of $y^2$ and $x^3$ gives $\beta^2 = \alpha^3$. Put $u = \alpha/\beta$; then $u^2 = \alpha^2/\beta^2 = \alpha^{-1}$ and $u^3 = \beta^{-1}$, so $x = u^2\varphi^* x' - u^2\rho$ and $y = u^3\varphi^* y' - u^3\gamma x - u^3\tau$, which after substituting $x$ is of the stated form.

(iii) If both models have rational coefficients, $u \in \mathbb{Q}^\times$ and $v_q(\Delta') = v_q(\Delta) - 12v_q(u)$, $v_q(c_4') = v_q(c_4) - 4v_q(u)$; so $3v_q(c_4') - v_q(\Delta') = 3v_q(c_4) - v_q(\Delta)$, which is $v_q(j)$ (with $v_q(0) = +\infty$ when $c_4 = 0$). Here $u \in \mathbb{Q}^\times$ because, by (ii), the isomorphism over $\mathbb{Q}$ between the two models is a substitution with coefficients in $\mathbb{Q}$. □

Part (iii) is what makes $v_q(j_E)$ usable as an invariant of $E$, while $v_q(\Delta)$ is an invariant only modulo 12: a rescaling with $v_q(u) = -1$ keeps an integral model integral and raises $v_q(\Delta)$ by 12. Choosing, among the integral models, one with the least $v_q(\Delta)$ is the minimal model problem of Chapter 10.

**Theorem 9.6** (the $j$-invariant)**.**

*(i) Two elliptic curves over $K$ are isomorphic over $\bar{K}$ if and only if they have the same $j$-invariant.*

*(ii) Every $j_0 \in K$ is the $j$-invariant of an elliptic curve over $K$: one can take $y^2 + y = x^3$ for $j_0 = 0$ (if $\operatorname{char} K \neq 3$), $y^2 = x^3 + x$ for $j_0 = 1728$ (if $\operatorname{char} K \neq 2$), and*

$$y^2 + xy = x^3 - \frac{36}{j_0 - 1728}x - \frac{1}{j_0 - 1728} \qquad (j_0 \neq 0, 1728),$$

*which has $c_4 = j_0/(j_0 - 1728)$ and $\Delta = j_0^2/(j_0 - 1728)^3$.*

*Proof.* (i) "Only if" is Proposition 9.5(i). For "if" we give the proof when $\operatorname{char} K \neq 2, 3$; the general case is [185, Appendix A]. By Lemma 9.3(iii) (a change of coordinates with $u = 1/6$) both curves have short models $y^2 = x^3 - 27c_4x - 54c_6$, $y^2 = x^3 - 27c_4'x - 54c_6'$, and the change $(u, 0, 0, 0)$ replaces $(c_4, c_6)$ by $(u^{-4}c_4, u^{-6}c_6)$. So it suffices to find $u \in \bar{K}^\times$ with $c_4 = u^4c_4'$ and $c_6 = u^6c_6'$. If $j = j' = 0$ then $c_4 = c_4' = 0$ and $c_6, c_6' \neq 0$ (as $1728\Delta = -c_6^2$); take $u$ with $u^6 = c_6/c_6'$. If $j = j' = 1728$ then $c_6 = c_6' = 0$ (as $1728\Delta = c_4^3 - c_6^2$ and $j = 1728$ means $c_4^3 = 1728\Delta$); take $u^4 = c_4/c_4'$. Otherwise all of $c_4, c_6, c_4', c_6'$ are nonzero. Choose $u$ with $u^4 = c_4/c_4'$. Then $j = j'$ gives $\Delta/\Delta' = c_4^3/c_4'^3 = u^{12}$, and $c_6^2 = c_4^3 - 1728\Delta = u^{12}(c_4'^3 - 1728\Delta') = u^{12}c_6'^2$, so $c_6 = \pm u^6c_6'$; if the sign is $-$, replace $u$ by $iu$ with $i^2 = -1$, which keeps $u^4$ and changes the sign of $u^6$.

(ii) For $y^2 + y = x^3$, $b_2 = b_4 = b_8 = 0$ and $b_6 = 1$, so $(c_4, \Delta) = (0, -27)$; for $y^2 = x^3 + x$, $b_2 = b_6 = 0$, $b_4 = 2$, $b_8 = -1$, so $(c_4, \Delta) = (-48, -64)$. Hence the first curve is smooth exactly when $\operatorname{char} K \neq 3$ and the second exactly when $\operatorname{char} K \neq 2$, with $j = 0$ and $j = (-48)^3/(-64) = 1728$. For the third, put $k = 1/(j_0 - 1728)$, so $a_1 = 1$, $a_2 = a_3 = 0$, $a_4 = -36k$, $a_6 = -k$. Then $b_2 = 1$, $b_4 = -72k$, $b_6 = -4k$, $b_8 = a_1^2a_6 - a_4^2 = -k - 1296k^2$, and

$$c_4 = 1 + 1728k = \frac{j_0}{j_0 - 1728},$$

$$\Delta = k + 1296k^2 - 27 \cdot 16k^2 + 9 \cdot 288k^2 + 8 \cdot 72^3k^3 = k(1 + 1728k)^2 = \frac{j_0^2}{(j_0 - 1728)^3},$$

using $72^3 = 373248$, $8 \cdot 373248 = 1728^2$ and $1296 - 432 + 2592 = 3456 = 2 \cdot 1728$. So $j = c_4^3/\Delta = j_0^3(j_0 - 1728)^{-3} \cdot (j_0 - 1728)^3 j_0^{-2} = j_0$, and $\Delta \neq 0$ because $j_0 \neq 0$. □

**Proposition 9.7** (automorphisms)**.** *Let* $\operatorname{char} K \neq 2,3$ *and let* $E$ *be an elliptic curve over* $K$*. The group* $\operatorname{Aut}_{\bar K}(E)$ *of automorphisms of* $E$ *fixing* $O$ *is cyclic of order* 2 *if* $j \neq 0, 1728$*, of order* 4 *if* $j = 1728$*, and of order* 6 *if* $j = 0$*. On a short model* $y^2 = x^3 + Ax + B$ *the automorphisms are* $(x, y) \mapsto (u^2x, u^3y)$ *with* $u^4A = A$*,* $u^6B = B$*; and* $[-1](x, y) = (x, -y - a_1x - a_3)$ *on a general model.*

*Proof.* By Proposition 9.5(ii) an automorphism of the short model is a substitution $(u, r, s, t)$ preserving the shape: $a_1' = 0$ forces $s = 0$, then $a_2' = 0$ forces $r = 0$, and then $a_3' = 0$ forces $t = 0$. So it is $(u, 0, 0, 0)$ with $A = u^4A$, $B = u^6B$ (here $x = u^2x'$, so the map $(x', y') \mapsto (x, y)$ is $(u^2x', u^3y')$). On the short model $b_2 = 0$, $b_4 = 2A$, $b_6 = 4B$, so $c_4 = -48A$, $c_6 = -864B$, $\Delta = -16(4A^3 + 27B^2)$ and

$$j = 1728\,\frac{4A^3}{4A^3 + 27B^2};$$

thus $j = 0$ iff $A = 0$, and $j = 1728$ iff $B = 0$. If $AB \neq 0$ ($j \neq 0, 1728$) the conditions are $u^4 = u^6 = 1$, which mean $u^2 = u^6/u^4 = 1$; if $B = 0$ ($j = 1728$), $u^4 = 1$; if $A = 0$ ($j = 0$), $u^6 = 1$. Distinct $u$ give distinct maps, since $u = u^3/u^2$ is read off the map. So $\operatorname{Aut}_{\bar K}(E) \simeq \mu_n(\bar K)$ with $n = 2, 4, 6$. Since $\operatorname{char} K \nmid n$, the polynomial $X^n - 1$ is separable and $\mu_n(\bar K)$ has $n$ elements; a finite subgroup of $\bar K^\times$ is cyclic (Chapter 1). The formula for $[-1]$ is the substitution $(u, r, s, t) = (-1, 0, -a_1, -a_3)$, which sends $(x, y)$ to the other point with the same $x$. □

**Example 9.8** (the invariants of the running examples)**.** The table lists the global minimal models of Cremona's tables [36] and their invariants, all recomputed from the $a$-invariants with PARI/GP.

| curve | $[a_1, a_2, a_3, a_4, a_6]$ | $c_4$ | $c_6$ | $\Delta$ | $j$ |
|---|---|---|---|---|---|
| `11a1` | $[0, -1, 1, -10, -20]$ | $2^4 \cdot 31$ | 20008 | $-11^5$ | $-2^{12}31^3/11^5$ |
| `11a3` | $[0, -1, 1, 0, 0]$ | $2^4$ | $-152$ | $-11$ | $-2^{12}/11$ |
| `20a1` | $[0, 1, 0, 4, 4]$ | $-2^4 \cdot 11$ | $-2368$ | $-2^8 \cdot 5^2$ | $2^4 \cdot 11^3/5^2$ |
| `99a1` | $[1, -1, 1, -2, 0]$ | $3^4$ | 135 | $3^3 \cdot 11$ | $3^9/11$ |
| `121a1` | $[1, 1, 1, -30, -76]$ | $11 \cdot 131$ | 54703 | $-11^2$ | $-11 \cdot 131^3$ |
| `121c1` | $[1, 1, 0, -2, -7]$ | $11^2$ | 5203 | $-11^4$ | $-11^2$ |
| `121c2` | $[1, 1, 0, -3632, 82757]$ | $11^3 \cdot 131$ | $-72809693$ | $-11^8$ | $-11 \cdot 131^3$ |
| `121d1` | $[0, -1, 1, -40, -221]$ | $2^4 \cdot 11^2$ | 202312 | $-11^7$ | $-2^{12}/11$ |
| `539c1` | $[1, 0, 1, 170, -3237]$ | $-7^2 \cdot 167$ | 2808827 | $-7^9 \cdot 11^2$ | $167^3/(7^3 \cdot 11^2)$ |
| `726a1` | $[1, 1, 0, -35, -51]$ | $5 \cdot 11 \cdot 31$ | 31339 | $2^6 \cdot 3^3 \cdot 11^3$ | $5^3 \cdot 31^3/(2^6 \cdot 3^3)$ |
| `980i1` | $[0, 0, 0, 1568, -72716]$ | $-2^9 \cdot 3 \cdot 7^2$ | 62826624 | $-2^8 \cdot 5 \cdot 7^{11}$ | $2^{19}3^3/(5 \cdot 7^5)$ |
| `77c1` | $[1, 1, 0, 4, 11]$ | $-167$ | $-8189$ | $-7^3 \cdot 11^2$ | $167^3/(7^3 \cdot 11^2)$ |
| `140b1` | $[0, 0, 0, 32, 212]$ | $-2^9 \cdot 3$ | $-183168$ | $-2^8 \cdot 5 \cdot 7^5$ | $2^{19}3^3/(5 \cdot 7^5)$ |
| `539d1` | $[0, 1, 1, -16, -66]$ | $2^4 \cdot 7^2$ | 52136 | $-7^6 \cdot 11$ | $-2^{12}/11$ |

For instance, for `11a1`: $b_2 = -4$, $b_4 = -20$, $b_6 = -79$, $b_8 = -21$, so $c_4 = 16 + 480 = 496$, $c_6 = 64 + 2880 + 17064 = 20008$ and $\Delta = -16 \cdot (-21) + 64000 - 27 \cdot 6241 + 9 \cdot (-4)(-20)(-79) = -161051 = -11^5$. The valuations $v_q(j)$ feed the paper's term $t_\ell(E, q)$: $v_7(j(\texttt{539c1})) = -3$, so $t_3(\texttt{539c1}, 7) = v_3(3) = 1$; $v_7(j(\texttt{980i1})) = -5$, so $t_5(\texttt{980i1}, 7) = 1$; and $v_{11}(j(\texttt{121a1})) = 1 \geq 0$, so $t_\ell(\texttt{121a1}, 11) = 0$. Curves with the same $j$ in the table (`539c1` and `77c1`; `980i1` and `140b1`; `11a3`, `121d1` and `539d1`; `121a1` and `121c2`) are quadratic twists of one another (Section 9.10).

*In the paper* 9.9. Definition def:terms of the paper defines $t_\ell(E, q) = v_\ell(-v_q(j_E))$ when $v_q(j_E) < 0$. That this is a property of $E$ and not of the chosen equation is Proposition 9.5(iii); the appendix

computes it from `vj = valuation(j, q)` with `j = E.j`. In the proof of Lemma lem:twist (ii) the twisted model "is minimal exactly when the result is less than 12, otherwise 12 is subtracted": that $v_q(\Delta)$ can only change by multiples of 12 is Proposition 9.5(iii), and the minimality criterion itself is Chapter 10.

## 9.2 The group law

The points of an elliptic curve form an abelian group, and every group the paper writes down is a group of points or a subgroup of one: $E(\mathbb{Q})_{\mathrm{tors}}$ in Conjecture conj:eis, $E[\ell]$ in Section 1.1, $E(\mathbb{F}_q)$ behind the numbers $a_q$. This section gives the chord-tangent construction with its explicit formulas, identifies it with the group $\mathrm{Pic}^0(E)$ of Chapter 8 (which gives associativity for free), and proves the translation invariance of the invariant differential, which is what makes multiplication by $m$ separable when $\mathrm{char}\, K \nmid m$.

**Construction 9.10** (the chord-tangent law)**.** Let $E$ be a smooth Weierstrass cubic over $K$ and $P, Q \in E(\bar{K})$. The line through $P$ and $Q$ (the tangent line at $P$ if $P = Q$) meets $E$ in exactly three points counted with multiplicity (Chapter 8, Bezout); let $R$ be the third. The line through $R$ and $O$ meets $E$ in a third point, which is defined to be $P + Q$. For an affine point $R = (x, y)$, the line through $R$ and $O = [0 : 1 : 0]$ is the vertical line through $x$, and its third point is

$$-R = (x, -y - a_1 x - a_3).$$

Hence $P + Q = -R$, and $P + Q + R = O$ whenever $P$, $Q$, $R$ are the three intersections of a line with $E$ (the line at infinity $Z = 0$ meets $E$ only at $O$, with multiplicity 3).

**Proposition 9.11** (the addition formulas)**.** *Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be affine points of $E$.*

*(i) If $x_1 = x_2$ and $y_2 = -y_1 - a_1 x_1 - a_3$, then $P_1 + P_2 = O$.*

*(ii) Otherwise put, if $x_1 \neq x_2$,*

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}, \qquad \nu = \frac{y_1 x_2 - y_2 x_1}{x_2 - x_1},$$

*and, if $P_1 = P_2$ (so $2y_1 + a_1 x_1 + a_3 \neq 0$),*

$$\lambda = \frac{3x_1^2 + 2a_2 x_1 + a_4 - a_1 y_1}{2y_1 + a_1 x_1 + a_3}, \qquad \nu = \frac{-x_1^3 + a_4 x_1 + 2a_6 - a_3 y_1}{2y_1 + a_1 x_1 + a_3}.$$

*Then $P_1 + P_2 = (x_3, y_3)$ with $x_3 = \lambda^2 + a_1\lambda - a_2 - x_1 - x_2$ and $y_3 = -(\lambda + a_1)x_3 - \nu - a_3$.*

*(iii) (Duplication.) If $2P \neq O$, then with $F$ as in Lemma 9.3,*

$$x(2P) = \frac{x^4 - b_4 x^2 - 2b_6 x - b_8}{4x^3 + b_2 x^2 + 2b_4 x + b_6} \quad (x = x(P)).$$

*(iv) The points $P \neq O$ with $2P = O$ are those with $2y + a_1 x + a_3 = 0$; when $\mathrm{char}\, K \neq 2$ their $x$-coordinates are the three distinct roots of $F(x) = 4x^3 + b_2 x^2 + 2b_4 x + b_6$, so $E[2]$ has four elements.*

*Proof.* (i) $P_2 = -P_1$ by the formula for $-R$. (ii) The line $y = \lambda x + \nu$ passes through $P_1$ and $P_2$. For $x_1 \neq x_2$ this is the two-point form of a line: $\lambda x_1 + \nu = (y_2x_1 - y_1x_1 + y_1x_2 - y_2x_1)/(x_2 - x_1) = y_1$, and similarly at $P_2$. For $P_1 = P_2$, write $\eta_1 = 2y_1 + a_1x_1 + a_3$ and $\xi_1 = 3x_1^2 + 2a_2x_1 + a_4 - a_1y_1$; implicit differentiation of (9.1) gives $\eta\,dy = \xi\,dx$, so $\lambda = \xi_1/\eta_1$ is the slope of the tangent, and $\nu = y_1 - \lambda x_1 = (y_1\eta_1 - x_1\xi_1)/\eta_1$. Here $y_1\eta_1 - x_1\xi_1 = 2(y_1^2 + a_1x_1y_1) + a_3y_1 - 3x_1^3 - 2a_2x_1^2 - a_4x_1$, and replacing $y_1^2 + a_1x_1y_1$ by $x_1^3 + a_2x_1^2 + a_4x_1 + a_6 - a_3y_1$ (the equation of $E$ at $P_1$) leaves $-x_1^3 + a_4x_1 + 2a_6 - a_3y_1$, the displayed numerator. Substituting $y = \lambda x + \nu$ into (9.1) gives a monic cubic in $x$,

$$x^3 + (a_2 - \lambda^2 - a_1\lambda)x^2 + \cdots = 0,$$

whose roots are $x_1, x_2, x_3$ (with multiplicity). Comparing the $x^2$-coefficients gives $x_1 + x_2 + x_3 = \lambda^2 + a_1\lambda - a_2$. The third point is $R = (x_3, \lambda x_3 + \nu)$, and $P_1 + P_2 = -R$. (iii) Substitute the tangent slope into $x_3 = \lambda^2 + a_1\lambda - a_2 - 2x_1$. When $\operatorname{char} K \neq 2$ the computation is short. Write $x = x_1$, $\eta = \eta_1$, $\xi = \xi_1$; then $\lambda^2 + a_1\lambda = (\lambda + \frac{a_1}{2})^2 - \frac{a_1^2}{4}$ and

$$\eta\Big(\lambda + \frac{a_1}{2}\Big) = \xi + \frac{a_1}{2}\eta = 3x^2 + \Big(2a_2 + \frac{a_1^2}{2}\Big)x + a_4 + \frac{a_1a_3}{2} = \frac{1}{4}F'(x),$$

the term $a_1y$ cancelling. With $\eta^2 = F(x)$ (Lemma 9.3(ii)) and $a_1^2/4 + a_2 = b_2/4$,

$$x(2P) = \frac{F'(x)^2}{16F(x)} - \frac{b_2}{4} - 2x = \frac{F'(x)^2 - 4(b_2 + 8x)F(x)}{16F(x)}.$$

Expanding, $F'(x)^2 = 144x^4 + 48b_2x^3 + (4b_2^2 + 48b_4)x^2 + 8b_2b_4x + 4b_4^2$ and $4(b_2 + 8x)F(x) = 128x^4 + 48b_2x^3 + (4b_2^2 + 64b_4)x^2 + (8b_2b_4 + 32b_6)x + 4b_2b_6$; the difference is $16x^4 - 16b_4x^2 - 32b_6x - 4(b_2b_6 - b_4^2) = 16(x^4 - b_4x^2 - 2b_6x - b_8)$, by Lemma 9.3(i). In characteristic 2 the same formula holds: the statement says $\xi^2 + a_1\xi\eta - (a_2 + 2x)\eta^2 = x^4 - b_4x^2 - 2b_6x - b_8$ in $\mathbb{Z}[a_1, \ldots, a_6][x, y]$ modulo the equation of $E$, a ring that is free with basis $1, y$ over $\mathbb{Z}[a_1, \ldots, a_6][x]$; the computation just made proves it after tensoring with $\mathbb{Q}$, hence over $\mathbb{Z}$ (a free module over a torsion-free ring has no torsion), hence after any specialisation (it was also checked by machine). On `11a1` at $P = (5, 5)$ it gives $x(2P) = 1936/121 = 16$. (iv) $2P = O$ means $P = -P$, i.e. $y = -y - a_1x - a_3$. In characteristic not 2 this is $\eta = 0$, so $F(x) = 0$ by Lemma 9.3(ii); conversely each root of $F$ gives exactly one point with $\eta = 0$. The roots are distinct because $\operatorname{disc} F = 16\Delta \neq 0$. □

**Proposition 9.12** (the chord-tangent law is the law of $\mathrm{Pic}^0$)**.** *Let $\kappa\colon E(\bar{K}) \to \mathrm{Pic}^0(E)$, $P \mapsto$ the class of $(P) - (O)$, be the bijection of Chapter 8. Then $\kappa(P + Q) = \kappa(P) + \kappa(Q)$ for the chord-tangent law of Construction 9.10.*

*Proof.* Let $P, Q$ be given, $R$ the third point of the line $L$ through them, and $P + Q = -R$ the third point of the line $L'$ through $R$ and $O$. Write $L$ and $L'$ as linear forms in $X, Y, Z$. The rational functions $L/Z$ and $L'/Z$ have divisors

$$\operatorname{div}(L/Z) = (P)+(Q)+(R)-3(O), \qquad \operatorname{div}(L'/Z) = (R)+(-R)+(O)-3(O) = (R)+(-R)-2(O),$$

because $L$ meets $E$ in $P, Q, R$, and $Z$ meets $E$ only at $O$ with multiplicity 3 (if $L$ or $L'$ is the line $Z = 0$ itself the corresponding statement is trivial). Hence $\operatorname{div}(L/L') = (P)+(Q)-(P+Q)-(O)$, so $(P) - (O) + (Q) - (O) \sim (P + Q) - (O)$, which is the claim. □

**Theorem 9.13** (the group law)**.**

*(i)* $(E(\bar K),+)$ *is an abelian group with identity* $O$ *and inverse* $P\mapsto -P$.

*(ii) For every field* $L$ *with* $K\subseteq L\subseteq \bar K$, $E(L)$ *is a subgroup, and* $\sigma(P+Q)=\sigma P+\sigma Q$ *for every automorphism* $\sigma$ *of* $\bar K$ *over* $K$.

*(iii) The maps* $E\times E\to E$, $(P,Q)\mapsto P+Q$, *and* $E\to E$, $P\mapsto -P$, *are morphisms (given by rational functions with coefficients in* $K$*); in particular each translation* $\tau_Q\colon P\mapsto P+Q$ *is an isomorphism of curves, defined over* $K$ *if* $Q\in E(K)$.

*Proof.* (i) Commutativity is built into the construction (the line through $P$ and $Q$ is the line through $Q$ and $P$); $P+O=P$ because the line through $P$ and $O$ meets $E$ again at $-P$, and the line through $-P$ and $O$ meets it at $P$; and $P+(-P)=O$ because the vertical line through $P$ meets $E$ in $P$, $-P$, $O$, and the tangent at $O$ is $Z=0$, which meets $E$ only at $O$. Associativity holds because $\kappa$ is a bijection onto the group $\mathrm{Pic}^0(E)$ that carries $+$ to the group law there (Proposition 9.12). A direct verification from the formulas is possible but long; see [185, Section III.2] and the references there. (ii) The formulas of Proposition 9.11 have coefficients in $K$, so they commute with $\sigma$ and preserve coordinates in $L$. (iii) The formulas of Proposition 9.11 are rational functions; that they define a morphism also at the excluded points (where $x_1=x_2$, or a point is $O$) is [185, Section III.3]; for $\tau_Q$ with $Q$ fixed it follows from Chapter 8 (a rational map from a smooth curve to a projective variety is a morphism), and $\tau_{-Q}$ is its inverse. □

**Corollary 9.14** (principal divisors)**.** *A divisor* $D=\sum n_P(P)$ *on* $E$ *is principal if and only if* $\deg D=0$ *and* $\sum[n_P]P=O$ *in* $E(\bar K)$.

*Proof.* For $\deg D=0$, $D=\sum n_P\big((P)-(O)\big)$ has class $\sum n_P\kappa(P)=\kappa\big(\sum[n_P]P\big)$ by Proposition 9.12. Since $\kappa$ is injective, the class is 0 iff $\sum[n_P]P=O$. Principal divisors have degree 0 (Chapter 8). □

**Proposition 9.15** (the invariant differential)**.** *Let* $\omega$ *be the invariant differential of* $E$.

*(i)* $\omega$ *is holomorphic and has no zeros:* $\operatorname{div}(\omega)=0$.

*(ii)* $\tau_Q^*\omega=\omega$ *for every* $Q\in E(\bar K)$.

*(iii) Let* $E'$ *be an elliptic curve and* $\varphi,\psi\colon E'\to E$ *morphisms with* $\varphi(O)=\psi(O)=O$ *(isogenies, Definition 9.17). Then* $(\varphi+\psi)^*\omega=\varphi^*\omega+\psi^*\omega$, *where* $\varphi+\psi$ *is the pointwise sum and the pullback by the zero map is* 0. *Hence* $[m]^*\omega=m\,\omega$ *for every* $m\in\mathbb{Z}$.

*Proof.* (i) is Chapter 8 (the divisor of $dx/(2y+a_1x+a_3)$ on a smooth Weierstrass cubic). For (ii) and (iii) put $\eta=2y+a_1x+a_3$ and $\xi=3x^2+2a_2x+a_4-a_1y$. Differentiating (9.1) gives $\eta\,dy=\xi\,dx$, and $\omega=dx/\eta$. For a morphism $\varphi$ from a smooth curve to $E$ we put $\varphi^*\omega=0$ if $\varphi$ is constant; if $\varphi$ is nonconstant, $\varphi^*\eta\neq 0$ (pullback is injective on function fields) and $\varphi^*\omega=d(\varphi^*x)/\varphi^*\eta$.

*Step 1 (an identity).* Let $x_1,y_1,x_2,y_2$ be indeterminates, $W(x,y)=y^2+a_1xy+a_3y-x^3-a_2x^2-a_4x-a_6$, and let $X_3$, $Y_3$ be the expressions for $x_3$, $y_3$ in the chord case of Proposition 9.11(ii), so that their denominators are powers of $x_2-x_1$. Put $H_3=2Y_3+a_1X_3+a_3$, and $\eta_i$, $\xi_i$ for $\eta$, $\xi$ evaluated at $(x_i,y_i)$. Then for $i=1,2$

$$\eta_i\,\frac{\partial X_3}{\partial x_i}+\xi_i\,\frac{\partial X_3}{\partial y_i}=H_3 \tag{9.2}$$

in the ring $\mathbb{Z}[a_1,a_2,a_3,a_4,a_6][x_1,y_1,x_2,y_2,(x_1-x_2)^{-1}]$ modulo the ideal generated by $W(x_1,y_1)$ and $W(x_2,y_2)$. Since $W$ is monic of degree 2 in $y$, the difference of the two sides can be reduced, by subtracting multiples of $W(x_1,y_1)$ and $W(x_2,y_2)$, to an expression of degree at most 1 in $y_1$ and in $y_2$, without new denominators. We carried out this reduction with PARI/GP, with $a_1,\dots,a_6$ indeterminate, and it gives 0. So (9.2) holds after any specialisation of the variables into a field in which $W(x_1,y_1)=W(x_2,y_2)=0$ and $x_1\neq x_2$, in every characteristic.

*Step 2 (sums of morphisms).* Let $C$ be a smooth curve over $\bar{K}$ and $\varphi,\psi\colon C\to E$ morphisms, neither of them the constant map to $O$, such that $x\circ\varphi\neq x\circ\psi$ and the pointwise sum $\varphi+\psi$ is a nonconstant morphism. We show $(\varphi+\psi)^*\omega=\varphi^*\omega+\psi^*\omega$. Put $x_1=\varphi^*x$, $y_1=\varphi^*y$, $x_2=\psi^*x$, $y_2=\psi^*y$ in $\bar{K}(C)$; then $W(x_i,y_i)=0$ and $x_1-x_2\neq 0$. The sets $\varphi^{-1}(O)$ and $\psi^{-1}(O)$ are finite, and $x_1-x_2$ has finitely many zeros, so at all but finitely many $P\in C$ the points $\varphi(P)$, $\psi(P)$ are affine with different $x$-coordinates, and there Proposition 9.11(ii) gives $(\varphi+\psi)(P)=(X_3,Y_3)\big(x_1(P),y_1(P),x_2(P),y_2(P)\big)$. Two rational functions on $C$ that agree at infinitely many points are equal, so $(\varphi+\psi)^*x=X_3(x_1,y_1,x_2,y_2)$ and $(\varphi+\psi)^*\eta=H_3(x_1,y_1,x_2,y_2)$, the latter nonzero because $\varphi+\psi$ is nonconstant. By the chain rule for the derivation $d$ of $\bar{K}(C)$,

$$d\big((\varphi+\psi)^*x\big)=\sum_{i=1,2}\Big(\frac{\partial X_3}{\partial x_i}\,dx_i+\frac{\partial X_3}{\partial y_i}\,dy_i\Big).$$

If $\varphi$ is nonconstant, $\eta_1=\varphi^*\eta\neq 0$ and pulling back $\eta\,dy=\xi\,dx$ gives $dy_1=(\xi_1/\eta_1)\,dx_1$; so the term $i=1$ is $(\eta_1\partial X_3/\partial x_1+\xi_1\partial X_3/\partial y_1)dx_1/\eta_1=H_3\,\varphi^*\omega$ by (9.2). If $\varphi$ is constant, $dx_1=dy_1=0$ and the term is $0=H_3\,\varphi^*\omega$. The same holds for $i=2$. Dividing by $H_3=(\varphi+\psi)^*\eta$ gives the claim.

(ii) For $Q=O$ there is nothing to prove. For $Q\neq O$ apply Step 2 with $C=E$, $\varphi$ the identity and $\psi$ the constant map to $Q$: then $\varphi+\psi=\tau_Q$ is an isomorphism (Theorem 9.13(iii)) and $x\neq x(Q)$, so $\tau_Q^*\omega=\omega+0=\omega$.

(iii) First, $x\circ[-1]=x$ and $y\circ[-1]=-y-a_1x-a_3$ (Construction 9.10), so $[-1]^*\eta=-\eta$ and $[-1]^*\omega=-\omega$. The maps $\varphi\pm\psi$ are morphisms sending $O$ to $O$ (Theorem 9.13(iii)), and such a morphism is either the zero map or nonconstant, in which case its fibres, in particular its kernel, are finite (Chapter 8). If $\varphi=0$ or $\psi=0$ there is nothing to prove. If $\psi=-\varphi$, then $(\varphi+\psi)^*\omega=0$ and $\psi^*\omega=\varphi^*[-1]^*\omega=-\varphi^*\omega$. If $\psi=\varphi$: the set $E[2]$ is finite, since its points other than $O$ are the zeros of the nonzero function $\eta$ (Proposition 9.11(iv)); choose $Q$ with $2Q\neq O$. Then $\tau_Q+\tau_{-Q}=[2]$ is nonconstant, and $x\circ\tau_Q\neq x\circ\tau_{-Q}$: otherwise for every $P$ the points $P+Q$ and $P-Q$ would be equal or opposite, so $2Q=O$ or $2P=O$, and $E(\bar{K})=E[2]$ would be finite. Step 2 and (ii) give $[2]^*\omega=\tau_Q^*\omega+\tau_{-Q}^*\omega=2\omega$, hence $(2\varphi)^*\omega=([2]\circ\varphi)^*\omega=\varphi^*[2]^*\omega=2\varphi^*\omega$. In the remaining case $\varphi,\psi\neq 0$ and $\psi\neq\pm\varphi$, so $\varphi+\psi$ is nonconstant; and $x\circ\varphi\neq x\circ\psi$, for otherwise $\varphi(P)=\pm\psi(P)$ for every $P$ (two points with the same $x$-coordinate in $\mathbb{P}^1$ are equal or opposite), and $E'(\bar{K})$, which is infinite, would be the union of the finite kernels of $\varphi-\psi$ and $\varphi+\psi$. Step 2 applies.

Finally $[0]^*\omega=0$, $[m+1]^*\omega=[m]^*\omega+\omega$ for $m\geq 0$ by what was just proved, so $[m]^*\omega=m\,\omega$ for $m\geq 0$, and $[-m]^*\omega=([-1]\circ[m])^*\omega=[m]^*(-\omega)=-m\,\omega$. Silverman [185, Section III.5] argues differently, proving (iii) through differentials on $E\times E$; the explicit identity of Step 1 replaces that argument here. □

**Example 9.16** (torsion points on the running examples)**.** All computed with Proposition 9.11 (and checked with PARI).

(a) On `11a1`, $P=(5,5)$: $2P=(16,-61)$, $3P=(16,60)$, $4P=(5,-6)$, $5P=O$. Indeed $3P=-2P$ because $-(16,-61)=(16,61-1)=(16,60)$. In detail, with $a_1=0$, $a_2=-1$,

$a_3 = 1$, $a_4 = -10$: the tangent at $P$ has $\lambda = (3 \cdot 25 - 10 - 10)/(10 + 1) = 5$ and $\nu = 5 - 25 = -20$, so $x(2P) = 25 + 0 + 1 - 10 = 16$ and $y(2P) = -5 \cdot 16 + 20 - 1 = -61$. The chord through $P$ and $2P$ has $\lambda = (-61 - 5)/(16 - 5) = -6$ and $\nu = (5 \cdot 16 + 61 \cdot 5)/11 = 35$, so $x(3P) = 36 + 1 - 5 - 16 = 16$ and $y(3P) = 6 \cdot 16 - 35 - 1 = 60$. Since $x(3P) = x(2P)$ and $3P \neq 2P$, $3P = -2P$, so $5P = O$; and $4P = -P = (5, -5 - 1)$.

(b) On `20a1` $= [0, 1, 0, 4, 4]$, that is $y^2 = (x+1)(x^2+4)$: $(-1, 0)$ has order 2 (it satisfies $2y = 0$), $(0, 2)$ has order 3, and $(4, 10)$ has order 6, with $2(4, 10) = (0, 2)$ and $3(4, 10) = (-1, 0)$. For the doubling, $\lambda = (48 + 8 + 4)/20 = 3$, $\nu = 10 - 12 = -2$, $x = 9 - 1 - 8 = 0$ and $y = 2$. For $(4, 10) + (0, 2)$, $\lambda = 8/4 = 2$, $\nu = (0 - 8)/(-4) = 2$, $x = 4 - 1 - 4 - 0 = -1$ and $y = 2 - 2 = 0$.

(c) On `14a1` $= [1, 0, 1, 4, -6]$: $(2, -5)$ has order 3 (with $2(2, -5) = (2, 2) = -(2, -5)$), and $(9, 23)$ has order 6, with $2(9, 23) = (2, 2)$, $3(9, 23) = (1, -1)$.

(d) On `27a1` $= [0, 0, 1, 0, -7]$: $(3, 4)$ has order 3, $2(3, 4) = (3, -5)$.

(e) On `99a1`, $(0, 0)$ has infinite order: its multiples $(3, 2)$, $(-8/9, 13/27)$, $(105/64, -897/512)$ have growing denominators, which torsion points cannot have (Proposition 9.71); its canonical height is $0.3025\ldots > 0$.

The group law makes the maps $[m]$ and, more generally, maps of curves preserving $O$ into the natural morphisms between elliptic curves. Two facts from this section drive the next one: $E(\bar{K}) \to \mathrm{Pic}^0(E)$ is a group isomorphism (so kernels and pullbacks of divisors can be computed with Corollary 9.14), and $[m]^*\omega = m\omega$ (so $[m]$ is separable when $\mathrm{char}\, K \nmid m$, and $E[m]$ can be counted).

## 9.3 Isogenies

The paper works with isogeny classes: the $X_0(N)$-optimal curve is one curve of a class, a rational $\ell$-isogeny is the Eisenstein case of Conjecture conj:eis, and Proposition prop:no9 is an argument about a cyclic isogeny of degree 27. This section defines isogenies and their degree, states the existence of quotients by finite subgroups and of the dual isogeny, and proves the consequences the book needs: $\deg[m] = m^2$, $E[m] \simeq (\mathbb{Z}/m)^2$ when $\mathrm{char}\, K \nmid m$, and the factorisation of an isogeny into a multiplication map and a cyclic isogeny.

**Definition 9.17** (isogenies)**.** Let $E_1$, $E_2$ be elliptic curves over $K$. An *isogeny* $\varphi\colon E_1 \to E_2$ is a morphism of curves with $\varphi(O) = O$. It is *defined over* $K$ if it is given by rational functions with coefficients in $K$; for perfect $K$ this holds iff $\varphi(\sigma P) = \sigma\varphi(P)$ for all $\sigma \in G_K$ and $P \in E_1(\bar{K})$ (Chapter 8). Its *degree* is $\deg\varphi = [\bar{K}(E_1) : \varphi^*\bar{K}(E_2)]$ if $\varphi$ is nonconstant and $\deg 0 = 0$; $\deg_s\varphi$ and $\deg_i\varphi$ are the separable and inseparable degrees, and $\varphi$ is *separable* if $\deg_i\varphi = 1$ (Chapter 8). $\mathrm{Hom}(E_1, E_2)$ is the set of isogenies (over $\bar{K}$ unless stated otherwise), $\mathrm{End}(E) = \mathrm{Hom}(E, E)$, and $[m]\colon E \to E$ is multiplication by $m \in \mathbb{Z}$. $E_1$ and $E_2$ are *isogenous over* $K$ if there is a nonconstant isogeny $E_1 \to E_2$ defined over $K$. For $m \geq 1$, $E[m] = \{P \in E(\bar{K}) : mP = O\}$.

**Proposition 9.18** (first properties)**.**

*(i) An isogeny is either constant (the zero map) or surjective on $\bar{K}$-points.*

*(ii)* $\mathrm{Hom}(E_1, E_2)$ *is an abelian group under* $(\varphi+\psi)(P) = \varphi(P)+\psi(P)$, $\mathrm{End}(E)$ *is a ring under composition, and* $\deg(\psi\circ\varphi) = \deg\psi\cdot\deg\varphi$.

*(iii)* *For* $m \neq 0$, $[m] \neq 0$. *Hence* $\mathrm{Hom}(E_1, E_2)$ *is torsion-free, and* $\mathrm{End}(E)$ *has characteristic* 0.

*Proof.* (i) is Chapter 8 (a morphism of curves is constant or surjective). (ii) $\varphi+\psi$ is the composite of $(\varphi,\psi)\colon E_1 \to E_2 \times E_2$ with the addition morphism (Theorem 9.13(iii)), hence a morphism, and it sends $O$ to $O$. The group and ring axioms hold pointwise. For nonconstant $\varphi$, $\psi$ the degree is multiplicative in the tower $\bar{K}(E_1) \supseteq \varphi^*\bar{K}(E_2) \supseteq \varphi^*\psi^*\bar{K}(E_3)$; if one of them is constant both sides are 0. (iii) Assume first $\mathrm{char}\, K \neq 2$ and write $m = \pm 2^a m'$ with $m'$ odd; as degrees multiply, it suffices to show $[2] \neq 0$ and $[m'] \neq 0$. By Proposition 9.11(iv) the points with $2P = O$ are $O$ and three points with $2y + a_1x + a_3 = 0$; since $E(\bar{K})$ is infinite, $[2] \neq 0$. Let $T$ be one of the three points of order 2 (a root of the cubic $F$). Then $m'T = T \neq O$, so $[m'] \neq 0$. In characteristic 2 the same argument works with [3] in place of [2] and a point of order 3 in place of $T$ (the $x$-coordinates of the points of order 3 are roots of $\psi_3$, of degree 4 in characteristic 2; Proposition 9.29(i)); see [185, Section III.4] for the details. Finally, if $n\varphi = 0$ with $n \neq 0$, then $[n]\circ\varphi = 0$; a nonzero $\varphi$ is surjective by (i), so $[n]$ would vanish on $E_2(\bar{K})$, which we have excluded. Hence $\varphi = 0$. □

**Theorem 9.19** (isogenies are homomorphisms; kernels)**.** *Let* $\varphi\colon E_1 \to E_2$ *be a nonconstant isogeny.*

*(i)* $\varphi(P+Q) = \varphi(P)+\varphi(Q)$ *for all* $P, Q$.

*(ii)* $\ker\varphi$ *is finite of order* $\deg_s\varphi$. *Every fibre* $\varphi^{-1}(Q)$ *has* $\deg_s\varphi$ *points, each with ramification index* $\deg_i\varphi$.

*(iii)* *If* $\varphi$ *is separable, then* $\#\ker\varphi = \deg\varphi$, $\varphi$ *is unramified, and* $\bar{K}(E_1)/\varphi^*\bar{K}(E_2)$ *is a Galois extension whose group is* $\{\tau_T^* : T \in \ker\varphi\} \simeq \ker\varphi$.

*Sketch of proof.* (i) The pushforward $\varphi_*$ on divisors maps principal divisors to principal divisors (Chapter 8), so it induces a homomorphism $\mathrm{Pic}^0(E_1) \to \mathrm{Pic}^0(E_2)$, and $\varphi_*((P)-(O)) = (\varphi P)-(O)$. Through the bijections $\kappa$ of Proposition 9.12 this says that $\varphi$ is a homomorphism. (ii) For $T \in \ker\varphi$, $\varphi\circ\tau_T = \varphi$ by (i), so $\tau_T$ permutes each fibre and preserves ramification indices: $e_\varphi(P+T) = e_\varphi(P)$. Hence all points of a fibre $\varphi^{-1}(Q) = P + \ker\varphi$ have the same index $e$, and the degree formula of Chapter 8 gives $\#\ker\varphi\cdot e = \deg\varphi$. Since all but finitely many fibres have exactly $\deg_s\varphi$ points (Chapter 8), $\#\ker\varphi = \deg_s\varphi$ and $e = \deg_i\varphi$. (iii) If $\varphi$ is separable, $e = 1$ everywhere and $\#\ker\varphi = \deg\varphi$. The $\deg\varphi$ automorphisms $\tau_T^*$, $T \in \ker\varphi$, of $\bar{K}(E_1)$ fix $\varphi^*\bar{K}(E_2)$ (as $\varphi\circ\tau_T = \varphi$) and are distinct; an extension of degree $n$ with $n$ automorphisms over the base is Galois with that group (Chapter 5). This is [185, Section III.4]. In characteristic 0 every morphism of genus-one curves is separable and unramified (Chapter 8, Riemann-Hurwitz), which gives (iii) directly. □

**Theorem 9.20** (quotients by finite subgroups)**.**

*(i)* *Let* $H \subset E(\bar{K})$ *be a finite subgroup. There are an elliptic curve* $E'$ *and a separable isogeny* $\varphi\colon E \to E'$ *with* $\ker\varphi = H$; *the pair is unique up to a unique isomorphism of* $E'$. *If* $H$ *is stable as a set under* $G_K$, *then* $E'$ *and* $\varphi$ *can be taken to be defined over* $K$. *We write* $E' = E/H$.

*(ii) (Factorisation.) Let $\varphi\colon E \to E_1$ and $\psi\colon E \to E_2$ be nonconstant isogenies with $\psi$ separable and $\ker\psi \subseteq \ker\varphi$. Then there is a unique isogeny $\lambda\colon E_2 \to E_1$ with $\varphi = \lambda\circ\psi$; if $\varphi$ and $\psi$ are defined over $K$, so is $\lambda$, and $\ker\lambda = \psi(\ker\varphi)$.*

We do not prove this; see [185, Section III.4 and Exercise 3.13]. The idea of (ii): by Theorem 9.19(iii), $\varphi^*\bar{K}(E_1)$ is fixed by the translations by $\ker\psi$, hence contained in $\psi^*\bar{K}(E_2)$, and an inclusion of function fields is a morphism (Chapter 8); $\lambda$ is unique because $\psi$ is surjective, and $\lambda^\sigma\circ\psi = \varphi^\sigma = \varphi$ gives $\lambda^\sigma = \lambda$. The quotient in (i) is the curve whose function field is the fixed field of the translations by $H$. Vélu's formulas [200] give an explicit Weierstrass equation for $E/H$ and for $\varphi$ in terms of the coordinates of the points of $H$ (PARI's `ellisogeny`).

**Theorem 9.21** (the dual isogeny)**.** *Let $\varphi\colon E_1 \to E_2$ be a nonconstant isogeny of degree $m$. There is a unique isogeny $\hat\varphi\colon E_2 \to E_1$ with $\hat\varphi\circ\varphi = [m]_{E_1}$. Moreover:*

*(i) $\varphi\circ\hat\varphi = [m]_{E_2}$, $\deg\hat\varphi = \deg\varphi$ and $\hat{\hat\varphi} = \varphi$;*

*(ii) $\widehat{\psi\circ\varphi} = \hat\varphi\circ\hat\psi$, $\widehat{\varphi+\psi} = \hat\varphi + \hat\psi$ (with $\hat 0 = 0$), and $\widehat{[m]} = [m]$;*

*(iii) if $\varphi$ is defined over $K$, so is $\hat\varphi$;*

*(iv) $\deg\colon \mathrm{Hom}(E_1,E_2) \to \mathbb{Z}$ is a positive definite quadratic form: $\deg(n\varphi) = n^2\deg\varphi$, $\deg\varphi \geq 0$ with equality only for $\varphi = 0$, and $\langle\varphi,\psi\rangle = \deg(\varphi+\psi) - \deg\varphi - \deg\psi$ is bilinear.*

*Sketch of proof.* $\hat\varphi$ is the composite $E_2 \xrightarrow{\kappa} \mathrm{Pic}^0(E_2) \xrightarrow{\varphi^*} \mathrm{Pic}^0(E_1) \xrightarrow{\kappa^{-1}} E_1$: since $\varphi^*((Q)) = \sum_{\varphi(P)=Q} e_\varphi(P)(P)$ and all points of a fibre $P + \ker\varphi$ have index $\deg_i\varphi$ (Theorem 9.19(ii)), $\varphi^*((\varphi P) - (O)) = \deg_i\varphi\sum_{T\in\ker\varphi}((P+T)-(T))$, whose image under $\kappa^{-1}$ is $[\deg_i\varphi\cdot\#\ker\varphi]P = [\deg\varphi]P$ by Corollary 9.14. Uniqueness: if $\psi\circ\varphi = \psi'\circ\varphi$ then $(\psi-\psi')\circ\varphi = 0$ and $\psi = \psi'$ as $\varphi$ is surjective. The additivity $\widehat{\varphi+\psi} = \hat\varphi + \hat\psi$ is the deepest property; see [185, Section III.6]. The rest follows formally: from $\varphi\circ\hat\varphi\circ\varphi = \varphi\circ[m] = [m]\circ\varphi$ and surjectivity of $\varphi$ one gets $\varphi\circ\hat\varphi = [m]$; then $\deg\varphi\deg\hat\varphi = \deg[m] = m^2$ (Corollary 9.22(i), whose proof uses only $\widehat{[m]} = [m]$, itself a consequence of additivity and $[\hat 1] = [1]$), so $\deg\hat\varphi = m$. For (iv), $[\deg(\varphi+\psi)] = (\widehat{\varphi+\psi})\circ(\varphi+\psi) = [\deg\varphi] + [\deg\psi] + \hat\psi\circ\varphi + \hat\varphi\circ\psi$, so $\langle\varphi,\psi\rangle$ corresponds to $\hat\psi\circ\varphi + \hat\varphi\circ\psi$, which is visibly biadditive; and $\mathrm{End}(E_1)$ is torsion-free (Proposition 9.18(iii)), so the identification of integers with multiplication maps is injective ($[a] = [b]$ forces $a = b$). The remaining parts of (iv) are elementary: $\deg(n\varphi) = \deg([n]\circ\varphi) = \deg[n]\deg\varphi = n^2\deg\varphi$ by multiplicativity and $\deg[n] = n^2$; and $\deg\varphi$ is a degree of a field extension, hence positive, for $\varphi \neq 0$, while $\deg 0 = 0$ by definition. A function $Q\colon A \to \mathbb{Z}$ on an abelian group with $Q(nx) = n^2Q(x)$ and $Q(x+y) - Q(x) - Q(y)$ biadditive is exactly a quadratic form. □

**Corollary 9.22** (multiplication by $m$ and the $m$-torsion)**.** *Let $m \geq 1$.*

*(i) $\deg[m] = m^2$.*

*(ii) If $\mathrm{char}\,K \nmid m$, then $[m]$ is separable and $E[m] \simeq \mathbb{Z}/m \times \mathbb{Z}/m$.*

*(iii) If $\mathrm{char}\,K = p > 0$, then either $E[p^e] = 0$ for all $e \geq 1$, or $E[p^e] \simeq \mathbb{Z}/p^e$ for all $e \geq 1$.*

*Proof.* (i) By Theorem 9.21, $\widehat{[m]} = [m]$ and $\widehat{[m]}\circ[m] = [\deg[m]]$, so $[m^2] = [\deg[m]]$ and $[m^2 - \deg[m]] = 0$; as $\mathrm{End}(E)$ is torsion-free (Proposition 9.18(iii)), $\deg[m] = m^2$. (ii) By Proposition 9.15(iii), $[m]^*\omega = m\omega \neq 0$, so $[m]$ is separable (Chapter 8: a morphism is separable iff

it pulls back a nonzero differential to a nonzero one). The same holds for every $d \mid m$, as $\operatorname{char} K \nmid d$; so by Theorem 9.19(iii) and (i), $\#E[d] = \#\ker[d] = \deg[d] = d^2$ for every $d \mid m$. Let $A = E[m]$; it is a finite abelian group killed by $m$. For each prime $p \mid m$, $A[p] = E[p]$ has $p^2$ elements. By the structure theorem (Chapter 1) the $p$-primary part of $A$ is a product $\mathbb{Z}/p^{a_1} \times \cdots \times \mathbb{Z}/p^{a_k}$ with all $a_i \geq 1$, whose $p$-torsion $(\mathbb{Z}/p)^k$ has $p^k$ elements; so $k = 2$, the $p$-primary part is $\mathbb{Z}/p^a \times \mathbb{Z}/p^b$, and $a, b \leq v_p(m)$ because $m$ kills $A$. Since $\#A = m^2$, $a + b = 2v_p(m)$, so $a = b = v_p(m)$, and $A \simeq (\mathbb{Z}/m)^2$. (iii) We do not prove this; see [185, Section III.6]. Over $\mathbb{C}$, Proposition 9.55(ii) gives a second proof of (i) and (ii). □

**Proposition 9.23** (cyclic factorisation)**.** *Let* $\operatorname{char} K = 0$ *and let* $\varphi\colon E \to E'$ *be a nonconstant isogeny.*

*(i)* $\ker\varphi \simeq \mathbb{Z}/a \times \mathbb{Z}/ab$ *for unique integers* $a, b \geq 1$*, and* $a$ *is the largest integer with* $E[a] \subseteq \ker\varphi$*.*

*(ii)* $\varphi = \psi \circ [a]$ *for a unique isogeny* $\psi\colon E \to E'$*, and* $\ker\psi = a \cdot \ker\varphi \simeq \mathbb{Z}/b$ *is cyclic. If* $\varphi$ *is defined over* $K$*, so is* $\psi$*.*

*(iii) An isogeny over* $K$ *with cyclic kernel of order* $n$ *is called a* cyclic $n$-isogeny*. Its kernel* $C$ *contains exactly one subgroup* $C_d$ *of each order* $d \mid n$*, and* $C_d$ *is* $G_K$*-stable. Consequently a cyclic* $n$*-isogeny over* $K$ *is a composite of cyclic isogenies over* $K$ *of prime degrees, and if* $\ell \mid n$ *then* $E$ *has a* $K$*-rational cyclic* $\ell$*-isogeny, with kernel* $C_\ell$*.*

*Proof.* (i) $H = \ker\varphi$ is a finite subgroup of $E[N] \simeq (\mathbb{Z}/N)^2$ for $N = \#H$ (Corollary 9.22(ii)). For each prime $p$, $H[p] \subseteq E[p] \simeq (\mathbb{Z}/p)^2$, so the $p$-primary part of $H$ has at most two cyclic factors, and $H \simeq \mathbb{Z}/d_1 \times \mathbb{Z}/d_2$ with $d_1 \mid d_2$ (invariant factors, Chapter 1); put $a = d_1$, $b = d_2/d_1$. Then $H[a] \simeq (\mathbb{Z}/a)^2$ has $a^2$ elements and lies in $E[a]$, which has $a^2$ elements, so $E[a] \subseteq H$; and if $E[a'] \subseteq H$ then $E[a'] \simeq (\mathbb{Z}/a')^2$ lies in $H[a'] \simeq \mathbb{Z}/\gcd(a', d_1) \times \mathbb{Z}/\gcd(a', d_2)$, a group of order at most $\gcd(a', d_1) \cdot a'$; comparing orders, $\gcd(a', d_1) = a'$, that is, $a' \mid d_1$. The integers $a$, $b$ are unique because the invariant factors $d_1 \mid d_2$ are. (ii) $[a]$ is separable with kernel $E[a] \subseteq \ker\varphi$, so Theorem 9.20(ii) gives $\psi$ with $\varphi = \psi \circ [a]$, unique, defined over $K$ if $\varphi$ is, and with kernel $[a](\ker\varphi) = a \cdot (\mathbb{Z}/a \times \mathbb{Z}/ab) \simeq \mathbb{Z}/b$. (iii) A cyclic group of order $n$ has exactly one subgroup of each order $d \mid n$. For $\sigma \in G_K$, $\sigma(C_d)$ is a subgroup of order $d$ of $\sigma(C) = C$, so $\sigma(C_d) = C_d$. For a prime $p \mid n$, the isogeny $E \to E/C_p$ is defined over $K$ (Theorem 9.20(i)), and $\varphi$ factors through it (Theorem 9.20(ii)) as $E \to E/C_p \to E'$, the second map having cyclic kernel $C/C_p$ of order $n/p$, $G_K$-stable. Induction on $n$ finishes the proof. □

*Warning* 9.24 (isogenous over which field). In the paper and in this book "isogenous" and "isogeny class" for curves over $\mathbb{Q}$ always mean over $\mathbb{Q}$. Two curves over $\mathbb{Q}$ can be isogenous over $\overline{\mathbb{Q}}$ without being isogenous over $\mathbb{Q}$: a curve and its quadratic twist by $\mathbb{Q}(\sqrt{d})$ are isomorphic over $\mathbb{Q}(\sqrt{d})$ (Definition 9.83), but in general not isogenous over $\mathbb{Q}$: `11a1` has $a_2 = -2$, while its twist `121d2` lies in the class of `121d1`, which has $a_2 = 2$ (Example 9.68), so the two are not isogenous over $\mathbb{Q}$ (Theorem 9.74(ii)).

The finite groups $E[m] \simeq (\mathbb{Z}/m)^2$ are now in place, but nothing so far says where their points lie. The next section writes down polynomials whose roots are the $x$-coordinates of the points of $E[m]$; for $m = 3$ this is the tool the paper uses to read off the mod 3 image.

## 9.4 Division polynomials

The paper decides whether the mod 3 image of $E$ is the normaliser $N_s(\mathbb{F}_3)$ of a split Cartan subgroup by factoring the 3-division polynomial (Proposition prop:ns and the function `ns3` of its appendix, which calls `elldivpol(E,3)`). This section defines the division polynomials, proves their degrees, and proves that the four roots of $\psi_3$ are the $x$-coordinates of the four lines of $E[3]$, Galois-equivariantly. The group-theoretic side (which orbit patterns occur for which subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$) is in Chapters 1 and 5; the deduction of Proposition prop:ns is in Chapter 11.

**Definition 9.25** (division polynomials)**.** Let $R = \mathbb{Z}[a_1, a_2, a_3, a_4, a_6]$, $F(x) = 4x^3 + b_2x^2 + 2b_4x + b_6 \in R[x]$, and let $A = R[x][\psi_2]/(\psi_2^2 - F(x))$, a free $R[x]$-module with basis $1, \psi_2$. Define $\psi_m \in A$ by

$$\psi_0 = 0, \quad \psi_1 = 1, \quad \psi_2 = \psi_2, \quad \psi_3 = 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8,$$
$$\psi_4 = \psi_2(2x^6 + b_2x^5 + 5b_4x^4 + 10b_6x^3 + 10b_8x^2 + (b_2b_8 - b_4b_6)x + b_4b_8 - b_6^2),$$

and recursively, for $m \geq 2$ and $m \geq 3$ respectively,

$$\psi_{2m+1} = \psi_{m+2}\psi_m^3 - \psi_{m-1}\psi_{m+1}^3, \qquad \psi_2\psi_{2m} = \psi_m(\psi_{m+2}\psi_{m-1}^2 - \psi_{m-2}\psi_{m+1}^2),$$

where the second formula defines $\psi_{2m}$ because its right side is $\psi_2$ times an element of $A$ (Proposition 9.26), and multiplication by $\psi_2$ is injective on $A$. On a curve $E$ we evaluate $\psi_m$ at the coefficients of $E$ and put $\psi_2 = 2y + a_1x + a_3$, which satisfies $\psi_2^2 = F(x)$ on $E$ (Lemma 9.3(ii)); so $\psi_m$ becomes a function on $E$. These are the conventions of [185, Exercise 3.7] and of PARI; [202, Section 3.2] uses short models.

The idea behind the definition: $[m]$ is a morphism $E \to E$, so $x(mP)$ is a rational function of the coordinates of $P$, and its denominator should vanish exactly at the points with $mP = O$. The polynomials $\psi_m$ are chosen to be these denominators (more precisely, $x(mP) = x - \psi_{m-1}\psi_{m+1}/\psi_m^2$, Theorem 9.27), and the recursions express $\psi_{2m}$ and $\psi_{2m+1}$ through the addition of $mP$ and $(m+1)P$; the first cases are the duplication formula ($\psi_2$, $\psi_3$, Proposition 9.29(i)) and $\psi_4$. As an example, for `11a1` $= [0, -1, 1, -10, -20]$, with $b_2 = -4$, $b_4 = -20$, $b_6 = -79$, $b_8 = -21$ (Example 9.8), one gets $\psi_2^2 = F(x) = 4x^3 - 4x^2 - 40x - 79$ and $\psi_3 = 3x^4 - 4x^3 - 60x^2 - 237x - 21$, the polynomial listed in Example 9.30.

**Proposition 9.26** (degrees)**.** *For odd $m \geq 1$, $\psi_m \in R[x]$ is a polynomial of degree $(m^2-1)/2$ with leading coefficient $m$. For even $m \geq 2$, $\psi_m = \psi_2 f_m$ with $f_m \in R[x]$ of degree $(m^2-4)/2$ and leading coefficient $m/2$.*

*Proof.* For $m \leq 4$ this is read off the definition ($f_2 = 1$; $f_4$ has degree 6 and leading coefficient 2). Assume it for all indices below the one considered; we use $\psi_2^2 = F$, of degree 3 with leading coefficient 4. Write $\psi_k$ for odd $k$ and $\psi_k = \psi_2 f_k$ for even $k$.

$\psi_{2n+1}$, $n \geq 2$. If $n$ is even, $\psi_{n+2}\psi_n^3 = \psi_2^4 f_{n+2} f_n^3 = F^2 f_{n+2} f_n^3$ has degree $6 + \frac{(n+2)^2-4}{2} + 3\frac{n^2-4}{2} = 2n^2 + 2n$ and leading coefficient $16 \cdot \frac{n+2}{2} \cdot \frac{n^3}{8} = (n+2)n^3$, while $\psi_{n-1}\psi_{n+1}^3$ (odd indices) has degree $\frac{(n-1)^2-1}{2} + 3\frac{(n+1)^2-1}{2} = 2n^2 + 2n$ and leading coefficient $(n-1)(n+1)^3$. If $n$ is odd the roles are exchanged: $\psi_{n+2}\psi_n^3$ has leading coefficient $(n+2)n^3$ and $\psi_{n-1}\psi_{n+1}^3 = F^2 f_{n-1} f_{n+1}^3$ has $16 \cdot \frac{n-1}{2} \cdot \frac{(n+1)^3}{8} = (n-1)(n+1)^3$, both of degree $2n^2 + 2n$. In both cases $\psi_{2n+1} \in R[x]$ has degree at most $2n^2 + 2n = \frac{(2n+1)^2-1}{2}$ and coefficient of that power

$$(n+2)n^3 - (n-1)(n+1)^3 = 2n + 1.$$

$\psi_{2n}$, $n \geq 3$. If $n$ is odd, $\psi_{n\pm1}^2 = Ff_{n\pm1}^2$, so the bracket is $F(\psi_{n+2}f_{n-1}^2 - \psi_{n-2}f_{n+1}^2)$ and $\psi_{2n} = \psi_2 \cdot \psi_n(\psi_{n+2}f_{n-1}^2 - \psi_{n-2}f_{n+1}^2)$. The two products in the bracket have degree $\frac{(n\pm2)^2-1}{2} + (n \mp 1)^2 - 4 = \frac{3n^2-3}{2}$ and leading coefficients $(n+2)\frac{(n-1)^2}{4}$ and $(n-2)\frac{(n+1)^2}{4}$, whose difference is $\frac{1}{4}((n+2)(n-1)^2 - (n-2)(n+1)^2) = \frac{4}{4} = 1$. So $f_{2n}$ has degree $\frac{n^2-1}{2} + \frac{3n^2-3}{2} = 2n^2 - 2 = \frac{(2n)^2-4}{2}$ and leading coefficient $n$. If $n$ is even, $\psi_n = \psi_2 f_n$ and $\psi_{n\pm2} = \psi_2 f_{n\pm2}$, so $\psi_{2n} = \psi_2 \cdot f_n(f_{n+2}\psi_{n-1}^2 - f_{n-2}\psi_{n+1}^2)$; the bracket has degree $\frac{(n\pm2)^2-4}{2} + (n \mp 1)^2 - 1 = \frac{3n^2}{2}$ and leading coefficient $\frac{1}{2}((n+2)(n-1)^2 - (n-2)(n+1)^2) = 2$, so $f_{2n}$ has degree $\frac{n^2-4}{2} + \frac{3n^2}{2} = 2n^2 - 2$ and leading coefficient $\frac{n}{2} \cdot 2 = n$. (For $n = 4$ the term $f_{n-2} = f_2 = 1$; the case $n - 2 = 0$ does not arise since $n \geq 3$.) In particular the right side of the even recursion is always $\psi_2$ times an element of $R[x]$, as Definition 9.25 requires. □

For `11a1` the degrees $0, 3, 4, 9, 12, 19, 24, 33, 40$ and leading coefficients $1, 4, 3, 8, 5, 12, 7, 16, 9$ of PARI's `elldivpol(E,m)`, $m = 1, \ldots, 9$, agree with Proposition 9.26: for even $m$ PARI returns $\psi_2\psi_m = Ff_m$, of degree $3 + (m^2 - 4)/2$ and leading coefficient $4 \cdot m/2$.

**Theorem 9.27** (multiplication by $m$)**.** *Let $m \geq 2$ with $\operatorname{char} K \nmid m$, and $P = (x, y) \in E(\bar{K})$. Then $mP = O$ if and only if $\psi_m(P) = 0$, and if $mP \neq O$,*

$$x(mP) = x - \frac{\psi_{m-1}\psi_{m+1}}{\psi_m^2}(P).$$

*There is a similar formula for $y(mP)$ with denominator $\psi_m^3$. Equivalently,*

$$\operatorname{div}(\psi_m) = \sum_{T \in E[m]} (T) - m^2(O).$$

*Sketch of proof.* The formulas are proved by induction on $m$ from the addition formulas of Proposition 9.11, which is what the recursions of Definition 9.25 encode; see [202, Section 3.2] for short models and [185, Exercise 3.7] in general. For $m = 2$ the formula is the duplication formula, as $x - \psi_1\psi_3/\psi_2^2 = x - \psi_3/F = x(2P)$ by Proposition 9.29(i); for $m = 3$ the statement "$3P = O$ iff $\psi_3(P) = 0$" is Proposition 9.29(i). The form with divisors: $\psi_m$ has poles only at $O$, of order $m^2 - 1$ (by Proposition 9.26, as $\operatorname{ord}_O x = -2$, $\operatorname{ord}_O y = -3$), and its zeros are the $m^2 - 1$ points of $E[m] \smallsetminus \{O\}$ (Corollary 9.22(ii)), each simple. □

**Corollary 9.28** (roots of $\psi_m$ for odd $m$)**.** *Let $m$ be odd with $\operatorname{char} K \nmid m$. The roots of the polynomial $\psi_m(x)$ in $\bar{K}$ are exactly the $x$-coordinates of the points of $E[m] \smallsetminus \{O\}$; they are simple, there are $(m^2-1)/2$ of them, and $P \mapsto x(P)$ is a $G_K$-equivariant bijection $(E[m] \smallsetminus \{O\})/\{\pm1\} \to \{$roots of $\psi_m\}$.*

*Proof.* By Theorem 9.27 every $x(P)$, $P \in E[m] \smallsetminus \{O\}$, is a root. Two points have the same $x$-coordinate iff they are equal or opposite (the vertical line meets $E$ in $P$, $-P$, $O$). No point of $E[m] \smallsetminus \{O\}$ has order 2 since $m$ is odd, so $P \neq -P$, and the number of distinct values $x(P)$ is $(m^2 - 1)/2$ (Corollary 9.22(ii)). This is the degree of $\psi_m$, whose leading coefficient $m$ is nonzero in $K$ (Proposition 9.26); so these values are all the roots, each simple. Equivariance: $\psi_m$ has coefficients in $K$ and $x(\sigma P) = \sigma(x(P))$. □

**Proposition 9.29** (the 3-division polynomial)**.** *Let $\psi_3 = 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8$.*

*(i) As rational functions on $E$, $x(2P) - x(P) = -\psi_3(x)/\psi_2(P)^2$. Hence for $P \neq O$ with $2P \neq O$: $3P = O$ if and only if $\psi_3(x(P)) = 0$.*

*(ii) Let* $\operatorname{char} K \neq 2,3$. *Then* $E[3]$ *is a two-dimensional* $\mathbb{F}_3$*-vector space with four lines* $\{O,P,-P\}$, *and* $\{O,P,-P\} \mapsto x(P)$ *is a* $G_K$*-equivariant bijection from the set of lines of* $E[3]$ *onto the set of the four (distinct) roots of* $\psi_3$. *Consequently the degrees of the irreducible factors of* $\psi_3$ *over* $K$ *are the sizes of the orbits of* $\bar\rho_{E,3}(G_K)$ *on* $\mathbb{P}^1(\mathbb{F}_3)$ *(Definition 9.36);* $\psi_3$ *has a root in* $K$ *iff* $E[3]$ *has a* $G_K$*-stable line, iff* $E$ *has a* 3*-isogeny defined over* $K$ *(Proposition 9.75).*

*(iii) Let* $K=\mathbb{Q}$. *Then* $\operatorname{disc}(\psi_3) \in -3\cdot(\mathbb{Q}^\times)^2$, *and the splitting field of* $\psi_3$ *contains* $\mathbb{Q}(\sqrt{-3})$.

*Proof.* (i) By the duplication formula (Proposition 9.11(iii)),

$$\begin{aligned} x(2P)-x &= \frac{x^4-b_4x^2-2b_6x-b_8-x(4x^3+b_2x^2+2b_4x+b_6)}{F(x)} \\ &= -\frac{3x^4+b_2x^3+3b_4x^2+3b_6x+b_8}{F(x)}, \end{aligned}$$

and $F(x)=\psi_2^2$ on $E$. If $P\neq O$ and $2P\neq O$, then $3P=O$ iff $2P=-P$ iff $x(2P)=x(P)$ (as $2P\neq P$), iff $\psi_3(x(P))=0$. (ii) $E[3]\simeq(\mathbb{Z}/3)^2$ by Corollary 9.22(ii); its lines are the four subgroups of order 3, each of the form $\{O,P,-P\}$. By Corollary 9.28 with $m=3$, $x$ gives a $G_K$-equivariant bijection from the lines onto the four distinct roots of $\psi_3$. For a separable polynomial over $K$, the irreducible factors over $K$ correspond to the $G_K$-orbits on its roots (Chapter 5); after a choice of basis, $G_K$ acts on the lines of $E[3]$ through $\bar\rho_{E,3}$ and the action of $\mathrm{GL}_2(\mathbb{F}_3)$ on $\mathbb{P}^1(\mathbb{F}_3)$. A root in $K$ is a fixed root, that is, a $G_K$-stable line. (iii) The roots $r_1,\dots,r_4$ are distinct, so $\delta=\prod_{i<j}(r_i-r_j)\neq 0$, $\operatorname{disc}\psi_3=3^6\delta^2$, and $\sigma(\delta)=\operatorname{sgn}(\pi_\sigma)\delta$ where $\pi_\sigma$ is the permutation of the roots induced by $\sigma\in G_\mathbb{Q}$. (The factor $3^6=3^{2\cdot4-2}$ is the leading coefficient of $\psi_3$ to the power $2\deg-2$.) By (ii), $\pi_\sigma$ is the permutation of the four lines of $E[3]$ induced by $\bar\rho_{E,3}(\sigma)$, whose sign is $\det\bar\rho_{E,3}(\sigma)\in\{\pm1\}=\mathbb{F}_3^\times$ (Chapter 1). To see this directly, note that $g\mapsto\operatorname{sgn}(\pi_g)$ and det are both homomorphisms $\mathrm{GL}_2(\mathbb{F}_3)\to\{\pm1\}$, so it suffices to compare them on generators. The group $\mathrm{SL}_2(\mathbb{F}_3)$ is generated by the transvections $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$; the first fixes the line $[1:0]$ and permutes $[0:1]\mapsto[1:1]\mapsto[2:1]\mapsto[0:1]$ cyclically, an even permutation, and likewise for the second. And $\mathrm{GL}_2(\mathbb{F}_3)$ is generated by $\mathrm{SL}_2(\mathbb{F}_3)$ and $\operatorname{diag}(1,-1)$, which fixes $[1:0]$ and $[0:1]$ and swaps $[1:1]$ and $[1:-1]$, an odd permutation, with determinant $-1$. (A direct enumeration of the 48 elements confirms this.) Finally $\det\bar\rho_{E,3}=\bar\chi_3=\chi_{-3}$ (Corollary 9.33). So $\sigma$ fixes $\delta$ iff $\sigma$ fixes $\sqrt{-3}$; by Galois theory $\mathbb{Q}(\delta)=\mathbb{Q}(\sqrt{-3})$, and as the nontrivial automorphism of this field sends $\delta$ to $-\delta$, $\delta=b\sqrt{-3}$ with $b\in\mathbb{Q}^\times$, so $\delta^2=-3b^2\in-3(\mathbb{Q}^\times)^2$ and $\operatorname{disc}\psi_3=3^6\delta^2\in-3(\mathbb{Q}^\times)^2$. □

**Example 9.30** (factorisations of $\psi_3$)**.** The following factorisations over $\mathbb{Q}$ and discriminants were computed with PARI (`elldivpol(E,3)`, `factor`, `poldisc`); the image of $\bar\rho_{E,3}$ is taken from Cremona's `galrep` file (labels of [191]; no label means surjective; their meaning is Chapter 11).

| curve | $\psi_3$ over $\mathbb{Q}$ | $\operatorname{disc}\psi_3$ | image at 3 |
|---|---|---|---|
| `11a1` | $3x^4-4x^3-60x^2-237x-21$ (irreducible) | $-3^3\cdot11^{10}$ | $\mathrm{GL}_2(\mathbb{F}_3)$ |
| `121a1` | $3x^4+5x^3-177x^2-909x-1249$ (irreducible) | $-3^3\cdot11^4$ | $\mathrm{GL}_2(\mathbb{F}_3)$ |
| `726a1` | $(x^2-x-74)(3x^2+8x+20)$ | $-2^{12}3^911^6$ | `3Ns` |
| `338d1` | $(x^2+29x+1858)(3x^2-82x-172)$ | $-2^63^313^{18}$ | `3Ns` |
| `6912d1` | $3(x^2-4x-2)(x^2+4x+6)$ | $-2^{18}3^9$ | `3Ns` |
| `20a1` | $x(3x^3+4x^2+24x+48)$ | $-2^{16}3^35^4$ | `3B.1.1` |
| `14a1` | $(x-2)(3x+1)(x^2+2x+13)$ | $-2^{12}3^37^6$ | `3Cs.1.1` |
| `27a1` (CM) | $3x(x-3)(x^2+3x+9)$ | $-3^{21}$ | `3Cs.1.1` |
| `121b1` (CM) | $(x^2+3x-6)(3x^2-13x+15)$ | $-3^3\cdot11^6$ | $N_s(\mathbb{F}_3)$, not listed |

Every discriminant is $-3$ times a square, as Proposition 9.29(iii) says. For `726a1` the quadratic factors have discriminants $297 = 3^3 \cdot 11$ and $-176 = -2^4 \cdot 11$, so their roots generate $\mathbb{Q}(\sqrt{33})$ and $\mathbb{Q}(\sqrt{-11})$, and the third quadratic subfield of the compositum is $\mathbb{Q}(\sqrt{-3})$. For `20a1` the root $x = 0$ is the $x$-coordinate of the rational point $(0, 2)$ of order 3 (Example 9.16); the cubic factor is irreducible, so the other three lines form one orbit. For `14a1` the root 2 is the $x$-coordinate of $(2, -5)$, of order 3; the root $-1/3$ gives a second rational line, whose points $(-1/3, y)$ satisfy $y^2 + \frac{2}{3}y + \frac{199}{27} = 0$, of discriminant $-\frac{784}{27} = -3 \cdot (\frac{28}{9})^2$, so they are defined over $\mathbb{Q}(\sqrt{-3})$ and not over $\mathbb{Q}$: this line is $\mu_3$ (Proposition 9.77). The curve `121b1` has complex multiplication by $\mathbb{Q}(\sqrt{-11})$, in which 3 splits. Its $\psi_3$ has no root in $\mathbb{Q}$, so $E[3]$ is irreducible (Proposition 9.29(ii)); the image has orbits of sizes $2, 2$ on $\mathbb{P}^1(\mathbb{F}_3)$ and contains the image of complex conjugation (determinant $-1$, trace 0), so by the classification recalled in Paper link 9.31 it is exactly $N_s(\mathbb{F}_3)$, the normaliser of the split Cartan subgroup attached to the two primes above 3 in $\mathbb{Q}(\sqrt{-11})$. The `galrep` file gives no label at 3 for this curve: for CM curves it often omits labels (it does give `3Cs.1.1` for the CM curves `27a1` and `27a3`).

*In the paper* 9.31. Proposition prop:ns says that, for $E[3]$ irreducible, the 3-division polynomial factors as a product of two irreducible quadratics when the image is $N_s(\mathbb{F}_3)$ and is irreducible otherwise. The bridge between the image and the polynomial is Proposition 9.29(ii): the orbits of $N_s(\mathbb{F}_3)$ on the four lines have sizes $2, 2$, and the other irreducible subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ containing an element of determinant $-1$ and trace 0 (complex conjugation, Corollary 9.39), namely the normaliser of a nonsplit Cartan subgroup and $\mathrm{GL}_2(\mathbb{F}_3)$ itself, act transitively (Chapter 1; the deduction is in Chapter 11). The paper's function `ns3` applies this to `elldivpol(E,3)`; a reducible $E[3]$ gives a linear factor. The CM curve `121b1` of the table has the pattern $2 + 2$ and image exactly $N_s(\mathbb{F}_3)$, consistently with Proposition prop:ns; the paper excludes CM curves from its data, and `ns3` would classify `121b1` correctly. Section sec:anomaly uses `726a1`. The proof of Proposition prop:ns itself is Chapter 11.

## 9.5 The Weil pairing, the Tate module and the Galois representations

Section 1.1 of the paper begins with $V = V_\ell E$, $T = T_\ell E$, $\rho = \rho_{E,\ell}$ and its reduction $\bar{\rho}$; every later statement is about these objects or about the adjoint $\mathrm{ad}^0 T$ built from them. This section constructs them, proves that $\det \rho_{E,\ell}$ is the cyclotomic character (through the Weil pairing), identifies $T_\ell E$ with its own dual twisted by $\mathbb{Z}_\ell(1)$, and fixes the Frobenius conventions. The paper uses these facts in Section 1.1, in Section 4.2 (the module $T^D$ and the identification $M_{f,\lambda} = H^1(E, \mathbb{Z}_\ell)$ after Proposition prop:pullback), in Lemma lem:h0 (complex conjugation, $\det \rho = \chi_\ell$), and in Proposition prop:ns ($\det \bar{\rho}_{E,3} = \chi_{-3}$).

**Theorem 9.32** (the Weil pairing)**.** *Let $m \geq 1$ with $\operatorname{char} K \nmid m$. There is a pairing $e_m \colon E[m] \times E[m] \to \mu_m$ with the following properties.*

*(i) Bilinear: $e_m(S_1 + S_2, T) = e_m(S_1, T)e_m(S_2, T)$ and $e_m(S, T_1 + T_2) = e_m(S, T_1)e_m(S, T_2)$.*

*(ii) Alternating: $e_m(T, T) = 1$; hence $e_m(S, T) = e_m(T, S)^{-1}$.*

*(iii) Nondegenerate: if $e_m(S, T) = 1$ for all $S$, then $T = O$.*

*(iv) Galois-equivariant: $e_m(\sigma S, \sigma T) = \sigma(e_m(S, T))$ for $\sigma \in G_K$.*

*(v) Compatible:* $e_{mm'}(S,T) = e_m(m'S,T)$ *for* $S \in E[mm']$ *and* $T \in E[m]$.

*(vi) Adjoint: for an isogeny* $\varphi\colon E \to E'$, $S \in E[m]$, $T \in E'[m]$: $e_m(S, \hat{\varphi}T) = e_m(\varphi S, T)$.

*Sketch of proof.* For $T \in E[m]$ there is $f \in \bar{K}(E)$ with $\operatorname{div}(f) = m(T) - m(O)$ (Corollary 9.14). Choose $T'$ with $mT' = T$; there is $g$ with $\operatorname{div}(g) = [m]^*(T) - [m]^*(O) = \sum_{R \in E[m]}\big((T'+R)-(R)\big)$, again by Corollary 9.14, since $\sum_R (T' + R - R) = [m^2]T' = O$. Then $f \circ [m]$ and $g^m$ have the same divisor, so after scaling $f \circ [m] = g^m$. For $S \in E[m]$ and any point $X$, $g(X+S)^m = f(mX + mS) = f(mX) = g(X)^m$, so $g(X+S)/g(X)$ is an $m$-th root of unity; as a function of $X$ it is a morphism $E \to \mathbb{P}^1$ that is not surjective, hence constant (Chapter 8). Define $e_m(S,T) = g(X+S)/g(X)$. Properties (i) to (vi) are proved in [185, Section III.8]; (iv) is immediate because $\sigma$ carries the functions $f, g$ for $T$ to functions with the same properties for $\sigma T$. Over $\mathbb{C}$, Proposition 9.55(iv) gives an explicit formula. □

The properties already determine $e_2$ completely when $\operatorname{char} K \neq 2$. Let $S \neq T$ be two points of order 2. By (iii) applied to $T$, together with (ii), some $S' \in E[2]$ has $e_2(S',T) = -1$; since $e_2(O,T) = e_2(T,T) = 1$, $S'$ is one of the two points $S$, $S+T$, and $e_2(S+T,T) = e_2(S,T)e_2(T,T) = e_2(S,T)$, so $e_2(S,T) = -1$. Thus $e_2(S,T) = -1$ for any two distinct points of order 2, and $e_2 = 1$ whenever one argument is $O$ or the two arguments are equal. For larger $m$ the pairing carries genuine information; its first use is that it computes the determinant of the Galois action.

**Corollary 9.33** (the determinant)**.** *Let* $m \geq 1$ *with* $\operatorname{char} K \nmid m$.

*(i) For* $\sigma \in G_K$ *acting on* $E[m]$ *by a matrix* $g_\sigma \in \mathrm{GL}_2(\mathbb{Z}/m)$ *in some basis,* $\det g_\sigma = \chi_m^{\mathrm{cyc}}(\sigma)$.

*(ii) If* $E[m] \subseteq E(K)$ *then* $\mu_m \subseteq K$.

*(iii) Let* $\ell$ *be prime and* $L \subset E[\ell]$ *a* $G_K$*-stable line. Then* $E[\ell]/L \simeq \operatorname{Hom}(L, \mu_\ell)$ *as* $G_K$*-modules. If* $G_K$ *acts on* $L$ *by* $\eta_1$ *and on* $E[\ell]/L$ *by* $\eta_2$*, then* $\eta_1\eta_2 = \bar{\chi}_\ell$. *In particular a* $K$*-rational point of order* $\ell$ *gives an exact sequence of* $G_K$*-modules* $0 \to \mathbb{Z}/\ell \to E[\ell] \to \mu_\ell \to 0$.

*Proof.* (i) Let $S, T$ be the basis and $\zeta = e_m(S,T)$. It is a primitive $m$-th root of unity: if $\zeta^k = 1$ then $e_m(kS,T) = 1 = e_m(kS,S)$, so $e_m(kS,\cdot) = 1$ on $E[m]$ and $kS = O$ by (iii) and (ii) of Theorem 9.32, so $m \mid k$. If $\sigma S = aS + cT$ and $\sigma T = bS + dT$, bilinearity and the alternating property give $e_m(\sigma S, \sigma T) = e_m(S,T)^{ad} e_m(T,S)^{cb} = \zeta^{ad-bc}$, while Galois-equivariance gives $e_m(\sigma S, \sigma T) = \sigma(\zeta) = \zeta^{\chi_m^{\mathrm{cyc}}(\sigma)}$. So $\det g_\sigma \equiv \chi_m^{\mathrm{cyc}}(\sigma) \pmod m$. (ii) If $G_K$ fixes $E[m]$ pointwise then $\chi_m^{\mathrm{cyc}}(\sigma) = \det 1 = 1$ for all $\sigma$, so $G_K$ fixes $\mu_m$. (iii) Consider $\theta\colon E[\ell] \to \operatorname{Hom}(L, \mu_\ell)$, $P \mapsto (Q \mapsto e_\ell(Q,P))$, a homomorphism by bilinearity. Its kernel is $L^\perp = \{P : e_\ell(L,P) = 1\}$, which contains $L$ because $e_\ell(Q,Q') = 1$ for $Q, Q'$ in the line $L$ (write $Q' = kQ$ and use (i), (ii)). It is not all of $E[\ell]$: for a generator $Q_0$ of $L$, nondegeneracy and the alternating property give some $P$ with $e_\ell(Q_0,P) = e_\ell(P,Q_0)^{-1} \neq 1$. So $L^\perp$ is a proper subspace containing the line $L$, hence $\ker\theta = L$, and $\theta$ is onto because both $E[\ell]/L$ and $\operatorname{Hom}(L,\mu_\ell)$ have $\ell$ elements. With the action $(\sigma h)(Q) = \sigma(h(\sigma^{-1}Q))$ on $\operatorname{Hom}(L,\mu_\ell)$, $(\sigma\theta(P))(Q) = \sigma e_\ell(\sigma^{-1}Q, P) = e_\ell(Q, \sigma P)$, so $\theta$ is $G_K$-equivariant. For $h \in \operatorname{Hom}(L,\mu_\ell)$ and $Q \in L$, $(\sigma h)(Q) = \sigma\big(h(\eta_1(\sigma)^{-1}Q)\big) = \sigma\big(h(Q)\big)^{\eta_1(\sigma)^{-1}} = h(Q)^{\bar{\chi}_\ell(\sigma)\eta_1(\sigma)^{-1}}$, so $G_K$ acts on $\operatorname{Hom}(L,\mu_\ell)$ by $\bar{\chi}_\ell\eta_1^{-1}$, and $\eta_2 = \bar{\chi}_\ell\eta_1^{-1}$. If $L$ is spanned by a $K$-rational point, $\eta_1 = 1$ and $\operatorname{Hom}(L,\mu_\ell) \simeq \mu_\ell$. □

**Definition 9.34** (Tate modules)**.** For a prime $\ell \neq \operatorname{char} K$, the *$\ell$-adic Tate module* of $E$ is $T_\ell E = \varprojlim_n E[\ell^n]$, the inverse limit along the maps $[\ell]\colon E[\ell^{n+1}] \to E[\ell^n]$, and $V_\ell E = T_\ell E \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.

Also $E[\ell^\infty] = \bigcup_n E[\ell^n]$. These are the $T_\ell J$ and $V_\ell J$ of Chapter 8 for $J = E$, an elliptic curve being its own Jacobian.

**Proposition 9.35** (structure of the Tate module)**.**

*(i)* *$T_\ell E$ is a free $\mathbb{Z}_\ell$-module of rank 2, and the projection $T_\ell E \to E[\ell^n]$ is surjective with kernel $\ell^n T_\ell E$; so $T_\ell E/\ell^n T_\ell E \simeq E[\ell^n]$.*

*(ii)* *The map $V_\ell E \to E[\ell^\infty]$, $\ell^{-n}t \mapsto t_n$ (the $n$-th component of $t \in T_\ell E$), is well defined, surjective, and has kernel $T_\ell E$; so $V_\ell E/T_\ell E \simeq E[\ell^\infty] \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^2$.*

*(iii)* *The pairings $e_{\ell^n}$ assemble into a pairing $e\colon T_\ell E \times T_\ell E \to \mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^n}$, $e((S_n),(T_n)) = \big(e_{\ell^n}(S_n,T_n)\big)_n$, which is bilinear, alternating, perfect and $G_K$-equivariant.*

*(iv)* *Let $\operatorname{char} K = p > 0$ and define $T_pE = \varprojlim_n E[p^n]$ by the same formula ($E[p^n]$ being the group of $\bar{K}$-points killed by $p^n$). Then $T_pE$ is $0$ or $\mathbb{Z}_p$; this rests on Corollary 9.22(iii), which we do not prove.*

*Proof.* (i) Choose a basis $(P_1, Q_1)$ of $E[\ell] \simeq (\mathbb{Z}/\ell)^2$ and, inductively, $P_{n+1}, Q_{n+1}$ with $\ell P_{n+1} = P_n$, $\ell Q_{n+1} = Q_n$ ($[\ell]$ is surjective). The subgroup $H = \langle P_{n+1}, Q_{n+1}\rangle$ of $E[\ell^{n+1}] \simeq (\mathbb{Z}/\ell^{n+1})^2$ satisfies $\ell H = \langle P_n, Q_n\rangle = E[\ell^n]$ (induction) and $H \supseteq \ell^n H = E[\ell]$, so $\#H = \#\ell H \cdot \#H[\ell] = \ell^{2n}\cdot\ell^2$ and $H = E[\ell^{n+1}]$. In these compatible bases each $E[\ell^n]$ is $(\mathbb{Z}/\ell^n)^2$ and the transition maps are the reductions, so $T_\ell E \simeq \varprojlim (\mathbb{Z}/\ell^n)^2 = \mathbb{Z}_\ell^2$ (Chapter 1), and the projection to the $n$-th term is reduction modulo $\ell^n$. (ii) If $\ell^{-n}t = \ell^{-n'}t'$ with $n' \geq n$ then $t' = \ell^{n'-n}t$, whose $n'$-th component is $\ell^{n'-n}t_{n'} = t_n$; so the map is well defined. It is surjective by (i), and $\ell^{-n}t \mapsto 0$ iff $t_n = 0$ iff $t \in \ell^n T_\ell E$ iff $\ell^{-n}t \in T_\ell E$. (iii) Compatibility of the components: by Theorem 9.32(v), $e_{\ell^{n+1}}(S_{n+1},T_{n+1})^\ell = e_{\ell^{n+1}}(S_{n+1},\ell T_{n+1}) = e_{\ell^{n+1}}(S_{n+1},T_n) = e_{\ell^n}(\ell S_{n+1},T_n) = e_{\ell^n}(S_n,T_n)$, which is the transition map $\zeta \mapsto \zeta^\ell$ of $\mathbb{Z}_\ell(1)$. Bilinearity, the alternating property and equivariance pass to the limit. For perfectness, choose a generator of $\mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$ and a $\mathbb{Z}_\ell$-basis $s, t$ of $T_\ell E$. Since $e$ is alternating, its Gram matrix is $\left(\begin{smallmatrix} 0 & u \\ -u & 0\end{smallmatrix}\right)$ with $u = e(s,t)$, of determinant $u^2$. Modulo $\ell$ the pairing is $e_\ell$ on the basis $s_1, t_1$ of $E[\ell]$, and $e_\ell(s_1,t_1)$ is a primitive $\ell$-th root of unity (proof of Corollary 9.33(i)); so $u \not\equiv 0 \pmod{\ell}$, $u \in \mathbb{Z}_\ell^\times$, and $e$ is perfect. (iv) Given Corollary 9.22(iii), the same argument as (i) with one generator: either all $E[p^n]$ vanish, or $E[p^n] \simeq \mathbb{Z}/p^n$ with surjective transition maps $[p]$, and the limit is $\mathbb{Z}_p$. □

**Definition 9.36** (the $\ell$-adic and mod $\ell$ representations)**.** $G_K$ acts on each $E[\ell^n]$ by group automorphisms (Theorem 9.13(ii)), compatibly with $[\ell]$, hence on $T_\ell E$ and $V_\ell E$. After a choice of $\mathbb{Z}_\ell$-basis of $T_\ell E$ this is a homomorphism

$$\rho_{E,\ell}\colon G_K \to \operatorname{Aut}_{\mathbb{Z}_\ell}(T_\ell E) \simeq \mathrm{GL}_2(\mathbb{Z}_\ell),$$

and its reduction $\bar\rho_{E,\ell}\colon G_K \to \operatorname{Aut}(E[\ell]) \simeq \mathrm{GL}_2(\mathbb{F}_\ell)$ is the action on $E[\ell] = T_\ell E/\ell T_\ell E$; more generally $\rho_{E,\ell} \bmod \ell^k$ is the action on $E[\ell^k]$. Different bases give conjugate homomorphisms.

**Proposition 9.37** (properties of $\rho_{E,\ell}$)**.**

*(i)* *$\rho_{E,\ell}$ is continuous. The field $K(E[\ell^n])$ generated by the coordinates of the points of $E[\ell^n]$ is a finite Galois extension of $K$, and $\rho_{E,\ell} \bmod \ell^n$ factors through $\mathrm{Gal}(K(E[\ell^n])/K)$, on which it is injective.*

*(ii)* *$\det \rho_{E,\ell} = \chi_\ell$, the $\ell$-adic cyclotomic character, and $\det \bar\rho_{E,\ell} = \bar\chi_\ell$.*

*(iii) An isogeny $\varphi\colon E \to E'$ defined over $K$ induces an injective $G_K$-equivariant map $\varphi_\ell\colon T_\ell E \to T_\ell E'$ with $\operatorname{coker}\varphi_\ell \simeq (\ker\varphi)[\ell^\infty]$. Hence $V_\ell E \simeq V_\ell E'$, and $T_\ell E \simeq T_\ell E'$ as $G_K$-modules when $\ell \nmid \deg\varphi$.*

*Proof.* (i) $E[\ell^n]$ is finite and $G_K$-stable (Theorem 9.13(ii)), so its coordinates generate a finite extension $L_n$ stable under $G_K$, hence Galois, provided it is separable over $K$. Separability: let $m = \ell^n$ and $P \in E[m] \smallsetminus \{O\}$. If $m$ is odd, $x(P)$ is a root of $\psi_m$, a polynomial over $K$ whose roots are all simple (Corollary 9.28). If $m$ is even ($\ell = 2 \neq \operatorname{char} K$), either $2P = O$ and $x(P)$ is a root of the cubic $F$, whose roots are simple (Proposition 9.11(iv)), or $\psi_2(P) \neq 0$, and then $x(P)$ is a root of $f_m = \psi_m/\psi_2$ (Theorem 9.27); the $(m^2-4)/2$ distinct values $x(P)$ for $P \in E[m] \smallsetminus E[2]$ (points paired with their negatives) are roots of $f_m$, which has that degree and leading coefficient $m/2 \neq 0$ in $K$ (Proposition 9.26), so they are simple roots. A simple root of a polynomial over $K$ is separable over $K$. Then $y(P)$ is a root of $g(Y) = Y^2 + (a_1x(P) + a_3)Y - (x(P)^3 + a_2x(P)^2 + a_4x(P) + a_6)$, whose derivative at $y(P)$ is $\psi_2(P)$; if $\psi_2(P) \neq 0$ it is a simple root, hence separable over $K(x(P))$, and if $\psi_2(P) = 0$ (so $\operatorname{char} K \neq 2$, since in characteristic 2 the prime $\ell$ is odd and $2P \neq O$) then $y(P) = -(a_1x(P) + a_3)/2 \in K(x(P))$. So the coordinates of the points of $E[\ell^n]$ lie in $K^{\mathrm{sep}}$; this argument does not assume $K$ perfect, as [185, Section III.7] does. An element of $G_K$ acts trivially on $E[\ell^n]$ iff it fixes $L_n$. So the kernel of $\rho_{E,\ell} \bmod \ell^n$ is the open subgroup $G_{L_n}$, and the inverse images of the basic open subgroups $\ker(\mathrm{GL}_2(\mathbb{Z}_\ell) \to \mathrm{GL}_2(\mathbb{Z}/\ell^n))$ are open: $\rho_{E,\ell}$ is continuous (Chapter 5). (ii) By Corollary 9.33(i) with $m = \ell^n$ for all $n$, $\det\rho_{E,\ell}(\sigma) \equiv \chi_\ell(\sigma) \pmod{\ell^n}$ for every $n$. (iii) $\varphi$ maps $E[\ell^n]$ to $E'[\ell^n]$ compatibly with $[\ell]$ and with $G_K$ (Theorem 9.19(i), Definition 9.17), so it induces $\varphi_\ell$, and $\varphi_V = \varphi_\ell \otimes \mathbb{Q}_\ell$. As $\hat\varphi \circ \varphi = [\deg\varphi]$ is invertible on $V_\ell E$, $\varphi_V$ is injective, hence an isomorphism by dimensions, and $\varphi_\ell$ is injective. Under the identifications $V/T \simeq E[\ell^\infty]$ of Proposition 9.35(ii), $\varphi_V$ induces $\varphi\colon E[\ell^\infty] \to E'[\ell^\infty]$, whose kernel is $\varphi_V^{-1}(T_\ell E')/T_\ell E \simeq T_\ell E'/\varphi_\ell(T_\ell E) = \operatorname{coker}\varphi_\ell$ and also equals $(\ker\varphi)[\ell^\infty]$. Finally $\#\ker\varphi$ divides $\deg\varphi$ (Theorem 9.19(ii)), so if $\ell \nmid \deg\varphi$ the cokernel is 0. □

**Corollary 9.38** ($T_\ell E$ is self-dual up to a twist)**.** *The Weil pairing gives $G_K$-isomorphisms*

$$T_\ell E \xrightarrow{\sim} \operatorname{Hom}_{\mathbb{Z}_\ell}(T_\ell E, \mathbb{Z}_\ell(1)) = (T_\ell E)^\vee(1), \quad t \mapsto e(\cdot, t), \qquad E[\ell^k] \simeq \operatorname{Hom}(E[\ell^k], \mu_{\ell^k}).$$

*Equivalently $(T_\ell E)^\vee \simeq T_\ell E(-1)$ and $\rho_{E,\ell}^\vee \simeq \rho_{E,\ell} \otimes \chi_\ell^{-1}$, which is the identity $V^\vee \simeq V \otimes \det^{-1}$ for two-dimensional $V$ (Chapter 4). Consequently $\mathrm{ad}^0((T_\ell E)^\vee) \simeq \mathrm{ad}^0(T_\ell E)$ as $G_K$-modules (Chapter 4): the trace-zero adjoint does not distinguish $T_\ell E$ from its dual, nor from any lattice homothetic to either.*

*Proof.* The map is $\mathbb{Z}_\ell$-linear, bijective because $e$ is perfect (Proposition 9.35(iii)), and equivariant by the computation in the proof of Corollary 9.33(iii) with $e$ in place of $e_\ell$. The finite-level statement is the same argument with $e_{\ell^k}$. Tensoring with $\mathbb{Z}_\ell(-1)$ gives the second form. Transposition $g \mapsto g^{\mathsf{T}}$ is a $\mathbb{Z}_\ell$-linear bijection $\mathrm{End}(T) \to \mathrm{End}(T^\vee)$ preserving traces, and it is $G_K$-equivariant because $\rho^\vee(\sigma)g^{\mathsf{T}}\rho^\vee(\sigma)^{-1} = (\rho(\sigma)g\rho(\sigma)^{-1})^{\mathsf{T}}$ for $\rho^\vee(\sigma) = (\rho(\sigma)^{-1})^{\mathsf{T}}$; so $\mathrm{ad}^0(T) \simeq \mathrm{ad}^0(T^\vee)$. Twisting by a character does not change End, hence not $\mathrm{ad}^0$ (Chapter 4). □

**Corollary 9.39** (complex conjugation)**.** *Let $E/\mathbb{Q}$, $\ell$ odd, and $c \in G_{\mathbb{Q}}$ a complex conjugation. Then $\bar\rho_{E,\ell}(c)$ is conjugate in $\mathrm{GL}_2(\mathbb{F}_\ell)$ to $\mathrm{diag}(1,-1)$: it has eigenvalues $1, -1$, trace 0 and determinant $-1$. The same holds for $\rho_{E,\ell}(c)$ in $\mathrm{GL}_2(\mathbb{Z}_\ell)$.*

*Proof.* $c^2 = 1$ and $\chi_\ell(c) = -1$, as $c$ inverts roots of unity. A matrix $g$ with $g^2 = 1$ over a ring in which 2 is invertible satisfies $M = M^+ \oplus M^-$ with $M^\pm = \frac{1\pm g}{2}M$, the $\pm1$-eigenspaces; over

$\mathbb{Z}_\ell$ (or $\mathbb{F}_\ell$) both are free, of ranks adding to 2. Since $\det g = -1$, $g \neq \pm 1$, so both ranks are 1. (Directly: $E(\mathbb{R})$ is a compact real Lie group of dimension one, $S^1$ or $S^1 \times \mathbb{Z}/2$, so its $\ell$-torsion $E[\ell]^c$ is cyclic of order $\ell$.) □

**Convention 9.40** (Frobenius)**.**

(a) For $E$ over a finite field $\mathbb{F}_\mathfrak{q}$, the *Frobenius endomorphism* $\pi = \pi_\mathfrak{q}\colon (x, y) \mapsto (x^\mathfrak{q}, y^\mathfrak{q})$ acts on $E(\overline{\mathbb{F}}_p)$ exactly as the arithmetic Frobenius $\mathrm{Frob}_\mathfrak{q} \in G_{\mathbb{F}_\mathfrak{q}}$, $t \mapsto t^\mathfrak{q}$.

(b) For $E/\mathbb{Q}$ with good reduction at $p \neq \ell$, $\mathrm{Frob}_p$ in a group-theoretic statement is an *arithmetic* Frobenius at $p$ (Chapter 6), and $\mathrm{Frob}_p^{-1}$ is a geometric one. $\rho_{E,\ell}(\mathrm{Frob}_p)$ has characteristic polynomial $X^2 - a_p X + p$ (Theorem 9.67); its roots $\alpha_p, \beta_p$ satisfy $\alpha_p \beta_p = p$.

(c) On the dual $V_\ell E^\vee = H^1$ the geometric Frobenius has the eigenvalues $\alpha_p, \beta_p$ (the inverse transpose of $\rho(\mathrm{Frob}_p)$ has eigenvalues $\alpha_p^{-1}, \beta_p^{-1}$, and $\mathrm{Frob}_p^{-1}$ inverts them again), while on $V_\ell E$ itself the geometric Frobenius has the eigenvalues $\alpha_p^{-1} = \beta_p/p$ and $\beta_p^{-1} = \alpha_p/p$. Local $L$-factors are defined with the geometric Frobenius (Appendix B, Appendix B): $L_p(M, s) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid M^{I_p})^{-1}$. Hence, at a prime $p$ of good reduction,

$$L_p(E, s) = \det(1 - \mathrm{Frob}_p\ p^{-s} \mid V_\ell E)^{-1} = L_p(V_\ell E^\vee, s) = (1 - a_p p^{-s} + p^{1-2s})^{-1},$$

$$L_p(V_\ell E, s) = L_p(E, s+1),$$

the last because $\det(1 - \mathrm{Frob}_p^{-1}\, X \mid V_\ell E) = (1 - \beta_p X/p)(1 - \alpha_p X/p)$, or because $V_\ell E \simeq V_\ell E^\vee(1)$ (Corollary 9.38).

(d) On $\mathrm{ad}^0 V_\ell E$ both Frobenii have the eigenvalues $\alpha_p/\beta_p, 1, \beta_p/\alpha_p$, so the good factor of the adjoint does not depend on the convention; at a multiplicative prime it does (on the invariant line the arithmetic Frobenius acts by $q$, the geometric one by $q^{-1}$), and the paper's factor $1 - q^{-1-s}$ is the geometric one (Chapters 7 and 19).

A numerical illustration of (b) to (d). For `11a1` at $p = 2$, $a_2 = -2$ (Example 9.68), so $X^2 + 2X + 2$ has roots $\alpha_2 = -1 + i$, $\beta_2 = -1 - i$, with $\alpha_2\beta_2 = 2$ and $|\alpha_2| = \sqrt{2}$. The arithmetic Frobenius has eigenvalues $-1 \pm i$ on $V_\ell E$; the geometric Frobenius has eigenvalues $\alpha_2^{-1} = \frac{-1-i}{2} = \beta_2/2$ and $\beta_2^{-1} = \alpha_2/2$ on $V_\ell E$, and $-1 \pm i$ on $V_\ell E^\vee$. So $L_2(E, s) = \big((1-\alpha_2 2^{-s})(1-\beta_2 2^{-s})\big)^{-1} = (1 + 2\cdot 2^{-s} + 2\cdot 2^{-2s})^{-1}$, while $L_2(V_\ell E, s)$ has the eigenvalues divided by 2, which is the shift $s \mapsto s+1$. On $\mathrm{ad}^0 V_\ell E$ the eigenvalues are $\alpha_2/\beta_2 = -i$, 1 and $\beta_2/\alpha_2 = i$ for either Frobenius. Mixing up the two conventions would replace $L_2(E, s)$ by $L_2(E, s+1)$, a shift that is invisible in the adjoint factor at good primes but not at multiplicative ones, where the paper's factor $1 - q^{-1-s}$ is the geometric one.

*In the paper* 9.41. Section 1.1 of the paper ("Let $V = V_\ell E$, $T = T_\ell E$, $\rho = \rho_{E,\ell}$, $\bar\rho$ its reduction") uses Definitions 9.34 and 9.36 and Propositions 9.35 and 9.37; the paper states the same Frobenius conventions as Convention 9.40. In Section 4.2 and after Proposition prop:pullback the paper identifies the lattice $M_{f,\lambda}$ of Diamond, Flach and Guo with $H^1(E, \mathbb{Z}_\ell)$, using its Proposition prop:integral, which identifies their integral structure with $H^1(X_0(N), \mathbb{Z}_\ell)$ with no localisation (Chapter 14). That module is $\mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell) = (T_\ell E)^\vee$, which by Corollary 9.38 is $T_\ell E(-1)$: the dual of $T_\ell E$, not $T_\ell E$ itself. Since the adjoint is insensitive to duals and twists, $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T_\ell E$, which is all the paper uses. Lemma lem:h0 uses that complex conjugation has distinct eigenvalues $\pm 1$ (Corollary 9.39), which makes an irreducible $\bar\rho$ absolutely irreducible (Chapter 11), and $\det \rho = \chi_\ell$ (Proposition 9.37(ii)) in the determinant argument; Proposition prop:ns uses $\det \bar\rho_{E,3} = \bar\chi_3 = \chi_{-3}$.

## 9.6 Endomorphisms and complex multiplication

Curves with complex multiplication are the exceptions of the paper: Conjecture conj:eis and the data sections exclude them, Proposition prop:no9 reaches a contradiction by producing complex multiplication, and the theorem of Calegari and Emerton quoted in Section sec:two has a CM alternative. This section records the structure of endomorphism rings, the Galois action on the endomorphisms of a CM curve over $\mathbb{Q}$, the thirteen rational CM $j$-invariants, and the facts about $a_p$ that recognise CM. The number-theoretic side (orders, class number one, splitting of primes) is Chapter 6.

**Theorem 9.42** (endomorphism rings)**.** $\mathrm{End}_{\bar{K}}(E)$ *has no zero divisors, and* deg *is a positive definite quadratic form on it. As a ring it is isomorphic to* $\mathbb{Z}$*, to an order in an imaginary quadratic field, or, only when* $\operatorname{char} K = p > 0$*, to a maximal order in the quaternion algebra over* $\mathbb{Q}$ *ramified exactly at* $p$ *and* $\infty$*. In characteristic* 0 *only the first two occur.*

The first sentence follows from Theorem 9.21(iv) and the multiplicativity of degrees. We do not prove the classification; see [185, Sections III.9 and V.3]. Over $\mathbb{C}$ the dichotomy $\mathbb{Z}$ or an imaginary quadratic order is proved in Proposition 9.55(iii), and every field of characteristic 0 over which a given $E$ and its endomorphisms are defined embeds into $\mathbb{C}$.

**Proposition 9.43** (normalised endomorphisms and the Galois action)**.** *Let* $\operatorname{char} K = 0$ *and* $\omega$ *the invariant differential of* $E$*.*

*(i) The map* $\mathrm{End}_{\bar{K}}(E) \to \bar{K}$*,* $\alpha \mapsto c_\alpha$ *where* $\alpha^*\omega = c_\alpha\omega$*, is an injective ring homomorphism. Its image* $\mathcal{O}$ *is* $\mathbb{Z}$ *or an order in an imaginary quadratic field* $F \subset \bar{K}$*; we write* $[c]$ *for the endomorphism with* $c_{[c]} = c$*, consistently with* $[m]$ *for* $m \in \mathbb{Z}$*.*

*(ii) An endomorphism defined over* $K$ *has* $c_\alpha \in K$*. In particular* $\mathrm{End}_{\mathbb{Q}}(E) = \mathbb{Z}$ *for every* $E/\mathbb{Q}$*.*

*(iii) Let* $E/\mathbb{Q}$ *and* $\mathrm{End}_{\overline{\mathbb{Q}}}(E) = \mathcal{O} \neq \mathbb{Z}$*, with fraction field* $F$*. For* $\sigma \in G_{\mathbb{Q}}$ *and* $c \in \mathcal{O}$*,* $\sigma\circ[c]\circ\sigma^{-1} = [\sigma(c)]$ *on* $E(\overline{\mathbb{Q}})$*. So every endomorphism is defined over* $F$*, and* $\sigma$ *acts on* $\mathcal{O}$ *by complex conjugation when* $\sigma|_F \neq 1$*. On the Tate module,* $\rho_{E,\ell}(\sigma)\,[c]_\ell\,\rho_{E,\ell}(\sigma)^{-1} = [\sigma(c)]_\ell$*.*

*Proof.* (i) Additivity is Proposition 9.15(iii), and $(\alpha\circ\beta)^*\omega = \beta^*(c_\alpha\omega) = c_\alpha c_\beta\omega$. Every holomorphic differential is a constant multiple of $\omega$ (Chapter 8), so $c_\alpha$ is defined. A nonzero $\alpha$ is separable (characteristic 0), so $\alpha^*\omega \neq 0$ (Chapter 8): the map is injective. Its image is a commutative ring isomorphic to $\mathrm{End}_{\bar{K}}(E)$, so by Theorem 9.42 it is $\mathbb{Z}$ or an imaginary quadratic order. (ii) If $\alpha$ is defined over $K$, then $\alpha^*\omega$ is a differential over $K$ and $\omega$ is one, so $c_\alpha \in K$. For $K = \mathbb{Q}$, $c_\alpha$ is a rational number integral over $\mathbb{Z}$ (it lies in an order), hence an integer $m$, and $\alpha = [m]$ by injectivity. (iii) For $\sigma \in G_{\mathbb{Q}}$ let $\alpha^\sigma$ be the endomorphism obtained by applying $\sigma$ to the coefficients of the rational functions defining $\alpha$; on points, $\alpha^\sigma(P) = \sigma(\alpha(\sigma^{-1}P))$. Applying $\sigma$ to $\alpha^*\omega = c_\alpha\omega$, and using that $\omega$ has rational coefficients, gives $(\alpha^\sigma)^*\omega = \sigma(c_\alpha)\omega$; so $[c]^\sigma = [\sigma(c)]$ by (i). If $\sigma|_F = 1$ this gives $[c]^\sigma = [c]$ for all $c$, so every endomorphism is defined over $F$ (Chapter 8). The statement on $T_\ell E$ is the same identity applied to $\ell$-power torsion points. □

**Definition 9.44** (complex multiplication)**.** An elliptic curve $E$ over a field of characteristic 0 has *complex multiplication* (*CM*) if $\mathrm{End}_{\bar{K}}(E) \neq \mathbb{Z}$. Then $\mathrm{End}_{\bar{K}}(E) \simeq \mathcal{O}$, an order in an imaginary quadratic field $F$ (Proposition 9.43), and $E$ has *CM by* $\mathcal{O}$, or *by* $F$.

**Lemma 9.45** (CM is an isogeny invariant)**.** *If $\varphi\colon E \to E'$ is a nonconstant isogeny of degree $n$, then $\alpha \mapsto \frac{1}{n}\varphi\circ\alpha\circ\hat{\varphi}$ is an isomorphism of $\mathbb{Q}$-algebras $\mathrm{End}(E)\otimes\mathbb{Q} \to \mathrm{End}(E')\otimes\mathbb{Q}$. In particular $E$ has CM iff $E'$ has, and then by the same field $F$.*

*Proof.* $\frac{1}{n^2}(\varphi\alpha\hat{\varphi})(\varphi\beta\hat{\varphi}) = \frac{1}{n^2}\varphi\alpha[n]\beta\hat{\varphi} = \frac{1}{n}\varphi\alpha\beta\hat{\varphi}$ (Theorem 9.21), additivity is clear, and $1 \mapsto \frac{1}{n}\varphi\hat{\varphi} = 1$. The map $\beta \mapsto \frac{1}{n}\hat{\varphi}\beta\varphi$ in the other direction is inverse to it, by the same computation. □

**Theorem 9.46** (the rational CM $j$-invariants)**.** *If $E/\mathbb{Q}$ has CM by the order $\mathcal{O}$, then $\mathcal{O}$ has class number one, so its discriminant $D_{\mathcal{O}}$ is one of the thirteen values in the table below, and $j(E)$ is the corresponding value; conversely each value is the $j$-invariant of a curve over $\mathbb{Q}$ with CM by that order. Here $j(E) = j(\tau)$ for $\tau = \frac{D_{\mathcal{O}}+\sqrt{D_{\mathcal{O}}}}{2}$ (Section 9.7).*

| $D_{\mathcal{O}}$ | $j$ | *example* | $D_{\mathcal{O}}$ | $j$ | *example* |
|---|---|---|---|---|---|
| $-3$ | $0$ | *27a1, 36a1* | $-19$ | $-884736 = -2^{15}3^3$ | |
| $-4$ | $1728$ | *32a1* | $-27$ | $-12288000 = -2^{15}\cdot 3\cdot 5^3$ | *27a2, 27a4* |
| $-7$ | $-3375 = -3^3 5^3$ | | $-28$ | $16581375 = 3^3 5^3 17^3$ | |
| $-8$ | $8000 = 2^6 5^3$ | | $-43$ | $-884736000 = -2^{18}3^3 5^3$ | |
| $-11$ | $-32768 = -2^{15}$ | *121b1* | $-67$ | $-147197952000 = -2^{15}3^3 5^3 11^3$ | |
| $-12$ | $54000 = 2^4 3^3 5^3$ | | $-163$ | $-262537412640768000 = -2^{18}3^3 5^3 23^3 29^3$ | |
| $-16$ | $287496 = 2^3 3^3 11^3$ | | | | |

We do not prove this; see [185, Appendix C, Section 11], [184, Chapter II] and [34, Sections 7 and 10 to 12]. The ingredients are: $j(E)$ generates the ring class field of $\mathcal{O}$ over $F$, of degree $h(\mathcal{O})$ (so $j(E)\in\mathbb{Q}$ forces $h(\mathcal{O}) = 1$), and the theorem of Heegner, Baker and Stark (Chapter 6) lists the orders of class number one. The thirteen values were checked numerically as $j(\tau)$ with PARI's `ellj`, and the examples' $j$-invariants with `ellinit`.

**Theorem 9.47** (CM and the numbers $a_p$)**.** *Let $E/\mathbb{Q}$ have CM by an order in the imaginary quadratic field $F$, and let $\chi_F$ be the quadratic character of $F$.*

*(i)* $\rho_{E,\ell} \simeq \rho_{E,\ell}\otimes\chi_F$ *for every prime $\ell$.*

*(ii)* $a_p(E) = 0$ *for every prime $p$ of good reduction that is inert in $F$ (the reduction is supersingular, Definition 9.64).*

*(iii) For a prime $p$ of good reduction that splits in $F$, there is $\pi_p$ in the ring of integers of $F$ with $\pi_p\bar{\pi}_p = p$ and $a_p(E) = \pi_p + \bar{\pi}_p \neq 0$.*

*(iv) Every prime ramified in $F$ divides the conductor $N$ of $E$. In particular, if $E$ has CM by $\mathbb{Q}(\sqrt{\ell^*})$ for an odd prime $\ell$, then $\ell \mid N$.*

*(v) Conversely, if $a_p(E) = 0$ for all primes $p$ inert in an imaginary quadratic field $F$ outside a set of density 0, then $E$ has CM by $F$.*

*Sketch of proof.* (i), (ii) and (iv) are proved in Chapter 6 from Proposition 9.43(iii) and Theorem 9.67; we recall the argument. Choose $0 \neq c \in \mathcal{O}$ with $\bar{c} = -c$; for instance $c = \sqrt{D_{\mathcal{O}}}$, which lies in $\mathcal{O} = \mathbb{Z}\big[\frac{D_{\mathcal{O}}+\sqrt{D_{\mathcal{O}}}}{2}\big]$ as $2\cdot\frac{D_{\mathcal{O}}+\sqrt{D_{\mathcal{O}}}}{2} - D_{\mathcal{O}}$. For $\sigma\in G_{\mathbb{Q}}$, $\sigma(c) = c$ if $\sigma$ is trivial on $F$ and $\sigma(c) = \bar{c} = -c$ otherwise, that is, $\sigma(c) = \chi_F(\sigma)c$. By Proposition 9.43(iii), $\rho(\sigma)[c]_\ell\rho(\sigma)^{-1} = [\chi_F(\sigma)c]_\ell = \chi_F(\sigma)[c]_\ell$, so the map $[c]_\ell$ on $V_\ell E$, invertible because $[\bar{c}]\circ[c] = [c\bar{c}]$ is multiplication by the nonzero integer $c\bar{c}$, satisfies $[c]_\ell\rho(\sigma) = \chi_F(\sigma)\rho(\sigma)[c]_\ell$: it is an isomorphism $\rho \to \rho\otimes\chi_F$. Taking traces, $\mathrm{tr}\,\rho(\sigma) = \chi_F(\sigma)\,\mathrm{tr}\,\rho(\sigma)$, and at a good $p\neq\ell$ inert in $F$,

$\chi_F(\mathrm{Frob}_p) = -1$ gives $a_p = \operatorname{tr}\rho(\mathrm{Frob}_p) = 0$ (Theorem 9.67). For (iv): if $p$ ramified in $F$ were a prime of good reduction, take $\ell \neq p$; $\rho$ is unramified at $p$ (Theorem 9.67), but some $\tau \in I_p$ has $\chi_F(\tau) = -1$, and then $[c]_\ell = \rho(\tau)[c]_\ell\rho(\tau)^{-1} = -[c]_\ell$, a contradiction. (iii) is Deuring's theorem [46]: the reduction is ordinary, its endomorphism ring contains the reduction of $\mathcal{O}$ and is an order of $F$, and the Frobenius endomorphism $\pi_p$ (Definition 9.60) is an element of norm $p$ with trace $a_p$; see [184, Chapter II, Section 10]. (v) follows from Serre's open image theorem (Chapter 11); see [178]. □

The checks: $a_p = 0$ at every good inert $p < 500$ and $a_p \neq 0$ at every good split $p < 500$ for `27a1`, `36a1` ($F = \mathbb{Q}(\sqrt{-3})$), `32a1` ($F = \mathbb{Q}(i)$) and `121b1` ($F = \mathbb{Q}(\sqrt{-11})$); and among the 462 curves of conductor below 10,000 with one of the thirteen $j$-invariants, every prime dividing the discriminant of $F$ divides the conductor, as (iv) says.

*Warning* 9.48 (the cyclotomic character is not a quadratic character). The CM condition of Theorem 9.47(i) involves the quadratic character $\chi_F$; for $F = \mathbb{Q}(\sqrt{\ell^*})$ this is $\chi_{\ell^*} = \bar{\chi}_\ell^{(\ell-1)/2}$. The cyclotomic character $\chi_\ell$ is not quadratic, and $\rho_{E,\ell} \simeq \rho_{E,\ell} \otimes \chi_\ell$ is impossible for every $E$ (determinants give $\chi_\ell^2 = 1$). Only for $\ell = 3$ do $\bar{\chi}_\ell$ and $\chi_{\ell^*}$ coincide: $\bar{\chi}_3 = \chi_{-3}$. The paper's Lemma lem:h0 works with $\bar{\chi}_\ell$. By Theorem 9.47(iv) a curve $E/\mathbb{Q}$ with $\ell \nmid N$ has no CM by $\mathbb{Q}(\sqrt{\ell^*})$.

*In the paper* 9.49. The paper excludes CM curves in Conjecture conj:eis and in its data ("not CM"). Proposition prop:no9 (Proposition 9.78 below) ends with "so $E'$, hence $E$, has complex multiplication": the passage from $E'$ to $E$ is Lemma 9.45. The CM alternative in the theorem of Calegari and Emerton (Section sec:two: "it has complex multiplication and $N \in \{27, 32, 49, 243\}$") concerns CM curves of these four conductors; the optimal curves `27a1`, `32a1`, `49a1` and `243b1`, of modular degrees 1, 1, 1 and 9 (`degphi`), have $j = 0$, 1728, $-3375$ and 0, that is, CM by the orders of discriminant $-3$, $-4$, $-7$, $-3$ (Theorem 9.46). The paper's examples are `27a1` and `243b1`.

## 9.7 Elliptic curves over the complex numbers

Section 1 of the paper describes the modular degree as "$\langle f, f\rangle$ divided by the area of the period lattice of $E$" (Zagier, Watkins; the exact formula, with the Manin constant, is in Chapter 14). This section constructs the period lattice: it proves that every elliptic curve over $\mathbb{C}$ is a torus $\mathbb{C}/\Lambda$, through the Weierstrass function $\wp$, that isomorphism classes of curves over $\mathbb{C}$ correspond to lattices up to homothety and to points of $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$, and computes the periods of `11a1`. The analysis is that of Appendix A: a doubly periodic function is, in the language of physics, a function on a torus, and the Weierstrass function plays for the torus the role that a Fourier series plays for the circle.

**Definition 9.50** (lattices)**.** A *lattice* is a subgroup $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 \subset \mathbb{C}$ with $\omega_1, \omega_2$ linearly independent over $\mathbb{R}$; the basis is *oriented* if $\mathrm{Im}(\omega_2/\omega_1) > 0$. Lattices $\Lambda$, $\Lambda'$ are *homothetic* if $\Lambda' = c\Lambda$ for some $c \in \mathbb{C}^\times$. The *covolume* $\mathrm{vol}(\Lambda) = |\mathrm{Im}(\bar{\omega}_1\omega_2)|$ is the area of a fundamental parallelogram. For $k \geq 2$ the *Eisenstein series* of $\Lambda$ is $G_{2k}(\Lambda) = \sum_{\omega\in\Lambda\smallsetminus\{0\}} \omega^{-2k}$, and

$$g_2(\Lambda) = 60G_4(\Lambda), \qquad g_3(\Lambda) = 140G_6(\Lambda), \qquad \Delta(\Lambda) = g_2^3 - 27g_3^2, \qquad j(\Lambda) = 1728\frac{g_2^3}{\Delta(\Lambda)}.$$

**Lemma 9.51** (lattice sums)**.** *$\sum_{\omega\in\Lambda\smallsetminus\{0\}}|\omega|^{-s}$ converges for every real $s>2$. Hence $G_{2k}(\Lambda)$ converges absolutely for $k\geq 2$, and $G_{2k}(c\Lambda)=c^{-2k}G_{2k}(\Lambda)$, $g_2(c\Lambda)=c^{-4}g_2(\Lambda)$, $g_3(c\Lambda)=c^{-6}g_3(\Lambda)$, $\Delta(c\Lambda)=c^{-12}\Delta(\Lambda)$, and $j(c\Lambda)=j(\Lambda)$ (once $\Delta(\Lambda)\neq 0$ is known, Theorem 9.52).*

*Proof.* The function $(a,b)\mapsto|a\omega_1+b\omega_2|$ is continuous and positive on the compact subset of $\mathbb{R}^2$ where $\max(|a|,|b|)=1$, so it has a minimum $\mu>0$ there; by homogeneity $|a\omega_1+b\omega_2|\geq\mu n$ when $\max(|a|,|b|)=n$. There are $8n$ pairs of integers with $\max(|a|,|b|)=n$, so the sum is at most $\sum_{n\geq1}8n(\mu n)^{-s}$, which converges for $s>2$. The homogeneity is clear from the definitions. □

**Theorem 9.52** (the Weierstrass function)**.** *Let $\Lambda$ be a lattice and*

$$\wp(z)=\wp(z;\Lambda)=\frac{1}{z^2}+\sum_{\omega\in\Lambda\smallsetminus\{0\}}\Big(\frac{1}{(z-\omega)^2}-\frac{1}{\omega^2}\Big).$$

*(i) The series converges absolutely and uniformly on compact subsets of $\mathbb{C}\smallsetminus\Lambda$; $\wp$ is meromorphic on $\mathbb{C}$, even, $\Lambda$-periodic, with a double pole at each point of $\Lambda$ and no other pole.*

*(ii) Near 0, $\wp(z)=z^{-2}+\sum_{k\geq1}(2k+1)G_{2k+2}(\Lambda)z^{2k}$.*

*(iii) $\wp'(z)^2=4\wp(z)^3-g_2(\Lambda)\wp(z)-g_3(\Lambda)$.*

*(iv) The values $e_1=\wp(\omega_1/2)$, $e_2=\wp(\omega_2/2)$, $e_3=\wp((\omega_1+\omega_2)/2)$ are the three roots of $4x^3-g_2x-g_3$, and they are distinct; hence $\Delta(\Lambda)=16\prod_{i<j}(e_i-e_j)^2\neq0$.*

*(v) A $\Lambda$-periodic meromorphic function whose poles lie in $\Lambda$ and have order at most 2 is $a\wp+b$ with $a,b\in\mathbb{C}$; if the poles have order at most 3, it is $a\wp'+b\wp+c$.*

*Proof.* (i) For $|z|\leq R$ and $|\omega|\geq 2R$, $|(z-\omega)^{-2}-\omega^{-2}|=|z(2\omega-z)|/|\omega^2(z-\omega)^2|\leq R\cdot\frac{5}{2}|\omega|\cdot|\omega|^{-2}\cdot4|\omega|^{-2}=10R|\omega|^{-3}$, and $\sum|\omega|^{-3}<\infty$ (Lemma 9.51). The finitely many terms with $|\omega|<2R$ are meromorphic. So the series converges normally on compact sets avoiding $\Lambda$ and $\wp$ is meromorphic (Appendix A, Weierstrass's theorem on uniform limits), with poles exactly at $\Lambda$, double. Replacing $\omega$ by $-\omega$ shows $\wp(-z)=\wp(z)$. Differentiating termwise, $\wp'(z)=-2\sum_{\omega\in\Lambda}(z-\omega)^{-3}$, which is visibly $\Lambda$-periodic; so $\wp(z+\omega_i)-\wp(z)$ is a constant $C_i$, and evaluating at $z=-\omega_i/2$ (not a pole) gives $C_i=\wp(\omega_i/2)-\wp(-\omega_i/2)=0$. (ii) For $|z|<\min|\omega|$, $(z-\omega)^{-2}-\omega^{-2}=\sum_{n\geq1}(n+1)z^n\omega^{-n-2}$; summing over $\omega$ (absolutely convergent) the odd powers cancel in pairs $\pm\omega$, and the coefficient of $z^{2k}$ is $(2k+1)G_{2k+2}$. (iii) From (ii), $\wp=z^{-2}+3G_4z^2+5G_6z^4+O(z^6)$ and $\wp'=-2z^{-3}+6G_4z+20G_6z^3+O(z^5)$, so $\wp'^2=4z^{-6}-24G_4z^{-2}-80G_6+O(z^2)$ and $4\wp^3=4(z^{-6}+9G_4z^{-2}+15G_6+O(z^2))=4z^{-6}+36G_4z^{-2}+60G_6+O(z^2)$. Hence $\wp'^2-4\wp^3=-60G_4z^{-2}-140G_6+O(z^2)$, and adding $60G_4\wp=60G_4z^{-2}+O(z^2)$ gives $\wp'^2-4\wp^3+60G_4\wp+140G_6=O(z^2)$, that is, $\wp'^2-4\wp^3+g_2\wp+g_3=O(z^2)$; this is where the normalisations $g_2=60G_4$, $g_3=140G_6$ come from. The left side is $\Lambda$-periodic and holomorphic away from $\Lambda$, and by this expansion holomorphic at 0 and vanishing there; a holomorphic doubly periodic function is bounded, hence constant (Liouville), and the constant is 0. (iv) $\wp'$ is odd and periodic, so $\wp'(\omega_i/2)=-\wp'(-\omega_i/2)=-\wp'(\omega_i/2)$ vanishes at each half-period, and (iii) shows that the $e_i$ are roots of $4x^3-g_2x-g_3$. On the compact Riemann surface $\mathbb{C}/\Lambda$, $\wp-e_i$ has a single pole, of order 2, so it has exactly two zeros counted with multiplicity (Chapter 8); the half-period $h_i$ with $\wp(h_i)=e_i$ is a zero of order at least 2 since $\wp'(h_i)=0$, so it is the only zero. As the three half-periods are distinct modulo $\Lambda$, the $e_i$ are distinct. The discriminant of a cubic $ax^3+cx+d$ is $-4ac^3-27a^2d^2$, which for $4x^3-g_2x-g_3$ is $16g_2^3-432g_3^2=16(g_2^3-27g_3^2)$; it is

also $a^4\prod_{i<j}(e_i-e_j)^2 = 4^4\prod_{i<j}(e_i-e_j)^2$. Dividing by 16 gives the formula for $\Delta(\Lambda)$. (v) If $f$ has poles of order at most 2 at the points of $\Lambda$, choose $a$ so that $f - a\wp$ has at most a simple pole at 0. The residues of a periodic meromorphic function on $\mathbb{C}/\Lambda$ sum to 0 (Chapter 8, residue theorem), so $f - a\wp$ has no pole; it is holomorphic and doubly periodic, hence constant. The case of order 3 is the same, subtracting first a multiple of $\wp'$, which has a triple pole at 0. □

**Theorem 9.53** (complex uniformisation)**.**

*(i) Let $\Lambda$ be a lattice and $E_\Lambda$ the curve $y^2 = 4x^3 - g_2(\Lambda)x - g_3(\Lambda)$ (a Weierstrass curve after $y \mapsto 2y$; it is smooth by Theorem 9.52(iv)). The map $\Theta\colon \mathbb{C}/\Lambda \to E_\Lambda(\mathbb{C})$, $z \mapsto (\wp(z), \wp'(z))$ for $z \notin \Lambda$ and $0 \mapsto O$, is a biholomorphic group isomorphism, and $\Theta^*(dx/y) = dz$.*

*(ii) Let $E$ be an elliptic curve over $\mathbb{C}$ with invariant differential $\omega$, and $\Lambda_E = \{\int_\gamma \omega : \gamma \in H_1(E(\mathbb{C}),\mathbb{Z})\}$ its lattice of periods. Then $E \simeq E_{\Lambda_E}$, by an isomorphism that matches $\omega$ with $dx/y$, so $E(\mathbb{C}) \simeq \mathbb{C}/\Lambda_E$ with $\omega \leftrightarrow dz$. Equivalently: for $A, B \in \mathbb{C}$ with $4A^3 + 27B^2 \neq 0$ there is a unique lattice $\Lambda$ with $g_2(\Lambda) = -4A$ and $g_3(\Lambda) = -4B$.*

*Proof.* (i) $\Theta$ is holomorphic (in the chart $(x/y, 1/y)$ near $O$ it is $z \mapsto (\wp/\wp', 1/\wp') = (-z/2 + O(z^5), -z^3/2 + O(z^7))$, holomorphic and sending 0 to $(0,0)$) and lands in $E_\Lambda$ by Theorem 9.52(iii). It is bijective: $O$ has the single preimage 0, and for an affine point $(x_0, y_0)$ the function $\wp - x_0$ has exactly two zeros $\pm z_0$ modulo $\Lambda$ (counted with multiplicity, as in the proof of Theorem 9.52(iv)); since $\wp'(-z_0) = -\wp'(z_0)$ and $\wp'(z_0)^2 = y_0^2$, exactly one of $z_0, -z_0$ maps to $(x_0, y_0)$, and if $z_0 \equiv -z_0$ then $y_0 = \wp'(z_0) = 0$. A holomorphic bijection of compact Riemann surfaces is biholomorphic (Chapter 8). Group law: let $z_1, z_2 \in \mathbb{C}$ with $\Theta(z_1)$, $\Theta(z_2)$ affine, distinct and not opposite, and let $y = \lambda x + \nu$ be the line through them. Then $f = \wp' - \lambda\wp - \nu$ is periodic with a single pole, of order 3, at 0 modulo $\Lambda$, so it has three zeros $z_1, z_2, z_3$ modulo $\Lambda$, and $z_1 + z_2 + z_3 \in \Lambda$ (the "only if" half of Abel's theorem, Chapter 8, for the torus: the sum of the zeros minus the sum of the poles of a periodic function lies in $\Lambda$). The three intersections of the line with $E_\Lambda$ are $\Theta(z_1), \Theta(z_2), \Theta(z_3)$, so $\Theta(z_1) + \Theta(z_2) = -\Theta(z_3) = \Theta(-z_3) = \Theta(z_1 + z_2)$ (Construction 9.10; on $E_\Lambda$, $-(x,y) = (x,-y)$, and $\wp$ is even, $\wp'$ odd). The remaining cases follow by continuity, since $\Theta(z_1+z_2) - \Theta(z_1) - \Theta(z_2)$ is continuous in $(z_1, z_2)$ and vanishes on a dense set. Finally $\Theta^*(dx/y) = \wp'(z)dz/\wp'(z) = dz$.

(ii) We use the Abel-Jacobi theorem of Chapter 8: $E(\mathbb{C})$ is a compact Riemann surface of genus one, $\omega$ spans its holomorphic differentials, $\Lambda_E$ is a lattice, and $\iota\colon E(\mathbb{C}) \to \mathbb{C}/\Lambda_E$, $P \mapsto \int_O^P \omega$, is biholomorphic. It is a homomorphism: $\iota(P)$ is the Abel-Jacobi image of $(P) - (O)$, the Abel-Jacobi map is additive on divisors of degree 0 and kills principal divisors, and $(P+Q) - (O) \sim (P) - (O) + (Q) - (O)$ (Proposition 9.12). Since $d\iota = \omega$, $(\iota^{-1})^*\omega = dz$. Put $\eta = 2y + a_1x + a_3$, so that $\eta^2 = F(x) = 4x^3 + b_2x^2 + 2b_4x + b_6$ (Lemma 9.3), and let $X(z) = x(\iota^{-1}(z))$, $H(z) = \eta(\iota^{-1}(z))$. These are $\Lambda_E$-periodic meromorphic functions with poles only at $\Lambda_E$, of orders 2 and 3 (as $x$ and $\eta$ have poles of orders 2 and 3 at $O$ only), and $dX/dz = H$ because $\omega = dx/\eta$ corresponds to $dz$. If $X = cz^{-2} + \cdots$ then $H = -2cz^{-3} + \cdots$, and comparing the $z^{-6}$ terms of $H^2 = F(X)$ gives $4c^2 = 4c^3$, so $c = 1$. By Theorem 9.52(v), $X = \wp + r$ for a constant $r$ (with $\wp = \wp(\cdot;\Lambda_E)$), and then $H = \wp'$. So $\wp'^2 = F(\wp + r)$ as functions; since $\wp$ takes infinitely many values, $F(x + r) = 4x^3 - g_2x - g_3$ as polynomials. Hence the change of coordinates $x \mapsto x - r$, $\eta \mapsto \eta$ (of the form of Proposition 9.5, with $u = 1$ after the normalisation $\eta = 2y$) maps $E$ to $E_{\Lambda_E}$, and $\Theta^{-1}$ composed with it is $\iota$; it matches $\omega = dx/\eta$ with $dx/y$ on $E_{\Lambda_E}$. For the second form, apply this to $E\colon y^2 = x^3 + Ax + B$, for which $\eta = 2y$ and $F(x) = 4x^3 + 4Ax + 4B$ has no $x^2$ term, so $r = 0$ and $g_2 = -4A$, $g_3 = -4B$. Uniqueness: if $\Lambda$, $\Lambda'$ have the same $g_2, g_3$, then $E_\Lambda = E_{\Lambda'}$, and $\Theta'^{-1} \circ \Theta$ is

a holomorphic group isomorphism $\mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ pulling $dz$ back to $dz$; by Proposition 9.55(i) it is $z \mapsto z$, so $\Lambda = \Lambda'$. □

A second route to (ii) avoids Abel-Jacobi: the function $j\colon \mathcal{H} \to \mathbb{C}$ is surjective (Chapter 12, valence formula), so for given $A, B$ there is a lattice with $j(\Lambda) = j$ of $y^2 = x^3 + Ax + B$, and a homothety adjusts $g_2, g_3$ [185, Section VI.5]. The book uses the route above, so that Chapter 12 may cite this theorem without circularity.

*Remark* 9.54 (translation invariance over $\mathbb{C}$). Theorem 9.53(ii) gives a second proof of Proposition 9.15(ii) and (iii) in characteristic 0. Under $E(\mathbb{C}) \simeq \mathbb{C}/\Lambda_E$ the group law is addition, $\tau_Q$ is $z \mapsto z + z_Q$, and $\omega$ is $dz$, which translations preserve. For isogenies $\varphi, \psi\colon E' \to E$, Proposition 9.55(i) below writes them as $z \mapsto \alpha z$ and $z \mapsto \beta z$; then $\varphi + \psi$ is $z \mapsto (\alpha + \beta)z$, which pulls $dz$ back to $(\alpha + \beta)\,dz$, the sum of the pullbacks. For a curve over any field of characteristic 0, each identity involves finitely many elements, which generate a field that embeds into $\mathbb{C}$. The proof of Theorem 9.53 used only Proposition 9.12 and Chapter 8, not Proposition 9.15.

**Proposition 9.55** (maps between complex tori)**.** *Let $\Lambda_1, \Lambda_2, \Lambda$ be lattices.*

*(i) The holomorphic maps $\mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$ sending 0 to 0 are the maps $z \mapsto \alpha z$ with $\alpha \in \mathbb{C}$, $\alpha\Lambda_1 \subseteq \Lambda_2$. They are group homomorphisms, and isomorphisms iff $\alpha\Lambda_1 = \Lambda_2$. Hence $E_{\Lambda_1} \simeq E_{\Lambda_2}$ over $\mathbb{C}$ iff $\Lambda_1$ and $\Lambda_2$ are homothetic, and isogenies $E_{\Lambda_1} \to E_{\Lambda_2}$ correspond to such $\alpha$.*

*(ii) For $\alpha \neq 0$ the map has degree $[\Lambda_2 : \alpha\Lambda_1]$ and kernel $\alpha^{-1}\Lambda_2/\Lambda_1$. For $\alpha = m$: $\deg[m] = m^2$ and $E[m] = \frac{1}{m}\Lambda/\Lambda \simeq (\mathbb{Z}/m)^2$.*

*(iii) $\mathrm{End}(\mathbb{C}/\Lambda) = \{\alpha \in \mathbb{C} : \alpha\Lambda \subseteq \Lambda\}$ is $\mathbb{Z}$ or an order in an imaginary quadratic field; the latter happens iff $\tau = \omega_2/\omega_1$ is an imaginary quadratic number, and then the order lies in $\mathbb{Q}(\tau)$.*

*(iv) For a basis $(\omega_1, \omega_2)$ of $\Lambda$, $e_m(a\frac{\omega_1}{m} + b\frac{\omega_2}{m}, c\frac{\omega_1}{m} + d\frac{\omega_2}{m}) = \zeta^{ad-bc}$, where $\zeta = e_m(\frac{\omega_1}{m}, \frac{\omega_2}{m})$ is a primitive $m$-th root of unity. For an oriented basis $\zeta = e^{\pm 2\pi i/m}$, the sign depending only on the normalisation of $e_m$ (Convention 9.1(f)); nothing in the book depends on it.*

*Proof.* (i) A holomorphic $f\colon \mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$ with $f(0) = 0$ lifts to a holomorphic $F\colon \mathbb{C} \to \mathbb{C}$ with $F(0) = 0$ ($\mathbb{C}$ is simply connected and $\mathbb{C} \to \mathbb{C}/\Lambda_2$ is a covering). For $\omega \in \Lambda_1$, $F(z+\omega) - F(z) \in \Lambda_2$ is continuous in $z$, hence constant; so $F'$ is $\Lambda_1$-periodic and entire, hence constant by Liouville, and $F(z) = \alpha z$ with $\alpha\Lambda_1 \subseteq \Lambda_2$. Conversely such $\alpha$ define holomorphic homomorphisms, with inverse of the same kind iff $\alpha\Lambda_1 = \Lambda_2$. By Theorem 9.53 and the fact that holomorphic maps of compact Riemann surfaces of smooth projective curves are algebraic (Chapter 8, Riemann existence and GAGA), isomorphisms and isogenies of the curves are the same as these maps. (ii) The kernel is $\{z : \alpha z \in \Lambda_2\}/\Lambda_1 = \alpha^{-1}\Lambda_2/\Lambda_1$, of order $[\alpha^{-1}\Lambda_2 : \Lambda_1] = [\Lambda_2 : \alpha\Lambda_1]$; a nonconstant holomorphic homomorphism of tori is an unramified covering whose fibres are cosets of the kernel, so its degree is the order of the kernel. For $\alpha = m$, $[\Lambda : m\Lambda] = m^2$. (iii) If $\alpha\Lambda \subseteq \Lambda$, write $\alpha\omega_1 = a\omega_1 + b\omega_2$ and $\alpha\omega_2 = c\omega_1 + d\omega_2$ with $a, b, c, d \in \mathbb{Z}$. Then $\alpha$ is an eigenvalue of $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with eigenvector $(\omega_1, \omega_2)$, so it is a root of the monic integral polynomial $X^2 - (a+d)X + (ad-bc)$. If $\alpha$ were real, the real and imaginary parts of the eigenvector, which span $\mathbb{R}^2$ because $\omega_1, \omega_2$ are $\mathbb{R}$-independent, would both be eigenvectors, so $M = \alpha \cdot 1$ and $\alpha \in \mathbb{Z}$. So either $\alpha \in \mathbb{Z}$, or $\alpha$ is a non-real quadratic integer, and then $b \neq 0$ and $\tau = (\alpha - a)/b \in \mathbb{Q}(\alpha)$ is imaginary quadratic. Conversely if $A\tau^2 + B\tau + C = 0$ with integers $A \neq 0$, $B$, $C$, then $A\tau\Lambda_\tau \subseteq \Lambda_\tau$ for $\Lambda_\tau = \mathbb{Z} + \mathbb{Z}\tau$ ($A\tau \cdot \tau = -B\tau - C$), and $A\tau \notin \mathbb{Z}$. The ring $\mathrm{End}(\mathbb{C}/\Lambda)$ is a subring of $\mathbb{C}$ containing $\mathbb{Z}$, of rank

at most 2 over $\mathbb{Z}$ (it embeds into $\Lambda \simeq \mathbb{Z}^2$ by $\alpha \mapsto \alpha\omega_1$), so it is $\mathbb{Z}$ or an order in $\mathbb{Q}(\tau)$. (iv) By Theorem 9.32(i), (ii) and the computation in the proof of Corollary 9.33(i). The value of $\zeta$ is computed in [185, Chapter VI, exercises]; it matches the intersection pairing on $H_1$ up to a universal sign (Chapter 8). □

**Theorem 9.56** (moduli of curves over $\mathbb{C}$)**.**

*(i) Every lattice is homothetic to $\Lambda_\tau = \mathbb{Z} + \mathbb{Z}\tau$ for some $\tau \in \mathcal{H}$, and $\Lambda_\tau$, $\Lambda_{\tau'}$ are homothetic iff $\tau' = \frac{a\tau+b}{c\tau+d}$ for some $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Hence the isomorphism classes of elliptic curves over $\mathbb{C}$, the homothety classes of lattices, and the points of $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$ are in bijection.*

*(ii) $j(\tau) := j(\Lambda_\tau)$ is holomorphic on $\mathcal{H}$ and $\mathrm{SL}_2(\mathbb{Z})$-invariant, and $j(i) = 1728$, $j(e^{2\pi i/3}) = 0$.*

*(iii) $j\colon \mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H} \to \mathbb{C}$ is a bijection, and $j(\tau) = q^{-1} + 744 + 196884q + \cdots$ with $q = e^{2\pi i\tau}$.*

*Proof.* (i) Given a lattice, choose an oriented basis (swap $\omega_1$, $\omega_2$ if necessary); then $\omega_1^{-1}\Lambda = \Lambda_\tau$ with $\tau = \omega_2/\omega_1 \in \mathcal{H}$. If $\Lambda_{\tau'} = c\Lambda_\tau$, then $(c, c\tau)$ and $(1, \tau')$ are two oriented bases of $\Lambda_{\tau'}$ (as $\mathrm{Im}(c\tau/c) = \mathrm{Im}\,\tau > 0$), so $c\tau = a\tau' + b$, $c = c'\tau' + d$ with $\left(\begin{smallmatrix} a & b \\ c' & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z})$; then $\tau = \frac{a\tau'+b}{c'\tau'+d}$, and $\mathrm{Im}\,\tau = (ad - bc')\,\mathrm{Im}\,\tau'/|c'\tau' + d|^2$ forces $ad - bc' = +1$; the inverse matrix expresses $\tau'$ in terms of $\tau$. Conversely $\Lambda_{\gamma\tau} = (c\tau + d)^{-1}\Lambda_\tau$ for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. The bijection with curves is Theorem 9.53 and Proposition 9.55(i). (ii) $G_{2k}(\Lambda_\tau) = \sum_{(m,n)\neq(0,0)} (m + n\tau)^{-2k}$ converges uniformly on compact subsets of $\mathcal{H}$ (the bound of Lemma 9.51 is locally uniform in $\tau$), so it is holomorphic (Appendix A); $\Delta(\Lambda_\tau) \neq 0$, so $j$ is holomorphic, and invariant by (i) and Lemma 9.51. For $\tau = i$, $i\Lambda_i = \Lambda_i$, so $G_6 = i^{-6}G_6 = -G_6$, $g_3 = 0$ and $j = 1728$; for $\tau = \zeta_3 = e^{2\pi i/3}$, $\zeta_3\Lambda = \Lambda$, so $G_4 = \zeta_3^{-4}G_4$, $g_2 = 0$ and $j = 0$. (iii) We do not prove this here; it is proved in Chapter 12 (fundamental domain, valence formula, $q$-expansions). Injectivity also follows from (i) and Theorem 9.6(i), surjectivity from Theorem 9.6(ii) and Theorem 9.53(ii). □

**Definition 9.57** (the period lattice of a curve over $\mathbb{Q}$)**.** Let $E/\mathbb{Q}$ be given by its global minimal model (Chapter 10) and $\omega = dx/(2y + a_1x + a_3)$ its Néron differential. The *period lattice* of $E$ is $\Lambda_E = \{\int_\gamma \omega : \gamma \in H_1(E(\mathbb{C}), \mathbb{Z})\}$, so that $E(\mathbb{C}) \simeq \mathbb{C}/\Lambda_E$ with $\omega \leftrightarrow dz$ (Theorem 9.53(ii)). Its *area* is $\mathrm{vol}(\Lambda_E)$, and $\omega_1(E)$ is the least positive real element of $\Lambda_E$ (the *real period*).

**Proposition 9.58** (periods of curves over $\mathbb{Q}$)**.** *Let $E/\mathbb{Q}$ be as in Definition 9.57.*

*(i) $\Lambda_E$ is stable under complex conjugation. If $\Delta > 0$, $\Lambda_E = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1 > 0$ and $\omega_2 \in i\mathbb{R}$ (a rectangular lattice; $E(\mathbb{R})$ has two components). If $\Delta < 0$, $\Lambda_E = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1 > 0$ and $\mathrm{Re}\,\omega_2 = \omega_1/2$ ($E(\mathbb{R})$ is connected).*

*(ii) $\mathrm{vol}(\Lambda_E) = \omega_1\,\mathrm{Im}\,\omega_2$ for such a basis with $\mathrm{Im}\,\omega_2 > 0$, and $\mathrm{vol}(\Lambda_E) = \frac{i}{2}\int_{E(\mathbb{C})} \omega \wedge \bar{\omega}$.*

*(iii) The periods are computed by the arithmetic-geometric mean $\mathrm{AGM}(a, b)$, the common limit of $a_{n+1} = (a_n + b_n)/2$, $b_{n+1} = \sqrt{a_n b_n}$ for $a_0 = a > 0$, $b_0 = b > 0$. If $\Delta < 0$, let $e_1$ be the real root of $F(x) = 4x^3 + b_2x^2 + 2b_4x + b_6$, $\gamma = 3e_1 + b_2/4$ and $\delta = \sqrt{3e_1^2 + b_2e_1/2 + b_4/2}$; then*

$$\omega_1 = \frac{2\pi}{\mathrm{AGM}(2\sqrt{\delta}, \sqrt{2\delta + \gamma})}, \qquad \mathrm{Im}\,\omega_2 = \frac{\pi}{\mathrm{AGM}(2\sqrt{\delta}, \sqrt{2\delta - \gamma})}$$

*for the basis of (i) with $\mathrm{Im}\,\omega_2 > 0$ [30, Section 7.4], [35, Section 3.7]. In PARI the periods are `E.omega`.*

*(iv) If $\varphi\colon E \to E'$ is an isogeny over $\mathbb{Q}$ and $\omega'$ is the Néron differential of $E'$, then $\varphi^*\omega' = c_\varphi\,\omega$ with $c_\varphi \in \mathbb{Q}^\times$, and $z \mapsto c_\varphi z$ maps $\Lambda_E$ into $\Lambda_{E'}$; so the period lattices of isogenous curves are commensurable.*

*Sketch of proof.* (i) Complex conjugation acts on $E(\mathbb{C})$ and on $H_1(E(\mathbb{C}),\mathbb{Z})$, and $\int_{\bar\gamma}\omega = \overline{\int_\gamma \omega}$ because $\omega$ has real coefficients; so $\bar\Lambda_E = \Lambda_E$. For a conjugation-stable lattice, $\Lambda \cap \mathbb{R} = \mathbb{Z}\omega_1$ and $\Lambda \cap i\mathbb{R} = \mathbb{Z}i\mu$ with $\omega_1, \mu > 0$, and $2\lambda = (\lambda + \bar\lambda) + (\lambda - \bar\lambda) \in \mathbb{Z}\omega_1 + \mathbb{Z}i\mu$ for every $\lambda \in \Lambda$, so $\Lambda$ contains $\mathbb{Z}\omega_1 + \mathbb{Z}i\mu$ with index 1 or 2: rectangular, or with basis $\omega_1$, $\frac{\omega_1}{2} + \frac{i\mu}{2}$. Which case occurs is decided by the number of real roots of $F$, that is, by the sign of $\Delta$ [35, Section 3.7]. (ii) The area of the parallelogram; and $\frac{i}{2}dz \wedge d\bar z = dx \wedge dy$ on $\mathbb{C}$. (iii) We do not prove the AGM formula; see the references given, and Exercise 9.114 for the numbers of `11a1`. (iv) $\varphi^*\omega'$ is a holomorphic differential defined over $\mathbb{Q}$, hence a rational multiple of $\omega$, nonzero because $\varphi$ is separable; under the uniformisations $\varphi$ becomes $z \mapsto c_\varphi z$ (Proposition 9.55(i)). For a modular parametrisation in place of $\varphi$ the analogous constant is the subject of the Manin constant (Chapter 14). □

**Example 9.59** (the periods of `11a1`)**.** For `11a1`, $\Delta = -11^5 < 0$, and PARI gives (to 20 digits)

$$\omega_1 = 1.2692093042795534217\ldots,$$

$$\omega_2 = \tfrac{1}{2}\omega_1 + 1.4588166169384952294\ldots\, i,$$

so that

$$\operatorname{vol}(\Lambda_E) = \omega_1 \operatorname{Im}\omega_2 = 1.8515436234559593178\ldots.$$

The value $\omega_1$ agrees with the column $\Omega = 1.26920930427955$ of Cremona's `allbsd` file. The point $\tau = \omega_2/\omega_1 = \frac{1}{2} + 1.1493901061232523807\ldots\, i$ is equivalent under $\tau \mapsto \tau - 1$ to $-\frac{1}{2} + 1.14939\ldots\, i$, which lies in the standard fundamental domain $|\operatorname{Re}\tau| \le \frac{1}{2}$, $|\tau| \ge 1$, and $j(\tau) = -757.672637860056752\ldots = -122023936/161051 = j(\texttt{11a1})$ to 18 significant digits, the precision to which $\tau$ is given. (PARI returns the basis $(\omega_1, \frac{1}{2}\omega_1 - 1.4588\ldots i)$, which is not oriented.) The area enters the formula $\deg\varphi_E = 4\pi^2 c_E^2\langle f, f\rangle / \operatorname{vol}(\Lambda_E)$ of Chapter 14, where for `11a1` the modular degree is 1.

## 9.8 Elliptic curves over finite fields

The numbers $a_q$ of the paper are point counts over finite fields, and its local factors at good primes and at the type $\mathrm{I}_0^*$ primes are built from the roots $\alpha, \beta$ of $X^2 - a_q X + q$. This section proves the Hasse bound, the characteristic polynomial of Frobenius on the Tate module, and the identity $(\mathfrak{q}+1)^2 - a^2 = \#E(\mathbb{F}_\mathfrak{q}) \cdot \#E^{\mathrm{tw}}(\mathbb{F}_\mathfrak{q})$ behind the type $\mathrm{I}_0^*$ row of the paper's table, and it connects $a_p$ with $\rho_{E,\ell}(\mathrm{Frob}_p)$ for a curve over $\mathbb{Q}$ with good reduction at $p$.

**Definition 9.60** (the Frobenius endomorphism and $a_\mathfrak{q}$)**.** Let $E$ be an elliptic curve over $\mathbb{F}_\mathfrak{q}$, $\mathfrak{q} = p^r$. The *Frobenius endomorphism* is $\pi = \pi_\mathfrak{q}\colon (x, y) \mapsto (x^\mathfrak{q}, y^\mathfrak{q})$, and

$$a_\mathfrak{q} = a_\mathfrak{q}(E) = \mathfrak{q} + 1 - \#E(\mathbb{F}_\mathfrak{q}).$$

**Proposition 9.61** (properties of $\pi$)**.**

*(i) $\pi$ is an endomorphism of $E$ of degree $\mathfrak{q}$, purely inseparable, and it acts on $E(\overline{\mathbb{F}}_p)$ as the arithmetic Frobenius* $\mathrm{Frob}_\mathfrak{q}$.

*(ii)* $E(\mathbb{F}_{\mathfrak{q}^n}) = \ker(1-\pi^n)$ *for* $n \geq 1$.

*(iii)* $1-\pi$ *is separable, so* $\#E(\mathbb{F}_{\mathfrak{q}}) = \deg(1-\pi)$.

*Proof.* (i) $\pi$ maps $E$ to itself because the coefficients of $E$ lie in $\mathbb{F}_{\mathfrak{q}}$ and are fixed by $t \mapsto t^{\mathfrak{q}}$; it fixes $O$. Its degree, inseparability and its action on points are Chapter 8 (the Frobenius morphism). (ii) A point of $E(\overline{\mathbb{F}}_p)$ is fixed by $\pi^n$ iff its coordinates are fixed by $t \mapsto t^{\mathfrak{q}^n}$, iff they lie in $\mathbb{F}_{\mathfrak{q}^n}$. (iii) $\pi^*(dx) = d(x^{\mathfrak{q}}) = \mathfrak{q}x^{\mathfrak{q}-1}dx = 0$, so $\pi^*\omega = 0$ and, by Proposition 9.15(iii), $(1-\pi)^*\omega = \omega \neq 0$; so $1-\pi$ is separable (Chapter 8) and $\#\ker(1-\pi) = \deg(1-\pi)$ (Theorem 9.19(iii)). □

**Theorem 9.62** (Hasse)**.** $|a_{\mathfrak{q}}| \leq 2\sqrt{\mathfrak{q}}$.

*Proof.* By Theorem 9.21(iv) the pairing $\langle \varphi, \psi \rangle = \deg(\varphi+\psi) - \deg\varphi - \deg\psi$ is bilinear on $\mathrm{End}(E)$. Taking $\varphi = 1$, $\psi = -\pi$, and using $\deg(-\pi) = \deg\pi = \mathfrak{q}$ (Proposition 9.61(i)) and $\langle 1, -\pi\rangle = -\langle 1, \pi\rangle$, one gets $\deg(1-\pi) = 1 + \mathfrak{q} - \langle 1, \pi\rangle$, so $\langle 1, \pi\rangle = 1 + \mathfrak{q} - \#E(\mathbb{F}_{\mathfrak{q}}) = a_{\mathfrak{q}}$ by Proposition 9.61(iii). For $m, n \in \mathbb{Z}$, expanding in the same way,

$$0 \leq \deg(m - n\pi) = m^2 \deg 1 + n^2 \deg\pi - mn\langle 1, \pi\rangle = m^2 - a_{\mathfrak{q}}mn + \mathfrak{q}n^2.$$

A binary quadratic form that is nonnegative on $\mathbb{Z}^2$ is nonnegative on $\mathbb{Q}^2$ (by homogeneity) and on $\mathbb{R}^2$ (by continuity). Taking $(m, n) = (a_{\mathfrak{q}}/2, 1)$ gives $\mathfrak{q} - a_{\mathfrak{q}}^2/4 \geq 0$, that is, $a_{\mathfrak{q}}^2 \leq 4\mathfrak{q}$. □

**Theorem 9.63** (the characteristic polynomial of Frobenius)**.** *Let* $\ell \neq p$.

*(i) For every* $\varphi \in \mathrm{End}(E)$*, the induced endomorphism* $\varphi_\ell$ *of* $T_\ell E$ *has* $\det\varphi_\ell = \deg\varphi$ *and* $\operatorname{tr}\varphi_\ell = 1 + \deg\varphi - \deg(1-\varphi)$*; both lie in* $\mathbb{Z}$.

*(ii)* $\pi_\ell$ *has characteristic polynomial* $X^2 - a_{\mathfrak{q}}X + \mathfrak{q}$*, and* $\pi^2 - a_{\mathfrak{q}}\pi + \mathfrak{q} = 0$ *in* $\mathrm{End}(E)$.

*(iii) Let* $\alpha, \beta \in \mathbb{C}$ *be the roots of* $X^2 - a_{\mathfrak{q}}X + \mathfrak{q}$*. Then* $\alpha + \beta = a_{\mathfrak{q}}$, $\alpha\beta = \mathfrak{q}$, $|\alpha| = |\beta| = \sqrt{\mathfrak{q}}$*, and for every* $n \geq 1$

$$\#E(\mathbb{F}_{\mathfrak{q}^n}) = \mathfrak{q}^n + 1 - \alpha^n - \beta^n; \quad \textit{in particular} \quad \#E(\mathbb{F}_{\mathfrak{q}^2}) = \mathfrak{q}^2 + 1 - (a_{\mathfrak{q}}^2 - 2\mathfrak{q}).$$

*Equivalently, the zeta function of* $E$ *is*

$$Z(E/\mathbb{F}_{\mathfrak{q}}, T) = \exp\Big(\sum_{n\geq 1} \#E(\mathbb{F}_{\mathfrak{q}^n})\frac{T^n}{n}\Big) = \frac{1 - a_{\mathfrak{q}}T + \mathfrak{q}T^2}{(1-T)(1-\mathfrak{q}T)}.$$

*Proof.* (i) The Weil pairing $e$ on $T_\ell E$ (Proposition 9.35(iii)) satisfies $e(\varphi s, \varphi t) = e(s, \hat\varphi\varphi t) = e(s,t)^{\deg\varphi}$ by Theorem 9.32(vi) and Theorem 9.21. For any $\mathbb{Z}_\ell$-linear $g$ on a free module of rank 2 and an alternating bilinear form $e$, $e(gs, gt) = e(s,t)^{\det g}$ (check on a basis, as in Corollary 9.33). Since $e$ is perfect, some $e(s,t)$ generates $\mathbb{Z}_\ell(1)$, so $\det\varphi_\ell = \deg\varphi$. For a $2\times 2$ matrix $g$, $\det(1-g) = 1 - \operatorname{tr} g + \det g$, and $(1-\varphi)_\ell = 1 - \varphi_\ell$. (ii) By (i) and Proposition 9.61(iii), $\det\pi_\ell = \deg\pi = \mathfrak{q}$ and $\operatorname{tr}\pi_\ell = 1 + \mathfrak{q} - \#E(\mathbb{F}_{\mathfrak{q}}) = a_{\mathfrak{q}}$. By Cayley-Hamilton, $\pi^2 - a_{\mathfrak{q}}\pi + \mathfrak{q}$ acts as 0 on $T_\ell E$, hence kills every $E[\ell^n]$; a nonzero endomorphism has finite kernel (Theorem 9.19(ii)), so it is 0. (iii) The discriminant $a_{\mathfrak{q}}^2 - 4\mathfrak{q} \leq 0$ (Theorem 9.62), so $\alpha, \beta$ are complex conjugate, and $\alpha\beta = \mathfrak{q}$ gives $|\alpha|^2 = \mathfrak{q}$. As $\pi^n$ is the Frobenius of $\mathbb{F}_{\mathfrak{q}^n}$, $\#E(\mathbb{F}_{\mathfrak{q}^n}) = \deg(1-\pi^n) = \det(1-\pi_\ell^n) = (1-\alpha^n)(1-\beta^n)$, the eigenvalues of $\pi_\ell^n$ being $\alpha^n, \beta^n$. For $n = 2$, $\alpha^2 + \beta^2 = a_{\mathfrak{q}}^2 - 2\mathfrak{q}$. Finally $\sum_n (1 + \mathfrak{q}^n - \alpha^n - \beta^n)T^n/n = -\log(1-T) - \log(1-\mathfrak{q}T) + \log(1-\alpha T) + \log(1-\beta T)$. □

For `11a1`, with $a_3=-1$, $a_5=1$, $a_7=-2$, $a_{13}=4$, the formula gives $\#E(\mathbb{F}_{p^2})=15,35,60,180$ for $p=3,5,7,13$ (for instance $9+1-((-1)^2-6)=15$ and $49+1-(4-14)=60$), and a direct count over $\mathbb{F}_{p^2}$ with PARI gives the same numbers.

**Definition 9.64** (supersingular and ordinary)**.** An elliptic curve $E$ over $\mathbb{F}_{\mathfrak{q}}$ is *supersingular* if $E[p]=0$, and *ordinary* otherwise.

**Proposition 9.65** (criteria for supersingularity)**.**

*(i) $E$ is supersingular iff $p \mid a_{\mathfrak{q}}$, iff $\mathrm{End}_{\overline{\mathbb{F}}_p}(E)$ is an order in a quaternion algebra.*

*(ii) If $\mathfrak{q}=p\geq 5$, then $E$ is supersingular iff $a_p=0$.*

*Proof.* (i) We do not prove this; see [185, Section V.3]. (ii) By (i) supersingularity means $p \mid a_p$, and $|a_p|\leq 2\sqrt{p}<p$ for $p\geq 5$ (Theorem 9.62), so $p \mid a_p$ iff $a_p=0$. □

For example, by (i) the reduction of `11a1` is supersingular at $p=19$ and $p=29$, where $a_p=0$, and also at $p=2$, where $a_2=-2$ is divisible by 2 (Example 9.68); it is ordinary at the other good primes $p<40$ (PARI's `ellissupersingular` agrees). The case $p=2$ shows why (ii) needs $p\geq 5$: for $p=2,3$ one has $2\sqrt{p}\geq p$, so $a_p=\pm p$ is allowed by the Hasse bound.

**Proposition 9.66** (point counts of a quadratic twist over $\mathbb{F}_{\mathfrak{q}}$)**.** *Let $\mathfrak{q}$ be odd, $E\colon y^2=f(x)$ with $f$ a cubic over $\mathbb{F}_{\mathfrak{q}}$ with distinct roots (so that $E$ is smooth), $\theta\in\mathbb{F}_{\mathfrak{q}}^{\times}$ a non-square, and $E^{\mathrm{tw}}\colon \theta y^2=f(x)$, which is isomorphic to $y^2=\theta^3 f(x/\theta)$, the twist of $E$ by $\theta$ in the sense of Definition 9.83. Let $\eta$ be the quadratic character of $\mathbb{F}_{\mathfrak{q}}^{\times}$, extended by $\eta(0)=0$. Then*

$$\#E(\mathbb{F}_{\mathfrak{q}})=\mathfrak{q}+1+\sum_{x\in\mathbb{F}_{\mathfrak{q}}}\eta(f(x)),\qquad a_{\mathfrak{q}}(E^{\mathrm{tw}})=-a_{\mathfrak{q}}(E),\qquad \#E(\mathbb{F}_{\mathfrak{q}})+\#E^{\mathrm{tw}}(\mathbb{F}_{\mathfrak{q}})=2\mathfrak{q}+2,$$

$$(\mathfrak{q}+1)^2-a_{\mathfrak{q}}(E)^2=\#E(\mathbb{F}_{\mathfrak{q}})\cdot\#E^{\mathrm{tw}}(\mathbb{F}_{\mathfrak{q}}).$$

*Proof.* For each $x$, the number of $y\in\mathbb{F}_{\mathfrak{q}}$ with $y^2=c$ is $1+\eta(c)$, and with $\theta y^2=c$ it is $1+\eta(c/\theta)=1-\eta(c)$. Adding the point $O$, $\#E(\mathbb{F}_{\mathfrak{q}})=1+\sum_x(1+\eta(f(x)))$ and $\#E^{\mathrm{tw}}(\mathbb{F}_{\mathfrak{q}})=1+\sum_x(1-\eta(f(x)))$. So $a_{\mathfrak{q}}(E)=-\sum_x\eta(f(x))=-a_{\mathfrak{q}}(E^{\mathrm{tw}})$, the sum of the two counts is $2\mathfrak{q}+2$, and their product is $(\mathfrak{q}+1-a_{\mathfrak{q}})(\mathfrak{q}+1+a_{\mathfrak{q}})$. The isomorphism: multiplying $\theta y^2=f(x)$ by $\theta^3$ and putting $Y=\theta^2 y$, $X=\theta x$ gives $Y^2=\theta^3 f(X/\theta)$. □

**Theorem 9.67** (Frobenius at a prime of good reduction)**.** *Let $E/\mathbb{Q}$ have good reduction at $p$ (its minimal model has $p\nmid\Delta$), let $\tilde{E}$ be the reduction, $a_p=a_p(E)=p+1-\#\tilde{E}(\mathbb{F}_p)$, and $\ell\neq p$.*

*(i) $\rho_{E,\ell}$ is unramified at $p$, and $\rho_{E,\ell}(\mathrm{Frob}_p)$ has characteristic polynomial $X^2-a_pX+p$, for every choice of arithmetic Frobenius $\mathrm{Frob}_p$; in particular $\operatorname{tr}\rho_{E,\ell}(\mathrm{Frob}_p)=a_p$ does not depend on $\ell$.*

*(ii) $\bar{\rho}_{E,\ell}(\mathrm{Frob}_p)$ has characteristic polynomial $X^2-a_pX+p$ modulo $\ell$, so $\operatorname{tr}\bar{\rho}_{E,\ell}(\mathrm{Frob}_p)\equiv a_p \pmod{\ell}$ and $\det\bar{\rho}_{E,\ell}(\mathrm{Frob}_p)\equiv p \pmod{\ell}$.*

*We write $\alpha_p$, $\beta_p$ for the roots of $X^2-a_pX+p$.*

*Sketch of proof.* Fix an embedding $\overline{\mathbb{Q}} \to \overline{\mathbb{Q}}_p$; it defines a decomposition group $D_p \simeq G_{\mathbb{Q}_p}$ and its inertia group $I_p$ (Chapters 6 and 7). The points of $E[\ell^n]$ have coordinates in the finite extension $L = \mathbb{Q}_p(E[\ell^n])$ of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}}_p$, and reduction modulo the maximal ideal of the valuation ring of $L$ (equivalently of $\overline{\mathbb{Q}}_p$, with residue field $\bar{\mathbb{F}}_p$) maps $E[\ell^n]$ injectively into $\tilde{E}[\ell^n]$ (Chapter 10: the kernel of reduction has no torsion prime to $p$); both have $\ell^{2n}$ elements (Corollary 9.22(ii)), so it is a bijection, equivariant for $D_p \to G_{\mathbb{F}_p}$. Hence $I_p$ acts trivially on $E[\ell^n]$, and $\mathrm{Frob}_p$ acts as $\mathrm{Frob}_p \in G_{\mathbb{F}_p}$ does on $\tilde{E}[\ell^n]$, that is, as $\pi$ (Proposition 9.61(i)). By Theorem 9.63(ii) the characteristic polynomial is $X^2 - a_p X + p$. Part (ii) is the reduction of (i). This is the easy half of the criterion of Néron, Ogg and Shafarevich (Chapter 11); see [185, Section VII.7]. □

**Example 9.68** (the numbers $a_p$ for the running examples)**.** Point counts on the global minimal models, $p < 40$; at a bad prime $a_p = p + 1 - \#\tilde{E}(\mathbb{F}_p)$ counting the singular point, which gives 1, $-1$ or 0 for split multiplicative, nonsplit multiplicative and additive reduction (Chapter 10).

| curve | $a_2$ | $a_3$ | $a_5$ | $a_7$ | $a_{11}$ | $a_{13}$ | $a_{17}$ | $a_{19}$ | $a_{23}$ | $a_{29}$ | $a_{31}$ | $a_{37}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| `11a1` | $-2$ | $-1$ | 1 | $-2$ | 1 | 4 | $-2$ | 0 | $-1$ | 0 | 7 | 3 |
| `20a1` | 0 | $-2$ | $-1$ | 2 | 0 | 2 | $-6$ | $-4$ | 6 | 6 | $-4$ | 2 |
| `99a1` | $-1$ | 0 | $-4$ | $-2$ | $-1$ | $-2$ | 2 | $-6$ | 4 | $-6$ | 4 | $-6$ |
| `121a1` | $-1$ | 2 | 1 | 2 | 0 | $-1$ | 5 | $-6$ | 2 | $-9$ | $-2$ | $-3$ |
| `121d1` | 2 | $-1$ | 1 | 2 | 0 | $-4$ | 2 | 0 | $-1$ | 0 | 7 | 3 |
| `539c1` | 1 | $-2$ | 2 | 0 | 1 | $-4$ | $-4$ | 0 | $-4$ | $-6$ | $-10$ | $-6$ |
| `726a1` | $-1$ | $-1$ | 0 | 0 | 0 | $-6$ | 6 | $-6$ | 6 | $-6$ | 4 | $-2$ |
| `980i1` | 0 | $-3$ | 1 | 0 | $-5$ | 3 | 1 | $-6$ | 6 | $-9$ | 4 | 2 |

Two patterns can be read off. For `726a1`, whose mod 3 image is $N_s(\mathbb{F}_3)$ and so satisfies $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$, $a_p \equiv 0 \pmod 3$ at the good $p \equiv 2 \pmod 3$ ($a_5 = 0$, $a_{17} = 6$, $a_{23} = 6$, $a_{29} = -6$), by Theorem 9.67(ii) and $\chi_{-3}(\mathrm{Frob}_p) = -1$. For `11a1`, $a_p \equiv 1 + p \pmod 5$ at every good $p$ (Proposition 9.77). The row of `121d1` is $\chi_{-11}(p)$ times that of `11a1` for $p \neq 11$ (Corollary 9.87). The coefficients of `11a1` agree with those of $q\prod_{n\geq 1}(1-q^n)^2(1-q^{11n})^2$, $q = e^{2\pi i\tau}$, up to $q^{20}$ (Chapter 12).

*In the paper* 9.69. In the table of Section 1 and in Lemma lem:euler, $E'$ is the twist of good reduction at a type $\mathrm{I}_0^*$ prime $q$, and $\alpha, \beta$ are the roots of $X^2 - a_q(E')X + q$; the proof uses "$\alpha\beta = q$ and $\alpha + \beta = a_q(E')$" (Theorem 9.63(iii)), and that $\mathrm{ad}^0 V$ has the Frobenius eigenvalues $\alpha/\beta, 1, \beta/\alpha$ for either Frobenius (Convention 9.40(d)). By Proposition 9.66, the factor $(q+1)^2 - a_q(E')^2$ of $e_\ell(E, q)$ is $\#E'(\mathbb{F}_q) \cdot \#E'^{\mathrm{tw}}(\mathbb{F}_q)$, the product of the numbers of points of the reduction of $E'$ and of its unramified quadratic twist; so $v_\ell((q+1)^2 - a_q(E')^2)$ is the sum of the $\ell$-adic valuations of two point counts. Lemma lem:old uses that the characteristic polynomial of $\bar{\rho}_h(\mathrm{Frob}_p)$ is $X^2 - a_p(h)X + p$, the analogue of Theorem 9.67(ii) for a newform $h$ (Chapter 15).

With Theorem 9.67 the local information at the good primes is complete: $\rho_{E,\ell}$ is unramified there and its Frobenius traces are the point counts $a_p$. The next section turns to global questions over $\mathbb{Q}$, where the same representations, now reduced modulo $\ell$, decide whether $E$ has rational torsion or rational isogenies.

## 9.9 Elliptic curves over the rationals: Mordell-Weil, torsion, isogenies

The paper's Eisenstein case (Conjecture conj:eis, Section sec:eis, Proposition prop:no9) is about rational torsion points and rational isogenies, and its remarks on optimality (Section 1, Section sec:oot) are about isogeny classes. This section states the theorems of Mordell and Weil, of Mazur and of Kenku, proves that $E[\ell]$ is reducible exactly when $E$ has a rational $\ell$-isogeny, describes the case $E[\ell] \simeq \mathbb{Z}/\ell \oplus \mu_\ell$, proves the paper's Proposition prop:no9, and proves that when $E[\ell]$ is irreducible nothing $\ell$-adic depends on which curve of the isogeny class one chooses.

**Theorem 9.70** (Mordell, Weil)**.** *For $E/\mathbb{Q}$, the group $E(\mathbb{Q})$ is finitely generated: $E(\mathbb{Q}) \simeq \mathbb{Z}^r \oplus E(\mathbb{Q})_{\mathrm{tors}}$ with $E(\mathbb{Q})_{\mathrm{tors}}$ finite. The integer $r$ is the* rank *of $E$.*

We do not prove this; see [185, Chapter VIII]. The proof has two halves: the weak Mordell-Weil theorem, that $E(\mathbb{Q})/2E(\mathbb{Q})$ is finite, which comes from the Kummer map into a Selmer group (Chapter 17); and a descent with the canonical height, which is a quadratic form on $E(\mathbb{Q})$ with finitely many points below any bound.

**Proposition 9.71** (torsion and reduction)**.** *Let $E/\mathbb{Q}$ be given by its minimal model and $p$ a prime of good reduction.*

*(i) Reduction modulo $p$ is injective on $E(\mathbb{Q})[m]$ for every $m$ prime to $p$, and, if $p \geq 3$, on all of $E(\mathbb{Q})_{\mathrm{tors}}$. Hence $\#E(\mathbb{Q})_{\mathrm{tors}}$ divides $\#\tilde{E}(\mathbb{F}_p)$ for every good prime $p \geq 3$.*

*(ii) (Nagell and Lutz.) A torsion point of a model with integral coefficients and $a_1 = a_3 = 0$ has integral coordinates.*

*(iii) For `11a1`, $E(\mathbb{Q})_{\mathrm{tors}} = \langle (5,5) \rangle \simeq \mathbb{Z}/5$; for `20a1`, $E(\mathbb{Q})_{\mathrm{tors}} = \langle (4,10) \rangle \simeq \mathbb{Z}/6$.*

*Proof.* (i) is Chapter 10 (the kernel of reduction is the formal group, which has no torsion prime to $p$, and no $p$-torsion for $p \geq 3$ over $\mathbb{Z}_p$); see [185, Sections VII.3 and VIII.7]. (ii) We do not prove this; see [185, Section VIII.7]. (iii) For `11a1`, $\#\tilde{E}(\mathbb{F}_3) = 5$ and $\#\tilde{E}(\mathbb{F}_7) = 10$ (from $a_3 = -1$, $a_7 = -2$, Example 9.68), so $\#E(\mathbb{Q})_{\mathrm{tors}}$ divides 5, and $(5,5)$ has order 5 (Example 9.16). For `20a1`, $\#\tilde{E}(\mathbb{F}_3) = 6$ and $\#\tilde{E}(\mathbb{F}_7) = 6$, so the torsion has order dividing 6, and $(4,10)$ has order 6. □

**Theorem 9.72** (Mazur)**.** *For $E/\mathbb{Q}$, $E(\mathbb{Q})_{\mathrm{tors}}$ is isomorphic to one of*

$$\mathbb{Z}/n \ (1 \leq n \leq 10 \text{ or } n = 12), \qquad \mathbb{Z}/2 \times \mathbb{Z}/2n \ (1 \leq n \leq 4),$$

*and each of these fifteen groups occurs for infinitely many $j$-invariants.*

We do not prove this; see [125] and [185, Section VIII.7]. In Cremona's tables for conductor below 10,000 the torsion orders that occur are $1, \ldots, 10$, 12 and 16 (the last only as $\mathbb{Z}/2 \times \mathbb{Z}/8$). Consequences used by the paper: $v_\ell(\#E(\mathbb{Q})_{\mathrm{tors}}) \leq 1$ for $\ell = 5, 7$, $E(\mathbb{Q})$ has no point of order $\ell \geq 11$, and $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) \leq 2$ with equality only for $\mathbb{Z}/9$.

**Definition 9.73** (isogeny classes)**.** Curves $E, E'$ over $\mathbb{Q}$ are *isogenous* if there is a nonconstant isogeny $E \to E'$ defined over $\mathbb{Q}$. The equivalence classes (Theorem 9.74(i)) are the *isogeny classes*, labelled by Cremona as $N$ followed by a letter code (Section 9.12). Cremona's *isogeny matrix* of a class lists, for each pair of curves, the least degree of a cyclic isogeny over $\mathbb{Q}$ between them.

**Theorem 9.74** (isogeny classes)**.**

*(i) Being isogenous over $\mathbb{Q}$ is an equivalence relation.*

*(ii) Isogenous curves $E$, $E'$ have $V_\ell E \simeq V_\ell E'$ for every $\ell$; hence the same $a_p$ at every prime of good reduction for both, the same $L$-function (Section 9.11), and the same rank. They have the same conductor. In general they have different torsion, discriminants, $j$-invariants, Tamagawa numbers and periods.*

*(iii) (Faltings.) Conversely, $E$ and $E'$ are isogenous over $\mathbb{Q}$ iff $a_p(E) = a_p(E')$ for almost all $p$, iff $V_\ell E \simeq V_\ell E'$ as $G_{\mathbb{Q}}$-modules for one $\ell$.*

*(iv) (Kenku.) An isogeny class over $\mathbb{Q}$ has at most 8 curves (up to isomorphism over $\mathbb{Q}$).*

*Proof.* (i) Reflexive by [1], transitive by composition, symmetric by the dual isogeny, defined over $\mathbb{Q}$ (Theorem 9.21). (ii) $V_\ell E \simeq V_\ell E'$ is Proposition 9.37(iii); traces of Frobenius give the $a_p$ (Theorem 9.67). For the rank: $\varphi\colon E(\mathbb{Q}) \to E'(\mathbb{Q})$ and $\hat\varphi$ compose to $[\deg\varphi]$, which has finite kernel, so $\operatorname{rank} E(\mathbb{Q}) \le \operatorname{rank} E'(\mathbb{Q})$, and symmetrically. The conductor depends only on $V_\ell E$ as a representation of the local Galois groups (Chapters 10 and 11); we do not prove this here. The examples of Section 9.12 show the differences. (iii) We do not prove this; see [63]. (iv) We do not prove this; see [96, Theorem 2]. □

Examples (Cremona's `allisog` file; rows and columns ordered by curve number): `11a` has matrix $\left(\begin{smallmatrix} 1 & 5 & 5 \\ 5 & 1 & 25 \\ 5 & 25 & 1 \end{smallmatrix}\right)$; `20a` has $\left(\begin{smallmatrix} 1 & 2 & 3 & 6 \\ 2 & 1 & 6 & 3 \\ 3 & 6 & 1 & 2 \\ 6 & 3 & 2 & 1 \end{smallmatrix}\right)$; `121a` and `121c` consist of two curves joined by an 11-isogeny; `121d` has $\left(\begin{smallmatrix} 1 & 5 & 25 \\ 5 & 1 & 5 \\ 25 & 5 & 1 \end{smallmatrix}\right)$; `99a`, `539c` and `726a` consist of two curves joined by a 2-isogeny; `980i` has one curve. Below conductor 10,000 the classes have $1, 2, 3, 4, 6$ or $8$ curves.

**Proposition 9.75** (rational $\ell$-isogenies and reducibility)**.** *Let $\ell$ be a prime, $\operatorname{char} K \neq \ell$, and $E/K$.*

*(i) The isogenies of degree $\ell$ from $E$ defined over $K$, up to isomorphism of the target, correspond bijectively to the $G_K$-stable lines of $E[\ell]$, by $\varphi \mapsto \ker\varphi$.*

*(ii) $E[\ell]$ is reducible as an $\mathbb{F}_\ell[G_K]$-module iff $E$ has a $K$-rational $\ell$-isogeny. A $K$-rational point of order $\ell$ spans such a line; the converse fails.*

*(iii) If $E[\ell]$ is reducible, with $G_K$ acting on a stable line by $\eta_1$ and on the quotient by $\eta_2$, then $\overline{\rho}^{\mathrm{ss}}_{E,\ell} \simeq \eta_1 \oplus \eta_2$ and $\eta_1\eta_2 = \bar\chi_\ell$.*

*Proof.* (i) An isogeny of degree $\ell$ is separable, since its inseparable degree is a power of $\operatorname{char} K$ dividing $\ell$; so its kernel has order $\ell$ (Theorem 9.19(iii)) and is a line in $E[\ell]$, $G_K$-stable if the isogeny is defined over $K$ (Definition 9.17). Conversely a $G_K$-stable line $L$ gives $E \to E/L$ over $K$, and two separable isogenies with the same kernel differ by an isomorphism of the targets (Theorem 9.20). (ii) $E[\ell]$ is two-dimensional, so it is reducible iff it has a stable line; a rational point of order $\ell$ is fixed by $G_K$, so its span is stable. For $K = \mathbb{Q}$: `11a2` has a rational 5-isogeny (its class is `11a`) and trivial torsion; `539a1` has a rational 3-isogeny (image `3B`) and trivial torsion. (iii) is Corollary 9.33(iii). □

As a check on Proposition 9.75 against the data: for $\ell = 3, 5, 7$ and curve number 1 of every class without CM below conductor 10,000, Cremona's `galrep` file lists a Borel or split Cartan image at $\ell$ (Chapter 11) exactly when the row of the isogeny matrix contains a degree divisible by $\ell$.

**Theorem 9.76** (Mazur, Kenku: rational cyclic isogenies)**.** *Let $E/\mathbb{Q}$.*

*(i) If $E$ has a rational isogeny of prime degree $\ell$, then $\ell \in \{2,3,5,7,11,13,17,19,37,43,67,163\}$.*

*(ii) If $E$ has a rational cyclic isogeny of degree $n$, then $n \le 19$ or $n \in \{21,25,27,37,43,67,163\}$.*

*(iii) The degrees $n = 19, 27, 43, 67, 163$ occur only for curves with complex multiplication. For $n = 27$ the curves have $j = -2^{15}\cdot 3\cdot 5^3$ and CM by the order of discriminant $-27$.*

We do not prove this. Part (i) is Mazur's theorem [126]; (ii) and (iii) combine it with Kenku's determinations of the rational points of $X_0(N)$ for the remaining $N$ [92, 94, 93, 95], and the case $N = 27$, where $X_0(27)$ has genus one, is Ligozat's determination of the rational points of the modular curves of genus one [114]. The complete list (ii) is [96, Theorem 1], which attributes the levels of genus one to Ligozat, Ogg and Joly; Kenku does not discuss complex multiplication, so (iii) is not in that paper. A summary table is in [120] (we have not checked its numbering). In Cremona's tables the cyclic 27-isogeny joins `27a2` and `27a4`, both with $j = -12288000 = -2^{15}\cdot 3\cdot 5^3$ (Theorem 9.46), and the isogeny degrees occurring below conductor 10,000 are 1 to 16, 18, 19, 21, 25, 27, 37, 43 and 67.

**Proposition 9.77** (the structure $\mathbb{Z}/\ell \oplus \mu_\ell$)**.** *Let $\ell$ be odd and let $E/\mathbb{Q}$ have a rational point $P_0$ of order $\ell$.*

*(i) In a basis $(P_0, Q)$ of $E[\ell]$, $\bar\rho_{E,\ell} = \left(\begin{smallmatrix} 1 & * \\ 0 & \bar\chi_\ell \end{smallmatrix}\right)$.*

*(ii) The following are equivalent: the sequence $0 \to \mathbb{Z}/\ell \to E[\ell] \to \mu_\ell \to 0$ of Corollary 9.33(iii) splits; $E[\ell] \simeq \mathbb{Z}/\ell \oplus \mu_\ell$ as $G_\mathbb{Q}$-modules; the image of $\bar\rho_{E,\ell}$ is conjugate to $\{\mathrm{diag}(1,d) : d \in \mathbb{F}_\ell^\times\}$ (Sutherland's label $\ell$`Cs.1.1`, Chapters 1 and 11); $E[\ell]$ has two $G_\mathbb{Q}$-stable lines, that is, $E$ has two rational $\ell$-isogenies, with kernels $\mathbb{Z}/\ell$ and $\mu_\ell$.*

*(iii) $a_p \equiv 1 + p \pmod{\ell}$ and $\#\tilde E(\mathbb{F}_p) \equiv 0 \pmod{\ell}$ for every good prime $p \ne \ell$.*

*Proof.* (i) is Corollary 9.33(iii) with $\eta_1 = 1$. (ii) The sequence splits iff there is a stable complement to $\langle P_0\rangle$, iff in a suitable basis the matrices are diagonal, $\mathrm{diag}(1, \bar\chi_\ell(\sigma))$; as $\bar\chi_\ell\colon G_\mathbb{Q} \to \mathbb{F}_\ell^\times$ is surjective, the image is then the whole group $\{\mathrm{diag}(1,d)\}$. A stable complement is a second stable line; since $\ell$ is odd, $1 \ne \bar\chi_\ell$, the two characters are distinct, and the stable lines are then exactly the two eigenlines, on which $G_\mathbb{Q}$ acts trivially ($\mathbb{Z}/\ell$) and by $\bar\chi_\ell$ ($\mu_\ell$). Conversely two distinct stable lines give a splitting. Rational $\ell$-isogenies correspond to stable lines by Proposition 9.75. (iii) By (i) and Theorem 9.67(ii), $a_p \equiv \mathrm{tr}\left(\begin{smallmatrix} 1 & * \\ 0 & p \end{smallmatrix}\right) = 1 + p$, and $\#\tilde E(\mathbb{F}_p) = p + 1 - a_p \equiv 0$. □

Examples (all checked with PARI and the `galrep` file). The curves `14a1`, `19a1`, `26a1` and the CM curve `27a1` have $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ (image `3Cs.1.1`); `20a1` has a rational point of order 3 with non-split $E[3]$ (image `3B.1.1`). The curve `11a1` has $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ (image `5Cs.1.1`), and $a_p \equiv 1 + p \pmod 5$ at every good $p < 500$. Its two 5-isogenies have kernels $\mathbb{Z}/5 = \langle(5,5)\rangle$ and $\mu_5$ (the points with $x$-coordinate a root of $5x^2 + 5x - 29$, a factor of $\psi_5$); Vélu's formulas give `11a1`$/\langle(5,5)\rangle \simeq$ `11a2`, with trivial torsion, and `11a1`$/\mu_5 \simeq$ `11a3`, which contains the rational group $E[5]/\mu_5 \simeq \mathbb{Z}/5$ and has torsion $\mathbb{Z}/5$.

**Proposition 9.78** (no rational point of order 9 beside $\mathbb{Z}/3 \oplus \mu_3$)**.** *Let $E/\mathbb{Q}$ be without complex multiplication, with $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$. Then $E(\mathbb{Q})$ has no point of order 9; so $E(\mathbb{Q})[3^\infty] \simeq \mathbb{Z}/3$. Consequently, for every $E/\mathbb{Q}$ without complex multiplication,*

$$v_3\big(\#E(\mathbb{Q})_{\mathrm{tors}}\big) + \big[E[3] \simeq \mathbb{Z}/3 \oplus \mu_3\big] \le 2,$$

*where* $[\cdot]$ *is* 1 *if the condition holds and* 0 *otherwise; and* $v_\ell(\#E(\mathbb{Q})_{\mathrm{tors}}) \le 1$ *for* $\ell = 5, 7$, $= 0$ *for* $\ell \ge 11$.

*Proof.* Suppose $P \in E(\mathbb{Q})$ has order 9.

*Step 1.* $3P$ is a rational point of order 3. The rational points of $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ are $\mathbb{Z}/3 \oplus \mu_3(\mathbb{Q}) = \mathbb{Z}/3 \oplus 0$, because $\mu_3(\mathbb{Q}) = \{1\}$ ($\chi_{-3} \neq 1$). So $\langle 3P\rangle$ is the line $\mathbb{Z}/3$, and $\langle P\rangle \cap E[3] = \langle 3P\rangle$ (the elements of order dividing 3 in a cyclic group of order 9) meets the line $\mu_3$ only in $O$.

*Step 2.* The line $\mu_3$ is $G_{\mathbb{Q}}$-stable, so $\varphi\colon E \to E' = E/\mu_3$ is defined over $\mathbb{Q}$ (Theorem 9.20), of degree 3; let $\hat\varphi\colon E' \to E$ be its dual, also over $\mathbb{Q}$, of degree 3, with $\varphi \circ \hat\varphi = [3]_{E'}$ (Theorem 9.21). For $Q \in E'[3]$, $\varphi(\hat\varphi(Q)) = 3Q = O$, so $\hat\varphi(E'[3]) \subseteq \ker\varphi = \mu_3$. The kernel of $\hat\varphi$ has order 3 and is killed by 3, so it lies in $E'[3]$, and $\#\hat\varphi(E'[3]) = 9/3 = 3$. Hence $\hat\varphi(E'[3]) = \mu_3$.

*Step 3.* Let $C = \hat\varphi^{-1}(\langle P\rangle) \subset E'(\overline{\mathbb{Q}})$. It is a subgroup, $G_{\mathbb{Q}}$-stable because $\langle P\rangle$ is (it consists of rational points) and $\hat\varphi$ is defined over $\mathbb{Q}$, and $\#C = \#\ker\hat\varphi \cdot \#\langle P\rangle = 27$, because $\hat\varphi$ is surjective on $\overline{\mathbb{Q}}$-points.

*Step 4.* $C$ is cyclic. If $E'[3] \subseteq C$, then $\mu_3 = \hat\varphi(E'[3]) \subseteq \hat\varphi(C) = \langle P\rangle$, contradicting Step 1. By Proposition 9.23(i), $C \simeq \mathbb{Z}/a \times \mathbb{Z}/ab$ with $a^2 b = 27$ and $E'[a] \subseteq C$; so $a = 1$ and $C \simeq \mathbb{Z}/27$.

*Step 5.* $E' \to E'/C$ is a rational cyclic isogeny of degree 27 (Theorem 9.20). By Theorem 9.76(iii), $E'$ has complex multiplication, hence so has $E$ (Lemma 9.45): a contradiction.

For the consequences: by Theorem 9.72, $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) \le 2$ with equality only for $E(\mathbb{Q})_{\mathrm{tors}} \simeq \mathbb{Z}/9$; in that case the bracket is 0 by what was just proved, and otherwise $v_3 \le 1$. The statements for $\ell \ge 5$ are read off Mazur's list. □

Data: the 264 curves of conductor below 10,000 with image `3Cs.1.1` have torsion of order 3 (244 curves) or 6 (20 curves); the six curves there with a rational point of order 9 (`54b3`, `714i1`, `1482l1`, `1554n1`, `1890r3`, `6942n1`) all have image `3B.1.1`.

*In the paper* 9.79. Proposition 9.78 is the paper's Proposition prop:no9, with every step expanded; it turns the bound "$D \le 2$ for $\ell = 3$" of Conjecture conj:eis into a consequence of the first bound $D \le v_\ell(\#E(\mathbb{Q})_{\mathrm{tors}}) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3]$, and gives "$D \le 1$ for $\ell \ge 5$" in the same way. The paper cites [114], Ligozat's study of $X_0(27)$, a curve of genus one, for the fact that a rational cyclic 27-isogeny forces complex multiplication (Theorem 9.76(iii)), and [96, Theorem 1] for the list of the levels $N$ with $Y_0(N)(\mathbb{Q}) \neq \emptyset$, which contains 27; Kenku credits the levels of genus one to Ligozat, Ogg and Joly. In Section sec:eis, "a rational $\ell$-isogeny" and "Borel type" refer to Proposition 9.75, and the condition $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$, which the paper identifies with the image `3Cs.1.1` (the subgroup of order 2 of a split Cartan subgroup fixing a nonzero vector), is Proposition 9.77(ii). The second bound of Conjecture conj:eis counts the curves of the isogeny class with a rational point of order $\ell$; in the class `11a` at $\ell = 5$ these are `11a1` and `11a3`.

**Lemma 9.80** (isogenies prime to $\ell$)**.** *Let* $E/\mathbb{Q}$ *and* $\ell$ *a prime with* $E[\ell]$ *irreducible.*

*(i) Every isogeny* $\varphi\colon E \to E'$ *over* $\mathbb{Q}$ *is* $\psi \circ [a]$ *with* $\psi\colon E \to E'$ *a cyclic isogeny over* $\mathbb{Q}$ *of degree prime to* $\ell$.

*(ii) Hence* $T_\ell E \simeq T_\ell E'$ *as* $G_{\mathbb{Q}}$*-modules,* $\rho_{E',\ell}$ *and* $\rho_{E,\ell}$ *are conjugate in* $\mathrm{GL}_2(\mathbb{Z}_\ell)$, $E'[\ell] \simeq E[\ell]$ *is irreducible, and every invariant built from the* $G_{\mathbb{Q}}$*-module* $T_\ell E$ *(the adjoint* $\mathrm{ad}^0 T_\ell E$*, the length* $h_\ell(E)$*, the* $\ell$*-adic Tamagawa factors and Selmer groups of* $\mathrm{ad}^0 T_\ell E$*) is the same for* $E$ *and* $E'$.

*(iii)* $t_\ell(E, q) = t_\ell(E', q)$ *for every prime* $q$.

*Proof.* (i) By Proposition 9.23, $\varphi = \psi \circ [a]$ with $\psi$ cyclic over $\mathbb{Q}$ of some degree $n$. If $\ell \mid n$, then by Proposition 9.23(iii) $E$ has a rational cyclic $\ell$-isogeny, so $E[\ell]$ has a $G_{\mathbb{Q}}$-stable line (Proposition 9.75), contradicting irreducibility. (ii) $\psi$ induces $T_\ell E \simeq T_\ell E'$ by Proposition 9.37(iii), and everything listed is a function of the $G_{\mathbb{Q}}$-module $T_\ell E$ (with its restrictions to decomposition groups). □

*Sketch of proof of (iii).* $t_\ell(E,q) = v_\ell(-v_q(j_E))$ when $v_q(j_E) < 0$. Over a finite extension of $\mathbb{Q}_q$, $E$ becomes isomorphic to a Tate curve with parameter $q_E$, $v_q(q_E) = -v_q(j_E)$ (Chapter 10). An isogeny of prime degree $r$ between Tate curves has kernel either $\mu_r$ or generated by an $r$-th root of the parameter, and the parameter of the target is $q_E^r$, resp. that $r$-th root; so $v_q(j)$ is multiplied or divided by $r$. Composing, $v_q(j_{E'}) = v_q(j_E) \cdot r_1^{\pm 1} \cdots r_k^{\pm 1}$ with $r_1 \cdots r_k = \deg \psi$ prime to $\ell$, and $v_\ell(-v_q(j))$ is unchanged. (The multiplication map $[a]$ does not change the target.) If $v_q(j_E) \geq 0$ then also $v_q(j_{E'}) \geq 0$, potentially good reduction being an isogeny invariant (Chapter 10). The facts about Tate curves used here are proved in Chapter 10. □

**Corollary 9.81** (trivial isogeny classes)**.** *If the isogeny class of $E/\mathbb{Q}$ consists of $E$ alone, then $E$ has no rational $\ell$-isogeny for any prime $\ell$, so $E[\ell]$ is irreducible for every $\ell$ (including $\ell = 2$), and $E$ is the $X_0(N)$-optimal curve of its class. Example:* `980i1`*.*

*Proof.* A rational $\ell$-isogeny $E \to E'$ would put $E'$ in the class, and $E' \not\simeq E$ over $\mathbb{Q}$, since an isogeny of degree $\ell$ from $E$ to a curve isomorphic to $E$ would be an endomorphism of degree $\ell$ defined over $\mathbb{Q}$, impossible as $\mathrm{End}_{\mathbb{Q}}(E) = \mathbb{Z}$ (Proposition 9.43(ii)) and $\deg[m] = m^2$. Irreducibility follows from Proposition 9.75. The optimal curve is a curve of the class (Chapter 14). □

*In the paper* 9.82. Section 1 of the paper: "When $E[\ell]$ is irreducible, every curve of the class is linked to $E$ by a cyclic isogeny of degree prime to $\ell$" is Lemma 9.80(i); the consequence for the modular degree is Chapter 14. By (ii) and (iii) of the lemma, the hypothesis of Theorem thm:main on $\bar{\rho}_{E,\ell}$, the right-hand side of (eq:identity), and the terms $e_\ell(E,q)$ (read off $V_\ell E$ at $q$), $t_\ell(E,q)$ and $h_\ell(E)$ of (eq:rule) in Conjecture conj:euler do not depend on which curve of the class is taken. The remaining term $w_3(E)$ of (eq:rule) is defined by the Kodaira type at 3, and Kodaira types are not isogeny invariants in general: `121c1` and `121c2`, joined by an 11-isogeny, have types IV and IV$^*$ at 11, and `11a1` and `11a3`, joined by a 5-isogeny, have $\mathrm{I}_5$ and $\mathrm{I}_1$ at 11. In both examples the isogeny has degree equal to the residue characteristic $q = 11$, or the reduction at $q$ is multiplicative. In the situation of (eq:rule) at $\ell = 3$ the type at 3 is an invariant of the class, and the paper proves it (Section sec:data): "when $E[3]$ is irreducible the class has no 3-isogeny, so every isogeny in it is a composite of isogenies of prime degree $p \neq 3$, and an isogeny of prime degree $p$ preserves the Kodaira type at a prime $q \neq p$ of potentially good reduction [56, Theorem 5.4(1)]; so $w_3$ is constant on the class." The first clause is Lemma 9.80(i) together with the factorisation of a cyclic isogeny into isogenies of prime degree; the theorem of Dokchitser and Dokchitser is applied with $q = 3$. On the tables, below conductor 10,000, among the classes with $3 \mid N$, at least two curves and potentially good reduction at 3, "$w_3$ is constant on all 1,295 classes without a 3-isogeny, as it must be" (27 of them have complex multiplication, which Conjecture conj:euler excludes). With a 3-isogeny, where (eq:rule) does not apply at $\ell = 3$, the type at 3 can vary: $w_3$ "varies in 828 of the 1,293 classes with one", for example `54a` with types IV$^*$, II$^*$, II and $w_3 = 2, 2, 1$; Conjecture conj:euler at $\ell = 3$ excludes these classes, as $E[3]$ is then reducible. The conjecture is in any case stated for the optimal curve. Section sec:oot keeps curves "with trivial isogeny class (so that the curve is optimal and $E[\ell]$ is irreducible for every $\ell$)": this is Corollary 9.81.

## 9.10 Quadratic twists

Quadratic twists run through the paper: Lemma lem:twist describes the twist $E^{(q)}$ of $E$ by $\mathbb{Q}(\sqrt{q^*})$, the table of Section 1 and Lemma lem:euler use the twist $E'$ of good reduction at a type $\mathrm{I}_0^*$ prime, Theorems thm:twist and thm:quant twist by $\chi_S$, and the Example of Section 3 compares `121a1` with the class `121c`, and `539c1` with `77c1`. This section defines twists for a general Weierstrass model, computes their invariants, proves $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi_d$ and $a_p(E^{(d)}) = \chi_d(p)a_p(E)$, and settles which of the two possible twists the paper means. The effect of twisting on Kodaira types and conductors (Lemma lem:twist (i), (ii)) is proved in Chapter 10.

The idea is simple. The curves $y^2 = f(x)$ and $dy^2 = f(x)$ become isomorphic over $K(\sqrt{d})$, by $y \mapsto y/\sqrt{d}$, but this isomorphism is not defined over $K$: an automorphism $\sigma$ with $\sigma(\sqrt{d}) = -\sqrt{d}$ changes it by the automorphism $[-1]$. So the two curves have the same geometry (the same $j$) but Galois acts on their points differently, through an extra sign $\chi_d(\sigma)$, and every arithmetic quantity built from the Galois action (the $a_p$, the representation $\rho_{E,\ell}$) is multiplied by $\chi_d$. Over a finite field $\mathbb{F}_p$, with $d$ a non-square, this is the count of Proposition 9.66: an $x$ with $f(x)$ a nonzero square gives two points on one curve and none on the other.

**Definition 9.83** (quadratic twists)**.** Let $\operatorname{char} K \neq 2$ and $d \in K^\times$.

(a) For a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ of $E$, the *quadratic twist* of $E$ by $d$ (or by $K(\sqrt{d})$, or by $\chi_d$) is
$$E^{(d)}\colon y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6,$$
which is isomorphic to $dy^2 = x^3 + a_2x^2 + a_4x + a_6$ by $(x, y) \mapsto (dx, d^2y)$.

(b) For a general model, complete the square first (Lemma 9.3(ii)): from $\eta^2 = 4x^3 + b_2x^2 + 2b_4x + b_6$ one gets $E^{(d)}\colon \eta^2 = 4x^3 + db_2x^2 + 2d^2b_4x + d^3b_6$, and with $X = 4x$, $Y = 4\eta$ the Weierstrass model
$$E^{(d)}\colon Y^2 = X^3 + db_2X^2 + 8d^2b_4X + 16d^3b_6,$$
which has integral coefficients when the $a_i$ and $d$ are integers. For $a_1 = a_3 = 0$ this agrees with (a) (up to the change of coordinates $u = 2$).

(c) For an odd prime $q$, the paper's $E^{(q)}$ is $E^{(q^*)}$, the twist by $\mathbb{Q}(\sqrt{q^*})$; and for a finite set $S$ of odd primes, $E_S = E^{(d_S)}$ with $d_S = \prod_{q\in S} q^*$ (Definition 9.89).

**Proposition 9.84** (first properties of twists)**.** *Let $E\colon y^2 = x^3 + a_2x^2 + a_4x + a_6$ and $d \in K^\times$, $\operatorname{char} K \neq 2$.*

*(i) Up to isomorphism over $K$, $E^{(d)}$ depends only on the class of $d$ in $K^\times/(K^\times)^2$; $E^{(1)} \simeq E$ and $(E^{(d)})^{(d')} \simeq E^{(dd')}$.*

*(ii) Fix $\sqrt{d} \in \bar{K}$. The map $\psi\colon E^{(d)} \to E$, $(x, y) \mapsto (x/d,\, y/(d\sqrt{d}))$, is an isomorphism defined over $K(\sqrt{d})$, and $\psi^\sigma = [\chi_d(\sigma)] \circ \psi$ for $\sigma \in G_K$, where $\psi^\sigma(P) = \sigma(\psi(\sigma^{-1}P))$. Equivalently $\psi(\sigma P) = \chi_d(\sigma)\,\sigma(\psi(P))$ for $P \in E^{(d)}(\bar{K})$.*

*(iii) Through $\psi$, the $G_K$-set $E^{(d)}(\bar{K})$ is $E(\bar{K})$ with the* twisted action *$\sigma * P = \chi_d(\sigma)\sigma(P)$. In particular $E^{(d)}(K) \simeq \{P \in E(K(\sqrt{d})) : \sigma P = -P\}$ for the generator $\sigma$ of $\mathrm{Gal}(K(\sqrt{d})/K)$ when $d$ is not a square.*

*(iv) The twists of $E$ in general (the curves over $K$ isomorphic to $E$ over $\bar{K}$) are classified by $H^1(G_K, \operatorname{Aut}_{\bar{K}} E)$; when $j \neq 0, 1728$, $\operatorname{Aut} E = \{\pm 1\}$ and every twist is quadratic.*

*Proof.* (i) For $d' = du^2$, the change of coordinates $(u, 0, 0, 0)$ of Proposition 9.5 transforms $E^{(du^2)}$ into $E^{(d)}$ (it divides the coefficients of $x^2$, $x$, 1 by $u^2$, $u^4$, $u^6$). The other two statements are immediate from the formula. (ii) Substituting $x = dx'$, $y = d\sqrt{d}\,y'$ into the equation of $E^{(d)}$ and dividing by $d^3$ gives the equation of $E$ in $(x', y')$; the inverse is $(x', y') \mapsto (dx', d\sqrt{d}y')$. For $\sigma \in G_K$, $\sigma(d\sqrt{d}) = \chi_d(\sigma)d\sqrt{d}$, so $\psi^\sigma(x, y) = (x/d, \chi_d(\sigma)y/(d\sqrt{d}))$, which is $\psi(x, y)$ or $-\psi(x, y) = (x/d, -y/(d\sqrt{d}))$ according as $\chi_d(\sigma) = \pm 1$. (iii) is (ii) restated; a point $P \in E(\bar{K})$ corresponds to a $K$-point of $E^{(d)}$ iff $\sigma * P = P$ for all $\sigma$, i.e. $\sigma P = \chi_d(\sigma)P$. (iv) We do not prove this; see [185, Sections X.2 and X.5]; for the automorphism groups see Proposition 9.7. □

**Proposition 9.85** (invariants of twists)**.**

*(i) The model of Definition 9.83(a) has* $c_4(E^{(d)}) = d^2c_4$, $c_6(E^{(d)}) = d^3c_6$, $\Delta(E^{(d)}) = d^6\Delta$ *and* $j(E^{(d)}) = j(E)$. *The integral model of Definition 9.83(b) has invariants* $2^4d^2c_4$, $2^6d^3c_6$, $2^{12}d^6\Delta$.

*(ii) Let* $\operatorname{char} K \neq 2, 3$. *If* $j(E) = j(E') \neq 0, 1728$, *then* $E' \simeq E^{(d)}$ *over* $K$ *with* $d = \dfrac{c_6(E')\,c_4(E)}{c_6(E)\,c_4(E')}$, *for any models of* $E$ *and* $E'$.

*(iii) Let* $q$ *be an odd prime. If* $q \equiv 1 \pmod 4$ *then* $q = q^*$. *If* $q \equiv 3 \pmod 4$, *the twists of* $E$ *by* $q$ *and by* $q^*$ *differ by the twist by* $-1$, *which multiplies* $c_6$ *by* $-1$, *and they are in general different curves, the twist by* $q$ *being ramified at* 2 *as well. The twist by* $q^*$ *has invariants* $(q^*)^2c_4$, $(q^*)^3c_6$, $(q^*)^6\Delta$; *their* $q$*-adic valuations are those of* $q^2c_4$, $q^3c_6$, $q^6\Delta$.

*Proof.* (i) For $y^2 = x^3 + Ax^2 + Bx + C$, $b_2 = 4A$, $b_4 = 2B$, $b_6 = 4C$, $b_8 = 4AC - B^2$. Replacing $(A, B, C)$ by $(dA, d^2B, d^3C)$ multiplies $b_i$ by $d^{i/2}$ for $i = 2, 4, 6, 8$; since $c_4$, $c_6$, $\Delta$ are homogeneous of weights 4, 6, 12 in this grading, they are multiplied by $d^2$, $d^3$, $d^6$, and $j$ is unchanged. For the integral model, the same computation with $(A, B, C) = (db_2, 8d^2b_4, 16d^3b_6)$ gives $2^4d^2c_4$, $2^6d^3c_6$, $2^{12}d^6\Delta$ (checked by machine); this is the change of coordinates with $u = 1/2$ applied to $d^2c_4$, $d^3c_6$, $d^6\Delta$. (ii) First, the class of $d$ modulo squares does not depend on the models: a change of coordinates of $E'$ with parameter $u$ replaces $(c_4', c_6')$ by $(u^{-4}c_4', u^{-6}c_6')$ and so multiplies $d$ by $u^{-2}$, and similarly for $E$ (it multiplies $d$ by $u^2$); by Proposition 9.84(i) a square factor does not change $E^{(d)}$ up to isomorphism. So we may take short models (Lemma 9.3(iii)), which have invariants $6^4c_4$, $6^6c_6$ (and likewise for $E'$); passing to them for both curves does not change the quotient defining $d$ at all. As $j \neq 0, 1728$, $c_4, c_6, c_4', c_6' \neq 0$, and $j = j'$ is equivalent to $c_6^2/c_4^3 = c_6'^2/c_4'^3$. Then $d^2c_4 = c_6'^2c_4^3/(c_6^2c_4'^2) = c_4'$ and $d^3c_6 = d^2c_4 \cdot dc_6/c_4 = c_4' \cdot c_6'/c_4' = c_6'$. By (i), $E^{(d)}$ has the same $c_4$, $c_6$ as $E'$, hence the same short model. (iii) $q = -q^*$ when $q \equiv 3 \pmod 4$, and $E^{(-q^*)} \simeq (E^{(q^*)})^{(-1)}$ by Proposition 9.84(i). By (i) the twist by $-1$ multiplies $c_6$ by $-1$; the field $\mathbb{Q}(\sqrt{-1})$ is ramified at 2. The last statement is (i) with $v_q(q^*) = v_q(q) = 1$. □

An example of (ii): `980i1` and `140b1` have the same $j = 2^{19}3^3/(5 \cdot 7^5)$ (Example 9.8), with $(c_4, c_6) = (-75264, 62826624)$ and $(-1536, -183168)$. The recipe gives

$$d = \frac{(-183168)(-75264)}{62826624 \cdot (-1536)} = -\frac{183168 \cdot 49}{62826624} = -\frac{1}{7},$$

using $75264 = 49 \cdot 1536$ and $62826624 = 7 \cdot 8975232 = 7 \cdot 183168 \cdot 49$. So `140b1` $\simeq$ `980i1`$^{(-1/7)} \simeq$ `980i1`$^{(-7)}$, as $-1/7 = -7 \cdot (1/7)^2$; this is the identification used in Warning 9.91(b).

**Theorem 9.86** (the Galois representation of a twist)**.** *Let* $\operatorname{char} K \neq 2$, $d \in K^\times$ *and* $\ell \neq \operatorname{char} K$. *Then* $\psi$ *of Proposition 9.84(ii) induces a* $G_K$*-isomorphism* $T_\ell E^{(d)} \simeq T_\ell E \otimes \chi_d$*; that is,* $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi_d$. *Consequently* $\bar\rho_{E^{(d)},\ell} \simeq \bar\rho_{E,\ell} \otimes \chi_d$, $E^{(d)}[2] \simeq E[2]$, $\det \rho_{E^{(d)},\ell} = \det \rho_{E,\ell}$, $\operatorname{ad}^0 T_\ell E^{(d)} \simeq \operatorname{ad}^0 T_\ell E$, *and* $E^{(d)}[\ell]$ *is irreducible iff* $E[\ell]$ *is.*

*Proof.* $\psi$ is an isomorphism of groups (Theorem 9.19(i)), so it maps $E^{(d)}[\ell^n]$ bijectively to $E[\ell^n]$, compatibly with $[\ell]$, and by Proposition 9.84(ii) $\psi(\sigma P) = \chi_d(\sigma)\sigma(\psi(P))$. This says that $\psi_\ell \colon T_\ell E^{(d)} \to T_\ell E \otimes \chi_d$ is $G_K$-equivariant. On $E[2]$, $\chi_d(\sigma) = \pm 1$ acts trivially. Determinants are multiplied by $\chi_d^2 = 1$, and $\operatorname{ad}^0(T \otimes \chi) = \operatorname{ad}^0 T$ (Chapter 4). Twisting by a character preserves the lattice of stable subspaces. □

**Corollary 9.87** ($a_p$ of a twist)**.** *Let* $E/\mathbb{Q}$, $d \in \mathbb{Q}^\times$ *not a square,* $D$ *the discriminant of* $\mathbb{Q}(\sqrt{d})$, *and* $p$ *a prime of good reduction for* $E$ *with* $p \nmid D$. *Then* $E^{(d)}$ *has good reduction at* $p$ *and*

$$a_p(E^{(d)}) = \chi_d(p)\, a_p(E), \qquad \chi_d(p) = \Big(\frac{D}{p}\Big).$$

*Proof.* Good reduction of $E^{(d)}$ at $p$: $E$ and $E^{(d)}$ become isomorphic over the extension $\mathbb{Q}_p(\sqrt{d})$, unramified at $p$, and good reduction is invariant under unramified extension (Chapter 10). By Theorem 9.86 and Theorem 9.67, for $\ell \neq p$, $a_p(E^{(d)}) = \operatorname{tr}(\rho_{E,\ell} \otimes \chi_d)(\mathrm{Frob}_p) = \chi_d(\mathrm{Frob}_p) a_p(E)$, and $\chi_d(\mathrm{Frob}_p) = \left(\frac{D}{p}\right)$ (Convention 9.1(d)). An elementary proof when $p$ is odd and $p \nmid d\Delta$: with $E \colon y^2 = f(x)$ and $E^{(d)} \colon y^2 = d^3 f(x/d)$, Proposition 9.66 gives $a_p(E^{(d)}) = -\sum_x \eta(d^3 f(x/d)) = -\eta(d) \sum_{x'} \eta(f(x')) = \eta(d) a_p(E)$: here $\eta(d^3) = \eta(d)^3 = \eta(d)$, and $x' = x/d$ runs over $\mathbb{F}_p$ as $x$ does. Finally $\eta(d) = \left(\frac{d}{p}\right) = \left(\frac{D}{p}\right)$, since $d/D$ is the square of a rational number whose numerator and denominator are prime to $p$ (as $p \nmid dD$). □

Checked for every good $p < 400$, $p = 2$ included where allowed: `121d1` against `11a3`, `121d2` against `11a1` (with $\chi_{-11}$); `121c2` against `121a1` ($\chi_{-11}$); `77c1` against `539c1` and `140b1` against `980i1` ($\chi_{-7}$).

**Proposition 9.88** (the twist of good reduction at a type $\mathrm{I}_0^*$ prime)**.** *Let* $q$ *be an odd prime at which* $E/\mathbb{Q}$ *has type* $\mathrm{I}_0^*$, *and* $E' = E^{(q^*)}$, *which has good reduction at* $q$ *(Chapter 10).*

*(i)* $V_\ell E \simeq V_\ell E' \otimes \chi_{q^*}$ *for every* $\ell \neq q$, *with* $V_\ell E'$ *unramified at* $q$.

*(ii) The twists of* $E$ *with good reduction at* $q$ *are the curves* $E^{(q^* d)}$ *with* $q \nmid D$, $D$ *the discriminant of* $\mathbb{Q}(\sqrt{d})$*; for them* $a_q(E^{(q^* d)}) = \chi_d(q) a_q(E') = \pm a_q(E')$. *Hence* $a_q(E')^2$, $(q+1)^2 - a_q(E')^2$ *and the set* $\{\alpha/\beta, \beta/\alpha\}$ *for the roots of* $X^2 - a_q(E')X + q$ *depend only on* $E$ *and* $q$.

*(iii) Example:* `539d1` *has type* $\mathrm{I}_0^*$ *at* 7, *and* `539d1`$^{(-7)} \simeq$ `11a3`, *with* $a_7 = -2$. *So* $(7+1)^2 - a_7^2 = 60$ *and* $e_3($`539d1`$, 7) = v_3(7-1) + v_3(60) = 2 = v_3(72) = v_3(\deg \varphi_E)$ *for* $E =$ `539d1`.

*Proof.* (i) is Theorem 9.86 (as $\chi_{q^*}^2 = 1$) and Theorem 9.67. (ii) $E^{(q^* d)} = (E')^{(d)}$ (Proposition 9.84(i)). If $q \nmid D$, it has good reduction at $q$ and $a_q = \chi_d(q) a_q(E')$ by Corollary 9.87. If $q \mid D$, it is the twist of the curve $E'$, of good reduction at $q$, by a character ramified at $q$, which has type $\mathrm{I}_0^*$ (Chapter 10), not good reduction. Replacing $a_q(E')$ by $-a_q(E')$ replaces $(\alpha, \beta)$ by $(-\alpha, -\beta)$ and does not change $\alpha/\beta$. (iii) Computed with PARI: the minimal model of the twist of `539d1` by $-7$ is $[0, -1, 1, 0, 0]$, which is `11a3`; the modular degree $\deg \varphi_E = 72$ of $E =$ `539d1` is from Cremona's `degphi` file. □

**Definition 9.89** (composite twists)**.** For a finite set $S$ of odd primes, $\chi_S = \prod_{q\in S}\chi_{q^*}$ is the quadratic character of $\mathbb{Q}(\sqrt{d_S})$, $d_S = \prod_{q\in S} q^*$, a fundamental discriminant $\equiv 1 \pmod 4$; it has conductor $\prod_{q\in S} q$ and is unramified outside $S$ (Chapters 2 and 6). The twist $E_S = E^{(d_S)}$ is obtained by twisting successively by the $q^*$, $q \in S$, in any order (Proposition 9.84(i)), $\rho_{E_S,\ell} \simeq \rho_{E,\ell}\otimes\chi_S$, and $a_p(E_S) = \chi_S(p)a_p(E)$ for every good $p \notin S$ (Corollary 9.87).

*Remark* 9.90 (what a twist changes). Under every quadratic twist the invariants $j$, $v_q(j)$, hence $t_\ell(E,q)$, and $\mathrm{ad}^0\,\rho_{E,\ell}$ are unchanged (Proposition 9.85(i), Theorem 9.86). The numbers $a_q$, the discriminant, the Kodaira type at a prime where the twist ramifies, and the naive Euler factor $L_q^{\mathrm{nv}}$ change. So $e_\ell(E,q)$, which compares the true adjoint factor (an invariant of $\mathrm{ad}^0\,V$) with the naive one, is invariant under twists unramified at $q$ (which change neither factor, by Chapter 10) but not under twists ramified at $q$: `11a1` has type $\mathrm{I}_5$ at 11 and $e_3 = 0$, while its twist $\texttt{11a1}^{(-11)} \simeq \texttt{121d2}$ has type $\mathrm{I}_5^*$ and $e_3 = v_3(11^2-1) = 1$; likewise `11a3` ($\mathrm{I}_1$) and $\texttt{11a3}^{(-11)} \simeq \texttt{121d1}$ ($\mathrm{I}_1^*$). The types are from Chapter 10.

*Warning* 9.91 ($q$ versus $q^*$; curve versus class). (a) For $q \equiv 3 \pmod 4$ the twist by $\mathbb{Q}(\sqrt{q})$ is not the paper's twist. For `121a1`: $\texttt{121a1}^{(-11)} \simeq \texttt{121c2}$, of conductor 121, while $\texttt{121a1}^{(11)} \simeq \texttt{1936k2}$, of conductor $1936 = 2^4\cdot 11^2$. Only $q$-adic valuations enter Lemma lem:twist (ii), and they agree (Proposition 9.85(iii)). (b) A twist of a curve is a curve, and a twist of a class is a class. With the minimal models of Cremona's tables, as checked with PARI:

$$\texttt{11a1}^{(-11)} \simeq \texttt{121d2},\quad \texttt{11a3}^{(-11)} \simeq \texttt{121d1},\quad \texttt{11a2}^{(-11)} \simeq \texttt{121d3},\quad \texttt{121a1}^{(-11)} \simeq \texttt{121c2},$$

$$\texttt{539c1}^{(-7)} \simeq \texttt{77c1},\quad \texttt{980i1}^{(-7)} \simeq \texttt{140b1},\quad \texttt{11a1}^{(-7)} \simeq \texttt{539d2},$$

$$\texttt{11a3}^{(-7)} \simeq \texttt{539d1},\quad \texttt{20a1}^{(-1)} \simeq \texttt{80b1}.$$

Statements about $a_p$, $L$-functions, $V_\ell$ and (for $\ell$ prime to the isogeny degrees) $T_\ell$ are statements about classes; statements about $j$, $\Delta$, Kodaira types, torsion and $v_q(j)$ are about curves. "The twist of `11a1` by $\chi_{-11}$ is in the class `121d`" is right; "it is `121d1`" is not.

*In the paper* 9.92. The proof of Lemma lem:twist (ii) of the paper: "the twist has a model with invariants $(q^*)^2c_4$, $(q^*)^3c_6$, $(q^*)^6\Delta$ and the same $j$-invariant" is Proposition 9.85(i), (iii), and its model $y^2 = x^3 + q^*a_2x^2 + (q^*)^2a_4x + (q^*)^3a_6$ is Definition 9.83(a). Lemma lem:euler: in the row $\mathrm{I}_n^*$, "$V = V'\otimes\chi_{q^*}$ with $V'$ Steinberg at $q$" and "$\mathrm{ad}^0\,V = \mathrm{ad}^0\,V'$" are Theorem 9.86 (the Steinberg property of $V'$ is Chapter 10); in the row $\mathrm{I}_0^*$, "$V = V'\otimes\chi_{q^*}$ with $V'$ unramified" is Proposition 9.88(i). Theorems thm:twist and thm:quant: $E_S$, $\chi_S$ and $a_p(f) = \chi_S(p)a_p(g)$ are Definition 9.89. The table of Section 1: "$E'$ the quadratic twist of $E$ by $\mathbb{Q}(\sqrt{q^*})$" is Proposition 9.88, and (ii) of that proposition is why only $a_q(E')^2$ matters. Definition def:terms: "The term $t_\ell$ is invariant under quadratic twist, and $e_\ell(E,q)$ under quadratic twists unramified at $q$" is Remark 9.90. The Example of Section 3 (ex:qual) twists `121a1` and `539c1`, `980i1` by $\chi_{-11}$ and $\chi_{-7}$; its identifications are at the level of classes, and Warning 9.91(b) gives the curves.

## 9.11 The $L$-function

The paper's newform $f = f_E$ is the modular form whose $L$-series is $L(E,s)$, and Lemma lem:old uses the multiplicativity of its coefficients and the recursion at good primes. This section defines $L(E,s)$ by its Euler product, including the factors at the bad primes, and proves what follows from the results above. Analytic continuation and the functional equation are Chapter 19, modularity is Chapter 14, and the adjoint $L$-function is Chapter 19.

**Definition 9.93** (the $L$-function of $E$)**.** Let $E/\mathbb{Q}$ have conductor $N$ (Chapter 10), given by its global minimal model. For every prime $p$ let $\tilde{E}$ be the reduction modulo $p$ and $a_p = p+1-\#\tilde{E}(\mathbb{F}_p)$, the singular point included when $p \mid N$; so for $p \mid N$, $a_p = 1$, $-1$ or $0$ according as the reduction is split multiplicative, nonsplit multiplicative or additive (Chapter 10). Define

$$L(E,s) = \prod_{p\nmid N} (1 - a_p p^{-s} + p^{1-2s})^{-1} \prod_{p\mid N} (1 - a_p p^{-s})^{-1} = \sum_{n\geq 1} a_n n^{-s}.$$

**Proposition 9.94** (properties of $L(E,s)$)**.**

*(i)* $|a_n| \leq d(n)\sqrt{n}$*, where* $d(n)$ *is the number of divisors of* $n$*; the series and the product converge absolutely and agree for* $\operatorname{Re} s > 3/2$*.*

*(ii)* $a_{mn} = a_m a_n$ *for coprime* $m, n$*;* $a_{p^{k+1}} = a_p a_{p^k} - p a_{p^{k-1}}$ *for* $p \nmid N$*,* $k \geq 1$*; and* $a_{p^k} = a_p^k$ *for* $p \mid N$*.*

*(iii) For* $p \neq \ell$ *the local factor is* $L_p(V_\ell E^\vee, s) = \det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid (V_\ell E^\vee)^{I_p})^{-1}$ *(Convention 9.40): at good* $p$ *by Theorem 9.67; at bad* $p$*, where* $\dim (V_\ell E^\vee)^{I_p}$ *is* $1$ *(multiplicative, with* $\mathrm{Frob}_p^{-1}$ *acting by* $a_p = \pm 1$*) or* $0$ *(additive), by Chapter 11. At a good prime this is also* $\det(1 - \mathrm{Frob}_p\, p^{-s} \mid V_\ell E)^{-1}$*, but not at a multiplicative prime: there the arithmetic Frobenius acts on the line* $(V_\ell E)^{I_p}$ *by* $p a_p$*, not by* $a_p$*. Isogenous curves have the same* $L$*-function (Theorem 9.74(ii) and the invariance of the conductor), and* $L(E^{(d)}, s)$ *has the same local factors as* $\sum \chi_d(n) a_n n^{-s}$ *at the primes* $p \nmid ND$ *(Corollary 9.87).*

*(iv) (Modularity.)* $L(E,s) = L(f_E, s)$ *for a newform* $f_E$ *of weight* $2$ *on* $\Gamma_0(N)$*;* $L(E,s)$ *extends to an entire function and satisfies a functional equation under* $s \mapsto 2 - s$*.*

*Proof.* (i) At a good prime, $(1 - a_p X + pX^2)^{-1} = \big((1-\alpha_p X)(1-\beta_p X)\big)^{-1}$, so $a_{p^k} = \sum_{i=0}^{k} \alpha_p^i \beta_p^{k-i}$ and $|a_{p^k}| \leq (k+1)p^{k/2}$ by Theorem 9.63(iii); at a bad prime $|a_{p^k}| = |a_p|^k \leq 1$. With (ii) this gives $|a_n| \leq d(n)\sqrt{n}$. Since $d(n) = O(n^c)$ for every $c > 0$, $\sum |a_n| n^{-\sigma}$ converges for $\sigma > 3/2$; the product of the absolutely convergent local series over $p \leq x$ contains every $a_n n^{-s}$ with all prime factors of $n$ at most $x$, and the difference from the full series tends to 0. (ii) Expanding the product, $a_n$ is the product of the coefficients $a_{p^k}$ of the local series for $p^k \parallel n$; the recursion is the coefficient of $X^{k+1}$ in $(1 - a_p X + pX^2) \sum_k a_{p^k} X^k = 1$. (iii) At a good prime $\det(1 - \mathrm{Frob}_p^{-1} X \mid V_\ell E^\vee) = \det(1 - \mathrm{Frob}_p X \mid V_\ell E) = 1 - a_p X + pX^2$ by Convention 9.40 and Theorem 9.67; the bad primes are Chapter 11. At a multiplicative prime $V_\ell E$ restricted to the decomposition group is an extension of $\varepsilon$ by $\varepsilon\chi_\ell$, with $\varepsilon$ unramified and $\varepsilon(\mathrm{Frob}_p) = a_p$ (Chapter 10), so the inertia invariants of $V_\ell E$ are the line $\varepsilon\chi_\ell$, with arithmetic Frobenius eigenvalue $p a_p$, and those of $V_\ell E^\vee \simeq V_\ell E \otimes \chi_\ell^{-1}$ are the line $\varepsilon$, with geometric Frobenius eigenvalue $\varepsilon(\mathrm{Frob}_p)^{-1} = a_p$. (iv) We do not prove this; it is the theorem of Wiles, Taylor-Wiles and Breuil, Conrad, Diamond and Taylor [207, 197, 16], with the analytic statements from the theory of modular forms; see Chapters 14 and 19. □

For `11a1`, $a_1, \ldots, a_{13} = 1, -2, -1, 2, 1, 2, -2, 0, -2, -2, 1, -2, 4$; for instance $a_4 = a_2^2 - 2 = 2$, $a_8 = a_2 a_4 - 2a_2 = 0$, $a_9 = a_3^2 - 3 = -2$, and $a_{11} = 1$ (11 is split multiplicative). The composite indices come from multiplicativity: $a_6 = a_2 a_3 = 2$, $a_{10} = a_2 a_5 = -2$, $a_{12} = a_4 a_3 = -2$. The local factor at 2 is $(1 + 2\cdot 2^{-s} + 2^{1-2s})^{-1} = \sum_k a_{2^k} 2^{-ks}$, and the recursion $a_{2^{k+1}} = -2a_{2^k} - 2a_{2^{k-1}}$ continues $1, -2, 2, 0, -4, 8, -8, 0, \ldots$; the bound of (i) reads $|a_{2^k}| \leq (k+1)2^{k/2}$, here $|a_{16}| = 4 \leq 5 \cdot 4$.

## 9.12 Labels and the running examples

The paper names curves by Cremona's labels and relies on Cremona's convention that the first curve of a class is the optimal one. This section explains the labels, gives their LMFDB equivalents, and collects in one table the data of the curves that the paper and this book use as running examples. The modular degrees, Kodaira types and images are proved or computed in Chapters 10, 11 and 14; here they are quoted from Cremona's files [36] and checked with PARI.

**Convention 9.95** (Cremona and LMFDB labels)**.**

(a) A Cremona label $N\,c\,k$ consists of the conductor $N$, an isogeny-class code $c$ (a, b, ..., z, ba, bb, ...) and the number $k$ of the curve in its class: `539c1` is the first curve of the third class of conductor 539. In Cremona's tables curve number 1 of each class is the $X_0(N)$-optimal curve (Chapter 14), with a single exception, the class `990h`, whose optimal curve is `990h3` $= [1,-1,1,-1568,-4669]$: the `degphi` file records the modular degree 1728 of the class `990h` with this equation. Optimality is not known for every class of the tables: Section sec:data of the paper says "Optimality is determined in the tables for every class of conductor at most 400000 and for 359,009 further classes up to 500000", while its tables go up to 500000. So below conductor 10,000, the range of this chapter, optimality is determined for every class. Cremona's book [35] and [3] print the class code in capitals: `99A1` is `99a1`.

(b) LMFDB labels `N.ck` [198] order the classes and the curves within a class differently, so the same curve can carry different letters and numbers, and the LMFDB's curve number 1 is in general not the optimal curve. The correspondence, from Cremona's `alllabels` file:

| Cremona | LMFDB | Cremona | LMFDB | Cremona | LMFDB | Cremona | LMFDB |
|---|---|---|---|---|---|---|---|
| 11a1 | 11.a2 | 20a1 | 20.a4 | 121a1 | 121.a2 | 539c1 | 539.d2 |
| 11a2 | 11.a1 | 27a1 | 27.a3 | 121c1 | 121.c2 | 539d1 | 539.a3 |
| 11a3 | 11.a3 | 77c1 | 77.c2 | 121c2 | 121.c1 | 726a1 | 726.b2 |
| 14a1 | 14.a6 | 99a1 | 99.a2 | 121d1 | 121.d3 | 980i1 | 980.b1 |
| 140b1 | 140.b1 | 990h3 | 990.i3 | 121d2 | 121.d2 | | |

**Example 9.96** (the paper's curves)**.** Data from Cremona's files (`allcurves`, `allisog`, `degphi`, `galrep`), rechecked with PARI; "images" lists the non-surjective $\bar\rho_{E,\ell}$ in the notation of [191]; types from PARI's `elllocalred` (Chapter 10).

| curve | $N$ | rank, torsion | class (degrees) | $\deg\varphi_E$ | images | types |
|---|---|---|---|---|---|---|
| 11a1 | 11 | 0, $\mathbb{Z}/5$ | 5, 5 | 1 | 5Cs.1.1 | $\mathrm{I}_5$ at 11 |
| 20a1 | $2^2\cdot 5$ | 0, $\mathbb{Z}/6$ | 2, 3, 6 | 1 | 2B, 3B.1.1 | $\mathrm{IV}^*$, $\mathrm{I}_2$ at 2, 5 |
| 121a1 | $11^2$ | 0, 0 | 11 | 6 | 11B.1.6 | II at 11 |
| 121c1 | $11^2$ | 0, 0 | 11 | 6 | 11B.1.7 | IV at 11 |
| 121d1 | $11^2$ | 0, 0 | 5, 25 | 24 | 5B.4.1 | $\mathrm{I}_1^*$ at 11 |
| 539c1 | $7^2\cdot 11$ | 1, $\mathbb{Z}/2$ | 2 | $288 = 2^5 3^2$ | 2B | $\mathrm{I}_3^*$, $\mathrm{I}_2$ at 7, 11 |
| 980i1 | $2^2\cdot 5\cdot 7^2$ | 0, 0 | trivial | $2880 = 2^6 3^2 5$ | none | $\mathrm{IV}^*$, $\mathrm{I}_1$, $\mathrm{I}_5^*$ at 2, 5, 7 |
| 726a1 | $2\cdot 3\cdot 11^2$ | 1, $\mathbb{Z}/2$ | 2 | $144 = 2^4 3^2$ | 2B, 3Ns | $\mathrm{I}_6$, $\mathrm{I}_3$, III at 2, 3, 11 |
| 99a1 | $3^2\cdot 11$ | 1, $\mathbb{Z}/2$ | 2 | 4 | 2B | III, $\mathrm{I}_1$ at 3, 11 |
| 77c1 | $7\cdot 11$ | 0, $\mathbb{Z}/2$ | 2 | 6 | 2B | $\mathrm{I}_3$, $\mathrm{I}_2$ at 7, 11 |
| 140b1 | $2^2\cdot 5\cdot 7$ | 0, 0 | trivial | 60 | none | $\mathrm{IV}^*$, $\mathrm{I}_1$, $\mathrm{I}_5$ at 2, 5, 7 |
| 539d1 | $7^2\cdot 11$ | 1, 0 | 5, 25 | 72 | 5B.4.1 | $\mathrm{I}_0^*$, $\mathrm{I}_1$ at 7, 11 |
| 14a1 | $2\cdot 7$ | 0, $\mathbb{Z}/6$ | 2, 3, 3, 6, 6 | 1 | 2B, 3Cs.1.1 | $\mathrm{I}_6$, $\mathrm{I}_3$ at 2, 7 |

The column "class" gives the first row of Cremona's isogeny matrix without the diagonal entry, that is, the degrees of the cyclic isogenies from the curve to the other curves of its class. `11a1` is the middle curve of its class, 5-isogenous to both others (which are 25-isogenous to each other). `121d1` and `539d1`, the twists of `11a3`, are at an end: the class matrices of `121d` and `539d` are $\left(\begin{smallmatrix} 1 & 5 & 25 \\ 5 & 1 & 5 \\ 25 & 5 & 1 \end{smallmatrix}\right)$, and the middle curves are `121d2` and `539d2`, the twists of `11a1`. Roles in the paper: `121a1`, `539c1` and `980i1` illustrate cases (b) and (c) of Theorem thm:qual in the Example of Section 3 (case (a) is illustrated there by `52a1`), with the congruent forms coming from `11a1` (twisted by $\chi_{-11}$, giving the class `121d`, and by $\chi_{-7}$, giving `539d`) and `20a1` (twisted by $\chi_{-7}$); `121c2` is `121a1`$^{(-11)}$, `77c1` is `539c1`$^{(-7)}$, `140b1` is `980i1`$^{(-7)}$, and `539d1` is `11a3`$^{(-7)}$ (Warning 9.91); `726a1` is the paper's example with image $N_s(\mathbb{F}_3)$ (Section sec:anomaly and the appendix); `99A1` is the curve of [3] with modular degree 4 and congruence number 12 (Remark after Theorem thm:qual; Chapter 14 recomputes $r_E = 12$); `14a1` has $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ (Section sec:eis). Among the first nine curves, `539c1`, `726a1` and `99a1` have rank 1; so has `539d1`. The curves `27a1`, `32a1`, `36a1`, `121b1` with CM, and `338d1`, `6912d1` with image `3Ns`, appear in Sections 9.4 and 9.6.

The PARI/GP commands behind these computations (Chapter 22 explains them):

```
E = ellinit([1,1,1,-30,-76]);   \\ the curve 121a1
[E.c4, E.c6, E.disc, E.j]       \\ invariants
ellglobalred(E)                  \\ conductor, change to the minimal model, Tamagawa
    product
elllocalred(E, 11)               \\ Kodaira code and Tamagawa number at 11
elltors(E)                       \\ torsion subgroup
ellisomat(E)                     \\ the curves of the isogeny class and the matrix
factor(elldivpol(E, 3))          \\ the 3-division polynomial over Q
vector(12, i, ellap(E, prime(i)))
```

## 9.13 Exercises

**Exercise 9.97.** For `99a1` $= [1,-1,1,-2,0]$ compute $b_2, b_4, b_6, b_8$, $c_4$, $c_6$, $\Delta$ and $j$, check $4b_8 = b_2b_6 - b_4^2$ and $1728\Delta = c_4^3 - c_6^2$, and find a short model $Y^2 = X^3 + AX + B$ with integral $A, B$. (Answer: $c_4 = 3^4$, $c_6 = 135$, $\Delta = 3^3 \cdot 11$, $j = 3^9/11$.)

**Exercise 9.98.** Use Proposition 9.5 to show that for a prime $q$ the residue of $v_q(\Delta)$ modulo 12 and the number $v_q(j)$ are the same for all models of $E/\mathbb{Q}$, and deduce that a model with integral coefficients and $v_q(\Delta) < 12$ has the least $v_q(\Delta)$ among integral models. Give an example of an integral model with $v_q(\Delta) \geq 12$ that is not of least $v_q(\Delta)$.

**Exercise 9.99.** On `14a1` $= [1,0,1,4,-6]$, compute the multiples of $(9,23)$. Using $\#\tilde{E}(\mathbb{F}_3) = \#\tilde{E}(\mathbb{F}_5) = 6$ and Proposition 9.71, show $E(\mathbb{Q})_{\mathrm{tors}} \simeq \mathbb{Z}/6$. Which point of order 2 is rational, and over which field are the other two points of order 2 defined?

**Exercise 9.100.** Let $B \subset \mathrm{GL}_2(\mathbb{F}_3)$ be the upper triangular subgroup and $B_1 = \{\left(\begin{smallmatrix} 1 & * \\ 0 & d \end{smallmatrix}\right)\}$. Show that $B_1$ fixes one point of $\mathbb{P}^1(\mathbb{F}_3)$ and is transitive on the other three. Explain with Proposition 9.29 why $\psi_3$ of `20a1` (image `3B.1.1`) has the pattern $1+3$, while that of `14a1` (image `3Cs.1.1`) has the pattern $1+1+2$, and identify the root of $\psi_3$ of `14a1` that belongs to the line $\mu_3$.

**Exercise 9.101.** For `726a1`, check that $\psi_3 = (x^2 - x - 74)(3x^2 + 8x + 20)$, that the splitting field of $\psi_3$ is $\mathbb{Q}(\sqrt{33}, \sqrt{-11})$, and that it contains $\mathbb{Q}(\sqrt{-3})$ as Proposition 9.29(iii) requires. Check

that $3 \mid a_p$ for the good primes $p \equiv 2 \pmod 3$ below 100, and explain why no such statement holds for $p \equiv 1 \pmod 3$.

**Exercise 9.102.** Twist `121a1` $= [1,1,1,-30,-76]$ by $d = -11$ using the integral model of Definition 9.83(b), and use Proposition 9.85(ii) to show that the result is isomorphic to `121c2` $= [1,1,0,-3632,82757]$ and not to `121c1`. Then twist by $d = 11$ and show that the result has bad reduction at 2, so it is not isomorphic to any curve of conductor 121.

**Exercise 9.103.** Show that $j(\texttt{121d1}) = j(\texttt{11a3}) \neq j(\texttt{11a1})$, so that `121d1` is a twist of `11a3` and not of `11a1`. Check $a_p(\texttt{121d1}) = \chi_{-11}(p)a_p(\texttt{11a1})$ for $p \leq 50$, $p \neq 11$, and explain why the numbers $a_p$ cannot distinguish `11a1` from `11a3`.

**Exercise 9.104.** Show that the twist of `539c1` by $-7$ has $j = 167^3/(7^3 \cdot 11^2)$ and is `77c1`. Compare $v_7(\Delta)$ for the two curves (9 against 3) and explain the difference of 6 with Proposition 9.85(i) and Proposition 9.5(iii).

**Exercise 9.105.** Let $s_n = \alpha^n + \beta^n$ for the roots of $X^2 - aX + \mathfrak{q}$. Show $s_{n+1} = as_n - \mathfrak{q}s_{n-1}$, and use Theorem 9.63 to compute $\#\texttt{11a1}(\mathbb{F}_{27})$ from $a_3 = -1$. (Answer: 20.) Show that $\#E(\mathbb{F}_{\mathfrak{q}})$ divides $\#E(\mathbb{F}_{\mathfrak{q}^n})$ for every $n$.

**Exercise 9.106.** For $E' =$ `11a3` and $q = 7$ compute $\#E'(\mathbb{F}_7)$ and the number of points of the quadratic twist of the reduction, and check that their product is $(7+1)^2 - a_7^2 = 60$ (Proposition 9.66). Deduce $e_3(\texttt{539d1}, 7) = 2$, and show that replacing $E'$ by the twist $\texttt{539d1}^{(-7d)}$ for any $d$ with $7 \nmid \operatorname{disc} \mathbb{Q}(\sqrt{d})$ gives the same value.

**Exercise 9.107.** Check that $a_p(\texttt{27a1}) = 0$ for the primes $p \equiv 2 \pmod 3$, $p \leq 50$, and that $a_p(\texttt{121b1}) = 0$ for the primes $p \leq 50$ inert in $\mathbb{Q}(\sqrt{-11})$; explain both with Theorem 9.47. Compute $j(\frac{1+\sqrt{-163}}{2})$ numerically, show that it is $-640320^3$, and explain why $e^{\pi\sqrt{163}}$ is within $10^{-12}$ of an integer.

**Exercise 9.108.** Use Corollary 9.33(ii) to show that if $E/\mathbb{Q}$ has $E[m] \subseteq E(\mathbb{Q})$ then $m \leq 2$. Find a curve over $\mathbb{Q}$ with $E[2] \subseteq E(\mathbb{Q})$, and show that if $E[3] \subseteq E(K)$ for a number field $K$ then $\sqrt{-3} \in K$.

**Exercise 9.109.** The class `11a` has isogeny matrix $\left(\begin{smallmatrix} 1 & 5 & 5 \\ 5 & 1 & 25 \\ 5 & 25 & 1 \end{smallmatrix}\right)$ and mod 5 images `5Cs.1.1`, `5B.1.2`, `5B.1.1` for `11a1`, `11a2`, `11a3`. Show that $T_\ell$ of the three curves are isomorphic for $\ell \neq 5$, but that $T_5(\texttt{11a1}) \not\simeq T_5(\texttt{11a2})$. Explain, using $\texttt{11a1}/\langle(5,5)\rangle \simeq \texttt{11a2}$ and $\texttt{11a1}/\mu_5 \simeq \texttt{11a3}$, why `11a3` has a rational point of order 5 and `11a2` has none.

**Exercise 9.110.** Adapt the proof of Proposition 9.78 to show, without Mazur's torsion theorem but with Theorem 9.76(ii), that a curve $E/\mathbb{Q}$ with $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ has no rational point of order 25. Where does the argument use that $\ell$ is odd?

**Exercise 9.111.** Let $\varphi\colon E \to E'$ be a cyclic $n$-isogeny over $K$ ($\operatorname{char} K = 0$). Show that $\hat{\varphi}$ is cyclic and that $\ker\hat{\varphi} = \varphi(E[n])$. Now let $E/\mathbb{Q}$ have no CM and let $E'$ be isogenous to it. Show that $\psi \mapsto \hat{\varphi} \circ \psi$ embeds $\operatorname{Hom}(E, E')$ into $\operatorname{End}(E) = \mathbb{Z}$, so that $\operatorname{Hom}(E, E') \simeq \mathbb{Z}$, and deduce that $E$ and $E'$ are joined by a cyclic isogeny of a unique degree: this is why the entries of Cremona's isogeny matrix are well defined.

**Exercise 9.112.** For $\Lambda = \mathbb{Z}+\mathbb{Z}i$ and $\Lambda = \mathbb{Z}+\mathbb{Z}\zeta_3$ show $g_3 = 0$, respectively $g_2 = 0$, and compute $\mathrm{End}(\mathbb{C}/\Lambda)$. Show that the curve $y^2 = x^3 - x$ (`32a2`, $\Delta = 64 > 0$) has a rectangular period lattice, and that the lattice of `11a1` (Example 9.59) is not rectangular. Show that for every lattice the map $z \mapsto \bar{z}$ carries $\Lambda$ to a lattice with the conjugate $j$-invariant.

**Exercise 9.113.** Let $E/\mathbb{Q}$, $\ell$ odd, and $c$ a complex conjugation. Using only Theorem 9.32 and $c(\zeta) = \zeta^{-1}$ on $\mu_\ell$, show that $\det \bar{\rho}_{E,\ell}(c) = -1$ (compute $e_\ell(cS, cT)$ for a basis $S, T$ of $E[\ell]$). Deduce from $c^2 = 1$ that $\bar{\rho}_{E,\ell}(c)$ is conjugate to $\mathrm{diag}(1, -1)$, and hence that $E(\mathbb{R})[\ell]$ has order $\ell$. For $\ell = 2$ show that the determinant gives no information, and that $c$ acts trivially on $E[2]$ if and only if $F$ has three real roots, if and only if $\Delta > 0$; check this on `32a2` and `11a1`.

**Exercise 9.114.** For `11a1` compute the real root $e_1$ of $F(x) = 4x^3 - 4x^2 - 40x - 79$, the numbers $\gamma$, $\delta$ of Proposition 9.58(iii), and the two arithmetic-geometric means, and check that the formula gives the periods of Example 9.59. How many iterations of the AGM give 20 correct digits? (Answer: $e_1 = 4.3463081582\ldots$, $\gamma = 12.0389244746\ldots$, $\delta = 6.1626753526\ldots$, $\omega_1 = 1.2692093042\ldots$, $\mathrm{Im}\,\omega_2 = 1.4588166169\ldots$; the convergence is quadratic.)

**Exercise 9.115.** Using the table of Convention 9.95, find the curves whose Cremona and LMFDB numbers differ, and check with the `alllabels` and `degphi` files of [36] that `990h3` is the optimal curve of `990h`. Why can the LMFDB's ordering not be used to identify the optimal curve?

## 9.14 Notes and references

**Sources followed.** Sections 9.1 to 9.5 follow Silverman's first book [185], Chapter III (Weierstrass equations in III.1, the group law in III.2 and III.3, isogenies in III.4 to III.6, the Tate module in III.7, the Weil pairing in III.8, endomorphism rings in III.9, automorphisms in III.10), with the division polynomials of [185, Exercise 3.7] and [202, Chapter 3]. Section 9.6 follows [184, Chapter II] and [34]; Deuring's theorem is [46]. Section 9.7 follows [185, Chapter VI] and [52, Chapter 1], with the period lattice of a curve over $\mathbb{Q}$ as in [35, Chapter III]; the AGM algorithm is [35, Section 3.7] and [30, Section 7.4]. Section 9.8 follows [185, Chapter V]; the proof of Hasse's theorem by the positivity of the degree is Silverman's. Section 9.9 follows [185, Chapters VII and VIII] for torsion and Mordell-Weil; the torsion theorem is [125], the prime-degree isogeny theorem [126], and the remaining cyclic isogenies are in Kenku's papers [92, 94, 93, 95]; the complete list of cyclic degrees is [96, Theorem 1] and the bound on the size of isogeny classes [96, Theorem 2] (both checked against the published text); the isogeny theorem is [63]. Vélu's formulas are [200]. Section 9.10 follows [185, Section X.2 and X.5]; Section 9.11 follows [185, Appendix C, Section 16] and [35, Chapter II]. Other textbook accounts: [86], [101], [134]; for the Galois representations, [178, Chapter I].

**Theorem and section numbers.** The chapter cites sections rather than theorem numbers. The section numbers of [185] (second edition), [184], [202], [34], [30], [35] and [52] given above are pointers recalled from those books; they have not been checked against the printed editions. The table of rational cyclic isogenies in [120] was not checked either.

**The proposition on points of order 9.** Proposition 9.78 is Proposition prop:no9 of the paper; the argument through the cyclic 27-isogeny of $E/\mu_3$ is elementary once Theorem 9.76(iii)

is known. The data of Cremona's tables below conductor 10,000 agree with it, as recorded after the proof.

**Computations.** All numbers in the chapter were computed with PARI/GP [148] through `cypari2`, on Cremona's tables [36] for conductor below 10,000: the invariants of Example 9.8, the multiples of points, the division polynomials and their factorisations and discriminants, the numbers $a_p$ and the point counts over $\mathbb{F}_{p^2}$ and $\mathbb{F}_{27}$, the Hasse bound for the good primes below 2000 of seven curves, the thirteen CM $j$-values and the CM checks, the twist identifications (by computing the minimal model of the twisted equation and comparing with the tables), the isogeny matrices, the LMFDB labels, the torsion data behind Proposition 9.78, the consistency of the `galrep` and `allisog` files (114,126 pairs of a curve and $\ell \in \{3, 5, 7\}$, with 59 discrepancies, all for CM curves, for which the `galrep` file gives no Borel label), the Kodaira types at 3 in the classes without a 3-isogeny (Paper link 9.82), and the periods of `11a1`, also by the AGM. The symbolic identities of Lemma 9.3, (9.2) in the proof of Proposition 9.15, Proposition 9.5, Proposition 9.11(iii), Proposition 9.29(i) and Proposition 9.85(i) were checked with the $a_i$ as indeterminates.

# Chapter 10

# Elliptic curves over local fields: reduction, Kodaira types, Tate curves

The paper works one bad prime at a time. At each bad prime $q$ of the optimal curve $E$ it reads off a handful of local invariants: the Kodaira symbol, the geometric component group $\Phi_q^{\mathrm{geom}}$ and the Tamagawa number $c_q$, the valuations $v_q(\Delta)$ and $v_q(j)$, the conductor exponent, the Tate parameter $q_E$ and its unit part $u$ at a potentially multiplicative prime, the order of the image of inertia at a potentially good prime, and the effect on all of these of the quadratic twist by $\mathbb{Q}(\sqrt{q^*})$. This chapter defines each of them, proves what can be proved with the tools of the book, states the rest precisely with references, and computes everything on the paper's curves. The local field theory (inertia, the tame character, Kummer theory, Artin conductors) is in Chapter 7; Weierstrass equations, the group law, torsion, Tate modules and quadratic twists of models are in Chapter 9; the Galois representations built from the present results are in Chapter 11.

## Core results at a glance

Throughout, $K$ is complete for a discrete valuation $v$ with $v(\varpi) = 1$, valuation ring $\mathcal{O}$, perfect residue field $k$ of characteristic $p$, and $\ell \neq p$ unless stated otherwise.

**10.1 Minimal models.** A model with all $a_i \in \mathcal{O}$ is integral; it is minimal if $v(\Delta)$ is least among integral models. Minimal models exist; two of them differ by $(u, r, s, t)$ with $u \in \mathcal{O}^\times$, $r, s, t \in \mathcal{O}$; the Néron differential is defined up to $\mathcal{O}^\times$, and $v(\Delta_{\min})$, $v(c_4)$, $v(c_6)$ of a minimal model and $v(j) = 3v(c_4) - v(\Delta)$ are invariants of $E/K$. *(proved here)* Definition 10.3, Proposition 10.4.

**10.2 Minimality criteria.** An integral model with $v(\Delta) < 12$, or $v(c_4) < 4$, or $v(c_6) < 6$ is minimal. For $p \geq 5$ it is minimal if and only if $v(\Delta) < 12$ or $v(c_4) < 4$, and otherwise $(\varpi, 0, 0, 0)$ applied to $y^2 = x^3 - 27c_4x - 54c_6$ lowers $v(\Delta)$ by 12. *(proved here)* Over $\mathbb{Z}$, $c_4, c_6$ with $\Delta = (c_4^3 - c_6^2)/1728 \in \mathbb{Z} \setminus \{0\}$ come from an integral model if and only if $v_3(c_6) \neq 2$ and either $c_6 \equiv -1 \pmod 4$ or $v_2(c_4) \geq 4$, $c_6 \equiv 0, 8 \pmod{32}$ (Kraus). *(stated; reference)* Proposition 10.5.

**10.3 Global minimal models.** Every $E/\mathbb{Q}$ has a model over $\mathbb{Z}$ minimal at every $p$, unique up to $(\pm 1, r, s, t)$, $r, s, t \in \mathbb{Z}$, and unique with $a_1, a_3 \in \{0, 1\}$, $a_2 \in \{-1, 0, 1\}$; its discriminant is $\Delta_E$. *(proof sketched here)* Theorem 10.6.

**10.4 Reduction types.** For a minimal model: good iff $v(\Delta) = 0$; multiplicative iff $v(\Delta) > 0 = v(c_4)$ (node; split iff the tangent slopes are in $k$); additive iff $v(\Delta) > 0$, $v(c_4) > 0$ (cusp). *(proved here)* Definition 10.9.

**10.5 Nonsingular points.** $\tilde{E}_{\mathrm{ns}}(k) \simeq k^+$ (cusp), $k^\times$ (split node), $\ker(\mathrm{N}\colon l^\times \to k^\times)$ (nonsplit node, $l/k$ quadratic); over $\mathbb{F}_p$ of orders $p$, $p-1$, $p+1$. Hence for $E/\mathbb{Q}$ with bad reduction at $p$, $a_p = p+1-\#\tilde{E}(\mathbb{F}_p)$ is $1$, $-1$, $0$ for split, nonsplit, additive. *(proved here)* Proposition 10.10, Corollary 10.11.

**10.6 Split versus nonsplit.** After moving the node to $(0,0)$, split iff $T^2 + a_1T - a_2$ has a root in $k$; for $p$ odd, split iff $-c_6$ (equivalently $-c_4/c_6$) is a square in $k^\times$; for $p$ odd and $k$ finite the unramified quadratic twist exchanges split and nonsplit, and over the unramified quadratic extension the reduction is split. *(proved here)* Proposition 10.12.

**10.7 The reduction filtration.** With $E_0(K) = \{P : \tilde{P} \text{ nonsingular}\}$ and $E_1(K) = \{P : \tilde{P} = \tilde{O}\}$, there is an exact sequence $0 \to E_1(K) \to E_0(K) \to \tilde{E}_{\mathrm{ns}}(k) \to 0$. *(proved here)* Definition 10.14, Proposition 10.15.

**10.8 Formal group.** $E_1(K) \simeq \hat{E}(\mathfrak{m})$ *(stated; reference)*; for $m$ prime to $p$, $[m]$ is bijective on $E_1(K)$, which has no prime-to-$p$ torsion *(proved here, given the formal group)*. Theorem 10.16.

**10.9 Torsion and reduction.** For $m$ prime to $p$, $E_0(K)[m] \hookrightarrow \tilde{E}_{\mathrm{ns}}(k)$. If $E$ has good reduction, $I_K$ acts trivially on $E[m]$ and on $T_\ell E$, and reduction $E[m] \to \tilde{E}[m](\bar{k})$ is a Galois-equivariant isomorphism. *(proved here)* Corollary 10.17.

**10.10 Index of $E_0$.** For $k$ finite, $E(K)/E_0(K)$ is finite, trivial for good reduction. *(proved here)* Proposition 10.18.

**10.11 Potential reduction.** $E$ has potentially good reduction iff $v(j) \geq 0$, and potentially multiplicative reduction iff $v(j) < 0$; then it is multiplicative over every extension where it is semistable. Multiplicative reduction has $v(j) = -v(\Delta_{\min})$. *(proof sketched here)* Definition 10.20, Theorem 10.21.

**10.12 Semistable reduction.** $E$ becomes semistable over a finite extension, for instance $K(E[m])$ with $m \geq 3$ prime to $p$. *(proof sketched here)* Theorem 10.22.

**10.13 The four classes.** The Kodaira symbols fall into: good reduction, $\mathrm{I}_0$; multiplicative, $\mathrm{I}_n$ ($n \geq 1$, $n = v(\Delta_{\min}) = -v(j)$); additive potentially multiplicative, $\mathrm{I}_n^*$ ($n \geq 1$); additive potentially good, II, III, IV, $\mathrm{I}_0^*$, IV$^*$, III$^*$, II$^*$, and at $p = 2$ only also some $\mathrm{I}_m^*$ ($m \geq 1$). *(proved here, given the classification)* Remark 10.23.

**10.14 Néron models.** The Néron model $\mathcal{E}$ (smooth separated group scheme of finite type over $\mathcal{O}$, generic fibre $E$, Néron mapping property) exists and is unique; it is the smooth locus of the minimal regular model. *(stated; reference)* Definition 10.26, Theorem 10.27.

**10.15 Component group.** $\mathcal{E}(\mathcal{O}) = E(K)$, $\mathcal{E}_k^0 \simeq \tilde{E}_{\mathrm{ns}}$, $\mathcal{E}^0(\mathcal{O}) = E_0(K)$, and the quotient $\mathcal{E}_k/\mathcal{E}_k^0$ is a finite étale group scheme over $k$, the component group scheme, whose groups of points are written $\Phi_E(k)$ and $\Phi_E(\bar{k}) = \Phi^{\mathrm{geom}}$ *(stated; reference)*; for $k$ finite $E(K)/E_0(K) \simeq \Phi_E(k)$ *(proof sketched here)*. Proposition 10.28.

**10.16** $\Phi^{\mathrm{geom}}$ **and** $c$**.** For $E/\mathbb{Q}$: $\Phi_p^{\mathrm{geom}} = \Phi_p(\bar{\mathbb{F}}_p)$ and $c_p = \#\Phi_p(\mathbb{F}_p) = [E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)]$, the Frobenius invariants; $c_p \mid \#\Phi_p^{\mathrm{geom}}$, with equality iff Frobenius acts trivially. The paper's statements are about $\#\Phi_q^{\mathrm{geom}}$. *(proved here)* Definition 10.29, Convention 10.30.

**10.17 Unramified base change.** Néron models commute with étale base change; type, Kodaira symbol, $\Phi^{\mathrm{geom}}$, $v(\Delta_{\min})$ and $f$ are invariant under unramified extension; $c$ is not. *(stated; reference)* Proposition 10.31.

**10.18 Tamagawa numbers of** $\mathrm{I}_n$**.** $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$; $c = n$ (split), $c = \gcd(2, n)$ (nonsplit); for $\ell$ odd the $\ell$-part of $c$ is $\ell^{v_\ell(n)}$ (split) or 1 (nonsplit). *(proved here, given Tate's theorem)* Remark 10.32, Proposition 10.65.

**10.19 Inertia invariants and component groups.** Let $K/\mathbb{Q}_q$ be finite, $\ell \neq q$. If the reduction is additive, $E[\ell^t]^{I_K} = E(K^{\mathrm{ur}})[\ell^t] \simeq \Phi^{\mathrm{geom}}[\ell^t]$, Galois-equivariantly, and $E(K)[\ell^\infty] \simeq \Phi_E(k)[\ell^\infty]$. If it is multiplicative, $0 \to \mu_{\ell^t} \otimes \varepsilon \to E[\ell^t]^{I_K} \to \Phi^{\mathrm{geom}}[\ell^t] \to 0$ with $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$, $\varepsilon$ trivial iff split. For $\ell$ odd, $\dim E[\ell]^{I_K}$ is $1 + [\ell \mid n]$ for $\mathrm{I}_n$, 1 for $\mathrm{IV}, \mathrm{IV}^*$ if $\ell = 3$, and 0 for every other additive type. *(proved here)* Proposition 10.33.

**10.20 Kodaira-Néron classification.** The special fibres and $(m, \Phi^{\mathrm{geom}})$ are: $\mathrm{I}_0$, $\mathrm{I}_1$, II: $m = 1$, 0; $\mathrm{I}_n$: $n$, $\mathbb{Z}/n$; III: 2, $\mathbb{Z}/2$; IV: 3, $\mathbb{Z}/3$; $\mathrm{I}_0^*$: 5, $(\mathbb{Z}/2)^2$; $\mathrm{I}_n^*$: $n+5$, $\mathbb{Z}/4$ ($n$ odd) or $(\mathbb{Z}/2)^2$ ($n$ even); $\mathrm{IV}^*$: 7, $\mathbb{Z}/3$; $\mathrm{III}^*$: 8, $\mathbb{Z}/2$; $\mathrm{II}^*$: 9, 0. Possible $c$: $\mathrm{IV}, \mathrm{IV}^*$: $1, 3$; $\mathrm{I}_0^*$: $1, 2, 4$; $\mathrm{I}_n^*$: $2, 4$; $\mathrm{III}, \mathrm{III}^*$: 2; $\mathrm{II}, \mathrm{II}^*$: 1. *(stated; reference)* Theorem 10.36.

**10.21 Tate's algorithm.** The eleven steps of Theorem 10.37 compute the type, $m$, $f = v(\Delta) + 1 - m$, $c$ and a minimal model; explicit translations for Step 2 at $p \geq 5$, $p = 3$, $p = 2$, and completing the square for Step 6 at odd $p$. *(proof sketched here)* Theorem 10.37.

**10.22 The tame table.** For $p \geq 5$ and a minimal model, $(v(c_4), v(c_6), v(\Delta))$ is $(0, 0, n)$ for $\mathrm{I}_n$; $(\geq 1, 1, 2)$ II; $(1, \geq 2, 3)$ III; $(\geq 2, 2, 4)$ IV; $(\geq 2, \geq 3, 6)$ $\mathrm{I}_0^*$; $(2, 3, n+6)$ $\mathrm{I}_n^*$; $(\geq 3, 4, 8)$ $\mathrm{IV}^*$; $(3, \geq 5, 9)$ $\mathrm{III}^*$; $(\geq 4, 5, 10)$ $\mathrm{II}^*$. So for potentially good reduction the type is determined by $v(\Delta) \in \{0, 2, 3, 4, 6, 8, 9, 10\}$, and $f = 2$ for every additive type. *(proved here, given Tate's algorithm, and Ogg's formula for the index of* $\mathrm{I}_n^*$*)* Corollary 10.38.

**10.23 The cubic criterion for** $\mathrm{I}_0^*$**.** For $p$ odd and $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with $\varpi \mid a_2$, $\varpi^2 \mid a_4$, $\varpi^3 \mid a_6$: type $\mathrm{I}_0^*$ iff $T^3 + a_{2,1}T^2 + a_{4,2}T + a_{6,3}$ has distinct roots mod $\varpi$, and then $c = 1 + \#\text{roots in } k$; every $\mathrm{I}_0^*$ at odd $p$ has such a model. The twist by $q^*$ of a model with good reduction at odd $q$ has cubic $T^3 + \epsilon a_2T^2 + a_4T + \epsilon a_6$, $\epsilon = q^*/q$, hence type $\mathrm{I}_0^*$. *(proved here)* Proposition 10.39.

**10.24 PARI's Kodaira codes.** 1: $\mathrm{I}_0$; $n+4$: $\mathrm{I}_n$ and $-(n+4)$: $\mathrm{I}_n^*$ *for* $n \geq 1$ *only*; $-1$: $\mathrm{I}_0^*$; $\pm 2, \pm 3, \pm 4$: II/II*, III/III*, IV/IV*. `elllocalred` returns $[f, k, [u, r, s, t], c]$. A session on `726a1`, `20a1`, `539c1`, `11a1` shows the output. *(proved here, as a dictionary)* Convention 10.41, Example 10.84.

**10.25 Conductor exponent.** $f = a(V_\ell E) = \dim V - \dim V^I + \mathrm{sw}(V)$, independent of $\ell$ *(stated; reference)*; it is 0 (good), 1 (multiplicative, $V^I$ a line), $2 + \mathrm{sw} \geq 2$ (additive, $V^I = 0$), and $\mathrm{sw} = 0$ for $p \geq 5$, and for $p$ odd at an additive potentially multiplicative prime *(proved here, given the criterion of Néron, Ogg and Shafarevich)*. Definition 10.45, Proposition 10.46.

**10.26 Ogg's formula.** $f = v(\Delta_{\min}) + 1 - m$. *(stated; reference)* Theorem 10.47.

**10.27 Conductor bounds.** $f_p \le 2$ for $p \ge 5$ *(proved here)*, $f_3 \le 5$, $f_2 \le 8$ *(stated; reference)*. Theorem 10.48.

**10.28 The conductor.** $N = \prod_p p^{f_p}$; $p \,\|\, N$ iff multiplicative, $p^2 \mid N$ iff additive, $p^3 \nmid N$ for $p \ge 5$; isogenous curves have the same conductor. *(proved here)* Definition 10.49, Proposition 10.50.

**10.29 Serre-Tate.** For potentially good reduction there is a smallest $L \supseteq K^{\mathrm{ur}}$ with good reduction, $L = K^{\mathrm{ur}}(E[m])$ ($m \ge 3$, $p \nmid m$); $G_{\mathrm{ss}} = \mathrm{Gal}(L/K^{\mathrm{ur}})$ is the image of $I_K$ in $\mathrm{Aut}(T_\ell E)$ for every $\ell \ne p$; it embeds in $\mathrm{SL}_2(\mathbb{Z}/m)$, its elements have orders $1, 2, 3, 4, 6$, and for $p \ge 5$ it is cyclic. *(proof sketched here)* Theorem 10.51.

**10.30 Kraus.** For $p \ge 5$ and potentially good reduction, $G_{\mathrm{ss}}$ is cyclic of order $12/\gcd(v(\Delta_{\min}), 12)$: $1, 6, 4, 3, 2, 3, 4, 6$ for $\mathrm{I}_0, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ *(proved here, given Serre-Tate)*. At $p = 3$: cyclic of order $1, 2, 3, 4, 6$ or $\mathbb{Z}/3 \rtimes \mathbb{Z}/4$; at $p = 2$: cyclic of order $1, 2, 3, 4, 6$, $Q_8$ or $\mathrm{SL}_2(\mathbb{F}_3)$ *(stated; reference)*. Theorem 10.52.

**10.31 Tame additive reduction.** Additive potentially good reduction over a finite $K/\mathbb{Q}_p$ is tame ($f = 2$) iff $p \ge 5$, or $p = 3$ and type $\mathrm{III}, \mathrm{III}^*, \mathrm{I}_0^*$, or $p = 2$ and type $\mathrm{IV}, \mathrm{IV}^*$. *(stated; reference)* Theorem 10.53.

**10.32 Types IV, IV$^*$ at 2.** Such a curve over $\mathbb{Q}_2$ has potentially good reduction, $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$, the image of $I_2$ on $T_\ell E$ ($\ell$ odd) is generated by $\sigma$ of order 3 with eigenvalues $\zeta_3^{\pm 1}$, unipotent on $E[3]$, $f_2 = 2$, and Frob $\sigma$ $\mathrm{Frob}^{-1} = \sigma^{-1}$ *(proved here)* Corollary 10.54. Consequently (the paper link after it, with Lemma lem:euler and the normalisation of $P_q$ in Chapter 19) $P_2(X) = 1 + 2X$ and $e_\ell(E, 2) = v_\ell(3/2)$ *(consequence; see Chapter 19)*.

**10.33 The Tate curve.** For $v(\mathfrak{q}) > 0$, $E_{\mathfrak{q}}\colon y^2 + xy = x^3 - 5s_3 x - (5s_3 + 7s_5)/12$ has $\Delta = \mathfrak{q}\prod(1 - \mathfrak{q}^n)^{24}$, $j = \mathfrak{q}^{-1} + 744 + 196884\mathfrak{q} + \cdots$, split multiplicative reduction of type $\mathrm{I}_{v(\mathfrak{q})}$ *(proved here)*, and $E_{\mathfrak{q}}(\bar{K}) \simeq \bar{K}^\times/\mathfrak{q}^{\mathbb{Z}}$ Galois-equivariantly *(stated; reference)*. Theorem 10.57.

**10.34 Inverting $j$.** $\mathfrak{q}(J) = J + 744J^2 + 750420J^3 + 872769632J^4 + \cdots \in J\mathbb{Z}[[J]]$ inverts $1/j(\mathfrak{q})$; for $v(j_0) < 0$, $\mathfrak{q}(1/j_0)$ is the unique $\mathfrak{q}_0$ with $v(\mathfrak{q}_0) > 0$, $j(E_{\mathfrak{q}_0}) = j_0$, and $v(\mathfrak{q}_0) = -v(j_0)$. *(proved here)* Proposition 10.58.

**10.35 Curves with $v(j) < 0$.** $E$ is the quadratic twist of $E_{q_E}$ by $\gamma = -c_4/c_6$: split iff $\gamma$ is a square, nonsplit iff $K(\sqrt{\gamma})/K$ is unramified quadratic, additive iff it is ramified; for multiplicative reduction $n = v(\Delta_{\min}) = -v(j) = v(q_E)$. *(proof sketched here)* Theorem 10.59.

**10.36 Tate parameter and unit part.** For $E/\mathbb{Q}_q$ with $v_q(j) < 0$: $q_E = \mathfrak{q}(1/j_E)$, $\nu = v_q(q_E) = -v_q(j_E)$ ($= n_q$ at a multiplicative prime), $u = q_E q^{-\nu} \in \mathbb{Z}_q^\times$, $c(u)$ the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$. These depend only on $j_E$, so $t_\ell(E, q) = v_\ell(\nu)$ is twist invariant *(proved here)*; under an isogeny of prime degree $d$, $\nu \mapsto d\nu$ or $\nu/d$, so $t_\ell$ is invariant under isogenies of degree prime to $\ell$ *(proof sketched here)*. Definition 10.60, Proposition 10.61.

**10.37 Torsion of the Tate curve.** $E_{\mathfrak{q}}[\ell^t] = \{\zeta^a(\mathfrak{q}^{1/\ell^t})^b\}$, an extension $0 \to \mu_{\ell^t} \to E_{\mathfrak{q}}[\ell^t] \to \mathbb{Z}/\ell^t \to 0$; on $T_\ell E_{\mathfrak{q}}$, $\sigma \mapsto \left(\begin{smallmatrix} \chi_\ell(\sigma) & \kappa(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ with $\kappa$ the Kummer cocycle of $\mathfrak{q}$; for the twist by $\varepsilon$, $\left(\begin{smallmatrix} \chi_\ell\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$, with corner $\varepsilon\kappa$. *(proved here, given Tate's theorem)* Proposition 10.62.

**10.38 Types at additive potentially multiplicative primes.** For $p$ odd the type is $\mathrm{I}_\nu^*$, $\nu = -v(j)$, with $f = 2$, $v(\Delta_{\min}) = \nu + 6$, and conversely $\mathrm{I}_n^*$ ($n \ge 1$) at odd $p$ forces

$v(j) = -n$ *(proof sketched here, given Ogg's formula).* For $K = \mathbb{Q}_2$ the type is $\mathrm{I}_m^*$ with $m = \nu + 2f_2 - 4 \in \{\nu + 4, \nu + 8\}$ *(stated; reference).* Theorem 10.66.

**10.39 Unramified twists (the paper's Lemma lem:twist (i)).** A twist by a quadratic character unramified at $p$ preserves at $p$ the type, $\Phi_p^{\mathrm{geom}}$, $v_p(\Delta_{\min})$, $f_p$ and $q_E$; $c_p$ may change. *(proved here)* Lemma 10.69.

**10.40 Ramified twists (the paper's Lemma lem:twist (ii)).** For odd $q$ the twist by $q^*$ exchanges $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ (same $n \geq 1$) and $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$; for $q \geq 5$ also $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$ with $v_q(\Delta_{\min})$ shifted by 6 mod 12; the conductor exponent of the twist at $q$ is 0 (good), 1 ($\mathrm{I}_n$) or 2 (additive). *(proved here; the equality of the indices in $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ uses Ogg's formula)* Lemma 10.70.

**10.41 The exchanges at 3.** The twist by $-3$ exchanges $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$ at 3 and preserves $f_3$. *(proved here, given Ogg's formula)* Proposition 10.72.

**10.42 Twist-symmetrised weight.** For odd $q$ and odd $\ell$,

$$w_\ell(E, q) = \max\bigl(v_\ell\#\Phi_q^{\mathrm{geom}}(E),\, v_\ell\#\Phi_q^{\mathrm{geom}}(E^{(q)})\bigr), \qquad w_\ell(E, 2) = v_\ell\#\Phi_2^{\mathrm{geom}}(E).$$

*(proved here)* Proposition 10.74.

**10.43 What changes at 2 and 3.** For $p \in \{2, 3\}$: the type determines neither $v(\Delta_{\min})$ nor $f$ (only $f = v(\Delta_{\min}) + 1 - m$); $f \leq 8$ at 2, $f \leq 5$ at 3; $G_{\mathrm{ss}}$ can be non-abelian, of order divisible by $p$; the type determines $G_{\mathrm{ss}}$ only for $\mathrm{IV}, \mathrm{IV}^*$ at 2 ($\mathbb{Z}/3$) and $\mathrm{III}, \mathrm{III}^*, \mathrm{I}_0^*$ at 3 ($\mathbb{Z}/4, \mathbb{Z}/4, \mathbb{Z}/2$); at 2 a symbol $\mathrm{I}_m^*$ may be potentially good, and if not, $m \neq -v_2(j)$. So the adjoint Euler factor is not a function of the type at 2, 3. *(summary of results proved or stated elsewhere in the chapter)* Remark 10.76.

**10.44 Inertia at 2.** For additive potentially good reduction over $\mathbb{Q}_2$, the ramification pattern of the roots of $\psi_3$ is $(1,1,1,1)$, $(1,3,3,3)$, $(2,2,2,2)$, $(4,4,4,4)$ according as $G_{\mathrm{ss}}$ is $\mathbb{Z}/2$ (a ramified twist of good reduction), cyclic of order 3 or 6, $\mathbb{Z}/4$, or $Q_8$ or $\mathrm{SL}_2(\mathbb{F}_3)$; order 3 occurs exactly for IV, IV$^*$. *(proved here, given Kraus's list of groups and Theorem 10.53)* Proposition 10.77.

**10.45 The prime 3.** Types $\mathrm{III}, \mathrm{III}^*, \mathrm{I}_0^*$ at 3 are tame with $G_{\mathrm{ss}}$ cyclic of order $4, 4, 2$; types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$ are wild with $3 \mid e$ and $f_3 \in \{3, 4, 5\}$; $\mathrm{I}_n^*$ is potentially multiplicative with $f_3 = 2$. *(proved here, given Theorem 10.53)* Remark 10.79.

**10.46 Symbols $\mathrm{I}_m^*$ at 2.** They are potentially good in 61% of the pairs below conductor 10,000; when potentially multiplicative, $t_\ell(E, 2) = v_\ell(\nu)$ with $\nu = m - 4$ or $m - 8$, not $v_\ell(m)$. *(proved here, with data)* Remark 10.81.

**10.47 Worked data.** Tate's algorithm by hand on `121a1`, `121c2`, `539c1`, `539d1`, `726a1`, `20a1`, `99a1`, `980i1`; minimal models of twists; semistability defects; Tate parameters; twist pairs; local data of the running examples; what Cremona's tables contain. Example 10.7, Example 10.42, Example 10.56, Example 10.67, Example 10.75, Example 10.85, Remark 10.83.

## Where the paper uses this chapter

- **Definition def:terms** ($t_\ell(E,q) = v_\ell(-v_q(j_E))$, potentially multiplicative primes, the symbols $\mathrm{I}_n^*$ at 2): Proposition 10.4, Theorem 10.21, Proposition 10.58, Definition 10.60, Proposition 10.61, Theorem 10.66, Remark 10.81.
- **The table of Section 1 and Lemma lem:euler** (types at $q \geq 5$, inertia images): Corollary 10.38, Theorem 10.51, Theorem 10.52, Lemma 10.70.
- **Section 1.4** ($\Phi_q^{\mathrm{geom}}$ versus $c_q$, the exchange of types, the weight $w_\ell$, the inequalities (eq:main), (eq:twist)): Proposition 10.28, Definition 10.29, Theorem 10.36, Lemma 10.70, Proposition 10.72, Proposition 10.74.
- **Lemma lem:twist (i), (ii)**: proved as Lemma 10.69 and Lemma 10.70, using Proposition 10.5, Proposition 10.39, Theorem 10.59 and Theorem 10.66.
- **Lemmas lem:tamagawa-exp and lem:tamfactor** (the Tate curve over the unramified quadratic extension, $q_E$, $u$, $c(u)$, the matrices with corner $\varepsilon\kappa$, the check with $c_q$): Proposition 10.12, Proposition 10.33, Theorem 10.59 to Proposition 10.65; the lemmas themselves are proved in Chapter 11 and Chapter 18.
- **Lemma lem:IV and Remark rem:types** ($E[3]^{I_q}$ a line or 0, Kraus's defect, the case $q = 2$): Proposition 10.33, Theorem 10.51, Theorem 10.52, Corollary 10.54.
- **Proposition prop:shapes and Lemma lem:tamadd** (the image of inertia at a potentially good prime is finite, its elements have orders $1, 2, 3, 4, 6$, it is cyclic for $q \geq 5$; a potentially multiplicative curve is a quadratic twist of a Tate curve): Theorem 10.51, Theorem 10.52, Theorem 10.59, Proposition 10.77.
- **Remark rem:wild, Section sec:wild, Proposition prop:forms**: Theorem 10.53, Corollary 10.54, Remark 10.76, Proposition 10.77, Remark 10.79, Remark 10.81.
- **Theorems thm:qual, thm:quant and the proof of thm:main** (twists at $\mathrm{I}_0^*$, $\mathrm{I}_n^*$, II, II$^*$ primes; conductor exponents; the prime 2 of type $\mathrm{I}_n^*$): Proposition 10.50, Lemma 10.70, Theorem 10.59.
- **The appendix** (`ellglobalred`, `R[4]`, `R[5]`, `K[i][2]`, the Kodaira codes, `valuation(j, q)`): Convention 10.41, Example 10.84.
- **The Example of Section 3, the Remark after it, Section sec:anomaly**: Example 10.42, Example 10.67, Example 10.75, Example 10.85.

## 10.1 Integral and minimal Weierstrass models

This section fixes the local setting and the one choice that every later statement depends on: a Weierstrass model of $E$ over a complete discretely valued field that is integral and has the smallest possible discriminant. Every local invariant the paper reads off at a bad prime $q$ (the Kodaira type, the valuations $v_q(\Delta)$ and $v_q(c_4)$, the Tamagawa number $c_q$) is defined through such a model. The paper uses minimality explicitly in the proof of Lemma lem:twist (ii), where the twisted model "is minimal exactly when the result is less than 12, otherwise 12 is subtracted": that sentence is Proposition 10.5(ii) below.

**Convention 10.1** (the local setting)**.** Throughout the chapter $K$ is a field of characteristic 0, complete for a discrete valuation $v$, with valuation ring $\mathcal{O}$, maximal ideal $\mathfrak{m} = \varpi\mathcal{O}$, perfect residue field $k = \mathcal{O}/\mathfrak{m}$ of characteristic $p$, and $v(\varpi) = 1$. In every application $K$ is $\mathbb{Q}_p$, a finite extension of $\mathbb{Q}_p$, or a finite unramified extension of one of these; $K^{\mathrm{ur}}$ is the maximal unramified extension and $I_K$ the inertia group (Chapter 7). When the paper's bad prime is meant we write $K = \mathbb{Q}_q$ and $p = q$. The letter $q$ is also used for the Tate parameter in the literature; in this chapter the variable of the Tate curve is written $\mathfrak{q}$ and the Tate curve $E_{\mathfrak{q}}$, and the Tate parameter of a given curve $E$ is always $q_E$ (Section 10.7). The arithmetic Frobenius is Frob; the geometric Frobenius at a named prime is written $\mathrm{Frob}_q^{-1}$, because $\Phi_q(\cdot)$ denotes a component group in this chapter (Appendix B, Appendix B, conventions B-C2 and B-C11).

**Convention 10.2** (models)**.** A *model* of an elliptic curve $E/K$ is a Weierstrass equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in K,$$

with the general coefficients $a_1, \dots, a_6$ and the quantities $b_2, b_4, b_6, b_8, c_4, c_6, \Delta, j$ of Chapter 9. Two models of $E$ are related by a change of coordinates $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ with $u \in K^\times$, $r, s, t \in K$, written $(u, r, s, t)$, and then

$$u^4c_4' = c_4, \qquad u^6c_6' = c_6, \qquad u^{12}\Delta' = \Delta, \qquad j' = j,$$

together with the formulas for the $a_i'$ and $b_i'$ recalled in Chapter 9. At $p = 2$ and $p = 3$ a short model $y^2 = x^3 + Ax + B$ is in general not minimal, so short models are used only when $p \geq 5$.

**Definition 10.3** (integral and minimal models)**.** A model of $E/K$ is *integral* if all $a_i \in \mathcal{O}$; then $b_2, b_4, b_6, b_8, c_4, c_6, \Delta \in \mathcal{O}$, as they are polynomials with integer coefficients in the $a_i$. An integral model is *minimal* if $v(\Delta)$ is smallest among all integral models of $E$; its discriminant is the *minimal discriminant* $\Delta_{\min}$, well defined up to $\mathcal{O}^\times$. For a minimal model the invariant differential $\omega = dx/(2y + a_1x + a_3)$ is the *Néron differential*.

The reason for minimising is that the reduction of a model modulo $\varpi$ depends on the model, and a careless choice of coordinates makes a perfectly good curve look singular. Let $p \geq 5$ and take $y^2 = x^3 + x + 1$ over $\mathbb{Q}_p$; its discriminant is $-16 \cdot 31 = -496$, so for $p \neq 31$ it reduces to a smooth cubic. The change $(p^{-1}, 0, 0, 0)$ multiplies $a_i$ by $p^i$ and gives the integral model $y^2 = x^3 + p^4x + p^6$ of the same curve, with discriminant $p^{12} \cdot (-496)$; it reduces to the cuspidal cubic $y^2 = x^3$. Since $u^{12}\Delta' = \Delta$, the valuations $v(\Delta)$ of the integral models of $E$ all lie in one class modulo 12, and a change of coordinates can only add multiples of 12. Minimising removes these spurious factors $\varpi^{12}$, and Proposition 10.4 below shows that what is left (the reduction of a minimal model, up to a change of coordinates over $k$) depends only on $E$.

**Proposition 10.4** (existence and uniqueness of minimal models)**.** *Let $E/K$ be an elliptic curve.*

*(i) $E$ has a minimal model.*

*(ii) Two minimal models of $E$ are related by a change $(u, r, s, t)$ with $u \in \mathcal{O}^\times$ and $r, s, t \in \mathcal{O}$.*

*(iii) The Néron differential is well defined up to multiplication by $\mathcal{O}^\times$. The integers $v(\Delta_{\min})$, $v(c_4)$ and $v(c_6)$ of a minimal model depend only on $E/K$, and $v(j) = 3v(c_4) - v(\Delta)$ is the same for every model.*

*Proof.* (i) Start from any model. The change $(u,0,0,0)$ multiplies $a_i$ by $u^{-i}$; with $u=\varpi^{-n}$ the new coefficients are $\varpi^{ni}a_i$, which lie in $\mathcal{O}$ for $n$ large. So integral models exist, the values $v(\Delta)$ of integral models are nonnegative integers, and a smallest one is attained.

(ii) Any two models of $E$ are related by some $(u,r,s,t)$ (Chapter 9). If both are minimal, then $v(\Delta)=v(\Delta')$ and $u^{12}\Delta'=\Delta$ give $v(u)=0$. The transformation formulas for $b_6$ and $b_8$ read

$$4r^3+b_2r^2+2b_4r+b_6=u^6b_6',$$
$$3r^4+b_2r^3+3b_4r^2+3b_6r+b_8=u^8b_8',$$

with right-hand sides in $\mathcal{O}$. Suppose $v(r)<0$. If $p\neq 3$, in the second equation the term $3r^4$ has valuation $4v(r)$, strictly less than the valuation of every other term ($\geq 3v(r)$, $2v(r)$, $v(r)$ or $0$), which is impossible. If $p=3$ then $4$ is a unit and the same argument applies to $4r^3$ in the first equation. So $r\in\mathcal{O}$. Next, $ua_1'=a_1+2s$ and $u^2a_2'=a_2-sa_1+3r-s^2$ show that $s$ is a root of the monic polynomial $X^2+a_1X+(u^2a_2'-a_2-3r)$ with coefficients in $\mathcal{O}$; as $\mathcal{O}$ is integrally closed, $s\in\mathcal{O}$. Likewise $u^6a_6'=a_6+ra_4+r^2a_2+r^3-ta_3-t^2-rta_1$ shows that $t$ is a root of a monic quadratic over $\mathcal{O}$, so $t\in\mathcal{O}$.

(iii) Under $(u,r,s,t)$ the invariant differential is multiplied by $u$ (Chapter 9), and $u$ is a unit by (ii). The relations $u^4c_4'=c_4$, $u^6c_6'=c_6$, $u^{12}\Delta'=\Delta$ with $u$ a unit give the invariance of the three valuations. Finally $j=c_4^3/\Delta$ does not depend on the model at all. □

**Proposition 10.5** (criteria for minimality)**.** *Let a model of $E/K$ be integral.*

*(i) (any $p$) If $v(\Delta)<12$, or $v(c_4)<4$, or $v(c_6)<6$, the model is minimal.*

*(ii) ($p\geq 5$) The model is minimal if and only if $v(\Delta)<12$ or $v(c_4)<4$. If it is not minimal, the model $y^2=x^3-27\varpi^{-4}c_4x-54\varpi^{-6}c_6$ is integral with discriminant of valuation $v(\Delta)-12$; repeating this step finitely often gives a minimal model.*

*(iii) (Kraus's conditions; $K=\mathbb{Q}$) Let $c_4,c_6\in\mathbb{Z}$ with $\Delta=(c_4^3-c_6^2)/1728\in\mathbb{Z}\setminus\{0\}$. There is a Weierstrass model with coefficients in $\mathbb{Z}$ whose invariants are $c_4,c_6$ if and only if $v_3(c_6)\neq 2$, and either $c_6\equiv -1 \pmod 4$, or $v_2(c_4)\geq 4$ and $c_6\equiv 0$ or $8 \pmod{32}$. Consequently, for $p\in\{2,3\}$, an integral model over $\mathbb{Z}$ with $p^4\mid c_4$, $p^6\mid c_6$ and $p^{12}\mid\Delta$ fails to be minimal at $p$ exactly when $(c_4/p^4,c_6/p^6)$ satisfies the condition at $p$ ($v_3(c_6/3^6)\neq 2$ for $p=3$; the congruence condition for $p=2$).*

*Proof.* (i) Suppose an integral model with invariants $c_4',c_6',\Delta'$ and $v(\Delta')<v(\Delta)$ exists, related by $(u,r,s,t)$. Then $u^{12}\Delta'=\Delta$ forces $v(u)\geq 1$, hence $v(\Delta)=v(\Delta')+12v(u)\geq 12$, $v(c_4)=v(c_4')+4v(u)\geq 4$ and $v(c_6)=v(c_6')+6v(u)\geq 6$, as $c_4',c_6',\Delta'$ are integral. Each of the three inequalities in the hypothesis contradicts one of these.

(ii) By (i) the condition is sufficient. Conversely let $v(c_4)\geq 4$ and $v(\Delta)\geq 12$. From $c_6^2=c_4^3-1728\Delta$ we get $v(c_6^2)\geq 12$, so $v(c_6)\geq 6$. The curve has the model $y^2=x^3-27c_4x-54c_6$ (Chapter 9), whose discriminant is $6^{12}\Delta$. (For a short model $y^2=x^3+Ax+B$ one has $b_2=0$, $b_4=2A$, $b_6=4B$, hence $c_4=-48A$ and $c_6=-864B$; with $A=-27c_4$, $B=-54c_6$ this gives the invariants $1296c_4=6^4c_4$ and $46656c_6=6^6c_6$, so the short model is obtained by a change $(u,r,s,t)$ with $u=1/6$, and its discriminant is $u^{-12}\Delta=6^{12}\Delta$.) Applying $(\varpi,0,0,0)$ to it gives $y^2=x^3-27\varpi^{-4}c_4x-54\varpi^{-6}c_6$, which is integral, with discriminant $6^{12}\varpi^{-12}\Delta$. As $p\geq 5$, $6$ is a unit and the valuation is $v(\Delta)-12$. So the model was not minimal. Each step lowers $v(\Delta)$ by 12, so the process stops.

(iii) We do not prove this; see Kraus [106], and [35, Section 3.2] for its use in algorithms. The consequence follows by applying the criterion to the invariants $c_4/p^4$, $c_6/p^6$ of the model obtained by $(p,0,0,0)$: the conditions at the other prime are unchanged, since $p$ is a unit there (for $p=3$: multiplication by $3^{-6}\equiv 9 \pmod{32}$ preserves the classes 0 and 8 modulo 32 and the class $-1$ modulo 4). □

We checked (iii) on the data: 20,000 random integral models with coefficients in $[-50,50]$ all satisfy Kraus's conditions, and in all 8,248 pairs (curve, $p\in\{2,3\}$) of Cremona's tables below conductor 10,000 whose minimal model has $p^4 \mid c_4$, $p^6 \mid c_6$, $p^{12} \mid \Delta$, the scaled invariants fail the condition at $p$, as they must.

Part (i) is only a sufficient condition when $p\in\{2,3\}$, and (iii) is what replaces the converse of (ii) there. For example `32a1` is $y^2 = x^3 + 4x$, with $c_4 = -48\cdot 4 = -2^6\cdot 3$, $c_6 = 0$ and $\Delta = -16\cdot 4\cdot 4^3 = -2^{12}$: all three valuations at 2 reach the thresholds of (i), and yet the model is minimal at 2. Indeed a model with smaller discriminant would come from $(2,r,s,t)$ and have invariants $c_4/2^4 = -12$, $c_6/2^6 = 0$ and $\Delta/2^{12} = -1$; these fail Kraus's condition at 2 ($0 \not\equiv -1 \pmod 4$, and $v_2(-12) = 2 < 4$), so no integral model has them.

**Theorem 10.6** (global minimal models)**.** *Every elliptic curve $E/\mathbb{Q}$ has a model with coefficients in $\mathbb{Z}$ that is minimal at every prime $p$. It is unique up to a change $(u,r,s,t)$ with $u=\pm 1$ and $r,s,t\in\mathbb{Z}$, and its discriminant $\Delta_E$ is the* minimal discriminant *of $E$. Imposing $a_1,a_3\in\{0,1\}$ and $a_2\in\{-1,0,1\}$ makes it unique.*

*Sketch of proof.* Start from a model over $\mathbb{Z}$. At each of the finitely many primes $p$ dividing its discriminant choose a change $(u_p,r_p,s_p,t_p)$ over $\mathbb{Q}_p$ to a minimal model. Since $\mathbb{Z}$ is a principal ideal domain the ideal $\prod_p p^{v_p(u_p)}$ is generated by $u = \prod_p p^{v_p(u_p)}$, and the Chinese remainder theorem lets one choose $r,s,t\in\mathbb{Z}$ approximating the $r_p,s_p,t_p$ closely enough that the change $(u,r,s,t)$ is integral and minimal at every $p$ simultaneously. Uniqueness follows from Proposition 10.4(ii) at every prime: $u$ is a unit of $\mathbb{Z}$ and $r,s,t\in\mathbb{Z}_p$ for all $p$. The normalisation of $a_1,a_3,a_2$ uses $s$, $t$ and $r$ modulo 2, 2 and 3. See [185, VIII.8] and [35, Section 3.2]. □

Cremona's tables [35, 36] and PARI's `ellminimalmodel` give exactly this normalised global minimal model. All models printed in this book are of this form.

**Example 10.7** (minimal models)**.** (a) The curve `121a1` is $[1,1,1,-30,-76]$ with $\Delta = -11^2$. Its twist by $-11$, written with the $b$-invariants as in Chapter 9, is $y^2 = x^3 + db_2x^2 + 8d^2b_4x + 16d^3b_6$ with $d=-11$; its discriminant is $2^{12}d^6\Delta$, so $v_{11}(\Delta) = 2+6 = 8 < 12$ and the model is minimal at 11 by Proposition 10.5(i). It is the curve `121c2`, with $\Delta = -11^8$. Twisting `121c2` back by $-11$ gives $v_{11}(\Delta) = 8+6 = 14$ and $v_{11}(c_4) = 3+2 = 5 \ge 4$, so that model is not minimal at 11; one step of Proposition 10.5(ii) gives $v_{11}(\Delta) = 2$, and the curve is `121a1` again.

(b) The curve `140b1` is $[0,0,0,32,212]$, with $v_2(\Delta) = 8$, $v_2(c_4) = 9$ and $v_2(j) = 27-8 = 19$. It is minimal at 2 by Proposition 10.5(i), although $c_4$ is very divisible by 2.

(c) The short model $y^2 = x^3 - 27c_4x - 54c_6$ of `11a1` has discriminant $6^{12}\Delta$, so $v_2(\Delta) = v_3(\Delta) = 12$; it is not minimal at 2 or 3, since the model $[0,-1,1,-10,-20]$ has $\Delta = -11^5$.

*In the paper* 10.8. Lemma lem:twist (ii), proof: "the twisted model has $v_q(\Delta)$ increased by 6, and is minimal exactly when the result is less than 12, otherwise 12 is subtracted". For $q\ge 5$ and potentially good reduction this is Proposition 10.5: if the result is at least 12, then $v_q(j)\ge 0$ gives $3v_q(c_4) \ge v_q(\Delta) \ge 12$, so $v_q(c_4)\ge 4$ and part (ii) applies once. The full argument is Lemma 10.70.

## 10.2 Reduction modulo the uniformiser

This section defines good, multiplicative and additive reduction, splits multiplicative reduction into split and nonsplit, and studies the reduction map on points. The paper uses the trichotomy everywhere (the numbers $\delta_p = \dim V^{I_p} = 2, 1, 0$ of its Section 1.1; the PARI test `k >= 5` of its appendix), the values $a_p \in \{1, -1, 0\}$ at bad primes (Lemma lem:old, Definition def:terms), and the split/nonsplit distinction in Lemmas lem:tamagawa-exp and lem:tamfactor. The injectivity of reduction on prime-to-$p$ torsion (Corollary 10.17) is the easy half of the criterion of Néron, Ogg and Shafarevich (Chapter 11) and the input of Proposition 10.33.

**Definition 10.9** (reduction types)**.** Let $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ be a minimal model of $E/K$ and let $\tilde{E}$ be the plane cubic over $k$ defined by the reduced coefficients $\tilde{a}_i$. The curve $E$ has

- *good reduction* if $\tilde{E}$ is smooth, that is, $v(\Delta) = 0$;
- *multiplicative reduction* if $\tilde{E}$ has a node, that is, $v(\Delta) > 0$ and $v(c_4) = 0$;
- *additive reduction* if $\tilde{E}$ has a cusp, that is, $v(\Delta) > 0$ and $v(c_4) > 0$.

Multiplicative reduction is *split* if the two tangent directions at the node are defined over $k$, and *nonsplit* otherwise. *Semistable* means good or multiplicative. For $E/\mathbb{Q}$ and a prime $p$ we say that $E$ has good (multiplicative, additive) reduction at $p$ if $E/\mathbb{Q}_p$ has.

The definition does not depend on the minimal model: two minimal models differ by $(u, r, s, t)$ with $u \in \mathcal{O}^\times$ and $r, s, t \in \mathcal{O}$ (Proposition 10.4), which reduces to a change of coordinates over $k$. A Weierstrass cubic is smooth if and only if its discriminant is nonzero, and if $\tilde{\Delta} = 0$ it has exactly one singular point, a node if $\tilde{c}_4 \neq 0$ and a cusp if $\tilde{c}_4 = 0$ (Chapter 9). The singular point is $k$-rational: it is unique, hence fixed by $\mathrm{Gal}(\bar{k}/k)$, and $k$ is perfect.

**Proposition 10.10** (the group of nonsingular points)**.** *Let $\tilde{E}/k$ be a Weierstrass cubic with a singular point $S$, and let $\tilde{E}_{\mathrm{ns}}(k)$ be the set of $k$-points other than $S$. The chord-tangent construction makes $\tilde{E}_{\mathrm{ns}}(k)$ an abelian group with neutral element $\tilde{O}$, and*

*(i) if $S$ is a cusp, $\tilde{E}_{\mathrm{ns}}(k) \simeq k^+$;*

*(ii) if $S$ is a split node, $\tilde{E}_{\mathrm{ns}}(k) \simeq k^\times$;*

*(iii) if $S$ is a nonsplit node and $l/k$ is the quadratic extension containing the tangent slopes, $\tilde{E}_{\mathrm{ns}}(k) \simeq \ker(\mathrm{N}_{l/k} \colon l^\times \to k^\times)$.*

*For $k = \mathbb{F}_p$ the orders are $p$, $p - 1$ and $p + 1$.*

*Proof.* A line through two nonsingular points does not pass through $S$: a line through $S$ meets the cubic there with multiplicity at least 2, hence in at most one further point. So the chord-tangent law, defined exactly as for a smooth cubic (Chapter 9), stays inside $\tilde{E}_{\mathrm{ns}}$; associativity can be checked as in the smooth case, and also follows from the isomorphisms below, which we now construct.

In both cases we first translate $S$ to $(0, 0)$ (a change $(1, r, 0, t)$ over $k$; $S$ is $k$-rational). Then $\tilde{a}_6 = 0$ because $(0, 0)$ lies on the curve, and $\tilde{a}_4 = \tilde{a}_3 = 0$ because the partial derivatives $\partial F/\partial x = a_1y - 3x^2 - 2a_2x - a_4$ and $\partial F/\partial y = 2y + a_1x + a_3$ of $F = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6$

vanish at $(0,0)$. So the equation is $y^2+a_1xy-a_2x^2=x^3$, and the quadratic form $y^2+a_1xy-a_2x^2$ is the tangent cone at $S$: a product of two distinct linear forms at a node, a square at a cusp.

(i) At a cusp the quadratic part is $(y-\alpha x)^2$, with $\alpha=-a_1/2$ if $\operatorname{char}k\neq 2$; if $\operatorname{char}k=2$ then $a_1=0$ (the discriminant $a_1^2+4a_2=a_1^2$ vanishes) and $\alpha=\sqrt{a_2}$, which lies in $k$ because $k$ is perfect. The change $y\mapsto y+\alpha x$ makes the single tangent direction the line $y=0$, and the curve becomes $y^2=x^3$, that is, $Y^2Z=X^3$ in projective coordinates. Its points other than $S=[0:0:1]$ have $Y\neq 0$ and are $[t:1:t^3]$, $t\in k$, with $\tilde{O}=[0:1:0]$ at $t=0$. A line $\alpha X+\beta Y+\gamma Z=0$ not through $S$ has $\gamma\neq 0$ and meets the curve at the points with $\gamma t^3+\alpha t+\beta=0$, counted with multiplicity. The coefficient of $t^2$ is 0, so the three values of $t$ add up to 0. Three points add up to $\tilde{O}$ in the group law exactly when they are the intersection of a line with the curve, so $P\mapsto t(P)$ is a homomorphism $\tilde{E}_{\mathrm{ns}}(k)\to k^+$, and it is bijective.

(ii) With $S$ at $(0,0)$ as above, the quadratic part factors as $(y-\alpha x)(y-\beta x)$ with $\alpha\neq\beta$ the roots of $T^2+a_1T-a_2$; split means $\alpha,\beta\in k$. A nonsingular point $P\neq\tilde{O}$ has $x\neq 0$ (the only point of the curve with $x=0$ is $S$, as $y^2=0$ there), so it lies on the line $y=\lambda x$ through $S$ with $\lambda=y/x$. Substituting $y=\lambda x$ in the equation gives $x^2(\lambda^2+a_1\lambda-a_2)=x^3$, so $x=(\lambda-\alpha)(\lambda-\beta)$ and $y=\lambda x$; conversely every $\lambda\in k\setminus\{\alpha,\beta\}$ gives such a point. So $P\mapsto\lambda$ is a bijection from $\tilde{E}_{\mathrm{ns}}(k)\setminus\{\tilde{O}\}$ to $k\setminus\{\alpha,\beta\}$, and $\tilde{O}$ corresponds to $\lambda=\infty$. Put $\phi(P)=(\lambda-\alpha)/(\lambda-\beta)$ and $\phi(\tilde{O})=1$; this is a bijection onto $k^\times$. A line $y=mx+c_0$ with $c_0\neq 0$ meets the curve at the points with $\lambda x=mx+c_0$, that is, at the roots of $(\lambda-\alpha)(\lambda-\beta)(\lambda-m)-c_0=\prod_{i=1}^3(\lambda-\lambda_i)$ (the line does not pass through $S$, which is why $c_0\neq 0$). Evaluating at $\lambda=\alpha$ and at $\lambda=\beta$ gives $\prod(\alpha-\lambda_i)=-c_0=\prod(\beta-\lambda_i)$, hence $\prod_i(\lambda_i-\alpha)=c_0=\prod_i(\lambda_i-\beta)$, and dividing, $\prod_i\phi(P_i)=1$. A vertical line $x=x_0$ ($x_0\neq 0$) meets the curve at $\tilde{O}$ and at the two roots of $(\lambda-\alpha)(\lambda-\beta)-x_0$, and the same evaluation gives $\phi(P_1)\phi(P_2)=1=\phi(\tilde{O})^{-1}$. So in all cases three collinear points have $\phi(P_1)\phi(P_2)\phi(P_3)=1$. Since $P$, $-P$, $\tilde{O}$ are collinear, $\phi(-P)=\phi(P)^{-1}$; and if $P,Q,R$ are collinear then $P+Q=-R$, so $\phi(P+Q)=\phi(R)^{-1}=\phi(P)\phi(Q)$. So $\phi$ is a homomorphism to $k^\times$. (The same two lines, with sums in place of products, turn "the three values of $t$ add up to 0" into the homomorphism property in (i).)

(iii) Apply (ii) over $l$, where $\alpha,\beta$ are conjugate under the generator $\sigma$ of $\mathrm{Gal}(l/k)$. For a point $P$ with slope $\lambda\in l$, the conjugate point has slope $\sigma(\lambda)$, and $\phi(\sigma P)=(\sigma\lambda-\alpha)/(\sigma\lambda-\beta)=\sigma\big((\lambda-\beta)/(\lambda-\alpha)\big)=\sigma(\phi(P))^{-1}$. So $P$ is $k$-rational if and only if $\phi(P)\sigma(\phi(P))=1$, that is, $\mathrm{N}_{l/k}\,\phi(P)=1$.

For $k=\mathbb{F}_p$: $\#\mathbb{F}_p=p$, $\#\mathbb{F}_p^\times=p-1$, and the norm $\mathbb{F}_{p^2}^\times\to\mathbb{F}_p^\times$ is surjective, so its kernel has $(p^2-1)/(p-1)=p+1$ elements. □

**Corollary 10.11** ($a_p$ at a bad prime)**.** *Let $E/\mathbb{Q}$ have bad reduction at $p$ and put $a_p=p+1-\#\tilde{E}(\mathbb{F}_p)$, the singular point included in the count. Then $a_p=1$, $-1$, $0$ according as the reduction is split multiplicative, nonsplit multiplicative or additive.*

*Proof.* $\#\tilde{E}(\mathbb{F}_p)=\#\tilde{E}_{\mathrm{ns}}(\mathbb{F}_p)+1$, and Proposition 10.10 gives $p$, $p-1$, $p+1$ for the three cases. □

**Proposition 10.12** (split and nonsplit)**.** *Let $E/K$ have multiplicative reduction.*

*(i) After a change $(1,r,0,t)$ with $r,t\in\mathcal{O}$ moving the node to $(0,0)$, the reduction is split if and only if $T^2+a_1T-a_2$ has a root in $k$.*

*(ii) If $p$ is odd, the reduction is split if and only if $-c_6$ is a square in $k^\times$ for one (equivalently every) minimal model; equivalently, if and only if $-c_4/c_6$ is a square in $k^\times$.*

*(iii) If $p$ is odd and $k$ is finite, the twist of $E$ by the unramified quadratic extension of $K$ exchanges split and nonsplit reduction, and over the unramified quadratic extension of $K$ the reduction is split.*

*Proof.* (i) is the definition, by the factorisation of the quadratic part in the proof of Proposition 10.10(ii). (ii) After moving the node to $(0,0)$, $a_3, a_4, a_6 \in \mathfrak{m}$, so $b_4 \equiv 0$ and $b_6 \equiv 0$, and $c_4 = b_2^2 - 24b_4 \equiv b_2^2$, $c_6 = -b_2^3 + 36b_2b_4 - 216b_6 \equiv -b_2^3$. The discriminant of $T^2 + a_1T - a_2$ is $a_1^2 + 4a_2 = b_2$, which is nonzero mod $\mathfrak{m}$ because $c_4$ is a unit. So split means $\tilde{b}_2$ is a square, and $-\tilde{c}_6 = \tilde{b}_2 \cdot \tilde{b}_2^2$ is a square if and only if $\tilde{b}_2$ is, as is $-\tilde{c}_4/\tilde{c}_6 = \tilde{b}_2^{-1}$. Another minimal model changes $c_6$ by $u^6$ and $c_4/c_6$ by $u^{-2}$, with $u$ a unit, which does not change the square classes. (iii) Since $k$ is finite, $k^\times/(k^\times)^2$ has order 2, so $K$ has a unique unramified quadratic extension $K' = K(\sqrt{d})$, with $d$ a unit whose reduction is not a square. For $p$ odd we may take a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$; its twist by $d$ is $y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6$ (Chapter 9), which multiplies $c_6$ by $d^3$. As $\tilde{d}$ is not a square, this changes the square class of $-\tilde{c}_6$, and the twisted model is still minimal ($v(\Delta)$ is unchanged). Over the unramified quadratic extension $K'$ every element of $k^\times$ becomes a square in the residue field $k'$, so the reduction is split there. □

We checked (ii) on all 127,592 pairs (curve, odd prime of multiplicative reduction) of Cremona's tables below conductor 10,000: in every case $a_p = \left(\frac{-c_6}{p}\right)$.

*In the paper* 10.13. Lemma lem:tamagawa-exp ends with "This holds for split and for nonsplit reduction", and its proof passes to "the unramified quadratic extension of $\mathbb{Q}_q$ (which does not change the inertia group)", over which $E$ is split by Proposition 10.12(iii) and a Tate curve by Theorem 10.59. In Lemma lem:tamfactor the character $\varepsilon$ is unramified of order at most 2 with $\varepsilon(\mathrm{Frob}_q) = a_q$, and in Lemma lem:old $\varepsilon(\mathrm{Frob}_p) = a_p(f) = \pm 1$ by Corollary 10.11.

**Definition 10.14** (the reduction filtration)**.** For a minimal model of $E/K$ and $P \in E(K)$, write $P = [X : Y : Z]$ with $X, Y, Z \in \mathcal{O}$ not all in $\mathfrak{m}$, and let $\tilde{P} = [\tilde{X} : \tilde{Y} : \tilde{Z}] \in \tilde{E}(k)$ be its reduction. Put

$$E_0(K) = \{P \in E(K) : \tilde{P} \in \tilde{E}_{\mathrm{ns}}(k)\}, \qquad E_1(K) = \{P \in E(K) : \tilde{P} = \tilde{O}\}.$$

The point of the definition is that reduction is a group homomorphism only where it lands in a group: $\tilde{E}(k)$ is not a group when $\tilde{E}$ is singular, but $\tilde{E}_{\mathrm{ns}}(k)$ is (Proposition 10.10). So $E_0(K)$, the set of points whose reduction avoids the singular point, is the natural domain of a reduction homomorphism, and $E_1(K)$, the points $p$-adically close to $O$ (in affine coordinates, those with $v(x) < 0$), is its kernel. The three pieces $E_1(K)$, $E_0(K)/E_1(K)$ and $E(K)/E_0(K)$ are then studied separately: the first by the formal group, the second by Proposition 10.10, the third by the component group of Section 10.4.

**Proposition 10.15** (the reduction map)**.** *$E_0(K)$ is a subgroup of $E(K)$, $E_1(K)$ is a subgroup of $E_0(K)$, and reduction is a surjective homomorphism $E_0(K) \to \tilde{E}_{\mathrm{ns}}(k)$ with kernel $E_1(K)$:*

$$0 \longrightarrow E_1(K) \longrightarrow E_0(K) \longrightarrow \tilde{E}_{\mathrm{ns}}(k) \longrightarrow 0.$$

*If $E$ has good reduction, $E_0(K) = E(K)$.*

*Proof.* Reduction is well defined, since the primitive coordinates are unique up to $\mathcal{O}^\times$. Clearly $\tilde{O} = [0 : 1 : 0]$ is nonsingular, and $-P = (x, -y - a_1x - a_3)$ reduces to $-\tilde{P}$. Let $P_1, P_2 \in E_0(K)$ and let $P_3$ be the third intersection of the line $L$ through them (the tangent if $P_1 = P_2$) with $E$. Write $L$ as $\alpha X + \beta Y + \gamma Z = 0$ with $\alpha, \beta, \gamma \in \mathcal{O}$ not all in $\mathfrak{m}$; its reduction $\tilde{L}$ is a line.

Parametrising $L$ linearly and substituting into the cubic gives a cubic form in two variables with coefficients in $\mathcal{O}$ whose three roots are $P_1, P_2, P_3$; reducing it shows that $\tilde{L}$ meets $\tilde{E}$ exactly in $\tilde{P}_1, \tilde{P}_2, \tilde{P}_3$, counted with multiplicity (the reduced form is not identically zero, as $\tilde{E}$ is irreducible and does not contain the line $\tilde{L}$). If $\tilde{P}_3$ were the singular point, $\tilde{L}$ would meet $\tilde{E}$ with multiplicity at least 2 there and also at the nonsingular points $\tilde{P}_1, \tilde{P}_2$, four intersections in all, which is impossible. So $\tilde{P}_3$ is nonsingular and $\tilde{P}_1 + \tilde{P}_2 + \tilde{P}_3 = \tilde{O}$ in $\tilde{E}_{\mathrm{ns}}(k)$. Hence $E_0(K)$ is closed under addition and negation, and reduction is a homomorphism on it; its kernel is $E_1(K)$ by definition.

Surjectivity: let $(\tilde{x}_0, \tilde{y}_0)$ be a nonsingular affine point of $\tilde{E}(k)$ (the point $\tilde{O}$ is the image of $O$). With $F(x,y) = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6$, one of $\partial F/\partial x$, $\partial F/\partial y$ is nonzero at the point. If $\partial F/\partial y(\tilde{x}_0, \tilde{y}_0) \neq 0$, choose any lift $x_0 \in \mathcal{O}$ of $\tilde{x}_0$; the polynomial $F(x_0, y)$ has the simple root $\tilde{y}_0$ modulo $\mathfrak{m}$, and Hensel's lemma (Chapter 7) lifts it to $y_0 \in \mathcal{O}$ with $F(x_0, y_0) = 0$. The other case is the same with the roles of $x$ and $y$ exchanged. The point $(x_0, y_0) \in E(K)$ reduces to $(\tilde{x}_0, \tilde{y}_0)$. □

**Theorem 10.16** (the formal group)**.** *Let a minimal model of $E/K$ be given.*

*(i) In the parameter $z = -x/y$ near $O$, the curve is described by a power series $w(z) = -1/y = z^3 + a_1zw + \cdots$ with coefficients in $\mathbb{Z}[a_1, \ldots, a_6]$, and the group law by a power series $F(z_1, z_2) \in \mathbb{Z}[a_1, \ldots, a_6][[z_1, z_2]]$; $F$ is a one-parameter commutative formal group $\hat{E}$, and $P \mapsto z(P)$ is an isomorphism $E_1(K) \simeq \hat{E}(\mathfrak{m})$, the set $\mathfrak{m}$ with the group law $F$.*

*(ii) For $r > v(p)/(p-1)$ the formal logarithm gives an isomorphism $\hat{E}(\mathfrak{m}^r) \simeq \mathfrak{m}^r$ with the additive group.*

*(iii) For every integer $m$ prime to $p$, multiplication by $m$ is a bijection of $E_1(K)$. In particular $E_1(K)$ has no nonzero torsion of order prime to $p$, and it is uniquely $m$-divisible.*

*Sketch of proof.* (i) and (ii) are [185, IV.1 to IV.6 and VII.2]: one solves the equation of the curve for $w$ as a power series in $z$ by iteration, computes the third intersection of a line with the curve in these coordinates, and checks that the resulting series converge on $\mathfrak{m}$ and match the reduction filtration. (iii) is a complete argument given (i): a formal group law has the form $F(z_1, z_2) = z_1 + z_2 +$ (terms of degree $\geq 2$), so by induction $[m](z) = F([m-1](z), z) = (m-1)z + z + \cdots$ is $mz +$ (terms of degree $\geq 2$) for $m \geq 1$, and $[-1](z) = -z + \cdots$ handles negative $m$; here $m \in \mathcal{O}^\times$ because $p \nmid m$. A power series $f(z) = mz + \cdots$ with coefficients in $\mathcal{O}$ and $m$ a unit has a compositional inverse $g(z) = m^{-1}z + \cdots$ with coefficients in $\mathcal{O}$: writing $g = \sum_{n \geq 1} g_n z^n$, the coefficient of $z^n$ in $g(f(z))$ is $g_n m^n$ plus a polynomial with integer coefficients in $g_1, \ldots, g_{n-1}$ and the coefficients of $f$, so the equations $g(f(z)) = z$ determine the $g_n$ recursively, dividing only by powers of $m$. As $f$ and $g$ have coefficients in $\mathcal{O}$ and no constant term, both converge on $\mathfrak{m}$ and map $\mathfrak{m}$ into $\mathfrak{m}$, and the identities $g \circ f = f \circ g = z$ of power series become identities of maps. So $[m]$ is a bijection of $\hat{E}(\mathfrak{m})$, hence of $E_1(K)$; in particular its kernel $E_1(K)[m]$ is 0. □

**Corollary 10.17** (torsion and reduction)**.** *Let $m$ be prime to $p$.*

*(i) Reduction $E_0(K)[m] \to \tilde{E}_{\mathrm{ns}}(k)$ is injective.*

*(ii) If $E$ has good reduction, $E(K)[m] \hookrightarrow \tilde{E}(k)$; the inertia group $I_K$ acts trivially on $E[m]$, so $E[m] \subset E(K^{\mathrm{ur}})$; and reduction is an isomorphism $E[m] \to \tilde{E}[m](\bar{k})$ compatible with the actions of $G_K$ and $\mathrm{Gal}(\bar{k}/k)$. Consequently $I_K$ acts trivially on $T_\ell E$ for every prime $\ell \neq p$.*

*Proof.* (i) The kernel is $E_1(K)[m]$, which is 0 by Theorem 10.16(iii). (ii) The first statement is (i) with $E_0(K) = E(K)$. Let $L = K(E[m])$, a finite extension; the given model has unit discriminant over $L$ too, so $E/L$ has good reduction and (i) over $L$ shows that $E(L)[m] = E[m]$ injects into $\tilde{E}(k_L)$. Let $\sigma \in I_K$ and $P \in E[m]$. Since $\sigma$ acts trivially on $k_L$, the point $\sigma P - P \in E[m]$ reduces to $\widetilde{\sigma P} - \tilde{P} = \tilde{\sigma}\tilde{P} - \tilde{P} = \tilde{O}$, so $\sigma P = P$. Hence $E[m] \subset E(K^{\mathrm{ur}})$. Reduction from $E[m]$ to $\tilde{E}[m](\bar{k})$ is injective, and both groups have $m^2$ elements ($m$ is prime to the characteristics; Chapter 9), so it is bijective. It commutes with the Galois actions because reduction commutes with automorphisms of $K^{\mathrm{ur}}$. Passing to the limit over $m = \ell^n$ gives the statement for $T_\ell E$. □

**Proposition 10.18** (the index of $E_0$)**.** *$E(K)/E_0(K)$ is finite when $k$ is finite; it is trivial when the reduction is good, and its order (the Tamagawa number) is computed in Proposition 10.28, Theorem 10.36 and Proposition 10.65.*

*Proof.* For good reduction $E_0(K) = E(K)$ by definition. In general $E_0(K)$ is open in the compact group $E(K)$ (reduction is locally constant on $E(K)$: two points whose primitive coordinates are congruent modulo $\mathfrak{m}$ have the same reduction), so its cosets form an open cover of the compact group $E(K)$, and the index is finite. Compactness: as $k$ is finite, $\mathcal{O} = \varprojlim \mathcal{O}/\mathfrak{m}^n$ is compact, so the set of primitive triples (those in $\mathcal{O}^3$ not all in $\mathfrak{m}$) is compact; it maps continuously onto $\mathbb{P}^2(K)$, which is therefore compact, and $E(K)$ is closed in it, being the zero set of a homogeneous polynomial. The finer statement that $E(K)/E_0(K)$ is the group of $k$-points of the component group of the Néron model is Proposition 10.28; see [185, VII.6] and [184, IV.9]. □

**An example: torsion outside $E_0$.** The curve `11a1`, $y^2 + y = x^3 - x^2 - 10x - 20$, has split multiplicative reduction of type $\mathrm{I}_5$ at 11 with $c_{11} = 5$ (Example 10.85), and $E(\mathbb{Q})_{\mathrm{tors}} \simeq \mathbb{Z}/5$, with nonzero points $(5,5)$, $(16,-61)$, $(16,60)$, $(5,-6)$ (the multiples $P, 2P, 3P, 4P$ of $P = (5,5)$, computed with PARI). Modulo 11 all four reduce to $(5,5)$, since $16 \equiv 5$, $-61 \equiv 60 \equiv -6 \equiv 5$. With $F = y^2 + y - x^3 + x^2 + 10x + 20$ one has $F(5,5) = 0$, $\partial F/\partial y = 2y + 1 = 11$ and $\partial F/\partial x = -3x^2 + 2x + 10 = -55$ at $(5,5)$; both partials vanish modulo 11, so $(5,5)$ is the node of $\tilde{E}$. Hence no nonzero torsion point lies in $E_0(\mathbb{Q}_{11})$, the subgroup $E(\mathbb{Q})_{\mathrm{tors}}$ meets $E_0(\mathbb{Q}_{11})$ trivially, and it maps isomorphically onto $E(\mathbb{Q}_{11})/E_0(\mathbb{Q}_{11})$, a group of order $c_{11} = 5$. This does not contradict Corollary 10.17(i), which constrains only torsion inside $E_0$.

*In the paper* 10.19. Section 1.1 of the paper: $\delta_p = \dim V^{I_p}$ is 2 at a good prime because $I_p$ acts trivially on $T_\ell E$ (Corollary 10.17(ii)); the values 1 and 0 at multiplicative and additive primes come from Proposition 10.62 and Proposition 10.46. Section 1.4: the Tamagawa number $c_q = \#\Phi_q(\mathbb{F}_q)$, which is $[E(\mathbb{Q}_q) : E_0(\mathbb{Q}_q)]$ (Proposition 10.28).

## 10.3 Potentially good and potentially multiplicative reduction

The reduction type can change under finite extensions of $K$, and the paper's local terms depend on which reduction a curve acquires: the term $t_\ell(E,q)$ of Definition def:terms lives at the potentially multiplicative primes, and the term $e_\ell(E,q)$ at a tame additive prime is read off the potentially good types. This section proves the criterion by $v(j)$ and states the semistable reduction theorem.

**Definition 10.20** (potential reduction)**.** $E/K$ has *potentially good* (resp. *potentially multiplicative*) reduction if there is a finite extension $L/K$ such that $E/L$ has good (resp. multiplicative) reduction.

The definition makes sense because a minimal model over $K$ need not stay minimal over an extension: the change $(u,0,0,0)$ with $u$ in the larger field can remove a factor of the discriminant that no $u \in K$ can remove. For example, let $p$ be odd and $E\colon y^2 = x^3 + px$ over $\mathbb{Q}_p$. Then $c_4 = -48p$ and $\Delta = -16 \cdot 4p^3 = -64p^3$, so $v(\Delta) = 3 < 12$, the model is minimal (Proposition 10.5(i)) and the reduction is additive ($v(c_4) > 0$); PARI gives type III at $p = 3$ and $p = 5$. Over $L = \mathbb{Q}_p(p^{1/4})$ the change $(u,0,0,0)$ with $u = p^{1/4}$, that is $x = p^{1/2}x'$, $y = p^{3/4}y'$, gives $p^{3/2}y'^2 = p^{3/2}x'^3 + p^{3/2}x'$, that is, $y'^2 = x'^3 + x'$, with discriminant $-64$, a unit: $E$ has good reduction over $L$. Accordingly $j = 1728$ is integral, $v(j) \geq 0$, as the criterion below requires.

**Theorem 10.21** (the $j$-invariant criterion)**.** *Let $E/K$ be an elliptic curve.*

*(i) $E$ has potentially good reduction if and only if $v(j) \geq 0$.*

*(ii) $E$ has potentially multiplicative reduction if and only if $v(j) < 0$; in that case $E$ has multiplicative reduction over every finite extension over which it is semistable, and never good reduction.*

*(iii) If $E$ has multiplicative reduction, $v(j) = -v(\Delta_{\min}) < 0$.*

*Sketch of proof.* (iii) For a minimal model $c_4$ is a unit, so $v(j) = 3v(c_4) - v(\Delta) = -v(\Delta) < 0$.

(i), the direction "$\Rightarrow$": if $E/L$ has good reduction, a minimal model over $L$ has $\Delta$ a unit and $c_4$ integral, so $j = c_4^3/\Delta$ is integral in $L$, hence $v(j) \geq 0$.

(i), the direction "$\Leftarrow$", for $p$ odd: completing the square (Chapter 9) gives a model $y^2 = g(x)$ with $g$ a monic cubic with distinct roots $e_1, e_2, e_3$, and over the finite extension $L = K(E[2]) = K(e_1,e_2,e_3)$ it reads $y^2 = (x-e_1)(x-e_2)(x-e_3)$. Over $L' = L(\sqrt{e_2 - e_1})$ the change $x = e_1 + (e_2 - e_1)x'$, $y = (e_2 - e_1)^{3/2}y'$ turns the three factors into $(e_2 - e_1)x'$, $(e_2 - e_1)(x' - 1)$ and $(e_2 - e_1)(x' - \lambda)$ with $\lambda = (e_3 - e_1)/(e_2 - e_1) \neq 0, 1$; dividing by $(e_2 - e_1)^3$ gives the Legendre model $y'^2 = x'(x' - 1)(x' - \lambda)$, for which

$$j = 2^8 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}, \qquad \Delta = 16\lambda^2(\lambda - 1)^2.$$

(Here $v$ denotes the valuation of $L'$ extending that of $K$; the sign of $v(j)$ does not depend on the normalisation, and $v(2) = 0$.) If $v(\lambda) < 0$ the numerator has valuation $6v(\lambda)$ and the denominator $4v(\lambda)$, so $v(j) = 2v(\lambda) < 0$. If $v(\lambda) > 0$ then $\lambda^2 - \lambda + 1$ and $\lambda - 1$ are units, so $v(j) = -2v(\lambda) < 0$, and likewise if $v(\lambda - 1) > 0$. So $v(j) \geq 0$ forces $\lambda$ and $\lambda - 1$ to be units (if $v(\lambda) = 0$ then $v(\lambda - 1) \geq 0$, and the remaining case is $v(\lambda - 1) = 0$), and then the Legendre model is integral with unit discriminant ($p$ odd): good reduction over $L'$. For $p = 2$ one uses a different cubic normal form; see [185, VII.5.5].

(ii) If $v(j) < 0$, $E$ never has good reduction, by (i). By the semistable reduction theorem (Theorem 10.22) it is semistable over some finite extension, hence multiplicative there, and over every larger extension. Conversely multiplicative reduction over $L$ gives $v_L(j) < 0$ by (iii), hence $v(j) < 0$. □

**Theorem 10.22** (semistable reduction)**.** *For every elliptic curve $E/K$ there is a finite extension $L/K$ such that $E/L$ is semistable. One may take $L = K(E[m])$ for any integer $m \geq 3$ prime to $p$.*

*Sketch of proof.* If $v(j) \geq 0$, Theorem 10.21(i) gives good reduction over a finite extension. If $v(j) < 0$, $E$ becomes isomorphic to a Tate curve, which has split multiplicative reduction, over

an extension of degree at most 2 (Theorem 10.59). The statement about $K(E[m])$ is the theorem of Serre and Tate [179, Section 2]; see also [185, VII.5] and Theorem 10.51 below. □

*Remark* 10.23 (the four classes). Combining Definition 10.9, Theorem 10.21 and the classification of Section 10.5, the Kodaira symbols fall into four classes:

- good reduction: $\mathrm{I}_0$;
- multiplicative reduction: $\mathrm{I}_n$, $n \geq 1$, with $n = v(\Delta_{\min}) = -v(j)$;
- additive, potentially multiplicative reduction: $\mathrm{I}_n^*$ with $n \geq 1$ (Theorem 10.66);
- additive, potentially good reduction: II, III, IV, $\mathrm{I}_0^*$, IV$^*$, III$^*$, II$^*$, and, at $p = 2$ only, also some symbols $\mathrm{I}_m^*$ with $m \geq 1$.

Until Section 10.5 the symbols are used only as names for these classes.

*Warning* 10.24. "Potentially multiplicative" does not mean "multiplicative": a curve of type $\mathrm{I}_n^*$ ($n \geq 1$) at an odd prime has additive reduction ($f_p \geq 2$, $a_p = 0$, $\delta_p = 0$), but $v(j) < 0$; at $p = 2$ a symbol $\mathrm{I}_m^*$ may also be potentially good (Remark 10.81). The paper's $n_q = v_q(\Delta_E)$ is defined at multiplicative primes; at an additive potentially multiplicative prime the relevant integer is $\nu = -v_q(j_E)$, which for odd $q$ is the index $n$ of the symbol $\mathrm{I}_n^*$ (Theorem 10.66) but for $q = 2$ is not.

*In the paper* 10.25. Definition def:terms: $t_\ell(E, q) = v_\ell(-v_q(j_E))$ if $v_q(j_E) < 0$ and 0 otherwise, "the $\ell$-part of the valuation of the Tate parameter at a potentially multiplicative prime". Lemma lem:euler: "The inertia group acts on $V$ through a finite cyclic quotient when the reduction is potentially good, and through a quadratic character times a unipotent when it is potentially multiplicative" (Theorem 10.51, Theorem 10.59). Remark rem:wild: a symbol $\mathrm{I}_n^*$ at 2 can be potentially good (Remark 10.81).

## 10.4 Néron models and component groups

The paper's inequalities (eq:main) and (eq:twist) are stated in terms of the geometric component group $\Phi_q^{\mathrm{geom}}$ of the Néron model of $E$ at $q$, not the Tamagawa number $c_q$. This section defines both, explains why they differ, and proves the fact that links component groups to Galois representations: the inertia invariants of $E[\ell^t]$ at an additive prime are the $\ell^t$-torsion of $\Phi_q^{\mathrm{geom}}$ (Proposition 10.33). That proposition is the decisive input of Lemma lem:IV and Remark rem:types (via Chapter 11), and of the treatment of the types IV, IV$^*$ at 2 in Corollary 10.54. We use scheme-theoretic language only as far as the statements need it.

A minimal Weierstrass model gives a scheme over $\mathcal{O}$ whose generic fibre is $E$ and whose special fibre is the cubic $\tilde{E}$. When the reduction is bad this model has two defects: its special fibre is singular, and its points over $\mathcal{O}$ do not see the group structure of $E(K)$ well (only $E_0(K)$ reduces into the group $\tilde{E}_{\mathrm{ns}}(k)$). Néron's idea is to look for the best smooth model: one that is a group scheme and whose $\mathcal{O}$-points are all of $E(K)$. Its special fibre is a smooth commutative algebraic group over $k$, usually disconnected; the identity component is $\tilde{E}_{\mathrm{ns}}$, and the group of components is a finite abelian group with an action of $\mathrm{Gal}(\bar{k}/k)$.

**Definition 10.26** (Néron model)**.** Let $R = \mathcal{O}$. A *Néron model* of $E/K$ is a smooth separated group scheme $\mathcal{E}$ of finite type over $R$ with generic fibre $\mathcal{E}_K = E$, such that for every smooth

$R$-scheme $S$ every $K$-morphism $S_K \to E$ extends uniquely to an $R$-morphism $S \to \mathcal{E}$ (the *Néron mapping property*).

**Theorem 10.27** (Néron)**.** *Every elliptic curve $E/K$ has a Néron model $\mathcal{E}$, unique up to unique isomorphism. It is the smooth locus of the minimal regular proper model of $E$ over $R$; if $E$ has good reduction it is the minimal Weierstrass model.*

We do not prove this; see Néron [141], [15, Sections 1.2, 1.4] and [184, IV.5, IV.6].

**Proposition 10.28** (the component group)**.** *Let $\mathcal{E}$ be the Néron model of $E/K$, $\mathcal{E}_k$ its special fibre and $\mathcal{E}_k^0$ the connected component of the identity.*

*(i) $\mathcal{E}(R) = E(K)$. For a minimal Weierstrass model, $\mathcal{E}_k^0 \simeq \tilde{E}_{\mathrm{ns}}$ and $\mathcal{E}^0(R) = E_0(K)$, where $\mathcal{E}^0$ is the open subgroup scheme with special fibre $\mathcal{E}_k^0$. The quotient $\mathcal{E}_k/\mathcal{E}_k^0$ is a finite étale group scheme over $k$, the* component group scheme *of $E$; we write its groups of points with an argument, $\Phi_E(k)$ and $\Phi_E(\bar{k})$, never as a bare letter. Concretely it is a finite abelian group $\Phi_E(\bar{k})$ with a continuous action of $\mathrm{Gal}(\bar{k}/k)$, and $\Phi_E(k) = \Phi_E(\bar{k})^{\mathrm{Gal}(\bar{k}/k)}$.*

*(ii) If $k$ is finite, $E(K)/E_0(K) \simeq \Phi_E(k)$.*

*Sketch of proof.* (i) is [184, IV.5, IV.9] and [15, Section 7.1]; the equality $\mathcal{E}(R) = E(K)$ is the mapping property applied to $S = \operatorname{Spec} R$. (ii) Since $\mathcal{E}$ is smooth, Hensel's lemma shows that reduction $\mathcal{E}(R) \to \mathcal{E}_k(k)$ is surjective. Since $k$ is finite and $\mathcal{E}_k^0$ is connected, Lang's theorem applies: a connected algebraic group $G$ over a finite field $k$ has a $k$-point on every principal homogeneous space (a variety over $k$ on which $G$ acts simply transitively over $\bar{k}$); in the language of Galois cohomology (Chapter 16), $H^1(k, G) = 0$. A component defined over $k$, that is a point of $\Phi_E(k)$, is such a space for $G = \mathcal{E}_k^0$, so it has a $k$-point, and $\mathcal{E}_k(k) \to \Phi_E(k)$ is surjective. The kernel of the composite $E(K) = \mathcal{E}(R) \to \Phi_E(k)$ is the set of points reducing into the identity component, $\mathcal{E}^0(R) = E_0(K)$. See [184, IV.9]. □

**Definition 10.29** (geometric component group and Tamagawa number)**.** Let $E/\mathbb{Q}$ and let $p$ be a prime. Write $\Phi_p(\mathbb{F}_p)$ and $\Phi_p(\bar{\mathbb{F}}_p)$ for the groups of points of the component group scheme over $\mathbb{F}_p$ of the Néron model of $E$ over $\mathbb{Z}_p$ (Proposition 10.28). The *geometric component group* is $\Phi_p^{\mathrm{geom}} = \Phi_p(\bar{\mathbb{F}}_p)$, and the *Tamagawa number* is

$$c_p = \#\Phi_p(\mathbb{F}_p) = [E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)].$$

At a good prime $\Phi_p^{\mathrm{geom}} = 0$ and $c_p = 1$. The product $\prod_p c_p$ is the Tamagawa factor in the conjecture of Birch and Swinnerton-Dyer; Cremona's file `allbsd` lists it (Chapter 22).

**Convention 10.30** ($\Phi^{\mathrm{geom}}$ versus $c$)**.** The component group is always written with an argument: $\Phi_p(\mathbb{F}_p)$ or $\Phi_p^{\mathrm{geom}} = \Phi_p(\bar{\mathbb{F}}_p)$ (over a general local field, $\Phi_E(k)$ or $\Phi^{\mathrm{geom}} = \Phi_E(\bar{k})$); a bare $\Phi$ is the geometric Frobenius (Appendix B). Since $\Phi_p(\mathbb{F}_p)$ is the subgroup of $\mathrm{Gal}(\bar{\mathbb{F}}_p/\mathbb{F}_p)$-invariants of $\Phi_p^{\mathrm{geom}}$, $c_p$ divides $\#\Phi_p^{\mathrm{geom}}$, with equality if and only if Frobenius acts trivially on $\Phi_p^{\mathrm{geom}}$. The paper's statements are about $\#\Phi_q^{\mathrm{geom}}$, never about $c_q$: for instance `121c1` and `121c2` at 11 and `980i1` at 2 have $\#\Phi^{\mathrm{geom}} = 3$ but $c = 1$ (Example 10.85). We never write $c_p$ for $\#\Phi_p^{\mathrm{geom}}$, and we never use a bare $c$ for the class of the unit part of the Tate parameter, which is $c(u)$ (Definition 10.60).

**Proposition 10.31** (unramified base change)**.** *The formation of the Néron model commutes with étale base change; in particular with base change from $K$ to a finite unramified extension $K'$, or to the completion of $K^{\mathrm{ur}}$. Consequently the reduction type, the Kodaira symbol, the group $\Phi^{\mathrm{geom}}$ with its structure, $v(\Delta_{\min})$ and the conductor exponent are unchanged under unramified extension of $K$, and a minimal model over $K$ stays minimal over $K'$. The Tamagawa number can change, since only the invariants of $\mathrm{Gal}(\bar{k}/k')$ on $\Phi^{\mathrm{geom}}$ are counted.*

We do not prove this; see [15, Section 7.2] and [184, IV.6]. (For the conductor exponent the invariance is also immediate from Definition 10.45, since $I_{K'} = I_K$.)

*Remark* 10.32 (Tamagawa numbers at multiplicative primes). At a prime of type $\mathrm{I}_n$ one has $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$, and $c = n$ for split, $c \in \{1, 2\}$ (according to the parity of $n$) for nonsplit reduction (Proposition 10.65). So for $\ell$ odd the $\ell$-part of $c$ is $\ell^{v_\ell(n)}$ in the split case and 1 in the nonsplit case, while the $\ell$-part of $\#\Phi^{\mathrm{geom}}$ is $\ell^{v_\ell(n)}$ in both. This is why the paper's inequality (eq:main) uses $\Phi_q^{\mathrm{geom}}$, and why the history paragraph of its Section 1.5 (Ribet and Takahashi) speaks of the $n_q$.

The facts $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ and the values of $c$ at $\mathrm{I}_n$ are proved in Section 10.7 (Proposition 10.65), from the Tate curve. That proof uses only the uniformisation of Theorem 10.59 and Proposition 10.31, not the proposition below, so the forward references to it in this remark and in the proof of Proposition 10.33(ii) are not circular.

**Proposition 10.33** (inertia invariants and component groups)**.** *Let $K/\mathbb{Q}_q$ be finite, with residue field $k$, let $\ell \neq q$ be a prime, $t \geq 1$ an integer, and let $E/K$ have bad reduction, with geometric component group $\Phi^{\mathrm{geom}} = \Phi_E(\bar{k})$.*

*(i) If the reduction is additive, there is a $\mathrm{Gal}(\bar{k}/k)$-equivariant isomorphism*

$$E[\ell^t]^{I_K} = E(K^{\mathrm{ur}})[\ell^t] \;\simeq\; \Phi^{\mathrm{geom}}[\ell^t].$$

*In particular $E(K)[\ell^\infty] \simeq \Phi_E(k)[\ell^\infty]$.*

*(ii) If the reduction is multiplicative, there is an exact sequence of $\mathrm{Gal}(\bar{k}/k)$-modules*

$$0 \longrightarrow \mu_{\ell^t} \otimes \varepsilon \longrightarrow E[\ell^t]^{I_K} \longrightarrow \Phi^{\mathrm{geom}}[\ell^t] \longrightarrow 0,$$

*where $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ ($n = v(\Delta_{\min})$) and $\varepsilon$ is the unramified character of order at most 2 which is trivial if and only if the reduction is split.*

*(iii) Hence, for $\ell$ odd, $\dim_{\mathbb{F}_\ell} E[\ell]^{I_K}$ is $1 + [\ell \mid n]$ for type $\mathrm{I}_n$ ($n \geq 1$), 1 for types IV and IV$^*$ when $\ell = 3$, and 0 for every other additive type.*

*Proof.* An element of $E[\ell^t] \subset E(\bar{K})$ is fixed by $I_K$ if and only if its coordinates lie in the fixed field $\bar{K}^{I_K} = K^{\mathrm{ur}}$; this is the first equality in (i). Let $K'$ run over the finite unramified extensions of $K$, with residue fields $k'$. By Proposition 10.31 a minimal model over $K$ stays minimal over each $K'$, the component group scheme of $E/K'$ is the base change of that of $E/K$ (so $\Phi_E(k')$ is the group of $k'$-points of the same scheme and $\Phi^{\mathrm{geom}}$ is unchanged), and by Proposition 10.15 and Proposition 10.28(ii) there are exact sequences

$$0 \to E_1(K') \to E_0(K') \to \tilde{E}_{\mathrm{ns}}(k') \to 0, \qquad 0 \to E_0(K') \to E(K') \to \Phi_E(k') \to 0,$$

compatible with the inclusions $K' \subset K''$ and with the action of $\mathrm{Gal}(K^{\mathrm{ur}}/K) = \mathrm{Gal}(\bar{k}/k)$. Taking the union over $K'$ (a direct limit of exact sequences is exact) gives

$$0 \to E_1^{\mathrm{ur}} \to E_0^{\mathrm{ur}} \to \tilde{E}_{\mathrm{ns}}(\bar{k}) \to 0, \qquad 0 \to E_0^{\mathrm{ur}} \to E(K^{\mathrm{ur}}) \to \Phi^{\mathrm{geom}} \to 0,$$

with $E_i^{\mathrm{ur}} = \bigcup_{K'} E_i(K')$. Now:

- $E_1^{\mathrm{ur}}$ is uniquely $\ell$-divisible, by Theorem 10.16(iii) applied over each $K'$.
- In the additive case $\tilde{E}_{\mathrm{ns}}(\bar{k}) \simeq \bar{k}^+$ (Proposition 10.10), which is uniquely $\ell$-divisible as $\ell \neq q$. In the multiplicative case $\tilde{E}_{\mathrm{ns}}(\bar{k}) \simeq \bar{k}^\times$, which is $\ell$-divisible ($\bar{k}$ is algebraically closed) with $\ell^t$-torsion $\mu_{\ell^t}(\bar{k}) \simeq \mathbb{Z}/\ell^t$. The Galois action is the natural one if the reduction is split, and is twisted by the nontrivial unramified quadratic character $\varepsilon$ if it is nonsplit, by the formula $\phi(\sigma P) = \sigma(\phi(P))^{-1}$ in the proof of Proposition 10.10(iii); reduction identifies $\mu_{\ell^t}(\bar{k})$ with $\mu_{\ell^t} \subset K^{\mathrm{ur}}$.

Apply the snake lemma (Chapter 3) to multiplication by $\ell^t$ on the first sequence. Since $E_1^{\mathrm{ur}}$ has no $\ell^t$-torsion and $E_1^{\mathrm{ur}}/\ell^t = 0$, we get $E_0^{\mathrm{ur}}[\ell^t] \simeq \tilde{E}_{\mathrm{ns}}(\bar{k})[\ell^t]$ and $E_0^{\mathrm{ur}}/\ell^t E_0^{\mathrm{ur}} \simeq \tilde{E}_{\mathrm{ns}}(\bar{k})/\ell^t = 0$. Then the snake lemma for $\ell^t$ on the second sequence gives

$$0 \to E_0^{\mathrm{ur}}[\ell^t] \to E(K^{\mathrm{ur}})[\ell^t] \to \Phi^{\mathrm{geom}}[\ell^t] \to E_0^{\mathrm{ur}}/\ell^t E_0^{\mathrm{ur}} = 0.$$

In the additive case $E_0^{\mathrm{ur}}[\ell^t] = 0$, which is (i); in the multiplicative case it is $\mu_{\ell^t} \otimes \varepsilon$, which is (ii), with $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ by Proposition 10.65. Every map is $\mathrm{Gal}(\bar{k}/k)$-equivariant. Taking $\mathrm{Gal}(\bar{k}/k)$-invariants in (i) gives $E(K)[\ell^t] = (E[\ell^t]^{I_K})^{G_K/I_K} \simeq \Phi^{\mathrm{geom}}[\ell^t]^{\mathrm{Gal}(\bar{k}/k)} = \Phi_E(k)[\ell^t]$.

(iii) In case (ii), $\#E[\ell]^{I_K} = \ell \cdot \#(\mathbb{Z}/n)[\ell]$, which is $\ell^2$ if $\ell \mid n$ and $\ell$ otherwise. In case (i), $\Phi^{\mathrm{geom}}$ is one of $0$, $\mathbb{Z}/2$, $\mathbb{Z}/3$, $(\mathbb{Z}/2)^2$, $\mathbb{Z}/4$ (Theorem 10.36), and for $\ell$ odd its $\ell$-torsion is nonzero only when $\ell = 3$ and $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/3$, that is, for the types IV and IV$^*$, where it is a line. □

*In the paper* 10.34. The paragraph before Lemma lem:IV in the paper states (i) with the proof above ("the snake lemma for multiplication by $\ell^k$ gives the isomorphism", citing [179, Section 1]). Lemma lem:IV: "$E[3]^{I_q}$ is a line" is (iii) for IV, IV$^*$. Remark rem:types: "$E[3]^{I_q} = 0$" for the other additive types is (iii). Lemma lem:tamfactor, the final check with $M = T$: the paper's module $(\mathbb{Z}/\ell^{v_\ell(n)})\bar{e}_1$ (the torsion of $H^1(I, T)$, generated by the class of $e_1$), with Frobenius acting by $\varepsilon$, is isomorphic as a Galois module to $\Phi^{\mathrm{geom}}[\ell^\infty]$, the quotient in (ii) generated by the image of $e_2$ (Frobenius acts on $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ by $\varepsilon$, Proposition 10.65); so its Frobenius invariants have the order of $\Phi_E(k)[\ell^\infty]$, the $\ell$-part of $c_q(E)$ (Remark 10.32). They are not the invariants of the sub $\mu_{\ell^t} \otimes \varepsilon$, on which Frobenius acts by $q\varepsilon$. Chapter 11 (Chapter 11, its item 2.2) cites this proposition instead of proving it again.

## 10.5 Kodaira symbols and Tate's algorithm

The paper names the reduction of $E$ at a bad prime by its Kodaira symbol, reads $\#\Phi_q^{\mathrm{geom}}$ off the symbol (Section 1.4), and for a tame prime $q \geq 5$ reads the omitted Euler factor off the symbol (the table of Section 1, Lemma lem:euler). It computes the symbols with Tate's algorithm, through PARI (Section 2, Section sec:data, the appendix). This section states the classification of Kodaira and Néron, states Tate's algorithm in full, derives from it the table for $p \geq 5$ and the criterion for $\mathrm{I}_0^*$ that Lemma lem:twist (ii) uses, fixes PARI's codes, and runs the algorithm by hand on the paper's curves.

**Regular models, blowing up, multiplicities.** The classification below is a statement about a surface over $\mathcal{O}$, and we explain the words it uses for a reader without algebraic geometry. A *model* of $E$ over $\mathcal{O}$ is a two-dimensional scheme, proper and flat over $\mathcal{O}$, with generic fibre $E$; the minimal Weierstrass model is one, given by one equation $F(x,y)=0$ in the coordinates $x, y$ over $\mathcal{O}$ (plus the point at infinity). Such a surface is *regular* at a point $P$ of its special fibre if the local ring there has a maximal ideal generated by two elements. For a surface given by one equation, and $P=(x_0,y_0)$ in the special fibre, this holds if and only if $F$ does not lie in $\mathfrak{n}_P^2$, where $\mathfrak{n}_P=(\varpi, x-\tilde{x}_0, y-\tilde{y}_0)$ with lifts $\tilde{x}_0, \tilde{y}_0$; informally, the three partial "derivatives" of $F$ in the directions $\varpi$, $x$, $y$ do not all vanish at $P$. A Weierstrass model is regular at every smooth point of $\tilde{E}$, and may fail to be regular at the singular point. *Blowing up* a point $P$ replaces it by the projective line of directions through it: in coordinates, one substitutes $x-\tilde{x}_0=\varpi x_1$, $y-\tilde{y}_0=\varpi y_1$ (and similarly in the other charts, where $x-\tilde{x}_0$ or $y-\tilde{y}_0$ is the divisor) and removes the factor of $\varpi$ that appears. Repeating this at the non-regular points, and then contracting the components that can be contracted without losing regularity, gives the *minimal regular proper model*, unique up to isomorphism (Lichtenbaum, Shafarevich; see [184, IV.7]). Its special fibre is the divisor of $\varpi$, a union of curves $C_i$ counted with *multiplicities* $m_i \geq 1$: $m_i$ is the order of vanishing of $\varpi$ along $C_i$. The *dual graph* has a vertex for each $C_i$ and an edge for each intersection point. A point of $E(K)$ extends, by properness, to an $\mathcal{O}$-point of the regular model, which meets the special fibre with intersection number 1; so it reduces to a smooth point of a component of multiplicity 1; this is why only those components contribute to the Néron model and to $\Phi^{\mathrm{geom}}$.

**Example 10.35** (blowing up a node: type $\mathrm{I}_2$)**.** Let $p\geq 5$ and $E\colon y^2=x^3+x^2+p^2$ over $\mathbb{Q}_p$. Here $b_2=4$, $b_4=0$, $b_6=4p^2$, $b_8=4p^2$, so $c_4=16$ is a unit and $\Delta=-16p^2(4+27p^2)$ has $v_p(\Delta)=2$: multiplicative reduction, with $\tilde{E}\colon y^2=x^3+x^2$ nodal at $(0,0)$, with tangents $y=\pm x$ (split). The equation $F=y^2-x^2-x^3-p^2$ lies in $(p,x,y)^2$, so the model is not regular at the node. Blow it up: with $x=px_1$, $y=py_1$ one gets $F=p^2(y_1^2-x_1^2-px_1^3-1)$, and the new surface $y_1^2-x_1^2-1-px_1^3=0$ is regular along the special fibre of this chart (the partial derivatives $2y_1$ and $-2x_1$ do not both vanish on $y_1^2-x_1^2=1$; the other charts are checked in the same way). Its special fibre in this chart is the conic $y_1^2-x_1^2=1$, a new component $C_1\simeq\mathbb{P}^1$ along which $\varpi=p$ vanishes to order 1. The strict transform $C_0$ of $\tilde{E}$ is the nodal cubic with its node pulled apart into the two points of $C_1$ at infinity in the directions $y_1=\pm x_1$, which are the tangent directions of the node. So the special fibre is two rational curves of multiplicity 1 meeting in two points: the cycle $\tilde{A}_1$ of type $\mathrm{I}_2$. Removing the two intersection points leaves two copies of $\mathbb{G}_m$, and $\Phi^{\mathrm{geom}}\simeq\mathbb{Z}/2$, as in Table 10.1. In the additive types higher multiplicities appear: in type $\mathrm{I}_0^*$ the central component has multiplicity 2, so it contributes nothing to the Néron model, and $\Phi^{\mathrm{geom}}$ comes from the four outer components.

**Theorem 10.36** (Kodaira, Néron)**.** *Let $E/K$ be an elliptic curve and let $m$ be the number of irreducible components (over $\bar{k}$) of the special fibre of the minimal regular proper model of $E$ over $\mathcal{O}$. The special fibre over $\bar{k}$ is one of the configurations of Table 10.1. The Néron model's special fibre is the union of the multiplicity-one components minus their intersection points with the other components, and $\Phi^{\mathrm{geom}}$ is the group of multiplicity-one components, with the structure given in the table. Over a finite residue field $k$ the Tamagawa number takes only the values in the last column.*

We do not prove Theorem 10.36; see Kodaira [103], Néron [141], Tate [193] and [184, IV.8, IV.9 and Table 4.1]. The orders $\#\Phi^{\mathrm{geom}}$ are the ones quoted in Section 1.4 of the paper ($n$; 1; 2; 3; 4 for $\mathrm{I}_n$; $\mathrm{II}, \mathrm{II}^*$; $\mathrm{III}, \mathrm{III}^*$; $\mathrm{IV}, \mathrm{IV}^*$; $\mathrm{I}_0^*, \mathrm{I}_n^*$). In the language of Lie theory (a physics reader's aside):

| symbol | $m$ | special fibre over $\bar{k}$ | $\Phi^{\mathrm{geom}}$ | $\#\Phi^{\mathrm{geom}}$ | possible $c$ |
|---|---|---|---|---|---|
| $\mathrm{I}_0$ | 1 | smooth curve of genus 1 | 0 | 1 | 1 |
| $\mathrm{I}_1$ | 1 | rational curve with a node | 0 | 1 | 1 |
| $\mathrm{I}_n$, $n \geq 2$ | $n$ | $n$ smooth rational curves of multiplicity 1 forming a cycle ($\tilde{A}_{n-1}$) | $\mathbb{Z}/n$ | $n$ | $n$ (split); 1 or 2 for $n$ odd or even (nonsplit) |
| II | 1 | rational curve with a cusp | 0 | 1 | 1 |
| III | 2 | two rational curves meeting at one point with a common tangent | $\mathbb{Z}/2$ | 2 | 2 |
| IV | 3 | three rational curves through one point, with distinct tangents | $\mathbb{Z}/3$ | 3 | 1 or 3 |
| $\mathrm{I}_0^*$ | 5 | one curve of multiplicity 2 meeting four of multiplicity 1 ($\tilde{D}_4$) | $(\mathbb{Z}/2)^2$ | 4 | 1, 2 or 4 |
| $\mathrm{I}_n^*$, $n \geq 1$ | $n+5$ | a chain of $n+1$ curves of multiplicity 2, with two curves of multiplicity 1 at each end ($\tilde{D}_{n+4}$) | $\mathbb{Z}/4$ ($n$ odd), $(\mathbb{Z}/2)^2$ ($n$ even) | 4 | 2 or 4 |
| $\mathrm{IV}^*$ | 7 | $\tilde{E}_6$, multiplicities $1,2,3,2,1,2,1$ | $\mathbb{Z}/3$ | 3 | 1 or 3 |
| $\mathrm{III}^*$ | 8 | $\tilde{E}_7$, multiplicities $1,2,3,4,3,2,1,2$ | $\mathbb{Z}/2$ | 2 | 2 |
| $\mathrm{II}^*$ | 9 | $\tilde{E}_8$, multiplicities $1,2,3,4,5,6,4,2,3$ | 0 | 1 | 1 |

Table 10.1: The Kodaira-Néron classification. The components are smooth rational curves except in the first, second and fourth rows; $\tilde{A}$, $\tilde{D}$, $\tilde{E}$ are the extended Dynkin diagrams formed by the dual graph (a vertex for each component, an edge for each intersection point).

$\Phi^{\mathrm{geom}}$ is the centre of the simply connected compact Lie group of type $A_{n-1}, D_{n+4}, E_6, E_7, E_8$ for $\mathrm{I}_n, \mathrm{I}_n^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ (III and IV correspond to $A_1$ and $A_2$; $\mathrm{I}_0$, $\mathrm{I}_1$ and II have trivial $\Phi^{\mathrm{geom}}$), which is the dual of the root lattice modulo the root lattice; the same diagrams classify the finite subgroups of $\mathrm{SU}(2)$, and the binary tetrahedral group $\mathrm{SL}_2(\mathbb{F}_3)$ met in Theorem 10.52 is one of them.

On Cremona's tables below conductor 10,000 (211,455 pairs of a curve and a bad prime) every value of $c$ in the last column occurs, and no other: for $\mathrm{IV}, \mathrm{IV}^*$ one finds $c \in \{1,3\}$, for $\mathrm{I}_0^*$ $c \in \{1,2,4\}$, for $\mathrm{I}_n^*$ $c \in \{2,4\}$, and for $\mathrm{I}_n$ the values predicted by Proposition 10.65 in all 153,343 multiplicative pairs.

**Theorem 10.37** (Tate's algorithm)**.** *Let an integral model of $E/K$ be given, with $\varpi$, $k$ as in Convention 10.1, and write $a_{i,j} = a_i/\varpi^j$. The following procedure terminates and returns the Kodaira symbol, the number $m$ of components, the conductor exponent $f = v(\Delta) + 1 - m$ (Theorem 10.47), the Tamagawa number $c$ and a minimal model (the model at the step where it stops).*

***Step 1.*** *If $\varpi \nmid \Delta$: type $\mathrm{I}_0$, $m = 1$, $f = 0$, $c = 1$.*

***Step 2.*** *Change coordinates by $(1, r, 0, t)$, $r, t \in \mathcal{O}$, so that the singular point of $\tilde{E}$ is $(0,0)$; then $\varpi \mid a_3, a_4, a_6$. If $\varpi \nmid b_2$: type $\mathrm{I}_n$ with $n = v(\Delta)$, $m = n$, $f = 1$; the reduction is split if $T^2 + a_1 T - a_2$ has a root in $k$, and then $c = n$; otherwise $c = 1$ ($n$ odd) or 2 ($n$ even).*

***Step 3.*** *If $\varpi^2 \nmid a_6$: type II, $m = 1$, $f = v(\Delta)$, $c = 1$.*

***Step 4.*** *If $\varpi^3 \nmid b_8$: type III, $m = 2$, $f = v(\Delta) - 1$, $c = 2$.*

***Step 5.*** *If $\varpi^3 \nmid b_6$: type* IV, $m = 3$, $f = v(\Delta) - 2$; $c = 3$ *if $Y^2 + a_{3,1}Y - a_{6,2}$ has a root in $k$, else $c = 1$.*

***Step 6.*** *Change coordinates by $(1, 0, s, t)$ so that $\varpi \mid a_1, a_2$, $\varpi^2 \mid a_3, a_4$, $\varpi^3 \mid a_6$. Let $P(T) = T^3 + a_{2,1}T^2 + a_{4,2}T + a_{6,3}$. If $\tilde{P}$ has distinct roots: type* $\mathrm{I}_0^*$, $m = 5$, $f = v(\Delta) - 4$, $c = 1 + \#\{$*roots of $\tilde{P}$ in $k$*$\}$.

***Step 7.*** *If $\tilde{P}$ has one simple and one double root: type $\mathrm{I}_n^*$ for some $n \geq 1$, $m = n + 5$, $f = v(\Delta) - 4 - n$. To find $n$, translate the double root to $0$ (a change $(1, \varpi r_0, 0, 0)$); then $\varpi \parallel a_2$, $\varpi^3 \mid a_4$, $\varpi^4 \mid a_6$. For $s = 1, 2, \ldots$ alternately consider the quadratics*

$$Y^2 + a_{3,(s+3)/2}Y - a_{6,s+3} \ (s \text{ odd}), \qquad a_{2,1}X^2 + a_{4,(s+4)/2}X + a_{6,s+3} \ (s \text{ even});$$

*if the current one has distinct roots in $\bar{k}$, stop with $n = s$, and $c = 4$ if its roots lie in $k$, else $c = 2$; if it has a double root, translate it to $0$ (by $t$ for $s$ odd, by $r$ for $s$ even, with the appropriate power of $\varpi$) and continue with $s + 1$.*

***Step 8.*** *If $\tilde{P}$ has a triple root, translate it to $0$; then $\varpi^2 \mid a_2$, $\varpi^3 \mid a_4$, $\varpi^4 \mid a_6$. If $Y^2 + a_{3,2}Y - a_{6,4}$ has distinct roots: type* $\mathrm{IV}^*$, $m = 7$, $f = v(\Delta) - 6$, $c = 3$ *if the roots are in $k$, else $c = 1$.*

***Step 9.*** *Otherwise translate its double root to $0$, so that $\varpi^3 \mid a_3$, $\varpi^5 \mid a_6$. If $\varpi^4 \nmid a_4$: type* $\mathrm{III}^*$, $m = 8$, $f = v(\Delta) - 7$, $c = 2$.

***Step 10.*** *If $\varpi^6 \nmid a_6$: type* $\mathrm{II}^*$, $m = 9$, $f = v(\Delta) - 8$, $c = 1$.

***Step 11.*** *Otherwise the model is not minimal: replace $a_i$ by $a_i/\varpi^i$ (the change $(\varpi, 0, 0, 0)$) and return to Step 1.*

*The translations can be made explicit. For $K = \mathbb{Q}_p$, Step 2 is achieved by the following $(r, t)$ modulo $p$:*

- *$p \geq 5$: $r = -b_2/12$ if $p \mid c_4$, $r = -(c_6 + b_2c_4)/(12c_4)$ otherwise, and $t = -(a_1r + a_3)/2$;*
- *$p = 3$: $r = -b_6$ if $3 \mid b_2$, $r = -b_2b_4$ otherwise, and $t = a_1r + a_3$;*
- *$p = 2$: $r = a_4$, $t = r(1 + a_2 + a_4) + a_6$ if $2 \mid b_2$, and $r = a_3$, $t = r + a_4$ otherwise.*

*For $p$ odd, Step 6 is completing the square: $s = -a_1/2$, $t = -a_3/2$, after which $a_1 = a_3 = 0$; at $p = 2$ one solves the congruences for $s$ and $t$ directly.*

*Sketch of proof.* Each step either recognises a configuration of the special fibre of the minimal regular model, by blowing up the singular point of the current model and examining the exceptional curves, or performs the change of coordinates that puts the next singularity at the origin. The conditions $\varpi^i \mid a_j$ record which monomials vanish to which order at the singular point. That the procedure stops and returns a minimal model is part of the theorem; the values of $f$ come from Ogg's formula. See Tate [193], [184, IV.9] and [35, Section 3.2]. The formulas for Step 2 are elementary: for $p \geq 5$ the singular point of $\tilde{E}$ has $x$-coordinate the multiple root of $4x^3 + b_2x^2 + 2b_4x + b_6$, which is $-b_2/12$ for a triple root and $-(c_6 + b_2c_4)/(12c_4)$ for a double root, and its $y$-coordinate is the zero of $\partial F/\partial y = 2y + a_1x + a_3$; the formulas at 3 and 2 come from the same computation in characteristic 3 and 2. We checked the three formulas on all 211,455 pairs (curve, bad prime) below conductor 10,000: in every case the translated model has $p \mid a_3, a_4, a_6$. □

**Corollary 10.38** (the tame table)**.** *Let $p \geq 5$ and let a minimal model of $E/K$ be given. Then*

*(i)* $v(\Delta) = 0$ *if and only if the type is* $\mathrm{I}_0$*;*

*(ii)* $v(c_4) = 0 < v(\Delta) = n$ *if and only if the type is* $\mathrm{I}_n$*;*

*(iii) if* $v(c_4) > 0$ *and* $v(j) \geq 0$*, the type is determined by* $v(\Delta) \in \{2, 3, 4, 6, 8, 9, 10\}$*, which gives* $\mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ *respectively;*

*(iv) if* $v(c_4) > 0$ *and* $v(j) < 0$*, then* $v(c_4) = 2$*,* $v(c_6) = 3$*,* $v(\Delta) = n + 6$ *with* $n = -v(j) \geq 1$*, and the type is* $\mathrm{I}_n^*$*.*

*More precisely, the triples* $(v(c_4), v(c_6), v(\Delta))$ *are as in Table 10.2. In particular* $f = 2$ *for every additive type.*

| type | $v(c_4)$ | $v(c_6)$ | $v(\Delta)$ | $m$ | $f$ |
|---|---|---|---|---|---|
| $\mathrm{I}_0$ | $\geq 0$ | $\geq 0$ | 0 | 1 | 0 |
| $\mathrm{I}_n$ | 0 | 0 | $n$ | $n$ | 1 |
| II | $\geq 1$ | 1 | 2 | 1 | 2 |
| III | 1 | $\geq 2$ | 3 | 2 | 2 |
| IV | $\geq 2$ | 2 | 4 | 3 | 2 |
| $\mathrm{I}_0^*$ | $\geq 2$ | $\geq 3$ | 6 | 5 | 2 |
| $\mathrm{I}_n^*$ | 2 | 3 | $n+6$ | $n+5$ | 2 |
| $\mathrm{IV}^*$ | $\geq 3$ | 4 | 8 | 7 | 2 |
| $\mathrm{III}^*$ | 3 | $\geq 5$ | 9 | 8 | 2 |
| $\mathrm{II}^*$ | $\geq 4$ | 5 | 10 | 9 | 2 |

Table 10.2: Valuations of a minimal model at $p \geq 5$. For $\mathrm{I}_0^*$ at least one of $v(c_4) = 2$, $v(c_6) = 3$ holds; $v(c_4) = \infty$ or $v(c_6) = \infty$ (when $c_4 = 0$ or $c_6 = 0$) is allowed wherever "$\geq$" appears.

*Proof.* As $p \geq 5$, the change $(1/6, *, *, *)$ to the short model $y^2 = x^3 + Ax + B$ with $A = -27c_4$, $B = -54c_6$ has $u$ a unit, so the short model is integral and minimal, with $v(A) = v(c_4)$, $v(B) = v(c_6)$ and the same $v(\Delta)$, where $\Delta = -16(4A^3 + 27B^2)$ up to a unit. For it $a_1 = a_2 = a_3 = 0$, $a_4 = A$, $a_6 = B$, $b_2 = 0$, $b_6 = 4B$, $b_8 = -A^2$. Run Theorem 10.37.

If $v(\Delta) = 0$ the type is $\mathrm{I}_0$. If $v(\Delta) > 0$ and $v(A) = 0$ the reduction is multiplicative, of type $\mathrm{I}_n$ with $n = v(\Delta)$; then $v(B) = 0$ too, since $4A^3 \equiv -27B^2$. Otherwise $v(A) \geq 1$, hence $v(B) \geq 1$, the singular point is already at $(0, 0)$ and $b_2 = 0$, so we pass to Step 3.

- Step 3: $\varpi^2 \nmid a_6$ means $v(B) = 1$: type II; then $v(27B^2) = 2 < 3 \leq v(4A^3)$, so $v(\Delta) = 2$.
- Step 4: $\varpi^3 \nmid b_8 = -A^2$ means $v(A) = 1$ (with $v(B) \geq 2$): type III, and $v(\Delta) = \min(3v(A), 2v(B)) = 3$, as $2v(B) \geq 4$. (Throughout, $v(\Delta) = v(4A^3 + 27B^2)$, which equals $\min(3v(A), 2v(B))$ whenever $3v(A) \neq 2v(B)$.)
- Step 5: $\varpi^3 \nmid b_6 = 4B$ means $v(B) = 2$ (with $v(A) \geq 2$): type IV, $v(\Delta) = 4$, as $3v(A) \geq 6 > 4$.
- Step 6: now $v(A) \geq 2$, $v(B) \geq 3$, and the conditions of Step 6 hold without change. $P(T) = T^3 + aT + b$ with $a = A/\varpi^2$, $b = B/\varpi^3$, whose discriminant is $-(4a^3 + 27b^2) = \Delta/(16\varpi^6)$ up to a unit. So $\tilde{P}$ has distinct roots if and only if $v(\Delta) = 6$: type $\mathrm{I}_0^*$.

- $\tilde{P}$ has a triple root $\rho$ if and only if $3\rho = 0$ (no $T^2$ term), so $\rho = 0$, that is, $v(A) \geq 3$ and $v(B) \geq 4$. If $v(\Delta) > 6$ and the root is not triple, then $v(A) = 2$ or $v(B) = 3$; but $v(A) = 2$, $v(B) \geq 4$ or $v(A) \geq 3$, $v(B) = 3$ would give $v(\Delta) = 6$. So Step 7 occurs exactly when $v(A) = 2$, $v(B) = 3$, $v(\Delta) > 6$; then $v(j) = 3v(c_4) - v(\Delta) = 6 - v(\Delta) < 0$, and by Theorem 10.66 the type is $\mathrm{I}_n^*$ with $n = -v(j) = v(\Delta) - 6$.

- Step 8: $v(A) \geq 3$, $v(B) \geq 4$, root already at 0; $Y^2 - a_{6,4}$ has distinct roots if and only if $a_{6,4} = B/\varpi^4$ is a unit ($p$ is odd), that is $v(B) = 4$: type $\mathrm{IV}^*$, and $v(\Delta) = \min(3v(A), 2v(B)) = 8$ since $3v(A) \geq 9$.

- Step 9: $v(B) \geq 5$; the double root of $Y^2 - a_{6,4}$ is already 0, so no translation is needed; $\varpi^4 \nmid a_4$ means $v(A) = 3$: type $\mathrm{III}^*$, and $v(\Delta) = 9$ since $3v(A) = 9 < 10 \leq 2v(B)$.

- Step 10: $v(A) \geq 4$; $\varpi^6 \nmid a_6$ means $v(B) = 5$: type $\mathrm{II}^*$, and $v(\Delta) = 10$ since $2v(B) = 10 < 12 \leq 3v(A)$.

- Step 11 would need $v(A) \geq 4$, $v(B) \geq 6$, impossible for a minimal model (Proposition 10.5(ii)).

This gives Table 10.2. In (iii), $v(j) \geq 0$ excludes $\mathrm{I}_n$ and $\mathrm{I}_n^*$ ($v(j) = -n$ there), and the seven remaining additive types have the distinct values $v(\Delta) = 2, 3, 4, 6, 8, 9, 10$. The values of $f$ follow from $f = v(\Delta) + 1 - m$. □

All 9,699 potentially good additive pairs (curve, $p \geq 5$) of Cremona's tables below conductor 10,000 have $v_p(\Delta)$ as in (iii), and all pairs with $p \geq 5$ fit Table 10.2. At $p = 2, 3$ no such table exists: Papadopoulos [147] gives the corresponding tables, which need congruence conditions on the unit parts of $c_4$, $c_6$, $\Delta$ besides the valuations.

**Proposition 10.39** (the cubic criterion for $\mathrm{I}_0^*$)**.** *Let $p$ be odd and let $E/K$ have a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with $\varpi \mid a_2$, $\varpi^2 \mid a_4$, $\varpi^3 \mid a_6$. Then $E$ has type $\mathrm{I}_0^*$ if and only if*

$$P(T) = T^3 + (a_2/\varpi)T^2 + (a_4/\varpi^2)T + a_6/\varpi^3$$

*has distinct roots modulo $\varpi$, and then $c = 1 + \#\{\text{roots of } \tilde{P} \text{ in } k\}$. Conversely every curve of type $\mathrm{I}_0^*$ at an odd $p$ has such a model. In particular, if $y^2 = x^3 + a_2x^2 + a_4x + a_6$ is a model over $\mathbb{Z}$ with good reduction at an odd prime $q$, its twist by $q^*$,*

$$y^2 = x^3 + q^*a_2x^2 + (q^*)^2a_4x + (q^*)^3a_6,$$

*has cubic $T^3 + \epsilon a_2T^2 + a_4T + \epsilon a_6$ with $\epsilon = q^*/q = \pm 1$, whose roots are $\epsilon$ times those of $x^3 + a_2x^2 + a_4x + a_6$; so it has type $\mathrm{I}_0^*$ at $q$.*

*Proof.* For the given model $a_1 = a_3 = 0$, so $b_2 = 4a_2$, $b_6 = 4a_6$, $b_8 = 4a_2a_6 - a_4^2$, and $(0, 0)$ is the singular point of $\tilde{E}$ (the reduced equation is $y^2 = x^3$). Run Theorem 10.37: Step 2 passes as $\varpi \mid b_2$; Step 3 passes as $v(a_6) \geq 3$; Step 4 passes as $v(b_8) \geq \min(v(a_2) + v(a_6), 2v(a_4)) \geq 4$; Step 5 passes as $v(b_6) \geq 3$; and the conditions of Step 6 already hold. So the type is $\mathrm{I}_0^*$ exactly when $\tilde{P}$ has distinct roots, with the stated $c$.

Conversely let $E$ have type $\mathrm{I}_0^*$ at odd $p$. Completing the square (a change $(1, 0, s, t)$ with $s = -a_1/2$, $t = -a_3/2$, integral since $p$ is odd) gives a minimal model $y^2 = f(x) = x^3 + a_2x^2 + a_4x + a_6$. The singular point of $\tilde{E}$ is $(x_0, 0)$ with $x_0$ a multiple root of $\tilde{f}$; translating $x$ by a lift of $x_0$ keeps the shape of the equation and gives $\varpi \mid a_4, a_6$. As the reduction is additive, $\tilde{f}$ has a triple root at

0, so $\varpi \mid a_2$. Since Steps 3, 4 and 5 are passed, $\varpi^2 \mid a_6$, $\varpi^3 \mid b_8 = 4a_2a_6 - a_4^2$ and $\varpi^3 \mid b_6 = 4a_6$. Then $v(4a_2a_6) \geq 3$ gives $v(a_4^2) \geq 3$, so $\varpi^2 \mid a_4$, and $\varpi^3 \mid a_6$: the model has the stated shape.

For the twist: $P(T) = T^3 + (q^*/q)a_2T^2 + (q^*/q)^2a_4T + (q^*/q)^3a_6 = T^3 + \epsilon a_2T^2 + a_4T + \epsilon a_6$, as $\epsilon^2 = 1$. If $\theta$ is a root of $g(x) = x^3 + a_2x^2 + a_4x + a_6$ then $(\epsilon\theta)^3 + \epsilon a_2(\epsilon\theta)^2 + a_4\epsilon\theta + \epsilon a_6 = \epsilon g(\theta) = 0$. The roots of $\tilde{g}$ are distinct because the model has good reduction at $q$ ($\Delta = 16\,\mathrm{disc}(g)$ and $q$ is odd), so those of $\tilde{P}$ are too. □

*Warning* 10.40. The sign $\epsilon = q^*/q$ matters when $q \equiv 3 \pmod 4$: the cubic of the twist is $T^3 + \epsilon a_2T^2 + a_4T + \epsilon a_6$, not the cubic of $E$. Its roots are $\epsilon$ times the old ones, so the conclusion "distinct roots" is unaffected, and so is the number of roots in $k$, which gives $c$. The twist must be by $q^*$, not by $q$; see Warning 10.71.

**Convention 10.41** (PARI's Kodaira codes)**.** For a curve `E` and a prime `p`, PARI's function `elllocalred(E,p)` returns the vector $[f_p, k_p, [u, r, s, t], c_p]$: the conductor exponent, the Kodaira code, the change of coordinates to a minimal model at $p$, and the Tamagawa number. The function `ellglobalred(E)` returns $[N, [u, r, s, t], \prod_p c_p, F, L]$ with $F$ the factorisation matrix of $N$ and $L$ the list of local data at the bad primes, in the order of $F$; in PARI 2.17.2 the entries of $L$ are $[f_p, k_p, 0, c_p]$. So the paper's `R[4]` is the factorised conductor and `R[4][,1]` its column of primes; with `K = R[5]`, the entry `K[i][2]` is the Kodaira code at the $i$-th bad prime. The codes are given in Table 10.3.

| code $k_p$ | symbol | $m$ | $\#\Phi^{\mathrm{geom}}$ | $f$ at $p \geq 5$ | $v(\Delta_{\min})$ at $p \geq 5$ |
|---|---|---|---|---|---|
| 1 | $\mathrm{I}_0$ | 1 | 1 | 0 | 0 |
| $n+4$ $(n \geq 1)$ | $\mathrm{I}_n$ | $n$ | $n$ | 1 | $n$ |
| 2 | II | 1 | 1 | 2 | 2 |
| 3 | III | 2 | 2 | 2 | 3 |
| 4 | IV | 3 | 3 | 2 | 4 |
| $-1$ | $\mathrm{I}_0^*$ | 5 | 4 | 2 | 6 |
| $-(n+4)$ $(n \geq 1)$ | $\mathrm{I}_n^*$ | $n+5$ | 4 | 2 | $n+6$ |
| $-4$ | $\mathrm{IV}^*$ | 7 | 3 | 2 | 8 |
| $-3$ | $\mathrm{III}^*$ | 8 | 2 | 2 | 9 |
| $-2$ | $\mathrm{II}^*$ | 9 | 1 | 2 | 10 |

Table 10.3: PARI's Kodaira codes. The formulas $n+4$ and $-(n+4)$ hold for $n \geq 1$ only: $-(0+4) = -4$ is $\mathrm{IV}^*$, and $\mathrm{I}_0^*$ is $-1$.

So `k >= 5` detects multiplicative reduction, `k <= -5` the symbols $\mathrm{I}_n^*$ with $n \geq 1$, and `abs(k) == 3` the types III, III$^*$. Since `ellglobalred` lists only the bad primes, the code 1 never occurs in its output.

**Example 10.42** (Tate's algorithm on the paper's curves)**.** All models are the global minimal models of Cremona's tables; every number below was recomputed with PARI.

(a) `121a1` $= [1, 1, 1, -30, -76]$ at 11: $v(\Delta) = 2$, $v(c_4) = 1$. Here $b_2 = a_1^2 + 4a_2 = 5$ and $11 \mid c_4$, so the formula $r = -b_2/12$ of the theorem applies: $12r \equiv -5 \pmod{11}$ gives $r \equiv 6$, and $t = -(a_1r + a_3)/2 = -7/2 \equiv 2 \pmod{11}$. Step 2 with $(r, t) = (6, 2)$ gives $[1, 19, 11, 88, -22]$, with $11 \mid a_3, a_4, a_6$ and $11 \mid b_2 = 1 + 4 \cdot 19 = 77$; Step 3: $v(a_6) = 1 < 2$. Type II, $c = 1$, $f = v(\Delta) = 2$.

(b) `121c2` $= [1,1,0,-3632,82757]$ at 11: $v(\Delta) = 8$. Step 2 with $(r,t) = (6,8)$ gives $[1,19,22,-3520,61105]$, with $v(a_6) = 2$, $v(b_8) \geq 3$, $v(b_6) \geq 3$, so Steps 3 to 5 are passed. Step 6 with $(s,t) = (5,110)$ gives $[11,-11,242,-4840,46585]$ and $P(T) = T^3 - T^2 - 40T + 35 \equiv (T-4)^3 \pmod{11}$: a triple root, so Step 8. Type IV$^*$, $c = 1$, $f = 2$.

(c) `539d1` $= [0,1,1,-16,-66]$ at 7: Step 2 with $(r,t) = (2,3)$ gives $[0,7,7,0,-98]$; Step 6 with $(s,t) = (0,21)$ gives $[0,7,49,0,-686]$ and $P(T) = T^3 + T^2 - 2 = (T-1)(T^2+2T+2)$, where $T^2+2T+2$ has discriminant $-4 \equiv 3$, a non-square modulo 7. Distinct roots, one in $\mathbb{F}_7$: type I$_0^*$, $c = 2$, $f = 2$.

(d) `539c1` $= [1,0,1,170,-3237]$ at 7: Step 2 with $(r,t) = (4,1)$, then Step 6 with $(s,t) = (3,21)$ give $[7,0,49,49,-3087]$ and $P(T) = T^3 + T - 9 \equiv (T-1)(T-3)^2 \pmod{7}$. Step 7: translating the double root by $r = 21$ gives $[7,63,196,1372,7203]$; the quadratic $Y^2 + 4Y - 3 \equiv (Y-5)^2$ has a double root, and $t = 245$ gives $[7,63,686,-343,-100842]$; the quadratic $2X^2 + 6X + 1 \equiv 2(X-2)^2$ has a double root, and $r = 98$ gives $[7,357,1372,40817,1411788]$; the quadratic $Y^2 + 4Y + 2 \equiv (Y-1)(Y-2)$ has distinct roots in $\mathbb{F}_7$. So $n = 3$: type I$_3^*$, $c = 4$, and $f = v(\Delta) - 4 - n = 9 - 7 = 2$.

(e) `726a1` $= [1,1,0,-35,-51]$: at 11, Step 2 with $(r,t) = (6,8)$ gives $[1,19,22,77,-121]$, with $v(a_6) = 2$ and $v(b_8) = 2 < 3$: type III, $c = 2$. At 2: $v(\Delta) = 6$, $v(c_4) = 0$, and the tangent cone gives nonsplit reduction: I$_6$, $c = 2$. At 3: I$_3$ nonsplit, $c = 1$.

(f) `20a1` $= [0,1,0,4,4]$ at 2: $v(\Delta) = 8$, $v(c_4) = 4$. The singular point is already at $(0,0)$, and Steps 3 to 5 are passed ($v(a_6) = 2$, $b_8 = 0$, $v(b_6) = 4$). Step 6 with $(s,t) = (1,2)$: for a change $(1,0,s,t)$ of a model with $a_1 = a_3 = 0$ the formulas of Chapter 9 reduce to $a_1' = 2s$, $a_2' = a_2 - s^2$, $a_3' = 2t$, $a_4' = a_4 - 2st$, $a_6' = a_6 - t^2$, which here give $[2,0,4,0,0]$, that is $y^2 + 2xy + 4y = x^3$. Now $2 \mid a_1, a_2$, $4 \mid a_3, a_4$, $8 \mid a_6$, and $P(T) = T^3$, a triple root; Step 8: $Y^2 + a_{3,2}Y - a_{6,4} = Y^2 + Y$ has the distinct roots $0, 1 \in \mathbb{F}_2$. Type IV$^*$, $c = 3$, $f = 8 + 1 - 7 = 2$.

(g) `99a1` $= [1,-1,1,-2,0]$ at 3: Step 2 with $(r,t) = (2,0)$ gives $[1,5,3,6,0]$, $v(b_2) = 1$, $a_6 = 0$, and $v(b_8) = 2 < 3$: type III, $c = 2$, $f = v(\Delta) - 1 = 2$.

(h) `980i1` $= [0,0,0,1568,-72716]$: at 2, type IV$^*$ with $c = 1$; at 5, I$_1$; at 7, the model already satisfies the conditions of Step 6 ($v(a_4) = 2$, $v(a_6) = 3$), with $P(T) = T^3 + 32T - 212 \equiv (T-2)(T+1)^2 \pmod{7}$; the subprocedure of Step 7 passes four double roots and stops at $s = 5$ with the quadratic $Y^2 + 1$, irreducible over $\mathbb{F}_7$: type I$_5^*$, $c = 2$, $v_7(\Delta) = 11 = 5 + 6$.

*In the paper* 10.43. The Example of Section 3 of the paper (types II of `121a1`, I$_3^*$ of `539c1`, I$_5^*$ of `980i1`), the Remark after it (`99a1`, `539d1`, `726a1`), Section sec:anomaly (`726a1`, type III at 11) and the appendix (`rule` on `121a1` and `726a1`) all use the types computed here. In the appendix, `k >= 5` selects the multiplicative primes (where the naive factor is removed), `abs(k) != 3` excludes III, III$^*$ unless `iii = 1`, and `k == 4 || k == 2`, `k == -4 || k == -2`, `k == -3` select IV, II, then IV$^*$, II$^*$, then III$^*$ for the weights $w_3$ at $q = 3$.

## 10.6 The conductor and the semistability defect

This section defines the conductor exponent $f_p$ and the conductor $N$, and studies how inertia acts on the Tate module at a potentially good prime: through a finite group $G_{\mathrm{ss}}$, whose order is

given by Kraus's formula $12/\gcd(v(\Delta),12)$ when $p \geq 5$. The paper uses the conductor exponents in Lemma lem:twist and Theorem thm:qual, the defect in Lemma lem:IV, Remark rem:types and Lemma lem:euler, and the wild primes in Remark rem:wild and Section sec:wild. The section ends with the theorem that types IV and IV$^*$ at 2 are tame with inertia image cyclic of order 3 (Corollary 10.54), which the paper uses in Lemma lem:IV and Proposition prop:forms.

**Convention 10.44** (wild part and defect)**.** The Swan conductor (wild part of the conductor) is written sw, as in Chapter 7; the letter $\delta_p$ is reserved for the paper's $\dim V^{I_p}$. For potentially good reduction the group measuring the defect of semistability is written $G_{\mathrm{ss}}$ (it is called $\Phi_K$ in some sources; we avoid that letter, which denotes component groups here), and $e = \#G_{\mathrm{ss}}$ is the *semistability defect*.

**Definition 10.45** (conductor exponent)**.** Let $E/K$ and $\ell \neq p$ a prime, and let $V = V_\ell E$ as a representation of $G_K$. The *conductor exponent* of $E$ is the Artin conductor of $V$ (Chapter 7):

$$f(E) = a(V) = \dim V - \dim V^{I_K} + \mathrm{sw}(V),$$

where $\mathrm{sw}(V) \geq 0$ is the Swan conductor, which vanishes if and only if the wild inertia group $P_K$ acts trivially on $V$. For $E/\mathbb{Q}$ and a prime $p$ we write $f_p = f_p(E)$ for the exponent of $E/\mathbb{Q}_p$.

The conductor exponent measures how far $V_\ell E$ is from being unramified. The tame part $\dim V - \dim V^{I_K}$ counts the dimensions that inertia moves, which is 0, 1 or 2; the Swan term is a weighted count of how deep in the ramification filtration the action of wild inertia reaches, and is nonzero exactly when the pro-$p$ group $P_K$ acts nontrivially, which forces $p$ to divide the order of the image of inertia. For `11a1` at 11 the action of inertia fixes a line (Proposition 10.46(ii) below), so $f_{11} = 1$; for `20a1` at 2, of type IV$^*$, inertia fixes no vector and acts through a group of order 3, prime to 2, so $f_2 = 2$ (Corollary 10.54).

That $f(E)$ does not depend on $\ell$ is a theorem of Ogg and of Serre and Tate [145, 179]; we do not prove it. The following values are proved here, given the criterion of Néron, Ogg and Shafarevich (Chapter 11: $E$ has good reduction if and only if $I_K$ acts trivially on $T_\ell E$) and two results of this chapter proved below.

**Proposition 10.46** (values of the conductor exponent)**.** *Let $\ell \neq p$.*

*(i) Good reduction: $V^{I_K} = V$ and $f = 0$.*

*(ii) Multiplicative reduction: $V^{I_K}$ is a line, $I_K$ acts tamely through a nontrivial unipotent, and $f = 1$.*

*(iii) Additive reduction: $V^{I_K} = 0$ and $f = 2 + \mathrm{sw}(V) \geq 2$.*

*(iv) If $p \geq 5$, $\mathrm{sw}(V) = 0$, so $f = 2$ for additive reduction. If $p$ is odd and the reduction is additive and potentially multiplicative, also $\mathrm{sw}(V) = 0$ and $f = 2$.*

*Proof.* (i) is Corollary 10.17(ii). (ii) By Proposition 10.62(iii), in a suitable basis $G_K$ acts on $T_\ell E$ by $\left(\begin{smallmatrix}\chi_\ell\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon\end{smallmatrix}\right)$ with $\varepsilon$ unramified. On $I_K$ the cyclotomic character $\chi_\ell$ is trivial ($\mu_{\ell^\infty} \subset K^{\mathrm{ur}}$), $\varepsilon$ is trivial, and $\kappa$ restricts to $\nu$ times the tame character $t_\ell\colon I_K \to \mathbb{Z}_\ell(1)$, with $\nu = v(q_E) \geq 1$ (Chapter 7); as in Appendix B, the tame character always carries its domain or an argument, and $t_\ell(E,q)$ always its two arguments. So $\sigma \in I_K$ acts by $\left(\begin{smallmatrix}1 & \nu t_\ell(\sigma) \\ 0 & 1\end{smallmatrix}\right)$: unipotent, nontrivial, trivial on $P_K$, with invariants the line spanned by $e_1$; so $f = 2 - 1 + 0 = 1$.

(iii) Suppose first that the reduction is potentially good. By Theorem 10.51 the image $\rho(I_K)$ is finite, and it lies in $\mathrm{SL}_2$ since $\det\rho=\chi_\ell$ is trivial on $I_K$. If $V^{I_K}\neq 0$, then in a basis $(v,w)$ with $v$ fixed every element of $\rho(I_K)$ has the form $\left(\begin{smallmatrix}1 & b\\ 0 & d\end{smallmatrix}\right)$ with $d=1$; a unipotent matrix of finite order in characteristic 0 is the identity, so $\rho(I_K)=1$ and the reduction would be good by the criterion of Néron, Ogg and Shafarevich. If the reduction is potentially multiplicative, then by Theorem 10.59 $E$ is the twist of a Tate curve by a ramified quadratic character $\chi$, and by Proposition 10.62 $\sigma\in I_K$ acts by $\chi(\sigma)\left(\begin{smallmatrix}1 & \nu t_\ell(\sigma)\\ 0 & 1\end{smallmatrix}\right)$. An element with $\chi(\sigma)=-1$ has the single eigenvalue $-1$ and fixes no nonzero vector. In both cases $V^{I_K}=0$ and $f=2+\mathrm{sw}(V)$.

(iv) If the reduction is potentially good, $\rho(I_K)\simeq G_{\mathrm{ss}}$ has order dividing 24 (Theorem 10.51(iii)); for $p\geq 5$ this is prime to $p$, so the pro-$p$ group $P_K$ acts trivially and $\mathrm{sw}(V)=0$. If it is potentially multiplicative and $p$ is odd, the quadratic character $\chi$ is trivial on the pro-$p$ group $P_K$, and so is $t_\ell\colon I_K\to\mathbb{Z}_\ell(1)$; so $P_K$ acts trivially again. □

**Theorem 10.47** (Ogg's formula)**.** *Let $E/K$, with $K/\mathbb{Q}_p$ finite, and let $m$ be the number of irreducible components over $\bar{k}$ of the special fibre of its minimal regular model (Table 10.1). Then*

$$f=v(\Delta_{\min})+1-m.$$

We do not prove this; it is due to Ogg [145], whose argument has a gap in some cases at $p=2$, and in general to Saito [169]; see [184, IV.11]. For $p\geq 5$ it can be read off Table 10.2. PARI computes $f$ through this formula, so a comparison with PARI's output is a consistency check and not an independent test.

**Theorem 10.48** (bounds for the conductor exponent)**.** *Let $E/\mathbb{Q}$. Then $f_p\leq 2$ for $p\geq 5$, $f_3\leq 5$ and $f_2\leq 8$.*

The first bound is Proposition 10.46; we do not prove the other two. They are special cases of the bounds of Brumer and Kramer [18] and of Lockhart, Rosen and Silverman [117] for abelian varieties. Below conductor 10,000 every value $f_2\in\{2,\dots,8\}$ and $f_3\in\{2,\dots,5\}$ occurs at an additive prime (4,750, 5,390, 7,067, 2,278, 3,946, 892, 388 pairs at 2; 13,194, 3,014, 914, 356 at 3).

**Definition 10.49** (the conductor)**.** The *conductor* of $E/\mathbb{Q}$ is $N=\prod_p p^{f_p}$.

**Proposition 10.50** (properties of the conductor)**.** *Let $E/\mathbb{Q}$ have conductor $N$.*

*(i) $p\mid N$ if and only if $E$ has bad reduction at $p$; $p\,\|\,N$ if and only if the reduction is multiplicative; $p^2\mid N$ if and only if it is additive; and $p^3\nmid N$ for $p\geq 5$.*

*(ii) Isogenous curves over $\mathbb{Q}$ have the same conductor.*

*Proof.* (i) is Proposition 10.46. (ii) An isogeny $E\to E'$ over $\mathbb{Q}$ induces a $G_{\mathbb{Q}}$-equivariant isomorphism $V_\ell E\simeq V_\ell E'$ (Chapter 9), and $f_p$ depends only on $V_\ell E$ restricted to $G_{\mathbb{Q}_p}$, for any $\ell\neq p$. □

The conductors of the paper's curves, all checked with PARI, are in the column $N$ of Example 10.85.

**Theorem 10.51** (Serre and Tate)**.** *Let $E/K$ have potentially good reduction, $K/\mathbb{Q}_p$ finite.*

*(i) There is a smallest extension $L$ of $K^{\mathrm{ur}}$ over which $E$ has good reduction. It is Galois over $K^{\mathrm{ur}}$, and $L=K^{\mathrm{ur}}(E[m])$ for every integer $m\geq 3$ prime to $p$.*

*(ii)* *Put* $G_{\mathrm{ss}} = \mathrm{Gal}(L/K^{\mathrm{ur}})$ *and* $e = \#G_{\mathrm{ss}}$. *For every prime* $\ell \neq p$ *the kernel of* $I_K \to \mathrm{Aut}(T_\ell E)$ *is* $\mathrm{Gal}(\bar{K}/L)$, *so the image of* $I_K$ *in* $\mathrm{Aut}(T_\ell E)$ *is isomorphic to* $G_{\mathrm{ss}}$, *independently of* $\ell$.

*(iii)* $G_{\mathrm{ss}}$ *embeds in* $\mathrm{SL}_2(\mathbb{Z}/m)$ *for every* $m \geq 3$ *prime to* $p$; *in particular* $e$ *divides* 24. *Every element of* $G_{\mathrm{ss}}$ *has order* 1, 2, 3, 4 *or* 6. *If* $p \geq 5$, $G_{\mathrm{ss}}$ *is cyclic of order* 1, 2, 3, 4 *or* 6.

*(iv)* $G_{\mathrm{ss}}$ *acts faithfully on the reduction* $\tilde{E}_L$ *of* $E$ *over* $L$; *if* $e > 2$ *then* $j(\tilde{E}_L) \in \{0, 1728\}$.

*Sketch of proof.* (ii) given (i): for an extension $L'$ of $K^{\mathrm{ur}}$ inside $\bar{K}$, $E$ has good reduction over $L'$ if and only if $\mathrm{Gal}(\bar{K}/L')$ (the inertia group of $L'$) acts trivially on $T_\ell E$, by the criterion of Néron, Ogg and Shafarevich over $L'$ (Chapter 11). So the fixed field of the kernel of $I_K \to \mathrm{Aut}(T_\ell E)$ is the smallest such $L'$; this proves (i) except for the description by $E[m]$, and (ii). For (iii): the image of $I_K$ in $\mathrm{Aut}(E[m])$ is $\mathrm{Gal}(K^{\mathrm{ur}}(E[m])/K^{\mathrm{ur}}) = G_{\mathrm{ss}}$ by (i), and its determinant is the cyclotomic character modulo $m$, trivial on $I_K$. So $e$ divides $\#\,\mathrm{SL}_2(\mathbb{Z}/m)$ for every $m \geq 3$ prime to $p$. If $p \neq 3$, take $m = 3$: $\#\,\mathrm{SL}_2(\mathbb{F}_3) = 24$. If $p = 3$, take $m = 4$ and $m = 5$: $e$ divides $\#\,\mathrm{SL}_2(\mathbb{Z}/4) = 48$ and $\#\,\mathrm{SL}_2(\mathbb{F}_5) = 120$, hence $\gcd(48, 120) = 24$. The characteristic polynomial of an element of $\rho(I_K)$ has integer coefficients independent of $\ell$ [179, Section 2], and its roots are roots of unity of degree at most 2 over $\mathbb{Q}$, of order $1, 2, 3, 4$ or 6. If $p \geq 5$, $p \nmid 24$, so $L/K^{\mathrm{ur}}$ is tamely ramified and $G_{\mathrm{ss}}$ is a quotient of the procyclic tame inertia group (Chapter 7), hence cyclic, of an order that is the order of one of its elements. The description $L = K^{\mathrm{ur}}(E[m])$ in (i) and statement (iv) are in [179, Section 2]; see also [171, Section 5.6]. We do not prove them. □

**Theorem 10.52** (Kraus)**.** *Let* $K/\mathbb{Q}_p$ *be finite and let* $E/K$ *have potentially good reduction.*

*(i)* *If* $p \geq 5$, $G_{\mathrm{ss}}$ *is cyclic of order* $e = 12/\gcd(v(\Delta_{\min}), 12)$. *By type:* $\mathrm{I}_0$, II, III, IV, $\mathrm{I}_0^*$, $\mathrm{IV}^*$, $\mathrm{III}^*$, $\mathrm{II}^*$ *give* $e = 1, 6, 4, 3, 2, 3, 4, 6$.

*(ii)* *If* $p = 3$, $G_{\mathrm{ss}}$ *is cyclic of order* $1, 2, 3, 4$ *or* 6, *or the semidirect product* $\mathbb{Z}/3 \rtimes \mathbb{Z}/4$ *of order* 12.

*(iii)* *If* $p = 2$, $G_{\mathrm{ss}}$ *is cyclic of order* $1, 2, 3, 4$ *or* 6, *or the quaternion group* $Q_8$, *or* $\mathrm{SL}_2(\mathbb{F}_3)$ *of order* 24. *For* $K = \mathbb{Q}_2$ *and* $K = \mathbb{Q}_3$, *Kraus determines* $G_{\mathrm{ss}}$ *from* $v(c_4)$, $v(c_6)$, $v(\Delta)$ *and congruences on their unit parts.*

*Proof.* (i), given Theorem 10.51. Put $e = 12/\gcd(v(\Delta), 12)$ for a minimal model, so that $ev(\Delta)/12$ is an integer, and let $K' = K^{\mathrm{ur}}(\varpi^{1/e})$, a cyclic extension of $K^{\mathrm{ur}}$ of degree $e$, totally ramified with uniformiser $\varpi' = \varpi^{1/e}$ ($K^{\mathrm{ur}}$ contains $\mu_e$ as $p \nmid e$). Let $v'$ be the valuation of $K'$, so $v' = ev$ on $K$. Apply $(u, 0, 0, 0)$ with $u = \varpi'^{\,ev(\Delta)/12}$. The new invariants satisfy

$$\begin{aligned} v'(\Delta') &= ev(\Delta) - 12 \cdot ev(\Delta)/12 = 0, \\ v'(c_4') &= ev(c_4) - 4ev(\Delta)/12 = e\,v(j)/3 \geq 0, \\ v'(c_6') &= ev(c_6) - 6ev(\Delta)/12 \geq 0, \end{aligned}$$

using $v(j) \geq 0$ (Theorem 10.21) for $c_4'$, and $2v(c_6) \geq \min(3v(c_4), v(\Delta)) = v(\Delta)$ (from $c_6^2 = c_4^3 - 1728\Delta$) for $c_6'$. As $p \geq 5$, the model $y^2 = x^3 - 27c_4'x - 54c_6'$ is integral with unit discriminant, so $E$ has good reduction over $K'$. Conversely, if $E$ has good reduction over an extension $L'$ of $K^{\mathrm{ur}}$ of ramification index $e'$, a minimal model over $L'$ has unit discriminant $u^{-12}\Delta$, so $e'v(\Delta) \equiv 0 \pmod{12}$ and $e \mid e'$. Apply this to the smallest extension $L$ of Theorem 10.51: $e$ divides $[L : K^{\mathrm{ur}}]$. But $L \subseteq K'$, as $E$ has good reduction over $K'$, so $[L : K^{\mathrm{ur}}] \leq e$. Hence

$L = K'$ and $G_{\mathrm{ss}} = \mathrm{Gal}(K'/K^{\mathrm{ur}}) \simeq \mu_e$ is cyclic of order $e$. The list by type is Corollary 10.38(iii): $v(\Delta) = 0, 2, 3, 4, 6, 8, 9, 10$ gives $e = 1, 6, 4, 3, 2, 3, 4, 6$.

(ii) and (iii) we do not prove; see Kraus [107] and [171, Section 5.6]. □

The theorem numbers inside [107] are quoted inconsistently in the secondary literature, so we cite the paper as a whole.

**Theorem 10.53** (tame additive reduction)**.** *Let $K/\mathbb{Q}_p$ be finite and let $E/K$ have additive, potentially good reduction. Then the reduction is tame (equivalently* $\mathrm{sw} = 0$*, equivalently $f = 2$) if and only if*

*(a) $p \geq 5$; or*

*(b) $p = 3$ and the type is* III*,* III$^*$ *or* $\mathrm{I}_0^*$*; or*

*(c) $p = 2$ and the type is* IV *or* IV$^*$*.*

We do not prove this; see Dokchitser and Dokchitser [56, Theorem 3.1], who deduce it from Kramer and Tunnell [105, Proposition 8.20] at $p = 2$ and from Kraus [107] at $p = 3$. The direction "type IV or IV$^*$ at 2 implies tame" is proved independently in Corollary 10.54. The data agree: below conductor 10,000, $f_2 = 2$ occurs exactly for the pairs of type IV or IV$^*$ at 2 and for no other additive type (Table 10.4), and at 3 the types III, III$^*$, $\mathrm{I}_0^*$ always have $f_3 = 2$ while II, II$^*$, IV, IV$^*$ always have $f_3 \in \{3, 4, 5\}$.

**Corollary 10.54** (types IV and IV$^*$ at 2)**.** *Let $E/\mathbb{Q}_2$ have Kodaira type* IV *or* IV$^*$*. Then:*

*(i) $E$ has potentially good reduction;*

*(ii) $G_{\mathrm{ss}}$ is cyclic of order 3, so for every odd prime $\ell$ the image of $I_2$ in $\mathrm{Aut}(T_\ell E)$ is cyclic of order 3, generated by an element $\sigma$ with $\det\sigma = 1$ and eigenvalues the two primitive cube roots of unity; on $E[3]$ the image is a nontrivial unipotent group and $E[3]^{I_2}$ is a line;*

*(iii) the reduction is tame and $f_2 = 2$;*

*(iv) every Frobenius lift* $\mathrm{Frob} \in G_{\mathbb{Q}_2}$ *satisfies* $\rho(\mathrm{Frob})\,\sigma\,\rho(\mathrm{Frob})^{-1} = \sigma^{-1}$*.*

*Proof.* By Proposition 10.33(i) and Theorem 10.36, $E[3]^{I_2} \simeq \Phi^{\mathrm{geom}}[3] \simeq \mathbb{Z}/3$ is a line.

(i) Suppose $v_2(j) < 0$. By Theorem 10.59, $E$ is the twist of the Tate curve $E_{q_E}$ by a quadratic character $\chi$, ramified since the reduction is additive, and by Proposition 10.62 an element $\sigma \in I_2$ acts on $T_3E$ by $\chi(\sigma)\left(\begin{smallmatrix} 1 & \kappa(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$, where $\kappa(\sigma) = \nu\, t_3(\sigma)$ is $\nu$ times the 3-adic tame character $t_3 \colon I_2 \to \mathbb{Z}_3(1)$ (Chapter 7; not the term $t_3(E, 2)$). On $I_2$ the cyclotomic character $\chi_3$ is trivial, so the cocycle relation $\kappa(gh) = \chi_3(g)\kappa(h) + \kappa(g)$ makes $\kappa|_{I_2}$ a homomorphism to $\mathbb{Z}_3$, and $H = \{\sigma \in I_2 : \kappa(\sigma) \equiv 0 \pmod 3\}$ is the kernel of its reduction $I_2 \to \mathbb{Z}/3$, a subgroup of index 1 or 3. The subgroup $\ker\chi|_{I_2}$ has index 2, because $\chi$ is ramified. If $H$ were contained in $\ker\chi$, the index 2 would divide $[I_2 : H] \in \{1, 3\}$; so $H \not\subseteq \ker\chi$, and some $\sigma \in H$ has $\chi(\sigma) = -1$. It acts on $E[3]$ by $-\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = -1$, which fixes no nonzero vector. This contradicts $E[3]^{I_2} \neq 0$. So $v_2(j) \geq 0$.

(ii) By Theorem 10.51 the image $\rho_3(I_2)$ is a finite subgroup of $\mathrm{SL}_2(\mathbb{Z}_3)$. Reduction modulo 3 is injective on it: if $g = 1 + 3^k A$ with $k \geq 1$ and $A \not\equiv 0 \pmod 3$ had finite order, a power of it

would have prime order $r$ and the same shape (with possibly larger $k$), and the binomial theorem (valid as 1 and $A$ commute) gives

$$g^r = 1 + r3^kA + \tbinom{r}{2}3^{2k}A^2 + \cdots \equiv 1 + r3^kA \pmod{3^{k+1}} \quad (r \neq 3),$$
$$g^3 = 1 + 3^{k+1}A + 3^{2k+1}A^2 + 3^{3k}A^3 \equiv 1 + 3^{k+1}A \pmod{3^{k+2}},$$

using $2k \geq k+1$, $2k+1 \geq k+2$ and $3k \geq k+2$ for $k \geq 1$. As $r$ is then a unit and $A \not\equiv 0 \pmod 3$, neither power is 1, a contradiction. The image modulo 3 is nontrivial (otherwise $\rho_3(I_2) = 1$ and the reduction would be good) and fixes the line $E[3]^{I_2}$ pointwise. In a basis $(v, w)$ of $E[3]$ with $v$ spanning that line, its elements are $\left(\begin{smallmatrix} 1 & b \\ 0 & d \end{smallmatrix}\right)$ with determinant $d = 1$: the group $\{\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)\}$ of order 3. So $\rho_3(I_2)$ is cyclic of order 3, hence $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$ and $\rho_\ell(I_2) \simeq \mathbb{Z}/3$ for every $\ell \neq 2$ (Theorem 10.51(ii)). A generator $\sigma$ has determinant 1 and $\sigma^3 = 1$, $\sigma \neq 1$; its eigenvalues $\lambda, \lambda^{-1}$ are cube roots of unity, and not both 1 (a unipotent element of finite order is trivial), so they are $\zeta_3^{\pm 1}$.

(iii) The order 3 is prime to 2, so the wild inertia group, a pro-2 group, acts trivially: $\mathrm{sw} = 0$. With $V^{I_2} = 0$ (Proposition 10.46(iii)), $f_2 = 2$.

(iv) As $\rho(P_2) = 1$, $\rho|_{I_2}$ factors through the tame quotient, and $\sigma = \rho(t)$ for a topological generator $t$ of it. The tame relation $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^2$ (Chapter 7) gives $\rho(\mathrm{Frob})\sigma\rho(\mathrm{Frob})^{-1} = \sigma^2 = \sigma^{-1}$. □

We checked the conclusions on all pairs of type IV or IV* at 2 below conductor 10,000 (the first two rows of Table 10.4): all have $v_2(j) \geq 0$ and $f_2 = 2$, and the four roots of the 3-division polynomial generate over $\mathbb{Q}_2$ fields of ramification indices $1, 3, 3, 3$, as (ii) predicts (Proposition 10.77).

*In the paper* 10.55. Lemma lem:IV of the paper is stated for every $q \neq 3$; its proof is the argument above (component group, then injectivity of reduction on finite subgroups of $\mathrm{GL}_2(\mathbb{Z}_3)$). For $q \geq 5$, Kraus's formula gives the order 3 directly: the paper notes after the lemma: "For $q \geq 5$ the order 3 is also the semistability defect $12/\gcd(v_q(\Delta_E), 12)$ of Kraus". The Galois-representation consequences, including the Serre conductor exponent 1 of $\bar{\rho}_{E,3}$ at $q$, are proved in Chapter 11. With (ii) and (iv), the computation of Lemma lem:euler for II, II*, IV, IV* applies verbatim at $q = 2$: the trace-zero endomorphisms commuting with $\sigma$ form the line spanned by $2\sigma + 1$, Frobenius inverts $\sigma$ and so acts on that line by $-1 = \chi_{-3}(2)$, so $L_2(A_f, s)^{-1} = 1 + 2^{-s}$, $P_2(X) = 1 + 2X$ and $e_\ell(E, 2) = v_\ell(3/2)$, which is 1 for $\ell = 3$ and 0 for $\ell \geq 5$ (Chapter 19). This is why Proposition prop:forms holds with no exception at $q = 2$ (the paper also reports PARI's factor $1 + 2X$ at 2 on the 92,812 working-set curves of these types), why Remark rem:wild can say that a type IV or IV* at 2 "always has $P_2(X) = 1 + 2X$", and why case (a) of Theorem thm:qual holds at $q = 2$.

**Example 10.56** (semistability defects of the paper's curves)**.** At the additive primes of the running examples: `121a1` at 11 (type II, $v(\Delta) = 2$): $e = 6$; `121c1` (IV, $v = 4$) and `121c2` (IV*, $v = 8$) at 11: $e = 3$; `539d1` at 7 ($\mathrm{I}_0^*$): $e = 2$; `726a1` at 11 (III, $v = 3$): $e = 4$. At the wild primes: `99a1` at 3 (type III, tame by Theorem 10.53) has $G_{\mathrm{ss}}$ cyclic of order 4 (Remark 10.79); `20a1`, `140b1` and `980i1` at 2 (type IV*) have $e = 3$ by Corollary 10.54; `80b1`, the twist of `20a1` by $-1$, has type $\mathrm{I}_0^*$ and $f_2 = 4$ at 2, and $G_{\mathrm{ss}}$ cyclic of order 6 (the order-3 group of `20a1` times the image of the wildly ramified character of $\mathbb{Q}_2(i)$; Proposition 10.77). The potentially multiplicative primes (`539c1` and `980i1` at 7, `121d1` at 11) have no defect: the image of inertia is infinite (Proposition 10.62).

## 10.7 The Tate curve

At a prime of multiplicative reduction, and at an additive prime with $v(j) < 0$, the curve is (a quadratic twist of) a Tate curve $\bar{K}^\times/q_E^{\mathbb{Z}}$. Everything the paper says at such a prime comes from this description: the term $t_\ell(E,q) = v_\ell(-v_q(j_E))$ of Definition def:terms is the $\ell$-part of the valuation of the Tate parameter; Lemma lem:tamagawa-exp reads the ramification of $E[\ell^t]$ off an $\ell^t$-th root of $q_E$; Lemma lem:tamfactor uses the unit part $u$ of $q_E$ and the matrices of the Galois action. This section constructs the Tate curve, defines $q_E$ and $u$, describes $E_{\mathfrak{q}}[\ell^t]$, computes the Tamagawa number, determines the type at an additive potentially multiplicative prime, and tabulates the Tate parameters of the paper's curves.

For the reader with a physics background: over $\mathbb{C}$, the exponential $z \mapsto e^{2\pi i z}$ identifies the torus $\mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$ with $\mathbb{C}^\times/\mathfrak{q}^{\mathbb{Z}}$, $\mathfrak{q} = e^{2\pi i\tau}$, the "multiplicative" picture of the same lattice. Tate's observation is that the second description makes sense over a $p$-adic field, where $\mathbb{C}/\Lambda$ does not.

**Theorem 10.57** (Tate)**.** *Let $\mathfrak{q} \in K^\times$ with $v(\mathfrak{q}) > 0$. Put*

$$s_k(\mathfrak{q}) = \sum_{n\geq 1} \frac{n^k \mathfrak{q}^n}{1-\mathfrak{q}^n}, \qquad a_4(\mathfrak{q}) = -5s_3(\mathfrak{q}), \qquad a_6(\mathfrak{q}) = -\frac{5s_3(\mathfrak{q}) + 7s_5(\mathfrak{q})}{12}.$$

*(i) The series $a_4, a_6$ have coefficients in $\mathbb{Z}$ and converge. The equation*

$$E_{\mathfrak{q}}\colon\ y^2 + xy = x^3 + a_4(\mathfrak{q})x + a_6(\mathfrak{q})$$

*defines an elliptic curve over $K$ with*

$$c_4 = 1 + 240 s_3(\mathfrak{q}), \qquad c_6 = -1 + 504 s_5(\mathfrak{q}), \qquad \Delta = \mathfrak{q}\prod_{n\geq 1}(1-\mathfrak{q}^n)^{24},$$

$$j(E_{\mathfrak{q}}) = \frac{1}{\mathfrak{q}} + 744 + 196884\,\mathfrak{q} + 21493760\,\mathfrak{q}^2 + \cdots.$$

*(ii) This model is minimal, and $E_{\mathfrak{q}}$ has split multiplicative reduction of type $\mathrm{I}_n$ with $n = v(\mathfrak{q}) = -v(j(E_{\mathfrak{q}}))$.*

*(iii) With*

$$X(w,\mathfrak{q}) = \sum_{n\in\mathbb{Z}} \frac{\mathfrak{q}^n w}{(1-\mathfrak{q}^n w)^2} - 2s_1(\mathfrak{q}), \qquad Y(w,\mathfrak{q}) = \sum_{n\in\mathbb{Z}} \frac{(\mathfrak{q}^n w)^2}{(1-\mathfrak{q}^n w)^3} + s_1(\mathfrak{q}),$$

*the map $\phi\colon \bar{K}^\times/\mathfrak{q}^{\mathbb{Z}} \to E_{\mathfrak{q}}(\bar{K})$, $w \mapsto (X(w,\mathfrak{q}), Y(w,\mathfrak{q}))$ for $w \notin \mathfrak{q}^{\mathbb{Z}}$ and $w \mapsto O$ for $w \in \mathfrak{q}^{\mathbb{Z}}$, is an isomorphism of groups commuting with the action of $G_K$. In particular $E_{\mathfrak{q}}(L) \simeq L^\times/\mathfrak{q}^{\mathbb{Z}}$ for every algebraic extension $L/K$.*

*Sketch of proof.* (i) $s_k(\mathfrak{q}) = \sum_{n\geq 1}\sigma_k(n)\mathfrak{q}^n$ with $\sigma_k(n) = \sum_{d\mid n} d^k$, so $s_k \in \mathfrak{q}\mathbb{Z}[[\mathfrak{q}]]$. For $a_6$ it suffices that $5d^3 + 7d^5 \equiv 0 \pmod{12}$ for every integer $d$: modulo 3, $d^3(5+7d^2) \equiv d^3 \cdot 12 \equiv 0$ if $3 \nmid d$, and trivially $d^3 \equiv 0$ if $3 \mid d$; modulo 4, if $d$ is odd then $d^2 \equiv 1$ and $5+7 = 12$, and if $d$ is even then $8 \mid d^3$. Power series with integer coefficients converge at $\mathfrak{q}$ with $v(\mathfrak{q}) > 0$. For the model, $a_1 = 1$ and $a_2 = a_3 = 0$ give $b_2 = a_1^2 + 4a_2 = 1$, $b_4 = a_1a_3 + 2a_4 = 2a_4$, $b_6 = a_3^2 + 4a_6 = 4a_6$; then $c_4 = b_2^2 - 24b_4 = 1 - 48a_4 = 1 + 240s_3$, and

$$c_6 = -b_2^3 + 36b_2b_4 - 216b_6 = -1 + 72a_4 - 864a_6 = -1 - 360s_3 + 72(5s_3 + 7s_5) = -1 + 504s_5;$$

$a_4 = -5s_3$ is chosen to make $c_4 = 1 + 240s_3$, and then $a_6 = -(5s_3 + 7s_5)/12$ is chosen so that the $s_3$ terms cancel in $c_6$, leaving $-1 + 504s_5$. So $c_4$ and $c_6$ are $E_4$ and $-E_6$, the normalised Eisenstein series of weights 4 and 6 in the variable $\mathfrak{q}$ (Chapter 12), and the classical identity $E_4^3 - E_6^2 = 1728\,\mathfrak{q}\prod(1-\mathfrak{q}^n)^{24}$ of power series with integer coefficients gives $\Delta$; $j = c_4^3/\Delta$ follows, and its coefficients are those of the $q$-expansion of the modular function $j$ (computed with PARI). As $v(\Delta) = v(\mathfrak{q})$ is finite, $\Delta \neq 0$.

(ii) $c_4 \in 1 + \mathfrak{m}$ is a unit and $v(\Delta) = v(\mathfrak{q})$, so the model is minimal (Proposition 10.5(i)) with multiplicative reduction, of type $\mathrm{I}_n$, $n = v(\Delta)$. The reduced curve is $y^2 + xy = x^3$ with the node at $(0,0)$, and the tangent cone $y^2 + xy = y(y + x)$ has the slopes 0 and $-1$ in $k$: split. Finally $v(j) = 3v(c_4) - v(\Delta) = -n$.

(iii) Over $\mathbb{C}$, with $\mathfrak{q} = e^{2\pi i\tau}$ and $w = e^{2\pi iz}$, one has $X = (2\pi i)^{-2}\wp(z;\mathbb{Z}+\mathbb{Z}\tau) - \frac{1}{12}$ and $Y = (2\pi i)^{-3}\frac{1}{2}\wp'(z) - \frac{1}{2}X$, so the identities $Y^2 + XY = X^3 + a_4X + a_6$ and the addition law hold as identities of formal series with rational coefficients (we also checked the first numerically to 40 digits). Tate's theorem is that they hold $p$-adically and that $\phi$ is bijective; see [184, V.3]. We do not prove it. □

**Proposition 10.58** (inverting $j$)**.** *There is a unique power series*

$$\mathfrak{q}(J) = J + 744J^2 + 750420J^3 + 872769632J^4 + 1102652742882J^5 + \cdots \in J\,\mathbb{Z}[[J]]$$

*such that $j(\mathfrak{q}(J)) = 1/J$ as formal series. For $j_0 \in K$ with $v(j_0) < 0$, the element $\mathfrak{q}_0 = \mathfrak{q}(1/j_0)$ is the unique $\mathfrak{q}_0 \in K^\times$ with $v(\mathfrak{q}_0) > 0$ and $j(E_{\mathfrak{q}_0}) = j_0$, and $v(\mathfrak{q}_0) = -v(j_0)$.*

*Proof.* Write $j(\mathfrak{q}) = \mathfrak{q}^{-1}U(\mathfrak{q})$ with $U \in 1 + \mathfrak{q}\mathbb{Z}[[\mathfrak{q}]]$. Then $J(\mathfrak{q}) := 1/j(\mathfrak{q}) = \mathfrak{q}\,U(\mathfrak{q})^{-1} \in \mathfrak{q} + \mathfrak{q}^2\mathbb{Z}[[\mathfrak{q}]]$. A power series $J(\mathfrak{q}) = \mathfrak{q} + \sum_{n\geq 2} b_n\mathfrak{q}^n$ with $b_n \in \mathbb{Z}$ has a compositional inverse $\mathfrak{q}(J) = J + \sum_{n\geq 2} c_nJ^n$ with $c_n \in \mathbb{Z}$: the coefficient of $J^n$ in $J(\mathfrak{q}(J))$ is $c_n$ plus a polynomial with integer coefficients in $c_2, \ldots, c_{n-1}$ and the $b_i$, so the equation $J(\mathfrak{q}(J)) = J$ determines the $c_n$ recursively in $\mathbb{Z}$. The coefficients printed were computed with PARI's `serreverse`. Since both series have integer coefficients, substituting an element of $\mathfrak{m}$ gives convergent series, and substitution respects composition. Put $\mathfrak{q}_0 = \mathfrak{q}(1/j_0)$; then $v(\mathfrak{q}_0) = v(1/j_0) = -v(j_0) > 0$ (the leading term dominates), and $J(\mathfrak{q}_0) = 1/j_0$, so $j(E_{\mathfrak{q}_0}) = j_0$. If also $v(\mathfrak{q}_1) > 0$ and $j(E_{\mathfrak{q}_1}) = j_0$, then $\mathfrak{q}_1 = \mathfrak{q}(J(\mathfrak{q}_1)) = \mathfrak{q}(1/j_0) = \mathfrak{q}_0$. □

**Theorem 10.59** (curves with $v(j) < 0$)**.** *Let $E/K$ have $v(j) < 0$, let $q_E = \mathfrak{q}(1/j_E)$ be as in Proposition 10.58, and let $\gamma = -c_4/c_6 \in K^\times/(K^\times)^2$, computed from any model of $E$. Then $E$ is the quadratic twist of $E_{q_E}$ by $\gamma$: $E \simeq E_{q_E}$ over $K(\sqrt{\gamma})$, and over $K$ if and only if $\gamma \in (K^\times)^2$. Moreover:*

*(i) $E$ has split multiplicative reduction if and only if $\gamma \in (K^\times)^2$;*

*(ii) $E$ has nonsplit multiplicative reduction if and only if $K(\sqrt{\gamma})/K$ is the unramified quadratic extension;*

*(iii) $E$ has additive reduction if and only if $K(\sqrt{\gamma})/K$ is ramified.*

*In the multiplicative case $n = v(\Delta_{\min}) = -v(j) = v(q_E)$.*

*Sketch of proof.* As $v(j) < 0$, $j \neq 0, 1728$, so $c_4c_6 \neq 0$. For $E_{\mathfrak{q}}$, $\gamma(E_{\mathfrak{q}}) = (1+240s_3)/(1-504s_5) = 1+24(10s_3+21s_5)/(1-504s_5)$ lies in $1+24\mathfrak{m}$, which consists of squares by Hensel's lemma (for $p$ odd, $1+\mathfrak{m}$ consists of squares; for $p=2$, $x^2 = 1+8m$ with $v(8m) > v(4)$ has a root near 1). Two

curves with the same $j \neq 0, 1728$ are twists of each other by $d = c_6(E)c_4(E_{\mathfrak{q}})/(c_6(E_{\mathfrak{q}})c_4(E)) = \gamma(E_{\mathfrak{q}})/\gamma(E)$ (Chapter 9), and $d \equiv \gamma(E)$ modulo squares. (Here is why: $j = 1728c_4^3/(c_4^3 - c_6^2)$ gives $c_4^3/c_6^2 = j/(j-1728)$, so $a = c_4(E)/c_4(E_{\mathfrak{q}})$ and $b = c_6(E)/c_6(E_{\mathfrak{q}})$ satisfy $a^3 = b^2$; for $d = b/a$ one gets $d^2 = a$ and $d^3 = b$, so $E$ and the twist of $E_{\mathfrak{q}}$ by $d$ have the same $c_4$ and $c_6$, hence are isomorphic over $K$.) This proves the first statement. If $\gamma$ is a square, $E \simeq E_{q_E}$ has split multiplicative reduction by Theorem 10.57(ii). If $K(\sqrt{\gamma})/K$ is unramified, $E$ acquires multiplicative reduction over it, hence has multiplicative reduction (Proposition 10.31); for $p$ odd, computing $\gamma$ from a minimal model makes it a unit ($c_4$ and $c_6$ are units), which is not a square, so by Hensel's lemma its reduction is not a square, and Proposition 10.12(ii) shows the reduction is nonsplit. If $E$ has multiplicative reduction and $p$ is odd, $c_4$ and $c_6$ are units for a minimal model, so $K(\sqrt{\gamma})$ is unramified; hence a ramified $\gamma$ gives additive reduction. This proves (i) to (iii) for $p$ odd; for $p = 2$ see [184, V.5]. The last sentence is Theorem 10.21(iii) and Proposition 10.58. □

**Definition 10.60** (Tate parameter and unit part)**.** Let $E/\mathbb{Q}_q$ have $v_q(j_E) < 0$. The *Tate parameter* of $E$ is $q_E = \mathfrak{q}(1/j_E) \in \mathbb{Q}_q^\times$, and

$$\nu = v_q(q_E) = -v_q(j_E) \geq 1;$$

at a multiplicative prime $\nu = n_q = v_q(\Delta_E)$ is the paper's $n$. The *unit part* is $u = q_E q^{-\nu} \in \mathbb{Z}_q^\times$, with reduction $\bar{u} \in \mathbb{F}_q^\times$; for a prime $\ell \neq q$, $c(u)$ denotes the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$ (the paper's $c$ in Lemma lem:tamfactor; the arithmetic of this class is in Chapter 2).

**Proposition 10.61** (invariance of the Tate parameter)**.** *Let $E/\mathbb{Q}_q$ have $v_q(j_E) < 0$.*

*(i) $q_E$, $\nu$, $u$ and $c(u)$ depend only on $j_E$; in particular they are the same for $E$ and for every quadratic twist of $E$, and $t_\ell(E, q) = v_\ell(\nu)$ is invariant under quadratic twist.*

*(ii) If $\varphi\colon E \to E'$ is an isogeny over $\mathbb{Q}_q$ of prime degree $d$, then $\nu(E') \in \{d\,\nu(E),\ \nu(E)/d\}$. Consequently $v_\ell(\nu)$ and $t_\ell(E, q)$ are unchanged by isogenies of degree prime to $\ell$.*

*Proof.* (i) is the definition: $q_E$ is a function of $j_E$, and twisting does not change $j$. (ii) *Sketch.* Over a finite extension $L$ of $\mathbb{Q}_q$ over which $E$ and $E'$ are both split, $E(\bar{L}) = \bar{L}^\times/q_E^{\mathbb{Z}}$, and $\ker\varphi$ is a subgroup $C$ of order $d$. If $C = \mu_d$, then $w \mapsto w^d$ identifies $\bar{L}^\times/\mu_d q_E^{\mathbb{Z}}$ with $\bar{L}^\times/q_E^{d\mathbb{Z}}$, so $E' \simeq E_{q_E^d}$ and $\nu(E') = d\nu(E)$. Otherwise $C$ is generated by the class of $\zeta q_E^{1/d}$ for some $\zeta \in \mu_d$, and $\bar{L}^\times/\langle \zeta q_E^{1/d}\rangle$ is the Tate curve with parameter $\zeta q_E^{1/d}$, so $\nu(E') = \nu(E)/d$. That an isogeny of Tate curves is induced by the corresponding map of multiplicative groups is part of Tate's theory [184, V.3, V.5]. For the consequence: since $d \neq \ell$, $v_\ell(d\nu) = v_\ell(\nu/d) = v_\ell(\nu)$. (A second proof: an isogeny of degree prime to $\ell$ induces an isomorphism $T_\ell E \simeq T_\ell E'$ of Galois modules, and by Lemma lem:tamagawa-exp (proved in Chapter 11) $v_\ell(\nu)$ is the largest $t$ such that $E[\ell^t]$ is unramified after twisting $E$ by the quadratic character of Theorem 10.59, which commutes with isogenies.) □

**Proposition 10.62** (torsion of the Tate curve)**.** *Let $\mathfrak{q} \in K^\times$ with $v(\mathfrak{q}) > 0$, $\ell$ a prime and $t \geq 1$.*

*(i) In $E_{\mathfrak{q}}(\bar{K}) = \bar{K}^\times/\mathfrak{q}^{\mathbb{Z}}$, fix an $\ell^t$-th root $\mathfrak{q}^{1/\ell^t}$ and a primitive $\ell^t$-th root of unity $\zeta$. Then $E_{\mathfrak{q}}[\ell^t] = \{\zeta^a(\mathfrak{q}^{1/\ell^t})^b : a, b \in \mathbb{Z}/\ell^t\}$, and there is an exact sequence of $G_K$-modules*

$$0 \longrightarrow \mu_{\ell^t} \longrightarrow E_{\mathfrak{q}}[\ell^t] \longrightarrow \mathbb{Z}/\ell^t \longrightarrow 0,$$

*the quotient being generated by the class of $\mathfrak{q}^{1/\ell^t}$.*

*(ii) Let $\ell \neq p$, choose compatible roots $\zeta_{\ell^k}$ and $\mathfrak{q}^{1/\ell^k}$, and let $e_1 = (\zeta_{\ell^k})_k$, $e_2 = (\mathfrak{q}^{1/\ell^k})_k$ be the corresponding basis of $T_\ell E_{\mathfrak{q}}$. Then $\sigma \in G_K$ acts by*

$$\begin{pmatrix} \chi_\ell(\sigma) & \kappa(\sigma) \\ 0 & 1 \end{pmatrix},$$

*where $\kappa$ is the Kummer cocycle of $\mathfrak{q}$: $\sigma(\mathfrak{q}^{1/\ell^k}) = \zeta_{\ell^k}^{\kappa(\sigma)} \mathfrak{q}^{1/\ell^k}$ for all $k$ (Chapter 5).*

*(iii) If $E$ is the twist of $E_{\mathfrak{q}}$ by a quadratic character $\varepsilon$ of $G_K$ (Theorem 10.59), then $G_K$ acts on $T_\ell E$ by*

$$\begin{pmatrix} \chi_\ell \varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{pmatrix}.$$

*Proof.* (i) A class $w \in \bar{K}^\times/\mathfrak{q}^{\mathbb{Z}}$ is killed by $\ell^t$ if and only if $w^{\ell^t} = \mathfrak{q}^b$ for some $b \in \mathbb{Z}$, that is, $w = \zeta^a \mathfrak{q}^{b/\ell^t}$ with $\mathfrak{q}^{b/\ell^t} = (\mathfrak{q}^{1/\ell^t})^b$. Changing $b$ by $\ell^t$ multiplies $w$ by $\mathfrak{q}$, so $b$ matters only modulo $\ell^t$, and $w$ is trivial if and only if $a \equiv b \equiv 0$. The map $w \mapsto b$ is a homomorphism onto $\mathbb{Z}/\ell^t$ with kernel $\mu_{\ell^t}$, and it commutes with $G_K$ because $\sigma(\mathfrak{q}^{1/\ell^t})$ is again an $\ell^t$-th root of $\mathfrak{q}$. By Theorem 10.57(iii) these are the $\ell^t$-torsion points of $E_{\mathfrak{q}}$, with their Galois action. (ii) In additive notation $\sigma(e_1) = \chi_\ell(\sigma)e_1$ by definition of $\chi_\ell$, and $\sigma(e_2) = e_2 + \kappa(\sigma)e_1$ by definition of $\kappa$; the columns of the matrix are the images of $e_1$ and $e_2$. (iii) $T_\ell E \simeq T_\ell E_{\mathfrak{q}} \otimes \varepsilon$ as $G_K$-modules (Chapter 9): every matrix of (ii) is multiplied by $\varepsilon(\sigma)$. In particular the upper right entry is $\varepsilon\kappa$, which satisfies the cocycle relation $c(gh) = \chi_\ell\varepsilon(g)c(h) + c(g)\varepsilon(h)$ for these matrices; $\kappa$ alone does not when $\varepsilon \neq 1$. □

*Warning* 10.63. In the nonsplit and the additive potentially multiplicative cases the corner of the matrix is $\varepsilon\kappa$, not $\kappa$. The proof of Lemma lem:tamfactor in the paper says so: "$\varepsilon\kappa$, not $\kappa$, satisfies the cocycle relation for the twisted matrices". In the nonsplit multiplicative case (the setting of Lemma lem:tamfactor) $\varepsilon$ is unramified, so nothing in the conclusion changes: $\varepsilon$ is trivial on inertia and only changes the Frobenius entry by a sign. In the additive potentially multiplicative case $\varepsilon$ is the ramified character of Theorem 10.59(iii), so it is not trivial on inertia; there the paper does not use the matrix of $T$ but applies the lemma to $E_{q_E}$, since $\mathrm{ad}^0 T$ is unchanged by a quadratic twist (Chapter 18).

*In the paper* 10.64. Lemma lem:tamagawa-exp: "The $\ell^t$-torsion is generated by $\mu_{\ell^t}$ and an $\ell^t$-th root of $q_E$" is Proposition 10.62(i), applied over the unramified quadratic extension, where a nonsplit curve becomes a Tate curve (Theorem 10.59) and the inertia group is unchanged. The proof of the lemma (inertia acts trivially on $E[\ell^t]$ if and only if $\ell^t \mid n_q$) is given in Chapter 11. Lemma lem:tamfactor uses the matrices of (iii), the decomposition $q_E = q^\nu u$, and the restriction of $\kappa$ to inertia, which is $\nu$ times the tame character (Chapter 7); the cohomology is done in Chapter 16, Chapter 17 and Chapter 18.

**Proposition 10.65** (Tamagawa numbers at multiplicative primes)**.** *Let $E/K$ have multiplicative reduction of type $\mathrm{I}_n$, $k$ finite.*

*(i) If the reduction is split, $E(K) \simeq K^\times/q_E^{\mathbb{Z}}$, $E_0(K)$ corresponds to $\mathcal{O}^\times q_E^{\mathbb{Z}}/q_E^{\mathbb{Z}}$, and the valuation induces $E(K)/E_0(K) \simeq \mathbb{Z}/n$; so $c = n$.*

*(ii) In both cases $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$. If the reduction is nonsplit, Frobenius acts on $\Phi^{\mathrm{geom}}$ by $-1$, so $c = \#\Phi_E(k)$ is 1 for $n$ odd and 2 for $n$ even.*

*Proof.* (i) By Theorem 10.59, $E \simeq E_{q_E}$, and $E(K) \simeq K^\times/q_E^{\mathbb{Z}}$ by Theorem 10.57(iii). Every class has a representative $w$ with $0 \le v(w) < n$, since multiplying by $q_E^{\pm 1}$ changes $v(w)$ by $\pm n$. Write $X(w,q_E) = \sum_{i\in\mathbb{Z}} X_i - 2s_1(q_E)$ and $Y(w,q_E) = \sum_i Y_i + s_1(q_E)$ termwise. For $i = -m \le -1$ one has $X_i = q_E^m w/(q_E^m - w)^2$ and $Y_i = q_E^m w^2/(q_E^m - w)^3$, and $s_1(q_E) \in \mathfrak{m}$. If $v(w) = 0$ and $\bar w \ne 1$, every term other than $X_0$, $Y_0$ lies in $\mathfrak{m}$, so $\phi(w)$ reduces to $\big(\bar w/(1-\bar w)^2,\ \bar w^2/(1-\bar w)^3\big)$, a point of $y^2 + xy = x^3$ other than the node $(0,0)$. (Directly: $y^2+xy = \bar w^4/(1-\bar w)^6 + \bar w^3/(1-\bar w)^5 = \bar w^3(\bar w+1-\bar w)/(1-\bar w)^6 = x^3$. Its slope is $\lambda = y/x = \bar w/(1-\bar w)$; the tangents at the node are $y = 0$ and $y = -x$, and the isomorphism of Proposition 10.10(ii) with $\alpha = 0$, $\beta = -1$ sends the point to $\lambda/(\lambda+1) = \bar w$. So on units the reduction map is $w \mapsto \bar w \in k^\times$.) If $\bar w = 1$, $X$ has a pole and $\phi(w)$ reduces to $\tilde O$. If $0 < v(w) < n$, then $v(X_0) = v(w)$, $v(Y_0) = 2v(w)$, $v(X_{-1}) = v(Y_{-1}) = n - v(w)$, and all the other terms have even larger valuation; all are positive, so $\phi(w)$ reduces to the node. Hence $E_0(K)$ is the image of $\mathcal{O}^\times q_E^{\mathbb{Z}}$, and $w \mapsto v(w) \bmod n$ gives $E(K)/E_0(K) \simeq \mathbb{Z}/n$.

(ii) Let $K'$ be the unramified quadratic extension, over which $E$ is split (Proposition 10.12(iii) for $p$ odd, Theorem 10.59 in general). By (i) over every finite unramified extension of $K'$ the component group has $\mathbb{Z}/n$ points, with trivial Galois action, so $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ (Proposition 10.31). If $E$ is nonsplit it is the twist of $E_{q_E}$ by the unramified quadratic character, and the nontrivial element $\sigma$ of $\mathrm{Gal}(K'/K)$ acts on $E(K') \simeq K'^\times/q_E^{\mathbb{Z}}$ by $w \mapsto \sigma(w)^{-1}$ (the twisted action, Chapter 9). On $E(K')/E_0(K') \simeq K'^\times/\mathcal{O}'^\times q_E^{\mathbb{Z}} \simeq \mathbb{Z}/n$ this is $a \mapsto -a$. The invariants of $-1$ on $\mathbb{Z}/n$ are $\{a : 2a = 0\}$, of order $\gcd(2,n)$. □

All 153,343 pairs (curve, prime of multiplicative reduction) of Cremona's tables below conductor 10,000 obey this rule.

**Theorem 10.66** (types at additive potentially multiplicative primes)**.** *Let $E/K$ have additive reduction with $v(j) < 0$, and put $\nu = -v(j) \ge 1$.*

*(i) If $p$ is odd, $E$ has type $\mathrm{I}_\nu^*$, $f = 2$ and $v(\Delta_{\min}) = \nu + 6$. Conversely, a curve of type $\mathrm{I}_n^*$ with $n \ge 1$ at an odd $p$ has $v(j) < 0$, hence $n = -v(j)$.*

*(ii) If $p = 2$ and $K = \mathbb{Q}_2$, $E$ has type $\mathrm{I}_m^*$ with $m = \nu + 2f_2 - 4$, where $f_2 = 2a(\chi) \in \{4,6\}$ and $a(\chi) \in \{2,3\}$ is the conductor exponent of the ramified quadratic character $\chi$ of Theorem 10.59; so $m = \nu+4$ or $m = \nu+8$.*

*Sketch of proof.* (i) By Theorem 10.59, $\gamma$ is in a ramified class, so $\gamma = \varpi w$ with $w$ a unit up to squares, and $E \simeq E''^{(\varpi)}$ where $E'' = E_{q_E}^{(w)}$ has multiplicative reduction (its $\gamma$ is the unramified class of $w$) with $v(\Delta''_{\min}) = \nu$. Take a minimal model of $E''$ of the form $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (complete the square; $p$ is odd) and translate the node to $(0,0)$, so that $\varpi \mid a_4, a_6$ and $\varpi \nmid a_2$ (the reduced cubic is $x^2(x + \tilde a_2)$ with a double, not triple, root). The twist by $\varpi$ is $y^2 = x^3 + \varpi a_2x^2 + \varpi^2 a_4 x + \varpi^3 a_6$. As in the proof of Proposition 10.39, Tate's algorithm passes Steps 2 to 5 and reaches Step 6 with $P(T) = T^3 + a_2T^2 + a_4T + a_6 \equiv T^2(T + \tilde a_2)$, which has a double and a simple root: the type is $\mathrm{I}_m^*$ for some $m \ge 1$, and the model is minimal, with $v(\Delta) = \nu + 6$. By Proposition 10.46(iv), $f = 2$, and Ogg's formula (Theorem 10.47) gives $m + 5 = v(\Delta) + 1 - f = \nu + 5$, so $m = \nu$.

For the converse, let $E$ have type $\mathrm{I}_n^*$, $n \ge 1$, at odd $p$. As in the converse part of the proof of Proposition 10.39, there is a minimal model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with $\varpi \mid a_2$, $\varpi^2 \mid a_4$, $\varpi^3 \mid a_6$ whose cubic $\tilde P$ has a double and a simple root. The model $y^2 = x^3 + (a_2/\varpi)x^2 + (a_4/\varpi^2)x + a_6/\varpi^3$ of the twist by $\varpi$ is integral and reduces to $y^2 = \tilde P(x)$, which has a node. So the twist has multiplicative reduction, and $v(j) = v(j(E^{(\varpi)})) < 0$.

(ii) We do not prove this. It is a special case of Lorenzini's description of the minimal regular model of a quadratic twist of a curve with multiplicative reduction [119] (we follow the citation in [154] and have not checked the numbering there); $f_2 = 2a(\chi)$ is the conductor exponent of the twist of the Steinberg representation by $\chi$ (Chapter 7), and $a(\chi) = 2$ for $\mathbb{Q}_2(\sqrt{-1})$, $\mathbb{Q}_2(\sqrt{3})$ and 3 for $\mathbb{Q}_2(\sqrt{\pm 2})$, $\mathbb{Q}_2(\sqrt{\pm 10})$. □

On Cremona's tables below conductor 10,000, all 17,761 additive pairs with $v_p(j) < 0$ have a symbol $\mathrm{I}_m^*$ (3,732 at $p = 2$, 7,805 at $p = 3$, the rest at $p \geq 5$); at odd $p$ always $m = \nu$; at $p = 2$, $m - \nu = 4$ with $f_2 = 4$ in 2,742 cases and $m - \nu = 8$ with $f_2 = 6$ in 990 cases. Examples at 2: `208a1` ($\mathrm{I}_5^*$, $\nu = 1$, $f_2 = 4$), `304b1` ($\mathrm{I}_7^*$, $\nu = 3$, $f_2 = 4$), `832c1` ($\mathrm{I}_9^*$, $\nu = 1$, $f_2 = 6$), `336d1` ($\mathrm{I}_{12}^*$, $\nu = 8$, $f_2 = 4$).

**Example 10.67** (Tate parameters of the paper's curves)**.** The Tate parameters $q_E = q^\nu u$ at the potentially multiplicative primes of the running examples, computed from the series of Proposition 10.58 (and, at odd $q$, also with PARI's `ellinit(E, O(q^k)).tate`, with the same result):

| curve, prime | type | $a_q$ | $\nu$ | $\bar{u} \bmod q$ | remark |
|---|---|---|---|---|---|
| `11a1`, 11 | $\mathrm{I}_5$ | 1 | 5 | $10 \equiv -1$ | $(\mathbb{F}_{11}^\times)^5 = \{\pm 1\}$: $\bar{u}$ is a 5th power |
| `11a3`, 11 | $\mathrm{I}_1$ | 1 | 1 | 8 | |
| `20a1`, 5 | $\mathrm{I}_2$ | $-1$ | 2 | 1 | nonsplit |
| `77c1`, 7 | $\mathrm{I}_3$ | $-1$ | 3 | 5 | same $q_E$ as `539c1` at 7 ($\mathrm{I}_3^*$) |
| `77c1` and `539c1`, 11 | $\mathrm{I}_2$ | 1 | 2 | 3 | |
| `140b1`, 7 | $\mathrm{I}_5$ | $-1$ | 5 | 1 | same $q_E$ as `980i1` at 7 ($\mathrm{I}_5^*$) |
| `140b1`, 5 | $\mathrm{I}_1$ | $-1$ | 1 | 2 | nonsplit |
| `980i1`, 5 | $\mathrm{I}_1$ | 1 | 1 | 2 | split: the twist by $-7$, $\chi_{-7}(5) = -1$ |
| `726a1`, 2 | $\mathrm{I}_6$ | $-1$ | 6 | 1 | |
| `726a1`, 3 | $\mathrm{I}_3$ | $-1$ | 3 | 2 | |
| `121d1`, 11 | $\mathrm{I}_1^*$ | 0 | 1 | 8 | same $q_E$ as `11a3` |
| `121d2`, 11 | $\mathrm{I}_5^*$ | 0 | 5 | 10 | same $q_E$ as `11a1` |

For instance $q_E = 10 \cdot 11^5 + 2 \cdot 11^6 + 6 \cdot 11^7 + \cdots$ for `11a1`. The values of $t_\ell$: $t_5(\texttt{11a1}, 11) = 1$; $t_3(\texttt{539c1}, 7) = t_3(\texttt{77c1}, 7) = v_3(3) = 1$; $t_5(\texttt{980i1}, 7) = 1$; $t_3(\texttt{726a1}, 2) = v_3(6) = 1$. At $q = 3$, `726a1` has $v_3(\nu) = 1$, which the appendix's `rule` counts as $t_3(E, 3)$; Definition def:terms defines $t_\ell(E, q)$ for every bad $q$, including $q = \ell$.

*Warning* 10.68 (PARI's Tate parameter at 2). In PARI/GP 2.17.2, the value `ellinit(E, O(2^k)).tate[3]` does not return the Tate parameter when $v_2(j) = -1$: in all 1,262 such pairs among the curves of conductor below 3,000 it returned $q_E^5$ (for instance $2^5 + 2^6 + \cdots$ for `208a1`, whose Tate parameter is $2 + 2^2 + 2^4 + \cdots$), while for every other pair with $v_p(j) < 0$ in that range ($\nu \geq 2$ at 2, and all odd $p$) it agreed with the series of Proposition 10.58. Before using PARI's value, check that $j(\mathsf{q}) = j_E$, or compute $q_E$ from the series. The paper's computations use $v_q(j)$ directly and are not affected.

## 10.8 Quadratic twists and reduction

The paper twists constantly: by $\chi_{q^*}$ at a prime of type $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$ (Theorems thm:qual (b), (c) and thm:quant, Lemma lem:euler), by a product $\chi_S$ of such characters (Theorems thm:twist and thm:quant), and in the definition of the twist-symmetrised weight $w_\ell(E, q)$ of its Section 1.4.

This section proves the paper's Lemma lem:twist (i) and (ii), which say what twisting does to Kodaira types, component groups and conductor exponents; part (iii) of that lemma (Atkin and Li) is about newforms and is stated in Chapter 13, with a sketch via Galois conductors (only the case $q^2 \mid M$ is proved there). Recall (Chapter 9) that for $d \in \mathbb{Q}^\times$ the twist $E^{(d)}$ of a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ is $y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6$, with invariants $d^2c_4$, $d^3c_6$, $d^6\Delta$ and the same $j$, and that $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi$, where $\chi$ is the quadratic character of $\mathbb{Q}(\sqrt{d})$; it is written $\chi_D$, indexed by the discriminant $D$ of $\mathbb{Q}(\sqrt{d})$ (for $d = 11$, $D = 44$ and $\chi = \chi_{-4}\chi_{-11}$). For an odd prime $q$, the paper's $E^{(q)}$ is the twist by $q^* = (-1)^{(q-1)/2}q$, that is, by $\chi_{q^*}$ and the field $\mathbb{Q}(\sqrt{q^*})$, which is unramified outside $q$ (Chapter 6).

**Lemma 10.69** (unramified twists; the paper's Lemma lem:twist (i))**.** *Let $E/\mathbb{Q}$, $p$ a prime and $d \in \mathbb{Q}^\times$ such that $\chi = \chi_D$, the quadratic character of $\mathbb{Q}(\sqrt{d})$ ($D$ its discriminant), is unramified at $p$ (for instance $\chi_{q^*}$ with $q \neq p$ an odd prime; at $p = 2$ this uses $q^* \equiv 1 \pmod 4$). Then $E$ and $E^{(d)}$ have at $p$ the same reduction type, the same Kodaira symbol, the same geometric component group $\Phi_p^{\mathrm{geom}}$, the same $v_p(\Delta_{\min})$, the same conductor exponent $f_p$, the same $v_p(j)$ and the same Tate parameter. The Tamagawa number $c_p$ can change.*

*Proof.* $E$ and $E^{(d)}$ become isomorphic over $\mathbb{Q}_p(\sqrt{d})$, which is $\mathbb{Q}_p$ or the unramified quadratic extension of $\mathbb{Q}_p$. By Proposition 10.31, the reduction type, the Kodaira symbol, $\Phi_p^{\mathrm{geom}}$ and $v_p(\Delta_{\min})$ are unchanged by this extension, so they agree for $E$ and $E^{(d)}$. The conductor exponents agree because $V_\ell E^{(d)} \simeq V_\ell E \otimes \chi$ and $\chi$ is trivial on $I_p$, so the two representations of $I_p$ coincide. The $j$-invariant, hence $v_p(j)$ and the Tate parameter, does not change under twisting. An example where $c_p$ changes: `11a1` has split multiplicative reduction of type $\mathrm{I}_5$ at 11, with $c_{11} = 5$; its twist by 13 (and 13 is not a square modulo 11, so $\chi_{13}$ is unramified at 11 with $\chi_{13}(\mathrm{Frob}_{11}) = -1$) is `1859a2`, of type $\mathrm{I}_5$ nonsplit at 11, with $c_{11} = 1$; so is its twist by $-3$, `99d2`. □

**Lemma 10.70** (ramified twists at odd primes; the paper's Lemma lem:twist (ii))**.** *Let $q$ be an odd prime and $E^{(q)} = E^{(q^*)}$.*

*(a) If $E$ has type $\mathrm{I}_n$ at $q$ with $n \geq 1$, then $E^{(q)}$ has type $\mathrm{I}_n^*$ at $q$, with the same $n$, and conversely.*

*(b) If $E$ has type $\mathrm{I}_0$ at $q$, then $E^{(q)}$ has type $\mathrm{I}_0^*$, and conversely.*

*(c) If $q \geq 5$, the twist exchanges $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$, and $v_q(\Delta_{\min})$ changes by $+6$ modulo 12.*

*(d) The conductor exponent $f_q(E^{(q)})$ is 0 if $E^{(q)}$ has good reduction, 1 if it has type $\mathrm{I}_n$ with $n \geq 1$, and 2 if it has additive reduction and is one of the curves in (a), (b), (c).*

*At every prime $p \neq q$ the two curves have the same type, $\Phi_p^{\mathrm{geom}}$ and $f_p$ (Lemma 10.69).*

*Proof.* Throughout, $q$ is odd and we use the twisted model of a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (complete the square first; for $q$ odd this does not change minimality at $q$). Its invariants are $(q^*)^2c_4$, $(q^*)^3c_6$, $(q^*)^6\Delta$: the twist by $q^*$, not by $q$, though only $q$-adic valuations are used below, and those are the same for $q$ and $q^*$.

(a) Here $v_q(j) < 0$. By Theorem 10.59 the reduction type of a curve with $v_q(j) < 0$ is determined by the square class of $\gamma = -c_4/c_6$: unramified for multiplicative, ramified for additive reduction. The twist multiplies $\gamma$ by $(q^*)^2/(q^*)^3 = 1/q^*$, which moves an unramified class to a ramified one and back: for odd $q$, $\mathbb{Q}_q(\sqrt{\gamma})/\mathbb{Q}_q$ is trivial or unramified exactly when $v_q(\gamma)$ is even (then $\gamma$ is a unit times a square), and ramified when it is odd, and $v_q(1/q^*) = -1$ changes the

parity. So $E$ is multiplicative if and only if $E^{(q)}$ is additive. By Theorem 10.66(i) the additive one has type $\mathrm{I}_\nu^*$, and the multiplicative one type $\mathrm{I}_\nu$, with the same $\nu = -v_q(j)$, as $j$ is unchanged. (The identification of the index of $\mathrm{I}_\nu^*$ with $\nu$ uses Ogg's formula, through Theorem 10.66.)

(b) If $E$ has good reduction at $q$, Proposition 10.39 shows that $E^{(q)}$ has type $\mathrm{I}_0^*$ (with the sign $\epsilon = q^*/q$ in the cubic). Conversely, if $E$ has type $\mathrm{I}_0^*$, Proposition 10.39 gives a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with $q \mid a_2$, $q^2 \mid a_4$, $q^3 \mid a_6$ and $\tilde{P}$ separable. The twist by $q^*$ is $y^2 = x^3 + q^*a_2x^2 + (q^*)^2a_4x + (q^*)^3a_6$, and the change $(q, 0, 0, 0)$ turns it into $y^2 = x^3 + \epsilon a_{2,1}x^2 + a_{4,2}x + \epsilon a_{6,3}$ with $\epsilon = q^*/q$ and $a_{i,j} = a_i/q^j$. This model is integral, and its cubic has the roots $\epsilon\theta$ for the roots $\theta$ of $P$, which are distinct modulo $q$; so its discriminant is a unit and $E^{(q)}$ has good reduction.

(c) Let $q \geq 5$ and let $E$ be potentially good and additive, with minimal $v = v_q(\Delta_{\min}) \in \{2, 3, 4, 6, 8, 9, 10\}$ (Corollary 10.38). The twisted model has $v_q(\Delta) = v + 6$ and $v_q(c_4)$ raised by 2. If $v + 6 < 12$ it is minimal (Proposition 10.5(i)). If $v + 6 \geq 12$, then, as $v_q(j) \geq 0$ is unchanged, $3v_q(c_4') \geq v_q(\Delta') \geq 12$ for the twisted model, so $v_q(c_4') \geq 4$ and one step of Proposition 10.5(ii) gives a minimal model with $v_q(\Delta) = v - 6 < 12$. So the new minimal valuation is $v + 6$ or $v - 6$, that is, $v + 6 \bmod 12$: $2 \leftrightarrow 8$, $3 \leftrightarrow 9$, $4 \leftrightarrow 10$ (and $0 \leftrightarrow 6$, which is (b)). By Corollary 10.38(iii) this is $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$.

(d) The values 0 and 1 are Proposition 10.46(i), (ii). In the additive cases of (a), (b), (c) we have $V_\ell E^{(q)} = V_\ell E \otimes \chi_{q^*}$, where $\chi_{q^*}$ is tamely ramified at $q$ (it is trivial on the pro-$q$ group $P_q$, as $q$ is odd), and $V_\ell E$ is tame at $q$: it is unramified (case (b)), multiplicative (case (a), Proposition 10.46(ii)), or additive with $q \geq 5$ (case (c), Proposition 10.46(iv)). So $P_q$ acts trivially on $V_\ell E^{(q)}$, $\mathrm{sw} = 0$, and $f_q(E^{(q)}) = 2$ by Proposition 10.46(iii). □

All 90,970 pairs (curve numbered 1 in its isogeny class, odd bad prime $q$) of Cremona's tables below conductor 10,000 behave as stated: the same $n$ in all 78,351 cases of (a), the shift of $v_q(\Delta_{\min})$ by 6 modulo 12 in all 5,264 potentially good additive cases with $q \geq 5$ other than $\mathrm{I}_0^*$, and the same type, conductor exponent and $v_p(\Delta_{\min})$ at every other bad prime.

*Warning* 10.71. The twist by $q^*$ has invariants $(q^*)^2c_4$, $(q^*)^3c_6$, $(q^*)^6\Delta$, as in the paper's proof of Lemma lem:twist (ii); the invariants $q^2c_4$, $q^3c_6$, $q^6\Delta$ are those of the twist by $q$. For $q \equiv 3 \pmod 4$ the two are different curves, differing by the twist by $-1$, which is ramified at 2: the twist of `121a1` by $-11$ is `121c2`, of conductor 121, while its twist by 11 is `1936k2`, of conductor $2^4 \cdot 11^2$ (type $\mathrm{I}_4^*$ at 2). Likewise the conductor exponents of the twist are the tame ones for the exchanges of (a), (b), (c), but not for general additive reduction at $q = 3$, which can be wild (Remark 10.79, Proposition 10.72).

**Proposition 10.72** (the exchanges at $q = 3$)**.** *Let $E/\mathbb{Q}$ have additive, potentially good reduction at 3 of type* II, III, IV, IV*, III* *or* II*, *and let $E^{(3)} = E^{(-3)}$. Then $f_3(E^{(3)}) = f_3(E)$, and the type of $E^{(3)}$ at 3 is* IV*, III*, II*, II, III, IV *respectively.*

*Proof.* The conductor: $V_\ell E^{(3)} = V_\ell E \otimes \chi_{-3}$ with $\chi_{-3}$ tamely ramified at 3, so $P_3$ acts on both through the same representation and the Swan conductors agree; $E^{(3)}$ is additive: it is not multiplicative, as $v_3(j) \geq 0$, and it does not have good reduction, since otherwise $E = (E^{(3)})^{(3)}$ would have type $\mathrm{I}_0^*$ by Lemma 10.70(b). So $V^{I_3} = 0$ for both and $f_3(E^{(3)}) = f_3(E) =: f$. The discriminants: by the formula of Chapter 9 for the twist of a general model, the twisted model has discriminant $2^{12}(-3)^6\Delta$, so $v_3(\Delta_{\min}(E^{(3)})) \equiv v_3(\Delta_{\min}(E)) + 6 \pmod{12}$. The components: both curves are potentially good ($j$ is unchanged and $v_3(j) \geq 0$), so neither has type $\mathrm{I}_n$ or, by the converse in Theorem 10.66(i), $\mathrm{I}_n^*$ with $n \geq 1$; so both have $1 \leq m \leq 9$, and Ogg's formula (Theorem 10.47) gives $v_3(\Delta_{\min}) = m + f - 1 \in [f, f + 8]$ for both. Two integers in an interval

of length 8 that are congruent modulo 12 after adding 6 to one of them differ by exactly 6; so $m(E^{(3)}) = m(E) \pm 6$. The types with $m \in \{1,2,3\}$ and additive potentially good reduction are II, III, IV, and those with $m \in \{7,8,9\}$ are IV$^*$, III$^*$, II$^*$. Since $m(E) \in \{1,2,3,7,8,9\}$ and $1 \le m(E^{(3)}) = m(E) \pm 6 \le 9$, $m(E^{(3)}) = m(E)+6$ if $m(E) \le 3$ and $m(E)-6$ if $m(E) \ge 7$. This gives the exchanges. □

The paper's proof of Lemma lem:twist (ii) treats $q = 3$ by the same argument, with the formula of Ogg and Saito. All 5,122 such pairs below conductor 10,000 (curves numbered 1) behave as stated. Examples: `54b1` (type II, $f_3 = 3$) and its twist `54a1` (type IV$^*$, $f_3 = 3$); `27a1` (IV$^*$) and `27a3` (II); `36a1` (III, $f_3 = 2$) and `36a3` (III$^*$).

*In the paper* 10.73. Section 1.4 of the paper lists the five exchanges, "all five at every odd $q$; at $q = 3$ the three middle exchanges keep the conductor exponent, which can exceed 2": these are Lemma 10.70 and Proposition 10.72. Lemma lem:euler ("$V = V' \otimes \chi_{q^*}$ with $V'$ Steinberg at $q$" for $\mathrm{I}_n^*$, and unramified for $\mathrm{I}_0^*$) uses (a) and (b). Theorems thm:qual (b), (c) and thm:quant twist a type II, II$^*$, $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$ prime to a type IV$^*$, IV, good or multiplicative one; Theorem thm:quant needs this at every odd $q$ in $S$, which (a) and (b) provide. The proof of Theorem thm:main uses the multiplicative twist at an $\mathrm{I}_n^*$ prime; at $q = 2$, where Lemma 10.70 does not apply, the local statement Theorem 10.59 (a ramified quadratic twist of a Tate curve) is what is needed, as in the proof of Lemma lem:tamfactor.

**Proposition 10.74** (the twist-symmetrised weight)**.** *For a bad prime $q$ and an odd prime $\ell$, let $w_\ell(E,q)$ be the paper's weight: $v_\ell(n)$ if the type at $q$ is $\mathrm{I}_n$, or $\mathrm{I}_n^*$ with $q$ odd; 1 if $\ell = 3$ and the type is* IV *or* IV$^*$*, or* II *or* II$^*$ *with $q$ odd; 0 otherwise. Then*

$$w_\ell(E,q) = \max\bigl(v_\ell(\#\Phi_q^{\mathrm{geom}}(E)),\, v_\ell(\#\Phi_q^{\mathrm{geom}}(E^{(q)}))\bigr) \text{ for odd } q, \qquad w_\ell(E,2) = v_\ell(\#\Phi_2^{\mathrm{geom}}(E)).$$

*Proof.* For odd $q$ use the exchanges of Lemma 10.70 and Proposition 10.72 and the orders of Table 10.1. These exchanges pair up all types at an odd $q$ ($\mathrm{I}_n^*$ with $\mathrm{I}_n$ for $n \ge 0$, and the six remaining additive types among themselves), so the type of $E$ at a bad prime $q$ lies in exactly one of the pairs below, in one order or the other. The maximum is symmetric in $E$ and $E^{(q)}$, and so is the definition of $w_\ell$ on each pair with both members bad (it gives $v_\ell(n)$ to $\mathrm{I}_n$ and $\mathrm{I}_n^*$, and $[\ell = 3]$ to II, IV$^*$, IV, II$^*$); for the pair $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ only the member of type $\mathrm{I}_0^*$ is bad at $q$, with $w_\ell = 0$. $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$: $\max(v_\ell(n), v_\ell(4)) = v_\ell(n)$, as $\ell$ is odd. $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$: $\max(0, v_\ell(4)) = 0$. II $\leftrightarrow$ IV$^*$ and IV $\leftrightarrow$ II$^*$: $\max(v_\ell(1), v_\ell(3)) = [\ell = 3]$. III $\leftrightarrow$ III$^*$: $v_\ell(2) = 0$. At $q = 2$: $\mathrm{I}_n$ gives $v_\ell(n)$, $\mathrm{I}_n^*$ and $\mathrm{I}_0^*$ give $v_\ell(4) = 0$, IV and IV$^*$ give $v_\ell(3) = [\ell = 3]$, the others 0, which is the definition at $q = 2$. □

**Example 10.75** (twist pairs)**.** Curve-level twists used in the paper, each identified by computing the twisted model, its minimal model and its label (all checked with PARI against Cremona's tables):

| $E$ | type at $q$ | $d$ | $E^{(d)}$ | type at $q$ | other bad primes |
|---|---|---|---|---|---|
| `11a1` | $\mathrm{I}_5$ at 11 | $-11$ | `121d2` | $\mathrm{I}_5^*$ ($v = 11$) | none |
| `11a3` | $\mathrm{I}_1$ at 11 | $-11$ | `121d1` | $\mathrm{I}_1^*$ ($v = 7$) | none |
| `121a1` | II ($v = 2$) at 11 | $-11$ | `121c2` | IV$^*$ ($v = 8$) | none |
| `121c1` | IV ($v = 4$) at 11 | $-11$ | `121a2` | II$^*$ ($v = 10$) | none |
| `539c1` | $\mathrm{I}_3^*$ at 7 | $-7$ | `77c1` | $\mathrm{I}_3$ | $\mathrm{I}_2$ at 11 on both |
| `539d1` | $\mathrm{I}_0^*$ at 7 | $-7$ | `11a3` | $\mathrm{I}_0$ | $\mathrm{I}_1$ at 11 on both |
| `980i1` | $\mathrm{I}_5^*$ at 7 | $-7$ | `140b1` | $\mathrm{I}_5$ | IV$^*$ at 2, $\mathrm{I}_1$ at 5 on both |
| `20a1` | IV$^*$ at 2 | $-4$ | `80b1` | $\mathrm{I}_0^*$, $f_2 = 4$ | $\mathrm{I}_2$ at 5 on both |

In the row of `980i1`, the type IV* at 2 is preserved because $-7 \equiv 1 \pmod 8$, so $\chi_{-7}$ is trivial on $G_{\mathbb{Q}_2}$. The last row is a wild twist at 2, to which no rule of Lemma 10.70 applies. The first row also shows that $e_\ell(E,q)$ is not invariant under twists ramified at $q$, as the paper notes after Definition def:terms with this pair: `11a1` has $e_3(E,11)=0$, while `121d2` has $e_3(E,11)=v_3(11^2-1)=1$ (Chapter 19); the Tate parameters at 11 coincide (Example 10.67).

## 10.9 The wild primes 2 and 3

At $p \geq 5$ every additive reduction is tame and the Kodaira symbol determines $v(\Delta_{\min})$, $f$ and the image of inertia (Corollary 10.38, Theorem 10.52). At $p=2$ and $p=3$ none of this holds. The paper meets this in Remark rem:wild ("At $q=2,3$ the Kodaira type does not determine $P_q$"), in Section sec:wild (the minima at 2 and the weights $w_3$ at 3), and in Definition def:terms (the index of a symbol $\mathrm{I}_n^*$ at 2 is not the valuation of the Tate parameter). This section collects what changes and gives the dictionary between the image of inertia at 2 and the factorisation of the 3-division polynomial over $\mathbb{Q}_2$, which is how the image is computed in practice.

*Remark* 10.76 (what changes at 2 and 3). Let $p \in \{2,3\}$.

(i) The type does not determine $v(\Delta_{\min})$ or $f$, although Ogg's formula $f = v(\Delta_{\min}) + 1 - m$ still links them (Theorem 10.47).

(ii) $f$ can exceed 2: up to 8 at 2 and 5 at 3 (Theorem 10.48).

(iii) $G_{\mathrm{ss}}$ can be non-abelian, and of order divisible by $p$ (Theorem 10.52(ii), (iii)).

(iv) The type does not determine $G_{\mathrm{ss}}$, with two exceptions: IV, IV* at 2 force $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$ (Corollary 10.54), and III, III*, $\mathrm{I}_0^*$ at 3 force $G_{\mathrm{ss}} \simeq \mathbb{Z}/4$, $\mathbb{Z}/4$, $\mathbb{Z}/2$ (Remark 10.79).

(v) At 2 a symbol $\mathrm{I}_m^*$ may be potentially good, and when it is potentially multiplicative $m \neq -v_2(j)$ (Theorem 10.66(ii), Remark 10.81).

Hence the Euler factor of the adjoint is not a function of the type at 2 and 3, as the paper says in Remark rem:wild; what determines it is the image of inertia (Chapter 19; the paper's Proposition prop:shapes).

**Proposition 10.77** (inertia at 2 and the 3-division polynomial)**.** *Let $E/\mathbb{Q}_2$ have additive, potentially good reduction. Let $x_1,\dots,x_4$ be the roots of the 3-division polynomial $\psi_3$ (the $x$-coordinates of the four lines of $E[3]$, Chapter 9) and $e_i$ the ramification index of $\mathbb{Q}_2(x_i)/\mathbb{Q}_2$. Then $G_{\mathrm{ss}}$ is one of $\mathbb{Z}/2$, $\mathbb{Z}/3$, $\mathbb{Z}/4$, $\mathbb{Z}/6$, $Q_8$, $\mathrm{SL}_2(\mathbb{F}_3)$, and the multiset $(e_1,\dots,e_4)$ is*

- $(1,1,1,1)$ *if $G_{\mathrm{ss}} \simeq \mathbb{Z}/2$; then $E$ is the twist of a curve with good reduction at 2 by a ramified quadratic character $\chi$, and $f_2 = 2a(\chi) \in \{4,6\}$;*
- $(1,3,3,3)$ *if $G_{\mathrm{ss}}$ is cyclic of order 3 or 6; order 3 occurs exactly for the types* IV, IV*, *with $f_2=2$, and order 6 is the twist of the order-3 case by a ramified quadratic character, with $f_2 \geq 3$;*
- $(2,2,2,2)$ *if $G_{\mathrm{ss}} \simeq \mathbb{Z}/4$;*
- $(4,4,4,4)$ *if $G_{\mathrm{ss}} \simeq Q_8$ or $\mathrm{SL}_2(\mathbb{F}_3)$.*

*Proof.* The list of groups is Theorem 10.52(iii) ($G_{\mathrm{ss}} \neq 1$ as the reduction is additive). By Theorem 10.51, $G_{\mathrm{ss}}$ is the image of $I_2$ in $\mathrm{Aut}(E[3])$, a subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$, and the $x_i$ correspond to the four lines of $E[3]$. For a root $x_i$, $e_i = [\mathbb{Q}_2^{\mathrm{ur}}(x_i) : \mathbb{Q}_2^{\mathrm{ur}}]$ (the residue degree is absorbed by $\mathbb{Q}_2^{\mathrm{ur}}$), which is the size of the orbit of $x_i$, that is of its line, under $I_2$. So $(e_i)$ is the multiset of orbit sizes of $G_{\mathrm{ss}}$ on the four lines, each orbit counted once for each of its lines. The element $-1$ acts trivially on lines, and $\mathrm{SL}_2(\mathbb{F}_3)/\{\pm 1\} \simeq A_4$ acts on the four lines as $A_4$ on four points. The unique element of order 2 of $\mathrm{SL}_2(\mathbb{F}_3)$ is $-1$; elements of order 3 and 6 map to 3-cycles; elements of order 4 map to double transpositions; $Q_8$ maps onto the Klein group, which is transitive; and $\mathrm{SL}_2(\mathbb{F}_3)$ maps onto $A_4$. This gives the four patterns. If $G_{\mathrm{ss}} = \{\pm 1\}$, the character $\psi\colon I_2 \to \{\pm 1\}$ through which inertia acts is invariant under conjugation by $G_{\mathbb{Q}_2}$ (a conjugate of an element of $I_2$ acts by the conjugate matrix, and $-1$ is central), and it extends to a quadratic character $\chi$ of $G_{\mathbb{Q}_2}$. Explicitly: choose a Frobenius lift $\phi \in G_{\mathbb{Q}_2}$. Its image generates $G_{\mathbb{Q}_2}/I_2 \simeq \hat{\mathbb{Z}}$ topologically, and since $\hat{\mathbb{Z}}$ is free as a profinite group on one generator, $a \mapsto \phi^a$ is a section, so $G_{\mathbb{Q}_2} = I_2 \rtimes \overline{\langle\phi\rangle}$ with $\overline{\langle\phi\rangle} \simeq \hat{\mathbb{Z}}$ and every element written uniquely as $i\phi^a$ ($i \in I_2$, $a \in \hat{\mathbb{Z}}$). Put $\chi(i\phi^a) = \psi(i)$. Then $\chi(i\phi^a \cdot j\phi^b) = \chi\big(i(\phi^a j \phi^{-a})\phi^{a+b}\big) = \psi(i)\psi(\phi^a j \phi^{-a}) = \psi(i)\psi(j)$ by the invariance of $\psi$, so $\chi$ is a continuous character extending $\psi$ (this is the construction of the paper's Proposition prop:shapes(ii)); then $\rho \otimes \chi$ is unramified, so $E \otimes \chi$ has good reduction (criterion of Néron, Ogg and Shafarevich), and $f_2(E) = a(V' \otimes \chi) = 2a(\chi)$ for $V'$ unramified (Chapter 7), with $a(\chi) \in \{2, 3\}$. The same argument applied to the character $g \mapsto g^3 \in \{\pm 1\}$ of a cyclic $G_{\mathrm{ss}}$ of order 6 shows that $E$ is the twist of a curve with $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$; order 6 is wild since $2 \mid 6$, so $f_2 \geq 3$. Order 3 is tame, so $f_2 = 2$, and by Theorem 10.53 and Corollary 10.54 this happens exactly for IV, IV$^*$. □

Table 10.4 records what occurs on Cremona's tables below conductor 10,000: all 20,979 pairs (curve, $p = 2$) with additive potentially good reduction at 2, by type, $f_2$ and pattern. The patterns were computed by factoring $\psi_3$ over $\mathbb{Q}$ and reading the ramification indices of the primes above 2 in each factor with PARI.

| type at 2 | $f_2$: pattern (number of pairs) |
|---|---|
| IV | 2: $(1,3,3,3)$ 2,094 |
| IV$^*$ | 2: $(1,3,3,3)$ 2,656 |
| II | 4: $(1,3,3,3)$ 469, $(4,4,4,4)$ 478; 6: $(1,1,1,1)$ 136, $(4,4,4,4)$ 395; 7: $(4,4,4,4)$ 236 |
| II$^*$ | 3: $(4,4,4,4)$ 1,171; 4: $(1,1,1,1)$ 291; 6: $(1,3,3,3)$ 108, $(4,4,4,4)$ 96 |
| III | 3: $(4,4,4,4)$ 1,062; 5: $(4,4,4,4)$ 644; 7: $(4,4,4,4)$ 210; 8: $(2,2,2,2)$ 98, $(4,4,4,4)$ 96 |
| III$^*$ | 3: $(4,4,4,4)$ 1,582; 5: $(4,4,4,4)$ 314; 7: $(4,4,4,4)$ 210; 8: $(2,2,2,2)$ 98, $(4,4,4,4)$ 96 |
| $\mathrm{I}_0^*$ | 4: $(1,3,3,3)$ 596, $(4,4,4,4)$ 720; 5: $(4,4,4,4)$ 792; 6: $(1,3,3,3)$ 196, $(4,4,4,4)$ 200 |
| $\mathrm{I}_m^*$ (potentially good) | 3: $(4,4,4,4)$ 1,575; 4: $(1,1,1,1)$ 466, $(4,4,4,4)$ 1,305; 5: $(4,4,4,4)$ 528; 6: $(1,1,1,1)$ 250, $(1,3,3,3)$ 146, $(4,4,4,4)$ 1,429; 7: $(4,4,4,4)$ 236 |

Table 10.4: Additive potentially good reduction at 2 below conductor 10,000: conductor exponent and ramification pattern of the roots of $\psi_3$ over $\mathbb{Q}_2$. The pattern $(1,3,3,3)$ with $f_2 = 2$ is $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$, with $f_2 \geq 4$ it is $\mathbb{Z}/6$.

In particular $f_2 = 2$ occurs only for IV and IV$^*$, in accordance with Theorem 10.53.

*In the paper* 10.78. The Euler factor $P_2(X)$ (Chapter 19; the paper's Proposition prop:shapes) is determined by $G_{\mathrm{ss}}$ and the action of Frobenius on $(\mathrm{ad}^0 V)^{I_2}$: cyclic of order 3 or 6 gives $P_2(X) = 1 + 2X$ and $e_3(E,2) = 1$ (the order-6 case reduces to the order-3 case, as $\mathrm{ad}^0 V$ is unchanged by a quadratic twist); $\mathbb{Z}/2$ gives the cubic $(1-2X)(1-(a^2-4)X+4X^2)$ of the twist $E'$ with good reduction, $a = a_2(E')$, and $e_3(E,2) = v_3(9-a^2)$; $\mathbb{Z}/4$ gives $1 \pm 2X$; $Q_8$ and $\mathrm{SL}_2(\mathbb{F}_3)$ act absolutely irreducibly, so $(\mathrm{ad}^0 V)^{I_2} = 0$ and $P_2 = 1$. So a type II or II$^*$ at 2 can have $e_3(E,2) \in \{0,1,2\}$: `336f1` has type II at 2, $f_2 = 4$, pattern $(1,3,3,3)$, $P_2 = 1+2X$, $e_3(E,2) = 1$ and $\deg \varphi_E = 24$; `1232l1`, the twist of `77a1` by $\chi_{-4}$ with $a_2(\texttt{77a1}) = 0$, has type II$^*$ at 2, $f_2 = 4$, pattern $(1,1,1,1)$, $P_2 = (1-2X)(1+2X)^2 = 1+2X-4X^2-8X^3$, $e_3(E,2) = 2$ and $\deg \varphi_E = 288 = 2^5 \cdot 3^2$. Remark rem:wild of the paper says that a type II or II$^*$ at 2 "usually has $P_2 = 1$", with these two curves as the exceptions. A potentially good $\mathrm{I}_m^*$ at 2 has the pattern $(4,4,4,4)$, hence $P_2 = 1$, in most cases (Table 10.4).

*Remark* 10.79 (the prime 3). Let $E/\mathbb{Q}_3$ have additive reduction.

(i) Types III, III$^*$, $\mathrm{I}_0^*$ are tame ($f_3 = 2$), with $G_{\mathrm{ss}}$ cyclic of order 4, 4, 2.

(ii) Types II, II$^*$, IV, IV$^*$ are wild: $f_3 \in \{3,4,5\}$ and $3 \mid e$.

(iii) Type $\mathrm{I}_n^*$ ($n \geq 1$) is potentially multiplicative, with $n = -v_3(j)$ and $f_3 = 2$.

*Justification.* (iii) is Theorem 10.66(i) and Proposition 10.46(iv). The tame/wild dichotomy in (i), (ii) is Theorem 10.53. In (ii), wild means that the pro-3 group $P_3$ acts nontrivially, so its image, a nontrivial 3-group, lies in $G_{\mathrm{ss}}$. In (i), $G_{\mathrm{ss}}$ is a quotient of the procyclic tame inertia of order prime to 3, so it is cyclic of order 2 or 4 (Theorem 10.51(iii)). It has order 2 if and only if $E$ is a ramified quadratic twist of a curve with good reduction (the argument of Proposition 10.77), that is, up to an unramified twist, the twist by $\chi_{-3}$ of such a curve, which has type $\mathrm{I}_0^*$ by Lemma 10.70(b) and Lemma 10.69; conversely type $\mathrm{I}_0^*$ is such a twist. So III, III$^*$ have order 4. □

Below conductor 10,000, at 3: III, III$^*$ (1,553 pairs each), $\mathrm{I}_0^*$ (2,283) and $\mathrm{I}_n^*$ (7,805) always have $f_3 = 2$; II and IV$^*$ both have $f_3 = 3,4,5$ in 879, 280, 107 pairs, and IV and II$^*$ both in 628, 177, 71 pairs. The equal distributions are Proposition 10.72: the twist by $\chi_{-3}$ is unramified outside 3, preserves the conductor, and exchanges these types.

*In the paper* 10.80. This is why Lemma lem:IV excludes $q = 3$ (at 3 the types IV, IV$^*$ are wild, and the conclusion that $\bar{\rho}_{E,3}$ is tamely ramified with Serre conductor exponent 1 at $q$ fails), and why the weights $w_3$ at $q = \ell = 3$ in Conjecture conj:euler are not explained by Lemma lem:euler. The Remark after the Example of Section 3 of the paper mentions $q = 3$ in cases (a) and (b) of Theorem thm:qual; there the obstruction is $q = \ell$, and the ramification at 3 of the types II, IV is also wild by (ii).

*Remark* 10.81 (symbols $\mathrm{I}_m^*$ at 2). At 2 a symbol $\mathrm{I}_m^*$ is potentially good in 5,935 of the 9,667 pairs below conductor 10,000 (61%; among curves numbered 1 in their class, which below conductor 10,000 are the optimal curves except in class `990h` (whose optimal curve is `990h3`; it is multiplicative at 2, as $2 \parallel 990$, and so does not enter this count), 2,340 of 4,037, that is 58%, the figure quoted in Remark rem:wild of the paper; the six curves with complex multiplication among them are all potentially good, so the optimal non-CM curves give 2,334 of 4,031, again 58%). When it is potentially multiplicative, $t_\ell(E,2) = v_\ell(\nu)$ with $\nu = -v_2(j) = m-4$ or $m-8$ (Theorem 10.66(ii)), so $v_\ell(m)$ is the wrong invariant: `304b1` has type $\mathrm{I}_7^*$ and $\nu = 3$, so $t_3(E,2) = 1$ although $v_3(7) = 0$; `832c1` has type $\mathrm{I}_9^*$ and $\nu = 1$, so $t_3(E,2) = 0$ although $v_3(9) = 2$.

*In the paper* 10.82. Definition def:terms: "a symbol $\mathrm{I}_n^*$ at $q=2$ can have potentially good reduction, and then $t_\ell=0$"; Section sec:wild: "the index $n$ of the symbol is not the valuation of the Tate parameter at 2, which is why $n$ was not the right invariant". An additive potentially multiplicative prime 2 is handled by Theorem 10.59 (a ramified quadratic twist of a Tate curve with $\nu=-v_2(j)$), as in the proof of Lemma lem:tamfactor, not by Lemma lem:twist, which is for odd $q$.

## 10.10 Cremona's tables and PARI

The paper's computations take the curves from Cremona's tables and recompute every local invariant with PARI (Section sec:data and the appendix). This short section says what the tables contain and what PARI computes, and collects the local data of the running examples in one table. File formats and the practical use of PARI are in Chapter 22.

*Remark* 10.83 (what the tables contain). Cremona's tables [35, 36] list, for every elliptic curve over $\mathbb{Q}$ up to their conductor bound, the global minimal model in the normalisation of Theorem 10.6, the conductor, the isogeny class, rank and torsion. The file `allbsd` adds the product $\prod_p c_p$ of the Tamagawa numbers, the real period and the other quantities of the conjecture of Birch and Swinnerton-Dyer; other files give the isogeny matrices, the modular degrees (`degphi`) and the mod $\ell$ images. Section sec:data of the paper describes the `ecdata` repository as containing every curve "with its Kodaira symbols, Tamagawa numbers, modular degree, isogeny class, torsion" and mod $\ell$ images, and adds: "The Kodaira types and discriminant exponents at the bad primes were recomputed with PARI/GP 2.15" from the $a$-invariants (its appendix calls `ellglobalred`). The files used in this book (`allcurves`, `allbsd`, `allisog`, `alllabels`, `degphi`, `galrep`, conductor below 10,000) are a subset of that repository and, unlike the paper's description of the full repository, do not list the Kodaira symbols or the Tamagawa numbers prime by prime: here these are recomputed with Tate's algorithm (`ellglobalred`), as in the paper's own check. File formats are in Chapter 22. The minimal discriminant and $j$ follow from the model.

**Example 10.84** (a PARI session)**.** With PARI/GP 2.17.2, for `726a1`, `20a1` and `11a1`:

```
E = ellinit([1,1,0,-35,-51]);        \\ 726a1
R = ellglobalred(E)
  \\ [726, [1, 0, 0, 0], 4, [2, 1; 3, 1; 11, 2], [[1, 10, 0, 2], [1, 7, 0, 1], [2, 3,
     0, 2]]]
R[4]                                  \\ [2, 1; 3, 1; 11, 2]
elllocalred(ellinit([0,1,0,4,4]), 2)  \\ 20a1: [2, -4, [1, 0, 0, 0], 3]
elllocalred(ellinit([1,0,1,170,-3237]), 7)   \\ 539c1: [2, -7, [1, 0, 0, 0], 4]
E11 = ellinit([0,-1,1,-10,-20]);     \\ 11a1
ellinit(E11, O(11^8)).tate[3]
  \\ 10*11^5 + 2*11^6 + 6*11^7 + 6*11^8 + 5*11^9 + ... + O(11^13)
valuation(E11.j, 11)                  \\ -5
```

The local data of `726a1` are $[f,k,*,c]=[1,10,*,2]$, $[1,7,*,1]$, $[2,3,*,2]$ at $2,3,11$: types $\mathrm{I}_6$, $\mathrm{I}_3$, III (codes $10=6+4$, $7=3+4$, 3), Tamagawa numbers $2,1,2$ with product 4, and conductor $2\cdot3\cdot11^2$. For `20a1` at 2 the code $-4$ is IV$^*$, $f_2=2$, $c_2=3$. For `539c1` at 7 the code $-7=-(3+4)$ is $\mathrm{I}_3^*$. For `11a1` the Tate parameter has $\nu=5$ and $\bar{u}=10$ (Example 10.67).

**Example 10.85** (local data of the running examples)**.** For each curve: conductor, minimal discriminant, and at each bad prime $q$ the type, $f_q$, $c_q$, $\#\Phi_q^{\mathrm{geom}}$ and $v_q(j)$ (all computed with PARI; the hand computations of Example 10.42 agree).

| curve | $N$ | $\Delta_{\min}$ | $q$: type, $f_q$, $c_q$, $\#\Phi_q^{\mathrm{geom}}$, $v_q(j)$ |
|---|---|---|---|
| 11a1 | 11 | $-11^5$ | 11: $\mathrm{I}_5$, 1, 5, 5, $-5$ |
| 20a1 | $2^2 \cdot 5$ | $-2^8 \cdot 5^2$ | 2: $\mathrm{IV}^*$, 2, 3, 3, 4; 5: $\mathrm{I}_2$, 1, 2, 2, $-2$ |
| 99a1 | $3^2 \cdot 11$ | $3^3 \cdot 11$ | 3: III, 2, 2, 2, 9; 11: $\mathrm{I}_1$, 1, 1, 1, $-1$ |
| 121a1 | $11^2$ | $-11^2$ | 11: II, 2, 1, 1, 1 |
| 121c1 | $11^2$ | $-11^4$ | 11: IV, 2, 1, 3, 2 |
| 121c2 | $11^2$ | $-11^8$ | 11: $\mathrm{IV}^*$, 2, 1, 3, 1 |
| 121d1 | $11^2$ | $-11^7$ | 11: $\mathrm{I}_1^*$, 2, 2, 4, $-1$ |
| 539c1 | $7^2 \cdot 11$ | $-7^9 \cdot 11^2$ | 7: $\mathrm{I}_3^*$, 2, 4, 4, $-3$; 11: $\mathrm{I}_2$, 1, 2, 2, $-2$ |
| 539d1 | $7^2 \cdot 11$ | $-7^6 \cdot 11$ | 7: $\mathrm{I}_0^*$, 2, 2, 4, 0; 11: $\mathrm{I}_1$, 1, 1, 1, $-1$ |
| 726a1 | $2 \cdot 3 \cdot 11^2$ | $2^6 \cdot 3^3 \cdot 11^3$ | 2: $\mathrm{I}_6$, 1, 2, 6, $-6$; 3: $\mathrm{I}_3$, 1, 1, 3, $-3$; 11: III, 2, 2, 2, 0 |
| 980i1 | $2^2 \cdot 5 \cdot 7^2$ | $-2^8 \cdot 5 \cdot 7^{11}$ | 2: $\mathrm{IV}^*$, 2, 1, 3, 19; 5: $\mathrm{I}_1$, 1, 1, 1, $-1$; 7: $\mathrm{I}_5^*$, 2, 2, 4, $-5$ |

The curves 121c1, 121c2 at 11 and 980i1 at 2 have $c_q = 1 < 3 = \#\Phi_q^{\mathrm{geom}}$; 726a1 at 2 and 3 has $c_q < \#\Phi_q^{\mathrm{geom}}$ because the reduction is nonsplit.

## 10.11 Exercises

The exercises go from routine to harder. Several ask the reader to check a claim of the paper; PARI is useful for the numerical ones (Chapter 22).

**Exercise 10.86.** Show that an integral model with $v(\Delta) < 12$ is minimal, and that for $p \geq 5$ an integral model with $v(c_4) \geq 4$ and $v(\Delta) \geq 12$ is not (Proposition 10.5). Decide whether the model $y^2 = x^3 - 27c_4x - 54c_6$ of 11a1 is minimal at 2 and at 3.

**Exercise 10.87.** For 11a1 at 11 and 20a1 at 5, decide whether the reduction is split, once from the polynomial $T^2 + a_1T - a_2$ after moving the node to $(0,0)$, and once from the Legendre symbol $\left(\frac{-c_6}{p}\right)$ (Proposition 10.12). Compare with $a_{11} = 1$ and $a_5 = -1$.

**Exercise 10.88.** Write down explicitly the isomorphisms $\tilde{E}_{\mathrm{ns}}(k) \simeq k^+$ for $y^2 = x^3$ and $\tilde{E}_{\mathrm{ns}}(k) \simeq k^\times$ for $y^2 = x^3 + x^2$, and check directly that three collinear points are sent to elements with sum 0, respectively product 1 (Proposition 10.10).

**Exercise 10.89.** Let $E =$ 11a1, which has good reduction at 3 and 7. Using Corollary 10.17, show that the prime-to-3 part of $E(\mathbb{Q})_{\mathrm{tors}}$ injects into $\tilde{E}(\mathbb{F}_3)$ and the prime-to-7 part into $\tilde{E}(\mathbb{F}_7)$. Count these groups (orders 5 and 10), deduce that $E(\mathbb{Q})_{\mathrm{tors}}$ has order dividing 5, and conclude that $E(\mathbb{Q})_{\mathrm{tors}} \simeq \mathbb{Z}/5$ by exhibiting a point of order 5 (for instance $(5,5)$ on $y^2 + y = x^3 - x^2 - 10x - 20$).

**Exercise 10.90.** Run Tate's algorithm by hand for 121a1 and 121c2 at 11 and for 539d1 and 539c1 at 7, and check the types used in the Example of Section 3 of the paper (Example 10.42).

**Exercise 10.91.** Check Lemma lem:twist (ii) of the paper on 121a1: its twist by $-11$ has type $\mathrm{IV}^*$ at 11. Then show that its twist by $+11$ has additive reduction at 2 (it is 1936k2, of type $\mathrm{I}_4^*$ at 2 with $v_2(j) = 0$), so that the twist by $q$ and the twist by $q^*$ really are different curves (Warning 10.71).

**Exercise 10.92.** Check the paper's argument for the case $n = 0$ of Lemma lem:twist (ii) (the cubic of Tate's algorithm of the twist is $T^3 + \epsilon a_2 T^2 + a_4 T + \epsilon a_6$, with distinct roots modulo $q$) for the twist of 11a3 by $-7$ (it is 539d1): compute a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ of 11a3, the cubic of the twist with the sign $\epsilon = -1$, and its roots modulo 7 (Proposition 10.39).

**Exercise 10.93.** Prove $w_\ell(E,q) = \max\big(v_\ell(\#\Phi_q^{\mathrm{geom}}(E)), v_\ell(\#\Phi_q^{\mathrm{geom}}(E^{(q)}))\big)$ for $q \geq 5$ from Table 10.1 and Lemma 10.70 (Proposition 10.74). Which curves of Example 10.85 have $c_q \neq \#\Phi_q^{\mathrm{geom}}$, and why?

**Exercise 10.94.** Verify Ogg's formula $f_q = v_q(\Delta_{\min}) + 1 - m_q$ at every bad prime of the curves of Example 10.85 (Theorem 10.47).

**Exercise 10.95.** Using Theorem 10.52, compute the semistability defect at each additive prime $q \geq 5$ of the running examples. Show directly, following the proof of Theorem 10.52, that `726a1` acquires good reduction at 11 over $\mathbb{Q}_{11}^{\mathrm{ur}}(11^{1/4})$ and not over $\mathbb{Q}_{11}^{\mathrm{ur}}(\sqrt{11})$.

**Exercise 10.96.** Write out the proof of Corollary 10.54 for `20a1`: identify $E[3]^{I_2}$ with $\Phi_2^{\mathrm{geom}}[3]$, and check that the 3-division polynomial of `20a1` has one root in $\mathbb{Q}_2$ and three roots generating a totally ramified cubic extension. Then check the paper's claim (Remark rem:wild, Proposition prop:forms) that a type IV or IV$^*$ at 2 has $P_2(X) = 1 + 2X$: deduce it from Corollary 10.54(ii), (iv) as in the paper link after it, and compare with PARI's `lfunsympow` for `20a1` and `52a1`.

**Exercise 10.97.** Show that $a_6(\mathsf{q}) = -(5s_3 + 7s_5)/12$ has integer coefficients, and compute $j(E_{\mathsf{q}})$ up to the term $196884\,\mathsf{q}$ (Theorem 10.57). Invert the series to obtain $\mathsf{q}(J) = J + 744J^2 + 750420J^3 + \cdots$ (Proposition 10.58).

**Exercise 10.98.** Check the paper's claim (Example of Section 3) that $t_3(\texttt{539c1}, 7) = 1$: compute $v_7(j)$, the Tate parameter of `539c1` at 7 to a few 7-adic digits and its unit part, and show that they coincide with those of `77c1` (Example 10.67).

**Exercise 10.99.** For `11a1` at 11, show that the unit part of $q_E$ is $\equiv -1 \pmod{11}$ and that it is a 5th power in $\mathbb{F}_{11}^\times$. This is the situation of the extra term of Lemma lem:tamfactor at $\ell = 5$ (although $\ell = 5$ is an Eisenstein prime for `11a1`, so the lemma applies and Theorem thm:main does not).

**Exercise 10.100.** Let $E/K$ have multiplicative reduction of type $\mathrm{I}_n$, with $k$ finite, and let $K'/K$ be the unramified quadratic extension, with $\mathrm{Gal}(K'/K) = \langle\sigma\rangle$. Using Theorem 10.59, $E(K) \simeq K^\times/q_E^{\mathbb{Z}}$ in the split case, and $E(K) \simeq \{w \in K'^\times/q_E^{\mathbb{Z}} : \sigma(w) = w^{-1}\}$ in the nonsplit case (the Frobenius action twisted by $w \mapsto w^{-1}$). Identify $E_0(K)$ with the classes represented by units, and compute $E(K)/E_0(K)$ in both cases for general $n$: it is $\mathbb{Z}/n$ (split), and $\mathbb{Z}/2$ or 0 according as $n$ is even or odd (nonsplit). This reproves Proposition 10.65(ii). Check $c_2 = 2$ and $c_3 = 1$ for `726a1`, which is nonsplit of type $\mathrm{I}_6$ at 2 and $\mathrm{I}_3$ at 3.

**Exercise 10.101.** Check the paper's statement (Definition def:terms, Section sec:wild) that the index of a symbol $\mathrm{I}_m^*$ at 2 is not the valuation of the Tate parameter, on `304b1` and `832c1`: compute $m$, $v_2(j)$, $f_2$ and $t_3(E,2)$ (Theorem 10.66, Remark 10.81). Then compute PARI's `ellinit(E, O(2^20)).tate[3]` for `832c1` and explain the result (Warning 10.68).

**Exercise 10.102.** Check Remark rem:wild of the paper, that a type II or II$^*$ at 2 "usually has $P_2 = 1$" but not always, on `336f1` and the type II$^*$ curve `1232l1`, using PARI's `lfunsympow` and the dictionary of Proposition 10.77. Explain the answer through the inertia images ($\mathbb{Z}/6$, respectively $\mathbb{Z}/2$), and verify that $v_3(\deg\varphi_E) = e_3(E,2) = 2$ for `1232l1` ($\deg\varphi_E = 288$).

## 10.12 Notes and references

**Sources followed.** Sections 10.1 to 10.3 follow Silverman's first book [185, Chapter VII] (minimal models in VII.1, reduction in VII.2 and VII.5, the formal group in Chapter IV and VII.2, torsion in VII.3 and VII.7, potential good reduction in VII.5); the proofs of Proposition 10.10 and Proposition 10.15 are the standard ones written out. Néron models, Kodaira symbols and Tate's algorithm follow [184, Chapter IV] (Sections IV.5 to IV.9, with the table of Kodaira types and Tamagawa numbers in IV.9 and Ogg's formula in IV.11), with [15] for the general theory and [35, Section 3.2] for the explicit form of the algorithm. The Tate curve follows [184, Chapter V] (V.3 for the uniformisation, V.5 for curves with $v(j) < 0$). The original sources are Néron [141], Kodaira [103], Tate [193], Ogg [145], Saito [169] and Serre and Tate [179]; Serre's paper [171, Section 5.6] discusses the image of inertia at 2 and 3.

**Theorem numbers.** We cite theorem numbers only where they have been checked: Dokchitser and Dokchitser [56, Theorem 3.1] and its source Kramer and Tunnell [105, Proposition 8.20] (the number was checked in the arXiv version of [56] when this book was planned). Elsewhere we cite sections. In particular Kraus's paper [107], which gives the semistability defect at all primes and the full 2- and 3-adic classification, is cited as a whole: the numbering of its theorems is quoted inconsistently in later papers (see Billerey [13] for the 2-adic statements and [152] for a recent restatement). Kraus's conditions on $c_4$, $c_6$ (Proposition 10.5(iii)) are in a different paper [106]. Papadopoulos [147] tabulates the Kodaira types at 2 and 3 in terms of valuations and congruences. The relation between Kodaira symbols and wild ramification for twists of curves with multiplicative reduction at 2 is in Lorenzini [119]; see also [154].

**The types IV, IV$^*$ at 2.** Corollary 10.54 has two independent proofs: the one given here, through the component group (Proposition 10.33), which is the proof of Lemma lem:IV in the paper; and one through the conductor, from Theorem 10.53: $f_2 = 2$ forces $\mathrm{sw} = 0$, so the image of inertia is a cyclic group of odd order dividing 24, hence of order 3.

**The exchanges at 3.** Proposition 10.72 is proved here from Ogg's formula, as in the paper's proof of Lemma lem:twist (ii); we have not found it stated elsewhere in the literature in this form, though it can presumably be read off the tables of [147].

**Computations.** All numbers in the chapter were computed with PARI/GP 2.17.2 [148] through `cypari2`, on Cremona's tables [36] for conductor below 10,000 (64,687 curves, 211,455 pairs of a curve and a bad prime). The population checks quoted in the text are: Kraus's conditions (Proposition 10.5); $a_p = \left(\frac{-c_6}{p}\right)$ (Proposition 10.12); the formulas for Step 2 of Tate's algorithm; the table for $p \geq 5$; the values of $c$ by type; the twist exchanges (90,970 pairs); the types at additive potentially multiplicative primes; the conductor exponents at 2 and 3; and the inertia patterns at 2 (20,979 pairs). The Tate parameters were computed from the series of Proposition 10.58; the discrepancy of PARI's `.tate` at 2 is recorded in Warning 10.68.

# Chapter 11

# Galois representations of elliptic curves

The paper works throughout with the $\ell$-adic representation $\rho = \rho_{E,\ell}$ of $G_{\mathbb{Q}}$ on the Tate module $T = T_\ell E$ and with its reduction $\bar\rho$ on $E[\ell]$. Every statement it makes about these representations, as opposed to the algebra of representations (Chapter 4), the field theory (Chapters 5 to 7) or the geometry of the curve (Chapters 9 and 10), is collected and proved here: the local representation at every prime $p \neq \ell$ by reduction type, the mod 3 picture of inertia behind Lemma lem:IV and Remark rem:types, the representation at $\ell$, the Serre conductor of $\bar\rho$, the possible images of $\bar\rho$, and the three statements on which the hypothesis of Theorem thm:main rests: Proposition prop:ns, Lemma lem:h0 and Proposition prop:Sf. The chapter supplies the elliptic-curve inputs; the representation theory it uses is proved in Chapter 4, the group theory of $\mathrm{GL}_2(\mathbb{F}_3)$ in Chapter 1, the local field theory in Chapter 7, and the reduction theory (Néron models, Kodaira types, Tate curves, Kraus's theorem) in Chapter 10.

## Core results at a glance

Throughout, $E/\mathbb{Q}$ has conductor $N$ and a global minimal model, $\ell$ is a prime (odd unless said otherwise), $T = T_\ell E$, $V = T \otimes \mathbb{Q}_\ell$, $\rho = \rho_{E,\ell}$, $\bar\rho = \bar\rho_{E,\ell}$, and $q \neq \ell$ is a prime; $\Phi_q^{\mathrm{geom}} = \Phi_q(\overline{\mathbb{F}}_q)$ is the geometric component group at $q$.

**11.1 Conventions.** $\mathrm{Frob}_p$ is an arithmetic Frobenius and $\mathrm{Frob}_p^{-1}$ a geometric one; $\chi_\ell$ is the $\ell$-adic cyclotomic character, $\chi_d$ the quadratic character of $\mathbb{Q}(\sqrt{d})$, $\chi_{q^*}$ the one of conductor $q$, and $\bar\chi_3 = \chi_{-3}$. *(convention)* Convention 11.1.

**11.2 The representations.** $\rho\colon G_{\mathbb{Q}} \to \mathrm{Aut}_{\mathbb{Z}_\ell}(T) \simeq \mathrm{GL}_2(\mathbb{Z}_\ell)$, $\rho \bmod \ell^k$ on $E[\ell^k] = T/\ell^k T$, $\bar\rho$ on $E[\ell]$, $A_f = \mathrm{ad}^0 V$, $\delta_p = \dim V^{I_p}$. *(definition)* Definition 11.2.

**11.3 Basic properties.** $\rho$ is continuous and $\rho \bmod \ell^k$ cuts out $\mathbb{Q}(E[\ell^k])$; $\det\rho = \chi_\ell$, so $\det\bar\rho = \bar\chi_\ell$ is onto $\mathbb{F}_\ell^\times$ and $\bar\rho(G_{\mathbb{Q}(\sqrt{\ell^*})})$ is the set of elements of $\bar\rho(G_{\mathbb{Q}})$ with square determinant; for $\ell = 3$, $\det\bar\rho_{E,3} = \chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$. For $\ell$ odd and $c$ a complex conjugation, $\rho(c)$ is conjugate in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ to $\mathrm{diag}(1,-1)$. $\rho$ is unramified outside $N\ell$. *(proved here, citing Chapters 5 and 9)* Proposition 11.3.

**11.4 Faltings.** $V_\ell E$ is a semisimple $\mathbb{Q}_\ell[G_{\mathbb{Q}}]$-module, and $\mathrm{Hom}(E,E') \otimes \mathbb{Z}_\ell \simeq \mathrm{Hom}_{G_{\mathbb{Q}}}(T_\ell E, T_\ell E')$. *(stated; reference)* Theorem 11.4.

**11.5 Twists and isogenies.** $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi_d$, so $\mathrm{ad}^0\,\rho$, the projective image of $\bar\rho$ and the irreducibility of $E[\ell]$ are twist invariant; twisting by $\chi_{-3}$ exchanges the mod 3 images `3B.1.1`, `3B.1.2` and fixes the others; twisting by $\chi_d$, $d \not\equiv 1, -3$ modulo squares, sends `3B.1.1`, `3B.1.2` to `3B` and `3Cs.1.1` to `3Cs`, and fixes `3Ns`, `3Nn` and the surjective image. If $E[\ell]$ is irreducible, every curve isogenous to $E$ has an isomorphic $T_\ell$. *(proved here)* Proposition 11.5.

**11.6 Irreducibility.** For $\ell$ odd: $\bar\rho$ is irreducible iff absolutely irreducible; it is reducible iff $E$ has a rational $\ell$-isogeny (for $\ell = 3$ iff $\psi_3$ has a rational root); $\bar\rho^{\mathrm{ss}}$ is determined by $a_p$ mod $\ell$ on any set of primes of density one, and so is $\bar\rho$ when it is irreducible. *(proved here, citing Chapters 4 and 5)* Proposition 11.6.

**11.7 Néron, Ogg, Shafarevich.** For $K/\mathbb{Q}_p$ finite and $\ell \neq p$: good reduction $\iff$ $I_K$ trivial on $T_\ell E$ $\iff$ $I_K$ trivial on $E[m]$ for all (equivalently infinitely many) $m$ prime to $p$ *(proved here)*; potentially good reduction $\iff$ $\rho(I_K)$ finite; potentially multiplicative $\iff$ $\rho(I_K)$ infinite *(proof sketched here)*. So $\rho_{E,\ell}$ is unramified at $p \neq \ell$ iff $p \nmid N$. Theorem 11.9.

**11.8 Inertia invariants and component groups.** For additive reduction at $q$, $E[\ell^k]^{I_q} \simeq \Phi_q^{\mathrm{geom}}[\ell^k]$; for multiplicative reduction $0 \to \mu_{\ell^k} \otimes \varepsilon \to E[\ell^k]^{I_q} \to \Phi_q^{\mathrm{geom}}[\ell^k] \to 0$ with $\Phi_q^{\mathrm{geom}} \simeq \mathbb{Z}/n_q$. *(stated; proved in Chapter 10)* Lemma 11.10.

**11.9 Serre-Tate.** For potentially good reduction at $q$: $\rho_{E,\ell}(I_q) \simeq \mathrm{Gal}(L/\mathbb{Q}_q^{\mathrm{ur}})$, $L$ the smallest extension of $\mathbb{Q}_q^{\mathrm{ur}}$ with good reduction, independently of $\ell \neq q$, with characteristic polynomials in $\mathbb{Z}[X]$ *(stated; reference)*; it lies in $\mathrm{SL}_2(\mathbb{Z}_\ell)$, injects into $\mathrm{GL}_2(\mathbb{F}_\ell)$ for $\ell$ odd, and $V^{I_q} = 0$ unless the reduction is good *(proved here)*; it is cyclic of order $12/\gcd(v_q(\Delta_{\min}), 12)$ for $q \geq 5$ *(proved here from Kraus's theorem)*. Theorem 11.12.

**11.10 Good primes.** For $p \nmid N\ell$, $\rho$ is unramified at $p$, $\mathrm{Frob}_p$ has characteristic polynomial $X^2 - a_p X + p$ on $V$, both Frobenii have eigenvalues $\alpha/\beta, 1, \beta/\alpha$ on $\mathrm{ad}^0\,V$, $\delta_p = 2$, and $\mathrm{tr}\,\bar\rho(\mathrm{Frob}_p) = a_p$ mod $\ell$. *(proved here, citing Chapter 9)* Proposition 11.14.

**11.11 The Tate shape.** At a multiplicative prime $q$, in a $\mathbb{Z}_\ell$-basis $e_1, e_2$ of $T$, $\rho(g) = \left(\begin{smallmatrix} \chi_\ell\varepsilon(g) & \varepsilon\kappa(g) \\ 0 & \varepsilon(g) \end{smallmatrix}\right)$ on $G_{\mathbb{Q}_q}$, with $\varepsilon$ unramified, $\varepsilon(\mathrm{Frob}_q) = a_q(E)$, and $\kappa$ the Kummer cocycle of the Tate parameter $q_E$. On $I_q$, $\rho(\sigma) = \left(\begin{smallmatrix} 1 & n_q t_\ell(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$; $V^{I_q} = \mathbb{Q}_\ell e_1$ ($\delta_q = 1$), the unique stable line; $(\mathrm{ad}^0\,V)^{I_q} = \mathbb{Q}_\ell e$, $e = e_1 \otimes e_2^\vee$, with $\mathrm{Frob}_q$ acting by $q$. *(proved here, given Chapter 10)* Theorem 11.15.

**11.12 Lemma lem:tamagawa-exp.** At a multiplicative prime $q$, $\rho$ mod $\ell^t$ is unramified at $q$ iff $\ell^t \mid n_q$, split or not; $\#E[\ell^t]^{I_q} = \ell^{t+\min(t, v_\ell(n_q))}$; if $\ell \nmid n_q$, $\bar\rho|_{D_q}$ is a non-split extension of $\varepsilon$ by $\varepsilon\bar\chi_\ell$. *(proved here, two proofs)* Lemma 11.17.

**11.13 Potentially multiplicative additive primes.** If $v_q(j) < 0$ and the reduction is additive ($q = 2$ allowed), $\rho|_{G_{\mathbb{Q}_q}} \simeq \rho' \otimes \chi$ with $\chi$ ramified quadratic and $\rho'$ of Tate shape with $v_q(q_E) = -v_q(j)$; $V^{I_q} = 0$, $\mathrm{ad}^0\,V \simeq \mathrm{ad}^0\,V'$, $\rho$ mod $\ell^t$ is ramified at $q$ for every $t \geq 1$, and $f_q = 2$ ($q$ odd), 4 or 6 ($q = 2$). *(proved here, given Chapter 10)* Proposition 11.19.

**11.14 $\delta_p$.** $\delta_p = 2, 1, 0$ for good, multiplicative, additive reduction at $p \neq \ell$, and $f_p = 2 - \delta_p + \mathrm{sw}_p(V)$. *(proved here)* Corollary 11.21.

**11.15 Additive potentially good reduction, $q \geq 5$.** $\rho(I_q) = \langle g \rangle$ with $g = -1$ ($\mathrm{I}_0^*$), $\sigma$ (IV, IV$^*$), $\tau$ (III, III$^*$), $-\sigma$ (II, II$^*$), where $\sigma^2 + \sigma + 1 = 0$ and $\tau^2 = -1$; $(\mathrm{ad}^0 V)^{I_q}$ is $\mathrm{ad}^0 V$, $\mathbb{Q}_\ell(2\sigma+1)$, $\mathbb{Q}_\ell\tau$, $\mathbb{Q}_\ell(2\sigma+1)$, and $\mathrm{Frob}_q$ (arithmetic or geometric) acts on the line by $\chi_{-3}(q)$, $\chi_{-4}(q)$, $\chi_{-3}(q)$; $V^{I_q} = 0$ and $f_q = 2$. *(proved here, given Kraus's theorem)* Theorem 11.22.

**11.16 Lemma lem:IV.** At a prime $q \neq 3$ of type IV or IV$^*$ the reduction is potentially good, $\rho_{E,\ell}(I_q)$ is cyclic of order 3 for every $\ell \neq q$, its image on $E[3]$ is a nontrivial unipotent group, $E[3]^{I_q}$ is a line, $\bar\rho_{E,3}$ is tame at $q$ with Serre exponent 1, $f_q = 2$, $T_3E \simeq \mathbb{Z}_3[\zeta_3]$, and $\mathrm{Frob}_q$ acts on $(\mathrm{ad}^0 V)^{I_q} = \mathbb{Q}_\ell(2\sigma+1)$ by $\chi_{-3}(q)$. *(proved here, two proofs)* Lemma 11.25.

**11.17 Types IV, IV$^*$ at 2.** $f_2 = 2$, $v_2(\Delta_{\min}) = 4$, resp. 8, the Serre exponent of $\bar\rho_{E,3}$ at 2 is 1, $L_2(A_f, s)^{-1} = 1 + 2^{-s}$, $P_2(X) = 1 + 2X$, $e_3(E,2) = 1$ and $e_\ell(E,2) = 0$ for $\ell \geq 5$. *(proved here)* Corollary 11.27.

**11.18 Inertia mod 3 at $q \geq 5$ (Remark rem:types).** $E[3]^{I_q} = 0$ and the Serre exponent of $\bar\rho_{E,3}$ at $q$ is $2 = f_q$ for every additive type except IV, IV$^*$ (exponent 1); the twist by $\chi_{q^*}$ lowers the Serre exponent exactly for II, II$^*$, $\mathrm{I}_n^*$, $\mathrm{I}_0^*$ and the conductor exponent of $E$ exactly for $\mathrm{I}_n^*$, $\mathrm{I}_0^*$. *(proved here)* Proposition 11.29.

**11.19 Good reduction at $\ell$.** For $\ell \nmid N$: $E[\ell^k]$ is the generic fibre of a finite flat group scheme over $\mathbb{Z}_\ell$ *(proof sketched here)*; $V$ is crystalline at $\ell$ with Hodge-Tate weights 0, 1, and the Serre weight of $\bar\rho$ is 2 *(stated; reference)*. Theorem 11.32.

**11.20 Fundamental characters.** $\omega\colon I_\ell \to \mathbb{F}_\ell^\times$ (equal to $\bar\chi_\ell|_{I_\ell}$) and $\omega_2\colon I_\ell \to \mathbb{F}_{\ell^2}^\times$ of order $\ell^2 - 1$ with $\omega_2^{\ell+1} = \omega$; conjugation by $\mathrm{Frob}_\ell$ sends $\omega_2$ to $\omega_2^\ell$, so $\omega_2$ does not extend to $G_{\mathbb{Q}_\ell}$. *(definition; proved here, given Chapter 7)* Definition 11.33, Proposition 11.34.

**11.21 Deligne and Fontaine.** For $\ell$ odd, $\ell \nmid N$: if $\ell \nmid a_\ell$, $\bar\rho|_{D_\ell} \simeq \begin{pmatrix} \bar\chi_\ell \mu(a_\ell^{-1}) & * \\ 0 & \mu(a_\ell) \end{pmatrix}$; if $\ell \mid a_\ell$, $\bar\rho|_{I_\ell} \otimes \mathbb{F}_{\ell^2} \simeq \omega_2 \oplus \omega_2^\ell$ and $\bar\rho|_{D_\ell}$ is irreducible. Hence $a_\ell \bmod \ell$ is determined by $\bar\rho|_{D_\ell}$. *(proof sketched here)* Theorem 11.35.

**11.22 Consequences at $\ell$.** For $\ell$ odd, $\ell \nmid N$: if $\bar\rho$ is reducible, $E$ is ordinary at $\ell$ and the two characters restrict to $I_\ell$ as $\omega$ and 1; if $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$, then $\ell = 3$ and $E$ is ordinary at 3. *(proved here, given Theorem 11.35)* Corollary 11.37.

**11.23 Serre conductor.** For $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_n(\overline{\mathbb{F}}_\ell)$ and $p \neq \ell$, $a_p(\bar\sigma) = n - \dim \bar\sigma^{I_p} + \mathrm{sw}_p(\bar\sigma)$ and $N(\bar\sigma) = \prod_{p \neq \ell} p^{a_p(\bar\sigma)}$; $a_p(\bar\sigma) = 0$ iff $\bar\sigma$ is unramified at $p$; $a_p$ is invariant under extension of scalars and twists unramified at $p$. *(definition; proved here)* Definition 11.38, Proposition 11.39.

**11.24 Serre conductor of $\bar\rho_{E,\ell}$.** For $p \neq \ell$, $a_p(\bar\rho_{E,\ell}) = f_p(E) - \dim_{\mathbb{F}_\ell} \Phi_p^{\mathrm{geom}}[\ell]$. So $N(\bar\rho_{E,\ell})$ divides the prime-to-$\ell$ part of $N$; the exponent at a multiplicative $p$ is 0 or 1 as $\ell \mid n_p$ or not; at an additive $p$ and $\ell$ odd it is $f_p - 1$ for $\ell = 3$ and types IV, IV$^*$, and $f_p$ otherwise. *(proved here)* Proposition 11.40.

**11.25 Open image.** For $E$ without CM, $\rho_{E,\ell}(G_{\mathbb{Q}})$ is open in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ and $\bar\rho_{E,\ell}$ is surjective for almost all $\ell$; with CM by an order in $K$ and $\ell$ odd and unramified in $K$ the image of $\bar\rho_{E,\ell}$ lies in the normaliser of a Cartan subgroup and not in the Cartan subgroup; a rational $\ell$-isogeny exists only for $\ell \in \{2, 3, 5, 7, 11, 13, 17, 19, 37, 43, 67, 163\}$. *(stated; reference)* Theorem 11.43.

**11.26 Constraints on images.** For $\ell$ odd, $H = \bar\rho(G_{\mathbb{Q}})$ has $\det H = \mathbb{F}_\ell^\times$ and contains a conjugate of $\mathrm{diag}(1,-1)$; so $H \not\subseteq \mathrm{SL}_2(\mathbb{F}_\ell)$, $H \not\subseteq C_{ns}(\mathbb{F}_\ell)$, and $H$ is $\mathrm{GL}_2(\mathbb{F}_\ell)$, or lies in a Borel subgroup, or in the normaliser of a Cartan subgroup (not in the Cartan subgroup if $\bar\rho$ is irreducible), or has projective image $A_4$, $S_4$ or $A_5$. *(proved here)* Proposition 11.44.

**11.27 The mod 3 images.** For every $E/\mathbb{Q}$, CM or not, $\bar\rho_{E,3}(G_{\mathbb{Q}})$ is conjugate to exactly one of $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ (`3Nn`), $N_s(\mathbb{F}_3)$ (`3Ns`), $\mathrm{B}(\mathbb{F}_3)$ (`3B`), $\left(\begin{smallmatrix}1&*\\0&*\end{smallmatrix}\right)$ (`3B.1.1`), $\left(\begin{smallmatrix}*&*\\0&1\end{smallmatrix}\right)$ (`3B.1.2`), $C_s(\mathbb{F}_3)$ (`3Cs`), $\{\mathrm{diag}(1,\pm1)\}$ (`3Cs.1.1`); never to $C_{ns}(\mathbb{F}_3)$. The orbits on the four lines are the factorisation pattern of $\psi_3$. *(proved here; counts below conductor* 10,000*)* Theorem 11.45.

**11.28 3-isogeny paths.** For every $E/\mathbb{Q}$, CM or not, the rational 3-isogenies make each isogeny class a disjoint union of paths; the ends of a path of length at least one have one stable line in $E[3]$, of characters $\eta$ and $\chi_{-3}\eta^{-1}$, the interior vertices two. So the ends pair the images `3B.1.1` and `3B.1.2`, which occur equally often in every isogeny class. *(proved here)* Proposition 11.46.

**11.29 Labels.** Sutherland's labels name the subgroup; the dictionary for $\ell = 3$ and the labels occurring at $\ell = 5, 7, 11, 13$ below conductor 10,000. *(convention; data)* Convention 11.48.

**11.30 Zywina.** For non-CM $E/\mathbb{Q}$ and $\ell \le 13$ the possible images of $\bar\rho_{E,\ell}$ are listed, with the corresponding families of $j$-invariants. *(stated; reference)* Theorem 11.49.

**11.31 Dihedral representations.** For $\ell$ odd, $\bar\rho$ irreducible and $K$ quadratic: $\bar\rho|_{G_K}$ not absolutely irreducible $\iff$ $\bar\rho \simeq \bar\rho \otimes \chi_K$ $\iff$ $\bar\rho \otimes \overline{\mathbb{F}}_\ell \simeq \mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}} \psi$ $\iff$ $\ell \mid a_p$ for almost all $p$ inert in $K$; then every isomorphism $\bar\rho \to \bar\rho \otimes \chi_K$ has trace 0 and $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho \otimes \chi_K)$ is a line. *(proved here)* Proposition 11.52.

**11.32 $\mathbb{Q}(\sqrt{\ell^*})$ is automatic for $\ell \ge 5$ (Proposition prop:Sf).** For $\ell$ odd, $\ell \nmid N$, $\bar\rho$ irreducible: if $\ell \ge 5$, $\bar\rho$ is absolutely irreducible on $G_{\mathbb{Q}(\sqrt{\ell^*})}$; if $\ell = 3$ and it is not, $E$ is ordinary at 3. *(proved here, given Theorem 11.35)* Theorem 11.53.

**11.33 Proposition prop:ns.** If $E[3]$ is irreducible, the image is $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$; $\bar\rho_{E,3}$ is absolutely irreducible on $G_{\mathbb{Q}(\sqrt{-3})}$ unless the image is $N_s(\mathbb{F}_3)$, and then $\bar\rho_{E,3} \otimes \mathbb{F}_9 \simeq \mathrm{Ind}\,\psi$ with $\psi$ of order 4, $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \chi_{-3}$, and $h_3(E) = 1$; $\psi_3$ is a product of two irreducible quadratics in that case and irreducible otherwise. CM allowed. *(proved here)* Proposition 11.55.

**11.34 Lemma lem:h0.** For $\ell$ odd and $\bar\rho$ irreducible, $H^0(\mathbb{Q}, \mathrm{ad}^0\,T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell) \simeq \mathbb{Z}/\ell^{h_\ell(E)}$ with $h_\ell(E)$ the largest $k$ such that $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$; $h_\ell(E) = 0$ for every $\ell \ge 5$ and $h_3(E) \le 1$, with $h_3(E) = 1$ iff $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \chi_{-3}$ iff the image is $N_s(\mathbb{F}_3)$; for every $E$, CM or not. *(proved here)* Lemma 11.58.

**11.35 Proposition prop:ns-structure.** If the image of $\bar\rho_{E,3}$ is $N_s(\mathbb{F}_3)$: $\bar\rho(I_p) \subseteq \langle J \rangle$ for $p \ne 3$; every potentially multiplicative $p \ne 3$ has $3 \mid v_p(j)$; no $p \ne 3$ has type IV, IV$^*$ and no $p \ge 5$ type II, II$^*$; the set $R$ of primes $p \ne 3$ with $\rho_{E,3}(I_p)$ cyclic of order 4 is nonempty and is the set of primes $p \ne 3$ ramified in either quadratic field of $\psi_3$; for $p \in R$, $\mathrm{Frob}_p$ acts on $(\mathrm{ad}^0 V)^{I_p}$ by $\chi_{-3}(p)$ and $e_3(E,p) = v_3(p - \chi_{-3}(p)) \ge 1$; if $p \in R$, $p \ge 5$, the type is III or III$^*$ and $p \equiv \pm 1 \pmod{12}$. *(proved here)* Proposition 11.60.

**11.36** $S_f$**.** For $\ell$ odd, $\ell \nmid N$: $\ell \notin S_f \iff E[\ell]$ irreducible and $h_\ell(E) = 0 \iff E[\ell]$ irreducible and not ($\ell = 3$ and image $N_s(\mathbb{F}_3)$). *(proved here, given Theorem 11.35 for $\ell \geq 5$)* Corollary 11.63.

**11.37 Reducible** $\bar\rho$**.** For $\ell$ odd, $\ell \nmid N$, $E[\ell]$ reducible: $\bar\rho^{\mathrm{ss}} = \eta_1 \oplus \eta_2$, $\eta_1\eta_2 = \bar\chi_\ell$, $E$ ordinary at $\ell$, $\{\eta_i|_{I_\ell}\} = \{\omega, 1\}$, both unramified at multiplicative primes; if $E$ is semistable, $\{\eta_1, \eta_2\} = \{1, \bar\chi_\ell\}$, $a_p \equiv 1 + p \pmod{\ell}$, and $E$ or an $\ell$-isogenous curve has a rational point of order $\ell$. *(proved here, given Theorem 11.35)* Proposition 11.64.

**11.38 Eisenstein labels at 3.** $3 \mid \#E(\mathbb{Q})_{\mathrm{tors}}$ iff the image is `3B.1.1` or `3Cs.1.1`; `3Cs.1.1` means $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$, not the split Cartan subgroup; without CM such a curve has no rational point of order 9. *(proved here, the last statement citing Chapter 9)* Proposition 11.65.

**11.39 Optimal level.** An odd irreducible $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ is modular of level $N(\bar\sigma)$ and weight $k(\bar\sigma)$ (Khare-Wintenberger); for $\ell \nmid N(E')$, $E'[\ell]$ irreducible, and image not $N_s(\mathbb{F}_3)$ if $\ell = 3$, $\bar\rho_{E',\ell}$ arises from a newform of weight 2 on $\Gamma_0(N(\bar\rho_{E',\ell}))$. *(stated; reference; the elliptic-curve inputs proved here)* Theorem 11.68.

## Where the paper uses this chapter

- **Section 1.1** ($V$, $T$, $\rho$, $\bar\rho$, $A_f = \mathrm{ad}^0 V$, $\delta_p = \dim V^{I_p}$, the arithmetic and geometric Frobenius): Convention 11.1, Definition 11.2, Proposition 11.14, Corollary 11.21.
- **The hypothesis of Theorem thm:main** ($\ell \notin S_f$) and **Proposition prop:Sf**: Proposition 11.6, Proposition 11.52, Theorem 11.53, Corollary 11.63.
- **Definition def:terms** ($t_\ell(E, q)$, potentially multiplicative primes, $\mathrm{I}_n^*$ at 2) and **Section 1.5, Remark rem:ko** (the Tamagawa exponent): Lemma 11.17, Proposition 11.19.
- **Lemma lem:tamagawa-exp**: Theorem 11.15, Lemma 11.17.
- **Lemma lem:IV** and **Remark rem:types**: Lemma 11.10, Theorem 11.12, Lemma 11.25, Corollary 11.27, Proposition 11.29.
- **Lemma lem:euler, Proposition prop:shapes, Lemma lem:tamadd** (the images of inertia and their invariants on $V$ and $\mathrm{ad}^0 V$): Theorem 11.12, Theorem 11.15, Proposition 11.19, Theorem 11.22, Lemma 11.25; the $L$-factors themselves are computed in Chapter 19.
- **Proposition prop:forms, Remark rem:wild, Section sec:wild** (the types IV, IV$^*$ at 2): Lemma 11.25, Corollary 11.27, Remark 11.31.
- **Lemma lem:old** (the local analysis at $p \nmid \ell N$, at $p = \ell$, at $p \| N$ ramified or not): Proposition 11.6(iii), Theorem 11.15, Lemma 11.17, Theorem 11.35.
- **The optimal-level paragraph of Section 3 and Theorem thm:qual** (cases (a), (b), (c), the level $M = N(\bar\rho)$): Theorem 11.32, Definition 11.38, Proposition 11.40, Lemma 11.25, Proposition 11.29, Theorem 11.68, Example 11.70.
- **The proof of Theorem thm:main** (the vanishing of $H^0(\mathbb{Q}, W^D)$; the potentially multiplicative primes, 2 included): Lemma 11.58, Proposition 11.19.

- **Theorem thm:quant** ($\mathrm{ad}^0 T$ is unchanged by the twist): Proposition 11.5.
- **Proposition prop:ns**, **Proposition prop:ns-structure**, **Remark rem:ns**, **Section sec:anomaly**, **Conjecture conj:euler** ($h_\ell(E)$): Theorem 11.45, Proposition 11.55, Lemma 11.58, Proposition 11.60.
- **Section sec:eis, Proposition prop:no9, Conjecture conj:eis**: Proposition 11.64, Proposition 11.65.
- **Section sec:data and the appendix** (the mod 3 images checked against $\psi_3$; the function `ns3`): Theorem 11.45, Remark 11.50.

## 11.1 The representations $\rho_{E,\ell}$ and $\bar\rho_{E,\ell}$

This section fixes the objects of the paper's Section 1.1 and collects their basic properties: the determinant, complex conjugation, the effect of twists and isogenies, and the meaning of irreducibility. Most proofs were given in Chapter 5 and Chapter 9; here they are assembled in the form the paper uses them. The paper uses these facts in the hypothesis of Theorem thm:main, in Proposition prop:ns ("$\det\bar\rho_{E,3}=\chi_{-3}$ cuts out $\mathbb{Q}(\sqrt{-3})$", complex conjugation), in Lemma lem:h0 and Lemma lem:old ("irreducible, hence absolutely irreducible") and in Theorem thm:quant (twists).

**Convention 11.1** (notation of this chapter)**.** (a) *Objects.* $E/\mathbb{Q}$ is given by a global minimal model and has conductor $N$; $p$, $q$, $\ell$ are primes, $\ell$ is the coefficient prime (odd unless said otherwise) and $q$ usually a bad prime; $\ell^*=(-1)^{(\ell-1)/2}\ell$, $q^*=(-1)^{(q-1)/2}q$ for odd $\ell,q$. The letter $q$ is never the variable of a $q$-expansion in this chapter; the Tate parameter of $E$ at $q$ is $q_E$ (never $u$), and $u=q_Eq^{-v_q(q_E)}$ is its unit part.

(b) *Local data.* $D_q\simeq G_{\mathbb{Q}_q}$ and $I_q$ are a decomposition and an inertia group at $q$ for a fixed embedding $\overline{\mathbb{Q}}\to\overline{\mathbb{Q}}_q$ (Chapter 6); $f_q=f_q(E)$ is the conductor exponent and $\mathrm{sw}_q$ the Swan conductor; at a multiplicative prime $n_q=v_q(\Delta_{\min})$; $\Phi_q(\cdot)$ is the component group scheme of the Néron model at $q$, always written with an argument, $\Phi_q^{\mathrm{geom}}=\Phi_q(\overline{\mathbb{F}}_q)$, and $c_q=\#\Phi_q(\mathbb{F}_q)$ divides $\#\Phi_q^{\mathrm{geom}}$ (Chapter 10).

(c) *Frobenius.* $\mathrm{Frob}_p$ is an *arithmetic* Frobenius at $p$ (inducing $x\mapsto x^p$ on residue fields), well defined modulo $I_p$ and up to conjugation; the geometric Frobenius is written $\mathrm{Frob}_p^{-1}$, never $\Phi_p$, because $\Phi_q(\cdot)$ is the component group here. Group-theoretic statements (conjugation, the Tate matrices, the tame relation $\mathrm{Frob}\ t\ \mathrm{Frob}^{-1}=t^q$) use Frob; $L$-factors use $\mathrm{Frob}^{-1}$ and are the business of Chapter 19, with $L_q(A_f,s)=\det(1-\mathrm{Frob}_q^{-1}q^{-s}\mid(\mathrm{ad}^0V)^{I_q})^{-1}$, $L_q(\mathrm{Sym}^2 f,s)=L_q(A_f,s-1)=P_q(q^{-s})^{-1}$ and $e_\ell(E,q)=v_\ell(L_q(A_f,1)^{-1}/L_q^{\mathrm{nv}}(A_f,1)^{-1})$.

(d) *Characters.* $\chi_\ell\colon G_{\mathbb{Q}}\to\mathbb{Z}_\ell^\times$ is the $\ell$-adic cyclotomic character and $\bar\chi_\ell$ its reduction; a letter subscript ($\chi_\ell$, $\chi_p$) always means a cyclotomic character, and at a numerical prime we write $\chi_3^{\mathrm{cyc}}$. For a non-square $d\in\mathbb{Q}^\times$, $\chi_d$ is the quadratic character of $\mathbb{Q}(\sqrt{d})$, written with the discriminant ($\chi_{-3}$, $\chi_{-4}$, $\chi_{-7}$, $\chi_{-11}$); the quadratic character of conductor $q$ is $\chi_{q^*}$, never $\chi_q$. Since $\mathbb{Q}(\sqrt{\ell^*})\subset\mathbb{Q}(\zeta_\ell)$, $\chi_{\ell^*}=\bar\chi_\ell^{(\ell-1)/2}$ (Chapter 5); in particular $\bar\chi_3=\chi_{-3}$, while $\chi_{\ell^*}\neq\bar\chi_\ell$ for $\ell\geq5$. The letter $\varepsilon$ denotes an unramified character of $G_{\mathbb{Q}_q}$ of order at most 2, and $\omega$, $\omega_2$ the fundamental characters of Definition 11.33. $E^{(d)}=E\otimes\chi_d$ is the quadratic twist, and $E^{(q)}=E^{(q^*)}$ for odd $q$.

(e) *Matrices.* As in Chapter 1: $J = \left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$ (the paper's $J$), $w = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, $u = \left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$, $h = \mathrm{diag}(1,-1)$; the subgroups $C_s$, $N_s$, $C_{ns}$, $N_{ns}$, B (upper triangular), $\mathrm{U} = \langle u\rangle$, $Q_8$ and $C_4 = \langle J\rangle$ of $\mathrm{GL}_2(\mathbb{F}_3)$; with $\mathbb{F}_9 = \mathbb{F}_3[i]$, $J$ acts as multiplication by $-i$. The letter $\sigma$ denotes an element of order 3 of an image of inertia (as in Lemmas lem:IV and lem:euler) and $\tau$ one with $\tau^2 = -1$.

(f) *Labels.* Curves carry Cremona's labels (`11a1`); mod $\ell$ images carry Sutherland's labels (`3Ns`, `3B.1.1`), always together with the subgroup they name (Convention 11.48). The basis of $\mathrm{ad}^0 V$ attached to a basis $e_1, e_2$ of $V$ is $e = e_1 \otimes e_2^\vee$, $h = e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee$, $f = e_2 \otimes e_1^\vee$; $G$ acts on $\mathrm{ad}^0 V$ by conjugation.

**Definition 11.2.** Let $E/\mathbb{Q}$ be an elliptic curve and $\ell$ a prime. The *$\ell$-adic Tate module* is $T = T_\ell E = \varprojlim_k E[\ell^k]$ (transition maps multiplication by $\ell$), free of rank 2 over $\mathbb{Z}_\ell$, and $V = V_\ell E = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ (Chapter 9). The group $G_\mathbb{Q}$ acts on $E[\ell^k] \subset E(\overline{\mathbb{Q}})$ through the action on coordinates, compatibly with the transition maps; this defines

$$\rho = \rho_{E,\ell} \colon G_\mathbb{Q} \to \mathrm{Aut}_{\mathbb{Z}_\ell}(T) \simeq \mathrm{GL}_2(\mathbb{Z}_\ell),$$

its reductions $\rho \bmod \ell^k \colon G_\mathbb{Q} \to \mathrm{Aut}(E[\ell^k]) \simeq \mathrm{GL}_2(\mathbb{Z}/\ell^k)$, $E[\ell^k] = T/\ell^k T$, and $\bar\rho = \bar\rho_{E,\ell} = \rho \bmod \ell$ on $E[\ell]$. The *adjoint* is $A = \mathrm{ad}^0 T$, the trace-zero endomorphisms of $T$ with $G_\mathbb{Q}$ acting by conjugation, and $A_f = \mathrm{ad}^0 V = A \otimes \mathbb{Q}_\ell$ (the paper's notation). For a prime $p \neq \ell$, $\delta_p = \dim_{\mathbb{Q}_\ell} V^{I_p}$.

Why these objects. A single $E[\ell^k]$ is a finite Galois module, and it forgets how the torsion of different levels fits together; the Tate module packages all levels at once into a lattice $T$ over the complete discrete valuation ring $\mathbb{Z}_\ell$, on which $G_\mathbb{Q}$ acts by $2 \times 2$ matrices. Passing to $V = T \otimes \mathbb{Q}_\ell$ gives a vector space, where semisimplicity and eigenvalues behave as in linear algebra over a field of characteristic 0; the lattice $T$ itself keeps the integral information that the paper's divisibility statements are about. The adjoint $\mathrm{ad}^0 V$ appears because the paper's $L$-function is $L(A_f, s) = L(\mathrm{ad}^0 V, s)$, and the integer $\delta_p$ appears because the local factor of an $L$-function at $p$ only sees the inertia invariants (Convention 11.1(c)); Corollary 11.21 computes $\delta_p$ for every reduction type.

A small example shows what the action looks like at the bottom level. Write $E$ as $y^2 = g(x)$ with $g \in \mathbb{Q}[x]$ a separable cubic (possible for every $E/\mathbb{Q}$ after a change of coordinates). Then $E[2] = \{O, (e_1, 0), (e_2, 0), (e_3, 0)\}$ with $e_1, e_2, e_3$ the roots of $g$, and $G_\mathbb{Q}$ acts on the three nonzero points by permuting the roots. Since $\mathrm{GL}_2(\mathbb{F}_2)$ is the group of all permutations of the three nonzero vectors of $\mathbb{F}_2^2$, the image of $\bar\rho_{E,2}$ is, as a permutation group, the Galois group of $g$: trivial if $g$ splits, of order 2 if it has exactly one rational root, and $C_3$ or $S_3$ if it is irreducible. For $\ell$ odd the analogue of $g$ is the $\ell$-division polynomial, and for $\ell = 3$ it is used throughout Section 11.7.

**Proposition 11.3** (basic properties)**.** *Let $E/\mathbb{Q}$ and $\ell$ be as above.*

*(i) $\rho$ is continuous, and the kernel of $\rho \bmod \ell^k$ is $G_{\mathbb{Q}(E[\ell^k])}$, where $\mathbb{Q}(E[\ell^k])$ is the finite Galois extension generated by the coordinates of the points of $E[\ell^k]$.*

*(ii)* $\det \rho = \chi_\ell$. *Hence $\mathbb{Q}(\zeta_\ell) \subseteq \mathbb{Q}(E[\ell])$, $\det \bar\rho = \bar\chi_\ell$ is surjective onto $\mathbb{F}_\ell^\times$, and*

$$\bar\rho(G_{\mathbb{Q}(\sqrt{\ell^*})}) = \{g \in \bar\rho(G_\mathbb{Q}) : \det g \in (\mathbb{F}_\ell^\times)^2\} \qquad (\ell \text{ odd}).$$

*For $\ell = 3$: $\det \bar\rho_{E,3} = \chi_{-3}$, the field it cuts out is $\mathbb{Q}(\sqrt{-3})$, and $\bar\rho_{E,3}(G_{\mathbb{Q}(\sqrt{-3})}) = \bar\rho_{E,3}(G_\mathbb{Q}) \cap \mathrm{SL}_2(\mathbb{F}_3)$.*

*(iii) Let $\ell$ be odd and $c \in G_{\mathbb{Q}}$ a complex conjugation. Then $T = T^{c=1} \oplus T^{c=-1}$ with both summands free of rank one, so $\rho(c)$ is conjugate in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ to $\mathrm{diag}(1,-1)$; in particular $\bar\rho(c)$ has determinant $-1$, trace $0$ and the two distinct eigenvalues $1$, $-1$.*

*(iv) $\rho$ is unramified at every prime $p \nmid N\ell$.*

*Proof.* (i) Each $E[\ell^k]$ is a finite set of points with algebraic coordinates, so the action of $G_{\mathbb{Q}}$ on it factors through the finite quotient $\mathrm{Gal}(\mathbb{Q}(E[\ell^k])/\mathbb{Q})$, and the stabiliser of every point is open. The preimage under $\rho$ of the basic open subgroup $1 + \ell^k M_2(\mathbb{Z}_\ell)$ of $\mathrm{GL}_2(\mathbb{Z}_\ell)$ is the kernel of $\rho \bmod \ell^k$, which is therefore open; this is continuity (Chapter 5). The kernel of the action on $E[\ell^k]$ consists of the $g$ fixing all coordinates of all $\ell^k$-torsion points, which is $G_{\mathbb{Q}(E[\ell^k])}$; the extension is Galois because $E[\ell^k]$ is $G_{\mathbb{Q}}$-stable.

(ii) The Weil pairing $e_{\ell^k}\colon E[\ell^k] \times E[\ell^k] \to \mu_{\ell^k}$ is bilinear, alternating, nondegenerate and Galois equivariant, and the pairings are compatible as $k$ varies (Chapter 9). For a basis $P, Q$ of $E[\ell^k]$, $\zeta = e_{\ell^k}(P,Q)$ is a primitive $\ell^k$-th root of unity, and for $g \in G_{\mathbb{Q}}$ with $\rho(g) = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ in that basis,

$$g(\zeta) = e_{\ell^k}(gP, gQ) = e_{\ell^k}(aP + cQ, bP + dQ) = \zeta^{ad-bc},$$

so $\det(\rho(g) \bmod \ell^k) = \chi_\ell(g) \bmod \ell^k$ for every $k$, whence $\det\rho = \chi_\ell$. An element fixing $E[\ell]$ pointwise fixes $\zeta_\ell = e_\ell(P,Q)$, so $\mathbb{Q}(\zeta_\ell) \subseteq \mathbb{Q}(E[\ell])$. The reduction $\bar\chi_\ell$ is the character $G_{\mathbb{Q}} \to \mathrm{Gal}(\mathbb{Q}(\zeta_\ell)/\mathbb{Q}) \simeq \mathbb{F}_\ell^\times$, which is surjective. For $\ell$ odd, $\mathbb{Q}(\sqrt{\ell^*})$ is the fixed field in $\mathbb{Q}(\zeta_\ell)$ of the squares of $\mathbb{F}_\ell^\times$ (Chapter 5); so $g \in G_{\mathbb{Q}(\sqrt{\ell^*})}$ iff $\bar\chi_\ell(g) = \det\bar\rho(g)$ is a square. Both inclusions of the displayed equality follow. If $g \in G_{\mathbb{Q}(\sqrt{\ell^*})}$, then $\bar\rho(g)$ is an element of $\bar\rho(G_{\mathbb{Q}})$ with square determinant. Conversely, an element $h$ of $\bar\rho(G_{\mathbb{Q}})$ is $\bar\rho(g)$ for some $g \in G_{\mathbb{Q}}$, and if $\det h$ is a square, then $\bar\chi_\ell(g) = \det h$ is a square, so this very $g$ lies in $G_{\mathbb{Q}(\sqrt{\ell^*})}$ and $h \in \bar\rho(G_{\mathbb{Q}(\sqrt{\ell^*})})$. (The point is that membership of $g$ in $G_{\mathbb{Q}(\sqrt{\ell^*})}$ depends only on $\det\bar\rho(g)$, so it does not matter which preimage of $h$ is chosen.) For $\ell = 3$, $\mathbb{F}_3^\times = \{\pm 1\}$, its squares are $\{1\}$, and $\bar\chi_3(g) = \pm 1$ according as $g(\zeta_3) = \zeta_3^{\pm 1}$, that is, according as $g$ fixes $\sqrt{-3} = 2\zeta_3 + 1$ or not: $\bar\chi_3 = \chi_{-3}$.

(iii) Since $c^2 = 1$, $\rho(c)^2 = 1$, and $\det\rho(c) = \chi_\ell(c) = -1$ because $c(\zeta) = \zeta^{-1}$ on all roots of unity. As $2 \in \mathbb{Z}_\ell^\times$, the endomorphisms $\mathrm{pr}_\pm = \frac12(1 \pm \rho(c))$ of $T$ are complementary idempotents, so $T = \mathrm{pr}_+ T \oplus \mathrm{pr}_- T$ with $\mathrm{pr}_\pm T = T^{c=\pm 1}$. Both are direct summands of a free $\mathbb{Z}_\ell$-module of rank 2, hence free; their ranks add to 2; and they cannot be 2 and 0, since then $\rho(c) = \pm 1$ would have determinant 1. So both have rank one, and a basis $e_+, e_-$ adapted to the decomposition gives $\rho(c) = \mathrm{diag}(1,-1)$. Reducing modulo $\ell$ gives the statement on $\bar\rho(c)$; $1 \neq -1$ in $\mathbb{F}_\ell$ as $\ell$ is odd.

(iv) The conductor exponent $f_p$ is defined in Chapter 10 as the Artin conductor of $V$, so $p \nmid N$ means $\dim V - \dim V^{I_p} + \mathrm{sw}_p(V) = 0$; both terms are nonnegative, so $V^{I_p} = V$, and $I_p$ acts trivially on $V$, hence on $T \subset V$ (see also Theorem 11.9 below). □

**Theorem 11.4** (Faltings)**.** *Let $E$, $E'$ be elliptic curves over $\mathbb{Q}$ and $\ell$ a prime. Then $V_\ell E$ is a semisimple $\mathbb{Q}_\ell[G_{\mathbb{Q}}]$-module, and the natural map $\mathrm{Hom}_{\mathbb{Q}}(E,E') \otimes \mathbb{Z}_\ell \to \mathrm{Hom}_{G_{\mathbb{Q}}}(T_\ell E, T_\ell E')$ is an isomorphism. In particular $E$ and $E'$ are isogenous over $\mathbb{Q}$ iff $V_\ell E \simeq V_\ell E'$ as $G_{\mathbb{Q}}$-modules, iff $a_p(E) = a_p(E')$ for almost all $p$.*

We do not prove this; see Faltings [63] (the Tate conjecture for abelian varieties over number fields). The last equivalence follows from the first two with Chebotarev's theorem and the Brauer-Nesbitt theorem (Chapter 5), since semisimple representations in characteristic 0 are determined by their traces. The book uses Theorem 11.4 only as background (for instance in Chapter 6); every argument about the paper's curves uses $\bar\rho$, for which the needed statements are proved below.

**Proposition 11.5** (twists and isogenies)**.** *Let $E/\mathbb{Q}$ and let $\ell$ be odd.*

*(i) For $d \in \mathbb{Q}^\times$ not a square, $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi_d$. Hence $\mathrm{ad}^0\,\rho_{E^{(d)},\ell} \simeq \mathrm{ad}^0\,\rho_{E,\ell}$, the images of $\bar\rho_{E^{(d)},\ell}$ and $\bar\rho_{E,\ell}$ in $\mathrm{PGL}_2(\mathbb{F}_\ell)$ coincide, and $E^{(d)}[\ell]$ is irreducible iff $E[\ell]$ is.*

*(ii) For $\ell = 3$ (the labels are those of Theorem 11.45): twisting by $\chi_{-3}$ exchanges `3B.1.1` and `3B.1.2` and fixes every other image; twisting by $\chi_d$ with $d \not\equiv 1, -3$ modulo $\mathbb{Q}^{\times 2}$ sends `3B.1.1` and `3B.1.2` to `3B`, sends `3Cs.1.1` to `3Cs`, and fixes `3Ns`, `3Nn` and the surjective image.*

*(iii) If $E[\ell]$ is irreducible, every curve $E'$ isogenous to $E$ over $\mathbb{Q}$ has $T_\ell E' \simeq T_\ell E$ as $\mathbb{Z}_\ell[G_\mathbb{Q}]$-modules; so $E$ and $E'$ have the same $\rho$, $\bar\rho$, image label at $\ell$, adjoint $\mathrm{ad}^0\,T$, and $h_\ell$ (Lemma 11.58).*

*Proof.* (i) The first statement is proved in Chapter 9: an isomorphism $E \to E^{(d)}$ defined over $\mathbb{Q}(\sqrt{d})$ transforms the action of $g \in G_\mathbb{Q}$ into $\chi_d(g)$ times it. Concretely, take $E\colon y^2 = x^3 + ax + b$ and $E^{(d)}\colon dy^2 = x^3 + ax + b$, and $\phi(x, y) = (x, y/\sqrt{d})$, an isomorphism $E \to E^{(d)}$ over $\mathbb{Q}(\sqrt{d})$. For $P = (x, y) \in E(\overline{\mathbb{Q}})$ and $g \in G_\mathbb{Q}$,

$$g(\phi(P)) = \Big(g(x), \frac{g(y)}{g(\sqrt{d})}\Big) = \Big(g(x), \frac{g(y)}{\chi_d(g)\sqrt{d}}\Big) = [\chi_d(g)]\,\phi(g(P)),$$

because $[-1](x, y) = (x, -y)$ on $E^{(d)}$. So, transported to $E$ by $\phi$, the action of $g$ on $E^{(d)}[\ell^k]$ is $\chi_d(g)\rho(g)$, compatibly in $k$. Conjugation by $\chi_d(g)\rho(g)$ equals conjugation by $\rho(g)$, so the adjoint representations agree; and $\chi_d(g)\bar\rho(g)$ and $\bar\rho(g)$ have the same image in $\mathrm{PGL}_2(\mathbb{F}_\ell)$. A line stable under $\bar\rho$ is stable under $\bar\rho \otimes \chi_d$, and conversely.

(ii) Write $H$ for the image of $\bar\rho = \bar\rho_{E,3}$ and $H'$ for that of $\bar\rho \otimes \chi_d$; then $H' \subseteq H \cdot \{\pm 1\}$ and $H \subseteq H' \cdot \{\pm 1\}$, because $(\bar\rho \otimes \chi_d)(s) = \chi_d(s)\bar\rho(s)$ with $\chi_d(s) = \pm 1$ for every $s \in G_\mathbb{Q}$. If $H$ contains an element $g$ with $g^2 = -1$, choose $s \in G_\mathbb{Q}$ with $\bar\rho(s) = g$: then $((\bar\rho \otimes \chi_d)(s))^2 = g^2 = -1$, so $-1 \in H'$, hence $H' \subseteq H\{\pm 1\} = H$ and $H \subseteq H'\{\pm 1\} = H'$, and $H' = H$. The groups $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ and $N_s(\mathbb{F}_3)$ all contain $J$, with $J^2 = -1$; so the surjective image, `3Nn` and `3Ns` are fixed by every twist. In the reducible cases write $\bar\rho \sim \left(\begin{smallmatrix} \eta_1 & * \\ 0 & \eta_2 \end{smallmatrix}\right)$ with $\eta_1\eta_2 = \chi_{-3}$ (Proposition 11.3(ii)); then $\bar\rho \otimes \chi_d \sim \left(\begin{smallmatrix} \eta_1\chi_d & * \\ 0 & \eta_2\chi_d \end{smallmatrix}\right)$, and the twist is split iff $\bar\rho$ is (by (i), the two representations have the same stable lines). By Theorem 11.45, `3B.1.1` is the non-split case with $\eta_1 = 1$, `3B.1.2` the non-split case with $\eta_1 = \chi_{-3}$, `3B` the non-split case with $\eta_1 \notin \{1, \chi_{-3}\}$, and `3Cs.1.1`, `3Cs` the split cases with $\{\eta_1, \eta_2\} = \{1, \chi_{-3}\}$, respectively $\{\eta_1, \eta_2\} \neq \{1, \chi_{-3}\}$. Twisting by $\chi_{-3}$ sends $\eta_1 = 1$ to $\chi_{-3}$ and $\chi_{-3}$ to 1, and sends a character outside $\{1, \chi_{-3}\}$ to another such; twisting by $\chi_d$ with $\chi_d \notin \{1, \chi_{-3}\}$ sends both 1 and $\chi_{-3}$ outside $\{1, \chi_{-3}\}$. This gives every case.

(iii) Over $\mathbb{Q}$ every isogeny between curves of the class is a composite of isogenies of prime degree with $G_\mathbb{Q}$-stable cyclic kernels (Chapter 9). An isogeny of degree $\ell$ would have a $G_\mathbb{Q}$-stable kernel of order $\ell$ inside $E[\ell]$ (or inside $E''[\ell]$ for an intermediate curve $E''$, whose $\ell$-torsion is isomorphic to $E[\ell]$ by induction), contradicting irreducibility. So $E$ and $E'$ are linked by isogenies $\varphi$ of degree prime to $\ell$, and such a $\varphi$ induces a $G_\mathbb{Q}$-equivariant injection $T_\ell E \to T_\ell E'$ whose cokernel has order the $\ell$-part of $\deg\varphi$, which is 1; so it is an isomorphism. In detail: $\varphi$ is defined over $\mathbb{Q}$, so the map $\varphi_*\colon T_\ell E \to T_\ell E'$ it induces on compatible systems of torsion points commutes with $G_\mathbb{Q}$; the dual isogeny satisfies $\hat\varphi \circ \varphi = [\deg\varphi]$ on $E$ and $\varphi \circ \hat\varphi = [\deg\varphi]$ on $E'$ (Chapter 9); and multiplication by the integer $\deg\varphi$, a unit of $\mathbb{Z}_\ell$, is an automorphism of both Tate modules. So $\hat\varphi_* \circ \varphi_*$ and $\varphi_* \circ \hat\varphi_*$ are invertible, and $\varphi_*$ is a $G_\mathbb{Q}$-isomorphism. The other objects are built from $T$ alone. □

**Proposition 11.6** (irreducibility)**.** *Let $\ell$ be odd.*

*(i) $\bar\rho_{E,\ell}$ is irreducible iff it is absolutely irreducible.*

*(ii) $\bar\rho_{E,\ell}$ is reducible iff $E$ has a rational isogeny of degree $\ell$; for $\ell = 3$, iff the 3-division polynomial $\psi_3$ has a root in $\mathbb{Q}$.*

*(iii) The semisimplification $\bar\rho^{\mathrm{ss}}$ is determined up to isomorphism by the classes $a_p(E) \bmod \ell$ for $p$ in any set of primes of density one not dividing $N\ell$; if $\bar\rho$ is irreducible, $\bar\rho$ itself is. If $g$ is a newform of weight 2 with coefficients in the ring of integers $\mathcal{O}$ of a number field, $\lambda \mid \ell$ a prime of $\mathcal{O}$ with residue field $k$, and $a_p(g) \equiv a_p(E) \pmod\lambda$ for a set of primes of density one, then $\bar\rho_{g,\lambda} \simeq \bar\rho_{E,\ell} \otimes_{\mathbb{F}_\ell} k$ over a common finite extension of $k$ whenever $\bar\rho_{E,\ell}$ is irreducible.*

*Proof.* (i) Suppose $\bar\rho$ is irreducible and $L \subset E[\ell] \otimes \overline{\mathbb{F}}_\ell$ is a line stable under $G_{\mathbb{Q}}$. Choose the basis of Proposition 11.3(iii), in which $\bar\rho(c) = \mathrm{diag}(1,-1)$. The line $L$ is stable under $\bar\rho(c)$, hence an eigenline of it; the eigenvalues $1 \neq -1$ are distinct, so the eigenlines are $\overline{\mathbb{F}}_\ell e_+$ and $\overline{\mathbb{F}}_\ell e_-$, and $L = \overline{\mathbb{F}}_\ell e_\pm$ for one sign. Then for every $g$, $\bar\rho(g)e_\pm \in L \cap E[\ell] = \mathbb{F}_\ell e_\pm$, so $\mathbb{F}_\ell e_\pm$ is a stable line of $E[\ell]$: a contradiction. The converse is trivial. (This is the criterion "distinct eigenvalues in the base field" of Chapter 4.)

(ii) A $G_{\mathbb{Q}}$-stable line of $E[\ell]$ is a $G_{\mathbb{Q}}$-stable subgroup $C$ of order $\ell$, and these are exactly the kernels of isogenies $E \to E/C$ defined over $\mathbb{Q}$ of degree $\ell$ (Chapter 9). For $\ell = 3$, a line is $\{O, P, -P\}$ and is stable iff $g(P) = \pm P$ for all $g$, iff the $x$-coordinate $x(P)$, which is the same for $P$ and $-P$ and different for different lines, is rational. The four roots of $\psi_3$ are the $x$-coordinates of the four lines (Chapter 9).

(iii) For $p \nmid N\ell$ the characteristic polynomial of $\bar\rho(\mathrm{Frob}_p)$ is $X^2 - a_pX + p \bmod \ell$ (Proposition 11.14 below). By Chebotarev's theorem every element of the finite group $\bar\rho(G_{\mathbb{Q}})$ is $\bar\rho(\mathrm{Frob}_p)$ for a set of primes of positive density, which meets every set of density one (the complement of a set of density one has density 0, so it cannot contain a set of positive density); so the characteristic polynomials of all elements of the image are determined by the $a_p \bmod \ell$ on the given set, and by the Brauer-Nesbitt theorem so is the semisimplification (Chapter 5, Chapter 4). If $\bar\rho$ is irreducible it is semisimple and equal to $\bar\rho^{\mathrm{ss}}$. For the newform, $\bar\rho_{g,\lambda}$ (Chapter 15) has characteristic polynomial $X^2 - a_p(g)X + p$ at $\mathrm{Frob}_p$ for $p$ prime to the level and $\ell$, so the same argument over $k$ gives $\bar\rho^{\mathrm{ss}}_{g,\lambda} \simeq (\bar\rho_{E,\ell} \otimes k)^{\mathrm{ss}}$ after extension of scalars to a common finite field; when $\bar\rho_{E,\ell}$ is irreducible it is absolutely irreducible by (i), so $\bar\rho_{E,\ell} \otimes k$ is irreducible and semisimple, and $\bar\rho_{g,\lambda}$, having it as its semisimplification and the same dimension, is isomorphic to it (a representation whose semisimplification is irreducible is irreducible). The statement over $\mathbb{F}_\ell$ alone needs the absolute irreducibility: Chapter 5 gives the details. □

*In the paper* 11.7. Lemma lem:old, the case $p \nmid \ell N$: "Consequently $\bar\rho_h \simeq \bar\rho$ over a common finite extension of $\mathbb{F}_\ell$, by Chebotarev and Brauer-Nesbitt, $\bar\rho$ being irreducible, hence absolutely irreducible." This is Proposition 11.6(iii) with (i), the paper's newform $h$ being the book's $g$. The same reading of "irreducible" is used in Lemma lem:h0 (Schur's lemma for $\bar\rho$) and in Proposition prop:Sf; for $\bar\rho_{E,\ell}$, $\ell$ odd, the two notions agree, but for the restriction of $\bar\rho$ to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$ they do not (Proposition 11.55).

**Example 11.8.** For `11a1`, $y^2 + y = x^3 - x^2 - 10x - 20$, the 3-division polynomial is $\psi_3 = 3x^4 - 4x^3 - 60x^2 - 237x - 21$, irreducible over $\mathbb{Q}$, with discriminant $-3^3 \cdot 11^{10}$ and Galois group $S_4$; so $E[3]$ is irreducible and, by Theorem 11.45, $\bar\rho_{\texttt{11a1},3}$ is surjective. At $\ell = 5$ the label is

`5Cs.1.1`: $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ (the rational point $(5,5)$ of order 5 spans the first summand), so $E[5]$ is reducible and `11a1` has two rational 5-isogenies, to `11a2` and `11a3`. All of this was recomputed with PARI [148].

Here is the work. The coefficients are $a_1 = 0$, $a_2 = -1$, $a_3 = 1$, $a_4 = -10$, $a_6 = -20$, so $b_2 = a_1^2 + 4a_2 = -4$, $b_4 = 2a_4 + a_1a_3 = -20$, $b_6 = a_3^2 + 4a_6 = -79$ and $b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2 = 80 - 1 - 100 = -21$, and $\psi_3 = 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8$ is the quartic above; the discriminant of the curve is $-11^5$. To see the Galois group, factor $\psi_3$ modulo small primes of good reduction not dividing $3\,\mathrm{disc}(\psi_3)$: modulo 5 it is irreducible, so Frobenius at 5 acts on the roots as a 4-cycle; modulo 7 it is a linear factor times an irreducible cubic, so Frobenius at 7 acts as a 3-cycle (Chapter 5). A subgroup of $S_4$ containing a 4-cycle and a 3-cycle has order divisible by 12 and contains an odd permutation, so it is $S_4$; in particular $\psi_3$ is irreducible, and no line of $E[3]$ is rational. For the point $P = (5,5)$: it lies on the curve ($5^2 + 5 = 30 = 5^3 - 5^2 - 10\cdot 5 - 20$), and PARI gives $5P = O$ with $P \neq O$, so $P$ has order 5.

## 11.2 Néron-Ogg-Shafarevich, component groups and inertia invariants

This section relates the action of inertia on the torsion of $E$ to the reduction of $E$. The criterion of Néron, Ogg and Shafarevich says when inertia acts trivially; the component-group lemma computes the invariants of inertia on $E[\ell^k]$ at every bad prime; the theorem of Serre and Tate describes the image of inertia at a prime of potentially good reduction. The paper uses all three: $\delta_p$ in Section 1.1, "potentially good" and the finiteness of $\rho_{E,3}(I_q)$ in Lemma lem:IV, the unramifiedness of $\bar{\rho}_h$ in Lemma lem:old, the sentence "$N(\bar{\sigma})$ is prime to every prime at which $\bar{\sigma}$ is unramified" in the optimal-level paragraph, and the remark before Lemma lem:IV ("At an additive prime $q \neq \ell$ the inertia invariants of the torsion are those of the component group").

**Theorem 11.9** (Néron, Ogg, Shafarevich)**.** *Let $K/\mathbb{Q}_p$ be a finite extension with residue field $k$, $E/K$ an elliptic curve and $\ell \neq p$ a prime. The following are equivalent:*

*(a) $E$ has good reduction over $K$;*

*(b) $I_K$ acts trivially on $T_\ell E$;*

*(c) $I_K$ acts trivially on $E[m]$ for every integer $m \geq 1$ prime to $p$;*

*(d) $I_K$ acts trivially on $E[m]$ for infinitely many integers $m$ prime to $p$.*

*Moreover:*

*(e) $E$ has potentially good reduction iff $\rho_{E,\ell}(I_K)$ is finite, for one, equivalently for every, $\ell \neq p$;*

*(f) $E$ has potentially multiplicative reduction ($v_K(j) < 0$) iff $\rho_{E,\ell}(I_K)$ is infinite, and then $\rho_{E,\ell}(I_K)$ contains a unipotent element $\neq 1$.*

*For $E/\mathbb{Q}$ and $p \neq \ell$: $\rho_{E,\ell}$ is unramified at $p$ iff $p \nmid N$.*

*Proof.* (a)⇒(c). Let $m$ be prime to $p$, $P \in E[m]$, and let $L/K$ be a finite Galois extension containing the coordinates of all points of $E[m]$. A minimal model of $E$ over $K$ with good reduction still has good reduction over $L$. Let $\sigma \in I_K$. Since $\sigma$ acts trivially on the residue field of $L$, $\sigma(P)$ and $P$ have the same reduction, so $\sigma(P) - P$ lies in $E_1(L)$, the kernel of reduction,

and in $E[m]$. But $E_1(L)$ has no torsion of order prime to $p$ (Chapter 10: it is isomorphic to the formal group, on which multiplication by $m$ is bijective). So $\sigma(P) = P$.

(c)⇒(b) and (b)⇒(d). $T_\ell E = \varprojlim E[\ell^k]$, so $I_K$ acts trivially on $T_\ell E$ iff it acts trivially on every $E[\ell^k]$; and $m = \ell^k$, $k \geq 1$, are infinitely many integers prime to $p$.

(d)⇒(a). Suppose $E$ has bad reduction, and let $m$ be prime to $p$ with $I_K$ acting trivially on $E[m]$. The points of $E[m]$ then have coordinates in the maximal unramified extension $K^{\mathrm{ur}}$, hence in a finite unramified extension $K'$ of $K$, with finite residue field $k'$. The reduction type, the identity component and the component group do not change under unramified extension (Chapter 10), so $E$ has bad reduction over $K'$ and there are exact sequences

$$0 \to E_1(K') \to E_0(K') \to \tilde{E}_{\mathrm{ns}}(k') \to 0, \qquad 0 \to E_0(K') \to E(K') \to \Phi_E(k') \to 0,$$

where $\Phi_E(k') \subseteq \Phi_E(\bar{k})$ is a subgroup of the geometric component group, a finite group whose order does not depend on $K'$ (Chapter 10). The group $E_1(K')$ has no $m$-torsion, so $E_0(K')[m]$ injects into $\tilde{E}_{\mathrm{ns}}(k')[m]$. Here $\tilde{E}_{\mathrm{ns}}(k')$ is $k'^{+}$ (additive reduction), which has no $m$-torsion as $p \nmid m$, or $k'^{\times}$ or the kernel of the norm from the quadratic extension of $k'$ (multiplicative reduction), subgroups of $\bar{k}^{\times}$, whose $m$-torsion is cyclic of order at most $m$ (it consists of roots of $X^m - 1$, which has at most $m$ roots in a field, and a finite subgroup of the multiplicative group of a field is cyclic). Hence $\#E_0(K')[m] \leq m$ and

$$m^2 = \#E[m] = \#E(K')[m] \leq \#E_0(K')[m] \cdot \#\Phi_E(k') \leq m \cdot \#\Phi_E(\bar{k}).$$

So $m \leq \#\Phi_E(\bar{k})$: only finitely many $m$ are possible, contradicting (d).

(e), (f): see the sketch below. For $E/\mathbb{Q}$, the conductor exponent $f_p$ is the Artin conductor $\dim V - \dim V^{I_p} + \mathrm{sw}_p(V)$ of $V = V_\ell E$ (its definition in Chapter 10), a sum of two nonnegative terms; so $f_p = 0$ iff $V^{I_p} = V$ (the Swan term then vanishes), iff $\rho_{E,\ell}$ is unramified at $p$, iff $E$ has good reduction at $p$ by (a)⇔(b) over $K = \mathbb{Q}_p$. □

*Sketch of proof of (e) and (f).* By the semistable reduction theorem (Chapter 10) there is a finite extension $L/K$ over which $E$ has good or multiplicative reduction, and which is good iff $v_K(j) \geq 0$. If it is good, (a)⇒(b) over $L$ shows that $I_L = I_K \cap G_L$, a subgroup of finite index of $I_K$, acts trivially on $T_\ell E$, so $\rho(I_K)$ is finite. If it is multiplicative, Theorem 11.15 over $L$ (after a further unramified quadratic extension, the curve is a Tate curve) shows that $\rho(I_L)$ consists of the matrices $\left(\begin{smallmatrix} 1 & n\,t_\ell(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ with $n = -v_L(j) > 0$ and $t_\ell$ onto $\mathbb{Z}_\ell$, an infinite group of unipotent elements. The two cases are exclusive, which gives (e) and (f); the condition on $j$ does not involve $\ell$. The references are Serre and Tate [179, Section 1 and Section 2] and [185, VII.7]. □

The following lemma is proved in Chapter 10; we restate it because every inertia computation of this chapter rests on it.

**Lemma 11.10** (inertia invariants and component groups)**.** *Let $q \neq \ell$ be primes, $E/\mathbb{Q}_q$ with bad reduction, $\Phi_q^{\mathrm{geom}} = \Phi_q(\overline{\mathbb{F}}_q)$ and $k \geq 1$.*

*(i) If the reduction is additive, there is a $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$-equivariant isomorphism $E[\ell^k]^{I_q} \simeq \Phi_q^{\mathrm{geom}}[\ell^k]$; in particular $E(\mathbb{Q}_q)[\ell^\infty] \simeq \Phi_q(\mathbb{F}_q)[\ell^\infty]$.*

*(ii) If the reduction is multiplicative, there is an exact sequence of $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$-modules $0 \to \mu_{\ell^k} \otimes \varepsilon \to E[\ell^k]^{I_q} \to \Phi_q^{\mathrm{geom}}[\ell^k] \to 0$, where $\Phi_q^{\mathrm{geom}} \simeq \mathbb{Z}/n_q$ and $\varepsilon$ is the unramified character of Theorem 11.15.*

*(iii) Consequently, for $\ell$ odd, $\dim_{\mathbb{F}_\ell} E[\ell]^{I_q}$ is $1 + [\ell \mid n_q]$ for type $\mathrm{I}_n$ ($n \geq 1$), 1 for types IV, IV$^*$ when $\ell = 3$, and 0 for every other additive type.*

*Recollection of the proof.* This is proved in Chapter 10. In one paragraph: $E[\ell^k]^{I_q} = E(\mathbb{Q}_q^{\mathrm{ur}})[\ell^k]$, and $\mathbb{Q}_q^{\mathrm{ur}}$ is the union of the finite unramified extensions $K'$. For each, $E(K')/E_0(K') \simeq \Phi_q(k')$, $E_1(K')$ is uniquely $\ell$-divisible, and $E_0(K')/E_1(K') \simeq \tilde{E}_{\mathrm{ns}}(k')$ is $k'^+$ (additive: uniquely $\ell$-divisible) or a torus (multiplicative: $\ell$-divisible after enlarging $k'$, with $\ell^k$-torsion $\mu_{\ell^k}$ twisted by the action of Frobenius on the tangent directions at the node). The snake lemma for multiplication by $\ell^k$ and the union over $K'$ give (i) and (ii). Part (iii) uses the orders of $\Phi_q^{\mathrm{geom}}$: $n$ for $\mathrm{I}_n$, $1, 2, 3, 4, 4, 3, 2, 1$ for $\mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{I}_n^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ (Chapter 10); for $\ell$ odd the only additive types with nonzero $\Phi_q^{\mathrm{geom}}[\ell]$ are IV and IV$^*$ at $\ell = 3$, with $\Phi_q^{\mathrm{geom}} \simeq \mathbb{Z}/3$. □

*Warning* 11.11. The lemma distinguishes additive from multiplicative reduction, and it concerns $\Phi_q^{\mathrm{geom}}$, not the Tamagawa number $c_q = \#\Phi_q(\mathbb{F}_q)$. For good reduction $E[\ell^k]^{I_q} = E[\ell^k]$ (Theorem 11.9). The Frobenius-invariant form $E(\mathbb{Q}_q)[\ell^\infty] \simeq \Phi_q(\mathbb{F}_q)[\ell^\infty]$ holds only for additive reduction: at a split multiplicative prime with $q \equiv 1 \pmod{\ell}$, $E(\mathbb{Q}_q)$ contains $\mu_\ell$ although $\ell$ may not divide $c_q$. The lemma holds at every $q \neq \ell$, including $q = 2, 3$; this is what makes it useful at the wild primes, where the Kodaira type does not determine the image of inertia.

**Theorem 11.12** (Serre and Tate; Kraus)**.** *Let $q$ be a prime, $E/\mathbb{Q}_q$ with potentially good reduction, and $\ell \neq q$.*

*(i) The kernel of $\rho_{E,\ell}|_{I_q}$ is $\mathrm{Gal}(\overline{\mathbb{Q}}_q/L)$, where $L = \mathbb{Q}_q^{\mathrm{ur}}(E[m])$ for any $m \geq 3$ prime to $q$ is the smallest extension of $\mathbb{Q}_q^{\mathrm{ur}}$ over which $E$ has good reduction. So $\rho_{E,\ell}(I_q) \simeq \mathrm{Gal}(L/\mathbb{Q}_q^{\mathrm{ur}})$ does not depend on $\ell$; moreover the characteristic polynomial of $\rho_{E,\ell}(\sigma)$, $\sigma \in I_q$, has coefficients in $\mathbb{Z}$ and does not depend on $\ell$.*

*(ii) $\rho_{E,\ell}(I_q)$ is a finite subgroup of $\mathrm{SL}_2(\mathbb{Z}_\ell)$. For $\ell$ odd, reduction modulo $\ell$ is injective on it. A finite subgroup of $\mathrm{SL}_2(\mathbb{Q}_\ell)$ with a nonzero fixed vector is trivial; so $V^{I_q} = 0$ unless the reduction is good.*

*(iii) $\rho_{E,\ell}(I_q)$ is cyclic of order 1, 2, 3, 4 or 6 if $q \geq 5$; one of these or $\mathbb{Z}/3 \rtimes \mathbb{Z}/4$ (order 12) if $q = 3$; one of these, $Q_8$ or $\mathrm{SL}_2(\mathbb{F}_3)$ if $q = 2$.*

*(iv) If $q \geq 5$, $\rho_{E,\ell}(I_q)$ has order prime to $q$, factors through the tame quotient of $I_q$, is cyclic, and has order $12/\gcd(v_q(\Delta_{\min}), 12)$, that is, $1, 6, 4, 3, 2, 3, 4, 6$ for $\mathrm{I}_0, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$.*

*Proof.* (i) is the theorem of Serre and Tate [179, Section 2], sketched in Chapter 10; we do not prove it here. (iii) is Kraus's classification [107], stated in Chapter 10.

(ii) Finiteness is (i) (or Theorem 11.9(e)). The determinant is $\chi_\ell$, which is trivial on $I_q$ because $\mu_{\ell^\infty}$ lies in the unramified extension $\mathbb{Q}_q^{\mathrm{ur}}$ ($q \neq \ell$); so the image lies in $\mathrm{SL}_2(\mathbb{Z}_\ell)$. The kernel of $\mathrm{GL}_2(\mathbb{Z}_\ell) \to \mathrm{GL}_2(\mathbb{F}_\ell)$ is $1 + \ell M_2(\mathbb{Z}_\ell)$, which for $\ell$ odd is torsion-free (Chapter 1); a finite subgroup meets it trivially, so reduction is injective on it. (The reason, in brief: write $g \neq 1$ in the kernel as $g = 1 + \ell^r A$ with $r \geq 1$ and $A \not\equiv 0 \pmod{\ell}$. For $m$ prime to $\ell$, $g^m \equiv 1 + m\ell^r A \not\equiv 1 \pmod{\ell^{r+1}}$; and the binomial expansion gives $g^\ell \equiv 1 + \ell^{r+1} A \pmod{\ell^{r+2}}$, because $\binom{\ell}{k}\ell^{kr}$ is divisible by $\ell^{2r+1}$ for $2 \leq k < \ell$ and $\ell^{r\ell}$ by $\ell^{r+2}$ when $\ell \geq 3$. Iterating, $g^{\ell^j} = 1 + \ell^{r+j} A_j$ with $A_j \equiv A \not\equiv 0 \pmod{\ell}$, and then $g^{\ell^j m} \neq 1$ for $m$ prime to $\ell$: no power of $g$ is 1. For $\ell = 2$ this fails: $-1 = 1 + 2 \cdot (-1)$ has order 2, which is why the injectivity is stated for $\ell$ odd.) Let $G \subseteq \mathrm{SL}_2(\mathbb{Q}_\ell)$ be finite and $v \neq 0$ with $gv = v$ for all $g \in G$. Each $g \in G$ has finite order, hence is diagonalisable over $\overline{\mathbb{Q}}_\ell$ (its

minimal polynomial divides $X^r-1$, which has distinct roots in characteristic 0); one eigenvalue is 1 and the determinant is 1, so both eigenvalues are 1 and $g=1$. Hence $V^{I_q}\neq 0$ forces $\rho(I_q)=1$, which is good reduction by Theorem 11.9.

(iv) The order of $\rho(I_q)$ divides 24 by (iii), hence is prime to $q\geq 5$. The wild inertia group $P_q$ is a pro-$q$ group, so its image is a $q$-group of order prime to $q$, hence trivial, and $\rho|_{I_q}$ factors through the tame quotient $I_q/P_q$. A finite quotient of the tame inertia group is cyclic (Chapter 7). The order is Kraus's semistability defect $12/\gcd(v_q(\Delta_{\min}),12)$ (Chapter 10, where it is deduced from (i)); the values by type follow from $v_q(\Delta_{\min})=0,2,3,4,6,8,9,10$ for the potentially good types at $q\geq 5$ (Chapter 10). □

*Remark* 11.13 (independence of $\ell$). Theorem 11.12(i), with Proposition 11.14, Theorem 11.15 and Proposition 11.19 below, shows that the Frobenius-semisimplified Weil-Deligne representation of $V_\ell E|_{G_{\mathbb{Q}_q}}$ $(q\neq\ell)$ is defined over $\mathbb{Q}$ and does not depend on $\ell$: at a good prime it is unramified with Frobenius polynomial $X^2-a_qX+q$; at a potentially multiplicative prime it is $\mathrm{sp}(2)$ twisted by an $\ell$-independent quadratic character; at a potentially good prime, inertia acts through the $\ell$-independent group $\mathrm{Gal}(L/\mathbb{Q}_q^{\mathrm{ur}})$ with integral characteristic polynomials, and Frobenius acts through its action on the good reduction of $E$ over $L$ ([179, Section 2]). This is the statement quoted in Chapter 7; we only sketch it, as the paper needs nothing beyond the explicit shapes proved below.

## 11.3 Good and multiplicative reduction; Lemma lem:tamagawa-exp

This section describes $\rho|_{G_{\mathbb{Q}_p}}$ at the primes $p\neq\ell$ of good and of potentially multiplicative reduction, and proves the paper's Lemma lem:tamagawa-exp. The matrices of Theorem 11.15 are the ones the paper writes in the proof of Lemma lem:tamfactor (Chapter 18) and quotes from [38, Theorem 3.1(e)] in Lemma lem:old; the potentially multiplicative additive case, including $q=2$, is what the proof of Theorem thm:main needs at the primes of type $\mathrm{I}_n^*$.

**Proposition 11.14** (good reduction)**.** *Let $p\nmid N\ell$ and $a_p=p+1-\#\tilde{E}(\mathbb{F}_p)$. Then $\rho$ is unramified at $p$; the arithmetic Frobenius $\mathrm{Frob}_p$ has characteristic polynomial $X^2-a_pX+p$ on $V$, with roots $\alpha$, $\beta$; the geometric Frobenius $\mathrm{Frob}_p^{-1}$ has roots $\alpha^{-1}$, $\beta^{-1}$, characteristic polynomial $X^2-\frac{a_p}{p}X+\frac{1}{p}$; on $\mathrm{ad}^0V$ both act with eigenvalues $\alpha/\beta,1,\beta/\alpha$; $\delta_p=2$. The characteristic polynomial of $\bar\rho(\mathrm{Frob}_p)$ is $X^2-a_pX+p \bmod \ell$, so $\mathrm{tr}\,\bar\rho(\mathrm{Frob}_p)\equiv a_p$ and $\det\bar\rho(\mathrm{Frob}_p)\equiv p \pmod\ell$. For $\ell=3$ and $p\nmid 3\,\mathrm{disc}(\psi_3)$, the orbits of $\bar\rho(\mathrm{Frob}_p)$ on the four lines of $E[3]$ have the lengths of the degrees of the irreducible factors of $\psi_3 \bmod p$.*

*Proof.* Unramified: Theorem 11.9. Reduction gives a $G_{\mathbb{Q}_p}$-equivariant isomorphism $T_\ell E\simeq T_\ell\tilde{E}$ (Theorem 11.9, proof of (a)⇒(c), and Chapter 10): reduction $E[\ell^k]\to\tilde{E}[\ell^k]$ is injective because its kernel lies in the kernel of reduction $E_1$, which has no $\ell$-torsion, and it is then bijective because both groups have $\ell^{2k}$ elements; it commutes with $G_{\mathbb{Q}_p}$, which acts on $\tilde{E}(\overline{\mathbb{F}}_p)$ through $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$. On $T_\ell\tilde{E}$ the element $\mathrm{Frob}_p$ acts as the Frobenius endomorphism $\pi_p$ of $\tilde{E}/\mathbb{F}_p$ (both raise coordinates to the $p$-th power), and the characteristic polynomial of $\pi_p$ on $T_\ell\tilde{E}$ is $X^2-a_pX+p$ (Chapter 9: $\deg(1-\pi_p)=\#\tilde{E}(\mathbb{F}_p)$ and $\deg\pi_p=p$). The inverse has the inverse eigenvalues. On $\mathrm{ad}^0V$, if $\rho(\mathrm{Frob}_p)$ has eigenvectors $v_\alpha$, $v_\beta$ (over $\overline{\mathbb{Q}}_\ell$; when $\alpha=\beta$ the eigenvalues on $\mathrm{ad}^0V$ are $1,1,1$ in any case), conjugation by it multiplies $v_\alpha\otimes v_\beta^\vee$, $v_\alpha\otimes v_\alpha^\vee-v_\beta\otimes v_\beta^\vee$, $v_\beta\otimes v_\alpha^\vee$ by $\alpha/\beta$, $1$, $\beta/\alpha$; the inverse gives the same set. $\delta_p=2$ as $I_p$ acts trivially. Reduction modulo $\ell$ of the characteristic

polynomial gives that of $\bar\rho(\mathrm{Frob}_p)$. The last statement is Chapter 5 (Frobenius acting on the roots of a polynomial whose discriminant is prime to $p$), applied to the four roots of $\psi_3$, which correspond to the four lines (Proposition 11.6(ii)). □

**Theorem 11.15** (the Tate shape)**.** *Let $q$ be a prime of multiplicative reduction, $\ell \neq q$, $n_q = v_q(\Delta_{\min}) = -v_q(j)$, $\varepsilon\colon G_{\mathbb{Q}_q} \to \{\pm 1\}$ the unramified character with $\varepsilon(\mathrm{Frob}_q) = a_q(E)$ ($+1$ for split, $-1$ for nonsplit reduction), $q_E \in \mathbb{Q}_q^\times$ the Tate parameter, and $\kappa\colon G_{\mathbb{Q}_q} \to \mathbb{Z}_\ell$ the Kummer cocycle of $q_E$ (after fixing a generator of $\mathbb{Z}_\ell(1)$), $g(q_E^{1/\ell^k}) = \zeta_{\ell^k}^{\kappa(g)} q_E^{1/\ell^k}$. There is a $\mathbb{Z}_\ell$-basis $e_1, e_2$ of $T$ in which*

$$\rho(g) = \begin{pmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa(g) \\ 0 & \varepsilon(g) \end{pmatrix} \qquad (g \in G_{\mathbb{Q}_q}).$$

*Consequently:*

*(i) on $I_q$, $\rho(\sigma) = \left(\begin{smallmatrix} 1 & n_q\, t_\ell(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$, where $t_\ell\colon I_q \to \mathbb{Z}_\ell$ is the $\ell$-adic tame character, which is onto; $\rho$ factors on $I_q$ through its maximal pro-$\ell$ quotient, and $\rho(I_q)$ is infinite;*

*(ii) $V^{I_q} = \mathbb{Q}_\ell e_1$, so $\delta_q = 1$, and $\mathrm{Frob}_q$ acts on it by $a_q(E)\, q$;*

*(iii) $\mathbb{Q}_\ell e_1$ is the unique $G_{\mathbb{Q}_q}$-stable line of $V$, and $V$ is a non-split extension of $\varepsilon$ by $\varepsilon\chi_\ell$;*

*(iv) the Weil-Deligne representation of $V|_{G_{\mathbb{Q}_q}}$ is $\mathrm{sp}(2) \otimes \mu_\varepsilon$, of conductor exponent 1;*

*(v) $(\mathrm{ad}^0 V)^{I_q} = \mathbb{Q}_\ell e$, $e = e_1 \otimes e_2^\vee$, on which $\mathrm{Frob}_q$ acts by $q$ and the geometric Frobenius by $q^{-1}$.*

*Proof.* By Chapter 10, $E$ over $\mathbb{Q}_q$ is the twist by $\varepsilon$ of the Tate curve $E_{q_E}$ (trivial twist when the reduction is split), and on $T_\ell E_{q_E}$, with $e_1 = (\zeta_{\ell^k})_k$ and $e_2 = (q_E^{1/\ell^k})_k$ (compatible systems in $\overline{\mathbb{Q}}_q^\times / q_E^{\mathbb{Z}}$), $g$ acts by $\left(\begin{smallmatrix} \chi_\ell(g) & \kappa(g) \\ 0 & 1 \end{smallmatrix}\right)$: indeed $g\zeta_{\ell^k} = \zeta_{\ell^k}^{\chi_\ell(g)}$ and $g(q_E^{1/\ell^k}) = \zeta_{\ell^k}^{\kappa(g)} q_E^{1/\ell^k}$. Twisting multiplies every matrix by the scalar $\varepsilon(g)$, which gives the displayed shape; the corner is $\varepsilon\kappa$, as the cocycle relation $c(gh) = \chi_\ell\varepsilon(g)c(h) + c(g)\varepsilon(h)$ requires. (The convention is that the columns of $\rho(g)$ are the images of the basis vectors: $ge_1 = \chi_\ell\varepsilon(g)e_1$ and $ge_2 = \varepsilon\kappa(g)e_1 + \varepsilon(g)e_2$. The Kummer cocycle satisfies $\kappa(gh) = \kappa(g) + \chi_\ell(g)\kappa(h)$, since $gh(q_E^{1/\ell^k}) = g(\zeta_{\ell^k}^{\kappa(h)} q_E^{1/\ell^k}) = \zeta_{\ell^k}^{\chi_\ell(g)\kappa(h)+\kappa(g)} q_E^{1/\ell^k}$; multiplying by $\varepsilon(gh) = \varepsilon(g)\varepsilon(h)$ gives exactly the upper right entry $\chi_\ell\varepsilon(g)\cdot\varepsilon\kappa(h) + \varepsilon\kappa(g)\cdot\varepsilon(h)$ of the product $\rho(g)\rho(h)$, so the displayed formula is a homomorphism.) The equalities $\varepsilon(\mathrm{Frob}_q) = a_q(E)$ and $n_q = v_q(q_E) = -v_q(j)$ are in Chapter 10.

(i) On $I_q$, $\varepsilon = 1$ and $\chi_\ell = 1$ ($q \neq \ell$). Write $q_E = q^{n_q} u$ with $u \in \mathbb{Z}_q^\times$; the Kummer cocycle is additive in its argument, $\kappa = n_q\kappa_q + \kappa_u$. The $\ell^k$-th roots of the unit $u$ lie in $\mathbb{Q}_q^{\mathrm{ur}}$ (Hensel's lemma, as $q \neq \ell$), so $\kappa_u$ vanishes on $I_q$; and $\kappa_q$ restricted to $I_q$ is the tame character $t_\ell$, which is onto $\mathbb{Z}_\ell$ (Chapter 7). Since $n_q \neq 0$, the image is infinite, and since $t_\ell$ factors through the maximal pro-$\ell$ quotient of $I_q$, so does $\rho|_{I_q}$.

(ii) A vector $xe_1 + ye_2$ is fixed by $\left(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}\right)$ iff $sy = 0$; some $s = n_q t_\ell(\sigma)$ is nonzero, so $V^{I_q} = \mathbb{Q}_\ell e_1$; $\mathrm{Frob}_q$ acts on $e_1$ by $\chi_\ell\varepsilon(\mathrm{Frob}_q) = q\, a_q(E)$.

(iii) A $G_{\mathbb{Q}_q}$-stable line is stable under $\rho(\sigma) = \left(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}\right)$ with $s \neq 0$, whose only eigenline is $\mathbb{Q}_\ell e_1$. So $\mathbb{Q}_\ell e_1$ is the only stable line, and $V$ is not a direct sum of two characters: the extension $0 \to \varepsilon\chi_\ell \to V \to \varepsilon \to 0$ is non-split.

(iv) The Weil-Deligne representation attached to this shape is computed in Chapter 7: the monodromy operator is $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ (from the logarithm of the unipotent action of $t_\ell$), Frobenius acts through $\varepsilon\chi_\ell$ and $\varepsilon$, and the conductor exponent is $\dim V - \dim V^{I_q} = 1$.

(v) In the basis $e, h, f$ of $\mathrm{ad}^0 V$ (Convention 11.1), conjugation by $\left(\begin{smallmatrix}1 & s\\ 0 & 1\end{smallmatrix}\right)$ sends $e \mapsto e$, $h \mapsto h - 2se$, $f \mapsto f + sh - s^2 e$. So $ae + bh + cf$ is fixed iff $(-2bs - cs^2)e + cs\,h = 0$, and for $s \neq 0$ this means $c = 0$ and then $b = 0$ (in $\mathbb{Q}_\ell$, 2 is invertible): $(\mathrm{ad}^0 V)^{I_q} = \mathbb{Q}_\ell e$. For an upper triangular $g = \left(\begin{smallmatrix}a & b\\ 0 & d\end{smallmatrix}\right)$ one has $geg^{-1} = \frac{a}{d}\,e$; for a Frobenius lift this is $\chi_\ell\varepsilon/\varepsilon = \chi_\ell(\mathrm{Frob}_q) = q$. The geometric Frobenius acts by $q^{-1}$. □

*In the paper* 11.16. Lemma lem:old, the cases $p \,\|\, N$: "$\rho_{E,\ell}|_{D_p}$ is an extension of an unramified character $\varepsilon$ by $\varepsilon\chi_\ell$ with $\varepsilon(\mathrm{Frob}_p) = a_p(f) = \pm 1$ [38, Theorem 3.1(e)]"; and "A non-split extension of two characters has a unique stable line". These are Theorem 11.15 and its part (iii) (and, modulo $\ell$, Lemma 11.17(iii)). In the proof of Lemma lem:tamfactor the paper uses the matrix of the arithmetic Frobenius, with corner $\varepsilon\kappa$; the factor $1 - q^{-1-s}$ of the table of Section 1 comes from the geometric Frobenius acting by $q^{-1}$ on $(\mathrm{ad}^0 V)^{I_q}$, (v). The two conventions are each right in their place; the paper's Section 1.1 says so ("The matrices in the proof of Lemma lem:tamfactor are those of the arithmetic Frobenius").

**Lemma 11.17** (the paper's Lemma lem:tamagawa-exp)**.** *Let $q$ be a prime of multiplicative reduction, $n_q = v_q(\Delta_E)$, and $\ell \neq q$ a prime. Then $\rho_{E,\ell} \bmod \ell^t$ is unramified at $q$ if and only if $\ell^t \mid n_q$. This holds for split and for nonsplit reduction. Moreover:*

*(i) the Tamagawa exponent $t_q = \max\{t \geq 0 : \rho \bmod \ell^t$ is unramified at $q\}$ equals $v_\ell(n_q) = v_\ell(-v_q(j)) = t_\ell(E, q)$;*

*(ii) $\#E[\ell^t]^{I_q} = \ell^{\,t+\min(t, v_\ell(n_q))}$;*

*(iii) if $\ell \nmid n_q$, then $\bar\rho|_{D_q}$ is ramified, a non-split extension of $\varepsilon$ by $\varepsilon\bar\chi_\ell$ with a unique stable line, and $\bar\rho(I_q)$ has order $\ell$;*

*(iv) if $\ell \mid n_q$, then $\bar\rho|_{D_q}$ is unramified and $\bar\rho(\mathrm{Frob}_q)$ has eigenvalues $a_q(E)q$ and $a_q(E)$ modulo $\ell$.*

*First proof (the paper's, expanded).* Let $K = \mathbb{Q}_q$ if the reduction is split and the unramified quadratic extension of $\mathbb{Q}_q$ if not. Then $I_K = I_q$ (an unramified extension does not change the inertia group), and over $K$ the curve is isomorphic to the Tate curve $E_{q_E}$ (Chapter 10: over the unramified quadratic extension the reduction becomes split, and $\varepsilon$ becomes trivial). By Tate's theorem $E_{q_E}(\overline{\mathbb{Q}}_q) \simeq \overline{\mathbb{Q}}_q^{\times}/q_E^{\mathbb{Z}}$ Galois-equivariantly, and $E[\ell^t]$ is the subgroup of classes $\zeta^a\alpha^b$, where $\zeta$ is a primitive $\ell^t$-th root of unity, $\alpha$ an $\ell^t$-th root of $q_E$, and $a, b \in \mathbb{Z}/\ell^t$. Since $q \neq \ell$, $\mu_{\ell^t} \subset \mathbb{Q}_q^{\mathrm{ur}}$, so $I_q$ fixes $\zeta$. For $\sigma \in I_q$, $\sigma(\alpha)/\alpha$ is an $\ell^t$-th root of unity; the class of $\alpha$ is fixed iff $\sigma(\alpha)/\alpha \in q_E^{\mathbb{Z}}$, and as $q_E^{\mathbb{Z}} \cap \mu_{\ell^t} = \{1\}$ ($v_q(q_E) > 0$), iff $\sigma(\alpha) = \alpha$. So $I_q$ acts trivially on $E[\ell^t]$ iff $\alpha \in \mathbb{Q}_q^{\mathrm{ur}}$ for some, equivalently every, choice of $\alpha$, that is (Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$, which contains $\mu_{\ell^t}$), iff $q_E$ is an $\ell^t$-th power in $\mathbb{Q}_q^{\mathrm{ur}}$. Write $q_E = q^{n_q}u$ with $u$ a unit. The unit $u$ is an $\ell^t$-th power in $\mathbb{Q}_q^{\mathrm{ur}}$: the polynomial $X^{\ell^t} - u$ is separable modulo $q$ and has a root modulo $q$ in $\overline{\mathbb{F}}_q$, which lifts by Hensel's lemma to a finite unramified extension (Chapter 7). So $q_E$ is an $\ell^t$-th power in $\mathbb{Q}_q^{\mathrm{ur}}$ iff $q^{n_q}$ is; and $q$ is a uniformiser of $\mathbb{Q}_q^{\mathrm{ur}}$, whose valuation group is $\mathbb{Z}$, so $q^{n_q} = y^{\ell^t}$ forces $\ell^t \mid n_q$, while conversely $\ell^t \mid n_q$ gives $q^{n_q} = (q^{n_q/\ell^t})^{\ell^t}$. Hence $\rho \bmod \ell^t$ is unramified iff $\ell^t \mid n_q$. Nothing depended on the reduction being split. □

*Second proof (component groups).* By Lemma 11.10(ii), $\#E[\ell^t]^{I_q} = \#\mu_{\ell^t} \cdot \#\Phi_q^{\mathrm{geom}}[\ell^t] = \ell^t \cdot \#(\mathbb{Z}/n_q)[\ell^t] = \ell^{t+\min(t, v_\ell(n_q))}$, which is (ii). So $E[\ell^t]^{I_q} = E[\ell^t]$, of order $\ell^{2t}$, iff $\min(t, v_\ell(n_q)) = t$, iff $\ell^t \mid n_q$. (A third, one-line proof: by Theorem 11.15(i), $\rho \bmod \ell^t$ on $I_q$ is $\left(\begin{smallmatrix}1 & n_q t_\ell\\ 0 & 1\end{smallmatrix}\right) \bmod \ell^t$, trivial iff $\ell^t \mid n_q$ since $t_\ell$ is onto $\mathbb{Z}_\ell$.) □

*Proof of the supplements.* (i) is the main statement, with $n_q = -v_q(j)$ (Theorem 11.15) and the definition of $t_\ell(E,q)$. (ii) is the second proof (or the matrices: $(x,y) \in (\mathbb{Z}/\ell^t)^2$ is fixed by all $\left(\begin{smallmatrix} 1 & n_q s \\ 0 & 1 \end{smallmatrix}\right)$, $s \in \mathbb{Z}_\ell$, iff $n_q y \equiv 0 \pmod{\ell^t}$, which leaves $\ell^t$ choices of $x$ and $\ell^{\min(t, v_\ell(n_q))}$ of $y$). (iii) If $\ell \nmid n_q$, $\bar\rho(\sigma) = \left(\begin{smallmatrix} 1 & n_q t_\ell(\sigma) \\ 0 & 1 \end{smallmatrix}\right) \bmod \ell$ runs through the whole unipotent group of order $\ell$; the extension of the unramified $\varepsilon$ by $\varepsilon\bar\chi_\ell$ (unramified too) is ramified, hence non-split, and as in Theorem 11.15(iii) the line $\mathbb{F}_\ell e_1$ is the only stable line. (iv) If $\ell \mid n_q$, $\bar\rho$ is trivial on $I_q$ and $\bar\rho(\mathrm{Frob}_q) = \left(\begin{smallmatrix} q\varepsilon(\mathrm{Frob}_q) & * \\ 0 & \varepsilon(\mathrm{Frob}_q) \end{smallmatrix}\right) \bmod \ell$ with $\varepsilon(\mathrm{Frob}_q) = a_q(E)$. □

*In the paper* 11.18. The paper's proof (quoted from Lemma lem:tamagawa-exp): "By Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$, which contains $\mu_{\ell^t}$ as $q \neq \ell$, inertia acts trivially on $E[\ell^t]$ if and only if $q_E$ is an $\ell^t$-th power in $\mathbb{Q}_q^{\mathrm{ur}}$; the units of its valuation ring are $\ell^t$-th powers by Hensel's lemma, so this holds if and only if $\ell^t \mid n_q$." The first proof above separates the two steps this sentence compresses: Kummer theory reduces the question to $q_E$ being an $\ell^t$-th power, and Hensel's lemma reduces it to the valuation. Section 1.5 uses part (i): "for multiplicative $q$ one has $t_q = v_\ell(n_q) = t_\ell(E,q)$ whether or not the reduction is split". Lemma lem:old uses (iii) and (iv) in its two cases $p \| N$, and the optimal-level paragraph uses the consequence that $N(\bar\rho)$ is prime to $q$ iff $\ell \mid n_q$ (Proposition 11.40).

A worked case. The curve `14a1`, $y^2 + xy + y = x^3 + 4x - 6$, has $\Delta_{\min} = -2^6 \cdot 7^3$, so it has type $\mathrm{I}_6$ at 2 and $\mathrm{I}_3$ at 7, with $a_2 = -1$ (nonsplit) and $a_7 = 1$ (split). Take $\ell = 3$. Since $3 \mid 6$ and $3 \mid 3$, Lemma 11.17 says that $\bar\rho_{E,3}$ is unramified at 2 and at 7, while $\rho \bmod 9$ is ramified at both ($v_3(6) = v_3(3) = 1$, so $t_2 = t_7 = 1$), and $\#E[3^t]^{I_q} = 3^{t+1}$ for $q = 2, 7$ and every $t \geq 1$. By (iv), $\bar\rho(\mathrm{Frob}_2)$ has eigenvalues $a_2 \cdot 2 = -2 \equiv 1$ and $a_2 = -1 \equiv 2 \pmod 3$, and $\bar\rho(\mathrm{Frob}_7)$ has eigenvalues $7 \equiv 1$ and 1. All of this agrees with the global structure $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ of `14a1` (image `3Cs.1.1`, Theorem 11.45): this module is unramified at every prime $q \neq 3$, and $\mathrm{Frob}_q$ acts on it with eigenvalues 1 and $q \bmod 3$. (Recomputed with PARI.) The reader can do the same for `11a1` and `11a3` at $\ell = 5$ in Exercise 11.72.

The next proposition treats the primes at which $E$ has additive reduction but $v_q(j) < 0$. They look additive, but after a ramified quadratic twist they become multiplicative, so everything just proved applies to them up to a sign.

**Proposition 11.19** (potentially multiplicative additive reduction)**.** *Let $q \neq \ell$ ($q = 2$ allowed), $\ell$ odd, and let $E$ have additive reduction at $q$ with $v_q(j) < 0$; put $\nu = -v_q(j)$. Then there is a ramified quadratic character $\chi$ of $G_{\mathbb{Q}_q}$ such that the twist of $E$ by $\chi$ has multiplicative reduction, and $\rho|_{G_{\mathbb{Q}_q}} \simeq \rho' \otimes \chi$ with $\rho'$ of the shape of Theorem 11.15 for a Tate parameter of valuation $\nu$. For $q$ odd one may take $\chi = \chi_{q^*}|_{G_{\mathbb{Q}_q}}$; then $E^{(q)}$ has type $\mathrm{I}_\nu$ at $q$ and $E$ has type $\mathrm{I}_\nu^*$. Moreover:*

*(i)* $\rho(I_q) = \{\chi(\sigma)\left(\begin{smallmatrix} 1 & \nu t_\ell(\sigma) \\ 0 & 1 \end{smallmatrix}\right)\}$ *is infinite and contains $-1$; $V^{I_q} = 0$, so $\delta_q = 0$;*

*(ii)* $\mathrm{ad}^0 V \simeq \mathrm{ad}^0 V'$ *as $G_{\mathbb{Q}_q}$-modules, so* $(\mathrm{ad}^0 V)^{I_q}$ *is a line on which* $\mathrm{Frob}_q$ *acts by $q$;*

*(iii)* $\rho \bmod \ell^t$ *is ramified at $q$ for every $t \geq 1$, while* $(\rho \otimes \chi) \bmod \ell^t$ *is unramified at $q$ iff $\ell^t \mid \nu$;*

*(iv) the conductor exponent is $f_q = 2a(\chi)$, where $a(\chi)$ is the conductor exponent of $\chi$: $f_q = 2$ for $q$ odd, and $f_2 \in \{4, 6\}$.*

*At $q = 2$ the index $m$ of the Kodaira symbol $\mathrm{I}_m^*$ is not $\nu$ (Chapter 10 gives $m = \nu + 2f_2 - 4$), and a symbol $\mathrm{I}_m^*$ at 2 can also occur with potentially good reduction.*

*Proof.* By Chapter 10, $E$ over $\mathbb{Q}_q$ is the quadratic twist of the Tate curve $E_{q_E}$ by $\gamma = -c_4/c_6$, and the reduction is additive iff $\mathbb{Q}_q(\sqrt{\gamma})/\mathbb{Q}_q$ is ramified; take $\chi$ the character of this extension, and $\rho' = \rho_{E_{q_E},\ell}$, of Tate shape with $\varepsilon = 1$ and $v_q(q_E) = \nu$. The twist of $E$ by $\chi$ is $E_{q_E}$, which is split multiplicative. For $q$ odd every ramified quadratic character of $G_{\mathbb{Q}_q}$ is $\chi_{q^*}$ times an unramified one (Chapter 7), and an unramified quadratic twist keeps the Tate shape (with $\varepsilon$ the unramified factor); the global twist $E^{(q)}$ has the local twist as its completion at $q$, of type $\mathrm{I}_\nu$, and $E$ has type $\mathrm{I}_\nu^*$ by the exchange of types (Chapter 10).

(i) The formula follows from Theorem 11.15(i). The pair $(t_\ell, \chi)\colon I_q \to \mathbb{Z}_\ell \times \{\pm 1\}$ is onto: its image is a closed subgroup projecting onto both factors, and since $\mathbb{Z}_\ell$ is pro-$\ell$ with $\ell$ odd while $\{\pm 1\}$ has order 2, such a subgroup is the whole product. (Explicitly: let $X$ be the image, and pick $(x, -1) \in X$. Its $\ell^k$-th power is $(\ell^k x, -1)$, as $\ell^k$ is odd; as $k \to \infty$ this tends to $(0, -1)$, which lies in $X$ because $X$ is closed. For any $y \in \mathbb{Z}_\ell$ some $(y, \pm 1)$ lies in $X$, and multiplying by $(0, -1)$ if necessary gives both $(y, 1)$ and $(y, -1)$.) So some $\sigma$ has $t_\ell(\sigma) = 0$, $\chi(\sigma) = -1$, and $\rho(\sigma) = -1$, which fixes no nonzero vector: $V^{I_q} = 0$. (ii) Conjugation by $\chi(g)\rho'(g)$ equals conjugation by $\rho'(g)$; then Theorem 11.15(v). (iii) $-1 \in \rho(I_q)$ and $-1 \not\equiv 1 \pmod{\ell^t}$ for $\ell$ odd; the second statement is Lemma 11.17 for $E_{q_E}$ (or its unramified twist), whose $\nu$ plays the role of $n_q$. (iv) The wild inertia group acts through $\chi$ only ($\rho'$ is tame on $I_q$, factoring through the pro-$\ell$ tame quotient), so on the wild inertia group $V$ is the sum of two copies of the character $\chi$. The Swan conductor is additive and depends only on the wild inertia group, and $a(\chi) = 1 + \mathrm{sw}(\chi)$ for a ramified character (Chapter 7); so $\mathrm{sw}(V) = 2\,\mathrm{sw}(\chi) = 2(a(\chi) - 1)$ and $f_q = \dim V - \dim V^{I_q} + \mathrm{sw}(V) = 2 + 2(a(\chi) - 1) = 2a(\chi)$. For $q$ odd $\chi$ is tame, $a(\chi) = 1$; for $q = 2$ the ramified quadratic characters of $\mathbb{Q}_2$ have conductor exponent 2 or 3 (Chapter 7). The last sentence is Chapter 10. □

*In the paper* 11.20. The proof of Theorem thm:main bounds $v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0 T))$ below by $t_\ell(E, p)$ "at every potentially multiplicative prime $p$, multiplicative or additive, $p = 2$ included, by Lemma lem:tamfactor". At an additive such prime, Lemma lem:tamfactor is applied to the Tate curve of which $E$ is a twist; this is legitimate because $\mathrm{ad}^0 T$ is unchanged by the twist, Proposition 11.19(ii) (integrally: conjugation by $\chi\rho'$ equals conjugation by $\rho'$ on $\mathrm{ad}^0 T$). The argument is local and needs no global twist, which is why it covers $q = 2$, where the global twist of Lemma lem:twist is not available. Lemma lem:euler's row $\mathrm{I}_n^*$ ("$V = V' \otimes \chi_{q^*}$ with $V'$ Steinberg at $q$") is Proposition 11.19 for $q$ odd.

**Corollary 11.21.** *For $p \neq \ell$, $\delta_p = \dim V^{I_p}$ is 2, 1, 0 according as $E$ has good, multiplicative or additive reduction at $p$, and the conductor exponent is $f_p = 2 - \delta_p + \mathrm{sw}_p(V)$.*

*Proof.* Good: Proposition 11.14. Multiplicative: Theorem 11.15(ii). Additive: Theorem 11.12(ii) if the reduction is potentially good (the image of inertia is then nontrivial, by Theorem 11.9), Proposition 11.19(i) if not. The conductor exponent of Chapter 10 is defined as the Artin conductor $a(V_\ell E) = \dim V - \dim V^{I_p} + \mathrm{sw}_p(V)$; that it agrees with Ogg's formula is the theorem of Ogg and Saito, stated there. □

## 11.4 Additive potentially good reduction: the image of inertia by Kodaira type

This section supplies the Kodaira-type column of the shapes table of Chapter 7: for each additive potentially good type it gives the image of inertia on $T$, its invariants on $\mathrm{ad}^0 V$ and the action of

Frobenius on them, and the reduction modulo 3 with its invariants and Serre exponent. It proves the paper's Lemma lem:IV (at every prime $q \neq 3$) and Remark rem:types, and gives the inertia half of Lemma lem:euler; the $L$-factors and the terms $e_\ell(E,q)$ are then computed in Chapter 19. The type IV at 2 is covered as well (Corollary 11.27).

**Theorem 11.22** (inertia at additive potentially good primes, $q \geq 5$)**.** *Let $q \geq 5$, $\ell \neq q$, and let $E$ have additive potentially good reduction at $q$. Then $\rho(I_q) = \langle g \rangle$ is cyclic, $g \in \mathrm{SL}_2(\mathbb{Z}_\ell)$, and, with $\sigma$ denoting an element with $\sigma^2+\sigma+1=0$ and $\tau$ one with $\tau^2=-1$:*

| *type* | $v_q(\Delta_{\min})$ | $\#\rho(I_q)$ | *generator* | $(\mathrm{ad}^0 V)^{I_q}$ | $\mathrm{Frob}_q$ *acts on it by* |
|---|---|---|---|---|---|
| $\mathrm{I}_0^*$ | 6 | 2 | $-1$ | $\mathrm{ad}^0 V$ | *eigenvalues* $\alpha'/\beta', 1, \beta'/\alpha'$ |
| IV, IV$^*$ | 4, 8 | 3 | $\sigma$ | $\mathbb{Q}_\ell(2\sigma+1)$ | $\chi_{-3}(q)$ |
| III, III$^*$ | 3, 9 | 4 | $\tau$ | $\mathbb{Q}_\ell\tau$ | $\chi_{-4}(q)$ |
| II, II$^*$ | 2, 10 | 6 | $-\sigma$ | $\mathbb{Q}_\ell(2\sigma+1)$ | $\chi_{-3}(q)$ |

*Here $\alpha', \beta'$ are the roots of $X^2 - a_q(E^{(q)})X + q$. In every case $V^{I_q} = 0$ and $f_q = 2$; for every Frobenius lift, $\rho(\mathrm{Frob}_q)\, g\, \rho(\mathrm{Frob}_q)^{-1} = g^q$; and the action on the line is the same for the arithmetic and the geometric Frobenius. Moreover $(2\sigma+1)^2 = -3$; $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[g] \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ (resp. $\mathbb{Z}_\ell \otimes \mathbb{Z}[i]$) for orders 3, 6 (resp. 4), which is not a domain when $\ell \equiv 1 \pmod 3$ (resp. $\pmod 4$); for $\mathrm{I}_0^*$, $\rho \otimes \chi_{q^*}$ is unramified at $q$ and equals $\rho_{E^{(q)},\ell}$ on $G_{\mathbb{Q}_q}$, $E^{(q)}$ having good reduction at $q$. For* III, III$^*$ *the line is spanned by $\tau$ itself, which is diagonalisable over $\mathbb{Q}_\ell$ only when $\ell \equiv 1 \pmod 4$.*

*Proof.* By Theorem 11.12(iv) the image is cyclic of the stated order $m \in \{2,3,4,6\}$ ($m \neq 1$ as the reduction is not good) and lies in $\mathrm{SL}_2(\mathbb{Z}_\ell)$. Let $g$ be a generator. It is diagonalisable over $\overline{\mathbb{Q}}_\ell$ with eigenvalues $\lambda, \lambda^{-1}$, roots of unity of the same order as each other, and the order of $g$ is the order of $\lambda$. So: for $m=2$, $\lambda=-1$ and $g=-1$; for $m=3$, $\lambda$ is a primitive cube root of unity and $g$ has characteristic polynomial $X^2+X+1$, so $g=\sigma$ satisfies $\sigma^2+\sigma+1=0$ (Cayley-Hamilton); for $m=4$, $\lambda=\pm i$ and $g=\tau$ has characteristic polynomial $X^2+1$; for $m=6$, $\lambda$ is a primitive sixth root of unity, $-\lambda$ a primitive cube root, and $-g=\sigma$ satisfies $\sigma^2+\sigma+1=0$. This proves the generator column; (Chapter 4 lists the same shapes.)

Invariants. For $m=2$ every endomorphism commutes with $-1$, so $(\mathrm{ad}^0 V)^{I_q} = \mathrm{ad}^0 V$. For $m \in \{3,4,6\}$, $g$ has distinct eigenvalues, so its commutant in $\mathrm{End}(V)$ is $\mathbb{Q}_\ell[g] = \mathbb{Q}_\ell \oplus \mathbb{Q}_\ell g$, of dimension 2 (its dimension over $\overline{\mathbb{Q}}_\ell$ is 2, and $\mathbb{Q}_\ell[g]$ is contained in it and two-dimensional). The trace-zero part is the line spanned by $g - g^{-1}$, whose trace is $(\lambda+\lambda^{-1}) - (\lambda^{-1}+\lambda) = 0$, and which is nonzero. For $g=\sigma$: $\sigma^{-1} = \sigma^2 = -\sigma-1$, so $g-g^{-1} = 2\sigma+1$, and $(2\sigma+1)^2 = 4\sigma^2+4\sigma+1 = -3$; for $g=-\sigma$, $g-g^{-1} = -(2\sigma+1)$, the same line; for $g=\tau$, $g-g^{-1}=2\tau$, the line $\mathbb{Q}_\ell\tau$, and $\tau$ is diagonalisable over $\mathbb{Q}_\ell$ iff its eigenvalues $\pm i$ lie in $\mathbb{Q}_\ell$, iff $\ell \equiv 1 \pmod 4$.

Frobenius. The image of inertia has order prime to $q$, so it is a quotient of the tame inertia group, and $g = \rho(t)$ for a topological generator $t$ of tame inertia. The tame relation $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^q$ (Chapter 7) gives $\rho(\mathrm{Frob}_q) g \rho(\mathrm{Frob}_q)^{-1} = g^q$ for every Frobenius lift. Then conjugation by $\rho(\mathrm{Frob}_q)$ sends $g - g^{-1}$ to $g^q - g^{-q}$. For $m=3$: $g^q = g$ if $q \equiv 1 \pmod 3$ and $g^q = g^2 = g^{-1}$ if $q \equiv 2 \pmod 3$, so the line is multiplied by $\chi_{-3}(q)$; for $m=6$: $q \equiv \pm 1 \pmod 6$, and $q \equiv 1 \pmod 6$ iff $q \equiv 1 \pmod 3$, again $\chi_{-3}(q)$; for $m=4$: $g^q = g^{\pm 1}$ according as $q \equiv \pm 1 \pmod 4$, that is, $\chi_{-4}(q)$. A sign $c = \pm 1$ is its own inverse, so the geometric Frobenius acts by the same sign. (Here the choice of Frobenius could have mattered: for the geometric Frobenius the tame relation reads $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q'}$ with $qq' = 1$ in the relevant quotient, so it raises $g$ to the power $q^{-1} \bmod m$. For $m \in \{3,4,6\}$ and $q$ prime to $m$ one has $q^2 \equiv 1 \pmod m$, so $q^{-1} \equiv q \pmod m$ and the two Frobenii conjugate $g$ to the same power. At a multiplicative prime there is

no such coincidence, Theorem 11.15(v).) For $m = 2$: the restriction of $\rho$ to $I_q$ is the quadratic character $\sigma \mapsto \pm 1$ of $I_q$; since $q$ is odd, $I_q$ has a unique continuous character of order 2 (it factors through the tame quotient $\prod_{r\neq q} \mathbb{Z}_r(1)$, and only the factor $\mathbb{Z}_2(1)$ has a quotient of order 2), namely $\chi_{q^*}|_{I_q}$. So $\rho \otimes \chi_{q^*}$ is unramified at $q$; it is $\rho_{E^{(q)},\ell}$ (Proposition 11.5), and $E^{(q)}$ has good reduction at $q$ by Theorem 11.9. Then $\mathrm{ad}^0 V = \mathrm{ad}^0 V_\ell E^{(q)}$, unramified with the eigenvalues of Proposition 11.14.

The rest. $V^{I_q} = 0$ by Theorem 11.12(ii); the wild inertia acts trivially, so $\mathrm{sw} = 0$ and $f_q = 2$. The lattice $T$ is free of rank one over $\mathbb{Z}_\ell[g] \simeq \mathbb{Z}_\ell[X]/(\Phi_m(X))$ for $m \in \{3, 4, 6\}$ (Chapter 6, where this is proved for every odd $\ell$; for $\ell = 2$ it follows in the same way when $\Phi_m$ stays irreducible modulo 2, and we need only $\ell$ odd), so $\mathrm{End}_{I_q}(T) = \mathrm{End}_{\mathbb{Z}_\ell[g]}(T) = \mathbb{Z}_\ell[g]$, and $\mathbb{Z}_\ell[X]/(\Phi_m) \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_m]$. If $\ell \equiv 1 \pmod m$, $\Phi_m$ has a root in $\mathbb{Z}_\ell$ by Hensel's lemma and $\mathbb{Z}_\ell[X]/(\Phi_m) \simeq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$ is not a domain. □

*In the paper* 11.23. Lemma lem:euler, types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$: "On $\mathrm{ad}^0 V$ the sign disappears and the invariants form the line spanned by $2\sigma + 1$, whose square is $-3$, inside $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ (not a field when $\ell \equiv 1 \pmod 3$)"; types III, III*: "the invariant line of $\mathrm{ad}^0 V$ is spanned by $\tau$ itself, which has trace 0 (it is diagonalisable only over $\mathbb{Q}_\ell(i)$)". These are Theorem 11.22. The paper's phrase "for either choice of Frobenius" is the statement that a sign is its own inverse; at a multiplicative prime the two Frobenii act by $q$ and $q^{-1}$ (Theorem 11.15(v)), and only the geometric one gives the factor $1 - q^{-1-s}$ of the table of Section 1. The $\mathrm{I}_0^*$ row holds at $q = 3$ as well, where the image of inertia is $\{\pm 1\}$ (Chapter 10).

*Remark* 11.24 (Kraus's table from component groups). At $q \geq 5$ the dictionary between types and orders in Theorem 11.22 can also be derived from the component groups alone, without Kraus's formula, given that the order $m$ lies in $\{2, 3, 4, 6\}$ and does not depend on $\ell$ (Theorem 11.12(i), (iv)). For a generator $g$ of order $m$, the fixed spaces of its reductions on $E[2]$ and $E[3]$ have dimensions $(2, 0)$, $(0, 1)$, $(1, 0)$, $(0, 0)$ for $m = 2, 3, 4, 6$: modulo 2, $-1$ is trivial, $\sigma$ has the irreducible characteristic polynomial $X^2 + X + 1$, and $\tau$ is a nontrivial unipotent (if $\tau = 1 + 2A$ with $A \in M_2(\mathbb{Z}_2)$, then $\tau^2 = -1$ gives $A + A^2 = -\frac{1}{2}$, impossible); modulo 3, $-1$ and $-\sigma$ have only the eigenvalue $-1$, $\sigma$ is a nontrivial unipotent, and $\tau$ has the irreducible characteristic polynomial $X^2 + 1$. By Lemma 11.10 at $\ell = 2$ and $\ell = 3$ these dimensions are those of $\Phi_q^{\mathrm{geom}}[2]$ and $\Phi_q^{\mathrm{geom}}[3]$, which are $(2, 0)$ for $\mathrm{I}_0^*$, $(0, 1)$ for IV, IV*, $(1, 0)$ for III, III* and $(0, 0)$ for II, II*. Since the four pairs are distinct, the type determines $m$: $\mathrm{I}_0^* \mapsto 2$, $\mathrm{IV}, \mathrm{IV}^* \mapsto 3$, $\mathrm{III}, \mathrm{III}^* \mapsto 4$, $\mathrm{II}, \mathrm{II}^* \mapsto 6$. The details are Exercise 11.77.

**Lemma 11.25** (the paper's Lemma lem:IV)**.** *Let $q \neq 3$ be a prime at which $E$ has reduction of type* IV *or* IV*. *Then:*

*(i) the reduction at $q$ is potentially good, and the image of $I_q$ in $\mathrm{Aut}(T_\ell E)$ is cyclic of order 3 for every prime $\ell \neq q$;*

*(ii) its image in $\mathrm{Aut}(E[3])$ is a nontrivial unipotent subgroup; $E[3]^{I_q}$ is a line; $\bar\rho_{E,3}$ is tamely ramified at $q$; the Serre conductor of $\bar\rho_{E,3}$ has exponent 1 at $q$; and the conductor of $E$ has exponent $f_q = 2$;*

*(iii) if $\sigma$ generates $\rho_{E,3}(I_q)$, then $T_3 E$ is free of rank one over $\mathbb{Z}_3[X]/(X^2 + X + 1) \simeq \mathbb{Z}_3[\zeta_3]$ with $X$ acting as $\sigma$, and $E[3] \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$ with $\pi = 1 - \zeta_3$, $\sigma - 1$ acting as multiplication by $-\pi$;*

*(iv) for $\ell \neq q$ and $\sigma$ a generator of $\rho_{E,\ell}(I_q)$, $(\mathrm{ad}^0 V_\ell E)^{I_q} = \mathbb{Q}_\ell(2\sigma+1)$ is a line on which $\mathrm{Frob}_q$ and $\mathrm{Frob}_q^{-1}$ act by $\chi_{-3}(q)$; so, with the conventions of Convention 11.1(c), $L_q(A_f,s)^{-1} = 1-\chi_{-3}(q)q^{-s}$, $P_q(X) = 1-\chi_{-3}(q)qX$ and $e_\ell(E,q) = v_\ell(q-\chi_{-3}(q))$.*

*Proof (the paper's, every step expanded).* (i) and (ii). By Lemma 11.10(i) (the reduction is additive, $q \neq 3$), $E[3]^{I_q} \simeq \Phi_q^{\mathrm{geom}}[3]$, and $\Phi_q^{\mathrm{geom}} \simeq \mathbb{Z}/3$ for the types IV, IV$^*$ (Chapter 10); so $E[3]^{I_q}$ is a line. Inertia acts nontrivially on $E[3]$: otherwise $E[3]^{I_q} = E[3]$ would have order 9, not 3.

The reduction is potentially good. Otherwise, by Proposition 11.19, $\rho_{E,3}(I_q)$ consists of the matrices $\chi(\sigma)\left(\begin{smallmatrix}1 & \nu t_3(\sigma)\\ 0 & 1\end{smallmatrix}\right)$ with $\chi$ ramified quadratic, and some $\sigma$ acts by $-1$ (proof of Proposition 11.19(i)), which fixes no line of $E[3]$: a contradiction. (The paper phrases the same argument as: the subgroup of $I_q$ on which the corner vanishes modulo 3 has index dividing 3, so it is not contained in the kernel of $\chi$, of index 2. Alternatively, types IV, IV$^*$ have $v_q(j) \geq 0$, Chapter 10.)

So $G = \rho_{E,3}(I_q)$ is finite (Theorem 11.9(e)), lies in $\mathrm{SL}_2(\mathbb{Z}_3)$, and reduction modulo 3 is injective on it (Theorem 11.12(ii)). Its image $\bar{G} \subseteq \mathrm{SL}_2(\mathbb{F}_3)$ is nontrivial and fixes a nonzero vector $v$. In a basis $(v,v')$ of $E[3]$ the elements of $\bar{G}$ are $\left(\begin{smallmatrix}1 & b\\ 0 & d\end{smallmatrix}\right)$ with $d = \det = 1$, so $\bar{G}$ lies in the unipotent group $\mathrm{U} = \{\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right)\}$ of order 3, and equals it (Chapter 1: the elements of $\mathrm{SL}_2(\mathbb{F}_3)$ fixing a nonzero vector form a group of order 3). Hence $G \simeq \bar{G}$ is cyclic of order 3, its image in $\mathrm{Aut}(E[3])$ is a nontrivial unipotent group, and by Theorem 11.12(i) the image of $I_q$ in $\mathrm{Aut}(T_\ell E)$ is cyclic of order 3 for every $\ell \neq q$. The order 3 is prime to $q$ (as $q \neq 3$), so the image is a quotient of the tame inertia group: the ramification is tame, $\mathrm{sw} = 0$. The Serre exponent of $\bar{\rho}_{E,3}$ at $q$ is $\dim E[3] - \dim E[3]^{I_q} + 0 = 2-1 = 1$, and the conductor exponent of $E$ is $\dim V - \dim V^{I_q} + 0 = 2-0 = 2$ (Theorem 11.12(ii)).

(iii) $\sigma$ has determinant $\chi_3^{\mathrm{cyc}}(\sigma) = 1$ and order 3, so its characteristic polynomial is $X^2+X+1$ (proof of Theorem 11.22) and $\sigma^2+\sigma+1 = 0$ on $T = T_3E$. So $T$ is a module over $\mathbb{Z}_3[X]/(X^2+X+1)$ with $X$ acting as $\sigma$, and this ring is $\mathbb{Z}_3[\zeta_3]$, a discrete valuation ring with uniformiser $\pi = 1-\zeta_3$ and residue field $\mathbb{F}_3$, $\pi^2 = -3\zeta_3$ (Chapter 6). The module $T$ is torsion-free of $\mathbb{Z}_3$-rank 2, hence a torsion-free finitely generated module of rank one over the DVR $\mathbb{Z}_3[\zeta_3]$, hence free of rank one (Chapter 6). Then $E[3] = T/3T \simeq \mathbb{Z}_3[\zeta_3]/(3) = \mathbb{Z}_3[\zeta_3]/(\pi^2)$, on which $\sigma-1$ acts as $\zeta_3 - 1 = -\pi$: nonzero, with square zero, a nontrivial unipotent whose fixed line is $\pi\mathbb{Z}_3[\zeta_3]/(\pi^2)$.

(iv) This is the computation of Theorem 11.22 for $m = 3$, which used only that $\rho(I_q)$ is cyclic of order 3, prime to $q$: the commutant of $\sigma$ meets $\mathrm{ad}^0 V$ in $\mathbb{Q}_\ell(2\sigma+1)$, and $\rho(\mathrm{Frob}_q)\sigma\rho(\mathrm{Frob}_q)^{-1} = \sigma^q$ by the tame relation, which is $\sigma^{\chi_{-3}(q)}$. So $\mathrm{Frob}_q^{-1}$ acts on $(\mathrm{ad}^0 V)^{I_q}$ by $\chi_{-3}(q)$, $L_q(A_f,s)^{-1} = 1-\chi_{-3}(q)q^{-s}$, $P_q(X) = L_q(A_f,s-1)^{-1}|_{X=q^{-s}} = 1-\chi_{-3}(q)qX$, and, the naive factor being 1 at an additive prime, $e_\ell(E,q) = v_\ell(1-\chi_{-3}(q)q^{-1}) = v_\ell(q-\chi_{-3}(q))$ since $v_\ell(q) = 0$. □

*Second proof of (i) and (ii) for $q \geq 5$.* By Kraus's theorem (Theorem 11.12(iv)) the image of $I_q$ is cyclic of order $12/\gcd(v_q(\Delta_{\min}),12) = 3$, as $v_q(\Delta_{\min}) \in \{4,8\}$; it is tame because $3 \neq q$. A generator $\sigma$ of $\rho_{E,3}(I_q)$ gives, as in (iii), $E[3] \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$ with $\sigma-1$ acting as $-\pi$: a nontrivial unipotent with a fixed line. The Serre exponent is $2-1 = 1$. □

*In the paper* 11.26. The paper states Lemma lem:IV for every $q \neq 3$, $q = 2$ included. What makes $q = 2$ accessible is the component-group argument, which Chapter 10 also gives for the inertia statement at 2; Kraus's formula, used in the second proof above, is available only for $q \geq 5$. At $q = 3$ the types IV, IV$^*$ are wildly ramified (Chapter 10), and the lemma does not apply: an image of order 3 at $q = 3$ is not tame. Theorem thm:qual (a) uses part (ii): "By Lemma lem:IV the Serre conductor $M = N(\bar{\rho})$ has exponent exactly 1 at $q$, so $M \mid N/q$."

A worked case of Lemma 11.25. The curve `121c1`, $y^2+xy = x^3+x^2-2x-7$, has $\Delta_{\min} = -11^4$ and type IV at $q = 11$. Kraus's formula gives an image of inertia of order $12/\gcd(4,12) = 3$, as the lemma predicts. Since $11 \equiv 2 \pmod 3$, $\chi_{-3}(11) = -1$: Frobenius inverts $\sigma$ and acts on $(\mathrm{ad}^0 V)^{I_{11}}$ by $-1$, so $L_{11}(A_f,s)^{-1} = 1+11^{-s}$, $P_{11}(X) = 1+11X$ (PARI's `lfuneuler(lfunsympow(E,2),11)` returns $1/(1+11X)$), and $e_\ell(E,11) = v_\ell(12)$, which is 1 for $\ell = 3$ and 0 for $\ell \geq 5$. The Serre exponent of $\bar\rho_{E,3}$ at 11 is $2-1 = 1$, one less than $f_{11} = 2$; this is why $N(\bar\rho_{\texttt{121c1},3}) = 11$ in Example 11.42.

**Corollary 11.27** (types IV, IV* at 2)**.** *Let $E/\mathbb{Q}$ have type* IV *or* IV* *at 2. Then $f_2(E) = 2$ (so $4 \parallel N$), $v_2(\Delta_{\min}) = 4$ for type* IV *and 8 for type* IV*, $\rho_{E,\ell}(I_2)$ is cyclic of order 3 for every odd $\ell$, $E[3]^{I_2}$ is a line, and the Serre conductor of $\bar\rho_{E,3}$ has exponent 1 at 2. Moreover* $\mathrm{Frob}_2$ *inverts the generator $\sigma$ of $\rho(I_2)$ and acts on $(\mathrm{ad}^0 V)^{I_2} = \mathbb{Q}_\ell(2\sigma+1)$ by $\chi_{-3}(2) = -1$; so $L_2(A_f,s)^{-1} = 1+2^{-s}$, $P_2(X) = 1+2X$, $e_3(E,2) = v_3(3/2) = 1$ and $e_\ell(E,2) = 0$ for $\ell \geq 5$.*

*Proof.* Everything except the discriminant is Lemma 11.25 at $q = 2$, with $\chi_{-3}(2) = -1$ ($2 \equiv -1 \pmod 3$, so $\sigma^2 = \sigma^{-1}$). Ogg's formula $f_2 = v_2(\Delta_{\min}) + 1 - m$ (Chapter 10), with $m$ the number of components of the special fibre of the minimal regular model, $m = 3$ for IV and $m = 7$ for IV*, gives $v_2(\Delta_{\min}) = f_2 - 1 + m = 4$, resp. 8. □

All 4,750 curves of conductor below 10,000 with type IV (2,094 curves) or IV* (2,656) at 2 were checked with PARI against Corollary 11.27: every one has $f_2 = 2$, $v_2(\Delta_{\min}) = 4$, resp. 8, $v_2(j) \geq 0$, and, computed from the fields of definition of the 3-torsion points, an image of $I_2$ on $E[3]$ of order 3 with an inertia-fixed point. On a random sample of 60 of them, and on `20a1`, `52a1`, `140b1` and `980i1`, `lfuneuler(lfunsympow(E,2),2)` returns $1/(1+2X)$, that is, $P_2(X) = 1+2X$.

*In the paper* 11.28. The proof of Proposition prop:forms treats the types IV, IV* at 2 by Lemma lem:IV ("of type IV, IV* and $\ell = 3$, $w_3(E,2) = 1 = e_3(E,2)$ by Lemma lem:IV"), and the paragraph after the proof reads: "On the 92,812 working-set curves with type IV or IV* at 2, PARI's factor at 2 is $1+2X$ for every one of them, as Lemma lem:IV requires." This is Corollary 11.27. The same corollary is what lets case (a) of Theorem thm:qual hold at $q = 2$: the Serre exponent of $\bar\rho_{E,3}$ at 2 is 1 while $4 \parallel N$, so the level drops at 2 exactly as at a prime $q \geq 5$; the paper's Example, case (a) at $q = 2$, is `52a1` (Example 11.70).

**Proposition 11.29** (inertia modulo 3 at $q \geq 5$; the paper's Remark rem:types)**.** *Let $q \geq 5$ and $\ell = 3$. The image $\bar\rho_{E,3}(I_q)$, the fixed space, the Serre exponent $a_q(\bar\rho_{E,3})$, the conductor exponent $f_q(E)$, and the Serre exponent after the twist by $\chi_{q^*}$ are as follows (types exchanged by the twist as in Chapter 10).*

| *type at $q$* | *$\bar\rho(I_q)$ generated by* | *order* | $\dim E[3]^{I_q}$ | $a_q(\bar\rho_{E,3})$ | $f_q(E)$ | $a_q(\bar\rho_{E^{(q)},3})$ |
|---|---|---|---|---|---|---|
| $\mathrm{I}_0$ | 1 | 1 | 2 | 0 | 0 | 2 |
| $\mathrm{I}_n$, $3 \nmid n$ | *unipotent* $\neq 1$ | 3 | 1 | 1 | 1 | 2 |
| $\mathrm{I}_n$, $3 \mid n$ | 1 | 1 | 2 | 0 | 1 | 2 |
| $\mathrm{I}_n^*$, $3 \nmid n$ | $-1$ *and a unipotent* $\neq 1$ | 6 | 0 | 2 | 2 | 1 |
| $\mathrm{I}_n^*$, $3 \mid n$ | $-1$ | 2 | 0 | 2 | 2 | 0 |
| $\mathrm{I}_0^*$ | $-1$ | 2 | 0 | 2 | 2 | 0 |
| II, II* | $-u$, $u$ *unipotent* $\neq 1$ | 6 | 0 | 2 | 2 | 1 |
| III, III* | $\bar\tau$, $\bar\tau^2 = -1$ | 4 | 0 | 2 | 2 | 2 |
| IV, IV* | *unipotent* $\neq 1$ | 3 | 1 | 1 | 2 | 2 |

*In particular: (a)* $E[3]^{I_q} = 0$ *and* $a_q(\bar\rho_{E,3}) = 2 = f_q$ *for every additive type other than* IV, IV$^*$ *(including* $\mathrm{I}_0^*$, III, III$^*$*), and* $E[3]^{I_q} = 0$ *for these types also at* $q = 2$*; (b) the twist by* $\chi_{q^*}$ *lowers the Serre exponent of* $\bar\rho_{E,3}$ *at* $q$ *exactly for* II, II$^*$, $\mathrm{I}_n^*$ *and* $\mathrm{I}_0^*$*, and lowers the conductor exponent of* $E$ *exactly for* $\mathrm{I}_n^*$ *and* $\mathrm{I}_0^*$ *(for* II $\leftrightarrow$ IV$^*$ *both curves have* $f_q = 2$*). For* $\ell \geq 5$ *and* $q \geq 5$*,* $q \neq \ell$*: every additive type has* $E[\ell]^{I_q} = 0$ *and Serre exponent* 2*, and* $\mathrm{I}_n$ *has exponent* 0 *or* 1 *according as* $\ell \mid n$ *or not.*

*Proof.* Reduce the generators of Theorem 11.15, Proposition 11.19 and Theorem 11.22 modulo 3. A unipotent $\left(\begin{smallmatrix}1 & nt\\ 0 & 1\end{smallmatrix}\right)$ with $t$ running through $\mathbb{Z}_3$ is trivial modulo 3 iff $3 \mid n$, and otherwise generates the group U of order 3; for $\mathrm{I}_n^*$ the image is $\{\pm 1\}$ times this (the pair $(t_3, \chi)$ is onto, proof of Proposition 11.19), of order 6 or 2. For the potentially good types reduction is injective (Theorem 11.12(ii)): $-1$ has order 2; $\sigma$ mod 3 is a nontrivial unipotent (Lemma 11.25(iii)); $-\sigma$ mod 3 is minus a nontrivial unipotent, of order 6; $\tau$ mod 3 has square $-1$ and characteristic polynomial $X^2 + 1$, irreducible over $\mathbb{F}_3$. The fixed spaces follow (an element with only the eigenvalue $-1$, or with irreducible characteristic polynomial, fixes no nonzero vector), and agree with Lemma 11.10(iii). At $q \geq 5$ every image of inertia has order prime to $q$ (unipotents modulo 3 have order 3), so the Swan term vanishes and $a_q = 2 - \dim E[3]^{I_q}$. The column $f_q(E)$ is Corollary 11.21 ($f_q = 1$ multiplicative, 2 additive at $q \geq 5$). The last column is the same computation for $E^{(q)}$, whose type is obtained by the exchanges $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$, $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$, II $\leftrightarrow$ IV$^*$, III $\leftrightarrow$ III$^*$, IV $\leftrightarrow$ II$^*$ (Chapter 10). Statements (a) and (b) are read off the table; the case $q = 2$ in (a) is Lemma 11.10, since $\Phi_2^{\mathrm{geom}}[3] = 0$ for all additive types other than IV, IV$^*$. For $\ell \geq 5$: $\Phi_q^{\mathrm{geom}}[\ell] = 0$ for every additive type, so $E[\ell]^{I_q} = 0$; and $\mathrm{I}_n$ is Lemma 11.17. □

All rows of the table were checked with PARI on curves of conductor below 10,000 with surjective mod 3 image: for each Kodaira type at a prime $q \geq 5$ and each residue of $n$ modulo 3, on up to 12 curves (all of them when fewer exist), the least common multiple of the ramification indices of $q$ in the fields $\mathbb{Q}(P)$, $P \in E[3]$, equals the order in the table, and an inertia-fixed point exists exactly in the rows with $\dim E[3]^{I_q} \geq 1$.

*In the paper* 11.30. The paper's Remark rem:types: "For all these types $E[3]^{I_q} = 0$ and the Serre conductor of $\bar\rho_{E,3}$ has exponent 2 at $q$. The twist by $\chi_{q^*}$ lowers that exponent for $\mathrm{I}_0^*$, $\mathrm{I}_n^*$, II, II$^*$; it lowers the conductor of $E$ only for $\mathrm{I}_0^*$ and $\mathrm{I}_n^*$." This is (a) and (b) of Proposition 11.29. The remark distinguishes the two exponents: in the case II $\leftrightarrow$ IV$^*$ the twist lowers the Serre exponent of $\bar\rho_{E,3}$ but not the conductor exponent of $E$. Among the additive rows, Theorem thm:qual treats exactly those in which, without the twist or with the twist by $\chi_{q^*}$, the Serre exponent at $q$ is strictly smaller than the conductor exponent at $q$ of the corresponding curve ($E$, respectively $E^{(q)}$): IV, IV$^*$ without the twist ($1 < 2$, case (a)); II, II$^*$ with it ($1 < 2$, case (b)); and $\mathrm{I}_n^*$ with $3 \mid n$ with it ($0 < 1$, case (c) at $\ell = 3$). In the rows $\mathrm{I}_0^*$ ($0 = 0$ after the twist), $\mathrm{I}_n^*$ with $3 \nmid n$ ($1 = 1$) and III, III$^*$ ($2 = 2$ either way) the Serre exponent never drops below the conductor exponent, and no level is gained.

*Remark* 11.31 (the wild primes). At $q = 2$ and $q = 3$ the Kodaira type does not determine $\rho(I_q)$ (Theorem 11.12(iii)). At $q = 2$ the image can be $Q_8$ or $\mathrm{SL}_2(\mathbb{F}_3)$ (non-abelian, so absolutely irreducible on $V$, $(\mathrm{ad}^0 V)^{I_2} = 0$ and $P_2 = 1$), cyclic of order 2, 4 or 6 (with a nontrivial wild part, since the tame quotient $\prod_{r \neq 2} \mathbb{Z}_r(1)$ of $I_2$ has no quotient of even order), or cyclic of order 3. The last happens exactly for the types IV, IV$^*$. These types have it by Corollary 11.27. Conversely, if $\rho(I_2)$ has order 3, the reduction is additive and potentially good (Theorem 11.9), reduction modulo 3 is injective on $\rho_{E,3}(I_2)$ (Theorem 11.12(ii)), and an element of order 3 of $\mathrm{SL}_2(\mathbb{F}_3)$ has characteristic polynomial $X^2 + X + 1 = (X - 1)^2$, so it is a nontrivial unipotent and fixes a line;

hence $E[3]^{I_2} \neq 0$, so $\Phi_2^{\mathrm{geom}}[3] \neq 0$ by Lemma 11.10(i), and among the additive types only IV and IV$^*$ have a component group of order divisible by 3 (Chapter 10). At $q = 3$ an image of order divisible by 3 is wild. What survives at every $q \neq \ell$ is Lemma 11.10 and the conductor formula Proposition 11.40. On samples of 40 curves per type at $q = 2$ (conductor below 10,000, surjective mod 3 image), for the potentially good types II, II$^*$, III, III$^*$, I$_0^*$ and I$_m^*$ the least common multiple of the ramification indices of 2 in the fields $\mathbb{Q}(P)$, $P \in E[3]$, was 2, 4, 6 or 8, for the potentially multiplicative symbols I$_m^*$ it was 2 or 6, and in all these cases $E[3]^{I_2} = 0$. The Euler factors at 2 and 3 are computed in Chapter 19; Remark rem:wild itself is discussed in Chapter 10.

## 11.5 The representation at $\ell$ when $E$ has good reduction at $\ell$

At $p = \ell$ the restriction $\rho|_{G_{\mathbb{Q}_\ell}}$ is not unramified even when $E$ has good reduction, and its description needs $p$-adic Hodge theory. The paper uses only three facts about it, all for $\ell \nmid N$: that $\bar\rho|_{D_\ell}$ comes from a finite flat group scheme ("good" in the optimal-level theorem), that $V$ is crystalline (through [49], Chapter 18), and that $a_\ell \bmod \ell$ is determined by $\bar\rho|_{D_\ell}$ (Lemma lem:old, case $p = \ell$). This section states them, sketches the elliptic-curve proofs, and draws the consequence that drives Proposition prop:Sf: a twist by $\chi_{\ell^*}$ can preserve $\bar\rho$ only when $\ell = 3$.

**Theorem 11.32** (good reduction at $\ell$)**.** *Let $\ell \nmid N$ and let $\mathcal{E}$ be the Néron model of $E$ over $\mathbb{Z}_\ell$, an elliptic curve over $\mathbb{Z}_\ell$.*

*(i) For every $k \geq 1$, $\mathcal{E}[\ell^k]$ is a finite flat group scheme over $\mathbb{Z}_\ell$ of order $\ell^{2k}$ whose generic fibre is $E[\ell^k]$; so $\bar\rho|_{D_\ell}$ is* finite *(flat), which is what "good" means in the optimal-level theorem [38, Theorem 3.15].*

*(ii) $V|_{G_{\mathbb{Q}_\ell}}$ is crystalline with Hodge-Tate weights* 0 *and* 1*, and the crystalline Frobenius on $\mathrm{D}_{\mathrm{cris}}(V^\vee)$ has characteristic polynomial $X^2 - a_\ell X + \ell$.*

*(iii) If $\bar\rho$ is irreducible, its Serre weight is $k(\bar\rho) = 2$.*

*Sketch of proof.* (i) $\mathcal{E}$ is an abelian scheme of relative dimension one over $\mathbb{Z}_\ell$ (Chapter 10: good reduction means the minimal model is smooth, and it is then the Néron model). Multiplication by $\ell^k$ on an abelian scheme is quasi-finite and proper, hence finite, and flat by the fibrewise criterion, of degree $\ell^{2k}$ on both fibres; its kernel is the finite flat group scheme $\mathcal{E}[\ell^k]$, with generic fibre $E[\ell^k]$. See [38, Section 2] for finite flat group schemes and the meaning of "finite" for $\bar\rho|_{D_\ell}$. (ii) is a theorem of Fontaine (and of Faltings for general abelian varieties); an introduction is Berger [11]; Chapter 18 uses it. We do not prove it. (iii) is Serre's recipe for the weight [176, Section 2], in the form of Edixhoven [61, Section 2]; for $\bar\rho_{E,\ell}$ with $\ell \nmid N$ the recipe gives 2 in both the ordinary and the supersingular case of Theorem 11.35. (For orientation only: at a multiplicative prime $\ell$ the weight is 2 or $\ell + 1$ according as $\ell \mid v_\ell(\Delta)$ or not; the paper never needs $\ell \mid N$ in a proof.) □

**Definition 11.33** (fundamental characters)**.** Let $\ell$ be a prime and $I_\ell \subset G_{\mathbb{Q}_\ell}$ the inertia group. For $n \geq 1$ choose a root $\varpi_n$ of $X^{\ell^n - 1} = \ell$ in $\overline{\mathbb{Q}}_\ell$, and define

$$\omega_n \colon I_\ell \to \mathbb{F}_{\ell^n}^\times, \qquad \omega_n(\sigma) = \frac{\sigma(\varpi_n)}{\varpi_n} \bmod \mathfrak{m},$$

where $\sigma(\varpi_n)/\varpi_n \in \mu_{\ell^n - 1}$ is reduced modulo the maximal ideal of $\bar{\mathbb{Z}}_\ell$ (which identifies $\mu_{\ell^n-1}$ with $\mathbb{F}_{\ell^n}^\times$). We write $\omega = \omega_1$ (the fundamental character of level 1) and $\omega_2$ (level 2).

Why these characters. Any continuous character of $I_\ell$ with values in a finite field of characteristic $\ell$ has order prime to $\ell$, so it kills the pro-$\ell$ wild inertia group and factors through the tame quotient, which is procyclic. The characters $\omega_n$ are the standard generators of the characters of tame inertia with values in $\mathbb{F}_{\ell^n}^\times$: every such character is a power of $\omega_n$ (by (i) of the next proposition, $\omega_n$ maps the tame quotient onto the cyclic group $\mathbb{F}_{\ell^n}^\times$). They are the language in which Theorem 11.35 describes $\bar\rho|_{I_\ell}$: level 1 in the ordinary case, level 2 in the supersingular case. For example, at $\ell = 3$, $\omega$ has order 2 and is the restriction of $\chi_{-3}$ to $I_3$, while $\omega_2 \colon I_3 \to \mathbb{F}_9^\times$ has order 8 and $\omega_2^4 = \omega$.

**Proposition 11.34.** *(i) $\omega_n$ is a well-defined continuous surjective homomorphism, independent of the choice of $\varpi_n$, and factors through the tame inertia group.*

*(ii) $\omega = \bar\chi_\ell|_{I_\ell}$, of order $\ell - 1$.*

*(iii) $\omega_2$ has order $\ell^2 - 1$ and $\omega_2^{\ell+1} = \omega$.*

*(iv) For a Frobenius lift $F \in G_{\mathbb{Q}_\ell}$ and $\sigma \in I_\ell$, $\omega_2(F\sigma F^{-1}) = \omega_2(\sigma)^\ell$. Hence $\omega_2$ does not extend to a character of $G_{\mathbb{Q}_\ell}$, and for any extension $\psi$ of $\omega_2$ to $G_{\mathbb{Q}_{\ell^2}}$ ($\mathbb{Q}_{\ell^2}$ the unramified quadratic extension) the induced representation $\operatorname{Ind}_{G_{\mathbb{Q}_{\ell^2}}}^{G_{\mathbb{Q}_\ell}} \psi$ is irreducible, with restriction $\omega_2 \oplus \omega_2^\ell$ to $I_\ell$.*

*Proof.* (i) $\mu_{\ell^n-1}$ lies in $\mathbb{Q}_\ell^{\mathrm{ur}}$, which is the fixed field of $I_\ell$; Kummer theory over $\mathbb{Q}_\ell^{\mathrm{ur}}$ (Chapter 7) shows that $\sigma \mapsto \sigma(\varpi_n)/\varpi_n$ is a homomorphism $I_\ell \to \mu_{\ell^n-1}$, independent of the root (two roots differ by a factor in $\mu_{\ell^n-1}$, fixed by $I_\ell$). The extension $\mathbb{Q}_\ell^{\mathrm{ur}}(\varpi_n)$ is totally and tamely ramified of degree $\ell^n - 1$ ($v(\varpi_n) = 1/(\ell^n - 1)$), so the map is onto, and it is trivial on wild inertia, a pro-$\ell$ group. (ii) Let $\varpi' = \zeta_\ell - 1$, a uniformiser of $\mathbb{Q}_\ell(\zeta_\ell)$. For $\sigma \in I_\ell$ with $\sigma(\zeta_\ell) = \zeta_\ell^a$, $a = \chi_\ell(\sigma) \bmod \ell$, $\sigma(\varpi')/\varpi' = 1 + \zeta_\ell + \cdots + \zeta_\ell^{a-1} \equiv a \pmod{\varpi'}$. Now $\ell = \prod_{a=1}^{\ell-1}(1 - \zeta_\ell^a)$ and $(1 - \zeta_\ell^a)/(1 - \zeta_\ell) \equiv a \pmod{\varpi'}$, with $(\ell-1)! \equiv -1 \pmod{\ell}$ (Wilson); as $\ell - 1$ is even, this gives $\varpi'^{\ell-1} = -\ell \cdot w$ with $w \equiv 1 \pmod{\varpi'}$, so $\varpi' = \zeta' \varpi_1 w'$ where $(\zeta')^{\ell-1} = -1$ and $(w')^{\ell-1} = w$; both $\zeta'$ and $w'$ lie in $\mathbb{Q}_\ell^{\mathrm{ur}}$ (Hensel's lemma, $\ell \nmid \ell - 1$) and are fixed by $I_\ell$. Hence $\omega(\sigma) = \sigma(\varpi')/\varpi' \bmod \mathfrak{m} = a$. (iii) $\varpi_2^{\ell+1}$ is a root of $X^{\ell-1} = \ell$, so $\omega_2^{\ell+1} = \omega$; the order is $\ell^2 - 1$ by (i). (iv) Put $F^{-1}(\varpi_2) = \zeta\varpi_2$ with $\zeta \in \mu_{\ell^2-1}$. Then $\sigma F^{-1}(\varpi_2) = \zeta\omega_2(\sigma)\varpi_2$ (lifting $\omega_2(\sigma)$ to $\mu_{\ell^2-1}$, fixed by $\sigma$), and applying $F$, which acts on $\mu_{\ell^2-1} \subset \mathbb{Q}_\ell^{\mathrm{ur}}$ by $x \mapsto x^\ell$, gives $F\sigma F^{-1}(\varpi_2) = F(\zeta)\omega_2(\sigma)^\ell F(\varpi_2) = \omega_2(\sigma)^\ell \varpi_2$, since $F(\zeta)F(\varpi_2) = F(\zeta\varpi_2) = \varpi_2$. A character of $G_{\mathbb{Q}_\ell}$ is invariant under conjugation, and $\omega_2^\ell \neq \omega_2$ because $\ell^2 - 1 \nmid \ell - 1$; so $\omega_2$ does not extend. The restriction of $\operatorname{Ind}\psi$ to $I_\ell \subset G_{\mathbb{Q}_{\ell^2}}$ is $\psi|_{I_\ell} \oplus \psi^F|_{I_\ell} = \omega_2 \oplus \omega_2^\ell$, two distinct characters, and a $G_{\mathbb{Q}_\ell}$-stable line would be stable under $I_\ell$, hence one of the two eigenlines, which $F$ exchanges. So $\operatorname{Ind}\psi$ is irreducible. □

**Theorem 11.35** (Deligne, Fontaine; the case of elliptic curves)**.** *Let $\ell$ be odd and $\ell \nmid N$. For $x \in \mathbb{F}_\ell^\times$ let $\mu(x)$ be the unramified character of $G_{\mathbb{Q}_\ell}$ with $\mu(x)(\mathrm{Frob}_\ell) = x$.*

*(i) (Ordinary case.) If $\ell \nmid a_\ell$, then*

$$\bar\rho|_{D_\ell} \simeq \begin{pmatrix} \bar\chi_\ell\,\mu(a_\ell^{-1}) & * \\ 0 & \mu(a_\ell) \end{pmatrix}, \qquad \bar\rho|_{I_\ell} \simeq \begin{pmatrix} \omega & * \\ 0 & 1 \end{pmatrix}.$$

*(ii) (Supersingular case.) If $\ell \mid a_\ell$ (then $a_\ell = 0$ for $\ell \geq 5$, and $a_3 \in \{0, \pm 3\}$), then $\bar\rho|_{I_\ell} \otimes \mathbb{F}_{\ell^2} \simeq \omega_2 \oplus \omega_2^\ell$ and $\bar\rho|_{D_\ell}$ is irreducible.*

*(iii) Hence $a_\ell \bmod \ell$ is determined by $\bar\rho|_{D_\ell}$: it is 0 if $\bar\rho|_{D_\ell}$ is irreducible, and otherwise it is the eigenvalue of $\mathrm{Frob}_\ell$ on the unique unramified one-dimensional quotient of $\bar\rho|_{D_\ell}$.*

*The same holds for a newform $g$ of weight 2 and level prime to $\ell$ with coefficients in $\mathcal{O}$, with $a_\ell(g) \bmod \lambda$ and $\bar\rho_{g,\lambda}$ (Deligne in the ordinary case, Fontaine in the supersingular case).*

*Sketch of proof.* (i) The finite flat group scheme $\mathcal{E}[\ell]$ over $\mathbb{Z}_\ell$ has a connected-étale sequence $0 \to \mathcal{E}[\ell]^0 \to \mathcal{E}[\ell] \to \mathcal{E}[\ell]^{\mathrm{et}} \to 0$. The étale quotient is determined by its special fibre, $\tilde E[\ell](\overline{\mathbb{F}}_\ell)$, which has order $\ell$ exactly when $\tilde E$ is ordinary, that is, when $\ell \nmid a_\ell$ (Chapter 9). On it $\mathrm{Frob}_\ell$ acts as the Frobenius endomorphism, by the unit root $\alpha_u$ of $X^2 - a_\ell X + \ell$ (the root which is an $\ell$-adic unit; the other root is $\ell/\alpha_u$), and $\alpha_u \equiv a_\ell \pmod\ell$ because $\alpha_u + \ell/\alpha_u = a_\ell$. So the quotient is $\mu(a_\ell)$, unramified, and the sub is $\det\bar\rho/\mu(a_\ell) = \bar\chi_\ell\mu(a_\ell^{-1})$, which restricts to $I_\ell$ as $\omega$ (Proposition 11.34(ii)). (ii) If $\tilde E$ is supersingular, $\mathcal{E}[\ell]$ is connected and the formal group of $\mathcal{E}$ has height 2; Serre [171, Section 1] shows that $I_\ell$ acts on $E[\ell] \otimes \mathbb{F}_{\ell^2}$ through $\omega_2 \oplus \omega_2^\ell$. By Proposition 11.34(iv) no line is stable under $D_\ell$ even over $\mathbb{F}_{\ell^2}$ (a stable line would be stable under $I_\ell$ and $F$), so $\bar\rho|_{D_\ell}$ is irreducible. The Hasse bound $|a_\ell| \le 2\sqrt{\ell} < \ell$ for $\ell \ge 5$ gives $a_\ell = 0$; for $\ell = 3$, $|a_3| \le 3$ and $3 \mid a_3$. (iii) If $\bar\rho|_{D_\ell}$ is irreducible we are in case (ii) and $a_\ell \equiv 0$. Otherwise we are in case (i) (in case (ii) the restriction is irreducible), and the one-dimensional quotients of $\bar\rho|_{D_\ell}$ are among the two characters of its semisimplification, $\bar\chi_\ell\mu(a_\ell^{-1})$ and $\mu(a_\ell)$; the first is ramified because $\omega \ne 1$ for $\ell$ odd, so $\mu(a_\ell)$ is the unique unramified one-dimensional quotient, and its value at $\mathrm{Frob}_\ell$ is $a_\ell \bmod \ell$. For newforms the description of $\bar\rho_{g,\lambda}|_{D_\ell}$ is due to Deligne (ordinary) and Fontaine (supersingular); see [61, Section 2], as cited by the paper, and Chapter 15. We do not prove these. (The paper writes $h$ for this newform; here it is $g$, and $h$ is the matrix $\mathrm{diag}(1,-1)$.) □

*In the paper* 11.36. Lemma lem:old, the case $p = \ell$: "By the theorems of Deligne and Fontaine describing $\bar\rho_f|_{D_\ell}$ and $\bar\rho_h|_{D_\ell}$ [61, §2], the residue class of $a_\ell$ is determined by the restriction to $D_\ell$: it is 0 when that restriction is irreducible, and otherwise the eigenvalue of Frobenius on the unramified quotient." This is Theorem 11.35(iii). The paper then says why that quotient is well defined: "the two characters of the semisimplification restrict to $I_\ell$ as $\bar\chi_\ell$ and 1, and $\bar\chi_\ell|_{I_\ell} \ne 1$ since $\ell$ is odd, so the unramified quotient is the module of $I_\ell$-coinvariants". In the notation of this chapter $\bar\chi_\ell|_{I_\ell} = \omega$, the fundamental character of level 1, which is nontrivial because it has order $\ell - 1 \ge 2$. Example: $\bar\rho_{\texttt{121c1},3} \simeq \bar\rho_{\texttt{11a1},3}$ (Example 11.42), both curves are ordinary at 3, and indeed $a_3(\texttt{121c1}) = 2 \equiv -1 = a_3(\texttt{11a1}) \pmod 3$.

**Corollary 11.37.** *Let $\ell$ be odd and $\ell \nmid N$.*

*(i) If $\bar\rho_{E,\ell}$ is reducible, then $E$ is ordinary at $\ell$, and the two characters of $\bar\rho^{\mathrm{ss}}$ restrict to $I_\ell$ as $\omega$ and 1.*

*(ii) If $\bar\rho_{E,\ell} \simeq \bar\rho_{E,\ell} \otimes \chi_{\ell^*}$, then $\ell = 3$ and $E$ is ordinary at 3.*

*Proof.* (i) In the supersingular case $\bar\rho|_{D_\ell}$, hence $\bar\rho$, is irreducible. In the ordinary case the characters of $\bar\rho^{\mathrm{ss}}$, restricted to $I_\ell$, are the composition factors of $\bar\rho|_{I_\ell}$, which are $\omega$ and 1 by Theorem 11.35(i) (Jordan-Hölder, Chapter 3).

(ii) Restrict to $I_\ell$ and compare semisimplifications over $\mathbb{F}_{\ell^2}$. As $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$ (Convention 11.1(d)), $\chi_{\ell^*}|_{I_\ell} = \omega^{(\ell-1)/2}$, which is the unique character of order 2 of the cyclic group $\omega(I_\ell) = \mathbb{F}_\ell^\times$: so $\omega^{(\ell-1)/2} \ne 1$. In the ordinary case the multisets $\{\omega, 1\}$ and $\{\omega^{(\ell+1)/2}, \omega^{(\ell-1)/2}\}$ must agree; as $\omega^{(\ell-1)/2} \ne 1$, the character 1 must equal $\omega^{(\ell+1)/2}$, so $\ell - 1 \mid (\ell+1)/2$, which for $\ell \ge 3$

forces $\ell-1 \le (\ell+1)/2$, that is, $\ell = 3$. In the supersingular case $\omega^{(\ell-1)/2} = \omega_2^{(\ell+1)(\ell-1)/2} = -1$ (the element of order 2 of $\mathbb{F}_{\ell^2}^\times$), and $\{\omega_2, \omega_2^\ell\} = \{-\omega_2, -\omega_2^\ell\}$ with $-\omega_2 \neq \omega_2$ would need $\omega_2^\ell = -\omega_2$, that is, $\omega_2^{\ell-1} = -1 = \omega_2^{(\ell^2-1)/2}$; as $\omega_2$ has order $\ell^2-1$ this means $\ell^2-1 \mid (\ell^2-1)/2-(\ell-1) = (\ell-1)^2/2$, that is, $\ell+1 \mid (\ell-1)/2$, impossible since $0 < (\ell-1)/2 < \ell+1$. So only the ordinary case with $\ell = 3$ remains. □

Both parts were checked on the data: every curve of conductor below 10,000 with a Borel or split Cartan label at an odd prime $\ell \nmid N$ (4,571 pairs of a curve and a prime: 3,628 at $\ell = 3$, 637 at 5, 238 at 7, 60 at 13, 8 at 37) has $\ell \nmid a_\ell$; and the 62 curves with label `3Ns` and $3 \nmid N$ all have $3 \nmid a_3$.

## 11.6 Conductors of mod $\ell$ representations

The optimal-level theorem lowers the level of a newform to the Serre conductor $N(\bar\rho)$, and every case of Theorem thm:qual is a computation of $N(\bar\rho)$, or of $N(\bar\rho \otimes \chi_{q^*})$, at one prime. This section defines the Serre conductor and proves the formula $a_p(\bar\rho_{E,\ell}) = f_p(E) - \dim \Phi_p^{\mathrm{geom}}[\ell]$, which contains the divisibility $N(\bar\rho_{E,\ell}) \mid N$ that the paper quotes from [22], and every local exponent the paper uses.

**Definition 11.38** (Serre conductor)**.** Let $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_n(\overline{\mathbb{F}}_\ell)$ be continuous (so its image is finite), on a space $W$, and let $p \neq \ell$. The *Artin exponent* of $\bar\sigma$ at $p$ is

$$a_p(\bar\sigma) = \dim W - \dim W^{I_p} + \mathrm{sw}_p(\bar\sigma),$$

where the Swan conductor $\mathrm{sw}_p$ is defined from the ramification groups of the finite image of $D_p$ exactly as for complex representations (Chapter 7). The *Serre conductor* is $N(\bar\sigma) = \prod_{p\neq\ell} p^{a_p(\bar\sigma)}$; it is prime to $\ell$ by definition. The *Serre weight* $k(\bar\sigma)$ of an odd irreducible two-dimensional $\bar\sigma$ is defined from $\bar\sigma|_{I_\ell}$ by Serre's recipe [176, Section 2]; we use it only through Theorem 11.32(iii).

The definition copies the conductor of an $\ell$-adic or complex representation, so that for $\bar\rho_{E,\ell}$ it can be compared term by term with the conductor exponent $f_p(E)$ of $V_\ell E$; the comparison is Proposition 11.40. The prime $\ell$ is left out because ramification at $\ell$ is measured by the weight instead. Small examples, all one-dimensional: a quadratic character $\chi_d$ at an odd prime $p \mid d$ has $W^{I_p} = 0$ and tame ramification, so $a_p(\chi_d) = 1$; at $p = 2$, $\chi_{-4}$ and $\chi_{\pm 8}$ have conductors 4 and 8, so $a_2(\chi_{-4}) = 2$ and $a_2(\chi_{\pm 8}) = 3$, the excess over 1 being the Swan conductor. For instance, $\chi_{-3}$ viewed modulo $\ell \neq 3$ has Serre conductor 3.

**Proposition 11.39.** *(i) $a_p(\bar\sigma) \ge 0$, and $a_p(\bar\sigma) = 0$ iff $\bar\sigma$ is unramified at $p$. So $N(\bar\sigma)$ is prime to every prime at which $\bar\sigma$ is unramified.*

*(ii) $a_p$ is unchanged by extension of scalars $\overline{\mathbb{F}}_\ell \supseteq k' \supseteq k$ and by twisting by a character unramified at $p$.*

*(iii) If the image of the wild inertia group $P_p$ acts trivially, $a_p(\bar\sigma) = \dim W - \dim W^{I_p}$.*

*Proof.* (i) Both $\dim W - \dim W^{I_p}$ and $\mathrm{sw}_p$ are nonnegative (Chapter 7). If $\bar\sigma$ is unramified, $W^{I_p} = W$ and every ramification group acts trivially, so $\mathrm{sw}_p = 0$ and $a_p = 0$. Conversely $a_p = 0$ forces $W^{I_p} = W$. (ii) The fixed spaces of the ramification groups commute with extension of scalars (they are kernels of linear maps defined over the smaller field), and they are unchanged by a twist which is trivial on $I_p$ and its subgroups. (iii) If $P_p$ acts trivially, so do all higher ramification groups, which lie in $P_p$, and $\mathrm{sw}_p = 0$. □

**Proposition 11.40** (Serre conductor of $\bar\rho_{E,\ell}$)**.** *Let $p \neq \ell$ be primes. Then*

$$a_p(\bar\rho_{E,\ell}) = f_p(E) - \dim_{\mathbb{F}_\ell} \Phi_p^{\mathrm{geom}}[\ell].$$

*Hence $N(\bar\rho_{E,\ell})$ divides the prime-to-$\ell$ part of $N$, with $v_p(N(\bar\rho_{E,\ell})) = v_p(N)$ unless $\ell \mid \#\Phi_p^{\mathrm{geom}}$. Explicitly the exponent is: $0$ at a good prime; $0$ or $1$ at a multiplicative prime according as $\ell \mid n_p$ or not; and, for $\ell$ odd, at an additive prime $f_p - 1$ if $\ell = 3$ and the type is* IV *or* IV$^*$*, and $f_p$ otherwise. In particular $\bar\rho_{E,\ell}$, $\ell$ odd, is ramified at every prime $p \neq \ell$ of additive reduction.*

*Proof.* The image $\rho(P_p)$ of the wild inertia group is a pro-$p$ subgroup of $\mathrm{GL}_2(\mathbb{Z}_\ell)$; it meets the pro-$\ell$ group $1 + \ell M_2(\mathbb{Z}_\ell)$ trivially, so it injects into $\mathrm{GL}_2(\mathbb{F}_\ell)$ and is a finite $p$-group, of order prime to $\ell$. For a finite group $H$ of order prime to $\ell$ acting on $T$, taking invariants commutes with reduction: the averaging operator $e_H = \frac{1}{\#H}\sum_{h\in H} h$ is defined over $\mathbb{Z}_\ell$, and $T^H = e_H T$, $(T/\ell T)^H = e_H(T/\ell T) = T^H/\ell T^H$, so $\dim_{\mathbb{F}_\ell}(T/\ell)^H = \mathrm{rank}_{\mathbb{Z}_\ell} T^H = \dim_{\mathbb{Q}_\ell} V^H$ (Chapter 4). (In more detail: $e_H$ is an idempotent commuting with $H$, so $T = e_H T \oplus (1 - e_H)T$; an element fixed by $H$ satisfies $e_H x = x$, and conversely $he_H = e_H$ for $h \in H$; the same holds on $T/\ell T$, and reducing the decomposition of $T$ modulo $\ell$ shows $e_H(T/\ell T) = e_H T/\ell e_H T$, a free module over $\mathbb{F}_\ell$ of rank $\mathrm{rank}\, e_H T$.) The Swan conductor is $\sum_{i\geq 1} \frac{\#G_i}{\#G_0} \dim(W/W^{G_i})$, where the $G_i$ are the ramification groups of a finite Galois extension $L/\mathbb{Q}_p$ whose own wild inertia group acts trivially on $T$ (Chapter 7); one such $L$ serves for $V$ and for $E[\ell]$ at once. For $i \geq 1$ the group $G_i$ lies in $G_1$, the image of $P_p$, and acts on $T$ through $\rho(P_p)$, of order prime to $\ell$, so the codimension of its fixed space is the same on $E[\ell]$ and on $V$. This gives $\mathrm{sw}_p(\bar\rho) = \mathrm{sw}_p(V)$. Then, with Corollary 11.21,

$$a_p(\bar\rho) = 2 - \dim E[\ell]^{I_p} + \mathrm{sw}_p(V) = f_p - (\dim E[\ell]^{I_p} - \delta_p).$$

It remains to show $\dim E[\ell]^{I_p} - \delta_p = \dim \Phi_p^{\mathrm{geom}}[\ell]$. Good reduction: $E[\ell]^{I_p} = E[\ell]$, $\delta_p = 2$, $\Phi_p^{\mathrm{geom}} = 0$. Multiplicative: by Lemma 11.10(ii) $\dim E[\ell]^{I_p} = 1 + \dim \Phi_p^{\mathrm{geom}}[\ell]$, and $\delta_p = 1$. Additive: by Lemma 11.10(i) $\dim E[\ell]^{I_p} = \dim \Phi_p^{\mathrm{geom}}[\ell]$, and $\delta_p = 0$. The explicit values follow from $\Phi_p^{\mathrm{geom}} \simeq \mathbb{Z}/n_p$ at multiplicative primes and Lemma 11.10(iii) at additive ones; $f_p \geq 2$ at an additive prime, and $\dim \Phi_p^{\mathrm{geom}}[\ell] \leq 1$ there for $\ell$ odd, so the exponent is at least 1. □

*In the paper* 11.41. The optimal-level paragraph of Section 3: "Recall also that $N(\bar\rho_{E',\ell})$ divides the conductor of $E'$ [22], and that $N(\bar\sigma)$ is prime to every prime at which $\bar\sigma$ is unramified." The second statement is Proposition 11.39(i), the first is Proposition 11.40 (Carayol proved it for the residual representations of all newforms; for elliptic curves the formula above is more precise, and is the form in which Livné [116] and Serre [176, Section 4] discuss it). Lemma lem:IV's "the Serre conductor of $\bar\rho_{E,3}$ has exponent 1 at $q$" is the case IV, IV$^*$; Lemma lem:old's hypothesis "$p \nmid M$ for every prime $p \,\|\, N$ at which $\bar\rho$ is unramified" is the multiplicative case with $\ell \mid n_p$.

**Example 11.42.** The following data were recomputed with PARI: the Kodaira types with the function `elllocalred`, and the exponents from Proposition 11.40. Every congruence was checked for all primes $p < 2000$ not dividing the two conductors and different from $\ell$, except at $\ell = 5$, where $p < 1224$, $p \neq 2, 5, 7$.

| curve | local data | $N(\bar\rho_{E,\ell})$ | residual congruence |
|---|---|---|---|
| $\ell=3$ | | | |
| 11a1 | $\mathrm{I}_5$ at 11 | 11 | |
| 121c1 | IV at 11, $f_{11}=2$ | 11 | $a_p(\texttt{121c1})\equiv a_p(\texttt{11a1})$ |
| 121a1 | II at 11 | $11^2$ | $a_p(\texttt{121a1})\equiv\chi_{-11}(p)a_p(\texttt{11a1})$ |
| 121c2 | $\mathrm{IV}^*$ at 11 (twist of 121a1 by $\chi_{-11}$) | 11 | |
| 77c1 | $\mathrm{I}_3$ at 7, $\mathrm{I}_2$ at 11 | 11 | $a_p(\texttt{77c1})\equiv a_p(\texttt{11a1})$ |
| 539c1 | $\mathrm{I}_3^*$ at 7, $\mathrm{I}_2$ at 11 | $7^2\cdot 11$ | $a_p(\texttt{539c1})\equiv\chi_{-7}(p)a_p(\texttt{77c1})$ |
| 52a1 | $\mathrm{IV}^*$ at 2, $\mathrm{I}_2$ at 13 | $2\cdot 13$ | $a_p(\texttt{52a1})\equiv a_p(\texttt{26b1})$ |
| 726a1 | $\mathrm{I}_6$ at 2, $\mathrm{I}_3$ at $3=\ell$, III at 11 | $11^2$ | |
| 6912d1 | III at 2 with $f_2=8$, II at $3=\ell$ | $2^8$ | |
| $\ell=5$ | | | |
| 11a1 | $\mathrm{I}_5$ at 11; $E[5]\simeq\mathbb{Z}/5\oplus\mu_5$ | 1 | |
| 140b1 | $\mathrm{IV}^*$ at 2, $\mathrm{I}_1$ at $5=\ell$, $\mathrm{I}_5$ at 7 | $2^2$ | $a_p(\texttt{140b1})\equiv a_p(\texttt{20a1})$ |
| 980i1 | $\mathrm{IV}^*$ at 2, $\mathrm{I}_1$ at $5=\ell$, $\mathrm{I}_5^*$ at 7 | $2^2\cdot 7^2$ | $a_p(\texttt{980i1})\equiv\chi_{-7}(p)a_p(\texttt{20a1})$ |

For instance, for 121c1 the exponent at 11 is $f_{11}-\dim\Phi_{11}^{\mathrm{geom}}[3]=2-1=1$, so $\bar\rho_{\texttt{121c1},3}$ has Serre conductor 11 and is isomorphic to $\bar\rho_{\texttt{11a1},3}$, irreducible (Example 11.8), by Proposition 11.6(iii); for 121a1 the exponent is 2, and the twist by $\chi_{-11}$ (to 121c2, type $\mathrm{IV}^*$) lowers it to 1. For 726a1, the multiplicative prime 2 has $3\mid n_2=6$ and drops out, and $3=\ell$ is excluded by definition. For 11a1 at $\ell=5$, $5\mid n_{11}=5$, so $\bar\rho_{\texttt{11a1},5}$ is unramified everywhere outside 5, as $E[5]\simeq\mathbb{Z}/5\oplus\mu_5$ requires. At $\ell=5$ the curves 140b1 and 980i1 have $5\mid N$, so they illustrate the mechanism of Theorem thm:qual (c), not the theorem.

## 11.7 The images of $\bar\rho_{E,\ell}$

Which subgroups of $\mathrm{GL}_2(\mathbb{F}_\ell)$ occur as images of $\bar\rho_{E,\ell}$ is a deep question in general and an elementary one at $\ell=3$. The paper needs the answer at $\ell=3$ in Proposition prop:ns (the three irreducible images), in Section sec:anomaly (the curves with image 3Ns), in Section sec:eis (the reducible images 3B.1.1 and 3Cs.1.1) and in Section sec:data ("the images of the mod $\ell$ Galois representations"). This section states the general theorems of Serre, Mazur and Zywina, proves the list at $\ell=3$ for every $E/\mathbb{Q}$ from the group theory of Chapter 1, fixes the dictionary with Sutherland's labels, and records how images are computed.

**Theorem 11.43** (Serre; Mazur). *(i) (Serre) If $E/\mathbb{Q}$ has no complex multiplication, $\rho_{E,\ell}(G_\mathbb{Q})$ is open in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ for every $\ell$, and $\bar\rho_{E,\ell}$ is surjective for all but finitely many $\ell$.*

*(ii) If $E$ has complex multiplication by an order in an imaginary quadratic field $K$ and $\ell$ is odd and unramified in $K$ (then $\ell$ is prime to the conductor of the order, which for $E/\mathbb{Q}$ is 1, 2 or 3, and 3 only for $K=\mathbb{Q}(\sqrt{-3})$, Chapter 9), the image of $\bar\rho_{E,\ell}$ lies in the normaliser of a Cartan subgroup, split or nonsplit according as $\ell$ splits or is inert in $K$, and is not contained in the Cartan subgroup itself.*

*(iii) (Mazur) $E/\mathbb{Q}$ has a rational isogeny of prime degree $\ell$ (equivalently, $\bar\rho_{E,\ell}$ is reducible, for $\ell$ odd) only for $\ell\in\{2,3,5,7,11,13,17,19,37,43,67,163\}$.*

*(iv) Whether $\bar\rho_{E,\ell}$ is surjective for every $\ell>37$ and every $E$ without complex multiplication (Serre's uniformity question) is open.*

We do not prove this; see Serre [171] for (i) and (ii) and Mazur [126] for (iii). For (ii): the endomorphisms defined over $K$ commute with $G_K$, so $\bar\rho(G_K)$ lies in the centraliser $(\mathcal{O}_K/\ell)^\times$ of the CM action, a Cartan subgroup; $G_\mathbb{Q}$ normalises it; and complex conjugation conjugates the CM action into its complex conjugate, so it is not in the Cartan subgroup. (iv) is stated only to mark the boundary of what is known.

**Proposition 11.44** (constraints on the image)**.** *Let $\ell$ be odd and $H = \bar\rho_{E,\ell}(G_\mathbb{Q})$. Then $\det H = \mathbb{F}_\ell^\times$ and $H$ contains an element conjugate to $\mathrm{diag}(1,-1)$. Hence $H \not\subseteq \mathrm{SL}_2(\mathbb{F}_\ell)$ and $H \not\subseteq C_{ns}(\mathbb{F}_\ell)$; and either $H = \mathrm{GL}_2(\mathbb{F}_\ell)$, or $H$ lies in a Borel subgroup ($\bar\rho$ reducible), or $H$ lies in the normaliser of a split or nonsplit Cartan subgroup and, if $\bar\rho$ is irreducible, not in the Cartan subgroup, or the image of $H$ in $\mathrm{PGL}_2(\mathbb{F}_\ell)$ is $A_4$, $S_4$ or $A_5$.*

*Proof.* $\det H = \mathbb{F}_\ell^\times$ and the element $\bar\rho(c) \sim \mathrm{diag}(1,-1)$ are Proposition 11.3(ii), (iii); $\mathrm{diag}(1,-1)$ has determinant $-1 \neq 1$. The nonsplit Cartan subgroup $C_{ns}(\mathbb{F}_\ell) \simeq \mathbb{F}_{\ell^2}^\times$ is cyclic, so its only involution is $-1$, of determinant 1; hence it does not contain a conjugate of $\mathrm{diag}(1,-1)$. By Dickson's classification (Chapter 1), if $\ell \mid \#H$ then $H$ lies in a Borel subgroup or contains $\mathrm{SL}_2(\mathbb{F}_\ell)$; in the second case $\det H = \mathbb{F}_\ell^\times$ forces $H = \mathrm{GL}_2(\mathbb{F}_\ell)$ ($H/\mathrm{SL}_2(\mathbb{F}_\ell)$ injects into $\mathbb{F}_\ell^\times$ by the determinant, and is onto). If $\ell \nmid \#H$, $H$ lies in the normaliser of a Cartan subgroup or has projective image $A_4$, $S_4$ or $A_5$. If $H$ lies in a split Cartan subgroup it is diagonal, so $\bar\rho$ is reducible; and it is not in a nonsplit one. □

The normaliser $N_s$ is not always a maximal subgroup: $N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$ up to conjugacy ($N_s(\mathbb{F}_3)$, of order 8, lies in a 2-Sylow subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$, and the 2-Sylow subgroups, of order 16, are the conjugates of $N_{ns}(\mathbb{F}_3)$), and $N_s(\mathbb{F}_5)$ lies in a subgroup with projective image $S_4$. So the large-$\ell$ picture of maximal subgroups must not be used at small $\ell$.

**Theorem 11.45** (the mod 3 images)**.** *For every $E/\mathbb{Q}$, with or without complex multiplication, the image $H = \bar\rho_{E,3}(G_\mathbb{Q})$ is conjugate in $\mathrm{GL}_2(\mathbb{F}_3)$ to exactly one of the eight subgroups in the following table; in particular never to $C_{ns}(\mathbb{F}_3)$ (label `3Cn`) and never to a subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$. The orbits of $H$ on the four lines of $E[3]$ correspond to the irreducible factors of $\psi_3$ over $\mathbb{Q}$, the orbit lengths being their degrees, and the group $H/(H \cap \{\pm 1\})$ acting on the lines is the Galois group of $\psi_3$.*

| *label* | *order* | *subgroup $H$* | *arithmetic meaning* | *degrees of factors of $\psi_3$* | $\mathrm{Gal}(\psi_3)$ |
|---|---|---|---|---|---|
| *(surjective)* | 48 | $\mathrm{GL}_2(\mathbb{F}_3)$ | *generic* | 4 | $S_4$ |
| `3Nn` | 16 | $N_{ns}(\mathbb{F}_3)$ | *irreducible, $\bar\rho \simeq \bar\rho \otimes \chi_d$, $d \neq -3$* | 4 | $D_4$ |
| `3Ns` | 8 | $N_s(\mathbb{F}_3)$ | *irreducible, $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$* | $2+2$ | $V_4$ |
| `3B` | 12 | $\mathrm{B}(\mathbb{F}_3)$ | *stable line, character $\neq 1, \chi_{-3}$* | $1+3$ | $S_3$ |
| `3B.1.1` | 6 | $\left(\begin{smallmatrix}1 & *\\ 0 & *\end{smallmatrix}\right)$ | *rational point of order 3, non-split* | $1+3$ | $S_3$ |
| `3B.1.2` | 6 | $\left(\begin{smallmatrix}* & *\\ 0 & 1\end{smallmatrix}\right)$ | *$\mu_3 \subset E[3]$, quotient $\mathbb{Z}/3$, non-split* | $1+3$ | $S_3$ |
| `3Cs` | 4 | $C_s(\mathbb{F}_3)$ | *$E[3] \simeq \eta \oplus \eta\chi_{-3}$, $\eta \neq 1, \chi_{-3}$* | $1+1+2$ | $C_2$ |
| `3Cs.1.1` | 2 | $\{\mathrm{diag}(1,\pm 1)\}$ | *$E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$* | $1+1+2$ | $C_2$ |

*Proof.* By Proposition 11.44, $H$ has surjective determinant and contains an element of determinant $-1$ and trace 0. By Chapter 1 the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ with surjective determinant fall into nine conjugacy classes, namely the eight of the table and $C_{ns}(\mathbb{F}_3)$, cyclic of order 8, and $C_{ns}(\mathbb{F}_3)$ is the only one of them without an element of determinant $-1$ and trace 0 (its elements of determinant $-1$ are its generators, of order 8, with nonzero trace). To see the last point, identify $C_{ns}(\mathbb{F}_3)$ with $\mathbb{F}_9^\times$ acting on $\mathbb{F}_9 \simeq \mathbb{F}_3^2$ by multiplication; then the determinant of multiplication by $x$ is the norm $x \cdot x^3 = x^4$ and its trace is $x + x^3$. If $x$ has order 8, then $x^4 = -1$, and

$x+x^3 = x(1+x^2) \neq 0$ because $x^2$ has order 4, so $x^2 \neq -1$. If $x$ has order dividing 4, then $x^4 = 1$. A subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$ does not have surjective determinant. So $H$ is conjugate to one of the eight. The four lines of $E[3]$ correspond bijectively and Galois-equivariantly to the four roots of $\psi_3$ (Proposition 11.6(ii)), so the orbits of $H$ on lines are the sets of roots of the irreducible factors of $\psi_3$, and the permutation group induced by $H$ on the lines, which is $H/(H \cap \{\pm 1\})$ ($\pm 1$ are the only elements fixing every line), is the Galois group of $\psi_3$. The orbit structures and groups in the table are read off from the subgroups (Chapter 1): $\mathrm{GL}_2(\mathbb{F}_3)$ and $N_{ns}(\mathbb{F}_3)$ are transitive, with quotients $S_4$ and $D_4$; $N_s(\mathbb{F}_3)$ has the two orbits $\{\langle e_1\rangle, \langle e_2\rangle\}$ and $\{\langle e_1+e_2\rangle, \langle e_1-e_2\rangle\}$, quotient $V_4$; the Borel groups fix $\langle e_1\rangle$ and permute the other three lines as $S_3$; the subgroups of $C_s$ fix $\langle e_1\rangle$, $\langle e_2\rangle$ and swap the other two. The arithmetic meanings: a stable line with character $\eta_1$ and quotient $\eta_2 = \chi_{-3}\eta_1^{-1}$ (determinant $\chi_{-3}$) gives image in B; it is $\left(\begin{smallmatrix}1 & *\\ 0 & *\end{smallmatrix}\right)$ iff $\eta_1 = 1$ (the points of the line are rational), $\left(\begin{smallmatrix}* & *\\ 0 & 1\end{smallmatrix}\right)$ iff $\eta_2 = 1$, i.e. $\eta_1 = \chi_{-3}$ (the line is $\mu_3$, by the Weil pairing); the image contains the unipotent group iff the extension is non-split; and in the split case $E[3] \simeq \eta_1 \oplus \eta_2$, with image $\{\mathrm{diag}(1, \pm 1)\}$ iff $\{\eta_1, \eta_2\} = \{1, \chi_{-3}\}$. The irreducible cases are Proposition 11.55 and Proposition 11.52: the image is $N_s(\mathbb{F}_3)$ iff $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$, and $N_{ns}(\mathbb{F}_3)$ is the normaliser of the Cartan subgroup $C_{ns}$, cut out by a quadratic field $\mathbb{Q}(\sqrt{d})$ with $\bar{\rho} \simeq \bar{\rho} \otimes \chi_d$ and $d \neq -3$ (as $C_{ns}(\mathbb{F}_3) \not\subseteq \mathrm{SL}_2(\mathbb{F}_3)$). □

Every case occurs. Among the 64,687 curves of conductor below 10,000 in Cremona's tables [36], of which 462 have complex multiplication, we classified the image of every curve from $\psi_3$ by the recipe of Remark 11.50 and compared it with the label in Cremona's file `galrep`, which follows [191]:

| label | non-CM | CM | examples |
|---|---|---|---|
| (surjective) | 53,760 | 0 | `11a1`, `15a1` |
| `3Nn` | 218 | 184 | `245a1`; CM: `32a1` |
| `3Ns` | 120 | 26 | `338d1`, `726a1`, `6912d1`; CM: `121b1`, `256a1` |
| `3B` | 4,072 | 154 | `50b1`, `539a1`; CM: `144a1` |
| `3B.1.1` | 2,772 | 39 | `14a4`, `20a1`; CM: `36a1` |
| `3B.1.2` | 2,772 | 39 | `14a3`, `20a3` |
| `3Cs` | 249 | 18 | `98a3`, `539a2` |
| `3Cs.1.1` | 262 | 2 | `14a1`, `19a1`, `26a1`; CM: `27a1`, `27a3` |

The classification from $\psi_3$ agrees with the `galrep` label for every curve that has one. The file gives no mod 3 label to 364 of the CM curves, those whose image is the one forced by the complex multiplication: the 26 curves with $j \in \{-32768, 8000\}$ (image `3Ns`), the 184 with $j \in \{1728, 287496, -3375, 16581375, -884736, -884736000, -147197952000\}$ (image `3Nn`) and the 154 with $j \in \{0, 54000, -12288000\}$ (image `3B`); the CM column counts all of them. Every count is for conductor below 10,000.

The equal counts of `3B.1.1` and `3B.1.2` in the table are not an accident. They hold class by class, because the rational 3-isogenies arrange each isogeny class into paths whose two ends carry the two labels.

**Proposition 11.46** (paths of 3-isogenies)**.** *Let $E/\mathbb{Q}$ be an elliptic curve, with or without complex multiplication. Let $\Gamma$ be the graph whose vertices are the $\mathbb{Q}$-isomorphism classes of the curves $\mathbb{Q}$-isogenous to $E$, two vertices being joined when there is an isogeny of degree 3 over $\mathbb{Q}$ between them (the dual isogeny also has degree 3, so the relation is symmetric).*

*(i) For each vertex $E'$, $E'[3]$ has at most two $G_{\mathbb{Q}}$-stable lines, and $C \mapsto E'/C$ is a bijection from the set of these lines to the set of neighbours of $E'$.*

*(ii)* *Each connected component of* $\Gamma$ *is a path* $E_0, E_1, \dots, E_k$ *(pairwise distinct vertices,* $E_{i-1}$ *joined to* $E_i$*). If* $k \geq 1$*, the two ends have exactly one stable line in* $E_0[3]$*, respectively* $E_k[3]$*, so their images are* `3B`*,* `3B.1.1` *or* `3B.1.2`*; the interior vertices have two, so their images are* `3Cs` *or* `3Cs.1.1`*.*

*(iii)* *If* $k \geq 1$ *and* $G_{\mathbb{Q}}$ *acts on the stable line of* $E_0[3]$ *by the character* $\eta$*, then it acts on the stable line of* $E_k[3]$ *by* $\chi_{-3}\eta^{-1}$*. Hence* $E_0$ *has image* `3B.1.1` *iff* $E_k$ *has image* `3B.1.2`*,* $E_0$ *has image* `3B.1.2` *iff* $E_k$ *has image* `3B.1.1`*, and* $E_0$ *has image* `3B` *iff* $E_k$ *has.*

*(iv)* *In every isogeny class over* $\mathbb{Q}$ *the number of curves with image* `3B.1.1` *equals the number with image* `3B.1.2`*.*

*Proof.* All isogenies and isomorphisms are over $\mathbb{Q}$. We use two facts from Chapter 9 and one from Chapter 10 and the literature. (F1) A finite $G_{\mathbb{Q}}$-stable subgroup $C$ of $A(\overline{\mathbb{Q}})$ is the kernel of an isogeny $A \to A/C$, and an isogeny is determined by its kernel up to an isomorphism of its target; for an isogeny $\varphi$ of degree 3, $\hat{\varphi}\varphi = [3]$, so $\ker\hat{\varphi} = \varphi(A[3])$, a line. (F2) Every $A/\mathbb{Q}$ has $\mathrm{End}_{\mathbb{Q}}(A) = \mathbb{Z}$ (the endomorphisms of a curve with complex multiplication by $K$ that are not integers are defined only over $K$); so $\mathrm{Aut}_{\mathbb{Q}}(A) = \{\pm 1\}$, and an endomorphism over $\mathbb{Q}$ of degree $3^m$ is $\pm 3^{m/2}$, whose kernel $A[3^{m/2}]$ is not cyclic when $m \geq 1$. (F3) An isogeny class over $\mathbb{Q}$ is finite: isogenous curves have the same conductor (Chapter 10), and there are finitely many curves over $\mathbb{Q}$ of given conductor (Shafarevich's theorem, [185, Section IX.6]).

*Step 1: cyclic composites.* Let $\varphi_i\colon A_{i-1} \to A_i$ $(1 \leq i \leq m)$ be isogenies of degree 3 with $\ker\varphi_{i+1} \neq \varphi_i(A_{i-1}[3])$ for $1 \leq i < m$. Then $\Phi_m = \varphi_m \circ \cdots \circ \varphi_1$ has cyclic kernel, of order $3^m$. For $m = 1$ this is clear. Suppose $\Phi_{m-1}$ has cyclic kernel and $\ker\Phi_m$ is not cyclic. A finite abelian 3-group that is not cyclic contains a subgroup $(\mathbb{Z}/3)^2$, and in $A_0$ that subgroup is $A_0[3]$; so $\Phi_m$ kills $A_0[3]$, that is, $\varphi_m$ kills $\Phi_{m-1}(A_0[3])$. As $\ker\Phi_{m-1}$ is cyclic, it meets $A_0[3]$ in a group of order 3, so $\Phi_{m-1}(A_0[3])$ has order 3; it lies in $\varphi_{m-1}(A_{m-2}[3])$, which also has order 3, so the two are equal, and $\ker\varphi_m = \varphi_{m-1}(A_{m-2}[3])$: a contradiction.

(i) If three lines of $E'[3]$ were stable, every element of $H = \bar{\rho}_{E',3}(G_{\mathbb{Q}})$ would have three eigenlines in a plane, hence be a scalar $\pm 1$, of determinant 1; but $\det H = \mathbb{F}_3^\times$ (Proposition 11.3(ii)). The neighbours of $E'$ are the curves $E'/C$, by (F1), since the kernel of an isogeny of degree 3 is a stable line. Suppose $C \neq C'$ are stable lines and $\iota\colon E'/C \to E'/C'$ is an isomorphism; let $\varphi\colon E' \to E'/C$ and $\psi\colon E' \to E'/C'$ be the quotient maps. If $\ker(\hat{\psi}\iota) = \varphi(E'[3]) = \ker\hat{\varphi}$, then by (F1) and (F2) $\hat{\psi}\iota = \pm\hat{\varphi}$; dualising (the dual of an isomorphism is its inverse), $\psi = \pm\iota\varphi$, so $C' = \ker\psi = \ker\varphi = C$, which is false. So Step 1 applies to $\varphi$ and $\hat{\psi}\iota$: the endomorphism $\hat{\psi}\iota\varphi$ of $E'$ has degree 9 and cyclic kernel, contradicting (F2). Hence $C \mapsto E'/C$ is injective.

(ii) $\Gamma$ has no cycle. Let $v_0, v_1, \dots, v_m = v_0$ be a cycle, $m \geq 3$, with $v_0, \dots, v_{m-1}$ distinct. Choose curves $A_i$ in the classes $v_i$, with $A_m = A_0$, and isogenies $\varphi_i\colon A_{i-1} \to A_i$ of degree 3. If $\ker\varphi_{i+1} = \varphi_i(A_{i-1}[3]) = \ker\hat{\varphi}_i$ for some $1 \leq i < m$, then $A_{i+1} \simeq A_i/\ker\hat{\varphi}_i \simeq A_{i-1}$ by (F1), so $v_{i+1} = v_{i-1}$, which is impossible in a cycle of length $m \geq 3$. By Step 1, $\varphi_m \circ \cdots \circ \varphi_1$ is an endomorphism of $A_0$ of degree $3^m$ with cyclic kernel, contradicting (F2). So each component is a tree, finite by (F3), in which every vertex has degree at most 2 by (i): a path. By (i) the degree of a vertex is its number of stable lines, 1 at the ends and 2 in the interior when $k \geq 1$. One stable line means $E[3]$ is a non-split extension (a split $E[3]$ has the two lines of its decomposition), two mean that it is split; the labels follow from Theorem 11.45.

(iii) Let $\varphi_i\colon E_{i-1} \to E_i$ $(1 \leq i \leq k)$ be isogenies of degree 3, $C_{i-1} = \ker\varphi_i \subset E_{i-1}[3]$ and $B_i = \varphi_i(E_{i-1}[3]) = \ker\hat{\varphi}_i \subset E_i[3]$, both stable lines. The map $\varphi_i$ induces a $G_{\mathbb{Q}}$-isomorphism $E_{i-1}[3]/C_{i-1} \simeq B_i$, and $\det\bar{\rho}_{E_{i-1},3} = \chi_{-3}$ (Proposition 11.3(ii)); so if $G_{\mathbb{Q}}$ acts on $C_{i-1}$ by $\alpha$, it

acts on $B_i$ by $\chi_{-3}\alpha^{-1}$. For $0 < i < k$, $E_i/B_i \simeq E_{i-1}$ and $E_i/C_i \simeq E_{i+1} \neq E_{i-1}$, so $B_i \neq C_i$ by (i), $E_i[3] = B_i \oplus C_i$, and $G_{\mathbb{Q}}$ acts on $C_i$ by $\chi_{-3}$ divided by its character on $B_i$. Starting from $\eta$ on $C_0$, induction gives $\chi_{-3}\eta^{-1}$ on every $B_i$ and $\eta$ on every $C_i$; at the end, $B_k$ is the stable line of $E_k[3]$. By Theorem 11.45 (the arithmetic meanings in its proof), a non-split $E'[3]$ whose stable line has character $\eta$ has image `3B.1.1` if $\eta = 1$, `3B.1.2` if $\eta = \chi_{-3}$, and `3B` otherwise; and $\eta \mapsto \chi_{-3}\eta^{-1}$ exchanges 1 and $\chi_{-3}$ and maps every other character to another character outside $\{1, \chi_{-3}\}$.

(iv) A curve with image `3B.1.1` or `3B.1.2` has exactly one stable line, so by (i) and (ii) it is an end of a component with $k \geq 1$, whose other end is a different curve. Sending each end to the other end of its component is an involution of the set of such ends in the class, and by (iii) it exchanges the two labels. □

The data agree. In Cremona's tables below conductor 10,000, the 3-isogenies read from the isogeny matrices of the file `allisog` form 4,924 components with at least one edge in the 38,042 classes; every one is a path whose interior vertices have image `3Cs` or `3Cs.1.1` and whose ends have the images `3B` and `3B`, or `3B.1.1` and `3B.1.2`; and in every class the two labels occur equally often.

*Remark* 11.47. The curves with a rational point of order 3 are exactly those with image `3B.1.1` or `3Cs.1.1` (Proposition 11.65): 2,811 and 264 curves below conductor 10,000, as the torsion orders in [36] confirm. The labels `3B.1.1` and `3B.1.2` occur equally often in the table (2,772 non-CM and 39 CM curves each), by Proposition 11.46(iv). The twist by $\chi_{-3}$ also exchanges them (Proposition 11.5(ii)), but it changes the conductor in general, so it does not explain the equality below a conductor bound; the pairing inside the class does. For example, the class `14a` consists of the two paths `14a3`, `14a1`, `14a4` and `14a5`, `14a2`, `14a6`: the ends `14a3`, `14a5` have image `3B.1.2`, the ends `14a4`, `14a6` image `3B.1.1`, and the middle curves `14a1`, `14a2` image `3Cs.1.1` ($E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$; the quotient by $\mu_3$ is the `3B.1.1` end, which has the rational point of order 3 coming from $E[3]/\mu_3$, and the quotient by $\mathbb{Z}/3$ is the `3B.1.2` end). The image `3Cs.1.1` is not the split Cartan subgroup, which is `3Cs`, but its subgroup $\{1, \mathrm{diag}(1,-1)\}$ of order 2. The image `3Nn` is dihedral too: $\bar{\rho}$ is induced from the quadratic field cut out by $\bar{\rho}^{-1}(C_{ns})$ (Proposition 11.52); what fails for `3Ns`, and only for it, is irreducibility over $\mathbb{Q}(\sqrt{-3})$ (Proposition 11.55).

**Convention 11.48** (Sutherland's labels)**.** A label such as `3B.1.1` names a conjugacy class of subgroups of $\mathrm{GL}_2(\mathbb{F}_\ell)$: the prime $\ell$; the type of the smallest standard subgroup containing it, `B` (Borel), `Cs`, `Cn` (split, nonsplit Cartan), `Ns`, `Nn` (their normalisers), or `A4`, `S4`, `A5` (exceptional projective image); then optional numbers identifying the subgroup inside that one. Surjective images have no label. We give the general rule only by reference [191] and always state the subgroup intrinsically; the dictionary at $\ell = 3$ is the table of Theorem 11.45, which we checked against Cremona's data. At $\ell = 5$: `5Cs.1.1` for `11a1` ($E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$), `5B.1.1` for `11a3` (a rational point of order 5, $E[5]$ non-split), `5B.1.2` for `11a2`. The labels at $\ell = 5, 7, 11, 13$ occurring for conductor below 10,000 are:

| $\ell$ | labels |
|---|---|
| 5 | `5B`, `5B.1.1`, `5B.1.2`, `5B.1.3`, `5B.1.4`, `5B.4.1`, `5B.4.2`, `5Cs.1.1`, `5Cs.1.3`, `5Cs.4.1`, `5Nn`, `5Ns`, `5Ns.2.1`, `5S4` |
| 7 | `7B`, `7B.1.1` to `7B.1.6`, `7B.2.1`, `7B.2.3`, `7B.6.1` to `7B.6.3`, `7Ns`, `7Ns.2.1`, `7Ns.3.1`, `7Ns.6.1.2` |
| 11 | `11B.1.3` to `11B.1.8`, `11B.10.3` to `11B.10.5`, `11Nn.1.4` |
| 13 | `13B`, `13B.3.1`, `13B.3.2`, `13B.4.1`, `13B.4.2`, `13B.5.1`, `13B.5.2`, `13Ns.5.1.4` |

**Theorem 11.49** (Zywina)**.** *For $E/\mathbb{Q}$ without complex multiplication and $\ell \in \{2,3,5,7,11,13\}$, the subgroups of $\mathrm{GL}_2(\mathbb{F}_\ell)$ that occur as $\bar\rho_{E,\ell}(G_\mathbb{Q})$ are known up to conjugacy, together with explicit families of $j$-invariants (rational functions on modular curves, and finitely many exceptional $j$) realising each of them.*

We do not prove this; see Zywina [210, Section 1]. At $\ell = 13$ the list in [210] depended on the rational points of certain modular curves, which later work determined; we give no reference for that. The book needs only $\ell = 3$, which Theorem 11.45 proves directly and for CM curves as well; the paper cites [210] for the three irreducible images at $\ell = 3$ and derives them directly.

*Remark* 11.50 (computing images). (a) At $\ell = 3$ the image is determined by $\psi_3$: its factorisation pattern, the Galois group of an irreducible quartic factor, and, for each rational root $x_0$, the character of the line, which is $\chi_D$ with $D = 4x_0^3 + b_2x_0^2 + 2b_4x_0 + b_6$ (the line is $\{O, \pm P\}$ with $(2y(P) + a_1x_0 + a_3)^2 = D$, and $g(P) = \pm P$ according as $g$ fixes $\sqrt{D}$). The following function returns the label of Theorem 11.45; on all 64,687 curves of conductor below 10,000 it agrees with Cremona's `galrep` file wherever that has a label.

```
mod3image(E) =
{
  my(F = factor(elldivpol(E, 3))[,1], d, f, r, ch);
  d = vecsort([poldegree(g) | g <- F]);
  if (d == [4], return (if (polgalois(F[1])[1] == 24, "GL2", "3Nn")));
  if (d == [2, 2], return ("3Ns"));
  f = 4*x^3 + E.b2*x^2 + 2*E.b4*x + E.b6;
  r = [subst(f, x, -polcoef(g, 0)/polcoef(g, 1)) | g <- F, poldegree(g) == 1];
  ch = [core(numerator(t)*denominator(t)) | t <- r];
  if (#r == 1, return (if (ch[1] == 1, "3B.1.1", if (ch[1] == -3, "3B.1.2", "3B"))));

  if (vecsort(ch) == [-3, 1], "3Cs.1.1", "3Cs");
}
```

The paper's appendix uses the shorter test `ns3`, which returns 1 iff $\psi_3$ is a product of two quadratics; by the table this detects $N_s(\mathbb{F}_3)$ exactly (a reducible $\bar\rho$ gives a rational root). (b) For general $\ell$, Sutherland's algorithm [191] computes the image from the statistics of the characteristic polynomials of $\bar\rho(\mathrm{Frob}_p)$ for many $p$ (Chebotarev), together with division polynomials and Zywina's families; we use its output through Cremona's files. (c) The inertia data of this chapter were computed from the fields $\mathbb{Q}(P)$, $P \in E[3]$: for each irreducible factor $g$ of $\psi_3$, the field generated by a root $x_0$ and by $\sqrt{D(x_0)}$, and the ramification indices of $q$ in it (`nfinit`, `idealprimedec`). When the image of inertia has order prime to $q$, its order is the least common multiple of these indices, and $E[3]^{I_q} \neq 0$ iff some prime above $q$ in some $\mathbb{Q}(P)$ is unramified.

*In the paper* 11.51. Section sec:data: "The mod 3 images of the tables were checked against the factorisation of the 3-division polynomial (Proposition prop:ns) on every curve for which it mattered." The appendix's `ns3`: "`ns3(E) = my(F = factor(elldivpol(E, 3))); (#F[,1] == 2 && poldegree(F[1,1]) == 2);`". Both rest on Theorem 11.45. Section sec:eis calls `3Cs.1.1` "the subgroup of order 2 of the split Cartan subgroup that fixes a point".

## 11.8 Absolute irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$, Proposition prop:ns and Lemma lem:h0

The hypothesis of Theorem thm:main is that $\bar\rho$ stay absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, which is the condition $\ell \notin S_f$ of [49, (31)]. This section shows what that condition means for elliptic curves: by Clifford theory its failure is $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$; by the local analysis at $\ell$ this happens, for $\ell \nmid N$, only at $\ell = 3$ and only for the image `3Ns`; and it is then measured by the group $H^0(\mathbb{Q}, \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ of Lemma lem:h0, whose length is at most one. It proves the paper's Proposition prop:ns, Lemma lem:h0, Proposition prop:ns-structure and Proposition prop:Sf, with every compressed step expanded, and explains Section sec:anomaly.

**Proposition 11.52** (dihedral representations)**.** *Let $\ell$ be odd, $\bar\rho = \bar\rho_{E,\ell}$ irreducible, $K$ a quadratic field and $\chi_K$ its character. The following are equivalent:*

*(a) $\bar\rho|_{G_K}$ is not absolutely irreducible;*

*(b) $\bar\rho \simeq \bar\rho \otimes \chi_K$ (over $\mathbb{F}_\ell$, equivalently over $\overline{\mathbb{F}}_\ell$);*

*(c) $\bar\rho \otimes \overline{\mathbb{F}}_\ell \simeq \mathrm{Ind}_{G_K}^{G_\mathbb{Q}} \psi$ for a character $\psi\colon G_K \to \overline{\mathbb{F}}_\ell^\times$;*

*(d) $a_p(E) \equiv 0 \pmod\ell$ for all primes $p \nmid N\ell$ inert in $K$ (equivalently, for those outside a set of density zero).*

*When they hold, every isomorphism $\phi\colon \bar\rho \to \bar\rho \otimes \chi_K$ has trace 0, and $H^0(\mathbb{Q}, \mathrm{ad}^0 \bar\rho \otimes \chi_K) = \mathrm{Hom}_{G_\mathbb{Q}}(\bar\rho, \bar\rho \otimes \chi_K)$ is a line.*

*Proof.* By Proposition 11.6(i), $\bar\rho$ is absolutely irreducible; so by Schur's lemma $\mathrm{End}_{G_\mathbb{Q}}(\bar\rho \otimes \overline{\mathbb{F}}_\ell) = \overline{\mathbb{F}}_\ell$.

(b)⇒(a). An isomorphism $\phi\colon \bar\rho \to \bar\rho \otimes \chi_K$ satisfies $\phi\bar\rho(g) = \chi_K(g)\bar\rho(g)\phi$, so it commutes with $\bar\rho(G_K)$, and it is not a scalar (a scalar would force $\chi_K = 1$). So $\mathrm{End}_{G_K}(\bar\rho \otimes \overline{\mathbb{F}}_\ell)$ has dimension at least 2, and $\bar\rho|_{G_K}$ is not absolutely irreducible.

(a)⇒(c). Let $L \subset \bar\rho \otimes \overline{\mathbb{F}}_\ell$ be a $G_K$-stable line, with character $\psi$. By Frobenius reciprocity, $\mathrm{Hom}_{G_\mathbb{Q}}(\mathrm{Ind}_{G_K}^{G_\mathbb{Q}} \psi, \bar\rho \otimes \overline{\mathbb{F}}_\ell) = \mathrm{Hom}_{G_K}(\psi, \bar\rho|_{G_K} \otimes \overline{\mathbb{F}}_\ell) \neq 0$ (Chapter 4). A nonzero map from the two-dimensional $\mathrm{Ind}\,\psi$ to the absolutely irreducible two-dimensional $\bar\rho \otimes \overline{\mathbb{F}}_\ell$ is surjective, hence an isomorphism.

(c)⇒(b). $\mathrm{Ind}\,\psi \otimes \chi_K \simeq \mathrm{Ind}(\psi \otimes \chi_K|_{G_K}) = \mathrm{Ind}\,\psi$, as $\chi_K$ is trivial on $G_K$. So $\mathrm{Hom}_{G_\mathbb{Q}}(\bar\rho, \bar\rho \otimes \chi_K) \otimes \overline{\mathbb{F}}_\ell \neq 0$; the space of $G_\mathbb{Q}$-maps is the kernel of a system of linear equations over $\mathbb{F}_\ell$, and its dimension does not change under extension of scalars, so there is a nonzero $\mathbb{F}_\ell$-linear $G_\mathbb{Q}$-map $\bar\rho \to \bar\rho \otimes \chi_K$, which is an isomorphism as both are irreducible of the same dimension.

(b)⇒(d). For $p \nmid N\ell$ inert in $K$, $\chi_K(\mathrm{Frob}_p) = -1$, and taking traces at $\mathrm{Frob}_p$ gives $a_p \equiv -a_p$, so $2a_p \equiv 0$ and $a_p \equiv 0 \pmod\ell$.

(d)⇒(b). The representations $\bar\rho$ and $\bar\rho \otimes \chi_K$ have the same determinant ($\chi_K^2 = 1$) and the same trace $\pm a_p$ at $\mathrm{Frob}_p$ for every $p \nmid N\ell d_K$: at split primes $\chi_K(p) = 1$, and at inert primes both traces are 0 outside a set of density zero. By Proposition 11.6(iii) (Chebotarev and Brauer-Nesbitt), $\bar\rho \simeq \bar\rho \otimes \chi_K$, both being irreducible.

Trace zero. Choose $g \in G_\mathbb{Q} \backslash G_K$; then $\phi\bar\rho(g) = -\bar\rho(g)\phi$, so $\bar\rho(g)\phi\bar\rho(g)^{-1} = -\phi$ and $\mathrm{tr}\,\phi = -\mathrm{tr}\,\phi$, whence $\mathrm{tr}\,\phi = 0$ ($\ell$ odd). So $\mathrm{Hom}_{G_\mathbb{Q}}(\bar\rho, \bar\rho \otimes \chi_K)$ lies in $(\mathrm{ad}^0 \bar\rho \otimes \chi_K)^{G_\mathbb{Q}}$, and conversely; it is a line because any two isomorphisms differ by an automorphism of $\bar\rho$, a scalar. □

For $K = \mathbb{Q}(\sqrt{\ell^*})$ condition (a) is exactly the failure of the hypothesis of Theorem thm:main. Not every dihedral $\bar\rho$ fails it: the image `3Nn` is induced from a quadratic field other than $\mathbb{Q}(\sqrt{-3})$, and those curves satisfy the hypothesis.

**Theorem 11.53** (the hypothesis on $\mathbb{Q}(\sqrt{\ell^*})$ is automatic for $\ell \geq 5$)**.** *Let $\ell$ be odd, $\ell \nmid N$, and $\bar\rho_{E,\ell}$ irreducible. If $\ell \geq 5$, then $\bar\rho_{E,\ell}$ is absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$. If $\ell = 3$ and $\bar\rho_{E,3}$ is not absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, then $E$ is ordinary at 3 (and the image is $N_s(\mathbb{F}_3)$, Proposition 11.55).*

*Proof.* If $\bar\rho$ is not absolutely irreducible on $G_{\mathbb{Q}(\sqrt{\ell^*})}$, then $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$ by Proposition 11.52, and Corollary 11.37(ii) gives $\ell = 3$ and ordinary reduction at 3. □

The data agree. Below conductor 10,000, the curves with a label $\ell$`Ns` or $\ell$`Nn`, $\ell \geq 5$, for which $a_p \equiv 0 \pmod{\ell}$ at each of the first 150 primes $p \nmid N\ell$ inert in $\mathbb{Q}(\sqrt{\ell^*})$ (all below 2,000) are 24 curves (labels `5Ns.2.1`, `7Ns.2.1`, `7Ns.3.1`, `7Ns.6.1.2`), and all of them have $\ell \mid N$. The 28 curves with label `5Ns`, `5Nn` or `7Ns` and $\ell \nmid N$ are induced from other quadratic fields: for instance `608b1` (`5Ns`) from $\mathbb{Q}(i)$, `1044a1` (`5Nn`) from $\mathbb{Q}(\sqrt{-3})$, and `9225a1` (`7Ns`) from $\mathbb{Q}(\sqrt{-3})$.

*In the paper* 11.54. The paper states this as Proposition prop:Sf: "Let $\ell$ be an odd prime with $\ell \nmid N$ and $E[\ell]$ irreducible. If $\ell \geq 5$, then $\ell \notin S_f$. If $\ell = 3$, then $3 \in S_f$ if and only if the image of $\bar\rho_{E,3}$ is conjugate to $N_s(\mathbb{F}_3)$, and then $E$ is ordinary at 3." Its proof is the argument of Corollary 11.37(ii). Accordingly the paper's abstract reads "(for $\ell \geq 5$ this only asks that $E[\ell]$ be irreducible)", and Section sec:forms records that there are no other failures of the hypothesis among the non-CM curves with $E[\ell]$ irreducible, "the curves whose mod $\ell$ representation is induced from $\mathbb{Q}(\sqrt{\ell^*})$ with $\ell \geq 5$ all having $\ell \mid N$".

**Proposition 11.55** (the paper's Proposition prop:ns)**.** *Let $E/\mathbb{Q}$, with or without complex multiplication, have $E[3]$ irreducible, and let $H$ be the image of $\bar\rho = \bar\rho_{E,3}$. Then $H$ is conjugate to $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$, and $\bar\rho$ is absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ unless $H \sim N_s(\mathbb{F}_3)$. In that case:*

*(i)* $\bar\rho(G_{\mathbb{Q}(\sqrt{-3})}) = C_4 = \langle J \rangle$, *and* $\bar\rho|_{G_{\mathbb{Q}(\sqrt{-3})}}$ *is irreducible over $\mathbb{F}_3$ but not absolutely irreducible;*

*(ii)* $\bar\rho \otimes \mathbb{F}_9 \simeq \mathrm{Ind}_{G_{\mathbb{Q}(\sqrt{-3})}}^{G_\mathbb{Q}} \psi$ *for a character $\psi$ of $G_{\mathbb{Q}(\sqrt{-3})}$ of order 4 with values in $\mathbb{F}_9^\times$;*

*(iii)* $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ *over $\mathbb{F}_3$, $H^0(\mathbb{Q}, \mathrm{ad}^0\,\bar\rho(1)) \neq 0$, and $h_3(E) = 1$ (Lemma 11.58).*

*The 3-division polynomial is a product of two irreducible quadratics in $\mathbb{Q}[x]$ if $H \sim N_s(\mathbb{F}_3)$, and irreducible otherwise.*

*Proof.* (1) By Proposition 11.3(ii), $\bar\rho(G_{\mathbb{Q}(\sqrt{-3})}) = H \cap \mathrm{SL}_2(\mathbb{F}_3)$.

(2) By Theorem 11.45 the irreducible images are $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ and $N_s(\mathbb{F}_3)$ (the others lie in Borel subgroups). The proof used only the determinant and complex conjugation, so it covers curves with complex multiplication.

(3) The determinant-one subgroups are $\mathrm{SL}_2(\mathbb{F}_3)$, $Q_8$ and $C_4 = \langle J \rangle$, respectively (Chapter 1).

(4) $\mathrm{SL}_2(\mathbb{F}_3)$ and $Q_8$ act absolutely irreducibly on $\mathbb{F}_3^2$. Being non-abelian and acting faithfully is not enough, since a Borel subgroup is both. The argument: a subgroup of $\mathrm{GL}_2(\overline{\mathbb{F}}_3)$ fixing a line lies in a Borel subgroup $\left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right)$, and its commutator subgroup lies in the unipotent radical $\left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right)$, abelian of exponent 3. The commutator subgroup of $Q_8$ is $\{\pm 1\}$, which has an element of order 2, and that of $\mathrm{SL}_2(\mathbb{F}_3)$ is $Q_8$, non-abelian (Chapter 1). So neither fixes a line over $\overline{\mathbb{F}}_3$.

(5) $C_4$ is abelian. Its generator $J$ has characteristic polynomial $X^2+1$, irreducible over $\mathbb{F}_3$, so $C_4$ fixes no line of $\mathbb{F}_3^2$ but over $\mathbb{F}_9=\mathbb{F}_3[i]$ it has the two eigenlines with eigenvalues $i$ and $-i$ ($J=m_{-i}$ in the model $\mathbb{F}_3^2=\mathbb{F}_9$; Chapter 1). So $\bar\rho|_{G_{\mathbb{Q}(\sqrt{-3})}}$ is irreducible over $\mathbb{F}_3$ and $\bar\rho|_{G_{\mathbb{Q}(\sqrt{-3})}}\otimes\mathbb{F}_9\simeq\psi\oplus\psi^{-1}$, where $\psi$ is the character on one eigenline; $\psi(G_{\mathbb{Q}(\sqrt{-3})})=\langle i\rangle$ has order 4. This proves (i).

(6) By Proposition 11.52 with $K=\mathbb{Q}(\sqrt{-3})$ ((a)⇒(c) and (a)⇒(b)), $\bar\rho\otimes\mathbb{F}_9\simeq\operatorname{Ind}\psi$ (the proof of (a)⇒(c) works over $\mathbb{F}_9$, where the line is defined), and $\bar\rho\simeq\bar\rho\otimes\chi_{-3}$ over $\mathbb{F}_3$, the isomorphism descending because spaces of homomorphisms commute with extension of scalars. This proves (ii) and the first part of (iii). Conversely, if $H$ is $\mathrm{GL}_2(\mathbb{F}_3)$ or $N_{ns}(\mathbb{F}_3)$, (4) shows that $\bar\rho|_{G_{\mathbb{Q}(\sqrt{-3})}}$ is absolutely irreducible.

(7) $\operatorname{Hom}_{G_\mathbb{Q}}(\bar\rho,\bar\rho\otimes\bar\chi_3)\neq0$ since $\bar\chi_3=\chi_{-3}$, and it lies in $\operatorname{ad}^0\bar\rho(1)$ by the trace-zero statement of Proposition 11.52; so $H^0(\mathbb{Q},\operatorname{ad}^0\bar\rho(1))\neq0$, and $h_3(E)=1$ by the equivalence $h_3(E)=1\iff\bar\rho\simeq\bar\rho\otimes\chi_{-3}$ of Lemma 11.58(iii), whose proof does not use the present proposition (only the last equivalence there, with the image $N_s(\mathbb{F}_3)$, does).

(8) The roots of $\psi_3$ correspond to the four lines of $E[3]$ (Proposition 11.6(ii)). $N_s(\mathbb{F}_3)$ has two orbits of size two on them, and $N_{ns}(\mathbb{F}_3)$ and $\mathrm{GL}_2(\mathbb{F}_3)$ are transitive (Theorem 11.45); $\psi_3$ has no rational root as $E[3]$ is irreducible. So $\psi_3$ is a product of two irreducible quadratics in the first case and irreducible in the other two. □

**Example 11.56.** For `726a1`, $\psi_3=(x^2-x-74)(3x^2+8x+20)$; the discriminants $1+4\cdot74=297=3^3\cdot11$ and $8^2-4\cdot3\cdot20=-176=-2^4\cdot11$ (a quadratic $ax^2+bx+c$ has its roots in $\mathbb{Q}(\sqrt{b^2-4ac})$, and square factors can be dropped) give the quadratic fields $\mathbb{Q}(\sqrt{33})$ and $\mathbb{Q}(\sqrt{-11})$, which with $\mathbb{Q}(\sqrt{-3})$ are the three quadratic subfields of the biquadratic splitting field $(33\cdot(-11)=-3\cdot11^2)$. For `6912d1`, $y^2=x^3-6x-8$ (the example of Chapter 5), $\psi_3=3(x^2-4x-2)(x^2+4x+6)$, with fields $\mathbb{Q}(\sqrt{6})$, $\mathbb{Q}(\sqrt{-2})$. For `338d1`, $\psi_3=(x^2+29x+1858)(3x^2-82x-172)$, with fields $\mathbb{Q}(\sqrt{-39})$, $\mathbb{Q}(\sqrt{13})$. For `11a1`, $\psi_3$ is irreducible with Galois group $S_4$ (Example 11.8). In the three `3Ns` cases, $a_p\equiv0\pmod3$ at every prime $p\equiv2\pmod3$ of good reduction: for `726a1`, $a_5=0$, $a_{17}=6$, $a_{23}=6$, $a_{29}=-6$, $a_{41}=-6$, $a_{47}=6$, $a_{53}=-12$, $a_{59}=-12$.

*In the paper* 11.57. The paper's proof of Proposition prop:ns contains steps (1) to (8) in compressed form: "The first two act absolutely irreducibly on $\mathbb{F}_3^2$: a subgroup of $\mathrm{GL}_2(\overline{\mathbb{F}}_3)$ fixing a line lies in a Borel subgroup, so its commutator subgroup is abelian of exponent 3, whereas the commutator subgroups $\{\pm1\}$ of $Q_8$ and $Q_8$ of $\mathrm{SL}_2(\mathbb{F}_3)$ are not." Being non-abelian and acting faithfully would not suffice, as step (4) notes. The restriction to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ in the `3Ns` case is irreducible over $\mathbb{F}_3$ and becomes reducible only over $\mathbb{F}_9$, where the induction of (ii) takes place.

**Lemma 11.58** (the paper's Lemma lem:h0)**.** *Let $\ell$ be odd and $\bar\rho=\bar\rho_{E,\ell}$ irreducible, $E$ with or without complex multiplication. Let $W^D=\operatorname{ad}^0T(1)\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$, so $W^D[\ell^k]=\operatorname{ad}^0(T/\ell^k)(1)$.*

*(i) $H^0(\mathbb{Q},W^D)[\ell^k]=H^0(\mathbb{Q},\operatorname{ad}^0(T/\ell^k)(1))$, and this group equals $\operatorname{Hom}_{G_\mathbb{Q}}(T/\ell^k,(T/\ell^k)(1))$: every $G_\mathbb{Q}$-map $T/\ell^k\to(T/\ell^k)(1)$ has trace 0.*

*(ii) $H^0(\mathbb{Q},W^D)$ is finite and cyclic, $H^0(\mathbb{Q},W^D)\simeq\mathbb{Z}/\ell^h$, where $h=h_\ell(E)$ is the largest $k\geq0$ such that $\rho\bmod\ell^k\simeq(\rho\otimes\chi_\ell)\bmod\ell^k$.*

*(iii) $h_\ell(E)=0$ for every $\ell\geq5$, and $h_3(E)\leq1$, with $h_3(E)=1$ iff $\bar\rho_{E,3}\simeq\bar\rho_{E,3}\otimes\chi_{-3}$, iff $\bar\rho_{E,3}$ is not absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, iff the image of $\bar\rho_{E,3}$ is $N_s(\mathbb{F}_3)$.*

*(iv) $h_\ell$ is invariant under quadratic twists and, when $E[\ell]$ is irreducible, under isogenies.*

*In particular* $H^0(\mathbb{Q}, W^D) = 0$ *whenever* $\ell \notin S_f$.

*Proof.* (i) For $\ell$ odd, $\mathrm{End}(T/\ell^k) = \mathrm{ad}^0(T/\ell^k) \oplus (\mathbb{Z}/\ell^k)\cdot 1$ (a scalar $a$ has trace $2a$, and 2 is a unit), so $\mathrm{ad}^0 T \otimes \mathbb{Z}/\ell^k = \mathrm{ad}^0(T/\ell^k)$ and $W^D[\ell^k] = \mathrm{ad}^0 T(1) \otimes \ell^{-k}\mathbb{Z}_\ell/\mathbb{Z}_\ell \simeq \mathrm{ad}^0(T/\ell^k)(1)$. Taking $G_\mathbb{Q}$-invariants commutes with taking $\ell^k$-torsion. The $G_\mathbb{Q}$-invariants of $\mathrm{End}(T/\ell^k)(1)$ are the $G_\mathbb{Q}$-maps $\phi\colon T/\ell^k \to (T/\ell^k)(1)$, and the trace is a $G_\mathbb{Q}$-map $\mathrm{End}(T/\ell^k)(1) \to (\mathbb{Z}/\ell^k)(1) = \mu_{\ell^k}$; so $\mathrm{tr}\,\phi \in H^0(\mathbb{Q}, (\mathbb{Z}/\ell^k)(1)) = 0$, as $\mathbb{Q}$ contains no nontrivial $\ell$-th root of unity for $\ell$ odd. (Equivalently, as in the paper: $\mathrm{tr}\,\phi = \chi_\ell(g)\,\mathrm{tr}\,\phi$ for every $g$, and $1 - \chi_\ell(c) = 2$ is a unit.)

(ii) By (i) with $k = 1$, $H^0(\mathbb{Q}, W^D)[\ell] = \mathrm{Hom}_{G_\mathbb{Q}}(\bar\rho, \bar\rho \otimes \bar\chi_\ell)$. Both are absolutely irreducible (Proposition 11.6(i)), so this space is 0 or, if they are isomorphic, isomorphic to $\mathrm{End}_{G_\mathbb{Q}}(\bar\rho) = \mathbb{F}_\ell$: its dimension is at most one. A submodule $X$ of $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^3$ with $\dim_{\mathbb{F}_\ell} X[\ell] \le 1$ is 0, $\mathbb{Z}/\ell^h$ or $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ (Chapter 3: its Pontryagin dual is a quotient of $\mathbb{Z}_\ell^3$ with at most one generator). An element of $X$ of exact order $\ell^k$ is a $G_\mathbb{Q}$-map $\phi\colon T/\ell^k \to (T/\ell^k)(1)$ not divisible by $\ell$, that is, whose reduction modulo $\ell$ is a nonzero $G_\mathbb{Q}$-map $\bar\rho \to \bar\rho \otimes \bar\chi_\ell$, hence an isomorphism (both irreducible); then $\phi$ is an isomorphism by Nakayama's lemma, and $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$. Conversely an isomorphism $T/\ell^k \to (T/\ell^k)(1)$ is a $G_\mathbb{Q}$-map whose reduction is nonzero, an element of exact order $\ell^k$ by (i). So $X$ has an element of order $\ell^k$ iff $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$; that $X$ is finite, hence $\simeq \mathbb{Z}/\ell^h$ with $h$ as stated, follows from (iii).

(iii) If $\phi\colon T/\ell^k \to (T/\ell^k)(1)$ is a $G_\mathbb{Q}$-isomorphism, $\phi\rho(g)\phi^{-1} = \chi_\ell(g)\rho(g)$ modulo $\ell^k$, and taking determinants, $\det\rho(g) = \chi_\ell(g)^2 \det\rho(g)$, so $\chi_\ell(g)^2 \equiv 1 \pmod{\ell^k}$ for every $g \in G_\mathbb{Q}$. The cyclotomic character is onto $\mathbb{Z}_\ell^\times$ (Chapter 5), and $2 \in \mathbb{Z}_\ell^\times$; so $4 \equiv 1 \pmod{\ell^k}$, that is, $\ell^k \mid 3$. For $\ell \ge 5$ this forces $k = 0$, and for $\ell = 3$ it forces $k \le 1$ (Chapter 2: $a^2 \equiv 1 \pmod{\ell^k}$ for all units $a$ iff $\ell^k = 3$). So $h_\ell(E) = 0$ for $\ell \ge 5$ and $h_3(E) \le 1$, with $h_3(E) = 1$ iff $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \bar\chi_3$, and $\bar\chi_3 = \chi_{-3}$ (Proposition 11.3(ii)). By Proposition 11.52 with $K = \mathbb{Q}(\sqrt{-3})$ this is equivalent to $\bar\rho_{E,3}$ not being absolutely irreducible on $G_{\mathbb{Q}(\sqrt{-3})}$, and by Proposition 11.55 to the image being $N_s(\mathbb{F}_3)$.

(iv) $W^D$ depends only on $\mathrm{ad}^0 T$, which is unchanged by twists (Proposition 11.5(i): conjugation by $\chi_d(g)\rho(g)$ equals conjugation by $\rho(g)$), and $T$ is unchanged by isogenies when $E[\ell]$ is irreducible (Proposition 11.5(iii)). Finally $\ell \notin S_f$ means that $\bar\rho$ is absolutely irreducible on $G_{\mathbb{Q}(\sqrt{\ell^*})}$ (and $\ell \nmid 2N$); for $\ell = 3$ this excludes $h_3(E) = 1$ by (iii), and for $\ell \ge 5$, $h_\ell(E) = 0$ anyway. □

*In the paper* 11.59. The paper's Lemma lem:h0 states (i) to (iii): $H^0(\mathbb{Q}, W^D)$ is "finite and cyclic, of length $h_\ell(E)$, the largest $k$ such that $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$", with the bound $h_3(E) \le 1$, the same chain of equivalences for $h_3(E) = 1$, and the clause "This holds for every $E$, with or without complex multiplication". Part (iv) is added here. The paper's proof is the one above in compressed form; the determinant argument of (iii) is what gives finiteness, and it needs no hypothesis on complex multiplication. (For $\ell \nmid N$ the CM case is anyway harmless: complex multiplication by $\mathbb{Q}(\sqrt{\ell^*})$ forces $\ell \mid N$, Chapter 9.) The criterion for $h_\ell(E) > 0$ uses $\bar\chi_\ell$, not $\chi_{\ell^*}$; they agree only for $\ell = 3$, and for $\ell \ge 5$ the isomorphism $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$ can hold only if $\ell \mid N$ (Theorem 11.53).

**Proposition 11.60** (the paper's Proposition prop:ns-structure)**.** *Let* $E/\mathbb{Q}$ *(complex multiplication allowed) have* $\bar\rho = \bar\rho_{E,3}$ *with image conjugate to* $N_s(\mathbb{F}_3) = \langle J, h\rangle$*, and let* $K_1$*,* $K_2$ *be the fields cut out by the two quadratic factors of* $\psi_3$*, the fixed fields of* $\bar\rho^{-1}(C_s)$ *and of* $\bar\rho^{-1}(\{\pm 1, \pm w\})$.

*(i)* $K_1$*,* $K_2$ *and* $\mathbb{Q}(\sqrt{-3})$ *are the three quadratic subfields of the splitting field of* $\psi_3$*, which is biquadratic.*

*(ii)* *For every prime $p \neq 3$, $\bar\rho(I_p) \subseteq C_4 = N_s(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3) = \langle J\rangle$. Consequently every prime $p \neq 3$ with $v_p(j) < 0$ has $3 \mid v_p(j)$ (so $t_3(E,p) \geq 1$); every prime $p \geq 5$ of type $\mathrm{I}_n^*$ has $3 \mid n$; no prime $p \geq 5$ has type* II *or* II$^*$*; no prime $p \neq 3$ has type* IV *or* IV$^*$.

*(iii)* *Let $R$ be the set of primes $p \neq 3$ ramified in $K_1$. Then $R$ is also the set of primes $p \neq 3$ ramified in $K_2$, it equals $\{p \neq 3 : \bar\rho(I_p) = C_4\}$, and it is nonempty. For $p \in R$ the reduction is potentially good and $\rho_{E,\ell}(I_p)$ is cyclic of order 4 for every $\ell \neq p$; if $p \geq 5$, $p$ has type* III *or* III$^*$*. Conversely every prime $p \geq 5$ of type* III *or* III$^*$ *lies in $R$.*

*(iv)* *For $p \in R$ and $\ell \neq p$, $\mathrm{Frob}_p$ (arithmetic or geometric) acts on $(\mathrm{ad}^0 V_\ell E)^{I_p} = \mathbb{Q}_\ell\tau$ by $\chi_{-3}(p)$; so $L_p(A_f,s)^{-1} = 1-\chi_{-3}(p)p^{-s}$, $P_p(X) = 1-\chi_{-3}(p)pX$ and $e_3(E,p) = v_3(p-\chi_{-3}(p)) \geq 1$. If $p \geq 5$, then $\chi_{-4}(p) = \chi_{-3}(p)$, that is, $p \equiv 1$ or $11 \pmod{12}$. If $p = 2$, then $P_2(X) = 1+2X$ and $e_3(E,2) = 1$.*

*(v)* *$a_p(E) \equiv 0 \pmod 3$ for every prime $p \nmid 3N$ with $p \equiv 2 \pmod 3$.*

*(vi)* *If $3 \nmid N$, $E$ is ordinary at 3.*

*(vii)* *$h_3(E) = 1$.*

*Proof.* Recall $h = \mathrm{diag}(1,-1)$, $w = hJ = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, and $N_s(\mathbb{F}_3) = \{\pm1, \pm J, \pm h, \pm w\}$, dihedral of order 8 with centre $\{\pm1\}$.

(i) The splitting field of $\psi_3$ is the fixed field of the kernel of the action on lines, $\bar\rho^{-1}(\{\pm1\})$, with Galois group $N_s(\mathbb{F}_3)/\{\pm1\} \simeq V_4$; so it is biquadratic, and its three quadratic subfields are the fixed fields of the three subgroups of index 2 of $N_s(\mathbb{F}_3)$ containing $-1$: $C_4 = \langle J\rangle$, $C_s = \{\pm1, \pm h\}$ and $\{\pm1, \pm w\}$. The first is the determinant-one subgroup, with fixed field $\mathbb{Q}(\sqrt{-3})$ (Proposition 11.3(ii)). The stabiliser in $N_s(\mathbb{F}_3)$ of the line $\langle e_1\rangle$ is $C_s$, and that of $\langle e_1+e_2\rangle$ is $\{\pm1, \pm w\}$ ($J(e_1+e_2) = e_1-e_2$, $h(e_1+e_2) = e_1-e_2$); so the roots of the factor of $\psi_3$ belonging to the orbit $\{\langle e_1\rangle, \langle e_2\rangle\}$ generate $K_1$, and those of the other factor generate $K_2$.

(ii) For $p \neq 3$, $\det\bar\rho = \chi_{-3}$ is unramified at $p$, so $\bar\rho(I_p) \subseteq H \cap \mathrm{SL}_2(\mathbb{F}_3) = C_4$, which has no element of order 3 or 6. If $v_p(j) < 0$, then on $I_p$, $\rho_{E,3}(\sigma) = \chi(\sigma)\left(\begin{smallmatrix}1&\nu t_3(\sigma)\\0&1\end{smallmatrix}\right)$ with $\nu = -v_p(j)$ and $\chi$ trivial or quadratic (Theorem 11.15, Proposition 11.19); on the kernel of $\chi$ (of index at most 2) $t_3$ is still onto $\mathbb{Z}_3$, so if $3 \nmid \nu$ the image modulo 3 contains a unipotent of order 3, which is impossible. So $3 \mid \nu$, and $t_3(E,p) = v_3(\nu) \geq 1$; at $p \geq 5$ the type $\mathrm{I}_n^*$ has $n = \nu$ (Proposition 11.19). At $p \geq 5$ of type II, II$^*$ the image of inertia is cyclic of order 6 and injects into $\mathrm{GL}_2(\mathbb{F}_3)$ (Theorem 11.12), so it contains an element of order 3 in $C_4$: impossible. At $p \neq 3$ of type IV, IV$^*$, $\bar\rho(I_p)$ has order 3 (Lemma 11.25): impossible.

(iii) The subgroups of $C_4$ are 1, $\{\pm1\}$ and $C_4$; the first two lie in $C_s$ and in $\{\pm1, \pm w\}$, and $C_4$ lies in neither. The prime $p$ ramifies in $K_1$ iff $I_p$ is not contained in the index-two normal subgroup $\bar\rho^{-1}(C_s)$, iff $\bar\rho(I_p) \not\subseteq C_s$, iff $\bar\rho(I_p) = C_4$; likewise for $K_2$. As $C_s \neq C_4$, $K_1 \neq \mathbb{Q}(\sqrt{-3})$; the only quadratic field unramified outside 3 is $\mathbb{Q}(\sqrt{-3})$ (the discriminant of $\mathbb{Q}(\sqrt{d})$, $d$ squarefree, is $d$ if $d \equiv 1 \pmod 4$ and $4d$ otherwise, and its prime factors are the ramified primes; so only 3 ramifies iff $d = \pm3$ and $d \equiv 1 \pmod 4$, that is, $d = -3$); so some $p \neq 3$ ramifies in $K_1$, and $R \neq \emptyset$. Let $p \in R$. If the reduction at $p$ were potentially multiplicative, $\bar\rho(I_p)$ would consist of elements $\pm u$ with $u$ unipotent, of orders 1, 2, 3 or 6; but $J$ has order 4. So the reduction is potentially good, $\rho_{E,3}(I_p)$ is finite and injects into $\mathrm{GL}_2(\mathbb{F}_3)$ (Theorem 11.12(ii)): it is cyclic of order 4, generated by an element $\tau$ with $\tau \equiv J^{\pm1}$, $\tau^2 = -1$. By Theorem 11.12(i), $\rho_{E,\ell}(I_p)$ is cyclic of order 4 for every $\ell \neq p$; for $p \geq 5$ this is the semistability defect 4, that is, $v_p(\Delta_{\min}) \in \{3, 9\}$,

types III, III* (Theorem 11.12(iv)). Conversely, at $p \geq 5$ of type III or III*, $\rho_{E,3}(I_p)$ is cyclic of order 4 and injects modulo 3, so $\bar{\rho}(I_p)$ has order 4 in $C_4$: $\bar{\rho}(I_p) = C_4$ and $p \in R$.

(iv) Let $p \in R$ and $\ell \neq p$. The commutant of $\tau$ in $\mathrm{ad}^0 V$ is $\mathbb{Q}_\ell \tau$ (Theorem 11.22, whose argument uses only that $\tau$ has the distinct eigenvalues $\pm i$). Frobenius normalises $\rho(I_p)$, so $\rho(\mathrm{Frob}_p)\tau\rho(\mathrm{Frob}_p)^{-1} = \tau^{\pm 1}$, and acts on $\mathbb{Q}_\ell\tau$ by that sign. The sign does not depend on $\ell$: it is the action of $\mathrm{Frob}_p$ by conjugation on the cyclic group $\mathrm{Gal}(L/\mathbb{Q}_p^{\mathrm{ur}}) \simeq \rho_{E,\ell}(I_p)$ (Theorem 11.12(i)); so it can be read at $\ell = 3$, and modulo 3, where reduction is injective. In $N_s(\mathbb{F}_3)$ the elements of $C_4$ commute with $J$, and those of the other coset $hC_4$ conjugate $J$ to $J^{-1}$ ($hJh^{-1} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = J^{-1}$). Since $\bar{\rho}(I_p) = C_4$, the coset of $\bar{\rho}(\mathrm{Frob}_p)$ modulo $C_4$ is well defined, and it is $C_4$ iff $\det \bar{\rho}(\mathrm{Frob}_p) = \chi_{-3}(p) = 1$. So $\mathrm{Frob}_p$, and also $\mathrm{Frob}_p^{-1}$, acts on $(\mathrm{ad}^0 V)^{I_p}$ by $\chi_{-3}(p)$; then $L_p(A_f, s)^{-1} = 1 - \chi_{-3}(p)p^{-s}$, $P_p(X) = 1 - \chi_{-3}(p)pX$, and, the naive factor being 1, $e_3(E,p) = v_3(1 - \chi_{-3}(p)p^{-1}) = v_3(p - \chi_{-3}(p)) \geq 1$ because $p \equiv \chi_{-3}(p) \pmod 3$. If $p \geq 5$, the image of inertia is tame and the tame relation gives $\rho(\mathrm{Frob}_p)\tau\rho(\mathrm{Frob}_p)^{-1} = \tau^p = \tau^{\chi_{-4}(p)}$ (as in Theorem 11.22); comparing, $\chi_{-4}(p) = \chi_{-3}(p)$: $p \equiv 1 \pmod{12}$ or $p \equiv 11 \pmod{12}$. If $p = 2$: $\chi_{-3}(2) = -1$, $P_2(X) = 1 + 2X$ and $e_3(E,2) = v_3(3) = 1$.

(v) $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{-3}$ (Proposition 11.55) and Proposition 11.52(d); the primes $p \equiv 2 \pmod 3$ are those inert in $\mathbb{Q}(\sqrt{-3})$. (vi) Theorem 11.53. (vii) Lemma 11.58(iii). □

These conclusions were checked on all 146 curves with image $N_s(\mathbb{F}_3)$ and conductor below 10,000 (120 without complex multiplication and 26 with): in every case $R$ computed from the discriminants of the two quadratic factors of $\psi_3$ is the same for both factors and agrees with the primes $p \neq 3$ at which the image of inertia on $E[3]$ is cyclic of order 4 with no fixed point; every multiplicative prime $p \neq 3$ has $3 \mid n_p$; no type II, II* occurs at $p \geq 5$ and no type IV, IV* at $p \neq 3$; $a_p \equiv 0 \pmod 3$ for $p \equiv 2 \pmod 3$, $p < 410$, $p \nmid 3N$; and $3 \nmid a_3$ whenever $3 \nmid N$. Of the 120 non-CM curves, 100 have primes $p \geq 5$ in $R$, all of type III or III* and $\equiv 1$ or $11 \pmod{12}$; for the other 20 (the classes `3840h`, `3840k`, `3840o`, `3840r`, `4864b`, `4864c`, `4864o`, `4864p`, `6912d`, `6912f`, `6912j`, `6912l`, `6912p`, `6912r`, `6912v`, `6912x`), $R = \{2\}$, and the type at 2 is III or III* with $f_2 = 8$. PARI's Euler factors are $1 + 11X$ for `726a1` at 11, $1 - 13X$ for `338d1` at 13, and $1 + 2X$ for `3840h1` and `6912d1` at 2, as (iv) predicts. Of the 26 curves with complex multiplication, 14 have $R = \{11\}$ and 12 have $R = \{2\}$.

For the optimal curves (the curve of Cremona's file `degphi` in each class, which is the first curve of the class except for `990h3`) we recomputed the counts: below conductor 10,000 there are 90 optimal curves without complex multiplication with image $N_s(\mathbb{F}_3)$, of which 44 have $3 \mid N$, 74 have a prime $p \geq 5$ in $R$ and 16 have $R = \{2\}$; and 13 optimal curves with complex multiplication with this image, namely `121b1`, `256a1`, `256d1`, `1089e1`, `1936f1`, `2304c1`, `2304k1`, `3025a1`, `5929a1`, `6400f1`, `6400r1`, `7744a1` and `7744s1`, of which 3 have $3 \mid N$, 7 (those with $j = -32768$) have $R = \{11\}$ and 6 (those with $j = 8000$) have $R = \{2\}$.

*In the paper* 11.61. Section sec:anomaly: "Conversely every 3Ns curve of the tables has a prime of type III or III* with a positive term $e_3$"; "The residue classes of the type III primes of the violating curves are $q \equiv 1$ and $q \equiv 11 \pmod{12}$ in comparable numbers, with some $q = 2$". Proposition 11.60 proves these observations, except that the Kodaira type at $p = 2$ of a prime of $R$ is not determined by the argument: the paper says so ("That the type at 2 is always III or III* when $R = \{2\}$ is not proved"), and the data show III or III* in every case. What is proved at 2 is the cyclic image of order 4 and the Euler factor $1 + 2X$. The paper's "the violation involves a prime $p \equiv -1 \pmod 3$ with $t_3(E,p) \geq 1$", recorded there as an observation, is consistent with (ii): every potentially multiplicative prime $p \neq 3$ of a `3Ns` curve has $t_3(E,p) \geq 1$.

The paragraph after the proof of Proposition prop:ns-structure says: "Among the 90 optimal curves without complex multiplication with image 3Ns of conductor below 10,000 (44 of them with $3 \mid N$), 74 have a prime $p \geq 5$ in $R$, of type III or III$^*$ and $p \equiv 1$ or $11 \pmod{12}$, and 16 have $R = \{2\}$"; it then lists the 13 optimal curves with complex multiplication and image $N_s(\mathbb{F}_3)$, 7 with $R = \{11\}$ and 6 with $R = \{2\}$. Our recomputation above agrees with all these counts.

*In the paper* 11.62. The paper's Proposition prop:ns-structure has the content of Proposition 11.60, with the parts in another order: its (iv) is (i) here; its (i) is (ii) here, where the clause on type $\mathrm{I}_n^*$ at $p \geq 5$ is the case $v_p(j) = -n$ of the clause on $v_p(j) < 0$; its (ii) and (iii) are (iii) and (iv) here; and its (v) to (vii) are (v) to (vii) here. The paper's proof takes its (iv) second, after its (i): "We prove (iv) next, because (ii) uses the fields $K_1$ and $K_2$ it describes." Putting that part first here reflects the same dependence.

**Corollary 11.63** ($S_f$ for elliptic curves)**.** *Let $\ell$ be odd and $\ell \nmid N$. The following are equivalent: (a) $\ell \notin S_f$, that is, $\bar{\rho}_{E,\ell}$ is absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$; (b) $E[\ell]$ is irreducible and $h_\ell(E) = 0$; (c) $E[\ell]$ is irreducible and, if $\ell = 3$, the image of $\bar{\rho}_{E,3}$ is not $N_s(\mathbb{F}_3)$. So for $\ell \nmid 2N$ the hypothesis of Theorem thm:main reads: $E[\ell]$ is irreducible, and not ($\ell = 3$ and image `3Ns`).*

*Proof.* By definition [49, (31)], $S_f$ consists of the primes dividing $2N$ and those $\ell$ for which $\bar{\rho}|_{G_{\mathbb{Q}(\sqrt{\ell^*})}}$ is not absolutely irreducible. If $E[\ell]$ is reducible, so is its restriction, and $\ell \in S_f$; so all three conditions require irreducibility. Given it, for $\ell \geq 5$ (a) holds by Theorem 11.53 and (b), (c) hold by Lemma 11.58(iii); for $\ell = 3$ all three are equivalent to the image not being $N_s(\mathbb{F}_3)$, by Proposition 11.55 and Lemma 11.58(iii). □

Below conductor 10,000 the curves with $3 \nmid N$, $E[3]$ irreducible and $3 \in S_f$ are the 62 curves with label `3Ns` and $3 \nmid N$ and the 20 curves with complex multiplication, image $N_s(\mathbb{F}_3)$ and $3 \nmid N$ (with $j = -32768$ or 8000).

## 11.9 The Eisenstein case

When $E[\ell]$ is reducible, the Hecke maximal ideal attached to $(f, \ell)$ is Eisenstein, multiplicity one and the theorems of Diamond, Flach and Guo are unavailable, and the modular degree can lose the local divisibility of Theorem thm:main. The paper studies this case in Section sec:eis and Conjecture conj:eis, where the controlling invariant is rational torsion. This section proves the structure of a reducible $\bar{\rho}$ at a prime $\ell \nmid N$ and the dictionary between the reducible mod 3 images and torsion.

**Proposition 11.64** (reducible $\bar{\rho}$)**.** *Let $\ell$ be odd, $\ell \nmid N$, and $E[\ell]$ reducible, with a stable line on which $G_{\mathbb{Q}}$ acts by $\eta_1$ and quotient character $\eta_2$.*

*(i)* $\bar{\rho}^{\mathrm{ss}} = \eta_1 \oplus \eta_2$ *and* $\eta_1\eta_2 = \bar{\chi}_\ell$.

*(ii) $E$ is ordinary at $\ell$, and* $\{\eta_1|_{I_\ell}, \eta_2|_{I_\ell}\} = \{\omega, 1\}$.

*(iii)* $\eta_1$ *and* $\eta_2$ *are unramified at every prime of multiplicative reduction.*

*(iv) If $E$ is semistable, then* $\{\eta_1, \eta_2\} = \{1, \bar{\chi}_\ell\}$, *so* $a_p \equiv 1 + p \pmod{\ell}$ *for every $p \nmid N\ell$, and $E$ or a curve $\ell$-isogenous to it over $\mathbb{Q}$ has a rational point of order $\ell$.*

*Proof.* (i) The composition factors of $\bar\rho$ are $\eta_1$ and $\eta_2$, and $\eta_1\eta_2 = \det\bar\rho = \bar\chi_\ell$ (Proposition 11.3(ii)). (ii) is Corollary 11.37(i). (iii) At a multiplicative prime $q$, $\bar\rho(I_q)$ consists of unipotent elements (Theorem 11.15(i)); a unipotent element has the single eigenvalue 1, so it acts trivially on every one-dimensional subquotient, in particular on the line and on the quotient. (iv) If $E$ is semistable, $\eta_1$ and $\eta_2$ are unramified at every $p \neq \ell$: at good primes by Theorem 11.9, at multiplicative primes by (iii). A character of $G_\mathbb{Q}$ unramified outside $\ell$ (and $\infty$) factors through $\mathrm{Gal}(\mathbb{Q}(\zeta_{\ell^\infty})/\mathbb{Q}) \simeq \mathbb{Z}_\ell^\times$ by the theorem of Kronecker and Weber (Chapter 6); if it takes values in $\overline{\mathbb{F}}_\ell^\times$, its order is prime to $\ell$, so it factors through the quotient $\mathrm{Gal}(\mathbb{Q}(\zeta_\ell)/\mathbb{Q}) \simeq \mathbb{F}_\ell^\times$ and is a power $\bar\chi_\ell^a$. As $\mathbb{Q}(\zeta_\ell)/\mathbb{Q}$ is totally ramified at $\ell$, $\bar\chi_\ell^a$ is determined by its restriction $\omega^a$ to $I_\ell$; by (ii) the two characters are $\bar\chi_\ell$ and 1. Then $a_p \equiv \mathrm{tr}\,\bar\rho(\mathrm{Frob}_p) = \eta_1(\mathrm{Frob}_p) + \eta_2(\mathrm{Frob}_p) = p+1 \pmod\ell$. If $\eta_1 = 1$ the line consists of rational points: a rational point of order $\ell$. If $\eta_1 = \bar\chi_\ell$, the line $C$ is isomorphic to $\mu_\ell$, and on $E' = E/C$ the image of $E[\ell]$ is $E[\ell]/C$, a subgroup of order $\ell$ on which $G_\mathbb{Q}$ acts by $\eta_2 = 1$: a rational point of order $\ell$ on the $\ell$-isogenous curve $E'$. □

We checked (iv) on every curve of conductor below 10,000 with square-free $N$, $\ell \nmid N$, and a Borel or split Cartan label at $\ell = 3$, 5, 7 (1,315, 158 and 26 curves): $a_p \equiv 1 + p \pmod\ell$ for every prime $p < 282$ with $p \nmid N\ell$.

**Proposition 11.65** (the reducible mod 3 images and torsion)**.** *(i) The mod 3 image of $E$ is* `3B.1.1` *if and only if $E$ has a rational point of order 3 and $E[3]$ is not split; it is* `3Cs.1.1` *iff $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$; and it is* `3B.1.2` *iff $\mu_3 \subset E[3]$ with quotient $\mathbb{Z}/3$ and $E[3]$ not split. The twist by $\chi_{-3}$ exchanges* `3B.1.1` *and* `3B.1.2`*.*

*(ii)* $3 \mid \#E(\mathbb{Q})_{\mathrm{tors}}$ *iff the image is* `3B.1.1` *or* `3Cs.1.1`*.*

*(iii) If $E$ has no complex multiplication and $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$, then $E(\mathbb{Q})$ has no point of order 9. Hence, for $E$ without complex multiplication, $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3] \le 2$.*

*(iv) At $\ell = 5$: the image* `5Cs.1.1` *means $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ (*`11a1`*), and* `5B.1.1` *a rational point of order 5 with $E[5]$ non-split (*`11a3`*).*

*Proof.* (i) is the table of Theorem 11.45, read through the arithmetic meanings given in its proof, and Proposition 11.5(ii). (ii) A rational point of order 3 spans a line on which $G_\mathbb{Q}$ acts trivially, so $H$ fixes a nonzero vector. Of the eight images, `3B.1.1` and `3Cs.1.1` fix $e_1$; `3B`, `3Cs`, `3Ns`, `3Nn` and $\mathrm{GL}_2(\mathbb{F}_3)$ contain $-1$, which fixes no nonzero vector; and `3B.1.2` $= \{\left(\begin{smallmatrix} a & b \\ 0 & 1 \end{smallmatrix}\right)\}$ contains $\mathrm{diag}(-1,1)$, whose fixed vectors are the multiples of $e_2$, and $u$, which moves $e_2$. (iii) This is proved in Chapter 9, and is the paper's Proposition prop:no9: if $P$ had order 9, then with $\phi\colon E \to E' = E/\mu_3$ the group $C = \hat\phi^{-1}(\langle P\rangle) \subset E'$ would be a $G_\mathbb{Q}$-stable cyclic subgroup of order 27 (it does not contain $E'[3]$, since $\hat\phi(E'[3]) = \mu_3 \not\subseteq \langle P\rangle$), and a rational cyclic 27-isogeny forces complex multiplication. The inequality follows from Mazur's theorem on torsion [125]: $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) \le 2$, with equality only for $\mathbb{Z}/9$. (iv) is read from Cremona's data as in Convention 11.48 (and Example 11.8). □

Below conductor 10,000 the curves with $3 \mid \#E(\mathbb{Q})_{\mathrm{tors}}$ are exactly the 2,811 curves with image `3B.1.1` (torsion orders 3, 6, 9, 12) and the 264 with image `3Cs.1.1` (torsion orders 3 and 6 only, as (iii) requires).

*In the paper* 11.66. Section sec:eis describes `3Cs.1.1` as "the subgroup of order 2 of the split Cartan subgroup that fixes a point", lists `14a1` (torsion $\mathbb{Z}/6$, image `3Cs.1.1`) among the curves with $D = 2$, and proves in Proposition prop:no9 that the two sources of a deficit 2 never combine, so that the bound $D \le 2$ of Conjecture conj:eis follows from its first bound: this is Proposition 11.65(iii).

*Remark* 11.67 (what reducibility costs). If $E[\ell]$ is reducible, the lattices $T_\ell E'$ of the curves $E'$ in the isogeny class are not all homothetic (Chapter 4: the stable lattices of a reducible residual representation), so $\mathrm{ad}^0 T$ and $h_\ell$ may change along the class, and the choice of the optimal curve matters at $\ell$. The maximal ideal of the Hecke algebra attached to $(f,\ell)$ is Eisenstein (Chapter 14), multiplicity one and the theorems of Diamond, Flach and Guo are unavailable, and the modular degree can fall short of the local bound: the deficit $D$ of Conjecture conj:eis. This is why the hypothesis "$E[\ell]$ irreducible" appears in Theorem thm:quant, Conjecture conj:euler and Proposition prop:forms.

## 11.10 Serre's conjecture for $\bar\rho_{E,\ell}$ and the optimal level

Theorem thm:qual lowers the level of $f_E$, or of a twist, to the Serre conductor, and then twists back. This section states the optimal-level theorem in the form the paper uses and lists the elliptic-curve inputs, all proved above; the general statement and the passage to trivial nebentypus are in Chapter 15.

**Theorem 11.68** (Serre's conjecture; the optimal level). *(i) (Khare and Wintenberger) Every odd irreducible continuous $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ arises from a newform of weight $k(\bar\sigma)$, level $N(\bar\sigma)$ and character determined by $\det\bar\sigma$.*

*(ii) (Ribet, Carayol, Diamond; [38, Theorem 3.15] as quoted by the paper) Let $\ell$ be odd and let $E'/\mathbb{Q}$ have $\ell \nmid N(E')$ and $\bar\rho = \bar\rho_{E',\ell}$ irreducible, and, if $\ell = 3$, image not $N_s(\mathbb{F}_3)$. Then there are a newform $g$ of weight 2 on $\Gamma_0(N(\bar\rho))$ and a prime $\lambda \mid \ell$ of its coefficient field with $\bar\rho_{g,\lambda} \simeq \bar\rho \otimes \overline{\mathbb{F}}_\ell$; in particular $a_p(g) \equiv a_p(E') \pmod\lambda$ for all $p \nmid \ell N(E')$.*

We do not prove this. Part (i) is [98], with Serre's formulation in [176]. Part (ii) is the level-lowering theorem of Ribet [156], Carayol [22] and Diamond [47], which needs only the modularity of $\bar\rho$; Chapter 15 states it in general and proves the specialisation to $\bar\rho_{E',\ell}$, including the triviality of the nebentypus. The inputs from elliptic curves are: $\bar\rho$ is modular of level $N(E')$ (modularity, [207, 16]; Chapter 14); it is odd, $\det\bar\rho(c) = -1$ (Proposition 11.3(iii)); it is absolutely irreducible, and absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ when $\ell = 3$ (Proposition 11.6(i), Proposition 11.55); its restriction to $D_\ell$ is finite flat, so $\delta(\bar\rho) = 0$ (Theorem 11.32(i)); its weight is 2 (Theorem 11.32(iii)); $\det\bar\rho = \bar\chi_\ell$ (Proposition 11.3(ii)); and $N(\bar\rho)$ is given by Proposition 11.40.

*In the paper* 11.69. The optimal-level paragraph of Section 3: "For $\bar\sigma = \bar\rho_{E',\ell}$ with $E'$ an elliptic curve of conductor prime to $\ell$, the restriction to $D_\ell$ comes from the finite flat group scheme $E'[\ell]$ and is good, and $\det\bar\sigma = \bar\chi_\ell$ forces the nebentypus, which is trivial modulo $\lambda$ and of order prime to $\ell$, to be trivial." The first clause is Theorem 11.32(i), the determinant is Proposition 11.3(ii), and the nebentypus argument is Chapter 15's. Theorem thm:qual states the extra hypothesis at $\ell = 3$ as the paper's "absolutely irreducible when restricted to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$"; by Proposition 11.55 this is equivalent to the image not being $N_s(\mathbb{F}_3)$, the form in which Theorem 11.68(ii) states it.

**Example 11.70** (the paper's Example, read through this chapter). The congruences of the Example of Section 3 of the paper (case (a) at $q = 2$, case (b), case (c) at $\ell = 3$ and at $\ell = 5$), together with one case (a) at $q = 11$ that the paper does not list and the book adds as an illustration (the curve `121c1` of Example 11.42), with the Serre conductors of Example 11.42 and the modular degrees from Cremona's file `degphi`:

| case | curve, $\deg\varphi_E$ | type at $q$ | level lowering | congruent form |
|---|---|---|---|---|
| (a), $q=11$ (added) | `121c1`, 6 | IV at 11 | $N(\bar\rho_3)=11$ | $f_{\texttt{11a}}$ |
| (a), $q=2$ | `52a1`, 3 | IV$^*$ at 2 | $N(\bar\rho_3)=26$ | $f_{\texttt{26b}}$ |
| (b) | `121a1`, 6 | II at 11 | twist by $\chi_{-11}$: `121c2`, $N(\bar\rho_3)=11$ | $f_{\texttt{11a}}$, twisted |
| (c), $\ell=3$ | `539c1`, $2^5\cdot3^2$ | $\mathrm{I}_3^*$ at 7 | twist by $\chi_{-7}$: `77c1`, $N(\bar\rho_3)=11$ | $f_{\texttt{11a}}$, twisted |
| (c), $\ell=5$ | `980i1`, $2^6\cdot3^2\cdot5$ | $\mathrm{I}_5^*$ at 7 | twist by $\chi_{-7}$: `140b1`, $N(\bar\rho_5)=4$ | $f_{\texttt{20a}}$, twisted |

The paper identifies the twisted forms: $f_{\texttt{11a}}\otimes\chi_{-11}=f_{\texttt{121d}}$, $f_{\texttt{11a}}\otimes\chi_{-7}=f_{\texttt{539d}}$ and $f_{\texttt{20a}}\otimes\chi_{-7}=f_{\texttt{980g}}$ (recomputed: the twists of `11a1` and `20a1` have the traces of Frobenius of `121d1`, `539d1`, `980g1`). In case (a) at $q=11$, $3\mid\deg\varphi_{\texttt{121c1}}=6$; at $q=2$, $3=\deg\varphi_{\texttt{52a1}}$, where the bound $e_3(E,2)=1$ of Corollary 11.27 is attained. In case (b) the Serre exponent 2 of $\bar\rho_{\texttt{121a1},3}$ at 11 drops to 1 after the twist (Proposition 11.29). In case (c) the type $\mathrm{I}_3^*$ has $3\mid n=3$, so the twist $\bar\rho\otimes\chi_{-7}$ is unramified at 7 (Lemma 11.17 for `77c1`, type $\mathrm{I}_3$) and the level drops by $7^2$. The congruences of Example 11.42 were checked for all $p<2000$ ($p<1224$ at $\ell=5$) away from the bad primes; Sturm's bound turns the paper's versions into identities for all coefficients (Chapter 12). At $\ell=5$, $5\mid980$, so this case illustrates the congruence, not Theorem thm:qual.

## 11.11 Exercises

**Exercise 11.71.** Let $\ell$ be odd. (a) Show directly that $\rho_{E,\ell}(c)$ is conjugate in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ to $\mathrm{diag}(1,-1)$ for a complex conjugation $c$, and deduce that $\bar\rho_{E,\ell}$ is irreducible iff it is absolutely irreducible. (b) Where does the argument fail for $\ell=2$? Show that $\bar\rho_{E,2}(c)$ is either trivial or a transvection, according as $E(\mathbb{R})$ has two components or one (consider the real 2-torsion points).

*Hint.* (b) The idempotents $\frac12(1\pm\rho(c))$ need 2 to be invertible, and modulo 2 the eigenvalues 1 and $-1$ coincide. In a model $y^2=g(x)$, $E(\mathbb{R})$ has two components iff $g$ has three real roots; then $c$ fixes $E[2]$ pointwise, and otherwise it fixes the one real point of order 2 and swaps the other two, which on $\mathbb{F}_2^2$ is a transvection.

**Exercise 11.72.** (The paper's Lemma lem:tamagawa-exp on `11a1` and `11a3`.) Both curves have multiplicative reduction at 11, with $n_{11}=5$, resp. 1. Show that $\bar\rho_{\texttt{11a1},5}$ is unramified at 11 and $\bar\rho_{\texttt{11a3},5}$ is ramified there, compute $\#E[5^t]^{I_{11}}$ for both curves and every $t\ge1$, and explain why $E[5]\simeq\mathbb{Z}/5\oplus\mu_5$ for `11a1` is consistent with the first statement.

*Answer.* By Lemma 11.17(ii), $\#E[5^t]^{I_{11}}=5^{t+1}$ for `11a1` and $5^t$ for `11a3`. The module $\mathbb{Z}/5\oplus\mu_5$ is unramified at every prime other than 5.

**Exercise 11.73.** Let $K/\mathbb{Q}_p$ be finite, $\ell\neq p$, and $E/K$. Using the filtration $E_1\subseteq E_0\subseteq E$ over the finite unramified extensions of $K$, show: (a) if the reduction is additive, $E(K^{\mathrm{ur}})[\ell^\infty]$ is finite, of order dividing $\#\Phi_E(\bar k)$; (b) if it is multiplicative, $E(K^{\mathrm{ur}})[\ell^\infty]$ is an extension of a finite group by $\mu_{\ell^\infty}$, so it is infinite; (c) if it is good, $E(K^{\mathrm{ur}})[\ell^\infty]\simeq(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^2$. Deduce once more the criterion of Néron, Ogg and Shafarevich, and the dichotomy of Theorem 11.9(e), (f) for $E$ with bad reduction over $K$ itself.

**Exercise 11.74.** Let $q\neq\ell$ and let $E$ have additive reduction at $q$. (a) Deduce from Lemma 11.10 that the $\ell$-part of $c_q$ is $\#E(\mathbb{Q}_q)[\ell^\infty]$. (b) Show by an example that (a) fails for split multiplicative reduction (take $q\equiv1\pmod\ell$ and $\ell\nmid n_q$). (c) Compare with the final check in the proof of the paper's Lemma lem:tamfactor, where the $\ell$-part of $c_q$ at a multiplicative prime is read from the quotient $\Phi_q^{\mathrm{geom}}[\ell^\infty]$ of $E[\ell^\infty]^{I_q}$, not from the sub $\mu_{\ell^\infty}\otimes\varepsilon$.

**Exercise 11.75.** (The paper's Lemma lem:IV for `121c1`.) The curve `121c1` has type IV at 11. Show that $N(\bar\rho_{\texttt{121c1},3}) = 11$ and check $a_p(\texttt{121c1}) \equiv a_p(\texttt{11a1}) \pmod 3$ for all primes $p < 100$, $p \neq 11$, including $p = 3$, where both curves are ordinary. Which statement of Theorem 11.35 does the case $p = 3$ illustrate?

*Answer.* The exponent at 11 is $f_{11} - \dim \Phi_{11}^{\mathrm{geom}}[3] = 2-1 = 1$. At $p = 3$ one has $a_3(\texttt{121c1}) = 2$ and $a_3(\texttt{11a1}) = -1$; the congruence is part (iii) of Theorem 11.35, since Proposition 11.6(iii) only covers primes not dividing $3N$.

**Exercise 11.76.** (The paper's Remark rem:wild on the type IV at 2.) Prove that a curve of type IV or IV$^*$ at 2 has $P_2(X) = 1 + 2X$ and $e_3(E, 2) = 1$ using only Lemma 11.10 at $q = 2$, Theorem 11.12(i), (ii) and the tame relation; do not use Kraus's classification. Check the result on `20a1` and `980i1` (type IV$^*$ at 2) with `lfuneuler(lfunsympow(E,2),2)`.

**Exercise 11.77.** Carry out Remark 11.24: for $q \geq 5$ derive the orders $\mathrm{I}_0^* \mapsto 2$, $\mathrm{IV}, \mathrm{IV}^* \mapsto 3$, $\mathrm{III}, \mathrm{III}^* \mapsto 4$, $\mathrm{II}, \mathrm{II}^* \mapsto 6$ of the image of inertia from the component groups, Lemma 11.10 at $\ell = 2$ and $\ell = 3$, and the independence of $\ell$. In particular prove that an element $\tau \in \mathrm{SL}_2(\mathbb{Z}_2)$ with $\tau^2 = -1$ is not congruent to 1 modulo 2, and that its reduction fixes a line of $\mathbb{F}_2^2$.

*Hint.* The reduction $\bar\tau$ satisfies $\bar\tau^2 = 1$, so its characteristic polynomial is $X^2 + 1 = (X + 1)^2$ over $\mathbb{F}_2$; an element of $\mathrm{GL}_2(\mathbb{F}_2)$ with this characteristic polynomial is 1 or a transvection, and it is not 1 by the argument of Remark 11.24.

**Exercise 11.78.** (The paper's Remark rem:types.) Verify the table of Proposition 11.29 row by row on the curves `121a1` (II), `121c2` (IV$^*$), `726a1` (III at 11), `539d1` ($\mathrm{I}_0^*$ at 7) and `539c1` ($\mathrm{I}_3^*$ at 7), by computing the ramification indices of the prime in the fields of definition of the 3-torsion points. Which statements of the remark concern $\bar\rho_{E,3}$ and which concern $E$, and in which row of the table do the twist's effects on the two differ?

**Exercise 11.79.** Let $\ell$ be odd. Show that $\bar\rho_{E,\ell}(G_{\mathbb{Q}})$ is never contained in $\mathrm{SL}_2(\mathbb{F}_\ell)$ and never conjugate to a subgroup of $C_{ns}(\mathbb{F}_\ell)$. Show that it can be contained in $C_s(\mathbb{F}_\ell)$, and characterise this case by the splitting of $E[\ell]$.

*Hint.* Use the determinant and complex conjugation (Proposition 11.44). The image lies in a conjugate of $C_s(\mathbb{F}_\ell)$ iff $E[\ell]$ is a direct sum of two $G_{\mathbb{Q}}$-stable lines; `14a1` at $\ell = 3$ is an example.

**Exercise 11.80.** (The paper's Proposition prop:ns for `726a1`.) Factor $\psi_3$ for `726a1`, identify the three quadratic subfields $\mathbb{Q}(\sqrt{33})$, $\mathbb{Q}(\sqrt{-11})$, $\mathbb{Q}(\sqrt{-3})$ of its splitting field, deduce that the image is $N_s(\mathbb{F}_3)$, and check $a_p \equiv 0 \pmod 3$ for the primes $p \equiv 2 \pmod 3$, $p < 60$, $p \neq 2, 11$.

**Exercise 11.81.** (The paper's Section sec:anomaly.) Let $E$ have image `3Ns`. (a) Show $h_3(E) = 1$ using Lemma 11.58 only. (b) Show without determinants that $\rho \bmod 9 \simeq (\rho \otimes \chi_3^{\mathrm{cyc}}) \bmod 9$ would give $a_p \equiv p a_p \pmod 9$ for $p \nmid 3N$, hence $9 \mid a_p$ for every such $p \equiv 2 \pmod 3$. (c) For `726a1`, $a_{17} = 6$: so the trace test of (b) also excludes $h_3 \geq 2$ for this curve. Why is such a test unnecessary?

*Hint.* (b) Take traces at $\mathrm{Frob}_p$: $\chi_3^{\mathrm{cyc}}(\mathrm{Frob}_p) = p$, and $1 - p$ is a unit modulo 3 when $p \equiv 2 \pmod 3$. (c) $9 \nmid 6$; the test is unnecessary because the determinant argument in the proof of Lemma 11.58(iii) gives $h_3(E) \leq 1$ for every curve.

**Exercise 11.82.** Prove parts (iii) and (iv) of Proposition 11.60 in the case of `726a1` ($R = \{11\}$), `338d1` ($R = \{13\}$) and `6912d1` ($R = \{2\}$): find $R$ from the discriminants of the quadratic factors of $\psi_3$, check the Kodaira types at the primes of $R$, the classes modulo 12, and the Euler factors $1 + 11X$, $1 - 13X$, $1 + 2X$ returned by `lfuneuler(lfunsympow(E,2),p)`.

**Exercise 11.83.** Let $\ell \geq 5$ and $\ell \nmid N$. (a) Using the fundamental characters, show that $\bar\rho_{E,\ell} \not\simeq \bar\rho_{E,\ell} \otimes \chi_{\ell^*}$, without appealing to Corollary 11.37 (redo its proof for the ordinary and the supersingular case separately). (b) The curve `608b1` ($N = 2^5 \cdot 19$) has mod 5 image `5Ns`. Find the quadratic field $K$ with $\bar\rho_{\texttt{608b1},5} \simeq \bar\rho_{\texttt{608b1},5} \otimes \chi_K$ by testing $a_p \equiv 0 \pmod 5$ at the primes inert in the quadratic fields of discriminant dividing $4N$, and check that $K \neq \mathbb{Q}(\sqrt{5})$.

**Exercise 11.84.** Let $E$ be semistable, $\ell$ odd, $\ell \nmid N$, and $E[\ell]$ reducible. Prove that $a_p \equiv 1 + p \pmod \ell$ for every $p \nmid N\ell$ and that $E$ or an $\ell$-isogenous curve has a rational point of order $\ell$. Check both statements for `14a1` and `26a1` at $\ell = 3$, and find the isogenous curve with a point of order 3 for `14a3`.

*Answer for the last part.* `14a3` has image `3B.1.2`, its stable line is $\mu_3$, and its torsion group has order 2. Its only 3-isogenous neighbour is `14a1` (the isogeny matrix in `allisog`), whose torsion group is $\mathbb{Z}/6$.

**Exercise 11.85.** Determine the mod 3 images of `14a1` and of `539a2` from $\psi_3$ and the characters of the rational lines. Explain why `3Cs.1.1`, the image of `14a1`, is not the full split Cartan subgroup, and which curve of the pair has the full split Cartan image.

**Exercise 11.86.** Compute $N(\bar\rho_{E,3})$ from Proposition 11.40 for `539c1`, `77c1`, `726a1` and `6912d1`, and $N(\bar\rho_{E,5})$ for `980i1` and `140b1`. Which of these curves occur in the paper's Example of Section 3, directly or as the twist of a curve there, and which are illustrations added here? Which congruences of that Example do the former explain, and why does the one at $\ell = 5$ not fall under Theorem thm:qual?

*Answer for the conductors.* $7^2 \cdot 11$, $11$, $11^2$, $2^8$ at $\ell = 3$, and $2^2 \cdot 7^2$, $2^2$ at $\ell = 5$; compare Example 11.42. At $\ell = 5$ both curves have $5 \mid N$.

**Exercise 11.87.** Let $E$ have image `3Ns` and $3 \nmid N$. (a) Show that $\omega_2$ has order 8 for $\ell = 3$, so that in the supersingular case $\bar\rho_{E,3}(I_3)$ would contain an element of order 8. (b) Show that $N_s(\mathbb{F}_3)$ has no element of order 8, and deduce that $E$ is ordinary at 3: a second proof of Proposition 11.60(vi). (c) Show that in the ordinary case $\bar\rho(I_3)$ has order 2 or 6, and use $N_s(\mathbb{F}_3)$ to show it has order 2.

*Hint.* (a) $\omega_2$ maps onto $\mathbb{F}_9^\times$. (b) $N_s(\mathbb{F}_3)$ is dihedral of order 8; its elements have orders 1, 2, 4. (c) In the ordinary case $\bar\rho(I_3)$ lies in $\left(\begin{smallmatrix}\omega & * \\ 0 & 1\end{smallmatrix}\right)$ with $\omega$ of order 2, a group of order 6, and $\bar\rho(I_3)$ surjects onto the values of $\omega$; $N_s(\mathbb{F}_3)$ has no element of order 3.

## 11.12 Notes and references

**Sources followed.** The Tate module, the Weil pairing and the criterion of Néron, Ogg and Shafarevich follow Silverman [185, Chapters III and VII] (III.7, III.8 and VII.7 are the relevant sections in the editions we know; these are unverified pointers) and Serre and Tate [179, Sections 1 and 2]; the proof of (d)⇒(a) in Theorem 11.9 is the one in [179, Section 1]. The Tate curve and its Galois module are in [184, Chapter V] (V.3 and V.5, unverified pointers), and the matrix form at a multiplicative prime is the elliptic-curve case of [38, Theorem 3.1(e)], the number the paper cites. The image of inertia at a potentially good prime is due to Serre and Tate [179, Section 2] and Kraus [107]; we cite Kraus's paper as a whole (Chapter 10 explains why). The fundamental characters and the description of $\bar\rho|_{I_\ell}$ in the ordinary and supersingular cases are in Serre [171, Section 1]; the description for newforms, due to Deligne and Fontaine, is quoted from Edixhoven [61, Section 2], as in the paper. Serre's conjecture, the Serre conductor and the weight are from Serre [176] (Sections 1 and 2 for the definitions and Section 4 for examples

from elliptic curves: unverified pointers); the divisibility $N(\bar\rho) \mid N$ is Carayol's [22] for newforms, and the conductor of a residual representation of a modular form is studied by Livné [116]. The open image theorem is Serre's [171]; the list of degrees of rational isogenies is Mazur's [126]; the images for non-CM curves at $\ell \le 13$ are Zywina's [210], and the labels Sutherland's [191]. Serre's conjecture is the theorem of Khare and Wintenberger [98]; the level-lowering theorem is due to Ribet [156], Carayol [22] and Diamond [47], in the form of [38, Theorem 3.15] quoted by the paper. Faltings's theorem is [63]; crystalline representations are introduced in Berger's survey [11].

**Theorem numbers.** We cite theorem numbers only where the paper cites them: [38, Theorem 3.1(e)], [38, Theorem 3.15], [61, Section 2], and the definition [49, (31)] of $S_f$. All other references are to sections, and the section numbers of books given above are pointers we have not checked against printed editions.

**What is new here.** Three statements of this chapter, also proved in the paper, are not standard in the literature as far as we know: that the hypothesis on $\mathbb{Q}(\sqrt{\ell^*})$ is automatic for $\ell \ge 5$, $\ell \nmid N$ (Theorem 11.53, the paper's Proposition prop:Sf); that the types IV, IV$^*$ at 2 have cyclic inertia of order 3 (Lemma 11.25, Corollary 11.27); and the structure of the curves with image `3Ns` (Proposition 11.60). The first may well be known; we found no reference. The formula $a_p(\bar\rho_{E,\ell}) = f_p - \dim \Phi_p^{\mathrm{geom}}[\ell]$ (Proposition 11.40) is standard in substance (it is how the Serre conductor of an elliptic curve is computed in practice) but we know no reference that states it in this form. The pairing of the images `3B.1.1` and `3B.1.2` by the ends of the 3-isogeny paths (Proposition 11.46) is elementary and surely known; we found no reference that states it.

**Computations.** All numbers were computed with PARI/GP 2.17.2 [148] through `cypari2`, on Cremona's files `allcurves`, `allisog`, `galrep` and `degphi` [36] for conductors below 10,000 (64,687 curves, 462 with complex multiplication, recognised by their $j$-invariant among the 13 rational CM $j$-invariants). The checks reported in the text are: the mod 3 image of every curve from $\psi_3$ against the `galrep` label (Theorem 11.45); the 3-isogeny paths of all 38,042 classes (Proposition 11.46); the image of inertia modulo 3 by Kodaira type at $q \ge 5$ and at $q = 2$ (Proposition 11.29, Remark 11.31); all 4,750 curves with type IV or IV$^*$ at 2 (Corollary 11.27); all 146 curves with image `3Ns`, and the 103 optimal ones among them, the optimal curve being the one listed in `degphi` (Proposition 11.60); the ordinarity at $\ell$ of the curves with reducible $\bar\rho_{E,\ell}$, $\ell \nmid N$ (Corollary 11.37); the congruences $a_p \equiv 1 + p$ (Proposition 11.64); the dihedral images at $\ell = 5, 7$ (Theorem 11.53); and the Serre conductors and congruences of Example 11.42. Every count is for conductor below 10,000; nothing beyond that range was checked.

# Part IV

# Modular forms and modular curves

# Chapter 12

# Modular forms and modular curves

This chapter builds the modular forms and modular curves of the paper: the curve $X_0(N)$ from which the modular parametrisation starts, the space $S_2(\Gamma_0(N))$ in which the newform $f = f_E$ lives, its $q$-expansions and their integrality, the Petersson product that defines "orthogonal to $f$" and the Petersson norm $\langle f, f\rangle$, and the Atkin-Lehner involution $w_N$ of the paper's Section 4.1. It assumes complex analysis and uses Chapter 1 (group actions, $\mathrm{SL}_2(\mathbb{Z}/N)$), Chapter 2 (congruences, the Chinese remainder theorem, Kronecker symbols), Chapter 3 (free modules and saturation), Chapter 4 (characters of finite abelian groups), Chapter 8 (Riemann surfaces, Riemann-Hurwitz, holomorphic differentials, Riemann-Roch, the cup product) and Chapter 9 (lattices, $j$, the curves `11a1` and others). Hecke operators, newforms and twists are Chapter 13; the Jacobian $J_0(N)$, the modular parametrisation and the modular degree are Chapter 14; Galois representations of newforms are Chapter 15.

## Core results at a glance

Throughout, $\mathcal{H}$ is the upper half plane, $q = e^{2\pi i z}$, $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$, the slash operator is $(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz+d)^{-k}f(\gamma z)$, $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$, and the Petersson product is unnormalised.

**12.1 Conventions.** $\mathcal{H}$, $\mathcal{H}^* = \mathcal{H} \cup \mathbb{P}^1(\mathbb{Q})$, $d\mu = dx\,dy/y^2$; $q = e^{2\pi i z}$ is never a prime here; slash operator with $(\det\gamma)^{k/2}$ (equal to Diamond-Shurman's in weight 2); $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$ is the paper's $w$; Petersson product unnormalised; "coefficientwise" means at all $n$. Convention 12.1.

**12.2 $\mathrm{GL}_2^+(\mathbb{R})$ on $\mathcal{H}$.** $\operatorname{Im}\gamma z = \det\gamma \operatorname{Im} z/|cz+d|^2$, $\frac{d}{dz}\gamma z = \det\gamma/(cz+d)^2$, the cocycle $j(\gamma\gamma', z) = j(\gamma, \gamma' z)j(\gamma', z)$, $\mathcal{H} = \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2)$, $d\mu$ invariant. *(proved here)* Proposition 12.2.

**12.3 Element types** (definition): elliptic, parabolic, hyperbolic by $|\operatorname{tr}\gamma| <, =, > 2$; elliptic elements of $\mathrm{SL}_2(\mathbb{Z})$ have order 3, 4, 6; the fixed points of parabolic elements of $\mathrm{SL}_2(\mathbb{Z})$ are $\mathbb{P}^1(\mathbb{Q})$. *(proved here)* Proposition 12.3.

**12.4 Fundamental domain.** $\mathcal{F} = \{|\operatorname{Re} z| \le \frac{1}{2}, |z| \ge 1\}$ meets every orbit, its interior points are inequivalent, the only nontrivial stabilisers are at $i$, $\zeta_3$, $-\bar{\zeta}_3$ (orders $2, 3, 3$ in $\mathrm{PSL}_2$), $\mathrm{SL}_2(\mathbb{Z}) = \langle S, T\rangle$, $\operatorname{vol}\mathcal{F} = \pi/3$. *(proved here)* Theorem 12.4.

**12.5** $X(1)$ is a sphere, obtained from $\mathcal{F}$ by gluing, with elliptic points $i$, $\zeta_3$ only. *(proved here)* Corollary 12.5.

**12.6 Congruence subgroups** (definition): $\Gamma(N) \subseteq \Gamma_1(N) \subseteq \Gamma_0(N)$. Definition 12.6.

**12.7 Indices.** $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N)$ is onto; $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = N^3 \prod_{p|N}(1-p^{-2})$, $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)] = N^2 \prod(1-p^{-2})$, $\Gamma_0(N)\backslash \mathrm{SL}_2(\mathbb{Z}) \simeq \mathbb{P}^1(\mathbb{Z}/N)$ and $\mu(N) = N\prod_{p|N}(1+\frac{1}{p})$; $\Gamma_0(N)/\Gamma_1(N) \simeq (\mathbb{Z}/N)^\times$. *(proved here)* Proposition 12.7.

**12.8 Fundamental domains of subgroups.** $\bigcup \gamma_i \mathcal{F}$ over coset representatives; $\mathrm{vol}(\Gamma_0(N)\backslash\mathcal{H}) = \pi\mu(N)/3$. *(proved here)* Proposition 12.8.

**12.9 Cusps and widths** (definition). Definition 12.9.

**12.10 Cusps as orbits.** Cusps of $\Gamma \ni -I$ are the $\langle T\rangle$-orbits on $\Gamma\backslash \mathrm{SL}_2(\mathbb{Z})$, widths are orbit lengths, $\sum h_s = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma]$. *(proved here)* Proposition 12.10.

**12.11 Cusps of $\Gamma_0(N)$.** $\gcd(\text{denominator}, N)$ is an invariant; cusps $a/c$, $c \mid N$, $a \bmod \gcd(c, N/c)$, width $N/\gcd(c^2, N)$, $\nu_\infty(N) = \sum_{c|N} \varphi(\gcd(c, N/c))$. *(proved here for the invariant, for $N$ squarefree and for $N = p^2$; general case sketched here; [52], [146])* Proposition 12.12.

**12.12 Elliptic points of $\Gamma_0(N)$** (definition and count): $\nu_2(N)$ and $\nu_3(N)$ are the numbers of roots of $u^2+1$ and $u^2-u+1$ modulo $N$, given by Kronecker symbols; the fibres of $Y_0(N) \to Y(1)$ over $i$ and $\zeta_3$. *(proved here)* Proposition 12.14.

**12.13 $Y(\Gamma)$ is a Riemann surface.** Proper discontinuity, Hausdorff quotient, finite cyclic stabilisers; charts $t = w^e$. *(proved here for (i); (ii) sketched here; [52], [180])* Theorem 12.15.

**12.14 $X(\Gamma)$ is a compact Riemann surface** (with charts $q_h$ at the cusps), algebraic by Riemann existence and GAGA. *(proof sketched here; [52], Chapter 8)* Theorem 12.16.

**12.15 Genus formula.** $X_0(N) \to X(1)$ has degree $\mu(N)$, ramified over $i$, $\zeta_3$, $\infty$, and $g(X_0(N)) = 1 + \frac{\mu}{12} - \frac{\nu_2}{4} - \frac{\nu_3}{3} - \frac{\nu_\infty}{2}$; table of genera ($g(X_0(11)) = 1$, $g(X_0(121)) = 6$, $g(X_0(539)) = 49$). *(proved here, given the charts of Theorem 12.15(ii) and Theorem 12.16, and Riemann-Hurwitz from Chapter 8)* Theorem 12.17.

**12.16 Moduli.** $Y_0(N)$ classifies $(E, C)$, $C$ cyclic of order $N$; $Y_1(N)$ classifies $(E, P)$; $Y(N)$ classifies $(E, (P, Q))$, $(P, Q)$ a basis of $E[N]$ with Weil pairing $e_N(P, Q) = \zeta_N$, where $\zeta_N = e_N(\frac{1}{N}, \frac{\tau}{N})$ on $\mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$ is $e^{2\pi i/N}$ or its inverse according to the normalisation of $e_N$; $w_N(E, C) = (E/C, E[N]/C)$; cusps are degenerate objects; $X_0(N)$ has a model over $\mathbb{Q}$. *(proved here for (i) to (iii), given Chapter 9; (iv), (v) stated; [43], [91], [52])* Theorem 12.18.

**12.17 Slash operator** (definition) $(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz+d)^{-k} f(\gamma z)$. Definition 12.20.

**12.18 Slash properties.** Action, scalars act by $\mathrm{sgn}(\lambda)^k$, $(f|_2\gamma)\,dz = \gamma^*(f\,dz)$, $y^{k/2}|f|$ invariant. *(proved here)* Lemma 12.21.

**12.19 Modular and cusp forms** (definition): $M_k(\Gamma) \supseteq S_k(\Gamma)$, holomorphy and vanishing at all cusps. Definition 12.22.

**12.20 First properties.** Odd weight vanishes for $\Gamma \ni -I$; products; $f(dz) \in M_k(\Gamma_0(Md))$ for $f \in M_k(\Gamma_0(M))$. *(proved here)* Proposition 12.23.

**12.21** **$q$-expansions.** $f = \sum_{n\geq 0} a_n q^n$ on all of $\mathcal{H}$, $a_n = \int_0^1 f(x+iy)e^{-2\pi in(x+iy)}dx$, expansions at the other cusps in $q_h$, cusp forms have no constant terms, and $f \mapsto (a_n)$ is injective. *(proved here)* Proposition 12.24.

**12.22** **Hecke bound.** For $f \in S_k(\Gamma_0(N))$, $y^{k/2}|f|$ is bounded and $a_n(f) = O(n^{k/2})$. *(proved here)* Proposition 12.26.

**12.23** **Ramanujan-Petersson.** For a newform and $p \nmid N$, $|a_p| \leq 2p^{(k-1)/2}$, so $\beta_p = \bar{\alpha}_p$ and $|\alpha_p/\beta_p| = 1$. *(stated; [44], [52])* Theorem 12.27.

**12.24** **Nebentypus** (definition): $M_k(\Gamma_0(N), \chi)$, diamond operators. Definition 12.29.

**12.25** **Decomposition by characters.** $M_k(\Gamma_1(N)) = \bigoplus_\chi M_k(\Gamma_0(N), \chi)$; $M_k(\Gamma_0(N), \chi) = 0$ unless $\chi(-1) = (-1)^k$. *(proved here)* Proposition 12.30.

**12.26** **Eisenstein series of level one.** $G_k = 2\zeta(k)E_k \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ for even $k \geq 4$, $E_k = 1 - \frac{2k}{B_k}\sum \sigma_{k-1}(n)q^n$; $E_2(-1/z) = z^2E_2(z) + \frac{6z}{\pi i}$ and $E_2^* = E_2 - \frac{3}{\pi y}$ is invariant. *(proved here for $k \geq 4$; for $E_2$, proved here given the transformation law of $\eta$)* Proposition 12.32.

**12.27** **$\Delta$ and $j$.** $\Delta = (E_4^3 - E_6^2)/1728 = \sum \tau(n)q^n \in S_{12}$ with integer coefficients, $\Delta(\Lambda_z) = (2\pi)^{12}\Delta(z)$; $j = E_4^3/\Delta = q^{-1} + 744 + 196884q + \cdots$, $j(z) = j(\Lambda_z)$; $\Delta = \eta^{24}$. *(proved here; $\Delta = \eta^{24}$ given the transformation law of $\eta$)* Proposition 12.33.

**12.28** **Valence formula, level one.** $v_\infty + \frac{1}{2}v_i + \frac{1}{3}v_{\zeta_3} + \sum v_p = \frac{k}{12}$. *(proved here)* Theorem 12.34.

**12.29** **Level one structure.** $M_2 = 0$, $M_k = \mathbb{C}E_k$ for $k = 4, \ldots, 10, 14$, $S_k = \Delta M_{k-12}$, dimension formula, $M_* = \mathbb{C}[E_4, E_6]$; zeros of $\Delta$, $E_4$, $E_6$. *(proved here)* Corollary 12.35.

**12.30** **$j$ is a bijection** $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H} \to \mathbb{C}$ and an isomorphism $X(1) \simeq \mathbb{P}^1(\mathbb{C})$; every $j_0$ is $j(\Lambda)$. *(proved here; the isomorphism $X(1) \simeq \mathbb{P}^1(\mathbb{C})$ given the charts of Theorem 12.15(ii) and Theorem 12.16, and Chapter 8)* Theorem 12.36.

**12.31** **The function field of $X_0(N)$.** $j_N(z) = j(Nz)$ is a meromorphic function on $X_0(N)$; $\mathbb{C}(X(1)) = \mathbb{C}(j)$; the conjugates of $j_N$ over $\mathbb{C}(j)$ are the $\mu(N)$ distinct functions $j(N\alpha_i z)$, $\mathrm{SL}_2(\mathbb{Z}) = \bigsqcup \Gamma_0(N)\alpha_i$, so $j_N$ has degree $\mu(N)$ over $\mathbb{C}(j)$ and $\mathbb{C}(X_0(N)) = \mathbb{C}(j, j_N)$. *(proved here, given the charts of Theorem 12.15(ii) and Theorem 12.16, and Chapter 8 for $[\mathbb{C}(X_0(N)) : \mathbb{C}(j)] = \deg \pi$)* Proposition 12.38.

**12.32** **Valence formula for $\Gamma_0(N)$.** $\sum_P v_P(f)/e_P = k\mu(N)/12$. *(proof sketched here; [52], [180])* Theorem 12.39.

**12.33** **Weight two forms are differentials.** $S_2(\Gamma_0(N)) \simeq H^0(X_0(N), \Omega^1)$ by $f \mapsto f(z)\,dz$; $M_2$ corresponds to differentials with simple poles at the cusps, residue $\frac{h}{2\pi i}b_0$; $\dim S_2 = g$, $\dim M_2 = g + \nu_\infty - 1$. *(proved here, given the charts of Theorem 12.15(ii) and Theorem 12.16, and Chapter 8)* Theorem 12.40.

**12.34** **Dimension formulas** for $S_k(\Gamma_0(N))$, $M_k(\Gamma_0(N))$. *(proved here for $k \leq 2$; sketched here for $k \geq 4$; [52])* Theorem 12.41.

**12.35** **Weight two Eisenstein series.** $E_2(z) - NE_2(Nz) \in M_2(\Gamma_0(N)) \smallsetminus S_2$, with $a_0 = 1 - N$; $M_2(\Gamma_0(p)) = \mathbb{C}E_2^{(p)} \oplus S_2(\Gamma_0(p))$. *(proved here, given the transformation law of $\eta$)* Proposition 12.42.

**12.36 The $\eta$-function.** Holomorphic without zeros, $\eta(z+1) = e^{\pi i/12}\eta(z)$, $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$, $\eta'/\eta = \frac{\pi i}{12}E_2$. *(transformation law stated; [6]; the rest proved here)* Proposition 12.45.

**12.37 Level 11.** $S_2(\Gamma_0(11)) = \mathbb{C}\,\eta(z)^2\eta(11z)^2$, zeros only at the two cusps, $f_{11}|_2 W_{11} = -f_{11}$; the coefficients are those of `11a1`. *(proved here given the transformation law of $\eta$, except (iv): stated, Chapter 14, checked for $n \le 2000$)* Theorem 12.46.

**12.38 Petersson product** (definition): the unnormalised integral $\langle f, g\rangle = \int_{\Gamma_0(N)\backslash\mathcal{H}} f\bar{g}y^k\,d\mu$. Definition 12.48.

**12.39 Petersson properties.** Well defined, convergent, positive definite; in weight 2, $\langle f, g\rangle = \frac{i}{2}\int_{X_0(N)} \omega_f \wedge \overline{\omega_g}$. *(proved here)* Proposition 12.49.

**12.40 Normalisations.** PARI's `mfpetersson` is the book's product divided by $\mu(N)$; other books divide by $\pi\mu(N)/3$; $\deg\varphi_E \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_E^2\langle f, f\rangle$. Convention 12.50, checked in Example 12.53.

**12.41 Change of group.** $\langle f, g\rangle_\Gamma = [\bar{\Gamma}_0(N) : \bar{\Gamma}]\langle f, g\rangle_{\Gamma_0(N)}$; invariance under normalising $\alpha$; for $\Gamma \subset \mathrm{SL}_2(\mathbb{Q})$ containing some $\Gamma(M)$ with finite index, $\langle f, g\rangle_\Gamma = [\bar{\Gamma} : \bar{\Gamma}_1]^{-1}\langle f, g\rangle_{\Gamma_1}$ ($\Gamma_1 \subseteq \Gamma \cap \mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup of finite index, $\bar{}$ the image in $\mathrm{PSL}_2(\mathbb{R})$) is well defined, and $\langle f|_k\alpha, g|_k\alpha\rangle_{\alpha^{-1}\Gamma\alpha} = \langle f, g\rangle_\Gamma$ for every $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. *(proved here)* Proposition 12.51.

**12.42 Orthogonal complement.** $S_k = \mathbb{C}f \oplus f^\perp$ with projection $g \mapsto \frac{\langle g,f\rangle}{\langle f,f\rangle}f$. *(proved here)* Proposition 12.52.

**12.43 Integral forms** (definition): $S_k(\Gamma_0(N); A)$ for $A \subseteq \mathbb{C}$, and $S_k(\Gamma_0(N); R) = S_k(\Gamma_0(N); \mathbb{Z}) \otimes R$ for any ring $R$. Definition 12.56.

**12.44 Integral bases.** $S_k(\Gamma_0(N))$ and $M_k(\Gamma_0(N))$ have bases of forms with integer coefficients. *(stated; [180], [52])* Theorem 12.57.

**12.45 Sturm's bound.** For $f \in M_k(\Gamma_0(N); \mathcal{O})$, $a_n(f) \in \lambda$ for $n \le k\mu(N)/12$ implies it for all $n$. *(proof sketched here; [190], [187])* Theorem 12.58.

**12.46 The integral lattice.** $S_2(\Gamma_0(N); \mathbb{Z})$ is free of rank $\dim S_2$ and spans; truncation at $B_2(N)$ is a split injection; $q$-expansion is injective over every ring, and over $\mathbb{Z}_\ell$ the image is the $\ell$-integral forms; congruences modulo $m$ or $\lambda$ are decided by $a_n$, $n \le B_2(N)$. *(proved here, given Theorems 12.57 and 12.58)* Corollary 12.59.

**12.47 Atkin-Lehner matrices** (definition): $W_d = \left(\begin{smallmatrix} dx & y \\ Nz & dw \end{smallmatrix}\right)$, $\det W_d = d$, for $d \,\|\, N$. Definition 12.63.

**12.48 Atkin-Lehner involutions.** The matrix $W_d$ normalises $\Gamma_0(N)$, $W_d^2 \in d\Gamma_0(N)$, $W_dW_{d'} \in \gcd(d, d')\mathcal{W}_{d*d'}$, where $\mathcal{W}_e$ is the set of Atkin-Lehner matrices for $e \,\|\, N$ and $d * d' = dd'/\gcd(d, d')^2$; commuting holomorphic involutions $w_d$ of $X_0(N)$ and of $M_k(\Gamma_0(N))$ ($k$ even); $w_N\colon z \mapsto -1/(Nz)$; $w_d^*\omega_f = \omega_{f|_2W_d}$. *(proved here; the statement on $X_0(N)$ given the charts of Theorem 12.15(ii) and Theorem 12.16)* Proposition 12.64.

**12.49 Atkin-Lehner and pairings.** $w_d$ is a self-adjoint isometric involution of $S_k(\Gamma_0(N))$; $\dim S_2^+ = g(X_0(N)/w_N)$; $w_N^*$ preserves the cup product and acts by $\varepsilon_N(f)$ on $\omega_f$, $\overline{\omega_f}$ when $f|_2W_N = \varepsilon_N(f)f$. *(proved here, given Chapter 8)* Proposition 12.65.

**12.50 Atkin-Lehner and cusps.** $w_N$ maps cusps of denominator invariant $c$ to $N/c$; for $N$ squarefree the $w_e$ act simply transitively on the cusps. *(proved here)* Proposition 12.67.

## Where the paper uses this chapter

- **Section 1** ("let $f = f_E \in S_2(\Gamma_0(N))$ be the newform attached to it"; $\varphi_E\colon X_0(N) \to E$; the Petersson norm): Definition 12.22, Proposition 12.24, Theorem 12.16, Theorem 12.18, Theorem 12.17, Theorem 12.40, Definition 12.48, Paper link 12.19, Paper link 12.55.

- **Section 3, opening paragraph** (the congruence number: "coefficientwise", "integer coefficients", "orthogonal to $f$"; the Hecke algebra as a subring of $\operatorname{End} S_2(\Gamma_0(N))$): Proposition 12.24, Definition 12.56, Proposition 12.52, Paper link 12.25.

- **Lemma lem:crit** (congruence criterion: $f^{\perp}$, $S_2(\Gamma_0(N);\mathbb{Z})$, the version over $\mathbb{Z}_\ell$): Proposition 12.52, Theorem 12.57, Corollary 12.59, Paper link 12.60; the perfect pairing with $\mathbb{T}$ is Chapter 13's, the lemma is proved in Chapter 15.

- **Lemma lem:old** (oldforms): the forms $h(dz)$, Proposition 12.23(e); orthogonality, Definition 12.48.

- **The optimal-level theorem in Section 3** ("nebentypus of order prime to $\ell$"): Definition 12.29, Proposition 12.30, Paper link 12.31.

- **Example ex:qual** (the four congruences, proved for all $n$ by Sturm's bound): Theorem 12.58, Corollary 12.59, Example 12.61, Paper link 12.62.

- **Section 4.1** (the operator $w = W_{M^\Sigma}$, "the Atkin-Lehner involution $w_{M^\Sigma}$ with no scalar factor"; $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$; the step on $w_N$ in Proposition prop:pullback): Definition 12.63, Proposition 12.64, Proposition 12.65, Paper link 12.66.

- **Remark rem:degree** ($\deg\varphi_E \cdot \operatorname{vol}(\Lambda_E) = 4\pi^2 c_E^2\langle f, f\rangle$, unnormalised product): Convention 12.50, Example 12.53.

- **Theorem thm:twist, the set-up of the proof of Theorem thm:quant and Lemma lem:step1** (integrality of twists, $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$, the projection by the idempotent $e_{\mathfrak{m}}$): Definition 12.56, Corollary 12.59, Paper link 12.60, Paper link 12.55.

- **Lemma lem:euler and Theorem thm:dfg14** ($|\alpha/\beta| = 1$ at good primes): Theorem 12.27, Paper link 12.28.

- **Conjecture conj:eis and Section sec:eis** (the Eisenstein case): Proposition 12.42, Example 12.43, Paper link 12.44.

- **Definition def:terms** ($v_q(j_E)$): Proposition 12.33, Theorem 12.36, Paper link 12.37.

- **The running example `11a1`**: Theorem 12.46, Example 12.43, Example 12.53.

## 12.1 The upper half plane

This section fixes the stage: the upper half plane, the action of $\mathrm{GL}_2^+(\mathbb{R})$ by Möbius transformations, and the invariant measure. Nothing here is specific to the paper, but everything later rests on it: the curve $X_0(N)$ of the paper's Section 1 is a quotient of $\mathcal{H}$, the newform $f$ is a function on $\mathcal{H}$, and the Petersson norm $\langle f, f\rangle$ is an integral against the measure $d\mu$ defined here. We also fix the conventions of the chapter. The reader with a physics background will meet three old friends on the way: $q$-expansions are Fourier series (Proposition 12.24), the valence formula is the argument principle (Theorem 12.34), and the Petersson product is a weighted $L^2$ inner product (Definition 12.48).

**Convention 12.1** (notation of this chapter)**.**

(a) *The upper half plane.* $\mathcal{H} = \{z = x+iy \in \mathbb{C} : y > 0\}$, as in Chapters 8 and 9; $\mathcal{H}^* = \mathcal{H} \cup \mathbb{P}^1(\mathbb{Q})$ with $\mathbb{P}^1(\mathbb{Q}) = \mathbb{Q} \cup \{\infty\}$; $d\mu = dx\,dy/y^2$. The variable of $q$-expansions is $q = e^{2\pi i z}$, and $q_h = e^{2\pi i z/h}$ for $h \geq 1$. In this chapter primes are called $p$ or $\ell$ and $q$ is always $e^{2\pi i z}$, except inside the environments "In the paper", which say so when they use the paper's $q$ for a bad prime (Appendix B, B-C1).

(b) *Matrices.* $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, and $\zeta_3 = e^{2\pi i/3}$. The letter $T$ for this matrix is local to Sections 12.1 and 12.2 and the exercises; elsewhere in the book $T$ is $T_\ell E$ (Appendix B, B-C15). For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ we write $j(\gamma, z) = cz + d$ (not to be confused with the $j$-invariant, which always appears as $j(z)$ or $j(\Lambda)$).

(c) *Indices.* For a subgroup $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ of finite index, $\bar{\Gamma}$ is its image in $\mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\{\pm I\}$. The degree of $X(\Gamma) \to X(1)$ (Theorem 12.17) is $[\mathrm{PSL}_2(\mathbb{Z}) : \bar{\Gamma}]$, which equals $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma]$ when $-I \in \Gamma$ (always the case for $\Gamma_0(N)$) and half of it otherwise.

(d) *The slash operator.* For $\gamma \in \mathrm{GL}_2^+(\mathbb{R})$, $k \in \mathbb{Z}$ and $f\colon \mathcal{H} \to \mathbb{C}$,

$$(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz+d)^{-k} f(\gamma z).$$

Diamond and Shurman [52] use $(\det\gamma)^{k-1}$ instead; the two agree in weight 2, the only weight the paper uses, and differ by a positive scalar otherwise. The choice $(\det\gamma)^{k/2}$ makes every Atkin-Lehner operator an exact involution and an isometry in every even weight (Propositions 12.64 and 12.65).

(e) *Atkin-Lehner.* $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$. This is exactly the paper's matrix $w$ of Section 4.1, with $M^\Sigma$ in place of $N$. The symbol $w_N$ denotes both the involution $z \mapsto -1/(Nz)$ of $X_0(N)$ and the operator $f \mapsto f|_k W_N$ on forms. Some books use $\left(\begin{smallmatrix} 0 & 1 \\ -N & 0 \end{smallmatrix}\right) = -W_N$; since $-I$ acts trivially on $\mathcal{H}$ and on forms of even weight, nothing changes.

(f) *The Petersson product* $\langle f, g\rangle = \int_{\Gamma_0(N)\backslash\mathcal{H}} f\bar{g}\,y^k\,d\mu$ is *unnormalised* and linear in the first variable (Definition 12.48, Convention 12.50, Appendix B, B-C8).

(g) *Coefficients.* $a_n(f)$ is the $n$-th coefficient of the expansion of $f$ at $\infty$. "$f \equiv g \pmod{\mathfrak{a}}$ coefficientwise" means $a_n(f) - a_n(g) \in \mathfrak{a}$ for *all* $n \geq 0$; a congruence "at all primes $p \notin \Sigma$" is a different and weaker statement, and the two are always distinguished, as the paper distinguishes them in Lemma lem:crit and Lemma lem:old.

(h) *Numbers attached to $N$.* $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$; $\nu_2(N)$, $\nu_3(N)$ the numbers of elliptic points of order 2 and 3 of $\Gamma_0(N)$; $\nu_\infty(N)$ its number of cusps; $g(N)$ the genus of $X_0(N)$; $B_k(N) = \lfloor k\mu(N)/12 \rfloor$ the Sturm bound, so $B_2(N) = \lfloor \mu(N)/6 \rfloor$.

(i) *What is not here.* Hecke operators, eigenforms, newforms, oldforms and twists of forms are Chapter 13's; this chapter uses them only in examples and in the environments "In the paper", with a forward reference. When $\dim S_2(\Gamma_0(N)) = 1$ the unique form with $a_1 = 1$ is called *the normalised generator*. The Jacobian $J_0(N)$, the modular parametrisation and the Manin constant are Chapter 14's.

**Proposition 12.2** (the action of $\mathrm{GL}_2^+(\mathbb{R})$ on $\mathcal{H}$)**.** *The group $\mathrm{GL}_2^+(\mathbb{R})$ of real $2 \times 2$ matrices of positive determinant acts on $\mathcal{H}$ by $\gamma z = \frac{az+b}{cz+d}$, and the scalar matrices act trivially. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$ and $z = x + iy \in \mathcal{H}$:*

*(i)* $\mathrm{Im}(\gamma z) = \det\gamma \cdot \mathrm{Im}\, z/|cz+d|^2$*; in particular $\gamma z \in \mathcal{H}$;*

*(ii)* $\frac{d}{dz}(\gamma z) = \det\gamma/(cz+d)^2$*;*

*(iii)* $j(\gamma\gamma', z) = j(\gamma, \gamma' z)\, j(\gamma', z)$ *for $\gamma, \gamma' \in \mathrm{GL}_2^+(\mathbb{R})$;*

*(iv)* $\mathrm{PSL}_2(\mathbb{R}) = \mathrm{SL}_2(\mathbb{R})/\{\pm I\}$ *acts faithfully and transitively on $\mathcal{H}$, the stabiliser of $i$ in $\mathrm{SL}_2(\mathbb{R})$ is $\mathrm{SO}(2)$, and $\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2) \to \mathcal{H}$, $g \mapsto gi$, is a bijection;*

*(v) the measure $d\mu = dx\,dy/y^2$ is $\mathrm{GL}_2^+(\mathbb{R})$-invariant: $\int_{\gamma U} \phi\, d\mu = \int_U \phi(\gamma z)\, d\mu(z)$ for measurable $U \subseteq \mathcal{H}$ and $\phi \geq 0$.*

*Proof.* (i) $\gamma z = \frac{(az+b)(c\bar z+d)}{|cz+d|^2}$, and the imaginary part of the numerator $ac|z|^2 + adz + bc\bar z + bd$ is $(ad-bc)y$. (ii) $\frac{d}{dz}\frac{az+b}{cz+d} = \frac{a(cz+d)-c(az+b)}{(cz+d)^2} = \frac{\det\gamma}{(cz+d)^2}$. (iii) If $\gamma' = \left(\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right)$, the bottom row of $\gamma\gamma'$ is $(ca'+dc',\, cb'+dd')$, and $j(\gamma, \gamma' z) j(\gamma', z) = \left(c\frac{a'z+b'}{c'z+d'} + d\right)(c'z+d') = (ca'+dc')z + (cb'+dd')$. The same computation for the top row shows $(\gamma\gamma')z = \gamma(\gamma' z)$, so we have an action; $\lambda I$ acts as the identity. (iv) $\left(\begin{smallmatrix} \sqrt{y} & x/\sqrt{y} \\ 0 & 1/\sqrt{y} \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$ sends $i$ to $x+iy$, so the action is transitive. If $\gamma i = i$ with $\gamma \in \mathrm{SL}_2(\mathbb{R})$ then $ai + b = -c + di$, so $a = d$, $b = -c$ and $a^2 + c^2 = 1$: $\gamma$ is a rotation, and conversely rotations fix $i$. If $\gamma z = z$ for all $z$, then $cz^2 + (d-a)z - b = 0$ identically, so $c = b = 0$, $a = d$, and $\gamma = \pm I$. The bijection is orbit-stabiliser (Chapter 1). (v) $z \mapsto \gamma z$ is holomorphic, so its real Jacobian is $|d(\gamma z)/dz|^2 = (\det\gamma)^2/|cz+d|^4$ by (ii), while $\mathrm{Im}(\gamma z)^{-2} = |cz+d|^4/((\det\gamma)^2 y^2)$ by (i). The product is $y^{-2}$, which is the change of variables formula for $d\mu$. □

The metric $ds^2 = (dx^2 + dy^2)/y^2$ is the hyperbolic metric, $d\mu$ is its area element, and the Möbius maps of $\mathrm{SL}_2(\mathbb{R})$ are its orientation-preserving isometries. We will not need the metric, only the measure.

**Proposition 12.3** (elliptic, parabolic and hyperbolic elements)**.** *Let $\gamma \in \mathrm{SL}_2(\mathbb{R})$, $\gamma \neq \pm I$. Call $\gamma$* elliptic *if $|\mathrm{tr}\,\gamma| < 2$,* parabolic *if $|\mathrm{tr}\,\gamma| = 2$ and* hyperbolic *if $|\mathrm{tr}\,\gamma| > 2$. Acting on $\mathcal{H} \cup \mathbb{P}^1(\mathbb{R})$:*

*(i) an elliptic $\gamma$ has exactly one fixed point in $\mathcal{H}$ (and its conjugate in the lower half plane); a parabolic $\gamma$ has exactly one fixed point, which lies in $\mathbb{P}^1(\mathbb{R})$; a hyperbolic $\gamma$ has two fixed points, both in $\mathbb{P}^1(\mathbb{R})$;*

*(ii) if $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ is elliptic, then $\mathrm{tr}\,\gamma \in \{0, \pm 1\}$, and $\gamma$ has order 4 (if $\mathrm{tr}\,\gamma = 0$), 3 (if $\mathrm{tr}\,\gamma = -1$) or 6 (if $\mathrm{tr}\,\gamma = 1$); its image in $\mathrm{PSL}_2(\mathbb{Z})$ has order 2 or 3;*

*(iii) the fixed points of the parabolic elements of* $\mathrm{SL}_2(\mathbb{Z})$ *are exactly the points of* $\mathbb{P}^1(\mathbb{Q})$.

*Proof.* (i) If $c \neq 0$, the fixed points in $\mathbb{C}$ are the roots of $cz^2+(d-a)z-b=0$, whose discriminant is $(d-a)^2+4bc=(a+d)^2-4$; so there are two complex conjugate roots, one in $\mathcal{H}$, if $|\operatorname{tr}\gamma|<2$, one real double root if $|\operatorname{tr}\gamma|=2$, and two real roots if $|\operatorname{tr}\gamma|>2$, and $\infty$ is not fixed. If $c=0$ then $\gamma=\left(\begin{smallmatrix} a & b \\ 0 & a^{-1}\end{smallmatrix}\right)$ fixes $\infty$; $|\operatorname{tr}\gamma| = |a+a^{-1}| \geq 2$, with equality iff $a=\pm 1$, in which case $b\neq 0$ and $\infty$ is the only fixed point; otherwise $b/(a^{-1}-a)$ is a second, real, fixed point. (ii) The trace is an integer of absolute value less than 2. The characteristic polynomial is $X^2+1$, $X^2+X+1$ or $X^2-X+1$, which divide $X^4-1$, $X^3-1$ and $X^6-1$ (more precisely $X^3+1$); by Cayley-Hamilton $\gamma^2=-I$, $\gamma^3=I$, $\gamma^3=-I$ respectively, and no smaller power is $\pm I$ since $\gamma\neq\pm I$ and the eigenvalues $\pm i$, $\zeta_3^{\pm 1}$, $-\zeta_3^{\pm 1}$ are not real. (iii) A parabolic $\gamma\in\mathrm{SL}_2(\mathbb{Z})$ fixes $\infty$ (if $c=0$) or the double root $(a-d)/(2c)\in\mathbb{Q}$. Conversely, $a/c\in\mathbb{Q}$ in lowest terms is $\sigma\infty$ for some $\sigma=\left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)\in\mathrm{SL}_2(\mathbb{Z})$ (solve $ad-bc=1$), and $\infty=\sigma\infty$ for $\sigma=I$; then $\sigma T\sigma^{-1}$ has trace 2, is not $I$, and fixes $\sigma\infty$. □

## 12.2 The modular group and its fundamental domain

The modular group $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathcal{H}$ with a very simple fundamental domain. This section proves the classical facts: the domain $\mathcal{F}$, the identifications on its boundary, the generation of $\mathrm{SL}_2(\mathbb{Z})$ by $S$ and $T$, and the area $\pi/3$. They are used three times later: the quotient $X(1)$ is a sphere (the base of the genus formula), the valence formula is a contour integral around $\partial\mathcal{F}$, and the fundamental domains of $\Gamma_0(N)$ over which the Petersson product is integrated are unions of translates of $\mathcal{F}$. Chapter 9 uses this section through the statement that $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$ classifies elliptic curves over $\mathbb{C}$.

**Theorem 12.4** (the fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$)**.** *Let* $\mathcal{F}=\{z\in\mathcal{H} : |\operatorname{Re} z|\leq\frac{1}{2},\ |z|\geq 1\}$ *and let* $G=\langle S,T\rangle\subseteq\mathrm{SL}_2(\mathbb{Z})$.

*(i) Every* $z\in\mathcal{H}$ *is* $G$*-equivalent to a point of* $\mathcal{F}$.

*(ii) If two distinct points* $z,z'$ *of* $\mathcal{F}$ *are* $\mathrm{SL}_2(\mathbb{Z})$*-equivalent, then either* $\operatorname{Re} z=\pm\frac{1}{2}$ *and* $z'=z\mp 1$, *or* $|z|=1$ *and* $z'=-1/z$.

*(iii) The stabiliser of* $z\in\mathcal{F}$ *in* $\mathrm{PSL}_2(\mathbb{Z})$ *is trivial, except for* $z=i$ *(generated by* $S$*, of order* 2*),* $z=\zeta_3$ *(generated by* $ST$*, of order* 3*) and* $z=-\bar{\zeta}_3=e^{\pi i/3}$ *(generated by* $TS$*, of order* 3*).*

*(iv)* $G=\mathrm{SL}_2(\mathbb{Z})$*, and* $S^2=(ST)^3=-I$.

*(v)* $\int_{\mathcal{F}} d\mu=\pi/3$.

*Proof.* (i) For $\gamma\in\mathrm{SL}_2(\mathbb{Z})$, $\operatorname{Im}(\gamma z)=y/|cz+d|^2$ by Proposition 12.2(i). Since $|cz+d|^2=(cx+d)^2+c^2y^2$, there are only finitely many pairs $(c,d)\in\mathbb{Z}^2$ with $|cz+d|\leq 1$; so $\operatorname{Im}(\gamma z)$, $\gamma\in G$, takes a largest value. Choose $\gamma\in G$ realising it and $n\in\mathbb{Z}$ with $|\operatorname{Re}(T^n\gamma z)|\leq\frac{1}{2}$, and put $z'=T^n\gamma z$; then $\operatorname{Im} z'=\operatorname{Im}\gamma z$ is still maximal. If $|z'|<1$, then $\operatorname{Im}(Sz')=\operatorname{Im}(-1/z')=\operatorname{Im} z'/|z'|^2>\operatorname{Im} z'$, contradicting maximality, as $ST^n\gamma\in G$. So $z'\in\mathcal{F}$.

(ii) and (iii). Let $z\in\mathcal{F}$ and $\gamma=\left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)\in\mathrm{SL}_2(\mathbb{Z})$, $\gamma\neq\pm I$, with $\gamma z\in\mathcal{F}$. Exchanging $(z,\gamma)$ with $(\gamma z,\gamma^{-1})$ if necessary we may assume $\operatorname{Im}\gamma z\geq\operatorname{Im} z$, that is, $|cz+d|\leq 1$. As $\operatorname{Im} z\geq\sqrt{3}/2$, $|c|\sqrt{3}/2\leq|cz+d|\leq 1$, so $c\in\{0,\pm 1\}$. If $c=0$ then $a=d=\pm 1$ and $\gamma z=z+b'$ with $b'=\pm b\neq 0$; both $z$ and $z+b'$ have real part in $[-\frac{1}{2},\frac{1}{2}]$, so $b'=\pm 1$ and $\operatorname{Re} z=-b'/2$. This is the first case

of (ii), and $\gamma$ fixes no point of $\mathcal{F}$. If $c=\pm1$, replace $\gamma$ by $-\gamma$ (the same transformation) to get $c=1$, so $|z+d|\le 1$. If $d\neq 0$ then $|x+d|\ge\frac12$ and $|z+d|^2=(x+d)^2+y^2\ge\frac14+\frac34=1$, with equality only for $d=1$, $z=\zeta_3$ or $d=-1$, $z=-\bar\zeta_3$. If $d=0$ then $|z|\le 1$, so $|z|=1$.

- $c=1$, $d=0$: then $b=-1$ and $\gamma z=a-1/z=a-\bar z$ (as $|z|=1$). For $\gamma z\in\mathcal{F}$ we need $|a-x|\le\frac12$ with $|x|\le\frac12$: either $a=0$ and $\gamma z=-\bar z=-1/z$ (the second case of (ii); a fixed point only for $z=i$, and then $\gamma=S$), or $a=1$, $x=\frac12$, $z=-\bar\zeta_3$ and $\gamma z=z$ ($\gamma=\left(\begin{smallmatrix}1&-1\\1&0\end{smallmatrix}\right)=TS$), or $a=-1$, $x=-\frac12$, $z=\zeta_3$ and $\gamma z=z$ ($\gamma=\left(\begin{smallmatrix}-1&-1\\1&0\end{smallmatrix}\right)=(ST)^2$).

- $c=d=1$, $z=\zeta_3$: then $a-b=1$ and $\gamma z=a-\frac{1}{\zeta_3+1}=a-e^{-\pi i/3}=a-\frac12+\frac{\sqrt3}{2}i$, which lies in $\mathcal{F}$ for $a=0$ ($\gamma=\left(\begin{smallmatrix}0&-1\\1&1\end{smallmatrix}\right)=ST$, fixing $\zeta_3$) and $a=1$ ($\gamma=\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$, sending $\zeta_3$ to $-\bar\zeta_3=\zeta_3+1$, an instance of the first case of (ii)).

- $c=1$, $d=-1$, $z=-\bar\zeta_3$: symmetric to the previous case (conjugate by $z\mapsto-\bar z$), giving $TS$ and the element sending $-\bar\zeta_3$ to $\zeta_3$.

Collecting the cases gives (ii) and the list (iii); the orders follow from $S^2=-I$ and $(ST)^3=-I$, checked by multiplying out: $ST=\left(\begin{smallmatrix}0&-1\\1&1\end{smallmatrix}\right)$, $(ST)^2=\left(\begin{smallmatrix}-1&-1\\1&0\end{smallmatrix}\right)$, $(ST)^3=-I$, and $TS=T(ST)T^{-1}$.

(iv) Let $\gamma\in\mathrm{SL}_2(\mathbb{Z})$ and $z_0=2i$, an interior point of $\mathcal{F}$ with trivial stabiliser in $\mathrm{PSL}_2(\mathbb{Z})$ by (iii). By (i) there is $g\in G$ with $g\gamma z_0\in\mathcal{F}$. By (ii), the equivalent points $z_0$ and $g\gamma z_0$ of $\mathcal{F}$ coincide (an interior point is equivalent to no other point of $\mathcal{F}$), so $g\gamma=\pm I$ by (iii), and $\gamma=\pm g^{-1}\in G$ since $-I=S^2\in G$.

(v) $\int_{\mathcal{F}}d\mu=\int_{-1/2}^{1/2}\int_{\sqrt{1-x^2}}^{\infty}\frac{dy}{y^2}\,dx=\int_{-1/2}^{1/2}\frac{dx}{\sqrt{1-x^2}}=2\arcsin\frac12=\frac\pi3$. □

For instance, $z=(17+i)/40$ is moved into $\mathcal{F}$ by $T^{-3}ST^2S$: $Sz=-40/(17+i)=-\frac{68}{29}+\frac{4}{29}i$, $T^2Sz=-\frac{10}{29}+\frac{4}{29}i$, which has absolute value less than 1, $ST^2Sz=\frac52+i$, and $T^{-3}ST^2Sz=-\frac12+i$, a point of the left edge of $\mathcal{F}$.

**Corollary 12.5** (the level one quotient)**.** *The quotient $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$ is obtained from $\mathcal{F}$ by gluing the two vertical sides by $T$ and folding the arc $\{|z|=1\}\cap\mathcal{F}$ at $i$ by $S$. Adding the single orbit $\mathbb{P}^1(\mathbb{Q})=\mathrm{SL}_2(\mathbb{Z})\infty$ gives $X(1)=\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}^*$, which is homeomorphic to a sphere. The only points of $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$ with nontrivial stabiliser are the orbits of $i$ and of $\zeta_3$ (which contains $-\bar\zeta_3$), with stabilisers of orders 2 and 3 in $\mathrm{PSL}_2(\mathbb{Z})$.*

*Proof.* The description of the identifications is Theorem 12.4(i), (ii); the stabilisers are (iii), and $-\bar\zeta_3=T\zeta_3$. That $\mathbb{P}^1(\mathbb{Q})$ is a single orbit is the transitivity of $\mathrm{SL}_2(\mathbb{Z})$ on fractions in lowest terms (Proposition 12.3(iii)). For the topology, compactify $\mathcal{F}$ by the point $\infty$, with the neighbourhoods $\{\mathrm{Im}\,z>C\}\cup\{\infty\}$ (Theorem 12.16); the result is a closed disc, a curvilinear triangle with vertices $\zeta_3$, $-\bar\zeta_3$, $\infty$. The gluings give a cell decomposition of $X(1)$ with three vertices (the class of $\zeta_3$ and $-\bar\zeta_3$, the point $i$, the point $\infty$), two edges (the class of the vertical sides, the class of the two halves of the arc) and one face. Its Euler characteristic is $3-2+1=2$, and a closed orientable surface of Euler characteristic 2 is a sphere. □

So $X(1)$ has genus 0; this is used in the genus formula (Theorem 12.17) and does not depend on the function $j$, which will give an explicit isomorphism $X(1)\simeq\mathbb{P}^1(\mathbb{C})$ in Theorem 12.36.

## 12.3 Congruence subgroups and their indices

The paper works with $\Gamma_0(N)$ throughout: $S_2(\Gamma_0(N))$, $X_0(N)$, $J_0(N)$. This section defines the three standard congruence subgroups, computes their indices, and builds a fundamental domain for $\Gamma_0(N)$ from the translates of $\mathcal{F}$. The index $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$ is the degree of $X_0(N) \to X(1)$, enters the genus formula, the dimension formulas and the Sturm bound, and is the factor by which PARI's Petersson product differs from the book's (Convention 12.50).

**Definition 12.6** (congruence subgroups)**.** For $N \geq 1$ let

$$\Gamma(N) = \ker\big(\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N)\big), \qquad \Gamma_1(N) = \Big\{\gamma \equiv \left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right) \bmod N\Big\},$$

$$\Gamma_0(N) = \Big\{\gamma \equiv \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right) \bmod N\Big\},$$

so that $\Gamma(N) \subseteq \Gamma_1(N) \subseteq \Gamma_0(N) \subseteq \mathrm{SL}_2(\mathbb{Z})$. A *congruence subgroup* is a subgroup of $\mathrm{SL}_2(\mathbb{Z})$ containing $\Gamma(N)$ for some $N$, its *level* being the least such $N$.

Why these three groups? Each is the stabiliser of a piece of level-$N$ structure on the lattice $\Lambda_\tau = \mathbb{Z} + \mathbb{Z}\tau$, as Theorem 12.18 makes precise: $\Gamma(N)$ fixes the basis $(\frac{1}{N}, \frac{\tau}{N})$ of $\frac{1}{N}\Lambda_\tau/\Lambda_\tau$ (a full level structure), $\Gamma_1(N)$ fixes the point $\frac{1}{N}$, and $\Gamma_0(N)$ only the cyclic subgroup $\langle\frac{1}{N}\rangle$ it generates. The weakest structure gives the largest group and the smallest curve, and it is the one the paper needs. A small example: for $N = 2$ the units modulo 2 are trivial, so $\Gamma_0(2) = \Gamma_1(2)$. At $N = 3$ the matrix $\left(\begin{smallmatrix} 1 & 0 \\ 3 & 1 \end{smallmatrix}\right)$ lies in $\Gamma_1(3)$, while $\left(\begin{smallmatrix} -1 & 0 \\ 3 & -1 \end{smallmatrix}\right)$ lies in $\Gamma_0(3)$ but not in $\Gamma_1(3)$ ($d = -1 \not\equiv 1 \pmod 3$); neither lies in $\Gamma(3)$, whose elements are $\equiv I$ entrywise.

**Proposition 12.7** (indices)**.** *Let $N \geq 1$.*

*(i) Reduction $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N)$ is surjective. Hence $\Gamma(N)$ is normal of index $\#\,\mathrm{SL}_2(\mathbb{Z}/N) = N^3\prod_{p\mid N}(1-p^{-2})$.*

*(ii) $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)] = N^2\prod_{p\mid N}(1-p^{-2})$, and $\Gamma_1(N)$ is normal in $\Gamma_0(N)$ with $\Gamma_0(N)/\Gamma_1(N) \simeq (\mathbb{Z}/N)^\times$ via $\gamma \mapsto d \bmod N$.*

*(iii) The map $\Gamma_0(N)\gamma \mapsto (c : d)$, $(c, d)$ the bottom row of $\gamma$, is a bijection $\Gamma_0(N)\backslash\,\mathrm{SL}_2(\mathbb{Z}) \to \mathbb{P}^1(\mathbb{Z}/N)$, the set of pairs $(c, d) \in (\mathbb{Z}/N)^2$ with $\gcd(c, d, N) = 1$ modulo multiplication by $(\mathbb{Z}/N)^\times$. Hence*

$$\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = \#\mathbb{P}^1(\mathbb{Z}/N) = N\prod_{p\mid N}\Big(1 + \frac{1}{p}\Big).$$

*(iv) $-I \in \Gamma_0(N)$ for every $N$, and $-I \in \Gamma_1(N)$ if and only if $N \leq 2$.*

*Proof.* (i) Let $\bar\gamma = \left(\begin{smallmatrix} \bar a & \bar b \\ \bar c & \bar d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z}/N)$. Lift $\bar c, \bar d$ to integers $c, d$ with $c \neq 0$ (take $c = N$ if $\bar c = 0$). Since $\gcd(c, d, N) = 1$, we can replace $d$ by $d + tN$ with $\gcd(c, d + tN) = 1$: let $t$ be the product of the primes dividing $c$ but not $d$; a prime $p \mid c$ that divides $d$ does not divide $N$ or $t$, so $p \nmid d + tN$, and a prime $p \mid c$ not dividing $d$ divides $t$, so again $p \nmid d + tN$. Now choose $a', b' \in \mathbb{Z}$ with $a'd - b'c = 1$. The solutions $(x, y) \in (\mathbb{Z}/N)^2$ of $xd - yc = 1$ are $(a', b') + k(c, d)$, $k \in \mathbb{Z}/N$. Indeed the matrix $g = \left(\begin{smallmatrix} a' & b' \\ c & d \end{smallmatrix}\right)$ has determinant 1, so its rows $(a', b')$ and $(c, d)$ form a basis of $(\mathbb{Z}/N)^2$, and every $(x, y)$ is $s(a', b') + k(c, d)$ with $s, k \in \mathbb{Z}/N$; then $xd - yc = s(a'd - b'c) + k(cd - dc) = s$,

so $xd - yc = 1$ exactly when $s = 1$. So $(\bar{a}, \bar{b}) = (a' + kc, b' + kd) \bmod N$ for some $k \in \mathbb{Z}$, and $\left(\begin{smallmatrix} a'+kc & b'+kd \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ lifts $\bar{\gamma}$. For the order: by the Chinese remainder theorem $\mathrm{SL}_2(\mathbb{Z}/N) = \prod \mathrm{SL}_2(\mathbb{Z}/p^e)$; reduction $\mathrm{GL}_2(\mathbb{Z}/p^e) \to \mathrm{GL}_2(\mathbb{F}_p)$ is onto with kernel $I + pM_2(\mathbb{Z}/p^e)$ of order $p^{4(e-1)}$, $\#\,\mathrm{GL}_2(\mathbb{F}_p) = (p^2-1)(p^2-p)$, and $\det\colon \mathrm{GL}_2(\mathbb{Z}/p^e) \to (\mathbb{Z}/p^e)^\times$ is onto, so $\#\,\mathrm{SL}_2(\mathbb{Z}/p^e) = p^{4(e-1)}(p^2-1)(p^2-p)/(p^{e-1}(p-1)) = p^{3e}(1-p^{-2})$. (ii) $\Gamma_1(N)$ is the preimage of the subgroup $\{\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)\}$ of order $N$, so its index is $\#\,\mathrm{SL}_2(\mathbb{Z}/N)/N$. The map $\Gamma_0(N) \to (\mathbb{Z}/N)^\times$, $\gamma \mapsto d$, is a homomorphism (the lower right entry of a product of upper triangular matrices is the product of the lower right entries), onto by (i) applied to $\left(\begin{smallmatrix} d^{-1} & 0 \\ 0 & d \end{smallmatrix}\right)$, with kernel $\Gamma_1(N)$ (if $c \equiv 0$ and $d \equiv 1$ then $a \equiv 1$). (iii) $\Gamma_0(N)$ is the preimage of the upper triangular group $B \subseteq \mathrm{SL}_2(\mathbb{Z}/N)$, so $\Gamma_0(N)\backslash \mathrm{SL}_2(\mathbb{Z}) = B \backslash \mathrm{SL}_2(\mathbb{Z}/N)$ by (i). Left multiplication by $\left(\begin{smallmatrix} u & b \\ 0 & u^{-1} \end{smallmatrix}\right)$ multiplies the bottom row by $u^{-1}$; conversely if $g, g' \in \mathrm{SL}_2(\mathbb{Z}/N)$ have bottom rows $(c, d)$ and $u(c, d)$ then $g'g^{-1}$ has bottom row $(0, u)$ and lies in $B$. Every row with $\gcd(c, d, N) = 1$ is the bottom row of an element of $\mathrm{SL}_2(\mathbb{Z}/N)$ (complete it as in (i)). So $\Gamma_0(N)\backslash \mathrm{SL}_2(\mathbb{Z}) \simeq \mathbb{P}^1(\mathbb{Z}/N)$, and $\#\mathbb{P}^1(\mathbb{Z}/N) = \prod_{p^e \| N} \#\mathbb{P}^1(\mathbb{Z}/p^e)$ by the Chinese remainder theorem. To count $\mathbb{P}^1(\mathbb{Z}/p^e)$: in a pair $(c, d)$ with $\gcd(c, d, p) = 1$ at least one entry is a unit. If $c$ is a unit, scaling by $c^{-1}$ gives a unique representative $(1 : d)$, $d \in \mathbb{Z}/p^e$: $p^e$ points. If $c$ is not a unit, then $d$ is, and scaling by $d^{-1}$ gives a unique representative $(c : 1)$ with $p \mid c$: $p^{e-1}$ points. So $\#\mathbb{P}^1(\mathbb{Z}/p^e) = p^e + p^{e-1} = p^e(1 + \frac{1}{p})$, and the product over $p^e \| N$ is the stated formula. For example $\mu(20) = \#\mathbb{P}^1(\mathbb{Z}/4) \cdot \#\mathbb{P}^1(\mathbb{Z}/5) = 6 \cdot 6 = 36$. (iv) $-I$ is upper triangular, and $-I \equiv I \bmod N$ iff $N \mid 2$. □

The values of $\mu(N)$ at the levels of the paper's examples are:

| $N$ | 11 | 20 | 37 | 52 | 77 | 99 | 121 | 140 | 539 | 726 | 980 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\mu(N)$ | 12 | 36 | 38 | 84 | 96 | 144 | 132 | 288 | 672 | 1584 | 2016 |

(computed from (iii); for $N \in \{11, 20, 37, 77, 99, 121, 140\}$ also by a direct count of $\mathbb{P}^1(\mathbb{Z}/N)$).

**Proposition 12.8** (fundamental domains from cosets)**.** *Let $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ be a subgroup of finite index containing $-I$, and $\mathrm{SL}_2(\mathbb{Z}) = \bigsqcup_{i=1}^{m} \Gamma\gamma_i$. Then $\mathcal{F}_\Gamma = \bigcup_i \gamma_i \mathcal{F}$ is a fundamental domain for $\Gamma$: every point of $\mathcal{H}$ is $\Gamma$-equivalent to a point of $\mathcal{F}_\Gamma$, and no two distinct interior points of $\mathcal{F}_\Gamma$ that lie in the interiors of the translates $\gamma_i\mathcal{F}$ are $\Gamma$-equivalent. Its area is $\int_{\mathcal{F}_\Gamma} d\mu = m\pi/3$; for $\Gamma_0(N)$, $\mathrm{vol}(\Gamma_0(N)\backslash\mathcal{H}) = \pi\mu(N)/3$. For $N = p$ prime one may take $\gamma_0 = I$ and $\gamma_{j+1} = ST^j$, $0 \le j \le p-1$.*

*Proof.* For $z \in \mathcal{H}$ write $z = \gamma w$ with $w \in \mathcal{F}$ and $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ (Theorem 12.4(i)), and $\gamma = \delta\gamma_i$ with $\delta \in \Gamma$; then $\delta^{-1}z = \gamma_i w \in \gamma_i\mathcal{F}$. If $\gamma_i w = \delta\gamma_j w'$ with $w, w'$ interior points of $\mathcal{F}$ and $\delta \in \Gamma$, then $w = \gamma_i^{-1}\delta\gamma_j w'$, so $w = w'$ and $\gamma_i^{-1}\delta\gamma_j = \pm I$ by Theorem 12.4(ii), (iii); as $-I \in \Gamma$, $\Gamma\gamma_i = \Gamma\gamma_j$, so $i = j$ and the two points coincide. The translates meet only in boundaries, which have measure zero, so the area is $m$ times that of $\mathcal{F}$ by the invariance of $d\mu$. For $N = p$ the bottom rows of $I$ and $ST^j = \left(\begin{smallmatrix} 0 & -1 \\ 1 & j \end{smallmatrix}\right)$ are $(0 : 1)$ and $(1 : j)$, the $p + 1$ points of $\mathbb{P}^1(\mathbb{F}_p)$. □

For $\Gamma_0(11)$ the domain consists of $\mathcal{F}$ and the eleven translates $ST^j\mathcal{F}$, $0 \le j \le 10$. The translate $ST^j\mathcal{F}$ is the image of the strip piece $T^j\mathcal{F}$ under $z \mapsto -1/z$, a curvilinear triangle bounded by three geodesics (two of them semicircles through 0) with one ideal vertex at the cusp 0. So eleven of the twelve triangles cluster at 0, and the domain has two ideal vertices, $\infty$ and 0, which are the two cusps of $\Gamma_0(11)$ (Proposition 12.12).

## 12.4 Cusps, widths and elliptic points

The compact curve $X_0(N)$ is $Y_0(N)$ together with finitely many cusps, and the map $X_0(N) \to X(1)$ is ramified only at the cusps and at the elliptic points. This section counts both. The counts feed the genus formula (Theorem 12.17), the dimension formulas (Theorem 12.41) and the action of the Atkin-Lehner involutions on cusps (Proposition 12.67). In the paper the cusp $\infty$ is the base point of the modular parametrisation $\varphi_E\colon X_0(N) \to E$ (Chapter 14), and the expansion of $f$ at $\infty$ is the $q$-expansion whose coefficients are the $a_n(f)$ of every congruence.

**Definition 12.9** (cusps and widths)**.** Let $\Gamma$ be a congruence subgroup. The *cusps* of $\Gamma$ are the $\Gamma$-orbits on $\mathbb{P}^1(\mathbb{Q})$. For a cusp $s = \sigma\infty$ ($\sigma \in \mathrm{SL}_2(\mathbb{Z})$) the *width* of $s$ is the least $h \geq 1$ such that $\sigma T^h \sigma^{-1} \in \pm\Gamma$. If $-I \notin \Gamma$, the cusp is *irregular* when $-\sigma T^h \sigma^{-1} \in \Gamma$ but $\sigma T^h \sigma^{-1} \notin \Gamma$, and *regular* otherwise.

The width exists because $\sigma T^N \sigma^{-1} \in \Gamma(N) \subseteq \Gamma$ when $\Gamma \supseteq \Gamma(N)$ ($\Gamma(N)$ is normal). It does not depend on the choices: $\sigma'\infty = \sigma\infty$ iff $\sigma' \in \sigma \cdot \{\pm T^m\}$, which commutes with $T^h$, and replacing $s$ by $\gamma s$ ($\gamma \in \Gamma$) conjugates $\sigma T^h \sigma^{-1}$ by $\gamma$.

**Proposition 12.10** (cusps as orbits)**.** *Let $\Gamma$ be a congruence subgroup containing $-I$. The map $\Gamma\sigma \mapsto \Gamma\sigma\infty$ induces a bijection between the orbits of $\langle T\rangle$ acting on the right on $\Gamma\backslash \mathrm{SL}_2(\mathbb{Z})$ and the cusps of $\Gamma$, and the width of the cusp $\Gamma\sigma\infty$ is the length of the orbit of $\Gamma\sigma$. In particular $\Gamma$ has finitely many cusps and $\sum_s h_s = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma]$. For $\Gamma_0(N)$, via Proposition 12.7(iii), the cusps are the orbits of $(c : d) \mapsto (c : c + d)$ on $\mathbb{P}^1(\mathbb{Z}/N)$.*

*Proof.* The map $\Gamma\backslash \mathrm{SL}_2(\mathbb{Z}) \to \Gamma\backslash\mathbb{P}^1(\mathbb{Q})$ is onto because $\mathrm{SL}_2(\mathbb{Z})$ is transitive on $\mathbb{P}^1(\mathbb{Q})$ (Proposition 12.3(iii)). Its fibres: $\Gamma\sigma\infty = \Gamma\sigma'\infty$ iff $\sigma'\infty = \gamma\sigma\infty$ for some $\gamma \in \Gamma$ iff $\sigma' \in \Gamma\sigma\,\mathrm{Stab}(\infty) = \Gamma\sigma\{\pm T^m\}$, that is, iff $\Gamma\sigma'$ lies in the $\langle T\rangle$-orbit of $\Gamma\sigma$ ($-I \in \Gamma$). The orbit of $\Gamma\sigma$ has length the least $h \geq 1$ with $\Gamma\sigma T^h = \Gamma\sigma$, that is, $\sigma T^h \sigma^{-1} \in \Gamma$. The right action of $T$ on bottom rows is $(c, d) \mapsto (c, c + d)$. □

The simplest case shows how the bookkeeping works. For $\Gamma_0(p)$, $p$ prime, $\mathbb{P}^1(\mathbb{F}_p)$ has the $p + 1$ points $(0 : 1)$ and $(1 : d)$, $d \in \mathbb{F}_p$. The map $(c : d) \mapsto (c : c + d)$ fixes $(0 : 1)$, which is the coset of $I$, so the cusp $\infty = I\infty$ has width 1; and it sends $(1 : d)$ to $(1 : d + 1)$, so the $p$ points $(1 : d)$ form a single orbit of length $p$, containing $(1 : 0)$, the coset of $S$, whose cusp is $S\infty = 0$. Hence $\Gamma_0(p)$ has the two cusps $\infty$ and 0, of widths 1 and $p$, and $1 + p = \mu(p)$. Geometrically: of the $p + 1$ translates of $\mathcal{F}$ making up the fundamental domain of Proposition 12.8, one touches $\infty$ and $p$ touch 0.

*Warning* 12.11 (irregular cusps). If $-I \notin \Gamma$, Proposition 12.10 holds with $\mathrm{PSL}_2$ in place of $\mathrm{SL}_2$, and some cusps can be irregular: for $\Gamma_1(4)$ and $\sigma = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$ ($s = \frac{1}{2}$), $\sigma T \sigma^{-1} = \left(\begin{smallmatrix} -1 & 1 \\ -4 & 3 \end{smallmatrix}\right)$ is congruent to $\left(\begin{smallmatrix} -1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$ modulo 4, so $-\sigma T \sigma^{-1} \in \Gamma_1(4)$ but $\sigma T \sigma^{-1} \notin \Gamma_1(4)$. At an irregular cusp a form of odd weight has an expansion in half-integral powers of $q_h$, and "$b_0 = 0$" is not the right vanishing condition. None of this happens for $\Gamma_0(N)$, which contains $-I$; the paper only needs $\Gamma_0(N)$.

**Proposition 12.12** (the cusps of $\Gamma_0(N)$)**.** *Let $N \geq 1$. Write a point of $\mathbb{P}^1(\mathbb{Q})$ as $x/y$ with $\gcd(x, y) = 1$, $y \geq 0$ (and $\infty = 1/0$).*

*(i) The integer $\gcd(y, N)$ depends only on the $\Gamma_0(N)$-class of $x/y$. The cusp $\infty \sim 1/N$ has width 1 and the cusp $0 = 0/1$ has width $N$.*

*(ii) Every cusp is equivalent to some $a/c$ with $c \mid N$, $c \geq 1$, $\gcd(a,c)=1$, and $a/c$, $a'/c'$ of this form are equivalent iff $c=c'$ and $a \equiv a' \pmod{\gcd(c, N/c)}$. Hence*

$$\nu_\infty(N) = \sum_{c \mid N} \varphi(\gcd(c, N/c)),$$

*and the cusp $a/c$ has width $N/\gcd(c^2, N)$.*

*(iii) If $N$ is squarefree, the cusps are the $2^{\omega(N)}$ classes of $1/c$, $c \mid N$, and $1/c$ has width $N/c$.*

*(iv) If $N = p^2$ with $p$ prime, the cusps are $\infty$, $0$ and $u/p$ for $1 \leq u \leq p-1$, of widths $1$, $p^2$ and $1$; so $\nu_\infty(p^2) = p+1$.*

*Proof.* (i) For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$, $\gamma(x/y) = (ax+by)/(cx+dy)$, and $(ax+by, cx+dy)$ is primitive because $\gamma \in \mathrm{SL}_2(\mathbb{Z})$; since $N \mid c$ and $\gcd(d,N)=1$, $\gcd(cx+dy, N) = \gcd(dy, N) = \gcd(y, N)$. For the widths: $T \in \Gamma_0(N)$, and $ST^hS^{-1} = \left(\begin{smallmatrix} 1 & 0 \\ -h & 1 \end{smallmatrix}\right) \in \Gamma_0(N)$ iff $N \mid h$. Finally $1/N = \left(\begin{smallmatrix} 1 & 0 \\ N & 1 \end{smallmatrix}\right)\infty$ with $\left(\begin{smallmatrix} 1 & 0 \\ N & 1 \end{smallmatrix}\right) \in \Gamma_0(N)$.

(iii) By the Chinese remainder theorem $\mathbb{P}^1(\mathbb{Z}/N) = \prod_{p \mid N} \mathbb{P}^1(\mathbb{F}_p)$, and $T$ acts on each factor by $(c:d) \mapsto (c : c+d)$, which fixes $(0:1)$ and permutes the $p$ points $(1:d)$ cyclically. For a point $P$ let $A(P)$ be the set of primes $p$ at which its component is not $(0:1)$. Then $T^jP = P$ iff $p \mid j$ for all $p \in A(P)$, and by the Chinese remainder theorem $j \bmod \prod_{p \in A} p$ moves the components at $A$ simultaneously through all of $\prod_{p \in A} \mathbb{F}_p$. So the $\langle T \rangle$-orbits are the sets $\{P : A(P) = A\}$, one for each subset $A$, of length $\prod_{p \in A} p$. For $c = \prod_{p \notin A} p$ the coset of $\sigma = \left(\begin{smallmatrix} 1 & 0 \\ c & 1 \end{smallmatrix}\right)$ has bottom row $(c:1)$, which has $A(P) = A$, and $\sigma\infty = 1/c$; its width is $\prod_{p \in A} p = N/c$ by Proposition 12.10.

(iv) $\mathbb{P}^1(\mathbb{Z}/p^2)$ consists of the $p^2$ points $(1:d)$ and the $p$ points $(pu:1)$, $u \in \mathbb{Z}/p$. The points $(1:d)$ form one $T$-orbit, which contains $(1:0)$, the coset of $S$; its cusp is $S\infty = 0$, of width $p^2$. Each $(pu:1)$ is fixed by $T$, since $(pu:1+pu) = ((1+pu)^{-1}pu : 1) = (pu:1)$. The point $(0:1)$ is the coset of $I$ (cusp $\infty$). For $1 \leq u \leq p-1$ choose $s,t$ with $us - tp = 1$ and $\sigma = \left(\begin{smallmatrix} u & t \\ p & s \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$; its bottom row $(p:s)$ equals $(ps^{-1}:1) = (pu:1)$ because $s^{-1} \equiv u \pmod{p}$; so the cusp of $(pu:1)$ is $\sigma\infty = u/p$, of width 1.

(ii) We do not prove the general case; see [52, Section 3.8] (where the statement is a proposition; number not verified) and [146]. Sketch: by (i) the class of $x/y$ determines $c = \gcd(y, N)$; one moves $x/y$ by an element of $\Gamma_0(N)$ to a fraction $a/c$ with the same $c$, and then shows that the residue of $a$ modulo $\gcd(c, N/c)$ is the remaining invariant, by the same orbit computation on $\mathbb{P}^1(\mathbb{Z}/N)$ as in (iii) and (iv), prime by prime. The counts and widths in (ii) agree with (iii), (iv), and with a direct computation of the $T$-orbits on $\mathbb{P}^1(\mathbb{Z}/N)$ for $N \in \{8, 12, 18, 20, 36, 72, 99, 121\}$. □

**Example 12.13** (cusps at the paper's levels)**.** The following lists come from (ii), and each was checked against the $T$-orbits on $\mathbb{P}^1(\mathbb{Z}/N)$:

- $N = 11$: $\infty$, $0$, of widths $1$, $11$ (sum $12 = \mu(11)$).
- $N = 20$: $\infty$, $\frac{1}{10}$, $\frac{1}{5}$, $\frac{1}{4}$, $\frac{1}{2}$, $0$, of widths $1, 1, 4, 5, 5, 20$ (sum 36).
- $N = 99$: the cusps $a/c$ for $c \in \{1, 3, 9, 11, 33, 99\}$, with $a$ taken modulo $\gcd(c, 99/c)$, which is $1, 3, 1, 1, 3, 1$ respectively. There are eight cusps, of widths $99$ ($c=1$), $11, 11$ ($c=3$: the cusps $\frac{1}{3}, \frac{2}{3}$), $11$ ($c=9$), $9$ ($c=11$), $1, 1$ ($c=33$), $1$ ($c=99$); sum 144.
- $N = 121$: twelve cusps, $\infty$, $0$ and $u/11$ for $1 \leq u \leq 10$, of widths $1$, $121$ and ten of width $1$ (sum 132).

- $\nu_\infty(539) = 16$, $\nu_\infty(726) = \nu_\infty(980) = 48$.

**Proposition 12.14** (elliptic points of $\Gamma_0(N)$)**.** *A point $z \in \mathcal{H}$ is an* elliptic point *of $\Gamma_0(N)$ if its stabiliser in $\bar\Gamma_0(N)$ is nontrivial; its order is the order of that stabiliser. The elliptic points of order 2 of $Y_0(N) = \Gamma_0(N)\backslash\mathcal{H}$ lie over the orbit of $i$ and are in bijection with the solutions $u \bmod N$ of $u^2+1 \equiv 0$; those of order 3 lie over the orbit of $\zeta_3$ and are in bijection with the solutions of $u^2-u+1 \equiv 0 \pmod N$. Hence*

$$\nu_2(N) = \begin{cases} 0, & 4 \mid N, \\ \prod_{p\mid N}\left(1+\left(\frac{-4}{p}\right)\right), & \text{otherwise}, \end{cases} \qquad \nu_3(N) = \begin{cases} 0, & 9 \mid N, \\ \prod_{p\mid N}\left(1+\left(\frac{-3}{p}\right)\right), & \text{otherwise}, \end{cases}$$

*with Kronecker symbols (Chapter 2), so $\left(\frac{-4}{2}\right) = \left(\frac{-3}{3}\right) = 0$. Moreover the fibre of $Y_0(N) \to Y(1)$ over the orbit of $i$ consists of the $\nu_2(N)$ elliptic points and $(\mu(N)-\nu_2(N))/2$ points with trivial stabiliser, and over the orbit of $\zeta_3$ of the $\nu_3(N)$ elliptic points and $(\mu(N)-\nu_3(N))/3$ points with trivial stabiliser.*

*Proof.* By Corollary 12.5 every elliptic point is $\Gamma_0(N)$-equivalent to $\gamma i$ or $\gamma\zeta_3$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, with stabiliser $\gamma\langle S\rangle\gamma^{-1}$, respectively $\gamma\langle ST\rangle\gamma^{-1}$, in $\mathrm{PSL}_2(\mathbb{Z})$. The points of $Y_0(N)$ over the orbit of $i$ are the classes $\Gamma_0(N)\gamma i$, and $\Gamma_0(N)\gamma i = \Gamma_0(N)\gamma' i$ iff $\gamma' \in \Gamma_0(N)\gamma\langle S\rangle$. So they correspond to the orbits of $\langle S\rangle$ (of order 2 in $\mathrm{PSL}_2$) acting on the right on $\Gamma_0(N)\backslash\mathrm{SL}_2(\mathbb{Z}) = \mathbb{P}^1(\mathbb{Z}/N)$. The point $\gamma i$ is elliptic for $\Gamma_0(N)$ iff $\gamma S\gamma^{-1} \in \Gamma_0(N)$ iff $\Gamma_0(N)\gamma S = \Gamma_0(N)\gamma$: the elliptic points are the fixed points of $S$, and the other points form orbits of length 2; this gives the count of the fibre. Now $(c,d)S = (d,-c)$, and $(d:-c) = (c:d)$ means $d = uc$, $-c = ud$ for a unit $u$; then $\gcd(c,N) = \gcd(c,d,N) = 1$, so $c$ is a unit and we may take $c = 1$: the fixed points are the $(1:u)$ with $u^2+1 \equiv 0$, distinct for distinct $u$. (Explicitly: $d = uc$ and $-c = ud = u^2c$ give $c(u^2+1) = 0$ with $c$ a unit, so $u^2+1 = 0$; conversely $(1,u)S = (u,-1) = u\cdot(1,u)$ when $u^2 = -1$.) Likewise $\langle ST\rangle$ has order 3 in $\mathrm{PSL}_2(\mathbb{Z})$, $ST = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ and $(c,d)ST = (d, d-c)$. The equation $(d : d-c) = (c:d)$ means $d = uc$ and $d-c = ud$ for a unit $u$, so $uc - c = u^2c$, that is, $c(u^2-u+1) = 0$; as before $c$ is a unit, we may take $c = 1$, and the fixed points are the $(1:u)$ with $u^2-u+1 \equiv 0$ (conversely $(1,u)ST = (u,u-1) = (u,u^2) = u\cdot(1,u)$ for such $u$). The points that are not fixed fall into orbits of length 3, which gives the count of the fibre over $\zeta_3$. For instance, modulo 13 the roots of $u^2+1$ are $5, 8$ and those of $u^2-u+1$ are $4, 10$, so the elliptic points of $\Gamma_0(13)$ are the classes of $(1:5)$, $(1:8)$ (order 2) and $(1:4)$, $(1:10)$ (order 3), and $\nu_2(13) = \nu_3(13) = 2$.

The number of roots is multiplicative in $N$ (Chinese remainder theorem). For $p$ odd, $u^2+1$ has $1+\left(\frac{-1}{p}\right) = 1+\left(\frac{-4}{p}\right)$ roots modulo $p$, simple, which lift uniquely modulo $p^e$ by Hensel's lemma (Chapter 2); modulo 2 there is one root, and modulo 4 none. For $p \neq 3$, $u^2-u+1$ has discriminant $-3$ and $1+\left(\frac{-3}{p}\right)$ roots modulo $p$, simple if $p \neq 3$, lifting uniquely (for $p = 2$: none, $\left(\frac{-3}{2}\right) = -1$); modulo 3 it is $(u+1)^2$, with one root, and modulo 9 it has none ($u \equiv 2, 5, 8$ give $3, 21, 57 \equiv 3 \pmod 9$). □

For example $\nu_2(11) = \nu_3(11) = 0$; $\nu_2(13) = \nu_3(13) = 2$; $\nu_2(37) = \nu_3(37) = 2$; $\nu_2(5) = \nu_2(10) = 2$, $\nu_3(5) = \nu_3(10) = 0$; and $\nu_2 = \nu_3 = 0$ at every level $N \in \{20, 52, 77, 99, 121, 140, 539, 726, 980\}$ of the paper's examples (for $\nu_2$: $4 \mid N$ or a prime $p \equiv 3 \pmod 4$ divides $N$; for $\nu_3$: $9 \mid N$ or a prime $p \equiv 2 \pmod 3$ divides $N$).

## 12.5 The modular curves as compact Riemann surfaces

This section builds the curve of the paper's title page: $X_0(N)$, the compact Riemann surface on which the newform $f$ is a differential and from which the modular parametrisation $\varphi_E\colon X_0(N)\to E$ starts. We construct $Y_0(N)$ and $X_0(N)$ with their charts, prove the genus formula from the Riemann-Hurwitz formula of Chapter 8, describe $Y_0(N)$ as the moduli space of elliptic curves with a cyclic subgroup of order $N$, and state the existence of the model over $\mathbb{Q}$. In Section 4.1 of the paper the cohomology $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$ is the cohomology of this compact Riemann surface.

**Theorem 12.15** ($Y(\Gamma)$ is a Riemann surface)**.** *Let $\Gamma$ be a congruence subgroup.*

*(i) $\Gamma$ acts properly discontinuously on $\mathcal{H}$: for every compact $K\subset\mathcal{H}$ the set $\{\gamma\in\Gamma:\gamma K\cap K\neq\emptyset\}$ is finite. Every $z\in\mathcal{H}$ has a neighbourhood $U$ such that $\gamma U\cap U\neq\emptyset$ only for $\gamma$ in the stabiliser $\Gamma_z$, which is finite and cyclic modulo $\pm I$. The quotient $Y(\Gamma)=\Gamma\backslash\mathcal{H}$ is a connected Hausdorff space.*

*(ii) $Y(\Gamma)$ is a Riemann surface: near the image of a point $z_0$ with $\#\bar\Gamma_{z_0}=e$, the function $t=\left(\frac{z-z_0}{z-\bar z_0}\right)^e$ is a chart, and the transition maps are holomorphic. A $\Gamma$-invariant holomorphic function on an open $\Gamma$-stable subset of $\mathcal{H}$ is a holomorphic function on its image in $Y(\Gamma)$.*

*Proof.* (i) Choose $0<y_1<y_2$ and $R$ with $K\subseteq\{y_1\le\operatorname{Im}z\le y_2,\ |z|\le R\}$. If $z\in K$ and $\gamma z\in K$ then $|cz+d|^2=\operatorname{Im}z/\operatorname{Im}\gamma z\le y_2/y_1$, so $|c|\le\sqrt{y_2/y_1}/y_1$ and $|d|\le\sqrt{y_2/y_1}+|c|R$. Applying this to $\gamma^{-1}=\left(\begin{smallmatrix}d&-b\\-c&a\end{smallmatrix}\right)$ and the point $\gamma z\in K$ bounds $|a|$. If $c\neq0$ then $b=(ad-1)/c$; if $c=0$ then $\gamma=\pm\left(\begin{smallmatrix}1&b\\0&1\end{smallmatrix}\right)$ with $|b|\le 2R$. So finitely many $\gamma$. The stabiliser $\Gamma_z$ is a discrete subgroup of the compact group $g\,\mathrm{SO}(2)g^{-1}$ (Proposition 12.2(iv)), hence finite, and modulo $\pm I$ it is a finite subgroup of the circle group, hence cyclic. Given $z$, take a closed disc $K$ around $z$; of the finitely many $\gamma$ with $\gamma K\cap K\neq\emptyset$, those not fixing $z$ move $z$ off itself, and a small enough open disc $U\ni z$ satisfies $\gamma U\cap U=\emptyset$ for them. Likewise, if $z_1,z_2$ are not $\Gamma$-equivalent, take a closed disc $K$ containing both; for the finitely many $\gamma$ with $\gamma K\cap K\neq\emptyset$ we have $\gamma z_1\neq z_2$, so small discs $U_1\ni z_1$, $U_2\ni z_2$ satisfy $\gamma U_1\cap U_2=\emptyset$ for all $\gamma\in\Gamma$; their images in $Y(\Gamma)$ are disjoint open sets. $Y(\Gamma)$ is connected as the image of $\mathcal{H}$. (ii) We sketch this; see [52, Sections 2.1, 2.2] and [180, Section 1.5]. The Cayley map $w=\frac{z-z_0}{z-\bar z_0}$ sends $z_0$ to 0 and conjugates $\bar\Gamma_{z_0}$ to the rotations $w\mapsto\zeta w$, $\zeta^e=1$, so $t=w^e$ identifies a small disc around $z_0$ modulo $\bar\Gamma_{z_0}$ with a disc around 0. Where two such charts overlap, at least one of the points involved has trivial stabiliser and the transition is a composite of Möbius maps and an $e$-th power or root, holomorphic away from $t=0$. A $\Gamma_{z_0}$-invariant holomorphic function of $w$ is a holomorphic function of $w^e$ (expand in powers of $w$: only exponents divisible by $e$ survive). □

**Theorem 12.16** ($X(\Gamma)$ is a compact Riemann surface)**.** *Give $\mathcal{H}^*=\mathcal{H}\cup\mathbb{P}^1(\mathbb{Q})$ the topology in which the neighbourhoods of $\infty$ are the sets $\{\operatorname{Im}z>C\}\cup\{\infty\}$ and those of $s=\sigma\infty$ are their images under $\sigma\in\mathrm{SL}_2(\mathbb{Z})$ (discs tangent to $\mathbb{R}$ at $s$, together with $s$). For a congruence subgroup $\Gamma$, $X(\Gamma)=\Gamma\backslash\mathcal{H}^*$ is a compact connected Hausdorff space, and with the chart $q_h=e^{2\pi i\sigma^{-1}(z)/h}$ near a cusp $s=\sigma\infty$ of width $h$ (Definition 12.9) it is a compact Riemann surface. We write $X_0(N)$, $X_1(N)$, $X(N)$ for $\Gamma_0(N)$, $\Gamma_1(N)$, $\Gamma(N)$, and $X(1)$ for $\mathrm{SL}_2(\mathbb{Z})$. By the Riemann existence theorem and GAGA (Chapter 8), $X_0(N)$ is the Riemann surface of a smooth projective curve over $\mathbb{C}$, and holomorphic maps between these curves are morphisms.*

*Sketch of proof.* Compactness: $\mathcal{H}^*=\mathrm{SL}_2(\mathbb{Z})\cdot(\mathcal{F}\cup\{\infty\})$ by Theorem 12.4(i), so $X(\Gamma)$ is the image of the finitely many compact sets $\gamma_i(\mathcal{F}\cup\{\infty\})$ of Proposition 12.8. Hausdorff: the horoball

neighbourhoods of inequivalent cusps, and of a cusp and a point of $\mathcal{H}$, can be chosen with disjoint $\Gamma$-orbits. Chart at a cusp: the stabiliser of $\infty$ in $\sigma^{-1}\Gamma\sigma$ is generated by $\pm T^h$, and $z \mapsto e^{2\pi i z/h}$ identifies $\{\operatorname{Im} z > C\}$ modulo $T^h$ with a punctured disc; adding the cusp fills the puncture. See [52, Section 2.4], [180, Section 1.5]. For Riemann existence and GAGA see Chapter 8. □

**Theorem 12.17** (the genus of $X_0(N)$)**.** *The map $\pi\colon X_0(N) \to X(1)$, $\Gamma_0(N)z \mapsto \mathrm{SL}_2(\mathbb{Z})z$, is a holomorphic map of degree $\mu(N)$, unramified outside the images of $i$, $\zeta_3$ and $\infty$. Over $i$ it has $\nu_2(N)$ points with $e = 1$ and $(\mu(N) - \nu_2(N))/2$ points with $e = 2$; over $\zeta_3$, $\nu_3(N)$ points with $e = 1$ and $(\mu(N) - \nu_3(N))/3$ with $e = 3$; over $\infty$ the $\nu_\infty(N)$ cusps, with $e$ equal to the width. Consequently*

$$g(X_0(N)) = 1 + \frac{\mu(N)}{12} - \frac{\nu_2(N)}{4} - \frac{\nu_3(N)}{3} - \frac{\nu_\infty(N)}{2}.$$

*Proof.* $\pi$ is continuous and, in the charts of Theorems 12.15 and 12.16, holomorphic. Ramification: let $z \in \mathcal{H}$ have stabilisers of orders $e_1$ in $\bar\Gamma_0(N)$ and $e_2$ in $\mathrm{PSL}_2(\mathbb{Z})$, so $e_1 \mid e_2$. With $w$ the Cayley coordinate at $z$, the chart of $X_0(N)$ is $t = w^{e_1}$ and that of $X(1)$ is $s = w^{e_2}$, so $s = t^{e_2/e_1}$ and the ramification index is $e_2/e_1$. At a cusp $s = \sigma\infty$ of width $h$, the chart of $X_0(N)$ is $q_h$ and that of $X(1)$ at $\infty$ is $q = e^{2\pi i \sigma^{-1}(z)} = q_h^h$, so the ramification index is $h$. By Corollary 12.5 only the orbits of $i$ and $\zeta_3$ have $e_2 > 1$, so $\pi$ is unramified over the rest of $Y(1)$, and the degree is the number of points over a generic point, which is $[\mathrm{PSL}_2(\mathbb{Z}) : \bar\Gamma_0(N)] = \mu(N)$ (cosets of $\Gamma_0(N)$ act simply on the orbit of a point with trivial stabiliser). The fibres over $i$ and $\zeta_3$ were counted in Proposition 12.14, and the widths add up to $\mu(N)$ (Proposition 12.10). Now $g(X(1)) = 0$ by Corollary 12.5, and the Riemann-Hurwitz formula of Chapter 8, $2g - 2 = \deg\pi \cdot (2g(X(1)) - 2) + \sum_P (e_P - 1)$, gives the following. Over $i$ the $(\mu - \nu_2)/2$ points with $e = 2$ contribute 1 each; over $\zeta_3$ the $(\mu - \nu_3)/3$ points with $e = 3$ contribute 2 each; over $\infty$ the cusps contribute $\sum_s (h_s - 1) = \mu - \nu_\infty$; all other points have $e = 1$. So

$$2g - 2 = \mu \cdot (-2) + \frac{\mu - \nu_2}{2} + \frac{2(\mu - \nu_3)}{3} + (\mu - \nu_\infty) = \frac{\mu}{6} - \frac{\nu_2}{2} - \frac{2\nu_3}{3} - \nu_\infty,$$

since $-2 + \frac12 + \frac23 + 1 = \frac16$; dividing by 2 and adding 1 gives the formula. □

Two instances. For $N = 11$: $g = 1 + \frac{12}{12} - 0 - 0 - \frac22 = 1$. For $N = 37$, where $u^2+1$ and $u^2-u+1$ both have two roots modulo 37 ($37 \equiv 1 \pmod{12}$): $g = 1 + \frac{38}{12} - \frac24 - \frac23 - \frac22 = \frac{12+38-6-8-12}{12} = 2$.

| $N$ | $\mu$ | $\nu_2$ | $\nu_3$ | $\nu_\infty$ | $g$ | $N$ | $\mu$ | $\nu_2$ | $\nu_3$ | $\nu_\infty$ | $g$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 0 | 37 | 38 | 2 | 2 | 2 | 2 |
| 11 | 12 | 0 | 0 | 2 | 1 | 43 | 44 | 0 | 2 | 2 | 3 |
| 13 | 14 | 2 | 2 | 2 | 0 | 77 | 96 | 0 | 0 | 4 | 7 |
| 14 | 24 | 0 | 0 | 4 | 1 | 99 | 144 | 0 | 0 | 8 | 9 |
| 20 | 36 | 0 | 0 | 6 | 1 | 121 | 132 | 0 | 0 | 12 | 6 |
| 23 | 24 | 0 | 0 | 2 | 2 | 140 | 288 | 0 | 0 | 12 | 19 |
| 25 | 30 | 2 | 0 | 6 | 0 | 539 | 672 | 0 | 0 | 16 | 49 |
| 27 | 36 | 0 | 0 | 6 | 1 | 726 | 1584 | 0 | 0 | 48 | 109 |
| 36 | 72 | 0 | 0 | 12 | 1 | 980 | 2016 | 0 | 0 | 48 | 145 |

Every entry was computed from the formulas of Propositions 12.7, 12.12 and 12.14; for $N \le 140$ the counts $\mu, \nu_2, \nu_3, \nu_\infty$ were also obtained by brute force on $\mathbb{P}^1(\mathbb{Z}/N)$ (fixed points of $S$ and $ST$,

orbits of $T$), and for every tabulated $N$ the genus agrees with PARI's `mfdim([N,2],1)`, which is $\dim S_2(\Gamma_0(N)) = g$ by Theorem 12.40. By the formula, among $N \le 2000$ the genus is 0 exactly for $N \in \{1,\dots,10,12,13,16,18,25\}$ and 1 exactly for $N \in \{11,14,15,17,19,20,21,24,27,32,36,49\}$; Exercise 12.73 shows that $g(X_0(N)) \ge 2$ for all larger $N$, so these lists are complete. In particular $X_0(11)$ has genus 1; it is the elliptic curve `11a1` (Chapter 14). These values agree with those quoted in Chapter 8.

**Theorem 12.18** (the moduli interpretation)**.** *For $\tau \in \mathcal{H}$ let $\Lambda_\tau = \mathbb{Z} + \mathbb{Z}\tau$ and $E_\tau = \mathbb{C}/\Lambda_\tau$.*

*(i) Every pair $(E, C)$ of an elliptic curve $E/\mathbb{C}$ and a cyclic subgroup $C \subset E$ of order $N$ is isomorphic to $(E_\tau, \langle \frac{1}{N} + \Lambda_\tau \rangle)$ for some $\tau \in \mathcal{H}$, and $\Gamma_0(N)\tau \mapsto (E_\tau, \langle \frac{1}{N} \rangle)$ is a bijection from $Y_0(N)$ onto the set of isomorphism classes of such pairs.*

*(ii) Likewise $\Gamma_1(N)\tau \mapsto (E_\tau, \frac{1}{N} + \Lambda_\tau)$ is a bijection from $Y_1(N)$ onto the isomorphism classes of pairs $(E, P)$ with $P \in E$ of exact order $N$. Let $\zeta_N = e_N(\frac{1}{N} + \Lambda_\tau, \frac{\tau}{N} + \Lambda_\tau)$, with $e_N$ the Weil pairing of Chapter 9; it is the same primitive $N$-th root of unity for every $\tau \in \mathcal{H}$, namely $e^{2\pi i/N}$ or $e^{-2\pi i/N}$ according to the normalisation of $e_N$ (Chapter 9). Then $\Gamma(N)\tau \mapsto (E_\tau, (\frac{1}{N} + \Lambda_\tau, \frac{\tau}{N} + \Lambda_\tau))$ is a bijection from $Y(N)$ onto the isomorphism classes of pairs $(E, (P, Q))$ with $(P, Q)$ a basis of $E[N]$ and $e_N(P, Q) = \zeta_N$.*

*(iii) Under (i) the map $\pi$ of Theorem 12.17 is $(E, C) \mapsto E$, and $(E, C) \mapsto (E/C, E[N]/C)$ is $\Gamma_0(N)\tau \mapsto \Gamma_0(N)(-1/(N\tau))$, the Atkin-Lehner involution $w_N$ (Proposition 12.64).*

*(iv) (Cusps.) The cusps correspond to degenerate objects: generalised elliptic curves (Néron polygons) with a cyclic subgroup of order $N$; over the Tate curve $\mathbb{G}_m/\mathsf{q}^{\mathbb{Z}}$ the cusp $\infty$ corresponds to the subgroup $\mu_N$ and the cusp $0$ to a subgroup generated by an $N$-th root of $\mathsf{q}$.*

*(v) ($\mathbb{Q}$-structure.) There is a smooth projective curve $X_0(N)_{\mathbb{Q}}$ over $\mathbb{Q}$ whose complex points are $X_0(N)$, such that the bijection (i) is compatible with the action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on pairs $(E, C)$ defined over $\overline{\mathbb{Q}}$. The cusps $\infty$ and $0$ are rational, and a cusp $a/c$, $c \mid N$, is defined over $\mathbb{Q}(\zeta_{\gcd(c, N/c)})$.*

*Proof.* (i) By Chapter 9 every $E/\mathbb{C}$ is isomorphic to some $E_\tau$, every isomorphism $E_\tau \to E_{\tau'}$ is $z \mapsto \alpha z$ with $\alpha\Lambda_\tau = \Lambda_{\tau'}$, and for every such $\alpha$ there is $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ with $\tau' = \gamma\tau$ and $\alpha = \pm(c\tau + d)^{-1}$. A cyclic subgroup of order $N$ of $\frac{1}{N}\Lambda_\tau/\Lambda_\tau \simeq (\mathbb{Z}/N)^2$ is generated by $(c\tau + d)/N$ with $\gcd(c, d, N) = 1$; by Proposition 12.7(i) we may take $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, and then $z \mapsto z/(c\tau + d)$ maps $(E_\tau, \langle \frac{c\tau+d}{N} \rangle)$ to $(E_{\gamma\tau}, \langle \frac{1}{N} \rangle)$. So every pair has the stated form. For injectivity, let $\tau' = \gamma\tau$ as above. From $\tau = \gamma^{-1}\tau' = \frac{d\tau' - b}{-c\tau' + a}$,

$$c\tau + d = \frac{c(d\tau' - b) + d(a - c\tau')}{a - c\tau'} = \frac{ad - bc}{a - c\tau'} = (a - c\tau')^{-1},$$

so the isomorphism $z \mapsto z/(c\tau + d)$ sends $\frac{1}{N}$ to $\frac{a - c\tau'}{N}$, which generates $\langle \frac{1}{N} + \Lambda_{\tau'} \rangle$ iff $N \mid c$ (and then $a$ is a unit modulo $N$). So $(E_\tau, \langle \frac{1}{N} \rangle) \simeq (E_{\tau'}, \langle \frac{1}{N} \rangle)$ iff $\tau' \in \Gamma_0(N)\tau$; the sign $\pm$ does not matter since $-I \in \Gamma_0(N)$. (ii) The same computation: $\frac{a - c\tau'}{N} \equiv \pm\frac{1}{N} \pmod{\Lambda_{\tau'}}$ iff $N \mid c$ and $a \equiv \pm 1 \pmod N$, that is, $\gamma \in \pm\Gamma_1(N)$, which acts on $\mathcal{H}$ as $\Gamma_1(N)$. For $Y(N)$: if $(P, Q)$ is a basis of $E_\tau[N]$, write $P = \frac{a + b\tau}{N}$, $Q = \frac{c + d\tau}{N}$ modulo $\Lambda_\tau$; bilinearity gives $e_N(P, Q) = \zeta_N^{ad - bc}$, so $e_N(P, Q) = \zeta_N$ iff $ad - bc \equiv 1 \pmod N$, and by Proposition 12.7(i) we may then take

$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Then $\omega_1 = a+b\tau$, $\omega_2 = c+d\tau$ is a basis of $\Lambda_\tau$ with $\tau' = \omega_2/\omega_1 = \frac{d\tau+c}{b\tau+a} \in \mathcal{H}$, and $z \mapsto z/\omega_1$ maps $(E_\tau, (P,Q))$ to $(E_{\tau'}, (\frac{1}{N}, \frac{\tau'}{N}))$: this is surjectivity. Conversely an isomorphism $z \mapsto \alpha z$ from $(E_\tau, (\frac{1}{N}, \frac{\tau}{N}))$ to $(E_{\tau'}, (\frac{1}{N}, \frac{\tau'}{N}))$ satisfies $\alpha = m_{11} + m_{12}\tau'$, $\alpha\tau = m_{21} + m_{22}\tau'$ with $\left(\begin{smallmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{smallmatrix}\right) \equiv I \pmod N$ of determinant 1 (it maps the oriented basis $(1,\tau)$ to an oriented basis of $\Lambda_{\tau'}$), so $\tau = \frac{m_{22}\tau'+m_{21}}{m_{12}\tau'+m_{11}}$ with $\left(\begin{smallmatrix} m_{22} & m_{21} \\ m_{12} & m_{11} \end{smallmatrix}\right) \in \Gamma(N)$: this is injectivity. That $\zeta_N$ does not depend on $\tau$ is part (iv) of the proposition on isogenies of complex tori in Chapter 9 ($e_N = \zeta^{\det}$ on $\frac{1}{N}\Lambda/\Lambda$ for an oriented basis). (iii) The point $\pi(\Gamma_0(N)\tau) = \mathrm{SL}_2(\mathbb{Z})\tau$ of $Y(1)$ corresponds to the isomorphism class of $E_\tau$ alone (the case $N = 1$ of (i)), so $\pi$ forgets $C$. For the second statement, $E_\tau/\langle\frac{1}{N}\rangle = \mathbb{C}/(\frac{1}{N}\mathbb{Z} + \mathbb{Z}\tau)$, and multiplication by $N$ identifies it with $\mathbb{C}/\Lambda_{N\tau}$, sending $E_\tau[N]/C$ to the subgroup generated by $\tau = \frac{N\tau}{N}$. Multiplication by $(N\tau)^{-1}$ identifies $(\mathbb{C}/\Lambda_{N\tau}, \langle\frac{N\tau}{N}\rangle)$ with $(\mathbb{C}/\Lambda_{-1/(N\tau)}, \langle\frac{1}{N}\rangle)$, since $(N\tau)^{-1}\Lambda_{N\tau} = \mathbb{Z}(N\tau)^{-1} + \mathbb{Z} = \Lambda_{-1/(N\tau)}$. (iv), (v) We do not prove these; see [43] and [91] for the moduli problem over $\mathbb{Z}$ and its compactification, [52, Sections 7.7 and 1.5] and [180, Section 6.7] for the model over $\mathbb{Q}$, and [146] for the fields of definition of the cusps (all section numbers not verified). □

*In the paper* 12.19. The paper's Section 1 speaks of "the $X_0(N)$-optimal curve, for which the modular parametrisation $\varphi_E\colon X_0(N) \to E$ has minimal degree". Here $X_0(N)$ is the curve of Theorem 12.16, defined over $\mathbb{Q}$ by Theorem 12.18(v), and $\varphi_E$ sends the rational cusp $\infty$ to the origin (Chapter 14). For $N = 11$ the curve itself has genus 1 (the table above), and is the curve `11a1`, so that $\deg\varphi_{\texttt{11a1}} = 1$. In Section 4.1 the paper uses $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$, the cohomology of the compact Riemann surface $X_0(M^\Sigma)$ (Chapter 8); its rank is $2g(X_0(M^\Sigma))$, for example $2g(X_0(121)) = 12$.

## 12.6 Modular forms, $q$-expansions and nebentypus

This section defines the objects of the paper's first sentence about modular forms, "let $f = f_E \in S_2(\Gamma_0(N))$ be the newform attached to it": the spaces $M_k(\Gamma_0(N))$ and $S_k(\Gamma_0(N))$, holomorphy and vanishing at the cusps, and the $q$-expansion $\sum a_n(f)q^n$, whose coefficients are the numbers compared in every congruence of the paper. It also proves the elementary growth bound for the $a_n$, states the Ramanujan-Petersson bound, and introduces forms with a character (nebentypus), which occur in the optimal-level theorem used in the paper's Section 3.

**Definition 12.20** (the slash operator)**.** For $k \in \mathbb{Z}$, $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$ and $f\colon \mathcal{H} \to \mathbb{C}$, put $(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz+d)^{-k}f(\gamma z)$ (Convention 12.1(d)).

**Lemma 12.21** (properties of the slash operator)**.** *For $\gamma, \gamma' \in \mathrm{GL}_2^+(\mathbb{R})$ and $f\colon \mathcal{H} \to \mathbb{C}$:*

*(i)* $(f|_k\gamma)|_k\gamma' = f|_k(\gamma\gamma')$*;*

*(ii)* $f|_k(\lambda I) = \mathrm{sgn}(\lambda)^k f$ *for* $\lambda \in \mathbb{R}^\times$*;*

*(iii) in weight 2,* $(f|_2\gamma)(z)\,dz = \gamma^*(f(z)\,dz)$*, the pullback of the differential* $f(z)\,dz$ *under* $z \mapsto \gamma z$*;*

*(iv)* $\mathrm{Im}(\gamma z)^{k/2}|f(\gamma z)| = \mathrm{Im}(z)^{k/2}|(f|_k\gamma)(z)|$*; so if* $f|_k\gamma = f$ *for all* $\gamma$ *in a group* $\Gamma$*, the function* $y^{k/2}|f|$ *is* $\Gamma$*-invariant.*

*Proof.* (i) $(f|_k\gamma)|_k\gamma'(z) = (\det\gamma')^{k/2}j(\gamma',z)^{-k}(\det\gamma)^{k/2}j(\gamma,\gamma' z)^{-k}f(\gamma\gamma' z)$, which is $f|_k(\gamma\gamma')(z)$ by the cocycle identity Proposition 12.2(iii) and $\det(\gamma\gamma') = \det\gamma\det\gamma'$. (ii) $(\lambda^2)^{k/2}\lambda^{-k} = |\lambda|^k\lambda^{-k}$.

(iii) $\gamma^*(f(z)\,dz) = f(\gamma z)\,d(\gamma z) = f(\gamma z)\det\gamma\,(cz+d)^{-2}\,dz$ by Proposition 12.2(ii). (iv) Take absolute values and use Proposition 12.2(i): $\operatorname{Im}(\gamma z)^{k/2} = (\det\gamma)^{k/2}y^{k/2}|cz+d|^{-k}$. □

With Diamond-Shurman's $(\det\gamma)^{k-1}$, (i), (ii) up to a power of $|\lambda|$, and (iii) hold, but (iv) and the isometry property of Proposition 12.65 need the exponent $k/2$. In weight 2 the two normalisations coincide, which is why the paper's operator $w$ of Section 4.1 is the Atkin-Lehner involution "with no scalar factor" in either convention.

**Definition 12.22** (modular forms and cusp forms)**.** Let $\Gamma$ be a congruence subgroup and $k \in \mathbb{Z}$. A *modular form of weight $k$ for* $\Gamma$ is a holomorphic function $f\colon \mathcal{H} \to \mathbb{C}$ such that

(a) $f|_k\gamma = f$ for all $\gamma \in \Gamma$, and

(b) $f$ is *holomorphic at the cusps*: for every $\sigma \in \mathrm{SL}_2(\mathbb{Z})$ the function $f|_k\sigma$ is bounded on $\{\operatorname{Im} z \geq 1\}$.

It is a *cusp form* if moreover $(f|_k\sigma)(z) \to 0$ as $\operatorname{Im} z \to \infty$, for every $\sigma \in \mathrm{SL}_2(\mathbb{Z})$. The modular forms form a complex vector space $M_k(\Gamma)$, the cusp forms a subspace $S_k(\Gamma)$. The *level* of a form for $\Gamma_0(N)$ is $N$; a cusp form of level $N$ in the paper is an element of $S_2(\Gamma_0(N))$.

Condition (a) says that $f(z)\,(dz)^{k/2}$ is invariant under $\Gamma$ (for $k=2$ this is Lemma 12.21(iii) below), so that $f$ lives on the quotient $Y(\Gamma)$ in this twisted sense. Condition (b) is the analogue of "holomorphic" at the points that $Y(\Gamma)$ is missing. A cusp $s = \sigma\infty$ is moved to $\infty$ by $\sigma^{-1}$, and the behaviour of $f$ near $s$ is the behaviour of $f|_k\sigma$ near $\infty$; so the condition must be imposed for every $\sigma$, not only for $\sigma = I$. It is not automatic: after Theorem 12.46 we exhibit a holomorphic function of weight 12 for $\Gamma_0(11)$ that vanishes at $\infty$ to order 21 but has a pole at the cusp 0. Since $\mathrm{SL}_2(\mathbb{Z})$ has the single cusp $\infty$, at level one only $\sigma = I$ matters.

**Proposition 12.23** (first properties)**.** *Let $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup.*

*(a) Condition (b) of Definition 12.22, and the cusp condition, at $\sigma$ depend only on the cusp $\Gamma\sigma\infty$.*

*(b) If $-I \in \Gamma$ and $k$ is odd, $M_k(\Gamma) = 0$. In particular only even weights occur for $\Gamma_0(N)$.*

*(c) If $\Gamma' \subseteq \Gamma$ then $M_k(\Gamma) \subseteq M_k(\Gamma')$ and $S_k(\Gamma) \subseteq S_k(\Gamma')$.*

*(d) $M_k(\Gamma)M_l(\Gamma) \subseteq M_{k+l}(\Gamma)$.*

*(e) Let $f \in M_k(\Gamma_0(M))$ and $d \geq 1$. The function $f(dz) = d^{-k/2}(f|_k\left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right))(z)$ lies in $M_k(\Gamma_0(Md))$, and in $S_k(\Gamma_0(Md))$ if $f$ is a cusp form.*

*Proof.* (a) If $\sigma'\infty = \gamma\sigma\infty$ with $\gamma \in \Gamma$, then $\sigma' = \gamma\sigma(\pm T^m)$, and $f|_k\sigma' = \pm f|_k\sigma|_k T^m$ is a translate of $\pm f|_k\sigma$, which preserves $\{\operatorname{Im} z \geq 1\}$. (b) $f = f|_k(-I) = (-1)^k f$ by Lemma 12.21(ii). (c) is clear. (d) $(fg)|_{k+l}\gamma = (f|_k\gamma)(g|_l\gamma)$, and products of bounded functions are bounded. (e) Let $\alpha = \left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right)$. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right) \in \Gamma_0(Md)$, $\alpha\gamma = \gamma'\alpha$ with $\gamma' = \left(\begin{smallmatrix} a & db \\ c/d & e \end{smallmatrix}\right) \in \Gamma_0(M)$, so $f|_k\alpha|_k\gamma = f|_k\gamma'|_k\alpha = f|_k\alpha$. For $\sigma \in \mathrm{SL}_2(\mathbb{Z})$ write $\alpha\sigma = \sigma'\beta$ with $\sigma' \in \mathrm{SL}_2(\mathbb{Z})$ and $\beta = \left(\begin{smallmatrix} a' & b' \\ 0 & d' \end{smallmatrix}\right)$ upper triangular with integer entries. To find them, write the first column of the integer matrix $\alpha\sigma$ as $a'v$ with $a' \geq 1$ and $v \in \mathbb{Z}^2$ primitive, complete $v$ to a matrix $\sigma' \in \mathrm{SL}_2(\mathbb{Z})$ with first column $v$ (possible because $v$ is primitive), and put $\beta = \sigma'^{-1}\alpha\sigma$. It has integer entries, its first column is $\sigma'^{-1}(a'v) = (a', 0)^{\mathsf{T}}$, and $\det\beta = \det\alpha = d$, so $\beta = \left(\begin{smallmatrix} a' & b' \\ 0 & d' \end{smallmatrix}\right)$ with $a'd' = d$ and $d' \geq 1$. Then $f|_k\alpha\sigma = (f|_k\sigma')|_k\beta$, and

$f|_k\sigma' = \sum_{n\ge 0} b_n q_h^n$ (Proposition 12.24(iii) below) is bounded on every half plane $\{\operatorname{Im} z \ge C\}$, $C > 0$, and tends to $b_0$; since $\beta$ maps $\{\operatorname{Im} z \ge 1\}$ to $\{\operatorname{Im} z \ge a'/d'\}$, $f|_k\alpha\sigma$ is bounded there, and tends to 0 if $b_0 = 0$. □

The forms $f(dz)$ of (e) are the raw material of Chapter 13's oldforms; the paper's Lemma lem:old works with "the forms $h(dz)$, $d \mid N/M$" for a newform $h$ of level $M$, which lie in $S_2(\Gamma_0(N))$ by (e).

**Proposition 12.24** ($q$-expansions)**.** *Let $f \in M_k(\Gamma_0(N))$.*

*(i) $f(z+1) = f(z)$, and $f(z) = \sum_{n\ge 0} a_n(f) q^n$, $q = e^{2\pi i z}$, for all $z \in \mathcal{H}$, the series converging absolutely and uniformly on $\{\operatorname{Im} z \ge y_0\}$ for every $y_0 > 0$. For every $y > 0$,*

$$a_n(f) = \int_0^1 f(x+iy)\, e^{-2\pi i n(x+iy)}\, dx.$$

*(ii) More generally, a holomorphic $g\colon \mathcal{H} \to \mathbb{C}$ of period 1 is $g(z) = G(q)$ for a holomorphic $G$ on the punctured disc $0 < |q| < 1$, with a Laurent expansion $G = \sum_{n\in\mathbb{Z}} a_n q^n$; $g$ is bounded on $\{\operatorname{Im} z \ge 1\}$ iff $G$ has a removable singularity at 0 iff $a_n = 0$ for all $n < 0$. Then we write $g(\infty) = a_0$.*

*(iii) For $\sigma \in \mathrm{SL}_2(\mathbb{Z})$ with $s = \sigma\infty$ of width $h$, $f|_k\sigma$ has period $h$ and an expansion $f|_k\sigma = \sum_{n\ge 0} b_n(\sigma) q_h^n$, the* expansion of $f$ at $s$. *The form $f$ is a cusp form iff $b_0(\sigma) = 0$ for a set of $\sigma$ representing all cusps.*

*(iv) ($q$-expansion principle, analytic form.) The map $f \mapsto (a_n(f))_{n\ge 0}$ is injective on $M_k(\Gamma_0(N))$.*

*Proof.* (ii) Since $z \mapsto q = e^{2\pi i z}$ maps $\mathcal{H}$ onto the punctured disc and identifies exactly the points that differ by integers, $G(q) = g(z)$ is well defined and holomorphic (locally $z = \frac{1}{2\pi i}\log q$). Its Laurent series converges absolutely and locally uniformly on the punctured disc. The set $\{\operatorname{Im} z \ge 1\}$ corresponds to $0 < |q| \le e^{-2\pi}$, and Riemann's theorem on removable singularities gives the equivalences. (i) $T \in \Gamma_0(N)$ gives the period, and (ii) the expansion with $n \ge 0$ (taking $\sigma = I$ in Definition 12.22(b)). A power series with radius of convergence at least 1 converges absolutely and uniformly on $|q| \le e^{-2\pi y_0} < 1$. The coefficient formula is Cauchy's formula $a_n = \frac{1}{2\pi i}\oint_{|q|=e^{-2\pi y}} G(q) q^{-n-1}\, dq$ with $q = e^{2\pi i(x+iy)}$, $dq = 2\pi i q\, dx$. (iii) $\sigma T^h \sigma^{-1} \in \pm\Gamma_0(N) = \Gamma_0(N)$, so $f|_k\sigma|_k T^h = f|_k\sigma$, that is, $f|_k\sigma$ has period $h$; apply (ii) to $z \mapsto (f|_k\sigma)(hz)$. The cusp condition at $\sigma$ is $b_0(\sigma) = 0$, and depends only on the cusp by Proposition 12.23(a). (iv) By (i) the coefficients determine $f$ on all of $\mathcal{H}$. □

A much stronger statement holds: $f$ is determined by $a_0, \dots, a_{B_k(N)}$ (Theorem 12.39, and Theorem 12.58 for congruences).

The expansion is a Fourier series. A function on $\mathcal{H}$ of period 1 in $x$ has a Fourier series $\sum_n c_n(y) e^{2\pi i n x}$; holomorphy (the Cauchy-Riemann equation $\partial_y f = i\partial_x f$) forces $c_n'(y) = -2\pi n c_n(y)$, so $c_n(y) = a_n e^{-2\pi n y}$, and the $n$-th mode is $a_n e^{2\pi i n x} e^{-2\pi n y} = a_n q^n$. These are the separated solutions of Laplace's equation on a half-strip that stay bounded as $y \to \infty$ exactly when $n \ge 0$; the condition "holomorphic at $\infty$" is the boundary condition that kills the growing modes. So a modular form is a Fourier series on the quotient $\mathcal{H}/\langle T\rangle$, a half-infinite cylinder, with one more symmetry, $z \mapsto -1/(Nz)$ or the rest of $\Gamma_0(N)$, that the coefficients must respect.

*In the paper* 12.25. Every congruence in the paper is a statement about the coefficients $a_n$ of Proposition 12.24(i). In the opening paragraph of Section 3 the congruence number is "the largest integer $r$ such that $f \equiv g \pmod r$ coefficientwise for some $g \in S_2(\Gamma_0(N))$ with integer coefficients orthogonal to $f$": coefficientwise means $a_n(f) \equiv a_n(g) \pmod r$ for all $n \geq 0$ (Convention 12.1(g)); integer coefficients means $g \in S_2(\Gamma_0(N); \mathbb{Z})$ (Definition 12.56); orthogonality is for the Petersson product (Definition 12.48). Lemma lem:crit asks for $a_n(g) \equiv a_n(f) \pmod \lambda$ "for every $n \geq 1$", while Lemma lem:old starts from a congruence at the primes $p \nmid \ell N$ only and passes to all $n$ through the Hecke recursion (Chapter 13).

**Proposition 12.26** (the Hecke bound)**.** *Let $f \in S_k(\Gamma_0(N))$. Then $y^{k/2}|f(z)|$ is bounded on $\mathcal{H}$, and there is a constant $C_f$ with $|a_n(f)| \leq C_f n^{k/2}$ for all $n \geq 1$. Consequently the Dirichlet series $\sum_{n\geq 1} a_n(f) n^{-s}$ converges absolutely for $\operatorname{Re} s > \frac{k}{2} + 1$.*

*Proof.* The function $\phi = y^{k/2}|f|$ is continuous and $\Gamma_0(N)$-invariant (Lemma 12.21(iv)), so it suffices to bound it on the fundamental domain $\bigcup_i \gamma_i \mathcal{F}$ of Proposition 12.8. For $w \in \mathcal{F}$, $\phi(\gamma_i w) = \operatorname{Im}(w)^{k/2}|(f|_k\gamma_i)(w)|$ by Lemma 12.21(iv). On $\mathcal{F}$, $\operatorname{Im} w \geq \sqrt{3}/2$, and $f|_k\gamma_i = \sum_{n\geq 1} b_n q_h^n$ without constant term, so $|(f|_k\gamma_i)(w)| \leq |q_h| \sum_{n\geq 1} |b_n| r_0^{n-1} = C_i e^{-2\pi \operatorname{Im} w/h}$ with $r_0 = e^{-\pi\sqrt{3}/h} < 1$. Since $y^{k/2}e^{-2\pi y/h}$ is bounded, $\phi$ is bounded, say by $C$. Then by Proposition 12.24(i) at height $y = 1/n$: $|a_n(f)| \leq \int_0^1 |f(x + i/n)| e^{2\pi}\, dx \leq C n^{k/2} e^{2\pi}$. Here $|e^{-2\pi i n(x+i/n)}| = e^{2\pi n \cdot (1/n)} = e^{2\pi}$, and $|f(x+iy)| \leq C y^{-k/2} = C n^{k/2}$ at $y = 1/n$. The choice $y = 1/n$ balances the two factors: a larger $y$ makes $e^{2\pi n y}$ grow exponentially in $n$, a smaller one makes $y^{-k/2}$ worse. The last statement follows by comparison with $\sum n^{k/2 - \operatorname{Re} s}$, which converges for $\operatorname{Re} s - \frac{k}{2} > 1$. □

For forms that are not cuspidal the bound fails: the Eisenstein series $E_k$ of Proposition 12.32 has $|a_n| = \frac{2k}{|B_k|}\sigma_{k-1}(n) \geq \frac{2k}{|B_k|} n^{k-1}$. The true size of the coefficients of an eigenform is much smaller.

**Theorem 12.27** (Ramanujan-Petersson; Eichler-Shimura, Igusa, Deligne)**.** *Let $f \in S_k(\Gamma_0(N))$ be a normalised newform (Chapter 13) and $p \nmid N$ a prime. Then $|a_p(f)| \leq 2p^{(k-1)/2}$; equivalently the roots $\alpha_p, \beta_p$ of $X^2 - a_p(f)X + p^{k-1}$ satisfy $|\alpha_p| = |\beta_p| = p^{(k-1)/2}$, so that $\beta_p = \bar{\alpha}_p$ and $|\alpha_p/\beta_p| = 1$. Moreover $a_n(f) = O(n^{(k-1)/2+\varepsilon})$ for every $\varepsilon > 0$.*

We do not prove this; for $k = 2$ it follows from the Eichler-Shimura relation and Weil's Riemann hypothesis for curves over finite fields (Chapter 8, Chapter 14), and for $f = f_E$ it is Hasse's bound $|a_p(E)| \leq 2\sqrt{p}$ (Chapter 9); in general it is Deligne's theorem [44]. The statement and references are in [52, Section 8.7] (section number not verified). Checks: $|a_p(\texttt{11a1})| \leq 2\sqrt{p}$ for all $p < 10^5$ (the largest ratio is 0.9940), and Ramanujan's $|\tau(p)| \leq 2p^{11/2}$ for all $p < 2000$ (largest ratio 0.9594). The equivalence: $\alpha_p\beta_p = p^{k-1}$ and $\alpha_p + \beta_p = a_p$ real, so the roots are complex conjugate of absolute value $p^{(k-1)/2}$ iff the discriminant $a_p^2 - 4p^{k-1}$ is $\leq 0$.

*In the paper* 12.28. Lemma lem:euler and Theorem thm:dfg14 use the roots $\alpha, \beta$ of $X^2 - a_p X + p$ at a good prime $p$ ($k = 2$). Wiles's level-raising factor $(1 - \frac{\alpha}{\beta}p^{-1})(1 - p^{-1})(1 - \frac{\beta}{\alpha}p^{-1})$ of the paragraph after Theorem thm:dfg14 is a nonzero real number because $\beta = \bar{\alpha}$ and $|\alpha/\beta| = 1$: then $\frac{\beta}{\alpha} = (\frac{\alpha}{\beta})^{-1} = \overline{\alpha/\beta}$, so the third factor is the complex conjugate of the first, and the product equals $(1 - p^{-1})|1 - \frac{\alpha}{\beta}p^{-1}|^2$, which is positive because $|\frac{\alpha}{\beta}p^{-1}| = p^{-1} < 1$.

**Definition 12.29** (nebentypus)**.** Let $\chi$ be a Dirichlet character modulo $N$. The space of modular forms of weight $k$, level $N$ and *character* (or *nebentypus*) $\chi$ is

$$M_k(\Gamma_0(N), \chi) = \Big\{ f \in M_k(\Gamma_1(N)) : f|_k\gamma = \chi(d) f \text{ for all } \gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N) \Big\},$$

and $S_k(\Gamma_0(N),\chi)$ is its subspace of cusp forms. For $d \in (\mathbb{Z}/N)^\times$ the *diamond operator* $\langle d\rangle$ on $M_k(\Gamma_1(N))$ is $f \mapsto f|_k\gamma$ for any $\gamma \in \Gamma_0(N)$ with lower right entry $\equiv d \pmod N$.

The reason for this definition is that $\Gamma_1(N)$, not $\Gamma_0(N)$, is the natural group for many constructions, and $\Gamma_0(N)$ acts on $M_k(\Gamma_1(N))$ through the finite abelian quotient $\Gamma_0(N)/\Gamma_1(N) \simeq (\mathbb{Z}/N)^\times$. Decomposing under a finite abelian group means decomposing by characters, and $M_k(\Gamma_0(N),\chi)$ is the $\chi$-isotypic piece; the trivial character gives back $M_k(\Gamma_0(N))$. Nontrivial characters genuinely occur: PARI gives $\dim S_2(\Gamma_1(13)) = 2$ while $S_2(\Gamma_0(13)) = 0$ (the genus of $X_0(13)$ is 0), the two dimensions coming from a character of order 6 modulo 13 and its complex conjugate, one dimension each.

**Proposition 12.30** (decomposition by characters)**.** *(i)* $M_k(\Gamma_0(N),\mathbf{1}) = M_k(\Gamma_0(N))$ *for the trivial character* $\mathbf{1}$*.*

*(ii)* $M_k(\Gamma_0(N),\chi) = 0$ *unless* $\chi(-1) = (-1)^k$*.*

*(iii) The diamond operators are well defined and give an action of* $(\mathbb{Z}/N)^\times$ *on* $M_k(\Gamma_1(N))$*, and* $M_k(\Gamma_1(N)) = \bigoplus_\chi M_k(\Gamma_0(N),\chi)$*, the sum over all Dirichlet characters modulo* $N$*; likewise for* $S_k$*.*

*Proof.* (i) $\chi(d) = 1$. (ii) Apply the definition to $\gamma = -I$: $(-1)^k f = \chi(-1)f$. (iii) $\Gamma_1(N)$ is normal in $\Gamma_0(N)$ with quotient $(\mathbb{Z}/N)^\times$ via $\gamma \mapsto d$ (Proposition 12.7(ii)). For $f \in M_k(\Gamma_1(N))$ and $\gamma \in \Gamma_0(N)$, $f|_k\gamma$ is again invariant under $\Gamma_1(N) = \gamma\Gamma_1(N)\gamma^{-1}$ and holomorphic at the cusps ($f|_k\gamma|_k\sigma = f|_k(\gamma\sigma)$ with $\gamma\sigma \in \mathrm{SL}_2(\mathbb{Z})$), and depends only on $\gamma\Gamma_1(N)$; so the finite abelian group $(\mathbb{Z}/N)^\times$ acts linearly on $M_k(\Gamma_1(N))$. The operators $e_\chi = \frac{1}{\varphi(N)}\sum_{d\in(\mathbb{Z}/N)^\times} \overline{\chi(d)}\langle d\rangle$ are orthogonal idempotents with $\sum_\chi e_\chi = 1$ and $\langle d\rangle e_\chi = \chi(d)e_\chi$ (orthogonality of characters, Chapter 4). In detail: substituting $d' \mapsto d^{-1}d'$,

$$\langle d\rangle e_\chi = \frac{1}{\varphi(N)}\sum_{d'} \overline{\chi(d')}\langle dd'\rangle = \frac{\chi(d)}{\varphi(N)}\sum_{d'} \overline{\chi(dd')}\langle dd'\rangle = \chi(d)e_\chi;$$

hence $e_\psi e_\chi = \frac{1}{\varphi(N)}\sum_d \overline{\psi(d)}\chi(d)e_\chi$, which is $e_\chi$ if $\psi = \chi$ and 0 otherwise; and $\sum_\chi e_\chi = \frac{1}{\varphi(N)}\sum_d(\sum_\chi \overline{\chi(d)})\langle d\rangle = \langle 1\rangle = 1$. So $M_k(\Gamma_1(N))$ is the direct sum of the images $e_\chi M_k(\Gamma_1(N))$. Each image is $M_k(\Gamma_0(N),\chi)$: the image consists of $\chi$-eigenvectors of every $\langle d\rangle$, which is the defining condition $f|_k\gamma = \chi(d)f$, and conversely $e_\chi f = f$ for such $f$. The operators preserve cusp forms. □

The twist of $f \in S_k(\Gamma_0(M))$ by a primitive Dirichlet character $\chi$ of conductor $r$ is a form of character $\chi^2$ and level dividing $\mathrm{lcm}(M, r^2)$ (Chapter 13); for the quadratic characters $\chi_{q^*}$ of the paper $\chi^2 = \mathbf{1}$, so twists stay in the spaces $S_2(\Gamma_0(\cdot))$ with trivial character.

*In the paper* 12.31. The optimal-level theorem quoted in the paper's Section 3 produces "a newform of weight 2 and level $N(\bar\sigma)\ell^{\delta(\bar\sigma)}$ with nebentypus of order prime to $\ell$", an element of $S_2(\Gamma_0(M),\chi)$ in the sense of Definition 12.29. The paper then observes that "$\det\bar\sigma = \bar\chi_\ell$ forces the nebentypus, which is trivial modulo $\lambda$ and of order prime to $\ell$, to be trivial", so that the newform lies in $S_2(\Gamma_0(M))$; the Galois-theoretic content of that sentence (the determinant of $\rho_f$ is $\chi\chi_\ell$) is Chapter 15's.

## 12.7 Level one: Eisenstein series, $\Delta$, $j$ and the valence formula

Level one is where every structural fact can be proved with a contour integral. This section constructs the Eisenstein series $E_k$, the cusp form $\Delta$ and the function $j$, proves the valence formula, and deduces the structure of $M_*(\mathrm{SL}_2(\mathbb{Z}))$ and the bijectivity of $j$. It discharges two debts of Chapter 9: the $q$-expansion of $j$ and the bijection $j\colon \mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H} \to \mathbb{C}$. The paper uses level one only through these objects: the $j$-invariant $j_E$ of Chapter 9, whose valuations enter Definition def:terms, and the Eisenstein series behind the Eisenstein case of Section sec:eis (Proposition 12.42).

**Proposition 12.32** (Eisenstein series of level one)**.** *For even $k \geq 4$ the series $G_k(z) = \sum_{(m,n)\in\mathbb{Z}^2\smallsetminus\{0\}}(mz+n)^{-k}$ converges absolutely and locally uniformly on $\mathcal{H}$, $G_k \in M_k(\mathrm{SL}_2(\mathbb{Z}))$, and $G_k = 2\zeta(k)E_k$ with*

$$E_k(z) = 1 - \frac{2k}{B_k}\sum_{n\geq 1}\sigma_{k-1}(n)q^n, \qquad E_4 = 1 + 240\sum_{n\geq 1}\sigma_3(n)q^n, \qquad E_6 = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n,$$

*where $\sigma_r(n) = \sum_{d|n} d^r$ and $B_k$ is the Bernoulli number. In the notation of Chapter 9, $G_k(z) = G_k(\Lambda_z)$. The series $E_2(z) = 1 - 24\sum_{n\geq 1}\sigma_1(n)q^n$ converges on $\mathcal{H}$ but is not modular: it satisfies*

$$E_2(-1/z) = z^2E_2(z) + \frac{6z}{\pi i},$$

*equivalently $E_2^*(z) = E_2(z) - \frac{3}{\pi y}$ satisfies $E_2^*|_2\gamma = E_2^*$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$.*

*Proof.* Absolute and locally uniform convergence of $\sum|mz+n|^{-k}$ for $k > 2$ is the lattice-sum lemma of Chapter 9, so $G_k$ is holomorphic. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$,

$$G_k(\gamma z) = \sum\Big(\frac{m(az+b)+n(cz+d)}{cz+d}\Big)^{-k} = (cz+d)^k\sum\big((ma+nc)z+(mb+nd)\big)^{-k},$$

which is $(cz+d)^kG_k(z)$ since $(m,n) \mapsto (ma+nc, mb+nd)$ permutes $\mathbb{Z}^2\smallsetminus\{0\}$. For the expansion we use the Lipschitz formula, valid for $k \geq 2$ and $z \in \mathcal{H}$:

$$\sum_{n\in\mathbb{Z}}(z+n)^{-k} = \frac{(-2\pi i)^k}{(k-1)!}\sum_{r\geq 1} r^{k-1}q^r.$$

It follows from the partial fraction expansion $\pi\cot\pi z = \frac{1}{z} + \sum_{n\neq 0}(\frac{1}{z+n} - \frac{1}{n})$ and $\pi\cot\pi z = \pi i\frac{q+1}{q-1} = -\pi i - 2\pi i\sum_{r\geq 1}q^r$ (for $|q| < 1$), by differentiating both $k-1$ times. In detail: $\pi\cot\pi z = \pi i\frac{e^{\pi iz}+e^{-\pi iz}}{e^{\pi iz}-e^{-\pi iz}} = \pi i\frac{q+1}{q-1}$ (multiply numerator and denominator by $e^{\pi iz}$), and $\frac{q+1}{q-1} = -(1+q)\sum_{r\geq 0}q^r = -1 - 2\sum_{r\geq 1}q^r$. Both expansions converge locally uniformly, so they may be differentiated termwise. The $(k-1)$-st derivative of $\frac{1}{z+n}$ is $(-1)^{k-1}(k-1)!\,(z+n)^{-k}$, the constants $-\frac{1}{n}$ disappear, and that of $q^r = e^{2\pi irz}$ is $(2\pi ir)^{k-1}q^r$. Hence $(-1)^{k-1}(k-1)!\sum_n(z+n)^{-k} = -2\pi i\sum_r(2\pi ir)^{k-1}q^r = -(2\pi i)^k\sum_r r^{k-1}q^r$, which is the Lipschitz formula. Separate the terms $m = 0$, which give $2\zeta(k)$, and use the symmetry $(m,n) \mapsto (-m,-n)$ ($k$ even):

$$\begin{aligned} G_k(z) &= 2\zeta(k) + 2\sum_{m\geq 1}\sum_{n\in\mathbb{Z}}(mz+n)^{-k} = 2\zeta(k) + \frac{2(2\pi i)^k}{(k-1)!}\sum_{m\geq 1}\sum_{r\geq 1} r^{k-1}q^{mr} \\ &= 2\zeta(k) + \frac{2(2\pi i)^k}{(k-1)!}\sum_{n\geq 1}\sigma_{k-1}(n)q^n. \end{aligned}$$

(In the first line, the inner sum for fixed $m$ is the Lipschitz formula at the point $mz$, with $e^{2\pi i r m z} = q^{mr}$ and $(-2\pi i)^k = (2\pi i)^k$ as $k$ is even; in the second, the pairs $(m,r)$ with $mr = n$ contribute $\sum_{r|n} r^{k-1} = \sigma_{k-1}(n)$.) With Euler's $\zeta(k) = -\frac{(2\pi i)^k B_k}{2\,k!}$ we get $2\zeta(k)\cdot(-\frac{2k}{B_k}) = \frac{2k(2\pi i)^k}{k!} = \frac{2(2\pi i)^k}{(k-1)!}$, so the displayed expansion is $2\zeta(k)E_k$. So $G_k$ is bounded on $\{\operatorname{Im} z \geq 1\}$, and since $G_k|_k\sigma = G_k$ for every $\sigma \in \mathrm{SL}_2(\mathbb{Z})$, it is holomorphic at the cusp; $B_4 = -\frac{1}{30}$ and $B_6 = \frac{1}{42}$ give 240 and $-504$. The transformation law of $E_2$ is Proposition 12.45 below, combined with $\frac{\eta'}{\eta} = \frac{\pi i}{12}E_2$: differentiating $\log\eta(-1/z) = \frac{1}{2}\log\frac{z}{i} + \log\eta(z)$ gives $z^{-2}\frac{\eta'}{\eta}(-1/z) = \frac{1}{2z} + \frac{\eta'}{\eta}(z)$, which is the stated identity. Then $\operatorname{Im}(-1/z) = y/|z|^2$ gives $E_2^*(-1/z) = z^2E_2^*(z) + \frac{6z}{\pi i} + \frac{3(z^2-|z|^2)}{\pi y} = z^2E_2^*(z)$, because $z^2 - |z|^2 = z(z-\bar z) = 2iyz$; and $E_2^*$ is invariant under $T$. As $S$ and $T$ generate $\mathrm{SL}_2(\mathbb{Z})$ (Theorem 12.4(iv)) and $\gamma \mapsto (\phi \mapsto \phi|_2\gamma)$ is an action (Lemma 12.21(i), valid for any function), $E_2^*|_2\gamma = E_2^*$ for all $\gamma$. □

Since $E_2$ rests on the transformation law of $\eta$, which we do not prove, the statements about $E_2$ carry the tag "given the transformation law of $\eta$" on the card.

**Proposition 12.33** ($\Delta$ and $j$). *(i)* $\Delta = (E_4^3 - E_6^2)/1728$ *is a nonzero cusp form of weight* 12 *for* $\mathrm{SL}_2(\mathbb{Z})$ *with integer coefficients,*

$$\Delta = \sum_{n\geq 1}\tau(n)q^n = q - 24q^2 + 252q^3 - 1472q^4 + 4830q^5 - 6048q^6 + \cdots,$$

*and* $\Delta(\Lambda_z) = (2\pi)^{12}\Delta(z)$, *where* $\Delta(\Lambda) = g_2^3 - 27g_3^2$ *is the discriminant of Chapter 9.*

*(ii)* $j = E_4^3/\Delta = 1728E_4^3/(E_4^3 - E_6^2)$ *is holomorphic on* $\mathcal{H}$, $\mathrm{SL}_2(\mathbb{Z})$*-invariant, equal to* $j(\Lambda_z)$ *of Chapter 9, and*

$$j(z) = q^{-1} + 744 + 196884q + 21493760q^2 + 864299970q^3 + 20245856256q^4 + \cdots,$$

*with all coefficients integers;* $j(i) = 1728$ *and* $j(\zeta_3) = 0$.

*(iii) (Jacobi)* $\Delta(z) = q\prod_{n\geq 1}(1-q^n)^{24} = \eta(z)^{24}$.

*Proof.* (i) $\Delta \in M_{12}$ by Proposition 12.23(d). Write $E_4 = 1 + 240A$, $E_6 = 1 - 504B$ with $A = \sum\sigma_3(n)q^n$, $B = \sum\sigma_5(n)q^n$. Then $E_4^3 - E_6^2 = 144(5A+7B) + 1728(100A^2 + 8000A^3 - 147B^2)$, since $3\cdot 240^2 = 1728\cdot 100$, $240^3 = 1728\cdot 8000$, $504^2 = 1728\cdot 147$ and $3\cdot 240 = 720$, $2\cdot 504 = 1008$. Moreover $d^5 - d^3 = d^3(d-1)(d+1)$ is divisible by 12 for every $d$, so $\sigma_5(n) \equiv \sigma_3(n) \pmod{12}$ and $5A + 7B \equiv 12A \equiv 0 \pmod{12}$. So $\Delta$ has integer coefficients, its constant term is 0 and $a_1 = (720 + 1008)/1728 = 1$; it is a cusp form because the only cusp of $\mathrm{SL}_2(\mathbb{Z})$ is $\infty$. The listed coefficients are computed from the expansions. In Chapter 9, $g_2 = 60G_4(\Lambda)$ and $g_3 = 140G_6(\Lambda)$; with $G_4 = 2\zeta(4)E_4 = \frac{\pi^4}{45}E_4$ and $G_6 = 2\zeta(6)E_6 = \frac{2\pi^6}{945}E_6$ this gives $g_2 = \frac{4}{3}\pi^4E_4$, $g_3 = \frac{8}{27}\pi^6E_6$ and $g_2^3 - 27g_3^2 = \frac{64}{27}\pi^{12}(E_4^3 - E_6^2) = \frac{64\cdot 1728}{27}\pi^{12}\Delta = (2\pi)^{12}\Delta$. (ii) $\Delta$ has no zero on $\mathcal{H}$ (Corollary 12.35(ii), which uses only the valence formula), so $j$ is holomorphic and, as a quotient of two forms of weight 12, invariant. $j(\Lambda_z) = 1728g_2^3/\Delta(\Lambda_z) = 1728\cdot\frac{64}{27}\pi^{12}E_4^3/((2\pi)^{12}\Delta) = E_4^3/\Delta$. Since $\Delta = q(1-24q+\cdots)$ has integer coefficients and leading coefficient 1, $1/\Delta \in q^{-1}\mathbb{Z}[[q]]$, so $j \in q^{-1}\mathbb{Z}[[q]]$; the coefficients shown were computed with PARI. $E_6(i) = 0$ because $E_6(-1/z) = z^6E_6(z)$ at $z = i$ gives $E_6(i) = -E_6(i)$; so $j(i) = 1728E_4(i)^3/E_4(i)^3 = 1728$, as $E_4(i) \neq 0$ (Corollary 12.35(ii)). Likewise $E_4(\zeta_3) = 0$: the element $ST = \left(\begin{smallmatrix}0 & -1\\ 1 & 1\end{smallmatrix}\right)$ fixes $\zeta_3$, so $E_4(\zeta_3) = (\zeta_3+1)^4E_4(\zeta_3)$, and $\zeta_3 + 1 = e^{\pi i/3}$ has $(\zeta_3+1)^4 = e^{4\pi i/3} \neq 1$. As $\Delta(\zeta_3) \neq 0$, $j(\zeta_3) = E_4(\zeta_3)^3/\Delta(\zeta_3) = 0$. (iii) Given the

transformation law of $\eta$ (Proposition 12.45), $\eta^{24}(z+1) = \eta^{24}(z)$ and $\eta^{24}(-1/z) = (z/i)^{12}\eta^{24}(z) = z^{12}\eta^{24}(z)$, so $\eta^{24} \in M_{12}(\mathrm{SL}_2(\mathbb{Z}))$ (as $S, T$ generate); its expansion $q\prod(1-q^n)^{24} = q - 24q^2 + \cdots$ has no constant term, so $\eta^{24} \in S_{12}(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}\Delta$ (Corollary 12.35(iii)), and comparing $a_1$ gives $\eta^{24} = \Delta$. □

**Theorem 12.34** (the valence formula for $\mathrm{SL}_2(\mathbb{Z})$)**.** *Let $f \neq 0$ be a meromorphic function on $\mathcal{H}$ with $f|_k\gamma = f$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, meromorphic at $\infty$ in the sense that $f(z) = \sum_{n\geq n_0} a_n q^n$ for $\operatorname{Im} z$ large. For $p \in \mathcal{H}$ let $v_p(f)$ be the order of $f$ at $p$ (it depends only on the orbit of $p$), and $v_\infty(f)$ the order in $q$ at $\infty$. Then*

$$v_\infty(f) + \frac{1}{2}v_i(f) + \frac{1}{3}v_{\zeta_3}(f) + \sum_{p\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H},\ p\neq i,\zeta_3} v_p(f) = \frac{k}{12}.$$

*Proof.* The order is constant on orbits because $f(\gamma z) = (cz+d)^k f(z)$ and $cz+d$ has no zeros on $\mathcal{H}$. Since $f = q^{v_\infty} \cdot$ (a nonzero power series) for $\operatorname{Im} z$ large, the zeros and poles of $f$ in $\mathcal{F}$ lie in a compact part $\operatorname{Im} z \leq Y_0$, so there are finitely many. Assume first that $f$ has no zeros or poles on the boundary of $\mathcal{F}$ except possibly at $i$, $\zeta_3$, $-\bar{\zeta}_3$. Let $Y > Y_0$ and let $\mathcal{C}$ be the boundary of the region obtained from $\{z \in \mathcal{F} : \operatorname{Im} z \leq Y\}$ by removing small discs of radius $r$ around $i$, $\zeta_3$ and $-\bar{\zeta}_3$, oriented counterclockwise. By the argument principle,

$$\frac{1}{2\pi i}\oint_{\mathcal{C}} \frac{f'}{f}\,dz = \sum_{p\neq i,\zeta_3} v_p(f),$$

the sum over the orbits with a representative in the interior of $\mathcal{F}$, one for each. We compute the left side piece by piece.

- The top segment, from $\frac{1}{2}+iY$ to $-\frac{1}{2}+iY$: in the variable $q$ it is a circle around 0 traversed clockwise, and $\frac{f'}{f}\,dz = \frac{dF/dq}{F}\,dq$ with $F(q) = f(z)$; the integral is $-v_\infty(f)$.
- The two vertical sides correspond under $T$ and are traversed in opposite directions, and $f(z+1) = f(z)$, so their contributions cancel.
- The two arcs of the unit circle, from $\zeta_3$ to $i$ (call it $A$) and from $i$ to $-\bar{\zeta}_3$ (call it $A'$), with small pieces removed at the ends: $S$ maps $A'$ onto $A$ with the orientation reversed. Differentiating $f(Sz) = z^k f(z)$ gives $\frac{f'}{f}(Sz)\,d(Sz) = \frac{f'}{f}(z)\,dz + k\frac{dz}{z}$, so $\int_{A'}\frac{f'}{f}\,dz = -\int_A(\frac{f'}{f}\,dz + k\frac{dz}{z})$ and the two arcs contribute $-k\int_A\frac{dz}{z}$. Along the unit circle from the argument $2\pi/3$ to $\pi/2$, $\int\frac{dz}{z} = i(\frac{\pi}{2} - \frac{2\pi}{3}) = -\frac{\pi i}{6}$; dividing by $2\pi i$ gives $\frac{k}{12}$ in the limit $r \to 0$.
- The small arc around $i$ is a half circle traversed clockwise; as $r \to 0$ it contributes $-\frac{1}{2}v_i(f)$. The small arcs around $\zeta_3$ and $-\bar{\zeta}_3$ each subtend the angle $\pi/3$ (the angle of $\mathcal{F}$ at its corners) and are traversed clockwise; they contribute $-\frac{1}{6}v_{\zeta_3}(f) - \frac{1}{6}v_{-\bar{\zeta}_3}(f) = -\frac{1}{3}v_{\zeta_3}(f)$.

Adding up and letting $r \to 0$ gives the formula. If $f$ has a zero or pole $p$ on a vertical side, it has one at the $T$-equivalent point on the other side: deform the contour by a small arc going into $\mathcal{F}$ around one of them and out of $\mathcal{F}$ around the other, so that exactly one of the two is counted and the $T$-symmetry of the contour is kept; likewise for an $S$-symmetric pair of points on the arc. □

Physically this is Gauss's law for the "charge density" $\frac{1}{2\pi i}\frac{f'}{f}$: the residue theorem counts the zeros inside the contour, and the symmetries $T$ and $S$ make the flux through the sides cancel, except for the defect $k\frac{dz}{z}$ produced by the factor of automorphy, which contributes the constant $\frac{k}{12}$.

**Corollary 12.35** (the structure of $M_*(\mathrm{SL}_2(\mathbb{Z}))$)**.** *(i)* $M_k(\mathrm{SL}_2(\mathbb{Z})) = 0$ *for* $k < 0$*, for* $k$ *odd, and for* $k = 2$*;* $M_0(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}$*;* $M_k(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}E_k$ *for* $k = 4, 6, 8, 10, 14$*.*

*(ii)* $\Delta$ *has no zero on* $\mathcal{H}$ *and a simple zero at* $\infty$*;* $E_4$ *vanishes only on the orbit of* $\zeta_3$*, simply, and* $E_6$ *only on the orbit of* $i$*, simply.*

*(iii) Multiplication by* $\Delta$ *is an isomorphism* $M_{k-12}(\mathrm{SL}_2(\mathbb{Z})) \to S_k(\mathrm{SL}_2(\mathbb{Z}))$*, and for even* $k \geq 0$*,* $\dim M_k(\mathrm{SL}_2(\mathbb{Z})) = \lfloor k/12 \rfloor$ *if* $k \equiv 2 \pmod{12}$ *and* $\lfloor k/12 \rfloor + 1$ *otherwise.*

*(iv)* $\bigoplus_k M_k(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}[E_4, E_6]$*, with* $E_4$ *and* $E_6$ *algebraically independent.*

*Proof.* For $f \in M_k$ nonzero all orders are $\geq 0$, so the valence formula reads $n + \frac{a}{2} + \frac{b}{3} = \frac{k}{12}$ with integers $n, a, b \geq 0$ ($n$ collecting $v_\infty$ and the orders at the other orbits, $a = v_i$, $b = v_{\zeta_3}$). (i) For $k < 0$ there is no solution, and for $k = 2$ there is none ($\frac{a}{2} + \frac{b}{3} = \frac{1}{6}$ is impossible). Odd $k$ is Proposition 12.23(b). For $k = 0$, $f - f(z_0)$ has a zero, so it is 0: $f$ is constant. For $k \in \{4, 6, 8, 10, 14\}$ every solution has $v_\infty = 0$ (for $k < 12$ because $v_\infty \leq k/12$, for $k = 14$ because $\frac{a}{2} + \frac{b}{3} = \frac{1}{6}$ is impossible), so $f \mapsto a_0(f)$ is injective on $M_k$, and $E_k \neq 0$. (ii) For $\Delta$, $v_\infty \geq 1$ and the formula with $k = 12$ forces $v_\infty = 1$ and no other zero. For $E_4$ the only solution of $n + \frac{a}{2} + \frac{b}{3} = \frac{1}{3}$ is $b = 1$; for $E_6$ the only solution of $= \frac{1}{2}$ is $a = 1$. (iii) If $f \in S_k$, then $f/\Delta$ is holomorphic on $\mathcal{H}$ by (ii), of weight $k-12$, and its order at $\infty$ is $v_\infty(f) - 1 \geq 0$; so $f/\Delta \in M_{k-12}$. For $k \geq 4$ even, $M_k = \mathbb{C}E_k \oplus S_k$ since $a_0(E_k) = 1$. The dimension formula follows by induction on $k$ from (i) and $\dim M_k = 1 + \dim M_{k-12}$ for $k \geq 4$ (with $M_{k-12}$ for $k = 12$ being $M_0 = \mathbb{C}$). (iv) For even $k \geq 4$ choose $a, b \geq 0$ with $4a + 6b = k$; then $E_4^a E_6^b \in M_k$ has constant term 1, so $f - a_0(f)E_4^a E_6^b \in S_k = \Delta M_{k-12}$, and $\Delta = (E_4^3 - E_6^2)/1728$; induction on $k$ shows $M_k \subseteq \mathbb{C}[E_4, E_6]$. If a nontrivial polynomial relation held, a homogeneous component of it of some weight $k$ would be a relation $\sum_{4a+6b=k} c_{ab} E_4^a E_6^b = 0$; dividing by a suitable power of $E_6$ (or $E_4$) gives a nonzero polynomial satisfied by the function $E_4^3/E_6^2$, which is then constant; but $E_4^3/E_6^2$ vanishes at $\zeta_3$ and is 1 at $\infty$. □

These dimensions agree with PARI's `mfdim([1,k],4)` for all even $k \leq 26$.

**Theorem 12.36** ($j$ is a bijection)**.** *The function* $j$ *induces a bijection* $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H} \to \mathbb{C}$*, and extends to an isomorphism of Riemann surfaces* $X(1) \to \mathbb{P}^1(\mathbb{C})$ *with* $j(\infty) = \infty$*. Consequently, for every* $j_0 \in \mathbb{C}$ *there is a lattice* $\Lambda \subset \mathbb{C}$ *with* $j(\Lambda) = j_0$*.*

*Proof.* For $\lambda \in \mathbb{C}$ the form $f_\lambda = E_4^3 - \lambda\Delta \in M_{12}$ has constant term 1, so $v_\infty(f_\lambda) = 0$, and the valence formula reads $n + \frac{a}{2} + \frac{b}{3} = 1$. Its solutions in nonnegative integers are $(1, 0, 0)$, $(0, 2, 0)$, $(0, 0, 3)$, so $f_\lambda$ vanishes on exactly one orbit. As $\Delta$ has no zero on $\mathcal{H}$, $j(z) = \lambda$ iff $f_\lambda(z) = 0$; so $j$ takes every value exactly once on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$. By Theorem 12.15(ii) the invariant holomorphic function $j$ is holomorphic on $Y(1)$, and at $\infty$ it has a simple pole in the chart $q$. So $j\colon X(1) \to \mathbb{P}^1(\mathbb{C})$ is a bijective holomorphic map of compact Riemann surfaces, hence of degree 1 and an isomorphism (Chapter 8). Finally $j(z) = j(\Lambda_z)$ (Proposition 12.33(ii)). □

*In the paper* 12.37. Chapter 9 proves that lattices up to homothety correspond to $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$ and to elliptic curves over $\mathbb{C}$, and left the bijectivity of $j$ and the $q$-expansion of $j$ to this chapter. They are Theorem 12.36 and Proposition 12.33(ii); nothing here uses the uniformisation theorem, so Chapter 9 may use Theorem 12.36 as a second proof that every elliptic curve over $\mathbb{C}$ is a complex torus. The paper uses $j$ only through $v_q(j_E)$ (Definition def:terms, with the paper's $q$ a prime there): the algebraic $j$-invariant of Chapter 9, which equals $j(\tau)$ at any $\tau$ with $E(\mathbb{C}) \simeq \mathbb{C}/\Lambda_\tau$. The Tate curve of Chapter 10 is the `q`-adic analogue of the expansion of $j$ in $q$.

The bijectivity of $j$ also describes the function field of $X_0(N)$, which Chapter 14 uses to put $X_0(N)$ over $\mathbb{Q}$. Meromorphic functions on $X_0(N)$ are identified with $\Gamma_0(N)$-invariant meromorphic functions on $\mathcal{H}$ that are meromorphic in the charts $q_h$ at the cusps, and by Riemann existence and GAGA (Chapter 8) they form the function field $\mathbb{C}(X_0(N))$ of the curve. Write $\delta_N = \left(\begin{smallmatrix} N & 0 \\ 0 & 1 \end{smallmatrix}\right)$, so that $j(Nz) = j(\delta_N z)$.

**Proposition 12.38** (the function field of $X_0(N)$)**.** *Let $N \geq 1$, $j_N(z) = j(Nz)$, and $\mathrm{SL}_2(\mathbb{Z}) = \bigsqcup_{i=1}^{\mu(N)} \Gamma_0(N)\alpha_i$ with $\alpha_1 = I$.*

*(i) $j_N$ is a meromorphic function on $X_0(N)$, holomorphic on $Y_0(N)$.*

*(ii) Every meromorphic function on $X(1)$ is a rational function of $j$: $\mathbb{C}(X(1)) = \mathbb{C}(j)$; through $\pi$ of Theorem 12.17, $\mathbb{C}(j) \subseteq \mathbb{C}(X_0(N))$.*

*(iii) The functions $f_i(z) = j(\delta_N \alpha_i z)$, $1 \leq i \leq \mu(N)$, are pairwise distinct and are permuted by $z \mapsto \gamma z$, $\gamma \in \mathrm{SL}_2(\mathbb{Z})$; the polynomial $\Phi(X) = \prod_i (X - f_i)$ has coefficients in $\mathbb{C}(j)$ and is the minimal polynomial of $j_N = f_1$ over $\mathbb{C}(j)$. So $[\mathbb{C}(j, j_N) : \mathbb{C}(j)] = \mu(N)$.*

*(iv) $\mathbb{C}(X_0(N)) = \mathbb{C}(j, j_N)$.*

*Proof.* (i) For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$, $\delta_N \gamma = \gamma' \delta_N$ with $\gamma' = \left(\begin{smallmatrix} a & Nb \\ c/N & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, so $j_N(\gamma z) = j(\gamma' \delta_N z) = j_N(z)$. So $j_N$ is a $\Gamma_0(N)$-invariant holomorphic function on $\mathcal{H}$, holomorphic on $Y_0(N)$ by Theorem 12.15(ii). At a cusp $\sigma\infty$ of width $h$, write $\delta_N \sigma = \sigma' \beta$ with $\sigma' \in \mathrm{SL}_2(\mathbb{Z})$ and $\beta = \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$, $ad = N$ (as in the proof of Proposition 12.23(e)). Then $j_N(\sigma z) = j(\beta z) = \sum_{n \geq -1} c(n) e^{2\pi i n b/d} e^{2\pi i n a z/d}$, with $c(n)$ the coefficients of $j$ (Proposition 12.33); this is invariant under $z \mapsto z + h$ (as $\pm \sigma T^h \sigma^{-1} \in \Gamma_0(N)$), so only exponents in $\frac{1}{h}\mathbb{Z}$ occur, and it is a Laurent series in $q_h$ with finitely many negative terms: $j_N$ is meromorphic at the cusp. (ii) By Theorem 12.36, $j \colon X(1) \to \mathbb{P}^1(\mathbb{C})$ is an isomorphism, and the meromorphic functions on $\mathbb{P}^1(\mathbb{C})$ are the rational functions (Chapter 8). Composition with $\pi$ embeds $\mathbb{C}(X(1))$ in $\mathbb{C}(X_0(N))$. (iii) Replacing $\alpha_i$ by $\gamma\alpha_i$, $\gamma \in \Gamma_0(N)$, does not change $f_i$, by (i). For $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, $\Gamma_0(N)\alpha_i\gamma = \Gamma_0(N)\alpha_{s(i)}$ for a permutation $s$ of the cosets, and $f_i(\gamma z) = f_{s(i)}(z)$; the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\Gamma_0(N)\backslash \mathrm{SL}_2(\mathbb{Z})$ by right multiplication is transitive. *Distinctness.* Suppose $f_i = f_l$. For each $z \in \mathcal{H}$, $j(\delta_N \alpha_i z) = j(\delta_N \alpha_l z)$, so by Theorem 12.36 there is $g \in \mathrm{SL}_2(\mathbb{Z})$ with $\delta_N \alpha_i z = g \delta_N \alpha_l z$. For fixed $g$ the set of such $z$ is closed, and $\mathrm{SL}_2(\mathbb{Z})$ is countable, so by Baire's theorem one of these sets has an interior point. Two Möbius transformations that agree on an open set are equal, and the matrices of $\mathrm{GL}_2^+(\mathbb{R})$ inducing a given one differ by a real scalar; comparing determinants, $\delta_N \alpha_i = \pm g \delta_N \alpha_l$. Then $\alpha_i \alpha_l^{-1} = \pm \delta_N^{-1} g \delta_N = \pm \left(\begin{smallmatrix} g_{11} & g_{12}/N \\ N g_{21} & g_{22} \end{smallmatrix}\right)$, which lies in $\mathrm{SL}_2(\mathbb{Z})$ and has lower left entry divisible by $N$, so $\alpha_i \alpha_l^{-1} \in \Gamma_0(N)$ and $i = l$. *Coefficients.* The coefficients of $\Phi$ are the elementary symmetric functions of the $f_i$: holomorphic on $\mathcal{H}$, invariant under $\mathrm{SL}_2(\mathbb{Z})$ (which permutes the $f_i$), and Laurent series in $e^{2\pi i z/N}$ with finitely many negative terms (by the computation in (i), $f_i(z) = j(\beta_i z)$ with $\beta_i = \left(\begin{smallmatrix} a_i & b_i \\ 0 & d_i \end{smallmatrix}\right)$, $a_i d_i = N$, and $a_i/d_i = a_i^2/N$), hence, being invariant under $z \mapsto z + 1$, Laurent series in $q$. So they are meromorphic functions on $X(1)$, and lie in $\mathbb{C}(j)$ by (ii). *Irreducibility.* Let $P \in \mathbb{C}(j)[X]$ with $P(f_1) = 0$. Composing with $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ fixes the coefficients of $P$, so $P(f_{s(1)}) = 0$; by transitivity every $f_i$ is a root, and as the $f_i$ are distinct, $\deg P \geq \mu(N)$. Hence $\Phi$ is the minimal polynomial of $f_1 = j_N$ and $[\mathbb{C}(j, j_N) : \mathbb{C}(j)] = \mu(N)$. (iv) The holomorphic map $j \circ \pi \colon X_0(N) \to \mathbb{P}^1(\mathbb{C})$ has degree $\deg \pi \cdot \deg j = \mu(N) \cdot 1$ (Theorems 12.17 and 12.36), and the degree of a nonconstant function $F$ on a curve $C$, viewed as a map to $\mathbb{P}^1$, is $[\mathbb{C}(C) : \mathbb{C}(F)]$ (Chapter 8). So $[\mathbb{C}(X_0(N)) : \mathbb{C}(j)] = \mu(N)$, and $\mathbb{C}(j, j_N) \subseteq \mathbb{C}(X_0(N))$ has the same degree over $\mathbb{C}(j)$ by (iii); the two fields are equal. $\square$

The polynomial $\Phi$, written as a polynomial $\Phi_N(j, X)$ in two variables, is the classical modular polynomial; for $N = 2, 3, 5$ PARI's `polmodular` has degree $3, 4, 6 = \mu(N)$ in each variable. That $\Phi_N \in \mathbb{Z}[X, Y]$ (Kronecker), and the resulting model of $X_0(N)$ over $\mathbb{Q}$ with function field $\mathbb{Q}(j, j_N)$, are in Chapter 14.

## 12.8 Forms on $\Gamma_0(N)$: valence, differentials and dimensions

This section passes from level one to $\Gamma_0(N)$. Its central statement is that weight-two cusp forms are exactly the holomorphic differentials on $X_0(N)$, so that $\dim S_2(\Gamma_0(N))$ is the genus. This is how the paper's newform $f$ becomes the differential $f(z)\,dz$ that the modular parametrisation pulls back from $E$ (Chapter 14), and how $S_2(\Gamma_0(N)) \oplus \overline{S_2(\Gamma_0(N))}$ becomes $H^1(X_0(N), \mathbb{C})$ in the proof of Proposition prop:pullback. We also state the valence formula and the dimension formulas in every weight, and construct the weight-two Eisenstein series, which gives the Eisenstein congruence for `11a1`.

**Theorem 12.39** (the valence formula for $\Gamma_0(N)$)**.** *Let $f \in M_k(\Gamma_0(N))$, $f \neq 0$. For $P \in X_0(N)$ let $v_P(f)$ be the order of $f$ at a representative $z \in \mathcal{H}$ (in the variable $z$) if $P \in Y_0(N)$, and the order in $q_h$ of the expansion of $f$ at the cusp $P$ if $P$ is a cusp; let $e_P \in \{1, 2, 3\}$ be the order of the stabiliser of $z$ in $\bar{\Gamma}_0(N)$, and $e_P = 1$ at the cusps. Then*

$$\sum_{P \in X_0(N)} \frac{v_P(f)}{e_P} = \frac{k\,\mu(N)}{12}.$$

*In particular $v_\infty(f) \leq k\mu(N)/12$, so $f$ is determined by $a_0(f), \ldots, a_{B_k(N)}(f)$.*

*Sketch of proof.* Let $\mathrm{SL}_2(\mathbb{Z}) = \bigsqcup_{i=1}^{\mu} \Gamma_0(N)\gamma_i$ and $F = \prod_i f|_k\gamma_i$. Replacing $\gamma_i$ by $\delta\gamma_i$ ($\delta \in \Gamma_0(N)$) does not change $f|_k\gamma_i$, and right multiplication by $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ permutes the cosets, so $F|_{k\mu}\gamma = F$; $F$ is holomorphic at $\infty$ because each $f|_k\gamma_i$ is, so $F \in M_{k\mu}(\mathrm{SL}_2(\mathbb{Z}))$ and Theorem 12.34 applies to it. Its order at a point $Q$ of $X(1)$ is the sum of the orders of the $f|_k\gamma_i$ there, and grouping the $\gamma_i$ according to the points $P$ of $X_0(N)$ over $Q$, each $P$ contributes $v_P(f)$ times $e_Q/e_P$ (at an elliptic point) or $v_P(f)$ (at a cusp, where the order in $q$ of $f|_k\sigma$ is $v_P(f)/h$ and the width $h$ is the number of cosets over $P$). Comparing with the weights $\frac{1}{2}$, $\frac{1}{3}$, 1 of the level-one formula gives the result. For $k = 2$ one can instead use Theorem 12.40 and $\deg \mathrm{div}(\omega) = 2g - 2$ (Chapter 8) with the corrections at the cusps and elliptic points. See [52, Chapter 3, exercises of Section 3.1], [180, Section 2.4] and [187, Chapter 9] (locations not verified). □

For example $f_{11} = \eta(z)^2\eta(11z)^2$ (Theorem 12.46) has simple zeros at the cusps $\infty$ and 0 and no other zeros: $1 + 1 = 2 \cdot 12/12$.

**Theorem 12.40** (weight-two forms are differentials)**.** *For $f\colon \mathcal{H} \to \mathbb{C}$ holomorphic put $\omega_f = f(z)\,dz$.*

*(i) $f \mapsto \omega_f$ is an isomorphism from $S_2(\Gamma_0(N))$ onto the space $H^0(X_0(N), \Omega^1)$ of holomorphic differentials on $X_0(N)$.*

*(ii) It maps $M_2(\Gamma_0(N))$ isomorphically onto the space of meromorphic differentials on $X_0(N)$ that are holomorphic on $Y_0(N)$ and have at most simple poles at the cusps; the residue at the cusp $s = \sigma_s\infty$ of width $h_s$ is $\mathrm{res}_s\,\omega_f = \frac{h_s}{2\pi i}\,b_0(f|_2\sigma_s)$.*

*(iii) Hence* $\dim S_2(\Gamma_0(N)) = g(X_0(N))$, $\dim M_2(\Gamma_0(N)) = g(X_0(N)) + \nu_\infty(N) - 1$ *(which is* 0 *for* $N = 1$*), and* $\sum_s h_s\, b_0(f|_2\sigma_s) = 0$ *for every* $f \in M_2(\Gamma_0(N))$*, the sum over the cusps.*

*Proof.* A meromorphic differential on $X_0(N)$ restricts to a $\Gamma_0(N)$-invariant meromorphic differential $g(z)\,dz$ on $\mathcal{H}$ minus the elliptic points, and by Lemma 12.21(iii), $\gamma^*(g\,dz) = g\,dz$ for all $\gamma \in \Gamma_0(N)$ iff $g|_2\gamma = g$. It remains to compare the local conditions. *Elliptic points.* Let $z_0$ have stabiliser of order $e$ in $\bar\Gamma_0(N)$, let $w$ be the Cayley coordinate at $z_0$, in which the stabiliser acts by $w \mapsto \zeta w$, $\zeta = e^{2\pi i/e}$, and $t = w^e$ the chart (Theorem 12.15). An invariant differential $F(w)\,dw$ with $F$ holomorphic at 0 satisfies $F(\zeta w)\zeta = F(w)$; writing $F = \sum c_n w^n$ gives $c_n\zeta^{n+1} = c_n$, so only $n \equiv -1 \pmod e$ occur and $F(w) = w^{e-1}G(w^e)$ with $G$ holomorphic. Then $F(w)\,dw = w^{e-1}G(t)\,dw = \frac{1}{e}G(t)\,dt$ is holomorphic in $t$. So in weight 2 there is no condition at the elliptic points: a holomorphic $f$ gives a differential holomorphic there. *Cusps.* At a cusp $s = \sigma\infty$ of width $h$, pull back by $\sigma$: $\sigma^*(\omega_f) = (f|_2\sigma)(z)\,dz$ and $dz = \frac{h}{2\pi i}\frac{dq_h}{q_h}$. If $f|_2\sigma = \sum_{n\ge n_0} b_n q_h^n$ then $\omega_f = \frac{h}{2\pi i}\sum b_n q_h^{n-1}\,dq_h$ near $s$. This is holomorphic iff $b_n = 0$ for $n \le 0$, that is, iff $f$ vanishes at $s$, and has at most a simple pole iff $b_n = 0$ for $n < 0$, that is, iff $f$ is holomorphic at $s$; the residue is $\frac{h}{2\pi i}b_0$. Together: holomorphic differentials correspond exactly to cusp forms, and differentials holomorphic on $Y_0(N)$ with at most simple poles at the cusps to modular forms. This is (i) and (ii). For (iii), $\dim H^0(X,\Omega^1) = g$ (Chapter 8). Let $D$ be the sum of the $\nu_\infty \ge 1$ cusps and $K$ a canonical divisor. By Riemann-Roch (Chapter 8) the space of meromorphic differentials $\omega$ with $\operatorname{div}\omega \ge -D$ has dimension $\deg(K+D) - g + 1 + \ell(-D) = (2g-2+\nu_\infty) - g + 1 + 0 = g + \nu_\infty - 1$, as $\deg(-D) < 0$. The relation between the constant terms is the residue theorem $\sum_s \operatorname{res}_s \omega_f = 0$ (Chapter 8). □

For instance $S_2(\mathrm{SL}_2(\mathbb{Z})) = 0$ and $g(X(1)) = 0$; $\dim S_2(\Gamma_0(11)) = 1$, $\dim S_2(\Gamma_0(37)) = 2$, $\dim S_2(\Gamma_0(121)) = 6$, $\dim S_2(\Gamma_0(539)) = 49$; PARI agrees.

**Theorem 12.41** (dimension formulas)**.** *Let* $g = g(X_0(N))$, $\nu_2 = \nu_2(N)$, $\nu_3 = \nu_3(N)$, $\nu_\infty = \nu_\infty(N)$. *For even* $k \ge 4$,

$$\dim S_k(\Gamma_0(N)) = (k-1)(g-1) + \Big(\frac{k}{2} - 1\Big)\nu_\infty + \Big\lfloor\frac{k}{4}\Big\rfloor\nu_2 + \Big\lfloor\frac{k}{3}\Big\rfloor\nu_3,$$

$$\dim M_k(\Gamma_0(N)) = \dim S_k(\Gamma_0(N)) + \nu_\infty.$$

*For* $k = 2$*:* $\dim S_2 = g$ *and* $\dim M_2 = g + \nu_\infty - 1$*. Moreover* $M_0(\Gamma_0(N)) = \mathbb{C}$*,* $S_0(\Gamma_0(N)) = 0$*, and* $M_k(\Gamma_0(N)) = 0$ *for* $k < 0$ *or* $k$ *odd.*

*Proof.* For $k = 2$ this is Theorem 12.40(iii). For $k$ odd use Proposition 12.23(b). For $k \le 0$: a nonzero $f \in M_k$ has all $v_P \ge 0$, so by Theorem 12.39 $k \ge 0$, and for $k = 0$ it has no zeros, so $f - f(z_0)$ vanishes identically. For even $k \ge 4$ we give only a sketch; see [52, Theorems 3.5.1 and 3.6.1] (numbers not verified) and [180, Section 2.6]. The map $f \mapsto f\,(dz)^{k/2}$ identifies $M_k(\Gamma_0(N))$ with the sections of a line bundle on $X_0(N)$: the $\frac{k}{2}$-fold differentials whose orders are at least $-\frac{k}{2}$ at each cusp and at least $-\lfloor\frac{k}{2}(1-\frac{1}{e})\rfloor$ at an elliptic point of order $e$ (the local computation of Theorem 12.40 in the $\frac{k}{2}$-th tensor power). This line bundle has degree $\frac{k}{2}(2g-2) + \frac{k}{2}\nu_\infty + \lfloor\frac{k}{4}\rfloor\nu_2 + \lfloor\frac{k}{3}\rfloor\nu_3$, which exceeds $2g-2$, so Riemann-Roch gives its dimension $\deg - g + 1$, and similarly for cusp forms with $\frac{k}{2} - 1$ in place of $\frac{k}{2}$ at the cusps. □

The formulas were checked against PARI's `mfdim([N,k],1)` and `mfdim([N,k],4)` for all $N \le 60$ and $N \in \{77, 99, 121, 539\}$, $k \in \{2, 4, \dots, 14\}$. Two instances by hand. At level one

($g=0$, $\nu_2=\nu_3=\nu_\infty=1$) and $k=12$: $11\cdot(-1)+5+3+4=1$, the line $\mathbb{C}\Delta$ of Corollary 12.35. At level 11 ($g=1$, $\nu_2=\nu_3=0$, $\nu_\infty=2$) and $k=4$: $3\cdot 0+1\cdot 2=2$, so $\dim S_4(\Gamma_0(11))=2$ and $\dim M_4(\Gamma_0(11))=4$. The floors $\lfloor\frac{k}{4}\rfloor$ and $\lfloor\frac{k}{3}\rfloor$ are the elliptic corrections: a form of weight $k$ is forced to vanish at an elliptic point of order $e$ to an order that depends on $k$ modulo $2e$, as the case $E_6(i)=0$ of Proposition 12.33 illustrates.

**Proposition 12.42** (the weight-two Eisenstein series of level $N$)**.** *For $N>1$ the function $E_2^{(N)}(z)=E_2(z)-NE_2(Nz)$ lies in $M_2(\Gamma_0(N))$, is not a cusp form, and*

$$E_2^{(N)}=(1-N)-24\sum_{n\geq 1}(\sigma(n)-N\sigma(n/N))q^n,\qquad \sigma=\sigma_1,\ \sigma(n/N)=0 \text{ if } N\nmid n.$$

*For $N=p$ prime, $M_2(\Gamma_0(p))=\mathbb{C}E_2^{(p)}\oplus S_2(\Gamma_0(p))$.*

*Proof.* With $E_2^*=E_2-\frac{3}{\pi y}$ (Proposition 12.32), $E_2^*(Nz)=E_2(Nz)-\frac{3}{\pi Ny}$, so $E_2^{(N)}=E_2^*-NE_2^*(Nz)$, and $NE_2^*(Nz)=(E_2^*|_2\alpha)(z)$ for $\alpha=\left(\begin{smallmatrix}N&0\\0&1\end{smallmatrix}\right)$ (Definition 12.20: $(\det\alpha)^1\cdot 1^{-2}=N$). For $\gamma\in\Gamma_0(N)$, $\alpha\gamma\alpha^{-1}=\left(\begin{smallmatrix}a&Nb\\c/N&d\end{smallmatrix}\right)\in\mathrm{SL}_2(\mathbb{Z})$, so $E_2^*|_2\alpha|_2\gamma=E_2^*|_2\alpha$ by the invariance of $E_2^*$; hence $E_2^{(N)}|_2\gamma=E_2^{(N)}$. At a cusp, for $\sigma\in\mathrm{SL}_2(\mathbb{Z})$ write $\alpha\sigma=\sigma'\beta$ with $\sigma'\in\mathrm{SL}_2(\mathbb{Z})$ and $\beta=\left(\begin{smallmatrix}a'&b'\\0&d'\end{smallmatrix}\right)$, $a'd'=N$, as in Proposition 12.23(e). Then $E_2^{(N)}|_2\sigma=E_2^*-E_2^*|_2\beta$, and $(E_2^*|_2\beta)(z)=\frac{a'}{d'}E_2(\frac{a'z+b'}{d'})-\frac{a'}{d'}\cdot\frac{3d'}{\pi a'y}$, so the non-holomorphic terms cancel and $E_2^{(N)}|_2\sigma=E_2(z)-\frac{a'}{d'}E_2(\frac{a'z+b'}{d'})$, a power series in $q^{1/d'}$ without negative powers, bounded on $\{\mathrm{Im}\,z\geq 1\}$. This is the step where $E_2^*$ earns its place: $E_2^*$ is modular but not holomorphic, $E_2$ is holomorphic but not modular, and the combination $E_2^*-E_2^*|_2\alpha$ is both, because the two correction terms $\frac{3}{\pi y}$ and $N\cdot\frac{3}{\pi Ny}$ cancel. The expansion comes from substituting $Nz$ for $z$ in that of $E_2$: $NE_2(Nz)=N-24N\sum_{m\geq 1}\sigma(m)q^{Nm}$, and the term $q^{Nm}$ is $q^n$ with $m=n/N$. So $a_0=1-N\neq 0$. For $N=p$, $\nu_\infty=2$, so $\dim M_2=g+1$ (Theorem 12.40), and $E_2^{(p)}\notin S_2$. □

At the two cusps of $\Gamma_0(p)$ the constant terms are $1-p$ at $\infty$ (width 1) and $1-\frac{1}{p}$ at 0 (width $p$; here $\sigma=S$, $\beta=\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)$), and $1\cdot(1-p)+p(1-\frac{1}{p})=0$, as Theorem 12.40(iii) requires.

**Example 12.43** (the Eisenstein congruence for `11a1`)**.** Let $f_{11}$ be the normalised generator of $S_2(\Gamma_0(11))$ (Theorem 12.46). Then for every $n\geq 1$,

$$a_n(f_{11})\equiv\sigma(n)-11\sigma(n/11)\pmod 5,$$

and in particular $a_\ell(\texttt{11a1})\equiv 1+\ell\pmod 5$ for every prime $\ell\neq 11$, and $a_{11}=1$. Indeed $h=12(f_{11}+\frac{1}{24}E_2^{(11)})=-5+12\sum_{n\geq 1}(a_n(f_{11})-\sigma(n)+11\sigma(n/11))q^n$ lies in $M_2(\Gamma_0(11);\mathbb{Z})$: the constant term is $\frac{12}{24}(1-11)=-5$, and the $n$-th coefficient is $12a_n(f_{11})+\frac{12}{24}\cdot(-24)(\sigma(n)-11\sigma(n/11))$. The multiple $\frac{1}{24}E_2^{(11)}$ is chosen so that the first coefficients cancel ($a_1=1-1=0$), and the factor 12 makes the constant term $\frac{1-11}{24}=-\frac{5}{12}$ an integer. Its coefficients at $n=0,1,2$ are $-5$, $12(1-1)=0$ and $12(-2-3)=-60$, all divisible by 5. The Sturm bound is $B_2(11)=\lfloor 12/6\rfloor=2$, so by Theorem 12.58 every coefficient of $h$ is divisible by 5, and $12(a_n-\sigma(n)+11\sigma(n/11))\equiv 0$ gives the claim since $5\nmid 12$. (Checked directly for $n\leq 5000$; the identification of $f_{11}$ with the form of `11a1` at every $n$ is Theorem 12.46(iv).) This congruence is the modular face of the rational point of order 5 on `11a1` (Chapter 9): if $E(\mathbb{Q})$ has a point of order 5, then $5\mid\#\tilde{E}(\mathbb{F}_\ell)=\ell+1-a_\ell$ for every good $\ell\neq 5$, because reduction is injective on torsion of order prime to $\ell$.

*In the paper* 12.44. The paper's Conjecture conj:eis and Section sec:eis treat the Eisenstein case, where $\bar\rho$ is reducible and the newform $f$ is congruent modulo $\ell$ to an Eisenstein series. Example 12.43 is the simplest instance: for `11a1` and $\ell = 5$, $\bar\rho_{E,5}$ is reducible, $\deg\varphi_E = 1$, the component group at 11 (type $\mathrm{I}_5$) has order 5, and the deficit $D = 1$ of Conjecture conj:eis is matched by $v_5(\#E(\mathbb{Q})_{\mathrm{tors}}) = 1$. Open question 4 of the paper asks to identify the deficit with an invariant of the Eisenstein ideal, for prime level the ideal of $\mathbb{T}$ generated by the $T_p - 1 - p$, $p \nmid N$, and $1 + w_N$ ([125], Chapter 14).

## 12.9 The Dedekind $\eta$-function and the level 11 example

The running example of the paper is the curve `11a1`, and its newform is the most explicit modular form there is: $\eta(z)^2\eta(11z)^2$. This section proves that this product spans $S_2(\Gamma_0(11))$, computes its Atkin-Lehner sign, and lists its first coefficients next to the numbers $a_p$ of `11a1`. The same form reappears in the paper's Example of Section 3 (the congruences with $f_{\mathtt{11a}} \otimes \chi_{-11}$ and $f_{\mathtt{11a}} \otimes \chi_{-7}$), in the Eisenstein congruence of Example 12.43, and in the Petersson computations of Example 12.53.

**Proposition 12.45** (the Dedekind $\eta$-function)**.** *The function* $\eta(z) = q^{1/24}\prod_{n\ge1}(1-q^n)$, $q^{1/24} = e^{2\pi iz/24}$, *is holomorphic and has no zeros on* $\mathcal{H}$, *and*

$$\eta(z+1) = e^{\pi i/12}\eta(z), \qquad \eta(-1/z) = \sqrt{z/i}\,\eta(z),$$

*with the branch of the square root that is positive on the imaginary axis. Its logarithmic derivative is* $\frac{\eta'}{\eta} = \frac{\pi i}{12}E_2$, *and* $\eta^{24} = \Delta$.

*Proof.* The product converges absolutely and locally uniformly since $\sum_n |q|^n < \infty$, and no factor vanishes, so $\eta$ is holomorphic without zeros; the first law is clear. The second we do not prove; see [6, Chapter 3], [102, Section III.2] (the usual proof applies the residue theorem to a suitable cotangent product, or uses the modularity of $E_2$ proved independently). The logarithmic derivative is $\frac{2\pi i}{24} - \sum_{n\ge1}\frac{2\pi i n q^n}{1-q^n} = \frac{2\pi i}{24}(1 - 24\sum_{n\ge1}\sum_{m\ge1} nq^{nm}) = \frac{\pi i}{12}E_2$. The identity $\eta^{24} = \Delta$ is Proposition 12.33(iii). □

**Theorem 12.46** ($S_2(\Gamma_0(11))$)**.** *Let* $f_{11}(z) = \eta(z)^2\eta(11z)^2 = q\prod_{n\ge1}(1-q^n)^2(1-q^{11n})^2$.

*(i)* $f_{11} \in S_2(\Gamma_0(11))$; *it has simple zeros at the cusps* $\infty$ *and* 0 *and no zeros on* $\mathcal{H}$.

*(ii)* $S_2(\Gamma_0(11)) = \mathbb{C}f_{11}$, *and* $f_{11}$ *is its normalised generator.*

*(iii)* $f_{11}|_2 W_{11} = -f_{11}$.

*(iv) The coefficients of* $f_{11}$ *are the numbers* $a_n$ *of the curve* `11a1`: $y^2 + y = x^3 - x^2 - 10x - 20$, *that is,* $a_p(f_{11}) = p + 1 - \#\tilde E(\mathbb{F}_p)$ *for* $p \ne 11$ *and* $a_{11}(f_{11}) = 1$, *and* $\sum_n a_n(f_{11})n^{-s} = L(\mathtt{11a1}, s)$.

*Proof.* (ii) and the zeros, for a generator $h$ of $S_2(\Gamma_0(11))$. By Theorem 12.17 and Theorem 12.40, $\dim S_2(\Gamma_0(11)) = g(X_0(11)) = 1$. Let $h \ne 0$ span it. The holomorphic differential $\omega_h$ on the curve $X_0(11)$ of genus 1 has a divisor of degree $2g - 2 = 0$ (Chapter 8) which is effective, so it is 0. Since $\nu_2(11) = \nu_3(11) = 0$, the proof of Theorem 12.40 shows that the order of $\omega_h$ at a point of $Y_0(11)$ is the order of $h$, and at a cusp it is the order of $h$ in $q_h$ minus 1. So $h$ has no zeros on $\mathcal{H}$, and simple zeros at the two cusps.

(i) Put $F = f_{11}^{12} = \eta(z)^{24}\eta(11z)^{24} = \Delta(z)\Delta(11z)$ (Proposition 12.45). By Proposition 12.23(d), (e), $F \in S_{24}(\Gamma_0(11))$. Its order at $\infty$ is $1 + 11 = 12$. At the cusp 0: with $\alpha = \left(\begin{smallmatrix} 11 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, $\Delta(11z) = 11^{-6}(\Delta|_{12}\alpha)(z)$ and $\alpha S = S\left(\begin{smallmatrix} 1 & 0 \\ 0 & 11 \end{smallmatrix}\right)$, so $\Delta(11z)|_{12}S = 11^{-6}\Delta|_{12}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 11 \end{smallmatrix}\right) = 11^{-12}\Delta(z/11)$ and $F|_{24}S = 11^{-12}\Delta(z)\Delta(z/11)$, of order $11 + 1 = 12$ in $q_{11} = e^{2\pi iz/11}$. So the weight-0 function $\phi = h^{12}/F$ is $\Gamma_0(11)$-invariant, holomorphic on $\mathcal{H}$ ($\eta$ has no zeros), and holomorphic and nonzero at both cusps (orders $12 - 12 = 0$). It is a holomorphic function on the compact connected Riemann surface $X_0(11)$, hence a nonzero constant $c$. Then $(h/f_{11})^{12} = c$ on $\mathcal{H}$, and the continuous function $h/f_{11}$ on the connected set $\mathcal{H}$ takes values in the finite set of twelfth roots of $c$, so it is constant. Hence $f_{11}$ is a nonzero multiple of $h$: it lies in $S_2(\Gamma_0(11))$ and has the zeros of $h$. As $a_1(f_{11}) = 1$, it is the normalised generator, which is (ii).

(iii) By Proposition 12.45, $f_{11}(-1/(11z)) = \eta(-1/(11z))^2\eta(-1/z)^2 = \frac{11z}{i}\eta(11z)^2 \cdot \frac{z}{i}\eta(z)^2 = -11z^2 f_{11}(z)$. With $W_{11} = \left(\begin{smallmatrix} 0 & -1 \\ 11 & 0 \end{smallmatrix}\right)$, $(f_{11}|_2 W_{11})(z) = 11 \cdot (11z)^{-2} f_{11}(-1/(11z)) = -f_{11}(z)$.

(iv) We do not prove this; it is the Eichler-Shimura theorem for $X_0(11)$, which is itself an elliptic curve isomorphic to `11a1` (Chapter 14, [52, Section 8.8], section number not verified). We checked $a_n(f_{11}) = a_n(\texttt{11a1})$ for all $n \le 2000$ with PARI's `ellan`. □

Two remarks on (iii). The fixed point of $w_{11}$ is $z_0 = i/\sqrt{11}$, and there the identity $f|_2 W_{11} = \varepsilon_{11}(f)f$ reads $-f(z_0) = \varepsilon_{11}(f)f(z_0)$; so $\varepsilon_{11}(f) = +1$ would force $f(z_0) = 0$. Numerically $f_{11}(i/\sqrt{11}) = 0.10287776677\ldots \neq 0$, which confirms $\varepsilon_{11}(f) = -1$ independently of the $\eta$-transformation law; PARI's `mfatkineigenvalues` also gives $-1$. Second, the sign agrees with Example 12.68(c): $a_{11}(f_{11}) = 1 = -\varepsilon_{11}(f)$.

The first coefficients are

$$f_{11} = q - 2q^2 - q^3 + 2q^4 + q^5 + 2q^6 - 2q^7 - 2q^9 - 2q^{10} + q^{11} \\ - 2q^{12} + 4q^{13} + 4q^{14} - q^{15} - 4q^{16} - 2q^{17} + 4q^{18} + 2q^{20} + \cdots,$$

and the $a_p$ for $p = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37$ are $-2, -1, 1, -2, 1, 4, -2, 0, -1, 0, 7, 3$. Reducing $y^2 + y = x^3 - x^2 - 10x - 20$ modulo $p$ and counting points gives $p + 1 - a_p$ (Chapter 9); for instance $\#\tilde{E}(\mathbb{F}_2) = 5$ and $\#\tilde{E}(\mathbb{F}_{13}) = 10$. The rational point of order 5 of `11a1` shows up as $5 \mid p + 1 - a_p$, the congruence of Example 12.43. Physically $f_{11}$ is a Fourier series on the half cylinder (Section 12.6) whose modes are dictated by the arithmetic of a cubic curve.

The computation in the proof of (i) also gives the example promised after Definition 12.22, of why holomorphy must be checked at every cusp. Let $g(z) = \Delta(11z)^2/\Delta(z)$. It is holomorphic on $\mathcal{H}$ ($\Delta$ has no zeros there), and $g|_{12}\gamma = g$ for $\gamma \in \Gamma_0(11)$, since $\Delta(11z)^2$ has weight 24, $\Delta$ has weight 12, and the slash operator is multiplicative in the sense $(F/G)|_{a-b}\gamma = (F|_a\gamma)/(G|_b\gamma)$. At $\infty$ its expansion is $q^{22-1}(1+\cdots)$, so $g$ is bounded on $\{\operatorname{Im} z \ge 1\}$ and even vanishes to order 21. But at the cusp 0, by the formula $\Delta(11z)|_{12}S = 11^{-12}\Delta(z/11)$ of the proof, $g|_{12}S = 11^{-24}\Delta(z/11)^2/\Delta(z)$, and in the local parameter $q_{11} = e^{2\pi iz/11}$ the numerator has order 2 and the denominator $\Delta(z) = q_{11}^{11} + \cdots$ order 11: $g$ has a pole of order 9 at 0, and $g \notin M_{12}(\Gamma_0(11))$. The valence count is consistent: $21 - 9 = 12 = \frac{12 \cdot 12}{12}$ (Theorem 12.39; its sketch of proof applies to meromorphic forms, since Theorem 12.34 does).

**Example 12.47** (small levels)**.** (a) For the twelve levels $N = 11$, 14, 15, 17, 19, 20, 21, 24, 27, 32, 36, 49 of genus 1, $S_2(\Gamma_0(N))$ is one-dimensional, spanned by the newform of the optimal curve of conductor $N$ (Chapter 14). Eight of these are $\eta$-products, each checked against Cremona's curve for $n \le 300$:

| $N$ | form | $N$ | form |
|---|---|---|---|
| 11 | $\eta(z)^2\eta(11z)^2$ | 24 | $\eta(2z)\eta(4z)\eta(6z)\eta(12z)$ |
| 14 | $\eta(z)\eta(2z)\eta(7z)\eta(14z)$ | 27 | $\eta(3z)^2\eta(9z)^2$ |
| 15 | $\eta(z)\eta(3z)\eta(5z)\eta(15z)$ | 32 | $\eta(4z)^2\eta(8z)^2$ |
| 20 | $\eta(2z)^2\eta(10z)^2$ | 36 | $\eta(6z)^4$ |

(curves `11a1`, `14a1`, `15a1`, `20a1`, `24a1`, `27a1`, `32a1`, `36a1`). That each product is a modular form for $\Gamma_0(N)$ follows from Ligozat's criterion for $\eta$-quotients [114] (stated, not proved here), or, as in Theorem 12.46, from the orders of its twelfth power at the cusps (Exercise 12.78).

(b) $\dim S_2(\Gamma_0(37)) = 2$, spanned by the newforms of `37a1` ($q - 2q^2 - 3q^3 + 2q^4 - 2q^5 + \cdots$) and `37b1` ($q + q^3 - 2q^4 + \cdots$), with $w_{37}$-eigenvalues $+1$ and $-1$ (PARI); `37a1` has rank 1 and root number $-1$, `37b1` rank 0 and root number $+1$.

(c) $\dim S_2(\Gamma_0(121)) = 6$: four newforms (the classes `121a` to `121d`) and the two forms $f_{11}(z)$, $f_{11}(11z)$ coming from level 11 (Chapter 13).

## 12.10 The Petersson inner product

The Petersson product is the inner product on cusp forms. It enters the paper in two ways. Qualitatively, through orthogonality: the congruence number $r_E$ is defined with forms "orthogonal to $f$", Lemma lem:crit uses "the orthogonal complement $f^{\perp}$", and Lemma lem:old uses forms "orthogonal to the newform $f$". Quantitatively, through the Petersson norm: "Up to explicit factors $\deg \varphi_E$ is $\langle f, f\rangle$ divided by the area of the period lattice of $E$" (the paper's Introduction), made exact in Remark rem:degree. This section defines the product, fixes its normalisation once and for all, proves its basic properties, and checks the normalisation numerically against the modular degrees of five curves.

**Definition 12.48** (the Petersson product)**.** For $f \in S_k(\Gamma_0(N))$ and $g \in M_k(\Gamma_0(N))$,

$$\langle f, g\rangle = \langle f, g\rangle_{\Gamma_0(N)} = \int_{\Gamma_0(N)\backslash\mathcal{H}} f(z)\overline{g(z)}\, y^k\, \frac{dx\, dy}{y^2},$$

the integral over any fundamental domain, for instance $\bigcup_i \gamma_i \mathcal{F}$ (Proposition 12.8). For a subgroup $\Gamma \subseteq \Gamma_0(N)$ of finite index the same integral over $\Gamma\backslash\mathcal{H}$ is written $\langle f, g\rangle_{\Gamma}$.

**Proposition 12.49** (basic properties)**.** *(i) The integrand $\phi(z) = f(z)\overline{g(z)}\, y^k$ is invariant under $\Gamma_0(N)$, so the integral does not depend on the fundamental domain.*

*(ii) The integral converges absolutely, and $\langle f, g\rangle = \sum_i \int_{\mathcal{F}} (f|_k\gamma_i)\overline{(g|_k\gamma_i)}\, y^k\, d\mu$.*

*(iii) On $S_k(\Gamma_0(N))$, $\langle\ ,\ \rangle$ is a positive definite Hermitian inner product, linear in the first variable.*

*(iv) In weight 2, $\langle f, g\rangle = \iint_{\Gamma_0(N)\backslash\mathcal{H}} f\bar{g}\, dx\, dy = \frac{i}{2}\int_{X_0(N)} \omega_f \wedge \overline{\omega_g}$ for $f, g \in S_2(\Gamma_0(N))$: the Petersson product is the Hodge inner product of the holomorphic differentials of Theorem 12.40.*

*Proof.* (i) $\phi(\gamma z) = \operatorname{Im}(\gamma z)^k f(\gamma z)\overline{g(\gamma z)} = y^k (f|_k\gamma)(z)\overline{(g|_k\gamma)(z)}$ for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, by the computation of Lemma 12.21(iv) without absolute values; for $\gamma \in \Gamma_0(N)$ this is $\phi(z)$. With the invariance of $d\mu$, two fundamental domains, which differ by cutting into pieces and moving the pieces by elements of $\Gamma_0(N)$, give the same integral. (ii) The same identity with $\gamma = \gamma_i$ and the change of variables $z = \gamma_i w$ give the formula. On $\mathcal{F}$, $\operatorname{Im} w \geq \sqrt{3}/2$, $|(f|_k\gamma_i)(w)| \leq C_i e^{-2\pi \operatorname{Im} w/h_i}$ (no constant term, as in Proposition 12.26) and $|(g|_k\gamma_i)(w)| \leq C_i'$ (holomorphy at the cusp); and $\int_{\mathcal{F}} e^{-2\pi y/h} y^{k-2}\, dx\, dy < \infty$. (iii) Sesquilinearity and $\langle g, f\rangle = \overline{\langle f, g\rangle}$ are clear; $\langle f, f\rangle = \int |f|^2 y^k\, d\mu > 0$ unless $f$ vanishes on an open set, hence everywhere. (iv) $y^2 \cdot y^{-2} = 1$, and $\frac{i}{2}\, dz \wedge d\bar{z} = dx \wedge dy$, because $dz \wedge d\bar{z} = (dx + i\, dy) \wedge (dx - i\, dy) = -2i\, dx \wedge dy$; also $\omega_f \wedge \overline{\omega_g} = f\bar{g}\, dz \wedge d\bar{z}$. The integral over $X_0(N)$ is the integral over a fundamental domain, the cusps and elliptic points having measure zero (Chapter 8). □

The Petersson product is the weighted $L^2$ inner product $\int f\bar{g}\, w\, d\mu$ of functions on a surface of finite area, with the weight $w = y^k$, just as Sturm-Liouville theory uses weighted inner products. Chapter 13 shows that the Hecke operators $T_n$, $(n, N) = 1$, are self-adjoint for it, so that their eigenforms are orthogonal, as the eigenfunctions of a self-adjoint operator are.

**Convention 12.50** (normalisations of the Petersson product)**.**

(a) The book's product is the *unnormalised* one of Definition 12.48 (Appendix B, B-C8).

(b) PARI's `mfpetersson` returns the book's product divided by $\mu(N)$.

(c) The "normalised" product of some books is the book's divided by $\operatorname{vol}(\Gamma_0(N)\backslash\mathcal{H}) = \pi\mu(N)/3$; it does not depend on the level at which $f$ and $g$ are viewed (Proposition 12.51).

(d) The formula $\deg \varphi_E \cdot \operatorname{vol}(\Lambda_E) = 4\pi^2 c_E^2 \langle f, f\rangle$ (Chapter 14, the paper's Remark rem:degree) holds with the book's product and with the Manin constant $c_E > 0$ defined by $\varphi_E^* \omega_E = c_E \cdot 2\pi i\, f(z)\, dz$ (Chapters 8 and 14).

A numerical value of a Petersson norm is never quoted without its normalisation.

**Proposition 12.51** (change of group and invariance)**.** *(i) If $\Gamma \subseteq \Gamma_0(N)$ has finite index, then $\langle f, g\rangle_\Gamma = [\bar{\Gamma}_0(N) : \bar{\Gamma}]\, \langle f, g\rangle_{\Gamma_0(N)}$. In particular, for $f \in S_k(\Gamma_0(N))$, $g \in M_k(\Gamma_0(N))$ viewed at level $NM$, $\langle f, g\rangle_{\Gamma_0(NM)} = \frac{\mu(NM)}{\mu(N)} \langle f, g\rangle_{\Gamma_0(N)}$.*

*(ii) If $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ normalises $\Gamma_0(N)$, then $\langle f|_k\alpha, g|_k\alpha\rangle = \langle f, g\rangle$ for $f \in S_k(\Gamma_0(N))$, $g \in M_k(\Gamma_0(N))$, provided $f|_k\alpha$ and $g|_k\alpha$ again lie in these spaces.*

*(iii) (Groups commensurable with $\mathrm{SL}_2(\mathbb{Z})$.) Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Q})$ be a subgroup that contains some $\Gamma(M)$ as a subgroup of finite index, and write $\bar{\Gamma}$ for its image in $\mathrm{PSL}_2(\mathbb{R})$. Define $M_k(\Gamma) \supseteq S_k(\Gamma)$ by the conditions of Definition 12.22 (invariance under $\Gamma$, behaviour under every $\sigma \in \mathrm{SL}_2(\mathbb{Z})$). For a congruence subgroup $\Gamma_1 \subseteq \mathrm{SL}_2(\mathbb{Z})$ write $\langle f, g\rangle_{\Gamma_1} = \int_{\mathcal{F}_{\Gamma_1}} f\bar{g} y^k\, d\mu$, with $\mathcal{F}_{\Gamma_1}$ the domain of Proposition 12.8 for $\pm\Gamma_1$. For $f \in S_k(\Gamma)$, $g \in M_k(\Gamma)$ the number*

$$\langle f, g\rangle_\Gamma = [\bar{\Gamma} : \bar{\Gamma}_1]^{-1} \langle f, g\rangle_{\Gamma_1},$$

*for $\Gamma_1 \subseteq \Gamma \cap \mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup of finite index in $\Gamma$, does not depend on $\Gamma_1$, and for $\Gamma \subseteq \Gamma_0(N)$ of finite index it is the product of Definition 12.48.*

*(iv) (Change of variables.) In the situation of (iii) let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then $\alpha^{-1}\Gamma\alpha$ is again a group as in (iii), $f \mapsto f|_k\alpha$ maps $M_k(\Gamma)$ onto $M_k(\alpha^{-1}\Gamma\alpha)$ and $S_k(\Gamma)$ onto $S_k(\alpha^{-1}\Gamma\alpha)$, and*

$$\langle f|_k\alpha, g|_k\alpha\rangle_{\alpha^{-1}\Gamma\alpha} = \langle f, g\rangle_\Gamma \qquad (f \in S_k(\Gamma),\ g \in M_k(\Gamma)).$$

*Proof.* (i) A fundamental domain of $\Gamma$ is the union of $[\bar\Gamma_0(N) : \bar\Gamma]$ translates of one of $\Gamma_0(N)$ (the proof of Proposition 12.8, with $\Gamma_0(N)$ in place of $\mathrm{SL}_2(\mathbb{Z})$), and the integrand is $\Gamma_0(N)$-invariant. Both $\Gamma_0(NM)$ and $\Gamma_0(N)$ contain $-I$, so the index is $\mu(NM)/\mu(N)$. (ii) Let $D$ be a fundamental domain. The identity of (i) of Proposition 12.49 holds for $\alpha \in \mathrm{GL}_2^+(\mathbb{R})$ (the factors $\det\alpha$ cancel against $\mathrm{Im}(\alpha z) = \det\alpha\, y/|cz+d|^2$), so $\int_D (f|_k\alpha)\overline{(g|_k\alpha)} y^k\, d\mu = \int_D \phi(\alpha z)\, d\mu(z) = \int_{\alpha D} \phi\, d\mu$, and $\alpha D$ is a fundamental domain for $\alpha\Gamma_0(N)\alpha^{-1} = \Gamma_0(N)$.

For (iii) and (iv) we use one measure-theoretic fact. Call a measurable set $D \subseteq \mathcal{H}$ a *fundamental set* for a congruence subgroup $\Gamma_1$ if the translates $\gamma D$, $\gamma \in \bar\Gamma_1$, cover $\mathcal{H}$ and any two distinct ones meet in a null set; $\mathcal{F}_{\Gamma_1}$ is one, by Proposition 12.8 (translates of $\mathcal{F}$ meet only in boundary arcs). If $D, D'$ are fundamental sets for $\Gamma_1$ and $\phi \ge 0$ is measurable and $\Gamma_1$-invariant, then, by the invariance of $\phi\, d\mu$,

$$\int_D \phi\, d\mu = \sum_{\gamma\in\bar\Gamma_1} \int_{D\cap\gamma D'} \phi\, d\mu = \sum_{\gamma\in\bar\Gamma_1} \int_{\gamma^{-1}D\cap D'} \phi\, d\mu = \int_{D'} \phi\, d\mu;$$

applied to $|\phi|$ and then to the positive and negative parts of $\mathrm{Re}\,\phi$ and $\mathrm{Im}\,\phi$, the same holds for every $\Gamma_1$-invariant $\phi$ integrable over $D'$. (iii) Since $f\bar g\, y^k$ is invariant under $-I$, it is invariant under $\pm\Gamma_1$, and the integral over $\mathcal{F}_{\Gamma_1}$ converges by the argument of Proposition 12.49(ii) (the expansions of $f|_k\sigma$ in $q_h$ at every cusp have no constant term). If $\Gamma_2 \subseteq \Gamma_1$ are two congruence subgroups as in (iii) and $\delta_1, \dots, \delta_r$ represent $\bar\Gamma_2\backslash\bar\Gamma_1$, then $\bigcup_j \delta_j\mathcal{F}_{\Gamma_1}$ is a fundamental set for $\Gamma_2$, so $\langle f, g\rangle_{\Gamma_2} = r\langle f, g\rangle_{\Gamma_1}$, with $r = [\bar\Gamma_1 : \bar\Gamma_2]$; as indices multiply, $[\bar\Gamma : \bar\Gamma_2]^{-1}\langle f, g\rangle_{\Gamma_2} = [\bar\Gamma : \bar\Gamma_1]^{-1}\langle f, g\rangle_{\Gamma_1}$. Two choices $\Gamma_1, \Gamma_1'$ are compared through $\Gamma_1 \cap \Gamma_1'$, which is again a congruence subgroup of finite index in $\Gamma$. For $\Gamma \subseteq \Gamma_0(N)$ take $\Gamma_1 = \Gamma$ (and $\Gamma_0(N)$ itself in Definition 12.48). (iv) Replacing $\alpha$ by a positive multiple changes neither $\alpha^{-1}\Gamma\alpha$ nor $f|_k\alpha$ (the factors $\lambda^k$ and $\lambda^{-k}$ cancel), so we may assume $\alpha \in M_2(\mathbb{Z})$; put $m = \det\alpha$ and $\alpha' = m\alpha^{-1} \in M_2(\mathbb{Z})$. For every $L \ge 1$ and $X \in M_2(\mathbb{Z})$,

$$\alpha(1+mLX)\alpha^{-1} = 1 + L\,\alpha X\alpha', \qquad \alpha^{-1}(1+mLX)\alpha = 1 + L\,\alpha' X\alpha,$$

so $\alpha\Gamma(mL)\alpha^{-1} \subseteq \Gamma(L)$ and $\alpha^{-1}\Gamma(mL)\alpha \subseteq \Gamma(L)$. Hence $\alpha^{-1}\Gamma\alpha \supseteq \alpha^{-1}\Gamma(M)\alpha \supseteq \Gamma(mM)$, and $[\alpha^{-1}\Gamma\alpha : \Gamma(mM)] \le [\alpha^{-1}\Gamma\alpha : \alpha^{-1}\Gamma(m^2M)\alpha] = [\Gamma : \Gamma(m^2M)] < \infty$ because $\alpha^{-1}\Gamma(m^2M)\alpha \subseteq \Gamma(mM)$. So $\alpha^{-1}\Gamma\alpha$ is as in (iii). For $\gamma \in \Gamma$, $(f|_k\alpha)|_k(\alpha^{-1}\gamma\alpha) = f|_k\gamma|_k\alpha = f|_k\alpha$ by Lemma 12.21. For $\sigma \in \mathrm{SL}_2(\mathbb{Z})$ write $\alpha\sigma = \sigma'\beta$ with $\sigma' \in \mathrm{SL}_2(\mathbb{Z})$ and $\beta$ upper triangular with integer entries, as in the proof of Proposition 12.23(e); then $(f|_k\alpha)|_k\sigma = (f|_k\sigma')|_k\beta$ is bounded on $\{\mathrm{Im}\, z \ge 1\}$, and tends to 0 there if $f$ is a cusp form, by the argument given there. So $f|_k\alpha \in M_k(\alpha^{-1}\Gamma\alpha)$, in $S_k$ if $f \in S_k(\Gamma)$; $\alpha^{-1}$ gives the inverse map. For the product take $\Gamma_1 = \Gamma(m^2M) \subseteq \Gamma$. Then $\Delta = \alpha^{-1}\Gamma_1\alpha$ satisfies $\Gamma(m^3M) \subseteq \Delta \subseteq \Gamma(mM)$ (the first inclusion because $\alpha\Gamma(m^3M)\alpha^{-1} \subseteq \Gamma(m^2M)$), so it is a congruence subgroup of finite index in $\alpha^{-1}\Gamma\alpha$, and conjugation by $\alpha$ gives $[\overline{\alpha^{-1}\Gamma\alpha} : \bar\Delta] = [\bar\Gamma : \bar\Gamma_1]$. It therefore suffices to show $\langle f|_k\alpha, g|_k\alpha\rangle_\Delta = \langle f, g\rangle_{\Gamma_1}$. With $\phi = f\bar g\, y^k$, the identity $\phi(\alpha z) = (f|_k\alpha)(z)\overline{(g|_k\alpha)(z)}\, y^k$ (as in (ii)) and the invariance of $d\mu$ give

$$\langle f|_k\alpha, g|_k\alpha\rangle_\Delta = \int_{\mathcal{F}_\Delta} \phi(\alpha z)\, d\mu(z) = \int_{\alpha\mathcal{F}_\Delta} \phi\, d\mu.$$

The set $\alpha\mathcal{F}_\Delta$ is a fundamental set for $\Gamma_1 = \alpha\Delta\alpha^{-1}$: its translates are $\gamma\alpha\mathcal{F}_\Delta = \alpha(\alpha^{-1}\gamma\alpha)\mathcal{F}_\Delta$, and the diffeomorphism $z \mapsto \alpha z$ preserves null sets. By the fact above (first for $|\phi|$, which shows integrability), the integral equals $\int_{\mathcal{F}_{\Gamma_1}} \phi\, d\mu = \langle f, g\rangle_{\Gamma_1}$. □

Chapter 13 uses (iv) in two ways. For the adjoints of double coset operators, $\Gamma = \Gamma_0(N) \cap \alpha\Gamma_0(N)\alpha^{-1}$ for a matrix $\alpha$ of a double coset; then $\Gamma$ and $\alpha^{-1}\Gamma\alpha = \alpha^{-1}\Gamma_0(N)\alpha \cap \Gamma_0(N)$ are both congruence subgroups of $\mathrm{SL}_2(\mathbb{Z})$, and (iv) is needed only because $\alpha$ normalises neither. For the norms of oldforms, $\Gamma = \Gamma_0(M)$ and $\alpha = \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$; then $\alpha^{-1}\Gamma\alpha$ does not lie in $\mathrm{SL}_2(\mathbb{Z})$, and (iii) is needed to define the product over it.

The factor in (i) is why the normalisation matters as soon as levels change: the paper passes from level $M$ to $M^\Sigma$ in Section 4.1, and Theorem thm:quant works at level $N$ with forms of level $N_S$. The normalised product of Convention 12.50(c) is level-independent; congruence questions do not see the factor at all, since they only use orthogonality.

**Proposition 12.52** (orthogonal complements)**.** *For $0 \neq f \in S_k(\Gamma_0(N))$ let $f^\perp = \{g \in S_k(\Gamma_0(N)) : \langle g, f\rangle = 0\}$. Then $S_k(\Gamma_0(N)) = \mathbb{C}f \oplus f^\perp$, and the projection onto $\mathbb{C}f$ along $f^\perp$ is $g \mapsto \frac{\langle g,f\rangle}{\langle f,f\rangle} f$.*

*Proof.* $\langle f, f\rangle > 0$, so $g - \frac{\langle g,f\rangle}{\langle f,f\rangle} f \in f^\perp$: its product with $f$ is $\langle g, f\rangle - \frac{\langle g,f\rangle}{\langle f,f\rangle}\langle f, f\rangle = 0$. So $g = \frac{\langle g,f\rangle}{\langle f,f\rangle} f + (g - \frac{\langle g,f\rangle}{\langle f,f\rangle} f)$ is a decomposition, and it is unique because $\mathbb{C}f \cap f^\perp = 0$ (if $cf \in f^\perp$ then $c\langle f, f\rangle = 0$, so $c = 0$). □

Equivalently, $f^\perp$ is the kernel of the nonzero linear functional $g \mapsto \langle g, f\rangle$, so it has codimension 1. For $N = 22$, with $f = f_{11}$ and $S_2(\Gamma_0(22))$ spanned by $f_{11}(z)$ and $f_{11}(2z)$ (Example 12.68(b)), the line $f^\perp$ is spanned by $f_{11}(2z) - \frac{\langle f_{11}(2z), f_{11}\rangle_{\Gamma_0(22)}}{\langle f_{11}, f_{11}\rangle_{\Gamma_0(22)}} f_{11}(z)$; with the values of Exercise 12.80 (where $\langle f_{11}(z), f_{11}(2z)\rangle_{\Gamma_0(22)} = -\langle f_{11}, f_{11}\rangle_{\Gamma_0(11)}$ and $\langle f_{11}, f_{11}\rangle_{\Gamma_0(22)} = 3\langle f_{11}, f_{11}\rangle_{\Gamma_0(11)}$), the ratio is $-\frac{1}{3}$ and $f^\perp = \mathbb{C}(f_{11}(2z) + \frac{1}{3} f_{11}(z))$. Its $q$-expansion has non-integral coefficients, and $3f_{11}(2z) + f_{11}(z)$ spans $f^\perp \cap S_2(\Gamma_0(22); \mathbb{Z})$: this is the kind of lattice on which the paper's congruence questions are posed (here $f_{11}$ is not a newform of level 22, so the example only illustrates the definition).

By itself $f^\perp$ is defined transcendentally. Chapter 13 shows, for a newform $f$ with rational coefficients, that $f^\perp$ is stable under all Hecke operators, $U_p$ included (the adjoint of $U_p$ is $w_N U_p w_N$, and $f$ is an eigenvector of $w_N$), that it is spanned by forms with rational coefficients, and that $f^\perp \cap S_2(\Gamma_0(N); \mathbb{Z})$ is a saturated sublattice of rank $\dim S_2(\Gamma_0(N)) - 1$. None of this is proved here.

**Example 12.53** (Petersson norms and modular degrees)**.** For five optimal curves of the paper's world, with $f$ the newform, $\Lambda_E$ the period lattice of the Néron differential of the minimal model (Chapter 9) and $\deg \varphi_E$ the modular degree (Cremona's tables, Chapter 14), PARI gives (unnormalised product, $\langle f, f\rangle = \mu(N) \cdot$ `mfpetersson`):

| curve | $N$ | $\mu(N)$ | $\langle f, f\rangle$ | $\mathrm{vol}(\Lambda_E)$ | $4\pi^2\langle f, f\rangle / \mathrm{vol}(\Lambda_E)$ | $\deg \varphi_E$ |
|---|---|---|---|---|---|---|
| `11a1` | 11 | 12 | 0.0469001478734952 | 1.85154362345596 | 1.0000000 | 1 |
| `20a1` | 20 | 36 | 0.0813494568253817 | 3.21154782844007 | 1.0000000 | 1 |
| `37a1` | 37 | 38 | 0.3717541475106961 | 7.33813274078958 | 2.0000000 | 2 |
| `99a1` | 99 | 144 | 0.5385195687108607 | 5.31497510542145 | 4.0000000 | 4 |
| `121a1` | 121 | 132 | 0.4282393600119572 | 2.81770204819580 | 6.0000000 | 6 |

(the ratio agrees with $\deg\varphi_E$ to 40 digits; for `11a1`, `mfpetersson` returns $0.0039083456561246 = 0.0469001478734952/12$). So $\deg\varphi_E\cdot\mathrm{vol}(\Lambda_E) = 4\pi^2\langle f,f\rangle$ for these curves, whose Manin constant is 1. The derivation is one line, given Chapter 14: if $\varphi_E^*\omega_E = c_E\cdot 2\pi i\, f(z)\,dz$, then pulling back the area form,

$$\deg\varphi_E\cdot\frac{i}{2}\int_E \omega_E\wedge\bar\omega_E = \frac{i}{2}\int_{X_0(N)}\varphi_E^*\omega_E\wedge\overline{\varphi_E^*\omega_E} = 4\pi^2c_E^2\cdot\frac{i}{2}\int_{X_0(N)}\omega_f\wedge\overline{\omega_f},$$

that is $\deg\varphi_E\cdot\mathrm{vol}(\Lambda_E) = 4\pi^2c_E^2\langle f,f\rangle$ by Proposition 12.49(iv) and Chapter 8 (degree of a pullback of a 2-form). The Manin constant stands in the numerator: for the degree-5 parametrisation $X_0(11)\to$ `11a3` one finds $\mathrm{vol}(\Lambda_{\texttt{11a3}}) = 5\,\mathrm{vol}(\Lambda_{\texttt{11a1}})$, so $5\cdot 5\,\mathrm{vol}(\Lambda_{\texttt{11a1}}) = 4\pi^2c^2\langle f,f\rangle$ forces the constant of that parametrisation to be 5 (Exercise 12.79).

*Remark* 12.54 (what other chapters add). (a) (Chapter 13.) For $(n,N)=1$ the Hecke operator $T_n$ is self-adjoint for $\langle\ ,\ \rangle$; for $p\mid N$, $U_p$ is not, its adjoint being $w_NU_pw_N$; newforms with different systems of eigenvalues are orthogonal, and oldforms are orthogonal to newforms.

(b) (Chapter 19.) Rankin-Selberg: for a newform $f$ of weight 2 and level $N$, $\langle f,f\rangle = N\,L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2)/(8\pi^3)$ with the naive symmetric square $L$-function ([49, Lemma 2.12], [183], [155]); we checked this to 40 digits for `11a1` and `37a1` with PARI's `lfunsympow` (at these squarefree levels the naive and the true $L$-function agree, so the check does not see the difference). At `121a1`, of additive reduction at 11 with $a_{11}=0$, it does: $\langle f,f\rangle = 0.4282393600\ldots = 121\,L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2)/(8\pi^3)$, while the true $L$-function gives $121\,L(\mathrm{Sym}^2 f,2)/(8\pi^3) = 0.3925527466\ldots$; the ratio is $\frac{12}{11}$ to 40 digits, the inverse of the Euler factor $(1+11^{1-s})^{-1}$ at $s=2$ that the true $L$-function has at 11 and the naive one omits (PARI's coefficients of `lfunsympow` at 11 and 121 are $-11$ and 121). The Rankin series $\sum a_n(f)^2n^{-s}$ is not $L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)$: it equals $\zeta^{(N)}(s-1)L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)/\zeta^{(N)}(2s-2)$, and at a good prime it carries an extra factor $1+p^{1-s}$ (Appendix B, B-C7).

(c) (Chapter 14.) The formula of Zagier and Watkins $\deg\varphi_E\cdot\mathrm{vol}(\Lambda_E) = 4\pi^2c_E^2\langle f,f\rangle$ in general.

*In the paper* 12.55. The Introduction of the paper says: "The classical route to the degree is the Petersson norm. Up to explicit factors $\deg\varphi_E$ is $\langle f,f\rangle$ divided by the area of the period lattice of $E$ (Zagier [209], Watkins [204])". Remark rem:degree makes this exact: "$\deg\varphi_E\cdot\mathrm{vol}(\Lambda_E) = 4\pi^2c_E^2\langle f,f\rangle$, with the unnormalised Petersson product and the Manin constant $c_E>0$ in the numerator". Both the normalisation and the place of $c_E$ are those checked in Example 12.53. The orthogonality in the definition of $r_E$, in Lemma lem:crit ($f^\perp$) and in Lemma lem:old is that of Proposition 12.52; in the proof of Theorem thm:twist the paper uses instead "the $\mathbb{T}$-stable complement of the $g$-line", which coincides with the Petersson complement for a newform $g$ (Chapter 13), and in the set-up of the proof of Theorem thm:quant it insists on the $\mathbb{T}$-stable complement of the line of $g^S$, "not the Petersson complement, which is not $\mathbb{T}$-stable when the $U_p$ are involved", because there the form $g^S$ is not a newform.

## 12.11 Integral forms, the Sturm bound and congruences

The paper's congruences live in $S_2(\Gamma_0(N);\mathbb{Z})$ and $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$: the congruence number is defined with an "integral cusp form", Lemma lem:crit uses the pairing "$S_2(\Gamma_0(N);\mathbb{Z})\times\mathbb{T}\to\mathbb{Z}$" and its version over $\mathbb{Z}_\ell$, Theorem thm:twist needs a twist map that "preserves integrality", and Lemma lem:step1 projects a congruence inside $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$. This section defines these modules,

states Shimura's theorem that they are lattices of full rank, proves Sturm's bound in the form needed, and deduces that a congruence between two forms of the same level can be decided on finitely many coefficients. It then proves the congruences of the paper's Example for all $n$. The perfect pairing with the Hecke algebra is Chapter 13's.

**Definition 12.56** (integral forms)**.** For a subring $A \subseteq \mathbb{C}$ let $M_k(\Gamma_0(N); A) = \{f \in M_k(\Gamma_0(N)) : a_n(f) \in A \text{ for all } n \geq 0\}$ and $S_k(\Gamma_0(N); A) = S_k(\Gamma_0(N)) \cap M_k(\Gamma_0(N); A)$, which are $A$-modules. For an arbitrary commutative ring $R$ (such as $\mathbb{Z}_\ell$, $\mathbb{F}_\ell$ or $\mathcal{O}_\lambda$, which are not subrings of $\mathbb{C}$ in a preferred way) put

$$S_k(\Gamma_0(N); R) = S_k(\Gamma_0(N); \mathbb{Z}) \otimes_{\mathbb{Z}} R,$$

with the *$q$-expansion map* $S_k(\Gamma_0(N); R) \to R[[q]]$, $f \otimes r \mapsto \sum_n r a_n(f) q^n$.

Three remarks. (a) $S_k(\Gamma_0(N); \mathbb{Z})$ is *saturated* in $\mathbb{Z}[[q]]$ by definition: if $g \in \mathbb{Z}[[q]]$ and $mg \in S_k(\Gamma_0(N); \mathbb{Z})$ for an integer $m \geq 1$, then $g \in S_k(\Gamma_0(N))$ and so $g \in S_k(\Gamma_0(N); \mathbb{Z})$. (b) Integrality refers to the expansion at $\infty$ only. Integrality at the other cusps is a different matter (for instance $f_{11}|_2 S = -\frac{1}{11} f_{11}(z/11)$ by Theorem 12.46(iii), which has non-integral coefficients in $q_{11}$); the paper never needs it. (c) For $R = \mathbb{Z}_\ell$ this is the paper's $S_2(\Gamma_0(N); \mathbb{Z}_\ell)$; Corollary 12.59(iii) shows that it is the module of forms in $S_2(\Gamma_0(N); \mathbb{Q}) \otimes \mathbb{Q}_\ell$ with $\ell$-integral $q$-expansion.

**Theorem 12.57** (Shimura; integral bases)**.** *$S_k(\Gamma_0(N))$ has a basis over $\mathbb{C}$ consisting of forms with integer coefficients; the same holds for $M_k(\Gamma_0(N))$.*

We do not prove this; see [180, Theorem 3.52] (number not verified) and [52, Section 6.5]. There are three standard proofs: through the rationality of the Hecke eigenvalues and the Galois action on newforms (Shimura), through the algebraic theory of modular forms and its $q$-expansion principle (Deligne-Rapoport and Katz, [43]), and for $k = 2$ through $H^1(X_0(N), \mathbb{Z})$ and the Eichler-Shimura isomorphism (Chapter 14). For $M_k(\mathrm{SL}_2(\mathbb{Z}))$ it is Corollary 12.35(iv): the monomials $E_4^a E_6^b$ have integer coefficients.

**Theorem 12.58** (Sturm's bound)**.** *Let $\mathcal{O}$ be the ring of integers of a number field $K \subset \mathbb{C}$, $\lambda$ a nonzero prime ideal of $\mathcal{O}$, and $f \in M_k(\Gamma_0(N); \mathcal{O})$. If $a_n(f) \in \lambda$ for all $n \leq \frac{k\mu(N)}{12}$, then $a_n(f) \in \lambda$ for all $n$. For $k = 2$ the condition is on $0 \leq n \leq B_2(N) = \lfloor \mu(N)/6 \rfloor$.*

*Sketch of proof. Level one.* Let $d$ be the dimension of $M_k(\mathrm{SL}_2(\mathbb{Z}))$. By Corollary 12.35 the module $M_k(\mathrm{SL}_2(\mathbb{Z}); \mathbb{Z})$ has a basis $g_0, \dots, g_{d-1}$ with $a_i(g_j) = \delta_{ij}$ for $0 \leq i, j < d$ (row-reduce the monomials $E_4^a E_6^b \Delta^c$, $4a + 6b + 12c = k$, whose expansions are triangular with leading coefficients 1), and the valence formula gives $v_\infty(f) \leq d - 1 \leq k/12$ for every nonzero $f \in M_k$. For $f \in M_k(\mathrm{SL}_2(\mathbb{Z}); \mathcal{O})$, $f - \sum_{j<d} a_j(f) g_j$ vanishes to order at least $d$ at $\infty$, so it is 0; hence if $a_j(f) \in \lambda$ for $j < d$, every coefficient of $f$ is in $\lambda$. *Level $N$.* Replace $f$ by the norm $F = \prod_i f|_k \gamma_i \in M_{k\mu}(\mathrm{SL}_2(\mathbb{Z}))$ of the proof of Theorem 12.39. The coefficients of the conjugates $f|_k \gamma_i$ are algebraic, and the delicate point, which rests on the integrality statements behind Theorem 12.57, is that after scaling by a suitable constant they become $\lambda$-integral and the $\lambda$-adic order of vanishing of $F$ at $\infty$ is at least that of $f$; then the level-one case in weight $k\mu$, with bound $k\mu/12$, applies to $F$. See [190] and [187, Chapter 9] (theorem number not verified). □

The values $B_2(N)$ at the levels of the paper's congruences are $B_2(11) = 2$, $B_2(20) = 6$, $B_2(37) = 6$, $B_2(52) = 14$, $B_2(99) = 24$, $B_2(121) = 22$, $B_2(539) = 112$, $B_2(980) = 336$; they agree with PARI's `mfsturm`. The bound $\frac{1}{6} N \prod_{p|N} (1 + \frac{1}{p})$ that the paper's Example calls Sturm's bound is $\mu(N)/6$, whose integer part is $B_2(N)$.

**Corollary 12.59** (the integral lattice and its $q$-expansions)**.** *Let $d = \dim S_2(\Gamma_0(N))$, $B = B_2(N)$ and $L = S_2(\Gamma_0(N);\mathbb{Z})$.*

*(i) $L$ is free of rank $d$, the map $L \otimes_{\mathbb{Z}} \mathbb{C} \to S_2(\Gamma_0(N))$ is an isomorphism, and $S_2(\Gamma_0(N);\mathbb{Q}) = L \otimes_{\mathbb{Z}} \mathbb{Q}$.*

*(ii) The truncation $\tau_B \colon L \to \mathbb{Z}^B$, $f \mapsto (a_1(f), \dots, a_B(f))$, is injective with torsion-free cokernel; so $\tau_B(L)$ is a direct summand of $\mathbb{Z}^B$.*

*(iii) For every commutative ring $R$ the $q$-expansion map $S_2(\Gamma_0(N);R) \to R[[q]]$ is injective, and so is its truncation $S_2(\Gamma_0(N);R) \to R^B$. For $R = \mathbb{Z}_\ell$ its image is $(S_2(\Gamma_0(N);\mathbb{Q}) \otimes \mathbb{Q}_\ell) \cap \mathbb{Z}_\ell[[q]]$; for $R = \mathbb{F}_\ell$, an element is $0$ iff its first $B$ coefficients are.*

*(iv) For $f, g \in L$ and an integer $m \geq 1$, $f \equiv g \pmod m$ coefficientwise iff $a_n(f) \equiv a_n(g) \pmod m$ for $1 \leq n \leq B$. For $f, g \in S_2(\Gamma_0(N);\mathcal{O})$ and a prime $\lambda$ of $\mathcal{O}$, $f \equiv g \pmod \lambda$ coefficientwise iff $a_n(f) \equiv a_n(g) \pmod \lambda$ for $n \leq B$.*

*(v) Forms of levels $M_1, M_2$ are compared in $S_2(\Gamma_0(N))$ with $N = \operatorname{lcm}(M_1, M_2)$ (Proposition 12.23(c)), with the bound $B_2(N)$. A congruence at the primes $p \notin \Sigma$ only is not of this kind: it must first be turned into a congruence at all $n$ (by depletion, Chapter 13), which raises the level and the bound.*

*Proof.* (i) By Theorem 12.57 there are $f_1, \dots, f_d \in L$ forming a basis of $S_2(\Gamma_0(N))$. The vectors $\tau_B(f_j) \in \mathbb{Z}^B$ are linearly independent: otherwise, since they are integer vectors, there are integers $c_j$, not all 0, with $\sum c_j \tau_B(f_j) = 0$, and $h = \sum c_j f_j \in L$ is a nonzero form (the $f_j$ are independent) with $a_n(h) = 0$ for $n \leq B$; by Theorem 12.58 with $\mathcal{O} = \mathbb{Z}$ every coefficient of $h$ is divisible by every prime, so $h = 0$, a contradiction. Hence $\tau_B$ is injective on $S_2(\Gamma_0(N))$ (which the $f_j$ span), so $L$ embeds in $\mathbb{Z}^B$ and is free of finite rank $r$. Elements of $L$ that are independent over $\mathbb{Z}$ have independent images in $\mathbb{Q}^B$, hence in $\mathbb{C}^B$, so $r \leq d$; and $r \geq d$ because of the $f_j$. So a $\mathbb{Z}$-basis of $L$ is a $\mathbb{C}$-basis of $S_2(\Gamma_0(N))$. If $f$ has rational coefficients, write $f = \sum c_j e_j$ in a $\mathbb{Z}$-basis $(e_j)$ of $L$; the $c_j$ solve a linear system $\sum c_j \tau_B(e_j) = \tau_B(f)$ with rational data and a unique solution, so $c_j \in \mathbb{Q}$. (ii) Injectivity was shown in (i). If $g \in L$ and $\tau_B(g) \in m\mathbb{Z}^B$ with $m > 1$, let $p$ be a prime dividing $m$. Then $a_n(g) \equiv 0 \pmod p$ for $n \leq B$ ($a_0 = 0$), so for all $n$ by Theorem 12.58, and $g/p \in \mathbb{Z}[[q]] \cap S_2(\Gamma_0(N)) = L$ with $\tau_B(g/p) \in \frac{m}{p}\mathbb{Z}^B$. By induction $g/m \in L$. So $\mathbb{Z}^B/\tau_B(L)$ is torsion-free, hence free, and the sequence $0 \to L \to \mathbb{Z}^B \to \mathbb{Z}^B/\tau_B(L) \to 0$ splits (Chapter 3). (iii) Tensoring a split injection with $R$ gives a split injection $L \otimes R \to R^B$; it is the composite of the $q$-expansion map with truncation, so both are injective. For $R = \mathbb{Z}_\ell$: an element of $L \otimes \mathbb{Z}_\ell$ has $q$-expansion in $\mathbb{Z}_\ell[[q]]$ and lies in $L \otimes \mathbb{Q}_\ell = S_2(\Gamma_0(N);\mathbb{Q}) \otimes \mathbb{Q}_\ell$. Conversely let $F = \sum c_j e_j$, $c_j \in \mathbb{Q}_\ell$, have coefficients in $\mathbb{Z}_\ell$. Since $\tau_B(e_1), \dots, \tau_B(e_d)$ extend to a basis of $\mathbb{Z}^B$ by (ii), the coordinates of $\tau_B(F) \in \mathbb{Z}_\ell^B$ in that basis are in $\mathbb{Z}_\ell$, and they are the $c_j$; so $F \in L \otimes \mathbb{Z}_\ell$. The statement for $\mathbb{F}_\ell$ is injectivity of the truncation. (iv) The first statement is the argument of (ii) applied to $h = f - g$: if $\tau_B(h) \in m\mathbb{Z}^B$ then $h/m \in L$. The second is Theorem 12.58 applied to $f - g \in M_2(\Gamma_0(N);\mathcal{O})$. (v) is a remark; for instance the depletion of a form of level 121 at 3 has level $121 \cdot 9 = 1089$ and $B_2(1089) = 264$. □

*In the paper* 12.60. Lemma lem:crit ends: "Every step holds verbatim over $\mathbb{Z}_\ell$, the pairing being perfect after tensoring with $\mathbb{Z}_\ell$." On the side of forms, the module there is $S_2(\Gamma_0(N);\mathbb{Z}_\ell) = S_2(\Gamma_0(N);\mathbb{Z}) \otimes \mathbb{Z}_\ell$ (Definition 12.56); Corollary 12.59(iii) says that it embeds in $\mathbb{Z}_\ell[[q]]$, so that congruences modulo $\ell^t$ coefficientwise make sense for its elements, and that it consists exactly

of the $\ell$-integral forms. The set-up of the proof of Theorem thm:quant uses the same module: it speaks of "$g^S \equiv h \pmod{\ell^t}$ coefficientwise with $h \in C \cap S_2(\Gamma_0(N);\mathbb{Z}_\ell)$", $C$ the $\mathbb{T}$-stable complement of the line of $g^S$, and the proof of Lemma lem:step1 continues: "The idempotent $e_{\mathfrak{m}}$ of $\mathbb{T}_\ell$ fixes $g^S$ (as $\pi$ factors through $\mathbb{T}_{\mathfrak{m}}$) and preserves $C$ and the integral forms, so $h' = e_{\mathfrak{m}}h$ lies in $C \cap S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ and $g^S \equiv h' \pmod{\ell^t}$." Here the idempotent $e_{\mathfrak{m}} \in \mathbb{T}_\ell$ acts on $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ (the Hecke operators preserve $S_2(\Gamma_0(N);\mathbb{Z})$, Chapter 13), and if $g^S - h \in \ell^t S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ then $e_{\mathfrak{m}}(g^S - h) = g^S - e_{\mathfrak{m}}h \in \ell^t S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ because $e_{\mathfrak{m}}g^S = g^S$; by the injectivity of (iii) this is a congruence of $q$-expansions. In Theorem thm:twist the twist map $h \mapsto h \otimes \chi_S = \sum \chi_S(n)a_n(h)q^n$ "maps $S_2(\Gamma_0(N_S))$ into $S_2(\Gamma_0(\mathrm{lcm}(N_S, D^2))) \subset S_2(\Gamma_0(N))$, preserves integrality": the inclusion of spaces is Chapter 13's, and integrality is clear on $q$-expansions because $\chi_S(n) \in \{0, \pm 1\}$.

**Example 12.61** (the paper's congruences hold for all $n$)**.** The paper's Example after the proof of Theorem thm:qual lists four congruences. In each case the two forms lie in the same space $S_2(\Gamma_0(N);\mathbb{Z})$ (for the twisted forms this uses the level of a twist, Chapter 13), so by Corollary 12.59(iv) a check of $a_n$ for $n \le B_2(N)$ proves the congruence for every $n$.

(a) $f_{\mathtt{52a}} \equiv f_{\mathtt{26b}}(z) - f_{\mathtt{26b}}(2z) \pmod 3$ in $S_2(\Gamma_0(52);\mathbb{Z})$ (the second form by Proposition 12.23(e)), $B_2(52) = 14$; here $a_2(\mathtt{26b1}) = 1$.

(b) $f_{\mathtt{121a}} \equiv f_{\mathtt{11a}} \otimes \chi_{-11} = f_{\mathtt{121d}} \pmod 3$ in $S_2(\Gamma_0(121);\mathbb{Z})$, $B_2(121) = 22$. At the primes $3 = \ell$ and $11 \mid N$: $a_3 = 2$ and $-1$, $a_{11} = 0$ and 0.

(c) $f_{\mathtt{539c}} \equiv f_{\mathtt{11a}} \otimes \chi_{-7} = f_{\mathtt{539d}} \pmod 3$ in $S_2(\Gamma_0(539);\mathbb{Z})$, $B_2(539) = 112$. Here $a_3 = -2$ and 1, $a_7 = 0$ and 0, $a_{11} = 1$ and 1.

(d) $f_{\mathtt{980i}} \equiv f_{\mathtt{20a}} \otimes \chi_{-7} = f_{\mathtt{980g}} \pmod 5$ in $S_2(\Gamma_0(980);\mathbb{Z})$, $B_2(980) = 336$. Here $a_2 = 0$ and 0, $a_5 = 1$ and 1, $a_7 = 0$ and 0.

We checked each congruence for all $n \le 3000$, far beyond the bounds, and the equalities $f_{\mathtt{11a}} \otimes \chi_{-11} = f_{\mathtt{121d}}$, $f_{\mathtt{11a}} \otimes \chi_{-7} = f_{\mathtt{539d}}$, $f_{\mathtt{20a}} \otimes \chi_{-7} = f_{\mathtt{980g}}$ of twisted and tabulated forms for all $n \le 3000$ (each of these is again a statement in one space $S_2(\Gamma_0(N);\mathbb{Z})$, so the check up to $B_2(N)$ proves it once the twist is known to lie there). The Eisenstein congruence of Example 12.43 is the same argument in $M_2(\Gamma_0(11);\mathbb{Z})$.

*In the paper* 12.62. The paper says, at the start of its Example: "in each case the two forms lie in the same space $S_2(\Gamma_0(N))$ and have integer coefficients, so a check up to Sturm's bound [190], $\frac{1}{6}N\prod_{p|N}(1+\frac{1}{p})$, proves the congruence for all $n$." This is Corollary 12.59(iv), and Example 12.61 supplies the numbers. A check of the $a_p$ alone would also suffice, provided the $a_p$ with $p \mid \ell N$ are compared as well, because the $a_n$ with $n \le B_2(N)$ are determined by the $a_p$ with $p \le B_2(N)$ through the Hecke recursion (Chapter 13; Exercise 12.81). In Lemma lem:old the paper passes from the $a_p$ to all $a_n$ by that recursion, not by Sturm's bound.

## 12.12 The Atkin-Lehner involutions

The Atkin-Lehner involutions are the symmetries of $X_0(N)$ that do not come from $\mathrm{SL}_2(\mathbb{Z})$. The paper meets the most important one, $w_N$, in Section 4.1: the pairing $\hat{\delta}_!$ of Diamond, Flach and Guo is the cup product composed with "the operator $[UwU]_\omega$ attached to $w = \left(\begin{smallmatrix} 0 & -1 \\ M^\Sigma & 0 \end{smallmatrix}\right)$ [49, §1.5.3, (13), (14)], which for weight 2 and trivial character is the Atkin-Lehner involution $w_{M^\Sigma}$ with no scalar factor". (In [49, Section 1.5.3], $w$ is the element with these entries of $\prod_{p|N}\mathrm{GL}_2(\mathbb{Z}_p)$

at level $N$, and $[UwU]_\omega$ "coincides with $\psi(-1_\infty)N^{k-2}[Uw^{-1}U]_\omega$"; for $k=2$ and trivial character that scalar is 1; checked.) Through $\hat\delta_!$ the involution enters Proposition prop:pullback, Lemma lem:hecke and Step 2 of the proof of Theorem thm:quant. This section defines $W_d$ for every exact divisor $d$ of $N$, proves that the $w_d$ are commuting involutions of $X_0(N)$ and of the spaces of forms, that they are isometries for the Petersson product and the cup product, computes their action on cusps, and explains why their action on $q$-expansions is not simple. The fact that "$w_{M^\Sigma}T_n w_{M^\Sigma}^{-1}$ is the adjoint of $T_n$ for the cup product" and the pairing $\hat\delta_!$ itself are Chapter 14's.

**Definition 12.63** (Atkin-Lehner matrices)**.** Let $d \parallel N$, that is, $d \mid N$ and $\gcd(d, N/d)=1$. An *Atkin-Lehner matrix* for $d$ is an integer matrix

$$W_d = \begin{pmatrix} dx & y \\ Nz & dw \end{pmatrix}, \qquad \det W_d = d^2xw - Nyz = d.$$

Such matrices exist: dividing by $d$, the condition is $dxw-(N/d)yz=1$, which has solutions since $\gcd(d,N/d)=1$. For $d=N$ we take $W_N=\left(\begin{smallmatrix}0&-1\\N&0\end{smallmatrix}\right)$ ($x=w=0$, $y=-1$, $z=1$), and for $d=1$, $W_1=I$. We write $w_d$ for the induced map of $X_0(N)$ and for the operator $f\mapsto f|_k W_d$ on forms.

**Proposition 12.64** (the Atkin-Lehner involutions)**.** *Let $d, d' \parallel N$, and let $\mathcal{W}_d$ be the set of all Atkin-Lehner matrices for $d$.*

*(i)* $\Gamma_0(N)\mathcal{W}_d = \mathcal{W}_d = \mathcal{W}_d\Gamma_0(N)$*, and any two elements of $\mathcal{W}_d$ differ by left (or right) multiplication by an element of $\Gamma_0(N)$. In particular $W_d$ normalises $\Gamma_0(N)$.*

*(ii)* $W_d^2 \in d\cdot\Gamma_0(N)$*; explicitly* $W_d^2 = d\left(\begin{smallmatrix} dx^2+(N/d)yz & (x+w)y \\ Nz(x+w) & (N/d)yz+dw^2\end{smallmatrix}\right)$.

*(iii)* $W_dW_{d'} \in \gcd(d,d')\cdot\mathcal{W}_{d*d'}$ *with* $d*d' = dd'/\gcd(d,d')^2$.

*(iv)* $z\mapsto W_dz$ *induces a holomorphic involution $w_d$ of $X_0(N)$, independent of the choice of $W_d$; the $w_d$ commute, $w_dw_{d'}=w_{d*d'}$, and they form a group of automorphisms of $X_0(N)$ that is a quotient of $(\mathbb{Z}/2)^{\omega(N)}$ ($\omega(N)$ the number of primes dividing $N$). The involution $w_N$ is $z\mapsto -1/(Nz)$, and in the moduli description of Theorem 12.18 it is $(E,C)\mapsto(E/C, E[N]/C)$.*

*(v) For even $k$, $f\mapsto f|_kW_d$ is a well-defined involution of $M_k(\Gamma_0(N))$ preserving $S_k(\Gamma_0(N))$; these operators commute, and $w_N=\prod_{p^e\parallel N} w_{p^e}$.*

*(vi) In weight 2, $w_d^*(\omega_f)=\omega_{f|_2W_d}$ for $f\in M_2(\Gamma_0(N))$.*

*Proof.* (i) For $\gamma=\left(\begin{smallmatrix}a&b\\Nc&e\end{smallmatrix}\right)\in\Gamma_0(N)$,

$$\gamma W_d = \begin{pmatrix} adx+bNz & ay+bdw \\ Ncdx+eNz & Ncy+edw\end{pmatrix},$$

whose upper left and lower right entries are divisible by $d$ (as $d\mid N$), whose lower left entry is divisible by $N$, and whose determinant is $d$; so $\gamma W_d\in\mathcal{W}_d$, and likewise $W_d\gamma\in\mathcal{W}_d$. For $W,W'\in\mathcal{W}_d$ with entries $(dx,y,Nz,dw)$ and $(dx',y',Nz',dw')$, the matrix $W'W^{-1}=\frac{1}{d}W'\left(\begin{smallmatrix}dw&-y\\-Nz&dx\end{smallmatrix}\right)$ has entries $dx'w-\frac{N}{d}y'z$, $xy'-x'y$, $N(z'w-w'z)$, $dxw'-\frac{N}{d}yz'$: integers, with the lower left divisible by $N$, and determinant 1. So $W'\in\Gamma_0(N)W$, and similarly $W'\in W\Gamma_0(N)$. Then $W_d\gamma W_d^{-1}\in\mathcal{W}_dW_d^{-1}\subseteq\Gamma_0(N)$. (ii) Multiply out: $W_d^2=\left(\begin{smallmatrix}d^2x^2+Nyz & d(x+w)y\\ dN(x+w)z & Nyz+d^2w^2\end{smallmatrix}\right)$, which is $d$ times

the displayed matrix; that matrix is integral with lower left entry divisible by $N$ and determinant $\det(W_d)^2/d^2 = 1$. (iii) If $\gcd(d,d')=1$ then $dd' \parallel N$, and with $W_{d'} = \left(\begin{smallmatrix} d'x' & y' \\ Nz' & d'w' \end{smallmatrix}\right)$,

$$W_d W_{d'} = \begin{pmatrix} dd'xx' + Nyz' & dxy' + d'yw' \\ N(d'zx' + dwz') & Nzy' + dd'ww' \end{pmatrix}.$$

The upper left and lower right entries are divisible by $dd'$ because $dd' \mid N$, the lower left entry is divisible by $N$, and the determinant is $dd'$: so $W_dW_{d'} \in \mathcal{W}_{dd'}$. In general write $g = \gcd(d,d')$, $d = ge$, $d' = ge'$. For a prime $p \mid g$, $v_p(d) = v_p(N) = v_p(d')$ (both are exact divisors), so $v_p(g) = v_p(d)$ and $g$, $e$, $e'$ are pairwise coprime exact divisors of $N$. By the coprime case and (i), $\mathcal{W}_d = \mathcal{W}_g\mathcal{W}_e = \mathcal{W}_e\mathcal{W}_g$ and $\mathcal{W}_{d'} = \mathcal{W}_g\mathcal{W}_{e'}$, so by (ii) $W_dW_{d'} \in \mathcal{W}_e\mathcal{W}_g\mathcal{W}_g\mathcal{W}_{e'} = g\,\mathcal{W}_e\Gamma_0(N)\mathcal{W}_{e'} = g\,\mathcal{W}_{ee'}$, and $ee' = d * d'$. (iv) A matrix of $\mathrm{GL}_2^+(\mathbb{Q})$ that normalises $\Gamma_0(N)$ induces a bijection of $\Gamma_0(N)\backslash\mathcal{H}^*$ (it maps $\mathbb{P}^1(\mathbb{Q})$ to itself), holomorphic in the charts of Theorems 12.15 and 12.16 because it conjugates stabilisers to stabilisers. By (i) the map does not depend on the choice of $W_d$, by (ii) its square is induced by $d\gamma$, which acts as $\gamma \in \Gamma_0(N)$, that is, trivially; and by (iii) $w_dw_{d'} = w_{d*d'} = w_{d'}w_d$. The map $S \mapsto w_{\prod_{p\in S} p^{v_p(N)}}$ from subsets of the primes dividing $N$, with symmetric difference as group law, is then a surjective homomorphism from $(\mathbb{Z}/2)^{\omega(N)}$. For $W_N$, $W_Nz = -1/(Nz)$, and the moduli statement is Theorem 12.18(iii). (v) By (i) two choices of $W_d$ differ by $\gamma \in \Gamma_0(N)$, which acts trivially on $M_k(\Gamma_0(N))$; for $\gamma \in \Gamma_0(N)$, $f|_kW_d|_k\gamma = f|_k\gamma'|_kW_d = f|_kW_d$ with $\gamma' = W_d\gamma W_d^{-1} \in \Gamma_0(N)$; and holomorphy at the cusps follows as in Proposition 12.23(e), writing $W_d\sigma = \sigma'\beta$ with $\sigma' \in \mathrm{SL}_2(\mathbb{Z})$ and $\beta$ upper triangular. By (ii) and Lemma 12.21(ii), $f|_kW_d^2 = f|_k(d\gamma) = f$ for $k$ even (with Diamond-Shurman's normalisation one would get $d^{k-2}f$). By (iii) the operators commute, and $w_N = \prod w_{p^e}$ since $N$ is the $*$-product of the $p^e$. (vi) Lemma 12.21(iii). □

The three normalisations of $W_N$ in use, $\left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 1 \\ -N & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 1/\sqrt{N} \\ -\sqrt{N} & 0 \end{smallmatrix}\right)$, differ by the scalars $-1$ and $\sqrt{N}$, which act trivially on $\mathcal{H}$ and, with the slash operator of Definition 12.20, on forms of even weight. So the paper's $w$ in Section 4.1 is exactly $w_{M^\Sigma}$, with no hidden scalar: in weight 2 the factor $(\det)^{k/2} = (\det)^{k-1} = \det$ is the same in all conventions.

**Proposition 12.65** (Atkin-Lehner and the Petersson and cup products)**.** *Let $k$ be even and $d \parallel N$.*

*(i) For $f,g \in S_k(\Gamma_0(N))$: $\langle f|_kW_d, g|_kW_d\rangle = \langle f,g\rangle$ and $\langle f|_kW_d, g\rangle = \langle f, g|_kW_d\rangle$. So $w_d$ is a self-adjoint isometric involution of $S_k(\Gamma_0(N))$, which is the orthogonal direct sum of its $\pm1$-eigenspaces $S_k(\Gamma_0(N))^\pm$, and the $w_d$ are simultaneously diagonalisable.*

*(ii) For $k=2$, $\dim S_2(\Gamma_0(N))^+$ (for $w_N$) is the genus of the quotient curve $X_0(N)/w_N$.*

*(iii) $w_N$ is an involutive automorphism of the compact Riemann surface $X_0(N)$, so $w_N^*$ preserves the cup product on $H^1(X_0(N),\mathbb{Z})$ and $(w_N^*)^2 = 1$. If $f \in S_2(\Gamma_0(N))$ and $f|_2W_N = \varepsilon_N(f)f$ with $\varepsilon_N(f) = \pm1$, then $w_N^*$ acts by $\varepsilon_N(f)$ on $\omega_f$, on $\overline{\omega_f}$, and on the real plane of $H^1(X_0(N),\mathbb{R})$ spanned by $\mathrm{Re}\,\omega_f$ and $\mathrm{Im}\,\omega_f$.*

*Proof.* (i) The first identity is Proposition 12.51(ii). Applying it to $f$ and $g|_kW_d$ and using $g|_kW_d|_kW_d = g$ gives the second. A self-adjoint involution of a finite-dimensional Hermitian space is diagonalisable with eigenvalues $\pm1$ and orthogonal eigenspaces, and commuting diagonalisable operators are simultaneously diagonalisable. (ii) The quotient $X^+ = X_0(N)/\langle w_N\rangle$ of a compact

Riemann surface by a finite group of automorphisms is a compact Riemann surface, and $\pi^+\colon X_0(N) \to X^+$ is holomorphic of degree 2 (stated; [136, Chapter III, Section 3], section not verified). Pullback identifies $H^0(X^+, \Omega^1)$ with the $w_N$-invariant holomorphic differentials on $X_0(N)$: it is injective, and an invariant holomorphic differential descends, by the local computation in the proof of Theorem 12.40 with $e = 2$ at the fixed points of $w_N$. By (vi) of Proposition 12.64 the invariant differentials are the $\omega_f$ with $f \in S_2(\Gamma_0(N))^+$, and $\dim H^0(X^+, \Omega^1) = g(X^+)$ (Chapter 8). (iii) An automorphism of a compact Riemann surface preserves the orientation, so it preserves the integral of a 2-form and hence the cup product (Chapter 8). By Proposition 12.64(vi), $w_N^*\omega_f = \omega_{f|_2 W_N} = \varepsilon_N(f)\omega_f$; conjugating, $w_N^*\overline{\omega_f} = \varepsilon_N(f)\overline{\omega_f}$ as $\varepsilon_N(f)$ is real; and the real and imaginary parts are combinations of $\omega_f$ and $\overline{\omega_f}$ with real coefficients. By Chapter 8, $H^1(X_0(N), \mathbb{C}) = H^0(\Omega^1) \oplus \overline{H^0(\Omega^1)}$, so these classes live in $H^1$. □

Check of (ii) by Riemann-Hurwitz: if $w_N$ has $r$ fixed points then $2g(X_0(N)) - 2 = 2(2g(X^+) - 2) + r$. The number of fixed points is given by class numbers (for $N > 4$ it is $h(-4N) + h(-N)$ if $N \equiv 3 \pmod 4$ and $h(-4N)$ otherwise, $h$ the class number of the order of that discriminant; stated, [146]). For $N = 11$: $r = h(-44) + h(-11) = 3 + 1 = 4$, so $g(X^+) = 0$ and $S_2(\Gamma_0(11))^+ = 0$, consistent with $f_{11}|_2 W_{11} = -f_{11}$. For $N = 23$: $r = 3 + 3 = 6$, $g(X^+) = 0$. For $N = 37$: $r = h(-148) = 2$, $g(X^+) = 1$, and indeed $\dim S_2(\Gamma_0(37))^+ = 1$ (the form of `37a1`). For $N = 43$: $r = h(-172) + h(-43) = 3 + 1 = 4$, $g(X^+) = 1 = \dim S_2(\Gamma_0(43))^+$. The dimensions of the $+$-spaces were computed from PARI's Atkin-Lehner matrices.

*In the paper* 12.66. The pairing of Section 4.1 is $\hat{\delta}_! = \frac{1}{2}\langle x, w_{M^\Sigma}^* y\rangle$ on classes from $X_0(M^\Sigma)$, "perfect and alternating", and "the Hecke operators, $U_p$ included, are self-adjoint for it". What this chapter supplies is: the paper's $w$ is the matrix $W_{M^\Sigma}$ of Definition 12.63, with no scalar (the remark after Proposition 12.64); $w_N$ is an involutive automorphism of $X_0(N)$ (Proposition 12.64(iv)), so $w_N^* = (w_N^{-1})^*$ and $w_N^*$ preserves the cup product (Proposition 12.65(iii)), which is the input needed for the pairing $\langle x, w_N^* y\rangle$ to be perfect on each local component (Chapter 8); and on the $f$-part, $w_N^*$ acts by the sign $\varepsilon_N(f)$, on $f\,dz$ and on its conjugate (Proposition 12.65(iii)), which is the step "on which it acts by $\varepsilon_N(f)$ ($f|w_N = \varepsilon_N(f)f$ and $\varepsilon_N(f)$ is real)" of the proof of Proposition prop:pullback. That a newform is an eigenvector of $w_N$ is Chapter 13's; for $f_{11}$ it is Theorem 12.46(iii). The identity $T_n^\dagger = w_N T_n w_N$ for the cup product, and $\hat{\delta}_!$ itself, are Chapter 14's.

**Proposition 12.67** (Atkin-Lehner involutions and cusps)**.** *(i) $w_N$ exchanges $\infty$ and 0, and maps the cusps with* gcd(*denominator*, $N$) $= c$ *to those with* gcd(*denominator*, $N$) $= N/c$ *(Proposition 12.12(i)).*

*(ii) If $N$ is squarefree and $e \mid N$, then $w_e$ maps the cusp $1/c$ ($c \mid N$) to the cusp $1/(ce/\gcd(c,e)^2)$. The group $\{w_e : e \mid N\}$ is isomorphic to $(\mathbb{Z}/2)^{\omega(N)}$ and acts simply transitively on the $2^{\omega(N)}$ cusps.*

*Proof.* (i) $W_N\infty = 0$ and $W_N 0 = \infty$. Let $a/c'$ be a cusp in lowest terms with $\gcd(c', N) = c$. Then $W_N(a/c') = -c'/(Na)$. Write $c' = cc''$ with $\gcd(c'', N/c) = 1$ and $\gcd(a, c') = 1$; then $-c'/(Na) = -c''/((N/c)a)$, and since $\gcd(c'', (N/c)a) = 1$ this is in lowest terms with denominator $(N/c)a$, and $\gcd((N/c)a, N) = (N/c)\gcd(a, c) = N/c$ (a prime dividing $a$ and $c$ would divide $a$ and $c'$). (ii) For a prime $p \mid N$ and a cusp $u/v$ in lowest terms let $\delta_p(u/v) = 1$ if $p \mid v$ and 0 otherwise; by Proposition 12.12(i), (iii) the vector $(\delta_p)_{p|N}$ is a well-defined invariant of the cusp which separates the $2^{\omega(N)}$ cusps, the cusp $1/c$ having $\delta_p = 1$ exactly for $p \mid c$. We show that $w_e$ changes $\delta_p$ exactly for $p \mid e$. Let $W_e = \left(\begin{smallmatrix} ex & y \\ Nz & ew \end{smallmatrix}\right)$ and $p \mid N$. If $p \nmid e$: $p$ divides the lower left entry

$Nz$ and not the determinant $e$, so it divides neither the upper left nor the lower right entry. For $u/v$ in lowest terms the image is $(exu+yv)/(Nzu+ewv)$; if $p \mid v$ then $p \nmid u$, the numerator is $\equiv exu \not\equiv 0$ and the denominator $\equiv 0 \pmod p$, so $p$ divides the reduced denominator; if $p \nmid v$ the denominator is $\equiv ewv \not\equiv 0$, so $p$ does not divide the reduced denominator. So $\delta_p$ is unchanged. If $p \mid e$ (so $p \nmid N/e$): from $exw - (N/e)yz = 1$, $p \nmid y$. If $p \nmid v$, the numerator $exu+yv \equiv yv \not\equiv 0$ and the denominator $\equiv 0 \pmod p$, so the image has $\delta_p = 1$. So $w_e$ maps the set $X$ of cusps with $\delta_p = 0$ into the set $Y$ with $\delta_p = 1$; both have $2^{\omega(N)-1}$ elements and $w_e$ is a bijection, so $w_e(X) = Y$ and $w_e(Y) = X$. Hence $w_e$ toggles $\delta_p$ for $p \mid e$ and fixes it for $p \nmid e$, which is the formula: the primes of the new denominator are those of $c$ and $e$ not common to both. The action of the $w_e$ on the vectors $(\delta_p)$ is by symmetric difference, simply transitive; in particular $w_e \neq 1$ for $e \neq 1$, so the group is $(\mathbb{Z}/2)^{\omega(N)}$. □

We checked (ii) for $N = 6$, 30, 77 with explicit matrices $W_e$. For $N$ not squarefree the action can have fixed points: on the six cusps of $X_0(20)$, $w_4$ fixes $\frac{1}{2}$ and $\frac{1}{10}$ (Exercise 12.83).

**Example 12.68** (the action on $q$-expansions)**.** (a) $f|_k W_N(z) = N^{k/2}(Nz)^{-k} f(-1/(Nz))$ is, up to the factor and the change of variable $z \mapsto Nz$, the expansion of $f$ at the cusp 0. It is determined by $f$ through the transformation law. Since $M_k(\Gamma_0(N))$ is finite-dimensional and $f \mapsto (a_0(f), \dots, a_B(f))$ is injective for $B = \lfloor k\mu(N)/12 \rfloor$ (Theorem 12.39), each $a_n(f|_k W_N)$ is a linear combination $\sum_{m \le B} c_{n,m} a_m(f)$ with constants $c_{n,m}$ depending on $N$, $k$ and $n$; but these constants depend on the whole space and can only be computed through the transformation law (for instance from a basis whose expansions at 0 are known). There is no uniform coefficientwise rule: $a_n(f|_k W_N) = \pm a_n(f)$ for all $n$ holds only for eigenvectors of $w_N$.

(b) (Oldforms; Chapter 13.) Let $h \in M_k(\Gamma_0(M))$, $p \nmid M$ prime, and $N = Mp$. Then $W_N = W_M \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) W_N = W_M \cdot pI$, so by Proposition 12.23(e)

$$(h|_k W_N)(z) = p^{k/2}(h|_k W_M)(pz), \qquad \big(h(pz)\big)\big|_k W_N = p^{-k/2}(h|_k W_M)(z).$$

So $w_N$ exchanges the lines of $h(z)$ and $h(pz)$ when $h$ is a $w_M$-eigenvector, and it is not diagonal in the basis of $q$-expansions. For $h = f_{11}$, $k = 2$, $p = 2$: in the basis $f_{11}(z), f_{11}(2z)$ of $S_2(\Gamma_0(22))$ (which they span, $g(X_0(22)) = 2$), $w_{22}$ has the matrix $\left(\begin{smallmatrix} 0 & -1/2 \\ -2 & 0 \end{smallmatrix}\right)$, whose square is $I$ (checked numerically at a point of $\mathcal{H}$). Indeed, with $h|_2 W_{11} = -h$ the two formulas give $f_{11}|_2 W_{22} = 2 \cdot (-f_{11})(2z) = -2f_{11}(2z)$ and $f_{11}(2z)|_2 W_{22} = \frac{1}{2} \cdot (-f_{11}(z))$; these are the two columns. The eigenvectors are $f_{11}(z) \mp 2f_{11}(2z)$, with eigenvalues $\pm 1$.

(c) (Chapters 13 and 19; stated.) A newform $f \in S_k(\Gamma_0(N))$ is an eigenvector of every $w_d$, $d \,\|\, N$, with eigenvalue $\varepsilon_d(f) = \pm 1$; for $k = 2$ and $p \,\|\, N$, $a_p(f) = -\varepsilon_p(f)$ ([8], Theorem 3; part not verified); and for $k = 2$ the sign of the functional equation of $L(f, s)$ is $-\varepsilon_N(f)$. Checks: `11a1` has $\varepsilon_{11} = -1$, $a_{11} = 1$, root number $+1$; `37a1` has $\varepsilon_{37} = +1$, $a_{37} = -1$, root number $-1$ and rank 1.

For (b): $W_M \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & -1 \\ Mp & 0 \end{smallmatrix}\right) = W_N$ and $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) W_N = \left(\begin{smallmatrix} 0 & -p \\ Mp & 0 \end{smallmatrix}\right) = W_M \cdot pI$; then $h|_k W_N = (h|_k W_M)|_k \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) = p^{k/2}(h|_k W_M)(pz)$, and $h(pz) = p^{-k/2} h|_k \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ gives $h(pz)|_k W_N = p^{-k/2} h|_k W_M|_k (pI) = p^{-k/2} h|_k W_M$.

## 12.13 Exercises

**Exercise 12.69.** Find $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with $\gamma z \in \mathcal{F}$ for $z = (2+i)/7$, following the proof of Theorem 12.4(i), and write $\gamma$ as a word in $S$ and $T$. *Answer:* $Sz = -\frac{14}{5} + \frac{7}{5}i$, then $T^3$: $\gamma = T^3S = \left(\begin{smallmatrix} 3 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\gamma z = \frac{1+7i}{5}$.

**Exercise 12.70.** Compute $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(11)]$ and $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(11)]$, and the degrees of $X_1(11) \to X(1)$ and $X(11) \to X(1)$. Explain why the degree is half the index. *Answer:* 120 and 1320; the degrees are 60 and 660, because $-I \notin \Gamma_1(11)$ and $\pm\gamma$ act in the same way (Convention 12.1(c)).

**Exercise 12.71.** Let $p$ be a prime. Using Proposition 12.12(ii), show that $\Gamma_0(p^3)$ has $2p$ cusps: one of width $p^3$, $p-1$ of width $p$, and $p$ of width 1. Check that the widths add up to $\mu(p^3)$. Deduce $g(X_0(27)) = 1$ and $g(X_0(125)) = 8$. *Hint:* $c = 1, p, p^2, p^3$ give $\varphi(\gcd(c, p^3/c)) = 1, p-1, p-1, 1$ cusps of widths $p^3, p, 1, 1$; the sum is $p^3 + p^2$. For 125: $\mu = 150$, $\nu_2 = 2$, $\nu_3 = 0$, $\nu_\infty = 10$.

**Exercise 12.72.** Find the elliptic points of $\Gamma_0(65)$ and of $\Gamma_0(91)$, by solving $u^2 + 1 \equiv 0$ and $u^2 - u + 1 \equiv 0$ modulo $N$, and compute $g(X_0(65))$ and $g(X_0(91))$. *Answer:* for 65, $u \in \{8, 18, 47, 57\}$ give $\nu_2 = 4$, and $\nu_3 = 0$ ($5 \equiv 2 \pmod 3$); $\mu = 84$, $\nu_\infty = 4$, $g = 5$. For 91, $\nu_2 = 0$ ($7 \equiv 3 \pmod 4$), and $u \in \{10, 17, 75, 82\}$ give $\nu_3 = 4$ ($7 \equiv 13 \equiv 1 \pmod 3$); $\mu = 112$, $\nu_\infty = 4$, $g = 7$.

**Exercise 12.73.** (a) Show that for a prime $p \geq 5$, $g(X_0(p)) = \lfloor \frac{p+1}{12} \rfloor - 1$ if $p \equiv 1 \pmod{12}$ and $\lfloor \frac{p+1}{12} \rfloor$ otherwise, and deduce that $g(X_0(p)) = 0$ exactly for $p \in \{2, 3, 5, 7, 13\}$. (b) Show $\nu_2(N), \nu_3(N) \leq 2^{\omega(N)} \leq \nu_\infty(N)$ and hence $g(X_0(N)) \geq 1 + \frac{\mu(N) - 13\nu_\infty(N)}{12}$. (c) Show $\nu_\infty(p^e) \leq 2p^{\lfloor e/2 \rfloor}$, so that $\nu_\infty(N)/\mu(N) \leq \prod_{p^e \| N} \frac{2p^{1+\lfloor e/2 \rfloor - e}}{p+1}$, and conclude that $\mu(N) - 13\nu_\infty(N) \geq 12$, hence $g(X_0(N)) \geq 2$, for all but finitely many $N$, all of which can be listed. *Answer to the last part:* there are 64 exceptions (63 with $N \geq 2$), the largest being $N = 169$; together with the table computed for $N \leq 2000$ this proves that the lists of levels of genus 0 and 1 in Section 12.5 are complete. (For (a): $\mu = p + 1$, $\nu_\infty = 2$, and $(\nu_2, \nu_3)$ is determined by $p$ mod 12.)

**Exercise 12.74.** (a) Use $\dim M_8(\mathrm{SL}_2(\mathbb{Z})) = 1$ to show $E_4^2 = E_8$ and deduce $\sigma_7(n) = \sigma_3(n) + 120 \sum_{m=1}^{n-1} \sigma_3(m)\sigma_3(n-m)$. (b) Do the same with $E_4E_6 = E_{10}$. *Answer to (b):* $11\sigma_9(n) = 21\sigma_5(n) - 10\sigma_3(n) + 5040 \sum_{m=1}^{n-1} \sigma_3(m)\sigma_5(n-m)$, using $E_{10} = 1 - 264 \sum \sigma_9(n) q^n$.

**Exercise 12.75.** Show $j - 1728 = E_6^2/\Delta$. Deduce that, as functions of $z$, $j - 1728$ has a double zero at $i$ and $j$ a triple zero at $\zeta_3$, and check that this is consistent with $j$ being an isomorphism $X(1) \to \mathbb{P}^1(\mathbb{C})$ in the charts of Theorem 12.15. *Hint:* $E_4^3 - 1728\Delta = E_6^2$; in the chart $t = w^2$ at $i$ (respectively $t = w^3$ at $\zeta_3$) the zero is simple.

**Exercise 12.76.** The group $\Gamma_0(4)$ has the cusps $\infty$, $\frac{1}{2}$, 0 of widths $1, 1, 4$, genus 0, and $\dim M_2(\Gamma_0(4)) = 2$. Compute the constant terms of $E_2^{(4)}$ at the three cusps by the method of the proof of Proposition 12.42, and verify the relation $\sum_s h_s b_0(f|_2\sigma_s) = 0$ of Theorem 12.40. Show that $E_2^{(2)}$ and $E_2^{(4)}$ form a basis of $M_2(\Gamma_0(4))$. *Answer:* $-3$ at $\infty$; at $\frac{1}{2}$, with $\sigma = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 4 & 0 \\ 0 & 1 \end{smallmatrix}\right)\sigma = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 2 & -1 \\ 0 & 2 \end{smallmatrix}\right)$, so the constant is $1 - \frac{2}{2} = 0$; at 0, $1 - \frac{1}{4}$. Then $1 \cdot (-3) + 1 \cdot 0 + 4 \cdot \frac{3}{4} = 0$. Likewise $E_2^{(2)}$ has constant terms $-1$, $-1$, $\frac{1}{2}$ (at $\frac{1}{2}$: $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)\sigma = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$). The two vectors $(-1, -1, \frac{1}{2})$ and $(-3, 0, \frac{3}{4})$ are not proportional, so the forms are independent.

**Exercise 12.77.** Let $f_{19}$ be the normalised generator of $S_2(\Gamma_0(19))$, $f_{19} = q - 2q^3 - 2q^4 + \cdots$. Prove that $a_n(f_{19}) \equiv \sigma(n) - 19\sigma(n/19) \pmod 3$ for all $n$, using $h = 4(f_{19} + \frac{1}{24}E_2^{(19)}) \in M_2(\Gamma_0(19); \mathbb{Z})$

and $B_2(19) = 3$. Deduce $a_\ell(\texttt{19a1}) \equiv 1 + \ell \pmod 3$ for primes $\ell \neq 19$, and relate it to the rational point of order 3 of `19a1`. *Hint:* $h = -3 + 4\sum(a_n - \sigma(n) + 19\sigma(n/19))q^n$ has coefficients $-3, 0, -12, -24$ for $n \le 3$. (The identification of $f_{19}$ with the form of `19a1` is Chapter 14; we checked the congruence for $n \le 3000$.)

**Exercise 12.78.** Prove that $f_{20} = \eta(2z)^2\eta(10z)^2 \in S_2(\Gamma_0(20))$ by the method of Theorem 12.46(i). You may use Ligozat's formula [114]: the order of $\prod_\delta \eta(\delta z)^{r_\delta}$ ($\delta \mid N$) at a cusp with denominator $c \mid N$, in the local parameter of $X_0(N)$, is $\frac{N}{24}\sum_\delta \frac{\gcd(c,\delta)^2 r_\delta}{\gcd(c, N/c)\, c\, \delta}$. *Hint:* $X_0(20)$ has genus 1 and no elliptic points, so a generator $h$ of $S_2(\Gamma_0(20))$ has a simple zero at each of the six cusps and no other zero. The form $f_{20}^{12} = \Delta(2z)\Delta(10z)$ lies in $S_{24}(\Gamma_0(20))$ by Proposition 12.23(e), and the formula gives order 12 at each of the six cusps $c = 1, 2, 4, 5, 10, 20$.

**Exercise 12.79.** (a) Compute $\langle f_{11}, f_{11}\rangle$ without `mfpetersson`, by integrating over the fundamental domain $\mathcal{F} \cup \bigcup_{j=0}^{10} ST^j\mathcal{F}$ of Proposition 12.8. Show first, from $W_{11} = S\left(\begin{smallmatrix}11&0\\0&1\end{smallmatrix}\right)$ and $f_{11}|_2 W_{11} = -f_{11}$, that $(f_{11}|_2 S)(z) = -\frac{1}{11} f_{11}(z/11)$, and deduce from Proposition 12.49(ii) that $\langle f_{11}, f_{11}\rangle = \int_{\mathcal{F}} (|f_{11}(z)|^2 + \frac{1}{121}\sum_{j=0}^{10} |f_{11}(\frac{z+j}{11})|^2)\, dx\, dy$. Evaluate the integral numerically with the first 400 coefficients. (b) The curves `11a2` and `11a3` receive parametrisations $X_0(11) \to$ `11a1` $\to$ `11a2`, `11a3` of degree 5. Compute $\mathrm{vol}(\Lambda_{\texttt{11a2}})$ and $\mathrm{vol}(\Lambda_{\texttt{11a3}})$ and use $\deg\varphi \cdot \mathrm{vol}(\Lambda) = 4\pi^2 c_\varphi^2 \langle f, f\rangle$ to find the constants $c_\varphi$ of these two parametrisations. *Answer:* (a) $f_{11}|_2 S = f_{11}|_2 W_{11}|_2 \left(\begin{smallmatrix}1/11&0\\0&1\end{smallmatrix}\right) = -\frac{1}{11} f_{11}(z/11)$, and $(f|_2 ST^j)(z) = (f|_2 S)(z + j)$; in weight 2, $y^2\, d\mu = dx\, dy$. Numerical integration (truncating at $y = 40$) gives 0.04690014787349, in agreement with $12 \cdot$ `mfpetersson` $=$ 0.0469001478734952 of Example 12.53. (b) The volumes are $\frac{1}{5}$ and 5 times $\mathrm{vol}(\Lambda_{\texttt{11a1}}) = 1.8515\ldots$, so $c_\varphi^2 = 1$ and 25: the constants are 1 and 5. With the Manin constant in the denominator one would get the impossible value $\frac{1}{5}$ for `11a3`.

**Exercise 12.80.** Let $f = f_{11}$. (a) Show $\langle f, f\rangle_{\Gamma_0(22)} = 3\langle f, f\rangle_{\Gamma_0(11)}$. (b) Deduce from Proposition 12.51(iv), with $\alpha = \left(\begin{smallmatrix}2&0\\0&1\end{smallmatrix}\right)$, that $\langle f(2z), f(2z)\rangle_{\Gamma_0(22)} = \frac{3}{4}\langle f, f\rangle_{\Gamma_0(11)}$. *Hint:* $\alpha\Gamma_0(22)\alpha^{-1} = \Gamma_0(11) \cap \Gamma^0(2)$, where $\Gamma^0(2)$ consists of the matrices with $b$ even; it has index 3 in $\Gamma_0(11)$, and $f(2z) = \frac{1}{2} f|_2\alpha$. (PARI gives $-\langle f, f\rangle_{\Gamma_0(11)}$ for $\langle f(z), f(2z)\rangle_{\Gamma_0(22)}$, which is $\frac{a_2(f)}{2}\langle f, f\rangle_{\Gamma_0(11)}$; Chapter 13 explains the factor.)

**Exercise 12.81.** Let $f, g \in S_2(\Gamma_0(N); \mathbb{Z})$ be normalised eigenforms of all the Hecke operators $T_n$, $n \ge 1$ (Chapter 13: then $a_{mn} = a_m a_n$ for $\gcd(m, n) = 1$, $a_{p^{r+1}} = a_p a_{p^r} - p\, a_{p^{r-1}}$ for $p \nmid N$, and $a_{p^r} = a_p^r$ for $p \mid N$). Show that if $a_p(f) \equiv a_p(g) \pmod\ell$ for all primes $p \le B_2(N)$, then $f \equiv g \pmod\ell$ coefficientwise. *Hint:* the recursions express $a_n$, $n \le B_2(N)$, through the $a_p$ with $p \le n$, by polynomials with integer coefficients; then use Corollary 12.59(iv). The hypothesis includes the primes $p \mid N$, whose $a_p$ are not determined by the others, and the prime $p = \ell$. For the congruence modulo 3 at level 121 these are the primes $p = \ell = 3$ and $p = 11$ dividing $N$, which is why Example 12.61(b) lists $a_3$ and $a_{11}$ separately.

**Exercise 12.82.** Find explicit Atkin-Lehner matrices $W_4$ and $W_5$ for $N = 20$, and check $W_4^2 \in 4\Gamma_0(20)$ and $W_4 W_5 \in \mathcal{W}_{20}$. *Answer:* for instance $W_4 = \left(\begin{smallmatrix}-4&1\\-20&4\end{smallmatrix}\right)$ ($x = -1$, $y = 1$, $z = -1$, $w = 1$) and $W_5 = \left(\begin{smallmatrix}5&1\\20&5\end{smallmatrix}\right)$ ($x = y = z = w = 1$). Then $W_4^2 = -4I = 4 \cdot (-I)$, and $W_4 W_5 = \left(\begin{smallmatrix}0&1\\-20&0\end{smallmatrix}\right) = -W_{20}$, an Atkin-Lehner matrix for 20.

**Exercise 12.83.** List the six cusps of $X_0(20)$ (Example 12.13) and compute the permutations induced by $w_4$, $w_5$, $w_{20}$. Show that the group $\{1, w_4, w_5, w_{20}\}$ has two orbits on the cusps, so that Proposition 12.67(ii) fails for $N$ not squarefree. *Answer:* $w_4$: $0 \leftrightarrow \frac{1}{4}$, $\frac{1}{5} \leftrightarrow \infty$, $\frac{1}{2}$ and $\frac{1}{10}$

fixed; $w_5$: $0 \leftrightarrow \frac{1}{5}$, $\frac{1}{2} \leftrightarrow \frac{1}{10}$, $\frac{1}{4} \leftrightarrow \infty$; $w_{20}$: $0 \leftrightarrow \infty$, $\frac{1}{2} \leftrightarrow \frac{1}{10}$, $\frac{1}{4} \leftrightarrow \frac{1}{5}$. The orbits are $\{0, \frac{1}{4}, \frac{1}{5}, \infty\}$ and $\{\frac{1}{2}, \frac{1}{10}\}$.

**Exercise 12.84.** Let $h \in S_2(\Gamma_0(M))$ with $h|_2 W_M = \varepsilon_M(h)h$, $p \nmid M$ prime, and $N = Mp^2$. Show that in the basis $h(z), h(pz), h(p^2z)$ the operator $w_N$ has the matrix $\varepsilon_M(h)\left(\begin{smallmatrix} 0 & 0 & p^{-2} \\ 0 & 1 & 0 \\ p^2 & 0 & 0 \end{smallmatrix}\right)$, and find its eigenvectors. For $h = f_{11}$, $p = 2$, which combinations of $f_{11}(z), f_{11}(2z), f_{11}(4z)$ lie in $S_2(\Gamma_0(44))^+$? *Hint:* $W_N = W_M\left(\begin{smallmatrix} p^2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)W_N = p\,W_M\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} p^2 & 0 \\ 0 & 1 \end{smallmatrix}\right)W_N = p^2 W_M$. The eigenvectors are $h(pz)$ (eigenvalue $\varepsilon_M(h)$) and $h \pm p^2 h(p^2 z)$ (eigenvalues $\pm\varepsilon_M(h)$). For $f_{11}$ ($\varepsilon_{11}(f_{11}) = -1$) only $f_{11}(z) - 4f_{11}(4z)$.

**Exercise 12.85.** (a) Show that $z_0 = i/\sqrt{N}$ is fixed by $w_N$, and that for $E = \mathbb{C}/\Lambda_{z_0}$ multiplication by $i\sqrt{N}$ is an endomorphism of degree $N$ with kernel $C = \langle \frac{1}{N} \rangle$; so the pair $(E, C)$ satisfies $(E/C, E[N]/C) \simeq (E, C)$, as Theorem 12.18(iii) predicts. (b) Using Riemann-Hurwitz for $X_0(N) \to X_0(N)/w_N$ and Proposition 12.65(ii), show that $w_N$ has $2g(X_0(N)) + 2 - 4\dim S_2(\Gamma_0(N))^+$ fixed points. Given $\dim S_2(\Gamma_0(121))^+ = 2$ (PARI), find the number of fixed points of $w_{121}$ and compare with the class number $h(-484) = 6$. *Answer:* (a) $i\sqrt{N} \cdot 1 = N \cdot \frac{i}{\sqrt{N}}$ and $i\sqrt{N} \cdot \frac{i}{\sqrt{N}} = -1$, so $i\sqrt{N}\Lambda_{z_0} \subseteq \Lambda_{z_0}$, of index $|i\sqrt{N}|^2 = N$, and $(i\sqrt{N})^{-1} \cdot \frac{i}{\sqrt{N}} = \frac{1}{N}$ generates the kernel. (b) $2 \cdot 6 + 2 - 4 \cdot 2 = 6$.

## 12.14 Notes and references

The chapter follows Diamond and Shurman [52] (Chapters 1 to 3 for the modular curves, the genus and the dimensions, Chapter 5 for the Petersson product and the Atkin-Lehner operators, Section 6.5 for integrality), Serre's *Course in Arithmetic* [172, Chapter VII] for level one (fundamental domain, valence formula, the structure of $M_*$, the bijectivity of $j$), and Shimura [180] (Chapters 1 to 3) for the general theory. Other good sources are Miyake [137] (Section 4.6 for the Atkin-Lehner operators), Koblitz [102, Chapter III], Knapp [101] (Chapters VIII, IX and XI), Apostol [6] (the $\eta$-function, Chapter 3) and Stein's computational book [187] (Chapter 9 for Sturm's bound). Atkin and Lehner's original paper is [8]; Sturm's theorem is [190]; Ligozat's criterion and his study of the curves of genus one are in [114]; the cusps of $X_0(N)$, their fields of definition and the fixed points of $w_N$ are in Ogg [146]; the moduli problem and the model of $X_0(N)$ over $\mathbb{Q}$ and over $\mathbb{Z}$ are in Deligne-Rapoport [43] and Katz-Mazur [91]. Deligne's proof of the Ramanujan-Petersson conjecture is [44]. For the formula of Zagier and Watkins see [209, 204] and Chapter 14; for the Rankin-Selberg formula [155, 183] and [49, Lemma 2.12], and Chapter 19. The curves and their invariants are Cremona's [35], and the computations use PARI/GP [148].

Pointers that were not checked against the printed books, and are given by chapter or section only: [52] Sections 1.1 to 1.5, 2.1 to 2.4, 3.1, 3.5 to 3.9, 5.4, 5.5, 6.5, 7.7, 8.7, 8.8, and the theorem numbers 3.5.1, 3.6.1 quoted for the dimension formulas; [180] Sections 1.5, 2.4, 2.6, 6.7 and Theorem 3.52 (integral bases); [172] Chapter VII; [187] Chapter 9; [136] Chapter III, Section 3 (quotients by finite groups); [6] Chapter 3; [102] Section III.2; [8] Theorem 3 for $a_p = -\varepsilon_p$. Every number printed in the chapter (indices, cusps and widths, elliptic points, genera, dimensions, Sturm bounds, $q$-expansions, Atkin-Lehner signs and matrices, Petersson norms, period areas, class numbers, congruences) was recomputed with PARI/GP 2.17 through `cypari2` and Cremona's tables for conductors below 10,000; the ranges of the checks are stated where the numbers appear.

The Petersson product has several normalisations in the literature (Convention 12.50); the book's is the unnormalised one, in which the paper's Remark rem:degree holds as stated. The slash operator is normalised with $(\det\gamma)^{k/2}$, as in Shimura [180] and Miyake [137] (not checked against their texts); Diamond and Shurman use $(\det\gamma)^{k-1}$. Neither choice matters in weight 2.

# Chapter 13

# Hecke operators, newforms and twists

The paper works throughout with the Hecke algebra $\mathbb{T}$ of $S_2(\Gamma_0(N))$, with its eigencharacters, with newforms and their oldforms, with the operators $U_p$ at primes dividing the level, with depleted forms and with quadratic twists. This chapter supplies all of that. It defines the Hecke operators by double cosets and computes them on $q$-expansions; proves commutativity, multiplicativity and the recursion; proves that $\mathbb{T}$ is free over $\mathbb{Z}$ of rank $\dim S_2(\Gamma_0(N))$ and that the pairing $(g,t) \mapsto a_1(tg)$ between integral forms and $\mathbb{T}$ is perfect; computes Petersson adjoints; states the theory of Atkin, Lehner and Li (newforms, multiplicity one, strong multiplicity one); works out $U_p$ on oldforms, the stabilised and depleted forms and the level $M^{\Sigma}$ of the paper's Section 4.1; proves the decomposition $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ used in Lemma lem:crit; and treats twisting, including the theorem of Atkin and Li and the transport of congruences along the twist map $\iota$ of Theorems thm:twist and thm:quant, with the consequence that the congruence number $r_E$ divides the Hecke congruence module of the depleted form $g^S$ of Step 1 of the proof of Theorem thm:quant at every prime. It ends with the $L$-function of a newform and the Rankin-Selberg residue. It uses Chapter 2 (Gauss sums, Dirichlet characters), Chapter 3 (lattices, saturation, duality of free modules, idempotents of finite algebras, generalised eigenspaces, congruence ideals), Chapter 6 (algebraic integers, orders) and, above all, Chapter 12 (the slash operator, $q$-expansions, the Petersson product, the Atkin-Lehner involutions, the integral structure $S_2(\Gamma_0(N);\mathbb{Z})$ and Sturm's bound). Galois representations of newforms (Chapter 15) appear only in sketches and remarks.

## Core results at a glance

Throughout, $N \geq 1$ is the level, $k \geq 2$ an even weight (the paper needs $k = 2$), $\Gamma = \Gamma_0(N)$, $S_k(N) = S_k(\Gamma_0(N))$, $M_k(N) = M_k(\Gamma_0(N))$, $(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz+d)^{-k}f(\gamma z)$, $V_d\colon h(z) \mapsto h(dz)$, $\langle\ ,\ \rangle$ the unnormalised Petersson product, $t^\dagger$ the adjoint of an operator, $\mathbb{T} = \mathbb{T}_N$ the full Hecke algebra (the $U_p$ included), $\mathbb{T}^{(N)}$ the anemic one, $\chi_{q^*}$ the quadratic character of conductor $q$ and $f^{\Sigma}$ the $\Sigma$-depleted form.

**13.1 Conventions.** The standing notation; the double coset operator, normalised as $[\Gamma\alpha\Gamma]_k f = (\det\alpha)^{k/2-1}\sum_i f|_k\alpha_i$, equal to Diamond-Shurman's; $T_p$ at $p \mid N$ is written $U_p$, and the level is written as a superscript when two levels occur, $T_p^{(M)}h = U_p^{(Mp)}h + p^{k-1}h(pz)$; naive twists $f \otimes \chi$ and depletion $f^{\Sigma}$. Convention 13.1, Convention 13.2, Convention 13.3, Convention 13.4.

**13.2 Double coset operators** (definition). For $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$, $\Gamma\alpha\Gamma = \bigsqcup_{i=1}^{r} \Gamma\alpha_i$ with $r = [\Gamma : \Gamma \cap \alpha^{-1}\Gamma\alpha]$ finite, and $[\Gamma\alpha\Gamma]_k$ is well defined and preserves $M_k(N)$ and $S_k(N)$. *(proved here)* Definition 13.5, Proposition 13.6.

**13.3 Coset decompositions.** For $p \nmid N$, $\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma = \bigsqcup_{j=0}^{p-1} \Gamma\left(\begin{smallmatrix}1&j\\0&p\end{smallmatrix}\right) \sqcup \Gamma\left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right) = \Gamma\left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right)\Gamma$; for $p \mid N$, $\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma = \bigsqcup_{j=0}^{p-1} \Gamma\left(\begin{smallmatrix}1&j\\0&p\end{smallmatrix}\right)$. $T_p = [\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma]_k$, written $U_p$ when $p \mid N$. *(proved here)* Proposition 13.7.

**13.4 $q$-expansions.** $a_n(T_p f) = a_{pn}(f) + p^{k-1}a_{n/p}(f)$ for $p \nmid N$ and $a_n(U_p f) = a_{pn}(f)$ for $p \mid N$, for all $n \geq 0$. *(proved here)* Proposition 13.8.

**13.5 The operators $\mathcal{U}_p$, $V_p$.** On $q$-series $\mathcal{U}_p V_p = 1$, $V_p\mathcal{U}_p$ is the projection onto the multiples of $p$, and the operators at different primes (and $V_d$ with $(d,p) = 1$) commute. *(proved here)* Lemma 13.9.

**13.6 The operators $T_n$.** With $T_{p^{r+1}} = T_p T_{p^r} - p^{k-1}T_{p^{r-1}}$ $(p \nmid N)$, $T_{p^r} = U_p^r$ $(p \mid N)$ and $T_n = \prod_{p^r \| n} T_{p^r}$: all $T_n$ commute; $T_m T_n = \sum_{d|(m,n),(d,N)=1} d^{k-1} T_{mn/d^2}$; $a_m(T_n f) = \sum_{d|(m,n),(d,N)=1} d^{k-1} a_{mn/d^2}(f)$, so $a_1(T_n f) = a_n(f)$; $\sum_r T_{p^r} X^r = (1 - T_p X + p^{k-1}X^2)^{-1}$ or $(1 - U_p X)^{-1}$. *(proved here)* Theorem 13.10.

**13.7 The Hecke algebra** (definition and proposition). $\mathbb{T} = \mathbb{Z}[T_n : n \geq 1]$ is commutative, equal to the $\mathbb{Z}$-span of the $T_n$, preserves $S_2(N;\mathbb{Z})$, and is a free $\mathbb{Z}$-module of finite rank; $\mathbb{T}^{(N)}$ is the $\mathbb{Z}$-span of the $T_n$ with $(n,N) = 1$. *(proved here, given the integral basis theorem of Chapter 12)* Definition 13.15, Proposition 13.16.

**13.8 Eigenforms** (definition and proposition). A simultaneous eigenvector of $\mathbb{T}$ has $a_1 \neq 0$; normalised ($a_1 = 1$) it satisfies $T_n f = a_n(f) f$, $a_{mn} = a_m a_n$ for $(m,n) = 1$, $a_{p^{r+1}} = a_p a_{p^r} - p^{k-1}a_{p^{r-1}}$ $(p \nmid N)$, $a_{p^r} = a_p^r$ $(p \mid N)$; $\phi_f \colon \mathbb{T} \to \mathbb{C}$, $T_n \mapsto a_n(f)$, is a ring homomorphism; distinct normalised eigenforms are linearly independent. *(proved here)* Definition 13.18, Proposition 13.19.

**13.9 Eigenform criterion.** A cusp form $g = \sum b_n q^n$ of level $L$ with $b_1 = 1$ is a normalised eigenform of $\mathbb{T}_L$ iff $(b_n)$ is multiplicative and satisfies the prime-power relations of level $L$. *(proved here)* Lemma 13.20.

**13.10 Coefficient fields.** For a normalised eigenform $f$ in $S_2(N)$ the $a_n(f)$ are algebraic integers, $K_f = \mathbb{Q}(a_n(f))$ is a number field, $\phi_f(\mathbb{T})$ is an order of $\mathcal{O}_{K_f}$, and $f^\sigma = \sum \sigma(a_n) q^n$ is again a normalised eigenform for $\sigma \in \mathrm{Aut}(\mathbb{C})$. *(proved here, given Chapter 6)* Proposition 13.22.

**13.11 The perfect pairing.** $S = S_2(N;\mathbb{Z})$, $d = \dim S_2(N)$: $\langle g, t\rangle = a_1(tg)$ is a perfect pairing $S \times \mathbb{T} \to \mathbb{Z}$ with $\langle ug, t\rangle = \langle g, ut\rangle$; $\mathbb{T}$ is free of rank $d$, $S \simeq \mathrm{Hom}(\mathbb{T},\mathbb{Z})$, $\mathbb{T} \simeq \mathrm{Hom}(S,\mathbb{Z})$, and the pairing stays perfect after $\otimes A$ for every ring $A$. $\mathbb{T}$ is spanned by $T_1, \ldots, T_B$, $B$ Sturm's bound, but in general not by $T_1, \ldots, T_d$. *(proved here, given the integral basis theorem; the last part given Sturm's bound)* Theorem 13.24.

**13.12 Eigenforms and homomorphisms.** For every commutative ring $A$, $S_2(N;A) \simeq \mathrm{Hom}_{\mathbb{Z}}(\mathbb{T}, A)$, and normalised eigenforms with coefficients in $A$ correspond to ring homomorphisms $\phi \colon \mathbb{T} \to A$, $\phi \mapsto \sum \phi(T_n) q^n$; maximal ideals of residue characteristic $\ell$ correspond to mod $\ell$ eigenforms up to conjugacy. *(proved here, given Theorem 13.24)* Theorem 13.26.

**13.13 Consequences of the pairing.** A form is determined by its pairing with $\mathbb{T}$; $S[\mathfrak{a}] \simeq \mathrm{Hom}(\mathbb{T}/\mathfrak{a}, \mathbb{Z})$ is saturated; $S[\ker\phi] = \mathbb{Z}f_\phi$ for $\phi\colon \mathbb{T} \to \mathbb{Z}$; if $I \cap J = 0$, $\mathbb{T}/I$ and $\mathbb{T}/J$ are torsion-free and $\mathbb{T}/(I+J)$ is finite, then $S/(S[I] \oplus S[J])$ has the order of $\mathbb{T}/(I+J)$. *(proved here)* Corollary 13.28.

**13.14 Congruence ideal and congruence number** (definition). $\eta(\phi) = \phi(\mathrm{Ann}_{\mathbb{T}_\mathcal{O}} \ker\phi)$; $r_f$ the largest $r$ with $f \equiv g \pmod r$ for an integral $g \perp f$; $r_f$ is finite. *(proved here)* Definition 13.29, Proposition 13.30.

**13.15 Adjoints of double cosets.** $\langle [\Gamma\alpha\Gamma]_k f, g\rangle = \langle f, [\Gamma\alpha'\Gamma]_k g\rangle$ with $\alpha' = (\det\alpha)\alpha^{-1}$. *(proved here)* Proposition 13.32.

**13.16 Self-adjointness.** $T_n$ is self-adjoint on $S_k(N)$ for $(n, N) = 1$; $S_k(N)$ is the orthogonal sum of the eigenspaces of $\mathbb{T}^{(N)}$, whose eigenvalues are real. *(proved here)* Theorem 13.33.

**13.17 The adjoint of $U_p$.** $U_p^\dagger = w_N U_p w_N$ for $p \mid N$; $U_p$ is in general neither self-adjoint nor normal; for a newform $h$ of level $M$ and $p \nmid M$, at level $Mp$, $\langle h(pz), h\rangle = \frac{p^{1-k}a_p(h)}{p+1}\langle h, h\rangle$ and $\langle h(pz), h(pz)\rangle = p^{-k}\langle h, h\rangle$. *(proved here)* Lemma 13.34, Proposition 13.35.

**13.18 Stable complements.** If $V$ is stable under $\mathbb{T}$ and $w_N$ then so is $V^\perp$; in particular $f^\perp$ is $\mathbb{T}$-stable for an eigenform $f$ of $\mathbb{T}$ that is an eigenvector of $w_N$. *(proved here)* Corollary 13.37.

**13.19 Oldforms and newforms** (definition and proposition). $S_k^{\mathrm{old}}(N) = \sum_{p|N}(S_k(N/p) + V_p S_k(N/p))$, $S_k^{\mathrm{new}}(N)$ its orthogonal complement; both are stable under $\mathbb{T}$ and $w_N$, and $w_N$ maps $V_d S_k(M)$ onto $V_{N/(Md)} S_k(M)$. *(proved here)* Definition 13.39, Proposition 13.40.

**13.20 Atkin-Lehner theory.** $S_k^{\mathrm{new}}(N)$ has an orthogonal basis of newforms; multiplicity one: the $\mathbb{T}^{(L)}$-eigenspace in $S_k(L)$ of the system of a newform $f$ of level $M \mid L$ is $\mathrm{span}\{f(dz) : d \mid L/M\}$; $S_k(N) = \bigoplus_{M|N}\bigoplus_{d|N/M} V_d S_k^{\mathrm{new}}(M)$ with basis the $h(dz)$; $f^\perp = \bigoplus_{h\neq f} V_h$. *(stated; reference, with a sketch)* Theorem 13.41.

**13.21 Strong multiplicity one.** Two newforms with $a_p(f) = a_p(g)$ for almost all $p$ (in weight 2: for a set of primes of density one) are equal; a generalised eigenform of $\mathbb{T}^{(L)}$ whose system differs from that of a newform $f$ at infinitely many $p$ is orthogonal to every $f(dz)$. *(stated; reference, with a sketch; the consequence proved here)* Theorem 13.42.

**13.22 Local behaviour of newforms.** A newform $f$ of level $N$ satisfies $f|_k W_N = \varepsilon_N(f) f$, $\varepsilon_N(f) = \pm 1$ *(proved here, given Theorem 13.41)*; $a_p(f)$ is real for $p \mid N$ *(proved here)*; $a_p(f) = 0$ if $p^2 \mid N$ and $a_p(f) = -p^{k/2-1}\varepsilon_p(f)$ if $p \,\|\, N$ *(stated; reference, with a sketch)*. Proposition 13.44.

**13.23 Galois conjugates.** The old and new subspaces are defined over $\mathbb{Q}$; conjugates of newforms are newforms; $K_f$ is totally real for a newform $f$. *(proved here, given Theorem 13.41)* Corollary 13.45.

**13.24 Dimensions.** With $\beta = \mu * \mu$:

$$\dim S_k(N) = \sum_{M|N} \sigma_0(N/M) \dim S_k^{\mathrm{new}}(M), \qquad \dim S_k^{\mathrm{new}}(N) = \sum_{M|N} \beta(N/M) \dim S_k(M).$$

*(proved here, given Theorem 13.41)* Corollary 13.46.

**13.25** $U_p$ **on oldforms.** For a newform $h$ of level $M$, on $V_{h,p,e} = \operatorname{span}\{h(p^j z) : 0 \le j \le e\}$: $U_p h(p^j z) = h(p^{j-1}z)$, $U_p h = a_p h - p^{k-1}h(pz)$ ($p \nmid M$) or $a_p h$ ($p \mid M$); $h(p^e z)$ is cyclic; characteristic polynomial $X^{e-1}(X^2 - a_p X + p^{k-1})$ or $X^e(X - a_p)$; eigenvectors: the stabilisations $h - \beta_p h(pz)$, $h - \alpha_p h(pz)$, $h$, and the depleted forms; semisimple iff the characteristic polynomial is squarefree. *(proved here, given Theorem 13.41)* Proposition 13.48.

**13.26** **Eigenvalues of** $U_p$**.** In weight 2 every eigenvalue of $U_p$ on $S_2(N)$ is 0, $\pm 1$ or a root of $X^2 - a_p(h)X + p$; the nonzero ones are units at every prime not above $p$. *(proved here, given Theorem 13.41 and Proposition 13.44)* Corollary 13.51.

**13.27** **Depletion** (definition and proposition). $\delta_p = 2, 1, 0$ for $p \nmid M$, $p \parallel M$, $p^2 \mid M$; $f^\Sigma = \prod_{p\in\Sigma} D_p f$; $f^\Sigma$ has least level $M^\Sigma = M\prod_{p\in\Sigma} p^{\delta_p}$, is a normalised eigenform of $\mathbb{T}_{M^\Sigma}$ with $U_p f^\Sigma = 0$ ($p \in \Sigma$), and its generalised eigenspace in $S_k(M^\Sigma)$ is $\mathbb{C}f^\Sigma$. *(proved here, given Theorem 13.41)* Definition 13.54, Proposition 13.55.

**13.28** **The structure of** $\mathbb{T}$**.** $S_k(N) = \bigoplus_h V_h$, $V_h \simeq \bigotimes_{p\mid N} V_{h,p,e_p}$ as a module over the $U_p$; $\mathbb{T}\otimes\overline{\mathbb{Q}} \simeq \prod_h \bigotimes_p \overline{\mathbb{Q}}[X]/(m_{h,p})$; $S_k(N)$ is free of rank one over $\mathbb{T}\otimes\mathbb{C}$; $\mathbb{T}\otimes\mathbb{Q}$ is reduced iff every $m_{h,p}$ is squarefree. *(proved here, given Theorem 13.41)* Theorem 13.59.

**13.29** **The newform splitting.** For a newform $f \in S_2(N)$ with rational coefficients, $f^\perp$ is $\mathbb{T}$-stable and $\mathbb{T}\otimes\mathbb{Q} = \mathbb{Q}\times\mathbb{T}'$, the first factor acting on $\mathbb{Q}f$ through $\phi_f$ and $\mathbb{T}'$ acting faithfully on $f^\perp$; the idempotent of the first factor lies in $\mathbb{T}\otimes\mathbb{Q}$ and kills $f^\perp$. *(proved here, given Theorem 13.41)* Theorem 13.60.

**13.30** $\mathbb{T}$**-stable versus Petersson complements.** For an eigencharacter $\pi$ whose generalised eigenspace is a line $\mathbb{C}g$: $S_k(N) = \mathbb{C}g \oplus g^c$ with $g^c$ the $\mathbb{T}$-stable complement, $\mathbb{T}\otimes\mathbb{Q} = \mathbb{Q}\times\mathbb{T}''$ when $k = 2$ and $\pi$ is $\mathbb{Q}$-valued, $g^c \cap S_2(N;\mathbb{Z})$ saturated of rank $d-1$; $g^c = g^\perp$ for newforms, not in general (level 44). *(proved here)* Proposition 13.62.

**13.31** **The congruence module of a rational eigencharacter.** Let $k = 2$ and $\pi\colon \mathbb{T} \to \mathbb{Q}$ a ring homomorphism whose generalised eigenspace is a line $\mathbb{C}g$, with $\mathbb{T}$-stable complement $g^c$ and $\Lambda = g^c \cap S_2(N;\mathbb{Z})$. Then $\pi(\mathbb{T}) = \mathbb{Z}$, $\mathbb{T}/(\ker\pi + \operatorname{Ann}_\mathbb{T}\ker\pi)$ is cyclic of order $r_\pi$ with $\eta(\pi) = r_\pi\mathbb{Z}$, $[S_2(N;\mathbb{Z}) : \mathbb{Z}g \oplus \Lambda] = r_\pi$, and $g \equiv h \pmod m$ for some $h \in \Lambda$ iff $m \mid r_\pi$; the same over $\mathbb{Z}_\ell$ for every $\ell$; for a newform $g$ of level $N$, $r_\pi = r_g$. *(proved here, given the integral basis theorem of Chapter 12)* Proposition 13.64.

**13.32** **Local components.** $\mathbb{T}_\ell = \prod_\mathfrak{m} \mathbb{T}_\mathfrak{m}$; the pairing is perfect on each $e_\mathfrak{m} S_2(N;\mathbb{Z}_\ell) \times \mathbb{T}_\mathfrak{m}$; a homomorphism $\phi\colon \mathbb{T}_\ell \to \mathcal{O}_\lambda$ factors through one $\mathbb{T}_\mathfrak{m}$, the other factors lie in $\ker\phi$, and $\operatorname{Ann}(\ker\phi)$ lies in $\mathbb{T}_\mathfrak{m}$, so that for surjective $\phi\colon \mathbb{T}_{\mathcal{O}_\lambda} \to \mathcal{O}_\lambda$ the congruence ideal $\eta(\phi)$ is computed in a local factor; $e_\mathfrak{m}$ preserves integrality, $\mathbb{T}$-stable subspaces and congruences. *(proved here)* Proposition 13.66.

**13.33** **Congruences.** For normalised eigenforms of $\mathbb{T}_N$: congruence at all $n$ $\iff$ at all primes $\iff$ $\phi_f \equiv \phi_g$ on $\mathbb{T}$; this implies, and when $\mathbb{T}/\mathfrak{m}_{f,\lambda} = \mathbb{F}_\ell$ is equivalent to, $\mathfrak{m}_{f,\lambda} = \mathfrak{m}_{g,\lambda}$; for newforms, congruence at the primes outside $\Sigma$ gives $f^\Sigma \equiv g^\Sigma$ at all $n$. *(proved here)* Proposition 13.68.

**13.34** **The twist operator.** For primitive $\chi$ of conductor $r$: $f\otimes\chi = \mathfrak{g}(\bar\chi)^{-1}\sum_u \bar\chi(u) f|_k\left(\begin{smallmatrix}1 & u/r\\ 0 & 1\end{smallmatrix}\right)$ lies in $S_k(\Gamma_0(\operatorname{lcm}(M, r^2)), \chi^2)$; for quadratic $\chi$ it preserves integrality, intertwines $T_p$ with

$\chi(p)T_p$ ($p \nmid r$), satisfies $U_p(f\otimes\chi)=0$ for $p \mid r$, and $(f\otimes\chi)\otimes\chi = f^{\{p\mid r\}}$; the paper's $\iota=\iota_S$. *(proved here)* Proposition 13.70.

**13.35 Atkin-Li.** For a newform $g$ of weight 2 on $\Gamma_0(M)$ and $\chi=\chi_{q^*}$ with $q^2\nmid M$, $g\otimes\chi$ is a newform on $\Gamma_0(\mathrm{lcm}(M,q^2))$; for $q^2\mid M$ it is the $q$-depletion of a newform of level dividing $M$. *(stated; reference, with a sketch; the case $q^2\mid M$ proved here)* Theorem 13.71.

**13.36 Transport along $\iota$.** $\iota$ maps $g^\perp$ into the forms orthogonal to every $f(dz)$, $f$ the newform with system $\chi_S(p)a_p(g)$; congruences $g\equiv h$ become $f\equiv\iota(h)$ when $f=\iota(g)$, so $r_g \mid r_f$ for rational $g$, $f$; conversely $\iota(f)=g^S$ and $\iota(f^\perp)$ lies in the $\mathbb{T}$-stable complement of $g^S$. *(proved here, given Theorem 13.41 and Theorem 13.42)* Proposition 13.74.

**13.37 $r_f$ divides the congruence module of $g^S$.** Let $g$ (level $M$) and $f$ (level $N$) be newforms of weight 2 with rational coefficients, $S$ a finite set of odd primes with $D^2\mid N$, $\iota(g)=f$ and $M^S=N$, and $\pi$ the eigencharacter of $g^S$ on $\mathbb{T}_N$. Then $r_f$ divides $r_\pi$, the positive generator of $\eta(\pi)=\eta^{\mathrm{Hecke}}(g^S)$. *(proved here, given Theorem 13.41 and Theorem 13.42)* Theorem 13.75.

**13.38 The case of the paper.** For $E/\mathbb{Q}$ of conductor $N$, $S$ the set of odd primes at which $E$ has type $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$, $E_S=E\otimes\chi_S$ and $g=f_{E_S}$: $r_E$ divides the positive generator of $\eta^{\mathrm{Hecke}}(g^S)$, as integers, so $v_\ell(r_E)\le v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ for every prime $\ell$. This is the inequality of the paper's Lemma lem:step1; with the reverse inequality, proved there for $\ell\notin S$, it is an equality at every prime $\ell\notin S$. *(proved here, given the twist lemmas of Chapter 10; the equality for $\ell\notin S$ in Chapter 20)* Corollary 13.76.

**13.39 Euler products.** $L(f,s)=\prod_{p\nmid N}(1-a_pp^{-s}+p^{k-1-2s})^{-1}\prod_{p\mid N}(1-a_pp^{-s})^{-1}$ for a normalised eigenform of $\mathbb{T}_N$; $L(f^\Sigma,s)=L(f,s)\prod_{p\in\Sigma}L_p(f,s)^{-1}$. *(proved here)* Proposition 13.78.

**13.40 Functional equation.** If $f|_kW_N=\varepsilon_Nf$ then $\Lambda(f,s)=N^{s/2}(2\pi)^{-s}\Gamma(s)L(f,s)$ is entire and $\Lambda(f,s)=i^k\varepsilon_N\Lambda(f,k-s)$; sign $-\varepsilon_N$ in weight 2. *(proof sketched here)* Theorem 13.79.

**13.41 Rankin-Selberg.** $\mathrm{Res}_{s=k}\sum|a_n(f)|^2n^{-s}=\frac{3(4\pi)^k}{\pi\Gamma(k)\mu(N)}\langle f,f\rangle$; for a rational newform of weight 2 and squarefree level $\langle f,f\rangle=N\,L(\mathrm{Sym}^2f,2)/(8\pi^3)$, and for every level $\langle f,f\rangle=N\,L^{\mathrm{nv}}(\mathrm{Sym}^2f,2)/(8\pi^3)$; the Rankin series is not the symmetric square $L$-function. *(proof sketched here; the local identities proved)* Theorem 13.80.

## Where the paper uses this chapter

- **Section 3, preamble** ($\mathbb{T}$ is generated by all $T_n$, "including the operators $U_p$ for $p\mid N$"): Proposition 13.7, Theorem 13.10, Proposition 13.16.
- **Lemma lem:crit** (congruence criterion): $\phi_f\colon\mathbb{T}\to\mathbb{Z}$, Proposition 13.19; the perfect pairing and its version over $\mathbb{Z}_\ell$, Theorem 13.24; the forms killed by $I$ and by $J$ and the index of $\mathbb{Z}f\oplus L$, Corollary 13.28; $\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$ and the $\mathbb{T}$-stability of $f^\perp$, Theorem 13.60, Corollary 13.37; congruences on all of $\mathbb{T}$ and the maximal ideal of $(g,\lambda)$, Proposition 13.68, Theorem 13.26. The lemma itself is proved in Chapter 15.
- **Lemma lem:old** (oldforms): the $\mathbb{T}$-stable span of the $h(dz)$ orthogonal to $f$, Theorem 13.41; every formula for $U_p$, the stabilisations, the depleted forms and the non-semisimplicity,

Proposition 13.48; $a_p = 0$ for $p^2 \mid M$, Proposition 13.44; the tensor structure, Theorem 13.59; the passage from primes to all $n$, Proposition 13.68. The lemma itself is proved in Chapter 15.

- **Lemma lem:twist (iii)** (Atkin-Li): Theorem 13.71.
- **Theorem thm:qual (b), (c)**: twists of newforms, $h \otimes \chi$ as a depletion, strong multiplicity one, Proposition 13.70, Theorem 13.71, Theorem 13.42.
- **Section 4.1**: $f^{\Sigma}$, $M^{\Sigma}$, $\delta_p$, Proposition 13.55; the Hecke action and $w_{M^{\Sigma}} T_n w_{M^{\Sigma}}^{-1}$ as adjoint, whose Petersson analogue is Proposition 13.35.
- **Theorem thm:twist**: the twist map $\iota$, generalised eigenspaces and the transport $g^{\perp} \to f^{\perp}$, Proposition 13.70, Proposition 13.74, Example 13.85. The theorem is proved in Chapter 20.
- **The set-up before Lemma lem:step1, and Lemma lem:step1** (Step 1 of the proof of Theorem thm:quant): strong multiplicity one, the eigenvalue 0 of $U_q$ with multiplicity one, units among the $U_q$-eigenvalues, the $\mathbb{T}$-stable complement of $g^S$, idempotents of $\mathbb{T}_\ell$, the $\mathbb{Z}_\ell$-version of Lemma lem:crit for $\pi$, Proposition 13.55, Corollary 13.51, Proposition 13.62, Proposition 13.64, Proposition 13.66, Proposition 13.74. The theorem is proved in Chapter 20.
- **Lemma lem:hecke and the paragraph after it** (the congruence ideal computed in $\mathbb{T}_{\mathfrak{m}}$): Proposition 13.66.
- **Section 1, Section 3 (preamble) and the remark after the Example of Section 3**: the congruence number $r_E$ (introduced in Section 1 and defined again in the preamble of Section 3), Definition 13.29, Example 13.84 (99a1, 539d1, in the unlabelled remark after the Example).
- **Section 1**: the Petersson norm, "$\langle f, f \rangle$ is the special value of the naive symmetric square $L$-function at $s = 2$", Theorem 13.80(iii).
- **Lemma lem:step1 and Remark rem:afterquant (2)**: Step 1 of the proof of Theorem thm:quant is an equality at every prime $\ell \notin S$; Proposition 13.64, Theorem 13.75, Corollary 13.76 and the remark after it.
- **The Example of Section 3**: $f_{11\mathrm{a}} \otimes \chi_{-11}$ is the form of 121d, Theorem 13.71, Example 13.86.

## 13.1 Introduction and conventions

This section fixes notation. Its purpose is to make every statement of the chapter unambiguous about three things that the paper handles silently: the normalisation of the Hecke operators, the level at which an operator acts, and the difference between a twisted $q$-series and a newform. The physics reader meets three familiar objects. The Hecke recursion $T_{p^{r+1}} = T_p T_{p^r} - p T_{p^{r-1}}$ is the three-term recursion of the Chebyshev polynomials of the second kind, and the product formula for $T_{p^a} T_{p^b}$ is the Clebsch-Gordan rule for characters of SU(2) (Remark 13.11). The commuting family of self-adjoint operators $T_n$, $(n, N) = 1$, is diagonalised by an orthogonal basis, as in the spectral theorem for commuting Hermitian operators (Theorem 13.33). The functional equation of $L(f, s)$ comes from the Mellin transform of $f(iy)$ (Theorem 13.79).

**Convention 13.1** (standing notation)**.** (a) $N \geq 1$ is the level, $k \geq 2$ an even integer, the weight; everything the paper needs is $k = 2$, and the general-$k$ form is given where it costs nothing. $S_k(N) = S_k(\Gamma_0(N))$ and $M_k(N) = M_k(\Gamma_0(N))$, with trivial character. For a commutative ring $A$, $S_2(N; A) = S_2(N; \mathbb{Z}) \otimes_{\mathbb{Z}} A$, where $S_2(N; \mathbb{Z})$ is the group of cusp forms with integer $q$-expansion at $\infty$ (Chapter 12). The letter $q$ is the variable $e^{2\pi iz}$ of $q$-expansions and also an odd prime (in Sections 13.9 and 13.11); the two never meet in one display. (b) $\mathbb{T} = \mathbb{T}_N = \mathbb{Z}[T_n : n \geq 1] \subset \mathrm{End}_{\mathbb{C}} S_k(N)$ is the *full* Hecke algebra of level $N$ (the paper's $\mathbb{T}$), $\mathbb{T}^{(N)} = \mathbb{Z}[T_n : (n, N) = 1]$ the *anemic* one, $\mathbb{T}_A = \mathbb{T} \otimes_{\mathbb{Z}} A$ and $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$. Unless said otherwise $\mathbb{T}$ acts on cusp forms. (c) An *eigenform* is a nonzero simultaneous eigenvector of all of $\mathbb{T}$ (the $U_p$ included); an *anemic eigenform* is one for $\mathbb{T}^{(N)}$ only; *normalised* means $a_1 = 1$; a *newform* is a normalised eigenform in the new subspace (Definition 13.39). A *system of eigenvalues* is a ring homomorphism $\phi\colon \mathbb{T} \to \overline{\mathbb{Q}}$ or $\mathbb{T}^{(N)} \to \overline{\mathbb{Q}}$; it is written with an upright $\phi$, the curly $\varphi$ being reserved for maps of curves. Its *generalised eigenspace* $V(\phi)$ is the set of $v$ killed by a power of $t - \phi(t)$ for every $t$ (Chapter 3); the $\mathbb{T}$-*stable complement* of $V(\phi)$ is the sum of the other generalised eigenspaces. (d) $\langle f, g\rangle = \int_{\Gamma_0(N)\backslash\mathcal{H}} f\bar{g}\, y^k\, dx\, dy/y^2$ is the Petersson product of Chapter 12, unnormalised and linear in the first variable; PARI's `mfpetersson` returns $\langle f, g\rangle/\mu(N)$ with $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = N\prod_{p|N}(1 + p^{-1})$. Adjoints of operators are written $t^\dagger$. The pairing $\langle g, t\rangle = a_1(tg)$ of Section 13.4 is $\mathbb{Z}$-bilinear; it is a different object, written with the same brackets only inside that section and always with one argument in $\mathbb{T}$. (e) "$f \equiv g \pmod{\mathfrak{a}}$" means $a_n(f) \equiv a_n(g) \pmod{\mathfrak{a}}$ for *all* $n \geq 1$; a congruence only at the primes outside a set is always written out as such. $\lambda$ is a prime of the ring of integers $\mathcal{O}$ of a number field, above $\ell$. (f) $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$ and $w_N$ is the operator $f \mapsto f|_k W_N$ (Chapter 12); for even $k$ it is an involution, since $W_N^2 = -N \cdot 1$ and scalars act trivially in even weight. $\varepsilon_N(f) = \pm 1$ denotes its eigenvalue on a newform.

**Convention 13.2** (double coset operators)**.** For $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ with $\Gamma\alpha\Gamma = \bigsqcup_i \Gamma\alpha_i$ we put

$$[\Gamma\alpha\Gamma]_k f = (\det\alpha)^{k/2-1} \sum_i f|_k \alpha_i,$$

with the slash operator $(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz + d)^{-k} f(\gamma z)$ of Chapter 12. Diamond and Shurman [52, Section 5.1] use the slash $(\det\gamma)^{k-1}(cz + d)^{-k} f(\gamma z)$ and no prefactor; since $(\det\alpha)^{k/2-1}(\det\alpha)^{k/2} = (\det\alpha)^{k-1}$, the two operators coincide. In weight 2 the prefactor is 1.

**Convention 13.3** (names of Hecke operators and their level)**.** At level $N$ the operators are $T_n$, $n \geq 1$. For a prime $p \mid N$ the operator $T_p$ is written $U_p$, as in the paper, and $T_{p^r} = U_p^r$. The operator $T_p$ depends on the level: for $h \in M_k(M)$ and $p \nmid M$, the operator $T_p$ of level $M$ and the operator $U_p$ of level $Mp$ differ,

$$T_p^{(M)} h = U_p^{(Mp)} h + p^{k-1} h(pz)$$

(Proposition 13.8). Whenever two levels are in play the level is written as a superscript. On formal $q$-series we use the operators

$$\mathcal{U}_p\colon \sum a_n q^n \mapsto \sum a_{pn} q^n, \qquad V_d\colon \sum a_n q^n \mapsto \sum a_n q^{dn},$$

so that $V_d h(z) = h(dz)$; by Proposition 13.8 the operator $U_p$ of any level divisible by $p$ is $\mathcal{U}_p$ on $q$-expansions, which is why the level of $U_p$ can often be left implicit.

**Convention 13.4** (twists and depletion)**.** (a) For a Dirichlet character $\chi$ of conductor $r$, the *(naive) twist* is $f \otimes \chi = \sum_n \chi(n) a_n(f) q^n$, with $\chi(n) = 0$ when $(n, r) > 1$. It is a $q$-series operation; whether $f \otimes \chi$ is a newform is a theorem (Theorem 13.71). (b) For an odd prime $q$, $q^* = (-1)^{(q-1)/2} q$ and $\chi_{q^*} = (\frac{q^*}{\cdot})$ is the quadratic character of conductor $q$, the character of $\mathbb{Q}(\sqrt{q^*})$, written $\chi_{q^*}$ as in the paper; the book never writes $\chi_q$, $\chi_\ell$ being the cyclotomic character. For a finite set $S$ of odd primes, $D = \prod_{q \in S} q$ and $\chi_S = \prod_{q \in S} \chi_{q^*}$, a primitive quadratic character of conductor $D$. Values used below: $\chi_{-7}(2) = 1$ and $\chi_{-11}(2) = -1$. For an elliptic curve, $E^{(q)} = E \otimes \chi_{q^*}$ is the twist by $\mathbb{Q}(\sqrt{q^*})$. (c) For a finite set $\Sigma$ of primes and a $q$-series $h = \sum a_n q^n$, the $\Sigma$-*depletion* is $h^\Sigma = \sum_{(n,\Sigma)=1} a_n q^n$, where $(n, \Sigma) = 1$ means that no prime of $\Sigma$ divides $n$; for one prime we write $h^{\{p\}}$. A parenthesised superscript never denotes depletion: it is reserved for twists of curves and for removed Euler factors ($\zeta^{(N)}$, $\mathbb{T}^{(N)}$).

## 13.2 Hecke operators: double cosets and $q$-expansions

This section defines the operators $T_n$ and $U_p$ and proves the formulas that the paper uses on every page of its Section 3: the action on $q$-expansions, $U_p h(p^j z) = h(p^{j-1} z)$, commutativity, multiplicativity and the recursion. The definition by double cosets is what makes the operators preserve modularity and what computes their Petersson adjoints (Section 13.5); the $q$-expansion formulas are what one computes with.

**Definition 13.5** (double coset operators)**.** Let $\Gamma = \Gamma_0(N)$ and $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. The *double coset* $\Gamma\alpha\Gamma = \{\gamma\alpha\gamma' : \gamma, \gamma' \in \Gamma\}$ is a union of right cosets $\Gamma\beta$. When it is a finite disjoint union $\Gamma\alpha\Gamma = \bigsqcup_{i=1}^r \Gamma\alpha_i$, the operator $[\Gamma\alpha\Gamma]_k$ on functions $\mathcal{H} \to \mathbb{C}$ is $[\Gamma\alpha\Gamma]_k f = (\det\alpha)^{k/2-1} \sum_{i=1}^r f|_k \alpha_i$ (Convention 13.2).

**Proposition 13.6.** *Let $\Gamma = \Gamma_0(N)$, $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ and $\Gamma_\alpha = \Gamma \cap \alpha^{-1}\Gamma\alpha$.*

*(i) $\Gamma_\alpha$ has finite index in $\Gamma$, and if $\Gamma = \bigsqcup_{i=1}^r \Gamma_\alpha \gamma_i$ then $\Gamma\alpha\Gamma = \bigsqcup_{i=1}^r \Gamma\alpha\gamma_i$. In particular $r = [\Gamma : \Gamma_\alpha]$.*

*(ii) $[\Gamma\alpha\Gamma]_k f$ does not depend on the choice of the representatives $\alpha_i$, and $[\Gamma\alpha\Gamma]_k$ maps $M_k(N)$ to $M_k(N)$ and $S_k(N)$ to $S_k(N)$.*

*Proof.* (i) Multiplying $\alpha$ by a positive integer changes neither $\Gamma_\alpha$ nor the cosets, so we may assume $\alpha$ integral; let $m = \det\alpha$. For $\gamma \in \Gamma(Nm)$ we have $\alpha\gamma\alpha^{-1} = \alpha\gamma \cdot m^{-1}\alpha'$ with $\alpha' = m\alpha^{-1}$ integral, and $\gamma \equiv 1 \pmod{Nm}$ gives $\alpha\gamma\alpha' \equiv \alpha\alpha' = m \pmod{Nm}$, so $\alpha\gamma\alpha^{-1} \equiv 1 \pmod{N}$ is an integral matrix of determinant one: it lies in $\Gamma(N) \subseteq \Gamma$. Hence $\Gamma(Nm) \subseteq \Gamma_\alpha$, and $\Gamma(Nm)$ has finite index in $\mathrm{SL}_2(\mathbb{Z})$. For the decomposition, the map $\Gamma_\alpha\gamma \mapsto \Gamma\alpha\gamma$ from $\Gamma_\alpha\backslash\Gamma$ to the set of right cosets in $\Gamma\alpha\Gamma$ is well defined (for $\delta \in \Gamma_\alpha$, $\alpha\delta\gamma = (\alpha\delta\alpha^{-1})\alpha\gamma$ with $\alpha\delta\alpha^{-1} \in \Gamma$), surjective (every element of $\Gamma\alpha\Gamma$ is $\gamma'\alpha\gamma$), and injective ($\Gamma\alpha\gamma = \Gamma\alpha\gamma'$ means $\alpha\gamma'\gamma^{-1}\alpha^{-1} \in \Gamma$, that is, $\gamma'\gamma^{-1} \in \Gamma_\alpha$).

(ii) If $\Gamma\alpha_i = \Gamma\beta_i$ then $\beta_i = \delta\alpha_i$ with $\delta \in \Gamma$ and $f|_k\beta_i = f|_k\alpha_i$ for $f$ invariant under $\Gamma$; so the sum is independent of the representatives. For $\gamma \in \Gamma$, the cosets $\Gamma\alpha_i\gamma$ are again the cosets of $\Gamma\alpha\Gamma$, permuted (right multiplication by $\gamma$ is a bijection of $\Gamma\alpha\Gamma$ preserving the partition into right cosets); hence $([\Gamma\alpha\Gamma]_k f)|_k\gamma = (\det\alpha)^{k/2-1} \sum_i f|_k\alpha_i\gamma = [\Gamma\alpha\Gamma]_k f$. Each $f|_k\alpha_i$ is holomorphic on $\mathcal{H}$. At the cusps: for $\sigma \in \mathrm{SL}_2(\mathbb{Z})$, write $\alpha_i\sigma = \sigma'\beta$ with $\sigma' \in \mathrm{SL}_2(\mathbb{Z})$ and $\beta$ upper triangular in $\mathrm{GL}_2^+(\mathbb{Q})$ (choose $\sigma'$ with first column proportional to the first column of $\alpha_i\sigma$). Then $f|_k\alpha_i\sigma = (f|_k\sigma')|_k\beta$, and $f|_k\sigma'$ has an expansion $\sum_{n \geq 0} b_n e^{2\pi i n z/h}$ (with $b_0 = 0$ if $f$ is cuspidal) by Chapter 12; slashing with $\beta = \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$ replaces $z$ by $(az + b)/d$ and multiplies by a constant, which keeps the function

bounded on $\operatorname{Im} z \geq 1$ (and tending to 0 if $b_0 = 0$). So $[\Gamma\alpha\Gamma]_k f$ is holomorphic (cuspidal) at every cusp. □

The next proposition computes the double coset that defines $T_p$. We use the set

$$\Delta_n^N = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{M}_2(\mathbb{Z}) : ad - bc = n,\ N \mid c,\ \gcd(a, N) = 1 \right\}.$$

It is stable under left and right multiplication by $\Gamma_0(N)$: if $\gamma = \left(\begin{smallmatrix} x & y \\ u & v \end{smallmatrix}\right) \in \Gamma_0(N)$ then $x$ is a unit modulo $N$ (as $xv \equiv 1 \pmod N$), the lower left entries of $\gamma\delta$ and $\delta\gamma$ are $\equiv 0 \pmod N$, and their upper left entries are $\equiv xa$ and $\equiv ax \pmod N$.

**Proposition 13.7** (coset decompositions)**.** *Let $p$ be a prime and $\Gamma = \Gamma_0(N)$.*

*(i) The right cosets $\Gamma\delta$ with $\delta \in \Delta_p^N$ are the $\Gamma\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)$, $0 \leq j < p$, which are distinct, together with $\Gamma\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ if $p \nmid N$.*

*(ii) If $p \nmid N$ then $\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma = \Gamma\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)\Gamma = \Delta_p^N = \bigsqcup_{j=0}^{p-1} \Gamma\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right) \sqcup \Gamma\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$, a union of $p+1$ cosets.*

*(iii) If $p \mid N$ then $\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma = \Delta_p^N = \bigsqcup_{j=0}^{p-1} \Gamma\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)$, a union of $p$ cosets.*

*We define $T_p = [\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma]_k$ on $M_k(N)$, and write $U_p$ for it when $p \mid N$ (Convention 13.3).*

*Proof.* (i) Let $\delta = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Delta_p^N$ and $g = \gcd(a, c)$. As $\gcd(a, N) = 1$, $g$ is prime to $N$, so $N \mid c/g$; the pair $(u, v) = (-c/g, a/g)$ is coprime with $N \mid u$, so it is the bottom row of some $\gamma \in \Gamma_0(N)$, and $\gamma\delta$ has lower left entry $ua + vc = 0$. So $\Gamma\delta$ contains an upper triangular matrix $\left(\begin{smallmatrix} a' & b' \\ 0 & d' \end{smallmatrix}\right)$, with $a'd' = p$; multiplying by $-1$ we may take $a', d' > 0$, and multiplying on the left by $\left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$ we may take $0 \leq b' < d'$. This gives $\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)$ with $0 \leq j < p$, or $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$; the latter has upper left entry $p$, which is prime to $N$ only if $p \nmid N$. Distinctness: $\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & j' \\ 0 & p \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 1 & (j-j')/p \\ 0 & 1 \end{smallmatrix}\right)$ is integral only if $j = j'$, and $\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 1/p & j \\ 0 & p \end{smallmatrix}\right)$ is not integral.

(ii) $\Delta_p^N$ contains $\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$ and is stable under $\Gamma$ on both sides, so it contains $\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma$, and each of the cosets of (i) lies in it. Conversely, $\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & j \\ 0 & 1 \end{smallmatrix}\right)$ lies in $\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma$; and since $\gcd(p, N) = 1$ there are integers $w, y$ with $pw - Ny = 1$, so $\gamma = \left(\begin{smallmatrix} p & y \\ N & w \end{smallmatrix}\right) \in \Gamma_0(N)$, and $\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\gamma = \left(\begin{smallmatrix} p & y \\ pN & pw \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & y \\ N & pw \end{smallmatrix}\right)\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ with $\left(\begin{smallmatrix} 1 & y \\ N & pw \end{smallmatrix}\right) \in \Gamma_0(N)$ (its determinant is $pw - Ny = 1$). So $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) \in \Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma$ and all $p+1$ cosets lie in the double coset. Two double cosets are equal or disjoint, so $\Gamma\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)\Gamma = \Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma$.

(iii) As in (ii), $\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma \subseteq \Delta_p^N$, which by (i) is the union of the $p$ cosets $\Gamma\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)$, each contained in $\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma$. □

**Proposition 13.8** ($q$-expansions)**.** *Let $f = \sum_{n \geq 0} a_n q^n \in M_k(N)$ and let $p$ be prime. For every $n \geq 0$,*

$$a_n(T_p f) = a_{pn}(f) + p^{k-1} a_{n/p}(f) \quad (p \nmid N), \qquad a_n(U_p f) = a_{pn}(f) \quad (p \mid N),$$

*with $a_{n/p}=0$ when $p\nmid n$. In weight 2, $a_n(T_pf)=a_{pn}+p\,a_{n/p}$. Equivalently $T_p=\mathcal{U}_p+p^{k-1}V_p$ ($p\nmid N$) and $U_p=\mathcal{U}_p$ ($p\mid N$) on q-expansions, and for $h\in M_k(M)$ with $p\nmid M$, $T_p^{(M)}h=U_p^{(Mp)}h+p^{k-1}h(pz)$.*

*Proof.* $f|_k\left(\begin{smallmatrix}1&j\\0&p\end{smallmatrix}\right)(z)=p^{k/2}p^{-k}f\left(\frac{z+j}{p}\right)$, so with the prefactor $p^{k/2-1}$ the coset $\left(\begin{smallmatrix}1&j\\0&p\end{smallmatrix}\right)$ contributes $p^{-1}f\left(\frac{z+j}{p}\right)=p^{-1}\sum_n a_ne^{2\pi in(z+j)/p}$. Summing over $j$, $\sum_{j=0}^{p-1}e^{2\pi inj/p}$ is $p$ if $p\mid n$ and 0 otherwise, so these cosets give $\sum_{p\mid n}a_nq^{n/p}=\sum_m a_{pm}q^m$. The coset $\left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right)$ gives $p^{k/2-1}\cdot p^{k/2}f(pz)=p^{k-1}\sum_m a_mq^{pm}$. Apply Proposition 13.7. The last statement compares the two formulas: at level $Mp$ the prime $p$ divides the level, so $U_p^{(Mp)}=\mathcal{U}_p$, while $T_p^{(M)}=\mathcal{U}_p+p^{k-1}V_p$. □

**Lemma 13.9** (the operators $\mathcal{U}_p$ and $V_p$)**.** *On formal q-series: $\mathcal{U}_pV_p=1$; $V_p\mathcal{U}_p$ is the projection $\sum a_nq^n\mapsto\sum_{p\mid n}a_nq^n$; and for primes $p\neq p'$ and integers $d$ prime to $p$, $\mathcal{U}_p$ and $V_p$ commute with $\mathcal{U}_{p'}$, $V_{p'}$ and $V_d$.*

*Proof.* Compare $n$-th coefficients, with the convention that $a_x=0$ when $x$ is not an integer. By definition the $n$-th coefficient of $\mathcal{U}_ph$ is $a_{pn}$ and that of $V_dh$ is $a_{n/d}$. So $\mathcal{U}_pV_p$ sends $(a_n)$ to $(a_{pn/p})=(a_n)$, and $V_p\mathcal{U}_p$ sends it to $(a_{p\cdot n/p})=(a_n)$ for $p\mid n$ and to 0 otherwise. For the commutation relations, with $p'\neq p$ prime and $(d,p)=1$:

- $\mathcal{U}_p\mathcal{U}_{p'}$ and $\mathcal{U}_{p'}\mathcal{U}_p$ both have $n$-th coefficient $a_{pp'n}$;
- $\mathcal{U}_pV_d$ has $n$-th coefficient $a_{pn/d}$ and $V_d\mathcal{U}_p$ has $a_{p(n/d)}$; both vanish unless $d\mid pn$, and since $(d,p)=1$ this is the condition $d\mid n$, under which they are equal; taking $d=p'$ gives $\mathcal{U}_pV_{p'}=V_{p'}\mathcal{U}_p$;
- $V_pV_d$ and $V_dV_p$ both have $n$-th coefficient $a_{n/(pd)}$; taking $d=p'$ gives $V_pV_{p'}=V_{p'}V_p$.

The remaining relation, $V_p\mathcal{U}_{p'}=\mathcal{U}_{p'}V_p$, is the second item with the roles of $p$ and $p'$ exchanged. □

**Theorem 13.10** (the operators $T_n$)**.** *Put $T_1=1$, $T_{p^{r+1}}=T_pT_{p^r}-p^{k-1}T_{p^{r-1}}$ for $p\nmid N$ and $r\geq1$, $T_{p^r}=U_p^r$ for $p\mid N$, and $T_n=\prod_{p^r\|n}T_{p^r}$. Then on $M_k(N)$:*

*(i) for $p\nmid N$, $T_{p^r}=\sum_{i=0}^r p^{i(k-1)}V_p^i\mathcal{U}_p^{r-i}$ on q-expansions, and for $p\mid N$, $T_{p^r}=\mathcal{U}_p^r$;*

*(ii) all the $T_n$ commute;*

*(iii) for $f\in M_k(N)$ and $m\geq1$,*

$$a_m(T_nf)=\sum_{d\mid(m,n),\ (d,N)=1}d^{k-1}a_{mn/d^2}(f),\qquad\text{in particular}\qquad a_1(T_nf)=a_n(f);$$

*(iv) $T_mT_n=\sum_{d\mid(m,n),\ (d,N)=1}d^{k-1}T_{mn/d^2}$, in particular $T_mT_n=T_{mn}$ when $(m,n)=1$;*

*(v) $\sum_{r\geq0}T_{p^r}X^r=(1-T_pX+p^{k-1}X^2)^{-1}$ for $p\nmid N$ and $(1-U_pX)^{-1}$ for $p\mid N$, as formal power series with coefficients in the commutative ring $\mathbb{Z}[T_p]$.*

*Proof.* (i) For $p\mid N$ this is Proposition 13.8. For $p\nmid N$ write $\mathcal{U}=\mathcal{U}_p$, $V=V_p$, $c=p^{k-1}$ and $R_r=\sum_{i=0}^r c^iV^i\mathcal{U}^{r-i}$. Then $R_0=1$, $R_1=\mathcal{U}+cV=T_p$, and, using $\mathcal{U}V^i=V^{i-1}$ for $i\geq1$ (Lemma 13.9),

$$T_pR_r=(\mathcal{U}+cV)\sum_{i=0}^r c^iV^i\mathcal{U}^{r-i}=\mathcal{U}^{r+1}+\sum_{i=1}^r c^iV^{i-1}\mathcal{U}^{r-i}+\sum_{i=0}^r c^{i+1}V^{i+1}\mathcal{U}^{r-i}=R_{r+1}+cR_{r-1}.$$

So $R_r$ satisfies the recursion defining $T_{p^r}$, with the same initial values, and $R_r = T_{p^r}$.

(ii) By (i), $T_{p^r}$ is a polynomial in $\mathcal{U}_p$ and $V_p$, and by Lemma 13.9 these commute with $\mathcal{U}_{p'}$, $V_{p'}$ for $p' \neq p$; so $T_{p^r}$ and $T_{p'^s}$ commute. Powers of one prime commute because $T_{p^r}$ is a polynomial in $T_p$. Hence all $T_n$ commute (and the order in the product defining $T_n$ is irrelevant).

(iii) For $n = p^r$ and $p \nmid N$: the $m$-th coefficient of $V^i\mathcal{U}^{r-i}f$ is $a_{p^{r-i}m/p^i}$ if $p^i \mid m$ and 0 otherwise; with $d = p^i$ this is $a_{mn/d^2}$ for $d \mid (m,n)$. So (i) gives the formula. For $p \mid N$ only $d = 1$ occurs and $a_m(\mathcal{U}_p^r f) = a_{p^r m}$. If $n = n_1 n_2$ with $(n_1, n_2) = 1$ and the formula holds for $n_1$ and $n_2$, then

$$a_m(T_{n_1}T_{n_2}f) = \sum_{d_1 \mid (m,n_1)} d_1^{k-1} \sum_{d_2 \mid (mn_1/d_1^2,\, n_2)} d_2^{k-1} a_{mn_1n_2/(d_1d_2)^2}(f)$$

(sums over divisors prime to $N$), and since $(n_1, n_2) = 1$ the condition $d_2 \mid (mn_1/d_1^2, n_2)$ is $d_2 \mid (m, n_2)$; the pairs $(d_1, d_2)$ correspond to the divisors $d = d_1d_2$ of $(m,n)$ prime to $N$. Induction on the number of prime factors gives (iii).

(iv) By (ii) and the definition, both sides are multiplicative in the coprime parts of $m$ and $n$, so it suffices to treat $m = p^a$, $n = p^b$. For $p \mid N$ this is $U_p^a U_p^b = U_p^{a+b}$. For $p \nmid N$ write $\tau_r = T_{p^r}$ (so $\tau_0 = T_1 = 1$ and $\tau_1 = T_p$), $c = p^{k-1}$, $R(a,b) = \sum_{i=0}^{\min(a,b)} c^i \tau_{a+b-2i}$, and prove $\tau_a\tau_b = R(a,b)$ for $b \geq a$ by induction on $a$; the case $b < a$ follows by commutativity, $R$ being symmetric. The cases $a = 0$ and $a = 1$ are the definition. For the step let $b \geq a+1$; then $\tau_{a+1}\tau_b = \tau_1\tau_a\tau_b - c\tau_{a-1}\tau_b = \tau_1 R(a,b) - cR(a-1,b)$. In $R(a,b)$ every index $a+b-2i \geq b-a \geq 1$, so $\tau_1\tau_{a+b-2i} = \tau_{a+b+1-2i} + c\tau_{a+b-1-2i}$, and

$$\tau_1 R(a,b) = \sum_{i=0}^{a} c^i \tau_{a+1+b-2i} + \sum_{i=1}^{a+1} c^i \tau_{a+b+1-2i}, \qquad cR(a-1,b) = \sum_{i=1}^{a} c^i \tau_{a+b+1-2i}.$$

The difference is $\sum_{i=0}^{a} c^i \tau_{a+1+b-2i} + c^{a+1}\tau_{b-a-1} = R(a+1,b)$.

(v) The coefficient of $X^{r+1}$ in $(\sum_r T_{p^r}X^r)(1 - T_pX + p^{k-1}X^2)$ is $T_{p^{r+1}} - T_pT_{p^r} + p^{k-1}T_{p^{r-1}} = 0$ for $r \geq 1$, that of $X$ is $T_p - T_p = 0$ and that of $X^0$ is 1; for $p \mid N$ the series is geometric. □

*Remark* 13.11 (Chebyshev polynomials and Clebsch-Gordan). Let $U_r$ be the Chebyshev polynomial of the second kind, $U_r(\cos\theta) = \sin((r+1)\theta)/\sin\theta$, with generating function $\sum_r U_r(x)X^r = (1 - 2xX + X^2)^{-1}$. Comparing with Theorem 13.10(v), $T_{p^r} = p^{r(k-1)/2}U_r(T_p/(2p^{(k-1)/2}))$ for $p \nmid N$. Indeed, put $c = p^{(k-1)/2}$ and replace $X$ by $cX$ in the generating function: $\sum_r c^r U_r(x)X^r = (1 - 2xcX + c^2X^2)^{-1}$, and with $2xc = T_p$ the right side is $(1 - T_pX + p^{k-1}X^2)^{-1} = \sum_r T_{p^r}X^r$; compare coefficients of $X^r$. If $T_p$ has the eigenvalue $2p^{(k-1)/2}\cos\theta$ (the Ramanujan-Petersson bound, Chapter 12, says $\theta$ can be taken real), then $T_{p^r}$ has the eigenvalue $p^{r(k-1)/2}\sin((r+1)\theta)/\sin\theta$: the character of the spin-$r/2$ representation of SU(2) at the rotation by $2\theta$. The product formula $T_{p^a}T_{p^b} = \sum_i p^{i(k-1)}T_{p^{a+b-2i}}$ of Theorem 13.10(iv) is then the Clebsch-Gordan rule $V_{a/2} \otimes V_{b/2} = \bigoplus_i V_{(a+b)/2-i}$ for these characters.

**Example 13.12** (Eisenstein series and $\Delta$)**.** (i) Level one. The Eisenstein series $E_k = 1 + c_k\sum_{n\geq 1}\sigma_{k-1}(n)q^n$ ($k \geq 4$) satisfies $T_pE_k = \sigma_{k-1}(p)E_k$. Indeed the constant term of $T_pE_k$ is $1 + p^{k-1} = \sigma_{k-1}(p)$, and for $n \geq 1$ one needs $\sigma_{k-1}(pn) + p^{k-1}\sigma_{k-1}(n/p) = \sigma_{k-1}(p)\sigma_{k-1}(n)$. Write $n = p^a m$ with $p \nmid m$; by multiplicativity of $\sigma_{k-1}$ both sides are $\sigma_{k-1}(m)$ times $\sigma_{k-1}(p^{a+1}) + p^{k-1}\sigma_{k-1}(p^{a-1})$, respectively $\sigma_{k-1}(p)\sigma_{k-1}(p^a)$, which agree. To see this put $c = p^{k-1}$, so that $\sigma_{k-1}(p^b) = \sum_{i=0}^{b} c^i$; then

$$\sigma_{k-1}(p)\sigma_{k-1}(p^a) = (1+c)\sum_{i=0}^{a} c^i = \sum_{i=0}^{a+1} c^i + \sum_{i=1}^{a} c^i = \sigma_{k-1}(p^{a+1}) + c\,\sigma_{k-1}(p^{a-1}),$$

where for $a = 0$ the middle sum is empty and $\sigma_{k-1}(p^{-1})$ is read as 0, matching the convention $a_{n/p} = 0$ for $p \nmid n$. Since $\dim S_{12}(1) = 1$ (Chapter 12), every $T_n$ preserves the line of $\Delta = \sum \tau(n)q^n$, so $T_n\Delta = \tau(n)\Delta$ by Theorem 13.10(iii), and $\tau$ is multiplicative and satisfies $\tau(p^{r+1}) = \tau(p)\tau(p^r) - p^{11}\tau(p^{r-1})$: Mordell's theorem.

(ii) Level 11. The weight-two Eisenstein series $E_2^{(11)} = -10 - 24\sum_{n\geq 1}(\sigma(n) - 11\sigma(n/11))q^n \in M_2(11)$ of Chapter 12 has coefficients $\sigma(n) - 11\sigma(n/11) = \sigma(m)$ for $n = 11^a m$, $11 \nmid m$ (as $\sigma(11^a) - 11\sigma(11^{a-1}) = 1$). By the computation in (i) with $k = 2$, $T_p E_2^{(11)} = (1+p)E_2^{(11)}$ for $p \neq 11$, and $U_{11}E_2^{(11)} = E_2^{(11)}$ because $\sigma(m)$ does not see the power of 11 (and the constant term is fixed by $\mathcal{U}_{11}$). By Chapter 12, $f_{11} \equiv -\frac{1}{24}E_2^{(11)} \pmod 5$ coefficientwise for $n \geq 1$; so the eigenvalues of $f_{11}$ satisfy $a_p \equiv 1 + p \pmod 5$ ($p \neq 11$) and $a_{11} = 1$. In terms of the algebra of the next section, $\mathbb{T}_{11} = \mathbb{Z}$ (acting on the line $\mathbb{C}f_{11}$), and its maximal ideal (5) contains $T_p - 1 - p$ ($p \neq 11$) and $U_{11} - 1$: it is an *Eisenstein* maximal ideal (Chapter 14). (Checked with PARI for $n \leq 2000$.) This is the case of reducible $\bar{\rho}$ that the paper sets aside in Section sec:eis.

*Remark* 13.13 (Hecke correspondences). In weight 2, $f \mapsto f\,dz$ identifies $S_2(N)$ with the holomorphic differentials on $X_0(N)$ (Chapter 12), and $T_p$ becomes the action on differentials of the Hecke correspondence $X_0(N) \leftarrow X_0(Np) \rightarrow X_0(N)$ given by $z \mapsto z$ and $z \mapsto pz$ (Chapter 8). This is how the same ring $\mathbb{T}$ acts on $J_0(N)$ and on $H^1(X_0(N), \mathbb{Z})$ (Chapter 14), and why the Hecke algebra of the paper's Section 4.1 is the one defined here. Nothing in this chapter depends on the remark [52, Section 6.3], [180, Section 7.2].

*In the paper* 13.14. The proof of Lemma lem:old writes "$U_p h(p^j z) = h(p^{j-1}z)$ for $j \geq 1$, and $U_p h = a_p(h)h - p\,h(pz)$ if $p \nmid M$, $U_p h = a_p(h)h$ if $p \mid M$". Here $U_p$ is the operator of the level $N$, $p \mid N$, which on $q$-expansions is $\mathcal{U}_p$ (Proposition 13.8). The first formula is $\mathcal{U}_p V_p = 1$ (Lemma 13.9) applied to $h(p^{j-1}z)$; the second is $\mathcal{U}_p h = T_p^{(M)}h - pV_p h = a_p(h)h - p\,h(pz)$; the third is $\mathcal{U}_p h = U_p^{(M)}h$. The last sentence of the first paragraph of that proof, "the congruence for all $n$ then follows from the multiplicativity of the coefficients, from $a_{p^k} = a_p^k$ for $p \mid N$, and from the recursion", uses Theorem 13.10(iv) in the form of Proposition 13.19(ii) below.

## 13.3 The Hecke algebra and eigenforms

This section defines the paper's Hecke algebra $\mathbb{T}$ ("the subring of $\mathrm{End}\, S_2(\Gamma_0(N))$ generated over $\mathbb{Z}$ by all $T_n$, $n \geq 1$, including the operators $U_p$ for $p \mid N$", Section 3), its eigenforms and the eigencharacter $\phi_f \colon \mathbb{T} \to \mathbb{Z}$ of Lemma lem:crit. The key point for the paper is that $\mathbb{T}$ is the $\mathbb{Z}$-span of the $T_n$, so that a statement about all $a_n$ is a statement about all of $\mathbb{T}$.

**Definition 13.15** (Hecke algebras)**.** $\mathbb{T} = \mathbb{T}_N$ is the subring of $\mathrm{End}_{\mathbb{C}}\, S_k(N)$ generated by the $T_n$, $n \geq 1$, and $\mathbb{T}^{(N)}$ the subring generated by the $T_n$ with $(n, N) = 1$. For a ring $A$, $\mathbb{T}_A = \mathbb{T} \otimes_{\mathbb{Z}} A$.

**Proposition 13.16.** *(i)* $\mathbb{T}$ *is commutative.*

*(ii)* $\mathbb{T}$ *is the* $\mathbb{Z}$*-span of* $\{T_n : n \geq 1\}$*, and* $\mathbb{T}^{(N)}$ *is the* $\mathbb{Z}$*-span of* $\{T_n : (n, N) = 1\}$*.*

*(iii) In weight* 2, $\mathbb{T}$ *preserves* $S_2(N;\mathbb{Z})$*. Given the integral basis theorem (*$S_2(N;\mathbb{Z})$ *is a lattice spanning* $S_2(N)$*, Chapter 12), restriction identifies* $\mathbb{T}$ *with a subring of* $\mathrm{End}_{\mathbb{Z}}\, S_2(N;\mathbb{Z}) \simeq \mathrm{M}_d(\mathbb{Z})$*,* $d = \dim S_2(N)$*; so* $\mathbb{T}$ *is a free* $\mathbb{Z}$*-module of finite rank, and* $\mathbb{T} \otimes A$ *acts on* $S_2(N;A)$ *for every ring* $A$*.*

*Proof.* (i) is Theorem 13.10(ii). (ii) By Theorem 13.10(iv) the $\mathbb{Z}$-span of the $T_n$ is closed under products, and it contains $T_1 = 1$; so it is the ring they generate. The same formula with $(m,n) = (m,n,N) = 1$ only involves $T_{mn/d^2}$ with $(mn/d^2, N) = 1$. (iii) By Theorem 13.10(iii) with $k = 2$, $a_m(T_n f)$ is an integer combination of the $a_j(f)$. Since $S_2(N;\mathbb{Z})$ spans $S_2(N)$, an operator that kills $S_2(N;\mathbb{Z})$ is zero; so restriction is injective, and $\mathbb{T}$ is a subgroup of the free group $\mathrm{M}_d(\mathbb{Z})$. □

*In the paper* 13.17. Lemma lem:crit concludes from "$a_n(g) \equiv a_n(f) \pmod{\lambda}$ for every $n \geq 1$" that "$\phi_g \equiv \phi_f \pmod{\lambda}$ on all of $\mathbb{T}$". This is Proposition 13.16(ii): every $t \in \mathbb{T}$ is $\sum c_n T_n$ with $c_n \in \mathbb{Z}$, so $\phi_g(t) - \phi_f(t) = \sum c_n(a_n(g) - a_n(f)) \in \lambda$. The same fact makes the identity $t^t = w^* t w^*$ of Chapter 8 hold for all $t \in \mathbb{T}$ once it holds for the $T_n$.

**Definition 13.18** (eigenforms and eigencharacters)**.** An *eigenform* (of level $N$) is a nonzero $f \in S_k(N)$ with $tf \in \mathbb{C}f$ for every $t \in \mathbb{T}$; it is *normalised* if $a_1(f) = 1$. Its *eigencharacter* is the map $\phi_f \colon \mathbb{T} \to \mathbb{C}$ with $tf = \phi_f(t) f$. An *anemic eigenform* is a nonzero simultaneous eigenvector of $\mathbb{T}^{(N)}$.

**Proposition 13.19.** *Let $f \in S_k(N)$ be an eigenform with $T_n f = \lambda_n f$.*

*(i) $a_n(f) = \lambda_n a_1(f)$ for all $n$; hence $a_1(f) \neq 0$, and $f/a_1(f)$ is a normalised eigenform with $T_n(f/a_1(f)) = a_n(f/a_1(f)) \cdot f/a_1(f)$ and $q$-expansion $\sum \lambda_n q^n$.*

*(ii) A normalised eigenform satisfies $a_{mn} = a_m a_n$ for $(m,n) = 1$, $a_{p^{r+1}} = a_p a_{p^r} - p^{k-1} a_{p^{r-1}}$ for $p \nmid N$, $r \geq 1$, and $a_{p^r} = a_p^r$ for $p \mid N$; more generally $a_m a_n = \sum_{d \mid (m,n),\,(d,N)=1} d^{k-1} a_{mn/d^2}$.*

*(iii) $\phi_f \colon \mathbb{T} \to \mathbb{C}$ is a ring homomorphism with $\phi_f(T_n) = a_n(f)$ if $f$ is normalised. The map $f \mapsto \phi_f$ is injective on normalised eigenforms, and distinct normalised eigenforms are linearly independent.*

*Proof.* (i) $a_n(f) = a_1(T_n f) = \lambda_n a_1(f)$ by Theorem 13.10(iii); if $a_1(f) = 0$ all coefficients vanish and $f = 0$. (ii) Apply $\phi_f$ to Theorem 13.10(iv) and use $\phi_f(T_n) = a_n$. (iii) $\phi_f(tu) f = tuf = \phi_f(u) tf = \phi_f(t)\phi_f(u) f$, and similarly for sums and 1. A normalised eigenform is $\sum \phi_f(T_n) q^n$, so it is determined by $\phi_f$. Distinct normalised eigenforms therefore have distinct characters. Suppose they were linearly dependent, and let $\sum_{i=1}^{s} c_i f_i = 0$ be a nontrivial relation with $s$ as small as possible; then every $c_i \neq 0$, and $s \geq 2$ because each $f_i \neq 0$. Pick $t$ with $\phi_{f_1}(t) \neq \phi_{f_2}(t)$. Applying $t$ to the relation gives $\sum_i c_i \phi_{f_i}(t) f_i = 0$, and subtracting $\phi_{f_1}(t)$ times the relation gives

$$\sum_{i=2}^{s} c_i \bigl(\phi_{f_i}(t) - \phi_{f_1}(t)\bigr) f_i = 0,$$

a relation with at most $s-1$ terms whose coefficient of $f_2$ is $c_2(\phi_{f_2}(t) - \phi_{f_1}(t)) \neq 0$. This contradicts the minimality of $s$. □

**Lemma 13.20** (eigenform criterion)**.** *Let $L \geq 1$ and let $g = \sum_{n \geq 1} b_n q^n \in S_k(L)$ with $b_1 = 1$. Then $g$ is a normalised eigenform of $\mathbb{T}_L$ if and only if $b_{mn} = b_m b_n$ for $(m,n) = 1$, $b_{p^{r+1}} = b_p b_{p^r} - p^{k-1} b_{p^{r-1}}$ for $p \nmid L$, $r \geq 1$, and $b_{p^r} = b_p^r$ for $p \mid L$. Its eigenvalues are then $T_n g = b_n g$.*

*Proof.* "Only if" is Proposition 13.19(ii). Conversely, the relations say that the function $n \mapsto b_n$ satisfies the same multiplicativity and prime-power recursions as $n \mapsto T_n$; the proof of Theorem 13.10(iv) used nothing else (it is a computation in the commutative ring generated by

the $T_{p^r}$, valid for any family satisfying these relations), so $b_m b_n = \sum_{d\mid(m,n),(d,L)=1} d^{k-1} b_{mn/d^2}$. By Theorem 13.10(iii), $a_m(T_n g) = \sum_d d^{k-1} b_{mn/d^2} = b_n b_m = a_m(b_n g)$ for all $m \geq 1$; both are cusp forms, so $T_n g = b_n g$. □

*Warning* 13.21 (anemic eigenforms). An anemic eigenform need not have $a_1 \neq 0$, and Proposition 13.19(i) fails for it: $f_{11}(2z) \in S_2(22)$ is an eigenvector of every $T_n$ with $(n, 22) = 1$ (they commute with $V_2$ by Lemma 13.9), and $a_1(f_{11}(2z)) = 0$. Only eigenforms of the full $\mathbb{T}$ can be normalised. This is why the paper insists on "a normalised eigenform $g \in S_2(\Gamma_0(N)) \otimes \overline{\mathbb{Q}}$ for all of $\mathbb{T}$" in Lemma lem:crit.

**Proposition 13.22** (coefficient fields)**.** *Let $f$ be a normalised eigenform in $S_2(N)$.*

*(i) Every $a_n(f)$ is an algebraic integer, $K_f = \mathbb{Q}(a_n(f) : n \geq 1)$ is a number field, and $\phi_f(\mathbb{T}) = \mathbb{Z}[a_n(f) : n \geq 1]$ is an order of $\mathcal{O}_{K_f}$.*

*(ii) For $\sigma \in \mathrm{Aut}(\mathbb{C})$, $f^\sigma = \sum \sigma(a_n(f)) q^n$ is again a normalised eigenform in $S_2(N)$, with $\phi_{f^\sigma} = \sigma \circ \phi_f$.*

*Proof.* (i) $\mathbb{T}$ is a commutative ring, finitely generated as a $\mathbb{Z}$-module (Proposition 13.16(iii)), and $\phi_f$ a ring homomorphism to $\mathbb{C}$; Chapter 6 shows that the image of such a homomorphism is an order in the ring of integers of the number field it spans. The image is $\mathbb{Z}[a_n(f)]$ by Proposition 13.16(ii). (ii) Let $g_1, \dots, g_d$ be a $\mathbb{Z}$-basis of $S_2(N;\mathbb{Z})$; the matrices $(m_{ji}^{(n)})$ of the $T_n$ in this basis are integral. Write $f = \sum_i c_i g_i$ and put $f^\sigma := \sum_i \sigma(c_i) g_i \in S_2(N)$; its coefficients are $\sigma(a_n(f))$ because those of the $g_i$ are integers. From $T_n f = a_n f$ we get $\sum_i m_{ji}^{(n)} c_i = a_n c_j$ for all $j$; applying $\sigma$, $T_n f^\sigma = \sigma(a_n) f^\sigma$. □

*Warning* 13.23 (non-real coefficients). A normalised eigenform of the full $\mathbb{T}$ that is not a newform can have non-real coefficients. At level 22 the operator $U_2$ acts on $\mathrm{span}\{f_{11}(z), f_{11}(2z)\}$ by the matrix $\left(\begin{smallmatrix} -2 & 1 \\ -2 & 0 \end{smallmatrix}\right)$ (columns: images of $f_{11}(z)$, $f_{11}(2z)$), with characteristic polynomial $X^2 + 2X + 2$; its eigenvectors $f_{11} - \beta f_{11}(2z)$, $\alpha, \beta = -1 \pm i$, are normalised eigenforms of $\mathbb{T}_{22}$ with $a_2 = \alpha \in \{-1+i, -1-i\}$ (Proposition 13.48). "$K_f$ is totally real" holds for newforms (Corollary 13.45), and for the eigenvalues of $\mathbb{T}^{(N)}$ (Theorem 13.33), not for these.

## 13.4 The integral structure: the perfect pairing

This section proves the statement that the proof of Lemma lem:crit cites as [159, Theorem 2.2], [1, Lemma 2.1]: "The pairing $S_2(\Gamma_0(N);\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$, $(g,t) \mapsto a_1(tg)$, is perfect", together with "Every step holds verbatim over $\mathbb{Z}_\ell$, the pairing being perfect after tensoring with $\mathbb{Z}_\ell$" and the pairing $S_2(\Gamma_0(N);\mathbb{Z}_\ell) \times \mathbb{T}_\ell \to \mathbb{Z}_\ell$ of Step 1 of the proof of Theorem thm:quant. The only input from outside is the integral basis theorem of Chapter 12: $S_2(N;\mathbb{Z})$ is a lattice spanning $S_2(N)$. Sturm's bound enters only in part (iv). Throughout the section $k = 2$, $S = S_2(N;\mathbb{Z})$ and $d = \dim S_2(N)$.

**Theorem 13.24** (the perfect pairing)**.** *Define $\langle g, t\rangle = a_1(tg)$ for $g \in S$ and $t \in \mathbb{T}$.*

*(i) $\langle\ ,\ \rangle$ is $\mathbb{Z}$-bilinear with values in $\mathbb{Z}$, and $\langle ug, t\rangle = \langle g, ut\rangle$ for $u \in \mathbb{T}$; moreover $\langle g, T_n\rangle = a_n(g)$.*

*(ii) The pairing is perfect: $S \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{T},\mathbb{Z})$, $g \mapsto \langle g, \cdot\rangle$, and $\mathbb{T} \to \mathrm{Hom}_{\mathbb{Z}}(S,\mathbb{Z})$, $t \mapsto \langle \cdot, t\rangle$, are isomorphisms.*

*(iii)* $\mathbb{T}$ *is a free* $\mathbb{Z}$*-module of rank* $d$*;* $S \simeq \mathrm{Hom}(\mathbb{T},\mathbb{Z})$ *and* $\mathbb{T} \simeq \mathrm{Hom}(S,\mathbb{Z})$ *as* $\mathbb{T}$*-modules; for every commutative ring* $A$ *the induced pairing* $S_2(N;A)\times\mathbb{T}_A \to A$ *is perfect, and* $\mathbb{T}_A$ *acts faithfully on* $S_2(N;A)$*. In particular this holds for* $A=\mathbb{Z}_\ell$*.*

*(iv) Let* $B$ *be Sturm's bound of Chapter 12. Then* $T_1,\dots,T_B$ *span* $\mathbb{T}$ *as a* $\mathbb{Z}$*-module. In general* $T_1,\dots,T_d$ *do not.*

*Proof.* (i) $\mathbb{T}$ preserves $S$ (Proposition 13.16(iii)), so the values are integers; bilinearity is clear; $\langle ug,t\rangle = a_1(tug) = a_1(utg) = \langle g,ut\rangle$ by commutativity; and $\langle g,T_n\rangle = a_1(T_ng) = a_n(g)$ by Theorem 13.10(iii).

(ii) *Injectivity on the left.* If $\langle g,t\rangle = 0$ for all $t$ then $a_n(g) = \langle g,T_n\rangle = 0$ for all $n$ and $g=0$. *Injectivity on the right.* If $\langle g,t\rangle = 0$ for all $g\in S$, then for every $g$ and $n$, $a_n(tg) = a_1(T_ntg) = \langle T_ng,t\rangle = 0$, so $tg=0$; as $S$ spans $S_2(N)$, $t=0$. *Ranks.* Tensoring with $\mathbb{Q}$, the two injections give $\dim_{\mathbb{Q}} S_{\mathbb{Q}} \le \dim_{\mathbb{Q}} \mathbb{T}_{\mathbb{Q}}$ and $\dim_{\mathbb{Q}} \mathbb{T}_{\mathbb{Q}} \le \dim_{\mathbb{Q}} S_{\mathbb{Q}}$, where $S_{\mathbb{Q}} = S\otimes\mathbb{Q}$ has dimension $d$; so $\mathbb{T}$ has rank $d$ and both maps become isomorphisms over $\mathbb{Q}$. *Surjectivity on the left.* Let $\psi\colon \mathbb{T}\to\mathbb{Z}$ be additive. By the isomorphism over $\mathbb{Q}$ there is $g\in S_{\mathbb{Q}}$, a form with rational coefficients, with $a_1(tg)=\psi(t)$ for all $t$. Then $a_n(g)=\psi(T_n)\in\mathbb{Z}$ for all $n$, so $g$ has integer coefficients: $g\in S$. Hence $S\to\mathrm{Hom}(\mathbb{T},\mathbb{Z})$ is an isomorphism. *Surjectivity on the right.* $\mathbb{T}$ is free (a subgroup of the free group $\mathrm{M}_d(\mathbb{Z})$, Proposition 13.16(iii)) of rank $d$, so the composite $\mathbb{T}\to\mathrm{Hom}(S,\mathbb{Z})\to\mathrm{Hom}(\mathrm{Hom}(\mathbb{T},\mathbb{Z}),\mathbb{Z})$, the second map being the dual of the isomorphism just proved, is the canonical isomorphism of a free module of finite rank with its double dual (Chapter 3); hence the first map is an isomorphism.

(iii) The first statements are (ii) and (i) (the actions match because $\langle ug,t\rangle = \langle g,ut\rangle$). A perfect pairing of free $\mathbb{Z}$-modules of finite rank has Gram matrix in $\mathrm{GL}_d(\mathbb{Z})$ in any bases, and so does its extension to $A$; hence the pairing $(S\otimes A)\times(\mathbb{T}\otimes A)\to A$ is perfect. If $t\in\mathbb{T}_A$ kills $S_2(N;A)$, then $\langle g,t\rangle = 0$ for all $g$, so $t=0$.

(iv) By (ii) the image of $\mathbb{T}$ in $\mathrm{Hom}(S,\mathbb{Z})$ is everything, and by (i) the image of $T_n$ is the functional $a_n$. Chapter 12 shows (from Sturm's bound) that the truncation $S\to\mathbb{Z}^B$, $g\mapsto(a_1(g),\dots,a_B(g))$, is injective with torsion-free cokernel, so it is a split injection and its dual $\mathbb{Z}^B\to\mathrm{Hom}(S,\mathbb{Z})$ is onto: the functionals $a_1,\dots,a_B$ span $\mathrm{Hom}(S,\mathbb{Z})$. So $T_1,\dots,T_B$ span $\mathbb{T}$. The examples below show that $T_1,\dots,T_d$ need not. □

**Example 13.25** (the pairing at small levels)**.** For each $N$ below we computed a $\mathbb{Z}$-basis of $S$ by saturating the lattice of $q$-expansions (PARI's `matrixqz`, with coefficients up to $B\cdot\max(B,41)$), the matrices of $T_1,\dots,T_B$ in that basis (all integral), the rank of their $\mathbb{Z}$-span (equal to $d$), the determinant of the pairing in $\mathbb{Z}$-bases (always $\pm1$, as Theorem 13.24 says) and the index of $\mathbb{Z}T_1+\cdots+\mathbb{Z}T_d$ in $\mathbb{T}$:

| $N$ | 11 | 23 | 33 | 37 | 44 | 77 | 88 | 99 | 121 | 539 |
|---|---|---|---|---|---|---|---|---|---|---|
| $d=\mathrm{rank}\,\mathbb{T}$ | 1 | 2 | 3 | 2 | 4 | 7 | 9 | 9 | 6 | 49 |
| $B$ | 2 | 4 | 8 | 6 | 12 | 16 | 24 | 24 | 22 | 112 |
| $\det\langle g_i,t_j\rangle$ | 1 | 1 | −1 | −1 | −1 | −1 | −1 | −1 | −1 | −1 |
| $[\mathbb{T}:\sum_{n\le d}\mathbb{Z}T_n]$ | 1 | 1 | 1 | 1 | 1 | 5 | 3 | 2 | $\infty$ | $\infty$ |

Here $\infty$ means that $T_1,\dots,T_d$ are linearly dependent. At level 23, $\mathbb{T}=\mathbb{Z}[T_2]$ with $T_2^2+T_2-1=0$, the ring of integers of $\mathbb{Q}(\sqrt{5})$ (Chapter 6).

Level 23 can be done by hand. Row reduction of the $q$-expansions gives the echelon basis

$$g_1 = q - q^3 - q^4 - 2q^6 + 2q^7 - q^8 + \cdots, \qquad g_2 = q^2 - 2q^3 - q^4 + 2q^5 + q^6 + 2q^7 - 2q^8 + \cdots$$

(integral to $n = 40$, far beyond Sturm's bound 4). It is a $\mathbb{Z}$-basis of $S$: a form $g \in S$ equals $a_1(g)g_1 + a_2(g)g_2$, because the difference lies in the two-dimensional space spanned by $g_1, g_2$ and has $a_1 = a_2 = 0$. In this basis the pairing with $T_1, T_2$ is $\langle g_i, T_j\rangle = a_j(g_i) = \delta_{ij}$, the identity matrix, which proves directly that $\mathbb{T} = \mathbb{Z}T_1 \oplus \mathbb{Z}T_2$. By Proposition 13.8, $a_n(T_2 g) = a_{2n}(g) + 2a_{n/2}(g)$, so $T_2 g_1$ has $a_1 = a_2(g_1) = 0$ and $a_2 = a_4(g_1) + 2a_1(g_1) = 1$, while $T_2 g_2$ has $a_1 = 1$ and $a_2 = a_4(g_2) + 2a_1(g_2) = -1$. Hence $T_2 g_1 = g_2$, $T_2 g_2 = g_1 - g_2$, the matrix of $T_2$ is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ (columns the images), and its characteristic polynomial is $X^2 + X - 1$.

**Theorem 13.26** (eigenforms are homomorphisms)**.** *Let $A$ be a commutative ring, and let $\mathbb{T}$ act on $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{T}, A)$ by $(t\psi)(u) = \psi(tu)$.*

*(i) The map $S_2(N; A) \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{T}, A)$, $g \mapsto (t \mapsto a_1(tg))$, is an isomorphism of $\mathbb{T}$-modules.*

*(ii) Under it the normalised eigenforms with coefficients in $A$ (elements $g$ with $a_1(g) = 1$ and $tg \in Ag$ for all $t \in \mathbb{T}$) correspond bijectively to the ring homomorphisms $\phi\colon \mathbb{T} \to A$, via $\phi \mapsto f_\phi := \sum_n \phi(T_n)q^n$, and $t f_\phi = \phi(t) f_\phi$.*

*(iii) For $A = \mathbb{C}$ (or $\overline{\mathbb{Q}}$): the normalised eigenforms in $S_2(N)$ are the $f_\phi$, $\phi \in \mathrm{Hom}(\mathbb{T}, \overline{\mathbb{Q}})$; there are finitely many, at most $d$. For $A = \overline{\mathbb{F}}_\ell$: the mod $\ell$ eigenforms $\sum \phi(T_n)q^n$ correspond to $\mathrm{Hom}(\mathbb{T}, \overline{\mathbb{F}}_\ell)$, the kernels of these homomorphisms are the maximal ideals of $\mathbb{T}$ of residue characteristic $\ell$, and a maximal ideal $\mathfrak{m}$ corresponds to the $[\mathbb{T}/\mathfrak{m} : \mathbb{F}_\ell]$ homomorphisms $\mathbb{T} \to \mathbb{T}/\mathfrak{m} \hookrightarrow \overline{\mathbb{F}}_\ell$, which are Frobenius conjugates. For $A = \mathcal{O}$, the ring of integers of a number field, and $f$ a normalised eigenform with coefficients in $\mathcal{O}$: $\phi_f\colon \mathbb{T} \to \mathcal{O}$, and the kernel $\mathfrak{m}_{f,\lambda}$ of its reduction $\mathbb{T} \to \mathcal{O}/\lambda$ is a maximal ideal of residue characteristic $\ell$,* the maximal ideal attached to $(f, \lambda)$.

*Proof.* (i) is Theorem 13.24(iii): $S \otimes A \simeq \mathrm{Hom}(\mathbb{T}, \mathbb{Z}) \otimes A = \mathrm{Hom}(\mathbb{T}, A)$, $\mathbb{T}$ being free of finite rank; the map is the one displayed. It is $\mathbb{T}$-linear by Theorem 13.24(i). (ii) Let $g \leftrightarrow \psi$. If $g$ is a normalised eigenform with $tg = \phi(t)g$, then $\psi(t) = a_1(tg) = \phi(t)$, and $\phi$ is a ring homomorphism (Proposition 13.19(iii) works over any $A$). Conversely, if $\psi$ is a ring homomorphism, then for $t, u \in \mathbb{T}$, $a_1(u\,tg) = \psi(ut) = \psi(t)\psi(u) = a_1(u\,\psi(t)g)$; by (i), $tg = \psi(t)g$, and $a_1(g) = \psi(1) = 1$. The $q$-expansion is $\sum \psi(T_n)q^n$ by Theorem 13.24(i). (iii) Ring homomorphisms $\mathbb{T} \to \overline{\mathbb{Q}}$ are determined by the images of a $\mathbb{Z}$-basis of $\mathbb{T}$, which are roots of their characteristic polynomials; there are at most $\dim \mathbb{T}_{\mathbb{Q}} = d$ of them (Chapter 3). For $\overline{\mathbb{F}}_\ell$: the kernel of $\mathbb{T} \to \overline{\mathbb{F}}_\ell$ is a prime ideal with finite quotient, hence maximal, of residue characteristic $\ell$; conversely $\mathbb{T}/\mathfrak{m}$ is a finite field of characteristic $\ell$, which has $[\mathbb{T}/\mathfrak{m} : \mathbb{F}_\ell]$ embeddings into $\overline{\mathbb{F}}_\ell$, permuted transitively by Frobenius. The last statement is the case $\mathbb{T} \to \mathcal{O} \to \mathcal{O}/\lambda$. □

*In the paper* 13.27. In the proof of Lemma lem:crit, "the kernel of $\mathbb{T} \to \mathcal{O}/\lambda$, a maximal ideal of residue characteristic $\ell$" is Theorem 13.26(iii) for $\phi_g$; in the paragraph after Lemma lem:hecke, "Let $\mathfrak{m}$ be the maximal ideal of $\mathbb{T} \otimes \mathbb{Z}_\ell$ containing $\ell$ and $\ker \phi_f$": this $\mathfrak{m}$ is $\mathfrak{m}_{f,\ell}$ (for rational $f$, $\lambda = \ell$), viewed in $\mathbb{T}_\ell$ (Proposition 13.66).

**Corollary 13.28** (consequences of the pairing)**.** *In the notation of Theorem 13.24:*

*(i) a form $g \in S_2(N; A)$ is determined by the values $a_1(tg)$, $t \in \mathbb{T}$;*

*(ii) for an ideal $\mathfrak{a} \subseteq \mathbb{T}$, $S[\mathfrak{a}] = \{g \in S : \mathfrak{a}g = 0\}$ corresponds to $\mathrm{Hom}(\mathbb{T}/\mathfrak{a}, \mathbb{Z})$; it is saturated in $S$, of rank $\mathrm{rank}_{\mathbb{Z}}(\mathbb{T}/\mathfrak{a})$;*

*(iii)* *for a ring homomorphism $\phi\colon \mathbb{T} \to \mathbb{Z}$, $S[\ker\phi] = \mathbb{Z}f_\phi$; in particular, for a newform $f$ with rational coefficients and $I = \ker\phi_f$, the forms in $S$ killed by $I$ are $\mathbb{Z}f$;*

*(iv)* *if $I, J \subseteq \mathbb{T}$ are ideals with $I \cap J = 0$, $\mathbb{T}/I$ and $\mathbb{T}/J$ torsion-free and $\mathbb{T}/(I+J)$ finite, then $S[I] \oplus S[J]$ has finite index in $S$, equal to $\#\mathbb{T}/(I+J)$.*

*Proof.* (i) is Theorem 13.26(i). (ii) $\mathfrak{a}g = 0$ iff $\langle \mathfrak{a}g, u\rangle = \langle g, \mathfrak{a}u\rangle = 0$ for all $u$ (by (i)), iff $\langle g, \cdot\rangle$ vanishes on $\mathfrak{a}\mathbb{T} = \mathfrak{a}$. Saturation: if $mg$ vanishes on $\mathfrak{a}$ then so does $g$. The rank of $\mathrm{Hom}(\mathbb{T}/\mathfrak{a}, \mathbb{Z})$ is that of $\mathbb{T}/\mathfrak{a}$. (iii) $\mathbb{T}/\ker\phi \simeq \mathbb{Z}$, so $S[\ker\phi]$ has rank one and corresponds to the functionals $c\phi$, $c \in \mathbb{Z}$; the form attached to $\phi$ is $f_\phi$ (Theorem 13.26(ii)). (iv) The map $\mathbb{T} \to \mathbb{T}/I \times \mathbb{T}/J$, $t \mapsto (t,t)$, is injective because $I \cap J = 0$, and its cokernel is $\mathbb{T}/(I+J)$ (send $(x,y)$ to $x-y$). Both sides are free (the quotients are torsion-free and finitely generated) and, the cokernel being finite, of the same rank; so in bases the map is a square integer matrix $M$ with $|\det M| = \#\mathbb{T}/(I+J)$ (Smith normal form, Chapter 3). The dual map $\mathrm{Hom}(\mathbb{T}/I, \mathbb{Z}) \oplus \mathrm{Hom}(\mathbb{T}/J, \mathbb{Z}) \to \mathrm{Hom}(\mathbb{T}, \mathbb{Z})$ has matrix $M^{\mathsf{T}}$, so its cokernel also has order $|\det M|$. Under $S \simeq \mathrm{Hom}(\mathbb{T}, \mathbb{Z})$ its image is $S[I] \oplus S[J]$, by (ii). □

The hypotheses of (iv) hold in Lemma lem:crit: $\mathbb{T}/I \simeq \mathbb{Z}$ for $I = \ker\phi_f$, and $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$ is the kernel of the action on $f^\perp$, so $\mathbb{T}/J$ is a subring of $\mathrm{End}(f^\perp)$, torsion-free; $I \cap J = 0$ and the finiteness of $\mathbb{T}/(I+J)$ come from Theorem 13.60. Torsion-freeness is what makes $\mathbb{T}/I \times \mathbb{T}/J$ free, so that the determinant argument applies.

**Definition 13.29** (congruence ideal and congruence number)**.** Let $\mathcal{O}$ be $\mathbb{Z}$, $\mathbb{Z}_{(\ell)}$, $\mathbb{Z}_\ell$ or the completion $\mathcal{O}_\lambda$ of the ring of integers of a number field, and $\phi\colon \mathbb{T}_{\mathcal{O}} \to \mathcal{O}$ a surjective ring homomorphism. Its *(Hecke) congruence ideal* is $\eta(\phi) = \phi(\mathrm{Ann}_{\mathbb{T}_{\mathcal{O}}}(\ker\phi)) \subseteq \mathcal{O}$ (Chapter 3). For a newform $f \in S_2(N)$ with rational coefficients, the *congruence number* $r_f$ is the largest integer $r$ such that $f \equiv g \pmod r$ for some $g \in S_2(N;\mathbb{Z})$ orthogonal to $f$ for the Petersson product. For an elliptic curve $E$ with newform $f_E$, $r_E = r_{f_E}$.

Both notions measure how far the eigenform is from being alone in $\mathbb{T}$. The ideal $\ker\phi$ consists of the operators that act on the form by 0, and $\mathrm{Ann}(\ker\phi)$ of the operators whose product with every such operator is 0. If $\mathbb{T}$ were a product $\mathcal{O} \times \mathbb{T}'$ with $\phi$ the first projection, then $\ker\phi = 0 \times \mathbb{T}'$, $\mathrm{Ann}(\ker\phi) = \mathcal{O} \times 0$ and $\eta(\phi) = \mathcal{O}$. A proper ideal $\eta(\phi)$ therefore measures the failure of $\mathbb{T}$ to split off the factor of $\phi$ integrally. A toy model: let $m \geq 1$, $\mathbb{T} = \{(a,b) \in \mathbb{Z}^2 : a \equiv b \pmod m\}$ and $\phi(a,b) = a$. Then $\ker\phi = \{(0,b) : m \mid b\}$; an element $(a,b)$ kills it iff $b = 0$, and $(a,0) \in \mathbb{T}$ iff $m \mid a$; so $\mathrm{Ann}(\ker\phi) = \{(a,0) : m \mid a\}$ and $\eta(\phi) = m\mathbb{Z}$. Here $\mathbb{T}$ has the two homomorphisms $(a,b) \mapsto a$ and $(a,b) \mapsto b$ to $\mathbb{Z}$, which agree modulo $m$ on all of $\mathbb{T}$ and are glued exactly to that extent; congruence numbers measure the same gluing on the side of forms. That $r_f\mathbb{Z} = \eta(\phi_f)$ for a rational newform is the paper's Lemma lem:crit (see the paragraph after Proposition 13.30).

**Proposition 13.30.** *For a newform $f \in S_2(N)$ with rational coefficients, $L = f^\perp \cap S_2(N;\mathbb{Z})$ is saturated of rank $d-1$, $\mathbb{Z}f \oplus L$ has finite index in $S_2(N;\mathbb{Z})$, and $r_f$ is finite: a congruence $f \equiv g \pmod r$ with $g \in L$ gives the element $(f-g)/r$ of $S_2(N;\mathbb{Z})$ whose class in $S_2(N;\mathbb{Z})/(\mathbb{Z}f \oplus L)$ has order $r$.*

*Proof.* By Theorem 13.60 below (whose proof does not use this proposition), $f^\perp$ is the kernel of an idempotent $e \in \mathbb{T} \otimes \mathbb{Q}$; hence $f^\perp$ is spanned by forms with rational coefficients, $f^\perp \cap S_2(N;\mathbb{Q})$ has dimension $d-1$, and $L$, the intersection of a lattice with a subspace, is saturated of rank

$d-1$. Also $f \notin f^{\perp}$, so $\mathbb{Z}f \oplus L$ has rank $d$, hence finite index. If $f \equiv g \pmod r$ with $g \in L$, then $h = (f-g)/r \in S_2(N;\mathbb{Z})$; if $mh \in \mathbb{Z}f \oplus L$ then, projecting to $\mathbb{Q}f$ along $f^{\perp}$, $m/r \in \mathbb{Z}$; so the class of $h$ has order exactly $r$, and $r$ is bounded by the index. □

The equality $r_f\mathbb{Z} = \eta(\phi_f)$ is the paper's Lemma lem:crit, proved in Chapter 15 from Theorem 13.24, Corollary 13.28 and Theorem 13.60; its relation with the modular degree (the theorem of Agashe, Ribet and Stein) is Chapter 14. The paper also uses $\eta(\pi)$ for the eigencharacter $\pi$ of a depleted form $g^S$, which is not a newform ("$\eta^{\mathrm{Hecke}}(g^S)$", Step 1 of the proof of Theorem thm:quant); Proposition 13.64 proves the conclusions of the lemma for such $\pi$, and for $\phi_f$ itself. Computed values of $r_f$ are in Example 13.84.

*In the paper* 13.31. The proof of Lemma lem:crit uses, in this order: the perfect pairing (Theorem 13.24); "under it the forms killed by $I$ are $\mathbb{Z}f$" (Corollary 13.28(iii)); "the forms killed by $J$ are $L = f^{\perp} \cap S_2(\Gamma_0(N);\mathbb{Z})$" (with $J = \mathrm{Ann}_{\mathbb{T}}(f^{\perp})$: $J$ kills $L$ by definition, and a form killed by $J$ is killed by the idempotent $e \in J \otimes \mathbb{Q}$ of Theorem 13.60(ii), so lies in $\ker e = f^{\perp}$); "Dualising the inclusion $\mathbb{T} \subset \mathbb{T}/I \times \mathbb{T}/J$ ... shows that $\mathbb{Z}f \oplus L$ has index $r'$" (Corollary 13.28(iv)); and "Every step holds verbatim over $\mathbb{Z}_\ell$, the pairing being perfect after tensoring with $\mathbb{Z}_\ell$" (Theorem 13.24(iii)). The full proof is in Chapter 15.

## 13.5 Petersson adjoints

This section computes the adjoints of the Hecke operators for the Petersson product. Its use in the paper is the sentence of the proof of Lemma lem:crit: "the $f$-line is $\mathbb{T}$-stable, and so is $f^{\perp}$, because $f$ is an eigenvector of the adjoints of the $T_n$ as well". For $(n,N)=1$ the operator $T_n$ is self-adjoint; $U_p$ is not, but its adjoint is $w_N U_p w_N$, and a newform is an eigenvector of $w_N$. The cohomological analogue, with the cup product in place of the Petersson product, is the statement of the paper's Section 4.1 that $w_{M^\Sigma} T_n w_{M^\Sigma}^{-1}$ is the adjoint of $T_n$ (Chapter 14).

For a congruence subgroup $\Gamma' \subseteq \mathrm{SL}_2(\mathbb{Z})$ containing $-1$, or a conjugate $\alpha^{-1}\Gamma'\alpha$ of one by $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$, and cusp forms $f, g$ of weight $k$ for it, we write $\langle f, g\rangle_{\Gamma'} = \int_{\Gamma'\backslash\mathcal{H}} f\bar{g}y^k \, d\mu$, $d\mu = dx\,dy/y^2$. The product is defined in Chapter 12 for these groups (each contains a conjugate of some $\Gamma(M)$, and the definition goes through such a subgroup). We use two facts from Chapter 12: the function $f\bar{g}\,y^k$ is $\Gamma'$-invariant, and for $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$, $\langle f|_k\alpha, g|_k\alpha\rangle_{\alpha^{-1}\Gamma'\alpha} = \langle f, g\rangle_{\Gamma'}$ (change of variables $z \mapsto \alpha z$, using $\mathrm{Im}(\alpha z)^{k/2}|f(\alpha z)| = (\mathrm{Im}\, z)^{k/2}|(f|_k\alpha)(z)|$), where $\alpha^{-1}\Gamma'\alpha$ need not lie in $\mathrm{SL}_2(\mathbb{Z})$. Every group used below is of this kind: $\Gamma_0(N)$, $\Gamma \cap \alpha^{-1}\Gamma\alpha$, and the conjugate $\alpha^{-1}\Gamma_0(M)\alpha$ in the proof of Proposition 13.35(ii).

**Proposition 13.32** (adjoints of double coset operators)**.** *Let* $\Gamma = \Gamma_0(N)$, $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ *and* $\alpha' = (\det\alpha)\alpha^{-1}$. *For* $f, g \in S_k(N)$,

$$\langle [\Gamma\alpha\Gamma]_k f, g\rangle = \langle f, [\Gamma\alpha'\Gamma]_k g\rangle.$$

*Proof.* Let $\Gamma_\alpha = \Gamma \cap \alpha^{-1}\Gamma\alpha$ and $\Gamma = \bigsqcup_{i=1}^{r} \Gamma_\alpha\gamma_i$, so that $\Gamma\alpha\Gamma = \bigsqcup \Gamma\alpha\gamma_i$ (Proposition 13.6(i)). The function $f|_k\alpha$ is invariant under $\alpha^{-1}\Gamma\alpha \supseteq \Gamma_\alpha$, so $F(z) = (f|_k\alpha)(z)\overline{g(z)}y^k$ is $\Gamma_\alpha$-invariant, and since $g|_k\gamma_i = g$, the $i$-th term of $[\Gamma\alpha\Gamma]_k f \cdot \bar{g}\,y^k$ is $(\det\alpha)^{k/2-1}F(\gamma_i z)$. If $\mathcal{F}$ is a fundamental domain for $\Gamma$, then $\bigcup_i \gamma_i\mathcal{F}$ is one for $\Gamma_\alpha$ (both groups contain $-1$), so

$$\langle [\Gamma\alpha\Gamma]_k f, g\rangle = (\det\alpha)^{k/2-1}\sum_i \int_{\mathcal{F}} F(\gamma_i z)\, d\mu = (\det\alpha)^{k/2-1}\langle f|_k\alpha, g\rangle_{\Gamma_\alpha}.$$

Now $\alpha'\alpha = (\det\alpha)\cdot 1$, and positive scalars act trivially in even weight, so $g = (g|_k\alpha')|_k\alpha$. By the change of variables, $\langle f|_k\alpha, (g|_k\alpha')|_k\alpha\rangle_{\Gamma_\alpha} = \langle f, g|_k\alpha'\rangle_{\alpha\Gamma_\alpha\alpha^{-1}}$, and $\alpha\Gamma_\alpha\alpha^{-1} = \alpha\Gamma\alpha^{-1}\cap\Gamma = \Gamma\cap\alpha'^{-1}\Gamma\alpha' = \Gamma_{\alpha'}$ (conjugation by a scalar is trivial). Applying the first computation to $\alpha'$ and $g$, $\langle[\Gamma\alpha'\Gamma]_k g, f\rangle = (\det\alpha')^{k/2-1}\langle g|_k\alpha', f\rangle_{\Gamma_{\alpha'}}$. Since $\det\alpha' = \det\alpha$ is real, taking complex conjugates gives $\langle f, [\Gamma\alpha'\Gamma]_k g\rangle = (\det\alpha)^{k/2-1}\langle f, g|_k\alpha'\rangle_{\Gamma_{\alpha'}} = \langle[\Gamma\alpha\Gamma]_k f, g\rangle$. □

**Theorem 13.33** (self-adjointness)**.** *For $(n,N)=1$ the operator $T_n$ is self-adjoint on $S_k(N)$. Consequently $\mathbb{T}^{(N)}\otimes\mathbb{C}$ is a commutative algebra of self-adjoint operators: the eigenvalues of every $T_n$, $(n,N)=1$, are real; $S_k(N)$ is the orthogonal direct sum of the simultaneous eigenspaces of $\mathbb{T}^{(N)}$; and the generalised eigenspaces of $\mathbb{T}^{(N)}$ are eigenspaces.*

*Proof.* For a prime $p\nmid N$, $\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)' = \left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right)$, and $\Gamma\left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right)\Gamma = \Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma$ (Proposition 13.7(ii)); so $T_p^\dagger = T_p$ by Proposition 13.32. For $(n,N)=1$, $T_n$ is a polynomial with integer coefficients in the $T_p$, $p\mid n$ (Theorem 13.10), and a real polynomial in commuting self-adjoint operators is self-adjoint. The rest is the spectral theorem for a finite commuting family of self-adjoint operators on a finite-dimensional Hermitian space: each is diagonalisable with real eigenvalues in an orthonormal basis, and commuting ones are simultaneously so; eigenspaces for distinct systems are orthogonal because $\lambda\langle v,w\rangle = \langle tv,w\rangle = \langle v,tw\rangle = \mu\langle v,w\rangle$ with $\lambda\neq\mu$ real. □

**Lemma 13.34** (the trace from level $Mp$ to level $M$)**.** *Let $p\nmid M$, $\Gamma_0(M) = \bigsqcup_{i=0}^{p}\Gamma_0(Mp)\gamma_i$, and $\mathrm{Tr}(g) = \sum_i g|_k\gamma_i$ for $g\in S_k(Mp)$. Then $\mathrm{Tr}(g)\in S_k(M)$, and for $h\in S_k(M)$: (i) $\langle h, g\rangle_{\Gamma_0(Mp)} = \langle h, \mathrm{Tr}\, g\rangle_{\Gamma_0(M)}$; (ii) $\mathrm{Tr}(h(pz)) = p^{1-k}T_p^{(M)}h$; (iii) $\langle h,h\rangle_{\Gamma_0(Mp)} = (p+1)\langle h,h\rangle_{\Gamma_0(M)}$.*

*Proof.* $\mathrm{Tr}(g)$ is $\Gamma_0(M)$-invariant because right multiplication by $\gamma\in\Gamma_0(M)$ permutes the cosets, and cuspidal by the argument of Proposition 13.6. (i) As in the proof of Proposition 13.32, $\bigcup\gamma_i\mathcal{F}$ is a fundamental domain for $\Gamma_0(Mp)$ when $\mathcal{F}$ is one for $\Gamma_0(M)$, and $h|_k\gamma_i = h$. (ii) Let $\alpha = \left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right)$; then $\Gamma_0(M)\cap\alpha^{-1}\Gamma_0(M)\alpha = \Gamma_0(Mp)$ (conjugation by $\alpha$ sends $\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)$ to $\left(\begin{smallmatrix}a&pb\\c/p&d\end{smallmatrix}\right)$), so by Proposition 13.6(i) and Proposition 13.7(ii), $p^{k/2-1}\sum_i h|_k\alpha\gamma_i = [\Gamma_0(M)\alpha\Gamma_0(M)]_k h = T_p^{(M)}h$. Since $h(pz) = p^{-k/2}h|_k\alpha$, $\mathrm{Tr}(h(pz)) = p^{-k/2}p^{1-k/2}T_p^{(M)}h$. (iii) is (i) with $g=h$, since $\mathrm{Tr}\, h = (p+1)h$. □

**Proposition 13.35** (the adjoint of $U_p$)**.** *Let $p\mid N$.*

*(i) $U_p^\dagger = w_N U_p w_N$ on $S_k(N)$.*

*(ii) Let $h$ be a newform of level $M$ and $p\nmid M$. In $S_k(Mp)$, with all products at level $Mp$,*

$$\langle h(pz), h\rangle = \frac{p^{1-k}a_p(h)}{p+1}\langle h,h\rangle, \qquad \langle h(pz), h(pz)\rangle = p^{-k}\langle h,h\rangle.$$

*(iii) $U_p$ is in general neither self-adjoint nor normal (Example 13.36).*

*Proof.* (i) With $W = W_N = \left(\begin{smallmatrix}0&-1\\N&0\end{smallmatrix}\right)$ and $W^{-1} = \frac{1}{N}\left(\begin{smallmatrix}0&1\\-N&0\end{smallmatrix}\right)$,

$$W\begin{pmatrix}1&0\\0&p\end{pmatrix}W^{-1} = \begin{pmatrix}0&-p\\N&0\end{pmatrix}\cdot\frac{1}{N}\begin{pmatrix}0&1\\-N&0\end{pmatrix} = \frac{1}{N}\begin{pmatrix}pN&0\\0&N\end{pmatrix} = \begin{pmatrix}p&0\\0&1\end{pmatrix},$$

and $W$ normalises $\Gamma$ (Chapter 12). Let $\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma = \bigsqcup_i\Gamma\alpha_i$. Conjugating by $W$ and using $W\Gamma W^{-1} = \Gamma$ gives $\Gamma\left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right)\Gamma = W(\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma)W^{-1} = \bigsqcup_i\Gamma W\alpha_i W^{-1}$, so the $W\alpha_i W^{-1}$ are coset

representatives for the double coset of $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$. Both double cosets have determinant $p$, hence the same prefactor $p^{k/2-1}$, and since the slash operator is a right action,

$$[\Gamma\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)\Gamma]_k g = p^{k/2-1}\sum_i ((g|_k W)|_k \alpha_i)|_k W^{-1} = w_N^{-1} U_p w_N g \qquad (w_N g = g|_k W).$$

As $w_N^{-1} = w_N$ (Convention 13.1(f)), Proposition 13.32 with $\alpha = \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$, $\alpha' = \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ gives $U_p^\dagger = w_N U_p w_N$.

(ii) By Lemma 13.34, $\langle h(pz), h\rangle_{Mp} = \langle \mathrm{Tr}(h(pz)), h\rangle_M = p^{1-k} a_p(h)\langle h, h\rangle_M$ and $\langle h, h\rangle_{Mp} = (p+1)\langle h, h\rangle_M$ (the first equality is the complex conjugate of Lemma 13.34(i)). For the second formula, $h(pz) = p^{-k/2} h|_k \alpha$ with $\alpha = \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and $\alpha^{-1}\Gamma_0(M)\alpha$ contains $\Gamma_0(Mp)$ with the same index $p+1$ as $\Gamma_0(Mp)$ in $\Gamma_0(M)$. Indeed $\alpha^{-1}\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\alpha = \left(\begin{smallmatrix} a & b/p \\ pc & d \end{smallmatrix}\right)$, so $\alpha^{-1}\Gamma_0(M)\alpha$ is the group of matrices $\left(\begin{smallmatrix} x & y \\ z & w \end{smallmatrix}\right)$ of determinant 1 with $x, w \in \mathbb{Z}$, $py \in \mathbb{Z}$ and $Mp \mid z$; it contains $\Gamma_0(Mp)$ (take $b = py$). For the index, $z \mapsto \alpha z$ is an isometry of $\mathcal{H}$ for $d\mu$, so $\alpha^{-1}\Gamma_0(M)\alpha$ and $\Gamma_0(M)$ have fundamental domains of the same hyperbolic area, and the index of a subgroup is the ratio of these areas (all groups here contain $-1$); so $\langle h|_k\alpha, h|_k\alpha\rangle_{\Gamma_0(Mp)} = (p+1)\langle h|_k\alpha, h|_k\alpha\rangle_{\alpha^{-1}\Gamma_0(M)\alpha} = (p+1)\langle h, h\rangle_{\Gamma_0(M)} = \langle h, h\rangle_{\Gamma_0(Mp)}$, and the factor $p^{-k}$ comes from $(p^{-k/2})^2$. □

**Example 13.36** (level 22)**.** Take $h = f_{11}$, $p = 2$, $N = 22$, and the basis $f_{11}(z), f_{11}(2z)$ of $S_2(22)$. By Proposition 13.35(ii) with $a_2(f_{11}) = -2$, the Gram matrix of the Petersson product is $\langle f_{11}, f_{11}\rangle_{22} \cdot G$, $G = \left(\begin{smallmatrix} 1 & -1/3 \\ -1/3 & 1/4 \end{smallmatrix}\right)$: the off-diagonal entry is $\frac{p^{1-k}a_p}{p+1} = \frac{2^{-1}\cdot(-2)}{3} = -\frac{1}{3}$ and the lower diagonal entry is $p^{-k} = \frac{1}{4}$. The matrix of $U_2$ (columns: images of the basis vectors) is $A = \left(\begin{smallmatrix} -2 & 1 \\ -2 & 0 \end{smallmatrix}\right)$, since $U_2 f_{11} = -2f_{11} - 2f_{11}(2z)$ and $U_2 f_{11}(2z) = f_{11}$ (Proposition 13.48(i) with $a_2 = -2$, $p^{k-1} = 2$). With $\langle x, y\rangle = x^{\mathsf{T}} G \bar{y}$ in coordinates, an operator with matrix $B$ is the adjoint of $A$ iff $x^{\mathsf{T}} A^{\mathsf{T}} G \bar{y} = x^{\mathsf{T}} G \bar{B} \bar{y}$ for all $x, y$, that is, $\bar{B} = G^{-1} A^{\mathsf{T}} G$; as $A$ and $G$ are real, the adjoint has matrix

$$G^{-1} A^{\mathsf{T}} G = \frac{36}{5}\begin{pmatrix} 1/4 & 1/3 \\ 1/3 & 1 \end{pmatrix}\begin{pmatrix} -4/3 & 1/6 \\ 1 & -1/3 \end{pmatrix} = \begin{pmatrix} 0 & -1/2 \\ 4 & -2 \end{pmatrix}.$$

The involution $w_{22}$ exchanges the two lines: $f_{11}|_2 W_{22} = -2f_{11}(2z)$ and $f_{11}(2z)|_2 W_{22} = -\frac{1}{2} f_{11}$ (Chapter 12, from $\varepsilon_{11}(f_{11}) = -1$), so $w_{22}$ has matrix $\left(\begin{smallmatrix} 0 & -1/2 \\ -2 & 0 \end{smallmatrix}\right)$, and indeed

$$\begin{pmatrix} 0 & -1/2 \\ -2 & 0 \end{pmatrix}\begin{pmatrix} -2 & 1 \\ -2 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1/2 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1/2 \\ 4 & -2 \end{pmatrix}.$$

Finally $AA^\dagger = \left(\begin{smallmatrix} 4 & -1 \\ 0 & 1 \end{smallmatrix}\right) \neq \left(\begin{smallmatrix} 1 & 0 \\ -4 & 4 \end{smallmatrix}\right) = A^\dagger A$: $U_2$ is not normal. (The Gram matrix was also computed with PARI's `mfpetersson`, to 40 digits.)

**Corollary 13.37** (stable orthogonal complements)**.** *Let $V \subseteq S_k(N)$ be a subspace stable under $\mathbb{T}$ and under $w_N$. Then $V^\perp$ is stable under $\mathbb{T}$ and $w_N$. In particular, if $f$ is an eigenform of $\mathbb{T}$ and an eigenvector of $w_N$, then $f^\perp$ is $\mathbb{T}$-stable.*

*Proof.* Since $\mathbb{T}$ is spanned by the $T_n$ (Proposition 13.16(ii)), it suffices to treat each $T_n$. Write $n = m\prod_{p|N} p^{r_p}$ with $(m, N) = 1$; by Theorem 13.10, $T_n = T_m \prod_{p|N} U_p^{r_p}$, a product of commuting

operators. The adjoint of a product is the product of the adjoints in the reverse order, and here the order does not matter, so

$$T_n^\dagger = T_m^\dagger \prod_{p|N} (U_p^\dagger)^{r_p} = T_m \prod_{p|N} (w_N U_p w_N)^{r_p}$$

by Theorem 13.33 and Proposition 13.35(i). Each factor maps $V$ into $V$: $T_m$ and $U_p$ because $V$ is $\mathbb{T}$-stable, and $w_N$ by hypothesis. So $T_n^\dagger V \subseteq V$, and for $g \in V^\perp$ and $v \in V$, $\langle T_n g, v\rangle = \langle g, T_n^\dagger v\rangle = 0$: $T_n V^\perp \subseteq V^\perp$. Also $w_N$ is self-adjoint (Chapter 12), so $\langle w_N g, v\rangle = \langle g, w_N v\rangle = 0$. For the last statement take $V = \mathbb{C}f$, which is stable under $\mathbb{T}$ and $w_N$ by hypothesis. □

*In the paper* 13.38. In the proof of Lemma lem:crit, "the $f$-line is $\mathbb{T}$-stable, and so is $f^\perp$, because $f$ is an eigenvector of the adjoints of the $T_n$ as well": by Proposition 13.35(i), $U_p^\dagger f = w_N U_p w_N f = \varepsilon_N(f)^2 a_p(f) f = a_p(f) f$, using $f|_2 W_N = \varepsilon_N(f) f$ (Proposition 13.44); this is Corollary 13.37 with $V = \mathbb{C}f$. A second proof, which needs no adjoints of $U_p$, is in Theorem 13.60: $f^\perp$ is the sum of the eigenspaces of $\mathbb{T}^{(N)}$ other than the line of $f$.

## 13.6 Oldforms and newforms: the theory of Atkin and Lehner

This section explains what $f = f_E$, "the newform of the isogeny class of $E$" of the paper's Section 3, is, and why the forms $h(dz)$ of Lemma lem:old span a subspace orthogonal to $f$. It states the theorems of Atkin and Lehner (newform bases, multiplicity one, the decomposition of $S_k(N)$) and strong multiplicity one, with sketches and references; and it proves the consequences the paper uses: $f$ is an eigenvector of $w_N$, its coefficients are real, its Galois conjugates are newforms, and the dimension of the new subspace.

**Definition 13.39** (oldforms and newforms)**.** For $M \mid N$ and $d \mid N/M$, $V_d\colon S_k(M) \to S_k(N)$, $h \mapsto h(dz) = d^{-k/2} h|_k \left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right)$ (Chapter 12). The *old subspace* is

$$S_k^{\mathrm{old}}(N) = \sum_{p|N} \left(S_k(N/p) + V_p S_k(N/p)\right),$$

equivalently the span of all $V_d S_k(M)$ with $M \mid N$, $M \neq N$ and $d \mid N/M$. The *new subspace* $S_k^{\mathrm{new}}(N)$ is its orthogonal complement for the Petersson product. A *newform of level $N$* is a normalised eigenform of $\mathbb{T}_N$ lying in $S_k^{\mathrm{new}}(N)$.

The two descriptions of the old subspace agree: $S_k(N/p) + V_p S_k(N/p)$ is of the second kind, and conversely, for $M \neq N$, $d \mid N/M$ and $h \in S_k(M)$, either $d = 1$ and $h \in S_k(N/p)$ for any $p \mid N/M$, or $p \mid d$ for some prime and $V_d h = V_p(V_{d/p} h)$ with $V_{d/p} h \in S_k(Md/p) \subseteq S_k(N/p)$.

**Proposition 13.40.** *(i) For $(n, N) = 1$, $M \mid N$, $d \mid N/M$ and $h \in S_k(M)$: $T_n^{(N)}(h(dz)) = (T_n^{(M)} h)(dz)$.*

*(ii) For $h \in S_k(M)$ and $d \mid N/M$: $(h(dz))|_k W_N = d^{-k/2} e^{k/2} (h|_k W_M)(ez)$ with $e = N/(Md)$. So $w_N$ maps $V_d S_k(M)$ onto $V_{N/(Md)} S_k(M)$.*

*(iii) $S_k^{\mathrm{old}}(N)$ and $S_k^{\mathrm{new}}(N)$ are stable under all of $\mathbb{T}$ (the $U_p$ included) and under $w_N$.*

*Proof.* (i) $T_n$ is a polynomial in the $T_p$, $p \mid n$, each equal to $\mathcal{U}_p + p^{k-1}V_p$ at both levels, and these commute with $V_d$ for $(d,p)=1$ (Lemma 13.9). (ii) Since $dMe = N$,

$$\begin{pmatrix} d & 0 \\ 0 & 1 \end{pmatrix} W_N = \begin{pmatrix} d & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix} = \begin{pmatrix} 0 & -d \\ N & 0 \end{pmatrix} = d \begin{pmatrix} 0 & -1 \\ Me & 0 \end{pmatrix} = d \cdot W_M \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix},$$

and the scalar matrix $d\cdot 1$ acts trivially in even weight ($f|_k(d\cdot 1) = d^k d^{-k} f$); so $(h|_k\left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right))|_k W_N = (h|_k W_M)|_k\left(\begin{smallmatrix} e & 0 \\ 0 & 1 \end{smallmatrix}\right)$. As $h|_k\left(\begin{smallmatrix} c & 0 \\ 0 & 1 \end{smallmatrix}\right) = c^{k/2}h(cz)$ and $h(dz) = d^{-k/2}h|_k\left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right)$, this gives $(h(dz))|_k W_N = d^{-k/2}(h|_k W_M)|_k\left(\begin{smallmatrix} e & 0 \\ 0 & 1 \end{smallmatrix}\right) = d^{-k/2}e^{k/2}(h|_k W_M)(ez)$. Since $w_M$ is a bijection of $S_k(M)$, the image of $V_d S_k(M)$ is $V_e S_k(M)$. (iii) The old subspace is stable under $T_n$, $(n,N)=1$, by (i), and under $w_N$ by (ii). For $p \mid N$, $U_p = \mathcal{U}_p$ on $q$-expansions (Proposition 13.8); for $h \in S_k(M)$ and $d \mid N/M$: if $p \mid d$, $\mathcal{U}_p V_d h = V_{d/p}h$; if $p \nmid d$, $\mathcal{U}_p V_d h = V_d \mathcal{U}_p h$, and $\mathcal{U}_p h$ is $U_p^{(M)}h$ if $p \mid M$, or $T_p^{(M)}h - p^{k-1}V_p h$ if $p \nmid M$ (then $p \mid N/M$ and $V_{dp}h$ is again old, as $dp \mid N/M$). So the old subspace is $\mathbb{T}$-stable, and the new subspace, its orthogonal complement, is $\mathbb{T}$-stable and $w_N$-stable by Corollary 13.37. □

**Theorem 13.41** (Atkin, Lehner, Li)**.** *Let $k \geq 2$ be even.*

*(i) (Newform basis.) $S_k^{\mathrm{new}}(N)$ has a basis of newforms; distinct newforms are orthogonal.*

*(ii) (Multiplicity one.) Let $f$ be a newform of level $M$ and $L$ a multiple of $M$. The subspace of $g \in S_k(L)$ with $T_n g = a_n(f)g$ for all $(n,L)=1$ is $\operatorname{span}\{f(dz) : d \mid L/M\}$, of dimension $\sigma_0(L/M)$. In particular two newforms of levels dividing $L$ with the same eigenvalues for all $T_n$, $(n,L)=1$, are equal.*

*(iii) (Decomposition.) $S_k(N) = \bigoplus_{M|N} \bigoplus_{d|N/M} V_d S_k^{\mathrm{new}}(M)$: the forms $h(dz)$, $h$ a newform of level $M \mid N$ and $d \mid N/M$, form a basis of $S_k(N)$, and the old subspace is spanned by those with $M \neq N$.*

*(iv) For a newform $f$ of level $M \mid N$ write $V_f = \operatorname{span}\{f(dz) : d \mid N/M\} \subseteq S_k(N)$. Then $S_k(N) = \bigoplus_f V_f$ orthogonally, $V_f$ is the eigenspace of $\mathbb{T}^{(N)}$ for the system of $f$, and for a newform $f$ of level $N$, $V_f = \mathbb{C}f$ and $f^\perp = \bigoplus_{h \neq f} V_h$.*

*We do not prove (i) to (iii); see [8] (for trivial character, the main theorems), [113] (any character), [52, Sections 5.6 to 5.8] and [137, Section 4.6]. Part (iv) follows: by (ii) the spaces $V_f$ are the eigenspaces of $\mathbb{T}^{(N)}$ for pairwise distinct systems, and they are orthogonal by Theorem 13.33.*

*Sketch of proof.* The route of Diamond and Shurman. The key is the *Main Lemma*: if $g \in S_k(N)$ has $a_n(g) = 0$ for every $n$ prime to $N$, then $g \in \sum_{p|N} V_p S_k(N/p)$. Granting it, let $g \in S_k^{\mathrm{new}}(N)$ be an anemic eigenform with $T_n g = \lambda_n g$ for $(n,N)=1$. If $a_1(g) = 0$, then $a_n(g) = a_1(T_n g) = 0$ for $(n,N)=1$, so $g$ is old by the Main Lemma, hence $g = 0$. So every nonzero anemic eigenform in the new subspace has $a_1 \neq 0$; two of them with the same system, normalised, have a difference with $a_1 = 0$, which is 0; this is multiplicity one in $S_k^{\mathrm{new}}(N)$. As $\mathbb{T}$ preserves $S_k^{\mathrm{new}}(N)$ and commutes with $\mathbb{T}^{(N)}$, each anemic eigenspace there is a line stable under $\mathbb{T}$, so its normalised generator is an eigenform of all of $\mathbb{T}$: a newform; the anemic eigenspaces exhaust $S_k^{\mathrm{new}}(N)$ by Theorem 13.33, which gives (i). Part (iii) follows by induction on $N$, applying the same argument to the old subspace, and the statement for other levels in (ii) needs in addition that newforms of different levels have different anemic systems, which comes out of the same induction. The Main Lemma itself is proved by an induction on the number of prime factors of $N$, using the operators $\mathcal{U}_p$, $V_p$ and the trace of Lemma 13.34 [52, Section 5.7]. □

**Theorem 13.42** (strong multiplicity one). *(i) If $f$ and $g$ are newforms of weight $k$ and levels $M$, $M'$ with $a_p(f) = a_p(g)$ for all but finitely many primes $p$, then $f = g$ (and $M = M'$).*

*(ii) (Weight 2.) The same holds if $a_p(f) = a_p(g)$ for all $p$ in a set of primes of Dirichlet density one.*

*(iii) Let $f$ be a newform of level $M$, $L$ a multiple of $M$, and $h \in S_k(L)$ an eigenvector of $\mathbb{T}^{(L)}$ for a system $\psi$ with $\psi(T_p) \neq a_p(f)$ for infinitely many $p$. Then $h$ is orthogonal to every $f(dz)$, $d \mid L/M$. If $h'$ is a newform of level dividing $L$ different from $f$, then its system satisfies this hypothesis.*

*Part (i) is stated with a reference: [137, Theorem 4.6.19], [113]; it is a consequence of (ii) of Theorem 13.41 applied at the level $L = \mathrm{lcm}(M, M')$ once one knows that finitely many $T_p$ may be ignored, which is the content of the theorem. Part (ii) is sketched below; (iii) is proved here.*

*Sketch of proof of (ii), and proof of (iii).* (ii) Let $\lambda$ be a prime above some $\ell$ of a number field containing the coefficients of $f$ and $g$. Their $\lambda$-adic Galois representations (Chapter 15) are unramified outside $\ell MM'$ and satisfy $\mathrm{tr}\,\rho_f(\mathrm{Frob}_p) = a_p(f)$ there. By Chebotarev (Chapter 5) the Frobenius elements at a set of primes of density one are dense in the Galois group, so $\mathrm{tr}\,\rho_f = \mathrm{tr}\,\rho_g$ everywhere by continuity; by Brauer-Nesbitt (Chapter 4) $\rho_f^{\mathrm{ss}} \simeq \rho_g^{\mathrm{ss}}$, so $a_p(f) = a_p(g)$ for all $p \nmid \ell MM'$, and (i) applies. (iii) By Theorem 13.33 $h$ and each $f(dz)$ are eigenvectors of the self-adjoint operators $T_p$, $p \nmid L$, with eigenvalues $\psi(T_p)$ and $a_p(f)$ (Proposition 13.40(i)); these differ for some $p \nmid L$, since they differ for infinitely many $p$. Eigenvectors of a self-adjoint operator for distinct eigenvalues are orthogonal. The last sentence is (i). □

*In the paper* 13.43. Strong multiplicity one is used in three places, and the Atkin-Lehner decomposition in a fourth. In the proof of Theorem thm:qual (b) and (c): "$g = f'$ by strong multiplicity one", two newforms with the same $a_p$ for $p \nmid qN$ (Theorem 13.42(i)). In the proof of Theorem thm:twist: "a system of $T_p$-eigenvalues ... differing from that of $g$ at infinitely many $p$", so that $\iota(h) \in f^{\perp}$ (Theorem 13.42(iii) and Proposition 13.74). In the set-up before Lemma lem:step1: "strong multiplicity one identifies the forms with the $T_p$-eigenvalues of $g$, $p \nmid N$, with the span of the $g(dz)$, $d \mid N/N_S$", which is Theorem 13.41(ii) with $L = N$, together with (i) to exclude newforms of other levels. The fourth is Lemma lem:old: "orthogonal to the newform $f$ (whether $M < N$, or $M = N$ and $h \neq f$)" is Theorem 13.41(iv).

**Proposition 13.44** (local behaviour of newforms). *Let $f$ be a newform of weight $k$ and level $N$.*

*(i) $f|_k W_N = \varepsilon_N(f) f$ with $\varepsilon_N(f) = \pm 1$. More generally (Atkin-Lehner) $f$ is an eigenvector of every $w_d$, $d \,\|\, N$, with eigenvalue $\varepsilon_d(f) = \pm 1$.*

*(ii) $a_p(f)$ is real for every $p \mid N$; hence all $a_n(f)$ are real.*

*(iii) If $p^2 \mid N$ then $a_p(f) = 0$; if $p \,\|\, N$ then $a_p(f) = -p^{k/2-1}\varepsilon_p(f)$, so in weight 2, $a_p(f) = -\varepsilon_p(f) \in \{\pm 1\}$.*

*(iv) For $f = f_E$, the newform of an elliptic curve of conductor $N$ (Chapter 14), $a_p(f) = 1$, $-1$, $0$ for split multiplicative, nonsplit multiplicative, additive reduction at $p$ (Chapter 9).*

*Part (i) for $w_N$ and part (ii) are proved here given Theorem 13.41; (i) for general $d$ and (iii) are stated with a reference [8], as cited by the paper ("a newform on $\Gamma_0(M)$ with $p^2 \mid M$ has $a_p = 0$ [8, Theorem 3]", proof of Lemma lem:old), and [52, Sections 5.8, 6.6].*

*Proof.* (i) For $p \nmid N$, $w_N T_p w_N^{-1} = [\Gamma \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) \Gamma]_k = T_p$ by the computation in the proof of Proposition 13.35(i) and Proposition 13.7(ii); so $w_N$ commutes with $\mathbb{T}^{(N)}$. By Proposition 13.40(iii) $w_N f$ lies in $S_k^{\mathrm{new}}(N)$, in the eigenspace of $\mathbb{T}^{(N)}$ for the system of $f$, which is $\mathbb{C}f$ by Theorem 13.41(ii). So $w_N f = \varepsilon_N(f) f$, and $\varepsilon_N(f)^2 = 1$ because $w_N$ is an involution. (ii) $a_p\langle f, f\rangle = \langle U_p f, f\rangle = \langle f, w_N U_p w_N f\rangle = \langle f, \varepsilon_N(f)^2 a_p f\rangle = \bar{a}_p \langle f, f\rangle$ by Proposition 13.35(i). The $a_p$, $p \nmid N$, are real by Theorem 13.33, and the recursion and multiplicativity (Proposition 13.19(ii)) give the rest. (iv) is the comparison of Chapter 14 (modularity, $a_p(f_E) = a_p(E)$ at all $p$) with the values of $a_p(E)$ at bad primes in Chapter 9. □

*Sketch of proof of (iii).* For $p \,\|\, N$: the trace of $f$ to level $N/p$ vanishes, since $\langle \mathrm{Tr}\, f, h\rangle_{N/p} = \langle f, h\rangle_N = 0$ for every $h \in S_k(N/p)$ (Lemma 13.34(i), $f$ being new). A coset computation like that of Lemma 13.34 expresses this trace as $\mathrm{Tr}\, f = f + p^{1-k/2} U_p(f|_k W_p)$ ([8], [113]; we do not reproduce the computation); with $f|_k W_p = \varepsilon_p f$ this gives $(1 + p^{1-k/2}\varepsilon_p a_p) f = 0$. Check: $f_{11}$, $p = 11$, $k = 2$: $\varepsilon_{11} = -1$, $a_{11} = 1$. For $p^2 \mid N$ we refer to [8, Theorem 3]. A conceptual reason, through Chapter 15: the local component of $f$ at $p$ has trivial central character and conductor exponent at least 2, so it is a ramified principal series, a ramified twist of the Steinberg representation or supercuspidal, and in each case its local $L$-factor is 1; as $L_p(f, s) = (1 - a_p p^{-s})^{-1}$ (Proposition 13.78), $a_p = 0$. □

Numerical checks of (iii) and (iv) (PARI, `ellap` on Cremona's curves): $a_{11}$(`11a1`) $= 1$ with $\varepsilon_{11} = -1$; $a_{37}$(`37a1`) $= -1$ with $\varepsilon_{37} = +1$ (root number $-\varepsilon_{37} = -1$, rank 1); $a_{11} = 0$ for `121a1`, `121b1`, `121c1`, `121d1`; $a_7 = 0$ for `539c1`, `539d1`; $a_2 = 0$ for `44a1`, `88a1`, `20a1`; $a_7$(`77c1`) $= -1$, $a_{11}$(`77c1`) $= 1$, $a_5$(`20a1`) $= -1$.

**Corollary 13.45** (Galois conjugates of newforms)**.** *For $g \in S_k(N)$ and $\sigma \in \mathrm{Aut}(\mathbb{C})$ let $g^\sigma$ be defined as in Proposition 13.22(ii) (apply $\sigma$ to the coordinates in a basis of forms with rational coefficients; in weight 2, $a_n(g^\sigma) = \sigma(a_n(g))$).*

*(i) $S_k^{\mathrm{old}}(N)$ and $S_k^{\mathrm{new}}(N)$ are stable under $g \mapsto g^\sigma$, and both are spanned by forms with rational coefficients.*

*(ii) If $f$ is a newform of level $N$, so is $f^\sigma$.*

*(iii) If $f$ is a newform, $K_f = \mathbb{Q}(a_n(f) : n \geq 1)$ is totally real.*

*Proof.* We give the proof in weight 2, where $S_2(N; \mathbb{Z})$ is available; the general case is the same with a rational basis of $S_k(N)$ (Chapter 12). (i) Each $S_2(N/p)$ has a basis of forms with integer coefficients, and $V_p$ preserves integrality; so the old subspace is spanned by forms with rational coefficients and is $\sigma$-stable. By Theorem 13.41(iii) and (ii), $S_2(N) = S_2^{\mathrm{old}}(N) \oplus S_2^{\mathrm{new}}(N)$, the systems of $\mathbb{T}^{(N)}$ occurring in the two summands are disjoint, and the new subspace is the sum of the eigenspaces of the systems that do not occur in the old one. The map $g \mapsto g^\sigma$ sends the eigenspace of a system $\psi$ to that of $\sigma \circ \psi$ (as in Proposition 13.22(ii)) and preserves the old subspace; so it permutes the systems occurring in the old subspace, hence also the others, and preserves the new subspace. A subspace $W$ of $\mathbb{C}^d$ (coordinates in a $\mathbb{Z}$-basis of $S_2(N; \mathbb{Z})$) stable under every $\sigma$ has a reduced row echelon basis, unique, hence fixed by every $\sigma$, hence rational. (ii) $f^\sigma$ is a normalised eigenform (Proposition 13.22(ii)) in the new subspace. (iii) An embedding $\tau\colon K_f \to \mathbb{C}$ extends to some $\sigma \in \mathrm{Aut}(\mathbb{C})$, and $\tau(a_n(f)) = a_n(f^\sigma)$ is real by (ii) and Proposition 13.44(ii). □

**Corollary 13.46** (dimensions)**.**

$$\dim S_k(N) = \sum_{M\mid N} \sigma_0(N/M)\dim S_k^{\mathrm{new}}(M), \qquad \dim S_k^{\mathrm{new}}(N) = \sum_{M\mid N} \beta(N/M)\dim S_k(M),$$

*where $\beta = \mu * \mu$ is the multiplicative function with $\beta(p) = -2$, $\beta(p^2) = 1$ and $\beta(p^r) = 0$ for $r \geq 3$ ($\mu$ the Moebius function).*

*Proof.* The first formula counts the basis of Theorem 13.41(iii). It says $D = \sigma_0 * D^{\mathrm{new}} = \mathbf{1}*\mathbf{1}*D^{\mathrm{new}}$ for the arithmetic functions $D(N) = \dim S_k(N)$ and $D^{\mathrm{new}}(N) = \dim S_k^{\mathrm{new}}(N)$; Moebius inversion twice gives $D^{\mathrm{new}} = \mu * \mu * D$ (Chapter 2). $\beta$ is multiplicative, and $\beta(p^r) = \sum_{i+j=r} \mu(p^i)\mu(p^j)$ gives the values. □

**Example 13.47** (dimensions in weight 2)**.** Computed with PARI's `mfdim` for the full and the new space, and checked against Corollary 13.46:

| $N$ | 11 | 22 | 33 | 44 | 77 | 88 | 99 | 121 | 242 | 539 | 980 | 1089 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\dim S_2(N)$ | 1 | 2 | 3 | 4 | 7 | 9 | 9 | 6 | 22 | 49 | 145 | 109 |
| $\dim S_2^{\mathrm{new}}(N)$ | 1 | 0 | 1 | 1 | 5 | 3 | 4 | 4 | 10 | 34 | 13 | 41 |

For instance $S_2(121)$ is spanned by the four rational newforms of the classes 121a to 121d and the two oldforms $f_{11}(z)$, $f_{11}(11z)$; and $S_2(88)$ by the three newforms of level 88, $f_{44}(z)$, $f_{44}(2z)$ and $f_{11}(2^j z)$, $0 \leq j \leq 3$ ($9 = 3 + 2 + 4$). The second formula of Corollary 13.46 can be checked by hand, using $\dim S_2(M) = 0$ for $M \leq 10$ and $\dim S_2(11) = 1$. At $N = 22$ only $M = 11$ and $M = 22$ contribute, and $\dim S_2^{\mathrm{new}}(22) = \beta(2)\cdot 1 + \beta(1)\cdot 2 = -2 + 2 = 0$: the space $S_2(22)$ is entirely old, spanned by $f_{11}(z)$ and $f_{11}(2z)$. At $N = 121$, $\dim S_2^{\mathrm{new}}(121) = \beta(121)\dim S_2(1) + \beta(11)\dim S_2(11) + \beta(1)\dim S_2(121) = 0 - 2 + 6 = 4$. At $N = 44$, the divisors with nonzero contributions are $M = 11, 22, 44$: $\beta(4)\cdot 1 + \beta(2)\cdot 2 + \beta(1)\cdot 4 = 1 - 4 + 4 = 1$.

## 13.7 $U_p$ on oldforms: stabilisation, depletion, non-semisimplicity

This section is the linear algebra behind the proof of Lemma lem:old and Step 1 of the proof of Theorem thm:quant, and behind the depleted forms $f^{\Sigma}$ and the levels $M^{\Sigma}$ of the paper's Section 4.1. The operator $U_p$ acts on the span of the $h(p^j z)$ as a shift with one modified column; its eigenvalues are the two roots of the Hecke polynomial, or $a_p(h)$, and 0; and it is not always semisimple.

**Proposition 13.48** ($U_p$ on the span of $h(p^j z)$)**.** *Let $h$ be a newform of weight $k$ and level $M$, $p$ a prime, $e \geq 0$, and $V_{h,p,e} = \mathrm{span}\{h(p^j z) : 0 \leq j \leq e\} \subseteq S_k(Mp^e)$, of dimension $e+1$ (Theorem 13.41(iii)). Let $U_p = \mathcal{U}_p$ be the operator of a level divisible by $p$ and by $Mp^e$ (assume $e \geq 1$ or $p \mid M$).*

*(i) $U_p h(p^j z) = h(p^{j-1}z)$ for $j \geq 1$; $U_p h = a_p(h)h - p^{k-1}h(pz)$ if $p \nmid M$; $U_p h = a_p(h)h$ if $p \mid M$.*

*(ii) $h(p^e z)$ is a cyclic vector for $U_p$ on $V_{h,p,e}$, so its characteristic and minimal polynomials coincide; they are*

$$X^{e-1}(X^2 - a_p(h)X + p^{k-1}) \quad (p \nmid M,\ e \geq 1), \qquad X^e(X - a_p(h)) \quad (p \mid M).$$

*(iii) Every eigenvalue has a one-dimensional eigenspace, spanned as follows. For $p \nmid M$, writing $X^2 - a_pX + p^{k-1} = (X - \alpha_p)(X - \beta_p)$: the* $p$-stabilisations *$h_\alpha = h - \beta_p h(pz)$ (eigenvalue $\alpha_p$) and $h_\beta = h - \alpha_p h(pz)$ (eigenvalue $\beta_p$), and, if $e \geq 2$, the* $p$-depleted form *$h^{\{p\}} = h - a_p h(pz) + p^{k-1}h(p^2z)$ (eigenvalue 0). For $p \parallel M$: $h$ (eigenvalue $a_p$, nonzero by Proposition 13.44(iii)) and, if $e \geq 1$, $h^{\{p\}} = h - a_p h(pz)$ (eigenvalue 0). For $p^2 \mid M$: $h = h^{\{p\}}$ (eigenvalue $a_p = 0$).*

*(iv) $U_p$ is semisimple on $V_{h,p,e}$ iff its characteristic polynomial is squarefree: iff $e \leq 2$ and $\alpha_p \neq \beta_p$ ($p \nmid M$); $e \leq 1$ ($p \parallel M$); $e = 0$ ($p^2 \mid M$). Otherwise the eigenvalue 0 has a single Jordan block, of size $e-1$, $e$ or $e+1$ respectively (or, if $p \nmid M$, $e \leq 2$ and $\alpha_p = \beta_p$, the eigenvalue $\alpha_p$ has a block of size 2).*

*(v) Each eigenvector in (iii) is a normalised eigenform of $\mathbb{T}_{Mp^e}$, with $U_p$-eigenvalue as stated and the eigenvalues of $h$ for the other $T_n$.*

*Proof.* (i) $\mathcal{U}_pV_p = 1$ (Lemma 13.9) gives the first formula; $\mathcal{U}_ph = T_p^{(M)}h - p^{k-1}V_ph$ for $p \nmid M$ and $\mathcal{U}_ph = U_p^{(M)}h$ for $p \mid M$ (Proposition 13.8). (ii) By (i), $U_p^j h(p^e z) = h(p^{e-j}z)$ for $0 \leq j \leq e$, so the iterates of $h(p^ez)$ span $V_{h,p,e}$: it is cyclic, and the minimal polynomial has degree $e+1$, hence equals the characteristic polynomial (Chapter 3). To find it, write $c = p^{k-1}$ and $v = h(p^ez)$, so $U_p^j v = h(p^{e-j}z)$ for $0 \leq j \leq e$. One more application of (i) gives, for $p \nmid M$ and $e \geq 1$,

$$U_p^{e+1}v = U_ph = a_ph - c\,h(pz) = a_pU_p^ev - c\,U_p^{e-1}v,$$

so $(U_p^{e+1} - a_pU_p^e + cU_p^{e-1})v = 0$; the monic polynomial $X^{e-1}(X^2 - a_pX + c)$ of degree $e+1$ kills the cyclic vector $v$, hence kills $V_{h,p,e}$ (it commutes with $U_p$), and being of the same degree as the minimal polynomial it is the minimal polynomial. For $p \mid M$, $U_p^{e+1}v = U_ph = a_pU_p^ev$ gives $X^e(X - a_p)$ in the same way. (In the basis $v_j = h(p^jz)$ the matrix is the shift $v_j \mapsto v_{j-1}$ with first column $(a_p, -p^{k-1}, 0, \dots)$ or $(a_p, 0, \dots)$, a companion matrix.) (iii) For a cyclic operator every eigenvalue has a single Jordan block (a second block would make the minimal polynomial of smaller degree than the characteristic polynomial), so a one-dimensional eigenspace, and the size of the block is the multiplicity of the eigenvalue as a root of the characteristic polynomial. The vectors are eigenvectors by (i): $U_p(h - \beta h(pz)) = (a_p - \beta)h - p^{k-1}h(pz) = \alpha(h - \beta h(pz))$ as $\alpha + \beta = a_p$ and $\alpha\beta = p^{k-1}$; $U_p(h - a_ph(pz) + p^{k-1}h(p^2z)) = a_ph - p^{k-1}h(pz) - a_ph + p^{k-1}h(pz) = 0$; $U_p(h - a_ph(pz)) = a_ph - a_ph = 0$ when $p \mid M$. (iv) A cyclic operator is semisimple iff its minimal polynomial, here equal to the characteristic polynomial, is squarefree. The root 0 of $X^{e-1}(X^2 - a_pX + p^{k-1})$ has multiplicity $e-1$ (the quadratic factor has constant term $p^{k-1} \neq 0$), that of $X^e(X - a_p)$ has multiplicity $e$ when $a_p \neq 0$ ($p \parallel M$) and $e+1$ when $a_p = 0$ ($p^2 \mid M$); this gives the conditions and the block sizes, by (iii). (v) The $T_n$ with $(n, Mp) = 1$ and the $U_{p'}$ with $p' \mid M$, $p' \neq p$, commute with $V_p$ (Lemma 13.9, Proposition 13.40(i)), so they act on $V_{h,p,e}$ by the eigenvalues of $h$; each listed vector has $a_1 = 1$ since $h(p^jz)$, $j \geq 1$, has $a_1 = 0$; and $T_{p^r} = U_p^r$ at a level divisible by $p$. □

**Example 13.49** (levels 22, 44, 88, 121)**.** For $h = f_{11}$ ($a_2 = -2$, $a_{11} = 1$) the matrices of $U_p$ on $V_{h,p,e}$, columns the images of $h(p^jz)$, are: at level 22, $\left(\begin{smallmatrix} -2 & 1 \\ -2 & 0 \end{smallmatrix}\right)$, characteristic polynomial $X^2 + 2X + 2$, $\alpha, \beta = -1 \pm i$; at level 44,

$$\begin{pmatrix} -2 & 1 & 0 \\ -2 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

characteristic polynomial $X(X^2+2X+2)$, with $f_{11}^{\{2\}} = f_{11} + 2f_{11}(2z) + 2f_{11}(4z)$ spanning the kernel (a vector $(x_0, x_1, x_2)$ is sent to $(-2x_0+x_1,\ -2x_0+x_2,\ 0)$, which vanishes iff $x_1 = x_2 = 2x_0$); at level 88,

$$\begin{pmatrix} -2 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad X^2(X^2+2X+2),$$

with a Jordan block of size 2 at 0; at level 121 ($p = 11 \,\|\, 11$), $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ ($U_{11}f_{11} = a_{11}f_{11} = f_{11}$ and $U_{11}f_{11}(11z) = f_{11}$), characteristic polynomial $X(X-1)$, eigenvectors $f_{11}$ and $f_{11} - f_{11}(11z)$. On the whole space, PARI's `mfheckemat` gives for $U_2$ on $S_2(22)$, $S_2(44)$, $S_2(88)$ the characteristic polynomials $X^2+2X+2$, $X^2(X^2+2X+2)$, $X^7(X^2+2X+2)$ and the minimal polynomials $X^2+2X+2$, $X(X^2+2X+2)$, $X^2(X^2+2X+2)$.

*In the paper* 13.50. The proof of Lemma lem:old asserts: "The eigenvalues of $U_p$ there are therefore the roots $\alpha_p, \beta_p$ of $X^2 - a_p(h)X + p$, together with 0 if $e_p \geq 2$, when $p \nmid M$; and $a_p(h)$, together with 0 if $e_p \geq 1$, when $p \mid M$"; "The operator $U_p$ is not semisimple on this span when 0 is a repeated eigenvalue, but each listed eigenvalue has an eigenvector"; and lists the stabilisations and depleted forms. All of this is Proposition 13.48(ii) to (iv) with $k = 2$. (When $p^2 \mid M$ the paper's "$a_p(h)$, together with 0" is the single eigenvalue 0, since $a_p(h) = 0$.) The eigenvalues $\alpha_p, \beta_p$ are those of the arithmetic Frobenius on $V_\ell$ of an elliptic curve with newform $h$, by the convention of Appendix B (Appendix B).

**Corollary 13.51** (eigenvalues of $U_p$ in weight 2)**.** *Let $p \mid N$. Every eigenvalue of $U_p$ on $S_2(N)$ is 0, or $\pm 1$, or a root of $X^2 - a_p(h)X + p$ for a newform $h$ of level $M \mid N$ with $p \nmid M$. For every prime $\lambda$ of $\overline{\mathbb{Q}}$ not above $p$, every nonzero eigenvalue of $U_p$ is a $\lambda$-adic unit. Hence a generalised eigenform of $\mathbb{T}$ whose $U_p$-eigenvalue $\mu$ satisfies $\mu \equiv 0 \pmod{\lambda}$ has $\mu = 0$.*

*Proof.* By Theorem 13.41(iii), $S_2(N) = \bigoplus_h V_h$, and on $V_h = \mathrm{span}\{h(dz)\}$ the operator $U_p$ acts through the factor $V_{h,p,e_p}$, $e_p = v_p(N/M_h)$ (write $d = p^j d'$ and use $\mathcal{U}_p V_{d'} = V_{d'}\mathcal{U}_p$). Its eigenvalues there are listed in Proposition 13.48: 0, $a_p(h) = \pm 1$ for $p \,\|\, M_h$ (Proposition 13.44(iii)), or the roots $\alpha, \beta$. These roots are algebraic integers with $\alpha\beta = p$, so at $\lambda \nmid p$ both are units; and $\pm 1$ are units. A nonzero eigenvalue is therefore $\not\equiv 0 \pmod{\lambda}$. □

The characteristic polynomials of $U_p$ on the whole space at the levels of the paper (PARI's `mfheckemat`):

- level 88, $U_2$: $X^7(X^2+2X+2)$; minimal polynomial $X^2(X^2+2X+2)$;
- level 99, $U_3$: $X^6(X+1)(X^2+X+3)$; minimal polynomial $X(X+1)(X^2+X+3)$;
- level 121, $U_{11}$: $X^5(X-1)$; minimal polynomial $X(X-1)$;
- level 539, $U_7$: $X^{42}(X-1)^3(X+1)^2(X^2+2X+7)$; minimal polynomial $X(X-1)(X+1)(X^2+2X+7)$;
- level 980, $U_7$: $X^{126}(X-1)^8(X+1)^9(X^2-2X+7)$; minimal polynomial $X(X-1)(X+1)(X^2-2X+7)$;
- level 1089, $U_3$: $X^{76}(X-1)^{10}(X+1)^{11}(X^2-2X+3)^2(X^2+X+3)^4$; minimal polynomial $X(X-1)(X+1)(X^2-2X+3)(X^2+X+3)$.

*In the paper* 13.52. In the proof of Lemma lem:step1 (the inequality $\geq$ for $\ell \notin S$), an eigencharacter $\phi$ congruent to $\pi$ has $\phi(U_q) \equiv 0$ for $q \in S$, and then $\phi(U_q) = 0$, because "Every eigenvalue of $U_q$ on $S_2(\Gamma_0(N))$ is 0, $\pm 1$ or a root of $X^2 - a_q(h_0)X + q$ for a newform $h_0$, hence 0 or a unit at every prime not above $q$; as $\ell \neq q$, $\phi(U_q) = 0$". The passage from $\phi(U_q) \equiv 0$ to $\phi(U_q) = 0$ is Corollary 13.51.

*Remark* 13.53 (non-semisimplicity and multiplicity). (i) At level 88 the minimal polynomial of $U_2$ is $X^2(X^2 + 2X + 2)$: $U_2$ is not semisimple, and $U_2(U_2^2 + 2U_2 + 2)$ is a nonzero nilpotent element of $\mathbb{T}_{88} \otimes \mathbb{Q}$. So the full Hecke algebra need not be reduced, and the step "$\wp \cap I = 0$" in the proof of the paper's Lemma lem:ddt417 must be proved, not deduced from reducedness (as the paper does, with this example, Chapter 20). The Jordan block comes from $V_{f_{11},2,3}$ ($e = 3$, Proposition 13.48(iv)); on $V_{f_{44},2,1}$ ($2^2 \mid 44$) $U_2$ is nilpotent of order 2 as well. (ii) A large multiplicity of the eigenvalue 0 does not by itself make $U_p$ non-semisimple: at level 539 the eigenvalue 0 of $U_7$ has multiplicity 42, but the minimal polynomial is squarefree, because no single $V_{h,7,e}$ has a repeated root ($539 = 7^2 \cdot 11$, so $e \leq 2$ for $7 \nmid M_h$, $e \leq 1$ for $7 \parallel M_h$, $e = 0$ for $7^2 \mid M_h$). The paper's "the eigenvalue 0 with multiplicity one" (Step 1 of the proof of Theorem thm:quant) is the algebraic multiplicity on the span of the $g(dz)$, not on all of $S_2(\Gamma_0(N))$. (iii) (Coleman-Edixhoven.) For a newform of weight 2 and level prime to $p$, $\alpha_p \neq \beta_p$ [32]; so $U_p$ is semisimple on the $p$-old part of level $Mp$. For rational $a_p$ this is elementary: $\alpha_p = \beta_p$ means that the discriminant $a_p^2 - 4p$ of $X^2 - a_pX + p$ vanishes, so $a_p^2 = 4p$, which is impossible for an integer $a_p$ because $4p$ is not a square ($v_p(4p) = 1$ for odd $p$, and $v_2(8) = 3$ for $p = 2$).

**Definition 13.54** (depletion and the level $M^\Sigma$)**.** Let $f$ be a newform of weight $k$ and level $M$, and $\Sigma$ a finite set of primes. For a prime $p$ put $\delta_p = \delta_p(f) = 2$ if $p \nmid M$, 1 if $p \parallel M$, and 0 if $p^2 \mid M$. The $\Sigma$-*depleted form* is $f^\Sigma = \sum_{(n,\Sigma)=1} a_n(f)q^n$, and $M^\Sigma = M \prod_{p \in \Sigma} p^{\delta_p}$.

**Proposition 13.55** (depleted forms)**.** *In the notation of Definition 13.54:*

*(i)* $\delta_p$ *is the number of roots of the Hecke polynomial at* $p$*, that is, the degree in* $p^{-s}$ *of* $L_p(f,s)^{-1}$ *(Proposition 13.78); for* $f = f_E$ *in weight* 2, $\delta_p = 2, 1, 0$ *for good, multiplicative and additive reduction, and* $\delta_p = \dim V_\ell E^{I_p}$ *for* $p \neq \ell$ *(Chapter 11), the paper's definition in Section 1.1.*

*(ii)* $f^\Sigma = (\prod_{p \in \Sigma} D_p) f$ *with* $D_p = 1 - a_pV_p + p^{k-1}V_p^2$ *(*$p \nmid M$*),* $D_p = 1 - a_pV_p$ *(*$p \parallel M$*),* $D_p = 1$ *(*$p^2 \mid M$*).*

*(iii)* $f^\Sigma \in S_k(M^\Sigma)$*, and* $M^\Sigma$ *is the least level* $L$ *with* $f^\Sigma \in S_k(L)$*.*

*(iv)* $f^\Sigma$ *is a normalised eigenform of the full Hecke algebra of level* $M^\Sigma$*, with* $U_pf^\Sigma = 0$ *for* $p \in \Sigma$ *and the eigenvalues of* $f$ *at the other* $T_n$*, and its generalised eigenspace in* $S_k(M^\Sigma)$ *is the line* $\mathbb{C}f^\Sigma$*.*

*(v) At a level* $N$ *with* $M^\Sigma \mid N$ *such that every prime dividing* $N$ *divides* $M^\Sigma$*,* $f^\Sigma$ *is still a normalised eigenform of* $\mathbb{T}_N$*, but its generalised eigenspace can be larger. Without the condition on the primes of* $N$ *it need not be an eigenform of* $\mathbb{T}_N$*.*

*(vi)* $L(f^\Sigma, s) = L(f,s) \prod_{p \in \Sigma} L_p(f,s)^{-1}$*.*

*Proof.* (i) is Proposition 13.78 and Proposition 13.44(iii),(iv); the Galois description is Chapter 11. (ii) The operators $D_p$ for different $p$ commute (Lemma 13.9), and $D_p$ only changes the coefficients at multiples of $p$. The $n$-th coefficient of $D_pf$, for $n = p^rm$ with $p \nmid m$, is $a_m$ times $a_{p^r} - a_pa_{p^{r-1}} + p^{k-1}a_{p^{r-2}}$ (good case), $a_{p^r} - a_pa_{p^{r-1}}$ (case $p \parallel M$) or $a_{p^r}$ (case $p^2 \mid M$), by multiplicativity, with

$a_{p^{-1}} = a_{p^{-2}} = 0$; for $r \geq 1$ this vanishes by Proposition 13.19(ii) and $a_p = 0$ when $p^2 \mid M$, and for $r = 0$ it is $a_m$. (iii) By (ii), $f^\Sigma$ is a combination of the $f(dz)$ with $d \mid \prod_{p\in\Sigma} p^{\delta_p}$, so it lies in $S_k(M^\Sigma)$. Its coefficient on $f(d_0 z)$, $d_0 = \prod_{p\in\Sigma} p^{\delta_p}$, is the product of the leading coefficients of the $D_p$, namely $p^{k-1}$ or $-a_p$, nonzero. If $f^\Sigma \in S_k(L)$, compare its expansions in the bases of Theorem 13.41(iii) at the levels $L$ and $\operatorname{lcm}(L, M^\Sigma)$: the basis at level $L$ is part of the basis at level $\operatorname{lcm}(L, M^\Sigma)$, and coordinates are unique, so $f(d_0 z)$ occurs at level $L$, that is, $Md_0 \mid L$. (iv) The coefficients of $f^\Sigma$ are multiplicative, vanish at $p^r$ ($r \geq 1$, $p \in \Sigma$), so that $b_{p^r} = 0^r$, and agree with those of $f$ elsewhere; by Lemma 13.20 at the level $M^\Sigma$, $f^\Sigma$ is a normalised eigenform with the stated eigenvalues. A generalised eigenvector for its eigencharacter is in particular an eigenvector of $\mathbb{T}^{(M^\Sigma)}$ for the system of $f$ (Theorem 13.33), so it lies in $V_f = \operatorname{span}\{f(dz) : d \mid d_0\}$ (Theorem 13.41(ii)). Writing $d = \prod_p p^{j_p}$, the map $\bigotimes_{p\in\Sigma} V_{f,p,\delta_p} \to V_f$, $\otimes_p h(p^{j_p} z) \mapsto f(dz)$ (with $h = f$ in each factor) is an isomorphism under which $U_p$ acts on the $p$-th factor (because $\mathcal{U}_p V_{d'} = V_{d'} \mathcal{U}_p$ for $(d', p) = 1$). On $V_{f,p,\delta_p}$ the characteristic polynomial of $U_p$ is $X(X^2 - a_p X + p^{k-1})$, $X(X - a_p)$ with $a_p \neq 0$, or $X$, so 0 is a simple root; the joint generalised eigenspace for the eigenvalues 0 is the tensor product of these lines, a line. (v) Every prime $p \mid N$ divides $M^\Sigma$, so either $p \in \Sigma$, where $b_{p^r} = 0 = b_p^r$, or $p \mid M$, $p \notin \Sigma$, where $b_{p^r} = a_{p^r}(f) = a_p(f)^r$ (Proposition 13.19(ii) at level $M$); the primes $p \nmid N$ satisfy the recursion of $f$. So the eigenform property is Lemma 13.20 at level $N$. The condition is needed: for $f = f_{11}$, $\Sigma = \{2\}$ and $N = 132 = 3 \cdot 44$, the coefficients of $f^\Sigma$ at 3 and 9 are $a_3 = -1$ and $a_9 = a_3^2 - 3 = -2 \neq a_3^2$, so $f^\Sigma$ is not an eigenvector of $U_3$ at level 132 (Lemma 13.20). For the second statement see Example 13.83: $f_{11}^{\{2\}}$ spans its generalised eigenspace at level $44 = M^{\{2\}}$, but lies in a Jordan block of $U_2$ at level 88. (vi) is Proposition 13.78 applied to $f^\Sigma$ and $f$. □

*Remark* 13.56 (why $\delta_p$). Depletion at $p$ removes the Euler factor at $p$ (Proposition 13.55(vi)), and each root of the Hecke polynomial removed costs one factor of $p$ in the level: $D_p$ is a polynomial of degree $\delta_p$ in $V_p$. In the language of Proposition 13.48, the eigenvalue 0 of $U_p$ appears in $V_{f,p,e}$ exactly when $e \geq \delta_p$, and with multiplicity one when $e = \delta_p$. For an additive prime nothing is removed and nothing is added: this is the paper's "adding an additive prime to $\Sigma$ changes neither the level nor the ideal" (paragraph after Theorem thm:dfg14).

**Example 13.57** (depleted forms)**.** $f_{11}^{\{2\}} = f_{11} + 2f_{11}(2z) + 2f_{11}(4z)$, of level 44 ($a_2 = -2$; checked to $n = 4000$); $f_{11}^{\{11\}} = f_{11} - f_{11}(11z)$, of level 121 ($a_{11} = 1$); for $f$ of class 121a, $\Sigma = \{11\}$: $f^\Sigma = f$, level 121 ($\delta_{11} = 0$); for $g$ of class 77c and $\Sigma = \{7\}$: $g^\Sigma = g - a_7(g)g(7z) = g + g(7z)$, level 539; for $g = f_{11}$ and $\Sigma = \{7\}$: $g^\Sigma = g + 2g(7z) + 7g(49z)$, level 539 ($a_7(f_{11}) = -2$). Each is read off from Proposition 13.55(ii). For $f_{11}$ at $p = 2 \nmid 11$, $D_2 = 1 - a_2 V_2 + 2V_2^2 = 1 + 2V_2 + 2V_2^2$ and $M^{\{2\}} = 11 \cdot 2^2$. At $p = 11 \parallel 11$, $D_{11} = 1 - a_{11} V_{11} = 1 - V_{11}$ and $M^{\{11\}} = 11 \cdot 11$. For 121a, $11^2 \mid 121$, so $D_{11} = 1$ and nothing changes. For 77c at $7 \parallel 77$, $D_7 = 1 - a_7 V_7 = 1 + V_7$ and $M^{\{7\}} = 77 \cdot 7$. For $f_{11}$ at $7 \nmid 11$, $D_7 = 1 + 2V_7 + 7V_7^2$ and $M^{\{7\}} = 11 \cdot 7^2$. As a check on the first one, the coefficient of $q^2$ in $f_{11} + 2f_{11}(2z) + 2f_{11}(4z)$ is $a_2 + 2a_1 = -2 + 2 = 0$, and that of $q^4$ is $a_4 + 2a_2 + 2a_1 = 2 - 4 + 2 = 0$.

*In the paper* 13.58. In Section 4.1 of the paper, "Put $M^\Sigma = M \prod_{p\in\Sigma} p^{\delta_p}$ and let $f^\Sigma = \sum_{(n,\Sigma)=1} a_n(f) q^n$ be the $\Sigma$-depleted eigenform, of level $M^\Sigma$" is Definition 13.54 and Proposition 13.55(iii), (iv). The set-up before Lemma lem:step1: "$N_S \prod_{q\in S} q^{\delta_q} = N$: the $S$-depleted form $g^S$ lives at level $N$", and "on that span the operators $U_q$, $q \in S$, have the eigenvalue 0 with multiplicity one", is Proposition 13.55(iv) for $g$ at level $N = N_S^S$.

## 13.8 The structure of $\mathbb{T}$, generalised eigenspaces and congruences

This section assembles the previous ones into a description of $\mathbb{T}\otimes\overline{\mathbb{Q}}$ and of the generalised eigenspaces of $\mathbb{T}$. It proves the decomposition $\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$ that the proof of Lemma lem:crit writes down without proof; it separates the $\mathbb{T}$-stable complement of an eigenform from its Petersson complement, the distinction on which Step 1 of the proof of Theorem thm:quant depends, and proves the congruence-module statement of Lemma lem:crit for every rational eigencharacter whose generalised eigenspace is a line; it describes the local components $\mathbb{T}_{\mathfrak{m}}$ of $\mathbb{T}_\ell$ and their effect on forms; and it explains how congruences at primes become congruences at all $n$.

**Theorem 13.59** (the structure of $\mathbb{T}$)**.** *Let $k$ be even and $N\geq 1$. For a newform $h$ of level $M_h\mid N$ put $e_p=e_p(h)=v_p(N/M_h)$ and $V_h=\operatorname{span}\{h(dz):d\mid N/M_h\}$.*

*(i) $S_k(N)=\bigoplus_h V_h$, the sum over the newforms $h$ of levels dividing $N$; $V_h$ is the eigenspace of $\mathbb{T}^{(N)}$ for the system of $h$, and as a module over $\mathbb{C}[U_p:p\mid N]$, $V_h\simeq\bigotimes_{p\mid N}V_{h,p,e_p}$, with $U_p$ acting on the $p$-th factor.*

*(ii) Let $m_{h,p}$ be the characteristic polynomial of $U_p$ on $V_{h,p,e_p}$ (Proposition 13.48(ii); it is $X-a_p(h)$ when $e_p=0$). Then*

$$\mathbb{T}\otimes\mathbb{C}\simeq\prod_h A_h,\qquad A_h\simeq\bigotimes_{p\mid N}\mathbb{C}[X]/(m_{h,p}(X)),$$

*where $A_h$ is the image of $\mathbb{T}\otimes\mathbb{C}$ in $\operatorname{End}(V_h)$; $\dim A_h=\dim V_h$, $\operatorname{rank}_{\mathbb{Z}}\mathbb{T}=\dim S_k(N)$, and $S_k(N)$ is free of rank one over $\mathbb{T}\otimes\mathbb{C}$.*

*(iii) The generalised eigenspaces of $\mathbb{T}$ in $S_k(N)$ are the joint generalised eigenspaces $V_h(\mu)$, $\mu=(\mu_p)_{p\mid N}$ a choice of a root $\mu_p$ of each $m_{h,p}$; $V_h(\mu)$ has dimension the product of the multiplicities of the $\mu_p$. $\mathbb{T}\otimes\mathbb{Q}$ is reduced iff every $m_{h,p}$ is squarefree.*

*Proof.* (i) is Theorem 13.41(iii),(iv); the tensor structure is the argument in the proof of Proposition 13.55(iv): $f(dz)\leftrightarrow\otimes_p h(p^{v_p(d)}z)$, and $\mathcal{U}_p$ commutes with $V_{d'}$ for $(d',p)=1$. (ii) On $V_h$ every $T_n$ with $(n,N)=1$ is a scalar, and every $T_n$ is a product of such an operator and powers of the $U_p$ (Theorem 13.10); so $A_h$ is the algebra generated by the $U_p$ on $\bigotimes_p V_{h,p,e_p}$. Each $U_p$ has a cyclic vector on its factor, so the algebra it generates there is $\mathbb{C}[X]/(m_{h,p})$ (Chapter 3); and the map $\bigotimes_p\mathbb{C}[X]/(m_{h,p})\to\bigotimes_p\operatorname{End}(V_{h,p,e_p})=\operatorname{End}(V_h)$ is injective, a tensor product of injective linear maps over a field. Hence $A_h\simeq\bigotimes_p\mathbb{C}[X]/(m_{h,p})$, of dimension $\prod_p\deg m_{h,p}=\prod_p(e_p+1)=\dim V_h$. The map $\mathbb{T}\otimes\mathbb{C}\to\prod_h A_h$ is injective because $\mathbb{T}\otimes\mathbb{C}$ acts faithfully on $S_k(N)=\bigoplus V_h$ ($\mathbb{T}$ preserves the lattice $S_k(N;\mathbb{Z})$ of forms with integer coefficients by Theorem 13.10(iii), which spans $S_k(N)$ in every even weight by the integral basis theorem of Chapter 12; so $\mathbb{T}\subseteq\operatorname{End}_{\mathbb{Z}}S_k(N;\mathbb{Z})$, and tensoring this inclusion with the flat $\mathbb{Z}$-module $\mathbb{C}$ gives $\mathbb{T}\otimes\mathbb{C}\subseteq\operatorname{End}_{\mathbb{C}}S_k(N)$). It is surjective: the systems of the $V_h$ are pairwise distinct (Theorem 13.41(ii)), so by Lagrange interpolation in finitely many $T_n$, $(n,N)=1$, which act semisimply (Theorem 13.33), there are idempotents $e_h\in\mathbb{T}^{(N)}\otimes\mathbb{C}$ acting as 1 on $V_h$ and 0 on the other $V_{h'}$; then $e_h(\mathbb{T}\otimes\mathbb{C})$ maps onto $A_h$ and to 0 in the other factors. The dimension count gives $\operatorname{rank}\mathbb{T}=\sum_h\dim V_h=\dim S_k(N)$ (for $k=2$ this is also Theorem 13.24). Each $V_h$ is cyclic over $A_h$, generated by $h((N/M_h)z)$ (a tensor product of cyclic vectors), so $S_k(N)=\bigoplus V_h$ is cyclic over $\prod A_h$, hence free of rank one by the dimension count. (iii) The generalised eigenspaces of the commutative algebra $A_h$ acting on $\bigotimes_p V_{h,p,e_p}$

are the tensor products of the generalised eigenspaces of the $U_p$; the algebra $\bigotimes_p \mathbb{C}[X]/(m_{h,p})$ is reduced iff each factor is, iff each $m_{h,p}$ is squarefree; and $\mathbb{T}\otimes\mathbb{Q}$ is reduced iff $\mathbb{T}\otimes\mathbb{C}$ is (Chapter 3: a finite algebra over a perfect field is reduced iff it stays reduced after extension of scalars). □

**Theorem 13.60** (the newform splitting)**.** *Let $f \in S_2(N)$ be a newform with rational coefficients, $\phi_f\colon \mathbb{T}\to\mathbb{Z}$ its eigencharacter, and $f^\perp$ its Petersson orthogonal complement.*

*(i) $f^\perp$ is the sum of the eigenspaces of $\mathbb{T}^{(N)}$ other than the line $\mathbb{C}f$; it is $\mathbb{T}$-stable and spanned by forms with rational coefficients.*

*(ii) There is $u\in\mathbb{T}$ with $uf^\perp = 0$ and $\phi_f(u)\neq 0$; then $e = u/\phi_f(u)\in\mathbb{T}\otimes\mathbb{Q}$ is an idempotent, acting as 1 on $f$ and 0 on $f^\perp$.*

*(iii) Let $\mathbb{T}'$ be the image of $\mathbb{T}\otimes\mathbb{Q}$ in $\mathrm{End}(f^\perp)$. Then $t\mapsto(\phi_f(t), t|_{f^\perp})$ is an isomorphism of $\mathbb{Q}$-algebras $\mathbb{T}\otimes\mathbb{Q}\simeq\mathbb{Q}\times\mathbb{T}'$, and $\mathbb{T}'$ acts faithfully on $f^\perp$.*

*(iv) With $I=\ker\phi_f$ and $J=\mathrm{Ann}_{\mathbb{T}}(f^\perp)$: $I=\mathbb{T}\cap(0\times\mathbb{T}')$, $J=\mathbb{T}\cap(\mathbb{Q}\times 0)$, $I\cap J=0$, $\phi_f(J)\neq 0$, and $\mathbb{T}/(I+J)\simeq\mathbb{Z}/\phi_f(J)$ is finite cyclic.*

*Proof.* (i) By Theorem 13.33, $S_2(N)$ is the orthogonal sum of the eigenspaces of $\mathbb{T}^{(N)}$, and the eigenspace of the system of $f$ is $\mathbb{C}f$ (Theorem 13.41(ii) with $L=M=N$). So $f^\perp$ is the sum of the others. Each eigenspace of $\mathbb{T}^{(N)}$ is $\mathbb{T}$-stable, as $\mathbb{T}$ is commutative. Rationality follows from (ii): $f^\perp=\ker e$ with $e\in\mathbb{T}\otimes\mathbb{Q}$, and $\mathbb{T}\otimes\mathbb{Q}$ preserves $S_2(N;\mathbb{Q})$.

(ii) Let $\psi_1,\dots,\psi_s$ be the systems of $\mathbb{T}^{(N)}$ occurring in $f^\perp$, all different from the system of $f$ on $\mathbb{T}^{(N)}$. For each $i$ choose a prime $p_i\nmid N$ with $\psi_i(T_{p_i})\neq a_{p_i}(f)$: two systems that agree on all $T_p$, $p\nmid N$, agree on $\mathbb{T}^{(N)}$ by Proposition 13.16(ii) and Proposition 13.19(ii). Let $P_i\in\mathbb{Z}[X]$ be a nonzero integer multiple of the minimal polynomial of the algebraic integer $\psi_i(T_{p_i})$ over $\mathbb{Q}$, and $u=\prod_i P_i(T_{p_i})\in\mathbb{T}$. On the eigenspace of $\psi_i$, $T_{p_i}$ is the scalar $\psi_i(T_{p_i})$, so $P_i(T_{p_i})$ kills it; the factors commute, so $u$ kills $f^\perp$. And $\phi_f(u)=\prod_i P_i(a_{p_i}(f))\neq 0$: $P_i$ is irreducible up to a constant, and a root $a_{p_i}(f)\in\mathbb{Q}$ would force $P_i$ to be linear with root $\psi_i(T_{p_i})=a_{p_i}(f)$. Then $e=u/\phi_f(u)$ acts as 1 on $f$ and 0 on $f^\perp$; as $S_2(N)=\mathbb{C}f\oplus f^\perp$, $e^2=e$ on $S_2(N)$, hence in $\mathbb{T}\otimes\mathbb{Q}$ (faithfulness).

(iii) $\mathbb{T}\otimes\mathbb{Q}=e(\mathbb{T}\otimes\mathbb{Q})\times(1-e)(\mathbb{T}\otimes\mathbb{Q})$. For $t\in\mathbb{T}\otimes\mathbb{Q}$, $te-\phi_f(t)e$ kills $f$ and $f^\perp$, so $te=\phi_f(t)e$ and $e(\mathbb{T}\otimes\mathbb{Q})=\mathbb{Q}e\simeq\mathbb{Q}$ via $\phi_f$. The element $1-e$ acts as 0 on $f$ and 1 on $f^\perp$, so $(1-e)(\mathbb{T}\otimes\mathbb{Q})$ maps isomorphically onto $\mathbb{T}'$ (injective because an element of it acting by 0 on $f^\perp$ also kills $f$, hence is 0). The resulting isomorphism is $t\mapsto(\phi_f(t),t|_{f^\perp})$.

(iv) $\mathbb{T}$ is torsion-free, so $\mathbb{T}\subseteq\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$. An element $t$ lies in $I$ iff its first coordinate vanishes, and in $J$ iff its second does; so $I\cap J=0$. As $u\in J$, $\phi_f(J)\neq 0$. The homomorphism $\phi_f$ is onto $\mathbb{Z}$ ($\phi_f(T_1)=1$), with kernel $I$, so $\mathbb{T}/(I+J)\simeq\mathbb{Z}/\phi_f(J)$, finite cyclic. □

*In the paper* 13.61. The proof of Lemma lem:crit begins: "Writing $\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$ with $\mathbb{T}'$ acting faithfully on $f^\perp$ (the $f$-line is $\mathbb{T}$-stable, and so is $f^\perp$, because $f$ is an eigenvector of the adjoints of the $T_n$ as well), one has $I=\mathbb{T}\cap(0\times\mathbb{T}')$ and $J=\mathbb{T}\cap(\mathbb{Q}\times 0)$, so $I\cap J=0$ and $\mathbb{T}/(I+J)\simeq\mathbb{Z}/\phi_f(J)$ is cyclic". Every clause is Theorem 13.60. The input that the paper leaves implicit is multiplicity one for the newform $f$ (Theorem 13.41(ii)): without it the local factor of $\mathbb{T}\otimes\mathbb{Q}$ through which $\phi_f$ factors could be larger than $\mathbb{Q}$, as happens for the depleted form $f_{11}^{\{2\}}$ at level 88 (Example 13.83).

**Proposition 13.62** (𝕋-stable versus Petersson complements)**.** *Let $g \in S_k(N)$ be a normalised eigenform of $\mathbb{T} = \mathbb{T}_N$ with eigencharacter $\pi$ whose generalised eigenspace is the line $\mathbb{C}g$ (a newform of level $N$; or $f^\Sigma$ at level $M^\Sigma$, Proposition 13.55(iv)). Let $g^c$ be the $\mathbb{T}$-stable complement of $\mathbb{C}g$, the sum of the other generalised eigenspaces.*

*(i) $S_k(N) = \mathbb{C}g \oplus g^c$, and $\mathbb{T} \otimes \mathbb{C} = \mathbb{C} \times \mathbb{T}''$ with $\mathbb{T}''$ acting faithfully on $g^c$. If $k = 2$ and $\pi$ takes values in $\mathbb{Q}$ (so that $g$ has rational coefficients), the idempotent of the first factor lies in $\mathbb{T} \otimes \mathbb{Q}$, $g^c$ is spanned by forms with rational coefficients, and $g^c \cap S_2(N;\mathbb{Z})$ is saturated of rank $d-1$. (Rationality of $\pi$ alone does not force $k = 2$: $\Delta$ has rational eigenvalues. The integral consequences are Proposition 13.64.)*

*(ii) If $g$ is a newform of level $N$, $g^c = g^\perp$.*

*(iii) If $g$ is not a newform, $g^c \neq g^\perp$ in general: for $g = f_{11}^{\{2\}}$ at level 44, the stabilisation $h = f_{11} - \beta f_{11}(2z)$ lies in $g^c$ and $\langle g, h\rangle = \frac{5}{12}\langle f_{11}, f_{11}\rangle_{44} \neq 0$.*

*Proof.* (i) The decomposition of a finite commutative algebra and its module into generalised eigenspaces (Chapter 3) gives an idempotent $e_g \in \mathbb{T}\otimes\mathbb{C}$ projecting onto the generalised eigenspace $\mathbb{C}g$ along $g^c$, and $e_g(\mathbb{T}\otimes\mathbb{C}) = \mathbb{C}e_g$ because that eigenspace is a line on which $\mathbb{T}$ acts by $\pi$. If $\pi$ is $\mathbb{Q}$-valued, the idempotent of $\mathbb{T}\otimes\mathbb{Q}$ attached to the maximal ideal $\ker(\pi\otimes\mathbb{Q})$ acts on $S_k(N)\otimes\mathbb{C}$ as the projection onto the sum of the generalised eigenspaces of the conjugates of $\pi$, which is $\mathbb{C}g$ ($\pi$ is its own only conjugate); so $e_g \in \mathbb{T}\otimes\mathbb{Q}$, and $g^c = \ker e_g$ is defined over $\mathbb{Q}$. Saturation and rank are proved as in Proposition 13.30: $g^c \cap S_2(N;\mathbb{Z})$ is the intersection of a lattice with a subspace of $S_2(N;\mathbb{Q})$ of dimension $d-1$. (ii) is Theorem 13.60(i): both are the sum of the other eigenspaces. (iii) $h$ is an eigenvector of $U_2$ with eigenvalue $\alpha \neq 0$, while $g$ has $U_2$-eigenvalue 0, so $h$ lies in another generalised eigenspace. The Gram matrix of $f_{11}(z)$, $f_{11}(2z)$, $f_{11}(4z)$ at level 44 is $\langle f_{11}, f_{11}\rangle_{44}$ times

$$\begin{pmatrix} 1 & -1/3 & 1/24 \\ -1/3 & 1/4 & -1/12 \\ 1/24 & -1/12 & 1/16 \end{pmatrix}$$

(PARI's `mfpetersson`, 40 digits; the $2\times 2$ block is Proposition 13.35(ii)), and with $g = (1,2,2)$ and $h = (1,-\beta,0)$ in these coordinates, $\langle g,h\rangle = g^{\mathsf{T}}G\bar{h}$ with $\bar{h} = (1,-\bar\beta,0)$. Now $G\bar{h} = (1+\frac{\bar\beta}{3}, -\frac{1}{3}-\frac{\bar\beta}{4}, \frac{1}{24}+\frac{\bar\beta}{12})$, and

$$g^{\mathsf{T}}G\bar{h} = \Big(1 - \frac{2}{3} + \frac{1}{12}\Big) + \bar\beta\Big(\frac{1}{3} - \frac{1}{2} + \frac{1}{6}\Big) = \frac{5}{12}.$$

So $\langle g,h\rangle = \frac{5}{12}\langle f_{11}, f_{11}\rangle_{44}$ for both $\beta = -1 \pm i$: the terms in $\beta$ cancel. □

*In the paper* 13.63. The proof of Theorem thm:twist speaks of "$g^\perp$, the $\mathbb{T}$-stable complement of the $g$-line (the sum of the generalised eigenspaces of the other systems of eigenvalues)": at level $N_S$, where $g$ is a newform, the two notions agree (Proposition 13.62(ii)). The set-up before Lemma lem:step1 uses "the $\mathbb{T}$-stable complement $C$ of the line of $g^S$ (the sum of the generalised eigenspaces of the other eigencharacters; not the Petersson complement, which is not $\mathbb{T}$-stable when the $U_p$ are involved)"; Proposition 13.62(iii) shows that the two differ there. The paper then applies the proof of Lemma lem:crit, "with $f^\perp$ replaced by" $C$ and carried out over $\mathbb{Z}_\ell$, to the eigencharacter $\pi$ of $g^S$, whose generalised eigenspace it has shown to be a line. The first step of that proof identifies the complement by Petersson adjoints and multiplicity one for a newform, which are not available for $g^S$; Proposition 13.64 below replaces that step by Proposition 13.62(i)

and proves the conclusions of the lemma for every $\mathbb{Q}$-valued eigencharacter of weight 2 whose generalised eigenspace is a line.

**Proposition 13.64** (the congruence module of a rational eigencharacter)**.** *Let $k = 2$, $S = S_2(N;\mathbb{Z})$, $S_{\mathbb{Q}} = S_2(N;\mathbb{Q})$, and let $\pi\colon \mathbb{T} \to \mathbb{Q}$ be a ring homomorphism whose generalised eigenspace in $S_2(N)$ is a line. Then $\pi(\mathbb{T}) = \mathbb{Z}$ and the line is $\mathbb{C}g$, where $g = \sum_n \pi(T_n)q^n \in S$ is the normalised eigenform of $\pi$. Let $g^c$ be the $\mathbb{T}$-stable complement of $\mathbb{C}g$, $W = g^c \cap S_{\mathbb{Q}}$, $\Lambda = g^c \cap S$, $I = \ker \pi$ and $J = \mathrm{Ann}_{\mathbb{T}}(I)$.*

*(i) $I \cap J = 0$, $J = \mathrm{Ann}_{\mathbb{T}}(g^c)$, $I = \mathrm{Ann}_{\mathbb{T}}(J)$, and $\mathbb{T}/(I+J)$ is a finite cyclic group. Writing $r_\pi = \#\mathbb{T}/(I+J)$, one has $\pi(J) = r_\pi\mathbb{Z}$, that is, $\eta(\pi) = r_\pi\mathbb{Z}$.*

*(ii) $S[I] = \mathbb{Z}g$, $S[J] = \Lambda$ and $[S : \mathbb{Z}g \oplus \Lambda] = r_\pi$. For an integer $m \geq 1$ there is $h \in \Lambda$ with $g \equiv h \pmod m$ if and only if $m \mid r_\pi$. So the moduli of congruences between $g$ and integral forms of the $\mathbb{T}$-stable complement are exactly the divisors of $r_\pi$.*

*(iii) Let $\ell$ be a prime, $S_\ell = S \otimes \mathbb{Z}_\ell$, $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$ and $\pi_\ell = \pi \otimes \mathbb{Z}_\ell\colon \mathbb{T}_\ell \to \mathbb{Z}_\ell$. Then $\pi_\ell(\mathrm{Ann}_{\mathbb{T}_\ell} \ker \pi_\ell) = \ell^{v_\ell(r_\pi)}\mathbb{Z}_\ell$, and for $t \geq 0$ there is $h \in S_\ell \cap (W \otimes \mathbb{Q}_\ell)$, a form of $S_\ell$ in the $\mathbb{T}$-stable complement, with $g \equiv h \pmod{\ell^t}$ if and only if $t \leq v_\ell(r_\pi)$.*

*(iv) If $g$ is a newform of level $N$, then $g^c = g^\perp$, $\Lambda = g^\perp \cap S$ and $r_\pi = r_g$, the congruence number of Definition 13.29.*

*Proof.* Each $\pi(t)$ is an eigenvalue of $t$, hence a root of the characteristic polynomial of $t$ acting on the lattice $S$ (Proposition 13.16), a monic integer polynomial, so $\pi(t)$ is a rational algebraic integer, an integer; and $\pi(T_1) = 1$, so $\pi(\mathbb{T}) = \mathbb{Z}$. By Theorem 13.26(ii), $g = f_\pi$ lies in $S$ and $tg = \pi(t)g$, so $g$ lies in the generalised eigenspace of $\pi$, which is therefore $\mathbb{C}g$.

*Step 1: the splitting.* By Proposition 13.62(i) there is an idempotent $e \in \mathbb{T} \otimes \mathbb{Q}$ acting as 1 on $g$ and as 0 on $g^c$, and $S_2(N) = \mathbb{C}g \oplus g^c$. As $e$ preserves $S_{\mathbb{Q}}$, we get $S_{\mathbb{Q}} = \mathbb{Q}g \oplus W$, $g^c = W \otimes \mathbb{C}$ and $W = \ker(e|_{S_{\mathbb{Q}}})$. The algebra $\mathbb{T} \otimes \mathbb{Q}$ acts faithfully on $S_{\mathbb{Q}}$ (Theorem 13.24(iii)). For $t \in \mathbb{T} \otimes \mathbb{Q}$, the elements $te$ and $\pi(t)e$ agree on $g$ and on $W$, so $te = \pi(t)e$. Put $\mathbb{T}' = (1-e)(\mathbb{T} \otimes \mathbb{Q})$. Then $t \mapsto (\pi(t), (1-e)t)$ identifies $\mathbb{T} \otimes \mathbb{Q}$ with $\mathbb{Q} \times \mathbb{T}'$ (the first factor being $\mathbb{Q}e$), and $\mathbb{T}'$ acts faithfully on $W$: an element $(1-e)t$ that kills $W$ also kills $g$, as $(1-e)g = 0$, hence is 0.

*Step 2: the ideals.* Since $\mathbb{Q}$ is flat over $\mathbb{Z}$, $I \otimes \mathbb{Q} = \ker(\pi \otimes \mathbb{Q}) = 0 \times \mathbb{T}'$. An element $t = (a, t')$ of $\mathbb{T} \subseteq \mathbb{Q} \times \mathbb{T}'$ kills $I$ if and only if it kills $I \otimes \mathbb{Q}$, if and only if $t'\mathbb{T}' = 0$, if and only if $t' = 0$ ($\mathbb{T}'$ contains its unit $1-e$). So $J = \mathbb{T} \cap (\mathbb{Q}e \times 0)$. By faithfulness on $W$, $t$ kills $g^c = W \otimes \mathbb{C}$ if and only if $t' = 0$; so $J = \mathrm{Ann}_{\mathbb{T}}(g^c)$. Clearly $I \cap J = 0$. Some multiple $me$, $m \geq 1$, lies in $\mathbb{T}$ (clear denominators), and $me \in J$ with $\pi(me) = m \neq 0$. An element $t \in \mathbb{T}$ kills $J$ iff $t \cdot me = m\pi(t)e = 0$, iff $\pi(t) = 0$; so $\mathrm{Ann}_{\mathbb{T}}(J) = I$. As $\pi$ maps $\mathbb{T}$ onto $\mathbb{Z}$ with kernel $I$ and is injective on $J$, $\mathbb{T}/(I+J) \simeq \mathbb{Z}/\pi(J)$, which is cyclic and finite because $m \in \pi(J)$. Hence $\pi(J) = r_\pi\mathbb{Z}$, and $\eta(\pi) = \pi(\mathrm{Ann}_{\mathbb{T}} \ker \pi) = \pi(J) = r_\pi\mathbb{Z}$ (Definition 13.29 with $\mathcal{O} = \mathbb{Z}$; $\pi$ is surjective). Finally $\mathbb{T}/I \simeq \mathbb{Z}$ and $\mathbb{T}/J$ are torsion-free: if $mt \in J$ with $m \neq 0$, then $m(1-e)t = 0$, so $(1-e)t = 0$ and $t \in J$.

*Step 3: the index.* By Corollary 13.28(iii), $S[I] = \mathbb{Z}f_\pi = \mathbb{Z}g$. The ideal $J$ kills $g^c \supseteq \Lambda$; conversely, if $x \in S$ and $Jx = 0$, then $me\,x = 0$, so $ex = 0$ and $x \in S \cap \ker e = \Lambda$. So $S[J] = \Lambda$. Step 2 gives the hypotheses of Corollary 13.28(iv), which yields $[S : \mathbb{Z}g \oplus \Lambda] = \#\mathbb{T}/(I+J) = r_\pi$.

*Step 4: the congruences.* The map $\bar{e}\colon S/(\mathbb{Z}g \oplus \Lambda) \to \mathbb{Q}g/\mathbb{Z}g$, $x \mapsto ex$, is well defined ($e$ fixes $g$, kills $\Lambda$ and maps $S$ into $S_{\mathbb{Q}} \cap \mathbb{C}g = \mathbb{Q}g$) and injective (if $ex = cg$ with $c \in \mathbb{Z}$, then $e(x - cg) = 0$, so $x - cg \in \Lambda$). So $S/(\mathbb{Z}g \oplus \Lambda)$ is isomorphic to a subgroup of order $r_\pi$ of $\mathbb{Q}g/\mathbb{Z}g \simeq \mathbb{Q}/\mathbb{Z}$, and the

only such subgroup is $r_\pi^{-1}\mathbb{Z}g/\mathbb{Z}g$. If $g \equiv h \pmod m$ with $h \in \Lambda$, then $x = (g-h)/m \in S$ and $ex = g/m$, whose class has order $m$ and lies in $r_\pi^{-1}\mathbb{Z}g/\mathbb{Z}g$; so $m \mid r_\pi$. Conversely, choose $x \in S$ with $\bar{e}(x)$ the class of $g/r_\pi$; replacing $x$ by $x - cg$ for a suitable $c \in \mathbb{Z}$ we may assume $ex = g/r_\pi$. Then $h_0 = g - r_\pi x$ lies in $S$ and $eh_0 = 0$, so $h_0 \in \Lambda$, and $g - h_0 = r_\pi x$: a congruence modulo $r_\pi$, hence modulo every divisor $m$ of $r_\pi$.

*Step 5: (iii).* By flatness of $\mathbb{Z}_\ell$ over $\mathbb{Z}$, $\ker \pi_\ell = I \otimes \mathbb{Z}_\ell$, and Step 2 run in $\mathbb{T}_\ell \subseteq \mathbb{Q}_\ell e \times (\mathbb{T}' \otimes \mathbb{Q}_\ell)$ shows $\mathrm{Ann}_{\mathbb{T}_\ell}(\ker \pi_\ell) = \mathbb{T}_\ell \cap \mathbb{Q}_\ell e$. As $\mathbb{T}/J$ is free, $J \otimes \mathbb{Z}_\ell$ is saturated in $\mathbb{T}_\ell$ with $\mathbb{Q}_\ell$-span $\mathbb{Q}_\ell e$, so $\mathbb{T}_\ell \cap \mathbb{Q}_\ell e = J \otimes \mathbb{Z}_\ell$, whose image under $\pi_\ell$ is $r_\pi \mathbb{Z}_\ell = \ell^{v_\ell(r_\pi)}\mathbb{Z}_\ell$. Likewise $\Lambda$ is saturated in $S$, so $\Lambda_\ell = \Lambda \otimes \mathbb{Z}_\ell$ equals $S_\ell \cap (W \otimes \mathbb{Q}_\ell)$. The group $S_\ell/(\mathbb{Z}_\ell g \oplus \Lambda_\ell) = (S/(\mathbb{Z}g \oplus \Lambda)) \otimes \mathbb{Z}_\ell$ is cyclic of order $\ell^{v_\ell(r_\pi)}$, and Step 4 runs verbatim with $\mathbb{Q}_\ell g/\mathbb{Z}_\ell g \simeq \mathbb{Q}_\ell/\mathbb{Z}_\ell$, whose finite subgroups are cyclic and determined by their order.

(iv) By Proposition 13.62(ii), $g^c = g^\perp$; so $\Lambda$ is the lattice $L$ of Proposition 13.30, and by (ii) the largest modulus of a congruence between $g$ and an element of $L$, which is $r_g$ by definition, is $r_\pi$. □

For a newform $f$ with rational coefficients, (i), (ii) and (iv) are the paper's Lemma lem:crit (whose full proof, with its own first step, is in Chapter 15); the point of the proposition is that only the line hypothesis is used. The hypothesis cannot be dropped: for $f_{11}^{\{2\}}$ at level 88 the generalised eigenspace has dimension 2 (Example 13.83), the local factor of $\mathbb{T}_{88} \otimes \mathbb{Q}$ through which $\pi$ factors is not $\mathbb{Q}$, and Step 1 fails.

*In the paper* 13.65. The set-up before Lemma lem:step1: the adapted proof of Lemma lem:crit "shows for every prime $\ell$ that $\ell^{v_\ell(\eta^{\mathrm{Hecke}}(g^S))}$ is the largest power $\ell^t$ for which $g^S \equiv h \pmod{\ell^t}$ coefficientwise with $h \in C \cap S_2(\Gamma_0(N); \mathbb{Z}_\ell)$", which is Proposition 13.64(iii) for the eigencharacter $\pi$ of $g^S$, whose generalised eigenspace at level $N = N_S^S$ is a line by Proposition 13.55(iv).

**Proposition 13.66** (local components)**.** *Let $\ell$ be a prime, $k = 2$, $S_\ell = S_2(N; \mathbb{Z}_\ell)$, and $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell = \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$ the decomposition over the maximal ideals $\mathfrak{m}$ of $\mathbb{T}_\ell$ (those of $\mathbb{T}$ containing $\ell$), with idempotents $e_{\mathfrak{m}}$ (Chapter 3).*

*(i) $S_\ell = \bigoplus_{\mathfrak{m}} e_{\mathfrak{m}} S_\ell$, and $e_{\mathfrak{m}} S_\ell \otimes \overline{\mathbb{Q}}_\ell$ is the sum of the generalised eigenspaces of the $\phi\colon \mathbb{T} \to \overline{\mathbb{Q}}_\ell$ whose reduction has kernel $\mathfrak{m}$.*

*(ii) A ring homomorphism $\phi\colon \mathbb{T}_\ell \to \mathcal{O}_\lambda$ factors through exactly one $\mathbb{T}_{\mathfrak{m}}$; the other factors lie in $\ker \phi$, and $\mathrm{Ann}_{\mathbb{T}_\ell}(\ker \phi) = \mathrm{Ann}_{\mathbb{T}_{\mathfrak{m}}}(\ker \phi \cap \mathbb{T}_{\mathfrak{m}})$. The same holds for the finite $\mathcal{O}_\lambda$-algebra $\mathbb{T}_{\mathcal{O}_\lambda} = \mathbb{T}_\ell \otimes_{\mathbb{Z}_\ell} \mathcal{O}_\lambda$ and its local factors; so for a surjective $\phi\colon \mathbb{T}_{\mathcal{O}_\lambda} \to \mathcal{O}_\lambda$ the congruence ideal $\eta(\phi) = \phi(\mathrm{Ann} \ker \phi)$ of Definition 13.29 is computed in the local factor through which $\phi$ factors.*

*(iii) $\langle e_{\mathfrak{m}} g, t\rangle = \langle g, e_{\mathfrak{m}} t\rangle$; the pairing of Theorem 13.24 restricts to perfect pairings $e_{\mathfrak{m}} S_\ell \times \mathbb{T}_{\mathfrak{m}} \to \mathbb{Z}_\ell$, and $e_{\mathfrak{m}} S_\ell \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(\mathbb{T}_{\mathfrak{m}}, \mathbb{Z}_\ell)$.*

*(iv) $e_{\mathfrak{m}}$ preserves $S_\ell$ and every $\mathbb{T}$-stable subspace of $S_2(N) \otimes \mathbb{Q}_\ell$. If $g \in e_{\mathfrak{m}} S_\ell$, $h \in S_\ell \cap W$ with $W$ a $\mathbb{T}$-stable subspace, and $g \equiv h \pmod{\ell^t}$, then $g \equiv e_{\mathfrak{m}} h \pmod{\ell^t}$ with $e_{\mathfrak{m}} h \in e_{\mathfrak{m}} S_\ell \cap W$.*

*Proof.* (i) The first statement holds because the $e_{\mathfrak{m}}$ are orthogonal idempotents with sum 1 acting on $S_\ell$. For the second, $S_\ell \otimes \overline{\mathbb{Q}}_\ell$ is the sum of the generalised eigenspaces $V(\phi)$, $\phi\colon \mathbb{T} \to \overline{\mathbb{Q}}_\ell$ (Chapter 3); such $\phi$ take values in the integral closure of $\mathbb{Z}_\ell$, since $\mathbb{T}$ is finite over $\mathbb{Z}$. On $V(\phi)$ the idempotent $e_{\mathfrak{m}}$ has the single eigenvalue $\phi(e_{\mathfrak{m}}) \in \{0, 1\}$, and an idempotent with one eigenvalue acts by it (its minimal polynomial divides $X^2 - X$). Finally $\phi(e_{\mathfrak{m}}) = 1$ iff $e_{\mathfrak{m}}$ is not in the maximal

ideal $\phi^{-1}(\mathfrak{m}_{\overline{\mathbb{Z}}_\ell})$ of $\mathbb{T}_\ell$, iff that ideal is $\mathfrak{m}$. (ii) $\phi(e_\mathfrak{m}) \in \{0,1\}$ and $\sum_\mathfrak{m} \phi(e_\mathfrak{m}) = 1$, so exactly one $\phi(e_\mathfrak{m})$ is 1, and $e_{\mathfrak{m}'} \in \ker\phi$ for $\mathfrak{m}' \neq \mathfrak{m}$, so $\mathbb{T}_{\mathfrak{m}'} = e_{\mathfrak{m}'}\mathbb{T}_\ell \subseteq \ker\phi$. If $a$ kills $\ker\phi$, it kills each $e_{\mathfrak{m}'}$, so $a = e_\mathfrak{m} a \in \mathbb{T}_\mathfrak{m}$; and an element of $\mathbb{T}_\mathfrak{m}$ kills $\ker\phi$ iff it kills $\ker\phi \cap \mathbb{T}_\mathfrak{m}$, since it kills the other factors. The argument uses only that $\mathbb{T}_\ell$ is a finite product of local rings with idempotents $e_\mathfrak{m}$; $\mathbb{T}_{\mathcal{O}_\lambda}$, finite over the complete local ring $\mathcal{O}_\lambda$, is such a product as well (Chapter 3). (iii) $\langle e_\mathfrak{m} g, t\rangle = a_1(te_\mathfrak{m} g) = \langle g, e_\mathfrak{m} t\rangle$. Hence $\langle e_\mathfrak{m} S_\ell, \mathbb{T}_{\mathfrak{m}'}\rangle = \langle S_\ell, e_\mathfrak{m} e_{\mathfrak{m}'}\mathbb{T}_\ell\rangle = 0$ for $\mathfrak{m} \neq \mathfrak{m}'$: the perfect pairing $S_\ell \times \mathbb{T}_\ell \to \mathbb{Z}_\ell$ (Theorem 13.24(iii)) is block diagonal for the two decompositions, so each block is perfect. (iv) $e_\mathfrak{m} \in \mathbb{T}_\ell$ acts on $S_\ell$, and a $\mathbb{T}$-stable $\mathbb{Q}_\ell$-subspace is $\mathbb{T}_\ell$-stable. If $g - h = \ell^t x$ with $x \in S_\ell$, then $g - e_\mathfrak{m} h = e_\mathfrak{m}(g-h) = \ell^t e_\mathfrak{m} x$. □

*In the paper* 13.67. In the paragraph after Lemma lem:hecke the paper says: "the congruence ideal is computed in $\mathbb{T}_\mathfrak{m}$: the other local factors of $\mathbb{T} \otimes \mathbb{Z}_\ell$ lie in the kernel $\wp$, $\phi_f$ factors through $\mathbb{T}_\mathfrak{m}$, and $\mathrm{Ann}(\wp)$ decomposes along the idempotents": Proposition 13.66(ii) (the other factors lie in the kernel, not in the annihilator). The proof of Lemma lem:step1: "The idempotent $e_\mathfrak{m}$ of $\mathbb{T}_\ell$ fixes $g^S$ (as $\pi$ factors through $\mathbb{T}_\mathfrak{m}$) and preserves $C$ and the integral forms": Proposition 13.66(iv). The perfectness of the cohomological pairing $\hat{\delta}_!$ on each local component is the same argument with $\hat{\delta}_!$ in place of $\langle g, t\rangle$ (Chapter 14).

**Proposition 13.68** (congruences)**.** *Let $\mathcal{O}$ be the ring of integers of a number field and $\lambda$ a prime of $\mathcal{O}$.*

*(i) For normalised eigenforms $f, g$ of $\mathbb{T}_N$ with coefficients in $\mathcal{O}$ the following are equivalent: (a) $a_n(f) \equiv a_n(g) \pmod\lambda$ for all $n \geq 1$; (b) the same for all primes $n = p$, including $p \mid N$; (c) $\phi_f(t) \equiv \phi_g(t) \pmod\lambda$ for all $t \in \mathbb{T}$. They imply (d) $\mathfrak{m}_{f,\lambda} = \mathfrak{m}_{g,\lambda}$, and are implied by it when $\mathbb{T}/\mathfrak{m}_{f,\lambda} = \mathbb{F}_\ell$ (for instance when $f$ has rational coefficients).*

*(ii) Let $f$, $g$ be newforms of weight $k$ and levels $M_f$, $M_g$ with coefficients in $\mathcal{O}$, and $\Sigma$ a finite set of primes containing every prime dividing $M_f M_g$. If $a_p(f) \equiv a_p(g) \pmod\lambda$ for all $p \notin \Sigma$, then $f^\Sigma \equiv g^\Sigma \pmod\lambda$, both forms lying in $S_k(L)$, $L = \mathrm{lcm}(M_f^\Sigma, M_g^\Sigma)$.*

*Proof.* (i) (a)⇒(b) is trivial. (b)⇒(a): by Proposition 13.19(ii) the $a_n$ of both forms are given by the same integer polynomials in the $a_p$ (multiplicativity, $a_{p^r} = a_p^r$ for $p \mid N$, the recursion with the same $p^{k-1}$ for $p \nmid N$). (a)⇔(c): $\mathbb{T}$ is the $\mathbb{Z}$-span of the $T_n$ (Proposition 13.16(ii)) and $\phi_f(T_n) = a_n(f)$. (c)⇒(d): the two reductions $\mathbb{T} \to \mathcal{O}/\lambda$ coincide, so they have the same kernel. If the kernels coincide and $\mathbb{T}/\mathfrak{m} = \mathbb{F}_\ell$, both reductions are the unique ring homomorphism $\mathbb{T} \to \mathbb{T}/\mathfrak{m} = \mathbb{F}_\ell \subseteq \mathcal{O}/\lambda$. (Without that hypothesis (d) only determines the reduction up to an automorphism of the residue field: Galois-conjugate forms $f$, $f^\sigma$ with $\sigma\lambda = \lambda$ have the same maximal ideal.) (ii) For $(n, \Sigma) = 1$ write $n = \prod p^r$ with $p \notin \Sigma$, so $p \nmid M_f M_g$; both $a_{p^r}(f)$ and $a_{p^r}(g)$ are given by the same recursion in $a_p$, so they are congruent, and by multiplicativity $a_n(f) \equiv a_n(g)$. For $(n, \Sigma) > 1$ both coefficients of the depleted forms vanish. The levels are Proposition 13.55(iii). □

*Remark* 13.69 (Sturm versus the recursion). Two tools turn finitely many congruences into congruences at all $n$, and they should not be confused. The recursion (Proposition 13.68(i)) transfers congruences at all primes, including those dividing the level, to all $n$, between eigenforms of the full $\mathbb{T}_N$. Sturm's bound (Chapter 12) transfers a finite check, $n \leq B$, to all $n$, between any two forms of the same space $S_k(N)$. Neither turns a congruence at the primes $p \notin \Sigma$ alone into one at all $n$: that requires depletion (Proposition 13.68(ii)) or a choice of $U_p$-eigenvalues (the proof of Lemma lem:old, Chapter 15). The congruences of the paper's Example of Section 3 hold at all $n$ by Sturm's bound (Chapter 12).

## 13.9 Twisting

This section treats the twist of a modular form by a Dirichlet character, in the generality the paper needs: quadratic characters $\chi_{q^*}$ and their products $\chi_S$. It proves the elementary properties of the twist operator (level, integrality, intertwining with Hecke operators, twisting twice), states the theorem of Atkin and Li (the paper's Lemma lem:twist (iii)), proves the transport of congruences along the twist map $\iota$ used in the proofs of Theorems thm:twist and thm:quant, and deduces that $r_E$ divides the Hecke congruence module of $g^S$ (Corollary 13.76). In this section $q$ denotes an odd prime; the $q$-expansion variable appears only as $e^{2\pi inz}$.

**Proposition 13.70** (the twist operator)**.** *Let $\chi$ be a primitive Dirichlet character of conductor $r$, $\mathfrak{g}(\bar\chi)=\sum_{u \bmod r}\bar\chi(u)e^{2\pi iu/r}$ the Gauss sum of $\bar\chi$, and $f=\sum a_n e^{2\pi inz}\in S_k(M)$.*

*(i)* $f\otimes\chi=\sum_n \chi(n)a_n e^{2\pi inz}=\mathfrak{g}(\bar\chi)^{-1}\sum_{u \bmod r}\bar\chi(u)\,f\big(z+\frac{u}{r}\big)$.

*(ii)* $f\otimes\chi\in S_k(\Gamma_0(L),\chi^2)$ *with* $L=\operatorname{lcm}(M,r^2)$*; for quadratic* $\chi$, $f\otimes\chi\in S_k(L)$.

*(iii) If $\chi$ is quadratic and $f\in S_2(M;\mathbb{Z})$, then $f\otimes\chi\in S_2(L;\mathbb{Z})$.*

*(iv) (Intertwining.) Let $\chi$ be quadratic. For a prime $p\nmid r$, $T_p^{(L)}(f\otimes\chi)=\chi(p)\,(T_p^{(M)}f)\otimes\chi$, where on each side $T_p$ means $U_p$ if $p$ divides the level; for $p\mid r$, $U_p^{(L)}(f\otimes\chi)=0$. Consequently $f\mapsto f\otimes\chi$ maps the (generalised) eigenspace of $\mathbb{T}^{(rM)}$ for a system $(\lambda_p)$ into the (generalised) eigenspace for $(\chi(p)\lambda_p)$, and a normalised eigenform of $\mathbb{T}_M$ to a normalised eigenform of $\mathbb{T}_L$.*

*(v) (Twisting twice.) $(f\otimes\chi)\otimes\bar\chi=\sum_{(n,r)=1}a_n e^{2\pi inz}=f^{\{p\mid r\}}$, the depletion at the primes dividing $r$. In particular $(g\otimes\chi_{q^*})\otimes\chi_{q^*}=g^{\{q\}}$, not $g$.*

*(vi) (The paper's $\iota$.) For a finite set $S$ of odd primes, $D=\prod_{q\in S}q$ and $\chi_S=\prod_{q\in S}\chi_{q^*}$, the map $\iota=\iota_S\colon S_2(M)\to S_2(\operatorname{lcm}(M,D^2))$, $h\mapsto h\otimes\chi_S$, is $\mathbb{C}$-linear, maps $S_2(M;\mathbb{Z})$ into $S_2(\operatorname{lcm}(M,D^2);\mathbb{Z})$, kills every $h(dz)$ with $(d,D)>1$, satisfies $\iota(h)=\iota(h^S)$ and $\iota\circ\iota(h)=h^S$, and maps the generalised eigenspace of $\mathbb{T}^{(DM)}$ for $(\lambda_p)$ into that for $(\chi_S(p)\lambda_p)$.*

*Proof.* (i) For a primitive character, $\sum_{u \bmod r}\bar\chi(u)e^{2\pi inu/r}=\chi(n)\mathfrak{g}(\bar\chi)$ for every integer $n$, including $(n,r)>1$, where both sides vanish, and $\mathfrak{g}(\bar\chi)\neq 0$ ($|\mathfrak{g}|^2=r$) (Chapter 2). Insert $f(z+u/r)=\sum_n a_n e^{2\pi inu/r}e^{2\pi inz}$.

(ii) Let $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma_0(L)$ and $u\in\mathbb{Z}$. Choose $u'\equiv d^2u \pmod r$. Then

$$\gamma'=\begin{pmatrix}1&u/r\\0&1\end{pmatrix}\gamma\begin{pmatrix}1&-u'/r\\0&1\end{pmatrix}=\begin{pmatrix}a+cu/r & b+(du-au')/r-cuu'/r^2\\ c & d-cu'/r\end{pmatrix}$$

is integral: $r^2\mid c$, and $du-au'\equiv du-ad^2u=du(1-ad)\equiv 0 \pmod r$, because $ad\equiv 1 \pmod r$ (as $r\mid c$). It has determinant 1 and lower left entry $c\equiv 0 \pmod M$, so $\gamma'\in\Gamma_0(M)$ and $f|_k\left(\begin{smallmatrix}1&u/r\\0&1\end{smallmatrix}\right)\gamma=f|_k\gamma'\left(\begin{smallmatrix}1&u'/r\\0&1\end{smallmatrix}\right)=f|_k\left(\begin{smallmatrix}1&u'/r\\0&1\end{smallmatrix}\right)$. As $u$ runs modulo $r$ so does $u'$ ($d$ is a unit mod $r$), and $\bar\chi(u)=\bar\chi(a^2u')=\chi(d)^2\bar\chi(u')$ since $u\equiv a^2u'$. Hence $(f\otimes\chi)|_k\gamma=\chi(d)^2(f\otimes\chi)$. Holomorphy and vanishing at the cusps hold for each $f|_k\left(\begin{smallmatrix}1&u/r\\0&1\end{smallmatrix}\right)$, by the argument of Proposition 13.6(ii). For quadratic $\chi$, $\chi(d)^2=1$ as $(d,r)=1$.

(iii) The coefficients $\chi(n)a_n$ are integers. (iv) By Proposition 13.8: if $p\nmid L$ (so $p\nmid M$), the $n$-th coefficient of $T_p^{(L)}(f\otimes\chi)$ is $\chi(pn)a_{pn}+p^{k-1}\chi(n/p)a_{n/p}=\chi(p)\chi(n)\big(a_{pn}+p^{k-1}a_{n/p}\big)$, using

$\chi(n/p) = \chi(n)\chi(p)^{-1} = \chi(n)\chi(p)$ for $p \mid n$ ($\chi(p) = \pm 1$); this is $\chi(p)a_n((T_p^{(M)}f)\otimes\chi)$. If $p \mid L$ and $p \nmid r$, then $p \mid M$ and both sides are $\mathcal{U}_p$: $\chi(pn)a_{pn} = \chi(p)\chi(n)a_{pn}$. If $p \mid r$, $\chi(pn) = 0$. For the consequence: $T_p\iota = \chi(p)\iota T_p$ gives $(T_p - \chi(p)\lambda_p)^m\iota = \chi(p)^m\iota(T_p - \lambda_p)^m$, and the normalised eigenform statement follows from Lemma 13.20 (the coefficients $\chi(n)a_n$ are again multiplicative and satisfy the recursions, with $b_{p^r} = 0$ for $p \mid r$).

(v) $\bar{\chi}(n)\chi(n)a_n$ is $a_n$ for $(n, r) = 1$ and 0 otherwise. (vi) Linearity and integrality are (iii); the coefficients of $h(dz)$ are supported on multiples of $d$, where $\chi_S$ vanishes if $(d, D) > 1$; $\iota(h) = \iota(h^S)$ and $\iota \circ \iota(h) = h^S$ are (v); the last statement is (iv) for $\chi_S$, which is primitive of conductor $D$ (Chapter 2). □

**Theorem 13.71** (Atkin and Li)**.** *Let $g$ be a newform of weight 2 on $\Gamma_0(M)$, $q$ an odd prime and $\chi = \chi_{q^*}$.*

*(i) If $q \nmid M$, $g \otimes \chi$ is a newform on $\Gamma_0(Mq^2)$.*

*(ii) If $q \parallel M$, $g \otimes \chi$ is a newform on $\Gamma_0(Mq)$.*

*(iii) If $q^2 \mid M$, there is a unique newform $g'$ with $a_p(g') = \chi(p)a_p(g)$ for all $p \neq q$; its level divides $M$ and has the same exponent as $M$ at every $p \neq q$, and $g \otimes \chi = (g')^{\{q\}}$, which equals $g'$ iff $a_q(g') = 0$.*

*In cases (i) and (ii) the level is $\operatorname{lcm}(M, q^2)$: this is the paper's Lemma lem:twist (iii). Parts (i), (ii) are stated with a reference, [9, Theorem 3.1] as cited by the paper, [113] for the case $q \nmid M$, [180, Section 3.6]; part (iii) is proved here.*

*Sketch of proof of (i), (ii), and proof of (iii).* (i), (ii): through Chapter 15. The level of a newform is the conductor of its $\lambda$-adic Galois representation $\rho_g$ (Carayol [21]), and the naive twist $g \otimes \chi$, an eigenform by Proposition 13.70(iv), has the system of eigenvalues of $\rho_g \otimes \chi$ at $p \neq q$. If $\rho_g$ is unramified at $q$ (case (i)) or its restriction to inertia at $q$ is unipotent and nontrivial (case (ii), Steinberg), then $\rho_g \otimes \chi$ has no inertia invariants at $q$ and is tamely ramified there, so its conductor exponent at $q$ is 2; at $p \neq q$ the conductor is unchanged. So the newform $g'$ attached to $\rho_g \otimes \chi$ has level $\operatorname{lcm}(M, q^2)$, hence $a_q(g') = 0$ (Proposition 13.44(iii)); at the primes $p \mid M$, $p \neq q$, $a_p(g') = \chi(p)a_p(g)$ by the argument given for (iii) below, run at the level $\operatorname{lcm}(M, q^2)$ in place of $M$; so $g'$ has the coefficients $\chi(n)a_n(g)$, and $g' = g \otimes \chi$. (iii) By Proposition 13.70(ii),(iv), $g \otimes \chi \in S_2(\operatorname{lcm}(M, q^2)) = S_2(M)$ is a normalised eigenform of $\mathbb{T}_M$, with $T_n$-eigenvalue $\chi(n)a_n(g)$ for $(n, M) = 1$, $U_p$-eigenvalue $\chi(p)a_p(g)$ for $p \mid M$, $p \neq q$, and $U_q$-eigenvalue 0. By Theorem 13.59(i) it lies in $V_{g'}$ for a newform $g'$ of level $M' \mid M$ with $a_p(g') = \chi(p)a_p(g)$ for $p \nmid M$, unique by Theorem 13.42(i). Symmetrically, $g' \otimes \chi$ lies in $S_2(\operatorname{lcm}(M', q^2)) \subseteq S_2(M)$ and is an eigenvector of $\mathbb{T}^{(M)}$ with the system of $g$, so it is a multiple of $g$ by Theorem 13.41(ii), in fact $g$ (compare $a_1$). Thus $g \in S_2(\operatorname{lcm}(M', q^2))$, and this forces $M \mid \operatorname{lcm}(M', q^2)$: a newform of level $M$ lies in $S_2(L)$ only if $M \mid L$, by the uniqueness of coordinates in the bases of Theorem 13.41(iii) (the basis at level $L$ is part of the basis at level $\operatorname{lcm}(L, M)$). Hence $v_p(M') = v_p(M)$ for every $p \neq q$. For $p \mid M$, $p \neq q$, the factor $V_{g',p,0}$ of $V_{g'} \simeq \bigotimes_p V_{g',p,e_p}$ is the line of $g'$, on which $U_p$ acts by $a_p(g')$; so $a_p(g') = \chi(p)a_p(g)$ there as well. At $q$, the $U_q$-eigenvalue of $g \otimes \chi$ is 0, and by Proposition 13.48(iii) the 0-eigenvector of $U_q$ on $V_{g',q,e_q}$ is the depleted form. A normalised eigenform of $\mathbb{T}_M$ in $V_{g'}$ is determined by its $U_p$-eigenvalues (each joint eigenspace is a tensor product of lines), so $g \otimes \chi = (g')^{\{q\}}$. Finally $(g')^{\{q\}} = g'$ iff $a_{q^r}(g') = 0$ for $r \geq 1$, iff $a_q(g') = 0$. □

**Example 13.72** (levels of twists)**.** Conductors of twisted curves, computed with PARI (`elltwist`, `ellglobalred`, minimal model matched to Cremona's tables); the newform of $E^{(q)}$ is the newform twist of $f_E$, equal to the naive twist in cases (i) and (ii):

| case | curve, character | twist | level |
|---|---|---|---|
| $q \nmid M$ | `11a1`, $\chi_{-7}$ | `539d2` | $11 \cdot 7^2$ |
| | `20a1`, $\chi_{-7}$ | `980g2` | $20 \cdot 7^2$ |
| | `37a1`, $\chi_5$ | `925e1` | $37 \cdot 5^2$ |
| $q \parallel M$ | `11a1`, $\chi_{-11}$ | `121d2` | $11^2$ |
| | `14a1`, $\chi_{-7}$ | `98a3` | $2 \cdot 7^2$ |
| | `77c1`, $\chi_{-7}$ | `539c1` | $11 \cdot 7^2$ |
| | `15a1`, $\chi_{-3}$ | `45a4` | $5 \cdot 3^2$ |
| $q^2 \mid M$ | `539c1`, $\chi_{-7}$ | `77c1` | level drops |
| | `539d1`, $\chi_{-7}$ | `11a3` | level drops |
| | `980i1`, $\chi_{-7}$ | `140b1` | level drops |
| | `121a1`, $\chi_{-11}$ | `121c2` | level kept |
| | `99a1`, $\chi_{-3}$ | `99c1` | level kept |
| | `49a1`, $\chi_{-7}$ | `49a3` | level kept |

The first rows can be seen in the first coefficients. From $a_2(\texttt{11a1}) = -2$, $a_3 = -1$, and $\chi_{-11}(2) = -1$, $\chi_{-11}(3) = 1$, the twist by $\chi_{-11}$ has $a_2 = 2$, $a_3 = -1$, $a_{11} = 0$, which are the coefficients of `121d1`; by $\chi_{-7}$, with $\chi_{-7}(2) = 1$, $\chi_{-7}(3) = -1$, it has $a_2 = -2$, $a_3 = 1$, $a_7 = 0$, the coefficients of `539d1` (PARI's `ellan`). The level in the first case is $11^2$ because $11 \parallel 11$ (case (ii)), and in the second $11 \cdot 7^2$ because $7 \nmid 11$ (case (i)). For all $n \le 3000$ we also checked the coefficient identities

$$a_n(\texttt{121d1}) = \chi_{-11}(n) a_n(\texttt{11a1}), \qquad a_n(\texttt{539d1}) = \chi_{-7}(n) a_n(\texttt{11a1}),$$
$$a_n(\texttt{539c1}) = \chi_{-7}(n) a_n(\texttt{77c1}), \qquad a_n(\texttt{121c1}) = \chi_{-11}(n) a_n(\texttt{121a1}),$$
$$a_n(\texttt{980g1}) = \chi_{-7}(n) a_n(\texttt{20a1});$$

the last fails with `980i1` in place of `980g1`. So $f_{20\mathrm{a}} \otimes \chi_{-7}$ is the form of 980g, and the paper's congruence "$f_{980\mathrm{i}} \equiv f_{20\mathrm{a}} \otimes \chi_{-7} = f_{980\mathrm{g}} \pmod 5$" (Example of Section 3) is a congruence between two different newforms of level 980, as the paper writes it. (As $5 \mid 980$, the paper uses this example only as an illustration.) In case $q^2 \mid M$ the naive twist is a depletion: $a_7(\texttt{77c1}) = -1$, while $f_{539\mathrm{c}} \otimes \chi_{-7}$ has $a_7 = 0$.

*In the paper* 13.73. Lemma lem:twist (iii) is Theorem 13.71(i),(ii); the paper cites [9, Theorem 3.1] and adds, in the proof of Lemma lem:twist, that "in the case $q \parallel M$, the local component of $g$ at $q$ is an unramified twist of the Steinberg representation, whose twist by a ramified quadratic character has conductor exponent 2", which is the sketch above in local language. The hypothesis $q^2 \nmid M$ cannot be dropped (Theorem 13.71(iii) and the last rows of Example 13.72). The proof of Theorem thm:qual (b) uses (ii) ("$h = g \otimes \chi$ is a newform on $\Gamma_0(Mq)$") and (c) uses (i) ("a newform on $\Gamma_0(Mq^2)$"); its parenthetical "Note that $h \otimes \chi$ is the $q$-depletion of $g$, not $g$" is Proposition 13.70(v).

**Proposition 13.74** (transport along the twist)**.** *Let $S$, $D$, $\chi_S$ and $\iota$ be as in Proposition 13.70(vi). Let $g$ be a newform of weight 2 and level $M$, $f$ a newform of level $N_f$ with $a_p(f) = \chi_S(p) a_p(g)$ for almost all $p$, and $L$ a common multiple of $\operatorname{lcm}(M, D^2)$ and $N_f$.*

(i) *If $h \in g^\perp$ (at level $M$), then $\iota(h) \in S_2(L)$ is orthogonal to every $f(dz)$, $d \mid L/N_f$. If $L = N_f$, then $\iota(h) \in f^\perp$.*

(ii) *If moreover $\iota(g) = f$ as $q$-series and $N_f = L$, a congruence $g \equiv h \pmod r$ with $h \in g^\perp \cap S_2(M;\mathbb{Z})$ gives $f \equiv \iota(h) \pmod r$ with $\iota(h) \in f^\perp \cap S_2(N_f;\mathbb{Z})$. Hence $r_g \mid r_f$ when both have rational coefficients.*

(iii) *Conversely, suppose $D^2 \mid N_f$ and $\iota(g) = f$. Then $\iota(f) = g^S$. If $g^S \in S_2(N_f)$ and its generalised eigenspace there is a line (Proposition 13.55(iv)), then $\iota$ maps $f^\perp$ into the $\mathbb{T}$-stable complement of $\mathbb{C}g^S$ in $S_2(N_f)$, and a congruence $f \equiv h \pmod r$ with $h \in f^\perp \cap S_2(N_f;\mathbb{Z})$ gives $g^S \equiv \iota(h) \pmod r$ with $\iota(h)$ integral in that complement.*

*Proof.* (i) By Theorem 13.41(iv) at level $M$, $g^\perp = \bigoplus_{h' \neq g} V_{h'}$, the sum over newforms $h' \neq g$ of levels dividing $M$; it suffices to treat $h \in V_{h'}$, an eigenvector of $\mathbb{T}^{(M)}$ for the system of $h'$. By Proposition 13.70(vi), $\iota(h)$ is an eigenvector of the $T_p$, $p \nmid L$, with eigenvalues $\chi_S(p)a_p(h')$. These differ from $a_p(f) = \chi_S(p)a_p(g)$ at infinitely many $p$, because $a_p(h') \neq a_p(g)$ for infinitely many $p$ (Theorem 13.42(i)). So Theorem 13.42(iii) applies. If $L = N_f$, the only $f(dz)$ is $f$. (ii) Apply $\iota$, which is linear and preserves integrality and congruences coefficientwise. With $r = r_g$ this gives a congruence modulo $r_g$ between $f$ and an integral form orthogonal to $f$; by Proposition 13.64(ii),(iv) for $\phi_f$ the moduli of such congruences are exactly the divisors of $r_f$, so $r_g \mid r_f$. (iii) $\iota(f) = \iota \circ \iota(g) = g^S$ by Proposition 13.70(v). At level $N_f$, $\operatorname{lcm}(N_f, D^2) = N_f$, so $\iota$ maps $S_2(N_f)$ to itself. By Theorem 13.41(iv), $f^\perp = \bigoplus_{h' \neq f} V_{h'}$ over newforms of levels dividing $N_f$, and $\iota(V_{h'})$ lies in the eigenspace of $\mathbb{T}^{(N_f)}$ for the system $\chi_S(p)a_p(h')$. If that were the system of $g$, then $a_p(h') = \chi_S(p)a_p(g) = a_p(f)$ for almost all $p$, and $h' = f$ by Theorem 13.42(i). So $\iota(f^\perp)$ lies in the sum of the eigenspaces of $\mathbb{T}^{(N_f)}$ for systems other than that of $g$, which is contained in the sum of the generalised eigenspaces of $\mathbb{T}$ other than that of $g^S$ (every generalised eigenspace of $\mathbb{T}$ lies in one eigenspace of $\mathbb{T}^{(N_f)}$), that is, in the $\mathbb{T}$-stable complement of $\mathbb{C}g^S$. □

Proposition 13.74(ii) is the proof of Theorem thm:twist. Part (iii), combined with Proposition 13.64, is the transport from $f$ to $g^S$ in the proof of the inequality $\leq$ of the paper's Lemma lem:step1, and gives a divisibility that holds at every prime.

**Theorem 13.75** ($r_f$ divides the congruence module of $g^S$)**.** *Let $g$ be a newform of weight 2 and level $M$ and $f$ a newform of weight 2 and level $N$, both with rational coefficients. Let $S$ be a finite set of odd primes, $D = \prod_{q\in S} q$, $\chi_S = \prod_{q\in S} \chi_{q^*}$ and $\iota = \iota_S$ (Proposition 13.70(vi)), and assume*

(a) *$D^2 \mid N$;*

(b) *$\iota(g) = f$, that is, $a_n(f) = \chi_S(n)a_n(g)$ for all $n \geq 1$;*

(c) *$M^S = N$, where $M^S = M\prod_{q\in S} q^{\delta_q(g)}$ (Definition 13.54).*

*Then $g^S \in S_2(N)$ is a normalised eigenform of $\mathbb{T} = \mathbb{T}_N$ whose generalised eigenspace is the line $\mathbb{C}g^S$; its eigencharacter $\pi\colon \mathbb{T} \to \mathbb{Z}$ satisfies the hypotheses of Proposition 13.64; and $r_f$ divides $r_\pi = \#\mathbb{T}/(\ker\pi + \operatorname{Ann}_{\mathbb{T}} \ker\pi)$, the positive generator of $\eta^{\mathrm{Hecke}}(g^S) = \pi(\operatorname{Ann}_{\mathbb{T}} \ker\pi)$. Equivalently, $v_\ell(r_f) \leq v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ for every prime $\ell$.*

*Proof.* By (c) and Proposition 13.55(iii),(iv), $g^S \in S_2(N)$ is a normalised eigenform of $\mathbb{T}_N$ and its generalised eigenspace is $\mathbb{C}g^S$. Its eigenvalues $\pi(T_n) = a_n(g^S)$ are integers, so $\pi$ is $\mathbb{Q}$-valued

and Proposition 13.64 applies. By the definition of $r_f$ there is $h \in f^{\perp} \cap S_2(N;\mathbb{Z})$ with $f \equiv h \pmod{r_f}$. The hypotheses of Proposition 13.74(iii) hold with $N_f = N$: $D^2 \mid N$ by (a), $\iota(g) = f$ by (b), and $g^S \in S_2(N)$ with a line as generalised eigenspace. So $g^S \equiv \iota(h) \pmod{r_f}$ with $\iota(h)$ an integral form in the $\mathbb{T}$-stable complement of $\mathbb{C}g^S$, and Proposition 13.64(ii) gives $r_f \mid r_\pi$. □

**Corollary 13.76** (the case of the paper)**.** *Let $E/\mathbb{Q}$ be an elliptic curve of conductor $N$ with newform $f = f_E$, let $S$ be the set of odd primes $q$ at which $E$ has Kodaira type $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$ ($n \geq 1$), and let $E_S = E \otimes \chi_S$, of conductor $N_S$, with newform $g = f_{E_S}$. Let $\pi\colon \mathbb{T}_N \to \mathbb{Z}$ be the eigencharacter of $g^S$. Then $r_E$ divides the positive generator of $\eta^{\mathrm{Hecke}}(g^S) = \pi(\mathrm{Ann}_{\mathbb{T}_N} \ker \pi)$, as integers: $v_\ell(r_E) \leq v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ for every prime $\ell$, including $\ell = 2$, the primes dividing $N$ and the primes of $S$.*

*Proof.* We check (a) to (c) of Theorem 13.75 with $M = N_S$, using the twist lemmas of Chapter 10 (the paper's Lemma lem:twist (i), (ii)). At $q \in S$, $E_S$ has good reduction if $E$ has type $\mathrm{I}_0^*$ and multiplicative reduction if $E$ has type $\mathrm{I}_n^*$, so $\delta_q(g) = 2$, resp. 1, $v_q(N_S) = 0$, resp. 1, and $v_q(N) = 2$ (the conductor exponent of the twist, Chapter 10). At a prime $p \notin S$ the character $\chi_S$ is unramified (at $p = 2$ because each $q^* \equiv 1 \pmod 4$), and $E$, $E_S$ have the same reduction type and conductor exponent. Hence $v_q(N) = 2$ for $q \in S$, which is (a), and $N_S \prod_{q \in S} q^{\delta_q(g)} = N$, which is (c). For (b): $V_\ell E_S \simeq V_\ell E \otimes \chi_S$, so at a prime $p \nmid D$ of good reduction $a_p(E_S) = \chi_S(p) a_p(E)$ (traces of Frobenius); at $p \nmid D$ of multiplicative reduction $a_p = \pm 1$ according as the reduction is split or not, and the unramified twist changes the splitting exactly when $\chi_S(p) = -1$ (Chapter 10); at $p \nmid D$ of additive reduction both are 0. With the modularity of $E$ and $E_S$ this is $a_p(f) = \chi_S(p) a_p(g)$ for every prime $p \nmid D$. Both $f$ and $\iota(g)$ are normalised with multiplicative coefficients satisfying the same prime-power relations at $p \nmid D$ (a prime $p \nmid D$ divides $N$ iff it divides $N_S$), and $a_{q^r}(f) = 0 = a_{q^r}(\iota(g))$ for $q \in S$, $r \geq 1$, by Proposition 13.44(iii) as $q^2 \mid N$. So $\iota(g) = f$. The theorem gives $r_E = r_f \mid r_\pi$. □

*Remark* 13.77 (Step 1 of the proof of Theorem thm:quant is an equality). The paper's Lemma lem:step1 states: "In this situation, $v_\ell(r_E) \leq v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ for every prime $\ell$", with equality at every prime $\ell \notin S$. The inequality $\leq$ is Corollary 13.76: it transports congruences from $f$ to $g^S$ along $\iota$. The opposite inequality, for $\ell \notin S$, transports congruences from $g^S$ to $f$ inside the $\mathfrak{m}$-component; its inputs from this chapter are Proposition 13.64, Proposition 13.66(iv) and Corollary 13.51, and it is proved in Chapter 20. Together, for every prime $\ell \notin S$,

$$v_\ell(r_E) = v_\ell(\eta^{\mathrm{Hecke}}(g^S)),$$

with no hypothesis on $\ell$ beyond $\ell \notin S$, in particular whether or not $\ell \in S_f$; under the hypotheses of Theorem thm:quant one has $\ell \nmid N$, so $\ell \notin S$. At level 539 the two modules are those of $(77\mathrm{c})^{\{7\}}$ and $(11\mathrm{a})^{\{7\}}$, of orders $288 = r(\texttt{539c1})$ and $504 = r(\texttt{539d1})$ (Example 13.85): the corollary explains why $r_E$ divides them. The equality at $\ell = 2$ is covered by the lemma; at $\ell = 7 \in S$ it is observed, not proved.

## 13.10 The $L$-function of a newform; Rankin-Selberg

This section attaches to a normalised eigenform its Dirichlet series and Euler product, states the functional equation, and states the Rankin-Selberg formula that relates the Petersson norm $\langle f, f \rangle$ to the symmetric square $L$-function. In the paper these appear in Section 1 ("The classical route to the degree is the Petersson norm", and "$\langle f, f \rangle$ is the special value of the naive symmetric

square $L$-function at $s=2$", which is Theorem 13.80(iii) up to the explicit factor $N/(8\pi^3)$; see also Warning 13.82) and in Section 1.1 ($\delta_p$ as the degree of a local factor). The relation with the true symmetric square, with all bad factors, is Chapter 19's.

**Proposition 13.78** (Euler products)**.** *Let $f$ be a normalised eigenform of $\mathbb{T}_N$ of weight $k$. The series $L(f,s)=\sum_{n\ge1}a_n(f)n^{-s}$ converges absolutely for $\operatorname{Re}s>k/2+1$ (Chapter 12), and there*

$$L(f,s)=\prod_{p\nmid N}(1-a_pp^{-s}+p^{k-1-2s})^{-1}\prod_{p\mid N}(1-a_pp^{-s})^{-1}.$$

*Write $L_p(f,s)$ for the factor at $p$. For a newform of weight 2: $L_p(f,s)^{-1}=(1-\alpha_pp^{-s})(1-\beta_pp^{-s})$ for $p\nmid N$, $(1\mp p^{-s})$ for $p\parallel N$ with $a_p=\pm1$, and 1 for $p^2\mid N$. Moreover $L(f^\Sigma,s)=L(f,s)\prod_{p\in\Sigma}L_p(f,s)^{-1}$ for a newform $f$, and for quadratic $\chi$ of conductor $r$ the factor of $L(f\otimes\chi,s)$ at $p\nmid rN$ is $(1-\chi(p)a_pp^{-s}+p^{k-1-2s})^{-1}$.*

*Proof.* By multiplicativity (Proposition 13.19(ii)) and absolute convergence,

$$L(f,s)=\prod_p\sum_{r\ge0}a_{p^r}p^{-rs}.$$

In detail: for $x>0$ the finite product $\prod_{p\le x}\sum_r a_{p^r}p^{-rs}$ of absolutely convergent series can be multiplied out, and by unique factorisation and $a_{mn}=a_ma_n$ for coprime $m,n$ it equals $\sum_n a_nn^{-s}$ over the $n$ all of whose prime factors are at most $x$; the difference from $L(f,s)$ is bounded by $\sum_{n>x}|a_nn^{-s}|$, which tends to 0 as $x\to\infty$. The series $\sum_r a_{p^r}X^r$ is the image under $\phi_f$ of $\sum_r T_{p^r}X^r$, which is $(1-a_pX+p^{k-1}X^2)^{-1}$ or $(1-a_pX)^{-1}$ by Theorem 13.10(v); put $X=p^{-s}$. The weight-2 factors use Proposition 13.44(iii). For $f^\Sigma$, whose coefficients at $p\in\Sigma$ vanish, the factor at $p\in\Sigma$ is 1, and the others are those of $f$. For $f\otimes\chi$, the coefficients $\chi(n)a_n$ are multiplicative, and

$$\sum_r\chi(p)^ra_{p^r}X^r=(1-a_p\chi(p)X+p^{k-1}\chi(p)^2X^2)^{-1},\qquad\chi(p)^2=1.$$

□

**Theorem 13.79** (functional equation)**.** *Let $f\in S_k(N)$ with $f|_kW_N=\varepsilon_Nf$, $\varepsilon_N=\pm1$ (for instance a newform, Proposition 13.44(i)). Then*

$$\Lambda(f,s)=N^{s/2}(2\pi)^{-s}\Gamma(s)L(f,s)=N^{s/2}\int_0^\infty f(iy)y^{s-1}\,dy$$

*extends to an entire function, bounded in vertical strips, with $\Lambda(f,s)=i^k\varepsilon_N\Lambda(f,k-s)$. In weight 2 the sign is $-\varepsilon_N$.*

*Sketch of proof.* For $\operatorname{Re}s$ large, integrate $f(iy)=\sum a_ne^{-2\pi ny}$ termwise against $y^{s-1}$, using $\int_0^\infty e^{-2\pi ny}y^{s-1}dy=(2\pi n)^{-s}\Gamma(s)$ (substitute $t=2\pi ny$ in the integral defining $\Gamma(s)$); the interchange of sum and integral is justified by absolute convergence, and gives

$$\int_0^\infty f(iy)y^{s-1}dy=(2\pi)^{-s}\Gamma(s)L(f,s).$$

Next, $W_N(iy)=-1/(Niy)=i/(Ny)$ and

$$(f|_kW_N)(iy)=N^{k/2}(Niy)^{-k}f\Big(\frac{i}{Ny}\Big)=N^{-k/2}i^{-k}y^{-k}f\Big(\frac{i}{Ny}\Big),$$

so the relation $f|_k W_N = \varepsilon_N f$ at $z = iy$ reads

$$f\Big(\frac{i}{Ny}\Big) = \varepsilon_N i^k N^{k/2} y^k f(iy).$$

Split the integral at $y = 1/\sqrt{N}$, the fixed point of $y \mapsto 1/(Ny)$. On $(0, 1/\sqrt{N})$ substitute $y = 1/(Nt)$, $dy = -dt/(Nt^2)$, with $t$ running over $(1/\sqrt{N}, \infty)$:

$$\int_0^{1/\sqrt{N}} f(iy) y^{s-1} dy = \int_{1/\sqrt{N}}^{\infty} \varepsilon_N i^k N^{k/2} t^k f(it)\,(Nt)^{1-s}\,\frac{dt}{Nt^2} = \varepsilon_N i^k N^{k/2-s} \int_{1/\sqrt{N}}^{\infty} f(it) t^{k-s-1} dt.$$

Multiplying by $N^{s/2}$,

$$\Lambda(f,s) = N^{s/2} \int_{1/\sqrt{N}}^{\infty} f(iy) y^{s-1} dy + i^k \varepsilon_N N^{(k-s)/2} \int_{1/\sqrt{N}}^{\infty} f(iy) y^{k-s-1} dy.$$

Both integrals converge for all $s$, because $f(iy)$ decays exponentially as $y \to \infty$ (its $q$-expansion starts at $n = 1$), and they define entire functions of $s$, bounded in vertical strips since $|y^{s-1}| = y^{\operatorname{Re} s - 1}$. This gives the continuation and the bound. For the symmetry, write the display as $\Lambda(f,s) = A(s) + i^k \varepsilon_N A(k-s)$ with $A(s) = N^{s/2} \int_{1/\sqrt{N}}^{\infty} f(iy) y^{s-1} dy$. Then $i^k \varepsilon_N \Lambda(f, k-s) = i^k \varepsilon_N A(k-s) + (i^k \varepsilon_N)^2 A(s) = \Lambda(f,s)$, as $(i^k \varepsilon_N)^2 = 1$ for even $k$. In weight 2, $i^2 = -1$ gives the sign $-\varepsilon_N$. This is the Mellin transform of RHB's integral transforms; see [52, Section 5.10], [137, Section 4.3]. □

Checks: `11a1` has $\varepsilon_{11} = -1$ (Chapter 12), sign $+1$ and rank 0; `37a1` has $\varepsilon_{37} = +1$, sign $-1$ and rank 1 (PARI's `ellrootno`).

**Theorem 13.80** (Rankin-Selberg)**.** *Let $f \in S_k(N)$ and $D(f,s) = \sum_{n \ge 1} |a_n(f)|^2 n^{-s}$.*

*(i) $D(f,s)$ converges for $\operatorname{Re} s > k$, continues meromorphically to $\mathbb{C}$, is holomorphic on $\operatorname{Re} s \ge k$ except for a simple pole at $s = k$, and*

$$\operatorname{Res}_{s=k} D(f,s) = \frac{3\,(4\pi)^k}{\pi\,\Gamma(k)\,\mu(N)}\,\langle f, f\rangle, \qquad \text{in weight } 2: \quad \operatorname{Res}_{s=2} D(f,s) = \frac{48\pi}{\mu(N)}\langle f, f\rangle,$$

*with the unnormalised Petersson product.*

*(ii) At a good prime $p$ of a newform $f$ of weight 2 with real coefficients,*

$$\sum_{r} a_{p^r}^2 X^r = \frac{1 + pX}{(1 - \alpha_p^2 X)(1 - pX)(1 - \beta_p^2 X)};$$

*at $p \,\|\, N$ the series is $(1-X)^{-1}$, and at $p^2 \mid N$ it is 1.*

*(iii) Hence, for a newform of weight 2 with rational coefficients,*

$$D(f,s) = \frac{\zeta^{(N)}(s-1)\, L^{(N)}(\operatorname{Sym}^2 f, s)}{\zeta^{(N)}(2s-2)} \prod_{p \mid N} \sum_{r} a_{p^r}^2 p^{-rs},$$

*where $L(\operatorname{Sym}^2 f, s)$ is normalised as in the paper and in PARI's `lfunsympow` (Appendix B, Appendix B) and the superscript $(N)$ removes the factors at $p \mid N$. The product over $p \mid N$*

*is the naive factor at $p$, so $D(f,s)=\zeta^{(N)}(s-1)L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)/\zeta^{(N)}(2s-2)$, with $L^{\mathrm{nv}}$ the naive symmetric square (the true factors at the good and multiplicative primes, 1 at the additive ones), and for every $N$*

$$\langle f,f\rangle=\frac{N\,L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2)}{8\pi^3}.$$

*If $N$ is squarefree, $L^{\mathrm{nv}}=L$ and $\langle f,f\rangle=N\,L(\mathrm{Sym}^2 f,2)/(8\pi^3)$.*

*(iv) In general $D(f,s)$ is* not *the symmetric square $L$-function: it has the extra factor $1+p^{1-s}$ at every good prime, and at the bad primes the relation with $L(\mathrm{Sym}^2 f,s)$ involves correction factors (Chapter 19; [49, Lemma 2.12], [183]).*

*Sketch of proof of (i); proof of (ii) and (iii).* (i) Let $E(z,s)=\sum_{\gamma\in\Gamma_\infty\backslash\Gamma_0(N)}\mathrm{Im}(\gamma z)^s$ be the non-holomorphic Eisenstein series of $\Gamma_0(N)$ at $\infty$ ($\Gamma_\infty$ the stabiliser of $\infty$, with $\pm1$). Unfolding the integral of $|f|^2y^kE(z,s)$ over $\Gamma_0(N)\backslash\mathcal{H}$ to the strip $\Gamma_\infty\backslash\mathcal{H}=\{0\le x\le 1\}$ and integrating in $x$ (Parseval) gives

$$\int_{\Gamma_0(N)\backslash\mathcal{H}}|f|^2y^kE(z,s)\,d\mu=\int_0^\infty\sum_n|a_n|^2e^{-4\pi ny}y^{s+k-2}dy=\frac{\Gamma(s+k-1)}{(4\pi)^{s+k-1}}D(f,s+k-1).$$

$E(z,s)$ continues meromorphically with a simple pole at $s=1$ of residue $1/\operatorname{vol}(\Gamma_0(N)\backslash\mathcal{H})=3/(\pi\mu(N))$, constant in $z$ (the fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$ has area $\pi/3$, and that of $\Gamma_0(N)$ is $\mu(N)$ times larger). Taking residues at $s=1$ on both sides, the left side gives $\frac{3}{\pi\mu(N)}\int|f|^2y^k\,d\mu=\frac{3}{\pi\mu(N)}\langle f,f\rangle$ and the right side $\frac{\Gamma(k)}{(4\pi)^k}\operatorname{Res}_{s=k}D(f,s)$; solving for the residue gives (i), and with $k=2$, $\frac{3(4\pi)^2}{\pi\Gamma(2)}=48\pi$ [155], [209, Section 1], [183]. (ii) Write $\alpha=\alpha_p$, $\beta=\beta_p$, $\alpha\beta=p$; then $a_{p^r}=(\alpha^{r+1}-\beta^{r+1})/(\alpha-\beta)$ (Remark 13.11), so that $a_{p^r}^2=\big(\alpha^{2(r+1)}-2p^{r+1}+\beta^{2(r+1)}\big)/(\alpha-\beta)^2$, using $(\alpha\beta)^{r+1}=p^{r+1}$. Summing three geometric series,

$$\sum_r a_{p^r}^2X^r=\frac{1}{(\alpha-\beta)^2}\Big(\frac{\alpha^2}{1-\alpha^2X}-\frac{2p}{1-pX}+\frac{\beta^2}{1-\beta^2X}\Big)=\frac{1+pX}{(1-\alpha^2X)(1-pX)(1-\beta^2X)}.$$

For the second equality, put $A=\alpha^2$, $B=\beta^2$, so $AB=p^2$, and bring the bracket to the common denominator. Its numerator is

$$A(1-pX)(1-BX)-2p(1-AX)(1-BX)+B(1-AX)(1-pX).$$

The constant term is $A+B-2p=(\alpha-\beta)^2$; the coefficient of $X$ is $-A(p+B)-B(p+A)+2p(A+B)=p(A+B)-2AB=p(A+B-2p)=p(\alpha-\beta)^2$; and the coefficient of $X^2$ is $pAB-2pAB+pAB=0$. So the numerator is $(\alpha-\beta)^2(1+pX)$. (The identity also holds when $\alpha=\beta$, by continuity in $a_p$.) At $p\,\|\,N$, $a_{p^r}^2=1$; at $p^2\mid N$, $a_{p^r}=0$ for $r\ge1$. (iii) At a good $p$, $L_p(\mathrm{Sym}^2 f,s)^{-1}=(1-\alpha^2p^{-s})(1-p^{1-s})(1-\beta^2p^{-s})$ (Appendix B), and $1+p^{1-s}=(1-p^{2-2s})/(1-p^{1-s})$ is the factor of $\zeta(s-1)/\zeta(2s-2)$; multiply over $p\nmid N$. By (ii) the local factor at $p\mid N$ is $(1-p^{-s})^{-1}$ if $p\,\|\,N$, which is $L_p(\mathrm{Sym}^2 f,s)$ at a multiplicative prime (Appendix B), and 1 if $p^2\mid N$, the naive factor at an additive prime; this gives the formula with $L^{\mathrm{nv}}$. The residue of $\zeta^{(N)}(s-1)$ at $s=2$ is $\prod_{p\mid N}(1-p^{-1})$ and $\zeta^{(N)}(2)=\frac{\pi^2}{6}\prod_{p\mid N}(1-p^{-2})$; since $(1-p^{-1})/(1-p^{-2})=(1+p^{-1})^{-1}$ and $\prod_{p\mid N}(1+p^{-1})=\mu(N)/N$,

$$\operatorname{Res}_{s=2}D=\frac{6}{\pi^2}\prod_{p\mid N}(1+p^{-1})^{-1}L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2)=\frac{6N}{\pi^2\mu(N)}L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2)$$

(the residue of $\zeta^{(N)}(s-1)$ times the value at $s=2$ of the other factors). Comparing with $\operatorname{Res}_{s=2} D = \frac{48\pi}{\mu(N)}\langle f,f\rangle$ from (i) gives $\langle f,f\rangle = \frac{6N}{48\pi^3} L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2) = \frac{N\,L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2)}{8\pi^3}$. For squarefree $N$ there are no additive primes. □

**Example 13.81** (numerical checks)**.** With PARI, to 30 to 40 digits ($\langle f,f\rangle$ is $\mu(N)$ times the value of `mfpetersson`; $L(\mathrm{Sym}^2 f,s)$ comes from `lfunsympow` and its bad factors from `lfuneuler`): for `11a1`, $\langle f,f\rangle = 0.0469001478734951878\ldots$ and $L(\mathrm{Sym}^2 f,2) = 1.0575992445909578\ldots$, and $11\,L(\mathrm{Sym}^2 f,2)/(8\pi^3)$ agrees with $\langle f,f\rangle$ to all digits; for `37a1`, $\langle f,f\rangle = 0.3717541475106960\ldots = 37L(\mathrm{Sym}^2 f,2)/(8\pi^3)$. The residue formula (i), with $D(f,s)$ written as in (iii) with the local factors at $p \mid N$ computed from (ii), holds to 30 digits for `11a1`, `37a1`, `20a1`, `121a1` and `99a1`; for the last three, whose levels are not squarefree, $N\,L(\mathrm{Sym}^2 f,2)/(8\pi^3)$ differs from $\langle f,f\rangle$ ($0.0542\ldots$ against $0.0813\ldots$ for `20a1`), while $N\,L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2)/(8\pi^3)$ agrees with it to 18 digits in all five cases (Exercise 13.100).

*Warning* 13.82 (the Rankin series is not the symmetric square)*.* The naive adjoint $L$-function is not the Euler product of $\sum a_n(f)^2 n^{-s}$. By Theorem 13.80(ii),(iii), at every good prime the Rankin series carries the extra factor $1+p^{1-s}$, and globally $\sum a_n^2 n^{-s} = \zeta^{(N)}(s-1)L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)/\zeta^{(N)}(2s-2)$, where $L^{\mathrm{nv}}$ is the naive symmetric square of Appendix B (Appendix B). The squarefree formula $\langle f,f\rangle = NL(\mathrm{Sym}^2 f,2)/(8\pi^3)$ uses the unnormalised Petersson product of Convention 13.1(d) and PARI's normalisation of $L(\mathrm{Sym}^2 f,s)$; the general formula with its bad factors is Chapter 19's.

## 13.11 Worked examples from the paper's world

This section puts the theory to work on the levels that the paper's examples live at: 88 (where the full Hecke algebra is not reduced), the small levels where congruence numbers and modular degrees can be compared, the twist pairs of Theorem thm:twist, and level 121. All numbers were computed with PARI/GP 2.17 through `cypari2`, against Cremona's tables for conductors below 10,000 [35].

**Example 13.83** (level 88)**.** $S_2(88)$ has dimension 9: three newforms of level 88, the newform $f_{44}$ of level 44 (old twice: $f_{44}(z)$, $f_{44}(2z)$), and $f_{11}$ (old four times: $f_{11}(2^j z)$, $0 \le j \le 3$). The operator $U_2$ has characteristic polynomial $X^7(X^2+2X+2)$ and minimal polynomial $X^2(X^2+2X+2)$. On $V_{f_{11},2,3}$ its matrix is the $4\times 4$ matrix of Example 13.49: a Jordan block of size 2 at 0, with kernel spanned by $f_{11}^{\{2\}} = f_{11} + 2f_{11}(2z) + 2f_{11}(4z)$ and generalised kernel spanned by it and $f_{11}^{\{2\}}(2z)$, and the two stabilisations $f_{11} - \beta f_{11}(2z)$, $\alpha,\beta = -1 \pm i$. On $V_{f_{44},2,1}$, $U_2$ is nilpotent ($a_2(f_{44}) = 0$, Proposition 13.48). So $\mathbb{T}_{88} \otimes \mathbb{Q}$ is not reduced (Theorem 13.59(iii)); the step $\wp \cap I = 0$ of Lemma lem:ddt417 must be proved (as the paper does, citing this level) rather than read off from reducedness; and the paper's "the eigenvalue 0 with multiplicity one" (Step 1 of the proof of Theorem thm:quant) is a statement about the minimal level $M^{\Sigma}$, which fails one level higher: $f_{11}^{\{2\}}$ spans its generalised eigenspace at level 44, not at level 88 (Proposition 13.55(v)). Also $\operatorname{rank}\mathbb{T}_{88} = 9$, the pairing has determinant $-1$, $[\mathbb{T}_{88} : \mathbb{Z}T_1 + \cdots + \mathbb{Z}T_9] = 3$, and the congruence number of the newform of 88a is 16, while $\deg \varphi_{\texttt{88a1}} = 8$ ($2^3 \mid 88$).

**Example 13.84** (congruence numbers)**.** For each rational newform $f$ below we computed $r_f$ as the index of $\mathbb{Z}f \oplus (f^c \cap S_2(N;\mathbb{Z}))$ in $S_2(N;\mathbb{Z})$ (Proposition 13.30, Proposition 13.62(ii)), with $f^c$ the image of $t - \phi_f(t)$ for a random $t \in \mathbb{T}$ for which $\phi_f(t)$ is a simple root of the characteristic polynomial of $t$; and compared with the modular degree of the optimal curve from Cremona's `degphi` file:

| curve | 11a1 | 33a1 | 37a1 | 37b1 | 44a1 | 77a1 | 77b1 |
|---|---|---|---|---|---|---|---|
| $r_f$ | 1 | 3 | 2 | 2 | 2 | 4 | 20 |
| $\deg\varphi_E$ | 1 | 3 | 2 | 2 | 2 | 4 | 20 |

| curve | 77c1 | 88a1 | 99a1 | 99b1 | 99c1 | 99d1 |
|---|---|---|---|---|---|---|
| $r_f$ | 6 | 16 | 12 | 12 | 12 | 6 |
| $\deg\varphi_E$ | 6 | 8 | 4 | 12 | 12 | 6 |

| curve | 121a1 | 121b1 | 121c1 | 121d1 | 539a1 | 539b1 | 539c1 | 539d1 |
|---|---|---|---|---|---|---|---|---|
| $r_f$ | 6 | 4 | 6 | 24 | 320 | 192 | 288 | 504 |
| $\deg\varphi_E$ | 6 | 4 | 6 | 24 | 320 | 192 | 288 | 72 |

The three differences (88a1 at 2, 99a1 at 3, 539d1 at 7) are at primes $p$ with $p^2 \mid N$, as the theorem of Agashe, Ribet and Stein allows (Chapter 14). The paper's remark after the Example of Section 3 quotes "the curve 99A1, of conductor $9 \cdot 11$, with modular degree 4 and congruence number 12" from [3, Table 1], and adds "539d1 has $\deg\varphi_E = 72$ and $r_E = 504 = 7 \cdot 72$, with $7^2 \mid 539$"; both agree with the table.

**Example 13.85** (the twist pairs of Theorem thm:twist)**.** (a) Case (c) ($N_S = N$): `121a1` has type II at 11, $f_{121a} \otimes \chi_{-11} = f_{121c}$ (Example 13.72; both have $a_{11} = 0$, so the naive twist is the newform), and $r(121a) = r(121c) = 6$, as Theorem thm:twist predicts. (b) Case $\mathrm{I}_n^*$: `539c1` has type $\mathrm{I}_3^*$ at 7, and $f_{539c} = f_{77c} \otimes \chi_{-7}$ (Atkin-Li, $7 \parallel 77$); $r(77c) = 6$ divides $r(539c) = 288$. At $\ell = 3$: $v_3(288) = 2 = v_3(6) + 1$, where $1 = v_3(7^2 - 1)$ is the Euler term $e_3(E, 7)$ of Theorem thm:quant (the paper's Definition def:terms; Chapter 19). (c) Case $\mathrm{I}_0^*$: `539d1` has type $\mathrm{I}_0^*$ at 7, $f_{539d} = f_{11a} \otimes \chi_{-7}$ ($7 \nmid 11$), $r(11a) = 1$ divides 504, and $v_3(504) = 2 = 0 + e_3(E, 7)$. (d) Step 1 of the proof of Theorem thm:quant at level 539, $S = \{7\}$: the depleted forms $g^S$ are $(77c)^{\{7\}} = g + g(7z)$ and $(11a)^{\{7\}} = f_{11} + 2f_{11}(7z) + 7f_{11}(49z)$ (Example 13.57), each spanning its generalised eigenspace in $S_2(539)$; their Hecke congruence modules, computed as the index of $\mathbb{Z}g^S \oplus ((g^S)^c \cap S_2(539;\mathbb{Z}))$ with the $\mathbb{T}$-stable complement (Proposition 13.62), have orders 288 and 504, equal to $r(539c)$ and $r(539d)$, consistent with Corollary 13.76 and Remark 13.77; and $\iota((77c)^{\{7\}}) = f_{539c}$, $\iota((11a)^{\{7\}}) = f_{539d}$ (checked to $n = 3000$; $\iota$ kills multiples of 7, so this is Example 13.72). At level 44 the module of $f_{11}^{\{2\}}$ has order 2, and at level 121 that of $f_{11}^{\{11\}} = f_{11} - f_{11}(11z)$ has order $24 = r(121d)$.

**Example 13.86** (level 121)**.** $S_2(121)$ has dimension 6: the newforms of 121a, 121b, 121c, 121d (all rational, with $a_{11} = 0$ as $11^2 \mid 121$), and $V_{f_{11},11,1} = \mathrm{span}\{f_{11}(z), f_{11}(11z)\}$. On the latter $U_{11}$ has matrix $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ ($a_{11}(f_{11}) = 1$, $11 \parallel 11$), eigenvectors $f_{11}$ (eigenvalue 1) and $f_{11}^{\{11\}} = f_{11} - f_{11}(11z)$ (eigenvalue 0, the depleted form, of level $M^{\{11\}} = 121$). On $S_2(121)$ the characteristic and minimal polynomials of $U_{11}$ are $X^5(X-1)$ and $X(X-1)$: $U_{11}$ is semisimple. The paper's Example of Section 3, case (b): "$f_{11a} \otimes \chi_{-11}$ is the form of 121d, and indeed $f_{121a} \equiv f_{121d} \pmod 3$ for all $n$ (Sturm's bound 22)". The first statement is Theorem 13.71(ii) ($11 \parallel 11$) and Example 13.72; the congruence is proved in Chapter 12 by Sturm's bound. Finally $T_1, \dots, T_6$ are linearly dependent although $\mathrm{rank}\,\mathbb{T}_{121} = 6$, while $T_1, \dots, T_{22}$ span $\mathbb{T}_{121}$ (Example 13.25).

```
\\ The computations of this chapter, in outline (PARI/GP 2.17).
mf = mfinit([88,2],1);  A = mfheckemat(mf,2);      \\ U_2 on S_2(88)
```

```
factor(charpoly(A))                                   \\ x^7 (x^2+2x+2)
factor(minpoly(A))                                    \\ x^2 (x^2+2x+2)
\\ integral basis: saturate the q-expansion lattice up to B*max(B,41)
B = mfsturm([88,2]); C = mfcoefs(mf, B*41)[2..B*41+1,]; Z = matrixqz(C,-2);
\\ Petersson Gram matrix at level 22 (mfpetersson divides by mu(22) = 36)
F = mfbasis(mfinit([11,2],1))[1]; mf22 = mfinit([22,2],1);
S = [mfsymbol(mf22,F), mfsymbol(mf22,mfbd(F,2))];
G = matrix(2,2,i,j,mfpetersson(S[i],S[j])); G/G[1,1]  \\ [1,-1/3;-1/3,1/4]
```

## 13.12 Exercises

**Exercise 13.87.** Let $p \mid N$ and $U_p^\star = [\Gamma_0(N)\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)\Gamma_0(N)]_2$. Show that $U_p^\star = w_N U_p w_N = U_p^\dagger$. Compute $U_2^\star$ on $f_{11}(z)$, $f_{11}(2z)$ at level 22, and show that $U_2^\star$ does not commute with $U_2$, so that $\mathbb{T}_{22}$ and $U_2^\dagger$ generate a noncommutative algebra. (Answer: $U_2^\star f_{11} = 4f_{11}(2z)$ and $U_2^\star f_{11}(2z) = -\frac12 f_{11} - 2f_{11}(2z)$; $U_2U_2^\star$ and $U_2^\star U_2$ have the matrices $\left(\begin{smallmatrix} 4 & -1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ -4 & 4 \end{smallmatrix}\right)$.)

**Exercise 13.88.** Using only Theorem 13.10 and $\dim S_{12}(1) = 1$, prove $\tau(mn) = \tau(m)\tau(n)$ for coprime $m, n$, and $\tau(p)^2 - \tau(p^2) = p^{11}$. Check with PARI that $\tau(2) = -24$, $\tau(4) = -1472$. (Answer: $576 + 1472 = 2048 = 2^{11}$.)

**Exercise 13.89.** For `11a1` check with PARI (`ellan`) that $a_{mn} = a_m a_n$ for coprime $m, n \le 200$, that $a_{p^{r+1}} = a_p a_{p^r} - p a_{p^{r-1}}$ for $p \neq 11$, and that $a_{11^r} = 1$. Explain why $a_{11^r} = a_{11}^r$ is a statement about the operator $U_{11}$ of level 11, and why $a_4 = a_2^2$ fails ($a_4(f_{11}) = 2$, $a_2(f_{11}) = -2$) although $f_{11}$ lies in $S_2(22)$: at level 22 the form $f_{11}$ is not an eigenvector of $U_2$, while its stabilisations are, with $a_{2^r} = \alpha^r$.

**Exercise 13.90.** Let $g_1, \dots, g_d$ be a $\mathbb{Z}$-basis of $S_2(N;\mathbb{Z})$. Show, using Theorem 13.24, that $T_1, \dots, T_d$ are linearly independent iff the matrix $(a_n(g_i))_{1\le i,n\le d}$ is invertible, and that then $[\mathbb{T} : \mathbb{Z}T_1 + \cdots + \mathbb{Z}T_d] = |\det(a_n(g_i))|$. At level 67 ($d = 5$) compute a $\mathbb{Z}$-basis from $q$-expansions and show that $T_1, \dots, T_5$ span a sublattice of index 8, while $T_1, \dots, T_6$ span $\mathbb{T}_{67}$. (Hint: under $\mathbb{T} \simeq \mathrm{Hom}(S_2(N;\mathbb{Z}),\mathbb{Z})$, $T_n$ is the functional $a_n$. Answer for level 67: index 8; the first $n$ with $\mathbb{Z}T_1 + \cdots + \mathbb{Z}T_n = \mathbb{T}_{67}$ is 6. Likewise at level 85, $d = 7$, the index is 7 and $n = 8$ suffices.)

**Exercise 13.91.** (Point of Lemma lem:hecke.) $S_2(37)$ is spanned by the newforms of 37a and 37b, both rational, with $r = 2$ (Example 13.84). Show that $t \mapsto (\phi_{37\mathrm{a}}(t), \phi_{37\mathrm{b}}(t))$ identifies $\mathbb{T}_{37}$ with $\{(a,b) \in \mathbb{Z}^2 : a \equiv b \pmod 2\}$. Decompose $\mathbb{T}_{37} \otimes \mathbb{Z}_\ell$ into local factors for $\ell = 2$ and $\ell = 3$, compute $\mathrm{Ann}(\ker \pi)$ and $\eta(\pi)$ for $\pi = \phi_{37\mathrm{a}}$, and decide in each case whether the factor through which $\pi$ does not factor lies in $\ker \pi$ or in $\mathrm{Ann}(\ker \pi)$. (Answer: $\mathbb{T}_2$ is local and $\eta(\pi) = 2\mathbb{Z}_2$; $\mathbb{T}_3 = \mathbb{Z}_3 \times \mathbb{Z}_3$, the second factor lies in $\ker \pi$, and $\eta(\pi) = \mathbb{Z}_3$. Use Theorem 13.60(iv) for the index.)

**Exercise 13.92.** At level 22, with the Gram matrix of Example 13.36, show that the two eigenvectors $f_{11} - \beta f_{11}(2z)$ of $U_2$ ($\beta = -1 \pm i$) are not orthogonal to each other, and conclude that no basis of $S_2(22)$ consists of orthogonal eigenvectors of $U_2$. (Hint: with $\beta_1 = -1+i$, $\beta_2 = -1-i$, the product is $(1 + \frac13(\beta_1 + \bar\beta_2) + \frac14\beta_1\bar\beta_2)\langle f_{11}, f_{11}\rangle_{22}$, and $\bar\beta_2 = \beta_1$ gives $(\frac13 + \frac{i}{6})\langle f_{11}, f_{11}\rangle_{22} \neq 0$.)

**Exercise 13.93.** Let $N = 176 = 2^4 \cdot 11$, so that $\dim S_2(176) = 19$ and $\dim S_2^{\mathrm{new}}(176) = 5$. Using Theorem 13.59 and Proposition 13.48, list the spaces $V_h$ and the Jordan blocks of $U_2$ on each, predict the characteristic and minimal polynomials of $U_2$ on $S_2(176)$, and check them with

PARI's `mfheckemat`. (Answer: $X^{17}(X^2+2X+2)$ and $X^3(X^2+2X+2)$; blocks of size 3 at 0 on $V_{f_{11},2,4}$ and $V_{f_{44},2,2}$, of size 2 on the three $V_{h,2,1}$ with $h$ new of level 88, of size 1 on the five newforms of level 176.)

**Exercise 13.94.** For $f = f_{11}$ compute $f^\Sigma$ and $M^\Sigma$ for $\Sigma = \{2\}, \{11\}, \{2,11\}, \{3,5\}$ (answers: 44, 121, 484, 2475), and check with PARI that $f^\Sigma$ lies in $S_2(M^\Sigma)$ but not in $S_2(M^\Sigma/p)$ for any $p \in \Sigma$. Show $L(f^\Sigma, s) = L(f,s)\prod_{p\in\Sigma} L_p(f,s)^{-1}$.

**Exercise 13.95.** (Step 1 of Theorem thm:quant.) For $g$ of class 77c and $g = f_{11}$, with $q = 7$ and $N = 539$, write the span of the $g(dz)$, $d \mid 539/M_g$, the matrix of $U_7$ on it, and show that the eigenvalue 0 has algebraic multiplicity one. Show that all other eigenvalues there are prime to 3. (Answer: $X(X+1)$ and $X(X^2+2X+7)$.)

**Exercise 13.96.** Compute $\dim S_2^{\mathrm{new}}(N)$ for $N = 88$, 121, 539 from the dimensions of $S_2(M)$, $M \mid N$, with Corollary 13.46, and compare with Example 13.47. Prove that $\dim S_2^{\mathrm{new}}(p^3) = \dim S_2(p^3) - 2\dim S_2(p^2) + \dim S_2(p)$.

**Exercise 13.97.** (Theorem thm:qual (b).) Let $g$ be a newform on $\Gamma_0(M)$ with $q \parallel M$, $\chi = \chi_{q^*}$, and $h = g\otimes\chi$, a newform on $\Gamma_0(Mq)$ (Theorem 13.71). Show that $h\otimes\chi = g^{\{q\}} \neq g$. Let $f$ and $f'$ be newforms with $a_p(f') = \chi(p)a_p(f)$ for almost all $p$, and suppose that the level of $f'$ is not $M$. Prove that $h \neq f$. (Answer: if $h = f$, then $a_p(g) = \chi(p)a_p(h) = a_p(f')$ for almost all $p$, so $g = f'$ by Theorem 13.42(i), and the levels would agree.)

**Exercise 13.98.** (Lemma lem:twist (iii).) Predict from Theorem 13.71 the conductor of each twist below, then check with PARI's `elltwist` and `ellglobalred` and identify the isogeny class from the $a_n$: `17a1` by $-3$, `21a1` by $-3$, `63a1` by $-3$, `27a1` by $-3$, `50a1` by 5. In which cases does the level drop, and in which is the naive twist a newform of the same level? (Answer: the classes 153c, 63a, 21a, 27a and 50b (the twist of `50a1` is `50b3`); the level is $\mathrm{lcm}(M, q^2)$ in the first two cases ($3 \nmid 17$, $3 \parallel 21$), drops in the third ($3^2 \mid 63$, and the twist undoes a twist), and is kept in the last two; `27a1` has complex multiplication by $\mathbb{Q}(\sqrt{-3})$, so its twist by $\chi_{-3}$ is isogenous to it.)

**Exercise 13.99.** (The remark after the Example of Section 3.) Compute $r_f$ for the newform of 99a by the lattice method of Example 13.84, compare with `ellmoddegree`, and explain why the discrepancy can only be at 3. Find the newforms of level 99 congruent to it modulo 3 at the primes $p \neq 3, 11$. (Answer: those of 99b and 99d; not those of 11a, 33a, 99c, which differ at $p = 2$.)

**Exercise 13.100.** (The naive symmetric square at an additive prime.) `20a1` has type IV$^*$ at 2 and multiplicative reduction at 5. Compute the Euler factors of PARI's `lfunsympow(E,2)` at 2 and 5 with `lfuneuler`, and check numerically that $\langle f, f\rangle = 20\,L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2)/(8\pi^3)$, whereas $20\,L(\mathrm{Sym}^2 f, 2)/(8\pi^3)$ is smaller by the factor $L_2(\mathrm{Sym}^2 f, 2)$. (Answer: $L_2(\mathrm{Sym}^2 f, s) = (1 - 2^{1-s})^{-1}$, so $L_2(\mathrm{Sym}^2 f, 2) = 2/3$; $L_5(\mathrm{Sym}^2 f, s) = (1-5^{-s})^{-1}$; $\langle f, f\rangle = 0.0813494568\ldots$ and $20L(\mathrm{Sym}^2 f, 2)/(8\pi^3) = 0.0542329712\ldots$.)

**Exercise 13.101.** Compute the sign of the functional equation of $L(f,s)$ for the newforms of `11a1`, `37a1`, `37b1`, `77c1` from Theorem 13.79 and the Atkin-Lehner eigenvalues $\varepsilon_N$ (PARI's `mfatkineigenvalues`), and compare with `ellrootno` and the ranks. (Answer: root numbers $+1$, $-1$, $+1$, $+1$.)

**Exercise 13.102.** At level 23 the Hecke algebra is $\mathbb{T}_{23} = \mathbb{Z}[T_2]$ with $T_2^2 + T_2 - 1 = 0$ (Chapter 6). Show that the two normalised eigenforms are Galois conjugate newforms with $K_f = \mathbb{Q}(\sqrt{5})$, and list the maximal ideals of $\mathbb{T}_{23}$ above 5 and above 11 together with the corresponding mod $\ell$ eigenforms $\sum \phi(T_n)q^n$ (Theorem 13.26). (Hint: 5 ramifies in $\mathbb{Q}(\sqrt{5})$, and $11 = \frac{7+\sqrt{5}}{2} \cdot \frac{7-\sqrt{5}}{2}$ splits; above 5 there is one maximal ideal, with residue field $\mathbb{F}_5$, and the two eigenforms become congruent modulo it.)

**Exercise 13.103.** (The eigenvectors in Lemma lem:old, at level 99.) Take $N = 99$ and $p = 3$. (a) For $h = f_{11}$ ($a_3 = -1$, $3 \nmid 11$, $e_3 = 2$), write the matrix of $U_3$ on $\mathrm{span}\{f_{11}(z), f_{11}(3z), f_{11}(9z)\}$, and check that $f_{11} - \beta f_{11}(3z)$ and $f_{11} - \alpha f_{11}(3z)$, $\alpha, \beta$ the roots of $X^2 + X + 3$, and the depleted form $f_{11} + f_{11}(3z) + 3f_{11}(9z)$ are eigenvectors with eigenvalues $\alpha$, $\beta$, 0. (b) For $h$ the newform of 33a ($a_3 = -1$, $3 \parallel 33$, $e_3 = 1$), show that $h$ and $h + h(3z)$ are eigenvectors with eigenvalues $-1$ and 0. (c) For $h$ the newform of 99a viewed at level 297 ($3^2 \mid 99$, $e_3 = 1$), show that $U_3$ is a nilpotent Jordan block on $\mathrm{span}\{h, h(3z)\}$ with eigenvector $h$. In each case show that every eigenvalue has a one-dimensional eigenspace, and check that the characteristic polynomial of $U_3$ on $S_2(99)$ is $X^6(X+1)(X^2+X+3)$. (Hint: Proposition 13.48; the six zeros come from the four newforms of level 99, from (a) and from (b).)

**Exercise 13.104.** (Primitivity in the Gauss-sum formula.) Let $\chi$ be the character modulo 9 induced by $\chi_{-3}$ (so $\chi(u) = \chi_{-3}(u)$ for $3 \nmid u$ and $\chi(u) = 0$ otherwise), and $G(n) = \sum_{u \bmod 9} \chi(u) e^{2\pi i n u/9}$. Show that $G(1) = 0$ and $G(3) = 3\,\mathfrak{g}(\chi_{-3}) = 3i\sqrt{3}$, so that $G(n) = \bar{\chi}(n)G(1)$ fails at $n = 3$ and the formula of Proposition 13.70(i) cannot be used with $\chi$. Show that nevertheless $f \otimes \chi = f \otimes \chi_{-3}$ as $q$-series. (Hint: group $u$ by its class modulo 3; the sum of $e^{2\pi i k/3}$ over $k$ mod 3 vanishes.)

**Exercise 13.105.** (Case (c) of Theorem thm:twist.) `75a1` has type IV at 5 and `75c1` type II. Show with PARI that the twist of `75a1` by $\chi_5$ is `75c2`, of type II$^*$ at 5, and that the twist of `75c1` is `75a2`, of type IV$^*$, in accordance with Chapter 10. Compute $r_f$ for the newforms of 75a and 75c by the lattice method of Example 13.84 and check $r(75a) = r(75c)$, as Theorem thm:twist predicts when $N_S = N$; do the same for 50a and 50b with $\chi_5$. (Answer: $r = 6$ for both 75a and 75c, equal to the modular degrees; $r = 2$ for 50a and 50b.)

## 13.13 Notes and references

Hecke operators on modular forms of level one go back to Hecke and, for the multiplicativity of $\tau$, to Mordell; the double coset formalism is Shimura's [180, Chapter 3]. We followed Diamond and Shurman [52, Chapter 5] for the double cosets, the $q$-expansion formulas, the Petersson adjoints and the theory of newforms, with the normalisation of Convention 13.2; Serre [172, Chapter VII] and Koblitz [102] treat level one, and Miyake [137, Chapter 4] the general theory with characters. The section numbers of Diamond-Shurman (5.1 to 5.10, 6.3), Shimura (3.1 to 3.6, 7.2), Miyake (4.3, 4.6), Agashe-Ribet-Stein (Section 3), Darmon-Diamond-Taylor (Section 1) and Zagier (Section 1) cited in the text are pointers we have not verified against the printed editions (for Diamond-Flach-Guo we use Section 1.7.3, the section the paper cites, which we checked: it defines $f^{\Sigma}$, $\eta_f^{\Sigma}$ and the maps of their Proposition 1.4), and the theorem numbers [137, Theorem 4.6.19], [9, Theorem 3.1] and [8, Theorem 3] are quoted as the paper (for the last two) and the literature cite them, not checked; nor is the exact form of the hypothesis in [9, Theorem 3.1].

The theory of newforms is due to Atkin and Lehner [8] for trivial character and to Li [113] in general; strong multiplicity one in the form of Theorem 13.42(i) is in Miyake's book and in

Li's paper; the density-one form uses the Galois representations of Chapter 15. The theorem on twists of newforms is Atkin and Li [9]. The semisimplicity of $U_p$ on $p$-old forms of weight 2 is Coleman and Edixhoven [32].

The perfect pairing $S_2(N;\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$ is cited by the paper from Ribet [159, Theorem 2.2] and Abbes-Ullmo [1, Lemma 2.1] (see also Agashe-Ribet-Stein [3, Section 3] and [38, Section 1]); the proof given here, from the integral basis theorem by a rank argument, is standard. The dictionary between eigenforms and homomorphisms (Theorem 13.26) is the starting point of Mazur's and Ribet's use of $\mathbb{T}$; see [38]. The notation $f^{\Sigma}$, $M^{\Sigma}$ follows Diamond, Flach and Guo [49, Section 1.7.3]. The Rankin-Selberg method is Rankin's [155]; the relation with the symmetric square and the bad factors is Shimura's [183] and [49, Lemma 2.12]; Zagier [209] gives the computation for elliptic curves. Stein's book [187] explains the algorithms, and PARI/GP [148] implements them (`mfinit`, `mfheckemat`, `mfpetersson`, `mftwist`, `mfcoefs`); the curve data are Cremona's [35].

Three results here supply proofs of statements the paper uses. Theorem 13.60 proves the decomposition $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ that the proof of Lemma lem:crit writes down, and shows that the $\mathbb{T}$-stability of $f^{\perp}$ needs no adjoints of the $U_p$. Proposition 13.64 proves the conclusions of Lemma lem:crit for every rational eigencharacter of weight 2 whose generalised eigenspace is a line, which is what the set-up before Lemma lem:step1 uses for $g^S$. Proposition 13.74(iii) gives the transport from $f$ to $g^S$ along $\iota$, and with it Corollary 13.76 proves $r_E \mid \eta^{\mathrm{Hecke}}(g^S)$ at every prime, the inequality $\leq$ of Lemma lem:step1 (Remark 13.77; the equality for $\ell \notin S$ is in Chapter 20).

# Chapter 14

# Modular curves over $\mathbb{Q}$, Jacobians and the modular degree

The paper is about one integer, the modular degree $\deg \varphi_E$ of the $X_0(N)$-optimal curve $E$ of an isogeny class. This chapter defines it and proves the facts about it that the paper uses. It puts the modular curve $X_0(N)$ of Chapter 12 over $\mathbb{Q}$, builds its Jacobian $J_0(N)$ and the action of the Hecke algebra of Chapter 13 on $J_0(N)$, on $H_1$ and on $H^1$ by correspondences, and proves the identity $T_n^t = w_N T_n w_N$ that makes the pairing $\hat{\delta}_!$ of Section 4.1 of the paper perfect, alternating and Hecke self-adjoint. It then constructs the optimal curve as a quotient of $J_0(N)$, characterises it, and proves the four descriptions of $\deg \varphi_E$ the paper uses: the degree of a map of curves, the integer $n$ with $E \to J_0(N) \to E$ equal to $[n]$, the Petersson norm divided by the area of the period lattice (the formula of Zagier and Watkins, with the Manin constant in the numerator), and the ideal of values of $\hat{\delta}_!$ on the lattice $M_{f,\lambda}$ of Diamond, Flach and Guo (Proposition prop:pullback of the paper, proved here with every step expanded, and also over $\mathbb{Z}$). From the last description the chapter rederives the divisibility $\deg \varphi_E \mid r_E$ of Agashe, Ribet and Stein at every prime, and explains why the equality can fail at primes $p$ with $p^2 \mid N$ (the curve `99a1`). The deep inputs (the $\mathbb{Q}$-structure of $X_0(N)$, the Eichler-Shimura relation, modularity, multiplicity one, the theorems of Mazur and of Calegari and Emerton) are stated precisely with references; the comparison of Ribet and Takahashi with Shimura curves is stated exactly, with its theorem numbers in the sources, and its $\ell$-adic form is derived from it.

## Core results at a glance

Throughout, $E/\mathbb{Q}$ has conductor $N$, $f = f_E$ is its newform, $\phi_f \colon \mathbb{T} \to \mathbb{Z}$ the eigencharacter, $I_f = \ker \phi_f$, $\mathbb{T}$ the full Hecke algebra of level $N$ ($U_p$ included), $\ell$ a prime, and $\varepsilon_N(f) = \pm 1$ the eigenvalue of $w_N$ on $f$.

**14.1 Conventions.** $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$; slash $(f|_k\gamma) = (\det\gamma)^{k/2}(cz+d)^{-k} f(\gamma z)$; $\mathbb{T}$ acts by pushforward $t_*$ on $J_0(N)$, $H_1$, $T_\ell J_0(N)$ and by pullback $t^*$ on $S_2$, $H^1$; unnormalised Petersson product; $\varphi^*\omega_E = c_\varphi \cdot 2\pi i f\, dz$ with $c_E > 0$; "optimal" means $X_0(N)$-optimal; Eisenstein means $\bar{\rho}_{\mathfrak{m}}$ reducible. *(convention)* Convention 14.1.

**14.2 $X_0(N)$ over $\mathbb{Q}$.** $X_0(N)$ is a smooth projective geometrically connected curve over $\mathbb{Q}$, a coarse moduli space for $(E, C)$ with $C$ cyclic of order $N$; $j$ and the $w_d$ are defined over $\mathbb{Q}$, $w_N(E, C) = (E/C, E[N]/C)$; good reduction at $p \nmid N$. *(stated; reference)* Theorem 14.2.

**14.3 Cusps over $\mathbb{Q}$.** Cusps $a/c$, $c \mid N$, $a \bmod \gcd(c, N/c)$; those of denominator $c$ are defined over $\mathbb{Q}(\zeta_{\gcd(c,N/c)})$ and permuted simply transitively; $\infty$, 0 are rational; all cusps are rational iff $\gcd(c, N/c) \le 2$ for all $c \mid N$. *(proof sketched here; reference)* Proposition 14.3.

**14.4 Rational points.** A $G_K$-stable cyclic $C$ gives a $K$-point of $Y_0(N)$, and conversely up to twist when $j \neq 0, 1728$; $Y_0(N)(\mathbb{Q}) \neq \emptyset$ iff $N \le 19$ or $N \in \{21, 25, 27, 37, 43, 67, 163\}$. *(proved here; the list stated, Mazur, Kenku)* Proposition 14.5.

**14.5 Hecke correspondences.** $\mathcal{C}_\alpha = (X(\Gamma_\alpha), \pi_1, \pi_\alpha)$, $\Gamma_\alpha = \Gamma \cap \alpha^{-1}\Gamma\alpha$, $\pi_1(\Gamma_\alpha z) = \Gamma z$, $\pi_\alpha(\Gamma_\alpha z) = \Gamma\alpha z$; $T_p$, $U_p = \mathcal{C}_{\mathrm{diag}(1,p)}$. *(definition)* Definition 14.8.

**14.6 Properties.** $\mathcal{C}_\alpha$ depends only on $\Gamma\alpha\Gamma = \bigsqcup \Gamma\alpha\gamma_i$; $X(\Gamma_{\mathrm{diag}(1,p)}) \simeq X_0(Np)$ with $\pi_1$, $\pi_\alpha$ becoming $z \mapsto pz$, $z \mapsto z$; on moduli $T_p(E, C) = \sum_D (E/D, (C + D)/D)$; defined over $\mathbb{Q}$. *(proved here; moduli sketched)* Proposition 14.9.

**14.7 Compatibility.** The form $\mathcal{C}_\alpha^*(f\,dz) = (\sum f|_2\alpha_i)\,dz$ is Chapter 13's $T_p$ or $U_p$; $t \mapsto t^* \in \mathrm{End}\, H^1(X_0(N), \mathbb{Z})$ and $t \mapsto t_* \in \mathrm{End}\, H_1$ are injective ring homomorphisms; $\mathbb{T}$ is finite free over $\mathbb{Z}$. *(proved here)* Proposition 14.10.

**14.8 The transpose.** $\mathcal{C}_\alpha^t \simeq \mathcal{C}_{(\det\alpha)\alpha^{-1}}$; $\mathcal{C}_{\beta\alpha\beta^{-1}} \simeq w_\beta \mathcal{C}_\alpha w_\beta^{-1}$; $W_N \mathrm{diag}(1, p) W_N^{-1} = \mathrm{diag}(p, 1)$; hence $T_p^t \simeq w_N T_p w_N$, $U_p^t \simeq w_N U_p w_N$, and $(t^*)^\dagger = w_N^* t^* w_N^*$ for the cup product, for every $t \in \mathbb{T}$; $T_p^t \simeq T_p$ for $p \nmid N$. *(proved here)* Theorem 14.11.

**14.9 The pairing $\hat{\delta}_!$.** $\hat{\delta}_!(x, y) = \langle x, w_N^* y\rangle$ is perfect and alternating on $H^1(X_0(N), R)$ ($R = \mathbb{Z}, \mathbb{Z}_\ell, \mathbb{Q}_\ell$), every $t^*$ is self-adjoint, and $H^1(X_0(N), \mathbb{Z}_\ell) = \bigoplus_{\mathfrak{m}} H^1_{\mathfrak{m}}$ orthogonally with $\hat{\delta}_!$ perfect on each summand. *(proved here)* Corollary 14.12.

**14.10 $J_0(N)$.** $J_0(N) = \mathrm{Jac}(X_0(N)_{\mathbb{Q}})$, $\iota(P) = [(P) - (\infty)]$. *(definition)* Definition 14.15.

**14.11 Structure.** Abelian variety over $\mathbb{Q}$ of dimension $g = \dim S_2$, good reduction at $p \nmid N$; $J_0(N)(\mathbb{C})$ is $S_2^\vee$ modulo $H_1(X_0(N), \mathbb{Z})$; universal property of $\iota$; dimensions 1, 1, 1, 2, 7, 9, 6, 19, 49, 109, 145 for $N = 11$, 14, 20, 37, 77, 99, 121, 140, 539, 726, 980. *(proved here, given Chapter 8; good reduction stated)* Theorem 14.16.

**14.12 Hecke on $J_0(N)$.** $t \mapsto t_*$ embeds $\mathbb{T}$ in $\mathrm{End}_{\mathbb{Q}} J_0(N)$, commuting with $G_{\mathbb{Q}}$ on $T_\ell J_0(N)$; $t^t = w_N t w_N$ (in general not in $\mathbb{T}$); $I_f J_0(N)$, $J_0(N)/I_f J_0(N)$, $H^1[I_f]$ are the same for the Albanese and the Picard actions. *(proved here)* Proposition 14.17.

**14.13 $T_\ell J_0(N)$ versus $H^1$.** $H^1(X_0(N), \mathbb{Z}_\ell) = (T_\ell J_0(N))^\vee \simeq T_\ell J_0(N)(-1)$; $\Psi = w_{N*} \circ \mathrm{PD}$ is a $\mathbb{T}_\ell$-isomorphism $H^1 \to T_\ell J_0(N)$, Galois-equivariant after the twist; freeness over $\mathbb{T}_{\mathfrak{m}}$ transfers, and $H^1(X_0(N), \mathbb{F}_\ell)_{\mathfrak{m}}[\mathfrak{m}] \simeq J_0(N)[\mathfrak{m}]$. *(proved here, given the étale statements of Chapter 8)* Proposition 14.18.

**14.14 The $f$-lattice.** $V_f = H^1(X_0(N), \mathbb{Q})[I_f]$, $M_{f,\lambda} = H^1(X_0(N), \mathbb{Z}_\ell)[I_f]$. *(definition)* Definition 14.20.

**14.15 Properties of $M_{f,\lambda}$.** $V_f \otimes \mathbb{C} = \mathbb{C}\omega_f \oplus \mathbb{C}\overline{\omega_f}$, also the eigenspace of the anemic algebra; $M_{f,\lambda} = H^1(X_0(N), \mathbb{Z})[I_f] \otimes \mathbb{Z}_\ell$ is saturated of rank 2; $w_N^*$ acts on it by $\varepsilon_N(f)$; it is $G_{\mathbb{Q}}$-stable. *(proved here, given Chapter 13)* Proposition 14.21.

**14.16 Shimura's quotient.** $J_f = J_0(N)/I_f J_0(N)$ is an abelian variety over $\mathbb{Q}$ of dimension $[K_f : \mathbb{Q}]$ with connected kernel $I_f J_0(N)$; for rational $f$, $E_f = J_f$ is an elliptic curve, $\mathbb{T}$ acts by $\phi_f$, $\varphi_{E_f}^* H^1(E_f, \mathbb{Q}) = V_f$; $L(E_f, s) = L(f, s)$. *(proved here for rational $f$; sketched in general; the L-function stated)* Theorem 14.23.

**14.17 Eichler-Shimura.** $(T_p)_* = \pi_p + \hat{\pi}_p$ on $J_0(N)_{\mathbb{F}_p}$ for $p \nmid N$; hence $\mathrm{Frob}_p^2 - T_p \mathrm{Frob}_p + p = 0$ on $T_\ell J_0(N)$, and $a_p(E_f) = a_p(f)$. *(congruence relation stated; consequences proved here)* Theorem 14.24.

**14.18 Modularity.** Every $E/\mathbb{Q}$ of conductor $N$ has a newform $f_E$ of level $N$ with $a_p(f_E) = a_p(E)$ for all $p$; equivalently $E$ is isogenous to $E_{f_E}$. *(stated; Wiles, Taylor-Wiles, BCDT, Carayol)* Theorem 14.25.

**14.19 Classes and newforms.** $E \mapsto f_E$ is a bijection from isogeny classes of conductor $N$ to rational newforms of level $N$. *(proof sketched here)* Corollary 14.26.

**14.20 Modular parametrisation.** A nonconstant $\varphi\colon X_0(N) \to E$ over $\mathbb{Q}$ with $\varphi(\infty) = O$; equivalently a nonzero $h\colon J_0(N) \to E$. *(definition)* Definition 14.29.

**14.21 Optimal curves.** Every $h\colon J_0(N) \to E$ ($E$ in the class of $f$) kills $I_f J_0(N)$; every parametrisation is $\psi \circ \varphi_{E_f}$ with $\psi$ an isogeny; $\psi$ an isomorphism $\Leftrightarrow \ker \varphi_*$ connected $\Leftrightarrow \varphi_*$ onto on $H_1 \Leftrightarrow \varphi^* H^1$ saturated $\Leftrightarrow \varphi^*$ injective on $E \Leftrightarrow \deg \varphi$ minimal; $E_f$ is the optimal curve. *(proved here, given Eichler-Shimura and modularity)* Theorem 14.30.

**14.22 The modular degree.** $\deg \varphi_E$ is the degree of $\varphi_E$; $E \to J_0(N) \to E$ is $[\deg \varphi_E]$; $A \cap B \simeq (\mathbb{Z}/\deg \varphi_E)^2$; the cup product on $\varphi_E^* H^1(E, \mathbb{Z})$ has ideal of values $\deg \varphi_E \mathbb{Z}$. *(proved here)* Proposition 14.31.

**14.23 Along the class.** If $E[\ell]$ is irreducible, every curve of the class has minimal parametrisation degree of the same $\ell$-adic valuation as $\deg \varphi_E$; false for `11a` at $\ell = 5$. *(proved here)* Proposition 14.32.

**14.24 Cremona's convention.** Curve 1 of each class is optimal, except `990h` (optimal curve `990h3`, $\deg \varphi_E = 1728$); optimality determined to conductor 400000 and for 359,009 classes beyond. *(convention; data)* Convention 14.34.

**14.25 Manin constant.** $\varphi^* \omega_E = c_\varphi \cdot 2\pi i f(z)\, dz$; $c_E = c_{\varphi_E} > 0$. *(definition)* Definition 14.37.

**14.26 $c_\varphi$.** $\varphi^* \omega_E \in \mathbb{C}\omega_f \setminus 0$, $c_\varphi \in \mathbb{Q}^\times$, $c_{\psi \circ \varphi_E} = c_E u_\psi$ with $u_\psi u_{\hat{\psi}} = \deg \psi$. *(proved here; integrality of $u_\psi$ stated)* Proposition 14.38.

**14.27 Known results.** $c_E \in \mathbb{Z}$ (Edixhoven); $c_E = 1$ conjecturally; $p \mid c_E \Rightarrow p = 2$ or $p^2 \mid N$; $c_E = 1$ for semistable $E$. *(stated; reference)* Theorem 14.39.

**14.28 Zagier-Watkins.** For every modular parametrisation,

$$\deg \varphi \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_\varphi^2 \langle f, f \rangle \qquad \text{(unnormalised Petersson product)}.$$

*(proved here)* Theorem 14.40.

**14.29 Congruence number.** $r_E$: the largest $r$ with $f \equiv g \pmod r$, $g \in S_2(\Gamma_0(N); \mathbb{Z}) \cap f^\perp$; $\eta_f = \phi_f(\mathrm{Ann}_{\mathbb{T}} I_f)$. *(definition)* Definition 14.45.

**14.30 Congruence ideals.** $\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$, $I_f\cap\operatorname{Ann} I_f=0$, $\eta_f\neq 0$; $\eta_f\mathbb{Z}_\ell$ is computed in $\mathbb{T}_{\mathfrak{m}}$; $r_E<\infty$. *(proved here)* Proposition 14.46.

**14.31 Lemma lem:crit.** $\mathbb{T}/(I_f+\operatorname{Ann}_{\mathbb{T}} f^{\perp})\simeq\mathbb{Z}/r_E$, so $r_E\mathbb{Z}=\eta_f$, also over $\mathbb{Z}_\ell$; a congruent eigenform in $f^{\perp}$ gives $\ell\mid r_E$. *(stated; proved in Chapter 15)* Lemma 14.47.

**14.32 Agashe-Ribet-Stein.** For optimal $E$: $\deg\varphi_E\mid r_E$, and $v_p(\deg\varphi_E)=v_p(r_E)$ when $p^2\nmid N$. *(stated; the divisibility proved here in Corollary 14.55)* Theorem 14.48.

**14.33 $M_{f,\lambda}$ is the pullback.** For optimal $E$ and every prime $\ell$: $\varphi_E^*$ is injective with saturated image, $\varphi_E^*H^1(E,\mathbb{Z}_\ell)=M_{f,\lambda}$, and $M_{f,\lambda}\simeq T_\ell E(-1)$ as Galois modules, so $\operatorname{ad}^0 M_{f,\lambda}\simeq\operatorname{ad}^0 T_\ell E$. *(proved here)* Proposition 14.52.

**14.34 $w_N$ on $M_{f,\lambda}$.** $\varphi_E\circ w_N=\varepsilon_N(f)\varphi_E+\varphi_E(0)$, so $w_N^*=\varepsilon_N(f)$ on $M_{f,\lambda}$. *(proved here)* Lemma 14.53.

**14.35 The degree as a congruence ideal.** For optimal $E$ and every prime $\ell$,

$$\hat{\delta}_!(\varphi_E^*a,\varphi_E^*b)=\varepsilon_N(f)\deg\varphi_E\,\langle a,b\rangle_E,$$

so the ideal of values of $\hat{\delta}_!$ on $M_{f,\lambda}$ is $\deg\varphi_E\mathbb{Z}_\ell$; with the paper's description of [49], $v_\lambda(\eta_{f,\lambda}^{\emptyset})=v_\ell(\deg\varphi_E)$. *(proved here)* Theorem 14.54.

**14.36 Divisibility at every prime.** For every prime $\ell$, $v_\ell(\deg\varphi_E)\le v_\ell(r_E)$, with equality if $H^1(X_0(N),\mathbb{Z}_\ell)_{\mathfrak{m}}$ is free over $\mathbb{T}_{\mathfrak{m}}$. *(proved here, given Lemma lem:crit)* Corollary 14.55.

**14.37 Lemma lem:hecke.** $v_\ell(\deg\varphi_E)=v_\ell(\text{values of }\hat{\delta}_!\text{ on }M_{f,\lambda})\le v_\ell(r_E)$ always; $=v_\ell(r_E)$ if $\ell^2\nmid N$; $=v_\lambda(\eta_{f,\lambda}^{\emptyset})$ if $\ell\nmid 2N$ and $\bar\rho$ irreducible. *(proved here, given Theorem 14.48 and the identification with [49])* Theorem 14.56.

**14.38 The maximal ideal of $(f,\ell)$.** $\mathfrak{m}_{f,\ell}\ni\ell,I_f$; $\bar\rho_{\mathfrak{m}}$ with $\operatorname{tr}\bar\rho_{\mathfrak{m}}(\operatorname{Frob}_p)=T_p$; $\mathfrak{m}_{f,\ell}$ is Eisenstein iff $E[\ell]$ is reducible. *(definition; $\bar\rho_{\mathfrak{m}}$ stated)* Definition 14.60.

**14.39 Multiplicity one.** For $\ell\nmid 2N$ and $\mathfrak{m}$ non-Eisenstein, $\dim_{\mathbb{T}/\mathfrak{m}} J_0(N)[\mathfrak{m}]=2$. *(stated; Mazur, Ribet, Ribet-Stein)* Theorem 14.61.

**14.40 Gorenstein.** Then $\mathbb{T}_{\mathfrak{m}}$ is Gorenstein and $H^1(X_0(N),\mathbb{Z}_\ell)_{\mathfrak{m}}\simeq\mathbb{T}_{\mathfrak{m}}^2\simeq T_\ell J_0(N)_{\mathfrak{m}}$; so $v_\ell(r_E)=v_\ell(\deg\varphi_E)$ for $\ell\nmid 2N$, $E[\ell]$ irreducible. *(proved here, given multiplicity one)* Theorem 14.62.

**14.41 Cuspidal subgroup.** $C_N\subseteq J_0(N)$ generated by cuspidal divisors of degree 0. *(definition)* Definition 14.65.

**14.42 Manin-Drinfeld.** $(T_p)_*(0)=(p+1)(0)$, $(T_p)_*(\infty)=(p+1)(\infty)$ for $p\nmid N$; $T_p-p-1$ is an isogeny; $[(0)-(\infty)]$ is a rational torsion point; $C_N$ is finite. *(proved here; the finiteness of $C_N$ stated)* Proposition 14.66.

**14.43 Mazur at prime level.** $J_0(N)(\mathbb{Q})_{\text{tors}}=C_N\simeq\mathbb{Z}/n$, $n=\operatorname{num}\frac{N-1}{12}$; $\mathbb{T}/I_{\text{Eis}}\simeq\mathbb{Z}/n$; Eisenstein maximal ideals at odd $\ell$ exist iff $\ell\mid n$. *(stated; reference)* Theorem 14.67.

**14.44 Shimura subgroup.** $\mathcal{S}_N=\ker(J_0(N)\to J_1(N))$. *(definition)* Definition 14.68.

**14.45 Its properties.** $\#\mathcal{S}_N$ is the saturation index of $\pi^*H^1(X_0(N),\mathbb{Z})$; $\mathcal{S}_N$ is finite, multiplicative and Eisenstein; $\mathcal{S}_N[\mathfrak{m}]=0$ for $\mathfrak{m}$ non-Eisenstein. *(proved here; (ii) stated)* Proposition 14.69.

**14.46 The image of 0.** $\varphi_E(0) \in E(\mathbb{Q})_{\mathrm{tors}}$ and $\varphi_E(0) \equiv c_E L(E,1) \pmod{\Lambda_E}$; its order is the denominator of $c_E L(E,1)/\Omega$. *(proved here, given the Mellin transform of Chapter 19)* Proposition 14.70.

**14.47 Shimura curves.** $X_0^D(M)$, its Jacobian isogenous to the $D$-new quotient of $J_0(DM)$; for $N = DM$ square-free, $E_D$ is the curve of the isogeny class that is an optimal quotient of $J_0^D(M)$ (in general not the $X_0(N)$-optimal one) and $\delta_D = \deg(\varphi_{E,D}\colon X_0^D(M) \to E_D)$. *(definition; stated)* Definition 14.74.

**14.48 Ribet-Takahashi.** For $N = dpqm$ square-free, $d$ with an even number of prime factors: $\delta^d(pqm)/\delta^{dpq}(m) = n_p(E_2)n_q(E_1)/i_q^2$, $n_r = v_r(\Delta) = \#\Phi_r^{\mathrm{geom}}$ (not the Tamagawa number), $i_q$ the order of the image of $\Phi_q^{\mathrm{geom}}(J_0^d(pqm)) \to \Phi_q^{\mathrm{geom}}(E_1)$; if $E$ is alone in its isogeny class, $\deg\varphi_E/\delta_D = \prod_{r|D} n_r$; if $E[\ell]$ is irreducible (any $\ell$), $v_\ell(\deg\varphi_E) = v_\ell(\delta_D) + \sum_{r|D} v_\ell(n_r)$. *(stated; [164, Theorems 1, 2], [192, Theorem 3.2, Corollary 3.5]; the $\ell$-adic form sketched here)* Theorem 14.75; sources, error term and example `57c1` in Remark 14.76.

**14.49 Watkins's conjecture.** $2^{\operatorname{rank} E(\mathbb{Q})} \mid \deg\varphi_E$; true for all 38,042 optimal curves of conductor below 10,000. *(conjecture; data checked)* Conjecture 14.77.

**14.50 Calegari-Emerton.** Odd $\deg\varphi_E$ implies at most two odd bad primes, even analytic rank, and a rational 2-torsion point, or prime conductor with supersingular reduction at 2 and $E(\mathbb{R})$ connected, or CM with $N \in \{27, 32, 49, 243\}$. *(stated; reference; data checked)* Theorem 14.78.

## Where the paper uses this chapter

- **Section 1 (Introduction) and the abstract.** The $X_0(N)$-optimal curve, the modular parametrisation $\varphi_E\colon X_0(N) \to E$ and "minimal degree" (Definition 14.29, Theorem 14.30, and Proposition 14.31); the sentence on the choice of curve when $E[\ell]$ is irreducible (Proposition 14.32); computability (Remark 14.79); the *abc* connection (Remark 14.49); the Petersson norm and Zagier, Watkins (Theorem 14.40); Agashe, Ribet and Stein and the congruence number $r_E$ (Definition 14.45, Theorem 14.48, and Corollary 14.55).

- **Section 1.5 (History).** The parametrisations by Shimura curves of Ribet and Takahashi (Definition 14.74, Theorem 14.75, and Remark 14.76).

- **Section sec:proof (proof of Theorem thm:qual).** The congruence number and Lemma lem:crit (Lemma 14.47, proved in Chapter 15); "it suffices to prove $\ell \mid r$" (Theorem 14.48); the remark on `99a1` and `726a1` (Examples 14.50 and 14.58).

- **Section 4.1 (sec:dfg).** $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$ with its Hecke action, the pairing $\hat{\delta}_!$ and the sentence "$w_{M^\Sigma} T_n w_{M^\Sigma}^{-1}$ is the adjoint of $T_n$ for the cup product" (Proposition 14.10, Theorem 14.11, and Corollary 14.12); the lattice $M_{f,\lambda}$ (Definition 14.20) and the checklist for the identification with Diamond, Flach and Guo's objects, including Proposition prop:integral and its Remark (Remark 14.57).

- **Proposition prop:pullback.** Proposition 14.52, Lemma 14.53 and Theorem 14.54, with a Commentary on the paper's statement and proof; the paragraph after the proposition is Proposition 14.52(d).

- **Lemma lem:hecke, Remark rem:free and the paragraph between them.** Theorem 14.56, the divisibility at every prime (Corollary 14.55), multiplicity one and the Gorenstein property (Definition 14.60, Theorems 14.61 and 14.62, and Remark 14.63), and the identification of $T_\ell J_0(N)$ with $H^1(X_0(N),\mathbb{Z}_\ell)$ as Hecke modules (Proposition 14.18).

- **Section 4.2 and the proof of Theorem thm:main.** $M_{f,\lambda} \simeq H^1(E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$ and $\mathrm{ad}^0 M_{f,\lambda} \simeq \mathrm{ad}^0 T_\ell E$ (Proposition 14.52(d)); the equality in (eq:identity) (Theorem 14.56).

- **Remark rem:degree.** The formula of Zagier and Watkins with $c_E^2$ in the numerator (Theorem 14.40 and Example 14.42).

- **Section sec:data.** Optimality in Cremona's tables and the class `990h` (Convention 14.34); `ellmoddegree` (Remark 14.79).

- **Conjecture conj:eis, Section sec:eis, Open question 4.** The cuspidal subgroup, Mazur's Eisenstein ideal, the Shimura subgroup and the image of the cusp 0 (Propositions 14.66 and 14.70, Theorem 14.67, Definition 14.68, and Remark 14.72).

- **Section sec:two, Open question 5.** Watkins's conjecture and the theorem of Calegari and Emerton (Conjecture 14.77 and Theorem 14.78).

## 14.1 Overview and standing notation

This section fixes the conventions of the chapter and says which section answers which sentence of the paper. The paper uses the chapter in six places: the optimal curve and its parametrisation (Sections 14.2 to 14.6); the Petersson norm and the Manin constant (Section 14.7); the congruence number and the theorem of Agashe, Ribet and Stein (Section 14.8); the cohomology of $X_0(N)$ with its Hecke action and the pairing $\hat{\delta}_!$ (Sections 14.3 and 14.4); the identification of the congruence ideal of Diamond, Flach and Guo with $\deg \varphi_E$ (Sections 14.9 and 14.10); and the Eisenstein case, the Shimura curves and the prime 2 (Sections 14.11 and 14.12).

**Convention 14.1** (standing conventions of this chapter)**.**

(C1) *Curves and levels.* $N \geq 1$; $X_0(N)$ is the compact Riemann surface of Chapter 12 and, by Theorem 14.2, a smooth projective curve over $\mathbb{Q}$; $Y_0(N) \subset X_0(N)$ is the complement of the cusps; $\infty$ and 0 are the cusps $\Gamma_0(N)\infty$ and $\Gamma_0(N)0$; $g = g(X_0(N))$; $\Gamma = \Gamma_0(N)$. $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$ and $w_N$ is the involution $z \mapsto -1/(Nz)$ of $X_0(N)$ and the operator $f \mapsto f|_2 W_N$ (Chapter 12). The slash operator is $(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz+d)^{-k} f(\gamma z)$, so that in weight 2 $(f|_2\gamma)(z)\,dz = \gamma^*(f(z)\,dz)$. The letter $q$ is the variable $e^{2\pi i z}$ of $q$-expansions and also a bad prime of $E$ (Appendix B, B-C1); the two never occur in one display.

(C2) *Hecke actions.* $\mathbb{T} = \mathbb{Z}[T_n : n \geq 1] \subset \mathrm{End}_{\mathbb{C}}\, S_2(\Gamma_0(N))$ is the full Hecke algebra of Chapter 13, the operators $U_p = T_p$ for $p \mid N$ included; $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$. A correspondence $\mathcal{C}$ acts on $J_0(N)$, on $H_1(X_0(N),\mathbb{Z})$ and on $T_\ell J_0(N)$ by pushforward $\mathcal{C}_*$ ("Albanese") and on $H^0(X_0(N),\Omega^1) = S_2(\Gamma_0(N))$, $H^1(X_0(N),\mathbb{Z})$ and $H^1(X_0(N),\mathbb{Z}_\ell)$ by pullback $\mathcal{C}^*$, the transpose of $\mathcal{C}_*$ (Chapter 8). For $t \in \mathbb{T}$ the two resulting actions are written $t_*$ and $t^*$ (Proposition 14.10); every statement "$\mathbb{T}$ acts on ..." names which.

(C3) *Duals and adjoints* (Appendix B, B-C4). $M^\vee = \mathrm{Hom}(M, \mathbb{Z}_\ell)$ (or $\mathrm{Hom}(M, \mathbb{Z})$); $A^\dagger$ is the adjoint of an operator for a pairing; a star after a map is a pullback. The paper's Pontryagin duals occur only inside `paperlink` environments.

(C4) *Petersson product, Manin constant* (B-C8, B-C9). $\langle f, g\rangle = \int_{\Gamma_0(N)\backslash\mathcal{H}} f\bar{g}\,dx\,dy$ is unnormalised in weight 2, and $\langle f, f\rangle = \frac{i}{2}\int_{X_0(N)} \omega_f \wedge \overline{\omega_f}$ with $\omega_f = f(z)\,dz$. For an elliptic curve $E/\mathbb{Q}$, $\omega_E$ is the Néron differential of a global minimal model, $\Lambda_E$ its period lattice and $\mathrm{vol}(\Lambda_E) = \frac{i}{2}\int_{E(\mathbb{C})} \omega_E \wedge \bar{\omega}_E$; $\varphi^*\omega_E = c_\varphi \cdot 2\pi i f(z)\,dz$ defines $c_\varphi$, and $c_E > 0$. PARI's `mfpetersson` divides by $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$. The Manin constant is never a bare $c$.

(C5) *Frobenius* (B-C2(g), B-C3). On $T_\ell J$ the arithmetic Frobenius at $p \nmid N\ell$ acts as the $p$-power Frobenius endomorphism $\pi_p$ of the reduction; $\hat{\pi}_p = p\pi_p^{-1}$ is the Verschiebung; on $H^1 = (T_\ell J)^\vee$ the geometric Frobenius has the same characteristic polynomial (Chapter 8).

(C6) *Optimal curves, labels, coefficients.* "Optimal" means $X_0(N)$-optimal (Theorem 14.30). Curve labels are Cremona's; [3] and [35] print `99A1` for `99a1`. The newform of an elliptic curve has rational coefficients, so $\lambda = \ell$ and $\mathcal{O} = \mathbb{Z}_\ell$ throughout, and the paper's $v_\lambda$ is $v_\ell$.

(C7) *Eisenstein.* A maximal ideal $\mathfrak{m}$ of $\mathbb{T}$ (or of $\mathbb{T}_\ell$) is *Eisenstein* if the residual representation $\bar{\rho}_\mathfrak{m}$ of Definition 14.60 is reducible. Mazur's *Eisenstein ideal* $I_{\mathrm{Eis}}$ at prime level is a specific ideal (Theorem 14.67). The letter $D$ is used for the quaternion discriminant of a Shimura curve (Section 14.12) and, in quotations, for the deficit of Conjecture conj:eis; $\mathcal{S}_N$ is the Shimura subgroup.

(C8) *Typography.* Double names take plain hyphens: Eichler-Shimura, Agashe-Ribet-Stein, Zagier-Watkins, Manin-Drinfeld, Ribet-Takahashi, Calegari-Emerton, Deligne-Rapoport, Boston-Lenstra-Ribet, Jacquet-Langlands.

**Standing notation.** Throughout, $E/\mathbb{Q}$ is an elliptic curve of conductor $N$, $f = f_E \in S_2(\Gamma_0(N))$ its newform (Theorem 14.25), $\phi_f\colon \mathbb{T} \to \mathbb{Z}$, $T_n \mapsto a_n(f)$, its eigencharacter (an upright $\phi$; a curly $\varphi$ is always a map of curves), $I_f = \ker\phi_f$ (the paper's $I$ in Lemma lem:crit and its $\wp$ in Lemma lem:ddt417), and $f^\perp$ the Petersson orthogonal complement of $f$ in $S_2(\Gamma_0(N))$, which is $\mathbb{T}$-stable (Chapter 13). $\varepsilon_N(f) = \pm 1$ is the eigenvalue of $w_N$ on $f$.

**Three conventions that matter silently.** (1) The factor $(\det\gamma)^{k/2}$ in the slash operator is chosen so that scalar matrices act trivially in weight 2: for $\gamma = -N\cdot 1$ one gets $(N^2)^1(-N)^{-2}f = f$. Since $W_N^2 = -N \cdot 1$, this is what makes $f \mapsto f|_2 W_N$ an involution of $S_2(\Gamma_0(N))$, so that its eigenvalues $\varepsilon_N(f)$ are $\pm 1$ and not $\pm\sqrt{\text{something}}$. It also makes $(f|_2\gamma)(z)\,dz$ the pullback of $f(z)\,dz$ along $z \mapsto \gamma z$, because $d(\gamma z)/dz = \det\gamma/(cz+d)^2$; every comparison below between Hecke operators on forms and on cohomology rests on that identity. (2) The sign $\varepsilon_N(f)$ is the eigenvalue of $w_N$, not the root number: the root number of $L(f, s)$ is $-\varepsilon_N(f)$ (Chapters 13 and 19). For `11a1`, of rank 0 and root number $+1$, $\varepsilon_{11}(f) = -1$; for `37a1`, of rank 1 and root number $-1$, $\varepsilon_{37}(f) = +1$ (PARI's `ellrootno`). (3) The Hecke algebra acts on two kinds of objects, covariantly (pushforward, on $J_0(N)$ and $H_1$) and contravariantly (pullback, on forms and $H^1$). The two actions are transposes of each other; for $T_n$ with $(n, N) = 1$ the distinction is harmless, but for $U_p$, $p \mid N$, the transpose is a different operator (Theorem 14.11), and this is the reason the pairing $\hat{\delta}_!$ of the paper contains $w_N$.

| sentence or object of the paper | where in this chapter |
|---|---|
| "the $X_0(N)$-optimal curve, for which the modular parametrisation $\varphi_E\colon X_0(N) \to E$ has minimal degree" | Theorem 14.30 and Proposition 14.31 |
| "$\deg \varphi_E$ is $\langle f, f\rangle$ divided by the area of the period lattice" | Theorem 14.40 |
| "$\deg \varphi_E$ divides the *congruence number* $r_E$" | Theorem 14.48 and Corollary 14.55 |
| $\hat{\delta}_!$ perfect, alternating, $\mathbb{T}$ self-adjoint | Theorem 14.11 and Corollary 14.12 |
| Proposition prop:pullback | Proposition 14.52 and Theorem 14.54 |
| Lemma lem:hecke, Remark rem:free | Theorems 14.56 and 14.62 and Remark 14.63 |
| `99A1`: degree 4, congruence number 12 | Examples 14.50 and 14.58 |
| Conjecture conj:eis, Open question 4 | Section 14.11 |
| Section 1.5, Section sec:two | Section 14.12 |

## 14.2 The modular curve $X_0(N)$ over $\mathbb{Q}$

Chapter 12 builds $X_0(N)$ as a compact Riemann surface. The paper needs more: the modular parametrisation $\varphi_E\colon X_0(N) \to E$ is a morphism of curves over $\mathbb{Q}$, its cohomology carries an action of $G_{\mathbb{Q}}$ (Section 4.1 of the paper works with $H^1$ of a curve over $\mathbb{Q}$), and the cusps $\infty$ and 0 are rational points (the base point of $\varphi_E$, and the point whose image is studied in Open question 4). This section states the existence of the $\mathbb{Q}$-model, describes the cusps over $\mathbb{Q}$, records what is known about rational points, and works out the smallest levels.

**Theorem 14.2** ($X_0(N)$ over $\mathbb{Q}$; stated)**.**

*(i) There is a smooth projective geometrically connected curve $X_0(N)_{\mathbb{Q}}$ over $\mathbb{Q}$ whose complex points are the Riemann surface $X_0(N)$, such that $Y_0(N)_{\mathbb{Q}}$ is a coarse moduli space for pairs $(E, C)$ of an elliptic curve and a cyclic subgroup of order $N$: for every algebraically closed field $K$ of characteristic 0, $Y_0(N)(K)$ is in bijection with the isomorphism classes of such pairs over $K$, compatibly with $\mathrm{Aut}(K)$, and for $K = \mathbb{C}$ the bijection is the one of Chapter 12, $\Gamma_0(N)\tau \mapsto (\mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau), \langle \frac{1}{N}\rangle)$.*

*(ii) The forgetful map $X_0(N) \to X(1) = \mathbb{P}^1_j$, $(E, C) \mapsto j(E)$, and the involutions $w_d$ for $d \parallel N$ are defined over $\mathbb{Q}$; on moduli $w_N(E, C) = (E/C, E[N]/C)$.*

*(iii) The function field of $X_0(N)_{\mathbb{Q}}$ is $\mathbb{Q}(j, j_N)$, $j_N(z) = j(Nz)$, and $\Phi_N(j, j_N) = 0$ for the modular polynomial $\Phi_N \in \mathbb{Z}[X, Y]$.*

*(iv) $X_0(N)_{\mathbb{Q}}$ has good reduction at every prime $p \nmid N$ (Igusa; Deligne and Rapoport). For $p \parallel N$ it has a semistable model over $\mathbb{Z}_p$ (Deligne and Rapoport) whose special fibre is two copies of $X_0(N/p)_{\mathbb{F}_p}$ meeting in ordinary double points at the supersingular points; at such a point the complete local ring over $W(\overline{\mathbb{F}}_p)$ is $W(\overline{\mathbb{F}}_p)[[x, y]]/(xy - p^k)$ with $k = \frac{1}{2}\#\,\mathrm{Aut}$ of the corresponding supersingular curve with its level structure, so $k \in \{1, 2, 3\}$ when $p \geq 5$ (for $N = p \in \{2, 3\}$ it can reach 12 or 6), and the model is regular except at the supersingular points whose pair has automorphisms other than $\pm 1$ (possible only for $j = 0$ or 1728), which Edixhoven resolves.*

We do not prove this; see [180, Sections 6.7, 7.2], [52, Sections 7.7, 8.2 to 8.6], [51, Sections 8 and 9], [43] and [91] (section numbers not verified), and [59] for the local rings at the supersingular

points and the minimal regular model. The idea of (i) and (iii): $j$ and $j_N$ are $\Gamma_0(N)$-invariant functions on $\mathcal{H}$ with $q$-expansions in $\mathbb{Z}((q))$, they generate the function field of $X_0(N)$ over $\mathbb{C}$ (Chapter 12), and the subfield $\mathbb{Q}(j, j_N)$ is linearly disjoint from $\mathbb{C}$ over $\mathbb{Q}$ (the $q$-expansion principle: a function in $\mathbb{C}(j, j_N)$ whose $q$-expansion has rational coefficients lies in $\mathbb{Q}(j, j_N)$); the smooth projective curve over $\mathbb{Q}$ with this function field is $X_0(N)_{\mathbb{Q}}$. The polynomial $\Phi_N$ has integer coefficients because the conjugates $j(\gamma z)$, $\gamma$ running over $\mathrm{SL}_2(\mathbb{Z})\backslash\{\alpha \in M_2(\mathbb{Z}) : \det\alpha = N,\ \alpha \text{ primitive}\}$, are permuted by $\mathrm{SL}_2(\mathbb{Z})$ and have $q$-expansions with coefficients in $\mathbb{Z}[\zeta_N]$, permuted by $\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ (Kronecker). From now on $X_0(N)$ also denotes the curve over $\mathbb{Q}$.

Two words in (i) deserve comment. A *coarse* moduli space has the right points over algebraically closed fields, but over a non-closed field $K$ a $K$-rational point need not come from a pair defined over $K$, and two pairs over $K$ with the same point need not be isomorphic over $K$; the obstruction is the automorphism group of the pair, which is $\{\pm 1\}$ except for $j = 0$ and $j = 1728$. Proposition 14.5 makes this precise. The simplest case is $N = 1$: $Y_0(1) = \mathbb{A}^1_j$ and $X_0(1) = \mathbb{P}^1_j$, and two elliptic curves over $\mathbb{Q}$ with the same $j$-invariant give the same rational point although they need not be isomorphic over $\mathbb{Q}$: for $j \neq 0, 1728$ they are quadratic twists of each other, and for example $y^2 = x^3 - x$ and its quadratic twist by 2, $y^2 = x^3 - 4x$, both with $j = 1728$, are not isomorphic over $\mathbb{Q}$ (an isomorphism would have to be $x \mapsto u^2 x$ with $u^4 = 4$). Part (iv) is needed in two places: good reduction at $p \nmid N$ gives the Eichler-Shimura relation (Theorem 14.24) and the unramifiedness of $T_\ell J_0(N)$ at $p$; the semistable model at $p \,\|\, N$ is what makes $J_0(N)$ semistable there, which is used for $X_0(37)$ in Example 14.6(c).

**Proposition 14.3** (cusps over $\mathbb{Q}$)**.**

*(i) The cusps of $X_0(N)$ are the classes of $a/c$ with $c \mid N$, $c \geq 1$, $\gcd(a, c) = 1$, $a$ taken modulo $\gcd(c, N/c)$; there are $\varphi(\gcd(c, N/c))$ of denominator $c$ ($\varphi$ is Euler's function in this proposition only), and $\sum_{c|N} \varphi(\gcd(c, N/c))$ in all.*

*(ii) The cusps of denominator $c$ are defined over $\mathbb{Q}(\zeta_{\gcd(c,N/c)})$, and $\mathrm{Gal}(\mathbb{Q}(\zeta_{\gcd(c,N/c)})/\mathbb{Q})$ permutes them simply transitively.*

*(iii) In particular $\infty$ ($c = N$) and $0$ ($c = 1$) are $\mathbb{Q}$-rational, and every cusp is $\mathbb{Q}$-rational if and only if $\gcd(c, N/c) \leq 2$ for every $c \mid N$.*

*Sketch of proof.* (i) is the description of the cusps of $\Gamma_0(N)$ in Chapter 12. (ii) Near the cusps the moduli problem degenerates to the Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$ over $\mathbb{Q}((q^{1/N}))$; the cusp of $a/c$ corresponds to a cyclic subgroup of order $N$ generated by an element $\zeta_N^u q^{v/N}$, and the field of definition of that subgroup modulo the automorphisms of the Tate curve with its subgroup is $\mathbb{Q}(\zeta_{\gcd(c,N/c)})$, the element $\sigma_t\colon \zeta \mapsto \zeta^t$ sending the cusp of $a/c$ to that of $a'/c$ with $a' \equiv t^{\pm 1} a \pmod{\gcd(c, N/c)}$ (the sign depends on the normalisation of the moduli interpretation at the cusps, and only the field of definition is used below). See [146], [187, Section 6.3] and [52, Chapter 3, exercises] (not verified). (iii) $\varphi(d) = 1$ iff $d \leq 2$; for $c = N$ and $c = 1$, $\gcd(c, N/c) = 1$. □

**Example 14.4** (cusps at three levels)**.** By (i), $X_0(20)$ has 6 cusps ($\gcd(c, 20/c) = 1, 2, 1, 1, 2, 1$ for $c = 1, 2, 4, 5, 10, 20$), all rational. $X_0(99)$ has 8 cusps: one for each of $c = 1, 9, 11, 99$, and two for each of $c = 3$ and $c = 33$ ($\gcd = 3$), each pair conjugate over $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$. $X_0(121)$ has 12 cusps: $\infty$, 0 and ten of denominator 11, conjugate over $\mathbb{Q}(\zeta_{11})$. These counts agree with Chapter 12.

**Proposition 14.5** (rational points of $Y_0(N)$)**.**

*(i) Let $K$ be a field of characteristic 0. If $E/K$ has a cyclic subgroup $C$ of order $N$ stable under $G_K$, the point $(E, C)$ of $Y_0(N)$ is $K$-rational. Conversely a $K$-rational point of $Y_0(N)$ with $j \neq 0, 1728$ is represented by such a pair over $K$, unique up to quadratic twist.*

*(ii) (Mazur; Kenku; stated.) $Y_0(N)(\mathbb{Q}) \neq \emptyset$ if and only if $N \leq 19$ or $N$ is one of 21, 25, 27, 37, 43, 67, 163. $Y_0(N)(\mathbb{Q})$ is infinite exactly when $X_0(N)$ has genus 0 ($N \leq 10$ or $N \in \{12, 13, 16, 18, 25\}$), and the rational points of $Y_0(27)$, $Y_0(43)$, $Y_0(67)$ and $Y_0(163)$ are CM points.*

*Proof.* (i) If $E$ and $C$ are defined over $K$, then for $\sigma \in \mathrm{Aut}(\bar{K}/K)$ the pair $(E^\sigma, C^\sigma)$ is $(E, C)$, so its point is fixed by $\sigma$ (Theorem 14.2(i)), and a point of a curve over $K$ fixed by $\mathrm{Aut}(\bar{K}/K)$ is $K$-rational. Conversely, let $P \in Y_0(N)(K)$ with $j(P) \neq 0, 1728$, and let $E_0/K$ be any curve with $j(E_0) = j(P)$ (Chapter 9). Over $\bar{K}$, $P$ is $(E_0, C)$ for a cyclic $C \subset E_0$ of order $N$. For $\sigma$ fixing $K$, $(E_0, C^\sigma) \simeq (E_0, C)$; as $\mathrm{Aut}(E_0) = \{\pm 1\}$ and $-1$ fixes every subgroup, $C^\sigma = C$. Two pairs over $K$ with the same point are isomorphic over $\bar{K}$ by an isomorphism unique up to sign, so they differ by a quadratic twist (Chapter 9). (ii) We do not prove this; see [126] for prime $N$, [92, 94, 93, 95] for the remaining levels and [114] for $N = 27$; the list is [96, Theorem 1] (checked against the published text: $Y_0(N)(\mathbb{Q})$ is empty except for $N \leq 10$, $N = 12$, 13, 16, 18, 25, of genus 0, and $N = 11$, 14, 15, 17, 19, 21, 27, 37, 43, 67, 163). Kenku credits the levels of genus one to Ligozat, Ogg and Joly and does not discuss complex multiplication; the CM statement is not in his paper. □

A consequence used in Chapter 9 (the proof of Proposition prop:no9 of the paper): a rational cyclic 27-isogeny is a rational point of $Y_0(27)$, so it exists only for CM curves [114].

**Example 14.6** (the smallest levels)**.**

(a) $X_0(11)$ has genus 1 and the rational point $\infty$, so it is an elliptic curve over $\mathbb{Q}$ with origin $\infty$, and $J_0(11) = X_0(11)$. Its period lattice is the lattice of periods of $2\pi i f_{11}(z)\,dz$, $f_{11} = \eta(z)^2\eta(11z)^2$ (Chapter 12), and its $j$-invariant, computed from the period ratio (Chapter 9), is $-122023936/161051$: $X_0(11)$ is the curve `11a1` $= [0, -1, 1, -10, -20]$ (the other two curves of the class have $j = -5289315910115737 6/11$ and $-4096/11$). The modular parametrisation of `11a1` is the identity, of degree 1.

(b) In the same way $X_0(14)$ and $X_0(20)$, of genus 1, are the curves `14a1` and `20a1`, with parametrisations of degree 1.

(c) $X_0(37)$ has genus 2, and $w_{37}$ has 2 fixed points (Chapter 12); by Riemann-Hurwitz for the quotient map of degree 2, $2g - 2 = 2(2g' - 2) + \sum_P (e_P - 1)$, and each fixed point of the involution is a ramification point with $e_P = 2$ while all other points are unramified; so $2 \cdot 2 - 2 = 2(2g' - 2) + 2$, and $X_0(37)/w_{37}$ has genus $g' = 1$. With origin the image of $\infty$, it is an elliptic curve over $\mathbb{Q}$ (the quotient map is defined over $\mathbb{Q}$ by Theorem 14.2(ii)), and the pullback of its holomorphic differential is $w_{37}$-invariant. $S_2(\Gamma_0(37))$ is spanned by the newforms $f_{37a}$ and $f_{37b}$ with $\varepsilon_{37} = +1$ and $-1$, so the pullback is a multiple of $f_{37a}\,dz$. The quotient has good reduction outside 37 (it is a quotient of $J_0(37)$, Theorem 14.16), and multiplicative reduction at 37: $J_0(37)$ has semistable reduction there, by the semistable model of Theorem 14.2(iv), and a quotient of an abelian variety with semistable reduction has semistable reduction (Grothendieck); so its conductor divides 37, and it is 37, no elliptic curve over $\mathbb{Q}$ having conductor 1; and the pullback of the differential of a curve of conductor

37 along a modular parametrisation is a multiple of its newform (Proposition 14.38). So the quotient lies in the class `37a` (Corollary 14.26). That class has one curve, `37a1`; so $X_0(37) \to X_0(37)/w_{37} \simeq$ `37a1` is a modular parametrisation of degree 2, and it is the optimal one (Example 14.36).

The identifications in (a) and (b) are computations from the periods; they are confirmed by Cremona's tables [35, 36], which give $\deg\varphi = 1$ for `11a1`, `14a1`, `20a1` and 2 for `37a1`.

*In the paper* 14.7. The abstract of the paper: "$\varphi_E\colon X_0(N) \to E$ its modular parametrisation" is a morphism of curves over $\mathbb{Q}$ sending the rational cusp $\infty$ to $O$ (Theorem 14.2, Proposition 14.3(iii)). Section 4.1 uses $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$ with its Galois action: this is the étale cohomology of the curve over $\mathbb{Q}$ (Proposition 14.18). The rational isogenies of Section sec:eis and the cyclic 27-isogeny of Proposition prop:no9 are rational points of $Y_0(\ell)$ and $Y_0(27)$ (Proposition 14.5).

The curve is now defined over $\mathbb{Q}$, with rational cusps $\infty$ and 0. What it still lacks is the structure that links it to modular forms beyond the identification $S_2 = H^0(X_0(N), \Omega^1)$: the Hecke operators, which so far act only on $q$-expansions. The next section realises them geometrically.

## 14.3 Hecke correspondences and their adjoints

The Hecke operators of Chapter 13 act on $q$-expansions. To act on $J_0(N)$, on $H_1(X_0(N), \mathbb{Z})$ and on $H^1(X_0(N), \mathbb{Z}_\ell)$, as Section 4.1 of the paper and Remark rem:free require, they must come from correspondences on the curve. This section defines the Hecke correspondences, shows that they induce Chapter 13's operators on differentials, so that the same ring $\mathbb{T}$ acts on forms and on cohomology, and proves the identity the paper uses in Section 4.1: *the adjoint of $T_n$ for the cup product is $w_N T_n w_N$*, with $U_p$ included. Chapter 8 supplies the general theory (correspondences, their actions, the adjoint of a correspondence is its transpose); the input specific to $X_0(N)$ is a two-line matrix computation.

For $\alpha \in M_2(\mathbb{Z})$ with $\det\alpha = m > 0$ put $\Gamma_\alpha = \Gamma \cap \alpha^{-1}\Gamma\alpha$. It has finite index in $\Gamma$: if $\gamma = 1 + NmX \in \Gamma(Nm)$, then $\alpha\gamma\alpha^{-1} = 1 + Nm\,\alpha X\alpha^{-1} = 1 + N\alpha X\alpha'$ with $\alpha' = m\alpha^{-1} \in M_2(\mathbb{Z})$ (the adjugate of $\alpha$), an integral matrix of determinant 1 congruent to 1 modulo $N$, hence in $\Gamma(N) \subseteq \Gamma$; so $\gamma \in \alpha^{-1}\Gamma\alpha$, and $\Gamma(Nm) \subseteq \Gamma_\alpha$. As $\Gamma(Nm)$ has finite index in $\mathrm{SL}_2(\mathbb{Z})$, so has $\Gamma_\alpha$.

Why correspondences? On the upper half plane the Hecke operator $T_p$ of Chapter 13 is a sum over the $p+1$ matrices $\left(\begin{smallmatrix}1 & j\\ 0 & p\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}p & 0\\ 0 & 1\end{smallmatrix}\right)$: it sends a point to a formal sum of $p+1$ points, and on moduli it sends an elliptic curve to the sum of its $p+1$ quotients by subgroups of order $p$. Such a "multivalued map" of a curve to itself is a correspondence: a third curve $C$ with two maps to $X_0(N)$, one used to pull points back (giving the several values) and the other to push them forward. Here $C = X(\Gamma_\alpha)$: a point of $C$ is a point $\Gamma z$ of $X_0(N)$ together with a choice of one of its values ($p+1$ of them for $T_p$, $p$ for $U_p$), and $\pi_1$, $\pi_\alpha$ forget the choice and take the value.

**Definition 14.8** (Hecke correspondences)**.** For $\alpha$ as above, the *Hecke correspondence* of $\alpha$ is the correspondence $\mathcal{C}_\alpha = (X(\Gamma_\alpha), \pi_1, \pi_\alpha)$ from $X_0(N)$ to itself in the sense of Chapter 8, where $X(\Gamma_\alpha)$ is the compactified quotient $\Gamma_\alpha\backslash\mathcal{H}^*$ and

$$\pi_1(\Gamma_\alpha z) = \Gamma z, \qquad \pi_\alpha(\Gamma_\alpha z) = \Gamma\alpha z.$$

($\pi_\alpha$ is well defined: for $\gamma \in \Gamma_\alpha$, $\alpha\gamma z = (\alpha\gamma\alpha^{-1})\alpha z$ with $\alpha\gamma\alpha^{-1} \in \Gamma$.) For a prime $p$ put $T_p = \mathcal{C}_{\mathrm{diag}(1,p)}$ if $p \nmid N$ and $U_p = \mathcal{C}_{\mathrm{diag}(1,p)}$ if $p \mid N$. For $\beta \in \mathrm{GL}_2^+(\mathbb{Q})$ normalising $\Gamma$, $w_\beta$ is the automorphism $\Gamma z \mapsto \Gamma\beta z$ of $X_0(N)$; for $\beta = W_N$ it is the involution $w_N$ ($W_N^2 = -N$ acts trivially on $\mathcal{H}$).

By Chapter 8, $\mathcal{C}_\alpha$ acts on divisors and on $J_0(N)$ by $(\mathcal{C}_\alpha)_* = \pi_{\alpha*}\pi_1^*$, on $H_1$ by $\pi_{\alpha*}\pi_1^!$, on $H^1$ by $\mathcal{C}_\alpha^* = \pi_{1*}\pi_\alpha^*$, and on holomorphic differentials by $\omega \mapsto \mathrm{Tr}_{\pi_1}(\pi_\alpha^*\omega)$.

**Proposition 14.9** (Hecke correspondences)**.** *Let $\alpha \in M_2(\mathbb{Z})$, $\det\alpha > 0$.*

*(i) If $\gamma_1, \dots, \gamma_r$ represent $\Gamma_\alpha\backslash\Gamma$, then $\Gamma\alpha\Gamma = \bigsqcup_i \Gamma\alpha\gamma_i$, and $(\mathcal{C}_\alpha)_*(\Gamma z) = \sum_i(\Gamma\alpha\gamma_i z)$ for $z \in \mathcal{H}$ (with multiplicities at the ramification points of $\pi_1$). If $\beta = \gamma\alpha\gamma'$ with $\gamma, \gamma' \in \Gamma$, then $\mathcal{C}_\beta \simeq \mathcal{C}_\alpha$: the map $\Gamma_\alpha z \mapsto \Gamma_\beta\gamma'^{-1}z$ is an isomorphism $X(\Gamma_\alpha) \to X(\Gamma_\beta)$ compatible with the two pairs of maps. So $\mathcal{C}_\alpha$ depends only on $\Gamma\alpha\Gamma$.*

*(ii) Let $p$ be prime and $\alpha = \mathrm{diag}(1,p)$. Then $\Gamma_\alpha = \Gamma_0(N) \cap \Gamma^0(p)$, $\alpha\Gamma_\alpha\alpha^{-1} = \Gamma_0(Np)$, and $\Gamma_\alpha z \mapsto \Gamma_0(Np)(z/p)$ is an isomorphism $X(\Gamma_\alpha) \simeq X_0(Np)$ under which $\pi_1$ becomes $z \mapsto pz$ and $\pi_\alpha$ becomes $z \mapsto z$. The degree of $\pi_1$ is $p+1$ if $p \nmid N$ and $p$ if $p \mid N$, with coset representatives giving*

$$\Gamma\alpha\Gamma = \bigsqcup_{j=0}^{p-1} \Gamma \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} \sqcup \Gamma \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \ (p \nmid N), \qquad \Gamma\alpha\Gamma = \bigsqcup_{j=0}^{p-1} \Gamma \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} \ (p \mid N).$$

*(iii) (Moduli; sketched.) Under Theorem 14.2(i), $(T_p)_*(E,C) = \sum_D (E/D, (C+D)/D)$ over the $p+1$ subgroups $D \subset E$ of order $p$ ($p \nmid N$), and $(U_p)_*(E,C) = \sum_{D \cap C = 0}(E/D, (C+D)/D)$ over the $p$ subgroups $D$ of order $p$ meeting $C$ trivially ($p \mid N$).*

*(iv) $T_p$, $U_p$ and the $w_d$ ($d \,\|\, N$) are correspondences defined over $\mathbb{Q}$.*

*Proof.* (i) $\Gamma\alpha\gamma = \Gamma\alpha\gamma'$ iff $\alpha\gamma'\gamma^{-1}\alpha^{-1} \in \Gamma$ iff $\gamma'\gamma^{-1} \in \Gamma \cap \alpha^{-1}\Gamma\alpha = \Gamma_\alpha$; so the cosets $\Gamma\alpha\gamma_i$ are distinct and exhaust $\Gamma\alpha\Gamma$. The fibre of $\pi_1$ over $\Gamma z$ is $\{\Gamma_\alpha\gamma_i z\}$ (with multiplicities), and $\pi_\alpha(\Gamma_\alpha\gamma_i z) = \Gamma\alpha\gamma_i z$; this is the formula for $\pi_{\alpha*}\pi_1^*$. For $\beta = \gamma\alpha\gamma'$: $\beta^{-1}\Gamma\beta = \gamma'^{-1}\alpha^{-1}\Gamma\alpha\gamma'$, so $\Gamma_\beta = \gamma'^{-1}\Gamma_\alpha\gamma'$ and $\Gamma_\alpha z \mapsto \Gamma_\beta\gamma'^{-1}z$ is a well-defined isomorphism. It satisfies $\pi_1(\Gamma_\beta\gamma'^{-1}z) = \Gamma z$ and $\pi_\beta(\Gamma_\beta\gamma'^{-1}z) = \Gamma\gamma\alpha\gamma'\gamma'^{-1}z = \Gamma\alpha z$.

(ii) For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma$, $\alpha\gamma\alpha^{-1} = \left(\begin{smallmatrix} a & b/p \\ pc & d \end{smallmatrix}\right)$, which lies in $\Gamma$ iff $p \mid b$; so $\Gamma_\alpha = \Gamma \cap \Gamma^0(p)$, and its conjugate $\alpha\Gamma_\alpha\alpha^{-1}$ consists of the $\left(\begin{smallmatrix} a & b' \\ c' & d \end{smallmatrix}\right)$ of determinant 1 with $Np \mid c'$, which is $\Gamma_0(Np)$. The map $z \mapsto \alpha z = z/p$ therefore induces $X(\Gamma_\alpha) \simeq X_0(Np)$, and $\Gamma z = \Gamma p(z/p)$, $\Gamma\alpha z = \Gamma(z/p)$ give the two maps. The index $[\Gamma_0(N) : \Gamma_0(N) \cap \Gamma^0(p)]$ is $p+1$ for $p \nmid N$ and $p$ for $p \mid N$. To see this, reduce modulo $p$: the condition $p \mid b$ only depends on $\gamma \bmod p$, so the index is that of the subgroup $\{b = 0\}$ in the image $G$ of $\Gamma_0(N)$ in $\mathrm{SL}_2(\mathbb{F}_p)$. If $p \nmid N$, then $G = \mathrm{SL}_2(\mathbb{F}_p)$ (already $\Gamma(N)$ surjects onto $\mathrm{SL}_2(\mathbb{F}_p)$, because $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/Np) = \mathrm{SL}_2(\mathbb{Z}/N) \times \mathrm{SL}_2(\mathbb{Z}/p)$ is onto), and $\{b = 0\}$ is the lower triangular subgroup, of order $p(p-1)$ in a group of order $p(p^2-1)$: index $p+1$. If $p \mid N$, then $c \equiv 0$ and $G$ is the upper triangular subgroup of order $p(p-1)$, in which $\{b = 0\}$ is the diagonal subgroup of order $p-1$: index $p$. The displayed matrices lie in $\Gamma\alpha\Gamma$ ($\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right) = \alpha\left(\begin{smallmatrix} 1 & j \\ 0 & 1 \end{smallmatrix}\right)$) and represent distinct cosets; see Chapter 13 and [52, Section 5.2]. For $p \nmid N$ the matrix $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ lies in $\Gamma\alpha\Gamma$ by the computation in the proof of Theorem 14.11(iv).

(iii) (Sketch.) The point $\Gamma z$ is $(\mathbb{C}/\Lambda_z, \langle\frac{1}{N}\rangle)$ with $\Lambda_z = \mathbb{Z}+\mathbb{Z}z$. The lattices $\Lambda_{(z+j)/p}$ $(0 \le j < p)$ and $\frac{1}{p}\mathbb{Z} + \mathbb{Z}z = \frac{1}{p}\Lambda_{pz}$ are the $p+1$ lattices containing $\Lambda_z$ with index $p$, that is, the subgroups $D$ of order $p$ of $E = \mathbb{C}/\Lambda_z$, and $\mathbb{C}/\Lambda' = E/D$; the image of $\frac{1}{N}$ generates $(C+D)/D$ (for $\Lambda' = \frac{1}{p}\Lambda_{pz}$ one rescales by $p$, and $\frac{1}{N}$, $\frac{1}{p}$ generate $\frac{1}{Np}\mathbb{Z}/\mathbb{Z}$ when $p \nmid N$). For $p \mid N$ the missing lattice $\frac{1}{p}\mathbb{Z} + \mathbb{Z}z$ is the one whose subgroup $D = \langle\frac{1}{p}\rangle$ lies in $C = \langle\frac{1}{N}\rangle$. See [52, Sections 5.2, 7.9].

(iv) $X_0(Np)$, $z \mapsto pz$ and $z \mapsto z$ are defined over $\mathbb{Q}$ by Theorem 14.2 (on moduli, with $C'$ the cyclic subgroup of order $Np$ attached to a point of $X_0(Np)$, $z \mapsto z$ is $(E, C') \mapsto (E, pC')$ and $z \mapsto pz$ is $(E, C') \mapsto (E/C'[p], C'/C'[p])$), and so are the $w_d$. □

Chapter 8 writes the Hecke correspondence for $p \nmid N$ as $(X_0(Np), z \mapsto z, z \mapsto pz)$. Since $\Gamma_{\mathrm{diag}(p,1)} = \Gamma_0(Np)$ (the computation in (ii)), that is $\mathcal{C}_{\mathrm{diag}(p,1)}$, the transpose of $\mathcal{C}_{\mathrm{diag}(1,p)}$ (Theorem 14.11(i)); for $p \nmid N$ the two are isomorphic (Theorem 14.11(iv)), but for $p \mid N$ the correspondence $(X_0(Np), z \mapsto z, z \mapsto pz)$ is $U_p^t$, not $U_p$, and $U_p$ itself is $(X_0(Np), z \mapsto pz, z \mapsto z)$.

**Proposition 14.10** (compatibility with the Hecke operators on forms)**.**

*(i) For $f \in S_2(\Gamma_0(N))$ and $\Gamma\alpha\Gamma = \bigsqcup_i \Gamma\alpha_i$: $\mathcal{C}_\alpha^*(f(z)\,dz) = \big(\sum_i f|_2\alpha_i\big)(z)\,dz$. For $\alpha = \mathrm{diag}(1,p)$ this is $(T_p f)(z)\,dz$ ($p \nmid N$), respectively $(U_p f)(z)\,dz$ ($p \mid N$), with Chapter 13's operators.*

*(ii) There is a unique ring homomorphism $\mathbb{T} \to \mathrm{End}_{\mathbb{Z}} H^1(X_0(N), \mathbb{Z})$, $t \mapsto t^*$, with $T_p \mapsto \mathcal{C}^*_{\mathrm{diag}(1,p)}$ for every prime $p$. It is injective, and its $\mathbb{C}$-linear extension to $H^1(X_0(N), \mathbb{C}) = S_2 \oplus \overline{S_2}$ is $t$ on $S_2$ and $\bar\omega \mapsto \overline{t\omega}$ on $\overline{S_2}$. The transposes $t_*$ on $H_1(X_0(N), \mathbb{Z}) = H^1(X_0(N), \mathbb{Z})^\vee$ define a second injective ring homomorphism, which is the action of the correspondences by pushforward. In particular $\mathbb{T}$ is a finite free $\mathbb{Z}$-module, and $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$ acts faithfully on $H^1(X_0(N), \mathbb{Z}_\ell)$ and on $H_1(X_0(N), \mathbb{Z}_\ell)$.*

*(iii) Since $\mathbb{T} \otimes \mathbb{C} \simeq \mathrm{Hom}_{\mathbb{C}}(S_2, \mathbb{C})$ by $(t, g) \mapsto a_1(tg)$ (Chapter 13), every $\psi \in \mathrm{Hom}_{\mathbb{Z}}(\mathbb{T}, \mathbb{Z})$ is $t \mapsto a_1(tg_\psi)$ for a unique $g_\psi \in S_2$ with $a_n(g_\psi) = \psi(T_n) \in \mathbb{Z}$; so $S_2(\Gamma_0(N); \mathbb{Z})$ has rank $g$. (This is the route from the integral structure of $H^1$ to Shimura's integral basis; Chapters 12 and 13 prove the statement differently.)*

*Proof.* (i) By definition $\mathcal{C}_\alpha^*\omega = \mathrm{Tr}_{\pi_1}(\pi_\alpha^*\omega)$. On $\mathcal{H}$, $\pi_\alpha$ is $z \mapsto \alpha z$, and $d(\alpha z)/dz = \det\alpha/(cz+d)^2$, so $\pi_\alpha^*(f(z)\,dz) = (f|_2\alpha)(z)\,dz$ (Convention 14.1(C1)). The local sections of $\pi_1$ near a point $\Gamma z$ off the branch locus are $s_i \colon \Gamma z \mapsto \Gamma_\alpha \gamma_i z$, so $\mathrm{Tr}_{\pi_1}(g\,dz) = \sum_i s_i^*(g\,dz) = \sum_i (g|_2\gamma_i)\,dz$ for a weight-two form $g$ on $\Gamma_\alpha$. With $g = f|_2\alpha$ and $\alpha_i = \alpha\gamma_i$ (part (i) of Proposition 14.9) the sum is $\sum_i f|_2\alpha_i$. For $\alpha = \mathrm{diag}(1,p)$ and the representatives of Proposition 14.9(ii): $f|_2\left(\begin{smallmatrix}1 & j\\ 0 & p\end{smallmatrix}\right) = p\cdot p^{-2} f(\frac{z+j}{p}) = \frac{1}{p} f(\frac{z+j}{p})$ and $f|_2\left(\begin{smallmatrix}p & 0\\ 0 & 1\end{smallmatrix}\right) = p \cdot 1^{-2} f(pz) = pf(pz)$. With $f = \sum_n a_n q^n$,

$$\sum_{j=0}^{p-1} \frac{1}{p} f\Big(\frac{z+j}{p}\Big) = \sum_n a_n e^{2\pi i n z/p} \cdot \frac{1}{p} \sum_{j=0}^{p-1} e^{2\pi i n j/p} = \sum_{p\mid n} a_n q^{n/p} = \sum_m a_{mp} q^m,$$

because the inner sum is 1 if $p \mid n$ and 0 otherwise, and $pf(pz) = \sum_n p a_n q^{pn} = \sum_m p a_{m/p} q^m$ (with $a_{m/p} = 0$ if $p \nmid m$). Summing gives $\sum_m (a_{mp} + p a_{m/p}) q^m$ (the last term for $p \nmid N$ only), which is Chapter 13's $T_p f$, respectively $U_p f$, in weight 2, where the normalising factor $(\det\alpha)^{k/2-1}$ is 1.

The class of a holomorphic $\omega$ is mapped by $\mathcal{C}^* = \pi_{1*}\pi_\alpha^*$ on $H^1$ to the class of $\mathrm{Tr}_{\pi_1}\pi_\alpha^*\omega$: Chapter 8 computes $\pi_{1*}$ on a representative $\theta$ vanishing near the branch points; if $\theta = \omega + dF$, then $\mathrm{Tr}\,\theta - \mathrm{Tr}\,\omega = d(\mathrm{Tr}\,F)$ on the complement of the branch points, so the two forms have the same integrals over loops avoiding them, and such loops span $H_1$.

(ii) Let $P$ be the polynomial ring $\mathbb{Z}[x_p : p \text{ prime}]$, let $\rho_1 \colon P \to \mathrm{End}_{\mathbb{C}} S_2$ send $x_p$ to $T_p$ (or $U_p$), and $\rho_2 \colon P \to \mathrm{End}_{\mathbb{Z}} H^1(X_0(N), \mathbb{Z})$, $x_p \mapsto \mathcal{C}^*_{\mathrm{diag}(1,p)}$. The image of $\rho_1$ is $\mathbb{T}$, because every $T_n$ is a polynomial in the $T_p$ and $U_p$ (Chapter 13). Each $\mathcal{C}^*$ on $H^1(X_0(N), \mathbb{C})$ preserves $S_2$ (pullbacks and traces of holomorphic forms are holomorphic, Chapter 8), acts there as $\rho_1(x_p)$ by (i), and commutes with complex conjugation, being defined over $\mathbb{Z}$. So for $F \in P$, $\rho_2(F)$ is

$\rho_1(F)$ on $S_2$ and its conjugate on $\overline{S_2}$, and $\rho_2(F)=0$ iff $\rho_1(F)=0$ ($H^1(X_0(N),\mathbb{Z})$ is a lattice in $H^1(X_0(N),\mathbb{C})$, which is $S_2\oplus\overline{S_2}$ by Chapter 8). Hence $\ker\rho_1=\ker\rho_2$, and $t\mapsto t^*$ is the isomorphism $\mathbb{T}=P/\ker\rho_1\to\rho_2(P)$. Transposition reverses products, and $\mathbb{T}$ is commutative, so $t\mapsto t_*$ is again a ring homomorphism; it is the pushforward action because $\mathcal{C}^*$ is the transpose of $\mathcal{C}_*$ (Chapter 8). $\mathbb{T}$ is a subring of the endomorphism ring of a lattice of rank $2g$, hence free of finite rank over $\mathbb{Z}$; $\mathbb{T}\otimes\mathbb{Z}_\ell\to\mathrm{End}(H^1(X_0(N),\mathbb{Z}))\otimes\mathbb{Z}_\ell$ is injective because $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$.

(iii) $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{T},\mathbb{Z})$ is a lattice in $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{T},\mathbb{C})=\mathrm{Hom}_{\mathbb{C}}(\mathbb{T}\otimes\mathbb{C},\mathbb{C})$, which the perfect pairing identifies with $S_2$; and $a_n(g_\psi)=a_1(T_ng_\psi)=\psi(T_n)$. □

Before the computation, the moduli picture shows what to expect. The transpose of a correspondence runs it backwards: $(T_p)^t$ sends $(E',C')$ to the sum of the pairs $(E,C)$ from which $T_p$ reaches $(E',C')$, that is, it goes back along the dual isogenies $E'=E/D\to E$. For $p\nmid N$ the kernel $E[p]/D$ of the dual is just another subgroup of order $p$ of $E'$, so going back is again of the form "divide by a subgroup of order $p$", and $T_p^t=T_p$. For $p\mid N$ the kernel of the dual of $E\to E/D$ with $D\cap C=0$ is $E[p]/D$, and since $C[p]+D=E[p]$ this is the subgroup of order $p$ of the level structure $(C+D)/D$: $U_p$ divides by subgroups meeting $C$ trivially, while its transpose divides by the subgroup contained in $C$. These are different operations, and $w_N$, which replaces $C$ by $E[N]/C$ on the quotient curve, exchanges "contained in the level structure" and "meeting it trivially". The theorem makes this precise with matrices.

**Theorem 14.11** (the transpose of a Hecke correspondence)**.**

*(i) For $\alpha\in M_2(\mathbb{Z})$ with $\det\alpha>0$ put $\alpha'=(\det\alpha)\alpha^{-1}$. Then $\mathcal{C}_\alpha^t\simeq\mathcal{C}_{\alpha'}$: the map $u\colon X(\Gamma_\alpha)\to X(\Gamma_{\alpha'})$, $\Gamma_\alpha z\mapsto\Gamma_{\alpha'}\alpha z$, is an isomorphism with $\pi_1\circ u=\pi_\alpha$ and $\pi_{\alpha'}\circ u=\pi_1$.*

*(ii) For $\beta\in\mathrm{GL}_2^+(\mathbb{Q})$ normalising $\Gamma$, $\mathcal{C}_{\beta\alpha\beta^{-1}}\simeq w_\beta\,\mathcal{C}_\alpha\,w_\beta^{-1}$ (whenever $\beta\alpha\beta^{-1}\in M_2(\mathbb{Z})$), so $\mathcal{C}^*_{\beta\alpha\beta^{-1}}=(w_\beta^{-1})^*\mathcal{C}^*_\alpha w^*_\beta$ on $H^1$.*

*(iii) $W_N\,\mathrm{diag}(1,p)\,W_N^{-1}=\mathrm{diag}(p,1)=\mathrm{diag}(1,p)'$. Hence, for every prime $p$,*

$$T_p^t\simeq w_NT_pw_N\ (p\nmid N),\qquad U_p^t\simeq w_NU_pw_N\ (p\mid N),$$

*and for every $t\in\mathbb{T}$, $(t^*)^\dagger=w_N^*t^*w_N^*$ for the cup product on $H^1(X_0(N),\mathbb{Z})$, and likewise for $t_*$ and the intersection pairing on $H_1(X_0(N),\mathbb{Z})$.*

*(iv) For $p\nmid N$, $\Gamma\,\mathrm{diag}(1,p)\,\Gamma=\Gamma\,\mathrm{diag}(p,1)\,\Gamma$; so $T_p^t\simeq T_p$, $T_p^*$ is self-adjoint for the cup product and commutes with $w_N^*$. For $p\mid N$, $U_p^*$ is in general not self-adjoint for the cup product (Exercise 14.82).*

*Proof.* (i) Scalars act trivially on $\mathcal{H}$ and by conjugation, so $\alpha'^{-1}\Gamma\alpha'=\alpha\Gamma\alpha^{-1}$ and $\Gamma_{\alpha'}=\Gamma\cap\alpha\Gamma\alpha^{-1}=\alpha\Gamma_\alpha\alpha^{-1}$. Hence $\Gamma_\alpha z\mapsto\Gamma_{\alpha'}\alpha z$ is well defined and bijective, an isomorphism of the compact Riemann surfaces (it is induced by an automorphism of $\mathcal{H}^*$). Then $\pi_1(u(\Gamma_\alpha z))=\Gamma\alpha z=\pi_\alpha(\Gamma_\alpha z)$, and $\pi_{\alpha'}(u(\Gamma_\alpha z))=\Gamma\alpha'\alpha z=\Gamma(\det\alpha)z=\Gamma z=\pi_1(\Gamma_\alpha z)$. So $u$ is an isomorphism from $\mathcal{C}_\alpha^t=(X(\Gamma_\alpha),\pi_\alpha,\pi_1)$ to $\mathcal{C}_{\alpha'}=(X(\Gamma_{\alpha'}),\pi_1,\pi_{\alpha'})$.

(ii) Put $\alpha_\beta=\beta\alpha\beta^{-1}$. As $\beta$ normalises $\Gamma$, $\Gamma_{\alpha_\beta}=\Gamma\cap\beta\alpha^{-1}\beta^{-1}\Gamma\beta\alpha\beta^{-1}=\beta\Gamma_\alpha\beta^{-1}$, and $v\colon\Gamma_\alpha z\mapsto\Gamma_{\alpha_\beta}\beta z$ is an isomorphism with $\pi_1(v(\Gamma_\alpha z))=\Gamma\beta z=w_\beta(\pi_1(\Gamma_\alpha z))$ and $\pi_{\alpha_\beta}(v(\Gamma_\alpha z))=\Gamma\beta\alpha z=w_\beta(\pi_\alpha(\Gamma_\alpha z))$. In the notation of Chapter 8, $(C,v^{-1}\circ\alpha,u\circ\beta)=u\,\mathcal{C}\,v$; here the triple is $(X(\Gamma_\alpha),w_\beta\pi_1,w_\beta\pi_\alpha)=w_\beta\,\mathcal{C}_\alpha\,w_\beta^{-1}$, which acts on $H^1$ by $(w_\beta^{-1})^*\mathcal{C}^*_\alpha w^*_\beta$ (Chapter 8).

(iii) $W_N \operatorname{diag}(1,p) = \left(\begin{smallmatrix} 0 & -p \\ N & 0 \end{smallmatrix}\right)$ and $W_N^{-1} = \frac{1}{N}\left(\begin{smallmatrix} 0 & 1 \\ -N & 0 \end{smallmatrix}\right)$, so

$$W_N \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} W_N^{-1} = \frac{1}{N}\begin{pmatrix} 0 & -p \\ N & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -N & 0 \end{pmatrix} = \frac{1}{N}\begin{pmatrix} pN & 0 \\ 0 & N \end{pmatrix} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix},$$

and $\operatorname{diag}(1,p)' = p\operatorname{diag}(1,p)^{-1} = \operatorname{diag}(p,1)$. By (i) and (ii), with $\alpha = \operatorname{diag}(1,p)$ and $\beta = W_N$ (which normalises $\Gamma_0(N)$, Chapter 12), $\mathcal{C}_\alpha^t \simeq \mathcal{C}_{\alpha'} = \mathcal{C}_{W_N\alpha W_N^{-1}} \simeq w_N \mathcal{C}_\alpha w_N^{-1}$, and $w_N^{-1} = w_N$. Isomorphic correspondences act identically, so by Chapter 8 (the adjoint of $\mathcal{C}^*$ for the cup product is $(\mathcal{C}^t)^*$) and (ii), $(T_p^*)^\dagger = (T_p^t)^* = w_N^* T_p^* w_N^*$, and the same for $U_p$. Let $\mathcal{A} = \{t \in \mathbb{T} : (t^*)^\dagger = w_N^* t^* w_N^*\}$. It contains 1 and every $T_p$, $U_p$, and it is closed under sums; it is closed under products because, with $w = w_N^*$, $w^2 = 1$ and $\mathbb{T}$ commutative, $((st)^*)^\dagger = (s^*t^*)^\dagger = (t^*)^\dagger (s^*)^\dagger = wt^*ws^*w = w(ts)^*w = w(st)^*w$. As $\mathbb{T}$ is generated as a ring by the $T_p$ and $U_p$, $\mathcal{A} = \mathbb{T}$. Poincare duality is an isometry carrying $\mathcal{C}^*$ to $(\mathcal{C}^t)_*$ (Chapter 8), which gives the statement on $H_1$.

(iv) Choose integers $b$, $d$ with $pd - bN = 1$ and put $\gamma_1 = \left(\begin{smallmatrix} p & b \\ N & d \end{smallmatrix}\right)$, $\gamma_2 = \left(\begin{smallmatrix} 1 & b \\ N & dp \end{smallmatrix}\right)$, both in $\Gamma_0(N)$. Then $\gamma_1 \operatorname{diag}(1,p) = \left(\begin{smallmatrix} p & bp \\ N & dp \end{smallmatrix}\right) = \operatorname{diag}(p,1)\,\gamma_2$, so $\operatorname{diag}(p,1) = \gamma_1 \operatorname{diag}(1,p)\gamma_2^{-1}$ and the double cosets agree. By Proposition 14.9(i) and (i) above, $T_p^t \simeq \mathcal{C}_{\operatorname{diag}(p,1)} \simeq T_p$, so $(T_p^*)^\dagger = T_p^*$; with (iii), $T_p^* = w_N^* T_p^* w_N^*$. □

Why a twisted pairing is wanted. The congruence modules of Chapter 3 are built from a perfect pairing on a lattice for which the whole Hecke algebra is self-adjoint: then the idempotents of $\mathbb{T}_\ell$ are self-adjoint, the local components are orthogonal, and the orthogonal complement of the $f$-part is the part killed by the annihilator of $I_f$. The cup product is perfect but, by Theorem 14.11, the $U_p^*$ are not self-adjoint for it. Composing with $w_N^*$ repairs this, at no cost: the new pairing is still perfect, and it stays alternating because $w_N^*$ is a self-adjoint isometry.

**Corollary 14.12** (the pairing $\hat{\delta}_!$)**.** *Let $R$ be $\mathbb{Z}$, $\mathbb{Z}_\ell$ or $\mathbb{Q}_\ell$ and $\langle\ ,\ \rangle$ the cup product on $H^1(X_0(N),R)$ (Chapter 8).*

*(i)* $(t^*)^\dagger = w_N^* t^* w_N^*$ *for every* $t \in \mathbb{T} \otimes R$.

*(ii) The pairing*

$$\hat{\delta}_!(x,y) = \langle x, w_N^* y\rangle$$

*is perfect and alternating on $H^1(X_0(N),R)$, and every $t^*$, $t \in \mathbb{T}\otimes R$ (the $U_p^*$ included), is self-adjoint for it.*

*(iii) Let $R = \mathbb{Z}_\ell$. Then every idempotent of $\mathbb{T}_\ell$ is self-adjoint for $\hat{\delta}_!$,*

$$H^1(X_0(N),\mathbb{Z}_\ell) = \bigoplus_{\mathfrak{m}} H^1(X_0(N),\mathbb{Z}_\ell)_{\mathfrak{m}}$$

*(over the maximal ideals $\mathfrak{m}$ of $\mathbb{T}_\ell$) is an orthogonal decomposition, and $\hat{\delta}_!$ is perfect on each summand.*

*Proof.* (i) is Theorem 14.11(iii), extended $R$-linearly. (ii) $w = w_N^*$ is an isometry of the cup product with $w^2 = 1$ (Chapter 8: an automorphism of a compact Riemann surface is holomorphic, hence orientation preserving of degree 1, and $\langle w^*x, w^*y\rangle = \deg(w_N)\langle x,y\rangle$), so $w^\dagger = w^{-1} = w$. The cup product is perfect, so $\hat{\delta}_!$, its composite with the automorphism $w$, is perfect: in matrix terms, if $G$ is the Gram matrix of the cup product in a basis and $W$ the matrix of

$w$, the Gram matrix of $\hat{\delta}_!$ is $GW$, and $\det(GW) = \det G \det W = \pm \det G$ is a unit. For the symmetry, use that the cup product on $H^1$ of a surface is antisymmetric and that $w$ is a self-adjoint isometry: $\hat{\delta}_!(y,x) = \langle y, wx\rangle = -\langle wx, y\rangle = -\langle x, w^\dagger y\rangle = -\langle x, wy\rangle = -\hat{\delta}_!(x,y)$; so $\hat{\delta}_!$ is antisymmetric, and alternating because $2\hat{\delta}_!(x,x) = 0$ and $H^1(X_0(N), R)$ is torsion-free. For $t \in \mathbb{T} \otimes R$, $\hat{\delta}_!(t^*x, y) = \langle x, (t^*)^\dagger wy\rangle = \langle x, wt^* wwy\rangle = \langle x, wt^* y\rangle = \hat{\delta}_!(x, t^*y)$. (iii) $\mathbb{T}_\ell$ is a finite free $\mathbb{Z}_\ell$-algebra (Proposition 14.10(ii)), so $\mathbb{T}_\ell = \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$ with orthogonal idempotents $e_{\mathfrak{m}}$ (Chapter 3). An element of $\mathbb{T}_\ell$ is self-adjoint by (ii); for an idempotent $e$, $\hat{\delta}_!(ex, (1-e)y) = \hat{\delta}_!(x, e(1-e)y) = 0$, so the decomposition is orthogonal, and a perfect pairing restricted to the summands of an orthogonal decomposition is perfect on each (Chapter 3). This is Chapter 8's corollary on twisted pairings, whose hypothesis $t^t = w^* t w^*$ is (i). □

The smallest case shows the sign. At level 11, $S_2(\Gamma_0(11)) = \mathbb{C} f_{11}$ and $\varepsilon_{11}(f_{11}) = -1$, so $w_{11}^*$ is $-1$ on $H^1(X_0(11), \mathbb{Z}) \simeq \mathbb{Z}^2$ (it is $-1$ on $\omega_f$ and on $\overline{\omega_f}$, which span $H^1(X_0(11), \mathbb{C})$), and $\hat{\delta}_! = -\langle\ ,\ \rangle$: in a symplectic basis its Gram matrix is $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, perfect and alternating, as it must be. The twist only becomes visible when $\mathbb{T}$ contains a $U_p$ that is not self-adjoint for the cup product (Exercise 14.82).

*Remark* 14.13 (the Petersson side). On $S_2(\Gamma_0(N))$, $T_n$ is self-adjoint for the Petersson product when $(n, N) = 1$, and $U_p^\dagger = w_N U_p w_N$ for $p \mid N$ (Chapter 13); a newform is a $w_N$-eigenvector with eigenvalue $\varepsilon_N(f) = \pm 1$. The two statements agree: on $S_2 \subset H^1(X_0(N), \mathbb{C})$, $\langle f, g\rangle = \frac{i}{2}\langle \omega_f, \overline{\omega_g}\rangle_{\text{cup}}$, and $\mathcal{C}^*$ commutes with complex conjugation, so $\frac{i}{2}\langle \mathcal{C}^*\omega_f, \overline{\omega_g}\rangle = \frac{i}{2}\langle \omega_f, \overline{(\mathcal{C}^t)^*\omega_g}\rangle$: the Petersson adjoint of $\mathcal{C}^*$ on $S_2$ is $(\mathcal{C}^t)^*$. This is why $f$ is an eigenvector of the adjoints of the $T_n$ (Lemma lem:crit), with the same eigenvalues.

*In the paper* 14.14. In Section 4.1 of the paper the pairing $\hat{\delta}_!$ "is perfect and alternating on $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$, and the Hecke operators, $U_p$ included, are self-adjoint for it, because $w_{M^\Sigma} T_n w_{M^\Sigma}^{-1}$ is the adjoint of $T_n$ for the cup product (this is what the composition with $w$ is for; see [49, §§1.6.1, 1.8.2] and [38, §4.4])". This is Theorem 14.11(iii) and Corollary 14.12(ii) at level $M^\Sigma$. The paper's $\hat{\delta}_!$ is that of [49], which on classes from $X_0(M^\Sigma)$ the paper describes as "$\frac{u}{2}$ times the cup product on $X_0(M^\Sigma)$ composed with $w_{M^\Sigma}$, for an $\ell$-adic unit $u$, so for odd $\ell$ only a unit separates the two"; the book's $\hat{\delta}_!$ (Appendix B, B-C13(f)) omits the factor $\frac{u}{2}$, a unit for odd $\ell$, so the two have the same ideal of values on every sublattice (Remark 14.57(3)). The sentence of the paragraph after Lemma lem:hecke, "the idempotent $e_{\mathfrak{m}}$ is self-adjoint for $\hat{\delta}_!$, so $H^1(X_0(N), \mathbb{Z}_\ell)$ splits orthogonally into its local components and the pairing is perfect on each", is Corollary 14.12(iii).

So far everything has been said in terms of $H^1$ and $H_1$ of the Riemann surface. The modular parametrisation, however, is a map to an elliptic curve, and the natural home of maps from a curve to elliptic curves is its Jacobian; the next section brings it in and sorts out which Hecke action lives where.

## 14.4 The Jacobian $J_0(N)$, its homology and the $f$-lattice

The modular parametrisation factors through the Jacobian $J_0(N)$, optimality is a statement about the kernel of $J_0(N) \to E$, Remark rem:free speaks of $J_0(N)[\mathfrak{m}]$ and of the Tate module, and Section 4.1 of the paper defines the lattice $M_{f,\lambda}$ inside $H^1(X_0(N), \mathbb{Z}_\ell)$. This section applies Chapter 8 to $X_0(N)$, keeps track of which Hecke action lives on which object, and defines $M_{f,\lambda}$.

**Definition 14.15** ($J_0(N)$)**.** $J_0(N) = \mathrm{Jac}(X_0(N)_{\mathbb{Q}})$, the Jacobian of the curve $X_0(N)$ over $\mathbb{Q}$ (Chapter 8); $\iota\colon X_0(N) \to J_0(N)$, $P \mapsto [(P) - (\infty)]$, is the Abel-Jacobi map with base point $\infty$.

**Theorem 14.16** (structure of $J_0(N)$)**.**

*(i)* *$J_0(N)$ is an abelian variety over $\mathbb{Q}$ of dimension $g = \dim S_2(\Gamma_0(N))$, with good reduction at every prime $p \nmid N$.*

*(ii)* *$J_0(N)(\mathbb{C}) = S_2(\Gamma_0(N))^\vee / H_1(X_0(N), \mathbb{Z})$, where $\gamma \in H_1$ is sent to $f \mapsto \int_\gamma f(z)\,dz$.*

*(iii)* *$\iota$ is defined over $\mathbb{Q}$, and every morphism $X_0(N) \to A$ over $\mathbb{Q}$ to an abelian variety sending $\infty$ to $0$ is $h \circ \iota$ for a unique homomorphism $h\colon J_0(N) \to A$ over $\mathbb{Q}$.*

*(iv)* *For $N = 11$, 14, 20, 37, 77, 99, 121, 140, 539, 726, 980 the dimension of $J_0(N)$ is 1, 1, 1, 2, 7, 9, 6, 19, 49, 109, 145.*

*Proof.* (i) The Jacobian of a smooth projective curve of genus $g$ over $\mathbb{Q}$ is an abelian variety of dimension $g$ over $\mathbb{Q}$ (Chapter 8, stated), and $g = \dim S_2$ because $f \mapsto f\,dz$ identifies $S_2(\Gamma_0(N))$ with the holomorphic differentials (Chapter 12). Good reduction of $J_0(N)$ at $p \nmid N$ follows from that of $X_0(N)$ (Theorem 14.2(iv)); we do not prove the implication (the Jacobian of a smooth proper curve over $\mathbb{Z}_p$ is an abelian scheme; see [15, Section 9.4], not verified). (ii) is the analytic Jacobian and the Abel-Jacobi theorem of Chapter 8 with $H^0(X_0(N), \Omega^1) = S_2$. (iii) The base point $\infty$ is rational (Proposition 14.3), so the universal property of Chapter 8 applies over $\mathbb{Q}$. (iv) is `mfdim([N,2],1)` in PARI, and equals the genus formula of Chapter 12. □

Two remarks on the statement. The choice of base point matters for (iii): a morphism $X_0(N) \to A$ factors through a homomorphism only after it has been normalised to send the base point to 0, and the base point must be rational for this to work over $\mathbb{Q}$; this is why modular parametrisations are required to send $\infty$ to $O$. And in genus one (ii) is concrete: for $N = 11$, $S_2(\Gamma_0(11))^\vee = \mathbb{C}$ by $\lambda \mapsto \lambda(f_{11})$, the lattice is $\{\int_\gamma f_{11}(z)\,dz : \gamma \in H_1\}$, and $\iota$ is an isomorphism $X_0(11) \to J_0(11)$, the Abel-Jacobi map of a curve of genus one with a rational point.

**Proposition 14.17** (the Hecke action on $J_0(N)$ and $T_\ell J_0(N)$)**.**

*(i)* *$t \mapsto t_*$ is an injective ring homomorphism $\mathbb{T} \to \mathrm{End}_{\mathbb{Q}}(J_0(N))$; on $J_0(N)(\mathbb{C}) = S_2^\vee / H_1$ the endomorphism $t_*$ is induced by the transpose of $t$ on $S_2$, and on the lattice $H_1$ it is $t_*$ of Proposition 14.10(ii).*

*(ii)* *On $T_\ell J_0(N) = H_1(X_0(N), \mathbb{Z}_\ell)$ the action is $t_*$, and it commutes with $G_{\mathbb{Q}}$.*

*(iii)* *For $t \in \mathbb{T}$ put $t^t = w_N t w_N \in \mathrm{End}_{\mathbb{C}} S_2$; on $H^1$ its action is $(t^t)^* = w_N^* t^* w_N^* = (t^*)^\dagger$ (Theorem 14.11(iii)), and $t \mapsto t^t$ is a ring isomorphism from $\mathbb{T}$ onto $w_N \mathbb{T} w_N$. In general $w_N \mathbb{T} w_N \neq \mathbb{T}$ (Exercise 14.82). The* Picard *action of $\mathbb{T}$ on $J_0(N)$ is $t \mapsto (t^t)_*$. A newform $f$ of level $N$ is an eigenvector of every $t^t$ with eigenvalue $\phi_f(t)$. Hence $I_f J_0(N)$, $J_0(N)/I_f J_0(N)$ and $H^1(X_0(N), \mathbb{Z}_\ell)[I_f]$ are the same whether $\mathbb{T}$ acts through $t_*$ (Albanese) or through $(t^t)_*$ (Picard).*

*Proof.* (i) A correspondence acts on $J_0(N)$ by $\mathcal{C}_*$, compatibly with Abel-Jacobi: by the transpose of $\mathcal{C}^*$ on differentials and by $\mathcal{C}_*$ on $H_1$ (Chapter 8). On generators these are $\mathbb{T}$'s operators, so the ring homomorphism of Proposition 14.10(ii) gives an action on the complex torus, injective because it is injective on $H_1$; it is defined over $\mathbb{Q}$ because the correspondences are (Proposition 14.9(iv)). (ii)

$T_\ell J = H_1 \otimes \mathbb{Z}_\ell$ naturally for homomorphisms of tori (Chapter 8); an endomorphism defined over $\mathbb{Q}$ commutes with $G_\mathbb{Q}$ on torsion points. (iii) The first assertions are Theorem 14.11(iii) and the fact that conjugation by an involution is a ring isomorphism. If $f$ is a newform, $f|_2 W_N = \varepsilon_N(f) f$ with $\varepsilon_N(f) = \pm 1$ (Chapter 13), so $w_N t w_N f = \varepsilon_N(f) w_N t f = \varepsilon_N(f)\phi_f(t) w_N f = \phi_f(t) f$. The Picard version of $I_f J_0(N)$ is $\sum_{t \in I_f} w_{N*} t_* w_{N*} J_0(N) = w_{N*}(I_f J_0(N))$. Its Lie algebra, in $S_2^\vee$, is the image under the transpose of $w_N^*$ of the Lie algebra of $I_f J_0(N)$; the latter is the annihilator of $S_2[I_f] = \mathbb{C} f$ (the Lie algebra of $t_* J_0(N)$ is the image of the transpose of $t$, whose annihilator is $\ker t$), and $w_N^*$ preserves $\mathbb{C} f$; a connected closed subgroup of a complex torus is determined by its Lie algebra, so $w_{N*}(I_f J_0(N)) = I_f J_0(N)$. On $H^1$, $x$ is killed by all $(t^t)^* - \phi_f(t)$, $t \in I_f$, iff $w_N^* x \in M_{f,\lambda}$, iff $x \in M_{f,\lambda}$, because $w_N^*$ preserves $M_{f,\lambda}$ (Proposition 14.21(iii)). □

**Proposition 14.18** ($T_\ell J_0(N)$ and $H^1(X_0(N), \mathbb{Z}_\ell)$ as Hecke modules)**.**

*(i)* $H^1(X_0(N), \mathbb{Z}_\ell) = \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell J_0(N), \mathbb{Z}_\ell)$*, with* $t^*$ *the transpose of* $t_*$*. As* $G_\mathbb{Q}$*-modules (étale cohomology of the curve over* $\mathbb{Q}$*),* $H^1_{\mathrm{et}}(X_0(N)_{\overline{\mathbb{Q}}}, \mathbb{Z}_\ell) \simeq (T_\ell J_0(N))^\vee \simeq T_\ell J_0(N)(-1)$*, and the cup product is* $G_\mathbb{Q}$*-equivariant with values in* $\mathbb{Z}_\ell(-1)$*.*

*(ii) The map* $\Psi = w_{N*} \circ \mathrm{PD} \colon H^1(X_0(N), \mathbb{Z}_\ell) \to H_1(X_0(N), \mathbb{Z}_\ell) = T_\ell J_0(N)$ *is an isomorphism of* $\mathbb{T}_\ell$*-modules for the actions* $t^*$ *on the source and* $t_*$ *on the target. Poincare duality* PD *alone intertwines* $t^*$ *with* $(t^t)_*$*. After the Tate twist by* $\mathbb{Z}_\ell(-1)$*,* $\Psi$ *is* $G_\mathbb{Q}$*-equivariant.*

*(iii) Hence, for every maximal ideal* $\mathfrak{m}$ *of* $\mathbb{T}_\ell$*, the module* $T_\ell J_0(N)_\mathfrak{m}$ *is free over* $\mathbb{T}_\mathfrak{m}$ *if and only if* $H^1(X_0(N), \mathbb{Z}_\ell)_\mathfrak{m}$ *is, and* $H^1(X_0(N), \mathbb{F}_\ell)_\mathfrak{m}[\mathfrak{m}] \simeq J_0(N)[\mathfrak{m}]$ *as* $\mathbb{T}/\mathfrak{m}$*-vector spaces.*

*Proof.* (i) is Chapter 8 (the Tate module and $H^1$; the Weil pairing and étale cohomology, stated there), with the comparison isomorphism compatible with correspondences. (ii) By Chapter 8, $\mathrm{PD} \circ \mathcal{C}^* = (\mathcal{C}^t)_* \circ \mathrm{PD}$, so $\mathrm{PD} \circ t^* = (t^t)_* \circ \mathrm{PD} = (w_N t w_N)_* \circ \mathrm{PD}$ by Theorem 14.11(iii) and Proposition 14.17(iii). Composing with $w_{N*}$ and using $w_N^2 = 1$: $\Psi \circ t^* = w_{N*}(w_N t w_N)_* \mathrm{PD} = t_* w_{N*} \mathrm{PD} = t_* \circ \Psi$. $\Psi$ is an isomorphism because both factors are. Over $\overline{\mathbb{Q}}$, Poincare duality is the Weil pairing $T_\ell J \simeq (T_\ell J)^\vee(1)$, which is $G_\mathbb{Q}$-equivariant, and $w_N$ is defined over $\mathbb{Q}$; so $\Psi$ is $G_\mathbb{Q}$-equivariant as a map $H^1 \to T_\ell J(-1)$. (iii) An isomorphism of $\mathbb{T}_\ell$-modules maps $\mathfrak{m}$-components to $\mathfrak{m}$-components. For the second statement reduce $\Psi$ modulo $\ell$: $H^1(X_0(N), \mathbb{F}_\ell) \simeq H_1(X_0(N), \mathbb{F}_\ell) = J_0(N)[\ell]$ (Chapter 8), $\mathbb{T}$-equivariantly, and take $\mathfrak{m}$-torsion. □

*Warning* 14.19 (four identifications that are easy to confuse)*.*

(1) $T_\ell J_0(N) = H_1(X_0(N), \mathbb{Z}_\ell)$: covariant, $\mathbb{T}$ acts by $t_*$.

(2) $H^1(X_0(N), \mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell J_0(N), \mathbb{Z}_\ell)$: contravariant, $\mathbb{T}$ acts by $t^*$; Galois-equivariantly it is $T_\ell J_0(N)(-1)$, not $T_\ell J_0(N)$.

(3) $\mathrm{PD} \colon H^1 \to H_1$ intertwines $t^*$ with $(t^t)_* = (w_N t w_N)_*$, not with $t_*$.

(4) $w_{N*} \circ \mathrm{PD}$ intertwines $t^*$ with $t_*$, and is Galois-equivariant only after the twist by $\mathbb{Z}_\ell(-1)$.

For an elliptic curve the difference between (1) and (2) is the difference between $T_\ell E$, of determinant $\chi_\ell$, and $H^1(E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$, of determinant $\chi_\ell^{-1}$. The adjoint does not see it: $\mathrm{ad}^0(T^\vee) \simeq \mathrm{ad}^0 T \simeq \mathrm{ad}^0(T(-1))$ (Chapter 4).

**Definition 14.20** (the $f$-lattice)**.** Let $f$ be a newform of level $N$ with rational coefficients and $\ell$ a prime. Put $V_f = H^1(X_0(N), \mathbb{Q})[I_f]$ and

$$M_{f,\lambda} = H^1(X_0(N), \mathbb{Z}_\ell)[I_f] = \{x : \ t^* x = \phi_f(t) x \text{ for all } t \in \mathbb{T}\}.$$

The plane $V_f$ is the part of the cohomology that "belongs to $f$"; over $\mathbb{Q}$ it is a 2-dimensional space, and the definition picks out a specific $\mathbb{Z}_\ell$-lattice in it. There are two natural lattices in $V_f \otimes \mathbb{Q}_\ell$: the intersection $M_{f,\lambda}$ with the integral cohomology, and the image of $H^1(X_0(N), \mathbb{Z}_\ell)$ under the projection onto the $f$-component $e_f H^1(X_0(N), \mathbb{Q}_\ell)$ (with $e_f$ the idempotent of $\mathbb{T} \otimes \mathbb{Q}_\ell$ of Proposition 14.46(i)). The first is contained in the second, with index measured by congruences between $f$ and the other forms; the paper and [49] use the first (Remark 14.57(4)). At level 11, where $\mathbb{T} = \mathbb{Z}$, the two agree and $M_{f,\lambda} = H^1(X_0(11), \mathbb{Z}_\ell)$.

**Proposition 14.21** (the $f$-lattice)**.** *In the situation of Definition 14.20:*

*(i)* $V_f \otimes \mathbb{C} = H^1(X_0(N), \mathbb{C})[I_f] = \mathbb{C}\,\omega_f \oplus \mathbb{C}\,\overline{\omega_f}$*, of dimension* 2*;* $V_f \otimes \mathbb{R}$ *is spanned by* $\operatorname{Re}\omega_f$ *and* $\operatorname{Im}\omega_f$*. The subspace killed by the anemic algebra* $\mathbb{T}^{(N)} = \mathbb{Z}[T_n : (n, N) = 1]$ *acting through* $\phi_f$ *is the same plane.*

*(ii)* $M_{f,\lambda} = H^1(X_0(N), \mathbb{Z})[I_f] \otimes \mathbb{Z}_\ell = H^1(X_0(N), \mathbb{Z}_\ell) \cap (V_f \otimes \mathbb{Q}_\ell)$*; it is a saturated* $\mathbb{Z}_\ell$*-sublattice of rank* 2*.*

*(iii)* $w_N^*$ *preserves* $V_f$ *and* $M_{f,\lambda}$*, and acts on both by* $\varepsilon_N(f)$*.*

*(iv)* $M_{f,\lambda}$ *is stable under* $G_\mathbb{Q}$ *(in étale cohomology).*

*Proof.* (i) Kernels of linear maps over $\mathbb{Q}$ commute with extension of scalars; so the complexification of $V_f$ is the space $H^1(X_0(N), \mathbb{C})[I_f]$. By Proposition 14.10(ii) this is $S_2[I_f] \oplus \overline{S_2[I_f]}$ (for $t \in I_f$ the class $t\bar{\omega}$, which is $\overline{t\omega}$, vanishes iff $t\omega$ does). The eigenspace of $\phi_f$ in $S_2$ is the line $\mathbb{C}f$: by the theory of newforms (Chapter 13), $S_2$ is the direct sum of the spaces spanned by the $g(dz)$, $g$ a newform of level $M \mid N$ and $d \mid N/M$, the anemic algebra acts on each by the eigenvalues of $g$, distinct newforms have distinct anemic eigenvalue systems (strong multiplicity one), and for $g = f$ of level $N$ only $d = 1$ occurs. This argument uses only $\mathbb{T}^{(N)}$, which gives the last sentence. (ii) The first equality is flatness of $\mathbb{Z}_\ell$ over $\mathbb{Z}$; the second holds because $x \in H^1(X_0(N), \mathbb{Z}_\ell)$ is killed by $I_f$ iff its image in $H^1(X_0(N), \mathbb{Q}_\ell)$ is. A kernel of endomorphisms of a torsion-free module is saturated (Chapter 3). The rank is $\dim V_f = 2$. (iii) $w_N^*$ commutes with every $t^*$, $t \in \mathbb{T}^{(N)}$ (Theorem 14.11(iv)), so it preserves the plane, which by (i) is the $\phi_f$-eigenspace of $\mathbb{T}^{(N)}$. On it $w_N^*\omega_f = (f|_2 W_N)\,dz = \varepsilon_N(f)\omega_f$, and $w_N^*\overline{\omega_f} = \overline{w_N^*\omega_f} = \varepsilon_N(f)\overline{\omega_f}$ since $\varepsilon_N(f)$ is real. $w_N^*$ preserves $H^1(X_0(N), \mathbb{Z}_\ell)$, hence $M_{f,\lambda}$. (iv) $t^*$ commutes with $G_\mathbb{Q}$, so the kernel of $I_f$ is $G_\mathbb{Q}$-stable. □

*In the paper* 14.22. Section 4.1 of the paper defines $M_{f^\Sigma,\lambda}$ as "the kernel on $M(M^\Sigma, 1)_{!,\lambda}$ of the kernel of the eigencharacter of $f^\Sigma$ on the full Hecke algebra, $U_p$ included"; with $\Sigma = \emptyset$ and $H^1(X_0(N), \mathbb{Z}_\ell)$ in place of the integral structure of Diamond, Flach and Guo this is Definition 14.20. In the proof of Proposition prop:pullback the paper notes that the plane $\mathbb{C}f\,dz \oplus \mathbb{C}\overline{f\,dz}$ "is defined over $\mathbb{Q}$: $H^1(X_0(N), \mathbb{Q})[I_f]$ is a two-dimensional $\mathbb{Q}$-form of it, and by flatness $H^1(X_0(N), \mathbb{Z}_\ell)[I_f] = H^1(X_0(N), \mathbb{Z})[I_f] \otimes \mathbb{Z}_\ell$ is a saturated lattice of rank two in $H^1(X_0(N), \mathbb{Q}_\ell)[I_f]$": this is Proposition 14.21(ii), with $V_f$ the $\mathbb{Q}$-form. It also says "By multiplicity one for newforms the plane is also the eigenspace of the $T_n$, $(n, N) = 1$, for the eigenvalues $a_n(f)$, and $w_N$ commutes with these $T_n$ (not with the $U_p$, $p \mid N$)": this is Proposition 14.21(i), (iii). Remark rem:free: "The identification of $T_\ell J_0(N)$ with $H^1(X_0(N), \mathbb{Z}_\ell)$ as Hecke modules uses Poincaré duality composed with $w_N$, which is not Galois equivariant; freeness transfers nevertheless" is Proposition 14.18(ii), (iii): $\Psi$ is Galois-equivariant after a Tate twist.

The lattice $M_{f,\lambda}$ has been defined purely in terms of Hecke operators. To connect it with an elliptic curve one needs a curve whose $H^1$ is $V_f$: this is Shimura's quotient, and the bridge between its Frobenius traces and the coefficients of $f$ is the congruence relation.

## 14.5 Eichler-Shimura and modularity

The paper starts from "the newform attached to" $E$. This section explains the two directions of that attachment: from a newform $f$ with rational coefficients Shimura's construction produces an elliptic curve $E_f$ as a quotient of $J_0(N)$, whose Frobenius traces are the $a_p(f)$ by the congruence relation of Eichler and Shimura; and the modularity theorem says that every elliptic curve over $\mathbb{Q}$ is isogenous to one of these. The quotient $E_f$ is the optimal curve of Section 14.6, and the congruence relation is the first step in the construction of the Galois representation of $f$ (Chapter 15).

**Theorem 14.23** (Shimura's quotient)**.** *Let $f$ be a newform of level $N$ with coefficient field $K_f$ and $I_f = \ker(\mathbb{T} \to K_f,\ T_n \mapsto a_n(f))$.*

*(i) $I_f J_0(N) = \sum_{t \in I_f} t_* J_0(N)$ is an abelian subvariety of $J_0(N)$ defined over $\mathbb{Q}$, and $J_f = J_0(N)/I_f J_0(N)$ is an abelian variety over $\mathbb{Q}$ of dimension $[K_f : \mathbb{Q}]$ on which $\mathbb{T}$ acts through $\mathbb{T}/I_f \hookrightarrow K_f$. The quotient map $\pi_f \colon J_0(N) \to J_f$ has connected kernel. (The book writes $J_f$ for Shimura's $A_f$, which is the paper's adjoint representation; Appendix B, B-C12(c).)*

*(ii) If $f$ has rational coefficients, $J_f =: E_f$ is an elliptic curve over $\mathbb{Q}$, $t \in \mathbb{T}$ acts on it as multiplication by $\phi_f(t)$, and $\varphi_{E_f} = \pi_f \circ \iota$ satisfies $\varphi_{E_f}^* H^1(E_f, \mathbb{Q}) = V_f$ and $\varphi_{E_f}^* \omega_{E_f} \in \mathbb{C}\,\omega_f$.*

*(iii) (Shimura; stated.) $L(J_f, s) = \prod_{\sigma \colon K_f \to \mathbb{C}} L(f^\sigma, s)$; in particular $L(E_f, s) = L(f, s)$.*

*Proof of (ii), and sketch of (i).* (i) Let $t_1, \dots, t_r$ generate $I_f$ as an abelian group. Then $I_f J_0(N)$, which is $\sum_i t_{i*} J_0(N)$, is the image of $J_0(N)^r$ under the homomorphism $(x_i) \mapsto \sum_i t_{i*} x_i$, defined over $\mathbb{Q}$; the image of an abelian variety under a homomorphism is an abelian subvariety (connected, as the image of a connected group), and the quotient of an abelian variety by an abelian subvariety exists over the field of definition (stated; see the references on abelian varieties in Chapter 8). $\mathbb{T}$ is commutative, so each $t_*$ preserves $I_f J_0(N)$ and acts on $J_f$; and $t - \phi_f(t)$, which lies in $I_f$ when $f$ is rational, maps $J_0(N)$ into $I_f J_0(N)$, so $t$ acts on $J_f$ as $\phi_f(t)$ in that case. The kernel $I_f J_0(N)$ is connected by construction. Over $\mathbb{C}$, the Lie algebra of $t_* J_0(N)$ is the image of the transpose of $t$ on $S_2^\vee$, whose annihilator in $S_2$ is $\ker t$; so the Lie algebra of $I_f J_0(N)$ is the annihilator of $S_2[I_f]$, which is spanned by the conjugates $f^\sigma$ and has dimension $[K_f : \mathbb{Q}]$ (Chapter 13). Hence $\dim J_f = [K_f : \mathbb{Q}]$. (ii) For rational $f$, $S_2[I_f] = \mathbb{C} f$ and $\dim J_f = 1$: $J_f$ is a connected projective group variety of dimension one over $\mathbb{Q}$, an elliptic curve with origin the image of 0. Let $x \in H^1(E_f, \mathbb{Q})$. As $t$ acts on $E_f$ as $\phi_f(t)$, and multiplication by an integer $m$ acts on $H^1$ of an elliptic curve as $m$ (Chapter 8), $t^* \varphi_{E_f}^* x = \varphi_{E_f}^*(\phi_f(t) x)$ (pullback along $\iota$ identifies $t_*$ on $J_0(N)$ with $t^*$ on $H^1(X_0(N))$, Proposition 14.17). So $\varphi_{E_f}^* H^1(E_f, \mathbb{Q}) \subseteq V_f$; both have dimension 2 ($\varphi_{E_f}^*$ is injective by Chapter 8), so they are equal. Then $\varphi_{E_f}^* \omega_{E_f}$ is a holomorphic form in $V_f \otimes \mathbb{C} = \mathbb{C}\omega_f \oplus \mathbb{C}\overline{\omega_f}$ (Proposition 14.21), hence in $\mathbb{C}\omega_f$. (iii) We do not prove this; see [180, Theorem 7.14 and Section 7.5] (theorem number not verified), [181], [52, Section 6.6]. At the primes $p \nmid N$ it follows from Theorem 14.24. □

**Theorem 14.24** (the congruence relation of Eichler and Shimura)**.** *Let $p \nmid N$ be prime.*

*(i) (Stated.) In* $\mathrm{End}(J_0(N)_{\mathbb{F}_p})$*, the reduction of* $(T_p)_*$ *is* $\pi_p + \hat{\pi}_p$*, where* $\pi_p$ *is the* $p$*-power Frobenius endomorphism and* $\hat{\pi}_p = p\pi_p^{-1}$ *the Verschiebung.*

*(ii) On* $T_\ell J_0(N)$*,* $\ell \neq p$*, the arithmetic Frobenius* $\mathrm{Frob}_p$ *(which acts as* $\pi_p$*, Chapter 8) satisfies* $\mathrm{Frob}_p^2 - (T_p)_* \mathrm{Frob}_p + p = 0$*. On each generalised eigenspace* $W_\psi$ *of* $\mathbb{T}$ *in* $T_\ell J_0(N) \otimes \overline{\mathbb{Q}}_\ell$ *(for a character* $\psi\colon \mathbb{T} \to \overline{\mathbb{Q}}_\ell$*) the eigenvalues of* $\mathrm{Frob}_p$ *are roots of* $X^2 - \psi(T_p)X + p$*.*

*(iii) If* $f$ *has rational coefficients,* $E_f$ *has good reduction at* $p$*,* $\det(X - \mathrm{Frob}_p \mid T_\ell E_f) = X^2 - a_p(f)X + p$*, and* $a_p(E_f) = p + 1 - \#\tilde{E}_f(\mathbb{F}_p) = a_p(f)$*; on* $H^1(E_f, \mathbb{Z}_\ell)$ *the geometric Frobenius has the same characteristic polynomial.*

*Proof.* (i) We do not prove this; see [180, Section 7.4], [52, Section 8.7], [51, Section 10] (numbering not verified). The idea is Kronecker's congruence $\Phi_p(X, Y) \equiv (X^p - Y)(X - Y^p) \pmod p$ read on moduli through Proposition 14.9(iii): over $\overline{\mathbb{F}}_p$, for an ordinary curve $E$ the quotient of $E$ by its connected subgroup of order $p$ is the Frobenius twist $E^{[p]}$, and the quotients by the $p$ étale subgroups are the curves whose Frobenius twist is $E$; so modulo $p$ the correspondence $T_p$ is the graph of the Frobenius morphism plus its transpose, which act on the Jacobian as $\pi_p$ and $\hat{\pi}_p$. (ii) As $J_0(N)$ has good reduction at $p$ and $\ell \neq p$, reduction modulo $p$ is injective on $\ell$-power torsion (the kernel of reduction is a pro-$p$ group, as for elliptic curves in Chapter 10), so it identifies $T_\ell J_0(N)$ with $T_\ell$ of the reduction, compatibly with all endomorphisms defined over $\mathbb{Q}$; under this identification the arithmetic Frobenius becomes $\pi_p$. Since $\pi_p\hat{\pi}_p = p$, (i) gives $\pi_p^2 - (T_p)_*\pi_p + p = \pi_p^2 - (\pi_p + \hat{\pi}_p)\pi_p + \hat{\pi}_p\pi_p = 0$. On $W_\psi$, $(T_p)_* - \psi(T_p)$ is nilpotent and commutes with $\mathrm{Frob}_p$; so $\mathrm{Frob}_p^2 - \psi(T_p)\,\mathrm{Frob}_p + p$ is nilpotent there, and every eigenvalue of $\mathrm{Frob}_p$ on $W_\psi$ is a root of $X^2 - \psi(T_p)X + p$. (iii) $E_f$ is a quotient of $J_0(N)$, which has good reduction at $p$, so $E_f$ has good reduction at $p$ (Néron-Ogg-Shafarevich, Chapter 11: $T_\ell E_f$ is a quotient of the unramified $T_\ell J_0(N) \otimes \mathbb{Q}_\ell$). The homomorphism $\pi_f$ reduces to a surjection $J_0(N)_{\mathbb{F}_p} \to (E_f)_{\mathbb{F}_p}$, which commutes with Frobenius and Verschiebung (both are natural for homomorphisms), and $T_p$ acts on $E_f$ as $a_p(f)$. Hence $a_p(f)\pi_f = \pi_f \circ (T_p)_* = (\pi_p + \hat{\pi}_p) \circ \pi_f$ after reduction, and as $\pi_f$ is surjective, $\pi_p + \hat{\pi}_p = a_p(f)$ in $\mathrm{End}(\tilde{E}_f)$. By Chapter 9, $\pi_p + \hat{\pi}_p = [p + 1 - \#\tilde{E}_f(\mathbb{F}_p)]$ and $\det(X - \mathrm{Frob}_p \mid T_\ell E_f) = X^2 - a_p(E_f)X + p$; $\mathrm{End}(\tilde{E}_f)$ is torsion-free, so $a_p(E_f) = a_p(f)$. The statement on $H^1$ is Chapter 8 (B-C3). □

**Theorem 14.25** (modularity; stated)**.** *For every elliptic curve* $E/\mathbb{Q}$ *of conductor* $N$ *there is a newform* $f_E \in S_2(\Gamma_0(N))$ *with rational coefficients such that* $a_p(f_E) = a_p(E)$ *for every prime* $p$ *(with* $a_p(E) \in \{0, \pm 1\}$ *the usual value at* $p \mid N$*). Equivalently,* $E$ *is isogenous over* $\mathbb{Q}$ *to* $E_{f_E}$*; and* $L(E, s) = L(f_E, s)$*.*

We do not prove this. It is the theorem of Wiles [207] and Taylor and Wiles [197] for semistable $E$, completed by Breuil, Conrad, Diamond and Taylor [16]; that the level of $f_E$ is the conductor is Carayol's theorem [21]. The conjecture goes back to Taniyama, Shimura and Weil; Weil showed that the functional equations of all twists characterise modular forms. The proof establishes an isomorphism $R = \mathbb{T}$ between a deformation ring of $\bar{\rho}_{E,\ell}$ and a localised Hecke algebra (Chapter 20), and it is in the course of that proof that the congruence ideals of Section 14.9 first appear. The equivalence of the two forms: $E_{f_E}$ has the same $a_p$ as $E$ for $p \nmid N$ (Theorem 14.24), so $T_\ell E \otimes \mathbb{Q}_\ell \simeq T_\ell E_{f_E} \otimes \mathbb{Q}_\ell$ (Chebotarev and Brauer-Nesbitt, Chapters 4 and 5; both are semisimple by Faltings, Chapter 11), and Faltings's isogeny theorem gives an isogeny; conversely isogenous curves have the same $a_p$ and the same conductor.

**Corollary 14.26** (isogeny classes and newforms)**.** $E \mapsto f_E$ *is a bijection from the* $\mathbb{Q}$*-isogeny classes of elliptic curves of conductor* $N$ *to the newforms of level* $N$ *with rational coefficients.*

*Sketch of proof.* Well defined: isogenous curves have the same $a_p$ (Chapter 9). Injective: curves with the same $a_p$ for almost all $p$ are isogenous (Faltings, as above). Surjective: for $f$ rational of level $N$, $E_f$ has conductor $N$ (Carayol [21]; stated) and $a_p(E_f) = a_p(f)$ for $p \nmid N$ (Theorem 14.24), so $f_{E_f}$ and $f$ are newforms of level $N$ with the same $a_p$ for almost all $p$, hence equal by strong multiplicity one (Chapter 13). □

**Example 14.27** (levels 11 and 99)**.** (a) The class `11a` corresponds to $f_{11} = \eta(z)^2\eta(11z)^2$. It has three curves, `11a1`, `11a2`, `11a3`, with isogeny degrees 5 (`11a1` to `11a2` and to `11a3`) and 25 (`11a2` to `11a3`) (`allisog` [36]). Since $X_0(11) = J_0(11)$ is `11a1` (Example 14.6) and $\mathbb{T} = \mathbb{Z}$ at level 11 (so $I_f = 0$ and $E_f = J_0(11)$), Theorem 14.24(iii) proves $\#\texttt{11a1}(\mathbb{F}_p) = p + 1 - a_p(f_{11})$ for every $p \neq 11$: the identity of Chapter 12, checked there numerically, is a theorem. The modular parametrisations of `11a2` and `11a3` are the two 5-isogenies out of `11a1`, of degree 5 each (Theorem 14.30(ii)).

(b) $\dim S_2(\Gamma_0(99)) = 9$. The newforms of level 99 with rational coefficients are those of the four classes `99a`, `99b`, `99c`, `99d`, and there are no others (`mfdim([99,2],0)` $= 4$); level 33 has one newform and level 11 one. By the decomposition of Chapter 13, a newform $g$ of level $M \mid N$ contributes the forms $g(dz)$, $d \mid N/M$, one for each divisor: the newforms of level 99 contribute one each, $f_{33}$ contributes $f_{33}(z)$, $f_{33}(3z)$ ($99/33 = 3$ has two divisors), and $f_{11}$ contributes $f_{11}(z)$, $f_{11}(3z)$, $f_{11}(9z)$ ($99/11 = 9$ has three). So $9 = 4 \cdot 1 + 1 \cdot 2 + 1 \cdot 3$. (Levels 1, 3 and 9 have no cusp forms of weight 2.) On the 3-dimensional old space of $f_{11}$ every $T_p$ with $p \nmid 99$ acts as the scalar $a_p(f_{11})$; this is why the anemic algebra alone cannot cut out the plane of an old form, while for a newform of level 99 it can (Proposition 14.21(i)).

*In the paper* 14.28. Section 1 of the paper: "Let $E/\mathbb{Q}$ be an elliptic curve of conductor $N$ and let $f = f_E \in S_2(\Gamma_0(N))$ be the newform attached to it" is Theorem 14.25; "the isogeny class of $E$" corresponds to $f$ by Corollary 14.26. Lemma lem:old uses the $a_p$ of newforms as traces of Frobenius, which is Theorem 14.24(iii) for rational newforms and its extension to all newforms in Chapter 15.

## 14.6 Optimal curves, the modular parametrisation and the modular degree

This section defines the objects of the paper's first sentence: the modular parametrisation, the $X_0(N)$-optimal curve and the modular degree. The main point is Theorem 14.30: every modular parametrisation of every curve of an isogeny class factors through Shimura's quotient $E_f$, so that "minimal degree", "connected kernel" and "saturated pullback on $H^1$" all single out the same curve. The section ends with Cremona's labelling convention and the table of modular degrees used in the rest of the book.

**Definition 14.29** (modular parametrisation)**.** Let $E/\mathbb{Q}$ have conductor $N$. A *modular parametrisation* of $E$ is a nonconstant morphism $\varphi\colon X_0(N) \to E$ over $\mathbb{Q}$ with $\varphi(\infty) = O$. By Theorem 14.16(iii), modular parametrisations correspond bijectively to nonzero homomorphisms $h\colon J_0(N) \to E$ over $\mathbb{Q}$, by $\varphi = h \circ \iota$ and $h = \varphi_*$ (the pushforward of divisor classes, Chapter 8).

**Theorem 14.30** (optimal curves)**.** *Let $\mathcal{I}$ be an isogeny class of elliptic curves over $\mathbb{Q}$ of conductor $N$, $f$ its newform (Corollary 14.26), $E_f = J_0(N)/I_f J_0(N)$ and $\varphi_{E_f} = \pi_f \circ \iota$ (Theorem 14.23).*

*(i) Every homomorphism $h\colon J_0(N) \to E$ over $\mathbb{Q}$ to a curve $E \in \mathcal{I}$ vanishes on $I_f J_0(N)$.*

*(ii) (Factorisation.)* $E_f \in \mathcal{I}$, *and for every* $E \in \mathcal{I}$ *and every modular parametrisation* $\varphi$ *of* $E$ *there is a unique isogeny* $\psi\colon E_f \to E$ *over* $\mathbb{Q}$ *with* $\varphi = \psi \circ \varphi_{E_f}$. *Hence* $\deg\varphi = \deg\psi \cdot \deg\varphi_{E_f}$, *and every isogeny* $\psi\colon E_f \to E$ *gives a modular parametrisation* $\psi \circ \varphi_{E_f}$.

*(iii) (Characterisations.) For* $E \in \mathcal{I}$ *and a modular parametrisation* $\varphi$ *of* $E$ *the following are equivalent: (a)* $\psi$ *in (ii) is an isomorphism; (b)* $\ker(\varphi_*\colon J_0(N) \to E)$ *is connected; (c)* $\varphi_*\colon H_1(X_0(N),\mathbb{Z}) \to H_1(E,\mathbb{Z})$ *is surjective; (d)* $\varphi^* H^1(E,\mathbb{Z})$ *is saturated in* $H^1(X_0(N),\mathbb{Z})$*; (e)* $\varphi^*\colon E = E^\vee \to J_0(N)$ *is injective; (f)* $\deg\varphi$ *is the minimum of the degrees of all modular parametrisations of all curves in* $\mathcal{I}$.

*(iv) (Uniqueness.) Up to unique isomorphism,* $(E_f, \varphi_{E_f})$ *is the only pair* $(E,\varphi)$ *with* $E \in \mathcal{I}$ *satisfying these conditions.* $E_f$ *is the* $X_0(N)$-optimal curve *(or* strong Weil curve*) of* $\mathcal{I}$, $\varphi_{E_f}$ *its* modular parametrisation $\varphi_E$, *and* $\deg\varphi_E$ *its* modular degree.

*Proof.* (i) We may assume $h \neq 0$; then $h$ is surjective and $h_*\colon V_\ell J_0(N) \to V_\ell E$ is a surjective $G_\mathbb{Q}$-equivariant map. Decompose $V = V_\ell J_0(N) \otimes \overline{\mathbb{Q}}_\ell = \bigoplus_\psi W_\psi$ into the generalised eigenspaces of $\mathbb{T}$ (Chapter 3); each $W_\psi$ is $G_\mathbb{Q}$-stable, because $G_\mathbb{Q}$ commutes with $\mathbb{T}$. Suppose $h_*(W_\psi) \neq 0$. It is a $G_\mathbb{Q}$-stable subspace of $V_\ell E \otimes \overline{\mathbb{Q}}_\ell$ and a quotient of $W_\psi$, so for every prime $p \nmid N\ell$ the arithmetic Frobenius has on it an eigenvalue $\alpha$ which is a root of $X^2 - a_p(E)X + p$ (Chapter 9) and of $X^2 - \psi(T_p)X + p$ (Theorem 14.24(ii)). Then $a_p(E) = \alpha + p/\alpha = \psi(T_p)$; and $a_p(E) = a_p(f)$ by Theorem 14.25. So $\psi(T_p) = a_p(f)$ for every prime $p \nmid N\ell$. The characters of $\mathbb{T}$ that occur in $V$ take algebraic values and are, after an identification of the algebraic numbers in $\overline{\mathbb{Q}}_\ell$ and in $\mathbb{C}$, the eigencharacters of eigenforms in $S_2(\Gamma_0(N))$ (Proposition 14.10(ii), with $\Psi$ of Proposition 14.18); by strong multiplicity one (Chapter 13), which allows any finite set of exceptional primes, an eigenform with $a_p = a_p(f)$ for all $p \nmid N\ell$ is a multiple of $f$ (Proposition 14.21(i)); so $\psi = \phi_f$. Hence $h_*$ kills $\bigoplus_{\psi \neq \phi_f} W_\psi$. That sum is $I_f V$: $I_f$ kills $W_{\phi_f}$, and for $\psi \neq \phi_f$ some $t \in I_f$ has $\psi(t) \neq 0$ (otherwise $\psi$ factors through $\mathbb{T}/I_f = \mathbb{Z}$ and equals $\phi_f$), and $t$ is invertible on $W_\psi$. (That $I_f$ kills the generalised eigenspace $W_{\phi_f}$, and not only the eigenspace: the anemic algebra, generated by the $T_n$ with $(n,N) = 1$, acts semisimply on $S_2(\Gamma_0(N))$, its operators being normal for the Petersson product (Chapter 13), hence semisimply on $V$; so $W_{\phi_f}$ lies in the eigenspace of the anemic algebra for the restriction of $\phi_f$, which corresponds to $\mathbb{C}\omega_f \oplus \mathbb{C}\overline{\omega_f}$ (Proposition 14.21(i)) and on which every $t \in \mathbb{T}$ acts as $\phi_f(t)$.) Now $V_\ell(I_f J_0(N)) = I_f V_\ell J_0(N)$, as $I_f J_0(N)$ is the image of $J_0(N)^r$ under $(t_i)$; so $h(I_f J_0(N))$ is a connected subgroup of $E$ with trivial rational Tate module, hence 0.

(ii) By (i) and the universal property of the quotient, $\varphi_* = \psi \circ \pi_f$ for a unique homomorphism $\psi\colon E_f \to E$ over $\mathbb{Q}$ ($\pi_f$ is surjective); $\psi \neq 0$, so $\psi$ is an isogeny, and $\varphi = \varphi_* \circ \iota = \psi \circ \varphi_{E_f}$. Degrees of nonconstant maps of curves multiply (Chapter 8). $E_f \in \mathcal{I}$: a curve $E \in \mathcal{I}$ is isogenous to $E_{f_E} = E_f$ by Theorem 14.25. Conversely $\psi \circ \varphi_{E_f}$ is nonconstant, defined over $\mathbb{Q}$, and sends $\infty$ to $\psi(O) = O$.

(iii) (b)⇔(c)⇔(d)⇔(e) is Chapter 8 (the proposition on the degree and the cup product, part (iii)), with $J(E) = E$. (a)⇒(b): $\ker\varphi_* = \ker(\psi \circ \pi_f) = \ker\pi_f = I_f J_0(N)$, connected. (b)⇒(a): $\ker\varphi_* = \pi_f^{-1}(\ker\psi)$ contains $I_f J_0(N)$ with finite quotient $\ker\psi$; both have dimension $g-1$, so if $\ker\varphi_*$ is connected it equals the connected closed subgroup $I_f J_0(N)$ of the same dimension, and $\ker\psi = 0$. (a)⇔(f): by (ii) the degrees of all modular parametrisations of curves in $\mathcal{I}$ are the numbers $\deg\psi \cdot \deg\varphi_{E_f}$, with minimum $\deg\varphi_{E_f}$, attained exactly when $\deg\psi = 1$.

(iv) If $(E,\varphi)$ satisfies (a), then $\psi\colon E_f \to E$ is the unique isomorphism with $\varphi = \psi \circ \varphi_{E_f}$. □

The logic of (i) in one sentence: a homomorphism $J_0(N) \to E$ sees only the part of $J_0(N)$ whose Frobenius eigenvalues are those of $E$, the congruence relation says that on the $\psi$-part

those eigenvalues are governed by $\psi(T_p)$, and strong multiplicity one says that only $\psi = \phi_f$ matches. The class `11a` shows the characterisations at work. There $J_0(11) = X_0(11) =$ `11a1` and $I_f = 0$, so $E_f =$ `11a1` with $\varphi_{E_f}$ the identity. The parametrisation of `11a3` is a 5-isogeny $\psi\colon$ `11a1` $\to$ `11a3`; the kernel of $\varphi_* = \psi$ is a group of order 5, five points, not connected, so (b) fails; on $H_1$ the image of $\psi_*$ has index 5 in $H_1(\texttt{11a3}, \mathbb{Z})$ (the index of $\psi_* H_1$ in $H_1$ is $\#\ker\psi$ for an isogeny of complex tori), so (c) fails; and its degree 5 is not minimal, so (f) fails.

The paper's definition (Section 1: "the $X_0(N)$-optimal curve, for which the modular parametrisation $\varphi_E\colon X_0(N) \to E$ has minimal degree") is (f); the proof of Proposition prop:pullback uses (b) and (c); and Section sec:data speaks of optimality as a property "determined in the tables", which is (a) checked by computation (Convention 14.34).

**Proposition 14.31** (descriptions of the modular degree)**.** *Let $E$ be optimal, $d = \deg\varphi_E$, $A = \varphi_E^*(E) \subseteq J_0(N)$ and $B = \ker\varphi_{E*} = I_f J_0(N)$.*

*(i) $d$ is the degree of the morphism of curves $\varphi_E$: the degree of the extension of function fields $\mathbb{Q}(X_0(N)) \supseteq \varphi_E^*\mathbb{Q}(E)$, the number of preimages of a general point, the factor by which $\varphi_E^*$ multiplies degrees of divisors, and the topological degree (Chapter 8).*

*(ii) $\varphi_{E*} \circ \varphi_E^* = [d]$ on $E$: the composite $E = E^\vee \to J_0(N) \to E$ is multiplication by $d$.*

*(iii) $A \simeq E$, and $A \cap B = \varphi_E^*(E[d]) \simeq (\mathbb{Z}/d)^2$; so $d$ is the exponent of the finite group $A \cap B$.*

*(iv) $\langle \varphi_E^* x, \varphi_E^* y\rangle_{X_0(N)} = d\,\langle x, y\rangle_E$ for $x, y \in H^1(E, \mathbb{Z})$; the cup product on $\varphi_E^* H^1(E, \mathbb{Z})$ has ideal of values $d\mathbb{Z}$ and Gram determinant $d^2$.*

*Proof.* (i) is Chapter 8. (ii) is the functoriality of Jacobians (Chapter 8): $\varphi_*\varphi^* = [\deg\varphi]$ on $J(E) = E$. (iii) $\varphi_E^*$ is injective by Theorem 14.30(iii)(e), so $A \simeq E$. For $y \in E$, $\varphi_E^* y \in B$ iff $\varphi_{E*}\varphi_E^* y = dy = 0$; so $A \cap B = \varphi_E^*(E[d]) \simeq E[d] \simeq (\mathbb{Z}/d)^2$, whose exponent is $d$. (iv) The projection formula (Chapter 8) says $\langle \varphi^* x, \varphi^* y\rangle_{X_0(N)} = \langle x, \varphi_*\varphi^* y\rangle_E$, and $\varphi_*\varphi^*$ is multiplication by $\deg\varphi = d$ on $H^1(E, \mathbb{Z})$ (dually, on $H_1$: the preimage of a loop on $E$ avoiding the branch points is a union of closed curves whose images cover the loop $d$ times). So the left side is $d\langle x, y\rangle_E$. The cup product on $H^1(E, \mathbb{Z}) \simeq \mathbb{Z}^2$ is unimodular: in a symplectic basis $a, b$ its Gram matrix is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, so its values are all of $\mathbb{Z}$ and its Gram determinant is 1. Multiplying by $d$ gives the Gram matrix $\left(\begin{smallmatrix} 0 & d \\ -d & 0 \end{smallmatrix}\right)$ for the basis $\varphi^* a$, $\varphi^* b$: values $d\mathbb{Z}$ and determinant $d^2$. □

**Proposition 14.32** (the valuation of the degree along the class)**.** *Let $E$ be optimal in $\mathcal{I}$, $\ell$ a prime with $E[\ell]$ irreducible, and $E' \in \mathcal{I}$. Every isogeny $E \to E'$ of minimal degree is cyclic of degree prime to $\ell$, and the minimal degree of a modular parametrisation of $E'$ has the same $\ell$-adic valuation as $\deg\varphi_E$. Without irreducibility this fails: in the class `11a` at $\ell = 5$ the minimal degrees are 1, 5, 5.*

*Proof.* If an isogeny $\psi\colon E \to E'$ has $E[m] \subseteq \ker\psi$ with $m > 1$, it factors as $\psi' \circ [m]$ with $\deg\psi' = \deg\psi/m^2$ ($[m]$ is surjective with kernel $E[m]$, so $\psi$ factors through it, and $\deg[m] = m^2$); $\psi'$ is defined over $\mathbb{Q}$ because $\psi$ and $[m]$ are. So an isogeny of minimal degree has a kernel containing no $E[m]$, $m > 1$, and a finite subgroup of $E \simeq (\mathbb{R}/\mathbb{Z})^2$ with that property is cyclic (a finite abelian group with at most 2 generators that is not cyclic contains some $(\mathbb{Z}/q)^2 = E[q]$, $q$ prime). The kernel is $G_\mathbb{Q}$-stable, $\psi$ being defined over $\mathbb{Q}$. If its order were divisible by $\ell$, the unique subgroup of order $\ell$ of this cyclic group would be $G_\mathbb{Q}$-stable too, contradicting the irreducibility of $E[\ell]$ (Chapter 9). By Theorem 14.30(ii) the modular parametrisations of $E'$ are the $\psi \circ \varphi_E$,

so the minimal degree is $\min_\psi \deg\psi \cdot \deg\varphi_E$, and $v_\ell(\min \deg\psi) = 0$. For `11a`: `11a1` is optimal with $\deg\varphi_E = 1$ (Example 14.6), and the isogenies of minimal degree to `11a2` and `11a3` have degree 5 (Example 14.27); $E[5]$ is reducible (`11a1` has a rational point of order 5). □

*In the paper* 14.33. Section 1 of the paper: "When $E[\ell]$ is irreducible, every curve of the class is linked to $E$ by a cyclic isogeny of degree prime to $\ell$, so $v_\ell$ of the degree of its minimal parametrisation is $v_\ell(\deg\varphi_E)$: at such an $\ell$ the choice of the optimal curve does not matter." This is Proposition 14.32. The hypothesis matters: the appendix of the paper notes that `rule` must be run on the optimal curve ("990h1 returns `[4, 3]` while the optimal curve 990h3 of its class returns `[3, 3]`"), and the class `990h` has 3-isogenies.

**Convention 14.34** (Cremona's labels and optimality)**.** In Cremona's tables [35, 36] curve number 1 of each class is the optimal curve, with the single exception of the class `990h`, whose optimal curve is `990h3` $= [1, -1, 1, -1568, -4669]$ (the `degphi` file lists the class with this equation and $\deg\varphi_E = 1728$; PARI's `ellmoddegree` also returns 1728 for it and $5184 = 3 \cdot 1728$ for `990h1`). Optimality is certified by comparing the period lattices of the curves of the class with the lattice of $\varphi^*$ computed from modular symbols [35, Chapter II]; Section sec:data of the paper says: "Optimality is determined in the tables for every class of conductor at most 400000 and for 359,009 further classes up to 500000." The LMFDB numbers curves differently (Chapter 9).

**Example 14.35** (the running table)**.** The optimal curves used in the book, with rank and torsion order from `allcurves`, $\deg\varphi_E$ from `degphi` [36] (in every case equal to PARI's `ellmoddegree`), and $\dim J_0(N)$:

| curve | $N$ | rank | $\#E(\mathbb{Q})_{\mathrm{tors}}$ | $\deg\varphi_E$ | $\dim J_0(N)$ |
|---|---|---|---|---|---|
| `11a1` | 11 | 0 | 5 | 1 | 1 |
| `20a1` | $2^2 \cdot 5$ | 0 | 6 | 1 | 1 |
| `37a1` | 37 | 1 | 1 | 2 | 2 |
| `99a1` | $3^2 \cdot 11$ | 1 | 2 | 4 | 9 |
| `121a1` | $11^2$ | 0 | 1 | 6 | 6 |
| `539c1` | $7^2 \cdot 11$ | 1 | 2 | $288 = 2^5 \cdot 3^2$ | 49 |
| `726a1` | $2 \cdot 3 \cdot 11^2$ | 1 | 2 | $144 = 2^4 \cdot 3^2$ | 109 |
| `980i1` | $2^2 \cdot 5 \cdot 7^2$ | 0 | 1 | $2880 = 2^6 \cdot 3^2 \cdot 5$ | 145 |

Further optimal curves that occur: `14a1` (1), `19a1` (1), `27a1` (1), `37b1` (2), `77c1` (6), `99b1` (12), `99c1` (12), `99d1` (6), `121b1` (4), `121c1` (6), `121d1` (24), `140b1` (60), `243b1` (9), `539d1` (72), `990h3` (1728). Checks: the curves of degree 1 are those of levels of genus one, as they must be (a map of degree one is an isomorphism, so $X_0(N) \simeq E$); `37a1` has degree 2 as in Example 14.36; `121a1` is the paper's case (b) of Theorem thm:qual with $e_3 = 1$ attained, `539c1` its case (c), `980i1` the $\ell = 5$ illustration of Example ex:qual, and `726a1` the curve with image 3Ns of the appendix.

**Example 14.36** ($X_0(37) \to$ `37a1` is optimal)**.** Let $\varphi\colon X_0(37) \to X_0(37)/w_{37} \simeq$ `37a1` be the quotient map of Example 14.6(c), of degree 2. By Theorem 14.30(ii), $\varphi = \psi \circ \varphi_E$ with $E$ the optimal curve of the class `37a`, which has only one curve; so $\psi$ is an endomorphism of `37a1` over $\mathbb{Q}$, that is, $[m]$ for an integer $m \neq 0$ (`37a1` has no complex multiplication, $j = 110592/37$), of degree $m^2$. As $2 = m^2 \deg\varphi_E$, $m = \pm 1$ and $\deg\varphi_E = 2$: the quotient by $w_{37}$ is the optimal parametrisation, without any appeal to tables.

The modular degree has now been described algebraically (degrees, kernels, lattices). The next section gives the analytic description that the paper quotes and that PARI uses to compute it, and isolates the one constant on which that description depends.

## 14.7 The Manin constant and the formula of Zagier and Watkins

The paper's introduction says: "Up to explicit factors $\deg\varphi_E$ is $\langle f,f\rangle$ divided by the area of the period lattice of $E$ (Zagier [209], Watkins [204])", and Remark rem:degree gives the exact formula. This section defines the constant that measures the difference between the Néron differential of $E$ and the differential of the newform, proves the exact formula in three lines, and checks it numerically; the warning after the proof records the normalisations that are easy to get wrong.

**Definition 14.37** (the Manin constant)**.** Let $\varphi$ be a modular parametrisation of $E$ (conductor $N$, newform $f$). The constant $c_\varphi$ is defined by

$$\varphi^*\omega_E = c_\varphi \cdot 2\pi i\, f(z)\,dz = c_\varphi\, f(q)\,\frac{dq}{q} \qquad (q = e^{2\pi i z}).$$

The *Manin constant* of $E$ is $c_E = c_{\varphi_E}$ for the optimal curve, the sign of $\omega_E$ being chosen so that $c_E > 0$ (Appendix B, B-C9).

Why such a constant must appear. On $X_0(N)$ there is a distinguished differential, $2\pi i f(z)\,dz = f(q)\,dq/q = (1 + a_2 q + \cdots)\,dq$, normalised by $a_1(f) = 1$; on $E$ there is another, the Néron differential, normalised by the integral structure of $E$ (it is $dx/(2y + a_1x + a_3)$ on a global minimal model, determined up to sign). Pulling the second back gives a differential on $X_0(N)$ in the line of the first (Proposition 14.38(i)), and nothing forces the two normalisations to agree; $c_\varphi$ is the ratio. The sign of $\omega_E$ is not determined by the minimal model (the change $y \mapsto -y - a_1x - a_3$ negates it), which is why one normalises $c_E > 0$. Example: for the optimal curve `11a1` the constant is 1, and for the parametrisation of `11a3` through the 5-isogeny it is 5 (Example 14.42).

**Proposition 14.38** (the constant $c_\varphi$)**.** *In Definition 14.37:*

*(i)* $\varphi^*\omega_E$ *is a nonzero multiple of* $\omega_f = f(z)\,dz$*, so* $c_\varphi$ *is well defined and nonzero.*

*(ii)* $c_\varphi \in \mathbb{Q}^\times$*.*

*(iii) If* $\varphi = \psi \circ \varphi_{E_0}$ *with* $E_0$ *optimal and* $\psi\colon E_0 \to E$ *an isogeny (Theorem 14.30(ii)), and* $\psi^*\omega_E = u_\psi\,\omega_{E_0}$*, then* $c_\varphi = c_{E_0}u_\psi$*;* $u_\psi$ *is a nonzero integer (stated), and* $u_\psi u_{\hat\psi} = \deg\psi$*.*

*Proof.* (i) By Theorem 14.30(ii), $\varphi = \psi\circ\varphi_{E_0}$. The pullback of an invariant differential by an isogeny is a constant multiple of the invariant differential (Chapter 9), so $\varphi^*\omega_E = u_\psi\varphi_{E_0}^*\omega_{E_0}$, and $\varphi_{E_0}^*\omega_{E_0} \in \mathbb{C}\omega_f$ by Theorem 14.23(ii). It is nonzero because $\varphi$ is nonconstant (a nonconstant holomorphic map is a local biholomorphism off a finite set). Also $2\pi i\,dz = dq/q$. (ii) The function $1/j$ lies in $\mathbb{Q}(X_0(N))$ and has $q$-expansion $q - 744q^2 + \cdots \in \mathbb{Q}[[q]]$ at the cusp $\infty$, which has width 1; so $q$ is a local parameter at the rational point $\infty$ for the $\mathbb{Q}$-structure, and the completed local ring there is $\mathbb{Q}[[q]]$. The differential $\varphi^*\omega_E$ is defined over $\mathbb{Q}$, so its expansion $c_\varphi\sum_n a_n(f)q^{n-1}dq$ has rational coefficients; as $a_1(f) = 1$, $c_\varphi \in \mathbb{Q}$. (iii) $c_\varphi$ is computed from $\varphi^*\omega_E = \varphi_{E_0}^*\psi^*\omega_E = u_\psi\varphi_{E_0}^*\omega_{E_0}$. That $u_\psi \in \mathbb{Z}$ uses Néron models: $\psi$ extends to the Néron models over $\mathbb{Z}$, and the Néron differential generates the invariant differentials of the Néron model (Chapter 10; [35, Chapter II]); we do not prove it here. From $\hat\psi\circ\psi = [\deg\psi]$ and $[m]^*\omega = m\omega$ (Chapter 9), $u_\psi u_{\hat\psi} = \deg\psi$. □

**Theorem 14.39** (what is known about the Manin constant; stated)**.** *Let* $E/\mathbb{Q}$ *be optimal.*

*(i) (Edixhoven [60].)* $c_E \in \mathbb{Z}$*; hence* $c_\varphi \in \mathbb{Z}$ *for every modular parametrisation, by Proposition 14.38(iii).*

*(ii) (Manin's conjecture.)* $c_E = 1$. *(The literature writes* $c_E = \pm 1$ *when the sign of* $\omega_E$ *is not normalised.)*

*(iii) If a prime* $p$ *divides* $c_E$*, then* $p = 2$ *or* $p^2 \mid N$*: Mazur [126] for* $p \nmid 2N$*, Abbes and Ullmo [1] for* $p \mid N$*,* $p^2 \nmid N$.

*(iv)* $c_E = 1$ *for every semistable optimal* $E$ *(Česnavičius [27], completing the cases left by [126, 1] at* $p = 2$*); further results relate* $c_E$ *to* $\deg \varphi_E$ *(Česnavičius, Neururer and Saha [28]), and computations of Cremona and of Agashe, Ribet and Stein [2] confirm* $c_E = 1$ *for all optimal curves in large ranges of conductors.*

*(v) For non-optimal parametrisations* $c_\varphi \neq 1$ *is possible (Example 14.42).*

We do not prove this. Items (iii) and (iv) are stated as we recall the sources; the exact hypotheses, in particular at $p = 2$ and the ranges of the computations, were not checked against them, and the book uses none of (i) to (iv) in a proof. The paper uses the Manin constant only in Remark rem:degree, where $c_E^2$ appears explicitly, and in the heuristic sentence of Section 1.

**Theorem 14.40** (the formula of Zagier and Watkins)**.** *For every modular parametrisation* $\varphi$ *of* $E$*, with the unnormalised Petersson product,*

$$\deg \varphi \cdot \operatorname{vol}(\Lambda_E) = 4\pi^2 c_\varphi^2 \, \langle f, f \rangle.$$

*In particular, for the optimal curve,*

$$\deg \varphi_E = \frac{4\pi^2 c_E^2 \langle f, f \rangle}{\operatorname{vol}(\Lambda_E)}, \quad \textit{which is } \frac{4\pi^2 \langle f, f \rangle}{\operatorname{vol}(\Lambda_E)} \textit{ if } c_E = 1.$$

*Proof.* For a nonconstant map $\varphi$ of compact Riemann surfaces and a smooth 2-form $\rho$ on the target, $\int_{X_0(N)} \varphi^* \rho = \deg \varphi \int_E \rho$ (Chapter 8: off the finitely many branch points $\varphi$ is a $\deg \varphi$-sheeted covering, each sheet maps diffeomorphically and orientation-preservingly onto its image, and a finite set has measure zero). With $\rho = \frac{i}{2} \omega_E \wedge \bar{\omega}_E$ the right side is $\deg \varphi \cdot \operatorname{vol}(\Lambda_E)$: under $E(\mathbb{C}) \simeq \mathbb{C}/\Lambda_E$ the form $\omega_E$ becomes $du$, and $\frac{i}{2} du \wedge d\bar{u}$ is the Lebesgue measure on the fundamental parallelogram. On the left, $\varphi^* \omega_E = 2\pi i c_\varphi \, \omega_f$ with $c_\varphi$ real (Proposition 14.38(ii)), so $\overline{\varphi^* \omega_E} = -2\pi i c_\varphi \, \overline{\omega_f}$ and

$$\varphi^* \rho = \frac{i}{2} (2\pi i c_\varphi)(-2\pi i c_\varphi) \, \omega_f \wedge \overline{\omega_f} = 4\pi^2 c_\varphi^2 \cdot \frac{i}{2} \, \omega_f \wedge \overline{\omega_f}.$$

Its integral over $X_0(N)$ is $4\pi^2 c_\varphi^2 \langle f, f \rangle$ by Convention 14.1(C4): on a fundamental domain, $\frac{i}{2} f \, dz \wedge \overline{f \, dz} = |f|^2 \cdot \frac{i}{2} dz \wedge d\bar{z} = |f|^2 \, dx \wedge dy$, since $dz \wedge d\bar{z} = (dx + i \, dy) \wedge (dx - i \, dy) = -2i \, dx \wedge dy$. □

For example, for `37a1`, with the numbers of Example 14.42, $4\pi^2 \cdot 0.37175\ldots / 7.33813\ldots = 2.0000\ldots$, the degree of the quotient map by $w_{37}$ (Example 14.36).

In the language of physics: the $L^2$ norm of a pulled-back one-form counts every sheet of the cover, so the area of a $d$-sheeted cover is $d$ times the area of the base.

*Warning* 14.41 (normalisations in the formula of Zagier and Watkins). (1) The Manin constant stands in the *numerator*: $\deg \varphi_E = 4\pi^2 c_E^2 \langle f, f \rangle / \operatorname{vol}(\Lambda_E)$. The check is the class `11a` (Example 14.42): the parametrisation of `11a3` has degree 5 and $c_\varphi = 5$, and $4\pi^2 \langle f, f \rangle / \operatorname{vol}(\Lambda_{\texttt{11a3}}) = 1/5 = 5/5^2$. (2) $\langle f, f \rangle$ is unnormalised; PARI's `mfpetersson` returns $\langle f, f \rangle / \mu(N)$, and books

that divide by $\mathrm{vol}(\Gamma_0(N)\backslash\mathcal{H}) = \pi\mu(N)/3$ change the constant $4\pi^2$. (3) $\mathrm{vol}(\Lambda_E)$ is the area of a fundamental parallelogram of the period lattice of the Néron differential, $|\mathrm{Im}(\bar{\omega}_1\omega_2)|$ for PARI's periods `E.omega` $= [\omega_1, \omega_2]$; it is not the real period $\Omega_E$. The normalisations of [209, 204] were not compared with these; the formula above is proved directly.

**Example 14.42** (numerical check)**.** Unnormalised Petersson norms ($\mu(N)$ times `mfpetersson`), period areas, and the quotient $4\pi^2\langle f, f\rangle/\,\mathrm{vol}(\Lambda_E) = \deg\varphi/c_\varphi^2$, computed with PARI to 40 digits (the last column agrees with a rational number to all of them):

| curve | $\langle f, f\rangle$ | $\mathrm{vol}(\Lambda_E)$ | $4\pi^2\langle f, f\rangle/\,\mathrm{vol}$ | $\deg\varphi$ | $c_\varphi$ |
|---|---|---|---|---|---|
| `11a1` | 0.04690014787349518782 | 1.85154362345595931771 | 1 | 1 | 1 |
| `11a2` | same | 0.37030872469119186354 | 5 | 5 | 1 |
| `11a3` | same | 9.25771811727979658854 | 1/5 | 5 | 5 |
| `20a1` | 0.08134945682538172857 | 3.21154782844006508758 | 1 | 1 | 1 |
| `37a1` | 0.37175414751069605028 | 7.33813274078957673907 | 2 | 2 | 1 |
| `99a1` | 0.53851956871086065179 | 5.31497510542145355032 | 4 | 4 | 1 |
| `121a1` | 0.42823936001195717858 | 2.81770204819580190671 | 6 | 6 | 1 |

In the class `11a` the parametrisations of `11a2` and `11a3` both have degree 5 (Theorem 14.30(ii)), so the formula forces $c_\varphi = 1$ for `11a2` and $c_\varphi = 5$ for `11a3`: the 5-isogeny $\psi$: `11a1` $\to$ `11a3` has $u_\psi = 5$, and its dual $u_{\hat{\psi}} = 1$. For the optimal curves of the table the computation gives $c_E^2 = 1$, in agreement with Manin's conjecture (the degrees being those of Cremona's tables).

*Remark* 14.43 (Rankin's method and the computation of $\deg\varphi_E$). (a) Rankin and Shimura express $\langle f, f\rangle$ through the special value $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2)$ of the naive symmetric square $L$-function: $\langle f, f\rangle = N\,L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2)/(8\pi^3)$ (Remark rem:degree of the paper, citing [155, 183] and [49, Lemma 2.12]; Chapter 19). Combined with Theorem 14.40 this is Watkins's method of computing $\deg\varphi_E$ [204], implemented in PARI's `ellmoddegree`, which assumes $c_E = 1$. (b) The Rankin series $\sum a_n^2 n^{-s}$ is not the naive symmetric square $L$-function (Chapter 19; Appendix B, B-C7). (c) In the book's normalisation the "explicit factors" of the paper's Section 1 are $4\pi^2 c_E^2$.

*In the paper* 14.44. Remark rem:degree: "By the formula of Zagier and Watkins [209, 204], $\deg\varphi_E \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_E^2\langle f, f\rangle$, with the unnormalised Petersson product and the Manin constant $c_E > 0$ in the numerator" is Theorem 14.40 for $\varphi = \varphi_E$. The remark's check "With $c_E = 1$ the first formula reproduces $\deg\varphi_E$ to 30 digits for 11a1, 121a1, 539d1, 1232l1 and 726a1" is of the same kind as Example 14.42.

The formula of Zagier and Watkins says how large $\deg\varphi_E$ is; it says nothing about which primes divide it. For that the paper uses congruences between $f$ and the other forms of level $N$, the subject of the next section.

## 14.8 The congruence number and the theorem of Agashe, Ribet and Stein

The paper reaches the modular degree through the congruence number: "By a theorem of Agashe, Ribet and Stein [3, Theorem 2.2], $\deg\varphi_E$ divides the *congruence number* $r_E$ ... with $v_p(\deg\varphi_E) = v_p(r_E)$ for every prime $p$ with $p^2 \nmid N$" (Section 1), and the proof of Theorem thm:qual reduces to "$\ell \mid r$". This section defines $r_E$ and the Hecke congruence ideal, states Lemma lem:crit (proved in Chapter 15) and the theorem of Agashe, Ribet and Stein, and computes $r_E$

directly for the curves of the book, `99a1` included. The divisibility half of the theorem is proved in Section 14.9 at every prime.

**Definition 14.45** (congruence number and congruence ideal)**.** Let $f = f_E$. The *congruence number* $r_E$ is the largest integer $r$ such that $f \equiv g \pmod r$ coefficientwise for some $g \in S_2(\Gamma_0(N);\mathbb{Z}) \cap f^\perp$. The *Hecke congruence ideal* of $f$ is $\eta_f = \phi_f(\mathrm{Ann}_{\mathbb{T}} I_f) \subseteq \mathbb{Z}$, and for a prime $\ell$, $\eta_f\mathbb{Z}_\ell = (\phi_f \otimes \mathbb{Z}_\ell)(\mathrm{Ann}_{\mathbb{T}_\ell}(I_f\mathbb{T}_\ell))$. (These are the definitions of Chapter 13 for $f = f_E$; $r_E$ depends only on the isogeny class.)

The two definitions measure the same phenomenon from the two sides of the duality between forms and Hecke operators. Over $\mathbb{Q}$ the space of cusp forms splits as $\mathbb{Q}f$ plus the complement $f^\perp$; over $\mathbb{Z}$ it does not, because an integral form can be congruent to $f$ modulo $r$ and to a form in $f^\perp$ at the same time. The congruence number is the largest such $r$, and $r_E > 1$ exactly when some prime $\ell$ makes $f$ congruent modulo $\ell$ to a combination of the other forms (for an eigenform, typically, to another eigenform, as in Lemma lem:crit below). The Hecke version asks the same question inside $\mathbb{T}$: $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$, and $\eta_f$ measures how far $\mathbb{T}$ is from the product of its two projections. At level 11, $S_2(\Gamma_0(11)) = \mathbb{C}f_{11}$, $f^\perp = 0$ and $\mathbb{T} = \mathbb{Z}$, so $r_E = 1$ and $\eta_f = \phi_f(\mathrm{Ann}_{\mathbb{Z}} 0) = \mathbb{Z}$. The same holds for every conductor below 26 (they are 11, 14, 15, 17, 19, 20, 21, 24, and $X_0(N)$ has genus one for each, so $f^\perp = 0$). At $N = 26$, where $X_0(26)$ has genus 2 and the classes `26a`, `26b` share it, $r_E = 2$ for both, as Theorem 14.48 gives from $\deg \varphi_E = 2$ (26 is square-free); level 37 is worked out by hand in Exercise 14.93.

**Proposition 14.46** (congruence ideals of a newform)**.**

*(i)* $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$*, with* $\phi_f \otimes \mathbb{Q}$ *the first projection; hence* $I_f \cap \mathrm{Ann}_{\mathbb{T}}(I_f) = 0$ *and* $\eta_f \neq 0$*.*

*(ii) Let* $\mathfrak{m} = \mathfrak{m}_{f,\ell}$ *be the maximal ideal of* $\mathbb{T}_\ell$ *containing* $\ell$ *and* $I_f$*. Then* $\eta_f\mathbb{Z}_\ell$ *is the congruence ideal of* $\phi_f$ *restricted to the local factor* $\mathbb{T}_\mathfrak{m}$*: the other local factors of* $\mathbb{T}_\ell$ *lie in* $\ker(\phi_f \otimes \mathbb{Z}_\ell)$*, not in its annihilator.*

*(iii)* $r_E$ *is finite:* $\mathbb{Z}f \oplus (f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z}))$ *has finite index in* $S_2(\Gamma_0(N);\mathbb{Z})$*.*

*Proof.* (i) $\mathbb{T} \otimes \mathbb{Q}$ is a finite-dimensional commutative $\mathbb{Q}$-algebra acting faithfully on $S_2(\Gamma_0(N))$ (Proposition 14.10); let $A$ be its local factor through which $\phi_f$ factors and $e$ its idempotent. Then $eS_2$ is the generalised eigenspace of $\phi_f$, which is $\mathbb{C}f$: the anemic algebra acts semisimply on $S_2(\Gamma_0(N))$, its operators being normal for the Petersson product (Chapter 13), so the generalised $\phi_f$-eigenspace of $\mathbb{T}$ lies in the eigenspace of the anemic algebra for the restriction of $\phi_f$, which is $\mathbb{C}f$ (Proposition 14.21(i)); and $A \otimes \mathbb{C}$ acts faithfully on the line $eS_2$, so $A \otimes \mathbb{C}$ embeds in $\mathbb{C}$ and $A = \mathbb{Q}$. The rest is Chapter 3 (the proposition on congruence ideals, parts (ii), (iii)). (ii) is part (v) of the same proposition, $\mathbb{Z}_\ell$ being complete. (iii) is Chapter 13: both summands are saturated and their ranks add up, and a congruence modulo $r$ gives the element $(f-g)/r$ of order $r$ in the quotient.

In more detail, (iii) also identifies $r_E$ with the index. Write $S = S_2(\Gamma_0(N);\mathbb{Z})$ and $L = f^\perp \cap S$. With $e$ the idempotent of the factor $\mathbb{Q}$ in (i), $eS_2 = \mathbb{C}f$ and $(1-e)S_2$ is the sum of the other generalised eigenspaces; as $f^\perp$ is $\mathbb{T}$-stable, $f^\perp = ef^\perp \oplus (1-e)f^\perp$ with $ef^\perp \subseteq \mathbb{C}f \cap f^\perp = 0$, so $f^\perp = (1-e)S_2$ by dimensions. Since $e$ has rational entries in a basis of $S$, $S \otimes \mathbb{Q} = \mathbb{Q}f \oplus (L \otimes \mathbb{Q})$, and the coefficient of $f$ is a linear map $P\colon S \to \mathbb{Q}$ with $P(f) = 1$ and kernel $L$. Its image is a finitely generated subgroup of $\mathbb{Q}$ containing 1, so it is $\frac{1}{r'}\mathbb{Z}$ for a positive integer $r'$, and $P$ induces $S/(\mathbb{Z}f \oplus L) \simeq \frac{1}{r'}\mathbb{Z}/\mathbb{Z} \simeq \mathbb{Z}/r'$. If $f \equiv g \pmod r$ with $g \in L$, then $h = (f-g)/r \in S$ and

$P(h) = 1/r$, so $r \mid r'$; conversely, if $P(h) = 1/r'$, then $g = f - r'h$ lies in $L$ and $f \equiv g \pmod{r'}$. Hence $r_E = r'$ is the index of $\mathbb{Z}f \oplus L$ in $S$, and the quotient is cyclic. This is the "cyclic-quotient argument" used in Example 14.50. □

**Lemma 14.47** (the paper's Lemma lem:crit; stated, proved in Chapter 15)**.** *Let $J = \mathrm{Ann}_{\mathbb{T}}(f^{\perp})$. Then $J = \mathrm{Ann}_{\mathbb{T}}(I_f)$ and $\mathbb{T}/(I_f + J) \simeq \mathbb{Z}/r_E$; equivalently $r_E\mathbb{Z} = \eta_f$. The same holds over $\mathbb{Z}_\ell$, with $r_E$ replaced by its $\ell$-part: $r_E\mathbb{Z}_\ell = \eta_f\mathbb{Z}_\ell$. If a normalised eigenform $g \in f^{\perp}$ with coefficients in the ring of integers of a number field satisfies $a_n(g) \equiv a_n(f) \pmod{\lambda}$ for every $n$, for a prime $\lambda \mid \ell$, then $\ell \mid r_E$.*

This chapter uses only $r_E\mathbb{Z}_\ell = \eta_f\mathbb{Z}_\ell$ (in Corollary 14.55 and Theorem 14.56).

**Theorem 14.48** (Agashe, Ribet and Stein; stated)**.** *Let $E$ be optimal of conductor $N$. Then $\deg\varphi_E \mid r_E$, and $v_p(\deg\varphi_E) = v_p(r_E)$ for every prime $p$ with $p^2 \nmid N$.*

We do not prove the theorem as a whole; it is [3, Theorem 2.2], as the paper cites it (the number is the paper's; we have not checked it against the published text). This chapter proves the divisibility at every prime (Corollary 14.55) and the equality at primes $\ell \nmid 2N$ for which $\mathfrak{m}_{f,\ell}$ is not Eisenstein, given multiplicity one (Theorem 14.62); the equality at $p \parallel N$, at $p = 2$ and in the Eisenstein case is theirs. The idea of their proof: with $A$ and $B$ as in Proposition 14.31, $\deg\varphi_E$ is the exponent of $A \cap B$, equivalently of $H_1(J_0(N),\mathbb{Z})/(H_1(A,\mathbb{Z}) + H_1(B,\mathbb{Z}))$, while $r_E$ is the exponent of $S_2(\mathbb{Z})/(S_2(\mathbb{Z})[I_f] + S_2(\mathbb{Z})[J])$; the perfect pairing $S_2(\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$ (Chapter 13) and the action of $\mathbb{T}$ on $H_1$ compare the two, and they agree at $p$ when $H_1(X_0(N),\mathbb{Z}_p)$ is locally free of rank two over $\mathbb{T} \otimes \mathbb{Z}_p$ at the relevant maximal ideal, which they establish for $p^2 \nmid N$ from results of Mazur, Ribet and others. Zagier [209] first compared $\deg\varphi_E$ with congruence primes, and Ribet [159] proved the divisibility $\deg\varphi_E \mid r_E$ in an earlier form.

*Remark* 14.49 (the size of the degree)*.* Section 1 of the paper: "the conjectured bound $\deg\varphi_E \ll_\varepsilon N^{2+\varepsilon}$ is equivalent to a form of the *abc* conjecture, see [3, §2] and the references there". The references there include Frey [71] and Mai and Murty [122]. Through Theorem 14.40 the bound is a bound for $\langle f, f\rangle/\mathrm{vol}(\Lambda_E)$; through Theorem 14.48 it bounds the congruence primes of $f$ (compare [140]). Nothing in the paper depends on it.

**Example 14.50** (congruence numbers computed directly)**.** The following algorithm computes $r_E$. (1) Take a $\mathbb{Q}$-basis of $S_2(\Gamma_0(N))$ with rational $q$-expansions (`mfbasis(mfinit([N,2],1))`), and compute the lattice $S_2(\Gamma_0(N);\mathbb{Z})$ of forms with integral coefficients as the saturation of the $\mathbb{Q}$-span of the truncated expansions in $\mathbb{Z}^M$, $M$ beyond Sturm's bound (`matrixqz(A,-2)`); a rational form whose first $M$ coefficients are integers is integral by Sturm's theorem (Chapter 12). (2) Compute the matrices of $T_p$ ($U_p$ for $p \mid N$) on this lattice from $a_n(T_pg) = a_{np}(g) + pa_{n/p}(g)$ (the second term for $p \nmid N$ only). (3) Find the primitive integral functional $\theta$ on the lattice with $\theta \circ T_p = a_p(f)\theta$ for $p \le 23$ (the common eigenline was already one-dimensional): its kernel is $f^{\perp} \cap S_2(\Gamma_0(N);\mathbb{Z})$, because $f^{\perp}$ is the sum of the other generalised eigenspaces. (4) Then $\theta\colon S_2(\Gamma_0(N);\mathbb{Z}) \to \mathbb{Z}$ is onto with kernel $L = f^{\perp} \cap S_2(\mathbb{Z})$, so $\mathbb{Z}f \oplus L$ has index $|\theta(f)|$ (indeed $\theta = \pm r'P$ for the map $P$ of the proof of Proposition 14.46, both having kernel $L$ and $\theta$ being onto $\mathbb{Z}$), and $r_E = |\theta(f)|$ by the cyclic-quotient argument given there. The results, with $\deg\varphi_E$ from `ellmoddegree` and `degphi`:

| curve | $N$ | $\deg\varphi_E$ | $r_E$ | primes $p$ with $p^2 \mid N$ |
|---|---|---|---|---|
| `11a1` | 11 | 1 | 1 | none |
| `20a1` | $2^2 \cdot 5$ | 1 | 1 | 2 |
| `37a1` | 37 | 2 | 2 | none |
| `77c1` | $7 \cdot 11$ | 6 | 6 | none |
| `88a1` | $2^3 \cdot 11$ | 8 | 16 | 2 |
| `99a1` | $3^2 \cdot 11$ | 4 | 12 | 3 |
| `99b1` | $3^2 \cdot 11$ | 12 | 12 | 3 |
| `99c1` | $3^2 \cdot 11$ | 12 | 12 | 3 |
| `99d1` | $3^2 \cdot 11$ | 6 | 6 | 3 |
| `121a1` | $11^2$ | 6 | 6 | 11 |
| `121b1` | $11^2$ | 4 | 4 | 11 |
| `121c1` | $11^2$ | 6 | 6 | 11 |
| `121d1` | $11^2$ | 24 | 24 | 11 |
| `140b1` | $2^2 \cdot 5 \cdot 7$ | 60 | 60 | 2 |
| `539c1` | $7^2 \cdot 11$ | 288 | 288 | 7 |
| `539d1` | $7^2 \cdot 11$ | 72 | 504 | 7 |
| `726a1` | $2 \cdot 3 \cdot 11^2$ | 144 | 1584 | 11 |
| `980i1` | $2^2 \cdot 5 \cdot 7^2$ | 2880 | 2880 | 2, 7 |

So `99a1` has $\deg\varphi_E = 4$ and $r_E = 12$, as [3, Table 1] and the remark after the paper's Example ex:qual say, and the discrepancy is at 3, with $3^2 \mid 99$. The congruence behind $3 \mid r_E$: $a_n(f_{99a}) \equiv a_n(f_{99b}) \equiv a_n(f_{99d}) \pmod 3$ for all $n \le 60$, beyond Sturm's bound 24 for $\Gamma_0(99)$ (which has index $99 \cdot \frac{4}{3} \cdot \frac{12}{11} = 144$ in $\mathrm{SL}_2(\mathbb{Z})$, and $2 \cdot 144/12 = 24$), so for all $n$ (Chapter 12). The first coefficients, from `ellan`,

| $n$ | 2 | 4 | 5 | 7 | 8 | 10 | 11 | 13 |
|---|---|---|---|---|---|---|---|---|
| $a_n(f_{99a})$ | $-1$ | $-1$ | $-4$ | $-2$ | 3 | 4 | $-1$ | $-2$ |
| $a_n(f_{99b})$ | $-1$ | $-1$ | 2 | 4 | 3 | $-2$ | $-1$ | $-2$ |
| $a_n(f_{99d})$ | 2 | 2 | $-1$ | $-2$ | 0 | $-2$ | $-1$ | 4 |

show the pattern (the coefficients $a_n$ with $3 \mid n$ vanish for all three, $3^2$ dividing the level): the differences are 0 or $\pm 3$, $\pm 6$. The forms $f_{99b}$, $f_{99d}$ are newforms of level 99 different from $f_{99a}$, hence in $f^{\perp}$, and Lemma lem:crit gives $3 \mid r_E$. ($f_{99c}$ is not congruent to $f_{99a}$ modulo 3: $a_2 = 1$ against $-1$.) The paper's curves `726a1` (ratio 11, $11^2 \mid 726$) and `539d1` (ratio 7, $7^2 \mid 539$) are further examples, and `88a1` (ratio 2, $2^3 \mid 88$) a fourth (Exercise 14.88). In every other case of the table the two numbers agree, also at primes $p$ with $p^2 \mid N$ (`121a1`, `539c1`, `980i1`, `99b1`, `99d1`): the equality can hold there; only the guarantee is lost.

*In the paper* 14.51. The remark after the Example following the proof of Theorem thm:qual: "The last step needs only $\ell^2 \nmid N$, and it cannot be dropped there: Table 1 of [3] lists the curve 99A1, of conductor $9 \cdot 11$, with modular degree 4 and congruence number 12; likewise 726a1 has $\deg\varphi_E = 144$ and $r_E = 1584 = 11 \cdot 144$, with $11^2 \mid 726$, and 539d1 has $\deg\varphi_E = 72$ and $r_E = 504 = 7 \cdot 72$, with $7^2 \mid 539$." All three pairs are recomputed in Example 14.50. The opening of Section sec:proof ("By [3, Theorem 2.2], $\deg\varphi_{E_0}$ divides $r$ and $v_p(\deg\varphi_{E_0}) = v_p(r)$ for every prime $p$ with $p^2 \nmid N$; as $\ell \nmid N$ it suffices to prove $\ell \mid r$") uses Theorem 14.48 at $p = \ell$, where $\ell^2 \nmid N$.

The theorem of Agashe, Ribet and Stein compares $\deg\varphi_E$, a geometric quantity, with $r_E$, a quantity about $q$-expansions. The next section explains the comparison from the cohomological side, where $\deg\varphi_E$ turns out to be exactly the congruence ideal attached to the pairing $\hat{\delta}_!$.

## 14.9 The modular degree as a congruence ideal

This section proves the paper's Proposition prop:pullback in the following form: for the optimal curve the lattice $M_{f,\lambda}$ is exactly $\varphi_E^* H^1(E,\mathbb{Z}_\ell)$, the pairing $\hat{\delta}_!$ has ideal of values $\deg\varphi_E\cdot\mathbb{Z}_\ell$ on it, and this holds for every prime $\ell$ with no hypothesis on $\bar\rho$ or on multiplicity one. The inputs are optimality (the kernel of $J_0(N)\to E$ is connected, so the pullback is saturated), Hecke equivariance, the Atkin-Lehner sign and the projection formula. Two consequences follow: the divisibility $\deg\varphi_E \mid r_E$ of Agashe, Ribet and Stein at every prime, by the inequality half of Lemma lem:ddt417 (Corollary 14.55); and Lemma lem:hecke of the paper (Theorem 14.56). The example `99a1` at $\ell=3$ shows that the Hecke-side inequality can be strict.

**Proposition 14.52** ($M_{f,\lambda}$ is the pullback of $H^1(E)$)**.** *Let $E$ be optimal with newform $f$, $\varphi=\varphi_E$, and $\ell$ any prime.*

*(a) (Hecke equivariance.) $\varphi_*\circ t_* = \phi_f(t)\varphi_*$ on $H_1(X_0(N),\mathbb{Z})$ and $t^*\circ\varphi^* = \phi_f(t)\varphi^*$ on $H^1(E,\mathbb{Z})$, for every $t\in\mathbb{T}$; so $\varphi^*H^1(E,\mathbb{Z})\subseteq H^1(X_0(N),\mathbb{Z})[I_f]$.*

*(b) (Saturation.) $\varphi^*\colon H^1(E,\mathbb{Z})\to H^1(X_0(N),\mathbb{Z})$ is injective with saturated image.*

*(c) $\varphi^*H^1(E,\mathbb{Z}) = H^1(X_0(N),\mathbb{Z})[I_f]$, and for every prime $\ell$*

$$M_{f,\lambda} = H^1(X_0(N),\mathbb{Z}_\ell)[I_f] = \varphi^*H^1(E,\mathbb{Z}_\ell).$$

*(d) (Galois structure.) In étale cohomology $\varphi^*$ is $G_{\mathbb{Q}}$-equivariant, so $M_{f,\lambda}\simeq H^1_{\mathrm{et}}(E_{\overline{\mathbb{Q}}},\mathbb{Z}_\ell)\simeq (T_\ell E)^\vee\simeq T_\ell E(-1)$ as $\mathbb{Z}_\ell[G_{\mathbb{Q}}]$-modules, and $\mathrm{ad}^0 M_{f,\lambda}\simeq \mathrm{ad}^0 T_\ell E$.*

*Proof.* (a) $E=E_f=J_0(N)/I_fJ_0(N)$ and $\varphi_*=\pi_f$ (Theorem 14.30(iv), Definition 14.29); $t$ acts on $E_f$ as $\phi_f(t)$ (Theorem 14.23(ii)), so $\pi_f\circ t_*=\phi_f(t)\pi_f$ on $J_0(N)$ and on $H_1$. Transposing, with $t^*$ the transpose of $t_*$ and $\varphi^*$ that of $\varphi_*$ (Proposition 14.10(ii)), gives $t^*\varphi^*=\phi_f(t)\varphi^*$; for $t\in I_f$ this is 0. (b) $\ker\varphi_*$ is connected (Theorem 14.30(iii)(b)), so $\varphi_*$ is surjective on $H_1$ and $\varphi^*H^1(E,\mathbb{Z})$ is saturated (Chapter 8, the proposition on the degree and the cup product). The mechanism is elementary: $H^1=\mathrm{Hom}(H_1,\mathbb{Z})$ and $\varphi^*=\mathrm{Hom}(\varphi_*,\mathbb{Z})$; a surjection $\varphi_*$ onto the free module $H_1(E,\mathbb{Z})$ has a section $s$, and then $s^*\circ\varphi^*=\mathrm{id}$, so $\varphi^*H^1(E,\mathbb{Z})$ is a direct summand of $H^1(X_0(N),\mathbb{Z})$, hence saturated. (For the parametrisation $\psi$ of `11a3`, $\psi_*H_1$ has index 5 and $\psi^*H^1(\texttt{11a3},\mathbb{Z})$ is not saturated: it contains $5x$ but not $x$ for a suitable class $x$.) $\varphi^*$ is injective because $\varphi_*\varphi^*=\deg\varphi\neq 0$ and $H^1(E,\mathbb{Z})$ is torsion-free. (c) $H^1(X_0(N),\mathbb{Z})[I_f]$ has rank 2 (Proposition 14.21), as has $\varphi^*H^1(E,\mathbb{Z})$, which it contains by (a). The quotient $H^1(X_0(N),\mathbb{Z})[I_f]/\varphi^*H^1(E,\mathbb{Z})$ is then finite, and it embeds in $H^1(X_0(N),\mathbb{Z})/\varphi^*H^1(E,\mathbb{Z})$, which is torsion-free by (b); so it is 0. Tensoring with the flat $\mathbb{Z}$-module $\mathbb{Z}_\ell$ and using Proposition 14.21(ii) gives the statement over $\mathbb{Z}_\ell$. (d) The comparison isomorphism between Betti and étale cohomology is compatible with pullbacks (Chapter 8), and $\varphi$ is defined over $\mathbb{Q}$; so $\varphi^*\colon H^1_{\mathrm{et}}(E_{\overline{\mathbb{Q}}},\mathbb{Z}_\ell)\to H^1_{\mathrm{et}}(X_0(N)_{\overline{\mathbb{Q}}},\mathbb{Z}_\ell)$ is $G_{\mathbb{Q}}$-equivariant, injective with image $M_{f,\lambda}$ by (c). $H^1_{\mathrm{et}}(E_{\overline{\mathbb{Q}}},\mathbb{Z}_\ell)\simeq(T_\ell E)^\vee\simeq T_\ell E(-1)$ by the Weil pairing (Chapter 8; Appendix B, B-C5(b)), and the trace-zero adjoint is insensitive to duals and Tate twists (Chapter 4; B-C12(b)). □

**Lemma 14.53** ($w_N$ on $M_{f,\lambda}$)**.** *Let $E$ be optimal with newform $f$. As morphisms $X_0(N)\to E$,*

$$\varphi_E\circ w_N = \varepsilon_N(f)\,\varphi_E + \varphi_E(0),$$

*and $w_N^*$ acts on $M_{f,\lambda}=\varphi_E^*H^1(E,\mathbb{Z}_\ell)$ by $\varepsilon_N(f)$. Hence $\hat{\delta}_!(x,y)=\varepsilon_N(f)\langle x,y\rangle$ for $x,y\in M_{f,\lambda}$.*

*Proof.* Write $\varepsilon_N = \varepsilon_N(f)$ and put $h(P) = \varphi_E(w_N P) - \varphi_E(0)$. Then $h$ is a morphism over $\mathbb{Q}$ (Theorem 14.2(ii)) with $h(\infty) = \varphi_E(0) - \varphi_E(0) = O$, as $w_N(\infty) = 0$. The differential $\omega_E$ is translation invariant, so

$$h^*\omega_E = w_N^*\varphi_E^*\omega_E = c_E \cdot 2\pi i\,(f|_2 W_N)(z)\,dz = \varepsilon_N\, c_E \cdot 2\pi i f(z)\,dz = \varepsilon_N\,\varphi_E^*\omega_E = (\varepsilon_N\varphi_E)^*\omega_E,$$

using Convention 14.1(C1) and $[-1]^*\omega_E = -\omega_E$. The homomorphisms $h_*$ and $(\varepsilon_N\varphi_E)_*\colon J_0(N) \to E$ are induced on $J_0(N)(\mathbb{C}) = S_2^\vee/H_1$ by the transposes of the pullbacks of $\omega_E$ (Chapter 8, functoriality of the Jacobian), which agree; so $h_* = (\varepsilon_N\varphi_E)_*$ and $h = h_* \circ \iota = \varepsilon_N\varphi_E$. On $H^1$, translations act trivially (they are homotopic to the identity) and $[\varepsilon_N]^* = \varepsilon_N$, so $w_N^*\varphi_E^* = (\varphi_E \circ w_N)^* = \varepsilon_N\varphi_E^*$, first on $H^1(E,\mathbb{Z})$ and then on $H^1(E,\mathbb{Z}_\ell)$; with Proposition 14.52(c) this is the action on $M_{f,\lambda}$. Finally $\hat\delta_!(x,y) = \langle x, w_N^* y\rangle = \varepsilon_N\langle x,y\rangle$. □

This proof does not use the description of $M_{f,\lambda}\otimes\mathbb{C}$ as a plane; Proposition 14.21(iii) gives a second proof.

**Theorem 14.54** (the modular degree is the ideal of values of $\hat\delta_!$)**.** *Let $E$ be optimal and $\ell$ any prime. For $a,b \in H^1(E,\mathbb{Z}_\ell)$,*

$$\hat\delta_!(\varphi_E^*a, \varphi_E^*b) = \varepsilon_N(f)\ \deg\varphi_E\ \langle a,b\rangle_E.$$

*Hence the ideal of values of $\hat\delta_!$ on $M_{f,\lambda}\times M_{f,\lambda}$ is $\deg\varphi_E\cdot\mathbb{Z}_\ell$, and the Gram determinant of $\hat\delta_!$ in a basis of $M_{f,\lambda}$ is $(\deg\varphi_E)^2$ up to a unit of $\mathbb{Z}_\ell$. The same holds over $\mathbb{Z}$: the ideal of values of $\hat\delta_!$ on $H^1(X_0(N),\mathbb{Z})[I_f]$ is $\deg\varphi_E\cdot\mathbb{Z}$. Consequently, if the ideal $\eta_{f,\lambda}^{\emptyset}$ of Diamond, Flach and Guo is the ideal of values of $\hat\delta_!$ (or of $\frac{u}{2}\hat\delta_!$, $u$ an $\ell$-adic unit) on $M_{f,\lambda}$, as Section 4.1 of the paper describes it for $\ell \nmid 2N$, then $v_\lambda(\eta_{f,\lambda}^{\emptyset}) = v_\ell(\deg\varphi_E)$, with no multiplicity-one or Gorenstein hypothesis.*

*Proof.* By Lemma 14.53 and the projection formula (Proposition 14.31(iv)), $\hat\delta_!(\varphi_E^*a,\varphi_E^*b) = \varepsilon_N(f)\langle\varphi_E^*a,\varphi_E^*b\rangle_{X_0(N)} = \varepsilon_N(f)\deg\varphi_E\langle a,b\rangle_E$. The cup product on $H^1(E,\mathbb{Z}_\ell)$ is perfect and alternating of rank two, so its ideal of values is $\mathbb{Z}_\ell$ and its Gram determinant a unit (Chapter 8); multiplying by the unit $\varepsilon_N(f)$ and by $\deg\varphi_E$ gives ideal of values $\deg\varphi_E\mathbb{Z}_\ell$ and Gram determinant $(\deg\varphi_E)^2$ times a unit. By Proposition 14.52(c), $\varphi_E^*H^1(E,\mathbb{Z}_\ell)$ is all of $M_{f,\lambda}$. The argument over $\mathbb{Z}$ is the same. The last sentence: $\frac{u}{2}$ is a unit for odd $\ell$. □

In the language of physics: a loop on $E$ lifts to $\deg\varphi_E$ loops on $X_0(N)$, and their intersection numbers add up.

Two checks. At level 11, $\varphi_E$ is the identity, $\deg\varphi_E = 1$ and $\varepsilon_{11}(f) = -1$, and the identity reads $\hat\delta_! = -\langle\ ,\ \rangle$ on $H^1(X_0(11),\mathbb{Z}_\ell)$, as computed after Corollary 14.12: the ideal of values is $\mathbb{Z}_\ell$. At level 37, $\varphi_E$ is the quotient by $w_{37}$, $\deg\varphi_E = 2$ and $\varepsilon_{37}(f_{37a}) = +1$, so $\hat\delta_!$ and the cup product agree on $M_{f,\lambda}$ and take the values $2\mathbb{Z}_\ell$ there; for $\ell = 2$ the cup product of the unimodular lattice $H^1(X_0(37),\mathbb{Z}_2)$ vanishes modulo 2 on the saturated sublattice $M_{f,\lambda}$, and by Corollary 14.55 this forces $2 \mid r_E$, which is the congruence $f_{37a} \equiv f_{37b} \pmod 2$ (Exercise 14.93).

*Commentary.* The paper states this as Proposition prop:pullback: "Let $E$ be $X_0(N)$-optimal with newform $f$, $\varphi = \varphi_E$, $I_f$ the kernel of the eigencharacter $\phi_f\colon \mathbb{T}\to\mathbb{Z}$ of $f$ on the full Hecke algebra, and $\ell$ a prime. Then $\varphi^*\colon H^1(E,\mathbb{Z}_\ell)\to H^1(X_0(N),\mathbb{Z}_\ell)$ is injective, with image the kernel lattice $H^1(X_0(N),\mathbb{Z}_\ell)[I_f]$, and" $\langle\varphi^*a,\ w_N^*\varphi^*b\rangle_{X_0(N)} = \varepsilon_N(f)\ \deg\varphi_E\ \langle a,b\rangle_E$ for $a,b\in H^1(E,\mathbb{Z}_\ell)$ (displayed there), "where $\langle\ ,\ \rangle$ denotes the cup product and $\varepsilon_N(f) = \pm1$ the eigenvalue of $w_N$ on $f$. In particular, for $\ell$ odd, the ideal of values of $\frac12\langle x, w_N^*y\rangle$ on $H^1(X_0(N),\mathbb{Z}_\ell)[I_f]$ is $\deg\varphi_E\,\mathbb{Z}_\ell$, and its

Gram determinant is $(\deg\varphi_E)^2$ up to a unit. If $\ell\nmid 2N$, then $M_{f,\lambda}=\varphi^*H^1(E,\mathbb{Z}_\ell)$, and" $v_\lambda(\eta^{\emptyset}_{f,\lambda})=v_\ell(\deg\varphi_E)$ (displayed). The book's Proposition 14.52, Lemma 14.53, and Theorem 14.54 prove this. The paper's "In particular" sentence is phrased for $\frac12\langle x,w_N^*y\rangle$ and so restricted to odd $\ell$; the book also states the ideal of values over $\mathbb{Z}$ (on $H^1(X_0(N),\mathbb{Z})[I_f]$ it is $\deg\varphi_E\,\mathbb{Z}$) and at $\ell=2$ for $\langle x,w_N^*y\rangle$, both immediate from the displayed identity. The two proofs differ in one step: the book derives the action of $w_N$ from the identity of maps $\varphi_E\circ w_N=\varepsilon_N(f)\varphi_E+\varphi_E(0)$, the paper from the fact that the plane is also the eigenspace of the $T_n$, $(n,N)=1$, which commute with $w_N$ (Proposition 14.21(i), (iii)). The Galois structure $M_{f,\lambda}\simeq T_\ell E(-1)$, which the paper states in the paragraph after the proof, is part (d) of Proposition 14.52. The last sentence of the paper's proof: "if $\ell\nmid 2N$, the integral structure of [49] is $H^1(X_0(N),\mathbb{Z}_\ell)$ (Proposition prop:integral), so $M_{f,\lambda}=H^1(X_0(N),\mathbb{Z}_\ell)[I_f]=\varphi^*H^1(E,\mathbb{Z}_\ell)$, and $\hat\delta_!$ is $\frac{u}{2}\langle x,w_N^*y\rangle$ with $u$ an $\ell$-adic unit". The identification of $H^1(X_0(N),\mathbb{Z}_\ell)$ with the integral structure of [49] holds for every prime $\ell$ with no localisation (Remark 14.57(1)); the hypothesis $\ell\nmid 2N$ enters through the standing assumption of Section 4.1 of the paper and the normalisation of $\hat\delta_!$ (Remark 14.57(3)), and no hypothesis on $\bar\rho$ is needed, in the paper's statement or in Theorem 14.54. Irreducibility of $\bar\rho$ enters only later, in the results of [49] that the paper applies to $\eta^{\emptyset}_{f,\lambda}$, such as their Proposition 1.4(c).

**Corollary 14.55** (the divisibility of Agashe, Ribet and Stein at every prime)**.** *Let $E$ be optimal. For every prime $\ell$, $v_\ell(\deg\varphi_E)\le v_\ell(r_E)$; that is, $\deg\varphi_E\mid r_E$. Equality holds at $\ell$ if $H^1(X_0(N),\mathbb{Z}_\ell)_{\mathfrak{m}}$ is free over $\mathbb{T}_{\mathfrak{m}}$ for $\mathfrak{m}=\mathfrak{m}_{f,\ell}$.*

*Proof.* Let $\mathcal{O}=\mathbb{Z}_\ell$, $\pi=\phi_f\otimes\mathbb{Z}_\ell\colon\mathbb{T}_\ell\to\mathbb{Z}_\ell$, $\wp=\ker\pi=I_f\mathbb{T}_\ell$ and $I=\mathrm{Ann}_{\mathbb{T}_\ell}(\wp)$. By Proposition 14.46 and Lemma lem:crit (Lemma 14.47), $\eta=\pi(I)=r_E\mathbb{Z}_\ell\neq 0$. By Proposition 14.46(ii) and Chapter 3, $\wp=\wp_{\mathfrak{m}}\times\mathbb{T}''$ with $\mathbb{T}''$ the product of the other local factors, so $\eta$ is the congruence ideal of $\pi|_{\mathbb{T}_{\mathfrak{m}}}$, and $L=H^1(X_0(N),\mathbb{Z}_\ell)_{\mathfrak{m}}$ satisfies $L[\wp_{\mathfrak{m}}]=H^1(X_0(N),\mathbb{Z}_\ell)[\wp]=M_{f,\lambda}$ (an element killed by $\wp$ is killed by $\mathbb{T}''$, hence lies in the $\mathfrak{m}$-component). $L$ is a faithful $\mathbb{T}_{\mathfrak{m}}$-module, free of finite rank over $\mathbb{Z}_\ell$, with the pairing $\hat\delta_!$, which is perfect on $L$ and for which $\mathbb{T}_{\mathfrak{m}}$ is self-adjoint (Corollary 14.12(iii)). By Chapter 3 (the module ingredients of Lemma lem:ddt417, and the Gram determinant of a saturated sublattice): $Q=L/(L[\wp_{\mathfrak{m}}]\oplus L[I_{\mathfrak{m}}])$ is killed by $\eta$, and $Q\simeq\mathrm{coker}(M_{f,\lambda}\to\mathrm{Hom}(M_{f,\lambda},\mathbb{Z}_\ell))$, the map induced by $\hat\delta_!$, so $\mathrm{length}\,Q$ is the valuation of the Gram determinant, $2v_\ell(\deg\varphi_E)$ by Theorem 14.54. $Q$ is a quotient of $\mathrm{Hom}(M_{f,\lambda},\mathbb{Z}_\ell)\simeq\mathbb{Z}_\ell^2$ killed by $r_E$, hence a quotient of $(\mathbb{Z}_\ell/r_E)^2$, and $2v_\ell(\deg\varphi_E)=\mathrm{length}\,Q\le 2v_\ell(r_E)$. If $L\simeq\mathbb{T}_{\mathfrak{m}}^2$, then $Q\simeq(\mathbb{T}_{\mathfrak{m}}/(\wp_{\mathfrak{m}}+I_{\mathfrak{m}}))^2\simeq(\mathbb{Z}_\ell/\eta)^2$ (Chapter 3), and equality holds. □

The two module facts used from Chapter 3 are short enough to see directly; write $M=M_{f,\lambda}$ and drop the subscript $\mathfrak{m}$. *$\eta$ kills $Q$:* for $t\in I$ and $x\in L$, $\pi(t)x=tx-(t-\pi(t))x$; here $tx\in L[\wp]$ because $\wp t=0$, and $(t-\pi(t))x\in\wp L\subseteq L[I]$ because $t-\pi(t)\in\wp$ and $I\wp=0$; so $\pi(t)x\in L[\wp]+L[I]$. *$Q$ is the cokernel of the Gram map:* the map $L\to\mathrm{Hom}(M,\mathbb{Z}_\ell)$, $x\mapsto\hat\delta_!(\,\cdot\,,x)|_M$, is onto, because $\hat\delta_!$ is perfect on $L$ and $M$ is saturated (so every functional on $M$ extends to $L$); its kernel is the orthogonal complement of $M$, which is $L[I]$ by the self-adjointness of $\mathbb{T}$ (Chapter 3); so $L/(M\oplus L[I])\simeq\mathrm{Hom}(M,\mathbb{Z}_\ell)/\{\hat\delta_!(\,\cdot\,,y)|_M : y\in M\}$. In a basis of $M$ the last group is $\mathbb{Z}_\ell^2$ modulo the column span of the Gram matrix $\left(\begin{smallmatrix}0 & \pm d\\ \mp d & 0\end{smallmatrix}\right)$, $d=\deg\varphi_E$ (Theorem 14.54), that is $(\mathbb{Z}_\ell/d)^2$, of length $2v_\ell(d)$. So in fact $Q\simeq(\mathbb{Z}_\ell/\deg\varphi_E)^2$, and the inequality is the statement that $(\mathbb{Z}_\ell/\deg\varphi_E)^2$ is killed by $r_E$.

This is the assembly of the inequality half of Lemma lem:ddt417 in the case at hand ($d=2$); Chapter 20 proves the lemma in general. The paper cites [3] for the divisibility; the corollary derives it from Theorem 14.54, so that the theorem of Agashe, Ribet and Stein is needed only for the reverse inequality.

**Theorem 14.56** (Lemma lem:hecke of the paper)**.** *Let $E$ be optimal and $\ell$ a prime.*

*(i)* $v_\ell(\deg\varphi_E) = v_\ell(\text{ideal of values of } \hat{\delta}_! \text{ on } M_{f,\lambda}) \le v_\ell(r_E)$*, with no further hypothesis.*

*(ii) If* $\ell^2 \nmid N$*, then* $v_\ell(\deg\varphi_E) = v_\ell(r_E)$ *(Theorem 14.48).*

*(iii) If* $\ell \nmid 2N$ *and* $\bar\rho_{E,\ell}$ *is irreducible, then, with the integral structure of [49] identified with* $H^1(X_0(N),\mathbb{Z}_\ell)$ *by the paper's Proposition prop:integral (Remark 14.57),*

$$v_\ell(\deg\varphi_E) = v_\ell(r_E) = v_\lambda(\eta^{\emptyset}_{f,\lambda}).$$

*Proof.* (i) is Theorem 14.54 and Corollary 14.55; (ii) is Theorem 14.48; (iii) combines (ii) with the last sentence of Theorem 14.54. □

*Commentary.* The paper states: "Let $E$ be $X_0(N)$-optimal and $\ell$ an odd prime with $\ell \nmid 2N$ and $\bar\rho$ irreducible. Then $v_\ell(\deg\varphi_E) = v_\ell(r_E) = v_\lambda(\eta^{\emptyset}_{f,\lambda})$." This is (iii); its proof uses only $\ell \nmid 2N$, so the hypothesis that $\bar\rho$ is irreducible, kept in (iii) to match the paper's statement, is not used. The book adds (i) and (ii), which hold at every prime: (i) says that the cohomological side of the identity needs no hypothesis on $\ell$ or $\bar\rho$, and it gives a proof of the inequality $v_\ell(\deg\varphi_E) \le v_\ell(r_E)$ at every prime that does not go through [3]; the divisibility itself is part of [3, Theorem 2.2] as the paper cites it, so the new content is the route, not the statement. The paragraph after the lemma in the paper notes that Lemma lem:ddt417 alone gives only $v_\ell(r_E) \ge v_\lambda(\eta^{\emptyset}_{f,\lambda})$, with equality under freeness. Consequences for the paper, assembled in Chapter 20: the identity (eq:identity) of Theorem thm:main holds with equality whenever $\ell \notin S_f$, with no multiplicity-one hypothesis; and in Step 4 of the proof of Theorem thm:quant, (iii) applied to the optimal curve of the class of the twist gives the comparison at level $N_S$ without freeness, so that the first inequality of Theorem thm:quant is an equality whenever $\ell \notin S_f$. Freeness is still the tool for forms not attached to an optimal quotient, such as the depleted form of Step 2 (Remark 14.63).

*Remark* 14.57 (the identification with the objects of Diamond, Flach and Guo). Theorem 14.54 is a theorem about the ideal of values of $\langle x, w_N^* y\rangle$ on $H^1(X_0(N),\mathbb{Z}_\ell)[I_f]$. That this ideal is the $\eta^{\emptyset}_{f,\lambda}$ of [49] rests on five points, which Section 4.1 of the paper establishes. Point (1) is the paper's Proposition prop:integral, which the paper proves and Chapter 20 proves in full (the proposition comparing the integral structure with $H^1(X_0(M),\mathbb{Z}_\ell)$ there); point (2) is proved below and point (5) in Chapter 3; Chapter 20 checks the normalisation in (3), the kernel lattice in (4) and the self-adjointness of the Hecke operators against the published text of [49] (Sections 1.5.2, 1.5.3, 1.6.1, 1.6.2, 1.8.2 there).

(1) *Which curve.* The integral structure $M(N,1)_{!,\lambda}$ of [49] is defined through a modular curve of full level. The paper's Proposition prop:integral: "Pull-back along the covering of $X_0(M^\Sigma)$ by the modular curve of full level identifies $H^1(X_0(M^\Sigma),\mathbb{Z})$ with the Betti lattice $M(M^\Sigma,1)_{!,B}$ of [49, §1.4.2], compatibly with the Hecke operators, $U_p$ included; in particular $M(M^\Sigma,1)_{!,\lambda} = H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$, with no localisation." It holds for every prime $\ell$, with no hypothesis on $\bar\rho$. The proof has four steps: compact support makes the invariants of the full-level cover equal to the cohomology of $Y_0(M^\Sigma)$ (Hochschild-Serre, and the equivariant Leray spectral sequence of the quotient map, whose $\mathcal{R}^1$ vanishes because a finite group has no nonzero homomorphism to $\mathbb{Z}$); the image of $H^1_c(Y_0)$ in $H^1(Y_0)$ is $H^1(X_0)$, which is point (2); Hecke equivariance is checked over $\mathbb{C}$; and the $\lambda$-adic structure is the Betti one tensored with $\mathbb{Z}_\ell$. The point matters because pulling back along a covering of degree $n$

multiplies the cup product by $n$ (Chapter 8): a lattice strictly between $\pi^*H^1(X_0(N),\mathbb{Z}_\ell)$ and its saturation would change the ideal of values by a non-unit. Compact support is what excludes this. With ordinary cohomology the invariants can be larger than the pull-back; in the paper's words (there $N$ is the full level and $M$ the level of $Y_0(M)$), "$H^1(Y(N),\mathbb{Z}_\ell)^G/\pi^*H^1(Y_0(M),\mathbb{Z}_\ell)$ is the group of characters $\Gamma_0(M)\to\mathbb{Q}_\ell/\mathbb{Z}_\ell$ of $\ell$-power order that factor through $G$, namely $\gamma\mapsto\psi(d \bmod p)$ for primes $p\mid M$ with $\ell\mid p-1$; for $M=11$ and $\ell=5$ it is the unramified covering $X_1(11)\to X_0(11)$". For $X_1(N)\to X_0(N)$ this excess is measured by the Shimura subgroup (Definition 14.68, Exercise 14.94); it plays no part in the comparison of integral structures.

(2) *Interior cohomology.* For constant coefficients, the image of $H^1_c(Y_0(N),\mathbb{Z}_\ell)$ in $H^1(Y_0(N),\mathbb{Z}_\ell)$ is $H^1(X_0(N),\mathbb{Z}_\ell)$, and $H^1(\Gamma_0(N),\mathbb{Z}_\ell)=H^1(Y_0(N),\mathbb{Z}_\ell)$. Proof: with $S$ the set of cusps, the exact sequence $H^0(S)\to H^1_c(Y)\to H^1(X)\to H^1(S)=0$ makes $H^1_c(Y)\to H^1(X)$ onto, and $H^1(X)\to H^1(Y)$ is injective because $H^1(X,Y)$ is, by excision, a sum of $H^1(D,D\setminus\{P\})=0$ over small discs $D$ around the cusps; so the image of $H^1_c(Y)\to H^1(Y)$ is $H^1(X)$. And $H^1(\Gamma_0(N),\mathbb{Z}_\ell)=\mathrm{Hom}(\Gamma_0(N),\mathbb{Z}_\ell)$ kills the elements of finite order ($\mathbb{Z}_\ell$ is torsion-free), and $\Gamma_0(N)$ modulo the normal subgroup they generate is $\pi_1(Y_0(N))$ (Armstrong's theorem on quotients by groups with fixed points), so $\mathrm{Hom}(\Gamma_0(N),\mathbb{Z}_\ell)=\mathrm{Hom}(\pi_1(Y_0(N)),\mathbb{Z}_\ell)=H^1(Y_0(N),\mathbb{Z}_\ell)$.

(3) *The operator.* $[UwU]_\omega$ is $w_N^*$ with no scalar in weight 2 and trivial character (in [49, Section 1.5.3] it "coincides with $\psi(-1_\infty)N^{k-2}[Uw^{-1}U]_\omega$"), and on classes pulled back from $X_0(N)$ the pairing $\hat\delta_!$ of [49] is (with $M^\Sigma=N$) "$\frac{u}{2}$ times the cup product on $X_0(M^\Sigma)$ composed with $w_{M^\Sigma}$, for an $\ell$-adic unit $u$". The reason, as the paper gives it: at the sufficiently small level $U'=U_0(N)\cap U_1(4)$ the normalised pairing of [49, Section 1.5.2] is Poincaré duality on $X_{U'}$ up to an $\ell$-adic unit when $\ell\nmid 2N$; it is normalised by dividing by the index, so on $U_0(N)$-invariants it is $[U_0(N):U']^{-1}$ times its value on $U'$-invariants; and the covering $X_{U'}\to X_0(N)$ has degree $[U_0(N):U']/2$, because $-1$ acts trivially, so by the projection formula the cup product on $X_{U'}$ of two pulled-back classes is $[U_0(N):U']/2$ times their cup product on $X_0(N)$. The book's $\hat\delta_!$ omits the factor $\frac{u}{2}$; this is harmless for odd $\ell$.

(4) *Kernel, not projection.* $M_{f,\lambda}$ is the kernel lattice $L[\wp]$, as the paper says and as [49] define it ($\mathcal{M}_f=\mathcal{M}(N,\psi)_![I_f]$ in their Section 1.6.2, where $\mathcal{M}[I]$ is the kernel of $(x_1,\dots,x_r)\colon\mathcal{M}\to\mathcal{M}^r$ for generators $x_i$ of $I$, Section 1.1.2), and not the image of $L$ under the projection to the $f$-component (the dual lattice, which would change the ideal of values by the congruence ideal).

(5) *Values versus Gram.* $\eta$ is the ideal of values, equivalently the square root of the Gram determinant (Chapter 3), as the paper says; in [49, Section 1.7.3] $\eta_f^\Sigma$ is defined by the image of $\wedge^2\mathcal{M}_{f^\Sigma}$ under the pairing.

Points (3) and (5) matter only up to units, and a unit changes nothing; points (1) and (4) are where a non-unit discrepancy could enter, and both hold exactly, so the paper's description of $\eta$ in Section 4.1 agrees with Theorem 14.54.

**Example 14.58** (`99a1` at $\ell=3$)**.** Let $E=$ `99a1` and $\ell=3$, so $3^2\mid N$ and the paper's theorems do not apply. By Theorem 14.54 the ideal of values of $\hat\delta_!$ on $M_{f,3}$ is $4\mathbb{Z}_3=\mathbb{Z}_3$, while $v_3(r_E)=1$ (Example 14.50). So the inequality of Corollary 14.55 is strict, and $L=H^1(X_0(99),\mathbb{Z}_3)_\mathfrak{m}$ is not free over $\mathbb{T}_\mathfrak{m}$ for $\mathfrak{m}=\mathfrak{m}_{f,3}$. Since $\mathrm{rank}_{\mathbb{Z}_3}L=2\,\mathrm{rank}_{\mathbb{Z}_3}\mathbb{T}_\mathfrak{m}$ (Theorem 14.62, step (b) of the proof,

which does not use $\ell \nmid N$), the freeness criterion of Chapter 3 shows $\dim_{\mathbb{F}_3} L/\mathfrak{m}L \neq 2$ (here $\mathbb{T}/\mathfrak{m} = \mathbb{F}_3$, since $\phi_f$ takes values in $\mathbb{Z}$); it is not 1, for then $L$ would be cyclic by Nakayama's lemma, a quotient of $\mathbb{T}_\mathfrak{m}$, of $\mathbb{Z}_3$-rank at most $\operatorname{rank}\mathbb{T}_\mathfrak{m} < 2\operatorname{rank}\mathbb{T}_\mathfrak{m}$; so it is at least 3, and by Proposition 14.18(iii) and the duality $(W/\mathfrak{m}W)^\vee \simeq W[\mathfrak{m}]$ of step (a) of the proof of Theorem 14.62 (valid for every $\ell$) $\dim_{\mathbb{F}_3} J_0(99)[\mathfrak{m}] \geq 3$: multiplicity one fails at $\mathfrak{m}$. Yet $\bar\rho_{\texttt{99a1},3}$ is irreducible: the class `99a` has only a 2-isogeny (`allisog`), and Cremona's `galrep` file lists no non-surjective mod 3 image for it. This is why the paper's Remark rem:free needs $\ell \nmid N$.

*In the paper* 14.59. The paragraph after Lemma lem:hecke in the paper: "Without the hypotheses the inequality of Lemma lem:ddt417 can be strict: at 99a1 and $\ell = 3$, where $3 \mid N$, $\deg\varphi_E = 4$ and $r_E = 12$, so $H^1(X_0(99),\mathbb{Z}_3)_\mathfrak{m}$ is not free over $\mathbb{T}_\mathfrak{m}$." This is Example 14.58. The same paragraph: "the congruence ideal is computed in $\mathbb{T}_\mathfrak{m}$: the other local factors of $\mathbb{T}\otimes\mathbb{Z}_\ell$ lie in the kernel $\wp$" is Proposition 14.46(ii). Section 4.2 of the paper: "by Proposition prop:pullback, $\varphi_E^*$ identifies the lattice $M_{f,\lambda}$ of [49] with $H^1(E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$, so its adjoint is $\mathrm{ad}^0 T$" is Proposition 14.52(d).

## 14.10 Multiplicity one, the Gorenstein property and Remark rem:free

The paper's Remark rem:free explains when $H^1(X_0(N),\mathbb{Z}_\ell)_\mathfrak{m}$ is free over the localised Hecke algebra: multiplicity one makes $\mathbb{T}_\mathfrak{m}$ Gorenstein and the module free of rank two. By Theorem 14.54 (the paper's Proposition prop:pullback), the paper's main theorems do not need this; it remains the conceptual reason for the equality in the theorem of Agashe, Ribet and Stein at $\ell \nmid 2N$, and the tool for forms that are not attached to an optimal quotient. This section defines the maximal ideal attached to $(f,\ell)$, states multiplicity one, proves the Gorenstein property from it with Chapter 3's criterion, and says what Remark rem:free is still for.

**Definition 14.60** (the maximal ideal of $(f,\ell)$)**.** For $f = f_E$ and a prime $\ell$, $\mathfrak{m} = \mathfrak{m}_{f,\ell}$ is the maximal ideal of $\mathbb{T}_\ell$ containing $\ell$ and $I_f$, the kernel of $\mathbb{T}_\ell \to \mathbb{Z}_\ell \to \mathbb{F}_\ell$, $t \mapsto \phi_f(t) \bmod \ell$; $\mathbb{T}_\ell = \prod_{\mathfrak{m}'} \mathbb{T}_{\mathfrak{m}'}$ over its maximal ideals (Chapter 3). For every maximal ideal $\mathfrak{m}$ of $\mathbb{T}$ (or $\mathbb{T}_\ell$) with residue characteristic $\ell$ there is a semisimple continuous representation $\bar\rho_\mathfrak{m}\colon G_\mathbb{Q} \to \mathrm{GL}_2(\mathbb{T}/\mathfrak{m})$, unramified outside $N\ell$, with $\operatorname{tr}\bar\rho_\mathfrak{m}(\mathrm{Frob}_p) = T_p \bmod \mathfrak{m}$ and $\det\bar\rho_\mathfrak{m} = \bar\chi_\ell$ (Chapter 15; stated). For $\mathfrak{m} = \mathfrak{m}_{f,\ell}$, $\bar\rho_\mathfrak{m} \simeq \bar\rho^{\mathrm{ss}}_{E,\ell}$, by Theorem 14.24(iii), Chebotarev and Brauer-Nesbitt (Chapters 4 and 5). $\mathfrak{m}$ is *Eisenstein* if $\bar\rho_\mathfrak{m}$ is reducible (Convention 14.1(C7)); so $\mathfrak{m}_{f,\ell}$ is Eisenstein iff $E[\ell]$ is reducible.

**Theorem 14.61** (multiplicity one; stated)**.** *Let $\ell \nmid 2N$ and let $\mathfrak{m}$ be a non-Eisenstein maximal ideal of $\mathbb{T}$ of residue characteristic $\ell$. Then $J_0(N)[\mathfrak{m}]$ is a 2-dimensional $\mathbb{T}/\mathfrak{m}$-vector space, isomorphic to $\bar\rho_\mathfrak{m}$ as a $G_\mathbb{Q}$-module.*

We do not prove this. For prime $N$ it is Mazur's [125, Chapter II]; in general the argument is Mazur's, through the $q$-expansion principle modulo $\ell$ and the good reduction of $J_0(N)$ at $\ell$, as written in [156, Section 5] (section number not verified). The isomorphism with $\bar\rho_\mathfrak{m}$ follows from the dimension by the theorem of Boston, Lenstra and Ribet. The paper's route, in Remark rem:free, is through $J_1(N)$: [163, Theorem 3.5] gives multiplicity one for $J_1(N)$, "the only possible exception there having Serre weight $\ell$, which does not occur in weight 2 with $\ell \nmid N$", and $J_0(N)[\mathfrak{m}] \to J_1(N)[\mathfrak{m}']$ is injective because its kernel lies in the Shimura subgroup, which is Eisenstein (Definition 14.68).

**Theorem 14.62** (Gorenstein and free of rank two)**.** *In the situation of Theorem 14.61, with* $\mathfrak{m}$ *a maximal ideal of* $\mathbb{T}_\ell$*:* $\mathbb{T}_\mathfrak{m}$ *is Gorenstein, and* $H^1(X_0(N),\mathbb{Z}_\ell)_\mathfrak{m} \simeq \mathbb{T}_\mathfrak{m}^2 \simeq T_\ell J_0(N)_\mathfrak{m}$ *as* $\mathbb{T}_\mathfrak{m}$*-modules. Consequently, for* $E$ *optimal,* $\ell \nmid 2N$ *and* $E[\ell]$ *irreducible,* $v_\ell(r_E) = v_\ell(\deg \varphi_E)$*.*

*Proof.* Let $L = H^1(X_0(N),\mathbb{Z}_\ell)_\mathfrak{m}$, with the pairing $\hat{\delta}_!$, perfect on $L$ with $\mathbb{T}_\mathfrak{m}$ self-adjoint (Corollary 14.12(iii)). We check the hypotheses of the freeness criterion of Chapter 3 with $d = 2$. (a) $\dim_{\mathbb{T}/\mathfrak{m}} L/\mathfrak{m}L = 2$. Put $W = L/\ell L = H^1(X_0(N),\mathbb{F}_\ell)_\mathfrak{m}$; $\hat{\delta}_!$ induces a perfect pairing on $W$ for which $\mathbb{T}$ is self-adjoint, so $(W/\mathfrak{m}W)^\vee \simeq W[\mathfrak{m}]$ (a functional vanishing on $\mathfrak{m}W$ is $\langle -, y\rangle$ with $\langle \mathfrak{m}W, y\rangle = \langle W, \mathfrak{m}y\rangle = 0$, that is $y \in W[\mathfrak{m}]$). Hence $\dim_{\mathbb{F}_\ell} L/\mathfrak{m}L = \dim_{\mathbb{F}_\ell} W[\mathfrak{m}] = \dim_{\mathbb{F}_\ell} J_0(N)[\mathfrak{m}]$ by Proposition 14.18(iii), and dividing by $[\mathbb{T}/\mathfrak{m} : \mathbb{F}_\ell]$, $\dim_{\mathbb{T}/\mathfrak{m}} L/\mathfrak{m}L = 2$ by Theorem 14.61. (b) $\operatorname{rank}_{\mathbb{Z}_\ell} L = 2\operatorname{rank}_{\mathbb{Z}_\ell} \mathbb{T}_\mathfrak{m}$. By Proposition 14.10(ii), (iii), $H^1(X_0(N),\mathbb{C}) = S_2 \oplus \overline{S_2}$ with $S_2 \simeq \operatorname{Hom}_\mathbb{Z}(\mathbb{T},\mathbb{C})$ as $\mathbb{T}$-modules, and $\overline{S_2} \simeq S_2$ as well (complex conjugation on the second factor of $\operatorname{Hom}_\mathbb{Z}(\mathbb{T},\mathbb{Z}) \otimes \mathbb{C}$ is $\mathbb{T}$-linear). So for $t \in \mathbb{T}$, $\operatorname{tr}(t \mid H^1(X_0(N),\mathbb{Q})) = 2\operatorname{tr}(t \mid \operatorname{Hom}_\mathbb{Z}(\mathbb{T},\mathbb{Q})) = 2\operatorname{tr}(t \mid \mathbb{T}\otimes\mathbb{Q})$, the trace of a transpose being the same. Both sides are $\mathbb{Q}_\ell$-linear in $t \in \mathbb{T}\otimes\mathbb{Q}_\ell$ and agree on $\mathbb{T}$, so they agree on the idempotent $e_\mathfrak{m}$, whose trace on a module is the dimension of its $\mathfrak{m}$-part: $\operatorname{rank} L = \dim e_\mathfrak{m} H^1(X_0(N),\mathbb{Q}_\ell) = 2\dim e_\mathfrak{m}(\mathbb{T}\otimes\mathbb{Q}_\ell) = 2\operatorname{rank}\mathbb{T}_\mathfrak{m}$. The criterion gives $L \simeq \mathbb{T}_\mathfrak{m}^2$ and $\mathbb{T}_\mathfrak{m}$ Gorenstein; $T_\ell J_0(N)_\mathfrak{m} \simeq L$ by Proposition 14.18(ii). For the last statement, $\mathfrak{m}_{f,\ell}$ is non-Eisenstein when $E[\ell]$ is irreducible (Definition 14.60), and Corollary 14.55 gives equality under freeness. □

This is the theorem the paper cites as [199] in Remark rem:free (see also [207, Chapter 2] and [38, Section 4]). Step (b) holds for every $\ell$ and every $\mathfrak{m}$; only (a) uses multiplicity one, and it is (a) that fails for `99a1` at $\ell = 3$ (Example 14.58).

*Remark* 14.63 (what Remark rem:free is still for)*.* After Theorems 14.54 and 14.56: (a) Remark rem:free is not needed for the equality in Lemma lem:hecke or in (eq:identity) of Theorem thm:main. (b) It is not needed for the equality in the first inequality of Theorem thm:quant when $\ell \notin S_f$ either: Theorem 14.56(iii) applies to the optimal curve of each of the two classes compared. It remains the natural tool for comparing the Hecke congruence ideal with the cohomological one for forms that are not attached to an optimal quotient, such as the depleted form $g^S$ of Step 2 of the paper's proof, where it turns an inequality into an equality. (c) It is the conceptual reason for the equality in the theorem of Agashe, Ribet and Stein at $\ell \nmid 2N$ with $E[\ell]$ irreducible (Theorem 14.62). It fails for $\ell \mid N$ in general (Example 14.58).

*In the paper* 14.64*.* Remark rem:free, sentence by sentence. "Freeness of $H^1(X_0(N),\mathbb{Z}_\ell)_\mathfrak{m}$ over $\mathbb{T}_\mathfrak{m}$ holds when $J_0(N)[\mathfrak{m}]$ is two-dimensional, since the localised Hecke algebra is then Gorenstein and the Tate module is free of rank two": Theorem 14.62, step (a) being the two-dimensionality and step (b) a rank count. "$J_0(N)[\mathfrak{m}]$ is two-dimensional for $\ell \nmid 2N$ and $\bar{\rho}$ irreducible": Theorem 14.61 (stated). "The identification of $T_\ell J_0(N)$ with $H^1(X_0(N),\mathbb{Z}_\ell)$ as Hecke modules uses Poincaré duality composed with $w_N$": Proposition 14.18. "After Proposition prop:pullback this is not needed for Theorem thm:main or for Theorem thm:quant": Remark 14.63.

## 14.11 Cuspidal subgroup, Eisenstein ideal, Shimura subgroup

When $E[\ell]$ is reducible every theorem of the paper is out of reach: the maximal ideal $\mathfrak{m}_{f,\ell}$ is Eisenstein, multiplicity one and the results of Diamond, Flach and Guo are unavailable, and the modular degree can fall short of the local bound (the deficit $D$ of Conjecture conj:eis). Open question 4 asks for an interpretation of that deficit "presumably through the cuspidal subgroup

of $J_0(N)$ [125]". This section defines the cuspidal subgroup, proves that the cusps 0 and $\infty$ differ by a torsion point and computes its image on the optimal curve, states Mazur's theorem at prime level, and defines the Shimura subgroup used in Remark rem:free.

**Definition 14.65** (the cuspidal subgroup)**.** The *cuspidal subgroup* $C_N \subseteq J_0(N)(\overline{\mathbb{Q}})$ is the subgroup generated by the classes of the divisors of degree 0 supported on the cusps. It is $G_{\mathbb{Q}}$-stable, because $G_{\mathbb{Q}}$ permutes the cusps (Proposition 14.3); the *rational cuspidal subgroup* is generated by the classes of the $G_{\mathbb{Q}}$-invariant cuspidal divisors of degree 0, and lies in $J_0(N)(\mathbb{Q})$.

**Proposition 14.66** (Manin-Drinfeld for 0 and $\infty$)**.**

*(i) For $p \nmid N$, $(T_p)_*(\infty) = (p+1)(\infty)$ and $(T_p)_*(0) = (p+1)(0)$ as divisors.*

*(ii) For $p \nmid N$ every eigenvalue of $T_p$ on $S_2(\Gamma_0(N))$ has absolute value at most $2\sqrt{p} < p+1$, so $(T_p - p - 1)_*$ is an isogeny of $J_0(N)$.*

*(iii) The class $x = [(0) - (\infty)]$ lies in $\ker(T_p - p - 1)_*$, so it is a torsion point of $J_0(N)(\mathbb{Q})$.*

*(iv) (Manin; Drinfeld; stated.) $C_N$ is finite.*

*Proof.* (i) By Proposition 14.9(i), $(T_p)_*(\Gamma c) = \sum_i (\Gamma \alpha_i c)$ for a cusp $c$, with the representatives $\alpha_i$ of Proposition 14.9(ii), counted with the multiplicities of $\pi_1$ (total $p+1$): every point of $X(\Gamma_\alpha)$ over $\Gamma c$ is $\Gamma_\alpha \gamma c$ for some $\gamma \in \Gamma$ and maps to $\Gamma \alpha \gamma c$, which is one of the $\Gamma \alpha_i c$. Now $\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)\infty = \infty$ and $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)\infty = \infty$, so all images of $\infty$ are $\infty$. And $\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)0 = j/p$, $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)0 = 0$; for $1 \le j < p$ the cusp $j/p$ has denominator $p$ with $\gcd(p, N) = 1$, so it is $\Gamma_0(N)$-equivalent to 0 (Proposition 14.3(i): the cusps of denominator prime to $N$ form the single class of 0). So all images of 0 are 0. (ii) By Theorem 14.24(ii) the eigenvalues $\alpha$ of Frobenius on $W_\psi$ are roots of $X^2 - \psi(T_p)X + p$, and $|\alpha| = \sqrt{p}$ by Weil's theorem (Chapter 8); so $\psi(T_p) = \alpha + p/\alpha$ has absolute value at most $2\sqrt{p}$ for every eigencharacter $\psi$ (this is the Ramanujan-Petersson bound in weight 2, Chapter 12). Hence $T_p - p - 1$ is invertible on $S_2$, its transpose on $S_2^\vee$ is invertible, and the induced endomorphism of $J_0(N)(\mathbb{C}) = S_2^\vee / H_1$ is surjective with finite kernel. (iii) By (i), $(T_p - p - 1)_* x = [(p+1)(0) - (p+1)(\infty)] - (p+1)x = 0$; an element of the finite kernel of an isogeny is torsion; 0 and $\infty$ are rational (Proposition 14.3). (iv) We do not prove this; see [125, Chapter II], [187, Section 6.6] and [146] (not verified). The proof is (i) to (iii) for every cusp, with the cusps not defined over $\mathbb{Q}$ handled by the Galois action. □

**Theorem 14.67** (Mazur at prime level; stated)**.** *Let $N$ be prime and $n$ the numerator of $(N-1)/12$.*

*(i) The class of $(0) - (\infty)$ has order $n$ and generates $C_N$.*

*(ii) $J_0(N)(\mathbb{Q})_{\mathrm{tors}} = C_N \simeq \mathbb{Z}/n$.*

*(iii) The* Eisenstein ideal *$I_{\mathrm{Eis}} \subseteq \mathbb{T}$, generated by the $T_p - 1 - p$ ($p \neq N$) and $1 + w_N$, satisfies $\mathbb{T}/I_{\mathrm{Eis}} \simeq \mathbb{Z}/n$; it annihilates $C_N$ and the Shimura subgroup $\mathcal{S}_N$, which has order $n$, is of multiplicative type, and is Cartier dual to $C_N$.*

*(iv) For $\ell$ odd, a maximal ideal $\mathfrak{m}$ of $\mathbb{T}$ of residue characteristic $\ell$ is Eisenstein in the sense of Convention 14.1(C7) iff $\mathfrak{m} \supseteq I_{\mathrm{Eis}}$, and such $\mathfrak{m}$ exist iff $\ell \mid n$.*

We do not prove this; see [125, Chapter II] (the numbering of Mazur's results, and the cases $\ell = 2$ and $\ell = N$ of (iv), were not checked against the text). For $N = 11$: $n = 5$, and $J_0(11)(\mathbb{Q}) = \texttt{11a1}(\mathbb{Q}) \simeq \mathbb{Z}/5$, generated by the cusp 0 (Example 14.6); $\ell = 5$ is Eisenstein, and indeed $a_p(f_{11}) \equiv 1 + p \pmod 5$ for $p \neq 11$ ($a_2 = -2$, $a_3 = -1$, $a_5 = 1$, $a_7 = -2$), the congruence of $f_{11}$ with the Eisenstein series of Chapter 12. For $N = 37$: $n = 3$, and the curve `37b1` has $E(\mathbb{Q}) \simeq \mathbb{Z}/3$.

**Definition 14.68** (the Shimura subgroup)**.** Let $\pi\colon X_1(N) \to X_0(N)$ be the forgetful map. The *Shimura subgroup* is $\mathcal{S}_N = \ker(\pi^*\colon J_0(N) \to J_1(N))$.

**Proposition 14.69** (the Shimura subgroup)**.**

*(i) Assume $g(X_0(N)) \geq 1$. Then $\#\mathcal{S}_N = [\mathrm{sat}(\pi^* H^1(X_0(N), \mathbb{Z})) : \pi^* H^1(X_0(N), \mathbb{Z})]$, the saturation taken in $H^1(X_1(N), \mathbb{Z})$; so $\mathcal{S}_N = 0$ iff $\pi^* H^1(X_0(N), \mathbb{Z})$ is saturated.*

*(ii) (Stated.) $\mathcal{S}_N$ is finite and of multiplicative type (Cartier dual to a constant group), and $T_p$ acts on it as $1 + p$ for $p \nmid N$: it is Eisenstein.*

*(iii) If $\mathfrak{m}$ is a non-Eisenstein maximal ideal of $\mathbb{T}$, then $\mathcal{S}_N[\mathfrak{m}] = 0$, so $\pi^*\colon J_0(N)[\mathfrak{m}] \to J_1(N)$ is injective.*

*Proof.* (i) is Chapter 8 (the proposition on the degree and the cup product, part (iii), (a)⇔(d) with the index formula) for $\pi$. (ii) We do not prove this; see [125, Chapter II] for prime $N$, and [115] and [161] in general (not verified). (iii) If $0 \neq x \in \mathcal{S}_N[\mathfrak{m}]$, then $T_p - 1 - p$ and $\mathfrak{m}$ both kill $x$, so $T_p \equiv 1 + p \pmod{\mathfrak{m}}$ for all $p \nmid N\ell$ ($\mathbb{T}/\mathfrak{m}$ acts faithfully on the nonzero $\mathbb{T}/\mathfrak{m}$-space spanned by the $\mathbb{T}$-translates of $x$). Then $\operatorname{tr} \bar{\rho}_{\mathfrak{m}}(\mathrm{Frob}_p) = \operatorname{tr}(1 \oplus \bar{\chi}_\ell)(\mathrm{Frob}_p)$ for those $p$, and by Chebotarev and Brauer-Nesbitt the semisimple $\bar{\rho}_{\mathfrak{m}}$ is $1 \oplus \bar{\chi}_\ell$, reducible: a contradiction. □

For the optimal curve, $\mathfrak{m}_{f,\ell}$ is Eisenstein exactly when $E$ has a rational $\ell$-isogeny (Definition 14.60; Chapter 11). Then Theorems 14.61 and 14.62, the theorems of [49] and the hypothesis of Lemma lem:hecke all fail, and the modular degree can lose local divisibility: the deficit $D$ of Conjecture conj:eis (Chapter 11).

**Proposition 14.70** (the image of the cusp 0)**.** *Let $E$ be optimal. Then $\varphi_E(0) \in E(\mathbb{Q})_{\mathrm{tors}}$, and under $E(\mathbb{C}) \simeq \mathbb{C}/\Lambda_E$, $P \mapsto \int_O^P \omega_E$,*

$$\varphi_E(0) \equiv c_E\, L(E, 1) \pmod{\Lambda_E}.$$

*Since $\Lambda_E \cap \mathbb{R} = \mathbb{Z}\Omega$ with $\Omega$ the least positive real period, $c_E L(E,1)/\Omega \in \mathbb{Q}$ and the order of $\varphi_E(0)$ is its denominator.*

*Proof.* $\varphi_E = \varphi_{E*} \circ \iota$ and $\iota(0) = x = [(0) - (\infty)]$, so $\varphi_E(0) = \varphi_{E*}(x)$, a rational torsion point by Proposition 14.66(iii). For $z \in \mathcal{H}$, the composite $\mathcal{H} \to X_0(N) \to E(\mathbb{C}) \to \mathbb{C}/\Lambda_E$ is $z \mapsto \int_{i\infty}^z \varphi_E^* \omega_E = c_E \int_{i\infty}^z 2\pi i f(\tau)\, d\tau$ (the integral converges at $i\infty$, $f$ being cuspidal), and it extends continuously to the cusp 0. Along the imaginary axis, $\tau = iy$:

$$\int_{i\infty}^0 2\pi i f(\tau)\, d\tau = \int_\infty^0 (-2\pi) f(iy)\, dy = 2\pi \int_0^\infty f(iy)\, dy = L(f, 1),$$

the last equality being the Mellin transform $(2\pi)^{-s}\Gamma(s)L(f,s) = \int_0^\infty f(iy) y^{s-1}\, dy$ at $s = 1$ (Chapter 19; stated there). With $L(f,1) = L(E,1)$ (Theorem 14.25) this gives the congruence. $\Lambda_E$ is stable under complex conjugation (E and $\omega_E$ are defined over $\mathbb{R}$), so $\Lambda_E \cap \mathbb{R}$ is a nonzero discrete subgroup $\mathbb{Z}\Omega$; a real number $r$ defines a point of order $m$ in $\mathbb{C}/\Lambda_E$ iff $m$ is least with $mr \in \mathbb{Z}\Omega$. □

**Example 14.71** ($L(E,1)/\Omega$ and torsion)**.** PARI (`lfun(E,1)/E.omega[1]`, recognised as a rational number to 40 digits) gives $L(E,1)/\Omega = 1/5,\ 1/6,\ 1/3,\ 1/6,\ 1/3,\ 2/3,\ 1,\ 1/3$ for `11a1`, `14a1`, `19a1`, `20a1`, `27a1`, `37b1`, `121a1`, `243b1`, whose torsion orders are 5, 6, 3, 6, 3, 3, 1, 3; and $L(E,1) = 0$ for the rank-one curves `37a1`, `99a1`, `539c1`, `726a1`, so $\varphi_E(0) = O$ for them. With $c_E = 1$ (Manin's conjecture, confirmed by computation in this range, Theorem 14.39(iv)) the order of $\varphi_E(0)$ is the denominator, and in each of these examples $\varphi_E(0)$ generates $E(\mathbb{Q})_{\mathrm{tors}}$. For `11a1`, $\varphi_E$ is the identity of $X_0(11)$ and the statement is that the cusp 0 has order 5 (Theorem 14.67).

*Remark* 14.72 (the Eisenstein case in the paper). Ihara's lemma, the homothety of lattices behind part (c) of Proposition 1.4 of [49] (Chapter 20) and multiplicity one (Theorem 14.61) all need $\bar{\rho}$ irreducible; Proposition 14.52 does not, and neither does the identification of the integral structure of [49] with $H^1(X_0(N),\mathbb{Z}_\ell)$ (Remark 14.57(1)), which holds for every prime $\ell$. Conjecture conj:eis bounds the deficit by the $\ell$-part of the torsion, and Chapter 9 proves that its bound $D \le 2$ at $\ell = 3$ follows from the first (Proposition prop:no9 of the paper). Proposition 14.70 links rational torsion with the cuspidal subgroup: $\varphi_E(0)$ is a rational torsion point, the image of the cuspidal class $[(0)-(\infty)]$, and it often generates $E(\mathbb{Q})_{\mathrm{tors}}$. This is a heuristic for Open question 4, not an argument.

*In the paper* 14.73. Open question 4: "Identify the deficit of Conjecture conj:eis with a natural invariant of the Eisenstein ideal at $\ell$, presumably through the cuspidal subgroup of $J_0(N)$ [125]." Definition 14.65, Propositions 14.66 and 14.70, and Theorem 14.67 are the objects named there. Remark rem:free: "$J_0(N)[\mathfrak{m}]$ injects into $J_1(N)[\mathfrak{m}']$ under the pull-back map, whose kernel, the Shimura subgroup, is Eisenstein" is Proposition 14.69(iii).

## 14.12 Shimura curves, the prime 2 and the computation of the modular degree

Three topics of the paper lie outside its theorems but next to them: the comparison of $\deg\varphi_E$ with the degree of a parametrisation by a Shimura curve (Section 1.5, the history of the multiplicative part of (eq:main)), the prime $\ell = 2$ (Section sec:two: Watkins's conjecture and the theorem of Calegari and Emerton), and the computation of $\deg\varphi_E$ on which all the paper's data rest. This section states what is used, with data checks.

**Definition 14.74** (Shimura curves; stated)**.** Let $D > 1$ be a product of an even number of distinct primes and $M \ge 1$ prime to $D$. The *Shimura curve* $X_0^D(M)$ is the curve attached to an Eichler order of level $M$ in the indefinite quaternion algebra over $\mathbb{Q}$ of discriminant $D$; it is a smooth projective curve over $\mathbb{Q}$ without cusps. By the Jacquet-Langlands correspondence, in Ribet's form [156], its Jacobian $J_0^D(M)$ is isogenous over $\mathbb{Q}$, compatibly with the Hecke operators $T_p$, $p \nmid DM$, to the $D$-new quotient of $J_0(DM)$. Now let $E$ be semistable of conductor $N = DM$ (so $N$ is square-free, the setting of [164, 192]). Among the curves isogenous to $E$ there is exactly one, written $E_D$, which is an optimal quotient of $J_0^D(M)$, that is, the target of a surjection $\xi_D\colon J_0^D(M) \to E_D$ with connected kernel, unique up to sign. The composite of $\xi_D$ with its dual $E_D \to J_0^D(M)$ is multiplication by a positive integer $\delta_D$ (written $\delta^D(M)$ in the sources), the degree of the parametrisation $\varphi_{E,D}\colon X_0^D(M) \to J_0^D(M) \to E_D$ (a base point being chosen; the degree does not depend on it). For $D = 1$, $E_1$ is the $X_0(N)$-optimal curve and $\delta_1 = \deg\varphi_{E_1}$. In general $E_D$ is not the $X_0(N)$-optimal curve: Takahashi notes "(This $E$ generally differs from $E$ in the optimal quotient $J_0(N) \to E$.)" [192, p. 75].

**Theorem 14.75** (Ribet and Takahashi; Takahashi; stated)**.** *Let $E$ be the $X_0(N)$-optimal curve in an isogeny class of semistable elliptic curves over $\mathbb{Q}$ of conductor $N$. For a prime $r \mid N$ and a curve $E'$ isogenous to $E$ write $n_r(E') = v_r(\Delta_{E'}) = \#\Phi_r^{\mathrm{geom}}(E')$, and $n_r = n_r(E)$.*

*(i) Let $N = dpqm$ with $p \neq q$ primes and $d$ a product of an even number of primes ($d = 1$ allowed). Let $E_1 = E_d$ and $\delta^d(pqm)$ be the optimal quotient of $J = J_0^d(pqm)$ and its degree, and $E_2 = E_{dpq}$ and $\delta^{dpq}(m)$ those of $J_0^{dpq}(m)$ (Definition 14.74). Let $i_q$ be the order of the image of the map $\Phi_q^{\mathrm{geom}}(J) \to \Phi_q^{\mathrm{geom}}(E_1)$ induced by the optimal quotient. Then $i_q$ divides $n_q(E_1)$ and $n_p(E_2)$, and*

$$\frac{\delta^d(pqm)}{\delta^{dpq}(m)} = \frac{n_p(E_2)\, n_q(E_1)}{i_q^2}.$$

*(ii) If $E$ is the only curve in its isogeny class, then $\deg \varphi_E/\delta_D = \prod_{r\mid D} n_r$ for every divisor $D$ of $N$ with an even number of prime factors.*

*(iii) If $\ell$ is a prime with $E[\ell]$ irreducible ($\ell = 2$ and $\ell \mid N$ are allowed), then $\ell \nmid i_q$ in (i), and for every divisor $D$ of $N$ with an even number of prime factors*

$$v_\ell(\deg \varphi_E) = v_\ell(\delta_D) + \sum_{r\mid D} v_\ell(n_r).$$

We do not prove (i) and (ii). Part (i) is [164, Theorems 1 and 2] (pages 11111 and 11112) in the sharpened form of [192, Theorem 3.2] (page 82), whose part (c) reads "$\delta/\delta' = (c_p'/i_q)\cdot j_q$. In particular, $\delta'$ divides $\delta$.", together with $j_q = c_q/i_q$ for the order $j_q$ of the cokernel of the same map (proof of [192, Theorem 2.3], page 80); divisibility of $n_p(E_2)$ by $i_q$ is [192, Theorem 3.2(b)]. Part (ii) is [192, Corollary 3.5] (page 83). In both sources $c_r$ is our $n_r$, not the Tamagawa number: Ribet and Takahashi write "Note that $c_p$, for instance, may be viewed as the order of the group of components of the fiber at $p$ of the Néron model for $A$." [164, page 11111], and Takahashi defines $c_r$ as the number of connected components of the fibre at $r$, that is $\mathrm{ord}_r(\Delta)$ [192, pages 75 and 76]. The Tamagawa number $c_r = \#\Phi_r^{\mathrm{geom}}(\mathbb{F}_r)$ of this book differs from $n_r$ at nonsplit primes (Chapter 10): for `57c2`, $n_{19} = 5$ (PARI `elllocalred`) while its Tamagawa number at 19 is 1, and Takahashi's Table I lists $c_{19}' = 5$ for this curve.

*Sketch of proof of (iii) from (i).* If $E'[\ell]$ is irreducible for one curve of the class, it is irreducible for all [164, page 11113], and any two curves of the class are then linked by an isogeny of degree prime to $\ell$, which induces isomorphisms of the $\ell$-primary parts of the component groups; so $v_\ell(n_r(E')) = v_\ell(n_r)$ for every $E'$ in the class and every $r \mid N$ [164, page 11111]. In (i) the prime $q$ divides the level $pqm$ of $J$, and [192, Theorem 2.7] gives $\mathrm{ord}_\ell(j_q) = \mathrm{ord}_\ell(n_q(E_1))$; since $i_q j_q = n_q(E_1)$, $\ell \nmid i_q$. Hence (i) gives $v_\ell(\delta^d(pqm)) - v_\ell(\delta^{dpq}(m)) = v_\ell(n_p) + v_\ell(n_q)$. Write $D = p_1q_1\cdots p_kq_k$ and apply this with $d = p_1q_1\cdots p_{j-1}q_{j-1}$, $(p,q) = (p_j, q_j)$ and $m = N/(p_1q_1\cdots p_jq_j)$ for $j = 1,\dots,k$; the sum telescopes from $\delta^1(N) = \deg\varphi_E$ to $\delta^D(N/D) = \delta_D$. □

*Remark* 14.76 (the sources, the hypotheses and the error term)*. History and hypotheses.* For an isogeny class with one element, Bertolini and Darmon showed that the conjecture of Birch and Swinnerton-Dyer implies $\deg\varphi_E/\delta_D = \prod_{r\mid D} n_r$ in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, and asked whether it holds in $\mathbb{Q}^\times$ [192, page 76]. Ribet and Takahashi [164, Theorem 1] prove the one-step formula in the shape $\delta^{pqD}(M) = \delta^D(pqM)\,\mathcal{E}(D,p,q,M)^2/(c_p'c_q)$, with an "error term" $\mathcal{E}(D,p,q,M)$ that is a positive divisor of $c_p'c_q$; they show that every prime $\ell$ dividing it has $A[\ell]$ reducible under the further hypothesis that $M$ is square-free but not a prime (the case of prime $M$ is excluded in

their footnote), and their Theorem 2 identifies it as the product of the order of the image of the map on component groups at $q$ and the order of the cokernel of the map at $p$. Takahashi proves that the map on component groups at a prime of the discriminant is surjective [192, Theorem 2.4], a consequence of a result of Bertolini and Darmon, so that the error term is $i_q$, and removes the condition on $M$: "In this paper, we present a proof of $(*)$ which works for any $D$ and $M$." [192, page 76], $(*)$ being the formula of (ii). The standing hypothesis of both sources is that $N$ is square-free; in (iii) no condition on $\ell$ other than the irreducibility of $E[\ell]$ enters. Pasten [151, Section 1] summarises the formula in the form of [164], with $M$ square-free and not a prime, and refines it in his Section 6 (a secondary pointer, not compared with the sources here).

*The error term.* Put $\gamma_D = \deg\varphi_E/(\delta_D \prod_{r\mid D} n_r) \in \mathbb{Q}^\times$. By (iii), $v_\ell(\gamma_D) = 0$ whenever $E[\ell]$ is irreducible: $\gamma_D$ is supported on the primes $\ell$ at which $E[\ell]$ is reducible. By (i) it is a product of the factors $i_q^{-2}$ and of ratios $n_r(E')/n_r$ for curves $E'$ isogenous to $E$. Ribet and Takahashi remark that the error term is not necessarily divisible by all the primes $\ell$ for which $A[\ell]$ is reducible [164, page 11111]. Example ([192, Section 4, Example 1 and Table I], with $N = 57$): for `57c1`, $\deg\varphi_E = 12$, $n_3 = 10$, $n_{19} = 1$, the curve $E_{57}$ is `57c2`, $\delta_{57} = 6$ and $i_3 = 5$; so $\gamma_{57} = 12/(6\cdot 10) = 1/5$, and $E[5]$ is reducible (`57c1` has a rational 5-isogeny to `57c2`). For the isolated curve `57a1`, $\deg\varphi_E = 4$, $n_3 = 2$, $n_{19} = 1$, and (ii) gives $\delta_{57} = 2$, as computed in [164, page 11111]. (The values of $\deg\varphi_E$ are those of `degphi` [36], the $n_r$ those of PARI's `elllocalred`; $\delta_{57}$ and $i_3$ are taken from the sources.)

*Use in the paper.* For $E[\ell]$ irreducible and $N$ square-free, (iii) gives $v_\ell(\deg\varphi_E) \geq \sum_{r\mid D} v_\ell(n_r) = \sum_{r\mid D} t_\ell(E,r)$, the last equality being $t_\ell(E,r) = v_\ell(n_r)$ at a multiplicative prime (Lemma lem:tamagawa-exp of the paper). This is the multiplicative part of (eq:main) at the primes of $D$, an even number of them; the paper's Theorem thm:main contains it, under its own hypotheses on $\ell$, for every multiplicative prime. The paper's own summary, in Section 1.5, is that they "compared, for semistable $E$, $\deg\varphi_E$ with the degree of a parametrisation by a Shimura curve and found the product of the $n_q$ over the primes of the quaternion discriminant, exactly when $E$ is the only curve in its isogeny class and otherwise up to primes where $E[\ell]$ is reducible"; this matches (ii) and (iii).

*The idea of the proof.* Proposition 1 of [164] (Proposition 3.1 of [192]) is an exact sequence $0 \to \mathcal{X}_p(J') \to \mathcal{X}_q(J) \to \mathcal{X}_q(J'')^2 \to 0$ of character groups of the tori of the reductions, with $J = J_0^d(pqm)$, $J' = J_0^{dpq}(m)$, $J'' = J_0^d(qm)$, compatible with the Hecke operators and the monodromy pairings; for $d = 1$ it is [156, Theorem 4.1], as both sources cite it. Grothendieck's exact sequence $0 \to \mathcal{X} \to \mathrm{Hom}(\mathcal{X},\mathbb{Z}) \to \Phi \to 0$ expresses each component group as the cokernel of the monodromy pairing, and comparing the $f$-isotypic lines on the two sides gives (i).

**Conjecture 14.77** (Watkins)**.** *For every optimal $E/\mathbb{Q}$, $2^{\operatorname{rank} E(\mathbb{Q})} \mid \deg\varphi_E$.*

Data: all 38,042 optimal curves of conductor below 10,000 satisfy it (`degphi` and `allcurves` [36]); for instance `37a1`, `99a1`, `121b1`, `539c1`, `726a1` have rank 1 and degrees 2, 4, 4, 288, 144. The conjecture is due to Watkins [204]; partial results are in [57, 20, 208, 24, 81, 12], which the paper cites without statements, and which this book does not state either.

**Theorem 14.78** (Calegari and Emerton; stated)**.** *Let $E/\mathbb{Q}$ be optimal of odd modular degree. Then $N$ has at most two odd prime factors, $E$ has even analytic rank, and either (a) $E$ has a rational point of order 2, or (b) $N$ is prime, $E$ is supersingular at 2 and $E(\mathbb{R})$ is connected, or (c) $E$ has complex multiplication and $N \in \{27, 32, 49, 243\}$.*

We do not prove this. The statement is the one of Section sec:two of the paper: "a curve of odd modular degree has at most two odd bad primes and even analytic rank, and either it

has a rational 2-torsion point, or its conductor is prime, it is supersingular at 2 and $E(\mathbb{R})$ is connected, or it has complex multiplication and $N \in \{27, 32, 49, 243\}$", from [20] (as Theorem 1.1 of the arXiv version; the published numbering was not checked). Data (`degphi`, `allcurves`, and PARI for $a_2$, the discriminant and $j$): 103 optimal curves of conductor below 10,000 have odd modular degree, and all satisfy the theorem; 55 have a rational point of order 2, 46 have prime conductor, no rational point of order 2, $a_2$ even and negative discriminant, and two, `27a1` ($\deg \varphi_E = 1$) and `243b1` ($\deg \varphi_E = 9$), both with $j = 0$ and torsion $\mathbb{Z}/3$, are covered only by (c). The odd-degree optimal curves of conductor below 100 are `11a1`, `14a1`, `15a1`, `17a1`, `19a1`, `20a1`, `21a1`, `24a1`, `27a1`, `32a1`, `33a1`, `36a1`, `46a1`, `49a1`, `52a1`, `57b1`, `67a1`, `73a1`, `89b1`.

*Remark* 14.79 (computing the modular degree). Three methods. (a) Zagier's [209]: compute $\varphi_{E*}$ on $H_1(X_0(N), \mathbb{Z})$ from modular symbols and read $\deg \varphi_E$ off the index of Proposition 14.31(iv), the cup product on $\varphi_E^* H^1(E, \mathbb{Z})$ having ideal of values $\deg \varphi_E \mathbb{Z}$ ([35, Chapter II]). (b) Watkins's [204]: compute $\langle f, f \rangle$ from the special value of the symmetric square $L$-function with the bad Euler factors and apply Theorem 14.40 with $c_E = 1$; PARI's `ellmoddegree` implements this, and, as Example 14.42 shows for `11a3`, returns $\deg \varphi / c_\varphi^2$ (here $1/5$) when applied to a non-optimal curve. (c) Cremona's comparison of the period lattices of the curves of a class with the lattice of $\varphi^*$ certifies optimality (Convention 14.34). The paper's appendix: "The function `ellmoddegree` needs about 32 MB of stack with PARI 2.17 for conductors in the thousands". Chapter 22 has the sessions.

*In the paper* 14.80. Section 1 of the paper: "The modular degree is computable [35, 204]" is Remark 14.79; Section sec:two quotes Watkins's conjecture (Conjecture 14.77) and the theorem of Calegari and Emerton (Theorem 14.78); Section 1.5 summarises Ribet and Takahashi (Theorem 14.75 and Remark 14.76).

## 14.13 Exercises

**Exercise 14.81.** List the cusps of $X_0(50)$ and of $X_0(72)$ by denominator, with their fields of definition, and find all $N \le 30$ for which some cusp of $X_0(N)$ is not rational. *Answer.* $X_0(50)$: 12 cusps; four each of denominators 5 and 10, conjugate over $\mathbb{Q}(\zeta_5)$, the other four rational. $X_0(72)$: 16 cusps; two each of denominators 3, 6, 12, 24, conjugate over $\mathbb{Q}(\sqrt{-3})$, the other eight rational. $N = 9, 16, 18, 25, 27$ (Proposition 14.3(iii)).

**Exercise 14.82.** Let $f = f_{11}$ and $V = \operatorname{span}\{f(z), f(2z)\} = S_2(\Gamma_0(22))$. (a) Using $a_2(f) = -2$ and $f|_2 W_{11} = -f$, show that in the basis $f(z)$, $f(2z)$ the matrices (columns are images) of $U_2$ and $w_{22}$ are $U = \left(\begin{smallmatrix} -2 & 1 \\ -2 & 0 \end{smallmatrix}\right)$ and $W = \left(\begin{smallmatrix} 0 & -1/2 \\ -2 & 0 \end{smallmatrix}\right)$. (b) Show that $\mathbb{T} = \mathbb{Z}[U_2]$ at level 22 (the $T_p$, $p \nmid 22$, act as $a_p(f)$ and $U_{11}$ as $a_{11}(f) = 1$), that $WUW = \left(\begin{smallmatrix} 0 & -1/2 \\ 4 & -2 \end{smallmatrix}\right)$ is not of the form $a + bU$, and deduce that $w_{22} \mathbb{T} w_{22} \neq \mathbb{T}$ (Proposition 14.17(iii)). (c) With $A = \langle f, f \rangle_{\Gamma_0(22)}$, the Petersson Gram matrix in this basis is $G = \left(\begin{smallmatrix} A & -A/3 \\ -A/3 & A/4 \end{smallmatrix}\right)$ (check the off-diagonal entry with PARI's `mfpetersson`). Verify $W^{\mathsf{T}} G W = G$ and $G^{-1} U^{\mathsf{T}} G = WUW$: the Petersson adjoint of $U_2$ is $w_{22} U_2 w_{22}$, and $U_2$ is not self-adjoint. *Hint.* $U_2 f = \sum a_{2n} q^n = a_2 f - 2 f(2z)$ and $U_2(f(2z)) = f$; $W_{22} = W_{11} \operatorname{diag}(2, 1)$ and $\operatorname{diag}(2, 1) W_{22} = 2 W_{11}$. For (b), $a + bU$ has entries $(a - 2b, b; -2b, a)$.

**Exercise 14.83.** Using Theorem 14.24(iii) and $X_0(11) =$ `11a1`, predict $\#\texttt{11a1}(\mathbb{F}_p)$ for $p = 2, 3, 5, 7, 13$ from $f_{11} = q - 2q^2 - q^3 + 2q^4 + q^5 + 2q^6 - 2q^7 - 2q^9 - 2q^{10} + q^{11} - 2q^{12} + 4q^{13} + \cdots$, and check by counting points on $y^2 + y = x^3 - x^2 - 10x - 20$. *Answer.* $5, 5, 5, 10, 10$, from $p + 1 - a_p$

with $a_p = -2,-1,1,-2,4$: $3+2$, $4+1$, $6-1$, $8+2$, $14-4$. All five are divisible by 5, as they must be for $p \neq 5, 11$: the rational 5-torsion of `11a1` injects into $\tilde{E}(\mathbb{F}_p)$, torsion of order prime to $p$ injecting under good reduction (Chapter 10).

**Exercise 14.84.** Let $\varphi\colon X_0(37) \to X_0(37)/w_{37} = $ `37a1` be the optimal parametrisation (Example 14.36). (a) Show $\varphi \circ w_{37} = \varphi$ and deduce from Lemma 14.53 that $\varepsilon_{37}(f_{37a}) = +1$ and $\varphi(0) = O$. (b) Deduce from Proposition 14.70 that $c_E L(\texttt{37a1}, 1) \in \mathbb{Z}\Omega$. (c) The functional equation of $L(f,s)$ has sign $-\varepsilon_N(f)$ (Chapters 13 and 19); conclude $L(\texttt{37a1}, 1) = 0$. *Hint for (a).* $w_{37}(\infty) = 0$, and $\varphi \circ w_{37} = \varepsilon_{37}\varphi + \varphi(0)$ with $\varphi \circ w_{37} = \varphi$ forces $\varepsilon_{37} = 1$ and $\varphi(0) = O$ (compare the pullbacks of $\omega_E$).

**Exercise 14.85.** In the class `11a` every isogeny `11a1` $\to$ `11a2` is $\psi_5 \circ [m]$ for the cyclic 5-isogeny $\psi_5$ and an integer $m \neq 0$ (the curves have no complex multiplication), and likewise for `11a3`. (a) Show that the degrees of the modular parametrisations of `11a2` and of `11a3` are the numbers $5m^2$. (b) Show that a degree-$d$ isogeny $\psi\colon E \to E'$ satisfies $u_\psi^2 \operatorname{vol}(\Lambda_E) = d \operatorname{vol}(\Lambda_{E'})$, and use the volumes of Example 14.42 to find $u_{\psi_5} = \pm 1$ for `11a2` and $\pm 5$ for `11a3`. (c) Deduce $c_\varphi = \pm m$, respectively $\pm 5m$, for the parametrisation of degree $5m^2$ (the sign is not normalised for a non-optimal curve), in agreement with Theorem 14.40. *Hint for (b).* Integrate $\frac{i}{2}\psi^*(\omega_{E'} \wedge \bar{\omega}_{E'})$ over $E$.

**Exercise 14.86.** Let $E$ be optimal, $\ell$ a prime with $E[\ell]$ irreducible, and $\varphi'$ a modular parametrisation of minimal degree of a curve $E'$ of the class. Show that $v_\ell(c_{\varphi'}) = v_\ell(c_E)$. *Hint.* $c_{\varphi'} = c_E u_\psi$ with $\psi$ of degree prime to $\ell$ (Proposition 14.32), and $u_\psi u_{\hat{\psi}} = \deg\psi$ with both factors integers.

**Exercise 14.87.** The class `77c` consists of the optimal curve `77c1` $= [1,1,0,4,11]$ and `77c2` $= [1,1,0,-51,110]$, linked by a cyclic 2-isogeny $\psi\colon$ `77c1` $\to$ `77c2`. (a) Compute $\deg\varphi_E$ for `77c1` from Theorem 14.40 with $c_E = 1$ (Theorem 14.39, 77 being square-free): take the newform of `77c` in PARI (the element of `mfeigenbasis(mfinit([77,2],0))` whose coefficients are those of `ellan`), compute `mfpetersson` of its symbol, multiply by $\mu(77) = 96$, and divide $4\pi^2$ times the result by $\operatorname{vol}(\Lambda_E) = |\operatorname{Im}(\bar{\omega}_1\omega_2)|$ for `E.omega`. (b) Compute $\operatorname{vol}(\Lambda_{\texttt{77c2}})$ and, with Exercise 14.85(b), find $u_\psi$; deduce the degree of a minimal parametrisation of `77c2` and its $c_\varphi$. What does `ellmoddegree` return for `77c2`? *Answer.* (a) $\langle f, f\rangle = 0.3610248954\ldots$, vol $= 2.3754485978\ldots$, and $4\pi^2\langle f,f\rangle/\operatorname{vol} = 6.0000\ldots$, so $\deg\varphi_E = 6$. (b) $\operatorname{vol}(\Lambda_{\texttt{77c2}}) = 1.1877242989\ldots$, half of the former, so $u_\psi^2 = 2 \cdot \frac{1}{2} = 1$ and $u_\psi = \pm 1$; the minimal parametrisation $\psi \circ \varphi_E$ has degree $2 \cdot 6 = 12$ (Theorem 14.30(ii)) and $c_\varphi = c_E u_\psi = \pm 1$ (Proposition 14.38); `ellmoddegree` returns $\deg\varphi / c_\varphi^2 = 12$.

**Exercise 14.88.** The optimal curve `88a1` $= [0,0,0,-4,4]$ has $\deg\varphi_E = 8$, and the algorithm of Example 14.50 gives $r_E = 16$. At which prime do the two differ, and why does this not contradict Theorem 14.48? Show that $H^1(X_0(88), \mathbb{Z}_2)_{\mathfrak{m}}$ is not free over $\mathbb{T}_{\mathfrak{m}}$ for $\mathfrak{m} = \mathfrak{m}_{f,2}$. *Answer.* At 2, with $2^2 \mid 88$; freeness would give equality in Corollary 14.55.

**Exercise 14.89.** Let $x = [(0) - (\infty)] \in J_0(N)$. (a) For $N = 11$, $T_2$ acts on $J_0(11)$ as $a_2 = -2$; deduce $5x = 0$ from $T_2 x = 3x$. (b) For $N = 37$, the newforms have $(a_2, a_3) = (-2,-3)$ and $(0,1)$. Show $(T_2+2)T_2 = 0$ and $(T_3+3)(T_3-1) = 0$ on $J_0(37)$, and deduce $15x = 21x = 0$, so $3x = 0$. Compare with Theorem 14.67. *Hint.* $\mathbb{T}$ acts faithfully on $J_0(N)$ (Proposition 14.17).

**Exercise 14.90.** Let $N = p$ be prime. Show $(U_p)_*(\infty) = p(\infty)$ and $(U_p)_*(0) = (0) + (p-1)(\infty)$, so that $(U_p - 1)_* x = 0$ for $x = [(0) - (\infty)]$. Using $U_p = -w_p$ on $S_2(\Gamma_0(p))$ (all forms are new,

and $a_p = -\varepsilon_p$ for a newform of prime level, Chapter 13), relate this to the generator $1 + w_p$ of Mazur's Eisenstein ideal. *Hint.* The representatives are $\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)$, $0 \le j < p$, and $j/p \sim \infty$ for $1 \le j < p$ when $N = p$.

**Exercise 14.91.** Let $\varphi\colon X_0(37) \to E = \texttt{37a1}$ be the quotient by $w_{37}$. Show that $\varphi^* H^1(E,\mathbb{Z})$ is the $(+1)$-eigenlattice of $w_{37}^*$ on $H^1(X_0(37),\mathbb{Z})$, that it equals $H^1(X_0(37),\mathbb{Z})[I_f]$ for $f = f_{37a}$, and that the cup product on it has ideal of values $2\mathbb{Z}$ and Gram determinant 4. *Hint.* $\varphi \circ w_{37} = \varphi$; the eigenlattice is saturated of rank 2; use Example 14.36 and Proposition 14.31(iv).

**Exercise 14.92.** With $E = \texttt{37a1}$, $A = \varphi_E^*(E)$ and $B = \ker \varphi_{E*}$ in $J_0(37)$: show that $B$ is an elliptic curve on which $\mathbb{T}$ acts through the eigencharacter of $f_{37b}$, that $A \cap B \simeq (\mathbb{Z}/2)^2$, and that addition $A \times B \to J_0(37)$ is an isogeny with kernel $\simeq (\mathbb{Z}/2)^2$. Deduce that $B$ lies in the class $\texttt{37b}$. *Hint.* Proposition 14.31(iii); the Lie algebra of $B$ is the annihilator of $\mathbb{C}f_{37a}$; Theorem 14.24(ii) on $T_\ell B$.

**Exercise 14.93.** At level 37 show that $\mathbb{T} = \mathbb{Z}[T_2] \simeq \mathbb{Z}[x]/(x(x+2))$: check $a_n(f_{37a}) \equiv a_n(f_{37b}) \pmod 2$ for $n \le 6$ (Sturm's bound) to get $\mathbb{T} \subseteq \{(a,b) \in \mathbb{Z}^2 : a \equiv b \bmod 2\}$ under $t \mapsto (\phi_{37a}(t), \phi_{37b}(t))$, and show that $\mathbb{Z}[T_2]$ is that whole ring. Deduce $\eta_{f_{37a}} = 2\mathbb{Z}$ and $r_{\texttt{37a1}} = 2 = \deg \varphi_E$. *Hint.* $T_2 \mapsto (-2, 0)$; $\mathrm{Ann}(\ker \phi_{37a}) = \mathbb{Z}T_2$.

**Exercise 14.94** (harder)**.** Let $\pi$ be the forgetful map $X_1(N) \to X_0(N)$, of degree $\#(\mathbb{Z}/N)^\times/\{\pm 1\}$ for $N \ge 3$. (a) Show $\langle \pi^* x, \pi^* y\rangle_{X_1(N)} = \deg \pi \, \langle x, y\rangle_{X_0(N)}$, and that $\pi^* H^1(X_0(N),\mathbb{Z})$ is saturated iff $\mathcal{S}_N = 0$. (b) For $N = 11$, both curves have genus 1, $\deg \pi = 5$ and $\#\mathcal{S}_{11} = 5$ (Theorem 14.67); show that the cup product on the saturation of $\pi^* H^1(X_0(11),\mathbb{Z})$ is unimodular, although on $\pi^* H^1(X_0(11),\mathbb{Z})$ it has values $5\mathbb{Z}$. (c) Relate this to point (1) of Remark 14.57: with ordinary cohomology the $G$-invariants of $H^1$ of a covering can be larger than the pull-back (here by the characters of order 5 of the paper's Remark after Proposition prop:integral), the ideal of values on the larger lattice is not the ideal of values on $X_0(N)$, and the difference is governed by the Shimura subgroup, which is Eisenstein. Explain why the integral structure of [49], built from compactly supported cohomology of a full-level cover, has no such excess. *Hint.* Chapter 8 (the proposition on the degree and the cup product, parts (i), (iii), (iv)) and Proposition 14.69(i).

## 14.14 Notes and references

*Modular curves over $\mathbb{Q}$ and Hecke correspondences.* The $\mathbb{Q}$-structure of $X_0(N)$, the Hecke correspondences and the congruence relation are in Shimura's book [180] (Chapters 6 and 7; we cite Sections 6.7, 7.2, 7.4 and 7.5 and Theorem 7.14 from memory, as unverified pointers), in Diamond and Shurman [52] (Sections 3.8, 5.2, 6.1 to 6.6, 7.7, 7.9, 8.2 to 8.7, again unverified pointers), and in the survey of Diamond and Im [51] (Sections 8 to 10). The integral models are Deligne and Rapoport's [43] and Katz and Mazur's [91]. The cusps over $\mathbb{Q}$ are in Ogg [146] and Stein's book [187]. The rational points of $Y_0(N)$ are Mazur's [126] and Kenku's [92, 94, 93, 95], with Ligozat [114] for $N = 27$. The identity $T_n^t = w_N T_n w_N$ (Theorem 14.11) is classical; the paper cites [49, §§1.6.1, 1.8.2] and [38, §4.4] for it; the proof here is the matrix computation $W_N \,\mathrm{diag}(1,p)\, W_N^{-1} = \mathrm{diag}(p,1)$ combined with Chapter 8's theorem that the adjoint of a correspondence is its transpose. Exercise 14.82 shows that $w_N \mathbb{T} w_N$ need not be $\mathbb{T}$; the book therefore treats $t \mapsto t^t$ as an isomorphism onto another ring of operators, not as an involution of $\mathbb{T}$.

*Optimal curves and the modular degree.* The factorisation Theorem 14.30 and Cremona's conventions are in [35, Chapter II]; Stevens [188] discusses optimal curves for $X_1(N)$ as well. The proof of Theorem 14.30(i) given here uses only the congruence relation, modularity and strong multiplicity one. Cremona's `ecdata` files [36] were used for every curve-level number in the chapter (conductor below 10,000), and every modular degree printed was recomputed with PARI's `ellmoddegree` [148].

*The Manin constant and Zagier-Watkins.* Zagier [209] computes $\deg \varphi_E$ from modular symbols and compares it with congruence primes; Watkins [204] computes it from the symmetric square $L$-value; Rankin and Shimura [155, 183] give $\langle f, f\rangle$ (Chapter 19). The proof of Theorem 14.40 given here is direct, so the normalisations of [209, 204] were not compared with the book's. The Manin constant: Manin's conjecture; Edixhoven [60]; Mazur [126]; Abbes and Ullmo [1]; Agashe, Ribet and Stein [2]; Česnavičius [27]; Česnavičius, Neururer and Saha [28]. The statements in Theorem 14.39(iii), (iv) are recalled, not checked against these texts.

*Congruence numbers.* Ribet [159] introduced congruence primes and proved $\deg \varphi_E \mid r_E$ in an early form; Agashe, Ribet and Stein [3] prove Theorem 14.48 (Theorem 2.2 and Table 1 as the paper cites them; numbers not verified) and discuss the *abc* connection (Section 2, with [71, 122]; see also [140]). The congruence numbers of Example 14.50 were computed for this book by the algorithm described there; they agree with the paper's values for `99a1`, `726a1` and `539d1`.

*The congruence ideal of Diamond, Flach and Guo.* Proposition 14.52, Lemma 14.53, and Theorem 14.54 are the book's version of the paper's Proposition prop:pullback; the argument is the projection formula (compare [209] and [3], whose arguments we have not compared in detail). The objects of [49] (Sections 1.5.3, 1.6.1, 1.6.2, 1.7.3, equations (13), (14), as the paper cites them; checked against the published text in Chapter 20) are compared with $H^1(X_0(N), \mathbb{Z}_\ell)$ in Chapter 20, which proves the paper's Proposition prop:integral; Remark 14.57 lists what that comparison involves. Lemma lem:ddt417 is [38, Lemma 4.17], inequality half.

*Multiplicity one, Gorenstein, Eisenstein.* Mazur [125] (Chapter II) for prime level and the Eisenstein ideal; Ribet [156] (Section 5, unverified) for multiplicity one in general; Ribet and Stein [163] (Theorem 3.5 as the paper cites it) for $J_1(N)$; Tilouine [199], Wiles [207, Chapter 2] and [38, Section 4] for the Gorenstein property; Ling and Oesterlé [115] and Ribet [161] for the Shimura subgroup. The proof of Theorem 14.62 from multiplicity one is Chapter 3's freeness criterion.

*Modularity, Shimura curves, the prime* 2. Modularity: [207, 197, 16], with Carayol [21] for the level. Shimura curves and the degree comparison: [156, 164, 192, 151]; Theorem 14.75 was checked against [164] (Theorems 1 and 2, Propositions 1 to 3) and [192] (Theorems 2.3, 2.4, 2.7, 3.2, Corollaries 3.4, 3.5, Table I); Pasten's summary (Section 1 of [151]) is a secondary pointer that we have not compared with them. Watkins's conjecture [204] and the results of [57, 20, 208, 24, 81, 12]; Theorem 14.78 is quoted from the paper and the arXiv version of [20]. The data checks of Conjecture 14.77 and Theorem 14.78 use all 38,042 optimal curves of conductor below 10,000.

# Chapter 15

# Galois representations of modular forms, level lowering and congruences

Section 3 of the paper proves Theorem thm:qual by a chain of four moves: attach a Galois representation to a newform, lower the level of its reduction modulo $\ell$, twist, and turn a congruence between eigenforms into a divisibility of the congruence number. Section 4 of the paper reads the same mechanism quantitatively: a depleted form is a form whose level has been raised, the change of the congruence ideal is the level-raising Euler factor, and the twist map transports congruences. This chapter supplies all of it. It constructs the representation $\rho_{f,\lambda}$ of a newform of weight 2 and compares it with $\rho_{E,\ell}$; it describes the restriction of $\rho_{f,\lambda}$ to a decomposition group at every prime (local-global compatibility, the local types, the conductor of a twist); it states Serre's conjecture and the theorems of Mazur, Ribet, Carayol and Diamond that lower the level, and Ribet's theorem that raises it; and it proves in full the paper's Lemma lem:crit, Lemma lem:old and Theorem thm:qual, with the paper's Example. The algebra is Chapter 3 and Chapter 4, the local Galois theory Chapter 7, everything specific to $\rho_{E,\ell}$ Chapter 11, the $q$-expansion side of Hecke operators Chapter 13, and the Jacobian $J_0(N)$, modularity and the modular degree Chapter 14. The deep imported theorems (Deligne, Carayol, Khare and Wintenberger, Ribet, Diamond, Diamond, Flach and Guo) are stated precisely, with the idea of the proof and a reference.

It helps to have the whole argument of Theorem thm:qual in view before the details. Start from $E$ of conductor $N$ and its newform $f = f_E$, and fix an odd prime $\ell \nmid N$. The reduction $\bar\rho = \bar\rho_{E,\ell}$ is a two-dimensional representation over $\mathbb{F}_\ell$ that remembers $a_p(f)$ mod $\ell$ for almost all $p$, but not the level $N$; its own "level" is the Serre conductor $N(\bar\rho)$, which divides $N$ and can be strictly smaller. The chain has six links:

(1) reduce: from $f$ to $\bar\rho$, or to its twist $\bar\rho \otimes \chi_{q^*}$ (Section 15.2);

(2) lower the level: find a newform $g$ of level $N(\bar\rho)$, resp. $N(\bar\rho \otimes \chi_{q^*})$, smaller than $N$, with the same reduction (Section 15.6, using the Serre conductor of Section 15.5 and the local description of $\rho_{f,\lambda}$ in Sections 15.3 and 15.4);

(3) twist back: $h = g$, resp. $h = g^{\chi_{q^*}}$, a newform $h \neq f$ of level dividing $N$ (the levels of twists, Section 15.3);

(4) choose an eigenform $g'$ in the span $V_h$ of the $h(dz)$ that is congruent to $f$ at *every* coefficient (the oldform lemma, Section 15.8);

(5) conclude $\ell \mid r_E$ (the congruence criterion, Section 15.8);

(6) conclude $\ell \mid \deg \varphi_{E_0}$ (Agashe, Ribet and Stein).

Each link loses or gains a specific piece of information (the level, the coefficients at $p \mid N\ell$, the sign of a twist), and most of the chapter is about keeping track of exactly that piece.

## Core results at a glance

Throughout, $f$ is a normalised newform of weight 2 on $\Gamma_0(N)$ with coefficient field $K_f$, $\lambda \mid \ell$ a prime of $K_f$, $\mathbb{T}$ the full Hecke algebra of level $N$, and $\phi_f \colon \mathbb{T} \to \mathcal{O}_f$ the eigencharacter; $\mathrm{Frob}_p$ is an arithmetic Frobenius; $q$ denotes a prime except in $q$-expansions.

**15.1 Conventions.** $\rho_{f,\lambda}$ is normalised by $\det(X - \rho_{f,\lambda}(\mathrm{Frob}_p)) = X^2 - a_p X + p$ $(p \nmid N\ell)$, $\det \rho_{f,\lambda} = \chi_\ell$; $H^1$ realises the dual. $\chi_{q^*}$ is the quadratic character of conductor $q$; $\chi_\ell$ is cyclotomic. *(convention)* Convention 15.1.

**15.2 Twists.** For $\chi$ quadratic of conductor $r$: the naive twist $g \otimes \chi = \sum \chi(n) a_n(g) q^n$, the newform twist $g^\chi$ $(\rho_{g^\chi} \simeq \rho_g \otimes \chi)$, the depletion $h^\Sigma$; $g \otimes \chi = (g^\chi)^{\Sigma(r)}$, and $g \otimes \chi = g^\chi$ iff $a_q(g^\chi) = 0$ for all $q \mid r$. *(convention)* Convention 15.2.

**15.3 Newforms.** $a_{mn} = a_m a_n$ for $(m,n) = 1$, $a_{p^{k+1}} = a_p a_{p^k} - p a_{p^{k-1}}$ $(p \nmid N)$, $a_{p^k} = a_p^k$ $(p \mid N)$; $a_p \in \{\pm 1\}$ if $p \,\|\, N$, $a_p = 0$ if $p^2 \mid N$; strong multiplicity one; $S_2(\Gamma_0(N)) = \bigoplus_h V_h$ with $V_h = \langle h(dz) : d \mid N/M \rangle$ $\mathbb{T}$-stable and mutually orthogonal. *(stated; proved in Chapter 13)* Proposition 15.4.

**15.4 Eichler-Shimura.** $J_f = J_0(N)/I_f J_0(N)$ has dimension $[K_f : \mathbb{Q}]$ and good reduction outside $N$; $V_\ell(J_f)$ is free of rank 2 over $K_f \otimes \mathbb{Q}_\ell$, and $\rho_{f,\lambda}$ on $V_\lambda(f)$ is continuous, unramified outside $N\ell$, with $\det(X - \rho_{f,\lambda}(\mathrm{Frob}_p)) = X^2 - a_p(f)X + p$ for $p \nmid N\ell$, $\det = \chi_\ell$, odd; $H^1_{\mathrm{et}}(X_0(N))[I_f]$ realises $V_\lambda(f)^\vee \simeq V_\lambda(f)(-1)$. *(proof sketched here; Shimura, Deligne)* Theorem 15.6.

**15.5 Irreducibility.** $\rho_{f,\lambda}$ is absolutely irreducible. *(proof sketched here; Ribet)* Proposition 15.7.

**15.6 Deligne.** A newform of weight $k \geq 2$ and character $\varepsilon$ has $\rho_{f,\lambda}$ unramified outside $N\ell$ with $\det(X - \rho(\mathrm{Frob}_p)) = X^2 - a_p X + \varepsilon(p) p^{k-1}$, $\det = \varepsilon \chi_\ell^{k-1}$, odd. *(stated; reference)* Theorem 15.8.

**15.7 Lattices.** Stable lattices exist; $\bar\rho^{\mathrm{ss}}$ is independent of the lattice; for $\ell$ odd $\bar\rho$ is irreducible iff absolutely irreducible; if $\bar\rho$ is irreducible all stable lattices are homothetic. *(proved here, given Chapters 4 and 5)* Proposition 15.9.

**15.8 Congruent newforms.** $a_p(f) \equiv a_p(h) \pmod{\lambda}$ on a density-one set gives $\bar\rho_f^{\mathrm{ss}} \simeq \bar\rho_h^{\mathrm{ss}}$, and $\bar\rho_h \simeq \bar\rho_f$ if $\ell$ is odd and $\bar\rho_f$ irreducible; conversely; and equality on a density-one set gives $f = h$. *(proved here, given Chapter 5)* Corollary 15.10.

**15.9 $\rho_{f_E,\ell}$ and $\rho_{E,\ell}$.** $J_f = E_0$ and $\rho_{f,\ell} \simeq \rho_{E,\ell} \otimes \mathbb{Q}_\ell$. The lattice $M_{f,\lambda} = H^1(X_0(N), \mathbb{Z}_\ell)[\ker \phi_f]$ equals $\varphi_{E_0}^* H^1(E_0, \mathbb{Z}_\ell)$, which is isomorphic to $T_\ell E_0(-1)$, without hypotheses; it is homothetic to $(T_\ell E)^\vee$ when $\bar\rho$ is irreducible; $\mathrm{ad}^0 M_{f,\lambda} \simeq \mathrm{ad}^0 T_\ell E_0$. *(proved here, given Chapter 14)* Theorem 15.12.

**15.10 The representation $\bar\rho_{\mathfrak m}$.** For a maximal ideal $\mathfrak m$ of $\mathbb T$ (equivalently of $\mathbb T_\ell$) of residue characteristic $\ell$: a normalised eigenform of all of $\mathbb T$ with eigencharacter $\theta\colon \mathbb T\to\mathcal O$ and $\ker(\mathbb T\to\mathcal O/\lambda)=\mathfrak m$ exists (Deligne-Serre lifting), lying in $V_h$ for a newform $h$ of level $M\mid N$; there is a semisimple continuous $\bar\rho_{\mathfrak m}\colon G_{\mathbb Q}\to \mathrm{GL}_2(\mathbb T/\mathfrak m)$, unramified outside $N\ell$, with $\det(X-\bar\rho_{\mathfrak m}(\mathrm{Frob}_p))=X^2-(T_p \bmod \mathfrak m)X+p$ ($p\nmid N\ell$), $\det\bar\rho_{\mathfrak m}=\bar\chi_\ell$, unique up to isomorphism (for $\ell$ odd the traces suffice), with $\bar\rho_{\mathfrak m}\otimes k_\lambda\simeq\bar\rho^{\mathrm{ss}}_{h,\lambda}$; for $\mathfrak m=\mathfrak m_{f,\lambda}$, $\bar\rho_{\mathfrak m}\otimes k_\lambda\simeq\bar\rho^{\mathrm{ss}}_{f,\lambda}$, and $\bar\rho_{\mathfrak m_{f_E,\ell}}\simeq\bar\rho^{\mathrm{ss}}_{E,\ell}$. *(proof sketched here; Deligne and Serre, Carayol)* Proposition 15.14.

**15.11 Eisenstein.** $\mathfrak m$ is Eisenstein if $\bar\rho_{\mathfrak m}$ is reducible over $\mathbb T/\mathfrak m$; for $\ell$ odd, equivalently absolutely reducible; $\mathfrak m_{f_E,\ell}$ is Eisenstein iff $E[\ell]$ is reducible. *(definition)* Definition 15.15.

**15.12 Local Langlands for $\mathrm{GL}_2(\mathbb Q_p)$.** A bijection $\pi\mapsto\mathrm{rec}(\pi)$ to two-dimensional Frobenius-semisimple Weil-Deligne representations, compatible with twists, preserving conductors and matching principal series, twists of Steinberg and supercuspidals with sums of characters, $\mathrm{sp}(2)\otimes\mu$ and irreducible representations. *(stated; reference)* Theorem 15.17.

**15.13 Local components.** $\Pi_f=\bigotimes'\Pi_v$; $\Pi_p$ has trivial central character and conductor exponent $v_p(N)$. *(definition; facts stated with references)* Definition 15.18.

**15.14 Carayol.** For $p\neq\ell$: $\mathrm{WD}_p(f)$ corresponds to $\Pi_p$ and is independent of $\ell$; $a(\rho_{f,\lambda}|_{G_{\mathbb Q_p}})=v_p(N)$, so the conductor of $\rho_{f,\lambda}$ is $N$ ($\ell\nmid N$); $L_p(f,s)=L(V_\lambda(f)|_{G_{\mathbb Q_p}},s-1)$. *(stated; reference)* Theorem 15.19.

**15.15 Local types.** Exactly one of (U), (St), (RSt), (RPS), (SC) holds at $p\neq\ell$; $\delta_p=2,1,0$ for $p\nmid N$, $p\,\|\,N$, $p^2\mid N$; $a_p=0$ and $V^{I_p}=0$ if $p^2\mid N$; in case (St), $\rho|_{G_{\mathbb Q_p}}\simeq\left(\begin{smallmatrix}\varepsilon\chi_\ell & *\\ 0 & \varepsilon\end{smallmatrix}\right)$ on every stable lattice with $\varepsilon(\mathrm{Frob}_p)=a_p=\pm1$, the reduction is non-split with a unique stable line when ramified, and $\bar\rho(\mathrm{Frob}_p)$ has eigenvalues $a_pp$, $a_p$ when unramified. *(proved here, given Theorem 15.19)* Theorem 15.20.

**15.16 Kodaira types.** For $f=f_E$ at $q\geq5$: type $\mathrm{I}_n$ gives (St), $\mathrm{I}_n^*$ gives (RSt), $\mathrm{I}_0^*$ gives (RPS); types II, II$^*$, IV, IV$^*$ give (RPS) or (SC) as $q\equiv1$ or $2\pmod3$, and III, III$^*$ give (RPS) or (SC) as $q\equiv1$ or $3\pmod4$. *(proved here, given Chapter 11)* Proposition 15.22.

**15.17 $\delta_p$ and the naive factor.** For $p\mid N$: $a_p=\pm1$ iff $\delta_p=1$, $a_p=0$ iff $\delta_p=0$; the naive factor equals the true one iff $\delta_p>0$; the true factor is twist-invariant, the naive one is not. *(proved here)* Corollary 15.25.

**15.18 Conductor of a quadratic twist.** For $q$ odd and $\chi$ ramified quadratic: (U)$\mapsto2$, (St)$\mapsto2$, (RSt)$\mapsto1$, (RPS) with $\mu_i|_I=\chi|_I\mapsto0$, tame (RPS)/(SC) of inertial order $3,4,6\mapsto2$. *(proved here, given Chapter 7)* Proposition 15.26.

**15.19 Atkin-Li (Lemma lem:twist(iii)).** If $q$ is odd and $q^2\nmid M$, then $g\otimes\chi_{q^*}=g^{\chi_{q^*}}$ is a newform of level $\mathrm{lcm}(M,q^2)$, with $a_q=0$. *(proved here, given Theorem 15.19)* Corollary 15.27.

**15.20 Twists of newforms.** $g^\chi$ exists and is unique, of level $M\prod_{q\mid r}q^{a(\mathrm{WD}_q\otimes\chi)-a(\mathrm{WD}_q)}$ dividing $\mathrm{lcm}(M,r^2)$; $a_n(g^\chi)=\chi(n)a_n(g)$ for $(n,r)=1$; $g\otimes\chi=(g^\chi)^{\Sigma(r)}$; $(g^\chi)^\chi=g$; $f_{E^{(d)}}=(f_E)^{\chi_d}$. *(proved here, given Theorem 15.19, Chapter 13)* Proposition 15.29.

**15.21 At $\ell\nmid N$.** $J_f[\ell^k]$ and its subquotients are finite flat, so $\bar\rho|_{D_\ell}$ is finite flat; $\rho|_{G_{\mathbb Q_\ell}}$ is crystalline with Hodge-Tate weights $0,1$ and crystalline Frobenius polynomial $X^2-a_\ell X+\ell$; Fontaine-Laffaille range for $\ell\geq3$; Serre weight 2 (for $\bar\rho$ irreducible). *(proof sketched here; references)* Theorem 15.31.

**15.22 Deligne, Fontaine.** For $\ell$ odd and $\ell \nmid N$: if $a_\ell \not\equiv 0$, $\bar\rho|_{D_\ell} \simeq \begin{pmatrix} \bar\chi_\ell \mu(\bar a_\ell^{-1}) & * \\ 0 & \mu(\bar a_\ell) \end{pmatrix}$; if $a_\ell \equiv 0$, $\bar\rho|_{I_\ell} \simeq \omega_2 \oplus \omega_2^\ell$ and $\bar\rho|_{D_\ell}$ is irreducible *(stated; reference)*; hence $a_\ell \bmod \lambda$ is 0 or the Frobenius eigenvalue on the unique unramified quotient, and is determined by $\bar\rho|_{D_\ell}$ *(proved here)*. Theorem 15.32.

**15.23 Serre's invariants.** Conductor $N(\bar\sigma) = \prod_{p\neq\ell} p^{a_p(\bar\sigma)}$, weight $k(\bar\sigma)$, character $\varepsilon(\bar\sigma)$; "arises from" and "modular". *(definition)* Definition 15.35.

**15.24** $N(\bar\rho_f) \mid N$. $a_p(\bar\rho^{\mathrm{ss}}) \le a_p(\bar\rho) = v_p(N) - (\dim \bar\rho^{I_p} - \delta_p) \le v_p(N)$ for $p \neq \ell$. *(proved here, given Theorem 15.19)* Proposition 15.36.

**15.25 Serre's conjecture (Khare, Wintenberger).** Every odd absolutely irreducible $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ arises from a newform of weight $k(\bar\sigma)$, level $N(\bar\sigma)$ and character lifting $\varepsilon(\bar\sigma)$. *(stated; reference)* Theorem 15.38.

**15.26 Mazur's principle.** $\ell$ odd, $\ell \nmid N$, $\bar\rho$ irreducible, $q \,\|\, N$, $\bar\rho$ unramified at $q$, $q \not\equiv 1 \pmod\ell$: $\bar\rho$ arises from level $N/q$. *(stated; idea of proof)* Theorem 15.39.

**15.27 Ribet.** $\ell$ odd, $\ell \nmid N$, $\bar\rho$ irreducible, $q \,\|\, N$, $\bar\rho$ unramified at $q$: $\bar\rho$ arises from level $N/q$. *(stated; idea of proof)* Theorem 15.40.

**15.28 The optimal level (DDT Theorem 3.15).** $\ell$ odd, $\bar\sigma$ irreducible and modular, absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ if $\ell = 3$: $\bar\sigma$ arises in weight 2, level $N(\bar\sigma)\ell^{\delta(\bar\sigma)}$, character of order prime to $\ell$, $\delta \in \{0,1\}$, $\delta = 0$ if $\bar\sigma|_{D_\ell}$ is finite flat. *(stated; reference)* Theorem 15.42.

**15.29 For elliptic curves.** $\ell$ odd, $\ell \nmid N'$, $\bar\rho_{E',\ell}$ irreducible, image not $N_s(\mathbb{F}_3)$ if $\ell = 3$: $\bar\rho_{E',\ell}$ arises from a newform on $\Gamma_0(N(\bar\rho))$, trivial character, and $N(\bar\rho) \mid N'$. *(proved here, given Theorem 15.42)* Corollary 15.43.

**15.30 Level raising (Ribet).** For $\ell$ odd, $\bar\rho = \bar\rho_{f,\lambda}$ irreducible and $q \nmid N\ell$: $\bar\rho$ arises from a form new at $q$ of level dividing $Nq$ iff $a_q^2 \equiv (q+1)^2 \pmod\lambda$; then $a_q(f) \equiv (q+1)a_q(g)$. *(necessity proved here; sufficiency stated, reference)* Theorem 15.47.

**15.31 The level-raising factor.** $(1 - \frac{\alpha}{\beta p})(1 - \frac{1}{p})(1 - \frac{\beta}{\alpha p}) = \frac{p-1}{p} \cdot \frac{(p+1)^2 - a_p^2}{p^2} = L_p(A_f, 1)^{-1}$ at good $p$; $1 - p^{-2}$ at $p \,\|\, N$. *(proved here)* Proposition 15.48.

**15.32 Theorem thm:dfg14.** For $f$ a newform on $\Gamma_0(M)$ with rational coefficients, $\ell \nmid 2M$, $\bar\rho_f$ irreducible, $\Sigma$ finite with $\ell \notin \Sigma$: $\eta_{f,\lambda}^{\Sigma} = \eta_{f,\lambda}^{\emptyset} \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}$. *(stated; proof idea in Chapter 20)* Theorem 15.49.

**15.33 Ihara's lemma.** For $q \nmid N\ell$ and $\mathfrak{m}$ non-Eisenstein (Definition 15.15), $H^1(X_0(N), \mathbb{F}_\ell)^2_{\mathfrak{m}} \to H^1(X_0(Nq), \mathbb{F}_\ell)$, $(x,y) \mapsto \alpha_1^* x + \alpha_q^* y$, is injective. *(stated; reference)* Lemma 15.51.

**15.34 The congruence number.** $r_f$ is the largest $r$ with $f \equiv g \pmod r$, $g \in f^\perp \cap S_2(\Gamma_0(N); \mathbb{Z})$. *(definition)* Definition 15.53.

**15.35 Agashe, Ribet, Stein.** $\deg \varphi_{E_0} \mid r$, with equal $p$-adic valuations for $p^2 \nmid N$; $r(\texttt{99a1}) = 12$, $\deg\varphi = 4$. *(stated; reference; data computed here)* Theorem 15.54.

**15.36 Lemma lem:crit.** $I = \ker \phi_f$, $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$: $I \cap J = 0$, $J = \mathrm{Ann}_{\mathbb{T}} I$, $I = \mathrm{Ann}_{\mathbb{T}} J$, $\mathbb{T}/(I+J) \simeq \mathbb{Z}/r$, $r\mathbb{Z} = \phi_f(J)$, $[S_2(\mathbb{Z}) : \mathbb{Z}f \oplus L] = r$, and the congruence moduli are the divisors of $r$; an eigenform $g \in W \otimes \overline{\mathbb{Q}}$ ($W = L \otimes \mathbb{Q}$, the rational forms in $f^\perp$) of all of $\mathbb{T}$ with $a_n(g) \equiv a_n(f) \pmod{\lambda}$ for every $n$ forces $\ell \mid r$; the same over $\mathbb{Z}_\ell$. *(proved here, given Chapter 13)* Lemma 15.55.

**15.37 $U_p$ on oldforms.** On $\langle h(p^j z)\rangle$: eigenvalues $\alpha_p, \beta_p$ (and 0 if $e_p \geq 2$) for $p \nmid M$, $a_p(h)$ and 0 for $p \parallel M$, 0 for $p^2 \mid M$, with explicit eigenvectors; any choice gives a normalised eigenform of all of $\mathbb{T}$ in $V_h$. *(proved here)* Lemma 15.57.

**15.38 Lemma lem:old.** $\ell$ odd, $\ell \nmid N$, $\bar\rho_{E,\ell}$ irreducible; a newform $h \neq f$ of level $M \mid N$ with $a_p(h) \equiv a_p(f)$ for $p \nmid \ell N$, and $p \nmid M$ at every $p \parallel N$ where $\bar\rho$ is unramified: $V_h$ contains an eigenform of all of $\mathbb{T}$ congruent to $f$ at every $n$, so $\ell \mid r$. *(proved here, given Theorems 15.20 and 15.32)* Lemma 15.58.

**15.39 Theorem thm:qual.** $\ell$ odd, $\ell \nmid N$, $\bar\rho_{E,\ell}$ irreducible (and not $N_s(\mathbb{F}_3)$ if $\ell = 3$), and at $q \neq \ell$ type (a) $\mathrm{IV}, \mathrm{IV}^*$, $q \geq 5$ or $q = 2$, $\ell = 3$; (b) $\mathrm{II}, \mathrm{II}^*$, $q \geq 5$, $\ell = 3$; or (c) $\mathrm{I}_n^*$, $\ell \mid n$, $q$ odd: then $\ell \mid \deg \varphi_{E_0}$. *(proved here, given the stated theorems Theorems 15.19, 15.32, 15.42 and 15.54)* Theorem 15.61.

**15.40 The twist map.** $\iota_S \colon h \mapsto h \otimes \chi_S$ twists generalised eigensystems by $\chi_S$, maps $g^\perp$ into $f^\perp$ when $f = g^{\chi_S}$ has level $N$ with $\mathrm{lcm}(M, D^2) \mid N$ ($M$ the level of $g$), and $\iota(g) = (g^{\chi_S})^S = \iota(g^S)$. *(proved here, given Chapter 13)* Proposition 15.68.

**15.41 Levels of $E_S$ and $g^S$.** For $S$ a set of odd primes $q$ with $q^2 \mid N$: $N_S = N/D$, $N/D^2$, $N$ for types $\mathrm{I}_n^*$, $\mathrm{I}_0^*$, potentially good (with $q \geq 5$) at every $q \in S$; $\iota(f_{E_S}) = f_E$; $g^S$ lies at level $N_S \prod q^{\delta_q} = N$. *(proved here)* Proposition 15.69.

**15.42 Theorem thm:twist.** If $q^2 \mid N$ for $q \in S$ and $\mathrm{lcm}(N_S, D^2) \mid N$, then $r_{E_S} \mid r_E$, with equality if also $\mathrm{lcm}(N, D^2) \mid N_S$. *(stated; proved in Chapter 20)* Theorem 15.71.

**15.43 Theorem thm:quant.** $\ell \nmid 2N$, $E[\ell]$ irreducible, $S$ the odd $\mathrm{I}_0^*$ and $\mathrm{I}_n^*$ primes: $v_\ell(\deg \varphi_E) \geq v_\ell(r_{E_S}) + \sum_{q \in S} e_\ell(E, q)$, with equality if $\ell \notin S_f$. *(stated; proved in Chapter 20)* Theorem 15.73.

**15.44 Tamagawa exponent.** $t_q$ is the largest $t$ with $\rho \bmod \lambda^t$ unramified at $q$. *(definition)* Definition 15.75.

**15.45 $t_q$ for $f_E$.** At $q \parallel N$, $t_q = v_\ell(n_q) = t_\ell(E, q)$. *(proved in Chapter 11; recalled)* Proposition 15.76.

**15.46 Pollack and Weston.** $\mathrm{ord}_\lambda \eta_f(aq, b) = t_q + \mathrm{ord}_\lambda \eta_f(a, qb)$ for $f$ of weight 2 and square-free level $N = aqb$ prime to $\ell$, if $\bar\rho$ is surjective, ramified at two primes or more and at every $r \mid qb$ with $r \equiv \pm 1 \pmod{\ell}$ (their hypothesis CR). *(stated; reference)* Theorem 15.78.

**15.47 Kim and Ota.** $\mathrm{ord}_\lambda \eta_f(N) = \mathrm{ord}_\lambda \eta_f(N^+, N^-) + \sum_{q \mid N^-} t_f(q)$ under their hypotheses *(stated; reference)*; hence $v_\ell(r_E) \geq \sum_{q \mid N^-} t_\ell(E, q)$ *(proved here)*. Theorem 15.79.

## Where the paper uses this chapter

- **Section 1.1** ($V$, $T$, $\rho$; $\delta_p = 2, 1, 0$; naive and true factors): Theorems 15.12 and 15.20 and Corollary 15.25.
- **Section 1.5** (history; Theorem thm:qual; Theorem thm:quant; Kim and Ota; "by level lowering and twisting rather than by the Taylor-Wiles method"): Theorems 15.61, 15.73, 15.78 and 15.79, Remarks 15.77 and 15.81.
- **Lemma lem:twist(iii)** (Atkin and Li): Corollary 15.27, from Proposition 15.26 and Theorem 15.19.
- **Lemma lem:euler** ("The local component of $f$ is an unramified twist of the Steinberg representation"; "$V = V' \otimes \chi_{q^*}$ with $V'$ Steinberg"): Proposition 15.22 and Theorem 15.20.
- **Section 3, Lemma lem:crit** (congruence criterion): Lemma 15.55, with Definition 15.53 and Theorem 15.54.
- **Section 3, Lemma lem:old** (oldforms): Lemmas 15.57 and 15.58, using Corollary 15.10 ($p \nmid \ell N$), Theorem 15.32 ($p = \ell$), Theorem 15.20 ($p^2 \mid N$; $p \,\|\, N$ via Theorem 3.1(e) of [38]).
- **Section 3, the optimal-level paragraph**: Theorem 15.42 and Corollary 15.43, Theorem 15.31 ("good"), Proposition 15.36 ($N(\bar\rho) \mid N$), Definition 15.35.
- **Section 3, cases (a), (b), (c)**: Theorem 15.61 and its proof (Section 15.9.1), with Theorem 15.40 in case (c) and Proposition 15.29.
- **Section 3, the Example and the remark after it** (`52a1`, `121a1`, `539c1`, `980i1`; 99A1): Examples 15.45 and 15.65 and Remark 15.67.
- **Section 4.1** ($M^\Sigma$, $f^\Sigma$) and **Theorem thm:dfg14** with the paragraph after it (Wiles's level-raising factor): Theorems 15.47 and 15.49 and Proposition 15.48.
- **Section 4.1, Proposition prop:pullback and the paragraph after it** (the lattice $M_{f,\lambda}$ of DFG04 is $H^1(E, \mathbb{Z}_\ell)$; $A = \mathrm{ad}^0 T$): Theorem 15.12.
- **Theorem thm:twist** and the data paragraph after it: Propositions 15.68 and 15.69 and Theorem 15.71.
- **Theorem thm:quant**, its proof (Steps 1 and 3) and Remark rem:afterquant: Proposition 15.69(iii), Lemma 15.57, Theorem 15.73, Remark 15.74.
- **Remark rem:ko(2)**: Theorem 15.79, Proposition 15.76.
- **Theorem thm:dfg37** (the hypothesis $\ell \notin S_f$): Remark 15.81.

## 15.1 Conventions and recollections

This section fixes the notation for newforms, Hecke algebras, Frobenius elements, characters and twists, and recalls from Chapter 13 the facts about newforms that every later proof uses. The one convention that matters for the paper is the distinction between the naive twist $f \otimes \chi$ of a

$q$-expansion and the newform twist $f^\chi$: the paper writes $f \otimes \chi$ for the naive twist, and its proof of Theorem thm:qual passes between the two either where they agree or by strong multiplicity one.

**Convention 15.1** (newforms, Hecke algebras, Frobenius, characters)**.** (a) *Newforms.* Unless said otherwise, $f = \sum_{n\geq 1} a_n(f) q^n$ is a normalised newform of weight 2 on $\Gamma_0(N)$ (trivial character); here $q = e^{2\pi i z}$, and a prime is written $q$ only when no $q$-expansion is in the same formula. $K_f = \mathbb{Q}(a_n(f) : n \geq 1)$ is its coefficient field, a totally real number field (Chapter 13), $\mathcal{O}_f$ its ring of integers, $\lambda \mid \ell$ a prime of $\mathcal{O}_f$, or of the ring of integers $\mathcal{O}$ of a finite extension of $K_f$ ("after enlarging $\mathcal{O}$"), $K_{f,\lambda}$ and $\mathcal{O}_\lambda$ the completions, $k_\lambda = \mathcal{O}_\lambda/\lambda$ the residue field (Chapter 6). For an elliptic curve $E/\mathbb{Q}$ of conductor $N$, $f_E$ is its newform (modularity, Chapter 14), and then $K_f = \mathbb{Q}$, $\lambda = \ell$.

(b) *Hecke algebras.* $\mathbb{T} = \mathbb{T}_N$ is the *full* Hecke algebra of level $N$, the subring of $\mathrm{End}_{\mathbb{C}} S_2(\Gamma_0(N))$ generated by all $T_n$, $n \geq 1$, the operators $U_p = T_p$ for $p \mid N$ included; $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$. The *eigencharacter* of $f$ is $\phi_f \colon \mathbb{T} \to \mathcal{O}_f$, $T_n \mapsto a_n(f)$ (upright $\phi$; a curly $\varphi$ is always a map of curves, such as the modular parametrisation $\varphi_E$). $J_f = J_0(N)/I_f J_0(N)$, $I_f = \ker \phi_f$, is Shimura's abelian variety attached to $f$; the letter $A_f$ is reserved for the paper's adjoint $\mathrm{ad}^0 V$.

(c) *Frobenius.* $\mathrm{Frob}_p$ is an *arithmetic* Frobenius at $p$ and $\mathrm{Frob}_p^{-1}$ the geometric one. $\chi_\ell(\mathrm{Frob}_p) = p$ for $p \neq \ell$. Local $L$-factors use the geometric Frobenius: $L(M, s) = \det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid M^{I_p})^{-1}$ for a representation $M$ of $G_{\mathbb{Q}_p}$ over a field of characteristic 0, $p \neq \ell$ (Chapter 7).

(d) *Normalisation of $\rho_{f,\lambda}$.* The representation $\rho_{f,\lambda}$ on $V_\lambda(f)$ is normalised by $\det(X - \rho_{f,\lambda}(\mathrm{Frob}_p)) = X^2 - a_p(f)X + p$ for $p \nmid N\ell$ and $\det \rho_{f,\lambda} = \chi_\ell$, so that $\rho_{f_E,\ell} \simeq \rho_{E,\ell}$ on $V_\ell E$ (Theorem 15.12). The cohomological realisation $H^1(X_0(N), \cdot)$ gives the dual $V_\lambda(f)^\vee \simeq V_\lambda(f)(-1)$, on which the *geometric* Frobenius at $p \nmid N\ell$ has trace $a_p(f)$.

(e) *Characters.* A Dirichlet character $\chi$ is also the Galois character with $\chi(\mathrm{Frob}_p) = \chi(p)$ for $p \nmid \mathrm{cond}\, \chi$. $\chi_\ell$ is the $\ell$-adic cyclotomic character and $\bar{\chi}_\ell$ its reduction; a letter subscript on $\chi$ always means a cyclotomic character. $\chi_d$ is the quadratic character of $\mathbb{Q}(\sqrt{d})$ for a fundamental discriminant $d$; for an odd prime $q$, $q^* = (-1)^{(q-1)/2} q$ and $\chi_{q^*}$ is the quadratic character of conductor $q$ (so $\chi_{-3}$, $\chi_{-7}$, $\chi_{-11}$ appear below, never $\chi_3$, $\chi_7$, $\chi_{11}$). For a finite set $S$ of odd primes, $\chi_S = \prod_{q\in S} \chi_{q^*}$, of conductor $D = \prod_{q\in S} q$.

(f) *Local components.* For a prime $p \neq \ell$, $\mathrm{WD}_p(f)$ is the Frobenius-semisimplification of the Weil-Deligne representation of $\rho_{f,\lambda}|_{G_{\mathbb{Q}_p}}$ (Chapter 7); in that context $N$ also denotes the monodromy operator, and the context always makes clear which $N$ is meant. $\delta_p = \dim V_\lambda(f)^{I_p}$.

(g) *Labels.* Curve labels are Cremona's: `11a1` is a curve, 11a an isogeny class, $f_{11a}$ its newform; the paper's (and ARS12's) 99A1 is `99a1`. "Optimal" means $X_0(N)$-optimal, and $E_0$ is the optimal curve in the class of $E$.

**Convention 15.2** (naive twists, newform twists and depletions)**.** Let $\chi$ be a quadratic Dirichlet character of conductor $r$, $\Sigma(r)$ the set of primes dividing $r$, and $g$ a newform of weight 2 with trivial character.

(a) The *naive twist* is the $q$-series $g \otimes \chi = \sum_n \chi(n) a_n(g) q^n$. This is the paper's $g \otimes \chi$ (Lemma lem:twist(iii), cases (b), (c) of the proof of Theorem thm:qual) and its twist map $\iota$ (Theorems thm:twist, thm:quant).

(b) The *newform twist* $g^\chi$ is the unique newform of weight 2 with trivial character such that $\rho_{g^\chi,\lambda} \simeq \rho_{g,\lambda} \otimes \chi$, equivalently $a_p(g^\chi) = \chi(p) a_p(g)$ for almost all $p$ (Proposition 15.29).

(c) For a set $\Sigma$ of primes the *depletion* of a $q$-series $h$ is $h^\Sigma = \sum_{(n,\Sigma)=1} a_n(h) q^n$ (the paper's $f^\Sigma$), and $h^{\{q\}}$ for one prime. A parenthesised superscript never means a depletion: $E^{(q)}$ is the twist of $E$ by $\mathbb{Q}(\sqrt{q^*})$.

Then (Proposition 15.29(ii))

$$g \otimes \chi = (g^\chi)^{\Sigma(r)}, \qquad g \otimes \chi = g^\chi \iff a_q(g^\chi) = 0 \text{ for every } q \mid r, \qquad (g \otimes \chi) \otimes \chi = g^{\Sigma(r)}.$$

The last identity holds because $\chi(n)^2 = 1$ for $(n, r) = 1$ and both sides vanish at $n$ with $(n, r) > 1$.

Where the choice of Frobenius matters, and where it does not. With the arithmetic Frobenius a Dirichlet character satisfies $\chi(\mathrm{Frob}_p) = \chi(p)$; with the geometric one it would be $\chi(p)^{-1}$. For the quadratic characters used in twisting, $\chi(p)^{-1} = \chi(p)$, so no statement about twists depends on the choice. For the cyclotomic character it does: $\chi_\ell(\mathrm{Frob}_p) = p$ but $\chi_\ell(\mathrm{Frob}_p^{-1}) = p^{-1}$. This is why $\rho_{f,\lambda}$ (arithmetic Frobenius, characteristic polynomial $X^2 - a_p X + p$) and its dual, the cohomological realisation (where the arithmetic Frobenius has characteristic polynomial $X^2 - a_p p^{-1} X + p^{-1}$ and the *geometric* one has $X^2 - a_p X + p$), carry the same information, and why every $L$-factor below is written with $\mathrm{Frob}_p^{-1}$.

To see the difference between the naive twist and the newform twist concretely, note that $g \otimes \chi$ has $a_n = 0$ whenever $(n, r) > 1$, because $\chi(n) = 0$ there. A newform of level divisible by $q^2$ has $a_q = 0$ anyway (Proposition 15.4(ii)), so for it nothing is lost; but a newform with $q \,\|\, \text{level}$ has $a_q = \pm 1$, and a $q$-series with $a_q = 0$ cannot be that newform. The example below shows both cases.

**Example 15.3** (the twist dictionary on two curves)**.** (a) The twist of `11a1` by $\mathbb{Q}(\sqrt{-11})$ is `121d2`, so $f_{11a}^{\chi_{-11}} = f_{121d}$, and $a_{11}(f_{121d}) = 0$ (type $\mathrm{I}_5^*$ at 11). Hence $f_{11a} \otimes \chi_{-11} = f_{121d}$ exactly, as the paper says in its Example. (b) The twist of `539c1` by $\mathbb{Q}(\sqrt{-7})$ is `77c1`, so $f_{539c}^{\chi_{-7}} = f_{77c}$. But $a_7(f_{77c}) = -1$ (type $\mathrm{I}_3$ at 7), so $f_{539c} \otimes \chi_{-7} = (f_{77c})^{\{7\}}$, the 7-depletion of $f_{77c}$, which is not a newform: it is an eigenform of level 539 for the full Hecke algebra there, with $U_7$-eigenvalue 0, in the span of $f_{77c}(z)$ and $f_{77c}(7z)$. Both statements were checked with PARI/GP on the coefficients $a_n$, $n \le 3000$ (for (b), $a_p(f_{539c})\chi_{-7}(p) = a_p(f_{77c})$ for all primes $p < 400$, $p \neq 7$).

**Proposition 15.4** (newforms; recollection)**.** *Let $f$ be as in Convention 15.1(a).*

*(i) $\phi_f(T_n) = a_n(f)$ for all $n$; $a_{mn} = a_m a_n$ for $(m, n) = 1$; $a_{p^{k+1}} = a_p a_{p^k} - p\, a_{p^{k-1}}$ for $p \nmid N$; and $a_{p^k} = a_p^k$ for $p \mid N$. In general $a_1(T_n g) = a_n(g)$ for every $g \in S_2(\Gamma_0(N))$, so an eigenform of all of $\mathbb{T}$ has $a_1 \neq 0$ and can be normalised.*

*(ii) $a_p(f) \in \{\pm 1\}$ if $p \,\|\, N$, and $a_p(f) = 0$ if $p^2 \mid N$ (Atkin and Lehner [8, Theorem 3]; reproved from the Galois side in Theorem 15.20).*

*(iii) (Strong multiplicity one.) Two newforms of weight 2 with trivial character, of any levels, with $a_p$ equal for all but finitely many $p$ are equal; in particular they have the same level. (Corollary 15.10(iv) relaxes "all but finitely many" to "a set of density one".)*

*(iv)* $S_2(\Gamma_0(N)) = \bigoplus_{M\mid N} \bigoplus_h V_h$, *where* $h$ *runs over the newforms of level* $M$ *and* $V_h$ *is the span of the* $h(dz)$, $d \mid N/M$, *which are linearly independent. Each* $V_h$ *is* $\mathbb{T}$*-stable,* $T_n$ *acts on* $V_h$ *by the scalar* $a_n(h)$ *for* $(n, N) = 1$, *and the* $V_h$ *are mutually orthogonal for the Petersson product. Consequently the generalised eigenspace in* $S_2(\Gamma_0(N))$ *of the operators* $T_p$, $p \nmid N$, *for the eigenvalues* $(a_p(h))_{p\nmid N}$ *is* $V_h$.

*(v)* $L(f, s) = \sum a_n(f) n^{-s} = \prod_{p\nmid N}(1 - a_p p^{-s} + p^{1-2s})^{-1} \prod_{p\mid N}(1 - a_p p^{-s})^{-1}$.

*Proof.* All of this is proved in Chapter 13 (Atkin-Lehner-Li theory, the Petersson product, the Hecke relations), except (ii) and (iii), which are quoted there from [8] and [52, Chapter 5]. The last sentence of (iv) follows from the rest of (iv) and from (iii): on $V_{h'}$ the $T_p$, $p \nmid N$, act by the scalars $a_p(h')$, and by (iii) these agree with the $a_p(h)$ for all $p \nmid N$ only if $h' = h$. □

*In the paper* 15.5. Lemma lem:old uses (iv) in its first sentence ("The forms $h(dz)$, $d \mid N/M$, span a $\mathbb{T}$-stable subspace $V$ of $S_2(\Gamma_0(N))$ orthogonal to the newform $f$"), and (i) and (ii) in its last paragraph and in the case $p^2 \mid N$. Lemma lem:crit uses (iii) and (iv) for the decomposition $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ (Lemma 15.55, Step 1), and Theorem thm:quant uses (iii) and (iv) in Step 1.

## 15.2 The Galois representation of a newform

This section attaches to a newform $f$ of weight 2 and a prime $\lambda$ of its coefficient field a two-dimensional $\lambda$-adic representation $\rho_{f,\lambda}$ of $G_{\mathbb{Q}}$, its lattices and its reduction $\bar{\rho}_{f,\lambda}$, and compares it with $\rho_{E,\ell}$ when $f = f_E$. The paper uses these objects everywhere in Section 3: the residual representation $\bar{\rho}_h$ of a newform $h$ in Lemma lem:old, the phrase "arises from a newform" in the optimal-level theorem, and "$\bar{\rho}_g \simeq \bar{\rho}$, that is, $a_p(g) \equiv a_p(f) \pmod{\lambda}$" in cases (a), (b), (c); and in Section 4.1 the lattice $M_{f,\lambda}$ of Diamond, Flach and Guo is compared with $T_\ell E$. Chapter 5 used the representations of newforms as a black box (its corollary on congruent traces, part (d)); Theorem 15.6 discharges it.

**Theorem 15.6** (Eichler, Shimura; Deligne and Rapoport)**.** *Let* $f$ *be a newform of weight* 2 *on* $\Gamma_0(N)$, $I_f = \ker \phi_f$, $J_f = J_0(N)/I_f J_0(N)$, $d = [K_f : \mathbb{Q}]$.

*(i)* $J_f$ *is an abelian variety over* $\mathbb{Q}$ *of dimension* $d$, *with good reduction at every* $p \nmid N$, *and* $\mathbb{T}/I_f$, *an order in* $K_f$, *acts on it by endomorphisms defined over* $\mathbb{Q}$.

*(ii) For every prime* $\ell$, $V_\ell(J_f) = T_\ell(J_f) \otimes \mathbb{Q}_\ell$ *is free of rank* 2 *over* $K_f \otimes \mathbb{Q}_\ell = \prod_{\lambda\mid\ell} K_{f,\lambda}$, *and* $G_{\mathbb{Q}}$ *acts* $K_f \otimes \mathbb{Q}_\ell$*-linearly. For* $\lambda \mid \ell$ *put* $V_\lambda(f) = V_\ell(J_f) \otimes_{K_f \otimes \mathbb{Q}_\ell} K_{f,\lambda}$, *a two-dimensional* $K_{f,\lambda}$*-space, and let* $\rho_{f,\lambda} \colon G_{\mathbb{Q}} \to \mathrm{GL}(V_\lambda(f)) \simeq \mathrm{GL}_2(K_{f,\lambda})$ *be the action. Then:*

  *(a)* $\rho_{f,\lambda}$ *is continuous and unramified at every* $p \nmid N\ell$;

  *(b) for* $p \nmid N\ell$, $\det(X - \rho_{f,\lambda}(\mathrm{Frob}_p)) = X^2 - a_p(f)X + p$;

  *(c)* $\det \rho_{f,\lambda} = \chi_\ell$;

  *(d)* $\rho_{f,\lambda}$ *is odd:* $\det \rho_{f,\lambda}(c) = -1$ *for every complex conjugation* $c$;

  *(e)* $\rho_{f,\lambda}$ *is absolutely irreducible (Proposition 15.7).*

*(iii) The* $I_f$*-torsion* $H^1_{\mathrm{et}}(X_0(N)_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)[I_f]$ *is free of rank* 2 *over* $K_f \otimes \mathbb{Q}_\ell$, *and its* $\lambda$*-component is* $V_\lambda(f)^\vee \simeq V_\lambda(f)(-1)$; *the geometric Frobenius at* $p \nmid N\ell$ *has trace* $a_p(f)$ *on it.*

*Sketch of proof.* (i) $J_0(N)$ has good reduction at every $p \nmid N$ (Igusa; Deligne and Rapoport [43]), hence so has its quotient $J_f$. The cotangent space of $J_f$ is the part of $S_2(\Gamma_0(N))$ killed by $I_f$, which is spanned by the $d$ Galois conjugates of $f$ (a newform occurs with multiplicity one, Proposition 15.4(iv)), so $\dim J_f = d$. These facts, and the action of $\mathbb{T}$ on $J_0(N)$ by endomorphisms over $\mathbb{Q}$, are in Chapter 14.

(ii) $H_1(J_f(\mathbb{C}), \mathbb{Q})$ is a vector space over the field $K_f = (\mathbb{T}/I_f) \otimes \mathbb{Q}$ of $\mathbb{Q}$-dimension $2d$, so it is free of rank 2 over $K_f$; tensoring with $\mathbb{Q}_\ell$ and using $V_\ell(J_f) \simeq H_1(J_f(\mathbb{C}), \mathbb{Q}) \otimes \mathbb{Q}_\ell$ ($\mathbb{T}$-equivariantly) gives the freeness, and the Galois action commutes with $\mathbb{T}$ because the Hecke correspondences are defined over $\mathbb{Q}$. Continuity is that of the action on Tate modules (Chapter 5). (a) For $p \nmid N\ell$, $J_f$ has good reduction at $p$, so inertia at $p$ acts trivially on $V_\ell(J_f)$ (the criterion of Néron, Ogg and Shafarevich for abelian varieties, due to Serre and Tate [179]; Chapter 11 proves the case of elliptic curves). (c) The Weil pairing attached to a polarisation of $J_f$ for which the Hecke operators are self-adjoint up to the Atkin-Lehner involution gives a $K_f \otimes \mathbb{Q}_\ell$-bilinear alternating perfect pairing $V_\ell(J_f) \times V_\ell(J_f) \to K_f \otimes \mathbb{Q}_\ell(1)$ (Shimura [180, Chapter 7]), so $\wedge^2_{K_{f,\lambda}} V_\lambda(f) \simeq K_{f,\lambda}(1)$ and $\det \rho_{f,\lambda} = \chi_\ell$. (b) Reduction modulo $p$ identifies $V_\ell(J_f)$ with $V_\ell((J_f)_{\mathbb{F}_p})$, $\mathrm{Frob}_p$ acting as the Frobenius endomorphism $\pi_p$. The Eichler-Shimura congruence relation $(T_p)_* = \pi_p + \hat{\pi}_p$ with $\pi_p \hat{\pi}_p = p$ (Chapter 14) gives $A^2 - a_p A + p = 0$ for $A = \rho_{f,\lambda}(\mathrm{Frob}_p)$, since $T_p$ acts on $V_\lambda(f)$ by $a_p = a_p(f)$. If $A$ is not a scalar, its characteristic polynomial is its minimal polynomial $X^2 - a_p X + p$. If $A = c$ is a scalar, then $c^2 = \det A = p$ by (c), and $c^2 - a_p c + p = 0$ gives $a_p = 2c = \mathrm{tr}\, A$; so again $\det(X - A) = X^2 - \mathrm{tr}\, A\, X + \det A = X^2 - a_p X + p$. (d) $\chi_\ell(c) = -1$, as $c$ inverts roots of unity. (iii) The étale cohomology $H^1_{\mathrm{et}}(X_0(N)_{\overline{\mathbb{Q}}}, \mathbb{Z}_\ell)$ is $\mathrm{Hom}(T_\ell J_0(N), \mathbb{Z}_\ell) \simeq T_\ell J_0(N)(-1)$, Hecke and Galois equivariantly (Chapter 8); its $I_f$-torsion is dual to the $I_f$-coinvariants $V_\ell(J_0(N))/I_f$, which is $V_\ell(J_f)$; and $V^\vee \simeq V \otimes \det^{-1} = V(-1)$ for two-dimensional $V$ with $\det = \chi_\ell$. An eigenvalue $\alpha$ of the arithmetic Frobenius on $V$ is the eigenvalue $\alpha$ of the geometric Frobenius on $V^\vee$, which gives the last claim. For details see Shimura [180, Chapter 7] (Theorem 7.14 in the edition we know; an unverified pointer), Diamond and Shurman [52, Chapter 9], Diamond and Im [51] and [38, Theorem 3.1]; the general case is Deligne's [44]. □

**Proposition 15.7** (irreducibility)**.** *For every newform $f$ of weight 2 with trivial character and every $\lambda$, $\rho_{f,\lambda}$ is absolutely irreducible.*

*Sketch of proof.* Suppose that $\rho_{f,\lambda} \otimes \overline{K}_{f,\lambda}$ has a stable line, with character $\psi_1$ and quotient character $\psi_2$; both are continuous characters $G_\mathbb{Q} \to \overline{\mathbb{Q}}_\ell^\times$ unramified outside $N\ell$, with $\psi_1\psi_2 = \chi_\ell$. They are subquotients of $V_\ell(J_f)$, which is Hodge-Tate with weights 0 and 1 (Faltings; Tate for abelian varieties with good reduction), so they are Hodge-Tate, and an abelian Hodge-Tate character of $G_\mathbb{Q}$ is of the form $\theta\chi_\ell^k$ with $\theta$ of finite order and $k \in \mathbb{Z}$ (Serre [178]). Write $\psi_i = \theta_i \chi_\ell^{k_i}$. The Hodge-Tate weight of $\psi_i$ is $k_i$ up to the sign convention, and the Hodge-Tate weights of a subquotient are among those of $V_\ell(J_f)$, so $k_i \in \{0, 1\}$; and $\psi_1\psi_2 = \chi_\ell$ gives $\theta_1\theta_2\chi_\ell^{k_1+k_2} = \chi_\ell$, so $k_1 + k_2 = 1$ ($\chi_\ell$ has infinite order, $\theta_1\theta_2$ finite order). Relabelling, $\{\psi_1, \psi_2\} = \{\theta_1, \theta_2\chi_\ell\}$. Evaluating the trace at $\mathrm{Frob}_p$ gives $a_p(f) = \theta_1(p) + \theta_2(p)p$ for all $p \nmid N\ell$ outside the conductors of the $\theta_i$, an identity between algebraic numbers. The $\theta_i(p)$ are roots of unity, so under every complex embedding $|a_p(f)| \geq |\theta_2(p)p| - |\theta_1(p)| = p - 1$. Now $p - 1 > 2\sqrt{p}$ is $(\sqrt{p} - 1)^2 > 2$, that is $\sqrt{p} > 1 + \sqrt{2}$, $p > 3 + 2\sqrt{2} \approx 5.83$, so it holds for every prime $p \geq 7$, and there are infinitely many admissible $p$. This contradicts the bound $|a_p(f)| \leq 2\sqrt{p}$ of Eichler, Shimura and Weil (Chapter 12). For the general case (any weight and character) see Ribet [158]. □

**Theorem 15.8** (Deligne; Deligne and Serre)**.** *Let $f$ be a normalised newform of weight $k \geq 2$, level $N$ and character $\varepsilon$, and $\lambda \mid \ell$ a prime of $K_f$. There is a continuous irreducible representation $\rho_{f,\lambda}\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(K_{f,\lambda})$, unramified outside $N\ell$, with $\det(X - \rho_{f,\lambda}(\mathrm{Frob}_p)) = X^2 - a_p(f)X + \varepsilon(p)p^{k-1}$ for $p \nmid N\ell$ and $\det \rho_{f,\lambda} = \varepsilon\chi_\ell^{k-1}$; it is odd. In weight 1 Deligne and Serre attach an Artin representation with the same properties ($k = 1$).*

We do not prove this; see Deligne [44] and [52, Chapter 9]. Only weight 2 and trivial character occur in the paper's proofs; the optimal-level theorem (Theorem 15.42) produces a priori a newform with a character, and the case of characters is needed only to see that this character is trivial (Corollary 15.43).

**Proposition 15.9** (lattices and reduction)**.** *Let $f$ be a newform of weight 2 with trivial character and $\lambda \mid \ell$.*

*(i) $\rho_{f,\lambda}$ stabilises an $\mathcal{O}_\lambda$-lattice $L \subset V_\lambda(f)$. The reduction $\bar\rho_L\colon G_{\mathbb{Q}} \to \mathrm{GL}(L/\lambda L) \simeq \mathrm{GL}_2(k_\lambda)$ has a semisimplification $\bar\rho_{f,\lambda}^{\mathrm{ss}}$ independent of $L$, and $\operatorname{tr}\bar\rho_L(\mathrm{Frob}_p) \equiv a_p(f)$, $\det\bar\rho_L(\mathrm{Frob}_p) \equiv p \pmod\lambda$ for $p \nmid N\ell$.*

*(ii) $\bar\rho_L$ is odd, and for $\ell$ odd it is irreducible over $k_\lambda$ if and only if it is absolutely irreducible.*

*(iii) If $\bar\rho_L$ is irreducible, every $G_{\mathbb{Q}}$-stable $\mathcal{O}_\lambda$-lattice in $V_\lambda(f)$ is homothetic to $L$, so $\bar\rho_{f,\lambda} := \bar\rho_L$ is well defined up to isomorphism.*

*Proof.* (i) $G_{\mathbb{Q}}$ is compact and acts continuously, so the $\mathcal{O}_\lambda$-span of the orbit of any lattice is a stable lattice (Chapter 5, the proposition on continuous representations). The independence of the semisimplification is the theorem of Brauer and Nesbitt in the form of Chapter 4: two stable lattices give reductions with the same characteristic polynomials at every element. The congruences follow from Theorem 15.6(ii)(b). (ii) $\bar\rho_L(c)$ has order dividing 2 and determinant $-1$, so for $\ell$ odd its eigenvalues are 1 and $-1$, which are distinct in $k_\lambda$. A line over $\bar k_\lambda$ stable under $\bar\rho_L(G_{\mathbb{Q}})$ is stable under $\bar\rho_L(c)$, hence is one of the two eigenlines of $\bar\rho_L(c)$, which are defined over $k_\lambda$; it is then a $k_\lambda$-rational stable line. (iii) This is the homothety lemma of Chapter 4. In detail: let $L'$ be another stable lattice and $\varpi$ a uniformiser of $\mathcal{O}_\lambda$. Since $L'$ is finitely generated and $L$ spans $V$, $\varpi^a L' \subseteq L$ for $a$ large, and $\varpi^a L' \not\subseteq \varpi L$ for $a$ small; take the largest $a$ with $\varpi^a L' \subseteq L$ and replace $L'$ by $\varpi^a L'$, so that $L' \subseteq L$, $L' \not\subseteq \lambda L$. The image of $L'$ in $L/\lambda L$ is a $G_{\mathbb{Q}}$-stable subspace (as $L'$ is stable) and nonzero, hence all of $L/\lambda L$ by irreducibility. So $L = L' + \lambda L$, and Nakayama's lemma for the finitely generated $\mathcal{O}_\lambda$-module $L/L'$ (which satisfies $\lambda(L/L') = L/L'$) gives $L = L'$. Hence every stable lattice is $\varpi^{-a}L$ for some $a$, and the reductions of $L$ and $\varpi^{-a}L$ are isomorphic via multiplication by $\varpi^{-a}$.

Without irreducibility (iii) fails, and the semisimplification in (i) is the most one can say: for $f = f_{11a}$ and $\ell = 5$ the lattices $T_5(\texttt{11a1})$ and $T_5(\texttt{11a3})$ are not homothetic, and their reductions have the same semisimplification but are not isomorphic: one is unramified at 11 and the other is not (Exercise 15.85). □

**Corollary 15.10** (congruent newforms)**.** *Let $f$, $h$ be newforms of weight 2 with trivial character, of levels $N_f$, $N_h$, with coefficients in the ring of integers $\mathcal{O}$ of a number field $K$, and $\lambda \mid \ell$ a prime of $\mathcal{O}$; write $\bar\rho_f$, $\bar\rho_h$ for reductions over $k_\lambda$ of lattices in $\rho_{f,\lambda}$, $\rho_{h,\lambda}$ (formed with the completion of $K$ at $\lambda$).*

*(i) If $a_p(f) \equiv a_p(h) \pmod\lambda$ for all $p$ in a set of primes of density one, then $\bar\rho_f^{\mathrm{ss}} \simeq \bar\rho_h^{\mathrm{ss}}$.*

*(ii) If moreover $\ell$ is odd and $\bar\rho_f$ is irreducible, then $\bar\rho_h \simeq \bar\rho_f$ for every choice of lattices.*

*(iii) Conversely, if $\bar\rho_f^{\mathrm{ss}} \simeq \bar\rho_h^{\mathrm{ss}}$, then $a_p(f) \equiv a_p(h) \pmod\lambda$ for every $p \nmid N_f N_h \ell$.*

*(iv) If $a_p(f) = a_p(h)$ for all $p$ in a set of density one, then $f = h$.*

*Proof.* Determinants agree: both are $\chi_\ell$ (Theorem 15.6(ii)(c)). (i), (ii), (iii) are the corollary on congruent traces of Chapter 5, parts (a), (b), (c), applied to $\rho_1 = \rho_{f,\lambda}$, $\rho_2 = \rho_{h,\lambda}$ with $S$ the primes dividing $N_f N_h \ell$; in (ii), $\bar\rho_f$ is absolutely irreducible by Proposition 15.9(ii), which is what that corollary needs. (iv) By the same corollary in characteristic 0 (Chebotarev and Brauer-Nesbitt), the semisimple representations $\rho_{f,\lambda}$, $\rho_{h,\lambda}$ (irreducible, Proposition 15.7) are isomorphic, so $a_p(f) = a_p(h)$ for all $p \nmid N_f N_h \ell$, and strong multiplicity one (Proposition 15.4(iii)) gives $f = h$. □

*In the paper* 15.11. Lemma lem:old, case $p \nmid \ell N$: "Consequently $\bar\rho_h \simeq \bar\rho$ over a common finite extension of $\mathbb{F}_\ell$, by Chebotarev and Brauer-Nesbitt, $\bar\rho$ being irreducible, hence absolutely irreducible." This is Corollary 15.10(ii) with $f = f_E$ and $\bar\rho = \bar\rho_{E,\ell}$ (so $\bar\rho_f \simeq \bar\rho \otimes_{\mathbb{F}_\ell} k_\lambda$ by Theorem 15.12); the common finite extension is $k_\lambda$. The irreducibility of $\bar\rho$ alone would give only $\bar\rho_h^{\mathrm{ss}} \simeq \bar\rho^{\mathrm{ss}}$; absolute irreducibility (Proposition 15.9(ii)) is what identifies $\bar\rho_h$ itself. The proof of Theorem thm:qual uses (i) to (iii) in the form "$\bar\rho_g \simeq \bar\rho$, that is, $a_p(g) \equiv a_p(f) \pmod\lambda$ for all $p \nmid 3N$".

**Theorem 15.12** ($\rho_{f_E,\ell}$ and $\rho_{E,\ell}$)**.** *Let $E/\mathbb{Q}$ have conductor $N$, $f = f_E$, $E_0$ the optimal curve of the isogeny class, $\varphi_{E_0}\colon X_0(N) \to E_0$ the modular parametrisation, and $\ell$ a prime.*

*(i) $J_f = E_0$, so $T_\ell E_0$ is a $G_\mathbb{Q}$-stable $\mathbb{Z}_\ell$-lattice in $V_\ell(f)$ and $\rho_{E_0,\ell} = \rho_{f,\ell}$ on it. For every $E$ in the class, $V_\ell E \simeq V_\ell E_0$ through an isogeny, so $\rho_{E,\ell} \otimes \mathbb{Q}_\ell \simeq \rho_{f,\ell}$.*

*(ii) The lattice $M_{f,\lambda} := H^1(X_0(N), \mathbb{Z}_\ell)[\ker \phi_f]$ is equal to the image $\varphi_{E_0}^* H^1(E_0, \mathbb{Z}_\ell)$, and $\varphi_{E_0}^*$ is a $G_\mathbb{Q}$-isomorphism from $H^1(E_0, \mathbb{Z}_\ell) = \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell E_0, \mathbb{Z}_\ell)$, which is isomorphic to $T_\ell E_0(-1)$, onto it. No irreducibility is needed.*

*(iii) If $\bar\rho_{E,\ell}$ is irreducible, every $G_\mathbb{Q}$-stable lattice in $V_\ell(E)^\vee$, in particular $M_{f,\lambda}$ and $(T_\ell E)^\vee$ for every $E$ in the class, is homothetic to $(T_\ell E)^\vee \simeq T_\ell E(-1)$.*

*(iv) In all cases $\mathrm{ad}^0 M_{f,\lambda} \simeq \mathrm{ad}^0 T_\ell E_0$ as $\mathbb{Z}_\ell[G_\mathbb{Q}]$-modules, and $\mathrm{ad}^0 M_{f,\lambda} \simeq \mathrm{ad}^0 T_\ell E$ for every $E$ in the class when $\bar\rho_{E,\ell}$ is irreducible.*

*Proof.* (i) That $J_f$ is the optimal quotient $E_0$ of $J_0(N)$ is Chapter 14 (modularity and the definition of optimality: the kernel of $J_0(N) \to E_0$ is $I_f J_0(N)$, connected). Then $V_\ell(f) = V_\ell(J_f) = V_\ell(E_0)$ with $K_f = \mathbb{Q}$. An isogeny $E \to E_0$ over $\mathbb{Q}$ induces a $G_\mathbb{Q}$-isomorphism $V_\ell E \to V_\ell E_0$. (ii) This is proved in Chapter 14 (the proposition identifying $M_{f,\lambda}$ with the pull-back, which is the paper's Proposition prop:pullback). In outline: the kernel of $J_0(N) \to E_0$ is connected, so $\varphi_{E_0 *}\colon H_1(X_0(N), \mathbb{Z}) \to H_1(E_0, \mathbb{Z})$ is surjective and $\varphi_{E_0}^*$ is injective with torsion-free cokernel; $\mathbb{T}$ acts on its image through $\phi_f$; the image and $M_{f,\lambda}$ are saturated of the same rank 2 in the same plane (the $\phi_f$-eigenspace of $H^1(X_0(N), \mathbb{Q})$ is two-dimensional, as a newform occurs once); so they are equal. Galois equivariance comes from the étale comparison (Chapter 8). (iii) $V_\ell(E)^\vee$ reduces to $\bar\rho_{E,\ell}^\vee \simeq \bar\rho_{E,\ell} \otimes \bar\chi_\ell^{-1}$, irreducible; apply Proposition 15.9(iii) (in the form of Chapter 4 for any irreducible residual representation). (iv) $\mathrm{ad}^0(L^\vee) \simeq \mathrm{ad}^0 L$ by $x \mapsto -x^{\mathsf{T}}$, and $\mathrm{ad}^0(L(n)) = \mathrm{ad}^0 L$, $\mathrm{ad}^0(cL) = \mathrm{ad}^0 L$ for a scalar $c$ (Chapter 4); apply (ii), and (iii) with the isomorphism $T_\ell E \simeq T_\ell E_0$ given by an isogeny of degree prime to $\ell$ when $E[\ell]$ is irreducible (Chapter 11). □

*In the paper* 15.13. In Section 4.1 of the paper, the paragraph after Proposition prop:pullback reads: "The lattice $M_{f,\lambda}$ of [49] is therefore $H^1(E,\mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$, by Galois-equivariance of $\varphi^*$ in étale cohomology, and its adjoint is $\mathrm{ad}^0 T_\ell E$, the adjoint being insensitive to dualising and twisting." This is Theorem 15.12(ii) and (iv) for the optimal curve $E = E_0$. By (iii), $M_{f,\lambda}$ is homothetic to $T_\ell E$ itself only after replacing $T_\ell E$ by its dual $T_\ell E(-1)$, and only when $\bar\rho$ is irreducible. Only the adjoint enters the paper's arguments (Theorem thm:main, with $A = \mathrm{ad}^0 T$), and it is the same for every one of these lattices.

The Hecke-algebra side of the same objects is the residual representation of a maximal ideal of $\mathbb{T}$. Chapter 14 uses it (multiplicity one, the Gorenstein property, the word "Eisenstein"), and Ihara's lemma (Lemma 15.51) is stated for it. Its construction combines the lifting lemma of Deligne and Serre with the representations of newforms constructed above.

**Proposition 15.14** (the residual representation of a maximal ideal)**.** *Let $N \geq 1$, $\mathbb{T} = \mathbb{T}_N$, and let $\mathfrak{m}$ be a maximal ideal of $\mathbb{T}$ with residue characteristic $\ell$; $k_\mathfrak{m} = \mathbb{T}/\mathfrak{m}$ is a finite field. (The maximal ideals of $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$ are the ideals $\mathfrak{m}\mathbb{T}_\ell$ for these $\mathfrak{m}$, with the same residue fields, so everything below applies to $\mathbb{T}_\ell$ as well.)*

*(i) (Lifting.) There are the ring of integers $\mathcal{O}$ of a number field, a prime $\lambda \mid \ell$ of $\mathcal{O}$, a ring homomorphism $\theta\colon \mathbb{T} \to \mathcal{O}$ with $\ker(\mathbb{T} \to \mathcal{O} \to \mathcal{O}/\lambda) = \mathfrak{m}$, and a normalised eigenform $g \in S_2(\Gamma_0(N))$ of all of $\mathbb{T}$ with eigencharacter $\theta$, so $a_n(g) = \theta(T_n) \in \mathcal{O}$. The form $g$ lies in $V_h$ for a newform $h$ of weight 2 on $\Gamma_0(M)$, $M \mid N$, and $\theta(T_p) = a_p(h)$ for $p \nmid N$.*

*(ii) There is a continuous semisimple representation $\bar\rho_\mathfrak{m}\colon G_\mathbb{Q} \to \mathrm{GL}_2(k_\mathfrak{m})$, unramified outside $N\ell$, with*

$$\det(X - \bar\rho_\mathfrak{m}(\mathrm{Frob}_p)) = X^2 - (T_p \bmod \mathfrak{m})X + p \qquad (p \nmid N\ell),$$

*and $\det\bar\rho_\mathfrak{m} = \bar\chi_\ell$. Among semisimple representations it is determined up to isomorphism by these characteristic polynomials; for $\ell$ odd the traces $T_p \bmod \mathfrak{m}$ alone determine it, while for $\ell = 2$ the determinant must be imposed as well. For every $(h,\lambda)$ as in (i), $\bar\rho_\mathfrak{m} \otimes_{k_\mathfrak{m}} k_\lambda \simeq \bar\rho^{\mathrm{ss}}_{h,\lambda}$, where $k_\mathfrak{m} \hookrightarrow k_\lambda$ is induced by $\theta$.*

*(iii) For a newform $f$ of weight 2 on $\Gamma_0(N)$ and a prime $\lambda \mid \ell$ of $\mathcal{O}_f$, $\mathfrak{m}_{f,\lambda} = \ker(\mathbb{T} \to \mathcal{O}_f \to k_\lambda)$, $t \mapsto \phi_f(t) \bmod \lambda$, is a maximal ideal and $\bar\rho_{\mathfrak{m}_{f,\lambda}} \otimes k_\lambda \simeq \bar\rho^{\mathrm{ss}}_{f,\lambda}$. For $f = f_E$, $\mathbb{T}/\mathfrak{m}_{f,\ell} = \mathbb{F}_\ell$ and $\bar\rho_{\mathfrak{m}_{f,\ell}} \simeq \bar\rho^{\mathrm{ss}}_{E,\ell}$.*

*Sketch of proof.* (i) $\mathbb{T}$ is free of finite rank over $\mathbb{Z}$ (a subring of $\mathrm{End}_\mathbb{Z} S_2(\Gamma_0(N);\mathbb{Z})$), so $\mathbb{T}_\ell = \prod_{\mathfrak{m}'} \mathbb{T}_{\mathfrak{m}'}$ over the maximal ideals containing $\ell$ (Chapter 3), and $\mathbb{T}_\mathfrak{m}$ is a nonzero direct factor, free of finite rank over $\mathbb{Z}_\ell$. Hence $\mathbb{T}_\mathfrak{m} \otimes \mathbb{Q}_\ell$ is a nonzero finite-dimensional $\mathbb{Q}_\ell$-algebra and has a homomorphism $\theta_\ell\colon \mathbb{T}_\mathfrak{m} \to \overline{\mathbb{Q}}_\ell$. Its image is integral over $\mathbb{Z}_\ell$, so lies in the integers of $\overline{\mathbb{Q}}_\ell$, and composing with the reduction map gives a homomorphism $\mathbb{T}_\mathfrak{m} \to \overline{\mathbb{F}}_\ell$ of the local ring $\mathbb{T}_\mathfrak{m}$, whose kernel is therefore $\mathfrak{m}\mathbb{T}_\mathfrak{m}$; restricted to $\mathbb{T}$, the kernel is $\mathfrak{m}$. The values $\theta_\ell(T_n)$ are roots of the characteristic polynomials of the $T_n$ on $S_2(\Gamma_0(N);\mathbb{Z})$, so they are algebraic; fixing $\overline{\mathbb{Q}} \subset \overline{\mathbb{Q}}_\ell$, they generate a number field, and $\theta$ is $\theta_\ell$ with values in its ring of integers $\mathcal{O}$, $\lambda$ the prime given by the embedding. Next, $\mathbb{T} \otimes \overline{\mathbb{Q}}$ acts faithfully on $S_2(\Gamma_0(N);\mathbb{Z}) \otimes \overline{\mathbb{Q}}$, so the generalised eigenspace for the character $\theta \otimes \overline{\mathbb{Q}}$ is nonzero (as in Step 1 of the proof of Lemma 15.55), and commuting operators on a nonzero space over an algebraically closed field have a common eigenvector $g$. By Proposition 15.4(i), $a_1(g) \neq 0$; normalised, $a_n(g) = a_1(T_n g) = \theta(T_n)$. By Proposition 15.4(iv), $g$ lies in the generalised eigenspace $V_h$ of the $T_p$, $p \nmid N$, for the eigenvalues $\theta(T_p)$, where $h$ is the newform with $a_p(h) = \theta(T_p)$. This is the lifting lemma of Deligne and Serre [45, Lemme

6.11] (checked): for a free module of finite type $M$ over a discrete valuation ring $\mathcal{O}$ and a set of pairwise commuting endomorphisms with a common (nonzero) eigenvector in $M/\mathfrak{m}_{\mathcal{O}}M$, there are a discrete valuation ring $\mathcal{O}' \supseteq \mathcal{O}$ with fraction field finite over that of $\mathcal{O}$ and maximal ideal meeting $\mathcal{O}$ in $\mathfrak{m}_{\mathcal{O}}$, and a nonzero common eigenvector in $\mathcal{O}' \otimes_{\mathcal{O}} M$ whose eigenvalues reduce to the given ones; Deligne and Serre stress that the eigenvalues lift, not the eigenvectors. They apply it in their 6.10 to the Hecke operators on modular forms with integral coefficients.

(ii) Enlarge $\mathcal{O}$ to contain $K_h$ and let $\bar\rho = \bar\rho^{\mathrm{ss}}_{h,\lambda}$ over $k_\lambda$ (Proposition 15.9(i)); it is unramified outside $M\ell$, and $\det(X - \bar\rho(\mathrm{Frob}_p)) \equiv X^2 - \theta(T_p)X + p \pmod\lambda$ for $p \nmid N\ell$, a polynomial with coefficients in the image of $k_{\mathfrak{m}}$. The map $\sigma \mapsto \det(X - \bar\rho(\sigma))$ is continuous with values in a finite set, and the Frobenius elements at $p \nmid N\ell$ are dense in $G_{\mathbb{Q}}$ (Chebotarev, Chapter 5); so every characteristic polynomial of $\bar\rho$ has coefficients in $k_{\mathfrak{m}}$. It remains to descend $\bar\rho \otimes \overline{\mathbb{F}}_\ell$, which is semisimple, to $k_{\mathfrak{m}}$. Let $F$ generate $\mathrm{Gal}(\overline{\mathbb{F}}_\ell/k_{\mathfrak{m}})$. If $\bar\rho \otimes \overline{\mathbb{F}}_\ell \simeq \psi_1 \oplus \psi_2$ is a sum of characters, its conjugate by $F$ has the same characteristic polynomials, so $\{\psi_1^F, \psi_2^F\} = \{\psi_1, \psi_2\}$ (Brauer-Nesbitt, Chapter 4). If $F$ fixes both, the $\psi_i$ take values in $k_{\mathfrak{m}}$ and $\bar\rho_{\mathfrak{m}} = \psi_1 \oplus \psi_2$. If $F$ exchanges them, $\psi_1$ takes values in the quadratic extension $k'$ of $k_{\mathfrak{m}}$ and $\psi_2 = \psi_1^F$; let $\bar\rho_{\mathfrak{m}}$ be $k'$, viewed as a two-dimensional $k_{\mathfrak{m}}$-space, with $\sigma$ acting by multiplication by $\psi_1(\sigma)$. It is irreducible, and over $\overline{\mathbb{F}}_\ell$ the eigenvalues of $\sigma$ are $\psi_1(\sigma)$ and $\psi_1(\sigma)^F$, so $\bar\rho_{\mathfrak{m}} \otimes \overline{\mathbb{F}}_\ell \simeq \psi_1 \oplus \psi_2$. If $\bar\rho \otimes \overline{\mathbb{F}}_\ell$ is absolutely irreducible, a representation over $\overline{\mathbb{F}}_\ell$ whose characteristic polynomials have coefficients in the finite field $k_{\mathfrak{m}}$ is realised over $k_{\mathfrak{m}}$, because the Brauer group of a finite field is trivial (Wedderburn); see Carayol [23, Théorème 2, Section 1.2] (checked: a representation over a semi-local extension $A'$ of a henselian local ring $A$ whose residue field has trivial Brauer group, with traces in $A$ and absolutely irreducible residual representations, is realised over $A$; here $A = k_{\mathfrak{m}}$, $A'$ a finite extension of it). Both cases are also covered at once by [45, Lemme 6.13] (checked): a semisimple representation over a finite field $k'$ is realised over any subfield $k$ containing the coefficients of all its characteristic polynomials. In each case $\bar\rho_{\mathfrak{m}}$ is semisimple and continuous (it has the same kernel as $\bar\rho \otimes \overline{\mathbb{F}}_\ell$). Uniqueness and the isomorphism with $\bar\rho^{\mathrm{ss}}_{h,\lambda}$ over $k_\lambda$ are Brauer-Nesbitt again, the characteristic polynomials being known on the dense set of Frobenius elements, hence everywhere; for $\ell$ odd, $2 < \ell$ and equal traces suffice (Chapter 4). The determinant is $\bar\chi_\ell$ because both are continuous and agree on $\mathrm{Frob}_p$, $p \nmid N\ell$.

(iii) The image of $\mathbb{T}$ in $k_\lambda$ is a finite subring of a field, hence a field, so $\mathfrak{m}_{f,\lambda}$ is maximal; in (i) one may take $\theta = \phi_f$, $g = h = f$, and (ii) gives the isomorphism. For $f = f_E$, $\mathcal{O}_f = \mathbb{Z}$, $k_\lambda = \mathbb{F}_\ell = \mathbb{T}/\mathfrak{m}_{f,\ell}$, and $\bar\rho^{\mathrm{ss}}_{f,\ell} \simeq \bar\rho^{\mathrm{ss}}_{E,\ell}$ because $T_\ell E_0$ is a stable lattice (Theorem 15.12(i)) and the semisimplified reduction does not depend on the lattice (Proposition 15.9(i)), nor on the curve in the isogeny class. □

**Definition 15.15** (Eisenstein maximal ideals)**.** A maximal ideal $\mathfrak{m}$ of $\mathbb{T}$ (or of $\mathbb{T}_\ell$) of residue characteristic $\ell$ is *Eisenstein* if $\bar\rho_{\mathfrak{m}}$ (Proposition 15.14) is reducible over $\mathbb{T}/\mathfrak{m}$, and *non-Eisenstein* otherwise.

For $\ell$ odd, $\bar\rho_{\mathfrak{m}}$ is odd ($\det\bar\rho_{\mathfrak{m}}(c) = \bar\chi_\ell(c) = -1$), so it is reducible over $\mathbb{T}/\mathfrak{m}$ if and only if it is absolutely reducible (the criterion of Chapter 4 by an element with two distinct eigenvalues in the field, as in Proposition 15.9(ii)). For $\ell = 2$ the two notions differ (Exercise 15.82), and some authors call $\mathfrak{m}$ Eisenstein when $\bar\rho_{\mathfrak{m}}$ is absolutely reducible; this book does not. By Proposition 15.14(iii), for $E/\mathbb{Q}$ of conductor $N$ and every prime $\ell$, $\mathfrak{m}_{f_E,\ell}$ is Eisenstein if and only if $E[\ell]$ is reducible. The name comes from Mazur's Eisenstein ideal at prime level (Chapter 14): it contains $T_p - 1 - p$ for $p \nmid N$, so a maximal ideal $\mathfrak{m}$ containing it has $\det(X - \bar\rho_{\mathfrak{m}}(\mathrm{Frob}_p)) = (X-1)(X-p)$ for $p \nmid N\ell$, hence $\bar\rho_{\mathfrak{m}} \simeq \mathbf{1} \oplus \bar\chi_\ell$ by Chebotarev and Brauer-Nesbitt, and $\mathfrak{m}$ is Eisenstein.

The smallest example is $N = 11$, $\ell = 5$. The curve `11a1` has a rational point of order 5, so $E[5]$ contains the trivial character $\mathbb{Z}/5$ and, by the Weil pairing, has quotient $\mu_5$; hence $a_p(f_{11a}) = \operatorname{tr}\bar\rho(\mathrm{Frob}_p) \equiv 1 + p \pmod 5$ for $p \neq 5, 11$ (for instance $a_2 = -2 \equiv 3$, $a_3 = -1 \equiv 4$, $a_7 = -2 \equiv 8$), and $\mathfrak{m}_{f_{11a},5}$ is Eisenstein. The distinction matters because the good properties used later (multiplicity one, the Gorenstein property in Chapter 14, Ihara's lemma Lemma 15.51) are proved only at non-Eisenstein maximal ideals, and the paper's hypothesis "$E[\ell]$ irreducible" is exactly the condition that $\mathfrak{m}_{f_E,\ell}$ be non-Eisenstein.

## 15.3 Local-global compatibility and local components

This section describes $\rho_{f,\lambda}|_{G_{\mathbb{Q}_p}}$ for every prime $p \neq \ell$. The paper uses this description in four places: $\delta_p = \dim V^{I_p} = 2, 1, 0$ in Section 1.1; the phrases "The local component of $f$ is an unramified twist of the Steinberg representation" and "$V = V' \otimes \chi_{q^*}$ with $V'$ Steinberg at $q$" in the proof of Lemma lem:euler; the proof of Lemma lem:twist(iii) (Atkin and Li) by the conductor of a twisted Steinberg representation; and the shape of $\bar\rho|_{D_p}$ at a prime $p \,\|\, N$ in Lemma lem:old (Theorem 3.1(e) of [38]). Chapter 7 has done all the Weil-Deligne computations; this section adds the automorphic names, the theorem of Carayol that connects the two sides, and the consequences for the levels of twists.

*Remark* 15.16 (recollection from Chapter 7). Let $K = \mathbb{Q}_p$, $p \neq \ell$, and let $V$ be a continuous representation of $G_K$ on a finite-dimensional space over a finite extension of $\mathbb{Q}_\ell$. Then: $V$ has a Weil-Deligne representation $\mathrm{WD}(V) = (r, N)$, with $r(I_K)$ finite and $N$ nilpotent, $r(w)Nr(w)^{-1} = \|w\|N$, $\|\mathrm{Frob}\| = p$, and $V \mapsto \mathrm{WD}(V)$ is an equivalence compatible with sums, tensor products, duals and twists; $V^{I_K} = (\ker N)^{r(I_K)}$; $L(V,s) = \det(1 - \mathrm{Frob}^{-1}\, p^{-s} \mid V^{I_K})^{-1}$; the conductor exponent is $a(V) = \dim V - \dim V^{I_K} + \mathrm{sw}(V)$, additive, invariant under unramified twists, zero exactly when $V$ is unramified, equal to $\dim V - \dim V^{I_K}$ when $V$ is tame. The special representation $\mathrm{sp}(2)$ has basis $e_0, e_1$ with $Ne_0 = e_1$, $Ne_1 = 0$, $r(w)e_i = \|w\|^i e_i$; a two-dimensional $(r, N)$ with $N \neq 0$ is $\mathrm{sp}(2)\otimes\mu$ for a unique character $\mu$ (the character on $V/\ker N$); $a(\mathrm{sp}(2)\otimes\mu) = 1$ for $\mu$ unramified and $2a(\mu)$ for $\mu$ ramified; a two-dimensional Frobenius-semisimple $(r, 0)$ is a sum of two characters or is irreducible, and for $p$ odd an irreducible one is induced from a quadratic extension. The $\epsilon$-factors of Langlands and Deligne are defined there as well, and they give root numbers: for a newform $f$ of weight 2 on $\Gamma_0(N)$ the sign $-\varepsilon_N(f)$ of the functional equation of $L(f,s)$ (Chapter 13) is the global root number, the product of the local root numbers (normalised $\epsilon$-factors) of the $\mathrm{WD}_p(f)$, which are 1 for $p \nmid N$ and $-a_p(f)$ for $p \,\|\, N$, and of $-1$ at the infinite place (Deligne [41]). The paper does not use $\epsilon$-factors. On the Galois side, $\mathrm{sp}(2) \otimes \mu$ is the representation $\left(\begin{smallmatrix} \mu\chi_\ell & * \\ 0 & \mu \end{smallmatrix}\right)$ of $G_K$ (after identifying characters of $W_K$ and of $G_K$ with values in $\mathbb{Q}_\ell^\times$); on $I_K$ it is $\mu$ times $\left(\begin{smallmatrix} 1 & c\,t_\ell \\ 0 & 1 \end{smallmatrix}\right)$ with $c \neq 0$ and $t_\ell$ the tame character, and it is non-split.

**Theorem 15.17** (the local Langlands correspondence for $\mathrm{GL}_2(\mathbb{Q}_p)$)**.** *Fix an isomorphism $\overline{\mathbb{Q}}_\ell \simeq \mathbb{C}$. There is a bijection $\pi \mapsto \mathrm{rec}(\pi)$ from isomorphism classes of irreducible admissible infinite-dimensional representations of $\mathrm{GL}_2(\mathbb{Q}_p)$ to isomorphism classes of two-dimensional Frobenius-semisimple Weil-Deligne representations of $W_{\mathbb{Q}_p}$; in the unitary normalisation it is characterised by the compatibility with twists by characters (identified through local class field theory) and the preservation of $L$- and $\epsilon$-factors of pairs. It preserves conductors, $a(\mathrm{rec}\,\pi) = a(\pi)$, where $p^{a(\pi)}$ is the conductor of Casselman's newvector theory [25], and it matches the types: principal series $\pi(\mu_1, \mu_2)$ with sums of two characters, twists $\mathrm{St}\otimes\mu$ of the Steinberg representation with $\mathrm{sp}(2) \otimes \mu$, and supercuspidal representations with irreducible representations ($N = 0$); in*

*each case up to one fixed unramified twist, so that "$\mu$ unramified" has the same meaning on both sides.*

We do not prove this; see Bushnell and Henniart [19], and Rohrlich [166] for the Weil-Deligne side of elliptic curves. In the unitary normalisation rec preserves $L$- and $\epsilon$-factors; the other normalisation in common use differs from it by the twist by $|\det|^{1/2}$ (and preserves these factors only up to the corresponding shift of $s$), and some sources attach to a newform the dual of our $\rho_{f,\lambda}$. The book needs no choice: conductors and types are invariant under unramified twists and duality, and the $L$-factors it uses are stated directly on the Galois side (Theorem 15.19(iii)). The automorphic names below are labels and enter no proof.

**Definition 15.18** (local components)**.** A newform $f$ of weight 2 on $\Gamma_0(N)$ generates a cuspidal automorphic representation $\Pi_f = \bigotimes'_v \Pi_v$ of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$ (Jacquet and Langlands [89]). Its factor $\Pi_p$ is the *local component of $f$ at $p$*. It has trivial central character and conductor exponent $a(\Pi_p) = v_p(N)$ (Casselman [25]), and the eigenvalue $a_p(f)$ is determined by $\Pi_p$ (it is, after normalisation, the eigenvalue of the Hecke operator at $p$ on the newvector). We say that $f$ has *type* (U), (St), (RSt), (RPS) or (SC) at $p$ according to the rows of Theorem 15.20; Loeffler and Weinstein [118] explain how to compute $\Pi_p$ in practice.

**Theorem 15.19** (local-global compatibility at $p \neq \ell$; Langlands, Deligne, Carayol)**.** *Let $f$ be a newform of weight 2 on $\Gamma_0(N)$, $p$ a prime and $\lambda \mid \ell$ with $\ell \neq p$; let $V = V_\lambda(f)|_{G_{\mathbb{Q}_p}}$.*

*(i)* $\mathrm{WD}_p(f)$ *corresponds to* $\Pi_p$ *under Theorem 15.17 (in the normalisation of the source). In particular* $\mathrm{WD}_p(f)$*, viewed over* $\mathbb{C}$*, does not depend on $\ell \neq p$ or on $\lambda$, and its type is that of* $\Pi_p$*.*

*(ii) The conductor exponent of $V$ is $a(V) = v_p(N)$. Hence the conductor of $\rho_{f,\lambda}$ is $N$ when $\ell \nmid N$, and its prime-to-$\ell$ part is that of $N$ in general.*

*(iii)* $L_p(f,s) = L(V, s-1) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{1-s} \mid V^{I_p})^{-1}$*, where $L_p(f,s)$ is the Euler factor of Proposition 15.4(v). Equivalently: for $p \nmid N$ the arithmetic Frobenius has eigenvalues $\alpha$, $\beta$ on $V$ with $\alpha + \beta = a_p(f)$, $\alpha\beta = p$; for $p \mid N$, either $V^{I_p} = 0$ and $a_p(f) = 0$, or $V^{I_p}$ is a line on which* $\mathrm{Frob}_p$ *acts by $p/a_p(f)$ and $a_p(f) \neq 0$ (equivalently by $p\,a_p(f)$, as $a_p(f) = \pm 1$ by Theorem 15.20).*

We do not prove (i), (ii), or (iii) at $p \mid N$; see Carayol [21] (the theorem stated in its introduction), after Langlands [111] and Deligne [41], and [38, Theorem 3.1]; (iii) is the equality of $L$-factors under Theorem 15.17, read through (i). The two forms of (iii) are equivalent by a one-line computation: for $p \nmid N$, the geometric Frobenius has eigenvalues $\alpha^{-1}$, $\beta^{-1}$, and $(1 - \alpha^{-1}p \cdot p^{-s})(1 - \beta^{-1}p \cdot p^{-s}) = (1 - \beta p^{-s})(1 - \alpha p^{-s})$ because $p/\alpha = \beta$; for a line on which $\mathrm{Frob}_p$ acts by $c$, $1 - c^{-1}p \cdot p^{-s} = 1 - a_p p^{-s}$ exactly when $c = p/a_p$, which is $p\,a_p$ when $a_p = \pm 1$. For $p \nmid N\ell$, (iii) is Theorem 15.6(ii)(b). On $V^\vee$, the cohomological realisation, (iii) reads: $a_p(f)$ is the trace of the geometric Frobenius on $(V^\vee)^{I_p}$ when $\delta_p \neq 1$, and its eigenvalue on that line when $\delta_p = 1$.

**Theorem 15.20** (the local types of a newform of weight 2)**.** *Let $f$ be a newform of weight 2 on $\Gamma_0(N)$, $p \neq \ell$, $V = V_\lambda(f)|_{G_{\mathbb{Q}_p}}$, $\mathrm{WD}_p(f) = (r, N)$ and $\delta_p = \dim V^{I_p}$. Exactly one of the following holds.*

| *type* | $v_p(N)$ | $\mathrm{WD}_p(f)$ | $\Pi_p$ | $\delta_p$ | $a_p(f)$ |
|---|---|---|---|---|---|
| *(U)* | 0 | $\mu_1 \oplus \mu_2$ *unramified,* $\mu_1\mu_2 = \|\cdot\|$ | *unramified principal series* | 2 | $\mu_1(\mathrm{Frob}) + \mu_2(\mathrm{Frob})$ |
| *(St)* | 1 | $\mathrm{sp}(2)\otimes\mu$, $\mu$ *unramified,* $\mu^2 = 1$ | *unramified twist of* St | 1 | $\mu(\mathrm{Frob}_p) = \pm 1$ |
| *(RSt)* | $2a(\chi)$ | $\mathrm{sp}(2)\otimes\mu\chi$, $\mu$ *unramified,* $\chi$ *ramified,* $(\mu\chi)^2 = 1$ | *ramified twist of* St | 0 | 0 |
| *(RPS)* | $2a(\mu_1)$ | $\mu_1 \oplus \mu_2$, *both ramified,* $\mu_1\mu_2 = \|\cdot\|$ | *ramified principal series* | 0 | 0 |
| *(SC)* | $\geq 2$ | *irreducible,* $N = 0$ | *supercuspidal* | 0 | 0 |

*In particular* $a_p(f) = 0$ *and* $V^{I_p} = 0$ *whenever* $p^2 \mid N$, *and* $\delta_p = 2, 1, 0$ *according as* $p \nmid N$, $p \| N$, $p^2 \mid N$. *In case (St), with* $\varepsilon = \mu$ *viewed as an unramified character of* $G_{\mathbb{Q}_p}$ *(*$\varepsilon(\mathrm{Frob}_p) = a_p(f)$*):*

*(a) every* $G_{\mathbb{Q}_p}$*-stable* $\mathcal{O}_\lambda$*-lattice* $L \subset V$ *has a basis in which* $\rho_{f,\lambda}|_{G_{\mathbb{Q}_p}} = \left(\begin{smallmatrix} \varepsilon\chi_\ell & * \\ 0 & \varepsilon \end{smallmatrix}\right)$*, the corner being nonzero on* $I_p$*;*

*(b) the reduction* $\bar\rho_L|_{D_p}$ *is an extension of* $\bar\varepsilon$ *by* $\bar\varepsilon\bar\chi_\ell$*; it is unramified if and only if the corner vanishes on* $I_p$ *modulo* $\lambda$*, and when it is ramified it is non-split and has exactly one stable line;*

*(c) if* $\bar\rho_L$ *is unramified at* $p$*, the eigenvalues of* $\bar\rho_L(\mathrm{Frob}_p)$ *are* $a_p(f)p$ *and* $a_p(f)$ *modulo* $\lambda$*.*

*Proof.* By Theorem 15.19(ii), $a(V) = v_p(N)$, and $\det V = \chi_\ell|_{G_{\mathbb{Q}_p}}$ is unramified, corresponding to $\|\cdot\|$ on $W_{\mathbb{Q}_p}$.

$a(V) = 0$. Then $V$ is unramified, $N = 0$, and $r(\mathrm{Frob})$ has the eigenvalues $\alpha, \beta$ of Theorem 15.19(iii); this is (U).

$a(V) = 1$. If $V^{I_p} = V$ then $a(V) = 0$. So $d = \dim V^{I_p} \leq 1$, and $1 = a(V) = 2 - d + \mathrm{sw}(V)$ gives $\mathrm{sw}(V) = d - 1 \leq 0$; as the Swan conductor is $\geq 0$, $\mathrm{sw}(V) = 0$ and $d = 1$, so $V$ is tame. If $N = 0$, then $V^{I_p} = V^{r(I_p)}$ and $r|_{I_p}$, a representation of the finite group $r(I_p)$ in characteristic 0, is semisimple (Maschke) with a fixed line: $r|_{I_p} \simeq 1 \oplus \psi$ with $\psi \neq 1$ (as $\dim V^{I_p} = 1$, not 2). But $\det r|_{I_p} = 1 \cdot \psi = \psi$ must be trivial, because $\det V = \chi_\ell$ is unramified at $p \neq \ell$. So $N \neq 0$, and $V \simeq \mathrm{sp}(2) \otimes \mu$ for the character $\mu$ on $V/\ker N$ (Remark 15.16). As $a(\mathrm{sp}(2)\otimes\mu) = 2a(\mu) \geq 2$ for ramified $\mu$, $\mu$ is unramified. Comparing determinants, $\|\cdot\|\mu^2 = \|\cdot\|$, so $\mu^2 = 1$. Now $V^{I_p} = \ker N$ is the line on which $W_{\mathbb{Q}_p}$ acts by $\|\cdot\|\mu$, so $\mathrm{Frob}_p$ acts on it by $p\,\mu(\mathrm{Frob}_p)$, and Theorem 15.19(iii) (first form: the eigenvalue is $p/a_p(f)$) gives $a_p(f) = \mu(\mathrm{Frob}_p)^{-1} = \mu(\mathrm{Frob}_p) = \pm 1$. This is (St).

$a(V) \geq 2$. $\dim V^{I_p} = 2$ is impossible ($V$ would be unramified). Suppose $\dim V^{I_p} = 1$. If $N = 0$, the argument just given applies verbatim (it used only $N = 0$ and $\dim V^{I_p} = 1$): $r|_{I_p} \simeq 1 \oplus \psi$ with $\psi = \det r|_{I_p} = 1$, so $r|_{I_p}$ is trivial and $V^{I_p} = V$, a contradiction. If $N \neq 0$, $V \simeq \mathrm{sp}(2) \otimes \mu$ and $V^{I_p} = (\ker N)^{r(I_p)}$ is a line only if $\mu$ is unramified, and then $a(V) = 1$, a contradiction. Hence $V^{I_p} = 0$, $\delta_p = 0$, $L(V, s) = 1$, and $a_p(f) = 0$ by Theorem 15.19(iii). If $N \neq 0$, $V \simeq \mathrm{sp}(2) \otimes \nu$ with $\nu$ ramified and $\nu^2 = 1$ (determinants again); writing $\nu = \mu\chi$ with $\chi$ a ramified character and $\mu$ unramified gives (RSt), with $a(V) = 2a(\nu) = 2a(\chi)$. If $N = 0$, $r$ is Frobenius-semisimple of dimension 2, so it is a sum of two characters, both ramified as $V^{I_p} = 0$, with product $\|\cdot\|$ (so $\mu_2|_{I_p} = \mu_1^{-1}|_{I_p}$ and $a(\mu_2) = a(\mu_1)$, $a(V) = 2a(\mu_1)$): this is (RPS); or it is irreducible: (SC). In (SC), $a(V) \geq 2$ since $V^{I_p} = 0$.

The types are mutually exclusive (they are distinguished by $\delta_p$, by $N$ and by reducibility), and the cases $v_p(N) = 0$, 1, $\geq 2$ are (U), (St), and the last three, which proves the statements on $a_p$ and $\delta_p$.

(a) In case (St), $V$ is the representation $\left(\begin{smallmatrix}\varepsilon\chi_\ell & * \\ 0 & \varepsilon\end{smallmatrix}\right)$ of Remark 15.16 with $\mu = \varepsilon$. The stable line $\ker N$ meets $L$ in a saturated rank-one submodule $L_1$, which is a direct summand of $L$; a basis $e_1$ of $L_1$ extended by $e_2$ to a basis of $L$ gives the matrix form, with entries in $\mathcal{O}_\lambda$: the first basis vector spans a stable line, so the first column of every $\rho(\sigma)$ is $(\varepsilon\chi_\ell(\sigma), 0)^{\mathsf{T}}$, and the determinant $\chi_\ell$ forces the lower right entry to be $\varepsilon(\sigma)$ (as $\varepsilon^2 = 1$). The corner is nonzero on $I_p$ because $V$ is ramified. (b) Reduce modulo $\lambda$; the corner restricted to $I_p$ is a homomorphism (the diagonal is trivial on $I_p$), so the reduction is unramified exactly when it vanishes on $I_p$ modulo $\lambda$. If the reduction is ramified it is non-split (a split extension of two unramified characters is unramified), and a non-split extension of two characters has a unique stable line (Chapter 4; this holds also when the two characters coincide, which happens when $p \equiv 1 \pmod{\ell}$). (c) Read off the diagonal: $\bar\varepsilon\bar\chi_\ell(\mathrm{Frob}_p) = a_p(f)p$ and $\bar\varepsilon(\mathrm{Frob}_p) = a_p(f)$. □

*In the paper* 15.21. Section 1.1 of the paper: "for a prime $p \neq \ell$ let $\delta_p = \dim V^{I_p}$, so $\delta_p = 2$, 1, 0 for good, multiplicative, additive reduction". For $f = f_E$ this is Theorem 15.20 (and, on the elliptic-curve side, Chapter 11). The case $p^2 \mid N$ of Lemma lem:old ("a newform on $\Gamma_0(M)$ with $p^2 \mid M$ has $a_p = 0$ [8, Theorem 3]") is the rows (RSt), (RPS), (SC) of the table, here derived from the Galois side; and the case $p \,\|\, N$ of Lemma lem:old ("$\rho_{E,\ell}|_{D_p}$ is an extension of an unramified character $\varepsilon$ by $\varepsilon\chi_\ell$ with $\varepsilon(\mathrm{Frob}_p) = a_p(f) = \pm 1$ [38, Theorem 3.1(e)]") is part (a), applied to $f$ and to the newform $h$ of that lemma.

**Proposition 15.22** (the local type of $f_E$ by Kodaira type)**.** *Let $E/\mathbb{Q}$ have bad reduction at a prime $q \geq 5$, $f = f_E$, $\ell \neq q$. The type of $f$ at $q$ is given by the Kodaira type of $E$ at $q$ as follows.*

| *Kodaira type at $q$* | $\rho_{E,\ell}(I_q)$ | *type of $f$ at $q$* |
|---|---|---|
| $\mathrm{I}_n$, $n \geq 1$ | *unipotent, infinite* | *(St)* |
| $\mathrm{I}_n^*$, $n \geq 1$ | $\{\pm 1\}\times$*unipotent, infinite* | *(RSt), with $\chi = \chi_{q^*}$ on $I_q$* |
| $\mathrm{I}_0^*$ | $\{\pm 1\}$ | *(RPS), $\mu_i = \chi_{q^*}\cdot$(unramified)* |
| $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$ | *cyclic of order* 6, 6, 3, 3 | *(RPS) if $q \equiv 1 \pmod 3$, (SC) if $q \equiv 2 \pmod 3$* |
| $\mathrm{III}, \mathrm{III}^*$ | *cyclic of order* 4 | *(RPS) if $q \equiv 1 \pmod 4$, (SC) if $q \equiv 3 \pmod 4$* |

*In the row $\mathrm{I}_0^*$, $V_\ell E|_{G_{\mathbb{Q}_q}} \simeq V' \otimes \chi_{q^*}$ with $V' = V_\ell E^{(q)}$ unramified, and the two characters are $\chi_{q^*}$ times the unramified characters sending $\mathrm{Frob}_q$ to the Frobenius eigenvalues of $E^{(q)}$. In the row $\mathrm{I}_n^*$, $V_\ell E|_{G_{\mathbb{Q}_q}} \simeq V' \otimes \chi_{q^*}$ with $V' = V_\ell E^{(q)}$ of type (St).*

*Proof.* The images of inertia are those of Chapter 11 (the Tate shape at multiplicative primes, the potentially multiplicative case, and the theorem on additive potentially good reduction at $q \geq 5$, from Kraus's theorem). Type $\mathrm{I}_n$: $\delta_q = 1$, so (St) by Theorem 15.20. Types $\mathrm{I}_n^*$ and $\mathrm{I}_0^*$: $E^{(q)}$ has type $\mathrm{I}_n$, resp. good reduction (the paper's Lemma lem:twist(ii), proved in Chapter 10), and $V_\ell E \simeq V_\ell E^{(q)} \otimes \chi_{q^*}$ (Chapter 9); so $\mathrm{WD}_q(f)$ is $(\mathrm{sp}(2) \otimes \mu) \otimes \chi_{q^*}$, resp. $(\nu_{\alpha'} \oplus \nu_{\beta'}) \otimes \chi_{q^*}$ with $\nu_x$ the unramified character sending $\mathrm{Frob}_q$ to $x$. The other rows: inertia acts through a finite cyclic group $\langle g \rangle$ of order $m \in \{3, 4, 6\}$, so $N = 0$; for $q \geq 5$ the action is tame (the wild inertia group is pro-$q$ and its image has order dividing $m$, prime to $q$, so it is trivial) and $\mathrm{Frob}_q\, g\, \mathrm{Frob}_q^{-1} = g^q$ (the tame relation, Chapter 7). The eigenvalues of $g$ are two distinct primitive $m$-th roots of unity $\zeta, \zeta^{-1}$: $g$ has finite order, so it is diagonalisable with root-of-unity eigenvalues $\zeta_1, \zeta_2$; $\zeta_1\zeta_2 = \det g = \chi_\ell(g) = 1$ because $\chi_\ell$ is unramified at $q \neq \ell$; if $\zeta_1 = 1$ then $\zeta_2 = 1$ and $g = 1$, which is absurd; so $\zeta_2 = \zeta_1^{-1}$ has the same order as $\zeta_1$, which is therefore the order $m$ of $g$, and $\zeta_1 \neq \zeta_1^{-1}$ since $m > 2$. As $q \geq 5$ is prime to $m$ and the units modulo 3, 4, 6 are $\pm 1$, either $q \equiv 1 \pmod m$ or $q \equiv -1 \pmod m$. In the first case $\mathrm{Frob}_q\, g\, \mathrm{Frob}_q^{-1} = g^q = g$, so $\mathrm{Frob}_q$ commutes

with $g$, hence preserves its two eigenlines (over $\overline{\mathbb{Q}}_\ell$), and $\mathrm{WD}_q(f)$ is a sum of two characters, ramified since $\zeta \neq 1$: (RPS). In the second case $\mathrm{Frob}_q$ conjugates $g$ to $g^q = g^{-1}$, so it carries the $\zeta$-eigenline of $g$ to the $\zeta$-eigenline of $g^{-1}$, which is the $\zeta^{-1}$-eigenline of $g$: it exchanges the two eigenlines, so no line is stable under $W_{\mathbb{Q}_q}$ (a stable line would be an eigenline of $g$), and $\mathrm{WD}_q(f)$ is irreducible: (SC). Finally $q \equiv 1 \pmod 6$ iff $q \equiv 1 \pmod 3$ for odd $q$. □

**Example 15.23** (local types of the paper's curves)**.** (Checked with PARI/GP's `elllocalred`.) `11a1` at 11: $\mathrm{I}_5$, (St), $a_{11} = 1$. `121d2` at 11: $\mathrm{I}_5^*$, (RSt). `539d1` at 7: $\mathrm{I}_0^*$, (RPS), the twist being `11a3`. `121c1` at 11: IV, and $11 \equiv 2 \pmod 3$, so (SC). `121a1` at 11: II, (SC). `726a1` at 11: III, and $11 \equiv 3 \pmod 4$, so (SC). `52a1` at 2 has type $\mathrm{IV}^*$; the table is for $q \geq 5$, but by Chapter 11 the image of $I_2$ is cyclic of order 3 here as well, and since $2 \equiv -1 \pmod 3$ the argument of the proof gives (SC) at 2.

*In the paper* 15.24. The proof of Lemma lem:euler says, for type $\mathrm{I}_n$: "The local component of $f$ is an unramified twist of the Steinberg representation", and for type $\mathrm{I}_n^*$: "$V = V' \otimes \chi_{q^*}$ with $V'$ Steinberg at $q$". These are the first two rows of Proposition 15.22; the proof of Lemma lem:euler uses them only through $\mathrm{WD}_q(f)$ and $(\mathrm{ad}^0 V)^{I_q}$, which Chapter 11 and Chapter 19 compute.

**Corollary 15.25** ($\delta_p$ and the naive factor)**.** *Let $f$ be a newform of weight 2 on $\Gamma_0(N)$ and $p \mid N$, $p \neq \ell$. Then $a_p(f) \in \{\pm 1\}$ iff $\delta_p = 1$, and $a_p(f) = 0$ iff $\delta_p = 0$. The naive local factor $L_p^{\mathrm{nv}}(A_f, s)$ of the paper, which equals the true factor $L_p(A_f, s)$ if $\delta_p > 0$ and 1 if $\delta_p = 0$ (equivalently $L_p^{\mathrm{nv}}(A_f, s)^{-1} = 1 - a_p(f)^2 p^{-1-s}$), is determined by $a_p(f)$; the true factor is determined by $\mathrm{WD}_p(f)$ and is unchanged when $\rho_{f,\lambda}$ is replaced by $\rho_{f,\lambda} \otimes \chi$ for a character $\chi$, because $\mathrm{ad}^0(\rho \otimes \chi) = \mathrm{ad}^0 \rho$. The naive factor is not twist-invariant: $f_{11a}$ at 11 has type (St) and $f_{121d} = f_{11a}^{\chi_{-11}}$ has type (RSt), so their naive factors at 11 are $1 - 11^{-1-s}$ and 1, while their true factors coincide.*

*Proof.* The first sentence is Theorem 15.20. If $\delta_p = 1$, $V = \mathrm{sp}(2) \otimes \mu$ and $(\mathrm{ad}^0 V)^{I_p}$ is the line spanned by $N$. Indeed, twisting by $\mu$ does not change $\mathrm{ad}^0 V$, and $r(I_p)$ is trivial on sp(2); so on the Galois side $\sigma \in I_p$ acts on sp(2) by $\exp(t_\ell(\sigma)N) = 1 + t_\ell(\sigma)N$ (Chapter 7), where $t_\ell$ is the surjective $\ell$-adic tame character of $I_p$. An endomorphism commutes with all these if and only if it commutes with $N$, so the invariants in $\mathrm{ad}^0 V$ (the trace-zero endomorphisms) are the trace-zero elements of the centraliser of $N$. For a nonzero nilpotent $2 \times 2$ matrix $N$ the centraliser is spanned by 1 and $N$, and its trace-zero part is the line spanned by $N$. The arithmetic Frobenius acts on $N$ by conjugation, $r(\mathrm{Frob}) N r(\mathrm{Frob})^{-1} = \|\mathrm{Frob}\| N = pN$, so the geometric Frobenius acts by $p^{-1}$, so $L_p(A_f, s)^{-1} = 1 - p^{-1-s} = 1 - a_p^2 p^{-1-s}$; if $\delta_p = 0$, $a_p = 0$ and the naive factor is 1 by definition. The twist-invariance of $\mathrm{ad}^0$ is Chapter 4. The computation of $L_p(A_f, s)$ in every case is Chapter 19. □

**Proposition 15.26** (the conductor of a quadratic twist, type by type)**.** *Let $q$ be an odd prime, $\chi$ a ramified quadratic character of $G_{\mathbb{Q}_q}$ (so $\chi$ is tame and $a(\chi) = 1$, and $\chi|_{I_q} = \chi_{q^*}|_{I_q}$), and $f$ a newform of weight 2 with trivial character, of type $t$ at $q$. Then:*

| *type of $f$ at $q$* | $a(\mathrm{WD}_q(f))$ | $a(\mathrm{WD}_q(f) \otimes \chi)$ | *type of* $\mathrm{WD}_q(f) \otimes \chi$ |
|---|---|---|---|
| *(U)* | 0 | 2 | *(RPS)* |
| *(St)* | 1 | 2 | *(RSt)* |
| *(RSt)* | 2 | 1 | *(St)* |
| *(RPS) with $\mu_i|_{I_q} = \chi|_{I_q}$* | 2 | 0 | *(U)* |
| *(RPS) or (SC), tame, inertia cyclic of order 3, 4 or 6* | 2 | 2 | *(RPS) or (SC)* |

*At a prime $p \neq q$ the twist by a character unramified at $p$ changes neither the type nor the conductor exponent.*

*Proof.* (U): $\mathrm{WD} \otimes \chi = \mu_1\chi \oplus \mu_2\chi$ with both characters ramified of conductor 1, so $a = 2$. (St): $\mathrm{sp}(2) \otimes \mu\chi$ with $\mu\chi$ ramified: $a = 2a(\chi) = 2$. (RSt): for $q$ odd, every ramified character $\nu$ with $\nu^2 = 1$ agrees with $\chi$ on $I_q$. Indeed, by local class field theory the image of $I_q$ in $G_{\mathbb{Q}_q}^{\mathrm{ab}}$ corresponds to $\mathbb{Z}_q^\times = \mu_{q-1} \times (1 + q\mathbb{Z}_q)$; a character of order dividing 2 is trivial on the pro-$q$ group $1 + q\mathbb{Z}_q$ ($q$ odd), so it factors through $\mathbb{Z}_q^\times \to \mathbb{F}_q^\times$, a cyclic group of even order with exactly one character of order 2, the Legendre symbol; hence all ramified quadratic characters of $\mathbb{Q}_q^\times$ restrict to the same character of $\mathbb{Z}_q^\times$, and differ by an unramified one. So $\mathrm{WD} = \mathrm{sp}(2) \otimes \nu$ with $\nu\chi$ unramified and $\mathrm{WD} \otimes \chi = \mathrm{sp}(2) \otimes \nu\chi$ has $a = 1$; in particular $v_q(N) = 2$ for type (RSt) at odd $q$. (RPS) with $\mu_i|_{I_q} = \chi|_{I_q}$: both $\mu_i\chi$ are unramified, $a = 0$. Last row: inertia acts on WD through $\langle g \rangle$, $g$ of order $m \in \{3, 4, 6\}$ with eigenvalues $\zeta^{\pm 1}$ primitive $m$-th roots of unity (as in the proof of Proposition 15.22), and on $\mathrm{WD} \otimes \chi$ through $\langle -g \rangle$: the tame inertia group is procyclic, $\chi$ is a nontrivial character of order 2 of it, so $\chi(t) = -1$ for a topological generator $t$, which may be chosen to map to $g$. If $\zeta$ has order 3, 4, 6, then $-\zeta$ has order 6, 4, 3 (for $m = 4$, $-\zeta = \zeta^{-1} = \pm i$); so $-g$ has order 6, 4, 3 and eigenvalues $-\zeta^{\pm 1} \neq 1$, so $(\mathrm{WD} \otimes \chi)^{I_q} = 0$, the twist is tame, and $a = 2$; its type is (RPS) or (SC) by the argument of Proposition 15.22. At $p \neq q$: an unramified twist preserves $a$ and the shape (Remark 15.16). □

**Corollary 15.27** (Atkin and Li; the paper's Lemma lem:twist(iii))**.** *Let $g$ be a newform of weight 2 on $\Gamma_0(M)$ and $q$ an odd prime with $q^2 \nmid M$; let $\chi = \chi_{q^*}$. Then $g^\chi$ has level $\mathrm{lcm}(M, q^2)$, type (RPS) or (RSt) at $q$, and $a_q(g^\chi) = 0$; consequently $g \otimes \chi = g^\chi$ is a newform on $\Gamma_0(\mathrm{lcm}(M, q^2))$: of level $Mq^2$ if $q \nmid M$ and $Mq$ if $q \parallel M$.*

*Proof.* The type of $g$ at $q$ is (U) or (St) (Theorem 15.20), so $\mathrm{WD}_q(g^\chi) = \mathrm{WD}_q(g) \otimes \chi$ has conductor exponent 2 and type (RPS) or (RSt) (Proposition 15.26); at $p \neq q$ the exponents are those of $g$. By Theorem 15.19(ii) the level of $g^\chi$ is $\mathrm{lcm}(M, q^2)$, and $a_q(g^\chi) = 0$ by Theorem 15.20. By Proposition 15.29(ii), $g \otimes \chi = (g^\chi)^{\{q\}} = g^\chi$. □

*In the paper* 15.28. The proof of Lemma lem:twist(iii) in the paper: "in the case $q \parallel M$, the local component of $g$ at $q$ is an unramified twist of the Steinberg representation, whose twist by a ramified quadratic character has conductor exponent 2". This is the row (St) of Proposition 15.26; the row (U) is the case $q \nmid M$, and Corollary 15.27 is the statement of Lemma lem:twist(iii) (Atkin and Li [9, Theorem 3.1], as the paper cites it). The hypothesis $q^2 \nmid M$ cannot be dropped: for $q^2 \mid M$ the rows (RSt) and (RPS) show that $g^\chi$ may have smaller level (`539c1`, `539d1` below), and then $g \otimes \chi = (g^\chi)^{\{q\}}$ differs from $g^\chi$ whenever $a_q(g^\chi) \neq 0$.

**Proposition 15.29** (twists of newforms)**.** *Let $g$ be a newform of weight 2 on $\Gamma_0(M)$ and $\chi$ a quadratic Dirichlet character of conductor $r$.*

*(i) $\rho_{g,\lambda} \otimes \chi$ is unramified outside $Mr\ell$ and $\mathrm{tr}(\rho_{g,\lambda} \otimes \chi)(\mathrm{Frob}_p) = \chi(p)a_p(g)$ for $p \nmid Mr\ell$.*

*(ii) There is a unique newform $g^\chi$ of weight 2 with trivial character such that $\rho_{g^\chi,\lambda} \simeq \rho_{g,\lambda} \otimes \chi$ (for one, equivalently every, $\lambda$). Its level divides $\mathrm{lcm}(M, r^2)$ and equals*

$$M \prod_{q \mid r} q^{a(\mathrm{WD}_q(g) \otimes \chi) - a(\mathrm{WD}_q(g))}.$$

*It satisfies $a_n(g^\chi) = \chi(n)a_n(g)$ for every $n$ prime to $r$, and $g \otimes \chi = (g^\chi)^{\Sigma(r)}$.*

*(iii)* $(g^\chi)^\chi = g$.

*(iv) For $E/\mathbb{Q}$ and a fundamental discriminant $d$, $f_{E^{(d)}} = (f_E)^{\chi_d}$.*

*Proof.* (i) For $p \nmid Mr\ell$ both $\rho_{g,\lambda}$ (Theorem 15.6) and $\chi$ (whose conductor is $r$) are trivial on $I_p$, hence so is $\rho_{g,\lambda} \otimes \chi$; and $(\rho_{g,\lambda} \otimes \chi)(\mathrm{Frob}_p) = \chi(\mathrm{Frob}_p)\rho_{g,\lambda}(\mathrm{Frob}_p) = \chi(p)\rho_{g,\lambda}(\mathrm{Frob}_p)$, whose trace is $\chi(p)a_p(g)$. (ii) Uniqueness: two such newforms have the same $a_p$ for almost all $p$, so they are equal (Proposition 15.4(iii)). Existence: by Chapter 13 (for $\chi = \chi_S$, the case used in this book; for any quadratic $\chi$ see Atkin and Li [9]) the naive twist $g \otimes \chi$ lies in $S_2(\Gamma_0(L))$, $L = \mathrm{lcm}(M, r^2)$ ($\chi^2$ is trivial on $(\mathbb{Z}/L)^\times$), and is an eigenform of $T_p$ with eigenvalue $\chi(p)a_p(g)$ for $p \nmid L$. Decompose it along Proposition 15.4(iv); its nonzero components are eigenforms with the same eigenvalues, so it lies in a single $V_h$ with $a_p(h) = \chi(p)a_p(g)$ for all $p \nmid L$, and $h$ is a newform of level dividing $L$. By (i) and Corollary 15.10 (Chebotarev and Brauer-Nesbitt in characteristic 0; both representations are irreducible), $\rho_{h,\lambda} \simeq \rho_{g,\lambda} \otimes \chi$; put $g^\chi = h$. Its level is the conductor of $\rho_{g,\lambda} \otimes \chi$ (Theorem 15.19(ii), with $\ell$ chosen prime to $Mr$), which differs from that of $\rho_{g,\lambda}$ only at the primes dividing $r$. Coefficients: for $p \nmid r$, $\mathrm{WD}_p(g^\chi) = \mathrm{WD}_p(g) \otimes \chi$ with $\chi$ unramified at $p$, so $\delta_p$ is the same for both and, by Theorem 15.19(iii) (applied with an $\ell \neq p$), $a_p(g^\chi) = \chi(p)a_p(g)$ for all $p \nmid r$: if $\delta_p = 2$ the Frobenius eigenvalues are multiplied by $\chi(p)$; if $\delta_p = 1$ the eigenvalue $p/a_p$ on $V^{I_p}$ is, and $\chi(p)^{-1} = \chi(p)$; if $\delta_p = 0$ both sides vanish. The multiplicativity and the recursions of Proposition 15.4(i), which $n \mapsto \chi(n)a_n(g)$ also satisfies on $n$ prime to $r$ (because $\chi(p)^2 = 1$), give $a_n(g^\chi) = \chi(n)a_n(g)$ for $(n, r) = 1$. Both $g \otimes \chi$ and $(g^\chi)^{\Sigma(r)}$ have $n$-th coefficient $\chi(n)a_n(g)$ for $(n, r) = 1$ and 0 otherwise, so they are equal. (iii) $\rho_{(g^\chi)^\chi} \simeq \rho_g \otimes \chi^2 = \rho_g$ and uniqueness. (iv) $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi_d$ (Chapter 9), and both $f_{E^{(d)}}$ and $(f_E)^{\chi_d}$ have this representation (Theorem 15.12(i)); apply uniqueness. □

**Example 15.30** (levels of twists)**.** (Each twist computed with PARI/GP's `elltwist`, reduced to a minimal model and identified in Cremona's tables; each newform identity checked on $a_p$, $p < 400$, $p \neq q$.) With $\chi = \chi_{-11}$ or $\chi_{-7}$ as appropriate:

| curve | type at $q$ | row | twist | level change | newform identity |
|---|---|---|---|---|---|
| `11a1` | $\mathrm{I}_5$ at 11, (St) | (St) | `121d2` | $11 \to 121$ | $f_{11a}^{\chi_{-11}} = f_{11a} \otimes \chi_{-11} = f_{121d}$ |
| `11a3` | $\mathrm{I}_1$ at 11, (St) | (St) | `121d1` | $11 \to 121$ | the same class |
| `11a1` | good at 7, (U) | (U) | `539d2` | $11 \to 539$ | $f_{11a}^{\chi_{-7}} = f_{11a} \otimes \chi_{-7} = f_{539d}$ |
| `20a1` | good at 7, (U) | (U) | `980g2` | $20 \to 980$ | $f_{20a}^{\chi_{-7}} = f_{20a} \otimes \chi_{-7} = f_{980g}$ |
| `121a1` | II at 11, (SC) | last | `121c2` | $121 \to 121$ | $f_{121a}^{\chi_{-11}} = f_{121a} \otimes \chi_{-11} = f_{121c}$ |
| `539c1` | $\mathrm{I}_3^*$ at 7, (RSt) | (RSt) | `77c1` | $539 \to 77$ | $f_{539c}^{\chi_{-7}} = f_{77c}$, $a_7(f_{77c}) = -1$ |
| `539d1` | $\mathrm{I}_0^*$ at 7, (RPS) | (RPS) | `11a3` | $539 \to 11$ | $f_{539d}^{\chi_{-7}} = f_{11a}$ |
| `980i1` | $\mathrm{I}_5^*$ at 7, (RSt) | (RSt) | `140b1` | $980 \to 140$ | $f_{980i}^{\chi_{-7}} = f_{140b}$, $a_7(f_{140b}) = -1$ |

In the three rows where the level drops, the naive twist is a depletion of the newform twist: $f_{539c} \otimes \chi_{-7} = f_{77c}^{\{7\}}$, $f_{980i} \otimes \chi_{-7} = f_{140b}^{\{7\}}$, $f_{539d} \otimes \chi_{-7} = f_{11a}^{\{7\}}$. The twist of the optimal curve `121c1` by $\mathbb{Q}(\sqrt{-11})$ is `121a2`, not `121a1`: the twist of a curve lands in the twisted class, but not necessarily on the curve with the same index.

## 15.4 The representation at $\ell$ when $\ell \nmid N$

At the prime $\ell$ itself the representation $\rho_{f,\lambda}$ is not described by a Weil-Deligne representation; when $\ell \nmid N$ it is crystalline, and its reduction is finite flat. The paper uses this twice: in the

optimal-level paragraph of Section 3 ("the restriction to $D_\ell$ comes from the finite flat group scheme $E'[\ell]$ and is good"), and in Lemma lem:old at $p=\ell$, where $a_\ell \bmod \lambda$ must be read off from $\bar\rho|_{D_\ell}$. For $\rho_{E,\ell}$ the same facts are in Chapter 11; here they are stated for newforms.

**Theorem 15.31** (the representation at $\ell \nmid N$)**.** *Let $f$ be a newform of weight 2 on $\Gamma_0(N)$, $\lambda \mid \ell$, $\ell \nmid N$.*

*(i) $J_f$ has good reduction at $\ell$; for every $k \geq 1$ the $G_{\mathbb{Q}_\ell}$-module $J_f[\ell^k]$ is the generic fibre of a finite flat group scheme over $\mathbb{Z}_\ell$ (the $\ell^k$-torsion of the Néron model, an abelian scheme), and so is every $G_{\mathbb{Q}_\ell}$-stable subquotient of it. In particular the reduction $\bar\rho_{f,\lambda}|_{D_\ell}$ of any stable lattice is* finite flat*, which is what Theorem 15.42 needs for $\delta = 0$ (the paper's word is "good").*

*(ii) $\rho_{f,\lambda}|_{G_{\mathbb{Q}_\ell}}$ is crystalline with Hodge-Tate weights 0 and 1, and the crystalline Frobenius on $\mathrm{D}_{\mathrm{cris}}(V_\lambda(f))$ has characteristic polynomial $X^2 - a_\ell(f)X + \ell$. For $\ell \geq 3$ the weights lie in the Fontaine-Laffaille range $[0, \ell-2]$, so that stable lattices correspond to Fontaine-Laffaille modules.*

*(iii) If $\bar\rho_{f,\lambda}$ is irreducible, its Serre weight is 2.*

*Sketch of proof.* (i) Good reduction of $J_0(N)$ at $\ell \nmid N$ is the theorem of Igusa and of Deligne and Rapoport [43], and it passes to the quotient $J_f$; the $\ell^k$-torsion of an abelian scheme is finite flat; scheme-theoretic closure in a finite flat group scheme shows that subquotients of finite flat Galois modules are finite flat [38]. (ii) Crystallinity of the Tate module of an abelian variety with good reduction is the crystalline comparison theorem (Fontaine, Messing, Faltings); see [11]. The characteristic polynomial of the crystalline Frobenius is that of the Frobenius endomorphism of the reduction, which is $X^2 - a_\ell X + \ell$ on the $\lambda$-part by the Eichler-Shimura relation, as in Theorem 15.6. The Fontaine-Laffaille correspondence is [68]. (iii) Serre's recipe for the weight [176], as completed by Edixhoven [61], gives weight 2 for an irreducible $\bar\rho$ whose restriction to $D_\ell$ is finite flat; Theorem 15.32 describes $\bar\rho|_{I_\ell}$ explicitly. We do not prove these statements. □

**Theorem 15.32** (Deligne; Fontaine)**.** *Let $\ell$ be odd, $f$ a newform of weight 2 on $\Gamma_0(N)$ with $\ell \nmid N$, $\lambda \mid \ell$, and $\bar\rho = \bar\rho_L$ for a stable lattice $L$. Let $\omega = \bar\chi_\ell|_{I_\ell}$ (the fundamental character of level 1) and $\omega_2$ a fundamental character of level 2 (Chapter 11). For $x \in k_\lambda^\times$ let $\mu(x)$ be the unramified character of $G_{\mathbb{Q}_\ell}$ with $\mu(x)(\mathrm{Frob}_\ell) = x$.*

*(i) (Deligne.) If $a_\ell(f) \not\equiv 0 \pmod\lambda$, then $\bar\rho|_{D_\ell} \simeq \begin{pmatrix} \bar\chi_\ell\,\mu(\bar a_\ell^{-1}) & * \\ 0 & \mu(\bar a_\ell) \end{pmatrix}$, where $\bar a_\ell = a_\ell(f) \bmod \lambda$.*

*(ii) (Fontaine.) If $a_\ell(f) \equiv 0 \pmod\lambda$, then $\bar\rho|_{I_\ell} \otimes \bar k_\lambda \simeq \omega_2 \oplus \omega_2^\ell$, and $\bar\rho|_{D_\ell}$ is irreducible, even over $\bar k_\lambda$.*

*(iii) Consequently $a_\ell(f) \bmod \lambda$ is determined by the isomorphism class of $\bar\rho|_{D_\ell}$: it is 0 if $\bar\rho|_{D_\ell}$ is irreducible; otherwise $\bar\rho|_{D_\ell}$ has exactly one one-dimensional unramified quotient, and $a_\ell(f) \bmod \lambda$ is the eigenvalue of $\mathrm{Frob}_\ell$ on it. In particular, if $h$ is a newform of weight 2 with trivial character, of level prime to $\ell$, with coefficients in $\mathcal{O}$, and $\bar\rho_h \simeq \bar\rho_f$ (over $k_\lambda$), then $a_\ell(h) \equiv a_\ell(f) \pmod\lambda$.*

*Proof.* We do not prove (i) and (ii); see Edixhoven [61, Section 2], where both are stated with references, Wiles [206] for the ordinary case, and Serre [171, Section 1] for the elliptic-curve case.

(iii) In case (ii) $\bar\rho|_{D_\ell}$ is irreducible and in case (i) it is not; so $\bar\rho|_{D_\ell}$ irreducible forces $a_\ell \equiv 0$. If $\bar\rho|_{D_\ell}$ is reducible we are in case (i). A one-dimensional quotient of $\bar\rho|_{D_\ell}$ is the quotient by a stable line; the stable lines are the given one, with quotient $\mu(\bar a_\ell)$, and, only if the extension splits, a second line, with quotient $\bar\chi_\ell\mu(\bar a_\ell^{-1})$. The latter is ramified, because $\omega$ has order $\ell-1\geq 2$ for $\ell$ odd. So the only unramified one-dimensional quotient is $\mu(\bar a_\ell)$, on which $\mathrm{Frob}_\ell$ acts by $\bar a_\ell$. Applying this to $f$ and to $h$, whose restrictions to $D_\ell$ are isomorphic, gives $a_\ell(h)\equiv a_\ell(f)$. □

**Example 15.33.** $a_3(f_{121c})=2$ and $a_3(f_{11a})=-1$, both nonzero modulo 3 and congruent modulo 3; this is the case $p=\ell=3$ of Lemma lem:old for the pair $(f,h)=(f_{121c},f_{11a})$ of Example 15.60. Likewise $a_3(f_{121a})=2\equiv -1=a_3(f_{121d})\pmod 3$, as the paper's Example needs. Both are instances of (i): $\bar\rho|_{D_3}$ is reducible with unramified quotient $\mu(\bar a_3)=\mu(-1)$, so $\mathrm{Frob}_3$ acts on that quotient by $-1$, and (iii) reads $a_3\equiv -1$ off it. For an instance of (ii), take $f=f_{11a}$ and $\ell=19$: $a_{19}(f_{11a})=0$ (Example 15.52), so $\bar\rho_{11a,19}|_{D_{19}}$ is irreducible, $\bar\rho|_{I_{19}}$ is a sum of the two fundamental characters of level 2, and `11a1` has supersingular reduction at 19.

*In the paper* 15.34. Lemma lem:old, case $p=\ell$: "By the theorems of Deligne and Fontaine describing $\bar\rho_f|_{D_\ell}$ and $\bar\rho_h|_{D_\ell}$ [61, §2], the residue class of $a_\ell$ is determined by the restriction to $D_\ell$: it is 0 when that restriction is irreducible, and otherwise the eigenvalue of Frobenius on the unramified quotient." This is Theorem 15.32(iii). The paper then says why the article is justified: "That quotient is well defined: the two characters of the semisimplification restrict to $I_\ell$ as $\bar\chi_\ell$ and 1, and $\bar\chi_\ell|_{I_\ell}\neq 1$ since $\ell$ is odd, so the unramified quotient is the module of $I_\ell$-coinvariants." This is the uniqueness in Theorem 15.32(iii), in the case (i) where $\bar\rho|_{D_\ell}$ is reducible. The remark after the paper's Example lists Lemma lem:old at $p=\ell$ among the uses of $\ell\nmid N$: Theorem 15.32 needs both levels prime to $\ell$.

## 15.5 Serre's conjecture: conductor, weight and character

This section defines the invariants that Serre attached to an odd irreducible mod $\ell$ representation and proves that the Serre conductor of $\bar\rho_{f,\lambda}$ divides the level of $f$. The paper uses the Serre conductor in the optimal-level paragraph and in cases (a), (b), (c) of the proof of Theorem thm:qual ("the Serre conductor $M=N(\bar\rho)$ has exponent exactly 1 at $q$, so $M\mid N/q$"); the divisibility $N(\bar\rho)\mid N$ is what puts the lowered level inside the original one.

**Definition 15.35** (Serre's invariants)**.** Let $\bar\sigma\colon G_{\mathbb{Q}}\to \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ be continuous, odd ($\det\bar\sigma(c)=-1$) and irreducible.

(a) The *Serre conductor* is $N(\bar\sigma)=\prod_{p\neq\ell}p^{a_p(\bar\sigma)}$, $a_p(\bar\sigma)=2-\dim\bar\sigma^{I_p}+\mathrm{sw}_p(\bar\sigma)$, the Artin conductor exponent of $\bar\sigma|_{G_{\mathbb{Q}_p}}$ (Chapter 11). It is prime to $\ell$ by definition and prime to every $p$ at which $\bar\sigma$ is unramified.

(b) The *Serre weight* $k(\bar\sigma)\geq 2$ is defined by Serre's recipe in terms of $\bar\sigma|_{I_\ell}$ [176, Section 2], [61]; it is 2 when $\bar\sigma|_{D_\ell}$ is finite flat.

(c) The *character* $\varepsilon(\bar\sigma)\colon(\mathbb{Z}/N(\bar\sigma))^\times\to\overline{\mathbb{F}}_\ell^\times$ is defined by $\det\bar\sigma=\varepsilon(\bar\sigma)\bar\chi_\ell^{k(\bar\sigma)-1}$.

(d) $\bar\sigma$ *arises from* a newform $g$ (of some weight, level and character) if $\bar\sigma\simeq\bar\rho_{g,\lambda}\otimes_{k_\lambda}\overline{\mathbb{F}}_\ell$ for some prime $\lambda\mid\ell$ of the coefficient field of $g$ and some embedding $k_\lambda\hookrightarrow\overline{\mathbb{F}}_\ell$; when $\bar\sigma$ is absolutely irreducible and either $\ell$ is odd or $\det\bar\sigma=\bar\chi_\ell$, this is equivalent, for $g$ of weight 2 with trivial character and level $M$, to $\mathrm{tr}\,\bar\sigma(\mathrm{Frob}_p)=a_p(g)\bmod\lambda$ for all $p\nmid M\ell N(\bar\sigma)$

(Brauer-Nesbitt, as in Corollary 15.10; for $\ell = 2$ traces do not determine a two-dimensional semisimple representation, and the determinant is needed). $\bar\sigma$ is *modular* if it arises from some newform.

**Proposition 15.36** ($N(\bar\rho_{f,\lambda}) \mid N$; Carayol, Livné)**.** *Let $f$ be a newform of weight 2 on $\Gamma_0(N)$, $\lambda \mid \ell$, $L$ a stable lattice, $\bar\rho_L$ its reduction, and $p \neq \ell$. Then*

$$a_p(\bar\rho_L^{\mathrm{ss}}) \;\le\; a_p(\bar\rho_L) \;=\; v_p(N) - \left(\dim \bar\rho_L^{I_p} - \delta_p\right) \;\le\; v_p(N).$$

*Hence the prime-to-$\ell$ part of $N$ is divisible by $N(\bar\rho_{f,\lambda})$ whenever $\bar\rho_{f,\lambda}$ is irreducible, and $N(\bar\rho_{f,\lambda}) \mid N$ if $\ell \nmid N$.*

*Proof.* Let $V = L \otimes K_{f,\lambda}$ and $P_p \subset I_p$ the wild inertia group, a pro-$p$ group. The kernel of $\mathrm{GL}_2(\mathcal{O}_\lambda) \to \mathrm{GL}_2(k_\lambda)$ is a pro-$\ell$ group, and a closed subgroup of it that is also a quotient of a pro-$p$ group is trivial; so $\rho(P_p)$ maps injectively to $\mathrm{GL}_2(k_\lambda)$, and is finite. The Swan conductor of $V$ (and of $\bar\rho_L$) is computed from the dimensions of the fixed spaces of the ramification groups $G_i$, $i \ge 1$, of a finite quotient through which $P_p$ acts (Chapter 7); these are $p$-groups, of order invertible in $\mathcal{O}_\lambda$. For such a group $G$ the averaging operator $e = \frac{1}{\#G}\sum_{g\in G} g$ is an idempotent of $\mathcal{O}_\lambda[G]$, so $L = eL \oplus (1-e)L$ with $eL = L^G$, and reduction modulo $\lambda$ gives $(L/\lambda L)^G = e(L/\lambda L) = L^G/\lambda L^G$. Hence $\dim V^{G_i} = \operatorname{rank} L^{G_i} = \dim (L/\lambda L)^{G_i}$ for all $i \ge 1$, and $\mathrm{sw}(V) = \mathrm{sw}(\bar\rho_L)$. Next, $L^{I_p} = L \cap V^{I_p}$ is saturated in $L$, so $L^{I_p} \cap \lambda L = \lambda L^{I_p}$ and $L^{I_p}/\lambda L^{I_p} \hookrightarrow (L/\lambda L)^{I_p}$; thus $\dim \bar\rho_L^{I_p} \ge \dim V^{I_p} = \delta_p$. Subtracting the two conductor formulas, $a(V) - a(\bar\rho_L) = \dim \bar\rho_L^{I_p} - \delta_p \ge 0$, and $a(V) = v_p(N)$ by Theorem 15.19(ii). For the first inequality, let $0 \to A \to B \to C \to 0$ be an exact sequence of $k_\lambda[G_{\mathbb{Q}_p}]$-modules with $B = \bar\rho_L$; the Swan conductor is additive (the fixed points of a group of order prime to $\ell$ form an exact functor), and $\dim B^{I_p} \le \dim A^{I_p} + \dim C^{I_p}$ (invariants are left exact), so $a(B) \ge a(A) + a(C) = a(A \oplus C)$. When $\bar\rho_{f,\lambda}$ is irreducible it does not depend on $L$ (Proposition 15.9(iii)), and multiplying over $p \neq \ell$ gives the divisibility. □

For $f = f_E$ the exact value is $a_p(\bar\rho_{E,\ell}) = f_p(E) - \dim_{\mathbb{F}_\ell} \Phi_p^{\mathrm{geom}}[\ell]$ (Chapter 11, with examples); Livné [116] lists the possible exponents in general. To see the drop $\dim \bar\rho_L^{I_p} - \delta_p$ of the proposition at work, take `77c1` and $\ell = 3$. At $p = 7$ the type is $\mathrm{I}_3$, so $\delta_7 = 1$; the component group has order 3, $\bar\rho$ is unramified at 7 and $\dim \bar\rho^{I_7} = 2$, so the exponent drops from $v_7(77) = 1$ to $1 - (2-1) = 0$. At $p = 11$ the type is $\mathrm{I}_2$, the component group has order 2, prime to 3, so $\bar\rho$ is ramified, $\dim \bar\rho^{I_{11}} = 1 = \delta_{11}$, and the exponent stays 1. Hence $N(\bar\rho_{77c1,3}) = 11$, the entry of Example 15.45.

*In the paper* 15.37. The optimal-level paragraph of Section 3: "Recall also that $N(\bar\rho_{E',\ell})$ divides the conductor of $E'$ [22], and that $N(\bar\sigma)$ is prime to every prime at which $\bar\sigma$ is unramified." The first clause is Proposition 15.36 for $f = f_{E'}$, the second is Definition 15.35(a). Both are used in cases (a), (b), (c) to get $M \mid N/q$ or $M \mid N/q^2$ and the hypothesis of Lemma lem:old.

**Theorem 15.38** (Serre's conjecture; Khare and Wintenberger)**.** *Every odd, absolutely irreducible, continuous $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ arises from a newform of weight $k(\bar\sigma)$, level $N(\bar\sigma)$ and a character lifting $\varepsilon(\bar\sigma)$.*

We do not prove this; see Khare and Wintenberger [98], Khare [97] for level one, Kisin [100] for an input at $\ell = 2$, and Serre's formulation [176]. The passage from the "weak" form (modular of some weight and level) to the "strong" form (the optimal weight and level) is due to Ribet,

Carayol, Diamond, Edixhoven, Buzzard and others; Section 15.6 states the part that concerns the level. The paper uses only that part, for $\bar{\rho}_{E',\ell}$, which is modular because $E'$ is (Chapter 14); it does not need the theorem of Khare and Wintenberger.

## 15.6 Lowering the level

A mod $\ell$ representation that arises from a newform of level $N$ may also arise at a smaller level. This section states the theorems that lower the level (Mazur's principle, Ribet's theorem, and the optimal-level theorem of Ribet, Carayol and Diamond in the form of [38, Theorem 3.15]), proves the specialisation to $\bar{\rho}_{E',\ell}$ that the paper uses (level exactly $N(\bar{\rho})$, trivial character), and works the paper's examples. This is the step of the proof of Theorem thm:qual that, in the words of the paragraph after that theorem, "lowers the level of $\bar{\rho}$, or of its twist $\bar{\rho} \otimes \chi_{q^*}$, and twists back" (the first half of that sentence).

**Theorem 15.39** (Mazur's principle)**.** *Let $\ell$ be odd, $f$ a newform of weight 2 on $\Gamma_0(N)$ with $\ell \nmid N$, $\lambda \mid \ell$ with $\bar{\rho} = \bar{\rho}_{f,\lambda}$ irreducible, and $q \,\|\, N$ a prime such that $\bar{\rho}$ is unramified at $q$ and $q \not\equiv 1 \pmod{\ell}$. Then $\bar{\rho}$ arises from a newform of weight 2 on $\Gamma_0(N/q)$.*

*Idea of proof.* $\bar{\rho}$ is realised in $J_0(N)[\mathfrak{m}]$ for the maximal ideal $\mathfrak{m} = \mathfrak{m}_{f,\lambda}$ of $\mathbb{T}_N$ attached to $(f, \lambda)$ (Proposition 15.14(iii)), which is non-Eisenstein. At $q$ the Néron model of $J_0(N)$ has semiabelian reduction whose toric part is governed by the $q$-new quotient (the character group of Grothendieck and Raynaud). If $\bar{\rho}$ arose only from $q$-new forms, $J_0(N)[\mathfrak{m}]$ would be controlled by the toric part, on which inertia acts unipotently and $\mathrm{Frob}_q$ acts by $\pm q$ on the subobject and $\pm 1$ on the quotient, with the monodromy pairing measuring the extension; "unramified at $q$" together with $q \not\equiv 1 \pmod{\ell}$ (so that the two Frobenius eigenvalues $\pm q$, $\pm 1$ are distinct modulo $\ell$) forces $\mathfrak{m}$ to be in the support of the $q$-old part, whose Hecke eigensystems come from level $N/q$. See Ribet [156] (where the principle is attributed to Mazur), [125] and [38]. We do not prove it. □

**Theorem 15.40** (Ribet's level-lowering theorem)**.** *Let $\ell$ be odd, $f$ a newform of weight 2 on $\Gamma_0(N)$ with $\ell \nmid N$, $\lambda \mid \ell$ with $\bar{\rho} = \bar{\rho}_{f,\lambda}$ irreducible, and $q \,\|\, N$ a prime at which $\bar{\rho}$ is unramified. Then $\bar{\rho}$ arises from a newform of weight 2 on $\Gamma_0(N/q)$.*

*Idea of proof.* When $q \not\equiv 1 \pmod{\ell}$ this is Mazur's principle. When $q \equiv 1 \pmod{\ell}$, Ribet chooses an auxiliary prime $r \nmid N\ell$, $r \not\equiv 1 \pmod{\ell}$, at which $\bar{\rho}$ satisfies the level-raising condition of Theorem 15.47, raises the level to $Nr$, and compares the character groups of the toric parts of $J_0(Nr)$ at $q$ and of the Jacobian of the Shimura curve attached to the quaternion algebra of discriminant $qr$ at $r$, using the uniformisations of Cerednik and Drinfeld ("Ribet's exact sequence"); multiplicity one then transfers $\bar{\rho}$ to the Shimura curve, and back to level $Nr/q$, from which Mazur's principle removes $r$. See Ribet [156] (which also treats $\ell \mid N$ under conditions given there; the case stated here, $\ell \nmid N$, is the one the paper uses), [162] for a proof without multiplicity one, and [38]. We do not prove it. □

*Remark* 15.41 (Carayol's lemma on characters). Carayol [22] shows that a representation arising from an eigenform of weight $k \geq 2$, level $M$ and character $\varepsilon$ also arises in the same weight and level with any character $\varepsilon'$ having the same reduction modulo $\lambda$ as $\varepsilon$ (under hypotheses stated in the source, which we have not reproduced). Taking $\varepsilon'$ to be the Teichmüller lift of $\bar{\varepsilon}$, which has order prime to $\ell$, explains the clause "with nebentypus of order prime to $\ell$" in Theorem 15.42. The underlying mechanism is the lifting lemma of Deligne and Serre: a system

of Hecke eigenvalues modulo $\lambda$ occurring in the reduction of a Hecke-stable lattice lifts to an eigenform in characteristic 0.

**Theorem 15.42** (the optimal level; Ribet, Carayol, Diamond)**.** *Let $\ell$ be odd and $\bar\sigma\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ irreducible and modular, and if $\ell = 3$ assume that $\bar\sigma|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))}$ is absolutely irreducible. Then $\bar\sigma$ arises from a newform of weight 2 and level $N(\bar\sigma)\ell^{\delta(\bar\sigma)}$ with character of order prime to $\ell$, where $\delta(\bar\sigma) \in \{0,1\}$ and $\delta(\bar\sigma) = 0$ when $\bar\sigma|_{D_\ell}$ is finite flat, that is, isomorphic to the generic fibre of a finite flat group scheme over $\mathbb{Z}_\ell$.*

We do not prove this; it is [38, Theorem 3.15], as the paper quotes it, combining Ribet [156], Carayol [22] and Diamond [47]. The paper says "good" for the condition giving $\delta = 0$; [38] state that condition in a more general form, which we have not reproduced, and the finite flat case stated here is the one the paper uses (Corollary 15.43). Two remarks on the hypothesis at $\ell = 3$, which is Diamond's. (a) For $\bar\sigma = \bar\rho_{E,3}$ irreducible it holds exactly when the image is not $N_s(\mathbb{F}_3)$, the normaliser of a split Cartan subgroup (Chapter 11, the proposition on the image 3Ns). (b) For $\ell \geq 5$, $\ell \nmid N(E)$ and $\bar\rho_{E,\ell}$ irreducible, the analogous condition on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$ holds automatically (Chapter 11), although the theorem does not need it.

**Corollary 15.43** (the specialisation to elliptic curves)**.** *Let $\ell$ be odd and $E'/\mathbb{Q}$ of conductor $N'$ with $\ell \nmid N'$, $\bar\sigma = \bar\rho_{E',\ell}$ irreducible and, if $\ell = 3$, of image not $N_s(\mathbb{F}_3)$. Then there are a newform $g$ of weight 2 on $\Gamma_0(N(\bar\sigma))$ (trivial character) and a prime $\lambda \mid \ell$ of its coefficient field with $\bar\rho_{g,\lambda} \simeq \bar\sigma \otimes_{\mathbb{F}_\ell} k_\lambda$; so $a_p(g) \equiv a_p(E') \pmod\lambda$ for all $p \nmid \ell N'$. Moreover $N(\bar\sigma) \mid N'$, and $N(\bar\sigma)$ is prime to every prime at which $\bar\sigma$ is unramified.*

*Proof.* $\bar\sigma$ is modular: $\bar\sigma \simeq \bar\rho_{f_{E'},\ell}$ by modularity of $E'$ and Theorem 15.12. It is odd, and irreducible, hence absolutely irreducible (Proposition 15.9(ii)). At $\ell = 3$ the hypothesis on the image is equivalent to absolute irreducibility on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ (Chapter 11). $E'$ has good reduction at $\ell$, so $\bar\sigma|_{D_\ell} = E'[\ell]$ is the generic fibre of the finite flat group scheme $\mathcal{E}'[\ell]$ of the Néron model, and $\delta(\bar\sigma) = 0$ (Theorem 15.31(i); Chapter 11). Theorem 15.42 gives a newform $g$ of weight 2, level $N(\bar\sigma)$ and character $\psi$ of order prime to $\ell$, with $\bar\sigma \simeq \bar\rho_{g,\lambda} \otimes \overline{\mathbb{F}}_\ell$. By Theorem 15.8 (weight 2), $\det\bar\rho_{g,\lambda} = \bar\psi\bar\chi_\ell$, while $\det\bar\sigma = \bar\chi_\ell$; so $\bar\psi = 1$. The values of $\psi$ are roots of unity of order prime to $\ell$, and reduction modulo $\lambda$ is injective on them ($X^m - 1$ is separable over $\mathbb{F}_\ell$ for $\ell \nmid m$); so $\psi = 1$ and $g$ is on $\Gamma_0(N(\bar\sigma))$. Both $\bar\rho_{g,\lambda}$ and $\bar\sigma \otimes k_\lambda$ are absolutely irreducible and become isomorphic over $\overline{\mathbb{F}}_\ell$, so they have the same characteristic polynomials and are isomorphic over $k_\lambda$ (Brauer-Nesbitt, Chapter 4); taking traces of Frobenius gives the congruences. Finally $N(\bar\sigma) \mid N'$ by Proposition 15.36 for $f_{E'}$, and the last claim is Definition 15.35(a). □

*In the paper* 15.44. The optimal-level paragraph of Section 3: "For $\bar\sigma = \bar\rho_{E',\ell}$ with $E'$ an elliptic curve of conductor prime to $\ell$, the restriction to $D_\ell$ comes from the finite flat group scheme $E'[\ell]$ and is good, and $\det\bar\sigma = \bar\chi_\ell$ forces the nebentypus, which is trivial modulo $\lambda$ and of order prime to $\ell$, to be trivial. So in that case $\bar\sigma$ arises from a newform of weight 2 on $\Gamma_0(N(\bar\sigma))$." This is Corollary 15.43; its proof spells out the two compressed steps (the determinant of $\bar\rho_g$, and injectivity of reduction on roots of unity of order prime to $\ell$). The paper states the extra hypothesis at $\ell = 3$ as absolute irreducibility on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, exactly as in Theorem 15.42.

**Example 15.45** (level lowering on the paper's curves)**.** The Serre conductors below are computed by the formula $a_p(\bar\rho_{E,\ell}) = f_p - \dim \Phi_p^{\mathrm{geom}}[\ell]$ of Chapter 11; each congruence $a_p(E) \equiv a_p(g) \pmod\ell$ (twisted by $\chi$ where indicated) was checked with PARI/GP for all primes $p < 1000$ not dividing $\ell N$.

| $E$, $\ell$ | reduction types | $N(\bar\rho_{E,\ell})$ | newform | mechanism |
|---|---|---|---|---|
| `121c1`, 3 | IV at 11 | 11 | $f_{11a}$ | additive $q^2 \to q$ (Lemma lem:IV) |
| `77c1`, 3 | $\mathrm{I}_3$ at 7, $\mathrm{I}_2$ at 11 | 11 | $f_{11a}$ | Ribet at $q = 7 \equiv 1 \pmod 3$ |
| `52a1`, 3 | $\mathrm{IV}^*$ at 2, $\mathrm{I}_2$ at 13 | 26 | $f_{26b}$ | additive $q^2 \to q$ at $q = 2$ |
| `121a1`, 3 | II at 11 | 121 | none lower | twist: $N(\bar\rho \otimes \chi_{-11}) = 11$, $f_{11a}$ |
| `539c1`, 3 | $\mathrm{I}_3^*$ at 7, $\mathrm{I}_2$ at 11 | 539 | none lower | twist: $N(\bar\rho \otimes \chi_{-7}) = 11$, $f_{11a}$ |
| `140b1`, 5 | $\mathrm{IV}^*$ at 2, $\mathrm{I}_1$ at 5, $\mathrm{I}_5$ at 7 | 4 | $f_{20a}$, level 20 | $5 \mid N$, $\bar\rho\vert_{D_5}$ not finite |
| `11a1`, 5 | $\mathrm{I}_5$ at 11 | 1 | none | $\bar\rho$ reducible |

For `52a1` the other newform of level 26, $f_{26a}$, is not congruent to $f_{52a}$ modulo 3; the lowering at $q = 2$ uses that the image of $I_2$ on $T_3E$ is cyclic of order 3 for type $\mathrm{IV}^*$ at 2 (Chapter 11), so $E[3]^{I_2}$ is a line and the Serre exponent at 2 is 1. For `140b1` at $\ell = 5$, $5 \mid N$ and $\bar\rho|_{D_5}$ is not finite ($5 \nmid v_5(\Delta) = 1$), so Corollary 15.43 does not apply; the newform has level $N(\bar\rho) \cdot 5 = 20$ (there is no newform of weight 2 and level 4), and indeed $a_p(f_{20a}) \equiv a_p(\texttt{140b1}) \pmod 5$ for $p \nmid 70$, while at $p = 7$ the congruence is the level-raising one, $a_7(f_{20a}) = 2 \equiv (7+1)a_7(\texttt{140b1}) = -8 \pmod 5$. For `11a1` at $\ell = 5$, $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ (image `5Cs.1.1`), so $a_p \equiv 1 + p \pmod 5$ for all $p \neq 5, 11$; there is no newform of level 1, consistent with Theorem 15.42, which needs irreducibility.

*Remark* 15.46 (history). Serre [176] conjectured that an odd irreducible $\bar\sigma$ arises from a newform of the optimal weight, level and character, and observed that the "epsilon conjecture" (lowering the level of the representation attached to the Frey curve of a putative solution of Fermat's equation) would, together with modularity, prove Fermat's Last Theorem. Mazur's principle handled $q \not\equiv 1 \pmod \ell$; Ribet [156] proved the epsilon conjecture, and with it the implication. Carayol [22] settled the conductor and character side, and Diamond [47] the remaining cases of the level, giving Theorem 15.42. Multiplicity one ($J_0(N)[\mathfrak{m}]$ two-dimensional for non-Eisenstein $\mathfrak{m}$, Chapter 14) was used in Ribet's original argument and removed in [162]. Khare and Wintenberger proved the existence part of Serre's conjecture (Theorem 15.38).

## 15.7 Raising the level

The converse operation adds a prime $q \nmid N\ell$ to the level. It is possible exactly when $a_q^2 \equiv (q+1)^2$, and the quantity $(q+1)^2 - a_q^2$ reappears in the paper as the level-raising Euler factor of Theorem thm:dfg14: depleting $f$ at a good prime is "raising its level", and the congruence ideal grows by the factor. The necessity of the congruence is also the step of Lemma lem:old at a prime $p \,\|\, N$ at which $\bar\rho$ is unramified, read backwards.

**Theorem 15.47** (Ribet's level-raising theorem)**.** *Let $f$ be a newform of weight 2 on $\Gamma_0(N)$, $\lambda \mid \ell$ with $\ell$ odd and $\bar\rho = \bar\rho_{f,\lambda}$ irreducible, and $q \nmid N\ell$ a prime.*

*(i) (Necessity.) If $\bar\rho$ arises from a newform $g$ of weight 2 on $\Gamma_0(M)$ with $q \mid M \mid Nq$, then $a_q(f) \equiv (q+1)a_q(g) \pmod{\lambda'}$ for a prime $\lambda' \mid \lambda$ of a common coefficient ring, with $a_q(g) = \pm 1$; hence $a_q(f)^2 \equiv (q+1)^2 \pmod \lambda$.*

*(ii) (Sufficiency.) Conversely, if $a_q(f)^2 \equiv (q+1)^2 \pmod \lambda$, then $\bar\rho$ arises from a newform of weight 2 on $\Gamma_0(M)$ with $q \mid M \mid Nq$.*

*Proof.* (i) $q \,\|\, M$, so $g$ has type (St) at $q$ (Theorem 15.20). As $\bar\rho_g \simeq \bar\rho$ is unramified at $q$, part (c) of that theorem says that $\bar\rho(\mathrm{Frob}_q)$ has the eigenvalues $a_q(g)q$ and $a_q(g)$ modulo $\lambda'$; its trace is $a_q(f)$ modulo $\lambda'$ (as $q \nmid N\ell$). So $a_q(f) \equiv (q+1)a_q(g)$, and squaring, with $a_q(g)^2 = 1$, $a_q(f)^2 \equiv (q+1)^2$

modulo $\lambda' \cap \mathcal{O}_f = \lambda$. (ii) We do not prove this; it is Ribet's theorem [160], whose proof uses Ihara's lemma (Lemma 15.51) to show that the image of the two degeneracy maps from level $N$ to level $Nq$ meets the $q$-new part in a module whose Hecke eigenvalues modulo $\lambda$ are those of $f$ exactly when $a_q^2 \equiv (q+1)^2$. Diamond and Taylor [53] prove a general form, including the raising of levels by $q^2$. We have not checked the precise hypotheses on $\ell$ in these two sources against the form stated here (the paper uses only the necessity (i), proved above, and the level-raising factor of Theorem thm:dfg14). □

The condition has a transparent meaning. With $\alpha, \beta$ the eigenvalues of $\rho_{f,\lambda}(\mathrm{Frob}_q)$, so $\alpha + \beta = a_q$ and $\alpha\beta = q$,

$$(1-\alpha)(1-\beta) = q+1-a_q, \quad (1+\alpha)(1+\beta) = q+1+a_q, \quad (q+1)^2 - a_q^2 = (1-\alpha)(1-\beta)(1+\alpha)(1+\beta).$$

So $a_q^2 \equiv (q+1)^2 \pmod{\lambda}$ says exactly that 1 or $-1$ is an eigenvalue of $\bar\rho(\mathrm{Frob}_q)$ (after enlarging $\mathcal{O}$); the other eigenvalue is then $q$, resp. $-q$, since the product is $q$. This is precisely the shape forced by part (c) of Theorem 15.20 on a form of type (St) at $q$ with unramified reduction: eigenvalues $a_q(g)$ and $a_q(g)q$ with $a_q(g) = \pm 1$. For example, for $f_{11a}$ at $q = 7$: $X^2+2X+7 \equiv X^2-X+1 = (X+1)^2 \pmod 3$, so $\bar\rho(\mathrm{Frob}_7)$ has the double eigenvalue $-1$, and $-7 \equiv -1$ as well; the matching form is $f_{77c}$ with $a_7(f_{77c}) = -1$ (Example 15.52).

**Proposition 15.48** (the level-raising factor)**.** *Let $p \nmid N$, and $\alpha$, $\beta$ the roots of $X^2 - a_p(f)X + p$. Then*

$$\Big(1 - \frac{\alpha}{\beta p}\Big)\Big(1 - \frac{1}{p}\Big)\Big(1 - \frac{\beta}{\alpha p}\Big) = \frac{p-1}{p} \cdot \frac{(p+1)^2 - a_p(f)^2}{p^2},$$

*a nonzero element of $K_f$. It is $L_p(A_f, 1)^{-1} = L_p^{\mathrm{nv}}(A_f, 1)^{-1}$, the eigenvalues of either Frobenius on $\mathrm{ad}^0 V$ being $\alpha/\beta$, 1, $\beta/\alpha$. For $\ell \neq p$ and $\lambda \mid \ell$, its $\lambda$-adic valuation is $v_\lambda(p-1) + v_\lambda\big((p+1)^2 - a_p^2\big)$, and the second term is positive exactly when the level-raising condition of Theorem 15.47 holds at $p$. At a prime $p \,\|\, N$ the corresponding factor is $L_p(A_f, 1)^{-1} = 1 - p^{-2}$, and at $p^2 \mid N$ the naive factor is 1.*

*Proof.* $\alpha\beta = p$ and $\alpha^2 + \beta^2 = a_p^2 - 2p$, so

$$\Big(1 - \frac{\alpha}{\beta p}\Big)\Big(1 - \frac{\beta}{\alpha p}\Big) = 1 - \frac{\alpha^2+\beta^2}{\alpha\beta p} + \frac{1}{p^2} = 1 - \frac{a_p^2 - 2p}{p^2} + \frac{1}{p^2} = \frac{p^2 + 2p + 1 - a_p^2}{p^2} = \frac{(p+1)^2 - a_p^2}{p^2},$$

and multiplying by $1 - \frac{1}{p} = \frac{p-1}{p}$ gives the identity. It is nonzero because $|a_p| \le 2\sqrt{p} < p+1$ under every complex embedding (Chapter 12). On $\mathrm{ad}^0 V$ the arithmetic Frobenius has eigenvalues $\alpha/\beta$, 1, $\beta/\alpha$, an inversion-stable set, so the geometric Frobenius has the same ones, and $L_p(A_f, s)^{-1} = (1 - \frac{\alpha}{\beta}p^{-s})(1 - p^{-s})(1 - \frac{\beta}{\alpha}p^{-s})$; at $s = 1$ this is the left side. The valuation follows since $p$ is a unit at $\lambda$. The factor at $p \,\|\, N$ is Corollary 15.25 at $s = 1$. □

A numerical instance, used several times below: for $f = f_{11a}$ and $p = 7$, $a_7 = -2$, so the factor is $\frac{6}{7} \cdot \frac{64-4}{49} = \frac{360}{343}$, with $v_3(360/343) = v_3(6) + v_3(60) = 1 + 1 = 2$. The two terms have different origins: $v_3(60) = 1$ is the level-raising congruence $3 \mid (7+1)^2 - a_7^2$ just discussed, and $v_3(6) = 1$ records $7 \equiv 1 \pmod 3$. At $p = 3 = \ell$ the same formula gives $\frac{2}{3} \cdot \frac{16-1}{9} = \frac{10}{9}$, of 3-adic valuation $-2$, which is why Theorem thm:dfg14 below needs $\ell \notin \Sigma$.

The term $v_\lambda(p-1)$ has a similar meaning, which we do not use: it is positive only when $p \equiv 1 \pmod \ell$, and then $\bar\rho$ can also arise at level $Np^2$ from forms of type (RPS) at $p$ whose inertial characters have $\ell$-power order (they reduce to the trivial character); this is part of the theory of non-optimal levels of Diamond and Taylor [53].

**Theorem 15.49** (Diamond, Flach and Guo; the paper's Theorem thm:dfg14)**.** *Let $f$ be a newform of weight 2 on $\Gamma_0(M)$ with rational coefficients, $\lambda = \ell$ a prime not dividing $2M$ with $\bar{\rho}_f$ irreducible, and $\Sigma$ a finite set of primes with $\ell \notin \Sigma$. Let $M^\Sigma = M\prod_{p\in\Sigma} p^{\delta_p}$, $f^\Sigma$ the $\Sigma$-depleted eigenform of level $M^\Sigma$, and $\eta^\Sigma_{f,\lambda}$ the congruence ideal of [49] attached to $f^\Sigma$ (defined in Chapter 20). Then*

$$\eta^\Sigma_{f,\lambda} = \eta^\emptyset_{f,\lambda} \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}.$$

We do not prove this here; it is [49, Proposition 1.4(c)] (checked against the published text), and Chapter 20 sketches the proof as Diamond, Flach and Guo give it (the degeneracy maps, the homothety of $G_{\mathbb{Q}}$-stable lattices when $\bar{\rho}_f$ is irreducible, and the primitivity $\gamma(f) = f^\Sigma$ read through the integral comparison isomorphism; Ihara's lemma is not used). The hypothesis $\ell \notin \Sigma$ cannot be dropped (for $f = f_{11\mathrm{a}}$, $\ell = 3$, $\Sigma = \{3\}$ the right side is $3^{-2}\mathbb{Z}_3$, which is not integral; see Chapter 20; in [49] it is a standing assumption of Section 1.7.3), and every use in the paper has $\ell \notin \Sigma$, as the paper notes after the theorem. By Proposition 15.48 and Corollary 15.25, adding to $\Sigma$ a prime with $\delta_p = 0$ changes neither the level nor the ideal, a good prime contributes $\frac{p-1}{p}\cdot\frac{(p+1)^2-a_p^2}{p^2}$, and a multiplicative prime $1-p^{-2}$. The reading for this chapter: $f^\Sigma$ is "$f$ with its level raised at $\Sigma$" (at a good $p$ it is the eigenform of level $Mp^2$ in the span of $f(z)$, $f(pz)$, $f(p^2z)$ with $U_p$-eigenvalue 0), and the change of the congruence ideal is the level-raising factor.

*In the paper* 15.50. The paragraph after Theorem thm:dfg14: "adding a good prime $p$ multiplies the ideal by $(1-\frac{\alpha}{\beta}p^{-1})(1-p^{-1})(1-\frac{\beta}{\alpha}p^{-1})$, Wiles's level-raising factor; adding a multiplicative prime multiplies it by $1-p^{-2}$." This is Proposition 15.48. The same identity, with $a_q(E')$ for the twist $E'$ of good reduction, gives the $\mathrm{I}_0^*$ entry of the table of Section 1 and of Lemma lem:euler, and Step 3 of the proof of Theorem thm:quant.

**Lemma 15.51** (Ihara's lemma)**.** *Let $q \nmid N\ell$ be prime, $\alpha_1, \alpha_q\colon X_0(Nq) \to X_0(N)$ the two degeneracy maps ($z \mapsto z$ and $z \mapsto qz$), and $\mathfrak{m}$ a maximal ideal of $\mathbb{T}_N$ of residue characteristic $\ell$ that is non-Eisenstein (Definition 15.15). Then the map $H^1(X_0(N), \mathbb{F}_\ell)^2_{\mathfrak{m}} \to H^1(X_0(Nq), \mathbb{F}_\ell)$, $(x, y) \mapsto \alpha_1^* x + \alpha_q^* y$, is injective.*

This follows from Ribet's form of the lemma: the kernel of $J_0(N)^2 \to J_0(Nq)$, $(x, y) \mapsto \alpha_1^* x + \alpha_q^* y$, is the Shimura subgroup $\mathcal{S}_N$ embedded antidiagonally, and only Eisenstein maximal ideals are in the support of $\mathcal{S}_N$. We prove neither statement; see Ribet [160] and Diamond and Taylor [53] (Ihara's original argument uses the congruence subgroup property of $\mathrm{SL}_2(\mathbb{Z}[1/q])$). It is an input of Theorem 15.47(ii). The proof of part (c) of Theorem thm:dfg14 sketched in Chapter 20, which follows the published proof of Diamond, Flach and Guo, does not use it.

**Example 15.52** (level raising for $f_{11a}$ and $f_{20a}$)**.** (Computed with PARI/GP on the new subspace of level $11q$, resp. $20q$: each newform given by `mfeigenbasis`, with coefficient field from `mffields`, was reduced modulo every prime above $\ell$ of that field and compared with $a_p(f)$ for all primes $p < 200$, $p \nmid \ell Nq$.) For $f = f_{11a}$ and $\ell = 3$:

| $q$ | $a_q(f_{11a})$ | $(q+1)^2 - a_q^2$ | $3 \mid ?$ | newforms of level $11q$ congruent to $f_{11a}$ modulo a prime above 3 |
|---|---|---|---|---|
| 2 | $-2$ | 5 | no | none (no newforms of level 22) |
| 5 | 1 | 35 | no | none |
| 7 | $-2$ | 60 | yes | $f_{77c}$ (rational), $a_7(f_{77c}) = -1$ |
| 13 | 4 | 180 | yes | one form with quartic coefficient field, at one of its two primes above 3 |
| 17 | $-2$ | 320 | no | none |
| 19 | 0 | 400 | no | none |
| 23 | $-1$ | 575 | no | none |
| 29 | 0 | 900 | yes | two Galois orbits (coefficient fields of degrees 3 and 7) |
| 31 | 7 | 975 | yes | one Galois orbit (coefficient field of degree 11) |
| 37 | 3 | 1435 | no | none |

So, for $q < 40$, $q \neq 3, 11$, the newforms congruent to $f_{11a}$ at level $11q$ occur exactly when $3 \mid (q+1)^2 - a_q^2$, as Theorem 15.47 predicts; at $q = 7$, $a_7(f_{11a}) = -2 \equiv (7+1)a_7(f_{77c}) = -8 \pmod 3$. At level 143 the congruent form is not the rational newform $f_{143a}$: $a_2(f_{143a}) = 0 \not\equiv -2 \pmod 3$. For $f = f_{20a}$ and $\ell = 5$ ($\bar{\rho}_{20a,5}$ is irreducible): $q = 3$ gives 12 and there are no newforms of level 60; $q = 7$ gives 60, and $f_{140b}$ is congruent to $f_{20a}$ modulo 5, with $a_7(f_{20a}) = 2 \equiv 8a_7(f_{140b}) = -8$; $q = 11$ and $q = 13$ give 144 and 192, and no newform of level 220 or 260 is congruent. The congruence $f_{11a} \equiv f_{77c}$ modulo 3 underlies the paper's Example for `539c1`, whose twist by $\chi_{-7}$ is `77c1`.

## 15.8 Congruences between eigenforms: the congruence criterion and the oldform lemma

This section proves the two lemmas with which Section 3 of the paper begins. Lemma lem:crit identifies the congruence number with the congruence ideal of $\phi_f$ in the full Hecke algebra and turns a congruence between $f$ and an eigenform orthogonal to $f$, *for every coefficient*, into the divisibility $\ell \mid r$. Lemma lem:old produces such an eigenform from a newform of lower level that is congruent to $f$ only at the primes $p \nmid \ell N$, by choosing the eigenvalues of the operators $U_p$ on the oldforms prime by prime. The gap between "every $n$" and "almost every prime $p$" is the whole point of the second lemma. The paper's statements are proved here in full (the book's Lemma 15.55 adds $I = \mathrm{Ann}_{\mathbb{T}}(J)$ and the description of all congruence moduli); Chapter 14 relates $r$ to the modular degree, and Chapter 20 restates Lemma lem:crit in the language of congruence modules.

**Definition 15.53** (the congruence number)**.** Let $f$ be a newform of weight 2 on $\Gamma_0(N)$ with rational coefficients (so $f = f_E$ for an elliptic curve $E$ of conductor $N$). $S_2(\Gamma_0(N); \mathbb{Z})$ is the lattice of forms with integral $q$-expansion (Chapter 12), $f^{\perp} \subset S_2(\Gamma_0(N))$ the orthogonal complement of $\mathbb{C}f$ for the Petersson product, and $L = f^{\perp} \cap S_2(\Gamma_0(N); \mathbb{Z})$. The *congruence number* $r = r_f = r_E$ is the largest integer $r \geq 1$ such that $f \equiv g \pmod r$ coefficientwise for some $g \in L$; by Lemma 15.55 it exists, and the moduli of such congruences are exactly the divisors of $r$. It depends only on $f$, that is, on the isogeny class.

**Theorem 15.54** (Agashe, Ribet and Stein)**.** *Let $E_0$ be the optimal curve of conductor $N$ and $r = r_{E_0}$. Then $\deg \varphi_{E_0} \mid r$, and $v_p(\deg \varphi_{E_0}) = v_p(r)$ for every prime $p$ with $p^2 \nmid N$.*

We do not prove this here; it is [3, Theorem 2.2], stated and discussed in Chapter 14. Example (computed with PARI/GP as the index of $\mathbb{Z}f \oplus L$ in $S_2(\Gamma_0(N);\mathbb{Z})$, by Lemma 15.55(i)): for `99a1` (the 99A1 of [3], $N = 3^2 \cdot 11$), $\deg\varphi = 4$ and $r = 12$, so the 3-adic valuations differ, as $9 \mid N$ allows; for `539d1`, $\deg\varphi = 72$ and $r = 504 = 7 \cdot 72$, with $7^2 \mid 539$; for `726a1`, $\deg\varphi = 144$ and $r = 1584 = 11 \cdot 144$, with $11^2 \mid 726$. For `11a1`, `52a1`, `77c1`, `121a1`, `121c1`, `539c1` the same computation gives $r = 1, 3, 6, 6, 6, 288$, equal to $\deg\varphi$ in each case.

**Lemma 15.55** (the congruence criterion; the paper's Lemma lem:crit)**.** *Let $f$, $L$, $r$ be as in Definition 15.53, $\mathbb{T}$ the full Hecke algebra of level $N$, $\phi_f\colon \mathbb{T} \to \mathbb{Z}$, $T_n \mapsto a_n(f)$, $I = \ker\phi_f$ and $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$. Write $S = S_2(\Gamma_0(N);\mathbb{Z})$, $S_{\mathbb{Q}} = S \otimes \mathbb{Q}$, and $W = L \otimes \mathbb{Q} = f^\perp \cap S_{\mathbb{Q}}$, the forms with rational coefficients orthogonal to $f$; Step 1 of the proof shows $f^\perp = W \otimes \mathbb{C}$.*

*(i) $I \cap J = 0$, $J = \mathrm{Ann}_{\mathbb{T}}(I)$, $I = \mathrm{Ann}_{\mathbb{T}}(J)$, and $\mathbb{T}/(I+J)$ is a finite cyclic group of order $r$; equivalently $r\mathbb{Z} = \phi_f(J) = \phi_f(\mathrm{Ann}_{\mathbb{T}} I)$, the congruence ideal of $\phi_f$. Moreover $[S_2(\Gamma_0(N);\mathbb{Z}) : \mathbb{Z}f \oplus L] = r$, and $f \equiv g \pmod m$ for some $g \in L$ if and only if $m \mid r$.*

*(ii) Let $\mathcal{O}$ be the ring of integers of a number field, $\lambda \mid \ell$ a prime of $\mathcal{O}$, and $g \in W \otimes \overline{\mathbb{Q}} = (S_{\mathbb{Q}} \otimes \overline{\mathbb{Q}}) \cap f^\perp$ (inside $S_{\mathbb{Q}} \otimes \mathbb{C} = S_2(\Gamma_0(N))$) a normalised eigenform of* all *of $\mathbb{T}$ with coefficients in $\mathcal{O}$ such that $a_n(g) \equiv a_n(f) \pmod\lambda$ for* every *$n \geq 1$. Then $\ell \mid r$.*

*(iii) The same holds over $\mathbb{Z}_\ell$: with $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$, $I_\ell = I \otimes \mathbb{Z}_\ell$, $J_\ell = J \otimes \mathbb{Z}_\ell$, one has $\mathbb{T}_\ell/(I_\ell + J_\ell) \simeq \mathbb{Z}/\ell^{v_\ell(r)}$, $\phi_f(\mathrm{Ann}_{\mathbb{T}_\ell} I_\ell) = \ell^{v_\ell(r)}\mathbb{Z}_\ell$, and $\ell^{v_\ell(r)}$ is the largest power $\ell^t$ such that $f \equiv h \pmod{\ell^t}$ for some $h \in L \otimes \mathbb{Z}_\ell = (S \otimes \mathbb{Z}_\ell) \cap (W \otimes \mathbb{Q}_\ell)$ (inside $S \otimes \mathbb{Q}_\ell$).*

*Proof.* With $S$, $S_{\mathbb{Q}}$ as in the statement, $S_2(\Gamma_0(N)) = S \otimes \mathbb{C}$ and $S$ is $\mathbb{T}$-stable (Chapters 12 and 13). $\mathbb{T}$ is a subring of $\mathrm{End}_{\mathbb{Z}}(S)$, so it is free of finite rank over $\mathbb{Z}$ and $\mathbb{T} \otimes \mathbb{Q} \subseteq \mathrm{End}_{\mathbb{Q}}(S_{\mathbb{Q}})$ acts faithfully.

*Step 1: $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$.* First, $f^\perp$ is $\mathbb{T}$-stable. The adjoint $t \mapsto t^\dagger$ for the Petersson product is additive and reverses products, so it suffices that $T_n^\dagger$ ($(n,N)=1$) and $U_p^\dagger$ ($p \mid N$) have $f$ as an eigenvector, these operators generating $\mathbb{T}$ as a ring (Proposition 15.4(i)). By Chapter 13, $T_n^\dagger = T_n$ for $(n,N) = 1$ and $U_p^\dagger = w_N U_p w_N$, where $w_N f = \varepsilon_N(f) f$ with $\varepsilon_N(f) = \pm 1$; so $U_p^\dagger f = \varepsilon_N(f)^2 a_p(f) f = a_p(f) f$. Hence for $t \in \mathbb{T}$ and $g \in f^\perp$, $\langle tg, f\rangle = \langle g, t^\dagger f\rangle = 0$. Next, $\mathbb{T} \otimes \mathbb{Q}$ is a finite-dimensional commutative $\mathbb{Q}$-algebra, so $\mathbb{T} \otimes \mathbb{Q} = \prod_{\mathfrak{n}} (\mathbb{T} \otimes \mathbb{Q})_{\mathfrak{n}}$ over its maximal ideals, and $S_{\mathbb{Q}} = \bigoplus_{\mathfrak{n}} (S_{\mathbb{Q}})_{\mathfrak{n}}$, the factor $(\mathbb{T} \otimes \mathbb{Q})_{\mathfrak{n}}$ acting faithfully on the generalised eigenspace $(S_{\mathbb{Q}})_{\mathfrak{n}}$ (Chapter 3). Let $\mathfrak{n}_f = \ker(\phi_f \otimes \mathbb{Q})$. The generalised eigenspace $(S_{\mathbb{Q}})_{\mathfrak{n}_f} \otimes \mathbb{C}$ lies in the generalised eigenspace of the $T_p$, $p \nmid N$, for the eigenvalues $a_p(f)$, which is $V_f = \mathbb{C}f$ (Proposition 15.4(iv), strong multiplicity one: $f$ has level $N$, so $V_f$ is spanned by $f$ alone). So $(S_{\mathbb{Q}})_{\mathfrak{n}_f} = \mathbb{Q}f$, and the factor $(\mathbb{T} \otimes \mathbb{Q})_{\mathfrak{n}_f}$ embeds in $\mathrm{End}_{\mathbb{Q}}(\mathbb{Q}f) = \mathbb{Q}$; it is therefore $\mathbb{Q}$, acting through $\phi_f$. Let $\mathbb{T}' = \prod_{\mathfrak{n} \neq \mathfrak{n}_f} (\mathbb{T} \otimes \mathbb{Q})_{\mathfrak{n}}$ and $W = \bigoplus_{\mathfrak{n} \neq \mathfrak{n}_f} (S_{\mathbb{Q}})_{\mathfrak{n}}$ (we show below that this is the $W$ of the statement): then $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$, $S_{\mathbb{Q}} = \mathbb{Q}f \oplus W$, and $\mathbb{T}'$ acts faithfully on $W$ with its unit acting as the identity. Let $e_f = (1,0)$. It projects $S_{\mathbb{Q}}$ onto $\mathbb{Q}f$ along $W$. Since $f^\perp$ is $\mathbb{T}$-stable, $e_f(f^\perp) \subseteq f^\perp \cap \mathbb{C}f = 0$, so $f^\perp \subseteq \ker e_f = W \otimes \mathbb{C}$, and the two have dimension $\dim S_{\mathbb{Q}} - 1$; hence $f^\perp = W \otimes \mathbb{C}$, $W = f^\perp \cap S_{\mathbb{Q}}$ and $L = S \cap W$, so $W = L \otimes \mathbb{Q}$.

*Step 2: the ideals.* In $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$, $\phi_f(a,t') = a$. An element $(a,t')$ kills $f^\perp = W \otimes \mathbb{C}$ iff $t' = 0$. Hence $I = \mathbb{T} \cap (0 \times \mathbb{T}')$ and $J = \mathbb{T} \cap (\mathbb{Q} \times 0)$, and $I \cap J = 0$. Since $\mathbb{T}$ is a lattice in $\mathbb{T} \otimes \mathbb{Q}$, $I \otimes \mathbb{Q} = 0 \times \mathbb{T}'$ and $J \otimes \mathbb{Q} = \mathbb{Q} \times 0$ (tensoring with $\mathbb{Q}$ is exact), and $J \neq 0$. An element $(a,t')$ kills $I$ iff $t'\mathbb{T}' = 0$ iff $t' = 0$, so $\mathrm{Ann}_{\mathbb{T}}(I) = J$; it kills $J$ iff $a = 0$, so $\mathrm{Ann}_{\mathbb{T}}(J) = I$. $\mathbb{T}/I \simeq \phi_f(\mathbb{T}) = \mathbb{Z}$ (as $\phi_f(1) = 1$), and $\phi_f$ is injective on $J$; so $\mathbb{T}/(I+J) \simeq \mathbb{Z}/\phi_f(J)$ is cyclic, and finite because

$\phi_f(J) \neq 0$. Let $r' = \#\mathbb{T}/(I+J)$. Both $I$ and $J$ are saturated in $\mathbb{T}$ (if $mt \in J$ with $t \in \mathbb{T}$, $m \neq 0$, then $t \in \mathbb{T} \cap (\mathbb{Q} \times 0)$), so $\mathbb{T}/I$ and $\mathbb{T}/J$ are free $\mathbb{Z}$-modules.

*Step 3: the forms killed by $I$ and by $J$.* The pairing $S \times \mathbb{T} \to \mathbb{Z}$, $(g,t) \mapsto a_1(tg)$, is perfect (Ribet [159, Theorem 2.2], Abbes and Ullmo [1, Lemma 2.1]; proved in Chapter 13). For an ideal $\mathfrak{a}$ of $\mathbb{T}$, the orthogonal of $\mathfrak{a}$ is $S[\mathfrak{a}] = \{g : \mathfrak{a}g = 0\}$: if $a_1(tg) = 0$ for all $t \in \mathfrak{a}$ then $a_n(tg) = a_1(T_n tg) = 0$ for all $n$, since $T_n t \in \mathfrak{a}$, so $tg = 0$. Under $S \simeq \mathrm{Hom}(\mathbb{T}, \mathbb{Z})$, $S[\mathfrak{a}] \simeq \mathrm{Hom}(\mathbb{T}/\mathfrak{a}, \mathbb{Z})$ (Chapter 3, the algebra-module duality). A form is killed by $I$ iff it is killed by $I \otimes \mathbb{Q} = 0 \times \mathbb{T}'$, iff its $W$-component vanishes (the unit of $\mathbb{T}'$ is the identity on $W$); so $S[I] = S \cap \mathbb{Q}f = \mathbb{Z}f$, because $a_1(f) = 1$. A form is killed by $J$ iff it is killed by $e_f$, iff it lies in $W$; so $S[J] = S \cap W = L$.

*Step 4: the index.* The sequence $0 \to \mathbb{T} \to \mathbb{T}/I \times \mathbb{T}/J \to \mathbb{T}/(I+J) \to 0$, with first map $t \mapsto (t \bmod I, t \bmod J)$ (injective as $I \cap J = 0$) and second map $(x,y) \mapsto x - y$, is exact. (The composite is $t \mapsto t - t = 0$; the second map is onto; and if $x - y \in I + J$ with $x = t_1 \bmod I$, $y = t_2 \bmod J$, write $t_1 - t_2 = i + j$, $i \in I$, $j \in J$; then $t = t_1 - i = t_2 + j$ maps to $(x,y)$.) Applying $\mathrm{Hom}(\cdot, \mathbb{Z})$ gives the exact sequence

$$0 \to \mathrm{Hom}(\mathbb{T}/(I+J), \mathbb{Z}) \to \mathrm{Hom}(\mathbb{T}/I, \mathbb{Z}) \oplus \mathrm{Hom}(\mathbb{T}/J, \mathbb{Z}) \to \mathrm{Hom}(\mathbb{T}, \mathbb{Z})$$
$$\to \mathrm{Ext}^1(\mathbb{T}/(I+J), \mathbb{Z}) \to \mathrm{Ext}^1(\mathbb{T}/I \times \mathbb{T}/J, \mathbb{Z}).$$

Here $\mathrm{Hom}(\mathbb{T}/(I+J), \mathbb{Z}) = 0$ (a finite group has no nonzero maps to $\mathbb{Z}$), $\mathrm{Ext}^1(\mathbb{T}/I \times \mathbb{T}/J, \mathbb{Z}) = 0$ (free modules), and $\#\,\mathrm{Ext}^1(A, \mathbb{Z}) = \#A$ for finite $A$ (for $A = \mathbb{Z}/m$, apply $\mathrm{Hom}(\cdot, \mathbb{Z})$ to $0 \to \mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m \to 0$; in general decompose $A$). So $\mathrm{Hom}(\mathbb{T}/I, \mathbb{Z}) \oplus \mathrm{Hom}(\mathbb{T}/J, \mathbb{Z}) \to \mathrm{Hom}(\mathbb{T}, \mathbb{Z})$ is injective with cokernel of order $r'$ (Chapter 3, dualising an inclusion of lattices). Under the pairing this map is the sum map $\mathbb{Z}f \oplus L \to S$. So $[S : \mathbb{Z}f \oplus L] = r'$.

*Step 5: the congruences.* The map $S/(\mathbb{Z}f \oplus L) \to \mathbb{Q}f/\mathbb{Z}f$, $g \mapsto e_f g$, is well defined ($e_f$ kills $L$ and fixes $f$) and injective (if $e_f g = cf$ with $c \in \mathbb{Z}$ then $g - cf \in S \cap W = L$). So $S/(\mathbb{Z}f \oplus L)$ is a finite subgroup of $\mathbb{Q}f/\mathbb{Z}f \simeq \mathbb{Q}/\mathbb{Z}$, hence cyclic of order $r'$, generated by the class of some $h \in S$ with $e_f h = f/r'$. Then $g_0 = r'h - f$ satisfies $e_f g_0 = 0$, so $g_0 \in L$, and $f = r'h - g_0 \equiv -g_0 \pmod{r'}$ with $-g_0 \in L$. Conversely, if $f \equiv g_1 \pmod m$ with $g_1 \in L$, then $h_1 = (f - g_1)/m \in S$ and $e_f h_1 = f/m$, whose class in $\mathbb{Q}f/\mathbb{Z}f$ has order exactly $m$; so $m \mid r'$. Hence the moduli of congruences between $f$ and elements of $L$ are exactly the divisors of $r'$, the largest is $r = r'$, and (i) is proved.

*Step 6: (ii).* Let $\phi_g \colon \mathbb{T} \to \mathcal{O}$, $t \mapsto$ the eigenvalue of $t$ on $g$; it is a ring homomorphism with $\phi_g(T_n) = a_n(g) \in \mathcal{O}$ (Proposition 15.4(i), $a_1(g) = 1$). It kills $J$: for $t \in J$, $\phi_g(t)g = tg = 0$ as $g \in W \otimes \overline{\mathbb{Q}} \subseteq f^{\perp}$. The two ring homomorphisms $\mathbb{T} \to \mathcal{O}/\lambda$ obtained from $\phi_g$ and $\phi_f$ agree on every $T_n$, hence on $\mathbb{T}$; so $\phi_g(t) \equiv \phi_f(t) = 0 \pmod \lambda$ for $t \in I$. Hence $I + J$ lies in the kernel $\mathfrak{m}$ of $\mathbb{T} \to \mathcal{O}/\lambda$, $t \mapsto \phi_g(t) \bmod \lambda$. Its image is a subring of the finite field $\mathcal{O}/\lambda$, hence a finite integral domain, hence a field of characteristic $\ell$; so $\mathfrak{m}$ is a maximal ideal with $\ell \in \mathfrak{m}$, and $\mathbb{T}/(I+J)$ surjects onto $\mathbb{T}/\mathfrak{m} \neq 0$, a group of exponent $\ell$. So $\ell \mid \#\mathbb{T}/(I+J) = r$.

*Step 7: (iii).* $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$, so $I_\ell = \ker(\phi_f \otimes \mathbb{Z}_\ell)$, $\mathbb{T}_\ell/(I_\ell + J_\ell) = (\mathbb{T}/(I+J)) \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^{v_\ell(r)}$, $\mathrm{Ann}_{\mathbb{T}_\ell}(I_\ell) = J_\ell$ (annihilators of finitely generated ideals commute with flat base change), and $\phi_f(J_\ell) = r\mathbb{Z}_\ell = \ell^{v_\ell(r)}\mathbb{Z}_\ell$. The pairing $S \otimes \mathbb{Z}_\ell \times \mathbb{T}_\ell \to \mathbb{Z}_\ell$ is perfect, $L \otimes \mathbb{Z}_\ell = (S \otimes \mathbb{Z}_\ell) \cap (W \otimes \mathbb{Q}_\ell)$ (a saturated sublattice stays saturated), and Steps 3 to 5 hold verbatim over $\mathbb{Z}_\ell$, with $\mathbb{Q}_\ell f/\mathbb{Z}_\ell f \simeq \mathbb{Q}_\ell/\mathbb{Z}_\ell$ in Step 5, whose finite subgroups are cyclic; the index is now $\ell^{v_\ell(r)}$ (Chapter 3, flat base change). □

*In the paper* 15.56. Lemma lem:crit of the paper states (i) without $I = \mathrm{Ann}_{\mathbb{T}}(J)$ and the list of moduli, (ii), and (iii) in the words "The same statements hold with $\mathbb{Z}$ replaced by $\mathbb{Z}_\ell$ and $r$ by

its $\ell$-part". Its proof is Steps 1 to 6 in compressed form: "the $f$-line is $\mathbb{T}$-stable, and so is $f^\perp$, because $f$ is an eigenvector of the adjoints of the $T_n$ as well" is the first paragraph of Step 1; the decomposition $\mathbb{T}\otimes\mathbb{Q}=\mathbb{Q}\times\mathbb{T}'$ needs only that the $\phi_f$-generalised eigenspace is the line of $f$, which is strong multiplicity one at level $N$ (the full Hecke algebra may contain nilpotents, but not on that line); the finiteness of $\mathbb{T}/(I+J)$ is $\phi_f(J)\neq 0$ (Step 2), which the paper's "cyclic, of order $r'$" takes for granted. The hypothesis that $g$ be an eigenform of all of $\mathbb{T}$ and congruent to $f$ at every $n$, the operators $U_p$ ($p\mid N$) included, is used in Step 6 and cannot be weakened to the primes $p\nmid \ell N$: that weakening is what Lemma 15.58 repairs.

The smallest case in which every object of the lemma can be seen is $N=37$. Here $S_2(\Gamma_0(37))$ has dimension 2, spanned by the two rational newforms $f=f_{37a}$ and $g=f_{37b}$, with

$$f=q-2q^2-3q^3+2q^4-2q^5+6q^6-\cdots,\qquad g=q+q^3-2q^4-q^7-\cdots.$$

Distinct newforms of the same level are orthogonal, so $W=\mathbb{Q}g$ and $L=S\cap\mathbb{Q}g=\mathbb{Z}g$ ($a_1(g)=1$). Every coefficient of $f-g$ is even (checked with PARI/GP for $n\le 3000$, far beyond the Sturm bound $\frac{1}{6}\cdot 38<7$), so $(f-g)/2\in S$ and $f\equiv g \pmod 2$. Conversely, if $f\equiv cg \pmod m$ with $c\in\mathbb{Z}$, then $a_1$ gives $c\equiv 1$ and $a_2$ gives $-2\equiv 0 \pmod m$, so $m\mid 2$. Hence $r=2$, which is the index $[S:\mathbb{Z}f\oplus\mathbb{Z}g]$ (computed as in the Notes). In the coordinates $t\mapsto(\phi_f(t),\phi_g(t))$, which embed $\mathbb{T}$ in $\mathbb{Z}\times\mathbb{Z}$, the lemma says $I=0\times 2\mathbb{Z}$, $J=2\mathbb{Z}\times 0$ and $\mathbb{T}/(I+J)\simeq\mathbb{Z}/2$, which forces $\mathbb{T}=\{(a,b):a\equiv b \pmod 2\}$: this is the toy model of Exercise 15.91 with $r=2$. As 37 is prime, Theorem 15.54 gives $\deg\varphi_{37a1}=2$, as Cremona's tables confirm.

**Lemma 15.57** (the operators $U_p$ on the span of an oldform)**.** *Let $h$ be a newform of weight 2 on $\Gamma_0(M)$, $M\mid N$, $V_h$ the span of the $h(dz)$, $d\mid N/M$, and for $p\mid N/M$ let $e_p=v_p(N/M)$ and $V_{h,p}$ the space with basis $h_j$, $0\le j\le e_p$ (to be thought of as $h(p^j z)$).*

*(i) $\bigotimes_{p\mid N/M}V_{h,p}\to V_h$, $\otimes_p h_{j_p}\mapsto h(\prod_p p^{j_p}z)$, is an isomorphism under which $U_p$ ($p\mid N/M$) acts on the $p$-th factor alone by $U_p h_j=h_{j-1}$ ($j\ge 1$) and $U_p h_0=a_p(h)h_0-p\,h_1$ if $p\nmid M$, $U_p h_0=a_p(h)h_0$ if $p\mid M$; $T_n$ ($(n,N)=1$) acts by $a_n(h)$, and $U_p$ for $p\mid M$ with $v_p(M)=v_p(N)$ by $a_p(h)$.*

*(ii) On $V_{h,p}$ the characteristic polynomial of $U_p$, its eigenvalues and one eigenvector for each are as follows, where $\alpha_p,\beta_p$ are the roots of $X^2-a_p(h)X+p$:*

| *situation* | *characteristic polynomial* | *eigenvalue* | *eigenvector* |
|---|---|---|---|
| $p\nmid M$ | $X^{e_p-1}(X^2-a_pX+p)$ | $\alpha_p$; $\beta_p$ | $h_0-\beta_p h_1$; $h_0-\alpha_p h_1$ |
| $p\nmid M$, $e_p\ge 2$ | | $0$ | $h_0-a_p(h)h_1+p\,h_2$ |
| $p\,\|\,M$ | $X^{e_p}(X-a_p)$ | $a_p(h)=\pm1$; $0$ | $h_0$; $h_0-a_p(h)h_1$ |
| $p^2\mid M$ | $X^{e_p+1}$ | $0$ | $h_0$ ($a_p(h)=0$) |

*$U_p$ is not semisimple on $V_{h,p}$ when $0$ is a repeated root of its characteristic polynomial.*

*(iii) For every choice of one eigenvalue $\mu_p$ from the table for each $p\mid N/M$, the image $g$ of the tensor product of the corresponding eigenvectors is an eigenform of all of $\mathbb{T}$ with $U_p g=\mu_p g$ ($p\mid N/M$), $U_p g=a_p(h)g$ (other $p\mid N$), $T_n g=a_n(h)g$ ($(n,N)=1$); it has $a_1(g)=1$, and its coefficients lie in $\mathcal{O}[\alpha_p,\beta_p : p\mid N/M]$ if those of $h$ lie in $\mathcal{O}$.*

*Proof.* (i) The $h(dz)$ are linearly independent (Proposition 15.4(iv)). On $q$-expansions $a_n(U_pF)=a_{pn}(F)$; for $F=h(p^j d'z)$ with $p\nmid d'$ this gives $U_pF=h(p^{j-1}d'z)$ if $j\ge 1$, and $(U_p h)(d'z)$ if

$j=0$, so $U_p$ acts on the $p$-th factor only. For $j=0$: if $p \mid M$, $U_p$ at level $N$ and at level $M$ have the same effect on $q$-expansions, so $U_p h = a_p(h)h$; if $p \nmid M$, $a_n(T_p h) = a_{pn}(h) + p\, a_{n/p}(h)$ (the Hecke operator at level $M$), so $U_p h = T_p h - p\, h(pz) = a_p(h)h - p\, h(pz)$. The statements about $T_n$ and about $U_p$ with $e_p = 0$ are Proposition 15.4(iv) and Chapter 13. (ii) In the basis $h_0, \dots, h_e$ ($e = e_p$) the matrix of $U_p$ is a companion-type matrix; expanding $\det(X - U_p)$ along the last row gives the polynomials in the table. The eigenvectors are checked directly. To see the characteristic polynomials, note that in this basis the matrix $A$ of $U_p$ has first column $(c_0, c_1, 0, \dots, 0)^{\mathsf{T}}$, with $(c_0, c_1) = (a_p, -p)$ if $p \nmid M$ and $(a_p, 0)$ if $p \mid M$ (here $e \geq 1$, as $p \mid N/M$), entries 1 just above the diagonal (from $U_p h_j = h_{j-1}$), and zeros elsewhere. Expanding $\det(X - A)$ along the last row, whose only nonzero entries are $X$ in the last place and possibly $-c_e$ in the first, gives $P_e(X) = XP_{e-1}(X) - c_e$, with $c_k = 0$ for $k \geq 2$ and $P_0 = X - c_0$ (the minor of the first entry is triangular with determinant $(-1)^e$, and its cofactor sign is $(-1)^e$); by induction, $\det(X - A) = X^{e+1} - c_0 X^e - c_1 X^{e-1}$. This is $X^{e-1}(X^2 - a_p X + p)$ if $p \nmid M$, $X^e(X - a_p)$ if $p \parallel M$, and $X^{e+1}$ if $p^2 \mid M$ (where $a_p = 0$). For instance, for $p \nmid M$ and $e = 2$,

$$A = \begin{pmatrix} a_p & 1 & 0 \\ -p & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \det(X - A) = X(X^2 - a_p X + p).$$

The eigenvectors: $U_p(h_0 - \beta h_1) = a_p h_0 - p\, h_1 - \beta h_0 = \alpha h_0 - \alpha\beta h_1 = \alpha(h_0 - \beta h_1)$, using $a_p = \alpha + \beta$, $p = \alpha\beta$; $U_p(h_0 - a_p h_1 + p\, h_2) = a_p h_0 - p\, h_1 - a_p h_0 + p\, h_1 = 0$; $U_p(h_0 - a_p h_1) = a_p h_0 - a_p h_0 = 0$ when $p \mid M$. If 0 is a repeated root, the 0-eigenspace is the line spanned by the listed eigenvector (the matrix has rank $e$), so $U_p$ is not diagonalisable. This is the shift-operator model of Chapter 3; compare Chapter 13. (iii) A tensor product of eigenvectors of operators acting on separate factors is a common eigenvector; $\mathbb{T}$ is generated by the $T_n$, $(n, N) = 1$, and the $U_p$, $p \mid N$, which act as stated. The coefficient of $h_0$ in each listed eigenvector is 1, and $h(dz)$ with $d > 1$ has $a_1 = 0$; so $a_1(g) = 1$. □

The idea of the next lemma is that the congruence $a_p(h) \equiv a_p(f)$ is given only for $p \nmid \ell N$, while Lemma 15.55(ii) needs it at every $n$. At $p = \ell$ the local theory at $\ell$ supplies it; at a prime $p \mid N$ one has the freedom to replace $h$ by another eigenform in $V_h$, and Lemma 15.57 lists the $U_p$-eigenvalues available. That freedom exists only where $p \mid N/M$. Where $v_p(M) = v_p(N)$ the operator $U_p$ acts on $V_h$ by the scalar $a_p(h)$, and one needs $a_p(h) \equiv a_p(f)$ outright; the hypothesis of the lemma excludes the one situation, $p \parallel N$ with $\bar\rho$ unramified at $p$ and $p \parallel M$, where this can fail (Exercise 15.93: the signs may differ when $p \equiv -1 \pmod{\ell}$).

**Lemma 15.58** (oldforms; the paper's Lemma lem:old)**.** *Let $f = f_E$ be as in Definition 15.53, $\ell$ an odd prime with $\ell \nmid N$, and $\bar\rho = \bar\rho_{E,\ell}$ irreducible. Let $h \neq f$ be a newform of weight 2 on $\Gamma_0(M)$, $M \mid N$, with coefficients in the ring of integers $\mathcal{O}$ of a number field, and $\lambda \mid \ell$ a prime of $\mathcal{O}$ with $a_p(h) \equiv a_p(f) \pmod{\lambda}$ for all primes $p \nmid \ell N$. Assume that $p \nmid M$ for every prime $p \parallel N$ at which $\bar\rho$ is unramified. Then, after enlarging $\mathcal{O}$ and $\lambda$, the span $V_h$ of the $h(dz)$, $d \mid N/M$, contains a normalised eigenform $g$ of all of $\mathbb{T}$ with coefficients in $\mathcal{O}$ and $a_n(g) \equiv a_n(f) \pmod{\lambda}$ for every $n \geq 1$; it lies in $W \otimes \overline{\mathbb{Q}}$ (notation of Lemma 15.55). Consequently $\ell \mid r$.*

*Proof. Step 1: orthogonality.* $V_h$ is $\mathbb{T}$-stable and orthogonal to $f$: if $M < N$ it is a different summand of Proposition 15.4(iv) from $V_f = \mathbb{C}f$, and if $M = N$ then $V_h = \mathbb{C}h$ with $h \neq f$. The forms $h(dz)$ have coefficients in $\mathcal{O} \subset \overline{\mathbb{Q}}$, so they lie in $S_{\mathbb{Q}} \otimes \overline{\mathbb{Q}}$ (their coordinates in a $\mathbb{Z}$-basis of $S$ solve a uniquely solvable linear system, given by finitely many coefficients, with entries in $\overline{\mathbb{Q}}$). So any $\overline{\mathbb{Q}}$-combination $g$ of the $h(dz)$ lies in $(S_{\mathbb{Q}} \otimes \overline{\mathbb{Q}}) \cap f^{\perp} = W \otimes \overline{\mathbb{Q}}$.

*Step 2: the residual representations.* By Theorem 15.12, $\bar\rho_f \simeq \bar\rho \otimes_{\mathbb{F}_\ell} k_\lambda$, which is absolutely irreducible (Proposition 15.9(ii)). By Corollary 15.10(ii), $\bar\rho_h \simeq \bar\rho \otimes k_\lambda$.

*Step 3: the choice of the $\mu_p$.* Enlarge $\mathcal{O}$ to contain the roots $\alpha_p, \beta_p$ of $X^2 - a_p(h)X + p$ for the primes $p \mid N/M$ with $p \nmid M$, and replace $\lambda$ by a prime above it (a congruence modulo the old $\lambda$ implies one modulo the new). For each $p \mid N/M$ choose $\mu_p$ as below, and let $g$ be the eigenform of Lemma 15.57(iii); its coefficients lie in $\mathcal{O}$ and $a_1(g) = 1$. We check $a_p(g) \equiv a_p(f) \pmod\lambda$ at every prime $p$.

- $p \nmid \ell N$: $a_p(g) = a_p(h) \equiv a_p(f)$ by hypothesis.
- $p = \ell$: $\ell \nmid N$, so $a_\ell(g) = a_\ell(h)$; $f$ and $h$ have weight 2 and level prime to $\ell$, and $\bar\rho_h \simeq \bar\rho_f$ (Step 2), so $a_\ell(h) \equiv a_\ell(f)$ by Theorem 15.32(iii).
- $p^2 \mid N$: $a_p(f) = 0$ (Theorem 15.20). If $v_p(M) = v_p(N)$, then $p^2 \mid M$ and $a_p(g) = a_p(h) = 0$ by the same theorem applied to $h$. Otherwise $e_p \geq 1$, and $e_p \geq 2$ if $p \nmid M$; in every case 0 is in the list of Lemma 15.57(ii): take $\mu_p = 0$, so $a_p(g) = 0 = a_p(f)$.
- $p \parallel N$, $\bar\rho$ unramified at $p$: by hypothesis $p \nmid M$, so $e_p = 1$ and the choices are $\alpha_p$, $\beta_p$. By Theorem 15.20(c) for $f$ (type (St)), $\bar\rho(\mathrm{Frob}_p)$ has eigenvalues $a_p(f)p$, $a_p(f)$. Since $p \nmid M\ell$, $\bar\rho_h(\mathrm{Frob}_p)$ has characteristic polynomial $X^2 - a_p(h)X + p$ modulo $\lambda$ (Proposition 15.9(i)), and $\bar\rho_h \simeq \bar\rho_f$; so $(X - \alpha_p)(X - \beta_p) \equiv (X - a_p(f)p)(X - a_p(f)) \pmod\lambda$. Reducing modulo $\lambda$ (where $\mathcal{O}/\lambda$ is a field), one of $\alpha_p, \beta_p$ is congruent to $a_p(f)$; take it as $\mu_p$.
- $p \parallel N$, $\bar\rho$ ramified at $p$: then $p \mid M$, since otherwise $\bar\rho_h$ would be unramified at $p$ (Theorem 15.6(ii)(a), $p \neq \ell$); so $v_p(M) = 1 = v_p(N)$ and $a_p(g) = a_p(h) = \pm 1$. By Theorem 15.20(a), (b) for $f$ and for $h$ (both of type (St) at $p$), $\bar\rho_f|_{D_p}$ and $\bar\rho_h|_{D_p}$ are extensions of $\bar\varepsilon_f$ by $\bar\varepsilon_f\bar\chi_\ell$, resp. of $\bar\varepsilon_h$ by $\bar\varepsilon_h\bar\chi_\ell$, with $\bar\varepsilon_f(\mathrm{Frob}_p) = a_p(f)$, $\bar\varepsilon_h(\mathrm{Frob}_p) = a_p(h)$, and both are ramified, hence non-split with a unique stable line. An isomorphism $\bar\rho_f|_{D_p} \simeq \bar\rho_h|_{D_p}$ carries the stable line to the stable line, hence induces an isomorphism of the quotients: $\bar\varepsilon_f = \bar\varepsilon_h$, and $a_p(h) \equiv a_p(f)$. (This holds also when $p \equiv 1 \pmod\ell$, where the two characters of each extension coincide.)

*Step 4: all $n$.* $f$ and $g$ are normalised eigenforms of all of $\mathbb{T}$, so both coefficient sequences satisfy $a_{mn} = a_m a_n$ for $(m, n) = 1$, $a_{p^k} = a_p^k$ for $p \mid N$ ($U_{p^k} = U_p^k$), and $a_{p^{k+1}} = a_p a_{p^k} - p\, a_{p^{k-1}}$ for $p \nmid N$, $p = \ell$ included (Proposition 15.4(i)). Congruence at every prime therefore gives congruence at every $n$. Indeed, for a prime $p$, $a_{p^k}$ is a polynomial in $a_p$ with integer coefficients ($a_p^k$ if $p \mid N$; for $p \nmid N$, by induction on $k$ from the recursion, starting with $a_1 = 1$), the same polynomial for $f$ and for $g$, so $a_p(g) \equiv a_p(f)$ gives $a_{p^k}(g) \equiv a_{p^k}(f) \pmod\lambda$; and for $n = \prod p^{k_p}$, $a_n = \prod a_{p^{k_p}}$ for both forms. Lemma 15.55(ii) now gives $\ell \mid r$. □

*In the paper* 15.59. The paper's proof of Lemma lem:old follows the same plan and the same case division; the eigenvalue lists and eigenvectors are those of Lemma 15.57. Two of its sentences are expanded here. "$\bar\rho_h \simeq \bar\rho$ over a common finite extension of $\mathbb{F}_\ell$, by Chebotarev and Brauer-Nesbitt, $\bar\rho$ being irreducible, hence absolutely irreducible" is Step 2 (irreducibility alone, without absolute irreducibility, would match only the semisimplifications over $\mathbb{F}_\ell$). "A non-split extension of two characters has a unique stable line" is Theorem 15.20(b) with Chapter 4. The weight-two fact that $\alpha_p \neq \beta_p$ (Coleman and Edixhoven [32]) is not needed: the argument uses only that each listed eigenvalue has an eigenvector.

**Example 15.60** (Lemma lem:old on three pairs)**.** (All congruences checked with PARI/GP for $n \le 3000$, beyond the Sturm bounds 16, 14, 22 at levels 77, 52, 121, so they hold for every $n$ by Chapter 12.)

(a) *A prime $p \,\|\, N$ at which $\bar\rho$ is unramified.* $f = f_{77c}$, $\ell = 3$, $h = f_{11a}$, $M = 11$, $N/M = 7$. `77c1` has type $\mathrm{I}_3$ at 7, and $3 \mid n_7 = 3$, so $\bar\rho_{77c1,3}$ is unramified at 7 (Chapter 11, Lemma lem:tamagawa-exp); at 11 it has type $\mathrm{I}_2$ and $\bar\rho$ is ramified, and $11 \mid M$. Here $a_7(h) = -2$, $X^2 + 2X + 7$ has roots $\alpha, \beta = -1 \pm \sqrt{-6}$, both congruent to $-1 = a_7(f_{77c})$ modulo the prime $(3, \sqrt{-6})$ of $\mathbb{Z}[\sqrt{-6}]$, the ring of integers of $\mathbb{Q}(\sqrt{-6})$, above 3 (which ramifies). So $g = h - \beta h(7z)$ has $U_7$-eigenvalue $\alpha$ and $g \equiv f_{77c}$ modulo $(3, \sqrt{-6})$ for every $n$: modulo that prime $g$ has coefficients $a_n(f_{11a}) + a_{n/7}(f_{11a})$. Hence $3 \mid r(\texttt{77c1})$; indeed $r = \deg\varphi = 6$.

(b) *An additive prime, at $q = 2$.* $f = f_{52a}$, $\ell = 3$, $h = f_{26b}$, $M = 26$, $N/M = 2$, $2 \,\|\, M$, $e_2 = 1$. Here $\mu_2 = 0$, $g = h - a_2(h)h(2z) = f_{26b}(z) - f_{26b}(2z)$ ($a_2(f_{26b}) = 1$), and $g \equiv f_{52a} \pmod 3$ for every $n$. At 13, $\bar\rho_{52a1,3}$ is ramified ($\mathrm{I}_2$) and $13 \mid M$. Hence $3 \mid r(\texttt{52a1})$; indeed $r = \deg\varphi = 3$. The other newform of level 26, $f_{26a}$, is not congruent to $f_{52a}$ modulo 3.

(c) *An additive prime, at $q = 11$.* $f = f_{121c}$, $\ell = 3$, $h = f_{11a}$, $M = 11$, $N/M = 11$, $11 \,\|\, M$, $e_{11} = 1$: $\mu_{11} = 0$, $g = f_{11a}(z) - f_{11a}(11z)$ ($a_{11}(f_{11a}) = 1$), the 11-depletion of $f_{11a}$, and $g \equiv f_{121c} \pmod 3$ for every $n$; $r(\texttt{121c1}) = \deg\varphi = 6$.

## 15.9 Proof of Theorem thm:qual; the Example; the remark on $\ell \mid N$

This section proves the paper's Theorem thm:qual in full, case by case, with the inputs of the previous sections: lower the level of $\bar\rho$ (case (a)) or of $\bar\rho \otimes \chi_{q^*}$ (cases (b), (c)) to the Serre conductor, twist back to a newform $h \neq f$ of level dividing $N$, and apply Lemma lem:old and Lemma lem:crit; the theorem of Agashe, Ribet and Stein passes from the congruence number to the modular degree. It then works the paper's Example and the remark that follows it.

**Theorem 15.61** (the paper's Theorem thm:qual)**.** *Let $E/\mathbb{Q}$ have conductor $N$ and let $\ell$ be an odd prime not dividing $N$ such that $\bar\rho_{E,\ell}$ is irreducible and, if $\ell = 3$, absolutely irreducible when restricted to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ (equivalently, the mod 3 image is not $N_s(\mathbb{F}_3)$). Let $q \neq \ell$ be a prime at which $E$ has additive reduction of one of the following kinds:*

*(a) type* IV *or* IV$^*$*, with $q \ge 5$ or $q = 2$, and $\ell = 3$;*

*(b) type* II *or* II$^*$*, with $q \ge 5$ and $\ell = 3$;*

*(c) type $\mathrm{I}_n^*$ with $n \ge 1$ and $\ell \mid n$, $q$ odd.*

*Then $\ell$ divides the modular degree of the $X_0(N)$-optimal curve $E_0$ in the isogeny class of $E$.*

### 15.9.1 The proof

*Proof.* Let $f = f_E$, $E_0$ the optimal curve and $r = r_{E_0} = r_f$ (Definition 15.53); write $\bar\rho = \bar\rho_{E,\ell}$, which is irreducible, hence absolutely irreducible (Proposition 15.9(ii)). Since $\ell \nmid N$, $\ell^2 \nmid N$, and Theorem 15.54 gives $v_\ell(\deg\varphi_{E_0}) = v_\ell(r)$; so it suffices to prove $\ell \mid r$. By Lemma 15.58 it suffices to find a newform $h \neq f$ of weight 2 on $\Gamma_0(M_h)$, $M_h \mid N$, and a prime $\lambda \mid \ell$ of its coefficient ring with

(H1) $a_p(h) \equiv a_p(f) \pmod{\lambda}$ for all primes $p \nmid \ell N$, and

(H2) $p \nmid M_h$ for every prime $p \,\|\, N$ at which $\bar\rho$ is unramified.

In all three cases $v_q(N) = 2$: types IV, IV$^*$, II, II$^*$ at $q \geq 5$ and IV, IV$^*$ at $q = 2$ have conductor exponent 2 (Chapter 11, Lemma lem:IV and its corollary at $q = 2$), and so has type $\mathrm{I}_n^*$ at an odd prime (Proposition 15.26, row (RSt), with Proposition 15.22).

*Case (a).* $\ell = 3$, type IV or IV$^*$ at $q$, $q \geq 5$ or $q = 2$.

(a1) By Lemma lem:IV (Chapter 11; at $q = 2$ by its corollary there) the image of $I_q$ on $E[3]$ is a nontrivial unipotent group, $E[3]^{I_q}$ is a line, $\bar\rho$ is tame at $q$, and the Serre exponent of $\bar\rho$ at $q$ is $2 - 1 = 1$.

(a2) Let $M = N(\bar\rho)$. Then $M \mid N$ (Proposition 15.36) and $v_q(M) = 1 < 2 = v_q(N)$, so $M \mid N/q$ and $M < N$.

(a3) Corollary 15.43 (with $E' = E$; its hypotheses are those of the theorem) gives a newform $g$ of weight 2 on $\Gamma_0(M)$ and $\lambda \mid 3$ with $\bar\rho_{g,\lambda} \simeq \bar\rho \otimes k_\lambda$, so $a_p(g) \equiv a_p(f) \pmod{\lambda}$ for $p \nmid 3N$.

(a4) Take $h = g$, $M_h = M$. Then $h \neq f$, since the levels differ (Proposition 15.4(iii)); (H1) holds; and (H2) holds because $N(\bar\rho)$ is prime to every prime at which $\bar\rho$ is unramified. In Lemma lem:old the prime $q$ falls under the case $p^2 \mid N$, with $v_q(M) < v_q(N)$, and the eigenvalue $\mu_q = 0$ of $U_q$ is used.

*Case (b).* $\ell = 3$, type II or II$^*$ at $q \geq 5$. Let $\chi = \chi_{q^*}$ and $E' = E^{(q)}$.

(b1) By Lemma lem:twist (i), (ii) (Chapter 10), $E'$ has type IV$^*$, resp. IV, at $q$, the same types as $E$ at the other primes, and conductor $N$; in particular $3 \nmid N(E')$.

(b2) $\bar\rho_{E',3} \simeq \bar\rho \otimes \chi$ (Chapter 9). It is irreducible, and, restricted to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, absolutely irreducible, since twisting by a character preserves both properties.

(b3) By Lemma lem:IV for $E'$, $M = N(\bar\rho \otimes \chi)$ has $v_q(M) = 1$, and $M \mid N(E') = N$; so $M \mid N/q$.

(b4) Corollary 15.43 for $E'$ gives a newform $g$ on $\Gamma_0(M)$ and $\lambda \mid 3$ with $\bar\rho_{g,\lambda} \simeq (\bar\rho \otimes \chi) \otimes k_\lambda$, so $a_p(g) \equiv \chi(p) a_p(f) \pmod{\lambda}$ for $p \nmid 3N$.

(b5) As $q \,\|\, M$, Corollary 15.27 says that $h = g \otimes \chi = g^\chi$ is a newform on $\Gamma_0(Mq)$, and $Mq \mid N$. Its coefficients are $\pm$ those of $g$ or 0, so $\lambda$ serves for $h$, and for $p \nmid 3N$, $a_p(h) = \chi(p) a_p(g) \equiv \chi(p)^2 a_p(f) = a_p(f) \pmod{\lambda}$: (H1).

(b6) $h \neq f$: otherwise $g = (g^\chi)^\chi = f^\chi = f_{E'}$ (Proposition 15.29(iii), (iv)), which has level $N \neq M$.

(b7) (H2): if $p \,\|\, N$ and $\bar\rho$ is unramified at $p$, then $p \neq q$ (as $q^2 \mid N$), so $\bar\rho \otimes \chi$ is unramified at $p$, $p \nmid M$, and $p \nmid Mq$.

*Case (c).* Type $\mathrm{I}_n^*$ at an odd $q \neq \ell$ with $\ell \mid n$. Let $\chi = \chi_{q^*}$ and $E' = E^{(q)}$.

(c1) By Lemma lem:twist(ii), $E'$ has type $\mathrm{I}_n$ at $q$, and by Lemma lem:twist(i) the same types as $E$ elsewhere; so its conductor is $N/q$, and $\ell \nmid N/q$.

(c2) By Lemma lem:tamagawa-exp for $E'$ (Chapter 11), $\bar\rho_{E',\ell} = \bar\rho \otimes \chi$ is unramified at $q$, because $\ell \mid n = n_q(E')$.

(c3) $M = N(\bar\rho \otimes \chi)$ is prime to $q$ and divides $N(E') = N/q$; so $M \mid N/q^2$.

(c4) Corollary 15.43 for $E'$ (irreducibility and the condition at $\ell = 3$ are twist-invariant, as in (b2)) gives a newform $g$ on $\Gamma_0(M)$ and $\lambda \mid \ell$ with $a_p(g) \equiv \chi(p)a_p(f) \pmod\lambda$ for $p \nmid \ell N$. At $q$ this lowering is the instance of Ribet's theorem (Theorem 15.40) in which $q$ exactly divides the level $N/q$ of $E'$ and $\bar\rho \otimes \chi$ is unramified at $q$.

(c5) As $q \nmid M$, Corollary 15.27 says that $h = g \otimes \chi = g^\chi$ is a newform on $\Gamma_0(Mq^2)$, and $Mq^2 \mid N$; (H1) follows as in (b5).

(c6) $h \neq f$: otherwise $g = f^\chi = f_{E'}$, of level $N/q \neq M$ (as $q \mid N/q$ and $q \nmid M$).

(c7) (H2) holds as in (b7).

In each case Lemma 15.58 gives $\ell \mid r$, and so $\ell \mid \deg\varphi_{E_0}$. □

*In the paper* 15.62. The paper's proof (Section sec:proof) is the one above. Three remarks on its wording, all about the twist dictionary of Convention 15.2. (1) Case (b): "its newform is $f' = f \otimes \chi$, a newform of weight 2 on $\Gamma_0(N)$ (both have $a_q = 0$)". The naive twist is the newform twist here because $a_q(f^\chi) = 0$, which holds since $q^2 \mid N$ (Theorem 15.20); Atkin and Li's theorem does not apply at this level ($q^2 \mid N$), and is used only for $h = g \otimes \chi$, where $q \parallel M$ ("As $q \parallel M$, Lemma lem:twist(iii) says that $h = g \otimes \chi$ is a newform on $\Gamma_0(Mq)$"). (2) The proof of $h \neq f$: the paper argues by strong multiplicity one and notes "Note that $h \otimes \chi$ is the $q$-depletion of $g$, not $g$, as $a_q(g) = \pm 1$"; the identity $g = h \otimes \chi$ fails as an identity of $q$-series, which is why the argument compares $a_p$ only for $p \nmid qN$. Steps (b6), (c6) give the argument with newform twists. (3) Case (c): the paper writes "let $f' = f_{E'}$ be its newform, of weight 2 on $\Gamma_0(N/q)$ with $q \parallel N/q$, $a_q(f') = \pm 1$ and $a_p(f') = \chi(p)a_p(f)$ for $p \neq q$" and notes that $f \otimes \chi$ is the $q$-depletion of $f'$; the naive twist $f \otimes \chi$ is not itself a newform on $\Gamma_0(N/q)$, since its $q$-th coefficient vanishes while $a_q(f_{E'}) = \pm 1$ (Example 15.3(b)).

*Remark* 15.63 (case (a) at $q = 2$, and the range of the hypotheses). In case (a) at $q = 2$ the only input specific to $q = 2$ is that for type IV or IV$^*$ at 2 the image of inertia on $T_3E$ is cyclic of order 3, so that the Serre exponent of $\bar\rho_{E,3}$ at 2 is 1 (Chapter 11). Case (b) is not claimed at $q = 2$: there the Kodaira type does not determine the image of inertia, and the twist by $\mathbb{Q}(\sqrt{q^*})$ is not available. In case (a) the hypothesis $3 \nmid N$ excludes $q = 3$. Below conductor 10,000 there are 1,051 optimal curves with type IV or IV$^*$ at 2 (PARI codes $\pm 4$), $3 \nmid N$ and mod 3 image neither reducible (labels `3B`, `3Cs`) nor 3Ns, and all of them have $3 \mid \deg\varphi$ (Cremona's `degphi` and `galrep` files; none of them has complex multiplication); the first are `52a1` ($\deg\varphi = 3$), `76a1` (6) and `92b1` (6). For $\ell \geq 5$ only the irreducibility of $\bar\rho$ is used, as the paper says after the theorem.

*Remark* 15.64 (why exactly these types). The proof needs a type at $q$ for which $\bar\rho$, or $\bar\rho \otimes \chi_{q^*}$, has Serre exponent at $q$ strictly smaller than the exponent of the curve it comes from, so that the level can be lowered and raised back by the twist to a *different* newform. By Chapter 11 (inertia modulo 3 by type) and Proposition 15.26:

(i) Types III, III$^*$ at $q \geq 5$: the Serre exponent of $\bar\rho_{E,3}$ and of its twist is 2; nothing can be lowered. These are the types relevant to the image 3Ns (Section sec:anomaly of the paper).

(ii) Type $\mathrm{I}_0^*$: $E' = E^{(q)}$ has good reduction at $q$ and $M = N(\bar\rho \otimes \chi)$ divides $N/q^2$. But Corollary 15.43 is existential and may return $g = f_{E'}$ itself (nothing excludes this when $N(\bar\rho \otimes \chi) = N/q^2$), and then $h = g^\chi = f$: no new congruence. For `539d1` at $\ell = 3$ ($\mathrm{I}_0^*$ at 7, twist `11a3`), $N(\bar\rho \otimes \chi_{-7}) = 11$ and $g = f_{11a}$, $h = f_{539d} = f$. The $\mathrm{I}_0^*$ contribution is reached by level raising instead (Theorem thm:quant and Theorem 15.73).

(iii) Type $\mathrm{I}_n^*$ with $\ell \nmid n$: $\bar\rho \otimes \chi$ is ramified at $q$ with exponent 1, so $M$ has $q^1$, $h = g^\chi$ has $q^2$, and Corollary 15.43 may return $g = f_{E'}$, giving $h = f$.

(iv) Types II, IV and their duals at $\ell \geq 5$: $\bar\rho_{E,\ell}$ has Serre exponent 2 at $q$ ($\Phi_q^{\mathrm{geom}}$ has no $\ell$-torsion), and so has every twist.

The paper's remark after Theorem thm:quant observes that the types of (ii) and (iii) are covered by Theorem thm:quant, with no hypothesis at $\mathbb{Q}(\sqrt{-3})$.

**Example 15.65** (the paper's Example)**.** The table lists, for the paper's four instances and the book's `121c1`, the case of Theorem thm:qual, the type at $q$, the twist at curve level (an `elltwist`, minimal model and table lookup), the Serre conductor that the proof lowers to and the newform there, the form $h$ of Lemma lem:old, and the modular degree; $e_\ell$, $t_\ell$ are the paper's local terms (Definition def:terms; Chapter 19).

| $E$ | case, $\ell$ | type at $q$ | twist | $M$, $g$ | $h$ | $\deg\varphi$ |
|---|---|---|---|---|---|---|
| `52a1` | (a), 3 | $\mathrm{IV}^*$ at 2 | none | 26, $f_{26b}$ | $f_{26b}$ | 3; $e_3(E,2) = 1$ |
| `121c1` | (a), 3 | IV at 11 | none | 11, $f_{11a}$ | $f_{11a}$ | 6; $e_3(E,11) = 1$ |
| `121a1` | (b), 3 | II at 11 | `121c2` ($\mathrm{IV}^*$) | 11, $f_{11a}$ | $f_{121d} = f_{11a} \otimes \chi_{-11}$ | 6; $e_3(E,11) = 1$ |
| `539c1` | (c), 3 | $\mathrm{I}_3^*$ at 7 | `77c1` ($\mathrm{I}_3$) | 11, $f_{11a}$ | $f_{539d} = f_{11a} \otimes \chi_{-7}$ | $2^5 \cdot 3^2$; $e_3 = t_3 = 1$ |
| `980i1` | (c), 5 | $\mathrm{I}_5^*$ at 7 | `140b1` ($\mathrm{I}_5$) | $4 \cdot 5$, $f_{20a}$ | $f_{980g} = f_{20a} \otimes \chi_{-7}$ | $2^6 \cdot 3^2 \cdot 5$; $t_5 = 1$, $e_5 = 0$ |

(i) *Every $n$.* Each congruence $f_E \equiv g'$ produced by Lemma lem:old holds for every $n$, not only at the primes $p \nmid \ell N$: both sides lie in $S_2(\Gamma_0(N))$ with integral coefficients, so a check up to Sturm's bound $\frac{1}{6}N\prod_{p\mid N}(1+\frac{1}{p})$ suffices (Chapter 12); the bounds are 14, 22, 22, 112, 336 for $N = 52$, 121, 121, 539, 980, and the congruences were checked with PARI/GP for all $n \leq 3000$. For the last three rows $M_h = N$, so $V_h = \mathbb{C}h$ and $g' = h$; for the first two $g' = h - h(qz)$, the $q$-depletion of $h$ (Example 15.60(b), (c)).

(ii) *Curves versus classes.* The twist of `121a1` by $\chi_{-11}$ is `121c2`, of type $\mathrm{IV}^*$; the optimal curve `121c1` has type IV. The class 121d of $f_{11a} \otimes \chi_{-11}$ contains `121d2`, the twist of `11a1` ($\mathrm{I}_5^*$), and `121d1`, the twist of `11a3` ($\mathrm{I}_1^*$). The class 539d of $f_{11a} \otimes \chi_{-7}$ contains `539d2`, the twist of `11a1`; the twist of `20a1` by $\chi_{-7}$ is `980g2`. Every quantity in the theorem depends only on the class: $r$ does, and $v_\ell(\deg\varphi)$ of the optimal curve is what the theorem concerns.

(iii) $5 \mid 980$. `980i1` has multiplicative reduction at 5, so $\ell = 5$ divides $N$ and the example lies outside the hypotheses of Theorem thm:qual (and of Theorem thm:main), as the paper says: "As $5 \mid 980$, this last example illustrates the congruence, not the theorem." Accordingly the lowered level is $N(\bar\rho_{140b1,5}) \cdot 5 = 20$, not 4 (Example 15.45), and Lemma lem:old at $p = \ell$ and Theorem 15.54 at $\ell$ are not available.

(iv) *Twists of newforms.* In the rows (b), (c), $h = g^{\chi} = g \otimes \chi$ by Corollary 15.27; the newform twists of $f_E$ itself are $f_{121a}^{\chi_{-11}} = f_{121c}$, $f_{539c}^{\chi_{-7}} = f_{77c}$, $f_{980i}^{\chi_{-7}} = f_{140b}$, and only the first equals the naive twist (Example 15.30).

(v) *Case (c) step by step for* `539c1`, $\ell = 3$, $q = 7$. $N = 7^2 \cdot 11$. (c1) $E' = E^{(7)}$ is `77c1`, of type $\mathrm{I}_3$ at 7 and conductor $77 = N/7$. (c2) $3 \mid n = 3$, so $\bar\rho \otimes \chi_{-7} = \bar\rho_{77c1,3}$ is unramified at 7. (c3) $M = N(\bar\rho_{77c1,3}) = 11$ (computed after Proposition 15.36), and indeed $11 \mid N/7^2 = 11$. (c4) The newform of level 11 is $g = f_{11a}$, and $a_p(f_{11a}) \equiv \chi_{-7}(p) a_p(f_{539c}) \pmod 3$ for $p \nmid 3 \cdot 539$. (c5) $7 \nmid 11$, so $h = f_{11a} \otimes \chi_{-7} = f_{539d}$, of level $11 \cdot 7^2 = 539$. (c6) $h = f_{539d} \neq f_{539c}$. (c7) The only prime exactly dividing $N$ is 11, where $\bar\rho$ is ramified (type $\mathrm{I}_2$, $3 \nmid 2$), so (H2) is empty. Since $M_h = N$, Lemma lem:old takes $g' = h$ itself, and Lemma 15.55(ii) gives $3 \mid r($`539c1`$) = 288$.

*In the paper* 15.66. The paper's Example lists the cases (a) at $q = 2$ (`52a1`), (b) (`121a1`), (c) at $\ell = 3$ (`539c1`) and (c) at $\ell = 5$ (`980i1`), with the Sturm bounds 14, 22, 112, 336; the row `121c1` is the book's. All data in the table were recomputed (PARI/GP 2.17, Cremona's tables).

*Remark* 15.67 (the paper's remark on $\ell \mid N$). The remark after the paper's Example says that $\ell \nmid N$ is used in three places. (i) In the optimal-level theorem: Corollary 15.43 needs $\bar\rho|_{D_\ell}$ finite flat, so that the level is $N(\bar\rho)$ rather than $N(\bar\rho)\ell$ (the triviality of the character uses only the determinant and the order prime to $\ell$); for `140b1` at $\ell = 5$ the level is $N(\bar\rho) \cdot 5$. (ii) In Lemma lem:old at $p = \ell$: Theorem 15.32 needs the levels prime to $\ell$. (iii) In passing from $r$ to $\deg \varphi$, which needs only $\ell^2 \nmid N$ and fails without it: for `99a1` (the 99A1 of [3, Table 1], $N = 9 \cdot 11$), $\deg \varphi = 4$ and $r = 12$; for `539d1`, $\deg \varphi = 72$ and $r = 504$; for `726a1`, $\deg \varphi = 144$ and $r = 1584$ (all three examples are the paper's, and all were recomputed here as indices of $\mathbb{Z}f \oplus L$). The remark ends: "so do they at $q = 3$ in cases (a) and (b), where $q = \ell$ and the theorem does not apply." Here $\ell = 3$, so $q = 3$ means $3 \mid N$: the inertia group at 3 then acts on $T_3 E$ through an infinite group (it contains the cyclotomic character on the determinant), and the local theory at $q = \ell$ is the $\ell$-adic one of Section 15.4, not that of Lemma lem:IV.

## 15.10 Twists and levels in the paper's twist theorems

Theorems thm:twist and thm:quant of the paper transport congruences from the newform $g = f_{E_S}$ of a twisted curve to $f = f_E$ by the twist map $\iota$, and Theorem thm:quant reads the result as level raising. This section proves the facts about levels and twists on which both rest and states the two theorems; their proofs are in Chapter 20, which uses the results below.

**Proposition 15.68** (the twist map)**.** *Let $S$ be a finite set of odd primes, $D = \prod_{q \in S} q$, $\chi_S = \prod_{q \in S} \chi_{q^*}$, and $\iota = \iota_S \colon h \mapsto h \otimes \chi_S = \sum_n \chi_S(n) a_n(h) q^n$. By Chapter 13, $\iota$ maps $S_2(\Gamma_0(M))$ into $S_2(\Gamma_0(\mathrm{lcm}(M, D^2)))$, preserves integrality, and $T_p \circ \iota = \chi_S(p)\, \iota \circ T_p$ for $p \nmid DM$. Moreover:*

*(i) $\iota$ sends a generalised eigenform of the $T_p$, $p \nmid DM$, with eigenvalues $(a_p)$ to a generalised eigenform with eigenvalues $(\chi_S(p) a_p)$.*

*(ii) Let $g$ be a newform of level $M$, and suppose that $f = g^{\chi_S}$ is a newform of level $N$ with $\mathrm{lcm}(M, D^2) \mid N$. Then $\iota$ maps $g^\perp = \bigoplus_{h \neq g} V_h \subseteq S_2(\Gamma_0(M))$ (the $\mathbb{T}_M$-stable complement of $\mathbb{C}g$) into $f^\perp \subseteq S_2(\Gamma_0(N))$.*

*(iii) On newforms, $\iota(g) = (g^{\chi_S})^S$; so $\iota(g) = g^{\chi_S}$ iff $a_q(g^{\chi_S}) = 0$ for every $q \in S$, and $\iota(g) = \iota(g^S)$.*

*(iv) On Galois representations $\iota$ is $\rho \mapsto \rho \otimes \chi_S$; it preserves* $\mathrm{ad}^0\rho$ *and the true local factors* $L_p(A_g, s)$.

*Proof.* (i) From $T_p\iota = \chi_S(p)\iota T_p$ and $\chi_S(p)^2 = 1$: $(T_p - \chi_S(p)a_p)\iota = \chi_S(p)\,\iota\,(T_p - a_p)$, so a power of $T_p - a_p$ killing $h$ gives the same power of $T_p - \chi_S(p)a_p$ killing $\iota(h)$. (ii) For a newform $h \neq g$ of level dividing $M$, $\iota(V_h)$ consists of eigenforms of the $T_p$, $p \nmid DN$, with eigenvalues $\chi_S(p)a_p(h)$, which are those of $h^{\chi_S}$; and $h^{\chi_S} \neq g^{\chi_S} = f$ by Proposition 15.29(iii). By Proposition 15.4(iv) at level $N$, $\iota(V_h)$ lies in $V_{h^{\chi_S}}$, which is orthogonal to $V_f = \mathbb{C}f$. (iii) is Proposition 15.29(ii) with $\Sigma(D) = S$, and $\iota$ kills the coefficients at $n$ with $(n, D) > 1$, where $g$ and $g^S$ differ. (iv) is Proposition 15.29(ii) and Corollary 15.25. □

**Proposition 15.69** (levels of the twisted curve and of the depleted form)**.** *Let $E/\mathbb{Q}$ have conductor $N$, $S$ a finite set of odd primes $q$ with $q^2 \mid N$, $D = \prod_{q\in S} q$, $E_S = E \otimes \chi_S$ of conductor $N_S$, $f = f_E$ and $g = f_{E_S}$.*

*(i) If $E$ has type $\mathrm{I}_n^*$ at every $q \in S$, then $N_S = N/D$; if type $\mathrm{I}_0^*$ at every $q \in S$, then $N_S = N/D^2$; if type* II, III, IV *or a dual at every $q \in S$, with $q \geq 5$, then $N_S = N$. In each case* $\mathrm{lcm}(N_S, D^2) \mid N$, *and in the third case also* $\mathrm{lcm}(N, D^2) \mid N_S$.

*(ii) $f = g^{\chi_S}$, and $\iota(g) = f$: the two $q$-series agree at every $n$.*

*(iii) In the first two cases the type of $g$ at $q \in S$ is (St), resp. (U), so $\delta_q(g) = 1$, resp. 2, and $N_S \prod_{q\in S} q^{\delta_q(g)} = N$: the $S$-depleted form $g^S$ lies in $S_2(\Gamma_0(N))$, and $\iota(g^S) = f$.*

*Proof.* (i) At $p \notin S$ the character $\chi_S$ is unramified, so the exponents agree (Lemma lem:twist(i)). At $q \in S$, $v_q(N) = 2$ ($q$ odd; Proposition 15.26), and by Lemma lem:twist(ii) the twist has type $\mathrm{I}_n$, $\mathrm{I}_0$, or a potentially good additive type, with exponent 1, 0, 2 (Proposition 15.26, rows (RSt), (RPS), last). The divisibilities are then immediate. (ii) $E = E_S \otimes \chi_S$, so $f = g^{\chi_S}$ by Proposition 15.29(iv); and $a_q(f) = 0$ for $q \in S$ as $q^2 \mid N$, so $\iota(g) = (g^{\chi_S})^S = f^S = f$ by Proposition 15.68(iii). (iii) $E_S$ is multiplicative, resp. good, at $q \in S$, so $g$ has type (St), resp. (U) (Proposition 15.22, Theorem 15.20), and the product is $N$ by (i). The depletion at $q$ of a form of level $N'$ with $q \,\|\, N'$ is $g - a_q(g)g(qz)$, of level $N'q$, and with $q \nmid N'$ it is $g - a_q(g)g(qz) + q\,g(q^2z)$, of level $N'q^2$ (Lemma 15.57); doing this for each $q \in S$ puts $g^S$ in $S_2(\Gamma_0(N))$. Finally $\iota(g^S) = \iota(g) = f$. □

**Example 15.70.** `539c1` ($\mathrm{I}_3^*$ at 7): $E_S$ = `77c1`, $N_S = 77 = 539/7$, and the depleted form $g^S = f_{77c}(z) + f_{77c}(7z)$ (as $a_7(f_{77c}) = -1$) lies at level 539. `539d1` ($\mathrm{I}_0^*$ at 7): $E_S$ = `11a3`, $N_S = 11 = 539/49$, and $g^S = f_{11a} + 2f_{11a}(7z) + 7f_{11a}(49z)$ ($a_7(f_{11a}) = -2$). `121a1` (II at 11): $N_S = 121$.

**Theorem 15.71** (the paper's Theorem thm:twist)**.** *Let $E/\mathbb{Q}$ have conductor $N$, $S$ a set of odd primes, $D = \prod_{q\in S} q$, $\chi_S = \prod_{q\in S} \chi_{q^*}$, and $E_S = E \otimes \chi_S$ of conductor $N_S$. If $q^2 \mid N$ for every $q \in S$ and* $\mathrm{lcm}(N_S, D^2) \mid N$, *then $r_{E_S} \mid r_E$; if moreover* $\mathrm{lcm}(N, D^2) \mid N_S$, *then $r_{E_S} = r_E$.*

The proof is in Chapter 20; in two lines: a congruence $g \equiv h \pmod m$ with $h \in g^\perp$ integral gives $f = \iota(g) \equiv \iota(h) \pmod m$ with $\iota(h) \in f^\perp$ integral (Proposition 15.68(ii), Proposition 15.69(ii)), so $m \mid r_E$ by Lemma 15.55(i); and the second hypothesis makes the situation symmetric. Proposition 15.69(i) shows that the hypotheses hold in the three cases listed after the theorem in the paper.

*In the paper* 15.72. The paragraph after Theorem thm:twist: "In cases (a) and (b) Theorem thm:quant predicts a difference equal to the Euler term, which is positive in case (a) at $\ell = 3$ for every $q \neq 3$, as $3 \mid q^2 - 1$: for instance $v_3$ of the degrees is 2 against 1 for 539c1 and 77c1, 2 against 1 for 980i1 and 140b1, 1 against 0 for 121d1 and 11a1, and 2 against 0 for 539d1 (type $\mathrm{I}_0^*$, $e_3(E,7) = 2$) and 11a1." Recomputed from Cremona's `degphi` file: $\deg\varphi = 288$, 6, 2880, 60, 24, 1, 72 for `539c1`, `77c1`, `980i1`, `140b1`, `121d1`, `11a1`, `539d1`, so the 3-adic valuations are 2, 1, 2, 1, 1, 0, 2, as stated, and each difference is $e_3(E,q)$: $v_3(7^2-1) = 1$, $v_3(7^2-1) = 1$, $v_3(11^2-1) = 1$, and $v_3(7-1) + v_3(8^2-(-2)^2) = 2$ (Proposition 15.48 with $a_7(f_{11a}) = -2$).

**Theorem 15.73** (the paper's Theorem thm:quant)**.** *Let $\ell$ be an odd prime with $\ell \nmid 2N$ and $E[\ell]$ irreducible, $S$ the set of odd primes $q$ at which $E$ has type $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$, and $E_S = E \otimes \chi_S$. Then*

$$v_\ell(\deg\varphi_E) \;\geq\; v_\ell(r_{E_S}) + \sum_{q\in S} e_\ell(E,q) \;\geq\; \sum_{q\in S} e_\ell(E,q).$$

*If moreover $\ell \notin S_f$ (automatic for $\ell \geq 5$), the first inequality is an equality, and $v_\ell(r_{E_S}) = v_\ell(\deg\varphi_{E'_S})$ for the optimal curve $E'_S$ of the class of $E_S$.*

The proof is in Chapter 20. Its four steps use, from this chapter: Proposition 15.69(iii) ($g^S$ at level $N$, $\iota(g^S) = f$), Proposition 15.68(i), (ii) (transport of complements), Lemma 15.55(iii) and its proof (the congruence ideal of the eigencharacter of $g^S$ as a largest congruence modulus, with the $\mathbb{T}$-stable complement in place of the Petersson complement), Lemma 15.57 (the $U_q$-eigenvalues on the span of the $g(dz)$: 0 occurs once, the others are $\pm 1$ or roots of $X^2 - a_q(g)X + q$, units at $\ell \neq q$), and Theorem 15.49 with Proposition 15.48 (Step 3: $L_q^{\mathrm{nv}}(A_g,1)^{-1} = L_q(A_f,1)^{-1}$, the factor that $f$ omits).

*Remark* 15.74 (Theorem thm:quant as level raising). Theorem thm:qual lowers the level of $\bar\rho \otimes \chi$ and twists back. Theorem thm:quant goes the other way, as the paper's remark after it says: "the level is raised, from $N_S$ to $N_S \prod q^{\delta_q} = N$, and the price is the level-raising Euler factor". The form $g = f_{E_S}$ of level $N_S$ is depleted at $S$, which places it at level $N$ (Proposition 15.69(iii)); by Theorem thm:dfg14 the congruence ideal grows by $\prod_{q\in S} L_q^{\mathrm{nv}}(A_g,1)^{-1}$, which is $1 - q^{-2}$ at the $\mathrm{I}_n^*$ primes and $\frac{q-1}{q} \cdot \frac{(q+1)^2 - a_q(E_S)^2}{q^2}$ at the $\mathrm{I}_0^*$ primes (Proposition 15.48), exactly the true adjoint factors that $f$ omits (Corollary 15.25); and $\iota$ transports the congruences to $f$. Examples at $\ell = 3$ (degrees from `degphi`, congruence numbers computed as in Theorem 15.54): `539d1`: $e_3(E,7) = v_3(6) + v_3(60) = 2$, $v_3(r_{11a}) = 0$, $v_3(72) = 2$; `539c1`: $e_3(E,7) = v_3(48) = 1$, $v_3(r_{77c}) = v_3(6) = 1$, $v_3(288) = 2$; `121d1`: $e_3(E,11) = v_3(120) = 1$, $v_3(r_{11a}) = 0$, $v_3(24) = 1$. Equality holds in all three, as the second part of the theorem predicts ($3 \notin S_f$ for these curves, whose mod 3 images are surjective). The potentially good types II, III, IV and their duals are not in $S$: there $N_S = N$, $f$ is its own depletion, Theorem thm:dfg14 gives no factor, and the omitted factor is reached only through Theorem thm:dfg37 (Chapter 20).

## 15.11 The literature on congruence ideals and Tamagawa exponents; Taylor-Wiles in one page

Section 1.5 of the paper places its results among theorems of Ribet and Takahashi, Pollack and Weston, and Kim and Ota, which compare congruence ideals at different levels with the "Tamagawa exponent" of $\rho$ at a multiplicative prime, and Remark rem:ko(2) derives one inequality of the paper from the formula of Kim and Ota. This section states those results with their

hypotheses, as far as the paper quotes them, and sketches the Taylor-Wiles method behind Theorem thm:dfg37.

**Definition 15.75** (the Tamagawa exponent)**.** Let $f$ be a newform of weight 2, $\lambda \mid \ell$ with $\bar\rho_{f,\lambda}$ irreducible, and $q \neq \ell$ a prime dividing the level. The *Tamagawa exponent* $t_q = t_q(\rho_{f,\lambda})$ is the largest $t \geq 0$ such that $\rho_{f,\lambda} \bmod \lambda^t$ (on any stable lattice; they are all homothetic, Proposition 15.9(iii)) is unramified at $q$. It is finite, as $\rho_{f,\lambda}$ is ramified at $q$. This is the paper's $t_q$ and Kim and Ota's $t_f(q)$ [99, Definition 2.12] (as the paper cites it).

**Proposition 15.76.** *For $f = f_E$ and a prime $q \,\|\, N$, $q \neq \ell$, $t_q = v_\ell(n_q) = t_\ell(E, q)$, whether the reduction is split or not.*

*Proof.* This is Lemma lem:tamagawa-exp of the paper, proved in Chapter 11: in the basis of the Tate shape, $\rho(\sigma) = \left(\begin{smallmatrix} 1 & n_q t_\ell(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ on $I_q$, with $t_\ell(I_q) = \mathbb{Z}_\ell$, so $\rho \bmod \ell^t$ is unramified at $q$ iff $\ell^t \mid n_q$; the twist by the unramified $\varepsilon$ does not change this. □

*Remark* 15.77 (Ribet and Takahashi). Ribet and Takahashi [164, 192] compare $\deg \varphi_E$ with the degree of a parametrisation of $E$ by the Shimura curve attached to an indefinite quaternion algebra of discriminant $N^-$ dividing $N$ ($N^-$ a product of an even number of primes), for $E$ semistable ($N$ square-free). The ratio is the product of the $n_q = \#\Phi_q^{\mathrm{geom}}$ (not of the Tamagawa numbers) over $q \mid N^-$, up to a rational factor supported on the primes $\ell$ at which $E[\ell]$ is reducible; exactly so when $E$ is alone in its isogeny class [192, Corollary 3.5], and one pair of primes at a time with an explicit error term [164, Theorems 1 and 2], [192, Theorem 3.2]. Chapter 14 states the comparison exactly, with these theorem numbers checked against the sources, and derives the $\ell$-adic form for $E[\ell]$ irreducible; Pasten [151] studies the error term further.

**Theorem 15.78** (Pollack and Weston)**.** *Let $\ell$ be an odd prime, $f$ a newform of weight 2, trivial character and square-free level $N = aqb$ prime to $\ell$, with $q$ prime, and $\lambda \mid \ell$. For $N = N_1N_2$ let $\eta_f(N_1, N_2)$ be the congruence ideal of $f$ in the Hecke algebra acting on the cusp forms of level $N$ that are new at every prime dividing $N_2$. Assume that $\bar\rho = \bar\rho_{f,\lambda}$ is surjective, that $\bar\rho$ is ramified at every prime $r \mid qb$ with $r \equiv \pm 1 \pmod{\ell}$, and that $\bar\rho$ is ramified at two primes at least. Then*

$$\operatorname{ord}_\lambda \eta_f(aq, b) = t_q + \operatorname{ord}_\lambda \eta_f(a, qb)\ :$$

*moving $q$ into the "new" part of the level changes the congruence ideal by exactly $\lambda^{t_q}$.*

We do not prove this. It is [153, Theorem 6.11], checked against the arXiv version (version 1 of `math/0610694`), whose numbering we cite (we have not compared it with the published version). There $p$ is our $\ell$, the assumptions on $\bar\rho$ other than the two ramified primes are their hypothesis CR for the pair $(\bar\rho_f, bq)$ (Section 1: $\bar\rho$ surjective, and ramified at every prime dividing $bq$ that is $\equiv \pm 1 \pmod{p}$), and the level is prime to $p$ by their standing assumptions (Notation). The identity for all $f$ of weight 2 and square-free level, with no hypothesis on $\bar\rho$, is the formula they propose in Section 6.6 and in display (2) of their introduction (it is not numbered as a conjecture; their Conjecture 6.1 is the anticyclotomic main conjecture). Their Definition 3.3 of the Tamagawa exponent is Definition 15.75. The paper's summary (Section 1.5) records both conditions, the two ramified primes and hypothesis CR.

**Theorem 15.79** (Kim and Ota)**.** *Let $N = N^+N^-$ with $N^-$ square-free, $\ell \geq 5$, $\ell \nmid N$, $\ell \notin S_f$, and $\bar\rho$ ramified at every $q \mid N^-$ with $q \equiv \pm 1 \pmod{\ell}$. Then*

$$\operatorname{ord}_\lambda \eta_f(N) = \operatorname{ord}_\lambda \eta_f(N^+, N^-) + \sum_{q \mid N^-} t_f(q),$$

*where $\eta_f(N)$ is the congruence ideal of $f$ in the full Hecke algebra of level $N$ and $\eta_f(N^+,N^-)$ that in the $N^-$-new quotient. Consequently, for $f=f_E$, $v_\ell(r_E)\geq\sum_{q\mid N^-}t_\ell(E,q)$.*

*Proof.* The formula is [99, Theorem 1.1], as the paper's Remark rem:ko(2) states it; we do not prove it. The consequence: $\eta_f(N^+,N^-)$ is an integral ideal, so $\operatorname{ord}_\lambda\eta_f(N^+,N^-)\geq 0$; $\operatorname{ord}_\lambda\eta_f(N)=v_\ell(r_E)$ by Lemma 15.55(iii) (the congruence ideal of $\phi_f$ in $\mathbb{T}_\ell$ is $\ell^{v_\ell(r_E)}\mathbb{Z}_\ell$); and $t_f(q)=t_\ell(E,q)$ by Proposition 15.76, the primes of $N^-$ being multiplicative. □

*In the paper* 15.80. Remark rem:ko(2): "Since $\eta_f(N^+,N^-)$ is integral, this gives $v_\ell(r_E)\geq\sum_{q\mid N^-}t_\ell(E,q)$, which (eq:thm) contains, without the conditions $\ell\geq 5$ and $q\not\equiv\pm1$; the equality is theirs." Theorem thm:main of the paper, which assumes $\ell\notin S_f$, contains this inequality for every multiplicative prime, including the primes $q\equiv\pm1 \pmod\ell$ and $\ell=3$; it does not give the equality, which is the content of the theorem of Kim and Ota.

*Remark* 15.81 (Taylor-Wiles and $R=\mathbb{T}$ in one page). Let $\bar\rho=\bar\rho_{f,\lambda}$ be absolutely irreducible, $\ell$ odd, and $\Sigma$ a finite set of primes not containing $\ell$. Mazur's *deformation ring* $R_\Sigma$ is the complete local $\mathcal{O}$-algebra that represents lifts of $\bar\rho$ (up to conjugation) which are minimally ramified outside $\Sigma\cup\{\ell\}$, unrestricted at the primes of $\Sigma$, and flat (or Fontaine-Laffaille, Theorem 15.31(ii)) at $\ell$, with determinant $\chi_\ell$. The localised Hecke algebra $\mathbb{T}_\Sigma=\mathbb{T}(N^\Sigma)_{\mathfrak{m}}$, at the level $N^\Sigma$ obtained by raising the level at $\Sigma$, carries such a lift (by Theorem 15.6 and Theorem 15.19), hence a surjection $R_\Sigma\to\mathbb{T}_\Sigma$. The theorem of Wiles and of Taylor and Wiles [207, 197], with Diamond's refinements, is that this map is an isomorphism of complete intersections. Its proof patches together Hecke modules at auxiliary levels $N^\Sigma Q$, where $Q$ is a set of *Taylor-Wiles primes* $q\equiv1 \pmod{\ell^n}$ at which $\bar\rho(\mathrm{Frob}_q)$ has distinct eigenvalues, chosen by Chebotarev so that the dual Selmer group is killed; finding them needs $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ absolutely irreducible. The *numerical criterion* then equates the length of $\wp/\wp^2$, the dual of the Selmer group $H^1_\Sigma(\mathbb{Q},W)$ of the adjoint, with the valuation of the congruence ideal $\eta^\Sigma$. What Theorem 3.7 of [49] (the paper's Theorem thm:dfg37) needs is exactly this: $\ell\notin S_f$, that is $\ell\nmid 2N$ and $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ absolutely irreducible, and $\Sigma\not\ni\ell$ with $M_{f,\lambda}$ minimally ramified outside $\Sigma$. For $\ell\geq5$, $\ell\nmid N$ and $E[\ell]$ irreducible the condition at $\mathbb{Q}(\sqrt{\ell^*})$ holds automatically (Chapter 11), so in the range of Theorem thm:main the hypothesis bites only for $\ell=3$ and image $N_s(\mathbb{F}_3)$. Theorem thm:qual avoids it: "Two statements hold under weaker hypotheses than Theorem thm:main, by level lowering and twisting rather than by the Taylor-Wiles method" (the paper's Section 1.5). Details are in Chapter 20 and [38].

## 15.12 Exercises

The exercises run from routine to harder. Those marked (machine) need PARI/GP and Cremona's tables; their answers were computed for this book.

**Exercise 15.82** ($\ell=2$)**.** Show that Proposition 15.9(ii) fails for $\ell=2$: for $E\colon y^2=x^3-3x+1$, the cubic has square discriminant 81, so $\mathbb{Q}(E[2])$ is a cyclic cubic field; deduce that $\bar\rho_{E,2}$ is irreducible over $\mathbb{F}_2$ but not absolutely irreducible. Which step of the proof of Proposition 15.9(ii) breaks? *Hint.* The image is the cyclic subgroup of order 3 of $\mathrm{GL}_2(\mathbb{F}_2)\simeq S_3$, which fixes no line of $\mathbb{F}_2^2$ but is diagonalisable over $\mathbb{F}_4$; complex conjugation acts trivially, as $\mathbb{Q}(E[2])$ is totally real, and in any case $1=-1$ in $\mathbb{F}_2$.

**Exercise 15.83** (naive and newform twists)**.** (machine) (a) Show that the twists of `11a1` by $\mathbb{Q}(\sqrt{-3})$ and by $\mathbb{Q}(\sqrt{5})$ are `99d2` and `275b2`, and deduce from Corollary 15.27 that $f_{11a}\otimes\chi_{-3}=$

$f_{99d}$ and $f_{11a} \otimes \chi_5 = f_{275b}$. (b) Show that $f_{77c} \otimes \chi_{-7} = f_{539c}$ but $f_{539c} \otimes \chi_{-7} \neq f_{77c}$, and compute $(f_{77c} \otimes \chi_{-7}) \otimes \chi_{-7}$. *Answer to (b).* `77c1` has type (St) at 7, so the first is Corollary 15.27; the second has $a_7 = 0 \neq -1$; and $(f_{77c} \otimes \chi_{-7}) \otimes \chi_{-7} = f_{77c}^{\{7\}}$.

**Exercise 15.84** (twists of curves and optimality)**.** (machine) Show that the twist of the optimal curve `19a1` by $\mathbb{Q}(\sqrt{-3})$ is `171b2`, not the optimal curve `171b1`, and that `171b1` is the twist of `19a3`; so twisting preserves optimality in neither direction. Explain why, when $E[\ell]$ is irreducible, $v_\ell$ of the degree of the minimal parametrisation $X_0(N) \to E$ of any curve $E$ of the class equals $v_\ell(\deg \varphi_{E_0})$, so that Theorem thm:qual can be stated for any $E$ in the class. Why does this argument say nothing at $\ell = 3$ for the classes 19a and 171b? *Hint.* When $E[\ell]$ is irreducible the isogenies in the class have degree prime to $\ell$ (Chapter 11). *Answer to the last question.* Both classes consist of three curves linked by isogenies of degrees 3 and 9 (Cremona's `allisog`), so $E[3]$ is reducible; $\deg \varphi_{171b1} = 8$ and $\deg \varphi_{19a1} = 1$.

**Exercise 15.85** (ramification of $\bar\rho$ at a Steinberg prime)**.** Let $f$ have type (St) at $p \neq \ell$. Using Theorem 15.20(a), show that $\bar\rho_L$ is unramified at $p$ for one stable lattice $L$ if and only if the corner of $\rho|_{I_p}$ is divisible by $\lambda$ on that lattice. For $f = f_{11a}$ and $p = 11$, find all $\ell$ for which some stable lattice has unramified reduction at 11, and compare $T_5$(`11a1`) with $T_5$(`11a3`). *Answer.* Only $\ell = 5$ ($n_{11} = 5$ for `11a1`); on $T_5$(`11a3`), where $n_{11} = 1$, the reduction is ramified. The two lattices are not homothetic, which is possible because $\bar\rho_{11a,5}$ is reducible.

**Exercise 15.86** (the conductor of twists)**.** (machine) Check Proposition 15.26 on the twists `14a1`$\otimes\chi_{-7}$ and `15a1`$\otimes\chi_{-3}$ (row (St)), `11a1`$\otimes\chi_{-7}$ (row (U)), `539c1`$\otimes\chi_{-7}$ (row (RSt)) and `539d1`$\otimes\chi_{-7}$ (row (RPS)) by identifying the twisted curves. *Answer.* `98a3`, `45a4`, `539d2`, `77c1`, `11a3`.

**Exercise 15.87** (Serre conductors)**.** Compute $N(\bar\rho_{E,\ell})$ and $N(\bar\rho_{E,\ell} \otimes \chi_{-7})$ for $(E, \ell) =$ (`539c1`, 3) and (`980i1`, 5), using the formula $a_p = f_p - \dim \Phi_p^{\mathrm{geom}}[\ell]$. *Answer.* 539 and 11; 196 and 4. Explain why in the second case the newform attached to $\bar\rho \otimes \chi_{-7}$ has level 20.

**Exercise 15.88** (characters of order divisible by $\ell$)**.** Let $\psi$ be a Dirichlet character and $\lambda \mid \ell$ a prime of $\mathbb{Z}[\psi]$ with $\psi \equiv 1 \pmod\lambda$. Show that $\psi = 1$ if the order of $\psi$ is prime to $\ell$. Give a character $\psi \neq 1$ of conductor 7 and a prime $\lambda \mid 3$ with $\psi \equiv 1 \pmod\lambda$. Where does the proof of Corollary 15.43 use that the character given by Theorem 15.42 has order prime to $\ell$? *Answer.* A cubic character modulo 7 takes values in $\mu_3$, and $\zeta_3 \equiv 1$ modulo $(1 - \zeta_3)$.

**Exercise 15.89** (level raising for $f_{37a}$)**.** (machine) For $f = f_{37a}$ and $\ell = 3$ ($\bar\rho_{37a,3}$ is surjective), find the primes $q < 25$, $q \neq 3, 37$, with $3 \mid (q+1)^2 - a_q^2$, and check that newforms of level $37q$ congruent to $f$ modulo a prime above 3 exist exactly for those $q$. *Answer.* $q = 7$ (63), 13 (192), 17 (324); at each of these levels there is a congruent rational newform (and at $q = 7$, 17 also forms with larger coefficient fields); for $q = 2, 5, 11, 19, 23$ there is none.

**Exercise 15.90** (the level-raising factor at $\ell = 3$)**.** Compute $v_3$ of $\frac{q-1}{q} \cdot \frac{(q+1)^2 - a_q(f_{11a})^2}{q^2}$ for $q = 7$ and $q = 13$. Use Theorem thm:quant to predict a lower bound for $v_3(\deg\varphi)$ for the optimal curves of the classes of `11a1`$\otimes\chi_{-7}$ and `11a1`$\otimes\chi_{13}$, and check it in the tables (machine). *Answer.* 2 and $1 + 2 = 3$; the classes are 539d ($\deg\varphi = 72$) and 1859a ($\deg\varphi = 432 = 2^4 \cdot 3^3$), so the bound is attained in both.

**Exercise 15.91** (a toy congruence module)**.** Let $r \geq 2$ and $\mathbb{T} = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod r\}$, a ring under componentwise operations, acting on $S = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{T}, \mathbb{Z})$ by $(t \cdot s)(t') = s(tt')$. The

two projections $\phi, \psi\colon \mathbb{T} \to \mathbb{Z}$ lie in $S$ and are eigenvectors of $\mathbb{T}$ (with eigencharacters $\phi$ and $\psi$), playing the roles of $f$ and of an eigenform orthogonal to it. Compute $I = \ker \phi$, $J = \mathrm{Ann}_{\mathbb{T}}(I)$, $\mathbb{T}/(I+J)$, the index $[S : \mathbb{Z}\phi \oplus \mathbb{Z}\psi]$, and the largest $m$ with $\phi - \psi \in mS$. *Answer.* $I = 0 \times r\mathbb{Z}$, $J = r\mathbb{Z} \times 0$, $\mathbb{T}/(I+J) \simeq \mathbb{Z}/r$, the index is $r$, and $(\phi - \psi)/r \in S$ because $a - b \in r\mathbb{Z}$ on $\mathbb{T}$, so $m = r$.

**Exercise 15.92** (the congruence number of `196a1`)**.** (machine) `196a1` is optimal, of conductor $196 = 2^2 \cdot 7^2$, with types IV at 2 and II at 7, and $\deg \varphi = 6$. Adapt the code of the Notes to compute $r = [S_2(\Gamma_0(196); \mathbb{Z}) : \mathbb{Z}f \oplus L]$ (the Notes use $T_5$, which works here, and $T_3$ gives the same index; what matters is that $a_p(f)$ is a simple eigenvalue of $T_p$). At which primes $p$ with $p^2 \mid N$ do $v_p(r)$ and $v_p(\deg \varphi)$ differ? *Answer.* $r = 42 = 7 \cdot 6$ (the same with $T_3$ and $T_5$); $v_7(42) = 1 \neq 0 = v_7(6)$, while $v_2(42) = 1 = v_2(6)$: the hypothesis $p^2 \nmid N$ of Theorem 15.54 allows a difference at $p$ but does not force one.

**Exercise 15.93** (the hypothesis of Lemma lem:old)**.** Let $f$, $h$, $\ell$, $\lambda$ be as in Lemma 15.58, $p \parallel N$ with $\bar\rho$ unramified at $p$, and suppose that $p \parallel M$ (the excluded case). Show that $a_p(h) \equiv a_p(f) \pmod{\lambda}$ unless $p \equiv -1 \pmod{\ell}$, and that when $p \equiv -1 \pmod{\ell}$ the two signs may differ. Why does this not contradict the lemma? *Hint.* Both $\{a_p(f)p, a_p(f)\}$ and $\{a_p(h)p, a_p(h)\}$ are the eigenvalues of $\bar\rho(\mathrm{Frob}_p)$; if $a_p(h) = -a_p(f)$ then $p \equiv -1$. When $p \mid M$ the operator $U_p$ acts on $V_h$ by the scalar $a_p(h)$, so no choice is left.

**Exercise 15.94** (case (a) at $q = 2$: `76a1`)**.** (machine) `76a1` has type IV$^*$ at 2 and $\mathrm{I}_1$ at 19, and surjective mod 3 image. Compute $N(\bar\rho_{76a1,3})$, find the newform $g$ given by Corollary 15.43, the eigenform of Lemma lem:old, and check $3 \mid \deg \varphi$. *Answer.* $N(\bar\rho) = 38$; $g = f_{38b}$ (not $f_{38a}$); $a_2(f_{38b}) = 1$, so the eigenform is $f_{38b}(z) - f_{38b}(2z)$; $\deg \varphi = 6$.

**Exercise 15.95** (the step $h \neq f$ by local types)**.** In cases (b) and (c) of the proof of Theorem thm:qual, give a second proof of $h \neq f$ by comparing the types at $q$ of $h = g^\chi$ and of $f$ (the type is determined by the newform, Theorem 15.19(i)). Then show that this argument cannot work for type $\mathrm{I}_0^*$, and compare with Remark 15.64(ii). *Answer.* In case (b), $q \parallel M$, so $g$ has type (St) at $q$ and $h$ has type (RSt) (Proposition 15.26), while $f$ has type (RPS) or (SC) (Proposition 15.22). In case (c), $q \nmid M$, so $g$ has type (U) and $h$ type (RPS), while $f$ has type (RSt) ($f$ is the twist by $\chi$ of $f_{E'}$, of type (St), so of type (RSt) by Proposition 15.26; this covers $q = 3$, where the Kodaira dictionary is not stated). For type $\mathrm{I}_0^*$, $f$ has type (RPS) with characters $\chi_{q^*}$ times unramified ones, which is also the type of $g^\chi$ for every $g$ of type (U) at $q$; so the types cannot separate $h$ from $f$, and indeed $h = f$ can occur.

**Exercise 15.96** (another instance of case (c))**.** (machine) `1175b1` has type $\mathrm{I}_3^*$ at 5 and surjective mod 3 image. Show that its twist by $\mathbb{Q}(\sqrt{5})$ is `235a1`, with types $\mathrm{I}_3$ at 5 and $\mathrm{I}_1$ at 47, so $M = N(\bar\rho \otimes \chi_5) = 47$. Show that the newform $g$ of level 47 (coefficient field $\mathbb{Q}(y)$, $y^4 - 4y^2 - y + 1 = 0$) satisfies $a_p(g) \equiv \chi_5(p) a_p(\texttt{1175b1})$ modulo the prime of residue degree 1 above 3, and not modulo the other. What is $h$, and what is its level? Check $3 \mid \deg \varphi$. *Answer.* $h = g \otimes \chi_5$, a newform of level $47 \cdot 25 = 1175$ with quartic coefficient field; $\deg \varphi = 288$.

**Exercise 15.97** (a type $\mathrm{I}_0^*$ curve)**.** (machine) Show that the twist of `37a1` by $\mathbb{Q}(\sqrt{-7})$ is `1813b1`, of type $\mathrm{I}_0^*$ at 7, and that its mod 3 representation is surjective (Cremona's `galrep` lists no exceptional image). Run the construction of the proof of Theorem thm:qual at $\ell = 3$ as if this were case (c): compute $N(\bar\rho \otimes \chi_{-7})$ and show that the construction returns $h = f$. Then compute $e_3(E, 7)$ from Proposition 15.48 with $a_7(f_{37a}) = -1$, and compare the bound of Theorem 15.73 with $\deg \varphi_{1813b1}$.

*Answer.* $\bar\rho\otimes\chi_{-7}\simeq\bar\rho_{37a1,3}$ is ramified at 37 (type $\mathrm{I}_1$, $3\nmid 1$), so $N(\bar\rho\otimes\chi_{-7})=37=N/7^2$; of the two newforms of level 37, $f_{37b}$ has reducible mod 3 representation (image `3Cs.1.1`), so $g=f_{37a}$ and $h=g^{\chi_{-7}}=f_{1813b}=f$. Next $e_3(E,7)=v_3(7-1)+v_3(8^2-1)=1+2=3$ and $v_3(r_{37a})=v_3(2)=0$, so $v_3(\deg\varphi)\geq 3$; and $\deg\varphi_{1813b1}=756=2^2\cdot 3^3\cdot 7$, so the bound is attained.

**Exercise 15.98** (Theorem thm:quant at $\ell=5$)**.** (machine) For `361b1` ($\mathrm{I}_1^*$ at 19) and `363a1` ($\mathrm{I}_1^*$ at 11), whose mod 5 representations are irreducible, find $E_S$, compute $e_5(E,q)=v_5(q^2-1)$, and compare $v_5(\deg\varphi_E)$ with $v_5(\deg\varphi_{E'_S})+e_5(E,q)$. *Answer.* $E_S$ in the classes 19a and 33a; $e_5=1$ in both; degrees 120, 1 and 180, 3: equality.

**Exercise 15.99** (Tamagawa exponents depend on the lattice when $\bar\rho$ is reducible)**.** Show that $t_q$ of Definition 15.75 does not depend on the stable lattice when $\bar\rho$ is irreducible. For $f=f_{11a}$, $\ell=5$, $q=11$ compute the largest $t$ with $\rho \bmod 5^t$ unramified at 11 on $T_5$(`11a1`) and on $T_5$(`11a3`). *Answer.* 1 and 0 ($n_{11}=5$, resp. 1).

## 15.13 Notes and references

**Sources followed.** The construction of $\rho_{f,\lambda}$ in weight 2 follows Shimura [180, Chapter 7] and Diamond and Shurman [52, Chapter 9], with the integral and residual facts as in Diamond and Im [51] and Darmon, Diamond and Taylor [38]; the residual representation of a maximal ideal of $\mathbb{T}$ rests on the lifting lemma of Deligne and Serre [45, Lemme 6.11] and, for its field of definition, on their Lemme 6.13 and on Carayol [23]; the general construction is Deligne's [44], and irreducibility in general is Ribet's [158]. The automorphic representation of a newform is from Jacquet and Langlands [89], conductors of representations of $\mathrm{GL}_2(\mathbb{Q}_p)$ from Casselman [25], the local Langlands correspondence from Bushnell and Henniart [19], and local-global compatibility from Langlands [111], Deligne [41] and Carayol [21]; Rohrlich [166] is the reference for Weil-Deligne representations of elliptic curves, and Loeffler and Weinstein [118] for computing local components. The local statements at $\ell$ are quoted from Edixhoven [61, Section 2] (as the paper cites them), Serre [171, 176], Wiles [206], Fontaine and Laffaille [68] and Berger [11]. Serre's conjecture is [176], proved by Khare and Wintenberger [98] (with Khare [97] and Kisin [100]); level lowering is due to Mazur [125], Ribet [156, 162], Carayol [22] and Diamond [47], and the conductor of a residual representation was studied by Livné [116]; level raising is Ribet's [160], in the general form of Diamond and Taylor [53]. The congruence criterion goes back to Ribet [159] and the perfect pairing to Ribet and to Abbes and Ullmo [1]; Agashe, Ribet and Stein [3] relate the congruence number to the modular degree. Twists of newforms are in Atkin and Li [9]. The congruence ideal at higher level is in Diamond, Flach and Guo [49]; Kim and Ota [99] and Pasten [151] are quoted as the paper summarises them; Pollack and Weston [153, Theorem 6.11] was checked against its arXiv version; Ribet and Takahashi [164] and Takahashi [192] were checked against the sources in Chapter 14. The Taylor-Wiles method is [207, 197], surveyed in [38].

**Theorem numbers.** The numbers printed are those the paper itself cites: [38, Theorems 3.1(e), 3.15], [159, Theorem 2.2], [1, Lemma 2.1], [8, Theorem 3], [9, Theorem 3.1], [3, Theorem 2.2], [61, Section 2], [49, Proposition 1.4(c)], [99, Theorem 1.1, Definition 2.12]. "Theorem 7.14" of [180] is an unverified pointer from the edition we know. The lifting lemma [45, Lemme 6.11], the descent lemma [45, Lemme 6.13] and [23, Théorème 2] were checked against the published

texts. All other references are cited by chapter, section or as a whole; we have not verified the numbering of the main theorems of [21], [160], [156] or [53], nor the precise hypotheses of [160] and [22] (Remark 15.41); the statements given here are those the paper uses, or special cases of the published ones.

**Normalisations.** Two normalisations of the local Langlands correspondence for $\mathrm{GL}_2$ are in use, and some sources attach to a newform the dual of $\rho_{f,\lambda}$; this chapter avoids the issue by stating everything it uses on the Galois side (Theorem 15.17 and the paragraph after it). Our $\rho_{f,\lambda}$ has arithmetic Frobenius trace $a_p$ and determinant $\chi_\ell$, as $\rho_{E,\ell}$ does; DFG04 work with cohomology, where the geometric Frobenius has trace $a_p$ (Theorem 15.6(iii)).

**Quotations and computations.** Quotations of the paper are verbatim, except that its internal cross-references are printed as label names ("Lemma lem:twist(iii)"). Every number in this chapter (coefficients, Kodaira types, twists, Serre conductors, congruences, Sturm bounds, congruence numbers, modular degrees, the level-raising tables and the count of 1,051 curves) was recomputed with PARI/GP 2.17.2 [148] through the Python package `cypari2` and Cremona's tables [35] for conductors below 10,000. The congruence numbers were computed as indices $[S_2(\Gamma_0(N);\mathbb{Z}) : \mathbb{Z}f \oplus L]$ (Lemma 15.55(i)), with $S_2(\Gamma_0(N);\mathbb{Z})$ the saturation of the $q$-expansion lattice of `mfinit([N,2],1)` up to twice the Sturm bound and $L$ the saturated image of $T_p - a_p(f)$ for a prime $p \nmid N$ at which $a_p(f)$ is a simple eigenvalue; for example:

```
mf = mfinit([99,2],1); n = 2*mfsturm([99,2]) + 20;
C = mfcoefs(mf, n); E = ellinit("99a1");
f = concat([0], ellan(E, n))~; M = mfheckemat(mf, 5);
S = matrixqz(C, -2); L = matrixqz(C*(M - ellap(E,5)), -2);
abs(matdet(lift(matinverseimage(S, matconcat([f, L])))))   \\ 12
```

Levels and twists of curves were identified by `elltwist`, `ellminimalmodel` and lookup in `allcurves`; newforms with irrational coefficients by `mfeigenbasis`, `mffields` and reduction modulo each prime above $\ell$ (`idealprimedec`, `nfeltval`).

# Part V

# Cohomology, Selmer groups and $L$-functions

# Chapter 16

# Homological algebra and group cohomology

The paper's Tamagawa factor $\mathrm{Tam}_q(\mathrm{ad}^0 T)$ is the Fitting ideal of the Frobenius-invariants of the torsion of $H^1(I_q, \mathrm{ad}^0 T)$, and its proof of Lemma lem:tamfactor computes that group in one paragraph: inflation-restriction, the identity $H^1(\langle t\rangle, M) = M/(t-1)M$, the conjugation action of Frobenius, an explicit $3\times 3$ matrix, the invariants. Around Lemma lem:dfg21 it uses long exact sequences, Pontryagin duals, the vanishing of the cohomology of the real place, and the formula $H^1_{\mathrm{ur}} = M^I/(\mathrm{Frob}-1)M^I$. This chapter supplies the group cohomology behind all of it, for abstract and for profinite groups, with continuous cochains and with the three kinds of coefficients the paper uses (finite or torsion discrete modules, finitely generated $\mathbb{Z}_\ell$-modules, $\mathbb{Q}_\ell$-vector spaces). Its centre is Section 16.10, where the group-cohomological part of Lemma lem:tamfactor is proved line by line. Galois cohomology proper (Kummer theory, local duality, Selmer groups) is Chapter 17; the Bloch-Kato conditions and the Tamagawa factors are Chapter 18.

## Core results at a glance

Throughout, $\ell$ is a prime; groups act on the left; all cohomology of profinite groups is continuous cochain cohomology with coefficients of type (D) (discrete, finite or $\ell$-primary torsion), (C) (finitely generated $\mathbb{Z}_\ell$-modules) or (V) (finite-dimensional $\mathbb{Q}_\ell$-spaces); Frob is an arithmetic Frobenius.

**16.1 Conventions.** Types (D), (C), (V); Frob arithmetic, Frob $t$ Frob$^{-1} = t^q$, Frob$^{-1}\, t\,$Frob $= t^{q^{-1}}$; conjugation action $(\sigma\cdot c)(h_1,\dots,h_n) = \sigma c(\sigma^{-1}h_1\sigma,\dots)$; $\ell$ odd is used only in Section 16.10 and for groups of order 2. *(convention)* Convention 16.1.

**16.2 Complexes.** Cochain complexes, cocycles, coboundaries, $H^n = Z^n/B^n$, morphisms and homotopies. *(definition)* Definition 16.3.

**16.3 Functoriality.** Morphisms induce maps on $H^n$ functorially; homotopic morphisms induce the same map. *(proved here)* Lemma 16.4.

**16.4 Snake lemma.** For a map of exact rows $A'\to A\to A''\to 0$, $0\to B'\to B\to B''$: $\ker f'\to\ker f\to\ker f''\xrightarrow{\delta}\mathrm{coker}\, f'\to\mathrm{coker}\, f\to\mathrm{coker}\, f''$ exact, $\delta$ by lift, apply $f$, pull back. *(stated; proved in Chapter 3)* Lemma 16.5.

**16.5 Long exact sequence.** A short exact sequence of complexes gives a natural long exact sequence with $\delta[c] = [a]$, $i(a) = db$, $p(b) = c$. *(proved here)* Theorem 16.6.

**16.6 Resolutions.** Projective and injective modules and resolutions. *(definition)* Definition 16.8.

**16.7 Existence and comparison.** Free and injective resolutions exist; comparison theorem; over a PID resolutions of length one, divisible modules injective. *(stated; reference)* Proposition 16.9.

**16.8 Ext and Tor.** $\operatorname{Ext}^n_R(M,N) = H^n \operatorname{Hom}_R(P_\bullet, N) = H^n \operatorname{Hom}_R(M, I^\bullet)$, $\operatorname{Tor}^R_n(M,N) = H_n(P_\bullet \otimes N)$. *(definition)* Definition 16.10.

**16.9 Properties.** Independence of resolutions, long exact sequences, vanishing; over a PID nothing in degree $\geq 2$; $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/m, N) = N/mN$, $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/m, N) = N[m]$, $\operatorname{Ext}^1_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^a, \mathbb{Z}/\ell^b) = \mathbb{Z}/\ell^{\min(a,b)}$. *(stated; reference; examples computed here)* Proposition 16.11.

**16.10 $\operatorname{Ext}^1$ and extensions.** $\operatorname{Ext}^1_R(M,N)$ classifies extensions of $M$ by $N$; 0 is the split one. *(stated; reference)* Theorem 16.12.

**16.11 $G$-modules.** $M^G$, $M_G = M/I_G M$, $N_G$; $\operatorname{Hom}_G(M,N) = \operatorname{Hom}(M,N)^G$; $\mathbb{Z}_\ell[G]$-modules give $\mathbb{Z}_\ell$-linear cohomology. *(definition)* Definition 16.14.

**16.12 Inhomogeneous cochains.** $C^n(G,M) = \operatorname{Map}(G^n, M)$ with the standard $d$; $(d^0 m)(g) = gm - m$, $(d^1\varphi)(g,h) = g\varphi(h) - \varphi(gh) + \varphi(g)$; $H^n(G,M) = H^n(C^\bullet(G,M))$. *(definition)* Definition 16.15.

**16.13 $d \circ d = 0$; functoriality in $M$.** *(proved here in degrees 0, 1; degree 2 an exercise; general degree by reference)* Lemma 16.16.

**16.14 Cohomology as Ext.** $H^n(G,M) \simeq \operatorname{Ext}^n_{\mathbb{Z}[G]}(\mathbb{Z}, M)$ via the bar resolution; $H^n(G,-)$ are the derived functors of $M \mapsto M^G$. *(stated; reference)* Theorem 16.17.

**16.15 $H^0$ and $H^1$.** $H^0 = M^G$; $H^1$ = crossed homomorphisms modulo principal ones, $\varphi(1) = 0$, $\varphi(g^{-1}) = -g^{-1}\varphi(g)$; trivial action: $H^1 = \operatorname{Hom}(G,M)$; $0 \to M'^G \to M^G \to M''^G \xrightarrow{\delta} H^1(M') \to H^1(M) \to H^1(M'')$ exact with $\delta(m'') = [g \mapsto gm - m]$. *(proved here)* Proposition 16.18.

**16.16 The cocycle in the corner.** $\rho = \left(\begin{smallmatrix} \eta_1 & \kappa \\ 0 & \eta_2 \end{smallmatrix}\right)$ is a homomorphism iff $\kappa\eta_2^{-1} \in Z^1(G, R(\eta_1\eta_2^{-1}))$; base change $e_2 \mapsto e_2 + re_1$ changes it by $d^0 r$; split iff the class is 0; every class occurs. For $\left(\begin{smallmatrix} \chi_\ell\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$ the cocycle is $\kappa \in Z^1(G, \mathbb{Z}_\ell(1))$, a homomorphism on $I$. *(proved here)* Proposition 16.21.

**16.17 $H^2$ and extensions.** $H^2(G,M)$ classifies extensions $1 \to M \to \mathcal{E} \to G \to 1$ inducing the action; cocycle $s(g)s(h)s(gh)^{-1}$. *(proof sketched here; Brown)* Theorem 16.23.

**16.18 Long exact sequence in $M$.** $0 \to M' \to M \to M'' \to 0$ gives $0 \to H^0(G,M') \to \cdots \to H^n(G,M'') \xrightarrow{\delta} H^{n+1}(G,M') \to \cdots$, natural, $\mathbb{Z}_\ell$-linear. *(proved here)* Proposition 16.24.

**16.19 $H^0$ and torsion.** $H^0(G, W[m]) = H^0(G,W)[m]$; for a $\mathbb{Z}_\ell[G]$-module $T$ without $\ell$-torsion, $W[\ell^k] \simeq T/\ell^k T$ for $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, so $H^0(G,W)[\ell^k] = H^0(G, T/\ell^k T)$; a finite $\ell$-group with $\#A[\ell] \leq \ell$ is cyclic of length the largest $k$ with an element of order $\ell^k$; if $\#H^0(G, T/\ell T) \leq \ell$,

$H^0(G,W)$ is finite cyclic or $\simeq \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and when finite its length is the largest $k$ with an invariant element of $T/\ell^k T$ outside $\ell T/\ell^k T$. *(proved here, given Chapter 3)* Lemma 16.25.

**16.20 Compatible pairs.** $(f\colon G' \to G,\ \varphi\colon M \to M')$ with $\varphi(f(g')m) = g'\varphi(m)$ induces $(f,\varphi)^*\colon H^n(G,M) \to H^n(G',M')$ on cochains. *(proved here)* Proposition 16.27.

**16.21 Res, Inf, Cor.** Restriction and inflation as compatible pairs; corestriction for finite index, $\mathrm{Cor}(m) = \sum g_i m$, $(\mathrm{Cor}\,\varphi)(g) = \sum_i g_{\tau(i)}\varphi(h_i(g))$; $\mathrm{Cor}\circ\mathrm{Res} = [G:H]$; for $H$ normal $\mathrm{Res}\circ\mathrm{Cor} = \sum_{\sigma\in G/H}\sigma_*$. *(proved here in degrees 0, 1; higher degrees stated)* Proposition 16.28.

**16.22 The order kills.** For $G$ finite of order $N$, $N\cdot H^n(G,M) = 0$ $(n \geq 1)$, explicitly $N\varphi = (-1)^n d\psi$, $\psi = \sum_g \varphi(\dots, g)$; so $H^n = 0$ if $N$ is invertible on $M$, $H^n$ finite for $M$ finitely generated, and for $\ell \nmid N$ and $\ell$-primary or $\mathbb{Z}_\ell$-modules $H^n = 0$ and $M^G$ is a direct summand. *(proved here)* Theorem 16.29.

**16.23 Conjugation action.** For $H \trianglelefteq G$, $(\sigma\cdot\varphi)(h_1,\dots) = \sigma\varphi(\sigma^{-1}h_1\sigma,\dots)$ is a $\mathbb{Z}_\ell$-linear left action on $H^n(H,M)$, trivial on $H$ $((h_0\cdot\varphi)-\varphi = d^0\varphi(h_0))$, so an action of $G/H$; Res lands in the invariants $(\sigma\cdot\mathrm{Res}\,c - \mathrm{Res}\,c = d^0c(\sigma))$. Same for profinite groups. *(proved here for $n\leq 1$; triviality on $H$ for $n\geq 2$ stated)* Proposition 16.30.

**16.24 Inflation-restriction.** $0 \to H^1(G/H,M^H) \to H^1(G,M) \to H^1(H,M)^{G/H}$ exact, extended by the transgression to $H^2(G/H,M^H) \to H^2(G,M)$; Inf is an isomorphism on $H^1$ if $H^1(H,M) = 0$; valid with continuous cochains, the five-term sequence for types (D) and, when $G/H \simeq \widehat{\mathbb{Z}}$, (C) and (V) (Proposition 16.61). *(three-term sequence and corollary proved here; five-term sequence and higher version stated, transgression sketched)* Theorem 16.31.

**16.25 Hochschild-Serre.** $E_2^{p,q} = H^p(G/H, H^q(H,M)) \Rightarrow H^{p+q}(G,M)$; if $H^q(H,M) = 0$ for $q \geq 1$, Inf is an isomorphism in all degrees. *(stated; reference)* Theorem 16.33.

**16.26 Finite cyclic groups.** $H^1(\langle t\rangle, M) = \ker N/(t-1)M$ via $\varphi \mapsto \varphi(t)$; $H^2 = M^G/NM$; period 2. *($H^1$ proved here; the rest stated)* Theorem 16.34.

**16.27 Tate cohomology.** $\hat H^0 = M^G/N_G M$, $\hat H^{-1} = \ker N_G/I_G M$, $\hat H^n = H^n$ $(n \geq 1)$. *(definition)* Definition 16.35.

**16.28 Properties of $\hat H$.** Long exact sequences in all degrees; killed by $\#G$; induced modules acyclic; period 2 for cyclic $G$. *(stated; reference)* Proposition 16.36.

**16.29 Groups of order 2.** $H^n(\mathbb{Z}/2,M)$ $(n\geq 1)$ is killed by 2; for $\ell$ odd and $\ell$-adic $M$ it is 0 and $M = M^+ \oplus M^-$; so $H^1(\mathbb{R},M)$ has exponent 2 and vanishes for $\ell$-adic $M$, $\ell$ odd. *(proved here)* Corollary 16.37.

**16.30 Discrete modules.** Continuous action on a discrete module. *(definition)* Definition 16.40.

**16.31 Discrete equivalences.** Continuous iff stabilisers open iff $M = \bigcup_U M^U$. *(proved here)* Lemma 16.41.

**16.32 Continuous cohomology.** $C^n_{\mathrm{cont}}(G,M)$; types (D), (C), (V); the convention that all profinite cohomology is continuous. *(definition)* Definition 16.42.

**16.33 Basic properties.** $H^1$ = continuous crossed homomorphisms mod principal; discrete $M$: $H^n(G,M) = \varinjlim_U H^n(G/U, M^U)$; type (C) continuity criterion; type (V) has a stable lattice. *(proved here)* Proposition 16.43.

**16.34 Long exact sequences, continuous.** Exist when $p$ has a continuous section, in particular for $M''$ discrete, surjections of finitely generated $\mathbb{Z}_\ell$-modules, $T \to V \to W$, $T \xrightarrow{\ell^k} T \to T/\ell^k$; and $\ker(H^1(G,T) \to H^1(G,V)) = H^1(G,T)_{\mathrm{tor}} = \delta(H^0(G,W))$. *(proved here)* Proposition 16.44.

**16.35 Tate's limit theorem.** $0 \to \varprojlim{}^1 H^{n-1}(G,T_k) \to H^n(G,T) \to \varprojlim H^n(G,T_k) \to 0$; $H^1(G,T) = \varprojlim H^1(G,T/\ell^k T)$ always; $H^n(G,V) = H^n(G,T) \otimes \mathbb{Q}_\ell$. *(stated; reference; the case $n=1$ and the statement for $V$ proved here)* Theorem 16.45.

**16.36 Finiteness of $H^1$.** $H^1(G,T) = \varprojlim H^1(G,T/\ell^k)$; if $H^1(G,A)$ is finite for finite $\ell$-primary $A$ (for instance $G$ finite, procyclic onto $\mathbb{Z}_\ell$, or $I_q$ with $q \neq \ell$), $H^1(G,T)$ is finitely generated and its torsion finite. *(proved here, given topological Nakayama, Remark 16.47)* Proposition 16.48.

**16.37 Supernatural orders.** $\#G$, pro-$\ell$ groups, order prime to $\ell$, $G^{(\ell)} = G/K_\ell$. *(definition)* Definition 16.49.

**16.38 Pro-$\ell$ quotients.** $G^{(\ell)}$ is the maximal pro-$\ell$ quotient; $\ell \nmid \#G \Rightarrow G^{(\ell)} = 1$, not conversely ($A_4$, $\ell = 2$); $K_\ell^{(\ell)} = 1$; closed subgroups of pro-$\ell$ groups and finitely generated $\mathbb{Z}_\ell$-modules are pro-$\ell$; Sylow subgroups. *(proved here; Sylow stated)* Lemma 16.50.

**16.39 Prime-to-$\ell$ vanishing.** If $\ell \nmid \#G$: $H^n(G,M) = 0$ ($n \geq 1$) for $M$ of type (D) $\ell$-primary, (C), (V), and $M^G$ is a direct summand; if $G^{(\ell)} = 1$ and the action is trivial, $\mathrm{Hom}_{\mathrm{cont}}(G,M) = 0$. *(proved here for $H^1$ and for type (D) in all degrees; type (C), (V) in degrees $\geq 2$ given Theorem 16.45)* Theorem 16.51.

**16.40 Through the pro-$\ell$ quotient.** $K$ normal acting trivially: $\mathrm{Inf}\colon H^1(G/K,M) \simeq H^1(G,M)$ if $K^{(\ell)} = 1$; in all degrees if $\ell \nmid \#K$; pro-$\ell$ image of $G$ gives $H^1(G,M) = H^1(G^{(\ell)},M)$. *(proved here, except the all-degrees statement, given Hochschild-Serre)* Corollary 16.52.

**16.41 Inertia at $q \neq \ell$.** With $K = \ker t_\ell$ ($\ell \nmid \#K$) and $M$ of type (C) or (V) with pro-$\ell$ inertia image: $H^1(I_q,M) \simeq H^1(I_q/K,M) \simeq M/(t-1)M$ via $c \mapsto c(t)$; $H^n(I_q,M) = 0$ for $n \geq 2$; for every $M$ of type (C), $H^1(I_q,M) \simeq M^K/(t-1)M^K$ is finitely generated with finite torsion. *(proved here, given Chapter 7; the vanishing for $n \geq 2$ given stated results)* Proposition 16.53.

**16.42 Procyclic groups.** Quotients of $\widehat{\mathbb{Z}}$; with a surjection $\pi$ onto $\mathbb{Z}_\ell$, $\pi(F)$ is a unit; continuous actions of $\widehat{\mathbb{Z}}$, $\mathbb{Z}_\ell$ on type (C) modules; unipotent elements generate $\mathbb{Z}_\ell$-actions. *(proved here)* Lemma 16.54.

**16.43 Cocycles on powers.** $c(F^k) = (1 + F + \cdots + F^{k-1})c(F)$, and $c(\gamma) \equiv \pi(\gamma)c(F)$ modulo $(F-1)M$; in particular $c(t^a) \equiv a\,c(t)$, $a \in \mathbb{Z}_\ell$. *(proved here)* Lemma 16.55.

**16.44 Procyclic, discrete coefficients (Serre).** $\Gamma = \widehat{\mathbb{Z}}$, $M$ discrete torsion, or $\Gamma = \mathbb{Z}_\ell$, $M$ discrete $\ell$-primary: $H^1(\Gamma,M) \simeq M/(F-1)M$ via $c \mapsto c(F)$, $\#H^1 = \#H^0$ for finite $M$, $H^n = 0$ for $n \geq 2$. *($H^0$, $H^1$ proved here; $n \geq 2$ stated)* Theorem 16.56.

**16.45 Geometric sums.** For $u \in \mathrm{Aut}(M)$, $M$ finitely generated over $\mathbb{Z}_\ell$: $(1+u+\cdots+u^{j-1})M \subseteq \ell^N M$ whenever $a_0\ell^N \mid j$, $a_0$ the order of $u$ mod $\ell$. *(proved here)* Lemma 16.57.

**16.46 Procyclic, compact coefficients.** $\Gamma$ procyclic with a surjection onto $\mathbb{Z}_\ell$ (e.g. $\mathbb{Z}_\ell$, $\mathbb{Z}_\ell(1)$, $\widehat{\mathbb{Z}}$, $I_q/P_q$), $M$ of type (C) or (V): $H^0 = \ker(F-1)$, $H^1(\Gamma, M) \simeq M/(F-1)M$ via $c \mapsto c(F)$, $H^n = 0$ for $n \geq 2$; $H^1(\Gamma, V) = 0$ iff $\det(1 - F \mid V) \neq 0$. *((i), (ii) proved here; (iii) stated)* Theorem 16.58.

**16.47 Inflation-restriction over $\widehat{\mathbb{Z}}$.** $G$ profinite, $H \trianglelefteq G$ closed, $G/H \simeq \widehat{\mathbb{Z}}$, $M$ of type (D) (torsion), (C) or (V): $0 \to H^1(G/H, M^H) \to H^1(G, M) \to H^1(H, M)^{G/H} \to 0$ is exact; with $H^2(G/H, M^H) = 0$ this is the five-term sequence of Theorem 16.31(b) for these coefficients. *(proved here; the vanishing of $H^2$ stated)* Proposition 16.61.

**16.48 The model setting.** $G \trianglerighteq I \supseteq K$, $\ell \nmid \#K$, $I/K \simeq \mathbb{Z}_\ell$ generated by $t$, Frob generating $G/I \simeq \widehat{\mathbb{Z}}$, Frob $t$ Frob$^{-1} \equiv t^q$, $M$ of type (C), $K$ acting trivially; dictionary with $G_{\mathbb{Q}_q}$, $I_q$, $\mathrm{ad}^0 T_\ell E$. *(construction; imports from Chapters 7 and 11)* Construction 16.63.

**16.49 Frobenius on $H^1$ of inertia.** $G$ acting on $M$ by $\rho_M$: $H^1(I, M) \simeq M/(t-1)M$; $\overline{\mathrm{Frob}}$ acts by $q^{-1}\rho_M(\mathrm{Frob})$ (for $M = \mathrm{ad}^0 T$: $q^{-1}\,\mathrm{Ad}\,\rho(\mathrm{Frob})$), well defined since $\rho_M(\mathrm{Frob})(t-1)M \subseteq (t-1)M$, independent of the lift; $(H^1(I, M)_{\mathrm{tor}})^{G/I} = \{\bar{m} \in (M/(t-1)M)_{\mathrm{tor}} : \rho_M(\mathrm{Frob})\bar{m} = q\bar{m}\}$. *(proved here)* Proposition 16.64.

**16.50 The Tate shape.** For $\rho(t) = \left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$, $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$ on $T$, and $\mathrm{Ad}\,\rho$ on $\mathrm{ad}^0 T$ (basis $e, h, f$): $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$; Frob $\cdot e = qe$, Frob $\cdot h = h - 2\varepsilon c' e$, Frob $\cdot f = q^{-1}(f + \varepsilon c' h - c'^2 e)$. *(proved here)* Lemma 16.66.

**16.51 The model computation.** $\ell$ odd, $w = v_\ell(n')$ (the paper's $v$ is $v_\ell(q-1)$): $(t-1)\,\mathrm{ad}^0 T = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$; $H^1(I, \mathrm{ad}^0 T) = (\mathbb{Z}_\ell/n')^2 \oplus \mathbb{Z}_\ell$, torsion $(\mathbb{Z}/\ell^w)^2$; $\overline{\mathrm{Frob}}$: $\bar{e} \mapsto \bar{e}$, $\bar{h} \mapsto q^{-1}(\bar{h} - 2\varepsilon c'\bar{e})$; invariants $\simeq \mathbb{Z}/\ell^w \oplus \mathbb{Z}/\ell^m$, $m = \min(w, v_\ell(q-1), v_\ell(c'))$, length $w + m$; $m$ depends on $c'$ only modulo $(n', q-1)$, the indeterminacy of $c'$; for $T$: invariants $\mathbb{Z}/\ell^w$ (split) or 0 (nonsplit). *(proved here)* Proposition 16.67.

**16.52 Fitting ideal of the invariants.** $\mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I, \mathrm{ad}^0 T)_{\mathrm{tor}})^{G/I}) = \ell^{w+m}\mathbb{Z}_\ell$, the Tamagawa factor; $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(n') + \min(v_\ell(n'), v_\ell(q-1), v_\ell(c'))$. *(proved here, given Chapter 3; identifications of $n'$, $c'$ in Chapters 11 and 17)* Corollary 16.68.

**16.53 Cup products.** $(a \cup b)(g_1, \ldots, g_{p+q}) = a(g_1, \ldots, g_p) \otimes g_1 \cdots g_p b(\ldots)$. *(definition)* Definition 16.71.

**16.54 Properties of $\cup$.** Leibniz rule; well defined on classes; associative, graded commutative, compatible with Res, Inf, $\delta$, Cor. *(degree $(1,1)$ proved here; the rest stated)* Proposition 16.72.

**16.55 Pontryagin duality.** $X^*$ for finite, compact and discrete $\ell$-primary modules: exact, reflexive, length-preserving; $(X^*)^G = (X_G)^*$; $H^n(G, A)^* = H_n(G, A^*)$. *(recalled from Chapter 3; duality of cohomology stated)* Proposition 16.74.

**16.56 Lengths.** length $A^*$ = length $A$; in $0 \to A \to B \to C \to D \to 0$, length $B$ = length $A$ + length $C$ − length $D$; hence the length identity of Lemma lem:dfg21. *(proved here, given Chapter 3)* Lemma 16.75.

**16.57 Galois cohomology.** $H^n(K,M) = H^n(G_K,M)$; $H^n(\mathbb{R},M)$; $H^1_{\mathrm{ur}}(\mathbb{Q}_p,M) = H^1(G_{\mathbb{Q}_p}/I_p, M^{I_p})$, which is the kernel of $H^1(\mathbb{Q}_p,M) \to H^1(I_p,M)$. *(convention; the equality proved here)* Convention 16.77.

**16.58 Unramified cohomology.** $H^1_{\mathrm{ur}}(\mathbb{Q}_p,M) = M^{I_p}/(\mathrm{Frob}_p - 1)M^{I_p}$ for $M$ of type (C) or (V); for $V$ it vanishes iff $\det(1 - \mathrm{Frob}_p \mid V^{I_p}) \neq 0$. *(proved here)* Proposition 16.78.

## Where the paper uses this chapter

- **Lemma lem:tamfactor** (the Tamagawa factor at a multiplicative prime), every group-cohomological step: the Tate shape as a cocycle (Proposition 16.21); "by inflation-restriction $H^1(I,M') = H^1(\langle t\rangle, M') = M'/(t-1)M'$" (Theorem 16.31, Corollary 16.52, Proposition 16.53, Theorem 16.58); the conjugation action of $G/I$ (Proposition 16.30); "$z(t^k) \equiv k\, z(t)$" (Lemma 16.55); "Frob acts on $M/(t-1)M$ as $q^{-1}\,\mathrm{Ad}(\rho(\mathrm{Frob}))$" (Proposition 16.64, Paper link 16.70); the explicit $(t-1)M$, the invariants and their Fitting ideal (Lemma 16.66, Proposition 16.67, Corollary 16.68); $H^1(G/I, \mathbb{Z}_\ell(1)^I) = \mathbb{Z}_\ell(1)/(q-1)$ (Example 16.59). The paperlink Paper link 16.70 matches each sentence of the proof.

- **Section 1.2** (the definition $\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p,M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}})$ and **Lemma lem:fpr**: the object $(H^1(I_p,M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}}$ is defined by Proposition 16.30 and computed by Proposition 16.64; the definition itself is Chapter 18's.

- **Lemma lem:tamadd**: inflation-restriction with $\mathrm{Hom}(K,M)$ (Theorem 16.31), "killed by $\#\Delta$" (Theorem 16.29), restriction to a Sylow subgroup (Proposition 16.28(d), Exercise 16.91), and $H^1$ of a free $\mathbb{Z}_3[C_3]$-module (Exercise 16.92).

- **Lemma lem:dfg21 and the paragraph after it**: long exact sequences and connecting maps (Propositions 16.24 and 16.44); Pontryagin duals and lengths (Proposition 16.74, Lemma 16.75); "$H^1(\mathbb{R},T^D)$ is killed by 2" (Corollary 16.37, Paper link 16.39); $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ and its vanishing (Proposition 16.78); "$H^1_f(\mathbb{Q}_p,T^D)$ is the torsion of $H^1(\mathbb{Q}_p,T^D)$" (Proposition 16.44(iii)).

- **Lemma lem:h0**: the $\ell^k$-torsion of $H^0(\mathbb{Q},W^D)$ and the length of a cyclic $H^0$ (Lemma 16.25, Paper link 16.26).

- **Proof of Theorem thm:main** and **Remark rem:ns**: the length bookkeeping along the sequence of Lemma lem:dfg21 (Lemma 16.75, Paper link 16.76).

## 16.1 Conventions and plan of the chapter

This section fixes the conventions that this chapter and Chapters 17 and 18 share. The paper uses group cohomology in one dense paragraph, the proof of Lemma lem:tamfactor, and in a handful of sentences around Lemma lem:dfg21, Lemma lem:h0 and the definition of $\mathrm{Tam}_p$ in its Section 1.2. Every one of those sentences rests on a convention (which Frobenius, which conjugation action, which topology on the coefficients), and a change of convention changes the formulas, though not the final answers. We fix them once.

**Convention 16.1.** (a) $\ell$ is a prime. Where $\ell$ must be odd we say so; this happens in exactly two places: the model computation of Section 16.10, where 2 must be a unit of $\mathbb{Z}_\ell$, and the vanishing of the cohomology of a group of order 2 on $\ell$-adic modules (Corollary 16.37).

(b) Groups act on the left. A cochain complex has differentials raising the degree. The connecting homomorphism of a short exact sequence of complexes is the one constructed in Theorem 16.6.

(c) All cohomology of a profinite group is *continuous cochain cohomology* (Definition 16.42). The coefficient modules are of three types: (D) discrete modules, finite or $\ell$-primary torsion, such as $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$; (C) finitely generated $\mathbb{Z}_\ell$-modules with the $\ell$-adic topology and a continuous action; (V) finite-dimensional $\mathbb{Q}_\ell$-vector spaces with a continuous action.

(d) Frob is always an *arithmetic* Frobenius: on $\overline{\mathbb{F}}_q$ it is $x \mapsto x^q$, and $\chi_\ell(\mathrm{Frob}_q) = q$ on $\mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}$ for $q \neq \ell$. For a prime $q \neq \ell$, $G_{\mathbb{Q}_q}/I_q = \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \simeq \widehat{\mathbb{Z}}$ is topologically generated by the image of $\mathrm{Frob}_q$.

(e) The conjugation action of $G/H$ on $H^n(H, M)$ is

$$(\sigma \cdot c)(h_1, \dots, h_n) = \sigma\, c(\sigma^{-1}h_1\sigma, \dots, \sigma^{-1}h_n\sigma),$$

which for $n = 1$, with the cocycle written $z$, is the paper's formula "$(\sigma \cdot z)(i) = \sigma z(\sigma^{-1}i\sigma)$" in the proof of Lemma lem:tamfactor (Proposition 16.30).

(f) The tame relation, proved in Chapter 7 and imported here: for a prime $q \neq \ell$ and a topological generator $t$ of the pro-$\ell$ tame inertia quotient of $G_{\mathbb{Q}_q}$,

$$\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q, \qquad \text{equivalently} \qquad \mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}, \quad q^{-1} \in \mathbb{Z}_\ell^\times.$$

(g) $\mathrm{ad}^0 T$ is the module of trace-zero endomorphisms of a representation $T$, with $g \cdot x = \rho(g)x\rho(g)^{-1}$. For $T$ free of rank 2 with basis $e_1, e_2$ its basis is $e = e_1 \otimes e_2^\vee$, $h = e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee$, $f = e_2 \otimes e_1^\vee$, as in the paper.

(h) $A^*$ is the Pontryagin dual of a module $A$; a star after a *map* of pairs, as in $(f, \varphi)^*$, is the induced map on cohomology. We write $A_{\mathrm{tor}}$ for the torsion submodule of a $\mathbb{Z}_\ell$-module, as the paper does, and length $= \mathrm{length}_{\mathbb{Z}_\ell}$.

*Warning* 16.2. Chapters 18 and 19 compute $L$-factors with the *geometric* Frobenius $\mathrm{Frob}^{-1}$: $L_p(M, s) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid M^{I_p})^{-1}$. An eigenvalue $\lambda$ of the arithmetic Frobenius is the eigenvalue $\lambda^{-1}$ of the geometric one. In this chapter only the arithmetic Frobenius occurs. The two conventions are both used in the paper: the proof of Lemma lem:tamfactor, with $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$, is written with the arithmetic Frobenius, and Section 1.1 uses the geometric one. The paragraph after Lemma lem:dfg21 uses both: the quotient $(V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ is written with the arithmetic Frobenius and the determinant "$\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1}$" with the geometric one. The quotient does not depend on the choice, because $\mathrm{Frob}_p^{-1} - 1 = -\mathrm{Frob}_p^{-1}(\mathrm{Frob}_p - 1)$ has the same image as $\mathrm{Frob}_p - 1$; and the vanishing statements we prove (Proposition 16.78) do not depend on the convention, since 1 is an eigenvalue of Frob if and only if it is an eigenvalue of $\mathrm{Frob}^{-1}$.

The chapter proceeds from the general to the particular. Sections 2 and 3 recall the homological algebra (complexes, the long exact sequence, derived functors in outline). Sections 4 to 6 define the cohomology of an abstract group by cochains and prove the facts the paper uses: the meaning of $H^0$, $H^1$, $H^2$, functoriality, restriction, inflation and corestriction, the conjugation action, inflation-restriction, the vanishing for coefficients on which the group order is invertible. Sections 7 to 9 pass to profinite groups with continuous cochains, which is the setting of Galois cohomology, and compute the cohomology of procyclic groups; the identity $H^1(\langle t\rangle, M) = M/(t-1)M$ is Theorem 16.58. Section 10 is the group-cohomological content of Lemma lem:tamfactor, done line by line. Section 11 treats cup products and lengths in exact sequences, Section 12 the notation of Galois cohomology and a preview of Chapter 17.

## 16.2 Complexes, cohomology and the long exact sequence

Every long exact sequence in the rest of the book (the Kummer sequences of Chapter 17, the sequences relating $T$, $V$ and $W$, the exact sequence of Lemma lem:dfg21 in Chapter 18) is an instance of Theorem 16.6. This section proves it, after recalling the snake lemma from Chapter 3. Throughout, $R$ is a commutative ring; in practice $R = \mathbb{Z}$ or $R = \mathbb{Z}_\ell$.

**Definition 16.3** (cochain complexes)**.** A *cochain complex* of $R$-modules is a sequence $C^\bullet = (C^n, d^n)_{n\in\mathbb{Z}}$ of $R$-modules $C^n$ and $R$-linear maps $d^n\colon C^n \to C^{n+1}$ with $d^{n+1}d^n = 0$ for all $n$. Its *cocycles* are $Z^n = \ker d^n$, its *coboundaries* $B^n = \operatorname{im} d^{n-1}$; since $d^n d^{n-1} = 0$, $B^n \subseteq Z^n$, and the *cohomology* is $H^n(C^\bullet) = Z^n/B^n$. A *morphism* $f\colon C^\bullet \to D^\bullet$ is a family of $R$-linear maps $f^n\colon C^n \to D^n$ with $d^n f^n = f^{n+1}d^n$. Two morphisms $f, g$ are *homotopic* if there are $R$-linear maps $s^n\colon C^n \to D^{n-1}$ with $f^n - g^n = d^{n-1}s^n + s^{n+1}d^n$ for all $n$. A sequence $A^\bullet \to B^\bullet \to C^\bullet$ of morphisms is *exact* if it is exact in every degree.

**Lemma 16.4.** *(i) A morphism $f$ maps $Z^n(C)$ into $Z^n(D)$ and $B^n(C)$ into $B^n(D)$, hence induces an $R$-linear map $H^n(f)\colon H^n(C) \to H^n(D)$, and $H^n(g\circ f) = H^n(g)\circ H^n(f)$, $H^n(\mathrm{id}) = \mathrm{id}$.*

*(ii) Homotopic morphisms induce the same map on cohomology.*

*Proof.* (i) If $dx = 0$ then $d(fx) = f(dx) = 0$, so $f(Z^n(C)) \subseteq Z^n(D)$; and $f(dy) = d(fy)$, so $f(B^n(C)) \subseteq B^n(D)$. Hence $H^n(f)[x] = [fx]$ is well defined: if $x' = x + dy$ is another representative, $fx' = fx + d(fy)$ has the same class. On representatives, $H^n(g\circ f)[x] = [g(f(x))] = H^n(g)(H^n(f)[x])$ and $H^n(\mathrm{id})[x] = [x]$. (ii) For $x \in Z^n(C)$, $f(x) - g(x) = d(s^n x) + s^{n+1}(dx) = d(s^n x) \in B^n(D)$, so $[f(x)] = [g(x)]$. Note that $s$ need not commute with $d$; only the combination $ds + sd$ enters. □

The definition of a homotopy may look unmotivated. Its role is exactly (ii): it is the weakest relation between $f$ and $g$ that forces equal maps on cohomology by the one-line argument above, and it is the relation that the comparison theorem (Proposition 16.9) produces between two lifts of the same map. This is why group cohomology, defined through a choice of resolution, does not depend on that choice.

We recall the snake lemma, proved in Chapter 3, because the construction of its connecting map is the template for every connecting map that follows.

**Lemma 16.5** (the snake lemma)**.** *Let*

$$\begin{array}{cccccccc} & A' & \xrightarrow{i} & A & \xrightarrow{p} & A'' & \to 0 \\ & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f''} & \\ 0 \to & B' & \xrightarrow{j} & B & \xrightarrow{r} & B'' & \end{array}$$

*be a commutative diagram of $R$-modules with exact rows. For $a'' \in \ker f''$ choose $a \in A$ with $p(a) = a''$; then $r(f(a)) = f''(a'') = 0$, so $f(a) = j(b')$ for a unique $b' \in B'$; put $\delta(a'') = b' + f'(A')$. Then $\delta\colon \ker f'' \to \operatorname{coker} f'$ is well defined and $R$-linear, and*

$$\ker f' \to \ker f \to \ker f'' \xrightarrow{\delta} \operatorname{coker} f' \to \operatorname{coker} f \to \operatorname{coker} f''$$

*is exact. If $i$ is injective the sequence starts with $0 \to \ker f'$; if $r$ is surjective it ends with $\operatorname{coker} f'' \to 0$.*

*Proof.* This is proved in Chapter 3; see also [205, Section 1.3]. □

**Theorem 16.6** (the long exact sequence)**.** *Let $0 \to A^\bullet \xrightarrow{i} B^\bullet \xrightarrow{p} C^\bullet \to 0$ be a short exact sequence of cochain complexes of $R$-modules.*

*(i) For each $n$ there is an $R$-linear map $\delta^n\colon H^n(C^\bullet) \to H^{n+1}(A^\bullet)$ defined as follows. Represent a class by $c \in Z^n(C)$; lift it to $b \in B^n$ with $p(b) = c$; then $p(db) = dc = 0$, so $db = i(a)$ for a unique $a \in A^{n+1}$; this $a$ is a cocycle, and $\delta^n[c] = [a]$.*

*(ii) The sequence*

$$\cdots \to H^n(A) \xrightarrow{i_*} H^n(B) \xrightarrow{p_*} H^n(C) \xrightarrow{\delta^n} H^{n+1}(A) \xrightarrow{i_*} H^{n+1}(B) \xrightarrow{p_*} \cdots$$

*is exact.*

*(iii) (Naturality.) A commutative diagram of short exact sequences of complexes $(0 \to A \to B \to C \to 0) \to (0 \to \tilde A \to \tilde B \to \tilde C \to 0)$, given by morphisms $\alpha, \beta, \gamma$, induces a commutative ladder between the two long exact sequences; in particular $\delta^n \circ H^n(\gamma) = H^{n+1}(\alpha) \circ \delta^n$.*

*Proof.* (i) Since $i$ is injective and $i(da) = d(i(a)) = ddb = 0$, $da = 0$. Independence of the choices: another lift is $b + i(x)$ with $x \in A^n$, which changes $a$ to $a + dx$; another representative is $c + dy$ with $y \in C^{n-1}$; lift $y$ to $\tilde y \in B^{n-1}$, then $b + d\tilde y$ lifts $c + dy$ and has the same differential $db$. So $[a]$ depends only on $[c]$. Linearity: if $(c_1, b_1, a_1)$ and $(c_2, b_2, a_2)$ are choices for $[c_1]$ and $[c_2]$, and $\lambda \in R$, then $(c_1 + \lambda c_2,\ b_1 + \lambda b_2,\ a_1 + \lambda a_2)$ is a valid choice for $[c_1] + \lambda[c_2]$, because $p$, $i$ and $d$ are $R$-linear; by independence of the choices, $\delta^n([c_1] + \lambda[c_2]) = [a_1] + \lambda[a_2]$.

(ii) A clean route is the snake lemma. For each of the three complexes $X^\bullet \in \{A^\bullet, B^\bullet, C^\bullet\}$ the differential induces a map $\bar d\colon X^n/dX^{n-1} \to Z^{n+1}(X)$, $x + dX^{n-1} \mapsto dx$ (well defined since $dd = 0$, and landing in $Z^{n+1}$ for the same reason). Its kernel is $\{x : dx = 0\}/dX^{n-1} = Z^n/B^n = H^n(X)$, and its cokernel is $Z^{n+1}(X)/dX^n = H^{n+1}(X)$. The rows $A^n/dA^{n-1} \to B^n/dB^{n-1} \to C^n/dC^{n-1} \to 0$ and $0 \to Z^{n+1}(A) \to Z^{n+1}(B) \to Z^{n+1}(C)$ are exact. In the first row, surjectivity at the right end holds because $p$ is surjective. For exactness in the middle, let $x \in B^n$ map to 0 in $C^n/dC^{n-1}$, that is $p(x) = dy$ with $y \in C^{n-1}$; lift $y$ to $\tilde y \in B^{n-1}$, so that $x - d\tilde y \in \ker p = i(A^n)$, and $x$ has the same class as $x - d\tilde y$, which comes from $A^n$. (The composite of the two maps is 0 because $pi = 0$.) In the second row, injectivity is inherited from $i$; and a cocycle $z$ of $B^\bullet$ with

$p(z) = 0$ is $i(a)$, with $da = 0$ because $i(da) = dz = 0$ and $i$ is injective. The two rows and the three vertical maps $\bar{d}$ form a commutative diagram, because $i$ and $p$ commute with $d$. The snake lemma applied to it gives the exact sequence

$$H^n(A) \to H^n(B) \to H^n(C) \xrightarrow{\delta} H^{n+1}(A) \to H^{n+1}(B) \to H^{n+1}(C),$$

and the snake construction of $\delta$ is exactly (i). Splicing these six-term sequences for all $n$ gives (ii).

(iii) The maps $i_*$, $p_*$ commute with $\alpha_*, \beta_*, \gamma_*$ by functoriality. For $\delta$: if $(c, b, a)$ are the choices in (i) for $[c]$, then $(\gamma c, \beta b, \alpha a)$ are valid choices for $[\gamma c]$, since $\tilde{p}\beta = \gamma p$ and $\tilde{i}\alpha = \beta i$ and $\beta$ commutes with $d$: indeed $\tilde{p}(\beta b) = \gamma(pb) = \gamma c$ and $d(\beta b) = \beta(db) = \beta(i(a)) = \tilde{i}(\alpha a)$. So $\delta^n[\gamma c] = [\alpha a] = H^{n+1}(\alpha)\delta^n[c]$. □

The recipe of (i) is worth memorising in the form "lift, differentiate, pull back": every connecting map in the book, from the Kummer map of Chapter 17 to the map $\delta\colon H^0(G, W) \to H^1(G, T)$ of Proposition 16.44, is computed this way. The class $\delta[c]$ is nonzero exactly when the lift $b$ cannot be chosen to be a cocycle. Proposition 16.18(iv) below is the first instance: there $\delta(m'')$ is the coboundary of a lift $m$ of $m''$, a coboundary in $M$ that takes values in $M'$, where it is a cocycle but in general not a coboundary. For example, for the group $G = \{1, \sigma\}$ acting on $M = \mathbb{Z}$ by $-1$ and the sequence $0 \to 2\mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2 \to 0$, the element $1 \in (\mathbb{Z}/2)^G$ lifts to $1 \in \mathbb{Z}$, and $\delta(1)$ is the class of $g \mapsto g \cdot 1 - 1$, which is $\sigma \mapsto -2$; this is not of the form $\sigma \mapsto \sigma x - x = -2x$ with $x \in 2\mathbb{Z}$, so $\delta(1) \neq 0$.

**Example 16.7.** Let $M$ be an $R$-module and $u$ an $R$-linear endomorphism of $M$. The complex $0 \to M \xrightarrow{u-1} M \to 0$ (in degrees 0 and 1) has $H^0 = \ker(u-1)$ and $H^1 = M/(u-1)M$. We shall see in Theorem 16.58 that, for a procyclic group $\Gamma$ with a surjection onto $\mathbb{Z}_\ell$ and generator $F$ acting by $u$ on a module $M$ of type (C), this complex computes $H^\bullet(\Gamma, M)$ in all degrees. For a finite cyclic group $\langle t \rangle$ of order $m$ the analogous complex is not this one: it is the infinite periodic complex $M \xrightarrow{t-1} M \xrightarrow{N} M \xrightarrow{t-1} M \xrightarrow{N} \cdots$ with $N = 1 + t + \cdots + t^{m-1}$, which is why $H^1(\langle t \rangle, M) = \ker N/(t-1)M$ (Theorem 16.34). The difference between the two answers is the subject of Warning 16.60.

## 16.3 Resolutions, derived functors, Ext and Tor in outline

This section is background. The paper never uses Ext or Tor by name; but $H^n(G, M)$ is a derived functor (Theorem 16.17), this is what makes it independent of the cochains chosen to compute it, and $\mathrm{Ext}^1$ is the conceptual home of the "cocycle in the corner" of a reducible two-dimensional representation, which the paper uses for the Tate curve (Proposition 16.21). We state the facts with references and compute the examples over $\mathbb{Z}$ and $\mathbb{Z}_\ell$ that occur later.

**Definition 16.8** (projective and injective resolutions)**.** An $R$-module $P$ is *projective* if $\mathrm{Hom}_R(P, -)$ is exact (equivalently, a direct summand of a free module); $I$ is *injective* if $\mathrm{Hom}_R(-, I)$ is exact. A *projective resolution* of $M$ is an exact sequence $\cdots \to P_1 \to P_0 \to M \to 0$ with every $P_i$ projective; an *injective resolution* is an exact sequence $0 \to M \to I^0 \to I^1 \to \cdots$ with every $I^i$ injective.

**Proposition 16.9** (existence and comparison)**.** *(i) Every $R$-module has a resolution by free modules, and an injective resolution.*

(ii) *(Comparison theorem.) A map $M \to N$ lifts to a morphism between given projective resolutions of $M$ and $N$, unique up to homotopy; dually for injective resolutions. Hence any two projective (resp. injective) resolutions of $M$ are homotopy equivalent.*

(iii) *If $R$ is a principal ideal domain (for instance $\mathbb{Z}$ or $\mathbb{Z}_\ell$), every module has a free resolution $0 \to P_1 \to P_0 \to M \to 0$ of length one (submodules of free modules are free), and the divisible modules, such as $\mathbb{Q}/\mathbb{Z}$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell$, are injective.*

*Proof.* We do not prove this; see [205, Sections 2.2, 2.3]. For (i) with free modules, map the free module on the elements of $M$ onto $M$ and repeat with the kernel. □

**Definition 16.10** (Ext and Tor)**.** For $R$-modules $M$, $N$, a projective resolution $P_\bullet \to M$ and an injective resolution $N \to I^\bullet$,

$$\operatorname{Ext}^n_R(M,N) = H^n(\operatorname{Hom}_R(P_\bullet, N)) = H^n(\operatorname{Hom}_R(M, I^\bullet)), \qquad \operatorname{Tor}^R_n(M,N) = H_n(P_\bullet \otimes_R N).$$

**Proposition 16.11** (properties of Ext and Tor)**.** (i) *The two descriptions of* Ext *agree and are independent of the resolutions;* $\operatorname{Ext}^0 = \operatorname{Hom}$ *and* $\operatorname{Tor}_0 = \otimes$.

(ii) *A short exact sequence in either variable gives a long exact sequence of* Ext *(resp.* Tor*).*

(iii) $\operatorname{Ext}^n_R(M,N) = 0$ *for* $n \geq 1$ *if $M$ is projective or $N$ is injective. Over a principal ideal domain* $\operatorname{Ext}^n = \operatorname{Tor}_n = 0$ *for* $n \geq 2$.

(iv) $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/m, N) \simeq N/mN$, $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/m, N) \simeq N[m]$, *and* $\operatorname{Ext}^1_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^a, \mathbb{Z}/\ell^b) \simeq \mathbb{Z}/\ell^{\min(a,b)}$.

*Proof.* (i) to (iii) are stated; see [205, Sections 2.4 to 2.7 and 3.1 to 3.3]. The vanishing for $n \geq 2$ over a principal ideal domain follows from resolutions of length one. (iv) Use the resolution $0 \to \mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m \to 0$. Applying $\operatorname{Hom}(-, N)$ gives the complex $N \xrightarrow{m} N$ in degrees 0, 1, whose $H^1$ is $N/mN$; tensoring with $N$ gives $N \xrightarrow{m} N$ in degrees 1, 0, whose $H_1$ is $N[m]$. Over $\mathbb{Z}_\ell$ the resolution $0 \to \mathbb{Z}_\ell \xrightarrow{\ell^a} \mathbb{Z}_\ell \to \mathbb{Z}/\ell^a \to 0$ gives $\operatorname{Ext}^1_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^a, N) = N/\ell^a N$, and $(\mathbb{Z}/\ell^b)/\ell^a = \mathbb{Z}/\ell^{\min(a,b)}$. □

**Theorem 16.12** ($\operatorname{Ext}^1$ classifies extensions)**.** $\operatorname{Ext}^1_R(M,N)$ *is in natural bijection with the set of equivalence classes of extensions* $0 \to N \to E \to M \to 0$ *of $R$-modules, two extensions being equivalent if there is an isomorphism $E \to E'$ inducing the identity on $N$ and on $M$. The split extension $N \oplus M$ corresponds to* 0*, and the group law corresponds to the Baer sum. The class of an extension is the image of* $\mathrm{id}_M$ *under the connecting map* $\operatorname{Hom}_R(M,M) \to \operatorname{Ext}^1_R(M,N)$ *of the long exact sequence of* $\operatorname{Hom}_R(M,-)$.

*Proof.* We do not prove this; see [205, Section 3.4]. Proposition 16.21 below proves by hand the case that the paper uses. □

A first example shows the bijection at work. By Proposition 16.11(iv), $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) = (\mathbb{Z}/2)/2(\mathbb{Z}/2) = \mathbb{Z}/2$ has two elements, and there are exactly two extensions $0 \to \mathbb{Z}/2 \to E \to \mathbb{Z}/2 \to 0$ up to equivalence: $E$ is a group of order 4, so $E \simeq \mathbb{Z}/2 \times \mathbb{Z}/2$ (the split extension, class 0) or $E \simeq \mathbb{Z}/4$ (with the sub $2\mathbb{Z}/4\mathbb{Z}$, the nonzero class). The extension class records whether a lift of the generator of the quotient can be chosen to be killed by 2. In the same way $\operatorname{Ext}^1_{\mathbb{Z}_\ell}(\mathbb{Z}/\ell^a, \mathbb{Z}/\ell^b) = \mathbb{Z}/\ell^{\min(a,b)}$ counts the ways of gluing a cyclic group of order $\ell^a$ on top of one of order $\ell^b$ $(a, b \geq 1)$; the cyclic extensions $E \simeq \mathbb{Z}/\ell^{a+b}$ are those whose class is a generator. The case the paper needs is the equivariant analogue: the module is $\mathbb{Z}_\ell^2$, the extension is one of Galois modules, and the class lives in a group cohomology $H^1$.

*Remark* 16.13. Over a discrete valuation ring such as $\mathbb{Z}_\ell$ the homological algebra of modules stops in degree one. This is one reason why the cohomology of a procyclic group with $\ell$-adic coefficients is so simple: in Section 16.9 the two-term complex of Example 16.7 computes it. Group cohomology does not stop in degree one in general: a finite cyclic group has nonzero cohomology in infinitely many degrees.

## 16.4 Group cohomology by cochains: $H^0$, $H^1$, $H^2$

This section defines $H^n(G, M)$ for an abstract group $G$ and proves what the paper uses about the low degrees. Lemma lem:tamfactor calls $\kappa$ "the Kummer cocycle" and reads a reducible two-dimensional representation as a 1-cocycle (Proposition 16.21); Lemma lem:h0 computes an $H^0$ through its $\ell^k$-torsion (Lemma 16.25); Lemma lem:dfg21 is a sequence of connecting maps (Proposition 16.24). The profinite version, with continuous cochains, is Section 16.7; every definition and proof of this section carries over to it with the word "continuous" added, and we say so there.

**Definition 16.14** ($G$-modules)**.** Let $G$ be a group. A *$G$-module* is an abelian group $M$ with a left action of $G$ by group automorphisms, that is, a left module over the group ring $\mathbb{Z}[G]$. The *invariants* are $M^G = \{m : gm = m \text{ for all } g\}$; the *coinvariants* are $M_G = M/I_G M$, where the augmentation ideal $I_G$ is the kernel of $\mathbb{Z}[G] \to \mathbb{Z}$, $\sum a_g g \mapsto \sum a_g$, generated by the elements $g - 1$. $M$ is *trivial* if $gm = m$ for all $g, m$. For finite $G$ the *norm* is $N_G = \sum_{g \in G} g$, and $N_G M \subseteq M^G$. For $G$-modules $M$, $N$, the group $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is a $G$-module by $(g\varphi)(m) = g\varphi(g^{-1}m)$, and $\mathrm{Hom}_G(M, N) = \mathrm{Hom}_{\mathbb{Z}}(M, N)^G$. A map of $G$-modules is a $\mathbb{Z}[G]$-linear map. If $M$ is a $\mathbb{Z}_\ell$-module and $G$ acts $\mathbb{Z}_\ell$-linearly, $M$ is a $\mathbb{Z}_\ell[G]$-module, and every cohomology group defined below is a $\mathbb{Z}_\ell$-module, every map $\mathbb{Z}_\ell$-linear.

The invariants and the coinvariants are the largest submodule and the largest quotient on which $G$ acts trivially; group cohomology measures how far $M \mapsto M^G$ is from being exact. A first example: let $G = \langle \sigma \rangle$ have order 2 and act on $\mathbb{Z}^2$ by swapping the coordinates. Then $M^G = \{(a, a)\}$, $I_G M = (\sigma - 1)M = \{(b, -b)\}$, $M_G = \mathbb{Z}^2/\{(b, -b)\} \simeq \mathbb{Z}$ via $(x, y) \mapsto x + y$, and $N_G M = \{(a + b, a + b)\} = M^G$. For the sign module $\mathbb{Z}^-$ (with $\sigma$ acting by $-1$) instead, $M^G = 0$, $M_G = \mathbb{Z}/2$ and $N_G = 0$. The examples of the paper are $\mathbb{Z}_\ell$-modules such as $T_\ell E$, $\mathrm{ad}^0 T$ and $\mathbb{Z}_\ell(1)$, with $G$ a Galois group acting through a representation.

**Definition 16.15** (inhomogeneous cochains)**.** For $n \geq 0$ let $C^n(G, M)$ be the group of all maps $G^n \to M$ ($C^0(G, M) = M$), and define $d^n \colon C^n(G, M) \to C^{n+1}(G, M)$ by

$$(d^n\varphi)(g_1, \dots, g_{n+1}) = g_1\varphi(g_2, \dots, g_{n+1}) + \sum_{i=1}^{n} (-1)^i \varphi(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) + (-1)^{n+1}\varphi(g_1, \dots, g_n).$$

In low degrees:

$$(d^0 m)(g) = gm - m,$$
$$(d^1\varphi)(g, h) = g\varphi(h) - \varphi(gh) + \varphi(g),$$
$$(d^2\psi)(g, h, k) = g\psi(h, k) - \psi(gh, k) + \psi(g, hk) - \psi(g, h).$$

The *cohomology* of $G$ with coefficients in $M$ is $H^n(G, M) = H^n(C^\bullet(G, M))$ (with $C^n = 0$ for $n < 0$). Elements of $Z^n(G, M) = \ker d^n$ are *$n$-cocycles*, of $B^n(G, M) = \mathrm{im}\, d^{n-1}$ *$n$-coboundaries*.

The formula for $d^n$ can be read off from the low degrees, which are the ones the book uses. In degree 0, $d^0 m = 0$ says that $m$ is invariant. In degree 1, $d^1\varphi = 0$ says $\varphi(gh) = \varphi(g) + g\varphi(h)$: $\varphi$ is a homomorphism "twisted by the action", the rule satisfied by the corner entry of a triangular representation (Proposition 16.21) and by the function $g \mapsto gm - m$. In degree 2, $d^2\psi = 0$ is the associativity constraint on the factor set of a group extension (Theorem 16.23). The general $d^n$ alternates between applying the action to the first variable, multiplying adjacent variables, and dropping the last variable; this pattern is what makes $d \circ d = 0$, by a cancellation in pairs. The adjective "inhomogeneous" distinguishes these cochains from the homogeneous cochains, the $G$-equivariant maps $G^{n+1} \to M$, which compute the same groups (Theorem 16.17; the two are related by $\varphi(g_1, \dots, g_n) = \Phi(1, g_1, g_1 g_2, \dots, g_1 \cdots g_n)$).

**Lemma 16.16.** *$d^{n+1} d^n = 0$ for all $n$, so $C^\bullet(G, M)$ is a cochain complex. A map of $G$-modules $f\colon M \to N$ induces a morphism of complexes $\varphi \mapsto f \circ \varphi$, hence maps $f_*\colon H^n(G, M) \to H^n(G, N)$; so $H^n(G, -)$ is a covariant additive functor.*

*Proof.* In degree 0: $(d^1 d^0 m)(g, h) = g(hm - m) - (ghm - m) + (gm - m) = 0$. In degree 1, expanding the four terms of $(d^2 d^1 \varphi)(g, h, k)$:

$$g(h\varphi(k) - \varphi(hk) + \varphi(h)) - (gh\varphi(k) - \varphi(ghk) + \varphi(gh)) \\ + (g\varphi(hk) - \varphi(ghk) + \varphi(g)) - (g\varphi(h) - \varphi(gh) + \varphi(g)) = 0,$$

every term cancelling against another. Degree 2 is Exercise 16.81. In general the identity follows from Theorem 16.17: $C^\bullet(G, M)$ is $\mathrm{Hom}_{\mathbb{Z}[G]}(F_\bullet, M)$ for a complex $F_\bullet$ (the bar resolution) whose differential squares to zero by a direct computation [17, Chapter III, Section 1], [143, Chapter I, Section 2]. Since $f$ is $G$-linear, $f(g\varphi(\dots)) = gf(\varphi(\dots))$, so $f \circ -$ commutes with $d$. □

**Theorem 16.17** (cohomology as a derived functor)**.** *Let $\mathbb{Z}$ have the trivial $G$-action. There is a resolution $F_\bullet \to \mathbb{Z}$ of $\mathbb{Z}$ by free $\mathbb{Z}[G]$-modules, $F_n$ free on $G^n$ (the bar resolution), with $\mathrm{Hom}_{\mathbb{Z}[G]}(F_\bullet, M) \simeq C^\bullet(G, M)$ as complexes. Hence $H^n(G, M) \simeq \mathrm{Ext}^n_{\mathbb{Z}[G]}(\mathbb{Z}, M)$, $H^n(G, -)$ is the $n$-th right derived functor of $M \mapsto M^G = \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, M)$, and any projective resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$ computes it.*

*Proof.* We do not prove this; see [17, Chapter I, Section 5 and Chapter III, Section 1] and [205, Section 6.5]. □

**Proposition 16.18** ($H^0$ and $H^1$)**.** *Let $M$ be a $G$-module.*

*(i)* $H^0(G, M) = M^G$.

*(ii)* $Z^1(G, M)$ *is the group of* crossed homomorphisms, *the maps $\varphi\colon G \to M$ with $\varphi(gh) = \varphi(g) + g\varphi(h)$, and $B^1(G, M)$ the group of* principal *ones, $g \mapsto gm - m$ for some $m \in M$. A 1-cocycle has $\varphi(1) = 0$ and $\varphi(g^{-1}) = -g^{-1}\varphi(g)$.*

*(iii) If $G$ acts trivially on $M$, then $B^1 = 0$ and $H^1(G, M) = \mathrm{Hom}(G, M) = \mathrm{Hom}(G^{\mathrm{ab}}, M)$. In particular $H^1(G, M) = 0$ if $G$ is finite, $M$ is torsion-free and the action is trivial.*

*(iv) For an exact sequence $0 \to M' \to M \xrightarrow{p} M'' \to 0$ of $G$-modules the sequence*

$$0 \to M'^G \to M^G \to M''^G \xrightarrow{\delta} H^1(G, M') \to H^1(G, M) \to H^1(G, M'')$$

*is exact, where $\delta(m'')$ is the class of $g \mapsto gm - m$ for any $m \in M$ with $p(m) = m''$ (this cocycle takes values in $M'$).*

*Proof.* (i) $C^{-1} = 0$ and $(d^0 m)(g) = gm - m$, so $H^0 = \ker d^0 = M^G$. (ii) $d^1\varphi = 0$ is the identity $\varphi(gh) = \varphi(g) + g\varphi(h)$. Putting $g = h = 1$ gives $\varphi(1) = 2\varphi(1)$, so $\varphi(1) = 0$; putting $h = g^{-1}$ gives $0 = \varphi(g) + g\varphi(g^{-1})$. (iii) With trivial action $gm - m = 0$ and the cocycle identity reads $\varphi(gh) = \varphi(g) + \varphi(h)$; a homomorphism to an abelian group factors through $G^{\mathrm{ab}}$. If $G$ is finite every $g$ has finite order $k$, so $k\varphi(g) = \varphi(g^k) = 0$ and $\varphi(g) = 0$ when $M$ is torsion-free. (iv) is the beginning of the long exact sequence of Proposition 16.24; the formula for $\delta$ is the recipe of Theorem 16.6(i): lift $m''$ to $m \in C^0(G, M) = M$ and apply $d^0$. Concretely, $p(gm - m) = gm'' - m'' = 0$ because $m''$ is invariant, so $gm - m$ lies in $M'$; the cochain $g \mapsto gm - m$ is a cocycle (it is $d^0 m$ in $M$, and $d^1 d^0 = 0$); another lift $m + m'$ changes it by $d^0 m'$, a coboundary in $M'$. The class $\delta(m'')$ is 0 if and only if $gm - m = gm' - m'$ for some $m' \in M'$, that is, if and only if $m''$ has the invariant lift $m - m'$; this is exactness at $M''^G$. □

*Remark* 16.19. The words come from topology. A reader who has met projective representations in quantum mechanics has seen 2-cocycles: a map $U\colon G \to \mathrm{U}(n)$ with $U(g)U(h) = \omega(g,h)U(gh)$ forces $\omega$ to be a 2-cocycle with values in the circle group (trivial action), and rescaling $U(g)$ by phases changes $\omega$ by a coboundary. Crossed homomorphisms appear in the same way as the off-diagonal entry of a triangular representation (Proposition 16.21).

*Remark* 16.20 (non-abelian $H^1$). If $A$ is a group, not necessarily abelian, on which $G$ acts by automorphisms, one still defines $Z^1(G, A)$ as the set of maps with $\varphi(gh) = \varphi(g)\,{}^g\varphi(h)$, and $H^1(G, A)$ as its quotient by the relation $\varphi \sim \psi$ if $\psi(g) = a^{-1}\varphi(g)\,{}^g a$ for some $a \in A$. This is a pointed set, not a group, and there is no $H^2$ in general. For an algebraic group $G_0$ over a field $k$ one writes $H^1(k, G_0) = H^1(\mathrm{Gal}(k^{\mathrm{sep}}/k), G_0(k^{\mathrm{sep}}))$; it classifies principal homogeneous spaces of $G_0$ over $k$ [177, Chapter I, Section 5 and Chapter III]. This is the sense of "$H^1(k, G) = 0$" (Lang's theorem for a connected group over a finite field) in Chapter 10. Nothing else in the book uses non-abelian cohomology.

**Proposition 16.21** (the cocycle in the corner)**.** *Let $R$ be a commutative ring, $G$ a group, $\eta_1, \eta_2\colon G \to R^\times$ characters, and let $R(\eta)$ denote $R$ with $g$ acting by multiplication by $\eta(g)$.*

*(i) A map $\rho\colon G \to \mathrm{GL}_2(R)$ of the form $\rho(g) = \begin{pmatrix} \eta_1(g) & \kappa(g) \\ 0 & \eta_2(g) \end{pmatrix}$ is a homomorphism if and only if $b(g) = \kappa(g)\eta_2(g)^{-1}$ is a 1-cocycle with values in $R(\eta_1\eta_2^{-1})$:*

$$b(gh) = b(g) + \eta_1\eta_2^{-1}(g)\, b(h).$$

*(ii) Replacing the basis vector $e_2$ by $e_2 + re_1$ ($r \in R$) replaces $b$ by $b + d^0 r$, where $(d^0 r)(g) = (\eta_1\eta_2^{-1}(g) - 1)r$. Scaling $e_1$ or $e_2$ by units multiplies $b$ by a unit.*

*(iii) $\rho$ is split, that is, $Re_1$ has a $G$-stable complement, if and only if the class of $b$ in $H^1(G, R(\eta_1\eta_2^{-1}))$ is 0.*

*(iv) Every class arises. Hence the isomorphism classes of extensions $0 \to R(\eta_1) \to E \to R(\eta_2) \to 0$ of $R[G]$-modules (with $E$ free of rank two over $R$, and isomorphisms inducing the identity on the sub and the quotient) are in bijection with $H^1(G, R(\eta_1\eta_2^{-1}))$; this is the case of Theorem 16.12 that we need.*

*(v) (The paper's shape.) Let $\rho(g) = \begin{pmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa(g) \\ 0 & \varepsilon(g) \end{pmatrix}$ with $\varepsilon\colon G \to \{\pm 1\}$ and $\chi_\ell\colon G \to \mathbb{Z}_\ell^\times$ characters, $R = \mathbb{Z}_\ell$. Then $\rho$ is a homomorphism if and only if $\kappa$ is a 1-cocycle with*

*values in $\mathbb{Z}_\ell(\chi_\ell)$, written $\mathbb{Z}_\ell(1)$ when $\chi_\ell$ is the cyclotomic character. On a subgroup $I$ on which $\chi_\ell$ and $\varepsilon$ are trivial, $\kappa|_I$ is a homomorphism $I \to \mathbb{Z}_\ell$, and the corner of $\rho|_I$ is $\kappa|_I$.*

*Proof.* (i) The upper right entry of $\rho(g)\rho(h)$ is $\eta_1(g)\kappa(h) + \kappa(g)\eta_2(h)$; the diagonal entries are always multiplicative. So $\rho$ is a homomorphism if and only if $\kappa(gh) = \eta_1(g)\kappa(h) + \kappa(g)\eta_2(h)$. Dividing by $\eta_2(gh) = \eta_2(g)\eta_2(h)$ gives the cocycle identity for $b$.

(ii) With $P = \left(\begin{smallmatrix} 1 & r \\ 0 & 1 \end{smallmatrix}\right)$, the matrix of $g$ in the basis $e_1, e_2 + re_1$ is

$$P^{-1}\rho(g)P = \begin{pmatrix} \eta_1(g) & \kappa(g) + r(\eta_1(g) - \eta_2(g)) \\ 0 & \eta_2(g) \end{pmatrix},$$

so $b$ becomes $b + r(\eta_1\eta_2^{-1} - 1)$. Scaling is clear.

(iii) A complement $L$ of $Re_1$ in $R^2$ maps isomorphically onto $R^2/Re_1 \simeq R$, so it is free of rank one, spanned by the unique $v \in L$ mapping to the class of $e_2$; and $v = e_2 + re_1$ for some $r \in R$. It is $G$-stable if and only if the corner vanishes in the basis $e_1, e_2 + re_1$, that is, $b + d^0 r = 0$ by (ii). So $\rho$ is split if and only if $b$ is a coboundary.

(iv) Given a cocycle $b$, put $\kappa = b\eta_2$; by (i), $\rho$ is a homomorphism, and $E = R^2$ with this action is an extension of the stated kind with class $[b]$. Conversely, an extension $0 \to R(\eta_1) \xrightarrow{\iota} E \to R(\eta_2) \to 0$ has the basis $e_1 = \iota(1)$, $e_2$ a lift of $1 \in R(\eta_2)$. In this basis $g$ fixes the line $Re_1$, acting on it by $\eta_1(g)$, and acts on $E/Re_1$ by $\eta_2(g)$, so its matrix has the shape of (i). The only freedom is the lift $e_2$, which can be replaced by $e_2 + re_1$; by (ii) this changes $b$ by the coboundary $d^0 r$, so $[b]$ depends only on $E$. If $E$, $E'$ with such bases have cocycles $b$ and $b' = b + d^0 r$, then in the basis $e_1, e_2 + re_1$ of $E$ the cocycle is $b'$ by (ii), so the $R$-linear map $E' \to E$ with $e_1' \mapsto e_1$, $e_2' \mapsto e_2 + re_1$ has the same matrices on both sides: it is $G$-equivariant, and it induces the identity on the sub and on the quotient. An isomorphism inducing the identity on the sub and the quotient maps $e_1$ to $e_1'$ and $e_2$ to a lift of 1, so it is of this form; hence the bijection.

(v) Here $\eta_1 = \chi_\ell\varepsilon$, $\eta_2 = \varepsilon$, so $\eta_1\eta_2^{-1} = \chi_\ell$ and $b = \varepsilon\kappa \cdot \varepsilon^{-1} = \kappa$. On $I$ the action on $\mathbb{Z}_\ell(\chi_\ell)$ is trivial, so the cocycle $\kappa|_I$ is a homomorphism by Proposition 16.18(iii). □

*In the paper* 16.22. The proof of Lemma lem:tamfactor writes the matrices of $G_{\mathbb{Q}_q}$ on $T_\ell E$ as $\left(\begin{smallmatrix} \chi_\ell\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$ "with $\varepsilon$ unramified of order at most 2 and $\kappa$ the Kummer cocycle of $q_E$", and adds that "$\varepsilon\kappa$, not $\kappa$, satisfies the cocycle relation for the twisted matrices". Proposition 16.21(i) and (v) make this precise: the corner divided by the lower character, $\varepsilon\kappa/\varepsilon = \kappa$, is the cocycle, with values in $\mathbb{Z}_\ell(1)$, and its restriction to inertia is a homomorphism. The uniqueness of the stable line in a non-split extension, used in Lemma lem:old, is proved in Chapter 4; (iii) above is its cohomological form. That $\kappa$ is a Kummer cocycle is Chapter 17's.

**Theorem 16.23** ($H^2$ and group extensions)**.** *Let $M$ be a $G$-module. The equivalence classes of group extensions $1 \to M \to \mathcal{E} \xrightarrow{\pi} G \to 1$ in which the conjugation action of $\mathcal{E}$ on the abelian normal subgroup $M$ induces the given action of $G$ are in bijection with $H^2(G, M)$. If $s\colon G \to \mathcal{E}$ is a set-theoretic section with $s(1) = 1$, the class is that of the 2-cocycle $\psi(g,h) = s(g)s(h)s(gh)^{-1} \in M$ (written additively in $M$). The semidirect product $M \rtimes G$ corresponds to 0.*

*Sketch of proof.* Since $\pi(s(g)s(h)) = \pi(s(gh))$, $\psi(g,h)$ lies in $M$. Write $\mathcal{E}$ multiplicatively, so that $s(g)s(h) = \psi(g,h)s(gh)$ and $s(g)\,m\,s(g)^{-1} = gm$ for $m \in M$. Expanding $s(g)s(h)s(k)$ in two ways,

$$\begin{aligned} &(s(g)s(h))s(k) = \psi(g,h)\psi(gh,k)\,s(ghk), \\ &s(g)(s(h)s(k)) = s(g)\psi(h,k)s(g)^{-1} \cdot s(g)s(hk) = (g\psi(h,k))\,\psi(g,hk)\,s(ghk), \end{aligned}$$

and comparing, in additive notation in $M$, $g\psi(h,k)+\psi(g,hk)=\psi(g,h)+\psi(gh,k)$, which is $d^2\psi=0$. Another section is $s'(g)=\varphi(g)s(g)$ with $\varphi\colon G\to M$, and

$$s'(g)s'(h)s'(gh)^{-1}=\varphi(g)\,s(g)\varphi(h)s(g)^{-1}\cdot s(g)s(h)s(gh)^{-1}\cdot\varphi(gh)^{-1},$$

which is $\varphi(g)+g\varphi(h)+\psi(g,h)-\varphi(gh)$ additively (the factors lie in the abelian group $M$): it changes $\psi$ by $d^1\varphi$. Conversely a cocycle $\psi$ (normalised so that $\psi(1,g)=\psi(g,1)=0$, which one can arrange by a coboundary) defines a group law $(m,g)(m',g')=(m+gm'+\psi(g,g'),gg')$ on $M\times G$. Details are in [17, Chapter IV, Section 3]. □

**Proposition 16.24** (functoriality in $M$ and the long exact sequence)**.** *Let $G$ be a group and $0\to M'\xrightarrow{i}M\xrightarrow{p}M''\to 0$ an exact sequence of $G$-modules. Then $0\to C^\bullet(G,M')\to C^\bullet(G,M)\to C^\bullet(G,M'')\to 0$ is an exact sequence of complexes, and there is a long exact sequence*

$$\begin{aligned}0\to H^0(G,M')\to H^0(G,M)\to H^0(G,M'')\xrightarrow{\delta}H^1(G,M')\\ \to H^1(G,M)\to H^1(G,M'')\xrightarrow{\delta}H^2(G,M')\to\cdots,\end{aligned}$$

*natural with respect to morphisms of short exact sequences. If the modules are $\mathbb{Z}_\ell[G]$-modules and the maps $\mathbb{Z}_\ell$-linear, every map in it is $\mathbb{Z}_\ell$-linear.*

*Proof.* For each $n$, $C^n(G,-)=\mathrm{Map}(G^n,-)$ takes the exact sequence to an exact sequence: injectivity and exactness in the middle are pointwise, and a map $G^n\to M''$ lifts pointwise because $p$ is surjective. The maps $i\circ-$, $p\circ-$ are morphisms of complexes (Lemma 16.16). Now apply Theorem 16.6. Naturality is Theorem 16.6(iii). $\mathbb{Z}_\ell$-linearity: for $\lambda\in\mathbb{Z}_\ell$, multiplication by $\lambda$ on $M'$, $M$, $M''$ is $G$-linear (the action is $\mathbb{Z}_\ell$-linear) and commutes with $i$ and $p$, so it is an endomorphism of the short exact sequence. It induces multiplication by $\lambda$ on each $C^n$, hence on each $H^n$, and naturality gives $\delta\circ\lambda=\lambda\circ\delta$. The maps $i_*$, $p_*$ are $\mathbb{Z}_\ell$-linear for the same reason. □

**Lemma 16.25** ($H^0$ and torsion)**.** *(a) For a $G$-module $W$ and an integer $m\geq 1$, $H^0(G,W[m])=H^0(G,W)[m]$.*

*(b) Let $T$ be a $\mathbb{Z}_\ell$-module without $\ell$-torsion on which $G$ acts $\mathbb{Z}_\ell$-linearly, and $W=T\otimes_{\mathbb{Z}_\ell}\mathbb{Q}_\ell/\mathbb{Z}_\ell$. Then $W[\ell^k]\simeq T/\ell^kT$ as $G$-modules, by $t\otimes\ell^{-k}\mapsto t \bmod \ell^k$. Hence $H^0(G,W)[\ell^k]\simeq H^0(G,T/\ell^kT)$, and $H^0(G,W)=\varinjlim_k H^0(G,T/\ell^kT)$ along the maps induced by multiplication by $\ell$, $T/\ell^kT\to T/\ell^{k+1}T$.*

*(c) A finite $\ell$-primary abelian group $A$ with $\#A[\ell]\leq\ell$ is cyclic, $A\simeq\mathbb{Z}/\ell^h$, where $h=\mathrm{length}_{\mathbb{Z}_\ell}A$ is the largest $k$ such that $A$ has an element of order $\ell^k$.*

*(d) Suppose in (b) that $H^0(G,W)$ is finite and $\#H^0(G,T/\ell T)\leq\ell$. Then $H^0(G,W)$ is cyclic, and $\mathrm{length}\,H^0(G,W)$ is the largest $k$ for which $H^0(G,T/\ell^kT)$ contains an element of exact order $\ell^k$, that is, an invariant element of $T/\ell^kT$ not in $\ell T/\ell^kT$.*

*(e) If $T$ in (b) is free of rank $r$ and $\#H^0(G,T/\ell T)\leq\ell$, then $H^0(G,W)$ is either finite cyclic or isomorphic to $\mathbb{Q}_\ell/\mathbb{Z}_\ell$.*

*Proof.* (a) Both sides are the set of $w\in W$ with $mw=0$ and $gw=w$ for all $g$.

(b) Since $T$ has no $\ell$-torsion, $T\to T\otimes\mathbb{Q}_\ell=T[1/\ell]$ is injective and $W=T[1/\ell]/T$. Then $W[\ell^k]=\ell^{-k}T/T$, and multiplication by $\ell^k$ is a $G$-isomorphism $\ell^{-k}T/T\to T/\ell^kT$. Apply (a).

As $W = \bigcup_k W[\ell^k]$, also $H^0(G,W) = \bigcup_k H^0(G, W[\ell^k])$; the inclusion $W[\ell^k] \subseteq W[\ell^{k+1}]$ becomes multiplication by $\ell$ under the isomorphisms.

(c) By the structure theorem $A \simeq \bigoplus_{i=1}^{s} \mathbb{Z}/\ell^{a_i}$ with all $a_i \geq 1$, and $A[\ell] \simeq (\mathbb{Z}/\ell)^s$, so $s \leq 1$. Then $A \simeq \mathbb{Z}/\ell^h$ with $h = \operatorname{length} A$, and $\mathbb{Z}/\ell^h$ has an element of order $\ell^k$ exactly for $k \leq h$.

(d) By (a), (b) the $\ell$-torsion of $H^0(G,W)$ is $H^0(G, T/\ell T)$, of order at most $\ell$, so (c) applies, and $H^0(G,W)$ has an element of order $\ell^k$ if and only if its subgroup $H^0(G,W)[\ell^k] \simeq H^0(G, T/\ell^k T)$ does. An element $\bar{x} \in T/\ell^k T$ has order $\ell^k$ if and only if $\ell^{k-1}x \notin \ell^k T$, if and only if $x \notin \ell T$, because $T$ has no $\ell$-torsion.

(e) $W \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$, and $X = H^0(G,W)$ is a submodule with $\#X[\ell] \leq \ell$ (by (a), (b), $X[\ell] = H^0(G, T/\ell T)$). By Pontryagin duality (Chapter 3), dualising the inclusion $X \hookrightarrow W$ makes $X^*$ a quotient of $((\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r)^* = \mathbb{Z}_\ell^r$, hence finitely generated. Dualising the exact sequence $0 \to X[\ell] \to X \xrightarrow{\ell} X$ gives the exact sequence $X^* \xrightarrow{\ell} X^* \to X[\ell]^* \to 0$ (the dual of multiplication by $\ell$ is multiplication by $\ell$), so $X^*/\ell X^* \simeq (X[\ell])^*$ has order at most $\ell$. By Nakayama's lemma $X^*$ is generated by one element: $X^* \simeq \mathbb{Z}_\ell$ or $\mathbb{Z}/\ell^h$, and $X \simeq X^{**}$ is $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ or $\mathbb{Z}/\ell^h$. □

Two examples show the two cases of (e). If $G$ acts trivially on $T = \mathbb{Z}_\ell$, then every $T/\ell^k T$ is invariant and $H^0(G,W) = W = \mathbb{Q}_\ell/\mathbb{Z}_\ell$: infinite. If $G$ acts on $T = \mathbb{Z}_\ell$ through a character $\chi$ with $\chi(G) \subseteq 1 + \ell^h\mathbb{Z}_\ell$, $\chi(G) \not\subseteq 1 + \ell^{h+1}\mathbb{Z}_\ell$, then the invariant elements of $T/\ell^k T$ are all of it for $k \leq h$ and lie in $\ell T/\ell^k T$ for $k > h$, so $H^0(G,W) \simeq \mathbb{Z}/\ell^h$ (Exercise 16.93). Part (d) is the tool for reading off $h$: find the largest level $k$ at which a primitive vector of $T/\ell^k T$ is invariant.

*In the paper* 16.26. In the proof of Lemma lem:h0: "The $\ell^k$-torsion of $H^0(\mathbb{Q}, W^D)$ is $H^0(\mathbb{Q}, \mathrm{ad}^0(T/\ell^k)(1))$" is Lemma 16.25(b) for the lattice $\mathrm{ad}^0 T(1)$, since $W^D = \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. The next step, that $H^0(\mathbb{Q}, W^D)$ "a submodule of $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^3$ with $\ell$-torsion of rank at most one, is finite cyclic or isomorphic to $\mathbb{Q}_\ell/\mathbb{Z}_\ell$", is (e), and the paper's determinant argument excludes $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. The identification of the length with "the largest $k$ such that $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$" is (d): an element of exact order $\ell^k$ is an invariant map $T/\ell^k \to (T/\ell^k)(1)$ not divisible by $\ell$. The lemma itself is proved in Chapter 11.

## 16.5 Changing the group

The proof of Lemma lem:tamfactor uses three operations that change the group: restriction from $G_{\mathbb{Q}_q}$ to the inertia group $I_q$, inflation from the quotient $I_q/K$ to $I_q$, and the conjugation action of $G_{\mathbb{Q}_q}/I_q$ on $H^1(I_q, M)$. The paper's "by inflation-restriction $H^1(I, M') = H^1(\langle t \rangle, M')$" is Theorem 16.31 with Corollary 16.52. Corestriction enters only through $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$, which gives Theorem 16.29, used for the real place after Lemma lem:dfg21. All of them are instances of one construction.

**Proposition 16.27** (compatible pairs)**.** *Let $f\colon G' \to G$ be a group homomorphism, $M$ a $G$-module, $M'$ a $G'$-module, and $\varphi\colon M \to M'$ a homomorphism of abelian groups with $\varphi(f(g')m) = g'\varphi(m)$ for all $g' \in G'$, $m \in M$ (a* compatible pair*). Then $c \mapsto \varphi \circ c \circ f^{\times n}$, that is*

$$c \longmapsto ((g'_1, \dots, g'_n) \mapsto \varphi(c(f(g'_1), \dots, f(g'_n)))),$$

*is a morphism of complexes $C^\bullet(G,M) \to C^\bullet(G',M')$. It induces maps $(f,\varphi)^*\colon H^n(G,M) \to H^n(G',M')$, compatible with the connecting maps of short exact sequences of compatible pairs, and $(f,\varphi)^* \circ (f_1,\varphi_1)^* = (f_1 \circ f, \varphi \circ \varphi_1)^*$ for composable pairs.*

*Proof.* Write $c' = \varphi \circ c \circ f^{\times n}$. In $(dc')(g'_1, \dots, g'_{n+1})$ the first term is $g'_1 \varphi(c(f(g'_2), \dots)) = \varphi(f(g'_1)c(f(g'_2), \dots))$ by compatibility; the middle terms are $\varphi(c(\dots, f(g'_i)f(g'_{i+1}), \dots))$ because $f$ is a homomorphism; the last is $\varphi(c(f(g'_1), \dots, f(g'_n)))$. The sum is $\varphi((dc)(f(g'_1), \dots, f(g'_{n+1})))$, so $dc' = (dc)'$. Compatibility with connecting maps follows from Theorem 16.6(iii), applied to the morphism of short exact sequences of complexes given by the three pairs. For the composition rule, let $(f_1 \colon G \to G_1, \varphi_1 \colon M_1 \to M)$ and $(f \colon G' \to G, \varphi \colon M \to M')$ be compatible. Then $(f_1 \circ f, \varphi \circ \varphi_1)$ is compatible, since $\varphi\varphi_1(f_1 f(g')m) = \varphi(f(g')\varphi_1(m)) = g'\varphi\varphi_1(m)$, and on a cochain $c$ both sides send $c$ to $(g'_1, \dots) \mapsto \varphi(\varphi_1(c(f_1 f(g'_1), \dots)))$. □

The definition of a compatible pair encodes the one requirement that makes "compose with $f$, then apply $\varphi$" commute with $d$: the first term of $d$ involves the action, so $\varphi$ must convert the action of $f(g')$ on $M$ into the action of $g'$ on $M'$. The variance is mixed (the group map goes from $G'$ to $G$, the module map from $M$ to $M'$), which is why restriction ($G'$ a subgroup) and inflation ($G$ a quotient, $M = M^H$ a submodule) both appear as special cases below.

**Proposition 16.28** (restriction, inflation, corestriction)**.** *Let $H \le G$ and $M$ a $G$-module.*

*(a)* Restriction $\mathrm{Res} \colon H^n(G, M) \to H^n(H, M)$ *is* $(H \hookrightarrow G, \mathrm{id}_M)^*$*: restrict cochains to $H^n$.*

*(b) If $H$ is normal,* inflation $\mathrm{Inf} \colon H^n(G/H, M^H) \to H^n(G, M)$ *is* $(G \to G/H,\ M^H \hookrightarrow M)^*$*: compose with the projection.*

*(c) If $[G : H] = r < \infty$, there are* corestriction *maps* $\mathrm{Cor} \colon H^n(H, M) \to H^n(G, M)$ *for all $n$, compatible with the long exact sequences. In degree 0,* $\mathrm{Cor}(m) = \sum_{i=1}^{r} g_i m$ *for left coset representatives $g_1, \dots, g_r$ ($G = \bigsqcup g_i H$), independent of the choice. In degree 1, writing $g g_i = g_{\tau(i)} h_i(g)$ with $\tau = \tau_g$ a permutation and $h_i(g) \in H$,*

$$(\mathrm{Cor}\,\varphi)(g) = \sum_{i=1}^{r} g_{\tau(i)}\, \varphi(h_i(g)).$$

*(d)* $\mathrm{Cor} \circ \mathrm{Res} = r$ *on $H^n(G, M)$ for all $n$. If $H$ is normal,* $\mathrm{Res} \circ \mathrm{Cor} = \sum_{\sigma \in G/H} \sigma_*$ *on $H^n(H, M)$, where $\sigma_*$ is the conjugation action of Proposition 16.30.*

*Proof.* (a), (b): the pairs are compatible: for (b), $m \in M^H$ and $\pi(g)m = gm$. Apply Proposition 16.27.

(c) Degree 0: for $m \in M^H$, $g_i h m = g_i m$, so $\sum g_i m$ does not depend on the representatives; it is $G$-invariant because $g$ permutes the cosets: $g \sum_i g_i m = \sum_i g_{\tau(i)} h_i(g) m = \sum_i g_{\tau(i)} m = \sum_j g_j m$. Degree 1: first $h_i(gg') = h_{\tau_{g'}(i)}(g) h_i(g')$ and $\tau_{gg'} = \tau_g \tau_{g'}$, since $gg' g_i = g g_{\tau_{g'}(i)} h_i(g') = g_{\tau_g \tau_{g'}(i)} h_{\tau_{g'}(i)}(g) h_i(g')$. For $\varphi \in Z^1(H, M)$,

$$(\mathrm{Cor}\,\varphi)(gg') = \sum_i g_{\tau_g \tau_{g'}(i)} \left[\varphi(h_{\tau_{g'}(i)}(g)) + h_{\tau_{g'}(i)}(g)\, \varphi(h_i(g'))\right].$$

Reindexing by $j = \tau_{g'}(i)$, the first part is $(\mathrm{Cor}\,\varphi)(g)$; in the second, $g_{\tau_g(j)} h_j(g) = g g_j$, so it is $g \sum_i g_{\tau_{g'}(i)} \varphi(h_i(g')) = g(\mathrm{Cor}\,\varphi)(g')$. So $\mathrm{Cor}\,\varphi$ is a cocycle. If $\varphi(h) = hm - m$, then $(\mathrm{Cor}\,\varphi)(g) = \sum_i (g_{\tau(i)} h_i(g) m - g_{\tau(i)} m) = \sum_i (g g_i m - g_{\tau(i)} m) = gm' - m'$ with $m' = \sum_i g_i m$, a coboundary. So Cor is defined on $H^1$. In higher degrees we do not prove the existence or the compatibility with long exact sequences; see [17, Chapter III, Section 9] and [143, Chapter I, Section 5].

(d) Degree 0: $\mathrm{Cor}(\mathrm{Res}\, m) = \sum g_i m = rm$ for $m \in M^G$. Degree 1: for $c \in Z^1(G,M)$, $h_i(g) = g_{\tau(i)}^{-1} g g_i$, and the cocycle identity gives $g_{\tau(i)} c(g_{\tau(i)}^{-1} g g_i) = -c(g_{\tau(i)}) + c(g) + g c(g_i)$: indeed $c(x^{-1}y) = c(x^{-1}) + x^{-1}c(y) = x^{-1}(c(y) - c(x))$ for $x = g_{\tau(i)}$ and $y = g g_i$, by Proposition 16.18(ii), and $c(gg_i) = c(g) + g c(g_i)$. Summing over $i$ (and using that $\tau$ is a permutation, so $\sum_i c(g_{\tau(i)}) = \sum_i c(g_i)$), $(\mathrm{Cor}\,\mathrm{Res}\, c)(g) = rc(g) + gm - m$ with $m = \sum_i c(g_i)$: so $\mathrm{Cor}\,\mathrm{Res}\, c = rc + d^0 m$. If $H$ is normal and $h \in H$, then $h g_i H = g_i (g_i^{-1} h g_i) H = g_i H$, so $\tau_h = \mathrm{id}$ and $h_i(h) = g_i^{-1} h g_i$; hence $(\mathrm{Res}\,\mathrm{Cor}\,\varphi)(h) = \sum_i g_i \varphi(g_i^{-1} h g_i) = \sum_i (g_i \cdot \varphi)(h)$. In degrees $n \geq 2$ (d) is stated; see the references for (c). □

The next theorem is the reason a group of order 2 is invisible to $\ell$-adic coefficients when $\ell$ is odd. We prove it in every degree with an explicit cochain, so that it does not depend on the corestriction in degrees $n \geq 2$.

**Theorem 16.29** (the order kills the cohomology)**.** *Let $G$ be a finite group of order $N$ and $M$ a $G$-module. Then $N \cdot H^n(G,M) = 0$ for every $n \geq 1$. More precisely, for $\varphi \in Z^n(G,M)$ put $\psi(g_1,\dots,g_{n-1}) = \sum_{g\in G} \varphi(g_1,\dots,g_{n-1},g)$; then $N\varphi = (-1)^n d^{n-1}\psi$. Consequently:*

*(i) if multiplication by $N$ is bijective on $M$, then $H^n(G,M) = 0$ for $n \geq 1$;*

*(ii) if $M$ is finitely generated as an abelian group, $H^n(G,M)$ is finite for $n \geq 1$;*

*(iii) if $\ell \nmid N$ and $M$ is an $\ell$-primary torsion group or a $\mathbb{Z}_\ell$-module, then $H^n(G,M) = 0$ for $n \geq 1$, and $M^G$ is a direct summand of $M$ as a $G$-module, cut out by the projector $N^{-1}N_G$.*

*Proof.* Write the cocycle identity $(d^n\varphi)(g_1,\dots,g_n,g) = 0$ and sum over $g \in G$:

$$g_1\psi(g_2,\dots,g_n) + \sum_{i=1}^{n-1}(-1)^i \psi(\dots, g_i g_{i+1}, \dots) + (-1)^n \sum_g \varphi(g_1,\dots,g_{n-1},g_n g) + (-1)^{n+1} N\varphi(g_1,\dots,g_n) = 0.$$

Here the first term comes from $g_1\varphi(g_2,\dots,g_n,g)$; the middle sum from the products $g_i g_{i+1}$ with $i \leq n-1$, in which the last variable $g$ is untouched; the third from the product $g_n g$ in position $n$; and the last from dropping the last variable $g$, which leaves $\varphi(g_1,\dots,g_n)$ for each of the $N$ values of $g$. As $g$ runs over $G$ so does $g_n g$, so the third sum is $(-1)^n\psi(g_1,\dots,g_{n-1})$. Comparing with the definition

$$(d^{n-1}\psi)(g_1,\dots,g_n) = g_1\psi(g_2,\dots,g_n) + \sum_{i=1}^{n-1}(-1)^i\psi(\dots,g_ig_{i+1},\dots) + (-1)^n\psi(g_1,\dots,g_{n-1}),$$

the first three terms add up to $(d^{n-1}\psi)(g_1,\dots,g_n)$, so $d^{n-1}\psi + (-1)^{n+1}N\varphi = 0$, that is $N\varphi = (-1)^n d^{n-1}\psi$. (For $n = 1$: with $m = \sum_g \varphi(g)$, summing $\varphi(\sigma g) = \varphi(\sigma) + \sigma\varphi(g)$ gives $m = N\varphi(\sigma) + \sigma m$.)

(i) Multiplication by $N$ on $H^n(G,M)$ is induced by the automorphism $N$ of $M$ (functoriality), so it is bijective; it is also 0. (ii) $C^n(G,M) = M^{G^n}$ is finitely generated, so $H^n(G,M)$ is a finitely generated abelian group killed by $N$. (iii) An integer prime to $\ell$ acts bijectively on an $\ell$-primary torsion group and on a $\mathbb{Z}_\ell$-module (it is a unit of $\mathbb{Z}_\ell$), so (i) applies. The map $e = N^{-1}N_G$ is $G$-linear, takes values in $M^G$ and is the identity on $M^G$; so $M = M^G \oplus \ker e$ as $G$-modules. □

**Proposition 16.30** (the conjugation action)**.** *Let $H \trianglelefteq G$ and $M$ a $G$-module. For $\sigma \in G$ the pair $(h \mapsto \sigma^{-1}h\sigma,\ m \mapsto \sigma m)$ is compatible and induces $\sigma_* \colon H^n(H,M) \to H^n(H,M)$, given on cochains by*

$$(\sigma \cdot \varphi)(h_1, \dots, h_n) = \sigma\, \varphi(\sigma^{-1}h_1\sigma, \dots, \sigma^{-1}h_n\sigma).$$

*(i) This is a left action of $G$ on $C^\bullet(H,M)$ by morphisms of complexes, hence on $H^n(H,M)$; it is $\mathbb{Z}_\ell$-linear when $M$ is a $\mathbb{Z}_\ell[G]$-module, so it preserves $H^n(H,M)_{\mathrm{tor}}$, and $(H^n(H,M)_{\mathrm{tor}})^G$ is defined.*

*(ii) $H$ acts trivially on $H^n(H,M)$; so the action factors through $G/H$, and the map induced by a class of $G/H$ does not depend on its representative. For $n = 1$ and $h_0 \in H$, $\varphi \in Z^1(H,M)$: $(h_0 \cdot \varphi) - \varphi = d^0(\varphi(h_0))$.*

*(iii) The restriction $H^n(G,M) \to H^n(H,M)$ lands in $H^n(H,M)^{G/H}$. For $n = 1$ and $c \in Z^1(G,M)$: $\sigma \cdot \operatorname{Res} c - \operatorname{Res} c = d^0(c(\sigma))$ on $H$.*

*Proof.* Compatibility: $\sigma\big((\sigma^{-1}h\sigma)m\big) = h\sigma m$. (i) The cochain formula is the one of Proposition 16.27. For the action, $(\sigma \cdot (\tau \cdot \varphi))(h) = \sigma\tau\, \varphi(\tau^{-1}\sigma^{-1}h\sigma\tau) = ((\sigma\tau) \cdot \varphi)(h)$, and similarly in several variables. Since $\varphi \mapsto \sigma \cdot \varphi$ is $\mathbb{Z}_\ell$-linear, it maps torsion classes to torsion classes.

(iii) and (ii) for $n = 1$. Let $c$ be a 1-cocycle on a group containing $H$ and $\sigma$ (that is, on $G$ for (iii); on $H$, with $\sigma = h_0 \in H$, for (ii)). By the cocycle identity and $c(\sigma^{-1}) = -\sigma^{-1}c(\sigma)$,

$$c(\sigma^{-1}h\sigma) = c(\sigma^{-1}) + \sigma^{-1}\big(c(h) + hc(\sigma)\big) = \sigma^{-1}\big(c(h) + hc(\sigma) - c(\sigma)\big),$$

so $(\sigma \cdot c)(h) = c(h) + (h-1)c(\sigma) = c(h) + (d^0c(\sigma))(h)$ for $h \in H$. Here $c(\sigma) \in M$, so $d^0c(\sigma)$ is a coboundary on $H$. For $n = 0$ both statements are clear ($\sigma \cdot m = \sigma m$, and $m \in M^H$, resp. $M^G$). For $n \geq 2$ the triviality of the action of $H$ and the invariance of the image of restriction are stated; see [17, Chapter III, Section 8] and [143, Chapter I, Section 5]. (They follow from the case $n = 0$ by dimension shifting, since both sides are $\delta$-functors.) The independence of the representative follows from the triviality on $H$. □

**Theorem 16.31** (inflation-restriction)**.** *Let $H \trianglelefteq G$ and $M$ a $G$-module.*

*(a) The sequence $0 \to H^1(G/H, M^H) \xrightarrow{\mathrm{Inf}} H^1(G,M) \xrightarrow{\mathrm{Res}} H^1(H,M)^{G/H}$ is exact.*

*(b) (Five-term exact sequence of Hochschild and Serre.) There is a homomorphism $\mathrm{tg} \colon H^1(H,M)^{G/H} \to H^2(G/H, M^H)$, the* transgression*, such that*

$$0 \to H^1(G/H, M^H) \xrightarrow{\mathrm{Inf}} H^1(G,M) \xrightarrow{\mathrm{Res}} H^1(H,M)^{G/H} \xrightarrow{\mathrm{tg}} H^2(G/H, M^H) \xrightarrow{\mathrm{Inf}} H^2(G,M)$$

*is exact.*

*(c) If $H^i(H,M) = 0$ for $1 \leq i \leq n-1$, then $0 \to H^n(G/H, M^H) \xrightarrow{\mathrm{Inf}} H^n(G,M) \xrightarrow{\mathrm{Res}} H^n(H,M)$ is exact.*

*(d) If $H^1(H,M) = 0$, then $\mathrm{Inf} \colon H^1(G/H, M^H) \to H^1(G,M)$ is an isomorphism. In particular, if $H$ acts trivially on $M$ and $\mathrm{Hom}(H,M) = 0$, then $\mathrm{Inf} \colon H^1(G/H, M) \to H^1(G,M)$ is an isomorphism.*

*These hold for abstract groups; and, with the conventions of Section 16.7, for profinite $G$, closed normal $H$ and continuous cochains with coefficients of type (D), (C) or (V) (in (b) and (c) for type (D); see Remark 16.32). For $G/H \simeq \widehat{\mathbb{Z}}$, (b) holds also for types (C) and (V), with $\mathrm{tg} = 0$ (Proposition 16.61).*

*Proof.* (a) Write $\pi\colon G \to G/H$. By Proposition 16.30(iii) the image of Res is $G/H$-invariant.

Inf is injective. Let $c \in Z^1(G/H, M^H)$ with $c \circ \pi = d^0 m$ for some $m \in M$, that is $c(\pi(g)) = gm - m$. For $h \in H$, $0 = c(1) = hm - m$, so $m \in M^H$, and then $c = d^0 m$ in $C^1(G/H, M^H)$.

$\operatorname{Res} \circ \operatorname{Inf} = 0$: $(c \circ \pi)(h) = c(1) = 0$.

$\ker \operatorname{Res} \subseteq \operatorname{im} \operatorname{Inf}$. Let $c \in Z^1(G, M)$ with $c(h) = hm - m$ for $h \in H$. Then $c' = c - d^0 m$ is cohomologous to $c$ and vanishes on $H$. For $g \in G$, $h \in H$: $c'(gh) = c'(g) + gc'(h) = c'(g)$, so $c'$ is constant on the cosets $gH$; and $c'(hg) = c'(h) + hc'(g) = hc'(g)$, while $hg = g(g^{-1}hg)$ gives $c'(hg) = c'(g)$. So $hc'(g) = c'(g)$: $c'$ takes values in $M^H$, and $c' = \bar{c} \circ \pi$ with $\bar{c}\colon G/H \to M^H$, which is a cocycle because $\pi$ is surjective: for $g, g' \in G$, $\bar{c}(\pi(g)\pi(g')) = c'(gg') = c'(g) + gc'(g') = \bar{c}(\pi(g)) + \pi(g)\bar{c}(\pi(g'))$, the action of $G/H$ on $M^H$ being that of any lift.

In words: a class on $G$ that dies on $H$ can be normalised to vanish on $H$, and a cocycle vanishing on a normal subgroup is automatically constant on its cosets with values in $M^H$, so it comes from the quotient. The sequence does not continue with a surjection in general; the obstruction to extending an invariant class from $H$ to $G$ is the transgression of (b). When $G/H \simeq \widehat{\mathbb{Z}}$ there is no obstruction (Proposition 16.61).

(d) follows from (a), since $H^1(H, M) = 0$ makes the invariants 0; with trivial action $M^H = M$ and $H^1(H, M) = \operatorname{Hom}(H, M)$ (Proposition 16.18(iii)).

(b) and (c) are stated; see [174, Chapter VII, Section 6], [143, Chapter I, Section 6 and Chapter II, Section 4] and [205, Section 6.8]. The transgression is constructed on cocycles: a class in $H^1(H, M)^{G/H}$ is represented by a cocycle $c$ on $H$ that extends to a cochain $\tilde{c}$ on $G$ with $\tilde{c}(gh) = \tilde{c}(g) + gc(h)$ and $\tilde{c}(hg) = c(h) + h\tilde{c}(g)$; then $d^1\tilde{c}$ is a 2-cocycle on $G$ that is inflated from a 2-cocycle on $G/H$ with values in $M^H$, and $\operatorname{tg}[c]$ is its class (Exercise 16.95). □

*Remark* 16.32 (continuous cochains). The proof of (a) uses only cochain manipulations, so it holds verbatim for continuous cochains (Definition 16.42) once one checks that $\bar{c}$ is continuous when $c'$ is. This holds because the projection $\pi\colon G \to G/H$ of topological groups is open: for an open set $U \subseteq M^H$, $\bar{c}^{-1}(U) = \pi(c'^{-1}(U))$ is open. Also $M^H$ is closed in $M$ (an intersection of closed sets $\{m : hm = m\}$, $M$ being Hausdorff), so for $M$ of type (C) or (V) it is again of type (C) or (V). This is the form in which Chapter 17 uses (a): $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = H^1(G_{\mathbb{Q}_p}/I_p, M^{I_p})$ is the kernel of the restriction to $I_p$ (Convention 16.77).

**Theorem 16.33** (the Hochschild-Serre spectral sequence)**.** *For $H \trianglelefteq G$ and a $G$-module $M$ there is a first-quadrant spectral sequence*

$$E_2^{p,q} = H^p(G/H, H^q(H, M)) \Longrightarrow H^{p+q}(G, M),$$

*with $G/H$ acting on $H^q(H, M)$ by conjugation; its edge maps are inflation and restriction, and its five-term exact sequence is Theorem 16.31(b). The same holds for profinite $G$, closed normal $H$ and $M$ of type (D). Consequently, if $H^q(H, M) = 0$ for all $q \geq 1$, then* $\operatorname{Inf}\colon H^p(G/H, M^H) \to H^p(G, M)$ *is an isomorphism for every $p$.*

*Proof.* We do not prove this; see [17, Chapter VII, Section 6], [205, Section 6.8] and [143, Chapter II, Section 4]. The consequence holds because the spectral sequence then has a single nonzero row $q = 0$. □

## 16.6 Finite groups

Two facts about finite groups reach the paper. The first is the formula for $H^1$ of a finite cyclic group, which is the model for the procyclic formula of Section 16.9 and also its main

counterexample. The second is that the cohomology of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$, a group of order 2, is killed by 2: the paper's "At $\infty$, $H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for $\ell$ odd" (the paragraph after Lemma lem:dfg21). Tate cohomology is recalled briefly, for orientation.

**Theorem 16.34** (cohomology of a finite cyclic group)**.** *Let $G = \langle t \rangle$ be cyclic of order $m$, $N = 1 + t + \cdots + t^{m-1}$, and $M$ a $G$-module.*

*(i) The map $\varphi \mapsto \varphi(t)$ induces an isomorphism $H^1(G, M) \simeq \ker(N\colon M \to M)/(t-1)M$. A cocycle is determined by $a = \varphi(t)$ through $\varphi(t^k) = (1 + t + \cdots + t^{k-1})a$ for $0 \le k < m$, and $a$ occurs if and only if $Na = 0$.*

*(ii) $H^2(G, M) \simeq M^G/NM$, and $H^{n+2}(G, M) \simeq H^n(G, M)$ for all $n \ge 1$.*

*(iii) For the trivial action on $\mathbb{Z}$: $H^1(G, \mathbb{Z}) = 0$, $H^2(G, \mathbb{Z}) = \mathbb{Z}/m$. For the trivial action on $\mathbb{Z}/m$: $H^1 = H^2 = \mathbb{Z}/m$.*

*Proof.* (i) Put $S_k = 1 + t + \cdots + t^{k-1}$ ($S_0 = 0$). By induction from $\varphi(t^{k+1}) = \varphi(t^k) + t^k\varphi(t)$, a cocycle satisfies $\varphi(t^k) = S_k a$ for all $k \ge 0$; at $k = m$, $0 = \varphi(1) = S_m a = Na$. Conversely, given $a$ with $Na = 0$, define $\varphi(t^k) = S_k a$ for $0 \le k < m$. For $0 \le j, k < m$ we have $S_{j+k} = S_j + t^j S_k$, and if $j + k \ge m$ then $S_{j+k}a = S_m a + t^m S_{j+k-m}a = S_{j+k-m}a$; so $\varphi(t^j t^k) = \varphi(t^j) + t^j\varphi(t^k)$ in all cases. Finally $(d^0 b)(t) = (t-1)b$, so the coboundaries correspond to $(t-1)M$.

(ii) is stated; see [174, Chapter VIII, Section 4] and [26, Chapter IV, Section 8]. It comes from the periodic free resolution $\cdots \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{t-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{t-1} \mathbb{Z}[G] \to \mathbb{Z} \to 0$ and Theorem 16.17: applying $\mathrm{Hom}_{\mathbb{Z}[G]}(-, M)$ gives $M \xrightarrow{t-1} M \xrightarrow{N} M \xrightarrow{t-1} M \xrightarrow{N} \cdots$.

(iii) For $\mathbb{Z}$ with trivial action $N = m$ is injective and $t - 1 = 0$, so $H^1 = 0$ and $H^2 = \mathbb{Z}/m\mathbb{Z}$. For $\mathbb{Z}/m$, $N = m = 0$ and $t - 1 = 0$. $\square$

**Definition 16.35** (Tate cohomology)**.** For a finite group $G$ and a $G$-module $M$ put $\hat{H}^n(G, M) = H^n(G, M)$ for $n \ge 1$, $\hat{H}^0(G, M) = M^G/N_G M$, $\hat{H}^{-1}(G, M) = \ker(N_G)/I_G M$, and $\hat{H}^{-n}(G, M) = H_{n-1}(G, M)$ (group homology) for $n \ge 2$.

**Proposition 16.36** (properties of Tate cohomology)**.** *Let $G$ be finite.*

*(i) A short exact sequence of $G$-modules gives a long exact sequence of the $\hat{H}^n$, infinite in both directions.*

*(ii) $\#G$ kills $\hat{H}^n(G, M)$ for every $n \in \mathbb{Z}$.*

*(iii) Induced modules $\mathbb{Z}[G] \otimes_{\mathbb{Z}} A$ have $\hat{H}^n = 0$ for every $n$.*

*(iv) For $G$ cyclic, $\hat{H}^n(G, M) \simeq \hat{H}^{n+2}(G, M)$ for every $n$; so $\hat{H}^{-1} = \ker N/(t-1)M \simeq H^1$ and $\hat{H}^0 = M^G/NM \simeq H^2$.*

*Proof.* We do not prove this; see [174, Chapter VIII, Sections 1 to 4] and [17, Chapter VI, Section 4]. For $n \ge 1$, (ii) is Theorem 16.29, and for $n = 0, -1$ it follows from $N_G \cdot m = \#G\, m$ for $m \in M^G$ and from $\#G\, m = N_G m - \sum_g (g-1)m$. $\square$

**Corollary 16.37** (groups of order 2 and $\ell$-adic coefficients)**.** *Let $G = \{1, \sigma\}$ have order 2, for instance $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$ generated by complex conjugation.*

*(i) For every $G$-module $M$ and $n \ge 1$, $H^n(G, M)$ is killed by 2.*

*(ii) Let $\ell$ be odd and $M$ a $\mathbb{Z}_\ell$-module with $\mathbb{Z}_\ell$-linear action (a module of type (D) that is $\ell$-primary, or of type (C) or (V)). Then $H^n(G,M)=0$ for all $n\geq 1$, $M=M^+\oplus M^-$ with $M^\pm=\{m : \sigma m=\pm m\}$, and $H^0(G,M)=M^+$.*

*In the notation $H^n(\mathbb{R},M):=H^n(\mathrm{Gal}(\mathbb{C}/\mathbb{R}),M)$: $H^1(\mathbb{R},M)$ is a group of exponent dividing 2, and it is 0 when $M$ is an $\ell$-adic module with $\ell$ odd.*

*Proof.* (i) is Theorem 16.29 with $N=2$. (ii) 2 is a unit of $\mathbb{Z}_\ell$, so Theorem 16.29(iii) applies; the projectors are $\frac{1}{2}(1\pm\sigma)$. □

**Example 16.38.** Let $\sigma$ act on $\mathbb{Z}$ by $-1$ (the module $\mathbb{Z}^-$). Then $N=1+\sigma=0$, so by Theorem 16.34, $H^1(G,\mathbb{Z}^-)=\mathbb{Z}/(\sigma-1)\mathbb{Z}=\mathbb{Z}/2\mathbb{Z}\neq 0$, and $H^2(G,\mathbb{Z}^-)=(\mathbb{Z}^-)^G/N\mathbb{Z}^-=0$. On $\mathbb{Z}_\ell^-$ with $\ell$ odd, $H^1(G,\mathbb{Z}_\ell^-)=\mathbb{Z}_\ell/2\mathbb{Z}_\ell=0$, as Corollary 16.37 predicts. On $\mathbb{Z}_2^-$ it is $\mathbb{Z}/2$: the hypothesis $\ell$ odd cannot be dropped.

*In the paper* 16.39. After Lemma lem:dfg21: "At $\infty$, $H^1(\mathbb{R},T^D)$ is killed by 2, so $H^1_f(\mathbb{R},T^D)=0$ for $\ell$ odd." The first clause is Corollary 16.37(i); since $H^1(\mathbb{R},T^D)$ is a $\mathbb{Z}_\ell$-module (Definition 16.14) and $2\in\mathbb{Z}_\ell^\times$, it is 0, and so is its submodule $H^1_f(\mathbb{R},T^D)$ (the local condition at $\infty$ is defined in Chapter 18). This is also the argument of Chapter 3. Consequently the term at $\infty$ in the exact sequence of Lemma lem:dfg21 vanishes, and Lemma 16.75 below has no contribution from the real place.

## 16.7 Profinite groups and continuous cohomology

Every cohomology group in the paper is the continuous cohomology of a profinite group: $H^1(G,\mathbb{Z}_\ell(1))$ and $H^1(I_q,\mathrm{ad}^0 T)$ in Lemma lem:tamfactor, $H^1(\mathbb{Q}_p,T^D)$ in Lemma lem:dfg21, $H^1_f(\mathbb{Q},W)$ in Theorem thm:main. This section defines it for the three types of coefficients, proves that the long exact sequences exist in the cases the book needs, and records what can go wrong. Profinite groups and the Krull topology are in Chapters 1 and 5.

**Definition 16.40** (discrete modules)**.** Let $G$ be a profinite group; its open normal subgroups form a basis of neighbourhoods of 1. A *discrete $G$-module* is a $G$-module $M$ such that the action $G\times M\to M$ is continuous for the discrete topology on $M$. Modules of *type (D)* are the discrete modules that are finite, or $\ell$-primary torsion, such as $W=T\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$.

**Lemma 16.41.** *For a $G$-module $M$ with the discrete topology the following are equivalent: (i) the action is continuous; (ii) every stabiliser $G_m=\{g : gm=m\}$ is open; (iii) $M=\bigcup_U M^U$, the union over the open normal subgroups $U$.*

*Proof.* (i)⇒(ii): $G_m$ is the preimage of $\{m\}$ under the continuous map $g\mapsto gm$. (ii)⇒(i): for $(g,m)$, the open set $gG_m\times\{m\}$ is mapped to $gm$. (ii)⇔(iii): an open subgroup has finite index, so it contains the intersection of its finitely many conjugates, an open normal subgroup; conversely $M^U\ni m$ means $U\subseteq G_m$. □

**Definition 16.42** (continuous cohomology)**.** Let $G$ be profinite and $M$ a Hausdorff topological abelian group with a continuous action $G\times M\to M$ by automorphisms. Let $C^n_{\mathrm{cont}}(G,M)\subseteq C^n(G,M)$ be the group of continuous maps $G^n\to M$. The differential of Definition 16.15 maps continuous cochains to continuous cochains, because the action and the group law of $M$ are continuous. The *continuous cohomology* is $H^n(G,M)=H^n(C^\bullet_{\mathrm{cont}}(G,M))$. The coefficient modules of the book are:

(D) discrete modules of type (D);

(C) finitely generated $\mathbb{Z}_\ell$-modules $M$ with the $\ell$-adic topology (the topology with basis of neighbourhoods of 0 the $\ell^k M$) and a continuous $\mathbb{Z}_\ell$-linear action;

(V) finite-dimensional $\mathbb{Q}_\ell$-vector spaces with their $\ell$-adic topology and a continuous linear action.

*Convention: all cohomology of profinite groups in this book is continuous cochain cohomology.* For a finite group with the discrete topology every cochain is continuous, and the definition agrees with Definition 16.15.

Why continuity? The Galois groups of the paper are profinite and its representations are continuous, and the cohomology must see exactly the continuous objects. The corner of a continuous triangular representation $\rho\colon G \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ is a continuous cocycle, and conversely a continuous cocycle gives a continuous $\rho$ by Proposition 16.21; so continuous $H^1$ classifies continuous extensions, which is what the paper's Kummer cocycle $\kappa$ is. Abstract cochains would also count discontinuous objects, which have no arithmetic meaning: for the profinite group $P = \prod_{n\geq 1} \mathbb{Z}/2$ the continuous homomorphisms $P \to \mathbb{Z}/2$ are those factoring through finitely many coordinates, a countable set, while the abstract ones form the dual of an $\mathbb{F}_2$-vector space of uncountable dimension, a much larger set. The three types match the three modules that occur together in the paper: a lattice $T$ (type (C)), the vector space $V = T \otimes \mathbb{Q}_\ell$ (type (V)) and the discrete module $W = V/T$ (type (D)).

**Proposition 16.43** (basic properties)**.** *Let $G$ be profinite.*

*(i)* $H^0(G, M) = M^G$*;* $H^1(G, M)$ *is the group of continuous crossed homomorphisms modulo the principal ones; if $G$ acts trivially, $H^1(G, M) = \mathrm{Hom}_{\mathrm{cont}}(G, M)$.*

*(ii) (Type (D).) For a discrete $G$-module $M$,*

$$H^n(G, M) = \varinjlim_{U} H^n(G/U, M^U),$$

*the limit over the open normal subgroups $U$, with the inflation maps $H^n(G/U, M^U) \to H^n(G/U', M^{U'})$ for $U' \subseteq U$ as transition maps.*

*(iii) (Type (C).) For $M$ finitely generated over $\mathbb{Z}_\ell$, a $\mathbb{Z}_\ell$-linear action is continuous if and only if each kernel $U_k$ of $G \to \mathrm{Aut}(M/\ell^k M)$ is open, if and only if $G \to \mathrm{Aut}_{\mathbb{Z}_\ell}(M) = \varprojlim_k \mathrm{Aut}(M/\ell^k M)$ is continuous.*

*(iv) (Type (V).) A continuous action of $G$ on $V$ of type (V) preserves a $\mathbb{Z}_\ell$-lattice $T$ (a finitely generated $\mathbb{Z}_\ell$-submodule spanning $V$), which is then of type (C), with $V = T \otimes \mathbb{Q}_\ell$.*

*Proof.* (i) As in Proposition 16.18.

(ii) Let $\varphi\colon G^n \to M$ be continuous. Its image is compact in the discrete space $M$, hence finite, and its fibres are open and closed. A clopen subset of the profinite space $G^n$ is a finite union of cosets of a subgroup $U^n$ with $U$ open normal (it is open, so a union of such cosets, and compact, so finitely many suffice; then take the intersection of the finitely many $U$). So $\varphi$ factors through $(G/U)^n$ for some $U$. Each of the finitely many values is fixed by an open normal subgroup (Lemma 16.41); shrinking $U$, $\varphi$ takes values in $M^U$. Hence $C^\bullet_{\mathrm{cont}}(G, M) = \bigcup_U C^\bullet(G/U, M^U)$, the

union of subcomplexes along the inflation maps, which are injective on cochains. Cohomology commutes with filtered unions (direct limits are exact), which gives (ii).

(iii) If the action is continuous, the stabiliser of each of the finitely many elements of $M/\ell^k M$ is open, so $U_k$ is open. Conversely, given $(g_0, m_0)$ and $k$, for $g = g_0 u$ with $u \in U_k$ and $m \in m_0 + \ell^k M$ we have $gm - g_0 m_0 = g(m - m_0) + g_0(um_0 - m_0) \in \ell^k M$. The last equivalence is the definition of the inverse-limit topology.

(iv) Choose a basis, so that $G \to \mathrm{GL}_n(\mathbb{Q}_\ell)$ is continuous. $\mathrm{GL}_n(\mathbb{Z}_\ell)$ is open in $\mathrm{GL}_n(\mathbb{Q}_\ell)$, so its preimage $H$ is an open subgroup, of finite index. The standard lattice $T_0 = \mathbb{Z}_\ell^n$ is $H$-stable, so $gT_0$ depends only on the coset $gH$, and $T = \sum_{g \in G/H} gT_0$ is a finite sum of lattices, hence a lattice; it is $G$-stable. □

**Proposition 16.44** (long exact sequences for continuous cohomology)**.** *Let $G$ be profinite and $0 \to M' \xrightarrow{i} M \xrightarrow{p} M'' \to 0$ an exact sequence of topological $G$-modules with continuous $G$-maps, $i$ a homeomorphism onto its closed image, and $p$ admitting a* continuous section*: a continuous map $s\colon M'' \to M$ with $ps = \mathrm{id}$, not necessarily additive.*

*(i)* $0 \to C^\bullet_{\mathrm{cont}}(G, M') \to C^\bullet_{\mathrm{cont}}(G, M) \to C^\bullet_{\mathrm{cont}}(G, M'') \to 0$ *is exact, and there is a natural long exact sequence*

$$\cdots \to H^n(G, M') \to H^n(G, M) \to H^n(G, M'') \xrightarrow{\delta} H^{n+1}(G, M') \to \cdots .$$

*(ii) A continuous section exists in each of the following cases, which are all the cases used in this book:*

- *(a) $M''$ is discrete;*
- *(b) $p$ is a surjection of finitely generated $\mathbb{Z}_\ell$-modules with their $\ell$-adic topologies;*
- *(c) $0 \to T \to V \to W \to 0$, with $T$ of type (C) without $\ell$-torsion, $V = T \otimes \mathbb{Q}_\ell$ and $W = V/T = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$;*
- *(d) $0 \to T \xrightarrow{\ell^k} T \to T/\ell^k T \to 0$ with $T$ of type (C) without $\ell$-torsion.*

*(iii) In case (c),* $\ker(H^1(G,T) \to H^1(G,V)) = H^1(G,T)_{\mathrm{tor}}$*, so the connecting map $\delta\colon H^0(G,W) \to H^1(G,T)$ has image $H^1(G,T)_{\mathrm{tor}}$ and kernel the image of $H^0(G,V)$.*

*Proof.* (i) Injectivity is clear. If $\varphi\colon G^n \to M$ is continuous and $p\varphi = 0$, then $\varphi$ takes values in $i(M')$, and $i^{-1} \circ \varphi$ is continuous because $i$ is a homeomorphism onto its image. If $\psi\colon G^n \to M''$ is continuous, $s \circ \psi$ is a continuous lift. Now apply Theorem 16.6.

(ii)(a) Every map out of a discrete space is continuous. (b) Write $M'' = \bigoplus_j \mathbb{Z}_\ell y_j$ with each summand free or $\simeq \mathbb{Z}/\ell^{a_j}$, choose $x_j \in M$ with $p(x_j) = y_j$, and put $s(\sum_j \lambda_j y_j) = \sum_j \tilde{\lambda}_j x_j$, where $\tilde{\lambda}_j = \lambda_j$ on a free summand and $\tilde{\lambda}_j \in \{0, 1, \dots, \ell^{a_j} - 1\} \subset \mathbb{Z}_\ell$ is the representative on a summand $\mathbb{Z}/\ell^{a_j}$. The coordinates are continuous for the $\ell$-adic topology, and a map out of the finite discrete set $\mathbb{Z}/\ell^{a_j}$ is continuous. The inclusion $i$ is an injective continuous map from a compact space to a Hausdorff one, hence a homeomorphism onto its closed image. (c) $T$ is open in $V$, so $W = V/T$ is discrete and (a) applies; $T$ is compact, so its image is closed. (d) $T/\ell^k T$ is finite and discrete, and $\ell^k\colon T \to \ell^k T$ is a continuous bijection of compact Hausdorff spaces.

(iii) A class in $H^1(G,V)$ is 0 if and only if it is $d^0 v$ for some $v \in V$. If $c \in Z^1(G,T)$ and $c = d^0 v$, choose $a$ with $\ell^a v \in T$; then $\ell^a c = d^0(\ell^a v)$ is a coboundary in $C^\bullet_{\mathrm{cont}}(G,T)$, so $[c]$ is torsion. Conversely, if $\ell^a[c] = 0$, then $\ell^a c = d^0 t$ with $t \in T$, and $c = d^0(\ell^{-a} t)$ in $V$. (Torsion in a

$\mathbb{Z}_\ell$-module means killed by a power of $\ell$, since other integers are units.) The rest is the exactness of $H^0(G,V)\to H^0(G,W)\to H^1(G,T)\to H^1(G,V)$. □

**Theorem 16.45** (Tate: inverse limits)**.** *Let $G$ be profinite and $T=\varprojlim_k T_k$ an inverse limit of finite discrete $G$-modules with surjective transition maps, with the inverse-limit topology (for example $T$ of type (C) and $T_k=T/\ell^kT$).*

*(i) For every $n\ge 0$ there is a natural exact sequence $0\to \varprojlim^1_k H^{n-1}(G,T_k)\to H^n(G,T)\to \varprojlim_k H^n(G,T_k)\to 0$.*

*(ii) In particular $H^1(G,T)=\varprojlim_k H^1(G,T_k)$ always (the $H^0(G,T_k)$ are finite, so the Mittag-Leffler condition holds and $\varprojlim^1=0$), and $H^n(G,T)=\varprojlim_k H^n(G,T_k)$ whenever all the $H^{n-1}(G,T_k)$ are finite.*

*(iii) For $T$ of type (C) and $V=T\otimes\mathbb{Q}_\ell$: $H^n(G,V)=H^n(G,T)\otimes_{\mathbb{Z}_\ell}\mathbb{Q}_\ell$ for all $n$.*

*Proof.* (i) is stated; see [196, Section 2], [143, Chapter II, Section 7] and [167, Appendix B]. The idea: a continuous map into $\varprojlim T_k$ is a compatible family of continuous maps into the $T_k$, so $C^\bullet_{\mathrm{cont}}(G,T)=\varprojlim_k C^\bullet(G,T_k)$, and the transition maps of these complexes are surjective (lift values through the finite discrete $T_{k+1}\to T_k$ by a section). For an inverse system of complexes with surjective transition maps, the cohomology of the limit sits in the stated $\varprojlim^1$ sequence [205, Theorem 3.5.8].

(ii) We prove the case $n=1$ directly. *Surjectivity.* Given compatible classes $[c_k]\in H^1(G,T_k)$, choose the cocycles inductively: if $c_{k+1}$ reduces to $c_k+d^0m$ with $m\in T_k$, lift $m$ to $\tilde m\in T_{k+1}$ and replace $c_{k+1}$ by $c_{k+1}-d^0\tilde m$. The compatible family $(c_k)$ is a continuous cocycle $c\colon G\to T$. *Injectivity.* Let $c\in Z^1_{\mathrm{cont}}(G,T)$ have every reduction $c_k$ a coboundary. The set $X_k=\{m\in T_k : c_k=d^0m\}$ is nonempty and finite (a coset of $T_k^G$), and reduction maps $X_{k+1}$ into $X_k$. An inverse limit of nonempty finite sets is nonempty (Chapter 5), so there is $m\in T$ with $c=d^0m$.

(iii) A continuous cochain $G^n\to V$ has compact image, covered by the increasing open sets $\ell^{-a}T$, hence contained in one of them. So $C^\bullet_{\mathrm{cont}}(G,V)=\bigcup_a \ell^{-a}C^\bullet_{\mathrm{cont}}(G,T)=C^\bullet_{\mathrm{cont}}(G,T)\otimes_{\mathbb{Z}_\ell}\mathbb{Q}_\ell$, and cohomology commutes with the exact functor $-\otimes_{\mathbb{Z}_\ell}\mathbb{Q}_\ell$. □

*Warning* 16.46 (pitfalls of continuous cohomology). (i) For $M$ of type (C), $H^n(G,M)$ is *not* $\varinjlim_U H^n(G/U,M^U)$. Example: $H^1(\mathbb{Z}_\ell,\mathbb{Z}_\ell)$ with trivial action is $\mathrm{Hom}_{\mathrm{cont}}(\mathbb{Z}_\ell,\mathbb{Z}_\ell)=\mathbb{Z}_\ell$ (the homomorphisms $a\mapsto\lambda a$), while every $H^1(\mathbb{Z}_\ell/\ell^k\mathbb{Z}_\ell,\mathbb{Z}_\ell)=\mathrm{Hom}(\mathbb{Z}/\ell^k,\mathbb{Z}_\ell)$ is 0, so the direct limit is 0. Equivalently: a continuous homomorphism from a compact group to $\mathbb{Z}_\ell$ with the discrete topology has finite, hence zero, image.

(ii) The long exact sequence needs the continuous section of Proposition 16.44; it exists in all the cases we use.

(iii) The $\varprojlim^1$ terms of Theorem 16.45 can be nonzero in general; they vanish under the finiteness assumptions stated there.

(iv) $H^n(G,V)$ for $V$ of type (V) is a $\mathbb{Q}_\ell$-vector space and sees no torsion; the torsion of $H^1(G,T)$ is the image of $H^0(G,W)$ (Proposition 16.44(iii)).

(v) For a finite group with the discrete topology continuous and abstract cohomology coincide.

(vi) The formula $H^1(\Gamma, M) = M/(F-1)M$ for a procyclic group has precise hypotheses (Warning 16.60): it fails for finite cyclic groups (where it is $\ker N/(F-1)M$) and for discrete modules that are not torsion: $H^1(\widehat{\mathbb{Z}}, \mathbb{Z}) = \mathrm{Hom}_{\mathrm{cont}}(\widehat{\mathbb{Z}}, \mathbb{Z}) = 0$ by the argument in (i), while $\mathbb{Z}/(F-1)\mathbb{Z} = \mathbb{Z}$.

*Remark* 16.47 (topological Nakayama, recalled). We use the following form of Nakayama's lemma, proved in Chapter 3: *let $X$ be a compact (Hausdorff) $\mathbb{Z}_\ell$-module, for instance a compact pro-$\ell$ abelian group; if $X/\ell X$ is finite, then $X$ is a finitely generated $\mathbb{Z}_\ell$-module, and any elements whose images generate $X/\ell X$ generate $X$.* It applies to $H^n(G, T)$ for $T$ of type (C) whenever this group is an inverse limit of finite groups (Theorem 16.45), and is used in Proposition 16.48 and in Chapter 17.

**Proposition 16.48** (finiteness of $H^1$ for type (C))**.** *Let $G$ be profinite and $T$ of type (C).*

*(i)* $H^1(G, T) = \varprojlim_k H^1(G, T/\ell^k T)$. *If the groups $H^1(G, T/\ell^k T)$ are finite, $H^1(G, T)$ is a compact $\mathbb{Z}_\ell$-module, an inverse limit of finite $\ell$-groups.*

*(ii) Suppose $H^1(G, A)$ is finite for every finite $G$-module $A$ of $\ell$-power order. Then $H^1(G, T)$ is a finitely generated $\mathbb{Z}_\ell$-module, $H^1(G, V) = H^1(G, T) \otimes \mathbb{Q}_\ell$ is finite-dimensional for $V = T \otimes \mathbb{Q}_\ell$, and $H^1(G, T)_{\mathrm{tor}}$ is finite.*

*(iii) The hypothesis of (ii) holds for finite groups, for procyclic groups admitting a continuous surjection onto $\mathbb{Z}_\ell$ (Theorem 16.58 gives $H^1(\Gamma, T) = T/(F-1)T$ directly), for the inertia group $I_q$ at a prime $q \neq \ell$ (Proposition 16.53(iv)), and for $G_{\mathbb{Q}_p}$ and $G_{\mathbb{Q},S}$ (Chapter 17).*

*Proof.* (i) The equality is Theorem 16.45(ii), proved there. An inverse limit of finite groups is compact.

(ii) First let $T$ have no $\ell$-torsion. The sequence $0 \to T \xrightarrow{\ell} T \to T/\ell T \to 0$ has a continuous section (Proposition 16.44(d)), and its long exact sequence embeds $H^1(G, T)/\ell H^1(G, T)$ into the finite group $H^1(G, T/\ell T)$. By (i) $H^1(G, T)$ is compact, and a compact $\mathbb{Z}_\ell$-module $A$ with $A/\ell A$ finite is finitely generated (Remark 16.47). In general $T_{\mathrm{tor}}$ is finite and $T/T_{\mathrm{tor}}$ has no $\ell$-torsion; the sequence $0 \to T_{\mathrm{tor}} \to T \to T/T_{\mathrm{tor}} \to 0$ has a continuous section (Proposition 16.44(b)), and its long exact sequence contains $H^1(G, T_{\mathrm{tor}}) \xrightarrow{\alpha} H^1(G, T) \xrightarrow{\beta} H^1(G, T/T_{\mathrm{tor}})$, exact at $H^1(G, T)$. The image of $\alpha$ is a quotient of the finite group $H^1(G, T_{\mathrm{tor}})$ (finite by hypothesis), and $H^1(G, T)/\operatorname{im}\alpha \simeq \operatorname{im}\beta$ is a submodule of the finitely generated module $H^1(G, T/T_{\mathrm{tor}})$ (by the first case), hence finitely generated since $\mathbb{Z}_\ell$ is noetherian. An extension of a finitely generated module by a finite one is finitely generated (lift generators of the quotient and add the finitely many elements of the sub), so $H^1(G, T)$ is finitely generated. The statement about $V$ is Theorem 16.45(iii), and the torsion of a finitely generated $\mathbb{Z}_\ell$-module is finite.

(iii) For a finite group, $H^1(G, A)$ is finite because the cochain groups are. For $I_q$, the proof of Proposition 16.53(iv) applies verbatim to a finite $\ell$-primary $A$, with Theorem 16.51(a) and Theorem 16.56 for $I/K \simeq \mathbb{Z}_\ell$, and gives $H^1(I_q, A) \simeq A^K/(t-1)A^K$, a finite group. The other cases are proved where cited. □

## 16.8 Groups of order prime to $\ell$ and pro-$\ell$ quotients

This section justifies the sentence of the proof of Lemma lem:tamfactor "the kernel of $I \to \langle t \rangle$ has trivial pro-$\ell$ quotient, so by inflation-restriction $H^1(I, M') = H^1(\langle t \rangle, M')$". Two properties

of a closed subgroup $K$ are in play: "order prime to $\ell$", under which all higher cohomology with $\ell$-adic coefficients vanishes, and the weaker "trivial maximal pro-$\ell$ quotient", under which $\mathrm{Hom}_{\mathrm{cont}}(K, M) = 0$ for trivial $\ell$-adic $M$. The paper needs only the weaker one, together with Theorem 16.31(d). Both hold for the kernel of the tame character.

**Definition 16.49** (supernatural orders and pro-$\ell$ groups)**.** A *supernatural number* is a formal product $\prod_p p^{n_p}$ over all primes with $n_p \in \{0, 1, 2, \dots, \infty\}$. The *order* of a profinite group $G$ is $\#G = \mathrm{lcm}_U \#(G/U)$ over the open normal subgroups $U$. $G$ is *pro-$\ell$* if $\#G$ is a power of $\ell$ (equivalently, every finite continuous quotient is an $\ell$-group), and has *order prime to $\ell$* if $\ell \nmid \#G$. The *maximal pro-$\ell$ quotient* of $G$ is $G^{(\ell)} = G/K_\ell$, where $K_\ell$ is the intersection of the open normal subgroups of $\ell$-power index.

**Lemma 16.50.** *Let $G$ be profinite.*

*(i) $G^{(\ell)}$ is pro-$\ell$, and every continuous homomorphism from $G$ to a pro-$\ell$ group factors through $G^{(\ell)}$. If $\ell \nmid \#G$ then $G^{(\ell)} = 1$; the converse fails (for $\ell = 2$, the alternating group $A_4$ has order divisible by 2 but no normal subgroup of index 2 or 4, so $A_4^{(2)} = 1$).*

*(ii) The kernel $K_\ell$ of $G \to G^{(\ell)}$ has $K_\ell^{(\ell)} = 1$.*

*(iii) A closed subgroup of a pro-$\ell$ group is pro-$\ell$. A finitely generated $\mathbb{Z}_\ell$-module is a pro-$\ell$ group, and so is the group $\{\left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right) : a \in \mathbb{Z}_\ell\} \simeq \mathbb{Z}_\ell$ of unipotent upper triangular matrices.*

*(iv) (Stated.) Sylow pro-$\ell$ subgroups exist and are conjugate; see [177, Chapter I, Section 1].*

*Proof.* (i) The open normal subgroups of $\ell$-power index are closed under finite intersection (the quotient by an intersection embeds in a product of $\ell$-groups), so $G/K_\ell$ is the inverse limit of the $\ell$-groups $G/U$; its open normal subgroups are images of open normal subgroups of $G$ containing some such $U$, so its finite quotients are $\ell$-groups. If $f\colon G \to P$ is continuous with $P$ pro-$\ell$, then for every open normal $V \subseteq P$, $f^{-1}(V)$ is open normal in $G$ with $G/f^{-1}(V)$ embedded in the $\ell$-group $P/V$, so $f^{-1}(V) \supseteq K_\ell$; as $\bigcap V = 1$, $f(K_\ell) = 1$. If $\ell \nmid \#G$, the only open normal subgroup of $\ell$-power index is $G$. For $A_4$: its normal subgroups are 1, $V_4$, $A_4$, of index 12, 3, 1.

(ii) Suppose $L_0$ is an open normal subgroup of $K_\ell$ of index $\ell^a$. Since $K_\ell$ is closed, $L_0 \supseteq K_\ell \cap U_0$ for some open normal subgroup $U_0$ of $G$. Every conjugate $gL_0g^{-1}$ is again normal of index $\ell^a$ in $K_\ell$ (as $K_\ell$ is normal in $G$) and contains $K_\ell \cap U_0$, so there are only finitely many, corresponding to subgroups of the finite group $K_\ell/(K_\ell \cap U_0)$. Their intersection $L$ is normal in $G$, and $K_\ell/L$ embeds in a finite product of groups of order $\ell^a$, so it is a finite $\ell$-group. In the profinite group $G/L$ choose an open normal subgroup meeting the finite subgroup $K_\ell/L$ trivially, and let $U$ be its preimage: $U$ is open normal in $G$ and $U \cap K_\ell = L$. Then $\#(G/U) = \#(UK_\ell/U) \cdot \#(G/UK_\ell)$, where $UK_\ell/U \simeq K_\ell/L$ is an $\ell$-group and $G/UK_\ell$ is a finite quotient of the pro-$\ell$ group $G/K_\ell$. So $G/U$ is an $\ell$-group, hence $U \supseteq K_\ell$ by the definition of $K_\ell$, so $L = K_\ell$ and $L_0 = K_\ell$.

(iii) If $H \le P$ is closed, the open normal subgroups $H \cap V$ ($V$ open normal in $P$) form a basis at 1 in $H$, and $H/(H \cap V)$ embeds in $P/V$. A finitely generated $\mathbb{Z}_\ell$-module is $\varprojlim M/\ell^k M$, with finite $\ell$-groups $M/\ell^k M$. The last group is isomorphic to $(\mathbb{Z}_\ell, +)$ by $a \mapsto \left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)$. □

**Theorem 16.51** (vanishing for groups of order prime to $\ell$)**.** *Let $G$ be profinite.*

*(a) If $\ell \nmid \#G$ and $M$ is a discrete $\ell$-primary torsion module, then $H^n(G, M) = 0$ for all $n \ge 1$.*

*(b) If $\ell \nmid \#G$ and $M$ is of type (C) or (V), then $H^1(G, M) = 0$; and $H^n(G, M) = 0$ for all $n \ge 1$.*

*(c) If $G$ acts trivially on $M$ of type (C), (V), or discrete $\ell$-primary, and $G^{(\ell)} = 1$, then $H^1(G, M) = \mathrm{Hom}_{\mathrm{cont}}(G, M) = 0$.*

*(d) If $\ell \nmid \#G$ and $M$ is as in (a) or (b), then $M^G$ is a direct summand of $M$ as a $G$-module.*

*Proof.* (a) By Proposition 16.43(ii), $H^n(G, M) = \varinjlim_U H^n(G/U, M^U)$. Each $\#(G/U)$ is prime to $\ell$ and $M^U$ is $\ell$-primary, so $H^n(G/U, M^U) = 0$ by Theorem 16.29(iii).

(b) For $M$ of type (C), $H^1(G, M) = \varprojlim_k H^1(G, M/\ell^k M) = 0$ by Theorem 16.45(ii) (proved for $n = 1$) and (a). For $n \geq 2$ the same argument uses Theorem 16.45(ii) in degree $n$, whose hypothesis holds because $H^{n-1}(G, M/\ell^k M)$ is finite (for $n = 2$ it is $H^1$, which is 0 by (a)); this part rests on the stated Theorem 16.45(i). For type (V) use Theorem 16.45(iii) with a stable lattice.

(c) A continuous homomorphism $f\colon G \to M$ has compact image $f(G)$. For $M$ of type (C), $f(G)$ is a closed subgroup of $M$, hence pro-$\ell$ by Lemma 16.50(iii); so $f$ factors through $G^{(\ell)} = 1$. For type (V), $f(G)$ lies in some $\ell^{-a}T$ and the same argument applies. For discrete $\ell$-primary $M$, $f(G)$ is a finite abelian $\ell$-group.

(d) For an open normal $U$ acting trivially on a finite or $\ell$-primary module $A$, put $e_U = \#(G/U)^{-1} \sum_{g \in G/U} g$ (the integer $\#(G/U)$ is prime to $\ell$, hence invertible on $A$). If $U' \subseteq U$ also acts trivially, $\sum_{g \in G/U'} g = \#(U/U') \sum_{g \in G/U} g$ on $A$, so $e_{U'} = e_U$: the projector $e$ does not depend on $U$. It is $G$-linear, has image in $A^G$ and is the identity on $A^G$ (Theorem 16.29(iii)). For $M$ of type (D) define $e$ on each element $m$ using any $U$ fixing $m$. For $M$ of type (C) the projectors on the $M/\ell^k M$ are compatible with reduction, and their limit is a projector on $M = \varprojlim M/\ell^k M$. For type (V), $e$ on a stable lattice extends $\mathbb{Q}_\ell$-linearly. In each case $M = M^G \oplus \ker e$. □

**Corollary 16.52** (cohomology through the pro-$\ell$ quotient)**.** *Let $G$ be profinite, $K \trianglelefteq G$ closed, and $M$ of type (D) ($\ell$-primary), (C) or (V), with $K$ acting trivially on $M$.*

*(a) If $K^{(\ell)} = 1$, then $\mathrm{Inf}\colon H^1(G/K, M) \to H^1(G, M)$ is an isomorphism.*

*(b) If $\ell \nmid \#K$, then $\mathrm{Inf}\colon H^n(G/K, M) \to H^n(G, M)$ is an isomorphism for every $n$ (for $M$ of type (D)).*

*(c) If the image of $G$ in the automorphism group of $M$ is pro-$\ell$ (for example, unipotent upper triangular in a basis of $M \simeq \mathbb{Z}_\ell^2$), the action factors through $G^{(\ell)}$, and $H^1(G, M) = H^1(G^{(\ell)}, M)$ by inflation.*

*Proof.* (a) By Theorem 16.51(c), $H^1(K, M) = \mathrm{Hom}_{\mathrm{cont}}(K, M) = 0$; apply Theorem 16.31(d) and Remark 16.32. (b) By Theorem 16.51(a), $H^q(K, M) = 0$ for $q \geq 1$; apply Theorem 16.33. This is stated, since the spectral sequence is. (c) The action is a continuous homomorphism to a pro-$\ell$ group, so it factors through $G^{(\ell)}$ by Lemma 16.50(i); then (a) applies with $K = K_\ell$, by Lemma 16.50(ii). □

**Proposition 16.53** (the inertia group at $q \neq \ell$)**.** *Let $q \neq \ell$ be primes, $I = I_q \subset G_{\mathbb{Q}_q}$ the inertia group, and $t_\ell\colon I \to \mathbb{Z}_\ell(1)$ the tame character of Chapter 7, with kernel $K$. Imported from Chapter 7: $t_\ell$ is surjective, it identifies $I^{(\ell)}$ with $\mathbb{Z}_\ell(1)$, and every finite quotient of $K$ has order prime to $\ell$. Let $t \in I$ map to a topological generator of $\mathbb{Z}_\ell(1)$. Then:*

*(i) $\ell \nmid \#K$, so $K^{(\ell)} = 1$, and $I/K \simeq \mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$ is procyclic, generated by the image of $t$.*

*(ii) For every $M$ of type (C) or (V) on which $I$ acts through a pro-$\ell$ group (for instance through unipotent matrices), $K$ acts trivially, and $c \mapsto c(t)$ induces*

$$H^1(I,M) \;\simeq\; H^1(I/K,M) \;\simeq\; M/(t-1)M.$$

*(iii) (Given Theorem 16.33 and Theorem 16.45(i), both stated.) $H^n(I,M)=0$ for $n\geq 2$, for $M$ as in (ii) of type (C).*

*(iv) For* every *$M$ of type (C) (resp. (V)), with no hypothesis on the image of $I$: $H^1(K,M)=0$, $M^K$ is of type (C) (resp. (V)), and*

$$\mathrm{Inf}\colon H^1(I/K,M^K)\xrightarrow{\sim} H^1(I,M), \qquad H^1(I,M)\simeq M^K/(t-1)M^K \quad \textit{via } [c]\mapsto c(t).$$

*In particular $H^1(I,M)$ is a finitely generated $\mathbb{Z}_\ell$-module (resp. a finite-dimensional $\mathbb{Q}_\ell$-space), and $H^1(I,M)_{\mathrm{tor}}$ is finite.*

*Proof.* (i) is the imported statement. (ii) The image of $K$ in the pro-$\ell$ image of $I$ is a quotient of $K$, so its finite quotients have order prime to $\ell$ and are $\ell$-groups: it is trivial. So $K$ acts trivially, Corollary 16.52(a) gives the first isomorphism, and Theorem 16.58 applied to $\Gamma = I/K \simeq \mathbb{Z}_\ell$ gives the second. Composing, a cocycle $c$ on $I$ that is inflated from $I/K$ is sent to $c(t)$; an arbitrary cocycle is cohomologous to an inflated one, and changing $c$ by $d^0 m$ changes $c(t)$ by $(t-1)m$, so the map $[c]\mapsto c(t) \bmod (t-1)M$ is defined on all of $H^1(I,M)$ by the same formula. (iii) By Theorem 16.51(a), $H^q(K, M/\ell^k M)=0$ for $q\geq 1$, so by Theorem 16.33 $H^n(I,M/\ell^k M) = H^n(I/K, M/\ell^k M)$, which is 0 for $n\geq 2$ (Theorem 16.56). Passing to the limit with Theorem 16.45 (the $H^1(I,M/\ell^k M)$ being finite) gives $H^n(I,M)=0$.

(iv) $H^1(K,M)=0$ by Theorem 16.51(b) (its proof for $H^1$ uses only Theorem 16.45(ii) in degree 1, which is proved). $M^K$ is closed in $M$ (Remark 16.32), so it is a $\mathbb{Z}_\ell$-submodule of type (C) (resp. a subspace, of type (V)), on which $I/K$ acts continuously. By Theorem 16.31(d), valid for continuous cochains by Remark 16.32, Inf is an isomorphism. By (i), $I/K\simeq\mathbb{Z}_\ell$ is procyclic, generated by the image of $t$, so Theorem 16.58 gives $H^1(I/K,M^K)\simeq M^K/(t-1)M^K$, $[c]\mapsto c(t)$, and the finiteness statements of its last paragraph. The composite is $[c]\mapsto c(t)$ as in (ii). □

## 16.9 Procyclic groups

The identity the paper uses, "$H^1(I,M')=H^1(\langle t\rangle,M')=M'/(t-1)M'$ for every $\mathbb{Z}_\ell$-module $M'$ of finite type with this action" (proof of Lemma lem:tamfactor), is Theorem 16.58. The same theorem, for the procyclic group $G_{\mathbb{Q}_p}/I_p\simeq\widehat{\mathbb{Z}}$, gives $H^1(G/I,\mathbb{Z}_\ell(1)^I)=\mathbb{Z}_\ell(1)/(q-1)$ (Lemma lem:tamfactor) and $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V^D)=(V^D)^{I_p}/(\mathrm{Frob}_p-1)$ (after Lemma lem:dfg21). The formula is false for finite cyclic groups, and it needs a different proof for discrete and for compact coefficients; we give both, with exact hypotheses.

**Lemma 16.54** (procyclic groups)**.** *A profinite group $\Gamma$ is* procyclic *if it is topologically generated by one element $F$ (the subgroup $\langle F\rangle$ is dense).*

*(i) $\Gamma$ is abelian, and $k\mapsto F^k$ extends to a continuous surjection $\widehat{\mathbb{Z}}\to\Gamma$, where $\widehat{\mathbb{Z}}=\varprojlim_n \mathbb{Z}/n\simeq\prod_p\mathbb{Z}_p$. So the procyclic groups are the quotients of $\widehat{\mathbb{Z}}$, the groups $\prod_p \mathbb{Z}/p^{a_p}$ with $a_p\in\{0,1,\dots,\infty\}$ and $\mathbb{Z}/p^\infty:=\mathbb{Z}_p$. Examples: $\widehat{\mathbb{Z}}=\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ with generator the Frobenius $x\mapsto x^q$; $\mathbb{Z}_\ell$; $\mathbb{Z}_\ell(1)$; the tame inertia group $I_q/P_q\simeq\prod_{p\neq q}\mathbb{Z}_p(1)$.*

*(ii) If $\pi\colon \Gamma \to \mathbb{Z}_\ell$ is a continuous surjection, then $\pi(F) \in \mathbb{Z}_\ell^\times$, and $k \mapsto F^k$ is injective on $\mathbb{Z}$. Replacing $\pi$ by $\pi(F)^{-1}\pi$ we may and do assume $\pi(F) = 1$; then $\pi(F^k) = k$, and for $a \in \mathbb{Z}_\ell$ we write $F^a$ for any element with $\pi(F^a) = a$ when $\pi$ is an isomorphism.*

*(iii) Let $M$ be of type (C). A continuous action of $\Gamma$ on $M$ is determined by $u = F|_M \in \mathrm{Aut}_{\mathbb{Z}_\ell}(M)$. For $\Gamma = \widehat{\mathbb{Z}}$ every $u$ arises. For $\Gamma = \mathbb{Z}_\ell$ exactly the $u$ with $u^{\ell^N} \to 1$ in $\mathrm{Aut}(M)$ as $N \to \infty$ arise; every unipotent $u$ (that is, $u - 1$ nilpotent) is one of them.*

*Proof.* (i) $\langle F\rangle$ is abelian and dense, so $\Gamma$ is abelian. For each open normal $U$, $\Gamma/U$ is a finite cyclic group generated by the image of $F$, of order $n_U$; the maps $\mathbb{Z}/n_U \to \Gamma/U$, $1 \mapsto F$, are compatible and give a continuous $\widehat{\mathbb{Z}} \to \varprojlim \Gamma/U = \Gamma$, whose image is compact and dense, hence everything. A quotient of $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ by a closed subgroup is the product of the quotients of the $\mathbb{Z}_p$, whose closed subgroups are $0$ and $p^a\mathbb{Z}_p$.

(ii) $\pi(\Gamma)$ is the closure of $\pi(F)\mathbb{Z}$, which is $\pi(F)\mathbb{Z}_\ell$; it is $\mathbb{Z}_\ell$ only if $\pi(F)$ is a unit. Then $\pi(F^k) = k\pi(F) \neq 0$ for $k \neq 0$.

(iii) $u$ determines the action on the dense subgroup $\langle F\rangle$, hence everywhere. $\mathrm{Aut}_{\mathbb{Z}_\ell}(M) = \varprojlim \mathrm{Aut}(M/\ell^k M)$ is profinite, so the closure of $\langle u\rangle$ is procyclic and $1 \mapsto u$ extends to $\widehat{\mathbb{Z}}$ by (i). If $\mathbb{Z}_\ell$ acts continuously then $u^{\ell^N} = \rho(\ell^N) \to \rho(0) = 1$. Conversely, if $u^{\ell^N} \to 1$, the map $k \mapsto u^k$ on $\mathbb{Z}$ is uniformly continuous for the $\ell$-adic topology (if $u^{\ell^N}$ acts trivially on $M/\ell^r M$, so does every power), and it extends to $\mathbb{Z}_\ell$ by completeness. If $u = 1 + X$ with $X^r = 0$, then $u^{\ell^N} = \sum_{i<r} \binom{\ell^N}{i} X^i$ and $v_\ell\binom{\ell^N}{i} = N - v_\ell(i) \to \infty$ for $1 \le i < r$. □

**Lemma 16.55** (cocycles on powers)**.** *Let $\Gamma$ be procyclic with generator $F$, $M$ a Hausdorff topological $\Gamma$-module and $c$ a continuous 1-cocycle. Put $S_k = 1 + F + \cdots + F^{k-1}$.*

*(i) $c(F^k) = S_k\, c(F)$ for $k \ge 0$, and $c(F^{-k}) = -F^{-k}c(F^k)$. So $c$ is determined by $c(F)$.*

*(ii) Suppose $\pi\colon \Gamma \to \mathbb{Z}_\ell$ is a continuous surjection with $\pi(F) = 1$, and $M$ is of type (C), (V), or finite of $\ell$-power order. Then $(F-1)M$ is closed, and for every $\gamma \in \Gamma$*

$$c(\gamma) \equiv \pi(\gamma)\, c(F) \pmod{(F-1)M},$$

*where $\pi(\gamma) \in \mathbb{Z}_\ell$ acts on the $\mathbb{Z}_\ell$-module $M/(F-1)M$. In particular $c(F^a) \equiv a\, c(F)$ modulo $(F-1)M$ for every $a \in \mathbb{Z}_\ell$ when $\pi$ is an isomorphism.*

*Proof.* (i) By induction, $c(F^{k+1}) = c(F^k) + F^k c(F)$; and $c(F^{-k}) = -F^{-k}c(F^k)$ by Proposition 16.18(ii). By continuity $c$ is determined on the closure of $\langle F\rangle$, which is $\Gamma$.

(ii) $(F-1)M$ is the image of a $\mathbb{Z}_\ell$-linear endomorphism: a finitely generated, hence compact, submodule for type (C), a subspace for type (V). So $M/(F-1)M$ is Hausdorff, and both sides of the congruence are continuous functions of $\gamma$ into it. On $M/(F-1)M$ the element $F$ acts as 1, so $S_k \equiv k$: explicitly $S_k - k = \sum_{i=0}^{k-1}(F^i - 1)$ and each $F^i - 1 = (F-1)(1 + F + \cdots + F^{i-1})$ maps $M$ into $(F-1)M$. By (i) the two sides agree at $\gamma = F^k$ for all $k \ge 0$, and for $k < 0$ too: $c(F^k) = -F^k c(F^{-k}) \equiv -(-k)c(F) = k\, c(F)$, since $F^k$ also acts as 1 on $M/(F-1)M$. The $F^k$, $k \in \mathbb{Z}$, form a dense subset of $\Gamma$, so the two sides agree everywhere. □

The congruence holds only modulo $(F-1)M$, and the paper needs no more. For $q^{-1} \in \mathbb{Z}_\ell$, which is not an integer, there is no exact formula for $c(F^{q^{-1}})$ as a finite sum; what (ii) says is that its class in $M/(F-1)M$ is $q^{-1}$ times that of $c(F)$, as if $F$ were 1. For instance, for $\ell = 5$ and $q = 11$ one has $q^{-1} \equiv 91 \pmod{125}$, so $c(F^{q^{-1}}) \equiv 91\, c(F)$ modulo $(F-1)M + 125M$.

**Theorem 16.56** (Serre: procyclic groups, discrete coefficients)**.** *Let either $\Gamma = \widehat{\mathbb{Z}}$ and $M$ a discrete torsion $\Gamma$-module, or $\Gamma = \mathbb{Z}_\ell$ and $M$ a discrete $\ell$-primary torsion $\Gamma$-module; let $F$ be the generator* 1*. Then $H^0(\Gamma, M) = \ker(F-1)$, and $c \mapsto c(F)$ induces an isomorphism*

$$H^1(\Gamma, M) \;\simeq\; M/(F-1)M.$$

*If $M$ is finite, $\#H^1(\Gamma, M) = \#H^0(\Gamma, M)$. Moreover $H^n(\Gamma, M) = 0$ for $n \geq 2$.*

*Proof.* $H^0$: an element fixed by $F$ is fixed by the closure of $\langle F \rangle$, since stabilisers are closed.

Injectivity of $c \mapsto c(F)$ modulo coboundaries. If $c(F) = (F-1)m$, then $c - d^0 m$ vanishes at $F$, hence on $\langle F \rangle$ by Lemma 16.55(i), hence everywhere, being continuous into a discrete (Hausdorff) space. Coboundaries map onto $(F-1)M$.

Surjectivity. Let $m \in M$. Its stabiliser is open, so it contains $n\Gamma$ for some $n \geq 1$ (an $\ell$-power if $\Gamma = \mathbb{Z}_\ell$): $F^n m = m$. Let $e$ be the order of $m$ (an $\ell$-power in the second case). Then, with $S_k$ as in Lemma 16.55,

$$S_{ne}\, m = S_n(1 + F^n + F^{2n} + \cdots + F^{(e-1)n})m = S_n(em) = 0,$$

using the factorisation $S_{ne}(F) = S_n(F)\, S_e(F^n)$ (group the $ne$ powers of $F$ into $e$ blocks of length $n$) and $F^{jn}m = m$. Define $c(F^k) = S_k m$ for $k \geq 0$. Since $S_{k+j} = S_k + F^k S_j$, $c(F^{k+ne}) = S_k m + F^k S_{ne} m = c(F^k)$, so $c$ is a well-defined function on $\mathbb{Z}/ne$, a quotient of $\Gamma$ (the quotient of $\widehat{\mathbb{Z}}$ or of $\mathbb{Z}_\ell$ by $ne$; in the second case $ne$ is a power of $\ell$); composed with $\Gamma \to \mathbb{Z}/ne$ it is continuous, since $\mathbb{Z}/ne$ is finite and discrete. The cocycle identity holds on $\mathbb{Z}/ne$, by the identity for $S$, hence on $\Gamma$. And $c(F) = m$.

If $M$ is finite, the exact sequence $0 \to M^{F=1} \to M \xrightarrow{F-1} M \to M/(F-1)M \to 0$ gives $\#H^0 = \#H^1$. The vanishing for $n \geq 2$ is stated; see [174, Chapter XIII, Section 1] and [177, Chapter I, Sections 3 and 4] (the groups $\widehat{\mathbb{Z}}$ and $\mathbb{Z}_\ell$ have cohomological dimension 1). □

The compact case needs one more ingredient: geometric sums of an automorphism of a finitely generated $\mathbb{Z}_\ell$-module become $\ell$-adically small along suitable arithmetic progressions.

**Lemma 16.57** (geometric sums)**.** *Let $M$ be a finitely generated $\mathbb{Z}_\ell$-module, $u \in \mathrm{Aut}_{\mathbb{Z}_\ell}(M)$, and $a_0$ the order of the image of $u$ in the finite group $\mathrm{Aut}(M/\ell M)$. For $j \geq 1$ put $S_j(u) = 1+u+\cdots+u^{j-1}$. Then $S_j(u)M \subseteq \ell^N M$ for every $N \geq 0$ and every $j$ divisible by $a_0 \ell^N$.*

*Proof.* We use $S_{ab}(x) = S_a(x)\, S_b(x^a)$, which holds because both sides are $\sum_{i<a,\, j<b} x^{i+aj}$. First, if $v = 1+Y$ with $YM \subseteq \ell M$, then $(1+Y)^i = 1+Y\cdot(\text{a polynomial in } Y)$, so $S_\ell(v) = \sum_{i=0}^{\ell-1}(1+Y)^i = \ell + YP(Y)$ for a polynomial $P$ with integer coefficients. Hence $S_\ell(v)$ maps $\ell^i M$ into $\ell^{i+1}M$ for every $i$: $\ell \cdot \ell^i M = \ell^{i+1}M$, and $YP(Y)\ell^i M = \ell^i YP(Y)M \subseteq \ell^i YM \subseteq \ell^{i+1}M$, since $Y$ is $\mathbb{Z}_\ell$-linear. Also $v^\ell = 1 + Y'$ with $Y' = (1+Y)^\ell - 1 = Y \cdot (\ell + \dots)$, so again $Y'M \subseteq \ell M$. Now $w = u^{a_0}$ satisfies $(w-1)M \subseteq \ell M$, because $u^{a_0}$ is trivial on $M/\ell M$ by the definition of $a_0$; so

$$S_{\ell^N}(w) = S_\ell(w)\, S_\ell(w^\ell)\, S_\ell(w^{\ell^2}) \cdots S_\ell(w^{\ell^{N-1}})$$

maps $M$ into $\ell^N M$ (each of the $N$ factors $S_\ell(w^{\ell^i})$ raises the level by one, by the first step applied to $v = w^{\ell^i}$). Hence so does $S_{a_0\ell^N}(u) = S_{a_0}(u) S_{\ell^N}(w)$, and so does $S_{a_0\ell^N j'}(u) = S_{a_0\ell^N}(u)\, S_{j'}(u^{a_0\ell^N})$ for every $j' \geq 1$ (all these operators are polynomials in $u$ and commute). □

For $M=\mathbb{Z}_\ell$ the lemma is a statement about numbers: $u\in\mathbb{Z}_\ell^\times$, and for $u\neq 1$, $S_j(u)=(u^j-1)/(u-1)$. Take $\ell=5$ and $u=2$. The order of 2 modulo 5 is $a_0=4$, and the lemma says $5^N \mid (2^j-1)/(2-1)$ whenever $4\cdot 5^N \mid j$. In fact $v_5(2^{4\cdot 5^N}-1)=N+1$ for $N=0,1,2,3$ (computed with PARI/GP), one more than the lemma guarantees, because $v_5(2^4-1)=v_5(15)=1$ already. The lemma needs $a_0$: for $j=5^N$ alone, $2^{5^N}\equiv 2 \pmod 5$, so $2^{5^N}-1$ is prime to 5.

**Theorem 16.58** (procyclic groups, compact coefficients)**.** *Let $\Gamma$ be a procyclic group admitting a continuous surjection $\pi$ onto $\mathbb{Z}_\ell$ (in $\Gamma\simeq\prod_p \mathbb{Z}/p^{a_p}$: $a_\ell=\infty$; examples $\mathbb{Z}_\ell$, $\mathbb{Z}_\ell(1)$, $\widehat{\mathbb{Z}}$, and $I_q/P_q$ for $q\neq\ell$), $F$ a topological generator, and $M$ of type (C) or (V). Then:*

*(i)* $H^0(\Gamma,M)=\ker(F-1)$.

*(ii) Every $m\in M$ is $c(F)$ for a unique continuous 1-cocycle $c$, and $c\mapsto c(F)$ induces an isomorphism of $\mathbb{Z}_\ell$-modules*

$$H^1(\Gamma,M)\ \simeq\ M/(F-1)M.$$

*(iii)* $H^n(\Gamma,M)=0$ *for $n\geq 2$.*

*Consequently $H^1(\Gamma,M)$ is a finitely generated $\mathbb{Z}_\ell$-module (resp. a finite-dimensional $\mathbb{Q}_\ell$-vector space), $H^1(\Gamma,M)_{\mathrm{tor}}=(M/(F-1)M)_{\mathrm{tor}}$ is finite, and for $V$ of type (V)*

$$H^1(\Gamma,V)=V/(F-1)V=0 \iff 1 \text{ is not an eigenvalue of } F \text{ on } V \iff \det(1-F\mid V)\neq 0.$$

*Proof.* (i) As in Theorem 16.56: the set of elements fixed by $F$ is fixed by the closure of $\langle F\rangle$, as $M$ is Hausdorff.

(ii), injectivity and uniqueness. The map $Z^1_{\mathrm{cont}}(\Gamma,M)\to M$, $c\mapsto c(F)$, is $\mathbb{Z}_\ell$-linear, and it is injective by Lemma 16.55(i): a continuous cocycle vanishing at $F$ vanishes on the dense subgroup $\langle F\rangle$, hence everywhere. Coboundaries $d^0m$ map onto $(F-1)M$. So we get an injection $H^1(\Gamma,M)\to M/(F-1)M$, and it remains to show that every $m\in M$ is $c(F)$ for some continuous cocycle $c$.

(ii), surjectivity, type (C). The idea: on the dense subgroup $\langle F\rangle\simeq\mathbb{Z}$ there is only one candidate, $F^k\mapsto S_k(u)m$, and the work is to show that it is uniformly continuous for the topology induced from $\Gamma$, so that it extends to $\Gamma$. Lemma 16.57 is exactly the needed estimate: $F^j$ close to 1 forces $S_j(u)m$ close to 0. Normalise $\pi(F)=1$ (Lemma 16.54(ii)); then $k\mapsto F^k$ is injective, and we may define $c_0\colon\langle F\rangle\to M$ by $c_0(F^k)=S_k(u)m$ for $k\geq 0$ and $c_0(F^{-k})=-F^{-k}c_0(F^k)$, where $u=F|_M$. The identity $S_{k+j}=S_k+u^kS_j$ (for all $k,j\in\mathbb{Z}$, with $S_{-k}=-u^{-k}S_k$) says that $c_0$ is a cocycle on $\langle F\rangle$. Let $a_0=m_0\ell^s$ ($\ell\nmid m_0$) be as in Lemma 16.57 and, for $N\geq 1$,

$$U_N=\ker\bigl(\Gamma\to\mathrm{Aut}(M/\ell^NM)\bigr)\cap\pi^{-1}(\ell^{s+N}\mathbb{Z}_\ell),$$

an open subgroup of $\Gamma$ (Proposition 16.43(iii)), decreasing in $N$. If $F^j\in U_N$, then $u^j$ is trivial on $M/\ell M$, so $a_0\mid j$, and $\ell^{s+N}\mid\pi(F^j)=j$; so $a_0\ell^N\mid j$ and $c_0(F^j)\in\ell^NM$ by Lemma 16.57 (for $j<0$ use $c_0(F^j)=-F^jc_0(F^{-j})$ and $u(\ell^NM)=\ell^NM$). Hence, for $F^k,F^{k'}$ in the same coset of $U_N$,

$$c_0(F^{k'})-c_0(F^k)=F^k\,c_0(F^{k'-k})\in\ell^NM.$$

Now define $c\colon\Gamma\to M$: for $\gamma\in\Gamma$ and $N\geq 1$ the coset $\gamma U_N$ meets the dense subgroup $\langle F\rangle$; let $c(\gamma)$ mod $\ell^NM$ be $c_0(F^k)$ mod $\ell^NM$ for any $F^k\in\gamma U_N$. This is independent of the choice by the display, compatible as $N$ grows because $U_{N+1}\subseteq U_N$, and so defines $c(\gamma)\in\varprojlim_N M/\ell^NM=M$.

It extends $c_0$, and $c(\gamma U_N) \subseteq c(\gamma) + \ell^N M$, so $c$ is continuous. For the cocycle identity, given $\gamma, \delta$ and $N$, choose $F^k \in \gamma U_N$, $F^{k'} \in \delta U_N$; then $F^{k+k'} \in \gamma\delta U_N$ and

$$c(\gamma\delta) \equiv c_0(F^{k+k'}) = c_0(F^k) + F^k c_0(F^{k'}) \equiv c(\gamma) + \gamma\, c(\delta) \pmod{\ell^N M},$$

because $F^k\gamma^{-1} \in U_N$ acts trivially on $M/\ell^N M$. As $N$ is arbitrary, $c$ is a continuous cocycle with $c(F) = m$.

(ii), type (V). Let $T$ be a $\Gamma$-stable lattice (Proposition 16.43(iv)) and $m \in V$; choose $a$ with $\ell^a m \in T$, let $c_T$ be the cocycle with $c_T(F) = \ell^a m$ just constructed, and put $c = \ell^{-a} c_T$.

(iii) is stated; see [177, Chapter I, Sections 3 and 4] ($\operatorname{cd}_\ell \Gamma \le 1$) and [196, Section 2]; it follows from Theorem 16.56 and Theorem 16.45 when $\Gamma$ is $\widehat{\mathbb{Z}}$ or $\mathbb{Z}_\ell$. For a general $\Gamma \simeq \prod_p \mathbb{Z}/p^{a_p}$ with $a_\ell = \infty$, first remove the prime-to-$\ell$ part: $\Gamma = \Gamma_\ell \times \Gamma'$ with $\Gamma_\ell \simeq \mathbb{Z}_\ell$ and $\ell \nmid \#\Gamma'$, so $H^q(\Gamma', A) = 0$ for $q \ge 1$ and finite $\ell$-primary $A$ (Theorem 16.51(a)), and Theorem 16.33 gives $H^n(\Gamma, A) = H^n(\Gamma_\ell, A^{\Gamma'})$, which vanishes for $n \ge 2$ by Theorem 16.56; then pass to the limit over $A = M/\ell^k M$ with Theorem 16.45.

The consequences: $M/(F-1)M$ is a quotient of a finitely generated module; for $V$, $F - 1$ is surjective if and only if it is injective, if and only if $\det(F-1) = \pm\det(1-F) \neq 0$. □

**Example 16.59.** Let $\Gamma$ and $F$ be as in Theorem 16.58.

(i) Trivial action on $M$ of type (C): $H^1(\mathbb{Z}_\ell, M) = \operatorname{Hom}_{\text{cont}}(\mathbb{Z}_\ell, M) = M$, the homomorphism with value $m$ at 1 being $a \mapsto am$; and $H^1(\widehat{\mathbb{Z}}, M) = M$, every continuous homomorphism $\widehat{\mathbb{Z}} \to M$ factoring through $\widehat{\mathbb{Z}} \to \mathbb{Z}_\ell$.

(ii) $\Gamma = \widehat{\mathbb{Z}} = \operatorname{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ acting on $\mathbb{Z}_\ell(1)$, $q \neq \ell$, so that $F$ acts by $q$. Then $H^0 = 0$ (as $q \neq 1$) and $H^1(\widehat{\mathbb{Z}}, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell/(q-1)\mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^{v_\ell(q-1)}$. On $\mu_{\ell^k}$, by Theorem 16.56, $H^0 = \{\zeta : \zeta^{q-1} = 1\}$ and $H^1 = \mu_{\ell^k}/\mu_{\ell^k}^{q-1}$ both have order $\ell^{\min(k, v_\ell(q-1))}$. This is the group $H^1(G/I, \mathbb{Z}_\ell(1)^I) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ of Lemma lem:tamfactor, for $G = G_{\mathbb{Q}_q}$, $I = I_q$ (on which $\chi_\ell$ is trivial).

(iii) $M = \mathbb{Z}_\ell e_1 \oplus \mathbb{Z}_\ell e_2$ with $F = \left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$, $n' \neq 0$: $(F-1)M = n'\mathbb{Z}_\ell e_1$, so $H^1 = (\mathbb{Z}_\ell/n')\bar{e}_1 \oplus \mathbb{Z}_\ell \bar{e}_2$, with torsion $\mathbb{Z}/\ell^{v_\ell(n')}$.

(iv) $\Gamma = \widehat{\mathbb{Z}}$ and $F$ acting on $M$ of type (C) through a finite quotient of order $m$ prime to $\ell$: then $M = M^F \oplus (F-1)M$ (the projector is $m^{-1}(1 + F + \cdots + F^{m-1})$, whose kernel is $(F-1)M$), and $H^1(\widehat{\mathbb{Z}}, M) = M/(F-1)M \simeq M^F$.

(v) $V$ of type (V): $H^1(\Gamma, V) = 0$ if and only if $\det(1 - F \mid V) \neq 0$.

*Verification.* (i) and (v) are special cases of Theorem 16.58. (ii) $(F-1)\mathbb{Z}_\ell(1) = (q-1)\mathbb{Z}_\ell(1)$, and $\mathbb{Z}_\ell/(q-1)$ has length $v_\ell(q-1)$; the finite case is the order of the kernel and cokernel of $\zeta \mapsto \zeta^{q-1}$ on the cyclic group $\mu_{\ell^k}$. (iii) $(F-1)e_1 = 0$, $(F-1)e_2 = n'e_1$. (iv) $N = 1 + F + \cdots + F^{m-1}$ satisfies $N(F-1) = 0$ and $N \equiv m$ modulo $F - 1$; so $e = m^{-1}N$ is a projector with image $M^F$, and $1 - e = m^{-1}(m - N)$ is divisible by $F - 1$, which shows $\ker e = (F-1)M$. □

*Warning* 16.60 (the hypotheses of the procyclic formula). The identity $H^1(\Gamma, M) = M/(F-1)M$ holds: for $\Gamma$ procyclic with a surjection onto $\mathbb{Z}_\ell$ and $M$ of type (C) or (V); for $\Gamma = \widehat{\mathbb{Z}}$ and $M$ discrete torsion; for $\Gamma = \mathbb{Z}_\ell$ and $M$ discrete $\ell$-primary. It fails in each of the following situations.

(i) $\Gamma$ finite cyclic: $H^1 = \ker N/(F-1)M$ (Theorem 16.34); with trivial action on $\mathbb{Z}_\ell$ this is 0, not $\mathbb{Z}_\ell$.

(ii) $\Gamma = \mathbb{Z}_p$ with $p \neq \ell$, acting on a $\mathbb{Z}_\ell$-module of type (C): $\Gamma$ has order prime to $\ell$, so $H^1(\Gamma, M) = 0$ by Theorem 16.51, while $M/(F-1)M$ can be nonzero (Exercise 16.94).

(iii) Discrete modules that are not torsion: $H^1(\widehat{\mathbb{Z}}, \mathbb{Z}) = 0$ (Warning 16.46(vi)).

The hypothesis of the paper, "every $\mathbb{Z}_\ell$-module $M'$ of finite type with this action", is the first case: $\Gamma = I/K \simeq \mathbb{Z}_\ell$, $M'$ of type (C).

The procyclic computations also give the five-term inflation-restriction sequence for compact and $\mathbb{Q}_\ell$-coefficients in the one case the book needs, $G_{\mathbb{Q}_p} \to G_{\mathbb{Q}_p}/I_p \simeq \widehat{\mathbb{Z}}$: Theorem 16.31(b) was stated only for discrete coefficients. The proof uses that $\widehat{\mathbb{Z}}$ is free on one generator, so that $G \to G/H$ splits.

**Proposition 16.61** (inflation-restriction over $\widehat{\mathbb{Z}}$)**.** *Let $G$ be profinite, $H \trianglelefteq G$ closed with $G/H \simeq \widehat{\mathbb{Z}}$, and $M$ of type (D) (discrete torsion), (C) or (V).*

*(i)* $\mathrm{Res}\colon H^1(G, M) \to H^1(H, M)^{G/H}$ *is surjective, so*

$$0 \to H^1(G/H, M^H) \xrightarrow{\mathrm{Inf}} H^1(G, M) \xrightarrow{\mathrm{Res}} H^1(H, M)^{G/H} \to 0$$

*is exact.*

*(ii) (Given Theorem 16.56 and Theorem 16.58(iii) for $n = 2$, stated.) $H^2(G/H, M^H) = 0$, so the five-term sequence of Theorem 16.31(b) holds for $M$, with* $\mathrm{tg} = 0$.

*Proof.* (i) Exactness on the left and in the middle is Theorem 16.31(a) with Remark 16.32. *A splitting.* Choose $F \in G$ mapping to the generator 1 of $G/H = \widehat{\mathbb{Z}}$, and let $\Gamma_0$ be the closure of $\langle F \rangle$. By Lemma 16.54(i) there is a continuous surjection $\widehat{\mathbb{Z}} \to \Gamma_0$, $1 \mapsto F$; composed with $\Gamma_0 \to G/H$ it is a continuous endomorphism of $\widehat{\mathbb{Z}}$ fixing 1, hence the identity (it fixes the dense subgroup $\mathbb{Z}$). So $\Gamma_0 \to G/H$ is an isomorphism, $H \cap \Gamma_0 = 1$, $G = H\Gamma_0$, and $(h, \gamma) \mapsto h\gamma$ is a continuous bijection $H \times \Gamma_0 \to G$ of compact Hausdorff spaces, hence a homeomorphism.

*Extending a cocycle.* Let $c \in Z^1(H, M)$ have $G/H$-invariant class. Then $F \cdot c - c = d^0 m$ for some $m \in M$, that is $Fc(F^{-1}hF) - c(h) = hm - m$ for $h \in H$; replacing $h$ by $FhF^{-1}$,

$$\gamma\, c(h) - c(\gamma h \gamma^{-1}) = (\gamma h \gamma^{-1} - 1)\, \tilde{c}(\gamma) \qquad (h \in H) \tag{$*$}$$

holds for $\gamma = F$ and $\tilde{c}(F) = m$. Let $\tilde{c}\colon \Gamma_0 \to M$ be the continuous cocycle with $\tilde{c}(F) = m$: it exists by Theorem 16.58(ii) for types (C), (V) and by Theorem 16.56 for type (D), since $\Gamma_0 \simeq \widehat{\mathbb{Z}}$. The set of $\gamma \in \Gamma_0$ satisfying $(*)$ for all $h$ is closed (both sides are continuous in $\gamma$, and $M$ is Hausdorff) and stable under products: if $(*)$ holds for $\gamma$ and $\delta$, then

$$\gamma\delta\, c(h) - c(\gamma\delta h \delta^{-1}\gamma^{-1}) = \gamma(\delta h \delta^{-1} - 1)\tilde{c}(\delta) + (\gamma\delta h \delta^{-1}\gamma^{-1} - 1)\tilde{c}(\gamma) = (\gamma\delta h \delta^{-1}\gamma^{-1} - 1)\big(\tilde{c}(\gamma) + \gamma\tilde{c}(\delta)\big),$$

which is $(*)$ for $\gamma\delta$ since $\tilde{c}(\gamma\delta) = \tilde{c}(\gamma) + \gamma\tilde{c}(\delta)$. It contains $F$, hence every $F^k$, $k \geq 1$, hence their closure, which is $\Gamma_0$ ($F^{-1}$ is the limit of $F^{k!-1}$, since $F^{k!} \to 1$ in the profinite group $\Gamma_0$). Now define $c^\sharp(h\gamma) = c(h) + h\tilde{c}(\gamma)$ for $h \in H$, $\gamma \in \Gamma_0$. It is continuous by the homeomorphism above, and it restricts to $c$ on $H$. For the cocycle identity, write $h_1\gamma_1 h_2 \gamma_2 = h_1(\gamma_1 h_2 \gamma_1^{-1}) \cdot \gamma_1\gamma_2$; then

$$c^\sharp(h_1\gamma_1 h_2\gamma_2) - c^\sharp(h_1\gamma_1) - h_1\gamma_1 c^\sharp(h_2\gamma_2) = h_1\Big(c(\gamma_1 h_2 \gamma_1^{-1}) + (\gamma_1 h_2 \gamma_1^{-1} - 1)\tilde{c}(\gamma_1) - \gamma_1 c(h_2)\Big),$$

after expanding $\tilde{c}(\gamma_1\gamma_2) = \tilde{c}(\gamma_1) + \gamma_1\tilde{c}(\gamma_2)$ and cancelling the terms $h_1\gamma_1 h_2 \tilde{c}(\gamma_2)$; the bracket is 0 by $(*)$ for $\gamma_1$. So $c^\sharp \in Z^1_{\mathrm{cont}}(G, M)$ and $\mathrm{Res}[c^\sharp] = [c]$.

(ii) $M^H$ is of the same type as $M$ (Remark 16.32; a submodule of a discrete torsion module is discrete torsion), and $H^2(\widehat{\mathbb{Z}}, M^H) = 0$ by the cited results. With $H^2(G/H, M^H) = 0$ the five-term sequence reduces to (i) followed by $0 \to 0 \to H^2(G, M)$. □

*Remark* 16.62. For $G = G_{\mathbb{Q}_p}$ and $H = I_p$ this is the exact sequence $0 \to H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) \to H^1(\mathbb{Q}_p, M) \to H^1(I_p, M)^{G_{\mathbb{Q}_p}/I_p} \to 0$ for $M$ of type (C) or (V), at every prime $p$ (the proof does not use $p \neq \ell$). Part (i) is proved here without any stated input; Chapter 17 reaches the same surjectivity through part (ii).

## 16.10 The model computation: inertia, Frobenius and the torsion of $H^1(I, \mathrm{ad}^0 T)$

This section is the group cohomology of Lemma lem:tamfactor, done line by line. We isolate the abstract setting (a profinite group with a normal subgroup whose $\ell$-adic part is procyclic, and an element normalising it by $t \mapsto t^q$), compute the conjugation action of Frobenius on $H^1$ of the inertia group, and then evaluate everything for the adjoint of a two-dimensional representation of the Tate shape. What is not done here: the identification of the datum $c'$ with the unit part of the Tate parameter, which is Kummer theory (Chapter 17), and the definition of $\mathrm{Tam}_q$ and its use in Theorem thm:main (Chapter 18).

**Construction 16.63** (the model setting)**.** The data are:

- a profinite group $G$, a closed normal subgroup $I \trianglelefteq G$, and a closed subgroup $K \subseteq I$, normal in $G$, of order prime to $\ell$, with $I/K \simeq \mathbb{Z}_\ell$ topologically generated by the image of an element $t \in I$; we fix the isomorphism $\pi\colon I/K \to \mathbb{Z}_\ell$ with $\pi(t) = 1$ and write $t^a$ for an element of $I$ with $\pi(t^a K) = a$, $a \in \mathbb{Z}_\ell$;
- an element $\mathrm{Frob} \in G$ whose image topologically generates $G/I \simeq \widehat{\mathbb{Z}}$;
- a positive integer $q$ prime to $\ell$ with $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} \equiv t^q \pmod K$, equivalently $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} \equiv t^{q^{-1}} \pmod K$ with $q^{-1} \in \mathbb{Z}_\ell^\times$;
- a finitely generated $\mathbb{Z}_\ell$-module $M$ with a continuous action $\rho_M\colon G \to \mathrm{Aut}_{\mathbb{Z}_\ell}(M)$ that is trivial on $K$.

The two forms of the relation are equivalent: conjugation by $\mathrm{Frob}^{-1}$ is a continuous automorphism $\alpha$ of $I/K \simeq \mathbb{Z}_\ell$, hence multiplication by a unit; the first form says $\alpha^{-1}(t) = t^q$, so $\alpha$ is multiplication by $q^{-1}$. We write $\rho_M$, not $\rho$, because in the application $\rho$ is the two-dimensional representation on $T$ and $M$ is often $\mathrm{ad}^0 T$, on which $g$ acts by $\rho_M(g) = \mathrm{Ad}\,\rho(g)\colon x \mapsto \rho(g) x \rho(g)^{-1}$; for $M = T$, $\rho_M = \rho$. On the cohomology we keep the dot notation $\sigma \cdot c$ of Proposition 16.30, and inside $M$ we also write $tm$ for $\rho_M(t)m$.

*Dictionary with the paper.* $G = G_{\mathbb{Q}_q}$ for a prime $q \neq \ell$, $I = I_q$, $K = \ker(t_\ell\colon I_q \to \mathbb{Z}_\ell(1))$ (of order prime to $\ell$ by Proposition 16.53), $t$ a lift of a generator of $\mathbb{Z}_\ell(1)$, Frob a lift of the arithmetic Frobenius; the relation is the tame relation of Chapter 7. With $T = T_\ell E$, the module is $M = \mathrm{ad}^0 T$ for $E$ with multiplicative or potentially multiplicative reduction at $q$, and $M = T$ only for multiplicative reduction. In these cases the action of $I_q$ on $M$ is unipotent in a suitable basis, hence has pro-$\ell$ image and is trivial on $K$ (Chapter 11; this is a hypothesis of the model, not proved here). At an additive potentially multiplicative prime $T$ itself does not qualify: $E$ is

a twist of a Tate curve by a ramified quadratic character, which is nontrivial on $I_q$ and, for $\ell$ odd, nontrivial on $K$ (its restriction to $I_q$ has order 2, prime to $\ell$, so it does not factor through the pro-$\ell$ quotient $I_q/K$); it cancels in $\mathrm{ad}^0 T$. The source of the definition of $\mathrm{Tam}_q$ that uses this computation is [69, I.4.2.2], as recalled in [49, p. 708].

**Proposition 16.64** (Frobenius on $H^1$ of inertia)**.** *In the model setting:*

*(a)* $\mathrm{Inf}\colon H^1(I/K, M) \to H^1(I, M)$ *is an isomorphism, and* $[c] \mapsto c(t) \bmod (t-1)M$ *is an isomorphism* $H^1(I, M) \simeq M/(t-1)M$.

*(b)* $\rho_M(\mathrm{Frob})((t-1)M) \subseteq (t-1)M$, *and under the isomorphism of (a) the conjugation action of* $G/I$ *on* $H^1(I, M)$ *is given by*

$$\overline{\mathrm{Frob}} \cdot \bar{m} = q^{-1}\rho_M(\mathrm{Frob})\,\bar{m};$$

*this does not depend on the lift* Frob *of the generator of* $G/I$. *For* $M = \mathrm{ad}^0 T$ *the operator is* $q^{-1}\,\mathrm{Ad}\,\rho(\mathrm{Frob})$.

*(c)* $H^1(I, M)_{\mathrm{tor}} = (M/(t-1)M)_{\mathrm{tor}}$ *is finite and* $G/I$*-stable, and*

$$\big(H^1(I, M)_{\mathrm{tor}}\big)^{G/I} = \{\bar{m} \in (M/(t-1)M)_{\mathrm{tor}} :\ \rho_M(\mathrm{Frob})\bar{m} = q\,\bar{m}\}.$$

*Proof.* (a) $K$ acts trivially and $K^{(\ell)} = 1$ (Lemma 16.50(i)), so Corollary 16.52(a) gives the first isomorphism, and Theorem 16.58 for $\Gamma = I/K \simeq \mathbb{Z}_\ell$ the second; the composite is $[c] \mapsto c(t)$, as in the proof of Proposition 16.53(ii).

(b) Since $K$ acts trivially, $\rho_M(\mathrm{Frob})\rho_M(t)\rho_M(\mathrm{Frob})^{-1} = \rho_M(\mathrm{Frob}\ t\ \mathrm{Frob}^{-1}) = \rho_M(t)^q$, so

$$\rho_M(\mathrm{Frob})(t-1) = (\rho_M(t)^q - 1)\rho_M(\mathrm{Frob}) = (t-1)(1 + t + \cdots + t^{q-1})\rho_M(\mathrm{Frob}),$$

and $\rho_M(\mathrm{Frob})(t-1)M \subseteq (t-1)M$. (We do not need, and do not claim, equality.) Let $c$ be a cocycle on $I$ inflated from $I/K$. By Proposition 16.30, $(\mathrm{Frob}\cdot c)(i) = \rho_M(\mathrm{Frob})\,c(\mathrm{Frob}^{-1}\,i\,\mathrm{Frob})$, again inflated from $I/K$ since $K$ is normal in $G$. At $i = t$: $\mathrm{Frob}^{-1}\,t\,\mathrm{Frob} \equiv t^{q^{-1}} \pmod K$, so by Lemma 16.55(ii), applied to $\Gamma = I/K$ with $\pi(t) = 1$,

$$(\mathrm{Frob}\cdot c)(t) = \rho_M(\mathrm{Frob})\,c(t^{q^{-1}}) \equiv \rho_M(\mathrm{Frob})(q^{-1}c(t) + x), \qquad x \in (t-1)M,$$

and $\rho_M(\mathrm{Frob})x \in (t-1)M$. So $\mathrm{Frob}\cdot[c]$ corresponds to $q^{-1}\rho_M(\mathrm{Frob})c(t)$ modulo $(t-1)M$. By Proposition 16.30(ii), $I$ acts trivially on $H^1(I, M)$, so the action of $\mathrm{Frob}\,i$ ($i \in I$) is that of Frob.

(c) The isomorphism of (a) is $\mathbb{Z}_\ell$-linear, so it identifies the torsion submodules; $(M/(t-1)M)_{\mathrm{tor}}$ is finite because $M/(t-1)M$ is finitely generated. The action of $G/I$ is $\mathbb{Z}_\ell$-linear, so it preserves the torsion. It remains to see that invariance under $G/I$ is invariance under Frob. The same computation as in (b) applies to any $\sigma \in G$: conjugation by $\sigma^{-1}$ acts on $I/K \simeq \mathbb{Z}_\ell$ by multiplication by a unit $a(\sigma)$, so $\sigma t \sigma^{-1} \equiv t^b \pmod K$ with $b = a(\sigma)^{-1} \in \mathbb{Z}_\ell^\times$, and $\rho_M(\sigma)(t-1) = (\rho_M(t)^b - 1)\rho_M(\sigma)$. Here $(\rho_M(t)^b - 1)M \subseteq (t-1)M$: for $b \in \mathbb{Z}$ the polynomial $X^b - 1$ is divisible by $X - 1$ (for $b < 0$ use $X^b - 1 = -X^b(X^{-b} - 1)$); for $b \in \mathbb{Z}_\ell$ take integers $b_j \to b$, so that $\rho_M(t)^{b_j} m \to \rho_M(t)^b m$ by continuity of the action of $I/K \simeq \mathbb{Z}_\ell$, and use that $(t-1)M$ is closed (Lemma 16.55(ii)). So $\rho_M(\sigma)(t-1)M \subseteq (t-1)M$, and as in (b), $\sigma$ acts on $M/(t-1)M$ by $a(\sigma)\rho_M(\sigma)$. Since $\sigma \mapsto a(\sigma) \in \mathbb{Z}_\ell^\times$ and $\rho_M$ are continuous, the stabiliser of an element of the Hausdorff space $M/(t-1)M$ is a closed subgroup of $G$. If it contains Frob it contains $I$ (which acts trivially) and the closure of the group generated by $I$ and Frob, which is $G$ because Frob generates $G/I$ topologically. Finally $q^{-1}\rho_M(\mathrm{Frob})\bar{m} = \bar{m}$ if and only if $\rho_M(\mathrm{Frob})\bar{m} = q\bar{m}$. □

The factor $q^{-1}$ is the point of (b), and two sanity checks show where it comes from. If $G$ acts trivially on $M=\mathbb{Z}_\ell$, then $H^1(I,\mathbb{Z}_\ell)=\mathrm{Hom}_{\mathrm{cont}}(I,\mathbb{Z}_\ell)$, and Frob acts on a homomorphism $c$ by $c\mapsto c(\mathrm{Frob}^{-1}\cdot\mathrm{Frob})$; since $\mathrm{Frob}^{-1}\,t\,\mathrm{Frob}\equiv t^{q^{-1}}$, the value at $t$ is multiplied by $q^{-1}$, in accordance with $q^{-1}\rho_M(\mathrm{Frob})=q^{-1}$. So $H^1(I,\mathbb{Z}_\ell)\simeq\mathbb{Z}_\ell(-1)$ as a $G/I$-module: inertia's $\ell$-part is $\mathbb{Z}_\ell(1)$, and homomorphisms out of it form its dual. If instead $M=\mathbb{Z}_\ell(1)$, with Frob acting by $q$, the operator is $q^{-1}q=1$, and $H^1(I,\mathbb{Z}_\ell(1))=\mathrm{Hom}(\mathbb{Z}_\ell(1),\mathbb{Z}_\ell(1))\simeq\mathbb{Z}_\ell$ with trivial Frobenius action. Only the second module has nonzero Frobenius-invariants; the same effect, the eigenvalue $q$ of $\rho_M(\mathrm{Frob})$ being the one that survives, drives the invariants in Proposition 16.67.

*Remark* 16.65 (the other convention). Some texts let $\sigma$ act by $c\mapsto\sigma^{-1}c(\sigma\cdot\sigma^{-1})$. With that convention the computation of (b) gives $q\,\rho_M(\mathrm{Frob})^{-1}$ instead of $q^{-1}\rho_M(\mathrm{Frob})$. The invariants are the same, since one operator is the inverse of the other; the formulas are not. We use the paper's convention throughout.

**Lemma 16.66** (the Tate shape and its adjoint)**.** *In the model setting let $T=\mathbb{Z}_\ell e_1\oplus\mathbb{Z}_\ell e_2$ and*

$$\rho(g)=\begin{pmatrix}\chi(g)\varepsilon(g) & b(g)\\ 0 & \varepsilon(g)\end{pmatrix},$$

*where $\varepsilon\colon G\to\{\pm1\}$ and $\chi\colon G\to\mathbb{Z}_\ell^\times$ are continuous characters trivial on $I$ with $\chi(\mathrm{Frob})=q$, and $b|_I\colon I\to\mathbb{Z}_\ell$ is a continuous homomorphism, trivial on $K$, with $b(t)=n'\neq0$. (In the paper $b=\varepsilon\kappa$, so $b|_I=\kappa|_I$, by Proposition 16.21(v).) Write $\varepsilon=\varepsilon(\mathrm{Frob})$ and $c'=b(\mathrm{Frob})$, so that*

$$\rho(t)=\begin{pmatrix}1 & n'\\ 0 & 1\end{pmatrix},\qquad \rho(\mathrm{Frob})=\begin{pmatrix}\varepsilon q & c'\\ 0 & \varepsilon\end{pmatrix}.$$

*Let $M=\mathrm{ad}^0T$, with $g$ acting by $\rho_M(g)=\mathrm{Ad}\,\rho(g)\colon x\mapsto\rho(g)x\rho(g)^{-1}$ (written $g\cdot x$), and basis $e=E_{12}$, $h=E_{11}-E_{22}$, $f=E_{21}$. Then*

$$\begin{aligned}&(t-1)e=0, && (t-1)h=-2n'e, && (t-1)f=n'h-n'^2e,\\ &\mathrm{Frob}\cdot e=qe, && \mathrm{Frob}\cdot h=h-2\varepsilon c'e, && \mathrm{Frob}\cdot f=q^{-1}(f+\varepsilon c'h-c'^2e).\end{aligned}$$

*Also $(t-1)^3=0$ on $\mathrm{ad}^0T$, and $(t-1)e_1=0$, $(t-1)e_2=n'e_1$ on $T$.*

*Proof.* $\rho(t)=1+n'e$, since $\chi$, $\varepsilon$ are trivial on $I$, and $\rho(t)^{-1}=1-n'e$ as $e^2=0$. The matrix units give $eh=-e$, $he=e$, $ehe=0$, $ef-fe=h$, $efe=e$. So $t\cdot x=(1+n'e)x(1-n'e)=x+n'(ex-xe)-n'^2exe$. For $x=e$: $e$. For $x=h$: $h+n'(-2e)$. For $x=f$: $f+n'h-n'^2e$.

(Here $eh=E_{12}(E_{11}-E_{22})=-E_{12}$, $he=E_{12}$, $ef=E_{11}$, $fe=E_{22}$, using $E_{ij}E_{kl}=\delta_{jk}E_{il}$.)

For $A=\left(\begin{smallmatrix}\alpha&\beta\\0&\delta\end{smallmatrix}\right)$, $A^{-1}=\left(\begin{smallmatrix}\alpha^{-1}&-\beta\alpha^{-1}\delta^{-1}\\0&\delta^{-1}\end{smallmatrix}\right)$, and a direct multiplication gives

$$\begin{aligned}AeA^{-1}&=\begin{pmatrix}0&\alpha\\0&0\end{pmatrix}A^{-1}=\begin{pmatrix}0&\alpha\delta^{-1}\\0&0\end{pmatrix}=\alpha\delta^{-1}e,\\ AhA^{-1}&=\begin{pmatrix}\alpha&-\beta\\0&-\delta\end{pmatrix}A^{-1}=\begin{pmatrix}1&-2\beta\delta^{-1}\\0&-1\end{pmatrix}=h-2\beta\delta^{-1}e,\\ AfA^{-1}&=\begin{pmatrix}\beta&0\\\delta&0\end{pmatrix}A^{-1}=\begin{pmatrix}\beta\alpha^{-1}&-\beta^2\alpha^{-1}\delta^{-1}\\\delta\alpha^{-1}&-\beta\alpha^{-1}\end{pmatrix}=\delta\alpha^{-1}f+\beta\alpha^{-1}h-\beta^2\alpha^{-1}\delta^{-1}e.\end{aligned}$$

With $\alpha=\varepsilon q$, $\beta=c'$, $\delta=\varepsilon$ and $\varepsilon^{-1}=\varepsilon$: $\alpha\delta^{-1}=q$, $\beta\delta^{-1}=\varepsilon c'$, $\delta\alpha^{-1}=q^{-1}$, $\beta\alpha^{-1}=\varepsilon c'q^{-1}$, $\beta^2\alpha^{-1}\delta^{-1}=c'^2q^{-1}$. Finally $(t-1)^2f=n'(t-1)h=-2n'^2e$ and $(t-1)^3f=0$. The symbolic computation was also checked with PARI/GP (Exercise 16.87). $\square$

**Proposition 16.67** (the model computation)**.** *Let $\ell$ be odd, $M = \mathrm{ad}^0 T$ as in Lemma 16.66, and $w = v_\ell(n')$. (We write $w$, not $v$, because in the statement of the paper's Lemma lem:tamfactor $v$ denotes $v_\ell(q-1)$.)*

*(a)* $(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$, *so*

$$H^1(I,M) = M/(t-1)M = (\mathbb{Z}_\ell/n')\bar{e} \oplus (\mathbb{Z}_\ell/n')\bar{h} \oplus \mathbb{Z}_\ell \bar{f},$$
$$H^1(I,M)_{\mathrm{tor}} = (\mathbb{Z}/\ell^w)\bar{e} \oplus (\mathbb{Z}/\ell^w)\bar{h} \simeq (\mathbb{Z}/\ell^w)^2.$$

*(b)* $\overline{\mathrm{Frob}}$ *acts on the torsion by* $\bar{e} \mapsto \bar{e}$ *and* $\bar{h} \mapsto q^{-1}(\bar{h} - 2\varepsilon c'\bar{e})$.

*(c) With* $m = \min(w,\, v_\ell(q-1),\, v_\ell(c'))$,

$$(H^1(I,M)_{\mathrm{tor}})^{G/I} = \{x\bar{e} + y\bar{h} :\ (q-1)y \equiv 0,\ c'y \equiv 0 \pmod{\ell^w}\} \ \simeq\ \mathbb{Z}/\ell^w \oplus \mathbb{Z}/\ell^m,$$

*of order* $\ell^{w+m}$, *length* $w+m$ *and Fitting ideal* $\ell^{w+m}\mathbb{Z}_\ell$.

*(d)* $m$ *depends on* $c'$ *only modulo* $\ell^{\min(w, v_\ell(q-1))}\mathbb{Z}_\ell = (n', q-1)\mathbb{Z}_\ell$. *This is exactly the indeterminacy of* $c' = b(\mathrm{Frob})$: *replacing* Frob *by* Frob $t^a$ *($a \in \mathbb{Z}_\ell$) changes $c'$ by $\varepsilon q n' a$, and replacing $e_2$ by $e_2 + re_1$ (which changes the cocycle $b/\varepsilon$ by a coboundary, Proposition 16.21(ii)) changes $c'$ by $\varepsilon r(q-1)$; rescaling $e_1$, $e_2$ multiplies $c'$ by a unit.*

*(e) (Check with* $M = T$.*)* $H^1(I,T) = (\mathbb{Z}_\ell/n')\bar{e}_1 \oplus \mathbb{Z}_\ell \bar{e}_2$, *with torsion* $(\mathbb{Z}/\ell^w)\bar{e}_1$, *on which* $\overline{\mathrm{Frob}}$ *acts by* $q^{-1}\cdot \varepsilon q = \varepsilon$. *So* $(H^1(I,T)_{\mathrm{tor}})^{G/I}$ *is* $(\mathbb{Z}/\ell^w)\bar{e}_1$ *if* $\varepsilon = 1$ *and* $0$ *if* $\varepsilon = -1$, *with Fitting ideal* $\ell^w\mathbb{Z}_\ell$, *resp.* $\mathbb{Z}_\ell$.

*Proof.* (a) By Lemma 16.66, $(t-1)M$ is spanned by $-2n'e$ and $n'h - n'^2 e$, that is $(t-1)M = n'(\mathbb{Z}_\ell \cdot 2e + \mathbb{Z}_\ell(h - n'e))$. As $\ell$ is odd, $2 \in \mathbb{Z}_\ell^\times$, so this is $n'(\mathbb{Z}_\ell e + \mathbb{Z}_\ell(h-n'e)) = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$. Since $e, h, f$ is a basis, $M/(t-1)M = (\mathbb{Z}_\ell/n')\bar{e} \oplus (\mathbb{Z}_\ell/n')\bar{h} \oplus \mathbb{Z}_\ell\bar{f}$, and $\mathbb{Z}_\ell/n'\mathbb{Z}_\ell = \mathbb{Z}/\ell^w$ (if $w = 0$ it is 0). Proposition 16.64(a) identifies this with $H^1(I,M)$.

(b) By Proposition 16.64(b), $\overline{\mathrm{Frob}}$ acts by $q^{-1}\,\mathrm{Ad}\,\rho(\mathrm{Frob})$, that is $x \mapsto q^{-1}\,\mathrm{Frob}\cdot x$; by Lemma 16.66, $q^{-1}\,\mathrm{Frob}\cdot e = e$ and $q^{-1}\,\mathrm{Frob}\cdot h = q^{-1}(h - 2\varepsilon c' e)$.

(c) $\overline{\mathrm{Frob}}(x\bar{e} + y\bar{h}) - (x\bar{e} + y\bar{h}) = -2\varepsilon c' q^{-1} y\,\bar{e} + (q^{-1}-1)y\,\bar{h}$. This vanishes if and only if $c'y \equiv 0$ and $(q-1)y \equiv 0 \pmod{\ell^w}$, because $2$, $\varepsilon$, $q$ are units and $\bar{e}, \bar{h}$ are independent generators of $(\mathbb{Z}/\ell^w)^2$. By Proposition 16.64(c) these are the $G/I$-invariants. The condition on $y \in \mathbb{Z}/\ell^w$ is $\ell^{\min(w, v_\ell(q-1), v_\ell(c'))}y = 0$, which cuts out the subgroup $\ell^{w-m}\mathbb{Z}/\ell^w \simeq \mathbb{Z}/\ell^m$; $x$ is free. In detail: for $a \in \mathbb{Z}_\ell$ with $\alpha = v_\ell(a)$ (and $\alpha = \infty$ if $a = 0$), the condition $ay \equiv 0 \pmod{\ell^w}$ on $y \in \mathbb{Z}/\ell^w$ is $\ell^{\min(\alpha, w)}y = 0$, because $a = \ell^\alpha \cdot$unit. Two such conditions, for $a = q-1$ and $a = c'$, hold together if and only if $\ell^\mu y = 0$ with $\mu$ the smaller exponent, $\mu = \min(w, v_\ell(q-1), v_\ell(c')) = m$. The elements of $\mathbb{Z}/\ell^w$ killed by $\ell^m$ are the multiples of $\ell^{w-m}$, a cyclic group of order $\ell^m$. The order and length follow, and $\mathrm{Fitt}_{\mathbb{Z}_\ell}(A) = \ell^{\mathrm{length}\,A}\mathbb{Z}_\ell$ for finite $A$ (Chapter 3).

(d) If $c'' = c' + \ell^\mu z$ with $\mu = \min(w, v_\ell(q-1))$ and $z \in \mathbb{Z}_\ell$, then $\min(\mu, v_\ell(c'')) = \min(\mu, v_\ell(c'))$, so $m$ is unchanged. The first change: $\rho(\mathrm{Frob}\ t^a) = \rho(\mathrm{Frob})\rho(t)^a = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & n'a \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} \varepsilon q & c' + \varepsilon q n' a \\ 0 & \varepsilon \end{smallmatrix}\right)$. The second: by the formula in the proof of Proposition 16.21(ii), the corner becomes $b + r(\chi\varepsilon - \varepsilon)$, whose value at Frob is $c' + \varepsilon r(q-1)$; on $I$ it is unchanged, so $n'$ is unchanged. In both cases the change is a multiple of $n'$ or of $q-1$, hence lies in $(n', q-1)\mathbb{Z}_\ell = \ell^\mu\mathbb{Z}_\ell$.

(e) By Lemma 16.66, $(t-1)T = n'\mathbb{Z}_\ell e_1$, which gives $H^1(I,T)$ by Proposition 16.64(a). Here $\rho_M = \rho$, and the action is $q^{-1}\rho(\mathrm{Frob})\bar{e}_1 = q^{-1}\varepsilon q \bar{e}_1$. If $\varepsilon = -1$ the invariants are the $x$ with $2x \equiv 0 \pmod{\ell^w}$, which is $x \equiv 0$ for $\ell$ odd. □

A numerical instance, with every step visible. Take $\ell = 3$, $q = 7$, $n' = 9$ and $c' = 3$, so $w = 2$, $v_3(q-1) = v_3(6) = 1$, $v_3(c') = 1$ and $m = 1$. In the basis $e, h, f$ the matrix of $t - 1$ has columns $0$, $(-18, 0, 0)$, $(-81, 9, 0)$; its elementary divisors over $\mathbb{Z}$ are $9, 18, 0$ (PARI/GP, `matsnf`), that is $9, 9, 0$ up to units of $\mathbb{Z}_3$, so $H^1(I, \mathrm{ad}^0 T)_{\mathrm{tor}} \simeq (\mathbb{Z}/9)^2$. The invariance conditions on $x\bar{e} + y\bar{h}$ are $6y \equiv 0$ and $3y \equiv 0 \pmod 9$, satisfied by $y \in \{0, 3, 6\}$, and $x$ is arbitrary: the invariants are $\mathbb{Z}/9 \oplus 3\mathbb{Z}/9 \simeq \mathbb{Z}/9 \oplus \mathbb{Z}/3$, of order $27 = 3^{w+m}$. Replacing $c'$ by a unit, say $c' = 1$, gives $m = 0$ and invariants $\mathbb{Z}/9$; replacing $q = 7$ by $q = 19$ ($v_3(18) = 2$) and keeping $c' = 3$ still gives $m = 1$, now limited by $c'$.

**Corollary 16.68** (the Fitting ideal of the invariants)**.** *In the model setting with $\ell$ odd, the ideal $\mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I, M)_{\mathrm{tor}})^{G/I})$ is defined (the module is finite), and equals $\ell^{w+m}\mathbb{Z}_\ell$ for $M = \mathrm{ad}^0 T$ and $\ell^w\mathbb{Z}_\ell$ or $\mathbb{Z}_\ell$ (as $\varepsilon = 1$ or $-1$) for $M = T$. For $G = G_{\mathbb{Q}_q}$ this ideal is the Tamagawa factor $\mathrm{Tam}_q(M)$ of the paper's Section 1.2, defined in Chapter 18; so*

$$v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(n') + \min(v_\ell(n'), v_\ell(q-1), v_\ell(c')),$$

*which is the formula of Lemma lem:tamfactor once $v_\ell(n') = v_\ell(n)$ (Chapters 7 and 11) and $\min(v_\ell(n), v_\ell(q-1), v_\ell(c')) = \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$ for the paper's class $c(u)$ of the unit part $u$ of the Tate parameter (Chapter 17). Lemma lem:tamfactor itself is assembled in Chapter 18; the Kummer identification of $c'$ and its indeterminacy modulo $(n, q-1)$ are proved in Chapter 17.*

*Proof.* Proposition 16.67(c), (e) and $\mathrm{Fitt}_{\mathbb{Z}_\ell}(\bigoplus \mathbb{Z}/\ell^{a_i}) = \ell^{\sum a_i}\mathbb{Z}_\ell$ (Chapter 3). The identifications of $v_\ell(n')$ and of $c'$ are proved in the chapters cited. □

*Remark* 16.69. (i) If $w = 0$ every torsion module above is 0 and every Fitting ideal is $\mathbb{Z}_\ell$, consistently with the formula ($m \le w = 0$).

(ii) The free part $\mathbb{Z}_\ell \bar{f}$ of $H^1(I, \mathrm{ad}^0 T)$ is not Frob-stable: $q^{-1}\,\mathrm{Frob}\cdot\bar{f} = q^{-2}(\bar{f} + \varepsilon c'\bar{h} - c'^2\bar{e})$ by Lemma 16.66. Only the torsion is needed, and it is stable (Proposition 16.64(c)).

(iii) For $M = T$ the answer $\ell^{v_\ell(n)}$ (split, $\varepsilon = 1$) or 1 (nonsplit, $\varepsilon = -1$) is the $\ell$-part of the Tamagawa number $c_q(E)$, which is $n$ for split and 1 or 2 for nonsplit multiplicative reduction (Chapter 10). This is the paper's "check".

(iv) The hypothesis $\ell$ odd enters only through $2 \in \mathbb{Z}_\ell^\times$ in (a), (c) and (e). For $\ell = 2$ the answer changes (Exercise 16.88).

*In the paper* 16.70. The proof of Lemma lem:tamfactor, sentence by sentence (quotations verbatim, from the proof):

- "the kernel of $I \to \langle t \rangle$ has trivial pro-$\ell$ quotient, so by inflation-restriction $H^1(I, M') = H^1(\langle t \rangle, M') = M'/(t-1)M'$ for every $\mathbb{Z}_\ell$-module $M'$ of finite type with this action": Corollary 16.52(a) (which is Theorem 16.31(d) with Theorem 16.51(c)) and Theorem 16.58; the identification is $[c] \mapsto c(t)$ (Proposition 16.64(a)).
- "$(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$, and as 2 is a unit, $(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$; so $H^1(I, M) = (\mathbb{Z}_\ell/n')\bar{e} \oplus (\mathbb{Z}_\ell/n')\bar{h} \oplus \mathbb{Z}_\ell \bar{f}$ and its torsion is $(\mathbb{Z}/\ell^{v_\ell(n)})^2$": Lemma 16.66 and Proposition 16.67(a), with $v_\ell(n') = v_\ell(n)$.
- "The group $G/I$ acts on $H^1(I, M)$ by $(\sigma \cdot z)(i) = \sigma z(\sigma^{-1} i \sigma)$ for a cocycle $z$": Proposition 16.30.

- "for the arithmetic Frobenius Frob, $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ and $z(t^k) \equiv k\, z(t)$ modulo $(t-1)M$, so Frob acts on $M/(t-1)M$ as $q^{-1}\,\mathrm{Ad}(\rho(\mathrm{Frob}))$, that is, by conjugation by $\rho(\mathrm{Frob})$ followed by multiplication by $q^{-1}$": the relation is Construction 16.63 (from Chapter 7), the congruence is Lemma 16.55(ii) with $k = q^{-1} \in \mathbb{Z}_\ell$, the conclusion is Proposition 16.64(b). Here $\rho$ is the representation on $T$, and $\mathrm{Ad}(\rho(\mathrm{Frob}))$ is its conjugation action on $M = \mathrm{ad}^0\, T$.
- "the adjoint action sends $e$ to $qe$ and $h$ to $h - 2c'\varepsilon e$, so on the torsion $\bar{e} \mapsto \bar{e}$ and $\bar{h} \mapsto q^{-1}(\bar{h} - 2c'\varepsilon\bar{e})$": Lemma 16.66 and Proposition 16.67(b).
- "The fixed points are the $x\bar{e} + y\bar{h}$ with $(q-1)y \equiv 0$ and $c'y \equiv 0 \pmod{\ell^{v_\ell(n)}}$": Proposition 16.67(c), where Proposition 16.64(c) supplies the fact, implicit in the paper, that invariance under $G/I$ is invariance under Frob.
- "in general it is well defined only modulo $\ell^{\min(v_\ell(n), v_\ell(q-1))}$ (changing the lift changes it by a multiple of $n'$, changing the cocycle by a multiple of $q-1$)": Proposition 16.67(d).
- "As a check, the same computation for $M = T$ gives $(\mathbb{Z}/\ell^{v_\ell(n)})\bar{e}_1$ with Frob acting by $\varepsilon$": Proposition 16.67(e).

The remaining steps (that $\kappa|_I$ is $n$ times the tame character, the splitting $\kappa = n\kappa_q + \kappa_u$, and the value of $c'$) are Kummer theory and are proved in Chapter 17; the assembly is in Chapter 18. The invariants are taken in the torsion $(\mathbb{Z}/\ell^{v_\ell(n)})^2$, as the definition of Tam requires.

## 16.11 Cup products, Pontryagin duality and lengths

The paper uses duality in one place: the exact sequence of Lemma lem:dfg21 has Pontryagin duals as its last two terms, and the proof of Theorem thm:main adds lengths along it. The duality that produces those dual terms is Tate's local duality, a cup product pairing proved in Chapter 17. Here we define cup products, recall what Pontryagin duality does to lengths, and prove the length bookkeeping.

**Definition 16.71** (cup products)**.** Let $G$ be a group (or a profinite group, with continuous cochains and modules of type (D), or of type (C) and (V) with the tensor product over $\mathbb{Z}_\ell$) and $A$, $B$ two $G$-modules, $A \otimes B$ with the diagonal action. For $a \in C^p(G, A)$, $b \in C^q(G, B)$ put

$$(a \cup b)(g_1, \dots, g_{p+q}) = a(g_1, \dots, g_p) \otimes g_1 \cdots g_p\, b(g_{p+1}, \dots, g_{p+q}) \in A \otimes B.$$

For $p = q = 1$: $(a \cup b)(g, h) = a(g) \otimes g\, b(h)$. A $G$-equivariant pairing $A \times B \to C$ composed with $\cup$ gives $C^p(G, A) \times C^q(G, B) \to C^{p+q}(G, C)$.

**Proposition 16.72** (properties of the cup product)**.** *(i) $d(a \cup b) = da \cup b + (-1)^p a \cup db$. Hence $\cup$ induces bilinear maps $H^p(G, A) \times H^q(G, B) \to H^{p+q}(G, A \otimes B)$.*

*(ii) For $p = q = 1$ directly: if $a$, $b$ are 1-cocycles then $a \cup b$ is a 2-cocycle; if $a = d^0 m$ then $a \cup b = d^1(m \otimes b)$ with $(m \otimes b)(g) = m \otimes b(g)$; if $b = d^0 m'$ then $a \cup b = -d^1\chi$ with $\chi(g) = a(g) \otimes gm'$.*

*(iii) (Stated.) The cup product is associative and graded commutative, $a \cup b = (-1)^{pq}\tau_*(b \cup a)$ with $\tau\colon B \otimes A \to A \otimes B$ the flip; it commutes with restriction and inflation; it is compatible with connecting homomorphisms (up to the sign $(-1)^p$ when the connecting map acts on the second factor); and $\mathrm{Cor}(\mathrm{Res}\, a \cup b) = a \cup \mathrm{Cor}\, b$.*

*Proof.* (i) in general is a direct computation; see [143, Chapter I, Section 4] and [17, Chapter V, Section 3]. We check the case used most, (ii). Let $\psi(g,h)=a(g)\otimes gb(h)$ with $a$, $b$ cocycles. In $(d^2\psi)(g,h,k)=g\psi(h,k)-\psi(gh,k)+\psi(g,hk)-\psi(g,h)$ substitute $a(gh)=a(g)+ga(h)$ and $b(hk)=b(h)+hb(k)$:

$$ga(h)\otimes ghb(k)-\big(a(g)+ga(h)\big)\otimes ghb(k)+a(g)\otimes\big(gb(h)+ghb(k)\big)-a(g)\otimes gb(h)=0.$$

If $a(g)=gm-m$: $d^1(m\otimes b)(g,h)=gm\otimes gb(h)-m\otimes b(gh)+m\otimes b(g)=gm\otimes gb(h)-m\otimes gb(h)=a(g)\otimes gb(h)$, so $a\cup b=d^1(m\otimes b)$. If $b(h)=hm'-m'$: $d^1\chi(g,h)=ga(h)\otimes ghm'-a(gh)\otimes ghm'+a(g)\otimes gm'=-a(g)\otimes ghm'+a(g)\otimes gm'=-(a\cup b)(g,h)$. So the cup product of classes is well defined in degrees $(1,1)$. (iii) is stated with the same references and [174, Chapter VIII, Section 3]. $\square$

*Remark* 16.73. In Chapter 17 the cup product $H^i(\mathbb{Q}_p,M)\times H^{2-i}(\mathbb{Q}_p,M^*(1))\to H^2(\mathbb{Q}_p,\mu)\simeq\mathbb{Q}/\mathbb{Z}$ is Tate's local duality pairing, from which the dual terms $H^1_f(\mathbb{Q}_p,T^D)^*$ and $H^0(\mathbb{Q},W^D)^*$ of Lemma lem:dfg21 come. In Chapter 8 the cup product on $H^1(\pi_1(X),\mathbb{Z})$ for a compact Riemann surface $X$ agrees with the topological one.

**Proposition 16.74** (Pontryagin duality, recalled and stated)**.** *(i) (Recalled from Chapter 3.)* $A^*=\mathrm{Hom}(A,\mathbb{Q}/\mathbb{Z})$ *for finite* $A$*, which is* $\mathrm{Hom}(A,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ *when* $A$ *is* $\ell$*-primary;* $T^*=\mathrm{Hom}_{\mathrm{cont}}(T,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ *for* $T$ *of type (C) is discrete* $\ell$*-primary* ($(\mathbb{Z}_\ell^r)^*=(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$)*, and* $W^*$ *is of type (C) for* $W$ *discrete* $\ell$*-primary with* $W[\ell]$ *finite. The functor* $X\mapsto X^*$ *is exact,* $X^{**}=X$*, and* $\operatorname{length}X^*=\operatorname{length}X$ *for finite* $X$*.*

*(ii) For a* $G$*-module* $X$ *of one of these kinds,* $X^*$ *is a* $G$*-module by* $(g\varphi)(x)=\varphi(g^{-1}x)$*, of the dual kind, and* $(X^*)^G=(X_G)^*$*.*

*(iii) (Stated.) For* $G$ *profinite and* $A$ *a discrete torsion* $G$*-module,* $H^n(G,A)^*\simeq H_n(G,A^*)$*, the homology of the profinite group* $G$ *with coefficients in the compact module* $A^*$*; for* $G$ *finite and* $A$ *finite,* $\hat H^n(G,A)^*\simeq\hat H^{-n-1}(G,A^*)$*.*

*Proof.* (i) is Chapter 3. (ii) $g\varphi$ is again a (continuous) homomorphism and the action is continuous because $G$ acts through finite quotients on each finite piece. A homomorphism $\varphi$ is invariant if and only if $\varphi(g^{-1}x)=\varphi(x)$ for all $g$, $x$, that is, if and only if it kills $I_GX$. (iii) We do not prove this; see [143, Chapter I, Section 3 and Chapter II, Section 7] and [17, Chapter VI, Section 7]. $\square$

**Lemma 16.75** (lengths in exact sequences)**.** *(i) For a finite* $\mathbb{Z}_\ell$*-module* $A$*,* $\operatorname{length}(A^*)=\operatorname{length}A$*; dualising an exact sequence of finite* $\mathbb{Z}_\ell$*-modules gives an exact sequence with the arrows reversed.*

*(ii) In an exact sequence* $0\to A\to B\to C\to D\to 0$ *of* $\mathbb{Z}_\ell$*-modules of finite length,* $\operatorname{length}B=\operatorname{length}A+\operatorname{length}C-\operatorname{length}D$*. More generally the alternating sum of the lengths in a finite exact sequence of modules of finite length is* $0$*.*

*(iii) Consequently, in the exact sequence of Lemma lem:dfg21,*

$$0\to H^1_f(\mathbb{Q},W)\to H^1_\Sigma(\mathbb{Q},W)\to\bigoplus_{p\in\Sigma\cup\{\infty\}}H^1_f(\mathbb{Q}_p,T^D)^*\to H^0(\mathbb{Q},W^D)^*\to 0,$$

*if all terms are finite,*

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma \cup \{\infty\}} \operatorname{length} H^1_f(\mathbb{Q}_p, T^D) - \operatorname{length} H^0(\mathbb{Q}, W^D),$$

*and the term at $\infty$ is 0 for $\ell$ odd (Paper link 16.39).*

*Proof.* (i) is Chapter 3. (ii) Length is additive on short exact sequences (Jordan-Holder, Chapter 3). Split $0 \to A \to B \to C \to D \to 0$ at the image $X$ of $B \to C$: $0 \to A \to B \to X \to 0$ and $0 \to X \to C \to D \to 0$ give $\operatorname{length} B = \operatorname{length} A + \operatorname{length} X$ and $\operatorname{length} C = \operatorname{length} X + \operatorname{length} D$; eliminating $\operatorname{length} X$ gives the formula. For a general exact sequence $0 \to A_1 \to \cdots \to A_r \to 0$, let $X_i$ be the image of $A_i \to A_{i+1}$; then $0 \to X_{i-1} \to A_i \to X_i \to 0$ is exact (with $X_0 = X_r = 0$), so $\operatorname{length} A_i = \operatorname{length} X_{i-1} + \operatorname{length} X_i$, and in $\sum_i (-1)^i \operatorname{length} A_i$ the terms $\operatorname{length} X_i$ cancel in pairs. (iii) Apply (ii), then (i) to the last two terms, and use that the length of a finite direct sum is the sum of the lengths. □

*In the paper* 16.76. The proof of Theorem thm:main in the paper uses (iii) with $H^0(\mathbb{Q}, W^D) = 0$ (Lemma lem:h0), and Lemma lem:fpr to evaluate each local term. Remark rem:ns uses it with $\operatorname{length} H^0(\mathbb{Q}, W^D) = h_\ell(E)$, which is the displayed identity there: "$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma} \big(v_\ell(L_p(A_f, 1)^{-1}) + v_\ell(\operatorname{Tam}_p(A))\big) - h_\ell(E)$". The paper's "* is the Pontryagin dual" in Lemma lem:dfg21 is the dual of finite groups, and the only properties used are those of (i). The finiteness of the terms is part of the input of Chapter 18 (DFG04's theorems), not a consequence of this lemma.

## 16.12 Galois cohomology: notation and the road ahead

This short section fixes the notation of Galois cohomology and proves the one formula that is purely group-theoretic: the unramified part of $H^1$ of a local field is $M^I/(\operatorname{Frob} - 1)M^I$. Everything arithmetic (Hilbert 90, Kummer theory, local duality, Selmer groups) is in Chapter 17.

**Convention 16.77** (Galois cohomology)**.** For a field $K$ with separable closure $\overline{K}$, $G_K = \operatorname{Gal}(\overline{K}/K)$ is profinite (Chapter 5); a *Galois module* is a $G_K$-module of type (D), (C) or (V), and $H^n(K, M) := H^n(G_K, M)$, continuous cochain cohomology. For $K = \mathbb{Q}_p$: $I_p \trianglelefteq G_{\mathbb{Q}_p}$, and $G_{\mathbb{Q}_p}/I_p = \operatorname{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \simeq \widehat{\mathbb{Z}}$ is generated by the arithmetic Frobenius $\operatorname{Frob}_p$ ($x \mapsto x^p$); $H^n(I_p, M)$ carries the conjugation action of $G_{\mathbb{Q}_p}/I_p$; $H^n(\mathbb{R}, M) := H^n(\operatorname{Gal}(\mathbb{C}/\mathbb{R}), M)$; $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$ with $\chi_\ell(\operatorname{Frob}_p) = p$ for $p \neq \ell$; $\operatorname{Frob}_p$ never denotes the geometric Frobenius $\operatorname{Frob}_p^{-1}$. The *unramified part* of $H^1$ is

$$H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) := H^1(G_{\mathbb{Q}_p}/I_p, M^{I_p}) \xrightarrow[\sim]{\mathrm{Inf}} \ker\big(\mathrm{Res}\colon H^1(\mathbb{Q}_p, M) \to H^1(I_p, M)\big),$$

the two descriptions agreeing by Theorem 16.31(a) and Remark 16.32.

**Proposition 16.78** (unramified cohomology, a preview)**.** *Let $p$ be a prime and $\Gamma = G_{\mathbb{Q}_p}/I_p \simeq \widehat{\mathbb{Z}}$.*

*(i) For $M$ of type (C): $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = M^{I_p}/(\operatorname{Frob}_p - 1)M^{I_p}$.*

*(ii) For $V$ of type (V): $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = V^{I_p}/(\operatorname{Frob}_p - 1)V^{I_p}$, and it is 0 if and only if $\det(1 - \operatorname{Frob}_p \mid V^{I_p}) \neq 0$, if and only if $\det(1 - \operatorname{Frob}_p^{-1} \mid V^{I_p}) \neq 0$.*

*Proof.* $M^{I_p}$ is a closed $\mathbb{Z}_\ell$-submodule of $M$, hence of type (C) (resp. a subspace, of type (V)), with a continuous action of $\Gamma$, and $\widehat{\mathbb{Z}}$ surjects onto $\mathbb{Z}_\ell$. Apply Theorem 16.58 with $F = \mathrm{Frob}_p$. The last equivalence holds because 1 is an eigenvalue of an invertible operator if and only if it is an eigenvalue of its inverse. □

*In the paper* 16.79. After Lemma lem:dfg21: "For $p \in \Sigma$ the group $H^1_f(\mathbb{Q}_p, V^D) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ vanishes, because $\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1} \neq 0$". The last equality in the chain and the vanishing are Proposition 16.78(ii). (The image of $\mathrm{Frob}_p - 1$ equals that of $\mathrm{Frob}_p^{-1} - 1 = -\mathrm{Frob}_p^{-1}(\mathrm{Frob}_p - 1)$, so the quotient does not depend on which Frobenius is meant.) The equality $H^1_f = H^1_{\mathrm{ur}}$ at $p \neq \ell$ is Chapter 18, and the value of the determinant is Chapter 19. The next sentences, "So $H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group", use Proposition 16.44(iii): the preimage of 0 under $H^1(\mathbb{Q}_p, T^D) \to H^1(\mathbb{Q}_p, V^D)$ is the torsion; finiteness is Proposition 16.48(ii) with the finiteness theorem of Chapter 17.

*Remark* 16.80 (the road ahead). Chapter 17 proves: Hilbert's theorem 90 and Kummer theory, $H^1(K, \mu_n) = K^\times/(K^\times)^n$ and $H^1(K, \mathbb{Z}_\ell(1)) = \varprojlim K^\times/(K^\times)^{\ell^k}$ (with Theorem 16.45); the Kummer class of the Tate parameter and the value of $c'$ in Proposition 16.67; finiteness and Tate local duality for $G_{\mathbb{Q}_p}$; the unramified classes as exact annihilators of each other; Selmer groups and the comparison sequence behind Lemma lem:dfg21. Chapter 18 defines the Bloch-Kato conditions $H^1_f$ and the Tamagawa factors $\mathrm{Tam}_p$, proves Lemma lem:fpr, and assembles Lemma lem:tamfactor from Proposition 16.67.

## 16.13 Exercises

**Exercise 16.81.** Verify $d^3 d^2 = 0$ on $C^2(G, M)$ by expanding $(d^3 d^2 \psi)(g_1, g_2, g_3, g_4)$; there are 16 terms, which cancel in pairs. *Hint.* Group the terms by the pattern of products of the $g_i$ in the arguments of $\psi$.

**Exercise 16.82.** (a) For $G$ cyclic of order $m$ acting trivially on $\mathbb{Z}/n$, show $H^1(G, \mathbb{Z}/n) \simeq H^2(G, \mathbb{Z}/n) \simeq \mathbb{Z}/\gcd(m, n)$. (b) Let $t$ generate $G = \mathbb{Z}/4$ and act on $\mathbb{Z}[i]$ by multiplication by $i$. Compute $H^0$, $H^1$, $H^2$. *Answer.* (a) $N = m$ and $t - 1 = 0$, so $H^1 = \ker(m \mid \mathbb{Z}/n) = (n/\gcd(m, n))\mathbb{Z}/n\mathbb{Z}$ and $H^2 = (\mathbb{Z}/n)/m(\mathbb{Z}/n) = \mathbb{Z}/\gcd(m, n)$, both cyclic of order $\gcd(m, n)$. (b) $H^0 = 0$; $N = 1 + i + i^2 + i^3 = 0$, so $H^1 = \mathbb{Z}[i]/(i - 1) \simeq \mathbb{Z}/2$ and $H^2 = 0/0 = 0$.

**Exercise 16.83.** Let $\sigma$ be complex conjugation, acting on $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$ by $\zeta \mapsto \zeta^{-1}$. Show $H^0(\mathbb{R}, \mathbb{Z}_\ell(1)) = 0$ and $H^1(\mathbb{R}, \mathbb{Z}_\ell(1)) \simeq \mathbb{Z}_\ell/2\mathbb{Z}_\ell$, which is $\mathbb{Z}/2$ for $\ell = 2$ and 0 for $\ell$ odd; and $H^1(\mathbb{R}, \mathbb{Z}_\ell) = 0$ for every $\ell$ (trivial action). Deduce that for $\ell$ odd $H^1(\mathbb{R}, M) = 0$ for every subquotient $M$ of $\mathrm{ad}^0 T(1)$, in accordance with the paragraph after Lemma lem:dfg21. *Answer.* Theorem 16.34(i): on $\mathbb{Z}_\ell(1)$, $N = 1 + \sigma = 0$ and $(\sigma - 1)\mathbb{Z}_\ell(1) = 2\mathbb{Z}_\ell(1)$; on $\mathbb{Z}_\ell$, $N = 2$ is injective. The last part is Corollary 16.37.

**Exercise 16.84.** (a) Show that $H^2(\mathbb{Z}/2, \mathbb{Z}/2) \simeq \mathbb{Z}/2$ (trivial action). (b) For the extension $0 \to 2\mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ and the section $s(0) = 0$, $s(1) = 1$, compute the cocycle $\psi$ of Theorem 16.23 and show it is not a coboundary. (c) Which extension corresponds to 0? *Answer.* (a) Theorem 16.34(ii): $M^G/NM = (\mathbb{Z}/2)/2(\mathbb{Z}/2)$. (b) $\psi(1, 1) = s(1) + s(1) - s(0) = 2$, the nonzero element, and $\psi$ vanishes elsewhere. For a coboundary, $(d^1\varphi)(0, 0) = \varphi(0)$ and $(d^1\varphi)(1, 1) = 2\varphi(1) - \varphi(0) = \varphi(0)$ in $\mathbb{Z}/2$, so a coboundary takes equal values at $(0, 0)$ and $(1, 1)$; as $\psi(0, 0) = 0 \neq \psi(1, 1)$, $\psi$ is not one. (c) $\mathbb{Z}/2 \times \mathbb{Z}/2$.

**Exercise 16.85.** Let $\widehat{\mathbb{Z}}$ act trivially on $T=\mathbb{Z}_\ell$, $V=\mathbb{Q}_\ell$, $W=\mathbb{Q}_\ell/\mathbb{Z}_\ell$. Compute $H^0$ and $H^1$ of all three, write down the long exact sequence of Proposition 16.44(c), and check that the connecting map $H^0(\widehat{\mathbb{Z}},W)\to H^1(\widehat{\mathbb{Z}},T)$ is zero, in accordance with $H^1(\widehat{\mathbb{Z}},T)_{\mathrm{tor}}=0$. Then do the same with $F$ acting on $T=\mathbb{Z}_\ell(1)$ by $q$, $q\neq\ell$, and check that the connecting map is onto $H^1(\widehat{\mathbb{Z}},\mathbb{Z}_\ell(1))=\mathbb{Z}/\ell^{v_\ell(q-1)}$. *Answer.* Trivial action: $\mathbb{Z}_\ell,\mathbb{Q}_\ell,\mathbb{Q}_\ell/\mathbb{Z}_\ell$ in degree 0 and $\mathbb{Z}_\ell,\mathbb{Q}_\ell,\mathbb{Q}_\ell/\mathbb{Z}_\ell$ in degree 1 (Theorems 16.56 and 16.58), and $H^0(V)\to H^0(W)$ is onto. Second case: $H^0(V)=0$, $H^0(W)=\mu_{\ell^\infty}^{\mathrm{Frob}}\simeq\mathbb{Z}/\ell^{v_\ell(q-1)}$, $H^1(V)=0$. So $\delta$ is injective (its kernel is the image of $H^0(V)=0$) between two groups of the same finite order $\ell^{v_\ell(q-1)}$, hence bijective.

**Exercise 16.86.** Let $\widehat{\mathbb{Z}}$, with generator $F$, act on $\mathbb{Z}_\ell(j)=\mathbb{Z}_\ell(1)^{\otimes j}$ ($j\in\mathbb{Z}$) by $q^j$, where $q\neq\ell$ is a prime. Show that $H^1(\widehat{\mathbb{Z}},\mathbb{Z}_\ell(j))\simeq\mathbb{Z}/\ell^{v_\ell(q^j-1)}$ for $j\neq0$, and compute $v_\ell(q^j-1)$ for $\ell=3$, $q=7$, $j=1,2,3$. *Answer.* Theorem 16.58; $v_3(6)=1$, $v_3(48)=1$, $v_3(342)=2$.

**Exercise 16.87.** Verify the six identities of Lemma 16.66 and $(t-1)^3=0$ on $\mathrm{ad}^0T$ by machine. In PARI/GP (this code was run):

```
e=[0,1;0,0]; h=[1,0;0,-1]; f=[0,0;1,0];
Ad(A,x)=A*x*A^(-1);
R=[1,n;0,1];
[Ad(R,e)-e, Ad(R,h)-h+2*n*e, Ad(R,f)-f-(n*h-n^2*e)]
D(x)=Ad(R,x)-x;
D(D(D(f)))
for(s=-1,1, if(s, F=[s*q,c;0,s]; print([Ad(F,e)-q*e, Ad(F,h)-(h-2*s*c*e), Ad(F,f)-(f
    +s*c*h-c^2*e)/q])))
```

Every output is a zero matrix. Then check the paper's claim that $H^1(I_q,\mathrm{ad}^0T)_{\mathrm{tor}}\simeq(\mathbb{Z}/\ell^{v_\ell(n)})^2$ for $\ell$ odd, by computing the elementary divisors of the $3\times3$ matrix of $t-1$ in the basis $e,h,f$. *Answer.* The columns are $0$, $(-2n',0,0)$, $(-n'^2,n',0)$; for $\ell$ odd the elementary divisors are $n',n',0$ up to units.

**Exercise 16.88.** Redo Proposition 16.67(a) for $\ell=2$. Show that $(t-1)M=n'(2\mathbb{Z}_2e\oplus\mathbb{Z}_2h')$ with $h'=h-n'e$, so that $H^1(I,\mathrm{ad}^0T)_{\mathrm{tor}}\simeq\mathbb{Z}/2^{w+1}\oplus\mathbb{Z}/2^w$, not $(\mathbb{Z}/2^w)^2$; and that for $M=T$ and $\varepsilon=-1$ the invariants are $\{x:2x\equiv0\}\simeq\mathbb{Z}/2$ when $w\geq1$. Where else does the proof of Proposition 16.67 use $\ell$ odd? *Answer.* $e,h',f$ is a basis, and $(t-1)h=-2n'e$, $(t-1)f=n'h'$. In (c) the factor 2 in $2\varepsilon c'q^{-1}y$ is no longer a unit.

**Exercise 16.89.** Let $\ell$ be odd and $\mathrm{End}\,T=\mathrm{ad}^0T\oplus\mathbb{Z}_\ell\cdot1$ in the model setting. Show that $H^1(I,\mathrm{End}\,T)_{\mathrm{tor}}=H^1(I,\mathrm{ad}^0T)_{\mathrm{tor}}$, compatibly with the action of $G/I$, so that the Fitting ideal of the invariants is the same for $\mathrm{End}\,T$ and for $\mathrm{ad}^0T$. *Hint.* The decomposition is $G$-stable for $\ell$ odd ($x=(x-\frac12\operatorname{tr}x)+\frac12\operatorname{tr}x$), and $H^1(I,\mathbb{Z}_\ell)=\mathbb{Z}_\ell$ is torsion-free (Example 16.59(i)).

**Exercise 16.90** (the curve `11a1` at $q=11$, $\ell=5$)**.** The curve $E=$ `11a1` has split multiplicative reduction at 11 with $n=v_{11}(\Delta_E)=5$ ($\Delta_E=-11^5$), Tamagawa number $c_{11}=5$, $j_E=-2^{12}\cdot31^3/11^5$, and its Tate parameter is $q_E=10\cdot11^5+O(11^6)$, so the unit part $u=q_E\,11^{-5}$ satisfies $u\equiv-1\pmod{11}$ (computed with PARI/GP: `ellinit(E,O(11^8)).tate`). (a) Check $u\equiv-1$ by hand from $q_E=j_E^{-1}(1+O(11^5))$, using $2^{12}\equiv4$ and $31^3\equiv3\pmod{11}$. (b) The fifth powers in $\mathbb{F}_{11}^\times$ are $\{\pm1\}$, so $\bar u$ is a fifth power; granting from Chapter 17 that $\min(v_5(n),v_5(q-1),v_5(c'))=\min(v_5(n),v_5(q-1),v_5(c))$ with $v_5(c)=v_5(q-1)=1$ here, deduce from Proposition 16.67 that $(H^1(I_{11},\mathrm{ad}^0T_5E)_{\mathrm{tor}})^{G/I}$ has order 25. (c) Check Proposition 16.67(e) against $c_{11}=5$. (d) The modular degree of `11a1` is 1. Explain why this does not contradict Theorem thm:main. *Answer.*

(b) $w=1$, $v_5(10)=1$, $m=1$, order $5^{1+1}$. (c) $\varepsilon=1$, invariants $\mathbb{Z}/5$. (d) Theorem thm:main requires $E[5]$ irreducible, and $E(\mathbb{Q})\simeq\mathbb{Z}/5$, so $E[5]$ is reducible; the computation illustrates Lemma lem:tamfactor, which needs no irreducibility.

```
E = ellinit([0,-1,1,-10,-20], O(11^8));
Q = E.tate[3];  \\ Tate parameter q_E
[valuation(Q,11), (Q/11^5) % 11]   \\ returns [5, 10]
```

**Exercise 16.91.** Let $G$ be finite, $P$ a Sylow $\ell$-subgroup and $M$ a $G$-module. Show that $\mathrm{Res}\colon H^1(G,M)\to H^1(P,M)$ is injective on the $\ell$-primary part of $H^1(G,M)$. This is the step "by restriction and corestriction" in the proof of Lemma lem:tamadd. *Hint.* $\mathrm{Cor}\circ\mathrm{Res}=[G:P]$ is prime to $\ell$ (Proposition 16.28(d), proved in degree 1).

**Exercise 16.92.** Let $C_3=\langle\sigma\rangle$. (a) Show $H^1(C_3,\mathbb{Z}_3[C_3])=0$, using Theorem 16.34. (b) Let $T=\mathbb{Z}_3^2$ with $\sigma$ acting with characteristic polynomial $X^2+X+1$. Show $H^1(C_3,T)=T/(\sigma-1)T\simeq\mathbb{F}_3$. These are the two facts behind Lemma lem:tamadd at $\ell=3$ and the sentence after it, "then $H^1(\langle\sigma\rangle,T)=T/(1-\zeta_3)T\simeq\mathbb{F}_3$". *Answer.* (a) On $\mathbb{Z}_3[C_3]$, $\ker N$ is the augmentation ideal, which equals $(\sigma-1)\mathbb{Z}_3[C_3]$. (b) $N=1+\sigma+\sigma^2=0$ on $T$, and $T\simeq\mathbb{Z}_3[\zeta_3]$ with $\sigma=\zeta_3$ (Chapter 6), so $T/(\sigma-1)T=\mathbb{Z}_3[\zeta_3]/(\zeta_3-1)\simeq\mathbb{F}_3$.

**Exercise 16.93.** Let $\chi\colon G\to\mathbb{Z}_\ell^\times$ be a character with $\chi(G)\subseteq 1+\ell^h\mathbb{Z}_\ell$ but $\chi(G)\not\subseteq 1+\ell^{h+1}\mathbb{Z}_\ell$, $T=\mathbb{Z}_\ell(\chi)$ and $W=T\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$. Show that $H^0(G,T/\ell^kT)$ is all of $T/\ell^kT$ for $k\le h$ and has no element of order $\ell^k$ for $k>h$, so that $H^0(G,W)\simeq\mathbb{Z}/\ell^h$ by Lemma 16.25(d). *Answer.* For $k\le h$, $g$ acts on $T/\ell^kT=\mathbb{Z}/\ell^k$ by $\chi(g)\equiv 1\pmod{\ell^k}$, trivially. For $k>h$ pick $g$ with $v_\ell(\chi(g)-1)=h$; for $x\notin\ell T$, $(\chi(g)-1)x$ has valuation $h<k$, so $gx\neq x$ in $T/\ell^kT$. By Lemma 16.25(b), $H^0(G,W)[\ell^{h+1}]\simeq H^0(G,T/\ell^{h+1}T)$ has no element of order $\ell^{h+1}$, so $H^0(G,W)$ is killed by $\ell^h$ and lies in the finite group $W[\ell^h]$; and $\#H^0(G,T/\ell T)\le\#(T/\ell T)=\ell$. So (d) applies, with largest $k$ equal to $h$.

**Exercise 16.94.** Let $p\neq\ell$ and let $\Gamma=\mathbb{Z}_p$ act on $M$ of type (C) through an element $u$ of order $p^a$. Show that $H^1(\Gamma,M)=0$, that $\ker N/(u-1)M=0$ for $N=\sum_{i<p^a}u^i$, and that $M/(u-1)M\neq 0$ in general (take $u=1$). So the formula of Theorem 16.58 needs the surjection onto $\mathbb{Z}_\ell$. *Hint.* Theorem 16.51(b); and Theorem 16.34 for the finite quotient, whose order is a unit of $\mathbb{Z}_\ell$.

**Exercise 16.95** (harder)**.** Prove Theorem 16.31(b). Let $c\in Z^1(H,M)$ have $G/H$-invariant class. (a) Show that $c$ can be changed by a coboundary and extended to a cochain $\tilde c$ on $G$ with $\tilde c(gh)=\tilde c(g)+gc(h)$ and $\tilde c(hg)=c(h)+h\tilde c(g)$ for $g\in G$, $h\in H$. (b) Show that $d^1\tilde c$ takes values in $M^H$ and factors through $G/H\times G/H$, defining a 2-cocycle whose class $\mathrm{tg}[c]\in H^2(G/H,M^H)$ is well defined. (c) Prove exactness at $H^1(H,M)^{G/H}$ and at $H^2(G/H,M^H)$. *Hint.* [143, Chapter I, Section 6]; for (a) fix coset representatives and use the invariance $\sigma\cdot c-c=d^0m_\sigma$.

## 16.14 Notes and references

The homological algebra of Sections 16.2 and 16.3 follows Weibel [205]: the long exact sequence, the snake lemma, $\mathrm{Ext}^1$ and extensions, and the Lyndon-Hochschild-Serre spectral sequence are in his Sections 1.3, 3.4 and 6.8; we recall the theorem numbers 1.3.1, 1.3.2, 3.4.3, 3.5.8 and 6.8.2 from memory and have not checked them against a printed copy. Group cohomology by cochains, restriction, corestriction, inflation and the conjugation action follow Brown [17, Chapters III to VI], Serre's *Local Fields* [174, Chapters VII and VIII], and Atiyah and Wall's chapter in

Cassels-Frohlich [26, Chapter IV]; the chapter and section pointers to these books are likewise unverified against the printed editions. The explicit cochain $\psi$ in the proof of Theorem 16.29 avoids the corestriction in higher degrees.

Continuous cohomology of profinite groups with discrete coefficients is in Serre's *Galois Cohomology* [177, Chapter I] and in Neukirch, Schmidt and Wingberg [143, Chapters I and II]. For compact coefficients the standard reference is Tate [196, Section 2], where the $\varprojlim^1$ sequence of Theorem 16.45 is proved; Rubin [167, Appendix B] gives a summary adapted to Galois representations. Our proof of Theorem 16.58 is direct: it constructs the cocycle from the geometric sums of Lemma 16.57 instead of passing to the limit over $M/\ell^k M$, so that the identity the paper uses does not depend on the stated part of Tate's theorem. The procyclic computation with discrete coefficients is Serre's [174, Chapter XIII, Section 1] (pointer unverified). The supernatural order and the maximal pro-$\ell$ quotient are in [177, Chapter I, Section 1]; Milne [133, Chapter II] is a readable alternative for inflation-restriction and Tate cohomology.

The spectral sequence of Theorem 16.33 was found by Lyndon [121] and in general form by Hochschild and Serre [84]; only its five-term exact sequence is used in the book, and only through Corollary 16.52(b) and Proposition 16.53(iii), neither of which the paper needs.

The definition of the Tamagawa factor as the Fitting ideal of the $G_{\mathbb{Q}_p}$-invariants of the torsion of $H^1(I_p, M)$, which Section 16.10 computes, is Fontaine and Perrin-Riou's [69, I.4.2.2], as recalled by Diamond, Flach and Guo [49, p. 708]. The shape of the Tate representation is [184, Chapter V] and [38, Theorem 3.1(e)]; the corner $\varepsilon\kappa$ is the paper's. The conjugation action is the paper's $(\sigma \cdot z)(i) = \sigma z(\sigma^{-1} i \sigma)$; with the other common convention the Frobenius acts on $H^1(I, M)$ by $q\rho_M(\mathrm{Frob})^{-1}$ (Remark 16.65).

# Chapter 17

# Galois cohomology, local and global, and Selmer groups

Chapter 16 built the machinery of group cohomology: cochains, inflation and restriction, continuous cohomology of profinite groups, procyclic groups and the group-theoretic core of Lemma lem:tamfactor. This chapter specialises it to Galois groups of fields, local and global, and develops the arithmetic the paper uses: Hilbert's Theorem 90 and Kummer theory in cohomological form; the cohomology of a lattice $T$, of $V = T \otimes \mathbb{Q}_\ell$ and of $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and their Kummer duals $T^D$, $V^D$, $W^D$; Tate's local duality and Euler characteristic formula; unramified cohomology and the exact orthogonality of unramified classes; the Kummer class of the Tate parameter, which identifies the corner entry of the Frobenius matrix in Lemma lem:tamfactor; the groups $G_{\mathbb{Q},S}$ and the real place; the classical Selmer group of an elliptic curve; Selmer structures and their duals; the Poitou-Tate sequence and the formula of Greenberg and Wiles; and the exact sequence comparing two Selmer structures, of which Lemma lem:dfg21 (Lemma 2.1 of Diamond, Flach and Guo) is a special case. The deep duality theorems are stated with references; everything the paper uses on top of them is proved. Chapter 18 then supplies the Bloch-Kato local conditions, the Tamagawa factors and the theorem of Diamond, Flach and Guo that the Selmer group of $V^D$ vanishes. The chapter uses Chapter 3 for $\mathbb{Z}_\ell$-modules and Pontryagin duality, Chapter 5 for infinite Galois theory and Kummer characters, Chapter 6 for decomposition and inertia groups, Chapter 7 for local fields, the tame character and Kummer cocycles, Chapters 9 to 11 for elliptic curves, the Tate curve and the shape of $T_\ell E$ at a multiplicative prime, and Chapter 16 throughout.

## Core results at a glance

Throughout, $\ell$ is a prime different from the characteristic, $T$ a free $\mathbb{Z}_\ell$-module of finite rank with a continuous Galois action, $V = T \otimes \mathbb{Q}_\ell$, $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $\iota\colon H^1(-,T) \to H^1(-,V)$, $\pi\colon H^1(-,V) \to H^1(-,W)$; Frob is an arithmetic Frobenius and $\Phi = \mathrm{Frob}^{-1}$; $^*$ is the Pontryagin dual and $^D$ the Kummer dual.

**17.1 Conventions.** $H^n(K,M) = H^n(G_K,M)$, continuous cochains, coefficients of type (D), (C), (V); Frob arithmetic in all cohomology, $\Phi$ only in $L$-factors, $L_p(V,s) = \det(1-\Phi p^{-s} \mid V^{I_p})^{-1}$; the Kummer connecting map is $a \mapsto [\sigma \mapsto \sigma(\alpha)/\alpha]$, $\alpha^n = a$, with no sign. Conventions 17.1 to 17.3.

**17.2 Galois modules.** $\overline{K}$, $\overline{K}^\times$, $\mu_n$, $\mathbb{Z}/n(r)$, $E[n]$, $E(\overline{K})$, $\mu_{\ell^\infty}$ are discrete; $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$ (action by $\chi_\ell$), $T_\ell E$ are of type (C); $G_{\mathbb{F}_q} \simeq \hat{\mathbb{Z}}$. Example 17.4.

**17.3 Hilbert 90.** For every Galois extension $L/K$ (finite or not), $H^1(\mathrm{Gal}(L/K), L^\times) = 0$ and $H^1(\mathrm{Gal}(L/K), L) = 0$; so $H^1(K, \overline{K}^\times) = H^1(K, \overline{K}) = 0$. *(proved here; $H^n(K, \overline{K}) = 0$ for $n \geq 2$ stated)* Theorem 17.5.

**17.4 Kummer theory.** For $n$ prime to $\mathrm{char}\, K$: $\delta\colon K^\times/(K^\times)^n \xrightarrow{\sim} H^1(K, \mu_n)$, $\delta(a) = [\sigma \mapsto \sigma(\alpha)/\alpha]$, natural in $K$; $H^2(K, \mu_n) \simeq \mathrm{Br}(K)[n]$; if $\mu_n \subset K$, $\delta(a)$ is the Kummer character. *(proved here; compatibility with corestriction stated)* Theorem 17.7.

**17.5 Kummer class, Kummer cocycle** (definition): $\delta(a)$, a cocycle $\sigma \mapsto \sigma(\alpha)/\alpha$, and the $\ell$-adic cocycle $\kappa_a(\sigma) = (\sigma(a^{1/\ell^k})/a^{1/\ell^k})_k \in \mathbb{Z}_\ell(1)$. Definition 17.9.

**17.6 Properties.** $\delta_K(a)|_{G_L} = 0$ iff $a \in (L^\times)^n$; for $\mu_n \subset K$, $\ker \chi_a = G_{K(\alpha)}$ and $[K(\alpha) : K]$ is the order of $a$ in $K^\times/(K^\times)^n$; $\kappa_a$ reduces modulo $\ell^k$ to a Kummer cocycle. *(proved here)* Proposition 17.10.

**17.7 Finite fields.** $H^1(\mathbb{F}_q, \mu_{\ell^k}) \simeq \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^k} \simeq \mu_{\ell^k}/\mu_{\ell^k}^{q-1}$, $u \mapsto u^{(q-1)/\ell^k}$ $(k \leq v = v_\ell(q-1))$; $H^1(\mathbb{F}_q, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^v$; $v_\ell$ of a class, with $v_\ell(0) = v$. *(proved here, given Chapter 16)* Example 17.11.

**17.8 Brauer groups.** $\mathrm{Br}(\mathbb{F}_q) = \mathrm{Br}(\mathbb{C}) = 0$, $\mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2$; $\mathrm{inv}_K\colon \mathrm{Br}(K) \simeq \mathbb{Q}/\mathbb{Z}$ for $K/\mathbb{Q}_p$ finite, so $H^2(K, \mu_n) \simeq \mathbb{Z}/n$; the sequence $0 \to \mathrm{Br}(\mathbb{Q}) \to \bigoplus_v \mathrm{Br}(\mathbb{Q}_v) \xrightarrow{\sum \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0$ is exact. *(stated; reference)* Theorem 17.12.

**17.9 Lattices and duals** (convention and definition): $W[\ell^k] \simeq T/\ell^k T$; $M^D = \mathrm{Hom}(M, \mu)$ for finite $M$; $T^D = \mathrm{Hom}(T, \mathbb{Z}_\ell(1))$, $V^D = T^D \otimes \mathbb{Q}_\ell$, $W^D = V^D/T^D$. Convention 17.14 and Definition 17.15.

**17.10 Kummer duality.** The pairings $T \times T^D \to \mathbb{Z}_\ell(1)$, $V \times V^D \to \mathbb{Q}_\ell(1)$, $W \times T^D \to \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)$, $T \times W^D \to \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)$, $M \times M^D \to \mu$ are perfect and equivariant; $W^D \simeq T^*(1)$, $(T/\ell^k)^D \simeq T^D/\ell^k$, $(T^D)^D = T$; $E[n]^D \simeq E[n]$ by the Weil pairing. *(proved here)* Proposition 17.16.

**17.11 The cohomology of $T$, $V$, $W$.** If $H^1(G, T/\ell)$ and $H^2(G, T/\ell)$ are finite: $H^n(G, T)$ is finitely generated $(n \leq 2)$ and $H^n(G, V) = H^n(G, T) \otimes \mathbb{Q}_\ell$; $H^1(G, T)_{\mathrm{tors}} = \ker \iota = \delta H^0(G, W) \simeq H^0(G, W)/H^0(G, W)_{\mathrm{div}}$, and $\simeq H^0(G, W)$ if $H^0(G, V) = 0$; $0 \to H^1(G, T) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \to H^1(G, W) \to H^2(G, T)_{\mathrm{tors}} \to 0$ with the first term $= \mathrm{im}\, \pi = H^1(G, W)_{\mathrm{div}}$; the finite-level sequences. *(proved here, given Chapter 16)* Proposition 17.17.

**17.12 $\ell$-adic Kummer theory.** $H^1(K, \mathbb{Z}_\ell(1)) \simeq \widehat{K^\times} = \varprojlim K^\times/(K^\times)^{\ell^k}$, $a \mapsto \kappa_a$; $H^1(K, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) \simeq K^\times \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. *(proved here, given Chapter 16)* Theorem 17.20.

**17.13 Local Kummer theory** ($K/\mathbb{Q}_p$ finite, $\ell \neq p$, residue field $\mathbb{F}_q$, $v = v_\ell(q-1)$): $0 \to \mathcal{O}_K^\times \otimes \mathbb{Z}_\ell \to \widehat{K^\times} \xrightarrow{v_K} \mathbb{Z}_\ell \to 0$, so $H^1(K, \mathbb{Z}_\ell(1)) \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^v$ with torsion the unit classes; $\#H^1(K, \mu_{\ell^k}) = \ell^{k+\min(k,v)}$; $H^1(K, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) \simeq \mathbb{Q}_\ell/\mathbb{Z}_\ell$; $H^1(K, \mathbb{Q}_\ell(1)) \simeq \mathbb{Q}_\ell$. *(proved here)* Proposition 17.21.

**17.14 Local finiteness.** For $K/\mathbb{Q}_p$ finite and $M$ finite, $H^n(K, M)$ is finite and 0 for $n \geq 3$; hence $H^n(K, T)$ is finitely generated and Proposition 17.17 applies to $G_K$. *(stated; reference; sketched for $H^1$ with $p \nmid \#M$)* Theorem 17.22.

**17.15 Local Euler characteristic.** $\#H^0(K,M)\#H^2(K,M)/\#H^1(K,M) = |\#M|_K$; for $V$: $\sum(-1)^i \dim H^i(K,V) = -[K:\mathbb{Q}_p]\dim V$ if $\ell = p$, 0 if $\ell \neq p$. *(stated; reference)* Theorem 17.23.

**17.16 Tate local duality.** $H^i(K,M) \times H^{2-i}(K,M^D) \to H^2(K,\mu) \simeq \mathbb{Q}/\mathbb{Z}$ is perfect; $H^i(K,W) \times H^{2-i}(K,T^D) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $H^i(K,T) \times H^{2-i}(K,W^D) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$ and $H^i(K,V) \times H^{2-i}(K,V^D) \to \mathbb{Q}_\ell$ are perfect; $\langle \pi x, y\rangle = \langle x, \iota y\rangle \bmod \mathbb{Z}_\ell$. *(stated; reference)* Theorem 17.24.

**17.17 Consequences.** $H^2(K,M) \simeq H^0(K,M^D)^*$; for $p \nmid \#M$, $\#H^1(K,M) = \#H^0(K,M)\#H^0(K,M^D)$; for $\ell \neq p$, $\dim H^1(K,V) = \dim H^0(K,V) + \dim H^0(K,V^D)$ and $H^2(K,T) \simeq H^0(K,W^D)^*$. *(proved here, given Theorems 17.23 and 17.24)* Corollary 17.25.

**17.18 Orders over $\mathbb{Q}_p$.** $\#H^i(\mathbb{Q}_p,\mathbb{Z}/\ell^k) = \ell^k, \ell^{k+m}, \ell^m$ and $\#H^i(\mathbb{Q}_p,\mu_{\ell^k}) = \ell^m, \ell^{k+m}, \ell^k$ ($m = \min(k, v_\ell(p-1))$, $p \neq \ell$); $\mu_p$ over $\mathbb{Q}_p$ has characteristic $p^{-1}$; $E[5] \subset E(\mathbb{Q}_{11})$ for `11a1`. *(proved here, given the stated theorems)* Example 17.26.

**17.19 Unramified classes** (definition; $K/\mathbb{Q}_p$ finite): $H^1_{\mathrm{ur}}(K,M) = \mathrm{im}(\mathrm{Inf}\colon H^1(G_K/I_K, M^{I_K}) \to H^1(K,M)) = \ker(H^1(K,M) \to H^1(I_K,M)) \simeq H^1(G_K/I_K, M^{I_K})$. Definition 17.27.

**17.20 Computation.** $H^1_{\mathrm{ur}}(\mathbb{Q}_p,M) \simeq M^I/(\mathrm{Frob}-1)M^I$; $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p,M) = \#H^0(\mathbb{Q}_p,M)$ for finite $M$; $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V) = 0$ iff $H^0(\mathbb{Q}_p,V) = 0$ iff $\det(1-\mathrm{Frob} \mid V^I) \neq 0$ iff $\det(1-\Phi \mid V^I) \neq 0$; for $p \neq \ell$, $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p,T) = \ell^{v_\ell \det(1-\mathrm{Frob}|V^I)} = \ell^{v_\ell \det(1-\Phi|V^I)}$ when finite. *(proved here, given Chapter 16)* Proposition 17.28.

**17.21 The inertia sequence.** $0 \to H^1_{\mathrm{ur}}(\mathbb{Q}_p,M) \to H^1(\mathbb{Q}_p,M) \to H^1(I,M)^{G/I} \to 0$ is exact (conjugation action), for $M$ of type (D), and for types (C), (V) at $p \neq \ell$. *(proved here, given Chapter 16 and Theorem 17.22; types (C), (V) by passage to the limit)* Proposition 17.30.

**17.22 Exact orthogonality** ($p \neq \ell$). $H^1_{\mathrm{ur}}(\mathbb{Q}_p,M)$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p,M^D)$ are exact annihilators ($M$ finite of $\ell$-power order); so are $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V)$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V^D)$. False for $p = \ell$. *(proved here, given Theorems 17.23 and 17.24)* Theorem 17.32.

**17.23 Propagation.** For $L_V \subseteq H^1(K,V)$: $L_T = \iota^{-1}L_V \supseteq H^1(K,T)_{\mathrm{tors}}$, $L_W = \pi L_V$ divisible; if $L_V^\perp = L_{V^D}$ then $L_W^\perp = L_{T^D}$ and $(H^1(K,W)/L_W)^* \simeq L_{T^D}$; for $T$ unramified, $p \neq \ell$, the propagation of $H^1_{\mathrm{ur}}$ is $H^1_{\mathrm{ur}}$. *(proved here, given Theorem 17.24)* Proposition 17.34.

**17.24 Vanishing unramified cohomology.** If $p \neq \ell$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V) = 0$: $H^0(\mathbb{Q}_p,V) = 0$, $\iota^{-1}(0) = H^1(\mathbb{Q}_p,T)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}_p,W)$ is finite and contains $H^1_{\mathrm{ur}}(\mathbb{Q}_p,T)$, and $\pi(0) = 0$. (The paper: "$H^1_f(\mathbb{Q}_p,T^D)$ is the torsion of $H^1(\mathbb{Q}_p,T^D)$, a finite group".) *(proved here, given Theorem 17.22)* Corollary 17.36.

**17.25 Examples.** $H^1_{\mathrm{ur}}(\mathbb{Q}_p,\mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(p-1)$; $H^1(\mathbb{Q}_p,\mathbb{Z}_\ell) = H^1_{\mathrm{ur}}$; Tate curve: $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V_\ell E) = 0$; $\mathrm{ad}^0 V_\ell E$ at a multiplicative prime: $\det(1-\Phi \mid (V^D)^I) = 1-p^{-2}$, $\det(1-\mathrm{Frob} \mid (V^D)^I) = 1-p^2$, $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p,T^D) = \ell^{v_\ell(p^2-1)}$; types III, III*: $\ell^{v_\ell(q-\chi_{-4}(q))}$. *(proved here, given Chapter 11; the last an illustration depending on Chapter 19)* Example 17.38.

**17.26 Unit classes.** For $K/\mathbb{Q}_p$ finite, $p \nmid n$: the Kummer isomorphism maps $\mathcal{O}_K^\times$ onto $H^1_{\mathrm{ur}}(K,\mu_n)$; the valuation sequence of $\widehat{K^\times}$ is the inertia sequence for $\mathbb{Z}_\ell(1)$, with $\mathrm{Res}\,\kappa_x = v_K(x)t_\ell$ and $\mathrm{Hom}(I_K,\mathbb{Z}_\ell(1))^{G/I} = \mathbb{Z}_\ell t_\ell$. *(proved here)* Proposition 17.39.

**17.27 Value of an unramified class.** $H^1(G_{\mathbb{Q}_q}/I_q,\mathbb{Z}_\ell(1)) \simeq \mathbb{Z}_\ell(1)/(q-1)$; for $u \in \mathbb{Z}_q^\times$, $\kappa_u$ is inflated as a cocycle and $\kappa_u(\mathrm{Frob})$ is the class of $\bar{u}$ in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$. *(proved here)* Proposition 17.40.

**17.28 The splitting given by $q$.** $H^1(\mathbb{Q}_q,\mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell[\kappa_q] \oplus H^1_{\mathrm{ur}}(\mathbb{Q}_q,\mathbb{Z}_\ell(1))$ canonically; $\mathrm{pr}_{\mathrm{ur}}[\kappa_x]$ is the class of the unit part of $x$. *(proved here)* Proposition 17.41.

**17.29 The corner of $\rho$(Frob).** For $E$ multiplicative at $q \neq \ell$, $\ell$ odd, with $\rho = \left(\begin{smallmatrix}\chi_\ell\varepsilon & \varepsilon\kappa\\ 0 & \varepsilon\end{smallmatrix}\right)$: $\kappa = n\kappa_q + \kappa_u$, $\mathrm{pr}_{\mathrm{ur}}[\kappa]$ is the class of $\bar{u}$; $\kappa|_I = nt_\ell$ (so $n' = n$); for the lift fixing the roots of $q$, $c' = \varepsilon\kappa_u(\mathrm{Frob})$, of class $\pm$ the class of $\bar{u}$; in general $c'$ is defined modulo $(n, q-1)\mathbb{Z}_\ell$; $\min(v_\ell(n), v_\ell(q-1), v_\ell(c')) = \min(v_\ell(n), v_\ell(q-1), v_\ell(c))$. *(proved here, given Chapters 7 and 11)* Theorem 17.43.

**17.30 The formula of Lemma lem:tamfactor** (a pointer). $\mathrm{length}((H^1(I,\mathrm{ad}^0 T)_{\mathrm{tors}})^{G/I}) = v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c))$: the model computation of Chapter 16 with the identification of $n'$, $c'$ above; the lemma is assembled in Chapter 18. *(proved here, given Chapter 16)* Corollary 17.44.

**17.31 Tate parameters at $\ell = 3$.** `77c1`, `791a1`, `1577a1` at $q = 7, 7, 19$: extra term $0, 1, 1$, bound $1, 2, 2 = v_3(\deg \varphi_E)$; `11a1` at $\ell = 5$: Eisenstein. *(computed)* Example 17.46.

**17.32 Lemma lem:tamagawa-exp, cohomologically.** $\rho \bmod \ell^t$ is unramified at $q$ iff $nt_\ell \equiv 0 \pmod{\ell^t}$ iff $\ell^t \mid n$ iff $q_E$ is an $\ell^t$-th power in $\mathbb{Q}_q^{\mathrm{ur}}$. *(proved here)* Remark 17.47.

**17.33** $G_{\mathbb{Q},S}$ (definition): $G_{\mathbb{Q}}/H_S$, $H_S$ generated by the inertia groups above $p \notin S$; localisation $\mathrm{Res}_v$ is independent of the embedding. Definition 17.48.

**17.34 Unramified outside $S$.** $H^1(G_{\mathbb{Q},S},M) = \{c \in H^1(\mathbb{Q},M) : \mathrm{Res}_p\, c \in H^1_{\mathrm{ur}}\ \forall p \notin S\}$; for finite $M$ every class is unramified almost everywhere. *(proved here)* Proposition 17.49.

**17.35 Finiteness for $G_{\mathbb{Q},S}$.** $H^1(G_{\mathbb{Q},S},M)$ and $H^2(G_{\mathbb{Q},S},M)$ are finite for finite $M$; so Proposition 17.17 applies to $G_{\mathbb{Q},S}$. *($H^1$ sketched, given Hermite-Minkowski; $H^2$ and* cd *stated; reference)* Theorem 17.50.

**17.36 The real place.** $2H^n(\mathbb{R},M) = 0$; for $\ell$ odd $H^n(\mathbb{R},M) = 0$ $(n \geq 1)$ and $\hat{H}^0(\mathbb{R},M) = 0$; so $H^1_f(\mathbb{R},T^D) = 0$; $H^1(\mathbb{R},\mu_2) = \mathbb{Z}/2$. *(proved here)* Proposition 17.51.

**17.37 Global Kummer theory.** $H^1(G_{\mathbb{Q},S},\mu_n) \simeq \mathbb{Z}_S^\times/(\mathbb{Z}_S^\times)^n$, finite. *(proved here)* Example 17.53.

**17.38 Kummer sequence of $E$.** $0 \to E(K)/nE(K) \xrightarrow{\delta} H^1(K,E[n]) \to H^1(K,E)[n] \to 0$, $\delta(P) = [\sigma \mapsto \sigma Q - Q]$. *(proved here)* Proposition 17.55.

**17.39 Local Kummer image** $(p \nmid n)$. $\#E(\mathbb{Q}_p)/n = \#E(\mathbb{Q}_p)[n] = \#H^1_{\mathrm{ur}}(\mathbb{Q}_p,E[n])$; for good reduction the image is $H^1_{\mathrm{ur}}$; it is its own annihilator (Tate); for $W = E[\ell^\infty]$ at a multiplicative prime, $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p,W)$ is the $\ell$-part of $c_p$. *(proved here; the self-duality stated)* Proposition 17.56.

**17.40 Selmer and Sha** (definition and proposition): $\mathrm{Sel}_n(E/\mathbb{Q})$, $\mathrm{Sha}(E/\mathbb{Q})$, and $0 \to E(\mathbb{Q})/n \to \mathrm{Sel}_n(E/\mathbb{Q}) \to \mathrm{Sha}(E/\mathbb{Q})[n] \to 0$. *(proved here)* Definition 17.57 and Proposition 17.58.

**17.41 Finiteness; weak Mordell-Weil.** $\mathrm{Sel}_n(E/\mathbb{Q}) \subseteq H^1(G_{\mathbb{Q},S}, E[n])$ is finite; $E(\mathbb{Q})/nE(\mathbb{Q})$ is finite. *(proved here, given Theorem 17.50)* Theorem 17.59 and Corollary 17.60.

**17.42 Selmer structures** (definition): local conditions $\mathcal{L}_v$, unramified almost everywhere; $H^1_{\mathcal{L}}(\mathbb{Q}, M)$; relaxed $\mathcal{L}^\Sigma$, strict $\mathcal{L}_\Sigma$, classical and propagated structures. Definition 17.62 and Example 17.63.

**17.43 Dual structures** (definition and proposition): $\mathcal{L}_v^* = \mathcal{L}_v^\perp$ on $M^D$, or on $T^D$ for structures on $W$; $\mathcal{L}^*$ is a Selmer structure, $\mathcal{L}^{**} = \mathcal{L}$, $(\mathcal{L}^\Sigma)^* = (\mathcal{L}^*)_\Sigma$, and the dual of a propagated structure is propagated. *(proved here, given Theorem 17.24)* Definition 17.64 and Proposition 17.65.

**17.44 $\ell$-adic Selmer groups.** $H^1_{\mathcal{L}}(\mathbb{Q}, W)$ is cofinitely generated; if $H^1_{L_V}(\mathbb{Q}, V) = 0$ then $H^1_{\mathcal{L}}(\mathbb{Q}, T) = H^1(G_{\mathbb{Q},S}, T)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}, W)/\mathrm{div}$; $\operatorname{corank} H^1_{\mathcal{L}}(\mathbb{Q}, W) = \dim H^1_{L_V}(\mathbb{Q}, V)$. *(proved here, given Theorem 17.50; the corank formula stated)* Proposition 17.66.

**17.45 Poitou-Tate.** The nine-term exact sequence for $M$ finite unramified outside $S$; the image of $H^1(G_S, M)$ in $\bigoplus_{v\in S} H^1(\mathbb{Q}_v, M)$ is the exact annihilator of that of $H^1(G_S, M^D)$. *(stated; reference)* Theorem 17.68.

**17.46 Reciprocity.** $\sum_v \mathrm{inv}_v(\mathrm{Res}_v\, x \cup \mathrm{Res}_v\, y) = 0$ for $x \in H^1(\mathbb{Q}, M)$, $y \in H^1(\mathbb{Q}, M^D)$. *(proved here, given Theorem 17.12)* Proposition 17.69.

**17.47 Greenberg-Wiles.** $\#H^1_{\mathcal{L}}(\mathbb{Q}, M)/\#H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D) = \frac{\#H^0(\mathbb{Q},M)}{\#H^0(\mathbb{Q},M^D)} \prod_v \frac{\#\mathcal{L}_v}{\#H^0(\mathbb{Q}_v,M)}$. *(stated; reference; finiteness proved)* Theorem 17.70.

**17.48 A check by hand.** For $M = \mathbb{Z}/\ell$ unramified everywhere both sides are 1; relaxed at $p$, the formula counts the characters of order $\ell$ and conductor $p$. *(proved here, given the stated theorems)* Example 17.71.

**17.49 Comparison of orders.** For $\mathcal{L} \subseteq \mathcal{L}'$: $\#H^1_{\mathcal{L}'}/\#H^1_{\mathcal{L}} = \prod_v \#\mathcal{L}'_v/\#\mathcal{L}_v \cdot \#H^1_{\mathcal{L}'^*}/\#H^1_{\mathcal{L}^*}$. *(proved here, given Theorem 17.70)* Corollary 17.72.

**17.50 The comparison sequence.** $0 \to H^1_{\mathcal{L}} \to H^1_{\mathcal{L}'} \to \bigoplus_v \mathcal{L}'_v/\mathcal{L}_v \to (H^1_{\mathcal{L}^*})^* \to (H^1_{\mathcal{L}'^*})^* \to 0$ is exact ($M$ finite). *(complex property proved here; exactness proved here, given Theorems 17.68 and 17.70)* Theorem 17.73.

**17.51 The comparison sequence for $(W, T^D)$.** The same with $W$ and the dual structures on $T^D$. *(sketched here; reference)* Theorem 17.74.

**17.52 Relaxing at $\Sigma$** (the abstract form of DFG04, Lemma 2.1). For propagated structures with $L^\perp_{V,v} = L_{V^D,v}$: the five-term sequence $0 \to H^1_{\mathcal{L}}(\mathbb{Q}, W) \to H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) \to \bigoplus_{p\in\Sigma}(\mathcal{L}^*_p)^* \to H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D)^* \to H^1_{(\mathcal{L}^*)_\Sigma}(\mathbb{Q}, T^D)^* \to 0$; if $H^1_{L_{V^D}}(\mathbb{Q}, V^D) = 0$ and $H^0(\mathbb{Q}_p, V^D) = 0$ for one $p \in \Sigma$, then $0 \to H^1_{\mathcal{L}}(\mathbb{Q}, W) \to H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) \to \bigoplus_{p\in\Sigma\cup\{\infty\}}(\mathcal{L}^*_p)^* \to H^0(\mathbb{Q}, W^D)^* \to 0$. *(proved here, given Theorem 17.74)* Theorem 17.75.

**17.53 The length identity.** $\operatorname{length} H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) = \operatorname{length} H^1_{\mathcal{L}}(\mathbb{Q}, W) + \sum_{p\in\Sigma} \operatorname{length} \mathcal{L}^*_p - \operatorname{length} H^0(\mathbb{Q}, W^D)$. *(proved here, given Theorem 17.75)* Corollary 17.77.

**17.54 A toy case.** $T = \mathbb{Z}_\ell$, $\Sigma = \{p\}$: the sequence identifies the characters of $\ell$-power order unramified outside $p$ with $(\mathbb{Z}_p^\times \otimes \mathbb{Z}_\ell)^*$. *(proved here, given the stated theorems)* Example 17.79.

**17.55 What Chapter 18 adds.** The Bloch-Kato conditions and their orthogonality at $\ell$, Theorem thm:dfg27, Lemma lem:fpr, and the value of the Euler factor. Remark 17.80.

## Where the paper uses this chapter

- *Lemma lem:tamfactor (the Tamagawa factor at a multiplicative prime), the end of its proof*: the group $H^1(G, \mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$ (Theorem 17.20 and Proposition 17.21); the splitting $\mathbb{Z}_\ell\kappa_q \oplus H^1_{\mathrm{ur}}$ and the class of the unit part (Propositions 17.39 to 17.41); the value $c'$ and its indeterminacy (Theorem 17.43); the assembled formula (Corollary 17.44, with Chapters 16 and 18); its check with $M = T$ (Proposition 17.56(iv)).

- *Lemma lem:tamagawa-exp*: "By Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$" (Proposition 17.10 and Remark 17.47).

- *The paragraph after Lemma lem:dfg21*: $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ and its vanishing (Proposition 17.28, Warning 17.29); "$H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group" (Proposition 17.17, Corollary 17.36); "$H^1(\mathbb{R}, T^D)$ is killed by 2" (Proposition 17.51).

- *Lemma lem:dfg21 (DFG04, Lemma 2.1)*: Tate local duality and propagation (Theorem 17.24 and Proposition 17.34), the exact orthogonality at $p \neq \ell$ (Theorem 17.32), Selmer structures and their duals (Definition 17.62 and Propositions 17.65 and 17.66), and the comparison sequences (Theorems 17.73 to 17.75).

- *The proof of Theorem thm:main and Remark rem:ns*: the length identity (Corollary 17.77), with $H^0(\mathbb{Q}, W^D) = 0$, resp. of length $h_3(E) = 1$.

- *The Selmer groups of Theorem thm:main and Section 4.2*: $H^1_f(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ and $H^1_\Sigma(\mathbb{Q}, W)$ as Selmer groups of a structure and its relaxation (Example 17.63).

- *Section 1.2 (*$\mathrm{Tam}_p$*) and Lemma lem:fpr, through Chapter 18*: the inertia sequence and the order of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T^D)$ (Propositions 17.28 and 17.30, Example 17.38(d), (e)).

- *Section sec:tam and Remark rem:ko(1)*: the extra term at a multiplicative prime (Theorem 17.43, Example 17.46).

## 17.1 Galois cohomology of a field; Hilbert 90; Kummer theory

This section specialises the cohomology of profinite groups of Chapter 16 to absolute Galois groups. Its two theorems, Hilbert's Theorem 90 in cohomological form and Kummer theory without the hypothesis $\mu_n \subset K$, are the reason the paper may write $H^1(G, \mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$ in the proof of Lemma lem:tamfactor and argue "by Kummer theory" in the proof of Lemma lem:tamagawa-exp. The finite field computation (Example 17.11) is the identification $\mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ that the same proof uses.

**Convention 17.1** (fields, Galois groups, coefficients)**.** $K$ is a field, $\overline{K}$ a separable closure and $G_K = \mathrm{Gal}(\overline{K}/K)$ with the Krull topology (Chapter 5). For a $G_K$-module $M$ we write $H^n(K,M) := H^n(G_K, M)$, continuous cochain cohomology as in Chapter 16, for the three types of coefficients used there: (D) discrete modules (every element has an open stabiliser); (C) finitely generated $\mathbb{Z}_\ell$-modules with the $\ell$-adic topology and a continuous action; (V) finite-dimensional $\mathbb{Q}_\ell$-vector spaces with a continuous action. Whenever $\ell$-adic coefficients occur, $\mathrm{char}\, K \neq \ell$. For $K = \mathbb{Q}_p$ the inertia group $I_p \trianglelefteq G_{\mathbb{Q}_p}$ has quotient $G_{\mathbb{Q}_p}/I_p = \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \simeq \hat{\mathbb{Z}}$; for $K = \mathbb{R}$, $G_{\mathbb{R}} = \{1, c\}$ with $c$ complex conjugation. For $K \subseteq L \subseteq \overline{K}$ the restriction $\mathrm{Res}\colon H^n(K,M) \to H^n(L,M)$ is induced by $G_L \subseteq G_K$; for $L/K$ Galois, inflation-restriction (Chapter 16) gives the exact sequence $0 \to H^1(\mathrm{Gal}(L/K), M^{G_L}) \to H^1(K,M) \to H^1(L,M)^{\mathrm{Gal}(L/K)}$.

**Convention 17.2** (Frobenius)**.** $\mathrm{Frob} = \mathrm{Frob}_p$ is an *arithmetic* Frobenius: an element of $G_{\mathbb{Q}_p}$ inducing $x \mapsto x^p$ on $\overline{\mathbb{F}}_p$, well defined modulo $I_p$. It acts on $\mu_n$ $(p \nmid n)$ by $\zeta \mapsto \zeta^p$, so $\chi_\ell(\mathrm{Frob}_p) = p$ for $\ell \neq p$, where $\chi_\ell$ is the $\ell$-adic cyclotomic character. $\Phi = \mathrm{Frob}^{-1}$ is the geometric Frobenius. Every cohomological statement of this chapter (conjugation actions, unramified cohomology, the matrices of Lemma lem:tamfactor) uses Frob; $\Phi$ appears only where a local $L$-factor is named, and then $L_p(V,s) = \det(1 - \Phi\, p^{-s} \mid V^{I_p})^{-1}$ (Chapter 7). An eigenvalue $\lambda$ of Frob is the eigenvalue $\lambda^{-1}$ of $\Phi$.

**Convention 17.3** (connecting maps)**.** For an exact sequence $0 \to A \to B \to C \to 0$ of $G$-modules the connecting map $H^0(G,C) \to H^1(G,A)$ sends $c$ to the class of $g \mapsto gb - b$, where $b \in B$ lifts $c$ (Chapter 16). For the sequence $1 \to \mu_n \to \overline{K}^\times \to \overline{K}^\times \to 1$, written multiplicatively, this is $a \mapsto [\sigma \mapsto \sigma(\alpha)/\alpha]$ with $\alpha^n = a$: the convention of the Kummer characters of Chapter 5 and of the Kummer cocycles of Chapter 7. No sign is introduced anywhere in this chapter.

Recall from Chapter 16 that for a discrete $G_K$-module $M$ every continuous cochain factors through a finite quotient, so

$$H^n(K,M) = \varinjlim_{L} H^n(\mathrm{Gal}(L/K), M^{G_L}), \tag{17.1}$$

the limit over the finite Galois extensions $L/K$ inside $\overline{K}$, with inflation as transition maps. In particular $H^0(K,M) = M^{G_K}$, and for $M$ with trivial action $H^1(K,M) = \mathrm{Hom}_{\mathrm{cont}}(G_K, M)$.

**Example 17.4** (Galois modules)**.** The following are discrete $G_K$-modules: the additive group $\overline{K}$; the multiplicative group $\overline{K}^\times$; $\mu_n = \mu_n(\overline{K})$ for $n$ prime to $\mathrm{char}\, K$; $\mathbb{Z}/n$ with trivial action and its twists $\mathbb{Z}/n(r) = \mu_n^{\otimes r}$; for an elliptic curve $E/K$, the group $E(\overline{K})$ and its subgroup $E[n]$, $\mathrm{char}\, K \nmid n$ (Chapter 9); the $\ell$-primary modules $\mu_{\ell^\infty} = \bigcup_k \mu_{\ell^k}$ and $E[\ell^\infty]$. The following are of type (C): $\mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}$ (transition maps $\zeta \mapsto \zeta^\ell$), on which $\sigma$ acts by multiplication by $\chi_\ell(\sigma)$; the Tate module $T_\ell E$; $\mathrm{ad}^0\, T_\ell E$. For a finite field $K = \mathbb{F}_q$ the group $G_K$ is procyclic, $G_K \simeq \hat{\mathbb{Z}}$, topologically generated by $\mathrm{Frob}_q\colon x \mapsto x^q$ (Chapter 5).

**Theorem 17.5** (Hilbert's Theorem 90)**.** *Let $K$ be a field.*

*(i) For every finite Galois extension $L/K$, $H^1(\mathrm{Gal}(L/K), L^\times) = 0$ and $H^1(\mathrm{Gal}(L/K), L) = 0$.*

*(ii) More generally, for every Galois extension $L/K$, finite or not, $H^1(\mathrm{Gal}(L/K), L^\times) = 0$ and $H^1(\mathrm{Gal}(L/K), L) = 0$ (continuous cochains, $L$ and $L^\times$ discrete).*

*(iii) In particular $H^1(K, \overline{K}^\times) = 0$ and $H^1(K, \overline{K}) = 0$.*

*(iv) (Stated.)* $H^n(K,\overline{K})=0$ *for all* $n\geq 1$.

*Proof.* (i) This is Hilbert's Theorem 90 of Chapter 5; we recall the argument because it is short. Let $G=\mathrm{Gal}(L/K)$ and let $(a_\sigma)$ be a 1-cocycle with values in $L^\times$, $a_{\sigma\tau}=a_\sigma\,\sigma(a_\tau)$. The distinct automorphisms $\tau\in G$ restrict to distinct homomorphisms $L^\times\to L^\times$, so by Dedekind's independence of characters (Chapter 5) no nontrivial $L$-linear combination of them vanishes identically; as every coefficient $a_\tau$ is nonzero, the map $c\mapsto\sum_{\tau\in G}a_\tau\tau(c)$ is not identically zero. Choose $c$ with $b=\sum_\tau a_\tau\tau(c)\neq 0$. The cocycle identity, rewritten as $\sigma(a_\tau)=a_\sigma^{-1}a_{\sigma\tau}$, gives

$$\sigma(b)=\sum_\tau \sigma(a_\tau)\,\sigma\tau(c)=\sum_\tau a_\sigma^{-1}a_{\sigma\tau}\,\sigma\tau(c)=a_\sigma^{-1}\sum_\rho a_\rho\,\rho(c)=a_\sigma^{-1}b,$$

where the third step substitutes $\rho=\sigma\tau$, which runs over $G$ as $\tau$ does. So $a_\sigma=b/\sigma(b)=\sigma(b^{-1})/b^{-1}$ is a coboundary. In the additive case, $a_{\sigma\tau}=a_\sigma+\sigma(a_\tau)$. As $L/K$ is separable, the trace $\mathrm{Tr}_{L/K}=\sum_\tau\tau$ is not identically zero (Dedekind again, all coefficients being 1), and it is $K$-linear with values in $K$, so it is surjective; choose $c$ with $\mathrm{Tr}_{L/K}(c)=1$ and put $b=\sum_\tau a_\tau\tau(c)$. Then

$$\sigma(b)=\sum_\tau\sigma(a_\tau)\,\sigma\tau(c)=\sum_\tau(a_{\sigma\tau}-a_\sigma)\,\sigma\tau(c)=\sum_\rho a_\rho\,\rho(c)-a_\sigma\,\mathrm{Tr}_{L/K}(c)=b-a_\sigma,$$

so $a_\sigma=b-\sigma(b)$ is a coboundary.

(ii) $L^\times$ is a discrete $\mathrm{Gal}(L/K)$-module, and for a finite Galois subextension $L'/K$ of $L/K$ the invariants of $\mathrm{Gal}(L/L')$ in $L^\times$ are $L'^\times$. By (17.1) for the profinite group $\mathrm{Gal}(L/K)$, $H^1(\mathrm{Gal}(L/K),L^\times)=\varinjlim_{L'}H^1(\mathrm{Gal}(L'/K),L'^\times)$, a direct limit of zero groups by (i). The additive case is the same. (iii) is (ii) with $L=\overline{K}$.

(iv) For finite $G$ the additive group $L$ is an induced $G$-module by the normal basis theorem, hence has vanishing cohomology in positive degrees; the limit argument of (ii) applies. We do not use this; see [174, Chapter X, Section 1] and [177, Chapter II, Section 1]. □

For a cyclic group the theorem takes its classical form. If $\mathrm{Gal}(L/K)=\langle\sigma\rangle$ has order $m$, a cocycle is determined by $a=a_\sigma$, and the cocycle identity forces $a_{\sigma^j}=a\,\sigma(a)\cdots\sigma^{j-1}(a)$ and $a_{\sigma^m}=a_1=1$, that is, $\mathrm{N}_{L/K}(a)=1$; conversely every element of norm 1 defines a cocycle in this way. So (i) says: *an element of norm* 1 *is of the form* $b/\sigma(b)$. For $L=\mathbb{C}$, $K=\mathbb{R}$, $\sigma$ complex conjugation, the elements of norm 1 are the complex numbers of absolute value 1, and for instance $(3+4i)/5=(2+i)/(2-i)$, with $b=2+i$.

*Remark* 17.6. Part (ii) applies to $L=\bar{k}(C)$, $K=k(C)$ for a curve $C$ over a perfect field $k$ (the extension $\bar{k}(C)/k(C)$ is Galois with group $G_k$): this is the form in which Chapter 8 uses Hilbert 90 to show that a $G_k$-invariant principal divisor is the divisor of a function defined over $k$.

**Theorem 17.7** (Kummer theory in cohomological form)**.** *Let* $n\geq 1$ *be prime to* $\mathrm{char}\,K$.

*(i) The sequence* $1\to\mu_n\to\overline{K}^\times\xrightarrow{x\mapsto x^n}\overline{K}^\times\to 1$ *is an exact sequence of discrete* $G_K$*-modules, and its connecting map*

$$\delta=\delta_K\colon K^\times/(K^\times)^n\longrightarrow H^1(K,\mu_n),\qquad \delta(a)=[\sigma\mapsto\sigma(\alpha)/\alpha]\quad(\alpha\in\overline{K},\ \alpha^n=a),$$

*is an isomorphism of groups.*

*(ii)* $H^2(K,\mu_n)\simeq\mathrm{Br}(K)[n]$*, the* $n$*-torsion of the* Brauer group $\mathrm{Br}(K):=H^2(K,\overline{K}^\times)$.

*(iii)* *If $\mu_n \subset K$, then $H^1(K,\mu_n) = \mathrm{Hom}_{\mathrm{cont}}(G_K,\mu_n)$ and $\delta(a)$ is the Kummer character $\chi_a$ of Chapter 5.*

*(iv)* *For a separable algebraic extension $L/K$, $\mathrm{Res}\circ\delta_K = \delta_L \circ (K^\times/K^{\times n} \to L^\times/L^{\times n})$. (Stated.) For $L/K$ finite, $\mathrm{Cor}\circ\delta_L = \delta_K \circ \mathrm{N}_{L/K}$.*

*Proof.* (i) The kernel of $x \mapsto x^n$ on $\overline{K}^\times$ is by definition $\mu_n$, which gives exactness at the left and in the middle; the three modules are discrete because each element of $\overline{K}$ lies in a finite extension of $K$ and so has an open stabiliser. For surjectivity, let $a \in \overline{K}^\times$: the polynomial $X^n - a$ has derivative $nX^{n-1}$, which is prime to $X^n - a$ because $n$ is invertible in $K$ and $a \neq 0$; so $X^n - a$ is separable and has a root in the separable closure $\overline{K}$, and $x \mapsto x^n$ is surjective. The long exact sequence (Chapter 16) begins

$$1 \to \mu_n(K) \to K^\times \xrightarrow{n} K^\times \xrightarrow{\delta} H^1(K,\mu_n) \to H^1(K,\overline{K}^\times) \xrightarrow{n} H^1(K,\overline{K}^\times) \to H^2(K,\mu_n) \to H^2(K,\overline{K}^\times) \xrightarrow{n} H^2(K,\overline{K}^\times).$$

Here $H^0(K,\overline{K}^\times) = K^\times$ because the fixed field of $G_K$ in the separable closure is $K$ (infinite Galois theory, Chapter 5), and the map $K^\times \to K^\times$ is $x \mapsto x^n$, with image $(K^\times)^n$. By Theorem 17.5(iii) the fifth term is zero, so $\delta$ is surjective with kernel $(K^\times)^n$, and induces an isomorphism $K^\times/(K^\times)^n \simeq H^1(K,\mu_n)$. By Convention 17.3 the class $\delta(a)$ is represented by $\sigma \mapsto \sigma(\alpha)/\alpha$ for any lift $\alpha$ of $a$, that is, any $n$-th root of $a$ in $\overline{K}$. (ii) The same sequence, with $H^1(K,\overline{K}^\times) = 0$, identifies $H^2(K,\mu_n)$ with the kernel of multiplication by $n$ on $H^2(K,\overline{K}^\times)$. (iii) If $\mu_n \subset K$ the action on $\mu_n$ is trivial, a cocycle is a continuous homomorphism, and coboundaries are zero; $\sigma \mapsto \sigma(\alpha)/\alpha$ is the Kummer character $\chi_a$ of Chapter 5. (iv) Choose $\overline{L} = \overline{K}$; for $\sigma \in G_L$ the restriction of the cocycle $\sigma \mapsto \sigma(\alpha)/\alpha$ is the cocycle that defines $\delta_L(a)$ with the same root $\alpha$. The corestriction formula is [174, Chapter XIV, Section 1] and [143, Chapter VI, Section 2]; we do not use it. □

*Remark* 17.8. The connecting map measures the failure of the choice of an $n$-th root to be Galois equivariant: a root $\alpha$ is defined only up to $\mu_n$, and $\sigma(\alpha)/\alpha$ records how $\sigma$ moves it, much as a phase is defined only up to a constant and its variation is what is observed. Changing $\alpha$ to $\zeta\alpha$ changes the cocycle by $\sigma \mapsto \sigma(\zeta)/\zeta$, a coboundary.

The simplest case is $K = \mathbb{Q}$, $n = 2$. Then $\mu_2 = \{\pm 1\}$ has trivial action, $H^1(\mathbb{Q},\mu_2) = \mathrm{Hom}_{\mathrm{cont}}(G_{\mathbb{Q}},\{\pm1\})$, and $\delta(a)$ is the character $\sigma \mapsto \sigma(\sqrt{a})/\sqrt{a}$, trivial exactly on $G_{\mathbb{Q}(\sqrt{a})}$. The theorem says that the quadratic characters of $G_{\mathbb{Q}}$ (together with the trivial character) correspond bijectively to the square classes $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, that is, to the quadratic fields $\mathbb{Q}(\sqrt{a})$ together with $\mathbb{Q}$ itself, and that the product of characters corresponds to the product of square classes: the character of $\mathbb{Q}(\sqrt{ab})$ is the product of those of $\mathbb{Q}(\sqrt{a})$ and $\mathbb{Q}(\sqrt{b})$.

**Definition 17.9** (Kummer map, Kummer class, Kummer cocycle)**.** The isomorphism $\delta$ of Theorem 17.7 is the *Kummer map* of $K$ (for $n$); $\delta(a) \in H^1(K,\mu_n)$ is the *Kummer class* of $a \in K^\times$, and a cocycle $\sigma \mapsto \sigma(\alpha)/\alpha$ with $\alpha^n = a$ is a *Kummer cocycle* of $a$. For $\ell \neq \mathrm{char}\,K$ and a compatible system $(a^{1/\ell^k})_{k\geq 0}$ of $\ell$-power roots of $a$ (that is, $(a^{1/\ell^{k+1}})^\ell = a^{1/\ell^k}$), the *$\ell$-adic Kummer cocycle* is

$$\kappa_a(\sigma) = \left(\sigma(a^{1/\ell^k})/a^{1/\ell^k}\right)_{k\geq 0} \in \varprojlim_k \mu_{\ell^k} = \mathbb{Z}_\ell(1),$$

the cocycle of Chapters 5 and 7.

**Proposition 17.10** (properties of Kummer classes)**.** *Let $n$ be prime to* $\operatorname{char} K$ *and $a \in K^\times$.*

*(i) For every separable algebraic extension $L/K$:* $\operatorname{Res}_{G_L} \delta_K(a) = 0$ *if and only if $a \in (L^\times)^n$.*

*(ii) If $\mu_n \subset K$ and $\alpha^n = a$, the kernel of $\delta(a) = \chi_a$ is $G_{K(\alpha)}$, and $K(\alpha)/K$ is cyclic of degree equal to the order of $a$ in $K^\times/(K^\times)^n$.*

*(iii) The reduction modulo $\ell^k$ of the $\ell$-adic Kummer cocycle $\kappa_a$, that is its image under $\mathbb{Z}_\ell(1) \to \mu_{\ell^k}$, is the Kummer cocycle of $a$ in $H^1(K, \mu_{\ell^k})$ attached to the root $a^{1/\ell^k}$.*

*Proof.* (i) By Theorem 17.7(iv) the restriction of $\delta_K(a)$ is $\delta_L(a)$, which is zero if and only if $a \in (L^\times)^n$ by Theorem 17.7(i) for $L$. (ii) $\chi_a(\sigma) = 1$ if and only if $\sigma(\alpha) = \alpha$, so $\ker \chi_a = G_{K(\alpha)}$ and $\mathrm{Gal}(K(\alpha)/K) \simeq \chi_a(G_K)$, a subgroup of the cyclic group $\mu_n$. Its order is the order of $\chi_a$ in $\mathrm{Hom}(G_K, \mu_n)$; since $\chi_a^m = \chi_{a^m}$ and $\delta$ is injective, $\chi_a^m = 1$ if and only if $a^m \in (K^\times)^n$. (iii) The projection $\mathbb{Z}_\ell(1) \to \mu_{\ell^k}$ takes the component of index $k$, which is $\sigma(a^{1/\ell^k})/a^{1/\ell^k}$. □

**Example 17.11** (Kummer theory over a finite field)**.** Let $K = \mathbb{F}_q$ with $q$ a power of a prime $p \neq \ell$, $v = v_\ell(q-1)$, and $\mathrm{Frob}_q$ the generator $x \mapsto x^q$ of $G_K \simeq \hat{\mathbb{Z}}$. Two computations of $H^1(\mathbb{F}_q, \mu_{\ell^k})$ agree.

(a) By Kummer theory it is $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^k}$, cyclic of order $\ell^{\min(k,v)}$, $\mathbb{F}_q^\times$ being cyclic of order $q-1$ (Chapter 2).

(b) By the procyclic computation of Chapter 16 ($G_K \simeq \hat{\mathbb{Z}}$, finite coefficients), $c \mapsto c(\mathrm{Frob}_q)$ identifies it with $\mu_{\ell^k}/(\mathrm{Frob}_q - 1)\mu_{\ell^k} = \mu_{\ell^k}/\mu_{\ell^k}^{q-1}$, of order $\gcd(\ell^k, q-1) = \ell^{\min(k,v)}$.

The composite isomorphism sends $u \in \mathbb{F}_q^\times$ to $\delta(u)(\mathrm{Frob}_q) = \mathrm{Frob}_q(\alpha)/\alpha = \alpha^{q-1}$, where $\alpha \in \overline{\mathbb{F}}_q$ and $\alpha^{\ell^k} = u$. For $k \le v$ this is $(\alpha^{\ell^k})^{(q-1)/\ell^k} = u^{(q-1)/\ell^k}$, the map of Chapter 5 (its Kummer proposition over finite fields). Passing to the limit over $k$ (Theorem 17.20 below, or directly the procyclic computation with $\mathrm{Frob}_q$ acting on $\mathbb{Z}_\ell(1)$ by $q$):

$$H^1(\mathbb{F}_q, \mathbb{Z}_\ell(1)) \;=\; \mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1) \;\simeq\; \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell = \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v} \;\simeq\; \mathbb{Z}/\ell^v.$$

The first equality is $c \mapsto c(\mathrm{Frob}_q)$; the second isomorphism is the Kummer isomorphism, canonical; the third depends on the choice of a generator. The class of $u$ has valuation at least $k$ (for $k \le v$) if and only if $u \in (\mathbb{F}_q^\times)^{\ell^k}$, where, as in Lemma lem:tamfactor, a class $c \in \mathbb{Z}/\ell^v$ has $v_\ell(c) = \min(v_\ell(\tilde{c}), v)$ for any lift $\tilde{c} \in \mathbb{Z}$; in particular $v_\ell(0) = v$.

For instance, $q = 7$, $\ell = 3$, $k = v = 1$: the map is $u \mapsto u^2$, from $\mathbb{F}_7^\times$ onto $\mu_3(\mathbb{F}_7) = \{1, 2, 4\}$, with kernel the cubes $\{1, 6\}$; so $3 \mapsto 2$ and $5 \mapsto 4$, and the classes of 3 and 5 are the two nonzero elements of $H^1(\mathbb{F}_7, \mu_3) \simeq \mathbb{Z}/3$.

The arithmetic Frobenius matters here, though only up to sign. Evaluating at the geometric Frobenius $\Phi = \mathrm{Frob}_q^{-1}$ instead, the cocycle identity $0 = c(\Phi\,\mathrm{Frob}_q) = c(\Phi) + \Phi\, c(\mathrm{Frob}_q)$ gives $c(\Phi) = -\Phi\, c(\mathrm{Frob}_q)$ (additive notation in $\mathbb{Z}_\ell(1)$); as $\Phi$ acts by $q^{-1}$ and $q^{-1} \equiv 1$ modulo $(q-1)\mathbb{Z}_\ell$, the class of $c(\Phi)$ is minus that of $c(\mathrm{Frob}_q)$. In multiplicative notation the value at $\Phi$ is the inverse class $u^{-(q-1)/\ell^k}$. The valuation of the class, the only thing Lemma lem:tamfactor uses, is the same for both.

**Theorem 17.12** (Brauer groups)**.** *(i)* $\mathrm{Br}(\mathbb{F}_q) = 0$, $\mathrm{Br}(\mathbb{C}) = 0$ *and* $\mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2$.

*(ii) Let $K$ be a finite extension of $\mathbb{Q}_p$. There is a canonical isomorphism* $\mathrm{inv}_K\colon \mathrm{Br}(K) \to \mathbb{Q}/\mathbb{Z}$, *the* invariant. *For a finite extension $L/K$,* $\mathrm{inv}_L \circ \mathrm{Res} = [L:K]\cdot \mathrm{inv}_K$ *and* $\mathrm{inv}_K \circ \mathrm{Cor} = \mathrm{inv}_L$. *With Theorem 17.7(ii), $H^2(K,\mu_n) \simeq \mathbb{Z}/n$ canonically, for every $n \geq 1$.*

*(iii) (Albert, Brauer, Hasse, Noether.) The sequence*

$$0 \to \mathrm{Br}(\mathbb{Q}) \to \bigoplus_v \mathrm{Br}(\mathbb{Q}_v) \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0$$

*is exact, the sum over all places $v$ of $\mathbb{Q}$, with* $\mathrm{inv}_\infty\colon \mathrm{Br}(\mathbb{R}) \simeq \frac{1}{2}\mathbb{Z}/\mathbb{Z}$. *In particular an element of* $\mathrm{Br}(\mathbb{Q})$ *has only finitely many nonzero local components, and the sum of its local invariants is* $0$.

We do not prove this; see [174, Chapters X, XII and XIII] for (i) and (ii), Serre's and Tate's chapters in [26, Chapters VI and VII] for (ii) and (iii), and [177, Chapter II, Section 5]. Part (ii) is the cohomological form of local class field theory (Chapter 7); it provides the target $H^2(K,\mu) \simeq \mathbb{Q}/\mathbb{Z}$ of the local pairing of Section 17.3, and part (iii) is the reciprocity law behind Proposition 17.69.

*In the paper* 17.13. The proof of Lemma lem:tamfactor writes the Galois action on $T_\ell E$ through matrices "with $\varepsilon$ unramified of order at most 2 and $\kappa$ the Kummer cocycle of $q_E$ in $H^1(G,\mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$". The equality is Theorem 17.20 below for $K=\mathbb{Q}_q$, which rests on Theorem 17.7 at each finite level; the class of $\kappa$ is the Kummer class of $q_E$ in the sense of Definition 17.9, and the cocycle itself is the $\ell$-adic Kummer cocycle of the same definition. The proof of Lemma lem:tamagawa-exp argues "By Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$, which contains $\mu_{\ell^t}$ as $q \neq \ell$"; that is Proposition 17.10(i), (iii) with $L = \mathbb{Q}_q^{\mathrm{ur}}$, and Remark 17.47 gives the cohomological reading.

So far the coefficients have been finite or discrete. The paper's coefficients are $\ell$-adic: $\mathbb{Z}_\ell(1)$ in Lemma lem:tamfactor, and the lattice $T^D$ with its rational and divisible companions in Lemma lem:dfg21. The next section passes from the finite levels $\mu_{\ell^k}$, $T/\ell^k T$ to these modules by limits, and introduces the Kummer dual, which is the module that local and global duality will pair with $T$.

## 17.2 $\ell$-adic coefficients: $T$, $V$, $W$ and Kummer duals

The paper works with three modules built from one lattice: $T^D$, $V^D$ and $W^D$ in Section 4.2 and Lemma lem:dfg21, and $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ in Theorem thm:main. This section defines the Kummer duals in general, proves the exact relation between the cohomology of $T$, $V$ and $W$ (in particular that the torsion of $H^1(G,T)$ comes from $H^0(G,W)$, the fact behind "$H^1_f(\mathbb{Q}_p,T^D)$ is the torsion of $H^1(\mathbb{Q}_p,T^D)$, a finite group" after Lemma lem:dfg21), and computes $H^1(K,\mathbb{Z}_\ell(1))$, the group in which the Kummer class of the Tate parameter lives in Lemma lem:tamfactor.

**Convention 17.14** (lattices and duals)**.** $T$ is a free $\mathbb{Z}_\ell$-module of finite rank with a continuous $\mathbb{Z}_\ell$-linear action of a profinite group $G$; $V = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ and $W = T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell/\mathbb{Z}_\ell = V/T$, so that $W[\ell^k] = \ell^{-k}T/T \simeq T/\ell^k T$ (Chapter 3). The maps induced by $T \subset V \to W$ are written

$$\iota\colon H^n(G,T) \to H^n(G,V), \qquad \pi\colon H^n(G,V) \to H^n(G,W).$$

$A^*$ is the *Pontryagin dual*: $\mathrm{Hom}(A,\mathbb{Q}/\mathbb{Z})$ for a finite group, $\mathrm{Hom}_{\mathrm{cont}}(A,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ for a finitely generated or a discrete $\ell$-primary $\mathbb{Z}_\ell$-module (Chapter 3); this is the paper's $^*$ in Lemma lem:dfg21.

$T^\vee = \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathbb{Z}_\ell)$ and $V^\vee$ are linear duals. $M(1) = M \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(1)$. "Length" means $\mathbb{Z}_\ell$-length, length $= \mathrm{length}_{\mathbb{Z}_\ell}$, and $\#X = \ell^{\mathrm{length}\, X}$ for a finite $\mathbb{Z}_\ell$-module $X$.

**Definition 17.15** (Kummer duals)**.** Let $\mu$ be the group of roots of unity in $\overline{K}$ of order prime to $\mathrm{char}\, K$. For a finite $G$-module $M$ of order prime to $\mathrm{char}\, K$, the *Kummer dual* is $M^D = \mathrm{Hom}(M, \mu)$; if $M$ has $\ell$-power order, $M^D = \mathrm{Hom}(M, \mu_{\ell^\infty}) = M^*(1)$. For $T$ as in Convention 17.14:

$$T^D := \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathbb{Z}_\ell(1)) = T^\vee(1), \qquad V^D := T^D \otimes \mathbb{Q}_\ell = \mathrm{Hom}_{\mathbb{Q}_\ell}(V, \mathbb{Q}_\ell(1)), \qquad W^D := V^D/T^D.$$

In all cases $G$ acts on homomorphisms by $(g\phi)(x) = g\,\phi(g^{-1}x)$.

These are the paper's $T^D$, $V^D$, $W^D$ of Section 4.2, where $T$ is the adjoint lattice $A = \mathrm{ad}^0\, T_\ell E$; the identification $T^D \simeq A(1)$ through the trace form is made in Chapter 18.

**Proposition 17.16** (Kummer duality for lattices)**.** *(i) The evaluation pairings*

$$T \times T^D \to \mathbb{Z}_\ell(1), \qquad V \times V^D \to \mathbb{Q}_\ell(1), \qquad M \times M^D \to \mu,$$
$$W \times T^D \to \mathbb{Q}_\ell/\mathbb{Z}_\ell(1), \qquad T \times W^D \to \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)$$

*are $G$-equivariant and perfect: each module is identified with the continuous homomorphisms from the other into the target.*

*(ii)* $W^D \simeq \mathrm{Hom}(T, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) = T^*(1)$*, and for each* $k$*,* $W[\ell^k]^D = (T/\ell^k T)^D \simeq T^D/\ell^k T^D \simeq W^D[\ell^k]$.

*(iii)* $(T^D)^D = T$ *canonically;* $W^D$ *is the Kummer dual of* $T$ *and* $T^D$ *that of* $W$ *in the sense of (i).*

*(iv)* $\mathbb{Z}_\ell^D = \mathbb{Z}_\ell(1)$ *and* $\mathbb{Z}_\ell(1)^D = \mathbb{Z}_\ell$*; for* $T = \mathbb{Z}_\ell$*,* $W^D = \mu_{\ell^\infty}$*, and the Kummer dual of* $W = \mathbb{Q}_\ell/\mathbb{Z}_\ell$ *in the sense of (iii) is* $\mathbb{Z}_\ell(1)$*. For an elliptic curve* $E/K$ *and* $n$ *prime to* $\mathrm{char}\, K$*, the Weil pairing gives* $E[n]^D \simeq E[n]$ *and* $(T_\ell E)^D \simeq T_\ell E$.

*Proof.* (i) Equivariance: $\langle gx, g\phi\rangle = (g\phi)(gx) = g\,\phi(g^{-1}gx) = g\langle x, \phi\rangle$. Perfectness of $T \times T^D \to \mathbb{Z}_\ell(1)$ is linear algebra over $\mathbb{Z}_\ell$ for free modules of finite rank ($\mathbb{Z}_\ell(1)$ is free of rank one), and tensoring with $\mathbb{Q}_\ell$ gives the second pairing. For the third, a choice of basis $e_1, \dots, e_r$ of $T$ and a generator of $\mathbb{Z}_\ell(1)$ identifies $T^D$ with $\mathbb{Z}_\ell^r$ by the dual basis and $W$ with $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$, and the pairing becomes $((x_i), (y_i)) \mapsto \sum x_i y_i \in \mathbb{Q}_\ell/\mathbb{Z}_\ell$; that this identifies each side with the continuous dual of the other is the duality between $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ of Chapter 3. The fourth pairing is the third for $T^D$ in place of $T$, by (iii). For finite $M$ the pairing $M \times \mathrm{Hom}(M, \mu) \to \mu$ is perfect because $\mu$ contains a cyclic group of order $\exp(M)$ and $\#\,\mathrm{Hom}(M, \mu) = \#M$. (ii) For $T$ free, $\mathrm{Hom}(T, \mathbb{Z}_\ell(1)) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell = \mathrm{Hom}(T, \mathbb{Z}_\ell(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$, and $V^D/T^D = T^D \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. Similarly $\mathrm{Hom}(T/\ell^k T, \mu_{\ell^k}) = \mathrm{Hom}(T, \mathbb{Z}_\ell(1)) \otimes \mathbb{Z}/\ell^k$, and $W^D[\ell^k] = \ell^{-k}T^D/T^D \simeq T^D/\ell^k T^D$. (iii) The map $T \to \mathrm{Hom}(\mathrm{Hom}(T, \mathbb{Z}_\ell(1)), \mathbb{Z}_\ell(1))$, $x \mapsto (\phi \mapsto \phi(x))$, is an isomorphism of free modules of the same rank (check on a basis) and is equivariant by (i). The rest is a restatement of (i). (iv) The first statements are immediate from the definitions. The Weil pairing $e_n \colon E[n] \times E[n] \to \mu_n$ is bilinear, alternating, perfect and Galois equivariant (Chapter 9), so $P \mapsto e_n(P, \cdot)$ is an isomorphism $E[n] \to E[n]^D$; the pairings for $n = \ell^k$ are compatible and give $T_\ell E \simeq \mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell(1))$. □

**Proposition 17.17** (the cohomology of $T$, $V$, $W$)**.** *Let* $G$ *be a profinite group and* $T$ *as in Convention 17.14, and suppose that* $H^1(G, T/\ell T)$ *and* $H^2(G, T/\ell T)$ *are finite. (This holds for* $G = G_{\mathbb{F}_q}$*,* $G_{\mathbb{R}}$*,* $G_{\mathbb{Q}_p}$ *and* $G_{\mathbb{Q},S}$*: Chapter 16, Theorem 17.22 and Theorem 17.50.)*

*(i)* $H^n(G,T)$ *is a finitely generated* $\mathbb{Z}_\ell$*-module for* $n \le 2$*, and* $H^n(G,V) = H^n(G,T) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$*, so that* $\iota$ *is* $x \mapsto x \otimes 1$*.*

*(ii) (Torsion of* $H^1(G,T)$*.) The sequence* $H^0(G,V) \to H^0(G,W) \xrightarrow{\delta} H^1(G,T) \xrightarrow{\iota} H^1(G,V)$ *is exact,* $\ker \iota = H^1(G,T)_{\mathrm{tors}}$*, and the image of* $H^0(G,V)$ *in* $H^0(G,W)$ *is* $H^0(G,T)\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$*, which is the maximal divisible subgroup* $H^0(G,W)_{\mathrm{div}}$*. Hence*

$$H^1(G,T)_{\mathrm{tors}} \;\simeq\; H^0(G,W)/H^0(G,W)_{\mathrm{div}},$$

*a finite group. If* $H^0(G,V) = 0$*, then* $H^0(G,W)$ *is finite and* $\delta\colon H^0(G,W) \to H^1(G,T)_{\mathrm{tors}}$ *is an isomorphism.*

*(iii) (Divisible part of* $H^1(G,W)$*.) The sequence* $0 \to H^1(G,T) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \to H^1(G,W) \to H^2(G,T)_{\mathrm{tors}} \to 0$ *is exact; the first term is the image of* $\pi$ *and equals* $H^1(G,W)_{\mathrm{div}}$*. So* $H^1(G,W)$ *is cofinitely generated, of corank* $\operatorname{rank}_{\mathbb{Z}_\ell} H^1(G,T) = \dim_{\mathbb{Q}_\ell} H^1(G,V)$*.*

*(iv) (Finite levels.) For* $n \ge 1$ *and* $k \ge 1$ *the sequences*

$$0 \to H^{n-1}(G,T)/\ell^k \to H^{n-1}(G,T/\ell^k T) \to H^n(G,T)[\ell^k] \to 0,$$
$$0 \to H^{n-1}(G,W)/\ell^k \to H^n(G,W[\ell^k]) \to H^n(G,W)[\ell^k] \to 0$$

*are exact.*

*Proof.* (iv) The sequences $0 \to T \xrightarrow{\ell^k} T \to T/\ell^k T \to 0$ and $0 \to W[\ell^k] \to W \xrightarrow{\ell^k} W \to 0$ (exact because $W$ is divisible) admit continuous sections (Chapter 16), so they give long exact sequences. The first contains the segment

$$H^{n-1}(G,T) \xrightarrow{\ell^k} H^{n-1}(G,T) \to H^{n-1}(G,T/\ell^k T) \to H^n(G,T) \xrightarrow{\ell^k} H^n(G,T),$$

because the map induced on cohomology by multiplication by $\ell^k$ on $T$ is multiplication by $\ell^k$. At the middle term, the kernel of the map to $H^n(G,T)$ is the image of the first $\ell^k$, which is $\ell^k H^{n-1}(G,T)$, and its image is the kernel of the second $\ell^k$, which is $H^n(G,T)[\ell^k]$; this is the first short exact sequence. The second comes in the same way from the segment $H^{n-1}(G,W) \xrightarrow{\ell^k} H^{n-1}(G,W) \to H^n(G,W[\ell^k]) \to H^n(G,W) \xrightarrow{\ell^k} H^n(G,W)$.

(i) First, all $H^n(G,T/\ell^k T)$, $n \le 2$, are finite, by induction on $k$. For $k = 1$ this is the hypothesis for $n = 1, 2$, and $H^0(G,T/\ell T) \subseteq T/\ell T$ is finite. For $k \ge 2$, the exact sequence $0 \to T/\ell^{k-1}T \xrightarrow{\ell} T/\ell^k T \to T/\ell T \to 0$ gives an exact segment $H^n(G,T/\ell^{k-1}T) \to H^n(G,T/\ell^k T) \to H^n(G,T/\ell T)$, so $\#H^n(G,T/\ell^k T) \le \#H^n(G,T/\ell^{k-1}T) \cdot \#H^n(G,T/\ell T)$. By Chapter 16 (the theorem of Tate on limits) $H^n(G,T) = \varprojlim_k H^n(G,T/\ell^k T)$ for $n \le 2$, a compact pro-$\ell$ $\mathbb{Z}_\ell$-module, and by (iv) its quotient by $\ell$ embeds in the finite group $H^n(G,T/\ell T)$. A compact pro-$\ell$ $\mathbb{Z}_\ell$-module $X$ with $X/\ell X$ finite is finitely generated, by the topological form of Nakayama's lemma (Chapter 3; [167, Appendix B]). The equality $H^n(G,V) = H^n(G,T) \otimes \mathbb{Q}_\ell$ is Chapter 16: a continuous cochain with values in $V$ has compact image, hence takes values in $\ell^{-a}T$ for some $a$.

(ii) The sequence $0 \to T \to V \to W \to 0$ has a continuous section (Chapter 16), whence the exact sequence $H^0(G,T) \to H^0(G,V) \to H^0(G,W) \xrightarrow{\delta} H^1(G,T) \xrightarrow{\iota} H^1(G,V)$. By (i), $\iota$ is $X \to X \otimes \mathbb{Q}_\ell$ for the finitely generated $\mathbb{Z}_\ell$-module $X = H^1(G,T)$, whose kernel is $X_{\mathrm{tors}}$; so $\delta(H^0(G,W)) = H^1(G,T)_{\mathrm{tors}}$, a finite group. Next, $T^G = T \cap V^G$, and $V^G = T^G \otimes \mathbb{Q}_\ell$ (if $x \in V^G$ then $\ell^a x \in T \cap V^G$ for some $a$). So the image of $V^G$ in $W = V/T$ is $V^G/(V^G \cap T) = V^G/T^G \simeq$

$T^G \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, a divisible group; call it $D$. Then $D \subseteq (W^G)_{\mathrm{div}}$, and $(W^G)_{\mathrm{div}}/D$ is divisible (a quotient of a divisible group) and finite (a subgroup of $W^G/D \simeq H^1(G,T)_{\mathrm{tors}}$), hence zero, since a finite divisible group is trivial. This proves $D = (W^G)_{\mathrm{div}}$ and the displayed isomorphism. If $V^G = 0$, then $D = 0$ and $W^G \simeq H^1(G,T)_{\mathrm{tors}}$ is finite.

(iii) The continuation of the same long exact sequence reads $H^1(G,T) \xrightarrow{\iota} H^1(G,V) \xrightarrow{\pi} H^1(G,W) \to H^2(G,T) \to H^2(G,V)$. The image of $\pi$ is $\operatorname{coker}\iota = (X\otimes\mathbb{Q}_\ell)/\iota(X) \simeq X\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$ (for $X$ finitely generated, $X\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell = (X/X_{\mathrm{tors}})\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$), and the cokernel of $\pi$ is $\ker(H^2(G,T) \to H^2(G,T)\otimes\mathbb{Q}_\ell) = H^2(G,T)_{\mathrm{tors}}$, finite by (i). So $H^1(G,W)$ is an extension of a finite group by $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$, $r = \operatorname{rank} X$, which is cofinitely generated of corank $r$; the image of $\pi$ is divisible of finite index, and the argument at the end of (ii) shows that it is the maximal divisible subgroup. □

*Warning* 17.18. $H^1(G,V)$ is a $\mathbb{Q}_\ell$-vector space and sees no torsion: two lattices with the same $V$ can have different $H^1(G,T)_{\mathrm{tors}}$, and it is $H^0(G,W)$, not $H^0(G,V)$, that controls it. The torsion is $H^0(G,W)$ modulo its divisible part, not $H^0(G,W)$ itself; the two agree when $H^0(G,V) = 0$, which is the case in every application in the paper (Corollary 17.36 and Theorem 17.75).

**Example 17.19** (checks of Proposition 17.17)**.** (a) $G = G_{\mathbb{F}_q}$, $q$ prime to $\ell$, $T = \mathbb{Z}_\ell(1)$: $\mathrm{Frob}_q$ acts on $V = \mathbb{Q}_\ell(1)$ by $q \neq 1$, so $H^0(G,V) = 0$; $H^0(G,W) = \mu_{\ell^\infty}(\mathbb{F}_q)$ has order $\ell^{v_\ell(q-1)}$; and $H^1(G,T) = \mathbb{Z}_\ell(1)/(q-1)$ (Example 17.11) is all torsion, isomorphic to $H^0(G,W)$, as (ii) predicts.

(b) $G = G_{\mathbb{R}}$, $\ell$ odd, any $T$: $H^n(\mathbb{R},T) = 0$ for $n \geq 1$ (Proposition 17.51), and $H^0(\mathbb{R},W) = W^{c=1} = T^{c=1}\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is divisible, so $H^1(\mathbb{R},T)_{\mathrm{tors}} = 0$ is consistent with (ii).

(c) $G = G_{\mathbb{Q}_p}$, $p \neq \ell$, $T = \mathbb{Z}_\ell(1)$: by Proposition 17.21, $H^1(\mathbb{Q}_p,\mathbb{Z}_\ell(1))_{\mathrm{tors}}$ is the image of $\mathbb{Z}_p^\times\otimes\mathbb{Z}_\ell$, of order $\ell^{v_\ell(p-1)} = \#\mu_{\ell^\infty}(\mathbb{Q}_p) = \#H^0(\mathbb{Q}_p,\mathbb{Q}_\ell/\mathbb{Z}_\ell(1))$.

**Theorem 17.20** ($\ell$-adic Kummer theory)**.** *Let $K$ be a field of characteristic different from $\ell$, and $\widehat{K^\times} := \varprojlim_k K^\times/(K^\times)^{\ell^k}$ (transition maps the projections).*

*(i) The Kummer maps induce an isomorphism $\widehat{K^\times} \simeq H^1(K,\mathbb{Z}_\ell(1))$; the class of $a \in K^\times$ is $\kappa_a := (\delta_k(a))_k$, represented by the $\ell$-adic Kummer cocycle of Definition 17.9 for any compatible system of roots.*

*(ii) The Kummer maps induce an isomorphism $K^\times\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell \simeq H^1(K,\mathbb{Q}_\ell/\mathbb{Z}_\ell(1))$.*

*(iii) $\kappa_{ab} = \kappa_a + \kappa_b$; $\kappa_a = 0$ if and only if $a \in \bigcap_k (K^\times)^{\ell^k}$; for a separable algebraic extension $L/K$ the restriction of $\kappa_a$ to $G_L$ is $\kappa_a$ computed over $L$.*

*Proof.* (i) The groups $H^0(K,\mu_{\ell^k})$ are finite, so by Chapter 16 (Tate's theorem on limits, the case $n = 1$) $H^1(K,\mathbb{Z}_\ell(1)) = \varprojlim_k H^1(K,\mu_{\ell^k})$, with transition maps induced by $\mu_{\ell^{k+1}} \to \mu_{\ell^k}$, $\zeta \mapsto \zeta^\ell$. On the Kummer side: if $\alpha^{\ell^{k+1}} = a$, the image of the cocycle $\sigma \mapsto \sigma(\alpha)/\alpha$ is $\sigma \mapsto \sigma(\alpha^\ell)/\alpha^\ell$, the Kummer cocycle of $a$ at level $k$ for the root $\beta = \alpha^\ell$. So the transition maps correspond to the projections $K^\times/(K^\times)^{\ell^{k+1}} \to K^\times/(K^\times)^{\ell^k}$, and Theorem 17.7(i) at each level gives (i). That $(\delta_k(a))_k$ is represented by $\kappa_a$ is Proposition 17.10(iii). (ii) Cohomology of discrete modules commutes with direct limits (by (17.1), a cochain of a profinite group with values in a direct limit factors through a finite quotient and takes finitely many values), so $H^1(K,\mu_{\ell^\infty}) = \varinjlim_k H^1(K,\mu_{\ell^k})$ along the inclusions $\mu_{\ell^k} \subset \mu_{\ell^{k+1}}$. If $\alpha^{\ell^k} = a$, then $\alpha^{\ell^{k+1}} = a^\ell$ and the cocycle $\sigma \mapsto \sigma(\alpha)/\alpha$ is also the Kummer cocycle of $a^\ell$ at level $k+1$; so the transition maps are $a \mapsto a^\ell$, and

$\varinjlim (K^\times/(K^\times)^{\ell^k}, a \mapsto a^\ell) = \varinjlim K^\times \otimes \mathbb{Z}/\ell^k = K^\times \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. (iii) The first two statements follow from (i), since $\widehat{K^\times}$ is the $\ell$-adic completion and $a$ maps to 0 if and only if it lies in every $(K^\times)^{\ell^k}$; the last is Theorem 17.7(iv) at each level. □

**Proposition 17.21** (local Kummer theory, $\ell \neq p$)**.** *Let $K$ be a finite extension of $\mathbb{Q}_p$, $\ell \neq p$, with ring of integers $\mathcal{O}_K$, uniformiser $\varpi$, residue field $\mathbb{F}_q$ and normalised valuation $v_K$; put $v = v_\ell(q-1)$.*

*(i) The sequence $0 \to \mathcal{O}_K^\times \otimes \mathbb{Z}_\ell \to \widehat{K^\times} \xrightarrow{v_K} \mathbb{Z}_\ell \to 0$ is exact and split (by $\varpi$), and $\mathcal{O}_K^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}/\ell^v$. Hence $H^1(K, \mathbb{Z}_\ell(1)) \simeq \mathbb{Z}_\ell \times \mathbb{Z}/\ell^v$, with torsion subgroup the classes of units.*

*(ii) $\#H^1(K, \mu_{\ell^k}) = \ell^k \cdot \#\mu_{\ell^k}(K) = \ell^{k+\min(k,v)}$.*

*(iii) $H^1(K, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) \simeq K^\times \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \simeq \mathbb{Q}_\ell/\mathbb{Z}_\ell$, through $v_K$.*

*(iv) $H^1(K, \mathbb{Q}_\ell(1)) \simeq \widehat{K^\times} \otimes \mathbb{Q}_\ell \simeq \mathbb{Q}_\ell$, spanned by the class of $\varpi$.*

*Proof.* (i) By Chapter 7, $K^\times = \varpi^{\mathbb{Z}} \times \mu_{q-1} \times U^{(1)}$ with $U^{(1)} = 1 + \varpi\mathcal{O}_K$ a pro-$p$ group, on which raising to an $\ell^k$-th power is bijective (it is a $\mathbb{Z}_p$-module, $x \mapsto x^a$ making sense for $a \in \mathbb{Z}_p$, and $\ell^k$ is a unit of $\mathbb{Z}_p$, with inverse $x \mapsto x^{\ell^{-k}}$). So $K^\times/(K^\times)^{\ell^k} \simeq \mathbb{Z}/\ell^k \times \mu_{q-1}/\mu_{q-1}^{\ell^k}$, the first factor given by $v_K$. The second has order $\gcd(\ell^k, q-1) = \ell^{\min(k,v)}$: in a cyclic group of order $q-1$ the $\ell^k$-th powers form the subgroup of index $\gcd(\ell^k, q-1)$. The projections from level $k+1$ to level $k$ are the reductions $\mathbb{Z}/\ell^{k+1} \to \mathbb{Z}/\ell^k$ on the first factor and, on the second, surjections between cyclic groups of orders $\ell^{\min(k+1,v)}$ and $\ell^{\min(k,v)}$, which are isomorphisms once $k \geq v$. So the limit is $\mathbb{Z}_\ell \times \mathbb{Z}/\ell^v$, and the second factor is $\mathcal{O}_K^\times \otimes \mathbb{Z}_\ell \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$. Apply Theorem 17.20(i). (ii) is the computation at level $k$, with Theorem 17.7(i). (iii) In $K^\times \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ the finite group $\mu_{q-1}$ and the uniquely $\ell$-divisible group $U^{(1)}$ contribute nothing, and $\varpi^{\mathbb{Z}} \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell$. (iv) By Chapter 16, $H^1(K, \mathbb{Q}_\ell(1)) = H^1(K, \mathbb{Z}_\ell(1)) \otimes \mathbb{Q}_\ell$, and the torsion dies. □

Numerical instances (with $v_3(6) = 1$, $v_3(18) = 2$, $v_5(10) = 1$, $v_3(4) = 0$): $H^1(\mathbb{Q}_7, \mathbb{Z}_3(1)) \simeq \mathbb{Z}_3 \times \mathbb{Z}/3$; $H^1(\mathbb{Q}_{19}, \mathbb{Z}_3(1)) \simeq \mathbb{Z}_3 \times \mathbb{Z}/9$; $H^1(\mathbb{Q}_{11}, \mathbb{Z}_5(1)) \simeq \mathbb{Z}_5 \times \mathbb{Z}/5$; $H^1(\mathbb{Q}_5, \mathbb{Z}_3(1)) \simeq \mathbb{Z}_3$, every unit of $\mathbb{Z}_5$ being a cube.

With the $\ell$-adic modules and the Kummer groups in hand, the next step is to say what is special about *local* fields: their cohomology groups with finite coefficients are finite, they satisfy an Euler characteristic formula, and they are paired with the cohomology of the Kummer dual. The computations just made for $\mu_{\ell^k}$ and $\mathbb{Z}_\ell(1)$ will serve as checks of these general theorems.

## 17.3 Local Galois cohomology: finiteness, Euler characteristic, Tate local duality

The three theorems of this section are the deep input of the chapter, and they are stated with references. Finiteness is behind the words "a finite group" in the paragraph after Lemma lem:dfg21; Tate local duality is what makes the dual terms $H^1_f(\mathbb{Q}_p, T^D)^*$ of Lemma lem:dfg21 the right objects (through Proposition 17.34); and the Euler characteristic, combined with duality, gives the counting arguments of Section 17.4. The examples at the end make the statements concrete over $\mathbb{Q}_p$.

Throughout the section $K$ is a finite extension of $\mathbb{Q}_p$ (the paper needs only $K = \mathbb{Q}_p$), $M$ is a finite discrete $G_K$-module, and $M^D = \mathrm{Hom}(M, \mu)$ its Kummer dual (Definition 17.15).

**Theorem 17.22** (finiteness; cohomological dimension). *(i) $H^n(K, M)$ is finite for every $n$, and $H^n(K, M) = 0$ for $n \geq 3$.*

*(ii) Consequently, for $T$ as in Convention 17.14 with a continuous action of $G_K$, the groups $H^n(K, T)$ are finitely generated $\mathbb{Z}_\ell$-modules, the $H^n(K, V)$ are finite-dimensional, the $H^n(K, W)$ are cofinitely generated, and Proposition 17.17 applies with $G = G_K$.*

We do not prove (i); see [177, Chapter II, Section 5], [143, Chapter VII, Section 2] and [131, Chapter I, Section 2]. (ii) The hypothesis of Proposition 17.17 is (i) for $M = T/\ell T$, so for $n \leq 2$ the statements about $T$, $V$ and $H^1(K, W)$ are Proposition 17.17(i), (iii). For $n \geq 3$: by (i) and induction on $k$, $H^n(K, T/\ell^k T) = 0$, so $H^n(K, T) = \varprojlim_k H^n(K, T/\ell^k T) = 0$ by Tate's limit theorem (Chapter 16; its hypothesis, finiteness of the $H^{n-1}(K, T/\ell^k T)$, holds by (i)), hence $H^n(K, V) = 0$; and $H^n(K, W) = \varinjlim_k H^n(K, W[\ell^k]) = 0$ since cohomology of discrete modules commutes with direct limits. For $H^2(K, W)$: the sequence of Proposition 17.17(iv) with $n = 2$ shows $H^2(K, W)[\ell]$ is a quotient of the finite group $H^2(K, W[\ell])$, and $H^2(K, W)$ is $\ell$-primary torsion, so it is cofinitely generated (Chapter 3). Alternatively, $H^2(K, W) \simeq H^0(K, T^D)^*$ by Theorem 17.24.

*Sketch of proof of (i) for $H^1$ and $p \nmid \#M$.* This is the case used at the primes of $\Sigma$; at $p = \ell$ the theorem is used as stated. The action of $G_K$ on $M$ factors through $\mathrm{Gal}(L/K)$ for a finite Galois extension $L/K$, and inflation-restriction gives $0 \to H^1(\mathrm{Gal}(L/K), M) \to H^1(K, M) \to \mathrm{Hom}_{\mathrm{cont}}(G_L, M)$. The first group is finite (finite group, finite module). A continuous homomorphism $G_L \to M$ kills the wild inertia group $P_L$, which is pro-$p$ while $M$ has order prime to $p$; so it factors through $\mathrm{Gal}(L^{\mathrm{tame}}/L)$, which is topologically generated by two elements, a lift of Frobenius and a generator of tame inertia (Chapter 7). Such a homomorphism is determined by two elements of $M$, so there are finitely many. □

**Theorem 17.23** (Tate's local Euler characteristic formula).

$$\frac{\#H^0(K, M) \cdot \#H^2(K, M)}{\#H^1(K, M)} = |\#M|_K = p^{-[K:\mathbb{Q}_p]\, v_p(\#M)}.$$

*In particular the left side is 1 when $p \nmid \#M$. For $V$ of type (V) over $\mathbb{Q}_\ell$: $\dim H^0(K, V) - \dim H^1(K, V) + \dim H^2(K, V)$ equals $-[K : \mathbb{Q}_p] \dim V$ if $\ell = p$, and $0$ if $\ell \neq p$.*

We do not prove this; see [177, Chapter II, Section 5], [131, Chapter I, Section 2] and [143, Chapter VII, Section 3]. The $\ell$-adic form follows from the finite one applied to $T/\ell^k T$, using Proposition 17.17(iv) to compare ranks.

**Theorem 17.24** (Tate local duality). *(i) For $0 \leq i \leq 2$ the cup product (Chapter 16) followed by the evaluation $M \otimes M^D \to \mu$ and the invariant (Theorem 17.12(ii), through $H^2(K, \mu) = \varinjlim_n H^2(K, \mu_n) \simeq \mathbb{Q}/\mathbb{Z}$),*

$$H^i(K, M) \times H^{2-i}(K, M^D) \longrightarrow H^2(K, \mu) \xrightarrow{\mathrm{inv}_K} \mathbb{Q}/\mathbb{Z},$$

*is a perfect pairing of finite groups. It is natural in $M$, compatible with restriction and corestriction, and compatible with connecting maps up to the sign $(-1)^i$.*

*(ii) ($\ell$-adic forms.) For $T$ as in Convention 17.14 with a continuous action of $G_K$ and $0 \le i \le 2$, the cup products with the pairings of Proposition 17.16(i) give perfect pairings*

$$H^i(K,W)\times H^{2-i}(K,T^D)\to \mathbb{Q}_\ell/\mathbb{Z}_\ell, \qquad H^i(K,T)\times H^{2-i}(K,W^D)\to \mathbb{Q}_\ell/\mathbb{Z}_\ell,$$
$$H^i(K,V)\times H^{2-i}(K,V^D)\to \mathbb{Q}_\ell;$$

*the first two identify the discrete group with the Pontryagin dual of the finitely generated one, the third is a perfect pairing of finite-dimensional $\mathbb{Q}_\ell$-vector spaces.*

*(iii) (Compatibility.) For $x\in H^1(K,V)$ and $y\in H^1(K,T^D)$, $\langle \pi x, y\rangle_{W,T^D} = \langle x, \iota y\rangle_{V,V^D} \bmod \mathbb{Z}_\ell$; symmetrically with the roles of $(T,V,W)$ and $(T^D,V^D,W^D)$ exchanged.*

We do not prove this; see [177, Chapter II, Section 5], [195], [131, Chapter I, Section 2] and [143, Chapter VII, Section 2] for (i), and [167, Section 1.4] for (ii) and (iii). Part (ii) is obtained from (i) applied to $T/\ell^kT$ and $(T/\ell^kT)^D = T^D/\ell^kT^D$ (Proposition 17.16(ii)) by passing to the limit. Part (iii) is the naturality of the cup product for the compatible pairings $V\times T^D\to\mathbb{Q}_\ell(1)$ and $W\times T^D\to\mathbb{Q}_\ell/\mathbb{Z}_\ell(1)$, together with the fact that $H^2(K,\mathbb{Q}_\ell(1))=\mathbb{Q}_\ell\to H^2(K,\mathbb{Q}_\ell/\mathbb{Z}_\ell(1))=\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is the projection. No sign matters for any order or length in this book, and we do not track signs.

**Corollary 17.25** (consequences of local duality)**.** *(i) $H^2(K,M)\simeq H^0(K,M^D)^*$. In particular $H^2(K,\mu_{\ell^k})\simeq\mathbb{Z}/\ell^k$ and $H^2(K,\mathbb{Z}/\ell^k)\simeq\mu_{\ell^k}(K)^*$.*

*(ii) If $p\nmid \#M$, then $\#H^1(K,M)=\#H^0(K,M)\cdot\#H^0(K,M^D)$. If $\ell\neq p$, then $\dim H^1(K,V)=\dim H^0(K,V)+\dim H^0(K,V^D)$.*

*(iii) If $\ell\neq p$, then $H^2(K,T)\simeq H^0(K,W^D)^*$; it is finite if $H^0(K,V^D)=0$.*

*Proof.* (i) is Theorem 17.24(i) with $i=2$; for $M=\mu_{\ell^k}$, $M^D=\mathbb{Z}/\ell^k$ with trivial action, and for $M=\mathbb{Z}/\ell^k$, $M^D=\mu_{\ell^k}$. (ii) Theorem 17.23 gives $\#H^1=\#H^0\cdot\#H^2$, since $|\#M|_K=1$ when $p\nmid\#M$, and (i) gives $\#H^2(K,M)=\#H^0(K,M^D)$. The dimension formula is the same argument with the $\ell$-adic forms of the two theorems: for $\ell\neq p$ the Euler characteristic gives $\dim H^1(K,V)=\dim H^0(K,V)+\dim H^2(K,V)$, and the third pairing of Theorem 17.24(ii) with $i=2$ gives $\dim H^2(K,V)=\dim H^0(K,V^D)$. (iii) is Theorem 17.24(ii) with $i=2$; if $H^0(K,V^D)=0$, then $H^0(K,W^D)$ is finite by Proposition 17.17(ii) applied to $T^D$. □

For instance, if $p\neq\ell$: $\dim H^1(\mathbb{Q}_p,\mathbb{Q}_\ell)=1+0=1$, because $H^0(\mathbb{Q}_p,\mathbb{Q}_\ell)=\mathbb{Q}_\ell$ (trivial action) while $H^0(\mathbb{Q}_p,\mathbb{Q}_\ell(1))=0$ (Frob acts on $\mathbb{Q}_\ell(1)$ by $p\neq 1$); and, exchanging the roles, $\dim H^1(\mathbb{Q}_p,\mathbb{Q}_\ell(1))=0+1=1$ (compare Proposition 17.21(iv)). For an elliptic curve $E/\mathbb{Q}_p$, $H^0(\mathbb{Q}_p,E[\ell])=E(\mathbb{Q}_p)[\ell]$ and $E[\ell]^D\simeq E[\ell]$ (Proposition 17.16(iv)), so (ii) gives $\#H^1(\mathbb{Q}_p,E[\ell])=\#E(\mathbb{Q}_p)[\ell]^2$.

**Example 17.26** (orders of local cohomology groups)**.** (a) Let $p\neq\ell$, $v=v_\ell(p-1)$, $k\ge1$ and $m=\min(k,v)$, so that $\#\mu_{\ell^k}(\mathbb{Q}_p)=\ell^m$. Then for $i=0,1,2$:

$$\#H^i(\mathbb{Q}_p,\mathbb{Z}/\ell^k)=\ell^k,\ \ell^{k+m},\ \ell^m; \qquad \#H^i(\mathbb{Q}_p,\mu_{\ell^k})=\ell^m,\ \ell^{k+m},\ \ell^k.$$

For $\mu_{\ell^k}$ this needs no duality: $H^0$ is $\mu_{\ell^k}(\mathbb{Q}_p)$, $H^1$ is Proposition 17.21(ii), and $H^2\simeq\mathrm{Br}(\mathbb{Q}_p)[\ell^k]=\mathbb{Z}/\ell^k$ by Theorem 17.7(ii) and Theorem 17.12(ii); the Euler characteristic $\ell^m\ell^k/\ell^{k+m}=1$ is visible. For $\mathbb{Z}/\ell^k$ the values come from Corollary 17.25(i), (ii): the action is trivial, so $H^0=\mathbb{Z}/\ell^k$; $H^2\simeq H^0(\mathbb{Q}_p,\mu_{\ell^k})^*$ has order $\ell^m$; and $\#H^1=\#H^0(\mathbb{Q}_p,\mathbb{Z}/\ell^k)\cdot\#H^0(\mathbb{Q}_p,\mu_{\ell^k})=\ell^{k+m}$. For $(p,\ell,k)=(7,3,1)$: $3,9,3$ and $3,9,3$; for $(19,3,2)$: $9,81,9$ and $9,81,9$; for $(5,3,2)$, where $m=0$: $9,9,1$ and $1,9,9$.

(b) $p$ odd, $M=\mu_p$ over $\mathbb{Q}_p$: $H^0=\mu_p(\mathbb{Q}_p)=1$ (the roots of unity in $\mathbb{Q}_p$ have order dividing $p-1$); $H^1=\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^p$ has order $p^2$, since $\mathbb{Q}_p^\times=p^{\mathbb{Z}}\times\mu_{p-1}\times(1+p\mathbb{Z}_p)$ and $1+p\mathbb{Z}_p\simeq\mathbb{Z}_p$; $H^2=\mathrm{Br}(\mathbb{Q}_p)[p]\simeq\mathbb{Z}/p$. The Euler characteristic is $1\cdot p/p^2=p^{-1}=|p|_p$, as Theorem 17.23 predicts for $\#M=p$.

(c) The curve `11a1` over $\mathbb{Q}_{11}$, $\ell=5$. The reduction at 11 is split multiplicative of type $\mathrm{I}_5$ ($a_{11}=1$), with Tate parameter $q_E=11^5u$, $u\in\mathbb{Z}_{11}^\times$, $\bar u=10=-1$ (computed with PARI, Example 17.46). As $-1=(-1)^5$, $\bar u$ is a fifth power in $\mathbb{F}_{11}^\times$, so $u$ is a fifth power in $\mathbb{Z}_{11}^\times$ (Hensel), and $q_E=(11u^{1/5})^5$ is a fifth power in $\mathbb{Q}_{11}$; and $\mu_5\subset\mathbb{Q}_{11}$ as $11\equiv1\pmod5$. The 5-torsion of the Tate curve is generated by $\mu_5$ and a fifth root of $q_E$ (Chapter 10), so $E[5]\subset E(\mathbb{Q}_{11})$: all twelve roots of the 5-division polynomial lie in $\mathbb{Q}_{11}$, and $4x^3+b_2x^2+2b_4x+b_6$ is a square in $\mathbb{Q}_{11}$ at each of them (checked with `polrootspadic`). Hence $\#H^1(\mathbb{Q}_{11},E[5])=\#E(\mathbb{Q}_{11})[5]^2=5^4$ by Corollary 17.25(ii). This is an Eisenstein example ($E(\mathbb{Q})_{\mathrm{tors}}=\mathbb{Z}/5$), used for illustration only.

Local duality pairs all of $H^1(\mathbb{Q}_p,M)$ with all of $H^1(\mathbb{Q}_p,M^D)$. Selmer groups, however, are cut out by *subgroups* of the local $H^1$, and what global duality needs is that the subgroups chosen for $M$ and for $M^D$ are each other's annihilators. The next section constructs the subgroups used at every prime $p\neq\ell$, the unramified classes, and proves exactly this.

## 17.4 Unramified cohomology

Unramified classes are the local conditions at every prime $p\neq\ell$ in the Selmer groups of the paper. This section computes them ($H^1_{\mathrm{ur}}(\mathbb{Q}_p,M)=M^{I_p}/(\mathrm{Frob}_p-1)M^{I_p}$), proves that for $p\neq\ell$ the unramified subgroups of $M$ and of its Kummer dual are exact annihilators, explains how local conditions on $V$ propagate to $T$ and $W$, and draws the consequence the paper states after Lemma lem:dfg21: at a prime $p\in\Sigma$ the propagated condition on $T^D$ is the torsion of $H^1(\mathbb{Q}_p,T^D)$, a finite group, isomorphic to $H^0(\mathbb{Q}_p,W^D)$. The torsion form of the exact sequence of Proposition 17.30 is the starting point of the Tamagawa factors of Chapter 18.

In this section $p$ is a prime, $G=G_{\mathbb{Q}_p}$, $I=I_p$, Frob an arithmetic Frobenius, and $M$ a $G$-module which is either discrete and torsion (type (D)), or of type (C), or of type (V). The parts that need $p\neq\ell$ say so.

The definition below is modelled on characters. For $M=\mathbb{Z}/n$ with trivial action, the group $H^1(\mathbb{Q}_p,\mathbb{Z}/n)$ is the group of continuous homomorphisms $G\to\mathbb{Z}/n$, and such a character is *unramified* if it kills $I$, that is, if the cyclic extension it cuts out is unramified; such characters are the same as homomorphisms $G/I\simeq\hat{\mathbb{Z}}\to\mathbb{Z}/n$, determined by their value at Frob, so there are exactly $n$ of them. For a general $M$ one keeps the two descriptions, "killed by restriction to $I$" and "coming from the quotient $G/I$", and checks that they agree. A global class that is unramified at almost every prime is the cohomological form of an extension unramified almost everywhere, and Selmer groups impose this condition at all but finitely many primes.

**Definition 17.27** (unramified classes)**.** Let $K$ be a finite extension of $\mathbb{Q}_p$, with inertia group $I_K\trianglelefteq G_K$, and $M$ a $G_K$-module of type (D), (C) or (V). The group of *unramified classes* is

$$H^1_{\mathrm{ur}}(K,M):=\mathrm{im}\big(\mathrm{Inf}\colon H^1(G_K/I_K,M^{I_K})\to H^1(K,M)\big)=\ker\big(\mathrm{Res}\colon H^1(K,M)\to H^1(I_K,M)\big);$$

for $K=\mathbb{Q}_p$, $G_K=G$ and $I_K=I$, this is $H^1_{\mathrm{ur}}(\mathbb{Q}_p,M)$. The results below are stated for $\mathbb{Q}_p$, the case the paper uses; their proofs apply verbatim to $K$, with Frob an arithmetic Frobenius of $K$

(acting by $x \mapsto x^q$ on the residue field $\mathbb{F}_q$ of $K$), $G_K/I_K \simeq \hat{\mathbb{Z}}$, and $|\cdot|_K$ in the Euler characteristic. The two descriptions agree, and inflation is injective, by inflation-restriction (Chapter 16, valid for the three types of coefficients); so $H^1_{\mathrm{ur}}(K,M) \simeq H^1(G_K/I_K, M^{I_K})$. The module $M$ is *unramified* if $I_K$ acts trivially on it. The definition is made for every $p$, including $p=\ell$, where it is *not* the right local condition (Warning 17.33; Chapter 18).

**Proposition 17.28** (computation of unramified cohomology). *(i) $c \mapsto c(\mathrm{Frob})$ induces an isomorphism $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) \simeq M^I/(\mathrm{Frob}-1)M^I$.*

*(ii) If $M$ is finite, $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = \#H^0(\mathbb{Q}_p, M)$.*

*(iii) For $V$ of type (V), $\dim H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = \dim H^0(\mathbb{Q}_p, V)$, and the following are equivalent: $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = 0$; $H^0(\mathbb{Q}_p, V) = 0$; $1$ is not an eigenvalue of Frob on $V^I$; $\det(1 - \mathrm{Frob} \mid V^I) \neq 0$; $\det(1 - \Phi \mid V^I) \neq 0$.*

*(iv) Let $p \neq \ell$ and $T$ be a lattice with $T \otimes \mathbb{Q}_\ell = V$. Then $T^I = T \cap V^I$ is a lattice in $V^I$, saturated in $T$, and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) = T^I/(\mathrm{Frob}-1)T^I$. If the conditions of (iii) hold, $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T)$ is finite of order*

$$\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) = \ell^{\,v_\ell(\det(1-\mathrm{Frob}\mid V^I))} = \ell^{\,v_\ell(\det(1-\Phi\mid V^I))}.$$

*Proof.* (i) $G/I \simeq \hat{\mathbb{Z}}$ is procyclic with topological generator the image of Frob, and $M^I$ is a $G/I$-module of the same type as $M$ ($M^I$ is closed in $M$ for types (C) and (V), being an intersection of kernels of continuous maps). By the procyclic computations of Chapter 16 (the discrete torsion case, and the compact case, which applies because $\hat{\mathbb{Z}}$ surjects onto $\mathbb{Z}_\ell$), $c \mapsto c(\mathrm{Frob})$ is an isomorphism $H^1(G/I, M^I) \simeq M^I/(\mathrm{Frob}-1)M^I$. Compose with the inverse of inflation. (ii) For an endomorphism $f$ of a finite group $A$, $\#\ker f \cdot \#\operatorname{im} f = \#A = \#\operatorname{im} f \cdot \#\operatorname{coker} f$, so $\ker f$ and $\operatorname{coker} f$ have the same order. Apply this to $f = \mathrm{Frob}-1$ on $A = M^I$: its kernel is $(M^I)^{\mathrm{Frob}=1} = M^G$, because $G$ is topologically generated by $I$ and Frob and the stabiliser of an element is closed, and its cokernel is $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)$ by (i). (iii) The same with dimensions; the kernel of $\mathrm{Frob}-1$ on $V^I$ is $V^G$. The equivalences of the last four conditions are linear algebra, and on $V^I$, of dimension $d$,

$$\det(1-\Phi) = \det(\mathrm{Frob}^{-1}(\mathrm{Frob}-1)) = (-1)^d \det(\Phi)\det(1-\mathrm{Frob}),$$

so one determinant vanishes if and only if the other does. (iv) $T^I = T \cap V^I$ is clear, and it is saturated in $T$ (if $\ell x \in T^I$ with $x \in T$, then $x$ is $I$-fixed, $T$ being torsion-free); every $x \in V^I$ has $\ell^a x \in T \cap V^I$, so $T^I$ spans $V^I$. Part (i) for $M = T$ gives $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) = T^I/(\mathrm{Frob}-1)T^I$. If $\det(1 - \mathrm{Frob} \mid V^I) \neq 0$, $\mathrm{Frob}-1$ is an injective endomorphism of the free $\mathbb{Z}_\ell$-module $T^I$, and its cokernel has order $\ell^{v_\ell(\det(\mathrm{Frob}-1))}$ by the elementary divisor theorem (Chapter 3): in suitable bases of $T^I \simeq \mathbb{Z}_\ell^d$ the matrix of $\mathrm{Frob}-1$ is $\operatorname{diag}(\ell^{a_1}, \dots, \ell^{a_d})$, the cokernel is $\bigoplus_i \mathbb{Z}/\ell^{a_i}$, of order $\ell^{\sum a_i}$, and the change of bases multiplies the determinant by a unit, so $\sum a_i = v_\ell(\det(\mathrm{Frob}-1))$. Finally $\det(\Phi \mid T^I)$ is a unit of $\mathbb{Z}_\ell$ ($\Phi$ is an automorphism of the lattice $T^I$), so the two determinants have the same valuation by the displayed identity. □

*Warning* 17.29 (the Frobenius convention). The *vanishing* of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V)$ and the *order* of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T)$ do not depend on whether one writes Frob or $\Phi$ (Proposition 17.28(iii), (iv)). The *value* of the determinant does. Only $\det(1 - \Phi \mid (V^D)^{I_p})$ equals $L_p(A_f, 1)^{-1}$ (Convention 17.2; Example 17.38(d) shows the two values at a multiplicative prime). The paper writes the geometric Frobenius $\mathrm{Frob}_p^{-1}$ in this determinant, in the paragraph after Lemma lem:dfg21.

**Proposition 17.30** (unramified classes and the action on $H^1(I,M)$)**.** *Let $M$ be of type (D), or of type (C) or (V) with $p \neq \ell$. The sequence*

$$0 \to H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) \to H^1(\mathbb{Q}_p, M) \xrightarrow{\mathrm{Res}} H^1(I,M)^{G/I} \to 0$$

*is exact, where $G/I$ acts on $H^1(I,M)$ by the conjugation action $(\sigma \cdot c)(i) = \sigma\, c(\sigma^{-1} i \sigma)$ of Chapter 16.*

*Proof.* Exactness on the left and in the middle is Definition 17.27; only the surjectivity of Res onto the invariants needs proof. The restriction lands in the invariants because conjugation by an element of $G$ acts trivially on $H^1(G,M)$ (Chapter 16).

*Type (D).* The five-term exact sequence of inflation-restriction for $I \trianglelefteq G$ and discrete coefficients (Chapter 16) reads $0 \to H^1(G/I, M^I) \to H^1(G,M) \to H^1(I,M)^{G/I} \to H^2(G/I, M^I)$, and the last group vanishes: $G/I \simeq \hat{\mathbb{Z}}$, and $H^2(\hat{\mathbb{Z}}, -) = 0$ for discrete torsion modules (Chapter 16, the procyclic computations).

*Type (C), $p \neq \ell$.* Put $M_k = M/\ell^k M$, a finite discrete module. By the case (D), each $\mathrm{Res}_k \colon H^1(G, M_k) \to A_k := H^1(I, M_k)^{G/I}$ is surjective. The groups $H^1(G, M_k)$ are finite (Theorem 17.22), and so are the $H^1(I, M_k)$: the wild inertia group $P$ is pro-$p$ and $M_k$ has $\ell$-power order, so $H^1(P, M_k) = 0$, inflation-restriction gives $H^1(I, M_k) = H^1(I/P, M_k^P)$, and $I/P$ is procyclic (Chapter 7), so this is a quotient of $M_k^P$ (Chapter 16). By Tate's limit theorem in degree 1 (Chapter 16, valid for every profinite group), $H^1(G,M) = \varprojlim_k H^1(G, M_k)$ and $H^1(I,M) = \varprojlim_k H^1(I, M_k)$, compatibly with restriction and with the action of $G/I$; and invariants commute with inverse limits, so $H^1(I,M)^{G/I} = \varprojlim_k A_k$. Let $(y_k)_k \in \varprojlim_k A_k$. The fibres $\mathrm{Res}_k^{-1}(y_k)$ are nonempty finite sets, and reduction maps $\mathrm{Res}_{k+1}^{-1}(y_{k+1})$ into $\mathrm{Res}_k^{-1}(y_k)$; an inverse limit of nonempty finite sets is nonempty (compactness), so there is a compatible family $(x_k)_k$ with $\mathrm{Res}_k x_k = y_k$, that is, an $x \in H^1(G,M)$ with $\mathrm{Res}\, x = (y_k)_k$.

*Type (V), $p \neq \ell$.* Let $T \subset M$ be a $G$-stable lattice (Chapter 16), so that $H^1(G,M) = H^1(G,T) \otimes \mathbb{Q}_\ell$ and $H^1(I,M) = H^1(I,T) \otimes \mathbb{Q}_\ell$ (Chapter 16). The group $X = H^1(I,T) = \varprojlim_k H^1(I, T/\ell^k T)$ is compact, and $X/\ell X$ embeds in the finite group $H^1(I, T/\ell T)$ (Proposition 17.17(iv), which holds for any profinite group); so $X$ is finitely generated by topological Nakayama (Chapter 3) and its torsion is killed by some $N \geq 1$. Let $y \in H^1(I,M)^{G/I}$ and write $y = \ell^{-a}\bar{x}$ with $x \in X$ and $\bar{x}$ its image. For $g \in G/I$, $gx - x$ maps to 0 in $H^1(I,M)$, so it is torsion and $g(Nx) = Nx$. By the case (C), $Nx = \mathrm{Res}\, z$ with $z \in H^1(G,T)$, and then $y = \mathrm{Res}\big((N\ell^a)^{-1} \iota z\big)$. □

Chapter 16 extends the five-term sequence to coefficients of types (C) and (V) when the quotient is $\hat{\mathbb{Z}}$, which gives the proposition at $p = \ell$ as well; the paper needs it only at $p \neq \ell$, where the argument above is self-contained.

*Remark* 17.31. For $p \neq \ell$ and $M$ of type (C), $H^1(I,M)$ is computed through the tame quotient $I \to \mathbb{Z}_\ell(1)$ (Chapter 16, the tame inertia example). Taking torsion subgroups in Proposition 17.30 for $M = T^D$ is the first step of the formula of Fontaine and Perrin-Riou, and the ideal $\mathrm{Fitt}_{\mathbb{Z}_\ell}\big((H^1(I_p, M)_{\mathrm{tors}})^{G/I}\big)$ (the Fitting ideal *of the invariants*) is the Tamagawa factor $\mathrm{Tam}_p(M)$ of Section 1.2 of the paper (Chapter 18).

**Theorem 17.32** (exact orthogonality of unramified classes)**.** *Let $p \neq \ell$.*

*(i) For a finite $G$-module $M$ of $\ell$-power order, $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M^D)$ are exact annihilators of each other under the pairing of Theorem 17.24(i).*

*(ii) For $V$ of type (V), $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V)$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D)$ are exact annihilators under the pairing of Theorem 17.24(ii).*

*Proof.* (i) *Orthogonality.* Let $x = \mathrm{Inf}(a)$ and $y = \mathrm{Inf}(b)$, where $a \in H^1(G/I, M^I)$ and $b \in H^1(G/I, (M^D)^I)$. The cup product commutes with inflation (Chapter 16), so $x \cup y$ is the inflation of $a \cup b$, and its image under $M \otimes M^D \to \mu_{\ell^\infty}$ is the inflation of a class in $H^2(G/I, \mu_{\ell^\infty}^I)$. That group is zero ($G/I \simeq \hat{\mathbb{Z}}$ and $\mu_{\ell^\infty}^I = \mu_{\ell^\infty}$ is discrete torsion), so $\langle x, y\rangle = 0$. *Counting.* By Proposition 17.28(ii), Corollary 17.25(i) and Theorem 17.23 (with $p \nmid \#M$),

$$\begin{aligned}\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) \cdot \#H^1_{\mathrm{ur}}(\mathbb{Q}_p, M^D) &= \#H^0(\mathbb{Q}_p, M) \cdot \#H^0(\mathbb{Q}_p, M^D) \\ &= \#H^0(\mathbb{Q}_p, M) \cdot \#H^2(\mathbb{Q}_p, M) = \#H^1(\mathbb{Q}_p, M).\end{aligned}$$

As the pairing is perfect, the annihilator of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)$ has order $\#H^1(\mathbb{Q}_p, M)/\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = \#H^1_{\mathrm{ur}}(\mathbb{Q}_p, M^D)$; it contains $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M^D)$, so it equals it, and symmetrically. (ii) The orthogonality argument is the same, with $H^2(\hat{\mathbb{Z}}, \mathbb{Q}_\ell(1)) = 0$. By Proposition 17.28(iii) and Corollary 17.25(ii), $\dim H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) + \dim H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = \dim H^0(\mathbb{Q}_p, V) + \dim H^0(\mathbb{Q}_p, V^D) = \dim H^1(\mathbb{Q}_p, V)$, and a perfect pairing of finite-dimensional spaces finishes the proof. □

A numerical instance: $p = 7$, $\ell = 3$, $M = \mathbb{Z}/3$, $M^D = \mu_3$, with $\mu_3 \subset \mathbb{Q}_7$. By Example 17.26(a) with $k = m = 1$, $H^1(\mathbb{Q}_7, \mathbb{Z}/3)$ has order 9. The unramified subgroup has order $\#H^0(\mathbb{Q}_7, \mathbb{Z}/3) = 3$: it consists of the three characters of $G_{\mathbb{Q}_7}$ that factor through $\mathrm{Gal}(\mathbb{Q}_{7^3}/\mathbb{Q}_7) \simeq \mathbb{Z}/3$, where $\mathbb{Q}_{7^3}$ is the unramified extension of degree 3. On the Kummer side $H^1_{\mathrm{ur}}(\mathbb{Q}_7, \mu_3)$ has order $\#H^0(\mathbb{Q}_7, \mu_3) = 3$ and is the image of the units (Proposition 17.39(i) below). The theorem says that these two groups of order 3 annihilate each other and that each is the full annihilator of the other: $3 \cdot 3 = 9$.

*Warning* 17.33 (the case $p = \ell$). Over $\mathbb{Q}_p$, $p$ odd: $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}/p) = \#H^0(\mathbb{Q}_p, \mathbb{Z}/p) = p$ and $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mu_p) = \#H^0(\mathbb{Q}_p, \mu_p) = 1$ (Proposition 17.28(ii), which holds for every $p$), while $\#H^1(\mathbb{Q}_p, \mathbb{Z}/p) = p \cdot \#H^2(\mathbb{Q}_p, \mathbb{Z}/p) \cdot p = p^2$ by Theorem 17.23 with $\#H^2(\mathbb{Q}_p, \mathbb{Z}/p) = \#\mu_p(\mathbb{Q}_p) = 1$. The annihilator of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}/p)$ has order $p$, not 1: the unramified subgroups are not exact annihilators when $p = \ell$. (By local class field theory the annihilator of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}/p)$ is the image of $\mathbb{Z}_p^\times$ in $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^p = H^1(\mathbb{Q}_p, \mu_p)$; we do not use this.) At $\ell$ the right local condition is that of Bloch and Kato (Chapter 18), and the book never writes $H^1_f = H^1_{\mathrm{ur}}$ at $p = \ell$.

The next proposition explains how a local condition on $V$ determines conditions on $T$ and on $W$, and why the dual of the quotient of $H^1(\mathbb{Q}_p, W)$ by the propagated condition is the propagated condition on $T^D$; this is the mechanism that turns the paper's definition $H^1_f(\mathbb{Q}_p, T^D) = \iota^{-1} H^1_f(\mathbb{Q}_p, V^D)$ into the dual terms of Lemma lem:dfg21.

**Proposition 17.34** (propagation of local conditions)**.** *Let $K$ be a finite extension of $\mathbb{Q}_p$ ($p = \ell$ allowed) and $T$ as in Convention 17.14, with a continuous action of $G_K$. For a $\mathbb{Q}_\ell$-subspace $L_V \subseteq H^1(K, V)$ put*

$$L_T := \iota^{-1}(L_V) \subseteq H^1(K, T), \qquad L_W := \pi(L_V) \subseteq H^1(K, W),$$

*the conditions* propagated *from $L_V$; likewise $L_{T^D}$, $L_{W^D}$ from a subspace $L_{V^D} \subseteq H^1(K, V^D)$.*

*(i) $L_T \supseteq H^1(K, T)_{\mathrm{tors}}$, and $L_W$ is divisible, so $L_W \subseteq H^1(K, W)_{\mathrm{div}}$.*

*(ii) If $L_V$ and $L_{V^D}$ are exact annihilators under the pairing $H^1(K, V) \times H^1(K, V^D) \to \mathbb{Q}_\ell$, then $L_W$ and $L_{T^D}$ are exact annihilators under $H^1(K, W) \times H^1(K, T^D) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$; hence $\big(H^1(K, W)/L_W\big)^* \simeq L_{T^D}$. Symmetrically $L_T$ and $L_{W^D}$ are exact annihilators.*

*(iii) If $p \neq \ell$, $T$ is unramified and $L_V = H^1_{\mathrm{ur}}(K,V)$, then $L_T = H^1_{\mathrm{ur}}(K,T)$ and $L_W = H^1_{\mathrm{ur}}(K,W)$.*

*Proof.* (i) $\ker\iota = H^1(K,T)_{\mathrm{tors}}$ by Proposition 17.17(ii), and $\pi$ is $\mathbb{Z}_\ell$-linear on a $\mathbb{Q}_\ell$-vector space, so its image is divisible. (ii) Let $y \in L_{T^D}$ and $x \in L_V$. By Theorem 17.24(iii), $\langle \pi x, y\rangle = \langle x, \iota y\rangle \bmod \mathbb{Z}_\ell = 0$, since $\iota y \in L_{V^D}$ is orthogonal to $L_V$. So $L_{T^D} \subseteq L_W^\perp$. Conversely let $y \in H^1(K,T^D)$ be orthogonal to $L_W$. For $x \in L_V$ and $k \geq 0$, $\ell^{-k}x \in L_V$ ($L_V$ is a $\mathbb{Q}_\ell$-subspace), so $\pi(\ell^{-k}x) \in L_W$ and, by the compatibility again, $0 = \langle \pi(\ell^{-k}x), y\rangle = \ell^{-k}\langle x, \iota y\rangle \bmod \mathbb{Z}_\ell$. Thus the element $\langle x, \iota y\rangle$ of $\mathbb{Q}_\ell$ lies in $\ell^k\mathbb{Z}_\ell$ for every $k$, whence $\langle x, \iota y\rangle = 0$. Thus $\iota y \in L_V^\perp = L_{V^D}$ and $y \in L_{T^D}$. This proves $L_W^\perp = L_{T^D}$. The pairing identifies the discrete group $H^1(K,W)$ with the Pontryagin dual of $H^1(K,T^D)$, and for a subgroup $A$ of a discrete group, $(A^\perp)^\perp = A$ (Pontryagin duality, Chapter 3); so $L_{T^D}^\perp = L_W$ as well, and $(H^1(K,W)/L_W)^* = L_W^\perp = L_{T^D}$. The symmetric statement is the same argument with the second pairing of Theorem 17.24(ii). (iii) As $I$ acts trivially, $H^1(I,T) = \mathrm{Hom}_{\mathrm{cont}}(I,T)$ and $H^1(I,V) = \mathrm{Hom}_{\mathrm{cont}}(I,V)$, and $\mathrm{Hom}(I,T) \to \mathrm{Hom}(I,V)$ is injective, $T$ being torsion-free. So for $x \in H^1(K,T)$, $\mathrm{Res}_I(\iota x) = 0$ if and only if $\mathrm{Res}_I\, x = 0$: $L_T = H^1_{\mathrm{ur}}(K,T)$. Inflation commutes with $\pi$, and, as $I$ acts trivially on $V$ and $W$, so that $V^I = V$ and $W^I = W$, Proposition 17.28(i) identifies $\pi$ on unramified classes with the map $H^1(G/I,V) = V/(\mathrm{Frob}-1)V \to H^1(G/I,W) = W/(\mathrm{Frob}-1)W$ induced by $V \to W$; it is surjective because $V \to W$ is. So $\pi(H^1_{\mathrm{ur}}(K,V)) = H^1_{\mathrm{ur}}(K,W)$. □

*Warning* 17.35. For $p \neq \ell$ and $T$ *ramified* at $p$, the propagated condition $\iota^{-1}(H^1_{\mathrm{ur}}(\mathbb{Q}_p,V))$ can be strictly larger than $H^1_{\mathrm{ur}}(\mathbb{Q}_p,T)$: it contains all of $H^1(\mathbb{Q}_p,T)_{\mathrm{tors}}$. The quotient is what the Tamagawa factor of Chapter 18 measures. The two coincide when $T$ is unramified at $p$ (Proposition 17.34(iii)); they must not be identified in general.

**Corollary 17.36** (vanishing unramified cohomology)**.** *Let $p \neq \ell$, let $T$ be a lattice with a continuous action of $G_{\mathbb{Q}_p}$, and suppose $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V) = 0$. Then $H^0(\mathbb{Q}_p,V) = 0$, and for the conditions propagated from $L_V = H^1_{\mathrm{ur}}(\mathbb{Q}_p,V) = 0$:*

$$L_T = \iota^{-1}(0) = H^1(\mathbb{Q}_p,T)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}_p,W), \qquad L_W = 0.$$

*$L_T$ is a finite group, and it contains the finite group $H^1_{\mathrm{ur}}(\mathbb{Q}_p,T)$.*

*Proof.* $H^0(\mathbb{Q}_p,V) = 0$ by Proposition 17.28(iii). Then $L_T = \ker\iota = H^1(\mathbb{Q}_p,T)_{\mathrm{tors}}$, which is $\delta(H^0(\mathbb{Q}_p,W)) \simeq H^0(\mathbb{Q}_p,W)$ and finite by Proposition 17.17(ii) and Theorem 17.22. $H^1_{\mathrm{ur}}(\mathbb{Q}_p,T)$ is finite by Proposition 17.28(iv), hence torsion, hence contained in $L_T$. $L_W = \pi(0) = 0$. □

*In the paper* 17.37. The paragraph after Lemma lem:dfg21 reads: "For $p \in \Sigma$ the group $H^1_f(\mathbb{Q}_p,V^D) = H^1_{\mathrm{ur}}(\mathbb{Q}_p,V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ vanishes, because $\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = L_p(A_f,1)^{-1} \neq 0$; for the same reason $H^0(\mathbb{Q}_p,V^D) = 0$, …", and, in the next sentence, "So $H^1_f(\mathbb{Q}_p,T^D)$ is the torsion of $H^1(\mathbb{Q}_p,T^D)$, a finite group." In the book's terms: the equality $H^1_f = H^1_{\mathrm{ur}}$ at $p \neq \ell$ is Chapter 18; the equality with $(V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ is Proposition 17.28(i); the vanishing and the vanishing of $H^0(\mathbb{Q}_p,V^D)$ are Proposition 17.28(iii), applied to $V^D$, with the nonvanishing of $L_p(A_f,1)^{-1}$ supplied by Chapter 19; and the last sentence is Corollary 17.36 applied to $T^D$, since $H^1_f(\mathbb{Q}_p,T^D)$ is defined in Lemma lem:dfg21 as the propagated condition $\iota^{-1}H^1_f(\mathbb{Q}_p,V^D)$. *Commentary.* The book proves slightly more than the paper states: $H^1_f(\mathbb{Q}_p,T^D) \simeq H^0(\mathbb{Q}_p,W^D)$, which is how Chapter 18 computes its length (Lemma lem:fpr). The determinant is written with $\mathrm{Frob}_p^{-1}$, the geometric Frobenius, as Warning 17.29 requires (with $\mathrm{Frob}_p$ it would be $1 - p^2$ instead of $1 - p^{-2}$ at a multiplicative prime); the vanishing, and the finiteness of $H^1_f(\mathbb{Q}_p,T^D)$, hold in either convention.

**Example 17.38** (unramified cohomology in the paper's situations)**.** Let $p \neq \ell$.

(a) $T = \mathbb{Z}_\ell(1)$: $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(p-1)\mathbb{Z}_\ell(1)$, cyclic of order $\ell^{v_\ell(p-1)}$; under Theorem 17.20 it is the image of $\mathbb{Z}_p^\times$ (Proposition 17.39).

(b) $T = \mathbb{Z}_\ell$ with trivial action: every continuous homomorphism $\psi\colon G_{\mathbb{Q}_p} \to \mathbb{Z}_\ell$ is unramified. Indeed $\psi|_I$ factors through the tame character (Chapter 7): $\psi$ kills the wild inertia group, a pro-$p$ group mapping to the pro-$\ell$ group $\mathbb{Z}_\ell$, and is then determined by its value on a topological generator $t$ of the $\ell$-part of tame inertia. The tame relation $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^p$ holds in the tame quotient, and $\psi$, a homomorphism to an abelian group, takes the same value on conjugate elements; so $\psi(t) = \psi(t^p) = p\psi(t)$, that is, $(p-1)\psi(t) = 0$, and $\psi(t) = 0$ because $\mathbb{Z}_\ell$ is torsion-free. Hence $H^1(\mathbb{Q}_p, \mathbb{Z}_\ell) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}_\ell) = \mathbb{Z}_\ell$, the dual $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) = \mu_{\ell^\infty}/\mu_{\ell^\infty}^{p-1} = 0$ (a divisible group), and Proposition 17.34(ii), (iii) are consistent.

(c) The Tate curve: if $E$ has multiplicative reduction at $p$, $T = T_\ell E$ has a basis $e_1, e_2$ with $I$ acting through $e_2 \mapsto e_2 + n'e_1$, $n' \neq 0$, and $\mathrm{Frob}\, e_1 = \varepsilon p e_1$, $\varepsilon = \pm 1$ (Chapter 11). So $V^I = \mathbb{Q}_\ell e_1$ with Frob-eigenvalue $\varepsilon p \neq 1$; $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V_\ell E) = 0$ and $H^1(\mathbb{Q}_p, T_\ell E)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}_p, E[\ell^\infty]) = E(\mathbb{Q}_p)[\ell^\infty]$.

(d) $V = \mathrm{ad}^0 V_\ell E$ at a multiplicative prime: $V^I = \mathbb{Q}_\ell e$, $e = e_1 \otimes e_2^\vee$, with $\mathrm{Frob}\, e = pe$ (Chapter 16, the model computation). In matrix terms, $\mathrm{ad}^0 V$ is the space of trace-zero endomorphisms of $V_\ell E$ with $g$ acting by $X \mapsto \rho(g) X \rho(g)^{-1}$, and $e$ is the matrix unit $E_{12}$ ($e_2 \mapsto e_1$, $e_1 \mapsto 0$). A generator of the image of inertia is $\left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$ with $n' \neq 0$; the matrices commuting with it are the combinations of 1 and $E_{12}$, and the trace-zero ones are the multiples of $E_{12}$. For an upper triangular $B = \left(\begin{smallmatrix} b_{11} & b_{12} \\ 0 & b_{22} \end{smallmatrix}\right)$ one has $BE_{12}B^{-1} = (b_{11}/b_{22})E_{12}$, and with $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} \varepsilon p & * \\ 0 & \varepsilon \end{smallmatrix}\right)$ this gives $\mathrm{Frob}\, e = pe$, whatever the sign $\varepsilon$. For $V^D = V(1)$ the eigenvalue of Frob on $(V^D)^I$ is $p^2$ and that of $\Phi$ is $p^{-2}$. So $\det(1 - \Phi \mid (V^D)^I) = 1 - p^{-2}$, the paper's $L_p(A_f, 1)^{-1}$ at a multiplicative prime, while $\det(1 - \mathrm{Frob} \mid (V^D)^I) = 1 - p^2$: different numbers, with the same $\ell$-adic valuation $v_\ell(p^2 - 1)$ and the same nonvanishing. By Proposition 17.28(iv), $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, T^D) = \ell^{v_\ell(p^2-1)}$.

(e) (An illustration that depends on Chapter 19.) Types III, III$^*$ at $q \geq 5$: by Lemma lem:euler (proved in Chapter 19) $(\mathrm{ad}^0 V)^{I_q}$ is a line on which Frob acts by $\chi_{-4}(q) = \pm 1$. Then $(V^D)^{I_q}$ is a line with Frob-eigenvalue $q\chi_{-4}(q)$, and $\#H^1_{\mathrm{ur}}(\mathbb{Q}_q, T^D) = \ell^{v_\ell(1 - q\chi_{-4}(q))} = \ell^{v_\ell(q - \chi_{-4}(q))}$, which is $\ell^{e_\ell(E,q)}$ in the notation of Definition def:terms.

(f) Good reduction at $p$: $V_\ell E$ is unramified and the eigenvalues of Frob have absolute value $\sqrt{p}$ (Hasse), so 1 is not one of them and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V_\ell E) = 0$.

The examples (c) and (d) concern a prime of multiplicative reduction, where the matrix of $\rho(\mathrm{Frob})$ has a corner entry $c'$ that the order of $H^1_{\mathrm{ur}}$ does not see but the Tamagawa factor of Lemma lem:tamfactor does. To compute that corner one needs the Kummer class of the Tate parameter, which is the subject of the next section.

## 17.5 The Kummer class of the Tate parameter

This section proves the last sentences of the paper's proof of Lemma lem:tamfactor: the splitting of $H^1(\mathbb{Q}_q, \mathbb{Z}_\ell(1))$ given by the uniformiser $q$, the identification of the unramified component of

the Kummer class of the Tate parameter with the class of the reduction of its unit part, and the exact indeterminacy of the corner entry $c'$ of $\rho(\mathrm{Frob})$. The field-theoretic facts (Hensel's lemma, the tame character $t_\ell$, the decomposition $\kappa_x = v_K(x)\kappa_\varpi + \kappa_u$, the value of $\kappa_u$ at Frobenius, the Frobenius lift fixing roots of the uniformiser) are proved in Chapter 7, and the group cohomology of $\mathrm{ad}^0 T$ is Chapter 16 (the model computation). What is added here is the cohomological reading and the identification of $c'$; the assembly of Lemma lem:tamfactor is Chapter 18.

Throughout, $q \neq \ell$ are primes, $G = G_{\mathbb{Q}_q}$, $I = I_q$, $v = v_\ell(q-1)$, and $t_\ell \colon I \to \mathbb{Z}_\ell(1)$ is the tame character of Chapter 7: for any compatible system of $\ell$-power roots of $q$, $t_\ell(\sigma) = \kappa_q(\sigma)$ for $\sigma \in I$, independently of the system.

**Proposition 17.39** (unramified Kummer classes are unit classes)**.** *Let $K$ be a finite extension of $\mathbb{Q}_p$, $p \neq \ell$, with residue field $k_K$, ring of integers $\mathcal{O}_K$ and uniformiser $\varpi$.*

*(i) For $n$ prime to $p$, the Kummer isomorphism $K^\times/(K^\times)^n \simeq H^1(K,\mu_n)$ maps the image of $\mathcal{O}_K^\times$ onto $H^1_{\mathrm{ur}}(K,\mu_n)$. In the limit, $\mathcal{O}_K^\times \otimes \mathbb{Z}_\ell \subseteq \widehat{K^\times}$ corresponds to $H^1_{\mathrm{ur}}(K,\mathbb{Z}_\ell(1))$.*

*(ii) The valuation sequence of Proposition 17.21(i) corresponds to the sequence of Proposition 17.30 for $M = \mathbb{Z}_\ell(1)$:*

$$\begin{array}{ccccccccc} 0 & \to & \mathcal{O}_K^\times \otimes \mathbb{Z}_\ell & \to & \widehat{K^\times} & \xrightarrow{v_K} & \mathbb{Z}_\ell & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & H^1_{\mathrm{ur}}(K,\mathbb{Z}_\ell(1)) & \to & H^1(K,\mathbb{Z}_\ell(1)) & \xrightarrow{\mathrm{Res}} & \mathrm{Hom}(I_K,\mathbb{Z}_\ell(1))^{G/I} & \to & 0 \end{array}$$

*with vertical isomorphisms the Kummer map and $a \mapsto a\,t_\ell$; in particular $\mathrm{Res}\,\kappa_x = v_K(x)\,t_\ell$ and $\mathrm{Hom}_{\mathrm{cont}}(I_K,\mathbb{Z}_\ell(1))^{G/I} = \mathbb{Z}_\ell\,t_\ell$.*

*Proof.* (i) A unit $u$ has an $n$-th root in $K^{\mathrm{ur}}$: $X^n - \bar{u}$ is separable over $k_K$, so it has a simple root in a finite extension of $k_K$, which lifts by Hensel's lemma to a root in the corresponding finite unramified extension of $K$ (Chapter 7). So $I_K$ fixes an $n$-th root of $u$, and the Kummer cocycle of $u$ built from it vanishes on $I_K$: $\delta(u) \in H^1_{\mathrm{ur}}(K,\mu_n)$. Conversely let $x = \varpi^a u$ with $\mathrm{Res}_{I_K}\delta(x) = 0$. By Proposition 17.10(i) for $L = K^{\mathrm{ur}}$ (whose absolute Galois group is $I_K$), $x$ is an $n$-th power in $K^{\mathrm{ur}}$; as $u$ is one, so is $\varpi^a$, and since $v_K$ extends to $K^{\mathrm{ur}}$ with the same value group, $n \mid a$. So $x \in \mathcal{O}_K^\times (K^\times)^n$. The limit statement follows by Theorem 17.20. (ii) The formula $\mathrm{Res}\,\kappa_x = v_K(x)t_\ell$ is Chapter 7 (the theorem on the Kummer cocycle on inertia), so the right square commutes. Every continuous homomorphism $I_K \to \mathbb{Z}_\ell(1)$ factors through $t_\ell$, which is surjective, so $\mathrm{Hom}_{\mathrm{cont}}(I_K,\mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell t_\ell$; and $t_\ell$ is invariant under the conjugation action, since $t_\ell(\sigma^{-1}i\sigma) = \chi_\ell(\sigma)^{-1}t_\ell(i)$ (the tame relation, Chapter 7) gives $(\sigma\cdot t_\ell)(i) = \chi_\ell(\sigma)t_\ell(\sigma^{-1}i\sigma) = t_\ell(i)$. The left square commutes by (i), and the outer vertical maps are isomorphisms, hence so is the middle one (as we knew from Theorem 17.20). □

**Proposition 17.40** (the value of an unramified Kummer class)**.** *Let $K = \mathbb{Q}_q$. Then $\mathbb{Z}_\ell(1)^I = \mathbb{Z}_\ell(1)$, and $c \mapsto c(\mathrm{Frob})$ gives*

$$H^1(G/I,\mathbb{Z}_\ell(1)) \;\simeq\; \mathbb{Z}_\ell(1)/(q-1)\mathbb{Z}_\ell(1) \;=\; \mu_{\ell^v}(\mathbb{F}_q) \qquad (\text{cyclic of order } \ell^v).$$

*Let $u \in \mathbb{Z}_q^\times$ and let $\kappa_u$ be a Kummer cocycle built from a compatible system of $\ell$-power roots of $u$ in $\mathbb{Q}_q^{\mathrm{ur}}$. Then $\kappa_u$ is inflated from $G/I$ as a cocycle (not only as a class), so $\kappa_u(\mathrm{Frob})$ depends only on the image of* Frob *in $G/I$, and its class in $\mathbb{Z}_\ell(1)/(q-1)$ is $\bar{u}^{(q-1)/\ell^v}$, that is, the class of the reduction $\bar{u}$ in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell = \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v}$. Hence the composite*

$$\mathbb{Z}_q^\times \otimes \mathbb{Z}_\ell \xrightarrow{\sim} H^1_{\mathrm{ur}}(\mathbb{Q}_q,\mathbb{Z}_\ell(1)) = H^1(G/I,\mathbb{Z}_\ell(1)) \xrightarrow{\sim} \mathbb{Z}_\ell(1)/(q-1)$$

*is the isomorphism "reduce modulo $q$ and take the class in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$".*

*Proof.* $I$ acts trivially on $\mathbb{Z}_\ell(1)$ because $\mathbb{Q}_q^{\mathrm{ur}}$ contains all $\ell$-power roots of unity, and Frob acts by $q$; the isomorphism is the procyclic computation of Chapter 16 (its example with Frob acting by an integer $q$ prime to $\ell$). For $\sigma \in I$ and $g \in G$, $\kappa_u(g\sigma) = \kappa_u(g) + \chi_\ell(g)\kappa_u(\sigma) = \kappa_u(g)$, because the chosen roots of $u$ lie in $\mathbb{Q}_q^{\mathrm{ur}}$ and are fixed by $\sigma$; so $\kappa_u$ factors through $G/I$. The class of $\kappa_u(\mathrm{Frob})$ is computed in Chapter 7 (the theorem on the Kummer cocycle on inertia and on Frobenius, part (iii)): it is the Teichmüller lift of $\bar{u}^{(q-1)/\ell^v}$. Finally $\mathbb{Z}_q^\times \otimes \mathbb{Z}_\ell = \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$ because $1 + q\mathbb{Z}_q$ is pro-$q$, and $y \mapsto y^{(q-1)/\ell^v}$ maps $\mathbb{F}_q^\times$ onto $\mu_{\ell^v}(\mathbb{F}_q)$ with kernel $(\mathbb{F}_q^\times)^{\ell^v}$. □

After a choice of a generator of $\mathbb{Z}_\ell(1)$, $\mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{Z}/\ell^v$, and the valuation of a class is read as in Example 17.11: $v_\ell(c) = \min(v_\ell(\tilde{c}), v)$, so $v_\ell(0) = v$; it does not depend on the generator.

**Proposition 17.41** (the canonical splitting attached to the uniformiser $q$)**.** *For $K = \mathbb{Q}_q$ the prime $q$ is a canonical uniformiser and $\mathbb{Q}_q^\times = q^{\mathbb{Z}} \times \mathbb{Z}_q^\times$ canonically. Hence*

$$H^1(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell\,[\kappa_q] \;\oplus\; H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Z}_\ell(1))$$

*canonically, and there is a canonical retraction*

$$\mathrm{pr}_{\mathrm{ur}} \colon H^1(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) \to H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) = H^1(G/I, \mathbb{Z}_\ell(1)^I)$$

*with $\mathrm{pr}_{\mathrm{ur}}[\kappa_x] = [\kappa_{xq^{-v_q(x)}}]$: it sends the class of $x$ to the class of its unit part.*

*Proof.* Tensor the decomposition $\mathbb{Q}_q^\times = q^{\mathbb{Z}} \times \mathbb{Z}_q^\times$ with $\mathbb{Z}_\ell$ after $\ell$-adic completion: $\widehat{\mathbb{Q}_q^\times} = \mathbb{Z}_\ell\,\hat{q} \oplus (\mathbb{Z}_q^\times \otimes \mathbb{Z}_\ell)$. Apply the Kummer isomorphism (Theorem 17.20) and Proposition 17.39(i). □

*Remark* 17.42. Two cautions. First, the retraction is not functorial: inflation goes from $H^1(G/I, -)$ to $H^1(G, -)$, not the other way, and over a general finite extension $K$ of $\mathbb{Q}_q$ the analogous splitting depends on the choice of uniformiser (replacing $\varpi$ by $\varpi w$, $w$ a unit, changes $\mathrm{pr}_{\mathrm{ur}}[\kappa_\varpi]$ from 0 to $-[\kappa_w]$, since $\varpi = \varpi' w^{-1}$; compare Exercise 17.86). Over $\mathbb{Q}_q$ the uniformiser $q$ is canonical, and the unramified component of $[\kappa]$ in the paper's splitting $\mathbb{Z}_\ell\kappa_q \oplus H^1_{\mathrm{ur}}$ has the precise meaning $\mathrm{pr}_{\mathrm{ur}}[\kappa]$. Second, the value $\kappa(\mathrm{Frob})$ of a cocycle at a Frobenius lift is a different object from the class $\mathrm{pr}_{\mathrm{ur}}[\kappa]$; Theorem 17.43(iii), (iv) compare them.

**Theorem 17.43** (the corner of $\rho(\mathrm{Frob})$ at a multiplicative prime)**.** *Let $E/\mathbb{Q}$ have multiplicative reduction at $q$, let $\ell \neq q$ be odd, $q_E \in \mathbb{Q}_q^\times$ the Tate parameter, $n = v_q(q_E) = n_q$, $u = q_E q^{-n} \in \mathbb{Z}_q^\times$ its unit part, and $\varepsilon \colon G \to \{\pm 1\}$ the unramified character with $\varepsilon(\mathrm{Frob}) = a_q(E)$. Fix compatible systems of $\ell$-power roots of $q$ and of $u$ (the latter in $\mathbb{Q}_q^{\mathrm{ur}}$), give $q_E$ the product system, and let $\kappa = \kappa_{q_E}$, $\kappa_q$, $\kappa_u$ be the Kummer cocycles. Fix a generator $\zeta$ of $\mathbb{Z}_\ell(1)$ and write $\mathbb{Z}_\ell(1) = \mathbb{Z}_\ell$. By Chapters 10 and 11, $T = T_\ell E$ has a basis $e_1, e_2$ in which*

$$\rho(g) = \begin{pmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa(g) \\ 0 & \varepsilon(g) \end{pmatrix} \qquad (g \in G),$$

*the corner being $\varepsilon\kappa$. Then:*

*(i) $\kappa = n\kappa_q + \kappa_u$ as cocycles; $[\kappa]$ is the Kummer class of $q_E$, and $\mathrm{pr}_{\mathrm{ur}}[\kappa] = [\kappa_u]$ is the class of $\bar{u}$ in $\mathbb{F}_q^\times \otimes \mathbb{Z}_\ell = \mathbb{Z}_\ell(1)/(q-1)$.*

*(ii)* $\kappa|_I = n\,t_\ell$. *For* $t \in I$ *with* $t_\ell(t) = \zeta$, $te_1 = e_1$ *and* $te_2 = e_2 + ne_1$: *the paper's* $n'$ *may be taken equal to* $n$, *and in any case* $v_\ell(n') = v_\ell(n)$.

*(iii) Let* $\mathrm{Frob}_0$ *be a Frobenius lift fixing the chosen* $\ell$*-power roots of* $q$. *Then* $\kappa_q(\mathrm{Frob}_0) = 0$ *and*

$$\rho(\mathrm{Frob}_0) = \begin{pmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{pmatrix}, \qquad \varepsilon = a_q(E), \quad c' = \varepsilon\,\kappa_u(\mathrm{Frob}_0),$$

*and the class of* $c'$ *in* $\mathbb{Z}_\ell(1)/(q-1)$ *is* $\varepsilon$ *times the class of* $\bar{u}$.

*(iv) For an arbitrary lift* $\mathrm{Frob} = \mathrm{Frob}_0\,\sigma$ *(*$\sigma \in I$*) and an arbitrary cocycle in the class of* $\kappa$ *(another choice of roots of* $q_E$*), the corner* $c'$ *of* $\rho(\mathrm{Frob})$ *changes by an element of the ideal* $(n, q-1)\mathbb{Z}_\ell$: $\kappa(\mathrm{Frob}_0\,\sigma) = \kappa(\mathrm{Frob}_0) + q\,n\,t_\ell(\sigma)$, *and a coboundary changes the value at* Frob *by* $(q-1)z$, $z \in \mathbb{Z}_\ell$. *A change of* $\zeta$ *multiplies* $c'$ *by a unit.*

*(v) Consequently, for every choice,*

$$\min(v_\ell(n), v_\ell(q-1), v_\ell(c')) = v_\ell((n, q-1, c')\mathbb{Z}_\ell) = \min(v_\ell(n), v_\ell(q-1), v_\ell(c)),$$

*where* $c \in \mathbb{Z}/\ell^v$ *is the class of* $\bar{u}$ *(for any generator) and* $v_\ell(0) = v$.

*Proof.* (i) The cocycle identity $\kappa = n\kappa_q + \kappa_u$ for the product system is Chapter 7 (part (i) of its theorem on the Kummer cocycle). That $[\kappa]$ is the Kummer class of $q_E$ is Theorem 17.20(i). By Proposition 17.41, $\mathrm{pr}_{\mathrm{ur}}[\kappa] = [\kappa_u]$, and by Proposition 17.40 its value is the class of $\bar{u}$. (ii) $\kappa_u$ vanishes on $I$ (its roots lie in $\mathbb{Q}_q^{\mathrm{ur}}$) and $\kappa_q|_I = t_\ell$, so $\kappa|_I = nt_\ell$; this is independent of the roots, since coboundaries $(\chi_\ell - 1)z$ vanish on $I$. As $\varepsilon|_I = 1$ and $\chi_\ell|_I = 1$, $\rho(t) = \left(\begin{smallmatrix} 1 & n \\ 0 & 1 \end{smallmatrix}\right)$ in the basis $e_1, e_2$. (iii) $\mathrm{Frob}_0$ fixes the chosen roots of $q$, so $\kappa_q(\mathrm{Frob}_0) = 0$ and $\kappa(\mathrm{Frob}_0) = \kappa_u(\mathrm{Frob}_0)$; also $\chi_\ell(\mathrm{Frob}_0) = q$ and $\varepsilon(\mathrm{Frob}_0) = a_q(E)$. The class of $\kappa_u(\mathrm{Frob}_0)$ modulo $(q-1)$ is the class of $\bar{u}$ by Proposition 17.40. Such a lift exists by Chapter 7 (its corollary on the Frobenius lift fixing roots). (iv) By the cocycle identity and (ii), $\kappa(\mathrm{Frob}_0\,\sigma) = \kappa(\mathrm{Frob}_0) + \chi_\ell(\mathrm{Frob}_0)\kappa(\sigma) = \kappa(\mathrm{Frob}_0) + q\,n\,t_\ell(\sigma)$, and $\varepsilon(\mathrm{Frob}_0\,\sigma) = \varepsilon(\mathrm{Frob}_0)$; so $c'$ changes by $\varepsilon q n t_\ell(\sigma) \in n\mathbb{Z}_\ell$. Another system of roots of $q_E$ changes $\kappa$ by the coboundary $g \mapsto (\chi_\ell(g) - 1)z$ (Chapter 7), whose value at any Frobenius lift is $(q-1)z$; it corresponds to the change of basis $e_2 \mapsto e_2 + ze_1$. Replacing $\zeta$ by $\zeta^a$, $a \in \mathbb{Z}_\ell^\times$, multiplies all coordinates of $\kappa$ by $a^{-1}$. (v) For elements $x_1, \dots, x_r$ of $\mathbb{Z}_\ell$, the ideal they generate is $\ell^m\mathbb{Z}_\ell$ with $m = \min_i v_\ell(x_i)$ (with $v_\ell(0) = \infty$). By (iv) the ideal $(n, q-1, c')$ does not depend on any choice, and by (iii) it equals $(n, q-1, \tilde{c})$ for any $\tilde{c} \in \mathbb{Z}_\ell$ lifting the class of $\bar{u}$ in $\mathbb{Z}_\ell/(q-1) = \mathbb{Z}/\ell^v$ (the sign $\varepsilon$ is a unit). Finally $\min(v_\ell(n), v, v_\ell(\tilde{c})) = \min(v_\ell(n), v, \min(v_\ell(\tilde{c}), v)) = \min(v_\ell(n), v, v_\ell(c))$. □

**Corollary 17.44** (the formula of Lemma lem:tamfactor)**.** *In the situation of Theorem 17.43, with* $M = \mathrm{ad}^0 T$, *the model computation of Chapter 16 gives*

$$\mathrm{length}\Big( (H^1(I, \mathrm{ad}^0 T)_{\mathrm{tors}})^{G/I} \Big) = v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c)).$$

*Proof.* Chapter 16 proves the length $v_\ell(n') + \min(v_\ell(n'), v_\ell(q-1), v_\ell(c'))$ for matrices of this shape with any $n'$ and $c'$; this chapter contributes only the identification of $n'$ and $c'$: $v_\ell(n') = v_\ell(n)$ by Theorem 17.43(ii), and the minimum by part (v). □

Lemma lem:tamfactor itself (the Fitting ideal of this module is $\mathrm{Tam}_q(\mathrm{ad}^0 T)$, the assembly of the proof, and the additive potentially multiplicative case) is Chapter 18; the corollary is the pointer from the two computations to that assembly.

*In the paper* 17.45. The end of the proof of Lemma lem:tamfactor reads: "Finally, $q$ being a uniformiser of $\mathbb{Q}_q$, $H^1(\mathbb{Q}_q,\mathbb{Z}_\ell(1))$ splits as $\mathbb{Z}_\ell\kappa_q \oplus H^1_{\mathrm{ur}}$, and $\kappa = n\kappa_q + \kappa_u$, where $\kappa_u$ is unramified, inflated from the class of $u$ in $H^1(G/I,\mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$. The value $c' = \varepsilon\kappa(\mathrm{Frob})$ is a cocycle value: it is $\varepsilon$ times the class of $u$ for the Frobenius lift fixing a compatible system of $\ell$-power roots of $q$, and in general it is well defined only modulo $\ell^{\min(v_\ell(n),v_\ell(q-1))}$ (changing the lift changes it by a multiple of $n'$, changing the cocycle by a multiple of $q-1$)." The splitting is Proposition 17.41; the decomposition and the inflation are Theorem 17.43(i) and Proposition 17.40; the value at the special lift is part (iii), and the indeterminacy, the ideal $(n, q-1)\mathbb{Z}_\ell = \ell^{\min(v_\ell(n),v_\ell(q-1))}\mathbb{Z}_\ell$, is part (iv). "The class of $u$" means the class of its reduction $\bar{u}$, as the statement of the lemma says. The check "the same computation for $M = T$ gives $(\mathbb{Z}/\ell^{v_\ell(n)})\bar{e}_1$ with Frob acting by $\varepsilon$, that is, the $\ell$-part of the Tamagawa number $c_q(E)$" is Proposition 17.56(iv) below, in its cohomological form.

**Example 17.46** (Tate parameters and the extra term at $\ell = 3$)**.** The table lists optimal curves with a multiplicative prime $q \equiv 1 \pmod 3$, $3 \mid n$, and $3 \nmid N$; $c$ is the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{3^{v_3(q-1)}}$, "extra" is $\min(v_3(n), v_3(q-1), v_3(c))$, and "bound" is $\sum_q t_3(E,q) + \text{extra}$, the right side of (eq:identity) without the Selmer term (these curves have no additive prime, so no term $e_3$). The last row is at $\ell = 5$. The Tate parameters were computed with PARI as `ellinit(E,O(q^30)).tate[3]`, the Kodaira symbols with `elllocalred`, the degrees with `ellmoddegree` and checked against Cremona's `degphi` file, and the mod $\ell$ images read from Cremona's `galrep` file.

| curve, $q$ | $a_q$ | $n$ | $\bar{u}$ | $\bar{u}$ a cube? | extra | bound | $v_3(\deg\varphi_E)$ |
|---|---|---|---|---|---|---|---|
| `77c1`, 7 | $-1$ | 3 | 5 | no | 0 | 1 | 1 ($\deg\varphi_E = 6$) |
| `791a1`, 7 | $-1$ | 3 | 6 | yes | 1 | 2 | 2 ($\deg\varphi_E = 144$) |
| `1577a1`, 19 | $+1$ | 3 | 8 | yes, not a ninth power | 1 | 2 | 2 ($\deg\varphi_E = 720$) |
| `11a1`, 11 ($\ell = 5$) | $+1$ | 5 | 10 | a fifth power | 1 | (Eisenstein) | $v_5(1) = 0$ |

The cubes in $\mathbb{F}_7^\times$ are $\{1, 6\}$; the cubes in $\mathbb{F}_{19}^\times$ are $\{1, 7, 8, 11, 12, 18\}$ and the ninth powers $\{1, 18\}$; the fifth powers in $\mathbb{F}_{11}^\times$ are $\{1, 10\}$. The other bad primes are 11 for `77c1`, 113 for `791a1` and 83 for `1577a1`, all of type $\mathrm{I}_2$, with $t_3 = 0$. In the first three rows the mod 3 image is all of $\mathrm{GL}_2(\mathbb{F}_3)$ (the `galrep` file lists no exceptional image at 3), so the hypothesis of Theorem thm:main holds, and the bound equals $v_3(\deg\varphi_E)$: the extra term of Remark rem:ko(1) is present exactly when $\bar{u}$ is a cube, and the Selmer term is 0. For `11a1` at $\ell = 5$ the local term $t_5 + \text{extra} = 2$ exceeds $v_5(\deg\varphi_E) = 0$: here $E[5]$ is reducible ($E(\mathbb{Q})_{\mathrm{tors}} = \mathbb{Z}/5$, image `5Cs.1.1`), and the example shows why the theorem needs $\bar{\rho}$ irreducible.

```
E = ellinit([1,0,1,-31,117]);            \\ 791a1
qE = ellinit(E, O(7^30)).tate[3];
n = valuation(qE, 7); ub = lift(Mod(truncate(qE/7^n), 7));
[n, ub, ispower(Mod(ub,7), 3), ellap(E,7), ellmoddegree(E)]
\\ [3, 6, 1, -1, 144]
```

*Remark* 17.47 (Lemma lem:tamagawa-exp in cohomological form). Let $E$ be multiplicative at $q \neq \ell$ and $t \geq 1$. By Theorem 17.43(ii) the action of $I$ on $T/\ell^t T$ is $\left(\begin{smallmatrix} 1 & nt_\ell \\ 0 & 1 \end{smallmatrix}\right)$ modulo $\ell^t$, for split and nonsplit reduction alike ($\varepsilon|_I = 1$). So $\rho \bmod \ell^t$ is unramified at $q$ if and only if $nt_\ell \equiv 0 \pmod{\ell^t}$ on $I$, if and only if $\ell^t \mid n$, $t_\ell$ being surjective. Equivalently, by Proposition 17.10(i) over $\mathbb{Q}_q^{\mathrm{ur}} \supset \mu_{\ell^t}$: $q_E \in \left((\mathbb{Q}_q^{\mathrm{ur}})^\times\right)^{\ell^t}$ if and only if $\ell^t \mid n$. The lemma itself is proved in Chapter 11; this remark only gives the cohomological reading of the paper's phrase "By Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$".

## 17.6 Global Galois cohomology: $G_{\mathbb{Q}}$, $G_{\mathbb{Q},S}$ and the real place

The Selmer groups of Theorem thm:main are subgroups of the cohomology of $G_{\mathbb{Q}}$ cut out by local conditions. This section introduces the group $G_{\mathbb{Q},S}$ through which the cohomology of a module unramified outside $S$ is computed, proves that classes unramified outside $S$ are exactly the classes of $G_{\mathbb{Q},S}$, states the finiteness theorem behind every finiteness statement of the paper, and disposes of the real place, which for $\ell$ odd contributes nothing (the paper's "At $\infty$, $H^1(\mathbb{R},T^D)$ is killed by 2, so $H^1_f(\mathbb{R},T^D)=0$ for $\ell$ odd").

**Definition 17.48** (places, $\mathbb{Q}_S$, $G_{\mathbb{Q},S}$)**.** $p$ runs over the primes and $v$ over the places of $\mathbb{Q}$, including $\infty$, with $\mathbb{Q}_\infty=\mathbb{R}$. For each $v$ fix an embedding $\overline{\mathbb{Q}}\to\overline{\mathbb{Q}}_v$; it identifies $G_{\mathbb{Q}_v}$ with a decomposition group in $G_{\mathbb{Q}}$ (Chapters 6 and 7), and gives the *localisation* $\mathrm{Res}_v\colon H^n(\mathbb{Q},M)\to H^n(\mathbb{Q}_v,M)$. A different embedding replaces $G_{\mathbb{Q}_v}$ by a conjugate, and the conjugation action of $G_{\mathbb{Q}}$ on its own cohomology is trivial (Chapter 16), so $\mathrm{Res}_v$ does not depend on the embedding. $S$ is a finite set of places containing $\infty$; $\mathbb{Q}_S\subseteq\overline{\mathbb{Q}}$ is the maximal extension of $\mathbb{Q}$ unramified at every prime $p\notin S$; and $G_{\mathbb{Q},S}=\mathrm{Gal}(\mathbb{Q}_S/\mathbb{Q})=G_{\mathbb{Q}}/H_S$, where $H_S$ is the closed normal subgroup generated by the inertia groups $I_{\mathfrak{p}}$ of all primes $\mathfrak{p}$ of $\overline{\mathbb{Q}}$ above primes $p\notin S$. A $G_{\mathbb{Q}}$-module $M$ (finite, of type (C) or (V), or discrete $\ell$-primary such as $W$) is *unramified outside $S$* if $H_S$ acts trivially; it is then a $G_{\mathbb{Q},S}$-module. In the paper $\Sigma$ is a finite set of primes (there, the primes dividing $N$) with $\ell\notin\Sigma$.

Examples: $\mu_n$ is unramified outside $\{\infty\}\cup\{p\mid n\}$; $E[n]$, $T_\ell E$ and $\mathrm{ad}^0 T_\ell E$ are unramified outside $\{\infty,\ell\}\cup\{p\mid N\}$, $N$ the conductor of $E$, by the criterion of Néron, Ogg and Shafarevich (Chapter 11).

**Proposition 17.49** (classes unramified outside $S$)**.** *Let $M$ be unramified outside $S$.*

*(i) Inflation identifies $H^1(G_{\mathbb{Q},S},M)$ with $\{c\in H^1(\mathbb{Q},M):\mathrm{Res}_p\,c\in H^1_{\mathrm{ur}}(\mathbb{Q}_p,M)$ for all $p\notin S\}$.*

*(ii) If $M$ is finite, every $c\in H^1(\mathbb{Q},M)$ is unramified at all but finitely many primes.*

*Proof.* (i) Inflation-restriction for $H_S\trianglelefteq G_{\mathbb{Q}}$ (Chapter 16), with $M^{H_S}=M$, gives the exact sequence $0\to H^1(G_{\mathbb{Q},S},M)\to H^1(\mathbb{Q},M)\to H^1(H_S,M)=\mathrm{Hom}_{\mathrm{cont}}(H_S,M)$. So $c$ is inflated from $G_{\mathbb{Q},S}$ if and only if the homomorphism $c|_{H_S}$ is zero. The set of inertia groups $I_{\mathfrak{p}}$, $\mathfrak{p}$ above $p\notin S$, is stable under conjugation ($\sigma I_{\mathfrak{p}}\sigma^{-1}=I_{\sigma\mathfrak{p}}$), so the closed subgroup it generates is normal and equals $H_S$; and the kernel of a continuous homomorphism is a closed subgroup. Hence $c|_{H_S}=0$ if and only if $c|_{I_{\mathfrak{p}}}=0$ for every such $\mathfrak{p}$. For $i\in I_{\mathfrak{p}}$, which acts trivially on $M$, and $\sigma\in G_{\mathbb{Q}}$, two applications of the cocycle identity $c(gh)=c(g)+g\,c(h)$ give

$$c(\sigma i\sigma^{-1})=c(\sigma)+\sigma\,c(i\sigma^{-1})=c(\sigma)+\sigma c(i)+\sigma i\,c(\sigma^{-1})=\sigma c(i),$$

because $i$ acts trivially, so $\sigma i\,c(\sigma^{-1})=\sigma c(\sigma^{-1})$, and $0=c(1)=c(\sigma\sigma^{-1})=c(\sigma)+\sigma c(\sigma^{-1})$. So $c$ vanishes on $I_{\sigma\mathfrak{p}}$ if and only if it vanishes on $I_{\mathfrak{p}}$, and it suffices to take the prime $\mathfrak{p}$ of the fixed embedding, where the condition is $\mathrm{Res}_p\,c\in H^1_{\mathrm{ur}}(\mathbb{Q}_p,M)$. (ii) Let $L$ be the finite Galois extension cut out by the action on $M$. The restriction of $c$ to $G_L$ is a continuous homomorphism to the finite group $M$; its kernel is an open subgroup of $G_L$, with fixed field $L'$ finite over $\mathbb{Q}$; let $L''$ be the Galois closure of $L'$. If $p$ is unramified in $L''$, every $I_{\mathfrak{p}}$ above $p$ lies in $G_{L''}\subseteq G_{L'}$, on which $c$ vanishes. Only finitely many primes ramify in $L''$. □

**Theorem 17.50** (finiteness for $G_{\mathbb{Q},S}$)**.** *Let $M$ be finite and unramified outside the finite set $S$.*

*(i) $H^1(G_{\mathbb{Q},S},M)$ is finite.*

*(ii) (Stated.)* $H^2(G_{\mathbb{Q},S}, M)$ *is finite.*

*(iii) (Stated.) If* $\ell$ *is odd,* $M$ *has* $\ell$*-power order and* $\ell \in S$*, then* $H^n(G_{\mathbb{Q},S}, M) = 0$ *for* $n \geq 3$*.*

*Consequently, for a lattice* $T$ *unramified outside* $S$*, Proposition 17.17 applies with* $G = G_{\mathbb{Q},S}$*:* $H^1(G_{\mathbb{Q},S}, T)$ *is finitely generated,* $H^1(G_{\mathbb{Q},S}, W)$ *is cofinitely generated, and* $H^1(G_{\mathbb{Q},S}, T)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}, W)/H^0(\mathbb{Q}, W)_{\mathrm{div}}$*.*

*Sketch of proof of (i).* Let $L$ be the field cut out by $M$; it lies in $\mathbb{Q}_S$, and $H = \mathrm{Gal}(\mathbb{Q}_S/L)$ is open and normal in $G_{\mathbb{Q},S}$ with trivial action on $M$. Inflation-restriction gives $0 \to H^1(\mathrm{Gal}(L/\mathbb{Q}), M) \to H^1(G_{\mathbb{Q},S}, M) \to \mathrm{Hom}_{\mathrm{cont}}(H, M)$. The first group is finite. A continuous homomorphism $H \to M$ has kernel $\mathrm{Gal}(\mathbb{Q}_S/L')$ with $L'/L$ Galois of degree dividing $\#M$ and unramified outside $S$, and by the theorem of Hermite and Minkowski (there are finitely many number fields of bounded degree unramified outside a given finite set of primes; stated, [185, Chapter VIII, Section 1]) there are finitely many such $L'$, each with finitely many homomorphisms. For (ii) and (iii) see [143, Chapter VIII, Section 3] and [131, Chapter I, Section 4]. The consequence is Proposition 17.17, whose hypothesis is (i) and (ii) for $M = T/\ell T$; note $H^0(G_{\mathbb{Q},S}, W) = H^0(\mathbb{Q}, W)$ since $H_S$ acts trivially. □

**Proposition 17.51** (the real place)**.** *Let* $G_{\mathbb{R}} = \{1, c\}$ *and* $M$ *a* $G_{\mathbb{R}}$*-module. Then* $2H^n(\mathbb{R}, M) = 0$ *for* $n \geq 1$*. If* $\ell$ *is odd and* $M$ *is discrete* $\ell$*-primary, of type (C) or of type (V), then* $H^n(\mathbb{R}, M) = 0$ *for* $n \geq 1$*, the Tate group* $\hat{H}^0(\mathbb{R}, M) = M^{c=1}/(1+c)M$ *is zero, and* $M = M^+ \oplus M^-$ *with* $M^{\pm} = \{m : cm = \pm m\}$*. In particular, for* $\ell$ *odd every local condition at* $\infty$ *is* $0 = H^1(\mathbb{R}, M)$*, and* $H^1_f(\mathbb{R}, T^D) \subseteq H^1(\mathbb{R}, T^D) = 0$*.*

*Proof.* A finite group of order 2 kills its cohomology in positive degrees (Chapter 16; for continuous cochains with coefficients of type (C) as well): the composite of restriction to the trivial subgroup and corestriction back is multiplication by the index 2, and it factors through $H^n(\{1\}, M) = 0$ for $n \geq 1$. If 2 is invertible on $M$ (which is the case for $\ell$ odd and $M$ discrete $\ell$-primary, of type (C) or of type (V)), multiplication by 2 on $M$ is an automorphism, so it induces an automorphism of $H^n(\mathbb{R}, M)$; that automorphism is also zero, so $H^n(\mathbb{R}, M) = 0$. If $cm = m$ then $m = (1+c)(m/2)$, so $\hat{H}^0 = 0$; and $m = \frac{m+cm}{2} + \frac{m-cm}{2}$ gives the decomposition. □

For $\ell = 2$ this fails: by Theorem 17.7(i), $H^1(\mathbb{R}, \mu_2) = \mathbb{R}^\times/\mathbb{R}^{\times 2} = \{\pm 1\}$.

*In the paper* 17.52. The last sentence of the paragraph after Lemma lem:dfg21, "At $\infty$, $H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for $\ell$ odd", is Proposition 17.51: the first half holds for every module, the second needs $\ell$ odd, and Example 17.54 shows that it fails for $\ell = 2$. The summand $p = \infty$ of the sequence of Lemma lem:dfg21 is therefore 0.

**Example 17.53** (global Kummer theory)**.** By Theorem 17.7, $H^1(\mathbb{Q}, \mu_n) = \mathbb{Q}^\times/(\mathbb{Q}^\times)^n$. For $p \nmid n$ the localisation of $\delta(x)$ is $\delta_p(x)$ (Theorem 17.7(iv)), which is unramified if and only if $x \in \mathbb{Z}_p^\times(\mathbb{Q}_p^\times)^n$, that is, $n \mid v_p(x)$ (Proposition 17.39(i)). So for $S \supseteq \{\infty\} \cup \{p \mid n\}$, by Proposition 17.49(i),

$$H^1(G_{\mathbb{Q},S}, \mu_n) = \{x \in \mathbb{Q}^\times : n \mid v_p(x)\ \forall p \notin S\}/(\mathbb{Q}^\times)^n \ \simeq\ \mathbb{Z}_S^\times/(\mathbb{Z}_S^\times)^n, \qquad \mathbb{Z}_S^\times = \pm \prod_{p \in S \setminus \{\infty\}} p^{\mathbb{Z}}.$$

(Unique factorisation in $\mathbb{Z}$: an $x$ with $n \mid v_p(x)$ for $p \notin S$ is a unit of $\mathbb{Z}_S = \mathbb{Z}[1/p : p \in S]$ times an $n$-th power, and a unit of $\mathbb{Z}_S$ that is an $n$-th power in $\mathbb{Q}$ is an $n$-th power of a unit.) The group is finite, of order $n^{\#S-1}$ times 2 or 1 according as $n$ is even or odd: the model case of

Theorem 17.50. Indeed $\mathbb{Z}_S^\times \simeq \{\pm1\} \times \mathbb{Z}^{\#S-1}$, the free part generated by the primes of $S$, so $\mathbb{Z}_S^\times/(\mathbb{Z}_S^\times)^n \simeq \{\pm1\}/\{\pm1\}^n \times (\mathbb{Z}/n)^{\#S-1}$, and $\{\pm1\}^n$ is $\{1\}$ for $n$ even and $\{\pm1\}$ for $n$ odd. For example, for $n=3$ and $S=\{\infty,3\}$, $H^1(G_{\mathbb{Q},S},\mu_3)$ has order 3, represented by 1, 3, 9.

**Example 17.54.** $H^1(\mathbb{R},\mathbb{Z}_2(1)) = \mathbb{Z}/2$: complex conjugation acts on $\mathbb{Z}_2(1)$ by $-1$, and for the group of order 2, $H^1 = \ker(1+c)/(c-1)M = M/2M$. In detail, with $c=-1$ on $M=\mathbb{Z}_2(1)$, the map $1+c$ is zero, so its kernel is all of $M$, and $(c-1)M = -2M = 2M$.

With the local conditions at every place understood (unramified at $p \neq \ell$, nothing at $\infty$ for $\ell$ odd) and the finiteness of $H^1(G_{\mathbb{Q},S},M)$ available, one can define Selmer groups and prove that they are finite. The next section does this first for the classical Selmer group of an elliptic curve, the model for all the others.

## 17.7 Elliptic curves: the Kummer map, Selmer groups and Sha

The classical Selmer group of an elliptic curve is the model of every Selmer group in the paper, and its finiteness is the weak Mordell-Weil theorem that Chapter 9 deferred to this chapter. The local computation Proposition 17.56(iv) is the classical shadow of the Tamagawa factors: for $W = E[\ell^\infty]$ at a multiplicative prime $p \neq \ell$, the unramified cohomology has the order of the $\ell$-part of the Tamagawa number $c_p$, which is the check the paper performs at the end of the proof of Lemma lem:tamfactor.

**Proposition 17.55** (the Kummer sequence of an elliptic curve)**.** *Let $E/K$ be an elliptic curve and $n$ prime to* $\operatorname{char} K$*. The sequence $0 \to E[n] \to E(\overline{K}) \xrightarrow{[n]} E(\overline{K}) \to 0$ is an exact sequence of discrete $G_K$-modules, and its long exact sequence gives the* Kummer map

$$\delta\colon E(K)/nE(K) \hookrightarrow H^1(K,E[n]), \qquad \delta(P) = [\sigma \mapsto \sigma Q - Q] \quad \text{with } nQ = P,$$

*and the exact sequence*

$$0 \to E(K)/nE(K) \to H^1(K,E[n]) \to H^1(K,E)[n] \to 0, \qquad H^1(K,E) := H^1(K,E(\overline{K})).$$

*Proof.* $[n]$ is a nonconstant isogeny, hence surjective on $\overline{K}$-points, with kernel $E[n]$ (Chapter 9); and every point of $E(\overline{K})$ has an open stabiliser. The long exact sequence $E(K) \xrightarrow{n} E(K) \to H^1(K,E[n]) \to H^1(K,E) \xrightarrow{n} H^1(K,E)$ gives the rest, and the connecting map is that of Convention 17.3. □

Replacing $E$ by the multiplicative group gives Kummer theory, and Hilbert 90 says that the analogue $H^1(K,\overline{K}^\times)$ of $H^1(K,E)$ vanishes; for elliptic curves $H^1(K,E)$ need not vanish, and this is where Sha comes from.

**Proposition 17.56** (the local Kummer image)**.** *Let $E/\mathbb{Q}_p$ be an elliptic curve and $n$ prime to $p$.*

*(i)* $\#E(\mathbb{Q}_p)/nE(\mathbb{Q}_p) = \#E(\mathbb{Q}_p)[n] = \#H^0(\mathbb{Q}_p,E[n]) = \#H^1_{\mathrm{ur}}(\mathbb{Q}_p,E[n])$.

*(ii) If $E$ has good reduction at $p$, then* $\delta(E(\mathbb{Q}_p)/nE(\mathbb{Q}_p)) = H^1_{\mathrm{ur}}(\mathbb{Q}_p,E[n])$.

*(iii) (Tate; stated.) For every reduction type, $\delta(E(\mathbb{Q}_p)/nE(\mathbb{Q}_p))$ is its own exact annihilator under the pairing of Theorem 17.24(i) with $E[n]^D \simeq E[n]$.*

*(iv) Let $\ell \neq p$ be odd and $W = E[\ell^\infty]$. The Kummer image $E(\mathbb{Q}_p)\otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ in $H^1(\mathbb{Q}_p, W)$ is 0, while $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) = W^I/(\mathrm{Frob}-1)W^I$; if $E$ has multiplicative reduction of type $\mathrm{I}_m$ at $p$, then $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) = \ell^{v_\ell(m)}$ if the reduction is split and 1 if it is nonsplit: the $\ell$-part of the Tamagawa number $c_p$.*

*Proof.* (i) Let $E_1(\mathbb{Q}_p)$ be the kernel of reduction; it is the group of the formal group on $p\mathbb{Z}_p$, on which $[n]$ is an isomorphism because $n$ is a unit (Chapter 10), and $F = E(\mathbb{Q}_p)/E_1(\mathbb{Q}_p)$ is finite. The snake lemma for $[n]$ on $0 \to E_1(\mathbb{Q}_p) \to E(\mathbb{Q}_p) \to F \to 0$ gives $E(\mathbb{Q}_p)[n] \simeq F[n]$ and $E(\mathbb{Q}_p)/n \simeq F/n$, and $\#F[n] = \#F/nF$ for a finite group. The last equality is Proposition 17.28(ii). (ii) Let $P \in E(\mathbb{Q}_p)$. The reduction $\tilde{E}(\overline{\mathbb{F}}_p)$ is $n$-divisible, so there is a point of $\tilde{E}(\overline{\mathbb{F}}_p)$ with $n$ times it equal to $\tilde{P}$; it lies in $\tilde{E}(\mathbb{F}_{p^r})$ for some $r$ and lifts, by Hensel's lemma for the smooth model, to $Q_0 \in E(K')$, $K'$ the unramified extension of degree $r$. Then $P - nQ_0$ lies in $E_1(K')$, on which $[n]$ is bijective, so $P = nQ$ with $Q \in E(K') \subseteq E(\mathbb{Q}_p^{\mathrm{ur}})$. Inertia fixes $Q$, so $\delta(P)$ vanishes on $I_p$: $\delta(E(\mathbb{Q}_p)/n) \subseteq H^1_{\mathrm{ur}}(\mathbb{Q}_p, E[n])$, and the orders agree by (i). (iii) See [131, Chapter I, Section 3]. (iv) $E(\mathbb{Q}_p)$ has the finite-index subgroup $E_1(\mathbb{Q}_p)$, a pro-$p$ group, which is uniquely $\ell$-divisible; so $E(\mathbb{Q}_p)\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell = 0$. The formula for $H^1_{\mathrm{ur}}$ is Proposition 17.28(i). For type $\mathrm{I}_m$, $T_\ell E$ has a basis with $\rho(t) = \left(\begin{smallmatrix}1 & m'\\ 0 & 1\end{smallmatrix}\right)$, $v_\ell(m') = v_\ell(m)$, and $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix}\varepsilon p & *\\ 0 & \varepsilon\end{smallmatrix}\right)$ (Theorem 17.43, Chapter 11). In $W = (\mathbb{Q}_\ell/\mathbb{Z}_\ell)e_1 \oplus (\mathbb{Q}_\ell/\mathbb{Z}_\ell)e_2$ an element $xe_1 + ye_2$ is $I$-invariant iff $m'y = 0$, so $W^I$ is an extension of $\mathbb{Z}/\ell^{v_\ell(m)}$ (the $y$-coordinate) by $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)e_1 \simeq \mu_{\ell^\infty}$. On the subgroup $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)e_1$, $\mathrm{Frob}-1$ is multiplication by $\varepsilon p - 1 \neq 0$, which is surjective on a divisible group; on the quotient it is multiplication by $\varepsilon - 1$. So $W^I/(\mathrm{Frob}-1)W^I \simeq (\mathbb{Z}/\ell^{v_\ell(m)})/(\varepsilon - 1)$, which is $\mathbb{Z}/\ell^{v_\ell(m)}$ if $\varepsilon = 1$ and 0 if $\varepsilon = -1$, 2 being a unit. The Tamagawa number is $c_p = m$ in the split case and 1 or 2 in the nonsplit case (Chapter 10). □

Part (iv) agrees with the check at the end of the proof of Lemma lem:tamfactor, which computes the invariants of the torsion of $H^1(I, T)$ instead (Chapter 16, the model computation for $M = T$): for $M = T_\ell E$, the propagated condition $\iota^{-1}H^1_{\mathrm{ur}}(\mathbb{Q}_p, V)$ is all of $H^1(\mathbb{Q}_p, T)_{\mathrm{tors}}$ (Example 17.38(c)), and the gap between it and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T)$ is measured by the same Tamagawa number.

**Definition 17.57** (the $n$-Selmer group and Sha)**.** For $E/\mathbb{Q}$ and $n \geq 1$,

$$\mathrm{Sel}_n(E/\mathbb{Q}) = \{c \in H^1(\mathbb{Q}, E[n]) : \mathrm{Res}_v\, c \in \delta_v(E(\mathbb{Q}_v)/nE(\mathbb{Q}_v)) \text{ for all } v\},$$
$$Ш(E/\mathbb{Q}) = \ker\Big(H^1(\mathbb{Q}, E) \to \prod_v H^1(\mathbb{Q}_v, E)\Big),$$

where $\delta_v$ is the Kummer map of $E/\mathbb{Q}_v$.

**Proposition 17.58.** *The sequence $0 \to E(\mathbb{Q})/nE(\mathbb{Q}) \xrightarrow{\delta} \mathrm{Sel}_n(E/\mathbb{Q}) \to Ш(E/\mathbb{Q})[n] \to 0$ is exact.*

*Proof.* Compare the sequence of Proposition 17.55 for $\mathbb{Q}$ with those for the $\mathbb{Q}_v$, through the localisation maps; the connecting maps commute with localisation, so the diagram

$$\begin{array}{ccc} H^1(\mathbb{Q}, E[n]) & \to & H^1(\mathbb{Q}, E)[n] \\ \downarrow{\scriptstyle \mathrm{Res}_v} & & \downarrow{\scriptstyle \mathrm{Res}_v} \\ H^1(\mathbb{Q}_v, E[n]) & \to & H^1(\mathbb{Q}_v, E)[n] \end{array}$$

commutes, and the kernel of the lower map is $\delta_v(E(\mathbb{Q}_v)/nE(\mathbb{Q}_v))$ by Proposition 17.55 over $\mathbb{Q}_v$. So a class $c \in H^1(\mathbb{Q}, E[n])$ satisfies the local condition at $v$ if and only if the image of $\mathrm{Res}_v\, c$

in $H^1(\mathbb{Q}_v, E)[n]$ is zero, that is, if and only if its image $\bar{c} \in H^1(\mathbb{Q}, E)[n]$ localises to zero at $v$. Hence $c \in \mathrm{Sel}_n$ if and only if $\bar{c} \in Ш$. So $\mathrm{Sel}_n$ is the preimage of $Ш[n]$ under the surjection $H^1(\mathbb{Q}, E[n]) \to H^1(\mathbb{Q}, E)[n]$; it therefore maps onto $Ш[n]$, and the kernel of this map is the kernel $\delta(E(\mathbb{Q})/n)$ of the surjection, which lies in $\mathrm{Sel}_n$ (the localisation of a global Kummer class is a local Kummer class). □

**Theorem 17.59** (finiteness of the Selmer group; weak Mordell-Weil)**.** *Let $E/\mathbb{Q}$ have conductor $N_E$, $n \geq 2$, and $S = \{\infty\} \cup \{p \mid nN_E\}$. Then $\mathrm{Sel}_n(E/\mathbb{Q}) \subseteq H^1(G_{\mathbb{Q},S}, E[n])$, and $\mathrm{Sel}_n(E/\mathbb{Q})$ is finite.*

*Proof.* For $p \notin S$, $E$ has good reduction and $p \nmid n$, so the local condition at $p$ is $H^1_{\mathrm{ur}}(\mathbb{Q}_p, E[n])$ (Proposition 17.56(ii)); $E[n]$ is unramified outside $S$, and Proposition 17.49(i) puts $\mathrm{Sel}_n$ inside $H^1(G_{\mathbb{Q},S}, E[n])$, which is finite by Theorem 17.50(i). □

**Corollary 17.60** (weak Mordell-Weil)**.** *$E(\mathbb{Q})/nE(\mathbb{Q})$ is finite for every $n \geq 1$.*

*Proof.* It injects into $\mathrm{Sel}_n(E/\mathbb{Q})$ by Proposition 17.58. □

With the descent by the canonical height (Chapter 9; [185, Chapter VIII]) this gives the theorem of Mordell and Weil that $E(\mathbb{Q})$ is finitely generated.

**Example 17.61** (`11a1` and $n = 5$)**.** For $E =$ `11a1`, Cremona's `allbsd` file gives rank 0, $E(\mathbb{Q}) \simeq \mathbb{Z}/5$, Tamagawa product 5 and analytic order of Ш equal to 1. The same value is printed in [35, Table 4] (row 11A, column $S$), where for rank 0, $S = L(E,1)\,\#E(\mathbb{Q})^2_{\mathrm{tors}}/(\Omega \prod_p c_p)$, with $\Omega$ Cremona's real period ($L(E,1)/\Omega = 1/5$ in that row), is defined as the number that the Birch and Swinnerton-Dyer conjecture predicts to be $\#Ш$; for rank 0 it is exact, but it is a statement about $L(E,1)$, not a proof about Ш, and the table does not claim otherwise. That Ш is finite for a curve of analytic rank 0 is a theorem of Kolyvagin [104] (with the modularity of $E$ and Gross and Zagier), but neither the table nor that finiteness gives $Ш[5] = 0$; here $\bar{\rho}_{E,5}$ is reducible ($E$ has a rational point of order 5), and a proof needs a separate argument, for instance a descent by the 5-isogenies of the class, which we do not give. Granting $Ш[5] = 0$, $\mathrm{Sel}_5(E/\mathbb{Q}) \simeq E(\mathbb{Q})/5E(\mathbb{Q}) \simeq \mathbb{Z}/5$. Over $\mathbb{Q}_{11}$, $E[5] \subset E(\mathbb{Q}_{11})$ (Example 17.26(c)), so the local condition at 11 is a subgroup of order $\#E(\mathbb{Q}_{11})[5] = 25$ (Proposition 17.56(i)) of $H^1(\mathbb{Q}_{11}, E[5])$, which has order 625 (Example 17.26(c)). It is not the unramified subgroup, although that has order 25 too. Indeed $E$ is a Tate curve over $\mathbb{Q}_{11}$ ($a_{11} = 1$), $E(\mathbb{Q}_{11}) \simeq \mathbb{Q}_{11}^\times / q_E^{\mathbb{Z}}$, and as $q_E$ is a fifth power, $E(\mathbb{Q}_{11})/5 \simeq \mathbb{Q}_{11}^\times/(\mathbb{Q}_{11}^\times)^5$; the Kummer map of $E$ restricted to this is the Kummer map of $\mathbb{G}_m$ followed by $\mu_5 \subset E[5]$. So the local condition is $H^1(\mathbb{Q}_{11}, \mu_5) = \mathrm{Hom}(G_{\mathbb{Q}_{11}}, \mu_5) \subset \mathrm{Hom}(G_{\mathbb{Q}_{11}}, E[5])$ (the action on $E[5]$ is trivial), which contains the ramified class $\delta(11)$. Its intersection with the unramified subgroup has order 5, in agreement with Proposition 17.56(iv) and $c_{11} = 5$.

The definition of $\mathrm{Sel}_n(E/\mathbb{Q})$ has a shape that recurs: a global cohomology group, a choice of subgroup of the local $H^1$ at every place, equal to the unramified subgroup at almost every place, and the classes whose localisations all lie in the chosen subgroups. The paper's Selmer groups have the same shape with other modules ($W$ built from $\mathrm{ad}^0 T$, and $T^D$) and other local conditions. The next section isolates this shape and its dual.

## 17.8 Selmer structures and dual Selmer structures

The paper's Selmer groups $H^1_f(\mathbb{Q}, W)$ and $H^1_\Sigma(\mathbb{Q}, W)$ are defined by local conditions: the Bloch-Kato condition at $\ell$, the unramified condition at the other primes, and, for $H^1_\Sigma$, no condition at the primes of $\Sigma$. This section gives the general language (Selmer structures, after Mazur and Rubin), the dual structure on the Kummer dual defined by local duality, and the two facts about $\ell$-adic Selmer groups that the proof of Lemma lem:dfg21 needs. The Bloch-Kato structure itself is defined in Chapter 18.

From now on $\ell$ is odd, so that every local condition at $\infty$ is $0 = H^1(\mathbb{R}, M)$ (Proposition 17.51).

**Definition 17.62** (Selmer structures)**.** Let $M$ be a $G_\mathbb{Q}$-module unramified outside a finite set $S$ of places (containing $\infty$ and $\ell$), of one of the following kinds: finite of $\ell$-power order; $W$, discrete and cofinitely generated; or $T$, of type (C). A *Selmer structure* $\mathcal{L}$ on $M$ is a family of subgroups $\mathcal{L}_v \subseteq H^1(\mathbb{Q}_v, M)$, one for each place $v$, closed when $M$ is of type (C), such that $\mathcal{L}_v = H^1_{\mathrm{ur}}(\mathbb{Q}_v, M)$ for all $v$ outside a finite set $S_\mathcal{L} \supseteq S$. Its *Selmer group* is

$$\begin{aligned} H^1_\mathcal{L}(\mathbb{Q}, M) &= \{c \in H^1(\mathbb{Q}, M) : \mathrm{Res}_v\, c \in \mathcal{L}_v \text{ for all } v\} \\ &= \ker\Big(H^1(G_{\mathbb{Q},S_\mathcal{L}}, M) \to \bigoplus_{v \in S_\mathcal{L}} H^1(\mathbb{Q}_v, M)/\mathcal{L}_v\Big), \end{aligned}$$

the equality by Proposition 17.49(i). We write $\mathcal{L} \subseteq \mathcal{L}'$ if $\mathcal{L}_v \subseteq \mathcal{L}'_v$ for all $v$.

**Example 17.63** (Selmer structures)**.** (a) The *unramified* structure, $\mathcal{L}_p = H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)$ at every prime.

(b) For a finite set $\Sigma$ of primes and a structure $\mathcal{L}$: the structure *relaxed* at $\Sigma$, $\mathcal{L}^\Sigma_p = H^1(\mathbb{Q}_p, M)$ for $p \in \Sigma$ and $\mathcal{L}^\Sigma_v = \mathcal{L}_v$ otherwise; and the structure *strict* at $\Sigma$, $\mathcal{L}_{\Sigma,p} = 0$ for $p \in \Sigma$ and $\mathcal{L}_{\Sigma,v} = \mathcal{L}_v$ otherwise.

(c) The classical structure $\mathcal{L}_v = \delta_v(E(\mathbb{Q}_v)/nE(\mathbb{Q}_v))$ on $E[n]$, $n = \ell^k$ ($\ell$ odd, as fixed above), a Selmer structure by Proposition 17.56(ii), with Selmer group $\mathrm{Sel}_n(E/\mathbb{Q})$.

(d) *Propagated structures.* Given a lattice $T$ unramified outside $S$ and $\mathbb{Q}_\ell$-subspaces $L_{V,v} \subseteq H^1(\mathbb{Q}_v, V)$ with $L_{V,v} = H^1_{\mathrm{ur}}(\mathbb{Q}_v, V)$ for $v \notin S$, the families $(\iota^{-1}L_{V,v})_v$ on $T$ and $(\pi L_{V,v})_v$ on $W$ (Proposition 17.34) are Selmer structures: at $p \notin S$, $p \neq \ell$, they are the unramified conditions by Proposition 17.34(iii).

The Bloch-Kato structure of Chapter 18 is of type (d), and the paper's $H^1_f(\mathbb{Q}, W)$ and $H^1_\Sigma(\mathbb{Q}, W)$ are the Selmer groups of the Bloch-Kato structure on $W$ and of its relaxation at $\Sigma$, the latter described in Section 4.2 of the paper as the group "with no condition at the primes of $\Sigma$, the Bloch-Kato condition at $\ell$ and the unramified condition elsewhere".

Global duality does not compare a Selmer group with the Selmer group of the same module, but with a Selmer group of the Kummer dual, whose local conditions are forced by local duality: the condition at $v$ on $M^D$ must be the annihilator of the condition at $v$ on $M$. This is the content of the next definition. For example, the dual of the unramified structure on $\mathbb{Z}/\ell$ is, at every $p \neq \ell$, the unramified structure on $\mu_\ell$ (Theorem 17.32); and the dual of the structure "everything at $p$" is "nothing at $p$".

**Definition 17.64** (dual Selmer structures)**.** (a) Let $M$ be finite of $\ell$-power order and $\mathcal{L}$ a Selmer structure on $M$. The *dual structure* $\mathcal{L}^*$ on $M^D$ is $\mathcal{L}_v^* = \mathcal{L}_v^\perp$, the annihilator under the local pairing of Theorem 17.24(i) (at $v = \infty$ both sides are 0). (b) Let $T$ be a lattice unramified outside $S$ and $\mathcal{L}$ a Selmer structure on $W$. The *dual structure* $\mathcal{L}^*$ on $T^D$ is $\mathcal{L}_v^* = \mathcal{L}_v^\perp \subseteq H^1(\mathbb{Q}_v, T^D)$, the annihilator under the first pairing of Theorem 17.24(ii); symmetrically for structures on $T$, with duals on $W^D$.

**Proposition 17.65** (properties of dual structures)**.** *(i) The dual of a Selmer structure is a Selmer structure: at every prime $p \notin S_\mathcal{L}$ (so $p \neq \ell$), $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)^\perp = H^1_{\mathrm{ur}}(\mathbb{Q}_p, M^D)$, and likewise for $(W, T^D)$.*

*(ii) $\mathcal{L}^{**} = \mathcal{L}$; $\mathcal{L} \subseteq \mathcal{L}'$ implies $\mathcal{L}'^* \subseteq \mathcal{L}^*$; and $(\mathcal{L}^\Sigma)^* = (\mathcal{L}^*)_\Sigma$: relaxed and strict are dual.*

*(iii) If $\mathcal{L}$ on $W$ is propagated from $(L_{V,v})_v$ and $L_{V,v}^\perp = L_{V^D,v}$ for every $v$, then $\mathcal{L}^*$ is the structure on $T^D$ propagated from $(L_{V^D,v})_v$.*

*Proof.* (i) For finite $M$ this is Theorem 17.32(i). For $(W, T^D)$ and $p \notin S$, $T$ and $T^D$ are unramified at $p$, so by Proposition 17.34(iii) $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T^D)$ are propagated from $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V)$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D)$, which are exact annihilators by Theorem 17.32(ii); apply Proposition 17.34(ii). (ii) For a perfect pairing between a finite (or discrete) group and its dual, $(A^\perp)^\perp = A$ and $A \subseteq A'$ implies $A'^\perp \subseteq A^\perp$; and $H^1(\mathbb{Q}_p, M)^\perp = 0$. The last fact gives the duality of relaxed and strict structures: at $p \in \Sigma$, $(\mathcal{L}_p^\Sigma)^\perp = H^1(\mathbb{Q}_p, M)^\perp = 0 = (\mathcal{L}^*)_{\Sigma,p}$, and at $v \notin \Sigma$ both structures are $\mathcal{L}_v^\perp = \mathcal{L}_v^*$. (iii) is Proposition 17.34(ii) at each place. □

**Proposition 17.66** ($\ell$-adic Selmer groups)**.** *Let $T$ be unramified outside $S$ and $\mathcal{L}$ the structures on $T$ and $W$ propagated from $(L_{V,v})_v$, and let $H^1_{L_V}(\mathbb{Q}, V) = \{c \in H^1(G_{\mathbb{Q},S}, V) : \mathrm{Res}_v\, c \in L_{V,v}\ \forall v\}$.*

*(i) $H^1_\mathcal{L}(\mathbb{Q}, W)$ is cofinitely generated and $H^1_\mathcal{L}(\mathbb{Q}, T)$ is finitely generated.*

*(ii) If $H^1_{L_V}(\mathbb{Q}, V) = 0$, then $H^1_\mathcal{L}(\mathbb{Q}, T) = H^1(G_{\mathbb{Q},S}, T)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}, W)/H^0(\mathbb{Q}, W)_{\mathrm{div}}$, and $H^1_\mathcal{L}(\mathbb{Q}, T) \simeq H^0(\mathbb{Q}, W)$ if moreover $H^0(\mathbb{Q}, V) = 0$.*

*(iii) (Stated.) $\mathrm{corank}_{\mathbb{Z}_\ell} H^1_\mathcal{L}(\mathbb{Q}, W) = \dim_{\mathbb{Q}_\ell} H^1_{L_V}(\mathbb{Q}, V)$; in particular $H^1_\mathcal{L}(\mathbb{Q}, W)$ is finite if and only if $H^1_{L_V}(\mathbb{Q}, V) = 0$.*

*Proof.* (i) $H^1_\mathcal{L}(\mathbb{Q}, W) \subseteq H^1(G_{\mathbb{Q},S}, W)$, which is cofinitely generated by Theorem 17.50, and a subgroup of a cofinitely generated group is cofinitely generated (its Pontryagin dual is a quotient of a finitely generated $\mathbb{Z}_\ell$-module). $H^1_\mathcal{L}(\mathbb{Q}, T)$ is a submodule of the finitely generated $H^1(G_{\mathbb{Q},S}, T)$, and $\mathbb{Z}_\ell$ is noetherian. (ii) Let $c \in H^1_\mathcal{L}(\mathbb{Q}, T)$; then $c \in H^1(G_{\mathbb{Q},S}, T)$ (the conditions outside $S$ are unramified, Example 17.63(d) and Proposition 17.49(i)), and $\mathrm{Res}_v(\iota c) = \iota(\mathrm{Res}_v\, c) \in L_{V,v}$ for all $v$, so $\iota c \in H^1_{L_V}(\mathbb{Q}, V) = 0$, and $c$ is torsion by Proposition 17.17(ii). Conversely, if $c \in H^1(G_{\mathbb{Q},S}, T)$ is torsion, so is each $\mathrm{Res}_v\, c$, which therefore lies in $\ker \iota \subseteq \iota^{-1} L_{V,v}$ (Proposition 17.34(i)). The isomorphisms are Proposition 17.17(ii) for $G = G_{\mathbb{Q},S}$ (Theorem 17.50). (iii) The image of $H^1_\mathcal{L}(\mathbb{Q}, T) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ in $H^1_\mathcal{L}(\mathbb{Q}, W)$ has finite index, and $H^1_\mathcal{L}(\mathbb{Q}, T) \otimes \mathbb{Q}_\ell = H^1_{L_V}(\mathbb{Q}, V)$; we do not give the details, see [167, Section 1.5]. □

*In the paper* 17.67. In Theorem thm:main the group $H^1_f(\mathbb{Q}, \mathrm{ad}^0\, T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is the Selmer group of the Bloch-Kato structure on $W$, and in Section 4.2 of the paper $H^1_\Sigma(\mathbb{Q}, W) \supseteq H^1_f(\mathbb{Q}, W)$ is the Selmer group of its relaxation at $\Sigma$ (Example 17.63(b), (d)). Lemma lem:dfg21 puts $H^1_f(\mathbb{Q}_p, T^D) = \iota^{-1} H^1_f(\mathbb{Q}_p, V^D)$, the dual structure on $T^D$ in the sense of Proposition 17.65(iii),

and its hypothesis $H^1_f(\mathbb{Q}, V^D) = 0$ is the hypothesis of Proposition 17.66(ii) for $T^D$; that proposition then says (as $H^0(\mathbb{Q}, V^D) = 0$) that the Selmer group $H^1_f(\mathbb{Q}, T^D)$ is $H^0(\mathbb{Q}, W^D)$, the group whose dual is the last term of the lemma.

What remains is to relate the Selmer group of a structure to that of its dual structure. Locally this is Tate duality; globally it needs a reciprocity law, which says that a global class of $M$ and a global class of $M^D$ are orthogonal under the *sum* of the local pairings, and the deeper statement that, at the places of $S$, the global classes are as large as this orthogonality allows. These are the subject of the next section.

## 17.9 Global duality: Poitou-Tate and the formula of Greenberg and Wiles

Global duality relates a Selmer group to the Selmer group of the dual structure on the Kummer dual. In the paper it enters only through Lemma lem:dfg21, the exact sequence of Diamond, Flach and Guo, which is a comparison of two Selmer structures; this section states the Poitou-Tate sequence and the formula of Greenberg and Wiles, proves the reciprocity law that makes the comparison maps well defined, and checks the formula by hand in the simplest case, where it recovers a piece of class field theory over $\mathbb{Q}$. Throughout, $\ell$ is odd, $M$ is finite of $\ell$-power order, and $M^D$ is its Kummer dual.

**Theorem 17.68** (the Poitou-Tate exact sequence)**.** *Let $M$ be unramified outside $S$, with $\infty, \ell \in S$, and $G_S = G_{\mathbb{Q},S}$. There is an exact sequence*

$$\begin{aligned} 0 \to H^0(G_S, M) &\to \textstyle\bigoplus_{v\in S} \hat{H}^0(\mathbb{Q}_v, M) \to H^2(G_S, M^D)^* \\ &\to H^1(G_S, M) \to \textstyle\bigoplus_{v\in S} H^1(\mathbb{Q}_v, M) \to H^1(G_S, M^D)^* \\ &\to H^2(G_S, M) \to \textstyle\bigoplus_{v\in S} H^2(\mathbb{Q}_v, M) \to H^0(G_S, M^D)^* \to 0, \end{aligned}$$

*where $\hat{H}^0(\mathbb{Q}_p, M) = H^0(\mathbb{Q}_p, M)$ for primes $p$, $\hat{H}^0(\mathbb{R}, M) = 0$ ($\ell$ odd), the maps $\bigoplus_v H^i(\mathbb{Q}_v, M) \to H^{2-i}(G_S, M^D)^*$ send $(c_v)_v$ to $y \mapsto \sum_v \mathrm{inv}_v(c_v \cup \mathrm{Res}_v\, y)$, and the other maps are localisations or their duals. In particular the image of $H^1(G_S, M)$ in $\bigoplus_{v\in S} H^1(\mathbb{Q}_v, M)$ is the exact annihilator of the image of $H^1(G_S, M^D)$ in $\bigoplus_{v\in S} H^1(\mathbb{Q}_v, M^D)$ under the sum of the local pairings.*

We do not prove this; see [131, Chapter I, Section 4], [143, Chapter VIII, Section 6], [195] and [167, Section 1.7]. The last sentence is exactness at $\bigoplus_{v\in S} H^1(\mathbb{Q}_v, M)$: the kernel of the map to $H^1(G_S, M^D)^*$ is by definition the annihilator of the image of $H^1(G_S, M^D)$, and the annihilator of that annihilator is the image itself, the local pairings being perfect.

**Proposition 17.69** (reciprocity)**.** *For $x \in H^1(\mathbb{Q}, M)$ and $y \in H^1(\mathbb{Q}, M^D)$, the local cup products $\mathrm{Res}_v\, x \cup \mathrm{Res}_v\, y$ are zero for all but finitely many $v$, and*

$$\sum_v \mathrm{inv}_v(\mathrm{Res}_v\, x \cup \mathrm{Res}_v\, y) = 0.$$

*Proof.* By Proposition 17.49(ii), for all but finitely many primes $p$ (say $p \neq \ell$, with $M$, $x$ and $y$ unramified at $p$) both $\mathrm{Res}_p\, x$ and $\mathrm{Res}_p\, y$ are inflated from $G_{\mathbb{Q}_p}/I_p$, so their cup product, pushed to $\mu_{\ell^\infty}$, is inflated from $H^2(\hat{\mathbb{Z}}, \mu_{\ell^\infty}) = 0$, as in the proof of Theorem 17.32. The cup product $x \cup y$, pushed to $\mu_{\ell^\infty}$, is a class in $H^2(\mathbb{Q}, \mu_{\ell^k})$ for some $k$, that is, by Theorem 17.7(ii), an element $\beta$ of

$\mathrm{Br}(\mathbb{Q})[\ell^k]$. Cup products commute with restriction, and the isomorphism of Theorem 17.7(ii) commutes with restriction (it comes from a connecting map), so the local components of $\beta$ are the classes $\mathrm{Res}_v x \cup \mathrm{Res}_v y$. The sum of the invariants of the local components of an element of $\mathrm{Br}(\mathbb{Q})$ is 0 by Theorem 17.12(iii). □

**Theorem 17.70** (the formula of Greenberg and Wiles)**.** *Let $\mathcal{L}$ be a Selmer structure on $M$ and $\mathcal{L}^*$ its dual on $M^D$. Then $H^1_{\mathcal{L}}(\mathbb{Q}, M)$ and $H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)$ are finite, and*

$$\frac{\#H^1_{\mathcal{L}}(\mathbb{Q}, M)}{\#H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)} = \frac{\#H^0(\mathbb{Q}, M)}{\#H^0(\mathbb{Q}, M^D)} \prod_v \frac{\#\mathcal{L}_v}{\#H^0(\mathbb{Q}_v, M)},$$

*the product over all places including $\infty$; all but finitely many factors are* 1*.*

The finiteness follows from Definition 17.62 and Theorem 17.50(i), and almost all factors are 1 by Proposition 17.28(ii). We do not prove the formula; it is due to Greenberg and to Wiles, see [207, Section 1] and [38, Section 2]. The proof combines Theorem 17.68 with Tate's global Euler characteristic formula $\#H^0(G_S, M)\,\#H^2(G_S, M)/\#H^1(G_S, M) = \#H^0(\mathbb{R}, M)/|\#M|_\infty$ (stated; [131, Chapter I, Section 5]), which supplies the factor $1/\#H^0(\mathbb{R}, M)$ at infinity.
How to read the formula: at a prime $v$, the factor $\#\mathcal{L}_v/\#H^0(\mathbb{Q}_v, M)$ compares the chosen local condition with the "natural size" of an unramified condition, which is $\#H^0(\mathbb{Q}_v, M)$ by Proposition 17.28(ii). Enlarging $\mathcal{L}_v$ by a factor $\ell^a$ either enlarges $H^1_{\mathcal{L}}(\mathbb{Q}, M)$ or shrinks $H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)$ (whose condition at $v$ shrinks by the same factor), and the formula says that the two effects together account for exactly $\ell^a$. The formula does not compute either Selmer group on its own; it computes their ratio, which is why it is used together with a separate vanishing statement for the dual group, as in the examples below and in Theorem 17.75.

**Example 17.71** (the formula by hand)**.** Let $M = \mathbb{Z}/\ell$, so $M^D = \mu_\ell$.

(a) Let $\mathcal{L}$ be unramified at every prime, $\ell$ included, and 0 at $\infty$. Then $H^1_{\mathcal{L}}(\mathbb{Q}, \mathbb{Z}/\ell)$ consists of the homomorphisms $G_\mathbb{Q} \to \mathbb{Z}/\ell$ unramified everywhere, which are 0 because $\mathbb{Q}$ has no nontrivial unramified extension (Minkowski; stated). The dual structure: $\mathcal{L}^*_p = H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mu_\ell)$ for $p \neq \ell$ (Theorem 17.32), and at $\ell$ the annihilator of $H^1_{\mathrm{ur}}(\mathbb{Q}_\ell, \mathbb{Z}/\ell)$, which is the image of $\mathbb{Z}_\ell^\times$ in $\mathbb{Q}_\ell^\times/(\mathbb{Q}_\ell^\times)^\ell$ (Warning 17.33; this uses local class field theory). By Example 17.53 and these conditions, $H^1_{\mathcal{L}^*}(\mathbb{Q}, \mu_\ell) = \{x \in \mathbb{Q}^\times : \ell \mid v_p(x) \text{ for all } p\}/(\mathbb{Q}^\times)^\ell = \{\pm 1\}(\mathbb{Q}^\times)^\ell/(\mathbb{Q}^\times)^\ell = 0$, as $-1 = (-1)^\ell$. The left side is 1. On the right, $\#H^0(\mathbb{Q}, \mathbb{Z}/\ell)/\#H^0(\mathbb{Q}, \mu_\ell) = \ell$; the factor at each prime is $\#H^1_{\mathrm{ur}}/\#H^0 = 1$ (Proposition 17.28(ii), valid at $\ell$ too); and the factor at $\infty$ is $\#\mathcal{L}_\infty/\#H^0(\mathbb{R}, \mathbb{Z}/\ell) = 1/\ell$. The product is 1.

(b) Relax $\mathcal{L}$ at one prime $p \neq \ell$ and put $m = 1$ if $p \equiv 1 \pmod{\ell}$, $m = 0$ otherwise. The dual structure becomes strict at $p$: $x$ with $\ell \mid v_q(x)$ for all primes $q$ and $x \in (\mathbb{Q}_p^\times)^\ell$; then $x = \pm y^\ell$ and the dual Selmer group is still 0. On the right only the factor at $p$ changes, to $\#H^1(\mathbb{Q}_p, \mathbb{Z}/\ell)/\#H^0(\mathbb{Q}_p, \mathbb{Z}/\ell) = \ell^{1+m}/\ell = \ell^m$ (Example 17.26(a)). So the formula says: the homomorphisms $G_\mathbb{Q} \to \mathbb{Z}/\ell$ unramified outside $p$ (and at $\ell$, and at $\infty$) number $\ell^m$. This is what class field theory gives: they factor through $\mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \simeq (\mathbb{Z}/p)^\times$ by the theorem of Kronecker and Weber, and $\mathrm{Hom}((\mathbb{Z}/p)^\times, \mathbb{Z}/\ell)$ has order $\ell^m$. The factor at $\infty$ is essential: without it the two sides would differ by $\ell$.

**Corollary 17.72** (comparison of two structures)**.** *For Selmer structures $\mathcal{L} \subseteq \mathcal{L}'$ on $M$,*

$$\frac{\#H^1_{\mathcal{L}'}(\mathbb{Q}, M)}{\#H^1_{\mathcal{L}}(\mathbb{Q}, M)} = \prod_v \frac{\#\mathcal{L}'_v}{\#\mathcal{L}_v} \cdot \frac{\#H^1_{\mathcal{L}'^*}(\mathbb{Q}, M^D)}{\#H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)}.$$

*Proof.* Divide the formula of Theorem 17.70 for $\mathcal{L}'$ by the one for $\mathcal{L}$; the global terms $\#H^0$ and the local terms $\#H^0(\mathbb{Q}_v, M)$ cancel. □

This is the numerical shadow of the exact sequence of Theorem 17.73; applied at each finite level to a structure and its relaxation at $\Sigma$, and passed to the limit, it gives the length identity in the proof of Theorem thm:main.

## 17.10 Comparing Selmer structures; the abstract form of Lemma 2.1 of Diamond, Flach and Guo

This section proves the exact sequence that compares the Selmer groups of two structures $\mathcal{L} \subseteq \mathcal{L}'$, first for finite modules and then for the pair $(W, T^D)$, and specialises it to the relaxation of a structure at a finite set $\Sigma$. The result, Theorem 17.75, is Lemma lem:dfg21 of the paper with an abstract local condition in place of the Bloch-Kato one, and Corollary 17.77 is the length identity used in the proof of Theorem thm:main and in Remark rem:ns. Chapter 18 supplies what is specific to the Bloch-Kato structure (Remark 17.80). Throughout, $\ell$ is odd.

**Theorem 17.73** (the comparison sequence)**.** *Let $M$ be finite of $\ell$-power order, unramified outside $S$, and $\mathcal{L} \subseteq \mathcal{L}'$ Selmer structures on $M$, so that $\mathcal{L}'^* \subseteq \mathcal{L}^*$ on $M^D$. Consider*

$$0 \to H^1_{\mathcal{L}}(\mathbb{Q}, M) \to H^1_{\mathcal{L}'}(\mathbb{Q}, M) \xrightarrow{\mathrm{loc}} \bigoplus_v \mathcal{L}'_v/\mathcal{L}_v \xrightarrow{\theta} H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)^* \xrightarrow{\mathrm{Res}^*} H^1_{\mathcal{L}'^*}(\mathbb{Q}, M^D)^* \to 0,$$

*where* loc *is localisation,* $\theta((c_v)_v)(y) = \sum_v \mathrm{inv}_v(c_v \cup \mathrm{Res}_v\, y)$*, and* $\mathrm{Res}^*$ *is the dual of the inclusion* $H^1_{\mathcal{L}'^*} \subseteq H^1_{\mathcal{L}^*}$*. The sum is finite ($\mathcal{L}'_v = \mathcal{L}_v$ for almost all $v$), and $\theta$ is well defined because* $\mathrm{Res}_v\, y \in \mathcal{L}^*_v = \mathcal{L}_v^{\perp}$*.*

*(i) The sequence is exact at $H^1_{\mathcal{L}}$ and at $H^1_{\mathcal{L}'}$,* $\mathrm{Res}^*$ *is surjective, and it is a complex: $\theta \circ \mathrm{loc} = 0$ and $\mathrm{Res}^* \circ \theta = 0$. In particular $\#H^1_{\mathcal{L}'}(\mathbb{Q}, M)/\#H^1_{\mathcal{L}}(\mathbb{Q}, M) \le \prod_v \#(\mathcal{L}'_v/\mathcal{L}_v)$.*

*(ii) The sequence is exact.*

*Proof.* (i) The kernel of loc consists of the classes of $H^1_{\mathcal{L}'}$ whose localisations lie in $\mathcal{L}_v$ for all $v$, which is $H^1_{\mathcal{L}}$. $\mathrm{Res}^*$ is the dual of an injection of finite groups, hence surjective. For $x \in H^1_{\mathcal{L}'}(\mathbb{Q}, M)$ and $y \in H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)$,

$$\theta(\mathrm{loc}\, x)(y) = \sum_{v:\, \mathcal{L}'_v \neq \mathcal{L}_v} \mathrm{inv}_v(\mathrm{Res}_v\, x \cup \mathrm{Res}_v\, y) = \sum_v \mathrm{inv}_v(\mathrm{Res}_v\, x \cup \mathrm{Res}_v\, y) = 0,$$

the middle equality because at the other places $\mathrm{Res}_v\, x \in \mathcal{L}_v$ and $\mathrm{Res}_v\, y \in \mathcal{L}_v^{\perp}$, the last by Proposition 17.69. For $y \in H^1_{\mathcal{L}'^*}$ and $c_v \in \mathcal{L}'_v$, $c_v \cup \mathrm{Res}_v\, y = 0$ because $\mathrm{Res}_v\, y \in \mathcal{L}'^*_v = \mathcal{L}'^{\perp}_v$; so $\mathrm{Res}^* \circ \theta = 0$. The inequality is the injectivity of $H^1_{\mathcal{L}'}/H^1_{\mathcal{L}} \to \bigoplus_v \mathcal{L}'_v/\mathcal{L}_v$.

(ii) *Exactness at the middle term.* Enlarge $S$ so that $\mathcal{L}_v = \mathcal{L}'_v = H^1_{\mathrm{ur}}(\mathbb{Q}_v, M)$ and $\mathcal{L}^*_v = H^1_{\mathrm{ur}}(\mathbb{Q}_v, M^D)$ for $v \notin S$. In $P = \bigoplus_{v \in S} H^1(\mathbb{Q}_v, M)$ let $B$ be the image of $H^1(G_S, M)$ and $A = \bigoplus_{v \in S} \mathcal{L}_v$; in $P' = \bigoplus_{v \in S} H^1(\mathbb{Q}_v, M^D)$ let $B'$ be the image of $H^1(G_S, M^D)$. By Theorem 17.68, $B^{\perp} = B'$, and $A^{\perp} = \bigoplus_{v \in S} \mathcal{L}^*_v$. Hence

$$(A + B)^{\perp} = A^{\perp} \cap B' = \mathrm{loc}(H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)),$$

since a class of $H^1(G_S, M^D)$ with localisations in $\mathcal{L}_v^*$ for $v \in S$ is exactly a class of $H^1_{\mathcal{L}^*}$ (Proposition 17.49(i)). Now let $(c_v) \in \bigoplus_v \mathcal{L}'_v/\mathcal{L}_v$ with $\theta((c_v)) = 0$, and lift it to $c = (c_v)_{v\in S} \in \bigoplus_{v\in S} \mathcal{L}'_v \subseteq P$. The condition $\theta((c_v)) = 0$ says that $\sum_{v\in S} \mathrm{inv}_v(c_v \cup \mathrm{Res}_v\, y) = 0$ for every $y \in H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)$, that is, that $c$ is orthogonal, under the sum of the local pairings, to $\mathrm{loc}(H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)) = (A+B)^\perp$. As the sum of the local pairings is a perfect pairing of the finite groups $P$ and $P'$, $c \in (A+B)^{\perp\perp} = A + B$: $c = a + \mathrm{loc}(x)$ with $a \in A$ and $x \in H^1(G_S, M)$. Then $\mathrm{Res}_v\, x = c_v - a_v \in \mathcal{L}'_v$ for $v \in S$, and $x$ is unramified outside $S$, so $x \in H^1_{\mathcal{L}'}(\mathbb{Q}, M)$ and $\mathrm{loc}(x) \equiv (c_v)$ modulo $\bigoplus \mathcal{L}_v$.

*Exactness at $H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)^*$.* By exactness at the middle term and at $H^1_{\mathcal{L}'}$,

$$\#\operatorname{im}\theta = \frac{\prod_v \#(\mathcal{L}'_v/\mathcal{L}_v)}{\#\operatorname{im}(\mathrm{loc})} = \prod_v \frac{\#\mathcal{L}'_v}{\#\mathcal{L}_v} \cdot \frac{\#H^1_{\mathcal{L}}(\mathbb{Q}, M)}{\#H^1_{\mathcal{L}'}(\mathbb{Q}, M)} = \frac{\#H^1_{\mathcal{L}^*}(\mathbb{Q}, M^D)}{\#H^1_{\mathcal{L}'^*}(\mathbb{Q}, M^D)} = \#\ker \mathrm{Res}^*,$$

the third equality by Corollary 17.72. In detail: by exactness at the middle term, $\ker\theta = \operatorname{im}(\mathrm{loc})$, so $\#\operatorname{im}\theta$ is the order of the source divided by $\#\operatorname{im}(\mathrm{loc})$; by exactness at $H^1_{\mathcal{L}'}$, $\#\operatorname{im}(\mathrm{loc}) = \#H^1_{\mathcal{L}'}/\#H^1_{\mathcal{L}}$; and $\#\ker\mathrm{Res}^*$ is the quotient of the orders of the two last terms, since $\mathrm{Res}^*$ is surjective and Pontryagin duality preserves orders. As $\operatorname{im}\theta \subseteq \ker\mathrm{Res}^*$ by (i), they are equal. □

So the comparison sequence is proved here given Theorem 17.68 (for the middle) and Theorem 17.70 (for the fourth term); only the complex property (i) is elementary. It can also be read off directly from the nine-term sequence; see [167, Section 1.7] and [128, Section 2.3].

For the pair $(W, T^D)$ we use the dual structures of Definition 17.64(b).

**Theorem 17.74** (the comparison sequence for $W$ and $T^D$)**.** *Let $T$ be a lattice unramified outside $S$, $\mathcal{L} \subseteq \mathcal{L}'$ Selmer structures on $W$, and $\mathcal{L}^* \supseteq \mathcal{L}'^*$ their dual structures on $T^D$. Then*

$$0 \to H^1_{\mathcal{L}}(\mathbb{Q}, W) \to H^1_{\mathcal{L}'}(\mathbb{Q}, W) \to \bigoplus_v \mathcal{L}'_v/\mathcal{L}_v \to H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D)^* \to H^1_{\mathcal{L}'^*}(\mathbb{Q}, T^D)^* \to 0$$

*is exact, with maps defined as in Theorem 17.73, and $\mathcal{L}'_v/\mathcal{L}_v \simeq (\mathcal{L}_v^*/\mathcal{L}_v'^*)^*$ by local duality.*

*Sketch of proof.* For each $k$ give $W[\ell^k] = T/\ell^k T$ the structures $\mathcal{L}_{(k),v}$ = preimage of $\mathcal{L}_v$ under $H^1(\mathbb{Q}_v, W[\ell^k]) \to H^1(\mathbb{Q}_v, W)$, and $T^D/\ell^k T^D = W[\ell^k]^D$ the structures given by the images of $\mathcal{L}_v^*$ under $H^1(\mathbb{Q}_v, T^D) \to H^1(\mathbb{Q}_v, T^D/\ell^k T^D)$; by the compatibility of local duality with these maps (Theorem 17.24) they are mutually dual, and likewise for $\mathcal{L}'$. Apply Theorem 17.73 at each level. Take the direct limit over $k$ for the terms on $W$: direct limits are exact, the Selmer groups of $W[\ell^k]$ tend to those of $W$ (the error terms $H^0(\mathbb{Q}, W)/\ell^k$ of Proposition 17.17(iv) die in the limit, $H^0(\mathbb{Q}, W)$ being torsion), and the local quotients tend to $\mathcal{L}'_v/\mathcal{L}_v$. Take the inverse limit for the terms on $T^D$: they are finite groups, so the limit is exact (Mittag-Leffler), and $\varprojlim_k H^1_{\mathcal{L}^*_{(k)}}(\mathbb{Q}, T^D/\ell^k) = H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D)$ by Chapter 16 (Tate's theorem on limits). Pontryagin duality exchanges the two limits. The local isomorphism is Proposition 17.34(ii) or, in general, the perfectness of the local pairing. See [167, Section 1.7 and Appendix B], [74, Section 4] and the proof of [49, Lemma 2.1]. □

**Theorem 17.75** (relaxing at $\Sigma$: the abstract form of Lemma 2.1 of Diamond, Flach and Guo)**.** *Let $\ell$ be odd and $T$ a lattice unramified outside $S$. Let $(L_{V,v})_v$ be $\mathbb{Q}_\ell$-subspaces $L_{V,v} \subseteq H^1(\mathbb{Q}_v, V)$ with $L_{V,v} = H^1_{\mathrm{ur}}(\mathbb{Q}_v, V)$ for $v \notin S$, and $(L_{V^D,v})_v$ subspaces of $H^1(\mathbb{Q}_v, V^D)$ with $L^\perp_{V,v} = L_{V^D,v}$ for every $v$. Let $\mathcal{L}$ be the structure on $W$ propagated from $(L_{V,v})$ and $\mathcal{L}^*$ the structure on $T^D$ propagated from $(L_{V^D,v})$, so $\mathcal{L}_p^* = \iota^{-1}(L_{V^D,p})$; by Proposition 17.65(iii), $\mathcal{L}^*$ is the dual of $\mathcal{L}$. Let $\Sigma \subseteq S$ be a finite set of primes $p \neq \ell$.*

*(i) There is an exact sequence*

$$0 \to H^1_{\mathcal{L}}(\mathbb{Q}, W) \to H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) \to \bigoplus_{p\in\Sigma} (\mathcal{L}_p^*)^* \to H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D)^* \to H^1_{(\mathcal{L}^*)_\Sigma}(\mathbb{Q}, T^D)^* \to 0.$$

*(ii) Assume (a)* $H^1_{L_{V^D}}(\mathbb{Q}, V^D) = 0$. *Then*

$$H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D) = H^1(G_{\mathbb{Q},S}, T^D)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}, W^D)/H^0(\mathbb{Q}, W^D)_{\mathrm{div}}.$$

*(iii) Assume (a) and (b)* $H^0(\mathbb{Q}_p, V^D) = 0$ *for at least one* $p \in \Sigma$. *Then* $H^0(\mathbb{Q}, V^D) = 0$, $H^0(\mathbb{Q}, W^D)$ *is finite,* $H^1_{(\mathcal{L}^*)_\Sigma}(\mathbb{Q}, T^D) = 0$, *and there is an exact sequence*

$$0 \to H^1_{\mathcal{L}}(\mathbb{Q}, W) \to H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) \to \bigoplus_{p\in\Sigma\cup\{\infty\}} (\mathcal{L}_p^*)^* \to H^0(\mathbb{Q}, W^D)^* \to 0,$$

*where* $\mathcal{L}^*_\infty = H^1(\mathbb{R}, T^D) = 0$.

*Proof.* (i) Apply Theorem 17.74 with $\mathcal{L}' = \mathcal{L}^\Sigma$. By Proposition 17.65(ii) its dual is $(\mathcal{L}^*)_\Sigma$. The local quotients are nonzero only at $p \in \Sigma$, where $\mathcal{L}^\Sigma_p/\mathcal{L}_p = H^1(\mathbb{Q}_p, W)/\mathcal{L}_p \simeq (\mathcal{L}^*_p)^*$ by Proposition 17.34(ii).

(ii) By Proposition 17.66(ii), applied to $T^D$ and the structure propagated from $(L_{V^D,v})$, hypothesis (a) gives $H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D) = H^1(G_{\mathbb{Q},S}, T^D)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}, W^D)/H^0(\mathbb{Q}, W^D)_{\mathrm{div}}$.

(iii) $H^0(\mathbb{Q}, V^D) \subseteq H^0(\mathbb{Q}_p, V^D) = 0$ for the prime $p$ of (b). Then $H^0(\mathbb{Q}, W^D)$ is finite and $\delta\colon H^0(\mathbb{Q}, W^D) \to H^1(G_{\mathbb{Q},S}, T^D)_{\mathrm{tors}} = H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D)$ is an isomorphism (Proposition 17.17(ii)). The strict Selmer group $H^1_{(\mathcal{L}^*)_\Sigma}(\mathbb{Q}, T^D)$ is the subgroup of the classes $\delta(x)$ with $\mathrm{Res}_p\,\delta(x) = 0$ for all $p \in \Sigma$. By naturality of connecting maps, $\mathrm{Res}_p\,\delta(x) = \delta_p(\mathrm{Res}_p\,x)$, where $\delta_p\colon H^0(\mathbb{Q}_p, W^D) \to H^1(\mathbb{Q}_p, T^D)$ is the local connecting map and $\mathrm{Res}_p\,x$ is $x$ viewed in $H^0(\mathbb{Q}_p, W^D) \supseteq H^0(\mathbb{Q}, W^D)$. At the prime $p$ of (b), $\delta_p$ is injective, its kernel being the image of $H^0(\mathbb{Q}_p, V^D) = 0$ (Proposition 17.17(ii) for $G_{\mathbb{Q}_p}$). So $\delta(x)$ in the strict group forces $x = 0$, and the strict group is 0. The sequence of (i) becomes the one displayed, after identifying $H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D)^*$ with $H^0(\mathbb{Q}, W^D)^*$ through $\delta$; the summand at $\infty$ is 0 by Proposition 17.51. □

*Remark* 17.76. Without (b) the conclusion of (iii) can fail only through the last term: the sequence of (i) is still exact, with $H^1_{\mathcal{L}^*}(\mathbb{Q}, T^D)$ as in (ii), and its last term is the dual of the group of $\delta(x)$, $x \in H^0(\mathbb{Q}, W^D)$, with $x \in H^0(\mathbb{Q}_p, W^D)_{\mathrm{div}}$ for every $p \in \Sigma$ (the kernel of $\delta_p$ is the image of $H^0(\mathbb{Q}_p, V^D)$, which is the divisible part by Proposition 17.17(ii)).

**Corollary 17.77** (the length identity)**.** *Under the hypotheses (a) and (b) of Theorem 17.75,*

$$\operatorname{length} H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) = \operatorname{length} H^1_{\mathcal{L}}(\mathbb{Q}, W) + \sum_{p\in\Sigma} \operatorname{length} \mathcal{L}^*_p - \operatorname{length} H^0(\mathbb{Q}, W^D),$$

*both sides being finite if and only if* $H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W)$ *is finite, and infinite otherwise. If* $L_{V^D,p} = 0$ *for* $p \in \Sigma$ *(as in the paper), every* $\mathcal{L}^*_p = H^1(\mathbb{Q}_p, T^D)_{\mathrm{tors}}$ *is finite, and the identity holds with all terms finite as soon as* $H^1_{\mathcal{L}}(\mathbb{Q}, W)$ *is finite.*

*Proof.* In an exact sequence $0 \to A \to B \to C \to D \to 0$ of $\mathbb{Z}_\ell$-modules with $D$ finite, $B$ is finite if and only if $A$ and $C$ are (the image of $B$ in $C$ has finite index), and then $\operatorname{length} B = \operatorname{length} A + \operatorname{length} C - \operatorname{length} D$ (Chapter 16, lengths in exact sequences). To see the formula, let $I$

be the image of $B \to C$; the sequence splits into $0 \to A \to B \to I \to 0$ and $0 \to I \to C \to D \to 0$, and additivity of length on short exact sequences gives $\operatorname{length} B = \operatorname{length} A + \operatorname{length} I$ and $\operatorname{length} C = \operatorname{length} I + \operatorname{length} D$. Pontryagin duality preserves lengths of finite modules (Chapter 3), so $\operatorname{length}(\mathcal{L}_p^*)^* = \operatorname{length} \mathcal{L}_p^*$ and $\operatorname{length} H^0(\mathbb{Q}, W^D)^* = \operatorname{length} H^0(\mathbb{Q}, W^D)$. If $L_{V^D,p} = 0$, $\mathcal{L}_p^* = \ker \iota = H^1(\mathbb{Q}_p, T^D)_{\mathrm{tors}}$, finite by Proposition 17.17(ii) and Theorem 17.22. □

*In the paper* 17.78. Lemma lem:dfg21 of the paper ([49, Lemma 2.1]) is Theorem 17.75(iii) for the Bloch-Kato structure: $L_{V,v} = H^1_f(\mathbb{Q}_v, V)$, $L_{V^D,v} = H^1_f(\mathbb{Q}_v, V^D)$, $\mathcal{L} = f$ on $W$, $\mathcal{L}^\Sigma$ the structure of $H^1_\Sigma(\mathbb{Q}, W)$, and $\mathcal{L}_p^* = H^1_f(\mathbb{Q}_p, T^D) = \iota^{-1} H^1_f(\mathbb{Q}_p, V^D)$ exactly as the lemma defines it. The hypotheses are discharged as follows. $L^\perp_{V,v} = L_{V^D,v}$: at $p \neq \ell$ this is Theorem 17.32(ii), at $\ell$ it is the theorem of Bloch and Kato (Chapter 18), and at $\infty$ both are 0. Hypothesis (a) is $H^1_f(\mathbb{Q}, V^D) = 0$, Theorem thm:dfg27. Hypothesis (b) holds at every $p \in \Sigma$ by the paragraph after the lemma (Paper link 17.37): $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = 0$, hence $H^0(\mathbb{Q}_p, V^D) = 0$ by Proposition 17.28(iii). The paper's hypothesis $H^0(\mathbb{Q}, V^D) = 0$ then follows, as the paper notes ("so the hypothesis $H^0(\mathbb{Q}, V^D) = 0$ is automatic here"). In the paper's situation $L_{V^D,p} = 0$ for $p \in \Sigma$, so $\mathcal{L}_p^* = H^1(\mathbb{Q}_p, T^D)_{\mathrm{tors}} \simeq H^0(\mathbb{Q}_p, W^D)$ (Corollary 17.36). *Commentary.* The paper states the lemma under the hypothesis $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$, quoting DFG04; the book proves the version with hypothesis (b) at one prime of $\Sigma$, which is what the derivation from global duality uses and what holds in the paper's application.

*Proof of Theorem thm:main, the Selmer step.* In the proof of Theorem thm:main the paper combines Theorem thm:dfg27, Lemma lem:dfg21 and the vanishing $H^0(\mathbb{Q}, W^D) = 0$ of Lemma lem:h0 into the identity $\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma} \operatorname{length} H^1_f(\mathbb{Q}_p, T^D)$. This is Corollary 17.77 with $H^0(\mathbb{Q}, W^D) = 0$; the finiteness of $H^1_\Sigma(\mathbb{Q}, W)$ is Theorem thm:dfg37 (Chapter 20).

*Remark rem:ns.* When $\ell = 3$ and the image of $\bar\rho$ is $N_s(\mathbb{F}_3)$, Lemma lem:h0 (proved in Chapter 11) gives $\operatorname{length} H^0(\mathbb{Q}, W^D) = h_3(E) = 1$; for $\ell \geq 5$, $h_\ell(E) = 0$, and $h_3(E) \leq 1$ always, with or without complex multiplication. If $H^1_f(\mathbb{Q}, V^D) = 0$, which the paper assumes for these curves, hypotheses (a) and (b) hold, and Corollary 17.77 with the local lengths of Lemma lem:fpr (Chapter 18) is the displayed identity of Remark rem:ns, with $-h_\ell(E)$ on the right.

**Example 17.79** (a toy case: class field theory of $\mathbb{Q}$ again)**.** Let $T = \mathbb{Z}_\ell$ with trivial action, $\ell$ odd: $V = \mathbb{Q}_\ell$, $W = \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $T^D = \mathbb{Z}_\ell(1)$, $W^D = \mu_{\ell^\infty}$. Take $S = \{\infty, \ell, p\}$, $\Sigma = \{p\}$ with $p \neq \ell$, and $L_{V,q} = H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Q}_\ell)$ at every prime $q$ (including $\ell$), $L_{V,\infty} = 0$.

- *The dual conditions.* For $q \neq \ell$, $L_{V^D,q} = H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Q}_\ell(1)) = 0$ (Theorem 17.32(ii); $\mathrm{Frob}_q$ acts on $\mathbb{Q}_\ell(1)$ by $q \neq 1$). At $\ell$, $H^1(\mathbb{Q}_\ell, \mathbb{Q}_\ell)$ has dimension 2 and $H^1(\mathbb{Q}_\ell, \mathbb{Q}_\ell(1)) = \widehat{\mathbb{Q}_\ell^\times} \otimes \mathbb{Q}_\ell$ has dimension 2 (Theorem 17.23 with $\ell = p$: $0 - \dim H^1 + 1 = -1$); the annihilator of the line of unramified characters is the line spanned by the unit classes (local class field theory; stated). So $L_{V^D,\ell} = \mathbb{Z}_\ell^\times \otimes \mathbb{Q}_\ell$.

- *Hypothesis (a).* $H^1(G_{\mathbb{Q},S}, \mathbb{Q}_\ell(1)) = (\mathbb{Z}_S^\times \otimes \mathbb{Z}_\ell) \otimes \mathbb{Q}_\ell$, with $\mathbb{Z}_S^\times = \pm p^{\mathbb{Z}} \ell^{\mathbb{Z}}$ (Example 17.53 at each level). The condition at $p$ kills the $p$-component (the class of $p$ in $H^1(\mathbb{Q}_p, \mathbb{Q}_\ell(1))$ is nonzero, Proposition 17.21(iv)), and the condition at $\ell$ kills the $\ell$-component (the class of $\ell$ is not a unit class). So $H^1_{L_{V^D}}(\mathbb{Q}, \mathbb{Q}_\ell(1)) = 0$.

- *Hypothesis (b).* $H^0(\mathbb{Q}_p, \mathbb{Q}_\ell(1)) = 0$, as $\mathrm{Frob}_p$ acts by $p \neq 1$.

- *The terms.* $H^1_{\mathcal{L}}(\mathbb{Q},W)=\mathrm{Hom}(G_{\mathbb{Q}},\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ unramified everywhere, which is 0 (Minkowski). $H^1_{\mathcal{L}^\Sigma}(\mathbb{Q},W)$ is the group of characters of $\ell$-power order unramified outside $p$. $\mathcal{L}_p^*=\iota^{-1}(0)=H^1(\mathbb{Q}_p,\mathbb{Z}_\ell(1))_{\mathrm{tors}}=\mathbb{Z}_p^\times\otimes\mathbb{Z}_\ell\simeq\mathbb{Z}/\ell^{v_\ell(p-1)}$ (Proposition 17.21). $H^0(\mathbb{Q},\mu_{\ell^\infty})=0$.

The sequence of Theorem 17.75(iii) reads $0\to0\to H^1_{\mathcal{L}^\Sigma}(\mathbb{Q},W)\to(\mathbb{Z}_p^\times\otimes\mathbb{Z}_\ell)^*\to0\to0$: the characters of $\ell$-power order of $G_{\mathbb{Q}}$ unramified outside $p$ form a group dual to the $\ell$-part of $(\mathbb{Z}/p)^\times$, as the theorem of Kronecker and Weber says. A variant with $H^0(\mathbb{Q},W^D)\neq0$, the toy model of $h_3(E)=1$, is Exercise 17.93.

*Remark* 17.80 (what Chapter 18 adds). To obtain the paper's Lemma lem:dfg21 and the proof of Theorem thm:main from Theorem 17.75 and Corollary 17.77, Chapter 18 supplies: the Bloch-Kato local conditions $H^1_f(\mathbb{Q}_v,V)$, with $H^1_f=H^1_{\mathrm{ur}}$ at $p\neq\ell$ and $H^1_f(\mathbb{R},-)=0$; their exact orthogonality at $\ell$ (Bloch and Kato); the vanishing $H^1_f(\mathbb{Q},V^D)=0$ (Theorem thm:dfg27, [49, Theorem 2.7]); and the splitting length $H^1_f(\mathbb{Q}_p,T^D)=\mathrm{length}\,H^0(\mathbb{Q}_p,W^D)=v_\ell(L_p(A_f,1)^{-1})+v_\ell(\mathrm{Tam}_p(A))$ of Lemma lem:fpr ([69, I.4.2.2]), whose Euler factor is Proposition 17.28(iv) and whose Tamagawa factor comes from Proposition 17.30. Chapter 19 identifies $\det(1-\mathrm{Frob}_p^{-1}\mid(V^D)^{I_p})$ with $L_p(A_f,1)^{-1}$. This chapter supplies the duality, the local structure (Section 17.4) and the global sequence.

## 17.11 Exercises

**Exercise 17.81** (a twist by an even quadratic character)**.** Let $\ell$ be odd, $d>1$ a fundamental discriminant, $\chi_d$ its (even) quadratic character and $T=\mathbb{Z}_\ell(\chi_d)$, so $T^D=\mathbb{Z}_\ell(1)(\chi_d)$. (a) For $p\nmid d\ell$ show $H^1_{\mathrm{ur}}(\mathbb{Q}_p,T)=\mathbb{Z}_\ell/(\chi_d(p)-1)$, which is $\mathbb{Z}_\ell$ or 0, and $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p,T^D)=\ell^{v_\ell(p\chi_d(p)-1)}$. (b) For $p\mid d$, $p\neq\ell$, show $H^1(\mathbb{Q}_p,T)=0$. (c) Show $H^0(\mathbb{Q},W)=0$ and $H^0(\mathbb{Q},W^D)=0$; contrast with Exercise 17.93. (d) Evaluate (a) for $d=5$, $\ell=3$ at $p=2,7,11$. *Answer:* (a) Proposition 17.28(i), (iv), Frob acting by $\chi_d(p)$ on $T$ and by $p\chi_d(p)$ on $T^D$. (b) $I_p$ acts by $\chi_d|_{I_p}\neq1$ on $V$ and on $V^D$, so both have no invariants, $H^1(\mathbb{Q}_p,V)=0$ by Corollary 17.25, and $H^1(\mathbb{Q}_p,T)=H^1(\mathbb{Q}_p,T)_{\mathrm{tors}}\simeq H^0(\mathbb{Q}_p,W)\subseteq W[2]=0$ (Proposition 17.17(ii)). (c) $H^0(\mathbb{Q},W)\subseteq W[2]=0$; an element of order $\ell$ of $W^D$ is fixed only if $\bar\chi_\ell\chi_d=1$, which forces $\bar\chi_\ell$ quadratic, so $\ell=3$ and $\chi_d=\chi_{-3}$, which is odd. (d) $\chi_5(2)=\chi_5(7)=-1$, $\chi_5(11)=1$: $H^1_{\mathrm{ur}}(\mathbb{Q}_p,T)=0,0,\mathbb{Z}_3$ and $\#H^1_{\mathrm{ur}}(\mathbb{Q}_p,T^D)=3^{v_3(3)},3^{v_3(8)},3^{v_3(10)}=3,1,1$.

**Exercise 17.82.** Compute $\#H^i(\mathbb{Q}_p,\mathbb{Z}/\ell^k)$ and $\#H^i(\mathbb{Q}_p,\mu_{\ell^k})$, $i=0,1,2$, for $(p,\ell,k)=(13,3,2)$, $(31,5,2)$ and $(17,3,1)$, and check the Euler characteristic and the duality of Corollary 17.25(i). *Answer:* with $m=\min(k,v_\ell(p-1))$ the orders are $\ell^k,\ell^{k+m},\ell^m$ and $\ell^m,\ell^{k+m},\ell^k$; here $m=1,1,0$, giving $9,27,3$ and $3,27,9$; $25,125,5$ and $5,125,25$; $3,3,1$ and $1,3,3$.

**Exercise 17.83.** For the group $\{1,c\}$ of order 2 the cohomology is 2-periodic: $H^{\mathrm{odd}}=\ker(1+c)/(c-1)M$ and $H^{\mathrm{even}}=M^{c=1}/(1+c)M$ in positive degrees (Chapter 16). Compute $H^1$ and $H^2$ of $G_{\mathbb{R}}$ with coefficients in $\mathbb{Z}_2$ (trivial action) and in $\mathbb{Z}_2(1)$, and check the second against $\mathrm{Br}(\mathbb{R})=\mathbb{Z}/2$ through the limit of the groups $H^2(\mathbb{R},\mu_{2^k})$. *Answer:* $H^1(\mathbb{R},\mathbb{Z}_2)=0$, $H^2(\mathbb{R},\mathbb{Z}_2)=\mathbb{Z}/2$; $H^1(\mathbb{R},\mathbb{Z}_2(1))=\mathbb{Z}/2$, $H^2(\mathbb{R},\mathbb{Z}_2(1))=0$, which is $\varprojlim\mathrm{Br}(\mathbb{R})[2^k]$ along the maps induced by $\zeta\mapsto\zeta^2$, multiplication by 2, hence 0. So the paper's "$H^1_f(\mathbb{R},T^D)=0$" genuinely needs $\ell$ odd.

**Exercise 17.84.** Check the paper's identification $H^1(G,\mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$ for $q=13$, $\ell=3$ by computing both sides, and identify the torsion with $H^0(\mathbb{Q}_{13},\mathbb{Q}_3/\mathbb{Z}_3(1))$ as Proposition 17.17(ii) predicts. *Answer:* $\mathbb{Z}_3\times\mathbb{Z}/3$ on both sides ($v_3(12)=1$); the torsion is the image of the units, of order $3=\#\mu_{3^\infty}(\mathbb{Q}_{13})$, since $13\equiv 1 \pmod 3$ and $13\not\equiv 1 \pmod 9$.

**Exercise 17.85.** Let $q=37$, $\ell=3$, so $v=v_3(36)=2$. For $u=2$, 8, 6 compute the class of $u$ in $\mathbb{F}_{37}^\times\otimes\mathbb{Z}_3=\mathbb{Z}_3(1)/(36)\simeq\mathbb{Z}/9$ and its valuation, in the sense of Lemma lem:tamfactor. *Answer:* the cubes in $\mathbb{F}_{37}^\times$ are $\{1,6,8,10,11,14,23,26,27,29,31,36\}$ and the ninth powers $\{1,6,31,36\}$; so the valuations are 0, 1 and 2, the last because the class of 6 is 0 and $v_3(0)$ is read as $v=2$.

**Exercise 17.86.** Let $K$ be a finite extension of $\mathbb{Q}_q$, $q\neq\ell$, and $\varpi$, $\varpi'=\varpi w$ two uniformisers ($w$ a unit). Let pr and pr$'$ be the retractions $H^1(K,\mathbb{Z}_\ell(1))\to H^1_{\mathrm{ur}}(K,\mathbb{Z}_\ell(1))$ attached to them as in Proposition 17.41. Show that $\mathrm{pr}'[\kappa_x]=\mathrm{pr}[\kappa_x]-v_K(x)[\kappa_w]$. Deduce that, with $n=v_K(x)$ and $c_1$, $c_2$ the classes of $\mathrm{pr}[\kappa_x]$ and $\mathrm{pr}'[\kappa_x]$ in $\mathbb{Z}/\ell^{v_\ell(\#k_K-1)}$, $\min(v_\ell(n),v_\ell(\#k_K-1),v_\ell(c_1))=\min(v_\ell(n),v_\ell(\#k_K-1),v_\ell(c_2))$: the quantity of Lemma lem:tamfactor does not depend on the uniformiser. *Hint:* $x=\varpi^n u=\varpi'^n(uw^{-n})$; the two classes differ by a multiple of $n$.

**Exercise 17.87.** Let $\varepsilon\colon G\to\{\pm1\}$ be a character, $\chi\colon G\to\mathbb{Z}_\ell^\times$ a character and $\kappa$ a cocycle with values in $\mathbb{Z}_\ell(\chi)$, so $\kappa(gh)=\kappa(g)+\chi(g)\kappa(h)$. Show that $g\mapsto\left(\begin{smallmatrix}\chi\varepsilon(g) & \varepsilon\kappa(g)\\ 0 & \varepsilon(g)\end{smallmatrix}\right)$ is a homomorphism and that $g\mapsto\left(\begin{smallmatrix}\chi\varepsilon(g) & \kappa(g)\\ 0 & \varepsilon(g)\end{smallmatrix}\right)$ is one only if $\kappa(gh)=\varepsilon(h)\kappa(g)+\chi(g)\varepsilon(g)\kappa(h)$ for all $g,h$, which fails for a Kummer cocycle when $\varepsilon\neq1$. What changes in $c'$ if the matrix with corner $\kappa$ is used? *Answer:* the upper right entry of the product is $\chi\varepsilon(g)\varepsilon\kappa(h)+\varepsilon\kappa(g)\varepsilon(h)=\varepsilon(gh)\kappa(gh)$; with the corner $\varepsilon\kappa$, $c'$ is multiplied by $\varepsilon(\mathrm{Frob})=\pm1$, which changes no valuation.

**Exercise 17.88.** (PARI) For `874a1` at $q=19$ and `1066a1` at $q=13$ ($\ell=3$), compute the Tate parameter with `ellinit(E,O(q^30)).tate`, its valuation $n$ and the reduction $\bar u$ of its unit part; decide whether $\bar u$ is a cube and, for $q=19$, a ninth power; deduce the extra term $\min(v_3(n),v_3(q-1),v_3(c))$ and compare $\sum_q t_3(E,q)+\mathrm{extra}$ with $v_3(\deg\varphi_E)$. *Answer:* `874a1` $=[1,-1,0,-19,-13]$: $n=3$, $\bar u=12$, a cube but not a ninth power, $a_{19}=-1$, extra 1; the other bad primes 2, 23 are of type $\mathrm{I}_1$; bound 2, and $\deg\varphi_E=144$. `1066a1` $=[1,0,1,-1693,26680]$: $n=3$, $\bar u=5$, a cube mod 13, extra 1; the other bad primes are 2 ($\mathrm{I}_7$) and 41 ($\mathrm{I}_2$); bound 2, and $\deg\varphi_E=504=2^3\cdot3^2\cdot7$. Both have surjective mod 3 image, and the bound is attained.

**Exercise 17.89.** Let $E$ have good reduction at $p\neq\ell$, $a=a_p(E)$, $V=\mathrm{ad}^0V_\ell E$ and $V^D=V(1)$. Show that $H^1_{\mathrm{ur}}(\mathbb{Q}_p,V^D)=0$ and that

$$\#H^1_{\mathrm{ur}}(\mathbb{Q}_p,T^D)=\ell^{\,v_\ell\left((p-1)\left((p+1)^2-a^2\right)\right)},$$

and compute $\det(1-\Phi\mid V^D)$ and $\det(1-\mathrm{Frob}\mid V^D)$. Evaluate for `11a1` at $p=2$, $\ell=5$. *Answer:* the arithmetic Frobenius has eigenvalues $p\alpha/\beta$, $p$, $p\beta/\alpha$ on $V^D$, of absolute value $p$; $\det(1-\mathrm{Frob})=(1-p)((p+1)^2-a^2)$ and $\det(1-\Phi)=(1-p^{-1})((p+1)^2-a^2)/p^2$, of the same $\ell$-adic valuation. For `11a1`, $a_2=-2$ and $(p-1)((p+1)^2-a^2)=5$: the group has order 5.

**Exercise 17.90.** Let $E$ have reduction of type IV or IV$^*$ at $q\neq3$, $\ell\neq q$ odd. Using Lemma lem:IV of the paper ($(\mathrm{ad}^0V_\ell E)^{I_q}$ is a line on which $\mathrm{Frob}_q$ acts by $\chi_{-3}(q)$), show that $\#H^1_{\mathrm{ur}}(\mathbb{Q}_q,T^D)=\ell^{v_\ell(q-\chi_{-3}(q))}$, and check the paper's value $e_3(E,2)=1$ at $q=2$. *Hint:* Proposition 17.28(iv) with eigenvalue $q\chi_{-3}(q)$; $\chi_{-3}(2)=-1$ and $v_3(3)=1$.

**Exercise 17.91.** Apply Theorem 17.70 to $M = \mathbb{Z}/\ell$ ($\ell$ odd) with $\mathcal{L}_p = H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Z}/\ell)$ for $p \neq \ell$, $\mathcal{L}_\ell = H^1(\mathbb{Q}_\ell, \mathbb{Z}/\ell)$ and $\mathcal{L}_\infty = 0$. Compute both sides and identify the cyclic extensions of $\mathbb{Q}$ of degree $\ell$ unramified outside $\ell$. *Answer:* the dual Selmer group is 0 (it is strict at $\ell$); the right side is $\ell \cdot \frac{1}{\ell} \cdot \frac{\#H^1(\mathbb{Q}_\ell,\mathbb{Z}/\ell)}{\#H^0(\mathbb{Q}_\ell,\mathbb{Z}/\ell)} = \ell^2/\ell = \ell$, with $\#H^1(\mathbb{Q}_\ell, \mathbb{Z}/\ell) = \ell^2$ by Theorem 17.23. So there are $\ell$ homomorphisms, and one such field: the subfield of degree $\ell$ of $\mathbb{Q}(\zeta_{\ell^2})$.

**Exercise 17.92.** In Example 17.79 take $\Sigma = \{p_1, p_2\}$, two primes different from $\ell$. Check the hypotheses of Theorem 17.75 and show that the characters of $G_\mathbb{Q}$ of $\ell$-power order unramified outside $\{p_1, p_2\}$ form a group isomorphic to $\mathbb{Z}/\ell^{v_\ell(p_1-1)} \times \mathbb{Z}/\ell^{v_\ell(p_2-1)}$.

**Exercise 17.93** (the toy model of $h_3(E)$)**.** Let $T = \mathbb{Z}_3(\chi_{-3})$, the module $\mathbb{Z}_3$ on which $G_\mathbb{Q}$ acts through the quadratic character $\chi_{-3}$. Show that $W^D = \mu_{3^\infty} \otimes \chi_{-3}$ and $H^0(\mathbb{Q}, W^D) \simeq \mathbb{Z}/3$: use $\bar{\chi}_3^{\mathrm{cyc}} = \chi_{-3}$ for the elements of order 3, and an element $\sigma$ with $\chi_3^{\mathrm{cyc}}(\sigma) = 4$ for those of order 9. Show that for $\ell \geq 5$ and every quadratic character $\psi$, $H^0(\mathbb{Q}, \mu_{\ell^\infty} \otimes \psi) = 0$. Relate this to Lemma lem:h0 ($h_\ell(E) = 0$ for $\ell \geq 5$, $h_3(E) \leq 1$). *Hint:* $\sigma$ fixes $\zeta_3$ (as $4 \equiv 1 \pmod 3$), so $\chi_{-3}(\sigma) = 1$, and $\sigma$ acts on $\mu_9 \otimes \chi_{-3}$ by $4 \not\equiv 1 \pmod 9$. For $\ell \geq 5$, $\bar{\chi}_\ell \psi$ is nontrivial because $\bar{\chi}_\ell$ has order $\ell - 1 \geq 4$; the determinant argument of Lemma lem:h0 is the same mechanism in rank two.

**Exercise 17.94.** ($\ell = p = 2$.) Show that $\#\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2 = 8$, and check Theorem 17.23 for $M = \mu_2$ over $\mathbb{Q}_2$. *Answer:* $\mathbb{Q}_2^\times = 2^\mathbb{Z} \times \{\pm 1\} \times (1 + 4\mathbb{Z}_2)$ and $1 + 4\mathbb{Z}_2 \simeq \mathbb{Z}_2$, so the quotient is $(\mathbb{Z}/2)^3$; $\#H^0 = \#H^2 = 2$, and $2 \cdot 2/8 = \frac{1}{2} = |2|_2$.

**Exercise 17.95.** Let $E =$ `77c1` (nonsplit multiplicative of type $\mathrm{I}_3$ at 7). Show that $E(\mathbb{Q}_7)[3] = 0$ and deduce that $H^i(\mathbb{Q}_7, E[3]) = 0$ for $i = 0, 1, 2$. *Hint:* $E(\mathbb{Q}_7)/E_0(\mathbb{Q}_7)$ has order $c_7 = 1$, $E_0/E_1 \simeq \tilde{E}_{\mathrm{ns}}(\mathbb{F}_7)$ has order $7+1 = 8$, $E_1(\mathbb{Q}_7)$ is pro-7; then use Corollary 17.25 with $E[3]^D \simeq E[3]$.

**Exercise 17.96.** Let $E$ have good reduction at $p \neq \ell$. Show that $H^1(\mathbb{Q}_p, T_\ell E) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, T_\ell E)$, finite of order $\ell^{v_\ell(\#\tilde{E}(\mathbb{F}_p))}$, and that it is isomorphic to $E(\mathbb{Q}_p)[\ell^\infty]$. Check with `11a1`, $p = 2$, $\ell = 5$. *Hint:* $\dim H^1(\mathbb{Q}_p, V) = 0$ by Corollary 17.25(ii) ($V$ and $V^D \simeq V$ have no invariants); $\det(1 - \mathrm{Frob} \mid V) = 1 - a_p + p$; for `11a1`, $\#\tilde{E}(\mathbb{F}_2) = 5$.

**Exercise 17.97.** (Harder.) Prove Theorem 17.32(i) for $M = \mathbb{Z}/\ell$, $M^D = \mu_\ell$, $p \neq \ell$, directly from local class field theory (Chapter 7): for a character $\chi\colon G_{\mathbb{Q}_p} \to \mathbb{Z}/\ell$ and $a \in \mathbb{Q}_p^\times$, $\mathrm{inv}(\chi \cup \delta(a)) = \pm\frac{1}{\ell}\chi(\mathrm{rec}(a))$, where rec is the reciprocity map. Deduce that unramified characters are orthogonal to unit classes, and count.

**Exercise 17.98** (Greenberg-Wiles on the Kummer side)**.** Let $\ell$ be odd, $M = \mu_\ell$, and $\mathcal{L}$ the structure with $\mathcal{L}_p = H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mu_\ell)$ for $p \neq \ell$, $\mathcal{L}_\ell = H^1(\mathbb{Q}_\ell, \mu_\ell)$, $\mathcal{L}_\infty = 0$. Using global Kummer theory and Proposition 17.39, show that $H^1_{\mathcal{L}}(\mathbb{Q}, \mu_\ell)$ is the image of $\ell^\mathbb{Z}$ in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^\ell$, of order $\ell$. Identify the dual structure on $\mathbb{Z}/\ell$ and show $H^1_{\mathcal{L}^*}(\mathbb{Q}, \mathbb{Z}/\ell) = 0$. Check Theorem 17.70, and compare with Exercise 17.91, where the roles of $M$ and $M^D$ are exchanged. *Answer:* $a \in \mathbb{Q}^\times$ is a unit times an $\ell$-th power at each $p \neq \ell$ iff $\ell \mid v_p(a)$ for all $p \neq \ell$, iff $a \in \pm\ell^\mathbb{Z}(\mathbb{Q}^\times)^\ell$, and $-1 = (-1)^\ell$. The dual structure is unramified at $p \neq \ell$ (Theorem 17.32) and 0 at $\ell$; a character in it is unramified everywhere, hence trivial (Minkowski). Right side: $\frac{\#H^0(\mathbb{Q},\mu_\ell)}{\#H^0(\mathbb{Q},\mathbb{Z}/\ell)} = \frac{1}{\ell}$, the factors at $p \neq \ell$ and at $\infty$ are 1 (Proposition 17.28(ii); $H^0(\mathbb{R}, \mu_\ell) = 1$), and at $\ell$ the factor is $\#H^1(\mathbb{Q}_\ell, \mu_\ell)/\#H^0(\mathbb{Q}_\ell, \mu_\ell) = \ell^2$ (Example 17.26(b)); so both sides equal $\ell$.

## 17.12 Notes and references

*Galois cohomology of fields.* Serre's *Galois Cohomology* [177] (Chapter I for profinite groups, Chapter II for number fields and local fields) and his *Local Fields* [174] (Chapters X to XIV: Hilbert 90, Kummer theory, the Brauer group, local class field theory in cohomological form) are the sources for Section 17.1; Neukirch, Schmidt and Wingberg [143] (Chapters I, II, VI to VIII) is the comprehensive reference, and the articles of Serre and Tate in Cassels and Fröhlich [26] remain the best short account of local and global class field theory with cohomology. Milne's *Class Field Theory* [133] is a free alternative.

*Continuous cohomology and $\ell$-adic coefficients.* The passage from finite to $\ell$-adic coefficients (Proposition 17.17) follows Tate [196, Section 2] and Rubin [167, Sections 1.1 to 1.3 and Appendix B]. Rubin's first chapter is the closest model for Sections 17.4, 17.8 and 17.10: unramified and propagated local conditions, Selmer groups defined by local conditions, and the comparison exact sequence. The term "Selmer structure", and the relaxed and strict structures, are from Mazur and Rubin [128, Chapter 2].

*Duality.* Tate announced local and global duality and the Euler characteristic formulas at the Stockholm congress [195]; complete proofs are in [177, Chapter II, Section 5] (local), and in Milne [131, Chapter I] and [143, Chapters VII and VIII] (local and global, including the Poitou-Tate sequence). The formula of Theorem 17.70 is due to Greenberg and to Wiles; it is stated in [207, Section 1] and proved in [38, Section 2]. Greenberg's survey [74] treats Selmer groups of $\ell$-adic representations and the comparison sequence in the form used by Diamond, Flach and Guo [49, Section 2].

*Elliptic curves.* Section 17.7 follows Silverman [185, Chapters VIII and X]; the Tate curve and its Galois representation are in [184, Chapter V] and [178, Chapter IV, Appendix]. Tate's theorem that the local Kummer image is its own annihilator is in [131, Chapter I, Section 3].

*The paper's sources.* Lemma lem:dfg21 is [49, Lemma 2.1] and Theorem thm:dfg27 is [49, Theorem 2.7], as the paper cites them; both numbers and statements were checked against the published text (Lemma 2.1 on p. 696, with the hypotheses that $\Sigma$ is nonempty, contains every prime at which $H^1_f(\mathbb{Q}_p, W) \neq H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$, and $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$; Theorem 2.7 on p. 699). The formula of Fontaine and Perrin-Riou is [69, I.4.2.2], and the Bloch-Kato conditions are in [14, Section 3]; these two numbers are the paper's.

*Citations.* Every reference in this chapter to a book is to a chapter or section, not to a numbered theorem. Theorem numbers we recall but have not checked against the printed editions, and which the reader may find useful as unverified pointers, are: Serre, *Galois Cohomology*, II.5.2 (local duality) and II.5.7 (Euler characteristic); Milne, *Arithmetic Duality Theorems*, I.2.3, I.2.6, I.2.8 and I.4.10; Wiles [207], Proposition 1.6; Darmon, Diamond and Taylor [38], Theorem 2.19; Rubin [167], Theorem 1.7.3 and Lemma 1.3.2; Bloch and Kato [14], Proposition 3.8; Silverman [185], X.4.2.

*Computations.* The numbers in Examples 17.26, 17.46 and 17.61 and in the exercises were computed with PARI/GP [148] 2.17.2 through `cypari2` (`ellinit`, `ellinit(E,O(q^30)).tate`, `elllocalred`, `ellap`, `ellmoddegree`, `elldivpol`, `polrootspadic`, `ispower`) and compared with Cremona's tables [35] (files `allcurves`, `allbsd`, `degphi`, `galrep`, conductor below 10,000).

# Chapter 18

# Bloch-Kato Selmer groups, Tamagawa factors and the adjoint motive

Theorem thm:main of the paper says that $v_\ell(\deg \varphi_E)$ is the length of a Bloch-Kato Selmer group plus Tamagawa factors plus omitted Euler factors. This chapter defines the first two, computes the second at every bad prime, and proves the Selmer half of the theorem: the comparison of the Bloch-Kato Selmer group with the $\Sigma$-imprimitive one of Diamond, Flach and Guo, with every local term identified. It proves the paper's Lemmas lem:fpr, lem:tamfactor and lem:tamadd and its Corollary cor:explicit, restates Lemma lem:dfg21 and Lemma lem:h0 in the language of Selmer groups, and places the whole in the framework of the Tamagawa number conjecture of Bloch and Kato.

## Core results at a glance

Throughout, $E/\mathbb{Q}$ has conductor $N$, $\ell$ is an odd prime, $T = T_\ell E$, $V = T \otimes \mathbb{Q}_\ell$, $A = \mathrm{ad}^0 T$, $A_f = \mathrm{ad}^0 V$, $W = A \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1))$, $V^D = T^D \otimes \mathbb{Q}_\ell$, $W^D = V^D/T^D$, and $\Sigma$ is the set of primes dividing $N$; $p \neq \ell$ in all local statements unless said otherwise.

**18.1 Conventions.** Linear dual $M^\vee$, Kummer dual $M^D = \mathrm{Hom}(M, \mathbb{Z}_\ell(1)) = M^\vee(1)$, Pontryagin dual $X^*$; $\mathrm{Frob}_p$ arithmetic, $L_p(M, s) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid M^{I_p})^{-1}$ with the geometric Frobenius; $\mathbb{Q}_\ell(1)$ has Hodge-Tate weight $+1$; Tate data $q_E$, $\nu = -v_q(j_E)$, $u = q_E q^{-\nu}$, $c(u)$ the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$, $v_\ell(c(u))$ the largest $k \le v_\ell(q-1)$ with $\bar{u}$ an $\ell^k$-th power. *(convention)* Convention 18.1.

**18.2 $\ell$-adic representations.** A continuous $G_K \to \mathrm{GL}(V)$ on a finite-dimensional $\mathbb{Q}_\ell$-space; a stable lattice $T$; $W = V/T$. *(definition)* Definition 18.4.

**18.3 Lattices.** Stable lattices exist; $W[\ell^k] \simeq T/\ell^k T$; lattices are commensurable, and all stable lattices are homothetic if $T/\ell T$ is irreducible. *(proved here)* Proposition 18.5.

**18.4 Tate twists.** $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$, $M(n) = M \otimes \mathbb{Z}_\ell(1)^{\otimes n}$. *(definition)* Definition 18.6.

**18.5 Twists at $p \neq \ell$.** $M(n)^{I_p} = M^{I_p}(n)$ and Frobenius eigenvalues are multiplied by $p^n$ (arithmetic) or $p^{-n}$ (geometric); $\wedge^2 T_\ell E \simeq \mathbb{Z}_\ell(1)$, $(T_\ell E)^\vee \simeq T_\ell E(-1)$. *(proved here, given Chapter 9)* Proposition 18.7.

**18.6 Kummer duals.** $V^D = V^\vee(1)$, $T^D = T^\vee(1)$, $W^D = V^D/T^D$; $X^* = \mathrm{Hom}_{\mathrm{cont}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$. *(definition)* Definition 18.8.

**18.7 Kummer duality.** $W^D \simeq \mathrm{Hom}(T, \mu_{\ell^\infty})$, $T^D \simeq \mathrm{Hom}(W, \mu_{\ell^\infty})$, perfect equivariant pairings; $(T^D)^D = T$; $X^*$ has the length of $X$ for finite $X$. *(proved here, given Chapter 3)* Proposition 18.9.

**18.8 The adjoint is self-dual.** For $\ell$ odd the trace form is perfect on $A$ (Gram determinant $-2$), so $T^D \simeq A(1)$, $V^D \simeq A_f(1)$, $W^D \simeq \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$; $\mathrm{ad}^0$ is unchanged by twisting by a character, by homothety and by dualising; so $\mathrm{ad}^0 M_{f,\lambda} \simeq \mathrm{ad}^0 T_\ell E$, and $\mathrm{ad}^0 T$ is the same $G_{\mathbb{Q}_q}$-module for $E$ and its twists. *(proved here, given Chapters 4 and 14)* Proposition 18.10.

**18.9 Geometric representations.** Unramified almost everywhere and de Rham at $\ell$. *(definition)* Definition 18.12.

**18.10 Purity.** $V_\ell E$ and its tensor constructions are geometric *(stated; reference)*; at $p \nmid N\ell$ the Frobenius eigenvalues on $V_\ell E$, $A_f$, $V^D$ have absolute values $p^{1/2}$, $1$, $p$ *(proved here, given Chapter 9)*. Proposition 18.13.

**18.11 Global invariants.** If $\bar\rho$ is absolutely irreducible, $H^0(\mathbb{Q}, A_f) = H^0(\mathbb{Q}, W) = 0$; $H^0(\mathbb{Q}, V^D) = 0$ for every $E$; $A_f$, $A_f(1)$, $A_f(-1)$ are pairwise non-isomorphic. *(proved here)* Proposition 18.14.

**18.12 The Frobenius dictionary.** $L_p(M(1), s) = L_p(M, s+1)$; $(F-1)X = (F^{-1}-1)X$; $\#X/(F-1)X = \ell^{v_\ell \det(1-F^{-1}|X)}$ for a lattice $X$; $\det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid V_\ell E)^{-1} = L_p(E, s+1)$. *(proved here)* Proposition 18.15.

**18.13 The adjoint at $p \neq \ell$.** $A_f^{I_p}$, the geometric Frobenius on it and $L_p(A_f, 1)^{-1} = \det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p})$ by reduction type: $1 - p^{-2}$ at potentially multiplicative primes, $(1 - \frac{\alpha}{\beta p})(1 - \frac{1}{p})(1 - \frac{\beta}{\alpha p})$ at good primes, $1 - cp^{-1}$ for cyclic inertia of order $3, 4, 6$, $\frac{p-1}{p} \cdot \frac{(p+1)^2 - a^2}{p^2}$ for $\rho(I_p) = \{\pm 1\}$, $1$ for non-abelian inertia. *(proved in Chapters 11 and 19; recalled here with the reasons)* Proposition 18.17.

**18.14 Local invariants.** $H^0(\mathbb{Q}_p, V^D) = 0$ and $L_p(A_f, 1)^{-1} \in \mathbb{Q}^\times$; $H^0(\mathbb{Q}_p, A_f)$ is $0$ or a line by type, nonzero at some bad primes. *(proved here)* Lemma 18.18.

**18.15 Local finiteness.** $H^1(\mathbb{Q}_p, T^D)$ has rank $\dim H^0(\mathbb{Q}_p, A_f)$; it is finite when that is 0; its torsion is always finite. *(proved here, given Chapter 17)* Corollary 18.20.

**18.16 Integrality.** $L_p(A_f, 1)^{-1} = \det(1 - \mathrm{Frob}_p^{-1} \mid (T^D)^{I_p}) \in \mathbb{Z}_{(\ell)} \cap \mathbb{Q}^\times$, $\#(T^D)^{I_p}/(\mathrm{Frob}_p - 1) = \ell^{v_\ell(L_p(A_f,1)^{-1})}$; the valuation is $v_\ell(p^2-1)$ at potentially multiplicative primes and $e_\ell(E,p)$ at additive primes. *(the lattice statements proved here; the value and valuation proved in Chapter 19)* Lemma 18.21.

**18.17 Period rings.** $B_{\mathrm{dR}} \supset B_{\mathrm{st}} \supset B_{\mathrm{cris}}$, $D_\bullet(V) = (B_\bullet \otimes V)^{G_{\mathbb{Q}_\ell}}$; de Rham, semistable, crystalline representations; Hodge-Tate weights. *(stated; reference)* Definition 18.23.

**18.18 Good reduction.** $V_\ell B$ is crystalline iff the abelian variety $B/\mathbb{Q}_\ell$ has good reduction; for $E$ good at $\ell$, $\mathrm{D}_{\mathrm{cris}}((V_\ell E)^\vee)$ has Frobenius polynomial $X^2 - a_\ell X + \ell$. *(stated; reference)* Theorem 18.24.

**18.19 Stability.** Crystalline and de Rham representations are stable under the tensor operations; for $\ell \nmid N$, $A_f$, $V^D$, $V^\vee$ are crystalline with weights $\{-1,0,1\}$, $\{0,1,2\}$, $\{-1,0\}$. *(stated; reference; the consequences proved here)* Proposition 18.25.

**18.20 Fontaine-Laffaille.** An exact equivalence between strongly divisible filtered modules with weights in an interval of length at most $\ell - 2$ and lattices in crystalline representations. *(stated; reference)* Theorem 18.26.

**18.21 Local dimensions at $\ell$.** $\dim H^1_f(\mathbb{Q}_\ell, V) = \dim H^0(\mathbb{Q}_\ell, V) + \#\{\text{positive Hodge-Tate weights}\}$; for $\ell \nmid N$, $\dim H^1_f(\mathbb{Q}_\ell, A_f) = \dim H^0(\mathbb{Q}_\ell, A_f) + 1$ and $\dim H^1_f(\mathbb{Q}_\ell, V^D) = 2$. *(proof sketched here; reference)* Proposition 18.28.

**18.22 Bloch-Kato local conditions.** $H^1_f(\mathbb{Q}_p, V) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V)$ for $p \neq \ell$; $H^1_f(\mathbb{Q}_\ell, V) = \ker(H^1(\mathbb{Q}_\ell, V) \to H^1(\mathbb{Q}_\ell, B_{\mathrm{cris}} \otimes V))$, with $H^1_e \subseteq H^1_f \subseteq H^1_g$; $H^1_f(\mathbb{R}, V) = 0$. *(definition)* Definition 18.29.

**18.23 Propagation.** $H^1_f(\mathbb{Q}_p, T) = \iota^{-1} H^1_f(\mathbb{Q}_p, V)$, $H^1_f(\mathbb{Q}_p, W) = \pi H^1_f(\mathbb{Q}_p, V)$. *(definition)* Definition 18.30.

**18.24 Propagated conditions.** $H^1_f(\mathbb{Q}_p, T) \supseteq H^1(\mathbb{Q}_p, T)_{\mathrm{tor}} = \ker \iota$; $H^1_f(\mathbb{Q}_p, W) \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^{\dim H^1_f(\mathbb{Q}_p, V)}$. *(proved here, given Chapter 17)* Proposition 18.31.

**18.25 The conditions at $p \neq \ell$.** $H^1_f(\mathbb{Q}_p, V) = V^{I_p}/(\mathrm{Frob}_p - 1)$; $0 \to H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) \to H^1(\mathbb{Q}_p, T) \to H^1(I_p, T)^{G_{\mathbb{Q}_p}} \to 0$; $H^1_f(\mathbb{Q}_p, T)/H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) \simeq (H^1(I_p, T)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}}$; $H^1_f(\mathbb{Q}_p, W)$ is the maximal divisible subgroup of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$; if $H^0(\mathbb{Q}_p, V) = 0$ then $H^1_f(\mathbb{Q}_p, T) = H^1(\mathbb{Q}_p, T)_{\mathrm{tor}}$ is finite. *(proved here, given Chapters 16 and 17)* Proposition 18.32.

**18.26 Duality.** $H^1_f(\mathbb{Q}_v, V)$ and $H^1_f(\mathbb{Q}_v, V^D)$ are exact annihilators; $H^1(\mathbb{Q}_v, W)/H^1_f(\mathbb{Q}_v, W) \simeq H^1_f(\mathbb{Q}_v, T^D)^*$. *(proved here for $v \neq \ell$, given Chapter 17; stated with reference at $v = \ell$)* Proposition 18.34.

**18.27 Bloch-Kato Selmer groups.** $H^1_f(\mathbb{Q}, M)$: classes whose localisations lie in $H^1_f(\mathbb{Q}_v, M)$ at every place. *(definition)* Definition 18.36.

**18.28 Finite level.** $H^1_f = H^1_{\mathrm{ur}}$ at unramified primes, and $H^1_f(\mathbb{Q}, M) \subseteq H^1(G_{\mathbb{Q},S}, M)$ is cut out by the conditions at $S$. *(proved here)* Lemma 18.37.

**18.29 Relations.** $H^1_f(\mathbb{Q}, T) = \iota^{-1} H^1_f(\mathbb{Q}, V)$ with torsion the image of $H^0(\mathbb{Q}, W)$; $H^1_f(\mathbb{Q}, W)$ is cofinitely generated with maximal divisible subgroup $\pi H^1_f(\mathbb{Q}, V)$, so it is finite iff $H^1_f(\mathbb{Q}, V) = 0$. *(proved here, given the finiteness theorems of Chapter 17)* Proposition 18.38.

**18.30 The $\Sigma$-imprimitive Selmer group.** $H^1_\Sigma(\mathbb{Q}, W)$: Bloch-Kato at $\ell$, unramified outside $\Sigma \cup \{\ell\}$, nothing on $\Sigma$. *(definition)* Definition 18.39.

**18.31 Finiteness of $H^1_\Sigma$.** If $\Sigma$ ($\ell \notin \Sigma$) contains the primes $p \neq \ell$ at which $W$ is ramified, $H^1_f(\mathbb{Q}, V) = 0$ and $H^0(\mathbb{Q}_p, V^D) = 0$ for $p \in \Sigma$, then $0 \to H^1_f(\mathbb{Q}, W) \to H^1_\Sigma(\mathbb{Q}, W) \to \bigoplus_{p \in \Sigma} H^1_f(\mathbb{Q}_p, T^D)^*$ is exact with finite outer terms. *(proved here)* Proposition 18.40.

**18.32 The paper's Selmer groups.** $H^1_f(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ and $H^1_\Sigma(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$. *(definition)* Definition 18.42.

**18.33 Euler characteristic.** $\dim H^1_f(\mathbb{Q}, V) - \dim H^1_f(\mathbb{Q}, V^D) = \dim H^0(\mathbb{Q}, V) - \dim H^0(\mathbb{Q}, V^D) + \sum_v (\dim H^1_f(\mathbb{Q}_v, V) - \dim H^0(\mathbb{Q}_v, V))$ *(proof sketched here; reference)*; for $A_f$ the right side is 0 when $\ell \nmid N$ and $\bar\rho$ is absolutely irreducible, so then $H^1_f(\mathbb{Q}, A_f) = 0 \iff H^1_f(\mathbb{Q}, V^D) = 0$ *(proved here, given the formula)*. Proposition 18.43.

**18.34 Tamagawa factor.** $\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}})$, the Fitting ideal of the invariants. *(definition)* Definition 18.45.

**18.35 Index description.** $\ell^{v_\ell(\mathrm{Tam}_p(M))} = [H^1_f(\mathbb{Q}_p, M) : H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)]$. *(proved here)* Proposition 18.47.

**18.36 Basic properties.** $\mathrm{Tam}_p(M) = \mathbb{Z}_\ell$ if $M$ is unramified or inertia acts through a finite group of order prime to $\ell$; $\mathrm{Tam}_p(\mathrm{ad}^0 T) = \mathrm{Tam}_p(\mathrm{ad}^0(T \otimes \chi))$; $\mathrm{Tam}_p(\mathrm{ad}^0 T_\ell E) = \mathbb{Z}_\ell$ at good primes and when $\rho(I_p) = \{\pm 1\}$. *(proved here)* Proposition 18.48.

**18.37 Fontaine and Perrin-Riou (Lemma lem:fpr).** If $H^0(\mathbb{Q}_p, X \otimes \mathbb{Q}_\ell) = 0$ then $0 \to X^{I_p}/(\mathrm{Frob}_p - 1) \to H^1(\mathbb{Q}_p, X)_{\mathrm{tor}} \to (H^1(I_p, X)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}} \to 0$ is exact and $\mathrm{Fitt}\, H^1_f(\mathbb{Q}_p, X) = \det(1 - \mathrm{Frob}_p^{-1} \mid X^{I_p})\, \mathrm{Tam}_p(X)$; for $X = T^D$, $\mathrm{Fitt}\, H^1_f(\mathbb{Q}_p, T^D) = L_p(A_f, 1)^{-1}\, \mathrm{Tam}_p(T^D)$ and $\mathrm{length}\, H^1_f(\mathbb{Q}_p, T^D) = \mathrm{length}\, H^0(\mathbb{Q}_p, W^D)$. *(proved here, given Chapter 17)* Theorem 18.49.

**18.38 $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$.** For the adjoint of $E$ at every $p \neq \ell$; at potentially multiplicative primes both invariant modules are $\mathbb{Z}/\ell^a \oplus \mathbb{Z}/\ell^m$, with $a = v_\ell(\nu)$ and $m = \min(a, v_\ell(p-1), v_\ell(c(u)))$. *(proved here, given Chapters 16 and 17)* Proposition 18.50.

**18.39 Tamagawa numbers.** $v_\ell(\mathrm{Tam}_p(T_\ell E)) = v_\ell(c_p)$. *(proved here, given Chapter 10)* Proposition 18.51.

**18.40 Tate data.** The Tate shape $\left(\begin{smallmatrix} \chi_\ell \varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$, $\kappa|_I = n\, t_\ell$, $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q$, $\kappa = n\kappa_q + \kappa_u$, $[\kappa_u] \leftrightarrow \bar u$. *(convention; recollection from Chapters 7, 11 and 17)* Convention 18.52.

**18.41 Lemma lem:tamfactor.** At a multiplicative prime $q \neq \ell$: $H^1(I_q, \mathrm{ad}^0 T)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2$ and

$$v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u))) \;\geq\; t_\ell(E, q).$$

*(proved here, given Chapters 11, 16 and 17)* Lemma 18.53.

**18.42 Consequences.** The excess $\min(\dots)$ is positive iff $\ell \mid n$, $q \equiv 1 \pmod{\ell}$ and $\bar u \in (\mathbb{F}_q^\times)^\ell$; it is 0 at $q = 2$ and independent of split or nonsplit. *(proved here)* Corollary 18.55.

**18.43 Potentially multiplicative additive primes.** At $q \neq \ell$, $q = 2$ included: $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(\nu) + \min(v_\ell(\nu), v_\ell(q-1), v_\ell(c(u)))$. *(proved here, given Chapters 10 and 11)* Proposition 18.57.

**18.44 Examples.** Tate data and $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T))$ for `11a1`, `14a1`, `539c1`, `791a1`, `1184f1`, `1748e1`. *(computed)* Example 18.58.

**18.45 Finite inertia image.** If $I_q$ acts through a finite $\Delta$, $H^1(I_q, M)_{\mathrm{tor}} = H^1(\Delta, M)$. *(proved here)* Lemma 18.59.

**18.46** $\mathrm{ad}^0 T$ **is free over** $\mathbb{Z}_3[C_3]$**.** For $\sigma \in \mathrm{GL}_2(\mathbb{Z}_3)$ of order 3, $\mathrm{ad}^0 T \simeq \mathbb{Z}_3[\langle\sigma\rangle]$, so $H^1(\langle\sigma\rangle, \mathrm{ad}^0 T) = 0$, while $H^1(\langle\sigma\rangle, T) \simeq \mathbb{F}_3$. *(proved here)* Proposition 18.60.

**18.47 Additive primes (Lemma lem:tamadd).** For $\ell$ odd and $q \neq \ell$ of potentially good additive reduction, $H^1(I_q, \mathrm{ad}^0 T)_{\mathrm{tor}} = 0$ and $\mathrm{Tam}_q(\mathrm{ad}^0 T) = \mathrm{Tam}_q(\mathrm{ad}^0 T(1)) = \mathbb{Z}_\ell$. *(proved here, given Chapter 11)* Theorem 18.61.

**18.48 The explicit identity (Corollary cor:explicit).** $\sum_{q|N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = \sum_{v_q(j)<0}\big(t_\ell + \min(t_\ell, v_\ell(q-1), v_\ell(c(u_q)))\big)$ *(proved here)*; hence, under the hypothesis of Theorem thm:main, $v_\ell(\deg \varphi_E) = \mathrm{length}\, H^1_f(\mathbb{Q}, W) + \sum_q e_\ell(E,q) + \sum_{v_q(j)<0}\big(t_\ell + \min(\dots)\big)$ *(given Theorem thm:main, Chapter 20)*. Corollary 18.63.

**18.49 The excess on the tables.** Below conductor 10,000 the excess is never negative and is 0 for 8,688 of 13,203, 19,019 of 23,361, 25,133 of 28,134 curves at $\ell = 3, 5, 7$. *(computed)* Remark 18.64.

**18.50 The exceptional set.** $S_f = \{\ell : \ell \mid 2N$ or $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ not absolutely irreducible$\}$. *(definition)* Definition 18.65.

**18.51 Theorem thm:dfg27.** If $\ell \notin S_f$ then $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0 = H^1_f(\mathbb{Q}, A_f)$. *(stated; reference; a derivation from Theorem thm:dfg37 sketched)* Theorem 18.66.

**18.52 Lemma lem:h0 in Selmer form.** If $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$, $H^0(\mathbb{Q}, W^D) \simeq H^1_f(\mathbb{Q}, T^D) = H^1(\mathbb{Q}, T^D)_{\mathrm{tor}}$ *(proved here)*; $H^0(\mathbb{Q}, W^D) \simeq \mathbb{Z}/\ell^{h_\ell(E)}$, $h_\ell = 0$ for $\ell \geq 5$, $h_3 \leq 1$ with equality iff the image is $N_s(\mathbb{F}_3)$ *(proved in Chapter 11)*. Proposition 18.67.

**18.53 Lemma lem:dfg21.** If $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$ and $\Sigma \neq \emptyset$, $\ell \notin \Sigma$, contains the ramified primes $p \neq \ell$:

$$0 \to H^1_f(\mathbb{Q}, W) \to H^1_\Sigma(\mathbb{Q}, W) \to \bigoplus_{p \in \Sigma \cup \{\infty\}} H^1_f(\mathbb{Q}_p, T^D)^* \to H^0(\mathbb{Q}, W^D)^* \to 0.$$

*(proved here, given Chapter 17)* Lemma 18.69.

**18.54 The length identity.** Under these hypotheses, $\mathrm{length}\, H^1_\Sigma(\mathbb{Q}, W) = \mathrm{length}\, H^1_f(\mathbb{Q}, W) + \sum_{p\in\Sigma}\big(v_\ell(L_p(A_f,1)^{-1}) + v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0 T))\big) - h_\ell(E)$. *(proved here, given Proposition 18.43 and Chapters 17 and 19)* Corollary 18.71.

**18.55 The Selmer identity.** For $\ell \notin S_f$, $H^1_f(\mathbb{Q}, W)$ and $H^1_\Sigma(\mathbb{Q}, W)$ are finite and

$$\mathrm{length}\, H^1_\Sigma(\mathbb{Q}, W) = \mathrm{length}\, H^1_f(\mathbb{Q}, W) + \sum_{q|N} v_\ell(L_q(A_f,1)^{-1}) + \sum_{v_q(j)<0}\Big(t_\ell(E,q) + \min\big(t_\ell(E,q), v_\ell(q-1), v_\ell(c(u_q))\big)\Big).$$

*(proved here, given Theorem thm:dfg27)* Theorem 18.73.

**18.56 Remark rem:ns.** For $\ell = 3$, image $N_s(\mathbb{F}_3)$, $3 \nmid N$: if $H^1_f(\mathbb{Q}, V^D) = 0$, the length identity holds with $-h_3(E) = -1$. *(proved here, given Proposition 18.43; conditional)* Proposition 18.77.

**18.57 Section sec:tam.** Below conductor 10,000 the refined inequality has no exception; equality rises from 596 to 714 of 1,050 curves at $\ell = 3$ and from 203 to 220 of 252 at $\ell = 5$. *(computed)* Example 18.80.

**18.58 The adjoint motive.** Realisations $A_f$, Hodge numbers $h^{1,-1} = h^{0,0} = h^{-1,1} = 1$, $L(\mathrm{ad}^0 f, s) = L(\mathrm{Sym}^2 f, s+1)$; $s = 1$ is critical and $L(\mathrm{ad}^0 f, 1) \neq 0$. *(definition; properties stated with reference, the nonvanishing given Chapter 19)* Definition 18.82.

**18.59 The Tamagawa number conjecture.** Schematically,

$$\frac{L(M,0)}{\Omega(M)} \sim_\ell \frac{\prod_{p\neq\ell} \#(\mathbb{Z}_\ell/\operatorname{Tam}_p(T)) \cdot \#H^1_f(\mathbb{Q}, W_M^D)}{\#H^0(\mathbb{Q}, W_M) \cdot \#H^0(\mathbb{Q}, W_M^D)}.$$

*(stated in outline; reference)* Conjecture 18.83.

**18.60 Diamond, Flach and Guo, Theorem 0.1.** For $\lambda \notin S_f$ the $\lambda$-part of the conjecture holds for the adjoint motive at $s = 1$. *(stated; reference)* Theorem 18.85.

**18.61 Diamond, Flach and Guo, Theorem 0.2 (thm:dfg37).** For $\ell \notin S_f$, $H^1_\Sigma(\mathbb{Q}, W)$ is finite of length $v_\lambda(\eta^\Sigma_{f,\lambda})$. *(stated; reference; see Chapter 20)* Theorem 18.86.

**18.62 Flach.** If the conductor of $E$ is minimal among its quadratic twists, $\ell \geq 5$ is a prime of good reduction and $\bar\rho_{E,\ell}$ is surjective, then the degree of any modular parametrisation $X_0(N) \to E$ annihilates $H^1_f(\mathbb{Q}, W^D)$ ($W^D \simeq \mathrm{Sym}^2 T_\ell E \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$) and $H^1_f(\mathbb{Q}, W)$; both are finite. *(stated; reference)* Theorem 18.87.

## Where the paper uses this chapter

- **Section 1.2** (the definition of $\operatorname{Tam}_p(M)$; "at a prime of good reduction it is 0"; the Bloch-Kato Selmer group): Definition 18.45, Proposition 18.48, Definition 18.36, Definition 18.42.
- **Section 1.1** (the geometric Frobenius in $L_p(A_f, s)$): Proposition 18.15, Proposition 18.17.
- **Section 4.2** ($A$, $W$, $T^D \simeq A(1)$ via the trace form, $M_{f,\lambda}$ and its adjoint, $H^1_f$, $H^1_\Sigma$, $S_f$): Proposition 18.10, Definition 18.39, Definition 18.65.
- **Theorem thm:dfg27**: Theorem 18.66, Proposition 18.43, Proposition 18.14.
- **Lemma lem:dfg21 and the paragraph after it**: Lemma 18.69, Proposition 18.32, Proposition 18.34, Lemma 18.18, Corollary 18.20.
- **Lemma lem:fpr**: Theorem 18.49, Proposition 18.50, Lemma 18.21, Proposition 18.15.
- **Lemma lem:h0** (as it is used in Selmer groups): Proposition 18.67.
- **Lemma lem:tamfactor** (all of it, including the additive potentially multiplicative case and $q = 2$): Convention 18.52, Lemma 18.53, Proposition 18.57, Proposition 18.51.
- **Lemma lem:tamadd and Corollary cor:explicit**: Lemma 18.59, Proposition 18.60, Theorem 18.61, Corollary 18.63.

- **The proof of Theorem thm:main** (the Selmer bookkeeping from Theorem thm:dfg27 to the local terms; the potentially multiplicative primes): Corollary 18.71, Theorem 18.73, Remark 18.74.
- **Remark rem:ko(1)** and the abstract's sentence on the Tamagawa factor: Corollary 18.55.
- **Remark rem:ns, Section 1.3, Conjecture conj:euler** (the term $h_\ell(E)$), **Open question 2**: Proposition 18.77, Remark 18.78, Remark 18.79.
- **Section sec:tam** and **Section sec:sharp** (the identity (eq:identity) "says what the excess is"): Example 18.80, Remark 18.81, Example 18.76.
- **Open question 3**: Theorem 18.61, Remark 18.64.
- **Section 1** (the Tamagawa number conjecture of Diamond, Flach and Guo; Flach's theorem; the Petersson norm): Theorem 18.85, Theorem 18.87, Remark 18.89.

## 18.1 Overview and conventions

The paper's Theorem thm:main is an identity between $v_\ell(\deg \varphi_E)$ and three kinds of terms: the length of a Bloch-Kato Selmer group, Tamagawa factors, and omitted Euler factors. This chapter owns the first two and the bookkeeping that joins them to the third. It defines the objects of the paper's Section 1.2 ($\ell$-adic representations, lattices, the three duals, the Bloch-Kato local conditions, the Selmer groups $H^1_f$ and $H^1_\Sigma$, the Tamagawa factor $\mathrm{Tam}_p$); proves Lemma lem:fpr at every $p \neq \ell$, Lemma lem:tamfactor in every step, and Lemma lem:tamadd; specialises the abstract comparison of Selmer structures of Chapter 17 to the paper's Lemma lem:dfg21; and proves the Selmer half of Theorem thm:main (Theorem 18.73), which Chapter 20 combines with the congruence ideals. It ends with Remark rem:ns, Section sec:tam, Open questions 2 and 3, and the Tamagawa number conjecture of Bloch and Kato, of which (eq:identity) is a reading through the modular degree. The group cohomology is Chapter 16's, the Galois cohomology (Tate local duality, the Euler characteristic formulas, unramified classes, Poitou-Tate) is Chapter 17's, and the local representations of $E$ are Chapter 11's; this chapter cites them and does not prove them again. Two facts are imported from later chapters and are used only in remarks and examples: the values $e_\ell(E,q)$ (Chapter 19) and the congruence-ideal theorems of Diamond, Flach and Guo (Chapter 20).

**Convention 18.1** (notation of this chapter)**.** (a) *Objects.* $E/\mathbb{Q}$ is given by a global minimal model and has conductor $N$; $\ell$ is an odd prime unless said otherwise; $p$ is any prime and $q$ usually a bad prime of $E$. $T = T_\ell E$, $V = T \otimes \mathbb{Q}_\ell$, $\rho = \rho_{E,\ell}$, $\bar\rho$ its reduction. The paper's objects are $A = \mathrm{ad}^0 T$ (trace zero), $A_f = \mathrm{ad}^0 V = A \otimes \mathbb{Q}_\ell$, $W = A \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and their Kummer duals $T^D$, $V^D$, $W^D$ (item (b)); $\Sigma$ is the set of primes dividing $N$, and $\delta_p = \dim V^{I_p}$. In the general statements of Sections 18.2 to 18.6 the letters $V$, $T$, $W = V/T$ denote an arbitrary $\ell$-adic representation, a stable lattice and the quotient; when such a statement is applied to the paper, the triple is $(A_f, A, W)$, with Kummer dual $(V^D, T^D, W^D)$. This is the paper's own usage (Section 4.2), and every application below names the triple.

(b) *Three duals* (Appendix B, convention B-C4). For a $\mathbb{Z}_\ell$-module $M$: the *linear dual* $M^\vee = \mathrm{Hom}_{\mathbb{Z}_\ell}(M, \mathbb{Z}_\ell)$; the *Kummer* (or Cartier) *dual* $M^D = \mathrm{Hom}_{\mathbb{Z}_\ell}(M, \mathbb{Z}_\ell(1)) = M^\vee(1)$; for a compact or discrete $\ell$-primary module $X$, the *Pontryagin dual* $X^* = \mathrm{Hom}_{\mathrm{cont}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$.

The symbol * never means a linear dual here. For the paper's objects $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1))$, not $\mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell(1))$.

(c) *Frobenius* (B-C2, B-C3). $\mathrm{Frob}_p$ is an *arithmetic* Frobenius at $p$, and $\mathrm{Frob}_p^{-1}$ the *geometric* one. Group theory and cohomology use $\mathrm{Frob}_p$: conjugation on $H^1(I_p, -)$, unramified classes $M^{I_p}/(\mathrm{Frob}_p - 1)M^{I_p}$, the Tate matrices of Lemma lem:tamfactor, the tame relation $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^q$. Local $L$-factors use the geometric Frobenius: for $p \neq \ell$ and a representation $M$ of $G_{\mathbb{Q}_p}$ on a finite-dimensional $\mathbb{Q}_\ell$-space,

$$L_p(M, s) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid M^{I_p})^{-1}.$$

We do not write $\Phi_p$ for the geometric Frobenius, because component groups occur in this chapter; $c_p = [E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)]$ is the Tamagawa number and $\Phi_p^{\mathrm{geom}}$ the geometric component group.

(d) *Twists and Hodge-Tate weights* (B-C5). $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^k}$, $M(n) = M \otimes \mathbb{Z}_\ell(1)^{\otimes n}$. The Hodge-Tate weight of $\mathbb{Q}_\ell(1)$ is $+1$; so $V_\ell E$ has weights $\{0, 1\}$, $A_f$ has $\{-1, 0, 1\}$ and $V^D = A_f(1)$ has $\{0, 1, 2\}$. Every formula below that depends on this sign says so.

(e) *Cohomology* (Chapter 16). Continuous cochain cohomology; $H^i(K, M) = H^i(G_K, M)$, $H^i(\mathbb{R}, M) = H^i(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), M)$; $G_{\mathbb{Q},S}$ is the Galois group of the maximal extension of $\mathbb{Q}$ unramified outside a finite set $S$ of places containing $\ell$ and $\infty$. "Length" is $\mathbb{Z}_\ell$-length; $\mathrm{Fitt}_{\mathbb{Z}_\ell}(X) = \ell^{\mathrm{length}\, X}\mathbb{Z}_\ell$ for a finite $\mathbb{Z}_\ell$-module $X$, and $v_\ell(\ell^a\mathbb{Z}_\ell) = a$ (B-C10).

(f) *Tate data* (B-C14). At a prime $q \neq \ell$ with $v_q(j_E) < 0$: $q_E \in \mathbb{Q}_q^\times$ is the Tate parameter (never $u$), $\nu = v_q(q_E) = -v_q(j_E)$ (equal to $n = n_q = v_q(\Delta_E)$ at a multiplicative prime), $u = q_E q^{-\nu} \in \mathbb{Z}_q^\times$ its unit part, $\bar{u} \in \mathbb{F}_q^\times$ its reduction, $c(u)$ the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$, and $v_\ell(c(u))$ the largest $k \leq v_\ell(q-1)$ with $\bar{u} \in (\mathbb{F}_q^\times)^{\ell^k}$ (so the paper's "$v_\ell(0)$" reads $v_\ell(q-1)$). The paper writes $c$ for this class in Lemma lem:tamfactor and $c(u)$ in Section sec:tam; $c_q$ is always the Tamagawa number. $\varepsilon$ is the unramified character of $G_{\mathbb{Q}_q}$ with $\varepsilon(\mathrm{Frob}_q) = a_q(E)$ at a multiplicative prime.

(g) *Characters* (B-C6). $\chi_\ell$ is the $\ell$-adic cyclotomic character and $\bar{\chi}_\ell$ its reduction; $\chi_d$ is the quadratic character of $\mathbb{Q}(\sqrt{d})$ for a fundamental discriminant $d$ ($\chi_{-3}$, $\chi_{-4}$, and $\chi_{q^*}$ of conductor $q$); $\bar{\chi}_3 = \chi_{-3}$.

*Warning* 18.2 (three duals). $M^\vee = \mathrm{Hom}(M, \mathbb{Z}_\ell)$ is linear (Bloch-Kato and Fontaine-Perrin-Riou write $V^*$ for it); $M^D = \mathrm{Hom}(M, \mathbb{Z}_\ell(1))$ is the paper's $^D$; $X^* = \mathrm{Hom}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is the paper's $^*$. Tate local duality pairs $H^1(\mathbb{Q}_p, W)$ with $H^1(\mathbb{Q}_p, T^D)$, and the only Pontryagin duals in Lemma lem:dfg21 are duals of finite groups, which have the same length (Chapter 3).

*Warning* 18.3 (two Frobenii). In cohomology the choice of Frobenius is invisible up to $\ell$-adic units: $(\mathrm{Frob} - 1)X = (\mathrm{Frob}^{-1} - 1)X$ for any Frob-stable $X$, and $\det(1 - \mathrm{Frob} \mid X)$, $\det(1 - \mathrm{Frob}^{-1} \mid X)$ differ by the unit $\det(-\mathrm{Frob}^{-1} \mid X)$ on a lattice $X$. As rational numbers they differ: at a multiplicative prime $\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = 1 - p^{-2}$, while $\det(1 - \mathrm{Frob}_p \mid (V^D)^{I_p}) = 1 - p^2$. Only the first is $L_p(A_f, 1)^{-1}$.

The paper's sentences and the results of this chapter that prove them are matched in the following table; the section *Where the paper uses this chapter* at the head of the chapter lists the same correspondence by result.

| paper | this chapter |
|---|---|
| Section 1.2: the definition of $\mathrm{Tam}_p(M)$, "at a prime of good reduction it is 0" | Definition 18.45, Proposition 18.48 |
| Section 1.2, Section 4.2: the Bloch-Kato Selmer group, $H^1_\Sigma$ | Definition 18.36, Definition 18.39, Definition 18.42 |
| Section 4.2: $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$; $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T$ | Proposition 18.10 |
| Theorem thm:dfg27 | Theorem 18.66 |
| Lemma lem:dfg21 and the paragraph after it | Lemma 18.69, Proposition 18.32, Lemma 18.18 |
| Lemma lem:fpr | Theorem 18.49, Proposition 18.50, Lemma 18.21 |
| Lemma lem:h0 (Selmer form) | Proposition 18.67 |
| Lemma lem:tamfactor | Lemma 18.53, Proposition 18.57 |
| Lemma lem:tamadd, Corollary cor:explicit | Theorem 18.61, Corollary 18.63 |
| proof of Theorem thm:main (the Selmer bookkeeping) | Corollary 18.71, Theorem 18.73 |
| Remark rem:ko(1) | Corollary 18.55 |
| Remark rem:ns, Open question 2, the term $h_\ell(E)$ | Proposition 18.77, Remark 18.78, Remark 18.79 |
| Section sec:tam | Example 18.80, Remark 18.81 |
| Open question 3 | Theorem 18.61, Remark 18.64 |

**How the pieces fit.** The Selmer half of Theorem thm:main is a chain of four length computations, and the chapter is organised along it.

(1) Lemma lem:dfg21 (Lemma 18.69) is an exact sequence of finite groups comparing $H^1_\Sigma(\mathbb{Q}, W)$ with $H^1_f(\mathbb{Q}, W)$. Lengths are additive along exact sequences of finite $\mathbb{Z}_\ell$-modules, and a finite group and its Pontryagin dual have the same length, so it gives

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma} \operatorname{length} H^1_f(\mathbb{Q}_p, T^D) - \operatorname{length} H^0(\mathbb{Q}, W^D).$$

(2) Lemma lem:fpr (Theorem 18.49) computes each local length: it is the sum $v_\ell(L_p(A_f, 1)^{-1}) + v_\ell(\mathrm{Tam}_p(T^D))$. Then Proposition 18.50 replaces $T^D$ by $A$ in the Tamagawa factor.

(3) Lemmas lem:tamfactor and lem:tamadd (Lemma 18.53, Proposition 18.57, Theorem 18.61) compute $v_\ell(\mathrm{Tam}_q(A))$ prime by prime from the reduction type and the Tate parameter.

(4) Lemma lem:h0 (Proposition 18.67) gives $\operatorname{length} H^0(\mathbb{Q}, W^D) = h_\ell(E)$, which is 0 when $\ell \notin S_f$.

Before any of this can be stated one needs the coefficient modules and their duals (Section 18.2), their local behaviour at $p \neq \ell$ (Section 18.3) and at $\ell$ (Section 18.4), the local conditions (Section 18.5) and the global Selmer groups (Section 18.6). Steps (2) and (3) are Sections 18.7 to 18.9; steps (1) and (4) are Section 18.10; the chain is closed in Section 18.11. The last three sections look beyond the hypothesis, at the numerical test of Section sec:tam, and at the conjecture of Bloch and Kato.

## 18.2 $\ell$-adic representations, lattices, twists and duals

This section sets up the coefficient modules of all the cohomology in the paper: an $\ell$-adic representation $V$, a stable lattice $T$, the discrete module $W = V/T$, their Tate twists and their three duals, and, for the adjoint, the identification $T^D \simeq A(1)$ that the paper makes in Section 4.2 "via the trace form". It ends with the global invariants of $A_f$, $W$ and $V^D$, which enter Theorem thm:dfg27 and Lemma lem:h0.

**Definition 18.4** ($\ell$-adic representations)**.** Let $K$ be a field with absolute Galois group $G_K$. An *$\ell$-adic representation* of $G_K$ is a finite-dimensional $\mathbb{Q}_\ell$-vector space $V$ with a continuous homomorphism $\rho_V\colon G_K \to \mathrm{GL}(V)$. A *lattice* in $V$ is a finitely generated $\mathbb{Z}_\ell$-submodule $T$ that spans $V$ (so $T \simeq \mathbb{Z}_\ell^{\dim V}$); it is *stable* if $\rho_V(G_K)T = T$. Given a stable lattice we put $W = V/T = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, a discrete $\ell$-primary $G_K$-module.

Two examples to keep in mind. For the trivial representation, $V = \mathbb{Q}_\ell$, $T = \mathbb{Z}_\ell$ and $W = \mathbb{Q}_\ell/\mathbb{Z}_\ell$. For an elliptic curve, $T = T_\ell E = \varprojlim_k E[\ell^k]$ is a stable lattice in $V = V_\ell E$, and $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell = \varinjlim_k T/\ell^k T = \varinjlim_k E[\ell^k] = E[\ell^\infty]$ (the transition maps of the direct limit being the inclusions $E[\ell^k] \subseteq E[\ell^{k+1}]$, which correspond to multiplication by $\ell$ on $T/\ell^k T \to T/\ell^{k+1}T$). The paper's lattice is $A = \mathrm{ad}^0 T_\ell E$, of rank 3. The three objects play different parts: $V$ is where dimensions are counted and linear algebra is done; $T$ carries the integral information (Fitting ideals, indices of lattices); and $W$ is discrete and torsion, so that its cohomology groups can be finite and have lengths. Every quantity in Theorem thm:main is the length of such a group.

**Proposition 18.5** (lattices)**.** *Let $V$ be an $\ell$-adic representation of $G_K$.*

*(i) $V$ has a stable lattice.*

*(ii) For a stable lattice $T$ the sequence $0 \to T \to V \to W \to 0$ is exact, $W \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^{\dim V}$ as a group, and $W[\ell^k] = \ell^{-k}T/T \simeq T/\ell^k T$ as $G_K$-modules; the projection $V \to W$ has a continuous set-theoretic section.*

*(iii) Any two lattices $T, T'$ are commensurable: $\ell^a T \subseteq T' \subseteq \ell^{-a}T$ for some $a \geq 0$. If $T$ is stable and $T/\ell T$ is an irreducible $\mathbb{F}_\ell[G_K]$-module, every stable lattice is $\ell^b T$ for some $b \in \mathbb{Z}$.*

*Proof.* (i) Let $T_0$ be any lattice. $\mathrm{GL}(T_0)$ is open in $\mathrm{GL}(V)$, so $H = \rho_V^{-1}(\mathrm{GL}(T_0))$ is an open subgroup of the compact group $G_K$ and has finite index. If $g_1, \dots, g_r$ are coset representatives of $G_K/H$, then $T = \sum_i \rho_V(g_i)T_0$ is a finite sum of lattices, hence a lattice, and it is stable because $gg_iH = g_{\pi(i)}H$ for a permutation $\pi$. (ii) Exactness is the definition of $W$; $T \simeq \mathbb{Z}_\ell^d$ gives $W \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^d$, and $x \mapsto \ell^{-k}x$ identifies $T/\ell^k T$ with $W[\ell^k] = \ell^{-k}T/T$ equivariantly. A section is given by choosing, in a basis of $T$, the representatives of $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ of the form $\sum_{i<0} a_i \ell^i$ with $0 \leq a_i < \ell$; it is continuous because $W$ is discrete (see Chapter 16 for the use of such sections in the long exact sequence of continuous cohomology). (iii) Choose a $\mathbb{Z}_\ell$-basis of $T$. Each of the finitely many generators of $T'$ has coordinates in $\mathbb{Q}_\ell$ in this basis; if $\ell^{-a}$ bounds their denominators, then $T' \subseteq \ell^{-a}T$. Since $T'$ spans $V$, the same argument with the roles exchanged gives $T \subseteq \ell^{-a'}T'$, and the larger of $a, a'$ works for both inclusions. If $T'$ is stable, replace it by $\ell^b T'$ with $b$ minimal such that $\ell^b T' \subseteq T$; such a $b$ exists, because the inclusion holds for $b \geq a$ and fails for $b$ very negative ($T' \neq 0$ and $\bigcap_k \ell^k T = 0$). Then $\ell^b T' \not\subseteq \ell T$, so its image in $T/\ell T$ is a nonzero stable subspace, hence all of $T/\ell T$, and $\ell^b T' + \ell T = T$ gives $\ell^b T' = T$ by Nakayama's lemma (Chapter 3). $\square$

The irreducibility hypothesis in (iii) cannot be dropped. If $G_K$ acts trivially on $V = \mathbb{Q}_\ell^2$, then $T = \mathbb{Z}_\ell^2$ and $T' = \mathbb{Z}_\ell \oplus \ell\mathbb{Z}_\ell$ are both stable, and $T'$ is not $\ell^b T$ for any $b$ (it has index $\ell$ in $T$, while a lattice $\ell^b T \subseteq T$ has index $\ell^{2b}$); here $T/\ell T$ is the trivial two-dimensional module, which is reducible. An arithmetic instance: the curves `11a1` and `11a2` are linked by an isogeny of degree 5 with cyclic kernel, which identifies $V_5$ of the two curves and maps one Tate module onto a stable sublattice of index 5 in the other, so the two lattices are not homothetic; correspondingly $E[5]$ is reducible for these curves. This is one reason why the paper needs $\bar\rho$ irreducible: only then is $T$, up to scaling, determined by $V$.

**Definition 18.6** (Tate twists)**.** $\mathbb{Z}_\ell(1) = \varprojlim_k \mu_{\ell^k}$ (transition maps $\zeta \mapsto \zeta^\ell$), on which $G_K$ acts through the cyclotomic character $\chi_\ell$; $\mathbb{Q}_\ell(1) = \mathbb{Z}_\ell(1) \otimes \mathbb{Q}_\ell$; $\mathbb{Z}_\ell(n) = \mathbb{Z}_\ell(1)^{\otimes n}$ for $n \geq 0$ and $\mathbb{Z}_\ell(-1) = \mathrm{Hom}(\mathbb{Z}_\ell(1), \mathbb{Z}_\ell)$, $\mathbb{Z}_\ell(n) = \mathbb{Z}_\ell(-1)^{\otimes(-n)}$ for $n < 0$. For a $\mathbb{Z}_\ell[G_K]$-module $M$, $M(n) = M \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(n)$, so $g$ acts on $M(n)$ as $\chi_\ell(g)^n$ times its action on $M$.

**Proposition 18.7** (twists)**.** *(i)* $M(n)(m) = M(n+m)$ *and* $\mathrm{Hom}(M(n), N) \simeq \mathrm{Hom}(M, N)(-n)$*, equivariantly.*

*(ii) Let* $K = \mathbb{Q}_p$ *with* $p \neq \ell$*. The inertia group* $I_p$ *acts trivially on* $\mathbb{Z}_\ell(1)$ *and* $\mathrm{Frob}_p$ *acts on it by* $p$*. Hence* $M(n)^{I_p} = M^{I_p}(n)$*, and the eigenvalues of* $\mathrm{Frob}_p$ *(resp.* $\mathrm{Frob}_p^{-1}$*) on* $M(n)^{I_p} \otimes \mathbb{Q}_\ell$ *are those on* $M^{I_p} \otimes \mathbb{Q}_\ell$ *multiplied by* $p^n$ *(resp.* $p^{-n}$*).*

*(iii) The Weil pairing gives* $\wedge^2 T_\ell E \simeq \mathbb{Z}_\ell(1)$*, hence* $\det \rho = \chi_\ell$*,* $(T_\ell E)^\vee \simeq T_\ell E(-1)$ *and* $(T_\ell E)^D \simeq T_\ell E$*.*

*Proof.* (i) Both are the formulas for $\chi_\ell^n \chi_\ell^m = \chi_\ell^{n+m}$ and for $(g\phi)(x) = g\phi(g^{-1}x)$. (ii) $\mathbb{Q}_p(\mu_{\ell^\infty})/\mathbb{Q}_p$ is unramified for $p \neq \ell$, and the arithmetic Frobenius raises roots of unity of order prime to $p$ to the $p$-th power, so $\chi_\ell(\mathrm{Frob}_p) = p$ (Chapter 7). As $I_p$ acts trivially on $\mathbb{Z}_\ell(n)$, $M(n)^{I_p} = M^{I_p} \otimes \mathbb{Z}_\ell(n)$, and $\mathrm{Frob}_p$ acts as $p^n$ times its action on $M^{I_p}$. (iii) The Weil pairing $T \times T \to \mathbb{Z}_\ell(1)$ is perfect, alternating and equivariant (Chapter 9); it identifies $\wedge^2 T$ with $\mathbb{Z}_\ell(1)$, so $\det\rho = \chi_\ell$, and $x \mapsto e(x, \cdot)$ is an isomorphism $T \to \mathrm{Hom}(T, \mathbb{Z}_\ell(1)) = T^D = T^\vee(1)$; twisting by $-1$ gives $T^\vee \simeq T(-1)$. $\square$

Two instances of (ii) and (iii). $\mathbb{Q}_\ell(1)$ is unramified at $p \neq \ell$ with $\mathrm{Frob}_p$ acting by $p$, so with the geometric Frobenius $\det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid \mathbb{Q}_\ell(1))^{-1} = (1 - p^{-(s+1)})^{-1}$, the Euler factor of $\zeta(s+1)$. And at a good prime $p \neq \ell$, where $\mathrm{Frob}_p$ has eigenvalues $\alpha, \beta$ on $V_\ell E$ with $\alpha\beta = p$, (iii) says that the eigenvalues on $(V_\ell E)^\vee \simeq V_\ell E(-1)$ are $\alpha/p = \beta^{-1}$ and $\beta/p = \alpha^{-1}$; this agrees with the fact that dualising inverts eigenvalues. A twist thus multiplies every Frobenius eigenvalue at $p$ by the same power of $p$ and leaves the action of inertia unchanged; it moves weights without changing the local shape.

**Definition 18.8** (Kummer duals)**.** For a lattice $T$ in an $\ell$-adic representation $V$ put $V^D = \mathrm{Hom}_{\mathbb{Q}_\ell}(V, \mathbb{Q}_\ell(1)) = V^\vee(1)$, $T^D = \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathbb{Z}_\ell(1)) = T^\vee(1)$, a stable lattice in $V^D$, and $W^D = V^D/T^D$. For a compact or discrete $\ell$-primary module $X$, $X^* = \mathrm{Hom}_{\mathrm{cont}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is its Pontryagin dual.

**Proposition 18.9** (Kummer duality)**.** *Let* $T$ *be a stable lattice in* $V$ *and* $W = V/T$*.*

*(i)* $W^D \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(T, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) = \mathrm{Hom}(T, \mu_{\ell^\infty})$ *and* $T^D \simeq \mathrm{Hom}_{\mathrm{cont}}(W, \mu_{\ell^\infty})$*; the evaluation pairings* $T \times W^D \to \mu_{\ell^\infty}$ *and* $W \times T^D \to \mu_{\ell^\infty}$ *are perfect and* $G_K$*-equivariant.*

*(ii)* $(T^D)^D = T$*,* $(V^D)^D = V$ *and* $(W^D)^D = W$ *(the last in the sense* $V^{DD}/T^{DD}$*).*

*(iii) For a finite $\mathbb{Z}_\ell$-module $X$, $X^*$ is finite of the same length as $X$.*

*(iv) Bloch and Kato and Fontaine and Perrin-Riou write $V^*(1)$ for $V^D$; Diamond, Flach and Guo write $T^D$, $V^D$, $W^D$ exactly as the paper.*

*Proof.* (i) $T$ is free, so $\mathrm{Hom}(T,-)$ is exact: applying it to $0 \to \mathbb{Z}_\ell(1) \to \mathbb{Q}_\ell(1) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell(1) \to 0$ gives $W^D = \mathrm{Hom}(T,\mathbb{Q}_\ell(1))/\mathrm{Hom}(T,\mathbb{Z}_\ell(1)) \simeq \mathrm{Hom}(T,\mathbb{Q}_\ell/\mathbb{Z}_\ell(1))$. Next, $\mathrm{Hom}_{\mathrm{cont}}(T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell, \mu_{\ell^\infty}) = \mathrm{Hom}(T, \mathrm{Hom}(\mathbb{Q}_\ell/\mathbb{Z}_\ell, \mu_{\ell^\infty})) = \mathrm{Hom}(T, \mathbb{Z}_\ell(1))$, as $\mathrm{Hom}(\mathbb{Q}_\ell/\mathbb{Z}_\ell, \mathbb{Q}_\ell/\mathbb{Z}_\ell) = \mathbb{Z}_\ell$. Perfectness is Pontryagin duality between $T$ and $T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (Chapter 3), twisted by $\mathbb{Z}_\ell(1)$; equivariance holds because all the maps are built from evaluation. The general statement is in Chapter 17. (ii) $\mathrm{Hom}(\mathrm{Hom}(T,\mathbb{Z}_\ell(1)),\mathbb{Z}_\ell(1)) = T$ for $T$ free of finite rank. (iii) is Chapter 3. (iv) is a matter of notation [14, Section 3], [69], [49, Section 2.1]. □

Examples. For $T = \mathbb{Z}_\ell$ with trivial action, $T^D = \mathbb{Z}_\ell(1)$ and $W^D = \mu_{\ell^\infty}$; for $T = \mathbb{Z}_\ell(1)$, $T^D = \mathbb{Z}_\ell$ and $W^D = \mathbb{Q}_\ell/\mathbb{Z}_\ell$. For $T = T_\ell E$, the Weil pairing gives $T^D \simeq T$ and $W^D \simeq E[\ell^\infty]$ (Proposition 18.7(iii)). The reason for twisting the linear dual by $\mathbb{Z}_\ell(1)$ is Tate local duality: the cup product pairs $H^1(\mathbb{Q}_p, W)$ with $H^1(\mathbb{Q}_p, T^D)$ into $H^2(\mathbb{Q}_p, \mu_{\ell^\infty}) \simeq \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and $\mu_{\ell^\infty}$, not $\mathbb{Q}_\ell/\mathbb{Z}_\ell$, is the module whose $H^2$ is canonically $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ (Chapter 17). So every local or global duality statement below pairs a module with its Kummer dual.

**Proposition 18.10** (the adjoint is self-dual)**.** *Let $T$ be a free $\mathbb{Z}_\ell$-module of rank 2 with a continuous action of a group $G$, $A = \mathrm{ad}^0 T$ the trace-zero endomorphisms with $g \cdot x = \rho(g) x \rho(g)^{-1}$.*

*(i) If $\ell$ is odd, the trace form $(x,y) \mapsto \mathrm{tr}(xy)$ is a perfect $G$-invariant pairing on $A$ (Gram matrix with determinant $-2$ in the basis $e, h, f$), so $A \simeq A^\vee$ and*

$$T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1), \qquad V^D \simeq A_f(1), \qquad W^D \simeq \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell.$$

*For $\ell = 2$ the image of $A$ in $A^\vee$ has index 2.*

*(ii) $\mathrm{ad}^0(T \otimes \chi) = \mathrm{ad}^0 T$ for a character $\chi\colon G \to \mathbb{Z}_\ell^\times$, $\mathrm{ad}^0(\ell^b T) = \mathrm{ad}^0 T$ inside $\mathrm{ad}^0 V$, and $x \mapsto x^\vee$ (the dual map) is a $G$-isomorphism $\mathrm{ad}^0 T \to \mathrm{ad}^0(T^\vee)$. In particular $\mathrm{ad}^0(T(n)) = \mathrm{ad}^0 T$ for $n \in \mathbb{Z}$.*

*(iii) Consequently: $\mathrm{ad}^0 M_{f,\lambda} \simeq \mathrm{ad}^0 T_\ell E$ for the lattice $M_{f,\lambda} = \varphi_E^* H^1(E, \mathbb{Z}_\ell) \simeq (T_\ell E)^\vee \simeq T_\ell E(-1)$ of Diamond, Flach and Guo; and for a quadratic twist $E'$ of $E$ (or any twist of $E$ over $\mathbb{Q}_q$ by a character of $G_{\mathbb{Q}_q}$) $\mathrm{ad}^0 T_\ell E' \simeq \mathrm{ad}^0 T_\ell E$ as $G_{\mathbb{Q}_q}$-modules.*

*Proof.* (i) is Chapter 4 (the trace form and $\mathrm{Hom}(A, \mathbb{Z}_\ell(\chi)) \simeq A(\chi)$ for $\ell$ odd); the Gram matrix in the basis $e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ is

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

of determinant $-2$, a unit of $\mathbb{Z}_\ell$ exactly when $\ell$ is odd, and its cokernel has order 2 for $\ell = 2$. (The entries: $ef = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $h^2 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ have traces 1 and 2, while $e^2 = f^2 = 0$, $eh = -e$ and $fh = f$ have trace 0; and $\mathrm{tr}(xy) = \mathrm{tr}(yx)$.) The form is $G$-invariant because $\mathrm{tr}(gxg^{-1} \cdot gyg^{-1}) = \mathrm{tr}(xy)$, so $x \mapsto \mathrm{tr}(x\,\cdot)$ is a $G$-equivariant map $A \to A^\vee$, an isomorphism when the Gram determinant is a unit. Twisting the isomorphism $A \to A^\vee$ by $\mathbb{Z}_\ell(1)$ gives $A(1) \simeq A^\vee(1) = T^D$; tensoring

with $\mathbb{Q}_\ell$ and $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ gives the other two. (ii) $\chi(g)$ is a scalar, so conjugation by $\chi(g)\rho(g)$ equals conjugation by $\rho(g)$; $\mathrm{End}(\ell^b T) = \mathrm{End}(T)$ inside $\mathrm{End}(V)$; and for $x \in \mathrm{End}(T)$ the dual map $x^\vee \in \mathrm{End}(T^\vee)$ satisfies $(\rho(g)x\rho(g)^{-1})^\vee = \rho^\vee(g)x^\vee\rho^\vee(g)^{-1}$ with $\rho^\vee(g) = (\rho(g)^{-1})^\vee$, and $\mathrm{tr}\, x^\vee = \mathrm{tr}\, x$ (Chapter 4, the invariance of the adjoint). (iii) The identification $M_{f,\lambda} = \varphi_E^* H^1(E, \mathbb{Z}_\ell)$ is the paper's Proposition prop:pullback (Chapter 14), and $H^1(E, \mathbb{Z}_\ell) = (T_\ell E)^\vee \simeq T_\ell E(-1)$ by Proposition 18.7(iii); apply (ii) twice. A twist $\rho \otimes \chi$ has the same adjoint by (ii). □

*In the paper* 18.11. Section 4.2 writes "$T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ (via the trace form)": this is (i), and it is one of the places where $\ell$ odd is used. In the same paragraph the paper says that, by Proposition prop:pullback, $\varphi_E^*$ identifies the lattice $M_{f,\lambda}$ of Diamond, Flach and Guo with $H^1(E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$, so that its adjoint is $\mathrm{ad}^0 T$: this is (iii). The lattices $M_{f,\lambda}$ and $T_\ell E$ are not homothetic (one is the dual of the other), but their adjoints agree, which is all that is used. The step at the primes of type $\mathrm{I}_n^*$ in the proof of Theorem thm:main and the statement of Lemma lem:tamfactor at additive potentially multiplicative primes use the last assertion of (iii).

**Definition 18.12** (geometric representations)**.** An $\ell$-adic representation of $G_\mathbb{Q}$ is *geometric* (Fontaine and Mazur) if it is unramified outside a finite set of primes and its restriction to $G_{\mathbb{Q}_\ell}$ is de Rham (Definition 18.23).

**Proposition 18.13** (geometricity and purity)**.** *(i) $V_\ell E$, $\mathbb{Q}_\ell(n)$, and every representation obtained from them by direct sums, tensor products, duals and subquotients ($A_f$, $V^D = A_f(1)$, $\mathrm{Sym}^2 V$) are geometric.*

*(ii) For $p \nmid N\ell$ the eigenvalues of $\mathrm{Frob}_p$ on $V_\ell E$ have complex absolute value $p^{1/2}$, those on $A_f$ absolute value 1, and those on $V^D = A_f(1)$ absolute value $p$ (and $p^{-1}$ for $\mathrm{Frob}_p^{-1}$).*

*Proof.* (i) $V_\ell E$ is unramified outside $N\ell$ (Chapter 11) and de Rham at $\ell$ (a theorem of Faltings; for good reduction at $\ell$ it is crystalline, Theorem 18.24); $\mathbb{Q}_\ell(n)$ is unramified outside $\ell$ and crystalline; and de Rham representations are stable under the listed operations (Proposition 18.25 and [11]). We do not prove the theorem of Faltings; see [11]. (ii) The eigenvalues of $\mathrm{Frob}_p$ on $V_\ell E$ are the roots $\alpha, \beta$ of $X^2 - a_p X + p$, and $|\alpha| = |\beta| = p^{1/2}$ by Hasse's bound (Chapter 9). On $A_f$ they are $\alpha/\beta, 1, \beta/\alpha$ (Chapter 11), of absolute value 1, and Proposition 18.7(ii) multiplies them by $p$ on $A_f(1)$. □

**Proposition 18.14** (global invariants)**.** *Let $\ell$ be odd.*

*(i) If $\bar\rho$ is absolutely irreducible, $H^0(\mathbb{Q}, A_f) = 0$ and $H^0(\mathbb{Q}, W) = 0$.*

*(ii) For every $E$, $H^0(\mathbb{Q}, V^D) = 0$.*

*(iii) If $\bar\rho$ is irreducible, $H^0(\mathbb{Q}, W^D)$ is finite and cyclic, of length $h_\ell(E)$; $h_\ell(E) = 0$ for $\ell \geq 5$ and $h_3(E) \leq 1$ (Proposition 18.67).*

*(iv) The $G_\mathbb{Q}$-modules $A_f$, $A_f(1) = V^D$ and $A_f(-1)$ are pairwise non-isomorphic.*

*Proof.* (i) $W[\ell] = A/\ell A = \mathrm{ad}^0(T/\ell T)$ (trace-zero matrices modulo $\ell$, as $\ell$ is odd). By Schur's lemma an endomorphism of $T/\ell T$ commuting with the absolutely irreducible $\bar\rho(G_\mathbb{Q})$ is a scalar $a$, and its trace $2a$ vanishes only for $a = 0$. So $H^0(\mathbb{Q}, W)[\ell] = 0$, and $H^0(\mathbb{Q}, W) = 0$ because a nonzero element of an $\ell$-primary group has a multiple of order $\ell$. If $0 \neq x \in H^0(\mathbb{Q}, A_f)$, scale it into $A \smallsetminus \ell A$; its image in $A/\ell A$ is a nonzero invariant, a contradiction. (ii) $V^D$ is unramified at a prime $p \nmid N\ell$, and an invariant vector would be fixed by $\mathrm{Frob}_p$; but no eigenvalue of $\mathrm{Frob}_p$ on

$V^D$ equals 1, by Proposition 18.13(ii). (iii) is the paper's Lemma lem:h0, proved in Chapter 11 and restated in Proposition 18.67. (iv) At a prime $p \nmid N\ell$ the eigenvalues of $\mathrm{Frob}_p$ on the three modules have absolute values 1, $p$ and $p^{-1}$, so their characteristic polynomials differ. □

By (iv), only $A_f(1)$ carries the local data (Tamagawa factors at $T^D$, Euler factors at $s=1$) that enter Lemma lem:fpr; this is why the paper writes $T^D$ and not $A$ in the local terms of Lemma lem:dfg21, and why the identity $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$ needs a proof (Proposition 18.50).

## 18.3 Local behaviour at $p \neq \ell$: invariants, Euler factors, weights

This section collects what the Selmer bookkeeping needs about the restriction of $A_f$ and $V^D$ to $G_{\mathbb{Q}_p}$ for a prime $p \neq \ell$: the Frobenius dictionary for $L$-factors, the table of inertia invariants and Euler factors by reduction type, the vanishing of $H^0(\mathbb{Q}_p, V^D)$, and the integrality of $L_p(A_f,1)^{-1}$. The paper uses these in the paragraph after Lemma lem:dfg21 and in Lemma lem:fpr. The local representations themselves are Chapter 7's and Chapter 11's; the values $e_\ell(E,p)$ are Chapter 19's.

**Proposition 18.15** (the Frobenius dictionary)**.** *Let $p \neq \ell$, $F = \mathrm{Frob}_p$, and let $M$ be a finite-dimensional representation of $G_{\mathbb{Q}_p}$ over $\mathbb{Q}_\ell$.*

*(a) $L_p(M(1),s) = L_p(M, s+1)$. If $L$-factors were defined with $F$ instead of $F^{-1}$, one would get $L_p(M, s-1)$ on the right.*

*(b) For any $\mathbb{Z}_\ell[F]$-module $X$ on which $F$ is invertible, $(F-1)X = (F^{-1}-1)X$.*

*(c) Let $X$ be a free $\mathbb{Z}_\ell$-module of finite rank $d$ with an automorphism $F$ such that $\det(1-F^{-1} \mid X) \neq 0$. Then $F-1$ is injective on $X$ and*

$$\#\, X/(F-1)X = \ell^{v_\ell(\det(1-F^{-1}\mid X))} = \ell^{v_\ell(\det(1-F\mid X))}.$$

*(d) On $V_\ell E$ at a good prime $p \neq \ell$, $\det(1-F^{-1}p^{-s} \mid V_\ell E)^{-1} = L_p(E, s+1)$, while $\det(1-F^{-1}p^{-s} \mid (V_\ell E)^\vee)^{-1} = L_p(E,s)$.*

*Proof.* (a) $M(1)^{I_p} = M^{I_p}(1)$ and $F^{-1}$ acts on it as $p^{-1}F^{-1}$ (Proposition 18.7(ii)), so $\det(1-F^{-1}p^{-s} \mid M(1)^{I_p}) = \det(1-F^{-1}p^{-(s+1)} \mid M^{I_p})$. With $F$ the factor $p$ would appear instead: $\det(1-Fp^{-s} \mid M(1)^{I_p}) = \det(1-Fp^{-(s-1)} \mid M^{I_p})$. (b) $F^{-1}-1 = -F^{-1}(F-1)$ and $F^{-1}$ is an automorphism of $X$. (c) $F-1 = F(1-F^{-1})$, so $\det(F-1) = \det F \cdot \det(1-F^{-1}) \neq 0$, and $\det F \in \mathbb{Z}_\ell^\times$ because $F$ is an automorphism of $X$. The cokernel of an injective endomorphism $\phi$ of $\mathbb{Z}_\ell^d$ has order $\ell^{v_\ell(\det\phi)}$ (Smith normal form, Chapter 3: $\phi = UDU'$ with $U, U' \in \mathrm{GL}_d(\mathbb{Z}_\ell)$ and $D = \mathrm{diag}(\ell^{a_1}, \dots, \ell^{a_d})$, so $\mathrm{coker}\,\phi \simeq \bigoplus_i \mathbb{Z}/\ell^{a_i}$, while $v_\ell(\det\phi) = \sum_i a_i$ because $\det U$, $\det U'$ are units). Similarly $1-F = -F(1-F^{-1})$. For instance, if $X = \mathbb{Z}_\ell$ and $F$ acts by $p$, then $X/(F-1)X = \mathbb{Z}_\ell/(p-1)\mathbb{Z}_\ell$ has order $\ell^{v_\ell(p-1)}$, and $\det(1-F^{-1}) = (p-1)/p$ has the same valuation. (d) The eigenvalues of $F$ on $V_\ell E$ are $\alpha, \beta$ with $\alpha+\beta = a_p$, $\alpha\beta = p$, so those of $F^{-1}$ are $\alpha^{-1}, \beta^{-1}$, with sum $a_p/p$ and product $1/p$, and $(1-\alpha^{-1}X)(1-\beta^{-1}X) = 1 - a_p p^{-1}X + p^{-1}X^2$; at $X = p^{-s}$ this is $1 - a_p p^{-(s+1)} + p \cdot p^{-2(s+1)} = L_p(E, s+1)^{-1}$. On the dual the geometric Frobenius has eigenvalues $\alpha, \beta$ and gives $1 - a_p p^{-s} + p^{1-2s}$ (Chapter 7, Appendix B, B-C3). □

*In the paper* 18.16. The paper's Section 1.1 defines "$L_p(A_f,s) = \det(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid A_f^{I_p})^{-1}$ is the true local factor, defined with the geometric Frobenius as in [49, §1.1.1]", and Lemma lem:fpr writes "$L_p(A_f,1)^{-1} = L_p(A_f(1),0)^{-1}$", which is (a) and holds only with the geometric Frobenius.

The paragraph after Lemma lem:dfg21 writes $(V^D)^{I_p}/(\mathrm{Frob}_p - 1)$, an arithmetic expression, next to $\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1}$, a geometric one; (b) and (c) show that the two are compatible. With the arithmetic Frobenius in the definition of $L_p(A_f, s)$ the multiplicative factor at $s = 1$ would be $1 - p^2$, not $1 - p^{-2}$.

**Proposition 18.17** (the adjoint at $p \neq \ell$)**.** *Let $p \neq \ell$. The inertia invariants of $A_f$ and of $V^D = A_f(1)$, the eigenvalues of the geometric Frobenius $\mathrm{Frob}_p^{-1}$ on them, and $L_p(A_f, 1)^{-1} = \det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p})$ are as follows ($\alpha, \beta$ the roots of $X^2 - a_p X + p$ at a good prime; $a = a_p(E')$ for a quadratic twist $E'$ of $E$ with good reduction at $p$ in the row $\mathrm{I}_0^*$):*

| *reduction at $p$* | $\delta_p$ | $\dim A_f^{I_p}$ | $\mathrm{Frob}_p^{-1}$ *on* $A_f^{I_p}$ | $\mathrm{Frob}_p^{-1}$ *on* $(V^D)^{I_p}$ | $L_p(A_f, 1)^{-1}$ |
|---|---|---|---|---|---|
| *good* | 2 | 3 | $\frac{\alpha}{\beta}, 1, \frac{\beta}{\alpha}$ | $\frac{\alpha}{\beta p}, \frac{1}{p}, \frac{\beta}{\alpha p}$ | $(1 - \frac{\alpha}{\beta p})(1 - \frac{1}{p})(1 - \frac{\beta}{\alpha p})$ |
| *multiplicative* | 1 | 1 | $p^{-1}$ | $p^{-2}$ | $1 - p^{-2}$ |
| *additive, $v_p(j) < 0$* | 0 | 1 | $p^{-1}$ | $p^{-2}$ | $1 - p^{-2}$ |
| $\rho(I_p) = \{\pm 1\}$ | 0 | 3 | *as for $E'$* | *as for $E'$* | $\frac{p-1}{p} \cdot \frac{(p+1)^2 - a^2}{p^2}$ |
| $\rho(I_p)$ *cyclic of order* $3, 4, 6$ | 0 | 1 | $c$ | $cp^{-1}$ | $1 - cp^{-1}$ |
| $\rho(I_p)$ *non-abelian* | 0 | 0 | *none* | *none* | 1 |

*Here $c = \pm 1$ is the eigenvalue of Frobenius (either one) on the line $(A_f)^{I_p}$. For $p \geq 5$: the row $\{\pm 1\}$ is the type $\mathrm{I}_0^*$; the cyclic row is the types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$ with $c = \chi_{-3}(p)$ and the types $\mathrm{III}, \mathrm{III}^*$ with $c = \chi_{-4}(p)$; the non-abelian row does not occur. For $p = 2$ the types $\mathrm{IV}, \mathrm{IV}^*$ are in the cyclic row with $c = \chi_{-3}(2) = -1$. At $p = 2, 3$ the Kodaira type does not determine the row in general.*

*Proof.* This is the table of Chapter 7 (the adjoint $L$-factors of the local shapes) and of the paper's Lemma lem:euler and Proposition prop:shapes, whose proofs are in Chapter 11 (the images of inertia) and Chapter 19 (the factors; the value of $L_p(A_f, 1)^{-1}$ is the $L$-part of Lemma lem:fpr proved there); we recall the reasons. At a good prime $A_f$ is unramified and Proposition 18.13(ii) gives the eigenvalues; the product formula is $(1 - \frac{\alpha}{\beta p})(1 - \frac{\beta}{\alpha p}) = 1 - \frac{\alpha^2 + \beta^2}{p^2} + \frac{1}{p^2} = \frac{(p+1)^2 - a_p^2}{p^2}$, using $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{p}$ and $\alpha^2 + \beta^2 = a_p^2 - 2p$. At a potentially multiplicative prime, over $\mathbb{Q}_p$, $\rho$ is a twist by a character (unramified, $\varepsilon$, if the reduction is multiplicative; ramified quadratic if it is additive) of the representation of a Tate curve, $\left(\begin{smallmatrix} \chi_\ell & \kappa \\ 0 & 1 \end{smallmatrix}\right)$, so by Proposition 18.10(ii) $A_f$ is the adjoint of the Tate representation; inertia acts through $\left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right)$ with $* \neq 0$, the invariants are the line of $e = e_1 \otimes e_2^\vee$, and $\mathrm{Frob}_p$ acts on $e$ by $\chi_\ell(\mathrm{Frob}_p) = p$ (the upper-left entry divided by the lower-right one), so $\mathrm{Frob}_p^{-1}$ acts by $p^{-1}$, and by $p^{-2}$ on $e \otimes \mathbb{Z}_\ell(1)$. In the potentially good cases $\rho(I_p)$ is finite (Chapter 11); for $\rho(I_p) = \{\pm 1\}$ the adjoint is unramified and equal to that of the twist $E'$ of good reduction; for a cyclic image generated by $g$ with eigenvalues $\zeta^{\pm 1}$, $\zeta \neq \zeta^{-1}$, the trace-zero commutant of $g$ is the line spanned by $g - g^{-1}$ (over $\overline{\mathbb{Q}}_\ell$ one may take $g = \mathrm{diag}(\zeta, \zeta^{-1})$; its commutant is the diagonal matrices, the trace-zero ones are the multiples of $\mathrm{diag}(1, -1) = (g - g^{-1})/(\zeta - \zeta^{-1})$, and $g - g^{-1}$ is defined over $\mathbb{Q}_\ell$). Frobenius normalises $\rho(I_p) = \langle g \rangle$, because $I_p$ is normal in $G_{\mathbb{Q}_p}$, so it conjugates $g$ to a generator of $\langle g \rangle$; a cyclic group of order 3, 4 or 6 has exactly the two generators $g^{\pm 1}$. Hence conjugation by Frobenius sends $g - g^{-1}$ to $\pm(g - g^{-1})$: it acts on the line by a sign $c$, the same for $\mathrm{Frob}_p$ and $\mathrm{Frob}_p^{-1}$; for $p \geq 5$ the tame relation identifies $c$ with $\chi_{-3}(p)$ or $\chi_{-4}(p)$ (Chapter 11); for a non-abelian image the faithful representation $\rho|_{I_p}$ is absolutely irreducible and the commutant consists of scalars. In each case the factor is $\prod(1 - \lambda p^{-1})$ over the eigenvalues $\lambda$ of $\mathrm{Frob}_p^{-1}$ on $A_f^{I_p}$, that is, $\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p})$ by Proposition 18.15(a). □

**Lemma 18.18** (local invariants)**.** *Let $p \neq \ell$.*

*(i)* $H^0(\mathbb{Q}_p, V^D) = 0$, *and* $\det(1-\mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = L_p(A_f,1)^{-1}$ *is a nonzero rational number.*

*(ii)* $H^0(\mathbb{Q}_p, A_f)$ *is the subspace of* $A_f^{I_p}$ *fixed by Frobenius. It is* 0 *at multiplicative primes, at additive potentially multiplicative primes, in the cyclic row with* $c=-1$ *and in the non-abelian row; it is a line at good primes and in the cyclic row with* $c=+1$*; in the row* $\{\pm 1\}$ *it contains the line of the eigenvalue* 1.

*Proof.* (i) By the table every eigenvalue of $\mathrm{Frob}_p^{-1}$ on $(V^D)^{I_p}$ has complex absolute value $p^{-1}$ or $p^{-2}$, so none is 1; hence $\mathrm{Frob}_p$ fixes no nonzero vector of $(V^D)^{I_p}$, that is, $H^0(\mathbb{Q}_p, V^D) = ((V^D)^{I_p})^{\mathrm{Frob}_p=1} = 0$, and the determinant is nonzero. It is rational by the last column of the table ($\alpha\beta = p$, $\alpha+\beta = a_p \in \mathbb{Z}$). (ii) $H^0(\mathbb{Q}_p, A_f) = (A_f^{I_p})^{\mathrm{Frob}_p=1}$; read off the table. At a good prime $\alpha/\beta \neq 1$, since $\alpha=\beta$ would give $a_p^2 = 4p$, which is impossible for an integer $a_p$; so the eigenvalue 1 has multiplicity one. □

*Warning* 18.19. $H^0(\mathbb{Q}_p, A_f) \neq 0$ does occur at bad primes: at every prime with $\rho(I_p) = \{\pm 1\}$ (type $\mathrm{I}_0^*$ for $p \geq 5$), and at a prime of type III with $p \equiv 1 \pmod 4$. There $H^1(\mathbb{Q}_p, T^D)$ is infinite (Corollary 18.20). Only (i) enters the paper's argument.

**Corollary 18.20** (local finiteness)**.** *Let $p \neq \ell$.*

*(i)* $H^1(\mathbb{Q}_p, T^D)$ *is a finitely generated* $\mathbb{Z}_\ell$*-module of rank* $\dim H^0(\mathbb{Q}_p, A_f)$.

*(ii) If* $H^0(\mathbb{Q}_p, A_f) = 0$ *(multiplicative and additive potentially multiplicative primes, the cyclic row with* $c=-1$*, the non-abelian row), then* $H^1(\mathbb{Q}_p, T^D)$ *is finite, and* $H^1(\mathbb{Q}_p, W) \simeq H^1(\mathbb{Q}_p, T^D)^*$ *is finite.*

*(iii) In all cases* $H^1(\mathbb{Q}_p, T^D)_{\mathrm{tor}}$ *is finite.*

*Proof.* $H^1(\mathbb{Q}_p, T^D)$ is finitely generated over $\mathbb{Z}_\ell$ and $H^1(\mathbb{Q}_p, T^D) \otimes \mathbb{Q}_\ell = H^1(\mathbb{Q}_p, V^D)$ (Chapter 17, the cohomology of $T$, $V$, $W$). For $p \neq \ell$ the local Euler characteristic of a $\mathbb{Q}_\ell$-representation is 0 (by Tate's formula, $\dim H^0 - \dim H^1 + \dim H^2 = 0$ at $p \neq \ell$; see Chapter 17). So the dimension of $H^1(\mathbb{Q}_p, V^D)$ is $\dim H^0(\mathbb{Q}_p, V^D) + \dim H^2(\mathbb{Q}_p, V^D)$, and Tate local duality gives $H^2(\mathbb{Q}_p, V^D) \simeq H^0(\mathbb{Q}_p, (V^D)^D)^\vee = H^0(\mathbb{Q}_p, A_f)^\vee$ (Chapter 17). With Lemma 18.18(i) this is (i). In (ii) a finitely generated $\mathbb{Z}_\ell$-module of rank 0 is finite, and Tate local duality for the pair $(W, T^D)$ (Chapter 17) gives the second statement. (iii) The torsion of a finitely generated $\mathbb{Z}_\ell$-module is finite. □

**Lemma 18.21** (integrality of the Euler factor)**.** *Let $p \neq \ell$ and $X = T^D$. Then $X^{I_p} = X \cap (V^D)^{I_p}$ is a* $\mathrm{Frob}_p$*-stable lattice in* $(V^D)^{I_p}$,

$$L_p(A_f,1)^{-1} = \det(1 - \mathrm{Frob}_p^{-1} \mid X^{I_p}) \in \mathbb{Z}_{(\ell)} \cap \mathbb{Q}^\times, \qquad \#\, X^{I_p}/(\mathrm{Frob}_p - 1)X^{I_p} = \ell^{\,v_\ell(L_p(A_f,1)^{-1})}.$$

*The valuation is* $v_\ell(p^2-1)$ *at multiplicative and at additive potentially multiplicative primes, and* $e_\ell(E,p)$ *at every additive prime.*

*Proof.* The value of $L_p(A_f,1)^{-1}$, its rationality and nonvanishing, and its valuation by reduction type are the $L$-part of Lemma lem:fpr, proved in Chapter 19; we do not repeat that proof. What this chapter needs in addition is the lattice. $X^{I_p}$ is the intersection of a lattice with a subspace, hence a lattice in $(V^D)^{I_p}$ (it is saturated in $X$ and spans the subspace, since every vector has a

multiple in $X$); it is $\mathrm{Frob}_p$-stable because $I_p$ is normal in $G_{\mathbb{Q}_p}$. So $\mathrm{Frob}_p^{-1}$ has a characteristic polynomial with coefficients in $\mathbb{Z}_\ell$ on it, and $\det(1-\mathrm{Frob}_p^{-1} \mid X^{I_p}) = \det(1-\mathrm{Frob}_p^{-1} \mid (V^D)^{I_p})$ lies in $\mathbb{Z}_\ell$; being rational and nonzero (Chapter 19, or Lemma 18.18(i)), it lies in $\mathbb{Z}_\ell \cap \mathbb{Q}^\times = \mathbb{Z}_{(\ell)} \cap \mathbb{Q}^\times$. The order of the cokernel is Proposition 18.15(c). □

*In the paper* 18.22. Lemma lem:fpr: "$L_p(A_f,1)^{-1} = L_p(A_f(1),0)^{-1}$ is a nonzero rational number which is an $\ell$-adic integer, of valuation $e_\ell(E,p)$ at an additive prime and $v_\ell(1-p^{-2})$ at a multiplicative one". This $L$-part of the lemma is proved in Chapter 19; the first equality is also Proposition 18.15(a), and Lemma 18.21 adds the lattice $(T^D)^{I_p}$ and the order of its Frobenius cokernel, which the Fitting-ideal half (Theorem 18.49) uses. The paragraph after Lemma lem:dfg21 says "for the same reason $H^0(\mathbb{Q}_p, V^D) = 0$" (Lemma 18.18(i)) and "So $H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group" (Corollary 18.20(iii) and Proposition 18.32(v) below). It does not say that $H^1(\mathbb{Q}_p, T^D)$ is finite, which would be false at the primes of Warning 18.19.

## 18.4 The representation at $\ell$: $p$-adic Hodge theory as a black box

Nothing in the paper's own proofs uses $p$-adic Hodge theory directly. It enters in two places only: in the definition of the Bloch-Kato local condition at $\ell$, which is part of the definition of $H^1_f(\mathbb{Q}, W)$ and of $H^1_\Sigma(\mathbb{Q}, W)$ in Section 1.2 and Section 4.2, and in the hypotheses and proofs of the theorems of Diamond, Flach and Guo that the paper imports. This section states what is needed, with references; every item is stated, not proved, and the reader may treat $H^1_f(\mathbb{Q}_\ell, -)$ as a black box with the properties of Proposition 18.28 and Proposition 18.34.

**Definition 18.23** (period rings and admissible representations)**.** Fontaine constructs topological $\mathbb{Q}_\ell$-algebras $B_{\mathrm{dR}} \supset B_{\mathrm{st}} \supset B_{\mathrm{cris}}$ with continuous actions of $G_{\mathbb{Q}_\ell}$, with the following structure: $B_{\mathrm{dR}}$ is the fraction field of a complete discrete valuation ring with residue field $\mathbb{C}_\ell$ and a uniformiser $t$ on which $G_{\mathbb{Q}_\ell}$ acts by $\chi_\ell$, and it carries the filtration $\mathrm{Fil}^i B_{\mathrm{dR}} = t^i B^+_{\mathrm{dR}}$ with $\mathrm{gr}^i B_{\mathrm{dR}} \simeq \mathbb{C}_\ell(i)$; $B_{\mathrm{cris}}$ and $B_{\mathrm{st}}$ carry a Frobenius $\varphi$, and $B_{\mathrm{st}}$ a monodromy operator $N$ with $N\varphi = \ell\varphi N$ and $B_{\mathrm{cris}} = B_{\mathrm{st}}^{N=0}$; and $B_{\mathrm{dR}}^{G_{\mathbb{Q}_\ell}} = B_{\mathrm{st}}^{G_{\mathbb{Q}_\ell}} = B_{\mathrm{cris}}^{G_{\mathbb{Q}_\ell}} = \mathbb{Q}_\ell$. For an $\ell$-adic representation $V$ of $G_{\mathbb{Q}_\ell}$ and $\bullet \in \{\mathrm{dR}, \mathrm{st}, \mathrm{cris}\}$ put $D_\bullet(V) = (B_\bullet \otimes_{\mathbb{Q}_\ell} V)^{G_{\mathbb{Q}_\ell}}$; then $\dim_{\mathbb{Q}_\ell} D_\bullet(V) \le \dim V$, and $V$ is called *de Rham*, *semistable* or *crystalline* when equality holds for $\bullet = \mathrm{dR}$, st, cris. Crystalline implies semistable implies de Rham. $\mathrm{D}_{\mathrm{dR}}(V)$ is a filtered vector space; the *Hodge-Tate weights* of a de Rham $V$ are the integers $-i$ with $\mathrm{gr}^i \mathrm{D}_{\mathrm{dR}}(V) \neq 0$, counted with multiplicity, so that $\mathbb{Q}_\ell(1)$, with $\mathrm{D}_{\mathrm{dR}}(\mathbb{Q}_\ell(1)) = t^{-1}\mathbb{Q}_\ell$, has the weight $+1$ (Convention 18.1(d)). We do not prove these facts; see Fontaine [67] and Berger's survey [11].

**Theorem 18.24** (good reduction and crystalline representations)**.** *Let $B$ be an abelian variety over $\mathbb{Q}_\ell$. Then $V_\ell B$ is crystalline if and only if $B$ has good reduction, and semistable if and only if $B$ has semistable reduction. If $E/\mathbb{Q}$ has good reduction at $\ell$, then $V_\ell E$ is crystalline with Hodge-Tate weights $0, 1$, and the crystalline Frobenius on $\mathrm{D}_{\mathrm{cris}}((V_\ell E)^\vee)$ has characteristic polynomial $X^2 - a_\ell X + \ell$.*

*Reference.* We do not prove this. The direct implications are theorems of Fontaine and of Faltings, and the converse for crystalline representations is due to Coleman and Iovita [31]; see Berger [11] for an introduction. The statement for $E$ is the one recorded in Chapter 11 (good reduction at $\ell$) and in Chapter 15. □

**Proposition 18.25** (stability)**.** *Crystalline (resp. de Rham) representations of $G_{\mathbb{Q}_\ell}$ are stable under direct sums, tensor products, duals, subquotients and Tate twists, and $\mathrm{D}_{\mathrm{cris}}$ (resp. $\mathrm{D}_{\mathrm{dR}}$) is compatible with these operations. Consequently, if $\ell \nmid N$, the representations $A_f$, $V^D = A_f(1)$ and $(V_\ell E)^\vee$ are crystalline, with Hodge-Tate weights $\{-1,0,1\}$, $\{0,1,2\}$ and $\{-1,0\}$.*

*Proof.* The formal properties are stated with references ([67], [11, Section 2]); we do not prove them. The consequences: $V_\ell E$ is crystalline with weights $\{0,1\}$ (Theorem 18.24), so its dual has weights $\{-1,0\}$ and $V_\ell E \otimes (V_\ell E)^\vee = \mathrm{End}(V_\ell E)$ has weights $\{-1,0,0,1\}$; $A_f$ is the complement of the scalars $\mathbb{Q}_\ell \cdot 1$ (weight 0), a direct summand because 2 is invertible ($x = \frac{1}{2}\operatorname{tr}(x) + (x - \frac{1}{2}\operatorname{tr}(x))$), so it has weights $\{-1,0,1\}$; and twisting by $\mathbb{Q}_\ell(1)$ adds 1. □

**Theorem 18.26** (Fontaine and Laffaille)**.** *Let $\ell$ be odd and $a \le b$ integers with $b - a \le \ell - 2$. The functor of Fontaine and Laffaille is an exact, fully faithful functor from the category of strongly divisible filtered Dieudonné modules over $\mathbb{Z}_\ell$ with filtration jumps in $[a,b]$ to the category of $\mathbb{Z}_\ell$-lattices in crystalline representations of $G_{\mathbb{Q}_\ell}$, whose essential image consists of the stable lattices in crystalline representations with Hodge-Tate weights in $[-b,-a]$ (up to the sign convention); it is compatible with subquotients and with reduction modulo $\ell^k$. For $V_\ell E$, $\ell \nmid N$, the weights $\{0,1\}$ lie in an interval of length $1 \le \ell - 2$ for every odd $\ell$, and the theory describes $T_\ell E$ and the finite flat group schemes $\mathcal{E}[\ell^k]$ of Chapter 11.*

*Reference.* We do not prove this; see Fontaine and Laffaille [68], and [14, Section 4] and [49, Sections 1 and 2] for the use made of it in the Bloch-Kato and Diamond-Flach-Guo theories. □

*Remark* 18.27 (the case $\ell = 3$). The adjoint has weights $\{-1,0,1\}$, an interval of length 2, which exceeds $\ell - 2 = 1$ when $\ell = 3$. So Theorem 18.26 cannot be applied to $A$ directly at $\ell = 3$. The paper states, after Theorem thm:dfg37, that the theorem of Diamond, Flach and Guo "covers $\ell = 3$ in weight 2: $S_f$ excludes only the primes dividing $2N$ and the induced case". At the level of statements this is checked in Chapter 20 against the published text: the exceptional set $S_f$ of [49] excludes only $\lambda \mid Nk!$ (for $k = 2$, $\lambda \mid 2N$) and the case where the residual representation is not absolutely irreducible over $\mathbb{Q}(\sqrt{\ell^*})$, and the main theorems are stated for every $\lambda \notin S_f$, so they include $\ell = 3$ when $3 \nmid N$ and $\bar\rho|_{G_{\mathbb{Q}(\sqrt{-3})}}$ is absolutely irreducible. The proof has also been checked there: Diamond, Flach and Guo apply Fontaine-Laffaille theory to $M_{f,\lambda}$ itself, of Hodge-Tate weights $0, 1$, and define the local condition at $\ell$ for the adjoint through extensions of such objects (their Section 1.7.1 notes that the adjoint itself would need $\ell - 1 > 2(k-1)$); the one extra input at $\ell = k + 1 = 3$ is in the Tamagawa factor at $\ell$ for $A_f(1)$, which enters Theorem 0.1 but not the theorems the paper uses (Chapter 20).

**Proposition 18.28** (local dimensions at $\ell$)**.** *Let $V$ be a de Rham representation of $G_{\mathbb{Q}_\ell}$. With the sign convention of Convention 18.1(d),*

$$\dim_{\mathbb{Q}_\ell} H^1_f(\mathbb{Q}_\ell, V) = \dim H^0(\mathbb{Q}_\ell, V) + \dim \mathrm{D}_{\mathrm{dR}}(V)/\mathrm{Fil}^0 \mathrm{D}_{\mathrm{dR}}(V),$$

*and $\dim \mathrm{D}_{\mathrm{dR}}(V)/\mathrm{Fil}^0 \mathrm{D}_{\mathrm{dR}}(V)$ is the number of positive Hodge-Tate weights of $V$. Examples: $H^1_f(\mathbb{Q}_\ell, \mathbb{Q}_\ell) = H^1_{\mathrm{ur}}(\mathbb{Q}_\ell, \mathbb{Q}_\ell)$ and $H^1_f(\mathbb{Q}_\ell, \mathbb{Q}_\ell(1))$ (the $\mathbb{Q}_\ell$-span of the Kummer classes of units of $\mathbb{Z}_\ell$) both have dimension 1. If $\ell \nmid N$, then $H^0(\mathbb{Q}_\ell, V^D) = 0$,*

$$\dim H^1_f(\mathbb{Q}_\ell, A_f) = \dim H^0(\mathbb{Q}_\ell, A_f) + 1, \qquad \dim H^1_f(\mathbb{Q}_\ell, V^D) = 2,$$

*and the sum of the two is $\dim H^1(\mathbb{Q}_\ell, A_f)$, as the duality of Proposition 18.34 requires.*

*Sketch of proof.* The dimension formula is due to Bloch and Kato [14, Section 3]; see also [11] and Rubin [167, Section 1.3]. We do not prove it. For $\ell \nmid N$: $A_f$ has exactly one positive weight, and $V^D$ has two. An invariant vector of $V^D$ would give a line in $\mathrm{D}_{\mathrm{cris}}(V^D)$ on which $\varphi = 1$; but the eigenvalues of $\varphi$ on $\mathrm{D}_{\mathrm{cris}}(V^D) = \mathrm{D}_{\mathrm{cris}}(A_f) \otimes \mathrm{D}_{\mathrm{cris}}(\mathbb{Q}_\ell(1))$ are $\ell^{-1}$ times those on $\mathrm{D}_{\mathrm{cris}}(A_f)$, which are of the form $\alpha/\beta, 1, \beta/\alpha$ with $\alpha\beta = \ell$ and $|\alpha| = |\beta| = \ell^{1/2}$ (Theorem 18.24), so none is 1. Finally Tate's local Euler characteristic formula at $\ell$ gives $\dim H^1(\mathbb{Q}_\ell, A_f) = \dim H^0(\mathbb{Q}_\ell, A_f) + \dim H^2(\mathbb{Q}_\ell, A_f) + 3$, and $H^2(\mathbb{Q}_\ell, A_f)$ is dual to $H^0(\mathbb{Q}_\ell, V^D) = 0$ (Chapter 17); so $\dim H^1(\mathbb{Q}_\ell, A_f) = \dim H^0(\mathbb{Q}_\ell, A_f) + 3$. □

The two basic examples show what the formula selects. The whole group $H^1(\mathbb{Q}_\ell, \mathbb{Q}_\ell) = \mathrm{Hom}_{\mathrm{cont}}(G_{\mathbb{Q}_\ell}, \mathbb{Q}_\ell)$ has dimension 2: by local class field theory the abelianised group is the profinite completion of $\mathbb{Q}_\ell^\times \simeq \ell^{\mathbb{Z}} \times \mu_{\ell-1} \times (1 + \ell\mathbb{Z}_\ell)$, and homomorphisms to $\mathbb{Q}_\ell$ are determined by their values on $\ell$ and on $1 + \ell\mathbb{Z}_\ell \simeq \mathbb{Z}_\ell$. The unramified line consists of the homomorphisms trivial on the units, and the formula gives $\dim H^1_f(\mathbb{Q}_\ell, \mathbb{Q}_\ell) = 1 + 0 = 1$: no positive weight, one invariant. Similarly $H^1(\mathbb{Q}_\ell, \mathbb{Q}_\ell(1)) = (\varprojlim_k \mathbb{Q}_\ell^\times/(\mathbb{Q}_\ell^\times)^{\ell^k}) \otimes \mathbb{Q}_\ell$ (Kummer theory) has dimension 2, spanned by the classes of $\ell$ and of a principal unit, and the formula gives $\dim H^1_f(\mathbb{Q}_\ell, \mathbb{Q}_\ell(1)) = 0 + 1 = 1$, the line of the units. In both cases $H^1_f$ is half of $H^1$, as the duality between $\mathbb{Q}_\ell$ and $\mathbb{Q}_\ell(1)$ requires, and in the second it is *not* the unramified subspace, which is 0 (inertia acts on $\mathbb{Q}_\ell(1)$ through $\chi_\ell(I_\ell) = \mathbb{Z}_\ell^\times$, so $\mathbb{Q}_\ell(1)^{I_\ell} = 0$): the Kummer classes of units are ramified at $\ell$. This is why the condition at $\ell$ cannot simply be "unramified".

For $A_f$ the weights are symmetric about 0, so the count "one positive weight" does not depend on the sign convention; for $V^D$ it does, and with the opposite convention the formula is stated with negative weights. This is the only place where the sign matters below.

## 18.5 Bloch-Kato local conditions

This section defines the local conditions $H^1_f(\mathbb{Q}_p, -)$ of Bloch and Kato on $V$, $T$ and $W$, computes them at $p \neq \ell$, and records their duality. The paper uses them in Section 1.2 ("the Bloch-Kato Selmer group"), in the statement of Lemma lem:dfg21 ("$H^1_f(\mathbb{Q}_p, T^D) = \iota^{-1} H^1_f(\mathbb{Q}_p, V^D)$") and in the paragraph after it; the difference between the Bloch-Kato and the unramified conditions on $W$, worked out in Example 18.35, is the origin of Tamagawa factors.

Why these conditions. A class in $H^1(\mathbb{Q}_p, V)$ is the same thing as an extension $0 \to V \to X \to \mathbb{Q}_\ell \to 0$ of $\ell$-adic representations of $G_{\mathbb{Q}_p}$ (Chapter 16). A Selmer group singles out the extensions that are as "geometric" at $p$ as $V$ itself. At $p \neq \ell$ the natural requirement is that inertia acts on $X$ no worse than on $V$; the unramified classes, inflated from $G_{\mathbb{Q}_p}/I_p$, are those for which the extension splits over $I_p$. At $p = \ell$ that requirement is wrong, as the Kummer classes of units show (the paragraphs after Proposition 18.28), and Bloch and Kato replace "unramified" by "crystalline", the $\ell$-adic analogue of good reduction (Theorem 18.24). A small example at $p \neq \ell$: $H^1(\mathbb{Q}_p, \mathbb{Q}_\ell(1)) = (\varprojlim \mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^{\ell^k}) \otimes \mathbb{Q}_\ell$ is the line of the Kummer class of $p$ (the units contribute only the finite group $\mathbb{F}_p^\times \otimes \mathbb{Z}_\ell$), and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, \mathbb{Q}_\ell(1)) = \mathbb{Q}_\ell/(p-1)\mathbb{Q}_\ell = 0$; so $H^1_f(\mathbb{Q}_p, \mathbb{Q}_\ell(1)) = 0$.

**Definition 18.29** (Bloch-Kato local conditions)**.** Let $V$ be an $\ell$-adic representation of $G_{\mathbb{Q}_p}$.

(i) For $p \neq \ell$: $H^1_f(\mathbb{Q}_p, V) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V) = \ker\big(H^1(\mathbb{Q}_p, V) \to H^1(I_p, V)\big)$.

(ii) For $p = \ell$: $H^1_e(\mathbb{Q}_\ell, V) = \ker\big(H^1(\mathbb{Q}_\ell, V) \to H^1(\mathbb{Q}_\ell, B_{\mathrm{cris}}^{\varphi=1} \otimes V)\big)$, $H^1_f(\mathbb{Q}_\ell, V) = \ker\big(H^1(\mathbb{Q}_\ell, V) \to H^1(\mathbb{Q}_\ell, B_{\mathrm{cris}} \otimes V)\big)$ and $H^1_g(\mathbb{Q}_\ell, V) = \ker\big(H^1(\mathbb{Q}_\ell, V) \to H^1(\mathbb{Q}_\ell, B_{\mathrm{dR}} \otimes V)\big)$;

so $H^1_e \subseteq H^1_f \subseteq H^1_g$, and $H^1_f(\mathbb{Q}_\ell, V)$ classifies the extensions $0 \to V \to X \to \mathbb{Q}_\ell \to 0$ with $X$ crystalline when $V$ is crystalline.

(iii) At the real place: $H^1_f(\mathbb{R}, V) = 0$, which is all of $H^1(\mathbb{R}, V)$ (a group killed by 2 and a $\mathbb{Q}_\ell$-vector space).

Only $H^1_f$ is used in the paper; $H^1_e$ and $H^1_g$ are recorded for orientation [14, Section 3].

**Definition 18.30** (propagation)**.** Let $T$ be a $G_{\mathbb{Q}_p}$-stable lattice in $V$, $W = V/T$, and $\iota\colon H^1(\mathbb{Q}_p, T) \to H^1(\mathbb{Q}_p, V)$, $\pi\colon H^1(\mathbb{Q}_p, V) \to H^1(\mathbb{Q}_p, W)$ the maps induced by $T \subset V \to W$. Put

$$H^1_f(\mathbb{Q}_p, T) = \iota^{-1}\big(H^1_f(\mathbb{Q}_p, V)\big), \qquad H^1_f(\mathbb{Q}_p, W) = \pi\big(H^1_f(\mathbb{Q}_p, V)\big),$$

and likewise $H^1_f(\mathbb{Q}_p, T^D)$, $H^1_f(\mathbb{Q}_p, W^D)$ from $H^1_f(\mathbb{Q}_p, V^D)$.

**Proposition 18.31** (propagated conditions)**.** *In the situation of Definition 18.30:*

*(a)* $\ker \iota = H^1(\mathbb{Q}_p, T)_{\mathrm{tor}}$*, it is the image of* $H^0(\mathbb{Q}_p, W)$ *under the connecting map, and* $H^1_f(\mathbb{Q}_p, T) \supseteq H^1(\mathbb{Q}_p, T)_{\mathrm{tor}}$*;*

*(b)* $\iota(H^1_f(\mathbb{Q}_p, T))$ *is a lattice in* $H^1_f(\mathbb{Q}_p, V)$*, so* $H^1_f(\mathbb{Q}_p, T)/H^1(\mathbb{Q}_p, T)_{\mathrm{tor}}$ *is free of rank* $\dim H^1_f(\mathbb{Q}_p, V)$*;*

*(c)* $H^1_f(\mathbb{Q}_p, W) \simeq H^1_f(\mathbb{Q}_p, V)/\iota H^1_f(\mathbb{Q}_p, T) \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^{\dim H^1_f(\mathbb{Q}_p, V)}$ *is divisible, and it is* $0$ *exactly when* $H^1_f(\mathbb{Q}_p, V) = 0$*.*

*Proof.* The long exact sequence of $0 \to T \to V \to W \to 0$ (Chapter 17) gives $H^0(\mathbb{Q}_p, W) \to H^1(\mathbb{Q}_p, T) \xrightarrow{\iota} H^1(\mathbb{Q}_p, V) \xrightarrow{\pi} H^1(\mathbb{Q}_p, W)$, and $H^1(\mathbb{Q}_p, T)$ is finitely generated with $H^1(\mathbb{Q}_p, T) \otimes \mathbb{Q}_\ell = H^1(\mathbb{Q}_p, V)$. (a) The kernel of $\iota$ is the image of the torsion group $H^0(\mathbb{Q}_p, W)$, so it consists of torsion elements; conversely a torsion element maps to a torsion element of a $\mathbb{Q}_\ell$-vector space, which is 0. The inclusion holds since $0 \in H^1_f(\mathbb{Q}_p, V)$. (b) $\iota H^1(\mathbb{Q}_p, T)$ is a lattice $L$ in $H^1(\mathbb{Q}_p, V)$; for a subspace $U$, $L \cap U$ is finitely generated and spans $U$ (every $u \in U$ has a multiple in $L$), so it is a lattice in $U$, and $\iota H^1_f(\mathbb{Q}_p, T) = L \cap H^1_f(\mathbb{Q}_p, V)$. (c) The kernel of $\pi$ is $\iota H^1(\mathbb{Q}_p, T) = L$, so $\pi$ restricted to $U = H^1_f(\mathbb{Q}_p, V)$ has kernel $L \cap U$, a lattice, and $U/(L \cap U) \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^{\dim U}$. □

**Proposition 18.32** (the conditions at $p \neq \ell$)**.** *Let* $p \neq \ell$*,* $G = G_{\mathbb{Q}_p}$*,* $I = I_p$*,* $F = \mathrm{Frob}_p$*, and let* $T$ *be a stable lattice in* $V$*,* $W = V/T$*.*

*(i)* $H^1_f(\mathbb{Q}_p, V) \simeq V^I/(F-1)V^I$*, of dimension* $\dim H^0(\mathbb{Q}_p, V)$*.*

*(ii) The sequence* $0 \to H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) \to H^1(\mathbb{Q}_p, T) \xrightarrow{\mathrm{res}} H^1(I, T)^{G/I} \to 0$ *is exact, with* $H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) \simeq T^I/(F-1)T^I$*; the same holds for* $W$*.*

*(iii)* $H^1_f(\mathbb{Q}_p, T) = \mathrm{res}^{-1}\big(H^1(I, T)_{\mathrm{tor}}\big)$*, and*

$$H^1_f(\mathbb{Q}_p, T)/H^1_{\mathrm{ur}}(\mathbb{Q}_p, T) \;\simeq\; \big(H^1(I, T)_{\mathrm{tor}}\big)^{G/I}.$$

*(iv)* $H^1_f(\mathbb{Q}_p, W)$ *is the maximal divisible subgroup of* $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) = W^I/(F-1)W^I$*.*

*(v) If $H^0(\mathbb{Q}_p,V)=0$, then $H^1_f(\mathbb{Q}_p,V)=0$, $H^1_f(\mathbb{Q}_p,W)=0$ and $H^1_f(\mathbb{Q}_p,T)=H^1(\mathbb{Q}_p,T)_{\mathrm{tor}}$, a finite group. If moreover $H^0(\mathbb{Q}_p,V^D)=0$, then $H^1(\mathbb{Q}_p,T)$ is finite and $H^1_f(\mathbb{Q}_p,T)=H^1(\mathbb{Q}_p,T)$.*

*Proof.* (i) and (ii) are Chapter 17 (unramified cohomology): inflation identifies $H^1_{\mathrm{ur}}(\mathbb{Q}_p,M)$ with $H^1(G/I,M^I)=M^I/(F-1)M^I$, $G/I\simeq\widehat{\mathbb{Z}}$ being procyclic, and the five-term inflation-restriction sequence gives the exactness of (ii) up to the last map. For the discrete module $W$ the sequence is Chapter 16's, and res is onto $H^1(I,W)^{G/I}$ because the next term $H^2(G/I,W^I)$ vanishes ($\widehat{\mathbb{Z}}$ has cohomological dimension 1 on torsion modules). For the compact module $T$ and for $V$ we use the extension of the five-term sequence to compact and $\mathbb{Q}_\ell$ coefficients when the quotient is isomorphic to $\widehat{\mathbb{Z}}$, proved in Chapter 16, and the surjectivity of $H^1(\mathbb{Q}_p,M)\to H^1(I,M)^{G/I}$ for such $M$ at $p\neq\ell$, proved in Chapter 17. Every later use of this surjectivity ((iii) below, Step 2 of Theorem 18.49) goes through (ii), at $p\neq\ell$ only. For $V$, $F-1$ is an endomorphism of the finite-dimensional space $V^I$, so its kernel $H^0(\mathbb{Q}_p,V)=(V^I)^{F=1}$ and its cokernel have the same dimension.

(iii) Let $\iota_I\colon H^1(I,T)\to H^1(I,V)$. Restriction commutes with $\iota$, so $c\in H^1(\mathbb{Q}_p,T)$ lies in $H^1_f(\mathbb{Q}_p,T)$ if and only if $\iota_I(\operatorname{res}c)=0$. As in Proposition 18.31(a), the kernel of $\iota_I$ is the image of $W^I$ under the connecting map, a torsion group, and it contains every torsion element, $H^1(I,V)$ being a $\mathbb{Q}_\ell$-vector space; so $\ker\iota_I=H^1(I,T)_{\mathrm{tor}}$. By (ii) res maps $H^1(\mathbb{Q}_p,T)$ onto $H^1(I,T)^{G/I}$ with kernel $H^1_{\mathrm{ur}}(\mathbb{Q}_p,T)$, so $H^1_f(\mathbb{Q}_p,T)/H^1_{\mathrm{ur}}(\mathbb{Q}_p,T)\simeq H^1(I,T)^{G/I}\cap H^1(I,T)_{\mathrm{tor}}=(H^1(I,T)_{\mathrm{tor}})^{G/I}$.

(iv) $\pi$ commutes with inflation, so $H^1_f(\mathbb{Q}_p,W)=\pi(H^1_{\mathrm{ur}}(\mathbb{Q}_p,V))$ is the image of $V^I$ in $W^I/(F-1)W^I$. Let $D\subseteq W^I$ be the image of $V^I$, that is, $V^I/T^I$ (note $T^I=T\cap V^I$), a divisible group of corank $\dim V^I$. It is the maximal divisible subgroup of $W^I$: $W^I$ is cofinitely generated (its $\ell$-torsion lies in the finite $W[\ell]$), so $(W^I)_{\mathrm{div}}\simeq(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^r$, and its Tate module is an $I$-invariant submodule of the Tate module $T$ of $W$, hence lies in $T^I$, so $r\le\operatorname{rank}T^I=\dim V^I$; thus $(W^I)_{\mathrm{div}}=D$ and $W^I/D$ is finite. The image of $D$ in $H^1_{\mathrm{ur}}(\mathbb{Q}_p,W)=W^I/(F-1)W^I$ is divisible with finite cokernel (a quotient of $W^I/D$), so it is the maximal divisible subgroup: a divisible subgroup maps to a divisible subgroup of a finite group, which is 0.

(v) By (i) $H^1_f(\mathbb{Q}_p,V)=0$; then $H^1_f(\mathbb{Q}_p,W)=\pi(0)=0$ and $H^1_f(\mathbb{Q}_p,T)=\ker\iota=H^1(\mathbb{Q}_p,T)_{\mathrm{tor}}$, finite because $H^1(\mathbb{Q}_p,T)$ is finitely generated. If also $H^0(\mathbb{Q}_p,V^D)=0$, then $\dim H^1(\mathbb{Q}_p,V)=\dim H^0(\mathbb{Q}_p,V)+\dim H^0(\mathbb{Q}_p,V^D)=0$ (local Euler characteristic and duality, as in the proof of Corollary 18.20), so $H^1(\mathbb{Q}_p,T)$ is torsion, hence finite and equal to $H^1_f(\mathbb{Q}_p,T)$. □

*In the paper* 18.33. The paragraph after Lemma lem:dfg21: "For $p\in\Sigma$ the group $H^1_f(\mathbb{Q}_p,V^D)=H^1_{\mathrm{ur}}(\mathbb{Q}_p,V^D)=(V^D)^{I_p}/(\mathrm{Frob}_p-1)$ vanishes, because $\det(1-\mathrm{Frob}_p^{-1}\mid(V^D)^{I_p})=L_p(A_f,1)^{-1}\neq0$". This is (i) applied to $V^D$, with Lemma 18.18(i) and Proposition 18.15(c) for the nonvanishing. Then (v), for the lattice $T^D$, is "$H^1_f(\mathbb{Q}_p,T^D)$ is the torsion of $H^1(\mathbb{Q}_p,T^D)$".

**Proposition 18.34** (duality of the Bloch-Kato conditions)**.** *Let $V$ be an $\ell$-adic representation of $G_{\mathbb{Q}_v}$, $T\subset V$ a stable lattice, $W=V/T$. Under the cup-product pairing $H^1(\mathbb{Q}_v,V)\times H^1(\mathbb{Q}_v,V^D)\to H^2(\mathbb{Q}_v,\mathbb{Q}_\ell(1))=\mathbb{Q}_\ell$ the subspaces $H^1_f(\mathbb{Q}_v,V)$ and $H^1_f(\mathbb{Q}_v,V^D)$ are exact annihilators of each other, for every place $v$ (for $v=\ell$ when $V$ is de Rham). Under the perfect pairing $H^1(\mathbb{Q}_v,W)\times H^1(\mathbb{Q}_v,T^D)\to\mathbb{Q}_\ell/\mathbb{Z}_\ell$ the subgroups $H^1_f(\mathbb{Q}_v,W)$ and $H^1_f(\mathbb{Q}_v,T^D)$ are exact annihilators of each other; hence*

$$H^1(\mathbb{Q}_v,W)/H^1_f(\mathbb{Q}_v,W)\ \simeq\ H^1_f(\mathbb{Q}_v,T^D)^*.$$

*Proof.* For $v = p \neq \ell$ both conditions are the unramified ones, and the exact orthogonality of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V)$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D)$ is Chapter 17 (the cup product of two inflated classes is inflated from $H^2(\widehat{\mathbb{Z}}, \mathbb{Q}_\ell(1)) = 0$, and $\dim H^1_{\mathrm{ur}}(V) + \dim H^1_{\mathrm{ur}}(V^D) = \dim H^0(V) + \dim H^0(V^D) = \dim H^1(\mathbb{Q}_p, V)$). For $v = \ell$ this is [14, Section 3], which we do not prove. For $v = \infty$ all the groups vanish ($\ell$ odd). The integral statement follows from the rational one for propagated conditions (Chapter 17, propagation of local conditions): if $y \in H^1_f(\mathbb{Q}_v, T^D)$ and $x \in H^1_f(\mathbb{Q}_v, V)$, then $\langle \pi x, y\rangle = \langle x, \iota y\rangle \bmod \mathbb{Z}_\ell = 0$; conversely, if $y$ annihilates $\pi H^1_f(\mathbb{Q}_v, V)$, then $\langle \ell^{-k} x, \iota y\rangle \in \mathbb{Z}_\ell$ for all $k$, so $\langle x, \iota y\rangle = 0$ for all $x \in H^1_f(\mathbb{Q}_v, V)$ and $\iota y \in H^1_f(\mathbb{Q}_v, V^D)$; the pairing between $H^1(\mathbb{Q}_v, W)$ and $H^1(\mathbb{Q}_v, T^D)$ is perfect (Tate local duality, Chapter 17), and for a perfect pairing between a discrete and a compact group the quotient by a subgroup is dual to its annihilator. □

**Example 18.35** (unramified is not $f$ for $W$)**.** Let $E$ have split multiplicative reduction at $p \neq \ell$ with $\ell \mid n = n_p$, and take $V = V_\ell E$, $T = T_\ell E$, $W = E[\ell^\infty]$. In the basis of the Tate shape (Chapter 11) inertia acts through $t$ with $te_1 = e_1$, $te_2 = e_2 + n' e_1$, $v_\ell(n') = v_\ell(n)$, and $\mathrm{Frob}_p$ by $\left(\begin{smallmatrix} p & c' \\ 0 & 1 \end{smallmatrix}\right)$. Then $V^{I_p} = \mathbb{Q}_\ell e_1$, on which $\mathrm{Frob}_p$ acts by $p \neq 1$, so $H^0(\mathbb{Q}_p, V) = 0$, $H^1_f(\mathbb{Q}_p, V) = 0$ and $H^1_f(\mathbb{Q}_p, W) = 0$ (Proposition 18.32(v)). On the other hand $xe_1 + ye_2 \bmod T$ is fixed by $t$ iff $yn'e_1 \in T$, so

$$W^{I_p} = (\mathbb{Q}_\ell/\mathbb{Z}_\ell)e_1 \oplus (\ell^{-v_\ell(n)}\mathbb{Z}_\ell/\mathbb{Z}_\ell)e_2, \qquad (\mathrm{Frob}_p - 1)(xe_1 + ye_2) = \big((p-1)x + c'y\big)e_1,$$

and since $(p-1)\mathbb{Q}_\ell/\mathbb{Z}_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $(\mathrm{Frob}_p - 1)W^{I_p} = (\mathbb{Q}_\ell/\mathbb{Z}_\ell)e_1$ and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) \simeq \mathbb{Z}/\ell^{v_\ell(n)}$. This is the $\ell$-part of the component group $\mathbb{Z}/n$, on which Frobenius acts trivially in the split case. The Kummer condition at $p \neq \ell$ in the classical Selmer group is $E(\mathbb{Q}_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell = 0$ (Example 18.41), which is the Bloch-Kato condition, not the unramified one; the difference is the Tamagawa phenomenon of Section 18.7. For the adjoint at a multiplicative prime $q$, $H^1_f(\mathbb{Q}_q, W) = 0$ always (Lemma 18.18(ii) gives $H^0(\mathbb{Q}_q, A_f) = 0$, and Proposition 18.32(v) then gives $H^1_f(\mathbb{Q}_q, W) = 0$), while $H^1_{\mathrm{ur}}(\mathbb{Q}_q, W) = 0$ when $\ell \nmid n_q$ (Exercise 18.93).

## 18.6 Global Bloch-Kato Selmer groups and the Σ-imprimitive Selmer group

This section defines the Selmer groups of Theorem thm:main: the Bloch-Kato Selmer group $H^1_f(\mathbb{Q}, W)$ (Section 1.2 of the paper) and the Σ-imprimitive group $H^1_\Sigma(\mathbb{Q}, W)$ of Diamond, Flach and Guo (Section 4.2), proves their basic finiteness properties, and records the Euler characteristic count that makes the two vanishing statements of Theorem thm:dfg27 equivalent.

**Definition 18.36** (Bloch-Kato Selmer groups)**.** Let $V$ be a geometric representation of $G_\mathbb{Q}$, $T$ a stable lattice, $W = V/T$, and let $M$ be one of $V$, $T$, $W$, $V^D$, $T^D$, $W^D$. The *Bloch-Kato Selmer group* is

$$H^1_f(\mathbb{Q}, M) = \{c \in H^1(\mathbb{Q}, M) : \ \mathrm{loc}_v(c) \in H^1_f(\mathbb{Q}_v, M) \text{ for every place } v\},$$

with the local conditions of Definition 18.29 and Definition 18.30.

**Lemma 18.37** (finite level)**.** *Let $S$ be a finite set of places containing $\ell$, $\infty$ and the primes at which $V$ is ramified. For $p \notin S$, $H^1_f(\mathbb{Q}_p, M) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)$ for each of the six modules, and $H^1_f(\mathbb{Q}, M)$ is the subgroup of $H^1(G_{\mathbb{Q},S}, M)$ defined by the conditions at the places of $S$.*

*Proof.* For $V$ this is the definition. If $T$ is unramified at $p$, then $H^1(I_p,T)=\mathrm{Hom}_{\mathrm{cont}}(I_p,T)$ is torsion-free (a homomorphism into the pro-$\ell$ group $T$ factors through the maximal pro-$\ell$ quotient $\mathbb{Z}_\ell(1)$ of $I_p$, and $\mathrm{Hom}(\mathbb{Z}_\ell(1),T)\simeq T(-1)$), so Proposition 18.32(iii) gives $H^1_f=H^1_{\mathrm{ur}}$. If $W$ is unramified at $p$, $H^1_{\mathrm{ur}}(\mathbb{Q}_p,W)=W/(F-1)W$ is a quotient of a divisible group, hence divisible, and equals $H^1_f(\mathbb{Q}_p,W)$ by Proposition 18.32(iv). Finally, a class of $H^1(\mathbb{Q},M)$ that is unramified at every $p\notin S$ is inflated from $G_{\mathbb{Q},S}$: by inflation-restriction its restriction to the kernel $H$ of $G_{\mathbb{Q}}\to G_{\mathbb{Q},S}$ is a homomorphism $H\to M$ that kills every inertia group at a prime outside $S$, and these inertia groups generate $H$ as a closed normal subgroup (the fixed field of the subgroup they generate is unramified outside $S$). □

**Proposition 18.38** (relations)**.** *In the situation of Definition 18.36:*

*(i)* $H^1_f(\mathbb{Q},T)$ *is the preimage of* $H^1_f(\mathbb{Q},V)$ *under* $\iota\colon H^1(G_{\mathbb{Q},S},T)\to H^1(G_{\mathbb{Q},S},V)$*, and* $H^1_f(\mathbb{Q},T)_{\mathrm{tor}}=H^1(G_{\mathbb{Q},S},T)_{\mathrm{tor}}=\ker\iota$ *is the image of* $H^0(\mathbb{Q},W)$*.*

*(ii)* $H^1_f(\mathbb{Q},W)$ *is a cofinitely generated* $\mathbb{Z}_\ell$*-module.*

*(iii) The maximal divisible subgroup of* $H^1_f(\mathbb{Q},W)$ *is* $\pi(H^1_f(\mathbb{Q},V))$*; so the corank of* $H^1_f(\mathbb{Q},W)$ *is* $\dim H^1_f(\mathbb{Q},V)$*, and* $H^1_f(\mathbb{Q},W)$ *is finite if and only if* $H^1_f(\mathbb{Q},V)=0$*.*

*Proof.* We use from Chapter 17 that $H^i(G_{\mathbb{Q},S},M)$ is finite for finite $M$ and finitely generated over $\mathbb{Z}_\ell$ for $M=T$, and the same for $\mathbb{Q}_p$. (i) The local condition on $T$ is the preimage of the one on $V$, place by place; and the argument of Proposition 18.31(a) applies to $G_{\mathbb{Q},S}$; torsion classes satisfy every local condition because they lie in $\ker\iota$. (ii) $H^1_f(\mathbb{Q},W)[\ell]$ lies in $H^1(G_{\mathbb{Q},S},W)[\ell]$, which is a quotient of the finite group $H^1(G_{\mathbb{Q},S},W[\ell])$ (from $0\to W[\ell]\to W\xrightarrow{\ell}W\to 0$, $W$ being divisible); a discrete $\ell$-primary module with finite $\ell$-torsion is cofinitely generated (Chapter 3). (iii) $D_0=\pi(H^1_f(\mathbb{Q},V))$ lies in $H^1_f(\mathbb{Q},W)$ (the local conditions on $W$ are the images of those on $V$) and is divisible of corank $\dim H^1_f(\mathbb{Q},V)$ (its kernel $\iota H^1_f(\mathbb{Q},T)$ is a lattice, as in Proposition 18.31); so $D_0\subseteq D$. The converse inclusion is the real content: a class of $H^1_f(\mathbb{Q},W)$ need not come from $H^1_f(\mathbb{Q},V)$, and we show that the classes that do not form, modulo $D_0$, a finite group, which cannot contain a nonzero divisible subgroup. Let $D$ be the maximal divisible subgroup of $X=H^1_f(\mathbb{Q},W)$. The cokernel of $\pi\colon H^1(G_{\mathbb{Q},S},V)\to H^1(G_{\mathbb{Q},S},W)$ injects into the finitely generated $\mathbb{Z}_\ell$-module $H^2(G_{\mathbb{Q},S},T)$, which has no nonzero divisible subgroup; so $D\subseteq\pi(H^1(G_{\mathbb{Q},S},V))$. Let $Y=\{y\in H^1(G_{\mathbb{Q},S},V):\pi(y)\in X\}$ and $\lambda\colon H^1(G_{\mathbb{Q},S},V)\to Q=\bigoplus_{p\in S}H^1(\mathbb{Q}_p,V)/H^1_f(\mathbb{Q}_p,V)$ the localisation. For $y\in Y$ and $p\in S$, $\pi_p(\mathrm{loc}_p\,y)\in H^1_f(\mathbb{Q}_p,W)=\pi_p(H^1_f(\mathbb{Q}_p,V))$, so $\mathrm{loc}_p\,y\in H^1_f(\mathbb{Q}_p,V)+\iota_pH^1(\mathbb{Q}_p,T)$. Hence $\lambda(Y)$ lies in the image $\Lambda$ of $\bigoplus_p\iota_pH^1(\mathbb{Q}_p,T)$, a finitely generated $\mathbb{Z}_\ell$-submodule of $Q$. Choose a complement $C$ of $\ker\lambda=H^1_f(\mathbb{Q},V)$ in $H^1(G_{\mathbb{Q},S},V)$; $\lambda|_C$ is injective, so $M_0=C\cap\lambda^{-1}(\Lambda)$ is finitely generated, and $Y\subseteq H^1_f(\mathbb{Q},V)+M_0$. Then $X\cap\pi(H^1(G_{\mathbb{Q},S},V))=\pi(Y)\subseteq D_0+\pi(M_0)$, and $\pi(M_0)$ is finitely generated and torsion, hence finite. So $D\subseteq D_0+\pi(M_0)$, and the image of $D$ in the finite group $(D_0+\pi(M_0))/D_0$ is divisible, hence 0: $D\subseteq D_0$. □

**Definition 18.39** (the $\Sigma$-imprimitive Selmer group)**.** For a finite set $\Sigma$ of primes not containing $\ell$, and $\ell$ odd,

$$H^1_\Sigma(\mathbb{Q},W)=\{c\in H^1(\mathbb{Q},W):\ \mathrm{loc}_\ell\,c\in H^1_f(\mathbb{Q}_\ell,W),\ \mathrm{loc}_p\,c\in H^1_{\mathrm{ur}}(\mathbb{Q}_p,W)\text{ for }p\notin\Sigma\cup\{\ell\}\},$$

with no condition at the primes of $\Sigma$ (none is needed at $\infty$, where $H^1(\mathbb{R}, W) = 0$). If $\Sigma$ contains every prime $p \neq \ell$ at which $W$ is ramified, then $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) = H^1_f(\mathbb{Q}_p, W)$ for $p \notin \Sigma \cup \{\ell\}$ (Lemma 18.37), so $H^1_\Sigma(\mathbb{Q}, W) \supseteq H^1_f(\mathbb{Q}, W)$ is "Bloch-Kato outside $\Sigma$, nothing on $\Sigma$".

For the paper $\Sigma$ is the set of primes dividing $N$. It may contain primes at which $W$ is unramified, namely the primes with $\rho(I_p) = \{\pm 1\}$ (type $\mathrm{I}_0^*$ for $p \geq 5$); dropping the condition there is harmless.

Why relax the conditions at $\Sigma$. The group that the Taylor-Wiles method of Diamond, Flach and Guo controls is $H^1_\Sigma(\mathbb{Q}, W)$, whose length is a congruence ideal for newforms allowed to vary at the primes of $\Sigma$ (Theorem 18.86). The group in the paper's identity is the smaller $H^1_f(\mathbb{Q}, W)$. The price of passing from one to the other is a finite local term at each $p \in \Sigma$, and computing those terms (Lemma lem:fpr and the Tamagawa factors) is most of the work of this chapter.

**Proposition 18.40** (finiteness of $H^1_\Sigma$)**.** *Let $\Sigma$ be as in Definition 18.39, containing the primes $p \neq \ell$ at which $W$ is ramified, and suppose $H^1_f(\mathbb{Q}, V) = 0$ and $H^0(\mathbb{Q}_p, V^D) = 0$ for every $p \in \Sigma$. Then $H^1_\Sigma(\mathbb{Q}, W)$ is finite: the sequence*

$$0 \to H^1_f(\mathbb{Q}, W) \to H^1_\Sigma(\mathbb{Q}, W) \to \bigoplus_{p \in \Sigma} H^1(\mathbb{Q}_p, W)/H^1_f(\mathbb{Q}_p, W) \simeq \bigoplus_{p \in \Sigma} H^1_f(\mathbb{Q}_p, T^D)^*$$

*is exact and its outer terms are finite.*

*Proof.* The kernel of the localisation map consists of the classes satisfying the Bloch-Kato condition at every place (at $p \notin \Sigma \cup \{\ell\}$ by Lemma 18.37, at $\infty$ trivially), that is, $H^1_f(\mathbb{Q}, W)$, finite by Proposition 18.38(iii). The isomorphism is Proposition 18.34. For $p \in \Sigma$, $H^1_f(\mathbb{Q}_p, V^D) = 0$ by Proposition 18.32(i), so $H^1_f(\mathbb{Q}_p, T^D) = H^1(\mathbb{Q}_p, T^D)_{\mathrm{tor}}$ is finite (Proposition 18.32(v)). □

**Example 18.41** (the classical Selmer group)**.** Take $V = V_\ell E$, $T = T_\ell E$, $W = E[\ell^\infty]$, $\ell$ odd. At $p \neq \ell$: $E(\mathbb{Q}_p)$ has an open subgroup of finite index isomorphic to $\mathbb{Z}_p$, which is $\ell$-divisible, so $E(\mathbb{Q}_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell = 0$; and $H^0(\mathbb{Q}_p, V) = 0$ because $E(\mathbb{Q}_p)[\ell^\infty]$ is finite, so $H^1_f(\mathbb{Q}_p, W) = 0$ by Proposition 18.32(v). At $\ell$ the image of the Kummer map $E(\mathbb{Q}_\ell) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \to H^1(\mathbb{Q}_\ell, W)$ is $H^1_f(\mathbb{Q}_\ell, W)$, by a theorem of Bloch and Kato [14, Section 3], which we do not prove. Hence $H^1_f(\mathbb{Q}, E[\ell^\infty]) = \mathrm{Sel}_{\ell^\infty}(E/\mathbb{Q})$, the classical Selmer group of Chapter 17. The Tamagawa numbers $c_p$ do not appear in the Selmer group itself: they appear in the comparison with relaxed conditions (Example 18.35, Section 18.10).

**Definition 18.42** (the paper's Selmer groups)**.** Let $\ell$ be odd. The *adjoint Bloch-Kato Selmer group* is $H^1_f(\mathbb{Q}, W)$ for $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (Definition 18.36 for the triple $(A_f, A, W)$), and $H^1_\Sigma(\mathbb{Q}, W)$ is Definition 18.39 with $\Sigma$ the set of primes dividing $N$. Their duals involve $W^D \simeq \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ (Proposition 18.10), $T^D$ and $V^D$.

The Selmer group with $\mathbb{Q}_\ell$ coefficients $H^1_f(\mathbb{Q}, A_f)$ vanishes under the hypothesis of Theorem thm:main (Theorem 18.66), so the length in (eq:identity) is that of $H^1_f(\mathbb{Q}, W)$ with discrete coefficients, as the paper writes it: $\mathrm{length}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$.

**Proposition 18.43** (Euler characteristic)**.** *Let $V$ be a geometric representation of $G_\mathbb{Q}$. Then*

$$\begin{aligned}\dim H^1_f(\mathbb{Q}, V) - \dim H^1_f(\mathbb{Q}, V^D) \\ = \dim H^0(\mathbb{Q}, V) - \dim H^0(\mathbb{Q}, V^D) + \sum_v \big(\dim H^1_f(\mathbb{Q}_v, V) - \dim H^0(\mathbb{Q}_v, V)\big),\end{aligned}$$

*the sum over all places, almost all terms being* 0. *For* $V = A_f$ *with* $\ell \nmid N$ *and* $\bar\rho$ *absolutely irreducible the terms are:*

| *term* | *value* | *reason* |
|---|---|---|
| $\dim H^0(\mathbb{Q}, A_f) - \dim H^0(\mathbb{Q}, V^D)$ | $0 - 0$ | *Proposition 18.14 ($\bar\rho$ abs. irreducible)* |
| $v = p \neq \ell$ | $0$ | *Proposition 18.32(i)* |
| $v = \ell$ | $+1$ | *Proposition 18.28* |
| $v = \infty$ | $-\dim A_f^{c=1} = -1$ | *$c$ has eigenvalues $1, -1, -1$ on $A_f$* |

*So, under these hypotheses,* $\dim H^1_f(\mathbb{Q}, A_f) = \dim H^1_f(\mathbb{Q}, V^D)$*; in particular* $H^1_f(\mathbb{Q}, A_f) = 0$ *if and only if* $H^1_f(\mathbb{Q}, V^D) = 0$.

*Sketch of proof.* The general formula is the $\mathbb{Q}_\ell$ form of the formula of Greenberg and Wiles, which Chapter 17 states for finite modules (with references to [207] and [38]; it is not proved in this book): apply it to $T/\ell^k T$ with the conditions induced from $H^1_f(\mathbb{Q}_v, V)$ (preimages and images, which are mutually dual by Proposition 18.34) and compare the growth in $k$ of the two sides, using Proposition 18.31 and Proposition 18.38; see [69] and [167, Section 1.7]. For the table: complex conjugation $c$ acts on $V_\ell E$ with eigenvalues $1, -1$ (Chapter 11), so in a basis where $\rho(c) = \mathrm{diag}(1, -1)$ it fixes $h$ and negates $e$ and $f$; and at $\infty$, $H^1_f(\mathbb{R}, A_f) = 0$ and $H^0(\mathbb{R}, A_f) = A_f^{c=1}$. The irreducibility of $\bar\rho$ enters only through $H^0(\mathbb{Q}, A_f) = 0$; without it one may use instead Faltings's theorem $\mathrm{End}_{\mathbb{Q}_\ell[G_\mathbb{Q}]} V_\ell E = \mathbb{Q}_\ell$ for $E/\mathbb{Q}$ (which gives $H^0(\mathbb{Q}, A_f) = 0$ directly), which we do not prove. □

*In the paper* 18.44. Theorem thm:dfg27 states "$H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$ (and the same for $A \otimes \mathbb{Q}_\ell$)". The parenthesis is the other half of the equivalence just proved, and Remark rem:ns uses the same equivalence: "The vanishing of $H^1_f(\mathbb{Q}, V^D)$, equivalently of $H^1_f(\mathbb{Q}, \mathrm{ad}^0 V_3 E)$, is not known for these curves".

## 18.7 Tamagawa factors

This section defines the Tamagawa factor of Fontaine and Perrin-Riou at a prime $p \neq \ell$, proves its index description and its basic properties, and proves the paper's Lemma lem:fpr: the local term $H^1_f(\mathbb{Q}_p, T^D)$ of Lemma lem:dfg21 has Fitting ideal $L_p(A_f, 1)^{-1}\,\mathrm{Tam}_p(T^D)$, and $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$. It also checks that for $T_\ell E$ the Tamagawa factor is the $\ell$-part of the Tamagawa number $c_p$, as the name promises. The paper uses the definition in Section 1.2 and the lemma in the proof of Theorem thm:main and in Remark rem:ns. A Tamagawa factor is the size of a local defect counted by torsion in a first cohomology group, much as the index of a sublattice measures a ratio of volumes.

**Definition 18.45** (Tamagawa factor)**.** Let $p \neq \ell$ and let $M$ be a free $\mathbb{Z}_\ell$-module of finite rank with a continuous action of $G_{\mathbb{Q}_p}$. The group $G_{\mathbb{Q}_p}/I_p$ acts on $H^1(I_p, M)$ by $(\sigma \cdot c)(i) = \sigma\, c(\sigma^{-1} i \sigma)$ (Chapter 16), and preserves its torsion subgroup. The *Tamagawa factor* of $M$ at $p$ is

$$\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}\Big( \big(H^1(I_p, M)_{\mathrm{tor}}\big)^{G_{\mathbb{Q}_p}} \Big),$$

the Fitting ideal of the $G_{\mathbb{Q}_p}$-invariants of the torsion of $H^1(I_p, M)$. It is a nonzero ideal $\ell^a \mathbb{Z}_\ell$, with $a = \mathrm{length}\big((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}}\big)$, and we write $v_\ell(\mathrm{Tam}_p(M)) = a$.

That the ideal is nonzero: $H^1(I_p, M)$ is a finitely generated $\mathbb{Z}_\ell$-module (Chapter 16: $I_p$ acts on the pro-$\ell$ module $M$ through a quotient that is an extension of $\mathbb{Z}_\ell(1)$ by a finite group, the wild inertia and the prime-to-$\ell$ part of tame inertia contributing nothing), so its torsion is finite, and so are its invariants; the Fitting ideal of a finite $\mathbb{Z}_\ell$-module $X$ is $\ell^{\operatorname{length} X}\mathbb{Z}_\ell$ (Chapter 3).

Why the name, in the simplest case. Let $E$ have split multiplicative reduction at $p \neq \ell$, with $n = v_p(\Delta_E)$, and take $M = T_\ell E$. In the notation of Example 18.35, inertia acts through $t$ with $te_1 = e_1$, $te_2 = e_2 + n'e_1$, so $H^1(I_p, T) = T/(t-1)T = T/n'\mathbb{Z}_\ell e_1 = (\mathbb{Z}_\ell/n')\bar{e}_1 \oplus \mathbb{Z}_\ell \bar{e}_2$ (the cohomology of the pro-$\ell$ part $\langle t \rangle$ of inertia, as in Steps 2 to 4 of Lemma 18.53), with torsion $(\mathbb{Z}/\ell^{v_\ell(n)})\bar{e}_1$. Frobenius acts on it by $p^{-1} \cdot p = 1$: the factor $p^{-1}$ comes from the tame relation $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^p$, and $p$ is the upper-left entry of $\rho(\mathrm{Frob}_p)$ in the split case. So $v_\ell(\mathrm{Tam}_p(T_\ell E)) = v_\ell(n)$, which is $v_\ell(c_p)$ because the Tamagawa number at a split multiplicative prime is $c_p = n$. Proposition 18.51 proves $v_\ell(\mathrm{Tam}_p(T_\ell E)) = v_\ell(c_p)$ at every $p \neq \ell$.

*In the paper* 18.46. Section 1.2 of the paper defines "$\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}})$, an ideal of $\mathbb{Z}_\ell$ [49, p. 708], [69, I.4.2.2]". This is Definition 18.45: the invariants are taken inside the Fitting ideal, as they must be (an ideal of $\mathbb{Z}_\ell$ has no Galois action), and the computation in Lemma lem:tamfactor uses them so.

**Proposition 18.47** (the index description)**.** *For $p \neq \ell$ and $M$ as in Definition 18.45,*

$$\ell^{v_\ell(\mathrm{Tam}_p(M))} = \#(H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}} = [H^1_f(\mathbb{Q}_p, M) : H^1_{\mathrm{ur}}(\mathbb{Q}_p, M)].$$

*Proof.* The first equality holds because a finite $\mathbb{Z}_\ell$-module $Y$ has order $\ell^{\operatorname{length} Y}$ and Fitting ideal $\ell^{\operatorname{length} Y}\mathbb{Z}_\ell$. The second is Proposition 18.32(iii), with $T = M$: it gives $H^1_f(\mathbb{Q}_p, M)/H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) \simeq (H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}/I_p}$, and $I_p$ acts trivially on $H^1(I_p, M)$ (inner automorphisms act trivially on cohomology, Chapter 16), so invariants under $G_{\mathbb{Q}_p}/I_p$ and under $G_{\mathbb{Q}_p}$ are the same. □

So the Tamagawa factor measures how much larger the Bloch-Kato condition on the lattice is than the naive unramified condition. For $W$ the inclusion goes the other way (Proposition 18.32(iv)): $H^1_f(\mathbb{Q}_p, W)$ is the maximal divisible subgroup of $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$. In the example above both defects have order $\ell^{v_\ell(n)}$: $H^1_{\mathrm{ur}}(\mathbb{Q}_p, E[\ell^\infty]) \simeq \mathbb{Z}/\ell^{v_\ell(n)}$ while $H^1_f(\mathbb{Q}_p, E[\ell^\infty]) = 0$ (Example 18.35).

**Proposition 18.48** (basic properties)**.** *Let $p \neq \ell$.*

*(i) If $I_p$ acts trivially on $M$, then $\mathrm{Tam}_p(M) = \mathbb{Z}_\ell$.*

*(ii) If $I_p$ acts on $M$ through a finite quotient of order prime to $\ell$, then $\mathrm{Tam}_p(M) = \mathbb{Z}_\ell$.*

*(iii) $\mathrm{Tam}_p(M)$ depends only on the isomorphism class of the $G_{\mathbb{Q}_p}$-module $M$; in particular $\mathrm{Tam}_p(\ell^b M) = \mathrm{Tam}_p(M)$ for a lattice $M$ in a representation.*

*(iv) $\mathrm{Tam}_p(\mathrm{ad}^0 T) = \mathrm{Tam}_p(\mathrm{ad}^0(T \otimes \chi))$ for every character $\chi$ of $G_{\mathbb{Q}_p}$ with values in $\mathbb{Z}_\ell^\times$.*

*In particular $\mathrm{Tam}_p(\mathrm{ad}^0 T_\ell E) = \mathbb{Z}_\ell$ at every prime $p \nmid N\ell$, and at every prime with $\rho(I_p) = \{\pm 1\}$.*

*Proof.* (i) $H^1(I_p, M) = \mathrm{Hom}_{\mathrm{cont}}(I_p, M)$, and a continuous homomorphism into the pro-$\ell$ group $M$ factors through the maximal pro-$\ell$ quotient of $I_p$, which is $\mathbb{Z}_\ell(1)$ for $p \neq \ell$ (Chapter 7); so $H^1(I_p, M) \simeq \mathrm{Hom}(\mathbb{Z}_\ell(1), M) = M(-1)$ is torsion-free. (ii) is the case $\#\Delta$ prime to $\ell$ of Lemma 18.59 below: the torsion is $H^1(\Delta, M)$, killed by $\#\Delta$, which is a unit. (iii) The definition uses only the module. (iv) The two modules are equal (Proposition 18.10(ii)). For the last sentence: at a good prime $\mathrm{ad}^0 T$ is unramified, and if $\rho(I_p) = \{\pm 1\}$ then $I_p$ acts trivially on $\mathrm{ad}^0 T$. □

**Theorem 18.49** (Fontaine and Perrin-Riou at $p \neq \ell$)**.** *Let $p \neq \ell$ and let $X$ be a free $\mathbb{Z}_\ell$-module of finite rank with a continuous action of $G_{\mathbb{Q}_p}$ such that $H^0(\mathbb{Q}_p, X \otimes \mathbb{Q}_\ell) = 0$. Then $H^1_f(\mathbb{Q}_p, X) = H^1(\mathbb{Q}_p, X)_{\mathrm{tor}}$ is finite, there is an exact sequence*

$$0 \to X^{I_p}/(\mathrm{Frob}_p - 1)X^{I_p} \to H^1(\mathbb{Q}_p, X)_{\mathrm{tor}} \to \left(H^1(I_p, X)_{\mathrm{tor}}\right)^{G_{\mathbb{Q}_p}} \to 0,$$

*and*

$$\begin{aligned} \mathrm{Fitt}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}_p, X) &= \det(1 - \mathrm{Frob}_p^{-1} \mid X^{I_p}) \cdot \mathrm{Tam}_p(X), \\ \mathrm{length}\, H^1_f(\mathbb{Q}_p, X) &= v_\ell \det(1 - \mathrm{Frob}_p^{-1} \mid X^{I_p}) + v_\ell(\mathrm{Tam}_p(X)). \end{aligned}$$

*Moreover $H^1(\mathbb{Q}_p, X)_{\mathrm{tor}} \simeq H^0(\mathbb{Q}_p, X \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$. For the paper's $X = T^D$ and every $p \neq \ell$:*

$$\begin{aligned} \mathrm{Fitt}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}_p, T^D) &= L_p(A_f, 1)^{-1} \cdot \mathrm{Tam}_p(T^D), \\ \mathrm{length}\, H^1_f(\mathbb{Q}_p, T^D) &= \mathrm{length}\, H^0(\mathbb{Q}_p, W^D) = v_\ell(L_p(A_f, 1)^{-1}) + v_\ell(\mathrm{Tam}_p(T^D)). \end{aligned}$$

*Proof.* Write $G = G_{\mathbb{Q}_p}$, $I = I_p$, $F = \mathrm{Frob}_p$, $U = X \otimes \mathbb{Q}_\ell$.

*Step 1 (the unramified part is finite).* $H^0(\mathbb{Q}_p, U) = (U^I)^{F=1} = 0$, so 1 is not an eigenvalue of $F$, nor of $F^{-1}$, on $U^I$, and $\det(1 - F^{-1} \mid X^I) \neq 0$. By Proposition 18.15(c) the group $X^I/(F-1)X^I$ is finite of order $\ell^{v_\ell \det(1 - F^{-1} \mid X^I)}$; its Fitting ideal is generated by $\det(1 - F^{-1} \mid X^I)$.

*Step 2 (inflation-restriction).* By Proposition 18.32(ii) the sequence $0 \to X^I/(F-1)X^I \to H^1(\mathbb{Q}_p, X) \to H^1(I, X)^{G/I} \to 0$ is exact.

*Step 3 (torsion).* Taking torsion subgroups keeps it exact. Exactness at the left and in the middle is clear (the left term is finite, hence torsion). For surjectivity, let $y \in (H^1(I, X)^{G/I})_{\mathrm{tor}}$, say $\ell^k y = 0$, and lift it to $x \in H^1(\mathbb{Q}_p, X)$. Then $\ell^k x$ lies in the finite left term, so $\ell^{k+m} x = 0$ for some $m$, and $x$ is torsion. Finally $(H^1(I, X)^{G/I})_{\mathrm{tor}} = (H^1(I, X)_{\mathrm{tor}})^{G/I}$.

*Step 4 (Fitting ideals).* Lengths are additive in short exact sequences of finite $\mathbb{Z}_\ell$-modules, and $\mathrm{Fitt}(Y) = \ell^{\mathrm{length}\, Y}\mathbb{Z}_\ell$ for finite $Y$ (Chapter 3). So $\mathrm{Fitt}\, H^1(\mathbb{Q}_p, X)_{\mathrm{tor}} = \det(1 - F^{-1} \mid X^I) \cdot \mathrm{Tam}_p(X)$.

*Step 5 (the Bloch-Kato condition).* By Proposition 18.32(v), $H^1_f(\mathbb{Q}_p, X) = H^1(\mathbb{Q}_p, X)_{\mathrm{tor}}$. The isomorphism with $H^0(\mathbb{Q}_p, X \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is the connecting map of $0 \to X \to U \to X \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \to 0$, injective because $H^0(\mathbb{Q}_p, U) = 0$, with image $\ker \iota$ (Proposition 18.31(a)).

For $X = T^D$: $H^0(\mathbb{Q}_p, V^D) = 0$ and $\det(1 - F^{-1} \mid (T^D)^I) = L_p(A_f, 1)^{-1}$ by Lemma 18.18 and Lemma 18.21. □

A first example. Let $p$ be a prime of multiplicative reduction with $\ell \nmid n_p$, and $X = T^D = A(1)$. The invariants $X^{I_p}$ are the line $\mathbb{Z}_\ell e \otimes \mathbb{Z}_\ell(1)$ (proof of Proposition 18.17), on which $\mathrm{Frob}_p$ acts by $p \cdot p = p^2$; so the unramified part is $\mathbb{Z}_\ell/(p^2 - 1)$, of length $v_\ell(p^2 - 1) = v_\ell(1 - p^{-2})$. The Tamagawa part is trivial, since $H^1(I_p, A)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n_p)})^2 = 0$ (Lemma 18.53). Hence length $H^1_f(\mathbb{Q}_p, T^D) = v_\ell(p^2 - 1)$. For `11a1` at $p = 11$ and $\ell = 3$ ($n_{11} = 5$): $H^1_f(\mathbb{Q}_{11}, T^D) \simeq \mathbb{Z}_3/120\mathbb{Z}_3 \simeq \mathbb{Z}/3$. When $\ell \mid n_p$ the Tamagawa part is no longer trivial, and computing it is the subject of Section 18.8.

**Proposition 18.50** ($\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$)**.** *Let $\ell$ be odd, $p \neq \ell$, $A = \mathrm{ad}^0 T_\ell E$ and $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$. Then $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$. More precisely, $H^1(I_p, A(1)) = H^1(I_p, A)(1)$ as $\mathbb{Z}_\ell$-modules with $G_{\mathbb{Q}_p}/I_p$-action, so the two torsion subgroups are the same group with Frobenius actions differing by the factor $p$; and:*

*(a) if the reduction at $p$ is good or potentially good, both torsion subgroups are 0;*

*(b) if the reduction at $p$ is potentially multiplicative, with the notation of Lemma 18.53 (and Proposition 18.57), both invariant modules are isomorphic to $\mathbb{Z}/\ell^a \oplus \mathbb{Z}/\ell^m$ with $a = v_\ell(\nu)$ and $m = \min(a, v_\ell(p-1), v_\ell(c')) = \min(a, v_\ell(p-1), v_\ell(c(u)))$.*

*Proof.* $I_p$ acts trivially on $\mathbb{Z}_\ell(1)$, so $A(1) = A \otimes \mathbb{Z}_\ell(1)$ is the same $I_p$-module as $A$ tensored with a free rank-one module with trivial $I_p$-action; cochains and cohomology commute with $\otimes\mathbb{Z}_\ell(1)$, and $\mathrm{Frob}_p$ acts on $\mathbb{Z}_\ell(1)$ by $p$. (a) At good and potentially good primes $H^1(I_p, A)_{\mathrm{tor}} = 0$ by Proposition 18.48 and Theorem 18.61, so both factors are trivial. (b) By Lemma 18.53 (applied, at an additive potentially multiplicative prime, to the Tate shape of Proposition 18.57), $H^1(I_p, A)_{\mathrm{tor}} = (\mathbb{Z}/\ell^a)\bar{e} \oplus (\mathbb{Z}/\ell^a)\bar{h}$ with $\mathrm{Frob}_p$ acting by $p^{-1}\,\mathrm{Ad}\,\rho(\mathrm{Frob}_p)$, so on the twist it acts by $\mathrm{Ad}\,\rho(\mathrm{Frob}_p)$: $\bar{e} \mapsto p\bar{e}$, $\bar{h} \mapsto \bar{h} - 2\varepsilon c'\bar{e}$. Thus $x\bar{e} + y\bar{h}$ is fixed iff $(p-1)x - 2\varepsilon c' y \equiv 0 \pmod{\ell^a}$, the kernel of the linear form $\phi(x,y) = (p-1)x - 2\varepsilon c' y$ on $(\mathbb{Z}/\ell^a)^2$. The image of $\phi$ is the ideal generated by $p-1$ and $c'$ in $\mathbb{Z}/\ell^a$ ($2\varepsilon$ is a unit), of order $\ell^{a-m}$, so the kernel has order $\ell^{2a-(a-m)} = \ell^{a+m}$. Its structure: if $m = v_\ell(p-1) \le v_\ell(c')$ (and $m < a$), write $\phi = \ell^m \phi'$ with $\phi'(x,y) = \gamma x - \delta y$, $\gamma$ a unit; the kernel is $\{(x,y) : \gamma x \equiv \delta y \pmod{\ell^{a-m}}\}$, and $y$ is free in $\mathbb{Z}/\ell^a$ while $x$ is determined modulo $\ell^{a-m}$, giving $\mathbb{Z}/\ell^a \oplus \mathbb{Z}/\ell^m$; the case $m = v_\ell(c') < v_\ell(p-1)$ is symmetric, and if $m = a$ the form is 0. The invariants for $A$ itself are $\{x\bar{e} + y\bar{h} : (p-1)y \equiv 0,\ c'y \equiv 0\} = \mathbb{Z}/\ell^a \oplus \mathbb{Z}/\ell^m$ (Lemma 18.53). □

The general statement $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$ is [49, (57)], which the paper cites in Lemma lem:fpr; for the adjoint of an elliptic curve the proposition proves it in every case that occurs. As a check, an exhaustive computation over $\ell \in \{3,5\}$, $a \in \{0,1,2\}$, all primes $p < 120$ with $p \ne \ell$, $\varepsilon = \pm 1$ and $c' \in \{0, \dots, \ell^3 - 1\}$ (26,448 parameter choices) found both invariant modules of order $\ell^{a + \min(a, v_\ell(p-1), v_\ell(c'))}$ in every case.

**Proposition 18.51** (the Tamagawa factor of $T_\ell E$)**.** *For $p \ne \ell$, $v_\ell(\mathrm{Tam}_p(T_\ell E)) = v_\ell(c_p)$, the valuation of the Tamagawa number $c_p = [E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)]$.*

*Proof.* Let $T = T_\ell E$, $V$, $W = E[\ell^\infty]$, $I = I_p$. As in the proof of Proposition 18.32(iii), the connecting map identifies $W^I/(W^I)_{\mathrm{div}}$ with $H^1(I,T)_{\mathrm{tor}} = \ker(H^1(I,T) \to H^1(I,V))$, equivariantly for $G_{\mathbb{Q}_p}/I$ (the image of $V^I$ in $W^I$ is $(W^I)_{\mathrm{div}}$, as shown there). Now $W^I = E(\mathbb{Q}_p^{\mathrm{ur}})[\ell^\infty]$, and by Chapter 10 the group $E(\mathbb{Q}_p^{\mathrm{ur}})$ is an extension of $\Phi_p^{\mathrm{geom}} = \Phi_p(\overline{\mathbb{F}}_p)$ by $E_0(\mathbb{Q}_p^{\mathrm{ur}})$, which is an extension of the group $\tilde{E}_{\mathrm{ns}}(\overline{\mathbb{F}}_p)$ of nonsingular points of the reduction (an elliptic curve, $\mathbb{G}_m$ or $\mathbb{G}_a$ over $\overline{\mathbb{F}}_p$) by the kernel of reduction $E_1(\mathbb{Q}_p^{\mathrm{ur}})$. The field $\mathbb{Q}_p^{\mathrm{ur}}$ is not complete, but it is the union of the finite unramified extensions $K$ of $\mathbb{Q}_p$, and each $E_1(K)$ is a pro-$p$ group (the formal group over the maximal ideal of $\mathcal{O}_K$), hence uniquely $\ell$-divisible; so $E_1(\mathbb{Q}_p^{\mathrm{ur}})$ is uniquely $\ell$-divisible. The reduction map $E_0(K) \to \tilde{E}_{\mathrm{ns}}(\mathbb{F}_K)$ is onto for each $K$ (Hensel's lemma), and $\tilde{E}_{\mathrm{ns}}(\overline{\mathbb{F}}_p)$ is $\ell$-divisible; so $E_0(\mathbb{Q}_p^{\mathrm{ur}})$ is $\ell$-divisible, and its $\ell$-power torsion maps isomorphically onto that of $\tilde{E}_{\mathrm{ns}}(\overline{\mathbb{F}}_p)$, a divisible group ($(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^2$, $\mu_{\ell^\infty}$ or 0). Hence $0 \to E_0(\mathbb{Q}_p^{\mathrm{ur}})[\ell^\infty] \to W^I \to \Phi_p^{\mathrm{geom}}[\ell^\infty] \to 0$ is exact (Chapter 11, inertia invariants and component groups), the first term is $(W^I)_{\mathrm{div}}$ (a divisible subgroup with finite quotient), and $H^1(I,T)_{\mathrm{tor}} \simeq \Phi_p^{\mathrm{geom}}[\ell^\infty]$ as Frobenius modules. The Frobenius invariants of $\Phi_p^{\mathrm{geom}}$ are $\Phi_p(\mathbb{F}_p)$, of order $c_p$. □

At a multiplicative prime this is also the last step of the proof of Lemma 18.53: $H^1(I,T)_{\mathrm{tor}} = (\mathbb{Z}/\ell^{v_\ell(n)})\bar{e}_1$ with Frobenius acting by $\varepsilon$, so $v_\ell(\mathrm{Tam}_p(T)) = v_\ell(n)$ if the reduction is split and 0 if it is nonsplit. On all curves of Cremona's tables of conductor below 3,000 (17,834 curves), at every multiplicative prime $q$ and $\ell \in \{3,5,7\}$, $\ell \ne q$ (101,336 pairs), $v_\ell(c_q)$ is $v_\ell(n_q)$ if $a_q = 1$ and 0 if $a_q = -1$, as it should be. At a prime $p \ne 3$ of type IV or IV$^*$ with $\ell = 3$,

$H^1(I,T)_{\mathrm{tor}} = H^1(C_3,T) \simeq \mathbb{F}_3$ (Proposition 18.60), and the proposition says that Frobenius acts on it trivially exactly when $3 \mid c_p$ (for example `50a1` at $p=5$, type IV, $c_5=3$).

## 18.8 The Tamagawa factor of $\mathrm{ad}^0 T$ at a multiplicative prime: Lemma lem:tamfactor

This section proves the paper's Lemma lem:tamfactor in full, with every step the paper compresses expanded, and extends it to the additive potentially multiplicative primes, $q=2$ included. The lemma is the source of the terms $t_\ell(E,q)$ in (eq:thm) and of the refinement tested in Section sec:tam. The work is divided between three chapters: Chapter 16 computes the group cohomology of the Tate shape (the model computation: $H^1(I,\mathrm{ad}^0 T)$, its torsion, the Frobenius action and the invariants, for arbitrary corner values $n'$ and $c'$); Chapter 17 identifies $c'$ through the Kummer cocycle of $q_E$ and its indeterminacy $(n, q-1)$; this chapter is the home of the lemma itself: the identification with the Tamagawa factor, the assembly, and the additive potentially multiplicative case. The other two chapters state their results as corollaries pointing here.

**Convention 18.52** (Tate data)**.** Let $q \neq \ell$ be a prime of multiplicative reduction, $n = n_q$, $G = G_{\mathbb{Q}_q}$, $I = I_q$, and Frob an arithmetic Frobenius in $G$. We use:

(a) (Chapter 11, the Tate shape) a $\mathbb{Z}_\ell$-basis $e_1, e_2$ of $T$ in which $\rho(g) = \left(\begin{smallmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa(g) \\ 0 & \varepsilon(g) \end{smallmatrix}\right)$ for $g \in G$, where $\kappa$ is a Kummer cocycle of $q_E$ with values in $\mathbb{Z}_\ell(1) \simeq \mathbb{Z}_\ell$ (the corner is $\varepsilon\kappa$, not $\kappa$: only $\varepsilon\kappa$ satisfies the cocycle relation for these matrices when $\varepsilon \neq 1$);

(b) (Chapter 7) $\kappa|_I = n\, t_\ell$ with $t_\ell$ the tame character, so a topological generator $t$ of the pro-$\ell$ tame quotient of $I$ acts by $te_1 = e_1$, $te_2 = e_2 + n'e_1$ with $v_\ell(n') = v_\ell(n)$, and the kernel of $I \to \langle t \rangle \simeq \mathbb{Z}_\ell$ has no nontrivial pro-$\ell$ quotient;

(c) (Chapter 7) the tame relation $\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q$, equivalently $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ with $q^{-1} \in \mathbb{Z}_\ell^\times$;

(d) (Chapter 17) $\kappa = n\kappa_q + \kappa_u$ for $q_E = q^n u$, with $\kappa_u$ unramified, and the canonical splitting $H^1(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell \kappa_q \oplus H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Z}_\ell(1))$ given by the uniformiser $q$;

(e) (Chapter 17) $H^1_{\mathrm{ur}}(\mathbb{Q}_q, \mathbb{Z}_\ell(1)) = H^1(G/I, \mathbb{Z}_\ell(1)) = \mathbb{Z}_\ell(1)/(q-1) \simeq \mathbb{F}_q^\times \otimes \mathbb{Z}_\ell$, under which the class of $\kappa_u$ is the class of $\bar{u}$.

We write $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$ with $\varepsilon = \varepsilon(\mathrm{Frob}) = a_q(E)$ and $c' = \varepsilon\kappa(\mathrm{Frob})$.

**Lemma 18.53** (the Tamagawa factor at a multiplicative prime)**.** *Let $q$ be a prime of multiplicative reduction, $n = n_q = v_q(\Delta_E)$, $\ell$ an odd prime different from $q$, $M = \mathrm{ad}^0 T$, and $q_E$, $u$, $c(u)$ as in Convention 18.1(f). Then $H^1(I_q, M)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2$ and*

$$v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(n) + \min(v_\ell(n),\, v_\ell(q-1),\, v_\ell(c(u))).$$

*In particular $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) \geq t_\ell(E,q) = v_\ell(n)$, with equality unless $q \equiv 1 \pmod{\ell}$, $\ell \mid n$ and $\bar{u}$ is an $\ell$-th power in $\mathbb{F}_q^\times$.*

*Proof. Step 1 (the shape on inertia).* By Convention 18.52(a), (b), $I$ acts on $T$ through $\langle t \rangle$, with $t = \left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$ in the basis $e_1, e_2$ ($\varepsilon$ is trivial on $I$).

*Steps 2 to 4 (the group cohomology).* These are the model computation of Chapter 16, which we quote for the matrices of Convention 18.52, where $v_\ell(n') = v_\ell(n)$. The kernel $K$ of $I \to \langle t\rangle$ has no nontrivial pro-$\ell$ quotient, so inflation-restriction gives $H^1(I,M) = H^1(\langle t\rangle, M) = M/(t-1)M$; here a class is determined by the value $c(t)$ of a cocycle, read in $M/(t-1)M$, as for any procyclic group (Chapter 16). In the basis $e = e_1 \otimes e_2^\vee = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $h = e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $f = e_2 \otimes e_1^\vee = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ of $M$, conjugation by $t = \left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$ gives

$$tet^{-1} = e, \qquad tht^{-1} = h - 2n'e, \qquad tft^{-1} = f + n'h - n'^2 e,$$

so $(t-1)(ae+bh+cf) = -(2n'b + n'^2 c)e + n'c\,h$. The image is spanned by $2n'e$ and $n'h - n'^2 e$; as 2 is a unit ($\ell$ odd) and $n'^2 e \in n'\mathbb{Z}_\ell e$, it is $(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$, so

$$H^1(I,M) = (\mathbb{Z}_\ell/n')\bar{e} \oplus (\mathbb{Z}_\ell/n')\bar{h} \oplus \mathbb{Z}_\ell \bar{f}, \qquad H^1(I,M)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2.$$

By the tame relation $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ and $c(t^a) \equiv a\, c(t)$ modulo $(t-1)M$, the *arithmetic* Frobenius acts on $M/(t-1)M$ as $q^{-1}\,\mathrm{Ad}\,\rho(\mathrm{Frob})$, that is, on the torsion by $\bar{e} \mapsto \bar{e}$ and $\bar{h} \mapsto q^{-1}(\bar{h} - 2\varepsilon c'\bar{e})$. In detail: the cocycle relation gives $c(t^a) = (1 + t + \cdots + t^{a-1})c(t) \equiv a\, c(t)$ modulo $(t-1)M$ for integers $a \geq 1$, and by continuity for every $a \in \mathbb{Z}_\ell$; with the action of Definition 18.45,

$$(\mathrm{Frob}\cdot c)(t) = \mathrm{Frob}\; c(\mathrm{Frob}^{-1}\, t\, \mathrm{Frob}) = \mathrm{Frob}\; c(t^{q^{-1}}) \equiv q^{-1}\; \mathrm{Frob}\; c(t).$$

Conjugation by $g = \rho(\mathrm{Frob}) = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$ sends $e \mapsto qe$ (the ratio $\varepsilon q/\varepsilon$ of the diagonal entries) and $h \mapsto h - 2\varepsilon c' e$ (multiply out, using $\varepsilon^2 = 1$). So $x\bar{e} + y\bar{h}$ goes to $x\bar{e} + q^{-1}y\bar{h} - 2q^{-1}\varepsilon c' y\,\bar{e}$, and it is fixed if and only if $(q^{-1}-1)y \equiv 0$ and $2q^{-1}\varepsilon c' y \equiv 0$ modulo $\ell^{v_\ell(n)}$; as $q$ and $2\varepsilon$ are units, this is $(q-1)y \equiv c'y \equiv 0$. The coordinate $x$ is free, and the admissible $y$ form the subgroup of $\mathbb{Z}/\ell^{v_\ell(n)}$ of elements divisible by $\ell^{v_\ell(n)-m}$ with $m = \min(v_\ell(n), v_\ell(q-1), v_\ell(c'))$, a cyclic group of order $\ell^m$. Hence

$$\begin{aligned}(H^1(I,M)_{\mathrm{tor}})^G = \{x\bar{e} + y\bar{h} :\ (q-1)y \equiv c'y \equiv 0 \pmod{\ell^{v_\ell(n)}}\} \\ \simeq \mathbb{Z}/\ell^{v_\ell(n)} \oplus \mathbb{Z}/\ell^{\min(v_\ell(n),\, v_\ell(q-1),\, v_\ell(c'))}.\end{aligned}$$

*Step 5 (the class $c'$).* The number $c' = \varepsilon\kappa(\mathrm{Frob})$ is a cocycle value, not a class: changing the Frobenius lift or the cocycle changes it by an element of $(n, q-1)\mathbb{Z}_\ell$, which does not change $\min(v_\ell(n), v_\ell(q-1), v_\ell(c'))$, and for the lift fixing a compatible system of $\ell$-power roots of $q$ its class modulo $q-1$ is $\varepsilon$ times the class of $\bar{u}$. Hence $\min(v_\ell(n), v_\ell(q-1), v_\ell(c')) = \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$ for every choice. This is the theorem of Chapter 17 on the corner of $\rho(\mathrm{Frob})$ at a multiplicative prime, with Convention 18.1(f) for the meaning of $v_\ell(c(u))$; that theorem is local (it uses only the $G_{\mathbb{Q}_q}$-module $T$ in its Tate shape and the Kummer cocycle of $q_E$), which matters in Proposition 18.57.

*Step 6 (conclusion).* By Definition 18.45, $\mathrm{Tam}_q(\mathrm{ad}^0 T)$ is the Fitting ideal of the finite module of Steps 2 to 4, whose length is the sum of the exponents; with Step 5, $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$. The excess $\min(\dots)$ is positive iff $\ell \mid n$, $\ell \mid q-1$ and $\bar{u} \in (\mathbb{F}_q^\times)^\ell$. As a check, the model computation for $M = T$ (Chapter 16) gives $H^1(I,T)_{\mathrm{tor}} = (\mathbb{Z}/\ell^{v_\ell(n)})\bar{e}_1$ with Frob acting by $q^{-1}\cdot\varepsilon q = \varepsilon$; the invariants have length $v_\ell(n)$ if $\varepsilon = 1$ and 0 if $\varepsilon = -1$ ($\ell$ odd), the $\ell$-part of $c_q$ (Proposition 18.51). □

*In the paper* 18.54. The proof of Lemma lem:tamfactor in the paper is Steps 1 to 6 in compressed form. The corner of the matrices is $\varepsilon\kappa$, not $\kappa$ (Step 1); the Frobenius in the matrices is arithmetic (Steps 2 to 4), and it acts on $M/(t-1)M$ by $q^{-1}\operatorname{Ad}\rho(\mathrm{Frob})$, so that "the adjoint action sends $e$ to $qe$". The value $c'$ is handled in Step 5: the unramified projection of $\kappa$ exists only through the splitting attached to the uniformiser $q$, and the indeterminacy of $c'$ is $(n, q-1)$. The paper's proof says the same: "it is $\varepsilon$ times the class of $u$ for the Frobenius lift fixing a compatible system of $\ell$-power roots of $q$, and in general it is well defined only modulo $\ell^{\min(v_\ell(n),v_\ell(q-1))}$". The identification of $\mathrm{Tam}_q$ with the Fitting ideal of the invariants of the torsion is Definition 18.45.

**Corollary 18.55** (consequences)**.** *In the situation of Lemma 18.53:*

*(i)* $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) - t_\ell(E,q) = \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u))) \in \{0,1,\dots,v_\ell(n)\}$;

*(ii) the excess is positive iff* $\ell \mid n$, $q \equiv 1 \pmod{\ell}$ *and* $\bar{u} \in (\mathbb{F}_q^\times)^\ell$; *it is* 0 *for* $q = 2$;

*(iii) (Remark rem:ko(1)) at such a prime* $v_\ell(\mathrm{Tam}_q) \geq t_\ell(E,q)+1$, *and the excess exceeds* 1 *only if* $\ell^2 \mid n$, $\ell^2 \mid q-1$ *and* $\bar{u} \in (\mathbb{F}_q^\times)^{\ell^2}$;

*(iv) the excess does not depend on whether the reduction is split.*

*Proof.* (i) to (iii) are read off the formula; $v_\ell(2-1) = 0$. (iv) The formula involves only $n$, $q$ and $\bar{u}$; $\varepsilon$ entered Steps 2 to 4 only through the unit $2\varepsilon$. □

*In the paper* 18.56. Remark rem:ko(1) says that at a potentially multiplicative prime $q \equiv 1 \pmod{\ell}$ with $\ell \mid n$ whose Tate parameter has an $\ell$-th power as unit part, Lemma lem:tamfactor gives $v_\ell(\mathrm{Tam}_q) \geq t_\ell(E,q)+1$, and that (eq:identity) raises the bound accordingly. This is (iii), and Proposition 18.57 covers the potentially multiplicative primes. The abstract's sentence on the Tamagawa factor ("by at least one", with the condition $\ell \mid n_q$) is (ii) and (iii): without $\ell \mid n_q$ the excess is 0 whatever the unit part.

**Proposition 18.57** (potentially multiplicative additive primes)**.** *Let* $q \neq \ell$ *($q = 2$ allowed) be a prime of additive, potentially multiplicative reduction,* $\nu = -v_q(j_E)$, $q_E$ *the Tate parameter (determined by* $j(q_E) = j_E$*), and* $u = q_E q^{-\nu}$. *Then*

$$v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(\nu) + \min\big(v_\ell(\nu), v_\ell(q-1), v_\ell(c(u))\big) \;\geq\; t_\ell(E,q).$$

*Proof.* Over $\mathbb{Q}_q$, $E$ is the twist of the Tate curve $E_{q_E}$ by a ramified quadratic character $\chi$ of $G_{\mathbb{Q}_q}$ (Chapter 10, Chapter 11), so $T_\ell E \simeq T_\ell E_{q_E} \otimes \chi$ and $\mathrm{ad}^0 T_\ell E \simeq \mathrm{ad}^0 T_\ell E_{q_E}$ as $G_{\mathbb{Q}_q}$-modules (Proposition 18.10(ii)). The Tate curve $E_{q_E}$ has split multiplicative reduction with $v_q(\Delta) = v_q(q_E) = \nu$ and Tate parameter $q_E$, and the proof of Lemma 18.53 uses only the $G_{\mathbb{Q}_q}$-module $\mathrm{ad}^0 T$ in its Tate shape (with $\varepsilon = 1$). The results of Chapters 16 and 17 quoted in Steps 2 to 5 are stated there for a curve over $\mathbb{Q}$, but their proofs are local: they use only the Tate shape of the $G_{\mathbb{Q}_q}$-module and the Kummer cocycle of the Tate parameter. So they apply verbatim to the Tate curve $E_{q_E}$ over $\mathbb{Q}_q$, and the formula holds with $n = \nu$. At $q = 2$, $v_\ell(q-1) = 0$ and the value is $t_\ell(E,2) = v_\ell(-v_2(j_E))$. □

For odd $q$ one may instead twist globally, by $\chi_{q^*}$, using Lemma lem:twist; for $q = 2$ that lemma does not apply, and the local twist is needed. The paper states the formula at additive potentially multiplicative primes in Lemma lem:tamfactor itself, with the local argument. At $q = 2$ the index of a Kodaira symbol $\mathrm{I}_m^*$ need not be $-v_2(j)$, and the reduction may be potentially good (Chapter 10); the proposition uses $\nu = -v_2(j_E)$.

**Example 18.58** (Tate data and Tamagawa factors)**.** The table lists, for some curves and primes $q$ of potentially multiplicative reduction, the Kodaira type, $\nu$, the reduction $\bar{u}$ of the unit part of the Tate parameter, $\ell$, $v = v_\ell(q-1)$, $k = v_\ell(c(u))$, the resulting $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T))$, and $t_\ell = v_\ell(\nu)$. The unit part was computed from $j_E$: since $q_E = j_E^{-1} + 744 j_E^{-2} + \cdots$ with integer coefficients and $v_q(j_E^{-1}) = \nu \geq 1$, $\bar{u}$ is the reduction of $j_E^{-1} q^{-\nu}$; for `11a1` and `791a1` this agrees with PARI's `ellinit(E,O(q^20)).tate`.

| curve, $q$ | type | $\nu$ | $\bar{u}$ | $\ell$ | $v$ | $k$ | $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T))$ | $t_\ell$ |
|---|---|---|---|---|---|---|---|---|
| `11a1`, 11 | $\mathrm{I}_5$ | 5 | 10 | 5 | 1 | 1 | 2 | 1 |
| `14a1`, 7 | $\mathrm{I}_3$ | 3 | 6 | 3 | 1 | 1 | 2 | 1 |
| `539c1`, 7 | $\mathrm{I}_3^*$ | 3 | 5 | 3 | 1 | 0 | 1 | 1 |
| `791a1`, 7 | $\mathrm{I}_3$ | 3 | 6 | 3 | 1 | 1 | 2 | 1 |
| `791a1`, 113 | $\mathrm{I}_2$ | 2 | 106 | 3 | 0 | 0 | 0 | 0 |
| `1184f1`, 37 | $\mathrm{I}_3$ | 3 | 36 | 3 | 2 | 2 | 2 | 1 |
| `1748e1`, 19 | $\mathrm{I}_6$ | 6 | 12 | 3 | 2 | 1 | 2 | 1 |

The first row in detail. For `11a1`, $j_E = -2^{12} \cdot 31^3/11^5$, so $\nu = 5$ and $\bar{u}$ is the reduction of $j_E^{-1} 11^{-5} = -1/(2^{12} \cdot 31^3)$ modulo 11. Since $2^{12} = 2^{10} \cdot 4 \equiv 4$ and $31^3 \equiv 9^3 = 729 \equiv 3$, the denominator is $\equiv 12 \equiv 1$, and $\bar{u} = -1 \equiv 10$. With $\ell = 5$: $v = v_5(10) = 1$, and the fifth powers in $\mathbb{F}_{11}^\times$ (cyclic of order 10) form the subgroup $\{\pm 1\}$, which contains 10; so $k = 1$, the excess is $\min(v_5(5), 1, 1) = 1$, and $v_5(\mathrm{Tam}_{11}(\mathrm{ad}^0 T)) = 1 + 1 = 2$. In the same way, for `1748e1` at 19: $\mathbb{F}_{19}^\times$ is cyclic of order 18, the cubes are the elements with $x^6 = 1$ and the ninth powers those with $x^2 = 1$; $12^6 \equiv 1$ but $12^2 \equiv 11$, so $k = 1$. For `11a1` at $\ell = 5$ and `14a1` at $\ell = 3$ the residual representation is reducible: the lemma applies (it is local and needs no irreducibility), but Theorem thm:main does not; `14a1` has $\deg \varphi_E = 1$. `539c1` has the same $j$-invariant, hence the same Tate parameter at 7, as its twist `77c1`; 5 is not a cube modulo 7. In `1184f1` the cap $v_\ell(\nu) = 1$ binds although $\bar{u} = -1$ is a ninth power modulo 37; in `1748e1`, 12 is a cube but not a ninth power modulo 19. The curves `791a1`, `1184f1`, `1748e1` have $3 \nmid N$, $E[3]$ irreducible, mod 3 image not `3Ns`, and no CM; they reappear in Example 18.76.

## 18.9 Tamagawa factors at additive primes, and Open question 3

This section computes the Tamagawa factors of $\mathrm{ad}^0 T$ at the additive primes: they are trivial at every potentially good additive prime, for every odd $\ell$, and given by Proposition 18.57 at the potentially multiplicative ones. These are the paper's Lemma lem:tamadd and, with Lemma lem:tamfactor, its Corollary cor:explicit; Open question 3 of the paper starts from them.

**Lemma 18.59** (finite inertia image)**.** *Let $q \neq \ell$ and let $M$ be a free $\mathbb{Z}_\ell$-module of finite rank on which $I = I_q$ acts continuously through a finite quotient $\Delta = I/K$. Then inflation identifies $H^1(\Delta, M)$ with $H^1(I, M)_{\mathrm{tor}}$, and $H^1(I, M)/H^1(\Delta, M)$ embeds in $\mathrm{Hom}(K, M)^\Delta \subseteq M(-1)$, which is torsion-free.*

*Proof.* Inflation-restriction (Chapter 16) gives $0 \to H^1(\Delta, M) \to H^1(I, M) \to \mathrm{Hom}_{\mathrm{cont}}(K, M)^\Delta$. The group $K$ is the inertia group of a finite extension of $\mathbb{Q}_q^{\mathrm{ur}}$, and a continuous homomorphism from $K$ to the pro-$\ell$ group $M$ factors through the maximal pro-$\ell$ abelian quotient of $K$, which is $\mathbb{Z}_\ell(1)$ as $q \neq \ell$ (Chapter 7); so $\mathrm{Hom}(K, M) \simeq M(-1)$ is torsion-free. $H^1(\Delta, M)$ is killed by $\#\Delta$, hence torsion, and it is therefore the whole torsion subgroup. □

**Proposition 18.60** (the adjoint lattice is free over $\mathbb{Z}_3[C_3]$)**.** *Let $\sigma \in \mathrm{GL}_2(\mathbb{Z}_3)$ have order 3, $T = \mathbb{Z}_3^2$ with $\sigma$ acting, and $P = \langle\sigma\rangle$. Then $\mathrm{ad}^0 T$, with $P$ acting by conjugation, is a free $\mathbb{Z}_3[P]$-module of rank one; hence $H^1(P, \mathrm{ad}^0 T) = 0$. By contrast $H^1(P, T) \simeq \mathbb{F}_3$.*

*Proof.* The eigenvalues of $\sigma$ are cube roots of unity; as $\mathbb{Q}_3$ contains no primitive cube root of unity, the characteristic polynomial, which has coefficients in $\mathbb{Z}_3$, is $(X-1)^2$ or $X^2+X+1$, and $(X-1)^2$ would make the semisimple element $\sigma$ (it has finite order) equal to 1. So $\mathbb{Z}_3[\sigma] \simeq \mathbb{Z}_3[X]/(X^2+X+1) \simeq \mathbb{Z}_3[\zeta_3]$, a discrete valuation ring with uniformiser $1-\zeta_3$ (Chapter 6), and $T$, torsion-free of $\mathbb{Z}_3$-rank 2, is free of rank one over it. Choosing a generator $e_1$ and the basis $e_1, \sigma e_1$ gives $\sigma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$. In the basis $e, h, f$ of $\mathrm{ad}^0 T$,

$$e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \qquad \sigma e \sigma^{-1} = \left(\begin{smallmatrix} 0 & 0 \\ -1 & 0 \end{smallmatrix}\right) = -f, \qquad \sigma^2 e \sigma^{-2} = \left(\begin{smallmatrix} -1 & 1 \\ -1 & 1 \end{smallmatrix}\right) = e - h - f,$$

and the matrix of coordinates $\left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 0 & -1 \\ 0 & -1 & -1 \end{smallmatrix}\right)$ has determinant $-1$. So $e, \sigma e, \sigma^2 e$ is a $\mathbb{Z}_3$-basis and $\mathrm{ad}^0 T = \mathbb{Z}_3[P]\, e \simeq \mathbb{Z}_3[P]$. For a cyclic group, $H^1(P, \mathbb{Z}_3[P]) = \ker N/(\sigma-1)\mathbb{Z}_3[P] = 0$, where $N = 1+\sigma+\sigma^2$: for $x = a + b\sigma + c\sigma^2$ one has $Nx = (a+b+c)N$, so $\ker N = \{a+b+c=0\}$, and such an $x$ equals $(\sigma-1)(b+c(1+\sigma))$. On $T$, $N = 0$ (as $\sigma^2+\sigma+1=0$), so $H^1(P,T) = T/(\sigma-1)T \simeq \mathbb{Z}_3[\zeta_3]/(1-\zeta_3) \simeq \mathbb{F}_3$. □

Conceptually: modulo 3, $\sigma$ is a nontrivial unipotent on $T/3T$, and $\mathrm{ad}^0(T/3T) \simeq \mathrm{Sym}^2(T/3T) \otimes \det^{-1}$ is the symmetric square of a $2\times 2$ Jordan block, a single $3\times 3$ Jordan block in characteristic 3, that is, the regular module $\mathbb{F}_3[C_3]$; Nakayama's lemma lifts this to $\mathbb{Z}_3$. The coordinates above were also checked in PARI.

**Theorem 18.61** (Tamagawa factors at additive primes)**.** *Let $\ell$ be odd, $q \neq \ell$ a prime of additive reduction of $E$, and $A = \mathrm{ad}^0 T_\ell E$.*

*(i) If the reduction at $q$ is potentially good, then $H^1(I_q, A)_{\mathrm{tor}} = 0$, so $\mathrm{Tam}_q(A) = \mathrm{Tam}_q(A(1)) = \mathbb{Z}_\ell$.*

*(ii) If it is potentially multiplicative, $\mathrm{Tam}_q(A)$ is given by Proposition 18.57.*

*Proof.* (ii) is Proposition 18.57. (i) The image $\rho(I_q)$ is finite and its elements have characteristic polynomials in $\mathbb{Z}[X]$ (Serre and Tate; Chapter 11), so their orders lie in $\{1,2,3,4,6\}$. Let $\Delta$ be the image of $I_q$ in $\mathrm{Aut}(A)$, that is, $\rho(I_q)/(\rho(I_q) \cap \{\pm 1\})$. By Lemma 18.59, $H^1(I_q, A)_{\mathrm{tor}} = H^1(\Delta, A)$, which is killed by $\#\Delta$, a product of powers of 2 and 3. If $\ell \geq 5$ it is 0. Let $\ell = 3$. Restriction followed by corestriction is multiplication by the index, so the 3-primary group $H^1(\Delta, A)$ injects into $H^1(P, A)$ for a Sylow 3-subgroup $P$ of $\Delta$ (Chapter 16). A Sylow 3-subgroup $P'$ of $\rho(I_q)$ maps isomorphically onto $P$ (the kernel $\{\pm 1\}$ has order prime to 3). $P'$ has exponent 3; if it had order at least 9 it would contain a subgroup $C_3 \times C_3$ (a group of order 9 is abelian), whose elements, being commuting and of finite order, are simultaneously diagonalisable over $\mathbb{C}$ inside $\mathrm{SL}_2$ ($\det\rho = \chi_\ell$ is trivial on $I_q$), hence of the form $\mathrm{diag}(a, a^{-1})$ with $a^3 = 1$, a group of order 3; a contradiction. So $P$ is trivial or generated by the image of an element $\sigma$ of order 3, and $H^1(P, A) = 0$ by Proposition 18.60. Finally $H^1(I_q, A(1)) = H^1(I_q, A)(1)$ (Proposition 18.50). □

So $\mathrm{Tam}_q(\mathrm{ad}^0 T)$ is trivial at every potentially good additive prime, of every Kodaira type, at every $q \neq \ell$ including $q = 2, 3$, even for $\ell = 3$ at the types II, II$^*$, IV, IV$^*$, where $\mathrm{Tam}_q(T)$ can be divisible by 3 (Proposition 18.51).

*In the paper* 18.62. Lemma lem:tamadd of the paper is (i), with the same proof: inflation-restriction through the finite image of inertia, element orders $1, 2, 3, 4, 6$, and the freeness of $\mathrm{ad}^0 T$ over $\mathbb{Z}_3[P]$ shown by the same three coordinate vectors.

**Corollary 18.63** (the explicit identity)**.** *Let $\ell$ be odd, $\ell \nmid N$. Then*

$$\sum_{q|N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = \sum_{q:\, v_q(j_E)<0} \Big(t_\ell(E,q) + \min\big(t_\ell(E,q), v_\ell(q-1), v_\ell(c(u_q))\big)\Big),$$

*with $u_q$ the unit part of the Tate parameter at $q$. Consequently, under the hypothesis of Theorem thm:main ($E$ optimal, $\ell \notin S_f$), (eq:identity) reads*

$$v_\ell(\deg \varphi_E) = \mathrm{length}\, H^1_f(\mathbb{Q}, W) + \sum_{q|N} e_\ell(E,q) + \sum_{q:\, v_q(j_E)<0} \Big(t_\ell(E,q) + \min\big(t_\ell(E,q), v_\ell(q-1), v_\ell(c(u_q))\big)\Big).$$

*Proof.* The first identity: Lemma 18.53 at the multiplicative primes, Proposition 18.57 at the additive potentially multiplicative ones, and Theorem 18.61(i) at the potentially good additive primes; $t_\ell(E,q) = v_\ell(\nu_q)$. The second: Theorem thm:main, whose proof is assembled in Chapter 20 from Theorem 18.73 below and the congruence-ideal theorems, with the equality of (eq:identity) for the optimal curve proved in Chapter 14 (Proposition prop:pullback). □

This is the paper's Corollary cor:explicit. The excess of $v_\ell(\deg \varphi_E)$ over the explicit local bound is the length of the adjoint Bloch-Kato Selmer group. What remains of Open question 3 is that Selmer group.

*Remark* 18.64 (the excess on the tables). We computed the excess $v_\ell(\deg \varphi_E) - [\sum_q e_\ell(E,q) + \sum_{v_q(j)<0}(t_\ell + \min(t_\ell, v_\ell(q-1), v_\ell(c(u_q))))]$ on the optimal curves of Cremona's tables of conductor below 10,000 without complex multiplication, with $\ell \nmid N$, $E[\ell]$ irreducible and, for $\ell = 3$, mod 3 image not `3Ns` (the optimal curve is the one listed in Cremona's `degphi` file; irreducibility and the image are read from `galrep`; $e_\ell$ from PARI's `lfunsympow`, as in the paper's `rule`):

| $\ell$ | curves | excess 0 | 1 | 2 | 3 | $\geq 4$ |
|---|---|---|---|---|---|---|
| 3 | 13,203 | 8,688 | 3,288 | 942 | 248 | 37 |
| 5 | 23,361 | 19,019 | 3,993 | 328 | 19 | 2 |
| 7 | 28,134 | 25,133 | 2,888 | 111 | 2 | 0 |

The excess is never negative. By Corollary 18.63 it is length $H^1_f(\mathbb{Q}, W)$; so the adjoint Selmer group vanishes for 65.8% of these curves at $\ell = 3$, 81.4% at $\ell = 5$ and 89.3% at $\ell = 7$. The Tamagawa factor at $q = \ell$ is not defined by Definition 18.45 (it needs $p$-adic Hodge theory) and belongs to Open question 1 (Chapter 20).

## 18.10 The comparison of Selmer groups: Lemma lem:dfg21 and $H^0(\mathbb{Q}, W^D)$

This section proves the paper's Lemma lem:dfg21 (Lemma 2.1 of Diamond, Flach and Guo) by specialising the abstract comparison of Selmer structures of Chapter 17 to the Bloch-Kato

conditions, identifies its last term with $H^0(\mathbb{Q}, W^D)$ (Lemma lem:h0), and derives the length identity that the proof of Theorem thm:main and Remark rem:ns use. It also states Theorem thm:dfg27, which supplies the hypothesis of the lemma.

**Definition 18.65** (the exceptional set)**.** For $f = f_E$ (weight 2, trivial character, rational coefficients) the *exceptional set* of Diamond, Flach and Guo [49, Section 0.2 and (31)] (there: $\lambda \mid Nk!$, or the residual representation is not absolutely irreducible on $G_F$, $F = \mathbb{Q}(\sqrt{\ell^*})$) is

$$S_f = \{\ell : \ \ell \mid 2N, \text{ or } \bar{\rho}_{E,\ell}|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))} \text{ is not absolutely irreducible}\}.$$

The hypothesis of Theorem thm:main is $\ell \notin S_f$.

For $\ell \nmid 2N$ the following are equivalent (the paper's Proposition prop:Sf, proved in Chapter 11): $\ell \notin S_f$; $E[\ell]$ is irreducible and $h_\ell(E) = 0$; $E[\ell]$ is irreducible and, if $\ell = 3$, the mod 3 image is not $N_s(\mathbb{F}_3)$. In particular for $\ell \geq 5$ the condition on $\mathbb{Q}(\sqrt{\ell^*})$ is automatic once $E[\ell]$ is irreducible.

**Theorem 18.66** (Diamond, Flach and Guo, Theorem 2.7)**.** *If $\ell \notin S_f$, then $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$ and $H^0(\mathbb{Q}, A_f) = H^1_f(\mathbb{Q}, A_f) = 0$.*

*Sketch of proof.* This is [49, Theorem 2.7], the paper's Theorem thm:dfg27; we do not prove it. The $H^0$ statements hold for every $E$ with $\bar{\rho}$ absolutely irreducible (Proposition 18.14). The content is the vanishing of the adjoint Selmer group with $\mathbb{Q}_\ell$ coefficients, which comes from the Taylor-Wiles method: the identification of a universal deformation ring with a Hecke algebra forces the tangent space $H^1_f(\mathbb{Q}, A_f)$ to vanish. The published proof ([49], pp. 699 to 700, checked) derives it from the other theorem of Diamond, Flach and Guo that the paper uses, as follows (Theorem thm:dfg37, Chapter 20): it says that $H^1_\Sigma(\mathbb{Q}, W)$ is finite, so $H^1_f(\mathbb{Q}, W) \subseteq H^1_\Sigma(\mathbb{Q}, W)$ is finite, so $H^1_f(\mathbb{Q}, A_f) = 0$ (Proposition 18.38(iii)), so $H^1_f(\mathbb{Q}, V^D) = 0$ (Proposition 18.43). □

**Proposition 18.67** (Lemma lem:h0 in Selmer form)**.** *Let $\ell$ be odd and $\bar{\rho}$ irreducible (equivalently, absolutely irreducible: complex conjugation has distinct eigenvalues $\pm 1$).*

*(i) If $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$, then $H^1_f(\mathbb{Q}, T^D) = H^1(\mathbb{Q}, T^D)_{\mathrm{tor}}$, and the connecting map is an isomorphism $H^0(\mathbb{Q}, W^D) \xrightarrow{\sim} H^1_f(\mathbb{Q}, T^D)$.*

*(ii) (Lemma lem:h0, Chapter 11) $H^0(\mathbb{Q}, W^D) = H^0(\mathbb{Q}, \mathrm{ad}^0\, T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is finite and cyclic, of length $h_\ell(E)$, the largest $k$ with $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$. Moreover $h_\ell(E) = 0$ for $\ell \geq 5$, and $h_3(E) \leq 1$ with equality iff $\bar{\rho}_{E,3} \simeq \bar{\rho}_{E,3} \otimes \chi_{-3}$, iff the image of $\bar{\rho}_{E,3}$ is conjugate to $N_s(\mathbb{F}_3)$. This holds for every $E$, with or without complex multiplication, and whether or not $\ell \mid N$.*

*Proof.* (i) By Proposition 18.38(i) for the lattice $T^D$, $H^1_f(\mathbb{Q}, T^D) = \iota^{-1}(H^1_f(\mathbb{Q}, V^D)) = \ker \iota = H^1(G_{\mathbb{Q},S}, T^D)_{\mathrm{tor}}$, the image of $H^0(\mathbb{Q}, W^D)$ under the connecting map; the connecting map is injective because $H^0(\mathbb{Q}, V^D) = 0$. (ii) is proved in Chapter 11. In brief: the $\ell$-torsion of $H^0(\mathbb{Q}, W^D)$ is the space of $G_\mathbb{Q}$-maps $\bar{\rho} \to \bar{\rho} \otimes \bar{\chi}_\ell$, of dimension at most one by Schur's lemma; an element of exact order $\ell^k$ is an isomorphism $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$ by Nakayama's lemma; taking determinants gives $\chi_\ell^2 \equiv 1 \pmod{\ell^k}$ on $G_\mathbb{Q}$, and as $\chi_\ell(G_\mathbb{Q}) = \mathbb{Z}_\ell^\times \ni 2$ this forces $k = 0$ for $\ell \geq 5$ and $k \leq 1$ for $\ell = 3$; and $\bar{\chi}_3 = \chi_{-3}$. □

*In the paper* 18.68. Lemma lem:h0 in the paper states (ii) with the criterion over $\mathbb{Q}(\sqrt{-3})$ ("if and only if $\bar{\rho}_{E,3}$ is not absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$") and proves it by the determinant argument; the equivalence with the image being $N_s(\mathbb{F}_3)$ is the paper's Proposition prop:ns (Chapter 11), and the lemma adds that it holds with or without complex multiplication. The criterion for $h_\ell(E) > 0$ involves the cyclotomic $\bar{\chi}_\ell$, not the quadratic $\chi_{\ell^*}$; the two agree only for $\ell = 3$. Conjecture conj:euler defines $h_\ell(E)$ as the length of $H^0(\mathbb{Q}, \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$, which is length $H^1_f(\mathbb{Q}, T^D)$ by (i) whenever $H^1_f(\mathbb{Q}, V^D) = 0$.

**Lemma 18.69** (Diamond, Flach and Guo, Lemma 2.1)**.** *Let $\ell$ be odd. Suppose $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$, and let $\Sigma$ be a nonempty finite set of primes, $\ell \notin \Sigma$, containing every prime $p \neq \ell$ at which $W$ is ramified. Then there is an exact sequence*

$$0 \to H^1_f(\mathbb{Q}, W) \to H^1_\Sigma(\mathbb{Q}, W) \to \bigoplus_{p \in \Sigma \cup \{\infty\}} H^1_f(\mathbb{Q}_p, T^D)^* \to H^0(\mathbb{Q}, W^D)^* \to 0,$$

*in which the term at $\infty$ is 0 and, for $p \in \Sigma$, $H^1_f(\mathbb{Q}_p, T^D) = H^1(\mathbb{Q}_p, T^D)_{\mathrm{tor}} \simeq H^0(\mathbb{Q}_p, W^D)$ is finite.*

*Proof.* Chapter 17 proves the following (its abstract form of the lemma): let $T$ be unramified outside a finite set $S$ of places containing $\ell$, $\infty$ and $\Sigma$, and let $(L_v)_v$ be local conditions on $V$ with $L_v = H^1_{\mathrm{ur}}(\mathbb{Q}_v, V)$ for $v \notin S$ and $L_v^\perp = L_v^D$ for every $v$, where $(L_v^D)$ are conditions on $V^D$; propagate them to $W$ and to $T^D$ (preimages and images), and let $\mathcal{L}^\Sigma$ be the structure on $W$ obtained by dropping the condition at the primes of $\Sigma$. Assume (a) the Selmer group of $V^D$ for $(L_v^D)$ vanishes, and (b) $H^0(\mathbb{Q}_p, V^D) = 0$ for some $p \in \Sigma$. Then

$$0 \to H^1_{\mathcal{L}}(\mathbb{Q}, W) \to H^1_{\mathcal{L}^\Sigma}(\mathbb{Q}, W) \to \bigoplus_{p \in \Sigma \cup \{\infty\}} (L^D_{T^D,p})^* \to H^0(\mathbb{Q}, W^D)^* \to 0$$

is exact, with $L^D_{T^D,p}$ the propagated condition on $T^D$ at $p$. We apply this to the triple $(A_f, A, W)$ with $L_v = H^1_f(\mathbb{Q}_v, A_f)$ and $L_v^D = H^1_f(\mathbb{Q}_v, V^D)$, and check each point.

- $L_v = H^1_{\mathrm{ur}}$ for $v \notin S$: Definition 18.29(i).
- $L_v^\perp = L_v^D$ at every place: Proposition 18.34 ($v = p \neq \ell$: unramified orthogonality; $v = \ell$: Bloch and Kato, $A_f$ being crystalline for $\ell \nmid N$, Proposition 18.25, and in general de Rham; $v = \infty$: all groups vanish).
- The propagated structures are the Bloch-Kato conditions on $W$ and $T^D$ (Definition 18.30), so $H^1_{\mathcal{L}}(\mathbb{Q}, W) = H^1_f(\mathbb{Q}, W)$ and $L^D_{T^D,p} = H^1_f(\mathbb{Q}_p, T^D)$.
- $\mathcal{L}^\Sigma$ is the structure of $H^1_\Sigma$: no condition on $\Sigma$, Bloch-Kato at $\ell$, and at $p \notin \Sigma \cup \{\ell\}$ the Bloch-Kato condition, which is the unramified one because $W$ is unramified there (Lemma 18.37); at $\infty$ there is nothing to impose.
- (a) is the hypothesis $H^1_f(\mathbb{Q}, V^D) = 0$.
- (b) holds at every $p \in \Sigma$ (Lemma 18.18(i)), and $\Sigma \neq \emptyset$.

The term at $\infty$ is $H^1(\mathbb{R}, T^D)^* = 0$, a group killed by 2 and by a power of $\ell$. The description of the local terms is Proposition 18.32(v) and Theorem 18.49. □

*In the paper* 18.70. The paper's Lemma lem:dfg21 includes the hypothesis $\ell \notin \Sigma$, as the lemma above does. It is needed: $H^1_\Sigma(\mathbb{Q}, W)$ keeps the Bloch-Kato condition at $\ell$, and the computation of the paragraph after the lemma ("For $p \in \Sigma$ the group $H^1_f(\mathbb{Q}_p, V^D) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ vanishes") would fail at $p = \ell$, where $H^1_f(\mathbb{Q}_\ell, V^D)$ is not the unramified group and has dimension 2 when $\ell \nmid N$ (Proposition 18.28), as that paragraph says. In the paper's application $\Sigma$ is the set of primes dividing $N$ and $\ell \nmid N$.

What each hypothesis does, in the abstract proof of Chapter 17: $H^1_f(\mathbb{Q}, V^D) = 0$ makes the dual Selmer group $H^1_f(\mathbb{Q}, T^D)$ torsion, hence equal to $H^0(\mathbb{Q}, W^D)$ (Proposition 18.67(i)); $\Sigma \neq \emptyset$ with $H^0(\mathbb{Q}_p, V^D) = 0$ at some $p \in \Sigma$ makes the strict dual Selmer group (classes of $H^1_f(\mathbb{Q}, T^D)$ that vanish at $\Sigma$) zero, because the local connecting map $H^0(\mathbb{Q}_p, W^D) \to H^1(\mathbb{Q}_p, T^D)$ is then injective, and this gives the surjection onto $H^0(\mathbb{Q}, W^D)^*$; and $\ell$ odd kills the archimedean terms. The paper's hypothesis $H^0(\mathbb{Q}, V^D) = 0$ is implied by (b), as the paragraph after Lemma lem:dfg21 observes: there it is automatic, and only $H^1_f(\mathbb{Q}, V^D) = 0$ needs Theorem thm:dfg27.

**Corollary 18.71** (the length identity)**.** *Under the hypotheses of Lemma 18.69 for the paper's objects, with $\Sigma$ the set of primes dividing $N$ and $\bar\rho$ irreducible, all terms are finite and*

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma} \Big( v_\ell(L_p(A_f, 1)^{-1}) + v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0 T)) \Big) - h_\ell(E).$$

*Proof.* The abstract length identity is Chapter 17's corollary to its comparison theorem: under the hypotheses checked in the proof of Lemma 18.69, $\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma} \operatorname{length} H^1_f(\mathbb{Q}_p, T^D) - \operatorname{length} H^0(\mathbb{Q}, W^D)$, with all terms finite as soon as $H^1_f(\mathbb{Q}, W)$ is finite. What this chapter adds is the identification of the terms. Finiteness of $H^1_f(\mathbb{Q}, W)$: $H^1_f(\mathbb{Q}, V^D) = 0$ gives $H^1_f(\mathbb{Q}, A_f) = 0$ by Proposition 18.43 ($\ell \nmid N$ as $\Sigma$ is the set of primes dividing $N$ and $\ell \notin \Sigma$; $\bar\rho$ irreducible, hence absolutely irreducible), so $H^1_f(\mathbb{Q}, W)$ is finite by Proposition 18.38(iii). The local lengths are $v_\ell(L_p(A_f, 1)^{-1}) + v_\ell(\mathrm{Tam}_p(T^D))$ (Theorem 18.49, with the value of the Euler factor from Chapter 19), with $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(\mathrm{ad}^0 T)$ (Proposition 18.50); and $\operatorname{length} H^0(\mathbb{Q}, W^D) = h_\ell(E)$ (Proposition 18.67(ii)). □

*In the paper* 18.72. The proof of Theorem thm:main contains the display "$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma} \operatorname{length} H^1_f(\mathbb{Q}_p, T^D)$" (with $H^0(\mathbb{Q}, W^D) = 0$), and then says that by Lemma lem:fpr each local term is $v_\ell(L_p(A_f, 1)^{-1}) + v_\ell(\mathrm{Tam}_p(A))$; Remark rem:ns contains the display with $-h_\ell(E)$. Both are Corollary 18.71.

## 18.11 The Selmer half of the proof of Theorem thm:main

This section states and proves the part of Theorem thm:main that is about Selmer groups: the identity between the lengths of $H^1_\Sigma(\mathbb{Q}, W)$ and $H^1_f(\mathbb{Q}, W)$ with every local term made explicit. Chapter 20 combines it with the congruence-ideal theorems of Diamond, Flach and Guo and with the identification of the congruence ideal with the modular degree (Chapter 14) to prove Theorem thm:main.

**Theorem 18.73** (the Selmer identity)**.** *Let $E/\mathbb{Q}$ be an elliptic curve of conductor $N$, $\ell \notin S_f$,*

*and $\Sigma$ the set of primes dividing $N$. Then $H^1_f(\mathbb{Q},W)$ and $H^1_\Sigma(\mathbb{Q},W)$ are finite and*

$$\operatorname{length} H^1_\Sigma(\mathbb{Q},W) = \operatorname{length} H^1_f(\mathbb{Q},W) + \sum_{q\mid N} v_\ell(L_q(A_f,1)^{-1}) + \sum_{q:\, v_q(j_E)<0} \Big( t_\ell(E,q) + \min\big(t_\ell(E,q), v_\ell(q-1), v_\ell(c(u_q))\big)\Big).$$

*Here $v_\ell(L_q(A_f,1)^{-1}) = v_\ell(q^2-1)$ at the potentially multiplicative primes and $e_\ell(E,q)$ at the potentially good additive primes. Optimality of $E$ is not used.*

*Proof.* $\ell \notin S_f$ means $\ell \nmid 2N$ and $\bar\rho$ absolutely irreducible over $\mathbb{Q}(\sqrt{\ell^*})$, hence over $\mathbb{Q}$. $\Sigma \neq \emptyset$ because no elliptic curve over $\mathbb{Q}$ has good reduction everywhere (Chapter 10), and $W$ is unramified outside $\Sigma \cup \{\ell\}$. By Theorem 18.66 the hypotheses of Lemma 18.69 hold, and $H^1_f(\mathbb{Q},A_f)=0$ directly, so the finiteness of $H^1_f(\mathbb{Q},W)$ (Proposition 18.38(iii)) does not need Proposition 18.43 here. By Proposition 18.67(ii) and the equivalences after Definition 18.65, $h_\ell(E)=0$. Corollary 18.71 gives the identity with $\sum_{p\in\Sigma} v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0 T))$, which Corollary 18.63 (its first part, which is proved in this chapter) makes explicit. The values of the Euler terms are Lemma 18.21. □

*Remark* 18.74 (the assembly, in Chapter 20). For the optimal curve and $\ell \notin S_f$: $v_\ell(\deg\varphi_E) = v_\lambda(\eta^{\emptyset}_{f,\lambda})$ (Proposition prop:pullback and Lemma lem:hecke, Chapter 14; an equality with no multiplicity-one hypothesis); $v_\lambda(\eta^{\Sigma}_{f,\lambda}) = v_\lambda(\eta^{\emptyset}_{f,\lambda}) + \sum_{p\in\Sigma} v_\ell(L_p^{\mathrm{nv}}(A_f,1)^{-1})$ (Theorem thm:dfg14); and $v_\lambda(\eta^{\Sigma}_{f,\lambda}) = \operatorname{length} H^1_\Sigma(\mathbb{Q},W)$ (Theorem thm:dfg37). With Theorem 18.73, the naive factors cancel the true ones at the multiplicative primes ($\delta_p=1$), and at the additive primes ($\delta_p=0$) the naive factor is 1, so the surviving Euler terms are $\sum_q e_\ell(E,q)$ (Definition def:terms). The result is (eq:identity), with equality, and (eq:thm) follows from $\operatorname{length} H^1_f(\mathbb{Q},W)\geq 0$ and $\min(\dots)\geq 0$, including the primes of type $\mathrm{I}^*_n$ at $q=2$ (Proposition 18.57). These steps are proved in Chapter 20; they are recalled here only to show where Theorem 18.73 enters.

*In the paper* 18.75. The proof of Theorem thm:main, from the sentence that invokes Theorem thm:dfg27 for the hypotheses of Lemma lem:dfg21 to the display of $v_\ell(\deg\varphi_E)$, is Theorem 18.73. The final step, at the potentially multiplicative primes of additive reduction, uses Proposition 18.57: the paper applies Lemma lem:tamfactor, whose last sentence covers these primes through the local twist, as here; a global quadratic twist lowering the level does not exist at $p=2$.

**Example 18.76** (four curves at $\ell=3$)**.** For the following optimal curves, $3\nmid N$, $E[3]$ is irreducible, the mod 3 image is not `3Ns`, and $E$ has no CM, so $3\notin S_f$. The explicit local bound is $\sum_q e_3(E,q) + \sum_{v_q(j)<0}(t_3 + \min(t_3, v_3(q-1), v_3(c(u_q))))$; by Corollary 18.63 the Selmer length is $v_3(\deg\varphi_E)$ minus it.

| curve | local data | bound | $\deg\varphi_E$ | length $H^1_f(\mathbb{Q},W)$ |
|---|---|---|---|---|
| `121a1` | II at 11, $e_3 = v_3(12) = 1$ | 1 | 6 | 0 |
| `791a1` | $\mathrm{I}_3$ at 7 (2), $\mathrm{I}_2$ at 113 (0) | 2 | 144 | 0 |
| `1184f1` | $\mathrm{III}^*$ at 2 (0), $\mathrm{I}_3$ at 37 (2) | 2 | 1728 | 1 |
| `1748e1` | IV at 2 ($e_3=1$), $\mathrm{I}_6$ at 19 (2), $\mathrm{I}_1$ at 23 (0) | 3 | 3240 | 1 |

For `791a1` the bound (eq:thm) is 1, and the Tamagawa excess at 7 is needed for equality. The numbers in parentheses are the Tamagawa contributions (Example 18.58).

The same data give the $\Sigma$-imprimitive group. For `791a1`, $\Sigma = \{7, 113\}$, both primes multiplicative, with $v_3(7^2-1) = v_3(48) = 1$ and $v_3(113^2-1) = v_3(2^5 \cdot 3 \cdot 7 \cdot 19) = 1$; so Theorem 18.73 gives

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + (1+1) + (2+0) = 0 + 4 = 4,$$

the value $\operatorname{length} H^1_f(\mathbb{Q}, W) = 0$ being the one in the table (it uses Theorem thm:main). By Theorem thm:dfg37 this is $v_3(\eta^{\Sigma}_{f,\lambda})$; it is consistent with the assembly of Remark 18.74, where $v_3(\eta^{\Sigma}_{f,\lambda}) = v_3(\deg \varphi_E) + \sum_{p \in \Sigma} v_3(L_p^{\mathrm{nv}}(A_f, 1)^{-1}) = 2 + 2$, the naive and true factors agreeing at multiplicative primes.

## 18.12 Beyond the hypothesis: Remark rem:ns, $h_\ell(E)$, Open question 2

When $\ell = 3$ and the mod 3 image is $N_s(\mathbb{F}_3)$, the hypothesis $\ell \notin S_f$ fails, $H^0(\mathbb{Q}, W^D)$ is not zero, and the paper's Remark rem:ns records what the exact sequence of Lemma lem:dfg21 then says. This section makes the conditional statement precise, identifies the correction term $h_\ell(E)$ of Conjecture conj:euler, and states what is missing for Open question 2.

**Proposition 18.77** (Remark rem:ns)**.** *Let $\ell = 3$, $3 \nmid N$, and suppose the image of $\bar\rho_{E,3}$ is conjugate to $N_s(\mathbb{F}_3)$. Then $H^0(\mathbb{Q}, V^D) = 0$, $h_3(E) = 1$, and, if $H^1_f(\mathbb{Q}, V^D) = 0$ (equivalently $H^1_f(\mathbb{Q}, \mathrm{ad}^0 V_3 E) = 0$), the groups $H^1_f(\mathbb{Q}, W)$ and $H^1_\Sigma(\mathbb{Q}, W)$ are finite and*

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma} \Big( v_3(L_p(A_f, 1)^{-1}) + v_3(\mathrm{Tam}_p(\mathrm{ad}^0 T)) \Big) - 1.$$

*Proof.* $\bar\rho$ is irreducible (the image $N_s(\mathbb{F}_3)$ fixes no line), $H^0(\mathbb{Q}, V^D) = 0$ (Proposition 18.14(ii)), $h_3(E) = 1$ (Proposition 18.67(ii)), and $\Sigma \neq \emptyset$. The equivalence is Proposition 18.43. Under the condition, Lemma 18.69 and Corollary 18.71 apply. □

Theorem thm:dfg27 does not supply the vanishing of $H^1_f(\mathbb{Q}, V^D)$ here, since $3 \in S_f$; the proposition states it as a hypothesis, as Remark rem:ns of the paper does ("we assume it here"). It is predicted by the Bloch-Kato conjecture (Conjecture 18.83), since $L(\mathrm{ad}^0 f, 1) \neq 0$; we know of no proof for these curves and have not searched the literature after Diamond, Flach and Guo for one. Section 1.3 of the paper also states the condition.

*Remark* 18.78 (the correction term of Conjecture conj:euler). In Conjecture conj:euler the correction is $h_\ell(E) = \operatorname{length} H^0(\mathbb{Q}, \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$: by Proposition 18.67 it is 0 for $\ell \geq 5$ and is 1 exactly for the curves with mod 3 image $N_s(\mathbb{F}_3)$, for every conductor. It is the term $H^0(\mathbb{Q}, W^D)^*$ of Lemma 18.69, and it enters Proposition 18.77 with the sign $-1$, the sign with which it appears in (eq:rule). The paper's function `ns3(E)` (two quadratic factors of the 3-division polynomial) computes it (Chapter 11). Example: `726a1` $= [1, 1, 0, -35, -51]$, with $3 \mid N$, so outside every theorem; its 3-division polynomial factors as $(x^2 - x - 74)(3x^2 + 8x + 20)$, so $h_3 = 1$; it has type III at 11 with $e_3 = v_3(12) = 1$, $\deg \varphi_E = 144$, and the paper's `rule(E,3)` and `rule(E,3,1)` both return $[2, 2]$: in the second the type III term and the correction cancel.

*Remark* 18.79 (Open question 2). Open question 2 asks to prove the corrected identity of Remark rem:ns for the curves with image $N_s(\mathbb{F}_3)$, that is, to establish $H^1_f(\mathbb{Q}, \mathrm{ad}^0 V_3 E) = 0$ and $v_\lambda(\eta^{\Sigma}_{f,\lambda}) = \operatorname{length} H^1_\Sigma(\mathbb{Q}, W)$ in this dihedral case. On the Selmer side Proposition 18.77 shows

what the identity would give and that the only missing input is $H^1_f(\mathbb{Q}, A_f) = 0$; on the congruence side the missing input is a Taylor-Wiles theorem for a residual representation induced from $\mathbb{Q}(\sqrt{-3})$ (Chapter 20). The data of Section sec:anomaly (failures of (eq:thm) by exactly one, never more) are consistent with the identity with $-h_3(E)$.

## 18.13 Section sec:tam: testing the Tamagawa factors on the tables

The paper's Section sec:tam tests the refinement of (eq:thm) by the Tamagawa excess of Corollary 18.55. This section reproduces the test for conductors below 10,000 and says what it does and does not test.

**Example 18.80** (the test)**.** The population: optimal curves without CM satisfying the hypothesis of Theorem thm:main ($\ell \nmid N$, $E[\ell]$ irreducible, image not `3Ns` at $\ell = 3$) that have a potentially multiplicative prime $q \equiv 1 \pmod{\ell}$ with $\ell \mid \nu_q$. For each, compare $v_\ell(\deg \varphi_E)$ with the bound $\sum_q e_\ell + \sum_q t_\ell$ of (eq:thm) and with the refined bound that adds the Tamagawa excesses. The paper's population is the working set (conductors up to 500,000) with primes of type $\mathrm{I}_n$ only; ours is conductors below 10,000 with primes of type $\mathrm{I}_n$ or $\mathrm{I}^*_n$ (Proposition 18.57):

| source | $\ell$ | curves | positive excess | equality, bound (eq:thm) | equality, refined bound |
|---|---|---|---|---|---|
| paper | 3 | 46,222 | 12,095 | 20,822 (45.0%) | 28,007 (60.6%) |
| paper | 5 | 10,940 | 1,646 | 7,333 (67.0%) | 8,610 (78.7%) |
| here | 3 | 1,050 | 186 | 596 (56.8%) | 714 (68.0%) |
| here | 5 | 252 | 24 | 203 (80.6%) | 220 (87.3%) |
| here | 7 | 18 | 1 | 17 | 18 |

No curve violates the refined inequality, in these populations or in the full populations of Remark 18.64 (13,203, 23,361 and 28,134 curves). The rates are not comparable with the paper's (smaller conductors; $\mathrm{I}^*_n$ primes included); the paper also reports these figures below conductor 10,000 at $\ell = 3, 5$.

*Remark* 18.81 (what Section sec:tam tests). Given Theorem 18.73 and its assembly with the equality $v_\ell(\deg \varphi_E) = v_\lambda(\eta^{\emptyset}_{f,\lambda})$ (Chapter 14, Chapter 20), a violation of the refined inequality is impossible, and the proportion of equality is the proportion of curves with $H^1_f(\mathbb{Q}, W) = 0$. So Section sec:tam is a consistency check of the Tamagawa term of (eq:identity) and of the computation of Tate parameters, not an independent test of a conjecture. The paper's phrasing ("the extra divisibility of the modular degree predicted by the Tamagawa factor of the adjoint is always present, the Selmer length accounting for the rest") says this: extra divisibility also comes from the Selmer group.

## 18.14 The Tamagawa number conjecture and the adjoint motive

This section places Theorem thm:main in the framework of Bloch and Kato. Nothing here is used in a proof. The adjoint motive of $f$, its $L$-function, the shape of the Tamagawa number conjecture, the theorem of Diamond, Flach and Guo that proves it for the adjoint at $\ell \notin S_f$, and Flach's earlier finiteness theorem are stated with references, and a labelled heuristic explains why (eq:identity) is the conjecture read through the modular degree.

**Definition 18.82** (the adjoint motive)**.** Let $f$ be a weight-2 newform with rational coefficients. The *adjoint motive* $\mathrm{ad}^0 M_f$ is the motive with $\ell$-adic realisations $A_f = \mathrm{ad}^0 V_\ell(f) \simeq \mathrm{Sym}^2 V_\ell(f) \otimes \det^{-1}$ (Chapter 4), and Betti and de Rham realisations built from those of $M_f$ in the same way; it is pure of weight 0 with Hodge numbers $h^{1,-1} = h^{0,0} = h^{-1,1} = 1$, and

$$L(\mathrm{ad}^0 f, s) = \prod_p L_p(A_f, s) = L(\mathrm{Sym}^2 f, s+1),$$

with the true Euler factors (at $p = \ell$ computed through $\mathrm{D}_{\mathrm{cris}}$, or with $V_{\ell'}$ for a prime $\ell' \neq p$) and the normalisation $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$ of Appendix B, B-C7. The point $s = 1$ is critical in the sense of Deligne [42], and $L(\mathrm{ad}^0 f, 1) \neq 0$.

The last assertion follows from Chapter 19: the formula $\langle f, f\rangle = N\, L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2)/(8\pi^3)$ of Rankin and Shimura, quoted in the paper's Remark rem:degree, shows $L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1) > 0$, and $L(\mathrm{ad}^0 f, 1)$ differs from it by the nonzero factors $L_q(A_f, 1)/L_q^{\mathrm{nv}}(A_f, 1)$ (Lemma 18.18). For the critical value and the Hodge numbers see [42] and [49, Section 1]; for the symmetric square, Shimura [183] and Gelbart and Jacquet [73].

**Conjecture 18.83** (Bloch and Kato, in outline)**.** *Let $M$ be a motive over $\mathbb{Q}$ whose $L$-function is critical and nonzero at $s = 0$, $\ell$ a prime, $T$ a stable lattice in the $\ell$-adic realisation of $M$, $W_M = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ and $W_M^D = T^D \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. Then, for a suitably normalised period $\Omega(M)$,*

$$\frac{L(M,0)}{\Omega(M)} \sim_\ell \frac{\prod_{p\neq\ell} \#(\mathbb{Z}_\ell/\operatorname{Tam}_p(T)) \cdot \#H^1_f(\mathbb{Q}, W_M^D)}{\#H^0(\mathbb{Q}, W_M) \cdot \#H^0(\mathbb{Q}, W_M^D)},$$

*where $\sim_\ell$ means equality up to an $\ell$-adic unit, and the contributions of $\ell$ and $\infty$ are absorbed in $\Omega(M)$ (the Tamagawa measure of Bloch and Kato).*

This is a schematic form of the conjecture of Bloch and Kato [14, Section 5], in the formulation of Fontaine and Perrin-Riou [69]; the precise normalisations of the period and of the terms at $\ell$ differ between these sources and [49, Section 2], and we do not reconcile them in general; for $M = \mathrm{ad}^0 M_f(1)$ in weight 2, Remark 18.89 computes the period of [49, Section 2.3] against the book's normalisation. The book never uses the conjecture in a proof. For $M = \mathrm{ad}^0 M_f(1)$, whose realisation is $A_f(1) = V^D$ with lattice $T^D$, one has $L(M,0) = L(\mathrm{ad}^0 f, 1)$, $W_M = W^D$ and $W_M^D = W$; so the right side is

$$\frac{\prod_{p\neq\ell} \#(\mathbb{Z}_\ell/\operatorname{Tam}_p(T^D)) \cdot \#H^1_f(\mathbb{Q}, W)}{\#H^0(\mathbb{Q}, W^D) \cdot \#H^0(\mathbb{Q}, W)} :$$

the adjoint Selmer group of Theorem thm:main, the Tamagawa factors of Lemma lem:fpr, and $h_\ell(E)$ in the denominator, with the same sign as in Remark rem:ns.

**Example 18.84** (Birch and Swinnerton-Dyer)**.** For $M = h^1(E)(1)$, with realisation $V_\ell E$ and $L(M,0) = L(E,1)$, Conjecture 18.83 is the $\ell$-part of the formula of Birch and Swinnerton-Dyer in rank 0:

| term of Conjecture 18.83 | term of the BSD formula |
|---|---|
| $\Omega(M)$ | the real period $\Omega_E$ |
| $\#(\mathbb{Z}_\ell/\operatorname{Tam}_p(T_\ell E))$ | $\ell$-part of $c_p$ (Proposition 18.51) |
| $\#H^1_f(\mathbb{Q}, W_M^D)$, $W_M^D \simeq E[\ell^\infty]$ | $\#\,\mathrm{Sha}(E/\mathbb{Q})[\ell^\infty]$ (rank 0; Example 18.41) |
| $\#H^0(\mathbb{Q}, W_M) \cdot \#H^0(\mathbb{Q}, W_M^D)$ | $\#E(\mathbb{Q})[\ell^\infty]^2$ |

Here $T^D \simeq T$ by the Weil pairing (Proposition 18.7(iii)). This is orientation only; see [14, Section 6] and Chapter 9.

**Theorem 18.85** (Diamond, Flach and Guo, Theorem 0.1)**.** *Let $f$ be a newform (of weight $k \geq 2$) and $\lambda$ a prime of its coefficient field not in the exceptional set $S_f$. Then the $\lambda$-part of the Tamagawa number conjecture of Bloch and Kato holds for $A_f$ and $A_f(1)$, that is, for $L(A_f, s)$ at $s = 0$ and at the critical point $s = 1$ [49, Theorem 0.1].*

*Reference.* We do not prove this; it is the theorem the paper cites when it says that its hypothesis "is the one under which Diamond, Flach and Guo prove the Tamagawa number conjecture for the adjoint motive of $f$". The period is Deligne's period, computed in [49, Section 2.3, Theorem 2.10] (the integral structures it uses are those of their Section 1; in weight 2 it is $w_E N \cdot 2\pi \operatorname{vol}(\Lambda_E)/(c_E^2 N)$ up to an $\ell$-adic unit, Remark 18.89), and the conjecture is formulated in [49, Section 2.4, Conjecture 2.14]; the theorem is their Theorem 2.15. The proof combines the identification of the congruence ideal with the special value (Hida's form of the Rankin-Shimura method, their Lemma 2.12), Theorem thm:dfg37, Theorem thm:dfg14, the triviality of the Tamagawa factor at $\ell$ (their Proposition 2.16) and the local computations of Lemma lem:fpr; Chapter 20 gives the parts the paper uses. □

**Theorem 18.86** (Diamond, Flach and Guo, Theorem 0.2)**.** *If $\ell \notin S_f$ and $\Sigma$ is a finite set of primes, $\ell \notin \Sigma$, outside which $M_{f,\lambda}$ is minimally ramified (for $f = f_E$: any $\Sigma$ containing the primes dividing $N$), then $H^1_\Sigma(\mathbb{Q}, W)$ is finite of length $v_\lambda(\eta^\Sigma_{f,\lambda})$.*

*Reference.* This is the paper's Theorem thm:dfg37, which the paper says is "the form given as Theorem 0.2" of [49]. As stated here, with minimal ramification outside $\Sigma$, it is [49, Theorem 3.7]; their Theorem 0.2 is the case in which $\Sigma$ contains the primes dividing $N$ (checked). We do not prove it; the statement, the meaning of "minimally ramified" and the outline of the Taylor-Wiles argument are in Chapter 20. It is recalled here only so that Theorem 18.73 can be read against it. □

**Theorem 18.87** (Flach)**.** *Let $E/\mathbb{Q}$ be an elliptic curve of conductor $N$ whose conductor is minimal among its quadratic twists, and let $\varphi\colon X_0(N) \to E$ be a modular parametrisation (of any degree). Let $\ell \geq 5$ be a prime at which $E$ has good reduction and such that $\bar\rho_{E,\ell}\colon G_\mathbb{Q} \to \mathrm{GL}_2(\mathbb{F}_\ell)$ is surjective. Then $\deg\varphi$ annihilates the Bloch-Kato Selmer groups $H^1_f(\mathbb{Q}, W^D)$, where $W^D \simeq \mathrm{Sym}^2 T_\ell E \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and $H^1_f(\mathbb{Q}, W)$; in particular both are finite. For the optimal curve one may take $\deg\varphi = \deg\varphi_E$.*

*Reference.* We do not prove this; see [66, Theorem 1 and the Remarks after it, p. 308; Propositions 1.1 and 2.2] (checked against the published text). Flach writes $p$ for our $\ell$, $V = \mathrm{Sym}^2 H^1(E_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)$ with its natural lattice $\mathrm{Sym}^2 H^1(E_{\overline{\mathbb{Q}}}, \mathbb{Z}_\ell)$, and $A$ for the quotient (his $A$, not ours). His Theorem 1 has exactly the two hypotheses above and states that $\deg\varphi$ annihilates $\mathrm{Sha}(\mathbb{Q}, A(2))$, the cotorsion quotient of his Selmer group $S(\mathbb{Q}, A(2))$; the first Remark after it says that the proof gives $\deg\varphi \cdot S(\mathbb{Q}, A(2)) = 0$, and the two groups are equal (p. 326). The statement for $A(1)$ is his Proposition 1.1 (p. 310), whose hypothesis, the existence of suitable classes $c(l)$, is supplied by his Proposition 2.2 (p. 318). His $S(\mathbb{Q}, -)$ is defined by the local conditions of Bloch and Kato (his Section 1). That the conductor of $E$ is minimal among its quadratic twists is a standing assumption stated just before Theorem 1: the Selmer groups do not change under a quadratic twist, while $\deg\varphi$ does. Since $H^1(E_{\overline{\mathbb{Q}}}, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$, his $A(2)$ is $\mathrm{Sym}^2 T_\ell E \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ and his $A(1)$ is $\mathrm{Sym}^2 T_\ell E(-1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \simeq W$. Note that $\mathrm{Sym}^2 V_\ell E \simeq A_f(1) = V^D$ (since

$\mathrm{ad}^0 V \simeq \mathrm{Sym}^2 V \otimes \det^{-1}$ and $\det = \chi_\ell$), so Theorem 1 of [66] is about $H^1_f(\mathbb{Q}, W^D)$ in the paper's notation; when $\bar\rho$ is absolutely irreducible its finiteness is equivalent to $H^1_f(\mathbb{Q}, V^D) = 0$ (Proposition 18.38(iii)), hence to $H^1_f(\mathbb{Q}, A_f) = 0$ (Proposition 18.43), hence to the finiteness of $H^1_f(\mathbb{Q}, W)$. □

*In the paper* 18.88. Section 1 of the paper: "Flach's finiteness theorem for the symmetric square [66] is the first place where the modular degree meets the Selmer group of $\mathrm{Sym}^2 f$".

*Remark* 18.89 (the modular degree as an adjoint $L$-value: Deligne's period). By the formula of Zagier and Watkins, $\deg\varphi_E \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_E^2 \langle f, f\rangle$, with the unnormalised Petersson product on $\Gamma_0(N)$ (Chapter 14); by the Rankin-Shimura method $\langle f, f\rangle = N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)/(8\pi^3)$ (Chapter 19). The paper's Remark rem:degree turns this into $v_\ell(\deg\varphi_E) = v_\ell(B_E) + \sum_q e_\ell(E, q)$ for $\ell \nmid N$, with $B_E = c_E^2 N\, L(\mathrm{ad}^0 f, 1)/(2\pi\,\mathrm{vol}(\Lambda_E))$, so the ratio of the naive to the true local factors produces the terms $e_\ell(E, q)$. We show that $2\pi\,\mathrm{vol}(\Lambda_E)/(c_E^2 N)$ is, up to the factor $w_E N$, Deligne's period of $\mathrm{ad}^0 M_f(1)$ measured against the integral structure of Diamond, Flach and Guo. This is a derivation from formulas of [49] that we quote (their numbers checked against the published text), not a proof from first principles. Their $B_f = A_f(1)$ is our $\mathrm{ad}^0 M_f(1)$. We specialise their Sections 1.5, 2.3 and 2.4 to weight $k = 2$, trivial character and coefficient field $\mathbb{Q}$: their parity index (the element of $\{0, 1\}$ congruent to $k$ modulo 2, called $\eta$ there; it is not a congruence ideal) is 0, so $i^{k-0} = -1$; $(k-2)! = (k-1)! = 1$; the Dirichlet structure $M_\psi$ is $\mathbb{Q}$ with $b_{\mathrm{dR}} = 1$ and epsilon factor 1; and every prime dividing $2N$ lies in $S_f$ (Definition 18.65).

(a) *The epsilon factor.* Their functional equation (48), written with $\Lambda(f, s) = (2\pi)^{-s}\Gamma(s)L(f, s)$, compared with $N^{s/2}\Lambda(f, s) = w_E N^{(2-s)/2}\Lambda(f, 2-s)$ (Chapter 19), gives $\varepsilon(M_f) = w_E N$, where $w_E = -\varepsilon_N(f)$. Their Lemma 2.11 then says that the operator $W$ of their (13) acts on $f$ by $i^2\varepsilon(M_f)/N = -w_E = \varepsilon_N(f)$: $W$ is the Fricke involution, and their pairing (16) is the pairing $\hat\delta_!$ of Chapter 20.

(b) *The Petersson products.* Their $(g, h)_\Gamma = (-2i)^{-1}\int_{\Gamma\backslash\mathcal{H}} g\bar h\, d\tau \wedge d\bar\tau$ is $\int_{\Gamma\backslash\mathcal{H}} g\bar h\, dx\, dy$ in weight 2, since $d\tau \wedge d\bar\tau = -2i\, dx \wedge dy$. Since $-1 \in \Gamma_0(N)$, while $-1 \in \Gamma_1(N)$ only for $N \le 2$, the covering $\Gamma_1(N)\backslash\mathcal{H} \to \Gamma_0(N)\backslash\mathcal{H}$ has degree $\varphi(N)\delta(N)/2$, with their $\delta(N) = 2$ for $N \le 2$ and 1 otherwise; so $(f, f)_{\Gamma_1(N)} = \frac12\varphi(N)\delta(N)\langle f, f\rangle$. Their Lemma 2.12 in weight 2, $(f, f)_{\Gamma_1(N)} = \delta(N)N\varphi(N)L^{\mathrm{nv}}(A_f, 1)/(16\pi^3)$, is therefore exactly $\langle f, f\rangle = N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)/(8\pi^3)$, and their naive factors (Section 1.7.2) are ours factor by factor.

(c) *Deligne's period of $A_f$.* Their de Rham class $f$ is $2\pi i f\, d\tau$ (the trivialisation (4)), and $H^2_B$ of the modular curve is $(2\pi i)^{-1}\mathbb{Z}$ (Section 1.1.3); hence the factor $(k-2)!(4\pi)^{k-1} = 4\pi = |2\pi i|^2 \cdot |-2i|/|2\pi i|$ (up to sign) in their comparison (10) of the de Rham pairing with $(\,,\,)_{\Gamma(N')}$. Their pairing at level $N$ is the one at an auxiliary level $N'$ divided by $[U : U_{N'}] = \varphi(N')\varphi(N)\delta(N)[\bar\Gamma_1(N) : \bar\Gamma(N')]$ (their (11) and (53)), which is twice the degree of $X(N') \to X_0(N)$ because $-1 \in \Gamma_0(N)$. With (a), their (44) and (51) to (53) give

$$\langle f, c^+(A_f)(f \otimes F_\infty)\rangle = \frac{4\pi\,(f, f)_{\Gamma_1(N)}}{\varphi(N)\delta(N)}\,(-w_E)\,\iota = -2\pi w_E\langle f, f\rangle\,\iota = -\frac{w_E N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)}{4\pi^2}\,\iota,$$

which is the identity displayed at the start of their proof of Theorem 2.10, for $k = 2$.

(d) *The twist.* By their Lemma 2.9, $c^+(B_f) = -2\pi^2\,\mathrm{tw}(c^+(A_f))$; here $-2\pi^2 = (2\pi i)^2/2$, the $\frac12$ coming from the projection to the $F_\infty$-eigenspaces. Hence, with the formula of Zagier and

Watkins,

$$\langle f, c^+(B_f)(f\otimes(2\pi i)^2\otimes\iota^2)\rangle = 4\pi^3 w_E\langle f,f\rangle\,\iota = \frac{w_E N}{2}\,L^{\mathrm{nv}}(\mathrm{ad}^0 f,1)\,\iota = \frac{w_E\deg\varphi_E}{2}\cdot\frac{2\pi\,\mathrm{vol}(\Lambda_E)}{c_E^2}\,\iota.$$

This agrees with their Theorem 2.10 and (43), using their (38) in weight 2, $L(A_f,0) = \varepsilon(A_f)L(A_f,1)/(2\pi^2)$.

(e) *The integral structure.* Let $\ell\notin S_f$ and let $e$ generate $\mathrm{Hom}_{\mathbb{Z}_\ell}(\theta(B_{f,\lambda}),\omega_B)$, the lattice of their (58). The argument in their proof of Theorem 2.15 (written there for $M_f^\Sigma$; it applies verbatim to $\Sigma=\emptyset$) gives $\mathbb{Z}_\ell\,\langle f, e(f\otimes(2\pi i)^2\otimes\iota^2)\rangle = \eta_{f,\lambda}^{\emptyset}\,\iota$. By Chapter 20 and Section 4.1 of the paper, on classes pulled back from $X_0(N)$ the pairing $\hat\delta_!$ is "$\frac{u}{2}$ times the cup product on $X_0(M^\Sigma)$ composed with $w_{M^\Sigma}$, for an $\ell$-adic unit $u$" (here $M^\emptyset = N$); on $M_{f,\lambda} = \varphi_E^* H^1(E,\mathbb{Z}_\ell)$ its values form the ideal $\frac{u}{2}\deg\varphi_E\,\mathbb{Z}_\ell = \deg\varphi_E\,\mathbb{Z}_\ell = \eta_{f,\lambda}^{\emptyset}$. So Deligne's period relative to their integral structure, $\Omega_{\mathrm{DFG}} = c^+(B_f)/e$, satisfies

$$\Omega_{\mathrm{DFG}}\in\mathbb{Z}_\ell^\times\cdot\frac{w_E N\,L^{\mathrm{nv}}(\mathrm{ad}^0 f,1)}{\deg\varphi_E} = \mathbb{Z}_\ell^\times\cdot w_E\,\frac{2\pi\,\mathrm{vol}(\Lambda_E)}{c_E^2} = \mathbb{Z}_\ell^\times\cdot w_E N\cdot\frac{2\pi\,\mathrm{vol}(\Lambda_E)}{c_E^2 N}.$$

The bookkeeping behind the unit: the $\frac12$ of (d) is matched by the $\frac12$ in $\frac{u}{2}$, and $\deg\varphi_E$ cancels between the Petersson norm and $\eta^\emptyset$. The constant $u$ is known only to be an $\ell$-adic unit, so the comparison is asserted up to $\mathbb{Z}_\ell^\times$ and not up to sign; nothing more is used. The only discrepancy with the book's normalisation $2\pi\,\mathrm{vol}(\Lambda_E)/(c_E^2N)$ is therefore the factor $w_E N$, a unit at every $\ell\nmid N$; no power of 2 or $\pi$ is left over.

(f) *The conjecture.* For $\ell\notin S_f$, their Conjecture 2.14 for $M=B_f$, proved as their Theorem 2.15 (Theorem 18.85), says that

$$\mathbb{Z}_\ell\cdot\frac{L(B_f,0)}{\Omega_{\mathrm{DFG}}} = \frac{\mathrm{Fitt}(\mathrm{Sha}(A_{f,\lambda}))\cdot\mathrm{Tam}(B_{f,\lambda})}{\mathrm{Fitt}\,H^0(\mathbb{Q},B_{f,\lambda}\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell)\cdot\mathrm{Fitt}\,H^0(\mathbb{Q},A_{f,\lambda}\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell)}.$$

The groups $H^0$ vanish, the factors at $\ell$ and $\infty$ are trivial (their Proposition 2.16; $\ell$ is odd), $\mathrm{Sha}(A_{f,\lambda}) = H^1_f(\mathbb{Q},W)$ because $H^1_f(\mathbb{Q},A_f)=0$ (Theorem 18.66), and $\mathrm{Tam}_p(B_{f,\lambda}) = \mathrm{Tam}_p(A_{f,\lambda})$ by their (57). As $L(B_f,0)=L(\mathrm{ad}^0 f,1)$ and $L(\mathrm{ad}^0 f,1)/\Omega_{\mathrm{DFG}}\in\mathbb{Z}_\ell^\times\cdot B_E$, this is

$$v_\ell(B_E) = \mathrm{length}\,H^1_f(\mathbb{Q},W) + \sum_{q\mid N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)),$$

which, with $v_\ell(\deg\varphi_E) = v_\ell(B_E) + \sum_q e_\ell(E,q)$, is (eq:identity). It is not an independent proof: their proof of Theorem 2.15 reduces to their Theorem 3.7, the paper's Theorem thm:dfg37, as Theorem thm:main does.

(g) *Numbers* (PARI/GP 2.17.2: `ellmoddegree`, `E.area`, `mspetersson` on the basis of the $f$-part of the integral modular symbols returned by `msfromell`, $\langle f,f\rangle = \mu(N)\cdot$`mfpetersson`, $L(\mathrm{ad}^0 f,1)$ from `lfunsympow`; $c_E=1$ for these optimal curves). The intersection matrix of the $f$-part is $\pm\deg\varphi_E\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$, and $\Omega_{\mathrm{DFG}} = w_E N\,L^{\mathrm{nv}}(\mathrm{ad}^0 f,1)/\deg\varphi_E$ divided by $2\pi\,\mathrm{vol}(\Lambda_E)/N$ is $w_E N$ to 20 digits, in each row ($L=L(\mathrm{ad}^0 f,1)$, $L^{\mathrm{nv}}=L^{\mathrm{nv}}(\mathrm{ad}^0 f,1)$; last column: the odd $\ell\nmid N$ in $S_f$):

| curve | $w_E N$ | $\deg \varphi_E$ | $L/L^{\mathrm{nv}}$ | $B_E$ | $L/\Omega_{\mathrm{DFG}}$ | $S_f$ |
|---|---|---|---|---|---|---|
| `11a1` | 11 | 1 | 1 | 1 | 1/11 | 5 |
| `37a1` | −37 | 2 | 1 | 2 | −2/37 | none |
| `121a1` | 121 | 6 | 11/12 | 11/2 | 1/22 | none |
| `121d1` | 121 | 24 | 121/120 | 121/5 | 1/5 | 5 |

For every odd $\ell \nmid N$ outside $S_f$, $v_\ell(L(\mathrm{ad}^0 f,1)/\Omega_{\mathrm{DFG}}) \geq 0$, and adding $\sum_q e_\ell(E,q) = v_\ell(L^{\mathrm{nv}}(\mathrm{ad}^0 f,1)/L(\mathrm{ad}^0 f,1))$ gives $v_\ell(\deg\varphi_E)$: 0 in every row except $\ell = 3$ for `121a1` and `121d1`, where $e_3(E,11) = 1 = v_3(6) = v_3(24)$. At $\ell = 5$, `121d1` has $v_5(B_E) = -1$, impossible for the right side of the conjecture; $5 \in S_f$ there ($E[5]$ is reducible), so nothing is predicted.

## 18.15 Exercises

**Exercise 18.90** (the two Frobenii)**.** Let $p \neq \ell$. Compute $\det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid \mathbb{Q}_\ell(1))^{-1}$ and $\det(1 - \mathrm{Frob}_p\, p^{-s} \mid \mathbb{Q}_\ell(1))^{-1}$. At a prime of multiplicative reduction, show that the geometric Frobenius gives $\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = 1 - p^{-2}$, which is the paper's $L_p(A_f,1)^{-1} = L_p(A_f(1),0)^{-1}$, while the arithmetic one gives $1 - p^2$, a number with the same $\ell$-adic valuation. *Answer:* $(1-p^{-1-s})^{-1}$ and $(1-p^{1-s})^{-1}$; on $(V^D)^{I_p} = \mathbb{Q}_\ell e \otimes \mathbb{Q}_\ell(1)$ the arithmetic Frobenius acts by $p \cdot p = p^2$; and $1 - p^2 = -p^2(1-p^{-2})$.

**Exercise 18.91** (the trace form at $\ell = 2$)**.** Compute the Gram matrix of the trace form of $\mathrm{ad}^0 T$ in the basis $e, h, f$. Show that for $\ell = 2$ the map $A \to A^\vee$, $x \mapsto \mathrm{tr}(x\,\cdot)$, is injective with cokernel of order 2, and find an element of $A^\vee$ not in its image. *Hint:* the dual basis vector $h^\vee$ is $\frac{1}{2}\mathrm{tr}(h\,\cdot)$. *Answer:* the Gram matrix is the one in the proof of Proposition 18.10; $\mathrm{tr}(e\,\cdot) = f^\vee$, $\mathrm{tr}(f\,\cdot) = e^\vee$ and $\mathrm{tr}(h\,\cdot) = 2h^\vee$, so the image is $\mathbb{Z}_2 e^\vee \oplus 2\mathbb{Z}_2 h^\vee \oplus \mathbb{Z}_2 f^\vee$, of index 2, and $h^\vee$ is not in it.

**Exercise 18.92** (local ranks)**.** For each row of the table of Proposition 18.17, find $\dim H^0(\mathbb{Q}_p, V^D)$, $\dim H^0(\mathbb{Q}_p, A_f)$ and, granting the local Euler characteristic formula, the $\mathbb{Z}_\ell$-rank of $H^1(\mathbb{Q}_p, T^D)$. Which bad primes have $H^1(\mathbb{Q}_p, T^D)$ infinite? Check in each case the paper's statement that $H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$. *Answer:* $H^0(\mathbb{Q}_p, V^D) = 0$ always; the rank is 1 in the row $\{\pm 1\}$ (the eigenvalue 1 has multiplicity one, as at a good prime) and in the cyclic row with $c = +1$, and 0 otherwise. The statement holds in all rows by Proposition 18.32(v).

**Exercise 18.93** (multiplicative primes)**.** Let $q$ be a prime of multiplicative reduction, $\ell \neq q$ odd, and $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. Show that $H^1_f(\mathbb{Q}_q, W) = 0$; that $W^{I_q} = (\mathbb{Q}_\ell/\mathbb{Z}_\ell)\, e$ if $\ell \nmid n_q$, so that $H^1_{\mathrm{ur}}(\mathbb{Q}_q, W) = 0$; and that if $\ell \mid n_q$, $W^{I_q}$ contains $\ell^{-a}\mathbb{Z}_\ell h/\mathbb{Z}_\ell h$ modulo $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)e$, with $a = v_\ell(n_q)$. For $W = E[\ell^\infty]$ with split reduction and $\ell \mid n_q$, show $H^1_{\mathrm{ur}}(\mathbb{Q}_q, W) \simeq \mathbb{Z}/\ell^a$ while $H^1_f(\mathbb{Q}_q, W) = 0$. *Hint:* for $x = ae + bh + cf$, $(t-1)x = -(2n'b + n'^2 c)e + n'c\,h$; and multiplication by $q - 1 \neq 0$ is surjective on $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. Compare Example 18.35.

**Exercise 18.94** (the lemma for `14a1`)**.** The curve `14a1` $= [1,0,1,4,-6]$ has split multiplicative reduction at 7 with $n_7 = 3$, and the unit part of its Tate parameter reduces to $6 \equiv -1 \pmod 7$. Check the paper's statement $H^1(I_q, \mathrm{ad}^0 T)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2$ for $q = 7$, $\ell = 3$; compute the Frobenius invariants and show $v_3(\mathrm{Tam}_7(\mathrm{ad}^0 T)) = 2$. Explain why $\deg\varphi_E = 1$ is no contradiction with (eq:identity). *Answer:* $-1 = 3^3$ in $\mathbb{F}_7$ is a cube and $v_3(6) = 1$, so the minimum is 1. The curve has $E[3]$ reducible (its torsion is $\mathbb{Z}/6$), so Theorem thm:main does not apply, while Lemma 18.53 needs no irreducibility.

**Exercise 18.95** (the indeterminacy of $c'$)**.** In the situation of Convention 18.52, take $\ell = 3$, $q = 19$ and $n = 6$ (as for `1748e1` at 19). Find a change of Frobenius lift that changes $c'$ by an element of valuation exactly 1, so that the class of $c'$ modulo $q-1$ is not well defined, and check that $\min(v_3(n), v_3(q-1), v_3(c'))$ does not change. Then let $K/\mathbb{Q}_q$ be ramified of degree $e$ with uniformiser $\varpi$: show that replacing $\varpi$ by $\varpi w$ ($w$ a unit) changes the unramified projection of the Kummer class of $q_E$ by the class of $w^{-v_K(q_E)}$, so that "the image of $\kappa$ in $H^1(G_K/I_K, \mathbb{Z}_\ell(1))$" depends on a choice. *Answer:* the change under $\mathrm{Frob} \mapsto \mathrm{Frob}\, t^a$ is $\varepsilon q a n'$, of valuation $v_3(n) = 1$ for $a = 1$, while $v_3(q-1) = 2$; the minimum is capped by $v_3(n) = 1$ either way. Over $K$, $q_E = \varpi^k u_K = (\varpi w)^k (w^{-k} u_K)$ with $k = v_K(q_E)$, so the unit part changes by $w^{-k}$.

**Exercise 18.96** ($\mathrm{Tam}(T)$ and $c_q$)**.** Check the last sentence of the proof of Lemma lem:tamfactor ("the same computation for $M = T$ gives ... the $\ell$-part of the Tamagawa number $c_q(E)$") for `11a1` at $q = 11$, $\ell = 5$ (split, $n = 5$, $c_{11} = 5$) and for `14a1` at $q = 2$, $\ell = 3$ (nonsplit, $n = 6$, $c_2 = 2$). *Answer:* $(\mathbb{Z}/5)\bar{e}_1$ with Frobenius acting by $\varepsilon = 1$: invariants of order 5; $(\mathbb{Z}/3)\bar{e}_1$ with Frobenius acting by $-1$: invariants 0, and $3 \nmid c_2$.

**Exercise 18.97** ($\mathrm{Tam}(A(1)) = \mathrm{Tam}(A)$ at $\ell = 5$)**.** Let $\ell = 5$ and $q$ a prime of split multiplicative reduction with $25 \mid n_q$. (a) For $q = 101$ (so $25 \mid q-1$) and $\bar{u}$ a 25-th power in $\mathbb{F}_{101}^\times$, show that the excess of Lemma 18.53 is 2 and $v_5(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = 4$, and that the invariant modules for $A$ and for $A(1)$ in $(\mathbb{Z}/25)\bar{e} \oplus (\mathbb{Z}/25)\bar{h}$ coincide. (b) For $q = 11$ and $c' = 5$, list both invariant modules and show that they are isomorphic to $\mathbb{Z}/25 \oplus \mathbb{Z}/5$ but are different subgroups. *Answer:* (a) $\min(2,2,2) = 2$; as $25 \mid q-1$, the conditions $(q-1)y \equiv 0$, $c'y \equiv 0$ for $A$ and $(q-1)x - 2c'y \equiv 0$ for $A(1)$ both reduce to $c'y \equiv 0 \pmod{25}$, which holds for all $y$. (b) For $A$: $10y \equiv 5y \equiv 0$, that is $5 \mid y$. For $A(1)$: $10x - 10y \equiv 0 \pmod{25}$, that is $x \equiv y \pmod 5$; the element $\bar{e} + \bar{h}$ lies in the second and not in the first.

**Exercise 18.98** (inertia of order 6, and type IV, at $\ell = 3$)**.** Let $\sigma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ and $\tau = -\sigma$, of order 6 (the image of inertia at the types II, II$^*$ for $q \geq 5$). Show that $H^1(\langle\tau\rangle, T) = 0$ and $H^1(\langle\tau\rangle, \mathrm{ad}^0 T) = 0$ for $T = \mathbb{Z}_3^2$, and compare with the Tamagawa numbers $c_q = 1$ at these types. The curve `50a1` $= [1,0,1,-1,-2]$ has type IV at 5 and $c_5 = 3$: deduce from Proposition 18.51 that Frobenius acts trivially on $H^1(I_5, T_3E)_{\mathrm{tor}} \simeq \mathbb{F}_3$, so $v_3(\mathrm{Tam}_5(T_3E)) = 1$, while $\mathrm{Tam}_5(\mathrm{ad}^0 T_3E) = \mathbb{Z}_3$. *Answer:* $\tau$ has characteristic polynomial $X^2 - X + 1$ and $\tau^6 = 1 \neq \tau$, so the norm $1 + \tau + \cdots + \tau^5$ is 0 on $T$ and $H^1(\langle\tau\rangle, T) = T/(\tau - 1)T$, of order $3^{v_3(\det(\tau-1))} = 1$, since $\det(\tau - 1) = 1 - 1 + 1 = 1$. On $\mathrm{ad}^0 T$, $\tau$ acts as $\sigma$; the subgroup $\langle\tau^3\rangle = \{\pm 1\}$ acts trivially and $\mathrm{Hom}(\{\pm 1\}, \mathrm{ad}^0 T) = 0$, so inflation-restriction gives $H^1(\langle\tau\rangle, \mathrm{ad}^0 T) = H^1(\langle\sigma\rangle, \mathrm{ad}^0 T) = 0$ (Proposition 18.60).

**Exercise 18.99** (the refined bound for `791a1`)**.** The curve `791a1` $= [1,0,1,-31,117]$ has conductor $7 \cdot 113$, type $\mathrm{I}_3$ at 7 and $\mathrm{I}_2$ at 113, and $\deg \varphi_E = 144$. Show that the unit part of its Tate parameter at 7 reduces to a cube in $\mathbb{F}_7^\times$, so $v_3(\mathrm{Tam}_7(\mathrm{ad}^0 T)) = 2$; that the bound (eq:thm) at $\ell = 3$ is 1 and the refined bound is $2 = v_3(144)$; and conclude, granting Theorem thm:main, that $H^1_f(\mathbb{Q}, \mathrm{ad}^0 T_3E \otimes \mathbb{Q}_3/\mathbb{Z}_3) = 0$. *Hint:* the unit part is the reduction of $j_E^{-1} 7^{-3} = (j_E 7^3)^{-1}$; it is $6 = -1$.

**Exercise 18.100** (an Euler term)**.** For `121a1` at $p = 11$ (type II) and $\ell = 3$, check the paper's statement in Lemma lem:fpr that $L_p(A_f, 1)^{-1}$ is an $\ell$-adic integer of valuation $e_\ell(E, p)$. *Answer:* $\chi_{-3}(11) = -1$, so $L_{11}(A_f, 1)^{-1} = 1 + \frac{1}{11} = \frac{12}{11}$, of 3-adic valuation 1.

**Exercise 18.101** (the right end of Lemma lem:dfg21)**.** Prove that the last map of the sequence of Lemma 18.69 is surjective, using the abstract sequence of Chapter 17 and the injectivity of $H^0(\mathbb{Q}_p, W^D) \to H^1(\mathbb{Q}_p, T^D)$ for $p \in \Sigma$. Then consider $\ell = 3$ and $T = \mathbb{Z}_3(\chi_{-3})$ (the character $\chi_{-3}$ on $\mathbb{Z}_3$): show $H^0(\mathbb{Q}, W^D) \simeq \mathbb{Z}/3$ for $W^D = \mu_{3^\infty} \otimes \chi_{-3}$, and explain why the sequence cannot be exact with $\Sigma = \emptyset$ and the Bloch-Kato conditions. *Hint:* $\chi_3^{\mathrm{cyc}} \chi_{-3} \equiv 1 \pmod 3$ but not modulo 9; with $\Sigma = \emptyset$ the middle map would be the identity of $H^1_f(\mathbb{Q}, W)$, and the sequence would force $H^0(\mathbb{Q}, W^D) = 0$.

**Exercise 18.102** ($h_\ell(E)$)**.** Show by the determinant argument that $H^0(\mathbb{Q}, W^D) = 0$ for $\ell \geq 5$ and has order at most 3 for $\ell = 3$, for every $E$ with $\bar\rho$ irreducible. For `726a1`, use the factorisation of its 3-division polynomial to show $h_3 = 1$. Why does an argument with $a_p = 0$ at the primes inert in $\mathbb{Q}(\sqrt{\ell^*})$ not give this? *Hint:* $\rho \simeq \rho \otimes \chi_\ell$ would give $pa_p = a_p$ at good $p$; the relevant twist is by the cyclotomic character, not by the quadratic character $\chi_{\ell^*}$.

**Exercise 18.103** (the Euler characteristic)**.** Verify the four rows of the table of Proposition 18.43 and deduce $H^1_f(\mathbb{Q}, A_f) = 0 \iff H^1_f(\mathbb{Q}, V^D) = 0$. Then apply the general formula to $V^D$ instead of $A_f$ and check that it gives the same equivalence. Where is the sign convention for Hodge-Tate weights used? *Hint:* for $V^D$ the term at $\ell$ is $\dim H^1_f(\mathbb{Q}_\ell, V^D) - 0 = 2$ and the term at $\infty$ is $-\dim (V^D)^{c=1} = -2$ ($c$ acts on $\mathbb{Z}_\ell(1)$ by $-1$).

**Exercise 18.104** (Section sec:tam on a small range)**.** Using the expansion $q_E = j^{-1} + 744 j^{-2} + \cdots$ to obtain the unit parts, reproduce the test of Example 18.80 for conductors at most 2,000 and $\ell = 3$: among the 2,158 optimal non-CM curves with $3 \nmid N$, $E[3]$ irreducible and image not `3Ns`, find the 128 with a potentially multiplicative prime $q \equiv 1 \pmod 3$, $3 \mid \nu_q$; check that equality rises from 87 to 102 curves, and that 19 have a positive Tamagawa excess.

**Exercise 18.105** (the conjecture and the identity)**.** (Harder.) Show that Conjecture 18.83 for $M = h^1(E)(1)$ reproduces the local and torsion terms of the Birch and Swinnerton-Dyer formula in rank 0, and that for $M = \mathrm{ad}^0 M_f(1)$ each term on the right corresponds to a term of (eq:identity) or of Remark rem:ns. Where does $h_\ell(E)$ appear, and why does $H^0(\mathbb{Q}, W)$ not appear in (eq:identity)? *Hint:* Example 18.84; Proposition 18.14(i).

## 18.16 Notes and references

**Sources followed.** The Bloch-Kato local conditions, Selmer groups and the conjecture are from Bloch and Kato [14] (Sections 3, 5 and 6) and the reformulation of Fontaine and Perrin-Riou [69]; the propagation of local conditions and the comparison sequences follow Rubin [167] (Chapter 1) and are proved, for this book, in Chapter 17; local and global duality are in Milne [131], Neukirch, Schmidt and Wingberg [143], Tate [195], Greenberg [74] and Wiles [207]. $p$-adic Hodge theory is quoted from Fontaine [67], Fontaine and Laffaille [68], Coleman and Iovita [31] and Berger's survey [11]. The adjoint motive and the theorems on it are those of Diamond, Flach and Guo [49] and Flach [66]; critical values are Deligne's [42]; the symmetric square is due to Shimura [183] and Gelbart and Jacquet [73]. The structure of $E(\mathbb{Q}_p^{\mathrm{ur}})$ used in Proposition 18.51 is in Silverman [184] (Chapter IV) and Grothendieck [76]; the images of inertia are due to Serre and Tate [179] and Kraus [107].

**Theorem numbers.** We cite theorem numbers where the paper cites them: [49, Theorems 0.1, 2.7, 3.7, Lemma 2.1, (31), (57), p. 708], all checked against the published text (as are the further

items of [49] named in the reference proofs and remarks above: Theorems 2.10, 2.15, Lemmas 2.9, 2.11, 2.12, Proposition 2.16, Conjecture 2.14 and their equations (10), (13), (16), (38), (43), (44), (48), (51) to (53), (57), (58)), and [69, I.4.2.2]. All other references are to sections, and the section numbers given for Bloch and Kato (Sections 3 to 6), Rubin (Sections 1.3 and 1.7), Berger (Section 2) and Silverman (Chapter IV) are pointers we have not checked against the printed texts. In particular we have not verified the numbers under which the dimension formula of Proposition 18.28, the duality of Proposition 18.34 at $\ell$ and the identification of the Kummer image at $\ell$ (Example 18.41) appear in [14]. The statement of Theorem 18.87 and the items [66, Theorem 1, Propositions 1.1, 2.2] were checked against the published text.

**History.** Bloch and Kato formulated their conjecture in 1990 [14]; Fontaine and Perrin-Riou recast it in terms of Galois cohomology and defined the local Tamagawa factors used here [69]. Flach proved the first finiteness theorem for the Selmer group of the symmetric square in 1992 [66], and Diamond, Flach and Guo proved the $\ell$-part of the conjecture for adjoint motives of modular forms in 2004 [49], using the Taylor-Wiles method.

**What the book adds.** Proposition 18.50 proves $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$ for the adjoint of an elliptic curve in every case, where the paper cites [49, (57)]; Proposition 18.51 gives a proof that $\mathrm{Tam}_p(T_\ell E)$ is the $\ell$-part of $c_p$ at every $p \neq \ell$; Proposition 18.38(iii) and Proposition 18.43 make explicit the step from Theorem thm:dfg27 to the finiteness of $H^1_f(\mathbb{Q}, W)$ and the equivalence used in Remark rem:ns. For the paper's Lemma lem:tamadd and Corollary cor:explicit, Theorem 18.61 and Corollary 18.63 give the same argument with every step expanded.

**Computations.** All numbers were computed with PARI/GP 2.17.2 [148] through `cypari2`, on Cremona's files `allcurves`, `degphi` and `galrep` [36] for conductors below 10,000. Curves with CM were recognised by their $j$-invariant among the 13 rational CM $j$-invariants and removed. The optimal curve of a class is the one listed in `degphi`; irreducibility of $E[\ell]$ means that `galrep` lists no label $\ell$B or $\ell$Cs; the image `3Ns` is read from `galrep`. The terms $e_\ell(E, q)$ were computed from `lfuneuler(lfunsympow(E,2),q)` exactly as in the paper's function `rule`; the unit parts of the Tate parameters as the reduction of $j_E^{-1} q^{v_q(j_E)}$, checked against `ellinit(E,O(q^20)).tate` on `11a1` and `791a1`; $v_\ell(c(u))$ by testing $\bar{u}^{(q-1)/\ell^k} = 1$. The counts are those of Remark 18.64 (13,203, 23,361, 28,134 curves), Example 18.80 and Exercise 18.104; the check of Proposition 18.50 (26,448 parameter choices) and of $v_\ell(c_q)$ (101,336 pairs on the 17,834 curves of conductor below 3,000) are described after Proposition 18.50 and Proposition 18.51; the coordinates of Proposition 18.60 and the outputs of `rule` for `121a1`, `726a1`, `791a1` ($[1,1]$, $[2,2]$, $[2,1]$) were recomputed. Nothing beyond conductor 10,000 was checked.

# Chapter 19

# $L$-functions: Euler products, local factors, the adjoint and the symmetric square

The paper's title names the object of this chapter: the local factors of the adjoint $L$-function. The paper's guiding sentence is that "the primes at which the naive and the true adjoint $L$-functions differ are exactly the primes at which the modular degree acquires extra local divisibility", and its local terms $e_\ell(E,q)$ are valuations of Euler factors. This chapter supplies everything about $L$-functions that the paper uses. It starts from Dirichlet series and Euler products (the Riemann zeta function, Dirichlet $L$-functions and Dedekind zeta functions), defines the $L$-function of an $\ell$-adic representation by its local factors with the geometric Frobenius, and specialises to $L(E,s)$, to the adjoint $L(\mathrm{ad}^0 f, s)$ and to the symmetric square $L(\mathrm{Sym}^2 f, s) = L(\mathrm{ad}^0 f, s-1)$. It then does three things the paper needs in detail. It separates the naive adjoint $L$-function from the true one and proves the exact relations with the Rankin series $\sum a_n^2 n^{-s}$ and with Shimura's series $\sum a_{n^2} n^{-s}$; it turns the Rankin-Selberg residue of Chapter 13 and the formula of Zagier and Watkins of Chapter 14 into the identity $v_\ell(\deg \varphi_E) = v_\ell(B_E) + \sum_q e_\ell(E,q)$, which is the conceptual reason why the omitted factors appear in the paper's theorem; and it computes every local factor at every bad prime, proving the paper's Lemma lem:euler in full at $q \geq 5$ and the paper's Proposition prop:shapes at every prime, 2 and 3 included. The local Galois theory (Weil-Deligne representations, local factors, the shapes of the adjoint) is Chapter 7; the images of inertia by Kodaira type are Chapters 10 and 11; the $L$-function of a newform, its functional equation and the Rankin-Selberg unfolding are Chapter 13; the Bloch-Kato side is Chapter 18; the congruence ideals are Chapter 20.

## Core results at a glance

Throughout, $E/\mathbb{Q}$ is given by a global minimal model, of conductor $N$, $f = f_E \in S_2(\Gamma_0(N))$ is its newform, $\ell$ is a prime, $V = V_\ell E$, $T = T_\ell E$, $A_f = \mathrm{ad}^0 V$ (trace zero), $\delta_p = \dim V^{I_p}$ for $p \neq \ell$; $\mathrm{Frob}_p$ is an arithmetic Frobenius and $\Phi_p = \mathrm{Frob}_p^{-1}$ the geometric one; $X = p^{-s}$ in local factors; $\chi_{-3}$, $\chi_{-4}$, $\chi_{q^*}$ are quadratic characters named by discriminant, $\chi_\ell$ is cyclotomic.

**19.1 Standing conventions.** Objects as above; $\langle\ ,\ \rangle$ is the unnormalised Petersson product, PARI's `mfpetersson` being $\langle\ ,\ \rangle/\mu(N)$, $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$; Kodaira codes of PARI

as in Appendix B. *(convention)* Convention 19.1.

**19.2 Frobenius and local factors.** For $p \neq \ell$ and a $G_{\mathbb{Q}_p}$-representation $M$ on a $\mathbb{Q}_\ell$-space, $L_p(M,s) = \det(1 - \Phi_p p^{-s} \mid M^{I_p})^{-1}$; so $L_p(M(n),s) = L_p(M, s+n)$, $L_p(\mathbb{Q}_\ell(1),s) = (1-p^{-1-s})^{-1}$, $L_p(V_\ell E^\vee, s) = L_p(E,s)$, $L_p(V_\ell E, s) = L_p(E, s+1)$. An arithmetic eigenvalue $\lambda$ is the geometric eigenvalue $\lambda^{-1}$. *(convention)* Convention 19.2.

**19.3 Normalisations.** $L(\mathrm{ad}^0 f, s) = \prod_p L_p(A_f, s)$ (value used at $s = 1$); $L(\mathrm{Sym}^2 f, s) = \prod_p L_p(\mathrm{Sym}^2 V^\vee, s)$, which is $L(\mathrm{ad}^0 f, s-1)$ (value used at $s = 2$); $P_q(X) \in \mathbb{Z}[X]$ with $L_q(\mathrm{Sym}^2 f, s) = P_q(q^{-s})^{-1}$, so $P_q(q^{-2}) = L_q(A_f, 1)^{-1}$; the naive factor $L_p^{\mathrm{nv}}(A_f, s)$ is $L_p(A_f, s)$ if $\delta_p > 0$ and 1 if $\delta_p = 0$; the Rankin series $D(f,s) = \sum a_n^2 n^{-s}$ is never called $L(\mathrm{Sym}^2 f, s)$. *(convention)* Convention 19.4.

**19.4 Dirichlet series.** $D(s) = \sum a_n n^{-s}$; if $\sum |a_n| n^{-\sigma_0} < \infty$ it converges absolutely and uniformly on $\mathrm{Re}\, s \geq \sigma_0$ and is holomorphic on $\mathrm{Re}\, s > \sigma_0$. *(definition)* Definition 19.5.

**19.5 Convergence, uniqueness, products.** $a_n = O(n^c)$ gives absolute convergence on $\mathrm{Re}\, s > c+1$; a Dirichlet series vanishing for all large real $s$ is zero; products of absolutely convergent Dirichlet series are Dirichlet series with the Dirichlet convolution. *(proved here)* Proposition 19.6.

**19.6 Euler products.** If $a$ is multiplicative and $\sum |a_n| n^{-\sigma} < \infty$, then on $\mathrm{Re}\, s \geq \sigma$ the series $\sum a_n n^{-s}$ equals $\prod_p \sum_k a_{p^k} p^{-ks}$, the product converging absolutely, and the sum is nonzero wherever every factor is. *(proved here)* Proposition 19.7.

**19.7 Gamma factors and the Mellin transform.** $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$, $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{C}}(s)$; $\int_0^\infty \sum a_n e^{-2\pi n y} y^{s-1} dy = (2\pi)^{-s}\Gamma(s) \sum a_n n^{-s}$ for $a_n = O(n^c)$, $\mathrm{Re}\, s > c+1$. *(proved here; duplication formula stated)* Lemma 19.8.

**19.8 The Riemann zeta function.** $\zeta(s) = \prod_p (1-p^{-s})^{-1} \neq 0$ on $\mathrm{Re}\, s > 1$; $\zeta(s) - \frac{1}{s-1}$ is holomorphic on $\mathrm{Re}\, s > 0$; $\zeta(2) = \pi^2/6$; $\xi(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ is meromorphic on $\mathbb{C}$ with simple poles at $0, 1$ only and $\xi(s) = \xi(1-s)$; $\zeta(1+it) \neq 0$. *(proved here for (i) to (iii); (iv) sketched; (v) stated)* Theorem 19.9.

**19.9 Partial zeta functions.** $\zeta^{(N)}(s) = \zeta(s) \prod_{p \mid N} (1 - p^{-s})$. *(definition)* Definition 19.10.

**19.10 The residue that turns Rankin into Petersson.** $\mathrm{Res}_{s=2}\, \zeta^{(N)}(s-1)/\zeta^{(N)}(2s-2) = \frac{6}{\pi^2} \prod_{p \mid N} (1+p^{-1})^{-1} = \frac{6N}{\pi^2 \mu(N)}$. *(proved here)* Proposition 19.11.

**19.11 Dirichlet $L$-functions.** $L(s,\chi) = \prod_p (1 - \chi(p) p^{-s})^{-1}$ on $\mathrm{Re}\, s > 1$; holomorphic on $\mathrm{Re}\, s > 0$ for $\chi \neq 1$; the functional equation of $\Lambda(s,\chi)$ with $W(\chi) = \tau(\chi)/(i^a\sqrt{m})$, $W = 1$ for real $\chi$; $L(1, \chi_{-4}) = \pi/4$, $L(1, \chi_{-3}) = \pi/(3\sqrt{3})$; $L(1,\chi) \neq 0$ and Dirichlet's theorem. *(proved here for (i), (ii), (iv)(a); (iii) sketched; (iv)(b), (v) stated)* Theorem 19.12.

**19.12 Dedekind zeta functions.** $\zeta_K = \zeta \cdot L(s, \chi_{D_K})$ for $K$ quadratic of discriminant $D_K$; $\zeta_K = \prod_\chi L(s,\chi)$ for $K$ abelian; Artin formalism. *(proved here for (i); (ii), (iii) stated)* Theorem 19.13.

**19.13 Local factors of $\ell$-adic representations.** $L_p(M,s)$ depends only on $\mathrm{WD}(M)$; it is additive and $L_p(M(n),s) = L_p(M, s+n)$; the arithmetic Frobenius gives the same factor

iff the eigenvalues on $M^{I_p}$ are stable under inversion (true for $\mathrm{ad}^0 V$ at a good prime, false at a multiplicative prime and for $V_\ell E$). *(definition)* Definition 19.15; *(proved here, given Chapter 7; the factor at $p=\ell$ stated)* Proposition 19.16.

**19.14 Compatible systems.** A weakly compatible system $(M_\ell)$ has $L(M,s)=\prod_p L_p(M_{\ell(p)},s)$, $\ell(p)\neq p$; independence of $\ell$ holds for $V_\ell E$, for every tensor construction from it and for $V_\lambda(f)$. *(definition; stated)* Definition 19.17, Theorem 19.18.

**19.15 Absolute convergence.** A system pure of weight $w$ (bad eigenvalues of absolute value at most $p^{w/2}$) has $L(M,s)$ absolutely convergent and nonzero on $\mathrm{Re}\, s>1+w/2$: $L(E,s)$ on $\mathrm{Re}\, s>3/2$, $L(\mathrm{ad}^0 f,s)$ on $\mathrm{Re}\, s>1$, $L(\mathrm{Sym}^2 f,s)$ on $\mathrm{Re}\, s>2$. *(proved here, given the Hasse bound)* Proposition 19.19.

**19.16 The $L$-function of $E$.** $L_p(E,s)^{-1}=1-a_p p^{-s}+p^{1-2s}$ (good), $1-a_p p^{-s}$ with $a_p=\pm1$ (multiplicative), $1$ (additive), with $a_p=p+1-\#\tilde{E}(\mathbb{F}_p)$ at every $p$. *(definition)* Definition 19.21. $L_p(E,s)=L_p(V_\ell E^\vee,s)$ for $p\neq\ell$, $\delta_p=\deg L_p(E,s)^{-1}$, and $L(E,s)$ is an isogeny invariant. *(proved here, given Chapters 9 to 11)* Proposition 19.22.

**19.17 Functional equation of $L(E,s)$.** $L(E,s)=L(f_E,s)$; $\Lambda(E,s)=N^{s/2}(2\pi)^{-s}\Gamma(s)L(E,s)$ is entire with $\Lambda(E,s)=w_E\Lambda(E,2-s)$, $w_E=-\varepsilon_N(f_E)$; $w_E=-\prod_{p|N}w_p$ with $w_p=-a_p$ at multiplicative $p$. *(modularity and local root numbers stated; the rest proved here given Chapter 13, where the functional equation of $L(f,s)$ is sketched)* Theorem 19.23.

**19.18 The adjoint and the symmetric square.** $\mathrm{Sym}^2 V^\vee\simeq A_f(-1)$, so $L_p(\mathrm{Sym}^2 f,s)=L_p(A_f,s-1)$ for every $p$; $P_p\in\mathbb{Z}[X]$, $P_p(0)=1$, $\deg P_p=\dim A_f^{I_p}\in\{0,1,3\}$, $P_p(p^{-2})=L_p(A_f,1)^{-1}\in\mathbb{Z}[1/p]$. *(definition)* Definition 19.24; *(proved here, given Chapter 4 and Sections 19.9, 19.10)* Proposition 19.25.

**19.19 The good factor.** For $p\nmid N\ell$: $L_p(A_f,s)^{-1}=(1-p^{-s})(1-(a_p^2-2p)p^{-1-s}+p^{-2s})$, $P_p(X)=(1-pX)(1-(a_p^2-2p)X+p^2X^2)$, $L_p(A_f,1)^{-1}=\frac{(p-1)((p+1)^2-a_p^2)}{p^3}>0$ (Wiles's level-raising factor); for odd $\ell\neq p$ it is an $\ell$-adic non-unit iff $p\equiv1\pmod\ell$ or $a_p\equiv\pm(p+1)\pmod\ell$. *(proved here)* Proposition 19.26.

**19.20 Analytic theory of $L(\mathrm{Sym}^2 f,s)$.** Meromorphic continuation (Shimura); $L(\mathrm{ad}^0 f,s)$ is the standard $L$-function of the Gelbart-Jacquet lift, entire, with gamma factor $\Gamma_{\mathbb{R}}(s+1)\Gamma_{\mathbb{C}}(s+1)$ and functional equation $s\leftrightarrow1-s$; $L(\mathrm{ad}^0 f,1)>0$. *(stated; positivity proved here, given Theorem 19.36)* Theorem 19.28.

**19.21 Conductor of the adjoint.** At $q\geq5$, $q\neq\ell$, the conductor exponent of $A_f$ is 0 for type $\mathrm{I}_0^*$ and 2 for every other bad type; the conductor of $L(\mathrm{Sym}^2 f,s)$ is $N_{\mathrm{ad}^0}=\prod q^2$ over the bad $q$ not of type $\mathrm{I}_0^*$ when $(N,6)=1$. *(proved here, given Chapter 7)* Proposition 19.29.

**19.22 The naive adjoint $L$-function.** $L_p^{\mathrm{nv}}=L_p$ for $p\nmid N$; for $p\mid N$, $L_p^{\mathrm{nv}}(A_f,s)^{-1}=1-a_p(f)^2p^{-1-s}$; so $L^{\mathrm{nv}}(\mathrm{ad}^0 f,s)$ is built from the $a_p(f)$ alone, and $L^{\mathrm{nv}}(\mathrm{ad}^0 f,s)=L(\mathrm{ad}^0 f,s)\prod_{p^2|N}L_p(A_f,s)^{-1}$. *(definition)* Definition 19.32; *(proved here, given Chapters 11 and 13)* Proposition 19.33.

**19.23 Rankin's and Shimura's series.** For $\mathrm{Re}\, s>2$, exactly: $\sum a_n(f)^2n^{-s}=\zeta^{(N)}(s-1)L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)/\zeta^{(N)}(2s-2)$ and $\sum a_{n^2}(f)n^{-s}=L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)/\zeta^{(N)}(2s-2)$. *(proved here; the first given Chapter 13)* Proposition 19.34.

**19.24 Petersson norm.** $\langle f,f\rangle = \frac{N}{8\pi^3}L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2) = \frac{N}{8\pi^3}L(\mathrm{ad}^0 f,1)\prod_{p^2\mid N}L_p(A_f,1)^{-1}$: only the additive primes contribute a correction. *(the first equality is Chapter 13's, where the unfolding is sketched; the true-L form proved here, given it)* Theorem 19.36.

**19.25 The degree formula.** For every modular parametrisation,

$$\deg\varphi\cdot\mathrm{vol}(\Lambda_E) = \frac{c_\varphi^2 N L^{\mathrm{nv}}(\mathrm{ad}^0 f,1)}{2\pi};\qquad B_E = \frac{c_E^2 N L(\mathrm{ad}^0 f,1)}{2\pi\,\mathrm{vol}(\Lambda_E)}\in\mathbb{Q}^\times$$

satisfies $\deg\varphi_E = B_E\prod_{q^2\mid N}L_q(A_f,1)^{-1}$, and $v_\ell(\deg\varphi_E) = v_\ell(B_E)+\sum_{q\mid N}e_\ell(E,q)$ for every $\ell\nmid N$, unconditionally. *(proved here, given Chapters 13 and 14)* Corollary 19.37.

**19.26 Flach's theorem.** If the conductor of $E$ is minimal among its quadratic twists, $p\geq 5$ is a prime of good reduction and $\bar\rho_{E,p}$ is surjective, the degree of any modular parametrisation $X_0(N)\to E$ annihilates the Bloch-Kato Selmer groups of the symmetric square of $E$ with $\mathbb{Q}_p/\mathbb{Z}_p$ coefficients (Tate twists 1 and 2), which are therefore finite. *(stated; reference)* Theorem 19.40.

**19.27 The local terms.** For a bad $q\neq\ell$: $e_\ell(E,q) = v_\ell(L_q(A_f,1)^{-1}/L_q^{\mathrm{nv}}(A_f,1)^{-1})$, which is $v_\ell(P_q(q^{-2})/(1-a_q^2q^{-2}))$; for every bad $q$: $t_\ell(E,q) = v_\ell(-v_q(j_E))$ if $v_q(j_E)<0$, else 0. *(definition; the paper's Definition def:terms)* Definition 19.41. $e_\ell(E,q)\geq 0$, $=0$ at multiplicative $q$, $=v_\ell(P_q(q^{-2}))$ at additive $q$; $t_\ell(E,q)=v_\ell(n)$ for $\mathrm{I}_n$ at any $q$ and $\mathrm{I}_n^*$ at odd $q$. *(proved here, given Chapter 10)* Proposition 19.42.

**19.28 Multiplicative primes.** $A_f^{I_q}$ is the line of $e$, $\Phi_q$ acts by $q^{-1}$, $P_q = 1-X$, $L_q(A_f,1)^{-1} = 1-q^{-2}$, $e_\ell(E,q)=0$; with $\mathrm{Frob}_q$ one would get $1-q^{1-s}$. *(proved here, given Chapter 11)* Proposition 19.43.

**19.29 Lemma lem:euler.** At a bad $q\geq 5$, $q\neq\ell$, the table of Lemma 19.45: $P_q$ is $1-X$ ($\mathrm{I}_n$, $\mathrm{I}_n^*$), $1-\chi_{-3}(q)qX$ (II, II*, IV, IV*), $1-\chi_{-4}(q)qX$ (III, III*), $(1-qX)(1-(a'^2-2q)X+q^2X^2)$ ($\mathrm{I}_0^*$, $a'=a_q(E^{(q^*)})$); $e_\ell$ is 0, $v_\ell(q^2-1)$, $v_\ell(q-\chi_{-3}(q))$, $v_\ell(q-\chi_{-4}(q))$, $v_\ell(q-1)+v_\ell((q+1)^2-a'^2)$. The rows $\mathrm{I}_n^*$, $\mathrm{I}_0^*$ hold at every odd $q\neq\ell$, the row $\mathrm{I}_n$ at every $q\neq\ell$. *(proved here, given Kraus's theorem via Chapter 11)* Lemma 19.45.

**19.30 Twists.** $L_q(A_f,s)$ and $L(\mathrm{ad}^0 f,s)$ are invariant under every quadratic twist, $t_\ell(E,q)$ too; $e_\ell(E,q)$ is invariant under twists unramified at $q$ but not in general (`11a1`: $e_3=0$; `121d2`: $e_3=1$). *(proved here)* Proposition 19.49.

**19.31 The $L$-part of Lemma lem:fpr.** For every prime $p\neq\ell$: $\det(1-\Phi_p\mid (V^D)^{I_p}) = L_p(A_f,1)^{-1} = P_p(p^{-2})$, a nonzero element of $\mathbb{Z}[1/p]$, so a nonzero $\ell$-adic integer, of $\ell$-adic valuation $v_\ell((p-1)((p+1)^2-a_p^2))$ (good $p$), $v_\ell(p^2-1)$ (multiplicative $p$, or additive with $v_p(j_E)<0$), $e_\ell(E,p)$ (additive $p$). *(proved here)* Proposition 19.52.

**19.32 The naive factors at $s=1$.** $L_p^{\mathrm{nv}}(A_f,1)^{-1}$ is $\frac{(p-1)((p+1)^2-a_p^2)}{p^3}$, $1-p^{-2}$, 1 for $p\nmid N$, $p\parallel N$, $p^2\mid N$; the true and naive valuations differ by $e_\ell(E,p)$. *(proved here)* Proposition 19.54.

**19.33 Every bad prime (the paper's Proposition prop:shapes).** At every bad $q\neq\ell$, 2 and 3 included, $P_q$ is $1-X$ (potentially multiplicative), $(1-qX)(1-(a'^2-2q)X+q^2X^2)$ ($\rho(I_q)=\{\pm1\}$, $a'=a_q$ of a quadratic twist with good reduction), $1-cqX$ with $c=\pm1$ ($\rho(I_q)$ cyclic of order 3, 4, 6), or 1 ($\rho(I_q)$ non-abelian, only at $q=2,3$); consequently

$e_3(E,2) \le 2$, $e_5(E,2) \le 1$, $e_\ell(E,2) = 0$ for $\ell \ge 7$, $e_5(E,3), e_7(E,3) \le 1$, $e_\ell(E,3) = 0$ for $\ell \ge 11$. *(proved here, given Chapters 7 and 11)* Proposition 19.56.

## Where the paper uses this chapter

- **Section 1 (introduction).** "The classical route to the degree is the Petersson norm": the Petersson norm as a special value (Theorem 19.36), the exact degree formula of Remark rem:degree (Corollary 19.37), the naive function $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = \zeta^{(N)}(2s-2)\sum a_{n^2}(f) n^{-s}$ of Shimura (Proposition 19.34), and Flach's theorem (Theorem 19.40).
- **Section 1.1 and Definition def:terms.** $\delta_p$, $L_p(A_f, s)$ with the geometric Frobenius, the naive factor, the shift $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$, PARI's `lfunsympow` (Conventions 19.2 and 19.4, Proposition 19.25, Remark 19.60); $e_\ell$, $t_\ell$ and their behaviour under twist (Definition 19.41, Proposition 19.49); the table of Section 1 (Lemma 19.45).
- **Lemma lem:euler, Proposition prop:shapes, Remark rem:wild, Section sec:wild.** Lemma 19.45, Proposition 19.56, Remark 19.59.
- **Theorem thm:dfg14 and the paragraph after it.** The naive factors at $s = 1$: Wiles's level-raising factor at a good prime, $1 - p^{-2}$ at a multiplicative prime, nothing at an additive prime (Proposition 19.26, Proposition 19.54).
- **Lemma lem:fpr and the paragraph after Lemma lem:dfg21.** "$L_p(A_f, 1)^{-1} = L_p(A_f(1), 0)^{-1}$ is a nonzero rational number which is an $\ell$-adic integer", with its valuation, and $\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1} \ne 0$ (Proposition 19.52).
- **The proof of Theorem thm:main, Theorem thm:quant (Step 3) and Remark rem:afterquant.** The cancellation of the naive factors and the values of the omitted ones (Proposition 19.54, Proposition 19.49, Remark 19.47).
- **Proposition prop:forms, Conjecture conj:euler, Sections sec:iso, sec:sharp and the appendix.** The values of $e_\ell$ by type and residue class (Remark 19.47), the factors at 2 and 3 (Proposition 19.56), the mathematics of `rule(E, l)` (Example 19.61) and the worked examples (Example 19.62).

## 19.1 Introduction and conventions

This section fixes the objects, the Frobenius convention and the four normalisations of the adjoint and symmetric square $L$-functions that the rest of the chapter uses. In the paper these are the conventions of Section 1.1 ("We follow the conventions of [49, §1.7.2]"), which are used in Definition def:terms, the table of Section 1, Lemma lem:euler, Lemma lem:fpr and the paragraph after Lemma lem:dfg21.

An $L$-function in this book is a Dirichlet series $\sum a_n n^{-s}$ with an Euler product $\prod_p L_p(s)$ whose factors are reciprocals of polynomials in $p^{-s}$. Three kinds occur. The *analytic* ones ($\zeta$, $L(s,\chi)$, $\zeta_K$) are defined by their coefficients. The *automorphic* ones ($L(f,s)$, the Rankin series) come from a modular form. The *arithmetic* ones ($L(E,s)$, $L(\mathrm{ad}^0 f, s)$) are defined by their local factors, which are characteristic polynomials of Frobenius on inertia invariants of a Galois representation. The paper lives on the comparison of the last two kinds: the naive adjoint

$L$-function is what the $q$-expansion of $f$ sees, the true one is what the Galois representation sees, and they differ exactly at the additive primes.

For the reader with a physics background three familiar objects reappear. The Mellin transform, which turns a $q$-expansion or a theta series into a Dirichlet series, is the Fourier transform on the multiplicative group $\mathbb{R}_{>0}$ ([165, Chapter 13]; Lemma 19.8). Poisson summation for the Gaussian, which is its own Fourier transform, gives the functional equation of $\zeta$ (Theorem 19.9). And, as an aside only, the Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ is the partition function at inverse temperature $s$ of a gas of free bosons with one mode of energy $\log p$ for each prime $p$ (the "primon gas"): each factor is a geometric series over occupation numbers, and unique factorisation says that the states are the integers.

**Convention 19.1** (standing conventions)**.** (a) $E/\mathbb{Q}$ is given by a global minimal model, $N$ is its conductor, and $f = f_E \in S_2(\Gamma_0(N))$ its newform (Chapter 14); $a_n = a_n(f)$, so $a_p = a_p(E)$ for every $p$. $\ell$ is a prime, $T = T_\ell E$, $V = V_\ell E = T \otimes \mathbb{Q}_\ell$, $\rho = \rho_{E,\ell}$, $A = \mathrm{ad}^0 T$ and $A_f = \mathrm{ad}^0 V$ (trace zero; the paper's notation, Appendix B, B-C12). For $p \neq \ell$, $\delta_p = \dim V^{I_p} \in \{2, 1, 0\}$ for good, multiplicative, additive reduction (Chapter 11); for $p = \ell$ the same symbol means $\dim V_{\ell'} E^{I_\ell}$ for any $\ell' \neq \ell$. Statements about $f$ hold for every newform of weight 2 with trivial character and rational coefficients, with $V = V_\lambda(f)$ (Chapter 15); where they need $E$ we say so.

(b) $p$, $q$, $\ell$ are primes; $q$ is usually a bad prime. In Lemma 19.8 and in Section 19.8 the variable of a $q$-expansion appears only as $e^{2\pi i z}$, never as $q$.

(c) Quadratic characters are named by discriminant (Appendix B, B-C6): $\chi_D$ is the character of $\mathbb{Q}(\sqrt{D})$ for a fundamental discriminant $D$, so $\chi_{-3}$, $\chi_{-4}$, $\chi_8$, $\chi_{-8}$ and, for an odd prime $q$ with $q^* = (-1)^{(q-1)/2} q$, the character $\chi_{q^*}$ of conductor $q$. A letter subscript $\chi_\ell$ is the $\ell$-adic cyclotomic character. A Dirichlet character is also the Galois character with $\chi(\mathrm{Frob}_p) = \chi(p)$. $E^{(d)} = E \otimes \chi_d$ is the quadratic twist by $\mathbb{Q}(\sqrt{d})$.

(d) $\langle f, g \rangle = \int_{\Gamma_0(N)\backslash \mathcal{H}} f \bar{g}\, dx\, dy$ is the unnormalised Petersson product of weight 2 (Appendix B, B-C8); PARI's `mfpetersson` returns $\langle f, g \rangle / \mu(N)$ with $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = N \prod_{p \mid N} (1 + p^{-1})$. $\Lambda_E$ is the period lattice of the Neron differential $\omega_E$ of the minimal model, $\mathrm{vol}(\Lambda_E)$ its covolume (PARI's `E.area`), and $c_E > 0$ the Manin constant (Appendix B, B-C9).

(e) Kodaira symbols are $\mathrm{I}_n$, $\mathrm{I}_n^*$ ($n \geq 1$), $\mathrm{I}_0^*$, II, III, IV and their duals. PARI's codes (the second entry of `elllocalred(E,q)`) are $n+4$ for $\mathrm{I}_n$ and $-(n+4)$ for $\mathrm{I}_n^*$ when $n \geq 1$, 1 for $\mathrm{I}_0$, $-1$ for $\mathrm{I}_0^*$, and $\pm 2, \pm 3, \pm 4$ for II/II$^*$, III/III$^*$, IV/IV$^*$ (Chapter 10).

(f) No component groups occur in this chapter, so the geometric Frobenius at $p$ is written $\Phi_p$ (Appendix B, B-C2(b)).

**Convention 19.2** (Frobenius and local factors)**.** $\mathrm{Frob}_p$ is an *arithmetic* Frobenius at $p$ (it induces $x \mapsto x^p$ on the residue field and acts on $\mu_n$, $p \nmid n$, by $\zeta \mapsto \zeta^p$); $\Phi_p = \mathrm{Frob}_p^{-1}$ is the *geometric* Frobenius. An eigenvalue $\lambda$ of $\mathrm{Frob}_p$ is the eigenvalue $\lambda^{-1}$ of $\Phi_p$. Group theory and cohomology use $\mathrm{Frob}_p$ (the tame relation $\mathrm{Frob}\ t\ \mathrm{Frob}^{-1} = t^p$, unramified cohomology); $L$-factors use $\Phi_p$: for $p \neq \ell$ and a continuous representation $M$ of $G_{\mathbb{Q}_p}$ on a finite-dimensional $\mathbb{Q}_\ell$-space,

$$L_p(M, s) = \det\left(1 - \Phi_p\, p^{-s} \mid M^{I_p}\right)^{-1}$$

(Chapter 7). Consequently

$$L_p(\mathbb{Q}_\ell(1),s)=(1-p^{-1-s})^{-1},\qquad L_p(M(n),s)=L_p(M,s+n),$$
$$L_p(V_\ell E^\vee,s)=L_p(E,s),\qquad L_p(V_\ell E,s)=L_p(E,s+1).$$

At a good prime $p\neq\ell$ the arithmetic Frobenius has eigenvalues $\alpha,\beta$ on $V_\ell E$, the roots of $X^2-a_pX+p$; the geometric one has $\alpha,\beta$ on $V_\ell E^\vee=H^1(E_{\overline{\mathbb{Q}}},\mathbb{Q}_\ell)$; on $\mathrm{ad}^0\,V$ both have $\alpha/\beta,1,\beta/\alpha$ (Appendix B, B-C3).

The displayed consequences are short computations, and they show where the choice of $\Phi_p$ matters. Let $p\neq\ell$. The group $G_{\mathbb{Q}_p}$ acts on $\mathbb{Q}_\ell(1)$ through the cyclotomic character $\chi_\ell$, which is unramified at $p$, and $\chi_\ell(\mathrm{Frob}_p)=p$ because $\mathrm{Frob}_p$ raises roots of unity of order prime to $p$ to the $p$-th power. So $\mathbb{Q}_\ell(1)^{I_p}=\mathbb{Q}_\ell(1)$, the geometric Frobenius acts on it by $p^{-1}$, and $\det(1-\Phi_p p^{-s})=1-p^{-1-s}$. For a general $M$, the twist $M(n)=M\otimes\mathbb{Q}_\ell(1)^{\otimes n}$ has the same inertia invariants as $M$ (the twisting character is unramified), and each eigenvalue $\mu$ of $\Phi_p$ on $M^{I_p}$ becomes $\mu p^{-n}$; since $1-\mu p^{-n}p^{-s}=1-\mu p^{-(s+n)}$, the factor of $M(n)$ at $s$ is the factor of $M$ at $s+n$. At a good prime $p$ the eigenvalues of $\Phi_p$ on $V_\ell E^\vee$ are $\alpha,\beta$, so

$$\det(1-\Phi_pX\mid V_\ell E^\vee)=1-(\alpha+\beta)X+\alpha\beta X^2=1-a_pX+pX^2,$$

which at $X=p^{-s}$ is $L_p(E,s)^{-1}$ of Definition 19.21; and the Weil pairing $V_\ell E\times V_\ell E\to\mathbb{Q}_\ell(1)$ gives $V_\ell E\simeq V_\ell E^\vee(1)$, whence $L_p(V_\ell E,s)=L_p(V_\ell E^\vee,s+1)=L_p(E,s+1)$. (The bad primes are Proposition 19.22.) So $L(E,s)$ is the $L$-function of the *dual* $V_\ell E^\vee$, the first $\ell$-adic cohomology of $E$, and not of the Tate module itself; with the Tate module one gets the same function shifted by one.

*Warning* 19.3 (the two Frobenii in the paper). The paper says in Section 1.1 that $L_p(A_f,s)=\det(1-\mathrm{Frob}_p^{-1}\,p^{-s}\mid A_f^{I_p})^{-1}$ "is the true local factor, defined with the geometric Frobenius as in [49, §1.1.1]", and that "The matrices in the proof of Lemma lem:tamfactor are those of the arithmetic Frobenius". Both are needed. The table of Section 1, Lemma lem:euler, PARI's factors and the identity $L_p(A_f,1)=L_p(A_f(1),0)$ of Lemma lem:fpr hold for $\Phi_p$ and fail for $\mathrm{Frob}_p$ at a multiplicative prime: there $\mathrm{Frob}_q$ acts on $A_f^{I_q}$ by $q$ and would give $1-q^{1-s}$ (Proposition 19.43). The rows of the potentially good types (II, II*, III, III*, IV, IV*, $\mathrm{I}_0^*$) do not see the difference, because there the eigenvalues on $A_f^{I_q}$ are stable under inversion; the rows $\mathrm{I}_n$ and $\mathrm{I}_n^*$ do, since for $\mathrm{I}_n^*$ too $A_f^{I_q}$ is the line of $e$ with $\mathrm{Frob}_q$ acting by $q$ (Proposition 19.16(iii)).

**Convention 19.4** (the four normalisations). (a) The *adjoint* $L$-function,

$$L(\mathrm{ad}^0\,f,s)=L(A_f,s)=\prod_p L_p(A_f,s),$$

has each local factor computed with some prime $\ell\neq p$ (Theorem 19.18). It is the paper's and DFG04's adjoint; its value is used at $s=1$; its functional equation relates $s$ and $1-s$.

(b) The *symmetric square* $L$-function is $L(\mathrm{Sym}^2\,f,s)=\prod_p L_p(\mathrm{Sym}^2\,V^\vee,s)$, which is $L(\mathrm{ad}^0\,f,s-1)$ (Proposition 19.25). It is the paper's $L(\mathrm{Sym}^2\,f,s)$ and PARI's `lfunsympow(E,2)`; its value is used at $s=2$; its functional equation relates $s$ and $3-s$. So $L(\mathrm{Sym}^2\,f,2)=L(\mathrm{ad}^0\,f,1)$, and we often write both.

(c) For every prime $q$, $P_q(X) \in \mathbb{Z}[X]$ is the polynomial with $L_q(\mathrm{Sym}^2 f, s) = P_q(q^{-s})^{-1}$; hence $P_q(q^{-2}) = L_q(A_f, 1)^{-1}$. PARI's `lfuneuler(lfunsympow(E,2),q)` returns the rational function $1/P_q(x)$ in $x = q^{-s}$.

(d) The *naive* factor is $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p > 0$ and 1 if $\delta_p = 0$; $L^{\mathrm{nv}}(\mathrm{ad}^0 f, s) = \prod_p L_p^{\mathrm{nv}}(A_f, s)$ and $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = L^{\mathrm{nv}}(\mathrm{ad}^0 f, s-1)$ (Definition 19.32).

(e) The *Rankin series* is $D(f, s) = \sum_{n \geq 1} a_n(f)^2 n^{-s}$. It is never called $L(\mathrm{Sym}^2 f, s)$; its relation with $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)$ is Proposition 19.34.

The objects and the normalisations used by the sources are compared in Remark 19.31.

Why so many normalisations? The adjoint $A_f$ has weight 0 (its good Frobenius eigenvalues $\alpha/\beta, 1, \beta/\alpha$ have absolute value 1), which makes it the natural object for Galois cohomology and for Bloch-Kato; the symmetric square of $V^\vee$ has weight 2 (eigenvalues $\alpha^2, p, \beta^2$), which is the classical normalisation of Shimura and of PARI. The two differ by the twist $\mathbb{Q}_\ell(-1)$, that is by the shift $s \mapsto s-1$. A concrete case: for `11a1` and $p = 2$ one has $a_2 = -2$, so $\alpha + \beta = -2$, $\alpha\beta = 2$ and $\alpha^2 + \beta^2 = a_2^2 - 2p = 0$; then (Proposition 19.26)

$$P_2(X) = (1-2X)(1-0\cdot X+4X^2) = (1-2X)(1+4X^2), \qquad L_2(A_f,1)^{-1} = P_2(\tfrac{1}{4}) = \tfrac{1}{2}\cdot\tfrac{5}{4} = \tfrac{5}{8},$$

and PARI's `1/lfuneuler(lfunsympow(E,2),2)` for $E =$ `11a1` is indeed $1 - 2x + 4x^2 - 8x^3$. The naive normalisation (d) differs from the true one only in which bad factors are kept, and the Rankin series (e) is a different Dirichlet series altogether (Proposition 19.34).

## 19.2 Dirichlet series and Euler products

This section collects the analytic facts about Dirichlet series that every later section uses: where they converge, why they are determined by their values, and when they factor over the primes. In the paper they are behind every Euler product, from Shimura's series $\sum a_{n^2}(f) n^{-s}$ in the introduction to the polynomial $P_q(X)$ of Definition def:terms.

**Definition 19.5** (Dirichlet series)**.** A *Dirichlet series* is a series $D(s) = \sum_{n \geq 1} a_n n^{-s}$ with $a_n \in \mathbb{C}$ and $s = \sigma + it \in \mathbb{C}$. Its *abscissa of absolute convergence* is $\sigma_a = \inf\{\sigma_0 \in \mathbb{R} : \sum_n |a_n| n^{-\sigma_0} < \infty\} \in [-\infty, +\infty]$.

**Proposition 19.6** (convergence, uniqueness, products)**.** *Let $D(s) = \sum a_n n^{-s}$ be a Dirichlet series.*

*(i) If $\sum_n |a_n| n^{-\sigma_0} < \infty$, then $D$ converges absolutely and uniformly on $\mathrm{Re}\, s \geq \sigma_0$, and $D$ is holomorphic on $\mathrm{Re}\, s > \sigma_0$, with $D'(s) = -\sum a_n (\log n) n^{-s}$.*

*(ii) If $a_n = O(n^c)$, then $\sigma_a \leq c + 1$. The same holds if $a_n = O(n^{c+r})$ for every real $r > 0$.*

*(iii) If $D(\sigma) = 0$ for all real $\sigma$ larger than some $\sigma_1 > \sigma_a$, then $a_n = 0$ for every $n$.*

*(iv) If $D(s) = \sum a_n n^{-s}$ and $G(s) = \sum b_n n^{-s}$ both converge absolutely at $s$, then $D(s)G(s) = \sum_n c_n n^{-s}$ with $c_n = \sum_{de=n} a_d b_e$, absolutely convergent.*

*Proof.* (i) On $\operatorname{Re} s \geq \sigma_0$, $|a_n n^{-s}| = |a_n| n^{-\sigma} \leq |a_n| n^{-\sigma_0}$, so the Weierstrass M-test gives absolute and uniform convergence. Each partial sum is entire, and a locally uniform limit of holomorphic functions is holomorphic, with the derivative the limit of the derivatives (Appendix A, Appendix A); on $\operatorname{Re} s \geq \sigma_0 + 2h$ with $h > 0$ the differentiated series is dominated by $\sum |a_n| (\log n) n^{-\sigma_0 - 2h}$, which converges because $\log n \leq C_h n^h$. (ii) If $|a_n| \leq C n^c$ then $\sum |a_n| n^{-\sigma} \leq C \sum n^{c-\sigma} < \infty$ for $\sigma > c + 1$. In the second case apply this with $c + r$ for each $r > 0$. (iii) Suppose not, and let $m$ be the least index with $a_m \neq 0$. For real $\sigma \geq \sigma_2 > \sigma_a$,

$$m^\sigma D(\sigma) = a_m + \sum_{n > m} a_n \Big(\frac{m}{n}\Big)^\sigma, \qquad \Big| a_n \Big(\frac{m}{n}\Big)^\sigma \Big| \leq |a_n| \Big(\frac{m}{n}\Big)^{\sigma_2} \quad (n > m),$$

and each term tends to 0 as $\sigma \to \infty$; by dominated convergence the sum tends to 0, so $m^\sigma D(\sigma) \to a_m \neq 0$, contradicting $D(\sigma) = 0$ for large $\sigma$. (iv) The double series $\sum_{d,e} a_d b_e (de)^{-s}$ converges absolutely, its absolute series being the product $\sum |a_d| d^{-\sigma} \cdot \sum |b_e| e^{-\sigma}$; so it may be summed in any order, in particular by grouping the pairs with $de = n$. □

**Proposition 19.7** (Euler products)**.** *Let $a \colon \mathbb{N} \to \mathbb{C}$ be multiplicative ($a_1 = 1$ and $a_{mn} = a_m a_n$ for coprime $m, n$) and let $\sigma \in \mathbb{R}$ with $\sum_n |a_n| n^{-\sigma} < \infty$. Then for* $\operatorname{Re} s \geq \sigma$

$$\sum_{n \geq 1} a_n n^{-s} = \prod_p \Big( \sum_{k \geq 0} a_{p^k} p^{-ks} \Big),$$

*where every factor converges absolutely and the product converges absolutely, that is, the sum over $p$ of $\big|\sum_{k \geq 1} a_{p^k} p^{-ks}\big|$ is finite. If every factor is nonzero at $s$, the sum is nonzero at $s$. If moreover $\sum_k a_{p^k} X^k = P_p(X)^{-1}$ with $P_p \in \mathbb{C}[X]$, $P_p(0) = 1$, $\deg P_p \leq d$ for every $p$, we say that the series has an* Euler product of degree $d$.

*Proof.* Each factor is a subseries of $\sum a_n n^{-s}$, so it converges absolutely. For $x \geq 2$ let $\Pi(x)$ be the finite product over $p \leq x$, say over $p_1, \dots, p_r$. A product of finitely many absolutely convergent series can be multiplied out term by term and the terms summed in any order (Proposition 19.6(iv), by induction on $r$), so

$$\Pi(x) = \sum_{k_1, \dots, k_r \geq 0} a_{p_1^{k_1}} \cdots a_{p_r^{k_r}} \, (p_1^{k_1} \cdots p_r^{k_r})^{-s} = \sum_{k_1, \dots, k_r \geq 0} a_{p_1^{k_1} \cdots p_r^{k_r}} \, (p_1^{k_1} \cdots p_r^{k_r})^{-s},$$

the second equality because the prime powers $p_i^{k_i}$ are pairwise coprime and $a$ is multiplicative. By unique factorisation each integer $n$ whose prime factors are all at most $x$ arises from exactly one tuple $(k_1, \dots, k_r)$; so $\Pi(x) = \sum_{n \in S(x)} a_n n^{-s}$, where $S(x)$ is the set of integers all of whose prime factors are at most $x$. Every $n \leq x$ lies in $S(x)$, so the difference between the full series and $\Pi(x)$ is a sum over integers $n > x$, and

$$\Big| \sum_{n \geq 1} a_n n^{-s} - \Pi(x) \Big| \leq \sum_{n > x} |a_n| n^{-\sigma} \longrightarrow 0 \qquad (x \to \infty).$$

For the absolute convergence, $\sum_p \big|\sum_{k \geq 1} a_{p^k} p^{-ks}\big| \leq \sum_{n \geq 2} |a_n| n^{-\sigma} < \infty$, the prime powers being distinct integers. A product $\prod (1 + u_p)$ with $\sum |u_p| < \infty$ and every $1 + u_p \neq 0$ has a nonzero limit: for $p$ larger than some $p_0$, $|u_p| \leq \frac{1}{2}$, the series $\sum_{p > p_0} \log(1 + u_p)$ (principal branch) converges absolutely because $|\log(1+u)| = \big|\sum_{k \geq 1} (-1)^{k+1} u^k / k\big| \leq |u| \sum_{k \geq 0} 2^{-k} = 2|u|$ for $|u| \leq \frac{1}{2}$, and the partial products over $p_0 < p \leq x$ are the exponentials of its partial sums, so they converge to exp of the full sum, which is not 0. The finitely many factors with $p \leq p_0$ are nonzero by hypothesis, so the whole product, which equals the sum, is nonzero. □

The hypothesis that *every* factor is nonzero cannot be dropped from the last assertion, and the absolute convergence cannot be dropped from the first: at $s = 0$ the product $\prod_p (1 - p^{-s})^{-1}$ has every factor infinite, while $\zeta(0) = -\frac{1}{2}$ (Exercise 19.66); an Euler product says nothing outside its region of absolute convergence.

The four examples of the chapter: $\zeta(s) = \sum n^{-s}$ and $L(s, \chi)$ have Euler products of degree 1 (Theorems 19.9 and 19.12); $L(f, s)$ has one of degree 2 (Chapter 13); $L(\mathrm{ad}^0 f, s)$ and $L(\mathrm{Sym}^2 f, s)$ have Euler products of degree 3 by definition (Section 19.6), and the point of Section 19.8 is that their coefficients are not simply $a_n^2$.

**Lemma 19.8** (gamma factors and the Mellin transform). *(i)* $\Gamma(s) = \int_0^\infty e^{-y} y^{s-1}\, dy$ *(*$\mathrm{Re}\, s > 0$*) satisfies* $\Gamma(s+1) = s\Gamma(s)$ *and* $\Gamma(\frac{1}{2}) = \sqrt{\pi}$*, and extends to a meromorphic function on* $\mathbb{C}$ *with simple poles at* $0, -1, -2, \ldots$ *and no zeros.*

*(ii) Put* $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ *and* $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$*. Then* $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{C}}(s)$*.*

*(iii) If* $a_n = O(n^c)$ *and* $F(y) = \sum_{n\geq 1} a_n e^{-2\pi n y}$ *for* $y > 0$*, then for* $\mathrm{Re}\, s > c + 1$

$$\int_0^\infty F(y)\, y^{s-1}\, dy = (2\pi)^{-s}\Gamma(s) \sum_{n\geq 1} a_n n^{-s}.$$

*Proof.* (i) is Appendix A (Appendix A). (ii) Legendre's duplication formula $\Gamma(z)\Gamma(z + \frac{1}{2}) = 2^{1-2z}\sqrt{\pi}\,\Gamma(2z)$ (see [39], the chapter on the Gamma function) with $z = s/2$ gives $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = \pi^{-s-\frac{1}{2}}\, 2^{1-s}\sqrt{\pi}\,\Gamma(s) = 2(2\pi)^{-s}\Gamma(s)$; we do not prove the duplication formula. (iii) For real $\sigma = \mathrm{Re}\, s > c + 1$, $\sum_n \int_0^\infty |a_n| e^{-2\pi n y} y^{\sigma-1} dy = (2\pi)^{-\sigma}\Gamma(\sigma) \sum |a_n| n^{-\sigma} < \infty$ by the substitution $y \mapsto y/(2\pi n)$; so Fubini allows integrating term by term, and the same substitution gives $\int_0^\infty e^{-2\pi n y} y^{s-1} dy = (2\pi n)^{-s}\Gamma(s)$. □

The Mellin transform $\phi \mapsto \int_0^\infty \phi(y) y^{s-1} dy$ is the Fourier transform on the multiplicative group $\mathbb{R}_{>0}$, with Haar measure $dy/y$: in the variable $u = \log y$ it is $\int \phi(e^u) e^{su} du$. Part (iii) is how the $L$-function of a modular form arises from its $q$-expansion (Chapter 13), and how $\zeta$ arises from a theta series below.

## 19.3 The zeta functions of Riemann and Dedekind, and Dirichlet $L$-functions

This section treats the three classical $L$-functions of degree one or two that the rest of the chapter uses as tools. The paper uses the residue of $\zeta$ at 1 and the value $\zeta(2)$ through the Rankin-Selberg formula of Remark rem:degree (Proposition 19.11), and the quadratic characters $\chi_{-3}$, $\chi_{-4}$, $\chi_{q^*}$ of the table of Section 1, of Lemma lem:twist and of Theorem thm:quant; the Dedekind zeta function explains why these characters appear in the local factors (Remark 19.14).

**Theorem 19.9** (the Riemann zeta function). *(i) For* $\mathrm{Re}\, s > 1$*,* $\zeta(s) = \sum_{n\geq 1} n^{-s} = \prod_p (1 - p^{-s})^{-1}$*, and* $\zeta(s) \neq 0$*.*

*(ii)* $\zeta(s) - \frac{1}{s-1}$ *extends to a holomorphic function on* $\mathrm{Re}\, s > 0$*. So* $\zeta$ *is meromorphic on* $\mathrm{Re}\, s > 0$ *with a single simple pole, at* $s = 1$*, of residue* 1*.*

*(iii)* $\zeta(2) = \pi^2/6$*.*

*(iv) (Riemann.)* $\xi(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ *extends to a meromorphic function on* $\mathbb{C}$ *whose only poles are simple poles at* $s = 0$ *and* $s = 1$*, and* $\xi(s) = \xi(1-s)$*.*

*(v) (Hadamard, de la Vallée Poussin.)* $\zeta(1+it) \neq 0$ *for every real* $t \neq 0$*.*

*Proof.* (i) $n \mapsto 1$ is multiplicative and $\sum n^{-\sigma} < \infty$ for $\sigma > 1$; apply Proposition 19.7, the factors $(1-p^{-s})^{-1}$ being nonzero. (ii) For $\operatorname{Re} s > 1$,

$$\zeta(s) - \frac{1}{s-1} = \sum_{n\geq 1} n^{-s} - \int_1^\infty x^{-s}\,dx = \sum_{n\geq 1} \int_n^{n+1} (n^{-s} - x^{-s})\,dx.$$

Here $\int_1^\infty x^{-s}dx = \frac{1}{s-1}$ for $\operatorname{Re} s > 1$, and both the sum and the integral have been cut into the pieces $[n, n+1]$, the term $n^{-s}$ being $\int_n^{n+1} n^{-s}dx$. For $n \leq x \leq n+1$, $|n^{-s} - x^{-s}| = |s\int_n^x u^{-s-1}du| \leq |s|\int_n^x u^{-\sigma-1}du \leq |s|\,n^{-\sigma-1}$, so the $n$-th term is bounded by $|s|n^{-\sigma-1}$. Each term $\int_n^{n+1}(n^{-s} - x^{-s})dx$ is an entire function of $s$, and on a compact set $K \subset \{\sigma > 0\}$, with $|s| \leq R$ and $\sigma \geq \delta > 0$ on $K$, the terms are bounded by $Rn^{-1-\delta}$, a convergent series; so the series converges uniformly on $K$ (M-test), its sum is holomorphic on $\sigma > 0$, and it is the required continuation of $\zeta(s) - \frac{1}{s-1}$. (iii) The Fourier series of the odd function $x$ on $(-\pi, \pi)$ has only sine terms, with coefficients $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin nx\,dx = \frac{2}{\pi}\Big[-\frac{x\cos nx}{n}\Big]_0^{\pi} + \frac{2}{\pi n}\int_0^{\pi}\cos nx\,dx = \frac{2(-1)^{n+1}}{n}$ (integration by parts). Parseval's identity $\frac{1}{\pi}\int_{-\pi}^{\pi} f^2dx = \sum_n b_n^2$ for odd $f$ ([165], Fourier series) gives $\frac{1}{\pi}\cdot\frac{2\pi^3}{3} = \sum_n \frac{4}{n^2}$, that is $\frac{2\pi^2}{3} = 4\zeta(2)$. (iv) and (v) are sketched below and stated respectively. □

*Sketch of proof of (iv).* Let $\theta(y) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 y}$ for $y > 0$. Poisson summation applied to the Gaussian $e^{-\pi x^2 y}$, whose Fourier transform is $y^{-1/2}e^{-\pi\xi^2/y}$ ([165], Fourier transforms), gives $\theta(1/y) = \sqrt{y}\,\theta(y)$. For $n \geq 1$ the substitution $u = \pi n^2 y$ gives $\int_0^\infty e^{-\pi n^2 y} y^{s/2}\frac{dy}{y} = (\pi n^2)^{-s/2}\Gamma(\frac{s}{2})$, the integral of Lemma 19.8(i) at $s/2$. Summing over $n \geq 1$ (Fubini, as in Lemma 19.8(iii)) and using $\sum_{n\geq 1} e^{-\pi n^2 y} = \frac{1}{2}(\theta(y) - 1)$, for $\operatorname{Re} s > 1$,

$$\xi(s) = \pi^{-s/2}\Gamma(\tfrac{s}{2})\sum_{n\geq 1} n^{-s} = \frac{1}{2}\int_0^\infty (\theta(y) - 1)\, y^{s/2}\,\frac{dy}{y}.$$

Split the integral at $y = 1$ and substitute $y \mapsto 1/y$ on $(0,1)$, which turns $\int_0^1 \phi(y)\frac{dy}{y}$ into $\int_1^\infty \phi(1/y)\frac{dy}{y}$. The functional equation of $\theta$ gives $\theta(1/y) - 1 = \sqrt{y}(\theta(y) - 1) + \sqrt{y} - 1$, so

$$\int_0^1 (\theta(y)-1)y^{s/2}\frac{dy}{y} = \int_1^\infty (\theta(y)-1)y^{(1-s)/2}\frac{dy}{y} + \int_1^\infty (y^{(1-s)/2} - y^{-s/2})\frac{dy}{y},$$

and the last integral is $\frac{2}{s-1} - \frac{2}{s}$ for $\operatorname{Re} s > 1$. Halving and adding the integral over $(1,\infty)$,

$$\xi(s) = \frac{1}{2}\int_1^\infty (\theta(y)-1)(y^{s/2} + y^{(1-s)/2})\frac{dy}{y} - \frac{1}{s} - \frac{1}{1-s}.$$

The integral converges for every $s$ (since $\theta(y) - 1 = O(e^{-\pi y})$ as $y \to \infty$) and, being an integral of an entire function of $s$ with locally uniform bounds, is entire; it is symmetric in $s \leftrightarrow 1-s$, and so is $-\frac{1}{s} - \frac{1}{1-s}$. So the right side is the meromorphic continuation of $\xi$, with simple poles at $s = 0$ and $s = 1$ only, and it is invariant under $s \mapsto 1-s$. See [39] and Exercise 19.66. □

Part (v) is the key input of the prime number theorem; we do not prove it; see [39]. It is not used in this book.

**Definition 19.10** (partial zeta functions)**.** For an integer $N \geq 1$, $\zeta^{(N)}(s) = \zeta(s)\prod_{p|N}(1-p^{-s}) = \prod_{p\nmid N}(1-p^{-s})^{-1}$, the zeta function with the Euler factors at the primes dividing $N$ removed. More generally, for an Euler product $L(s) = \prod_p L_p(s)$, $L^{(N)}(s) = \prod_{p\nmid N} L_p(s)$.

**Proposition 19.11** (the Rankin-Selberg residue)**.** *$\zeta^{(N)}(s-1)$ has a simple pole at $s=2$ with residue $\prod_{p|N}(1-p^{-1})$; $\zeta^{(N)}(2) = \frac{\pi^2}{6}\prod_{p|N}(1-p^{-2})$; and*

$$\operatorname{Res}_{s=2}\frac{\zeta^{(N)}(s-1)}{\zeta^{(N)}(2s-2)} = \frac{6}{\pi^2}\prod_{p|N}(1+p^{-1})^{-1} = \frac{6N}{\pi^2\mu(N)}.$$

*Proof.* The finite product $\prod_{p|N}(1-p^{1-s})$ is holomorphic and equals $\prod(1-p^{-1})$ at $s=2$, and $\zeta(s-1)$ has residue 1 at $s=2$ by Theorem 19.9(ii). The value $\zeta^{(N)}(2)$ is Theorem 19.9(iii) times the removed factors; it is nonzero, so the quotient has the residue $\prod(1-p^{-1})\cdot\frac{6}{\pi^2}\prod(1-p^{-2})^{-1} = \frac{6}{\pi^2}\prod(1+p^{-1})^{-1}$, and $\mu(N) = N\prod_{p|N}(1+p^{-1})$ (Chapter 12). □

**Theorem 19.12** (Dirichlet $L$-functions)**.** *Let $\chi$ be a Dirichlet character modulo $m$ (Chapter 2) and $L(s,\chi) = \sum_{n\geq 1}\chi(n)n^{-s}$.*

*(i) For $\operatorname{Re} s > 1$, $L(s,\chi) = \prod_p(1-\chi(p)p^{-s})^{-1} \neq 0$.*

*(ii) If $\chi$ is not principal, $L(s,\chi)$ converges (conditionally) on $\operatorname{Re} s > 0$, locally uniformly, and is holomorphic there.*

*(iii) If $\chi$ is primitive of conductor $m > 1$ and $\chi(-1) = (-1)^a$ with $a \in \{0,1\}$, then $\Lambda(s,\chi) = (m/\pi)^{(s+a)/2}\Gamma(\frac{s+a}{2})L(s,\chi)$ extends to an entire function and $\Lambda(s,\chi) = W(\chi)\Lambda(1-s,\bar\chi)$, with $W(\chi) = \tau(\chi)/(i^a\sqrt{m})$ of absolute value 1, $\tau(\chi)$ the Gauss sum (Chapter 2). For real $\chi$, $W(\chi) = 1$.*

*(iv) (a) $L(1,\chi_{-4}) = 1 - \frac13 + \frac15 - \cdots = \frac{\pi}{4}$ and $L(1,\chi_{-3}) = 1 - \frac12 + \frac14 - \frac15 + \cdots = \frac{\pi}{3\sqrt3}$. (b) $L(1,\chi) \neq 0$ for every non-principal $\chi$.*

*(v) (Dirichlet.) For $\gcd(a,m) = 1$ there are infinitely many primes $p \equiv a \pmod m$.*

*Proof.* (i) $\chi$ is completely multiplicative and bounded; apply Proposition 19.7. (ii) The partial sums $S(x) = \sum_{n\leq x}\chi(n)$ are bounded by $m$, since $\chi$ sums to 0 over a period (orthogonality, Chapter 2). Partial summation gives, for $\operatorname{Re} s > 0$ and $M < M'$,

$$\sum_{M<n\leq M'}\chi(n)n^{-s} = S(M')M'^{-s} - S(M)M^{-s} + s\int_M^{M'} S(x)x^{-s-1}dx,$$

which is at most $2mM^{-\sigma} + |s|\,m\,M^{-\sigma}/\sigma$ in absolute value: each boundary term is at most $mM^{-\sigma}$, and the integral is at most $|s|m\int_M^\infty x^{-\sigma-1}dx = |s|mM^{-\sigma}/\sigma$. On a compact subset of $\operatorname{Re} s > 0$, where $|s| \leq R$ and $\sigma \geq \delta > 0$, this is at most $m(2+R/\delta)M^{-\delta}$, which tends to 0 as $M\to\infty$ independently of $s$; so the series is uniformly Cauchy on compact subsets of $\operatorname{Re} s > 0$, and its sum, a locally uniform limit of entire functions, is holomorphic. (For the principal character the partial sums $S(x)$ grow like $x\varphi(m)/m$, and the argument fails, as it must: $L(s,\chi_0)$

is $\zeta(s)\prod_{p\mid m}(1-p^{-s})$, which has a pole at 1.) (iv)(a) For $0\le x<1$, $\sum_{k<K}(-1)^k x^{2k}=\frac{1-(-x^2)^K}{1+x^2}$; integrating over $[0,1]$,

$$\sum_{k<K}\frac{(-1)^k}{2k+1}=\int_0^1\frac{dx}{1+x^2}\mp\int_0^1\frac{x^{2K}}{1+x^2}dx,$$

where the sign $\mp$ is $-(-1)^K$, and the last integral is at most $\int_0^1 x^{2K}dx=\frac{1}{2K+1}$ since $1+x^2\ge 1$; so $L(1,\chi_{-4})=\int_0^1\frac{dx}{1+x^2}=\arctan 1=\frac{\pi}{4}$ (the series converges by (ii)). Likewise, since $\chi_{-3}(n)$ is $1,-1,0$ for $n\equiv 1,2,0 \pmod 3$, the series for $L(1,\chi_{-3})$ grouped in pairs is $\sum_k(\frac{1}{3k+1}-\frac{1}{3k+2})$, and $\frac{1}{3k+1}-\frac{1}{3k+2}=\int_0^1(x^{3k}-x^{3k+1})dx$. The geometric sum is $\sum_{k<K}(x^{3k}-x^{3k+1})=\frac{(1-x)(1-x^{3K})}{1-x^3}$, and $\frac{1-x}{1-x^3}=\frac{1}{1+x+x^2}$; integrating, and completing the square $1+x+x^2=(x+\frac12)^2+\frac34$ to find the primitive,

$$\sum_{k<K}\Big(\frac{1}{3k+1}-\frac{1}{3k+2}\Big)=\int_0^1\frac{dx}{1+x+x^2}-\int_0^1\frac{x^{3K}}{1+x+x^2}dx$$
$$\longrightarrow\Big[\frac{2}{\sqrt3}\arctan\frac{2x+1}{\sqrt3}\Big]_0^1=\frac{2}{\sqrt3}\Big(\frac{\pi}{3}-\frac{\pi}{6}\Big)=\frac{\pi}{3\sqrt3};$$

the partial sums grouped in pairs converge to $L(1,\chi_{-3})$, because $\chi_{-3}(n)/n\to 0$ and the series converges by (ii). □

*Sketch of proof of (iii).* For $\chi$ primitive put $\theta_\chi(y)=\sum_{n\in\mathbb{Z}}\chi(n)n^a e^{-\pi n^2 y/m}$. Poisson summation twisted by $\chi$, with the Gauss sum expressing $\chi$ through additive characters (primitivity is used here), gives $\theta_\chi(1/y)=W(\chi)\,y^{a+1/2}\,\theta_{\bar\chi}(y)$, and the Mellin transform of $\theta_\chi$ is $2\Lambda(s,\chi)$ up to the normalising power of $m$; there is no constant term, so no pole. For real primitive $\chi$, Gauss's determination of the sign of the quadratic Gauss sum gives $\tau(\chi)=\sqrt m$ if $\chi(-1)=1$ and $i\sqrt m$ if $\chi(-1)=-1$, so $W(\chi)=1$. See [39] and [201, Chapter 4]. □

Parts (iv)(b) and (v) are Dirichlet's theorems; we do not prove them; see [172, Chapter VI] and [39]. For real $\chi$, (iv)(b) follows from Theorem 19.13(i) and the class number formula. Numerically, PARI's `lfun(lfuncreate(-4),1)` and `lfun(lfuncreate(-3),1)` agree with $\pi/4$ and $\pi/(3\sqrt3)$ to 19 digits.

**Theorem 19.13** (Dedekind zeta functions)**.** *For a number field $K$, $\zeta_K(s)=\sum_{\mathfrak a}N\mathfrak a^{-s}=\prod_{\mathfrak P}(1-N\mathfrak P^{-s})^{-1}$ for $\operatorname{Re}s>1$, the sum over the nonzero ideals and the product over the nonzero prime ideals of $\mathcal O_K$ (Chapter 6).*

*(i) If $K$ is quadratic of discriminant $D_K$, then $\zeta_K(s)=\zeta(s)L(s,\chi_{D_K})$; the Euler factor of $\zeta_K$ at $p$ is $(1-p^{-s})^{-1}(1-\chi_{D_K}(p)p^{-s})^{-1}$. For example $\zeta_{\mathbb Q(i)}=\zeta\cdot L(s,\chi_{-4})$ and $\zeta_{\mathbb Q(\sqrt{-3})}=\zeta\cdot L(s,\chi_{-3})$.*

*(ii) If $K/\mathbb Q$ is abelian, $\zeta_K(s)=\prod_\chi L(s,\chi)$, the product over the primitive Dirichlet characters $\chi$ corresponding to the characters of $\operatorname{Gal}(K/\mathbb Q)$.*

*(iii) (Artin formalism.) Artin $L$-functions of representations of $\operatorname{Gal}(\overline{\mathbb Q}/\mathbb Q)$ with finite image are additive in direct sums and invariant under induction; $\zeta_K(s)=L(\operatorname{Ind}_{G_K}^{G_{\mathbb Q}}\mathbf 1,s)$.*

*Proof.* The Euler product of $\zeta_K$ is unique factorisation of ideals (Chapter 6) together with Proposition 19.7. (i) Group the factors by $p$. By the splitting law (Chapter 6), $p\mathcal O_K$ is a product of two primes of norm $p$ if $\chi_{D_K}(p)=1$, one prime of norm $p^2$ if $\chi_{D_K}(p)=-1$, and the square of a

prime of norm $p$ if $\chi_{D_K}(p) = 0$; the local factor is $(1-p^{-s})^{-2}$, $(1-p^{-2s})^{-1} = (1-p^{-s})^{-1}(1+p^{-s})^{-1}$ or $(1-p^{-s})^{-1}$, which is $(1-p^{-s})^{-1}(1-\chi_{D_K}(p)p^{-s})^{-1}$ in each case. Both sides are absolutely convergent Euler products on $\operatorname{Re} s > 1$, so they agree. (ii) and (iii) we do not prove; see [201, Theorem 4.3] for (ii) (theorem number unverified) and [142, Chapter VII, Sections 5 and 10] for (iii) (section numbers unverified). □

*Remark* 19.14 (why $\chi_{-3}$ and $\chi_{-4}$ appear in the table). The factor $(1-\chi_{-3}(q)q^{-s})^{-1}$ of the paper's table at a prime $q \geq 5$ of type II, II$^*$, IV or IV$^*$ is the Euler factor at $q$ of $L(s,\chi_{-3})$, hence of $\zeta_{\mathbb{Q}(\sqrt{-3})}/\zeta$. This is not a coincidence: the inertia-invariant line of $\mathrm{ad}^0 V$ is spanned by $2\sigma+1$, $\sigma$ a generator of $\rho(I_q)$ of order 3, and the arithmetic Frobenius acts on it as on $\sqrt{-3} = 2\zeta_3 + 1$ (Lemma 19.45); so, as a $G_{\mathbb{Q}_q}$-module, the line is the restriction of the global character $\chi_{-3}$. Likewise the line of $\tau$ at a prime of type III or III$^*$ is the restriction of $\chi_{-4}$, the character of $\mathbb{Q}(i) = \mathbb{Q}(\sqrt{-1})$.

## 19.4 $L$-functions of $\ell$-adic representations

This section defines the $L$-function of a Galois representation by its local factors, recalls from Chapter 7 why the factor at $p \neq \ell$ does not depend on $\ell$, and proves the absolute convergence that every later statement about values needs. In the paper it is the definition of $L_p(A_f, s)$ in Section 1.1 ("the true local factor"), and the sentence "PARI's `lfunsympow(E,2)` returns these factors at every prime".

**Definition 19.15** (local factors)**.** Let $p$ be a prime and $M$ a continuous representation of $G_{\mathbb{Q}_p}$ (or of $G_{\mathbb{Q}}$, restricted to a decomposition group at $p$) on a finite-dimensional $\mathbb{Q}_\ell$-vector space, or on a finite-dimensional vector space over a finite extension of $\mathbb{Q}_\ell$.

(a) For $p \neq \ell$, $L_p(M,s) = \det(1 - \Phi_p\, p^{-s} \mid M^{I_p})^{-1}$ (Convention 19.2; Chapter 7).

(b) For $p = \ell$ and $M$ crystalline, $L_\ell(M,s) = \det(1 - \varphi\, \ell^{-s} \mid \mathrm{D}_{\mathrm{cris}}(M))^{-1}$, where $\varphi$ is the crystalline Frobenius, normalised so that $\mathrm{D}_{\mathrm{cris}}(\mathbb{Q}_\ell(1))$ has $\varphi = \ell^{-1}$ (then $L_\ell(M(n),s) = L_\ell(M, s+n)$ as at $p \neq \ell$).

Two features of (a) deserve a word. First, why inertia invariants: a Frobenius element is only defined up to multiplication by $I_p$, so the operator $\Phi_p$ is well defined only on a space on which $I_p$ acts trivially; $M^{I_p}$ is the largest such subspace, and it is stable under $\Phi_p$ because $I_p$ is normal in $G_{\mathbb{Q}_p}$. When $M$ is unramified at $p$ nothing is lost, and the factor is the reciprocal characteristic polynomial of $\Phi_p$ on all of $M$, of degree $\dim M$; ramification lowers the degree. Second, the factor is the reciprocal of a polynomial in $p^{-s}$ with constant term 1, so it expands as a power series in $p^{-s}$ and the product over $p$ is a Dirichlet series. The smallest examples are $M = \mathbb{Q}_\ell$ (trivial action), with $L_p = (1-p^{-s})^{-1}$ and $L(\mathbb{Q}_\ell, s) = \zeta(s)$, and $M = \mathbb{Q}_\ell(\chi)$ for a Dirichlet character $\chi$ of conductor $m$ viewed as a Galois character, $\chi$ primitive. With $\chi(\mathrm{Frob}_p) = \chi(p)$ (Convention 19.1(c)), $\Phi_p$ acts on the line by $\chi(p)^{-1} = \bar{\chi}(p)$ for $p \nmid m$, while for $p \mid m$ inertia acts nontrivially on the line (primitivity) and the invariants are 0; so $L(\mathbb{Q}_\ell(\chi), s) = L(s, \bar{\chi})$. For a quadratic character, the only case this chapter needs, $\bar{\chi} = \chi$ and the choice of Frobenius is invisible; for $\mathbb{Q}_\ell(1)$, which is not of finite order, it is not (Convention 19.2).

**Proposition 19.16** (properties of local factors)**.** *Let $p \neq \ell$.*

*(i) $L_p(M,s)$ is well defined and depends only on the isomorphism class of the Weil-Deligne representation $\mathrm{WD}(M)$; in terms of it, $L_p(M,s) = \det(1 - r(\Phi)p^{-s} \mid (\ker N)^{r(I_p)})^{-1}$.*

*(ii)* $L_p(M \oplus M', s) = L_p(M, s)L_p(M', s)$ *and* $L_p(M(n), s) = L_p(M, s + n)$ *for every* $n \in \mathbb{Z}$.

*(iii) Write* $L_p^{\mathrm{Frob}}(M, s) = \det(1 - \mathrm{Frob}_p\, p^{-s} \mid M^{I_p})^{-1}$. *Then* $L_p^{\mathrm{Frob}}(M, s) = L_p(M, s)$ *for all* $s$ *iff the multiset of eigenvalues of* $\mathrm{Frob}_p$ *on* $M^{I_p}$ *is stable under* $\lambda \mapsto \lambda^{-1}$. *This holds for* $\mathrm{ad}^0 V$ *at a good prime and at a prime of potentially good reduction, and fails for* $\mathrm{ad}^0 V$ *at a prime of potentially multiplicative reduction (multiplicative, or additive with* $v_p(j_E) < 0$*) and for* $V_\ell E$ *at a good prime.*

*(iv) (At* $p = \ell$*.) If* $E$ *has good reduction at* $\ell$*, then* $V_\ell E$ *is crystalline and the crystalline Frobenius on* $\mathrm{D}_{\mathrm{cris}}(V_\ell E^\vee)$ *has characteristic polynomial* $X^2 - a_\ell X + \ell$*; so* $L_\ell(V_\ell E^\vee, s) = L_\ell(E, s)$*, and* $L_\ell(A_f, s)$ *is given by the formula of Proposition 19.26 at* $p = \ell$.

*Proof.* (i) and (ii) are Chapter 7 (the definition of $L$-factors and the proposition after it). (iii) If $\lambda_1, \dots, \lambda_d$ are the eigenvalues of $\mathrm{Frob}_p$ on $M^{I_p}$ (with multiplicity), then $\det(1 - \mathrm{Frob}_p X) = \prod(1 - \lambda_i X)$ and $\det(1 - \Phi_p X) = \prod(1 - \lambda_i^{-1} X)$; two polynomials with constant term 1 are equal iff their multisets of inverse roots are. On $\mathrm{ad}^0 V$ at a good prime the arithmetic eigenvalues are $\alpha/\beta, 1, \beta/\alpha$, a set stable under inversion. At a prime of potentially good reduction $A_f^{I_p}$ is either 0, a line on which Frobenius acts by $\pm 1$, or all of $A_f \simeq \mathrm{ad}^0 V_\ell E'$ for a twist $E'$ of good reduction (Proposition 19.56); each is stable. At a prime of potentially multiplicative reduction, multiplicative or additive, $A_f^{I_p}$ is a line with arithmetic eigenvalue $p \neq p^{-1}$ (Proposition 19.43). On $V_\ell E$ at a good prime the arithmetic eigenvalues are $\alpha, \beta$ with $\alpha\beta = p$; stability would need $\{\alpha, \beta\} = \{\alpha^{-1}, \beta^{-1}\} = \{\beta/p, \alpha/p\}$, impossible since $\alpha \neq \alpha/p$ and $\alpha = \beta/p$, $\beta = \alpha/p$ would give $p^2 = 1$. (iv) We do not prove this; it is the comparison theorem of Fontaine and Messing (crystalline cohomology of the reduction) together with the computation of the Frobenius on the crystalline cohomology of $\tilde{E}$; see [67] and [11] for $\mathrm{D}_{\mathrm{cris}}$, and [194, Section 4] for the definition of local factors at $p = \ell$ (section numbers unverified). □

The paper uses $L_p$ only at $p \neq \ell$, and says so ("At $p = \ell$ the definitions use $D_{\mathrm{cris}}$; we need them only for $p \neq \ell$"); every local factor in the rest of the chapter is at a prime $p \neq \ell$, or is computed with an auxiliary prime $\ell' \neq p$, which is legitimate by the next theorem.

The issue is this. A single $\ell$-adic representation $M_\ell$ gives factors $L_p(M_\ell, s)$ at all $p \neq \ell$ but not, by the recipe of Definition 19.15(a), at $p = \ell$. To get a full Euler product from the representations themselves one uses the whole family $(V_\ell E)_\ell$: the factor at $p$ is taken from any $M_{\ell'}$ with $\ell' \neq p$, and this makes sense only if the answer does not depend on $\ell'$. At a good prime of $E$ this is the statement that $X^2 - a_p X + p$ does not involve $\ell'$; at the bad primes it is a genuine theorem. The next definition and theorem make this precise.

**Definition 19.17** (compatible systems)**.** A *weakly compatible system* of $\ell$-adic representations of $G_{\mathbb{Q}}$ is a family $(M_\ell)_\ell$, $M_\ell$ a continuous representation of $G_{\mathbb{Q}}$ on a finite-dimensional $\mathbb{Q}_\ell$-space (or on a vector space over the completion at a prime above $\ell$ of a fixed number field $K$), unramified outside a finite set $S \cup \{\ell\}$, such that for every $p \notin S \cup \{\ell\}$ the characteristic polynomial of $\Phi_p$ on $M_\ell$ has coefficients in $\mathbb{Q}$ (resp. $K$) independent of $\ell$. Its *$L$-function* is

$$L(M, s) = \prod_p L_p(M_{\ell(p)}, s), \qquad \ell(p) \neq p,$$

which is well defined when, at each $p \in S$, the Frobenius-semisimplified Weil-Deligne representation $\mathrm{WD}(M_\ell|_{G_{\mathbb{Q}_p}})^{\mathrm{F\text{-}ss}}$ is independent of $\ell \neq p$ (as a representation over $\overline{\mathbb{Q}}$ after fixing embeddings).

**Theorem 19.18** (independence of $\ell$)**.** *The following are weakly compatible systems with Weil-Deligne representations independent of $\ell \neq p$ at every prime $p$, so that their L-functions are well defined:*

*(i)* $(V_\ell E)_\ell$, $(V_\ell E^\vee)_\ell$, *and every tensor construction from them (*$\mathrm{Sym}^2 V_\ell E$, $\mathrm{ad}^0 V_\ell E$, *Tate twists);*

*(ii)* $(V_\lambda(f))_\lambda$ *for a newform $f$ of weight 2, and every tensor construction from it.*

*Indication of the references.* We do not prove this. (i) At a good prime it is the independence of the characteristic polynomial $X^2 - a_p X + p$ (Chapter 9); at a potentially multiplicative prime the Weil-Deligne representation is read off the Tate curve and a quadratic character, independently of $\ell$ (Chapter 11); at a potentially good prime the kernel of $\rho_{E,\ell}|_{I_p}$ is independent of $\ell$ and the characteristic polynomials of the elements of the inertia image and of Frobenius have integer coefficients independent of $\ell$ [179, Section 2]; the $\ell$-independence of WD in this case is sketched in Chapter 11. Tensor constructions commute with WD (Chapter 7). (ii) is local-global compatibility for newforms, due to Carayol [21] (Chapter 15). In general (for the cohomology of an arbitrary variety) the independence is a conjecture. □

**Proposition 19.19** (absolute convergence)**.** *Let $(M_\ell)$ be a weakly compatible system of dimension $d$ which is* pure of weight $w$ *in the following sense: for $p \notin S$ the eigenvalues of $\Phi_p$ are algebraic numbers of absolute value $p^{w/2}$ under every complex embedding, and for $p \in S$ the eigenvalues of $\Phi_p$ on $M^{I_p}$ have absolute value at most $p^{w/2}$. Then the Euler product $L(M,s)$ converges absolutely, and is nonzero, on* $\mathrm{Re}\, s > 1 + w/2$. *In particular:*

*(a)* $L(E,s) = L(V_\ell E^\vee, s)$ *($w = 1$) on* $\mathrm{Re}\, s > \frac{3}{2}$;

*(b)* $L(\mathrm{ad}^0 f, s)$ *($w = 0$) on* $\mathrm{Re}\, s > 1$, *hence* $L(\mathrm{Sym}^2 f, s)$ *on* $\mathrm{Re}\, s > 2$, *and likewise the naive versions;*

*(c)* $L(\mathbb{Q}_\ell(1), s) = \zeta(s+1)$ *($w = -2$) on* $\mathrm{Re}\, s > 0$.

*Proof.* Write $L_p(M,s)^{-1} = \prod_{i=1}^{d_p}(1 - \lambda_i p^{-s})$ with $d_p \le d$ and $|\lambda_i| \le p^{w/2}$. Expanding each factor as a geometric series, $\prod_i (1-\lambda_i X)^{-1} = \prod_i \sum_{j\ge 0} \lambda_i^j X^j$, the coefficient of $X^k = p^{-ks}$ in $L_p(M,s)$ is the complete symmetric polynomial $h_k(\lambda_1,\dots,\lambda_{d_p}) = \sum_{j_1+\cdots+j_{d_p}=k} \lambda_1^{j_1}\cdots\lambda_{d_p}^{j_{d_p}}$. It has $\binom{k+d_p-1}{d_p-1} \le \binom{k+d-1}{d-1}$ monomials, each of absolute value at most $p^{kw/2}$, so $|h_k| \le \binom{k+d-1}{d-1} p^{kw/2}$; and $\binom{k+d-1}{d-1}p^{kw/2}$ is exactly the coefficient of $p^{-ks}$ in $(1-p^{w/2-s})^{-d}$, by the same expansion with all $\lambda_i = p^{w/2}$. Multiplying over $p$ (the coefficients of $L(M,s)$ are multiplicative, $c_n = \prod_{p^k \| n} h_k$), the Dirichlet series $L(M,s) = \sum c_n n^{-s}$ has $|c_n| \le r_d(n) n^{w/2}$, where $r_d(n) = \prod_{p^k\|n}\binom{k+d-1}{d-1}$ is the number of ordered factorisations $n = n_1 \cdots n_d$ and $\sum r_d(n) n^{-s} = \zeta(s)^d$; so $\sum |c_n| n^{-\sigma} \le \zeta(\sigma - w/2)^d < \infty$ for $\sigma > 1 + w/2$, and Proposition 19.7 applies, every factor being nonzero there (a factor $1 - \lambda p^{-s}$ vanishes only when $|p^s| = |\lambda| \le p^{w/2}$). (a) At a good $p$ the eigenvalues of $\Phi_p$ on $V_\ell E^\vee$ are $\alpha, \beta$, of absolute value $\sqrt{p}$ by the Hasse bound (Chapter 9); at a bad $p$ the eigenvalue is $a_p = \pm 1$ or there is none (Proposition 19.22). (b) At a good $p$ the eigenvalues on $A_f$ are $\alpha/\beta, 1, \beta/\alpha$, of absolute value 1; at a bad $p$ they are $p^{-1}$, $\pm 1$, or $\alpha'/\beta', 1, \beta'/\alpha'$ for a twist with good reduction (Sections 19.9 and 19.10), all of absolute value at most 1. The naive factors are a subset of the true ones. (c) $\Phi_p$ acts on $\mathbb{Q}_\ell(1)$ by $p^{-1}$. □

So the two points the paper uses, $s=1$ for $L(\mathrm{ad}^0 f,s)$ and $s=2$ for $L(\mathrm{Sym}^2 f,s)$, lie on the boundary of absolute convergence: the values in the paper are values of the analytic continuation (Theorem 19.28), or of the Rankin integral (Theorem 19.36). The local factors themselves, at $s=1$, are finite nonzero numbers; nothing analytic is needed for them.

**Example 19.20** (finite image of inertia)**.** Let $V$ be two-dimensional with $\det V$ unramified at $q$ and $\rho(I_q)$ finite cyclic of order $m\in\{2,3,4,6\}$, $q\nmid m$ (the three Kodaira cases at $q\geq 5$: $\mathrm{I}_0^*$; $\mathrm{II},\mathrm{II}^*,\mathrm{IV},\mathrm{IV}^*$; $\mathrm{III},\mathrm{III}^*$). Then $V^{I_q}=0$, so $L_q(V,s)=1$ in every case, while $L_q(\mathrm{ad}^0 V,s)$ is nontrivial in every case: for $m=2$ the representation is an unramified one twisted by a ramified quadratic character and $L_q(\mathrm{ad}^0 V,s)$ is the cubic factor of the unramified one; for $m=3,6$, $L_q(\mathrm{ad}^0 V,s)=(1-\chi_{-3}(q)q^{-s})^{-1}$; for $m=4$, $L_q(\mathrm{ad}^0 V,s)=(1-\chi_{-4}(q)q^{-s})^{-1}$. This is Chapter 7 (the shapes at a tame prime), and Lemma 19.45 attaches the Kodaira types. The moral for the paper: the $L$-function of $E$ sees nothing at an additive prime, the adjoint $L$-function always sees something, and the naive adjoint $L$-function, which is built from $a_q(E)=0$, cannot.

## 19.5 The $L$-function of an elliptic curve and of its newform

This section defines $L(E,s)$ with its local factors at every prime and states its functional equation. In the paper, $L(E,s)$ appears through $\delta_p$ (the degree of its local factor, Section 1.1) and through the newform $f$; its functional equation is background, used in the book for $L(f,1)$ in Chapter 14 and for root numbers (Chapter 7 points here for epsilon factors).

**Definition 19.21** (the $L$-function of $E$)**.** For every prime $p$ let $a_p=p+1-\#\tilde{E}(\mathbb{F}_p)$, where $\tilde{E}$ is the reduction of the global minimal model, the singular point included when the reduction is bad. Put

$$L_p(E,s)^{-1}=\begin{cases}1-a_p p^{-s}+p^{1-2s} & p\nmid N,\\ 1-a_p p^{-s} & p\parallel N,\\ 1 & p^2\mid N,\end{cases}\qquad L(E,s)=\prod_p L_p(E,s).$$

At a bad prime $\#\tilde{E}(\mathbb{F}_p)=\#\tilde{E}_{\mathrm{ns}}(\mathbb{F}_p)+1$, and $\tilde{E}_{\mathrm{ns}}$ is $\mathbb{G}_m$, its nonsplit form, or $\mathbb{G}_a$ (Chapter 10); so $a_p=1,-1,0$ for split multiplicative, nonsplit multiplicative, additive reduction, and the three lines of the definition are one formula: $L_p(E,s)^{-1}=1-a_p p^{-s}+\mathbf{1}_{p\nmid N}\,p^{1-2s}$.

For example, $E=$ `11a1`: $y^2+y=x^3-x^2-10x-20$. Modulo 2 the equation is $y^2+y=x^3+x^2$; for $x=0$ and $x=1$ the right side is 0, and $y^2+y=0$ has the two solutions $y=0,1$; with the point at infinity $\#\tilde{E}(\mathbb{F}_2)=5$ and $a_2=2+1-5=-2$, so $L_2(E,s)^{-1}=1+2\cdot 2^{-s}+2^{1-2s}$. At $p=11$ the reduction is split multiplicative, $a_{11}=1$ and $L_{11}(E,s)^{-1}=1-11^{-s}$. The first few coefficients $a_2,a_3,a_5,a_7=-2,-1,1,-2$ (PARI's `ellap`) are those of the newform $\eta(z)^2\eta(11z)^2$ of Chapter 13, as Theorem 19.23(i) predicts.

**Proposition 19.22** ($L(E,s)$ and $V_\ell E$)**.** *(i) For every $p\neq\ell$, $L_p(E,s)$ equals $L_p(V_\ell E^\vee,s)$ and $L_p(V_\ell E,s-1)$, and $\delta_p$ is the degree of $L_p(E,s)^{-1}$ as a polynomial in $p^{-s}$.*

*(ii) $L(E,s)$ converges absolutely, and is nonzero, on $\mathrm{Re}\,s>\frac{3}{2}$.*

*(iii) Isogenous curves have the same $L$-function.*

*Proof.* (i) At a good prime this is Chapter 7 (the proposition on local factors, part (v)). At a multiplicative prime, by the Tate shape (Chapter 11) $V^{I_p}$ is the line of $e_1$, on which $\mathrm{Frob}_p$ acts by $\chi_\ell(\mathrm{Frob}_p)\varepsilon(\mathrm{Frob}_p) = pa_p$; since $V^\vee \simeq V(-1)$ (Weil pairing), $(V^\vee)^{I_p} = V^{I_p}(-1)$, on which $\mathrm{Frob}_p$ acts by $a_p$ and $\Phi_p$ by $a_p^{-1} = a_p$; so $L_p(V^\vee, s) = (1 - a_p p^{-s})^{-1}$. At an additive prime $V^{I_p} = 0$ (Chapter 11), so both sides are 1. The second equality is $V^\vee \simeq V(-1)$ and Proposition 19.16(ii). The degree of $L_p(V^\vee, s)^{-1}$ in $p^{-s}$ is $\dim(V^\vee)^{I_p} = \delta_p$. (ii) is Proposition 19.19(a). (iii) An isogeny $E \to E'$ induces a $G_{\mathbb{Q}}$-isomorphism $V_\ell E \simeq V_\ell E'$ (Chapter 9); for each $p$ choose $\ell \neq p$ and apply (i). □

**Theorem 19.23** (functional equation of $L(E,s)$)**.** *Let $E/\mathbb{Q}$ have conductor $N$ and newform $f_E$.*

*(i) (Modularity; Wiles, Taylor-Wiles, Breuil-Conrad-Diamond-Taylor.) $L(E,s) = L(f_E, s)$.*

*(ii) $\Lambda(E,s) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(E,s) = N^{s/2}\int_0^\infty f_E(iy)\, y^{s-1}\, dy$ extends to an entire function, bounded in vertical strips, and $\Lambda(E,s) = w_E\,\Lambda(E, 2-s)$ with $w_E = -\varepsilon_N(f_E) \in \{\pm 1\}$, where $f_E|W_N = \varepsilon_N(f_E) f_E$.*

*(iii) $w_E = -\prod_{p|N} w_p(E)$, where $w_p(E) \in \{\pm 1\}$ is the local root number (Chapter 7, epsilon factors); $w_p(E) = -a_p$ at a multiplicative prime.*

*Proof.* (i) is modularity, stated in Chapter 14 [207, 197, 16]; we do not prove it. Given (i), (ii) is the functional equation of $L(f,s)$ for a newform of weight 2, proved (in sketch) in Chapter 13 from Lemma 19.8(iii) and $f|W_N = \varepsilon_N f$; the sign there is $i^2\varepsilon_N = -\varepsilon_N$. (iii) is the product formula for epsilon factors over all places (the factor at $\infty$ of a weight-2 form is $-1$) together with the computation of the local constants; we do not prove it; see [166] for the local root numbers of elliptic curves, including the additive primes. At a multiplicative prime the local component is an unramified twist $\mathrm{St}\otimes\mu$ of the Steinberg representation, with $w_p = -\mu(\mathrm{Frob}_p) = -a_p$ [166]. □

Examples (PARI's `ellrootno`; Chapter 13): `11a1` has $\varepsilon_{11} = -1$, $w = +1$ and analytic rank 0; `37a1` has $\varepsilon_{37} = +1$, $w = -1$ and analytic rank 1. By (iii), $w_E = -(-a_{11}) = a_{11} = +1$ for `11a1` and $w_E = a_{37} = -1$ for `37a1`, as it must be.

## 19.6 The adjoint and the symmetric square: definitions and the good primes

This section defines $L(\mathrm{ad}^0 f, s)$ and $L(\mathrm{Sym}^2 f, s)$, proves that they differ by a shift of one in $s$, and computes the factor at a good prime. In the paper this is Section 1.1 ("In the normalisation of the symmetric square, $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$"), Definition def:terms ("$P_q(X)$ is the polynomial with $L_q(\mathrm{Sym}^2 f, s) = P_q(q^{-s})^{-1}$") and the paragraph after Theorem thm:dfg14 ("adding a good prime $p$ multiplies the ideal by $(1-\frac{\alpha}{\beta}p^{-1})(1-p^{-1})(1-\frac{\beta}{\alpha}p^{-1})$, Wiles's level-raising factor").

**Definition 19.24** (the adjoint and symmetric square $L$-functions)**.** Let $f$ be a newform of weight 2 with trivial character and rational coefficients, and $V = V_\ell E$ (or $V_\lambda(f)$). Then

$$L(\mathrm{ad}^0 f, s) = L(A_f, s) = \prod_p L_p(A_f, s), \qquad L(\mathrm{Sym}^2 f, s) = \prod_p L_p(\mathrm{Sym}^2 V^\vee, s),$$

each local factor computed with some $\ell \neq p$ (Theorem 19.18). For every prime $p$, $P_p(X)$ is defined by $L_p(\mathrm{Sym}^2 f, s) = P_p(p^{-s})^{-1}$.

**Proposition 19.25** (the shift and the polynomials $P_p$)**.** *(i)* $\mathrm{Sym}^2 V^\vee \simeq A_f(-1)$ *as representations of* $G_{\mathbb{Q}}$*. Hence* $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$ *for every* $p$*, and* $L(\mathrm{Sym}^2 f, s) = L(\mathrm{ad}^0 f, s-1)$*.*

*(ii) For every* $p$*,* $P_p(X) \in \mathbb{Z}[X]$*,* $P_p(0) = 1$*,* $\deg P_p = \dim A_f^{I_p} \in \{0,1,3\}$*, and* $P_p(p^{-2}) = L_p(A_f,1)^{-1}$*, a nonzero element of* $\mathbb{Z}[1/p]$*.*

*(iii) At a good* $p$ *the eigenvalues of* $\Phi_p$ *on* $\mathrm{Sym}^2 V^\vee$ *are* $\alpha^2, \alpha\beta = p, \beta^2$*, and* $P_p(X) = (1-\alpha^2 X)(1-pX)(1-\beta^2 X)$*.*

*Proof.* (i) For a two-dimensional $W$, $\mathrm{Sym}^2 W \simeq \mathrm{ad}^0 W \otimes \det W$ (Chapter 4). Apply it to $W = V^\vee$: $\mathrm{ad}^0 V^\vee \simeq \mathrm{ad}^0 V$ (the map $x \mapsto -x^{\mathsf{T}}$ on trace-zero endomorphisms, or Appendix B, B-C12) and $\det V^\vee = (\det V)^{-1} = \chi_\ell^{-1}$, so $\mathrm{Sym}^2 V^\vee \simeq A_f \otimes \chi_\ell^{-1} = A_f(-1)$. Then $L_p(\mathrm{Sym}^2 V^\vee, s) = L_p(A_f(-1), s) = L_p(A_f, s-1)$ by Proposition 19.16(ii). (ii) By (i), $P_p(X) = \det(1 - \Phi_p X \mid (\mathrm{Sym}^2 V^\vee)^{I_p})$, of degree $\dim(\mathrm{Sym}^2 V^\vee)^{I_p} = \dim A_f^{I_p}$, the twist being unramified. The explicit lists, Proposition 19.26 at good $p$ and Proposition 19.56 at bad $p$, show that $P_p$ is one of $(1-pX)(1-(a^2-2p)X+p^2X^2)$ with $a \in \mathbb{Z}$, $1-X$, $1 \mp pX$, $1$: integral, of degree 3, 1, 1, 0. At $s=2$, $L_p(\mathrm{Sym}^2 f, 2) = L_p(A_f, 1)$, so $P_p(p^{-2}) = L_p(A_f,1)^{-1}$; it lies in $\mathbb{Z}[1/p]$, and it is nonzero by the same list (Proposition 19.26(ii) at good $p$, Proposition 19.56 at bad $p$). (iii) $\Phi_p$ has eigenvalues $\alpha, \beta$ on $V^\vee$ (Convention 19.2), hence $\alpha^2, \alpha\beta, \beta^2$ on $\mathrm{Sym}^2 V^\vee$. □

The isomorphism $\mathrm{Sym}^2 W \simeq \mathrm{ad}^0 W \otimes \det W$ used in (i) is easy to remember on eigenvalues: if $g$ has eigenvalues $a, b$ on $W$, it has $a^2, ab, b^2$ on $\mathrm{Sym}^2 W$, and $\frac{a}{b}, 1, \frac{b}{a}$ on $\mathrm{ad}^0 W$, which multiplied by $\det g = ab$ give $a^2, ab, b^2$ again. This checks the characters only; the isomorphism itself, for every $g$ at once, is Chapter 4. With $W = V^\vee$ and $g = \Phi_p$ at a good prime, $a, b = \alpha, \beta$ and $ab = p$: this is (iii), and it shows the shift by one in the simplest case, since $1 - p \cdot \frac{\alpha}{\beta} X$ is $1 - \alpha^2 X$.

**Proposition 19.26** (the factor at a good prime)**.** *Let* $p \nmid N$*,* $p \neq \ell$*,* $a = a_p(f)$*, and* $\alpha, \beta$ *the roots of* $X^2 - aX + p$*.*

*(i)* $\Phi_p$ *and* $\mathrm{Frob}_p$ *both have the eigenvalues* $\alpha/\beta, 1, \beta/\alpha$ *on* $A_f$*, and*

$$L_p(A_f, s)^{-1} = \Big(1 - \frac{\alpha}{\beta} p^{-s}\Big)(1 - p^{-s})\Big(1 - \frac{\beta}{\alpha} p^{-s}\Big) = (1-p^{-s})(1-(a^2-2p)p^{-1-s} + p^{-2s}),$$

$$P_p(X) = (1-pX)(1-(a^2-2p)X + p^2X^2).$$

*(ii)* $L_p(A_f,1)^{-1} = \dfrac{p-1}{p} \cdot \dfrac{(p+1)^2 - a^2}{p^2}$*, and* $(p+1)^2 - a^2 \geq (p-1)^2 > 0$*.*

*(iii) For an odd prime* $\ell \neq p$*:* $v_\ell(L_p(A_f,1)^{-1}) > 0$ *iff* $p \equiv 1 \pmod{\ell}$ *or* $a \equiv \pm(p+1) \pmod{\ell}$*.*

*Proof.* (i) The arithmetic Frobenius acts on $V$ with eigenvalues $\alpha, \beta$ and eigenvectors $v_1, v_2$ (if $\alpha = \beta$, use a triangular basis; only characteristic polynomials matter). On $\mathrm{End}\, V = V \otimes V^\vee$ it acts by conjugation, and $v_i \otimes v_j^\vee$ has eigenvalue $\lambda_i \lambda_j^{-1}$; the trace-zero part has $\alpha/\beta, 1, \beta/\alpha$. This set is stable under inversion, so $\Phi_p$ has the same eigenvalues. Then $(1 - \frac{\alpha}{\beta}X)(1 - \frac{\beta}{\alpha}X) = 1 - \frac{\alpha^2+\beta^2}{\alpha\beta}X + X^2 = 1 - \frac{a^2-2p}{p}X + X^2$. Replacing $X = p^{-s}$ by $p^{1-s} = pX$ (the shift of Proposition 19.25(i)) gives $P_p$. (ii) At $s=1$: $(1-\frac{1}{p})(1 - \frac{a^2-2p}{p^2} + \frac{1}{p^2}) = \frac{p-1}{p} \cdot \frac{p^2+2p+1-a^2}{p^2}$. By the Hasse bound $|a| \leq 2\sqrt{p}$ (Chapter 9), $(p+1)^2 - a^2 \geq (p+1)^2 - 4p = (p-1)^2$. (iii) $\ell \neq p$, so $v_\ell$ of the denominator is 0, and $\ell \mid (p-1)((p+1)^2 - a^2)$ iff $\ell \mid p-1$ or $\ell \mid (p+1-a)(p+1+a)$. □

The factor in (ii) is the one that Wiles's level-raising and the paper's Theorem thm:dfg14 attach to a good prime added to $\Sigma$ [38], and the condition in (iii) on $a$ is Ribet's level-raising condition $a_p \equiv \pm(p+1) \pmod{\ell}$ (Chapter 15). The same formula with $a = a_q(E')$ is the omitted factor at a prime of type $\mathrm{I}_0^*$ (Lemma 19.45).

A numerical illustration of (ii) and (iii). For `11a1`, with $a_2 = -2$ and $a_3 = -1$,

$$L_2(A_f,1)^{-1} = \frac{1}{2}\cdot\frac{9-4}{4} = \frac{5}{8}, \qquad L_3(A_f,1)^{-1} = \frac{2}{3}\cdot\frac{16-1}{9} = \frac{10}{9}.$$

The factor at 3 has $v_5 = 1$ although $3 \not\equiv 1 \pmod 5$: here $a_3 = -1 \equiv 3+1 \pmod 5$. This is no accident. `11a1` has a rational point of order 5, which injects into $\tilde{E}(\mathbb{F}_p)$ for every good $p \neq 5$ (reduction is injective on torsion of order prime to $p$, Chapter 9); so $5 \mid \#\tilde{E}(\mathbb{F}_p) = p+1-a_p$, that is $a_p \equiv p+1 \pmod 5$, and by (iii) *every* good factor $L_p(A_f,1)^{-1}$ with $p \neq 5$ is a 5-adic non-unit (PARI confirms the congruence for all good $p < 200$). This is the typical behaviour of a reducible residual representation $E[\ell]$, the case that the paper's main theorem excludes.

*In the paper* 19.27. Section 1.1: "In the normalisation of the symmetric square, $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$" is Proposition 19.25(i); "PARI's `lfunsympow(E,2)` returns these factors at every prime" means that `lfunsympow(E,2)` is $L(\mathrm{Sym}^2 f, s)$ in the normalisation of Convention 19.4(b), with the true local factors (Remark 19.60). The paragraph after Theorem thm:dfg14: "adding a good prime $p$ multiplies the ideal by $(1-\frac{\alpha}{\beta}p^{-1})(1-p^{-1})(1-\frac{\beta}{\alpha}p^{-1})$, Wiles's level-raising factor" is $L_p^{\mathrm{nv}}(A_f,1)^{-1} = L_p(A_f,1)^{-1}$ of Proposition 19.26(i) at $s=1$; the theorem itself is Chapter 20's.

## 19.7 Analytic theory, normalisations and critical values

This section states what is known about the analytic continuation and functional equation of $L(\mathrm{Sym}^2 f, s)$, computes the conductor of the adjoint, and compares the normalisations of the sources. The paper uses only the special value $L(\mathrm{Sym}^2 f, 2) = L(\mathrm{ad}^0 f, 1)$ (Section 1 and Remark rem:degree) and PARI's implementation of $L(\mathrm{Sym}^2 f, s)$; nothing in its proofs depends on the functional equation. The book needs the comparison of normalisations to read Shimura, Gelbart-Jacquet, DFG04 and PARI side by side.

**Theorem 19.28** (analytic theory of the symmetric square)**.** *Let $f$ be as in Definition 19.24.*

*(i) (Shimura [182].) The naive symmetric square $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = \zeta^{(N)}(2s-2)\sum_{n\geq 1} a_{n^2}(f) n^{-s}$ (Proposition 19.34) extends to a meromorphic function on $\mathbb{C}$; Shimura proves it by writing $\sum a_{n^2} n^{-s}$ as a Rankin-Selberg integral of $f$ against a theta series of weight $\frac{1}{2}$ and an Eisenstein series of half-integral weight, and proves its holomorphy except at explicitly controlled points. Since $L(\mathrm{Sym}^2 f, s)$ differs from it by finitely many factors $P_q(q^{-s})^{-1}$ (Proposition 19.33), $L(\mathrm{Sym}^2 f, s)$ is meromorphic on $\mathbb{C}$.*

*(ii) (Gelbart and Jacquet [73].) Let $\pi_f$ be the automorphic representation of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$ attached to $f$. There is an automorphic representation $\mathrm{Ad}(\pi_f)$ of $\mathrm{GL}_3(\mathbb{A}_{\mathbb{Q}})$, cuspidal unless $f$ has complex multiplication, whose standard L-function, in the unitary normalisation (centre $\frac{1}{2}$), is $L(s, \mathrm{Ad}(\pi_f)) = L(\mathrm{ad}^0 f, s)$, the local factors at the bad primes included. Consequently $L(\mathrm{ad}^0 f, s)$ extends to an entire function, and with*

$$\Lambda(\mathrm{ad}^0 f, s) = N_{\mathrm{ad}^0}^{s/2}\,\Gamma_{\mathbb{R}}(s+1)\,\Gamma_{\mathbb{C}}(s+1)\,L(\mathrm{ad}^0 f, s), \qquad N_{\mathrm{ad}^0} \text{ the conductor of } A_f,$$

*one has* $\Lambda(\mathrm{ad}^0 f, s) = w(\mathrm{ad}^0 f)\,\Lambda(\mathrm{ad}^0 f, 1-s)$ *with* $w(\mathrm{ad}^0 f) = \pm 1$ *($A_f$ is self-dual). Equivalently, the function* $\Lambda(\mathrm{Sym}^2 f, s) = N_{\mathrm{ad}^0}^{s/2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{C}}(s)L(\mathrm{Sym}^2 f, s)$ *satisfies* $\Lambda(\mathrm{Sym}^2 f, s) = w(\mathrm{ad}^0 f)\Lambda(\mathrm{Sym}^2 f, 3-s)$.

*(iii)* $L(\mathrm{Sym}^2 f, 2) = L(\mathrm{ad}^0 f, 1)$ *is a positive real number.*

*Proof.* (i) and (ii) we do not prove; see [182] and [73]. Two comments on (ii). The agreement of the local factors at the bad primes with the Galois side uses the compatibility of the Gelbart-Jacquet lift with the local Langlands correspondence and the local-global compatibility of Carayol (Chapter 15); we have not checked in which section of [73] the local factors at the ramified places are matched. In the CM case, $\rho_f \simeq \mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}}\psi$ for an imaginary quadratic $K$ and a character $\psi$ of $G_K$, and $\mathrm{ad}^0\rho_f \simeq \chi_K \oplus \mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}}(\psi/\psi^c)$ with $\psi^c$ the conjugate character; so $L(\mathrm{ad}^0 f, s) = L(s, \chi_K)\,L(s, \psi/\psi^c)$, with $\chi_K$ the quadratic character of $K$ and $\psi/\psi^c$ a Hecke character of $K$ of infinite order, and both factors are entire (Hecke). The value $w(\mathrm{ad}^0 f) = +1$ is what PARI's `lfunrootres` returns for every curve of Example 19.62 and for `11a1`; we do not claim it in general. (iii) By Proposition 19.33, $L(\mathrm{ad}^0 f, 1) = L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)\prod_{q^2\mid N} L_q(A_f, 1)$, an identity of meromorphic functions evaluated at a point where every factor is finite. $L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1) = L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2) = 8\pi^3\langle f, f\rangle/N > 0$ by Theorem 19.36, and each $L_q(A_f, 1)^{-1}$ is one of $1-q^{-2}$, $1 \mp q^{-1}$, $\frac{(q-1)((q+1)^2-a'^2)}{q^3}$, $1$ (Proposition 19.56), all positive by the Hasse bound. □

**Proposition 19.29** (the conductor of the adjoint)**.** *Let $q \geq 5$ be a bad prime, $q \neq \ell$. The conductor exponent of $A_f$ at $q$ is* 0 *if the type is* $\mathrm{I}_0^*$ *and* 2 *for each of the types* $\mathrm{I}_n$, $\mathrm{I}_n^*$ *($n \geq 1$),* II, II*, III, III*, IV, IV*. *More generally, at any bad $q \neq \ell$ the exponent is* 0 *when $\rho(I_q) = \{\pm 1\}$ and* 2 *when $A_f$ is tamely ramified with* $\dim A_f^{I_q} = 1$. *Hence if $(N, 6) = 1$,*

$$N_{\mathrm{ad}^0} = \prod_{q\mid N,\ q \text{ not of type } \mathrm{I}_0^*} q^2,$$

*which is not $N^2$: it is the square of the radical of $N$ with the primes of type* $\mathrm{I}_0^*$ *removed.*

*Proof.* For a tame Weil-Deligne representation $(r, N_0)$ the conductor exponent is the dimension minus $\dim(\ker N_0)^{r(I_q)}$ (Chapter 7). At a prime of type $\mathrm{I}_n$ or $\mathrm{I}_n^*$, $\mathrm{ad}^0 V$ is the adjoint of an unramified twist of the Steinberg representation (Proposition 19.43), $r$ is unramified on it and $\ker N_0$ is the line of $e$: exponent $3-1=2$. At the potentially good types with $\rho(I_q)$ cyclic of order 3, 4 or 6, $N_0 = 0$, the action is tame and the invariants are a line (Lemma 19.45): exponent $3-1=2$. If $\rho(I_q) = \{\pm 1\}$, $\mathrm{ad}^0 V$ is unramified: exponent 0. For $q \geq 5$ every bad type is one of these (Chapter 11), and $q \geq 5$ is tame. □

PARI agrees. `lfunparams(lfunsympow(E,2))` is `[N_sym, 3, [0, 0, 1]]`: motivic weight 3 (so the functional equation is $s \leftrightarrow 3-s$) and gamma factor $\Gamma_{\mathbb{R}}(s)^2\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{C}}(s)$ (Lemma 19.8(ii)), with $N_{\mathrm{sym}} = 121$ for `11a1`, `121a1`, `121d1`, `539d1` and `275b1` (the $\mathrm{I}_0^*$ primes 7 and 5 contribute nothing), $441 = 21^2$ for `147b1`, $5929 = 77^2$ for `539c1`. At $q = 2, 3$ the proposition still covers some cases: $100 = 2^2\cdot 5^2$ for `20a1` (type IV* at 2, inertia cyclic of order 3), $1764 = 42^2$ for `336f1`, and $5929 = 77^2$ for `1232l1` (inertia $\{\pm 1\}$ at 2, no contribution); wild cases give larger exponents, $81 = 3^4$ for `27a1` and $64 = 2^6$ for `64a1`. The sign returned by `lfunrootres` is $+1$ in all these cases.

*Remark* 19.30 (critical values and Deligne's conjecture). For a motive $M$ over $\mathbb{Q}$ with $L$-function $L(M,s)$ and gamma factor $L_\infty(M,s)$, Deligne calls an integer $n$ *critical* if neither $L_\infty(M,s)$ nor $L_\infty(M^\vee,1-s)$ has a pole at $s=n$, and conjectures that then $L(M,n)/c^{\pm}(M(n)) \in \mathbb{Q}^\times$ for an explicit period $c^{\pm}$ [42] (when $L(M,n)\neq 0$). For $A_f$, which is self-dual with $L_\infty = \Gamma_{\mathbb{R}}(s+1)\Gamma_{\mathbb{C}}(s+1)$, the poles of $L_\infty(s)$ are at $s=-1,-2,\dots$ and those of $L_\infty(1-s)$ at $s=2,3,\dots$; so the critical integers are $s=0$ and $s=1$, and for $L(\mathrm{Sym}^2 f,s)$ they are $s=1$ and $s=2$. At $s=1$, Theorem 19.36 gives

$$\frac{L(\mathrm{ad}^0 f,1)}{\pi^3\langle f,f\rangle} = \frac{8}{N\prod_{q^2\mid N} L_q(A_f,1)^{-1}} \in \mathbb{Q}^\times,$$

which is Deligne's conjecture at $s=1$ once his period is identified with $\pi^3\langle f,f\rangle$ up to $\mathbb{Q}^\times$; we have not checked that identification. Sturm proved the analogous rationality for the critical values of the symmetric square of forms of every weight [189], and Zagier computed them in level one [209] (the precise statements in these papers are not reproduced here). The paper uses only the case $s=1$, in the form of Corollary 19.37.

*Remark* 19.31 (normalisations in the sources). The objects that occur, with their relation to $L(A_f,s) = L(\mathrm{ad}^0 f,s)$ (every identity in the third column is proved in this chapter; the attributions in the second column are to the sources as cited by the paper and by Chapters 13 and 20):

| object and source | relation to $L(A_f,s)$ | functional equation | value used | factor at $p^2 \mid N$ |
|---|---|---|---|---|
| $L(\mathrm{ad}^0 f,s)$; paper, DFG04 ($L(\mathrm{ad}^0 M_f,s)$) | itself | $s\leftrightarrow 1-s$ | $s=1$ | true |
| $L(s,\mathrm{Ad}(\pi_f))$; Gelbart-Jacquet, unitary | $L(A_f,s)$ | $s\leftrightarrow 1-s$ | $s=1$ | true |
| $L(\mathrm{Sym}^2 f,s)$; paper, PARI's `lfunsympow(E,2)` | $L(A_f,s-1)$ | $s\leftrightarrow 3-s$ | $s=2$ | true |
| $\zeta^{(N)}(2s-2)\sum a_{n^2}n^{-s}$; Shimura [182] | $L^{\mathrm{nv}}(A_f,s-1)$ | via $L(\mathrm{Sym}^2 f,s)$ | $s=2$ | 1 |
| $L^{\mathrm{nv}}(\mathrm{ad}^0 f,s)$; paper, DFG04 (naive factors) | $L(A_f,s)\prod_{p^2\mid N} L_p(A_f,s)^{-1}$ | none | $s=1$ | 1 |
| $D(f,s)=\sum a_n^2 n^{-s}$; Rankin [155], Selberg [170] | $\zeta^{(N)}(s-1)L^{\mathrm{nv}}(A_f,s-1)/\zeta^{(N)}(2s-2)$ | via $E(z,s)$ | residue at 2 | 1 |
| $L(f\times f,s)=\zeta^{(N)}(2s-2)D(f,s)$; Rankin-Selberg convolution | $\zeta^{(N)}(s-1)L^{\mathrm{nv}}(A_f,s-1)$ | | pole at 2 | 1 |

For a newform of weight $k$ the symmetric square of the classical normalisation and the adjoint differ by $k-1$: $L(\mathrm{ad}^0 f,s) = L(\mathrm{Sym}^2 f, s+k-1)$, so "$L(\mathrm{Sym}^2 f,k)$" in papers on weight $k$ (Hida, Sturm, Zagier) is $L(\mathrm{ad}^0 f,1)$. Watkins [204] and Coates and Schmidt [29] work with the symmetric square of an elliptic curve; we have not checked their normalisation of $s$ against the table.

## 19.8 Naive versus true; Rankin-Selberg; the Petersson norm and the modular degree

This section is the heart of the chapter's contribution to the paper's introduction. It defines the naive adjoint $L$-function, shows that it is the $L$-function that the $q$-expansion of $f$ sees, proves its

exact relations with the series of Rankin and of Shimura, and turns the Rankin-Selberg residue into an exact formula for $\langle f, f\rangle$ and for $\deg \varphi_E$. In the paper this is Section 1 ("The $L$-function that appears there is the *naive* one"; "The classical route to the degree is the Petersson norm"), Remark rem:degree, and the role of the omitted factors in Theorem thm:main.

**Definition 19.32** (the naive adjoint $L$-function)**.** For $p \nmid N$ put $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$. For $p \mid N$ put $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p > 0$ and $L_p^{\mathrm{nv}}(A_f, s) = 1$ if $\delta_p = 0$ (the paper, Section 1.1, after [49, §1.7.2]). Then $L^{\mathrm{nv}}(\mathrm{ad}^0 f, s) = \prod_p L_p^{\mathrm{nv}}(A_f, s)$ and $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = L^{\mathrm{nv}}(\mathrm{ad}^0 f, s-1)$.

**Proposition 19.33** (what "naive" means)**.** *(i) For $p \mid N$, $L_p^{\mathrm{nv}}(A_f, s)^{-1} = 1 - a_p(f)^2 p^{-1-s}$. So $L^{\mathrm{nv}}(\mathrm{ad}^0 f, s)$ is determined by the Fourier coefficients $a_p(f)$ alone, through Proposition 19.26(i) at $p \nmid N$ and this formula at $p \mid N$.*

*(ii) $L^{\mathrm{nv}}(\mathrm{ad}^0 f, s) = L(\mathrm{ad}^0 f, s) \prod_{p^2 \mid N} L_p(A_f, s)^{-1}$; the two functions differ exactly at the additive primes, by the* omitted factors $L_p(A_f, s)^{-1}$.

*Proof.* (i) For $p \mid N$ either $p \,\|\, N$, $\delta_p = 1$ and $a_p = \pm 1$, or $p^2 \mid N$, $\delta_p = 0$ and $a_p = 0$ (Chapters 11 and 13). In the first case $L_p(A_f, s)^{-1} = 1 - p^{-1-s}$ (Proposition 19.43), which is $1 - a_p^2 p^{-1-s}$; in the second the naive factor is $1 = 1 - 0$. (ii) The factors agree at $p \nmid N$ and at $p \,\|\, N$. □

The omitted factors are invisible in the $q$-expansion: at an additive prime $a_p(f) = 0$ whatever the local representation is, and the local representation is not determined by $a_p$ (Chapter 15; Lemma 19.45 lists five different factors with $a_p = 0$). This is why the constructions of Diamond, Flach and Guo, which start from $q$-expansions and depleted forms, see $L^{\mathrm{nv}}$, while the Bloch-Kato conjecture sees $L$.

**Proposition 19.34** (the series of Rankin and of Shimura)**.** *Let $f$ be as in Definition 19.24. For $\mathrm{Re}\, s > 2$, both series below converge absolutely, and*

$$(A) \qquad \sum_{n \geq 1} a_n(f)^2 n^{-s} = \frac{\zeta^{(N)}(s-1)}{\zeta^{(N)}(2s-2)}\, L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s),$$

$$(B) \qquad \sum_{n \geq 1} a_{n^2}(f)\, n^{-s} = \frac{L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)}{\zeta^{(N)}(2s-2)},$$

*exactly, with no correction at the bad primes. In terms of local factors, with $X = p^{-s}$: at $p \nmid N$,*

$$\sum_{k \geq 0} a_{p^k}^2 X^k = \frac{1 + pX}{P_p(X)}, \qquad \sum_{k \geq 0} a_{p^{2k}} X^k = \frac{1 - p^2 X^2}{P_p(X)},$$

*and at $p \mid N$ both local series equal $(1 - a_p^2 X)^{-1} = L_p^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)$.*

*Proof.* Absolute convergence: $|a_n| \leq d(n)\sqrt{n}$ (Hasse at primes, multiplicativity and the recursion, Chapter 13) and $d(n) = O(n^r)$ for every $r > 0$, so $a_n^2$ and $a_{n^2}$ are $O(n^{1+r})$ (Proposition 19.6(ii)). Both coefficient sequences are multiplicative, so both sides are Euler products (Proposition 19.7) and it suffices to compare local factors. Write $\alpha, \beta$ for the roots of $X^2 - a_p X + p$, so that $a_{p^k} = (\alpha^{k+1} - \beta^{k+1})/(\alpha - \beta)$ (Chapter 13; when $\alpha = \beta$ read the identities below by continuity). The first local identity is Chapter 13 (the Rankin-Selberg theorem, part (ii)); for completeness, squaring gives $a_{p^k}^2 = (\alpha^{2k+2} - 2p^{k+1} + \beta^{2k+2})/(\alpha - \beta)^2$, and summing three geometric series,

$$\sum_k a_{p^k}^2 X^k = \frac{1}{(\alpha - \beta)^2}\Big(\frac{\alpha^2}{1 - \alpha^2 X} - \frac{2p}{1 - pX} + \frac{\beta^2}{1 - \beta^2 X}\Big) = \frac{1 + pX}{(1 - \alpha^2 X)(1 - pX)(1 - \beta^2 X)},$$

the last step by bringing to a common denominator (the numerator is $(\alpha-\beta)^2(1+pX)$; this identity of rational functions was also checked symbolically with PARI). For the second,

$$\sum_k a_{p^{2k}}X^k = \frac{1}{\alpha-\beta}\Big(\frac{\alpha}{1-\alpha^2X}-\frac{\beta}{1-\beta^2X}\Big) = \frac{(\alpha-\beta)+\alpha\beta(\alpha-\beta)X}{(\alpha-\beta)(1-\alpha^2X)(1-\beta^2X)} = \frac{1+pX}{(1-\alpha^2X)(1-\beta^2X)},$$

which is $(1-p^2X^2)/P_p(X)$ by Proposition 19.25(iii). The Euler factor of $\zeta(s-1)/\zeta(2s-2)$ at $p$ is $(1-p^{2-2s})/(1-p^{1-s}) = (1-p^2X^2)/(1-pX) = 1+pX$, and that of $1/\zeta(2s-2)$ is $1-p^2X^2$; this gives (A) and (B) at $p\nmid N$. At $p\mid N$, $a_{p^k}=a_p^k$ with $a_p\in\{0,\pm1\}$ (Chapter 13), so both local series are $\sum_k(a_pX)^k=(1-a_pX)^{-1}$; and $L_p^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)^{-1}=L_p^{\mathrm{nv}}(A_f,s-1)^{-1}=1-a_p^2p^{-s}$ by Proposition 19.33(i); the factors of $\zeta^{(N)}$ at $p$ are 1. □

Two consequences. (a) The Rankin series is *not* the naive symmetric square: at every good $p$ it carries the extra factor $1+p^{1-s}$. For `11a1`, with $a_2=-2$, the coefficient of $2^{-s}$ is $a_2^2=4$ in the Rankin series and $\alpha^2+2+\beta^2=a_2^2-2=2$ in $L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)$. (b) Shimura's series, completed by $\zeta^{(N)}(2s-2)$, *is* the naive symmetric square, on the nose. Numerically, at $s=6$ with 20,000 terms of (A) and 1,000 terms of (B), both identities hold for `11a1`, `20a1` and `121a1` to within $4\cdot10^{-15}$ (the truncation error), with $L^{\mathrm{nv}}$ computed from `lfunsympow` and `lfuneuler`; the two local identities were also checked as power series to order 20 for $(p,a_p)=(5,2),(7,-3),(11,0),(13,5)$.

*Warning* 19.35 (the Rankin series is not the naive symmetric square). By Proposition 19.34(A) the Euler product of $\sum a_n(f)^2n^{-s}$ is $\zeta^{(N)}(s-1)L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)/\zeta^{(N)}(2s-2)$, not $L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)$. The paper's introduction accordingly writes "The $L$-function that appears there is the *naive* one, $L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)=\zeta^{(N)}(2s-2)\sum_{n\ge1}a_{n^2}(f)n^{-s}$ (Shimura [182]), which has the true local factor at the primes of good or multiplicative reduction and the trivial factor at the additive primes", which is Proposition 19.34(B). Equivalently, with $L(f\times f,s)=\zeta^{(N)}(2s-2)\sum a_n^2n^{-s}$, one has $L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)=L(f\times f,s)/\zeta^{(N)}(s-1)$; dividing by $\zeta^{(N)}(2s-2)$ instead would be wrong.

**Theorem 19.36** (the Petersson norm as a special value)**.** *Let $f$ be as in Definition 19.24, of level $N$, and $\langle\ ,\ \rangle$ the unnormalised Petersson product. Then $L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)$ and $L(\mathrm{Sym}^2 f,s)$ are holomorphic at $s=2$ and*

$$\langle f,f\rangle = \frac{N}{8\pi^3}L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2) = \frac{N}{8\pi^3}L(\mathrm{Sym}^2 f,2)\prod_{p^2\mid N}P_p(p^{-2}) = \frac{N}{8\pi^3}L(\mathrm{ad}^0 f,1)\prod_{p^2\mid N}\frac{1}{L_p(A_f,1)}.$$

*Proof.* The first equality, at every level $N$, is the Rankin-Selberg theorem of Chapter 13, stated and proved there from the residue of $D(f,s)=\sum a_n^2n^{-s}$ at $s=2$ (the unfolding of $|f|^2y^2E(z,s)$, due to Rankin [155] and Selberg [170], is sketched there); we do not prove it again. In the notation of this chapter: $D(f,s)$ has a simple pole at $s=2$ with residue $\frac{48\pi}{\mu(N)}\langle f,f\rangle$ and is holomorphic elsewhere on $\mathrm{Re}\,s\ge2$, and $L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)=D(f,s)\zeta^{(N)}(2s-2)/\zeta^{(N)}(s-1)$ (Proposition 19.34(A)), so $L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)$ is holomorphic at $s=2$ and the residue of $\zeta^{(N)}(s-1)/\zeta^{(N)}(2s-2)$ there, $\frac{6N}{\pi^2\mu(N)}$ (Proposition 19.11), converts one residue into the other: near $s=2$, $D(f,s)=\frac{\zeta^{(N)}(s-1)}{\zeta^{(N)}(2s-2)}L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)$ with $L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)$ holomorphic, so comparing residues,

$$L^{\mathrm{nv}}(\mathrm{Sym}^2 f,2)=\frac{\mathrm{Res}_{s=2}D(f,s)}{\mathrm{Res}_{s=2}\zeta^{(N)}(s-1)/\zeta^{(N)}(2s-2)}=\frac{48\pi\langle f,f\rangle}{\mu(N)}\cdot\frac{\pi^2\mu(N)}{6N}=\frac{8\pi^3}{N}\langle f,f\rangle,$$

the factor $\mu(N)$ cancelling. What this chapter adds is the true value. By Proposition 19.33(ii) and Proposition 19.25, $L(\mathrm{Sym}^2 f, s)$ is $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)$ times $\prod_{p^2\mid N} P_p(p^{-s})^{-1}$, a finite product of rational functions of $p^{-s}$ which are finite and nonzero at $s = 2$ (Proposition 19.52); this gives holomorphy of $L(\mathrm{Sym}^2 f, s)$ at 2 and the second and third equalities, with $P_p(p^{-2}) = L_p(A_f, 1)^{-1}$. □

So the true value $L(\mathrm{Sym}^2 f, 2)$ gives $\langle f, f\rangle$ up to $N/(8\pi^3)$ and the omitted factors at the *additive* primes only; the multiplicative primes contribute nothing beyond the factor $N$. For square-free $N$ there is no correction. Chapter 13 states and proves the first (naive) equality at every level, and the true value for square-free $N$; the second and third equalities, with the correction $\prod_{p^2\mid N} P_p(p^{-2})$ at the additive primes, are what this chapter adds. The values for fifteen curves are in Example 19.63. Diamond, Flach and Guo [49, Lemma 2.12] express the Petersson norm through the naive $L$-value, as the paper says in Remark rem:degree ("$\langle f, f\rangle = N\,L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2)/(8\pi^3)$ at every level"). Their constant agrees with ours: their lemma is $(f, f)_{\Gamma_1(N)} = (k-1)!\,\delta(N)N\varphi(N)L^{\mathrm{nv}}(A_f, 1)/(4^k\pi^{k+1})$ for the unnormalised product, with $\delta(N) = 1$ for $N > 2$ and 2 for $N \le 2$, and $[\bar\Gamma_0(N) : \bar\Gamma_1(N)] = \delta(N)\varphi(N)/2$ turns it, for $k = 2$, into $N\,L^{\mathrm{nv}}(A_f, 1)/(8\pi^3)$ on $\Gamma_0(N)$ (Chapter 20). The proof above does not use their lemma.

**Corollary 19.37** (the degree formula)**.** *(i) For every modular parametrisation $\varphi\colon X_0(N) \to E$ with Manin constant $c_\varphi$,*

$$\deg\varphi \cdot \mathrm{vol}(\Lambda_E) = \frac{c_\varphi^2\, N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)}{2\pi}.$$

*(ii) For the optimal curve,* $\deg\varphi_E = \dfrac{c_E^2\, N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)}{2\pi\,\mathrm{vol}(\Lambda_E)}$.

*(iii) Put* $B_E = \dfrac{c_E^2\, N\, L(\mathrm{ad}^0 f, 1)}{2\pi\,\mathrm{vol}(\Lambda_E)}$. *Then* $\deg\varphi_E = B_E \prod_{q^2\mid N} L_q(A_f, 1)^{-1}$, $B_E \in \mathbb{Q}^\times$, *and for every prime $\ell \nmid N$*

$$v_\ell(\deg\varphi_E) = v_\ell(B_E) + \sum_{q\mid N} e_\ell(E, q).$$

*In particular the paper's inequality (eq:thm) is equivalent to $v_\ell(B_E) \ge \sum_{q\mid N} t_\ell(E, q)$.*

*Proof.* (i) is the formula of Zagier and Watkins, $\deg\varphi \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_\varphi^2\langle f, f\rangle$ (Chapter 14), combined with Theorem 19.36: $4\pi^2 \cdot \frac{N}{8\pi^3} = \frac{N}{2\pi}$, and $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2) = L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)$. (ii) is (i) for $\varphi_E$. (iii) By Proposition 19.33(ii) at $s = 1$, $L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1) = L(\mathrm{ad}^0 f, 1)\prod_{q^2\mid N} L_q(A_f, 1)^{-1}$; substitute in (ii). So $B_E = \deg\varphi_E \prod_{q^2\mid N} L_q(A_f, 1)$, a product of a positive integer and nonzero rational numbers (Proposition 19.52). For $\ell \nmid N$ every bad $q$ differs from $\ell$; at an additive $q$ the naive factor is 1, so $v_\ell(L_q(A_f, 1)^{-1}) = e_\ell(E, q)$, and at a multiplicative $q$, $e_\ell(E, q) = 0$ (Proposition 19.42). The sum over $q^2 \mid N$ is therefore the sum over all $q \mid N$. The last sentence: (eq:thm) reads $v_\ell(\deg\varphi_E) \ge \sum e_\ell + \sum t_\ell$. □

With $c_E = 1$, (ii) reproduces $\deg\varphi_E$ to at least 15 digits on the curves of Examples 19.62 and 19.63; the paper states the same formula in Remark rem:degree. The corollary is unconditional; it is the conceptual reason why the omitted factors appear in Theorem thm:main: the Petersson norm, hence $\deg\varphi_E$, sees $L^{\mathrm{nv}}$, and the ratio $\deg\varphi_E/B_E$ is exactly the product of the omitted factors. $v_\ell(B_E)$ can be negative when the hypotheses of Theorem thm:main fail: $B_E = 343/5$ for

`539d1` and 121/5 for `121d1` ($E[5]$ reducible), 49/3 for `98a1` and 2/3 for `20a1` ($E[3]$ reducible), 11/3 for `121b1` (complex multiplication by $\mathbb{Q}(\sqrt{-11})$, mod 3 image $N_s(\mathbb{F}_3)$) (Example 19.62).

*In the paper* 19.38. Remark rem:degree displays $\deg\varphi_E = \frac{c_E^2\,N\,L^{\mathrm{nv}}(\mathrm{ad}^0 f,1)}{2\pi\,\mathrm{vol}(\Lambda_E)} = B_E\prod_{q^2\mid N} L_q(A_f,1)^{-1}$ and states "$v_\ell(\deg\varphi_E) = v_\ell(B_E) + \sum_{q\mid N} e_\ell(E,q)$ for every $\ell \nmid N$, with no hypothesis" are Corollary 19.37(ii), (iii). The introduction says that "Up to explicit factors $\deg\varphi_E$ is $\langle f,f\rangle$ divided by the area of the period lattice of $E$"; the explicit factor is $4\pi^2c_E^2$ (Chapter 14).

*Remark* 19.39 (a Bloch-Kato reading). This is a reading, not a proof. Let $\ell \nmid 2N$ with $\ell \notin S_f$. The paper's identity (eq:identity) (an equality, Chapters 14 and 20) reads

$$v_\ell(\deg\varphi_E) = \mathrm{length}\,H^1_f(\mathbb{Q},W) + \sum_{q\mid N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) + \sum_{q\mid N} e_\ell(E,q), \qquad W = \mathrm{ad}^0 T\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell.$$

Subtracting Corollary 19.37(iii) gives $v_\ell(B_E) = \mathrm{length}\,H^1_f(\mathbb{Q},W) + \sum_q v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T))$: the $\ell$-part of a Bloch-Kato formula for the true value $L(\mathrm{ad}^0 f,1)$, with the period $2\pi\,\mathrm{vol}(\Lambda_E)/(c_E^2N)$. This is the theorem of Diamond, Flach and Guo [49, Theorem 0.1] read in the book's normalisation: their period of $\mathrm{ad}^0 M_f(1)$ (Deligne's period, computed in their Section 2.3 from their Lemmas 2.9, 2.11 and 2.12, and measured against their integral structure) is $w_EN\cdot 2\pi\,\mathrm{vol}(\Lambda_E)/(c_E^2N)$ up to an $\ell$-adic unit for $\ell\notin S_f$, and up to sign with the integral normalisation by $\varphi_E^*H^1(E,\mathbb{Z})$. In the derivation (Chapter 18) the $\frac12$ of their pairing cancels the $\frac12$ of the projection to the $F_\infty$-eigenspaces, $\deg\varphi_E$ cancels between the Petersson norm and the congruence ideal $\eta^{\emptyset}_{f,\lambda}$ (Chapter 20), and the only discrepancy left is the factor $w_EN = \varepsilon(M_f)$, a unit at every $\ell\nmid N$; numerically the ratio is $w_EN$ to 20 digits for `11a1`, `37a1`, `121a1` and `121d1`. The reading is not an independent proof, since their proof of Theorem 0.1 goes through their Theorem 3.7, as Theorem thm:main does. It explains the paper's sentence "the primes at which the naive and the true adjoint $L$-functions differ are exactly the primes at which the modular degree acquires extra local divisibility, that the amount is the valuation of the omitted factor": the Petersson norm sees $L^{\mathrm{nv}}$, Bloch-Kato sees $L$, and the ratio is the product of the omitted factors.

**Theorem 19.40** (Flach)**.** *Let $E/\mathbb{Q}$ be an elliptic curve (modular, as every $E/\mathbb{Q}$ is) of conductor $N$, minimal among its quadratic twists, and let $\varphi\colon X_0(N)\to E$ be a modular parametrisation. Let $p\ge 5$ be a prime at which $E$ has good reduction and such that $\bar\rho_{E,p}\colon G_{\mathbb{Q}}\to \mathrm{GL}_2(\mathbb{F}_p)$ is surjective, and put $\mathcal{W} = \mathrm{Sym}^2 H^1(E_{\overline{\mathbb{Q}}},\mathbb{Z}_p)\otimes\mathbb{Q}_p/\mathbb{Z}_p$. Then $\deg\varphi$ annihilates the Bloch-Kato Selmer groups $H^1_f(\mathbb{Q},\mathcal{W}(1))$ and $H^1_f(\mathbb{Q},\mathcal{W}(2))$; in particular both are finite.*

We do not prove this; see [66, Theorem 1 and the Remarks after it, p. 308; Propositions 1.1 and 2.2], checked against the published text. Flach's Theorem 1 has exactly the two hypotheses above (he writes $A$ for our $\mathcal{W}$) and states that $\deg\varphi$ annihilates $Ш(\mathbb{Q},\mathcal{W}(2))$, the cotorsion quotient of his Selmer group $S(\mathbb{Q},\mathcal{W}(2))$; the first Remark after it and p. 326 give the same for $S(\mathbb{Q},\mathcal{W}(2))$ itself, and his Proposition 1.1 (p. 310), whose hypothesis his Proposition 2.2 (p. 318) supplies, gives $\deg\varphi\cdot S(\mathbb{Q},\mathcal{W}(1)) = 0$. Minimality among quadratic twists is his standing assumption, stated just before Theorem 1 (the Selmer groups are unchanged by a quadratic twist, $\deg\varphi$ is not), and he remarks that under the surjectivity hypothesis the statement is unchanged by a $\mathbb{Q}$-isogeny. For the optimal curve one may take $\deg\varphi = \deg\varphi_E$. Langer [110, Theorem B] extends the finiteness, without an annihilator, to the primes of multiplicative reduction: if $E$ is semistable and $p\ge 3$ divides $N$, the Selmer group $S(\mathbb{Q},A)$ of $A = H^2(\overline{X},\mathbb{Q}_p/\mathbb{Z}_p(2))$, $X = E\times E$, is finite, with no hypothesis on $\bar\rho_{E,p}$; by his first Remarks it coincides with Flach's $S(\mathbb{Q},\mathcal{W}(2))$ (checked against the published text; his hypothesis H 1, the Gersten conjecture for

$\mathcal{K}_2$ on a regular model, is needed only for his Theorem A on zero-cycles). Since $\operatorname{Sym}^2 V^\vee(1) \simeq A_f$ (Proposition 19.25(i), with $V = V_pE$), and the isomorphism $\operatorname{Sym}^2 W' \simeq \operatorname{ad}^0 W' \otimes \det W'$ holds for lattices when 2 is invertible (Chapter 4), the group $H^1_f(\mathbb{Q}, \mathcal{W}(1))$ is the $H^1_f(\mathbb{Q}, W)$ of Remark 19.39 with $\ell = p$, and $H^1_f(\mathbb{Q}, \mathcal{W}(2))$ is the Selmer group of $\operatorname{Sym}^2 T_pE \otimes \mathbb{Q}_p/\mathbb{Z}_p$. The proof constructs classes in $H^1(X_0(N) \times X_0(N), \mathcal{K}_2)$ (Zariski cohomology of the sheaf $\mathcal{K}_2$) from functions on the curves $X_0(Nl)$, $l \nmid N$ prime, mapped to $X_0(N) \times X_0(N)$ (Flach's Proposition 2.1), and runs Kolyvagin's argument ("there will be no Euler system in our proof", p. 308); these classes are the ancestors of the Beilinson-Flach elements. The paper cites it as "the first place where the modular degree meets the Selmer group of $\operatorname{Sym}^2 f$", and does not use it. Its precise statement in Selmer language belongs to Chapter 18.

## 19.9 The local terms and Lemma lem:euler

This section computes the true and the naive local factors of the adjoint at every bad prime $q \geq 5$, defines the paper's local terms $e_\ell(E,q)$ and $t_\ell(E,q)$, and proves the paper's Lemma lem:euler in full. The inertia images by Kodaira type are Chapter 11's (through Kraus's theorem, stated in Chapter 10), and the Weil-Deligne shapes with their adjoints are Chapter 7's; this section assembles them, adds the column $P_q(X)$, the naive factor and the valuations. It also treats the behaviour under quadratic twist, the $L$-part of Lemma lem:fpr and the naive factors of Theorem thm:dfg14. Throughout, $q$ is a bad prime and $q \neq \ell$ unless stated otherwise.

**Definition 19.41** (the local terms; the paper's Definition def:terms)**.** For a bad prime $q$ put

$$t_\ell(E,q) = \begin{cases} v_\ell(-v_q(j_E)) & \text{if } v_q(j_E) < 0, \\ 0 & \text{otherwise,} \end{cases}$$

and, if $q \neq \ell$,

$$e_\ell(E,q) = v_\ell\Big(\frac{L_q(A_f,1)^{-1}}{L_q^{\mathrm{nv}}(A_f,1)^{-1}}\Big) = v_\ell\Big(\frac{P_q(q^{-2})}{1 - a_q(E)^2 q^{-2}}\Big).$$

The two expressions for $e_\ell$ agree by Proposition 19.25(ii) and Proposition 19.33(i) at $s = 1$. Note that $1 - a_q^2 q^{-2}$ is the naive factor at $s = 1$ only because $q$ is bad; at a good prime the naive factor is the cubic of Proposition 19.26. In words, $e_\ell(E,q)$ counts how many more powers of $\ell$ the true factor $L_q(A_f,1)^{-1}$ carries than the naive one, and Corollary 19.37 will show that this is exactly the extra $\ell$-divisibility of $\deg \varphi_E$ coming from $q$. The smallest example is `121a1`, of type II at 11: the naive factor is 1 (since $a_{11} = 0$), the true one is $P_{11}(11^{-2}) = 1 + \frac{1}{11} = \frac{12}{11}$ (Lemma 19.45), and $e_3(E,11) = v_3(\frac{12}{11}) = 1$, while $\deg \varphi_E = 6$. The term $t_\ell(E,q)$ is defined at every bad prime, $q = \ell$ included, as the paper's appendix and (eq:rule) use it.

**Proposition 19.42** (first properties of the local terms)**.** *Let $q$ be a bad prime.*

*(i) If $q \neq \ell$, $e_\ell(E,q)$ is a nonnegative integer. It is 0 if the reduction at $q$ is multiplicative, and $v_\ell(P_q(q^{-2})) = v_\ell(L_q(A_f,1)^{-1})$ if it is additive.*

*(ii) $t_\ell(E,q) = v_\ell(n)$ for a Kodaira symbol $\mathrm{I}_n$ at any $q$, and for $\mathrm{I}_n^*$ at odd $q$. At $q = 2$ a symbol $\mathrm{I}_m^*$ can have $-v_2(j) \neq m$, or $v_2(j) \geq 0$ (potentially good reduction), and then $t_\ell(E,2) = 0$.*

*(iii) The behaviour under quadratic twist is Proposition 19.49: $t_\ell$ is invariant, $e_\ell$ is not.*

*Proof.* (i) At a multiplicative prime the naive factor is the true one (Definition 19.32), so the ratio is 1. At an additive prime $a_q = 0$ and the naive factor is 1; then $e_\ell(E,q) = v_\ell(P_q(q^{-2}))$ with $P_q(q^{-2}) \in \mathbb{Z}[1/q] \setminus \{0\}$ (Proposition 19.25(ii)), whose $\ell$-adic valuation is a nonnegative integer since $\ell \neq q$. (ii) At a prime of type $\mathrm{I}_n$, $v_q(j) = -v_q(\Delta) = -n$; at an odd prime of type $\mathrm{I}_n^*$, $v_q(j) = -n$ (Chapter 10). The statements at $q = 2$ are Chapter 10 and the data of Remark 19.59. □

**Proposition 19.43** (multiplicative primes)**.** *Let $q \neq \ell$ be a prime with $v_q(j_E) < 0$ (potentially multiplicative reduction: multiplicative, split or not, or additive of type $\mathrm{I}_n^*$ at odd $q$, or an additive symbol at $q = 2$ with $v_2(j) < 0$). Then $A_f^{I_q}$ is the line spanned by $e = e_1 \otimes e_2^\vee$ in a Tate basis, $\Phi_q$ acts on it by $q^{-1}$ (and $\mathrm{Frob}_q$ by $q$), and*

$$L_q(A_f, s)^{-1} = 1 - q^{-1-s}, \qquad P_q(X) = 1 - X, \qquad L_q(A_f, 1)^{-1} = 1 - q^{-2}.$$

*If the reduction is multiplicative, the naive factor equals the true one ($a_q^2 = 1$) and $e_\ell(E,q) = 0$; if it is additive, $e_\ell(E,q) = v_\ell(1-q^{-2}) = v_\ell(q^2-1)$. With $\mathrm{Frob}_q$ in place of $\Phi_q$ one would get $1 - q^{1-s}$.*

*Proof.* Over $\mathbb{Q}_q$, $\rho \simeq \rho' \otimes \chi$ with $\chi$ a character of order at most 2 and $\rho'$ of Tate shape: in a basis $e_1, e_2$ of $V$, $\rho'(g) = \left(\begin{smallmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa(g) \\ 0 & \varepsilon(g) \end{smallmatrix}\right)$ with $\varepsilon$ unramified of order at most 2 and $\kappa$ the Kummer cocycle of the Tate parameter (Chapter 11; Appendix B, B-C14). Twisting by a character does not change $\mathrm{ad}^0$: $\mathrm{ad}^0(\rho' \otimes \chi) = \mathrm{ad}^0 \rho'$. On $\mathrm{ad}^0 V$ with basis $e = e_1 \otimes e_2^\vee$, $h$, $e_2 \otimes e_1^\vee$, conjugation by $u = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$ fixes $e$ and sends $h \mapsto h - 2xe$ and $e_2 \otimes e_1^\vee \mapsto e_2 \otimes e_1^\vee + xh - x^2 e$. On inertia $\varepsilon = \chi_\ell = 1$ and $\kappa$ is $v_q(q_E)$ times the $\ell$-adic tame character, which is nonzero; so some $x \neq 0$ occurs, and the invariants are the line of $e$ (characteristic 0, so $2x \neq 0$). Explicitly, in matrix terms $e$, $h$, $e_2 \otimes e_1^\vee$ are $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, and $u\,(ae + bh + c\,e_2 \otimes e_1^\vee)\,u^{-1}$ equals $(a - 2xb - x^2 c)e + (b + xc)h + c\,e_2 \otimes e_1^\vee$; invariance forces $xc = 0$, so $c = 0$, and then $2xb = 0$, so $b = 0$. For $g = \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$, $geg^{-1} = \frac{a}{d} e$; with $g = \rho'(\mathrm{Frob}_q)$ this is $\chi_\ell(\mathrm{Frob}_q) = q$, the sign $\varepsilon(\mathrm{Frob}_q)$ cancelling. So $\mathrm{Frob}_q$ acts by $q$, $\Phi_q$ by $q^{-1}$, and $\det(1 - \Phi_q q^{-s} \mid A_f^{I_q}) = 1 - q^{-1-s}$; then $P_q(X) = L_q(A_f, s-1)^{-1}|_{q^{-s}=X} = 1 - X$. If the reduction is multiplicative, $a_q = \pm 1$ and the naive factor at $s = 1$ is $1 - q^{-2}$, equal to the true one; if additive, $a_q = 0$ and the naive factor is 1. With $\mathrm{Frob}_q$: $1 - q\,q^{-s}$. □

*In the paper* 19.44. The row $\mathrm{I}_n$ of the table of Section 1; the proof of Lemma lem:euler ("$(\mathrm{ad}^0 V)^{I_q}$ is a line on which the geometric Frobenius has eigenvalue $q^{-1}$, so $L_q(A_f,s)^{-1} = 1 - q^{-1-s}$, which is also the naive factor since $a_q^2 = 1$: $e_\ell = 0$"); Definition def:terms ("it is 0 at a multiplicative prime, where the naive factor is the true one"); the paragraph after Theorem thm:dfg14 ("adding a multiplicative prime multiplies it by $1 - p^{-2}$"); Proposition prop:shapes(i); and Section sec:iso ("Multiplicative primes show no Euler term"). Lemma lem:fpr's "$v_\ell(1-p^{-2})$ at a multiplicative one" is $v_\ell(L_q(A_f,1)^{-1})$ here.

**Lemma 19.45** (the omitted factors at tame primes; the paper's Lemma lem:euler)**.** *Let $q \geq 5$ be a bad prime, $q \neq \ell$. Let $E' = E^{(q^*)} = E \otimes \chi_{q^*}$ (which has good reduction at $q$ when the type is $\mathrm{I}_0^*$) and $a' = a_q(E')$. Let $\sigma$ denote an element of order 3 and $\tau$ an element of order 4 of $\mathrm{SL}_2(\mathbb{Z}_\ell)$. Then:*

| *type* | $\rho(I_q)$ | $A_f^{I_q}$ | $\Phi_q$ *on it* | $L_q(A_f,s)^{-1}$ |
|---|---|---|---|---|
| $\mathrm{I}_n$ | *unipotent* | $\langle e\rangle$ | $q^{-1}$ | $1-q^{-1-s}$ |
| $\mathrm{I}_n^*$ | $\pm$*unipotent* | $\langle e\rangle$ | $q^{-1}$ | $1-q^{-1-s}$ |
| $\mathrm{IV},\mathrm{IV}^*$; $\mathrm{II},\mathrm{II}^*$ | $\langle\sigma\rangle$; $\langle-\sigma\rangle$ | $\langle 2\sigma+1\rangle$ | $\chi_{-3}(q)$ | $1-\chi_{-3}(q)q^{-s}$ |
| $\mathrm{III},\mathrm{III}^*$ | $\langle\tau\rangle$ | $\langle\tau\rangle$ | $\chi_{-4}(q)$ | $1-\chi_{-4}(q)q^{-s}$ |
| $\mathrm{I}_0^*$ | $\{\pm1\}$ | $A_f$ | $\frac{\alpha'}{\beta'},1,\frac{\beta'}{\alpha'}$ | $(1-\frac{\alpha'}{\beta'}q^{-s})(1-q^{-s})(1-\frac{\beta'}{\alpha'}q^{-s})$ |

| *type* | $P_q(X)$ | $P_q(q^{-2})=L_q(A_f,1)^{-1}$ | $e_\ell(E,q)$ |
|---|---|---|---|
| $\mathrm{I}_n$ | $1-X$ | $1-q^{-2}$ | $0$ |
| $\mathrm{I}_n^*$ | $1-X$ | $1-q^{-2}$ | $v_\ell(q^2-1)$ |
| $\mathrm{IV},\mathrm{IV}^*$; $\mathrm{II},\mathrm{II}^*$ | $1-\chi_{-3}(q)qX$ | $1-\chi_{-3}(q)q^{-1}$ | $v_\ell(q-\chi_{-3}(q))$ |
| $\mathrm{III},\mathrm{III}^*$ | $1-\chi_{-4}(q)qX$ | $1-\chi_{-4}(q)q^{-1}$ | $v_\ell(q-\chi_{-4}(q))$ |
| $\mathrm{I}_0^*$ | $(1-qX)(1-(a'^2-2q)X+q^2X^2)$ | $\frac{(q-1)((q+1)^2-a'^2)}{q^3}$ | $v_\ell(q-1)+v_\ell((q+1)^2-a'^2)$ |

*Here $\alpha',\beta'$ are the roots of $X^2-a'X+q$, and $(2\sigma+1)^2=-3$. The rows $\mathrm{I}_n^*$ and $\mathrm{I}_0^*$ hold at every odd $q\neq\ell$, and the row $\mathrm{I}_n$ at every $q\neq\ell$.*

*Proof. Step 1: the images of inertia.* By Chapter 11 (Serre and Tate; Kraus [107]), for $q\geq 5$: at $\mathrm{I}_n$ the image is unipotent and infinite; at $\mathrm{I}_n^*$ it is $\chi_{q^*}$ times an infinite unipotent group; at the potentially good types it is finite, tame (its order divides 24, prime to $q$), hence cyclic, contained in $\mathrm{SL}_2(\mathbb{Z}_\ell)$ (the determinant $\chi_\ell$ is unramified at $q$), and of order 3 ($\mathrm{IV},\mathrm{IV}^*$), 6 ($\mathrm{II},\mathrm{II}^*$), 4 ($\mathrm{III},\mathrm{III}^*$) or 2 ($\mathrm{I}_0^*$). An element of order 6 in $\mathrm{SL}_2$ is $-\sigma$ with $\sigma=(-\sigma)^4$ of order 3; the only element of order 2 in $\mathrm{SL}_2$ is $-1$.

*Step 2: the invariants.* $A_f^{I_q}$ is the trace-zero part of the commutant of $\rho(I_q)$ in $\operatorname{End}V$. For a cyclic group $\langle g\rangle$ with $g\neq\pm1$ of finite order, the commutant of $g$ in $\operatorname{End}V$ is $\mathbb{Q}_\ell[g]$, of dimension 2 (the eigenvalues of $g$ are distinct, Chapter 4), and its trace-zero part is the line spanned by $2g-\operatorname{tr}g$ (Exercise 19.73). For $g=\sigma$, with $\sigma^2+\sigma+1=0$ and $\operatorname{tr}\sigma=-1$, this is $2\sigma+1$, and $(2\sigma+1)^2=4\sigma^2+4\sigma+1=-3$; for $g=-\sigma$ the commutant is the same, so the sign disappears in $\mathrm{ad}^0$. For $g=\tau$, $\operatorname{tr}\tau=0$ and the line is spanned by $\tau$ itself. Here $\operatorname{End}_{I_q}(T)=\mathbb{Z}_\ell[\sigma]\simeq\mathbb{Z}_\ell\otimes\mathbb{Z}[\zeta_3]$ and $\operatorname{End}_{I_q}(V)=\mathbb{Q}_\ell\otimes\mathbb{Q}(\zeta_3)$, which is a field only for $\ell\not\equiv1\pmod 3$; likewise $\mathbb{Q}_\ell[\tau]\simeq\mathbb{Q}_\ell\otimes\mathbb{Q}(i)$, a field only for $\ell\not\equiv1\pmod 4$. In particular $\tau$ is diagonalisable over $\mathbb{Q}_\ell$ only when $\ell\equiv1\pmod 4$, so the invariant line must be described as the line of $\tau$, not through a diagonal form. For the unipotent shapes the invariants are the line of $e$ (Proposition 19.43). For $\{\pm1\}$ they are all of $A_f$.

*Step 3: the Frobenius.* The arithmetic Frobenius normalises $\rho(I_q)$ and acts on tame inertia by $g\mapsto g^q$: $\rho(\mathrm{Frob}_q)g\rho(\mathrm{Frob}_q)^{-1}=g^q$ for $g$ in a tame image (Chapter 7, the tame relation). Since $\mathrm{Frob}_q$ acts on $\mathrm{ad}^0V$ by conjugation, it sends the invariant vector $2g-\operatorname{tr}g$ to $2g^q-\operatorname{tr}g$, the trace being conjugation-invariant. For $g=\sigma$: $\sigma^q$ depends only on $q$ mod 3, so $\sigma^q=\sigma$ if $q\equiv1\pmod 3$ and $\sigma^q=\sigma^{-1}=\sigma^2$ if $q\equiv2\pmod 3$, and $2\sigma^{-1}+1=2\sigma^2+1=2(-\sigma-1)+1=-(2\sigma+1)$ by $\sigma^2+\sigma+1=0$; so $\mathrm{Frob}_q$ acts on $2\sigma+1$ by $+1$ or $-1$ according as $q\equiv1$ or $2\pmod 3$, that is by $\chi_{-3}(q)$ (the character of $\mathbb{Q}(\sqrt{-3})$, for which $\chi_{-3}(q)=1$ iff $q\equiv1\pmod 3$). For $g=\tau$: $\tau^q=\tau$ or $\tau^{-1}=-\tau$ as $q\equiv1$ or $3\pmod 4$; so $\mathrm{Frob}_q$ acts on $\tau$ by $\chi_{-4}(q)$. The eigenvalue is $\pm1$, so $\Phi_q=\mathrm{Frob}_q^{-1}$ acts by the same sign. For $\mathrm{I}_0^*$: $V\simeq V_\ell E'\otimes\chi_{q^*}$ with $V_\ell E'$ unramified at $q$ (Chapter 10, the paper's Lemma lem:twist(ii), valid at every odd $q$; Neron-Ogg-Shafarevich, Chapter 11), so $\mathrm{ad}^0V=\mathrm{ad}^0V_\ell E'$ is unramified with the eigenvalues $\alpha'/\beta',1,\beta'/\alpha'$ for either Frobenius (Proposition 19.26). For $\mathrm{I}_n^*$: $V\simeq V'\otimes\chi_{q^*}$ with $V'$ multiplicative at $q$, and $\mathrm{ad}^0V=\mathrm{ad}^0V'$ (Proposition 19.43).

*Step 4: $P_q$.* By Proposition 19.25(i), $P_q(X)$ is $L_q(A_f,s)^{-1}$ with $q^{-s}$ replaced by $qX$; at $s=1$, $P_q(q^{-2})=L_q(A_f,1)^{-1}$. For instance in the rows II to IV*, $1-\chi_{-3}(q)q^{-s}$ becomes $1-\chi_{-3}(q)qX$,

whose value at $X = q^{-2}$ is $1 - \chi_{-3}(q)q^{-1}$; in the rows $\mathrm{I}_n$, $\mathrm{I}_n^*$, $1 - q^{-1}q^{-s}$ becomes $1 - X$. For $\mathrm{I}_0^*$ this is Proposition 19.26(i), (ii) for $E'$.

*Step 5: the valuations.* The naive factor is 1 at the additive types ($a_q = 0$) and $1 - q^{-2}$ at $\mathrm{I}_n$ (Proposition 19.33). So $e_\ell = 0$ for $\mathrm{I}_n$, and for the additive types $e_\ell = v_\ell(P_q(q^{-2}))$: $v_\ell(1 - q^{-2}) = v_\ell(q^2 - 1)$; $v_\ell(1 - cq^{-1}) = v_\ell(q - c)$ for $c = \chi_{-3}(q), \chi_{-4}(q)$; and $v_\ell(q - 1) + v_\ell((q + 1)^2 - a'^2)$, the powers of $q$ being $\ell$-adic units.

*Well-definedness and the range of $q$.* For $\mathrm{I}_0^*$ the other ramified quadratic character of $\mathbb{Q}_q$ (the product of $\chi_{q^*}$ with the unramified one) gives a twist with $a_q = -a'$; only $a'^2$ enters. Only the exchanges $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ and $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ under $\chi_{q^*}$ were used in the rows $\mathrm{I}_n^*$ and $\mathrm{I}_0^*$, and they hold at every odd $q$ (Chapter 10); the row $\mathrm{I}_n$ used only the Tate shape, valid at every $q$. The hypothesis $q \geq 5$ enters only through Step 1 for the potentially good rows, where it makes the image tame and determined by the type. □

Verification: on all 64,687 curves of conductor below 10,000 in Cremona's tables [36], all 118,232 pairs (curve, bad prime $q \geq 5$) have PARI's $P_q$ equal to the table ($\mathrm{I}_n$: 102,309; $\mathrm{I}_n^*$: 6,224; $\mathrm{I}_0^*$: 2,241, with $a'$ from `elltwist`; II: 1,575; II*: 760; III and III*: 1,394 each; IV: 760; IV*: 1,575), with no exception.

*In the paper* 19.46. Lemma lem:euler and the table of Section 1 are Lemma 19.45. The proof in the paper writes "inside $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ (not a field when $\ell \equiv 1 \pmod 3$)" and "the invariant line of $\mathrm{ad}^0 V$ is spanned by $\tau$ itself, which has trace 0 (it is diagonalisable only over $\mathbb{Q}_\ell(i)$)" (Step 2). "Frobenius centralises $\sigma$ if $q \equiv 1 \pmod 3$ and inverts it if $q \equiv 2$, for either choice of Frobenius" is Step 3. The abstract's list of omitted factors is the last column.

*Remark* 19.47 (when the omitted factor is an $\ell$-adic non-unit). Let $q \geq 5$, $q \neq \ell$. For $\ell = 3$: $e_3(E, q) \geq 1$ always for the types II, II*, IV, IV* (since $q \equiv \chi_{-3}(q) \pmod 3$) and $\mathrm{I}_n^*$ (since $3 \mid q^2 - 1$); for III, III* iff $q \equiv \pm 1 \pmod{12}$; for $\mathrm{I}_0^*$ iff $q \equiv 1 \pmod 3$ or $a' \equiv \pm(q + 1) \pmod 3$. For $\ell \geq 5$: iff $q \equiv \chi(q) \pmod \ell$ with $\chi = \chi_{-3}$ or $\chi_{-4}$ (rows 3, 4), $q \equiv \pm 1 \pmod \ell$ ($\mathrm{I}_n^*$), $q \equiv 1 \pmod \ell$ or $a' \equiv \pm(q + 1) \pmod \ell$ ($\mathrm{I}_0^*$). These are elementary (Exercise 19.78). Examples, all recomputed with PARI (Example 19.62): `121a1` (type II at 11, $P_{11} = 1 + 11X$, $e_3 = v_3(12) = 1$, $\deg \varphi_E = 6$); `147b1` (type IV* at 7, $\chi_{-3}(7) = 1$, $e_3 = v_3(6) = 1$); `726a1` (type III at $11 \equiv -1 \pmod{12}$, $e_3 = v_3(12) = 1$); `539c1` (type $\mathrm{I}_3^*$ at 7, $e_3 = v_3(48) = 1$); `539d1` (type $\mathrm{I}_0^*$ at 7, $E' =$ `11a3`, $a' = -2$, $e_3 = v_3(6) + v_3(60) = 2$, $e_5 = 1$); `275b1` (type $\mathrm{I}_0^*$ at 5, $a' = \pm 1$, $e_7 = v_7(4 \cdot 35) = 1$, $\deg \varphi_E = 28$).

*In the paper* 19.48. Proposition prop:forms (proof): "At $q \geq 5$ of type $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$ and $\ell = 3$, $e_3(E, q) = v_3(q - \chi_{-3}(q)) \geq 1 = w_3(E, q)$ since $q \equiv \chi_{-3}(q) \pmod 3$" is the first case of Remark 19.47. Remark rem:afterquant (1), "a prime of type $\mathrm{I}_0^*$ with $q \equiv 1 \pmod 3$ does too", is the case $v_3(q - 1) \geq 1$ of the $\mathrm{I}_0^*$ row.

**Proposition 19.49** (quadratic twists)**.** *Let $\chi$ be a quadratic Dirichlet character and $E^\chi = E \otimes \chi$, with newform $f^\chi$.*

*(i) $A_{f^\chi} \simeq A_f$ as $G_\mathbb{Q}$-representations. Hence $L_q(A_f, s)$, $P_q$ and $L(\mathrm{ad}^0 f, s)$ are invariant under every quadratic twist.*

*(ii) If $\chi$ is unramified at $q$, then $\delta_q$, $a_q^2$, the naive factor at $q$ and (for $q \neq \ell$) $e_\ell(E, q)$ are the same for $E$ and $E^\chi$.*

*(iii) Let $q$ be odd, $q \neq \ell$, and $\chi = \chi_{q^*}$. If $q \geq 5$, the exchanges $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$ leave $e_\ell(E, q)$ unchanged ($\delta_q = 0$ on both sides). The exchange $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ changes it from 0 to*

*$v_\ell(q^2-1)$; under $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ the term is undefined on the good side and $v_\ell(q-1)+v_\ell((q+1)^2-a_q^2)$ on the $\mathrm{I}_0^*$ side. In every case in which $q$ is bad on both sides, $e_\ell(E,q)+v_\ell(L_q^{\mathrm{nv}}(A_f,1)^{-1}) = v_\ell(L_q(A_f,1)^{-1})$, and the right side is invariant in all cases.*

*(iv) $t_\ell(E,q)$ is invariant under every quadratic twist.*

*Proof.* (i) $V_\ell E^\chi \simeq V_\ell E \otimes \chi$, and conjugation by $\rho(g)\chi(g)$ equals conjugation by $\rho(g)$; so $\mathrm{ad}^0(V\otimes\chi) = \mathrm{ad}^0 V$. (ii) Over $\mathbb{Q}_q$, $\chi$ is unramified, so $V$ and $V\otimes\chi$ have the same inertia invariants, and $\chi(\mathrm{Frob}_q) = \pm 1$ changes $a_q$ only by a sign; the naive factor depends on $\delta_q$ and $a_q^2$ only, and the true one by (i). (iii) The exchanges of types are Chapter 10 (the paper's Lemma lem:twist(ii)). For the potentially good types $\delta_q = 0$ on both sides, so $e_\ell(E,q) = v_\ell(L_q(A_f,1)^{-1})$ on both sides, equal by (i). For $\mathrm{I}_n$ the naive factor is $1-q^{-2}$ and $e_\ell = 0$; for $\mathrm{I}_n^*$ it is 1 and $e_\ell = v_\ell(1-q^{-2})$. The last sentence is the definition of $e_\ell$ together with (i). (iv) $j(E^\chi) = j(E)$. □

**Example 19.50** (the twist of `11a1` by $\chi_{-11}$)**.** `11a1` has type $\mathrm{I}_5$ at 11 and $e_3(E,11) = 0$. Its twist by $\chi_{-11}$ is `121d2`, of type $\mathrm{I}_5^*$ at 11, with $P_{11} = 1-X$ (the same factor) and $e_3 = v_3(1-11^{-2}) = v_3(120/121) = 1$; $t_5 = v_5(5) = 1$ for both. Definition def:terms says "The term $t_\ell$ is invariant under quadratic twist, and $e_\ell(E,q)$ under quadratic twists unramified at $q$; a twist ramified at $q$ can change $e_\ell(E,q)$", with this example, which is Proposition 19.49(ii), (iii), (iv).

*In the paper* 19.51. Theorem thm:quant, Step 3: "the last equality because $A_g = A_f$ and, at $q\in S$, $g$ has $\delta_q \geq 1$, so that its naive factor is the true one, which is the factor that $f$ (with $a_q(f) = 0$) omits". Here $g = f_{E_S}$ with $E_S = E\otimes\chi_S$; $A_g = A_f$ is Proposition 19.49(i), and at $q\in S$ the curve $E_S$ is good or multiplicative, so $v_\ell(L_q^{\mathrm{nv}}(A_g,1)^{-1}) = v_\ell(L_q(A_f,1)^{-1}) = e_\ell(E,q)$ by (iii) and Lemma 19.45.

**Proposition 19.52** (the $L$-part of Lemma lem:fpr)**.** *Let $p\neq\ell$ be any prime, good or bad, and $V^D = \mathrm{Hom}(A_f,\mathbb{Q}_\ell(1)) \simeq A_f(1)$ (the trace form, nondegenerate on trace-zero matrices in characteristic 0, makes $A_f$ self-dual, Chapter 4). Then*

$$\det(1-\Phi_p \mid (V^D)^{I_p}) = L_p(A_f(1),0)^{-1} = L_p(A_f,1)^{-1} = P_p(p^{-2}),$$

*a nonzero element of $\mathbb{Z}[1/p]\subset\mathbb{Z}_{(\ell)}$, hence a nonzero $\ell$-adic integer. Its $\ell$-adic valuation is $v_\ell((p-1)((p+1)^2-a_p^2))$ if $p\nmid N$, $v_\ell(1-p^{-2}) = v_\ell(p^2-1)$ if $p$ is multiplicative or additive with $v_p(j_E)<0$, and $e_\ell(E,p)$ if $p$ is additive. If $\Phi_p$ is replaced by $\mathrm{Frob}_p$, the determinant changes by the $\ell$-adic unit $\det(-\mathrm{Frob}_p \mid (V^D)^{I_p})$, so its vanishing is unaffected, but the equality with $L_p(A_f,1)^{-1}$ needs $\Phi_p$.*

*Proof.* The first equality is the definition at $s=0$, the second Proposition 19.16(ii), the third Proposition 19.25(ii). By Proposition 19.56, $P_p(p^{-2})$ is one of $1-p^{-2}$, $1-c/p$ with $c=\pm1$, $\frac{(p-1)((p+1)^2-a'^2)}{p^3}$ with $|a'|\leq 2\sqrt{p}$, and 1; each is a nonzero element of $\mathbb{Z}[1/p]$ (the cubic by the Hasse bound, Proposition 19.26(ii)); at a good $p$ the list is Proposition 19.26(ii) itself, with $a' = a_p$. The valuations are Proposition 19.26(ii), Proposition 19.42(i) and Proposition 19.43. For the last statement: if $u$ has eigenvalues $\mu_i$ on a space of dimension $d$, then $\det(1-u^{-1}) = \prod(1-\mu_i^{-1}) = \prod(-\mu_i^{-1})\prod(1-\mu_i)$, and $\prod(-\mu_i^{-1}) = \det(-u)^{-1}$. On $(V^D)^{I_p} = A_f^{I_p}(1)$ the arithmetic eigenvalues are $p$ times those on $A_f^{I_p}$, that is $p\cdot p = p^2$ (multiplicative), $\pm p$, or $p\alpha'/\beta', p, p\beta'/\alpha'$; their product is a power of $p$ up to sign, a unit at $\ell$. □

*In the paper* 19.53. Lemma lem:fpr: "here $L_p(A_f,1)^{-1} = L_p(A_f(1),0)^{-1}$ is a nonzero rational number which is an $\ell$-adic integer, of valuation $e_\ell(E,p)$ at an additive prime and $v_\ell(1-p^{-2})$ at a multiplicative one" is Proposition 19.52. The paragraph after Lemma lem:dfg21: "For $p \in \Sigma$ the group $H^1_f(\mathbb{Q}_p, V^D) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$ vanishes, because $\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = L_p(A_f,1)^{-1} \neq 0$" uses the proposition with $\mathrm{Frob}_p^{-1} = \Phi_p$; the unramified cohomology is written with the arithmetic $\mathrm{Frob}_p$, and $\mathrm{Frob}_p - 1$ is invertible on $(V^D)^{I_p}$ iff $\Phi_p - 1$ is. The Fitting-ideal half of Lemma lem:fpr is Chapter 18's.

**Proposition 19.54** (the naive factors at $s=1$)**.** *(i)* $L_p^{\mathrm{nv}}(A_f,1)^{-1} = \frac{(p-1)((p+1)^2 - a_p^2)}{p^3}$ *if* $p \nmid N$, $1-p^{-2}$ *if* $p \,\|\, N$, *and* $1$ *if* $p^2 \mid N$.

*(ii) For a finite set* $\Sigma$ *of primes dividing* $N$*, none equal to* $\ell$*,*

$$\sum_{p \in \Sigma} \Big( v_\ell\big(L_p(A_f,1)^{-1}\big) - v_\ell\big(L_p^{\mathrm{nv}}(A_f,1)^{-1}\big) \Big) = \sum_{p \in \Sigma} e_\ell(E,p).$$

*(iii) With* $S$ *and* $g = f_{E_S}$ *as in the paper's Theorem thm:quant (*$S$ *the odd primes of type* $\mathrm{I}_0^*$ *or* $\mathrm{I}_n^*$*,* $\ell \notin S$*), for* $q \in S$*:* $v_\ell(L_q^{\mathrm{nv}}(A_g,1)^{-1}) = v_\ell(L_q(A_f,1)^{-1}) = e_\ell(E,q)$.

*Proof.* (i) is Proposition 19.26(ii) and Proposition 19.33(i) at $s=1$. (ii) is Definition 19.41 summed. (iii) At $q \in S$, $E_S$ has good or multiplicative reduction (Chapter 10), so $\delta_q(g) \geq 1$ and its naive factor is its true one, which is $L_q(A_f,s)$ by Proposition 19.49(i); and $e_\ell(E,q) = v_\ell(L_q(A_f,1)^{-1})$ because $q$ is additive for $E$. □

*In the paper* 19.55. The paragraph after Theorem thm:dfg14: "Only the primes with $\delta_p \geq 1$ contribute: adding an additive prime to $\Sigma$ changes neither the level nor the ideal" is (i) for $p^2 \mid N$. In the proof of Theorem thm:main, the congruence ideal $\eta^\Sigma$ carries the naive factors at the multiplicative primes of $\Sigma$, the Bloch-Kato side carries the true factors at every prime of $\Sigma$ (Lemma lem:fpr), and (ii) is the cancellation that leaves $\sum e_\ell(E,p)$. Theorem thm:quant, Step 3, is (iii). Remark rem:afterquant (3), "there the quadratic twist does not lower the level, $f$ is its own depletion, and Theorem thm:dfg14 gives no factor", is (i) for $p^2 \mid N$ at the potentially good types, where no twist makes $\delta_q \geq 1$ (Proposition 19.49(iii)). Theorem thm:dfg14 itself is Chapter 20's.

## 19.10 Every bad prime, including 2 and 3

This section computes $P_q$ at every bad prime $q \neq \ell$, without assuming $q \geq 5$. At $q = 2, 3$ the Kodaira type does not determine the image of inertia, but the image does determine $P_q$, and only four shapes occur. In the paper this is Proposition prop:shapes, Remark rem:wild and Section sec:wild ("Definition def:terms with PARI's factor explains every one of these minima"); it also covers the terms at 2 and 3 of Conjecture conj:euler, the types IV, IV* at 2 in Proposition prop:forms, and the $\mathrm{I}_0^*$ formula at $q=3$ used in Theorem thm:quant.

**Proposition 19.56** (the four shapes; the paper's Proposition prop:shapes)**.** *Let* $q$ *be a bad prime and* $\ell \neq q$ *any prime. Exactly one of the following holds.*

*(a)* Potentially multiplicative *(*$v_q(j_E) < 0$*):* $P_q(X) = 1 - X$ *and* $L_q(A_f,1)^{-1} = 1 - q^{-2}$*;* $e_\ell(E,q) = 0$ *if the reduction is multiplicative and* $v_\ell(q^2-1)$ *if it is additive.*

*(b)* Potentially good with $\rho(I_q) = \{\pm 1\}$. *Equivalently, some quadratic twist $E^{(d)}$ has good reduction at $q$; for $q$ odd $d = q^*$ works, and for $q = 2$ one of $d = -4, 8, -8$ works. With $a' = a_q(E^{(d)})$ (whose square does not depend on the choice),*

$$P_q(X) = (1 - qX)(1 - (a'^2 - 2q)X + q^2X^2), \qquad e_\ell(E, q) = v_\ell(q - 1) + v_\ell((q + 1)^2 - a'^2).$$

*(c)* Potentially good with $\rho(I_q)$ cyclic of order 3, 4 or 6. *$A_f^{I_q}$ is the line spanned by $x = 2g - \operatorname{tr} g$ for a generator $g$ of $\rho(I_q)$, $\Phi_q$ and $\mathrm{Frob}_q$ act on it by the same sign $c \in \{\pm 1\}$, $P_q(X) = 1 - cqX$ and $e_\ell(E, q) = v_\ell(q - c)$. For $q \geq 5$, $c = \chi_{-3}(q)$ (orders 3, 6) or $\chi_{-4}(q)$ (order 4); for the types* IV, IV$^*$ *at $q = 2$, $c = \chi_{-3}(2) = -1$; at $q = 2, 3$ the type does not determine $c$ in general.*

*(d)* Potentially good with $\rho(I_q)$ non-abelian. *Then $q \in \{2, 3\}$, $A_f^{I_q} = 0$, $P_q = 1$ and $e_\ell(E, q) = 0$.*

*Consequently, for $\ell$ odd: $e_3(E, 2) \in \{0, 1, 2\}$, $e_5(E, 2) \in \{0, 1\}$ and $e_\ell(E, 2) = 0$ for $\ell \geq 7$; $e_5(E, 3), e_7(E, 3) \in \{0, 1\}$ and $e_\ell(E, 3) = 0$ for $\ell \geq 11$. More precisely $e_3(E, 2) = 1$ exactly in case (a) additive and in case (c) with $c = -1$, $e_3(E, 2) = 2$ exactly in case (b) with $a' = 0$, $e_5(E, 2) = 1$ exactly in case (b) with $a' = \pm 2$; $e_5(E, 3) = 1$ exactly in case (b) with $a' = \pm 1$, and $e_7(E, 3) = 1$ exactly in case (b) with $a' = \pm 3$.*

*Proof.* (a) Over $\mathbb{Q}_q$, $E$ is a quadratic twist (possibly trivial) of a curve of Tate shape, at every $q$ including 2 (Chapter 11); Proposition 19.43 applies. In the other cases the reduction is potentially good, $\rho(I_q)$ is finite (Chapter 11, Serre and Tate), contained in $\mathrm{SL}_2(\mathbb{Z}_\ell)$ because $\chi_\ell$ is unramified at $q$, and the reduction is additive, so $\delta_q = 0$, $a_q = 0$ and the naive factor is 1; then $e_\ell(E, q) = v_\ell(P_q(q^{-2}))$.

*The trichotomy.* Let $G = \rho(I_q)$, finite, in $\mathrm{SL}_2(\overline{\mathbb{Q}}_\ell)$. If $G$ fixes a line of $\overline{\mathbb{Q}}_\ell^2$ it is upper triangular in a suitable basis; a finite group of upper triangular matrices in characteristic 0 meets the unipotent radical $\{\left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)\}$ trivially (a nontrivial unipotent has infinite order), so it injects into the diagonal torus and is abelian. (Explicitly, $\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) \mapsto (a, d)$ is a homomorphism whose kernel consists of unipotent matrices, so on a finite group it is injective, with abelian image.) Hence a non-abelian $G$ fixes no line: $V \otimes \overline{\mathbb{Q}}_\ell$ is an absolutely irreducible representation of $G$, its commutant is the scalars (Schur, Chapter 4), and $A_f^{I_q} = 0$, which is (d). A finite abelian subgroup of $\mathrm{SL}_2(\overline{\mathbb{Q}}_\ell)$ consists of commuting diagonalisable elements (an element of finite order $n$ is annihilated by $X^n - 1$, which has distinct roots in characteristic 0), hence is simultaneously diagonalisable. In a common eigenbasis each element is $\mathrm{diag}(\zeta, \zeta^{-1})$, and $g \mapsto \zeta$ is an injective homomorphism into the roots of unity; a finite group of roots of unity is cyclic. If $\zeta$ has order $n$, then $\zeta + \zeta^{-1} = \operatorname{tr} g$ is an integer, because the characteristic polynomials of the elements of $G$ have integer coefficients [179, Section 2] (Chapter 11); so $\zeta$ is a root of $X^2 - (\operatorname{tr} g)X + 1 \in \mathbb{Z}[X]$, its degree $\varphi_{\mathrm{Eu}}(n)$ over $\mathbb{Q}$ is at most 2, and $n \in \{1, 2, 3, 4, 6\}$. Order 1 would mean good reduction. If $G \neq \{\pm 1\}$, let $g$ generate it; $g$ has distinct eigenvalues $\zeta^{\pm 1}$, its commutant is $\overline{\mathbb{Q}}_\ell[g]$, of dimension 2, and the trace-zero part is the line of $x = 2g - \operatorname{tr} g$, defined over $\mathbb{Q}_\ell$; and $x^2 = 4g^2 - 4g \operatorname{tr} g + (\operatorname{tr} g)^2 = (\operatorname{tr} g)^2 - 4$ by Cayley-Hamilton ($g^2 = g \operatorname{tr} g - 1$), a nonzero scalar because $g \neq \pm 1$. The Frobenius normalises $I_q$, hence $G$, and so preserves the line of $x$: $\rho(\mathrm{Frob}_q) x \rho(\mathrm{Frob}_q)^{-1} = cx$; conjugation preserves $x^2$, so $c^2x^2 = x^2$ and $c = \pm 1$, the same for $\Phi_q$. Then $L_q(A_f, s)^{-1} = 1 - cq^{-s}$, $P_q(X) = 1 - cqX$ and $e_\ell = v_\ell(1 - c/q) = v_\ell(q - c)$: (c). For $q \geq 5$ the value of $c$ is Lemma 19.45; for types IV, IV$^*$ at 2 the image is cyclic of order 3 (Chapter 11, the paper's Lemma lem:IV), tame at 2, and the computation of Lemma 19.45 gives $c = \chi_{-3}(2) = -1$. That $c$ is not determined by the type at $q = 2, 3$ is shown by the data below (for instance types III at 2 occur with both signs).

*Case (b).* Suppose $G = \{\pm 1\}$, and let $\theta = \rho|_{I_q} \colon I_q \to \{\pm 1\}$. Since $-1$ is central, $\theta(\sigma i \sigma^{-1}) = \theta(i)$ for $\sigma \in G_{\mathbb{Q}_q}$. Choose a Frobenius lift $\phi \in G_{\mathbb{Q}_q}$; the closed subgroup it generates is a complement $\hat{\mathbb{Z}}$ of $I_q$, and $G_{\mathbb{Q}_q} = I_q \rtimes \overline{\langle \phi \rangle}$ (Chapter 7). Put $\chi(i\phi^k) = \theta(i)$; by the invariance of $\theta$ this is a continuous character of $G_{\mathbb{Q}_q}$ of order 2, ramified, with $\chi|_{I_q} = \theta$. Then $\rho \otimes \chi$ is unramified, so by Neron-Ogg-Shafarevich (Chapter 11) the twist of $E$ over $\mathbb{Q}_q$ by $\chi$ has good reduction. Every quadratic character of $G_{\mathbb{Q}_q}$ is the restriction of a global quadratic character $\chi_d$: for $q$ odd the ramified ones are $\chi_{q^*}$ and $\chi_{q^*}$ times the unramified quadratic character, which is the restriction of $\chi_D$ for any fundamental discriminant $D$ prime to $q$ with $\chi_D(q) = -1$; for $q = 2$ the ramified classes of $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$ are those of $-1, \pm 2$ and their products with the unramified class 5, and they are the restrictions of $\chi_{-4}$, $\chi_8$, $\chi_{-8}$ times 1 or $\chi_5$ (Chapter 7). An unramified quadratic twist does not change the reduction type, so one of $d = q^*$ ($q$ odd), $d \in \{-4, 8, -8\}$ ($q = 2$) gives good reduction. Conversely, if $E^{(d)}$ has good reduction then $\rho \otimes \chi_d$ is unramified, so $\rho(I_q) = \chi_d(I_q) \subseteq \{\pm 1\}$, and it is not trivial because $E$ is bad. Then $\mathrm{ad}^0 V = \mathrm{ad}^0 V_\ell E^{(d)}$ is unramified, and Proposition 19.26 gives $P_q$ and $e_\ell$; two choices of $d$ differ by an unramified twist, which changes $a'$ by a sign.

*The consequences.* At $q = 2$, $|a'| \le 2\sqrt{2}$ gives $a' \in \{0, \pm 1, \pm 2\}$ and $(q+1)^2 - a'^2 \in \{9, 8, 5\}$, while $q - 1 = 1$; so in case (b) $e_3 = 2$ for $a' = 0$, $e_5 = 1$ for $a' = \pm 2$, and nothing else. In case (a) additive $e_\ell = v_\ell(3)$; in case (c) $e_\ell = v_\ell(2 - c) \in \{v_\ell(1), v_\ell(3)\}$. At $q = 3$, $|a'| \le 2\sqrt{3}$ gives $a' \in \{0, \pm 1, \pm 2, \pm 3\}$ and $(q+1)^2 - a'^2 \in \{16, 15, 12, 7\}$, with $q - 1 = 2$; case (a) additive gives $v_\ell(8)$, case (c) gives $v_\ell(3 - c) \in \{v_\ell(2), v_\ell(4)\}$. Read off the odd primes $\ell \ne q$. □

Verification: on all 64,687 curves of conductor below 10,000, counting pairs (curve, $q$) by Kodaira type and PARI's $P_q$ (`lfuneuler(lfunsympow(E,2),q)`):

| $q$ | type | (a) $1 - X$ | (b) cubic | (c) $1 + qX$ | (c) $1 - qX$ | (d) 1 |
|---|---|---|---|---|---|---|
| 2 | $\mathrm{I}_n$ | 25,751 | | | | |
| 2 | $\mathrm{I}_m^*$ | 3,732 | 716 | 146 | | 5,073 |
| 2 | $\mathrm{I}_0^*$ | | | 792 | | 1,712 |
| 2 | II / II* | | 136 / 291 | 469 / 108 | | 1,109 / 1,267 |
| 2 | III / III* | | | 42 / 42 | 56 / 56 | 2,012 / 2,202 |
| 2 | IV / IV* | | | 2,094 / 2,656 | | |
| 3 | $\mathrm{I}_n$ / $\mathrm{I}_n^*$ | 25,283 / 7,805 | | | | |
| 3 | $\mathrm{I}_0^*$ | | 2,283 | | | |
| 3 | II / II* | | | 145 / 91 | 135 / 86 | 986 / 699 |
| 3 | III / III* | | | 1,553 / 1,553 | | |
| 3 | IV / IV* | | | 91 / 145 | 86 / 135 | 699 / 986 |

Every cubic factor (1,143 at 2, 2,283 at 3) equals the formula of (b) for a twist $E^{(d)}$ with good reduction found with `elltwist`, and no curve whose $P_q$ is not cubic has such a twist; at 2, a symbol $\mathrm{I}_m^*$ has $P_2 = 1 - X$ exactly when $v_2(j) < 0$ (3,732 of 9,667 pairs); no $P_q$ of degree 2 occurs. At $q = 2$ the sign $c = +1$ occurs below conductor 10,000 only for the types III, III*. The rows III, III* at $q = 3$ show only $1 + 3X$ below conductor 10,000; this is an observation, not a theorem. The distribution of $e_\ell$ at 2 and 3 over the same pairs: $e_3(E, 2) = 1$ for 10,081 pairs and 2 for 212; $e_5(E, 2) = 1$ for 215; $e_5(E, 3) = 1$ for 626, $e_7(E, 3) = 1$ for 184; all other values 0, as the proposition says.

Examples. `1232l1` is the twist of `77a1` by $\chi_{-4}$, with $a_2$(`77a1`) $= 0$; so $P_2 = (1-2X)(1+2X)^2 = 1 + 2X - 4X^2 - 8X^3$, $P_2(\frac{1}{4}) = \frac{9}{8}$, $e_3(E, 2) = 2$, and $\deg \varphi_E = 288 = 2^5 \cdot 3^2$. `592c1` is the twist of `37a1` by $\chi_{-4}$, with $a_2$(`37a1`) $= -2$; so $P_2 = (1 - 2X)(1 + 4X^2)$, $P_2(\frac{1}{4}) = \frac{5}{8}$, $e_5(E, 2) = 1$, and $\deg \varphi_E = 80 = 2^4 \cdot 5$: the prime 2 contributes at $\ell = 5$. `336f1` has type II at 2 and case (c) with

$c = -1$: $P_2 = 1 + 2X$, $e_3(E, 2) = 1$, $\deg \varphi_E = 24$. Every type IV or IV* at 2 is case (c) with $c = -1$ (4,750 pairs), as the proposition proves.

*In the paper* 19.57. Proposition prop:shapes states (i) to (iv) for $\ell$ odd; its proof is the argument above, with "Every quadratic character of $G_{\mathbb{Q}_q}$ is the restriction of a quadratic Dirichlet character $\chi'$ (for $q$ odd a product of $\chi_{q^*}$ and a character unramified at $q$; for $q = 2$ a product of $\chi_{-4}$, $\chi_8$ and $\chi_5$)".

*Remark* 19.58 (commentary: the book's version and the paper's). Proposition 19.56 states slightly more than Proposition prop:shapes: it allows any $\ell \neq q$, characterises case (b) by the existence of a quadratic twist of good reduction with the explicit choices of $d$, gives the value of $c$ for types IV, IV* at 2, and lists the possible values of $e_\ell(E, 2)$ and $e_\ell(E, 3)$. The paper states the four shapes and the formulas for $e_\ell$ in each case; nothing in the paper's arguments changes.

*Remark* 19.59 (reading Remark rem:wild). Remark rem:wild opens: "At $q = 2, 3$ the Kodaira type does not determine $P_q$, but PARI does, and Definition def:terms applies verbatim." In the terms of this section:

(i) The Kodaira type does not determine $P_q$ (the table above); what determines it is the image of inertia and $c$ or $a'$ (Proposition 19.56).

(ii) That a type IV or IV* at 2 always has $P_2(X) = 1 + 2X$ is a theorem (Proposition 19.56(c); Chapter 11; the paper's Lemma lem:IV).

(iii) For types II and II* at 2 all three shapes 1, $1 + 2X$ and cubic occur; `336f1` and `1232l1`, the paper's examples, show the last two.

(iv) For a type $\mathrm{I}_n^*$ at 2: below conductor 10,000, 5,935 of the 9,667 pairs (61%) are potentially good over all curves (2,340 of 4,037, 58%, over the optimal curves), and among the potentially good ones the cubic occurs only 716 times ($P_2 = 1$ for 5,073, $1 + 2X$ for 146); in all 716 the twist has $a' = \pm 1$, so by Proposition 19.56 this factor contributes nothing at odd $\ell$. Then $t_\ell = 0$.

That a cubic $P_q$ has shape (ii) holds for every curve, by Proposition prop:shapes (Proposition 19.56), as the paper notes ("that a cubic factor has shape (ii) is automatic, (ii) being the only cubic shape"). What the computation checks is that each cubic returned by PARI equals $(1 - qX)(1 - (a'^2 - 2q)X + q^2X^2)$ with $a'$ computed independently from a quadratic twist of good reduction (Remark 19.60): 1,143 cubics at 2 and 2,283 at 3 over all 64,687 curves, 534 and 1,308 over the optimal non-CM curves. The last sentence of the remark, "At $q = \ell$ the Euler factor is not the relevant quantity, and the weights $w_3$ of Conjecture conj:euler are empirical", is consistent with Definition 19.41, which defines $e_\ell$ only for $q \neq \ell$, while $t_\ell$ is used at $q = \ell$ as well.

## 19.11 Computing the factors: PARI and `rule(E, l)`

This section says what PARI's symmetric square $L$-function is in the normalisations of this chapter, and what the paper's appendix function `rule(E, l)` computes, term by term. In the paper this is Section 1.1 ("PARI's `lfunsympow(E,2)` returns these factors at every prime, including 2 and 3"), Section sec:data and the appendix. How to install and call PARI, precision and memory, and the line-by-line reading of the code are Chapter 22's.

*Remark* 19.60 (PARI's symmetric square). `S = lfunsympow(E, 2)` is the $L$-function $L(\mathrm{Sym}^2 f, s)$ of Convention 19.4(b), with the true local factors at every prime: `lfunparams(S)` is `[N_sym, 3, [0, 0, 1]]` (Proposition 19.29); `lfuneuler(S, q)` is $1/P_q(x)$ with $x = q^{-s}$; `lfun(S, 2)` is $L(\mathrm{Sym}^2 f, 2) = L(\mathrm{ad}^0 f, 1)$. Hence, for $q \neq \ell$, $e_\ell(E, q)$ is

```
valuation(subst(1/lfuneuler(S,q), x, 1/q^2) / (1 - ellap(E,q)^2/q^2), l).
```

We checked, on all 211,455 pairs (curve, bad prime) below conductor 10,000, that PARI's $P_q$ has one of the four shapes of Proposition 19.56, that it equals the factor of Lemma 19.45 at every $q \geq 5$, and that every cubic $P_q$ at $q = 2, 3$ is the formula of Proposition 19.56(b). The method by which PARI finds the bad factors from the curve is Watkins's [204] (as we understand it; we have not checked the implementation), and the numerical evaluation of $L$-values uses Dokchitser's algorithm [55].

```
E = ellinit([1,1,1,-30,-76]);        \\ 121a1
S = lfunsympow(E, 2);
lfunparams(S)                        \\ [121, 3, [0, 0, 1]]
1/lfuneuler(S, 11)                   \\ 11*x + 1, so P_11(X) = 1 + 11X
subst(1/lfuneuler(S,11), x, 1/121)   \\ 12/11 = L_11(A_f,1)^(-1)
lfun(S, 2)                           \\ 0.80473382314698593...
```

**Example 19.61** (the mathematics of `rule(E, l)`)**.** The appendix of the paper defines `rule(E, l, {iii = 0})`, which returns $[v_\ell(\deg \varphi_E), \mathrm{bound}]$. Term by term, for each bad prime $q$ with Kodaira code `k` (Convention 19.1(e)):

(a) for $q \neq \ell$, `r` $= P_q(q^{-2}) = L_q(A_f, 1)^{-1}$ (Remark 19.60), divided by $1 - q^{-2}$ when `k` $\geq 5$ (type $\mathrm{I}_n$, $n \geq 1$: the naive factor is the true one, Proposition 19.43), and `e` $= v_\ell($`r`$) = e_\ell(E, q)$ (Definition 19.41), except that the codes $\pm 3$ (types III, III$^*$) are skipped unless `iii` $= 1$; at an additive $q$ no division is made, because the naive factor is 1;

(b) for $q = \ell = 3$, the weight $w_3(E)$ of Conjecture conj:euler by code;

(c) at every $q$, $q = \ell$ included, `t` $= t_\ell(E, q)$ from `valuation(j, q)`;

(d) the correction $h_3(E)$, computed by `ns3(E)` (image $N_s(\mathbb{F}_3)$, Chapter 11), is subtracted when `iii` $= 1$ and $\ell = 3$.

Values, recomputed with PARI 2.17.2 and the paper's code verbatim: for `121a1`, with `E = ellinit([1,1,1,-30,-76])`, `rule(E, 3)` returns `[1, 1]`: at $q = 11$, type II, `r` $= \frac{12}{11}$, `e` $= 1$, `t` $= 0$, and $\deg \varphi_E = 6$. For `726a1` $= [1, 1, 0, -35, -51]$ (image 3Ns), `rule(E,3)` and `rule(E,3,1)` both return `[2, 2]`: at $q = 2$, type $\mathrm{I}_6$, `r` $= \frac{3}{4}/(1 - \frac{1}{4}) = 1$, `e` $= 0$, `t` $= v_3(6) = 1$; at $q = 3 = \ell$, type $\mathrm{I}_3$, $w_3 = 0$, `t` $= 1$; at $q = 11$, type III, `r` $= \frac{12}{11}$, `e` $= 1$ counted only with `iii` $= 1$, and then cancelled by `ns3` $= 1$; and $\deg \varphi_E = 144 = 2^4 \cdot 3^2$.

## 19.12 Worked examples

This section collects the numbers behind the statements of the chapter, all recomputed with PARI 2.17.2 (through `cypari2`) and checked against Cremona's tables [36]. The curves are those of the paper's Example after the proof of Theorem thm:qual, of its Remarks rem:wild and rem:degree, and of its appendix, together with the counterexamples of the boundary.

**Example 19.62** (local factors, degrees and $B_E$)**.** For each curve: the bad primes with Kodaira type, $P_q(X)$ and, at the additive primes, $P_q(q^{-2}) = L_q(A_f,1)^{-1}$; the nonzero $e_\ell(E,q)$ for odd $\ell \neq q$ with $\ell \le 13$; the modular degree (`ellmoddegree`, equal to Cremona's); and $B_E = \deg\varphi_E / \prod_{q^2\mid N} P_q(q^{-2})$, which agrees with $N\,L(\mathrm{ad}^0 f,1)/(2\pi\,\mathrm{vol}(\Lambda_E))$ computed from `lfun` and `E.area` to relative precision $2\cdot 10^{-16}$ in every row (so $c_E = 1$ for all these curves, as Chapter 14 expects).

| curve | bad primes: type, $P_q(X)$, $P_q(q^{-2})$ | nonzero $e_\ell$ | $\deg\varphi_E$ | $B_E$ |
|---|---|---|---|---|
| `11a1` | 11: $\mathrm{I}_5$, $1-X$ | none | 1 | 1 |
| `121a1` | 11: II, $1+11X$, $\frac{12}{11}$ | $e_3 = 1$ | 6 | $\frac{11}{2}$ |
| `121c1` | 11: IV, $1+11X$, $\frac{12}{11}$ | $e_3 = 1$ | 6 | $\frac{11}{2}$ |
| `147b1` | 3: $\mathrm{I}_1$; 7: $\mathrm{IV}^*$, $1-7X$, $\frac{6}{7}$ | $e_3(7) = 1$ | 42 | 49 |
| `200a1` | 2: $\mathrm{II}^*$, 1, 1; 5: $\mathrm{IV}^*$, $1+5X$, $\frac{6}{5}$ | $e_3(5) = 1$ | 120 | 100 |
| `350e1` | 2: $\mathrm{I}_{11}$; 5: $\mathrm{II}^*$, $1+5X$, $\frac{6}{5}$; 7: $\mathrm{I}_2$ | $e_3(5) = 1$ | 1320 | 1100 |
| `726a1` | 2: $\mathrm{I}_6$; 3: $\mathrm{I}_3$; 11: III, $1+11X$, $\frac{12}{11}$ | $e_3(11) = 1$ | 144 | 132 |
| `539c1` | 7: $\mathrm{I}_3^*$, $1-X$, $\frac{48}{49}$; 11: $\mathrm{I}_2$ | $e_3(7) = 1$ | 288 | 294 |
| `539d1` | 7: $\mathrm{I}_0^*$, $(1-7X)(1+10X+49X^2)$, $\frac{360}{343}$; 11: $\mathrm{I}_1$ | $e_3(7) = 2$, $e_5(7) = 1$ | 72 | $\frac{343}{5}$ |
| `980i1` | 2: $\mathrm{IV}^*$, $1+2X$, $\frac{3}{2}$; 5: $\mathrm{I}_1$; 7: $\mathrm{I}_5^*$, $1-X$, $\frac{48}{49}$ | $e_3(2) = e_3(7) = 1$ | 2880 | 1960 |
| `275b1` | 5: $\mathrm{I}_0^*$, $(1-5X)(1+9X+25X^2)$, $\frac{28}{25}$; 11: $\mathrm{I}_1$ | $e_7(5) = 1$ | 28 | 25 |
| `1232l1` | 2: $\mathrm{II}^*$, $(1-2X)(1+2X)^2$, $\frac{9}{8}$; 7: $\mathrm{I}_2$; 11: $\mathrm{I}_1$ | $e_3(2) = 2$ | 288 | 256 |
| `592c1` | 2: $\mathrm{II}^*$, $(1-2X)(1+4X^2)$, $\frac{5}{8}$; 37: $\mathrm{I}_1$ | $e_5(2) = 1$ | 80 | 128 |
| `336f1` | 2: II, $1+2X$, $\frac{3}{2}$; 3: $\mathrm{I}_1$; 7: $\mathrm{I}_2$ | $e_3(2) = 1$ | 24 | 16 |
| `52a1` | 2: $\mathrm{IV}^*$, $1+2X$, $\frac{3}{2}$; 13: $\mathrm{I}_2$ | $e_3(2) = 1$ | 3 | 2 |
| `98a1` | 2: $\mathrm{I}_2$; 7: $\mathrm{I}_1^*$, $1-X$, $\frac{48}{49}$ | $e_3(7) = 1$ | 16 | $\frac{49}{3}$ |
| `121b1` | 11: III, $1+11X$, $\frac{12}{11}$ | $e_3 = 1$ | 4 | $\frac{11}{3}$ |
| `121d1` | 11: $\mathrm{I}_1^*$, $1-X$, $\frac{120}{121}$ | $e_3 = e_5 = 1$ | 24 | $\frac{121}{5}$ |

Reading. For every row and every odd $\ell \nmid N$ with $E[\ell]$ irreducible and $\ell \notin S_f$, $v_\ell(B_E) \ge 0$, as Remark 19.39 leads one to expect. The rows with $v_\ell(B_E) < 0$ for an odd $\ell \nmid N$ are exactly those where the hypothesis of Theorem thm:main fails at that $\ell$: `539d1` and `121d1` at $\ell = 5$ ($E[5]$ reducible: they are isogenous to twists of `11a1`), `98a1` at $\ell = 3$ ($E[3]$ reducible), and `121b1` at $\ell = 3$ (complex multiplication by $\mathbb{Q}(\sqrt{-11})$, mod 3 image $N_s(\mathbb{F}_3)$). The identity of Corollary 19.37(iii) holds in every row at every $\ell \nmid N$: for instance `539d1` at $\ell = 3$ has $v_3(72) = 2 = v_3(\frac{343}{5}) + e_3(E,7)$.

How one row is computed, for `539d1`. At 7 the type is $\mathrm{I}_0^*$ and the twist by $\chi_{-7}$ is `11a3`, with $a' = a_7(\texttt{11a3}) = -2$ (isogenous curves have the same $a_p$, and $a_7(\texttt{11a1}) = -2$). So $a'^2 - 2q = 4 - 14 = -10$ and $P_7(X) = (1-7X)(1+10X+49X^2)$. At $X = 7^{-2}$,

$$P_7(7^{-2}) = \Big(1-\frac{1}{7}\Big)\Big(1+\frac{10}{49}+\frac{1}{49}\Big) = \frac{6}{7}\cdot\frac{60}{49} = \frac{360}{343}, \qquad 360 = 2^3\cdot 3^2\cdot 5,$$

which gives $e_3(E,7) = 2$ and $e_5(E,7) = 1$ (the formula $v_\ell(q-1) + v_\ell((q+1)^2 - a'^2)$ with $q-1 = 6$ and $64 - 4 = 60$ says the same). The prime 11 is multiplicative, with naive factor equal to the true one, and contributes nothing. Then $B_E = \deg\varphi_E / P_7(7^{-2}) = 72\cdot\frac{343}{360} = \frac{343}{5}$.

**Example 19.63** (the Petersson norm)**.** For fifteen curves, $\langle f,f\rangle$ (unnormalised: $\mu(N)$ times PARI's `mfpetersson`), $L(\mathrm{Sym}^2 f,2) = L(\mathrm{ad}^0 f,1)$ (`lfun(lfunsympow(E,2),2)`), the correction $\prod_{p^2\mid N} P_p(p^{-2})$ of Theorem 19.36, and the degree from Corollary 19.37(ii) with $c_E = 1$. In every case the ratio of $\langle f,f\rangle$ to $\frac{N}{8\pi^3} L(\mathrm{Sym}^2 f,2)\prod P_p(p^{-2})$ is 1 to within $10^{-15}$, and the degree is `ellmoddegree`'s.

| curve | $N$ | $\langle f,f\rangle$ | $L(\mathrm{Sym}^2 f,2)$ | $\prod_{p^2\mid N} P_p(p^{-2})$ | $\deg\varphi_E$ |
|---|---|---|---|---|---|
| 11a1 | 11 | 0.0469001478734952 | 1.05759924459096 | 1 | 1 |
| 14a1 | 14 | 0.0665237252389808 | 1.17865887460805 | 1 | 1 |
| 20a1 | 20 | 0.0813494568253817 | 0.67262500429864 | $\frac{3}{2}$ | 1 |
| 27a1 | 27 | 0.0684646511192122 | 0.62898782679185 | 1 | 1 |
| 36a1 | 36 | 0.1293900826782061 | 0.44576718924526 | 2 | 1 |
| 49a1 | 49 | 0.1252457646564030 | 0.55477211885316 | $\frac{8}{7}$ | 1 |
| 50a1 | 50 | 0.2164473907376362 | 0.89483035785868 | $\frac{6}{5}$ | 2 |
| 64a1 | 64 | 0.1741504912107096 | 0.67496978931117 | 1 | 2 |
| 99a1 | 99 | 0.5385195687108607 | 1.01196889364877 | $\frac{4}{3}$ | 4 |
| 121a1 | 121 | 0.4282393600119572 | 0.80473382314699 | $\frac{12}{11}$ | 6 |
| 121c1 | 121 | 0.4282393600119572 | 0.80473382314699 | $\frac{12}{11}$ | 6 |
| 121d1 | 121 | 0.5116379768017657 | 1.05759924459096 | $\frac{120}{121}$ | 24 |
| 162a1 | 162 | 1.0047768618552920 | 1.15386627335735 | $\frac{4}{3}$ | 12 |
| 52a1 | 52 | 0.1800459808053415 | 0.57256979446424 | $\frac{3}{2}$ | 3 |
| 80a1 | 80 | 0.3036483388848129 | 0.94150044089760 | 1 | 4 |

Three things to notice. The correction is 1 for 27a1, 64a1 and 80a1 although their levels are not square-free: their additive primes are of shape (d) of Proposition 19.56, with $P_q = 1$. $L(\mathrm{Sym}^2 f,2)$ is the same for 121a1 and its twist 121c1 by $\chi_{-11}$, and for 11a1 and 121d1 (a curve in the isogeny class of the twist of 11a1 by $\chi_{-11}$), as Proposition 19.49(i) says; the Petersson norms differ by the factor $N\prod P_p(p^{-2})$. And $\langle f,f\rangle$ for 121a1 is $\frac{12}{11}$ times $\frac{121}{8\pi^3}L(\mathrm{Sym}^2 f,2) = 0.3925527466\ldots$, as Chapter 12 also found.

## 19.13 Exercises

**Exercise 19.64.** Prove from Euler products that $\zeta(s)\zeta(s-1) = \sum_n \sigma(n)n^{-s}$, $\zeta(s)^2 = \sum_n d(n)n^{-s}$ and $\zeta(s-1)/\zeta(s) = \sum_n \varphi_{\mathrm{Eu}}(n)n^{-s}$, where $\sigma$, $d$ and $\varphi_{\mathrm{Eu}}$ are the sum of divisors, the number of divisors and Euler's totient function; say where each identity holds. (Hint: compare the local factors; for the last, $\sum_k \varphi_{\mathrm{Eu}}(p^k)X^k = \frac{1-X}{1-pX}$. All three hold on $\mathrm{Re}\, s > 2$.)

**Exercise 19.65.** Write down the Euler factors of $\zeta_{\mathbb{Q}(i)}$ at 2, 3, 5 and 13 and check that they are those of $\zeta(s)L(s,\chi_{-4})$. Deduce from Theorem 19.13(i) and Theorem 19.12(iv)(a) that the residue of $\zeta_{\mathbb{Q}(i)}$ at $s=1$ is $\pi/4$, and compare with the number of ideals of norm at most $x$, which is the number of lattice points of $\mathbb{Z}[i]$ in a disc of area $\pi x$, divided by 4. (Answer: $(1-2^{-s})^{-1}$, $(1-3^{-2s})^{-1}$, $(1-5^{-s})^{-2}$, $(1-13^{-s})^{-2}$.)

**Exercise 19.66.** Prove $\theta(1/y) = \sqrt{y}\,\theta(y)$ from the Poisson summation formula and the self-duality of the Gaussian, and complete the sketch of Theorem 19.9(iv). Deduce that $\xi$ has residue 1 at $s=1$ and $-1$ at $s=0$, and that $\zeta(0) = -\frac{1}{2}$. (Hint: $\Gamma_{\mathbb{R}}(s) \sim \frac{2}{s}$ as $s \to 0$.)

**Exercise 19.67.** Compute $L_p(\mathbb{Q}_\ell(n),s)$ for every $n \in \mathbb{Z}$ with the geometric Frobenius, and show that with the arithmetic Frobenius one would get $L_p(M(1),s) = L_p(M,s-1)$. At a multiplicative prime $p$, compute both $\det(1-\Phi_p p^{-1} \mid A_f^{I_p})$ and $\det(1-\mathrm{Frob}_p\, p^{-1} \mid A_f^{I_p})$, and both $\det(1-\Phi_p \mid (A_f(1))^{I_p})$ and $\det(1-\mathrm{Frob}_p \mid (A_f(1))^{I_p})$. Which pair satisfies the paper's identity $L_p(A_f,1) = L_p(A_f(1),0)$? (Answer: $(1-p^{-n-s})^{-1}$; $1-p^{-2}$ and 0; $1-p^{-2}$ and $1-p^2$; only the geometric pair, and the arithmetic one even gives $L_p^{\mathrm{Frob}}(A_f,1)^{-1} = 0$.)

**Exercise 19.68.** Without using Chapter 4, check $L_p(\mathrm{Sym}^2 f,s) = L_p(A_f,s-1)$ at a good prime by computing the eigenvalues of $\Phi_p$ on $A_f(-1)$: the eigenvalues $\alpha/\beta, 1, \beta/\alpha$ of $A_f$ multiplied by

the eigenvalue of $\Phi_p$ on $\mathbb{Q}_\ell(-1)$. Which value of $s$ for $L(\mathrm{Sym}^2 f, s)$ corresponds to the centre of the functional equation of $L(\mathrm{ad}^0 f, s)$? (Answer: $p \cdot \frac{\alpha}{\beta} = \alpha^2,\ p,\ \beta^2$; $s = \frac{3}{2}$.)

**Exercise 19.69.** Let $p \nmid N$ with $a_p(f) = 0$ (for `11a1`, $p = 19$). Show that $a_{p^{2k}} = (-p)^k$ and $a_{p^{2k+1}} = 0$, and check both local identities of Proposition 19.34 in this case. Show that the coefficient of $p^{-s}$ in $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s)$ is $a_p^2 - p$ in general, so that it is negative here, while every coefficient of the Rankin series is nonnegative. (Hint: $P_p(X) = (1 - pX)(1 + pX)^2$ when $a_p = 0$.)

**Exercise 19.70.** From the factors $P_2 = 1 + 2X$, $P_3 = 1 + 3X$ of `36a1`, and $P_5 = 1 + 5X$ of `50a1`, compute the corrections $\prod_{p^2 \mid N} P_p(p^{-2})$ of Theorem 19.36, and say which case of Proposition 19.56 each prime is in. Then check the values $\langle f, f \rangle$ of Example 19.63 for these two curves with PARI. (Answer: $\frac{3}{2} \cdot \frac{4}{3} = 2$; $\frac{6}{5}$; all three primes are in case (c) with $c = -1$.)

**Exercise 19.71.** For `147b1` ($N = 3 \cdot 7^2$, $\deg \varphi_E = 42$, $B_E = 49$) check the identity $v_\ell(\deg \varphi_E) = v_\ell(B_E) + \sum_q e_\ell(E, q)$ at $\ell = 3$, and show that it fails at $\ell = 7$ for the naive reading in which the term at $q = 7$ is omitted. Why does Corollary 19.37 exclude $\ell \mid N$? (Answer: $1 = 0 + 1$; $v_7(42) = 1 \neq 2 = v_7(49)$; for $q = \ell$ the term $e_\ell(E, q)$ is not defined, and $v_7(L_7(A_f, 1)^{-1}) = v_7(\frac{6}{7}) = -1$ is what makes the identity $v_7(\deg \varphi_E) = v_7(B_E) + v_7(L_7(A_f, 1)^{-1})$ hold.)

**Exercise 19.72.** Let $E =$ `275b1`, of type $\mathrm{I}_0^*$ at 5. Show that the two ramified quadratic twists of $E$ at 5 (by $\chi_5$ and by $\chi_5$ times an unramified quadratic character) have $a_5 = \pm a'$, and that $L_5(A_f, 1)^{-1}$ depends only on $a'^2$. Find $a'$ (with PARI, `ellap` of the minimal model of `elltwist(E,5)` at 5) and compute $P_5$, $L_5(A_f, 1)^{-1}$ and $e_\ell(E, 5)$ for $\ell = 3, 7$. (Answer: the twist by $\chi_5$ is `11a3`, of conductor 11, with $a_5 = 1$; $P_5 = (1 - 5X)(1 + 9X + 25X^2)$, $\frac{28}{25}$, $e_3 = 0$, $e_7 = 1$; $\deg \varphi_E = 28$.)

**Exercise 19.73.** Let $g \in \mathrm{SL}_2(\overline{\mathbb{Q}}_\ell)$ have finite order, $g \neq \pm 1$. Prove that the trace-zero endomorphisms commuting with $g$ form the line spanned by $x = 2g - \mathrm{tr}\, g$, that $x^2 = (\mathrm{tr}\, g)^2 - 4 \neq 0$, and that every element normalising $\langle g \rangle$ acts on this line by $\pm 1$. Deduce the III row of Lemma 19.45 without diagonalising $\tau$, and show that for $\ell \equiv 3 \pmod 4$ there is no basis of $V$ over $\mathbb{Q}_\ell$ in which $\tau$ is diagonal. (Hint: $g$ has distinct eigenvalues; for the last part, the eigenvalues $\pm i$ are not in $\mathbb{Q}_\ell$.)

**Exercise 19.74.** Show that `1232l1` is the twist of `77a1` by $\chi_{-4}$ (compare $j$-invariants and conductors), that $a_2($`77a1`$) = 0$, and compute $P_2$ and $e_3(E, 2)$ from Proposition 19.56(b). For `336f1`, show that none of the twists by $\chi_{-4}, \chi_8, \chi_{-8}$ has good reduction at 2, and conclude from $P_2 = 1 + 2X$ which case of the proposition it is in. (Answer: $P_2 = (1 - 2X)(1 + 2X)^2$, $e_3 = v_3(\frac{9}{8}) = 2$; `336f1` is in case (c) with $c = -1$.)

**Exercise 19.75.** The twist of `37a1` by $\chi_{37}$ is `1369d1`, of type $\mathrm{I}_1^*$ at 37. Compute $e_\ell(E, 37)$ for every odd $\ell$, the value $B_E = \deg \varphi_E \cdot L_{37}(A_f, 1)$ from $\deg \varphi_E = 2736$, and check Corollary 19.37(iii) at $\ell = 3$ and $\ell = 19$. Compare with $e_\ell($`37a1`$, 37)$. (Answer: $1 - 37^{-2} = \frac{1368}{1369}$, $1368 = 2^3 \cdot 3^2 \cdot 19$, so $e_3 = 2$ and $e_{19} = 1$; $B_E = 2738 = 2 \cdot 37^2$; $v_3(2736) = 2 = 0 + 2$, $v_{19}(2736) = 1 = 0 + 1$; for `37a1`, $e_\ell = 0$.)

**Exercise 19.76.** For `980i1`, compute $P_q(q^{-2})$ at each bad prime and its $\ell$-adic valuation for $\ell = 3, 5$, and identify which of them are the values of Lemma lem:fpr at an additive and at a multiplicative prime. Where is the Hasse bound used in Proposition 19.52, and why is it not needed for this curve? (Answer: $\frac{3}{2}$, $\frac{24}{25}$, $\frac{48}{49}$; at $\ell = 3$ the valuations are $1, 1, 1$, at $\ell = 5$ they are $0$, $-2$ (excluded: $5 \mid N$, $p = \ell$), $0$; the Hasse bound is needed only for the cubic factor of case (b).)

**Exercise 19.77.** For `11a1` and $p = 7$ ($a_7 = -2$) compute $L_7^{\mathrm{nv}}(A_f, 1)^{-1}$ and the primes $\ell$ dividing its numerator. For which of them is it Ribet's condition $a_7 \equiv \pm(7+1) \pmod{\ell}$ that holds, and for which is it $7 \equiv 1 \pmod{\ell}$? Why does Ribet's level-raising theorem not apply at $\ell = 5$ for this curve? (Answer: $\frac{360}{343}$; $\ell = 2, 3, 5$; $\ell = 3$: both; $\ell = 5$: $-2 \equiv 8$; $E[5]$ is reducible for `11a1`.)

**Exercise 19.78.** For $\ell = 3$ list the residue classes of primes $q \geq 5$ modulo 12 for which each additive type gives $e_3(E, q) \geq 1$, and prove the statements of Remark 19.47 for $\ell \geq 5$. Show that at a prime of type $\mathrm{I}_0^*$ with $q \equiv 2 \pmod 3$, $e_3(E, q) \geq 1$ iff $3 \mid a'$. (Answer: II to IV$^*$ and $\mathrm{I}_n^*$: all classes; III, III$^*$: $q \equiv 1, 11$; $\mathrm{I}_0^*$: $q \equiv 1, 7$ always, otherwise iff $a' \equiv \pm(q+1) \equiv 0 \pmod 3$.)

**Exercise 19.79.** Show that the critical integers of $L(\mathrm{Sym}^2 f, s)$, with gamma factor $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{C}}(s)$ and functional equation $s \leftrightarrow 3-s$, are 1 and 2. Compute the rational number $L(\mathrm{ad}^0 f, 1)/(\pi^3 \langle f, f \rangle)$ for `11a1`, `20a1` and `121a1`. (Answer: $\frac{8}{11}$, $\frac{8}{30} = \frac{4}{15}$, $\frac{8 \cdot 11}{121 \cdot 12} = \frac{2}{33}$.)

**Exercise 19.80.** `121c1` is the twist of `121a1` by $\chi_{-11}$. Using Proposition 19.49(i), Corollary 19.37 and Example 19.62, show that the two curves have the same $\mathrm{vol}(\Lambda_E)$ (given $c_E = 1$ for both), and check it with PARI's `E.area`. Is $\mathrm{vol}(\Lambda_E)$ invariant under every quadratic twist? (Answer: yes for this pair, $\mathrm{vol}(\Lambda_E) = 2.8177020481\ldots$ for both; no in general: `11a1` and its twist `121d2` have $\mathrm{vol}(\Lambda_E) = 1.8515436234\ldots$ and $0.1683221475\ldots$, the true factors at 11 being equal but the naive ones not.)

**Exercise 19.81.** Run the paper's `rule` on `121a1` and `726a1`, with and without `iii`, and account for each term using Example 19.61. Then run it on `1323l1` at $\ell = 3$ and on `592c1` at $\ell = 5$, and say which bad prime produces the bound. (Answer: `[1, 1]`; `[2, 2]` twice; for `1323l1` the bound is 2, from $e_3(E, 2) = 2$, with $v_3(288) = 2$; for `592c1` the bound is 1, from $e_5(E, 2) = 1$, with $v_5(80) = 1$.)

## 19.14 Notes and references

*Dirichlet series, $\zeta$, Dirichlet L-functions.* Sections 19.2 and 19.3 follow the classical treatment of Davenport [39] (the chapters on Dirichlet's theorem, on the functional equations of $\zeta$ and $L(s, \chi)$, on the Gamma function and on the zero-free region) and Serre [172, Chapter VI]; for Dedekind zeta functions of abelian fields, Washington [201, Chapter 4] (we cite Theorem 4.3 from memory) and Neukirch [142, Chapter VII] (the sections on Dedekind zeta functions and Artin $L$-functions; section numbers unverified). The chapter numbers of Davenport and Washington given here are pointers we have not verified against the printed editions.

*L-functions of Galois representations.* The definition with the geometric Frobenius and the Weil-Deligne formalism are Deligne's [41] and Tate's [194], as presented in Chapter 7; the independence of $\ell$ for elliptic curves goes back to Serre and Tate [179], and for newforms to Carayol [21]. The local factor at $p = \ell$ through $\mathrm{D}_{\mathrm{cris}}$ is Fontaine's [67]; see Berger's survey [11]. The root numbers of elliptic curves at every prime are computed by Rohrlich [166]. Modularity is [207, 197, 16], and the functional equation of $L(f, s)$ is in Diamond and Shurman [52, Section 5.10].

*The symmetric square.* Shimura [182] proved the meromorphic continuation of $\sum a_{n^2} n^{-s}$ by an integral representation with a theta series of weight $\frac{1}{2}$; Gelbart and Jacquet [73] constructed the lift $\mathrm{Ad}(\pi)$ to $\mathrm{GL}_3$, from which the analytic properties of the true $L(\mathrm{ad}^0 f, s)$ follow. The local factors of the symmetric square at the primes of potentially good reduction were computed by Coates and Schmidt [29], and Watkins [204] describes how the bad factors are found in practice;

we have not checked their normalisations against Remark 19.31. Deligne's conjecture is [42]; Sturm [189] and Zagier [209] treat the critical values of the symmetric square. Flach's finiteness theorem is [66, Theorem 1]; its hypotheses and annihilator in Theorem 19.40 were checked against the original (Theorem 1, Propositions 1.1 and 2.2 there). The Bloch-Kato conjecture is [14], and the theorem of Diamond, Flach and Guo is [49] (Chapters 18 and 20).

*Rankin-Selberg.* The method is due to Rankin [155] and Selberg [170]; Shimura [183] treats the special values. The exact relation of Proposition 19.34, with no correction at the bad primes, and the resulting Theorem 19.36 at every level are elementary consequences of the local computation (the naive equality at every level is Chapter 13's); the paper states the latter in Remark rem:degree, citing [155, 183] and [49, Lemma 2.12]. The degree formula combines it with the formula of Zagier [209] and Watkins [204] (Chapter 14).

*The local factors at the bad primes.* Lemma lem:euler at $q \geq 5$ rests on the images of inertia of Serre and Tate [179] and Kraus [107] (Chapters 10 and 11); the classification at every prime (Proposition 19.56, the paper's Proposition prop:shapes) needs only the finiteness of the image of inertia, its integrality, Schur's lemma and the extension of a quadratic character of inertia to $G_{\mathbb{Q}_q}$. The data in Sections 19.10 to 19.12 were computed with PARI/GP 2.17.2 [148] through `cypari2` on all 64,687 curves of conductor below 10,000 in Cremona's tables [35, 36], using `lfunsympow`, `lfuneuler`, `elltwist`, `ellap`, `elllocalred`, `ellmoddegree`, `mffromell` and `mfpetersson`.

# Part VI

# Congruences, the modular degree and the paper

# Chapter 20

# Hecke algebras, congruence ideals and the theorem of Diamond, Flach and Guo

This chapter is where Section 4 of the paper is proved. It joins three threads that earlier chapters have prepared separately. The first is the commutative algebra of congruences: a surjection $\pi\colon \mathbb{T} \twoheadrightarrow \mathcal{O}$ from a finite free algebra has a congruence ideal $\eta(\pi)$ that measures how far $\mathbb{T}$ is from splitting off the factor $\mathcal{O}$, and a lemma of Darmon, Diamond and Taylor compares $\eta(\pi)$ with the Gram determinant of a pairing on a module (Chapter 3 proves every ingredient; the lemma is proved here). The second is the arithmetic of modular forms: the Hecke algebra $\mathbb{T}$ of level $N$, the congruence number $r_E$, the depleted forms $f^{\Sigma}$ and the twist map $\iota$ (Chapters 13 to 15). The third is the theorem of Diamond, Flach and Guo [49]: the length of an imprimitive adjoint Selmer group is the valuation of a naive congruence ideal, and that ideal changes with the level by the naive Euler factors of the adjoint $L$-function. With the Selmer bookkeeping of Chapter 18 and the Euler factors of Chapter 19, these three threads give the paper's Theorem thm:main, as an equality, and Theorems thm:twist and thm:quant.

The chapter assembles; it does not re-derive what other chapters prove. Its own full proofs are: Lemma lem:ddt417 (Lemma 20.21); the comparison of integral structures, the paper's Proposition prop:integral (Proposition 20.27); the assembly of Theorem thm:main (Theorem 20.60); Theorem thm:twist (Theorem 20.69); Theorem thm:quant with its equality clause (Theorem 20.72); and the equality $v_\ell(r_E) = v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ in Step 1 of the proof of Theorem thm:quant, which is the paper's Lemma lem:step1 (Lemma 20.74). The deep theorems (the Taylor-Wiles method, the theorems of Diamond, Flach and Guo, Hida's theorem, the formula of Kim and Ota) are stated precisely, with an outline of their proofs and the references; for part (c) of the proposition of Diamond, Flach and Guo we sketch the deduction from parts (a) and (b) along the lines of their published proof, given the Fontaine-Laffaille comparison isomorphism that is part of their integral structures, which we cite (Theorem 20.39).

## Core results at a glance

Throughout, $f$ is a newform of weight 2 on $\Gamma_0(M)$ (from Section 20.9 on, $f = f_E$ with $E$ optimal of conductor $N$), $\mathbb{T}$ the full Hecke algebra ($U_p$ included), $\ell$ an odd prime, $\pi\colon \mathbb{T}_{\mathcal{O}} \twoheadrightarrow \mathcal{O}$ a surjection with $\wp = \ker \pi$, $I = \mathrm{Ann}(\wp)$, $\eta(\pi) = \pi(I)$, and $\mathrm{Frob}_p^{-1}$ the geometric Frobenius.

**20.1 Conventions and warnings.** Notation C(a) to C(h); the paper's $I, J$ of Lemma lem:crit are our $\wp, I$; the Hecke ideal $\eta^{\mathrm{Hecke}}$ and the cohomological ideal $\eta^{\Sigma}_{f,\lambda}$ are different objects. *(convention)* Convention 20.1, Warning 20.2, Warning 20.3.

**20.2 Local components.** $\mathbb{T}_\ell = \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$; $\mathfrak{m}_{f,\ell} = (\ell, T_n - a_n(f))$; $e_{\mathfrak{m}} S_2 \otimes \overline{\mathbb{Q}}_\ell = \bigoplus_{\mathfrak{m}_\phi = \mathfrak{m}} V(\phi)$, and forms of $V_h$ in $\mathfrak{m}_{f,\ell}$ have $\bar\rho_h^{\mathrm{ss}} \simeq \bar\rho_f^{\mathrm{ss}}$; the pairings $S_2(\mathbb{Z}_\ell) \times \mathbb{T}_\ell \to \mathbb{Z}_\ell$ and $\hat\delta_!$ are perfect on each component, the latter orthogonally. *(proved here by citation of Chapters 3, 13 and 14)* Proposition 20.4.

**20.3 Level 539.** At $\ell = 3$ the newforms 539c, 539d share one maximal ideal and the depleted $(77\mathrm{c})^{\{7\}}$, $(11\mathrm{a})^{\{7\}}$ another; the congruence modules have orders 288, 504, 288, 504. *(computed)* Example 20.6.

**20.4 Congruence ideal, module, cotangent module** (definition): $\eta(\pi) = \pi(I)$, $\mathcal{C}(\pi) = \mathbb{T}/(\wp + I)$, $\mathrm{Cot}(\pi) = \wp/\wp^2$. Definition 20.7.

**20.5 Properties.** $\mathcal{C}(\pi) \simeq \mathcal{O}/\eta$; $\eta \neq 0 \iff \wp \cap I = 0 \iff$ the factor of $\mathbb{T}_K$ at $\pi$ is $K$; then $I \simeq \eta$, $\mathrm{Ann}(I) = \wp$, $\eta$ is the denominator of the idempotent; $\eta$ is computed in the local factor of $\pi$ (the other factors lie in $\wp$); $\mathrm{Cot}(\pi)$ is finite iff the factor is $K$. *(proved here, given Chapter 3)* Proposition 20.8.

**20.6 Toy algebras.** $\mathbb{T}_r$: $\operatorname{length}\mathrm{Cot} = v(\eta) = r$; $\mathbb{T}_3^\flat$: $2 > 1$; $\mathcal{O}[x]/(x^2)$: $\eta = 0$. *(proved here)* Example 20.9.

**20.7 Hecke congruence ideal** (definition): $\eta_f^{\mathrm{Hecke}} = \eta(\phi_f)$, and $\eta^{\mathrm{Hecke}}(g)$ for an eigenform $g$. It is nonzero whenever the generalised eigenspace is a line (newforms, depleted forms). *(proved here, given Chapter 13)* Definition 20.10, Proposition 20.11.

**20.8 Lemma lem:crit as congruence modules.** $\mathbb{T}/(\wp + I) \simeq \mathbb{Z}/r_E$, $\eta_f^{\mathrm{Hecke}} = r_E\mathbb{Z}$, the congruence moduli are the divisors of $r_E$, also over $\mathbb{Z}_\ell$; and $S_2(\mathbb{Z})/(\mathbb{Z}f \oplus L) \simeq \mathrm{Hom}(\mathcal{C}(\phi_f), \mathbb{Q}/\mathbb{Z})$. *(stated; proved in Chapter 15; the duality proved here)* Theorem 20.12.

**20.9 Line eigenspaces.** For $\pi\colon \mathbb{T}_N \to \mathbb{Z}$ with generalised eigenspace $\mathbb{C}g$ and $\mathbb{T}$-stable complement $g^c$: $\eta(\pi) = r(\pi)\mathbb{Z}$ with $r(\pi) = [S_2(\mathbb{Z}) : \mathbb{Z}g \oplus (g^c \cap S_2(\mathbb{Z}))]$ the largest congruence modulus; the Petersson complement does not work in general. *(proved in Chapter 13; recalled)* Proposition 20.14, Warning 20.15.

**20.10 Hida.** Outside an explicit finite set, $\lambda$ divides $\langle f, f\rangle/\Omega_f^+\Omega_f^-$ iff $\lambda$ is a congruence prime. *(stated; reference)* Theorem 20.16, Remark 20.17.

**20.11 Three ideals.** For optimal $E$, $\ell \nmid 2N$, $E[\ell]$ irreducible: $\eta_f^{\mathrm{Hecke}}\mathbb{Z}_\ell = r_E\mathbb{Z}_\ell$, $v_\ell(r_E) = v_\ell(\deg\varphi_E) = v_\ell(\eta_{f,\ell}^{\emptyset})$; at `99a1`, $\ell = 3$: $1 \neq 0$. *(proved here, given the cited theorems)* Remark 20.18.

**20.12 Wiles's numerical criterion.** For local $\mathbb{T}$ finite free over $\mathcal{O}$ and $\eta \neq 0$: $\operatorname{length}\mathrm{Cot}(\pi) \geq v(\eta)$, with equality iff $\mathbb{T}$ is a complete intersection; $R \twoheadrightarrow \mathbb{T}$ with $\operatorname{length}\mathrm{Cot}(\pi_R) \leq v(\eta)$ is an isomorphism. *(proof sketched here; Wiles, Lenstra)* Theorem 20.19.

**20.13 Lemma lem:ddt417.** $\mathbb{T}$ finite free over a discrete valuation ring $\mathcal{O}$, acting self-adjointly on $L$ free with a perfect pairing; $L[\wp]$ of rank $d$ with Gram determinant $\Delta \neq 0$: then $d\,v(\eta) \geq v(\Delta)$, with equality if $L$ is free over $\mathbb{T}$. *(proved here, given Chapter 3)* Lemma 20.21;

Commentary on the paper's statement Remark 20.22; hypotheses Remark 20.23; test cases Example 20.24.

**20.14 Depletion** (definition, recalled): $\delta_p$, $M^\Sigma = M\prod_{p\in\Sigma} p^{\delta_p}$, $f^\Sigma$, the naive factor $L_p^{\mathrm{nv}}(A_f,s)$. Definition 20.25.

**20.15 Integral structure** (definition, after Diamond, Flach and Guo) and its comparison: pull-back identifies $H^1(X_0(M),\mathbb{Z})$ with the Betti lattice, Hecke-equivariantly ($U_p$ included), so $M(M,1)_{!,\lambda} = H^1(X_0(M),\mathbb{Z}_\ell)$ for every prime $\ell$, with no localisation; equivariant cohomology with compact support, Hochschild-Serre and equivariant Leray. *(proved here)* Definition 20.26, Proposition 20.27.

**20.16 The pairing** $\hat\delta_!$ (definition) and its properties: on pulled-back classes it is $\frac{u}{2}\langle x, w_M^* y\rangle$, $u$ an $\ell$-adic unit; perfect, alternating, $\mathbb{T}$ self-adjoint, perfect on components. *(proved here, given Chapter 14; the normalisation of Diamond, Flach and Guo as the paper describes it)* Definition 20.29, Proposition 20.30.

**20.17** $M_{f^\Sigma,\lambda}$ **and** $\eta^\Sigma_{f,\lambda}$ (definition): the kernel lattice of $\ker\theta_{f^\Sigma}$ and the ideal of values of $\hat\delta_!$ on it. For $\bar\rho_f$ irreducible: saturated of rank 2, in the $\mathfrak{m}$-component, Gram determinant $(\eta^\Sigma_{f,\lambda})^2$; for $\Sigma=\emptyset$ and optimal $E$, $M_{f,\lambda} = \varphi_E^* H^1(E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$. *(proved here, given Chapter 14 and Proposition 20.27)* Definition 20.31, Proposition 20.32.

**20.18 Lemma lem:hecke.** For optimal $E$, $\ell$ odd, $\ell\nmid N$, $E[\ell]$ irreducible: $v_\ell(\deg\varphi_E) = v_\ell(r_E) = v_\lambda(\eta^\emptyset_{f,\lambda})$, with no freeness hypothesis. *(proved here, given [3, Theorem 2.2], Chapter 14 and Proposition 20.27; the definitions of [49] it uses are checked against the published text)* Theorem 20.34; values Example 20.37.

**20.19 Remark rem:free.** Multiplicity one gives $H^1(X_0(N),\mathbb{Z}_\ell)_{\mathfrak{m}} \simeq \mathbb{T}^2_{\mathfrak{m}}$; not needed for Theorems thm:main and thm:quant. *(stated; proved in Chapter 14 given multiplicity one)* Remark 20.36.

**20.20 The map** $\gamma$ (definition): $\gamma = \sum_m \epsilon_m\phi_m$, $\gamma(f) = f^\Sigma$. Definition 20.38.

**20.21 Theorem thm:dfg14** (Diamond, Flach and Guo, Proposition 1.4). For $\ell\nmid 2M$, $\ell\notin\Sigma$: $\gamma^t\gamma = u\prod_p L_p^{\mathrm{nv}}(A_f,1)^{-1}$ on $M_f$; if $\bar\rho_f$ is irreducible, $\eta^\Sigma_{f,\lambda} = \eta^\emptyset_{f,\lambda}\prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f,1)^{-1}$; false for $\ell\in\Sigma$ (`11a1`, $\ell=3$). *(proof sketched here, (c) given (a), (b) and the cited integral comparison isomorphism, as in [49, Proposition 1.4]; reference)* Theorem 20.39, Remark 20.40.

**20.22 The three cases.** $L_p^{\mathrm{nv}}(A_f,1)^{-1}$ is 1 (additive), $\frac{p-1}{p}\cdot\frac{(p+1)^2-a_p^2}{p^2}$ (good; Wiles's factor), $1-p^{-2}$ (multiplicative). *(proved here)* Proposition 20.41; interpretation Remark 20.42; Ihara and the Eisenstein case Remark 20.43; refined structures Remark 20.44.

**20.23 The exceptional set** (definition): $\lambda \mid Nk!$ or $\bar\rho|_{G_{\mathbb{Q}(\sqrt{\ell^*})}}$ not absolutely irreducible. For $f_E$ and $\ell\nmid 2N$: $\ell\notin S_f$ iff $E[\ell]$ irreducible and not ($\ell=3$, image 3Ns), iff moreover $h_\ell(E)=0$. *(definition checked against [49]; the proposition proved in Chapter 11)* Definition 20.45, Proposition 20.46.

**20.24** $H^1_\Sigma(\mathbb{Q},W)$ (definition): unramified outside $\Sigma\cup\{\ell\}$, Bloch-Kato at $\ell$, free on $\Sigma$; equal to the Bloch-Kato conditions outside $\Sigma$ when $\Sigma$ contains the ramified primes. Definition 20.47.

**20.25 Minimal ramification** (definition); the primes dividing $N$ form an admissible $\Sigma$. Definition 20.48.

**20.26 Theorem thm:dfg37** (Diamond, Flach and Guo, Theorem 3.7): for $\lambda \notin S_f$ and $\Sigma \not\ni \ell$ with $M_{f,\lambda}$ minimally ramified outside $\Sigma$, length $H^1_\Sigma(\mathbb{Q}, A_{f,\lambda} \otimes K_\lambda/\mathcal{O}_\lambda) = v_\lambda(\eta^\Sigma_{f,\lambda})$. *(stated; reference)* Theorem 20.49.

**20.27 Theorem 0.2 of Diamond, Flach and Guo** (quoted): the same for $\Sigma \supseteq \{p \mid N\}$, $\lambda \nmid \Sigma$. *(stated; checked against [49])* Theorem 20.50.

**20.28 Theorem thm:dfg27.** For $\ell \notin S_f$: $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0 = H^0(\mathbb{Q}, A_f) = H^1_f(\mathbb{Q}, A_f)$. *(stated; reference)* Theorem 20.51.

**20.29 Theorem 0.1 of Diamond, Flach and Guo** (quoted): the $\lambda$-part of the Bloch-Kato conjecture for $A_f$ and $A_f(1)$, $\lambda \notin S_f$. *(stated; checked against [49])* Theorem 20.52.

**20.30 $\ell = 3$.** As published, the theorems cover $\ell = 3$ in weight 2 ($S_f$ excludes only $\lambda \mid 2N$ and the induced case); the proof treats $\ell = 3$ explicitly (Fontaine-Laffaille theory for $M_{f,\lambda}$, Section 1.8, Proposition 2.16). *(checked against [49])* Remark 20.53.

**20.31 Deformation rings** (definition) and Mazur's theorem: the ring $R_\Sigma$ exists, and the Pontryagin dual of its cotangent module at a point $\rho$ is the Selmer group $H^1_\Sigma(\mathbb{Q}, \mathrm{ad}^0 M_\rho \otimes K/\mathcal{O})$. *(proof sketched here; [38])* Definition 20.54, Theorem 20.55.

**20.32 The Hecke side** (definition): $\mathbb{T}_\Sigma$ and $R_\Sigma \twoheadrightarrow \mathbb{T}_\Sigma$. Definition 20.56.

**20.33 Taylor-Wiles.** For $\rho_0$ absolutely irreducible over $\mathbb{Q}(\sqrt{\ell^*})$, twist-minimal, Serre weight in $[2, \ell-1]$: $R_\Sigma \simeq \mathbb{T}_\Sigma$ complete intersections, cohomology free, and length $H^1_\Sigma = v(\eta^\Sigma)$. *(proof sketched here; references)* Theorem 20.57; hypotheses Remark 20.58; why length equals valuation Remark 20.59.

**20.34 Theorem thm:main.** For optimal $E$, $\ell$ odd, $\ell \nmid N$, $\ell \notin S_f$:

$$v_\ell(\deg \varphi_E) = \text{length}\, H^1_f(\mathbb{Q}, W) + \sum_{q \mid N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) + \sum_{q \mid N} e_\ell(E, q),$$

$$v_\ell(\deg \varphi_E) \geq \sum_{q \mid N} e_\ell(E, q) + \sum_{q \mid N} t_\ell(E, q),$$

an equality with no multiplicity-one hypothesis. *(proved here, given the theorems of Diamond, Flach and Guo and Chapters 14, 18 and 19)* Theorem 20.60; hypotheses Remark 20.63.

**20.35 Corollary cor:explicit.** $v_\ell(\deg \varphi_E) = \text{length}\, H^1_f(\mathbb{Q}, W) + \sum_q e_\ell(E, q) + \sum_{v_q(j)<0}(t_\ell + \min(t_\ell, v_\ell(q-1), v_\ell(c(u_q))))$. *(proved here, given Theorem 20.60 and Chapter 18)* Corollary 20.61; examples (`11a1`, `121a1`, `539c1` at 3, all with $H^1_f = 0$) Example 20.64.

**20.36 Remark rem:ko(1).** The Tamagawa excess $\min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$ raises the bound by exactly that amount when $\ell \notin S_f$. *(proved here, given Chapter 18)* Remark 20.65.

**20.37 Remark rem:ko(2).** Kim and Ota's formula gives $v_\ell(r_E) \geq \sum_{q \mid N^-} t_\ell(E, q)$, contained in (eq:thm), and with Corollary 20.61 a Selmer formula for the $N^-$-new congruence ideal. *(proved here, given [99])* Proposition 20.66, Remark 20.67.

**20.38 The twist map.** $\iota_S$ preserves integrality, twists eigensystems by $\chi_S$, maps $g^\perp$ into $f^\perp$ when $\iota(g) = f$; levels $N_S = N/D$, $N/D^2$, $N$ for types $\mathrm{I}_n^*$, $\mathrm{I}_0^*$, potentially good. *(proved in Chapters 13 and 15; recalled)* Proposition 20.68.

**20.39 Theorem thm:twist.** If $q^2 \mid N$ $(q \in S)$ and $\mathrm{lcm}(N_S, D^2) \mid N$, then $r_{E_S} \mid r_E$; if also $\mathrm{lcm}(N, D^2) \mid N_S$, then $r_{E_S} = r_E$. *(proved here, given Chapter 13)* Theorem 20.69; cases and data (25,477 and 14,880 pairs below conductor 10,000) Example 20.70.

**20.40 Theorem thm:quant.** For optimal $E$, $\ell$ odd, $\ell \nmid N$, $E[\ell]$ irreducible, $S$ the odd primes of type $\mathrm{I}_0^*$, $\mathrm{I}_n^*$ ($q = 3$ allowed): $v_\ell(\deg \varphi_E) \geq v_\ell(r_{E_S}) + \sum_{q \in S} e_\ell(E, q)$, with equality if $\ell \notin S_f$. *(proved here, given Theorem 20.39, Theorem 20.60 and the cited chapters)* Theorem 20.72.

**20.41 The set-up.** $g^S$ lives at level $N$ with generalised eigenspace $\mathbb{C}g^S$, and $\iota(g^S) = f$. *(proved here, given Chapters 10 and 13)* Lemma 20.73.

**20.42 Step 1 is an equality.** For every $E$ of conductor $N$, with $S$ and $g^S$ as in the set-up: $r_E \mid r_\pi$ and $v_\ell(r_E) = v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ for every prime $\ell \notin S$, $\ell = 2$ included, with no other hypothesis. *(proved here, given Chapter 13)* Lemma 20.74 (the paper's Lemma lem:step1); Commentary Remark 20.75.

**20.43 Steps 2 to 4.** $v_\ell(\eta^{\mathrm{Hecke}}(g^S)) \geq v_\ell(\eta^S_{g,\ell}) = v_\ell(\eta^\emptyset_{g,\ell}) + \sum_{q \in S} e_\ell(E, q)$ and $v_\ell(\eta^\emptyset_{g,\ell}) = v_\ell(r_{E_S})$. *(proved here, given Theorem 20.39)* Lemma 20.76.

**20.44 All equalities.** For $\ell \notin S_f$: $v_\ell(r_E) = v_\ell(\eta^{\mathrm{Hecke}}(g^S)) = v_\ell(\eta^S_{g,\ell}) = v_\ell(r_{E_S}) + \sum_{q \in S} e_\ell(E, q)$. *(proved here)* Corollary 20.77; remarks Remark 20.78, Remark 20.79.

**20.45 Beyond the hypothesis.** For image 3Ns at 3, (eq:identity) with $-h_3(E)$ would follow from $H^1_f(\mathbb{Q}, V^D) = 0$ and $v_3(\eta^\Sigma) = \mathrm{length}\, H^1_\Sigma$; what Open questions 1 and 2 need. *(conditional)* Remark 20.80, Remark 20.81, Remark 20.82.

**20.46 The citations of Diamond, Flach and Guo,** with their status. Remark 20.83.

## Where the paper uses this chapter

- **Section 1** ("The congruence number is the congruence ideal of $f$ in the full Hecke algebra ... from Hida ... to Diamond, Flach and Guo"; the hypothesis of Theorem thm:main and [49, Theorem 0.1]): Theorem 20.12, Theorem 20.16, Theorem 20.52, Definition 20.45.
- **Lemma lem:crit**, restated as a statement about congruence modules: Theorem 20.12, Proposition 20.14.
- **Section 4.1** ($M^\Sigma$, $f^\Sigma$, the integral structure, $M_{f^\Sigma,\lambda}$, $\hat{\delta}_!$, $\eta^\Sigma_{f,\lambda}$): Definition 20.25 to Proposition 20.32.
- **Theorem thm:dfg14 and the paragraph after it**: Theorem 20.39, Proposition 20.41.
- **Proposition prop:pullback and Lemma lem:hecke**: Proposition 20.32(iii), Theorem 20.34.
- **Lemma lem:ddt417**: Lemma 20.21, Remark 20.22. **Remark rem:free**: Remark 20.36.

- **Section 4.2** ($S_f$, Proposition prop:Sf, Theorems thm:dfg37 and thm:dfg27, the minimal-ramification paragraph): Definition 20.45 to Remark 20.53.
- **Lemmas lem:dfg21, lem:fpr, lem:h0**: used in Step 4 of Theorem 20.60, proved in Chapter 18.
- **Section 4.3** (the proof of Theorem thm:main, Corollary cor:explicit): Theorem 20.60, Corollary 20.61.
- **Remark rem:ko**: Remark 20.65, Proposition 20.66.
- **Theorem thm:twist and the paragraph after it**: Theorem 20.69, Example 20.70.
- **Theorem thm:quant, its proof and Remark rem:afterquant**: Theorem 20.72 to Remark 20.79.
- **Lemma lem:step1**: Lemma 20.74, Remark 20.75.
- **Remark rem:ns, Open questions 1 and 2**: Remark 20.80, Remark 20.81, Remark 20.82.
- **Every citation of [49]**: Remark 20.83.

## 20.1 Overview and conventions

This section fixes notation and gives a map of Section 4 of the paper. Every sentence of that section is either a citation of an imported theorem, a result proved in an earlier chapter, or a step proved here; the table below says which. The two warnings concern the two places where the paper's notation can mislead: the letters $I$ and $J$ of Lemma lem:crit, and the two different congruence ideals of Section 4.

**Convention 20.1** (standing notation)**.** (a) *Newforms and coefficients.* $f$ is a newform of weight 2 on $\Gamma_0(M)$ with trivial character. In Sections 20.2 to 20.8 its coefficients may lie in a number field $K$ with a prime $\lambda \mid \ell$; then $\mathcal{O} = \mathcal{O}_{K,\lambda}$, with uniformiser $\varpi$ and valuation $v$, $v(\varpi) = 1$. From Section 20.9 on, $f = f_E$ has rational coefficients, $K = \mathbb{Q}$, $\lambda = \ell$ and $\mathcal{O} = \mathbb{Z}_\ell$ (the paper's "$\lambda = \ell$"). The prime $\ell$ is odd and $\ell \nmid M$ unless we say otherwise.

(b) *Hecke algebras.* $\mathbb{T} = \mathbb{T}_M = \mathbb{Z}[T_n : n \geq 1] \subset \operatorname{End} S_2(\Gamma_0(M))$ is the *full* Hecke algebra, the operators $U_p = T_p$ for $p \mid M$ included (Chapter 13). $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$, $\mathbb{T}_\mathcal{O} = \mathbb{T} \otimes \mathcal{O}$; for a maximal ideal $\mathfrak{m}$ of $\mathbb{T}_\ell$ (or $\mathbb{T}_\mathcal{O}$), $\mathbb{T}_\mathfrak{m}$ is the localisation, a direct factor cut out by an idempotent $e_\mathfrak{m}$. $\mathbb{T}$ acts on $H^1(X_0(M), \mathbb{Z}_\ell)$ by pull-back $t^*$ (Chapter 14).

(c) *Congruence data.* For a surjective homomorphism of $\mathcal{O}$-algebras $\pi\colon \mathbb{T}_\mathcal{O} \twoheadrightarrow \mathcal{O}$ we write $\wp = \ker \pi$, $I = \operatorname{Ann}(\wp)$, $\eta(\pi) = \pi(I)$ (the *congruence ideal*), $\mathcal{C}(\pi) = \mathbb{T}_\mathcal{O}/(\wp + I)$ (the *congruence module*) and $\operatorname{Cot}(\pi) = \wp/\wp^2$ (the *cotangent module*). The letters $\mathcal{C}$ and Cot are this chapter's; the literature writes $\Omega$ and $\Phi$, which the book reserves for other objects (Appendix B).

(d) *Two congruence ideals.* $\eta^{\mathrm{Hecke}}(\pi)$ is $\eta(\pi)$ of (c), an invariant of the algebra $\mathbb{T}$; $\eta^{\Sigma}_{f,\lambda}$ is the naive $\Sigma$-finite congruence ideal of Diamond, Flach and Guo (Definition 20.31), an invariant of a lattice with a pairing. For a newform at its level, $\eta^{\mathrm{Hecke}}_f = \eta^{\mathrm{Hecke}}(\phi_f)$.

(e) *Lengths.* All counts of finite $\mathcal{O}$-modules are lengths $\mathrm{length}_{\mathcal{O}}$; for $\mathcal{O} = \mathbb{Z}_\ell$, $\#X = \ell^{\mathrm{length}\, X}$ (Appendix B, B-C10).

(f) *Frobenius and duals.* $\mathrm{Frob}_p$ is an arithmetic Frobenius and $\mathrm{Frob}_p^{-1}$ the geometric one; $L$-factors use the geometric Frobenius, $L_p(V,s) = \det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid V^{I_p})^{-1}$ (B-C2). $M^\vee = \mathrm{Hom}(M, \mathcal{O})$ is the linear dual, $X^*$ the Pontryagin dual, $V^D = \mathrm{Hom}(V, \mathbb{Q}_\ell(1))$, and $t^\dagger$ the adjoint of an operator. With $T = T_\ell E$: $A = \mathrm{ad}^0 T$, $A_f = \mathrm{ad}^0 V_\ell E$, $W = A \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $T^D \simeq A(1)$, $V^D$, $W^D$ as in Chapter 18.

(g) *Twists.* For a finite set $S$ of odd primes, $D = \prod_{q \in S} q$, $\chi_S = \prod_{q \in S} \chi_{q^*}$, $\iota = \iota_S \colon h \mapsto h \otimes \chi_S = \sum \chi_S(n) a_n(h) q^n$ (the naive twist, Chapter 13); $E_S = E \otimes \chi_S$ of conductor $N_S$, $g = f_{E_S}$, and $E^{(q)} = E \otimes \chi_{q^*}$. In this chapter $D$ has no other meaning.

(h) *Depletion.* $\delta_p = 2, 1, 0$ according as $p \nmid M$, $p \,\|\, M$, $p^2 \mid M$; $M^\Sigma = M \prod_{p \in \Sigma} p^{\delta_p}$; $f^\Sigma = \sum_{(n,\Sigma)=1} a_n(f) q^n$.

*Warning* 20.2 (the letters of Lemma lem:crit). In the paper's Lemma lem:crit, $I = \ker \phi_f$ and $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$. In Lemma lem:ddt417 and in this chapter, $\wp = \ker \pi$ and $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$. So for $\pi = \phi_f$ the paper's $I$ is our $\wp$, and the paper's $J$ is our $I$ (that $J = \mathrm{Ann}_{\mathbb{T}}(\ker \phi_f)$ is part of Lemma lem:crit, Theorem 20.12). We use the paper's letters only when we quote Lemma lem:crit.

*Warning* 20.3 (two congruence ideals). The Hecke congruence ideal $\eta^{\mathrm{Hecke}}(\pi)$ and the cohomological ideal $\eta^{\Sigma}_{f,\lambda}$ of Diamond, Flach and Guo are different objects. The first is defined from the ring $\mathbb{T}$ and the homomorphism $\pi$ alone; the second from the pairing $\hat{\delta}_!$ on a rank-two lattice in cohomology. Lemma lem:ddt417 compares them ($v(\eta^{\mathrm{Hecke}}) \geq v(\eta^{\mathrm{coh}})$), with equality when the cohomology is free over the Hecke algebra; for the newform of an optimal curve at $\ell \nmid 2N$ they agree without any freeness hypothesis (Remark 20.18). The paper's Section 4 uses both, and Step 2 of the proof of Theorem thm:quant is exactly the comparison.

The paper's Section 4, sentence by sentence, against this chapter and the chapters that supply the inputs:

| paper (Section 4) | here | inputs proved in |
|---|---|---|
| 4.1: $M^\Sigma$, $f^\Sigma$, the integral structure, $M_{f^\Sigma,\lambda}$, $\hat{\delta}_!$, $\eta^\Sigma_{f,\lambda}$ | Definition 20.25 to Definition 20.31 | Chapters 3, 13 and 14 |
| Theorem thm:dfg14 and the paragraph after it | Theorem 20.39, Proposition 20.41 | Chapters 15 and 19 |
| Proposition prop:pullback, Lemma lem:hecke | Theorem 20.34 | Chapter 14 |
| Lemma lem:ddt417 | Lemma 20.21 | Chapter 3 |
| Remark rem:free | Remark 20.36 | Chapters 3 and 14 |
| 4.2: $S_f$, Proposition prop:Sf | Definition 20.45 | Chapters 11 and 18 |
| Theorems thm:dfg37, thm:dfg27 | Theorem 20.49, Theorem 20.51 | imported [49] |
| Lemmas lem:dfg21, lem:fpr, lem:h0 | Step 4 of Theorem 20.60 | Chapters 11, 17 and 18 |
| 4.3: proof of Theorem thm:main, Corollary cor:explicit | Theorem 20.60 | Chapters 14, 18 and 19 |
| Remark rem:ko | Remark 20.65, Proposition 20.66 | Chapters 15 and 18 |
| 4.4: Theorem thm:twist | Theorem 20.69 | Chapters 13 and 15 |
| Theorem thm:quant and its proof | Theorem 20.72 | Chapters 10, 13 and 19 |
| Remark rem:afterquant | Lemma 20.74, Remark 20.79 | Chapter 13 |

It may help to see the whole proof of Theorem thm:main before its parts. With $\Sigma$ the set of primes dividing $N$, it is a chain of four equalities of $\ell$-adic valuations,

$$v_\ell(\deg \varphi_E) \overset{(1)}{=} v_\ell(\eta^{\emptyset}_{f,\ell}) \overset{(2)}{=} v_\ell(\eta^{\Sigma}_{f,\ell}) - \sum_{q\in\Sigma} v_\ell(L_q^{\mathrm{nv}}(A_f,1)^{-1})$$
$$\overset{(3)}{=} \operatorname{length} H^1_\Sigma(\mathbb{Q},W) - \sum_{q\in\Sigma} v_\ell(L_q^{\mathrm{nv}}(A_f,1)^{-1}),$$

followed by a purely Galois-cohomological step (4) that rewrites the length of $H^1_\Sigma(\mathbb{Q},W)$ as that of $H^1_f(\mathbb{Q},W)$ plus local terms. Equality (1) says that the modular degree is a congruence ideal (Section 20.5); (2) says how that ideal grows when the level is raised by depletion (Section 20.6); (3) is the theorem of Diamond, Flach and Guo that a congruence ideal measures a Selmer group (Sections 20.7 and 20.8); and (4) is Chapter 18. The sections below follow this order, after two sections of algebra that make the words "congruence ideal" precise. Theorem thm:quant (Section 20.12) runs a similar chain at a different pair of levels.

For a reader with a physics background, two images may help, one sentence each. The congruence module measures how far a lattice fails to split as the direct sum of its intersections with two complementary eigenspaces, as the unit cell of a crystal need not be a product of cells of two sublattices. The numerical criterion of Wiles (Theorem 20.19) is an index theorem: the size of a tangent space bounds a defect, with equality exactly in the most rigid (complete intersection) case.

## 20.2 Hecke algebras over $\mathbb{Z}_\ell$ and their local components

This section recalls, with references, the structure of the Hecke algebra over $\mathbb{Z}_\ell$ that every later section uses: the splitting into local components, the maximal ideal attached to a newform, and

which forms live in a given component. The paper uses these facts in the paragraph after Lemma lem:hecke ("Let $\mathfrak{m}$ be the maximal ideal of $\mathbb{T}\otimes\mathbb{Z}_\ell$ containing $\ell$ and $\ker\phi_f$"), in Remark rem:free and in Steps 1 and 2 of the proof of Theorem thm:quant. Nothing here is new; the point is to have the statements in one place, in the form in which they are used.

**Proposition 20.4** (local components, recollection)**.** *Let $M\geq 1$, $\mathbb{T}=\mathbb{T}_M$, $d=\dim S_2(\Gamma_0(M))$ and $\ell$ a prime.*

*(i)* *$\mathbb{T}$ is a free $\mathbb{Z}$-module of rank $d$; so $\mathbb{T}_\ell$ is a finite free $\mathbb{Z}_\ell$-algebra, it has finitely many maximal ideals, all containing $\ell$, and $\mathbb{T}_\ell=\prod_{\mathfrak{m}}\mathbb{T}_{\mathfrak{m}}$ with $\mathbb{T}_{\mathfrak{m}}=e_{\mathfrak{m}}\mathbb{T}_\ell$ local and finite free over $\mathbb{Z}_\ell$.*

*(ii)* *Let $\phi\colon\mathbb{T}\to\overline{\mathbb{Q}}_\ell$ be a ring homomorphism (a system of eigenvalues); its values lie in the ring of integers $\mathcal{O}$ of a finite extension of $\mathbb{Q}_\ell$, and $\mathfrak{m}_\phi=\ker(\mathbb{T}_\ell\to\mathcal{O}\to\mathcal{O}/\varpi)$ is a maximal ideal. If $\phi=\phi_f$ for a newform $f$ with rational coefficients, $\mathfrak{m}_{f,\ell}=\mathfrak{m}_{\phi_f}$ is generated by $\ell$ and the elements $T_n-a_n(f)$.*

*(iii)* *$e_{\mathfrak{m}}(S_2(\Gamma_0(M);\mathbb{Z}_\ell)\otimes\overline{\mathbb{Q}}_\ell)$ is the sum of the generalised eigenspaces $V(\phi)$ of the systems $\phi$ with $\mathfrak{m}_\phi=\mathfrak{m}$, that is, with $\phi\equiv\phi'$ modulo the maximal ideal of $\overline{\mathbb{Z}}_\ell$ for one (hence every) $\phi'$ with $\mathfrak{m}_{\phi'}=\mathfrak{m}$. If $V(\phi)$ lies in the old space $V_h$ of a newform $h$ and $\mathfrak{m}_\phi=\mathfrak{m}_{f,\ell}$, then $a_p(h)\equiv a_p(f)$ for all $p\nmid M\ell$, so $\bar\rho_h^{\mathrm{ss}}\simeq\bar\rho_f^{\mathrm{ss}}$.*

*(iv)* *The pairing $S_2(\Gamma_0(M);\mathbb{Z}_\ell)\times\mathbb{T}_\ell\to\mathbb{Z}_\ell$, $(h,t)\mapsto a_1(th)$, is perfect and restricts to a perfect pairing $e_{\mathfrak{m}}S_2(\Gamma_0(M);\mathbb{Z}_\ell)\times\mathbb{T}_{\mathfrak{m}}\to\mathbb{Z}_\ell$. The pairing $\hat\delta_!(x,y)=\langle x,w_M^*y\rangle$ on $H^1(X_0(M),\mathbb{Z}_\ell)$ is perfect and alternating, every $t\in\mathbb{T}$ is self-adjoint for it, and $H^1(X_0(M),\mathbb{Z}_\ell)=\bigoplus_{\mathfrak{m}}e_{\mathfrak{m}}H^1(X_0(M),\mathbb{Z}_\ell)$ is an orthogonal decomposition on whose summands $\hat\delta_!$ is perfect.*

*Proof.* (i) The freeness of $\mathbb{T}$ is proved in Chapter 13 (the perfect pairing with the integral forms); the splitting of a finite free algebra over the complete ring $\mathbb{Z}_\ell$ is the idempotent-lifting theorem of Chapter 3, which also shows that the maximal ideals contain $\ell$. (ii) The eigenvalues of $T_n$ are algebraic integers (Chapter 13), so $\phi(\mathbb{T})$ is a finitely generated $\mathbb{Z}$-module of integral elements and $\phi(\mathbb{T}_\ell)$ an order in a finite extension; its image in the residue field is a field, so $\mathfrak{m}_\phi$ is maximal. (In detail: $\phi(\mathbb{T}_\ell)$ is a $\mathbb{Z}_\ell$-submodule of $\mathcal{O}$ spanned by the finitely many values $\phi(t_i)$ on a $\mathbb{Z}$-basis $t_i$ of $\mathbb{T}$; the composite $\mathbb{T}_\ell\to\mathcal{O}/\varpi$ has as image a subring of the finite field $\mathcal{O}/\varpi$, and a subring of a finite field is a finite integral domain, hence a field. A ring homomorphism onto a field has maximal kernel.) If $\phi=\phi_f$ is $\mathbb{Z}$-valued, $\mathbb{T}_\ell/(\ell,T_n-a_n(f):n\geq1)$ is a quotient of $\mathbb{Z}/\ell$ (each $T_n$ becomes a scalar), and it is nonzero because it maps onto $\mathbb{F}_\ell$ through $\phi_f$; so the ideal generated is $\mathfrak{m}_{f,\ell}$. (iii) The first sentence is the theorem on generalised eigenspaces over a complete discrete valuation ring of Chapter 3, applied to $L=S_2(\Gamma_0(M);\mathbb{Z}_\ell)$, which is a faithful $\mathbb{T}_\ell$-module. For the second, $\phi(T_p)=a_p(h)$ for $p\nmid M$ on $V_h$ (multiplicity one, Chapter 13), so $\mathfrak{m}_\phi=\mathfrak{m}_{f,\ell}$ gives $a_p(h)\equiv a_p(f)$ modulo the maximal ideal for $p\nmid M\ell$, and Brauer-Nesbitt with Chebotarev gives the isomorphism of semisimplifications (Chapter 15). (iv) The first statement is Chapter 13 (the pairing is perfect over $\mathbb{Z}$ and stays perfect after $\otimes\mathbb{Z}_\ell$), and its restriction to components is Chapter 13 again (the local components proposition). The second is Chapter 14: $t^\dagger=w_M^*t^*w_M^*$ for the cup product, so $\hat\delta_!$ makes $\mathbb{T}$ self-adjoint, and an idempotent of a self-adjoint commutative algebra gives an orthogonal decomposition on whose summands a perfect pairing stays perfect (Chapter 3). □

*In the paper* 20.5. The paragraph after the paper's Lemma lem:hecke says that "the other local factors of $\mathbb{T}\otimes\mathbb{Z}_\ell$ lie in the kernel $\wp$, $\phi_f$ factors through $\mathbb{T}_{\mathfrak{m}}$, and $\mathrm{Ann}(\wp)$ decomposes along the

idempotents; and the idempotent $e_{\mathfrak{m}}$ is self-adjoint for $\hat{\delta}_!$, so $H^1(X_0(N),\mathbb{Z}_\ell)$ splits orthogonally into its local components and the pairing is perfect on each". These are Proposition 20.4(iv) and Proposition 20.8(iv) below.

**Example 20.6** (level 539 at $\ell = 3$)**.** This example runs through Sections 20.3 and 20.12. At level $539 = 7^2 \cdot 11$ consider four eigenforms of $\mathbb{T}_{539}$ with integer coefficients: the newforms $f_{539\mathrm{c}}$ and $f_{539\mathrm{d}}$, and the depleted forms $(f_{77\mathrm{c}})^{\{7\}}$ and $(f_{11\mathrm{a}})^{\{7\}}$, which live at level 539 because $\delta_7 = 1$ for `77c1` (multiplicative at 7, so $77 \cdot 7 = 539$) and $\delta_7 = 2$ for `11a1` (good at 7, so $11 \cdot 49 = 539$). The curve `539c1` has type $\mathrm{I}_3^*$ at 7 and is the twist of `77c1` by $\chi_{-7}$; `539d1` has type $\mathrm{I}_0^*$ at 7 and is the twist of `11a1` by $\chi_{-7}$. So $\iota_{\{7\}}$ sends the two depleted forms to the two newforms. Computed with PARI/GP (all $n \le 130$, beyond Sturm's bound 112 for $\Gamma_0(539)$): $a_n(f_{539\mathrm{c}}) \equiv a_n(f_{539\mathrm{d}}) \pmod 3$ for every $n$, and $a_n((f_{77\mathrm{c}})^{\{7\}}) \equiv a_n((f_{11\mathrm{a}})^{\{7\}}) \pmod 3$ for every $n$; but $a_3(f_{539\mathrm{c}}) = -2$ while $a_3((f_{77\mathrm{c}})^{\{7\}}) = 2$, and $-2 \not\equiv 2 \pmod 3$. (The sign is the twist at work: $a_3(f_{539\mathrm{c}}) = \chi_{-7}(3)\, a_3(f_{77\mathrm{c}})$ with $\chi_{-7}(3) = \left(\frac{-7}{3}\right) = \left(\frac{2}{3}\right) = -1$.) So $\mathbb{T}_{539} \otimes \mathbb{Z}_3$ has a maximal ideal $\mathfrak{m}_1$ containing the two newforms' systems and another, $\mathfrak{m}_2$, containing the two depleted systems; $\iota$ exchanges them (it multiplies $T_p$ by $\chi_{-7}(p)$). The Hecke congruence modules, computed as the index $[S_2(539;\mathbb{Z}) : \mathbb{Z}g \oplus (g^c \cap S_2(539;\mathbb{Z}))]$ with $g^c$ the $\mathbb{T}$-stable complement (Proposition 20.14), are

$$\#\mathcal{C}(f_{539\mathrm{c}}) = 288, \quad \#\mathcal{C}(f_{539\mathrm{d}}) = 504, \quad \#\mathcal{C}((f_{77\mathrm{c}})^{\{7\}}) = 288, \quad \#\mathcal{C}((f_{11\mathrm{a}})^{\{7\}}) = 504,$$

with $288 = 2^5 3^2$ and $504 = 2^3 3^2 7$. The modular degrees are $\deg \varphi_{\texttt{539c1}} = 288$ and $\deg \varphi_{\texttt{539d1}} = 72$ (Chapter 14): at $\ell = 7$, where $7^2 \mid 539$, the congruence number $r(\texttt{539d1}) = 504$ and the degree differ. That the depleted modules equal the congruence numbers of the twisted newforms is Lemma 20.74 at every $\ell \neq 7$. (Computational note: the index must be computed with the integral structure cut out by Sturm's bound $\frac{2}{12}[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(539)] = 112$, which is what `mfsturm([539,2])` returns; applied to a space created by `mfinit`, `mfsturm` returns 88 or 50 according to the space, too few coefficients to test integrality. Truncating at fewer than 112 coefficients gives wrong orders that depend on the truncation: with $a_1, \dots, a_{50}$ the four orders come out as 1152, 2016, 5760, 10080, with $a_1, \dots, a_{51}$ as 288, 504, 1440, 2520; with $a_1, \dots, a_{112}$ they are the correct 288, 504, 288, 504.)

## 20.3 Congruence modules and congruence ideals

This section defines the congruence ideal of a surjection from a finite free algebra, records its basic properties (proved in Chapter 3), and reads the paper's Lemma lem:crit in this language: the congruence number $r_E$ is the congruence ideal of $\phi_f$ in the full Hecke algebra, and the same holds for the depleted form $g^S$ of Theorem thm:quant once the right complement is used. In the paper this is the sentence of Section 1 "The congruence number is the congruence ideal of $f$ in the full Hecke algebra of level $N$ (Lemma lem:crit), and the theory of such ideals, from Hida ... to Diamond, Flach and Guo ..., expresses them through the adjoint Selmer group and the adjoint $L$-value", and it underlies Lemma lem:hecke and Step 1 of the proof of Theorem thm:quant. Hida's theorem, which started the subject, is stated at the end as context; the paper does not use it in a proof.

Before the definition, the idea in the smallest case. Suppose two eigenforms have integer coefficients and are congruent modulo 5 but not modulo 25, and nothing else is around. The Hecke algebra they generate is then, up to isomorphism, $\mathbb{T} = \{(a,b) \in \mathbb{Z}^2 : a \equiv b \pmod 5\}$,

the pair recording the eigenvalues of an operator on the two forms. Over $\mathbb{Q}$ the algebra splits as $\mathbb{Q}\times\mathbb{Q}$, with idempotent $e=(1,0)$ projecting onto the first form; but $e\notin\mathbb{T}$, and the least multiple of $e$ that lies in $\mathbb{T}$ is $5e=(5,0)$. So the integral algebra fails to split off the first factor, and the size of the failure is 5, the modulus of the congruence. The definition below extracts this number intrinsically: $\wp=\ker\pi$ is $0\times5\mathbb{Z}$, its annihilator $I$ is the set of elements of $\mathbb{T}$ supported on the first factor, here $5\mathbb{Z}\times0$, and $\pi(I)=5\mathbb{Z}$. Proposition 20.8(iii) shows that this is always the ideal of denominators of the idempotent, and Example 20.9(a) is the same example over any discrete valuation ring.

**Definition 20.7** (congruence ideal, congruence module, cotangent module)**.** Let $\mathcal{O}$ be a discrete valuation ring (in practice complete) with fraction field $K$, $\mathbb{T}$ a commutative $\mathcal{O}$-algebra which is free of finite rank as an $\mathcal{O}$-module, and $\pi\colon\mathbb{T}\twoheadrightarrow\mathcal{O}$ a surjective homomorphism of $\mathcal{O}$-algebras. Put $\wp=\ker\pi$ and $I=\mathrm{Ann}_{\mathbb{T}}(\wp)$. The *congruence ideal* of $\pi$ is $\eta(\pi)=\pi(I)\subseteq\mathcal{O}$, the *congruence module* is $\mathcal{C}(\pi)=\mathbb{T}/(\wp+I)$, and the *cotangent module* is $\mathrm{Cot}(\pi)=\wp/\wp^2$.

**Proposition 20.8** (properties of congruence ideals)**.** *In the situation of Definition 20.7:*

*(i)* $\pi$ *induces an isomorphism* $\mathcal{C}(\pi)\simeq\mathcal{O}/\eta(\pi)$.

*(ii)* $\eta(\pi)\neq0$ *if and only if* $\wp\cap I=0$*, if and only if the local factor of the finite* $K$*-algebra* $\mathbb{T}_K=\mathbb{T}\otimes K$ *through which* $\pi_K$ *factors is* $K$ *itself.*

*(iii) If* $\eta(\pi)\neq0$*, then* $\pi$ *restricts to an isomorphism* $I\simeq\eta(\pi)$*, so* $I$ *is free of rank one over* $\mathcal{O}$*;* $\mathrm{Ann}_{\mathbb{T}}(I)=\wp$*; and* $\eta(\pi)=\{a\in\mathcal{O}:ae_\pi\in\mathbb{T}\}$*, where* $e_\pi\in\mathbb{T}_K$ *is the idempotent of the factor* $K$ *of (ii).*

*(iv) If* $\mathbb{T}=\prod_{\mathfrak{m}}\mathbb{T}_{\mathfrak{m}}$ *is the decomposition into local factors (which exists when* $\mathcal{O}$ *is complete) and* $\pi$ *factors through* $\mathbb{T}_{\mathfrak{m}_0}$*, then the factors* $\mathbb{T}_{\mathfrak{m}}$*,* $\mathfrak{m}\neq\mathfrak{m}_0$*, lie in* $\wp$*,* $I\subseteq\mathbb{T}_{\mathfrak{m}_0}$*, and* $\eta(\pi)=\eta(\pi|_{\mathbb{T}_{\mathfrak{m}_0}})$*: the congruence ideal is computed in the local factor of* $\pi$*.*

*(v)* $\mathrm{Cot}(\pi)$ *is a finitely generated* $\mathcal{O}$*-module, and it is finite (of finite length) if and only if the factor of* $\mathbb{T}_K$ *through which* $\pi_K$ *factors is* $K$*.*

*Proof.* (i) The structure map $\mathcal{O}\to\mathbb{T}$ is a section of $\pi$, so $\mathbb{T}=\mathcal{O}\cdot1\oplus\wp$ as $\mathcal{O}$-modules, $t=\pi(t)+(t-\pi(t))$. Hence $\pi$ induces $\mathbb{T}/\wp\simeq\mathcal{O}$, which carries the image of $I$ onto $\pi(I)$, and $\mathbb{T}/(\wp+I)\simeq\mathcal{O}/\pi(I)$. (ii) and (iii) are the proposition on congruence ideals of Chapter 3; as Lemma lem:ddt417 uses them over a discrete valuation ring that need not be complete, we give the argument, which uses no completeness and only that $\mathbb{T}$ is torsion-free (not that it is reduced). The finite $K$-algebra $\mathbb{T}_K$ is a product of local artinian $K$-algebras; write $\mathbb{T}_K=A\times B$ with $A$ the factor through which $\pi_K$ factors and $\mathfrak{n}$ the maximal ideal of $A$, so that $\wp_K=\wp\otimes K=\mathfrak{n}\times B$ and $\wp=\mathbb{T}\cap\wp_K$. As $\mathbb{T}$ is torsion-free, $t\in\mathbb{T}$ kills $\wp$ iff it kills $\wp_K$, so $I=\mathbb{T}\cap\mathrm{Ann}_{\mathbb{T}_K}(\wp_K)=\mathbb{T}\cap(\mathrm{Ann}_A(\mathfrak{n})\times0)$. If $A=K$, then $\mathfrak{n}=0$ and $I=\mathbb{T}\cap Ke_\pi$ with $e_\pi=(1,0)$; since $\pi(ae_\pi)=a$, $\pi$ maps $I$ isomorphically onto $\eta(\pi)=\{a\in K:ae_\pi\in\mathbb{T}\}\subseteq\mathcal{O}$, which is nonzero because a nonzero multiple of $e_\pi$ lies in $\mathbb{T}$; so $\wp\cap I=0$, and $\mathrm{Ann}_{\mathbb{T}}(I)=\mathbb{T}\cap(0\times B)=\wp$. If $A\neq K$, then $\mathfrak{n}\neq0$ and $\mathrm{Ann}_A(\mathfrak{n})\subseteq\mathfrak{n}$ (a unit of $A$ kills nothing), so $I\subseteq\mathbb{T}\cap(\mathfrak{n}\times0)\subseteq\wp$ and $\eta(\pi)=0$; moreover $\mathrm{Ann}_A(\mathfrak{n})\neq0$ (it contains $\mathfrak{n}^{k-1}$ for the least $k$ with $\mathfrak{n}^k=0$), so $I\neq0$ and $\wp\cap I=I\neq0$. This proves (ii) and (iii). For (iv), note that the idempotent $1-e_{\mathfrak{m}_0}$ maps to 0 under $\pi$ (the image of an idempotent in the domain $\mathcal{O}$ is 0 or 1, and $\pi(e_{\mathfrak{m}_0})=1$), so $\prod_{\mathfrak{m}\neq\mathfrak{m}_0}\mathbb{T}_{\mathfrak{m}}=(1-e_{\mathfrak{m}_0})\mathbb{T}\subseteq\wp$; an element of $I$ kills $1-e_{\mathfrak{m}_0}$, hence lies in $e_{\mathfrak{m}_0}\mathbb{T}$, and inside $\mathbb{T}_{\mathfrak{m}_0}$ it kills $\wp\cap\mathbb{T}_{\mathfrak{m}_0}$; conversely an element of $\mathbb{T}_{\mathfrak{m}_0}$ killing $\wp\cap\mathbb{T}_{\mathfrak{m}_0}$ kills $\wp=(\wp\cap\mathbb{T}_{\mathfrak{m}_0})\times\prod_{\mathfrak{m}\neq\mathfrak{m}_0}\mathbb{T}_{\mathfrak{m}}$. So $I=\mathrm{Ann}_{\mathbb{T}_{\mathfrak{m}_0}}(\wp\cap\mathbb{T}_{\mathfrak{m}_0})$. (v) $\wp$

is a submodule of the finitely generated module $\mathbb{T}$ over the noetherian ring $\mathcal{O}$. Localisation is exact, so $\mathrm{Cot}(\pi)\otimes K=\wp_K/\wp_K^2$, where $\wp_K=\ker\pi_K$. Write $\mathbb{T}_K=A\times B$ with $A$ the local factor through which $\pi_K$ factors, so $\wp_K=\mathfrak{n}\times B$ with $\mathfrak{n}$ the maximal ideal of $A$, and $\wp_K/\wp_K^2=\mathfrak{n}/\mathfrak{n}^2$ (as $B^2=B$). By Nakayama's lemma in the artinian local ring $A$, $\mathfrak{n}/\mathfrak{n}^2=0$ iff $\mathfrak{n}=0$ iff $A=K$. A finitely generated $\mathcal{O}$-module is finite iff it is killed by $\otimes K$. □

**Example 20.9** (three toy algebras)**.** (a) For $r\geq 1$ let $\mathbb{T}_r=\{(a,b)\in\mathcal{O}^2 : a\equiv b \pmod{\varpi^r}\}$ and $\pi(a,b)=a$. Then $\wp=0\times\varpi^r\mathcal{O}$, $I=\varpi^r\mathcal{O}\times 0$, $\eta=\varpi^r\mathcal{O}$, $\mathcal{C}(\pi)\simeq\mathcal{O}/\varpi^r$, and $\wp^2=0\times\varpi^{2r}\mathcal{O}$, so $\mathrm{Cot}(\pi)\simeq\mathcal{O}/\varpi^r$: $\operatorname{length}\mathrm{Cot}(\pi)=v(\eta)=r$. In detail: $(a,b)\in\mathbb{T}_r$ has $\pi(a,b)=0$ iff $a=0$, and then $b\equiv 0 \pmod{\varpi^r}$, which gives $\wp$; an element $(a,b)$ kills $(0,\varpi^r)$ iff $b\varpi^r=0$, that is $b=0$, and $(a,0)\in\mathbb{T}_r$ iff $\varpi^r\mid a$, which gives $I$. As an $\mathcal{O}$-module $\mathbb{T}_r$ has basis $(1,1)$, $(0,\varpi^r)$, so $\mathbb{T}_r=\mathcal{O}\oplus\wp$ and $\mathbb{T}_r/(\wp+I)=\mathcal{O}/\varpi^r$ as in Proposition 20.8(i). This is the algebra of two eigenforms congruent modulo $\varpi^r$, and $\mathbb{T}_r\simeq\mathcal{O}[x]/(x(x-\varpi^r))$ by $x\mapsto(0,\varpi^r)$ is a complete intersection: the map $\mathcal{O}[x]\to\mathbb{T}_r$ is onto because $1\mapsto(1,1)$ and $x\mapsto(0,\varpi^r)$ are the basis above, it kills $x(x-\varpi^r)$, whose image is $(0,\varpi^r)\big((0,\varpi^r)-\varpi^r(1,1)\big)=(0,\varpi^r)(-\varpi^r,0)=0$, and both $\mathcal{O}[x]/(x(x-\varpi^r))$ and $\mathbb{T}_r$ are free of rank 2, so the surjection is an isomorphism. One equation in one variable: a complete intersection. (b) Let $\mathbb{T}_3^\flat=\{(a,b,c)\in\mathcal{O}^3 : a\equiv b\equiv c \pmod{\varpi}\}$ and $\pi(a,b,c)=a$. Then $\wp=0\times\varpi\mathcal{O}\times\varpi\mathcal{O}$, $I=\varpi\mathcal{O}\times 0\times 0$, $\eta=\varpi\mathcal{O}$, but $\wp^2=0\times\varpi^2\mathcal{O}\times\varpi^2\mathcal{O}$ and $\mathrm{Cot}(\pi)\simeq(\mathcal{O}/\varpi)^2$: $\operatorname{length}\mathrm{Cot}(\pi)=2>1=v(\eta)$. (Here $I$ consists of the $(a,b,c)\in\mathbb{T}_3^\flat$ killing $(0,\varpi,0)$ and $(0,0,\varpi)$, so $b=c=0$ and then $\varpi\mid a$; and $\wp^2$ is spanned by the products $(0,\varpi y,\varpi z)(0,\varpi y',\varpi z')=(0,\varpi^2yy',\varpi^2zz')$, which fill $0\times\varpi^2\mathcal{O}\times\varpi^2\mathcal{O}$. The defect comes from the two directions $(0,\varpi,0)$, $(0,0,\varpi)$ of $\wp$: the cotangent module sees both congruent partners of the first form, while $\eta$ only sees the single worst congruence.) Three pairwise congruent forms; $\mathbb{T}_3^\flat$ is not a complete intersection (it is not even Gorenstein, Chapter 3). (c) Let $\mathbb{T}=\mathcal{O}[x]/(x^2)$ and $\pi(x)=0$. Then $\wp=\mathcal{O}x=I$, $\eta=0$ and $\mathrm{Cot}(\pi)=\mathcal{O}x$ is infinite (indeed $(a+bx)x=ax$ vanishes iff $a=0$, so $I=\mathcal{O}x$, which $\pi$ kills; and $\wp^2=\mathcal{O}x^2=0$, so $\mathrm{Cot}(\pi)=\wp\simeq\mathcal{O}$): the factor of $\mathbb{T}_K$ at $\pi$ is $K[x]/(x^2)$, not $K$. This is the shape of a Jordan block of a Hecke operator, such as $U_2$ at level 88 on the forms $f_{11\mathrm{a}}(dz)$, $d\mid 8$ (Chapter 13). These three cases are the whole content of Wiles's numerical criterion (Theorem 20.19) in miniature: $\operatorname{length}\mathrm{Cot}\geq v(\eta)$, with equality for the complete intersection (a), strict inequality for (b), and both sides degenerate in (c).

**Definition 20.10** (the Hecke congruence ideal of a newform)**.** Let $f$ be a newform of weight 2 on $\Gamma_0(M)$, $K_f$ its coefficient field, $\lambda$ a prime of $K_f$ above $\ell$, and $\mathcal{O}$ the completion of the ring of integers of $K_f$ at $\lambda$. The eigencharacter $\phi_f\colon\mathbb{T}\to\mathcal{O}_{K_f}$ extends to a surjection $\phi_f\colon\mathbb{T}_{\mathcal{O}}\twoheadrightarrow\mathcal{O}$ (its image is an $\mathcal{O}$-submodule containing 1, hence $\mathcal{O}$). Its congruence ideal $\eta_f^{\mathrm{Hecke}}=\eta(\phi_f)\subseteq\mathcal{O}$ is the *Hecke congruence ideal* of $f$. More generally, for an eigenform $g$ of $\mathbb{T}_M$ with eigencharacter $\pi$, we write $\eta^{\mathrm{Hecke}}(g)=\eta(\pi)$.

**Proposition 20.11** (the Hecke congruence ideal is nonzero)**.** *In Definition 20.10, $\eta_f^{\mathrm{Hecke}}\neq 0$. More generally $\eta(\pi)\neq 0$ for every eigencharacter $\pi$ whose generalised eigenspace in $S_2(\Gamma_0(M))\otimes\overline{K}$ is a line, in particular for $\pi$ the eigencharacter of a depleted form $f^\Sigma$ at level $M^\Sigma$.*

*Proof.* $S_2(\Gamma_0(M))\otimes\mathbb{C}$ is free of rank one over $\mathbb{T}\otimes\mathbb{C}$ (Chapter 13, the structure of $\mathbb{T}$), so the local factor of $\mathbb{T}\otimes\overline{K}$ at $\pi$ has the dimension of the generalised eigenspace of $\pi$ (a local artinian factor $A$ of $\mathbb{T}\otimes\mathbb{C}$ acts on the corresponding piece of $S_2$, which is free of rank one over $A$, so the two have the same dimension). If that is a line, the factor is $\overline{K}$, so the factor of $\mathbb{T}_K$ through which $\pi$ factors is $K$, and $\eta(\pi)\neq 0$ by Proposition 20.8(ii). The descent from $\overline{K}$ to $K$ goes as follows. Let $A_K$ be the local factor of $\mathbb{T}_K$ through which $\pi_K$ factors; its residue field is $K$,

because $\pi_K\colon A_K \to K$ is onto. Then $A_K \otimes_K \overline{K}$ has the single residue field $\overline{K}$, so it is local, hence it is the factor of $\mathbb{T}\otimes\overline{K}$ at $\pi$, of dimension 1 by hypothesis. So $\dim_K A_K = 1$ and $A_K = K$. For a newform the generalised eigenspace is the line $\mathbb{C}f$ (multiplicity one and strong multiplicity one, Chapter 13); for a depleted form at level $M^\Sigma$ it is the line $\mathbb{C}f^\Sigma$ (Chapter 13, depletion). □

**Theorem 20.12** (Lemma lem:crit as a statement about congruence modules)**.** *Let $E$ have conductor $N$, $f = f_E$, $\mathbb{T} = \mathbb{T}_N$, $\wp = \ker\phi_f$ and $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$. Then $I = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$, the congruence module $\mathcal{C}(\phi_f) = \mathbb{T}/(\wp + I)$ is cyclic of order $r_E$, $\eta_f^{\mathrm{Hecke}} = r_E\mathbb{Z}$, and the integers $m$ for which $f \equiv h \pmod m$ with $h \in f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z})$ are exactly the divisors of $r_E$; the same holds over $\mathbb{Z}_\ell$ with $r_E$ replaced by its $\ell$-part. Moreover, with $L = f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z})$, the perfect pairing $S_2(\Gamma_0(N);\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$ induces*

$$S_2(\Gamma_0(N);\mathbb{Z})/(\mathbb{Z}f \oplus L) \;\simeq\; \mathrm{Hom}(\mathcal{C}(\phi_f), \mathbb{Q}/\mathbb{Z})\,:$$

*the congruence module is dual to the failure of the integral forms to split along the decomposition $S_2 = \mathbb{C}f \oplus f^\perp$.*

*Proof.* Everything except the last display is the paper's Lemma lem:crit, which Chapter 15 proves in full (with the decomposition $\mathbb{T}\otimes\mathbb{Q} = \mathbb{Q}\times\mathbb{T}'$ of Chapter 13); we do not repeat it. For the display: the forms killed by $\wp$ are $\mathbb{Z}f$ and those killed by $I$ are $L$ (Chapter 15, in the proof of Lemma lem:crit), and for a perfect pairing $S\times\mathbb{T}\to\mathbb{Z}$ with $\mathbb{T}$ acting compatibly, ideals $\mathfrak{a},\mathfrak{b}$ with $\mathfrak{a}\cap\mathfrak{b} = 0$, $\mathbb{T}/\mathfrak{a}$ and $\mathbb{T}/\mathfrak{b}$ torsion-free of complementary ranks, one has $S/(S[\mathfrak{a}]\oplus S[\mathfrak{b}]) \simeq (\mathbb{T}/(\mathfrak{a}+\mathfrak{b}))^*$ (Chapter 3, the duality between an algebra and a module). Take $\mathfrak{a} = \wp$, $\mathfrak{b} = I$; $\wp\cap I = 0$ by Proposition 20.11 and Proposition 20.8(ii), $\mathbb{T}/\wp \simeq \mathbb{Z}$ has rank 1 and $\mathbb{T}/I$ embeds in $\mathbb{T}'$, of rank $\mathrm{rank}\,\mathbb{T} - 1$. (Concretely: by the proof of Proposition 20.8, $I = \mathbb{T}\cap\mathbb{Q}e_\pi$ with $e_\pi$ the idempotent of the factor $\mathbb{Q}$, so $t\mapsto(1-e_\pi)t$ maps $\mathbb{T}$ into the factor $\mathbb{T}'$ of $\mathbb{T}\otimes\mathbb{Q}$ with kernel exactly $I$; hence $\mathbb{T}/I$ is a finitely generated subgroup of a $\mathbb{Q}$-vector space of dimension $\mathrm{rank}\,\mathbb{T}-1$ which spans it, torsion-free of that rank.) In words: the pairing identifies a form $h\in S$ with the functional $t\mapsto a_1(th)$ on $\mathbb{T}$; $h$ is killed by $\wp$ iff its functional factors through $\mathbb{T}/\wp$, and killed by $I$ iff it factors through $\mathbb{T}/I$. Since $\wp\cap I = 0$, $\mathbb{T}$ embeds in $\mathbb{T}/\wp\times\mathbb{T}/I$ with cokernel $\mathbb{T}/(\wp+I)$, and $S[\wp]\oplus S[I]$ is the set of functionals on $\mathbb{T}$ that extend to this larger lattice. The quotient $S/(S[\wp]\oplus S[I])$ is therefore the cokernel of the restriction $\mathrm{Hom}(\mathbb{T}/\wp\times\mathbb{T}/I,\mathbb{Z})\to\mathrm{Hom}(\mathbb{T},\mathbb{Z})$, which is $\mathrm{Ext}^1(\mathbb{T}/(\wp+I),\mathbb{Z}) \simeq \mathrm{Hom}(\mathbb{T}/(\wp+I),\mathbb{Q}/\mathbb{Z})$ by the long exact sequence of $\mathrm{Hom}(-,\mathbb{Z})$. □

*In the paper* 20.13. Lemma lem:crit is stated in the paper's Section 3 with the letters $I = \ker\phi_f$ and $J$ the annihilator of $f^\perp$ (Warning 20.2); the paper's proof says "Dualising the inclusion $\mathbb{T}\subset\mathbb{T}/I\times\mathbb{T}/J$, whose cokernel is $\mathbb{T}/(I+J)$, shows that $\mathbb{Z}f\oplus L$ has index $r'$". That sentence is the display of Theorem 20.12, read as a statement about orders. Its use in Section 4 is the identification $\eta_f^{\mathrm{Hecke}}\mathbb{Z}_\ell = r_E\mathbb{Z}_\ell$ in Lemma lem:hecke and in Step 1 of the proof of Theorem thm:quant.

**Proposition 20.14** (Lemma lem:crit for an eigencharacter whose generalised eigenspace is a line)**.** *Let $\pi\colon\mathbb{T}_N\to\mathbb{Z}$ be a ring homomorphism whose generalised eigenspace in $S_2(\Gamma_0(N))\otimes\mathbb{C}$ is a line $\mathbb{C}g$ ($g$ normalised), and let $g^c$ be the $\mathbb{T}$-stable complement of $\mathbb{C}g$, the sum of the generalised eigenspaces of the other systems of eigenvalues. Put $\Lambda = g^c\cap S_2(\Gamma_0(N);\mathbb{Z})$. Then $\mathbb{T}/(\ker\pi + \mathrm{Ann}_{\mathbb{T}}\ker\pi)$ is cyclic of order $r(\pi) = [S_2(\Gamma_0(N);\mathbb{Z}) : \mathbb{Z}g\oplus\Lambda]$, $\eta(\pi) = r(\pi)\mathbb{Z}$, and $g\equiv h \pmod m$ for some $h\in\Lambda$ if and only if $m\mid r(\pi)$. The same holds over $\mathbb{Z}_\ell$ for every prime $\ell$, with $r(\pi)$ replaced by its $\ell$-part. If $g$ is a newform of level $N$, then $g^c = g^\perp$ and $r(\pi) = r_g$.*

*Proof.* This is the proposition on the congruence module of a rational eigencharacter in Chapter 13, whose proof is that of Lemma lem:crit with $g^c$ in place of $f^\perp$. □

*Warning* 20.15 (the complement must be $\mathbb{T}$-stable). For $g$ not a newform at level $N$ (for instance a depleted form), the Petersson orthogonal complement of $\mathbb{C}g$ need not be $\mathbb{T}$-stable, because the operators $U_p$, $p \mid N$, are not normal (Chapter 13 gives an example at level 44). Proposition 20.14 is false with the Petersson complement in general. The set-up paragraph before the paper's Lemma lem:step1 accordingly uses "the $\mathbb{T}$-stable complement $C$ of the line of $g^S$ (the sum of the generalised eigenspaces of the other eigencharacters; not the Petersson complement, which is not $\mathbb{T}$-stable when the $U_p$ are involved)"; Theorem thm:twist uses $g^\perp$ for a newform $g$ at its own level, where the two complements agree.

**Theorem 20.16** (Hida)**.** *Let $f$ be a normalised newform of weight $k \geq 2$ and level $N$, and let $\Omega_f^\pm$ be its periods, normalised by integrality of the modular symbols. Then $\langle f, f\rangle$ divided by $\Omega_f^+\Omega_f^-$ is an algebraic number, and for primes $\lambda$ outside an explicit finite set (containing the primes dividing $N$ and small primes depending on $k$), $\lambda$ divides this ratio if and only if $\lambda$ is a congruence prime of $f$, that is, $\lambda$ divides $\eta_f^{\mathrm{Hecke}}$. Through the Rankin-Shimura method, $\langle f, f\rangle$ is a multiple, by explicit factors, of the naive adjoint $L$-value $L^{\mathrm{nv}}(A_f, 1)$.*

*Reference.* We do not prove this. The statement combines two results of Hida from 1981: [82] proves that a prime dividing the normalised Petersson norm (outside the excluded set) is a congruence prime, and a companion paper of the same year proves the converse; we cite only the first, have not checked the precise excluded set, and state the theorem as context. See [83] for an exposition; [159] treats the weight-2 case from the side of modular curves. The Petersson norm as a naive $L$-value is proved in the book's normalisation in Chapter 19: $\langle f, f\rangle = N\, L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2)/(8\pi^3)$ for a rational newform of weight 2. □

*Remark* 20.17 (from Hida to Diamond, Flach and Guo). Hida's theorem detects congruence primes; it does not compute the exact power. For an elliptic curve the exact power is the subject of this chapter: the modular degree is a period ratio (the formula of Zagier and Watkins, Chapter 14), and Theorem 20.60 identifies its $\ell$-adic valuation with the length of a Selmer group plus local terms. The paper cites [49, Lemma 2.12] for the formula of the Petersson norm on $\Gamma_1(N)$ as a naive $L$-value, which Diamond, Flach and Guo deduce from Hida's Rankin-Shimura computation; in the book's normalisation (unnormalised product on $\Gamma_0(N)$) it is the formula recalled in the proof above, proved in Chapter 19. The constants agree. [49, Lemma 2.12] gives $(f,f)_{\Gamma_1(N)} = (k-1)!\,\delta(N)\,N\,\varphi(N)\,L^{\mathrm{nv}}(A_f,1)/(4^k\pi^{k+1})$, with $\delta(N) = 1$ for $N > 2$ and $\delta(N) = 2$ for $N \leq 2$, for the product $(g,h)_\Gamma = (-2i)^{-1}\int_{\Gamma\backslash\mathcal{H}} g\bar{h}\, y^{k-2}\, dz \wedge d\bar{z} = \int_{\Gamma\backslash\mathcal{H}} g\bar{h}\, y^k\, d\mu$ of their Section 1.5.1, which is unnormalised. As $[\bar{\Gamma}_0(N) : \bar{\Gamma}_1(N)] = \delta(N)\varphi(N)/2$, the product on $\Gamma_0(N)$ is $2/(\delta(N)\varphi(N))$ times theirs, and for $k = 2$ and trivial character their formula becomes $\langle f, f\rangle = N\, L^{\mathrm{nv}}(A_f, 1)/(8\pi^3) = N\, L^{\mathrm{nv}}(\mathrm{Sym}^2 f, 2)/(8\pi^3)$ at every level.

*Remark* 20.18 (the three ideals of this chapter, at once). Let $E$ be $X_0(N)$-optimal, $f = f_E$, $\ell$ an odd prime with $\ell \nmid N$ and $E[\ell]$ irreducible. Then

$$\begin{aligned} \eta_f^{\mathrm{Hecke}}\mathbb{Z}_\ell &= r_E\mathbb{Z}_\ell && \text{(Theorem 20.12)},\\ v_\ell(r_E) &= v_\ell(\deg\varphi_E) && \text{([3, Theorem 2.2])},\\ v_\ell(\deg\varphi_E) &= v_\lambda(\eta_{f,\lambda}^{\emptyset}) && \text{(Theorem 20.34)}, \end{aligned}$$

so all three ideals agree. Lemma lem:ddt417 (Lemma 20.21) gives $v(\eta^{\mathrm{Hecke}}) \geq v(\eta^{\mathrm{coh}})$ in general, with equality when the cohomology is free over the Hecke algebra; the inequality is strict where freeness fails. Outside the hypotheses it can fail to be an equality: for `99a1` at $\ell = 3$, with $3 \mid 99$, the Hecke ideal has valuation $v_3(r_E) = v_3(12) = 1$ while the ideal of values of $\hat{\delta}_!$ on $M_{f,\lambda}$ is $\deg \varphi_E \, \mathbb{Z}_3 = 4\mathbb{Z}_3 = \mathbb{Z}_3$ (Chapter 14; $r = 12$ and $\deg \varphi_E = 4$ recomputed with PARI/GP).

## 20.4 Wiles's numerical criterion and Lemma lem:ddt417

This section proves the paper's Lemma lem:ddt417, the piece of commutative algebra that compares the Hecke congruence ideal with the Gram determinant of a pairing on a module. The paper uses it in Step 2 of the proof of Theorem thm:quant (with $d = 2$, for a depleted form) and, in the paragraph after Lemma lem:hecke, for the newform itself, where without the projection formula it gives only an inequality. To put it in context we first state Wiles's numerical criterion, of which the lemma is the half that survives in the presence of a module; the criterion itself is used in this book only in the outline of the Taylor-Wiles method (Section 20.8).

**Theorem 20.19** (Wiles's numerical criterion; Lenstra)**.** *Let $\mathcal{O}$ be a complete discrete valuation ring with finite residue field, $\mathbb{T}$ a local $\mathcal{O}$-algebra, finite and free as an $\mathcal{O}$-module, and $\pi\colon \mathbb{T} \twoheadrightarrow \mathcal{O}$ a surjection with $\eta = \eta(\pi) \neq 0$. Then:*

*(i)* $\mathrm{length}_{\mathcal{O}} \mathrm{Cot}(\pi) \geq v(\eta)$*;*

*(ii) equality holds if and only if $\mathbb{T}$ is a complete intersection over $\mathcal{O}$, that is, a quotient of a power series ring $\mathcal{O}[[x_1, \dots, x_n]]$ by $n$ elements;*

*(iii) if $R \twoheadrightarrow \mathbb{T}$ is a surjection of complete local noetherian $\mathcal{O}$-algebras compatible with $\pi$ (so that $\pi$ extends to $\pi_R\colon R \twoheadrightarrow \mathcal{O}$) and* $\mathrm{length}\,\mathrm{Cot}(\pi_R) \leq v(\eta) < \infty$*, then $R \to \mathbb{T}$ is an isomorphism and both rings are complete intersections.*

*Sketch of proof.* For a complete intersection $\mathbb{T} = \mathcal{O}[[x_1, \dots, x_n]]/(f_1, \dots, f_n)$ with $\pi(x_i) = 0$, both $\mathrm{length}\,\mathrm{Cot}(\pi)$ and $v(\eta)$ equal $v(\det(\partial f_i/\partial x_j)(0))$: the cotangent module is the cokernel of the Jacobian matrix at 0, and Tate's description of the annihilator of $\wp$ in a complete intersection (by the determinant of a matrix expressing the $f_i$ in terms of the $x_j$) gives $\eta$. The general inequality (i) and the converse in (ii) reduce to this case by writing $\mathbb{T}$ as a quotient of a complete intersection with the same cotangent module and comparing the two congruence ideals; Lenstra's proof uses the Gorenstein property and a flat deformation. Part (iii) follows from (i) for $\mathbb{T}$ and the surjection $\mathrm{Cot}(\pi_R) \twoheadrightarrow \mathrm{Cot}(\pi)$: $\mathrm{length}\,\mathrm{Cot}(\pi_R) \geq \mathrm{length}\,\mathrm{Cot}(\pi) \geq v(\eta) \geq \mathrm{length}\,\mathrm{Cot}(\pi_R)$ forces equalities, so $\mathbb{T}$ is a complete intersection, and then a complete intersection is not a proper quotient of a ring with the same cotangent length. We do not give the details; see [207, Appendix], [112], and [38, Section 5] (Wiles's numerical criterion). The examples of Example 20.9 show each case. □

*Remark* 20.20 (why the paper needs a version with a module). The ideal that Diamond, Flach and Guo control is not $\eta(\pi)$, an invariant of the ring, but the ideal of values of a pairing on the $\pi$-part of a module $L$ (the cohomology of a modular curve, Definition 20.31). Lemma lem:ddt417 compares the two. It is an equality when $L$ is free over $\mathbb{T}$, which is what a multiplicity-one theorem gives (the freeness criterion of Chapter 3 and the Gorenstein theorem of Chapter 14). The paper says: "We also need the following piece of commutative algebra, which is the inequality half of [38, Lemma 4.17]."

**Lemma 20.21** (Lemma lem:ddt417)**.** *Let $\mathcal{O}$ be a discrete valuation ring, $\mathbb{T}$ a commutative $\mathcal{O}$-algebra that is free of finite rank as an $\mathcal{O}$-module, and $L$ a $\mathbb{T}$-module that is free of finite rank over $\mathcal{O}$, with a perfect $\mathcal{O}$-bilinear pairing $\langle\ ,\ \rangle\colon L\times L\to\mathcal{O}$ for which every $t\in\mathbb{T}$ is self-adjoint. Let $\pi\colon\mathbb{T}\twoheadrightarrow\mathcal{O}$ be a surjective homomorphism of $\mathcal{O}$-algebras, $\wp=\ker\pi$, $I=\mathrm{Ann}_{\mathbb{T}}(\wp)$ and $\eta=\pi(I)$. Suppose that $L[\wp]$ has rank $d$ over $\mathcal{O}$ and that the Gram determinant $\Delta$ of the pairing on $L[\wp]$ is nonzero. Then*

$$\eta^d\subseteq\Delta\mathcal{O},\qquad\text{that is,}\qquad d\cdot v(\eta)\ \geq\ v(\Delta),$$

*and equality holds if $L$ is free over $\mathbb{T}$.*

*Proof.* Here $L[\mathfrak{a}]$ denotes the submodule of $L$ killed by an ideal $\mathfrak{a}$; it is free over $\mathcal{O}$ as a submodule of a free module over a principal ideal domain, so its Gram determinant is defined up to the square of a unit, and $v(\Delta)$ does not depend on the basis (Chapter 3).

*Step 0: the case $\eta=0$, and $\wp\cap I=0$.* If $\eta=0$ the inequality says $\infty\geq v(\Delta)$ and there is nothing to prove. Assume $\eta\neq0$ and fix $t\in I$ with $\pi(t)$ a generator of $\eta$. Then $\wp\cap I=0$ by Proposition 20.8(ii) (the argument there uses only that $\mathbb{T}$ is torsion-free, not that it is reduced).

*Step 1: $Q=L/(L[\wp]+L[I])$ is killed by $\eta$.* Let $x\in L$. Since $\wp I=0$, every element of $\wp$ kills $tx$, so $tx\in L[\wp]$. For $s\in I$, $s(\pi(t)-t)\in I\wp=0$, so $(\pi(t)-t)x\in L[I]$. Hence $\pi(t)x=tx+(\pi(t)-t)x\in L[\wp]+L[I]$, and $\pi(t)$, a generator of $\eta$, kills $Q$.

*Step 2: the sum is direct.* If $y\in L[\wp]\cap L[I]$, then $y$ is killed by $\wp+I$, which contains $\pi(t)=t+(\pi(t)-t)$ ($t\in I$, $\pi(t)-t\in\wp$). So $\pi(t)y=0$, and $y=0$ since $L$ is torsion-free and $\pi(t)\neq0$. Thus $Q=L/(L[\wp]\oplus L[I])$.

*Step 3: $L[\wp]$ is saturated and $L[\wp]^\perp=L[I]$.* $L[\wp]$ is saturated: if $ax\in L[\wp]$ with $0\neq a\in\mathcal{O}$, then $a(sx)=s(ax)=0$ for $s\in\wp$, so $sx=0$. Let $y\in L[I]$ and $x\in L[\wp]$. Since $\pi(t)-t\in\wp$ kills $x$ and $t\in I$ kills $y$,

$$\pi(t)\langle x,y\rangle=\langle tx,y\rangle+\langle(\pi(t)-t)x,y\rangle=\langle x,ty\rangle+0=0,$$

using self-adjointness of $t$, and $\langle x,y\rangle=0$. Thus $L[I]\subseteq L[\wp]^\perp$. Conversely let $y\perp L[\wp]$. For $s\in I$ and $z\in L$, $sz\in L[\wp]$ (as in Step 1), so $\langle sy,z\rangle=\langle y,sz\rangle=0$ for all $z$; the pairing being perfect, $sy=0$. So $y\in L[I]$. (This step is the only one that uses self-adjointness.)

*Step 4: $Q$ is the cokernel of the Gram map.* The map $L\to L[\wp]^\vee=\mathrm{Hom}_{\mathcal{O}}(L[\wp],\mathcal{O})$, $y\mapsto\langle-,y\rangle$, is surjective because $L[\wp]$ is saturated and the pairing is perfect: a saturated submodule of a finitely generated free module over a principal ideal domain is a direct summand ($L/L[\wp]$ is torsion-free, hence free, so the projection splits), so every $\lambda\in L[\wp]^\vee$ extends to some $\tilde\lambda\in L^\vee$, and perfectness writes $\tilde\lambda=\langle-,y\rangle$ for some $y\in L$. Its kernel is $L[\wp]^\perp=L[I]$ (Chapter 3, the Gram cokernel theorem). So $L/L[I]\simeq L[\wp]^\vee$, and under this isomorphism the image of $L[\wp]$ is the image of the Gram map $L[\wp]\to L[\wp]^\vee$. Hence $Q\simeq\mathrm{coker}(L[\wp]\to L[\wp]^\vee)$, a module of length $v(\Delta)$ (the cokernel of a square matrix over a discrete valuation ring has length the valuation of its determinant: by the elementary divisor theorem the matrix is $U\,\mathrm{diag}(\varpi^{a_1},\dots,\varpi^{a_d})\,V$ with $U,V$ invertible, the cokernel is $\bigoplus\mathcal{O}/\varpi^{a_i}$, of length $\sum a_i=v(\det)$). The nonvanishing $\Delta\neq0$ is what makes $Q$ finite.

*Step 5: the inequality.* $L[\wp]^\vee\simeq\mathcal{O}^d$ surjects onto $Q$, so $Q$ is generated by $d$ elements; by Step 1 it is killed by $\eta$. So $Q$ is a quotient of $(\mathcal{O}/\eta)^d$ and $v(\Delta)=\mathrm{length}\,Q\leq d\,v(\eta)$.

*Step 6: equality for free $L$.* Let $L\simeq\mathbb{T}^r$ as $\mathbb{T}$-modules. Then $L[\wp]\simeq(\mathbb{T}[\wp])^r=I^r$ (an element of $\mathbb{T}^r$ is killed by $\wp$ iff each coordinate is, and $\mathbb{T}[\wp]=\mathrm{Ann}_{\mathbb{T}}(\wp)=I$ by definition) and $L[I]\simeq\mathrm{Ann}_{\mathbb{T}}(I)^r=\wp^r$ (Proposition 20.8(iii)), and $I\simeq\eta$ is free of rank one over $\mathcal{O}$, so $d=r$. Then $Q\simeq(\mathbb{T}/(\wp+I))^r\simeq(\mathcal{O}/\eta)^d$ by Proposition 20.8(i), of length $d\,v(\eta)$. $\square$

*Remark* 20.22 (Commentary: the paper's statement of Lemma lem:ddt417). The paper states the lemma for $\mathcal{O}$ "a discrete valuation ring with finite residue field", and its proof concludes "$\#\mathcal{O}/\Delta \leq \#(\mathcal{O}/\eta)^d$". The book's Lemma 20.21 is stated with lengths over any discrete valuation ring; for a finite residue field of order $\kappa$ the two forms agree, since $\#X = \kappa^{\operatorname{length} X}$. Three steps that the paper's proof compresses are expanded above. (a) The paper's "First $\wp \cap I = 0$: ... $\mathbb{T}$ being torsion-free" is Step 0; the paper adds that this does not follow from reducedness, since the full Hecke algebra can have nilpotents (Example 20.9(c)). (b) The paper's "The sum $L[\wp] + L[I]$ is direct: $\pi(t) = t + (\pi(t) - t)$ lies in $I + \wp$ and is a nonzero scalar" is Step 2. (c) In the paper's proof of $L[\wp]^{\perp} = L[I]$, the converse inclusion shows that the single element $t$ kills $y$; this suffices because $I = \mathcal{O}t$ (Proposition 20.8(iii)), and Step 3 above uses every $s \in I$ instead. In the paper's direct inclusion the identity "$\pi(t)\langle x, y\rangle = \langle tx, y\rangle = \langle x, ty\rangle = 0$" uses $(\pi(t) - t)x = 0$ for $x \in L[\wp]$, which is how Step 3 reads it.

*Remark* 20.23 (on the hypotheses). (a) Without $\eta \neq 0$ the intersection $\wp \cap I$ can be nonzero: for $\mathbb{T} = \mathcal{O}[x]/(x^2)$ acting on $L = \mathbb{T}$, $\wp = I = \mathcal{O}x$. The lemma survives because then $\eta = 0$. (b) If $\eta = 0$ and $L$ is free over $\mathbb{T}$, then $\Delta = 0$ (Chapter 3): $I \subseteq \wp$ gives $I^2 = 0$ and the pairing vanishes on $L[\wp] = I^r$. So in the free case the hypothesis $\Delta \neq 0$ forces $\eta \neq 0$. (c) Self-adjointness is used only in Step 3. (d) For the Hecke algebra of a modular curve the module $L$ is $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$, the algebra $\mathbb{T}_{\mathfrak{m}}$ and the pairing $\hat{\delta}_!$ restricted to the $\mathfrak{m}$-component, which is perfect there (Proposition 20.4(iv)); by Proposition 20.8(iv) the congruence ideal of $\pi$ on $\mathbb{T}_\ell$ is the one computed on $\mathbb{T}_{\mathfrak{m}}$.

**Example 20.24** (the lemma on test cases)**.** (a) $L = \mathbb{T}_r$ (Example 20.9(a)) with $\langle s, u\rangle = \lambda_0(su)$, $\lambda_0(a, b) = (a - b)/\varpi^r$, a perfect pairing for which $\mathbb{T}_r$ is self-adjoint: $L[\wp] = I = \mathcal{O}(\varpi^r, 0)$, $d = 1$, $\Delta = \lambda_0((\varpi^r, 0)^2) = \lambda_0(\varpi^{2r}, 0) = \varpi^r$; equality, $L$ being free. (Perfectness: in the basis $u_1 = (1, 1)$, $u_2 = (0, \varpi^r)$ of $\mathbb{T}_r$ one has $u_1^2 = u_1$, $u_1u_2 = u_2$, $u_2^2 = \varpi^r u_2$, so the Gram matrix is $\left(\begin{smallmatrix} 0 & -1 \\ -1 & -\varpi^r \end{smallmatrix}\right)$, of determinant $-1$, a unit. Self-adjointness: $\langle ts, u\rangle = \lambda_0(tsu) = \langle s, tu\rangle$, the algebra being commutative.) (b) $L = \mathcal{O}^2$ with $\mathbb{T}_r$ acting diagonally and the standard pairing: $L[\wp] = \mathcal{O}e_1$, $\Delta = 1$, so $0 = v(\Delta) < r = v(\eta)$; strict, and $L$ is not free over $\mathbb{T}_r$. (c) $L = \mathbb{T}_r^2$ with $\langle x, y\rangle = \lambda_0(x_1y_2 - x_2y_1)$, alternating: $d = 2$, $\Delta = \varpi^{2r} = \eta^2$. (Here $L[\wp] = I^2$ has basis $(\tau, 0)$, $(0, \tau)$ with $\tau = (\varpi^r, 0)$, and $\langle(\tau, 0), (0, \tau)\rangle = \lambda_0(\tau^2) = \varpi^r$, so the Gram matrix is $\left(\begin{smallmatrix} 0 & \varpi^r \\ -\varpi^r & 0 \end{smallmatrix}\right)$; the pairing on $L$ is perfect because its Gram matrix is built from the unimodular one of (a).) This is the shape of Step 2 of the proof of Theorem thm:quant. (These computations are in Chapter 3.) (d) The modular case outside the hypotheses: for `99a1` and $\ell = 3$, $L = H^1(X_0(99), \mathbb{Z}_3)_{\mathfrak{m}}$, $d = 2$, $v_3(\eta^{\mathrm{Hecke}}) = v_3(r_E) = 1$ and the ideal of values on $L[\wp]$ is $4\mathbb{Z}_3 = \mathbb{Z}_3$, so $v_3(\Delta) = 0 < 2$: $L$ is not free over $\mathbb{T}_{\mathfrak{m}}$ (Remark 20.18).

## 20.5 The lattices and ideals of Diamond, Flach and Guo; Lemma lem:hecke

This section sets up the objects of the paper's Section 4.1: the depleted form $f^\Sigma$ at level $M^\Sigma$, the integral structure on cohomology, the lattice $M_{f^\Sigma,\lambda}$, the pairing $\hat{\delta}_!$ and the naive $\Sigma$-finite congruence ideal $\eta^\Sigma_{f,\lambda}$, in the form in which Diamond, Flach and Guo define them for weight 2 and trivial character [49, Sections 1.5.3, 1.6.2, 1.7.3], and compares them with the cohomology of $X_0(M^\Sigma)$, as the paper does. It then states Lemma lem:hecke as an equality, which Chapter 14 proves, and Remark rem:free. These are the inputs of Step 1 of the proof of Theorem thm:main and of Steps 2 to 4 of the proof of Theorem thm:quant.

**Definition 20.25** (depletion and naive factors, recalled)**.** Let $f$ be a newform of weight 2 on $\Gamma_0(M)$ and $\Sigma$ a finite set of primes. For a prime $p$ put $\delta_p = 2$, 1, 0 according as $p \nmid M$, $p \,\|\, M$, $p^2 \mid M$; for $f = f_E$ these are good, multiplicative and additive reduction, and $\delta_p = \dim V_\ell E^{I_p}$ for $p \neq \ell$ (Chapter 15). Put $M^\Sigma = M \prod_{p\in\Sigma} p^{\delta_p}$ and $f^\Sigma = \sum_{(n,\Sigma)=1} a_n(f) q^n$. The *naive* local factor of the adjoint $L$-function is $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p > 0$ and 1 if $\delta_p = 0$ (Chapter 19).

By Chapter 13 (depletion), $f^\Sigma$ is a normalised eigenform of the full Hecke algebra $\mathbb{T}_{M^\Sigma}$, with $U_p f^\Sigma = 0$ for $p \in \Sigma$, and its generalised eigenspace in $S_2(\Gamma_0(M^\Sigma))$ is the line $\mathbb{C} f^\Sigma$; $M^\Sigma$ is the least level at which $f^\Sigma$ is an eigenform of all $T_n$. The paper's Section 1.1 says it follows "the conventions of [49, §1.7.2]" for $\delta_p$ and the naive factors; there $\delta_p$ is the dimension of the inertia invariants of the lattice $M_{f,\lambda}$ of Definition 20.31, which is the same number because $M_{f,\lambda}$ is dual to $T_\ell E$ up to homothety.

**Definition 20.26** (the integral structure)**.** Let $M \geq 1$ and $U = U_0(M) \subset \mathrm{GL}_2(\widehat{\mathbb{Z}})$. For weight 2 and trivial character the coefficient sheaf of [49, Section 1.2.2] is the constant sheaf $\mathbb{Z}$. Choose an auxiliary $N' \geq 3$ with $M \mid N'$ and every prime factor of $N'$ in the set $S$ of [49, Section 1.4.1] (for instance $N' = 4M$); then $U_{N'} \subseteq U$. Let $Y(N')$ be the analytic modular curve of full level $N'$, a disjoint union of copies of $\Gamma(N')\backslash\mathcal{H}$ indexed by $(\mathbb{Z}/N'\mathbb{Z})^\times$, on which the finite group $G = U/U_{N'}$ acts through its image in $\mathrm{GL}_2(\mathbb{Z}/N'\mathbb{Z})$ [49, (5), Proposition 1.3]. Diamond, Flach and Guo put [49, Section 1.4.1]

$$M(M,1)_{c,B} = H^1_c(Y(N'),\mathbb{Z})^G, \qquad M(M,1)_{\mathrm{tf},B} = H^1(Y(N'),\mathbb{Z})^G$$

(the second is the torsion-free quotient of the invariants of ordinary cohomology; $H^1$ of an open surface is already torsion-free), and the *Betti lattice* $M(M,1)_{!,B}$ is the image of the forgetful map $M(M,1)_{c,B} \to M(M,1)_{\mathrm{tf},B}$ [49, Section 1.4.2], with the Hecke operators, $U_p$ included, acting by double cosets. None of this depends on $N'$ [49, Lemma 1.1]. For a prime $\ell$ the $\lambda$-adic structure $M(M,1)_{!,\lambda}$ is defined in the same way from $\ell$-adic cohomology, and $\mathbb{Z}_\ell \otimes M(M,1)_{?,B} \simeq M(M,1)_{?,\lambda}$ [49, Section 1.2.3].

The comparison with $X_0(M)$ uses cohomology with compact support of a space with an action of a finite group, and two spectral sequences. We recall what is needed; the general theory is in Chapter V of [77], and the group cohomology in [17].

*Equivariant cohomology with compact support.* Let a finite group $G$ act continuously on a locally compact Hausdorff space $X$. A *$G$-sheaf* on $X$ is a sheaf $\mathcal{F}$ of abelian groups together with isomorphisms $g^*\mathcal{F} \simeq \mathcal{F}$ compatible with the group law; the constant sheaf $\mathbb{Z}$, with $g$ acting on sections by $s \mapsto s \circ g^{-1}$, is one. The $G$-sheaves form an abelian category with enough injectives, and $\Gamma_c(X,\mathcal{F})$ is a $G$-module. The *$G$-equivariant cohomology with compact support* $H^n_{G,c}(X,\mathcal{F})$ is the $n$-th right derived functor of the left exact functor $\mathcal{F} \mapsto \Gamma_c(X,\mathcal{F})^G$. (For $G$ trivial it is $H^n_c(X,\mathcal{F})$.)

*Hochschild-Serre.* The functor $\Gamma_c(X,-)^G$ is the composite of $\Gamma_c(X,-)$, from $G$-sheaves to $G$-modules, and of invariants $(-)^G$; the first sends injective $G$-sheaves to $G$-modules acyclic for $(-)^G$. The spectral sequence of a composite functor (Grothendieck, loc. cit.) is therefore

$$E_2^{p,q} = H^p\big(G, H^q_c(X,\mathcal{F})\big) \implies H^{p+q}_{G,c}(X,\mathcal{F}),$$

with the edge map $H^n_{G,c}(X,\mathcal{F}) \to H^n_c(X,\mathcal{F})^G$ forgetting the group action; compare [17, Chapter VII, Section 6] for the case of groups.

*Equivariant Leray.* Let $\pi\colon X \to Y = X/G$ be the quotient map, $Y$ with the quotient topology. As $G$ is finite, $\pi$ is proper with finite fibres, and $G$ permutes each fibre transitively. For a $G$-sheaf $\mathcal{F}$ put $\pi_*^G\mathcal{F} = (\pi_*\mathcal{F})^G$, a sheaf on $Y$. As $\pi$ is proper, $\Gamma_c(X,\mathcal{F})^G = \Gamma_c(Y,\pi_*^G\mathcal{F})$, and $\pi_*^G$ sends injective $G$-sheaves to flabby sheaves, which are acyclic for $\Gamma_c$ (Grothendieck, loc. cit.). The spectral sequence of this composite is the *equivariant Leray spectral sequence*

$$E_2^{p,q} = H_c^p(Y,\mathcal{R}^q) \implies H_{G,c}^{p+q}(X,\mathcal{F}), \qquad \mathcal{R}^q = R^q\pi_*^G\mathcal{F},$$

with the edge map $H_c^n(Y,\pi_*^G\mathcal{F}) \to H_{G,c}^n(X,\mathcal{F})$ induced by pull-back of sections. Its stalks are computed as follows. The functor $\pi_*$ is exact, $\pi$ being finite, so $\mathcal{R}^q$ is the sheaf associated to $V \mapsto H^q(G,\mathcal{F}(\pi^{-1}V))$; cohomology of a finite group commutes with filtered colimits ($\mathbb{Z}$ has a resolution by finitely generated free $\mathbb{Z}[G]$-modules), so the stalk at $y$ is $H^q(G,(\pi_*\mathcal{F})_y)$, and $(\pi_*\mathcal{F})_y = \bigoplus_{x\in\pi^{-1}(y)}\mathcal{F}_x$ is the $G$-module induced from the $G_x$-module $\mathcal{F}_x$, for a point $x$ above $y$ with stabiliser $G_x$. By Shapiro's lemma [17, Chapter III, Section 6],

$$(\mathcal{R}^q)_y \simeq H^q(G_x,\mathcal{F}_x).$$

**Proposition 20.27** (comparison with $H^1(X_0(M))$)**.** *Let $M \geq 1$ and let $\ell$ be any prime. Pull-back along the covering $\pi\colon Y(N') \to Y_0(M)$ identifies $H^1(X_0(M),\mathbb{Z})$, viewed inside $H^1(Y_0(M),\mathbb{Z})$, with the Betti lattice $M(M,1)_{!,B}$ of Definition 20.26, compatibly with the Hecke operators of $\mathbb{T}_M$, $U_p$ included. Consequently*

$$M(M,1)_{!,\lambda} = H^1(X_0(M),\mathbb{Z}_\ell)$$

*as $\mathbb{T}_M \otimes \mathbb{Z}_\ell$-modules, with no localisation; in particular $M(M,1)_{!,\lambda,\mathfrak{m}} = H^1(X_0(M),\mathbb{Z}_\ell)_\mathfrak{m}$ for every maximal ideal $\mathfrak{m}$ of $\mathbb{T}_\ell$, Eisenstein or not.*

*Proof. Set-up.* Since $\det U = \widehat{\mathbb{Z}}^\times$ maps onto $(\mathbb{Z}/N'\mathbb{Z})^\times$, the group $G$ permutes the components of $Y(N')$ transitively; the stabiliser of the component $\Gamma(N')\backslash\mathcal{H}$ acts on it through the image of $\Gamma_0(M)$ in $\mathrm{SL}_2(\mathbb{Z}/N'\mathbb{Z})$, which is $\Gamma_0(M)/\Gamma(N')$ ($\Gamma(N') \subseteq \Gamma_0(M)$ as $M \mid N'$, and $\Gamma(N')$ is the kernel of reduction modulo $N'$). So the quotient is

$$Y(N')/G = \Gamma_0(M)\backslash\mathcal{H} = Y_0(M),$$

the coarse modular curve, and $\pi$ is the quotient map; it is finite and proper, and the stabiliser $G_x$ of a point is a finite group. On the constant sheaf $\mathbb{Z}$ every $G_x$ acts trivially on the stalk $\mathbb{Z}$, since $G$ acts on $Y(N')$ by homeomorphisms and on $\mathbb{Z}$ through its action on the space.

*Step 1: $\pi^*\colon H_c^1(Y_0(M),\mathbb{Z}) \to H_c^1(Y(N'),\mathbb{Z})^G$ is an isomorphism.* Take $X = Y(N')$ and $\mathcal{F} = \mathbb{Z}$ above. In the Hochschild-Serre spectral sequence the row $q = 0$ vanishes: a section of the constant sheaf over a connected space is a constant function, so its support is empty or the whole component, and every component of $Y(N')$ is a non-compact connected surface; hence $H_c^0(Y(N'),\mathbb{Z}) = 0$. The exact sequence of terms of low degree,

$$0 \to E_2^{1,0} \to H_{G,c}^1(Y(N'),\mathbb{Z}) \to E_2^{0,1} \to E_2^{2,0},$$

with $E_2^{1,0} = E_2^{2,0} = 0$, gives $H_{G,c}^1(Y(N'),\mathbb{Z}) \simeq H_c^1(Y(N'),\mathbb{Z})^G$ by the edge map. In the equivariant Leray spectral sequence the stalks are $(\mathcal{R}^0)_y = H^0(G_x,\mathbb{Z}) = \mathbb{Z}$, and indeed $\mathcal{R}^0 = \pi_*^G\mathbb{Z}$ is the constant sheaf $\mathbb{Z}$ on $Y_0(M)$ (a $G$-invariant function on a fibre is constant, $G$ acting transitively on it); and $(\mathcal{R}^1)_y = H^1(G_x,\mathbb{Z}) = \mathrm{Hom}(G_x,\mathbb{Z}) = 0$, because a homomorphism from a finite group to $\mathbb{Z}$ is zero. The exact sequence of terms of low degree,

$$0 \to H_c^1(Y_0(M),\mathcal{R}^0) \to H_{G,c}^1(Y(N'),\mathbb{Z}) \to H_c^0(Y_0(M),\mathcal{R}^1) = 0,$$

gives $H^1_c(Y_0(M),\mathbb{Z}) \simeq H^1_{G,c}(Y(N'),\mathbb{Z})$ by the edge map. On sections the composite of the two edge maps is pull-back of sections along $\pi$, so the composite isomorphism $H^1_c(Y_0(M),\mathbb{Z}) \to H^1_c(Y(N'),\mathbb{Z})^G$ is $\pi^*$.

*Step 2: the image.* As $\pi$ is proper, pull-back commutes with the forgetful maps from compactly supported to ordinary cohomology, and the square

$$\begin{array}{ccc} H^1_c(Y_0(M),\mathbb{Z}) & \longrightarrow & H^1(Y_0(M),\mathbb{Z}) \\ \downarrow \pi^* & & \downarrow \pi^* \\ H^1_c(Y(N'),\mathbb{Z})^G & \longrightarrow & H^1(Y(N'),\mathbb{Z})^G \end{array}$$

commutes. The bottom map is $M(M,1)_{c,B} \to M(M,1)_{\mathrm{tf},B}$, so its image is $M(M,1)_{!,B}$; the left map is onto by Step 1; so $M(M,1)_{!,B}$ is $\pi^*$ of the image of the top map. That image is $H^1(X_0(M),\mathbb{Z})$ (Chapter 14, interior cohomology): with $C$ the finite set of cusps, the exact sequence $H^0(C) \to H^1_c(Y_0(M)) \to H^1(X_0(M)) \to H^1(C) = 0$ shows that the map from $H^1_c(Y_0(M),\mathbb{Z})$ to $H^1(X_0(M),\mathbb{Z})$ is onto, and restriction from $H^1(X_0(M),\mathbb{Z})$ to $H^1(Y_0(M),\mathbb{Z})$ is injective, because $H^1(X_0(M),Y_0(M))$ is, by excision, a sum of groups $H^1(D, D\setminus\{P\}) = 0$ over small discs around the cusps. It remains to see that $\pi^*$ is injective on $H^1(X_0(M),\mathbb{Z})$. Restrict to one component $Y_{\Gamma(N')} = \Gamma(N')\backslash\mathcal{H}$ and its compactification $X_{\Gamma(N')}$; then $\pi^*$ on $H^1(X_0(M),\mathbb{Z})$ is the composite of $\bar\pi^*\colon H^1(X_0(M),\mathbb{Z}) \to H^1(X_{\Gamma(N')},\mathbb{Z})$, for the extension $\bar\pi$ of $\pi$ to the compact curves, with the restriction to $Y_{\Gamma(N')}$, which is injective by the excision argument just given. The map $\bar\pi$ is a nonconstant map of compact Riemann surfaces, of some degree $d \geq 1$, so pull-back multiplies the cup product by $d$ (Chapter 8): $\langle \bar\pi^* a, \bar\pi^* b\rangle = d\langle a,b\rangle$. If $\bar\pi^* a = 0$ then $d\langle a,b\rangle = 0$ for every $b$, so $a = 0$, the cup product on $H^1(X_0(M),\mathbb{Z})$ being perfect (Poincaré duality) and $\mathbb{Z}$ torsion-free. Hence $\bar\pi^*$ is injective, and $\pi^*$ maps $H^1(X_0(M),\mathbb{Z})$ isomorphically onto $M(M,1)_{!,B}$.

*Step 3: Hecke operators.* Both lattices are torsion-free and $\pi^*$ is injective, so for $t \in \mathbb{T}_M$ the identity $\pi^*(tx) = t\,\pi^*(x)$ may be checked in $M(M,1)_{!,B}\otimes\mathbb{C}$. There, by [49, Proposition 1.3], the Hecke operators of Diamond, Flach and Guo act on $\mathrm{Fil}^1 M(M,1)_{!,\mathrm{dR}}\otimes\mathbb{C} = S_2(\Gamma_0(M))$ as the classical $T_p$, and as $U_p$ for $p \mid M$, and they commute with complex conjugation $F_\infty$. The map $\pi^*$ is compatible with the comparison isomorphism (5) of [49] and with the Eichler-Shimura isomorphism $H^1(X_0(M),\mathbb{C}) = S_2(\Gamma_0(M)) \oplus \overline{S_2(\Gamma_0(M))}$, both being pull-back of $f(\tau)\,d\tau$; and the Hecke action on $H^1(X_0(M),\mathbb{Z})$ is the one for which the Eichler-Shimura isomorphism is Hecke equivariant (Convention 20.1(b), Chapter 14). So the two actions agree on the holomorphic summand, and, both commuting with $F_\infty$, which exchanges the two summands, on the antiholomorphic one. Hence $\pi^*$ is Hecke equivariant, $U_p$ included.

*Step 4: the $\lambda$-adic structure.* The ring $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$; invariants under a finite group (a kernel of a map to a finite product) and images commute with flat base change, so $M(M,1)_{!,\lambda} = \mathbb{Z}_\ell \otimes M(M,1)_{!,B}$ by [49, Section 1.2.3]. By Step 2 this is $\mathbb{Z}_\ell \otimes H^1(X_0(M),\mathbb{Z}) = H^1(X_0(M),\mathbb{Z}_\ell)$ (universal coefficients, $H_1(X_0(M),\mathbb{Z})$ being free), $\mathbb{T}_M$-equivariantly by Step 3. Localisation at $\mathfrak{m}$ is exact, which gives the last statement. □

*In the paper* 20.28. Proposition 20.27 is the paper's Proposition prop:integral (Section 4.1): "Pull-back along the covering of $X_0(M^\Sigma)$ by the modular curve of full level identifies $H^1(X_0(M^\Sigma),\mathbb{Z})$ with the Betti lattice $M(M^\Sigma,1)_{!,B}$ of [49, §1.4.2], compatibly with the Hecke operators, $U_p$ included; in particular $M(M^\Sigma,1)_{!,\lambda} = H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$, with no localisation." The paper's proof has the four steps above; the book adds the definitions, the stalks of $\mathcal{R}^q$ and the reduction of the injectivity of $\pi^*$ to compact curves. The paper uses the proposition in Proposition prop:pullback and, through Proposition 20.32(i), in Step 2 of the proof of Theorem thm:quant. The remark after it reads: "Compact support is what makes the comparison integral: $H^0_c$ of the covering vanishes,

so taking invariants under the covering group loses nothing, as in [49, Lemma 1.1]." With ordinary cohomology the invariants can be larger than the pull-back; in the paper's words (there $N$ is the full level), "$H^1(Y(N),\mathbb{Z}_\ell)^G/\pi^*H^1(Y_0(M),\mathbb{Z}_\ell)$ is the group of characters $\Gamma_0(M)\to\mathbb{Q}_\ell/\mathbb{Z}_\ell$ of $\ell$-power order that factor through $G$, namely $\gamma\mapsto\psi(d \bmod p)$ for primes $p\mid M$ with $\ell\mid p-1$; for $M=11$ and $\ell=5$ it is the unramified covering $X_1(11)\to X_0(11)$". In the Hochschild-Serre sequence for ordinary cohomology this is visible: the row $q=0$ is $H^p(G,H^0(Y(N'),\mathbb{Z}_\ell))$, and $H^0(Y(N'),\mathbb{Z}_\ell)$ is induced from the trivial module of the stabiliser $G_1$ of a component, so the cokernel of $H^1_G\to H^1(Y(N'),\mathbb{Z}_\ell)^G$ injects into $H^2(G,H^0(Y(N'),\mathbb{Z}_\ell))\simeq H^2(G_1,\mathbb{Z}_\ell)\simeq\mathrm{Hom}(G_1,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ (Shapiro). These characters, and the Eisenstein Hecke action on them (for $X_1(M)\to X_0(M)$ they are measured by the Shimura subgroup, Chapter 14), play no part in the proof: compact support removes them.

**Definition 20.29** (the pairing $\hat\delta_!$)**.** Diamond, Flach and Guo define $\hat\delta_!$ on $M(M,1)_!$ as their normalised pairing $\bar\delta$ (a cup product on a modular curve of sufficiently small level $U'$, restricted to invariants and divided by the index of $U'$) composed with the operator $[UwU]_\omega$ attached to $w=\left(\begin{smallmatrix}0&-1\\M&0\end{smallmatrix}\right)$ [49, Sections 1.5.2, 1.5.3, (13), (14)]; for weight 2 and trivial character this operator is the Atkin-Lehner involution $w_M$, with no scalar factor. On $H^1(X_0(M),\mathbb{Z}_\ell)$ the book writes $\hat\delta_!(x,y)=\langle x,w_M^*y\rangle$, the cup product composed with $w_M^*$ (Appendix B, B-C13(f); Chapter 14).

**Proposition 20.30** (properties of $\hat\delta_!$)**.** *Let $\ell$ be odd.*

*(i) If $\ell\nmid M$, then on classes pulled back from $X_0(M)$ the pairing of Diamond, Flach and Guo is $\frac{u}{2}\langle x,w_M^*y\rangle$ for an $\ell$-adic unit $u$; so $\frac{u}{2}$ is a unit in $\mathbb{Z}_\ell$.*

*(ii) $\langle x,w_M^*y\rangle$ is perfect and alternating on $H^1(X_0(M),\mathbb{Z}_\ell)$, every $t\in\mathbb{T}$, the $U_p$ included, is self-adjoint for it, and its restriction to each $\mathfrak{m}$-component is perfect.*

*So, by Proposition 20.27, the two pairings agree up to a unit on all of $H^1(X_0(M),\mathbb{Z}_\ell)$ when $\ell\nmid 2M$, and the ideals of values and Gram determinants of their restrictions to any sublattice have the same valuations.*

*Proof.* (i) Take a sufficiently small level $U'\subseteq U_0(M)$ with $\det U'=\widehat{\mathbb{Z}}^\times$, such as $U'=U_0(M)\cap U_1(4)$, and let $X_{U'}\to X_0(M)$ be the corresponding covering. For $\ell\nmid 2M$ the normalised pairing $\bar\delta$ of [49, Section 1.5.2] is Poincaré duality on $X_{U'}$ up to an $\ell$-adic unit ([49, Section 1.5.2], before Lemma 1.2). It is normalised by dividing by the index ("normalizing by dividing by $[U:U_N]$", [49, Section 1.5.2]), so on $U_0(M)$-invariants it is $[U_0(M):U']^{-1}$ times its value on $U'$-invariants. A finite map of curves multiplies the fundamental class in $H^2$ by its degree, so the cup product on $X_{U'}$ of classes pulled back from $X_0(M)$ is $\deg(X_{U'}\to X_0(M))$ times their cup product on $X_0(M)$. The group $U_0(M)/U'$ acts on $X_{U'}$ over $X_0(M)$ through its quotient by the image of $\pm1$, because $-1\in\mathrm{GL}_2(\mathbb{Q})$ acts trivially on the upper and lower half planes; so the degree is $[U_0(M):U']/2$, and on pulled-back classes $\bar\delta$ is $\frac{u}{2}$ times the cup product on $X_0(M)$, $u$ an $\ell$-adic unit. As $\ell$ is odd, $\frac{u}{2}$ is a unit. In [49, Section 1.5.3], $w$ is the element with entries $\left(\begin{smallmatrix}0&-1\\M&0\end{smallmatrix}\right)$ of $\prod_{p\mid M}\mathrm{GL}_2(\mathbb{Z}_p)$, and $[UwU]_\omega$ "coincides with $\psi(-1_\infty)N^{k-2}[Uw^{-1}U]_\omega$" (their $N$ is our $M$); for $k=2$ and $\psi=1$ there is no scalar, and $[UwU]_\omega$ is the involution induced by the global matrix $\left(\begin{smallmatrix}0&-1\\M&0\end{smallmatrix}\right)$, which lies in $w\prod_{p\nmid M}\mathrm{GL}_2(\mathbb{Z}_p)$, that is, by $w_M$. (ii) This is the corollary on $\hat\delta_!$ in Chapter 14: the transpose of the correspondence $t$ is $w_Mtw_M$, so $(t^*)^\dagger=w_M^*t^*w_M^*$ for the cup product, whence $\langle tx,w_M^*y\rangle=\langle x,w_M^*ty\rangle$; $w_M^*$ is an involution adjoint to itself, and the cup product is alternating and perfect on a smooth projective curve (Poincaré duality, Chapter 8), so $\langle x,w_M^*y\rangle$ is perfect, and alternating because $\langle x,w_M^*x\rangle=-\langle w_M^*x,x\rangle=-\langle w_M^*x,w_M^{*2}x\rangle=-\langle x,w_M^*x\rangle$, as $w_M^*$ preserves the cup product, so $2\langle x,w_M^*x\rangle=0$. The statement on components is Proposition 20.4(iv). □

**Definition 20.31** (the kernel lattice and the naive $\Sigma$-finite congruence ideal)**.** Let $f$ be a newform of weight 2 on $\Gamma_0(M)$ with rational coefficients, $\ell \nmid 2M$, and $\Sigma$ a finite set of primes. For an eigenform $h$ of $\mathbb{T}_{M'}$ with eigencharacter $\theta_h$ put

$$M_{h,\lambda} = M(M',1)_{!,\lambda}[\ker \theta_h],$$

the submodule killed by the kernel of $\theta_h$ on the full Hecke algebra [49, Sections 1.1.2, 1.6.2]. With $h = f^\Sigma$ at level $M^\Sigma$, the restriction of $\hat{\delta}_!$ to the rank-two lattice $M_{f^\Sigma,\lambda}$ is alternating; its ideal of values, the image of $\wedge^2 M_{f^\Sigma,\lambda}$ under $\hat{\delta}_!$, is an ideal of $\mathbb{Z}_\ell$, written $\eta^\Sigma_{f,\lambda}$: the *naive $\Sigma$-finite congruence ideal* of $f$ [49, Section 1.7.3]. For $\Sigma = \emptyset$, $f^\emptyset = f$ and $M_{f,\lambda}$ is the lattice of $f$ at level $M$.

**Proposition 20.32** (properties of $M_{f^\Sigma,\lambda}$ and $\eta^\Sigma_{f,\lambda}$)**.** *In Definition 20.31 assume that $\bar{\rho}_f$ is irreducible, and let $\mathfrak{m}$ be the maximal ideal of $\mathbb{T}_{M^\Sigma} \otimes \mathbb{Z}_\ell$ attached to $f^\Sigma$.*

*(i) $M_{f^\Sigma,\lambda} = H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)[\ker \theta_{f^\Sigma}]$; it is saturated, of rank 2, stable under $G_\mathbb{Q}$, and contained in the $\mathfrak{m}$-component.*

*(ii) The Gram determinant of $\hat{\delta}_!$ on $M_{f^\Sigma,\lambda}$ is $(\eta^\Sigma_{f,\lambda})^2$ up to a unit: if $a$ generates the ideal of values, the Gram matrix in a suitable basis is $\left(\begin{smallmatrix} 0 & a \\ -a & 0 \end{smallmatrix}\right)$.*

*(iii) For $\Sigma = \emptyset$ and $f = f_E$ with $E$ optimal, $M_{f,\lambda} = \varphi_E^* H^1(E,\mathbb{Z}_\ell)$, which is $\mathrm{Hom}(T_\ell E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$ as a Galois module; hence $\mathrm{ad}^0 M_{f,\lambda} \simeq \mathrm{ad}^0 T_\ell E$.*

*Proof.* (i) $\theta = \theta_{f^\Sigma}$ factors through $\mathbb{T}_\mathfrak{m}$, so $\theta(e_\mathfrak{m}) = 1$ and $1 - e_\mathfrak{m} \in \ker\theta$; an element $x$ killed by $\ker\theta$ satisfies $x = e_\mathfrak{m} x$. So $M_{f^\Sigma,\lambda}$ lies in the $\mathfrak{m}$-component; and Proposition 20.27 identifies the integral structure $M(M^\Sigma,1)_{!,\lambda}$ with $H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$ Hecke-equivariantly, with no localisation, so the two kernel lattices coincide. A submodule killed by an ideal is saturated (Step 3 of the proof of Lemma 20.21). By the Eichler-Shimura isomorphism $H^1(X_0(M^\Sigma),\mathbb{C}) \simeq S_2 \oplus \overline{S_2}$, which is Hecke-equivariant, the $\theta$-eigenspace is $\mathbb{C}f^\Sigma \oplus \mathbb{C}\overline{f^\Sigma}$, because the eigenspace of $\theta$ in $S_2$ lies in its generalised eigenspace, the line $\mathbb{C}f^\Sigma$; so the rank is 2. The Hecke correspondences are defined over $\mathbb{Q}$, so the étale realisation of $M_{f^\Sigma,\lambda}$ is $G_\mathbb{Q}$-stable (Chapter 14). (ii) An alternating form on $\mathbb{Z}_\ell^2$ with basis $e_1, e_2$ has Gram matrix $\left(\begin{smallmatrix} 0 & a \\ -a & 0 \end{smallmatrix}\right)$ with $a = \hat{\delta}_!(e_1,e_2)$, and its values $\hat{\delta}_!(xe_1+ye_2, ze_1+we_2) = (xw-yz)a$ fill $a\mathbb{Z}_\ell$; the determinant is $a^2$ (Chapter 3). (iii) is the theorem "$M_{f,\lambda}$ is the pull-back" of Chapter 14 (the paper's Proposition prop:pullback), with the Weil pairing and the comparison of étale and Betti cohomology. □

*In the paper* 20.33. The paper's Section 4.2 says that, by Proposition prop:pullback, $\varphi_E^*$ identifies $M_{f,\lambda}$ with $H^1(E,\mathbb{Z}_\ell) \simeq T_\ell E(-1)$, the dual of $T_\ell E$, "so its adjoint is $\mathrm{ad}^0 T$". Proposition 20.32(iii) is that sentence; the adjoint is insensitive to dualising and twisting (Chapter 18), so nothing in the assembly depends on the difference between $M_{f,\lambda}$ and $T_\ell E$.

**Theorem 20.34** (Lemma lem:hecke, with Proposition prop:pullback)**.** *Let $E$ be $X_0(N)$-optimal, $f = f_E$, $\ell$ an odd prime with $\ell \nmid N$ and $E[\ell]$ irreducible. Then*

$$v_\ell(\deg\varphi_E) = v_\ell(r_E) = v_\lambda(\eta^\emptyset_{f,\lambda}).$$

*No freeness or multiplicity-one hypothesis is needed. The first equality, and the fact that the ideal of values of $\langle x, w_N^* y\rangle$ on $\varphi_E^* H^1(E,\mathbb{Z}_\ell)$ is $\deg\varphi_E\,\mathbb{Z}_\ell$, are proved (given the cited theorem*

*of Agashe, Ribet and Stein); the identification of that ideal with the ideal $\eta^{\emptyset}_{f,\lambda}$ of [49] uses the normalisation of $[UwU]_\omega$ (Definition 20.29, Proposition 20.30), the kernel lattice of [49, Sections 1.1.2, 1.6.2] (Definition 20.31) and the definition of $\eta^{\Sigma}_{f}$ in [49, Section 1.7.3], all checked against the published text (Remark 20.83), and the comparison of integral structures, which is proved (Proposition 20.27).*

*Proof.* The first equality is the theorem of Agashe, Ribet and Stein, [3, Theorem 2.2], since $\ell^2 \nmid N$ (Chapter 14). For the second: by the theorem "the degree as a congruence ideal" of Chapter 14 (the paper's Proposition prop:pullback), for every $a, b \in H^1(E, \mathbb{Z}_\ell)$, $\langle \varphi_E^* a, w_N^* \varphi_E^* b \rangle_{X_0(N)} = \varepsilon_N(f) \deg \varphi_E \langle a, b \rangle_E$, and the cup product on $H^1(E, \mathbb{Z}_\ell)$ is unimodular, so the ideal of values of $\langle x, w_N^* y \rangle$ on $\varphi_E^* H^1(E, \mathbb{Z}_\ell) = M_{f,\lambda}$ is $\deg \varphi_E \, \mathbb{Z}_\ell$. By Proposition 20.30(i) and Proposition 20.27, the pairing of Diamond, Flach and Guo on their lattice $M_{f,\lambda}$ (which is the same lattice, by Proposition 20.32(i)) differs from it by the unit $\frac{u}{2}$. So $\eta^{\emptyset}_{f,\lambda} = \deg \varphi_E \, \mathbb{Z}_\ell$. □

*In the paper* 20.35. The paper's proof of Lemma lem:hecke is these two sentences. The paragraph after the lemma records the other route: Lemma 20.21 applied to $L = H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ and $\pi = \phi_f$, with $d = 2$ and $\Delta = (\eta^{\emptyset}_{f,\lambda})^2$, gives without the projection formula only $v_\ell(r_E) \geq v_\lambda(\eta^{\emptyset}_{f,\lambda})$, with equality when $L$ is free over $\mathbb{T}_{\mathfrak{m}}$; it is the argument of Step 2 of the proof of Theorem thm:quant, and the paragraph names the two points that make it correct: the congruence ideal is computed in $\mathbb{T}_{\mathfrak{m}}$, the other local factors lying in the *kernel* $\wp$ (not in its annihilator, Proposition 20.8(iv)); and $\hat{\delta}_!$ is perfect on the $\mathfrak{m}$-component (Proposition 20.4(iv)).

*Remark* 20.36 (Remark rem:free). For $\ell \nmid 2N$ and $\mathfrak{m}$ non-Eisenstein, $J_0(N)[\mathfrak{m}]$ is two-dimensional over $\mathbb{T}/\mathfrak{m}$ (Mazur, Ribet; [163, Theorem 3.5] for $J_1(N)$, transported to $J_0(N)$ through the Shimura subgroup, which is Eisenstein); then $\mathbb{T}_{\mathfrak{m}}$ is Gorenstein and $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}} \simeq \mathbb{T}_{\mathfrak{m}}^2$ [199]. Both steps are in Chapter 14 (multiplicity one stated, Gorenstein and freeness proved given it), and the freeness criterion is in Chapter 3. After Theorem 20.34 this is *not* needed for Theorem thm:main, nor for any step of Theorem thm:quant (Theorem 20.72), nor for the equality clauses of either. It remains the conceptual reason why the Hecke and the cohomological congruence ideals agree at $\ell \nmid 2N$ (Lemma 20.21 with equality), and with Lemma 20.21 it gives a second proof of $v_\ell(r_E) = v_\ell(\deg \varphi_E)$ for $\ell \nmid 2N$, $E[\ell]$ irreducible. Freeness fails in general when $\ell \mid N$: `99a1` at $\ell = 3$ (Example 20.24(d)).

**Example 20.37** (values of $\eta^{\emptyset}$)**.** By Theorem 20.34, $v_\ell(\eta^{\emptyset}_{f,\ell}) = v_\ell(\deg \varphi_E)$ whenever $\ell \nmid 2N$ and $E[\ell]$ is irreducible. At $\ell = 3$ (all curves below have $E[3]$ irreducible and $3 \nmid N$; degrees from PARI/GP and Cremona's tables): `11a1`, $\deg \varphi_E = 1$, $v_3 = 0$; `77c1`, 6, $v_3 = 1$; `121a1`, 6, $v_3 = 1$; `121d1`, 24, $v_3 = 1$; `539c1`, 288, $v_3 = 2$; `539d1`, 72, $v_3 = 2$. For the depleted forms of Example 20.6, Theorem 20.39 below gives $v_3(\eta^{\{7\}}_{77\mathrm{c},3}) = v_3(\eta^{\emptyset}_{77\mathrm{c},3}) + v_3(1 - 7^{-2}) = 1 + 1 = 2$ and $v_3(\eta^{\{7\}}_{11\mathrm{a},3}) = 0 + v_3(\frac{6}{7} \cdot \frac{64-4}{49}) = 0 + 2 = 2$, equal to $v_3(288)$ and $v_3(504)$: this is Theorem thm:quant read backwards (Theorem 20.72). The two factors are those of Proposition 20.41: `77c1` is multiplicative at 7, so the factor is $1 - 7^{-2} = \frac{48}{49}$ and $v_3(48) = 1$; `11a1` is good at 7 with $a_7 = -2$, so the factor is $\frac{7-1}{7} \cdot \frac{(7+1)^2 - (-2)^2}{7^2} = \frac{6}{7} \cdot \frac{60}{49}$, and $v_3(6) + v_3(60) = 1 + 1 = 2$. In the second case both parts of Wiles's factor contribute, $7 \equiv 1 \pmod 3$ and $a_7 \equiv -(7+1) \pmod 3$.

## 20.6 The variation of the congruence ideal with Σ

This section treats the paper's Theorem thm:dfg14, [49, Proposition 1.4(c)]: adding a prime $p$ to $\Sigma$ multiplies the naive congruence ideal by the naive Euler factor $L_p^{\mathrm{nv}}(A_f, 1)^{-1}$. The paper

uses it twice: in the proof of Theorem thm:main, to pass from level $N$ to $\Sigma = \{p \mid N\}$, and in Step 3 of the proof of Theorem thm:quant, to pass from the newform $g = f_{E_S}$ at level $N_S$ to its depletion at level $N$. We state it, explain the three cases of the paragraph after it, and show how part (c) follows from parts (a) and (b), following the published proof: homothety of lattices, and primitivity of $f^\Sigma$ read through the integral comparison isomorphism.

**Definition 20.38** (the map $\gamma$)**.** Let $f$ be a newform of weight 2 on $\Gamma_0(M)$ and $\Sigma$ a finite set of primes. For $m \mid M^\Sigma/M = \prod_{p\in\Sigma} p^{\delta_p}$ let $\epsilon_m \colon H^1(X_0(M)) \to H^1(X_0(M^\Sigma))$ be the pull-back along the degeneracy map $z \mapsto mz$ (on forms, $h \mapsto h(mz)$; [49, Section 1.7.3] normalise $\epsilon_m$ by the factor $m^{1-k} = m^{-1}$, a unit when $\ell \nmid m$), and let $\phi_m$ be the endomorphism of the cohomology at level $M$ which is multiplicative in coprime $m$ with $\phi_1 = 1$, $\phi_p = -T_p$ and $\phi_{p^2} = p$. Put

$$\gamma = \sum_{m \mid M^\Sigma/M} \epsilon_m \phi_m,$$

and let $\gamma^t$ be its adjoint for the pairings $\hat\delta_!$ at the two levels. On $q$-expansions $\gamma(f) = \prod_{p\in\Sigma,\,\delta_p=2}(1 - a_pV_p + pV_p^2)\prod_{p\in\Sigma,\,\delta_p=1}(1 - a_pV_p)\, f = f^\Sigma$, with $V_p \colon h(z) \mapsto h(pz)$ (Chapter 13, depletion).

The $q$-expansion formula is a direct check, which also shows why the coefficients $\phi_m$ are the right ones. As $V_p$ sends $\sum a_n q^n$ to $\sum a_n q^{pn}$, the $n$-th coefficient of $(1 - a_pV_p + pV_p^2)f$ is $a_n - a_p a_{n/p} + p\, a_{n/p^2}$ (terms with non-integral index read as 0). For $p \nmid n$ this is $a_n$. For $n = pm$ it is $a_{pm} - a_p a_m + p\, a_{m/p}$, which vanishes by the Hecke recursion $a_{pm} = a_p a_m - p\, a_{m/p}$ at a good prime. At a multiplicative prime the recursion is $a_{pm} = a_p a_m$, and $(1 - a_pV_p)f$ has $n$-th coefficient $a_n - a_p a_{n/p}$, again 0 for $p \mid n$. So the operator kills exactly the coefficients at multiples of $p$, which is depletion at $p$; on cohomology, $V_p$ is $\epsilon_p$, the scalar $a_p$ is $-\phi_p = T_p$ acting on $f$, and $p$ is $\phi_{p^2}$. For example, for $f = f_{11a}$ and $p = 7$ ($a_7 = -2$) one gets $f^{\{7\}} = (1 + 2V_7 + 7V_7^2)f$ at level $11 \cdot 7^2 = 539$, the depleted form of Example 20.6.

**Theorem 20.39** (Diamond, Flach and Guo, Proposition 1.4; the paper's Theorem thm:dfg14)**.** *Let $f$ be a newform of weight 2 on $\Gamma_0(M)$ with rational coefficients, $\ell$ a prime with $\ell \nmid 2M$, and $\Sigma$ a finite set of primes with $\ell \notin \Sigma$.*

*(a) $\gamma$ restricts to an isomorphism $M_f \otimes \mathbb{Q} \to M_{f^\Sigma} \otimes \mathbb{Q}$ of the rational eigenspaces, compatible with $f \mapsto f^\Sigma$.*

*(b) On $M_f$, $\gamma^t \circ \gamma = u \cdot \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}$ with $u = \prod_{p\in\Sigma,\,\delta_p=1}(-a_p)\prod_{p\in\Sigma,\,\delta_p=2} p$, an $\ell$-adic unit; in particular $\hat\delta_!$ is non-degenerate on $M_{f^\Sigma}$.*

*(c) If $\bar\rho_f$ is irreducible, then $\gamma(M_{f,\lambda}) = M_{f^\Sigma,\lambda}$ and*

$$\eta^\Sigma_{f,\lambda} = \eta^\emptyset_{f,\lambda} \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}.$$

*Sketch of proof.* (a) The maps $\epsilon_m$ are injective on the $f$-isotypic part with images the $f(mz)$-parts of the old space (multiplicity one, Chapter 13), and $\gamma$ maps the $f$-part onto the line of the depletion and its conjugate, by the $q$-expansion formula; it commutes with $G_\mathbb{Q}$ because the degeneracy maps are defined over $\mathbb{Q}$.

(b) This is a double-coset computation (Diamond, Flach and Guo refer to [207, Chapter 2] and to page 121 of the 1995 printing of [38] in *Current Developments in Mathematics*); we

sketch it in the case of one good prime $p$. The pairings $\hat{\delta}_!(\epsilon_i x, \epsilon_j y)$ for $i,j \in \{1,p,p^2\}$ are the cup products at level $M$ of $x$ with $y$ transformed by the composites $\epsilon_i^t\epsilon_j$, which are Hecke operators at level $M$ ($\epsilon_1^t\epsilon_1$ is the degree $(p+1)p$ of the covering, $\epsilon_1^t\epsilon_p$ is $T_p$, and so on); on $M_f$ each acts by a scalar polynomial in $a_p$ and $p$, and expanding $\gamma^t\gamma = \sum_{i,j}\phi_i^t\epsilon_i^t\epsilon_j\phi_j$ gives a scalar, which equals $p\,(p-1)((p+1)^2 - a_p^2)/p^3$ up to the normalisations of the $\epsilon_m$, that is $u$ times Wiles's factor. The multiplicative case is the $2\times 2$ analogue. The non-degeneracy follows since every factor is nonzero (Proposition 20.41). We do not give the details; see [49, Section 1.7.3].

(c) The lattices $\gamma(M_{f,\lambda})$ and $M_{f^\Sigma,\lambda}$ are $G_{\mathbb{Q}}$-stable lattices in the same two-dimensional representation, by (a), and $\gamma(M_{f,\lambda}) \subseteq M_{f^\Sigma,\lambda}$ because $\gamma$ preserves integral cohomology ($p \neq \ell$ for $p \in \Sigma$, so the normalising factors of the $\epsilon_m$ are units) and maps the $f$-eigenspace into the $f^\Sigma$-eigenspace by (a), and $M_{f^\Sigma,\lambda}$ is the intersection of that eigenspace with the integral cohomology. As $\bar{\rho}_f$ is irreducible, every $G_{\mathbb{Q}}$-stable lattice in it is homothetic to a given one (Chapter 15, lattices), so $\gamma(M_{f,\lambda}) = \ell^n M_{f^\Sigma,\lambda}$ for some $n \geq 0$. Diamond, Flach and Guo get $n=0$ from the de Rham side, not from a statement about the degeneracy maps modulo $\ell$; their proof reads: "If $\bar{\mathcal{M}}_{f,\lambda}$ is irreducible, then the image of $\mathcal{M}_{f,\lambda}$ must be of the form $\lambda^n\mathcal{M}_{f^\Sigma,\lambda}$ for some $n \geqslant 0$; since $\gamma_{\mathrm{dR}}(f) = f^\Sigma$, we see that $n=0$." The input is the integral comparison isomorphism that is part of their integral structures at $\lambda$ [49, Sections 1.1.2, 1.2.5], which we cite and do not prove (it is the theory of Fontaine and Laffaille [68] applied to the modular curve, which has good reduction at $\ell \nmid M^\Sigma$): $M_{f,\lambda}$ and $M_{f^\Sigma,\lambda}$ are the images under the fully faithful Fontaine-Laffaille functor of filtered modules $D_f$ and $D_{f^\Sigma}$ whose step $\mathrm{Fil}^1$ is $\mathbb{Z}_\ell f$, respectively $\mathbb{Z}_\ell f^\Sigma$ (for a normalised eigenform the first filtration step of the de Rham realisation is spanned by the form, [49, (15)]), and $\gamma$ is a morphism of these structures. If $\gamma(M_{f,\lambda}) = \ell^n M_{f^\Sigma,\lambda}$, full faithfulness gives an isomorphism $D_f \to \ell^n D_{f^\Sigma}$ of filtered modules, which maps $\mathrm{Fil}^1 D_f = \mathbb{Z}_\ell f$ onto $\ell^n\mathbb{Z}_\ell f^\Sigma$; as it maps $f$ to $\gamma(f) = f^\Sigma$, $n=0$. (Saturation statements of the type of Ihara's lemma for the degeneracy maps modulo $\ell$ would give another route; the published proof does not use them, and neither does the book.) Now for $x, y \in M_{f,\lambda}$,

$$\hat{\delta}_!(\gamma x, \gamma y) = \hat{\delta}_!(x, \gamma^t\gamma y) = u \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f,1)^{-1}\,\hat{\delta}_!(x,y),$$

so the ideal of values on $M_{f^\Sigma,\lambda} = \gamma(M_{f,\lambda})$ is $\eta_{f,\lambda}^{\emptyset}$ times $\prod_p L_p^{\mathrm{nv}}(A_f,1)^{-1}$, $u$ being a unit ($a_p = \pm 1$ at $\delta_p = 1$, and $p \neq \ell$). □

*Remark* 20.40 (status of the statement). The paper cites only part (c), as "[49, Proposition 1.4(c)]", under the standing assumptions of its Section 4.1: $f$ with rational coefficients, $\lambda = \ell \nmid 2M$, $\Sigma$ a finite set of primes; and it states the theorem with $\ell \notin \Sigma$. The hypothesis is needed: for $\ell \in \Sigma$ the statement of (c) is false, by the example that the paragraph after the paper's theorem gives, which we write out. Take $f = f_{11a}$, $\ell = 3$ (so $\ell \nmid 2M = 22$) and $\Sigma = \{3\}$. Here $\delta_3 = 2$ and $a_3 = -1$, so by Proposition 20.41(ii)

$$L_3^{\mathrm{nv}}(A_f,1)^{-1} = \frac{2}{3}\cdot\frac{16-1}{9} = \frac{2\cdot 15}{27} = \frac{10}{9}, \qquad v_3\Big(\frac{10}{9}\Big) = -2$$

(recomputed with PARI/GP, also from the Euler factor $(1+2X-6X^2-27X^3)^{-1}$ of the symmetric square at 3, evaluated at $X = 3^{-2}$). As $\eta_{f,3}^{\emptyset} = \deg\varphi_E\,\mathbb{Z}_3 = \mathbb{Z}_3$ (Example 20.37), the right side of (c) would be $3^{-2}\mathbb{Z}_3$, which is not contained in $\mathbb{Z}_3$; but however $\eta_{f,3}^{\{3\}}$ is read at the level $M^\Sigma = 99$, which 3 divides (for instance as the ideal of values of $\langle x, w_{99}^* y\rangle$ on $H^1(X_0(99),\mathbb{Z}_3)[\ker\theta_{f^\Sigma}]$), it is an ideal of values of an integral pairing on an integral lattice, so it lies in $\mathbb{Z}_3$. Every use in the

paper has $\ell \notin \Sigma$ ($\Sigma = \{p \mid N\}$ with $\ell \nmid N$, and $\Sigma = S$ with $\ell \nmid N$), so we state the theorem with $\ell \notin \Sigma$. In [49] the hypothesis is a standing assumption: in their Section 1.7.3, $\Sigma$ is a finite subset of $S = S_{\mathbf{f}}(K) \setminus \{\lambda\}$, the finite primes of $K$ other than $\lambda$. Part (b) is as printed: [49, Proposition 1.4(b)] reads $\gamma^t \circ \gamma = \phi_{N^\Sigma/N_f} \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f,1)^{-1}$ on $M_f$, with $\phi_p = -T_p$ and $\phi_{p^2} = \psi(p)p^{k-1}$, so for weight 2 and trivial character $\phi_{N^\Sigma/N_f}$ acts on $M_f$ by $u$; the proof of their Theorem 2.15 records that this element, $\prod_{\delta_p=1}(-a_p)\prod_{\delta_p=2}\psi(p)p^{k-1}$, is a unit at $\lambda$. The book uses only (c).

**Proposition 20.41** (the three cases)**.** *Let $f$ be a newform of weight 2 on $\Gamma_0(M)$ with rational coefficients and $p$ a prime. The factor $L_p^{\mathrm{nv}}(A_f,1)^{-1}$ of Theorem 20.39 is:*

*(i)* $1$ *if $\delta_p = 0$ ($p^2 \mid M$, additive for $f = f_E$): adding $p$ to $\Sigma$ changes neither the level nor the ideal;*

*(ii) if $\delta_p = 2$ ($p \nmid M$, good), with $\alpha, \beta$ the roots of $X^2 - a_pX + p$,*

$$\Big(1 - \frac{\alpha}{\beta}p^{-1}\Big)\Big(1 - p^{-1}\Big)\Big(1 - \frac{\beta}{\alpha}p^{-1}\Big) = \frac{p-1}{p}\cdot\frac{(p+1)^2 - a_p^2}{p^2},$$

*Wiles's level-raising factor, of $\ell$-adic valuation $v_\ell(p-1) + v_\ell((p+1)^2 - a_p^2)$ for $\ell \neq p$;*

*(iii)* $1 - p^{-2}$ *if $\delta_p = 1$ ($p \,\|\, M$, multiplicative), of valuation $v_\ell(p^2-1)$.*

*So only the primes with $\delta_p \geq 1$ contribute.*

*Proof.* (i) is the definition of the naive factor. (ii) The geometric Frobenius acts on $A_f = \mathrm{ad}^0 V$ with eigenvalues $\alpha/\beta$, $1$, $\beta/\alpha$ (Appendix B, B-C3(d); this set does not depend on the choice of Frobenius), so $L_p(A_f,1)^{-1}$ is the displayed product. Expanding the outer factors, $(1 - \frac{\alpha}{\beta}p^{-1})(1 - \frac{\beta}{\alpha}p^{-1}) = 1 - \frac{\alpha^2+\beta^2}{\alpha\beta}p^{-1} + p^{-2} = 1 - \frac{a_p^2-2p}{p^2} + \frac{1}{p^2} = \frac{(p+1)^2-a_p^2}{p^2}$, using $\alpha+\beta = a_p$, $\alpha\beta = p$. (iii) At a multiplicative prime $(\mathrm{ad}^0 V)^{I_p}$ is a line on which the geometric Frobenius acts by $p^{-1}$ (Chapter 19), so $L_p(A_f,s)^{-1} = 1 - p^{-1-s}$, which is $1 - p^{-2}$ at $s = 1$; equivalently $L_p^{\mathrm{nv}}(A_f,s)^{-1} = 1 - a_p^2 p^{-1-s}$ with $a_p^2 = 1$. The valuations follow, $p$ being prime to $\ell$. □

*Remark* 20.42 (what the factors count). At a good prime $p \neq \ell$, $v_\ell((p+1)^2 - a_p^2) > 0$ is Ribet's level-raising condition $a_p \equiv \pm(p+1) \pmod{\ell}$ for a newform of level $Mp$ congruent to $f$ (Chapter 15), and $v_\ell(p-1) > 0$ allows congruences with forms of level $Mp^2$ whose local component at $p$ is a ramified principal series of $\ell$-power order. The valuation of the factor is the number of new congruences that $f^\Sigma$ acquires at the higher level. At a multiplicative prime the factor $1 - p^{-2}$ is the analogue for raising $p \,\|\, M$ to $p^2$. The paper's paragraph after Theorem thm:dfg14 ("adding a good prime $p$ multiplies the ideal by … Wiles's level-raising factor; adding a multiplicative prime multiplies it by $1 - p^{-2}$") is Proposition 20.41.

*Remark* 20.43 (Ihara's lemma and the Eisenstein case). In the proof of part (c), as in [49], irreducibility of $\bar\rho_f$ enters through the homothety of lattices: without it $\gamma(M_{f,\lambda})$ need not be a multiple of $M_{f^\Sigma,\lambda}$, and the primitivity argument does not fix the index. (It does not enter through the comparison of integral structures, which holds with no localisation, Proposition 20.27.) Ihara's lemma, the injectivity of degeneracy maps modulo $\ell$ on non-Eisenstein components, is not used. In the Eisenstein case the factor relating $\eta^\Sigma$ and $\eta^\emptyset$ can differ from the Euler factor by a non-unit; the paper's Conjecture conj:eis and Section sec:eis concern that case, which lies outside the theorems of the paper's Section 4.

*Remark* 20.44 (refined integral structures). Diamond, Flach and Guo also use refined integral structures [49, Section 1.8], which change the level at exceptional primes $p \equiv -1 \pmod{\ell}$ where the local representation is induced from the unramified quadratic extension, and add an auxiliary prime $r$ at which the Euler factor is a unit, so that the relevant cover has degree prime to $\ell$ and the pairing is perfect integrally. With them the naive factors are replaced by the true ones; this is how the true Euler factors enter their proof of the Tamagawa number conjecture. The paper never uses the refined ideals: it combines the naive ideal (Theorem 20.39) with Lemma lem:dfg21, whose local terms carry the true factors, and the difference between the two is exactly $\sum_q e_\ell(E,q)$ (Step 5 of Theorem 20.60).

## 20.7 The exceptional set and the theorems of Diamond, Flach and Guo

This section states the theorems of Diamond, Flach and Guo that the paper imports in its Section 4.2: Theorem thm:dfg37 (the length of the $\Sigma$-imprimitive adjoint Selmer group is the valuation of the naive congruence ideal), Theorem thm:dfg27 (vanishing of global invariants and Selmer groups), and their main theorem on the Tamagawa number conjecture, together with the exceptional set $S_f$ and the notion of minimal ramification that enter the hypotheses. It also records what has been checked about the case $\ell = 3$, which Chapter 18 left open. The proofs are by the Taylor-Wiles method, outlined in Section 20.8.

**Definition 20.45** (the exceptional set)**.** Let $f$ be a newform of weight $k \geq 2$ and level $N$ with coefficients in $K$. Diamond, Flach and Guo define $S_f$ as the set of finite primes $\lambda$ of $K$ such that either $\lambda \mid Nk!$, or the residual representation $M_{f,\lambda}/\lambda M_{f,\lambda}$ is not absolutely irreducible when restricted to $G_F$, where $F = \mathbb{Q}(\sqrt{(-1)^{(\ell-1)/2}\ell})$ and $\lambda \mid \ell$ [49, Section 0.2]; the definition is repeated as the display [49, (31)] at the start of their Section 2 (p. 695), which is the number the paper cites. For $k = 2$ and $K = \mathbb{Q}$: $\ell \in S_f$ iff $\ell \mid 2N$ or $\bar{\rho}|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ is not absolutely irreducible.

Why this particular field. Write $\ell^* = (-1)^{(\ell-1)/2}\ell$, so $\ell^* = -3, 5, -7, -11, 13$ for $\ell = 3, 5, 7, 11, 13$; the sign is chosen so that $\ell^* \equiv 1 \pmod 4$, and then $F = \mathbb{Q}(\sqrt{\ell^*})$ is the unique quadratic field ramified only at $\ell$, the quadratic subfield of $\mathbb{Q}(\mu_\ell)$ (for $\ell = 3$, $F = \mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\mu_3)$). The Taylor-Wiles method needs auxiliary primes found inside $\mathbb{Q}(\mu_{\ell^n})$ (Remark 20.58(c)), so what matters is irreducibility after restriction to $G_{\mathbb{Q}(\mu_\ell)}$, and for an absolutely irreducible $\bar{\rho}$ of $G_\mathbb{Q}$ the condition in the definition is the one the method needs [49, Section 3.1]. An absolutely irreducible representation of $G_\mathbb{Q}$ that becomes reducible on the index-two subgroup $G_F$ is induced from a character of $G_F$ (Clifford theory for a subgroup of index 2, Chapter 4), so its projective image is dihedral; for elliptic curves with $\ell \nmid N$ this happens only at $\ell = 3$, for the normaliser of split Cartan image 3Ns of Proposition 20.46.

**Proposition 20.46** (the paper's Proposition prop:Sf)**.** *Let $E/\mathbb{Q}$ have conductor $N$, $f = f_E$, and $\ell$ an odd prime with $\ell \nmid N$. Then $\ell \notin S_f$ if and only if $E[\ell]$ is irreducible and, when $\ell = 3$, the mod 3 image is not conjugate to $N_s(\mathbb{F}_3)$ (Sutherland's label* 3Ns*); equivalently, $E[\ell]$ is irreducible and $h_\ell(E) = 0$. In particular, for $\ell \geq 5$ with $\ell \nmid N$, $\ell \notin S_f$ as soon as $E[\ell]$ is irreducible.*

*Proof.* This is the paper's Proposition prop:Sf, proved in Chapter 11 (with the fundamental characters of Chapter 15 and the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ of Chapter 1); the equivalence with

$h_\ell(E) = 0$ is Lemma lem:h0 (Chapters 11 and 18). In outline: an irreducible $\bar\rho$ is absolutely irreducible (complex conjugation has eigenvalues $\pm 1$), and its restriction to $G_F$ fails to be absolutely irreducible iff $\bar\rho \simeq \bar\rho \otimes \chi_{\ell^*}$; restricted to inertia at $\ell$ this is impossible for $\ell \geq 5$ (compare the fundamental characters in the ordinary and supersingular cases), and for $\ell = 3$ it happens exactly for the image $N_s(\mathbb{F}_3)$. □

**Definition 20.47** (the $\Sigma$-imprimitive Selmer group)**.** Let $\Sigma$ be a finite set of primes not containing $\ell$ and $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. Following [49, Section 2.1], $H^1_\Sigma(\mathbb{Q}, W)$ is the subgroup of $H^1(\mathbb{Q}, W)$ of classes that are unramified at every $p \notin \Sigma \cup \{\ell\}$ (that is, lie in $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$), satisfy the Bloch-Kato condition $H^1_f(\mathbb{Q}_\ell, W)$ at $\ell$, and are unrestricted at $p \in \Sigma$ and at $\infty$ (Chapter 18).

If $\Sigma$ contains every prime $p \neq \ell$ at which $W$ is ramified, then at $p \notin \Sigma \cup \{\ell\}$ the module $W$ is unramified, $W^{I_p} = W$ is divisible, and $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) = H^1_f(\mathbb{Q}_p, W)$ (Chapter 18); so $H^1_\Sigma(\mathbb{Q}, W)$ is also the group with the Bloch-Kato conditions outside $\Sigma$, and $H^1_f(\mathbb{Q}, W) \subseteq H^1_\Sigma(\mathbb{Q}, W)$. Diamond, Flach and Guo write the unramified condition in general because $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$ need not be divisible at a ramified prime (Exercise 20.92). At $\ell$ the condition is the Bloch-Kato one; for the finite modules $W_n = W[\lambda^n]$ they define it through Fontaine-Laffaille theory, as the extensions that lie in the essential image of the Fontaine-Laffaille functor, and [49, Proposition 2.2] shows that the direct limit over $n$ is the Bloch-Kato condition.

**Definition 20.48** (minimal ramification)**.** Let $\rho_0 \colon G_\mathbb{Q} \to \mathrm{GL}_2(k)$ be a residual representation and $p \neq \ell$. A lift $\rho$ of $\rho_0$ (to $\mathcal{O}$, or to an artinian $\mathcal{O}$-algebra $A$) is *minimally ramified at $p$* if $\rho(I_p) \simeq \rho_0(I_p)$ (reduction is an isomorphism on the images of inertia) when $\#\rho_0(I_p) \neq \ell$, and if the quotient $M/M^{I_p}$ of the module $M$ of $\rho$ by its inertia invariants is free of rank one over $A$ (for $\rho$ over $\mathcal{O}$: $\dim (M \otimes K)^{I_p} = 1$) when $\#\rho_0(I_p) = \ell$ [49, Section 3.1, p. 715, and p. 717]. (The case distinction of [49] is $\#\rho_0(I_p) = \ell$ or not, not whether $\ell$ divides $\#\rho_0(I_p)$.) $M_{f,\lambda}$ is *minimally ramified outside* $\Sigma$ if it is so at every prime $p \notin \Sigma \cup \{\ell\}$.

If $\Sigma$ contains every prime $p \neq \ell$ at which $\rho_f$ is ramified, then $M_{f,\lambda}$ is minimally ramified outside $\Sigma$: at $p \notin \Sigma \cup \{\ell\}$, $\rho_f(I_p) = 1 = \rho_0(I_p)$. For $f = f_E$ and $\Sigma = \{p \mid N\}$, $\rho_{E,\ell}$ is unramified outside $N\ell$ (Chapter 11), so the paper's choice qualifies. Diamond, Flach and Guo also use smaller sets $\Sigma$ internally (the minimal deformation problem at the twist-minimal level); the paper never needs them.

An example shows what the condition says at a prime that is kept out of $\Sigma$. Let $E$ have multiplicative reduction at $p \neq \ell$ with $n = -v_p(j_E)$. By the Tate parametrisation a generator of tame inertia acts on $T_\ell E$ by a matrix $\left(\begin{smallmatrix} 1 & n\cdot c \\ 0 & 1 \end{smallmatrix}\right)$ with $c \in \mathbb{Z}_\ell^\times$ (Chapter 11), so on $E[\ell]$ it acts trivially if $\ell \mid n$ and with order exactly $\ell$ if $\ell \nmid n$. For `11a1`, $p = 11$, $n = 5$ and $\ell = 3$, we are in the second case, $\#\rho_0(I_{11}) = 3 = \ell$, and the condition asks that $(V_3 E)^{I_{11}}$ be a line, which it is. So $M_{f,3}$ is minimally ramified at 11, and the empty set already satisfies the hypothesis of Theorem 20.49 for this curve. When $\ell \mid n$ the first clause asks that the lift be unramified at $p$, because $\rho_0(I_p) = 1$, and $\rho_{E,\ell}$ is not; so $p$ must then be put into $\Sigma$. Taking $\Sigma = \{p \mid N\}$ avoids the case distinction altogether.

**Theorem 20.49** (Diamond, Flach and Guo, Theorem 3.7; the paper's Theorem thm:dfg37)**.** *Let $f$ be a newform of weight $k \geq 2$ and level $N$ with coefficients in $K$, $\lambda \notin S_f$, and $\Sigma$ a finite set of primes not containing $\ell$ such that $M_{f,\lambda}$ is minimally ramified outside $\Sigma$. Then $H^1_\Sigma(\mathbb{Q}, A_{f,\lambda} \otimes K_\lambda/\mathcal{O}_\lambda)$ is finite, of length $v_\lambda(\eta^\Sigma_{f,\lambda})$. For $f = f_E$, $\ell \notin S_f$ and $\Sigma = \{p \mid N\}$:*

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = v_\ell(\eta^\Sigma_{f,\ell}), \qquad W = \mathrm{ad}^0 T_\ell E \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell.$$

*Reference.* We do not prove this; it is [49, Theorem 3.7] (p. 723; checked: the hypotheses are $\lambda \notin S_f$ and $M_{f,\lambda}$ minimally ramified outside $\Sigma$, $\ell \notin \Sigma$, and $\eta_f^{\Sigma}$ is the ideal "defined before Proposition 1.4"); see [49, Section 3] and the outline in Section 20.8. For $f = f_E$ the module $A_{f,\lambda} = \mathrm{ad}^0 M_{f,\lambda}$ is $\mathrm{ad}^0 T_\ell E$ by Proposition 20.32(iii), and $\Sigma = \{p \mid N\}$ satisfies the hypothesis by the remark after Definition 20.48. □

**Theorem 20.50** (Diamond, Flach and Guo, Theorem 0.2)**.** *The introduction of [49] states the theorem in the form the paper uses: "Let $f$ be a newform of weight $k \geq 2$ and level $N$ with coefficients in $K$. Suppose $\Sigma$ is a finite set of rational primes containing those dividing $N$. Suppose that $\lambda$ is a prime of $K$ which is not in $S_f$ and does not divide any prime in $\Sigma$. Then the $\mathcal{O}_{K,\lambda}$-module $H^1_{\Sigma}(G_{\mathbb{Q}}, A_{f,\lambda}/\mathcal{A}_{f,\lambda})$ has length $v_\lambda(\eta_f^{\Sigma})$." It is labelled there "Theorem 0.2 (= Theorem 3.7)". Here $A_{f,\lambda}/\mathcal{A}_{f,\lambda}$ is the divisible module $A_{f,\lambda} \otimes K_\lambda/\mathcal{O}_\lambda$, that is $W$ for $f = f_E$.*

*Reference.* We do not prove this; it is the case of Theorem 20.49 in which $\Sigma$ contains the primes dividing $N$, which suffices for the paper (Chapter 18 states it as a pointer). □

**Theorem 20.51** (Diamond, Flach and Guo, Theorem 2.7; the paper's Theorem thm:dfg27)**.** *If $\ell \notin S_f$, then $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$, and the same holds for $A_f = \mathrm{ad}^0 V$: $H^0(\mathbb{Q}, A_f) = H^1_f(\mathbb{Q}, A_f) = 0$. Here $V^D = A_f(1)$.*

*Reference.* We do not prove this; it is [49, Theorem 2.7] (p. 699, checked), stated there for $A_f$ and $B_f = A_f(1)$ together with $\mathrm{ord}_{s=0} L(A_f, s) = \mathrm{ord}_{s=0} L(B_f, s) = 0$. Chapter 18 states it and sketches the derivation, which is the published proof: the $H^0$ vanish by absolute irreducibility, $\dim H^1_f(\mathbb{Q}, A_f) = \dim H^1_f(\mathbb{Q}, B_f)$ by the Euler characteristic formula of Fontaine and Perrin-Riou, and $H^1_f(\mathbb{Q}, A_f) = 0$ because the divisible part of $H^1_f(\mathbb{Q}, W)$ lies in the finite group $H^1_{\Sigma}(\mathbb{Q}, W)$ of Theorem 20.49. Only $H^1_f(\mathbb{Q}, V^D) = 0$ is needed in the paper; $H^0(\mathbb{Q}, V^D) = 0$ holds for every $E$ (Chapter 18). □

**Theorem 20.52** (Diamond, Flach and Guo, Theorem 0.1)**.** *"Let $f$ be a newform of weight $k \geq 2$ and level $N$ with coefficients in $K$. If $\lambda$ is not in $S_f$, then the $\lambda$-part of the Bloch-Kato conjecture holds for $A_f$ and $A_f(1)$." [49, Theorem 0.1], which is labelled "(= Theorem 2.15)" there.*

*Reference.* We do not prove this; see [49, Sections 2.3, 2.4]. Its proof (that of their Theorem 2.15, pp. 709 to 711) takes for $\Sigma$ a nonempty set of primes $p \neq \ell$ containing those where $M_\lambda$ is ramified, and combines Theorem 20.39 (with the unit $u$), the comparison of $H^1_{\Sigma}$ with $H^1_f$ (their Lemma 2.1, the paper's Lemma lem:dfg21), the local formula of Fontaine and Perrin-Riou (the paper's Lemma lem:fpr) with [49, (57)], the triviality of the Tamagawa factor at $\ell$ [49, Proposition 2.16] and Theorem 20.49, with the Petersson norm ([49, Lemma 2.12]) and Deligne's period [49, Theorem 2.10] in place of the modular degree: it is the same assembly as the proof of Theorem thm:main (Theorem 20.60). The statement of the conjecture is in Chapter 18. □

*Remark* 20.53 (does Diamond, Flach and Guo's theorem cover $\ell = 3$ in weight 2?). Chapter 18 noted that the adjoint has Hodge-Tate weights $\{-1, 0, 1\}$, so that Fontaine-Laffaille theory cannot be applied to $\mathrm{ad}^0 T$ directly when $\ell = 3$, and left open how Diamond, Flach and Guo treat that case. What has been checked against the published text is this. The abstract of [49] says that $\lambda$ "is any prime of $K$ not dividing $Nk!$" such that the residual representation is absolutely irreducible over $\mathbb{Q}(\sqrt{(-1)^{(\ell-1)/2}\ell})$; the definition of $S_f$ in the introduction excludes only $\lambda \mid Nk!$ and the non-absolutely-irreducible case; and Theorems 0.1 and 0.2, quoted above, are stated for every $\lambda \notin S_f$. For $k = 2$, $Nk! = 2N$, so as *statements* these theorems include $\ell = 3$ whenever

$3 \nmid N$ and $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))}$ is absolutely irreducible; this is what the paper means by "It covers $\ell = 3$ in weight 2: $S_f$ excludes only the primes dividing $2N$ and the induced case". How the proof treats $\ell = 3$ has also been checked against the published text.

(1) Fontaine-Laffaille theory is applied to $M_{f,\lambda}$, not to the adjoint. A crystalline representation is *short* if $\mathrm{Fil}^0 D = D$, $\mathrm{Fil}^\ell D = 0$ and no nonzero quotient $V'$ has $V' \otimes \mathbb{Q}_\ell(\ell-1)$ unramified [49, Section 1.1.2], and $M_{f,\lambda\text{-crys}}$ is an object of the relevant category "if $\ell \nmid N$ and $\ell > k$" (p. 713), which holds for $k = 2$, $\ell = 3$. Section 1.7.1 of [49] records the obstruction of Chapter 18: the adjoint gives such an object only after twisting "if $\ell - 1 > 2(k-1)$", which fails for $k = 2$, $\ell = 3$.

(2) The local condition at $\ell$ for $W = \mathrm{Hom}(V_1, V_2) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ with $V_1, V_2$ short is defined at finite level by extensions in the essential image of the Fontaine-Laffaille functor and compared with the Bloch-Kato condition in [49, Proposition 2.2]; the dimension count of [49, Corollary 2.3] needs only $\ell > 2$.

(3) The refined integral structures assume that $\bar\rho$ is irreducible and, "if $3 \in \lambda$", absolutely irreducible on $G_{\mathbb{Q}(\mu_3)}$ [49, Section 1.8]; this is the condition $3 \notin S_f$.

(4) The Taylor-Wiles axioms ask for absolute irreducibility over the quadratic subfield of $\mathbb{Q}(\mu_\ell)$ and Serre weight $2 \le k \le \ell - 1$ [49, Section 3.1], which holds for $k = 2$, $\ell = 3$.

(5) The case $\ell = k + 1$ needs one extra input, in the triviality of the Tamagawa factor at $\ell$ for $A_f(1)$ [49, Proposition 2.16]: when $a_\ell \equiv 0 \pmod\lambda$ their Lemma 2.17 suffices; in the ordinary case ($a_\ell \not\equiv 0 \pmod\lambda$ and $\ell = k+1$) they appeal to Perrin-Riou's results on her conjecture $C_{EP}(V)$ for ordinary representations, with Colmez's proof of the reciprocity law $\mathrm{Rec}(V)$. This enters only Theorem 0.1, not Theorems 2.7 and 3.7, which are what Theorem thm:main uses.

In the same direction, [49, Lemma 2.5] (an irreducible residual representation is absolutely irreducible over $F$) assumes $\lambda \nmid N(2k-1)(2k-3)k!$, which for $k = 2$ excludes $\ell = 3$: this is the room for the image 3Ns. The book therefore treats Theorem thm:main at $\ell = 3$ (with $3 \nmid N$ and image not 3Ns) as a theorem.

## 20.8 The Taylor-Wiles method in outline

This section explains, without proofs, why the length of a Selmer group should equal the valuation of a congruence ideal, which is the content of Theorem 20.49. The reader's natural question, "why should a Selmer group know about congruences?", has a short answer: the Selmer group is the dual of the cotangent space of a deformation ring, the congruence ideal is the congruence ideal of a Hecke algebra, the two rings are isomorphic, and Wiles's numerical criterion converts the isomorphism into an equality of lengths. The paper does not use the method beyond Theorem 20.49, but its hypotheses are exactly those of the method, and Section 1.5 of the paper contrasts it with the "level lowering and twisting" of Theorems thm:qual and thm:quant. The standard reference is [38]; [207, 197, 48] are the original papers.

**Definition 20.54** (deformation rings)**.** Let $k$ be a finite field of characteristic $\ell$, $\mathcal{O}$ the ring of integers of a finite extension of $\mathbb{Q}_\ell$ with residue field $k$, and $\rho_0 \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(k)$ continuous, odd and absolutely irreducible. For a finite set $\Sigma$ of primes not containing $\ell$, a *deformation of type*

$\Sigma$ of $\rho_0$ to a complete local noetherian $\mathcal{O}$-algebra $A$ with residue field $k$ is a strict equivalence class of lifts $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(A)$ that are unramified outside $\Sigma \cup \{\ell\} \cup \{\text{primes where } \rho_0 \text{ ramifies}\}$, minimally ramified (Definition 20.48) at the ramified primes outside $\Sigma \cup \{\ell\}$, unrestricted at $\Sigma$, of fixed determinant $\chi_\ell$, and of Fontaine-Laffaille ("short crystalline") type at $\ell$.

**Theorem 20.55** (Mazur; representability and the tangent space)**.** *In the situation of Definition 20.54 the functor of deformations of type $\Sigma$ is represented by a complete local noetherian $\mathcal{O}$-algebra $R_\Sigma$. If $\pi\colon R_\Sigma \twoheadrightarrow \mathcal{O}$ corresponds to a lift $\rho$ with module $M_\rho$, there is a natural isomorphism*

$$\mathrm{Hom}_{\mathcal{O}}(\mathrm{Cot}(\pi), K/\mathcal{O}) \simeq H^1_\Sigma(\mathbb{Q}, \mathrm{ad}^0 M_\rho \otimes K/\mathcal{O}),$$

*the Selmer group with the local conditions of type $\Sigma$.*

*Sketch of proof.* Representability follows from Schlessinger's criteria, using absolute irreducibility (so that the deformations have only scalar automorphisms) and the finiteness of $H^1(G_{\mathbb{Q},S}, \mathrm{ad}^0\,\rho_0)$. Infinitesimal deformations of $\rho$ to the ring $(\mathcal{O}/\varpi^n)[\varepsilon]$, $\varepsilon^2 = 0$, correspond to cohomology classes in $H^1(\mathbb{Q}, \mathrm{ad}^0\, M_\rho/\varpi^n)$, a lift $(1+\varepsilon c)\rho$ being determined by a cocycle $c$. The computation behind this is short. Write a lift as $\rho'(g) = (1+\varepsilon c(g))\rho(g)$ with $c(g)$ a $2\times 2$ matrix over $\mathcal{O}/\varpi^n$. Since $\varepsilon^2 = 0$,

$$\rho'(g)\rho'(h) = \big(1 + \varepsilon\,(c(g) + \rho(g)c(h)\rho(g)^{-1})\big)\rho(g)\rho(h),$$

so $\rho'$ is a homomorphism iff $c(gh) = c(g) + g\cdot c(h)$, the cocycle condition for the adjoint action $g\cdot X = \rho(g)X\rho(g)^{-1}$; the determinant is $(1+\varepsilon\,\mathrm{tr}\,c(g))\det\rho(g)$, so fixing it forces $\mathrm{tr}\,c = 0$; and conjugating $\rho'$ by $1+\varepsilon X$ replaces $c(g)$ by $c(g) + X - g\cdot X$, a coboundary, so strict equivalence classes of lifts are classes in $H^1$. The conditions of type $\Sigma$ become the local conditions defining $H^1_\Sigma$, and passing to the limit over $n$ gives the isomorphism. We do not give the details; see [38, Sections 2.2 to 2.8]. □

**Definition 20.56** (the Hecke side)**.** With $\rho_0 = \bar\rho_{f,\lambda}$ and $\Sigma$ as above, let $\mathbb{T}_\Sigma$ be the localisation at the maximal ideal of $\rho_0$ of the Hecke algebra acting on the cohomology $L_\Sigma$ of the modular curve of level $N(\rho_0)\prod_{p\in\Sigma} p^{\delta_p}$ (with the minimal conductor $N(\rho_0)$ at the twist-minimal level), and let $\pi\colon \mathbb{T}_\Sigma \twoheadrightarrow \mathcal{O}$ be the eigencharacter of the depletion of $f$.

The Galois representation with coefficients in $\mathbb{T}_\Sigma$ built from the cohomology (Carayol) is a deformation of type $\Sigma$, so it induces $R_\Sigma \to \mathbb{T}_\Sigma$, which is surjective because $\mathbb{T}_\Sigma$ is generated by traces of Frobenius elements. The point $\pi$ of $\mathbb{T}_\Sigma$ pulls back to a point of $R_\Sigma$.

**Theorem 20.57** (Taylor-Wiles, in the axiomatic form of Diamond, Flach and Guo)**.** *Let $\rho_0$ be odd, absolutely irreducible on $G_F$ with $F = \mathbb{Q}(\sqrt{(-1)^{(\ell-1)/2}\ell})$, of minimal conductor among its twists, with Serre weight $k$, $2 \le k \le \ell - 1$. Then for every $\Sigma$ the map $R_\Sigma \to \mathbb{T}_\Sigma$ is an isomorphism of complete intersections, $L_\Sigma$ is free over $\mathbb{T}_\Sigma$, and for every lift $\rho$ of type $\Sigma$ with values in $\mathcal{O}$ (all of which are modular),*

$$\mathrm{length}_{\mathcal{O}} H^1_\Sigma(\mathbb{Q}, \mathrm{ad}^0 M_\rho \otimes K/\mathcal{O}) = v(\eta(\mathbb{T}_\Sigma, \pi)) = v(\eta^\Sigma),$$

*with $\eta^\Sigma$ the ideal of values of the pairing on $L_\Sigma[\ker\pi]$.*

*Sketch of proof.* For $\Sigma = \emptyset$: choose, for each $n$, a set $Q_n$ of auxiliary primes $q \equiv 1 \pmod{\ell^n}$ at which $\rho_0$ is unramified with $\rho_0(\mathrm{Frob}_q)$ having distinct eigenvalues, of a fixed cardinality $r = \dim H^1_\emptyset(\mathbb{Q}, \mathrm{ad}^0\,\rho_0)$, such that the dual Selmer group with the relaxed conditions at $Q_n$

vanishes. Allowing ramification at $Q_n$ makes $R_{Q_n}$ a quotient of a power series ring in $r$ variables and $\mathbb{T}_{Q_n}$ free over the group ring of an $\ell$-group $\Delta_{Q_n}$ of rank $r$; patching over $n$ produces a power series ring mapping onto a module free over a power series ring in the same number of variables, whence $R_\emptyset \simeq \mathbb{T}_\emptyset$ is a complete intersection and the cohomology is free. For general $\Sigma$, both sides change by the same local factors as $\Sigma$ grows: on the Selmer side by the comparison of local conditions (the paper's Lemma lem:dfg21 and Chapter 17), on the Hecke side by Theorem 20.39; with the numerical criterion (Theorem 20.19(iii)) this gives the equality at every $\Sigma$. Diamond, Flach and Guo formulate this as an axiomatic theorem [49, Section 3]; see [38] (Chapters 3 to 5) for the case of semistable elliptic curves, and [48] for the freeness and the role of the cohomology. We do not give the details. □

*Remark* 20.58 (where each part of the hypothesis $\ell \notin S_f$ is used). (a) $\ell \nmid N$: the representation is crystalline at $\ell$ with Hodge-Tate weights $0, 1$, Fontaine-Laffaille theory applies to $M_{f,\lambda}$, and the levels are prime to $\ell$. (b) $\ell \nmid k!$, that is $\ell \neq 2$ for $k = 2$: the weights $0, 1$ must lie in an interval of length at most $\ell - 2$. (c) Absolute irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$: the auxiliary primes $q \in Q_n$ are found by Chebotarev in $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\mu_{\ell^n}))$, and one needs elements $g$ there whose image $\rho_0(g)$ has distinct eigenvalues and which detect any given class of the dual Selmer group. When $\rho_0$ is induced from $\mathbb{Q}(\sqrt{\ell^*}) \subseteq \mathbb{Q}(\mu_\ell)$, its restriction to $G_{\mathbb{Q}(\mu_\ell)}$ is reducible, and the argument fails. For elliptic curves with $\ell \nmid 2N$ this bites only at $\ell = 3$ with image 3Ns (Proposition 20.46). The paper's Section 1.5 and Section sec:anomaly turn on (c): Theorems thm:qual and thm:quant avoid it, and the curves with image 3Ns are where the identity needs a correction (Section 20.13).

*Remark* 20.59 (why the length equals the valuation). In three lines, for $\ell \notin S_f$: (1) $H^1_\Sigma(\mathbb{Q}, W)^* \simeq \mathrm{Cot}(\pi_{R_\Sigma})$ (Theorem 20.55); (2) $R_\Sigma \simeq \mathbb{T}_\Sigma$ is a complete intersection, so $\mathrm{length}\,\mathrm{Cot}(\pi) = v(\eta(\mathbb{T}_\Sigma, \pi))$ (Theorem 20.57, Theorem 20.19(ii)); (3) $L_\Sigma$ is free over $\mathbb{T}_\Sigma$, so $\eta(\mathbb{T}_\Sigma, \pi) = \eta^\Sigma$ (Lemma 20.21, equality case, with $d = 2$). So the Selmer group "knows" the congruences because its dual is the cotangent space of a ring that is isomorphic to the Hecke algebra, and in a complete intersection the cotangent space has exactly the size of the congruence module. The toy algebras of Example 20.9 show what would go wrong otherwise.

## 20.9 The assembly: proof of Theorem thm:main

This section proves the paper's main theorem. The proof is an assembly: every input is a theorem proved or stated elsewhere, and the work here is to check that the hypotheses match and that the local terms combine as claimed. The paper's Section 4.3 gives the proof in one paragraph; we give it in six numbered steps, each naming its input and the hypothesis it uses. The result is an equality, with no multiplicity-one hypothesis, because Lemma lem:hecke is an equality (Theorem 20.34). From now on $f = f_E$ has rational coefficients and $\lambda = \ell$.

**Theorem 20.60** (the paper's Theorem thm:main)**.** *Let $E/\mathbb{Q}$ be $X_0(N)$-optimal, $f = f_E$, $T = T_\ell E$, $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and let $\ell$ be an odd prime with $\ell \nmid N$ and $\ell \notin S_f$ (equivalently, by Proposition 20.46, $E[\ell]$ irreducible and, if $\ell = 3$, mod 3 image not 3Ns). Then*

$$v_\ell(\deg \varphi_E) = \mathrm{length}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}, W) + \sum_{q \mid N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) + \sum_{q \mid N} e_\ell(E, q), \tag{20.1}$$

*which is the paper's (eq:identity), and consequently*

$$v_\ell(\deg \varphi_E) \geq \sum_{q \mid N} e_\ell(E, q) + \sum_{q \mid N} t_\ell(E, q), \tag{20.2}$$

*which is the paper's (eq:thm).*

*Proof.* Let $\Sigma$ be the set of primes dividing $N$; it is nonempty (no elliptic curve over $\mathbb{Q}$ has good reduction everywhere, Chapter 10) and $\ell \notin \Sigma$.

*Step 1 (the degree is the congruence ideal; uses optimality, $\ell \nmid 2N$, $E[\ell]$ irreducible).* By Theorem 20.34,

$$v_\ell(\deg \varphi_E) = v_\ell(\eta_{f,\ell}^{\emptyset}).$$

*Step 2 (vary $\Sigma$; uses $\bar\rho$ irreducible, $\ell \nmid 2N$, $\ell \notin \Sigma$).* By Theorem 20.39(c),

$$v_\ell(\eta_{f,\ell}^{\emptyset}) = v_\ell(\eta_{f,\ell}^{\Sigma}) - \sum_{q\in\Sigma} v_\ell(L_q^{\mathrm{nv}}(A_f,1)^{-1}).$$

By Proposition 20.41 only the multiplicative primes contribute, each with $v_\ell(q^2-1)$.

*Step 3 (Taylor-Wiles; uses $\ell \notin S_f$).* $M_{f,\ell}$ is minimally ramified outside $\Sigma$ (the remark after Definition 20.48) and $\mathrm{ad}^0 M_{f,\ell} \simeq \mathrm{ad}^0 T$ (Proposition 20.32(iii)), so Theorem 20.49 gives

$$v_\ell(\eta_{f,\ell}^{\Sigma}) = \operatorname{length} H^1_{\Sigma}(\mathbb{Q}, W).$$

*Step 4 (the Selmer half; uses $\ell \notin S_f$, $\ell$ odd).* By the Selmer identity of Chapter 18 (which packages the paper's Lemma lem:dfg21 in the form proved there, Lemma lem:fpr at every $q \in \Sigma$, Lemma lem:h0 in the form $h_\ell(E) = 0$ for $\ell \notin S_f$, and Theorem thm:dfg27 for the hypothesis $H^1_f(\mathbb{Q}, V^D) = 0$, the hypothesis $H^0(\mathbb{Q}, V^D) = 0$ being automatic),

$$\operatorname{length} H^1_{\Sigma}(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{q\in\Sigma} \Big( v_\ell(L_q(A_f,1)^{-1}) + v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) \Big).$$

The term at $\infty$ vanishes because $H^1(\mathbb{R}, T^D)$ is killed by 2 and $\ell$ is odd, and the term $H^0(\mathbb{Q}, W^D)^*$ of Lemma lem:dfg21 vanishes because $h_\ell(E) = 0$.

*Step 5 (cancellation of the naive factors; Definition def:terms).* Substituting Step 3 into Step 2 and then Step 4 into the result,

$$\begin{aligned} v_\ell(\deg \varphi_E) &= \operatorname{length} H^1_{\Sigma}(\mathbb{Q}, W) - \sum_{q\in\Sigma} v_\ell(L_q^{\mathrm{nv}}(A_f,1)^{-1}) \\ &= \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{q\in\Sigma} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) \\ &\quad + \sum_{q\in\Sigma} \Big( v_\ell(L_q(A_f,1)^{-1}) - v_\ell(L_q^{\mathrm{nv}}(A_f,1)^{-1}) \Big). \end{aligned}$$

The true Euler factor enters through the Selmer side (Step 4) and the naive one through the Hecke side (Step 2), so what survives is their ratio. With $\Sigma = \{q \mid N\}$ this reads

$$v_\ell(\deg \varphi_E) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{q\mid N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) + \sum_{q\mid N} v_\ell\Big( \frac{L_q(A_f,1)^{-1}}{L_q^{\mathrm{nv}}(A_f,1)^{-1}} \Big).$$

At a multiplicative $q$ the true and naive factors coincide ($\delta_q = 1 > 0$) and the term is 0; at an additive $q$ the naive factor is 1 and the term is $v_\ell(L_q(A_f,1)^{-1})$. In both cases the term is $e_\ell(E,q)$ by Definition def:terms (Chapter 19). This is (20.1).

*Step 6 (the inequality).* $\operatorname{length} H^1_f(\mathbb{Q}, W) \geq 0$. At every prime $q \mid N$ with $v_q(j_E) < 0$ (multiplicative, or additive of type $\mathrm{I}_n^*$, $q = 2$ included), $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = t_\ell(E,q) + \min(t_\ell(E,q), v_\ell(q-$

$1), v_\ell(c(u_q))) \geq t_\ell(E,q)$, by Lemma lem:tamfactor and its extension to potentially multiplicative additive primes (Chapter 18; at an additive prime one applies Lemma lem:tamfactor to the local quadratic twist that is multiplicative, $\mathrm{ad}^0 T$ being unchanged, with $n = -v_q(j_E)$, not the index of the Kodaira symbol, which differs at $q = 2$). At the other bad primes $t_\ell(E,q) = 0$ and $\mathrm{Tam}_q(\mathrm{ad}^0 T) = \mathbb{Z}_\ell$ (Lemma lem:tamadd, Chapter 18). This gives (20.2). □

**Corollary 20.61** (the paper's Corollary cor:explicit)**.** *Under the hypotheses of Theorem 20.60,*

$$v_\ell(\deg \varphi_E) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{q \mid N} e_\ell(E,q) + \sum_{q:\, v_q(j_E)<0} \Big( t_\ell(E,q) + \min\big(t_\ell(E,q), v_\ell(q-1), v_\ell(c(u_q))\big)\Big),$$

*with $u_q$ the unit part of the Tate parameter at $q$ and $c(u_q)$ the class of its reduction (Appendix B, B-C14). So* $\operatorname{length} H^1_f(\mathbb{Q}, W)$ *is computable from* $\deg \varphi_E$ *and local data.*

*Proof.* Substitute the values of the Tamagawa factors of Chapter 18 (Step 6 above) into (20.1). □

*In the paper* 20.62. The paper's proof of Theorem thm:main (Section 4.3), sentence by sentence. "By Lemma lem:hecke, $v_\ell(\deg \varphi_E) = v_\lambda(\eta^{\emptyset}_{f,\lambda})$": Step 1. "By Theorem thm:dfg14, ...": Step 2. "By Theorem thm:dfg37, $v_\lambda(\eta^{\Sigma}_{f,\lambda}) = \operatorname{length} H^1_\Sigma(\mathbb{Q}, W)$": Step 3. "By Theorem thm:dfg27 the hypotheses of Lemma lem:dfg21 hold, and $H^0(\mathbb{Q}, W^D) = 0$ by Lemma lem:h0", with the display and "by Lemma lem:fpr each local term is ...": Step 4. The display of $v_\ell(\deg \varphi_E)$ and "the last sum is $\sum_{p \mid N} e_\ell(E,p)$ by Definition def:terms": Step 5. "For (eq:thm), the Selmer length is nonnegative, $v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0 T)) \geq t_\ell(E,p)$ at every potentially multiplicative prime $p$, multiplicative or additive, $p = 2$ included, by Lemma lem:tamfactor": Step 6. Like Step 6, the paper argues locally at the additive potentially multiplicative primes, so that $p = 2$ needs no separate treatment.

*Remark* 20.63 (the hypotheses, and what fails without each).

| hypothesis | used in | without it |
|---|---|---|
| $E$ optimal | Step 1 | for another curve of the class the parametrisation has degree $\deg\psi \cdot \deg\varphi_E$; harmless for $v_\ell$ when $E[\ell]$ is irreducible (Chapter 14) |
| $\ell \nmid N$ | Steps 1, 3, 4 | Agashe-Ribet-Stein fails at $\ell^2 \mid N$ (`99a1`); Fontaine-Laffaille and the integral structures need $\ell \nmid N$; Lemma lem:fpr needs $q \neq \ell$ |
| $\ell$ odd | Steps 1, 4 | the factor $\frac{u}{2}$ in $\hat{\delta}_!$; the term at $\infty$; self-duality of $\mathrm{ad}^0 T$ |
| $E[\ell]$ irreducible | Steps 1, 2 | non-Eisenstein $\mathfrak{m}$; homothety of lattices, also in Theorem 20.39(c) |
| irreducible over $\mathbb{Q}(\sqrt{\ell^*})$ | Steps 3, 4 | Taylor-Wiles (Remark 20.58); $h_\ell(E) = 0$ and $H^1_f(\mathbb{Q}, V^D) = 0$ |
| multiplicity one | nowhere | Remark 20.36 |

**Example 20.64** (three curves at $\ell = 3$)**.** Each curve below has $3 \nmid N$ and $E[3]$ irreducible with image not 3Ns (Cremona's `galrep` file), so $3 \notin S_f$ and Theorem 20.60 applies; the Selmer lengths are therefore *computed*. Data from PARI/GP. (a) `11a1`: type $\mathrm{I}_5$ at 11, $\deg \varphi_E = 1$. Here $e_3(E,11) = 0$ (multiplicative), $t_3(E,11) = v_3(5) = 0$, so $v_3(\mathrm{Tam}_{11}(\mathrm{ad}^0 T)) = 0$, and

$0 = \text{length}\, H^1_f(\mathbb{Q}, W)$. (b) `121a1`: type II at 11 (potentially good), $\deg \varphi_E = 6$. PARI's symmetric square Euler factor at 11 is $(1+11X)^{-1}$, so $L_{11}(A_f, 1)^{-1} = 1+\frac{11}{121} = \frac{12}{11}$ and $e_3(E, 11) = v_3(12) = 1$; $\mathrm{Tam}_{11}(\mathrm{ad}^0 T) = \mathbb{Z}_3$ (Lemma lem:tamadd). So $1 = v_3(6) = \text{length}\, H^1_f(\mathbb{Q}, W) + 0 + 1$ and $H^1_f(\mathbb{Q}, \mathrm{ad}^0 T_3 E \otimes \mathbb{Q}_3/\mathbb{Z}_3) = 0$. (c) `539c1`: type $\mathrm{I}_3^*$ at 7 ($v_7(j) = -3$) and $\mathrm{I}_2$ at 11, $\deg \varphi_E = 288$, $v_3 = 2$. At 7: $e_3 = v_3(1 - 7^{-2}) = v_3(48) = 1$, $t_3 = v_3(3) = 1$, and the Tate parameter of the multiplicative twist `77c1` is $q_E = 5 \cdot 7^3 + O(7^4)$, so $\bar{u} = 5$, which is not a cube in $\mathbb{F}_7^\times$ (the cubes are $\pm 1$: $1^3 = 2^3 = 4^3 = 1$ and $3^3 = 5^3 = 6^3 = -1$ in $\mathbb{F}_7$): $\min(1, v_3(6), v_3(c(u))) = 0$ and $v_3(\mathrm{Tam}_7(\mathrm{ad}^0 T)) = 1$. At 11: $t_3 = v_3(2) = 0$, $e_3 = 0$. So $2 = \text{length}\, H^1_f(\mathbb{Q}, W) + 1 + 1$, and the Selmer group vanishes again. Had $\bar{u}$ been a cube, the identity would have forced $v_3(\deg \varphi_E) \geq 3$. Further examples with nonzero Selmer groups are in Chapter 18.

## 20.10 Remark rem:ko: the Tamagawa excess and the formula of Kim and Ota

This section treats the paper's Remark rem:ko, which compares Theorem 20.60 with two finer statements: the excess in the Tamagawa factor at a multiplicative prime (part (1), tested in the paper's Section sec:tam), and the formula of Kim and Ota for the change of the congruence ideal when primes are moved into the new part of the level (part (2), quoted in the paper's Section 1.5).

*Remark* 20.65 (Remark rem:ko(1)). Let $q \neq \ell$ be a prime with $v_q(j_E) < 0$, $n = -v_q(j_E)$, and $u$ the unit part of the Tate parameter. The Tamagawa factor is $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = t_\ell(E, q) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$, with $t_\ell(E, q) = v_\ell(n)$ (Chapter 18). The excess is positive exactly when $\ell \mid n$, $q \equiv 1 \pmod{\ell}$ and $\bar{u}$ is an $\ell$-th power in $\mathbb{F}_q^\times$, and then it is at least 1 and can be larger. By (20.1), for $\ell \notin S_f$ the lower bound for $v_\ell(\deg \varphi_E)$ rises by exactly that excess:

$$v_\ell(\deg \varphi_E) \;\geq\; \sum_{q \mid N} e_\ell(E, q) + \sum_{q:\, v_q(j_E) < 0} \Big( t_\ell(E, q) + \min\big(t_\ell(E, q), v_\ell(q-1), v_\ell(c(u_q))\big)\Big),$$

at multiplicative primes and at additive potentially multiplicative primes alike. The paper's "gives $v_\ell(\mathrm{Tam}_q) \geq t_\ell(E, q) + 1$" is the case of a positive excess; the abstract's "raises the bound by at least one" is the same statement. In Example 20.64(c) the excess at 7 vanishes because $\bar{u} = 5$ is not a cube.

**Proposition 20.66** (Remark rem:ko(2): Kim and Ota against Theorem thm:main)**.** *Let $E$ be optimal with conductor $N = N^+N^-$, $N^-$ square-free, $\ell \geq 5$, $\ell \nmid N$, $\ell \notin S_f$, $\bar{\rho}$ ramified at every $q \mid N^-$ with $q \equiv \pm 1 \pmod{\ell}$, so that the formula of Kim and Ota holds (Chapter 15):*

$$\mathrm{ord}_\lambda\, \eta_f(N) = \mathrm{ord}_\lambda\, \eta_f(N^+, N^-) + \sum_{q \mid N^-} t_\ell(E, q).$$

*(i) $v_\ell(r_E) \geq \sum_{q \mid N^-} t_\ell(E, q)$, and this is contained in (20.2), which needs neither $\ell \geq 5$ nor the condition at $q \equiv \pm 1$.*

*(ii) If $\eta_f(N)$ is either the Hecke congruence ideal $\eta_f^{\mathrm{Hecke}}\mathbb{Z}_\ell$ of Definition 20.10 or the ideal $\eta_{f,\lambda}^{\emptyset}$*

*of Definition 20.31, then*

$$\operatorname{ord}_\lambda \eta_f(N^+,N^-) = \operatorname{length} H^1_f(\mathbb{Q},W) + \sum_{q\mid N} e_\ell(E,q) + \sum_{q\mid N^-} \min\bigl(t_\ell, v_\ell(q-1), v_\ell(c(u_q))\bigr) + \sum_{q\mid N^+,\, v_q(j_E)<0} \bigl(t_\ell + \min(t_\ell, v_\ell(q-1), v_\ell(c(u_q)))\bigr),$$

*with $t_\ell = t_\ell(E,q)$: a Selmer-theoretic reading of the congruence ideal of the $N^-$-new quotient.*

*Proof.* (i) is in Chapter 15; it is contained in (20.2) because the primes of $N^-$ are multiplicative and all other terms of (20.2) are nonnegative. (ii) In either case $\operatorname{ord}_\lambda \eta_f(N) = v_\ell(\deg \varphi_E)$: by Theorem 20.12 and Agashe-Ribet-Stein in the first, by Theorem 20.34 in the second. Subtract the formula of Kim and Ota from Corollary 20.61; the primes of $N^-$ have $v_q(j_E) < 0$, and their terms $t_\ell(E,q)$ cancel. □

*Remark* 20.67 (on the identification in (ii)). We have not checked in [99] whether their $\eta_f(N)$ is defined as the congruence ideal of the full Hecke algebra (as the paper's Remark rem:ko(2) says: "concerns the same congruence ideal") or as a cohomological ideal; under the hypotheses of the proposition the two have the same valuation (Remark 20.18), so (ii) holds in either reading. The theorem numbers [99, Theorem 1.1, Definition 2.12] are cited as the paper cites them.

## 20.11 Twist monotonicity: Theorem thm:twist

This section proves the paper's Theorem thm:twist: under a quadratic twist that does not raise the level, the congruence number can only drop. The proof transports congruences along the twist map $\iota$ and uses nothing beyond Chapter 13; in particular it needs no hypothesis on $\ell$. It is the model for Step 1 of the proof of Theorem thm:quant in the next section, and its data paragraph is where the paper first sees the Euler terms that Theorem thm:quant explains.

**Proposition 20.68** (the twist map, recollection)**.** *Let $S$ be a finite set of odd primes, $D = \prod_{q\in S} q$, $\chi_S$ and $\iota = \iota_S$ as in Convention 20.1(g).*

*(i) $\iota$ maps $S_2(\Gamma_0(M))$ into $S_2(\Gamma_0(\operatorname{lcm}(M, D^2)))$ and integral forms to integral forms; $\iota(h) = \iota(h^S)$, and $\iota\circ\iota(h) = h^S$.*

*(ii) If $h$ lies in the generalised eigenspace of the operators $T_p$, $p \nmid MD$, for a system $(\lambda_p)$, then $\iota(h)$ lies in the generalised eigenspace for the system $(\chi_S(p)\lambda_p)$.*

*(iii) Let $f$ be a newform of level $N$. If $\iota(h)$ has level dividing $N$ and $h$ lies in a sum of generalised eigenspaces whose twisted systems $(\chi_S(p)\lambda_p)$ differ from the system of $f$ at infinitely many $p$, then $\iota(h) \in f^\perp$; in particular, if $g$ is a newform with $\iota(g) = f$, then $\iota$ maps the forms of level dividing $N$ in $g^\perp$ into $f^\perp$.*

*(iv) (Levels of twists.) Let $E$ have conductor $N$, $S$ a set of odd primes with $q^2 \mid N$ for $q \in S$, and suppose $E$ has, at every $q \in S$, type $\mathrm{I}_n^*$ ($n \geq 1$), resp. $\mathrm{I}_0^*$, resp. $\mathrm{II}, \mathrm{III}, \mathrm{IV}$ or a dual with $q \geq 5$. Then $N_S = N/D$, resp. $N/D^2$, resp. $N$, and in every case $\iota(f_{E_S}) = f_E$.*

*Proof.* (i) and (ii) are Chapter 13 (the twist operator); (iii) is Chapter 13 (transport along $\iota$), which rests on strong multiplicity one: a generalised eigenform of the $T_p$, $p \nmid N$, for a system different from that of $f$ at infinitely many $p$ is orthogonal to every $f(dz)$. (iv) is Chapter 15 (levels of $E_S$), with the twist lemma of Chapter 10 for the types, and the case of the paper's Lemma lem:twist(iii) for $\iota(f_{E_S}) = f_E$ (Chapter 13, the case of the paper). □

**Theorem 20.69** (the paper's Theorem thm:twist)**.** *Let $E/\mathbb{Q}$ have conductor $N$, let $S$ be a set of odd primes, $D = \prod_{q\in S} q$, $\chi_S = \prod_{q\in S} \chi_{q^*}$, and $E_S = E \otimes \chi_S$ of conductor $N_S$. If $q^2 \mid N$ for every $q \in S$ and $\operatorname{lcm}(N_S, D^2) \mid N$, then $r_{E_S} \mid r_E$. If moreover $\operatorname{lcm}(N, D^2) \mid N_S$, then $r_{E_S} = r_E$.*

*Proof.* Let $f = f_E$, $g = f_{E_S}$, $\iota = \iota_S$.

*(1) $\iota(g) = f$.* By Proposition 20.68(i), $\iota(g) \in S_2(\Gamma_0(\operatorname{lcm}(N_S, D^2))) \subseteq S_2(\Gamma_0(N))$. Both $f$ and $\iota(g)$ have multiplicative coefficients ($\chi_S$ is multiplicative). For a prime $p \nmid D$ the character $\chi_S$ is unramified at $p$ (also at $p = 2$, as $D^* = \prod q^* \equiv 1 \pmod 4$), so $V_\ell E \simeq V_\ell E_S \otimes \chi_S$ gives $a_p(f) = \chi_S(p) a_p(g)$ at primes of good reduction, and the same holds at multiplicative primes (the unramified twist changes split to nonsplit exactly when $\chi_S(p) = -1$) and at additive primes (both coefficients vanish). The prime-power coefficients at $p \nmid D$ satisfy the same recursion on both sides, $p$ dividing $N$ if and only if it divides $N_S$. For $q \in S$, $a_{q^k}(f) = 0$ because $q^2 \mid N$ (Chapter 13), and $a_{q^k}(\iota(g)) = \chi_S(q^k) a_{q^k}(g) = 0$. So $\iota(g) = f$.

*(2) $\iota$ maps $g^\perp$ into $f^\perp$.* At level $N_S$, $g$ is a newform, so $g^\perp$ is $\mathbb{T}$-stable and is the sum of the generalised eigenspaces of the systems of eigenvalues different from that of $g$ (Chapter 13); each such system differs from that of $g$ at infinitely many primes (strong multiplicity one). Twisting multiplies the eigenvalues by $\chi_S(p)$, so the twisted systems differ from that of $f = \iota(g)$ at infinitely many primes, and Proposition 20.68(iii) gives $\iota(g^\perp) \subseteq f^\perp$, the images having level dividing $N$.

*(3) Transport.* By definition of $r_{E_S}$ there is $h \in g^\perp \cap S_2(\Gamma_0(N_S); \mathbb{Z})$ with $g \equiv h \pmod{r_{E_S}}$ coefficientwise. Applying $\iota$, which preserves integrality and congruences (it multiplies each coefficient by 0 or $\pm 1$), $f = \iota(g) \equiv \iota(h) \pmod{r_{E_S}}$ with $\iota(h) \in f^\perp \cap S_2(\Gamma_0(N); \mathbb{Z})$. By Lemma lem:crit (Theorem 20.12) the moduli of such congruences are exactly the divisors of $r_E$, so $r_{E_S} \mid r_E$. (The definition of $r_E$ as the largest modulus gives only $r_{E_S} \le r_E$; the divisibility needs more than the definition: the moduli are closed under lcm, which is the cyclicity step of Lemma lem:crit, its elementary half.)

*(4) Symmetry.* $E = (E_S) \otimes \chi_S$, and if $\operatorname{lcm}(N, D^2) \mid N_S$ then $q^2 \mid N_S$ for $q \in S$; so the hypotheses hold with the roles of $E$ and $E_S$ exchanged, and $r_E \mid r_{E_S}$. □

**Example 20.70** (the three cases, and the data)**.** By Proposition 20.68(iv) the hypotheses of Theorem 20.69 hold in three cases (the paper's list after the theorem). (a) Type $\mathrm{I}_n^*$ at every $q \in S$: $N_S = N/D$, $r_{E_S} \mid r_E$. For instance `539c1` ($\mathrm{I}_3^*$ at 7) and `77c1`: $r = 288$ and 6; `980i1` ($\mathrm{I}_5^*$ at 7) and `140b1` ($\mathrm{I}_5$ at 7): $\deg \varphi_E = 2880$ and 60; `121d1` ($\mathrm{I}_1^*$ at 11) and `11a1`: $r = 24$ and 1. (b) Type $\mathrm{I}_0^*$ at every $q \in S$: $N_S = N/D^2$, $E_S$ good at $S$. For instance `539d1` and `11a1`: $r = 504$ and 1. (c) Potentially good types II, III, IV or duals at $q \ge 5$: $N_S = N$ and $r_{E_S} = r_E$. For instance `121a1` (type II at 11) and its twist by $\chi_{-11}$, which is `121c2` (type IV$^*$; the twist exchanges II and IV$^*$): $r = 6 = 6$. (The curve `121c1`, of type IV, is isogenous to `121c2` and is the twist of `121a2`, of type II$^*$; $r$ depends only on the newform, so it is 6 for all four curves.) (Congruence numbers computed with PARI/GP as the index of Proposition 20.14, degrees with `ellmoddegree`.) At $\ell = 3$ the 3-adic valuations of the modular degrees in (a) and (b) are 2 against 1, 2 against 1, 1 against 0, and 2 against 0: the difference is $e_3(E, q)$, namely $v_3(7^2 - 1) = 1$, $v_3(48) = 1$, $v_3(11^2 - 1) = 1$, and, for the $\mathrm{I}_0^*$ prime 7 of `539d1`, $v_3(\frac{6}{7} \cdot \frac{64-4}{49}) = 2$. This is Theorem thm:quant. *The data, recomputed.* For every $X_0(N)$-optimal curve without complex multiplication of conductor below

10,000 (Cremona's `degphi` file), let $S$ be the set of odd primes of type $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$ (PARI codes $-1$ and $\le -5$), and consider the pairs $(E,\ell)$ with $S\neq\emptyset$, $\ell\in\{3,5,7,11,13\}$, $\ell\nmid N$ and $E[\ell]$ irreducible (no label $\ell$B or $\ell$Cs in Cremona's `galrep` file). There are 25,477 such pairs (15,607 with $3\in S$; 8 with $\ell=3$ and image 3Ns), and in every one $v_\ell(\deg\varphi_E)=v_\ell(\deg\varphi_{E'_S})+\sum_{q\in S}e_\ell(E,q)$, with $E'_S$ the optimal curve of the class of $E_S$, located by its conductor and its $a_p$ for $p<80$, and $e_\ell(E,q)=v_\ell(q^2-1)$ (type $\mathrm{I}_n^*$) or $v_\ell(q-1)+v_\ell((q+1)^2-a_q(E_S)^2)$ (type $\mathrm{I}_0^*$). For case (c), with $S$ the set of primes $q\ge5$ of type II, III, IV or a dual and the same conditions on $\ell$, there are 14,880 pairs (42 with image 3Ns at $\ell=3$), and $v_\ell(\deg\varphi_E)=v_\ell(\deg\varphi_{E'_S})$ in every one.

*In the paper* 20.71. The paragraph after Theorem thm:twist in the paper reports, on the working set, the pairs (curve, prime of type $\mathrm{I}_n^*$) and (curve, prime of potentially good additive type) with $v_\ell(\deg\varphi_E)\ge v_\ell(\deg\varphi_{E_0^{(q)}})$, $E_0^{(q)}$ the optimal curve of the isogeny class of $E^{(q)}$, and "equality in every pair of case (c)", and then: "In cases (a) and (b) Theorem thm:quant predicts a difference equal to the Euler term, which is positive in case (a) at $\ell=3$ for every $q\neq3$, as $3\mid q^2-1$"; and "Below conductor 10,000 the difference equals the Euler term in all 25,477 pairs". The count 25,477 agrees with Example 20.70; the working-set counts are the paper's and were not recomputed here. The difference is positive in case (a) as well as in case (b): `539c1`, `980i1`, `121d1`.

## 20.12 Theorem thm:quant

This section proves the paper's Theorem thm:quant: at the odd primes of type $\mathrm{I}_0^*$ and $\mathrm{I}_n^*$ the omitted Euler factor divides the modular degree, under the sole hypothesis that $E[\ell]$ is irreducible, and with equality (after removing the congruence number of the twist) when $\ell\notin S_f$. The proof raises the level from $N_S$ to $N$ by depleting the newform of the twist, and moves congruences between levels with $\iota$. Step 1 of the proof is an equality at every prime $\ell\notin S$ (Lemma 20.74, the paper's Lemma lem:step1), with no other hypothesis on $\ell$.

**Theorem 20.72** (the paper's Theorem thm:quant)**.** *Let $E$ be $X_0(N)$-optimal and let $\ell$ be an odd prime with $\ell\nmid N$ and $E[\ell]$ irreducible. Let $S$ be the set of odd primes $q$ at which $E$ has type $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$ ($n\ge1$), $\chi_S=\prod_{q\in S}\chi_{q^*}$, $E_S=E\otimes\chi_S$, and $E'_S$ the optimal curve of the isogeny class of $E_S$. Then*

$$v_\ell(\deg\varphi_E)\ \ge\ v_\ell(r_{E_S})+\sum_{q\in S}e_\ell(E,q)\ \ge\ \sum_{q\in S}e_\ell(E,q). \tag{20.3}$$

*If moreover $\ell\notin S_f$ (automatic for $\ell\ge5$, Proposition 20.46), the first inequality is an equality and $v_\ell(r_{E_S})=v_\ell(\deg\varphi_{E'_S})$:*

$$v_\ell(\deg\varphi_E)=v_\ell(\deg\varphi_{E'_S})+\sum_{q\in S}e_\ell(E,q).$$

**Lemma 20.73** (the set-up)**.** *In Theorem 20.72, let $f=f_E$, $g=f_{E_S}$ (of level $N_S$), $D=\prod_{q\in S}q$ and $\iota=\iota_S$. Then $\ell\notin S$, and:*

*(i) $E_S$ has good reduction at the $\mathrm{I}_0^*$ primes of $S$ and multiplicative reduction at the $\mathrm{I}_n^*$ primes; so $\delta_q(g)=2$, resp. 1, and $N_S^S=N_S\prod_{q\in S}q^{\delta_q(g)}=N$;*

*(ii) $D^2\mid N$, and at $q\notin S$ the curves $E$ and $E_S$ have the same reduction type, the same Euler factors (true and naive) and the same $e_\ell(E,q)$;*

*(iii)* $g^S \in S_2(\Gamma_0(N))$ *is a normalised eigenform of* $\mathbb{T} = \mathbb{T}_N$ *with* $U_q g^S = 0$ *for* $q \in S$*, its generalised eigenspace is the line* $\mathbb{C}g^S$*, and* $\iota(g^S) = \iota(g) = f$*.*

*Proof.* $\ell \notin S$ because $S$ consists of primes dividing $N$; in particular $\ell \neq 3$ if $3 \in S$. (i) is the twist lemma of Chapter 10 (the paper's Lemma lem:twist, valid at every odd $q$ for these exchanges: $\mathrm{I}_0^* \leftrightarrow \mathrm{I}_0$, $\mathrm{I}_n^* \leftrightarrow \mathrm{I}_n$), with the levels of Proposition 20.68(iv). (ii) $v_q(N) = 2$ for $q \in S$ (types $\mathrm{I}_0^*$, $\mathrm{I}_n^*$ at odd $q$ have conductor exponent 2, Chapter 10). At $q \notin S$ the character $\chi_S$ is unramified (at $q = 2$ too, as $\prod_{q\in S} q^* \equiv 1 \pmod 4$), so the twist does not change the type, $a_q(E_S)^2 = a_q(E)^2$, and the local factors of $\mathrm{ad}^0 V$, which are twist-invariant, agree (Chapter 19). (iii) By (i) and Chapter 13 (depletion) $g^S$ is an eigenform of $\mathbb{T}_N$ with generalised eigenspace $\mathbb{C}g^S$; $\iota$ kills the coefficients at multiples of $D$, so $\iota(g^S) = \iota(g)$, which is $f$ by Proposition 20.68(iv). □

**Lemma 20.74** (Step 1, as an equality)**.** *Let* $E/\mathbb{Q}$ *be any elliptic curve of conductor* $N$ *(optimal or not),* $S$ *the set of odd primes at which* $E$ *has type* $\mathrm{I}_0^*$ *or* $\mathrm{I}_n^*$*, and* $f$*,* $g$*,* $D$*,* $\iota$ *as in Lemma 20.73, whose parts (i) and (iii) use neither a prime* $\ell$ *nor optimality. Let* $\pi\colon \mathbb{T}_N \to \mathbb{Z}$ *be the eigencharacter of* $g^S$ *and* $\eta^{\mathrm{Hecke}}(g^S) = \pi(\mathrm{Ann}_{\mathbb{T}_N} \ker \pi) = r_\pi\mathbb{Z}$ *(Proposition 20.14). Then* $r_E \mid r_\pi$*, and for every prime* $\ell \notin S$*,*

$$v_\ell(r_E) = v_\ell(\eta^{\mathrm{Hecke}}(g^S)).$$

*No hypothesis on* $\ell$ *other than* $\ell \notin S$ *is needed: neither* $\ell$ *odd, nor* $\ell \nmid N$*, nor* $E[\ell]$ *irreducible, nor* $\ell \notin S_f$*.*

*Proof.* $r_E \mid r_\pi$. This is Chapter 13 ("$r_f$ divides the congruence module of $g^S$", and its corollary for the paper's situation): take $h \in f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z})$ with $f \equiv h \pmod{r_E}$; then $g^S = \iota(f) \equiv \iota(h) \pmod{r_E}$, because $\iota \circ \iota(g) = g^S$ and $\iota(f) = \iota(\iota(g))$; and $\iota(f^\perp)$ lies in the $\mathbb{T}$-stable complement of $\mathbb{C}g^S$ (the systems in $f^\perp$ differ from that of $f$ at infinitely many primes, so their twists differ from that of $g$, by strong multiplicity one). Proposition 20.14 gives $r_E \mid r_\pi$. So $v_\ell(r_E) \leq v_\ell(r_\pi)$ for every $\ell$.

$v_\ell(r_E) \geq v_\ell(r_\pi)$ *for* $\ell \notin S$ *(the inequality used in Step 1).* Let $t = v_\ell(r_\pi)$ and $\mathfrak{m}$ the maximal ideal of $\mathbb{T}_\ell$ attached to $\pi$. By Proposition 20.14 over $\mathbb{Z}_\ell$ there is $h \in g^c \cap S_2(\Gamma_0(N);\mathbb{Z}_\ell)$, $g^c$ the $\mathbb{T}$-stable complement of $\mathbb{C}g^S$, with $g^S \equiv h \pmod{\ell^t}$. Since $\pi$ factors through $\mathbb{T}_\mathfrak{m}$, $e_\mathfrak{m} g^S = g^S$; the idempotent $e_\mathfrak{m} \in \mathbb{T}_\ell$ preserves $S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ and $g^c$, so $h' = e_\mathfrak{m} h \in g^c \cap S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ and $g^S - h' = e_\mathfrak{m}(g^S - h) \in \ell^t S_2(\Gamma_0(N);\mathbb{Z}_\ell)$. By Proposition 20.4(iii), $h'$ lies in the sum of the generalised eigenspaces $V(\phi)$ with $\phi \neq \pi$ and $\phi \equiv \pi$ modulo the maximal ideal of $\overline{\mathbb{Z}}_\ell$. For such $\phi$ and $q \in S$, $\phi(U_q) \equiv \pi(U_q) = 0$; every eigenvalue of $U_q$ on $S_2(\Gamma_0(N))$ is 0 or a unit at every prime not above $q$ (Chapter 13: it is 0, $\pm 1$ or a root of $X^2 - a_q(h)X + q$ for a newform $h$), and $\ell \neq q$, so $\phi(U_q) = 0$. (The unit claim for the roots: they are algebraic integers $\alpha, \beta$ with $\alpha\beta = q$, so at a prime not above $q$ neither can have positive valuation. A nonzero eigenvalue is therefore an $\ell$-adic unit, and cannot be $\equiv 0$ modulo the maximal ideal; the congruence $\phi(U_q) \equiv 0$ forces $\phi(U_q) = 0$. This is where the hypothesis $\ell \notin S$ enters.) Each $V(\phi)$ lies in the old space $V_h$ of a newform $h$ of level dividing $N$. If $h = g$, then $\phi$ agrees with $\pi$ on the $T_p$, $p \nmid N$, and on the $U_q$, $q \in S$; the operators $U_p$ for $p \mid N_S$, $p \notin S$, act on $V_g$ by the scalar $a_p(g)$ (no oldforms at those primes, $v_p(N) = v_p(N_S)$); so $\phi = \pi$, which is excluded. Hence $h \neq g$. The twist $\iota(V(\phi))$ lies in $S_2(\Gamma_0(N))$, because the level of a twist by $\chi_S$ divides $\mathrm{lcm}(N, D^2) = N$, and in the generalised eigenspace of the $T_p$, $p \nmid N$, for the system $\chi_S(p)a_p(h)$; this differs from the system $\chi_S(p)a_p(g)$ of $f$ at infinitely many $p$ (otherwise $h = g$ by strong multiplicity one), so $\iota(V(\phi)) \subseteq f^\perp$ (Proposition 20.68(iii)). Therefore $\iota(h') \in f^\perp \cap S_2(\Gamma_0(N);\mathbb{Z}_\ell)$ and

$$f = \iota(g^S) \equiv \iota(h') \pmod{\ell^t}.$$

By Lemma lem:crit over $\mathbb{Z}_\ell$ (Theorem 20.12), $t \le v_\ell(r_E)$. □

*Remark* 20.75 (Commentary: Lemma lem:step1 and Remark rem:afterquant (2)). Lemma 20.74 is the paper's Lemma lem:step1, with the same proof, and the paper's Remark rem:afterquant (2) records: "Step 1 is an equality at every prime $\ell \notin S$, with no hypothesis on $\ell$ at all (Lemma lem:step1): the twist map sends $f$ to $g^S$ and $f^\perp$ into the $\mathbb{T}$-stable complement of $g^S$." So the lemma goes beyond the hypotheses of Theorem thm:quant: $\ell = 2$ is allowed, $\ell$ may divide $N$, $E[\ell]$ may be reducible, and $\ell$ may lie in $S_f$. The reason is that the transport along $\iota$ works in both directions (Chapter 13 supplies the direction from $f$ to $g^S$), and the proof of the direction from $g^S$ to $f$ above uses only $\ell \neq q$ for $q \in S$. This settles, at $\ell = 2$, the point left by Chapter 13, which proves the equality under the hypotheses of Theorem thm:quant and records the equality at $\ell = 2$ at level 539 as observed: the lemma proves it. At level 539, Example 20.6 gives $r(\texttt{539c1}) = 288$ and $r(\texttt{539d1}) = 504$, equal to the two depleted modules as integers; the 2-parts and 3-parts are covered by the lemma, while the 7-parts ($7 \in S$) are not, and there the equality ($7^0$ and $7^1$) is observed, not proved.

**Lemma 20.76** (Steps 2 to 4)**.** *In the situation of Lemma 20.73, with $\ell$ odd, $\ell \nmid N$ and $E[\ell]$ irreducible:*

*(i) (Step 2)* $v_\ell(\eta^{\mathrm{Hecke}}(g^S)) \ge v_\ell(\eta^S_{g,\ell})$*;*

*(ii) (Step 3)* $v_\ell(\eta^S_{g,\ell}) = v_\ell(\eta^\emptyset_{g,\ell}) + \sum_{q\in S} e_\ell(E,q)$*;*

*(iii) (Step 4)* $v_\ell(\eta^\emptyset_{g,\ell}) = v_\ell(\deg \varphi_{E'_S}) = v_\ell(r_{E_S})$.

*Proof.* The residual representation of $g$ is $\bar\rho \otimes \chi_S$, irreducible, and $\ell \nmid 2N_S$, $N_S$ dividing $N$.

(i) Apply Lemma 20.21 with $\mathcal{O} = \mathbb{Z}_\ell$, $\mathbb{T}$ replaced by $\mathbb{T}_\mathfrak{m}$ ($\mathfrak{m}$ the maximal ideal of $\pi$), $L = H^1(X_0(N),\mathbb{Z}_\ell)_\mathfrak{m}$, the pairing $\hat\delta_!$ (perfect on the $\mathfrak{m}$-component, with $\mathbb{T}_\mathfrak{m}$ self-adjoint, Proposition 20.4(iv)), and $\pi\colon \mathbb{T}_\mathfrak{m} \twoheadrightarrow \mathbb{Z}_\ell$. The congruence ideal of $\pi$ on $\mathbb{T}_\mathfrak{m}$ is $\eta^{\mathrm{Hecke}}(g^S)\mathbb{Z}_\ell$ (Proposition 20.8(iv)). The module $L[\ker\pi]$ is $H^1(X_0(N),\mathbb{Z}_\ell)[\ker\pi]$, which is $M_{g^S,\ell}$ of Definition 20.31 (level $N = N_S^S$; Proposition 20.32(i), $\mathfrak{m}$ being non-Eisenstein as its residual representation is $\bar\rho\otimes\chi_S$): rank $d = 2$, Gram determinant $(\eta^S_{g,\ell})^2$ up to a unit (Proposition 20.32(ii)), nonzero by (ii) and (iii) below. So $2v_\ell(\eta^{\mathrm{Hecke}}(g^S)) \ge 2v_\ell(\eta^S_{g,\ell})$. Freeness of $L$ over $\mathbb{T}_\mathfrak{m}$ is not needed.

(ii) Apply Theorem 20.39(c) to $g$ (level $N_S$, $\ell \nmid 2N_S$, $\ell \notin S$, $\bar\rho_g$ irreducible) with $\Sigma = S$: $v_\ell(\eta^S_{g,\ell}) = v_\ell(\eta^\emptyset_{g,\ell}) + \sum_{q\in S} v_\ell(L_q^{\mathrm{nv}}(A_g,1)^{-1})$. At $q \in S$, $\delta_q(g) \ge 1$, so $L_q^{\mathrm{nv}}(A_g,1) = L_q(A_g,1)$; $A_g \simeq A_f$, the adjoint being unchanged by twisting (Chapter 19); and $f$ has $\delta_q(f) = 0$, so $e_\ell(E,q) = v_\ell(L_q(A_f,1)^{-1})$ by Definition def:terms. The values are those of Lemma lem:euler: $v_\ell(q^2-1)$ at an $\mathrm{I}_n^*$ prime and $v_\ell(q-1) + v_\ell((q+1)^2 - a_q(E_S)^2)$ at an $\mathrm{I}_0^*$ prime, $q = 3$ included (Chapter 19; for $q = 3$, $\ell \neq 3$).

(iii) Theorem 20.34 applied to $E'_S$, which is optimal of conductor $N_S$, with $\ell \nmid 2N_S$ and $E'_S[\ell] \simeq E_S[\ell]$ irreducible (the isogeny between $E_S$ and $E'_S$ has degree prime to $\ell$, as $E_S[\ell]$ is irreducible). Its newform is $g$. No freeness is used. □

*Proof of Theorem 20.72. The inequalities.* As $\ell^2 \nmid N$, $v_\ell(\deg\varphi_E) = v_\ell(r_E)$ [3, Theorem 2.2]. By Lemma 20.74 (only the inequality $\ge$ is needed here) and Lemma 20.76,

$$v_\ell(\deg\varphi_E) = v_\ell(r_E) = v_\ell(\eta^{\mathrm{Hecke}}(g^S)) \ge v_\ell(\eta^S_{g,\ell}) = v_\ell(r_{E_S}) + \sum_{q\in S} e_\ell(E,q),$$

and the second inequality of (20.3) holds because $r_{E_S}$ is an integer.

*The equality.* Let $\ell \notin S_f$. Then $\ell \notin S_{f_{E'_S}}$: $\ell \nmid 2N_S$, and $\bar\rho_{E'_S} \simeq \bar\rho \otimes \chi_S$, whose restriction to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$ is absolutely irreducible iff that of $\bar\rho$ is. Apply Theorem 20.60 to $E$ (conductor $N$) and to $E'_S$ (conductor $N_S$). The $G_{\mathbb{Q}}$-modules $\mathrm{ad}^0 T_\ell E$ and $\mathrm{ad}^0 T_\ell E'_S$ are isomorphic (twisting does not change the adjoint, and $T_\ell E_S \simeq T_\ell E'_S$ by an isogeny of degree prime to $\ell$), so the Selmer groups $H^1_f(\mathbb{Q}, W)$ coincide and, at every prime $q$, so do the Tamagawa factors $\mathrm{Tam}_q(\mathrm{ad}^0 T)$, which are local invariants of $\mathrm{ad}^0 T$ (at a prime where $\mathrm{ad}^0 T$ is unramified they are trivial, so the sums over $q \mid N$ and over $q \mid N_S$ agree). By Lemma 20.73(ii) the Euler terms agree at $q \notin S$; at $q \in S$, $e_\ell(E'_S, q) = 0$, the reduction of $E'_S$ being good or multiplicative. Subtracting the two identities (20.1),

$$v_\ell(\deg \varphi_E) - v_\ell(\deg \varphi_{E'_S}) = \sum_{q \in S} e_\ell(E, q),$$

and $v_\ell(\deg \varphi_{E'_S}) = v_\ell(r_{E_S})$ by Lemma 20.76(iii). □

**Corollary 20.77** (every step is an equality when $\ell \notin S_f$)**.** *In Theorem 20.72, if $\ell \notin S_f$ then*

$$v_\ell(r_E) = v_\ell(\eta^{\mathrm{Hecke}}(g^S)) = v_\ell(\eta^S_{g,\ell}) = v_\ell(r_{E_S}) + \sum_{q \in S} e_\ell(E, q).$$

*In particular the Hecke and cohomological congruence ideals of the depleted form $g^S$ at level $N$ have the same valuation, a freeness-type statement obtained without any multiplicity-one theorem.*

*Proof.* The chain in the proof of Theorem 20.72 begins and ends with $v_\ell(\deg \varphi_E) = v_\ell(r_E)$ when the first inequality of (20.3) is an equality, so every inequality in it is an equality. □

*Remark* 20.78 (the outer inequality, and the word "elementary"). The outer inequality, $v_\ell(\deg \varphi_E)$ at least $\sum_{q \in S} e_\ell(E, q)$, needs only Step 1 (the paper's direction), Steps 2 and 3, and the integrality of $\eta^{\emptyset}_{g,\ell}$; neither Agashe-Ribet-Stein at level $N_S$ nor any multiplicity-one theorem. It is still not elementary: Step 3 is the proposition of Diamond, Flach and Guo and Step 2 is Lemma lem:ddt417. The paper's abstract says "by a twisting argument and the level-raising formula of Diamond, Flach and Guo", and the paper calls elementary only the transport of congruences along the twist, that is Step 1, which is.

*Remark* 20.79 (Remark rem:afterquant (1) and (3)). (1) Theorem thm:quant is level raising: from $N_S$ to $N_S^S = N$, at the price of the level-raising Euler factors of Proposition 20.41. For $\ell = 3$, a single prime of type $\mathrm{I}_n^*$ (necessarily $q \geq 5$, since $3 \nmid N$) forces $3 \mid \deg \varphi_E$, because $3 \mid q^2 - 1$; so does a prime of type $\mathrm{I}_0^*$ with $q \equiv 1 \pmod 3$, because then $3 \mid q - 1$. Neither needs the hypothesis on $\mathbb{Q}(\sqrt{-3})$, and neither is in Theorem thm:qual. (3) For the potentially good types $\mathrm{II}, \mathrm{III}, \mathrm{IV}$ and their duals the quadratic twist does not lower the level ($N_S = N$ in Proposition 20.68(iv)), $f$ is its own depletion at $S$, and Theorem 20.39 gives no factor; Step 1 has nothing to transport. The omitted factor at such primes is reached only through Theorem 20.49, that is with $\ell \notin S_f$, and the paper's Section sec:anomaly shows that for $\ell = 3$ and image 3Ns the bound can fail through a type III term. The data of Example 20.70(c) (equality of $v_\ell$ of the degrees in all 14,880 pairs) are consistent with this: twisting at potentially good primes moves neither side.

## 20.13 Beyond the hypothesis: Remark rem:ns and Open questions 1 and 2

This section records what the assembly of Section 20.9 gives, and what it would need, at the two boundaries of the hypothesis of Theorem thm:main: the curves with $\ell = 3$ and image 3Ns, where

$3 \in S_f$ (the paper's Remark rem:ns, Section sec:anomaly and Open question 2), and the primes $\ell \mid N$ (Open question 1).

*Remark* 20.80 (Remark rem:ns: the conditional identity for image 3Ns). Let $\ell = 3$, $3 \nmid N$, and let the image of $\bar{\rho}_{E,3}$ be $N_s(\mathbb{F}_3)$. Then $3 \in S_f$ and neither Theorem 20.49 nor Theorem 20.51 applies. Steps 1 and 2 of Theorem 20.60 still hold ($E[3]$ is irreducible). Step 4 holds in a corrected form *if* $H^1_f(\mathbb{Q}, V^D) = 0$ (equivalently $H^1_f(\mathbb{Q}, \mathrm{ad}^0\, V_3E) = 0$, Chapter 18); the condition $H^0(\mathbb{Q}, V^D) = 0$ is automatic. Then Lemma lem:dfg21 gives, with $h_3(E) = 1$ (Lemma lem:h0; Chapter 11),

$$\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = \operatorname{length} H^1_f(\mathbb{Q}, W) + \sum_{p \in \Sigma} \left( v_3(L_p(A_f, 1)^{-1}) + v_3(\mathrm{Tam}_p(\mathrm{ad}^0\, T)) \right) - h_3(E)$$

(Chapter 18 proves this, conditionally). If in addition $v_3(\eta^{\Sigma}_{f,3}) = \operatorname{length} H^1_\Sigma(\mathbb{Q}, W)$ held, as in Theorem 20.49, Steps 1, 2 and 5 would give (eq:identity) with $h_3(E)$ subtracted on the right, which is the shape of Conjecture conj:euler. The paper states the assumption $H^1_f(\mathbb{Q}, V^D) = 0$ explicitly ("we assume it here"); Diamond, Flach and Guo's Theorem 2.7 does not supply it for these curves, because it is proved only for $\lambda \notin S_f$.

*Remark* 20.81 (Open question 2). For the curves with image 3Ns at $\ell = 3$ two inputs are missing. (a) On the Hecke side, $v_3(\eta^{\Sigma}_{f,3}) = \operatorname{length} H^1_\Sigma(\mathbb{Q}, W)$ would need the Taylor-Wiles method for a residual representation induced from $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\mu_3)$, where the auxiliary primes cannot kill the dual Selmer group (Remark 20.58(c)). (b) On the Selmer side, $H^1_f(\mathbb{Q}, \mathrm{ad}^0\, V_3E) = 0$. The paper's data (failures of (eq:thm) by exactly one, never more, for these curves) and the equality in all 8 pairs with image 3Ns in Example 20.70, which lie outside Theorem 20.72's equality clause (the paper's Open question 2 records the same 8 pairs), are consistent with the corrected identity. We do not survey later work on residually dihedral representations, which we have not checked against these questions.

*Remark* 20.82 (Open question 1: primes $\ell$ dividing $N$). At $\ell \mid N$ three steps of Theorem 20.60 fail. Step 1: Agashe-Ribet-Stein can fail when $\ell^2 \mid N$, and $r_E \neq \deg \varphi_E$ occurs (`99a1`: $r = 12$, $\deg \varphi_E = 4$ at $\ell = 3$; `539d1`: $r = 504$, $\deg \varphi_E = 72$ at $\ell = 7$; Chapter 14). Step 3: $\ell \mid N$ puts $\ell$ in $S_f$; the integral structures and the Fontaine-Laffaille condition at $\ell$ need $\ell \nmid N$. Step 5: the Euler factor at $q = \ell$ is not the relevant local quantity (the paper's Remark rem:wild). A replacement would be a deformation condition at $\ell$ (finite flat, ordinary or semistable) whose local index plays the role of $e_\ell(E, \ell)$; the paper's empirical weights $w_3$ at $q = \ell = 3$ are the target. We record the question without an answer.

## 20.14 The imported statements

This section lists, for reference, every statement of [49] that the paper cites, what it says, where the paper uses it, and what this book has checked. It is the place that Chapters 18 and 19 point to for the status of these citations.

*Remark* 20.83 (the citations of Diamond, Flach and Guo). Every entry was compared with the published text [49] for this chapter; the column "where" gives the page of the journal, and the content column what the check found.

| citation | content (book's notation) | paper's use | where |
|---|---|---|---|
| Theorem 0.1 (= Theorem 2.15) | $\lambda$-part of Bloch-Kato for $A_f$, $A_f(1)$, $\lambda \notin S_f$ | Section 1.2 | pp. 666, 709 |
| Theorem 0.2 (= Theorem 3.7) | length $H^1_\Sigma = v_\lambda(\eta_f^\Sigma)$, $\Sigma \supseteq \{p \mid N\}$, $\lambda \notin S_f$, $\lambda \nmid \Sigma$ | thm:dfg37 | p. 666 |
| definition of $S_f$ (Section 0.2 and display (31)) | $\lambda \mid Nk!$ or residually not absolutely irreducible over $\mathbb{Q}(\sqrt{\ell^*})$ | Section 4.2 | pp. 666, 695 |
| Theorem 3.7 | as Theorem 0.2, with "minimally ramified outside $\Sigma$", $\ell \notin \Sigma$ | thm:dfg37 | p. 723 |
| Proposition 1.4(c) | $\eta^\Sigma = \eta^\emptyset \prod L_p^{\mathrm{nv}}(A_f,1)^{-1}$ if $\bar{\mathcal{M}}_{f,\lambda}$ is irreducible; proved by homothety and $\gamma_{\mathrm{dR}}(f) = f^\Sigma$ | thm:dfg14 | p. 687 |
| Theorem 2.7 | $H^0 = H^1_f = 0$ for $A_f$ and $A_f(1)$ | thm:dfg27 | p. 699 |
| Lemma 2.1 | the exact sequence of Lemma lem:dfg21, for $\Sigma$ nonempty | lem:dfg21 | p. 696 |
| Lemma 2.12 | $(f,f)_{\Gamma_1(N)}$ as a naive adjoint $L$-value; $N/(8\pi^3)$ on $\Gamma_0(N)$ for $k=2$ | Section 1, rem:degree | p. 706 |
| (57), p. 708 | $\mathrm{Tam}_p(M) = \mathrm{Tam}_p(M^D)$; definition of $\mathrm{Tam}_p$ | lem:fpr, Section 1.2 | p. 708 |
| Sections 1.5.3, 1.6.2, 1.7.2, 1.7.3; (13), (14) | $\hat{\delta}_!$ through $[UwU]_\omega$, $M_f = M(N,\psi)_![I_f]$, $\delta_p$, $L_p^{\mathrm{nv}}$, $\eta_f^\Sigma$ | Section 4.1 | pp. 682 to 687 |
| Sections 1.6.1, 1.8.2 | Hecke action compatible with $\hat{\delta}_!$ (Proposition 1.3); self-adjoint for the pairing (25) | Section 4.1 | pp. 682, 689 |
| Section 2.1 | $H^1_f$ and $H^1_\Sigma$ | Section 4.2 | p. 695 |
| Lemma 1.1, Sections 1.2.2, 1.2.3, 1.4.1, 1.4.2, Proposition 1.3, (5) | the Betti and $\lambda$-adic structures as invariants of full-level cohomology, independent of the level; Hecke action | Proposition prop:integral | Sections 1.2, 1.4 |

Two differences of wording matter. Lemma 2.1 of [49] asks that $\Sigma$ contain every prime at which $H^1_f(\mathbb{Q}_p, W) \neq H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$; the paper's "every prime $p \neq \ell$ at which $W$ is ramified" implies it, since the two groups agree at unramified $p \neq \ell$. And $H^1_\Sigma$ of [49] uses the unramified condition outside $\Sigma \cup \{\ell\}$, which is the Bloch-Kato condition when $\Sigma \supseteq \{p \mid N\}$ (Section 20.7). A longer version of [49] is now available [50], dated "August 2001, with minor update December 2025"; it numbers differently: according to a reading of arXiv:2512.02348 not re-checked for this chapter, its Theorem 0.1 is labelled "= Theorem 8.10", its Theorem 0.2 "= Theorem 7.15", and its exceptional set is called $T_f$. The book, like the paper, cites the published numbering.

## 20.15 Exercises

**Exercise 20.84.** Let $1 \leq r \leq s$ and $\mathbb{T} = \{(a,b,c) \in \mathcal{O}^3 : a \equiv b \pmod{\varpi^r},\ a \equiv c \pmod{\varpi^s}\}$, with $\pi(a,b,c) = a$. Compute $\wp$, $I$, $\eta(\pi)$ and $\mathrm{Cot}(\pi)$, and show that $\mathrm{length}\,\mathrm{Cot}(\pi) > v(\eta)$, so that $\mathbb{T}$ is not a complete intersection. Do the same for the projections to $b$ and to $c$. (Answer: $\wp = 0 \times \varpi^r\mathcal{O} \times \varpi^s\mathcal{O}$, $I = \varpi^s\mathcal{O} \times 0 \times 0$, $\eta = \varpi^s\mathcal{O}$, $\mathrm{Cot} \simeq \mathcal{O}/\varpi^r \oplus \mathcal{O}/\varpi^s$, of length $r+s > s$. For the projection to $b$: $\wp$ has basis $(\varpi^r, 0, \varpi^r)$, $(0,0,\varpi^s)$, $I = 0 \times \varpi^r\mathcal{O} \times 0$ and $\eta = \varpi^r\mathcal{O}$; the products of the basis elements span $\varpi^r\wp$, so $\mathrm{Cot} = \wp/\varpi^r\wp \simeq (\mathcal{O}/\varpi^r)^2$, of length $2r > r$. For the

projection to $c$: $\wp = \varpi^s\mathcal{O}\times\varpi^r\mathcal{O}\times 0$, $I = 0\times 0\times\varpi^s\mathcal{O}$, $\eta=\varpi^s\mathcal{O}$ and $\mathrm{Cot}\simeq\mathcal{O}/\varpi^s\oplus\mathcal{O}/\varpi^r$, of length $r+s>s$. The inequality is strict for every projection, as it must be by Theorem 20.19(ii): $\mathbb{T}$ is local and not a complete intersection, so no surjection to $\mathcal{O}$ gives equality.)

**Exercise 20.85.** Let $\mathbb{T}=\mathcal{O}[x]/(x^2(x-\varpi))$ and $\pi(x)=0$. Show that $\mathrm{Ann}_{\mathbb{T}}(\wp)=x(x-\varpi)\mathbb{T}\subseteq\wp$, so $\eta=0$ and $\wp\cap I\neq 0$, and that the factor of $\mathbb{T}_K$ at $\pi$ is $K[x]/(x^2)$. Then let $\pi'(x)=\varpi$ and show $\eta(\pi')=\varpi^2\mathcal{O}$. Why does Lemma lem:ddt417 survive the first case? (Hint: Remark 20.23(a). For $\pi'$: $I=x^2\mathbb{T}$ and $\pi'(x^2)=\varpi^2$.)

**Exercise 20.86.** Let $\mathbb{T}_3^\flat$ act diagonally on $L=\mathcal{O}^3$ with the standard pairing. Run the proof of Lemma 20.21: find $L[\wp]$, $L[I]$, $Q$, and compare $v(\Delta)$ with $v(\eta)$. Is $L$ free over $\mathbb{T}_3^\flat$? (Answer: $L[\wp]=\mathcal{O}e_1$, $L[I]=\mathcal{O}e_2\oplus\mathcal{O}e_3$, $Q=0$, $v(\Delta)=0<1=v(\eta)$; $L$ is not free, since by Step 6 a free module would give $v(\Delta)=d\,v(\eta)$.)

**Exercise 20.87.** (a) Prove that for an alternating form on a free module of rank two over $\mathcal{O}$ the image of $\wedge^2$ is the ideal of values and the Gram determinant is its square (Proposition 20.32(ii)). (b) Show that in rank four the ideal of values does not determine the Gram determinant: compare the block-diagonal forms with blocks $\left(\begin{smallmatrix}0&a\\-a&0\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}0&b\\-b&0\end{smallmatrix}\right)$ for $(a,b)=(1,1)$ and $(1,\varpi)$. (Answer to (b): both have ideal of values $\mathcal{O}$; the determinants are 1 and $\varpi^2$. So for a lattice of rank $d>2$ the Gram determinant, not the ideal of values, is the right invariant; Diamond, Flach and Guo only need rank two.)

**Exercise 20.88.** Let $\mathbb{T}$ be a commutative $\mathbb{Z}_\ell$-algebra of operators on a lattice $L$ with a perfect pairing for which every $t\in\mathbb{T}$ is self-adjoint. Show that every idempotent $e\in\mathbb{T}$ is self-adjoint, that $eL\perp(1-e)L$, and that the pairing restricted to $eL$ is perfect. Deduce that for $\mathfrak{m}=\mathfrak{m}_{f,\ell}$ the lattice $M_{f,\lambda}$ lies in $e_{\mathfrak{m}}H^1(X_0(N),\mathbb{Z}_\ell)$ and that $\hat{\delta}_!$ is perfect there. (Hint: $\langle ex,(1-e)y\rangle=\langle x,e(1-e)y\rangle=0$; a perfect pairing on $eL\oplus(1-e)L$ with the summands orthogonal is perfect on each.)

**Exercise 20.89.** Let $f=f_{11\mathrm{a}}$ and $\ell=3$. For the primes $p<40$, $p\neq 3,11$, compute $v_3$ of Wiles's factor $\frac{(p-1)((p+1)^2-a_p^2)}{p^3}$ and say which part, $p-1$ or $(p+1)^2-a_p^2$, contributes. Which primes satisfy Ribet's level-raising condition $a_p\equiv\pm(p+1)\pmod 3$? (Answer, with $a_p=-2,1,-2,4,-2,0,-1,0,7,3$ for $p=2,5,7,13,17,19,23,29,31,37$: the valuation is positive for $p=7$ $(1+1)$, 13 $(1+2)$, 19 $(2+0)$, 29 $(0+2)$, 31 $(1+1)$, 37 $(2+0)$; the level-raising condition holds for $p=7,13,29,31$; at $p=19,37$ only $p\equiv 1\pmod 3$ contributes, which corresponds to congruences at level $11p^2$.)

**Exercise 20.90.** For $f=f_{11\mathrm{a}}$ compute $L_2^{\mathrm{nv}}(A_f,1)^{-1}$ and $L_7^{\mathrm{nv}}(A_f,1)^{-1}$, and deduce $v_3(\eta_{f,3}^{\{2\}})$ and $v_3(\eta_{f,3}^{\{7\}})$ from Theorem 20.39. Compare the second with the Euler factor of the symmetric square of `539d1` at 7, which PARI gives as $(1+3X-21X^2-343X^3)^{-1}$, and explain the coincidence. (Answer: $\frac12\cdot\frac{9-4}{4}=\frac58$, a 3-adic unit, so $v_3(\eta^{\{2\}})=0$; $\frac67\cdot\frac{64-4}{49}=\frac{360}{343}$, so $v_3(\eta^{\{7\}})=0+2=2$. And $1+\frac{3}{49}-\frac{21}{49^2}-\frac{343}{49^3}=\frac{360}{343}$: `539d1` is the twist of `11a1` by $\chi_{-7}$, the true adjoint factor is twist-invariant, and `11a1` is good at 7, so its true factor is Wiles's factor. This is Step 3 of the proof of Theorem thm:quant, and $v_3(\eta^{\{7\}})=2=v_3(504)$.)

**Exercise 20.91.** Let $E$ have good reduction at $\ell\geq 5$ and $E[\ell]$ irreducible. Using the description of $\bar\rho|_{I_\ell}$ by fundamental characters (ordinary: $\omega\oplus 1$ up to semisimplification; supersingular: $\omega_2\oplus\omega_2^\ell$), show that $\bar\rho\not\simeq\bar\rho\otimes\chi_{\ell^*}$, so $\ell\notin S_f$. Where does the argument fail for $\ell=3$? (Hint: $\chi_{\ell^*}|_{I_\ell}=\omega^{(\ell-1)/2}$; in the ordinary case $\{\omega,1\}=\{\omega^{(\ell+1)/2},\omega^{(\ell-1)/2}\}$ forces $(\ell-1)\mid\frac{\ell+1}{2}$. For $\ell=3$ it holds: $\omega^2=1$ and $\omega^1=\omega$.)

**Exercise 20.92.** (a) Let $p \neq \ell$ and $W = T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ with $T$ unramified at $p$. Show that $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W) = W/(\mathrm{Frob}_p - 1)W$ is divisible and equals $H^1_f(\mathbb{Q}_p, W)$. (b) Let $T = \mathbb{Z}_\ell^2$ with inertia acting through the tame quotient by $t \mapsto \left(\begin{smallmatrix} 1 & \ell^a \\ 0 & 1 \end{smallmatrix}\right)$ and a Frobenius acting trivially on $T^{I_p}$ (possible when $p \equiv 1 \pmod{\ell^a}$). Show that $W^{I_p} \simeq \mathbb{Q}_\ell/\mathbb{Z}_\ell \oplus \mathbb{Z}/\ell^a$ and that $H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$ is not divisible, while $H^1_f(\mathbb{Q}_p, W)$, its maximal divisible subgroup (Chapter 18), is $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. This is why Diamond, Flach and Guo write the unramified condition only outside a set $\Sigma$ that contains the ramified primes. (Hint for (b): $(x, y)$ is fixed by $t$ iff $\ell^a y = 0$ in $\mathbb{Q}_\ell/\mathbb{Z}_\ell$.)

**Exercise 20.93.** Write out Theorem 20.60 for `77c1` at $\ell = 3$: $N = 77$, type $\mathrm{I}_3$ at 7 and $\mathrm{I}_2$ at 11, $\deg \varphi_E = 6$, $E[3]$ irreducible with image not 3Ns, and Tate parameter at 7 equal to $5 \cdot 7^3 + O(7^4)$. Compute every term of Corollary 20.61 and deduce length $H^1_f(\mathbb{Q}, \mathrm{ad}^0\, T_3E \otimes \mathbb{Q}_3/\mathbb{Z}_3)$. (Answer: $e_3 = 0$ at both primes (semistable); at 7, $t_3 = 1$ and $\bar{u} = 5$ is not a cube in $\mathbb{F}_7^\times$, so the Tamagawa term is 1; at 11, $t_3 = v_3(2) = 0$. So $1 = v_3(6) = \text{length}\, H^1_f + 1$ and the Selmer group is 0. Compare Example 20.64(c): `539c1` is the twist of `77c1` by $\chi_{-7}$, with the same Selmer group and Tamagawa factors and one more Euler term.)

**Exercise 20.94.** In the situation of Remark 20.65 with $\ell = 3$, $n = 9$ and $q \equiv 1 \pmod 9$, show that the excess $\min(v_3(9), v_3(q-1), v_3(c(u)))$ is 2 if $\bar{u}$ is a ninth power in $\mathbb{F}_q^\times$, 1 if it is a cube but not a ninth power, and 0 otherwise; so the bound of (eq:thm) can be exceeded by 2 at one prime. (Hint: $c(u)$ is the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^9$, a cyclic group of order 9, and $v_3$ of a class is read as in Appendix B, B-C10.)

**Exercise 20.95.** Show that $\iota_{\{11\}}(f_{121\mathrm{c}}) = f_{121\mathrm{a}}$ (both sides lie in $S_2(\Gamma_0(121))$, so it suffices to compare $a_n$ for $n \leq 22$, Sturm's bound; for instance $a_2, a_3, a_5, a_7, a_{13}$ are $-1, 2, 1, 2, -1$ for `121a1` and $1, 2, 1, -2, 1$ for `121c1`), and deduce $r($`121a1`$) = r($`121c1`$)$ from Theorem 20.69. Why does the same twist not lower the level? (Answer: $\chi_{-11}(p)$ is $-1, +1, +1, -1, -1$ for $p = 2, 3, 5, 7, 13$; the twist by $\chi_{-11}$ exchanges the types II and IV$^*$ at 11 (the twist of `121a1`, type II, is `121c2`, type IV$^*$; `121c1`, type IV, is only isogenous to it, being the twist of `121a2`, type II$^*$), all of conductor exponent 2, so $N_S = N = 121$ and both hypotheses of the theorem hold. The congruence number depends only on the newform, hence only on the isogeny class: $r = 6$ for both classes.)

**Exercise 20.96.** (Theorem thm:quant at $q = 3$.) The curves `333d1` and `477a1` have type $\mathrm{I}_0^*$ at 3 and $\deg \varphi_E = 28$. PARI gives the Euler factor of the symmetric square at 3 as $(1 - 6X + 18X^2 - 27X^3)^{-1}$ for both. Compute $L_3(A_f, 1)^{-1}$, compare with the formula $\frac{2}{3} \cdot \frac{16 - a_3(E')^2}{9}$ for a twist $E'$ with good reduction at 3, find $a_3(E')^2$, and check the inequality of Theorem 20.72 at $\ell = 7$. (Answer: $1 - \frac{6}{9} + \frac{18}{81} - \frac{27}{729} = \frac{14}{27}$, so $a_3(E')^2 = 9$; $e_7(E, 3) = v_7(\frac{14}{27}) = 1 \leq v_7(28) = 1$. As $7 \notin S_f$, the equality clause gives $v_7(\deg \varphi_{E'_S}) = 0$.)

**Exercise 20.97.** For `539d1` and $\ell = 3$ compute each term of the chain of Corollary 20.77: $v_3(r_E)$, $v_3(\eta^{\mathrm{Hecke}}(g^S))$ with $g = f_{11\mathrm{a}}$, $v_3(\eta^{\{7\}}_{g,3})$, $v_3(r_{E_S})$ and $e_3(E, 7)$, using Example 20.6 and Exercise 20.90. Then do the same at $\ell = 2$ for Step 1 only, which Lemma 20.74 covers although $\ell = 2$ is outside the hypotheses of Theorem thm:quant. (Answer: $2 = 2 = 2 = 0 + 2$. At $\ell = 2$: $v_2(504) = 3 = v_2(504)$.)

**Exercise 20.98.** Explain where the proof of the inequality $v_\ell(r_E) \geq v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ (the second half of the proof of Lemma 20.74) breaks down for $\ell = q \in S$, when $E$ has type $\mathrm{I}_0^*$ at $q$ and $E_S$ is ordinary at $q$ ($q \nmid a_q(g)$). (Answer: the roots $\alpha, \beta$ of $X^2 - a_q(g)X + q$ satisfy $\alpha\beta = q$ with $\alpha$ a $q$-adic unit, so $\beta$ is not a unit; the stabilisation $g(z) - \alpha g(qz)$, with $U_q$-eigenvalue $\beta \equiv 0$, lies in the same maximal ideal as $g^S$ at $\ell = q$. It belongs to the old space of $g$ itself, and $\iota$ sends it to a

form with the system of $f$, which is not in $f^{\perp}$: the transport fails. Congruences between $g^S$ and this stabilisation contribute to $\eta^{\mathrm{Hecke}}(g^S)$ but not to $r_E$.)

## 20.16 Notes and references

*Congruence modules.* Congruence primes go back to Hida [82], who related them to the Petersson norm and the special value of the symmetric square; [83] is an exposition, and [159] treats weight 2 through modular curves. The algebraic formalism of Definition 20.7 is that of [38, Section 4]; Lenstra's paper [112] proves the numerical criterion in the form of Theorem 20.19 and its relation to the Gorenstein property. Lemma lem:ddt417 is the inequality half of [38, Lemma 4.17], as the paper cites it; we have not compared the numbering with the published text of [38], and the book proves only the part the paper states. Wiles's numerical criterion is in [207, Appendix] and [38, Section 5]; the theorem number sometimes quoted for it in [38] is not verified here.

*Diamond, Flach and Guo.* The paper cites the published version [49]; so does the book. Every item of it that the paper or the book cites (the abstract, the definition of $S_f$ in Section 0.2 and as display (31), Theorems 0.1, 0.2, 2.7, 2.15, 3.7, Propositions 1.4, 2.2 and 2.16, Lemmas 2.1, 2.5 and 2.12, the displays (13), (14), (15), (57), the definition of minimal ramification and the sections listed in Remark 20.83) was checked against the published text for this chapter, with the page numbers given there; the items of Sections 1.2 and 1.4 used in the proof of Proposition prop:integral (Lemma 1.1, Sections 1.2.2, 1.2.3, 1.4.1, 1.4.2, display (5)) are cited as the paper cites them, by section, without page numbers. Equivariant cohomology with compact support and its two spectral sequences are in Chapter V of [77]; Shapiro's lemma and the Hochschild-Serre spectral sequence for groups are in [17]. The longer version [50] ("August 2001, with minor update December 2025") contains the foundational details omitted from the published paper, with different numbering. The Taylor-Wiles method is [207, 197]; [48] gives the freeness of the cohomology and the version without the Gorenstein hypothesis; [38] (Chapters 2 to 5) is the standard account, and we followed its organisation in Section 20.8. Level raising and Ihara's lemma are [160, 156] and [53]; Fontaine-Laffaille theory is [68], and the local formula of Fontaine and Perrin-Riou is [69, I.4.2.2], as recalled in [49].

*The modular degree.* Agashe, Ribet and Stein [3, Theorem 2.2] (theorem number as cited by the paper and checked in Chapter 14); the multiplicity-one and Gorenstein results behind Remark rem:free are [125], [163, Theorem 3.5] (as cited by the paper) and [199]; the Shimura subgroup is [115, 161]. The formula of Kim and Ota is [99, Theorem 1.1], and Pollack and Weston's earlier result is [153, Theorem 6.11] (numbering of the arXiv version; hypotheses in Chapter 15); Ribet and Takahashi's comparison with Shimura curves is [164, Theorems 1 and 2], sharpened in [192, Theorem 3.2, Corollary 3.5] (checked against the sources in Chapter 14).

*What is new here.* Relative to the paper: the length form of Lemma lem:ddt417 over any discrete valuation ring; and the recomputation of the data of the paragraph after Theorem thm:twist (Example 20.70). All numbers in this chapter were computed with PARI/GP [148] (version 2.17, through the Python interface `cypari2`) and Cremona's files [36] for conductors below 10,000; how to reproduce such computations is explained in Chapter 22, and the notation is collected in Appendix B. The paper's own account of these results, with the book's commentary, is in Chapter 21.

# Chapter 21

# Reading the paper: a guided tour

This chapter reads the research note "Local factors of the adjoint $L$-function and the $\ell$-adic valuations of the modular degree" section by section, in the paper's own order. It is written last, from the finished book. For each section and each numbered result of the paper the tour does three things: it states the result (in the paper's words when they are quoted, in the book's notation otherwise); it says in which chapter of the book each ingredient is proved or explained; and it expands the steps that the paper compresses, with short guidance on the hard ones. The chapter ends with a map of the paper (dependencies, hypotheses, imported statements) and a table of worked verifications.

## Core results at a glance

The paper's results are listed in the paper's order, with the paper's labels. Throughout, $E/\mathbb{Q}$ has conductor $N$, $f = f_E$, $\ell$ is an odd prime, $T = T_\ell E$, $V = V_\ell E$, $A_f = \mathrm{ad}^0 V$, $W = \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, and $\mathrm{Frob}_p^{-1}$ is the geometric Frobenius.

**21.1 Conventions.** Standing notation; $\mathrm{Frob}_p$ arithmetic, $L_p(M,s) = \det(1 - \mathrm{Frob}_p^{-1} p^{-s} \mid M^{I_p})^{-1}$, $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$, $L_q(\mathrm{Sym}^2 f, s) = P_q(q^{-s})^{-1}$; duals $M^\vee$, $X^*$, $M^D$, and $T^D \simeq A(1)$ for $\ell$ odd. *(convention)* Conventions 21.1 to 21.3.

**21.2 Definition def:terms.** $t_\ell(E,q) = v_\ell(-v_q(j))$ if $v_q(j) < 0$, else 0 (every bad $q$); $e_\ell(E,q) = v_\ell(P_q(q^{-2})/(1 - a_q^2 q^{-2}))$ $(q \neq \ell)$, a nonnegative integer, 0 at multiplicative $q$; $t_\ell$ is twist invariant, $e_\ell(E,q)$ invariant under twists unramified at $q$ only (`11a1`, `121d2`). *(definition; properties proved here.)* Definition 21.4.

**21.3 Tamagawa factor** (definition). $\mathrm{Tam}_p(M) = \mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p, M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}})$; $H^1_f(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ the Bloch-Kato Selmer group. *(definition.)* Definition 21.5.

**21.4 Theorem thm:main.** $E$ optimal, $\ell \nmid 2N$ odd, $\ell \notin S_f$: $v_\ell(\deg \varphi_E) = \mathrm{length}\, H^1_f(\mathbb{Q}, W) + \sum_{q \mid N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) + \sum_{q \mid N} e_\ell(E,q)$, hence $v_\ell(\deg \varphi_E) \geq \sum_q (e_\ell(E,q) + t_\ell(E,q))$. *(proved here, given the theorems of Diamond, Flach and Guo, [3, Theorem 2.2] and Proposition prop:integral, which is proved here.)* Theorem 21.6.

**21.5 Conjecture conj:euler.** For $E$ optimal, non-CM, $E[\ell]$ irreducible: $v_\ell(\deg \varphi_E) \geq \sum_{q \neq \ell} e_\ell + \sum_q t_\ell + [\ell = 3] w_3(E) - h_\ell(E)$; $h_\ell = 0$ for $\ell \geq 5$, $h_3 \in \{0, 1\}$ with $h_3 = 1$ iff image 3Ns. *(stated; the facts on $h_\ell$ proved here.)* Conjecture 21.7.

**21.6 Component groups and weights** (definition): $\#\Phi_q^{\mathrm{geom}}$ by type; $w_\ell(E,q)$; (eq:main), (eq:twist); $w_\ell = \max(v_\ell\#\Phi_q^{\mathrm{geom}}(E), v_\ell\#\Phi_q^{\mathrm{geom}}(E^{(q)}))$ for odd $q$. Definition 21.8.

**21.7 Proposition prop:forms.** Under the hypothesis of Theorem thm:main, (eq:main) and (eq:twist) hold; likewise under (eq:rule) with $h_\ell = 0$. *(proved here, given Theorem 21.6 and Lemma 21.13.)* Proposition 21.9.

**21.8 Conjecture conj:eis.** Eisenstein deficit $D \le v_\ell(\#E(\mathbb{Q})_{\mathrm{tors}}) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3]$ and $\le$ the number of curves of the class with a rational $\ell$-torsion point; $D \le 2$ follows from the first bound. *(stated.)* Conjecture 21.10.

**21.9 Lemma lem:tamagawa-exp.** At multiplicative $q \ne \ell$: $\rho \bmod \ell^t$ unramified at $q$ iff $\ell^t \mid n_q$. *(proved here, given the Tate curve, Chapter 10.)* Lemma 21.11.

**21.10 Inertia invariants.** At additive $q \ne \ell$: $E[\ell^k]^{I_q} \simeq \Phi_q^{\mathrm{geom}}[\ell^k]$. *(proved here, given the Néron filtration, Chapter 10.)* Lemma 21.12.

**21.11 Lemma lem:IV.** At $q \ne 3$ of type IV, IV$^*$: potentially good, $\rho(I_q)$ cyclic of order 3, $E[3]^{I_q}$ a line, Serre exponent 1, $f_q = 2$, $P_q = 1 - \chi_{-3}(q)qX$, $e_\ell = v_\ell(q - \chi_{-3}(q))$; at $q = 2$, $P_2 = 1 + 2X$, $e_3 = 1$. *(proved here, given [179] and Chapters 6 and 7.)* Lemma 21.13.

**21.12 Lemma lem:twist.** (i) Same type, $\Phi_p^{\mathrm{geom}}$, $f_p$ at $p \ne q$; (ii) $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ $(n \ge 0)$, II $\leftrightarrow$ IV$^*$, III $\leftrightarrow$ III$^*$, IV $\leftrightarrow$ II$^*$ at every odd $q$, with $f_3$ unchanged by the last three at $q = 3$; (iii) Atkin-Li. *((i), (ii) proved here, given Tate's algorithm and, at $q = 3$, the Ogg-Saito formula; (iii) stated, reference [9].)* Lemma 21.15.

**21.13 Lemma lem:euler.** The table: $L_q(A_f, s)^{-1}$ and $e_\ell$ for $\mathrm{I}_n$ (every $q \ne \ell$), $\mathrm{I}_n^*$, $\mathrm{I}_0^*$ (odd $q$), II, II$^*$, IV, IV$^*$ and III, III$^*$ $(q \ge 5)$. *(proved here, given the inertia images of Chapter 11.)* Lemma 21.16.

**21.14 Proposition prop:shapes.** At every bad $q \ne \ell$, $P_q$ is $1 - X$, the $\mathrm{I}_0^*$ cubic $(1 - qX)(1 - (a^2 - 2q)X + q^2X^2)$, $1 - cqX$ $(c = \pm 1)$, or 1 (nonabelian inertia, $q = 2, 3$ only). *(proved here, given Chapters 11 and 19.)* Proposition 21.17.

**21.15 Lemma lem:tamfactor.** At potentially multiplicative $q \ne \ell$ ($q = 2$ included): $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(n) + \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u))) \ge t_\ell(E,q)$. *(proved here, given Chapters 16 and 17.)* Lemma 21.19.

**21.16 Lemma lem:tamadd.** At potentially good additive $q \ne \ell$: $H^1(I_q, \mathrm{ad}^0 T)_{\mathrm{tor}} = 0$, so $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = 0$. *(proved here, given Chapters 6 and 16.)* Lemma 21.20.

**21.17 Congruence number** (definition) and Agashe-Ribet-Stein: $\deg \varphi_E \mid r_E$, equal $p$-parts for $p^2 \nmid N$. *(the theorem stated; reference [3, Theorem 2.2].)* Definition 21.21.

**21.18 Lemma lem:crit.** $\mathbb{T}/(\ker \phi_f + \mathrm{Ann}(f^\perp)) \simeq \mathbb{Z}/r_E$; a congruent eigenform orthogonal to $f$ gives $\ell \mid r_E$. *(proved in Chapter 15; walked through here.)* Lemma 21.22.

**21.19 Lemma lem:old.** A congruent newform $h \ne f$ of level $M \mid N$, with $p \nmid M$ at the $p \,\|\, N$ where $\bar\rho$ is unramified, gives $\ell \mid r_E$. *(proved in Chapter 15; walked through here.)* Lemma 21.23.

**21.20 Optimal-level theorem** (Ribet, Carayol, Diamond; [38, Theorem 3.15]). *(stated; reference.)* Theorem 21.24.

**21.21 Theorem thm:qual.** $\ell \nmid N$ odd, $\bar\rho$ irreducible (absolutely over $\mathbb{Q}(\sqrt{-3})$ if $\ell = 3$), and a prime of type $\mathrm{IV}, \mathrm{IV}^*$ ($q \geq 5$ or $q = 2$, $\ell = 3$), $\mathrm{II}, \mathrm{II}^*$ ($q \geq 5$, $\ell = 3$) or $\mathrm{I}_n^*$ ($n \geq 1$, $\ell \mid n$, $q$ odd): then $\ell \mid \deg \varphi_{E_0}$. *(proved here, given Theorem 21.24 and [3].)* Theorem 21.25.

**21.22 Objects of Diamond, Flach and Guo** (definition): $M^\Sigma$, $f^\Sigma$, the integral structure, $M_{f^\Sigma,\lambda}$, $\hat\delta_!$, $\eta^\Sigma_{f,\lambda}$. Definition 21.27.

**21.23 Proposition prop:integral.** For every prime $\ell$, pull-back identifies $H^1(X_0(M^\Sigma), \mathbb{Z})$ with the Betti lattice $M(M^\Sigma, 1)_{!,B}$, Hecke equivariantly, $U_p$ included; so $M(M^\Sigma, 1)_{!,\lambda} = H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)$, with no localisation. *(proved here; expanded proof in Chapter 20.)* Proposition 21.28.

**21.24 Theorem thm:dfg14** (Diamond, Flach and Guo, Proposition 1.4(c)). For $\ell \nmid 2M$, $\ell \notin \Sigma$, $\bar\rho_f$ irreducible: $\eta^\Sigma = \eta^\emptyset \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}$. *(stated; reference.)* Theorem 21.29.

**21.25 Proposition prop:pullback.** For optimal $E$: $\varphi^* H^1(E, \mathbb{Z}_\ell) = H^1(X_0(N), \mathbb{Z}_\ell)[I_f]$, $\langle \varphi^* a, w_N^* \varphi^* b\rangle = \varepsilon_N \deg \varphi_E \langle a, b\rangle$; for $\ell \nmid 2N$, $v_\lambda(\eta^\emptyset_{f,\lambda}) = v_\ell(\deg \varphi_E)$. *(proved here, given Chapter 14 and Proposition prop:integral.)* Proposition 21.30.

**21.26 Lemma lem:ddt417.** $d\, v(\eta) \geq v(\Delta)$, with equality if $L$ is free over $\mathbb{T}$. *(proved here, given Chapter 3.)* Lemma 21.31.

**21.27 Lemma lem:hecke.** $v_\ell(\deg \varphi_E) = v_\ell(r_E) = v_\lambda(\eta^\emptyset_{f,\lambda})$ for $E$ optimal, $\ell \nmid 2N$, $\bar\rho$ irreducible; no freeness. *(proved here, given [3] and Proposition 21.30.)* Lemma 21.32.

**21.28 Selmer set-up** (definition): $W$, $T^D$, $V^D$, $W^D$, $H^1_\Sigma$, $S_f$. Definition 21.34.

**21.29 Proposition prop:Sf.** For $\ell \nmid N$ odd and $E[\ell]$ irreducible: $\ell \notin S_f$ if $\ell \geq 5$; $3 \in S_f$ iff image $N_s(\mathbb{F}_3)$ (then ordinary at 3); $\ell \notin S_f$ iff $h_\ell = 0$. *(proved here, given [171, Section 1] and Chapter 4.)* Proposition 21.35.

**21.30 Theorems thm:dfg37 and thm:dfg27** (Diamond, Flach and Guo, Theorems 3.7 and 2.7). For $\ell \notin S_f$: $\operatorname{length} H^1_\Sigma(\mathbb{Q}, W) = v_\lambda(\eta^\Sigma)$; $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$. *(stated; reference; they cover $\ell = 3$.)* Theorems 21.36 and 21.37.

**21.31 Lemma lem:dfg21.** For $\ell \notin \Sigma$: $0 \to H^1_f \to H^1_\Sigma \to \bigoplus H^1_f(\mathbb{Q}_p, T^D)^* \to H^0(\mathbb{Q}, W^D)^* \to 0$; $H^0(\mathbb{Q}, V^D) = 0$ automatic; $H^1_f(\mathbb{Q}_p, T^D)$ the torsion of $H^1(\mathbb{Q}_p, T^D)$. *(stated; reference; the paragraph after it proved here.)* Lemma 21.38.

**21.32 Lemma lem:fpr.** $\operatorname{Fitt} H^1_f(\mathbb{Q}_p, T^D) = L_p(A_f, 1)^{-1} \operatorname{Tam}_p(T^D)$ and $\operatorname{Tam}_p(T^D) = \operatorname{Tam}_p(A)$. *(first identity proved here; second stated, reference [49, (57)], and checked for the elliptic adjoint.)* Lemma 21.39.

**21.33 Lemma lem:h0.** $H^0(\mathbb{Q}, W^D)$ finite cyclic of length $h_\ell(E)$; $h_\ell = 0$ for $\ell \geq 5$; $h_3 \leq 1$, with equality iff $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$ iff image $N_s(\mathbb{F}_3)$; CM allowed. *(proved here, given Chapters 4 and 6.)* Lemma 21.40.

**21.34 Corollary cor:explicit.** $v_\ell(\deg \varphi_E) = \operatorname{length} H^1_f + \sum e_\ell + \sum_{v_q(j)<0}(t_\ell + \min(t_\ell, v_\ell(q-1), v_\ell(c(u_q))))$. *(proved here, given Theorem 21.6.)* Corollary 21.41.

**21.35 The degree formula** (Remark rem:degree). $\deg \varphi_E = c_E^2 N\, L^{\mathrm{nv}}(\mathrm{ad}^0 f, 1)/(2\pi \operatorname{vol}(\Lambda_E)) = B_E \prod_{q^2 \mid N} L_q(A_f, 1)^{-1}$; $v_\ell(\deg \varphi_E) = v_\ell(B_E) + \sum e_\ell$ for $\ell \nmid N$. *(proved here, given the formula of Zagier and Watkins and the Petersson formula.)* Proposition 21.42.

**21.36 Theorem thm:twist.** $r_{E_S} \mid r_E$ if $q^2 \mid N$ $(q \in S)$ and $\operatorname{lcm}(N_S, D^2) \mid N$; equality if also $\operatorname{lcm}(N, D^2) \mid N_S$. *(proved here, given Chapter 13.)* Theorem 21.44.

**21.37 Lemma lem:step1.** For every $E$, with $S$ its odd $\mathrm{I}_0^*$, $\mathrm{I}_n^*$ primes: $v_\ell(r_E) \le v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ at every prime $\ell$, with equality at every $\ell \notin S$. *(proved here and in Chapter 20.)* Lemma 21.45.

**21.38 Theorem thm:quant.** For $\ell \nmid 2N$, $E[\ell]$ irreducible, $S$ the odd $\mathrm{I}_0^*$, $\mathrm{I}_n^*$ primes: $v_\ell(\deg \varphi_E) \ge v_\ell(r_{E_S}) + \sum_S e_\ell$, with equality if $\ell \notin S_f$. *(proved here, given Theorems 21.6 and 21.29; also in Chapter 20.)* Theorem 21.46.

**21.39 Proposition prop:ns.** $E[3]$ irreducible: image $\mathrm{GL}_2(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ or $N_s(\mathbb{F}_3)$; absolutely irreducible over $\mathbb{Q}(\sqrt{-3})$ unless $N_s$; then $\bar\rho \otimes \mathbb{F}_9 = \operatorname{Ind} \psi$, $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$, $h_3 = 1$; $\psi_3$ is two quadratics or irreducible. *(proved here, given Chapters 1 and 4.)* Proposition 21.47.

**21.40 Proposition prop:ns-structure.** For image $N_s(\mathbb{F}_3)$: (i) $\bar\rho(I_p) \subseteq C_4$ $(p \ne 3)$, so $t_3 \ge 1$ at potentially multiplicative $p \ne 3$; (ii) the order-four primes are the $p \ne 3$ ramified in $K_1$, a nonempty set, with $e_3 = v_3(p - \chi_{-3}(p)) \ge 1$ there; (iii) at $p \ge 5$ they are the primes of type III, III$^*$, and $p \equiv \pm 1 \pmod{12}$; (iv) biquadratic splitting field; (v) $3 \mid a_p$ for $p \equiv 2 \pmod 3$, $p \nmid 3N$; (vi) ordinary at 3 if $3 \nmid N$; (vii) $h_3 = 1$. *(proved here, given Kraus's theorem and Proposition 21.35.)* Proposition 21.48.

**21.41 Remark rem:ns** (conditional proposition). For image 3Ns, if $H^1_f(\mathbb{Q}, V^D) = 0$: length $H^1_\Sigma =$ length $H^1_f + \sum_p (v(L_p^{-1}) + v(\mathrm{Tam}_p)) - 1$; with $v(\eta^\Sigma) =$ length $H^1_\Sigma$ this gives (eq:identity) minus $h_3$. *(proved here, conditionally, given Lemma 21.38.)* Proposition 21.49.

**21.42 Proposition prop:no9.** A non-CM $E/\mathbb{Q}$ with $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ has no rational point of order 9; so $D \le 2$ in Conjecture conj:eis follows from its first bound. *(proved here, given [114] and [125].)* Proposition 21.50.

**21.43 The map and the verifications.** Dependencies, hypotheses and imported statements (Section 21.17); recomputed curve data (Section 21.18).

## Where the paper uses this chapter

All of it: this chapter is the paper, read with the book open beside it. Each numbered section below covers one section of the paper (or a part of one) and opens by saying which. Sections 21.17 and 21.18 are devices of the tour: the dependency map and the table of hypotheses (used to see which inputs each statement needs), and the worked verifications (used to check every number quoted from a curve). A reader who wants only the proofs can read Chapter 20 for Section 4 of the paper, Chapters 10 and 11 for Sections 2 and 5, Chapter 15 for Section 3, Chapters 18 and 19 for the local terms and Chapter 22 for Section 6 and the appendix; this chapter says, result by result, which of these to open.

## 21.1 How to read the tour

This section fixes the conventions the tour uses and explains how it is marked. It corresponds to the paper's Section 1.1 (conventions) and to the opening of every section of the paper, which use the same notation.

**Convention 21.1** (standing notation)**.** $E/\mathbb{Q}$ is an elliptic curve of conductor $N$; unless said otherwise it is the $X_0(N)$-optimal curve of its isogeny class, $\varphi_E\colon X_0(N)\to E$ its modular parametrisation and $\deg\varphi_E$ its modular degree; $f=f_E\in S_2(\Gamma_0(N))$ is its newform and $\phi_f\colon\mathbb{T}\to\mathbb{Z}$, $T_n\mapsto a_n(f)$, the eigencharacter of $f$ on the full Hecke algebra $\mathbb{T}$ (generated by all $T_n$, $n\geq 1$, the $U_p$ for $p\mid N$ included). The prime $\ell$ is odd; $T=T_\ell E$, $V=V_\ell E$, $\rho=\rho_{E,\ell}$, $\bar\rho$ its reduction on $E[\ell]$; $A_f=\mathrm{ad}^0 V$ and $A=\mathrm{ad}^0 T$ are the trace-zero adjoints (the macro already contains the superscript 0); $A_f$ is never an abelian variety. A bad prime is written $q$, any prime $p$. For $p\neq\ell$, $\delta_p=\dim V^{I_p}$ is $2,1,0$ for good, multiplicative, additive reduction. $v_\ell$ is the $\ell$-adic valuation, extended to $\mathbb{Q}^\times$, to ideals of $\mathbb{Z}_\ell$, and to a class $c\in\mathbb{Z}/\ell^m$ by $v_\ell(c)=\min(v_\ell(\tilde c),m)$ for any lift $\tilde c$; length is $\mathbb{Z}_\ell$-length. Characters: $\chi_\ell$ is the $\ell$-adic cyclotomic character and $\bar\chi_\ell$ its reduction; a numerical subscript means the quadratic character of a quadratic field of that discriminant ($\chi_{-3}$, $\chi_{-4}$, $\chi_{-7}$, $\chi_{-11}$); for an odd prime $q$, $q^*=(-1)^{(q-1)/2}q$ and $\chi_{q^*}$ is the quadratic character of conductor $q$; at $\ell=3$ the cyclotomic character is $\chi_3^{\mathrm{cyc}}$, and $\bar\chi_3^{\mathrm{cyc}}=\chi_{-3}$. $E^{(q)}$ is the twist of $E$ by $\chi_{q^*}$ and $E_S=E\otimes\chi_S$ with $\chi_S=\prod_{q\in S}\chi_{q^*}$. The Tamagawa number is $c_q=\#\Phi_q(\mathbb{F}_q)$ and $\Phi_q^{\mathrm{geom}}=\Phi_q(\overline{\mathbb{F}}_q)$ the geometric component group; the class of the unit part of a Tate parameter is $c(u)$; the Manin constant is $c_E>0$. Depletion is $f^\Sigma$; the naive twist of a form is $h\otimes\chi=\sum\chi(n)a_n(h)q^n$, the newform twist $h^\chi$. These are the conventions of Appendix B, and the paper uses the same ones.

**Convention 21.2** (Frobenius and $L$-factors)**.** $\mathrm{Frob}_p$ is an arithmetic Frobenius: it induces $x\mapsto x^p$ on the residue field and $\chi_\ell(\mathrm{Frob}_p)=p$. On the tame quotient of inertia, $\mathrm{Frob}\,t\,\mathrm{Frob}^{-1}=t^p$, equivalently $\mathrm{Frob}^{-1}\,t\,\mathrm{Frob}=t^{p^{-1}}$. The geometric Frobenius is $\mathrm{Frob}_p^{-1}$ (the letter $\Phi$ is not used for it in this chapter, since $\Phi_q^{\mathrm{geom}}$ is a component group). For $p\neq\ell$ and a $\mathbb{Q}_\ell$-representation $M$ of $G_{\mathbb{Q}_p}$,

$$L_p(M,s)=\det\bigl(1-\mathrm{Frob}_p^{-1}\,p^{-s}\mid M^{I_p}\bigr)^{-1},$$

the convention of [49, Sections 1.1.1, 1.7.2]. On $V$ at a good prime the arithmetic Frobenius has the roots $\alpha,\beta$ of $X^2-a_pX+p$; on $\mathrm{ad}^0 V$ both Frobenii have the eigenvalues $\alpha/\beta,1,\beta/\alpha$; at a multiplicative prime the arithmetic Frobenius acts on $(\mathrm{ad}^0 V)^{I_q}$ by $q$ and the geometric one by $q^{-1}$, and only the latter gives $L_q(A_f,s)^{-1}=1-q^{-1-s}$. Further, $L_p(\mathrm{Sym}^2 f,s)=L_p(A_f,s-1)$, $L_q(\mathrm{Sym}^2 f,s)=P_q(q^{-s})^{-1}$ defines $P_q\in\mathbb{Z}[X]$, and $L_p(M(1),s)=L_p(M,s+1)$. These facts are proved in Chapters 7 and 19; the paper states the same convention in Section 1.1 and says that the matrices in the proof of Lemma lem:tamfactor are those of the arithmetic Frobenius.

**Convention 21.3** (duals)**.** $M^\vee=\mathrm{Hom}(M,\mathbb{Z}_\ell)$ is the linear dual, $X^*=\mathrm{Hom}_{\mathrm{cont}}(X,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ the Pontryagin dual (the paper's $^*$ in Lemma lem:dfg21), $M^D=\mathrm{Hom}(M,\mathbb{Z}_\ell(1))$ the Kummer dual; a star after the name of a map is a pullback ($\varphi_E^*$). For $A=\mathrm{ad}^0 T$: $W=A\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$, $T^D=\mathrm{Hom}(A,\mathbb{Z}_\ell(1))$, $V^D=T^D\otimes\mathbb{Q}_\ell$, $W^D=V^D/T^D$. For $\ell$ odd the trace form is perfect on $\mathrm{ad}^0 T$ (Chapter 4), so $T^D\simeq A(1)$ and $W^D\simeq\mathrm{ad}^0 T(1)\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$. The adjoint does not see duals or twists: $\mathrm{ad}^0(T^\vee)=\mathrm{ad}^0 T=\mathrm{ad}^0(T(n))=\mathrm{ad}^0(T\otimes\chi)$.

**How the tour is marked.** Every result of the paper is restated in a numbered environment whose title carries the paper's label (the paper's labels are quoted as text, for instance Lemma

lem:tamfactor). After the statement comes a line saying where the book proves each ingredient, and then the expansion of the paper's argument, with guidance where a step is hard. Where the book proves more than the paper needs, a one-line remark says so. Sentences of the paper are quoted verbatim, between quotation marks, with the paper's own symbols; everything else is paraphrase.

**What the tour does not claim.** The paper's large-range counts (conductor up to 500000 and the curves outside the tables) were not recomputed for this book; the tour reads the paper's tables from their printed rows and quotes their entries as printed. The book's own computations cover conductor below 10,000 (38,042 optimal curves, Cremona's files [36], PARI/GP [148] version 2.17.2 through the Python interface `cypari2`); they are always labelled as the book's, with their range, and they were made with the paper in hand, so they are checks, not blind tests. A theorem number of another author's paper is said to be *verified* only when it was compared with the published text; otherwise it is given *as cited* by the paper.

**How the paper was produced.** The paper ends with a statement on the use of an AI system, which says that the work was carried out with an AI system at every stage, under the author's direction, that several of its results and proofs were worked out with the same system while this book was being written, that the note has not yet been refereed and "the author has to review it before it is circulated", and that the author takes full responsibility for all statements. This book was itself drafted with the assistance of an AI system, under human direction (see the preface). Neither fact changes what is proved: every argument below is meant to be checked line by line, and every count can be rerun.

## 21.2 The abstract and the introduction

This section reads the abstract and the first two paragraphs of the paper's Section 1. Their job is to announce the identity (eq:identity), to say where the modular degree meets the adjoint $L$-function, and to name the inputs. Nothing is proved there; the tour says where each claim is proved.

**The abstract, claim by claim.** The abstract makes nine claims. (1) For $E$ optimal and $\ell \nmid N$ odd with $\bar{\rho}$ absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, the identity

$$v_\ell(\deg \varphi_E) = \mathrm{length}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell) + \sum_{q \mid N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) + \sum_{q \mid N} e_\ell(E, q)$$

holds: this is Theorem thm:main, proved in Section 21.11 below and in Chapter 20. The parenthesis "(for $\ell \geq 5$ this only asks that $E[\ell]$ be irreducible)" is Proposition prop:Sf (Proposition 21.35). (2) The values of $e_\ell(E, q)$ at a tame $q \geq 5$ are Lemma lem:euler (Lemma 21.16). (3) The Tamagawa factors are trivial at potentially good primes (Lemma lem:tamadd) and at least $t_\ell(E, q)$ at potentially multiplicative ones (Lemma lem:tamfactor). (4) The proof is an assembly of four inputs from Diamond, Flach and Guo and Fontaine and Perrin-Riou, the local computations and the projection formula; the abstract adds that "it also uses an identification of their integral structure with the cohomology of $X_0(N)$, which we prove": that is Proposition prop:integral (Proposition 21.28), which holds integrally, for every prime $\ell$, with no localisation. (5) "The Tamagawa factor raises the bound by at least one whenever a Steinberg prime $q \equiv 1$

(mod $\ell$) with $\ell \mid n_q$ has a Tate parameter whose unit part is an $\ell$-th power": a Steinberg prime is a prime of multiplicative reduction (the local representation is an unramified twist of the Steinberg representation), and the claim is the case $\min(v_\ell(n), v_\ell(q-1), v_\ell(c(u))) \geq 1$ of Lemma lem:tamfactor. (6) The $\mathrm{I}_0^*$ and $\mathrm{I}_n^*$ terms hold without the hypothesis over $\mathbb{Q}(\sqrt{\ell^*})$: Theorem thm:quant, which still needs $E[\ell]$ irreducible and $\ell \nmid 2N$. (7) The failure of the inequality at $\ell = 3$ for image $N_s(\mathbb{F}_3)$, and (8) its validity with the correction term "for every curve of conductor at most 500000 that the tables determine to be optimal, with $E[\ell]$ irreducible and without complex multiplication", are data statements (Section 21.14); the two hypotheses matter (`11a1` at the Eisenstein prime $\ell = 5$ violates the bound, Section 21.16). (9) In the Eisenstein case the abstract claims only "that the bound 2 for the deficit at $\ell = 3$ follows from a conjectured bound by the rational torsion": that is Proposition prop:no9 (Proposition 21.50); the torsion bound itself is Conjecture conj:eis.

**The first paragraph of the introduction.** "When $E[\ell]$ is irreducible, every curve of the class is linked to $E$ by a cyclic isogeny of degree prime to $\ell$, so $v_\ell$ of the degree of its minimal parametrisation is $v_\ell(\deg \varphi_E)$". This is proved in Chapter 9 (an isogeny of degree divisible by $\ell$ between curves of the class would give a rational subgroup of order $\ell$, that is, $E[\ell]$ reducible somewhere in the class, hence everywhere) and Chapter 14 (the degree of $\varphi_{E'} = \psi \circ \varphi_E$ is $\deg \psi \cdot \deg \varphi_E$). The equivalence of the bound $\deg \varphi_E \ll N^{2+\varepsilon}$ with a form of the *abc* conjecture is quoted from [3, Section 2] and is not used.

**The second paragraph: naive and true $L$-functions.** Three facts are used. First, the formula of Zagier and Watkins and the special value $\langle f, f \rangle$; the exact form is Remark rem:degree, read in Section 21.11 (Proposition 21.42). Second, Agashe, Ribet and Stein: $\deg \varphi_E \mid r_E$ with equal $p$-parts when $p^2 \nmid N$ (Definition 21.21). Third, the naive symmetric square $L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) = \zeta^{(N)}(2s-2) \sum_{n \geq 1} a_{n^2}(f) n^{-s}$, Shimura's series [182], whose local factors are the true ones at good and multiplicative primes and 1 at the additive primes. A reader should not confuse it with the Rankin series $\sum_n a_n(f)^2 n^{-s}$, which differs: at a good prime $p$, with $X = p^{-s}$,

$$\sum_{k \geq 0} a_{p^k}^2 X^k = \frac{1 + pX}{(1-\alpha^2 X)(1-pX)(1-\beta^2 X)}, \qquad \sum_{k \geq 0} a_{p^{2k}} X^k = \frac{1 + pX}{(1-\alpha^2 X)(1-\beta^2 X)},$$

so that $\sum_n a_n^2 n^{-s} = \zeta^{(N)}(s-1) L^{\mathrm{nv}}(\mathrm{Sym}^2 f, s) / \zeta^{(N)}(2s-2)$ exactly; at a multiplicative prime both local series are $1/(1-X)$, at an additive prime both are 1 (Chapter 19, Exercise 21.59). The paragraph ends with the paper's thesis: the primes where the naive and the true adjoint $L$-functions differ are exactly the primes where the modular degree acquires extra local divisibility, and the amount is the valuation of the omitted factor. Everything after this is the proof of that sentence and the description of its boundary.

## 21.3 Section 1.1 of the paper: the local terms

This section reads the definition of the two local terms $e_\ell(E, q)$ and $t_\ell(E, q)$ and the table of omitted Euler factors. They are the right-hand side of every inequality in the paper. The table is proved by Lemma lem:euler (Section 21.7); here we fix the definitions and their elementary properties.

**Definition 21.4** (the paper's Definition def:terms)**.** For a bad prime $q$ put $t_\ell(E,q) = v_\ell(-v_q(j_E))$ if $v_q(j_E) < 0$ and $t_\ell(E,q) = 0$ otherwise; and, if $q \neq \ell$,

$$e_\ell(E,q) = v_\ell\Big(\frac{L_q(A_f,1)^{-1}}{L_q^{\mathrm{nv}}(A_f,1)^{-1}}\Big) = v_\ell\Big(\frac{P_q(q^{-2})}{1-a_q(E)^2q^{-2}}\Big),$$

where $L_q^{\mathrm{nv}}(A_f,s) = L_q(A_f,s)$ if $\delta_q > 0$ and $L_q^{\mathrm{nv}}(A_f,s) = 1$ if $\delta_q = 0$.

*Where proved.* $L_q$ and $P_q$: Chapter 19; $v_q(j)$ and Kodaira types: Chapter 10; the Tate parameter: Chapter 10.

*Why these two terms.* The paper compares two objects. The modular degree is computed by the Petersson norm of $f$, and the Petersson norm sees the *naive* adjoint $L$-value, in which the Euler factor at an additive prime is replaced by 1 (Remark rem:degree, Proposition 21.42). The Selmer side of the theorem of Diamond, Flach and Guo sees the *true* $L$-value. The term $e_\ell(E,q)$ is the $\ell$-part, at one prime, of the ratio between the two: it records what is lost at $q$ when the naive value replaces the true one. The term $t_\ell(E,q)$ has a different origin: when $v_q(j) < 0$ the curve is, up to a twist, a Tate curve with parameter of valuation $n = -v_q(j)$, and $\ell^{t_\ell(E,q)}$ is the $\ell$-part of $n$, the $\ell$-part of the order of the component group of that Tate curve; it enters through the Tamagawa factor (Lemma lem:tamfactor below). A small example, with the values of Section 21.18: `726a1` has type $\mathrm{I}_6$ at 2, $\mathrm{I}_3$ at 3 and III at 11, with $P_{11}(X) = 1+11X$. At $\ell = 3$: $e_3(E,2) = 0$ (multiplicative), $t_3(E,2) = v_3(6) = 1$; $t_3(E,3) = v_3(3) = 1$, while $e_3(E,3)$ is not defined ($q = \ell$); and $e_3(E,11) = v_3(1+11\cdot 11^{-2}) = v_3(12/11) = 1$, $t_3(E,11) = 0$ ($v_{11}(j) \geq 0$).

*Expansion.* (i) The second expression equals the first: $L_q^{\mathrm{nv}}(A_f,1)^{-1}$ is $1-a_q^2q^{-2}$ in every case, since $a_q = \pm 1$ at a multiplicative prime, where the true factor is $1-q^{-2}$, and $a_q = 0$ at an additive prime, where the naive factor is 1; and $L_q(A_f,1)^{-1} = P_q(q^{-2})$ because $L_q(A_f,s) = L_q(\mathrm{Sym}^2 f, s+1)$. (ii) $e_\ell(E,q) = 0$ at a multiplicative prime (naive equals true), and $e_\ell(E,q) = v_\ell(P_q(q^{-2}))$ at an additive prime. (iii) $e_\ell(E,q)$ is a finite nonnegative integer: $P_q$ has integer coefficients and constant term 1, $q^{-2}$ is an $\ell$-adic unit, so $P_q(q^{-2}) \in \mathbb{Z}[1/q]$ has nonnegative valuation; it is nonzero because each eigenvalue $\lambda$ of the geometric Frobenius on $(\mathrm{ad}^0 V)^{I_q}$ has complex absolute value 1 or $q^{-1}$ (Chapter 19), so $1-\lambda q^{-1} \neq 0$. (iv) $t_\ell(E,q) = v_\ell(n)$ for type $\mathrm{I}_n$ at any $q$ and for $\mathrm{I}_n^*$ at odd $q$ (where $v_q(j) = -n$); at $q = 2$ a symbol $\mathrm{I}_m^*$ can have $v_2(j) \geq 0$ (potentially good reduction), and then $t_\ell(E,2) = 0$, or $-v_2(j) \neq m$ (Chapter 10). (v) Twists: $j$ and $\mathrm{ad}^0 V$ are invariant under every quadratic twist, so $t_\ell(E,q)$ and the true factor are. The naive factor at $q$ is invariant under twists unramified at $q$ (the type at $q$ does not change), and so then is $e_\ell(E,q)$. A twist ramified at $q$ that moves the reduction between multiplicative and $\mathrm{I}_n^*$, or between good and $\mathrm{I}_0^*$, changes the naive factor and with it $e_\ell$: `11a1` has type $\mathrm{I}_5$ at 11 and $e_3 = 0$, its twist `121d2` by $\chi_{-11}$ has type $\mathrm{I}_5^*$ and $e_3 = v_3(120) = 1$; the twist `539d2` of `11a1` by $\chi_{-7}$ lies in the class `539d`, of type $\mathrm{I}_0^*$ at 7, with $e_3(\texttt{539d1}, 7) = v_3(6) + v_3(60) = 2$, while `11a1` has no term at 7.

*Reading guidance.* The paper's sentence "The term $t_\ell$ is invariant under quadratic twist, and $e_\ell(E,q)$ under quadratic twists unramified at $q$" is (v); the asymmetry is the whole point of the paper, since the true factor is twist invariant and the naive one is not. Note also that $t_\ell$ is defined at every bad $q$, $q = \ell$ included, as (eq:rule) and the appendix's function need, while $e_\ell(E,q)$ is defined only for $q \neq \ell$.

**The table.** For a bad prime $q \geq 5$, $q \neq \ell$, with $E'$ the twist of $E$ by $\mathbb{Q}(\sqrt{q^*})$ and $\alpha, \beta$ the roots of $X^2 - a_q(E')X + q$:

| type at $q$ | $L_q(A_f,s)^{-1}$ | $e_\ell(E,q)$ |
|---|---|---|
| $\mathrm{I}_n$ | $1-q^{-1-s}$ | $0$ |
| $\mathrm{I}_n^*$ | $1-q^{-1-s}$ | $v_\ell(q^2-1)$ |
| $\mathrm{II},\mathrm{II}^*,\mathrm{IV},\mathrm{IV}^*$ | $1-\chi_{-3}(q)q^{-s}$ | $v_\ell(q-\chi_{-3}(q))$ |
| $\mathrm{III},\mathrm{III}^*$ | $1-\chi_{-4}(q)q^{-s}$ | $v_\ell(q-\chi_{-4}(q))$ |
| $\mathrm{I}_0^*$ | $(1-\frac{\alpha}{\beta}q^{-s})(1-q^{-s})(1-\frac{\beta}{\alpha}q^{-s})$ | $v_\ell(q-1)+v_\ell((q+1)^2-a_q(E')^2)$ |

The table is the content of Lemma lem:euler (Lemma 21.16). The sentence after it reads: "The row $\mathrm{I}_n$ holds at every $q\neq\ell$, $q=2$ included, the rows $\mathrm{I}_n^*$ and $\mathrm{I}_0^*$ at every odd $q\neq\ell$, and the row IV, IV$^*$ at every $q\neq 3,\ell$". The rows II, III need $q\geq 5$, where inertia acts tamely; at $q=2,3$ the factor is given by Proposition prop:shapes and computed by PARI.

The last column follows from the middle one and Definition 21.4 by evaluating at $s=1$ and comparing with the naive factor. For $\mathrm{I}_n$ the naive factor is the true one, so $e_\ell=0$. In the other rows the reduction is additive, the naive factor is 1, and $e_\ell$ is the valuation of the true factor at $s=1$: $v_\ell(1-q^{-2})=v_\ell(q^2-1)$ for $\mathrm{I}_n^*$ ($q$ is an $\ell$-adic unit), $v_\ell(1-\chi q^{-1})=v_\ell(q-\chi)$ with $\chi=\chi_{-3}(q)$ or $\chi_{-4}(q)$ for the next two rows, and the $\mathrm{I}_0^*$ value is computed in the proof of Lemma lem:euler. One consequence is used repeatedly: $\chi_{-3}(q)$ is the Legendre symbol $(\frac{q}{3})$, so $\chi_{-3}(q)\equiv q \pmod 3$ and $3\mid q-\chi_{-3}(q)$; every prime $q\geq 5$ of type II, II$^*$, IV or IV$^*$ contributes at least 1 at $\ell=3$. Likewise $\chi_{-4}(q)\equiv q \pmod 4$, so $q-\chi_{-4}(q)$ is divisible by 4, which matters only at $\ell=2$.

## 21.4 Sections 1.2 and 1.3 of the paper: the theorem and the conjecture

This section reads the statement of the main theorem, the definition of the Tamagawa factor that enters it, and the conjecture that extends its inequality beyond the hypothesis. The proofs are read in Section 21.11.

**Definition 21.5** (Tamagawa factor and Selmer group; Section 1.2 of the paper)**.** For a prime $p\neq\ell$ and a $G_{\mathbb{Q}_p}$-stable $\mathbb{Z}_\ell$-lattice $M$, $\mathrm{Tam}_p(M)=\mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p,M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}})$, where $G_{\mathbb{Q}_p}/I_p$ acts on $H^1(I_p,M)$ by conjugation; it is an ideal $\ell^m\mathbb{Z}_\ell$ and $v_\ell(\mathrm{Tam}_p(M))=m$. $H^1_f(\mathbb{Q},W)$, $W=\mathrm{ad}^0\,T\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$, is the Bloch-Kato Selmer group [49, Section 2.1].

*Where proved.* The definition and the identification with $c_p(E)$ for $M=T$: Chapter 18; the conjugation action: Chapter 16; the Bloch-Kato conditions: Chapters 17 and 18. Read the parentheses with care: the invariants are taken of the torsion submodule, and the Fitting ideal of the resulting finite module, as in [69, I.4.2.2].

*Why this definition.* At a prime $p\neq\ell$ the Bloch-Kato local condition on $M\otimes\mathbb{Q}_\ell$ is the unramified one. For the lattice $M$ there are two candidate conditions: classes that are unramified integrally, and classes whose image in $H^1(\mathbb{Q}_p,M\otimes\mathbb{Q}_\ell)$ is unramified. Their difference is measured by the classes that become trivial on $I_p$ after inverting $\ell$, that is, by the torsion of $H^1(I_p,M)$; and the restriction map $H^1(\mathbb{Q}_p,M)\to H^1(I_p,M)^{G_{\mathbb{Q}_p}/I_p}$ is onto (inflation-restriction, $G_{\mathbb{Q}_p}/I_p\simeq\widehat{\mathbb{Z}}$ having cohomological dimension one), so the part of the torsion fixed by Frobenius is exactly what classes over $\mathbb{Q}_p$ can produce. The ideal $\mathrm{Tam}_p(M)$ is the size of that discrepancy. Two cases fix the intuition. If $M$ is unramified at $p$, then $H^1(I_p,M)=\mathrm{Hom}(I_p,M)$ is torsion-free and $\mathrm{Tam}_p(M)=\mathbb{Z}_\ell$. If $M=T$ and $E$ has split multiplicative reduction of type $\mathrm{I}_n$ at $p$, then $\mathrm{Tam}_p(T)=\ell^{v_\ell(n)}\mathbb{Z}_\ell$, the $\ell$-part of the Tamagawa number $c_p(E)=n$ (step (7) of the proof of Lemma lem:tamfactor below); this is why the ideal carries the name.

**Theorem 21.6** (the paper's Theorem thm:main)**.** *Let $E$ be $X_0(N)$-optimal and $\ell$ an odd prime, $\ell \nmid 2N$, such that $\bar{\rho}_{E,\ell}$ is absolutely irreducible on* $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$*. Then*

$$v_\ell(\deg \varphi_E) = \mathrm{length}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell) + \sum_{q|N} v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) + \sum_{q|N} e_\ell(E,q), \tag{21.1}$$

*and in particular*

$$v_\ell(\deg \varphi_E) \geq \sum_{q|N} e_\ell(E,q) + \sum_{q|N} t_\ell(E,q). \tag{21.2}$$

*Where proved.* The assembly is Chapter 20; the proof is read in Section 21.11. Its inputs are the theorems of Diamond, Flach and Guo (stated with references), [3, Theorem 2.2], and the identification of the integral structure of [49] with $H^1(X_0(N), \mathbb{Z}_\ell)$ (Proposition prop:integral, Proposition 21.28). The introduction says of this identification that it is not a published statement in the form needed, and that the paper proves it: "which we prove; it holds integrally, with no localisation". The book proves it as well (Chapter 20). The equality in (eq:identity), rather than an inequality, comes from Proposition prop:pullback (Proposition 21.30): no freeness or multiplicity-one hypothesis enters.

*Reading the identity.* All three terms on the right of (eq:identity) are nonnegative integers: a length, the valuation of an ideal of $\mathbb{Z}_\ell$, and the integers of Definition 21.4. The global term length $H^1_f$ is not computable from local data; the other two are, by the lemmas of Section 2 of the paper. So (eq:thm) is obtained by dropping the Selmer length and bounding each Tamagawa term from below: $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) \geq t_\ell(E,q)$ at a potentially multiplicative $q$ (Lemma lem:tamfactor), and $t_\ell(E,q) = 0$ at the other primes. An example where every term is visible: `791a1` has type $\mathrm{I}_3$ at 7 and $\mathrm{I}_2$ at 113, $\deg \varphi_E = 144$, and surjective mod 3 image (Cremona's `galrep` lists only 2B), so the theorem applies at $\ell = 3$. Both primes are multiplicative, so $e_3 = 0$ at both; at 113, $v_3(2) = 0$ and the Tamagawa term is 0; at 7, $v_3(3) = 1$, $v_3(7-1) = 1$ and the unit part of the Tate parameter reduces to $\bar{u} = 6 = (-1)^3$, a cube in $\mathbb{F}_7^\times$, so Lemma lem:tamfactor gives $v_3(\mathrm{Tam}_7(\mathrm{ad}^0 T)) = 1 + \min(1,1,1) = 2$. Then (eq:identity) reads $v_3(144) = 2 = \mathrm{length}\, H^1_f + 2 + 0$, so the Selmer group vanishes, while (eq:thm) gives only $2 \geq t_3(E,7) = 1$.

The paragraph after the theorem says that the hypothesis is $\ell \notin S_f$ in the notation of [49, (31)] (the set $S_f$ is defined in Definition 21.34 below), the one under which they prove the Tamagawa number conjecture for the adjoint motive [49, Theorem 0.1] (both verified against the published text, Chapter 20), and that for $\ell \nmid 2N$ it means: $E[\ell]$ irreducible and, if $\ell = 3$, mod 3 image not $N_s(\mathbb{F}_3)$ (Proposition prop:Sf, Proposition 21.35). It then lists what the paper adds to [49]: the identification of the congruence ideal with the modular degree, the local terms, the twisting theorems, the boundary of the hypothesis and the computations, each read below.

**Conjecture 21.7** (the paper's Conjecture conj:euler)**.** *Let $E$ be $X_0(N)$-optimal without complex multiplication, $\ell$ an odd prime with $E[\ell]$ irreducible, and $h_\ell(E) = \mathrm{length}\, H^0(\mathbb{Q}, \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$. Then*

$$v_\ell(\deg \varphi_E) \geq \sum_{q|N,\, q \neq \ell} e_\ell(E,q) + \sum_{q|N} t_\ell(E,q) + [\ell = 3]\, w_3(E) - h_\ell(E), \tag{21.3}$$

*where $w_3(E) = 1$ for the types $\mathrm{II}, \mathrm{IV}, \mathrm{III}^*$ at 3, 2 for $\mathrm{II}^*, \mathrm{IV}^*$, and 0 for $\mathrm{III}, \mathrm{I}_0^*$; and $w_3(E) = 0$ when $3 \nmid N$ or the reduction at 3 is potentially multiplicative.*

*What is proved about it.* (i) $h_\ell(E) = 0$ for every $\ell \geq 5$, and $h_3(E) \leq 1$ with equality exactly for image $N_s(\mathbb{F}_3)$ (Lemma lem:h0 and Proposition prop:ns, Lemma 21.40 and Proposition 21.47); this

holds with or without complex multiplication. (ii) Under the hypothesis of Theorem thm:main, (eq:rule) is (eq:thm), so it is a theorem there. (iii) For image $N_s(\mathbb{F}_3)$ the correction is what the sequence of Lemma lem:dfg21 produces *if* $H^1_f(\mathbb{Q}, V^D) = 0$, which is not known for these curves (Proposition 21.49); the paper says so ("which is assumed and not known for these curves"). (iv) The weights $w_3$ are empirical; the paper's Remark rem:oq1, read in Section 21.14, relates them to a twist relation at $q = \ell$ observed on the data. In the paragraph after the conjecture, the condition for $h_\ell(E) > 0$ is $\bar\rho \simeq \bar\rho \otimes \bar\chi_\ell$, a twist by the cyclotomic character (not by the quadratic character $\chi_{\ell^*}$); it can hold only at $\ell = 3$, where $\bar\chi_3^{\mathrm{cyc}} = \chi_{-3}$.

*The shape of the right side.* The first two sums are the right side of (eq:thm), with the term $e_\ell(E, \ell)$ omitted because it is not defined at $q = \ell$. The weight $w_3(E)$ stands in for the missing local term at $q = \ell = 3$ (it is read off the Kodaira type at 3, and is 0 when $3 \nmid N$). The correction $h_\ell(E)$ is the length of $H^0(\mathbb{Q}, W^D)$, the last term of the four-term sequence of Lemma lem:dfg21: when that term is nonzero, the Selmer group $H^1_\Sigma$ is shorter by $h_\ell(E)$ than the sum of its local pieces (Proposition 21.49), and the conjecture subtracts exactly that. A worked instance at a curve outside the theorem: `726a1`, with the local terms computed after Definition 21.4, has image 3Ns at $\ell = 3$, so $h_3(E) = 1$; its reduction at 3 is multiplicative, so $w_3(E) = 0$. The right side of (eq:rule) is $e_3(E, 11) + t_3(E, 2) + t_3(E, 3) + 0 - 1 = 1 + 1 + 1 - 1 = 2$, and $v_3(\deg \varphi_E) = v_3(144) = 2$: equality. Without the correction the bound would be 3, and the inequality would fail.

## 21.5 Sections 1.4 and 1.5 of the paper: component groups, history, the other statements

This section reads the two component-group inequalities, Proposition prop:forms, the historical paragraph, and the statements of Theorems thm:qual and thm:quant, Proposition prop:ns and Conjecture conj:eis, whose proofs come later in the paper.

**Definition 21.8** (component groups and the twist-symmetrised weight; Section 1.4)**.** For a bad prime $q$, $\#\Phi_q^{\mathrm{geom}}$ is $n$ for $\mathrm{I}_n$, 1 for $\mathrm{II}, \mathrm{II}^*$, 2 for $\mathrm{III}, \mathrm{III}^*$, 3 for $\mathrm{IV}, \mathrm{IV}^*$ and 4 for $\mathrm{I}_0^*$, $\mathrm{I}_n^*$. The weight $w_\ell(E, q)$ is $v_\ell(n)$ if the type is $\mathrm{I}_n$, or $\mathrm{I}_n^*$ with $q$ odd; 1 if $\ell = 3$ and the type is IV or IV$^*$, or II or II$^*$ with $q$ odd; and 0 otherwise. The two inequalities are

$$\text{(eq:main)} \quad v_\ell(\deg \varphi_E) \geq \sum_{q \mid N} v_\ell(\#\Phi_q^{\mathrm{geom}}), \qquad \text{(eq:twist)} \quad v_\ell(\deg \varphi_E) \geq \sum_{q \mid N} w_\ell(E, q).$$

*Where proved.* The orders of $\Phi_q^{\mathrm{geom}}$: Chapter 10. The paper's identity $w_\ell(E, q) = \max(v_\ell(\#\Phi_q^{\mathrm{geom}}(E)), v_\ell(\#\Phi_q^{\mathrm{geom}}(E^{(q)})))$ for odd $q$ follows from the type exchanges of Lemma lem:twist(ii) by inspection. Written out, with $\ell$ odd and the orders of the component groups listed above: $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ gives $\max(v_\ell(n), v_\ell(4)) = v_\ell(n)$; $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ gives $\max(0, v_\ell(4)) = 0$; II $\leftrightarrow$ IV$^*$ and IV $\leftrightarrow$ II$^*$ give $\max(0, v_\ell(3)) = v_\ell(3)$, which is 1 for $\ell = 3$ and 0 otherwise; III $\leftrightarrow$ III$^*$ gives $\max(v_\ell(2), v_\ell(2)) = 0$. These are the values of the definition, case by case. The point of $w_\ell$ is that it does not change under the twist by $\chi_{q^*}$, whereas $v_\ell(\#\Phi_q^{\mathrm{geom}})$ does (`121a1`, of type II at 11, has $\#\Phi_{11}^{\mathrm{geom}} = 1$, and its twist `121c2`, of type IV$^*$, has $\#\Phi_{11}^{\mathrm{geom}} = 3$). At $q = 3$ the exchanges II $\leftrightarrow$ IV$^*$, III $\leftrightarrow$ III$^*$, IV $\leftrightarrow$ II$^*$ rest on the formula of Ogg and Saito (Lemma 21.15, Chapter 10); the paper's parenthesis "(all five at every odd $q$; at $q = 3$ the three middle exchanges keep the conductor exponent, which can exceed 2)" says exactly this. Since $w_\ell \geq v_\ell(\#\Phi_q^{\mathrm{geom}})$ termwise, (eq:twist) implies (eq:main).

**Proposition 21.9** (the paper's Proposition prop:forms)**.** *Under the hypothesis of Theorem thm:main, (eq:main) and (eq:twist) hold; the same is true for every curve satisfying (eq:rule) with $h_\ell(E) = 0$.*

*Proof.* It suffices to bound $\sum_q w_\ell(E,q)$ by the right side of (eq:thm), respectively (eq:rule), prime by prime; every term of those right sides is nonnegative. Type $\mathrm{I}_n$: $w_\ell = v_\ell(n) = t_\ell$. Type $\mathrm{I}_n^*$, $q$ odd: $w_\ell = v_\ell(n) = t_\ell$ ($v_q(j) = -n$). Type $\mathrm{I}_n^*$, $q = 2$: $w_\ell = 0$. Types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$ at $q \geq 5$: for $\ell \geq 5$, $w_\ell = 0$; for $\ell = 3$, $e_3(E,q) = v_3(q - \chi_{-3}(q)) \geq 1 = w_3$, because $\chi_{-3}(q) \equiv q \pmod 3$ ($\chi_{-3}(q) = (\frac{q}{3})$ is 1 for $q \equiv 1$ and $-1$ for $q \equiv 2 \pmod 3$); values of $e_3$ from Lemma 21.16). The same types at $q = 3$ occur only under (eq:rule) (under Theorem thm:main $3 \nmid N$ when $\ell = 3$, and $w_\ell = 0$ when $\ell \geq 5$), and there $w_3(E) \geq 1 = w_3(E,3)$. Types $\mathrm{II}, \mathrm{II}^*$ at $q = 2$: $w_\ell = 0$. Types $\mathrm{IV}, \mathrm{IV}^*$ at $q = 2$, $\ell = 3$: $w_3(E,2) = 1 = e_3(E,2)$ by Lemma lem:IV (Lemma 21.13). Types $\mathrm{III}, \mathrm{III}^*, \mathrm{I}_0^*$: $w_\ell = 0$. □

The only delicate case is the type IV or IV* at $q = 2$, where Kraus's theorem does not apply; Lemma lem:IV covers it, and the paper's sentence after the proof checks it on data ("On the 92,812 working-set curves with type IV or IV* at 2, PARI's factor at 2 is $1 + 2X$ for every one of them").

**Section 1.5: the history.** Three statements of other authors are summarised. (i) Ribet and Takahashi compare, for semistable $E$, $\deg \varphi_E$ with the degree of a Shimura-curve parametrisation, exactly when $E$ is alone in its isogeny class and otherwise up to primes where $E[\ell]$ is reducible ([164, 192]; checked against the sources, Chapter 14). (ii) Pollack and Weston, under their hypothesis CR (surjective $\bar\rho$, ramified at every prime of the relevant part of the level that is $\pm 1$ modulo $\ell$; [153, Section 1 and Theorem 6.11], numbering of the arXiv version). (iii) Kim and Ota [99, Theorem 1.1], as cited, under the hypotheses recalled in Remark rem:ko; the identification of their Tamagawa exponent with $t_\ell(E,q)$ is Lemma lem:tamagawa-exp (Lemma 21.11). The final sentence, that Theorem thm:main contains $v_\ell(\deg \varphi_E) \geq \sum_{q \| N} t_\ell(E,q)$ including the primes $q \equiv \pm 1 \pmod \ell$ and $\ell = 3$, is correct under its hypothesis.

**The statements proved later.** Theorem thm:qual is Theorem 21.25 (Section 21.9); Theorem thm:quant is Theorem 21.46 (Section 21.12), whose proof rests on Lemma lem:step1; Proposition prop:ns is Proposition 21.47 (Section 21.13); Conjecture conj:eis is Conjecture 21.10 below. The sentence after Theorem thm:qual says that for $\ell \geq 5$ the theorem needs only the irreducibility of $\bar\rho$: for $\ell \geq 5$ the condition over $\mathbb{Q}(\sqrt{-3})$ is not asked, and the optimal-level theorem needs only irreducibility.

**Conjecture 21.10** (the paper's Conjecture conj:eis)**.** *Let $E$ be optimal, without complex multiplication, with a rational $\ell$-isogeny, $\ell$ odd, and $D = \sum_q v_\ell(\#\Phi_q^{\mathrm{geom}}) - v_\ell(\deg \varphi_E)$. Then $D \leq v_\ell(\#E(\mathbb{Q})_{\mathrm{tors}}) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3]$, and $D$ is at most the number of curves in the class with a rational point of order $\ell$.*

The consequences $D \leq 0$ without rational $\ell$-torsion, $D \leq 1$ for $\ell \geq 5$ and $D \leq 2$ for $\ell = 3$ follow from the first bound by Proposition prop:no9 (Proposition 21.50) and Mazur's theorem on torsion; only the first bound, and the second one, are conjectural. The section closes with a plan of the paper: Section 2 the local facts, Section 3 Theorem thm:qual, Section 4 Theorems thm:main and thm:quant, Section 5 the boundary of the hypothesis, Section 6 the computations, Section 7 the prime 2; the tour follows it.

## 21.6 Section 2 of the paper, part 1: inertia at a bad prime

This section reads the first half of the paper's Section 2 ("Local invariants"): the Tamagawa exponent at a multiplicative prime, the inertia invariants at an additive prime, Lemma lem:IV and Remark rem:types. They feed Theorem thm:qual (cases (a) to (c)), the table of Euler factors, and Proposition prop:forms.

**Lemma 21.11** (the paper's Lemma lem:tamagawa-exp)**.** *Let $q$ be a prime of multiplicative reduction, $n_q = v_q(\Delta_E)$, and $\ell \neq q$ a prime. Then $\rho_{E,\ell} \bmod \ell^t$ is unramified at $q$ if and only if $\ell^t \mid n_q$, for split and for nonsplit reduction.*

*Where proved.* Tate curves and their twists: Chapter 10; Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$: Chapter 7; the lemma: Chapter 11.

*Expanded proof.* Let $K$ be the unramified quadratic extension of $\mathbb{Q}_q$ ($K = \mathbb{Q}_q$ if the reduction is split). Over $K$, $E$ is isomorphic to the Tate curve $\overline{K}^\times/q_E^{\mathbb{Z}}$ with $v_q(q_E) = n_q$, and $I_K = I_q$ since $K/\mathbb{Q}_q$ is unramified. The group $E[\ell^t](\overline{K})$ is generated by $\mu_{\ell^t}$ and one $\ell^t$-th root $\theta$ of $q_E$. As $\ell \neq q$, $\mu_{\ell^t} \subset \mathbb{Q}_q^{\mathrm{ur}}$, so $I_q$ fixes $\mu_{\ell^t}$ and sends $\theta$ to $\zeta(\sigma)\theta$ with $\zeta(\sigma) \in \mu_{\ell^t}$; inertia acts trivially exactly when every $\ell^t$-th root of $q_E$ lies in $\mathbb{Q}_q^{\mathrm{ur}}$, that is, when $q_E \in (\mathbb{Q}_q^{\mathrm{ur}\times})^{\ell^t}$ (Kummer theory over $\mathbb{Q}_q^{\mathrm{ur}}$, which contains $\mu_{\ell^t}$: one root lies in $\mathbb{Q}_q^{\mathrm{ur}}$ iff all do, since they differ by elements of $\mu_{\ell^t}$; and the fixed field of $I_q$ in $\overline{\mathbb{Q}}_q$ is $\mathbb{Q}_q^{\mathrm{ur}}$, so $I_q$ fixes $\theta$ iff $\theta \in \mathbb{Q}_q^{\mathrm{ur}}$). Write $q_E = q^{n_q}u$ with $u$ a unit. Units of the valuation ring of $\mathbb{Q}_q^{\mathrm{ur}}$ are $\ell^t$-th powers: their reductions are, $\overline{\mathbb{F}}_q^\times$ being divisible, and Hensel's lemma lifts roots since the derivative $\ell^t X^{\ell^t-1}$ of $X^{\ell^t} - u$ is a unit at a root when $\ell \neq q$. So $q_E$ is an $\ell^t$-th power iff $q^{n_q}$ is. The valuation of $\mathbb{Q}_q^{\mathrm{ur}}$ takes the value 1 on $q$ (the extension is unramified), so an $\ell^t$-th power has valuation divisible by $\ell^t$; hence $q^{n_q}$ is not an $\ell^t$-th power unless $\ell^t \mid n_q$, and if $n_q = \ell^t m$ it is $(q^m)^{\ell^t}$. So $q_E$ is an $\ell^t$-th power iff $\ell^t \mid n_q$. □

The letters matter: $q_E$ is the Tate parameter and $u = q_E q^{-n}$ its unit part, the letter Lemma lem:tamfactor uses. The proof shows why the lemma holds for nonsplit reduction too: the unramified quadratic twist does not change the inertia group.

**Lemma 21.12** (inertia invariants at an additive prime; the paragraph before Lemma lem:IV)**.** *Let $q \neq \ell$ be a prime of additive reduction. Then $E[\ell^k]^{I_q} \simeq \Phi_q^{\mathrm{geom}}[\ell^k]$ for every $k \geq 1$.*

*Where proved.* The Néron model and the filtration $E_1 \subset E_0 \subset E$: Chapter 10; the argument: Chapter 11.

*Expanded proof.* $E[\ell^k]^{I_q} = E(\mathbb{Q}_q^{\mathrm{ur}})[\ell^k]$, since $\mathbb{Q}_q^{\mathrm{ur}}$ is the fixed field of $I_q$ in $\overline{\mathbb{Q}}_q$. Let $\mathcal{O}$ be the valuation ring of $\mathbb{Q}_q^{\mathrm{ur}}$ (or of its completion; torsion is the same). The Néron model gives exact sequences $0 \to E_0(\mathbb{Q}_q^{\mathrm{ur}}) \to E(\mathbb{Q}_q^{\mathrm{ur}}) \to \Phi_q^{\mathrm{geom}} \to 0$ and $0 \to E_1(\mathbb{Q}_q^{\mathrm{ur}}) \to E_0(\mathbb{Q}_q^{\mathrm{ur}}) \to \tilde{E}_{\mathrm{ns}}(\overline{\mathbb{F}}_q) \to 0$, and for additive reduction $\tilde{E}_{\mathrm{ns}}(\overline{\mathbb{F}}_q) \simeq \overline{\mathbb{F}}_q^+$. The kernel of reduction $E_1$ is the formal group, a pro-$q$ group; $\overline{\mathbb{F}}_q^+$ has exponent $q$. Both are uniquely $\ell$-divisible ($\ell \neq q$), hence so is $E_0(\mathbb{Q}_q^{\mathrm{ur}})$ (apply the snake lemma to $\ell^k$ on the second sequence: kernels and cokernels vanish on both ends, so they vanish in the middle). Apply the snake lemma to multiplication by $\ell^k$ on the first sequence; it gives the exact sequence

$$0 \to E_0(\mathbb{Q}_q^{\mathrm{ur}})[\ell^k] \to E(\mathbb{Q}_q^{\mathrm{ur}})[\ell^k] \to \Phi_q^{\mathrm{geom}}[\ell^k] \to E_0(\mathbb{Q}_q^{\mathrm{ur}})/\ell^k,$$

whose two outer terms vanish, so $E(\mathbb{Q}_q^{\mathrm{ur}})[\ell^k] \to \Phi_q^{\mathrm{geom}}[\ell^k]$ is an isomorphism. (The component group over $\mathbb{Q}_q^{\mathrm{ur}}$ is the geometric one, $\Phi_q^{\mathrm{geom}}$, since the residue field $\overline{\mathbb{F}}_q$ is algebraically closed.) □

This is the paragraph of the paper between Lemma lem:tamagawa-exp and Lemma lem:IV; it is what makes Lemma lem:IV work at $q = 2$, where Kraus's theorem is not available.

**Lemma 21.13** (the paper's Lemma lem:IV)**.** *Let $q \neq 3$ be a prime at which $E$ has reduction of type* IV *or* IV$^*$*. Then the reduction is potentially good; $\rho_{E,\ell}(I_q)$ is cyclic of order 3 for every $\ell \neq q$; $\bar\rho_{E,3}(I_q)$ is a nontrivial unipotent subgroup, $E[3]^{I_q}$ is a line, $\bar\rho_{E,3}$ is tamely ramified at $q$ with Serre conductor exponent 1, and the conductor exponent of $E$ at $q$ is 2. For $\ell \neq q$, $(\mathrm{ad}^0 V_\ell E)^{I_q}$ is a line on which $\mathrm{Frob}_q$ (either one) acts by $\chi_{-3}(q)$, so $P_q(X) = 1 - \chi_{-3}(q)qX$ and $e_\ell(E,q) = v_\ell(q - \chi_{-3}(q))$; at $q = 2$, $P_2(X) = 1 + 2X$ and $e_3(E,2) = 1$.*

*Where proved.* Chapter 11 (with the case $q = 2$); the commutant of an element of order 3: Chapter 6; the tame relation: Chapter 7; the conductor: Chapters 10 and 11.

*Expanded proof. Step 1: a line of invariants.* By Lemma 21.12 with $\ell = 3$, $k = 1$: $E[3]^{I_q} \simeq \Phi_q^{\mathrm{geom}}[3] = \mathbb{Z}/3$, since $\#\Phi_q^{\mathrm{geom}} = 3$. Inertia acts nontrivially on $E[3]$, otherwise $\Phi_q^{\mathrm{geom}}[3]$ would have order 9.

*Step 2: potentially good.* If $v_q(j) < 0$, over $\mathbb{Q}_q$ the curve is the twist of a Tate curve by a quadratic character $\chi$, ramified as the reduction is additive (Chapter 10). On $I_q$, in a suitable basis, $\rho_{E,3}(\sigma) = \chi(\sigma)\left(\begin{smallmatrix} 1 & n\theta(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ with $\theta$ the 3-adic tame character and $n = -v_q(j)$. The subgroup $H \subseteq I_q$ on which $n\theta \equiv 0 \pmod 3$ has index 1 or 3 (it is the kernel of the homomorphism $n\theta \bmod 3$ to $\mathbb{Z}/3$), so it is not contained in $\ker\chi$, of index 2 (if $H \subseteq \ker\chi$, then $2 = [I_q : \ker\chi]$ would divide $[I_q : H] \in \{1, 3\}$); an element $\sigma \in H \smallsetminus \ker\chi$ has $\bar\rho_{E,3}(\sigma) = \chi(\sigma)\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = -1$, which fixes no nonzero vector of $E[3]$, contradicting Step 1. So the reduction is potentially good and $\rho_{E,3}(I_q)$ is finite [179, Section 2].

*Step 3: the image.* $\det\rho_{E,3} = \chi_3^{\mathrm{cyc}}$ is unramified at $q \neq 3$, so $\rho_{E,3}(I_q) \subset \mathrm{SL}_2(\mathbb{Z}_3)$. The kernel of $\mathrm{GL}_2(\mathbb{Z}_3) \to \mathrm{GL}_2(\mathbb{F}_3)$ is $1 + 3\,\mathrm{M}_2(\mathbb{Z}_3)$, torsion-free (if $g = 1 + 3^a X$ with $a \geq 1$ and $X \not\equiv 0 \pmod 3$, then $g^p \equiv 1 + p3^a X \pmod{3^{a+1}}$ for a prime $p \neq 3$, and $g^3 \equiv 1 + 3^{a+1}X \pmod{3^{a+2}}$, so no power of $g$ of prime order is 1), so reduction is injective on the finite group $\rho_{E,3}(I_q)$. Its image fixes a nonzero vector of $\mathbb{F}_3^2$ and has determinant 1; in a basis $(v, w)$ with $v$ fixed such an element is $\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)$, so the image lies in a group of order 3 of unipotents, and being nontrivial, equals it. Hence $\rho_{E,3}(I_q)$ is cyclic of order 3. By [179, Section 2] the kernel of $\rho_{E,\ell}|_{I_q}$ does not depend on $\ell \neq q$, so $\rho_{E,\ell}(I_q)$ is cyclic of order 3 for every $\ell \neq q$.

*Step 4: tame, conductors.* 3 is prime to $q$, so the image is a quotient of tame inertia: the ramification is tame, with no Swan term. The Serre conductor exponent of $\bar\rho_{E,3}$ is $\dim E[3] - \dim E[3]^{I_q} = 1$, and the conductor exponent of $E$ is $\dim V - \dim V^{I_q} = 2$ ($V^{I_q} = 0$: the generator has eigenvalues $\zeta_3^{\pm 1} \neq 1$).

*Step 5: the Euler factor.* Let $\sigma$ generate $\rho(I_q)$ on $V = V_\ell E$; $\sigma$ has order 3 and determinant 1, so its eigenvalues are $\zeta_3^{\pm 1}$ and $\sigma^2 + \sigma + 1 = 0$. An endomorphism commuting with $\sigma$ is a polynomial in $\sigma$ (Chapter 6: $\sigma$ has distinct eigenvalues), $a + b\sigma$, of trace $2a - b$; so the trace-zero ones form the line spanned by $1 + 2\sigma = \sigma - \sigma^{-1}$. That line is $(\mathrm{ad}^0 V)^{I_q}$. On the tame quotient the arithmetic Frobenius acts by $x \mapsto x^q$, so $\rho(\mathrm{Frob}_q)\sigma\rho(\mathrm{Frob}_q)^{-1} = \sigma^q = \sigma^{\chi_{-3}(q)}$ ($q \equiv \chi_{-3}(q) \pmod 3$). Conjugation sends $\sigma - \sigma^{-1}$ to $\pm(\sigma - \sigma^{-1})$ with the sign $\chi_{-3}(q)$, and a sign is its own inverse, so the geometric Frobenius acts by the same sign. Hence $L_q(A_f, s)^{-1} = 1 - \chi_{-3}(q)q^{-s}$ and $P_q(X) = L_q(A_f, s-1)^{-1}|_{X=q^{-s}} = 1 - \chi_{-3}(q)qX$. The naive factor is 1, so $e_\ell(E,q) = v_\ell(1 - \chi_{-3}(q)q^{-1}) = v_\ell(q - \chi_{-3}(q))$. At $q = 2$, $\chi_{-3}(2) = -1$, $P_2 = 1 + 2X$, $e_3(E,2) = v_3(3/2) = 1$. □

**The paragraph after the lemma.** It records that $T_3E$ is free of rank one over $\mathbb{Z}_3[X]/(X^2+X+1) \simeq \mathbb{Z}_3[\zeta_3]$ at every $q \neq 3$ where $\rho_{E,3}(I_q)$ is cyclic of order 3, $q=2$ included: $T_3E$ is a torsion-free module of $\mathbb{Z}_3$-rank 2 over that discrete valuation ring, hence free of rank one (Chapter 6). Lemma lem:tamadd uses this, also at $q=2$. Only the identification of the order 3 with Kraus's semistability defect needs $q \geq 5$. At $q=3$ the types IV, IV$^*$ are wildly ramified (the image of inertia has order divisible by $3=q$), and the lemma does not apply.

*Remark* 21.14 (the paper's Remark rem:types). At $q \geq 5$, $\ell = 3$, the paper lists the image of inertia mod 3 for the other additive types: $\{\pm 1\}$ ($\mathrm{I}_0^*$), cyclic of order 4 (III, III$^*$), minus a unipotent of order 6 (II, II$^*$), a ramified quadratic character times a unipotent ($\mathrm{I}_n^*$). Each has $E[3]^{I_q} = 0$: $-1$, an element $\tau$ with $\tau^2 = -1$, and $-u$ with $u$ unipotent all fix no nonzero vector (for $-u$: $-uv = v$ gives $uv = -v$, but $u$ has the single eigenvalue 1). So the Serre exponent is 2. The twist by $\chi_{q^*}$ multiplies the generator by $-1$; it lowers the Serre exponent for $\mathrm{I}_0^*$ (to 0), $\mathrm{I}_n^*$ (to 1, or 0 when $3 \mid n$), II, II$^*$ (to 1, the twist being of type IV$^*$, IV), and the conductor of $E$ only for $\mathrm{I}_0^*$ and $\mathrm{I}_n^*$ (for II $\leftrightarrow$ IV$^*$ the exponent stays 2). These are the facts the remark states (Chapter 11). They explain why Theorem thm:qual treats exactly the types IV, IV$^*$ (no twist), II, II$^*$ (after the twist) and $\mathrm{I}_n^*$ with $n \geq 1$ and $\ell \mid n$: those are the rows where, after a twist or none, the Serre exponent drops below the conductor exponent.

## 21.7 Section 2 of the paper, part 2: twists and the omitted Euler factors

This section reads Lemma lem:twist, Lemma lem:euler (the table), Proposition prop:shapes (the four shapes of $P_q$ at every bad prime) and Remark rem:wild. These give the values of $e_\ell(E,q)$ everywhere. The method is the one of Step 5 of Lemma lem:IV, applied type by type: find the image of inertia, compute the inertia invariants of $\mathrm{ad}^0 V$ (the trace-zero endomorphisms commuting with that image), and read off how Frobenius acts on them. The quadratic twist enters because it does not change $\mathrm{ad}^0 V$, so a type that twists to a good or multiplicative one has the Euler factor of the twisted curve.

**Lemma 21.15** (the paper's Lemma lem:twist)**.** *Let $q$ be odd, $\chi = \chi_{q^*}$, $E^{(q)}$ the twist of $E$ by $\chi$.*

*(i) At every $p \neq q$, $E$ and $E^{(q)}$ have the same Kodaira type, the same $\Phi_p^{\mathrm{geom}}$ and the same conductor exponent.*

*(ii) If $E$ has type $\mathrm{I}_n$ at $q$ ($n \geq 0$), $E^{(q)}$ has type $\mathrm{I}_n^*$, and conversely. If $E$ has type* II, III, IV *at $q$, then $E^{(q)}$ has type* IV$^*$, III$^*$, II$^*$, *and conversely. For the exchanges $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$, $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ at every odd $q$ and for all the exchanges at $q \geq 5$, the conductor exponent of $E^{(q)}$ at $q$ is 2 when its reduction is additive and 1 when it is of type $\mathrm{I}_n$, $n \geq 1$. At $q=3$ the exchanges* II $\leftrightarrow$ IV$^*$, III $\leftrightarrow$ III$^*$, IV $\leftrightarrow$ II$^*$ *keep $f_3$ unchanged, and $f_3$ can exceed 2 (wild ramification).*

*(iii) (Atkin and Li) If $g$ is a newform of weight 2 on $\Gamma_0(M)$ with $q^2 \nmid M$, then $g \otimes \chi$ is a newform on $\Gamma_0(\mathrm{lcm}(M, q^2))$.*

*Where proved.* (i), (ii) for $q \geq 5$ and the $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$, $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ exchanges at every odd $q$: Chapter 10; the exchanges at $q=3$: Chapter 10, given the formula of Ogg and Saito; (iii) is stated with a reference [9, Theorem 3.1] and sketched in Chapter 13.

*Expanded proof of (i) and (ii).* (i) $\chi$ is unramified at $p$, so $E \simeq E^{(q)}$ over an unramified extension of $\mathbb{Q}_p$; Kodaira type, $\Phi_p^{\mathrm{geom}}$ and conductor exponent are invariant under unramified base change (Chapter 10). (ii) A minimal model with invariants $c_4, c_6, \Delta$ twists to a model with $(q^*)^2 c_4$, $(q^*)^3 c_6$, $(q^*)^6\Delta$ and the same $j$. If $v_q(j) < 0$, the type is $\mathrm{I}_n$ or $\mathrm{I}_n^*$ with $n = -v_q(j)$ according as the reduction is multiplicative or not, and a ramified quadratic twist exchanges the two (over $\mathbb{Q}_q$ the curve is a Tate curve or its ramified twist, Chapter 10). For $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$: a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with good reduction twists to $y^2 = x^3 + q^*a_2x^2 + (q^*)^2a_4x + (q^*)^3a_6$; Tate's algorithm, after dividing out the powers of $q$ (substitute $x = qT$ and divide the cubic by $q^3$, using $q^* = \epsilon q$ and $\epsilon^2 = 1$), looks at $T^3 + \epsilon a_2T^2 + a_4T + \epsilon a_6$, $\epsilon = q^*/q = \pm 1$, whose roots are $\epsilon$ times those of $x^3 + a_2x^2 + a_4x + a_6$, distinct modulo $q$; distinct roots is the criterion for $\mathrm{I}_0^*$; conversely a type $\mathrm{I}_0^*$ model at odd $q$ has this shape and twists to a good one. For $q \geq 5$ and potentially good reduction the type is read off $v_q(\Delta_{\min}) \in \{0, 2, 3, 4, 6, 8, 9, 10\}$; twisting adds 6, and subtracting 12 when the result is at least 12 gives the minimal model: $2 \leftrightarrow 8$, $3 \leftrightarrow 9$, $4 \leftrightarrow 10$. The conductor exponents are the tame ones, $\dim V - \dim V^{I_q}$, since $V_\ell E^{(q)} = V_\ell E \otimes \chi$ with $\chi$ tame and $V_\ell E$ good, multiplicative or (for $q \geq 5$) tame. At $q = 3$, for the types $\mathrm{II}, \mathrm{III}, \mathrm{IV}$ and their duals: $\chi_{-3}$ is tame, so the Swan conductor does not change, and neither $V_\ell E$ nor its twist has inertia invariants (the image of inertia is not contained in $\{\pm 1\}$, so it contains an element of order 3, 4 or 6, which has no eigenvalue $\pm 1$), so $f_3$ is unchanged; $v_3(\Delta_{\min})$ changes by 6 modulo 12; by Ogg's formula $v_3(\Delta_{\min}) = f_3 + m - 1$ with $m$ the number of components of the minimal regular model, $1 \leq m \leq 9$ for the potentially good types other than $\mathrm{I}_0^*$, which forces the exchange $m \mapsto m \pm 6$, that is the three exchanges listed (Chapter 10). □

*Reading guidance.* Two points in (ii) are easy to misread. The sentence on conductor exponents applies to the exchanges it names, not to the exchanges at $q = 3$: there $f_3$ is unchanged and can be 3 or more, by wild ramification; for example `54a1`, of type $\mathrm{IV}^*$ at 3, and its twist by $\chi_{-3}$, `54b1`, of type II at 3, both have $f_3 = 3$ (recomputed). And the exchanges at $q = 3$ are where the proof leaves Tate's algorithm: it combines the invariance of the Swan conductor under a tame twist with Ogg's formula, and the bound $1 \leq m \leq 9$ on the number of components then forces the shift $m \mapsto m \pm 6$. The appendix of the paper reports a numerical check on 5,122 pairs below conductor 10,000, which the book reproduced (Chapter 10).

**Lemma 21.16** (the paper's Lemma lem:euler)**.** *Let $q \geq 5$ be a bad prime, $q \neq \ell$. Then $L_q(A_f, s)^{-1}$ and $e_\ell(E, q)$ are as in the table of Section 21.3. The rows $\mathrm{I}_n^*$ and $\mathrm{I}_0^*$ hold at every odd $q \neq \ell$, and the row $\mathrm{I}_n$ at every $q \neq \ell$.*

*Where proved.* Chapter 19 (in its own language); the inertia images: Chapter 11 (Kraus's theorem [107] for $q \geq 5$); the $\mathrm{I}_0^*$ identity: Chapter 2.

*Expanded proof.* In every case $L_q(A_f, s)^{-1} = \prod(1 - \lambda q^{-s})$ over the eigenvalues $\lambda$ of $\mathrm{Frob}_q^{-1}$ on $(\mathrm{ad}^0 V)^{I_q}$ (Convention 21.2).

*Type $\mathrm{I}_n$, any $q \neq \ell$.* In a basis $e_1, e_2$ of $T$, $\rho|_{G_{\mathbb{Q}_q}} = \left(\begin{smallmatrix} \chi_\ell\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$ (Lemma 21.19), and a generator $t$ of tame $\ell$-inertia acts by $te_1 = e_1$, $te_2 = e_2 + n'e_1$, $n' \neq 0$. In the basis $e = e_1 \otimes e_2^\vee$, $h = e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee$, $f = e_2 \otimes e_1^\vee$ of $\mathrm{ad}^0 V$ (as matrices, $e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$), $t$ acts by conjugation $X \mapsto tXt^{-1}$ with $t = \left(\begin{smallmatrix} 1 & n' \\ 0 & 1 \end{smallmatrix}\right)$, $t^{-1} = \left(\begin{smallmatrix} 1 & -n' \\ 0 & 1 \end{smallmatrix}\right)$. Multiplying out, $tet^{-1} = e$, $tht^{-1} = \left(\begin{smallmatrix} 1 & -2n' \\ 0 & -1 \end{smallmatrix}\right) = h - 2n'e$ and $tft^{-1} = \left(\begin{smallmatrix} n' & -n'^2 \\ 1 & -n' \end{smallmatrix}\right) = f + n'h - n'^2e$; so $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2e$. A vector $xe + yh + zf$ is fixed iff $-2n'y - n'^2z = 0$ and $n'z = 0$, that is $y = z = 0$: $(\mathrm{ad}^0 V)^{I_q} = \mathbb{Q}_\ell e$. The arithmetic Frobenius sends $e$ to $(\varepsilon q)\varepsilon^{-1}e = qe$,

so the geometric one acts by $q^{-1}$ and $L_q(A_f,s)^{-1} = 1 - q^{-1-s}$. This is also the naive factor ($\delta_q = 1$), so $e_\ell = 0$.

*Type* $\mathrm{I}_n^*$, *$q$ odd.* By Lemma 21.15(ii), $V = V' \otimes \chi_{q^*}$ with $V' = V_\ell E^{(q)}$ multiplicative; $\mathrm{ad}^0 V = \mathrm{ad}^0 V'$, so the true factor is $1 - q^{-1-s}$; the naive factor is 1, and $e_\ell = v_\ell(1-q^{-2}) = v_\ell(q^2-1)$.

*Types* $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$, $q \geq 5$. By Kraus's theorem the image of $I_q$ on $V$ is cyclic of order 3 or 6, generated by $\sigma$ or $-\sigma$ with $\sigma$ of order 3. On $\mathrm{ad}^0 V$ the sign disappears; the computation is Step 5 of Lemma 21.13: the invariant line is spanned by $2\sigma + 1$, of square $-3$, inside $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$ (not a field when $\ell \equiv 1 \pmod 3$), and either Frobenius acts on it by $\chi_{-3}(q)$; $e_\ell = v_\ell(q - \chi_{-3}(q))$.

*Types* $\mathrm{III}, \mathrm{III}^*$, $q \geq 5$. The image is cyclic of order 4, generated by $\tau$ with $\tau^2 = -1$ (determinant 1, eigenvalues $\pm i$). The commutant is $\mathbb{Q}_\ell[\tau]$, and its trace-zero part is the line $\mathbb{Q}_\ell\tau$ ($\tau$ has trace 0; it is diagonalisable only over $\mathbb{Q}_\ell(i)$, so no diagonal basis exists over $\mathbb{Q}_\ell$ when $\ell \equiv 3 \pmod 4$). Frobenius sends $\tau$ to $\tau^q = \tau^{\chi_{-4}(q)}$, so it acts on the line by $\chi_{-4}(q)$ (either Frobenius), and $e_\ell = v_\ell(q - \chi_{-4}(q))$.

*Type* $\mathrm{I}_0^*$, *$q$ odd.* $V = V' \otimes \chi_{q^*}$ with $V' = V_\ell E'$ unramified, arithmetic Frobenius eigenvalues $\alpha, \beta$; $\mathrm{ad}^0 V = \mathrm{ad}^0 V'$ unramified with eigenvalues $\alpha/\beta, 1, \beta/\alpha$ for either Frobenius. With $\alpha\beta = q$ and $\alpha + \beta = a = a_q(E')$,

$$\Big(1 - \frac{\alpha}{\beta q}\Big)\Big(1 - \frac{1}{q}\Big)\Big(1 - \frac{\beta}{\alpha q}\Big) = \frac{q-1}{q}\Big(1 + \frac{1}{q^2} - \frac{\alpha^2+\beta^2}{q^2}\Big) = \frac{q-1}{q} \cdot \frac{(q+1)^2 - a^2}{q^2},$$

since $\alpha/\beta + \beta/\alpha = (\alpha^2+\beta^2)/q = (a^2-2q)/q$ and $q^2 + 1 - a^2 + 2q = (q+1)^2 - a^2$. So $e_\ell = v_\ell(q-1) + v_\ell((q+1)^2 - a^2)$. Only the exchanges $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ and $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$ were used in the last two cases, valid at every odd $q$; at $q = 3$ the values are $L_3(A_f,1)^{-1} = \frac{8}{9}$ ($\mathrm{I}_n^*$) and $\frac{2}{3} \cdot \frac{16 - a_3(E')^2}{9}$ ($\mathrm{I}_0^*$). □

*Reading guidance.* Two details of the paper's proof deserve attention. The commutant of the image of inertia is taken in $\mathrm{End}_{I_q}(T) = \mathbb{Z}_\ell[\sigma] \simeq \mathbb{Z}_\ell \otimes \mathbb{Z}[\zeta_3]$, which is not a field when $\ell \equiv 1 \pmod 3$; only the trace-zero line inside it matters. And for the types III, III$^*$ the generator $\tau$ is itself the trace-zero invariant, with no diagonalisation over $\mathbb{Q}_\ell$ needed. The sign by which Frobenius acts is the same for the arithmetic and the geometric Frobenius, as a sign is its own inverse. The closing sentence of the paper's proof, that the values agree with PARI's `lfunsympow` on every curve of the tables, was reproduced by the book on all 67,745 pairs (optimal curve, bad $q \geq 5$) below conductor 10,000 (Chapter 22).

**Proposition 21.17** (the paper's Proposition prop:shapes)**.** *Let $q$ be a bad prime and $\ell \neq q$ odd. Then $P_q(X)$ is (i) $1 - X$ if $v_q(j) < 0$, and then $e_\ell(E,q) = 0$ or $v_\ell(q^2-1)$ according as the reduction is multiplicative or additive; (ii) $(1-qX)(1-(a^2-2q)X+q^2X^2)$ if the reduction is potentially good and $\rho(I_q) = \{\pm 1\}$, with $a = a_q(E')$ for a quadratic twist $E'$ with good reduction at $q$, and then $e_\ell(E,q) = v_\ell(q-1) + v_\ell((q+1)^2 - a^2)$; (iii) $1 - cqX$ with $c = \pm 1$ if the reduction is potentially good and $\rho(I_q)$ is cyclic of order 3, 4 or 6, and then $e_\ell(E,q) = v_\ell(q-c)$; (iv) 1 if the reduction is potentially good and $\rho(I_q)$ is not abelian, which happens only for $q = 2, 3$, and then $e_\ell(E,q) = 0$.*

*Where proved.* Chapter 19; finite subgroups of $\mathrm{SL}_2$: Chapters 1 and 4; potentially good reduction and finite inertia: Chapter 11.

*Expanded proof.* The cases are exhaustive: if $v_q(j) < 0$ we are in (i); otherwise $\rho(I_q)$ is a finite subgroup of $\mathrm{SL}_2(\mathbb{Z}_\ell)$ ($\det = \chi_\ell$ is trivial on $I_q$) whose elements have characteristic polynomials

in $\mathbb{Z}[X]$ independent of $\ell$ [179, Section 2], so their orders are in $\{1,2,3,4,6\}$. (An element $g$ of finite order $m$ and determinant 1 is diagonalisable over $\overline{\mathbb{Q}}_\ell$ with eigenvalues $\zeta, \zeta^{-1}$, and $m$ is the order of the root of unity $\zeta$. The cyclotomic polynomial $\Phi_m$, the minimal polynomial of $\zeta$ over $\mathbb{Q}$, divides the characteristic polynomial of $g$, of degree 2 in $\mathbb{Z}[X]$; so $\deg \Phi_m \le 2$, which holds exactly for $m \in \{1,2,3,4,6\}$.) A finite abelian subgroup of $\mathrm{SL}_2$ over a field of characteristic 0 is cyclic (simultaneously diagonalisable over the algebraic closure, and $\mathrm{diag}(x, x^{-1}) \mapsto x$ embeds it in the multiplicative group). Order 1 is good reduction; order 2 is $\{\pm 1\}$, case (ii); orders $3, 4, 6$ are case (iii); otherwise (iv).

(i) Over $\mathbb{Q}_q$, $\rho \simeq \rho' \otimes \chi$ with $\rho'$ a Tate curve's representation up to an unramified twist and $\chi$ quadratic; $\mathrm{ad}^0 V = \mathrm{ad}^0 V'$, and the $\mathrm{I}_n$ computation of Lemma 21.16 gives $L_q(A_f, s)^{-1} = 1 - q^{-1-s}$, so $P_q(X) = 1 - X$. The naive factor is $1 - q^{-2}$ or 1.

(ii) $\psi = \rho|_{I_q}$ is a quadratic character of $I_q$, invariant under conjugation by $G_{\mathbb{Q}_q}$ (its values are $\pm 1$, central). Choose a Frobenius lift $\phi$ topologically generating a complement to $I_q$; $\chi(i\phi^k) = \psi(i)$ defines a quadratic character of $G_{\mathbb{Q}_q}$ extending $\psi$. Every quadratic character of $G_{\mathbb{Q}_q}$ is the local component of a quadratic Dirichlet character $\chi'$ (local class field theory: $\mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^2$ has order 4 for $q$ odd and 8 for $q = 2$, and the local components of $\chi_{q^*}$ and of a character unramified at $q$, respectively of $\chi_{-4}$, $\chi_8$, $\chi_5$, give that many distinct characters), and $E' = E \otimes \chi'$ has $\rho_{E'}|_{I_q} = \psi^2 = 1$: good reduction. Then $\mathrm{ad}^0 V = \mathrm{ad}^0 V_\ell E'$ has eigenvalues $\alpha/\beta, 1, \beta/\alpha$ and $P_q(X) = (1 - \alpha^2 X)(1 - qX)(1 - \beta^2 X)$ (the geometric eigenvalues times $q$, as $L_q(A_f, s-1)$), which is the stated polynomial as $\alpha^2 + \beta^2 = a^2 - 2q$ and $\alpha^2\beta^2 = q^2$. Only $a^2$ enters, so the sign ambiguity in the choice of $E'$ does not matter.

(iii) Let $g$ generate $\rho(I_q)$, of order $m \in \{3,4,6\}$, eigenvalues $\zeta^{\pm 1}$, $\zeta \ne \zeta^{-1}$. Trace-zero endomorphisms commuting with $g$ form the line spanned by $g - g^{-1}$ (commutant $\mathbb{Q}_\ell[g]$, two-dimensional). Frobenius normalises $\rho(I_q)$, so it conjugates $g$ to a generator $g^k$ with the same eigenvalue set $\{\zeta, \zeta^{-1}\}$, so to $g^{\pm 1}$; it acts on the line by a sign $c$, the same for both Frobenii. Hence $P_q(X) = 1 - cqX$ and $e_\ell = v_\ell(1 - c/q) = v_\ell(q - c)$.

(iv) A faithful two-dimensional representation of a nonabelian finite group in characteristic 0 is absolutely irreducible (a reducible one would be, after semisimplification, a sum of two characters, hence abelian image), so its commutant is the scalars and $(\mathrm{ad}^0 V)^{I_q} = 0$. For $q \ge 5$ the image is a quotient of tame inertia (its order divides 24, prime to $q$), hence cyclic; so (iv) needs $q \in \{2,3\}$. □

*Reading guidance.* The proposition is what lets the paper use PARI's Euler factor at $q = 2, 3$ without a Kodaira-type table: whatever the wild inertia does, the factor has one of four shapes, and $e_\ell(E, q)$ is read off it. The only cubic shape is (ii).

*Remark* 21.18 (the paper's Remark rem:wild). The remark says that at $q = 2, 3$ the Kodaira type does not determine $P_q$, gives `336f1` (type II at 2, $P_2 = 1 + 2X$, $e_3 = 1$, $\deg \varphi_E = 24$), `123211` (type II* at 2, the twist of `77a1` by $\chi_{-4}$, $P_2 = (1 - 2X)(1 + 2X)^2$, $e_3(E, 2) = 2$, $\deg \varphi_E = 288$: the bound of Theorem thm:main at $\ell = 3$ attained by the prime 2 alone) and `592c1` (type II* at 2, the twist of `37a1` by $\chi_{-4}$, $P_2 = (1 - 2X)(1 + 4X^2)$, $e_5(E, 2) = 1$, $\deg \varphi_E = 80$). All these values were recomputed for the book (Section 21.18). The two counts of the remark name their populations: the cubic factors at 2 (1,143) and at 3 (2,283) are counted over all curves below conductor 10,000, and 534 and 1,308 over the optimal non-CM curves (Chapter 19); since (ii) is the only cubic shape, the empirical content is that each cubic equals the formula of (ii) for a twist of good reduction found independently. The share of potentially good reduction among the types $\mathrm{I}_n^*$ at 2 is 58% (2,340 of 4,037) among optimal curves and 61% (5,935 of 9,667) among all curves (Chapter 10). The book reproduced all four counts.

## 21.8 Section 2 of the paper, part 3: the Tamagawa factors of the adjoint

This section reads Lemma lem:tamfactor (potentially multiplicative primes) and Lemma lem:tamadd (potentially good additive primes). With them every term of (eq:identity) other than the Selmer length is explicit (Corollary cor:explicit). Both proofs compute $H^1(I_q, \mathrm{ad}^0 T)$ from the image of inertia found in the previous two sections: at a potentially multiplicative prime inertia acts through a unipotent, and its first cohomology is the coinvariants, which have torsion; at a potentially good prime it acts through a finite group of order dividing 24, the torsion of the cohomology comes from that finite group, so it is killed by the group's order; it vanishes at $\ell \geq 5$ for that reason, and at $\ell = 3$ because $\mathrm{ad}^0 T$ is a free module over the group ring of a Sylow 3-subgroup.

**Lemma 21.19** (the paper's Lemma lem:tamfactor)**.** *Let $q$ be a prime of multiplicative reduction, $n = n_q$, $\ell$ odd, $\ell \neq q$, $M = \mathrm{ad}^0 T$. Let $q_E$ be the Tate parameter, $u = q_E q^{-n} \in \mathbb{Z}_q^\times$, and $c(u)$ the class of the reduction $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^v} \simeq \mathbb{Z}/\ell^v$, $v = v_\ell(q-1)$ (so $v_\ell(c(u)) \geq k$ iff $\bar{u}$ is an $\ell^k$-th power, for $k \leq v$). Then $H^1(I_q, M)_{\mathrm{tor}} \simeq (\mathbb{Z}/\ell^{v_\ell(n)})^2$ and*

$$v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) = v_\ell(n) + \min\big(v_\ell(n), v_\ell(q-1), v_\ell(c(u))\big).$$

*In particular $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T)) \geq t_\ell(E,q)$, with equality unless $q \equiv 1 \pmod{\ell}$, $\ell \mid n$ and $\bar{u}$ is an $\ell$-th power. The same formula holds at an additive potentially multiplicative prime $q \neq \ell$, $q = 2$ included, with $n = -v_q(j_E)$ and $q_E$ the Tate parameter with $j(q_E) = j_E$.*

*Where proved.* Chapter 18 (the lemma), Chapter 16 (the cohomology of $\langle t \rangle$ and the conjugation action), Chapter 17 (the Kummer class of $q_E$ and the identification of $c'$), Chapter 3 (Fitting ideals of finite modules).

*Expanded proof.* (1) *The Tate matrices.* Over the unramified quadratic extension $E$ is a Tate curve; $T$ has a basis $e_1, e_2$ in which $g \in G = G_{\mathbb{Q}_q}$ acts by $\left(\begin{smallmatrix} \chi_\ell \varepsilon(g) & \varepsilon\kappa(g) \\ 0 & \varepsilon(g) \end{smallmatrix}\right)$, $\varepsilon$ unramified of order at most 2 with $\varepsilon(\mathrm{Frob}) = a_q(E)$, $\kappa$ the Kummer cocycle of $q_E$ in $H^1(G, \mathbb{Z}_\ell(1)) = \varprojlim \mathbb{Q}_q^\times/(\mathbb{Q}_q^\times)^{\ell^k}$. The corner is $\varepsilon\kappa$: the nonsplit curve is the Tate curve twisted by $\varepsilon$, and $\left(\begin{smallmatrix} \chi_\ell & \kappa \\ 0 & 1 \end{smallmatrix}\right) \otimes \varepsilon = \left(\begin{smallmatrix} \chi_\ell \varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$; a corner $c$ must satisfy $c(gh) = \chi_\ell \varepsilon(g) c(h) + c(g)\varepsilon(h)$, which $\varepsilon\kappa$ does and $\kappa$ does not when $\varepsilon \neq 1$.

(2) *Inertia.* $I = I_q$ acts through its maximal pro-$\ell$ quotient, $\mathbb{Z}_\ell(1)$, generated by $t$; $\varepsilon|_I = 1$ and $\kappa|_I = n$ times the tame character, so $te_1 = e_1$, $te_2 = e_2 + n' e_1$ with $v_\ell(n') = v_\ell(n)$. The kernel of $I \to \langle t \rangle$ has no pro-$\ell$ quotient, so inflation-restriction gives $H^1(I, M') = H^1(\langle t \rangle, M') = M'/(t-1)M'$ via $c \mapsto c(t)$, for every finitely generated $\mathbb{Z}_\ell$-module $M'$ with this action (Chapter 16).

(3) *The torsion.* In the basis $e, h, f$ of $M$: $(t-1)e = 0$, $(t-1)h = -2n'e$, $(t-1)f = n'h - n'^2 e$. As 2 is a unit, $(t-1)M = n'(\mathbb{Z}_\ell e \oplus \mathbb{Z}_\ell h)$, so $H^1(I, M) = (\mathbb{Z}_\ell/n')\bar{e} \oplus (\mathbb{Z}_\ell/n')\bar{h} \oplus \mathbb{Z}_\ell \bar{f}$, with torsion $(\mathbb{Z}/\ell^{v_\ell(n)})^2$.

(4) *Frobenius.* $G/I$ acts by $(\sigma \cdot c)(i) = \sigma c(\sigma^{-1} i \sigma)$. With $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ and $c(t^k) \equiv k c(t)$ modulo $(t-1)M$, the arithmetic Frobenius acts on $M/(t-1)M$ by $q^{-1}\,\mathrm{Ad}(\rho(\mathrm{Frob}))$, conjugation by $\rho(\mathrm{Frob})$ followed by multiplication by $q^{-1}$, as the paper says. The factor $q^{-1}$ comes from the twist $\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$ of the tame generator, and it is what makes the count depend on $q-1$.

(5) *The fixed points.* Write $\rho(\mathrm{Frob}) = \left(\begin{smallmatrix} \varepsilon q & c' \\ 0 & \varepsilon \end{smallmatrix}\right)$, $\varepsilon = \varepsilon(\mathrm{Frob}) = \pm 1$, $c' = \varepsilon\kappa(\mathrm{Frob})$. Conjugation sends $e \mapsto qe$ and $h \mapsto h - 2c'\varepsilon e$; so on the torsion $\bar{e} \mapsto \bar{e}$ and $\bar{h} \mapsto q^{-1}(\bar{h} - 2c'\varepsilon\bar{e})$. The element

$x\bar{e}+y\bar{h}$ is fixed iff $(q-1)y\equiv 0$ and $c'y\equiv 0$ modulo $\ell^{v_\ell(n)}$ (compare the $\bar{h}$ and $\bar{e}$ coefficients, using that $q$, 2, $\varepsilon$ are units: the image of $x\bar{e}+y\bar{h}$ is $(x-2c'\varepsilon q^{-1}y)\bar{e}+q^{-1}y\bar{h}$). There are $\ell^{v_\ell(n)}$ choices of $x$ and $\ell^{\min(v_\ell(n),v_\ell(q-1),v_\ell(c'))}$ of $y$. (Count in $\mathbb{Z}/\ell^m$, $m=v_\ell(n)$: for $a=v_\ell(q-1)$ and $b=v_\ell(c')$, the conditions $\ell^a y\equiv 0$ and $\ell^b y\equiv 0$ modulo $\ell^m$ say $v_\ell(y)\geq m-\min(a,b)$, and the number of such $y$ is $\ell^{\min(m,a,b)}$.) The Fitting ideal of a finite $\mathbb{Z}_\ell$-module of order $\ell^m$ is $\ell^m\mathbb{Z}_\ell$ (Chapter 3), and $\mathrm{Tam}_q(M)$ is that Fitting ideal [69, I.4.2.2], recalled in [49, p. 708].

(6) *The value of $c'$.* $q$ being a uniformiser, $H^1(\mathbb{Q}_q,\mathbb{Z}_\ell(1))=\mathbb{Z}_\ell\kappa_q\oplus H^1_{\mathrm{ur}}$, and $\kappa=n\kappa_q+\kappa_u$ with $\kappa_u$ inflated from the class of $u$ in $H^1(G/I,\mathbb{Z}_\ell(1))=\mathbb{Z}_\ell(1)/(q-1)\simeq\mathbb{F}_q^\times\otimes\mathbb{Z}_\ell$. For the Frobenius lift fixing a compatible system of $\ell$-power roots of $q$ (which exists, $\mathbb{Q}_q(q^{1/\ell^\infty})$ being totally ramified), $\kappa_q(\mathrm{Frob})=0$ and $c'=\varepsilon\kappa_u(\mathrm{Frob})$, the class of $\bar{u}$ up to the sign $\varepsilon$. Another lift $\mathrm{Frob}\,t^k$ changes $c'$ by $\varepsilon qkn'$; another choice of roots changes $\kappa$ by a coboundary, hence $c'$ by a multiple of $q-1$. So $c'$ is well defined modulo $\ell^{\min(v_\ell(n),v_\ell(q-1))}$, which does not change the minimum, and $v_\ell(c')$ may be replaced by $v_\ell(c(u))$ (Chapter 17).

(7) *Check for $M=T$.* The same computation gives $H^1(I,T)_{\mathrm{tor}}=(\mathbb{Z}/\ell^{v_\ell(n)})\bar{e}_1$ with Frob acting by $q^{-1}\cdot\varepsilon q=\varepsilon$: all invariant if split, none if nonsplit ($\ell$ odd), which is the $\ell$-part of $c_q(E)$, as it should be.

(8) *The "in particular".* The extra term is positive iff $v_\ell(n)\geq 1$, $v_\ell(q-1)\geq 1$ and $v_\ell(c(u))\geq 1$. At $q=2$, $v_\ell(q-1)=0$, so the extra term vanishes.

(9) *Additive potentially multiplicative $q$.* Over $\mathbb{Q}_q$, $E$ is the twist of the Tate curve $E_{q_E}$ by a ramified quadratic character; $\mathrm{ad}^0 T$ is unchanged as a $G_{\mathbb{Q}_q}$-module, and the computation used only that module, with $n=v_q(q_E)=-v_q(j_E)$ (not the index of the Kodaira symbol at $q=2$). □

*Reading guidance.* Three points are easy to get wrong: the corner entry is $\varepsilon\kappa$, not $\kappa$ (step (1) and Exercise 21.55); the value $c'$ is determined only modulo $\ell^{\min(v_\ell(n),v_\ell(q-1))}$, which is harmless (step (6)); and at an additive potentially multiplicative prime the argument is local, so it covers $q=2$ (step (9)). Worked values, recomputed for the book with PARI's $q$-adic Tate parameter: `791a1` at 7 ($\mathrm{I}_3$, $\bar{u}=6$, a cube): $v_3(\mathrm{Tam}_7(\mathrm{ad}^0 T))=2$, $t_3=1$; `539c1` at 7 ($\mathrm{I}_3^*$, the same $q_E$ as `77c1`, $\bar{u}=5$, not a cube): 1; `1577a1` at 19 ($\bar{u}=8$, a cube but not a ninth power): extra term 1; `11a1` at 11, $\ell=5$ ($\bar{u}=10$, a fifth power): $v_5(\mathrm{Tam})=2$, although $E[5]$ is reducible, so only the lemma, not Theorem thm:main, applies there.

**Lemma 21.20** (the paper's Lemma lem:tamadd)**.** *Let $\ell$ be odd and $q\neq\ell$ a prime of additive potentially good reduction. Then $H^1(I_q,\mathrm{ad}^0 T)_{\mathrm{tor}}=0$; in particular $v_\ell(\mathrm{Tam}_q(\mathrm{ad}^0 T))=0$.*

*Where proved.* Chapter 18; inflation-restriction and restriction-corestriction: Chapter 16; $T_3E$ over $\mathbb{Z}_3[\zeta_3]$: Chapter 6.

*Expanded proof.* Let $M=\mathrm{ad}^0 T$, $K\subseteq I=I_q$ the kernel of the action on $M$, and $\Delta=I/K$, the image of the finite group $\rho(I)$ modulo $\rho(I)\cap\{\pm1\}$ (conjugation kills $\pm1$). Inflation-restriction: $0\to H^1(\Delta,M)\to H^1(I,M)\to\mathrm{Hom}(K,M)^\Delta$. $K$ is the inertia group of a finite extension of $\mathbb{Q}_q^{\mathrm{ur}}$; a continuous homomorphism $K\to M$ (a pro-$\ell$ group) factors through the maximal pro-$\ell$ abelian quotient of $K$, which is $\mathbb{Z}_\ell(1)$ as $q\neq\ell$; so $\mathrm{Hom}(K,M)\simeq M(-1)$ is torsion-free, and the torsion of $H^1(I,M)$ lies in $H^1(\Delta,M)$, which is killed by $\#\Delta$. Orders of elements of $\rho(I)$ lie in $\{1,2,3,4,6\}$ and $\#\rho(I)\mid 24$, so $\#\Delta$ is a product of 2s and 3s: $H^1(\Delta,M)=0$ for $\ell\geq 5$. Let $\ell=3$. Restriction to a Sylow 3-subgroup $P$ of $\Delta$ followed by corestriction is multiplication by the index, prime to 3, so the 3-part of $H^1(\Delta,M)$ injects into $H^1(P,M)$. As $\#\rho(I)$ divides 24 ($\rho(I)$ is cyclic of order $1,2,3,4$ or 6, the quaternion group or $\mathrm{SL}_2(\mathbb{F}_3)$, Chapter 11), a Sylow 3-subgroup of $\Delta$ has

order 1 or 3: $P$ is trivial or generated by the image of an element $\sigma \in \rho(I)$ of order 3. Then $T$ is free of rank one over $\mathbb{Z}_3[X]/(X^2+X+1)$ with $X$ acting as $\sigma$, so in a suitable basis $\sigma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$. With $e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: $e$, $\sigma e \sigma^{-1} = \left(\begin{smallmatrix} 0 & 0 \\ -1 & 0 \end{smallmatrix}\right)$ and $\sigma^2 e \sigma^{-2} = \left(\begin{smallmatrix} -1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ have coordinates $(1,0,0)$, $(0,0,-1)$, $(1,-1,-1)$ in the basis $e, h, f$ (a matrix $\left(\begin{smallmatrix} b & a \\ c & -b \end{smallmatrix}\right)$ has coordinates $(a,b,c)$); the determinant of these three vectors is $-1$, a unit. So $\mathrm{ad}^0 T \simeq \mathbb{Z}_3[P]$ is free over $\mathbb{Z}_3[P]$ and $H^1(P, \mathrm{ad}^0 T) = 0$. Hence $H^1(I,M)_{\mathrm{tor}} = 0$, and so are its $G_{\mathbb{Q}_q}$-invariants and their Fitting ideal's valuation. □

The decisive step at $\ell = 3$ is that $\mathrm{ad}^0 T$ is a free $\mathbb{Z}_3[P]$-module: a regular representation has no cohomology. The book checked the three coordinate vectors and their determinant in PARI. The sentence after the lemma, that $\mathrm{Tam}_q(T)$ can be divisible by 3 at a type IV prime because $H^1(\langle\sigma\rangle, T) = T/(1-\zeta_3)T \simeq \mathbb{F}_3$, is right: on $T$ the norm $1+\sigma+\sigma^2$ is 0, so $H^1 = \ker N/(\sigma-1)T = T/(1-\zeta_3)T$.

## 21.9 Section 3 of the paper: the proof of Theorem thm:qual

This section reads the paper's Section 3: the congruence number, the congruence criterion (Lemma lem:crit), the oldform lemma (Lemma lem:old), the optimal-level theorem, the three cases of Theorem thm:qual, the Example and the Remark after it. The method is qualitative: produce a congruent newform of lower level, and conclude that $\ell$ divides the congruence number, hence the degree.

**Definition 21.21** (the congruence number; the opening of Section 3)**.** For $E$ optimal with newform $f$ of level $N$, $r_E$ is the largest integer $r$ such that $f \equiv g \pmod r$ coefficientwise for some $g \in S_2(\Gamma_0(N);\mathbb{Z})$ orthogonal to $f$ for the Petersson product. Agashe, Ribet and Stein [3, Theorem 2.2]: $\deg\varphi_E \mid r_E$, and $v_p(\deg\varphi_E) = v_p(r_E)$ for every prime $p$ with $p^2 \nmid N$.

*Where proved.* The theorem of Agashe, Ribet and Stein is stated in Chapter 14 (checked against the source there); the divisibility is reproved there from the projection formula, and the equality at $\ell \nmid 2N$ with $E[\ell]$ irreducible follows from Proposition 21.30 below together with Lemma lem:crit. The examples where $r_E \neq \deg\varphi_E$ are computed in Chapters 13 and 14: `99a1` ($\deg\varphi_E = 4$, $r_E = 12$), `726a1` (144 and $1584 = 11 \cdot 144$), `539d1` (72 and $504 = 7 \cdot 72$); in each the gap is at a prime whose square divides $N$.

**Lemma 21.22** (the paper's Lemma lem:crit)**.** *Let $I = \ker\phi_f$ and $J = \mathrm{Ann}_{\mathbb{T}}(f^\perp)$. Then $\mathbb{T}/(I+J)$ is cyclic of order $r_E$; equivalently $r_E$ generates $\phi_f(J) = \phi_f(\mathrm{Ann}_{\mathbb{T}} I)$. Consequently, if $g \in S_2(\Gamma_0(N)) \otimes \overline{\mathbb{Q}}$ is a normalised eigenform for all of $\mathbb{T}$, orthogonal to $f$, with coefficients in the ring of integers $\mathcal{O}$ of a number field, and $a_n(g) \equiv a_n(f) \pmod\lambda$ for all $n$ and a prime $\lambda \mid \ell$, then $\ell \mid r_E$. The same holds over $\mathbb{Z}_\ell$ with the $\ell$-part of $r_E$.*

*Where proved.* In full in Chapter 15; restated as a statement about congruence modules in Chapter 20; the perfect pairing $S_2(\Gamma_0(N);\mathbb{Z}) \times \mathbb{T} \to \mathbb{Z}$ and the Hecke adjoints in Chapter 13; the lattice duality in Chapter 3.

*Walk through the paper's proof.* (1) $\mathbb{T} \otimes \mathbb{Q} = \mathbb{Q} \times \mathbb{T}'$ with $\mathbb{T}'$ faithful on $f^\perp$: the line of $f$ and $f^\perp$ are $\mathbb{T}$-stable. For $(n,N) = 1$, $T_n$ is self-adjoint for the Petersson product; for $p \mid N$, $U_p^\dagger = w_N U_p w_N^{-1}$ and $f$ is an eigenvector of $w_N$; so $f$ is an eigenvector of every $T_n^\dagger$, and the orthogonal complement of a line stable under all $T_n$ and all $T_n^\dagger$ is $\mathbb{T}$-stable (the paper's "because $f$ is an eigenvector of the adjoints of the $T_n$ as well"). Hence $I = \mathbb{T} \cap (0 \times \mathbb{T}')$, $J = \mathbb{T} \cap (\mathbb{Q} \times 0)$,

$I \cap J = 0$, and $\mathbb{T}/(I+J) \simeq \mathbb{Z}/\phi_f(J)$, of order $r'$ say ($r'$ is finite because $\phi_f(J) \neq 0$: $J \otimes \mathbb{Q} = \mathbb{Q} \times 0$). (2) Under the perfect pairing $(g,t) \mapsto a_1(tg)$ the forms killed by $I$ are $\mathbb{Z}f$ and those killed by $J$ are $L = f^{\perp} \cap S_2(\Gamma_0(N);\mathbb{Z})$. (3) Dualising $\mathbb{T} \hookrightarrow \mathbb{T}/I \times \mathbb{T}/J$, with cokernel $\mathbb{T}/(I+J)$, gives $[S_2(\Gamma_0(N);\mathbb{Z}) : \mathbb{Z}f \oplus L] = r'$. In detail: the sequence $0 \to \mathbb{T} \to \mathbb{T}/I \times \mathbb{T}/J \to \mathbb{T}/(I+J) \to 0$, $t \mapsto (t,t)$, $(a,b) \mapsto a-b$, is exact because $I \cap J = 0$. Apply $\mathrm{Hom}(-,\mathbb{Z})$. The groups $\mathbb{T}/I \simeq \mathbb{Z}$ and $\mathbb{T}/J$ are free ($I$ and $J$ are saturated in $\mathbb{T}$, being intersections of $\mathbb{T}$ with subspaces of $\mathbb{T} \otimes \mathbb{Q}$), and $\mathrm{Hom}(\mathbb{T}/(I+J),\mathbb{Z}) = 0$, so

$$0 \to \mathrm{Hom}(\mathbb{T}/I,\mathbb{Z}) \oplus \mathrm{Hom}(\mathbb{T}/J,\mathbb{Z}) \to \mathrm{Hom}(\mathbb{T},\mathbb{Z}) \to \mathrm{Ext}^1(\mathbb{T}/(I+J),\mathbb{Z}) \to 0,$$

and $\mathrm{Ext}^1(\mathbb{Z}/r',\mathbb{Z}) \simeq \mathbb{Z}/r'$. Under the perfect pairing, $\mathrm{Hom}(\mathbb{T},\mathbb{Z})$ is $S_2(\Gamma_0(N);\mathbb{Z})$ and $\mathrm{Hom}(\mathbb{T}/I,\mathbb{Z})$, $\mathrm{Hom}(\mathbb{T}/J,\mathbb{Z})$ are the forms killed by $I$ and by $J$, that is $\mathbb{Z}f$ and $L$ by (2). (4) The quotient embeds in $\mathbb{Q}f/\mathbb{Z}f$ by projection along $f^{\perp}$, so it is cyclic, generated by a form $h$ with projection $f/r'$; $r'h = f + g_0$ with $g_0 \in L$ gives a congruence modulo $r'$, so $r' \leq r_E$; conversely $f \equiv g_1 \pmod{r_E}$ with $g_1 \in L$ makes $(f-g_1)/r_E$ integral, of order $r_E$ in the quotient, so $r_E \mid r'$. (5) $\phi_g$ kills $J$ ($g \in f^{\perp}$) and $\phi_g \equiv \phi_f \pmod{\lambda}$, so $I+J$ lies in the kernel of $\mathbb{T} \to \mathcal{O}/\lambda$, a maximal ideal of residue characteristic $\ell$, and $\ell \mid r_E$. (6) Every step holds over $\mathbb{Z}_\ell$, the pairing being perfect after $\otimes\mathbb{Z}_\ell$. The hard step is (3): the index of $\mathbb{Z}f \oplus L$ is computed by dualising, and the cyclicity of $\mathbb{T}/(I+J)$ is what turns the congruence number into a single generator of an ideal.

**Lemma 21.23** (the paper's Lemma lem:old)**.** *Let $h \neq f$ be a newform of weight 2 on $\Gamma_0(M)$, $M \mid N$, with coefficients in $\mathcal{O}$, and $\lambda \mid \ell$ with $a_p(h) \equiv a_p(f) \pmod{\lambda}$ for all primes $p \nmid \ell N$. Assume $p \nmid M$ for every $p \,\|\, N$ at which $\bar\rho$ is unramified. Then, after enlarging $\mathcal{O}$ and $\lambda$, the span of the $h(dz)$, $d \mid N/M$, contains an eigenform $g$ as in Lemma lem:crit; so $\ell \mid r_E$.*

*Where proved.* Chapter 15; the $U_p$-structure of old spaces in Chapter 13; Chebotarev and Brauer-Nesbitt in Chapters 4 and 5; the local description at $\ell$ (Deligne and Fontaine, as in [61, Section 2]) and at a multiplicative prime [38, Theorem 3.1(e)] in Chapters 11 and 15.

*Walk through.* The span $V$ of the $h(dz)$ is $\mathbb{T}$-stable and orthogonal to $f$. On it $T_n$, $(n,N)=1$, acts by $a_n(h)$, and for $p \mid N/M$ with $e_p = v_p(N/M)$, $U_p$ acts on the span of $h(p^j z)$, $0 \leq j \leq e_p$, by $U_p h(p^j z) = h(p^{j-1}z)$ ($j \geq 1$) and $U_p h = a_p(h)h - ph(pz)$ ($p \nmid M$) or $a_p(h)h$ ($p \mid M$). Its eigenvalues are the roots $\alpha_p, \beta_p$ of $X^2 - a_p(h)X + p$, with 0 if $e_p \geq 2$ ($p \nmid M$), or $a_p(h)$ with 0 if $e_p \geq 1$ ($p \mid M$); each has an eigenvector (the $p$-stabilisations and the depleted forms), though $U_p$ is not semisimple when 0 is repeated. As the local spaces tensor together and $\mathbb{T}$ is commutative, any choice of one eigenvalue $\mu_p$ per $p \mid N/M$ gives a common eigenform $g$; one checks $a_p(g) \equiv a_p(f)$ prime by prime, and then for all $n$ by multiplicativity and the Hecke recursions. The five cases: $p \nmid \ell N$ (hypothesis); $p = \ell$ (both forms have level prime to $\ell$ and weight 2; $a_\ell \bmod \lambda$ is read off $\bar\rho|_{D_\ell}$, 0 if irreducible, the eigenvalue of Frobenius on the unramified quotient otherwise, a quotient that is well defined because the characters of the semisimplification restrict to $I_\ell$ as $\bar\chi_\ell$ and 1, and $\bar\chi_\ell|_{I_\ell} \neq 1$ as $\ell$ is odd, so that it is the module of $I_\ell$-coinvariants); $p^2 \mid N$ ($a_p(f) = 0$; $a_p(h) = 0$ if $v_p(M) = v_p(N)$ [8, Theorem 3], else 0 is an available eigenvalue); $p \,\|\, N$ with $\bar\rho$ unramified ($p \nmid M$ by hypothesis, $\bar\rho(\mathrm{Frob}_p)$ has eigenvalues $a_p(f)p$ and $a_p(f)$, so one of $\alpha_p, \beta_p$ is congruent to $a_p(f)$); $p \,\|\, N$ with $\bar\rho$ ramified ($p \mid M$, and two non-split extensions of characters match their unique stable lines, so $a_p(h) \equiv a_p(f)$). In the first case the paper adds: "$\bar\rho_h \simeq \bar\rho$ over a common finite extension of $\mathbb{F}_\ell$, by Chebotarev and Brauer-Nesbitt, $\bar\rho$ being irreducible, hence absolutely irreducible" (complex conjugation has the distinct eigenvalues $\pm 1$, so an irreducible $\bar\rho$ is absolutely irreducible, and Brauer-Nesbitt then gives an isomorphism, not only of semisimplifications). The

last case is the subtle one: the two local representations are matched through their unique stable lines, which needs the extensions to be non-split, that is, $\bar\rho$ ramified at $p$.

**Theorem 21.24** (the optimal-level theorem, as the paper uses it)**.** *Let $\ell$ be odd and $\bar\sigma\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ irreducible and modular, absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ if $\ell = 3$. Then $\bar\sigma$ arises from a newform of weight 2 and level $N(\bar\sigma)\ell^{\delta(\bar\sigma)}$ with nebentypus of order prime to $\ell$, $N(\bar\sigma)$ the Serre conductor and $\delta(\bar\sigma) = 0$ when $\bar\sigma|_{D_\ell}$ is good [38, Theorem 3.15] (Ribet, Carayol, Diamond [156, 22, 47]). For $\bar\sigma = \bar\rho_{E',\ell}$ with $E'$ of conductor prime to $\ell$ it arises from a newform on $\Gamma_0(N(\bar\sigma))$; and $N(\bar\rho_{E',\ell}) \mid N(E')$.*

*Where.* Stated with references and sketched in Chapter 15; we do not prove it. The deduction of trivial character is the paper's: $\det\bar\sigma = \bar\chi_\ell$ forces the nebentypus $\epsilon$ to reduce to 1; reduction is injective on roots of unity of order prime to $\ell$, so $\epsilon = 1$. The theorem number is as cited by the paper.

**Theorem 21.25** (the paper's Theorem thm:qual)**.** *Let $E/\mathbb{Q}$ have conductor $N$ and $\ell$ an odd prime, $\ell \nmid N$, with $\bar\rho_{E,\ell}$ irreducible and, if $\ell = 3$, absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$. Let $q \neq \ell$ be a prime of additive reduction of one of the kinds: (a) type IV or IV$^*$, $q \geq 5$ or $q = 2$, $\ell = 3$; (b) type II or II$^*$, $q \geq 5$, $\ell = 3$; (c) type $\mathrm{I}_n^*$ with $n \geq 1$ and $\ell \mid n$, $q$ odd. Then $\ell$ divides the modular degree of the optimal curve of the class of $E$.*

*Where proved.* Chapter 15, with the inputs Lemmas 21.11, 21.13, 21.15 and 21.23, Theorem 21.24 and [3, Theorem 2.2].

*Expanded proof.* Let $E_0$ be the optimal curve and $r = r_{E_0}$. As $\ell \nmid N$, $v_\ell(\deg\varphi_{E_0}) = v_\ell(r)$, so it suffices to find $h$ as in Lemma 21.23. In all three cases $q^2 \parallel N$: the conductor exponent at $q$ is 2 (Lemma 21.13 in case (a), Lemma 21.15(ii) in cases (b) and (c)). Notation: $h^\chi$ is the newform twist and $h \otimes \chi$ the naive twist; when $q \parallel \mathrm{level}(h)$ or $q \nmid \mathrm{level}(h)$, $h \otimes \chi$ is the newform $h^\chi$ (Lemma 21.15(iii)); for $h = f_E$ with $q^2 \parallel N$ and $E^{(q)}$ also additive at $q$ (case (b)), $f \otimes \chi$ is the newform of $E^{(q)}$, as both have $a_q = 0$; and $(g \otimes \chi) \otimes \chi = g^{\{q\}}$ is the $q$-depletion of $g$.

*Case (a).* By Lemma 21.13, the Serre conductor $M = N(\bar\rho)$ has $v_q(M) = 1$; as $M \mid N$ (Carayol), $M \mid N/q$. Theorem 21.24 gives a newform $g$ on $\Gamma_0(M)$ with $\bar\rho_g \simeq \bar\rho$, so $a_p(g) \equiv a_p(f) \pmod\lambda$ for $p \nmid 3N$. Take $h = g$: $h \neq f$ since the levels differ, and $p \nmid M$ for every $p$ at which $\bar\rho$ is unramified. At $p = q$ the eigenvalue 0 of $U_q$ is available ($v_q(M) = 1 < 2$). Lemma 21.23 gives $3 \mid r$.

*Case (b).* Let $\chi = \chi_{q^*}$, $E' = E^{(q)}$: type IV$^*$ or IV at $q$, the same types elsewhere, conductor $N$. Its newform is $f' = f \otimes \chi$: both $f'$ and $f \otimes \chi$ have $a_q = 0$ (additive reduction) and $a_p(f') = \chi(p)a_p(f)$ for $p \neq q$; this is not Atkin-Li, since $q^2 \mid N$. The representation $\bar\rho \otimes \chi = \bar\rho_{E',3}$ satisfies the hypotheses of Theorem 21.24 (irreducibility and absolute irreducibility over $\mathbb{Q}(\sqrt{-3})$ survive a twist). By Lemma 21.13 for $E'$, $M = N(\bar\rho \otimes \chi)$ has $v_q(M) = 1$, $M \mid N/q$; there is a newform $g$ on $\Gamma_0(M)$ with $a_p(g) \equiv \chi(p)a_p(f) \pmod\lambda$ for $p \nmid 3N$. As $q \parallel M$, $h = g \otimes \chi$ is a newform on $\Gamma_0(Mq)$, $Mq \mid N$, with $a_p(h) = \chi(p)a_p(g) \equiv \chi(p)^2 a_p(f) = a_p(f)$. If $h = f$, then $a_p(g) = \chi(p)a_p(h) = a_p(f')$ for all $p \nmid qN$, so $g = f'$ by strong multiplicity one (Chapter 13), impossible as their levels are $M < N$ and $N$. (Here $h \otimes \chi$ is the $q$-depletion of $g$, not $g$, since $a_q(g) = \pm 1$.) If $p \parallel N$ and $\bar\rho$ is unramified at $p$, then $p \neq q$, so $\bar\rho \otimes \chi$ is unramified at $p$ and $p \nmid Mq$. Lemma 21.23 gives $3 \mid r$.

*Case (c).* $E' = E^{(q)}$ has type $\mathrm{I}_n$ at $q$ and conductor $N/q$; $f' = f_{E'}$ has level $N/q$, $q \parallel N/q$, $a_q(f') = \pm 1$ and $a_p(f') = \chi(p)a_p(f)$ for $p \neq q$, so $f \otimes \chi$ is the $q$-depletion of $f'$, not $f'$ (for example `539c1` twisted by $\chi_{-7}$ lies in `77c`, and $a_7$(`77c1`) $= -1$). By Lemma 21.11 for $E'$, $\ell \mid n$

makes $\bar\rho\otimes\chi$ unramified at $q$, so $M=N(\bar\rho\otimes\chi)$ is prime to $q$ and $M\mid N/q^2$; Ribet's theorem (the case of Theorem 21.24 with $q$ exactly dividing the level of $E'$ and $\bar\rho\otimes\chi$ unramified at $q$) gives a newform $g$ on $\Gamma_0(M)$ with $a_p(g)\equiv\chi(p)a_p(f)$ for $p\nmid\ell N$. As $q\nmid M$, $h=g\otimes\chi$ is a newform on $\Gamma_0(Mq^2)$, $Mq^2\mid N$, congruent to $f$ at $p\nmid\ell N$. If $h=f$, then $g$ and $f'$ have equal $a_p$ for $p\nmid qN$, so $g=f'$, impossible since their levels $M$ and $N/q$ differ. The hypothesis of Lemma 21.23 holds as in case (b), and $\ell\mid r$. □

*Reading guidance.* The step to watch in cases (b) and (c) is $h\neq f$: it is proved by comparing $g$ with $f'$ through strong multiplicity one and their levels, and the distinction between the naive twist $f\otimes\chi$ and the newform of $E'$ matters there (in case (c), $f\otimes\chi$ is the $q$-depletion of $f'$). Why $\mathrm{I}_0^*$, and $\mathrm{I}_n^*$ with $\ell\nmid n$, are not in the theorem: after the twist the lowered form may be $f_{E'}$ itself, so $h=f$; Theorem thm:quant reaches these terms by raising the level instead.

**Example 21.26** (the paper's Example ex:qual, verified)**.** The paper's congruences, each between two forms in the same $S_2(\Gamma_0(N);\mathbb{Z})$, so that Sturm's bound $\frac{1}{6}N\prod_{p\mid N}(1+\frac{1}{p})$ [190] (Chapter 12) proves them for all $n$. (a) at $q=2$: `52a1`, type IV$^*$ at 2, Serre conductor 26 at $\ell=3$; $f_{52a}\equiv f_{26b}(z)-f_{26b}(2z)\pmod 3$ (Sturm bound 14). Here $2\parallel 26$ and $a_2(\texttt{26b1})=1$, so $f_{26b}(z)-a_2f_{26b}(2z)$ is the $U_2$-eigenvector of eigenvalue 0 in Lemma lem:old; $\deg\varphi_{\texttt{52a1}}=3$ and $e_3(E,2)=1$. (b) `121a1`, type II at 11 ($v_{11}(\Delta)=2$); its twist by $\chi_{-11}$ is `121c2`, of type IV$^*$ ($v_{11}(\Delta)=8$ on the minimal model; the optimal curve `121c1` of that class has type IV), whose mod 3 representation has Serre conductor 11 and comes from $f_{11a}$; twisting back, $f_{11a}\otimes\chi_{-11}=f_{121d}$, and $f_{121a}\equiv f_{121d}\pmod 3$ (Sturm bound 22); $\deg\varphi_E=6$, $e_3(E,11)=v_3(12)=1$. (c) at $\ell=3$: `539c1`, type $\mathrm{I}_3^*$ at 7; its twist by $\chi_{-7}$ is `77c1` (type $\mathrm{I}_3$, conductor 77); $f_{539c}\equiv f_{11a}\otimes\chi_{-7}=f_{539d}\pmod 3$ (Sturm bound 112); $\deg\varphi_E=2^5\cdot3^2$, and $e_3(E,7)=v_3(48)=1$, $t_3(E,7)=1$ are both attained. (c) at $\ell=5$: `980i1`, type $\mathrm{I}_5^*$ at 7; its twist by $\chi_{-7}$ is `140b1`; $f_{980i}\equiv f_{20a}\otimes\chi_{-7}=f_{980g}\pmod 5$ (Sturm bound 336); $\deg\varphi_E=2^6\cdot3^2\cdot5$. Since $5\mid 980$, the last illustrates the congruence, not the theorem, as the paper says. The book verified each congruence to $n\le400$ or $n\le1000$, each twist identification, each degree, and that PARI's `mfsturm` returns 14, 22, 112, 336 (Section 21.18). The level-raising conditions are visible: $a_7(\texttt{11a1})=-2$ and $(7+1)^2-4=60\equiv0\pmod3$; $a_7(\texttt{20a1})=2$ and $60\equiv0\pmod5$.

Sturm's bound is what makes the congruences proofs rather than observations: both forms lie in the same space $S_2(\Gamma_0(N))$ with integer coefficients. The class `980g`, the twist of `20a`, must not be confused with `980i`.

**The Remark after the Example.** It says that $\ell\nmid N$ is used in the optimal-level theorem, in Lemma lem:old at $p=\ell$, and to pass from $r_E$ to $\deg\varphi_E$, the last needing only $\ell^2\nmid N$, and gives `99a1` ($N=9\cdot11$, $\deg\varphi_E=4$, $r_E=12$; [3, Table 1]), `726a1` and `539d1` (above). All three were recomputed in Chapters 13 and 14; at `99a1` and $\ell=3$, moreover, $H^1(X_0(99),\mathbb{Z}_3)_{\mathfrak{m}}$ is not free over $\mathbb{T}_{\mathfrak{m}}$ (Chapter 14). The last sentence, that the data show the conclusion "at $q=3$ in cases (a) and (b), where $q=\ell$ and the theorem does not apply", names the obstruction: at $q=\ell$ the optimal-level theorem and Lemma lem:old at $p=\ell$ are not available.

## 21.10 Section 4.1 of the paper: congruence ideals and the modular degree

This section reads the paper's Section 4.1: the objects of Diamond, Flach and Guo for weight 2, the comparison of their integral structure with the cohomology of $X_0(M^\Sigma)$ (Proposition

prop:integral), their formula for the variation of the congruence ideal with $\Sigma$ (Theorem thm:dfg14), the identification of the congruence ideal at level $N$ with the modular degree (Proposition prop:pullback), the commutative algebra lemma (Lemma lem:ddt417), Lemma lem:hecke and Remark rem:free. This is Step 1 of the proof of Theorem thm:main, and the place where no multiplicity-one hypothesis is needed. Section 3 used congruences one at a time, to show that $\ell$ divides $r_E$; from here on the paper measures the whole congruence module, as an ideal $\eta$ of $\mathbb{Z}_\ell$, because that is the quantity that Diamond, Flach and Guo relate to a Selmer group.

**Definition 21.27** (the objects of Diamond, Flach and Guo in weight 2; Section 4.1)**.** Let $f$ be a newform of weight 2 on $\Gamma_0(M)$ with rational coefficients, $\lambda = \ell \nmid 2M$, $\Sigma$ a finite set of primes, $M^\Sigma = M\prod_{p\in\Sigma} p^{\delta_p}$ and $f^\Sigma = \sum_{(n,\Sigma)=1} a_n(f)q^n$, an eigenform of the full Hecke algebra of level $M^\Sigma$ with $U_p f^\Sigma = 0$ for $p \in \Sigma$. The integral structure $M(M^\Sigma,1)_{!,\lambda}$ is the image of the $U_0(M^\Sigma)$-invariants of the compactly supported cohomology of a modular curve of full level in its cohomology [49, Sections 1.2, 1.4]. The kernel lattice $M_{f^\Sigma,\lambda}$ is the subgroup killed by the kernel of the eigencharacter of $f^\Sigma$ on the full Hecke algebra. The pairing $\hat\delta_!$ is the cup product composed with $[UwU]_\omega$, which in weight 2 with trivial character is the Atkin-Lehner involution $w_{M^\Sigma}$; on classes pulled back from $X_0(M^\Sigma)$ it is $\frac{u}{2}\langle x, w^*_{M^\Sigma} y\rangle$ for an $\ell$-adic unit $u$. The naive $\Sigma$-finite congruence ideal $\eta^\Sigma_{f,\lambda} \subseteq \mathbb{Z}_\ell$ is the ideal of values of $\hat\delta_!$ on $M_{f^\Sigma,\lambda}$, the image of $\wedge^2 M_{f^\Sigma,\lambda}$; equivalently the Gram determinant on a basis is $(\eta^\Sigma_{f,\lambda})^2$ up to a unit.

*Where proved.* Chapter 20, where the definitions are compared with the published text of [49] (Sections 1.5.3, 1.6.1, 1.6.2, 1.7.3, 1.8.2, displays (13), (14)); the cup product, $w_N$ and the Hecke adjoints in Chapters 8 and 14; a rank-two alternating form has Gram matrix $\left(\begin{smallmatrix} 0 & a \\ -a & 0\end{smallmatrix}\right)$, so its Gram determinant is the square of a generator of its ideal of values (Chapter 3).

**Proposition 21.28** (the paper's Proposition prop:integral)**.** *Let $\ell$ be a prime. Pull-back along the covering of $X_0(M^\Sigma)$ by the modular curve of full level identifies $H^1(X_0(M^\Sigma),\mathbb{Z})$ with the Betti lattice $M(M^\Sigma,1)_{!,B}$ of [49, Section 1.4.2], compatibly with the Hecke operators, $U_p$ included; in particular $M(M^\Sigma,1)_{!,\lambda} = H^1(X_0(M^\Sigma),\mathbb{Z}_\ell)$, with no localisation.*

*Where proved.* In the paper, in four steps; the book's expanded proof is the comparison proposition of Chapter 20, which also explains equivariant cohomology with compact support, the Hochschild-Serre spectral sequence and the equivariant Leray spectral sequence that the paper uses without comment. Riemann surfaces, their compactly supported cohomology and the cup product: Chapter 8; the Eichler-Shimura isomorphism and the Hecke action on $H^1(X_0(N),\mathbb{Z})$: Chapter 14; group cohomology of finite groups: Chapter 16.

*The paper's proof, read.* Write $M$ for $M^\Sigma$ and $U = U_0(M)$. In weight 2 with trivial character the coefficient sheaf of [49, Section 1.2.2] is the constant sheaf $\mathbb{Z}$. The paper chooses an auxiliary $N \geq 3$ with $M \mid N$ and every prime factor of $N$ in the set $S$ of [49, Section 1.4.1], for instance $N = 4M$. Then $U_N \subseteq U$, and by [49, Section 1.4.1] $M(M,1)_{?,B} = H^1_?(Y(N),\mathbb{Z})^G$ for $? \in \{c, \mathrm{tf}\}$, where $Y(N)$ is the analytic modular curve of full level $N$, a disjoint union of copies of $\Gamma(N)\backslash\mathcal{H}$ indexed by $(\mathbb{Z}/N\mathbb{Z})^\times$, and $G = U/U_N$, a finite group acting through its image in $\mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$; the lattice $M(M,1)_{!,B}$ is the image of the compactly supported invariants in the torsion-free invariants, and nothing depends on $N$ [49, Lemma 1.1]. Since $\det U = \widehat{\mathbb{Z}}^\times$, $G$ permutes the components transitively, the stabiliser of one component acts on it through the image of $\Gamma_0(M)$, and the quotient map $\pi\colon Y(N) \to Y(N)/G = Y_0(M)$ is a finite map onto the coarse curve; point stabilisers are finite.

(1) *Compact support and invariants.* $\pi^*\colon H^1_c(Y_0(M),\mathbb{Z}) \to H^1_c(Y(N),\mathbb{Z})^G$ is an isomorphism. The paper passes through the $G$-equivariant cohomology with compact support $H^\bullet_{G,c}(Y(N),\mathbb{Z})$ and two spectral sequences converging to it. In the Hochschild-Serre spectral sequence $H^p(G, H^q_c(Y(N),\mathbb{Z})) \Rightarrow H^{p+q}_{G,c}(Y(N),\mathbb{Z})$ the row $q = 0$ is zero, because each component is a connected non-compact surface, so $H^0_c(Y(N),\mathbb{Z}) = 0$; in degree 1 only $E_2^{0,1} = H^1_c(Y(N),\mathbb{Z})^G$ survives. In the equivariant Leray spectral sequence of $\pi$, $H^p_c(Y_0(M),\mathcal{R}^q) \Rightarrow H^{p+q}_{G,c}(Y(N),\mathbb{Z})$, the stalk of $\mathcal{R}^q$ at $y$ is $H^q(G_x,\mathbb{Z})$ for a point $x$ over $y$ (Shapiro's lemma); so $\mathcal{R}^0 = \mathbb{Z}$ and $\mathcal{R}^1 = 0$, because $H^1(C,\mathbb{Z}) = \mathrm{Hom}(C,\mathbb{Z}) = 0$ for a finite group $C$. In degree 1 only $H^1_c(Y_0(M),\mathbb{Z})$ survives. The composite of the two edge isomorphisms is $\pi^*$.

(2) *The image.* Under these identifications DFG's map from compactly supported to torsion-free invariants is the forgetful map $H^1_c(Y(N),\mathbb{Z})^G \to H^1(Y(N),\mathbb{Z})^G$; as $\pi$ is proper, its image is $\pi^*$ of the image of $H^1_c(Y_0(M),\mathbb{Z}) \to H^1(Y_0(M),\mathbb{Z})$. Since $Y_0(M)$ is the compact surface $X_0(M)$ minus finitely many cusps, $H^1_c(Y_0(M),\mathbb{Z}) \to H^1(X_0(M),\mathbb{Z})$ is onto and $H^1(X_0(M),\mathbb{Z}) \to H^1(Y_0(M),\mathbb{Z})$ is injective, so that image is $H^1(X_0(M),\mathbb{Z})$. Finally $\pi^*$ is injective on the torsion-free group $H^1(Y_0(M),\mathbb{Z})$, since $\pi_*\pi^*$ is multiplication by $\deg\pi$.

(3) *Hecke operators.* Both lattices are torsion-free and $\pi^*$ is injective, so equivariance may be checked after $\otimes\mathbb{C}$. There DFG's operators act on $\mathrm{Fil}^1 M(M,1)_{!,\mathrm{dR}} \otimes \mathbb{C} = S_2(\Gamma_0(M))$ as the classical $T_p$, and as $U_p$ for $p \mid M$ [49, Proposition 1.3], commuting with complex conjugation; $\pi^*$ is compatible with DFG's comparison isomorphism and with the Eichler-Shimura isomorphism, both being pull-back of $f(\tau)\,d\tau$.

(4) *The $\lambda$-adic structure.* $\mathbb{Z}_\ell \otimes M_{?,B} \simeq M_{?,\ell}$ for every prime $\ell$ [49, Section 1.2.3], and invariants under a finite group and images commute with the flat base change $\mathbb{Z} \to \mathbb{Z}_\ell$; so $M(M,1)_{!,\lambda} = \mathbb{Z}_\ell \otimes M(M,1)_{!,B} = H^1(X_0(M),\mathbb{Z}_\ell)$. □

*Reading guidance.* Step (1) is the one to master, and its point is compact support. Both vanishing statements are elementary: $H^0_c$ of a connected non-compact surface is 0 (a compactly supported locally constant function on it is 0), and a finite group has no nonzero homomorphism to $\mathbb{Z}$. Together they say that taking $G$-invariants of $H^1_c$ loses nothing and that the finite stabilisers (the elliptic points) contribute nothing in degree 1. Step (2) is the familiar description of $H^1(X_0(M),\mathbb{Z})$ as the interior cohomology of $Y_0(M)$ (Chapter 14). Steps (3) and (4) are bookkeeping: equivariance between torsion-free lattices can be tested over $\mathbb{C}$, and Betti-to-$\ell$-adic comparison is a base change. The statement holds for every prime $\ell$, $\ell = 2$ and $\ell \mid M^\Sigma$ included; the hypothesis $\ell \nmid 2M^\Sigma$ of the surrounding text is needed for the pairing, not for the comparison.

*Reading remark* (the paper's Remark after the proposition). "Compact support is what makes the comparison integral": with ordinary cohomology the analogue of step (1) fails, because $H^0(Y(N),\mathbb{Z}) \neq 0$. The paper identifies the cokernel: $H^1(Y(N),\mathbb{Z}_\ell)^G/\pi^*H^1(Y_0(M),\mathbb{Z}_\ell)$ is the group of characters $\Gamma_0(M) \to \mathbb{Q}_\ell/\mathbb{Z}_\ell$ of $\ell$-power order that factor through $G$, namely $\gamma \mapsto \psi(d \bmod p)$ for primes $p \mid M$ with $\ell \mid p-1$; "for $M = 11$ and $\ell = 5$ it is the unramified covering $X_1(11) \to X_0(11)$". These characters cut out abelian unramified subcoverings such as the Shimura covering, on which the Hecke operators act through an Eisenstein ideal [115, 161]; the proof does not pass through them, since the compactly supported lattice never sees this cokernel. Proposition prop:pullback (last sentence), Lemma lem:hecke, Step 1 of the proof of Theorem thm:main and Step 2 of Theorem thm:quant use the proposition.

**The pairing.** The operator in $\hat{\delta}_!$ is $[UwU]_\omega$, attached to $w = \left(\begin{smallmatrix} 0 & -1 \\ M^\Sigma & 0 \end{smallmatrix}\right)$ [49, Section 1.5.3, (13), (14)]; in weight 2 with trivial character it is the Atkin-Lehner involution with no scalar factor. On

classes pulled back from $X_0(M^\Sigma)$, $\hat\delta_!$ is $\frac{u}{2}$ times the cup product on $X_0(M^\Sigma)$ composed with $w_{M^\Sigma}$, for an $\ell$-adic unit $u$, "so for odd $\ell$ only a unit separates the two". The paper's justification: for a sufficiently small level $U' \subseteq U_0(M^\Sigma)$ with $\det U' = \widehat{\mathbb{Z}}^\times$, such as $U_0(M^\Sigma) \cap U_1(4)$, the normalised pairing $\bar\delta$ of [49, Section 1.5.2] is Poincaré duality on $X_{U'}$ up to an $\ell$-adic unit when $\ell \nmid 2M^\Sigma$; it is normalised by dividing by the index, so on $U_0(M^\Sigma)$-invariants it is $[U_0(M^\Sigma) : U']^{-1}$ times its value on $U'$-invariants; and the covering $X_{U'} \to X_0(M^\Sigma)$ has degree $[U_0(M^\Sigma) : U']/2$, because $-1$ acts trivially. By the projection formula the two factors combine to $\frac{1}{2}$, up to the unit $u$. Only the ideal of values of the pairing enters the congruence ideal, so the unit $u$ never matters. Self-adjointness of the Hecke operators, $U_p$ included, for $\hat\delta_!$ comes from $w_{M^\Sigma} T_n w_{M^\Sigma}^{-1}$ being the adjoint of $T_n$ for the cup product ([49, Section 1.8.2, (25)]; Chapter 14). This is the reason for composing with $w$: the plain cup product is alternating and perfect, but the $U_p$ are not self-adjoint for it.

**Theorem 21.29** (Diamond, Flach and Guo, Proposition 1.4(c); the paper's Theorem thm:dfg14)**.** *Let $\ell \nmid 2M$ and $\ell \notin \Sigma$. If $\bar\rho_f$ is irreducible, then $\eta^\Sigma_{f,\lambda} = \eta^\emptyset_{f,\lambda} \prod_{p\in\Sigma} L_p^{\mathrm{nv}}(A_f, 1)^{-1}$.*

*Where.* Stated with the reference [49, Proposition 1.4(c)] (verified), and sketched in Chapter 20 from parts (a) and (b), given the integral comparison isomorphism of [49], which is cited. We do not prove it here. The three factors (Chapter 20): an additive $p$ contributes 1 (neither level nor ideal changes); a good $p$ contributes $(1 - \frac{\alpha}{\beta}p^{-1})(1-p^{-1})(1-\frac{\beta}{\alpha}p^{-1}) = \frac{p-1}{p} \cdot \frac{(p+1)^2 - a_p^2}{p^2}$, Wiles's level-raising factor; a multiplicative $p$ contributes $1 - p^{-2}$.
*Reading guidance.* The hypothesis $\ell \notin \Sigma$ cannot be dropped, and the paper's paragraph after the theorem gives the example: for $f = f_{11a}$, $\ell = 3$, $\Sigma = \{3\}$, $a_3 = -1$ and $L_3^{\mathrm{nv}}(A_f, 1)^{-1} = \frac{2}{3} \cdot \frac{16-1}{9} = \frac{10}{9}$, of 3-adic valuation $-2$, so the right side would be $3^{-2}\mathbb{Z}_\ell$ while $\eta^\Sigma_{f,\lambda}$ is an ideal of $\mathbb{Z}_\ell$ (Exercise 21.53 gives a second instance). In [49, Section 1.7.3] $\ell \notin \Sigma$ is a standing assumption, and every use in the paper has it ($\Sigma$ is a set of primes dividing the level, and $\ell$ is prime to the level).

**Proposition 21.30** (the paper's Proposition prop:pullback)**.** *Let $E$ be $X_0(N)$-optimal with newform $f$, $\varphi = \varphi_E$, $I_f = \ker \phi_f$, and $\ell$ any prime. Then $\varphi^*\colon H^1(E, \mathbb{Z}_\ell) \to H^1(X_0(N), \mathbb{Z}_\ell)$ is injective with image the kernel lattice $H^1(X_0(N), \mathbb{Z}_\ell)[I_f]$, and $\langle \varphi^* a, w_N^* \varphi^* b\rangle_{X_0(N)} = \varepsilon_N(f) \deg\varphi_E \langle a, b\rangle_E$, with $\varepsilon_N(f) = \pm 1$ the eigenvalue of $w_N$ on $f$. For $\ell$ odd the ideal of values of $\frac{1}{2}\langle x, w_N^* y\rangle$ on $H^1(X_0(N), \mathbb{Z}_\ell)[I_f]$ is $\deg\varphi_E\, \mathbb{Z}_\ell$ and its Gram determinant is $(\deg\varphi_E)^2$ up to a unit. If $\ell \nmid 2N$, then $M_{f,\lambda} = \varphi^* H^1(E, \mathbb{Z}_\ell)$ and $v_\lambda(\eta^\emptyset_{f,\lambda}) = v_\ell(\deg\varphi_E)$.*

*Where proved.* Chapter 14 (the image and the projection formula), Chapter 8 (cup product, pullback and degree), Chapter 20 (the identification with $\eta^\emptyset_{f,\lambda}$, given the comparison of Proposition 21.28).

*Expanded proof.* (1) *Injectivity and saturation.* By optimality the kernel $B$ of $J_0(N) \to E$ is connected; the long exact sequence of $0 \to B \to J_0(N) \to E \to 0$ in homology ends with $H_1(J_0(N), \mathbb{Z}) \to H_1(E, \mathbb{Z}) \to \pi_0(B) = 0$, so $\varphi_*$ is onto on $H_1$, and its dual $\varphi^*\colon H^1(E, \mathbb{Z}) \to H^1(X_0(N), \mathbb{Z})$ is injective with torsion-free cokernel; tensor with $\mathbb{Z}_\ell$. (2) *The image is the kernel lattice.* $\varphi^*\omega_E = c_E \cdot 2\pi i\, f(z)\, dz$, so $\varphi^* H^1(E, \mathbb{C}) = \mathbb{C} f\, dz \oplus \mathbb{C}\overline{f\, dz}$. This plane is $H^1(X_0(N), \mathbb{C})[I_f]$, because $f$ is new: the $\phi_f$-eigenspace in $S_2(\Gamma_0(N))$ is $\mathbb{C}f$. The plane is defined over $\mathbb{Q}$: $H^1(X_0(N), \mathbb{Q})[I_f]$ is a two-dimensional $\mathbb{Q}$-form of it, and by flatness $H^1(X_0(N), \mathbb{Z}_\ell)[I_f] = H^1(X_0(N), \mathbb{Z})[I_f] \otimes \mathbb{Z}_\ell$ is a saturated lattice of rank 2 in $H^1(X_0(N), \mathbb{Q}_\ell)[I_f]$. The image of $\varphi^*$ is a saturated rank-two lattice in the same $\mathbb{Q}_\ell$-plane, so the two coincide.

(Two saturated lattices of the same rank in the same $\mathbb{Q}_\ell$-plane are equal; this is the point where optimality, through the torsion-free cokernel of step (1), enters.) (3) *Projection formula.* $\varphi^*(a \cup b) = \varphi^* a \cup \varphi^* b$ and $\varphi^*$ multiplies the fundamental class of $H^2$ by $\deg \varphi_E$, so $\langle \varphi^* a, \varphi^* b \rangle_{X_0(N)} = \deg \varphi_E \langle a, b \rangle_E$. (4) *The involution.* $w_N$ preserves the plane: by multiplicity one for newforms the plane is also the eigenspace of the $T_n$ with $(n, N) = 1$ for the eigenvalues $a_n(f)$, and $w_N$ commutes with those $T_n$; it does not commute with the $U_p$, $p \mid N$, which is why the argument passes through the $T_n$ with $(n, N) = 1$, as the paper does. On the plane $w_N$ acts by $\varepsilon_N(f) = \pm 1$, which gives the displayed formula. (5) *Values.* The cup product on $H^1(E, \mathbb{Z})$ is unimodular, so the ideal of values of $\frac{1}{2}\langle x, w_N^* y \rangle$ on the image is $\deg \varphi_E \mathbb{Z}_\ell$ ($\frac{1}{2}$ and $\varepsilon_N$ are units), and the Gram determinant is its square. (6) *DFG's ideal.* If $\ell \nmid 2N$, DFG's integral structure is $H^1(X_0(N), \mathbb{Z}_\ell)$, Hecke equivariantly with the $U_p$ included (Proposition 21.28), so $M_{f,\lambda} = H^1(X_0(N), \mathbb{Z}_\ell)[I_f] = \varphi^* H^1(E, \mathbb{Z}_\ell)$ by (2); and $\hat{\delta}_!$ is $\frac{u}{2}\langle x, w_N^* y \rangle$ with $u$ an $\ell$-adic unit (the paragraph on the pairing above). By (5), with the unit $u$ changing nothing, $\eta_{f,\lambda}^{\emptyset} = \deg \varphi_E \mathbb{Z}_\ell$. □

*Reading guidance.* The proposition is the heart of the identification of $\deg \varphi_E$ with a congruence ideal, and it uses no multiplicity-one hypothesis, and no irreducibility of $\bar{\rho}$: the paper notes that "for the first part no irreducibility" is used, and the last sentence needs none either, because Proposition 21.28 holds without localisation. Only the last sentence uses Proposition prop:integral. The consequence the paper draws after the proof, $M_{f,\lambda} = H^1(E, \mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell E, \mathbb{Z}_\ell) \simeq T_\ell E(-1)$ by Galois equivariance of $\varphi^*$ in étale cohomology, so that its adjoint is $\mathrm{ad}^0 T_\ell E$, is what lets the Selmer groups of [49] be read as Selmer groups of $\mathrm{ad}^0 T_\ell E$: $M_{f,\lambda}$ lives in the dual representation, of determinant $\chi_\ell^{-1}$, but the adjoint does not see duals or twists (Convention 21.3).

**Lemma 21.31** (the paper's Lemma lem:ddt417, the inequality half of [38, Lemma 4.17])**.** *Let $\mathcal{O}$ be a discrete valuation ring with finite residue field, $\mathbb{T}$ a commutative $\mathcal{O}$-algebra, finite and free as an $\mathcal{O}$-module, acting on an $\mathcal{O}$-module $L$ free of finite rank with a perfect $\mathcal{O}$-bilinear pairing for which $\mathbb{T}$ is self-adjoint; $\pi \colon \mathbb{T} \to \mathcal{O}$ a surjective homomorphism, $\wp = \ker \pi$, $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$, $\eta = \pi(I)$; and suppose $L[\wp]$ free of rank $d$ with Gram determinant $\Delta \neq 0$. Then $\eta^d \subseteq \Delta\mathcal{O}$, that is $d\, v(\eta) \geq v(\Delta)$, with equality if $L$ is free over $\mathbb{T}$.*

*Where proved.* Chapter 20 (in length form, over any discrete valuation ring); saturation, the dual map and Gram determinants: Chapter 3.

*Expanded proof.* If $\eta = 0$ there is nothing to prove. Otherwise take $t \in I$ with $\pi(t)$ a generator of $\eta$. (1) $\wp \cap I = 0$: for $x \in \wp \cap I$, $tx = 0$ (as $t$ kills $\wp$) and $(\pi(t) - t)x = 0$ (as $\pi(t) - t \in \wp$ and $x$ kills $\wp$), so $\pi(t)x = 0$ and $x = 0$, $\mathbb{T}$ being torsion-free. Reducedness of $\mathbb{T}$ is not the reason: the full Hecke algebra can have nilpotents ($U_2$ at level 88 on the span of the $f_{11a}(dz)$, $d \mid 8$; Chapter 13). (2) $I = \mathcal{O}t$: $\pi|_I$ is injective by (1), with image $\eta = \pi(t)\mathcal{O}$. (3) $L[\wp] \cap L[I] = 0$: an element killed by $\wp$ and by $I$ is killed by $\pi(t) = t + (\pi(t) - t) \in I + \wp$, a nonzero scalar, so it is 0, $L$ being torsion-free. This is the paper's argument; note that the torsion-freeness of $L$ alone would not suffice, the nonzero scalar $\pi(t)$ in $I + \wp$ is what is used. (4) $Q = L/(L[\wp] \oplus L[I])$ is killed by $\eta$: for $x \in L$, $tx \in L[\wp]$ (as $\wp I = 0$) and $(\pi(t) - t)x \in L[I]$ (for $s \in I$, $s(\pi(t) - t) \in I \cap \wp = 0$), and their sum is $\pi(t)x$. (5) $L[\wp]^\perp = L[I]$: if $y \in L[I]$, $x \in L[\wp]$, then $\pi(t)\langle x, y \rangle = \langle tx, y \rangle = \langle x, ty \rangle = 0$. Conversely, if $y \perp L[\wp]$, then for every $z$, $\langle ty, z \rangle = \langle y, tz \rangle = 0$ as $tz \in L[\wp]$; so $ty = 0$, and by (2) $Iy = 0$. (6) $L[\wp]$ is saturated, so the pairing maps $L$ onto $\mathrm{Hom}(L[\wp], \mathcal{O})$ with kernel $L[\wp]^\perp = L[I]$; hence $Q$ is the cokernel of $L[\wp] \to \mathrm{Hom}(L[\wp], \mathcal{O})$, $x \mapsto \langle x, \cdot \rangle$. In a basis of $L[\wp]$ and the dual basis this map has the Gram matrix as its matrix, and the cokernel of a square matrix over a discrete valuation ring with nonzero determinant has length the valuation of the

determinant (Smith normal form, Chapter 3): length $v(\Delta)$. $Q$ is generated by $d$ elements (a quotient of $\mathrm{Hom}(L[\wp],\mathcal{O}) \simeq \mathcal{O}^d$) and killed by $\eta$, so its length is at most $d\,v(\eta)$. (7) If $L \simeq \mathbb{T}^d$, then $Q \simeq (\mathbb{T}/(\wp+I))^d \simeq (\mathcal{O}/\eta)^d$, of length $d\,v(\eta)$. □

*Reading guidance.* Steps (1) and (6) are the ones to master. Step (1) shows $\wp \cap I = 0$ from torsion-freeness of $\mathbb{T}$, not from reducedness, which fails for the full Hecke algebra. Step (6) identifies the cokernel $Q$ with the discriminant module of the pairing on $L[\wp]$, whose length is $v(\Delta)$; the inequality is then the trivial bound on a module generated by $d$ elements and killed by $\eta$. The finite residue field is used only to count orders; in length form the lemma holds over any discrete valuation ring (Chapter 20).

**Lemma 21.32** (the paper's Lemma lem:hecke)**.** *Let $E$ be $X_0(N)$-optimal and $\ell$ odd with $\ell \nmid 2N$ and $\bar\rho$ irreducible. Then $v_\ell(\deg\varphi_E) = v_\ell(r_E) = v_\lambda(\eta^{\emptyset}_{f,\lambda})$.*

*Proof.* $v_\ell(\deg\varphi_E) = v_\ell(r_E)$ is [3, Theorem 2.2] ($\ell^2 \nmid N$), and $v_\ell(\deg\varphi_E) = v_\lambda(\eta^{\emptyset}_{f,\lambda})$ is Proposition 21.30. □

The paragraph after the lemma describes a second route, Lemma lem:ddt417 applied to $L = H^1(X_0(N),\mathbb{Z}_\ell)_{\mathfrak{m}}$, which without the projection formula gives only $v_\ell(r_E) \geq v_\lambda(\eta^{\emptyset}_{f,\lambda})$, with equality when $L$ is free over $\mathbb{T}_{\mathfrak{m}}$; it is the argument of Step 2 of Theorem thm:quant, for a depleted form not attached to an optimal quotient. Two facts make it work: the other local factors of $\mathbb{T}\otimes\mathbb{Z}_\ell$ lie in the kernel $\wp$, so the congruence ideal is computed in $\mathbb{T}_{\mathfrak{m}}$; and $e_{\mathfrak{m}}$ is self-adjoint, so the pairing is perfect on each local component. The paper's example: at `99a1` and $\ell = 3$ the Hecke ideal has valuation $v_3(12) = 1$ and the cohomological ideal $v_3(4) = 0$, so the inequality of Lemma lem:ddt417 is strict ($2 > 0$) and $H^1(X_0(99),\mathbb{Z}_3)_{\mathfrak{m}}$ is not free (Chapter 14).

*Remark* 21.33 (the paper's Remark rem:free). For $\ell \nmid 2N$ and $\bar\rho$ irreducible, $J_0(N)[\mathfrak{m}]$ is two-dimensional ([163, Theorem 3.5] for $J_1(N)$, as cited; the exceptional Serre weight $\ell$ does not occur in weight 2 with $\ell \nmid N$; the kernel of the pullback $J_0(N) \to J_1(N)$ is the Shimura subgroup, which is Eisenstein; the references [115, 161] for it are the book's); then $\mathbb{T}_{\mathfrak{m}}$ is Gorenstein and the Tate module is free of rank two [199]. The identification of $T_\ell J_0(N)$ with $H^1(X_0(N),\mathbb{Z}_\ell)$ uses Poincaré duality composed with $w_N$, not Galois equivariant, but freeness transfers. After Proposition 21.30 this is not used by Theorems thm:main or thm:quant; it gives a second proof of $v_\ell(r_E) = v_\ell(\deg\varphi_E)$ in this range and explains why the Hecke and the cohomological congruence ideals agree there (Chapter 14).

## 21.11 Sections 4.2 and 4.3 of the paper: the adjoint Selmer group and the proof of Theorem thm:main

This section reads the set-up of the theorem of Diamond, Flach and Guo, Proposition prop:Sf, the three imported results, Lemmas lem:dfg21, lem:fpr and lem:h0, the proof of Theorem thm:main, Corollary cor:explicit, Remark rem:degree and Remark rem:ko.

**Definition 21.34** (the Selmer set-up; Section 4.2)**.** Let $f = f_E$, $\Sigma$ the set of primes dividing $N$, $\lambda = \ell \nmid 2N$, $A = \mathrm{ad}^0 T$ (by Proposition 21.30, $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T$), $W$, $T^D \simeq A(1)$, $V^D$, $W^D$ as in Convention 21.3. $H^1_\Sigma(\mathbb{Q},W) \supseteq H^1_f(\mathbb{Q},W)$ has no condition at $p \in \Sigma$, the Bloch-Kato condition at $\ell$ and the unramified condition elsewhere [49, Section 2.1]. $S_f$ is the set of $\lambda$ with $\lambda \mid 2N$ or $\bar\rho|_{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ not absolutely irreducible [49, (31)] (verified; it is the weight-2 case of $\lambda \mid Nk!$).

**Proposition 21.35** (the paper's Proposition prop:Sf)**.** *Let $\ell$ be odd, $\ell \nmid N$, $E[\ell]$ irreducible. If $\ell \geq 5$ then $\ell \notin S_f$. If $\ell = 3$ then $3 \in S_f$ iff the image of $\bar{\rho}_{E,3}$ is conjugate to $N_s(\mathbb{F}_3)$, and then $E$ is ordinary at 3. In all cases $\ell \notin S_f$ iff $h_\ell(E) = 0$.*

*Where proved.* Chapter 11 (with the description of $\bar{\rho}|_{I_\ell}$ for good reduction, [171, Section 1]), Clifford theory in Chapter 4.

*Expanded proof.* Let $K = \mathbb{Q}(\sqrt{\ell^*})$. $\bar{\rho}$ is absolutely irreducible (complex conjugation has eigenvalues $1 \neq -1$). For an absolutely irreducible $\bar{\rho}$ and an index-two subgroup $G_K$, $\bar{\rho}|_{G_K}$ fails to be absolutely irreducible iff $\bar{\rho} \simeq \bar{\rho} \otimes \chi_{\ell^*}$ (Clifford theory, Chapter 4; this needs $\bar{\rho}$ irreducible). Suppose so and restrict to $I_\ell$, where $\chi_{\ell^*} = \omega^{(\ell-1)/2}$, $\omega$ the fundamental character of level one. Good reduction at $\ell$ gives: if $E$ is ordinary, $\bar{\rho}|_{I_\ell}^{\mathrm{ss}} = \omega \oplus 1$; then $\{\omega, 1\} = \{\omega^{(\ell+1)/2}, \omega^{(\ell-1)/2}\}$, and $\omega^{(\ell-1)/2} = -1 \neq 1$ forces $\omega^{(\ell+1)/2} = 1$, i.e. $(\ell-1) \mid (\ell+1)/2$, so $\ell = 3$. If $E$ is supersingular, $\bar{\rho}|_{I_\ell} \otimes \mathbb{F}_{\ell^2} = \omega_2 \oplus \omega_2^\ell$ with $\omega_2$ of order $\ell^2 - 1$ and $\omega_2^{\ell+1} = \omega$, so $\omega^{(\ell-1)/2} = \omega_2^{(\ell^2-1)/2} = -1$; then $\{\omega_2, \omega_2^\ell\} = \{-\omega_2, -\omega_2^\ell\}$ needs $\omega_2^\ell = -\omega_2$, i.e. $\omega_2^{\ell-1} = \omega_2^{(\ell^2-1)/2}$, i.e. $(\ell^2-1) \mid (\ell-1)^2/2$, i.e. $(\ell+1) \mid (\ell-1)/2$: impossible. So $\ell \geq 5$ never occurs, and at $\ell = 3$ the curve is ordinary; the image is then $N_s(\mathbb{F}_3)$ by Proposition 21.47. The last statement is Lemma 21.40. □

*Reading guidance.* The whole proof is a comparison of characters of inertia at $\ell$: twisting by $\chi_{\ell^*} = \omega^{(\ell-1)/2}$ must permute the characters of $\bar{\rho}|_{I_\ell}$, and the fundamental characters leave no room for this except in the ordinary case at $\ell = 3$. So for $\ell \nmid N$ the residual representations induced from $\mathbb{Q}(\sqrt{\ell^*})$ with $\ell \geq 5$ do not occur; Section 6.6 of the paper notes that such curves all have $\ell \mid N$.

**Theorem 21.36** (Diamond, Flach and Guo, Theorem 3.7; the paper's Theorem thm:dfg37)**.** *If $\ell \notin S_f$, then $H^1_\Sigma(\mathbb{Q}, W)$ is finite of length $v_\lambda(\eta^\Sigma_{f,\lambda})$.*

**Theorem 21.37** (Diamond, Flach and Guo, Theorem 2.7; the paper's Theorem thm:dfg27)**.** *If $\ell \notin S_f$, then $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$, and the same for $A \otimes \mathbb{Q}_\ell$.*

*Where.* Both are stated with references in Chapter 20 (Taylor-Wiles method in the axiomatic form of [49], sketched there); we do not prove them. Theorem 3.7 is stated in [49] for $\Sigma \not\ni \ell$ with $M_{f,\lambda}$ minimally ramified outside $\Sigma$, which holds for the primes dividing $N$; this is their Theorem 0.2. It covers $\ell = 3$ in weight 2: $S_f$ excludes only $\lambda \mid 2N$ and the induced case, and the proof treats $\ell = 3$ explicitly (both checked against the published text, Chapter 20). For Theorem 0.1 at $\ell = 3$ with $a_3 \not\equiv 0 \pmod 3$ Diamond, Flach and Guo also use Perrin-Riou's and Colmez's results [49, Proposition 2.16]; Theorems 2.7 and 3.7 do not need them. The imported theorems are the deep input of the paper, and the only place where the Taylor-Wiles method enters.

**Lemma 21.38** (Diamond, Flach and Guo, Lemma 2.1; the paper's Lemma lem:dfg21)**.** *Suppose $H^0(\mathbb{Q}, V^D) = H^1_f(\mathbb{Q}, V^D) = 0$, and let $\Sigma$ be a nonempty finite set of primes, $\ell \notin \Sigma$, containing every prime $p \neq \ell$ at which $W$ is ramified. With $H^1_f(\mathbb{Q}_p, T^D) = \iota^{-1} H^1_f(\mathbb{Q}_p, V^D)$ for $\iota\colon H^1(\mathbb{Q}_p, T^D) \to H^1(\mathbb{Q}_p, V^D)$, there is an exact sequence*

$$0 \to H^1_f(\mathbb{Q}, W) \to H^1_\Sigma(\mathbb{Q}, W) \to \bigoplus_{p \in \Sigma \cup \{\infty\}} H^1_f(\mathbb{Q}_p, T^D)^* \to H^0(\mathbb{Q}, W^D)^* \to 0.$$

*Where.* Stated with the reference [49, Lemma 2.1] (verified), and derived in Chapters 17 and 18 from Poitou-Tate duality (the comparison of Selmer groups for two sets of local conditions).

The paper's condition on $\Sigma$ implies the one of [49] ($\Sigma$ contains every prime where $H^1_f(\mathbb{Q}_p, W) \neq H^1_{\mathrm{ur}}(\mathbb{Q}_p, W)$), as the paper says.

*The paragraph after the lemma, expanded.* For $p \in \Sigma$, $p \neq \ell$: $H^1_f(\mathbb{Q}_p, V^D) = H^1_{\mathrm{ur}}(\mathbb{Q}_p, V^D) = (V^D)^{I_p}/(\mathrm{Frob}_p - 1)$, and $\mathrm{Frob}_p - 1$ is injective on $(V^D)^{I_p}$ because $\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = L_p(A_f(1), 0)^{-1} = L_p(A_f, 1)^{-1} \neq 0$ (geometric Frobenius; with the arithmetic one the determinant differs by a unit at the primes in question, and injectivity is the same statement for Frob and $\mathrm{Frob}^{-1}$); so the quotient is 0, and $H^0(\mathbb{Q}_p, V^D) = \ker(\mathrm{Frob}_p - 1) = 0$, which makes the hypothesis $H^0(\mathbb{Q}, V^D) = 0$ automatic. Hence $H^1_f(\mathbb{Q}_p, T^D)$ is the torsion of $H^1(\mathbb{Q}_p, T^D)$, a finite group (the word "torsion" matters: $H^1(\mathbb{Q}_p, T^D)$ itself can be infinite). At $p = \ell$ the computation fails ($\dim H^1_f(\mathbb{Q}_\ell, V^D) = 2$), which is why $\ell \notin \Sigma$ is part of the lemma, as the paper says; it is harmless, as $\ell \nmid N$. At $\infty$, $H^1(\mathbb{R}, T^D)$ is killed by 2, so $H^1_f(\mathbb{R}, T^D) = 0$ for $\ell$ odd.

**Lemma 21.39** (the paper's Lemma lem:fpr)**.** *For $p \in \Sigma$: $\mathrm{Fitt}_{\mathbb{Z}_\ell} H^1_f(\mathbb{Q}_p, T^D) = L_p(A_f, 1)^{-1} \mathrm{Tam}_p(T^D)$ and $\mathrm{Tam}_p(T^D) = \mathrm{Tam}_p(A)$; and $L_p(A_f, 1)^{-1} = L_p(A_f(1), 0)^{-1}$ is a nonzero $\ell$-integral rational number, of valuation $e_\ell(E, p)$ at an additive prime and $v_\ell(1 - p^{-2})$ at a multiplicative one.*

*Where proved.* The first identity is [69, I.4.2.2] as recalled in the proof of [49, Theorem 2.15]; it is proved in Chapter 18 (the $L$-part in Chapter 19). The second is [49, (57)] in general (verified), and for the elliptic adjoint it follows from the explicit computations: both sides are trivial at potentially good primes (Lemma 21.20 and the same argument for $A(1)$), and at a potentially multiplicative prime the count of Lemma 21.19, with the Frobenius action twisted by $\chi_\ell(\mathrm{Frob}) = q$, gives the same number (Chapter 18; Exercise 21.54).

*Expansion of the first identity.* Inflation-restriction with $G/I \simeq \widehat{\mathbb{Z}}$ gives $0 \to X^I/(\mathrm{Frob} - 1)X^I \to H^1(G, X) \to H^1(I, X)^{G/I} \to 0$ for $X = T^D$ (Chapter 16). $\mathrm{Frob} - 1$ is injective on $X^I$ (previous paragraph), and $X^I/(\mathrm{Frob} - 1)X^I$ is finite of length $v_\ell(\det(\mathrm{Frob} - 1 \mid X^I)) = v_\ell(L_p(A_f, 1)^{-1})$. Taking torsion: the torsion of $H^1(G, X)$ is an extension of $H^1(I, X)^{G/I}_{\mathrm{tor}} = (H^1(I, X)_{\mathrm{tor}})^{G/I}$ by the finite group $X^I/(\mathrm{Frob} - 1)X^I$; lengths add, and $\mathrm{Fitt} = \ell^{\mathrm{length}}$ for finite $\mathbb{Z}_\ell$-modules. Integrality: $P_p$ has integral coefficients and $p^{-2}$ is an $\ell$-adic unit.

**Lemma 21.40** (the paper's Lemma lem:h0)**.** *Let $\ell$ be odd with $\bar\rho$ irreducible. Then $H^0(\mathbb{Q}, W^D) = H^0(\mathbb{Q}, \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is finite cyclic of length $h_\ell(E)$, the largest $k$ with $\rho \bmod \ell^k \simeq (\rho \otimes \chi_\ell) \bmod \ell^k$. Moreover $h_\ell(E) = 0$ for $\ell \geq 5$, and $h_3(E) \leq 1$, with equality iff $\bar\rho_{E,3} \simeq \bar\rho_{E,3} \otimes \chi_{-3}$, iff $\bar\rho_{E,3}$ is not absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, iff the image of $\bar\rho_{E,3}$ is conjugate to $N_s(\mathbb{F}_3)$. This holds with or without complex multiplication.*

*Where proved.* Chapter 11; Schur's lemma, Clifford theory and Frobenius reciprocity in Chapter 4; $\chi_\ell(G_\mathbb{Q}) = \mathbb{Z}_\ell^\times$ in Chapter 6.

*Expanded proof.* (1) $\bar\rho$ is absolutely irreducible. (2) $H^0(\mathbb{Q}, W^D)[\ell^k] = H^0(\mathbb{Q}, \mathrm{ad}^0(T/\ell^k)(1))$, the trace-zero $G_\mathbb{Q}$-maps $T/\ell^k \to (T/\ell^k)(1)$; every $G_\mathbb{Q}$-map $\phi$ has trace 0, since $\mathrm{tr}\,\phi = \chi_\ell(g)\,\mathrm{tr}\,\phi$ for all $g$ (equivariance says $\phi\rho(g) = \chi_\ell(g)\rho(g)\phi$, so $\phi = \chi_\ell(g)\rho(g)\phi\rho(g)^{-1}$, and the trace is invariant under conjugation) and $1 - \chi_\ell(c) = 2$ is a unit for complex conjugation $c$. (3) For $k = 1$ these are the maps $\bar\rho \to \bar\rho \otimes \bar\chi_\ell$, at most a line by Schur. So the $\ell$-torsion of the submodule $H^0(\mathbb{Q}, W^D)$ of $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^3$ has rank at most one, and $H^0(\mathbb{Q}, W^D)$ is finite cyclic or isomorphic to $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. (4) An element of exact order $\ell^k$ is a map $\phi\colon T/\ell^k \to (T/\ell^k)(1)$ nonzero mod $\ell$, hence an isomorphism mod $\ell$ (Schur), hence an isomorphism (Nakayama); conversely. (5) Determinants: taking determinants in $\phi\rho(g)\phi^{-1} = \chi_\ell(g)\rho(g)$ modulo $\ell^k$ gives $\det\rho(g) = \chi_\ell(g)^2 \det\rho(g)$, and

$\det\rho(g)=\chi_\ell(g)$ is a unit, so $\chi_\ell(g)^2\equiv 1 \pmod{\ell^k}$ for all $g$. Since $\chi_\ell(G_\mathbb{Q})=\mathbb{Z}_\ell^\times \ni 2$ and $4\not\equiv 1$ modulo $\ell$ for $\ell\geq 5$ and modulo 9, $k=0$ for $\ell\geq 5$ and $k\leq 1$ for $\ell=3$; in particular the $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ alternative of (3) is excluded and the group is finite. (6) At $\ell=3$, $\bar\chi_3^{\mathrm{cyc}}=\chi_{-3}$. If $\phi\colon \bar\rho\to\bar\rho\otimes\chi_{-3}$ is an isomorphism, it commutes with $\bar\rho(G_K)$, $K=\mathbb{Q}(\sqrt{-3})$, and is not scalar, so $\bar\rho|_{G_K}$ is not absolutely irreducible; conversely a $G_K$-stable line over $\overline{\mathbb{F}}_3$ with character $\psi$ gives, by Frobenius reciprocity, a nonzero map $\operatorname{Ind}\psi\to\bar\rho\otimes\overline{\mathbb{F}}_3$, an isomorphism by absolute irreducibility, and $\operatorname{Ind}\psi\otimes\chi_{-3}\simeq\operatorname{Ind}\psi$; the isomorphism descends to $\mathbb{F}_3$, Hom spaces commuting with extension of scalars. (7) The last equivalence is Proposition 21.47. □

*Reading guidance.* Two steps carry the proof. Step (5), the determinant argument, is what makes $H^0(\mathbb{Q},W^D)$ finite and bounds $k$: before it, the module could a priori be $\mathbb{Q}_\ell/\mathbb{Z}_\ell$, which is why the paper says "finite cyclic or isomorphic to $\mathbb{Q}_\ell/\mathbb{Z}_\ell$" first and excludes the second case afterwards. Step (6) is Clifford theory for the index-two subgroup $G_{\mathbb{Q}(\sqrt{-3})}$. The last equivalence, with the image $N_s(\mathbb{F}_3)$, is the first statement of Proposition prop:ns, whose proof uses this lemma only for the value $h_3(E)=1$; there is no circularity. The clause "in particular $H^0(\mathbb{Q},W^D)=0$ if $\ell\notin S_f$" combines the lemma with Proposition 21.35.

**The proof of Theorem thm:main (Section 4.3).** The proof uses, besides the lemmas above, Theorems 21.29, 21.36 and 21.37 and Lemma 21.38, [3, Theorem 2.2] and the comparison of integral structures (Proposition 21.28). Each step is listed with the hypothesis it uses.

*Proof of Theorem 21.6, expanded.* Let $\Sigma=\{q\mid N\}$; then $\ell\notin\Sigma$. *Step 1* (optimality, $\ell\nmid 2N$, $\bar\rho$ irreducible): $v_\ell(\deg\varphi_E)=v_\lambda(\eta_{f,\lambda}^{\emptyset})$, Lemma 21.32. *Step 2* ($\bar\rho$ irreducible, $\ell\notin\Sigma$): $v_\lambda(\eta_{f,\lambda}^{\emptyset})=v_\lambda(\eta_{f,\lambda}^{\Sigma})-\sum_{p\in\Sigma}v_\ell(L_p^{\mathrm{nv}}(A_f,1)^{-1})$, Theorem 21.29. *Step 3* ($\ell\notin S_f$): $v_\lambda(\eta_{f,\lambda}^{\Sigma})=\operatorname{length}H^1_\Sigma(\mathbb{Q},W)$, Theorem 21.36. *Step 4* ($\ell\notin S_f$, $\ell$ odd): by Theorem 21.37 only $H^1_f(\mathbb{Q},V^D)=0$ is needed in Lemma 21.38, $H^0(\mathbb{Q},V^D)=0$ being automatic; $H^0(\mathbb{Q},W^D)=0$ by Lemma 21.40 (as $h_\ell(E)=0$ for $\ell\notin S_f$, Proposition 21.35); the term at $\infty$ is 0. Lengths add along the exact sequence (Pontryagin duality preserves length): $\operatorname{length}H^1_\Sigma(\mathbb{Q},W)=\operatorname{length}H^1_f(\mathbb{Q},W)+\sum_{p\in\Sigma}\operatorname{length}H^1_f(\mathbb{Q}_p,T^D)$. *Step 5* ($p\neq\ell$): $\operatorname{length}H^1_f(\mathbb{Q}_p,T^D)=v_\ell(L_p(A_f,1)^{-1})+v_\ell(\mathrm{Tam}_p(A))$, Lemma 21.39. *Step 6* (assembly): combining, $v_\ell(\deg\varphi_E)=\operatorname{length}H^1_f(\mathbb{Q},W)+\sum_p v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0T))+\sum_p v_\ell(L_p(A_f,1)^{-1}/L_p^{\mathrm{nv}}(A_f,1)^{-1})$, and the last sum is $\sum_{p\mid N}e_\ell(E,p)$ by Definition 21.4: the naive factors at multiplicative primes cancel against the true ones, and at additive primes the naive factor is 1. This is (eq:identity). *Step 7* (eq:thm): $\operatorname{length}H^1_f\geq 0$; $v_\ell(\mathrm{Tam}_p(\mathrm{ad}^0T))\geq t_\ell(E,p)$ at every potentially multiplicative $p$, multiplicative or additive, $p=2$ included (Lemma 21.19, step (9)); $t_\ell(E,p)=0$ at the other primes. □

*Reading guidance.* The bookkeeping in Step 6 is the point of the paper: the naive factors that Theorem thm:dfg14 removes and the true factors that Lemma lem:fpr puts back agree at the multiplicative primes and differ exactly at the additive ones, and the difference is $\sum e_\ell$. Step 7 is local at each prime, so it covers $p=2$, where $t_\ell(E,2)$ can be positive although no quadratic twist of conductor dividing $N$ need make the reduction multiplicative.

**Corollary 21.41** (the paper's Corollary cor:explicit)**.** *Under the hypothesis of Theorem 21.6,*

$$v_\ell(\deg\varphi_E) = \operatorname{length} H^1_f(\mathbb{Q}, \operatorname{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell) + \sum_{q\mid N} e_\ell(E,q) + \sum_{q:\, v_q(j_E)<0} \Big(t_\ell(E,q) + \min\big(t_\ell(E,q), v_\ell(q-1), v_\ell(c(u_q))\big)\Big).$$

*Proof.* (eq:identity) with $\operatorname{Tam}_q(\operatorname{ad}^0 T)$ trivial at good primes (unramified $\operatorname{ad}^0 T$: $H^1(I_q, \operatorname{ad}^0 T) = \operatorname{Hom}(I_q, \operatorname{ad}^0 T)$ is torsion-free) and at potentially good additive primes (Lemma 21.20), and given by Lemma 21.19 at the potentially multiplicative ones, where $v_\ell(n) = t_\ell(E,q)$. □

So the excess of $v_\ell(\deg\varphi_E)$ over an explicit local bound is exactly the length of the adjoint Selmer group. The paper's Table tab:excess gives its distribution on the optimal non-CM curves below conductor 10,000 with $\ell \nmid N$, $E[\ell]$ irreducible and, at $\ell = 3$, image not 3Ns, for $\ell = 3, 5, 7$; at $\ell = 3$ the excess is 0, 1, 2, 3, at least 4 for 8,688, 3,288, 942, 248, 37 of 13,203 curves, and it is never negative, as the corollary requires. The book reproduced the row $\ell = 3$ (Chapter 18). Examples, with the Tate data recomputed: `121a1` ($e_3(E,11) = 1$, $\deg\varphi_E = 6$): Selmer length 0; `791a1` (explicit bound $1+1$ at 7, $\deg\varphi_E = 144$): 0; `1184f1` (explicit bound 2, $\deg\varphi_E = 1728$): 1; `1748e1` (explicit bound 3, $\deg\varphi_E = 3240$): 1.

**Proposition 21.42** (the degree formula; the paper's Remark rem:degree)**.** *For $E$ optimal,* $\deg\varphi_E \cdot \operatorname{vol}(\Lambda_E) = 4\pi^2 c_E^2 \langle f, f\rangle$ *(unnormalised Petersson product) and* $\langle f, f\rangle = N\, L^{\mathrm{nv}}(\operatorname{Sym}^2 f, 2)/(8\pi^3)$ *give*

$$\deg\varphi_E = \frac{c_E^2\, N\, L^{\mathrm{nv}}(\operatorname{ad}^0 f, 1)}{2\pi \operatorname{vol}(\Lambda_E)} = B_E \prod_{q^2\mid N} L_q(A_f, 1)^{-1}, \qquad B_E = \frac{c_E^2\, N\, L(\operatorname{ad}^0 f, 1)}{2\pi \operatorname{vol}(\Lambda_E)} \in \mathbb{Q}^\times;$$

*so* $v_\ell(\deg\varphi_E) = v_\ell(B_E) + \sum_{q\mid N} e_\ell(E,q)$ *for every $\ell \nmid N$, and (eq:thm) is equivalent to* $v_\ell(B_E) \geq \sum_q t_\ell(E,q)$.

*Where proved.* The formula of Zagier and Watkins (with $c_E^2$ in the numerator) in Chapter 14; the Petersson norm at every level in Chapters 13 and 19; [49, Lemma 2.12] uses the naive value (verified).

*Proof.* Divide the first formula by $\operatorname{vol}(\Lambda_E)$ and substitute the second: $\deg\varphi_E = 4\pi^2 c_E^2 N\, L^{\mathrm{nv}}(\operatorname{Sym}^2 f, 2)/(8\pi^3 \operatorname{vol})$, and $L^{\mathrm{nv}}(\operatorname{Sym}^2 f, 2) = L^{\mathrm{nv}}(\operatorname{ad}^0 f, 1)$. The naive and true $L$-functions differ exactly at the additive primes ($q^2 \mid N$), where the naive factor is 1: $L^{\mathrm{nv}}(\operatorname{ad}^0 f, 1) = L(\operatorname{ad}^0 f, 1) \prod_{q^2\mid N} L_q(A_f, 1)^{-1}$. $B_E$ is rational because $\deg\varphi_E$ and the $L_q(A_f,1)^{-1}$ are; taking $v_\ell$ with $\ell \nmid N$ (so each $L_q(A_f,1)^{-1}$ has valuation $e_\ell(E,q)$, and $e_\ell = 0$ at multiplicative $q$) gives the rest. □

The paper checks the first formula to 30 digits on five curves (with $c_E = 1$). The formula isolates the role of the omitted factors in one line: (eq:thm) is a statement about the true adjoint $L$-value.

*Remark* 21.43 (the paper's Remark rem:ko). (1) At a potentially multiplicative $q \equiv 1 \pmod\ell$ with $\ell \mid n$ and $\bar{u}$ an $\ell$-th power, Lemma 21.19 gives $v_\ell(\operatorname{Tam}_q) \geq t_\ell + 1$, and (eq:identity) raises the bound by exactly $\min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$ when $\ell \notin S_f$ (Corollary 21.41). (2) The formula of Kim and Ota [99, Theorem 1.1, Definition 2.12] (as cited; their hypotheses as the paper lists them) gives $v_\ell(r_E) \geq \sum_{q\mid N^-} t_\ell(E,q)$, contained in (eq:thm) under its hypothesis; the equality is theirs. Whether their $\eta_f(N)$ is the Hecke or the cohomological congruence ideal does not matter at $\ell \nmid N$ with $E[\ell]$ irreducible, since the two agree there (Lemma 21.32, Chapter 20).

## 21.12 Section 4.4 of the paper: twist monotonicity and Theorem thm:quant

This section reads the paper's Section 4.4: Theorem thm:twist (comparison of congruence numbers along a twist), the data paragraph after it, the set-up and Lemma lem:step1 (the twist transport), the proof of Theorem thm:quant in four steps with its equality clause, and Remark rem:afterquant. Theorem thm:quant is the only result of the paper on the $\mathrm{I}_0^*$ and $\mathrm{I}_n^*$ terms that does not need the hypothesis over $\mathbb{Q}(\sqrt{\ell^*})$.

**Theorem 21.44** (the paper's Theorem thm:twist)**.** *Let $E/\mathbb{Q}$ have conductor $N$, $S$ a set of odd primes, $D=\prod_{q\in S}q$, $E_S=E\otimes\chi_S$ of conductor $N_S$. If $q^2\mid N$ for every $q\in S$ and $\operatorname{lcm}(N_S,D^2)\mid N$, then $r_{E_S}\mid r_E$; if moreover $\operatorname{lcm}(N,D^2)\mid N_S$, then $r_{E_S}=r_E$.*

*Where proved.* Chapter 20; the twist map, strong multiplicity one and the $\mathbb{T}$-stable complement in Chapter 13. The twist map is written $\iota_S$ in the book, since $\iota$ is also the map $H^1(\mathbb{Q}_p,T^D)\to H^1(\mathbb{Q}_p,V^D)$.

*Expanded proof.* Let $f=f_E$, $g=f_{E_S}$, $\iota_S(h)=h\otimes\chi_S=\sum\chi_S(n)a_n(h)q^n$. (1) $\iota_S$ maps $S_2(\Gamma_0(N_S))$ into $S_2(\Gamma_0(\operatorname{lcm}(N_S,D^2)))\subseteq S_2(\Gamma_0(N))$ (Shimura's level of a twist, Chapter 13), preserves integrality, and intertwines $T_p$ with $\chi_S(p)T_p$ for $p\nmid DN_S$. (2) $\iota_S(g)=f$: both sides are multiplicative in $n$; $a_p(f)=\chi_S(p)a_p(g)$ for $p\nmid D$, the twist being unramified there; and $a_{q^k}(f)=0$ for $q\in S$ as $q^2\mid N$. (3) Let $g^\perp$ be the $\mathbb{T}$-stable complement of the $g$-line at level $N_S$, the sum of the generalised eigenspaces of the other systems of eigenvalues (at level $N_S$, $g$ is new, and this is also the Petersson complement). A generalised eigenform $h$ of level dividing $N_S$ whose $T_p$-eigenvalues ($p\nmid N_S$) differ from those of $g$ at infinitely many $p$ is sent to one whose eigenvalues differ from those of $f$ at infinitely many $p$, hence (strong multiplicity one) into $f^\perp$. (4) A congruence $g\equiv h\pmod r$ with $h\in g^\perp$ integral gives $f\equiv\iota_S(h)\pmod r$ with $\iota_S(h)\in f^\perp$ integral, so $r\mid r_E$ (Lemma 21.22); with $r=r_{E_S}$ this is the first claim. (5) Under the second hypothesis $\iota_S$ also maps level $N$ to level $N_S$ and the roles exchange. □

The hypotheses hold when $E$ has type $\mathrm{I}_n^*$ at each $q\in S$ ($N_S=N/D$; case (a)), type $\mathrm{I}_0^*$ ($N_S=N/D^2$; case (b)), or a potentially good type $\mathrm{II},\mathrm{III},\mathrm{IV}$ or a dual at each $q\in S$, $q\geq 5$ ($N_S=N$, equality; case (c)). The step to watch is (3): the complement of the $g$-line must be the $\mathbb{T}$-stable one, the sum of the other generalised eigenspaces, and the twist map respects it because it moves systems of Hecke eigenvalues by $\chi_S$.

**The data paragraph after the theorem.** The paper reports 157,938 and 292,580 working-set pairs with $v_\ell(\deg\varphi_E)\geq v_\ell(\deg\varphi_{E_0^{(q)}})$, $E_0^{(q)}$ the optimal curve of the class of $E^{(q)}$, with no exception, equality in every pair of case (c), and a difference equal to the Euler term in cases (a) and (b), with the examples `539c1/77c1`, `980i1/140b1`, `121d1/11a1` ($v_3$: 2 against 1, 2 against 1, 1 against 0) and `539d1/11a1` (2 against 0, type $\mathrm{I}_0^*$, $e_3(E,7)=2$). The examples were recomputed for the book (Section 21.18); below 10,000 the difference equals the Euler term in all 25,477 pairs (curve, $\ell$) with $S\neq\emptyset$, $\ell\nmid N$ and $E[\ell]$ irreducible, as the paper says (Chapters 20 and 22). In case (a) at $\ell=3$ the difference is always positive, since $3\mid q^2-1$ for every $q\neq 3$.

**The set-up.** For any $E$ of conductor $N$ let $S$ be the set of odd primes at which $E$ has type $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$, $D=\prod_{q\in S}q$, $E_S=E\otimes\chi_S$ of conductor $N_S$, $g=f_{E_S}$ and $\iota_S$ the twist map (the book

writes $\iota_S$, since $\iota$ is also the map $H^1(\mathbb{Q}_p, T^D) \to H^1(\mathbb{Q}_p, V^D)$). By Lemma 21.15, valid at every odd $q$ for these exchanges, $E_S$ is good at the $\mathrm{I}_0^*$ primes of $S$ and multiplicative at the $\mathrm{I}_n^*$ primes; so $\delta_q(g) = 2$, resp. 1, and $N_S \prod_{q\in S} q^{\delta_q} = N$: the depleted form $g^S$ lives at level $N$, and $\iota_S(g^S) = f$, $\iota_S(f) = g^S$ ($\iota_S$ kills the coefficients at multiples of $D$). Let $\pi\colon \mathbb{T} \to \mathbb{Z}$ be the eigencharacter of $g^S$ on the full Hecke algebra of level $N$ and $\eta^{\mathrm{Hecke}}(g^S) = \pi(\mathrm{Ann}_{\mathbb{T}} \ker \pi)$. Its generalised eigenspace is the line of $g^S$: the forms with the $T_p$-eigenvalues of $g$ ($p \nmid N$) span the $g(dz)$, $d \mid N/N_S$, on which $U_q$, $q \in S$, has the eigenvalue 0 with multiplicity one. So the proof of Lemma 21.22, with $f^\perp$ replaced by the $\mathbb{T}$-stable complement $C$ of the $g^S$-line (not the Petersson complement, which is not $\mathbb{T}$-stable when the $U_p$ are involved; Chapter 13), shows that $\ell^{v_\ell(\eta^{\mathrm{Hecke}}(g^S))}$ is the largest $\ell^t$ with $g^S \equiv h \pmod{\ell^t}$ for some $h \in C \cap S_2(\Gamma_0(N); \mathbb{Z}_\ell)$.

**Lemma 21.45** (the paper's Lemma lem:step1)**.** *In this situation, $v_\ell(r_E) \le v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ for every prime $\ell$, and $v_\ell(r_E) = v_\ell(\eta^{\mathrm{Hecke}}(g^S))$ for every prime $\ell \notin S$. No other hypothesis on $\ell$ is needed: $\ell = 2$, $\ell \mid N$, $E[\ell]$ reducible and $\ell \in S_f$ are allowed, and $E$ need not be optimal.*

*Where proved.* Chapter 20; the twist map, strong multiplicity one and the $\mathbb{T}$-stable complement in Chapter 13.

*Expanded proof. The inequality $\le$.* By Lemma 21.22 there is an integral $h \in f^\perp$ with $f \equiv h \pmod{r_E}$. As $f$ is new, $f^\perp$ is the sum of the generalised eigenspaces of the $T_p$, $p \nmid N$, for systems differing from that of $f$ at infinitely many $p$. The map $\iota_S$ preserves integrality, maps $S_2(\Gamma_0(N))$ into itself (the level of a twist by $\chi_S$ divides $\mathrm{lcm}(N, D^2) = N$), and intertwines $T_p$ with $\chi_S(p)T_p$ for $p \nmid N$; so $\iota_S(h) \in C$ is integral, and $g^S = \iota_S(f) \equiv \iota_S(h) \pmod{r_E}$.

*The inequality $\ge$ for $\ell \notin S$.* Let $t = v_\ell(\eta^{\mathrm{Hecke}}(g^S))$, $\mathfrak{m}$ the maximal ideal of $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$ containing $\ker \pi$ and $\ell$; it contains $U_q$ for $q \in S$. Take $h \in C$ integral with $g^S \equiv h \pmod{\ell^t}$; the idempotent $e_{\mathfrak{m}}$ fixes $g^S$ and preserves $C$ and integrality, so $h' = e_{\mathfrak{m}} h$ does as well. The form $h'$ lies in the generalised eigenspaces of eigencharacters $\phi \ne \pi$ congruent to $\pi$, and such a $\phi$ has $\phi(U_q) \equiv 0$. Every eigenvalue of $U_q$ on $S_2(\Gamma_0(N))$ is 0, $\pm 1$ or a root of $X^2 - a_q(h_0)X + q$ for a newform $h_0$, hence 0 or a unit at every prime not above $q$; as $\ell \ne q$, $\phi(U_q) = 0$. This is the only place where $\ell \notin S$ is used. Each such $\phi$ comes from a newform $h_0 \ne g$ (if $h_0 = g$, $\phi$ would agree with $\pi$ everywhere), so $\iota_S$ maps its generalised eigenspace into that of the system $\chi_S(p)a_p(h_0)$, which differs from that of $f$ at infinitely many $p$; so $\iota_S(h') \in f^\perp$, and $f = \iota_S(g^S) \equiv \iota_S(h') \pmod{\ell^t}$ gives $t \le v_\ell(r_E)$. □

*Reading guidance.* The lemma is elementary: no Galois representation enters, only the Hecke action and strong multiplicity one. The step that needs $\ell \notin S$ is the vanishing $\phi(U_q) = 0$ from $\phi(U_q) \equiv 0$. The paper's sentence after the lemma says what happens at $\ell \in S$: at level 539 the congruence modules of $f_{539c}$, $f_{539d}$ and of the depleted forms $(f_{77c})^{\{7\}}$, $(f_{11a})^{\{7\}}$ have orders 288, 504, 288, 504, and their 7-parts, $7 \in S$, are not covered by the lemma (Chapter 20).

**Theorem 21.46** (the paper's Theorem thm:quant)**.** *Let $\ell$ be an odd prime with $\ell \nmid 2N$ and $E[\ell]$ irreducible, $S$ the set of odd primes at which $E$ has type $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$, $E_S = E \otimes \chi_S$ and $E'_S$ the optimal curve of its class.*

*(i)* $v_\ell(\deg \varphi_E) \ge v_\ell(r_{E_S}) + \sum_{q\in S} e_\ell(E, q) \ge \sum_{q\in S} e_\ell(E, q)$.

*(ii) If moreover $\ell \notin S_f$, the first inequality is an equality, and $v_\ell(r_{E_S}) = v_\ell(\deg \varphi_{E'_S})$:* $v_\ell(\deg \varphi_E) = v_\ell(\deg \varphi_{E'_S}) + \sum_{q\in S} e_\ell(E, q)$.

*Where proved.* Chapter 20, given Theorem 21.29 and Theorem 21.6. (The paper numbers the two statements as one theorem; the book splits them as (i) and (ii) for reference.)

*Expanded proof.* Use the set-up above; $\ell \notin S$ as $\ell \nmid N$, so $\ell \neq 3$ when $3 \in S$.

*Step 1 (the twist map; elementary).* By Lemma 21.45, $v_\ell(r_E) \geq v_\ell(\eta^{\mathrm{Hecke}}(g^S))$, and the congruence ideal is computed in $\mathbb{T}_{\mathfrak{m}}$: the other local factors of $\mathbb{T}_\ell$ lie in $\ker \pi$, and the annihilator decomposes along the idempotents.

*Step 2 (Hecke versus cohomological ideal).* Lemma 21.31 with $L = H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ (the pairing of Definition 21.27 is perfect on the $\mathfrak{m}$-component, $e_{\mathfrak{m}}$ being self-adjoint) and $\pi$: $L[\wp] = M_{g^S,\lambda}$ has rank 2 and Gram determinant $(\eta^S_{g,\lambda})^2$ (Proposition 21.28; it lies in the $\mathfrak{m}$-component, as the paper says). Hence $v_\ell(\eta^{\mathrm{Hecke}}(g^S)) \geq v_\lambda(\eta^S_{g,\lambda})$, with no freeness hypothesis.

*Step 3 (Diamond, Flach and Guo).* Theorem 21.29 for $g$ ($\bar\rho \otimes \chi_S$ irreducible, $\ell \nmid 2N_S$, $\ell \notin S$): $v_\lambda(\eta^S_{g,\lambda}) = v_\lambda(\eta^\emptyset_{g,\lambda}) + \sum_{q\in S} v_\ell(L_q^{\mathrm{nv}}(A_g,1)^{-1})$. Since $A_g = A_f$ and $\delta_q(g) \geq 1$, the naive factor of $g$ at $q$ is the true factor, the one that $f$ (with $a_q(f) = 0$) omits: the sum is $\sum_{q\in S} e_\ell(E,q)$ (values from Lemma 21.16, $q = 3$ included).

*Step 4 (level $N_S$).* Lemma 21.32 for $E'_S$ ($\ell \nmid 2N_S$, $\bar\rho_g$ irreducible): $v_\lambda(\eta^\emptyset_{g,\lambda}) = v_\ell(\deg \varphi_{E'_S}) = v_\ell(r_{E_S})$, with no freeness hypothesis. With $v_\ell(\deg\varphi_E) = v_\ell(r_E)$ this proves (i); the second inequality of (i) holds since $r_{E_S}$ is an integer.

*The equality (ii).* If $\ell \notin S_f$ then $\ell \notin S_{f_{E_S}}$ (twisting preserves absolute irreducibility over $\mathbb{Q}(\sqrt{\ell^*})$, and $N_S \mid N$). Apply Corollary 21.41 to $E$ and to $E'_S$: $\mathrm{ad}^0 T_\ell E \simeq \mathrm{ad}^0 T_\ell E'_S$ (twist and prime-to-$\ell$ isogenies do not change the adjoint), so the Selmer groups and the Tamagawa factors coincide; at $q \mid N$, $q \notin S$, the twist is unramified and the $e_\ell$ agree; at $q \in S$, $e_\ell(E_S, q) = 0$ ($E_S$ good or multiplicative). The difference is $\sum_{q\in S} e_\ell(E,q)$, and $v_\ell(\deg\varphi_{E'_S}) = v_\ell(r_{E_S})$ as $\ell^2 \nmid N_S$. □

*Reading guidance.* The inequality (i) uses, from the Selmer side, only Theorem thm:dfg14, Lemma lem:ddt417 and Lemma lem:hecke, never Theorem thm:dfg37; that is why it needs no hypothesis over $\mathbb{Q}(\sqrt{\ell^*})$. The equality (ii) squeezes the chain of Steps 1 to 4 between two applications of Theorem thm:main, so every inequality of the proof is then an equality, Step 2 included, with no multiplicity one at level $N$ or $N_S$; Step 1 is an equality at every $\ell \notin S$ in any case, by Lemma 21.45.

*A worked instance.* `539c1` has type $\mathrm{I}_3^*$ at 7 and $\mathrm{I}_2$ at 11, $\deg\varphi_E = 288 = 2^5 \cdot 3^2$, and surjective mod 3 image (Cremona's `galrep` lists only 2B), so $3 \nmid 2N$, $E[3]$ is irreducible and $3 \notin S_f$ (Proposition 21.35). Here $S = \{7\}$, $E_S$ is the twist by $\chi_{-7}$, which lies in the class `77c` (type $\mathrm{I}_3$ at 7), and $\deg\varphi_{\texttt{77c1}} = 6$. The table gives $e_3(E,7) = v_3(7^2-1) = v_3(48) = 1$. Part (ii) predicts $v_3(288) = v_3(6) + 1 = 2$, which is right. The inequality of (i) needs none of the Selmer theorems, only Theorem thm:dfg14: the depleted form $g^S$ of the newform of `77c1` lives at level 539, and depleting at the multiplicative prime 7 multiplies the congruence ideal by the naive factor $1 - 7^{-2}$, whose 3-part is 3.

**Remark rem:afterquant.** (1) Theorem thm:quant is the twist-and-lower argument of Section 3 read the other way: the level is raised from $N_S$ to $N$ at the price of the level-raising Euler factor; for $\ell = 3$ a single $\mathrm{I}_n^*$ prime (necessarily $q \geq 5$, since $3 \nmid N$) forces $3 \mid \deg\varphi_E$ because $3 \mid q^2 - 1$, and so does an $\mathrm{I}_0^*$ prime with $q \equiv 1 \pmod 3$ (the term $v_3(q-1)$); neither is in Theorem thm:qual, and both need $E[3]$ irreducible, the standing hypothesis of the theorem. (2) The equalities, as in the reading guidance above. (3) For the potentially good types the twist does not lower the level, $f$ is its own depletion, and Theorem thm:dfg14 gives no factor; these terms come only through

Theorem thm:dfg37, and Section 6.7 of the paper shows the hypothesis cannot be dropped there (the 3Ns curves fail through a type III term).

## 21.13 Section 5 of the paper: the boundary of the hypothesis

This section reads the proof of Proposition prop:ns, Proposition prop:ns-structure and the count after it, and Remark rem:ns. They describe the curves at which the hypothesis of Theorem thm:main fails at $\ell = 3$ while $E[3]$ is irreducible, and the correction term of Conjecture conj:euler.

**Proposition 21.47** (the paper's Proposition prop:ns)**.** *Let $E/\mathbb{Q}$ have $E[3]$ irreducible. Then $\bar\rho_{E,3}$ is absolutely irreducible on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ unless its image is conjugate to $N_s(\mathbb{F}_3)$; in that case $\bar\rho_{E,3}\otimes\mathbb{F}_9$ is induced from a character $\psi$ of order 4 of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$, $\bar\rho_{E,3}\simeq\bar\rho_{E,3}\otimes\chi_{-3}$, $H^0(\mathbb{Q},\mathrm{ad}^0\,\bar\rho_{E,3}(1))\neq 0$ and $h_3(E)=1$. The 3-division polynomial is then a product of two irreducible quadratics (times the leading coefficient 3), and otherwise irreducible.*

*Where proved.* Chapter 11; the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ and the commutator argument in Chapter 1; Clifford theory, Frobenius reciprocity and descent of isomorphisms in Chapter 4; the division polynomial in Chapter 9; the quartic and its lines in Chapter 5.

*Expanded proof.* (1) $\det\bar\rho_{E,3}=\chi_{-3}$ (Weil pairing), which cuts out $\mathbb{Q}(\sqrt{-3})$; so the image of $G_{\mathbb{Q}(\sqrt{-3})}$ is $H\cap\mathrm{SL}_2(\mathbb{F}_3)$, $H$ the image of $\bar\rho$. (2) $H$ has surjective determinant, acts irreducibly on $\mathbb{F}_3^2$, and contains $\bar\rho(c)$ of determinant $-1$ and trace 0. Inspection of the subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$ (Chapter 1) leaves $\mathrm{GL}_2(\mathbb{F}_3)$ (order 48), $N_{ns}(\mathbb{F}_3)$ (order 16) and $N_s(\mathbb{F}_3)$ (order 8); the nonsplit Cartan subgroup $\mathbb{F}_9^\times$ is excluded, its elements of determinant $-1$ having order 8 and nonzero trace. (Why $\bar\rho(c)$ has trace 0: $c^2=1$ and $\det\bar\rho(c)=-1$, so $\bar\rho(c)$ has eigenvalues 1 and $-1$. In $\mathbb{F}_9^\times$, acting on $\mathbb{F}_9\simeq\mathbb{F}_3^2$ by multiplication, the determinant is the norm $x\mapsto x^{1+3}=x^4$; $x^4=-1$ means $x$ has order 8; then $x^2$ has order 4, so $x^2\neq-1$, and the trace $x+x^3=x(1+x^2)$ is nonzero.) Complex multiplication plays no role; the paper cites [210] for the list and derives it directly. (3) The determinant-one subgroups are $\mathrm{SL}_2(\mathbb{F}_3)$, $Q_8$ and $C_4=\langle J\rangle$, $J=\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$. (4) The first two act absolutely irreducibly: a subgroup of $\mathrm{GL}_2(\overline{\mathbb{F}}_3)$ fixing a line lies in a Borel subgroup, whose commutator subgroup is unipotent, so abelian of exponent 3; the commutator subgroups $\{\pm1\}$ of $Q_8$ and $Q_8$ of $\mathrm{SL}_2(\mathbb{F}_3)$ are not. (5) $C_4$ is abelian; $J$ has characteristic polynomial $X^2+1$, irreducible over $\mathbb{F}_3$, so the restriction is irreducible over $\mathbb{F}_3$ and splits over $\mathbb{F}_9$ as $\psi\oplus\psi^{-1}$, $\psi(J)=i$ of order 4. (6) Frobenius reciprocity gives a nonzero, hence bijective, map $\mathrm{Ind}\,\psi\to\bar\rho\otimes\mathbb{F}_9$; an induced representation from an index-two subgroup is invariant under twisting by that subgroup's character, so $\bar\rho\otimes\mathbb{F}_9\simeq(\bar\rho\otimes\chi_{-3})\otimes\mathbb{F}_9$, and the isomorphism descends to $\mathbb{F}_3$ (Hom spaces commute with extension of scalars). Then $H^0(\mathbb{Q},\mathrm{ad}^0\,\bar\rho(1))\neq0$ and $h_3=1$ by Lemma 21.40. (7) The four roots of $\psi_3$ are the $x$-coordinates of the four lines of $E[3]$; $N_s(\mathbb{F}_3)$ has two orbits of size two on the lines (the two Cartan lines and the two others), $N_{ns}(\mathbb{F}_3)$ and $\mathrm{GL}_2(\mathbb{F}_3)$ are transitive; $\psi_3$ has no rational root ($E[3]$ irreducible); so $\psi_3$ factors as two irreducible quadratics in the first case and is irreducible otherwise. □

*Reading guidance.* Step (4) is the one to check: being non-abelian and acting faithfully does not exclude a fixed line (a Borel subgroup is both), and it is the commutator subgroup that does. Example (recomputed): `726a1`, $\psi_3=(x^2-x-74)(3x^2+8x+20)$. Below conductor 10,000 the function `ns3` of the appendix agrees with Cremona's label 3Ns on all non-CM optimal curves (Chapter 22). $N_s(\mathbb{F}_3)$ is not a maximal subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$ (it has index 2 in $N_{ns}(\mathbb{F}_3)$ in the

standard models; Chapter 1); the paper does not say otherwise, but a reader coming from large $\ell$ should not assume it.

**Proposition 21.48** (the paper's Proposition prop:ns-structure)**.** *Let $E/\mathbb{Q}$, with or without complex multiplication, have mod 3 image conjugate to $N_s(\mathbb{F}_3) = \{\pm 1, \pm J, \pm h, \pm w\}$, with $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $h = \mathrm{diag}(1,-1)$ and $w = hJ$; let $C_s = \{\pm 1, \pm h\}$ be the split Cartan subgroup, $C_4 = \langle J \rangle = N_s(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3)$, and $K_1$, $K_2$ the fixed fields of $\bar\rho^{-1}(C_s)$ and $\bar\rho^{-1}(\{\pm 1, \pm w\})$, the fields of definition of the roots of the two quadratic factors of $\psi_3$. Let $R$ be the set of primes $p \neq 3$ with $\bar\rho(I_p) = C_4$.*

*(i) For every $p \neq 3$, $\bar\rho(I_p) \subseteq C_4$. Consequently every $p \neq 3$ with $v_p(j) < 0$ has $3 \mid v_p(j)$, so $t_3(E,p) \geq 1$; no $p \neq 3$ has type* IV, IV$^*$*; no $p \geq 5$ has type* II, II$^*$*.*

*(ii) $R$ is the set of primes $p \neq 3$ ramified in $K_1$, and also the set of primes $p \neq 3$ ramified in $K_2$; it is nonempty. At $p \in R$ the reduction is potentially good, $\rho_{E,\ell}(I_p)$ is cyclic of order 4 for every $\ell \neq p$, and $(\mathrm{ad}^0 V_\ell E)^{I_p}$ is a line on which either Frobenius acts by $\chi_{-3}(p)$; $P_p(X) = 1 - \chi_{-3}(p)pX$ and $e_3(E,p) = v_3(p - \chi_{-3}(p)) \geq 1$; in particular $P_2(X) = 1 + 2X$ and $e_3(E,2) = 1$ if $2 \in R$.*

*(iii) If $p \in R$ and $p \geq 5$, then $E$ has type* III *or* III$^*$ *at $p$ and $p \equiv \pm 1 \pmod{12}$. Conversely every $p \geq 5$ of type* III *or* III$^*$ *is in $R$.*

*(iv) The splitting field of $\psi_3$ is biquadratic, with quadratic subfields $K_1$, $K_2$ and $\mathbb{Q}(\sqrt{-3})$.*

*(v) $a_p(E) \equiv 0 \pmod 3$ for every $p \nmid 3N$ with $p \equiv 2 \pmod 3$.*

*(vi) If $3 \nmid N$, $E$ is ordinary at 3.*

*(vii) $h_3(E) = 1$.*

*Where proved.* Chapter 11; Kraus's theorem [107] for the inertia images at $p \geq 5$.

*Expanded proof.* Write $\bar\rho = \bar\rho_{E,3}$. (i) For $p \neq 3$, $\chi_{-3} = \det \bar\rho$ is unramified at $p$, so $\bar\rho(I_p) \subseteq N_s(\mathbb{F}_3) \cap \mathrm{SL}_2(\mathbb{F}_3) = C_4$, which has no element of order 3 or 6. If $v_p(j) < 0$, on $I_p$ the representation is $\chi(\sigma)\left(\begin{smallmatrix} 1 & n\theta(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ with $n = -v_p(j)$, $\theta$ the 3-adic tame character, $\chi$ quadratic (Lemma 21.19); on $\ker\chi$, where $\theta$ takes a unit value, this is a unipotent of order 3 mod 3 unless $3 \mid n$. Type IV, IV$^*$ at $p \neq 3$ gives $\bar\rho(I_p)$ of order 3 (Lemma 21.13); type II, II$^*$ at $p \geq 5$ gives $\rho(I_p)$ cyclic of order 6 (Kraus), injecting mod 3. Neither fits in $C_4$.

(iv) is proved before (ii), as the paper signposts: "We prove (iv) next, because (ii) uses the fields $K_1$ and $K_2$ it describes." The splitting field of $\psi_3$ is cut out by the kernel of the action on lines, $\bar\rho^{-1}(\{\pm 1\})$, with group $N_s(\mathbb{F}_3)/\{\pm 1\} \simeq (\mathbb{Z}/2)^2$; its quadratic subfields are the fixed fields of the three index-two subgroups containing $-1$: $C_4$ (fixed field $\mathbb{Q}(\sqrt{-3})$, the determinant), $C_s$ (the stabiliser of the Cartan lines) and $\{\pm 1, \pm w\}$ (the stabiliser of the other two lines).

(ii) For $p \neq 3$: $\bar\rho(I_p) \subseteq C_4$ and $C_4 \cap C_s = C_4 \cap \{\pm 1, \pm w\} = \{\pm 1\}$, so $\bar\rho(I_p) \not\subseteq C_s$ iff $\bar\rho(I_p) = C_4$ iff $\bar\rho(I_p) \not\subseteq \{\pm 1, \pm w\}$; that is, $p$ is ramified in $K_1$ iff $p \in R$ iff $p$ is ramified in $K_2$. $K_1 \neq \mathbb{Q}(\sqrt{-3})$, because $C_s \neq C_4$; every quadratic field other than $\mathbb{Q}(\sqrt{-3})$ is ramified at some prime $p \neq 3$; so $R \neq \emptyset$. (This does not use $3 \nmid N$, as the paper notes.) For $p \in R$ the reduction is not potentially multiplicative ($\bar\rho(I_p)$ would consist of $\pm$ unipotents, while $C_4$ has elements of order 4), so $\rho(I_p)$ is finite, injects into $\mathrm{GL}_2(\mathbb{F}_3)$, and is cyclic of order 4, generated by $\tau$ with $\tau^2 = -1$; the same holds on $T_\ell E$ for every $\ell \neq p$ (Serre and Tate). The trace-zero commutant of $\tau$ is $\mathbb{Q}_\ell\tau$. Frobenius

normalises $\langle\tau\rangle$ and conjugates $\tau$ to $\tau^{\pm1}$; the sign can be read mod 3, where $\tau$ reduces to $J^{\pm1}$. The elements of $N_s(\mathbb{F}_3)$ outside $C_4$ (the coset $hC_4$, of determinant $-1$) conjugate $J$ to $J^{-1}$, those in $C_4$ commute with it; and $\det\bar\rho(\mathrm{Frob}_p)=\chi_{-3}(p)$. So Frobenius acts on the line by $\chi_{-3}(p)$; the Euler factor and $e_3$ follow as in Proposition 21.17(iii), $e_3\ge1$ because $p\equiv\chi_{-3}(p)\pmod3$. At $p=2$, $\chi_{-3}(2)=-1$.

(iii) For $p\ge5$ the image is tame and $\rho(\mathrm{Frob}_p)\tau\rho(\mathrm{Frob}_p)^{-1}=\tau^p$, which by (ii) equals $\tau^{\chi_{-3}(p)}$; so $p\equiv\chi_{-3}(p)\pmod4$, i.e. $\chi_{-4}(p)=\chi_{-3}(p)$, i.e. $p\equiv\pm1\pmod{12}$. A cyclic image of order 4 means semistability defect 4, $v_p(\Delta)\in\{3,9\}$, types III, III$^*$ (Kraus). Conversely, type III or III$^*$ at $p\ge5$ gives $\rho(I_p)$ cyclic of order 4, injecting mod 3 onto a subgroup of order 4 of $C_4$, so $p\in R$.

(v) For $p\equiv2\pmod3$, $p\nmid3N$, $\det\bar\rho(\mathrm{Frob}_p)=-1$, so $\bar\rho(\mathrm{Frob}_p)\in hC_4=\{\pm h,\pm w\}$, all of trace 0; and $a_p\equiv\operatorname{tr}\bar\rho(\mathrm{Frob}_p)$. (vi) is in Proposition 21.35. (vii) is Proposition 21.47. □

*Reading guidance.* Everything follows from two facts: the determinant $\chi_{-3}$ is unramified away from 3, which confines inertia to the cyclic group $C_4$; and the three index-two subgroups of $N_s(\mathbb{F}_3)$ containing $-1$ correspond to the three quadratic subfields of the splitting field of $\psi_3$. The prime $p=2$ is the delicate case of (iii): Kraus's theorem is not available there, and the paper leaves open whether the type at 2 is always III or III$^*$ when $R=\{2\}$.

**The count after the proposition.** Among the 90 optimal non-CM curves with image 3Ns below conductor 10,000 (44 of them with $3\mid N$), 74 have a prime $p\ge5$ in $R$, of type III or III$^*$ with $p\equiv1$ or $11\pmod{12}$, and 16 have $R=\{2\}$, of type III or III$^*$ at 2 with $P_2=1+2X$. The proposition also covers the 13 optimal CM curves with image $N_s(\mathbb{F}_3)$ in that range ($j=-32768$ or 8000: `121b1`, `256a1`, `256d1`, `1089e1`, `1936f1`, `2304c1`, `2304k1`, `3025a1`, `5929a1`, `6400f1`, `6400r1`, `7744a1`, `7744s1`), 7 with $R=\{11\}$ and 6 with $R=\{2\}$, of type III at 2. All these counts were recomputed for the book (PARI 2.17.2, Cremona's files). That the type at 2 is always III or III$^*$ when $R=\{2\}$ is not proved, as the paper says (its last open question).

**Proposition 21.49** (the paper's Remark rem:ns, as a conditional statement)**.** *Let $\ell=3$ and the image of $\bar\rho_{E,3}$ be $N_s(\mathbb{F}_3)$, $3\nmid N$, $\Sigma=\{q\mid N\}$. Suppose $H^1_f(\mathbb{Q},V^D)=0$ (equivalently $H^1_f(\mathbb{Q},\mathrm{ad}^0\,V_3E)=0$). Then*

$$\operatorname{length}H^1_\Sigma(\mathbb{Q},W)=\operatorname{length}H^1_f(\mathbb{Q},W)+\sum_{p\in\Sigma}\bigl(v_3(L_p(A_f,1)^{-1})+v_3(\mathrm{Tam}_p(A))\bigr)-h_3(E),\qquad h_3(E)=1.$$

*If moreover $v_\lambda(\eta^\Sigma_{f,\lambda})=\operatorname{length}H^1_\Sigma(\mathbb{Q},W)$, then (eq:identity) holds with $h_3(E)$ subtracted on the right, which is the shape of Conjecture conj:euler.*

*Proof.* Lemma 21.38 needs only $H^0(\mathbb{Q},V^D)=0$, automatic as above, and the assumed $H^1_f(\mathbb{Q},V^D)=0$ (it is a statement of Poitou-Tate duality, valid for any $\ell\notin\Sigma$). Lengths add along the exact sequence; the last term is $H^0(\mathbb{Q},W^D)^*$, of length $h_3(E)=1$ (Lemma 21.40); the local terms are given by Lemma 21.39. For the second statement, Steps 1 and 2 of the proof of Theorem thm:main need only $\bar\rho$ irreducible, and Step 3 is the added assumption. □

*Reading remark.* The remark is a conditional statement, and the paper says so: "The vanishing of $H^1_f(\mathbb{Q},V^D)$, equivalently of $H^1_f(\mathbb{Q},\mathrm{ad}^0\,V_3E)$, is not known for these curves: we assume it here". Neither assumption is proved: [49, Theorem 2.7] gives the vanishing only for $\ell\notin S_f$, and the 3Ns curves at $\ell=3$ are exactly in $S_f$; the equality of Step 3 would need a Taylor-Wiles argument for a residually dihedral representation induced from $\mathbb{Q}(\sqrt{-3})$ (Chapter 20).

## 21.14 Section 6 of the paper: the computations

This section reads the paper's Section 6, subsection by subsection. Its six data tables (Tables tab:uniform, tab:iso, tab:counts, tab:ns, tab:isolated and tab:oot) are read from their printed rows: for each, what the columns mean, which rows carry the paper's claims, and whether the rows agree with the text; table entries are copied exactly as printed. Table tab:excess was read with Corollary cor:explicit. Every count below is the paper's unless it is labelled as the book's, and the book's counts (conductor below 10,000) are checks made with the statements known, not blind tests. How to rerun everything is Chapter 22; here we say what each statement means and what it tests.

**6.1 Data and protocol.** The paper uses Cremona's tables [35] (the `ecdata` repository at commit `25cec5e`), which list the curves of conductor below 500000 with per-prime local data, modular degrees, isogeny classes, torsion and mod $\ell$ images in the notation of [191]; optimality is determined for every class up to 400000 and for 359,009 further classes. Four ranges: the working set ($N \le 300000$), the hold-out, the extension and the curves outside the tables. The text gives the order of discovery, from which the status of each statement can be read off:

| statement | status, as the paper describes it |
|---|---|
| (eq:main), Conjecture conj:eis | found on the working set, then tested once on the hold-out and on the extension: a blind test |
| (eq:twist) | replaced a weaker form on the working set, then tested once on the hold-out and the extension; the out-of-table curves are expressly not counted as a blind test of it |
| Conjecture conj:euler (Euler terms) | formulated on the working set; hold-out and extension are a confirmation only; the out-of-table curves, generated before the rule, are its blind test |
| the weights $w_3$ | fixed afterwards from PARI's factors and the minima at $q = 3$; the out-of-table curves are not counted as a blind test of them |
| the correction $h_\ell$ | identified last, from the exceptions; never blind-tested (no out-of-table curve has image 3Ns) |
| Lemmas lem:euler, lem:twist, Proposition prop:forms, $h_3 = 1$ | theorems; the data are checks of computations |

Two further points in the protocol paragraph deserve attention. On the extension range, optimality is determined only for the 359,009 named classes, and for the others the paper shows that the right side of (eq:rule) at $\ell = 3$ is the same for every curve of the class. "Indeed, when $E[3]$ is irreducible the class has no 3-isogeny, so every isogeny in it is a composite of isogenies of prime degree $p \neq 3$, and an isogeny of prime degree $p$ preserves the Kodaira type at a prime $q \neq p$ of potentially good reduction" [56, Theorem 5.4(1)]; so $w_3$, read off the type at $q = 3$ when the reduction there is potentially good, is constant on the class (Chapter 9). With a 3-isogeny, where (eq:rule) does not apply at $\ell = 3$, $w_3$ can vary; the paper's count below 10,000 ($w_3$ constant on all 1,295 classes without a 3-isogeny, "as it must be", varying in 828 of the 1,293 with one, for instance `54a`, of types IV*, II*, II at 3) was reproduced for the book. The other terms of (eq:rule) are invariant under isogenies of degree prime to $\ell$ (Chapter 9). And the PARI versions: the paper used PARI/GP 2.15 for the large ranges and 2.17.2 for the checks below 10,000, and it records that 2.17.2 returns $q_E^5$ in place of $q_E$ for `ellinit(E, O(2^k)).tate` when $v_2(j_E) = -1$; no count is affected, as the Tate parameter at 2 enters none (Chapters 10 and 22).

**6.2 The rule on whole ranges (Table tab:uniform).** The table has twenty rows, one for each of the four ranges (working set, hold-out, extension, out of table) and each $\ell \in \{3, 5, 7, 11, 13\}$. How to read the columns: "curves" is the population, the optimal curves of the range with $E[\ell]$ irreducible and not CM (so it depends on $\ell$); "(eq:rule) $> 0$" counts those with a positive bound; "viol." the violations of (eq:rule); "equality" the share of the curves with a positive bound at which $v_\ell(\deg \varphi_E)$ equals the bound; "viol. without $h$" the violations once the correction $h_\ell$ is dropped; and the last column gives the same three numbers (positive bounds, violations, equality) for the "III-free" rule, with the type III, III$^*$ terms left out and no correction. The first row reads: working set, $\ell = 3$, 1,233,729 curves, 867,578 with a positive bound, 0 violations, equality 57.5%, 364 violations without $h$, III-free 861,074 / 0 / 56.8%. Two columns carry the paper's claims. The column "viol." is 0 in all twenty rows. The column "viol. without $h$" is 0 at every $\ell \geq 5$ and reads 364, 68, 43 and 0 at $\ell = 3$ on the four ranges, the failing curves of Table tab:ns (6.7 below); the III-free column has 0 violations in every row. The equality column is the sharpness of Section sec:sharp: on the working set 57.5%, 75.6%, 84.1%, 91.6% and 92.4% at $\ell = 3, 5, 7, 11, 13$; at $\ell = 3$ it reads 57.5%, 54.6%, 54.0% and 52.5% on the four ranges, the last for the 828 out-of-table curves with a positive bound. The III-free rule bounds fewer curves (861,074 against 867,578 on the working set at $\ell = 3$) and is attained slightly less often: leaving out the type III terms costs a little sharpness and removes the need for a correction. These rows agree with the text, which quotes 867,578, the five equality rates and the 52.5% of 828. That the III-free rule needs no correction is explained by Proposition 21.48: every 3Ns curve has a prime of $R$, whose term $e_3 \geq 1$ is a type III term at every $p \geq 5$. Book's range ($\ell = 3$): in the base population of 33,851 optimal non-CM curves below 10,000 with $E[3]$ irreducible, of which 20,317 have a positive bound, there is no violation of (eq:rule) or of the III-free rule (Chapter 22).

**6.3 Calibration and sharpness.** The argument: among working-set curves with bound 0, 53.2% have $3 \nmid \deg \varphi_E$; if $v_3(\deg \varphi_E)$ were independent of the local data, the 867,578 curves with a positive bound would give about 605,000 violations; there are none. (A sanity check, the book's arithmetic: the share 53.2% alone, applied to the 867,578 curves, gives the lower bound $0.532 \times 867{,}578 \approx 461{,}551$.) The paper adds the right reading: "This shows that the statement is strong, far from vacuous, not that it is valid; the figure depends on the reference population." On the book's range the same calculation gives about 14,275 expected violations, with a stratified permutation null of about 7,600 (Chapter 22): the number changes by a factor of two with the reference population, the conclusion not at all. Under the hypothesis of Theorem thm:main the excess over the rule is the Selmer length plus the Tamagawa excess (Corollary 21.41), as the paper says. The odd-part statistic (1.85% on the working set, 17.1% below 10,000) is range dependent, as the paper notes; below 10,000 the smallest prime factor of the quotient is 3 in 58.7% of the cases, it divides $N$ in 47.5% of the cases with a nontrivial quotient, and some prime factor of the quotient divides $N$ in 51.8%. The book reproduced these three figures.

**6.4 The Tamagawa factors.** The test of the refined bound with the extra term $\min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$: at $\ell = 3$, 46,222 working-set curves, 12,095 with a positive extra term, no exception, equality rising from 45.0% to 60.6%; at $\ell = 5$, 10,940, 1,646, from 67.0% to 78.7%; including the $\mathrm{I}_n^*$ primes below 10,000: 1,050 curves at $\ell = 3$ (from 56.8% to 68.0%) and 252 at $\ell = 5$. The book reproduced the small-range numbers (Chapters 18 and 22). Reading: under the hypothesis, (eq:identity) is an equality, so a violation is impossible and the equality rate is the proportion of curves with $H^1_f(\mathbb{Q}, W) = 0$; the test checks the Tamagawa term

of the identity, as the paper says. At $q = 2$ the extra term vanishes ($v_\ell(2-1) = 0$), which is why the Tate parameter at 2 is never needed.

**6.5 Each term in isolation (Table tab:iso).** The table is printed for $\ell = 3$ on the working set. Rows are the Kodaira types II, II*, IV, IV*, III, III*, $\mathrm{I}_0^*$, $\mathrm{I}_n^*$ of the one additive prime $q \geq 5$; the column pairs are $e = e_3(E, q) = 1, 2, 3, 4$, and in each pair "min" is the minimum of $v_3(\deg \varphi_E)$ over the cell and "$n$" its number of curves (curves with $E[3]$ irreducible and no multiplicative 3-contribution). The claim is carried by the "min" columns: in every printed cell the minimum equals $e$, that is 1, 2, 3, 4 along each row (III and III* have no curve at $e = 4$, and the cell is printed as a dash). The "$n$" columns say how much each cell weighs: the largest are $\mathrm{I}_n^*$ at $e = 1$ (30,497 curves), II at $e = 1$ (8,327) and IV* at $e = 1$ (8,221); the cells at $e = 4$ are small (II and IV*: 2; II* and IV: 1; $\mathrm{I}_0^*$: 247; $\mathrm{I}_n^*$: 13). The rows come in pairs: II and IV* have 821, 59, 2 curves at $e = 2, 3, 4$ and 8,327, 8,221 at $e = 1$; II* and IV have 296, 12, 1 and 4,184, 4,290; III and III* have 1,271, 76, 9 in both. These are the pairs exchanged by the twist at $q$ in Lemma lem:twist(ii), which keeps the conductor exponent at $q \geq 5$ (a reading of the book; the paper does not comment on it). What the table does not print, the text supplies: the cells with $e = 0$ (III: 5,534 curves, $\mathrm{I}_0^*$: 5,394), the $\mathrm{I}_0^*$ cells up to $e = 6$, the values at $\ell = 5, 7$ ("The same holds at $\ell = 5$ and 7", in the caption) and, for $\mathrm{I}_n^*$, that the excess over $e_\ell$ has minimum $v_\ell(n)$; the printed cells agree with the text's "In every cell, for $\ell = 3, 5, 7$, the minimum equals $e_\ell$". The book found the same pattern cell by cell below 10,000 (Chapter 22). Multiplicative primes alone show no Euler term, for every value of $v_\ell(q^2 - 1)$ up to 6; at $\ell = 3$ the value 0 occurs only for $q = \ell = 3$, as the paper notes, since $3 \mid q^2 - 1$ for every other $q$. This is the data side of the paper's thesis: the naive factor at a multiplicative prime is the true one, so nothing is omitted. The two-prime observation ($9 \mid \deg \varphi_E$ for all 2,502 curves with two odd primes of type II or II*; it is the entry 2 (2,502) of Table tab:isolated, 6.8 below) is consistent with the additivity of the $e_3$ terms.

**6.6 The two corollary forms (Table tab:counts).** The table has eighteen rows: $\ell = 3, 5, 7, 11, 13$ on the working set ($N \leq 300000$), the hold-out ($300000 < N \leq 400000$) and the extension ($400000 < N \leq 500000$), and $\ell = 3, 5, 7$ out of table ($500000 < N \leq 4 \cdot 10^6$). How to read the columns: "curves" is the size of the range, the same for every $\ell$ (1,311,066, 429,936, 359,009, 893; it is not the population of Table tab:uniform, which keeps only the curves with $E[\ell]$ irreducible and not CM); "(eq:main) $> 0$" and "(eq:twist) $> 0$" count the curves with $E[\ell]$ irreducible, not CM, with a positive right-hand side; the next two columns are the violations; the last column gives the Eisenstein curves with a positive bound in (eq:main) and the violations of Conjecture conj:eis. The claim is carried by the three violation columns, which are 0 in all eighteen rows. The first row: 1,311,066 curves, 598,833 with (eq:main) $> 0$, 705,563 with (eq:twist) $> 0$, 0, 0, and 67,845 / 0. The twisted form bounds more curves than (eq:main) in every row (out of table at $\ell = 7$: 1 against 0), and the rule bounds more still (867,578 in Table tab:uniform). The Eisenstein column on the working set reads 67,845, 4,454, 759, 0, 26 at $\ell = 3, \ldots, 13$, and 0 / 0 out of table. The equality rates of the two forms (39.4%, 71.1%, 81.2%, 91.5% for (eq:twist) at $\ell = 3, 5, 7, 13$, and 36.9%, 69.7%, 80.4%, 91.2% for (eq:main)) are in the text, not in the table. The rows agree with the text: "There are none" for the violations, "867,578 curves instead of 705,563 at $\ell = 3$", and the 67,845 Eisenstein curves of Section sec:eis. With Proposition 21.9 the absence of violations under the hypothesis of Theorem thm:main is a theorem; the data test the forms outside it, chiefly at the curves with $\ell \mid N$. The paragraph lists the failures of the hypothesis on the working set: the 1,170 non-CM curves with image 3Ns at

$\ell = 3$ and the curves with $\ell \mid N$; by Proposition prop:Sf there are no others among the non-CM curves with $E[\ell]$ irreducible. Its parenthesis on the CM curves with $j \in \{-32768, 8000\}$, which also have image $N_s(\mathbb{F}_3)$ and fail at $\ell = 3$ when $3 \nmid N$ (20 below 10,000) and which the `galrep` file leaves unlabelled, matches Chapter 11.

**6.7 The boundary: the** 3Ns **curves (Table tab:ns).** The table has one row per range. How to read the columns: "3Ns" is the number of optimal non-CM curves with mod 3 image 3Ns and $E[3]$ irreducible; "with III term" those with a positive type III term; "failing" those violating (eq:thm) with all types and no correction, with their share; "others failing" the curves without image 3Ns that violate it; "$h_3 = 1$" the 3Ns curves with $h_3(E) = 1$; the next column the residues modulo 12 of the type III primes of the failing curves, or $q = 2$; and "shortfall 1" the failing curves that miss the bound by exactly one. The rows:

| set | 3Ns | III term | failing | others | $h_3 = 1$ | $q \equiv 1$ / 11 (12) / $q = 2$ | shortfall 1 |
|---|---|---|---|---|---|---|---|
| working set | 1,170 | 1,170 | 364 (31%) | 0 | 1,170 of 1,170 | 140 / 160 / 72 | 364 of 364 |
| hold-out | 290 | 290 | 68 (23%) | 0 | 290 of 290 | 34 / 30 / 4 | 68 of 68 |
| extension | 237 | 237 | 43 (18%) | 0 | 237 of 237 | 16 / 25 / 2 | 43 of 43 |
| out of table | 0 | 0 | 0 (0%) | 0 | 0 of 0 | 0 / 0 / 0 | 0 of 0 |

Every column carries a claim: "III term" equals "3Ns" (every 3Ns curve has a positive type III term); "others" is 0 (only 3Ns curves fail, which is what "viol. without $h$" of Table tab:uniform counts); "$h_3 = 1$" is the whole population (Lemma lem:h0, a theorem); and the shortfall is always 1, so the correction $h_3 = 1$ is exactly enough. The residue column counts primes, not curves: on the working set $140 + 160 + 72 = 372$ exceeds the 364 failing curves, so some failing curves have two such primes. The table lists only the classes 1 and 11 modulo 12 at $q \geq 5$, and Proposition prop:ns-structure shows that no other class can occur ($p \equiv \pm 1 \pmod{12}$). The rows agree with the text. The paper: 364, 68, 43 and 0 violations of (eq:thm), with all types and no correction, on the four ranges, all at $\ell = 3$ and all at 3Ns curves; every 3Ns curve has a type III prime with positive $e_3$; the inequality fails for 31%, 23%, 18% of them, always by one; each violation involves a prime $p \equiv -1 \pmod 3$ with $t_3(E, p) \geq 1$, which the paper marks as "an observation, not proved". Proposition prop:ns-structure explains part of it (every potentially multiplicative $p \neq 3$ of an 3Ns curve has $t_3 \geq 1$). For the curves with $3 \mid N$ the count depends on the term $t_3(E, 3)$: below 10,000, 36 violations with it, as (eq:thm) is written, and 24 without, the twelve curves that differ (`726a1`, `726f1`, `2178b1`, `2178h1`, `3840h1`, `3840k1`, `3840o1`, `3840r1`, `5070h1`, `5070p1`, `5808y1`, `5808z1`) all having $t_3(E, 3) = 1$ from a type $\mathrm{I}_n$ or $\mathrm{I}_n^*$ at 3; the weights $w_3$ change nothing. The book reproduced these counts. Example: `726a1`, types $\mathrm{I}_6$ at 2, $\mathrm{I}_3$ at 3, III at 11, $\deg \varphi_E = 144$: the bound of (eq:thm) as written is $1 + 1 + 1 = 3 > 2$, by one, through $p = 2 \equiv -1 \pmod 3$ with $t_3(E, 2) = 1$; without $t_3(E, 3)$ it is 2 (Exercise 21.58 treats `3840h1`). The value $h_3(E) = 1$ for every 3Ns curve is Lemma 21.40, a theorem.

**6.8 The wild primes (Table tab:isolated).** The table isolates the additive types at $\ell = 3$ on the working set. Each entry is the minimum of $v_3(\deg \varphi_E)$ with the number of curves in parentheses. The header reads "weight 1, 2, 3" for every row, but the caption gives it two meanings: in the five untwisted rows (IV, IV*, III, III*, $\mathrm{I}_0^*$, at any $q$) it is the number of primes of that type, all additive primes having that type; in the four twisted rows it is the total weight of (eq:twist) from the primes named. The rows:

- untwisted: IV: 1 (25,414), 2 (969), 3 (11); IV*: 1 (30,168), 2 (2,268), 3 (45); III: 0 (34,904), 0 (2,208), 0 (34); III*: 0 (25,574), 0 (1,307), 0 (29); $\mathrm{I}_0^*$: 0 (42,924), 0 (3,093), 0 (53);

- twisted, odd $q$: II, II*: 1 (56,509), 2 (2,502), 3 (12); $\mathrm{I}_n^*$ with weight $v_3(n)$: 1 (28,175), 2 (3,453), 3 (89);
- twisted, $q = 2$: II, II*: 0 (44,672), with no curve of weight 2 or 3 (a dash in the table); $\mathrm{I}_n^*$ with weight $v_3(n)$: 0 (21,176), 1 (5,329), 1 (342).

Reading guidance. Where the minimum equals the column number (IV, IV* and both odd-$q$ twisted rows), each prime, or each unit of weight, forces one factor 3 in $\deg \varphi_E$, as the terms of (eq:twist) say. The rows with minimum 0 (III, III*, $\mathrm{I}_0^*$) carry no forced factor when the curves are grouped by the number of primes; their term is the Euler term $e_3$, which can vanish, and Table tab:iso shows the minimum $e$ once the curves are grouped by $e$. The paper's claims about the prime 2 are in the last two rows: types II, II* at 2 force nothing, and a type $\mathrm{I}_n^*$ at 2 forces nothing when $3 \parallel n$ and only one factor 3 when $9 \mid n$ or $27 \mid n$, where the odd-$q$ weights would give 2 and 3. The rows agree with the text: "$v_3(\deg \varphi_E) = 0$ for 14,182 of the 44,672 curves whose only candidate is such a prime" (the 14,182 is in the text only), "although with $9 \mid n$ the minimum is 1", and the 2,502 curves of Section sec:iso. Its explanation is Definition def:terms with PARI's factor, the factor having one of the four shapes of Proposition 21.17: a type II or II* at 2 *usually* has $P_2 = 1$ (among the optimal non-CM curves below 10,000, 1,092 of the 1,537 with type II at 2 and 514 of the 779 with type II*; Chapter 22); a type IV or IV* at 2 has $e_3 = 1$ by Lemma lem:IV; and at $q = 2$ the right invariant of a type $\mathrm{I}_n^*$ is $-v_2(j)$, not the index $n$. The weights $w_3$ at $q = 3$ are the observed minima (1 for $\mathrm{II}, \mathrm{IV}, \mathrm{III}^*$; 2 for $\mathrm{II}^*, \mathrm{IV}^*$; 0 for $\mathrm{III}, \mathrm{I}_0^*$), reproduced exactly on the book's range (Chapter 22). For $\ell = 5, 7$ no term is needed at $q = \ell$ or at $q = 2, 3$ beyond Definition def:terms, and the term at 2 can be positive (`592c1`); the paper adds that this is "true but not sharp" at $q = \ell$, by Remark rem:oq1.

**Remark rem:oq1 (a twist relation at $q = \ell$; data only).** For optimal non-CM curves below 10,000 with $\ell \mid N$, $\ell \in \{3, 5, 7, 11, 13\}$, the paper compares $v_\ell(\deg \varphi_E)$ with $v_\ell$ of the degree of the optimal curve of the class of the twist $E^{(\ell)}$ by $\chi_{\ell^*}$: (a) for type II*, III* or IV* at $\ell$ the difference is exactly 1 in all 4,412 cases (4,251 with $E[\ell]$ irreducible); (b) for type $\mathrm{I}_n^*$ at $\ell$ it is 0; (c) for type $\mathrm{I}_0^*$ at $\ell$ with $E[\ell]$ irreducible it is $v_\ell((\ell+1)^2 - a_\ell(E^{(\ell)})^2)$ in all 1,915 cases, failing in 13 cases with $E[\ell]$ reducible (`126a1` at $\ell = 3$, `275b1` at $\ell = 5$). How to read it: at $\ell = 3$, (a) is the pattern of the weights $w_3$, each starred type weighing one more than its partner under the exchanges of Lemma lem:twist(ii); in (b) and (c) the difference is the numerator of the table of Section 1.1 evaluated at $q = \ell$. Nothing here is proved, the observations were made on the whole range with no hold-out, and the paper offers no explanation; the book reproduced the counts (Chapter 22).

**6.9 Out of table (Table tab:oot).** 893 curves with prescribed local types and $500000 < N \leq 4 \cdot 10^6$, kept when the isogeny class is trivial (then the curve is optimal and $E[\ell]$ is irreducible for every $\ell$, since a rational $\ell$-subgroup would give an isogeny; Chapter 9) and not CM. Conjecture conj:euler and both forms hold for all of them, with equality in (eq:rule) for 52.5% of the 828 with a positive bound at $\ell = 3$. The table itself gives, per family of prescribed types, the number of curves, the largest conductor and the median time in seconds to compute one modular degree with `ellmoddegree`: II, II: 98, 3,968,800, 0.4; II, II, II: 1, 2,832,200, 0.2; IV: 334, 3,988,012, 1.1; IV*: 104, 3,994,600, 4.9; "IVmixed" (the text's "mixed types"): 266, 3,983,504, 1.3; random: 90, 2,710,716, 32.1; all: 893, 3,994,600, 1.5. Reading guidance: the table describes the sample, not the test; the results on these curves are the last rows of Tables tab:uniform and tab:counts

("out of table"), and the last row of Table tab:ns records that none of them has image 3Ns. The families add up to 893, the largest conductor is below $4 \cdot 10^6$, and the cost column shows why the sample is small: a random curve costs about twenty times the median. These rows agree with the text. These are the blind test of the Euler terms of the rule, and only of them, as the paper says.

**6.10 The Eisenstein case.** At $\ell = 3$ on the working set, 75,430 optimal non-CM curves with a rational 3-isogeny, 2,777 with $D > 0$, $D = 2$ for 22: seventeen with $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ (image 3Cs.1.1, the subgroup of order 2 of the split Cartan subgroup that fixes a point) and five with a point of order 9 (image 3B.1.1). The seventeen have torsion $\mathbb{Z}/3$ or $\mathbb{Z}/6$ (for instance `14a1`, torsion $\mathbb{Z}/6$, types $\mathrm{I}_6$ at 2 and $\mathrm{I}_3$ at 7, $\deg \varphi_E = 1$, so $D = v_3(6) + v_3(3) - 0 = 2$); with a rational point of order 3, Borel image and no point of order 9, $D \leq 1$ (`714i1`, torsion $\mathbb{Z}/9$, $\deg \varphi_E = 1080 = 2^3 \cdot 3^3 \cdot 5$, has $D = 2$). The book recomputed `14a1`, `714i1`, `20a1` ($D = 1 = v_3(\#E(\mathbb{Q})_{\mathrm{tors}})$) and the other examples (Section 21.18). The labels are used as in Cremona's `galrep` file, with their intrinsic descriptions; the paper says it has not compared them with the tables of [191] themselves. The two sources of a deficit of 2 never combine, and the paper turns this into a theorem:

**Proposition 21.50** (the paper's Proposition prop:no9)**.** *Let $E/\mathbb{Q}$ be without complex multiplication, with $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$. Then $E(\mathbb{Q})$ has no point of order 9. Consequently, for every non-CM $E/\mathbb{Q}$, $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3] \leq 2$, and $v_\ell(\#E(\mathbb{Q})_{\mathrm{tors}}) \leq 1$ for $\ell \geq 5$.*

*Where proved.* Chapter 9; the input that a rational cyclic 27-isogeny forces complex multiplication is Ligozat's determination of the rational points of $X_0(27)$ [114], stated there; Mazur's theorem on torsion [125] is stated in Chapter 9.

*Expanded proof.* Suppose $P \in E(\mathbb{Q})$ has order 9. Then $3P$ is a rational point of order 3, so $\langle 3P \rangle$ is the line $\mathbb{Z}/3$ of $E[3]$ ($\mu_3$ has no nonzero rational point), and $\langle P \rangle \cap E[3] = \langle 3P \rangle$ meets $\mu_3$ trivially. Let $\phi\colon E \to E' = E/\mu_3$ (rational, as $\mu_3$ is $G_{\mathbb{Q}}$-stable) and $\hat{\phi}$ its dual. From $\phi \circ \hat{\phi} = [3]$, $\hat{\phi}(E'[3]) \subseteq \ker \phi = \mu_3$, with equality since $\#\hat{\phi}(E'[3]) = 9/\#\ker \hat{\phi} = 3$. The subgroup $C = \hat{\phi}^{-1}(\langle P \rangle) \subset E'$ is $G_{\mathbb{Q}}$-stable of order 27 ($\hat{\phi}$ has degree 3 and is onto on points). If $C \supseteq E'[3]$ then $\mu_3 = \hat{\phi}(E'[3]) \subseteq \langle P \rangle$, a contradiction; a subgroup of order 27 of $E'[27] \simeq (\mathbb{Z}/27)^2$ not containing $E'[3]$ is cyclic. So $E' \to E'/C$ is a rational cyclic 27-isogeny. By Ligozat, $X_0(27)$ has genus 1 and its non-cuspidal rational points have $j = -2^{15} \cdot 3 \cdot 5^3$, CM by the order of discriminant $-27$; so $E'$, hence $E$, has CM, a contradiction. The consequences: by Mazur's list, $v_3(\#E(\mathbb{Q})_{\mathrm{tors}}) \leq 2$ with equality only for $\mathbb{Z}/9$, and $v_\ell \leq 1$ for $\ell \geq 5$. □

*Reading guidance.* The proof is a descent of a cyclic subgroup through the dual isogeny: the point of order 9 and the subgroup $\mu_3$ together produce a rational cyclic 27-isogeny on $E/\mu_3$, and such an isogeny exists only with complex multiplication. The paper cites [114] for that fact and [96, Theorem 1] for the list of levels $N$ with $Y_0(N)(\mathbb{Q}) \neq \emptyset$. The remaining paragraph of Section 6.10 (deficits at $\ell = 5, 7$ only with rational $\ell$-torsion; the deficit against (eq:twist) not controlled by torsion, which is why Conjecture conj:eis is stated for (eq:main)) is data. In the Eisenstein case the additive terms do not contribute reliably: `539a1` (image 3B, type $\mathrm{I}_2^*$ at 7) has $e_3 = 1$ and $v_3(\deg \varphi_E) = v_3(320) = 0$.

## 21.15 Sections 7 and 8 of the paper: the prime 2 and the open questions

This section reads the paper's Section 7 (the prime 2) and its list of open questions, and says where the book bears on each.

**Section 7: the prime 2.** Every theorem of the paper needs $\ell$ odd: Lemma lem:tamfactor uses that 2 is a unit, the trace form needs $\ell \neq 2$, the real place contributes $H^1(\mathbb{R}, T^D)$ killed by 2, and $S_f$ contains the prime 2. Section 7 is data and pointers: a 2-adic inequality with the types $\mathrm{III}, \mathrm{III}^*, \mathrm{I}_0^*, \mathrm{I}_n^*$ counted once, holding on the three ranges, and failing when $\mathrm{I}_0^*$ is counted twice; the book's range shows the same (Chapter 22). The theorem of Calegari and Emerton [20] is summarised in the numbering of its arXiv version, as the paper says: odd modular degree implies at most two odd bad primes, even analytic rank, and either a rational 2-torsion point, or prime conductor with supersingular reduction at 2 and $E(\mathbb{R})$ connected, or complex multiplication with $N \in \{27, 32, 49, 243\}$. Watkins's conjecture $2^{\operatorname{rank} E(\mathbb{Q})} \mid \deg \varphi_E$ is stated in Chapter 14.

**Open question 1 (the weights at $q = \ell$).** The paper asks to extend Theorem thm:main to $\ell \mid N$ and to identify the weights $w_3$ with a local term at $\ell$. What fails there: at $\ell \mid N$, Agashe-Ribet-Stein can fail ($\ell^2 \mid N$: `99a1`), $\ell \in S_f$ (Fontaine-Laffaille needs $\ell \nmid N$), and the Euler factor at $q = \ell$ is not the relevant local quantity. The paper's lead is Remark rem:oq1 (read in Section 21.14): it reproduces the pattern of the $w_3$ and suggests a weight 1 for the types $\mathrm{II}^*$, $\mathrm{III}^*$, $\mathrm{IV}^*$ at $q = \ell \geq 5$; added to (eq:rule), that weight gives no violation below 10,000 at $\ell = 5, 7, 11, 13$ and is attained by most of these curves.

**Open question 2 (the 3Ns curves).** Two inputs are missing, as Proposition 21.49 shows: $H^1_f(\mathbb{Q}, \mathrm{ad}^0 V_3 E) = 0$ and the equality $v_\lambda(\eta^{\Sigma}_{f,\lambda}) = \operatorname{length} H^1_{\Sigma}(\mathbb{Q}, W)$ for a residual representation induced from $\mathbb{Q}(\sqrt{-3})$. The data are consistent: failures by exactly one, and equality in Theorem thm:quant in the 8 pairs with image 3Ns below 10,000, which lie outside its equality clause.

**Open question 3 (the Selmer group).** With Lemmas lem:tamfactor and lem:tamadd, the excess of $v_\ell(\deg \varphi_E)$ over the explicit local bound is the Selmer length (Corollary 21.41, Table tab:excess), and what remains is to compute that Selmer group independently, for instance for `1184f1` and `1748e1` at $\ell = 3$ (Chapter 18).

**Open questions 4 to 6.** The Eisenstein deficit and the cuspidal subgroup ([125]; Chapter 14); the prime 2 and Watkins's conjecture; and the smaller point whether, in Proposition prop:ns-structure, the type at 2 is always III or $\mathrm{III}^*$ when $R = \{2\}$, true for all 16 non-CM and all 6 CM optimal curves below 10,000 where it occurs (Chapter 11). The book adds nothing beyond the statements and Proposition 21.50.

## 21.16 The appendix of the paper: the function `rule`

This section reads the paper's appendix (reproducibility and the PARI functions) and its statement on the use of an AI system. The program itself is read line by line in Chapter 22; here we say what it computes and how to use it.

```
rule(E, l, {iii = 0}) =
{ my(R = ellglobalred(E), P = R[4][,1], K = R[5], S = lfunsympow(E, 2), b = 0, j = E.
    j);
  for(i = 1, #P,
    my(q = P[i], k = K[i][2], e = 0, t = 0, vj = valuation(j, q));
    if(q != l,
      my(F = 1/lfuneuler(S, q), r = subst(F, x, 1/q^2));
      if(k >= 5, r /= (1 - 1/q^2));
      if(abs(k) != 3 || iii, e = valuation(r, l)),
      if(l == 3, e = if(k == 4 || k == 2, 1,
                     if(k == -4 || k == -2, 2, if(k == -3, 1, 0)))));
    if(vj < 0, t = valuation(-vj, l));
    b += e + t);
  if(iii && l == 3, b -= ns3(E));
  [valuation(ellmoddegree(E), l), b]; }
ns3(E) = my(F = factor(elldivpol(E, 3))); (#F[,1] == 2 && poldegree(F[1,1]) == 2);
```

**What it computes.** (The program is copied from the paper, comments omitted; the paper's comment line after `ns3` explains the test: $\psi_3$ has degree 4 and is squarefree, so two factors, one of them quadratic, means two quadratics.) $P$ is the list of bad primes and $k$ PARI's Kodaira code at $q$. For $q \neq \ell$, $F = 1/$`lfuneuler`$(S, q)$ is $P_q(x)$ (for `121a1`, `lfuneuler(S, 11)` is $1/(11x+1)$), $r = P_q(q^{-2}) = L_q(A_f, 1)^{-1}$, divided at a multiplicative prime ($k \geq 5$) by the naive factor $1 - q^{-2}$; so $e = e_\ell(E, q)$, except that the type III terms ($|k| = 3$) are left out unless `iii` is set. At $q = \ell = 3$ the code puts $e = w_3(E)$ (codes $4, 2, -3$ give 1; $-4, -2$ give 2; others 0), even when `iii` is not set. $t = t_\ell(E, q)$ at every bad $q$, $q = \ell$ included. With `iii` set and $\ell = 3$, `ns3`$(E)$ is subtracted; it is 1 exactly when $\psi_3$ is a product of two irreducible quadratics, that is, for $E[3]$ irreducible, exactly for image $N_s(\mathbb{F}_3)$ (Proposition 21.47), and then it equals $h_3(E)$. So `rule(E, l, 1)` returns $[v_\ell(\deg \varphi_E)$, right side of (eq:rule)] and `rule(E, l)` the III-free bound, provided $E$ is the optimal curve: `ellmoddegree` returns the degree of the given model. The paper says that the curve must be optimal with $E[\ell]$ irreducible, with the examples `990h1` (returns `[4, 3]`) against the optimal `990h3` (`[3, 3]`) and `11a1` at $\ell = 5$ (`[0, 1]`); the book reran them, and `11a3` at $\ell = 5$ returns `[-1, 0]` (its degree is 1/5 relative to the optimal curve). PARI's Kodaira codes: $n+4$ for $\mathrm{I}_n$ and $-(n+4)$ for $\mathrm{I}_n^*$ when $n \geq 1$, 1 for $\mathrm{I}_0$ and $-1$ for $\mathrm{I}_0^*$, $\pm 2$, $\pm 3$, $\pm 4$ for II/II$^*$, III/III$^*$, IV/IV$^*$; the qualifier "when $n \geq 1$" matters, as $\mathrm{I}_0$ and $\mathrm{I}_0^*$ have the codes $\pm 1$. The book's runs (PARI 2.17.2) of `rule` on the paper's and the book's example curves are in Section 21.18; on `726a1` both modes return `[2, 2]`, the type III term at 11 and the correction cancelling, as the paper says.

**Reproducibility and the statement on AI.** The repository named in the appendix holds the scripts it names and the result files from which the six data tables are generated; the book compared the printed tables with those files (Chapter 22, in its section on reproducibility) but did not rerun the large-range computations. The book's own reproduction is in Chapter 22 (Lemma lem:twist checked on twists of the book's range, Lemma lem:euler on 67,745 pairs, `ellmoddegree` against Cremona's `degphi` on all 38,042 optimal curves below 10,000, no mismatch). The statement on the use of an AI system is part of the paper; Section 21.1 summarises it.

## 21.17 The map of the paper

This section collects three devices that the section-by-section reading cannot show: which result feeds which (the dependency map), which hypothesis is used where and what fails without it (the hypothesis table), and the status of every statement imported from the literature.

**Dependencies.** An arrow means "is used in the proof of". Imported statements are in brackets.

| input | used in |
|---|---|
| [Tate curves, Kummer theory] | Lemma lem:tamagawa-exp → Theorem thm:qual (c), Section 1.5, Remark rem:ko |
| [Néron models] → inertia invariants (Lemma 21.12) | Lemma lem:IV at every $q \neq 3$, $q = 2$ included → Theorem thm:qual (a), (b), Proposition prop:forms, the table |
| [Tate's algorithm]; [Ogg-Saito] (at $q = 3$) | Lemma lem:twist (i), (ii) → Lemma lem:euler, Theorem thm:qual (b), (c), Theorem thm:twist, Theorem thm:quant |
| [[9, Theorem 3.1]] | Lemma lem:twist (iii) → Theorem thm:qual (b), (c) |
| [[107]; [179]] | Lemma lem:euler, Proposition prop:shapes → Definition def:terms values, Theorem thm:quant Step 3, Proposition prop:ns-structure |
| [[69, I.4.2.2]] + group cohomology | Lemma lem:tamfactor, Lemma lem:tamadd → Theorem thm:main Step 7, Corollary cor:explicit, Remark rem:ko |
| [[159, 1]: the perfect pairing] | Lemma lem:crit → Lemma lem:old, Theorem thm:twist, Theorem thm:quant Step 1 |
| [[61, 8]; [38, Theorem 3.1(e)]; Chebotarev; Brauer-Nesbitt] | Lemma lem:old → Theorem thm:qual |
| [[38, Theorem 3.15]]; [[3, Theorem 2.2]] | Theorem thm:qual |
| [[49, Sections 1.2, 1.4]; Hochschild-Serre; equivariant Leray] → Proposition prop:integral; projection formula | Proposition prop:pullback → Lemma lem:hecke → Theorem thm:main Step 1, Theorem thm:quant Step 4 |
| [[38, Lemma 4.17]] = Lemma lem:ddt417 | Theorem thm:quant Step 2 (and the second route to Lemma lem:hecke, Remark rem:free) |
| [[49, Proposition 1.4(c)]] = Theorem thm:dfg14 | Theorem thm:main Step 2, Theorem thm:quant Step 3 |
| [[49, Theorem 3.7]] = Theorem thm:dfg37 | Theorem thm:main Step 3 |
| [[49, Theorem 2.7]] = Theorem thm:dfg27 | $H^1_f(\mathbb{Q}, V^D) = 0$ in Lemma lem:dfg21 ($H^0(\mathbb{Q}, V^D) = 0$ automatic) |
| [[49, Lemma 2.1]] = Lemma lem:dfg21; Lemma lem:fpr | Theorem thm:main Steps 4, 5; Remark rem:ns (conditionally) |
| [Clifford theory; subgroups of $\mathrm{GL}_2(\mathbb{F}_3)$] | Proposition prop:ns ↔ Lemma lem:h0 → Theorem thm:main Step 4, Conjecture conj:euler ($h_\ell$), Section 6.7 |
| [inertia at $\ell$, Serre] | Proposition prop:Sf: $\ell \notin S_f$ automatic for $\ell \geq 5$ |
| Theorem thm:main | Proposition prop:forms, Corollary cor:explicit, Theorem thm:quant (ii) |
| Theorem thm:twist; the twist map → Lemma lem:step1 | Theorem thm:quant Step 1 |
| [[114]: $X_0(27)$]; [[125]: torsion] | Proposition prop:no9 → $D \leq 2$ in Conjecture conj:eis, given its first bound |

| input | used in |
|---|---|
| data (Section 6) | Conjecture conj:euler ($w_3$, $h_\ell$), Conjecture conj:eis (first bound), Section 7 |

Three spines: local (Lemmas lem:tamagawa-exp, lem:IV, lem:twist, lem:euler, lem:tamfactor, lem:tamadd), congruences (Lemmas lem:crit, lem:old, the optimal-level theorem, Theorem thm:qual) and Selmer (Propositions prop:integral, prop:pullback, Theorems thm:dfg14, thm:dfg37, thm:dfg27, Lemmas lem:dfg21, lem:fpr, lem:h0, Theorem thm:main). Theorem thm:quant uses the first two and, from the third, only Theorem thm:dfg14, Lemma lem:ddt417 and Lemma lem:hecke; it never uses Theorem thm:dfg37, which is why its inequality needs no hypothesis over $\mathbb{Q}(\sqrt{\ell^*})$, while its equality uses Theorem thm:main for $E$ and $E'_S$. Multiplicity one (Remark rem:free) feeds nothing.

**Hypotheses.** Aligned with the table of Chapter 20.

| hypothesis | where it is used | what fails without it |
|---|---|---|
| $\ell$ odd | Lemma lem:tamfactor (2 a unit); trace form; real place; $2 \in S_f$; Lemma lem:h0 (complex conjugation); optimal-level theorem; Lemma lem:old at $p = \ell$ | everything in Sections 2 to 4 at $\ell = 2$; Section 7 shows a different 2-adic pattern |
| $\ell \nmid N$ ($\ell^2 \nmid N$) | ARS12 (only $\ell^2 \nmid N$); $S_f$ and Fontaine-Laffaille; Lemma lem:old at $p = \ell$; optimal-level theorem; the local terms need $q \neq \ell$ | `99a1` ($\deg \varphi_E = 4$, $r_E = 12$ at $\ell = 3$; $H^1(X_0(99), \mathbb{Z}_3)_{\mathfrak{m}}$ not free); `726a1` and `539d1` at 11, 7; at $q = \ell = 3$ the empirical weights $w_3$ |
| $\ell \notin S_f$ ($\bar{\rho}$ absolutely irreducible over $\mathbb{Q}(\sqrt{\ell^*})$) | Theorems thm:dfg37, thm:dfg27; Lemma lem:h0 ($h_\ell = 0$); Theorem thm:qual at $\ell = 3$; the equality in Theorem thm:quant | automatic for $\ell \geq 5$, $\ell \nmid N$, $E[\ell]$ irreducible; at $\ell = 3$ fails exactly for image 3Ns, where (eq:thm) fails for 31%, 23%, 18% of the curves, by one (`726a1`) |
| $E[\ell]$ irreducible | Theorem thm:dfg14; non-Eisenstein $\mathfrak{m}$; Lemma lem:old; Theorems thm:qual, thm:quant | `539a1` (image 3B, $e_3 = 1$, $v_3(\deg \varphi_E) = 0$); `539d1` at $\ell = 5$ ($e_5 = 1$, $v_5(\deg \varphi_E) = 0$); `98a1` ($e_3 = 1$, $\deg \varphi_E = 16$); only (eq:main) survives, with the deficit of Conjecture conj:eis |
| optimality | ARS12; the definition of $\deg \varphi_E$; Proposition prop:pullback; the function `rule` | nothing at $\ell$ when $E[\ell]$ is irreducible ($v_\ell$ of the degree is a class invariant); matters for Conjecture conj:eis and for `rule` (`990h1: [4, 3]`) |
| multiplicity one | nowhere (Proposition prop:pullback and the squeeze in Theorem thm:quant (ii) replace it) | nothing; it holds in the paper's range (Remark rem:free) and fails for $\ell \mid N$ (`99a1` at 3) |

**Imported statements.** "Verified" means compared with the published text (for [49]: Ann. Sci. École Norm. Sup. (4) 37 (2004); a longer arXiv text, arXiv:2512.02348, numbers its sections and results differently, and every number in the paper and in this book refers to the published version). "As cited" means taken from the paper and not checked for this book.

| citation | content used | status |
|---|---|---|
| [49, Theorem 0.1] (= 2.15) | Bloch-Kato for $A_f$, $A_f(1)$ at $\lambda \notin S_f$ | verified |
| [49, Theorem 3.7] (= 0.2) | length $H^1_\Sigma(\mathbb{Q}, W) = v_\lambda(\eta^\Sigma)$ for $\ell \notin S_f$; covers $\ell = 3$ in weight 2 | verified |
| [49, Theorem 2.7] | $H^0 = H^1_f = 0$ for $A_f$, $A_f(1)$ | verified |
| [49, Proposition 1.4(c)] | $\eta^\Sigma = \eta^\emptyset \prod L_p^{\mathrm{nv}}(A_f, 1)^{-1}$, with $\ell \notin \Sigma$ standing | verified |
| [49, Lemma 2.1, (31), (57)] | the four-term sequence; $S_f$; $\mathrm{Tam}_p(M) = \mathrm{Tam}_p(M^D)$ | verified |
| [49, Lemma 2.12, p. 708, Sections 1.1.1 to 2.1, (13), (14)] | naive Petersson formula; Tam; Frobenius, $[UwU]_\omega$, kernel lattice, ideal of values | verified |
| [69, I.4.2.2] | the local Tamagawa formula | as recalled in DFG04 (verified there) |
| [3, Theorem 2.2] | $\deg \varphi_E \mid r_E$, equality at $p^2 \nmid N$ | checked in Chapter 14 |
| [38, 3.1(e), 3.15, 4.17, Section 4.4] | local shape at $q \parallel N$; optimal level; numerical criterion; adjoints | as cited |
| [9, Theorem 3.1]; [8, Theorem 3] | level of a twist; $a_p = 0$ for $p^2 \mid M$ | as cited |
| [159, Theorem 2.2]; [1, Lemma 2.1]; [61, Section 2] | perfect pairing; $a_\ell$ from $\bar\rho\|_{D_\ell}$ | as cited (pairing proved in Chapter 13) |
| [163, Theorem 3.5]; [199] | multiplicity one; Gorenstein | as cited |
| [99, Theorem 1.1, Definition 2.12] | variation of congruence ideals; Tamagawa exponent | as cited |
| [153, Theorem 6.11] | their identity, under CR | checked (arXiv version) |
| [164, 192] | Shimura-curve degrees, semistable $E$ | checked in Chapter 14 |
| [114]; [96] | rational points of $X_0(27)$; the list of levels | Kenku checked; Ligozat as cited |
| [107]; [179] | inertia images at $q \geq 5$; independence of $\ell$ | standard |
| [210]; [191] | mod 3 images; labels | by section; labels via Cremona's files |
| [20] | odd modular degree | arXiv version read; published numbering unchecked |

## 21.18 Worked verifications

This section lists the curve-level numbers the tour quotes, as recomputed for this chapter with PARI/GP 2.17.2 through `cypari2` and Cremona's files: Kodaira codes from `elllglobalred`, $P_q$ from `lfuneuler(lfunsympow(E,2),q)`, degrees from `ellmoddegree` (equal to Cremona's `degphi` on every optimal curve below 10,000), the outputs of the paper's `rule` (the code of Section 21.16), and Tate parameters from `ellinit(E, O(q^40)).tate`. Congruence numbers are those computed in Chapters 13 and 14.

| curve | local data and $P_q$ | degree, `rule`, remarks |
|---|---|---|
| `121a1` | II at 11, $P_{11} = 1 + 11X$, $e_3 = v_3(12/11) = 1$ | 6; `[1, 1]` in both modes; $r_E = 6$; twist `121c2` ($\mathrm{IV}^*$) |
| `539c1` | $\mathrm{I}_3^*$ at 7, $P_7 = 1 - X$, $e_3 = 1$, $t_3 = 1$; $\mathrm{I}_2$ at 11; $\bar{u} = 5$ at 7 | $288 = 2^5 3^2$; `[2, 2]` at 3, `[0, 0]` at 5; twist `77c1` ($\deg \varphi_E = 6$) |
| `980i1` | $\mathrm{IV}^*$ at 2 ($P_2 = 1 + 2X$), $\mathrm{I}_1$ at 5, $\mathrm{I}_5^*$ at 7 | $2880 = 2^6 3^2 5$; `[2, 2]` at 3, `[1, 1]` at 5; twist `140b1` (60) |
| `726a1` | $\mathrm{I}_6$ at 2, $\mathrm{I}_3$ at 3, III at 11 ($P_{11} = 1 + 11X$); 3Ns, `ns3` $= 1$ | 144; `[2, 2]` in both modes; $r_E = 1584$ |
| `99a1` | III at 3 ($P_3 = 1 + 3X$, not used at $q = \ell$), $\mathrm{I}_1$ at 11 | 4; `[0, 0]`; $r_E = 12$ |
| `121d1` | $\mathrm{I}_1^*$ at 11, $e_3 = v_3(120) = 1$ | 24; `[1, 1]`; twist of `11a3` |
| `121d2` | $\mathrm{I}_5^*$ at 11, $e_3 = 1$ | 120; twist of `11a1` ($e_3($`11a1`$, 11) = 0$) |
| `539d1` | $\mathrm{I}_0^*$ at 7, $P_7 = 1 + 3X - 21X^2 - 343X^3$, $e_3 = 2$; $\mathrm{I}_1$ at 11 | 72; `[2, 2]`; $r_E = 504$; at 5: `[0, 1]` (5B.4.1) |
| `539a1` | $\mathrm{I}_2^*$ at 7, $e_3 = 1$; image 3B | 320; `[0, 1]`: Eisenstein |
| `98a1` | $\mathrm{I}_2$ at 2, $\mathrm{I}_1^*$ at 7, $e_3 = 1$; image 3B | 16; `[0, 1]`: Eisenstein |
| `52a1` | $\mathrm{IV}^*$ at 2 ($P_2 = 1 + 2X$), $\mathrm{I}_2$ at 13 | 3; `[1, 1]`; $f_{52a} \equiv f_{26b} - f_{26b}(2z) \bmod 3$ |
| `336f1` | II at 2, $P_2 = 1 + 2X$, $e_3 = 1$ | 24; `[1, 1]` |
| `1232l1` | $\mathrm{II}^*$ at 2, $P_2 = 1 + 2X - 4X^2 - 8X^3$, $e_3 = 2$ | 288; `[2, 2]`; $=$ `77a1` $\otimes \chi_{-4}$, $a_2($`77a1`$) = 0$ |
| `592c1` | $\mathrm{II}^*$ at 2, $P_2 = 1 - 2X + 4X^2 - 8X^3$, $e_5 = 1$ | 80; `[1, 1]` at 5; $=$ `37a1` $\otimes \chi_{-4}$, $a_2 = -2$ |
| `791a1` | $\mathrm{I}_3$ at 7, $\bar{u} = 6$ a cube: Tamagawa excess 1 | 144; `[2, 1]`; explicit bound 2, Selmer 0 |
| `1184f1` | $\mathrm{III}^*$ at 2 ($P_2 = 1$), $\mathrm{I}_3$ at 37, $\bar{u} = 36$ a cube; image 3Nn | 1728; `[3, 1]`; explicit bound 2, Selmer 1 |
| `1748e1` | IV at 2 ($e_3 = 1$), $\mathrm{I}_6$ at 19 ($\bar{u} = 12$ a cube), $\mathrm{I}_1$ at 23 | 3240; `[4, 2]`; explicit bound 3, Selmer 1 |
| `11a1` | $\mathrm{I}_5$ at 11, torsion $\mathbb{Z}/5$, image 5Cs.1.1, $\bar{u} = 10$ | 1; `[0, 1]` at 5: Eisenstein |
| `14a1` | $\mathrm{I}_6$ at 2, $\mathrm{I}_3$ at 7, torsion $\mathbb{Z}/6$, 3Cs.1.1 | 1; $D = 2$ |
| `714i1` | $\mathrm{I}_9$ at 2 and 3, $\mathrm{I}_3$ at 7, $\mathrm{I}_1$ at 17, torsion $\mathbb{Z}/9$, 3B.1.1 | 1080; $D = 5 - 3 = 2$ |
| `20a1` | $\mathrm{IV}^*$ at 2, $\mathrm{I}_2$ at 5, torsion $\mathbb{Z}/6$, 3B.1.1 | 1; $D = 1$ |
| `990h1`, `990h3` | the same class; `990h3` is optimal | `[4, 3]` against `[3, 3]` |

The twist identifications (`11a1` by $-11$, $-7$ to `121d2`, `539d2`; `20a1` by $-7$ to `980g2`; `539c1` by $-7$ to `77c1`; `121a1` by $-11$ to `121c2`; `980i1` by $-7$ to `140b1`; `37a1`, `77a1` by $-4$ to `592c1`, `1232l1`) were checked by isomorphism of minimal models; the congruences of Example 21.26 to $n \le 400$ and $n \le 1000$; `mfsturm` returns 14, 22, 112, 336. Also recomputed: $j = -2^{15} \cdot 3 \cdot 5^3 = -12288000$ for discriminant $-27$; the three coordinate vectors and the determinant $-1$ in the proof of Lemma 21.20; $L_3^{\mathrm{nv}}(A_f, 1)^{-1} = 10/9$ for $f_{11a}$ (Theorem 21.29); the 3Ns counts of Section 5 and Section 6.7 of the paper (90 non-CM, 44 with $3 \mid N$, 74 and 16, 13 CM; 36 and 24 violations).

## 21.19 Exercises

Data for the curves are in Section 21.18; computations by hand, or with PARI, are allowed, and the hints say which.

**Exercise 21.51.** The curve `121a1` is $y^2+xy+y=x^3+x^2-30x-76$, with $\Delta=-11^2$ and $c_4=1441=11\cdot131$. (a) Show that it has additive, potentially good reduction of type II at 11. (b) Compute $e_3(E,11)$ and $e_5(E,11)$ from the table. (c) Using $\deg\varphi_E=6$, Corollary 21.41 and Lemma 21.20, determine length $H^1_f(\mathbb{Q},\mathrm{ad}^0\,T_3E\otimes\mathbb{Q}_3/\mathbb{Z}_3)$ (the mod 3 image is surjective). *Hint.* (a) $v_{11}(j)=3v_{11}(c_4)-v_{11}(\Delta)=1\geq0$ and $v_{11}(\Delta)=2$. (b) $\chi_{-3}(11)=-1$: $e_3=v_3(12)=1$, $e_5=0$. (c) $1=\mathrm{length}\,H^1_f+1+0$, so the Selmer length is 0.

**Exercise 21.52.** (a) The curve `37a1` has type $\mathrm{I}_1$ at 37 and $\deg\varphi_E=2$; its twist by $\chi_{37}$ is `1369d1`, of type $\mathrm{I}_1^*$ at 37, alone in its class. Compute $e_3$ and $e_{19}$ at 37 for both curves, explain why they differ although $\mathrm{ad}^0V$ does not change, and use Theorem 21.46(ii) to predict $v_3$ and $v_{19}$ of $\deg\varphi_{\texttt{1369d1}}$; check against $\deg\varphi_{\texttt{1369d1}}=2736$. (b) For the class `539d` (type $\mathrm{I}_0^*$ at 7, the twist of `11a` by $\chi_{-7}$) compute $e_3$ and $e_5$ at 7 from $a_7(\texttt{11a1})=-2$, compare with $\deg\varphi_E=72$, and explain the behaviour at $\ell=5$. *Hint.* (a) For `37a1` the naive and true factors are both $1-37^{-2}$, so $e_\ell=0$; for `1369d1` the naive factor is 1 and $e_\ell=v_\ell(37^2-1)=v_\ell(2^3\cdot3^2\cdot19)$: $e_3=2$, $e_{19}=1$. The hypotheses of (ii) hold for $\ell=3,19$ (the mod $\ell$ images of `37a1` are surjective, $\ell\nmid2\cdot1369$, and $\ell\notin S_f$ by Proposition 21.35, the image at 3 not being 3Ns), with $S=\{37\}$ and $E'_S=$ `37a1`; so $v_3=0+2$ and $v_{19}=0+1$, as $2736=2^4\cdot3^2\cdot19$. (b) $e_\ell=v_\ell(6)+v_\ell(64-4)$: $e_3=2=v_3(72)$, $e_5=1$ but $v_5(72)=0$; the mod 5 image of `539d1` is reducible (5B.4.1), outside Theorem thm:quant.

**Exercise 21.53.** Let $f=f_{37a}$, with $a_3=-3$, and $\ell=3$. Compute $L_3^{\mathrm{nv}}(A_f,1)^{-1}$ and show that the formula of Theorem 21.29 with $\Sigma=\{3\}$ cannot hold. Which ingredient of the theory of [49] needs $\ell\notin\Sigma$? *Answer.* $\frac{2}{3}\cdot\frac{16-9}{9}=\frac{14}{27}$, of 3-adic valuation $-3$, so the right side is not in $\mathbb{Z}_3$. Adding $\ell$ to the level destroys the condition $\ell\nmid$ level under which the integral structures and the Fontaine-Laffaille description at $\ell$ are defined (Chapter 20).

**Exercise 21.54.** Lemma lem:tamfactor in coordinates. Let $\ell=3$, $n=9$, $q=19$ ($v_3(q-1)=2$), and let Frob act on $(\mathbb{Z}/9)\bar e\oplus(\mathbb{Z}/9)\bar h$ by $\bar e\mapsto\bar e$, $\bar h\mapsto q^{-1}(\bar h-2c'\varepsilon\bar e)$, $\varepsilon=\pm1$. (a) For each $c'\in\mathbb{Z}/9$ count the fixed points and check that the count is $9\cdot3^{\min(2,2,v_3(c'))}$. (b) Replace the action by $q\cdot\mathrm{Frob}$ (the twist by $\chi_3(\mathrm{Frob})=q$, which is the action on $H^1(I,A(1))$) and show that the count is unchanged, so $\mathrm{Tam}_q(A(1))=\mathrm{Tam}_q(A)$ here. (c) Explain why (b) holds for every $n$, $q$, $\ell$. *Hint.* (a) Fixed iff $(q-1)y\equiv0$ and $c'y\equiv0$. (b) $x\bar e+y\bar h$ is fixed by $q\,\mathrm{Frob}$ iff $(q-1)x\equiv2c'\varepsilon y$; here $q-1\equiv0\pmod9$, so the condition is $c'y\equiv0$ with $x$ free. (c) In general the two conditions are $(q-1)y\equiv0$, $c'y\equiv0$ and $(q-1)x\equiv2c'\varepsilon y$; count the solutions of the second system by first choosing $y$ and compare.

**Exercise 21.55.** Let $G$ be a group, $\chi,\varepsilon\colon G\to\mathbb{Z}_\ell^\times$ characters with $\varepsilon^2=1$, and $c\colon G\to\mathbb{Z}_\ell$ a function. (a) Show that $g\mapsto\left(\begin{smallmatrix}\chi\varepsilon(g) & c(g)\\ 0 & \varepsilon(g)\end{smallmatrix}\right)$ is a homomorphism iff $c(gh)=\chi\varepsilon(g)c(h)+c(g)\varepsilon(h)$. (b) If $\kappa$ satisfies $\kappa(gh)=\chi(g)\kappa(h)+\kappa(g)$, show that $c=\varepsilon\kappa$ satisfies (a), and that $c=\kappa$ does not unless $(\varepsilon(g)-1)\chi(g)\kappa(h)+(\varepsilon(h)-1)\kappa(g)=0$ for all $g,h$. (c) Deduce the description of the nonsplit Tate curve used in Lemma 21.19. *Hint.* (a) The diagonal entries of the product of the matrices for $g$ and $h$ are $\chi\varepsilon(gh)$ and $\varepsilon(gh)$; the upper right entry is $\chi\varepsilon(g)c(h)+c(g)\varepsilon(h)$. (b) Multiply the cocycle relation for $\kappa$ by $\varepsilon(gh)=\varepsilon(g)\varepsilon(h)$. For $c=\kappa$, subtract the relation for $\kappa$ from the one required in (a). (c) The nonsplit curve is the Tate curve twisted by the unramified quadratic character $\varepsilon$; its representation is $\left(\begin{smallmatrix}\chi_\ell & \kappa\\ 0 & 1\end{smallmatrix}\right)\otimes\varepsilon$.

**Exercise 21.56.** Let $\mathcal{O}=\mathbb{Z}_\ell$, $\mathbb{T}=\{(a,b)\in\mathbb{Z}_\ell^2:a\equiv b\pmod{\ell^r}\}$ with $r\geq1$, and $\pi(a,b)=a$. (a) Compute $\wp$, $I=\mathrm{Ann}_{\mathbb{T}}(\wp)$ and $\eta$, and find $t$ with $I=\mathbb{Z}_\ell t$. (b) Let $\mathbb{T}$ act on $L'=\mathbb{Z}_\ell^2$ by

componentwise multiplication, with the pairing $\langle (a,b),(c,d)\rangle = ac+bd$. Check the hypotheses of Lemma 21.31, compute $L'[\wp]$ and $\Delta$, and show that the inequality is strict; is $L'$ free over $\mathbb{T}$? (c) Let $L=\mathbb{T}$ with $\langle (a,b),(c,d)\rangle = (ac-bd)/\ell^r$. Show that this pairing is integral, perfect and $\mathbb{T}$-self-adjoint, and that Lemma 21.31 holds with equality. *Answer.* (a) $\wp = 0\times \ell^r\mathbb{Z}_\ell$, $I=\ell^r\mathbb{Z}_\ell\times 0$, $\eta=\ell^r\mathbb{Z}_\ell$, $t=(\ell^r,0)$. (b) $L'[\wp]=\mathbb{Z}_\ell(1,0)$, $\Delta=1$, $v(\eta)=r>0$; $L'$ is not free over $\mathbb{T}$ (it is the normalisation of $\mathbb{T}$). (c) In the basis $(1,1)$, $(\ell^r,0)$ of $\mathbb{T}$ the Gram matrix is $\left(\begin{smallmatrix}0 & 1\\ 1 & \ell^r\end{smallmatrix}\right)$, of determinant $-1$; $L[\wp]=\mathbb{Z}_\ell(\ell^r,0)$ has $\Delta=\ell^r$, and $v(\eta)=r=v(\Delta)$.

**Exercise 21.57.** (a) Show that an alternating form on $\mathcal{O}^2$ has Gram matrix $\left(\begin{smallmatrix}0 & a\\ -a & 0\end{smallmatrix}\right)$ in every basis, so its Gram determinant is $a^2$ and its ideal of values is $a\mathcal{O}$. (b) Deduce from the projection formula that the Gram determinant of $\frac{1}{2}\langle x, w_N^* y\rangle$ on $\varphi_E^* H^1(E,\mathbb{Z}_\ell)$ is $(\deg\varphi_E)^2$ up to a unit, for $\ell$ odd. (c) For `11a1`, where $X_0(11)\to E$ has degree 1, check this directly. *Hint.* (a) $\langle x,x\rangle=0$ for all $x$ forces zero diagonal entries and $\langle e_2,e_1\rangle=-\langle e_1,e_2\rangle$; a change of basis by $P$ multiplies $a$ by $\det P$, a unit; and $\langle x,y\rangle = a\det(x,y)$ takes every value in $a\mathcal{O}$. (b) On the image $w_N^*$ acts by $\varepsilon_N(f)=\pm 1$, so the form is $\frac{\varepsilon_N(f)}{2}\deg\varphi_E$ times the pullback of the unimodular cup product of $E$; it is alternating, and (a) applies with $a=\pm\frac{1}{2}\deg\varphi_E$. (c) The parametrisation is an isomorphism, so $\varphi_E^*$ is an isomorphism of lattices carrying one unimodular cup product to the other; the Gram determinant is $1=1^2$.

**Exercise 21.58.** The curve `3840h1` has types III at 2 (with $P_2=1+2X$), $\mathrm{I}_3$ at 3 and $\mathrm{I}_6$ at 5, mod 3 image 3Ns, and $\deg\varphi_E = 1152 = 2^7\cdot 3^2$. (a) Compute the right side of (eq:thm) at $\ell=3$ as written, and the same without the terms at $q=3$; which prime $p\equiv -1 \pmod 3$ carries the failure, and by how much? (b) Compute `rule(E, 3)` and `rule(E, 3, 1)` by hand. (c) Identify the order-four prime of Proposition 21.48 and check part (ii) there. (d) Why is $w_3(E)=0$? *Answer.* (a) $e_3(E,2)=v_3(P_2(1/4))=v_3(3/2)=1$, $t_3(E,3)=1$, $t_3(E,5)=v_3(6)=1$: the bound is $3>2$, a failure by one, carried by $p=5$ with $t_3(E,5)=1$; without $t_3(E,3)$ the bound is 2. (b) Without `iii` the type III term at 2 is left out: $0+1+1=2$; with it, $1+1+1-\mathtt{ns3}(E)=2$; both return `[2, 2]`. (c) The prime 2, of type III: 5 is potentially multiplicative, so $\bar\rho(I_5)$ consists of $\pm$ unipotents, while $R\neq\emptyset$; by (ii), $P_2=1-\chi_{-3}(2)2X=1+2X$ and $e_3(E,2)=1$, as computed. (d) The reduction at 3 is multiplicative.

**Exercise 21.59.** Let $p$ be a good prime, $a_{p^k}$ the coefficients of $f$, $\alpha+\beta=a_p$, $\alpha\beta=p$, $X=p^{-s}$. (a) Using $a_{p^k}=(\alpha^{k+1}-\beta^{k+1})/(\alpha-\beta)$, prove $\sum_k a_{p^k}^2X^k = (1+pX)/((1-\alpha^2X)(1-pX)(1-\beta^2X))$ and $\sum_k a_{p^{2k}}X^k=(1+pX)/((1-\alpha^2X)(1-\beta^2X))$. (b) Deduce $\sum_n a_n^2 n^{-s} = \zeta^{(N)}(s-1)L^{\mathrm{nv}}(\mathrm{Sym}^2 f,s)/\zeta^{(N)}(2s-2)$, checking the multiplicative and additive primes separately. *Hint.* (a) $(\alpha-\beta)^2a_{p^k}^2=\alpha^{2k+2}-2p^{k+1}+\beta^{2k+2}$; summing geometric series, $(\alpha-\beta)^2\sum_k a_{p^k}^2X^k = \frac{\alpha^2}{1-\alpha^2X}-\frac{2p}{1-pX}+\frac{\beta^2}{1-\beta^2X}$; put everything over the common denominator and use $\alpha\beta=p$. For the second series, $(\alpha-\beta)a_{p^{2k}}=\alpha^{2k+1}-\beta^{2k+1}$ and $\alpha(1-\beta^2X)-\beta(1-\alpha^2X)=(\alpha-\beta)(1+pX)$. (If $\alpha=\beta$, argue by continuity in the coefficients.) (b) At a good prime the local factors of $\zeta^{(N)}(s-1)$ and $\zeta^{(N)}(2s-2)$ are $(1-pX)^{-1}$ and $(1-p^2X^2)^{-1}$; at $p\mid N$ they are 1, and $a_{p^k}=a_p^k$ with $a_p\in\{0,\pm1\}$.

**Exercise 21.60.** Compute the deficit $D$ of Conjecture conj:eis at $\ell=3$ for the optimal curves `26a1` (types $\mathrm{I}_3$ at 2 and 13, torsion $\mathbb{Z}/3$, image 3Cs.1.1, $\deg\varphi_E=2$), `30a1` ($\mathrm{I}_4$ at 2, $\mathrm{I}_3$ at 3, $\mathrm{I}_1$ at 5, torsion $\mathbb{Z}/6$, image 3B.1.1, $\deg\varphi_E=2$) and `6942n1` ($\mathrm{I}_{18}$ at 2, $\mathrm{I}_9$ at 3, $\mathrm{I}_3$ at 13, $\mathrm{I}_1$ at 89, torsion $\mathbb{Z}/9$, image 3B.1.1, $\deg\varphi_E=65664$), and say which summand of the first bound is responsible in each case. Is `6942n1` consistent with Proposition 21.50? *Answer.* `26a1`:

$D = 1 + 1 - 0 = 2 = v_3(3) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3]$, both summands. `30a1`: $D = 1 - 0 = 1 = v_3(6)$, the torsion. `6942n1`: $65664 = 2^7 \cdot 3^3 \cdot 19$, so $D = (2 + 2 + 1 + 0) - 3 = 2 = v_3(9)$, the torsion. Yes: its image 3B.1.1 means $E[3] \not\simeq \mathbb{Z}/3 \oplus \mu_3$, and the proposition excludes a point of order 9 only when $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$.

**Exercise 21.61.** Let $E$ have mod 3 image $N_s(\mathbb{F}_3)$. (a) Show that $a_p \equiv 0 \pmod 3$ for every prime $p \nmid 3N$ with $p \equiv 2 \pmod 3$. (b) Check it on `726a1` for $p = 5, 17, 23, 29, 41$ ($a_p = 0, 6, 6, -6, -6$). (c) Show that (a) also holds when the image is contained in the split Cartan subgroup $C_s(\mathbb{F}_3)$ (then $E[3]$ is reducible), but not for the image $N_{ns}(\mathbb{F}_3)$. *Hint.* (a) $\det \bar\rho(\mathrm{Frob}_p) = \chi_{-3}(p) = -1$ puts $\bar\rho(\mathrm{Frob}_p)$ in the coset of determinant $-1$, whose elements $\pm h$, $\pm w$ have trace 0. (c) The elements of determinant $-1$ of $C_s(\mathbb{F}_3)$ are $\pm h$; those of the nonsplit Cartan subgroup $\mathbb{F}_9^\times \subset N_{ns}(\mathbb{F}_3)$ have order 8 and nonzero trace (proof of Proposition 21.47), and by Chebotarev they occur as $\bar\rho(\mathrm{Frob}_p)$.

**Exercise 21.62.** The curves `848e1` and `1072a1` are the twists by $\chi_{-4}$ of `53a1` and `67a1`, with $a_2$(`53a1`) $= -1$ and $a_2$(`67a1`) $= 2$. PARI gives `848e1` the symbol $\mathrm{I}_4^*$ at 2, with $v_2(j) = 0$, and `1072a1` the type $\mathrm{II}^*$ at 2. (a) Using Proposition 21.17(ii), derive $P_2$ for both curves. (b) Show that $e_\ell$(`848e1`$, 2) = 0$ for every odd $\ell$ and $t_\ell$(`848e1`$, 2) = 0$, although the symbol at 2 is $\mathrm{I}_4^*$. (c) Compute $e_5$(`1072a1`$, 2)$ and compare with $\deg \varphi_E = 200$. *Answer.* (a) $(1 - 2X)(1 - (a^2 - 4)X + 4X^2)$: for `848e1`, $a^2 = 1$, $P_2 = (1 - 2X)(1 + 3X + 4X^2)$; for `1072a1`, $a^2 = 4$, $P_2 = (1 - 2X)(1 + 4X^2)$. (b) $P_2(1/4) = \frac{1}{2}(1 + \frac{3}{4} + \frac{1}{4}) = 1$, so $e_\ell = 0$; $v_2(j) \geq 0$, so the reduction is potentially good and $t_\ell = 0$: at $q = 2$ a symbol $\mathrm{I}_m^*$ does not imply potentially multiplicative reduction. (c) $P_2(1/4) = \frac{1}{2} \cdot \frac{5}{4} = \frac{5}{8}$, $e_5 = 1$, while $v_5(200) = 2$: the bound is not attained (see Exercise 21.63).

**Exercise 21.63.** Using Corollary 21.41, compute the explicit bound and the Selmer length length $H^1_f(\mathbb{Q}, \mathrm{ad}^0 T \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$: (a) for `1577a1` at $\ell = 3$ and at $\ell = 5$ (types $\mathrm{I}_3$ at 19 with $\bar u = 8$, and $\mathrm{I}_2$ at 83; $\deg \varphi_E = 720$; mod $\ell$ images surjective for every $\ell$); (b) for `1072a1` and `67a1` at $\ell = 5$ (Exercise 21.62; `67a1` has type $\mathrm{I}_1$ at 67 and $\deg \varphi_E = 5$; both have surjective mod 5 image). Explain why the two Selmer lengths in (b) must agree. *Answer.* (a) At $\ell = 3$: $t_3(19) = 1$, $v_3(18) = 2$, and 8 is a cube but not a ninth power in $\mathbb{F}_{19}^\times$, so the Tamagawa excess is $\min(1, 2, 1) = 1$; $t_3(83) = 0$; bound $2 = v_3(720)$, Selmer length 0. At $\ell = 5$: $t_5 = 0$ at both primes and $v_5(18) = v_5(82) = 0$; bound 0, $v_5(720) = 1$, Selmer length 1. (b) `1072a1`: bound $e_5(2) = 1$, $v_5(200) = 2$, Selmer length 1; `67a1`: bound 0, $v_5(5) = 1$, Selmer length 1. The modules $\mathrm{ad}^0 T_5 E$ of a curve and of its quadratic twist are isomorphic as $G_\mathbb{Q}$-modules, and the Bloch-Kato conditions depend only on the module, so the Selmer groups coincide; the degrees differ by the omitted factor $e_5$(`1072a1`$, 2)$.

## 21.20 Notes and references

**What was read.** The paper in full, as it accompanies this book, including the printed rows of its tables; the entries quoted in Section 21.14 are copied from them. The repository named in its appendix was read for the result files behind the six data tables (Chapter 22, in its section on reproducibility); its large-range computations were not rerun.

**Sources for the imported statements.** The theorems of Diamond, Flach and Guo are cited from the published version [49]; their numbers were compared with it (Chapter 20), and a later arXiv version (arXiv:2512.02348) numbers them differently. Agashe-Ribet-Stein [3],

Ribet-Takahashi [164] and Takahashi [192], Pollack-Weston [153] (arXiv version) and Kenku [96] were checked for the book (Chapters 9, 14 and 15); the other theorem numbers are as cited by the paper, and the list in Section 21.17 says which. The standard references behind the local computations are [184] (Tate's algorithm, Tate curves), [179], [107] and [193]; behind the Hecke side [38], [8], [9], [159], [1]; behind the Selmer side [69] and [14]; for the $L$-functions [182, 183, 155]. For the Eisenstein case and the prime 2: [125, 204, 20].

**Computations.** Every curve-level number printed in this chapter was recomputed with PARI/GP [148] 2.17.2 through `cypari2`, on Cremona's files [36] (commit `25cec5e`, conductor below 10,000); the scripts are in the style of Chapter 22. The population counts attributed to the book (the 3Ns counts, the 36 and 24 violations, the class-invariance of $w_3$) were recomputed for this chapter; the others are taken from the chapters cited at each place. None of these computations is a blind test: they were made with the statements known.

**The paper and the book.** The paper's statement on the use of an AI system says that several of its results and proofs were worked out while writing a companion textbook on the note; among them are Propositions prop:pullback, prop:Sf, prop:shapes, prop:ns-structure and prop:no9, Lemmas lem:tamadd and lem:step1, Corollary cor:explicit, the equality in Theorem thm:quant and the exchanges at $q = 3$ in Lemma lem:twist. In this book they are proved in the chapters named at each result; this chapter only reads them in the paper's order.

# Chapter 22

# Computation, data and the statistical protocol

The paper's theorems are proved; its conjectures, its sharpness claims and a dozen of its smaller statements rest on computations with PARI/GP on Cremona's tables, organised by a protocol of working sets and hold-outs. This chapter makes that side of the paper as checkable as the other. It teaches PARI/GP and the `ecdata` files from zero, with exactly the functions the paper uses; reads the paper's appendix code line by line and proves what it computes; runs it on the running examples; explains the protocol and judges the statistics, including the calibration argument of Section sec:sharp; and reproduces every table of the paper that the data of this range allow on all optimal curves of conductor below 10,000, from the very data commit the paper names, with scripts that run in minutes.

## Core results at a glance

Throughout, $E$ is an optimal curve given by Cremona's minimal model, $\ell$ an odd prime, $q$ a bad prime, $v = v_\ell(\deg \varphi_E)$; outputs are those of PARI/GP 2.17.2 through `cypari2` 2.2.4, and "below 10,000" means the 38,042 optimal curves of conductor at most 9,999 in the files of `ecdata` commit `25cec5e`. Items tagged *(proved here, by computation)* are established by the computation printed or described in the item, and rely on PARI computing what its documentation says. Because this chapter is computational, its card uses five further tags: *(convention)* and *(definition)* for a choice of notation, units or populations, and for a description of a software output or file format; *(construction)* for code printed in the chapter; *(a reading of the paper's text)* for a classification that interprets the paper's sentences and could be overturned by its author; *(an observation, by computation; no proof)* for a regularity seen in the data and not explained.

**22.1 Versions and data.** PARI/GP 2.17.2 via `cypari2` 2.2.4 (the paper: 2.15, and 2.17.2 for its checks below 10,000); the six files `allcurves`, `allbsd`, `allisog`, `alllabels`, `degphi`, `galrep` for conductors 0 to 9,999, identical (SHA-256) with commit `25cec5e` of `ecdata`, the paper's; four further files of that commit for cross-checks. *(convention)* Conventions 22.1 and 22.69.

**22.2 The five bounds.** $B_\ell^{\mathrm{nc}} = \sum_{q\neq\ell} e_\ell(E,q) + \sum_q t_\ell(E,q) + [\ell=3]w_3(E)$; $B_\ell^{\mathrm{rule}} = B_\ell^{\mathrm{nc}} - h_\ell(E)$ (the right side of (eq:rule)); $B_\ell^{\mathrm{III}}$ is $B_\ell^{\mathrm{nc}}$ without the $e_\ell$ of the types III, III$^*$ at $q \neq \ell$;

$B_\ell^{\mathrm{main}} = \sum_q v_\ell(\#\Phi_q^{\mathrm{geom}})$; $B_\ell^{\mathrm{tw}} = \sum_q w_\ell(E,q)$. Violation: $v < b$; equality: $v = b > 0$; excess $v - b$; rates among $b > 0$. *(convention)* Convention 22.2.

**22.3 Populations.** CM by the thirteen CM $j$-invariants; $E[\ell]$ reducible iff a `galrep` label begins with $\ell$B or $\ell$Cs; base population at $\ell$: non-CM, $E[\ell]$ irreducible; Eisenstein: non-CM, reducible; hypothesis population: base, $\ell \nmid N$, at $\ell = 3$ image not 3Ns. Below 10,000: 38,042 optimal curves, 189 CM; base populations 33,851, 37,319, 37,691, 37,839, 37,823 at $\ell = 3, 5, 7, 11, 13$. *(convention; counts proved here, by computation)* Convention 22.3.

**22.4 Running PARI; the language; pitfalls.** `gp` or `cypari2`; without `elldata`, `ellsearch` fails, so files are read directly; default stack 8,000,000 bytes, maximum 8,003,584; `factor` drops the content; a GP file must be loaded with `read`; `mfsturm` of an `mfinit` space is smaller than Sturm's bound $k\mu(N)/12$ (50 against 112 at level 539). *(stated; reference [149]; behaviour checked here)* Remarks 22.4 and 22.5, Example 22.6, and Warning 22.7.

**22.5 `ellglobalred`.** Returns $[N, [u, r, s, t], \prod c_p, F, L]$ with $F$ the factorisation of $N$ (so `R[4][,1]` is the column of bad primes) and $L[i] = [f_p, k_p, 0, c_p]$, $k_p$ PARI's Kodaira code; for `726a1`, $L = [[1, 10, 0, 2], [1, 7, 0, 1], [2, 3, 0, 2]]$ (types $\mathrm{I}_6$, $\mathrm{I}_3$, III). *(definition)* Definition 22.8.

**22.6 Kodaira codes.** 1: $\mathrm{I}_0$; $n + 4$: $\mathrm{I}_n$ and $-(n + 4)$: $\mathrm{I}_n^*$, for $n \geq 1$ only; $-1$: $\mathrm{I}_0^*$; $\pm 2, \pm 3, \pm 4$: II, III, IV and their duals. `k >= 5` iff multiplicative; `abs(k) == 3` iff III or III$^*$. *(convention)* Convention 22.9.

**22.7 The symmetric square.** `lfuneuler(lfunsympow(E, 2), q)` $= 1/P_q(X)$, so $P_q(q^{-2}) = L_q(A_f, 1)^{-1}$ is `subst(1/lfuneuler(S, q), x, 1/q^2)`; for `121a1`, `lfunparams(S)` = `[121, 3, [0, 0, 1]]` and $P_{11} = 1 + 11X$, $P_{11}(11^{-2}) = 12/11$. *(proved here, by computation)* Remark 22.10.

**22.8 Lemma lem:euler against PARI.** The factor read off the Kodaira type equals PARI's $P_q(q^{-2})$ at all 67,745 pairs (optimal curve below 10,000, bad $q \geq 5$). *(proved here, by computation)* Example 22.11.

**22.9 Tate parameters.** `ellinit(E, O(q^k)).tate[3]` is $q_E$ (not `ellinit(E, q)`, which reduces modulo $q$, and not `elltatepairing`); $\bar{u} = 10$ for `11a1` at 11, 5 for `77c1` and `539c1` at 7, 6 for `791a1` at 7, 8 for `1577a1` at 19; wrong at $q = 2$ when $v_2(j) = -1$ (PARI 2.17.2; checked on `208a1`). *(proved here, by computation)* Example 22.12.

**22.10 `ns3` and `galrep`.** For `726a1`, $\psi_3 = (x^2 - x - 74)(3x^2 + 8x + 20)$. Among the 37,853 non-CM optimal curves below 10,000, `ns3` $= 1$ exactly on the 90 labelled `3Ns`; among the CM ones on 13 curves with $j \in \{-32768, 8000\}$, with no 3-adic label in `galrep`. *(proved here, by computation)* Example 22.13.

**22.11 Lemma lem:twist against PARI.** All 85,848 pairs (optimal curve below 10,000, odd bad $q$) agree with Lemma lem:twist(i), (ii); the exchanges at $q = 3$ hold in all 5,122 pairs. *(proved here, by computation)* Example 22.14.

**22.12 `ellmoddegree` and `ellweilcurve`.** `ellmoddegree(E)` $= \deg\varphi / c_\varphi^2$ (1, 5, 1/5 on `11a1`, `11a2`, `11a3`); `ellweilcurve` marks the optimal curve by the Smith invariants $[1, 1]$ (`11a1`, `990h3`); the largest degrees below 10,000 need a stack of 16 to 32 MB. *(stated; reference [149]; examples proved here, by computation)* Remark 22.15.

**22.13 The `ecdata` files.** Formats of `allcurves`, `allbsd`, `allisog`, `alllabels`, `degphi`, `galrep`; these six do not contain per-prime Kodaira symbols or Tamagawa numbers, the repository's `curvedata` does. *(definition)* Definition 22.17 and Example 22.19.

**22.14 Cross-checks with `curvedata`, `opt_man`, `alldegphi`.** Kodaira codes, $c_p$, $f_p$ agree with `ellglobalred` on all 64,687 curves; exactly one optimal curve per class, the one of `degphi`, with Manin constant 1. *(proved here, by computation)* Example 22.18.

**22.15 Optimality.** Curve number 1 is optimal in every class below 10,000 except `990h`, whose optimal curve is `990h3`; for $E[\ell]$ irreducible, $v_\ell$ of the degree does not depend on the curve of the class, nor, at $\ell = 3$, does $w_3$ (1,295 classes checked below 10,000; it varies in 828 classes with a 3-isogeny). *(proved here, by computation; the second statement stated, reference Chapters 9 and 14)* Remark 22.20.

**22.16 Labels.** Cremona and LMFDB labels (`11a1` = `11.a2`); Sutherland's labels, reducible iff $\ell$B or $\ell$Cs; counts of the 3-adic labels among non-CM optimal curves below 10,000. *(convention)* Convention 22.21.

**22.17 The appendix code.** The functions `rule` and `ns3`, verbatim, with a line-by-line reading. *(construction)* Construction 22.23 and Remark 22.24.

**22.18 What `rule` computes.** Given PARI's documented behaviour: `e` $= e_\ell(E, q)$ for $q \neq \ell$, `t` $= t_\ell(E, q)$ for all $q$, `e` $= w_3(E)$ at $q = \ell = 3$; `ns3(E)` $= 1$ iff $\psi_3$ is a product of two irreducible quadratics, and `ns3` $= h_3$ when $E[3]$ is irreducible; for the optimal curve (and $E[3]$ irreducible if $\ell = 3$), `rule(E, l, 1)` $= [v, B_\ell^{\mathrm{rule}}]$ and `rule(E, l)` $= [v, B_\ell^{\mathrm{III}}]$. *(proved here, given PARI's documented behaviour and Chapter 11)* Proposition 22.25.

**22.19 Where `rule` misleads.** Non-optimal models (`11a3`: `[-1, 0]`; `990h1`: `[4, 3]`), CM curves, Eisenstein curves (`11a1` at 5: `[0, 1]`), $\ell \mid N$, codes at $n = 0$, versions. *(proved here, by computation)* Warning 22.26.

**22.20 Sessions.** `rule` returns `[1, 1]` on `121a1`; `[2, 2]` on `726a1` with and without `iii` ($B_3^{\mathrm{nc}} = 3$, `ns3` $= 1$); `[2, 2]` on `539c1`; `[2, 2]` and `[1, 1]` on `980i1` at 3, 5; `[0, 0]` on `99a1` ($\deg \varphi_E = 4$, $r_E = 12$); the prime 2 contributes through PARI's factor on `336f1`, `1232l1` ($\ell = 3$) and `592c1` ($\ell = 5$); `11a1` at 5 gives `[0, 1]`, outside every statement; the Tamagawa refinement is attained on `791a1` and `1577a1`. *(proved here, by computation)* Examples 22.28 to 22.35.

**22.21 Modular degrees.** Zagier's, Cremona's and Watkins's methods; `ellmoddegree` implements Watkins's; it agrees with `degphi` on all 38,042 optimal curves below 10,000. PARI does not compute $r_E$. *(methods stated, reference [204, 35]; the check proved here, by computation)* Remarks 22.36 and 22.38 and Example 22.37.

**22.22 Units and rates.** One optimal curve per class at a fixed $\ell$; equality rate and mean excess among positive bounds; base rates. *(definition)* Definitions 22.39 and 22.40.

**22.23 Base rates and pooling.** At $\ell = 3$ below 10,000, $3 \mid \deg \varphi_E$ for 76.1% of the base population, for 100% of the curves with a prime $q \geq 5$ of type II to IV$^*$, I$_n^*$ or I$_n$ with $3 \mid n$, and for 73% to 76% for the other types; pooled rates mix ranges. *(proved here, by computation)* Example 22.41 and Remark 22.42.

**22.24 The protocol.** The four ranges; the status of each statement: (eq:main), Conjecture conj:eis blind-tested on the hold-out; (eq:twist) blind-tested if fixed before the hold-out counts of (eq:main) were seen; the Euler terms blind-tested only on the out-of-table curves; the weights $w_3$, the correction $h_3$ and the Ш-free form never blind-tested. *(a reading of the paper's text)* Remarks 22.43 to 22.45 and Paper link 22.48.

**22.25 Multiple comparisons.** $m$ false candidates surviving with probabilities $\pi$, $\pi'$ leave about $m\pi$ and $m\pi\pi'$; the risk sits in the choice of form; features read off the data are validated only by other data. *(proved here)* Remark 22.46.

**22.26 The protocol in miniature.** The minima of $v_3(\deg \varphi_E)$ at $q = 3$ on $N < 5000$ are the paper's $w_3$ and hold on $5000 \le N < 10000$ (11,699 curves with $3 \mid N$, no violation). *(proved here, by computation)* Example 22.47.

**22.27 The null model.** Under $H_0$ ($v$ independent of the bound, common law $F$): $E[X] = \sum_{b>0} F(v < b)$ without independence; with independence $\operatorname{Var} X \le E[X]$ and $P(X = 0) \le e^{-E[X]}$; with pairwise independence $P(X = 0) \le 1/E[X]$; the pooled and the bound-0 laws of $v$ agree under $H_0$. *(proved here)* Proposition 22.49.

**22.28 Calibration.** The paper: 605,000 expected violations, 0 observed. Below 10,000 at $\ell = 3$: $\hat{E} = 14{,}275$ (standard deviation 61.5), stratified 14,177, permutation mean 7,638; 0 observed; $v = 0$ for 59.7% of bound-0 curves and 23.9% of all. It shows strength, not validity. *(the paper's numbers stated, reference Section sec:sharp; the book's proved here, by computation)* Examples 22.50 and 22.51 and Remark 22.52.

**22.29 Sharpness.** Below 10,000: 14,420 equalities at $\ell = 3$ against 4,548 expected under $H_0$; equality rates 71.0%, 84.5%, 90.7%, 96.5%, 96.0% at $\ell = 3, \dots, 13$; the odd part of $\deg \varphi_E$ equals the product of the twist bounds for 17.1% of 33,137 curves, and the quotient is always an integer. *(proved here, by computation)* Examples 22.53 and 22.54.

**22.30 The tables in small.** Below 10,000: no violation of (eq:rule), of the Ш-free form, of (eq:main), of (eq:twist) or of Conjecture conj:eis at $\ell = 3, 5, 7, 11, 13$; the 36 failures without the correction are 3Ns curves, by exactly one; (eq:thm) as written, with the terms $t_3(E, 3)$, gives the same 36, and 24 if the terms at $q = 3$ are dropped; the minima of Section sec:iso equal $e_\ell$ cell by cell; the Tamagawa refinement has no exception (1,050 and 252 curves at $\ell = 3, 5$, as in the paper); strict twist inequalities in cases (a) and (b). *(proved here, by computation)* Examples 22.55 to 22.61 and 22.63 and Remark 22.64.

**22.31 A twist relation at $q = \ell$.** Below 10,000, for $\ell \le 13$: a curve of type II$^*$, III$^*$ or IV$^*$ at $\ell$ has $v_\ell$ of its degree exactly one more than its twist by $\chi_{\ell^*}$ (4,412 cases); type $\mathrm{I}_n^*$ gives 0; type $\mathrm{I}_0^*$ gives $v_\ell((\ell+1)^2 - a_\ell^2)$ when $E[\ell]$ is irreducible. *(an observation, by computation; no proof)* Example 22.62.

**22.32 Reproducibility.** The paper's repository (its scripts and result files, with which the printed tables were compared; nothing in it was rerun); the book's scripts `sessions.gp`, `table_small.gp` and the Python scripts; a checklist: pin the data, record versions, separate data and results, rerun, cross-check, state what was not checked. *(construction; convention)* Remarks 22.65 and 22.68, Constructions 22.66 and 22.67, and Convention 22.69.

## Where the paper uses this chapter

- Section 1.1 (“PARI’s `lfunsympow(E,2)` returns these factors at every prime, including 2 and 3”): Remark 22.10 says what the function returns and how $P_q(q^{-2})$ and $e_\ell(E,q)$ are read off; Example 22.33 runs it at $q=2$.
- Definition def:terms and Conjecture conj:euler: the five bounds of Convention 22.2; the term $t_\ell$ at $q=\ell$ as the code computes it (Remark 22.24(g)).
- Section 1.4 (component groups; optimality): Definition 22.8, Convention 22.9, and Remark 22.20.
- Lemma lem:euler (“In each case the values were confirmed against PARI’s `lfunsympow` on every curve of the tables”): Example 22.11.
- Lemma lem:twist and the appendix’s `twist_lemma_check.gp`: Example 22.14.
- Remark rem:wild and Proposition prop:shapes at $q=2,3$: Example 22.33.
- Proposition prop:ns (the 3-division polynomial) and the function `ns3`: Example 22.13, Proposition 22.25(ii).
- Proposition prop:forms (“PARI’s factor at 2 is $1+2X$ for every one of them”): Example 22.63(c).
- Lemma lem:tamfactor and Section sec:tam (the Tate parameter): Examples 22.12, 22.35 and 22.59.
- The Example of Section sec:proof and the Remark after it (99A1, 726a1, 539d1): Examples 22.28 and 22.30 to 22.32 and Remark 22.38.
- Section sec:data and all its subsections and tables: Sections 22.9 to 22.11; the status of each data statement is Remark 22.44; the calibration argument of Section sec:sharp is Proposition 22.49.
- The paragraph after Theorem thm:twist (the twist pairs): Example 22.61.
- Section sec:two (the 2-adic inequality; Calegari-Emerton and CM curves): Example 22.63(d), (e).
- The appendix (the functions `rule` and `ns3`, the scripts, the memory remark): Sections 22.5, 22.6 and 22.12.

The mathematics behind every computation is in earlier chapters: Tate’s algorithm and the Kodaira codes in Chapter 10, the mod 3 images in Chapter 11, optimality, Manin constants and the congruence number in Chapter 14, the symmetric square and its local factors in Chapter 19, the Tamagawa factor of the adjoint in Chapter 18, what each data statement of the paper means in Chapter 21, and the vocabulary of elementary probability and statistics in Appendix A. This chapter computes, counts and judges; it cites the mathematics and does not repeat it.

## 22.1 Introduction and conventions

The paper rests on computations as much as on proofs. Its Section sec:data reports tests of its inequalities on every optimal curve of conductor up to 500000 whose optimality the tables determine, and on 893 curves beyond; its appendix prints a PARI/GP function, `rule(E, l, iii)`, that checks one curve; a dozen sentences elsewhere appeal to PARI ("PARI's `lfunsympow(E,2)` returns these factors at every prime", "the values were confirmed against PARI's `lfunsympow` on every curve of the tables"). This chapter has four jobs. It teaches the software and the data from zero: PARI/GP with exactly the functions the paper uses, and Cremona's tables in the form of the `ecdata` files (Sections 22.2 to 22.4). It reads the appendix code line by line, proves that it computes what the paper says under stated conditions, and runs it on the running examples, including the cases where it gives an answer that is not the paper's quantity (Sections 22.5 to 22.7). It explains the paper's protocol and judges its statistics: which statement was blind-tested and which only confirmed, what the calibration argument shows and what it does not (Sections 22.8 to 22.10). And it reproduces in small every table of the paper that the data of this range allow (not the table of out-of-table curves, whose curves lie outside Cremona's tables and are listed only in the paper's repository, and of the table of isolated types only the rows at $q = 3$), on all 38,042 optimal curves of conductor below 10,000, with scripts the reader can rerun in minutes (Sections 22.11 and 22.12).

Two kinds of numbers appear. Numbers the book computed are printed with the range they cover and were produced by the code printed or described here; numbers the paper reports for its large ranges are quoted as the paper's and marked as such: they cannot be rerun from the data used in this book. The book's ranges were not blind (every statement was known before the book's data were looked at), so the book's counts are reproductions, not tests.

**Convention 22.1** (software, data and outputs)**.** (a) *Software.* The paper used PARI/GP 2.15 [148] on its large ranges and PARI/GP 2.17.2 [150] for its checks below conductor 10,000 (Section sec:data). Every output printed in this chapter was produced with PARI/GP 2.17.2 (`version()` returns `[2, 17, 2]`) through the Python interface `cypari2` 2.2.4 [40], under Python 3.13 on Linux; no stand-alone `gp` binary was used. Every GP function printed here was loaded with `read` and run as printed.

(b) *Data.* Six files of the `ecdata` repository [36] for conductors 0 to 9,999: `allcurves`, `allbsd`, `allisog`, `alllabels`, `degphi`, `galrep` (each with the suffix `.00000-09999`). Their SHA-256 sums are listed in Convention 22.69; they agree with the files of commit `25cec5e` of the repository (full hash `25cec5ecfec8b9f016eb1631ac633194c2bed39f`), the commit the paper names. Four further files of the same commit and range (`alldegphi`, `curvedata`, `classdata`, `opt_man`) and the repository's `docs/file-format.txt` were used for the cross-checks of Example 22.18. All eleven files are fetched from that commit, by its full hash, into the directory `data/ecdata` of the book's source repository by the script `tools/setup_env.sh`, which checks them against the SHA-256 sums of `tools/ecdata.sha256`; the directory itself is not under version control.

(c) *Outputs.* A listing that shows output shows it exactly as PARI 2.17.2 prints it; inputs are preceded by `?` as in a `gp` session. Where PARI 2.15 might behave differently we say so (Warning 22.26(f)); we have not run 2.15.

(d) *Models and labels.* Curves are given by Cremona's reduced global minimal models

$[a_1, a_2, a_3, a_4, a_6]$ and named by Cremona's labels, `121a1`; "optimal" means $X_0(N)$-optimal (Chapter 14). The LMFDB's labels appear only in Convention 22.21.

(e) *Letters.* As in the paper, $q$ is a bad prime and $\ell$ an odd prime; $q_E$ is a Tate parameter. In code, `x` is PARI's polynomial variable, which in the output of `lfuneuler` stands for $q^{-s}$; we write $X$ for it in the text, as the paper does in $P_q(X)$.

**Convention 22.2** (the five bounds; violation, equality, excess)**.** Let $E$ be an optimal curve, $\ell$ an odd prime, and $v = v_\ell(\deg \varphi_E)$ the *observed valuation*. With $e_\ell(E,q)$, $t_\ell(E,q)$ ($q = \ell$ included), $w_3(E)$ and $h_\ell(E)$ as in Definition def:terms and Conjecture conj:euler (Chapters 19 and 21), with $\#\Phi_q^{\mathrm{geom}}$ the order of the geometric component group at $q$ and $w_\ell(E,q)$ the twist-symmetrised weight of the paper's Section 1.4 (Chapter 10), put

$$B_\ell^{\mathrm{nc}}(E) = \sum_{q\mid N,\, q\neq\ell} e_\ell(E,q) + \sum_{q\mid N} t_\ell(E,q) + [\ell = 3]\, w_3(E), \qquad B_\ell^{\mathrm{rule}}(E) = B_\ell^{\mathrm{nc}}(E) - h_\ell(E),$$

$$B_\ell^{\mathrm{main}}(E) = \sum_{q\mid N} v_\ell(\#\Phi_q^{\mathrm{geom}}), \qquad B_\ell^{\mathrm{tw}}(E) = \sum_{q\mid N} w_\ell(E,q),$$

and let $B_\ell^{\mathrm{III}}(E)$ be $B_\ell^{\mathrm{nc}}(E)$ with the terms $e_\ell(E,q)$ of the primes $q \neq \ell$ of type III or III$^*$ left out. So $B^{\mathrm{rule}}$ is the right side of (eq:rule), $B^{\mathrm{nc}}$ the same with no correction ("all types, no correction"), $B^{\mathrm{III}}$ the "III-free" form, $B^{\mathrm{main}}$ and $B^{\mathrm{tw}}$ the right sides of (eq:main) and (eq:twist). The right side of (eq:thm), $\sum_{q\mid N} e_\ell(E,q) + \sum_{q\mid N} t_\ell(E,q)$ with $e_\ell(E,q)$ defined for $q \neq \ell$ only and $t_\ell(E,q)$ at every bad $q$, $q = \ell$ included (Definition def:terms), is $B_\ell^{\mathrm{thm}}(E) = \sum_{q\mid N,\, q\neq\ell} e_\ell(E,q) + \sum_{q\mid N} t_\ell(E,q) = B_\ell^{\mathrm{nc}}(E) - [\ell = 3]\, w_3(E)$; it equals $B_\ell^{\mathrm{nc}}(E)$ when $\ell \nmid N$ or $\ell \geq 5$. For a bound $b$: a *violation* is $v < b$; an *equality* is $v = b$ with $b > 0$; the *excess* is $v - b$. Rates of violation and of equality, and mean excesses, are always taken among the curves with $b > 0$, as in the paper.

The five bounds differ in which local information they use, and one curve shows them side by side. Take `539c1` at $\ell = 3$ (its session is Example 22.30). Its bad primes are 7, of type $\mathrm{I}_3^*$, and 11, of type $\mathrm{I}_2$. At 7 the Euler term is $e_3(E,7) = v_3(48/49) = 1$ and, since $v_7(j) = -3$, the Tamagawa term is $t_3(E,7) = v_3(3) = 1$; at 11 the Euler term is 0 (multiplicative reduction) and $t_3(E,11) = v_3(2) = 0$. As $3 \nmid N$, there is no weight $w_3$, and the mod 3 image is not 3Ns (`ns3(E)` $= 0$; the only `galrep` label is `2B`, so $E[3]$ is irreducible and $h_3(E) = 0$). Hence $B_3^{\mathrm{nc}} = B_3^{\mathrm{rule}} = B_3^{\mathrm{thm}} = 2$, and $B_3^{\mathrm{III}} = 2$ since no prime has type III or III$^*$. The component groups have orders $\#\Phi_7^{\mathrm{geom}} = 4$ (every type $\mathrm{I}_n^*$) and $\#\Phi_{11}^{\mathrm{geom}} = 2$, so $B_3^{\mathrm{main}} = 0$; the twist-symmetrised weights are $w_3(E,7) = v_3(3) = 1$ and $w_3(E,11) = v_3(2) = 0$, so $B_3^{\mathrm{tw}} = 1$. With $v = v_3(288) = 2$ the curve satisfies all five bounds, with equality in the first four and excesses 2 and 1 over $B_3^{\mathrm{main}}$ and $B_3^{\mathrm{tw}}$. The order $B_\ell^{\mathrm{main}} \leq B_\ell^{\mathrm{tw}} \leq B_\ell^{\mathrm{nc}}$ seen here holds for every curve (Exercise 22.74); what (eq:rule) adds over (eq:main) on this curve is the Euler term at 7 and the full Tamagawa term $v_3(n)$ in place of $v_3(\#\Phi_7^{\mathrm{geom}}) = 0$.

**Convention 22.3** (populations and ranges)**.** (a) A curve has *CM* if its $j$-invariant is one of the thirteen rational CM $j$-invariants 0, 1728, −3375, 8000, −32768, 54000, 287496, −884736, −12288000, 16581375, −884736000, −147197952000, −262537412640768000 (Chapter 9).

(b) $E[\ell]$ is *reducible* if and only if a label of `galrep` for $E$ begins with "$\ell$B" or "$\ell$Cs", the prime part compared exactly (Convention 22.21).

(c) The *base population at* $\ell$ is the set of optimal curves without CM with $E[\ell]$ irreducible; the *Eisenstein population at* $\ell$ is the set of optimal curves without CM with $E[\ell]$ reducible;

the *hypothesis population at* $\ell$ is the base population cut down by $\ell \nmid N$ and, at $\ell = 3$, by "image not 3Ns": the curves to which Theorem thm:main applies (Proposition prop:Sf).

(d) The paper's ranges are the working set $N \le 300000$, the hold-out $300000 < N \le 400000$, the extension $400000 < N \le 500000$ and the out-of-table curves, $500000 < N \le 4 \cdot 10^6$. The book's ranges are $N < 5000$ (the *small working set*), $5000 \le N < 10000$ (the *small hold-out*) and their union; the split is for teaching only.

Below conductor 10,000 there are 38,042 optimal curves (one per line of `degphi`), of which 189 have CM. The base populations at $\ell = 3, 5, 7, 11, 13$ have 33,851, 37,319, 37,691, 37,839, 37,823 curves, and the Eisenstein populations 4,002, 534, 162, 14, 30; each pair adds up to the 37,853 curves without CM.

Appendix A (Appendix A) defines base rates, expectation, the inequalities of Markov and Chebyshev, hypothesis tests, hold-out sets and pre-registration, and works the paper's calibration example in words; we use that vocabulary without repeating it. A reader with a physics background knows the situation from experimental practice: the hold-out is a blinded signal region, the one-time test on it is the unblinding, and the multiple-comparisons caveat of Remark 22.46 is the look-elsewhere effect.

Several earlier chapters send the reader here for the practical side of a computation: Chapter 9 for the order of the output of `ellinit` and the session at the end of its examples, Chapter 10 for the formats of Cremona's files and the use of `ellglobalred`, Chapters 8 and 14 for `ellmoddegree` and its memory needs, Chapter 19 for installing and calling PARI, Chapter 1 for the test `ns3`, and Appendix A for the protocol and its counts. Each of these is met below.

## 22.2 PARI/GP from zero

This section is for the reader who has never used PARI/GP. It covers how to run it, the part of its language that the paper's code and the book's scripts use, a first session on `121a1` and `726a1`, and the mistakes that produce wrong answers silently. In the paper it is the background of the appendix.

*Remark* 22.4 (running PARI). (i) There are two ways to run PARI. The interpreter `gp` is distributed by the PARI group [148] and packaged by the usual Linux distributions; its manual [149] documents every function used here. The Python package `cypari2` [40] gives the same library inside Python: after `import cypari2; pari = cypari2.Pari()`, the call `pari('...')` evaluates any GP expression and `pari('read("file.gp")')` loads a file of GP code. All outputs in this chapter come from the second route; the results are those of the library and print exactly as `gp` prints them.

(ii) PARI has optional data packages. One of them, `elldata`, contains Cremona's tables inside PARI and is needed by `ellsearch` and `ellidentify`. It is not installed in the book's environment, where `ellsearch(11)` fails with "error opening elldata file". This is why the book reads the `ecdata` files directly (Section 22.4); a reader with `elldata` can replace the file loops by `forell` or `ellsearch`.

(iii) PARI works on a stack. Two settings matter: `parisize`, the current size, and `parisizemax`, the size up to which PARI may enlarge it when a computation needs more. In the book's installation a fresh `cypari2` session has `default(parisize)` = 8,000,000 and

`default(parisizemax)` = 8,003,584 bytes. In `gp` one writes `default(parisizemax, 10^9)`; in Python, `pari.allocatemem(s, smax)` sets both. A computation that needs more than `parisizemax` stops with "the PARI stack overflows": an error, never a wrong answer. The function `ellmoddegree` is the only one here that needs more than the default (Remark 22.15(iv)).

*Remark* 22.5 (the part of the GP language used here). Integers and rationals are exact, of arbitrary size. The letter `x` is PARI's default polynomial variable: never assign a value to it, since `lfuneuler` returns rational functions in `x`. Vectors `[a, b, c]` are indexed from 1; `#v` is the length and `v[2..#v]` a slice; `v~` is the transpose (a column vector prints with a trailing `~`). Matrices are written `[1, 2; 3, 4]`, with entry `M[i, j]` and column `M[, j]`; a $1 \times 2$ matrix prints as `Mat([a, b])`. A function is defined by `f(a, {b = 0}) = ...`, the braces marking an optional argument with its default; local variables are introduced by `my(...)`; a definition that spans several lines of a file is enclosed in braces `{ ... }`; a comment runs from `\\` to the end of the line. Control flow: `if(c, a, b)`, `for(i = 1, n, ...)`, `foreach(v, s, ...)`, `while(c, ...)`, `next`, `return`. A `Map()` is a dictionary (`mapput`, `mapisdefined`). Strings: `Str` concatenates, `strsplit(s, " ")` splits, `readstr(file)` returns the vector of the lines of a file, and `eval` turns a string such as `"[0,-1,1,-10,-20]"` into the vector it denotes (all available in 2.17.2). Arithmetic: `valuation(r, l)` is $v_\ell(r)$ for a rational $r$, and `valuation(0, l)` is `+oo`, so that a test `vj < 0` is false when $j = 0$; `factor` factors integers, rationals and polynomials; `kronecker(D, n)` is the Kronecker symbol; `O(q^k)` is the error term of $q$-adic precision $k$. Of the functions that Chapter 2 names as counterparts of its algebra, `valuation` and `kronecker` are used here; `znprimroot`, `ffgen` and `polisirreducible` are not needed by the paper.

**Example 22.6** (a first session)**.** On `121a1` $= [1, 1, 1, -30, -76]$ and `726a1` $= [1, 1, 0, -35, -51]$ (PARI/GP 2.17.2):

```
? version()
[2, 17, 2]
? E = ellinit([1,1,1,-30,-76]);
? [E.a1, E.a2, E.a3, E.a4, E.a6]
[1, 1, 1, -30, -76]
? [E.b2, E.c4, E.c6, E.disc, E.j]
[5, 1441, 54703, -121, -24729001]
? vector(13, i, E[i])
[1, 1, 1, -30, -76, 5, -59, -303, -1249, 1441, 54703, -121, -24729001]
? ellan(E, 12)
[1, -1, 2, -1, 1, -2, 2, 3, 1, -1, 0, -2]
? ellap(E, 2)
-1
? E = ellinit([1,1,0,-35,-51]);
? elltors(E)
[2, [2], [[6, -3]]]
? ellisomat(E)[2]
[1, 2; 2, 1]
? ellisomat(ellinit([0,-1,1,-10,-20]))[2]
[1, 5, 5; 5, 1, 25; 5, 25, 1]
```

The member functions `E.a1`, ..., `E.j` return the Weierstrass data of Chapter 9; for `121a1`, $\Delta = -121 = -11^2$ and $j = -24729001$. The first thirteen components of the vector `ellinit(E)` are, in this order, $a_1, a_2, a_3, a_4, a_6, b_2, b_4, b_6, b_8, c_4, c_6, \Delta, j$ (`E[1]` to `E[13]`; here $b_4 = -59$, $b_6 = -303$,

$b_8 = -1249$); the further components depend on the base field and the version, so the member functions are the safe way to read them. The vector `ellan(E, 12)` lists $a_1, \dots, a_{12}$; $a_{11} = 0$ because the reduction at 11 is additive. For `726a1`, `elltors` returns the order of $E(\mathbb{Q})_{\text{tors}}$, its structure and a generator: $\mathbb{Z}/2$, generated by $(6, -3)$. The second component of `ellisomat` is the matrix of isogeny degrees between the curves of the class: the class of `726a1` has two curves joined by a 2-isogeny, and that of `11a1` three curves joined by 5-isogenies, with the degree-25 isogeny between the second and the third. These are the matrices of the `allisog` file (Definition 22.17).

*Warning* 22.7 (pitfalls). (a) `ellinit` accepts any Weierstrass model. Local data (`elllocalred`, `ellglobalred`) are always those of a minimal model, and `ellglobalred` reports the change of variables to one; `ellminimalmodel` returns the minimal model itself. The member functions, on the other hand, are those of the model given.

(b) Never assign to `x` (Remark 22.5).

(c) A $q$-adic object carries its precision `O(q^k)`; results are correct only to that precision.

(d) A stack overflow is an error, not a wrong answer: raise `parisizemax` and rerun.

(e) GP indexes from 1 and Python from 0. In `cypari2`, the GP entry `R[5][i][2]` is `R[4][i-1][1]`, and the GP matrix entry `R[4][i, 1]` is `R[3][i-1, 0]`. `cypari2` does not accept Python slices of a PARI matrix (the expression `R[3][:,0]` raises a `TypeError`), so whole columns are best taken in GP, for instance with `pari('(g) -> g[4][,1]~')(R)`.

(f) `factor` of a polynomial with integer coefficients returns its irreducible factors in $\mathbb{Z}[x]$, primitive and with positive leading coefficient, and drops the content, sign included: `factor(3*x^4 + 6)` is `Mat([x^4 + 2, 1])` (Example 22.13). This is why the paper's `ns3` does not see the leading coefficient 3 of $\psi_3$.

(g) A file of GP code must be loaded with `read`. Pasted into `pari('...')` as one string, the paper's file `euler_rule.gp` does not produce two functions: in our test the definition of `ns3` became part of the body of `rule`, and `rule(E, 3)` returned a function instead of a pair. Reading the file is the reliable route.

(h) `mfsturm` depends on its argument. On a pair $[N, k]$ it returns Sturm's bound $k\mu(N)/12$, $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$ (Chapter 12): `mfsturm([539, 2])` is 112. On a space created by `mfinit` it returns a smaller number adapted to that space: `mfsturm(mfinit([539, 2]))` is 88 and, for the cuspidal space, `mfsturm(mfinit([539, 2], 1))` is 50. The smaller numbers are not valid bounds for the integrality of an arbitrary rational combination of $q$-expansions; used as truncation points in the congruence-module computation of Chapter 14 at level 539, they give wrong modules that depend on the truncation: for the four forms of Chapter 20, the coefficients $a_1, \dots, a_{50}$ give the orders 1152, 2016, 5760, 10080, and $a_1, \dots, a_{51}$ give 288, 504, 1440, 2520, while Sturm's bound 112 gives the correct 288, 504, 288, 504 (Chapter 20). For congruences and integrality, use $k\mu(N)/12$.

## 22.3 Invariants of a curve in PARI

This section goes through the PARI functions that compute the local invariants the paper uses: the local data of `ellglobalred`, the Euler factors of the symmetric square, the Tate parameter,

the 3-division polynomial, quadratic twists and the modular degree. For each it says what the output is, prints an exact session, and checks a lemma of the paper against PARI on every optimal curve below conductor 10,000. In the paper this is the content of the sentence of Section sec:data that lists what was recomputed with PARI, and of the checks claimed in Lemma lem:euler, Lemma lem:twist and the appendix.

**Definition 22.8** (the output of `ellglobalred` and `elllocalred`)**.** For an elliptic curve $E$ over $\mathbb{Q}$ given by any model, `elllocalred(E, p)` is the vector $[f_p, k_p, [u, r, s, t], c_p]$: the conductor exponent at $p$, PARI's Kodaira code $k_p$ (Convention 22.9), the change of variables to a model minimal at $p$, and the Tamagawa number $c_p = \#\Phi_p(\mathbb{F}_p)$. The vector `R = ellglobalred(E)` is $[N, [u, r, s, t], \prod_p c_p, F, L]$: the conductor; the change of variables to a global minimal model; the product of the Tamagawa numbers; the factorisation matrix $F$ of $N$, whose first column `R[4][,1]` lists the bad primes in increasing order; and the vector $L$ with $L[i]$ equal to `elllocalred(E, p_i)` for the $i$-th bad prime, except that over $\mathbb{Q}$ the local change of variables is replaced by 0 [149]. So the paper's `P = R[4][,1]` is the column of bad primes and its `K[i][2]`, with `K = R[5]`, the Kodaira code at the $i$-th of them.

For `726a1` (PARI/GP 2.17.2):

```
? E = ellinit([1,1,0,-35,-51]);
? R = ellglobalred(E)
[726, [1, 0, 0, 0], 4, [2, 1; 3, 1; 11, 2], [[1, 10, 0, 2], [1, 7, 0, 1], [2, 3, 0,
    2]]]
? R[4][,1]
[2, 3, 11]~
? R[5][3][2]
3
? elllocalred(E, 11)
[2, 3, [1, 0, 0, 0], 2]
? elllocalred(ellinit([0,1,0,4,4]), 2)
[2, -4, [1, 0, 0, 0], 3]
```

The model is already minimal (the change of variables is the identity $[1, 0, 0, 0]$), $N = 2 \cdot 3 \cdot 11^2$, the types are $\mathrm{I}_6$, $\mathrm{I}_3$, III (codes 10, 7, 3) with $c_2 = 2$, $c_3 = 1$, $c_{11} = 2$ and product 4. For `20a1` $= [0, 1, 0, 4, 4]$ at 2 the code $-4$ is $\mathrm{IV}^*$, with $f_2 = 2$ and $c_2 = 3$. The mathematics (Tate's algorithm, why $c_q$ divides $\#\Phi_q^{\mathrm{geom}}$ and can be smaller) is in Chapter 10; this is the data structure only. Cremona's file `curvedata` stores the same bad primes, codes, Tamagawa numbers and conductor exponents, and they agree with `ellglobalred` on every curve below conductor 10,000 (Example 22.18).

**Convention 22.9** (Kodaira codes in code)**.** PARI's code $k$ is 1 for $\mathrm{I}_0$; $n+4$ for $\mathrm{I}_n$ and $-(n+4)$ for $\mathrm{I}_n^*$, *for* $n \geq 1$ *only*; $-1$ for $\mathrm{I}_0^*$; $2, 3, 4$ for $\mathrm{II}, \mathrm{III}, \mathrm{IV}$ and $-2, -3, -4$ for $\mathrm{II}^*, \mathrm{III}^*, \mathrm{IV}^*$ (Chapter 10). The paper's appendix states the codes "$n+4$ for $\mathrm{I}_n$ and $-(n+4)$ for $\mathrm{I}_n^*$ when $n \geq 1$"; read without the last clause, $n+4$ at $n = 0$ would give 4, the code of IV. The tests used in the paper's code and in the book's scripts:

| test on $k$ | types selected | use |
|---|---|---|
| `k >= 5` | $\mathrm{I}_n$, $n = k-4 \geq 1$ | multiplicative: divide by the naive factor; $\#\Phi_q^{\mathrm{geom}} = n$ |
| `k <= -5` | $\mathrm{I}_n^*$, $n = -k-4 \geq 1$ | $t_\ell = v_\ell(n)$ when $q$ is odd |
| `k == -1` | $\mathrm{I}_0^*$ | $\#\Phi_q^{\mathrm{geom}} = 4$ |
| `abs(k) == 3` | III, III$^*$ | left out unless `iii` is set |
| `k == 4 \|\| k == 2` | IV, II | $w_3 = 1$ at $q = \ell = 3$ |
| `k == -4 \|\| k == -2` | IV$^*$, II$^*$ | $w_3 = 2$ at $q = \ell = 3$ |
| `k == -3` | III$^*$ | $w_3 = 1$ at $q = \ell = 3$ |

*Remark* 22.10 (the symmetric square in PARI). `S = lfunsympow(E, 2)` is PARI's $L$-function object for $L(\mathrm{Sym}^2 f, s)$, in the normalisation of Chapter 19, with the true local factors at every prime. `lfunparams(S)` returns the conductor of $L(\mathrm{Sym}^2 f, s)$, the number $k$ of the functional equation $s \leftrightarrow k-s$, and the gamma shifts. `lfuneuler(S, q)` is the Euler factor at $q$ as a rational function of $X = q^{-s}$ (printed in the variable `x`): it is $1/P_q(X)$. The point $X = q^{-2}$ is forced by the normalisation: in the book's conventions $L(\mathrm{Sym}^2 f, s) = L(\mathrm{ad}^0 f, s-1)$ (Chapter 19), so the adjoint factor at $s = 1$ is the symmetric-square factor at $s = 2$, that is, at $X = q^{-2}$; a reader who substitutes $X = q^{-1}$ gets a different, meaningless number. Hence $P_q(q^{-2}) = L_q(A_f, 1)^{-1}$ is the exact rational `subst(1/lfuneuler(S, q), x, 1/q^2)`, and $e_\ell(E, q)$ is the $\ell$-adic valuation of its quotient by $1 - a_q^2 q^{-2}$ (Definition def:terms). For `121a1` (PARI/GP 2.17.2):

```
? S = lfunsympow(ellinit([1,1,1,-30,-76]), 2);
? lfunparams(S)
[121, 3, [0, 0, 1]]
? lfuneuler(S, 11)
1/(11*x + 1)
? lfuneuler(S, 2)
1/(-8*x^3 - 2*x^2 + x + 1)
? subst(1/lfuneuler(S, 11), x, 1/11^2)
12/11
```

The conductor of $L(\mathrm{Sym}^2 f, s)$ is 121, the functional equation relates $s$ and $3-s$, and the gamma factor has shifts $0, 0, 1$. At the additive prime 11 (type II, $11 \equiv 2 \pmod 3$) the factor is $P_{11}(X) = 1 + 11X$, the row II of the paper's table with $\chi_{-3}(11) = -1$, and $P_{11}(11^{-2}) = 12/11$, so $e_3(E, 11) = 1$. At the good prime 2, with $a_2 = -1$, $P_2(X) = 1 + X - 2X^2 - 8X^3 = (1-2X)(1+3X+4X^2)$, the factor $(1-\alpha^2 X)(1-2X)(1-\beta^2 X)$ with $\alpha+\beta = -1$, $\alpha\beta = 2$: indeed $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 1 - 4 = -3$ and $\alpha^2\beta^2 = 4$, so $(1-\alpha^2 X)(1-\beta^2 X) = 1 + 3X + 4X^2$, and the middle factor is $1 - \alpha\beta X = 1 - 2X$. This is a useful check on the variable: the good factors are known in closed form, so a wrong reading of `x` shows up at once. The bad factors are computed when `S` is created; creating `S` and reading all bad factors took about 0.08 milliseconds per curve on the first 3,000 optimal curves of the tables. The general constructor `lfuncreate` is not needed. `lfun(S, 2)` is the value $L(\mathrm{Sym}^2 f, 2)$ that `ellmoddegree` uses.

**Example 22.11** (the omitted factors by hand, and the check of Lemma lem:euler)**.** For a bad prime $q \geq 5$, Lemma lem:euler reads $L_q(A_f, 1)^{-1}$ off the Kodaira type (Chapter 19). The function `euler_by_type` of the script `sessions.gp` (Construction 22.66) implements the table: $1-q^{-2}$ for $\mathrm{I}_n$ and $\mathrm{I}_n^*$; $1-\chi_{-3}(q)/q$ for II, II$^*$, IV, IV$^*$; $1-\chi_{-4}(q)/q$ for III, III$^*$; and $(q-1)((q+1)^2 - a'^2)/q^3$ for $\mathrm{I}_0^*$, where $a'$ is $a_q$ of the twist by $q^*$, computed as `ellap(ellinit(elltwist(E, qs)), q)` with `qs` $= q^*$. The function `eulercheck(E)` returns the bad primes $q \geq 5$ at which this differs from PARI's factor. For `539d1` $= [0, 1, 1, -16, -66]$, of type $\mathrm{I}_0^*$ at 7, with $a' = -2$ (PARI/GP 2.17.2):

```
? E = ellinit([0,1,1,-16,-66]);
? euler_by_type(E, 7)
360/343
? subst(1/lfuneuler(lfunsympow(E, 2), 7), x, 1/7^2)
360/343
? eulercheck(E)
[]
```

So $e_3(E,7) = v_3(360/343) = 2$: $(7-1)(64-4)/7^3 = 360/343$. The other rows of the table are as easy to test by hand. For `121a1` at 11 (type II, $11 \equiv 2 \pmod 3$, so $\chi_{-3}(11) = -1$) the table gives $1 - \chi_{-3}(11)/11 = 12/11$, and PARI's $P_{11}(X) = 1 + 11X$ of Remark 22.10 gives $P_{11}(11^{-2}) = 1 + 11/121 = 12/11$; at a multiplicative prime the table's $1 - q^{-2}$ is PARI's $1 - X$ at $X = q^{-2}$. *Check.* On all 67,745 pairs (optimal curve of conductor below 10,000, bad prime $q \geq 5$) the two values agree: 58,860 pairs of type $\mathrm{I}_n$, 2,709 of type $\mathrm{I}_n^*$, 3,588 of types II, II$^*$, IV, IV$^*$, 1,676 of types III, III$^*$ and 912 of type $\mathrm{I}_0^*$, with no mismatch. This is the reader's version of the paper's sentence "In each case the values were confirmed against PARI's `lfunsympow` on every curve of the tables" (proof of Lemma lem:euler); Chapter 19 reports the same check over all curves.

**Example 22.12** (the Tate parameter)**.** `Eq = ellinit(E, O(q^k))` is $E$ over $\mathbb{Q}_q$ with $q$-adic precision $k$, and when $v_q(j) < 0$ its member `Eq.tate` has as third component the Tate parameter $q_E \in \mathbb{Q}_q$, with $v_q(q_E) = -v_q(j)$ (Chapter 10). The function `tatedata(E, q, l)` of `sessions.gp` returns $[n, q_E, \bar{u}, k]$: $n = v_q(q_E)$; $q_E$ to precision $q^{30}$; $\bar{u}$, the reduction modulo $q$ of the unit part $u = q_E q^{-n}$, as an integer in $[0, q-1]$; and the largest $k \leq v_\ell(q-1)$ such that $\bar{u}$ is an $\ell^k$-th power in $\mathbb{F}_q^\times$, found by the test $\bar{u}^{(q-1)/\ell^k} = 1$. In the notation of the paper and of Chapter 18, $k = v_\ell(c(u))$, the valuation of the class of $\bar{u}$; we write $c(u)$ and never $c_q$, which is a Tamagawa number. Sessions (PARI/GP 2.17.2):

```
? E = ellinit([0,-1,1,-10,-20]);
? ellinit(E, O(11^8)).tate[3]
10*11^5 + 2*11^6 + 6*11^7 + 6*11^8 + 5*11^9 + 4*11^10 + 4*11^11 + 11^12 + O(11^13)
? T = tatedata(E, 11, 5); [T[1], T[3], T[4]]
[5, 10, 1]
? T = tatedata(ellinit([1,1,0,4,11]), 7, 3); [T[1], T[3], T[4]]
[3, 5, 0]
? T = tatedata(ellinit([1,0,1,170,-3237]), 7, 3); [T[1], T[3], T[4]]
[3, 5, 0]
? tatedata(ellinit([1,1,0,4,11]), 7, 3)[2] == tatedata(ellinit([1,0,1,170,-3237]), 7,
    3)[2]
1
? T = tatedata(ellinit([1,0,1,-31,117]), 7, 3); [T[1], T[3], T[4]]
[3, 6, 1]
? T = tatedata(ellinit([0,-1,1,30,-335]), 19, 3); [T[1], T[3], T[4]]
[3, 8, 1]
? ellinit(ellinit([1,1,0,4,11]), 7)
[]
```

For `11a1` at 11: $n = 5$, $q_E = 10 \cdot 11^5 + 2 \cdot 11^6 + \cdots$, $\bar{u} = 10 \equiv -1$, a fifth power in $\mathbb{F}_{11}^\times$ (the fifth powers are $\pm 1$), so $k = 1$ at $\ell = 5$. For `77c1` $= [1,1,0,4,11]$ at 7 (type $\mathrm{I}_3$, $a_7 = -1$, nonsplit): $n = 3$, $\bar{u} = 5$, not a cube in $\mathbb{F}_7^\times$ (the cubes are $\pm 1$), $k = 0$. Its twist `539c1` by $\chi_{-7}$ (type $\mathrm{I}_3^*$ at

7) has the same $j$, hence the same Tate parameter: PARI returns the same series. For `791a1` $= [1,0,1,-31,117]$ at 7, $\bar{u} = 6 \equiv -1$ is a cube; for `1577a1` $= [0,-1,1,30,-335]$ at 19, $\bar{u} = 8 = 2^3$ is a cube but not a ninth power, so $k = 1 < 2 = v_3(18)$. These are the examples of Chapters 17 and 18.

Why the test $\bar{u}^{(q-1)/\ell^k} = 1$ decides whether $\bar{u}$ is an $\ell^k$-th power: $\mathbb{F}_q^\times$ is cyclic of order $m = q-1$; let $g$ generate it and write $\bar{u} = g^a$. If $\ell^k \mid m$, then $\bar{u} = (g^b)^{\ell^k}$ is solvable exactly when $a \equiv b\ell^k \pmod m$ is, that is, when $\ell^k \mid a$; and $\bar{u}^{m/\ell^k} = g^{am/\ell^k} = 1$ exactly when $m \mid am/\ell^k$, that is, again when $\ell^k \mid a$. So the $\ell^k$-th powers are the $m/\ell^k$ roots of $u^{m/\ell^k} = 1$: in $\mathbb{F}_{11}^\times$ the fifth powers are the roots of $u^2 = 1$, namely $\pm 1$; in $\mathbb{F}_7^\times$ the cubes are $\pm 1$; in $\mathbb{F}_{19}^\times$ the cubes are the six roots of $u^6 = 1$ and the ninth powers are $\pm 1$. The tests behind the four values of $\bar{u}$ above (PARI/GP 2.17.2):

```
? Mod(10,11)^2
Mod(1, 11)
? Mod(5,7)^2
Mod(4, 7)
? Mod(6,7)^2
Mod(1, 7)
? Mod(8,19)^6
Mod(1, 19)
? Mod(8,19)^2
Mod(7, 19)
```

So 10 is a fifth power modulo 11, 5 is not a cube modulo 7 while 6 is, and 8 is a cube modulo 19 but not a ninth power. The loop `while` in `tatedata` runs exactly these tests for $k = 1, 2, \ldots$ until one fails or $k$ reaches $v_\ell(q-1)$. Three warnings. First, `elltatepairing` is the Tate pairing on torsion points and has nothing to do with the Tate parameter. Second, `ellinit(E, q)` with an integer prime $q$ reduces $E$ modulo $q$: at the bad prime 7 of `77c1` it returns `[]`, and at a good prime the curve over $\mathbb{F}_q$, never the curve over $\mathbb{Q}_q$, which needs the precision `O(q^k)`. Third, at $q = 2$ PARI 2.17.2 returns a wrong parameter when $v_2(j) = -1$ (Chapter 10: the output is $q_E^5$ instead of $q_E$; for `208a1` $= [0,-1,0,8,-16]$, of type $\mathrm{I}_5^*$ at 2 (code $-9$) with $v_2(j) = -1$, PARI returns a parameter of valuation 5, equal to the fifth power of the one obtained from $j$); there one inverts the series $j = \mathsf{q}^{-1} + 744 + 196884\mathsf{q} + \cdots$ of Chapter 10 instead. The paper's Tamagawa test never needs $q = 2$, since $q \equiv 1 \pmod \ell$ fails for $q = 2$ and every odd $\ell$; the $t_\ell$ terms use only $v_q(j)$.

**Example 22.13** (the mod 3 image from the 3-division polynomial)**.** `elldivpol(E, 3)` is the 3-division polynomial $\psi_3$, of degree 4 and leading coefficient 3, whose roots are the $x$-coordinates of the eight points of order 3. For `726a1` (PARI/GP 2.17.2):

```
? E = ellinit([1,1,0,-35,-51]);
? elldivpol(E, 3)
3*x^4 + 5*x^3 - 210*x^2 - 612*x - 1480
? factor(elldivpol(E, 3))
[x^2 - x - 74, 1; 3*x^2 + 8*x + 20, 1]
? ns3(E)
1
? factor(3*x^4 + 6)
Mat([x^4 + 2, 1])
? ns3(ellinit([0,-1,1,-7,10]))
1
```

The pattern 2+2 is that of image $N_s(\mathbb{F}_3)$ when $E[3]$ is irreducible (Proposition prop:ns; Chapter 11 has the table of patterns and images), and the paper's `ns3(E)` returns 1. `polgalois` requires an irreducible polynomial: on this $\psi_3$ it stops with "not an irreducible polynomial", so it can be used only when $\psi_3$ is irreducible. *Check.* Among the 37,853 optimal curves without CM below conductor 10,000, `ns3(E)` $= 1$ exactly for the 90 curves whose `galrep` entry contains `3Ns`, with no mismatch either way. Among the 189 optimal CM curves, `ns3` is 1 for 13: `121b1`, `256a1`, `256d1`, `1089e1`, `1936f1`, `2304c1`, `2304k1`, `3025a1`, `5929a1`, `6400f1`, `6400r1`, `7744a1`, `7744s1`, all with $j = -32768$ or 8000 (the last session line is `121b1`). Their image is $N_s(\mathbb{F}_3)$, but `galrep` gives them no 3-adic label: Cremona's file omits the normaliser-of-Cartan images forced by complex multiplication (Chapter 11), while it does list reducible images of CM curves, for instance `11B.1.3` for `121b1` and `3Cs.1.1` for `27a1`; 97 of the 189 optimal CM curves have at least one label. This is why CM is decided by $j$ (Convention 22.3).

**Example 22.14** (quadratic twists, and the check of Lemma lem:twist)**.** `elltwist(E, d)`, for a discriminant $d$, is a model of the quadratic twist of $E$ by $\mathbb{Q}(\sqrt{d})$, not necessarily minimal; `ellminimalmodel` returns Cremona's reduced minimal model, which can be looked up in `allcurves`. The function `twistcheck(E)` of `sessions.gp` lists, for each odd bad prime $q$, the code of $E$ at $q$, the code at $q$ of the twist $E^{(q)}$ by $q^*$, and the conductor of $E^{(q)}$ (PARI/GP 2.17.2):

```
? twistcheck(ellinit([1,1,1,-30,-76]))
[[11, 2, -4, 121]]
? twistcheck(ellinit([1,0,1,170,-3237]))
[[7, -7, 7, 77], [11, 6, -6, 5929]]
? twistcheck(ellinit([0,0,0,1568,-72716]))
[[5, 5, -5, 4900], [7, -9, 9, 140]]
? E = ellminimalmodel(ellinit(elltwist(ellinit([1,1,1,-30,-76]), -11))); [E.a1, E.a2,
    E.a3, E.a4, E.a6]
[1, 1, 0, -3632, 82757]
```

`121a1` (II at 11) twists to a curve of type IV$^*$ and the same conductor, `121c2` $= [1,1,0,-3632,82757]$; `539c1` goes to `77c` at 7 ($\mathrm{I}_3^* \to \mathrm{I}_3$) and to conductor $5929 = 7^2 \cdot 11^2$ at 11 ($\mathrm{I}_2 \to \mathrm{I}_2^*$); `980i1` goes to conductor 4900 at 5 and to `140b` at 7 ($\mathrm{I}_5^* \to \mathrm{I}_5$). *Check.* For every optimal curve below conductor 10,000 and every odd bad prime $q$ we compared the code and the conductor exponent of $E^{(q)}$ at $q$ with Lemma lem:twist(ii): $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ and $\mathrm{I}_0^* \to \mathrm{I}_0$ at every odd $q$, and the exchanges II $\leftrightarrow$ IV$^*$, III $\leftrightarrow$ III$^*$, IV $\leftrightarrow$ II$^*$ at $q \geq 5$. All 85,848 pairs agree (72,003 of type $\mathrm{I}_n$, 6,348 of type $\mathrm{I}_n^*$, 2,233 of type $\mathrm{I}_0^*$, 5,264 potentially good at $q \geq 5$), and in every pair the code and the conductor exponent at every other bad prime, 2 included, are unchanged, which is Lemma lem:twist(i). At $q = 3$ the same three exchanges, with $f_3$ unchanged, hold in all 5,122 pairs where $E$ has potentially good additive reduction at 3 of one of these six types; Chapter 10 proves this. The paper's own script `twist_lemma_check.gp` checked 4,577 random pairs (the appendix).

*Remark* 22.15 (the modular degree in PARI). (i) `ellmoddegree(E)` returns the modular degree of the given curve divided by the square of its Manin constant [149]: the degree $\deg\varphi$ of the minimal parametrisation $X_0(N) \to E$ divided by $c_\varphi^2$ (Chapter 14). On the optimal curve, where $c_E = 1$ in all of Cremona's tables, it is $\deg\varphi_E$; on another curve of the class it can be a fraction. For the class `11a`:

```
? [ellmoddegree(ellinit([0,-1,1,-10,-20])), ellmoddegree(ellinit
    ([0,-1,1,-7820,-263580])), ellmoddegree(ellinit([0,-1,1,0,0]))]
[1, 5, 1/5]
? W = ellweilcurve(ellinit([0,-1,1,0,0])); W[2]
[[1, 1/5], [1, 1], [5, 1]]
? apply(e -> [e.a4, e.a6], W[1])
[[0, 0], [-10, -20], [-7820, -263580]]
? W = ellweilcurve(ellinit([1,-1,1,-96608,-11533373])); W[2]
[[3, 1], [1, 1], [6, 1], [2, 1]]
? apply(e -> [e.a4, e.a6], W[1])
[[-96608, -11533373], [-1568, -4669], [-95528, -11804669], [6112, -41533]]
? [ellmoddegree(ellinit([1,-1,1,-96608,-11533373])), ellmoddegree(ellinit
    ([1,-1,1,-1568,-4669]))]
[5184, 1728]
```

(ii) `ellweilcurve(E)` returns $[vE, vS]$: minimal models of the curves of the class (in the order of `ellisomat`) and, for each, the Smith invariants of its lattice of modular symbols inside the lattice of the newform; the strong Weil (optimal) curve is the one with invariants $[c_E, c_E]$, $c_E$ its Manin constant: the documentation of 2.17.2 says "the strong Weil curve amongst the curves isogenous to $E$ is the one whose Smith invariants are $[c, c]$, where $c$ is the Manin constant, conjecturally equal to 1", and illustrates it on `11a3` with $[1, 1]$ [149]. Called on `11a3` it lists `11a3`, `11a1`, `11a2`, and the entry $[1, 1]$ marks `11a1`; called on `990h1` it lists `990h1`, `990h3`, `990h2`, `990h4`, and $[1, 1]$ marks `990h3`. This is how PARI identifies the optimal curve for moderate conductors.

(iii) The method is Watkins's [204]: compute $L(\mathrm{Sym}^2 f, 2)$ with the true bad factors, deduce the Petersson norm, and divide by the area of the period lattice (Chapters 14 and 19). The `lfun` machinery prints "`_lfuninit_worker: Warning: increasing stack size`" when it enlarges the stack.

(iv) *Cost and memory* (PARI/GP 2.17.2). A random sample of 50 optimal curves below conductor 10,000 took 0.135 seconds per curve on a two-core machine that was running the full check of Example 22.37 at the same time. With the default stack (Remark 22.4) `ellmoddegree` succeeds on `9999m1` ($\deg \varphi_E = 5632$) but overflows on the curves of largest degree below 10,000: `9510e1` ($\deg \varphi_E = 100{,}254{,}000$) succeeds with a 16 MB stack, `9690m1` (68,217,600) needs more than 16 MB and succeeds with 32 MB, and `7910b1` (31,272,384) overflows the default and succeeds with 16 MB; so of these only `9690m1` needs more than 16 MB (each run in a fresh process). This agrees with the paper's appendix: "The function `ellmoddegree` needs about 32 MB of stack with PARI 2.17 for conductors in the thousands (with PARI 2.15 and larger conductors, `parisizemax` of a few hundred megabytes)."

(v) *Checks.* `ellmoddegree` agrees with `degphi` on all 38,042 optimal curves below 10,000 (Example 22.37); and on all 2,650 non-optimal curves of conductor below 1000, `ellmoddegree(E)` equals the degree of `alldegphi` divided by the square of the Manin constant of `opt_man`, the value being a fraction for 33 of them (`11a3`, `14a4`, `14a6`, `15a3`, ...).

*In the paper* 22.16. Section sec:data: "The Kodaira types and discriminant exponents at the bad primes were recomputed with PARI/GP 2.15 [148] from the $a$-invariants, the Euler factors of $\mathrm{Sym}^2 f$ at every bad prime were computed with `lfunsympow` and `lfuneuler`, the Tate parameters with `ellinit` over $\mathbb{Q}_q$" (with a parenthesis on $q = 2$, Example 22.12), "and fifteen modular

degrees spanning the configurations were recomputed with `ellmoddegree`, all agreeing with the tables." Each of these computations is one of the items above (Definition 22.8, Remarks 22.10 and 22.15, and Example 22.12); below conductor 10,000 every one of them has been run on every optimal curve.

The functions of this section act on one curve at a time; the paper's counts run over hundreds of thousands. The next section describes where the curves, their optimal models and their degrees come from, so that a loop over the tables can feed these functions.

## 22.4 Cremona's tables and the `ecdata` files

The paper's data are Cremona's tables of elliptic curves over $\mathbb{Q}$ [35], in the form of the `ecdata` repository [36]. This section describes the files used in this book, shows how to read them in GP and in Python, records what the files say about optimality, and fixes the labels. In the paper it is the background of the first paragraph of Section sec:data and of Section 1.4.

**Definition 22.17** (the `ecdata` files)**.** The repository consists of plain text files, one per range of 10,000 conductors, grouped in directories by kind; each line describes one curve or one isogeny class, with fields separated by single spaces, and $a$-invariants written `[a1,a2,a3,a4,a6]` without spaces. A Cremona label $N$, class, number is written in three fields (`11 a 1`) or as one word (`11a1`). The six files used for the counts of this chapter, for conductors 0 to 9,999, have the following formats (read off the files and the repository's description `docs/file-format.txt`):

| file | lines | one line per | fields |
|---|---|---|---|
| `allcurves` | 64,687 | curve | $N$, class, number, $[a_i]$, rank, $\#E(\mathbb{Q})_{\mathrm{tors}}$ |
| `allbsd` | 64,687 | curve | as `allcurves`, then $\prod_p c_p$, $\Omega$, $L^{(r)}(E,1)/r!$, regulator, analytic order of Sha |
| `allisog` | 38,042 | class | $N$, class, 1, $[a_i]$ of the optimal curve, the list of $[a_i]$ of the class, the isogeny matrix |
| `alllabels` | 64,687 | curve | Cremona's $N$, class, number, then the LMFDB's $N$, class, number |
| `degphi` | 38,042 | class | $N$, class, 1, $\deg\varphi_E$, the set of primes dividing it, $[a_i]$ of the optimal curve |
| `galrep` | 64,687 | curve | label, then the labels of the non-surjective mod $\ell$ images |

Sample lines, in order from `allcurves`, `allbsd`, `allisog`, `alllabels`, `degphi` (three lines) and `galrep` (two lines):

```
11 a 1 [0,-1,1,-10,-20] 0 5
11 a 1 [0,-1,1,-10,-20] 0 5 5 1.26920930427955 0.253841860855911 1.00000000000000 1
11 a 1 [0,-1,1,-10,-20] [[0,-1,1,-10,-20],[0,-1,1,-7820,-263580],[0,-1,1,0,0]]
    [[1,5,5],[5,1,25],[5,25,1]]
11 a 2 11 a 1
11 a 1 1 {} [0,-1,1,-10,-20]
726 a 1 144 {2,3} [1,1,0,-35,-51]
990 h 1 1728 {2,3} [1,-1,1,-1568,-4669]
11a1 5Cs.1.1
726a1 2B 3Ns
```

So `11a1` has rank 0, torsion of order 5, $\prod_p c_p = 5$, $\Omega = 1.2692\ldots$ and $L(E,1) = \Omega/5$; its class has three curves; its modular degree is 1; and its mod 5 image is `5Cs.1.1`. A `galrep` line lists an image only for the primes $\ell$ at which it is not $\mathrm{GL}_2(\mathbb{F}_\ell)$: 20,930 of the 64,687 lines have no label at all. In the `degphi` and `allisog` files the third field is always 1 and the equation is that of the optimal curve, which is curve number 1 of its class except in one class (Remark 22.20).

The repository holds more: among the directories at commit `25cec5e` are `allgens` (generators), `aplist` ($a_p$ for $p < 100$), `intpts` (integral points), `alldegphi` (one line per curve: the degree of the minimal parametrisation of *every* curve of the class), `opt_man` (one line per curve: a flag marking the optimal curve, and the Manin constant), `curvedata` (one line per curve, with many fields, among them the bad primes, the Tamagawa numbers, the Kodaira symbols and the conductor exponents), `classdata` (one line per class: the isogeny matrix and the first coefficients of the newform), and `2adic`. The six files of the table do not contain the Kodaira symbols or the local Tamagawa numbers (only $\prod_p c_p$, in `allbsd`); `curvedata` does. So the paper's description of the repository ("which contain every elliptic curve over $\mathbb{Q}$ of conductor below 500000 with its Kodaira symbols, Tamagawa numbers, modular degree, isogeny class, torsion and the images of the mod $\ell$ Galois representations") is accurate for the repository as a whole.

**Example 22.18** (cross-checks against the further files)**.** At commit `25cec5e`, for conductors 0 to 9,999:

(a) the bad primes, the Kodaira codes (stored in PARI's coding), the Tamagawa numbers and the conductor exponents of `curvedata` agree with `ellglobalred` on all 64,687 curves;

(b) `opt_man` flags exactly one curve in each of the 38,042 classes as optimal, always the curve whose equation is in `degphi`, and gives it Manin constant 1; over all curves the Manin constants are 1 (64,592 curves), 2 (70), 3 (22), 4 (2) and 5 (1, the curve `11a3`);

(c) the degree that `alldegphi` gives for the optimal curve of each class equals the degree of `degphi`, in all 38,042 classes.

**Example 22.19** (reading the files)**.** In GP, `readstr(file)` returns the vector of lines, `strsplit(s, " ")` the fields of a line, and `eval` the vector written in a field (PARI/GP 2.17.2):

```
? Vec(strsplit("11 a 1 [0,-1,1,-10,-20] 0 5", " "))
["11", "a", "1", "[0,-1,1,-10,-20]", "0", "5"]
? eval("[0,-1,1,-10,-20]")
[0, -1, 1, -10, -20]
```

The function `readgal` of Construction 22.67 builds a `Map` from labels to lists of image labels, and its loop `tablerow` runs over `degphi`. In Python the same is `line.split()` and `ast.literal_eval` (or `eval`, on trusted data). One caution: the label of the optimal curve must not be assembled from the three label fields of a `degphi` line, since the number field is 1 even for the class `990h`, whose optimal curve is `990h3`. The book's Python script looks the equation up in `allcurves`; the GP script treats the one exception below 10,000 explicitly.

*Remark* 22.20 (optimality in the files). (i) In Cremona's labelling, curve number 1 of a class is the optimal curve with exactly one exception below conductor 10,000: of the 38,042 equations of `degphi`, exactly one is not curve number 1 of `allcurves`, namely `990h3` $= [1,-1,1,-1568,-4669]$ (Chapter 9). The consequence for computation: `ellmoddegree` gives 1728 on `990h3` and 5184 on `990h1`, and `rule` gives `[3, 3]` and `[4, 3]` at $\ell = 3$ (Warning 22.26).

(ii) The paper: "Optimality is determined in the tables for every class of conductor at most 400000 and for 359,009 further classes up to 500000" (Section sec:data). What "determined" means, and how Cremona certifies it, is in Chapter 14. For $E[\ell]$ irreducible the choice of curve in a class does not affect $v_\ell$ of the degree, by (iv), nor the terms $e_\ell$, $t_\ell$, $h_\ell$; and for $E[3]$ irreducible the weight $w_3$ is the same for all curves of the class, because the Kodaira type at 3 is (Chapter 9). The paper uses this for its extension range: "for a class whose optimal curve is not determined, the right side of (eq:rule) at $\ell = 3$ is nevertheless the same for every curve of the class". Indeed "when $E[3]$ is irreducible the class has no 3-isogeny, so every isogeny in it is a composite of isogenies of prime degree $p \neq 3$, and an isogeny of prime degree $p$ preserves the Kodaira type at a prime $q \neq p$ of potentially good reduction" [56, Theorem 5.4(1)], applied with $q = 3$. With a 3-isogeny, where (eq:rule) does not apply at $\ell = 3$, the weight can vary. On the book's data, below conductor 10,000, among the classes with $3 \mid N$, at least two curves and potentially good reduction at 3, the Kodaira type at 3 is constant on all 1,295 classes without a 3-isogeny; on the 1,293 classes with one, the type varies in 1,078 and $w_3$ in 828, the paper's count (for `54a` the types are IV*, II*, II). *(Proved here, by computation, from* `allisog` *and* `ellglobalred`*.)*

(iii) `ellweilcurve` identifies the optimal curve inside PARI for moderate conductors (Remark 22.15(ii)), and `opt_man` records it (Example 22.18).

(iv) When $E[\ell]$ is irreducible, every isogeny in the class has degree prime to $\ell$, so $v_\ell$ of the degree of the minimal parametrisation is the same for every curve of the class; this is the paper's remark in Section 1 and is proved in Chapter 14. When $E[\ell]$ is reducible it is not: the minimal parametrisations of `11a1`, `11a2`, `11a3` have degrees 1, 5, 5 (`alldegphi`), and `rule`, which uses $\deg\varphi/c_\varphi^2$, returns `[0, 1]`, `[1, 0]`, `[-1, 0]` at $\ell = 5$.

**Convention 22.21** (labels)**.** (i) A Cremona label is the conductor, a class code (`a`, ..., `z`, `ba`, `bb`, ...) and the number of the curve in its class: `11a1`, `539c1`; `121c` and `121d` denote classes. The capitalised form `99A1` of [35] and [3] is `99a1`.

(ii) An LMFDB label [198] is $N$.class number, with classes and curves ordered differently; the file `alllabels` is the dictionary. For example `11a1` is `11.a2` and `726a1` is `726.b2`.

(iii) The images in `galrep` carry Sutherland's labels [191], whose general scheme and table at $\ell = 3$ are in Chapter 11. What this chapter needs: an image is reducible if and only if its label begins with "$\ell$B" (contained in a Borel subgroup) or "$\ell$Cs" (contained in a split Cartan subgroup), and the prime part of a label must be compared exactly (the label `37B.8.1` is a 37-adic label and must not be read as a 3-adic one). The three labels at $\ell = 3$ that the paper names are `3Ns` (image $N_s(\mathbb{F}_3)$), `3Cs.1.1` ($E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$, image the subgroup of order 2 of the split Cartan subgroup that fixes a point) and `3B.1.1` (a rational point of order 3, image in a Borel subgroup, $E[3]$ not split). Among the 37,853 optimal curves without CM below 10,000 the 3-adic labels occur as follows: `3B.1.1` 1,858, `3B` 1,567, `3B.1.2` 503, `3Nn` 170, `3Ns` 90, `3Cs.1.1` 49, `3Cs` 25. These are counts of optimal curves; counts over all curves of the tables, as in Chapter 11, are larger.

(iv) `galrep` omits the normaliser-of-Cartan images forced by complex multiplication, though it lists reducible images of CM curves (Example 22.13); CM curves are therefore recognised by $j$, never by the labels.

*In the paper* 22.22. Section sec:data: "We used Cremona's tables [35] in the form of the `ecdata` repository at commit `25cec5e`". The six files used here are byte for byte those of that commit (their SHA-256 sums are in Convention 22.69), so the book's counts below conductor 10,000 are computed from the paper's own data. Section 1.4 reads the component groups off the Kodaira types; the paper recomputes the types with PARI, and Example 22.18(a) shows that, below conductor 10,000, the repository's own `curvedata` agrees.

With the functions of Section 22.3 and the files of this section, everything is in place to read the one piece of code the paper prints, which combines the two.

## 22.5 The paper's function `rule(E, l, iii)`, line by line

The paper's appendix prints two GP functions, `rule` and `ns3`, which check one curve against (eq:rule). This section prints them verbatim, reads them line by line as code, proves that they compute the right side of (eq:rule) and of its III-free form under stated conditions, and lists the cases in which they return something other than the paper's quantity. The mathematics of each term is in Chapters 19 and 21; here we read the program.

**Construction 22.23** (the appendix code)**.** The functions of the paper's appendix, which the appendix says "are the file `scripts/euler_rule.gp`", copied verbatim, with line numbers added:

```
1  rule(E, l, {iii = 0}) =
2  { my(R = ellglobalred(E), P = R[4][,1], K = R[5], S = lfunsympow(E, 2), b = 0, j = E.j);
3    for(i = 1, #P,
4      my(q = P[i], k = K[i][2], e = 0, t = 0, vj = valuation(j, q));
5      if(q != l,
6        my(F = 1/lfuneuler(S, q), r = subst(F, x, 1/q^2)); \\ P_q(q^{-2}): omitted factor
7        if(k >= 5, r /= (1 - 1/q^2));                      \\ multiplicative: naive one out
8        if(abs(k) != 3 || iii, e = valuation(r, l)),       \\ III, III* (codes 3,-3) if iii
9        if(l == 3, e = if(k == 4 || k == 2, 1,
10                     if(k == -4 || k == -2, 2, if(k == -3, 1, 0))));   \\ w_3 at q = 3
11     if(vj < 0, t = valuation(-vj, l));                    \\ Tamagawa exponent
12     b += e + t);
13   if(iii && l == 3, b -= ns3(E));                         \\ the correction term h_3
14   [valuation(ellmoddegree(E), l), b]; }
15 ns3(E) = my(F = factor(elldivpol(E, 3))); (#F[,1] == 2 && poldegree(F[1,1]) == 2);
16 \\ psi_3 has degree 4 and is squarefree, so two factors, one quadratic, means two quadratics
```

The sixteen lines contain no double hyphen. They are loaded with `read("euler_rule.gp")` in `gp`, or with `pari('read("euler_rule.gp")')` in Python (Warning 22.7(g)). The appendix adds: "Here `K[i][2]` is PARI's Kodaira code: $n+4$ for $\mathrm{I}_n$ and $-(n+4)$ for $\mathrm{I}_n^*$ when $n \geq 1$, 1 for $\mathrm{I}_0$ and $-1$ for $\mathrm{I}_0^*$, $\pm 2$ for II/II$^*$, $\pm 3$ for III/III$^*$, $\pm 4$ for IV/IV$^*$. The function `rule(E,l)` leaves the type III terms out and `rule(E,l,1)` is (eq:rule)".

*Remark* 22.24 (reading the code). (a) Lines 1 and 2. `rule` takes a curve, an odd prime and an optional flag `iii`, 0 by default. `R` is the output of `ellglobalred` (Definition 22.8); `P = R[4][,1]` is the column of bad primes in increasing order, and `K = R[5]` the local data in the same order, so that `K[i]` refers to `P[i]`. `S` is the symmetric square (Remark 22.10), `b` the bound being accumulated, `j` the $j$-invariant.

(b) Lines 3 and 4. A loop over the bad primes; for each, `k` is the Kodaira code, `e` and `t` the two terms (both 0 unless set), and `vj` $= v_q(j)$, which is `+oo` when $j = 0$.

(c) Line 6, for $q \neq \ell$. `F = 1/lfuneuler(S, q)` is the polynomial $P_q(X)$ in the variable `x`, and `r` the exact rational $P_q(q^{-2}) = L_q(A_f, 1)^{-1}$.

(d) Line 7. The test `k >= 5` selects the multiplicative primes (Convention 22.9). There the naive factor at $s = 1$ is $1 - a_q^2 q^{-2} = 1 - q^{-2}$, since $a_q = \pm 1$, and the division makes `r` the quotient of Definition def:terms; it equals 1, because $P_q(X) = 1 - X$ at a multiplicative prime, so $e_\ell(E, q) = 0$ there. At an additive prime $a_q = 0$, the naive factor is 1, and no division is needed.

(e) Line 8. The valuation of `r` is taken unless the type is III or III$^*$ (`abs(k) == 3`) and `iii` is 0: so `rule(E, 1)` leaves the type III terms at $q \neq \ell$ out, and `rule(E, 1, 1)` keeps them.

(f) Lines 9 and 10, the `else` branch of line 5, for $q = \ell$. No Euler term is taken. For $\ell = 3$ the variable `e` is set to the weight $w_3$ read off the code: 1 for IV and II (codes 4, 2), 2 for IV$^*$ and II$^*$ ($-4$, $-2$), 1 for III$^*$ ($-3$), and 0 for every other type (III, $\mathrm{I}_0^*$, and the potentially multiplicative types, whose contribution is $t_3$). For $\ell \geq 5$ nothing is added at $q = \ell$. The flag `iii` does not act here: `rule(E, 3)` counts the weight 1 of a type III$^*$ at $q = 3$.

(g) Line 11. For every bad prime, $q = \ell$ included, `t` $= v_\ell(-v_q(j))$ if $v_q(j) < 0$; this is $t_\ell(E, q)$ of Definition def:terms, which the paper defines at $q = \ell$ too. When $j = 0$, `vj` is `+oo`, the test fails and `t` $= 0$, as it should: $j = 0$ means potentially good reduction everywhere.

(h) Line 12 adds the two terms; line 13 subtracts `ns3(E)` when `iii` is set and $\ell = 3$; for $\ell \geq 5$ nothing is subtracted, as $h_\ell(E) = 0$ for $\ell \geq 5$ (Chapter 11).

(i) Line 14 returns the pair [$v_\ell$(`ellmoddegree(E)`), $b$]. `ellmoddegree` is by far the slowest step (Remark 22.15(iv)).

(j) Line 15. `ns3(E)` factors $\psi_3$ and returns 1 if the factorisation has exactly two distinct irreducible factors, the first of degree 2, and 0 otherwise. Line 16 is a comment giving the reason why this test detects the pattern $2 + 2$: $\psi_3$ has degree 4 and no repeated factor, so two factors, one of them quadratic, means two quadratics; Proposition 22.25(ii) proves both facts.

**Proposition 22.25** (what `rule` computes)**.** *Let $E/\mathbb{Q}$ be an elliptic curve given by any Weierstrass model, and $\ell$ an odd prime. Assume that `lfuneuler(lfunsympow(E, 2), q)` returns $1/P_q(X)$ and that `ellmoddegree(E)` returns $\deg\varphi/c_\varphi^2$ for the minimal parametrisation $\varphi$ of the given curve, as PARI's documentation states.*

*(i) For every bad prime $q \neq \ell$, the number `e` that `rule(E, l, 1)` computes at $q$ is $e_\ell(E, q)$; at every bad prime, the number `t` is $t_\ell(E, q)$; at $q = \ell = 3$, `e` is $w_3(E)$.*

*(ii) `ns3(E)` $= 1$ if and only if $\psi_3$ is a product of two irreducible quadratic polynomials over $\mathbb{Q}$. If $E[3]$ is irreducible, then `ns3(E)` $= h_3(E)$, and both are 1 exactly when the mod 3 image is $N_s(\mathbb{F}_3)$.*

*(iii) Hence, if $E$ is the optimal curve of its class (with Manin constant 1) and, when $\ell = 3$, $E[3]$ is irreducible, then `rule(E, l, 1)` returns $[v_\ell(\deg\varphi_E), B_\ell^{\mathrm{rule}}(E)]$ and `rule(E, l)` returns $[v_\ell(\deg\varphi_E), B_\ell^{\mathrm{III}}(E)]$; and $B_\ell^{\mathrm{nc}}(E)$ is the second entry of `rule(E, l, 1)` plus `ns3(E)` at $\ell = 3$, and equal to it at $\ell \geq 5$.*

*Proof.* (i) The local data are those of a minimal model whatever model is given (Definition 22.8), and the code `k` is $\geq 5$ exactly at the multiplicative primes, the code 1 of good reduction never occurring at a bad prime. At a multiplicative prime $a_q^2 = 1$ and the naive factor at $s = 1$ is $1 - q^{-2}$;

at an additive prime $a_q = 0$ and the naive factor is 1 (Chapter 19). So, by Remark 22.24(c), (d), after line 7 `r` is $P_q(q^{-2})/(1 - a_q^2 q^{-2})$, and line 8 with `iii` $= 1$ takes its $\ell$-adic valuation: this is the definition of $e_\ell(E, q)$. Line 11 is the definition of $t_\ell(E, q)$ word for word, the case $j = 0$ being handled by `valuation(0, q)` $=$ `+oo`. Lines 9 and 10 are the definition of $w_3(E)$ in Conjecture conj:euler: the weight is defined for potentially good reduction at 3, and the code gives 0 to the potentially multiplicative types, which are not potentially good.

(ii) PARI factors an integer polynomial into irreducible primitive factors and drops the content (Warning 22.7(f)), so the leading coefficient 3 of $\psi_3$ produces no extra factor. The eight points of order 3 fall into four pairs $\{P, -P\}$; the two points of a pair have the same $x$-coordinate, and points of different pairs have different ones, because a point of a Weierstrass curve is determined up to sign by its $x$-coordinate. So the four roots $x(P)$ of the quartic $\psi_3$ are distinct, and its factorisation has no repeated factor. If there are exactly two factors and the first has degree 2, the other has degree $4 - 2 = 2$; conversely, a factorisation $2 + 2$ has exactly two factors and both have degree 2, whatever order PARI lists them in. So `ns3(E)` $= 1$ if and only if the pattern is $2 + 2$. If $E[3]$ is irreducible, $\psi_3$ has no rational root. Indeed, if $x_0 \in \mathbb{Q}$ is a root and $P$ a point of order 3 with $x(P) = x_0$, then for every $\sigma \in G_{\mathbb{Q}}$ the point $\sigma(P)$ has order 3 and $x$-coordinate $\sigma(x_0) = x_0$, so $\sigma(P) = \pm P$; the subgroup $\{0, P, -P\}$ of order 3 is then Galois-stable and $E[3]$ is reducible. So the possible patterns are 4 and $2 + 2$, and the pattern is $2 + 2$ exactly when the image is $N_s(\mathbb{F}_3)$ (Proposition prop:ns; Chapter 11). For $E[3]$ irreducible, $h_3(E) \le 1$, with equality exactly when $\bar\rho \simeq \bar\rho \otimes \chi_{-3}$, which happens exactly for image $N_s(\mathbb{F}_3)$, with or without CM (Chapter 11, where Lemma lem:h0 and Proposition prop:ns are proved).

(iii) By (i) the sum accumulated in `b` by line 12 is $B_\ell^{\mathrm{nc}}(E)$ when `iii` $= 1$ and $B_\ell^{\mathrm{III}}(E)$ when `iii` $= 0$. Line 13 subtracts `ns3(E)` $= h_3(E)$ in the first case at $\ell = 3$, by (ii), giving $B_3^{\mathrm{rule}}(E)$; at $\ell \ge 5$, $h_\ell(E) = 0$ (Chapter 11), so $B_\ell^{\mathrm{rule}} = B_\ell^{\mathrm{nc}}$. Finally, for the optimal curve with $c_E = 1$, `ellmoddegree(E)` $= \deg \varphi_E$, so the first entry is $v_\ell(\deg \varphi_E)$. □

The proposition is about PARI's functions as documented; the rest of this chapter checks them against independent data. Below conductor 10,000 the book's Python implementation of the five bounds (Remark 22.68), which reads the stored factors and codes and never calls `rule`, agrees with the pure GP loop of Construction 22.67 on every optimal curve of conductor below 2,000 at $\ell = 3, 5, 7$ (Example 22.55), and `ellmoddegree` agrees with Cremona's degrees on all 38,042 optimal curves (Example 22.37).

*Warning* 22.26 (where `rule` does not compute the paper's quantity). (a) *A non-optimal curve.* `ellmoddegree` returns $\deg \varphi / c_\varphi^2$ for the curve given, so the first entry is not $v_\ell(\deg \varphi_E)$: on `11a3` at $\ell = 5$, `rule` returns `[-1, 0]` (a negative "valuation of a degree"), and on `990h1` at $\ell = 3$ it returns `[4, 3]` where the optimal `990h3` returns `[3, 3]`. Curve number 1 is not always the optimal curve (Remark 22.20).

(b) *CM curves.* `ns3` is 1 on 13 optimal CM curves below 10,000 (Example 22.13). Conjecture conj:euler excludes CM, so these outputs concern no statement of the paper.

(c) *Eisenstein curves.* `11a1` at $\ell = 5$ returns `[0, 1]`, an apparent violation; but $E[5]$ is reducible (`5Cs.1.1`) and the curve is outside every statement of the paper (Example 22.34). The population filter is part of the statement.

(d) $\ell \mid N$. `rule` runs, and the data satisfy (eq:rule) there, but no theorem applies: Theorem thm:main needs $\ell \nmid N$.

(e) *Codes.* $n+4$ and $-(n+4)$ are the codes of $\mathrm{I}_n$ and $\mathrm{I}_n^*$ only for $n \geq 1$ (Convention 22.9).

(f) *Versions.* In 2.17.2 `ellmoddegree` returns an integer on an optimal curve and a fraction on some non-optimal ones (type `t_INT` and `t_FRAC`); `valuation` accepts both. The appendix says the functions run with "version 2.15 or later"; we have not run 2.15. The book's own scripts also use `readstr` and `strsplit`, which exist in 2.17.2 and whose presence in 2.15 we have not checked.

(g) *Speed.* For a table, take $\deg \varphi_E$ from `degphi` and compute only the bound: the function `rulebound` of Construction 22.67.

*In the paper* 22.27. The appendix: "A single curve can be checked against (eq:rule) in PARI/GP (version 2.15 or later) with the following functions, which return $[v_\ell(\deg \varphi_E), \text{bound}]$ ... The curve must be the optimal curve of its class and $E[\ell]$ must be irreducible: for instance 990h1 returns `[4, 3]` while the optimal curve 990h3 of its class returns `[3, 3]`, and 11a1 at the Eisenstein prime $\ell = 5$ returns `[0, 1]`." Proposition 22.25 is the precise form of this, and Warning 22.26(a) and (c) reproduce the two examples. Optimality is what makes the first entry $v_\ell(\deg \varphi_E)$; irreducibility of $E[3]$ is what makes `ns3` the correction at $\ell = 3$; at $\ell \geq 5$ irreducibility is needed not for the computation but for any statement of the paper to apply.

The proposition says what `rule` computes in general; the next section watches it compute, prime by prime, on the curves that the paper and the earlier chapters use as running examples, including the ones on which its output is not the paper's quantity.

## 22.6 Worked sessions

Each session below gives the input, the exact output of PARI/GP 2.17.2, the per-prime breakdown printed by `rule_terms(E, 1, iii)` (Construction 22.66), whose rows are $[q, k, P_q(X), P_q(q^{-2}), e, w, t]$ (the Euler term $e$ for $q \neq \ell$, the weight $w$ at $q = \ell = 3$, the term $t$; `iii` defaults to 1), and a short reading. The mathematics of the terms is in Chapters 19 and 21. In the paper these are the examples of the appendix, the Example of Section sec:proof, the Remark after it, Remark rem:wild and Section sec:tam.

**Example 22.28** (121a1).

```
? read("euler_rule.gp"); read("sessions.gp");
? E = ellinit([1,1,1,-30,-76]);
? [rule(E, 3), rule(E, 3, 1), ellmoddegree(E)]
[[1, 1], [1, 1], 6]
? rule_terms(E, 3)
[[11, 2, 11*x + 1, 12/11, 1, 0, 0]]
```

The only bad prime is 11, of type II (code 2), with $P_{11}(X) = 1+11X$ and $e_3(E, 11) = v_3(12/11) = 1$; $t_3 = 0$ since $j = -11 \cdot 131^3$ (`factor(E.j)`), so $v_{11}(j) = 1 \geq 0$. The row of `rule_terms` reads accordingly: $q = 11$, code 2, $P_{11}(X)$, $P_{11}(11^{-2})$, then $e = 1$, $w = 0$ (as $11 \neq \ell$), $t = 0$. So the bound is 1, and $\deg \varphi_E = 6 = 2 \cdot 3$ attains it. The paper's appendix: "For example `rule(ellinit([1,1,1,-30,-76]), 3)` (the curve 121a1) returns `[1, 1]`". Here $B_3^{\mathrm{main}} = 0$ ($\#\Phi_{11}^{\mathrm{geom}} = 1$ for type II) while $B_3^{\mathrm{tw}} = 1$: this curve is the simplest case where (eq:twist) is stronger than (eq:main).

**Example 22.29** (726a1, the correction term).

```
? E = ellinit([1,1,0,-35,-51]);
? [rule(E, 3), rule(E, 3, 1), ns3(E), ellmoddegree(E)]
[[2, 2], [2, 2], 1, 144]
? rule_terms(E, 3)
[[2, 10, -x + 1, 3/4, 0, 0, 1], [3, 7, -x + 1, 8/9, 0, 0, 1], [11, 3, 11*x + 1,
    12/11, 1, 0, 0]]
? rule_terms(E, 3, 0)
[[2, 10, -x + 1, 3/4, 0, 0, 1], [3, 7, -x + 1, 8/9, 0, 0, 1], [11, 3, 11*x + 1,
    12/11, 0, 0, 0]]
```

At 2, type $\mathrm{I}_6$: $e = 0$, $t_3 = v_3(6) = 1$. At $3 = \ell$, type $\mathrm{I}_3$: no Euler term, $w = 0$, $t_3(E,3) = v_3(3) = 1$. At 11, type III, $P_{11} = 1+11X$, $e_3 = v_3(12/11) = 1$, kept only with `iii`. So $B_3^{\mathrm{III}} = 2$, $B_3^{\mathrm{nc}} = 3$, `ns3` $= 1$ ($\psi_3 = (x^2 - x - 74)(3x^2 + 8x + 20)$, Example 22.13) and $B_3^{\mathrm{rule}} = 2$; and $\deg \varphi_E = 144 = 2^4 3^2$. The appendix: "for 726a1, a curve with image 3Ns and type III at 11, `rule(E,3)` and `rule(E,3,1)` both return `[2, 2]`, the type III term and the correction cancelling." Without the correction the bound $B_3^{\mathrm{nc}} = 3$ fails by exactly one, and the failure involves the prime $2 \equiv -1 \pmod 3$ with $t_3(E,2) = 1$, as Section sec:anomaly says of every such failure. The curve has $3 \mid N$, so it is outside Theorem thm:main in any case. The right side of (eq:thm) as written includes $t_3(E,3)$: it is $B_3^{\mathrm{thm}} = t_3(E,2) + t_3(E,3) + e_3(E,11) = 1 + 1 + 1 = 3$, again a violation by one; only if the terms at $q = 3$ are dropped is the bound 2 and the violation gone (Example 22.57). The congruence number $r_E = 1584 = 11 \cdot 144$ (Chapter 14) is not computed by `rule`.

**Example 22.30** (539c1).

```
? E = ellinit([1,0,1,170,-3237]);
? [rule(E, 3), rule(E, 5), ellmoddegree(E)]
[[2, 2], [0, 0], 288]
? rule_terms(E, 3)
[[7, -7, -x + 1, 48/49, 1, 0, 1], [11, 6, -x + 1, 120/121, 0, 0, 0]]
```

At 7, code $-7$, type $\mathrm{I}_3^*$: $P_7 = 1 - X$, $P_7(7^{-2}) = 48/49$, no division (the prime is additive), $e_3 = v_3(48) = 1$, and $t_3 = v_3(3) = 1$. At 11, type $\mathrm{I}_2$: nothing. So the bound is $2 = v_3(288)$. The Tate parameter at 7 is that of the twist `77c1`, with $\bar{u} = 5$ not a cube in $\mathbb{F}_7^\times$ (Example 22.12), so the Tamagawa factor of the adjoint adds nothing beyond $t_3$ (Lemma lem:tamfactor). The paper (Example of Section sec:proof): "$\deg \varphi_{539\mathrm{c}1} = 2^5 \cdot 3^2$, and the two terms $e_3(E,7) = v_3(48) = 1$ and $t_3(E,7) = v_3(3) = 1$ are both attained."

**Example 22.31** (980i1).

```
? E = ellinit([0,0,0,1568,-72716]);
? [rule(E, 3), rule(E, 5), ellmoddegree(E)]
[[2, 2], [1, 1], 2880]
? rule_terms(E, 3)
[[2, -4, 2*x + 1, 3/2, 1, 0, 0], [5, 5, -x + 1, 24/25, 0, 0, 0], [7, -9, -x + 1,
    48/49, 1, 0, 0]]
? rule_terms(E, 5)
[[2, -4, 2*x + 1, 3/2, 0, 0, 0], [5, 5, -x + 1, 24/25, 0, 0, 0], [7, -9, -x + 1,
    48/49, 0, 0, 1]]
```

At $\ell = 3$: type $\mathrm{IV}^*$ at 2 with $P_2 = 1 + 2X$ and $e_3 = v_3(3/2) = 1$; type $\mathrm{I}_5^*$ at 7 with $e_3 = v_3(48/49) = 1$. At $\ell = 5$: $t_5(E,7) = v_5(5) = 1$, all $e_5 = 0$. $\deg \varphi_E = 2880 = 2^6 \cdot 3^2 \cdot 5$, and both bounds are attained. The contribution of 2 at $\ell = 3$ is a theorem, not an observation: a type IV or $\mathrm{IV}^*$ at 2 always has $P_2 = 1 + 2X$ (Lemma lem:IV; Chapter 19), and the check of

Example 22.63(c) confirms it on every optimal curve below 10,000. At $\ell = 5$ the prime 5 divides $N$, so the curve illustrates the congruence of the Example of Section sec:proof, not Theorem thm:qual (as the paper says).

**Example 22.32** (`99a1`).

```
? E = ellinit([1,-1,1,-2,0]);
? [rule(E, 3), ellmoddegree(E)]
[[0, 0], 4]
? rule_terms(E, 3)
[[3, 3, 3*x + 1, 4/3, 0, 0, 0], [11, 5, -x + 1, 120/121, 0, 0, 0]]
```

$N = 3^2 \cdot 11$. At $3 = \ell$, type III (code 3): no Euler term, $w_3 = 0$, and $t_3 = 0$ since $v_3(j) = 9 \geq 0$; PARI's $P_3 = 1 + 3X$ is printed by `rule_terms` but not used. At 11, type $\mathrm{I}_1$. The bound is 0 and $\deg \varphi_E = 4$. This is the curve `99A1` of [3], whose congruence number is $r_E = 12$ (Chapter 14): the gap at 3 between $r_E$ and $\deg \varphi_E$ is allowed by the theorem of Agashe, Ribet and Stein because $9 \mid N$ (Remark 22.38).

**Example 22.33** (the prime 2 through PARI's factor).

```
? E = ellinit([0,1,0,-1,2]);
? [rule(E, 3), ellmoddegree(E)]
[[1, 1], 24]
? rule_terms(E, 3)[1]
[2, 2, 2*x + 1, 3/2, 1, 0, 0]
? E = ellinit([0,0,0,32,-16]);
? [rule(E, 3), ellmoddegree(E)]
[[2, 2], 288]
? rule_terms(E, 3)[1]
[2, -2, -8*x^3 - 4*x^2 + 2*x + 1, 9/8, 2, 0, 0]
? factor(-8*x^3 - 4*x^2 + 2*x + 1)
[2*x - 1, 1; 2*x + 1, 2]
? E = ellinit([0,0,0,-16,-16]);
? [rule(E, 5), ellmoddegree(E)]
[[1, 1], 80]
? rule_terms(E, 5)
[[2, -2, -8*x^3 + 4*x^2 - 2*x + 1, 5/8, 1, 0, 0], [37, 5, -x + 1, 1368/1369, 0, 0,
    0]]
? factor(-8*x^3 + 4*x^2 - 2*x + 1)
[2*x - 1, 1; 4*x^2 + 1, 1]
```

`336f1` $= [0, 1, 0, -1, 2]$ has type II at 2 and nevertheless $P_2 = 1 + 2X$, so $P_2(2^{-2}) = 1 + \frac{2}{4} = \frac{3}{2}$ and $e_3(E, 2) = 1$, bound 1, $\deg \varphi_E = 24$. `1232l1` $= [0, 0, 0, 32, -16]$ has type $\mathrm{II}^*$ at 2 and the cubic $P_2 = (1 - 2X)(1 + 2X)^2$, so $P_2(\frac{1}{4}) = \frac{1}{2} \cdot (\frac{3}{2})^2 = \frac{9}{8}$ and $e_3(E, 2) = v_3(9/8) = 2$, bound 2, $\deg \varphi_E = 288$: the prime 2 alone attains the bound. (The prime 2 is additive, so $a_2 = 0$ and there is no naive factor to divide by. PARI's factorisation lists $2x - 1$, which is $-(1 - 2X)$: the product of the listed factors is $-P_2$, the sign being part of the content that `factor` drops, Warning 22.7(f).) `592c1` $= [0, 0, 0, -16, -16]$ has type $\mathrm{II}^*$ at 2, $P_2 = (1 - 2X)(1 + 4X^2)$, so $P_2(\frac{1}{4}) = \frac{1}{2} \cdot \frac{5}{4} = \frac{5}{8}$ and $e_5(E, 2) = v_5(5/8) = 1$, bound 1 at $\ell = 5$, $\deg \varphi_E = 80 = 2^4 \cdot 5$: the prime 2 contributes at $\ell = 5$. At 2 the Kodaira type does not determine $P_2$; the four possible shapes are proved in Chapter 19 (Proposition prop:shapes). Among the optimal curves without CM below conductor 10,000, type II at 2 gives $P_2 = 1$ for 1,092 curves, $1 + 2X$ for 371 and a cubic for 74; type $\mathrm{II}^*$ gives 514, 74 and 191. Remark rem:wild counts the cubic factors: "Among all curves of conductor below 10,000, every cubic factor at 2 (1,143) and at 3 (2,283) equals the formula of (ii)

for a twist of good reduction found independently (534 and 1,308 among the optimal non-CM curves)". The book recomputed both pairs of counts with `lfunsympow` (Chapter 19 found the same), and also the remark's shares of potentially good types $\mathrm{I}_n^*$ at 2: 58% among the optimal curves (2,340 of 4,037) and 61% over all curves (5,935 of 9,667). As the remark says, that a cubic factor has shape (ii) is automatic; the empirical content is the agreement with the twist, which Chapter 19 checks.

**Example 22.34** (`11a1` at an Eisenstein prime)**.**

```
? E = ellinit([0,-1,1,-10,-20]);
? [rule(E, 5), rule_terms(E, 5)]
[[0, 1], [[11, 9, -x + 1, 120/121, 0, 0, 1]]]
```

The bound is $t_5(E,11) = v_5(5) = 1$ and $\deg \varphi_E = 1$: an apparent violation. But `galrep` gives `5Cs.1.1`, so $E[5] \simeq \mathbb{Z}/5 \oplus \mu_5$ is reducible, and the curve belongs to no statement of the paper except Conjecture conj:eis. There the deficit is $D = v_5(\#\Phi_{11}^{\mathrm{geom}}) - v_5(\deg \varphi_E) = 1 - 0 = 1 \le v_5(\#E(\mathbb{Q})_{\mathrm{tors}}) = 1$. The Tate data (Example 22.12) say more: $\bar{u} = 10 \equiv -1$ is a fifth power in $\mathbb{F}_{11}^{\times}$, so the formula of Lemma lem:tamfactor would give $v_5(\mathrm{Tam}_{11}(\mathrm{ad}^0 T)) = 1 + \min(1,1,1) = 2$, while $v_5(\deg \varphi_E) = 0$. The identity (eq:identity) fails here because its hypothesis fails: the population filter is part of every statement, not an afterthought (Chapters 16 and 18 make the same point in cohomology).

**Example 22.35** (the Tamagawa refinement on three curves)**.** At $\ell = 3$, for a multiplicative prime $q \equiv 1 \pmod 3$ with $3 \mid n$, Lemma lem:tamfactor adds $\min(v_3(n), v_3(q-1), v_3(c(u)))$ to the Tamagawa exponent (Section sec:tam):

```
? E = ellinit([1,0,1,-31,117]);
? [rule(E, 3), ellmoddegree(E), ellglobalred(E)[4][,1]~]
[[2, 1], 144, [7, 113]]
? E = ellinit([1,1,0,4,11]);
? [rule(E, 3), ellmoddegree(E), ellglobalred(E)[4][,1]~]
[[1, 1], 6, [7, 11]]
? E = ellinit([0,-1,1,30,-335]);
? [rule(E, 3), ellmoddegree(E), ellglobalred(E)[4][,1]~]
[[2, 1], 720, [19, 83]]
```

In each case both bad primes are multiplicative, so every Euler term is 0, and the second entry of `rule` is the Tamagawa term alone: $v_3(3) = 1$ from the prime of type $\mathrm{I}_3$ and $v_3(2) = 0$ from the other, of type $\mathrm{I}_2$. Lemma lem:tamfactor applies at the first prime, which is $\equiv 1 \pmod 3$, and the first entry shows whether the extra term is needed. `791a1` at 7: $n = 3$, $\bar{u} = 6$ is a cube (Example 22.12), the extra term is $\min(1,1,1) = 1$, so the refined bound is $1 + 1 = 2 = v_3(144)$: attained. `77c1` at 7: $\bar{u} = 5$ is not a cube, the extra term is 0, and the bound 1 is attained by $\deg \varphi_E = 6$. `1577a1` at 19: $\bar{u} = 8$ is a cube but not a ninth power, $v_3(18) = 2$, the extra term is $\min(1,2,1) = 1$, and the refined bound $2 = v_3(720)$ is attained. The cohomology is in Chapters 17 and 18.

## 22.7 Computing modular degrees and congruence numbers

This short section says how the modular degree is computed, why it is the expensive step, how the book checked Cremona's degrees, and where the congruence number, which PARI does not compute, comes from. In the paper it is the sentence "The modular degree is computable

[35, 204]" of Section 1, the recomputations of Section sec:data and Section sec:oot, and the Remark after the Example of Section sec:proof.

*Remark* 22.36 (three methods). *Zagier's method* [209] computes the map $\varphi_{E*}$ on $H_1(X_0(N),\mathbb{Z})$ with modular symbols and reads $\deg\varphi_E$ off an index in homology; *Cremona's* [35, Chapter II] computes the period lattice of $E$ from modular symbols and compares it with the lattice of $\varphi_E^*$, which also certifies optimality. *Watkins's method* [204] uses the formula $\deg\varphi_E\cdot\mathrm{vol}(\Lambda_E) = 4\pi^2 c_E^2\langle f,f\rangle$ together with the expression of $\langle f,f\rangle$ through $L(\mathrm{Sym}^2 f,2)$ and the bad Euler factors; it is the method of `ellmoddegree`. The mathematics of all three is in Chapters 14 and 19. Why the computation is expensive (a heuristic, not a statement about PARI's code): $L(\mathrm{Sym}^2 f,2)$ is evaluated by Dokchitser's method [55], which needs a number of Dirichlet coefficients roughly proportional to the square root of the conductor of $L(\mathrm{Sym}^2 f,s)$; that conductor is at most $N^2$ when $(N,6)=1$ (Chapter 19), so the work grows roughly linearly with $N$, and the coefficients themselves cost more as $N$ grows. Cremona's tables contain the degrees of all optimal curves up to conductor 500000; the paper recomputed fifteen of them inside the tables and 893 outside (Section sec:oot). The congruence module of Chapter 14 is a Smith normal form computation (`matsnf`, `mathnf` in PARI; Chapter 3), and the modular-forms functions `mfinit`, `mfbasis`, `mfcoefs`, `mfheckemat`, `mfsturm`, `mfpetersson` used in Chapters 12 to 15 are explained there with their normalisations. One trap is recorded in Warning 22.7(h): `mfsturm` applied to a space returns less than Sturm's bound.

**Example 22.37** (`ellmoddegree` against Cremona's degrees)**.** The script `moddeg_check.py` (Remark 22.68) runs `ellmoddegree` on the equation of each of the 38,042 lines of `degphi` and compares with the degree written there, in two processes, with `parisizemax` $2\cdot10^9$, catching errors. All 38,042 curves were checked (19,021 in each process), with 0 mismatches and 0 errors; each process took about 42 minutes on a two-core machine (PARI/GP 2.17.2). The paper's sentence "fifteen modular degrees spanning the configurations were recomputed with `ellmoddegree`, all agreeing with the tables" (Section sec:data) is thereby extended, in the book's range, to every optimal curve, as the paper's own check below 10,000 also states ("`ellmoddegree` agrees with the tables on all of them").

*Remark* 22.38 (the congruence number). PARI has no function for the congruence number $r_E$. Chapter 14 computes it from `mfinit`, `mfbasis` and the Hecke action on the lattice of integral forms, and finds $r_E=12$ for `99a1` against $\deg\varphi_E=4$, $r_E=1584=11\cdot144$ for `726a1`, and $r_E=504=7\cdot72$ for `539d1`. The theorem of Agashe, Ribet and Stein [3, Theorem 2.2] gives $\deg\varphi_E \mid r_E$, with equal $p$-parts when $p^2\nmid N$; in the three examples the extra prime (3, 11, 7) has square dividing $N$. This is the Remark after the Example of Section sec:proof ("Table 1 of [3] lists the curve 99A1, of conductor $9\cdot11$, with modular degree 4 and congruence number 12").

## 22.8 Populations, violations and descriptive statistics

Before any test, one must say what is counted, in which population, and what a rate is a rate of. This section fixes the units and the descriptive statistics of the paper's tables (equality rates, mean excess, base rates) and computes them on the book's range. In the paper it is the background of every table of Section sec:data.

**Definition 22.39** (units and populations)**.** (i) The unit of the paper's tables is a curve, one per isogeny class (the optimal one), at a fixed $\ell$. Other units occur: pairs $(E,\ell)$ (Section

sec:forms: "913,244 working-set pairs $(E, \ell)$"), pairs (curve, bad prime $q$) (Lemma lem:twist and the paragraph after Theorem thm:twist), triples (curve, $q$, $\ell$). Counts in different units are not comparable.

(ii) The populations are those of Convention 22.3, and each filter is part of a statement: CM curves are excluded by Conjecture conj:euler and Conjecture conj:eis; a reducible $E[\ell]$ moves a curve to the Eisenstein population, where Conjecture conj:eis makes a different statement; and in the hypothesis population, where Theorem thm:main applies, the data check a theorem rather than test a conjecture.

(iii) Only one curve per class is counted because $v_\ell(\deg \varphi_E)$ is attached to the optimal curve; counting every curve of a class would count one newform several times. For $E[\ell]$ irreducible, $v_\ell$ of the degree and the terms $e_\ell$, $t_\ell$, $h_\ell$ are invariants of the class (Chapters 9 and 14), and at $\ell = 3$ so is the weight $w_3$ (Remark 22.20(ii)).

*CM and the hypothesis population.* Section sec:forms: "The hypothesis of Theorem thm:main fails on the working set for the 1,170 non-CM curves with image 3Ns at $\ell = 3$" and for the curves with $\ell \mid N$, with no other failures among the non-CM curves with $E[\ell]$ irreducible (Proposition prop:Sf). A CM curve with $j = -32768$ or 8000 and $3 \nmid N$ has image $N_s(\mathbb{F}_3)$ and fails the hypothesis at $\ell = 3$ too, and `galrep` gives it no 3-adic label: below conductor 10,000 there are 20 such curves in the tables, 10 of them optimal; the paper's parenthesis in Section sec:forms gives the 20. Likewise the count after Proposition prop:ns-structure, "Among the 90 optimal curves without complex multiplication with image 3Ns of conductor below 10,000 (44 of them with $3 \mid N$)", is over the population without CM (14 of the 44 with $3 \parallel N$, 6 with $9 \parallel N$, 24 with $27 \mid N$), and the paper lists separately the 13 optimal CM curves with image $N_s(\mathbb{F}_3)$ of Example 22.13, which the proposition also covers.

**Definition 22.40** (rates)**.** For a population and a bound $B$: the number of curves with $B > 0$; the number of violations; the *equality rate*, the number of equalities divided by the number of curves with $B > 0$; the *mean excess*, the average of $v - B$ over the curves with $B > 0$; and the *base rate* of an attribute, its proportion in the population (Appendix A).

Three remarks on these. Curves with $B = 0$ cannot violate the bound, and including them would inflate every rate; this is why all rates are taken among positive bounds. An equality rate is a property of a population and a range, not of "elliptic curves": at $\ell = 3$ it is 75.1% on $N < 5000$ and 68.0% on $5000 \le N < 10000$ (Example 22.55), against the paper's 57.5% on $N \le 300000$; degrees grow with $N$, and with them the room for extra divisibility. The mean excess over $B_3^{\mathrm{rule}}$ is 0.31, 0.42 and 0.37 on the two small ranges and their union, against the paper's 0.63 on its working set. Under the hypothesis of Theorem thm:main the excess is not noise: (eq:identity) is an equality there (Chapter 21), the Tamagawa factor of $\mathrm{ad}^0 T$ is trivial at the potentially good additive primes (Lemma lem:tamadd), and at the potentially multiplicative primes it exceeds $t_\ell$ by the explicit term of Lemma lem:tamfactor; so the excess is the length of the adjoint Selmer group plus that explicit term (Corollary cor:explicit). The data measure the excess; they do not explain it (Open question 3).

**Example 22.41** (base rates by Kodaira type)**.** In the base population at $\ell = 3$ below conductor 10,000 (33,851 curves), $3 \mid \deg \varphi_E$ for 76.1%. Conditionally on having at least one prime $q \ge 5$ of a given type (a curve with several types is counted in several rows):

| type at some $q \geq 5$ | curves | share with $3 \mid \deg \varphi_E$ |
|---|---|---|
| II / II$^*$ | 946 / 418 | 100% / 100% |
| IV / IV$^*$ | 468 / 896 | 100% / 100% |
| $\mathrm{I}_n^*$ | 2,233 | 100% |
| $\mathrm{I}_n$ with $3 \mid n$ | 5,134 | 100% |
| III / III$^*$ | 813 / 807 | 75.4% / 75.5% |
| $\mathrm{I}_0^*$ | 692 | 73.1% |
| $\mathrm{I}_n$ with $3 \nmid n$ | 31,185 | 74.6% |

The rows at 100% are exactly the types whose term is always positive at $\ell = 3$: $e_3 \geq 1$ for II to IV$^*$ ($v_3(q - \chi_{-3}(q)) \geq 1$) and for $\mathrm{I}_n^*$ ($v_3(q^2 - 1) \geq 1$) at $q \geq 5$, and $t_3 \geq 1$ for $\mathrm{I}_n$ with $3 \mid n$. The rows near the base rate are the types whose term is positive only for some $q$: III, III$^*$ for $q \equiv \pm 1 \pmod{12}$, $\mathrm{I}_0^*$ when $q \equiv 1 \pmod 3$ or $3 \mid (q+1)^2 - a'^2$. A base-rate table is the first thing to compute and the weakest evidence: it is the shadow of the inequality on a coarse partition. Section sec:iso of the paper refines it cell by cell (Example 22.58).

*Remark* 22.42 (pooling). Rates pooled over ranges or over types can mislead. The pooled equality rate at $\ell = 3$ below 10,000, 71.0%, is the average of the rates of the two small ranges, 75.1% and 68.0%, weighted by their numbers of positive bounds, 8,510 and 11,807: the equalities are 6,392 and 8,028, and $(6{,}392 + 8{,}028)/(8{,}510 + 11{,}807) = 14{,}420/20{,}317 = 71.0\%$. The pooled rate is pulled towards the range with more bounded curves. A comparison of the paper's 57.5% on $N \leq 300000$ with the book's 71.0% says nothing about a difference in the arithmetic of the curves; it says that the ranges differ, and the trend is visible inside the book's range already. When the composition of the population changes with the range (more bad primes, larger degrees), a pooled rate can even move in the opposite direction to every one of its parts (Simpson's paradox). A made-up illustration with two strata: population $A$ has 100 curves in a stratum with equality rate 90% and 900 in a stratum with rate 50%, so its pooled rate is $(90 + 450)/1000 = 54\%$; population $B$ has 900 curves in the first stratum at 80% and 100 in the second at 40%, pooled $(720 + 40)/1000 = 76\%$. Population $B$ is worse in each stratum and better pooled, only because it has more curves in the easy stratum. The fair comparison is like with like: the same statement, the same population, the same range. The paper's equality rate on its out-of-table curves, 52.5% of the 828 curves with a positive bound at $\ell = 3$ (Section sec:oot), is a rate on a population built by a different procedure (Remark 22.43).

## 22.9 The protocol

The paper describes a protocol: statements formulated on a working set, written down, tested once on a hold-out, then on further ranges. This section describes each range, classifies each data statement of the paper as blind-tested, confirmed only, or a theorem checked, explains what a blind test of a universal inequality establishes, discusses multiple comparisons, and runs the protocol in miniature on the book's data. In the paper it is the protocol paragraph of Section sec:data, with Sections sec:wild and sec:oot.

*Remark* 22.43 (the four ranges). The paper's ranges (Convention 22.3) are: the working set, $N \leq 300000$, where every class has determined optimality; the hold-out, $300000 < N \leq 400000$, the same; the extension, $400000 < N \leq 500000$, where optimality is determined for 359,009 classes only (for the other classes the bound at $\ell = 3$ is the same for every curve of the class, since $w_3$ is a class invariant when $E[3]$ is irreducible, Remark 22.20); and the out-of-table curves:

893 curves (the abstract) with $500000 < N \le 4 \cdot 10^6$; "we generated curves with prescribed local types" ("families with a type IV or IV$^*$ prime, with two or three type II primes, with mixed types, and random curves"), kept when the isogeny class is trivial "so that the curve is optimal and $E[\ell]$ is irreducible for every $\ell$", not CM, and with modular degrees computed by `ellmoddegree` (Section sec:oot). The vocabulary (working set, hold-out, pre-registration, blind test, confirmation) is that of Appendix A.

*Remark* 22.44 (the status of each data statement). We say that a statement is *blind-tested* on a range if it was written down before any quantity of that range entering the statement was looked at. Generating data before a statement is formulated is necessary for this and not sufficient: the data must also not have been examined. Reading the protocol paragraph of Section sec:data, with Sections sec:wild, sec:tam and sec:oot, the statements of the paper have the following status.

| statement | formulated on | blind test | confirmed only | remark |
|---|---|---|---|---|
| (eq:main) | working set | hold-out (once), then extension | | "nothing was adjusted after the hold-out test" |
| Conjecture conj:eis | working set | hold-out (once), then extension | | $D \le 2$ is a theorem (Proposition prop:no9) |
| (eq:twist) | working set, replacing a weaker form | hold-out and extension, "tested once" | | blind only if fixed before the hold-out counts of (eq:main) were seen: the order is not stated; the paper does not count the out-of-table curves as a blind test of it |
| Conjecture conj:euler, Euler terms | working set (minima by type) | out-of-table | hold-out, extension | the terms at 2 and 3 "were fixed afterwards by PARI's Euler factors" |
| weights $w_3$ | working set (minima at $q = 3$) | none | hold-out, extension | "empirical" (Section sec:wild); the paper does not count the out-of-table curves as a blind test of them |
| correction $h_3$ | from the exceptions | none | all ranges | "fitted on, and only confirmed by, data already seen"; $h_3 = 1$ on 3Ns curves is a theorem, the repaired inequality is not |
| III-free form | after the 3Ns exceptions | none | all ranges | chosen so as to have no exceptions |
| Tamagawa refinement | from Lemma lem:tamfactor | not stated | | a prediction of a theorem; the order of formulation and computation is not stated; tests the Tamagawa term of (eq:identity) |
| Lemmas lem:euler, lem:twist; Prop. prop:forms; $h_3 = 1$ | | | | theorems: the data check computations |

Two of these entries need a qualification that the text does not settle. The out-of-table curves, "generated before the rule was formulated", are a blind test of the Euler terms only if their modular degrees were not examined while the terms at 2 and 3 were being fixed; the paper does not say. And for (eq:twist), "then tested once on the hold-out and the extension" is a blind test only if (eq:twist) was written down before the hold-out counts of (eq:main) were seen.

The decisive sentences are these (Section sec:data): "Since the hold-out and the extension had already been used for (eq:main) and (eq:twist), they are a confirmation and not a blind test for Conjecture conj:euler; the out-of-table curves, generated before the rule was formulated,

are. They contain no curve with image 3Ns, so the correction term $h_3$ was fitted on, and only confirmed by, data already seen; and we do not count the out-of-table curves as a blind test of (eq:twist). Nor do we count them as a blind test of the weights $w_3$, which they could test only through curves with $3 \mid N$ and potentially good reduction at 3."

*Remark* 22.45 (what a blind test of an inequality establishes). A single violation refutes a universal inequality. Zero violations on $n$ fresh curves is evidence whose weight depends on two things: how many violations would have been expected if the statement were unrelated to the degree (the calibration of Section 22.10), and whether the fresh curves exercise the statement. A rough measure of the first: if a false statement were violated independently by each fresh curve with probability $p$, the chance of seeing no violation on $n$ curves would be $(1-p)^n \le e^{-np}$; with $p = 0.01$ and the paper's $n = 893$ out-of-table curves this is $0.99^{893} \approx 1.3 \cdot 10^{-4}$, while with $p = 0.001$ it is about 0.41. So the same zero count is strong evidence against a statement that would fail often and weak evidence against one that would fail rarely, and the calibration is what says which case one is in. The out-of-table families were built around the types IV, IV$^*$, two or three II primes, mixed types and random curves; they contain no 3Ns curve, so they cannot test the correction term, and they test the III terms only if such primes occur among them. The paper's "nothing was adjusted after the hold-out test" means that no term, weight or population filter of (eq:main) and Conjecture conj:eis was changed after their hold-out counts were seen; it does not cover the statements formulated later.

*Remark* 22.46 (multiple comparisons).

(i) The paper's statements came out of a search: the AI statement describes "the pre-registered search, which produced the candidate relation" and then "the conjectures and their successive refinements": from the Tamagawa number to $\Phi_q^{\mathrm{geom}}$, to the twist-symmetrised weights, to the Euler factors, to the weights $w_3$, to the correction term.

(ii) If $m$ false candidate statements each survive a working-set check with probability $\pi$, about $m\pi$ survive; a hold-out check with survival probability $\pi'$ for a false statement leaves about $m\pi\pi'$. The word "about" means "in expectation": the number of survivors is a sum of $m$ indicators, each of expectation $\pi$, so its expectation is $m\pi$ by linearity, with no independence needed; a false candidate survives both checks with probability $\pi\pi'$ when the two checks are independent, which is what a hold-out disjoint from the working set is meant to secure. This is the whole case for holding data back (Exercise 22.82).

(iii) For a universal inequality, $\pi'$ is tiny when the calibration says that thousands of violations would be expected if the bound were unrelated to the degree (Section 22.10). The risk of multiple comparisons therefore sits mostly in the *choice of form* (which weights, which correction, which population filter), where the working set was used repeatedly, and not in the bare existence of a relation.

(iv) Features chosen by looking at the data are validated only by data not used to choose them. The weights $w_3$ were defined as observed minima, so the working set cannot violate them; the III-free form was chosen to have no exceptions. This is the statistician's multiple-comparisons problem [203, chapter on multiple testing] and the physicist's look-elsewhere effect.

**Example 22.47** (the protocol in miniature)**.** Freeze the small working set $N < 5000$. Following the paper's recipe for $w_3$ (Section sec:wild), take the curves of the base population at $\ell = 3$ with $3 \mid N$, potentially good reduction at 3, and no other contribution ($e_3(E,q) = 0$ at every

bad $q \neq 3$, III terms included, and $t_3(E,q) = 0$ at every bad $q$), and read off the minimum of $v_3(\deg \varphi_E)$ by the type at 3. The filter "no other contribution" is what makes the minimum meaningful: on these curves every term of $B_3^{\mathrm{nc}}$ other than $w_3(E)$ vanishes, so a weight $w$ for a type is consistent with the data exactly when $v_3(\deg \varphi_E) \geq w$ on all of them, and the largest consistent weight is the minimum. On curves with other contributions the minimum would mix the weight with the other terms.

| type at 3 | II | II* | III | III* | IV | IV* | $\mathrm{I}_0^*$ |
|---|---|---|---|---|---|---|---|
| $N < 5000$: minimum (curves) | 1 (239) | 2 (156) | 0 (221) | 1 (221) | 1 (156) | 2 (239) | 0 (299) |
| $5000 \leq N < 10000$ | 1 (255) | 2 (186) | 0 (251) | 1 (251) | 1 (186) | 2 (255) | 0 (347) |

The minima on the small working set are exactly the paper's weights. Test them once on the small hold-out: the minima are the same, and the rule with these frozen weights has no violation among the 11,699 curves of the base population with $3 \mid N$ and $5000 \leq N < 10000$. In all, 3,239 curves of the base population at $\ell = 3$ below 10,000 have a positive weight $w_3(E)$. This replicates a recipe; it discovers nothing, and the small working set was not blind to the paper: the weights were known in advance. It shows the procedure, nothing more.

*In the paper* 22.48. The protocol paragraph of Section sec:data and the AI statement ("a frozen snapshot of the tables, a working set, a range held back for one-time tests, and a list of tests fixed before any was run"). Remark 22.44 classifies the statements; its two qualifications are questions to the author, not errors.

A blind test with zero violations is only as informative as the number of violations one would have expected without the statement being true. The next section computes that number.

## 22.10 The calibration argument

Section sec:sharp of the paper answers an obvious objection: if 3 divides most modular degrees anyway, zero violations of a lower bound at $\ell = 3$ might mean nothing. Its answer is a calibration, an expected number of violations under a null hypothesis, compared with the observed zero. This section makes the argument exact, reproduces it on the book's range with two other choices of reference population, says what it shows and what it does not, and adds the paper's two descriptive statistics of sharpness.

**Proposition 22.49** (the null model)**.** *Let a finite population of curves carry a bound $b(E) \in \{0,1,2,\dots\}$ and an observed valuation $v(E)$. Let $H_0$ be the hypothesis that the $v(E)$ are random variables, each with the same distribution $F$ and independent of the bounds. Let $X$ be the number of violations, $X = \#\{E : b(E) > 0,\ v(E) < b(E)\}$, and $p_E = F(v < b(E))$.*

*(i) Under $H_0$, $E[X] = \sum_{b(E)>0} p_E$, with no independence between curves needed. If $\hat{F}$ is the empirical distribution of $v$ on the curves with $b = 0$, then $\hat{E} = \sum_{b(E)>0} \hat{F}(v < b(E))$ is the plug-in estimate of $E[X]$.*

*(ii) If moreover the $v(E)$ are independent, then $\operatorname{Var} X = \sum p_E(1-p_E) \leq E[X]$ and $P(X = 0) = \prod_{b(E)>0}(1-p_E) \leq e^{-E[X]}$; and if they are only pairwise independent, $P(X=0) \leq \operatorname{Var} X / E[X]^2 \leq 1/E[X]$ whenever $E[X] > 0$.*

*(iii) Under $H_0$ the distribution of $v$ over the whole population, bound 0 and positive together, is also $F$. So a large difference between the empirical distributions of $v$ on all curves and on the bound-0 curves is itself evidence against $H_0$. A permutation test makes this exact: shuffle the observed values of $v$ among the curves and recount the violations; under $H_0$ (with exchangeable $v$) the observed count is a draw from the distribution of the recounts.*

*Proof.* (i) $X = \sum_{b(E)>0} 1_{\{v(E)<b(E)\}}$, and $E[1_{\{v(E)<b(E)\}}] = P(v(E) < b(E)) = p_E$, since $b(E)$ is a fixed number and $v(E)$ has distribution $F$; linearity of expectation (Appendix A) needs no independence. (ii) Write $X = \sum_E I_E$ with $I_E = 1_{\{v(E)<b(E)\}}$, the sum over the curves with $b(E) > 0$. An indicator satisfies $I_E^2 = I_E$, so $\operatorname{Var} I_E = E[I_E] - E[I_E]^2 = p_E(1 - p_E)$. In general $\operatorname{Var} X = \sum_E \operatorname{Var} I_E + \sum_{E \neq E'} \operatorname{Cov}(I_E, I_{E'})$; for independent (indeed for pairwise independent) indicators every covariance vanishes, so $\operatorname{Var} X = \sum_E p_E(1 - p_E)$, and this is at most $\sum_E p_E = E[X]$ because $1 - p_E \leq 1$. The event $X = 0$ is the intersection of the events $v(E) \geq b(E)$, of probabilities $1 - p_E$; under full independence its probability is the product $\prod_E (1 - p_E)$. Since the exponential function is convex, its graph lies above its tangent line at 0, so $1 - p \leq e^{-p}$ for every real $p$; hence $\prod_E (1 - p_E) \leq \prod_E e^{-p_E} = e^{-E[X]}$. Under pairwise independence only, the product formula is not available, but the variance formula is. If $X = 0$ then $|X - E[X]| = E[X]$, so Chebyshev's inequality $P(|X - E[X]| \geq t) \leq \operatorname{Var} X / t^2$ with $t = E[X] > 0$ gives $P(X = 0) \leq \operatorname{Var} X / E[X]^2 \leq E[X]/E[X]^2 = 1/E[X]$. (iii) Every $v(E)$ has distribution $F$ whatever $b(E)$ is. If the values are exchangeable, the vector of values is equal in distribution to any permutation of it, and the recount after a random permutation has the same distribution as $X$. □

The proposition is elementary; its content is in the choice of $H_0$ and of the reference population, which the next examples vary.

**Example 22.50** (the paper's numbers)**.** Section sec:sharp: "Among the 366,151 working-set curves with $E[3]$ irreducible, not CM, and right-hand side 0 in (eq:rule), the prime 3 does not divide $\deg \varphi_E$ for 53.2%. If the valuation of the degree were independent of the local data, the 867,578 working-set curves with a positive bound at $\ell = 3$ would produce roughly 605,000 violations (summing over them the proportion of bound-0 curves whose valuation is smaller than the bound); there are none." The arithmetic: the curves with $v = 0$ alone contribute at least $0.532 \cdot 867{,}578 \approx 461{,}550$ to $\hat{E}$, since $\hat{F}(v < b) \geq \hat{F}(v = 0)$ for $b \geq 1$; the stated 605,000 means an average $\hat{F}(v < b)$ of 0.697 over the bounded curves, close to the book's 0.703 (Example 22.51). Under independence the standard deviation of $X$ is at most $\sqrt{605000} \approx 778$, so the observed 0 lies about 778 standard deviations below the mean, and $P(X = 0) \leq e^{-605000}$. A number like this is not a meaningful $p$-value: it is far below any doubt about the model itself (independence across curves is an idealisation). That is why the paper, and Appendix A, compare 605,000 with 0 and state no $p$-value. The paper adds: "This shows that the statement is strong, far from vacuous, not that it is valid; the figure depends on the reference population."

**Example 22.51** (the calibration on the book's range)**.** In the base population at $\ell = 3$ below conductor 10,000 (33,851 curves, bound $B_3^{\mathrm{rule}}$): 13,534 curves have bound 0, with $v = 0, 1, 2, 3, 4, 5, 6$ for 8,084 (59.7%), 4,079, 1,081, 251, 33, 5, 1 of them; 20,317 have a positive bound (1: 13,553; 2: 5,534; 3: 1,109; 4: 112; 5: 9). The plug-in expectation is $\hat{E} = 14{,}275$ violations (70.3% of the bounded curves), with standard deviation 61.5 under independence; observed: 0.

In detail, the empirical law of $v$ on the 13,534 bound-0 curves gives

$$\hat{F}(v<1)=\tfrac{8084}{13534}\approx 0.597,\quad \hat{F}(v<2)=\tfrac{12163}{13534}\approx 0.899,\quad \hat{F}(v<3)=\tfrac{13244}{13534}\approx 0.979,$$
$$\hat{F}(v<4)=\tfrac{13495}{13534}\approx 0.997,\quad \hat{F}(v<5)=\tfrac{13528}{13534}\approx 0.9996,$$

and Proposition 22.49(i) weights these by the numbers of curves with bound $1,\dots,5$: the contributions are about 8,095.3, 4,973.4, 1,085.2, 111.7 and 9.0, of total 14,274.7. More than half of the expectation comes from the curves of bound 1, which a bound unrelated to the degree would get wrong whenever $v=0$. The standard deviation is $\left(\sum_E p_E(1-p_E)\right)^{1/2}$ with $p_E=\hat{F}(v<b(E))$, by Proposition 22.49(ii); it is below $\sqrt{\hat{E}}\approx 119.5$, as (ii) says it must be, and the observed 0 lies about $14{,}275/61.5\approx 232$ standard deviations below the expectation. By range, with $\hat{F}$ taken in each range: $\hat{E}=6{,}091$ on $N<5000$ ($\hat{F}(v=0)=0.630$) and 8,119 on $5000\le N<10000$ ($\hat{F}(v=0)=0.566$). At $\ell=5,7,11,13$ (union of the ranges) $\hat{E}=5{,}901$, 3,476, 1,187, 698; observed 0 each time. Two other reference populations, at $\ell=3$:

(a) *Stratified.* Split the curves into 16 strata by conductor (bins of width 2500) and by the number of prime factors of $N$, capped at 4, and take $\hat{F}$ from the bound-0 curves of each stratum: $\hat{E}=14{,}177$.

(b) *Permutation.* Shuffle $v$ among all the curves of each stratum, bound 0 and positive together, and recount: over 200 shuffles the number of violations has mean 7,638, standard deviation 49.3 (sample, divisor 199; 49.2 with divisor 200), minimum 7,517 and maximum 7,779. To reproduce these exactly one needs, besides Python's `random.seed(0)` set once before the 200 shuffles, the order of the curves (that of the lines of `degphi`) and the order of the strata (that of their first curve in this list); in each shuffle the list of $v$ of each stratum is shuffled with `random.shuffle`, stratum by stratum in that order.

The reference population changes the expected count by a factor of two and the conclusion not at all. The factor of two has a visible cause: $v=0$ for 59.7% of the bound-0 curves but only for 23.9% of all curves, a difference that Proposition 22.49(iii) says should not exist under $H_0$. The bound-0 curves are those with no local reason for divisibility, and their degrees are less divisible; the pooled distribution already contains the curves the bound singles out.

*Remark* 22.52 (what the calibration shows and what it does not). It shows that the zero count is not an artefact of 3 dividing most modular degrees (76.1% below 10,000, Example 22.41): the bound picks out, among all curves, those whose degree is divisible, and a bound unrelated to the degree would fail thousands of times in the book's range and hundreds of thousands of times in the paper's. It does not show:

(a) that the inequality holds beyond the data; that is the business of Theorem thm:main under its hypotheses, and of the conjecture outside them;

(b) anything about validity on the working set: the rule was fitted there (weights read off minima, terms at 2 and 3 fixed after the minima were seen), so on the working set the calibration measures the *strength* of the statement, as the paper says; its *validity* rests on the blind tests of Remark 22.44;

(c) a precise level of significance: independence across curves is an idealisation (twists and congruent newforms tie curves together), so the standard deviations are indicative only;

(d) that the null is the right one: "independent of the local data" is crude, and conditioning on the conductor and the number of bad primes, as in Example 22.51(a), is a sharper null.

The equality rates and the mean excess are descriptive statistics of the population, not tests (Definition 22.40).

**Example 22.53** (sharpness against chance)**.** Under the null of Proposition 22.49 the expected number of equalities among the 20,317 bounded curves at $\ell = 3$ below 10,000 is $\sum \hat{F}(v = b(E)) \approx$ 4,548; observed: 14,420 (71.0%). With the law of $v$ on the bound-0 curves of Example 22.51, the expected equalities are $13{,}553 \cdot \frac{4079}{13534} + 5{,}534 \cdot \frac{1081}{13534} + 1{,}109 \cdot \frac{251}{13534} + 112 \cdot \frac{33}{13534} + 9 \cdot \frac{5}{13534} \approx 4{,}084.7 + 442.0 + 20.6 + 0.3 + 0.0 = 4{,}547.6$: under the null, a curve of bound 1 attains its bound with probability 0.30 and a curve of bound 2 with probability 0.08, while the data attain the bound more than three times as often overall. Over the union of the small ranges, the equality rates of (eq:rule) at $\ell = 3, 5, 7, 11, 13$ are 71.0%, 84.5%, 90.7%, 96.5%, 96.0%, and the mean excesses 0.37, 0.17, 0.10, 0.04, 0.04. The paper (working set): "$v_\ell(\deg \varphi_E)$ equals the right-hand side of (eq:rule) in 57.5% of the cases at $\ell = 3$, 75.6% at $\ell = 5$, 84.1% at $\ell = 7$, 91.6% at $\ell = 11$ and 92.4% at $\ell = 13$, and the mean excess at $\ell = 3$ is 0.63." The ordering by $\ell$ is the same; the levels differ with the range (Remark 22.42).

**Example 22.54** (the odd part of the degree)**.** The paper's last statistic concerns the optimal curves without CM and with no isogeny of odd degree (so that $E[\ell]$ is irreducible for every odd $\ell$; decided here from the row of the optimal curve in the `allisog` matrix): how often the odd part of $\deg \varphi_E$ equals $\prod_\ell \ell^{B_\ell^{\mathrm{tw}}(E)}$, the product over odd $\ell$. Below conductor 10,000 there are 33,137 such curves. The quotient (odd part of $\deg \varphi_E$)$/\prod_\ell \ell^{B_\ell^{\mathrm{tw}}}$ is always an integer, which is (eq:twist) at every odd $\ell$ at once: the quotient is an integer exactly when $v_\ell(\deg \varphi_E) \geq B_\ell^{\mathrm{tw}}(E)$ for every odd $\ell$, and since $B_\ell^{\mathrm{tw}}(E) = 0$ unless $\ell = 3$ or $\ell$ divides some index $n$ of a type $\mathrm{I}_n$ or $\mathrm{I}_n^*$, only finitely many $\ell$ are involved, but they include primes above 13 that the tables of Section 22.11 do not examine. The quotient is 1 for 5,675 curves (17.1%; 25.5% of 14,666 curves on $N < 5000$ and 10.4% of 18,471 on $5000 \leq N < 10000$). The most common quotients are 3 (5,926 curves), 1, 5 (2,978), 9 (2,532), 15 (2,024), 7 (1,918), 21 (1,088) and 11 (967). Among the nontrivial quotients, the smallest prime factor is 3 in 58.7% of the cases (27,462 curves). The paper (working set, 1,226,216 curves): equality in 1.85% ("the proportion is range dependent: it is 17.1% below conductor 10,000"), quotients "most often 9, 3, 15, 45, 27 or 21", "its smallest prime factor is 3 in 63% of the cases (58.7% below conductor 10,000)". Below 10,000 the paper also gives the shares of nontrivial quotients whose smallest prime factor divides $N$, 47.5%, and with some prime factor dividing $N$, 51.8%; the book's records give the same two numbers. The local terms account for a sharp part of $\deg \varphi_E$ and for less and less of the rest as $N$ grows; the rest is the adjoint Selmer group together with the size of the Petersson norm (Chapter 19).

## 22.11 Reproducing the paper's tables below conductor 10,000

This section reproduces, on all 38,042 optimal curves of conductor below 10,000, every table and every count of the paper's Section sec:data that the data of this range allow. The paper's numbers quoted here are taken from its text and from its printed tables (read row by row in Chapter 21); they concern its large ranges and are marked as the paper's. The book's counts are reproductions on data that were not blind (Section 22.1). All counts are over the base population at $\ell$ (Convention 22.3) unless another is named.

**Example 22.55** (Table tab:uniform in small)**.** For each range and $\ell$: the base population; the curves with $B_\ell^{\mathrm{rule}} > 0$, the violations of (eq:rule) and the equality rate; the violations of $B_\ell^{\mathrm{nc}}$ (the rule without the correction); and the curves with $B_\ell^{\mathrm{III}} > 0$, the violations and the equality rate of the III-free form.

| range | $\ell$ | curves | rule $> 0$ | viol. | eq. | viol. without $h$ | III-free $> 0$ | viol. | eq. |
|---|---|---|---|---|---|---|---|---|---|
| $N < 5000$ | 3 | 15,102 | 8,510 | 0 | 75.1% | 16 | 8,430 | 0 | 75.0% |
| | 5 | 16,863 | 3,078 | 0 | 88.4% | 0 | 3,068 | 0 | 88.3% |
| | 7 | 17,107 | 1,644 | 0 | 91.5% | 0 | 1,642 | 0 | 91.5% |
| | 11 | 17,190 | 515 | 0 | 97.5% | 0 | 515 | 0 | 97.5% |
| | 13 | 17,180 | 296 | 0 | 97.3% | 0 | 296 | 0 | 97.3% |
| $5000 \le N < 10000$ | 3 | 18,749 | 11,807 | 0 | 68.0% | 20 | 11,697 | 0 | 67.6% |
| | 5 | 20,456 | 4,398 | 0 | 81.7% | 0 | 4,368 | 0 | 81.7% |
| | 7 | 20,584 | 2,291 | 0 | 90.2% | 0 | 2,287 | 0 | 90.2% |
| | 11 | 20,649 | 741 | 0 | 95.8% | 0 | 741 | 0 | 95.8% |
| | 13 | 20,643 | 432 | 0 | 95.1% | 0 | 428 | 0 | 95.1% |
| $N < 10000$ | 3 | 33,851 | 20,317 | 0 | 71.0% | 36 | 20,127 | 0 | 70.7% |
| | 5 | 37,319 | 7,476 | 0 | 84.5% | 0 | 7,436 | 0 | 84.5% |
| | 7 | 37,691 | 3,935 | 0 | 90.7% | 0 | 3,929 | 0 | 90.8% |
| | 11 | 37,839 | 1,256 | 0 | 96.5% | 0 | 1,256 | 0 | 96.5% |
| | 13 | 37,823 | 728 | 0 | 96.0% | 0 | 724 | 0 | 96.0% |

The pure GP loop of Construction 22.67 and the Python scripts agree on $N < 2000$: `tablerow(2000, 1, 1, dir)` returns `[4951, 2517, 0, 2046]`, `[5622, 915, 0, 845]`, `[5735, 468, 0, 446]` at $\ell = 3, 5, 7$ (curves, positive bounds, violations, equalities), and `tablerow(2000, 3, 0, dir)` returns `[4951, 2499, 0, 2028]` for the III-free form; the Python counts are the same. The paper (Section sec:data): "no violation at any $\ell$" on the four ranges; the violations without the correction "are curves with image 3Ns"; the III-free form "holds for every curve of the tables without any correction"; equality 57.5% at $\ell = 3$ on the working set. The book's table agrees in kind on every point: no violation, the uncorrected failures are exactly 3Ns curves (Example 22.57), and the III-free form never fails.

**Example 22.56** (Table tab:counts in small)**.** For each range and $\ell$: the curves with $B_\ell^{\mathrm{main}} > 0$, violations and equality rate; the same for $B_\ell^{\mathrm{tw}}$; and for the Eisenstein population: its size, the curves with $B_\ell^{\mathrm{main}} > 0$, those with a positive deficit $D = B_\ell^{\mathrm{main}} - v$, the violations of the torsion bound $D \le v_\ell(\#E(\mathbb{Q})_{\mathrm{tors}}) + [\ell = 3,\ E[3] \simeq \mathbb{Z}/3 \oplus \mu_3]$ and of the class bound ($D \le$ the number of curves of the class with a rational point of order $\ell$), and the curves with $D = 2$.

| range | $\ell$ | main $> 0$ | viol. | eq. | twist $> 0$ | viol. | eq. | Eis. | main $> 0$ | $D > 0$ | viol. | viol. (class) | $D = 2$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $N < 5000$ | 3 | 5,908 | 0 | 59.0% | 6,996 | 0 | 61.3% | 2,096 | 1,725 | 284 | 0 | 0 | 11 |
| | 5 | 2,478 | 0 | 86.5% | 2,681 | 0 | 87.2% | 335 | 199 | 45 | 0 | 0 | 0 |
| | 7 | 1,351 | 0 | 89.8% | 1,452 | 0 | 90.6% | 91 | 35 | 9 | 0 | 0 | 0 |
| | 11 | 450 | 0 | 96.4% | 472 | 0 | 97.2% | 8 | 0 | 0 | 0 | 0 | 0 |
| | 13 | 272 | 0 | 96.3% | 286 | 0 | 97.2% | 18 | 2 | 0 | 0 | 0 | 0 |
| $5000 \le N < 10000$ | 3 | 8,069 | 0 | 49.1% | 9,545 | 0 | 51.6% | 1,906 | 1,624 | 147 | 0 | 0 | 3 |
| | 5 | 3,307 | 0 | 76.6% | 3,636 | 0 | 78.6% | 199 | 139 | 8 | 0 | 0 | 0 |
| | 7 | 1,719 | 0 | 86.9% | 1,894 | 0 | 87.8% | 71 | 32 | 3 | 0 | 0 | 0 |
| | 11 | 579 | 0 | 94.3% | 631 | 0 | 94.9% | 6 | 0 | 0 | 0 | 0 | 0 |
| | 13 | 369 | 0 | 93.8% | 389 | 0 | 94.1% | 12 | 2 | 0 | 0 | 0 | 0 |
| $N < 10000$ | 3 | 13,977 | 0 | 53.3% | 16,541 | 0 | 55.7% | 4,002 | 3,349 | 431 | 0 | 0 | 14 |
| | 5 | 5,785 | 0 | 80.8% | 6,317 | 0 | 82.2% | 534 | 338 | 53 | 0 | 0 | 0 |
| | 7 | 3,070 | 0 | 88.1% | 3,346 | 0 | 89.0% | 162 | 67 | 12 | 0 | 0 | 0 |
| | 11 | 1,029 | 0 | 95.2% | 1,103 | 0 | 95.9% | 14 | 0 | 0 | 0 | 0 | 0 |
| | 13 | 641 | 0 | 94.9% | 675 | 0 | 95.4% | 30 | 4 | 0 | 0 | 0 | 0 |

The paper (Section sec:forms, working set): equality in (eq:twist) for "39.4% at $\ell = 3$, 71.1% at $\ell = 5$, 81.2% at $\ell = 7$ and 91.5% at $\ell = 13$, and in (eq:main) for 36.9%, 69.7%, 80.4% and 91.2%"; the rule "bounds 867,578 curves instead of 705,563 at $\ell = 3$" (in the book's range 20,317 instead of 16,541). The ordering is the same in both: main below twist below rule, for the number of bounded curves and for the equality rates.

**Example 22.57** (Table tab:ns in small)**.** The optimal curves without CM and with image 3Ns, the violations of $B_3^{\mathrm{nc}}$ (all types, no correction), and the residues modulo 12 of the III-type primes $q \neq 3$ with $e_3(E,q) \geq 1$ of the violating curves (or $q = 2$):

| range | 3Ns curves | positive III term | violate $B_3^{\mathrm{nc}}$ | other curves violating | `ns3` $= 1$ | residues of the III primes | shortfall |
|---|---|---|---|---|---|---|---|
| $N < 5000$ | 30 | 30 | 16 (53%) | 0 | 30 | 1: 4, 11: 8, $q = 2$: 4 | always 1 |
| $5000 \leq N < 10000$ | 60 | 60 | 20 (33%) | 0 | 60 | 1: 8, 11: 12 | always 1 |
| $N < 10000$ | 90 | 90 | 36 (40%) | 0 | 90 | 1: 12, 11: 20, $q = 2$: 4 | always 1 |

Every one of the 36 violators has a prime $p \equiv -1 \pmod 3$ with $t_3(E,p) \geq 1$, and 12 of them have $3 \mid N$. Of the 90 curves, 74 have a prime $q \geq 5$ of type III or III$^*$ with $e_3 \geq 1$ (22 and 52 on the two ranges) and the other 16 get their positive III term only at $q = 2$, from PARI's factor; 44 have $3 \mid N$. Proposition prop:ns-structure (proved in Chapter 11) explains the first two observations; the data check them.

*Which inequality.* Section sec:anomaly says that Table tab:ns "records what happens to the inequality (eq:thm), with all types included and no correction". The book counts violations of $B_3^{\mathrm{nc}}$, the rule with every term and no correction. Read as written, (eq:thm) has the Euler terms at $q \neq \ell$ and the terms $t_\ell(E,q)$ at every bad $q$, $q = \ell$ included, since Definition def:terms defines $t_\ell$ there; its right side is $B_3^{\mathrm{thm}} = B_3^{\mathrm{nc}} - w_3$ (Convention 22.2), and it has the same 36 violators (16 and 20 on the two ranges): dropping the weights $w_3$ changes nothing. Only if the terms at $q = 3$ are dropped altogether, $t_3(E,3)$ included, are there 24 violators (8 and 16), and still no curve without image 3Ns violates. The paper records the same: "Below conductor 10,000 there are 36 violations with them, as (eq:thm) is written, and 24 without", the twelve curves that differ (`726a1`, `726f1`, `2178b1`, `2178h1`, `3840h1`, `3840k1`, `3840o1`, `3840r1`, `5070h1`, `5070p1`, `5808y1`, `5808z1`) all having $t_3(E,3) = 1$ from a type $\mathrm{I}_n$ (codes 7, 10) or $\mathrm{I}_n^*$ (code $-7$) at 3, and "the weights $w_3$ change nothing". The paper (D): 364, 68, 43 and 0 violations on its four ranges, 31%, 23%, 18% of the 3Ns curves. The falling rate with $N$ is visible inside the book's range already (53%, then 33%).

**Example 22.58** (Section sec:iso in small)**.** Curves of the base population at $\ell$ with exactly one additive prime $q$, with $q \geq 5$, $q \neq \ell$, and no multiplicative prime with $\ell \mid n$: the minimum of $v_\ell(\deg \varphi_E)$ by the type at $q$ and the value of $e = e_\ell(E,q)$, with the number of curves. At $\ell = 3$:

| type | $e$: minimum (curves) | | |
|---|---|---|---|
| II | 1: 1 (264); 2: 2 (27) | II$^*$ | 1: 1 (135); 2: 2 (11) |
| IV | 1: 1 (150); 2: 2 (11) | IV$^*$ | 1: 1 (249); 2: 2 (27) |
| III | 0: 0 (193); 1: 1 (44); 2: 2 (2) | III$^*$ | 0: 0 (191); 1: 1 (44); 2: 2 (2) |
| $\mathrm{I}_0^*$ | 0: 0 (146); 1: 1 (42); 2: 2 (88); 3: 3 (50) | $\mathrm{I}_n^*$ | 1: 1 (813); 2: 2 (57); 3: 3 (3) |

In every $\mathrm{I}_n^*$ cell the excess $v_3(\deg\varphi_E) - e - t_3$ has minimum 0, also cell by cell in $(e, t_3)$; that is, the excess over $e$ has minimum exactly $t_3(E,q) = v_3(n)$, as the paper says. At $\ell = 5$ every cell's minimum equals $e$: $\mathrm{I}_0^*$ $e = 0, 1, 2$ (94, 75, 10 curves), II 0, 1 (210, 12), II$^*$ (94, 5), III (152, 11), III$^*$ (150, 11), IV (95, 5), IV$^*$ (209, 12), $\mathrm{I}_n^*$ (427, 141); and at $\ell = 7$ likewise, with $e \le 1$ (the largest cell $\mathrm{I}_n^*$ with $e = 0$, 806 curves). *Multiplicative primes alone:* for the curves with no additive prime and exactly one multiplicative prime $q$ with $\ell \mid n$, $\min(v_\ell(\deg\varphi_E) - v_\ell(n)) = 0$ for every value of $v_\ell(q^2-1)$: at $\ell = 3$, 4,367 curves, with $v_3(q^2-1) = 1, 2, 3, 4$ for 3,094, 268, 36, 3 curves and $q = 3 = \ell$ for 966; at $\ell = 5$, 2,588 curves. *Two primes:* the curves of the base population at $\ell = 3$ whose only contributions are two odd primes of type II or II$^*$ are 8 (`2450d1`, `3675k1`, `6050bg1`, `7350e1`, `7350ba1`, `7350bw1`, `8281h1`, `9800bj1`), and $9 \mid \deg\varphi_E$ for all 8. The paper (D, working set): the same pattern in every cell; the cells with $e = 0$ have 5,534 (III) and 5,394 ($\mathrm{I}_0^*$) curves; $\mathrm{I}_0^*$ runs up to $e = 6$; 2,502 curves with two primes of type II or II$^*$. In the multiplicative statement, "for every value of $v_\ell(q^2-1)$ up to 6", the paper notes that "at $\ell = 3$ the value 0 occurs only for $q = \ell = 3$", as in the book's count above.

**Example 22.59** (Section sec:tam in small)**.** The hypothesis population at $\ell$ (Convention 22.3), restricted to the curves with at least one prime $q$ of type $\mathrm{I}_n$ with $\ell \mid n$ and $q \equiv 1 \pmod{\ell}$; the extra term $\sum_q \min(v_\ell(n), v_\ell(q-1), v_\ell(c(u)))$ over such primes, from `ellinit(E, O(q^30)).tate` as in Example 22.12; and the refined bound $B_\ell^{\mathrm{rule}}$+ (extra term).

| | curves | positive extra term | violations | equality before | equality after |
|---|---|---|---|---|---|
| $\ell = 3$, type $\mathrm{I}_n$ | 989 | 179 | 0 | 550 (55.6%) | 662 (66.9%) |
| $\ell = 3$, with $\mathrm{I}_n^*$ | 1,050 | 186 | 0 | 596 (56.8%) | 714 (68.0%) |
| $\ell = 5$, type $\mathrm{I}_n$ | 246 | 24 | 0 | 197 (80.1%) | 214 (87.0%) |
| $\ell = 5$, with $\mathrm{I}_n^*$ | 252 | 24 | 0 | 203 (80.6%) | 220 (87.3%) |
| $\ell = 7$ | 18 | 1 | 0 | 17 | 18 |

The extra term is never more than 1 in this range. The rows "with $\mathrm{I}_n^*$" also use the odd primes of type $\mathrm{I}_n^*$ with $n \ge 1$ and $\ell \mid n$ and $q \equiv 1 \pmod{\ell}$, whose Tate parameter is that of the twist by $q^*$, which is multiplicative at $q$ (Example 22.12); the prime 2 never qualifies. The first curves with a positive extra term are `791a1` ($q = 7$), `805b1` (7, $\bar{u} = 1$), `805c1`, `805d1`, `874a1` (19, $\bar{u} = 12$), `874b1` (19, $\bar{u} = 18$) at $\ell = 3$; `4136e1`, `4136f1` (11, $\bar{u} = 10$), `5673a1` (61), `6507a1`, `6507b1` (241) at $\ell = 5$; and `4171a1` (43, $\bar{u} = 1$) at $\ell = 7$. The paper (D, working set, type $\mathrm{I}_n$ only): at $\ell = 3$, 46,222 curves, 12,095 with a positive extra term, equality from 45.0% to 60.6%; at $\ell = 5$, 10,940, 1,646, from 67.0% to 78.7%; and, including $\mathrm{I}_n^*$ below conductor 10,000, "1,050 such curves at $\ell = 3$, with equality rising from 56.8% to 68.0%, and 252 at $\ell = 5$, with equality rising from 203 to 220 curves", which the table reproduces exactly.

**Example 22.60** (Section sec:eis in small)**.** At $\ell = 3$: 4,002 Eisenstein curves, 3,349 with $B_3^{\mathrm{main}} > 0$, 431 with $D > 0$, $D = 2$ for 14. No curve without a rational point of order 3 has $D > 0$, and the torsion bound is attained with $D > 0$ in 427 cases. $D = 2$ occurs for twelve curves with image `3Cs.1.1` (`14a1`, `26a1`, `35a1`, `38a1`, `77b1`, `158d1`, `278b1`, `326c1`, `485a1`, `4385b1`, `7094c1`, `8027a1`), all of rank 0, with torsion $\mathbb{Z}/3$ except `14a1` (torsion $\mathbb{Z}/6$), and for two curves with a point of order 9 (`714i1`, `6942n1`, image `3B.1.1`). Of the 49 optimal curves without CM with image `3Cs.1.1`, none has a point of order 9 (torsion $\mathbb{Z}/3$ for 48 and $\mathbb{Z}/6$ for `14a1`); the four optimal curves with a point of order 9 (`714i1`, `1482l1`, `1554n1`, `6942n1`) have image `3B.1.1`. That the two sources of a deficit 2 never combine is a theorem (Proposition prop:no9). At $\ell = 5$ and 7, $D \le 1$, and $D > 0$ only with a rational point of order $\ell$ (53 and 12 curves). Measured against (eq:twist), the deficit $B_\ell^{\mathrm{tw}} - v$ exceeds the torsion bound for 180 curves at $\ell = 3$, 35 at

$\ell = 5$ and 10 at $\ell = 7$. The paper (D, working set): 75,430 Eisenstein curves, 67,845 with a positive bound, 2,777 with $D > 0$, $D = 2$ for 22 ("seventeen curves, all of rank 0, with torsion $\mathbb{Z}/3$ or $\mathbb{Z}/6$, for instance 14a1", and five with a point of order 9); the deficit against (eq:twist) exceeds the bound for 1,148 curves at $\ell = 3$ and 141 at $\ell = 5$. The book's twelve `3Cs.1.1` curves with $D = 2$ fit this description, `14a1` being the one with torsion $\mathbb{Z}/6$.

**Example 22.61** (the twist pairs after Theorem thm:twist)**.** For each optimal curve without CM below 10,000 and each odd $q$ with $q^2 \parallel N$ in case (a) (type $\mathrm{I}_n^*$), (b) (type $\mathrm{I}_0^*$) or (c) (potentially good, $q \geq 5$), let $E_q$ be the twist by $q^*$, found in `allcurves` through `ellminimalmodel`, and compare $v_\ell$ of the two optimal degrees for each $\ell \in \{3, 5, 7, 11, 13\}$ with $\ell \neq q$ and $\ell^2 \nmid N$. Pairs (curve, $q$): (a) 6,348, (b) 2,183, (c) 5,150; every twist was found in the tables. Triples (curve, $q$, $\ell$): (a) 24,373, (b) 8,348, (c) 19,554. In all of them $v_\ell(\deg \varphi_E) \geq v_\ell(\deg \varphi_{E_q})$; the inequality is strict in 2,779 triples of case (a), in 1,308 of case (b) and in none of case (c). Examples in case (a) at $\ell = 3$: `121d1` against `11a1` ($\deg \varphi_E$ 24 against 1), `75b1` against `15a1` (6 against 1), `150c1` against `30a1` (48 against 2), `539c1` against `77c1` (288 against 6), `980i1` against `140b1` (2880 against 60); in case (b), `275b1` against `11a1` at $\ell = 7$ (28 against 1) and `539d1` against `11a1` at $\ell = 3$ (72 against 1). When moreover $\ell \nmid N$ and $E[\ell]$ is irreducible, the difference equals the Euler term $e_\ell(E, q)$ in all 19,691 triples of case (a) and all 6,701 of case (b), as Theorem thm:quant predicts (Chapter 20). Chapter 20 counts in another unit, pairs (curve, $\ell$) with the twist by all odd primes of type $\mathrm{I}_0^*$ or $\mathrm{I}_n^*$ at once, $\ell \in \{3, 5, 7, 11, 13\}$, $\ell \nmid N$ and $E[\ell]$ irreducible: 25,477 such pairs (15,607 with 3 in the twisting set, 8 with $\ell = 3$ and image 3Ns), with the difference equal to the sum of the Euler terms in every one, and 14,880 pairs of case (c) (42 with image 3Ns at $\ell = 3$), all with equal valuations; the numbers of pairs were recounted here from the book's records, the equalities are checked there. The paper (D, working set): 157,938 and 292,580 pairs, "with no exception, with equality in every pair of case (c)"; it notes that in case (a) at $\ell = 3$ the predicted difference, the Euler term, is positive for every $q \neq 3$, and that "Below conductor 10,000 the difference equals the Euler term in all 25,477 pairs (curve, $\ell$)" of the twisting-set form.

**Example 22.62** (a twist relation at $q = \ell$, observed)**.** The weights $w_3$ of Conjecture conj:euler have a pattern: under the twist by $\mathbb{Q}(\sqrt{-3})$, which exchanges $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$ at 3 (Example 22.14), the starred type always has weight one more than its partner ($2 = 1 + 1$, $1 = 0 + 1$, $2 = 1 + 1$). The book's data show the same phenomenon at every $\ell$, as a relation between the degrees of a curve and of its twist at $q = \ell$. For each optimal curve $E$ without CM below conductor 10,000 with $\ell \mid N$, $\ell \in \{3, 5, 7, 11, 13\}$, let $E^{(\ell)}$ be its twist by $\chi_{\ell^*}$ and compare $v_\ell(\deg \varphi_E)$ with $v_\ell$ of the degree of the optimal curve of the class of $E^{(\ell)}$:

(a) if $E$ has type $\mathrm{II}^*$, $\mathrm{III}^*$ or $\mathrm{IV}^*$ at $\ell$, the difference is exactly 1 in all 4,412 cases (2,141, 1,210, 659, 239, 163 at $\ell = 3, 5, 7, 11, 13$; 4,251 of them with $E[\ell]$ irreducible);

(b) if $E$ has type $\mathrm{I}_n^*$ at $\ell$, the difference is 0 in all cases at $\ell = 3, 5, 7, 11, 13$ (3,639, 1,360, 677, 240, 173);

(c) if $E$ has type $\mathrm{I}_0^*$ at $\ell$ and $E[\ell]$ is irreducible, the difference is $v_\ell((\ell+1)^2 - a_\ell(E^{(\ell)})^2)$ in all 1,915 cases (it fails in 13 of the cases with $E[\ell]$ reducible, for instance `126a1` at $\ell = 3$ and `275b1` at $\ell = 5$).

In (b) and (c) the difference is what the numerator form of the $e_\ell$ column of the table of Section sec:localterms (to which Lemma lem:euler refers), $v_\ell(q^2-1)$ and $v_\ell(q-1)+v_\ell((q+1)^2-a'^2)$, gives at $q = \ell$ ($v_\ell(\ell^2-1) = 0$ and $v_\ell(\ell-1) = 0$); the rational forms $1-q^{-2}$ and $(q-1)((q+1)^2-a'^2)/q^3$

would give negative valuations at $q = \ell$. In (a) it is not, since at $q \neq \ell$ a potentially good pair has difference 0 (Example 22.61). All counts are over the curves whose twist has conductor below 10,000, so that its degree is in the tables. At $\ell \geq 5$, (a) suggests a weight 1 for the types II$^*$, III$^*$, IV$^*$ at $q = \ell$; added to (eq:rule), it gives no violation below 10,000 at $\ell = 5, 7, 11, 13$ and is attained by most of these curves. These are observations made on the whole of the book's range at once, with no hold-out, and no explanation is offered here. The paper records the same observations, with the same counts, in Remark rem:oq1 ("That no term is needed at $q = \ell \geq 5$ is true but not sharp, by the following observation"), and they bear on its first open question, which asks for the weights at $q = \ell$.

**Example 22.63** (further checks)**.** Below conductor 10,000, on the optimal curves:

(a) Lemma lem:euler and Lemma lem:twist: Examples 22.11 and 22.14.

(b) Proposition prop:shapes at $q = 2, 3$: the cubic factors, Example 22.33.

(c) Proposition prop:forms: all 3,226 optimal curves with type IV or IV$^*$ at 2 (3,194 of them without CM) have $P_2(X) = 1 + 2X$. The paper, after Proposition prop:forms: "On the 92,812 working-set curves with type IV or IV$^*$ at 2, PARI's factor at 2 is $1 + 2X$ for every one of them, as Lemma lem:IV requires." The data check a theorem.

(d) Section sec:two: for the 25,086 optimal curves without CM and without a label `2B` or `2Cs` (no rational 2-isogeny), the inequality $v_2(\deg \varphi_E) \geq \sum_{q \parallel N} v_2(n_q) + \#\{q : \text{type III}, \text{III}^*, \text{I}_0^* \text{ or } \text{I}_n^*\}$ holds with no exception, while counting $\text{I}_0^*$ with weight 2 fails 6 times (`99d1`, `931b1`, `2299d1`, `2475h1`, `5819a1`, `5931b1`), as the paper's 123, 26 and 20 exceptions on its ranges lead one to expect.

(e) The optimal CM curves of odd modular degree are `27a1`, `32a1`, `36a1`, `49a1` (degree 1) and `243b1` (degree 9); `27a1` and `243b1` have torsion $\mathbb{Z}/3$, no rational point of order 2 and non-prime conductor, so they fall under the CM case of the theorem of Calegari and Emerton quoted in Section sec:two (Chapter 14 states it).

*Remark* 22.64 (what agrees and what does not). The book's range is about fifty times smaller in conductor than the paper's working set, and every rate that depends on $N$ differs: the equality rates, the share of 3Ns curves that fail without the correction, the odd-part statistic, the mean excess. All of them move with $N$ inside the book's range already, in the direction of the paper's values. Every qualitative claim of Section sec:data reproduces: no violation of (eq:rule), (eq:main), (eq:twist) or of Conjecture conj:eis; the uncorrected failures are exactly 3Ns curves and fail by exactly one; the minima equal $e_\ell$ cell by cell; the weights $w_3$ are the minima at $q = 3$; the Tamagawa refinement has no exception and its counts below 10,000 are the paper's; the strict twist inequalities occur in cases (a) and (b). The paper's own counts below conductor 10,000 (Section 22.8, Examples 22.33, 22.54, 22.57, 22.59 and 22.62) are reproduced exactly.

## 22.12 Reproducibility

This last section describes the repository that the paper's appendix names, prints the book's own scripts, and states a reproducibility checklist applied to the book's runs. In the paper it is the appendix and the last sentences of the AI statement.

*Remark* 22.65 (the paper's repository). The appendix says: "All scripts, the pinned data commit and the result tables are in the repository `https://github.com/dsr001001/mcptrial` (directory `ecmine`)." It names `scripts/step10_sym2_euler.py` for the symmetric-square Euler factors, `scripts/step10_rule.py` and `scripts/step11_tw_boundary.py` for the rule on the whole ranges ("correction term, image statistics"), `scripts/step12_tamagawa.py` for the Tamagawa factors, `scripts/conj_check.py` for the two corollary forms, `scripts/euler_rule.gp` for the functions of Construction 22.23, and `twist_lemma_check.gp`; it adds that "each runs on the full tables in minutes to hours" and that "The pipeline was rerun from a clean clone and reproduced every count of Table tab:counts."

The repository exists, and its directory `ecmine` (branch `ccr-f42a94ab-alw7cu`, commit `f57c783`) holds these scripts, the result files and, in `paper/tables`, the six table files that the paper prints. The tables are written by scripts from result files. Tables tab:uniform and tab:ns come from `scripts/step11_tw_boundary.py`, which also writes `results/step11_summary.csv` and `results/step11_ns3.csv`; `scripts/step10_rule.py` writes `results/step10_rule_{work,holdout,ext,oot}_summary.csv`, the rule without the correction, and `scripts/step10_make_table.py` builds from them an older layout of Table tab:uniform with the III-free rule in the main columns, which is not the printed one. Tables tab:counts, tab:isolated and tab:oot come from `scripts/make_tables.py`, the first from `results/check_{work,holdout,ext,oot}_summary.csv` (written by `scripts/conj_check.py`), the second from `results/step6_u1_fit_work.txt` (untwisted rows) and per-curve files `results/step7_twist2_*.parquet` (twisted rows), the third from `results/oot_table.parquet`, which `scripts/oot_convert.py` builds from `results/oot_1.tsv` and `results/oot_2.tsv`. Table tab:iso was copied, as the repository's notes say, from `results/step9_euler_l3.txt`.

For this book the printed tables were compared with these files, by reading them; no script of the repository was run, and no large-range count was recomputed. This is a consistency check of the tables against the repository's output files, not a reproduction. The result: the six files in `paper/tables` agree character for character with the tables printed in the paper. Every entry of Table tab:uniform agrees with `results/step11_summary.csv`, whose rates are rounded to four decimals (two of them, 0.9155 and 0.5395, lie on a rounding boundary and are printed 91.6% and 54.0%); its columns of curves, positive bounds, violations without $h$ and the III-free counts agree with `results/step10_rule_*_summary.csv` as well. All eighteen rows of Table tab:counts agree with `results/check_*_summary.csv`, and all four rows of Table tab:ns with `results/step11_ns3.csv`, which also records, in a column the table does not print, that each of the 364, 68 and 43 failing curves has a potentially multiplicative prime $p \equiv 2 \pmod 3$ with $3 \mid v_p(j_E)$, the paper's observation on $t_3(E,p) \geq 1$. Every cell of Table tab:iso agrees with `results/step9_euler_l3.txt`. The five untwisted rows of Table tab:isolated agree with `results/step6_u1_fit_work.txt`; its four twisted rows could not be compared, because the per-curve `.parquet` files are excluded from the repository by its `.gitignore`. Table tab:oot was recounted from `results/oot_1.tsv` and `results/oot_2.tsv` (1,261 lines, 893 curves after removing repeated curves as `oot_convert.py` does), and every family's count, largest conductor and median time agrees. The scripts below are the book's own.

**Construction 22.66** (the script `sessions.gp`)**.** The functions used in the sessions of Sections 22.3 and 22.6; `rule_terms` repeats the loop of `rule` and records each row. `eulercheck` returned `[]` on every optimal curve below 10,000 (Example 22.11).

```
\\ sessions.gp: per-prime breakdown of the rule, Tate data, and the twist check.
```

```
\\ rule_terms(E, l, iii): one row [q, code, P_q(x), P_q(q^-2), e, w, t] per bad prime q.
rule_terms(E, l, {iii = 1}) =
{ my(R = ellglobalred(E), P = R[4][,1], K = R[5], S = lfunsympow(E, 2), M = []);
  for(i = 1, #P,
    my(q = P[i], k = K[i][2], F = 1/lfuneuler(S, q), r = subst(F, x, 1/q^2),
       e = 0, w = 0, t = 0, vj = valuation(E.j, q));
    if(q != l,
      my(rr = if(k >= 5, r / (1 - 1/q^2), r));
      if(abs(k) != 3 || iii, e = valuation(rr, l)),
      if(l == 3, w = if(k == 4 || k == 2, 1, if(k == -4 || k == -2, 2, if(k == -3, 1, 0)))));
    if(vj < 0, t = valuation(-vj, l));
    M = concat(M, [[q, k, F, r, e, w, t]]));
  M; }
\\ tatedata(E, q, l): [n, q_E, ubar, k], k the largest k <= v_l(q-1) with ubar an l^k-th power
tatedata(E, q, l) =
{ my(Eq = ellinit(E, O(q^30)), qE = Eq.tate[3], n = valuation(qE, q),
     u = lift(qE / q^n + O(q)), v = valuation(q - 1, l), k = 0);
  while(k < v && Mod(u, q)^((q - 1) / l^(k + 1)) == 1, k++);
  [n, qE, u, k]; }
\\ twistcheck(E): for each odd bad prime q, [q, code of E at q, code of the twist by q* at q, conductor
     of the twist]
twistcheck(E) =
{ my(P = ellglobalred(E)[4][,1], V = []);
  for(i = 1, #P, my(q = P[i]); if(q == 2, next);
    my(qs = if(q % 4 == 1, q, -q), Et = ellinit(elltwist(E, qs)));
    V = concat(V, [[q, elllocalred(E, q)[2], elllocalred(Et, q)[2], ellglobalred(Et)[1]]]));
  V; }
\\ euler_by_type(E, q): L_q(A_f,1)^(-1) from the Kodaira type, bad prime q >= 5 (Lemma lem:euler)
euler_by_type(E, q) =
{ my(k = elllocalred(E, q)[2]);
  if(k >= 5 || k <= -5, return(1 - 1/q^2));
  if(abs(k) == 2 || abs(k) == 4, return(1 - kronecker(-3, q)/q));
  if(abs(k) == 3, return(1 - kronecker(-4, q)/q));
  if(k == -1, my(a = ellap(ellinit(elltwist(E, if(q % 4 == 1, q, -q))), q));
    return((q - 1)*((q + 1)^2 - a^2)/q^3));
  error("not a bad prime"); }
\\ eulercheck(E): the bad primes q >= 5 where PARI's factor and the table differ (expected: [])
eulercheck(E) =
{ my(S = lfunsympow(E, 2), P = ellglobalred(E)[4][,1], bad = []);
  for(i = 1, #P, my(q = P[i]); if(q < 5, next);
    if(subst(1/lfuneuler(S, q), x, 1/q^2) != euler_by_type(E, q), bad = concat(bad, q)));
  bad; }
```

**Construction 22.67** (the script `table_small.gp`)**.** One row of Example 22.55 in pure GP, from the files `degphi` and `galrep` in the directory `dir`; it needs `euler_rule.gp` for `ns3`. `tablerow(Nmax, l, iii, dir)` returns [curves, positive bounds, violations, equalities] for the base population at $\ell$ with $N <$ `Nmax`. Below 2,000 each call takes a few seconds.

```
\\ table_small.gp: the paper's rule on Cremona's optimal curves of conductor < Nmax.
\\ Needs euler_rule.gp (the paper's appendix) for ns3; reads the ecdata files degphi.* and galrep.*.
rulebound(E, l, iii) =
{ my(R = ellglobalred(E), P = R[4][,1], K = R[5], S = lfunsympow(E, 2), b = 0, j = E.j);
  for(i = 1, #P,
    my(q = P[i], k = K[i][2], e = 0, t = 0, vj = valuation(j, q));
    if(q != l,
      my(r = subst(1/lfuneuler(S, q), x, 1/q^2));
      if(k >= 5, r /= (1 - 1/q^2));
      if(abs(k) != 3 || iii, e = valuation(r, l)),
      if(l == 3, e = if(k == 4 || k == 2, 1, if(k == -4 || k == -2, 2, if(k == -3, 1, 0)))));
    if(vj < 0, t = valuation(-vj, l));
    b += e + t);
  if(iii && l == 3, b -= ns3(E));
  b; }
\\ the thirteen rational CM j-invariants
```

```
{ CMJ = Set([0, 1728, -3375, 8000, -32768, 54000, 287496, -884736, -12288000, 16581375,
         -884736000, -147197952000, -262537412640768000]); }
\\ galrep: a Map from a curve label to its list of non-surjective image labels
readgal(file) =
{ my(M = Map()); foreach(readstr(file), s, my(w = strsplit(s, " "));
    mapput(M, w[1], w[2..#w])); M; }
strprefix(s, p) = #s >= #p && Vecsmall(s)[1..#p] == Vecsmall(p);
\\ E[l] reducible iff a label begins with "lB" or "lCs" (l followed by a letter, so 37B is not 3B)
reducible(G, lab, l) =
{ my(L); if(!mapisdefined(G, lab, &L), return(0));
  for(i = 1, #L, my(s = L[i]);
    if(strprefix(s, Str(l, "B")) || strprefix(s, Str(l, "Cs")), return(1)));
  0; }
\\ one row of the table: [curves, positive bounds, violations, equalities]
tablerow(Nmax, l, iii, dir) =
{ my(G = readgal(Str(dir, "/galrep.00000-09999")), c = [0, 0, 0, 0]);
  foreach(readstr(Str(dir, "/degphi.00000-09999")), s,
    my(w = strsplit(s, " "), N = eval(w[1]));
    if(N >= Nmax, next);
    my(E = ellinit(eval(w[6])), deg = eval(w[4]), lab);
    if(setsearch(CMJ, E.j), next);
    \\ the label of the optimal curve is N class number, except in class 990h (990h3)
    lab = Str(w[1], w[2], if(N == 990 && w[2] == "h", 3, w[3]));
    if(reducible(G, lab, l), next);
    my(b = rulebound(E, l, iii), v = valuation(deg, l));
    c[1]++; if(b > 0, c[2]++; if(v < b, c[3]++); if(v == b, c[4]++)));
  c; }
```

*Remark* 22.68 (the book's Python scripts). The counts of Sections 22.8, 22.10 and 22.11 come from five Python scripts using `cypari2`.

(i) `compute.py` writes one JSON record per optimal curve: the label (looked up in `allcurves` from the equation of `degphi`), $N$, $\deg\varphi_E$, $j$, CM, the `galrep` labels, `ns3`, rank, torsion, the row of the isogeny matrix and the torsion orders of the class, and for each bad prime $q$: $f_q$, the code, $c_q$, $P_q$, $P_q(q^{-2})$, $a_q$, $v_q(j)$, the value of `euler_by_type` for $q \geq 5$, and for odd $q$ the code, conductor exponent and minimal model of the twist by $q^*$. It calls one GP function per curve and runs in about 13 seconds on all 38,042 curves. Its core:

```
pari = cypari2.Pari(); pari.allocatemem(64*10**6, 2*10**9, silent=True)
for f in ['euler_rule.gp', 'sessions.gp', 'compute.gp']: pari(f'read("{f}")')
locd = pari('locdata')          # [N, j, ns3, per-prime data], one GP call per curve
for line in open(D + 'degphi.00000-09999'):
    w = line.split(); a = eval(w[5]); lab = bya[w[5]]   # bya: equation -> label, from allcurves
    N, j, ns3, V = locd(pari.ellinit(a))
    ...                          # convert to Python integers and fractions, write one JSON line
```

(ii) `common.py` computes the terms and the five bounds from the records exactly as `rule` does:

```
def terms(E, l, iii=1):
    out = []
    for p in E['primes']:
        q, k = p['q'], p['k']; e = w = t = 0
        if q != l:
            r = p['rf'] / (1 - Fraction(1, q * q)) if k >= 5 else p['rf']
            if abs(k) != 3 or iii: e = vl(r, l)
        elif l == 3: w = W3(k)
        if p['vj'] < 0: t = vl(-p['vj'], l)
        out.append((q, k, e, w, t))
    return out
def bounds(E, l):
    T1 = terms(E, l, 1); T0 = terms(E, l, 0)
    nc = sum(e + w + t for _, _, e, w, t in T1)
```

```
    h = E['ns3'] if l == 3 else 0
    main = sum(vl(phig(p['k']), l) for p in E['primes'])
    tw = 0
    for p in E['primes']:
        q, k = p['q'], p['k']
        if k >= 5 or (k <= -5 and q != 2): tw += vl(nidx(k), l)
        elif l == 3 and (abs(k) == 4 or (abs(k) == 2 and q != 2)): tw += 1
    thm = sum(e + t for _, _, e, w, t in T1)       # e = 0 at q = l: (eq:thm) as written
    return dict(rule=nc - h, nc=nc, iii=sum(e + w + t for _, _, e, w, t in T0),
                main=main, tw=tw, thm=thm)
```

Here `vl` is the $\ell$-adic valuation of a fraction, `W3` the weight of Convention 22.9, `phig` the order of $\Phi_q^{\mathrm{geom}}$ read off the code and `nidx` the index $n$ of $\mathrm{I}_n$ or $\mathrm{I}_n^*$. The population filter compares the prime part of each `galrep` label exactly, and CM is decided by $j$. The tables of Section 22.11 are loops over the records with these functions.

(iii) `calib.py` computes Examples 22.51 and 22.53; its core:

```
def calib(P, l):
    data = [(bounds(E, l)['rule'], v(E, l)) for E in P]
    zero = [vv for b, vv in data if b == 0]; F = Counter(zero); n0 = len(zero)
    pos = [(b, vv) for b, vv in data if b > 0]
    ps = [sum(F[x] for x in F if x < b) / n0 for b, vv in pos]     # Fhat(v < b)
    Ehat = sum(ps); sd = math.sqrt(sum(p * (1 - p) for p in ps))
    Eeq = sum(F[b] / n0 for b, vv in pos)                          # expected equalities
    return Ehat, sd, Eeq, sum(vv < b for b, vv in pos), sum(vv == b for b, vv in pos)
```

The stratified version applies `calib` in each stratum; the permutation version shuffles $v$ within strata with `random.seed(0)`, in the orders stated in Example 22.51(b).

(iv) `tam.py` computes Example 22.59, calling `ellinit(E, 0(q^30)).tate` at each qualifying prime (through the twist by $q^*$ at a prime of type $\mathrm{I}_n^*$) and testing $\bar{u}^{(q-1)/\ell^k} = 1$.

(v) `moddeg_check.py` compares `ellmoddegree` with `degphi` line by line (Example 22.37), and `curvedata_check.py` runs the cross-checks of Example 22.18.

**Convention 22.69** (a reproducibility checklist)**.** Applied to the book's runs:

(1) *Pin the data.* The commit of `ecdata` (first line) and the SHA-256 sums of the files read:

```
commit 25cec5ecfec8b9f016eb1631ac633194c2bed39f
b078b5226d276fbc64289833abfd8430e3f5969c657bc32f94db546e8f96e8fd  allbsd.00000-09999
259f3846329395b371e8079c77a6f1097adaebc98a054974573e241416efa968  allcurves.00000-09999
9e033f1d092509f280d23e2f2666577c96af9593911568ca9e9678c2e35b1d82  allisog.00000-09999
bc06d9bd10dbba36ddb84ad2fc407d2a71c8c89f22361befae8eda08aa18a377  alllabels.00000-09999
65d96cbab0de966409810b666e55b731515728c4d19904ea95b3750d48f6dd68  degphi.00000-09999
7faf8a6a92d85eba62dda9d025474a543bcd3ae61b8c489f581124f2ec34742d  galrep.00000-09999
89f4c055d668d7e7e55936bd5d391aaaa60df3133be8effc03394abcbc6ab7d5  alldegphi.00000-09999
b42836e901ac75874d13207faef0bdc47a3799571937b1099df56c57897e8040  curvedata.00000-09999
f1b7d7c310eb897ed4aad4da3e809b754e95be837cc4c1b404e1ed104b913223  classdata.00000-09999
11e4ebd69dbef2c6d9fce35c989d374f64a179e9d6ec13b5916cf453acd9298a  opt_man.00000-09999
1d523b06ba4920e27026224dc35f9d2c433432b645e26712a3f3defb45d42e29  file-format.txt
```

The first six are the files of the counts, the next four those of the cross-checks, the last the repository's description of the formats. All eleven are fetched from the commit by `tools/setup_env.sh` into `data/ecdata` (not under version control) and checked against these sums, listed in `tools/ecdata.sha256`. The commit is identified by its hash; we do not record its date.

(2) *Record the software and the settings.* PARI/GP 2.17.2, `cypari2` 2.2.4, Python 3.13; `parisize` $6.4 \cdot 10^7$ and `parisizemax` $2 \cdot 10^9$ bytes unless stated; random seed 0 for the permutations, with the orders of curves and strata of Example 22.51(b).

(3) *Separate data, code and results*, and write results as plain files with the counts (the JSON records), not only as summaries.

(4) *Rerun from a clean copy* and compare the counts.

(5) *Cross-check independently.* Here: the GP loop against the Python bounds; `ellmoddegree` against `degphi`; `ns3` against `galrep`; PARI's Euler factors against Lemma lem:euler; `ellglobalred` against `curvedata`; optimality flags against `degphi`.

(6) *State what was not checked.* Here: everything above conductor 10,000, where the paper's printed tables were only compared with the result files of its repository (Remark 22.65); the paper's out-of-table curves, whose modular degrees were not recomputed; PARI 2.15; the scripts of the paper's repository, none of which was run.

*In the paper* 22.70. The AI statement: "The scripts, the result files and the pinned data commit are in the repository named in the appendix, so that every count can be rerun". Below conductor 10,000 the pinned commit is the one used here, and every count of the paper that falls in this range has been rerun from it: the Tamagawa counts of Section sec:tam (Example 22.59), the two readings of the 3Ns violations (Example 22.57), the odd-part statistic (Example 22.54), the twist comparisons after Theorem thm:twist in their one-prime form (Example 22.61), the twist relation at $q = \ell$ of Remark rem:oq1 (Example 22.62) and the data quoted in Remark rem:wild and Section sec:boundary (Section 22.8 and Example 22.33).

## 22.13 Exercises

The exercises go from routine to harder. Each numerical answer was computed with PARI/GP 2.17.2 and the files of Convention 22.69.

**Exercise 22.71.** Run `ellglobalred` on `20a1` $= [0, 1, 0, 4, 4]$ and on `1232l1` $= [0, 0, 0, 32, -16]$. Read off the conductor, the bad primes, the Kodaira codes and types, the conductor exponents and the $c_p$, and say which entries are the paper's `R[4][,1]` and `K[i][2]`. Why does the code 1 never occur in `R[5]`? (Answer: `20a1`: $N = 20$, $[[2, -4, 0, 3], [1, 6, 0, 2]]$, so IV$^*$ at 2 with $f_2 = 2$, $c_2 = 3$ and $\mathrm{I}_2$ at 5 with $c_5 = 2$; `1232l1`: $N = 2^4 \cdot 7 \cdot 11$, II$^*$ at 2 with $f_2 = 4$, $\mathrm{I}_2$ at 7, $\mathrm{I}_1$ at 11, $\prod c_p = 2$. The code 1 is good reduction, and `R[5]` lists only the bad primes.)

**Exercise 22.72.** With `ellan`, check that $a_p(\texttt{121a1}) \equiv \chi_{-11}(p) a_p(\texttt{11a1}) \pmod 3$ for every prime $p < 500$, $p \neq 3, 11$ (use `kronecker(-11, p)`), and that $a_n(\texttt{121a1}) \equiv a_n(\texttt{121d1}) \pmod 3$ for all $n \leq 500$, with `121d1` $= [0, -1, 1, -40, -221]$. Explain why the second check, which is far more than needed, proves the congruence for every $n$ (`mfsturm([121, 2])` returns $22 = 2\mu(121)/12$; Chapter 12; see Warning 22.7(h)), and why the first does not by itself.

**Exercise 22.73.** Compute $P_q(X)$, $P_q(q^{-2})$ and $e_3(E, q)$ with `lfunsympow` and `lfuneuler` for `147b1` $= [0, 1, 1, -114, 473]$ at 7, `539d1` at 7 and `726a1` at 11, and compare with `euler_by_type`. (Answer: $1 - 7X$, $6/7$, 1 (type IV$^*$, $\chi_{-3}(7) = 1$); $1 + 3X - 21X^2 - 343X^3$, $360/343$, 2; $1 + 11X$, $12/11$, 1.)

**Exercise 22.74.** Write a GP function `allbounds(E, l)` that returns, as a vector and in this order, $v_\ell(\deg\varphi_E)$ and the bounds $B_\ell^{\mathrm{rule}}$, $B_\ell^{\mathrm{nc}}$, $B_\ell^{\mathrm{III}}$, $B_\ell^{\mathrm{main}}$, $B_\ell^{\mathrm{tw}}$ of an optimal curve, reading $\#\Phi_q^{\mathrm{geom}}$ and $w_\ell(E,q)$ off the codes. Check that it returns $[1,1,1,1,0,1]$ for `121a1` and $[2,2,3,2,2,2]$ for `726a1` at $\ell=3$. Then show from the term-by-term comparison in the proof of Proposition prop:forms that $B_\ell^{\mathrm{main}} \le B_\ell^{\mathrm{tw}} \le B_\ell^{\mathrm{nc}}$ for every curve and every odd $\ell$, and confirm it on the 38,042 optimal curves below 10,000 at $\ell = 3,5,7,11,13$.

**Exercise 22.75.** Using the files `alldegphi` and `opt_man` of Example 22.18, explain the outputs `[1, 0]` and `[-1, 0]` of `rule` on `11a2` and `11a3` at $\ell=5$: what is the degree of the minimal parametrisation of each, what is its Manin constant, and what does `ellmoddegree` return? Which of these runs concerns a statement of the paper? (Answer: degrees 5 and 5, Manin constants 1 and 5, `ellmoddegree` 5 and $5/25 = 1/5$; none, since $E[5]$ is reducible and the curves are not optimal.)

**Exercise 22.76.** Show that `ns3(E)` $= 0$ whenever $E[3]$ is reducible, whatever the factorisation pattern of $\psi_3$. Find, below conductor 500, an optimal curve with `ns3(E)` $= 1$ and complex multiplication, and say why Conjecture conj:euler and Table tab:ns exclude it. (Hint: a Galois-stable subgroup of order 3 gives a rational root of $\psi_3$, so a factor of degree 1. Answer: `121b1`, with $j=-32768$; `256a1`, with $j=8000$.)

**Exercise 22.77.** Using `table_small.gp`, compute the rows $\ell=5$ and $\ell=7$ for $N<2000$ and compare with the values in Example 22.55. Then modify `tablerow` to count, at $\ell=3$, the curves that violate the bound without the correction, and list them for $N<2000$. (Answer: `[5622, 915, 0, 845]` and `[5735, 468, 0, 446]`; the eight curves `338d1`, `338e1`, `726a1`, `726f1`, `1210a1`, `1210d1`, `1210h1`, `1210k1`.)

**Exercise 22.78.** For the eight curves of Exercise 22.77, check that the shortfall is 1 and that each has a prime $p \equiv -1 \pmod 3$ with $t_3(E,p) \ge 1$. Which primes are they? Ten optimal curves with image 3Ns have conductor below 2000; how many of them violate the bound without the correction, and what fraction is that? (Answer: $p=2$ for all eight, and also $p=5$ for the four of conductor 1210; 8 of 10.)

**Exercise 22.79.** Compute the Tate parameters of `805b1` $= [1,-1,1,-163,-758]$ at 7 and of `874a1` $= [1,-1,0,-19,-13]$ at 19, decide whether their unit parts are cubes, and check that the refined bound of Section sec:tam equals $v_3(\deg\varphi_E)$ in both cases. (Answer: $\bar u = 1$ and $\bar u = 12$, both cubes; the rule gives 1, the extra term is 1 in both, and $\deg\varphi_E = 144$ in both.)

**Exercise 22.80** (dependence)**.** Let $A_1,\dots,A_n$ be events of probability $p$ and $X$ the number that occur, so $E[X]=np$ in all cases. (a) If the $A_i$ are independent, show $P(X=0)=(1-p)^n \le e^{-np}$. (b) If $A_1=\cdots=A_n$, show $P(X=0)=1-p$. (c) Conclude that the expected count of Proposition 22.49(i) needs no independence but that every bound on $P(X=0)$ does, and say which kind of dependence among elliptic curves (twists, congruent newforms) pushes the truth towards (b).

**Exercise 22.81.** The share of $v_3(\deg\varphi_E)=0$ among the bound-0 curves at $\ell=3$ is 0.630 on $N<5000$, 0.566 on $5000\le N<10000$ and 0.532 on the paper's working set. Explain the trend, and why the paper's expected count of violations (605,000, about 70% of the bounded curves) is close to the book's proportion (70.3%) although $\hat F(v=0)$ is smaller in the paper. (Hint: $\hat F(v<b)$ for $b\ge 2$ includes $v=1$, whose share grows with $N$.)

**Exercise 22.82** (multiple comparisons)**.** Suppose $m = 200$ false candidate inequalities each survive a working-set check with probability $\pi = 0.05$, and a hold-out check with probability $\pi' = 0.05$, independently. How many false survivors are expected after the working set, and after the hold-out? If the same hold-out is then reused to choose among $m' = 200$ new candidates, how many false survivors does the reused hold-out let through, and what does a fresh hold-out achieve? (Answer: 10, then 0.5; a reused hold-out acts as a working set and lets through about 10; a fresh one brings this back to about 0.5.)

**Exercise 22.83.** Write `twist_lemma_check.gp`: for every optimal curve below conductor 2000 and every odd bad prime $q$, compare the code and the conductor exponent of the twist by $q^*$ at $q$ with Lemma lem:twist(ii), and check Lemma lem:twist(i) at the other primes. Extend the check to the potentially good types at $q = 3$ and compare with Example 22.14.

**Exercise 22.84.** The paper's correction term $h_3$ was never blind-tested. Using the paper's numbers (68 violations without the correction on the hold-out, 23% of the 3Ns curves there), estimate how many 3Ns curves a fresh range of the same size would contain, and how many failures of the uncorrected bound it would produce. Is that enough for a convincing test of "the correction of Conjecture conj:euler is exactly one on exactly these curves"? What else would the test need to contain? (Answer: about 300 3Ns curves and 60 to 70 failures; the test would also need curves with image 3Ns and $3 \nmid N$, where Theorem thm:main does not apply only because of the image.)

**Exercise 22.85.** Redo the Eisenstein analysis of Example 22.60 at $\ell = 5$. Among the optimal curves without CM below conductor 10,000 with $E[5]$ reducible, compute the deficit $D = B_5^{\mathrm{main}} - v_5(\deg \varphi_E)$; show from the data that $D \le 1$, list the curves with $D = 1$ and conductor below 300, and check with `elltors` that each has a rational point of order 5. How many of the Eisenstein curves at 5 with a rational point of order 5 have $D = 1$? Find one below conductor 300 with a point of order 5 and $D = 0$, and explain why the torsion bound of Conjecture conj:eis allows but does not force a deficit. (Answer: 534 Eisenstein curves at 5, 53 with $D = 1$ and none with $D \ge 2$; below 300: `11a1`, `38b1`, `50b1`, `57c1`, `58b1`, `66c1`, `75c1`, `110a1`, `118b1`, `158c1`, `186b1`, `203a1`, `246b1`, `286d1`, all with torsion $\mathbb{Z}/5$ except `66c1` ($\mathbb{Z}/10$); 53 of the 137 curves with a point of order 5 have $D = 1$ (50 with torsion $\mathbb{Z}/5$, 3 with $\mathbb{Z}/10$); `123a1` and `155a1` have $B_5^{\mathrm{main}} = 1$ and $\deg \varphi_E = 20$, so $D = 0$. The conjecture is an upper bound on $D$.)

**Exercise 22.86** (the protocol at $\ell = 5$)**.** Repeat Example 22.47 at $\ell = 5$: on $N < 5000$, read off the minimum of $v_5(\deg \varphi_E)$ over the curves of the base population with $5 \mid N$, potentially good reduction at 5 and no other contribution, by the type at 5; freeze the result; test it once on $5000 \le N < 10000$; and report honestly whether it survived. Compare with the paper's "That no term is needed at $q = \ell \ge 5$ is true but not sharp" (Section sec:wild). (Answer: minimum 1 for II$^*$, III$^*$, IV$^*$ (99, 156, 212 curves) and 0 for II, III, IV, I$_0^*$; the same on the hold-out (144, 172, 248 curves for the starred types); see Example 22.62.)

**Exercise 22.87.** Restrict the odd-part statistic of Example 22.54 to $\ell \in \{3, 5\}$: on the same 33,137 curves, how often does the $\{3,5\}$-part of $\deg \varphi_E$ equal $3^{B_3^{\mathrm{tw}}} 5^{B_5^{\mathrm{tw}}}$, and how often does each factor alone match? Do it on $N < 5000$ and on $5000 \le N < 10000$, and compare with the 17.1% of the full odd part. Which prime loses equality faster as $N$ grows, and why is the product of the two rates a poor guide to the joint rate? (Answer: joint 43.1% and 33.5% (37.7% on the union); at 3 alone 56.2% and 47.5% (51.4%), at 5 alone 79.4% and 73.5% (76.1%). The rate at 3 falls faster; the products 44.6% and 34.9% are close to the joint rates here, but nothing forces this: $v_3$ and $v_5$ of one degree need not behave independently.)

## 22.14 Notes and references

*PARI/GP.* The functions are documented in the user's guide [149]; the descriptions of `ellglobalred`, `elllocalred`, `ellmoddegree`, `ellweilcurve`, `lfuneuler`, `readstr` and `strsplit` used here were read from the documentation strings that `cypari2` 2.2.4 carries for PARI 2.17.2, not from the printed manual of 2.15, and we cite no section numbers. The paper cites PARI/GP 2.15 [148]; the book's runs are 2.17.2 [150] through `cypari2` [40]. Dokchitser's algorithm for $L$-values is [55]; the method of `ellmoddegree` is Watkins's [204].

*Cremona's tables.* The book [35] describes the algorithms and contains the tables to conductor 1000; the `ecdata` repository [36] extends them to conductor 500000, and its file `docs/file-format.txt` documents the formats of most files (not `curvedata`, `classdata` and `opt_man`, whose columns we read from the repository's scripts and from the data). The LMFDB [198] uses the same curves with its own labels. Sutherland's labels of mod $\ell$ images are defined in [191], and Zywina's classification [210] is the source of the list of images at $\ell = 3$ that Proposition prop:ns uses. Modular degrees and their computation: [209, 204] and [35, Chapter II]; the congruence number of `99A1` is in [3, Table 1]. Tate's algorithm and the Kodaira types are in [184, Chapter IV].

*Statistics.* The vocabulary is that of Appendix A (Appendix A); for expectation, Chebyshev's inequality and the law of large numbers see Feller [65] (the chapters on random variables and expectation and on the law of large numbers); for hypothesis tests, permutation tests and multiple testing see Wasserman [203] (the chapters on hypothesis testing and $p$-values); Riley, Hobson and Bence [165] cover the reader's background. Chapter titles are cited from memory and are pointers we have not checked against the printed editions.

*What was verified, and how.* Every number marked as the book's was produced with PARI/GP 2.17.2 through `cypari2` 2.2.4 and Python 3.13 on Linux, from the files of commit `25cec5e` of `ecdata` listed in Convention 22.69; the printed GP code was loaded with `read` and run as printed, and every printed output was copied from the run. The paper's numbers for its large ranges are quoted from its text and its printed tables and were not reproduced; the printed tables were compared with the result files of the paper's repository, a consistency check of the tables against those files, not a recomputation (Remark 22.65). Not checked: PARI 2.15; the scripts of the paper's repository, which were not run; the out-of-table curves; anything above conductor 10,000.

*Questions of protocol.* The text of Section sec:data leaves two questions of protocol open: whether the modular degrees of the out-of-table curves were examined while the terms at 2 and 3 were being fixed, and whether (eq:twist) was written down before the hold-out counts of (eq:main) were seen (Remark 22.44).

# Appendix A

# Prerequisites: what we assume and where to look

This appendix lists, as precise statements without proofs, what the book assumes from the reader's first courses: real analysis, linear algebra, complex analysis, the group theory, representation theory and probability of Riley, Hobson and Bence [165], and elementary statistics. Each item is a card: a definition, or a theorem tagged *(stated; reference)*, followed by the chapter of the book where it is first needed and a reference where a proof can be found. Nothing here is proved; the proofs are in the textbooks of Appendix A.6. A reader who recognises every card can skip the appendix. A reader who does not recognise one should read the corresponding part of the reference before the chapter of the book that needs it.

The book assumes nothing beyond this appendix. In particular it does not assume abstract algebra beyond the group theory of Appendix A.4: rings, fields, modules, Galois theory and everything after them are built from the beginning in Chapter 2, Chapter 3 and Chapter 5. Two warnings apply throughout. First, the reader's courses did linear algebra over $\mathbb{R}$ and $\mathbb{C}$; every statement of Appendix A.2 holds over an arbitrary field $k$ unless the field is named, and the book uses them over $\mathbb{F}_q$, $\mathbb{Q}_\ell$, $\mathbb{Q}$ and finite extensions of these. Second, the word "character" has two meanings in the book, RHB's (the trace function of a representation) and the number-theoretic one (a one-dimensional representation, such as a Dirichlet character or the cyclotomic character $\chi_\ell$); Appendix A.4 says which is which.

## A.1 Real analysis

This section is used from Chapter 3 on, where the ring $\mathbb{Z}_\ell$ is constructed by the recipe that constructs $\mathbb{R}$ from $\mathbb{Q}$, and in earnest in Chapter 9, Chapter 12 and Chapter 19, where lattice sums, $q$-expansions and Dirichlet series must be shown to converge.

**Definition A.1** (the real numbers; convergence)**.** $\mathbb{R}$ is an ordered field with the *least upper bound property*: every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound (supremum) in $\mathbb{R}$. A sequence $(x_n)$ of real or complex numbers *converges* to $x$ if for every $\varepsilon > 0$ there is $N$ with $|x_n - x| < \varepsilon$ for all $n \geq N$; it is *Cauchy* if for every $\varepsilon > 0$ there is $N$ with $|x_m - x_n| < \varepsilon$ for all $m, n \geq N$.

**Theorem A.2** (consequences of completeness (stated; reference))**.** *(a) Every bounded monotone sequence of real numbers converges.*

*(b) (Bolzano-Weierstrass) Every bounded sequence in $\mathbb{R}$ or $\mathbb{C}$ has a convergent subsequence.*

*(c) (Cauchy criterion) A sequence in $\mathbb{R}$ or $\mathbb{C}$ converges if and only if it is Cauchy. In the language of metric spaces: $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{R}^n$ with the Euclidean distance are complete.*

*(d) $\mathbb{Q}$ is not complete: a Cauchy sequence of rationals need not have a rational limit. $\mathbb{R}$ is, up to isomorphism, the completion of $\mathbb{Q}$: the set of Cauchy sequences of rationals modulo the sequences converging to 0, with the operations performed termwise.*

*First needed:* Chapter 3 and Chapter 7, where $\mathbb{Z}_\ell$ and $\mathbb{Q}_p$ are built from $\mathbb{Z}$ and $\mathbb{Q}$ by (d) with the $p$-adic absolute value in place of the usual one. *Reference:* [168, Chapters 1 and 3, Theorem 3.11].

*Two Cauchy sequences.* The decimal truncations $x_1 = 1.4$, $x_2 = 1.41$, $x_3 = 1.414, \dots$ of $\sqrt{2}$ form a Cauchy sequence of rationals: for $m, n \geq N$ both $x_m$ and $x_n$ lie in the interval $[x_N, x_N + 10^{-N}]$, so $|x_m - x_n| \leq 10^{-N}$. Its limit $x$ in $\mathbb{R}$ satisfies $x^2 = \lim x_n^2 = 2$, so the sequence has no limit in $\mathbb{Q}$; this is (d) in its simplest form. The same recipe with another absolute value gives a different completion. Let $|x|_3 = 3^{-v_3(x)}$, where $v_3(x)$ is the exponent of 3 in the rational number $x$, and let $s_n = 1 + 3 + \cdots + 3^n = (3^{n+1} - 1)/2$. For $m > n$ the difference $s_m - s_n$ is divisible by $3^{n+1}$, so $|s_m - s_n|_3 \leq 3^{-n-1}$ and $(s_n)$ is Cauchy for $|\cdot|_3$. In this case the limit is even rational: $s_n + \frac{1}{2} = 3^{n+1}/2$ has $|s_n + \frac{1}{2}|_3 = 3^{-n-1} \to 0$, so $1 + 3 + 9 + \cdots = -\frac{1}{2}$ is a true statement 3-adically. Completing $\mathbb{Q}$ for $|\cdot|_3$ gives $\mathbb{Q}_3$; Chapter 7 does this for every prime.

**Definition A.3** (series)**.** A series $\sum_{n \geq 1} a_n$ of complex numbers *converges* to $s$ if its partial sums $\sum_{n \leq N} a_n$ converge to $s$; it *converges absolutely* if $\sum |a_n|$ converges.

**Theorem A.4** (series (stated; reference))**.** *(a) An absolutely convergent series converges, and $|\sum a_n| \leq \sum |a_n|$.*

*(b) (Comparison) If $|a_n| \leq b_n$ for all large $n$ and $\sum b_n$ converges, then $\sum a_n$ converges absolutely.*

*(c) (Ratio and root tests) If $\limsup |a_{n+1}/a_n| < 1$, or if $\limsup |a_n|^{1/n} < 1$, then $\sum a_n$ converges absolutely; if $\limsup |a_n|^{1/n} > 1$ it diverges.*

*(d) (Rearrangement) Every rearrangement of an absolutely convergent series converges absolutely to the same sum.*

*(e) (Double series) If the sums $\sum |a_{m,n}|$ over finite sets of pairs $(m, n)$ are bounded, then $\sum a_{m,n}$ taken in any order converges absolutely to the same value, which equals both iterated sums $\sum_m \sum_n a_{m,n}$ and $\sum_n \sum_m a_{m,n}$. The same holds for sums indexed by any countable set, such as the nonzero elements of a lattice in $\mathbb{C}$.*

*(f) (Cauchy product) If $\sum a_n$ and $\sum b_n$ converge absolutely, then $\sum_n \left( \sum_{k=0}^{n} a_k b_{n-k} \right)$ converges absolutely to the product of the two sums.*

*(g) $\sum_{n \geq 1} n^{-\sigma}$ converges for real $\sigma > 1$ and diverges for $\sigma \leq 1$; for a lattice $\Lambda \subset \mathbb{C}$, $\sum_{0 \neq \omega \in \Lambda} |\omega|^{-\sigma}$ converges if and only if $\sigma > 2$.*

*First needed:* Chapter 9 (the Weierstrass $\wp$-function and the Eisenstein series $G_{2k}(\Lambda) = \sum' \omega^{-2k}$ are lattice sums, absolutely convergent for $2k \geq 3$ by (g) and summed in any order by (e)); Chapter 19 (Dirichlet series and Euler products). *Reference:* [168, Chapter 3]; [5, Chapter 8].

*Why the lattice sum converges exactly for $\sigma > 2$.* The second half of (g) reduces to the first by counting lattice points. Let $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1, \omega_2$ linearly independent over $\mathbb{R}$. For an integer $r \geq 1$, the points $m\omega_1 + n\omega_2$ with $\max(|m|, |n|) = r$ form the boundary of a square of lattice points, and there are $(2r+1)^2 - (2r-1)^2 = 8r$ of them. The function $(x, y) \mapsto |x\omega_1 + y\omega_2|$ is continuous and nowhere zero (by linear independence) on the compact set $\max(|x|, |y|) = 1$, so by Theorem A.10(b) it has a minimum $c > 0$ and a maximum $C$ there; by homogeneity every point of the $r$-th square satisfies $cr \leq |\omega| \leq Cr$. The contribution of the $r$-th square to $\sum'|\omega|^{-\sigma}$ therefore lies between $8C^{-\sigma}r^{1-\sigma}$ and $8c^{-\sigma}r^{1-\sigma}$, and the whole sum converges if and only if $\sum_r r^{1-\sigma}$ does, that is, if and only if $\sigma - 1 > 1$. The terms are positive, so by (e) the order of summation does not matter. In particular $G_{2k}(\Lambda)$ converges absolutely for $2k \geq 4$, while the sum for $G_2$ does not converge absolutely.

**Definition A.5** (uniform convergence)**.** Let $S$ be a set and $f_n, f\colon S \to \mathbb{C}$. Then $f_n \to f$ *uniformly on* $S$ if $\sup_{x\in S}|f_n(x) - f(x)| \to 0$. A series $\sum f_n$ converges uniformly if its partial sums do. For functions on an open subset $U$ of $\mathbb{C}$ or $\mathbb{R}^n$, *locally uniform* convergence means uniform convergence on every compact subset of $U$.

**Theorem A.6** (uniform convergence (stated; reference))**.** *(a) A uniform limit of continuous functions on a metric space is continuous.*

*(b) (Weierstrass M-test) If $|f_n(x)| \leq M_n$ for all $x \in S$ and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly on $S$.*

*(c) If $f_n \to f$ uniformly on $[a, b]$ and each $f_n$ is Riemann integrable, then $f$ is Riemann integrable and $\int_a^b f_n \to \int_a^b f$; a uniformly convergent series may be integrated term by term.*

*(d) If the $f_n$ are differentiable on $[a, b]$, $f_n(x_0)$ converges for one point $x_0$, and $f_n'$ converges uniformly, then $f_n$ converges uniformly to a differentiable $f$ with $f' = \lim f_n'$. For holomorphic functions the much stronger Theorem A.39 holds.*

*First needed:* Chapter 9; Chapter 12 ($q$-expansions and Eisenstein series converge locally uniformly on the upper half plane, by (b) on each compact set); Chapter 19. *Reference:* [168, Chapter 7, Theorems 7.10 and 7.12].

*The M-test for a q-expansion.* Suppose $|a_n| \leq Cn^k$ for all $n \geq 1$ and put $f(z) = \sum_{n\geq 1} a_n e^{2\pi inz}$ on the upper half plane. A compact set $K$ in the upper half plane has $\operatorname{Im} z \geq y_0$ on $K$ for some $y_0 > 0$ (the continuous function $\operatorname{Im} z$ attains a positive minimum on $K$, Theorem A.10(b)). On $K$, $|a_n e^{2\pi inz}| = |a_n| e^{-2\pi n \operatorname{Im} z} \leq Cn^k e^{-2\pi n y_0} = M_n$, and $\sum M_n$ converges by the ratio test, since $M_{n+1}/M_n = (1 + 1/n)^k e^{-2\pi y_0} \to e^{-2\pi y_0} < 1$. So by (b) the series converges absolutely and uniformly on $K$: it converges locally uniformly on the upper half plane, and Theorem A.39 then makes $f$ holomorphic. This is the argument used, without further comment, for every $q$-expansion in the book.

**Definition A.7** (power series)**.** A power series is $\sum_{n\geq 0} c_n(z - a)^n$ with $c_n, a \in \mathbb{C}$. Its *radius of convergence* is $R = 1/\limsup|c_n|^{1/n} \in [0, \infty]$.

**Theorem A.8** (power series (stated; reference))**.** *(a) The series converges absolutely for $|z - a| < R$, diverges for $|z - a| > R$, and converges uniformly on every closed disc $|z - a| \leq r$ with $r < R$.*

*(b) Its sum $f$ is complex differentiable on the open disc of radius $R$, with derivative the term-by-term derivative $\sum nc_n(z - a)^{n-1}$, which has the same radius of convergence.*

(c) *The coefficients are determined by $f$: $c_n = f^{(n)}(a)/n!$. Two power series centred at $a$ whose sums agree on a sequence of points $z_k \neq a$ converging to $a$ have the same coefficients.*

(d) *$\sum z^n/n!$ has $R = \infty$; $\sum z^n$ has $R = 1$ and sum $1/(1-z)$.*

*First needed:* Chapter 12 (a $q$-expansion $\sum a_n q^n$ is a power series in $q = e^{2\pi i z}$, $|q| < 1$). *Warning:* the $p$-adic expansions $\sum a_n p^n$ of Chapter 7 converge for a different absolute value, in which $|p^n| \to 0$; the analogy with power series is useful, the theorems above do not transfer. *Reference:* [168, Chapter 3, Theorem 3.39, and Chapter 8].

*Example.* If $|c_n| \leq Cn^k$, then $|c_n|^{1/n} \leq C^{1/n} n^{k/n} \to 1$, so the radius of convergence is at least 1: coefficients of polynomial growth, such as those of every $q$-expansion in the book, always give a power series in $q$ converging on $|q| < 1$. The bound 1 can be attained: $\sum_{n \geq 1} nq^n$ has radius exactly 1, and differentiating $\sum q^n = 1/(1-q)$ term by term by (b) and multiplying by $q$ gives its sum $q/(1-q)^2$.

**Definition A.9** (compactness)**.** A subset $K$ of $\mathbb{R}^n$ or $\mathbb{C}$ is *compact* if every cover of $K$ by open sets has a finite subcover; equivalently (Heine-Borel) if it is closed and bounded; equivalently if every sequence in $K$ has a subsequence converging to a point of $K$.

**Theorem A.10** (continuity and compactness (stated; reference))**.** *Let $K$ be compact and $f$ continuous on $K$.*

(a) *$f(K)$ is compact.*

(b) *If $f$ is real valued, it attains its maximum and its minimum on $K$.*

(c) *$f$ is uniformly continuous on $K$.*

(d) *A continuous bijection from a compact space onto a metric space is a homeomorphism.*

*First needed:* Chapter 8 (a holomorphic function on a compact connected Riemann surface is constant: (b) applied to $|f|$ with the maximum modulus principle); Chapter 12 (fundamental domains, the compactification of $Y_0(N)$ by the cusps). *Reference:* [168, Theorems 2.41, 4.14, 4.16 and 4.19].

*How (b) is used on a Riemann surface.* Let $X$ be a compact connected Riemann surface and $f$ holomorphic on $X$. The continuous function $|f|$ attains its maximum at some point $P$ by (b), which holds for continuous real functions on any compact space. In a coordinate chart around $P$, $f$ is a holomorphic function of one variable whose modulus has a local maximum, so it is constant near $P$ by the maximum modulus principle (Theorem A.36(e)). The set of points of $X$ near which $f$ is identically $f(P)$ is open by definition, nonempty, and closed by the identity theorem (Theorem A.36(d), applied in charts); since $X$ is connected it is all of $X$. Chapter 8 gives the details.

**Definition A.11** (the Riemann integral)**.** For a bounded $f\colon [a,b] \to \mathbb{R}$ and a partition $a = x_0 < \cdots < x_n = b$, the upper and lower sums are $\sum (x_i - x_{i-1}) \sup_{[x_{i-1},x_i]} f$ and $\sum (x_i - x_{i-1}) \inf_{[x_{i-1},x_i]} f$. The function is *Riemann integrable* if the infimum of the upper sums equals the supremum of the lower sums, and $\int_a^b f$ is the common value. Complex-valued functions are integrated by real and imaginary parts.

**Theorem A.12** (the Riemann integral (stated; reference))**.** *(a) Every continuous function on $[a,b]$ is integrable, and so is every monotone function and every bounded function with finitely many discontinuities.*

(b) *The integral is linear and monotone,* $\left|\int_a^b f\right| \le \int_a^b |f|$, *and* $\int_a^b = \int_a^c + \int_c^b$.

(c) *(Fundamental theorem of calculus) If* $f$ *is continuous on* $[a,b]$, *then* $F(x) = \int_a^x f$ *is differentiable with* $F' = f$; *if* $F$ *is differentiable on* $[a,b]$ *and* $F'$ *is integrable, then* $\int_a^b F' = F(b) - F(a)$.

(d) *Integration by parts and the substitution rule hold for* $C^1$ *functions.*

(e) *For a* $C^1$ *curve* $\gamma\colon [a,b] \to \mathbb{C}$ *and* $f$ *continuous on its image, the line integral is* $\int_\gamma f(z)\,dz = \int_a^b f(\gamma(t))\gamma'(t)\,dt$; *it does not depend on the parametrisation of* $\gamma$ *and changes sign when the orientation is reversed.*

*First needed:* Chapter 8 (line integrals define Cauchy's theorem and the periods of a differential); Chapter 9 (the periods of an elliptic curve over $\mathbb{C}$ are integrals of $dx/y$ over closed paths); Chapter 12 (the Petersson inner product is a double integral over a fundamental domain). *Reference:* [168, Chapter 6, Theorems 6.8, 6.20 and 6.21].

*The basic line integral.* Parametrise the unit circle by $\gamma(t) = e^{it}$, $0 \le t \le 2\pi$, so $\gamma'(t) = ie^{it}$. For an integer $n$, (e) gives

$$\int_{|z|=1} z^n\,dz = \int_0^{2\pi} e^{int}\,ie^{it}\,dt = i\int_0^{2\pi} e^{i(n+1)t}\,dt = \begin{cases} 2\pi i & n = -1, \\ 0 & n \ne -1, \end{cases}$$

since $e^{i(n+1)t}$ has an antiderivative $e^{i(n+1)t}/(i(n+1))$ of period $2\pi$ when $n \ne -1$. This one computation is behind the coefficient formula for Laurent series and the residue theorem of Appendix A.3: integrating a Laurent series term by term around a circle keeps only the coefficient of $z^{-1}$.

**Definition A.13** (improper integrals)**.** $\int_a^\infty f = \lim_{b\to\infty} \int_a^b f$ when the limit exists; for $f$ unbounded near 0, $\int_0^1 f = \lim_{\varepsilon\to 0^+} \int_\varepsilon^1 f$; $\int_0^\infty$ is the sum of $\int_0^1$ and $\int_1^\infty$. The integral *converges absolutely* if the corresponding integral of $|f|$ converges.

**Theorem A.14** (improper integrals (stated; reference))**.**

(a) *(Comparison) Suppose that* $|f| \le g$ *on* $[a,\infty)$ *and that* $\int_a^\infty g$ *converges. Then* $\int_a^\infty f$ *converges absolutely; and absolutely convergent improper integrals converge.*

(b) $\int_1^\infty t^{-\sigma}\,dt$ *converges if and only if* $\sigma > 1$; $\int_0^1 t^{\sigma-1}\,dt$ *converges if and only if* $\sigma > 0$; $\int_0^\infty e^{-ct}t^m\,dt$ *converges for every* $c > 0$ *and* $m > -1$.

(c) *(Parameters) If* $f(t,s)$ *is continuous on* $[a,\infty) \times K$ *with* $K$ *compact, and* $|f(t,s)| \le g(t)$ *with* $\int_a^\infty g < \infty$, *then* $s \mapsto \int_a^\infty f(t,s)\,dt$ *is continuous on* $K$; *if moreover* $\partial f/\partial s$ *exists, is continuous and is dominated in the same way, the integral may be differentiated under the integral sign.*

**Definition A.15** (the Gamma function)**.** For real $s > 0$, $\Gamma(s) = \int_0^\infty e^{-t}t^{s-1}\,dt$. The integral converges absolutely: near 0 by Theorem A.14(b), and near $\infty$ because $e^{-t}t^{s-1} \le t^{-2}$ for large $t$.

**Theorem A.16** (the Gamma function, real form (stated; reference))**.** (a) $\Gamma(s+1) = s\Gamma(s)$ *for* $s > 0$ *(integration by parts).*

(b) $\Gamma(1) = 1$, *hence* $\Gamma(n+1) = n!$ *for integers* $n \ge 0$; $\Gamma(1/2) = \sqrt{\pi}$.

*(c) For $n \geq 1$ and $s > 0$, $\int_0^\infty e^{-nt} t^{s-1}\,dt = n^{-s}\Gamma(s)$ (substitute $t \mapsto t/n$).*

*(d) $\Gamma$ is continuous and positive on $(0,\infty)$, and $\log\Gamma$ is convex there.*

*First needed:* Chapter 19 (the completed $L$-function $\Lambda(E,s) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(E,s)$ and the Mellin transform of Remark A.46); the complex form is Theorem A.45. *Reference:* [168, Chapter 8]; [165, Chapter 18].

*The computations behind (a) and (b), and behind the definition.* The bound $e^{-t}t^{s-1} \leq t^{-2}$ for large $t$ in Definition A.15 holds because $t^{s+1}e^{-t} \to 0$: choose an integer $m > s+1$; then $e^t \geq t^m/m!$ gives $t^{s+1}e^{-t} \leq m!\,t^{s+1-m} \to 0$. For (a), integrate by parts on $[\varepsilon, b]$:

$$\int_\varepsilon^b e^{-t}t^s\,dt = \left[-e^{-t}t^s\right]_\varepsilon^b + s\int_\varepsilon^b e^{-t}t^{s-1}\,dt,$$

and the boundary terms $e^{-\varepsilon}\varepsilon^s$ and $e^{-b}b^s$ tend to 0 as $\varepsilon \to 0^+$ (because $s > 0$) and $b \to \infty$. For (b), $\Gamma(1) = \int_0^\infty e^{-t}\,dt = 1$, and induction on (a) gives $\Gamma(n+1) = n\,\Gamma(n) = \cdots = n!$. The substitution $t = u^2$, $dt = 2u\,du$, turns $\Gamma(1/2)$ into $2\int_0^\infty e^{-u^2}\,du = \int_{-\infty}^\infty e^{-u^2}\,du$, which is $\sqrt{\pi}$ (the Gaussian integral, computed by squaring it and passing to polar coordinates).

With the analytic background in place, the next section turns to linear algebra, which the book uses far more heavily and over fields the reader's courses did not consider.

## A.2 Linear algebra

This section is used from Chapter 1 on. Throughout, $k$ is a field and all vector spaces are finite-dimensional over $k$; the reader's courses did $k = \mathbb{R}$ and $k = \mathbb{C}$, and the book uses $k = \mathbb{F}_q$, $\mathbb{Q}_\ell$, $\mathbb{Q}$ and their finite extensions, and the same constructions over the rings $\mathbb{Z}$ and $\mathbb{Z}_\ell$ (Chapter 3 says what survives over a ring).

**Definition A.17** (vector spaces, bases, dimension)**.** A *vector space* over $k$ is an abelian group $V$ with a scalar multiplication $k \times V \to V$ satisfying the usual axioms. A family $(v_i)$ is *linearly independent* if $\sum a_i v_i = 0$ forces all $a_i = 0$, *spanning* if every vector is a linear combination of it, and a *basis* if both. A *linear map* $T\colon V \to W$ satisfies $T(av+bw) = aTv + bTw$; its kernel and image are subspaces; $\mathrm{End}(V)$ is the ring of linear maps $V \to V$ and $\mathrm{GL}(V)$ its group of units.

**Theorem A.18** (bases and dimension (stated; reference))**.** *(a) Every finitely generated vector space has a basis, and any two bases have the same number of elements, the* dimension $\dim V$*. Over a finite field $\mathbb{F}_q$, a space of dimension $n$ has $q^n$ elements.*

*(b) Every linearly independent family extends to a basis; every spanning family contains a basis.*

*(c) For a subspace $U \subseteq V$:* $\dim U \leq \dim V$*, with equality if and only if $U = V$, and* $\dim V = \dim U + \dim(V/U)$.

*(d) (Rank-nullity) For a linear map $T\colon V \to W$,* $\dim V = \dim\ker T + \dim\mathrm{im}\,T$*. In particular an injective, or a surjective, linear map between spaces of the same dimension is an isomorphism.*

*(e)* $\dim(U+U') + \dim(U\cap U') = \dim U + \dim U'$ *for subspaces $U, U'$ of $V$.*

*First needed:* Chapter 1 (the lines of $\mathbb{F}_q^2$: there are $q+1$ of them, the points of $\mathbb{P}^1(\mathbb{F}_q)$); Chapter 4. *Reference:* [10, Chapters 2 and 3]; [58, Chapter 11].

*Counting lines.* A line (one-dimensional subspace) of $\mathbb{F}_q^2$ is spanned by any of its nonzero vectors, and it contains $q-1$ of them by (a). Two distinct lines meet only in 0, so the $q^2-1$ nonzero vectors of $\mathbb{F}_q^2$ fall into $(q^2-1)/(q-1) = q+1$ lines. For $q=3$ these are the four lines spanned by $(1,0)$, $(0,1)$, $(1,1)$, $(1,2)$.

**Definition A.19** (matrices of linear maps; change of basis)**.** Given bases $e = (e_1, \dots, e_n)$ of $V$ and $f = (f_1, \dots, f_m)$ of $W$, the *matrix* of $T\colon V \to W$ is the $m \times n$ matrix $A$ whose $j$-th column holds the coordinates of $Te_j$ in the basis $f$. Composition of maps corresponds to the product of matrices. If $e' = eP$ is another basis of $V$ (the columns of the invertible $P$ are the coordinates of the $e'_j$ in $e$) and $f' = fQ$ another basis of $W$, the matrix of $T$ in the new bases is $Q^{-1}AP$; for an endomorphism and a single change of basis it is $P^{-1}AP$. The *trace* of a square matrix is the sum of its diagonal entries.

The convention in Definition A.19 matters: books that record the old basis in terms of the new one write $PAP^{-1}$ instead, with $P$ replaced by its inverse. A check of the book's convention: in $V = k^2$ with the standard basis $e$, let $T$ have matrix $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, and take the new basis $e'_1 = e_1$, $e'_2 = e_1 + e_2$, so $P = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Then

$$P^{-1}AP = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix},$$

and indeed $Te'_1 = e_1 = e'_1$ and $Te'_2 = e_1 + 2e_2 = -e'_1 + 2e'_2$, whose coordinates $(1,0)$ and $(-1,2)$ are the columns found.

**Theorem A.20** (determinants (stated; reference))**.** *The determinant* $\det\colon \mathrm{M}_n(k) \to k$ *is the unique function of the columns that is multilinear, alternating, and takes the value* 1 *at the identity matrix.*

*(a)* $\det(AB) = \det A \det B$ *and* $\det A^{\mathsf{T}} = \det A$.

*(b)* $A$ *is invertible if and only if* $\det A \neq 0$*; then* $A^{-1} = (\det A)^{-1} \operatorname{adj} A$*, where* $\operatorname{adj} A$ *is the transpose of the matrix of cofactors. Over a commutative ring the same formula shows that* $A$ *is invertible if and only if* $\det A$ *is a unit.*

*(c) The determinant of a block upper triangular matrix is the product of the determinants of the diagonal blocks.*

*(d)* $\det(P^{-1}AP) = \det A$ *and* $\operatorname{tr}(P^{-1}AP) = \operatorname{tr} A$ *(as* $\operatorname{tr}(AB) = \operatorname{tr}(BA)$*), so* $\det T$ *and* $\operatorname{tr} T$ *are defined for an endomorphism* $T$ *independently of the basis.*

*(e)* det *and* tr *are compatible with extension of the field: the determinant of* $A$ *viewed over a larger field is the same element of* $k$.

*First needed:* Chapter 1 ($\mathrm{GL}_2(\mathbb{F}_q)$, $\mathrm{SL}_2(\mathbb{F}_q)$ and their orders, which are computed there); Chapter 9 (the determinant of the Galois action on $E[m]$ is the cyclotomic character, via the Weil pairing). *Reference:* [10, Chapter 10]; [109, Chapter XIII].

*Invertibility over a ring.* The ring version of (b) is the one the book uses over $\mathbb{Z}$ and $\mathbb{Z}_\ell$. The matrix $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ has determinant 2; its inverse $\left(\begin{smallmatrix} 1/2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ has entries in $\mathbb{Q}$ and in $\mathbb{Z}_\ell$ for every odd $\ell$, where 2 is a unit, but not in $\mathbb{Z}$ or $\mathbb{Z}_2$. So the same matrix is in $\mathrm{GL}_2(\mathbb{Z}_3)$ and not in $\mathrm{GL}_2(\mathbb{Z})$ or $\mathrm{GL}_2(\mathbb{Z}_2)$. By contrast $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has determinant 1, and its inverse $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ has integer entries.

**Definition A.21** (eigenvalues; characteristic and minimal polynomials)**.** Let $T \in \mathrm{End}(V)$, $n = \dim V$. A scalar $\lambda \in k$ is an *eigenvalue* of $T$ if $\ker(T - \lambda) \neq 0$; the nonzero elements of this kernel are the *eigenvectors* and the kernel is the *eigenspace*. The *characteristic polynomial* is $\chi_T(X) = \det(X \cdot 1 - A) \in k[X]$ for any matrix $A$ of $T$; it is monic of degree $n$ and does not depend on the basis. The *minimal polynomial* $\mu_T$ is the monic polynomial of least degree with $\mu_T(T) = 0$.

**Theorem A.22** (eigenvalues (stated; reference))**.** *(a) The eigenvalues of $T$ in $k$ are exactly the roots of $\chi_T$ in $k$.*

*(b) $\chi_T(X) = X^n - (\operatorname{tr} T)X^{n-1} + \cdots + (-1)^n \det T$; for $n = 2$, $\chi_T(X) = X^2 - (\operatorname{tr} T)X + \det T$.*

*(c) Over a field containing all the roots of $\chi_T$ (for instance an algebraic closure of $k$), $\operatorname{tr} T$ is the sum and $\det T$ the product of the roots, counted with multiplicity; the eigenvalues of $T^m$, and of $p(T)$ for a polynomial $p$, are the $m$-th powers, respectively the values $p(\lambda)$, of the eigenvalues $\lambda$, with multiplicity.*

*(d) (Cayley-Hamilton) $\chi_T(T) = 0$. Consequently $\mu_T$ divides $\chi_T$; every polynomial $p$ with $p(T) = 0$ is a multiple of $\mu_T$; and $\mu_T$ and $\chi_T$ have the same irreducible factors in $k[X]$, hence the same roots in an algebraic closure.*

*(e) Eigenvectors belonging to distinct eigenvalues are linearly independent.*

*First needed:* Chapter 1 (elements of $\mathrm{GL}_2(\mathbb{F}_3)$ of trace 0 and determinant $-1$); Chapter 11 (the characteristic polynomial $X^2 - a_pX + p$ of Frobenius on $T_\ell E$). *Reference:* [10, Chapters 5 and 8]; [109, Chapter XIV]; [58, Section 12.2].

*One matrix over several fields.* The paper's matrix $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ has trace 0 and determinant 1, so $\chi_J = X^2 + 1$ by (b), and $J^2 = -1$ by Cayley-Hamilton. Whether $J$ has eigenvalues depends on the field. Over $\mathbb{R}$ it has none; over $\mathbb{C}$ it has $\pm i$. Over $\mathbb{F}_p$ with $p$ odd it has eigenvalues exactly when $-1$ is a square modulo $p$: over $\mathbb{F}_5$ they are 2 and 3 ($2^2 = 4 = -1$), while over $\mathbb{F}_3$ the squares are 0 and 1, so there are none, and one must pass to $\mathbb{F}_9 = \mathbb{F}_3(i)$ to find $\pm i$. Over $\mathbb{F}_2$, $\chi_J = X^2 + 1 = (X + 1)^2$ has the double root 1, but $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) \neq 1$, so by Theorem A.24(c) $J$ is conjugate to $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and is not diagonalisable. Remark A.51 returns to the case of $\mathbb{F}_3$.

**Theorem A.23** (diagonalisation and generalised eigenspaces (stated; reference))**.** *Let $T \in \mathrm{End}(V)$, $n = \dim V$.*

*(a) $T$ is* diagonalisable *over $k$ (that is, $V$ has a basis of eigenvectors) if and only if $\mu_T$ is a product of distinct linear factors in $k[X]$, if and only if $V$ is the direct sum of the eigenspaces. In particular $T$ is diagonalisable if $\chi_T$ has $n$ distinct roots in $k$.*

*(b) Suppose $k$ contains all the roots of $\chi_T$. Then $V = \bigoplus_\lambda V_\lambda$ with $V_\lambda = \ker(T - \lambda)^n$ the* generalised eigenspace*, each $V_\lambda$ is $T$-stable, $\dim V_\lambda$ is the multiplicity of $\lambda$ as a root of $\chi_T$, and $T$ is diagonalisable if and only if every $V_\lambda$ is the eigenspace $\ker(T - \lambda)$.*

*(c) (Commuting operators) A family of pairwise commuting diagonalisable endomorphisms of $V$ is simultaneously diagonalisable: $V$ has a basis of common eigenvectors. More generally, if $k$ is algebraically closed, a family of pairwise commuting endomorphisms has a common eigenvector, and $V$ is the direct sum of the common generalised eigenspaces, each stable under the whole family.*

(d) *$T$ is called* semisimple *if it is diagonalisable over an algebraic closure of $k$,* nilpotent *if $T^m = 0$ for some $m$, and* unipotent *if $T - 1$ is nilpotent. An endomorphism of finite order $m$, with $m$ not divisible by the characteristic of $k$, is semisimple with eigenvalues $m$-th roots of unity, because $\mu_T$ divides $X^m - 1$, which has distinct roots.*

*First needed:* Chapter 4 (cyclic groups of order prime to the characteristic act semisimply; groups of order divisible by the characteristic need not); Chapter 13 (the Hecke operators commute; the $T_n$ with $(n, N) = 1$ are simultaneously diagonalisable on $S_2(\Gamma_0(N))$, the $U_p$ need not be, and the generalised eigenspaces of the Hecke algebra are the objects of the paper's Theorem thm:twist); Chapter 3 (generalised eigenspaces over a discrete valuation ring). *Reference:* [10, Chapters 5 and 8]; [109, Chapter XIV].

*The hypothesis in (d) cannot be dropped.* Over $\mathbb{F}_\ell$ the matrix $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ satisfies $u^m = \left(\begin{smallmatrix} 1 & m \\ 0 & 1 \end{smallmatrix}\right)$, so it has order exactly $\ell$, the characteristic. Its minimal polynomial is $(X-1)^2$ (as $u \neq 1$ and $(u-1)^2 = 0$), which is not a product of distinct linear factors, so $u$ is not diagonalisable over any field of characteristic $\ell$, although it has finite order. The argument of (d) breaks exactly where expected: $X^\ell - 1 = (X-1)^\ell$ in characteristic $\ell$ has a repeated root. Over $\mathbb{Q}$ the same matrix has infinite order. This $u$ is the shape of the inertia action at a prime of type IV on $E[3]$ (Chapter 11).

**Theorem A.24** (Jordan normal form (stated; reference))**.** *Suppose $k$ contains all the roots of $\chi_T$. Then $V$ has a basis in which the matrix of $T$ is block diagonal with* Jordan blocks $J_m(\lambda)$*: the $m \times m$ matrix with $\lambda$ on the diagonal, 1 just above the diagonal and 0 elsewhere. The multiset of blocks is uniquely determined by $T$, and $T$ is diagonalisable if and only if every block has size 1. Consequences for* $\dim V = 2$*:*

(a) *A unipotent $T \neq 1$ is conjugate to $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$; its fixed space* $\ker(T-1)$ *is a line, $(T-1)^2 = 0$, and over $k = \mathbb{F}_\ell$ it has order $\ell$. Likewise $-T$ ("minus a unipotent") is conjugate to $\left(\begin{smallmatrix} -1 & -1 \\ 0 & -1 \end{smallmatrix}\right)$ and has no nonzero fixed vector when the characteristic is odd.*

(b) *If $\chi_T = (X-\lambda)(X-\mu)$ with $\lambda \neq \mu$ in $k$, then $T$ is diagonalisable with eigenvalues $\lambda, \mu$.*

(c) *If $\chi_T = (X-\lambda)^2$, then either $T = \lambda$ or $T$ is conjugate to $\left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)$.*

*First needed:* Chapter 1 (unipotent elements of $\mathrm{GL}_2(\mathbb{F}_q)$ and the Borel subgroup); Chapter 4; Chapter 11 (the image of inertia at a prime of type IV is a nontrivial unipotent of order 3 on $E[3]$, with a line of fixed points: the paper's Lemma lem:IV and Remark rem:types); Chapter 13 ($U_p$ with the repeated eigenvalue 0 is a Jordan block, not a scalar). *Reference:* [10, Chapter 8]; [58, Section 12.3].

*The fixed vectors in (a), computed.* For $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $v = (v_1, v_2)$ one has $(T-1)v = (v_2, 0)$, so $\ker(T-1)$ is the line spanned by $e_1$. The matrix $-T$ fixes $v$ if and only if $(T+1)v = 0$; but $T + 1 = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$ has determinant 4, which is nonzero when the characteristic is odd, so then $-T$ fixes only 0. In characteristic 2, $-T = T$ and the line of fixed points returns.

**Definition A.25** (bilinear forms; Gram matrices)**.** A *bilinear form* on $V \times W$ is a map $B\colon V \times W \to k$ linear in each variable. Its *Gram matrix* with respect to bases $e$ of $V$ and $f$ of $W$ is $G = (B(e_i, f_j))$. The form is *perfect* (or nondegenerate) if $v \mapsto B(v, \cdot)$ is an isomorphism $V \to W^* = \mathrm{Hom}(W, k)$ and $w \mapsto B(\cdot, w)$ is an isomorphism $W \to V^*$; when $\dim V = \dim W$ each condition is equivalent to $\det G \neq 0$. For $V = W$ the form is *symmetric* if $B(v, w) = B(w, v)$ and *alternating* if $B(v, v) = 0$ for all $v$, which implies $B(v, w) = -B(w, v)$. A *sesquilinear form* on a

complex space is linear in the first variable and conjugate-linear in the second, and *Hermitian* if $B(w,v)=\overline{B(v,w)}$.

**Theorem A.26** (Gram matrices under change of basis (stated; reference))**.**

*(a) If the bases change as $e'=eP$ and $f'=fQ$, then the Gram matrix in the new bases is $G'=P^{\mathsf T}GQ$. For $V=W$ and a single change of basis, $G'=P^{\mathsf T}GP$ and $\det G'=(\det P)^2\det G$. So the determinant of the Gram matrix of a bilinear form on $V$ is well defined up to multiplication by the square of a unit: of $k^\times$ over a field, of the unit group over a ring. Its $\ell$-adic valuation, for a form on a free $\mathbb{Z}_\ell$-module, is therefore an invariant of the form.*

*(b) An alternating form on a two-dimensional space with basis $e_1,e_2$ is $B(v,w)=a\det(v\mid w)$ with $a=B(e_1,e_2)$; its Gram matrix is $\left(\begin{smallmatrix}0&a\\-a&0\end{smallmatrix}\right)$ and its Gram determinant is $a^2$, a square. More generally the Gram determinant of an alternating form on a space of even dimension $2m$ is the square of a polynomial in the entries (the Pfaffian), and an alternating form on a space of odd dimension is degenerate.*

*(c) A symmetric bilinear form over a field of characteristic not 2 has an orthogonal basis (a basis in which $G$ is diagonal).*

*First needed:* Chapter 3 (perfect pairings over $\mathbb{Z}_\ell$; the Gram determinant of a saturated sublattice and the order of the cokernel of $L\to\operatorname{Hom}(L,\mathcal{O})$); Chapter 8 (the cup product on $H^1$ of a curve is perfect and alternating); Chapter 20 (the pairing $\hat\delta_!$ restricted to the rank-two lattice $M_{f,\lambda}$ has Gram determinant $(\eta_{f,\lambda}^{\Sigma})^2$ up to a unit, by (b)). *Reference:* [109, Chapter XV]; [78].

*Where (a) and (b) come from.* Write $e_i'=\sum_k P_{ki}e_k$ and $f_j'=\sum_l Q_{lj}f_l$. Bilinearity gives $B(e_i',f_j')=\sum_{k,l}P_{ki}B(e_k,f_l)Q_{lj}=(P^{\mathsf T}GQ)_{ij}$, which is (a), and $\det G'=\det P\det G\det Q$ by Theorem A.20(a). For (b), an alternating form has $B(e_1,e_1)=B(e_2,e_2)=0$ and $B(e_2,e_1)=-B(e_1,e_2)=-a$, so for $v=v_1e_1+v_2e_2$ and $w=w_1e_1+w_2e_2$,

$$B(v,w)=v_1w_2\,a+v_2w_1\,(-a)=a\,(v_1w_2-v_2w_1)=a\det(v\mid w).$$

For example, the alternating form on $\mathbb{Z}_3^2$ with $a=3$ has Gram determinant 9, of 3-adic valuation 2, and no change of basis over $\mathbb{Z}_3$ can change that valuation, because $\det P$ is then a unit of $\mathbb{Z}_3$.

*In the paper* A.27. The paper's Lemma lem:ddt417 is stated for a free $\mathcal{O}$-module $L$ with a perfect bilinear pairing for which the algebra $\mathbb{T}$ is self-adjoint, and $L[\wp]$ free of rank $d$ with Gram determinant $\Delta$; its proof uses that the cokernel of $L[\wp]\to\operatorname{Hom}(L[\wp],\mathcal{O})$ has order $\#\mathcal{O}/\Delta$. In the application (the paragraph after Lemma lem:hecke and Step 2 of the proof of Theorem thm:quant) $d=2$, the pairing is alternating, and $\Delta=(\eta)^2$ up to a unit by Theorem A.26(b).

The cokernel statement is linear algebra over the discrete valuation ring $\mathcal{O}$. In a basis of the free module $L[\wp]$ and the dual basis of $\operatorname{Hom}(L[\wp],\mathcal{O})$, the map $v\mapsto B(v,\cdot)$ has as matrix the transpose $G^{\mathsf T}$ of the Gram matrix, so its cokernel is $\mathcal{O}^d/G^{\mathsf T}\mathcal{O}^d$. By the Smith normal form (proved in Chapter 3) this is $\bigoplus_i\mathcal{O}/d_i$ with $d_1\cdots d_d=\det G$ up to a unit, of order $\prod\#(\mathcal{O}/d_i)=\#\mathcal{O}/\Delta$. In the rank-two alternating case no Smith form is needed: the Gram matrix is $\left(\begin{smallmatrix}0&\eta\\-\eta&0\end{smallmatrix}\right)$, its image is $\eta\mathcal{O}^2$, and the cokernel is $(\mathcal{O}/\eta)^2$, of order $(\#\mathcal{O}/\eta)^2=\#\mathcal{O}/\eta^2$.

**Definition A.28** (inner products; adjoints)**.** An *inner product* on a real vector space is a symmetric bilinear form with $\langle v,v\rangle>0$ for $v\neq0$; on a complex vector space it is a Hermitian sesquilinear form with the same positivity. It defines the norm $\|v\|=\langle v,v\rangle^{1/2}$, orthogonality, and the *orthogonal complement* $U^\perp=\{v:\langle v,u\rangle=0\text{ for all }u\in U\}$ of a subspace $U$. More

generally, for a perfect bilinear form $B$ on $V$ (over any field, with no positivity), the *$B$-adjoint* of $T \in \mathrm{End}(V)$ is the unique $T^\dagger$ with $B(Tv, w) = B(v, T^\dagger w)$ for all $v, w$, and $T$ is *$B$-self-adjoint* if $T^\dagger = T$.

**Theorem A.29** (inner product spaces (stated; reference))**.** *Let $V$ be a finite-dimensional real or complex inner product space.*

*(a) (Cauchy-Schwarz) $|\langle v, w\rangle| \le \|v\|\|w\|$. $V$ has an orthonormal basis (Gram-Schmidt).*

*(b) For a subspace $U$: $V = U \oplus U^\perp$, $(U^\perp)^\perp = U$ and $\dim U^\perp = \dim V - \dim U$.*

*(c) Every $T \in \mathrm{End}(V)$ has a unique adjoint $T^*$ with $\langle Tv, w\rangle = \langle v, T^*w\rangle$; its matrix in an orthonormal basis is the conjugate transpose of that of $T$; $(ST)^* = T^*S^*$; and a subspace $U$ is $T$-stable if and only if $U^\perp$ is $T^*$-stable.*

*(d) (Spectral theorem) If $T = T^*$, then $V$ has an orthonormal basis of eigenvectors of $T$ and all eigenvalues are real; over $\mathbb{C}$ the same conclusion, with complex eigenvalues, holds for normal $T$ ($TT^* = T^*T$).*

*(e) A family of pairwise commuting self-adjoint (or normal) operators has a common orthonormal basis of eigenvectors.*

*(f) For a perfect bilinear form $B$ and a $B$-self-adjoint $T$, the $B$-orthogonal complement of a $T$-stable subspace is $T$-stable, and $T$ need not be diagonalisable.*

*First needed:* Chapter 12 and Chapter 13 (the Petersson inner product; the $T_n$ with $(n, N) = 1$ are self-adjoint for it, hence $S_2(\Gamma_0(N))$ has a basis of eigenforms by (e); $U_p$ is not self-adjoint; $f^\perp$ is Hecke-stable by (c) because $f$ is an eigenvector of the adjoints too); Chapter 3 (adjoints with respect to a pairing); Chapter 14 (the adjoint of $T_n$ for the cup product is $w_N T_n w_N$); Chapter 20. *Reference:* [10, Chapters 6 and 7]; [78].

*Item (f), and why it is weaker than (d).* If $U$ is $T$-stable, $w$ lies in the $B$-orthogonal complement of $U$ and $u \in U$, then $B(u, Tw) = B(Tu, w) = 0$ because $Tu \in U$; so $Tw$ is again orthogonal to $U$. What fails without positivity is the rest of the proof of the spectral theorem, and the failure is real. Writing $B(v, w) = v^{\mathsf{T}} G w$, a matrix $T$ is $B$-self-adjoint if and only if $T^{\mathsf{T}} G = GT$. Take $G = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, a perfect symmetric form over any field, and $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Then

$$T^{\mathsf{T}} G = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = GT,$$

so $T$ is $B$-self-adjoint, and it is not diagonalisable. The line $ke_1$ is $T$-stable and is its own $B$-orthogonal complement ($B(e_1, e_1) = 0$), so the complement does not split it off.

**Definition A.30** (dual spaces)**.** The *dual* of $V$ is $V^* = \mathrm{Hom}_k(V, k)$. A basis $e_1, \dots, e_n$ of $V$ has the *dual basis* $e_1^*, \dots, e_n^*$ of $V^*$ with $e_i^*(e_j) = 1$ if $i = j$ and 0 otherwise. The *transpose* of $T\colon V \to W$ is $T^*\colon W^* \to V^*$, $\varphi \mapsto \varphi \circ T$. The *annihilator* of a subspace $U \subseteq V$ is $U^0 = \{\varphi \in V^* : \varphi(U) = 0\}$.

**Theorem A.31** (duality (stated; reference))**.** *(a) $\dim V^* = \dim V$, and the map $V \to V^{**}$, $v \mapsto (\varphi \mapsto \varphi(v))$, is an isomorphism.*

*(b) $\dim U^0 = \dim V - \dim U$, $(U^0)^0 = U$ under (a), $(V/U)^* = U^0$ and $U^* = V^*/U^0$.*

(c) *The matrix of $T^*$ in the dual bases is the transpose of the matrix of $T$;* $\ker T^* = (\operatorname{im} T)^0$ *and* $\operatorname{im} T^* = (\ker T)^0$.

(d) *A perfect bilinear form $B\colon V \times W \to k$ identifies $W$ with $V^*$ and $V$ with $W^*$; under this identification the $B$-adjoint of Definition A.28 is the transpose.*

*First needed:* Chapter 3 ($\operatorname{Hom}(L, \mathcal{O})$ for a lattice $L$; Pontryagin duality is the analogue for torsion modules); Chapter 4 (the contragredient representation $V^*$); Chapter 8 ($H^1$ and $H_1$ of a curve are dual). *Reference:* [10, Chapter 3]; [78].

**Definition A.32** (tensor products)**.** The *tensor product* $V \otimes_k W$ is a vector space with a bilinear map $V \times W \to V \otimes W$, $(v, w) \mapsto v \otimes w$, that is universal: every bilinear map $V \times W \to U$ factors as a unique linear map $V \otimes W \to U$ composed with it. It exists and is unique up to unique isomorphism. The *symmetric square* $\operatorname{Sym}^2 V$ is the quotient of $V \otimes V$ by the span of the $v \otimes w - w \otimes v$, and the *exterior square* $\bigwedge^2 V$ the quotient by the span of the $v \otimes v$.

**Theorem A.33** (tensor products (stated; reference))**.** *(a) If $(e_i)$ is a basis of $V$ and $(f_j)$ a basis of $W$, then $(e_i \otimes f_j)$ is a basis of $V \otimes W$; so* $\dim(V \otimes W) = \dim V \cdot \dim W$. *Every element of $V \otimes W$ is a finite sum of elementary tensors, not in general a single one.*

(b) $\operatorname{Hom}(V, W) \simeq V^* \otimes W$ *via* $(\varphi \otimes w)(v) = \varphi(v)w$. *In particular* $\operatorname{End}(V) \simeq V^* \otimes V$, *and under this isomorphism the trace* $\operatorname{End}(V) \to k$ *is the contraction* $\varphi \otimes v \mapsto \varphi(v)$.

(c) *(Extension of scalars) For a field extension $k'/k$, $V_{k'} = V \otimes_k k'$ is a $k'$-vector space with the same basis; a linear map $T$ extends to $T_{k'}$ with the same matrix, the same characteristic and minimal polynomials, and the same trace and determinant.*

(d) *(Functoriality) Linear maps $S\colon V \to V'$ and $T\colon W \to W'$ induce $S \otimes T\colon V \otimes W \to V' \otimes W'$ with $(S \otimes T)(v \otimes w) = Sv \otimes Tw$; for endomorphisms,* $\operatorname{tr}(S \otimes T) = \operatorname{tr} S \cdot \operatorname{tr} T$, *and over a field containing the eigenvalues $\lambda_i$ of $S$ and $\mu_j$ of $T$ the eigenvalues of $S \otimes T$ are the products $\lambda_i \mu_j$, with multiplicity.*

(e) $\dim \operatorname{Sym}^2 V = n(n+1)/2$ *and* $\dim \bigwedge^2 V = n(n-1)/2$ *for* $n = \dim V$. *For $n = 2$:* $\operatorname{Sym}^2 V$ *has dimension* 3, $\bigwedge^2 V$ *has dimension* 1 *and $T$ acts on it by* $\det T$; *if $T$ has eigenvalues $\lambda, \mu$ then it has eigenvalues $\lambda^2, \lambda\mu, \mu^2$ on* $\operatorname{Sym}^2 V$, *and $\lambda/\mu$, 1, 1, $\mu/\lambda$ on $V^* \otimes V = \operatorname{End}(V)$ (conjugation action), hence $\lambda/\mu$, 1, $\mu/\lambda$ on the trace-zero part when $\lambda\mu \neq 0$.*

(f) *The same definition, with "bilinear" meaning $R$-bilinear, gives the tensor product of modules over a commutative ring $R$; (a) holds for free modules, (b) for finitely generated free modules, and (d) in general. The modules $T \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell/\mathbb{Z}_\ell$ of the paper are of this kind.*

*First needed:* Chapter 4 (tensor products of representations, the twist $\rho \otimes \chi$ by a character, the adjoint representation $\operatorname{End}(V) = V^* \otimes V$ and its trace-zero part $\operatorname{ad}^0 \rho$, the identity $\operatorname{ad}^0 \rho = \operatorname{Sym}^2 \rho \otimes \det \rho^{-1}$ for two-dimensional $\rho$); Chapter 3 (tensor products of modules); Chapter 19 (the eigenvalues $\alpha/\beta$, 1, $\beta/\alpha$ of Frobenius on $\operatorname{ad}^0 V$ at a good prime are (e)); Chapter 18. *Reference:* [109, Chapter XVI]; [58, Sections 10.4 and 11.5]; [78].

*The eigenvalues in (e), computed.* Suppose $T$ is invertible with an eigenbasis $v_1, v_2$, $Tv_1 = \lambda v_1$, $Tv_2 = \mu v_2$ (for a $T$ that is not diagonalisable, the same computation in a triangular basis gives triangular matrices with the same diagonal entries). On $V^* \otimes V = \operatorname{End}(V)$, $T$ acts by conjugation, $X \mapsto TXT^{-1}$: this is the convention that $T$ acts on $V^*$ by the transpose of $T^{-1}$

(the contragredient), so that the pairing $V^* \times V \to k$ is preserved. Let $E_{ij}$ be the map $v_j \mapsto v_i$, $v_{j'} \mapsto 0$ for $j' \neq j$. Then $TE_{ij}T^{-1}v_j = TE_{ij}(\lambda_j^{-1}v_j) = (\lambda_i/\lambda_j)v_i$, where $\lambda_1 = \lambda$ and $\lambda_2 = \mu$, so $E_{12}$, $E_{21}$, $E_{11}$, $E_{22}$ are eigenvectors with eigenvalues $\lambda/\mu$, $\mu/\lambda$, 1, 1. The trace-zero part is spanned by $E_{12}$, $E_{21}$ and $E_{11} - E_{22}$, with eigenvalues $\lambda/\mu$, $\mu/\lambda$, 1. On $\operatorname{Sym}^2 V$, the images of $v_1 \otimes v_1$, $v_1 \otimes v_2$, $v_2 \otimes v_2$ form a basis with eigenvalues $\lambda^2$, $\lambda\mu$, $\mu^2$. Dividing these by $\det T = \lambda\mu$ gives $\lambda/\mu$, 1, $\mu/\lambda$ again, which is the identity $\operatorname{ad}^0 \rho = \operatorname{Sym}^2 \rho \otimes \det \rho^{-1}$ seen on eigenvalues. For Frobenius at a good prime $p$, with $\alpha\beta = p$, the eigenvalues on the trace-zero part are $\alpha/\beta = \alpha^2/p$, 1, $\beta/\alpha = \beta^2/p$.

**Theorem A.34** (the trace form (stated; reference))**.** *On* $\mathrm{M}_n(k)$ *the* trace form $(A, B) \mapsto \operatorname{tr}(AB)$ *is a symmetric bilinear form, invariant under conjugation (*$\operatorname{tr}(gAg^{-1} \cdot gBg^{-1}) = \operatorname{tr}(AB)$*), and perfect: its Gram matrix with respect to the elementary matrices* $E_{ij}$ *is a permutation matrix. On the trace-zero matrices* $\mathfrak{sl}_2(k) = \{A \in \mathrm{M}_2(k) : \operatorname{tr} A = 0\}$*, with the basis*

$$e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \qquad h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right), \qquad f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right),$$

*one has* $\operatorname{tr}(ef) = 1$*,* $\operatorname{tr}(hh) = 2$*, and* $\operatorname{tr}(ee) = \operatorname{tr}(ff) = \operatorname{tr}(eh) = \operatorname{tr}(fh) = 0$*; the Gram matrix with respect to* $(e, h, f)$ *is*

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \textit{of determinant } -2.$$

*Hence the trace form on* $\mathfrak{sl}_2$ *is perfect over every field of characteristic not* 2*, and perfect on the free* $\mathbb{Z}_\ell$*-module* $\mathfrak{sl}_2(\mathbb{Z}_\ell)$ *for every odd prime* $\ell$ *(the determinant* $-2$ *is a unit of* $\mathbb{Z}_\ell$*); over* $\mathbb{F}_2$ *it is degenerate,* $h$ *being in its radical. For* $A \in \mathfrak{sl}_2(k)$ *one has* $\operatorname{tr}(A^2) = -2 \det A$*, and in general* $\operatorname{tr}(A^2) = (\operatorname{tr} A)^2 - 2 \det A$ *for* $A \in \mathrm{M}_2(k)$*.*

*First needed:* Chapter 4 (the trace form makes $\operatorname{ad}^0 \rho$ isomorphic to its own dual when 2 is invertible); Chapter 18 (for $A = \operatorname{ad}^0 T$ and $\ell$ odd, $T^D = \operatorname{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$ via the trace form, which the paper uses in Section 4.2); Chapter 3. *Reference:* [109, Chapter XV]; the computation above is a direct check.

*The direct check, written out.* Multiplying the basis matrices: $ef = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $fe = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, so $\operatorname{tr}(ef) = 1$; $h^2 = 1$, so $\operatorname{tr}(hh) = 2$; $e^2 = f^2 = 0$; $eh = \left(\begin{smallmatrix} 0 & -1 \\ 0 & 0 \end{smallmatrix}\right) = -e$ and $fh = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right) = f$, both of trace 0. Expanding the Gram determinant along its first row gives $1 \cdot (0 \cdot 0 - 2 \cdot 1) = -2$. Modulo 2 the middle row of the Gram matrix becomes 0, which says that $h$ pairs to 0 with every element: it lies in the radical. The last identity is Cayley-Hamilton: $A^2 - (\operatorname{tr} A)A + (\det A) \cdot 1 = 0$, and taking traces (with $\operatorname{tr} 1 = 2$) gives $\operatorname{tr}(A^2) = (\operatorname{tr} A)^2 - 2 \det A$.

This ends the linear algebra. The next section collects the complex analysis, which enters through Riemann surfaces, the Weierstrass $\wp$-function, modular forms and $L$-functions.

## A.3 Complex analysis

This section is used from Chapter 8 on (Riemann surfaces, orders of meromorphic functions), in Chapter 9 (the Weierstrass $\wp$-function), in Chapter 12 (the upper half plane, $q$-expansions, the $\eta$-function) and in Chapter 19 (Euler products, the Gamma function, the Mellin transform). Throughout, $U$ is an open subset of $\mathbb{C}$.

**Definition A.35** (holomorphic and meromorphic functions)**.** $f\colon U \to \mathbb{C}$ is *holomorphic* if it is complex differentiable at every point of $U$; *entire* if holomorphic on $\mathbb{C}$; *meromorphic* on $U$ if holomorphic on $U$ minus a discrete set of points at each of which it has a pole (Theorem A.37). A $C^1$ function $u + iv$ is holomorphic if and only if $u_x = v_y$ and $u_y = -v_x$ (Cauchy-Riemann).

**Theorem A.36** (Cauchy; Taylor (stated; reference))**.** *(a) (Cauchy's theorem) If $f$ is holomorphic on $U$ and $\gamma$ is a closed piecewise $C^1$ curve in $U$ that can be contracted to a point inside $U$ (always the case when $U$ is a disc or is simply connected), then $\int_\gamma f(z)\,dz = 0$.*

*(b) (Cauchy's integral formula) If $f$ is holomorphic on an open set containing the closed disc $|z-a| \le r$, then for $|z-a| < r$ and all $n \ge 0$, $f^{(n)}(z) = \frac{n!}{2\pi i}\int_{|w-a|=r} \frac{f(w)}{(w-z)^{n+1}}\,dw$.*

*(c) (Taylor) If $f$ is holomorphic on the disc $|z-a| < R$, then $f(z) = \sum_{n\ge 0} \frac{f^{(n)}(a)}{n!}(z-a)^n$ on that disc. Holomorphic functions are infinitely differentiable, and (Morera) a continuous $f$ with $\int_{\partial\Delta} f = 0$ for every triangle $\Delta \subset U$ is holomorphic.*

*(d) (Identity theorem) If $U$ is connected and $f, g$ holomorphic on $U$ agree on a set with an accumulation point in $U$, then $f = g$. The zeros of a nonconstant holomorphic function on a connected $U$ are isolated, and each has a finite* order $\operatorname{ord}_a f$*, the least $n$ with $f^{(n)}(a) \ne 0$; then $f(z) = (z-a)^{\operatorname{ord}_a f} g(z)$ with $g$ holomorphic and $g(a) \ne 0$.*

*(e) (Liouville) A bounded entire function is constant. (Maximum modulus) If $U$ is connected and $|f|$ has a local maximum in $U$, then $f$ is constant.*

*First needed:* Chapter 8. *Reference:* [4, Chapter 4]; [186, Chapters 2 and 3].

**Theorem A.37** (Laurent series; poles and residues (stated; reference))**.** *If $f$ is holomorphic on the annulus $r < |z-a| < R$, then*

$$f(z) = \sum_{n\in\mathbb{Z}} c_n (z-a)^n, \qquad c_n = \frac{1}{2\pi i}\int_{|w-a|=s} f(w)(w-a)^{-n-1}\,dw \quad (r < s < R),$$

*the series converging absolutely and locally uniformly on the annulus, and the $c_n$ are unique. When $r = 0$ (an* isolated singularity *at $a$) exactly one of the following holds: $c_n = 0$ for all $n < 0$ and $f$ extends holomorphically to $a$ (*removable*; equivalently $f$ is bounded near $a$); there is $m \ge 1$ with $c_{-m} \ne 0$ and $c_n = 0$ for $n < -m$ ($a$* pole of order $m$*; equivalently $|f(z)| \to \infty$ as $z \to a$); or neither (*essential*). In the first two cases* $\operatorname{ord}_a f$ *is the least $n$ with $c_n \ne 0$ (negative at a pole), and $\operatorname{ord}_a(fg) = \operatorname{ord}_a f + \operatorname{ord}_a g$, $\operatorname{ord}_a(f+g) \ge \min(\operatorname{ord}_a f, \operatorname{ord}_a g)$. The* residue *is $\operatorname{Res}_a f = c_{-1}$; at a simple pole it is $\lim_{z\to a}(z-a)f(z)$. The meromorphic functions on a connected $U$ form a field.*

*First needed:* Chapter 8 (the order of a meromorphic function at a point of a Riemann surface, and the divisor of a function); Chapter 12 (the $q$-expansion at a cusp is a Laurent series in $q$; a cusp form has no terms of order $\le 0$). *Reference:* [4, Chapter 5, Section 1]; [186, Chapter 3].

*Example.* The coefficient formula is the line integral computed after Theorem A.12: integrating $\sum_m c_m (w-a)^{m-n-1}$ term by term around the circle (allowed by the uniform convergence there) keeps only $m = n$. For a concrete Laurent series, take $f(z) = 1/(z(z-1))$ on $0 < |z| < 1$. Expanding $1/(z-1) = -\sum_{n\ge 0} z^n$ gives $f(z) = -z^{-1} - 1 - z - z^2 - \cdots$, so $f$ has a simple pole at 0, $\operatorname{ord}_0 f = -1$, and $\operatorname{Res}_0 f = -1$, in agreement with $\lim_{z\to 0} z f(z) = \lim_{z\to 0} 1/(z-1) = -1$.

**Theorem A.38** (the residue theorem (stated; reference))**.** *Let $f$ be holomorphic on $U$ except at isolated points $a_1, \dots, a_k$, and let $\gamma$ be a closed piecewise $C^1$ curve in $U$, not passing through any $a_j$, contractible in $U$. Then*

$$\frac{1}{2\pi i}\int_\gamma f(z)\,dz = \sum_j n(\gamma, a_j)\,\mathrm{Res}_{a_j} f,$$

*where $n(\gamma, a)$ is the winding number of $\gamma$ about $a$ ($1$ for a simple closed curve traversed counterclockwise with $a$ inside, $0$ with $a$ outside). (Argument principle) If $f$ is meromorphic on $U$ and has no zeros or poles on a simple closed curve $\gamma$, then $\frac{1}{2\pi i}\int_\gamma f'/f$ equals the number of zeros minus the number of poles inside $\gamma$, counted with multiplicity, that is, $\sum_a \mathrm{ord}_a f$ over the points inside.*

*First needed:* Chapter 8 (a meromorphic function on a compact Riemann surface has as many zeros as poles: the degree of a principal divisor is 0); Chapter 12 (the valence formula is the argument principle on a fundamental domain). *Reference:* [4, Chapter 4, Section 5]; [186, Chapter 3].

**Theorem A.39** (Weierstrass: uniform limits of holomorphic functions (stated; reference))**.** *If $f_n$ are holomorphic on $U$ and $f_n \to f$ uniformly on every compact subset of $U$, then $f$ is holomorphic on $U$ and $f_n^{(k)} \to f^{(k)}$ uniformly on every compact subset, for every $k \geq 1$. Consequently a series $\sum f_n$ of holomorphic functions that converges locally uniformly (for instance by the $M$-test on each compact set) has a holomorphic sum and may be differentiated term by term.*

*First needed:* Chapter 9 ($\wp(z) = z^{-2} + \sum'\big((z-\omega)^{-2} - \omega^{-2}\big)$ is meromorphic with double poles at the lattice points); Chapter 12 (Eisenstein series and $q$-expansions define holomorphic functions on the upper half plane); Chapter 19 (a Dirichlet series is holomorphic in its half-plane of absolute convergence). *Reference:* [4, Chapter 5, Section 1]; [186, Chapter 2].

*The estimate for $\wp$.* The subtraction of $\omega^{-2}$ is what makes the series converge. Combining the fractions,

$$\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} = \frac{\omega^2 - (z-\omega)^2}{\omega^2(z-\omega)^2} = \frac{2z\omega - z^2}{\omega^2(z-\omega)^2}.$$

Fix $r > 0$. For $|z| \leq r$ and $|\omega| \geq 2r$ one has $|z-\omega| \geq |\omega|/2$ and $|2z\omega - z^2| \leq 2r|\omega| + r^2 \leq \frac{5}{2}r|\omega|$, so the term has modulus at most $\frac{5}{2}r|\omega| \cdot 4/|\omega|^4 = 10r|\omega|^{-3}$. The lattice sum of $|\omega|^{-3}$ converges by Theorem A.4(g), so the $M$-test gives uniform convergence on the disc $|z| \leq r$ of the sum over $|\omega| \geq 2r$; the finitely many remaining terms are meromorphic with double poles at the lattice points. Without the subtraction the terms would be of size $|\omega|^{-2}$, and the lattice sum of these diverges.

**Definition A.40** (the Riemann sphere; Mobius transformations)**.** The *Riemann sphere* is $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$, with the neighbourhoods of $\infty$ the complements of compact subsets of $\mathbb{C}$ and the coordinate $w = 1/z$ near $\infty$; a function is holomorphic or meromorphic at $\infty$ if $f(1/w)$ is so at $w = 0$. A *Mobius transformation* is $z \mapsto (az+b)/(cz+d)$ with $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$, extended to $\widehat{\mathbb{C}}$ by $-d/c \mapsto \infty$ and $\infty \mapsto a/c$ (and $\infty \mapsto \infty$ if $c = 0$).

**Theorem A.41** (Mobius transformations (stated; reference))**.** *(a) Under composition the Mobius transformations form a group, and the map $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \mapsto \left(z \mapsto \frac{az+b}{cz+d}\right)$ is a surjective homomorphism from $\mathrm{GL}_2(\mathbb{C})$ onto it with kernel the scalar matrices; so the group is $\mathrm{PGL}_2(\mathbb{C}) = \mathrm{GL}_2(\mathbb{C})/\mathbb{C}^\times = \mathrm{SL}_2(\mathbb{C})/\{\pm 1\}$.*

*(b) The action on $\widehat{\mathbb{C}}$ is sharply triply transitive: for any two triples of distinct points there is exactly one Mobius transformation taking the first to the second.*

*(c) Mobius transformations map the family of circles and lines in $\mathbb{C}$ (lines being the circles through $\infty$) to itself.*

*(d) A Mobius transformation $g \neq 1$ with matrix in $\mathrm{SL}_2(\mathbb{C})$ has two fixed points on $\widehat{\mathbb{C}}$ if $(\operatorname{tr} g)^2 \neq 4$ and one if $(\operatorname{tr} g)^2 = 4$.*

*(e) The meromorphic functions on $\widehat{\mathbb{C}}$ are exactly the rational functions, and the biholomorphic maps $\widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ are exactly the Mobius transformations.*

*First needed:* Chapter 8 ($\mathbb{P}^1(\mathbb{C}) = \widehat{\mathbb{C}}$ is the first compact Riemann surface; the degree of a rational function); Chapter 12 (the action of $\mathrm{SL}_2(\mathbb{Z})$ on the cusps $\mathbb{P}^1(\mathbb{Q}) = \mathbb{Q} \cup \{\infty\}$). *Reference:* [4, Chapter 3, Section 3]; [186, Chapter 8].

*Where the trace in (d) comes from.* A finite $z$ is fixed by $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ if and only if $az + b = z(cz + d)$, that is, $cz^2 + (d - a)z - b = 0$. If $c \neq 0$, then $\infty \mapsto a/c$ is not fixed, and the quadratic has discriminant

$$(d - a)^2 + 4bc = (a + d)^2 - 4(ad - bc) = (\operatorname{tr} g)^2 - 4,$$

so there are two fixed points or one according as $(\operatorname{tr} g)^2 \neq 4$ or $= 4$. If $c = 0$, then $\infty$ is fixed, $ad = 1$, and the remaining equation $(d - a)z = b$ has one solution when $a \neq d$, which is when $(\operatorname{tr} g)^2 - 4 = (a - d)^2 \neq 0$; when $a = d = \pm 1$, $g$ is the translation $z \mapsto z + b/a$ with $b \neq 0$, and $\infty$ is its only fixed point.

**Theorem A.42** (the upper half plane (stated; reference))**.** *Let $\mathcal{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ and, for $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$, $gz = (az + b)/(cz + d)$.*

*(a) $\operatorname{Im}(gz) = \operatorname{Im}(z)/|cz + d|^2$, so $g$ maps $\mathcal{H}$ to itself, $(gh)z = g(hz)$, and $\mathrm{SL}_2(\mathbb{R})$ acts on $\mathcal{H}$; the kernel of the action is $\{\pm 1\}$. More generally $\mathrm{GL}_2(\mathbb{R})^+$, the matrices of positive determinant, acts by the same formula, with $\operatorname{Im}(gz) = \det(g) \operatorname{Im}(z)/|cz + d|^2$.*

*(b) The action is transitive, and the stabiliser of $i$ is $\mathrm{SO}_2(\mathbb{R}) = \{\left(\begin{smallmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{smallmatrix}\right)\}$.*

*(c) Every biholomorphic map $\mathcal{H} \to \mathcal{H}$ is $z \mapsto gz$ for some $g \in \mathrm{SL}_2(\mathbb{R})$, unique up to sign: $\mathrm{Aut}(\mathcal{H}) = \mathrm{PSL}_2(\mathbb{R})$. The Cayley map $z \mapsto (z - i)/(z + i)$ is a biholomorphic map of $\mathcal{H}$ onto the unit disc.*

*(d) $\frac{d}{dz}(gz) = (cz + d)^{-2}$, and the measure $dx\,dy/y^2$ on $\mathcal{H}$ ($z = x + iy$) is invariant under $\mathrm{SL}_2(\mathbb{R})$.*

*(e) An element $g \neq \pm 1$ of $\mathrm{SL}_2(\mathbb{R})$ is* elliptic *if $|\operatorname{tr} g| < 2$ (one fixed point in $\mathcal{H}$),* parabolic *if $|\operatorname{tr} g| = 2$ (one fixed point, in $\mathbb{R} \cup \{\infty\}$), and* hyperbolic *if $|\operatorname{tr} g| > 2$ (two fixed points in $\mathbb{R} \cup \{\infty\}$). The group also acts on $\mathbb{R} \cup \{\infty\}$, and $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathbb{Q} \cup \{\infty\}$.*

*First needed:* Chapter 9 (lattices in $\mathbb{C}$ up to homothety correspond to the points of $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$); Chapter 12 (everything: the modular group, fundamental domains, cusps, elliptic points, the Petersson inner product via (d)). *Reference:* [4, Chapter 3, Section 3]; [52, Chapter 1].

*The computations behind (a), (d) and (e).* For (a), multiply numerator and denominator by $\overline{cz + d}$:

$$gz = \frac{(az + b)(c\bar{z} + d)}{|cz + d|^2} = \frac{ac|z|^2 + adz + bc\bar{z} + bd}{|cz + d|^2},$$

and with $a, b, c, d$ real the imaginary part of the numerator is $(ad-bc)\operatorname{Im} z$. For (d), the quotient rule gives $\frac{d}{dz}(gz) = (a(cz+d) - c(az+b))/(cz+d)^2 = (ad-bc)/(cz+d)^2$. A holomorphic map multiplies areas by $|g'(z)|^2 = |cz+d|^{-4}$, and by (a) $1/y^2$ is multiplied by $|cz+d|^4$, so $dx\,dy/y^2$ is unchanged. For (e), the fixed points of a real $g$ with $c \neq 0$ are the roots of the real quadratic above, of discriminant $(\operatorname{tr} g)^2 - 4$: when $|\operatorname{tr} g| < 2$ they are a pair of complex conjugates, exactly one of them in $\mathcal{H}$, and otherwise they are real.

**Theorem A.43** (the exponential and the logarithm (stated; reference))**.**

*(a) The function* $\exp(z) = \sum_{n\geq 0} z^n/n!$ *is entire,* $\exp(z+w) = \exp z \exp w$, $\exp' = \exp$, $|\exp(x+iy)| = e^x$, *and* $\exp\colon \mathbb{C} \to \mathbb{C}^\times$ *is surjective with kernel* $2\pi i\mathbb{Z}$.

*(b) The map* $z \mapsto q = e^{2\pi i z}$ *sends* $\mathcal{H}$ *onto the punctured disc* $0 < |q| < 1$, *with* $q(z) = q(z')$ *if and only if* $z - z' \in \mathbb{Z}$, *and* $|q| = e^{-2\pi \operatorname{Im} z} \to 0$ *as* $\operatorname{Im} z \to \infty$. *Consequently a holomorphic* $f$ *on* $\mathcal{H}$ *with* $f(z+1) = f(z)$ *is* $f(z) = F(q)$ *for a unique holomorphic* $F$ *on the punctured disc, and by Theorem A.37*

$$f(z) = \sum_{n\in\mathbb{Z}} a_n q^n, \qquad a_n = \int_0^1 f(x+iy)\, e^{-2\pi i n(x+iy)}\, dx \quad \text{for any } y > 0,$$

*converging absolutely and locally uniformly on* $\mathcal{H}$*: this is the* $q$-expansion, *or Fourier expansion, of* $f$, *and "holomorphic at* $\infty$*" means* $a_n = 0$ *for* $n < 0$.

*(c) On a simply connected open set not containing* 0 *there is a holomorphic branch of the logarithm, unique up to adding a multiple of* $2\pi i$*; the principal branch on* $\mathbb{C} \setminus (-\infty, 0]$ *is* $\operatorname{Log}(re^{i\theta}) = \log r + i\theta$ *with* $-\pi < \theta < \pi$, *and* $z^s = \exp(s \operatorname{Log} z)$ *for* $s \in \mathbb{C}$. *For a positive integer* $n$ *and* $s \in \mathbb{C}$, $n^{-s} = \exp(-s \log n)$ *with the real logarithm, and* $|n^{-s}| = n^{-\operatorname{Re} s}$.

*(d)* $q^{1/24} = e^{2\pi i z/24}$ *is a well-defined holomorphic function on* $\mathcal{H}$ *(a function of* $z$, *not of* $q$*); it is the factor in front of the* $\eta$*-function.*

*First needed:* Chapter 12; Chapter 19 (Dirichlet series). *Reference:* [4, Chapter 2, Section 3]; [52, Chapter 1].

*The coefficient formula in (b).* On the horizontal segment $z = x + iy$, $0 \leq x \leq 1$, the series $\sum_m a_m e^{2\pi i m z}$ converges uniformly (the segment is compact), so it may be multiplied by $e^{-2\pi i n z}$ and integrated term by term (Theorem A.6(c)):

$$\int_0^1 f(x+iy) e^{-2\pi i n(x+iy)}\, dx = \sum_m a_m e^{-2\pi(m-n)y} \int_0^1 e^{2\pi i(m-n)x}\, dx = a_n,$$

because the last integral is 1 for $m = n$ and 0 otherwise. The answer does not depend on $y$, as the theorem says.

**Theorem A.44** (infinite products (stated; reference))**.** *(a) If* $\sum |a_n| < \infty$, *then the partial products of* $\prod_{n\geq 1}(1+a_n)$ *converge, the limit is* 0 *if and only if some factor is* 0, *and the limit does not depend on the order of the factors. (One says the product converges absolutely.)*

*(b) If the* $a_n$ *are holomorphic on* $U$ *and* $\sum |a_n|$ *converges uniformly on every compact subset of* $U$, *then* $P = \prod(1+a_n)$ *is holomorphic on* $U$, $P(z_0) = 0$ *if and only if some* $1 + a_n(z_0) = 0$, $\operatorname{ord}_{z_0} P = \sum_n \operatorname{ord}_{z_0}(1+a_n)$, *and where* $P \neq 0$ *its logarithmic derivative is* $P'/P = \sum a_n'/(1+a_n)$.

(c) *For $|q|<1$ the product $\prod_{n\geq 1}(1-q^n)$ converges absolutely, uniformly on $|q|\leq r<1$, to a nonvanishing holomorphic function of $q$; so $\eta(z)=q^{1/24}\prod_{n\geq 1}(1-q^n)$, $q=e^{2\pi iz}$, is holomorphic and nonvanishing on $\mathcal{H}$.*

(d) *For $\operatorname{Re}s>1$, $\prod_p(1-p^{-s})^{-1}$ over the primes converges absolutely, uniformly on $\operatorname{Re}s\geq 1+\delta$ for every $\delta>0$, and equals $\sum_{n\geq 1}n^{-s}$ (Euler). More generally, if $\sum_p|b_p|$ converges uniformly on compact sets then $\prod_p(1+b_p)$ is holomorphic and nonvanishing where no factor vanishes.*

*First needed:* Chapter 12 (the form of `11a1` is $\eta(z)^2\eta(11z)^2$); Chapter 19 (Euler products of $L(E,s)$, $L(\mathrm{ad}^0 f,s)$ and $L(\mathrm{Sym}^2 f,s)$). *Reference:* [4, Chapter 5, Section 2]; [186, Chapter 5].

*The form of `11a1` as a q-series.* By (c), $\eta(z)^2\eta(11z)^2$ is holomorphic on $\mathcal{H}$, and the fractional powers of $q$ combine to an integral one: $\eta(11z)$ has the factor $e^{2\pi i\cdot 11z/24}$, so the product has the factor $q^{2/24}q^{22/24}=q$, and

$$\begin{aligned}\eta(z)^2\eta(11z)^2 &= q\prod_{n\geq 1}(1-q^n)^2(1-q^{11n})^2\\ &= q-2q^2-q^3+2q^4+q^5+2q^6-2q^7-2q^9-2q^{10}+q^{11}-2q^{12}+\cdots.\end{aligned}$$

These twelve coefficients agree with the numbers $a_n$ of the curve $y^2+y=x^3-x^2-10x-20$ (Cremona label `11a1`) for $n\leq 12$, as computed with PARI/GP; Chapter 12 explains why they agree for every $n$.

*Why Euler's product equals the sum in (d).* Let $\sigma=\operatorname{Re}s>1$ and $x\geq 2$. Each factor is a geometric series, $(1-p^{-s})^{-1}=\sum_{k\geq 0}p^{-ks}$, absolutely convergent because $|p^{-s}|=p^{-\sigma}<1$. The product of the finitely many factors with $p\leq x$ may be expanded and rearranged by Theorem A.4(e) and (f), and by unique factorisation each $n$ whose prime factors are all at most $x$ appears exactly once:

$$\prod_{p\leq x}(1-p^{-s})^{-1}=\sum_{n:\ p\mid n\Rightarrow p\leq x}n^{-s}.$$

Every $n\leq x$ occurs on the right, so the difference from $\sum_{n\geq 1}n^{-s}$ is at most $\sum_{n>x}n^{-\sigma}$, which tends to 0 as $x\to\infty$ by Theorem A.4(g).

**Theorem A.45** (the Gamma function, complex form (stated; reference))**.**

(a) *The integral $\Gamma(s)=\int_0^\infty e^{-t}t^{s-1}\,dt$ converges absolutely and locally uniformly for $\operatorname{Re}s>0$, and $\Gamma$ is holomorphic there.*

(b) *(Functional equation) $\Gamma(s+1)=s\,\Gamma(s)$. Used backwards, it extends $\Gamma$ to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at $s=0,-1,-2,\dots$, with residue $(-1)^n/n!$ at $s=-n$. $\Gamma$ has no zeros, so $1/\Gamma$ is entire.*

(c) *$\Gamma(s)\Gamma(1-s)=\pi/\sin(\pi s)$; $\Gamma(1/2)=\sqrt{\pi}$; $\Gamma(s)\Gamma(s+\tfrac12)=2^{1-2s}\sqrt{\pi}\,\Gamma(2s)$ (Legendre's duplication formula).*

(d) *(Stirling) In every sector $|\arg s|\leq\pi-\delta$, $\Gamma(s)\sim\sqrt{2\pi}\,s^{s-1/2}e^{-s}$ as $|s|\to\infty$; on a vertical line, $|\Gamma(\sigma+it)|\sim\sqrt{2\pi}\,|t|^{\sigma-1/2}e^{-\pi|t|/2}$ as $|t|\to\infty$.*

*First needed:* Chapter 19 (the Gamma factors of $\Lambda(E,s)$ and of the completed symmetric square $L$-function; the poles of $\Gamma$ account for the trivial zeros). *Reference:* [4, Chapter 5, Section 2]; [186, Chapter 6].

*Two checks on the statements.* The residues in (b) follow from the functional equation used $n+1$ times: $\Gamma(s) = \Gamma(s+n+1)/(s(s+1)\cdots(s+n))$ for $\operatorname{Re} s > -n-1$ away from the poles, and at $s=-n$ the factor $s+n$ vanishes while the others give

$$\operatorname{Res}_{s=-n}\Gamma = \frac{\Gamma(1)}{(-n)(-n+1)\cdots(-1)} = \frac{(-1)^n}{n!}.$$

The vertical-line form of (d) can be checked exactly at $\sigma = 1$. By (b) and (c), $\Gamma(1+it)\Gamma(1-it) = it\,\Gamma(it)\Gamma(1-it) = it\pi/\sin(\pi it) = \pi t/\sinh(\pi t)$, and $\Gamma(1-it) = \overline{\Gamma(1+it)}$ because $\Gamma$ is real on the real axis. Hence $|\Gamma(1+it)|^2 = \pi t/\sinh(\pi t) \sim 2\pi|t|e^{-\pi|t|}$, which is the square of $\sqrt{2\pi}\,|t|^{1/2}e^{-\pi|t|/2}$, as (d) predicts.

*Remark* A.46 (the Mellin transform). For $\phi\colon (0,\infty)\to\mathbb{C}$, the *Mellin transform* is $(\mathcal{M}\phi)(s) = \int_0^\infty \phi(t)\,t^s\,\frac{dt}{t}$, defined on the vertical strip where the integral converges absolutely and holomorphic there by Theorem A.14(c). By definition $\Gamma(s) = \mathcal{M}(e^{-t})(s)$ for $\operatorname{Re} s > 0$, and the substitution $t \mapsto t/n$ gives $\mathcal{M}(e^{-nt})(s) = n^{-s}\Gamma(s)$. Hence, if $f(iy) = \sum_{n\ge 1} a_n e^{-2\pi n y}$ with coefficients of polynomial growth, then for $\operatorname{Re} s$ large, interchanging sum and integral by absolute convergence,

$$\int_0^\infty f(iy)\,y^s\,\frac{dy}{y} = (2\pi)^{-s}\Gamma(s)\sum_{n\ge 1} a_n n^{-s}.$$

In detail: the substitution $y = u/(2\pi n)$ gives $\int_0^\infty e^{-2\pi n y}y^s\,dy/y = (2\pi n)^{-s}\int_0^\infty e^{-u}u^s\,du/u = (2\pi n)^{-s}\Gamma(s)$ (the measure $dy/y$ is unchanged by scaling, which is why the Mellin transform is written with it). The interchange of sum and integral is justified because the sum of the integrals of the absolute values is finite: with $\sigma = \operatorname{Re} s$ and $|a_n| \le Cn^k$, it equals $\sum_n |a_n|(2\pi n)^{-\sigma}\Gamma(\sigma) \le C(2\pi)^{-\sigma}\Gamma(\sigma)\sum_n n^{k-\sigma}$, finite for $\sigma > k+1$ by Theorem A.4(g); this is the meaning of "$\operatorname{Re} s$ large". The Mellin transform of a cusp form along the imaginary axis is its $L$-function completed by the Gamma factor; the transformation law of $f$ under $z \mapsto -1/(Nz)$ becomes, through the substitution $y \mapsto 1/(Ny)$, the functional equation of $\Lambda(f,s)$, and the rapid decay of $f(iy)$ at both ends gives the analytic continuation. This is the mechanism of Chapter 19, and the one by which Riemann continued $\zeta(s)$ using the theta function. For the physics reader: the Mellin transform is the Fourier transform of [165, Chapter 13] in the variable $\log t$.

## A.4 Group theory and representation theory as in Riley, Hobson and Bence

This section is what Chapter 1 and Chapter 4 assume and then rebuild with proofs. The reader has met it in [165, Chapters 28 and 29], over $\mathbb{C}$ and for finite groups; the remark at the end says what changes in the book.

**Definition A.47** (groups)**.** A *group* is a set with an associative multiplication, an identity and inverses; it is *abelian* if the multiplication is commutative, and its *order* is its number of elements. A *subgroup* is a subset closed under multiplication and inverses; the *order of an element* $g$ is the order of the cyclic subgroup $\langle g\rangle$ it generates. A (left) *coset* of $H$ is a set $gH$; the *index* $[G:H]$ is the number of cosets. $H$ is *normal* if $gHg^{-1} = H$ for all $g$; then the cosets form the *quotient group* $G/H$. A *homomorphism* $\varphi\colon G\to G'$ satisfies $\varphi(gh) = \varphi(g)\varphi(h)$; its *kernel* is $\varphi^{-1}(1)$. Two elements are *conjugate* if $h = gxg^{-1}$ for some $g$; the *conjugacy classes* partition $G$; the *centraliser* $C_G(x)$ is the subgroup of elements commuting with $x$.

**Theorem A.48** (finite groups (stated; reference))**.** *(a) (Lagrange) The order of a subgroup divides the order of the group; hence the order of every element divides $\#G$, and $g^{\#G} = 1$.*

*(b) The kernel of a homomorphism is a normal subgroup, the image is a subgroup, and $G/\ker\varphi \simeq \operatorname{im}\varphi$.*

*(c) The conjugacy class of $x$ has $\#G/\#C_G(x)$ elements.*

*(d) A group of prime order is cyclic; a cyclic group of order $n$ is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ and has exactly one subgroup of each order dividing $n$.*

*(e) The examples to keep in mind: $\mathbb{Z}/n\mathbb{Z}$; the symmetric groups $S_n$; the dihedral group $D_n$ of order $2n$ (the symmetries of a regular $n$-gon); the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$; the matrix groups $\mathrm{GL}_n(\mathbb{R})$, $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$.*

*First needed:* Chapter 1, which proves all of this and continues with group actions, Sylow's theorems, semidirect products, finite abelian groups, profinite groups and the groups $\mathrm{GL}_2(\mathbb{F}_q)$, which RHB does not treat. *Reference:* [165, Chapter 28]; [58, Chapters 1 to 3].

*Example.* In $S_3$ the centraliser of the transposition $(12)$ is $\{1, (12)\}$, since $(12)$ commutes with neither $(13)$, $(23)$ nor the 3-cycles; so by (c) its conjugacy class has $6/2 = 3$ elements, the three transpositions. The centraliser of $(123)$ is the cyclic group $\langle(123)\rangle$ of order 3, and its class $\{(123), (132)\}$ has 2 elements. With the class $\{1\}$ this accounts for all $1 + 3 + 2 = 6$ elements.

**Definition A.49** (representations and characters over $\mathbb{C}$)**.** A *representation* of $G$ of degree $n$ over $\mathbb{C}$ is a homomorphism $D\colon G \to \mathrm{GL}_n(\mathbb{C})$ (RHB's "matrix representation"); equivalently, an action of $G$ by linear maps on an $n$-dimensional complex vector space $V$, a basis of $V$ turning one into the other. Two representations are *equivalent* if $D' = P^{-1}DP$ for an invertible $P$ (the same action in another basis). A subspace $U \subseteq V$ is *invariant* if $gU \subseteq U$ for all $g$; $V$ is *irreducible* if it has no invariant subspace other than 0 and $V$, and *completely reducible* (or *semisimple*) if it is a direct sum of irreducible invariant subspaces. The *character* of $D$ is $\chi_D(g) = \operatorname{tr} D(g)$: a class function with $\chi_D(1) = n$ and, for $G$ finite, $\chi_D(g^{-1}) = \overline{\chi_D(g)}$. Direct sums and tensor products of representations have characters the sum and the product. The *regular representation* is $V = \mathbb{C}[G]$, with basis $(e_h)_{h\in G}$ and $g \cdot e_h = e_{gh}$; its character is $\#G$ at $g = 1$ and 0 elsewhere.

**Theorem A.50** (representations of finite groups over $\mathbb{C}$ (stated; reference))**.** *Let $G$ be finite.*

*(a) (Maschke) Every representation over $\mathbb{C}$, or over any field whose characteristic does not divide $\#G$, is completely reducible: every invariant subspace has an invariant complement.*

*(b) (Schur) A $G$-equivariant linear map between irreducible representations is 0 or an isomorphism, and the $G$-equivariant endomorphisms of an irreducible complex representation are the scalars.*

*(c) (Orthogonality) For irreducible characters $\chi, \chi'$, $\frac{1}{\#G}\sum_{g\in G}\chi(g)\overline{\chi'(g)}$ is 1 if $\chi = \chi'$ and 0 otherwise. The irreducible characters form an orthonormal basis of the space of class functions, so their number is the number of conjugacy classes; and $\sum_\chi \chi(1)^2 = \#G$.*

*(d) A complex representation is determined up to equivalence by its character; the multiplicity of the irreducible $\chi$ in a representation of character $\psi$ is $\frac{1}{\#G}\sum_g \psi(g)\overline{\chi(g)}$; a representation is irreducible if and only if $\frac{1}{\#G}\sum_g|\chi(g)|^2 = 1$. The regular representation contains each irreducible representation with multiplicity equal to its degree.*

*(e) (Column orthogonality)* $\sum_\chi \chi(g)\overline{\chi(h)}$ *is* $\#C_G(g)$ *if* $g$ *and* $h$ *are conjugate and* 0 *otherwise.*

*(f) If* $G$ *is abelian, every irreducible complex representation has degree* 1*; the degree-one representations ("characters" in the second sense) of* $G$ *form a group isomorphic to* $G$*, and for* $G = \mathbb{Z}/n\mathbb{Z}$ *they are* $k \mapsto e^{2\pi ijk/n}$*,* $0 \le j < n$*. The degree of an irreducible representation of any finite* $G$ *divides* $\#G$*.*

*First needed:* Chapter 4. *Reference:* [165, Chapter 29]; [173, Part I]; [58, Chapter 18].

*Example:* $S_3$. There are three conjugacy classes (the example after Theorem A.48), hence three irreducible characters by (c): the trivial character, the sign, and the character $\chi$ of the two-dimensional representation on $\{x \in \mathbb{C}^3 : x_1 + x_2 + x_3 = 0\}$ by permuting coordinates, with values 2, 0, $-1$ on the classes of 1, the transpositions and the 3-cycles. The check of (c) for $\chi$, weighting each class by its size, is $\frac{1}{6}(1 \cdot 2^2 + 3 \cdot 0^2 + 2 \cdot (-1)^2) = 1$, so $\chi$ is irreducible by (d), and $1^2 + 1^2 + 2^2 = 6 = \#S_3$. The regular representation is the sum of the trivial representation, the sign and two copies of $\chi$.

*Remark* A.51 (what changes in the book). Four things change when the paper's representations arrive, all in Chapter 4. (1) *The field.* The representations of the paper live over $\mathbb{F}_\ell$, $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$, their algebraic closures, and rings of integers of number fields. When the characteristic $\ell$ divides the order of the group, as it does for $\mathrm{GL}_2(\mathbb{F}_3)$ in characteristic 3 and for an inertia group acting on $E[\ell]$ through a unipotent element of order $\ell$, Maschke's theorem fails: an invariant subspace need not have an invariant complement, and a representation need not be semisimple. For instance $\mathbb{Z}/3\mathbb{Z}$ acting on $\mathbb{F}_3^2$ through $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ fixes the line $\mathbb{F}_3 e_1$, and this line has no invariant complement: a complement would be a second invariant line, hence a second line of eigenvectors of $u$, but the only eigenvalue is 1 and its eigenspace is $\mathbb{F}_3 e_1$ (the computation after Theorem A.24). The invariants are then the composition factors (Jordan-Holder), not direct summands, and a character determines only the semisimplification. The right statement is Brauer-Nesbitt: two semisimple representations with the same characteristic polynomial of $\rho(g)$ for every $g$ are isomorphic; the traces determine the characteristic polynomials in characteristic 0 or when the characteristic exceeds the dimension, but not in general. An example of the failure: over $\mathbb{F}_3$, let $\mathbb{Z}/2\mathbb{Z}$ act on $\mathbb{F}_3^3$ trivially, and on another copy of $\mathbb{F}_3^3$ by $-1$. Both are semisimple (2 is prime to 3), and their traces are $3 = 0$ and $3 = 0$ at the identity and $3 = 0$ and $-3 = 0$ at the generator, so the traces agree everywhere; but the characteristic polynomials of the generator are $(X-1)^3$ and $(X+1)^3$, and the representations are not isomorphic. (2) *Irreducible versus absolutely irreducible.* Over a field that is not algebraically closed an irreducible representation can become reducible after extending scalars, and Schur's lemma in the form "equivariant endomorphisms are scalars" holds only for absolutely irreducible ones. The paper's example: the cyclic group of order 4 generated by $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ acts irreducibly on $\mathbb{F}_3^2$ ($X^2 + 1$ has no root in $\mathbb{F}_3$) but reducibly on $\mathbb{F}_9^2$, as $\psi \oplus \psi^{-1}$ for a character $\psi$ of order 4. This is why the hypothesis of the paper's main theorem says "absolutely irreducible". (3) *The group.* The Galois groups $G_\mathbb{Q}$, $G_{\mathbb{Q}_p}$, $I_p$ are infinite profinite groups, their representations are required to be continuous, and the finite theory is applied through finite quotients. (4) *Lattices.* A representation on a $\mathbb{Q}_\ell$-vector space $V$ with a stable $\mathbb{Z}_\ell$-lattice $T$ can be reduced modulo $\ell$ to a representation on $T/\ell T$ over $\mathbb{F}_\ell$; the reduction depends on $T$, its semisimplification does not, and when the reduction is irreducible all stable lattices are homothetic (Ribet's lemma). Finally, the word "character" in the book almost always means a degree-one representation of a group, usually a profinite one (the cyclotomic character $\chi_\ell$, the quadratic characters $\chi_{-3}$, $\chi_{-4}$, $\chi_{q^*}$, Dirichlet characters), not a trace function; Chapter 4 says "the character of $\rho$" when the trace is meant.

## A.5 Elementary probability and statistics

This section is used only in Chapter 22, which explains the paper's data protocol (Section sec:data of the paper) and its calibration argument (Section sec:sharp). The reader's course in statistics is more than enough; the cards fix the vocabulary.

**Definition A.52** (finite probability spaces; base rates)**.** A *finite probability space* is a finite set $\Omega$ with a function $P\colon \Omega \to [0,1]$ of total sum 1; an *event* is a subset $A \subseteq \Omega$, with $P(A) = \sum_{\omega \in A} P(\omega)$; a *random variable* is a function $X\colon \Omega \to \mathbb{R}$, with *expectation* $E[X] = \sum_\omega X(\omega)P(\omega)$ and *variance* $\operatorname{Var} X = E[(X - E[X])^2]$. The *indicator* $1_A$ of an event has $E[1_A] = P(A)$. Events $A, B$ are *independent* if $P(A \cap B) = P(A)P(B)$, and the *conditional probability* is $P(A \mid B) = P(A \cap B)/P(B)$. When $\Omega$ is a population with the uniform distribution $P(\omega) = 1/\#\Omega$, $P(A) = \#A/\#\Omega$ is the *proportion* of $A$ in the population, also called the *base rate* of the attribute $A$, and $E[X]$ is the population mean of $X$. If $A$ and $B$ were independent, the proportion of $A$ within $B$ would equal the base rate of $A$; comparing the two is the simplest test of association.

**Theorem A.53** (expectation (stated; reference))**.** *(a) (Linearity) $E[aX + bY] = aE[X] + bE[Y]$, with no assumption of independence. In particular the expected number of events among $A_1, \dots, A_n$ that occur is $\sum_i P(A_i)$, whether or not the $A_i$ are independent.*

*(b) For independent $X, Y$: $E[XY] = E[X]E[Y]$ and $\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y$.*

*(c) (Markov, Chebyshev) For $X \geq 0$ and $t > 0$, $P(X \geq t) \leq E[X]/t$; for any $X$, $P(|X - E[X]| \geq t) \leq \operatorname{Var} X/t^2$.*

*(d) (Law of large numbers) If $X_1, \dots, X_n$ are independent with the same distribution, of mean $m$ and finite variance, then $P(|(X_1 + \cdots + X_n)/n - m| \geq \varepsilon) \to 0$ as $n \to \infty$, for every $\varepsilon > 0$.*

*First needed:* Chapter 22. *Reference:* [65, Chapters I, V and IX]; [165, Chapter 30].

*How (d) follows from (b) and (c).* Let $S_n = X_1 + \cdots + X_n$ and $v = \operatorname{Var} X_1$. By (a) $E[S_n/n] = m$, and by (b), applied repeatedly, $\operatorname{Var} S_n = nv$, so $\operatorname{Var}(S_n/n) = v/n$. Chebyshev's inequality then gives $P(|S_n/n - m| \geq \varepsilon) \leq v/(n\varepsilon^2) \to 0$. (Repeated use of (b) needs $X_1 + \cdots + X_j$ and $X_{j+1}$ to be independent, which holds when the $X_i$ are independent as a family.)

**Definition A.54** (hypothesis tests)**.** A *hypothesis test* consists of a *null hypothesis* $H_0$, a precise probabilistic model for the data (for instance: two attributes are independent); a *test statistic* $T$, a number computed from the data; and a *rejection rule*, fixed before the data are examined, of the form "reject $H_0$ if $T \in R$", where $R$ is chosen so that $P_{H_0}(T \in R) \leq \alpha$ for a prescribed *significance level* $\alpha$. The *p-value* of an observed value $t$ is $P_{H_0}(T$ at least as extreme as $t)$. Rejecting $H_0$ says that the data would be very unlikely under $H_0$; it does not say that a particular alternative is true, and it says nothing at all if the rule was chosen after looking at the data.

**Example A.55** (the paper's calibration argument)**.** The population is the set of working-set curves ($N \leq 300000$) with $E[3]$ irreducible and without complex multiplication. Among the 366151 of them for which the right-hand side of the paper's rule (its inequality eq:rule) is 0, the base rate of $v_3(\deg \varphi_E) = 0$ is 53.2%. Take as null hypothesis that $v_3(\deg \varphi_E)$ is independent of the local data. Then a curve whose bound is $b$ would have $v_3(\deg \varphi_E) < b$ with the probability that a bound-0 curve has $v_3(\deg \varphi_E) < b$, and summing these probabilities over the 867578 working-set curves with a positive bound gives, by Theorem A.53(a), an expected number of about

605000 violations. In symbols: let $p_0(b)$ be the proportion of bound-0 curves with $v_3(\deg \varphi_E) < b$, and let $A_C$ be the event that the curve $C$ violates its bound $b(C)$. Under the null hypothesis $P(A_C) = p_0(b(C))$, and the expected number of violations is

$$\sum_{C:\ b(C)>0} p_0\big(b(C)\big),$$

whatever the dependence between the events $A_C$; linearity of expectation needs no independence. The observed number is 0. The force of the argument lies in the comparison of these two numbers, not in a $p$-value (which would be astronomically small): the inequality is not an artefact of most modular degrees being divisible by 3. What the argument does not show is that the inequality holds beyond the data; that is the business of the theorem under its hypotheses and of the conjecture outside them. The equality rates (57.5% at $\ell = 3$) and the mean excess (0.63) that the paper reports are descriptive statistics, proportions and means in the population, not tests.

**Definition A.56** (hold-out sets; pre-registration)**.** When a data set is searched for a relation among many candidates, some candidate will fit the data by chance, and the more thorough the search the more surely (the problem of *multiple comparisons*); a relation found on a data set therefore cannot be tested on the same data. The remedy is to split the data, before the search, into a *working set*, used for the search and for everything else, and a *hold-out set*, which is not examined; to write down the statement and the test, with its rejection rule, before looking at the hold-out (*pre-registration*); and then to test once. A statement that passes such a test has been *blind-tested*. A statement formulated after the hold-out has been used can only be *confirmed* on it, which is weaker, and a fresh hold-out is needed to blind-test it.

*How large the effect of multiple comparisons is.* Suppose 20 candidate relations, all false, are each tested at significance level $\alpha = 0.05$, with independent data. By Theorem A.53(a) the expected number of false rejections is $20 \cdot 0.05 = 1$, and by independence the probability that at least one candidate is rejected, and so looks like a discovery, is $1 - 0.95^{20} \approx 0.64$. A search over thousands of candidates makes a chance fit practically certain. Testing the survivor on the same data says nothing; testing it on a hold-out that played no part in the search restores the meaning of $\alpha$.

*In the paper* A.57. The paper's ranges are the working set ($N \le 300000$), the hold-out ($300000 < N \le 400000$), the extension ($400000 < N \le 500000$) and the out-of-table curves ($500000 < N \le 4 \cdot 10^6$). The component-group inequality (eq:main) and Conjecture conj:eis were found on the working set, written down, tested once on the hold-out, then on the extension, and nothing was adjusted afterwards; the twist-symmetrised form (eq:twist) was likewise tested once on the hold-out and the extension. The Euler-factor rule (Conjecture conj:euler) was formulated last, after the hold-out and the extension had been used, so for it they are a confirmation and the out-of-table curves, generated before the rule was formulated, are the blind test. Chapter 22 explains each range and each count of the paper's tables. *Reference:* [203]; [165, Chapter 31].

## A.6 Where to look

*Real analysis.* Rudin [168] is the reference used above and contains everything in Appendix A.1 with proofs; Apostol [5] treats double series, improper integrals and the Gamma function in more detail. RHB [165, Chapter 4] covers series and convergence tests at the level of a first course.

*Complex analysis.* Ahlfors [4] and Stein-Shakarchi [186] both contain all of Appendix A.3; the second has a chapter on the Gamma and zeta functions and one on the upper half plane. The first chapter of Diamond-Shurman [52] does the upper half plane, the action of $\mathrm{SL}_2(\mathbb{Z})$ and $q$-expansions in exactly the form Chapter 12 needs. RHB [165, Chapters 24 and 25] covers residues and contour integration.

*Linear algebra.* Axler [10] (over $\mathbb{R}$ and $\mathbb{C}$, with the spectral theorem and generalised eigenvectors done carefully) and Halmos [78] (bilinear forms, duality, tensor products, in a coordinate-free way) between them contain Appendix A.2 over $\mathbb{R}$ and $\mathbb{C}$. For arbitrary fields, and for the same constructions over rings, use Lang [109, Chapters XIII to XVI] or Dummit-Foote [58, Chapters 11 and 12, and Section 10.4]. RHB [165, Chapter 8] is the reader's starting point.

*Groups and representations.* RHB [165, Chapters 28 and 29] is what we assume. Serre [173] is the standard reference: Part I for characteristic 0, Part III for the modular theory that Chapter 4 needs. Dummit-Foote [58, Chapters 18 and 19] and Lang [109, Chapter XVIII] cover the same ground, and [58, Chapters 1 to 6] is the book's own reference for the group theory of Chapter 1.

*Probability and statistics.* RHB [165, Chapters 30 and 31] is what we assume. Feller [65] is the classical text on discrete probability; Wasserman [203] is a concise account of statistical inference, including hypothesis tests and multiple comparisons, at the level needed to read Chapter 22.

*After this appendix.* The book's own references for the material it builds are named in the "Notes and references" section of each chapter; the ones used most are Lang [109] and Dummit-Foote [58] for algebra, Atiyah-Macdonald [7] for commutative algebra, Serre [174] and Neukirch [142] for local fields and number fields, Silverman [185, 184] for elliptic curves, and Diamond-Shurman [52] for modular forms.

# Appendix B

# Notation, conventions and index of symbols

This appendix is a reference. It states the sixteen book-wide conventions B-C1 to B-C16, which bind every chapter, and it gives an index of the symbols of the book and of the paper with the chapter where each is defined. It proves nothing: every identity quoted in a convention is proved in the chapter named next to it.

## B.1 How to use this appendix

The book writes the paper's notation wherever the paper has one (the trace-zero adjoint $\mathrm{ad}^0 V$, the depletion $f^{\Sigma}$, the congruence ideal $\eta^{\Sigma}_{f,\lambda}$, the geometric component group $\Phi_q^{\mathrm{geom}}$, ...). Where the paper is ambiguous, or uses one letter for two things, the conventions below fix the book's reading, and the book's text follows it; quotations of the paper keep the paper's own symbols and are followed by the book's reading. The conventions are numbered B-C1 to B-C16, and the chapters cite them by these names. Each convention box is followed by the reason for it, the chapters it binds, and, where a chapter writes a permitted variant, which chapter and which variant. No number of the paper changes under any of these conventions: they change how statements are read, never what is computed.

The index of symbols (Appendix B.9) has one row for each symbol or group of related symbols, in thirteen groups ordered as the book is: numbers and modules, groups and representations, fields and characters, local fields, curves and reduction, Galois representations, modular forms, cohomology, $L$-functions, congruence ideals, the paper's quantities, computation, decorations. Inside a group the rows follow the order in which the symbols are introduced. The column "defined in" names the chapter where the symbol is defined, and a second chapter where it is specialised or first used in earnest; "see also" points to the convention that resolves a collision, and to the other meanings of the same letter, which are collected in Appendix B.10. The entries for every chapter, Chapter 21 included, have been checked against the written texts.

Cross-references go to chapters only, never to numbered items of other chapters: the chapter's own convention box (near its beginning) restates what it uses, and its card of core results locates the definitions.

## B.2 Letters for primes; valuations and lengths

These two conventions are used in every chapter. The letter $q$ has four meanings in the book and the paper; the table after B-C1 lists them.

**Convention B.1** (B-C1: letters for primes, and the collisions of $q$ and $\ell$)**.** $p$, $q$, $\ell$ denote primes. $\ell$ is the coefficient prime (odd in every use of the paper); $q$ is a bad prime of $E$ ($q \mid N$); $p$ is any prime. $\ell^* = (-1)^{(\ell-1)/2}\ell$ and $q^* = (-1)^{(q-1)/2}q$ for odd $\ell$, $q$. The letter $q$ also denotes the size of a finite field $\mathbb{F}_q$ (when it need not be prime, Chapter 9 writes $\mathfrak{q}$); the variable $q = e^{2\pi iz}$ of $q$-expansions; and, written $\mathsf{q}$, the variable of the Tate curve. The Tate parameter of $E$ is always $q_E$, never $u$ and never $\mathsf{q}$ alone. A chapter that uses two of these meanings says so in its first convention box and never puts two of them in one display.

*Reason.* The paper uses $q$ for bad primes and for $q$-expansions, and the algebra chapters need $\mathbb{F}_q$ for prime powers. *Binds:* every chapter. *In the chapters:* Chapter 6 writes the size of a residue field $N\mathfrak{p}$, never $q$; Chapter 8 uses $q$ only for field sizes; Chapter 12 uses it only for $e^{2\pi iz}$ outside its paper environments; Chapter 7 lets $q = \#k_K$ and the bad prime coincide, as they do for $K = \mathbb{Q}_q$.

| written | meaning | chapters |
|---|---|---|
| $q$ | a bad prime of $E$, $q \mid N$ | the paper; Chapters 10, 11, 18, 19 and 21 |
| $q$, $\mathbb{F}_q$ | size of a finite field, a prime power | Chapters 1, 2, 5, 7 and 8 |
| $\mathfrak{q}$ | size of a finite field, not necessarily prime | Chapter 9 |
| $q = e^{2\pi iz}$ | variable of $q$-expansions | Chapters 12 to 15 |
| $\mathsf{q}$, $E_{\mathsf{q}}$ | variable of the Tate curve, the Tate curve | Chapter 10 |
| $q_E$ | Tate parameter of $E$ at a prime with $v_q(j_E) < 0$ | Chapters 10, 17 and 18 |

**Convention B.2** (B-C10: valuations, lengths, orders, Fitting ideals)**.** $v_\ell$ is the $\ell$-adic valuation on $\mathbb{Q}$ (and on $\mathbb{Q}_\ell$), normalised by $v_\ell(\ell) = 1$, with $v_\ell(0) = +\infty$; for a nonzero ideal $\ell^a\mathbb{Z}_\ell$, $v_\ell(\ell^a\mathbb{Z}_\ell) = a$; for a discrete valuation ring $\mathcal{O}$ with uniformiser $\varpi$, $v(\varpi) = 1$, and the paper's $v_\lambda$ for $\lambda = \ell$ is $v_\ell$. For a class $c \in \mathbb{Z}/\ell^m$, $v_\ell(c) := \min(v_\ell(\tilde{c}), m)$ for any lift $\tilde{c} \in \mathbb{Z}$ (so, as in the paper's Lemma lem:tamfactor, "$v_\ell(0)$ is read as $v$", $v = v_\ell(q-1)$). "Length" without subscript means $\mathbb{Z}_\ell$-length, length $=$ length$_{\mathbb{Z}_\ell}$, and length$(\mathbb{Z}_\ell/\ell^a) = a$; for finite $\mathbb{Z}_\ell$-modules $\#X = \ell^{\text{length}\,X}$. Orders $\#\mathcal{O}/\Delta$ are used only for $\mathcal{O}$ with finite residue field; otherwise lengths. $\text{Fitt}_{\mathbb{Z}_\ell}(X) = \ell^{\text{length}\,X}\mathbb{Z}_\ell$ for finite $X$. $\text{Tam}_p(M) = \text{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p, M)_{\text{tor}})^{G_{\mathbb{Q}_p}})$: the Fitting ideal *of the invariants*.

*Reason.* The paper's Lemma lem:tamfactor evaluates $v_\ell$ at a class that may be 0, and its $\text{Tam}_p$ is the Fitting ideal of the invariants, not the invariants of a Fitting ideal. *Binds:* every chapter from Chapter 2 on. *Where proved:* the independence of the lift is in Chapter 2; lengths and Fitting ideals are in Chapter 3; $\text{Tam}_p$ is defined in Chapter 18.

## B.3 Frobenius

The paper uses the arithmetic Frobenius in group theory and cohomology (the proof of Lemma lem:tamfactor) and the geometric Frobenius in local $L$-factors (Section 1.1, its table, Lemma lem:euler, the sentence after Lemma lem:dfg21, Lemma lem:fpr). The paper says so in Section

1.1: "Throughout, $\mathrm{Frob}_p$ is an arithmetic Frobenius at $p$ and $\mathrm{Frob}_p^{-1}$ a geometric one". B-C2 fixes the symbols, B-C3 the eigenvalues.

**Convention B.3** (B-C2: arithmetic Frob, geometric $\mathrm{Frob}^{-1}$)**.**

(a) $\mathrm{Frob}_p$ (or Frob when the prime is clear) is an *arithmetic* Frobenius at $p$: an element of a decomposition group $D_p \simeq G_{\mathbb{Q}_p}$ inducing $x \mapsto x^p$ on the residue field, well defined modulo $I_p$ and up to conjugation. It acts on $\mu_n$ $(p \nmid n)$ by $\zeta \mapsto \zeta^p$, so $\chi_\ell(\mathrm{Frob}_p) = p$ for $\ell \neq p$. For a finite field $\mathbb{F}_q$, $\mathrm{Frob}_q \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ is $x \mapsto x^q$.

(b) $\Phi_p = \mathrm{Frob}_p^{-1}$ (or $\Phi$) is the *geometric* Frobenius. An eigenvalue $\lambda$ of Frob is the eigenvalue $\lambda^{-1}$ of $\Phi$. In a chapter where component groups $\Phi_q(\cdot)$ (B-C11) or cyclotomic polynomials $\Phi_n$ occur, the geometric Frobenius at a named prime is written $\mathrm{Frob}_q^{-1}$, never $\Phi_q$; $\Phi$ without subscript is always the geometric Frobenius of a local field. The symbols $\mathrm{Frob}^{\mathrm{geom}}$, $\mathrm{Frob}^{\mathrm{arith}}$ and the phrase "$\mathrm{Frob}_p$, the geometric one" are not used.

(c) *Group theory and cohomology use* Frob: the tame relation, conjugation actions on $H^1(I_p, -)$, unramified cohomology $H^1_{\mathrm{ur}}(\mathbb{Q}_p, M) = M^{I_p}/(\mathrm{Frob} - 1)M^{I_p}$, the matrices in the proof of Lemma lem:tamfactor, Frobenius on Kummer classes.

(d) *L-factors use the geometric Frobenius*: for a representation $M$ of $G_{\mathbb{Q}_p}$ on a finite-dimensional $\mathbb{Q}_\ell$-space $(p \neq \ell)$,

$$L_p(M, s) = \det\bigl(1 - \mathrm{Frob}_p^{-1}\, p^{-s} \mid M^{I_p}\bigr)^{-1},$$

so $L_p(\mathbb{Q}_\ell(1), s) = (1 - p^{-1-s})^{-1}$ and $L_p(M(n), s) = L_p(M, s+n)$. Diamond, Flach and Guo [49, 1.1.1, 1.7.2] use the geometric Frobenius for $L$-factors and call it $\mathrm{Frob}_p$.

(e) *The tame relation.* For a topological generator $t$ of the tame (or pro-$\ell$ tame) inertia quotient of $G_{\mathbb{Q}_q}$:

$$\mathrm{Frob}\; t\; \mathrm{Frob}^{-1} = t^q, \qquad \text{equivalently} \qquad \mathrm{Frob}^{-1}\, t\; \mathrm{Frob} = t^{q^{-1}}\ (q^{-1} \in \mathbb{Z}_\ell),$$

and $\Phi\, t\, \Phi^{-1} = t^{q^{-1}}$ for the geometric Frobenius. The paper's "$\mathrm{Frob}^{-1}\, t\, \mathrm{Frob} = t^{q^{-1}}$" is the second form, with the arithmetic Frobenius.

(f) *The Weil-Deligne relation.* With $\|w\| = q^{d(w)}$, $w|_{K^{\mathrm{ur}}} = \mathrm{Frob}^{d(w)}$ (so $\|\mathrm{Frob}\| = q$, $\|\Phi\| = q^{-1}$, and $\|w\| = \chi_\ell(w)$ for $\ell \neq p$), a Weil-Deligne representation $(r, N)$ satisfies

$$r(w)\, N\, r(w)^{-1} = \|w\|\, N, \quad \text{that is} \quad r(\mathrm{Frob})\, N\, r(\mathrm{Frob})^{-1} = q\, N, \quad r(\Phi)\, N\, r(\Phi)^{-1} = q^{-1} N.$$

(g) *Frobenius endomorphisms and the Artin map.* The $q$-power Frobenius morphism of a curve or abelian variety over $\mathbb{F}_q$ is $\pi_q$ (or $\pi$); it acts on $\overline{\mathbb{F}}_q$-points as $\mathrm{Frob}_q$ and on $T_\ell$ as $\mathrm{Frob}_q$; its dual isogeny (the Verschiebung) is $\hat{\pi}_q = q\pi_q^{-1}$. The Eichler-Shimura relation reads $(T_p)_* = \pi_p + \hat{\pi}_p$ on $J_0(N)_{\mathbb{F}_p}$. The local Artin map sends a uniformiser to an arithmetic Frobenius; sources that send it to the geometric one (Deligne [41]) are flagged where cited.

*Reason.* The paper uses both: the geometric Frobenius in Section 1.1, its table, Lemma lem:euler and Lemma lem:fpr, and the arithmetic one in Lemma lem:tamfactor; the factor at a multiplicative prime depends on which. A clash rule is needed because the paper's $\Phi_q$ is a component group. *Binds:* Chapter 4 to Chapter 10, Chapter 11, Chapter 14 to Chapter 22. *In the chapters:*

the geometric Frobenius is written $\Phi$, $\Phi_p$ in Chapter 7 ($\Phi$ alone), Chapter 17 and Chapter 19 (no component groups there), and $\mathrm{Frob}_p^{-1}$ in Chapters 5, 6, 8 to 11, 15, 16 and 18 and in Chapter 21. Chapter 4, which has no Galois groups, writes an abstract arithmetic Frobenius $F$ with $FgF^{-1} = g^q$ on an inertia image and $F^{-1}$ for the geometric one. Chapter 8 writes $\pi_r \colon C \to C^{[r]}$ for the $r$-power Frobenius morphism onto the Frobenius twist $C^{[r]}$. *Where proved:* the tame relation in Chapter 7; $L_p(M(1), s) = L_p(M, s+1)$ in Chapter 7; the Artin map in Chapters 6 and 7; Eichler-Shimura in Chapter 14.

**Convention B.4** (B-C3: which Frobenius has the eigenvalues $\alpha$, $\beta$)**.** Let $E/\mathbb{Q}$ have good reduction at $p \neq \ell$, $V = V_\ell E$, $a_p = p + 1 - \#\tilde{E}(\mathbb{F}_p)$, and let $\alpha, \beta$ be the roots of $X^2 - a_p X + p$ (so $\alpha + \beta = a_p$, $\alpha\beta = p$, $|\alpha| = |\beta| = \sqrt{p}$).

(a) On $V = V_\ell E$ the *arithmetic* Frobenius has characteristic polynomial $X^2 - a_p X + p$, eigenvalues $\alpha, \beta$; the geometric one has $\alpha^{-1} = \beta/p$, $\beta^{-1} = \alpha/p$ (a different set).

(b) On $V^\vee = H^1(E_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell) \simeq V(-1)$ the *geometric* Frobenius has eigenvalues $\alpha, \beta$ (and trace $a_p$).

(c) Hence $L_p(E, s) = (1 - a_p p^{-s} + p^{1-2s})^{-1} = L_p(V^\vee, s)$, while $L_p(V, s) = L_p(E, s+1)$. The same holds for $\rho_{f,\lambda}$ normalised by $\det(X - \rho_{f,\lambda}(\mathrm{Frob}_p)) = X^2 - a_p(f)X + p$: $L_p(f, s) = L_p(V_\lambda(f), s-1)$.

(d) On $\mathrm{ad}^0 V$ both Frobenii have the eigenvalues $\alpha/\beta, 1, \beta/\alpha$ (an inversion-stable set), so the good factor $L_p(A_f, s)^{-1} = (1 - \frac{\alpha}{\beta}p^{-s})(1 - p^{-s})(1 - \frac{\beta}{\alpha}p^{-s})$ does not depend on the convention; neither do the lines on which Frobenius acts by $\chi_{-3}(q)$ or $\chi_{-4}(q) = \pm 1$. It does at a prime of type $\mathrm{I}_n$ or $\mathrm{I}_n^*$: on $(\mathrm{ad}^0 V)^{I_q}$, the line of $e$, the arithmetic Frobenius acts by $q$, the geometric one by $q^{-1}$, and only the latter gives the paper's $1 - q^{-1-s}$ (and, for $\mathrm{I}_n^*$, the paper's $P_q$).

(e) In Lemma lem:old the paper's $\alpha_p, \beta_p$ are the roots of $X^2 - a_p(h)X + p$ for the newform $h$; they are the $U_p$-eigenvalues of the $p$-stabilisations $h - \beta_p h(pz)$, $h - \alpha_p h(pz)$.

*Reason.* It is easy to assume that both Frobenii have the eigenvalues $\alpha, \beta$ on $V_\ell E$; they do not, and the factor at a multiplicative prime depends on the choice. *Binds:* Chapters 6 to 9, 11, 13 to 15 and 18 to 21. *In the chapters:* Chapter 9 writes the roots $\alpha_p, \beta_p$. *Where proved:* (a) to (c) in Chapter 9; (d) in Chapters 7 and 19; (e) in Chapter 13.

*Warning* B.5 (where the paper uses which Frobenius). Geometric: Section 1.1 ($L_p(A_f, s)$, $\delta_p$), the table of Section 1, Definition def:terms ($P_q$, $e_\ell$), Lemma lem:euler, Proposition prop:shapes, the sentence after Lemma lem:dfg21 ("$\det(1 - \mathrm{Frob}_p^{-1} \mid (V^D)^{I_p}) = L_p(A_f, 1)^{-1}$"), Lemma lem:fpr. Arithmetic: the proof of Lemma lem:tamfactor (the matrices, $c' = \varepsilon\kappa(\mathrm{Frob})$, the action "$q^{-1}\,\mathrm{Ad}(\rho(\mathrm{Frob}))$"), unramified cohomology, the tame relation. Either: every statement about the lines of $(\mathrm{ad}^0 V)^{I_q}$ at the potentially good primes, where Frobenius acts by $\pm 1$.

| space | eigenvalues of Frob (arithmetic) | eigenvalues of $\mathrm{Frob}^{-1}$ (geometric) |
|---|---|---|
| $\mathbb{Z}_\ell(1)$ | $p$ | $p^{-1}$ |
| $V_\ell E$, $p$ good | $\alpha$, $\beta$ | $\alpha^{-1}=\beta/p$, $\beta^{-1}=\alpha/p$ |
| $V_\ell E^\vee = H^1(E_{\overline{\mathbb{Q}}},\mathbb{Q}_\ell)$, $p$ good | $\alpha^{-1}$, $\beta^{-1}$ | $\alpha$, $\beta$ |
| $\mathrm{ad}^0 V$, $p$ good | $\alpha/\beta$, 1, $\beta/\alpha$ | $\alpha/\beta$, 1, $\beta/\alpha$ |
| $(V_\ell E)^{I_q}$, type $\mathrm{I}_n$ | $qa_q$ | $q^{-1}a_q$ |
| $(\mathrm{ad}^0 V)^{I_q}$, type $\mathrm{I}_n$ or $\mathrm{I}_n^*$ | $q$ | $q^{-1}$ |
| $(\mathrm{ad}^0 V)^{I_q}$, types II, IV and duals, $q\geq 5$ | $\chi_{-3}(q)$ | $\chi_{-3}(q)$ |
| $(\mathrm{ad}^0 V)^{I_q}$, types III, III$^*$, $q\geq 5$ | $\chi_{-4}(q)$ | $\chi_{-4}(q)$ |

## B.4 Duals, adjoints, pullbacks and twists

Four different duals occur in the paper, and the star is used for pullbacks throughout the theory of modular curves. B-C4 gives each its own symbol; B-C5 fixes the signs of Tate twists and of the other sign-dependent constructions.

**Convention B.6** (B-C4: one symbol for each kind of dual)**.**

(a) *Linear dual*: the module $M^\vee=\mathrm{Hom}_R(M,R)$ ($R=\mathbb{Z}$, $\mathbb{Z}_\ell$, $\mathcal{O}$, $\mathbb{Q}_\ell$, a field); for a representation, the contragredient $V^\vee$ with $(g\lambda)(v)=\lambda(g^{-1}v)$. Never $M^*$, never $M'$.

(b) *Pontryagin dual* $X^*=\mathrm{Hom}_{\mathrm{cont}}(X,\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ (or $\mathrm{Hom}(X,\mathbb{Q}/\mathbb{Z})$ for a finite group), for $X$ finite, compact (finitely generated over $\mathbb{Z}_\ell$) or discrete $\ell$-primary. This is the paper's $^*$ in Lemma lem:dfg21.

(c) *Kummer (Cartier) dual* $M^D$: for a lattice or vector space, $M^D=\mathrm{Hom}(M,\mathbb{Z}_\ell(1))=M^\vee(1)$ (resp. $\mathrm{Hom}(V,\mathbb{Q}_\ell(1))$); for a finite $\ell$-primary module, $M^D=\mathrm{Hom}(M,\mu_{\ell^\infty})=M^*(1)$. Hence $V^D=V^\vee(1)$. For the discrete module $W=T\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$ the symbol $W^D$ is *notation* for $V^D/T^D=T^D\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell\simeq\mathrm{Hom}(T,\mu_{\ell^\infty})=T^*(1)$, a discrete module; it is *not* $\mathrm{Hom}(W,\mu_{\ell^\infty})$, which is the compact module $T^D$. So $W^D$ is the Kummer dual of $T$ and $T^D$ that of $W$, as in Chapter 17. In the paper (Section 4.2) and in Chapters 18, 20 and 21 the symbols $T^D,V^D,W^D$ are built from the *adjoint* lattice $A=\mathrm{ad}^0 T$: $T^D=\mathrm{Hom}(A,\mathbb{Z}_\ell(1))\simeq A(1)$, not $\mathrm{Hom}(T_\ell E,\mathbb{Z}_\ell(1))$.

(d) *Adjoint of an operator* $A$ for a pairing $\langle\ ,\ \rangle$: $A^\dagger$, defined by $\langle Ax,y\rangle=\langle x,A^\dagger y\rangle$ (Petersson product, cup product, abstract perfect pairings alike). Never $A^*$.

(e) *Pullback and pushforward* along a morphism or correspondence: $\varphi^*$, $\varphi_*$, $\mathcal{C}^*$, $\mathcal{C}_*$, and for $t\in\mathbb{T}$ the two actions $t^*$ (on $H^1$, $S_2$, $H^0(\Omega^1)$) and $t_*$ (on $H_1$, $J_0(N)$, $T_\ell J_0(N)$). A star after the name of a *map* is a pullback; a star after the name of a *module* is a Pontryagin dual. Every other star is one of the fixed exceptions: the Kodaira symbols II$^*$ etc., $q^*$, $\ell^*$, $\mathcal{H}^*$, $E_2^*$, the dual Selmer structure $\mathcal{L}^*$.

(f) *Transpose* of a matrix: $A^{\mathsf{T}}$ or $A^t$; of a correspondence: $\mathcal{C}^t$. *Dual isogeny*: $\hat{\varphi}$; dual abelian variety: $A^\vee$ (so $E^\vee=E$ via the principal polarisation).

*Reason.* The paper's * in Lemma lem:dfg21 is a Pontryagin dual, the literature on modular curves writes pullbacks with a star, and an adjoint written $A^*$ would collide with both. *Binds:* every chapter. *In the chapters:* Chapter 16 writes $(f,\varphi)^*$ for the map induced on cohomology by a map of pairs (a pullback, as (e) allows); Chapter 14 keeps the paper's Pontryagin duals inside its paper environments only.

| symbol | meaning | example |
|---|---|---|
| $M^\vee$ | linear dual, contragredient | $V_\ell E^\vee = H^1(E_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)$; $\mathrm{ad}^0(T^\vee) = \mathrm{ad}^0 T$ |
| $X^*$ | Pontryagin dual of a module | $H^1_\Sigma(\mathbb{Q}, W)^*$ in Lemma lem:dfg21 |
| $M^D$ | Kummer dual, $M^\vee(1)$, or $M^*(1)$ for finite $M$; $W^D := V^D/T^D$ | $T^D = \mathrm{Hom}(\mathrm{ad}^0 T, \mathbb{Z}_\ell(1))$, $W^D = \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ |
| $A^\dagger$ | adjoint for a pairing | $T_n^\dagger = w_N T_n w_N$ for the Petersson product |
| $\varphi^*$, $\varphi_*$ | pullback, pushforward | $\varphi_E^* \omega_E = c_E \cdot 2\pi i f(z)\, dz$; $t^*$, $t_*$ for $t \in \mathbb{T}$ |
| $M(n)$ | Tate twist | $V^D = A_f(1)$ |

**Convention B.7** (B-C5: Tate twists, Hodge-Tate weights and other signs)**.**

(a) The Tate module of $\mu$ is $\mathbb{Z}_\ell(1) = T_\ell \mu = \varprojlim_k \mu_{\ell^k}$ (transition maps $\zeta \mapsto \zeta^\ell$), on which $G_K$ acts through $\chi_\ell$; $\mathbb{Z}_\ell(n) = \mathbb{Z}_\ell(1)^{\otimes n}$, $\mathbb{Z}_\ell(-1) = \mathrm{Hom}(\mathbb{Z}_\ell(1), \mathbb{Z}_\ell)$, and $M(n) = M \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(n)$ for every $n \in \mathbb{Z}$; $\mathbb{Q}_\ell/\mathbb{Z}_\ell(1) = \mu_{\ell^\infty}$. So $\mathrm{Frob}_p$ acts on $\mathbb{Z}_\ell(1)$ by $p$, $\mathrm{Frob}_p^{-1}$ by $p^{-1}$, and $L_p(M(n), s) = L_p(M, s+n)$.

(b) The Weil pairing (Silverman's $e_m$ [185, III.8]) gives $\wedge^2 T_\ell E \simeq \mathbb{Z}_\ell(1)$, $\det \rho_{E,\ell} = \chi_\ell$, $T_\ell E \simeq (T_\ell E)^\vee(1)$, and $H^1(E, \mathbb{Z}_\ell) = (T_\ell E)^\vee \simeq T_\ell E(-1)$; for a curve $X$, $H^1_{\text{ét}}(X, \mathbb{Z}_\ell) \simeq T_\ell J(X)(-1)$.

(c) *Hodge-Tate weights*: $\mathbb{Q}_\ell(1)$ has Hodge-Tate weight $+1$; so $V_\ell E$ has weights $\{0,1\}$, $\mathrm{ad}^0 V$ has $\{-1,0,1\}$, $V^D = \mathrm{ad}^0 V(1)$ has $\{0,1,2\}$, $V_\ell E^\vee$ has $\{-1,0\}$. Sources with the opposite sign are flagged where cited.

(d) Other fixed signs: the cup product $\langle \alpha, \beta \rangle = \int_X \alpha \wedge \beta$ with the complex orientation (the paper needs only sign-invariant properties); the Kummer map $\delta(a)(\sigma) = \sigma(\alpha)/\alpha$, $\alpha^n = a$; the conjugation action $(\sigma \cdot c)(h) = \sigma\, c(\sigma^{-1} h \sigma)$ of $G/H$ on $H^n(H, M)$, the paper's (written there for a cocycle $z$ of $I$); the slash operator $(f|_k \gamma)(z) = (\det \gamma)^{k/2} (cz+d)^{-k} f(\gamma z)$; and $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$.

*Reason.* Every dimension formula of Bloch-Kato type depends on the sign of the Hodge-Tate weight, and the paper's cocycle computations depend on the Kummer and conjugation signs. *Binds:* Chapters 5, 7 to 9, 11, 12, 14 to 19 and 21. *In the chapters:* Chapters 12 and 13 compare the slash operator with Diamond and Shurman's $(\det \gamma)^{k-1}$, which agrees in weight 2. *Where proved:* (a) in Chapters 5 and 7; (b) in Chapters 8 and 9; (c) in Chapter 18; (d) in Chapters 8, 12, 16 and 17.

## B.5 Characters

The paper uses $\chi_\ell$ for the cyclotomic character and names its quadratic characters by their discriminants ($\chi_{-3}$, $\chi_{-11}$, $\chi_{q^*}$). B-C6 makes this a rule for the whole book: every quadratic character is named by its discriminant, and a letter subscript is reserved for the cyclotomic character.

**Convention B.8** (B-C6: characters)**.**

(a) $\chi_\ell\colon G_{\mathbb{Q}} \to \mathbb{Z}_\ell^\times$ is the $\ell$-adic *cyclotomic* character, $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$ on $\mu_{\ell^\infty}$; $\bar\chi_\ell\colon G_{\mathbb{Q}} \to \mathbb{F}_\ell^\times$ its reduction; $\omega_\ell\colon a \mapsto a \bmod \ell$ the Dirichlet character that corresponds to $\bar\chi_\ell$. A letter subscript ($\chi_\ell$, $\chi_p$) always means the cyclotomic character. At a numerical prime the cyclotomic character is written $\chi_3^{\mathrm{cyc}}$, $\chi_5^{\mathrm{cyc}}$; its reduction $\bar\chi_3$ (with the bar) is unambiguous, quadratic characters never carrying a bar.

(b) For a non-square $d \in \mathbb{Q}^\times$, $\chi_d$ is the *quadratic* character of $\mathbb{Q}(\sqrt{d})$, $\sigma(\sqrt{d}) = \chi_d(\sigma)\sqrt{d}$; as a Dirichlet character it is the Kronecker symbol $(\frac{D}{\cdot})$, $D$ the discriminant of $\mathbb{Q}(\sqrt{d})$, of conductor $|D|$. A *numerical* subscript always means $\chi_d$ with $d$ a fundamental discriminant written with its sign: $\chi_{-3}$, $\chi_{-4}$, $\chi_{-7}$, $\chi_{-11}$, $\chi_5$, $\chi_8$, $\chi_{13}$. The symbols $\chi_3$, $\chi_7$, $\chi_{11}$ are not used; for a discriminant that is not fundamental the Kronecker symbol is written out (so $(\frac{-12}{\cdot})$, never $\chi_{-12}$).

(c) For an odd prime $q$, the quadratic character of conductor $q$ is $\chi_{q^*}$, the character of $\mathbb{Q}(\sqrt{q^*})$, *never* $\chi_q$. Never write $\chi_\ell$ for a quadratic character: $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$ is the quadratic character of $\mathbb{Q}(\sqrt{\ell^*})$, and $\chi_{\ell^*} = \bar\chi_\ell$ if and only if $\ell = 3$, where $\bar\chi_3 = \chi_{-3}$.

(d) $\chi_S = \prod_{q\in S}\chi_{q^*}$ for a finite set $S$ of odd primes, of conductor $D = \prod_{q\in S} q$. $\varepsilon$ is the character of $G_{\mathbb{Q}_q}$ of order at most 2 in the Tate shape (B-C14): at a multiplicative prime it is unramified, with $\varepsilon(\mathrm{Frob}_q) = a_q(E)$; at an additive prime with $v_q(j) < 0$ (type $\mathrm{I}_n^*$ for odd $q$) it is a ramified quadratic character, so that $\rho$ is the multiplicative shape twisted by a ramified quadratic character, which $\mathrm{ad}^0\,\rho$ does not see. $\psi$ is the character of order 4 of $C_4 = \langle J\rangle$ with $\psi(J) = i \in \mathbb{F}_9^\times$ (the character of order 4 of Proposition prop:ns).

(e) A Dirichlet character $\chi$ is also the Galois character with $\chi(\mathrm{Frob}_p) = \chi(p)$ for $p \nmid \mathrm{cond}\,\chi$ (arithmetic Frobenius). The principal character modulo $n$ is $\chi_0$; the $p$-component of a Dirichlet character $\chi$ is $\chi^{(p)}$ (superscript, never $\chi_p$); the primitive character inducing $\chi$ is $\chi^{\mathrm{prim}}$.

*Reason.* Without (a) to (c), $\chi_3$ would be both the cyclotomic character at 3 and the quadratic character of conductor 3, and $\chi_{11}$ the character of $\mathbb{Q}(\sqrt{11})$ or of $\mathbb{Q}(\sqrt{-11})$. *Binds:* Chapters 2, 4 to 7, 9 to 11, 13, 15 and 17 to 22. *In the chapters:* Chapter 9 also writes $\chi_m^{\mathrm{cyc}}\colon G_K \to (\mathbb{Z}/m)^\times$ for the mod $m$ cyclotomic character; Chapter 5 writes $\chi_a$ for the Kummer character of $a$ (with values in $\mu_n$, B-C14 for the cocycle $\kappa$). *Where proved:* $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$ in Chapter 5; primitivity of $\chi_D$ for fundamental $D$ in Chapter 2; $\chi(\mathrm{Frob}_p) = \chi(p)$ in Chapter 6.

| character | field cut out | conductor | value at $p$ (arithmetic $\mathrm{Frob}_p$) |
|---|---|---|---|
| $\chi_\ell$ | $\mathbb{Q}(\mu_{\ell^\infty})$ | $\ell$-adic | $\chi_\ell(\mathrm{Frob}_p) = p$, $p \neq \ell$ |
| $\bar\chi_\ell$, $\omega_\ell$ | $\mathbb{Q}(\zeta_\ell)$ | $\ell$ | $p \bmod \ell$ |
| $\chi_{-3} = \bar\chi_3$ | $\mathbb{Q}(\sqrt{-3})$ | 3 | 1 if $p \equiv 1$, $-1$ if $p \equiv 2 \pmod 3$ |
| $\chi_{-4}$ | $\mathbb{Q}(i)$ | 4 | 1 if $p \equiv 1$, $-1$ if $p \equiv 3 \pmod 4$ |
| $\chi_{q^*}$ ($q$ odd) | $\mathbb{Q}(\sqrt{q^*})$ | $q$ | $\left(\frac{p}{q}\right)$ |
| $\chi_{\ell^*} = \bar\chi_\ell^{(\ell-1)/2}$ | $\mathbb{Q}(\sqrt{\ell^*}) \subset \mathbb{Q}(\zeta_\ell)$ | $\ell$ | $\left(\frac{p}{\ell}\right)$ |
| $\chi_S$ | $\mathbb{Q}(\sqrt{\prod_{q\in S} q^*})$ | $D = \prod_{q\in S} q$ | $\prod_{q\in S}\left(\frac{p}{q}\right)$ |
| $\varepsilon$ (local at $q$) | order $\le 2$; unramified iff $q$ is multiplicative | 1 or $q$ | $\varepsilon(\mathrm{Frob}_q) = a_q(E) = \pm 1$ at type $\mathrm{I}_n$ |
| $\psi$ (on $C_4$) | not a Galois character | | $\psi(J) = i \in \mathbb{F}_9^\times$ |

The value $\chi_{q^*}(p) = \left(\frac{q^*}{p}\right) = \left(\frac{p}{q}\right)$ for every prime $p \neq q$ (the Legendre symbol $\left(\frac{2}{q}\right)$ at $p = 2$) is quadratic reciprocity with its supplements, in the form proved in Chapter 2.

## B.6 $L$-functions, the Petersson product and the Manin constant

The adjoint and symmetric square $L$-functions differ by a shift, and the Petersson product and the Manin constant are normalised differently in different sources; the paper's Remark rem:degree, its Lemma lem:hecke and the numbers of its Section sec:data depend on these choices.

**Convention B.9** (B-C7: normalisations of the adjoint and symmetric square $L$-functions)**.** Let $f$ be a newform of weight 2 on $\Gamma_0(N)$, $V = V_\lambda(f)$ (for $f = f_E$, $V = V_\ell E$), $A_f = \mathrm{ad}^0 V$. Then

$$L(\mathrm{ad}^0 f, s) = L(A_f, s) = \prod_p L_p(A_f, s), \qquad L(\mathrm{Sym}^2 f, s) = \prod_p L_p(\mathrm{Sym}^2 V^\vee, s) = L(\mathrm{ad}^0 f, s-1),$$

with $L_p$ as in B-C2(d) ($p \neq \ell$; at $p = \ell$ through $\mathrm{D}_{\mathrm{cris}}$, or with $V_{\ell'}$ for $\ell' \neq p$). So $L_p(\mathrm{Sym}^2 f, s) = L_p(A_f, s-1)$, and the special values are $L(\mathrm{Sym}^2 f, 2) = L(\mathrm{ad}^0 f, 1)$. $P_q(X) \in \mathbb{Z}[X]$ is defined by $L_q(\mathrm{Sym}^2 f, s) = P_q(q^{-s})^{-1}$; hence $P_q(q^{-2}) = L_q(A_f, 1)^{-1}$. The *naive* factor is $L_p^{\mathrm{nv}}(A_f, s) = L_p(A_f, s)$ if $\delta_p = \dim V^{I_p} > 0$ and 1 if $\delta_p = 0$ (equivalently $L_p^{\mathrm{nv}}(A_f, s)^{-1} = 1 - a_p(f)^2 p^{-1-s}$ for $p \mid N$); $L^{\mathrm{nv}}$ is their product. The normalisations of the sources:

| object | source | equals | functional equation | point used |
|---|---|---|---|---|
| $L(\mathrm{ad}^0 f, s)$, $L(\mathrm{ad}^0 M_f, s)$ | the paper; [49, 1.7.2] | $L(A_f, s)$ | $s \leftrightarrow 1-s$ | $s = 1$ |
| $L(s, \pi_f, \mathrm{Ad})$ | Gelbart-Jacquet [73] (unitary) | $L(A_f, s)$ | $s \leftrightarrow 1-s$ | $s = 1$ |
| $L(\mathrm{Sym}^2 f, s)$ | the paper; PARI's `lfunsympow(E,2)`; [204] | $L(A_f, s-1)$ | $s \leftrightarrow 3-s$ | $s = 2$ |
| $\zeta^{(N)}(2s-2)\sum a_{n^2} n^{-s}$ | Shimura [182] | $L^{\mathrm{nv}}(A_f, s-1)$ | via $L(\mathrm{Sym}^2 f, s)$ | $s = 2$ |
| $D(f,s) = \sum a_n^2 n^{-s}$ | Rankin [155] | $\zeta^{(N)}(s-1)L^{\mathrm{nv}}(A_f, s-1)/\zeta^{(N)}(2s-2)$ | via $E(z,s)$ | residue at 2 |

The Rankin series is never called $L(\mathrm{Sym}^2 f, s)$. $e_\ell(E,q) = v_\ell(L_q(A_f,1)^{-1}/L_q^{\mathrm{nv}}(A_f,1)^{-1}) = v_\ell(P_q(q^{-2})/(1-a_q^2 q^{-2}))$ (Definition def:terms).

*Reason.* The two normalisations are both in the paper ($L(\mathrm{ad}^0 f, 1)$ in its theorems, $L(\mathrm{Sym}^2 f, 2)$ in its code), and the Rankin series has been mistaken for the symmetric square. *Binds:* Chapters 4, 7, 13, 15 and 18 to 22. *Where proved:* every identity of the table is proved in Chapter 19, which also compares the sources in more detail.

**Convention B.10** (B-C8: the Petersson product)**.** For $f, g \in S_k(\Gamma_0(N))$,

$$\langle f, g \rangle = \int_{\Gamma_0(N)\backslash\mathcal{H}} f(z)\overline{g(z)}\, y^k \, \frac{dx\, dy}{y^2},$$

*unnormalised* (not divided by the index or the volume), linear in the first variable; in weight 2, $\langle f, g\rangle = \frac{i}{2}\int_{X_0(N)} \omega_f \wedge \overline{\omega_g}$ with $\omega_f = f(z)\, dz$. PARI's `mfpetersson` returns $\langle f, g\rangle/\mu(N)$, $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = N\prod_{p\mid N}(1+p^{-1})$; other books divide by $\mathrm{vol}(\Gamma_0(N)\backslash\mathcal{H}) = \pi\mu(N)/3$. A numerical value is never quoted without its normalisation.

*Reason.* The formula of Zagier and Watkins (B-C9) holds with the unnormalised product only. *Binds:* Chapters 12 to 14, 19, 21 and 22. *Where proved:* Chapter 12, which checks the normalisation of `mfpetersson` numerically.

**Convention B.11** (B-C9: the Manin constant and the formula of Zagier and Watkins)**.** For a modular parametrisation $\varphi\colon X_0(N) \to E$ (sending $\infty$ to $O$) and the Néron differential $\omega_E$ of a global minimal model, $\varphi^*\omega_E = c_\varphi \cdot 2\pi i\, f(z)\, dz = c_\varphi\, f(q)\, dq/q$; the *Manin constant* of the optimal curve is $c_E = c_{\varphi_E}$. The sign of $\omega_E$ is chosen so that $c_E > 0$; Manin's conjecture reads $c_E = 1$. Then, with the product of B-C8,

$$\deg\varphi \cdot \mathrm{vol}(\Lambda_E) = 4\pi^2 c_\varphi^2\, \langle f, f\rangle, \qquad \mathrm{vol}(\Lambda_E) = \tfrac{i}{2}\int_{E(\mathbb{C})} \omega_E \wedge \bar\omega_E,$$

with $c^2$ *in the numerator* of $\deg\varphi_E = 4\pi^2 c_E^2 \langle f, f\rangle / \mathrm{vol}(\Lambda_E)$. (Check: the parametrisation of degree 5 of `11a3`, through `11a1`, has $c_\varphi = 5$, since $\mathrm{vol}(\Lambda_{\texttt{11a3}}) = 5\,\mathrm{vol}(\Lambda_{\texttt{11a1}})$ and `11a1` has $\deg\varphi_E = 1$, $c_E = 1$.) The Manin constant is never written as a bare $c$.

*Reason.* The letter $c$ already names the class $c(u)$ of Lemma lem:tamfactor (B-C14), cocycles and complex conjugation, and $c_q$ is a Tamagawa number. Only $c_E^2$ enters a formula, so the sign normalisation changes no statement. *Binds:* Chapters 9, 12, 14, 18, 19, 21 and 22. *In the chapters:* Chapter 9 writes $c_\varphi$ also for the constant of an isogeny, $\varphi^*\omega' = c_\varphi\omega$; Chapter 8 writes $m_0$ for the order of a component group so as not to use $c$. *Where proved:* the formula is in Chapter 14.

*Warning* B.12 (two normalisations that are easy to confuse). (1) PARI's `mfpetersson` is the book's $\langle f, f\rangle$ divided by $\mu(N)$; the formula of B-C9 needs the book's product, so a value from `mfpetersson` must be multiplied by $\mu(N)$ first. (2) The Rankin series $\sum a_n^2 n^{-s}$ is not $L(\mathrm{Sym}^2 f, s)$: it differs by $\zeta^{(N)}(s-1)/\zeta^{(N)}(2s-2)$ and by the naive factors at $p^2 \mid N$.

## B.7 Local data at a bad prime

At a bad prime $q$ the paper uses the Kodaira type, the component group, the Tamagawa number, the PARI code of the type, and, when $v_q(j) < 0$, the Tate parameter. B-C11 and B-C14 fix their names.

**Convention B.13** (B-C11: Kodaira symbols, component groups, PARI codes)**.**

(a) Kodaira symbols are typeset $\mathrm{I}_0$, $\mathrm{I}_n$, II, III, IV, $\mathrm{I}_0^*$, $\mathrm{I}_n^*$, IV$^*$, III$^*$, II$^*$; "type $\mathrm{I}_n$" assumes $n \geq 1$ unless $\mathrm{I}_0$ is meant; at $q = 2$ the index $m$ of a symbol $\mathrm{I}_m^*$ need not be $-v_2(j)$ and the reduction may be potentially good.

(b) $\Phi_q$ is the component group of the special fibre of the Néron model of $E$ at $q$, a finite étale group scheme over $\mathbb{F}_q$; it is always written *with an argument*: $\Phi_q(\mathbb{F}_q)$, $\Phi_q(\overline{\mathbb{F}}_q)$. The *geometric component group* $\Phi_q(\overline{\mathbb{F}}_q)$ is $\Phi_q^{\mathrm{geom}}$ (the paper's symbol), and $\Phi_q^{\mathrm{geom}}[\ell^k]$, $\#\Phi_q^{\mathrm{geom}}$ are written with it; the *Tamagawa number* is $c_q = \#\Phi_q(\mathbb{F}_q) = [E(\mathbb{Q}_q) : E_0(\mathbb{Q}_q)]$, which divides $\#\Phi_q^{\mathrm{geom}}$. A bare "$\Phi_q$" for a group is not used. The group measuring the defect of semistability is $G_{\mathrm{ss}}$, never $\Phi$.

(c) PARI's Kodaira codes (`elllocalred(E,p)[2]`, the paper's `K[i][2]`) are given in the table below; the formula $\pm(n+4)$ holds *for* $n \geq 1$ *only*. Codes appear only in tables, code listings and paper environments.

(d) The conductor exponent is $f_p = f_p(E)$, the Swan conductor $\mathrm{sw}_p(E)$; the letter $\delta_p$ is reserved for $\dim V^{I_p}$.

*Reason.* The paper's statements are about $\#\Phi_q^{\mathrm{geom}}$, not about $c_q$ (`121c1` at 11 has $\#\Phi^{\mathrm{geom}} = 3$ and $c = 1$), and $\Phi$ is also the geometric Frobenius and the cyclotomic polynomial. *Binds:* Chapters 10, 11, 14, 15, 18, 19, 21 and 22. *In the chapters:* over a general local field Chapter 10 writes $\Phi_E(k)$ and $\Phi^{\mathrm{geom}} = \Phi_E(\bar{k})$. *Where proved:* Chapter 10, which also gives the order of $\Phi_q^{\mathrm{geom}}$ for every type.

| type | $\mathrm{I}_0$ | $\mathrm{I}_n$ $(n \geq 1)$ | II | III | IV | $\mathrm{I}_0^*$ | $\mathrm{I}_n^*$ $(n \geq 1)$ | $\mathrm{IV}^*$ | $\mathrm{III}^*$ | $\mathrm{II}^*$ |
|---|---|---|---|---|---|---|---|---|---|---|
| PARI code | 1 | $n+4$ | 2 | 3 | 4 | $-1$ | $-(n+4)$ | $-4$ | $-3$ | $-2$ |
| $\#\Phi_q^{\mathrm{geom}}$ | 1 | $n$ | 1 | 2 | 3 | 4 | 4 | 3 | 2 | 1 |

**Convention B.14** (B-C14: the Tate data at a prime with $v_q(j) < 0$)**.** $q \neq \ell$; $q_E \in \mathbb{Q}_q^\times$ the Tate parameter (never $u$); $\nu = v_q(q_E) = -v_q(j_E)$, equal to $n = n_q = v_q(\Delta_E)$ at a multiplicative prime; $u = q_E q^{-\nu} \in \mathbb{Z}_q^\times$ its unit part, $\bar{u} \in \mathbb{F}_q^\times$ its reduction; $c(u)$ the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$ (as in the paper's Lemma lem:tamfactor), with $v_\ell(c(u))$ read as in B-C10. $\kappa = \kappa_{q_E}$ is the Kummer cocycle of $q_E$, $\kappa = \nu\kappa_q + \kappa_u$; $\varepsilon$ is a character of order at most 2, unramified with $\varepsilon(\mathrm{Frob}_q) = a_q(E)$ at a multiplicative prime and ramified quadratic at an additive prime (type $\mathrm{I}_n^*$ for odd $q$; B-C6(d)). At a multiplicative prime, $t$ is a topological generator of the pro-$\ell$ tame inertia quotient, acting by $te_1 = e_1$, $te_2 = e_2 + n'e_1$ with $v_\ell(n') = v_\ell(n)$; at type $\mathrm{I}_n^*$ the same holds for $\mathrm{Ad}\,\rho$, which does not see $\varepsilon$. In a basis $e_1, e_2$ of $T_\ell E$:

$$\rho(g) = \begin{pmatrix} \chi_\ell(g)\varepsilon(g) & \varepsilon(g)\kappa(g) \\ 0 & \varepsilon(g) \end{pmatrix},$$

*with corner* $\varepsilon\kappa$, not $\kappa$; $c' = \varepsilon(\mathrm{Frob})\kappa(\mathrm{Frob})$ for the arithmetic Frobenius, well defined modulo $\ell^{\min(v_\ell(n), v_\ell(q-1))}$.

*Reason.* Only $\varepsilon\kappa$ satisfies the cocycle relation for these matrices when $\varepsilon \neq 1$; the letter $u$ is the unit part, so the Tate parameter needs its own name. *Binds:* Chapters 2, 7, 10, 11, 16 to 18 and 20 to 22. *In the chapters:* Chapter 16 writes $w = v_\ell(n')$ in its model computation (the letter $v$ being the paper's $v_\ell(q-1)$) and $\rho_M$ for the action of $G$ on a module $M$ (so $\rho_M = \mathrm{Ad}\,\rho$ on $M = \mathrm{ad}^0 T$). *Where proved:* the Tate shape in Chapter 11; $\kappa|_I = n\,t_\ell$ in Chapter 7; the splitting of $\kappa$ and the indeterminacy of $c'$ in Chapter 17; the computation of Lemma lem:tamfactor in Chapters 16 and 18.

## B.8 The adjoint, Hecke algebras, twists and depletion

The paper's $A_f$ is the adjoint representation, while Shimura's notation uses $A_f$ for an abelian variety; the paper uses $I$ for two different ideals; parenthesised superscripts are used for twists and, in some sources, for depletions. B-C12 and B-C13 fix one reading.

**Convention B.15** (B-C12: the adjoint $A_f$ and the abelian variety $J_f$)**.**

(a) For a representation $V$ (or lattice $T$), $\mathrm{End}(V) = V \otimes V^\vee$ with $g \cdot x = \rho(g)x\rho(g)^{-1}$ is written $\mathrm{End}(V)$ or $\mathrm{Ad}\,V$; the *trace-zero adjoint* is $\mathrm{ad}^0 V$ (the paper's symbol; the superscript 0 is part of it); $\mathrm{Sym}^2 V$ is the symmetric square. For $\dim V = 2$ the basis of $\mathrm{ad}^0 V$ is $e = e_1 \otimes e_2^\vee$, $h = e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee$, $f = e_2 \otimes e_1^\vee$. $\mathrm{ad}^0 V \simeq \mathrm{Sym}^2 V \otimes \det^{-1}$ and, for $\ell$ odd, the trace form makes $\mathrm{ad}^0 T$ self-dual, $\mathrm{ad}^0 T \simeq (\mathrm{ad}^0 T)^\vee$.

(b) The paper's objects: $V = V_\ell E$, $T = T_\ell E$, $\rho = \rho_{E,\ell}$, $\bar\rho$; $A = \mathrm{ad}^0 T$, $A_f = \mathrm{ad}^0 V = A \otimes \mathbb{Q}_\ell$, $W = A \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$, $T^D = \mathrm{Hom}(A, \mathbb{Z}_\ell(1)) \simeq A(1)$, $V^D = T^D \otimes \mathbb{Q}_\ell$, $W^D = V^D/T^D = \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. $\mathrm{ad}^0$ is insensitive to duals and twists: $\mathrm{ad}^0(T^\vee) \simeq \mathrm{ad}^0 T = \mathrm{ad}^0(T(n)) = \mathrm{ad}^0(T \otimes \chi)$, so $\mathrm{ad}^0 M_{f,\lambda} = \mathrm{ad}^0 T_\ell E$ although $M_{f,\lambda} \simeq T_\ell E(-1)$.

(c) *$A_f$ is the adjoint, never an abelian variety.* The quotient $J_0(N)/I_f J_0(N)$ attached to a newform $f$ (Shimura's $A_f$) is written $J_f$, with the quotient map $\pi_f \colon J_0(N) \to J_f$; for rational $f$, $J_f = E_f$ is the optimal curve of the class.

*Reason.* The paper's notation is the book's, so the abelian variety is renamed. *Binds:* Chapters 4, 11, 14, 15 and 18 to 21. *In the chapters:* Chapter 4 names the isomorphism $\Xi \colon \mathrm{Sym}^2 V \to \mathrm{ad}^0 V$ (up to det). *Where proved:* (a) in Chapter 4; (c) in Chapter 14.

**Convention B.16** (B-C13: Hecke algebras, eigencharacters, twists and depletion)**.**

(a) The *full* Hecke algebra is $\mathbb{T} = \mathbb{T}_N = \mathbb{Z}[T_n : n \geq 1]$, the subring of $\mathrm{End}_{\mathbb{C}} S_2(\Gamma_0(N))$ generated by all $T_n$ (the operators $U_p = T_p$ for $p \mid N$ included); $\mathbb{T}^{(N)} = \mathbb{Z}[T_n : (n, N) = 1]$ the anemic one; $\mathbb{T}_\ell = \mathbb{T} \otimes \mathbb{Z}_\ell$; for a maximal ideal $\mathfrak{m} \ni \ell$, $\mathbb{T}_\mathfrak{m} = (\mathbb{T} \otimes \mathbb{Z}_\ell)_\mathfrak{m}$, a direct factor of $\mathbb{T}_\ell$ cut out by an idempotent $e_\mathfrak{m}$; the algebraic localisation is $\mathbb{T}_{(\mathfrak{m})}$.

(b) $\phi_f \colon \mathbb{T} \to \mathcal{O}_f$ (upright $\phi$) is the *eigencharacter* of $f$, $T_n \mapsto a_n(f)$; a general system of eigenvalues is also written $\phi$. $\varphi$ (curly) is reserved for *maps of curves*: $\varphi_E \colon X_0(N) \to E$, isogenies $\varphi$. The one fixed exception is Euler's function $\varphi(n) = \#(\mathbb{Z}/n)^\times$, always with an integer argument.

(c) The paper's two uses of $I$: in Lemma lem:crit $I = \ker \phi_f$ and $J = \mathrm{Ann}_\mathbb{T}(f^\perp)$; in Lemma lem:ddt417 $\wp = \ker \pi$ and $I = \mathrm{Ann}_\mathbb{T}(\wp)$. So the $J$ of lem:crit is the $I$ of lem:ddt417 (for $\pi = \phi_f$). The book keeps each lemma's letters inside that lemma and writes $\ker \phi_f$ (or $\wp$, or $I_f$) and $\mathrm{Ann}_\mathbb{T}(\wp)$ in its own text.

(d) *Twists*: for a Dirichlet character $\chi$ of conductor $r$, $f \otimes \chi = \sum \chi(n) a_n(f) q^n$ is the *naive twist* (the paper's $f \otimes \chi$ and its twist map $\iota$); $f^\chi$ is the *newform twist* ($\rho_{f^\chi} \simeq \rho_f \otimes \chi$); then $f \otimes \chi = (f^\chi)^{\Sigma(r)}$, the depletion at the set $\Sigma(r)$ of primes dividing $r$. $E^{(d)} = E \otimes \chi_d$ is the quadratic twist of a curve by $\mathbb{Q}(\sqrt{d})$; $E^{(q)} := E^{(q^*)}$ for odd $q$; $E_S = E \otimes \chi_S$.

(e) *Depletion*: $f^\Sigma = \sum_{(n,\Sigma)=1} a_n q^n$ for a set $\Sigma$ of primes (the paper's notation), $f^{\{p\}}$ for one prime, $f^S$ for the set $S$. A parenthesised superscript is reserved for *twists* of curves ($E^{(q)}$), for *removing Euler factors* ($\zeta^{(N)}$, $\mathbb{T}^{(N)}$), for the *level* of a Hecke operator when two levels are in play ($T_p^{(M)}$, $U_p^{(Mp)}$), and for the components $\chi^{(p)}$ of a Dirichlet character; it never denotes depletion. The Frobenius twist of a curve is $C^{[r]}$.

(f) $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$, $w_N$ the involution of $X_0(N)$ and the operator $f \mapsto f|_k W_N$; $\varepsilon_N(f) = \pm 1$ its eigenvalue on a newform. $\hat\delta_!$ (always with the subscript) is the pairing $\langle x, w_N y\rangle$ on $H^1(X_0(N), \mathbb{Z}_\ell)$. For odd $\ell$ it is DFG04's $\hat\delta_!$ in weight 2, trivial character, up to an $\ell$-adic unit: on classes pulled back from $X_0(N)$ theirs is $\frac{u}{2}\langle x, w_N y\rangle$ with $u$ an $\ell$-adic unit (the paragraph after the paper's Proposition prop:integral). $M_{f,\lambda} = H^1(X_0(N), \mathbb{Z}_\ell)[\ker \phi_f]$ (with $\lambda = \ell$ for rational $f$); $\eta_{f,\lambda}^\Sigma$ its ideal of values on $M_{f^\Sigma,\lambda}$.

*Reason.* Depletion written $h^{(p)}$ would collide with the twist $E^{(q)}$; a curly $\varphi$ for eigenvalue systems or an upright $\phi$ for the modular parametrisation would collide with the paper's $\phi_f$ and $\varphi_E$. *Binds:*

Chapters 3, 12 to 15 and 20 to 22. *In the chapters:* Chapter 13 writes $\mathcal{U}_p$, $V_d$ for the operators $\sum a_n q^n \mapsto \sum a_{pn} q^n$, $\sum a_n q^n \mapsto \sum a_n q^{dn}$ on $q$-series, and $\iota_S$ for the twist map; Chapter 14 writes $I_f = \ker \phi_f$. *Where proved:* (d) and (e) in Chapters 13 and 15; (f) in Chapters 12 and 14.

| | ker of the eigencharacter | its annihilator | congruence module |
|---|---|---|---|
| paper, Lemma lem:crit | $I = \ker \phi_f$ | $J = \mathrm{Ann}_{\mathbb{T}}(f^{\perp}) = \mathrm{Ann}_{\mathbb{T}}(I)$ | $\mathbb{T}/(I+J)$ |
| paper, Lemma lem:ddt417 | $\wp = \ker \pi$ | $I = \mathrm{Ann}_{\mathbb{T}}(\wp)$ | $\mathbb{T}/(\wp + I) \simeq \mathcal{O}/\eta$ |
| the book's own text | $\wp$, $\ker \phi_f$, $I_f$ | $\mathrm{Ann}_{\mathbb{T}}(\wp)$ | $\mathcal{O}/\eta$ |

## B.9 Index of symbols

The thirteen tables below list the symbols of the book and of the paper. "Defined in" names the chapter that defines the symbol and, where there is one, a second chapter that specialises it or uses it in earnest. Earlier chapters often mention a symbol before its definition, in their cards or in their paper environments, with a forward reference; the index names the defining chapter. "See also" gives the convention that governs the symbol and the other meanings of the same letter (Appendix B.10). Labels of the paper (lem:tamfactor, def:terms, ...) are the paper's own.

### B.9.1 Numbers, valuations, rings and modules

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $p$, $q$, $\ell$ | primes; $q$ a bad prime, $\ell$ the coefficient prime | Chapter 2 | B-C1; $q$ also a field size, $e^{2\pi i z}$, $\mathfrak{q}$ |
| $q^*$, $\ell^*$ | $(-1)^{(q-1)/2}q$, $(-1)^{(\ell-1)/2}\ell$ | Chapters 2 and 5 | star exception, B-C4(e) |
| $\mathfrak{q}$ | size of a finite field, not necessarily prime | Chapter 9 | B-C1 |
| $N\mathfrak{p}$ | size of the residue field of a prime $\mathfrak{p}$ | Chapter 6 | |
| $\mathbb{F}_q$, $\mathbb{F}_q^\times$, $(\mathbb{F}_q^\times)^{\ell^k}$ | finite field, its units, its $\ell^k$-th powers; $[\mathbb{F}_q^\times : (\mathbb{F}_q^\times)^{\ell^k}] = \ell^{\min(k, v_\ell(q-1))}$ | Chapter 2 | |
| $\mathbb{Z}/n$, $(\mathbb{Z}/n)^\times$, $R^\times$ | integers modulo $n$, units; unit group of a ring | Chapter 2 | |
| $\mu_n$, $\mu_n(K)$, $\mu_{\ell^\infty}$ | roots of unity | Chapter 2 | $\mu(N)$, Appendix B.10 |
| $\zeta_n$, $\zeta_3$, $\sqrt{-3}$ | $e^{2\pi i/n}$; $\zeta_3 = (-1+\sqrt{-3})/2$, $\sqrt{-3} := 1 + 2\zeta_3 = i\sqrt{3}$ | Chapters 2 and 6 | never $\omega$ or $\rho$ for $\zeta_3$ |
| $\Phi_n(x)$ | $n$-th cyclotomic polynomial | Chapter 2 | B-C2(b), B-C11(b) |
| $\theta_d$ | $(1+\sqrt{d})/2$ for $d \equiv 1 \pmod 4$ | Chapter 6 | $\theta_m$ (tame characters) |
| $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_3]$, $\mathbb{Z}_3[\zeta_3]$ | Gaussian and Eisenstein integers; a discrete valuation ring | Chapters 2 and 6 | |
| $\pi = 1 - \zeta_3$ | uniformiser of $\mathbb{Z}_3[\zeta_3]$ (local name of Chapter 6; the paper writes $1-\zeta_3$) | Chapter 6 | B-C15(a) |
| $v_p$, $v_\ell$, $\lvert x\rvert_p$ | $p$-adic valuation, $v_\ell(\ell) = 1$; $\lvert x\rvert_p = p^{-v_p(x)}$ | Chapter 2 | B-C10 |
| $v_\ell(c)$, $c \in \mathbb{Z}/\ell^m$ | $\min(v_\ell(\tilde{c}), m)$ | Chapter 2 | B-C10 |

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $\left(\frac{a}{p}\right)$, $\left(\frac{a}{n}\right)$ | Legendre, Jacobi and Kronecker symbols | Chapter 2 | |
| $\mathcal{O}$, $\varpi$, $k$, $K$, $v$ | discrete valuation ring, uniformiser, residue field, fraction field, valuation | Chapters 3 and 4 | uniformisers are $\varpi$, B-C15(a) |
| $\mathcal{O}_K$, $\bar{\mathbb{Z}}$ | ring of integers of a number field $K$; all algebraic integers | Chapter 6 | |
| $K_f$, $\mathcal{O}_f$ | coefficient field of a newform $f$, its ring of integers | Chapters 13 and 15 | |
| $\lambda$, $v_\lambda$, $k_\lambda$ | prime of $\mathcal{O}_f$ above $\ell$ ($\lambda = \ell$ for rational $f$), its valuation, residue field | Chapters 6 and 15 | eigenvalues $\lambda$ (local) |
| $E_\lambda$ | finite extension of $\mathbb{Q}_\ell$ (coefficient field) | Chapter 5 | $E$ an elliptic curve |
| $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$, $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ | $\ell$-adic integers, $\ell$-adic numbers, the divisible group | Chapter 3 | |
| length, $\mathrm{length}_R$, $\#X$ | length (length $=$ $\mathrm{length}_{\mathbb{Z}_\ell}$); order, $\#X = \ell^{\mathrm{length}\, X}$ | Chapter 3 | B-C10 |
| $\mathrm{Fitt}_R(M)$ | Fitting ideal | Chapter 3 | B-C10 |
| $\mathrm{Ann}_R(M)$, $M[\mathfrak{a}]$, $M[\ell^\infty]$, $M_{\mathrm{tor}}$ | annihilator, $\mathfrak{a}$-torsion, $\ell$-primary part, torsion | Chapter 3 | Mordell-Weil $E(\mathbb{Q})_{\mathrm{tors}}$ |
| saturated | $L \subseteq M$ with $M/L$ torsion-free | Chapter 3 | |
| $S^\perp$, ${}^\perp S$ | right and left orthogonal for a pairing | Chapter 3 | $f^\perp$ |
| $G$ (Gram matrix), $\Delta$ (Gram determinant) | $(\langle e_i, f_j\rangle)$; $\det G$, in Lemma lem:ddt417 of $L[\wp]$ | Chapter 3 | $\Delta$ discriminant, Appendix B.10 |
| $L$, $L[\wp]$, $L[I]$ | lattice with a perfect pairing; its $\wp$- and $I$-torsion | Chapter 3 | $L$-functions |
| $\wp$, $\eta$, $\mathcal{O}/\eta$ | $\ker(\pi\colon \mathbb{T} \to \mathcal{O})$; congruence ideal $\pi(\mathrm{Ann}_{\mathbb{T}}\, \wp)$; congruence module | Chapters 3 and 20 | B-C13(c); Dedekind $\eta(z)$ |
| $\mathbb{T}_{\mathfrak{m}}$, $e_{\mathfrak{m}}$, $L_{\mathfrak{m}}$, $\mathbb{T}_{(\mathfrak{m})}$ | factor $(\mathbb{T} \otimes \mathbb{Z}_\ell)_{\mathfrak{m}}$, its idempotent, $e_{\mathfrak{m}}(L \otimes \mathbb{Z}_\ell)$; algebraic localisation | Chapter 3 | B-C13(a) |
| $V(\phi)$ | generalised eigenspace of a system $\phi$ | Chapters 3 and 13 | B-C13(b) |
| Gorenstein | $\mathrm{Hom}_{\mathcal{O}}(\mathbb{T}, \mathcal{O})$ free of rank one over $\mathbb{T}$ | Chapter 3 | |
| $k[x]/(x^2)$ | dual numbers (not $k[\varepsilon]$, except locally in Chapter 20) | Chapter 3 | B-C15(k) |

## B.9.2 Groups and representations

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $C_n$, $D_n$, $Q_8$, $SD_{16}$, $V_4$, $S_n$, $A_n$ | cyclic, dihedral of order $2n$, quaternion, semidihedral, Klein, symmetric, alternating | Chapter 1 | $D_4$ has order 8 |
| $\mathrm{GL}_2$, $\mathrm{SL}_2$, $\mathrm{PGL}_2$, $\mathrm{PSL}_2$ | matrix groups | Chapter 1 | |
| B, U | Borel (upper triangular) and unipotent subgroups | Chapter 1 | |
| $C_s$, $N_s$, $C_{ns}$, $N_{ns}$ | split and nonsplit Cartan subgroups, their normalisers (standard models) | Chapter 1 | $N_s(\mathbb{F}_3) \subset N_{ns}(\mathbb{F}_3)$ |
| $J$, $u$, $w$, $a$, $\sigma$ (or $h$) | $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $u = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $a = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$, $\mathrm{diag}(1,-1)$ in $\mathrm{GL}_2(\mathbb{F}_3)$ | Chapters 1 and 4 | $h$ in Chapters 4 and 11; $w$, $J$, Appendix B.10 |

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $\mathbb{F}_9 = \mathbb{F}_3[i]$, $m_x$ | the field of 9 elements; multiplication by $x$, so $J = m_{-i}$ | Chapters 1 and 4 | |
| $\varepsilon$ (in Chapter 1) | the non-square of $\mathbb{F}_q$ in the standard models ($\varepsilon = -1$ at $q = 3$) | Chapter 1 | B-C15(k) |
| $\mathbb{P}^1(F)$, $[v]$ | lines of $F^2$ | Chapter 1 | |
| 3Ns, 3Nn, 3Cs, 3Cs.1.1, 3B, 3B.1.1, 3B.1.2 | Sutherland's labels of mod 3 images | Chapters 1 and 11 | B-C16; `\Ns` prints 3Ns |
| $\rho$, $\rho(g)$, $G$-map | a representation; an intertwiner | Chapter 4 | $\rho_{E,\ell}$ |
| $V^\vee$, $E_{ij}$ | contragredient; matrix units $e_i \otimes e_j^\vee$ | Chapter 4 | B-C4(a) |
| $\mathrm{Hom}_G(V, W)$ | $G$-maps, $= (V^\vee \otimes W)^G$ | Chapter 4 | |
| $\mathrm{tr}_V$ | trace function (the character in the sense of Riley, Hobson and Bence) | Chapter 4 | never $\chi_V$ |
| $R(\chi)$, $M(\chi)$, $\rho \otimes \chi$ | twist by a character | Chapter 4 | |
| $\rho^{ss}$, $\bar{\rho}^{ss}$ | semisimplification | Chapter 4 | |
| $\mathrm{Ind}_H^G$, $\mathrm{Res}_H^G$ | induction, restriction | Chapter 4 | Res in cohomology |
| $\psi$ | character of order 4 of $C_4$, $\psi(J) = i$ | Chapter 4 | B-C15(m) |
| $\mathrm{End}\, V$, $\mathrm{Ad}\, V$, $\mathrm{ad}^0 V$, $\mathrm{ad}^0 T$ | full adjoint; trace-zero adjoint | Chapter 4 | B-C12 |
| $\mathrm{Sym}^2 V$, $\Xi$ | symmetric square; the isomorphism $\mathrm{Sym}^2 V \otimes \det^{-1} \to \mathrm{ad}^0 V$ | Chapter 4 | |
| $e$, $h$, $f$ | basis $e_1 \otimes e_2^\vee$, $e_1 \otimes e_1^\vee - e_2 \otimes e_2^\vee$, $e_2 \otimes e_1^\vee$ of $\mathrm{ad}^0 V$ | Chapters 4 and 16 | Appendix B.10 |
| trace form | $(x, y) \mapsto \mathrm{tr}(xy)$ on $\mathrm{ad}^0 T$, perfect for $\ell$ odd | Chapter 4 | |
| $\sigma$, $\tau$ | inertia generators of orders 3 and 4 (Lemmas lem:IV, lem:euler) | Chapters 4 and 11 | Chapter 1's $\sigma$ is $h$ |
| $F$ | abstract arithmetic Frobenius, $FgF^{-1} = g^q$ on an inertia image | Chapter 4 | B-C2 |
| $\eta_1$, $\eta_2$ | the two characters of a reducible representation (Ribet's lemma, an extension) | Chapters 4 and 11 | B-C15(h) |
| homothetic | $L' = cL$ for lattices | Chapter 4 | |

### B.9.3 Fields, Galois groups, Frobenius, characters

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $\overline{\mathbb{Q}}$, $G_{\mathbb{Q}}$, $G_K$, $F^{\mathrm{sep}}$ | algebraic closure in $\mathbb{C}$; absolute Galois groups; separable closure | Chapter 5 | |
| $G_{\mathbb{Q},S}$ | Galois group of the maximal extension unramified outside $S$ | Chapter 17 | $S$, Appendix B.10 |
| $c$ (complex conjugation) | an element of order 2 of $G_{\mathbb{Q}}$ | Chapter 5 | $c(u)$, $c_q$, $c_E$, $c'$ are always decorated |
| $D_p$, $I_p$, $P_p$ | decomposition, inertia, wild inertia groups at $p$ | Chapters 6 and 7 | $D$, Appendix B.10 |
| $\mathrm{Frob}_p$, $\mathrm{Frob}_q$, Frob | arithmetic Frobenius | Chapters 5 and 6 | B-C2 |
| $\mathrm{Frob}_p^{-1}$, $\Phi_p$, $\Phi$ | geometric Frobenius | Chapters 6 and 7 | B-C2(b); $\Phi_q(\cdot)$, $\Phi_n$ |

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $\pi_q$, $\hat{\pi}_q$, $\pi_r$ | $q$-power Frobenius endomorphism, Verschiebung; Frobenius morphism $C \to C^{[r]}$ | Chapters 8 and 9 | B-C2(g); Appendix B.10 |
| $\chi_\ell$, $\bar{\chi}_\ell$ | $\ell$-adic cyclotomic character, its reduction | Chapter 5 | B-C6(a) |
| $\chi_3^{\mathrm{cyc}}$, $\chi_m^{\mathrm{cyc}}$ | cyclotomic character at a numerical prime; modulo $m$ | Chapters 5 and 9 | B-C6(a) |
| $\omega_\ell$ | Dirichlet character $a \mapsto a \bmod \ell$ | Chapter 2 | fundamental characters $\omega$ |
| $\chi_0$, $\chi^{(p)}$, $\chi^{\mathrm{prim}}$ | principal character; $p$-component; primitive character inducing $\chi$ | Chapter 2 | B-C6(e) |
| $\chi_d$, $\chi_D$ | quadratic character of $\mathbb{Q}(\sqrt{d})$, indexed by a fundamental discriminant | Chapters 2 and 5 | B-C6(b) |
| $\chi_{-3}$, $\chi_{-4}$, $\chi_8$ | characters of $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ | Chapters 2 and 6 | $\chi_{-3} = \bar{\chi}_3$ |
| $\chi_{q^*}$ | quadratic character of conductor $q$, never $\chi_q$ | Chapters 2 and 5 | B-C6(c) |
| $\chi_{\ell^*}$ | character of $\mathbb{Q}(\sqrt{\ell^*})$, $= \bar{\chi}_\ell^{(\ell-1)/2}$ | Chapter 5 | $= \bar{\chi}_\ell$ iff $\ell = 3$ |
| $\chi_S$, $D$ | $\prod_{q\in S} \chi_{q^*}$, its conductor $\prod_{q\in S} q$ | Chapters 2 and 13 | B-C6(d); $D$, Appendix B.10 |
| $\varepsilon$ | character of order $\le 2$ of $G_{\mathbb{Q}_q}$ in the Tate shape; unramified at a multiplicative $q$, ramified at type $\mathrm{I}_n^*$ | Chapters 7 and 11 | B-C14, B-C15(k) |
| $\eta_{\mathrm{ur}}$ | unramified quadratic character of $G_K$ | Chapter 7 | B-C15(h) |
| $\chi(\mathrm{Frob}_p) = \chi(p)$ | a Dirichlet character as a Galois character | Chapters 6 and 15 | B-C6(e) |
| $\mathbb{Z}_\ell(1)$, $M(n)$ | $\varprojlim \mu_{\ell^k}$; Tate twist | Chapters 5 and 18 | B-C5(a) |
| $\mathbb{Q}(\sqrt{\ell^*})$ | the quadratic subfield of $\mathbb{Q}(\zeta_\ell)$; $\bar{\rho}$ is required absolutely irreducible on its Galois group | Chapters 5 and 6 | |
| $\chi_a$, $\kappa_x$ | Kummer character of $a$ (values in $\mu_n$); Kummer cocycle of $x \in K^\times$ | Chapters 5 and 7 | B-C14 |
| $b(\sigma)$, $u(\sigma)$ | entries of the affine representation of a Kummer extension | Chapter 5 | |
| $\theta_m$, $\theta_2$ | tame characters $I_K \to \mu_m$; the character of order two of inertia | Chapter 7 | $\theta_d$ |
| Art | local Artin map, uniformiser $\mapsto$ arithmetic Frobenius | Chapters 6 and 7 | B-C2(g) |
| density | Dirichlet density | Chapter 5 | |

### B.9.4 Local fields and Weil-Deligne representations

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $K$, $\mathcal{O}_K$, $\mathfrak{m}_K$, $\varpi_K$, $k_K$, $v_K$, $\lvert\cdot\rvert_K$ | a $p$-adic field and its data | Chapter 7 | uniformisers $\varpi$ |
| $K^{\mathrm{ur}}$, $K^{\mathrm{tame}}$, $\mathbb{Q}_p^{\mathrm{ur}}$ | maximal unramified and tamely ramified extensions | Chapter 7 | |
| $I_K$, $P_K$ | inertia and wild inertia groups | Chapter 7 | |
| $t_\ell \colon I_K \to \mathbb{Z}_\ell(1)$, $t_\zeta$ | tame character; its $\mathbb{Z}_\ell$-valued version | Chapter 7 | $t_\ell(E, q)$, B-C15(i) |

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $t$ | topological generator of tame (pro-$\ell$) inertia | Chapters 7 and 16 | B-C14 |
| $[a]$ | Teichmüller lift of $a$ | Chapter 7 | |
| $W_K$, $d(w)$, $\|w\|$ | Weil group, degree map, $q^{d(w)}$ | Chapter 7 | B-C2(f) |
| $(r, N)$, $\mathrm{WD}(\rho)$, $\mathrm{WD}_p(f)$ | Weil-Deligne representation; of $\rho$; of $f$ at $p$ | Chapters 7 and 15 | $N$ monodromy, Appendix B.10 |
| $\mathrm{sp}(n)$ | special (Steinberg) representation | Chapter 7 | |
| $a(V)$, $\mathrm{sw}(V)$ | Artin conductor exponent, Swan conductor | Chapter 7 | $a_p$ coefficients |
| $L(V,s)$, $L_p(V,s)$ | local $L$-factor, with the geometric Frobenius ($\Phi$ in Chapter 7) | Chapter 7 | B-C2(d) |
| $\epsilon(r,\psi,dx)$ | local epsilon factor | Chapter 7 | $\varepsilon$ characters |
| $\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mu_{p-1} \times (1 + p\mathbb{Z}_p)$ | structure of $\mathbb{Q}_p^\times$ ($p$ odd) | Chapter 7 | |
| $\omega$, $\omega_2$ | fundamental characters of levels 1 and 2 at $\ell$ | Chapter 11 | periods $\omega_1, \omega_2$, B-C15(b) |
| $\varpi'$ | $\zeta_\ell - 1$, a uniformiser of $\mathbb{Q}_\ell(\zeta_\ell)$ | Chapter 11 | |
| $\Pi_p$, $\Pi_f$ | local component of $f$ at $p$; the automorphic representation $\Pi_f = \bigotimes' \Pi_v$ | Chapter 15 | not $\pi_p$ (Frobenius), not $\pi_f$ (quotient map), Appendix B.10 |
| $\mathrm{D}_{\mathrm{cris}}$, $B_{\mathrm{cris}}$, $B_{\mathrm{dR}}$, $B_{\mathrm{st}}$ | Fontaine's period rings and functor | Chapter 18 | |

### B.9.5 Curves, elliptic curves, reduction, Tate curves

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $g$, $l(D)$, $K_C$ | genus; dimension of a Riemann-Roch space; canonical divisor | Chapter 8 | $g$ newform; $\ell$ a prime |
| $\varphi^*$, $\varphi_*$ | pullback, pushforward | Chapter 8 | B-C4(e) |
| $\mathcal{C}$, $\mathcal{C}^t$, $\mathcal{C}^*$, $\mathcal{C}_*$ | correspondence, its transpose, its two actions | Chapter 8 | B-C4(e), B-C4(f) |
| $A^\dagger$ | adjoint for a pairing | Chapters 3 and 8 | B-C4(d) |
| $H_1(X,\mathbb{Z})$, $H^1(X,\mathbb{Z})$, $H^1(X,\mathbb{Z}_\ell)$, $\Lambda_X$ | homology, cohomology of a curve; period lattice | Chapter 8 | |
| $\langle\ ,\ \rangle$ (cup product), PD | $\int_X \alpha \wedge \beta$; Poincaré duality $H^1 \to H_1$ | Chapter 8 | B-C5(d), B-C8 |
| $\mathrm{Pic}^0$, $J(X)$, $T_\ell J$, AJ | Picard group, Jacobian, Tate module, Abel-Jacobi map | Chapter 8 | |
| $C^{[r]}$ | Frobenius twist of a curve (coefficients raised to the $r$-th power) | Chapter 8 | B-C13(e) |
| $m_0$ | order of the component group of $\ker(J(X) \to E)$ | Chapter 8 | B-C9 |
| $\deg \varphi$ | degree of a morphism of curves | Chapter 8 | |
| $a_1, \ldots, a_6$; $b_2, \ldots, b_8$; $c_4, c_6$; $\Delta$; $j$ | Weierstrass data | Chapter 9 | $\Delta$, Appendix B.10 |
| $(u, r, s, t)$ | change of Weierstrass coordinates | Chapter 10 | $u$ the unit part (B-C14) |
| $\omega$, $\omega_E$ | invariant differential $dx/(2y + a_1x + a_3)$; the Néron differential of a global minimal model | Chapters 9 and 14 | B-C15(b) |

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $E[m]$, $E(K)$, $E(\mathbb{Q})_{\mathrm{tors}}$ | torsion, points, torsion subgroup | Chapter 9 | |
| $\psi_m$ | division polynomials, $\psi_2 = 2y + a_1x + a_3$ | Chapter 9 | B-C15(m) |
| $e_m$ | Weil pairing (Silverman's) | Chapter 9 | B-C5(b) |
| $\hat{\varphi}$, $[m]$ | dual isogeny, multiplication by $m$ | Chapter 9 | B-C4(f) |
| $T_\ell E$, $V_\ell E$ | Tate module, $T_\ell E \otimes \mathbb{Q}_\ell$ | Chapter 9 | the paper's $T$, $V$ |
| $\Lambda_E$, $\mathrm{vol}(\Lambda_E)$, $\omega_1, \omega_2, \Theta$ | period lattice of $\omega_E$, its covolume (PARI's `E.area`), periods; the uniformisation $\mathbb{C}/\Lambda \to E(\mathbb{C})$ | Chapter 9 | B-C9 |
| $\pi$, $\pi_{\mathfrak{q}}$ | Frobenius endomorphism of $E/\mathbb{F}_{\mathfrak{q}}$ | Chapter 9 | B-C2(g) |
| $a_p(E)$ | $p + 1 - \#\tilde{E}(\mathbb{F}_p)$ | Chapter 9 | $a(V)$ conductor |
| $\alpha$, $\beta$, $\alpha_p$, $\beta_p$ | roots of $X^2 - a_pX + p$; arithmetic eigenvalues on $V_\ell E$ | Chapter 9 | B-C3 |
| $E^{(d)}$, $E^{(q)}$ | quadratic twist by $\mathbb{Q}(\sqrt{d})$; by $\mathbb{Q}(\sqrt{q^*})$ | Chapters 9 and 10 | B-C13(d) |
| $E_S$, $N_S$ | $E \otimes \chi_S$, its conductor | Chapters 9 and 15 | B-C13(d) |
| $L(E,s)$, $L_p(E,s)$ | Hasse-Weil $L$-function and its factors | Chapters 9 and 19 | $L_p(E,s) = L_p(V_\ell E^\vee, s)$, B-C3 |
| CM | complex multiplication; the thirteen CM $j$-invariants | Chapter 9 | |
| 11a1, 121c, 99A1, 11.a2 | Cremona, Cremona's book and Agashe-Ribet-Stein, LMFDB labels | Chapter 9 | B-C16 |
| minimal model, $\Delta_{\min}$ | (global) minimal Weierstrass model | Chapter 10 | |
| $E_0(K)$, $E_1(K)$, $\tilde{E}_{\mathrm{ns}}$ | filtration of $E(K)$; nonsingular points of the reduction | Chapter 10 | $E_0$ the optimal curve |
| $\mathcal{E}$, $\Phi_q(\cdot)$, $\Phi_E(k)$ | Néron model; its component group scheme | Chapter 10 | B-C11(b) |
| $\Phi_q^{\mathrm{geom}}$, $\Phi^{\mathrm{geom}}$ | geometric component group $\Phi_q(\overline{\mathbb{F}}_q)$ | Chapter 10 | B-C11(b) |
| $c_q$, $c_p$ | Tamagawa number $\#\Phi_q(\mathbb{F}_q) = [E(\mathbb{Q}_q) : E_0(\mathbb{Q}_q)]$ | Chapter 10 | never $\#\Phi_q^{\mathrm{geom}}$, never $c(u)$ |
| $\mathrm{I}_n, \ldots, \mathrm{II}^*$, Kodaira code | Kodaira symbols; PARI's code `K[i][2]` | Chapter 10 | B-C11(a), (c) |
| $m$ | number of irreducible components of the special fibre of the minimal regular model (Ogg's formula) | Chapter 10 | $m_0$ |
| $f_p$, $N$, $\mathrm{sw}_p(E)$ | conductor exponent, conductor, wild part | Chapter 10 | $f$ newform; $N$ monodromy |
| $G_{\mathrm{ss}}$, $e$ | semistability group, semistability defect $\#G_{\mathrm{ss}}$ | Chapter 10 | never $\Phi$; $e$, Appendix B.10 |
| $E_{\mathfrak{q}}$, $\mathfrak{q}$ | Tate curve, the variable of its parameter | Chapter 10 | B-C1 |
| $q_E$ | Tate parameter of $E$ | Chapter 10 | B-C14 |
| $n$, $n_q$, $\nu$ | $v_q(\Delta_E)$ at a multiplicative prime; $\nu = v_q(q_E) = -v_q(j_E)$ | Chapter 10 | B-C14 |
| $u$, $\bar{u}$, $c(u)$ | unit part $q_E q^{-\nu}$, its reduction, its class in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$ | Chapters 2 and 10 | B-C14 (the paper's $c(u)$) |
| $w_\ell(E,q)$ | twist-symmetrised weight $\max(v_\ell\#\Phi_q^{\mathrm{geom}}(E), v_\ell\#\Phi_q^{\mathrm{geom}}(E^{(q)}))$ for odd $q$; $w_\ell(E,2) = v_\ell\#\Phi_2^{\mathrm{geom}}(E)$ | Chapter 10 | $w$, Appendix B.10 |

### B.9.6 Galois representations of elliptic curves and of newforms

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $\rho = \rho_{E,\ell}$, $\bar\rho = \bar\rho_{E,\ell}$ | action of $G_{\mathbb{Q}}$ on $T_\ell E$, on $E[\ell]$ | Chapters 9 and 11 | |
| $\rho \bmod \ell^k$ | action on $E[\ell^k] = T/\ell^k T$ | Chapters 5 and 11 | |
| $V$, $T$ | $V_\ell E$, $T_\ell E$ (the paper's) | Chapter 11 | B-C15(j) |
| $A$, $A_f$ | $\mathrm{ad}^0 T$, $\mathrm{ad}^0 V$ | Chapters 11 and 18 | B-C12(b), (c) |
| $\delta_p$ | $\dim V^{I_p}$: $2, 1, 0$ for good, multiplicative, additive $p \neq \ell$ | Chapters 7 and 11 | B-C11(d) |
| absolutely irreducible | irreducible over $\bar k$ | Chapter 4 | |
| $N(\bar\rho)$, $k(\bar\rho)$ | Serre conductor, Serre weight | Chapter 11 | |
| $\bar\rho_{\mathfrak{m}}$ | residual representation of a maximal ideal of $\mathbb{T}$ | Chapters 14 and 15 | |
| $\delta(\bar\sigma)$, $\bar\sigma$ | the exponent at $\ell$ and the representation in the optimal level theorem of Diamond, Darmon and Taylor | Chapter 15 | B-C15(l) |
| $\rho_{f,\lambda}$, $\bar\rho_{f,\lambda}$, $V_\lambda(f)$ | Galois representation of a newform, $\det(X - \rho(\mathrm{Frob}_p)) = X^2 - a_p X + p$ | Chapter 15 | B-C3(c) |
| $h_\ell(E)$ | length of $H^0(\mathbb{Q}, \mathrm{ad}^0 T(1) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)$; 0 for $\ell \geq 5$, $h_3 \leq 1$ | Chapter 11 | |
| $S_f$ | exceptional set of Diamond, Flach and Guo | Chapters 11 and 18 | $S$, Appendix B.10 |
| $\iota\colon H^1(K,T) \to H^1(K,V)$, $\pi\colon H^1(K,V) \to H^1(K,W)$ | maps induced by $T \subset V \to W$ | Chapter 17 | twist map $\iota$, B-C15(c) |

### B.9.7 Modular forms, Hecke algebras, modular curves, the modular degree

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $\mathcal{H}$, $\mathcal{H}^*$, $d\mu$ | upper half plane, with the cusps; $dx\,dy/y^2$ | Chapter 12 | star exception |
| $S$, $T$ (matrices) | $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ (local to the first sections of Chapter 12) | Chapter 12 | B-C15(j) |
| $\Gamma_0(N)$, $\Gamma_1(N)$, $\Gamma(N)$, $\bar\Gamma$ | congruence subgroups; image in $\mathrm{PSL}_2(\mathbb{Z})$ | Chapter 12 | |
| $\mu(N)$ | $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = N \prod_{p\mid N}(1 + p^{-1})$ | Chapter 12 | B-C15(g) |
| $\nu_2(N)$, $\nu_3(N)$, $\nu_\infty(N)$, $g(N)$ | elliptic points, cusps, genus of $X_0(N)$ | Chapter 12 | $\nu$ (B-C14) |
| $Y_0(N)$, $X_0(N)$, $X_1(N)$ | modular curves | Chapters 12 and 14 | |
| $q = e^{2\pi i z}$, $q_h$ | $q$-expansion variable, $e^{2\pi i z/h}$ | Chapter 12 | B-C1 |
| $f\vert_k\gamma$, $j(\gamma, z)$ | slash operator with $(\det\gamma)^{k/2}$; $cz + d$ | Chapter 12 | B-C5(d) |
| $M_k$, $S_k$, $S_k(N)$, $S_2(N; A)$ | modular and cusp forms; forms with coefficients in $A$ | Chapters 12 and 13 | |
| $a_n(f)$ | $n$-th coefficient at $\infty$ | Chapter 12 | |
| $\langle f, g\rangle$ | Petersson product, unnormalised | Chapter 12 | B-C8 |
| $f^\perp$ | Petersson orthogonal complement | Chapters 12 and 13 | not the $\mathbb{T}$-stable complement when $U_p$ is involved |

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $E_k$, $E_2^*$, $\Delta(z)$, $j(z)$, $\eta(z)$ | Eisenstein series, the nonholomorphic $E_2^*$, discriminant form, $j$, Dedekind eta | Chapter 12 | B-C15(h) |
| $B_k(N)$ | Sturm bound $\lfloor k\mu(N)/12 \rfloor$ | Chapter 12 | $B_E$ |
| $W_N$, $w_N$ | Atkin-Lehner matrix $\left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$ and involution | Chapter 12 | B-C13(f) |
| $\varepsilon_N(f)$ | eigenvalue of $w_N$ on a newform | Chapters 13 and 14 | B-C15(k) |
| $T_n$, $T_p$, $U_p$, $T_p^{(M)}$ | Hecke operators; $U_p = T_p$ for $p \mid N$; the level when two are in play | Chapter 13 | B-C13(e) |
| $\mathcal{U}_p$, $V_d$ | $\sum a_n q^n \mapsto \sum a_{pn} q^n$; $\sum a_n q^n \mapsto \sum a_n q^{dn}$ | Chapter 13 | $V$, B-C15(j) |
| $[\Gamma\alpha\Gamma]_k$ | double coset operator | Chapter 13 | |
| $\mathbb{T}$, $\mathbb{T}_N$, $\mathbb{T}^{(N)}$, $\mathbb{T}_\ell$, $\mathbb{T}_A$ | full Hecke algebra, anemic algebra, scalar extensions | Chapter 13 | B-C13(a) |
| $\phi_f$, $\phi$ | eigencharacter $\mathbb{T} \to \mathcal{O}_f$; a system of eigenvalues | Chapters 13 and 15 | B-C13(b) |
| $I$, $J$ (Lemma lem:crit) | $\ker \phi_f$, $\mathrm{Ann}_{\mathbb{T}}(f^\perp)$ | Chapter 15 | B-C13(c) |
| $I_f$ | $\ker \phi_f$ | Chapter 14 | |
| $r_E$ | congruence number of $f_E$ | Chapters 14 and 15 | |
| $f^\Sigma$, $f^{\{p\}}$, $f^S$, $\Sigma(r)$ | depletion; the set of primes dividing $r$ | Chapters 13 and 15 | B-C13(e) |
| $M^\Sigma$ | $M \prod_{p\in\Sigma} p^{\delta_p}$, level of $f^\Sigma$ | Chapter 13 | |
| $f \otimes \chi$, $f^\chi$ | naive twist, newform twist | Chapters 13 and 15 | B-C13(d) |
| $\iota$, $\iota_S$ | twist map $h \mapsto h \otimes \chi_S$ | Chapters 13 and 15 | B-C15(c) |
| $L(f,s)$, $\Lambda(f,s)$ | $L$-function of a newform, completed | Chapter 13 | |
| $\alpha_p$, $\beta_p$, $\mu_p$ | roots of $X^2 - a_p(h)X + p$; chosen $U_p$-eigenvalue (Lemma lem:old) | Chapters 13 and 15 | B-C3(e) |
| $J_0(N)$, $J_1(N)$ | Jacobians of $X_0(N)$, $X_1(N)$ | Chapter 14 | |
| $t^*$, $t_*$ | the two actions of $t \in \mathbb{T}$ (on $H^1$, on $H_1$) | Chapter 14 | B-C4(e) |
| $J_f$, $\pi_f$, $E_f$ | $J_0(N)/I_f J_0(N)$, the quotient map; the optimal curve for rational $f$ | Chapter 14 | B-C12(c); automorphic $\Pi_f$ |
| $\varphi_E$, $\deg \varphi_E$ | modular parametrisation, modular degree | Chapter 14 | B-C13(b) |
| $E_0$ | the optimal curve of the class ($E_f$ of Chapter 14) | Chapter 15 | $E_0(K)$ |
| $c_\varphi$, $c_E$ | Manin constants, $c_E > 0$ | Chapter 14 | B-C9 |
| $\hat{\delta}_!$ | $\langle x, w_N y\rangle$ on $H^1(X_0(N), \mathbb{Z}_\ell)$ | Chapter 14 | B-C13(f) |
| $M_{f,\lambda}$, $M_{f^\Sigma,\lambda}$ | $H^1(X_0(M^\Sigma), \mathbb{Z}_\ell)[\ker \phi_{f^\Sigma}]$, of rank two | Chapter 14 | B-C12(b), B-C13(f) |
| $\mathfrak{m}$ | maximal ideal of $\mathbb{T} \otimes \mathbb{Z}_\ell$ attached to $(f,\ell)$; Eisenstein if $\bar{\rho}_{\mathfrak{m}}$ is reducible | Chapters 14 and 15 | $\mathfrak{m}_K$ |
| $I_{\mathrm{Eis}}$, $C_N$, $\mathcal{S}_N$ | Eisenstein ideal, cuspidal subgroup, Shimura subgroup | Chapter 14 | $\Sigma$ |
| $\delta_D$, $E_D$, $D$ | degree of the optimal parametrisation $X_0^D(M) \to E_D$ by a Shimura curve; quaternion discriminant | Chapter 14 | $D$, Appendix B.10 |
| $t_q$ | Tamagawa exponent (Kim-Ota): the largest $t$ with $\rho \bmod \ell^t$ unramified at $q$ | Chapter 15 | $= t_\ell(E,q)$ at multiplicative $q$; Kim and Ota's $t_f(q)$ |

### B.9.8 Cohomology, Selmer groups, Tamagawa factors

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $C^n(G,M)$, $Z^1$, $B^1$ | inhomogeneous cochains, cocycles, coboundaries | Chapter 16 | $C_n$ cyclic group |
| $H^n(G,M)$, $H^n(K,M)$ | continuous group and Galois cohomology, coefficients of type (D), (C), (V) | Chapters 16 and 17 | |
| $\hat{H}^n$ | Tate cohomology | Chapter 16 | |
| Res, Inf, Cor | restriction, inflation, corestriction | Chapter 16 | |
| $(\sigma\cdot c)(h)=\sigma c(\sigma^{-1}h\sigma)$ | conjugation action of $G/H$ on $H^n(H,M)$ | Chapter 16 | B-C5(d) |
| $\rho_M$ | the action of $G$ on a module $M$ ($\mathrm{Ad}\,\rho$ on $\mathrm{ad}^0 T$) | Chapter 16 | |
| $n'$, $w$ | the unipotent entry of $\rho(t)$, $v_\ell(n')=v_\ell(n)$; $w=v_\ell(n')$ | Chapters 16 and 18 | B-C14; $w$, Appendix B.10 |
| $\kappa$, $\kappa_q$, $\kappa_u$ | Kummer cocycle of $q_E$, of $q$, of $u$ | Chapter 17 | B-C14 |
| $c'$ | $\varepsilon(\mathrm{Frob})\kappa(\mathrm{Frob})$, well defined modulo $\ell^{\min(v_\ell(n),v_\ell(q-1))}$ | Chapters 17 and 18 | B-C14 |
| $H^1_{\mathrm{ur}}(\mathbb{Q}_p,M)$ | $M^{I_p}/(\mathrm{Frob}-1)M^{I_p}$ | Chapters 16 and 17 | B-C2(c) |
| $T$, $V$, $W$ | lattice, $T\otimes\mathbb{Q}_\ell$, $T\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$ | Chapter 17 | in the paper $W=A\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$, B-C12(b) |
| $T^D$, $V^D$, $W^D$ | Kummer duals; in the paper and Chapters 18, 20, 21 built from $A=\mathrm{ad}^0 T$ | Chapters 17 and 18 | B-C4(c) |
| $M^*(1)$ | $\mathrm{Hom}(M,\mu_{\ell^\infty})$; $=M^D$ for finite $M$; $T^*(1)=W^D$ for a lattice $T$, while $\mathrm{Hom}(W,\mu_{\ell^\infty})=T^D$ | Chapter 17 | B-C4(c) |
| $\mathrm{Sel}_n(E/\mathbb{Q})$, Sha | $n$-Selmer group, Shafarevich-Tate group | Chapter 17 | |
| $\mathcal{L}$, $\mathcal{L}^*$, $H^1_{\mathcal{L}}$, $\mathcal{L}^\Sigma$, $\mathcal{L}_\Sigma$ | Selmer structure, dual structure, Selmer group; relaxed and strict at $\Sigma$ | Chapter 17 | star exception |
| $H^1_f(\mathbb{Q}_p,M)$, $H^1_f(\mathbb{Q},M)$, $H^1_g$, $H^1_e$ | Bloch-Kato local conditions and Selmer groups | Chapter 18 | |
| $H^1_\Sigma(\mathbb{Q},W)$ | $\Sigma$-imprimitive Selmer group of Diamond, Flach and Guo | Chapter 18 | |
| $\mathrm{Tam}_p(M)$ | $\mathrm{Fitt}_{\mathbb{Z}_\ell}((H^1(I_p,M)_{\mathrm{tor}})^{G_{\mathbb{Q}_p}})$ | Chapter 18 | B-C10 |
| $\Sigma$ | the paper's finite set of primes (in Theorem thm:main, the primes dividing $N$) | Chapters 17 and 18 | $\mathcal{S}_N$; in Theorem thm:quant $\Sigma=S$ |
| $G_{\mathbb{R}}$, $H^1(\mathbb{R},M)$ | the real place | Chapter 17 | killed by 2 |
| $X^*$ | Pontryagin dual | Chapter 3 | B-C4(b) |
| $\mathrm{ad}^0 M_f$ | the adjoint motive | Chapter 18 | |

### B.9.9 $L$-functions and the paper's local terms

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $\zeta(s)$, $\zeta^{(N)}(s)$ | Riemann zeta; with the factors at $p \mid N$ removed | Chapter 19 | $\zeta_n$ |
| $L(s,\chi)$, $\zeta_K(s)$ | Dirichlet and Dedekind $L$-functions | Chapter 19 | |
| $\Gamma_{\mathbb{R}}$, $\Gamma_{\mathbb{C}}$ | $\pi^{-s/2}\Gamma(s/2)$, $2(2\pi)^{-s}\Gamma(s)$ | Chapter 19 | $\Gamma_0(N)$ |
| $L_p(M,s)$ | $\det(1-\mathrm{Frob}_p^{-1}\,p^{-s} \mid M^{I_p})^{-1}$ | Chapters 7 and 19 | B-C2(d) |
| $L(\mathrm{ad}^0 f,s)$, $L(A_f,s)$ | adjoint $L$-function | Chapter 19 | B-C7 |
| $L(\mathrm{Sym}^2 f,s)$ | symmetric square, $= L(\mathrm{ad}^0 f, s-1)$ | Chapter 19 | B-C7; never the Rankin series |
| $L_p^{\mathrm{nv}}$, $L^{\mathrm{nv}}$ | naive factors, naive $L$-function | Chapter 19 | B-C7 |
| $D(f,s)$ | Rankin series $\sum a_n(f)^2 n^{-s}$ | Chapter 19 | B-C7; $D$, Appendix B.10 |
| $P_q(X)$ | $L_q(\mathrm{Sym}^2 f,s) = P_q(q^{-s})^{-1}$ | Chapter 19 | B-C7 |
| $e_\ell(E,q)$ | $v_\ell(P_q(q^{-2})/(1-a_q^2q^{-2}))$, the omitted Euler factor | Chapter 19 | not invariant under ramified twists |
| $t_\ell(E,q)$ | $v_\ell(-v_q(j_E))$ if $v_q(j_E)<0$, else 0 | Chapter 19 | tame character $t_\ell$, B-C15(i) |
| $w_3(E)$ | weight of the prime 3 in Conjecture conj:euler | Chapter 19; Chapter 21 | $w_\ell(E,q)$ |
| $N_{\mathrm{ad}^0}$ | conductor of $A_f$ | Chapter 19 | |
| $\Lambda(\mathrm{ad}^0 f,s)$, $\Lambda(E,s)$, $w_E$ | completed $L$-functions; root number of $E$ | Chapter 19 | $\Lambda_E$; $w$, Appendix B.10 |
| $B_E$ | $c_E^2 N\, L(\mathrm{ad}^0 f,1)/(2\pi\,\mathrm{vol}(\Lambda_E))$ | Chapter 19 | $B_k(N)$ |

### B.9.10 Congruence ideals and the theorems of Diamond, Flach and Guo

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $\pi\colon \mathbb{T}_{\mathcal{O}} \to \mathcal{O}$, $\wp$, $I = \mathrm{Ann}_{\mathbb{T}}\,\wp$, $\eta(\pi)=\pi(I)$ | the data of Lemma lem:ddt417 | Chapters 3 and 20 | B-C13(c); $\pi$, B-C15(a) |
| $\mathcal{C}(\pi)$, $\mathrm{Cot}(\pi)$ | congruence module $\mathbb{T}_{\mathcal{O}}/(\wp+I) \simeq \mathcal{O}/\eta(\pi)$; cotangent module $\wp/\wp^2$ | Chapter 20 | the literature's $\Omega$, $\Phi$; correspondences $\mathcal{C}$ |
| $d$, $\Delta$, $Q$ | rank of $L[\wp]$, its Gram determinant, $L/(L[\wp]\oplus L[I])$ | Chapters 3 and 20 | $\Delta$ discriminant |
| $\eta^{\Sigma}_{f,\lambda}$, $\eta^{\emptyset}_{f,\lambda}$ | naive $\Sigma$-finite congruence ideal of Diamond, Flach and Guo; at level $N$ | Chapters 15 and 20 | $\eta(z)$ |
| $\eta^{\mathrm{Hecke}}$ | congruence ideal in the Hecke algebra (of $g^S$ in Theorem thm:quant) | Chapters 13 and 20 | |
| $\eta_f$, $\eta_f^{\mathrm{Hecke}}$, $\eta_f(N)$ | one ideal, $\phi_f(\mathrm{Ann}_{\mathbb{T}}\ker\phi_f) = r_E\mathbb{Z}$ for rational $f$: the names in Chapter 14, in Chapter 20, and Kim and Ota's (Chapter 15) | Chapters 14 and 20 | $\eta^{\Sigma}_{f,\lambda}$ |
| $M(M,1)_{!,\lambda}$ | the integral structure of Diamond, Flach and Guo at level $M$ (in the paper $M = M^{\Sigma}$), $= H^1(X_0(M),\mathbb{Z}_\ell)$ with no localisation (Proposition prop:integral) | Chapter 20 | |
| $R_\Sigma$, $\mathbb{T}_\Sigma$ | deformation ring, Hecke algebra of Taylor-Wiles | Chapters 15 and 20 | |

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $g$, $g^S$, $N_S^S$ | $f_{E_S}$, its $S$-depletion, $N_S \prod_{q\in S} q^{\delta_q(g)} = N$ | Chapters 15 and 20 | genus $g$ |

### B.9.11 The paper's named quantities, statements and data

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $D$ (Conjecture conj:eis) | deficit $\sum_q v_\ell(\#\Phi_q^{\mathrm{geom}}) - v_\ell(\deg\varphi_E)$ | Chapter 22; Chapter 21 | $D$, Appendix B.10 |
| $[\,\cdot\,]$ | Iverson bracket (1 if true, 0 if false); used from Chapter 2 on, notably in Chapters 10 and 22 | Chapter 2 | |
| (eq:main), (eq:twist), (eq:rule), (eq:identity), (eq:thm) | the paper's displayed statements, cited by these names | Chapter 21 | |
| $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ | the image 3Cs.1.1 (not "the split Cartan image") | Chapters 9 and 11 | |
| working set, hold-out, extension, out-of-table | the paper's four ranges of conductors | Chapter 22; Chapter 21 | |
| $B_\ell^{\mathrm{rule}}$, $B_\ell^{\mathrm{nc}}$, $B_\ell^{\mathrm{III}}$, $B_\ell^{\mathrm{main}}$, $B_\ell^{\mathrm{tw}}$; $B_\ell^{\mathrm{thm}}$ | the five bounds compared with $v_\ell(\deg\varphi_E)$; the literal right side of (eq:thm), $B_\ell^{\mathrm{thm}} = B_\ell^{\mathrm{nc}} - [\ell=3]\,w_3(E)$ (it includes $t_\ell(E,\ell)$) | Chapter 22 | $B_E$, $B_k(N)$ |
| violation, equality, excess | $v < b$; $v = b > 0$; $v - b$, for a bound $b$ and $v = v_\ell(\deg\varphi_E)$ | Chapter 22 | |

### B.9.12 Computation

| symbol | meaning | defined in | see also |
|---|---|---|---|
| `rule(E,l,iii)` | the paper's function: `rule(E,l,1)` $= [v_\ell(\deg\varphi_E), B_\ell^{\mathrm{rule}}]$, `rule(E,l)` $= [v_\ell(\deg\varphi_E), B_\ell^{\mathrm{III}}]$ | Chapters 19 and 22 | |
| `ns3(E)` | 1 if and only if $\psi_3$ is a product of two irreducible quadratics; equal to $h_3(E)$ when $E[3]$ is irreducible | Chapters 5 and 22 | the correction $h_3$ |
| `R[4]`, `R[5]`, `K[i][2]` | `ellglobalred` output: factored conductor, local data, Kodaira code | Chapter 10 | B-C11(c) |
| `elllocalred(E,p)` | $[f_p, k_p, [u,r,s,t], c_p]$ | Chapter 10 | |
| `lfunsympow(E,2)`, `lfuneuler` | $L(\mathrm{Sym}^2 f, s)$ and $1/P_q(x)$, $x = q^{-s}$ | Chapter 19 | B-C7 |
| `mfpetersson` | Petersson product divided by $\mu(N)$ | Chapter 12 | B-C8 |
| `E.area` | $\mathrm{vol}(\Lambda_E)$ | Chapter 19 | |
| `ellmoddegree`, `elldivpol`, `ellinit` | $\deg\varphi / c_\varphi^2$ (the modular degree for the optimal curve with $c_E = 1$; $1/5$ on `11a3`), division polynomial, curve | Chapters 14 and 22 | |
| `ecdata` files | Cremona's tables (`allcurves`, `degphi`, `galrep`, ...) | Chapters 9 and 22 | |

### B.9.13 Decorations and operators

| symbol | meaning | defined in | see also |
|---|---|---|---|
| $M^\vee$ | linear dual | Chapter 4 | B-C4(a) |
| $X^*$ | Pontryagin dual (modules) | Chapter 3 | B-C4(b), (e) |
| $M^D$ | Kummer dual $M^\vee(1)$, or $M^*(1)$ for finite $M$; $W^D := V^D/T^D$ | Chapters 17 and 18 | B-C4(c) |
| $A^\dagger$ | adjoint of an operator | Chapter 8 | B-C4(d) |
| $\varphi^*$, $\varphi_*$, $t^*$, $t_*$ | pullback, pushforward | Chapters 8 and 14 | B-C4(e) |
| $A^\mathsf{T}$, $A^t$, $\mathcal{C}^t$ | transpose | Chapters 3 and 8 | B-C4(f) |
| $\hat\varphi$, $\hat\pi_q$ | dual isogeny, Verschiebung | Chapters 9 and 14 | $\hat\delta_!$, $\hat H$ |
| $M(n)$ | Tate twist | Chapters 5 and 18 | B-C5(a) |
| $\bar\rho$, $\bar\chi_\ell$, $\bar u$ | reduction modulo $\ell$ (or modulo $q$) | Chapter 5 | complex conjugate $\bar z$ |
| $M^G$, $M^{I_p}$ | invariants | Chapter 16 | |
| $f^\Sigma$ | depletion | Chapter 13 | B-C13(e) |
| $(\cdot)^{(N)}$ | factors at $N$ removed; anemic; a level | Chapters 13 and 19 | B-C13(e) |
| $E^{(d)}$, $\chi^{(p)}$ | twist of a curve; component of a character | Chapters 2 and 9 | B-C13(d), B-C6(e) |
| $C^{[r]}$ | Frobenius twist | Chapter 8 | B-C13(e) |
| $f\otimes\chi$, $f^\chi$ | naive twist, newform twist | Chapters 13 and 15 | B-C13(d) |
| $S^\perp$, $f^\perp$ | orthogonal complement | Chapters 3 and 12 | |
| $Ш^*$, $q^*$, $\ell^*$, $\mathcal{H}^*$, $E_2^*$, $\mathcal{L}^*$ | the fixed exceptions to the star rule | Appendix B | B-C4(e) |
| $(\cdot)_{\mathrm{tor}}$, $(\cdot)_{\mathrm{tors}}$ | torsion submodule (the second in Chapter 3) | Chapters 3 and 16 | |

## B.10 Letters with several meanings

Many letters carry more than one meaning across the book, and some carry two in the paper itself. B-C15 states the rules; the table after it lists every meaning in use, with the chapters where it occurs. A meaning marked "local" is used only inside the named chapter or lemma, which says so in its convention box.

**Convention B.17** (B-C15: letters with several meanings)**.**

(a) The uniformisers of local fields and of discrete valuation rings are written $\varpi$; the paper's $\pi\colon \mathbb{T}\to\mathcal{O}$ (Lemma lem:ddt417) keeps its name inside that lemma, and $\pi = 1-\zeta_3$ is a local name of Chapter 6; $\pi_f$ is the quotient map to $J_f$; $\pi_q$ the Frobenius endomorphism (B-C2(g)).

(b) $\zeta_n = e^{2\pi i/n}$; a primitive cube root of unity is always $\zeta_3$, never $\omega$ or $\rho$. $\omega$ is a differential: $\omega_E$ (Néron or invariant differential), $\omega_f = f(z)\,dz$; $\omega_1,\omega_2$ are periods; the fundamental characters of Chapters 11 and 15 are $\omega$ (level 1) and $\omega_2$ (level 2), used only there; $\omega_\ell$ is the Dirichlet character of B-C6(a).

(c) $\iota$ is the twist map of Theorem thm:twist and the map $H^1(K,T)\to H^1(K,V)$ (both are the paper's); a chapter using both writes $\iota_S$ for the twist map.

(d) $D$: the conductor $\prod_{q\in S} q$ of $\chi_S$ (Theorem thm:twist); the deficit of Conjecture conj:eis; the discriminant of a quadratic field; the quaternion discriminant of a Shimura curve; a divisor

(Chapter 8). $D_p$ is a decomposition group, $\mathrm{D}_{\mathrm{cris}}$ Fontaine's functor, $^D$ the Kummer dual, $D(f,s)$ the Rankin series. Each chapter uses at most two of these and says which.

(e) $\Sigma$ is the paper's finite set of primes; the Shimura subgroup of $J_0(N)$ is $\mathcal{S}_N$; $S$ is the set of primes of Theorem thm:quant, $S_f$ the exceptional set of Diamond, Flach and Guo, $S$ a set of places in $G_{\mathbb{Q},S}$.

(f) $N$ is the conductor or level; the monodromy operator of a Weil-Deligne representation is $N$ only in local statements (Chapters 7 and 15); $N_s, N_{ns}$ are normalisers of Cartan subgroups; Norm is a norm map.

(g) $\mu_n$ are roots of unity; $\mu(N) = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$; an unramified character $\mu$ (Chapter 7) and the $U_p$-eigenvalue $\mu_p$ of Lemma lem:old are local names.

(h) $\eta_{f,\lambda}^{\Sigma}$, $\eta$ are congruence ideals; $\eta(z)$ is the Dedekind eta function; $\eta_1, \eta_2$ are the two characters of an extension (local, always with subscripts); $\eta_{\mathrm{ur}}$ is the unramified quadratic character (Chapter 7). $\Delta$ is the discriminant, $\Delta(z)$ the cusp form of level one, $\Delta$ the Gram determinant of Lemma lem:ddt417 (local). $\Lambda_E$, $\Lambda_X$ are period lattices, $\Lambda(\cdot, s)$ completed $L$-functions.

(i) $t_\ell(E,q)$ (the paper's term) always carries its arguments; the tame character $t_\ell\colon I \to \mathbb{Z}_\ell(1)$ is written with its domain or an argument $t_\ell(\sigma)$; $t$ alone is a generator of tame inertia.

(j) $V$ is $V_\ell E$ or a representation; the Verschiebung is $\hat{\pi}_p$; $V_d\colon h(z) \mapsto h(dz)$ (Chapter 13) and $V_f$ (Chapter 14, the $f$-part of $H^1$) are local names. $T$ is $T_\ell E$; $\mathbb{T}$ the Hecke algebra; $T_n$ Hecke operators; the matrix $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$ is $T$ only in the first two sections of Chapter 12.

(k) $\varepsilon$: the character of order at most 2 of the Tate shape (B-C14, B-C6(d)); $\varepsilon_N(f)$ the Atkin-Lehner sign; a root number is $w$ or $\varepsilon(\cdot)$ with its argument; the non-square of the standard models is $\varepsilon$ only in Chapter 1; $\epsilon$ (not $\varepsilon$) is a local epsilon factor. Three further uses are fixed exceptions: the nebentypus $\varepsilon$ of a newform with character (Chapter 15); the small positive real $\varepsilon$ of analytic estimates such as $O(n^{(k-1)/2+\varepsilon})$ (Chapters 12 and 14); and the variable $\varepsilon$, $\varepsilon^2 = 0$, of the dual numbers $(\mathcal{O}/\varpi^n)[\varepsilon]$ in the sketch of Mazur's theorem in Chapter 20 (local; elsewhere the dual numbers are $k[x]/(x^2)$).

(l) $\delta_p = \dim V^{I_p}$; $\hat{\delta}_!$ the pairing; $\delta$ a connecting map; $\delta(\bar{\sigma})$ the exponent in the optimal level theorem; $\delta_D$ the degree of a Shimura curve parametrisation.

(m) $\psi$: the character of order 4 (Proposition prop:ns); $\psi_m$ division polynomials; an isogeny $\psi$ (Chapters 9 and 14).

*Reason.* The collisions are those of the index; renaming the paper's letters is not allowed, so the book localises its own. *Binds:* the chapters named.

| letter | meanings | where |
|---|---|---|
| $A$ | $\mathrm{ad}^0 T$ (B-C12(b)); $A_f = \mathrm{ad}^0 V$, never an abelian variety (B-C12(c)); abelian varieties $A$, $A^\vee$; matrices $A$, $A^\dagger$, $A^{\mathsf{T}}$; the curves $A_0, A_1, \ldots$ of an isogeny chain (local) | Chapters 3, 4, 11 and 14 |
| $a$ | coefficients $a_n(f)$, $a_p(E)$; Weierstrass coefficients $a_1, \ldots, a_6$; the Artin conductor $a(V)$; the matrix $\left(\begin{smallmatrix}1 & -1\\ 1 & 1\end{smallmatrix}\right)$ of $\mathrm{GL}_2(\mathbb{F}_3)$ | Chapters 1, 7, 9 and 12 |

| letter | meanings | where |
|---|---|---|
| $\alpha$, $\beta$ | roots of $X^2 - a_p X + p$ (B-C3); classes in the cup product $\int_X \alpha \wedge \beta$ (B-C5(d)) | Chapters 8, 9 and 13 |
| $B$ | Bernoulli numbers $B_k$ (no argument); the Sturm bound $B_k(N)$ (with the level); $B_E$; the bounds $B_\ell^{\mathrm{rule}}, \ldots, B_\ell^{\mathrm{thm}}$ (superscript); the Borel subgroup B; coboundaries $B^1$ | Chapters 1, 12, 16, 19 and 22 |
| $C$ | cyclic groups $C_n$; Cartan subgroups $C_s$, $C_{ns}$; cochains $C^n$; a curve $C$ and its twist $C^{[r]}$; correspondences $\mathcal{C}$; the congruence module $\mathcal{C}(\pi)$; the cuspidal subgroup $C_N$ | Chapters 1, 8, 14, 16 and 20 |
| $c$ | complex conjugation; a cocycle or cohomology class $c$ (B-C5(d)); the class $c(u)$ of Lemma lem:tamfactor; $c_q$ Tamagawa number; $c'$ the corner of $\rho$(Frob); $c_E$, $c_\varphi$ Manin constants; $c_4, c_6$ Weierstrass invariants | B-C9, B-C11, B-C14; bare only for complex conjugation (Chapters 5 and 17), and for cocycles and auxiliary elements inside proofs (Chapters 16 to 18 and 20) |
| $\chi$ | characters (B-C6): $\chi_\ell$ cyclotomic, $\chi_d$ quadratic, $\chi_S$, $\chi_0$, $\chi^{(p)}$; Kummer characters $\chi_a$; never the trace function of a representation | Chapters 2, 5 and 9 |
| $D$ | see B-C15(d); also the dihedral groups $D_n$ (of order $2n$) | Chapters 1, 2, 6, 8, 14, 19, 21 and 22 |
| $d$ | a non-square or discriminant in $\chi_d$, $E^{(d)}$; the degree map $d(w)$ of the Weil group; the rank $d$ of $L[\wp]$ (Lemma lem:ddt417); the operator $V_d$; coboundary maps $d^0$; degrees of isogenies | Chapters 2, 3, 7, 13, 16 and 20 |
| $\Delta$ | discriminant; $\Delta(z)$; Gram determinant; a finite quotient $\Delta = I/K$ of inertia (local) | Chapters 3, 9, 12 and 18 |
| $e$ | basis vector $e_1 \otimes e_2^\vee$ of $\mathrm{ad}^0 V$; ramification index; semistability defect; idempotent $e_{\mathfrak{m}}$; $e_m$ Weil pairing; $e_\ell(E,q)$ | Chapters 3, 4, 7, 9, 10 and 19 |
| $\epsilon$, $\varepsilon$ | see B-C15(k): Tate-shape character; Atkin-Lehner sign $\varepsilon_N(f)$; non-square (Chapter 1); local epsilon factor $\epsilon$; nebentypus; analytic $\varepsilon > 0$; dual numbers (local) | Chapters 1, 7, 12 to 15, 18 and 20 |
| $\eta$ | see B-C15(h) | Chapters 3, 4, 7, 11, 12 and 15 |
| $f$ | newform; basis vector $e_2 \otimes e_1^\vee$ of $\mathrm{ad}^0 V$; residue degree; conductor exponent $f_p$; conductor $f_\chi$ of a character | Chapters 2, 4, 7, 10 and 13 |
| $g$ | genus; newform $f_{E_S}$; group element | Chapters 8 and 15 |
| $h$ | basis vector of $\mathrm{ad}^0 V$ and the matrix $\mathrm{diag}(1,-1)$; newform $h$ of Lemma lem:old; $h_\ell(E)$ | Chapters 4, 11 and 13 |
| $I$ | inertia groups $I_p$, $I_K$; $\ker \phi_f$ in Lemma lem:crit; $\mathrm{Ann}_{\mathbb{T}}(\wp)$ in Lemma lem:ddt417; $I_f$; $I_{\mathrm{Eis}}$; the augmentation ideal $I_G$ of $\mathbb{Z}[G]$ | B-C13(c); Chapter 16 |
| $\iota$ | see B-C15(c) | Chapters 13, 15, 17 and 21 |
| $j$ | the $j$-invariant $j_E$; the modular function $j(z)$ and $j_N = j(Nz)$; the automorphy factor $j(\gamma, z) = cz + d$ | Chapters 9, 12 and 14 |
| $J$ | the matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$; $\mathrm{Ann}_{\mathbb{T}}(f^\perp)$ in Lemma lem:crit; Jacobians $J(X)$, $J_0(N)$; $J_f$ | Chapters 1, 8, 14 and 15 |
| $\kappa$ | Kummer cocycle (B-C14); a bijection in Chapter 9 (local) | Chapters 9 and 17 |
| $L$ | a lattice with a perfect pairing, $L[\wp]$; $L$-functions $L(\cdot, s)$, $L_p$; Selmer structures $\mathcal{L}$; $l(D)$ in Riemann-Roch | Chapters 3, 8, 17 and 19 |
| $\lambda$ | prime above $\ell$; an eigenvalue (local) | Chapters 6 and 15 |
| $M$ | a module; a level, $M^\Sigma$; $M_{f,\lambda}$; spaces $M_k$ of modular forms; Tate twists $M(n)$; the integral structure $M(M,1)_{!,\lambda}$ | Chapters 3, 12 to 14 and 20 |
| $\mu$ | see B-C15(g); the Moebius function $\mu(n)$, only in Chapter 2 (local; elsewhere $\mu(N)$ is the index of $\Gamma_0(N)$) | Chapters 2, 7, 12 and 13 |

| letter | meanings | where |
|---|---|---|
| $N$ | see B-C15(f); $N\mathfrak{p}$ the norm of a prime | Chapters 6, 7 and 10 |
| $\nu$ | $v_q(q_E)$; $\nu_2, \nu_3, \nu_\infty$ for $\Gamma_0(N)$; $u-1$ in Chapter 4 (local) | Chapters 4, 12 and 18 |
| $\omega$ | see B-C15(b); a bilinear form $\omega(x,y) = \det[x \mid y]$ in Chapter 4 (local) | Chapters 2, 4, 9 and 11 |
| $\Omega$ | a coefficient field in Chapter 7 (local); the periods $\Omega_f^{\pm}$ of a newform; $\Omega^1$ differentials; never a congruence module (written $\mathcal{C}(\pi)$ in Chapter 20) | Chapters 7, 8 and 20 |
| $\phi$, $\varphi$ | upright: eigencharacters $\phi_f$; curly: maps of curves (B-C13(b)), and Euler's function $\varphi(n)$ with an integer argument | Chapters 8 and 13; $\varphi(n)$ in Chapters 1, 2, 5 to 7 and 12 |
| $\Phi$ | geometric Frobenius; component groups $\Phi_q(\cdot)$; cyclotomic polynomials $\Phi_n$; a composite isogeny $\Phi_m = \varphi_m \circ \cdots \circ \varphi_1$ (local); the modular polynomial $\Phi_p(X,Y)$; the minimal polynomial $\Phi(X)$ of $j_N$ (local) | B-C2(b), B-C11(b); Chapters 11, 12 and 14 |
| $\pi$ | see B-C15(a); the real number; the automorphic local component of $f$ is $\Pi_p$ and $\Pi_f = \bigotimes' \Pi_v$ (Chapter 15), never $\pi_p$, $\pi_f$ | Chapters 3, 6, 14 and 15 |
| $\psi$ | see B-C15(m); additive character in $\epsilon(r,\psi,dx)$ | Chapters 4, 7 and 9 |
| $q$ | see B-C1 | all |
| $r$ | the congruence number $r_E$ and $r_\pi$; the conductor $r$ of a twisting character; the entry of a change of coordinates $(u,r,s,t)$; a Weil-Deligne representation $(r,N)$; the rotation $r$ of $D_n$ | Chapters 1, 7, 10, 13, 14 and 20 |
| $\rho$ | representations; never $e^{2\pi i/3}$ | Chapters 4 and 11 |
| $S$ | the set of primes of Theorem thm:quant; $S_f$; a set of places in $G_{\mathbb{Q},S}$; cusp forms $S_k$; the matrix $\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$ (local); symmetric groups $S_n$; the Shimura subgroup $\mathcal{S}_N$ | B-C15(e), (j); Chapters 1 and 12 |
| $\sigma$ | element of order 3 of an inertia image; $\mathrm{diag}(1,-1)$ in Chapter 1; a Galois element; divisor sums $\sigma_{k-1}(n)$ | Chapters 1, 4 and 12 |
| $\Sigma$ | see B-C15(e) | Chapters 14, 17 and 18 |
| $t$ | see B-C15(i); $t \in \mathbb{T}$ | Chapters 7, 14 and 19 |
| $T$, $V$ | see B-C15(j) | Chapters 11 to 13 |
| $\tau$ | inertia generator of order 4; the Gauss sum $\tau(\chi)$; Ramanujan's $\tau(n)$; a point $\tau \in \mathcal{H}$ | Chapters 2, 4, 9, 11 and 12 |
| $\theta$ | tame characters $\theta_m$; $\theta_d = (1+\sqrt{d})/2$; $\theta = \rho\vert_{I_q}$ (local) | Chapters 6, 7 and 19 |
| $u$ | unit part of $q_E$; the matrix $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$; the first entry of a change of coordinates $(u,r,s,t)$ | B-C14; Chapters 1 and 10 |
| $w$ | the matrix $\left(\begin{smallmatrix}0 & 1\\ 1 & 0\end{smallmatrix}\right)$; $w_N$; $w_\ell(E,q)$, $w_3(E)$; root number $w_E$; $w = v_\ell(n')$ in Chapter 16 (local) | Chapters 1, 10, 12, 16 and 19 |
| $\Xi$ | the map $\mathrm{Sym}^2 V \otimes \det^{-1} \to \mathrm{ad}^0 V$ | Chapter 4 |
| $\zeta$ | roots of unity $\zeta_n$; the Riemann zeta function $\zeta(s)$, $\zeta^{(N)}(s)$; Dedekind zeta functions $\zeta_K(s)$ | Chapters 2 and 19 |

## B.11 Labels, typography and references

**Convention B.18** (B-C16: labels, names and typography)**.** Curve labels are Cremona's, as printed by the paper: `11a1` (a curve), `121c` (an isogeny class); the label `99A1` of Agashe, Ribet and Stein [3] and of Cremona's book [35] is Cremona's `99a1`; LMFDB labels are written `11.a2`. Mod $\ell$ image labels are Sutherland's [191] (`3Ns`, `3Cs.1.1`, `3B.1.1`, ...; the paper's symbol 3Ns prints

3Ns), always with the intrinsic description; the groups are $N_s(\mathbb{F}_3)$, $N_{ns}(\mathbb{F}_3)$ in their standard models. "Optimal" means $X_0(N)$-optimal. Double names take plain hyphens (Bloch-Kato, Diamond-Flach-Guo, Fontaine-Perrin-Riou, Atkin-Lehner); no dash ligatures are used anywhere.

*In the chapters:* Chapter 9 gives the dictionary between Cremona's and the LMFDB's labels for the curves of the book, and records that curve number 1 of each class in Cremona's tables is the optimal curve except in the class `990h`, whose optimal curve is `990h3`; Chapter 11 gives the dictionary of Sutherland's labels at $\ell = 3$.

## B.12 Notes and references

The conventions follow the sources the paper relies on wherever those agree, and say so where they do not. For the Frobenius: the local Artin map of Chapters 6 and 7 sends a uniformiser to an arithmetic Frobenius, while Deligne [41] uses the opposite normalisation and writes local $L$-factors with the geometric Frobenius (see Serre [174] and Neukirch [142] for local class field theory); Tate [194] normalises the Weil group as in B-C2(f); Diamond, Flach and Guo [49, 1.1.1, 1.7.2] use the geometric Frobenius in $L$-factors, under the name $\mathrm{Frob}_p$. For duals: Bloch and Kato [14] and Fontaine and Perrin-Riou [69] write $V^*(1)$ for the book's $V^D$, with $V^*$ the linear dual, while the paper's $^*$ in Lemma lem:dfg21 is a Pontryagin dual; Fontaine [67] is the reference for the period rings, and the sign of the Hodge-Tate weights of each source is checked where it is cited (Chapter 18). The Weil pairing is Silverman's [185, III.8]; Kodaira symbols, Néron models and component groups follow Silverman [184, IV.9] and Tate's algorithm [193]. The slash operator and the Petersson product are compared with Diamond and Shurman's [52] in Chapter 12; the normalisations of the adjoint and symmetric square $L$-functions follow Gelbart and Jacquet [73], Shimura [182] and Rankin [155], as compared in Chapter 19; the formula of B-C9 is that of Zagier [209] and Watkins [204]. The PARI functions and their outputs are those of the PARI/GP manual [149]; the curve data are Cremona's [35, 36], the image labels Sutherland's [191], and the LMFDB labels those of [198]. Theorem and section numbers of these sources quoted here are the ones given in the chapters that use them, which say where they were checked.

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